Annotations for Ravi Vakil's The Rising Sea: Foundations of Algebraic Geometry

Exercises Solutions and Commentary

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Date: February, 2025

Version: July 25, 2026

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Github: *https://github.com/Infinity547/Annotation-of-Ravi-Vakil-s-FOAG-The-Rising-Sea*

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Chapter 1 Preface

These are the notes and exercise solutions I've written while reading Ravi Vakil's FOAG (I will keep updating). For easier reading, I've also transcribed the content that was originally included in FOAG. I didn't do the double-starred exercises, and I didn't read the double-starred sections, but I've completed almost all the rest. My level is quite basic, so there are bound to be errors in the exercise solutions. If you spot any issues, feel free to provide feedback, you can reach me at this email address: wusiqi2001@outlook.com.

I will update exercise solutions on this website: https://github.com/Infinity547/Annotation-of-Ravi-Vakil-s-FOAG-The-Rising-Sea.

Part I

Preliminaries

Chapter 2 Category Theory

Example 2.1 (Product). For example, we will define the notion of **product** of schemes. As a motivation, we revisit the notion of product in a situation we know well: the category of sets. One way to define the product of sets U and V is as the set of ordered pairs $\{(u, v) : u \in U, v \in V\}$. But someone from a different mathematical culture might resonably define it as the set of symbols $\{{}^u_v : u \in U, v \in V\}$. These notions are "obviously the same". Better: there is "an obvious bijection between the two".

This can be made precise by giving a better definition of product, in terms of a **universal property**.

Definition 2.0.1 (Product)

Given two sets M and N, a ***product*** *is a set P, along with maps $\mu : P \to M$ and $\nu : P \to N$, such that for any set P' with maps $\nu‘ : P' \to M$ and $\nu' : P' \to N$, these maps must factor* ***uniquely*** *through P:*

$$\begin{tikzcd}
P' \arrow[dr, dashed, "\exists!"] \arrow[drr, bend left, "\nu'"] \arrow[ddr, bend right, "\mu'"'] & & \\
& P \arrow[r, "\nu"'] \arrow[d, "\mu"] & N \\
& M &
\end{tikzcd}$$

Thus a ***product*** *is a diagram:*

$$\begin{tikzcd}
P \arrow[r, "\nu"] \arrow[d, "\mu"'] & N \\
M &
\end{tikzcd}$$

and not just a set P, although the maps μ and ν are often left implicit.

This definition agrees with the traditional definition, with one twist: there isn't just a single product, but any two products come with a **unique** isomorphism between them.

2.1 Categories and functors

2.1.1 Categories

Definition 2.1.1 (Category)

A ***category*** *consists of a collection of* ***objects****, and for each pair of objects, a set of* ***morphism*** *(or arrows) between them. Morphsims are often informally called* ***maps****. The collection of objects of a category $\mathscr{C}$ is often denoted $\mathrm{obj}(\mathscr{C})$, but we will usually denote the collection also by $\mathscr{C}$. If $A, B \in \mathscr{C}$, then the set of morphisms from A to B is denoted by $\mathrm{Mor}(A, B)$. A morphism is often written $f : A \to B$, and A is said to be the* ***source*** *of f, and B the* ***target*** *of f.*

Morphisms compose as expected: there is a composition $\mathrm{Mor}(B, C) \times \mathrm{Mor}(A, B) \to \mathrm{Mor}(A, C)$, and if $f \in \mathrm{Mor}(A, B)$ and $g \in \mathrm{Mor}(B, C)$, then their composition is demoted $g \circ f$.

(1) *(**Composition is associative**) If $f \in \mathrm{Mor}(A, B)$, $g \in \mathrm{Mor}(B, C)$, and $h \in \mathrm{Mor}(C, D)$, then $h \circ (g \circ f) = (h \circ g) \circ f$.*

(2) *(**Identity morphism**) For each obeject $A \in \mathscr{C}$, there is always an* ***identity morphism*** *$\mathrm{id}_A : A \to A$, for any morphism $f : A \to B$ and $g : B \to C$, $\mathrm{id}_B \circ f = f$ and $g \circ \mathrm{id}_B = g$.*

Definition 2.1.2 (Isomorphism)

We called morphism f (or g) is an isomorphism, if $f : A \to B$ such that there exists some — necessarily unique — morphism $g : B \to A$, where $f \circ g$ and $g \circ f$ are the identity on B and A respectively.

Example 2.2 (Category of sets). The prototypical example to keep in mind is the category of sets, denoted $\mathbf{Sets}$. The objects are sets, and the morphisms are maps of sets.

Example 2.3 (Category of vector spaces). The category of vector spaces over a given field k, denoted $\mathbf{Vec}_k$. The objects are k-vector spaces, and the morphisms are linear transformations.

✍ **Exercise 2.1** A category in which each morphism is an isomorphism is called a **groupoid**.

(a) A perverse definition of a **group** is: a groupoid with one object.

(b) Describe a groupoid that is not a group.

Solution

(a) Look group G as a category with one obeject G and its morphism is the element of group. Since each element belong to a group is invertible, category G is a groupoid. Then we defined a group.

(b) A groupoid with two distinct object is not a group.

✍ **Exercise 2.2**

(1) If A is an object in category $\mathscr{C}$, show that the invertible elements of $\mathrm{Mor}(A, A)$ form a group, called the **automorphism group of** A, denoted $\mathrm{Aut}(A)$.

(2) What are the automorphism groups of the objects in Example 2.2 and Example 2.3 ?

(3) Show that two isomorphic objects have isomorphic automorphism groups. (Topological background: if X is a topological space, then the fundamental groupoid is the category where the objects are points of X, and the morphisms $x \to y$ are paths from x to y, up to homotopy. Then the automorphism group of x_0 is the pointed fundamental group $\pi_1(X, x_0)$. In the case where X is connected, and $\pi_1(X)$ is not abelian, this illustrates the fact that for a connected groupoid, the automorphism groups of the obejects are all isomorphic, but not canonically isomorphic.)

Proof

(1) $\mathrm{Aut}(A) = \{f \in \mathrm{Mor}(A, A) : f \text{ is invertible}\}$. Since each element in $\mathrm{Aut}(A)$ is invertible, and its identity is id_A, $\mathrm{Aut}(A)$ is a group.

(2) Let $A \in \mathbf{Sets}$, $\mathrm{Aut}(A) = \{f \in \mathrm{Mor}(A, A) : f \text{ is bijection}\}$. Let $V \in \mathbf{Vec}$, $\mathrm{Aut}(V) = \mathrm{GL}(V)$.

(3) Suppose that $X, Y \in \mathrm{obj}(\mathscr{C})$, where X is isomorphic to Y. Then exists morphisms $f : X \to Y$ and $g : Y \to X$ such that $f \circ g = \mathrm{id}_Y$ and $g \circ f = \mathrm{id}_X$. Define $\varphi : \mathrm{Aut}(X) \to \mathrm{Aut}(Y)$ by setting $\varphi(\alpha) = f \circ \alpha \circ g$. It is easy to see that φ is a group homomorphism, and its inverse is $\varphi^{-1}(\beta) = g \circ \beta \circ f$. Hence, $\mathrm{Aut}(X) \cong \mathrm{Aut}(Y)$.

□

Example 2.4 (Category of abelian groups). The abelian groups, along with group homomorphisms, form a category $\mathbf{Ab}$.

Example 2.5 (Category of modules over a ring). If A is a ring, then the A-modules form a category $\mathbf{Mod}_A$. (This category has additional structure; it will be the prototypical example of an **abelian category**.) Taking $A = k$, we obtain Example 2.3 ; taking $A = \mathbb{Z}$, we obtain Example 2.4 .

Example 2.6 (Category of rings). There is a category $\mathbf{Rings}$, where the objects are rings, and the morphisms are maps of rings in the usual sense (maps of sets which respect adition and multiplication, and which send 1 to 1)

Example 2.7 (Category of topological spaces). The topological spaces, along with continuous maps, form a category $\mathbf{Top}$. The isomorphisms are homeomorphisms.

In all of the above examples, the obejects of the categories were in obvious ways sets with additional structure. This needn't be the case, as the next example shows.

Definition 2.1.3 (Partially ordered sets)

*A **partially ordered set**, (or **poset**), is a set S along with a binary relation $\geq$ on S satisfying:*

(i) $x \geq x$ (reflexivity),

(ii) $x \geq y$ and $y \geq z$ imply $x \geq z$ (transitivity),

(iii) if $x \geq y$ and $y \geq x$ then $x = y$ (antisymmetry).

A partially ordered set $(S, \geq)$ can be interpreted as a category whose objects are the elements of S, and with a single morphism from x to y if and only if $x \geq y$ (and no morphism otherwise).

Example 2.8

(1) A trivial example is $(S, \geq)$ where $x \geq y$ if and only if $x = y$.

(2) Another example is

$$\begin{tikzcd} & \bullet \arrow[d] \\ \bullet \arrow[r] & \bullet \end{tikzcd} \tag{2.1}$$

Here there are three objects. The identity morphisms are omitted for convenience, and the two non-identity morphisms are depicted.

(3) A third example is

$$\begin{tikzcd} \bullet \arrow[r] \arrow[d] & \bullet \arrow[d] \\ \bullet \arrow[r] & \bullet \end{tikzcd} \tag{2.2}$$

Here the "obvious" morphisms are again omitted: the identity morphisms, and the morphism from the upper left to the lower right.

(4)

$$\cdots \longrightarrow \bullet \longrightarrow \bullet \longrightarrow \bullet \tag{2.3}$$

depicts a partially ordered set, where again, only the "generating morphisms" are depicted.

Example 2.9 (The category of subsets of a set, and the category of open subsets of a topological space.) If X is a set, then the subsets form a partially ordered set, where arrows are given by inclusion. Informally, if $U \subseteq V$, then we have exactly one morphism $U \to V$ in the category (and otherwise none).

Similarly, if X is a topological space, then the open sets form a partially ordered set, where the maps are given by inclusions.

Definition 2.1.4 (Subcategory)

*A **subcategory** $\mathscr{A}$ of a category $\mathscr{B}$ has as its objects some of the objects of $\mathscr{B}$, and some of the morphisms of $\mathscr{B}$, such that the objects of $\mathscr{A}$ include the sources and targets of the morphisms of $\mathscr{A}$, and are preserved by composition.*

Remark 2.1 Also we have an obvious "inclusion" $i : \mathscr{A} \to \mathscr{B}$, which will soon be an example of a functor.

Example 2.10 (2.1) is in an obvious way a subcategory of (2.2).

2.1.2 Functors

Definition 2.1.5 (Covariant functor)

*A **covariant functor** F from a category $\mathscr{A}$ to a category $\mathscr{B}$, denoted $F : \mathscr{A} \to \mathscr{B}$, is the following data.*

(1) *It is a map of obejects $F : \mathrm{obj}(\mathscr{A}) \to \mathrm{obj}(\mathscr{B})$,*

(2) *For each $A_1, A_2 \in \mathscr{A}$ and morphism $m \in \mathrm{Mor}_{\mathscr{A}}(A_1, A_2)$, $F(m) \in \mathrm{Mor}_{\mathscr{B}}(F(A_1), F(A_2))$,*

(3) *F preserves identity morphisms: for $A \in \mathscr{A}$, $F(\mathrm{id}_A) = \mathrm{id}_{F(A)}$,*

(4) *F preserves composition: $F(m_2 \circ m_1) = F(m_2) \circ F(m_1)$.*

Example 2.11 (Identity functors). A trivial example is the **identity functors** $\mathrm{id} : \mathscr{A} \to \mathscr{A}$.

Example 2.12 (Forgetful functors). Consider the functor from the category of vector space (over a field k) $\mathbf{Vec}_k$ to $\mathbf{Sets}$, that associates to each vector space its underlying set. The functor sends a linear transformation to its underlying map of sets. This is an example of a **forgetful functor** is $\mathbf{Mod}_A \to \mathbf{Ab}$ from A-modules to abelian groups, remembering only the abelian group structure of the A-module.

Example 2.13 (Topological examples). Examples of covariant functors include the fundamental group functor π_1, which sends a topological space X with choice of a point $x_0 \in X$ to a group $\pi_1(X, x_0)$, and the i-th homology functor $\mathbf{Top} \to \mathbf{Ab}$, which sends a topological space X to its i-th homology group $H_i(X, \mathbb{Z})$. The covariance corresponds to the fact that a continuous morphism of pointed topological spaces $\varphi : X \to Y$ with $\varphi(x_0) = y_0$ induces a map of fundamental groups $\pi_1(X, x_0) \to \pi_1(Y, y_0)$, and similarly for homology groups.

Example 2.14 (Hom functor). Suppose A is an object in a category $\mathscr{C}$. Then there is a functor $h^A : \mathscr{C} \to \mathbf{Sets}$ sending $B \in \mathscr{C}$ to $\mathrm{Mor}(A, B)$, and sending $f : B_1 \to B_2$ to $\mathrm{Mor}(A, B_1) \to \mathrm{Mor}(A, B_2)$ described by

$$[g : A \to B_1] \mapsto [f \circ g : A \to B_1 \to B_2].$$

Definition 2.1.6 (Composition)

*If $F : \mathscr{A} \to \mathscr{B}$ and $G : \mathscr{B} \to \mathscr{C}$ are covariant functors, then we define a functor $G \circ F : \mathscr{A} \to \mathscr{C}$ (the **composition** of G and F) in the obvious way.*

Definition 2.1.7 (Faithful, full, fully faithful and full subcategory)

*A covariant functor $F : \mathscr{A} \to \mathscr{B}$ is **faithful** if for all $A, A' \in \mathscr{A}$, the map $\mathrm{Mor}_{\mathscr{A}}(A, A') \to \mathrm{Mor}_{\mathscr{B}}(F(A), F(A'))$ is injective, and **full** if it is surjective. A functor that is full and faithful is **fully faithful**.*

*A subcategory $i : \mathscr{A} \to \mathscr{B}$ is a **full subcategory** if i is full. Thus a subcategory $\mathscr{A}'$ of $\mathscr{A}$ is full if and only if for all $A, B \in \mathrm{obj}(\mathscr{A}')$, $\mathrm{Mor}_{\mathscr{A}'}(A, B) = \mathrm{Mor}_{\mathscr{A}}(A, B)$.*

Remark 2.2

(1) For various philosophical reasons, the notion of "full" functor on its own is unimportant; "fully faithful" is the useful notion.

(2) Inclusions are always faithful, so there is no need for the phrase "faithful subcategory".

Example 2.15

(1) The forgetful functor $\mathbf{Vec}_k \to \mathbf{Sets}$ is faithful, but not full.

(2) If A is a ring, the category of f.g. A-modules is a full subcategory of the category $\mathbf{Mod}_A$ of A-modules.

Definition 2.1.8 (Contravariant functor)

*A **contravariant functor** is defined in the same way as a covariant functor, except the arrows switch directions: in the above langrage (Definition 2.1.5), $F(A_1 \to A_2)$ is now an arrow from $F(A_2)$ to $F(A_1)$, i.e., $F(A_2) \to F(A_1)$.*

Remark 2.3 Thus $F(m_2 \circ m_1) = F(m_1) \circ F(m_2)$, not $F(m_2) \circ F(m_1)$

Sometimes people describe a contravariant functor $\mathscr{C} \to \mathscr{D}$ as a covariant functor $\mathscr{C}^{opp} \to \mathscr{D}$, where $\mathscr{C}^{opp}$ is said to be the **opposite category** to $\mathscr{C}$.

Example 2.16 (Linear algebra example.) If $\mathbf{Vec}_k$ is the category of k-vector spaces, then taking duals gives a contravariant functor $(\cdot)^\vee : \mathbf{Vec}_k \to \mathbf{Vec}_k$. Indeed, to each linear transformation $f : V \to W$, we have a dual transformation $f^\vee : W^\vee \to V^\vee$, and $(f \circ g)^\vee = g^\vee \circ f^\vee$.

Example 2.17 (Topological example.) The i-th cohomology functor $H^i(\cdot, \mathbb{Z}) : \mathbf{Top} \to \mathbf{Ab}$ is a contravariant functor.

Example 2.18 There is a contravariant functor $\mathbf{Top} \to \mathbf{Rings}$ taking a topological space X to the ring of real-valued continuous functions on X. A morphism of topological spaces $X \to Y$ (a continuous map) induces the pullback map from functions on Y to functions on X.

Example 2.19 (The functor of points.) Suppose A is an object of a category $\mathscr{C}$. Then there is a cotravariant functor $h_A : \mathscr{C} \to \mathbf{Sets}$ sending $B \in \mathscr{C}$ to $\mathrm{Mor}(B, A)$, and sending the morphism $f : B_1 \to B_2$ to the morphism $\mathrm{Mor}(B_2, A) \to \mathrm{Mor}(B_1, A)$ via

$$[g : B_2 \to A] \mapsto [g \circ f : B_1 \to B_2 \to A].$$

This functor might reasonably be called the **functor of maps** (to A), but is actually known as the **functor of points**.

Definition 2.1.9 (Natural transformation)

*Suppose F and G are two covariant functors from $\mathscr{A}$ to $\mathscr{B}$. A **natural transformation of covariant functors** $F \to G$ is the data of a morphism $m_A : F(A) \to G(A)$ for each $A \in \mathscr{A}$ such that for each $f : A \to A'$ in $\mathscr{A}$, the diagram*

$$\begin{tikzcd} F(A) \arrow[r, "F(f)"] \arrow[d, "m_A"'] & F(A') \arrow[d, "m_{A'}"] \\ G(A) \arrow[r, "G(f)"'] & G(A') \end{tikzcd}$$

*commutes. A **natural isomorphism** of functors is a natural transformation such that each m_A is an isomorphism. (We make analogous definition when F and G are both contravariant.)*

Definition 2.1.10 (Equivalence of categories)

*The data of functors $F : \mathscr{A} \to \mathscr{B}$ and $F' : \mathscr{B} \to \mathscr{A}$ such that $F \circ F'$ is naturally isomorphic to the identity functor $\mathrm{id}_{\mathscr{B}}$ on $\mathscr{B}$ and $F' \circ F$ is naturally isomorphic to $\mathrm{id}_{\mathscr{A}}$ is said to be an **equivalence of categories**.*

Remark 2.4 The right notion of when two categories are "essentially the same" is not isomorphism (a functor giving bijections of objects and morphisms) but equivalence.

Two example might make this strangeconcept more comprehensible. The double dual of a finite-dimensional vector space V is not V, but we learn early to say that it is canonically isomorphic to V. We can make that precise as follows. Let $\mathbf{f.d.Vec}_k$ be the category of finite-dimensional vector spaces over k. Note that this category contains oodles of vector spaces of each dimension.

Exercise 2.3 Let $(\cdot)^{\vee\vee} : \mathbf{f.d.Vec}_k \to \mathbf{f.d.Vec}_k$ be the double dual functior from the category of finite-dimensional vector spaces over k to itself. Show that $(\cdot)^{\vee\vee}$ is natually isomorphic to $\mathrm{id}_{\mathbf{f.d.Vec}_k}$. (Without the finite-dimensionality hypothesis, we only get a natural transformation of functors from id to $(\cdot)^{\vee\vee}$.)

Proof Since $V \cong (V)^{\vee\vee}$, then there is an isomorphism $m_V : V \to (V)^{\vee\vee}$. By the definition of double dual vector space, the diagram

$$\begin{tikzcd}
V \arrow[r, "\mathrm{id}(f)=f"] \arrow[d, "m_V"'] & W \arrow[d, "m_W"] \\
(V)^{\vee\vee} \arrow[r, "(f)^{\vee\vee}"'] & (W)^{\vee\vee}
\end{tikzcd}$$

commutes. This implies that $(\cdot)^{\vee\vee}$ is natually isomorphic to $\mathrm{id}_{\mathbf{f.d.Vec}_k}$. $\square$

Example 2.20 Let $\mathscr{V}$ be the category whose objects are the k-vector spaces k^n for each $n \geq 0$ (there is one vector space for each n), and whose morphsims are linear transformations. The objects of $\mathscr{V}$ can be thought of as vector spaces with bases, and the morphisms as matrices. There is an obvious functor $\mathscr{V} \to \mathbf{f.d.Vec}_k$, as each k^n is a finite-dimensional vector space.

By linear algebra, $\mathscr{V} \to \mathbf{f.d.Vec}_k$ gives an equivalence of categories.

2.2 Universal properties determine an object up to unique isomorphsim

Products were defined by a universal property.

2.2.1 Product and coproduct

Definition 2.2.1 (Product)

Let $\mathscr{C}$ be a category, and let $\{A_i\}_{i\in I}$ be a family of objects in $\mathscr{C}$ indexed by a set I. A ***product*** *is an ordered pair $(C, \{p_i : C \to A_i\}_{i\in I})$, consisting of an object C and a family $\{p_i : C \to A_i\}_{i\in I}$ of* ***projections****, that is a solution to the following universal mapping problem: for every object X equipped with morphisms $f_i : X \to A_i$, there exists a unique morphism $\theta : X \to C$ making the diagram commute for each i.*

$$\begin{tikzcd}
X \arrow[dr, dashed, "\exists!\theta"] \arrow[drr, "f_i"] \arrow[ddr, "f_j"'] & & \\
& C \arrow[r, "p_i"] \arrow[d, "p_j"] & A_i \\
& A_j &
\end{tikzcd}$$

Proposition 2.2.1

If $\{A_i\}$ is a family of left R-modules, then the direct product $C = \prod_{i\in I} A_i$ is their product in ${}_R\mathbf{Mod}$.

Proof The statement of the proposition is incomplete, for a product requires projections. For each $j \in I$, define $p_j : C \to A_j$ by $p_j : (a_i) \mapsto a_j \in A_j$. Note that each p_j is an R-map.

Now let X be a module and, for each $i \in I$, let $f_i : X \to A_i$ be an R-map. Define $\theta : X \to C$ by $\theta : x \mapsto (f_i(x))$. First, the diagram commutes: if $x \in X$, then $p_i\theta(x) = f_i(x)$. Second, θ is unique. If

$\psi : X \to C$ makes the diagram commute, then $p_i\psi(x) = f_i(x)$ for all i, that is, for each i, the i-th coordinate of $\psi(x)$ is $f_i(x)$, which is also the i-th coordinate of $\theta(x)$. Hence, $\psi = \theta$. □

Coproducts were defined by a universal property.

Definition 2.2.2 (Coproduct)

Let $\mathscr{C}$ be a category, and let $\{A_i\}_{i\in I}$ be a family of obejects in $\mathscr{C}$ indexed by a set I. A ***coproduct*** *is an ordered pair $(C, \{\alpha_i : A_i \to C\}_{i\in I})$, consisting of an object C and a family $\{\alpha_i : A_i \to C\}_{i\in I}$ of morphisms, called* ***injections****, that is a solution to the following universal mapping problem: for every object X equipped with morphisms $\{f_i : A_i \to X\}_{i\in I}$, there exists a unique morphism $\theta : C \to X$ making the diagram commute for each i.*

$$\begin{tikzcd}
 & A_i \arrow[d, "\alpha_i"'] \arrow[dr, "f_i"] & \\
A_j \arrow[r, "\alpha_j"] \arrow[rr, bend right, "f_j"'] & C \arrow[r, dashed, "\exists!\theta"] & X
\end{tikzcd}$$

Coproduct is denoted by $\coprod$.

Proposition 2.2.2

If $\{A_i\}_{i\in I}$ is a family of left R-modules, then the direct sum $\bigoplus_{i\in I} A_i$ is their coproduct in ${}_R\mathbf{Mod}$.

Proof The statement of the proposition is incomplete, for a coproduct requires injections α_i. Write $C = \bigoplus_{i\in I} A_i$, and define $\alpha_i : A_i \to C$ by $a_i \mapsto (0, \cdots, 0, a_i, 0, \cdots)$. Note that each α_i is an R-map.

Let X be a module and for each $i \in I$, let $f_i : A_i \to X$ be an R-map. If $(a_i) \in C = \bigoplus_i A_i$, then only finitely many a_i are nonzero, and $(a_i) = \sum_i \alpha_i a_i$. Define $\theta : C \to X$ by $\theta : (a_i) \mapsto \sum_i f_i a_i$. The coproduct diagram does commute: if $a_i \in A_i$, then $\theta\alpha_i a_i = f_i a_i$. We now prove that θ is unique. If $\psi : C \to X$ makes the coproduct diagram commute, then

$$\psi((a_i)) = \psi\left(\sum_i \alpha_i a_i\right) = \sum_i \psi\alpha_i(a_i) = \sum_i f_i(a_i) = \theta((a_i)).$$

Therefore, $\psi = \theta$. □

Here are some simple but useful concepts.

Definition 2.2.3 (Initial, final, and zero objects)

An object A of a category $\mathscr{C}$ is called an ***initial object****, if for every object X in $\mathscr{C}$, there exists a unique morphism $A \to X$.*

An object Ω of a category $\mathscr{C}$ is called a ***final object****, if for every object X in $\mathscr{C}$, there exists a unique morphism $X \to \Omega$.*

An object is called a ***zero object*** *if it is both an initial object and a final object.*

✍ **Exercise 2.4** Show that any two initial objects are uniquely isomorphic . Show that any two final objects are uniquely isomorphic.

Proof Let A, A' be initial objects, then we have arrows $A \to A'$ and $A' \to A$. Note that $\mathrm{id}_A : A \to A$, the arrow $A \to A' \to A$ must be id_A, by the definition of initial object. Similarly, $A' \to A \to A'$ must be $\mathrm{id}_{A'}$. Hence, A is isomorphic to A'.

Similarly, we can prove that any two final objects are isomorphic. □

Remark 2.5 In other words, if an initial object exists, it is unique up to unique isomorphism, and similarly for finial objects.

Remark 2.6 Convention: we often say “the”, not “a”, for anything defined up to unique isomorphism.

✍ **Exercise 2.5** What are the initial and final objects in **Sets**, **Rings**, and **Top** (if exist)? How about in Example 2.9 ?

Solution *The initial object in* **Sets** *and* **Top** *is* $\varnothing$*. The final object in* **Sets** *and* **Top** *is the point set* $\{*\}$*. The initial object in* **Rings** *is* $\mathbb{Z}$*, and the final object is* 0*-ring.*

Consider the category of subset of a set X*, the object of this category is* $\mathscr{P}(X)$*, hence, the initial object is* $\varnothing$ *and the final object is* X*.*

2.2.2 Localization

Another important example of a definition by universal property is the notion of **localization** of a ring.

Definition 2.2.4 (Localization of ring)

A ***multiplicative subset*** S *of a ring* A *is a subset closed under multiplication containing* 1*. We define a ring* $S^{-1}A$ *as follow:*

$$S^{-1}A = \{a/s : a \in A, s \in S\}/\sim,$$

where $\sim$*:* $a_1/s_1 = a_2/s_2$ *if and only if for some* $s \in S$*,* $s(s_2a_1 - s_1a_2) = 0$*. We define* $(a_1/s_1) + (a_2/s_2) = (s_2a_1 + s_1a_2)/(s_1s_2)$*, and* $(a_1/s_1) \times (a_2/s_2) = (a_1a_2)/(s_1s_2)$*.*

We have a canonical ring map

$$A \to S^{-1}A$$

given by $a \mapsto a/1$.

Remark 2.7 If $0 \in S$, $S^{-1}A$ is the 0-ring. (Since, each element is equivalence by the definiton of $\sim$ if $0 \in S$.)

There are two particularly important flavors of multiplicative subsets.

Example 2.21 The first is $S = \{1, f, f^2, \cdots\}$, where $f \in A$. This localization is denoted A_f. Note that $A[t]/(tf-1) = A[\frac{1}{f}]$, we have an isomorphism $A_f \cong A[t]/(tf-1)$.

Example 2.22 The second is $A - \mathfrak{p}$, where $\mathfrak{p}$ is a prime ideal. This localization $S^{-1}A$ is denoted $A_{\mathfrak{p}}$.

Remark 2.8 Sometimes localization is first introduced in the special case where A is an integral domain and $0 \notin S$. In that case, $A \hookrightarrow S^{-1}A$, but this isn't always ture, as shown by the following exercise.

✍ **Exercise 2.6** Show that $A \to S^{-1}A$ is injective if and only if S contains no zero-divisors

Proof Let $a \in A$ and $a/1 = 0$, then $s(a-0) = sa = 0$ for some $s \in S$. Hence, $A \to S^{-1}A$ is an injective iff S contains no zero-divisor. □

If A is an integral domain and $S = A \setminus \{0\}$, then $S^{-1}A$ is called the **fraction field of** A, which we denote $K(A)$. The Exercise 2.6 shows that A is a subring of its fraction field $K(A)$.

We now return to the case where A is a general commutative ring.

Proposition 2.2.3 (Universal property of localization)

Let $g : A \to B$ *be a ring homomorphism such that* $g(s)$ *is a unit in* B *for all* $s \in S$*. Then there exists a unique ring homomorphism* $h : S^{-1}A \to B$ *such that* $g = h \circ f$*, that is, the following diagram is* *commute:*

$$\begin{tikzcd} A \arrow[r, "g"] \arrow[d, "f"'] & B \\ S^{-1}A, \arrow[ur, dashed, "\exists! h"'] & \end{tikzcd}$$

where $f : A \to S^{-1}A$ defined by $a \mapsto a/1$

Proof **Uniquness.** If h satisfies the conditions, then $h(a/1) = hf(a) = g(a)$ for all $a \in A$, hence, if $s \in S$,

$$h(1/s) = h((s/1)^{-1}) = h(s/1)^{-1} = g(s)^{-1}$$

and therefore $h(a/s) = h(a/1)h(1/s) = g(a)g(s)^{-1}$, so that h is uniquely determined by g.

Existence. Let $h(a/s) = g(a)g(s)^{-1}$. We shall to show that h is well-defined. If $a_1/s_1 = a_2/s_2$, then $s(a_1s_2 - a_2s_1) = 0$ for all $s \in S$, hence, $g(s)(g(a_1)g(s_2) - g(a_2)g(s_2)) = 0$. Since $g(s)$, $g(s_1)$ and $g(s_2)$ are units, we have $g(a_1)g(s_1)^{-1} = g(a_2)g(s_2)^{-1}$, that is, $h(a_1/s_1) = h(a_2/s_2)$. This implies that h is well-defined. By the definition of h, clearly, h is a ring homomorphism, which proved existence. □

Corollary 2.2.1

$S^{-1}A$ is the initial among A-algebras B where every element of S is sent to an invertible element in B.

Proof In fact, the data of "an A-algebra B" and "a ring map $A \to B$" are the same. Apply Proposition 2.2.3, we done immediately. □

Corollary 2.2.2

Any map $A \to B$ where every element of S is sent to an invertible element must factor uniquely through $A \to S^{-1}A$.

Proof Proposition 2.2.3, obviously! □

Corollary 2.2.3

A ring map out of $S^{-1}A$ is the same thing as a ring map from A that sends every element of S to an invertible element.

Proof Proposition 2.2.3, obviously! □

Corollary 2.2.4

An $S^{-1}A$-module is the same thing as an A-module for which $s \times \cdot : M \to M$ is an A-module isomorphism for all $s \in S$.

Proof $s \times \cdot : M \to M$ is an A-module isomorphism for all $s \in S$ iff each $s \in S$ is a unit iff M is $S^{-1}A$ module. □

In fact, it is cleaner to define $A \to S^{-1}A$ by the universal property to check various properties $S^{-1}A$ has. Now, let's define localizations of modules by universal property.

Definition 2.2.5 (Localization of module)

Suppose M is an A-module. We define the A-module map $\varphi : M \to S^{-1}M$ as being initial among A-module maps $M \to N$ such that elements of S are invertible in N ($s \times \cdot : N \to N$ is an isomorphism *for all $s \in S$). More precisely, any such map $\alpha : M \to N$ factors uniquely through φ:*

$$\begin{tikzcd} M \arrow[r, "\alpha"] \arrow[d, "\varphi"'] & N \\ S^{-1}M \arrow[ur, dashed, "\exists!"'] & \end{tikzcd}$$

Remark 2.9 Translation: $M \to S^{-1}M$ is universal initial among A-module maps from M to modules that are actually $S^{-1}A$-modules.

Note

(i) This determines $\varphi : M \to S^{-1}M$ up to unique isomorphism.

(ii) We are defining not only $S^{-1}M$, but also the map φ at the same time.

(iii) Essentially by definition the A-module structure on $S^{-1}M$ extends to an $S^{-1}A$-module structure.

Exercise 2.7 Show that $\varphi : M \to S^{-1}M$ exists, by constructing something satisfying the universal property.

Proof Let $S^{-1}M = \{m/s : m \in M, s \in S\}/\sim$, where $\sim$: $m_1/s_1 = m_2/s_2$ iff for some $s \in S, s(s_2m_1 - s_1m_2) = 0$. Define the additive structure by $(m_1/s_1) + (m_2/s_2) = (s_2m_1 + s_1m_2)/(s_1s_2)$, and the $S^{-1}A$-module structure is given by $(a_1/s_1) \cdot (m_2/s_2) = (a_1m_2)/(s_1s_2)$.

Let $\varphi : M \to S^{-1}M$ defined by $\varphi(m) = m/1$. Let N be any A-module with $s \times \cdot : N \to N$ is an isomorphism for all $s \in S$, and $f : M \to N$ be an A-module morphism, define $h : S^{-1}M \to N$ by setting $h(m/s) = f(m)f^{-1}(s)$. By the same way as Proposition 2.2.3, h unique exists, then we done. □

Exercise 2.8

(a) Show that localization commutes with finite products, or equivalently, with finite direct sums. In other words, if $M_1, \cdots, M_n$ are A-modules, describe an isomorphism of A-modules, and of $S^{-1}A$-modules $S^{-1}(M_1 \times \cdots \times M_n) \to S^{-1}M_1 \times \cdots \times S^{-1}M_n$.

(b) Show that localization commutes with arbitrary direct sums.

(c) Show that "localization does not necessarily commute with infinite products": the obvious map $S^{-1}(\prod_i M_i) \to \prod_i S^{-1}M_i$ induced by the universal property of localization is not always an isomorphism.

Proof

(a) Let $\varphi : S^{-1}(M_1\times\cdots\times M_n) \to S^{-1}M_1\times\cdots\times S^{-1}M_n$ by setting $\varphi\left(\frac{(m_1, \cdots, m_n)}{s}\right) = \left(\frac{m_1}{s}, \cdots, \frac{m_n}{s}\right)$.

The inverse of φ, by setting $\psi\left(\frac{m_1}{s_1}, \cdots, \frac{m_n}{s_n}\right) = \frac{\left(m_1\prod_{i\neq 1} s_i, \cdots, m_n\prod_{i\neq n} s_i\right)}{\prod_i s_i}$ (reduction of a fraction if possible). φ and ψ are $S^{-1}A$-module homomorphism, note that $\varphi\circ\psi = \mathrm{id}$ and $\psi\circ\varphi = \mathrm{id}$, $S^{-1}(M_1 \times \cdots \times M_n) \cong S^{-1}M_1 \times \cdots \times S^{-1}M_n$.

(b) Let $M = \bigoplus_i M_i$ and $N = \bigoplus_i S^{-1}M_i$, $f : M \to S^{-1}M$ and $f_i : M_i \to S^{-1}M_i$ be the localization map, $\iota_i : M_i \to M$ be embedding. Then we have a map $f \circ \iota_i : M_i \to S^{-1}M$. Appy the universal property of localization to following diagram

$$\begin{tikzcd} M_i \arrow[r, "f_i"] \arrow[dr, "f\circ\iota_i"'] & S^{-1}M_i \arrow[d, dashed, "\exists! h_i"] \\ & S^{-1}M \end{tikzcd} \tag{2.4}$$

there exists a unique homomorphism $h_i \to S^{-1}M$ such that (2.4) commute. Consider the following diagram

$$\begin{tikzcd} S^{-1}M_i \arrow[r, "h_i"] \arrow[dr, "e_i"'] & S^{-1}M \arrow[d, dashed, "\psi"] \\ & N \end{tikzcd} \tag{2.5}$$

where $e_i : S^{-1}M_i \to N$ be the embedding. We shall to show (2.5) is commute.
M is a coproduct, by Proposition 2.2.2. Consider the diagram

$$\begin{tikzcd} M_i \arrow[r, "\iota_i"] \arrow[dr, "e_i \circ f_i"'] & M \arrow[d, dashed, "\varphi"] \\ & N \end{tikzcd} \tag{2.6}$$

apply the universal property of coproduct, there exists unique homomorphism $\varphi : M \to N$ such that (2.6) is commute. Consider the diagram

$$\begin{tikzcd} M_i \arrow[r, "h_i \circ f_i"] \arrow[dr, "e_i \circ f_i"'] & S^{-1}M \arrow[d, dashed, "\widetilde{\varphi}"] \\ & N \end{tikzcd} \tag{2.7}$$

by the universal property of localization, there exists unique homomorphism $\widetilde{\varphi} : S^{-1}M \to N$ such that (2.7) is commute, that is, $\widetilde{\varphi} \circ h_i \circ f_i = e_i \circ f_i$. Consider the diagram

$$\begin{tikzcd} M_i \arrow[r, "f_i"] \arrow[dr, "f \circ \iota_i"] \arrow[ddr, "e_i \circ f_i"'] & S^{-1}M_i \arrow[d, "h_i"] \arrow[dd, bend left, "e_i"] \\ & S^{-1}M \arrow[d, "\widetilde{\varphi}"] \\ & N \end{tikzcd} \tag{2.8}$$

by the universal property, $e_i = \widetilde{\varphi} \circ h_i$. Let $\psi = \widetilde{\varphi}$, then (2.5) commute.
By above discussion, $\psi : S^{-1}M \to N$ is uniquely defined by $\psi((m_i)/s) = (m_i/s)$. And it is easy to see ψ is an isomorphism.

(c) Consider $\mathbb{Z}^{-1}\mathbb{Q} \times \cdots$, $(1, 1/2, 1/3, \cdots, 1/n, \cdots)$ has no preimage in $\prod_i \mathbb{Z}^{-1}\mathbb{Q}$. This implies that $\mathbb{Z}^{-1}(\prod_i \mathbb{Q})$ not isomorphic to $\prod_i \mathbb{Z}^{-1}\mathbb{Q}$.

□

Remark 2.10 Localization does not always commute with Hom.

2.2.3 Tensor product

Another important example of a universal property construction is the notion of **tensor product** of A-modules.

Definition 2.2.6 (Tensor product of A-modules)

*The **tensor product** is often defined as follows:*

$$\begin{aligned} \otimes_A : \mathrm{obj}(\mathbf{Mod}_A) \times \mathrm{obj}(\mathbf{Mod}_A) &\to \mathrm{obj}(\mathbf{Mod}_A) \\ (M, N) &\mapsto M \otimes_A N \end{aligned}$$

Suppose M, N are A-modules. Then elements of the tensor product $M \otimes_A N$ are finite A-linear combinations of symbols $m \otimes n$ ($m \in M$, $n \in N$), subject to relations:

$$(m_1 + m_2) \otimes n = m_1 \otimes n + m_2 \otimes n,$$
$$m \otimes (n_1 + n_2) = m \otimes n_1 + m \otimes n_2,$$
$$a(m \otimes n) = (am) \otimes n = m \otimes (an),$$

where $a \in A$, $m_1, m_2 \in M$, $n_1, n_2 \in N$.
More formally, $M \otimes_A N$ is the free A-module generated by $M \times N$, quotiented by the submodule generated by

$$(m_1 + m_2, n) - (m_1, n) - (m_2, n),$$
$$(m, n_1 + n_2) - (m, n_1) - (m, n_2),$$
$$a(m, n) - (am, n)$$
$$a(m, n) - (m, an),$$

for $a \in A$, $m, m_1, m_2 \in M$, $n, n_1, n_2 \in N$. The image of (m, n) in this quotient is $m \otimes n$.

Remark 2.11 If A is a field k, we recover the tensor product of vector space.

✍ **Exercise 2.9** Show that $\mathbb{Z}/(10) \otimes_{\mathbb{Z}} \mathbb{Z}/(12) \cong \mathbb{Z}/(2)$.

Proof Note that $\mathbb{Z}/(10) \otimes_{\mathbb{Z}} \mathbb{Z}/(12)$ is generated by $\{1 \otimes_{\mathbb{Z}} 1, 1 \otimes_{\mathbb{Z}} 0, 0 \otimes_{\mathbb{Z}} 1, 0 \otimes_{\mathbb{Z}} 0\}$, since $1 \otimes_{\mathbb{Z}} 0 = 0 \otimes_{\mathbb{Z}} 1 = 0 \otimes 0 = 0$, we have $\mathbb{Z}/(10) \otimes_{\mathbb{Z}} \mathbb{Z}/(12) \cong \mathbb{Z}/(2)$. ☐

Proposition 2.2.4 (Right-exactness of $(\cdot) \otimes_A N$)

$(\cdot) \otimes_A N$ gives a covariant functor $\mathbf{Mod}_A \to \mathbf{Mod}_A$. $(\cdot) \otimes_A N$ is a ***right-exact functor****, i.e., if*

$$M' \longrightarrow M \longrightarrow M'' \longrightarrow 0$$

is an exact sequence of A-modules, then the induced sequence

$$M' \otimes_A N \longrightarrow M \otimes_A N \longrightarrow M'' \otimes_A N \longrightarrow 0$$

is also exact.

Proof

(1) $(\cdot) \otimes_A N$ gives a covariant functior.
$(\cdot) \otimes_A N$ maps object in $\mathbf{Mod}_A$ to $\mathbf{Mod}_A$. Let $f \in \mathrm{Mor}(M, M')$, where $A, B \in \mathbf{Mod}_A$, define $(f) \otimes_A N = f \otimes_A \mathrm{id}_N : M \otimes_A N \to M' \otimes_A N$, then $f \otimes_A N \in \mathrm{Mor}(M \otimes_A N, M' \otimes_A N)$. By the definition of $f \otimes_A N$, $(\cdot) \otimes_A N$ preserves identity morphism and preserves composition.

(2) Say

$$\mathrm{Im}\, M' \xrightarrow{i} M \xrightarrow{p} M'' \longrightarrow 0.$$

is an exact sequence.
We shall to show that the sequence

$$M' \otimes_A N \xrightarrow{i \otimes \mathrm{id}_N} M \otimes_A N \xrightarrow{p \otimes \mathrm{id}_N} M'' \otimes_A N \longrightarrow 0$$

is exact.

(a) $\mathrm{Im}\, i \otimes \mathrm{id}_N = \mathrm{Ker}\, p \otimes \mathrm{id}_N$.
For $\mathrm{Im}\, i \otimes \mathrm{id}_N \subseteq \mathrm{Ker}\, p \otimes \mathrm{id}_N$, it suffices to prove that the composite is 0. Note that

$$(p \otimes \mathrm{id}_N)(i \otimes \mathrm{id}_N) = pi \otimes \mathrm{id}_N = 0 \otimes \mathrm{id}_N = 0,$$

then we done.

Let $E = \operatorname{Im}(i \otimes \mathrm{id}_N)$, then $E \subseteq \operatorname{Ker} p \otimes \mathrm{id}_N$, and therefore $i \otimes \mathrm{id}_N$ induces a map $\widehat{p} : (M \otimes N)/E \to M'' \otimes N$ with

$$\widehat{p} : m \otimes n + E \mapsto pm \otimes n,$$

where $m \in M$ and $n \in N$. Now if $\pi : M \otimes N \to (M \otimes N)/E$ is the natural nap, then

$$\widehat{p}\pi = p \otimes \mathrm{id}_N,$$

for both send $a \otimes b \mapsto pa \otimes b$.

$$\begin{tikzcd}
M \otimes N \arrow[rr, "\pi"] \arrow[dr, "p\otimes \mathrm{id}_N"'] & & (M \otimes N)/E \arrow[dl, "\widehat{p}"] \\
& M'' \otimes N &
\end{tikzcd}$$

Suppose we show that $\widehat{p}$ is an isomorphism. Then

$$\operatorname{Ker}(p \otimes \mathrm{id}_N) = \operatorname{Ker} \widehat{p}\pi = \operatorname{Ker} \pi = E = \operatorname{Im}(i \otimes \mathrm{id}_N),$$

and we done. To see that $\widehat{P}$ is an isomorphism, we construct its inverse $M'' \otimes N \to (M \otimes N)/E$. Define

$$f : M'' \times N \to (M \otimes N)/E$$

as follows. If $m'' \in M''$, there is $m \in M$ such that $p(m) = m''$, since p is surjective, let

$$f : (m'', n) \mapsto m \otimes n + E.$$

Now f is well-defined: if $pm_1 = m''$, then $p(m - m_1) = 0$ and $m - m_1 \in \operatorname{Ker} p = \operatorname{Im} i$. Thus there is $m' \in M'$ with $i(m') = m - m_1$, and hence $(m - m_1) \otimes n = i(m') \otimes n \in \operatorname{Im}(i \otimes \mathrm{id}_N) = E$, that is, $m \otimes n + E = m_1 \otimes n + E$. Clearly, f is A-biadditive, and so the definition of tensor product gives a homomorphism $\widehat{f} : M'' \otimes N \to (M \otimes N)/E$ with $\widehat{f}(m'' \otimes n) = m \otimes n + E$. It is easy to check that $\widehat{f}$ is the inverse of $\widehat{p}$, as desired.

Hence, we have $\operatorname{Im} i \otimes \mathrm{id}_N = \operatorname{Ker} p \otimes \mathrm{id}_N$.

(b) $p \otimes \mathrm{id}_N$ is surjective.

Since, p is surjective, if $\sum m_i'' \otimes n_i \in M'' \otimes N$, then there exist $m_i \in M$ with $pm_i = m_i''$ for all i. Hence, $(p \otimes \mathrm{id}_N)(\sum m_i \otimes n_i) = \sum m_i'' \otimes n_i$, which implies that p is surjective.

□

Remark 2.12 Tensor product is not left-exact: tensor the exact sequence of $\mathbb{Z}$-modules

$$0 \longrightarrow \mathbb{Z} \xrightarrow{\times 2} \mathbb{Z} \longrightarrow \mathbb{Z}/(2) \longrightarrow 0$$

with $\mathbb{Z}/(2)$.

The constructive definiton of $\otimes$ is a weird definition, and really the "wrong" definition.

Definition 2.2.7 (A-biadditive (or A-bilinear))

*Let A be a ring, and $M, N, P \in \mathbf{Mod}_A$, a map $f : M \times N \to P$ is called A-**bilinear** if*

$$\begin{aligned}
f(m_1 + m_2, n) &= f(m_1, n) + f(m_2, n), \\
f(m, n_1 + n_2) &= f(m, n_1) + f(m, n_2), \\
f(am, n) &= f(m, an) = af(m, n).
\end{aligned} \tag{2.9}$$

Any A-bilinear map $M \times N \to P$ factors through the tensor product uniquely: $M \times N \to M \otimes_A N \to P$.

Definition 2.2.8 (Tensor product)

*Given a ring R and modules $M, N \in \mathbf{Mod}_A$, then their **tensor product** is an A-module T and an A-bilinear fuction*

$$t : M \times N \to T,$$

such that given any A-bilinear map $t' : M \times N \to T'$, there is a unique A-linear map $f : T \to T'$ such that $t' = f \circ t$.

$$\begin{tikzcd} M \times N \arrow[rr, "t"] \arrow[dr, "t'"'] & & T \arrow[dl, dashed, "\exists! f"] \\ & T' & \end{tikzcd}$$

Proposition 2.2.5

Tensor product $(T, t : M \times N \to T)$ is unique up to unique isomorphism, i.e., if $(T', t' : M \times N \to T)$ another tensor product, $T \cong T'$.

Proof Since T' is a tensor product, for every A-module G and every A-bilinear map $f : M \times N \to G$, there exists a unique A-linear map $f' : T \to G$ making the following diagram commute.

$$\begin{tikzcd} M \times N \arrow[rr, "t'"] \arrow[dr, "f"'] & & T' \arrow[dl, dashed, "f'"] \\ & G & \end{tikzcd}$$

Setting $G = T$ and $f = t$, there is a homomorphism $g : T' \to T$ with $gt' = t$, setting $G = T'$ and $f = t'$ in the diagram defining T, there is a homomorphsim $\widetilde{g} : T \to T'$ with $\widetilde{g}t = t'$.

Consider the following diagram.

$$\begin{tikzcd} & T \arrow[d, "\widetilde{g}"] \arrow[dd, bend left=60, "\mathrm{id}_T"] \\ M \times N \arrow[ur, "t"] \arrow[r, "t'"] \arrow[dr, "t"'] & T' \arrow[d, "g"] \\ & T \end{tikzcd}$$

Now $g\widetilde{g}$ makes the big triangle with vertices $M \times N$, T, and T commute. But identity id_T also makes this diagram commute. By the definition of tensor product, $g\widetilde{g} = \mathrm{id}_T$. A similar argument shows that $\widetilde{g}g = \mathrm{id}_{T'}$. Hence, $\widetilde{g} : T \to T'$ is an isomorphism. $\square$

Corollary 2.2.5

The construction of Definition 2.2.6 satisfies the universal property of tensor product.

Definition 2.2.9 (Restriction of scalars)

*Let $f : A \to B$ be a homomorphism of rings and let N be a B-module. Then N has an A-module structure defined as follows: if $a \in A$ and $x \in N$, then ax is defined to be $f(a)x$. This A-module is said to be obtained from N by **restriction of scalars**. In particular, f defines in this way an A-module structure on B.*

Definition 2.2.10 (Extension of scalars)

*Let M be an A-module. By Definition 2.2.9, B can be regarded as an A-module, we can form the A-module $B \otimes_A M$. In fact, $B \otimes_A M$ carries a B-module structure such that $b(b' \otimes x)$ for all $b, b' \in B$ and all $x \in M$. The B-module $B \otimes_A M$ is said to be obtained from M by **extension of scalars**.*

Remark 2.13

(1) Extension of scalars describes a functor $(\cdot) \otimes_A M : \mathbf{Mod}_A \to \mathbf{Mod}_B$.

(2) Let $A \to B$ and $A \to C$ be morphisms of rings, then $B \otimes_A C$ has a natural structure of a ring. (Multiplication will be given by $(b_1 \otimes c_1)(b_2 \otimes c_2) = (b_1 b_2) \otimes (c_1 c_2)$.)

Proposition 2.2.6

Let M be an A-module. Then the $S^{-1}A$-modules $S^{-1}M$ and $S^{-1}A \otimes_A M$ are isomorphism; more precisely, there exist a unique isomorphism $f : S^{-1}A \otimes_A M \to S^{-1}M$ for which

$$f((a/s) \otimes m) = am/s$$

for all $a \in A, m \in M, s \in S$

Proof The mapping $S^{-1}A \times M \to S^{-1}M$ defined by

$$(a/s, m) \mapsto am/s$$

is A-bilinear, and therefore by the universal property of the tensor product induces an A-homomorphism

$$f : S^{-1}A \otimes_A M \to S^{-1}M \tag{2.10}$$

satisfying $f((a/s) \otimes m) = am/s$ for all $a \in A$, $m \in M$, $s \in S$.

Clearly, f is surjective, and is A-linear. Also f is uniquely defined by (2.10). We shall to prove f is injective.

Let $\sum_i (a_i/s_i) \otimes m_i$ be any element of $S^{-1}A \otimes M$. If $s = \prod_i s_i \in S$, let $t_i = \prod_{j \neq i} s_j$, we have

$$\sum_i \frac{a_i}{s_i} \otimes m_i = \sum_i \frac{a_i t_i}{s} \otimes m_i = \sum_i \frac{1}{s} \otimes a_i t_i m_i = \frac{1}{s} \otimes \sum_i a_i t_i m_i,$$

so that every element of $S^{-1}A \otimes M$ is of the form $(1/s) \otimes m$. Suppose that $f((1/s) \otimes m) = 0$, thus $m/s = 0$, hence, $tm = 0$ for some $t \in S$, and therefore

$$\frac{1}{s} \otimes m = \frac{t}{st} \otimes m = \frac{1}{st} \otimes tm = \frac{1}{st} \otimes 0 = 0.$$

Hence, f is injective, and therefore an isomorphism. □

Proposition 2.2.7 ($\otimes$ commutes with $\oplus$)

Tensor products commute with arbitrary direct sums: if M and $\{N_i\}$ are all A-modules, there is a A-isomorphism

$$\tau : M \otimes_A (\bigoplus_{i \in I} N_i) \to \bigoplus_{i \in I} (M \otimes_A N_i)$$

with $\tau : m \otimes (n_i) \mapsto (m \otimes n_i)$.

Proof Since the function $f : M \times (\bigoplus_i N_i) \to \bigoplus_i (M \otimes N_i)$ given by $f : (m, (n_i)) \mapsto (m \otimes n_i)$ is A-bilinear, by the universal property of tensor product, there is a A-homomorphism

$$\tau : M \otimes (\bigoplus_i N_i) \to \bigoplus (M \otimes N_i)$$

with $\tau : m \otimes (n_i) \mapsto (m \otimes n_i)$.

To see that τ is an isomorphism, we give its inverse. Denote the injection $N_k \to \bigoplus_i N_i$ by ι_k, where $\iota_k(n_k) \in \bigoplus_i N_i$ has k-th coordinates n_k and all other coordinates 0, so that $\mathrm{id}_M \otimes \iota_k : M \otimes N_k \to M \otimes (\bigoplus_i N_i)$. By Proposition 2.2.2, direct sum is the coproduct in ${}_R\mathbf{Mod}$ gives a homomorphism $\theta : \bigoplus_i (M \otimes N_i) \to M \otimes (\bigoplus_i N_i)$ with $\theta : (m \otimes n_i) \mapsto m \otimes \sum_i \iota_i(n_i)$. It is easy to check that θ is the inverse of τ, and therefore τ is an isomorphism. □

2.2.4 Fibered product and fibered coproduct

Definition 2.2.11 (Fibered product)

*Suppose morphsims $\alpha : X \to Z$ and $\beta : Y \to Z$ (in any category). Then the **fibered product** is an object $X \times_Z Y$ along with morphisms $\mathrm{pr}_X : X \times_Z Y \to X$ and $\mathrm{pr}_Y : X \times_Z Y \to Y$, where the two compositions $\alpha \circ \mathrm{pr}_X = \beta \circ \mathrm{pr}_Y$, such that given any object W with maps to X and Y whose compositions to Z agree (commute), these maps factor through some unique $W \to X \times_Z Y$:*

$$\begin{tikzcd} W \arrow[dr, dashed, "\exists!"] \arrow[drr, bend left] \arrow[ddr, bend right] & & \\ & X \times_Z Y \arrow[r, "\mathrm{pr}_Y"'] \arrow[d, "\mathrm{pr}_X"] & Y \arrow[d, "\beta"] \\ & X \arrow[r, "\alpha"'] & Z \end{tikzcd}$$

Remark 2.14 By the usual universal property argument, if it exists, it is unique up to unique isomorphism.

Definition 2.2.12 (Diagonal morphism)

*If $\pi : X \to Y$ is a morphism, and the fibered product $X \times_Y X$ exists, then determines a **diagonal morphism** $\delta_\pi : X \to X \times_Y X$ (or use notation $\delta_{X/Y}$), which satisfy $\delta \circ \mathrm{pr} = \mathrm{id}_{X \times_Y X}$ and $\mathrm{pr} \circ \delta = \mathrm{id}_X$.*

Note *Depending on your religion, the diagram*

$$\begin{tikzcd} X \times_Z Y \arrow[r, "\mathrm{pr}_Y"] \arrow[d, "\mathrm{pr}_X"'] & Y \arrow[d, "\beta"] \\ X \arrow[r, "\alpha"] & Z \end{tikzcd}$$

*is called a **fibered / pullback / Cartesian diagram / square**.*

The right way to interpret the notion of fibered product is first to think about what it means in the category of sets.

Exercise 2.10 Show that in $\mathbf{Sets}$,

$$X \times_Z Y = \{(x, y) \in X \times Y : \alpha(x) = \beta(y)\}.$$

More precisely, show that the right side, equipped with its evident maps to X and Y, satisfies the universal property of the fibered product.

Proof Let W be an object with $\eta_X : W \to X$ and $\eta_Y : W \to Y$ whose compositions to Z agree $\alpha\eta_X = \beta\eta_Y$, we need to show there exists unique homomorphism maps W to $X \times_Z Y$.

$$\begin{tikzcd}
W \arrow[dr, dashed, "\theta"] \arrow[drr, "\eta_Y"] \arrow[ddr, "\eta_X"'] & & \\
& X \times_Z Y \arrow[r, "\mathrm{pr}_Y"'] \arrow[d, "\mathrm{pr}_X"] & Y \arrow[d, "\beta"] \\
& X \arrow[r, "\alpha"'] & Z
\end{tikzcd}$$

Define the map $\theta : W \to X \times_Z Y$ by $\theta(w) = (\eta_X(w), \eta_Y(w))$. The values of θ do lie in $X \times_Z Y$, for $\alpha\eta_X = \beta\eta_Y$. We need to show that the diagram commutes and that θ is unique.

To show commutative, let $w \in W$, note that

$$\mathrm{pr}_X(\theta(w)) = \mathrm{pr}_X(\eta_X(w), \eta_Y(w)) = \eta_X(w)$$

and

$$\mathrm{pr}_Y(\theta(w)) = \mathrm{pr}_Y(\eta_X(w), \eta_Y(w)) = \eta_Y(w),$$

this implies the diagram commutes.

To show uniqueness, let $\psi : W \to X \times_Z Y$ be another maps which is the solution to the universal mapping problem. Let $w \in W$, say $\psi(w) = (s(w), t(w))$, note that $\mathrm{pr}_X \theta = \eta_X$ and $\mathrm{pr}_Y \theta = \eta_Y$, we have $\mathrm{pr}_X \theta(w) = s(w) = \eta_X(w)$ and $\mathrm{pr}_Y \theta(w) = t(w) = \eta_Y(w)$. Hence, $\psi = \theta$. □

✍ **Exercise 2.11** If X is a topological space, show that fibered products always exists in the category of open sets of X, by describing what a fibered product is.

Proof Let U, V be open subsets in the category of open sets of X, we claim that $U \cap V$ is the fibered product. Let Z be a open subset with inclusion $i : U \to Z$ and $i : V \to Z$. Let W be any open subset of X, which satisfy $W \subseteq U$ and $W \subseteq V$, then $W \subseteq U \cap V$. Hence, $U \cap V$ is the fibered product. □

✍ **Exercise 2.12** If Z is the final object in a category $\mathbb{C}$, and $X, Y \in \mathscr{C}$, show that $X \times_Z Y = X \times Y$: the fibered product over Z is uniquely isomorphic to the product. Assume all relevant fibered products exists.

Proof It suffice to show that product satisfy the universal property of fibered product. Consider the following diagram.

$$\begin{tikzcd}
W \arrow[dr, dashed] \arrow[drr, "\beta'"] \arrow[ddr, "\alpha'"'] & & \\
& X \times Y \arrow[r, "\mathrm{pr}_Y"'] \arrow[d, "\mathrm{pr}_X"] & Y \arrow[d, "\beta"] \\
& X \arrow[r, "\alpha"'] & Z
\end{tikzcd} \tag{2.11}$$

By the universal property of product, there exists unique $\theta : W \to X \times Y$ making the diagram

$$\begin{tikzcd}
W \arrow[dr, dashed, "\theta"] \arrow[drr, "\beta'"] \arrow[ddr, "\alpha'"'] & & \\
& X \times Y \arrow[r, "\mathrm{pr}_Y"'] \arrow[d, "\mathrm{pr}_X"] & Y \\
& X &
\end{tikzcd}$$

commute. Then 2.11 commute, which implies that the fibered product over Z is isomorphic to the product. □

✍ **Exercise 2.13** (Towers of cartesian diagrams are cartesian diagrams.) If the two squares in the following commutative diagram are Cartesian diagrams, show that the "outside rectangle" (involving U, V, Y, and Z) is also a Cartesian diagram.

$$\begin{tikzcd}
U \arrow[r] \arrow[d] & V \arrow[d] \\
W \arrow[r] \arrow[d] & X \arrow[d] \\
Y \arrow[r] & Z
\end{tikzcd}$$

Proof Consider the diagram.

$$\begin{tikzcd}
K \arrow[drr] \arrow[dr, dashed] \arrow[ddr, dashed] \arrow[dddr] & & \\
& U \arrow[r] \arrow[d] & V \arrow[d] \\
& W \arrow[r] \arrow[d] & X \arrow[d] \\
& Y \arrow[r] & Z
\end{tikzcd}$$

We need to define a morphism from K to U such that above diagram commute. Since W is the fibered product over Z, we have:

$$\begin{tikzcd}
K \arrow[ddrr] \arrow[ddr, dashed, "\theta"] \arrow[dddr] & & \\
& & V \arrow[d] \\
& W \arrow[r] \arrow[d] & X \arrow[d] \\
& Y \arrow[r] & Z.
\end{tikzcd}$$

Since U is the fibered product over X, we have:

$$\begin{tikzcd}
K \arrow[drr] \arrow[dr, dashed, "\psi"] \arrow[ddr, "\theta"] & & \\
& U \arrow[r] \arrow[d] & V \arrow[d] \\
& W \arrow[r] \arrow[d] & X \arrow[d] \\
& Y \arrow[r] & Z.
\end{tikzcd}$$

Then, we have

$$\begin{tikzcd}
K \arrow[drr] \arrow[dr, dashed, "\psi"] \arrow[ddr] \arrow[dddr] & & \\
& U \arrow[r] \arrow[d] & V \arrow[d] \\
& W \arrow[r] \arrow[d] & X \arrow[d] \\
& Y \arrow[r] & Z,
\end{tikzcd}$$

as desired. □

Exercise 2.14 Given morphisms $X_1 \to Y$, $X_2 \to Y$, and $Y \to Z$, show that there is a natural morphism $X_1 \times_Y X_2 \to X_1 \times_Z X_2$, assuming that both fibered products exist.

Proof Since both fibered products exist, we have following commutative diagram immediately.

$$\begin{tikzcd}
X_1 \times_Y X_2 \arrow[dr, dashed] \arrow[drr] \arrow[ddr] & & & \\
& X_1 \times_Z X_2 \arrow[r] \arrow[d] & X_2 \arrow[d] \arrow[ddr] & \\
& X_1 \arrow[r] \arrow[drr] & Z & \\
& & & Y \arrow[ul]
\end{tikzcd}$$

□

Exercise 2.15 (The diagonal-base-change diagram) Suppose we are given morphisms $X_1 \to Y$, $X_2 \to Y$, and $Y \to Z$. Show that the following diagram is a Cartesian square.

$$\begin{tikzcd}
X_1 \times_Y X_2 \arrow[r] \arrow[d] & X_1 \times_Z X_2 \arrow[d] \\
Y \arrow[r] & Y \times_Z Y
\end{tikzcd}$$

Assume all relevant (fibered) product exist.

Proof By condition, we have following Cartesian diagrams:

$$\begin{tikzcd}
X_1 \times_Y X_2 \arrow[r, "\mathrm{pr}^Y_{X_2}"] \arrow[d, "\mathrm{pr}^Y_{X_1}"'] & X_2 \arrow[d, "f_2"] \\
X_1 \arrow[r, "f_1"] & Y
\end{tikzcd}
\qquad
\begin{tikzcd}
X_1 \times_Z X_2 \arrow[r, "\mathrm{pr}^Z_{X_2}"] \arrow[d, "\mathrm{pr}^Z_{X_1}"'] & X_2 \arrow[d, "g\circ f_2"] \\
X_1 \arrow[r, "g\circ f_1"] & Z
\end{tikzcd}
\qquad
\begin{tikzcd}
Y \times_Z Y \arrow[r, "\mathrm{pr}^Z_{Y}"] \arrow[d, "\mathrm{pr}^Z_{Y}"'] & Y \arrow[d, "g"] \\
Y \arrow[r, "g"] & Z
\end{tikzcd}$$

By Exercise 2.14, exists a morphism $\theta : X_1 \times_Y X_2 \to X_1 \times_Z X_2$. By the universal property of $Y \times_Z Y$,

$$\begin{tikzcd}
X_1 \times_Z X_2 \arrow[dr, dashed, "\eta"] \arrow[drr, "f_2\circ \mathrm{pr}^Z_{X_2}"] \arrow[ddr, "f_1\circ \mathrm{pr}^Z_{X_1}"'] & & \\
& Y \times_Z Y \arrow[r, "\mathrm{pr}^Z_Y"'] \arrow[d, "\mathrm{pr}^Z_Y"] & Y \arrow[d, "g"] \\
& Y \arrow[r, "g"] & Z,
\end{tikzcd}$$

there exists a morphism $\eta : X_1 \times_Z X_2 \to Y \times_Z Y$ such that above diagram commute.

Now, we have a diagram,

$$\begin{tikzcd}
X_1 \times_Y X_2 \arrow[r, "\theta"] \arrow[d, "f_1\circ \mathrm{pr}^Y_{X_1}"'] & X_1 \times_Z X_2 \arrow[d, "\eta"] \\
Y \arrow[r, "\iota"] & Y \times_Z Y,
\end{tikzcd} \tag{2.12}$$

where ι be an inclusion and $\iota = (\mathrm{pr}^Z_Y)^{-1}$.

We need to show two things: (1) (2.12) commute; (2) (2.12) is Cartesian square.

(1) (2.12) is commute.

Note that $\eta = (\mathrm{pr}^Z_Y)^{-1} \circ f_1 \circ \mathrm{pr}^Z_{X_1}$ and $\mathrm{pr}^Z_{X_1} \circ \theta = \mathrm{pr}^Y_{X_1}$, hence

$$\eta \circ \theta = (\mathrm{pr}^Z_Y)^{-1} \circ f_1 \circ \mathrm{pr}^Z_{X_1} \circ \theta = \iota \circ (f_1 \circ \mathrm{pr}^Y_{X_1}),$$

which implies that (2.12) commute.

(2) (2.12) is Cartesian square.

Consider the diagram,

$$\begin{tikzcd}
W \arrow[drr, "\beta"] \arrow[dr, dashed, "\varphi"] \arrow[ddr, dashed, "\alpha'"'] \arrow[dddr, "\alpha"'] & & & \\
& X_1 \times_Y X_2 \arrow[r, "\theta"'] \arrow[d, "\mathrm{pr}^Y_{X_1}"] & X_1 \times_Z X_2 \arrow[dl, "\mathrm{pr}^Z_{X_1}"'] \arrow[dd, "\eta=(\mathrm{pr}^Z_Y)^{-1}\circ f_1\circ \mathrm{pr}^Z_{X_1}"'] \arrow[dddr, bend left, "\mathrm{pr}^Z_{X_2}"] & \\
& X_1 \arrow[d, "f_1"] & & \\
& Y \arrow[r, "\iota=(\mathrm{pr}^Z_Y)^{-1}"] \arrow[dl, "g"] & Y \times_Z Y & \\
Z & & & X_2, \arrow[ull, "f_2"]
\end{tikzcd} \tag{2.13}$$

where W any object with $\eta\beta = \iota\alpha$.

Let $\alpha' = \mathrm{pr}^Z_{X_1} \circ \beta$. We shall to show that $f_2 \circ \mathrm{pr}^Z_{X_2} \circ \beta = f_1 \circ \alpha'$, i.e., show the following diagram commute.

$$\begin{tikzcd}
W \arrow[r, "\mathrm{pr}^Z_{X_2}\circ\beta"] \arrow[d, "\alpha'"'] & X_2 \arrow[d, "f_2"] \\
X_1 \arrow[r, "f_1"] & Y
\end{tikzcd}$$

Since $f_1 \circ \mathrm{pr}^Z_{X_1} = f_2 \circ \mathrm{pr}^Z_{X_2}$, $f_2 \circ \mathrm{pr}^Z_{X_2} \circ \beta = f_1 \circ \alpha'$, hence above diagram commute. By the universal property of $X_1 \times_Y X_2$, there exists unique $\varphi : W \to X_1 \times_Y X_2$ such that (2.13) commute, as we desired. $\square$

✍ **Exercise 2.16** Show that coproduct for **Sets** is disjoint union. This why we use the notation $\coprod$ for disjoint union.

Proof Let $A, B \in$ **Sets**, let $A' = A \times \{1\}$ and $B' = B \times \{2\}$, then $A' \cap B' = \varnothing$. Define the function $\theta : A \coprod B = A' \cup B' \to X$ by

$$\theta(u) = \begin{cases} f(a) & \text{if } u = (a,1) \in A', \\ g(b) & \text{if } u = (b,2) \in B'. \end{cases} \tag{2.14}$$

The function θ is well-defined because $A' \cap B' = \varnothing$.

We shall to show that the disjoint union $A \coprod B = A' \cup B' \subseteq (A \cup B) \times \{1,2\}$ is a coproduct in **Sets**. If X is any set and $f : A \to X$ and $g : B \to X$ are functions. θ makes the following diagram commute,

$$\begin{tikzcd}
& B \arrow[d, "\beta"'] \arrow[dr, "g"] & \\
A \arrow[r, "\alpha"] \arrow[drr, "f"'] & A \coprod B \arrow[dr, "\theta"] & \\
& & X,
\end{tikzcd}$$

where $\alpha : A \to A \coprod B$ defined by $\alpha(a) = (a,1)$ and $\beta : B \to A \coprod B$ defined by $\beta(b) = (b,2)$.

Now, let us prove uniqueness of θ. If $\psi : A \coprod B \to X$ satisfies $\psi\alpha = f$ and $\psi\beta = g$, then $\psi(a,1) = f(a) = \theta(a,1)$ and $\psi(b,2) = g(b) = \theta(b,2)$. Hence, $\theta = \psi$. $\square$

Definition 2.2.13 (Fibered coproduct)

Suppose morphisms $\alpha : Z \to X$ and $\beta : Z \to Y$ (in any category). Then the ***fibered coproduct*** *is an object $X \cup_Z Y$ along with morphisms $\iota_X : X \to X \cup_Z Y$ and $\iota_Y : Y \to X \cup_Z Y$, where the two compositions $\iota_X \circ \alpha = \iota_Y \circ \beta$, such that given any object W with maps from X and Y whos compositions from Z agree, these maps factor through some unique $X \cup_Z Y \to W$:*

$$\begin{tikzcd} Z \arrow[r, "\beta"] \arrow[d, "\alpha"'] & Y \arrow[d, "\iota_Y"'] \arrow[ddr, bend left] & \\ X \arrow[r, "\iota_X"] \arrow[drr, bend right] & X \cup_Z Y \arrow[dr, dashed, "\exists!"] & \\ & & W. \end{tikzcd}$$

✍ **Exercise 2.17** Suppose $A \to B$ and $A \to C$ are two ring homomorphisms, so in particular B and C are A-modules. $B \otimes_A C$ has a natural structure of a ring. (Multiplication will be given by $(b_1 \otimes c_1)(b_2 \otimes c_2) = (b_1 b_2) \otimes (c_1 c_2)$.) Show that there is a natural morphism $B \to B \otimes_A C$ given by $b \mapsto b \otimes 1$. Similarly, there is a natural morphism $C \to B \otimes_A C$. Show that this givs a fibered coproduct on rings, i.e., that

$$\begin{tikzcd} A \arrow[r, "\beta"] \arrow[d, "\alpha"'] & C \arrow[d, "\iota_Y"'] \\ B \arrow[r, "\iota_X"] & B \otimes_A C \end{tikzcd}$$

satisfies the universal property of fibered coproduct.

Proof We first show that the diagram

$$\begin{tikzcd} A \arrow[r, "\beta"] \arrow[d, "\alpha"'] & C \arrow[d, "\iota_Y"'] \\ B \arrow[r, "\iota_X"] & B \otimes_A C \end{tikzcd}$$

is commute. Let $a \in A$, then

$$\iota_X \circ \alpha(a) = \alpha(a) \otimes_A 1 = a(1 \otimes_A 1)$$

and

$$\iota_Y \circ \beta(a) = 1 \otimes_A \beta(a) = a(1 \otimes_A 1).$$

Hence, the diagram commute.

Let W be an object with $\eta_B : B \to W$ and $\eta_C : C \to W$, which implies that $\eta_B \alpha = \eta_C \beta$.

$$\begin{tikzcd} A \arrow[r, "\beta"] \arrow[d, "\alpha"'] & C \arrow[d, "\iota_Y"'] \arrow[ddr, bend left, "\eta_C"] & \\ B \arrow[r, "\iota_X"] \arrow[drr, bend right, "\eta_B"'] & B \otimes_A C \arrow[dr, dashed, "\theta"] & \\ & & W \end{tikzcd}$$

Define $\theta : B \otimes_A C \to W$ by setting $\theta(b \otimes_A c) = \eta_B(b)\eta_C(c)$, then above diagram is commute.

To show the uniqueness of θ, we may suppose there exists $\psi : B \otimes_A C \to W$ such that above diagram commute. Note that $\psi(b \otimes_A c) = \psi(b \otimes 1)\psi(1 \otimes c) = \eta_B(b)\eta_C(c) = \theta(b \otimes_A c)$ for all $b \otimes_A c \in B \otimes_A C$, $\psi = \theta$. □

2.2.5 Monomorphisms and epimorphsims

Definition 2.2.14 (Monomorphism)

A morphism $\pi : X \to Y$ is a ***monomorphism*** *if any two morphisms $\mu_1 : Z \to X$ and $\mu_2 : Z \to X$ such that $\pi \circ \mu_1 = \pi \circ \mu_2$ must satisfy $\mu_1 = \mu_2$. In other words, there is at most one way of filling in the dotted arrow such that the diagram*

$$\begin{tikzcd} Z \arrow[d, dotted, "\leq 1"'] \arrow[dr] & \\ X \arrow[r, "\pi"] & Y \end{tikzcd}$$

commutes — for any object Z, the natural map (induced by π) $\operatorname{Mor}(Z, X) \to \operatorname{Mor}(Z, Y)$ is an injection.

Intuitively, it is the categorical version of an injective map, and indeed this notion generalizes the familiar notion of injective maps of sets.

Remark 2.15 The reason we don't use the word "injective" is that in some contexts, "injective" will have an intuitive meaning which may not agree with "monomorphism".

Example 2.23 In the category of divisible groups, the map $\mathbb{Q} \to \mathbb{Q}/\mathbb{Z}$ is a monomorphism but not injective.

Proof Clearly, $\mathbb{Q} \to \mathbb{Q}/\mathbb{Z}$ is not injective.

Let $\mu_1 : G \to \mathbb{Q}$ and $\mu_2 : G \to \mathbb{Q}$ be any two group homomorphism. Say $\pi : \mathbb{Q} \to \mathbb{Q}/\mathbb{Z}$. Let $\pi \circ \mu_1 = \pi \circ \mu_2$, then $\pi(\mu_1(g)) = \pi(\mu_2(g))$ for all $g \in G$, hence, $(\mu_1 - \mu_2)(g) \in \mathbb{Z}$. We claim that $\mu_1 - \mu_2 = 0$, say $h = \mu_1 - \mu_2$, then $h : G \to \mathbb{Z}$. Since G is divisible, we have $\operatorname{Im} h$ is divisible. The only divisible subgroup of $\mathbb{Z}$ is 0, hence $\operatorname{Im} h = 0$, i.e., $\mu_1 = \mu_2$. □

✍ **Exercise 2.18** Show that the composition of two monomorphisms is a monomorphism.

Proof Let $\pi_1 : X \to Y$ and $\pi_2 : Y \to Z$ be two monomorphism. Let $\pi_2 \circ \pi_1 \circ \mu_1 = \pi_2 \circ \pi_1 \circ \mu_2$, where $\mu_1 : Z \to X$ and $\mu_2 : Z \to X$. Since π_2 is monomorphism, $\pi_1 \circ \mu_1 = \pi_1 \circ \mu_2$. Since π_1 is monomorphism, $\mu_1 = \mu_2$. □

✍ **Exercise 2.19** Prove that a morphism $\pi : X \to Y$ is a monomorphism if and only if the fibered product $X \times_Y X$ exists, and the induced diagonal morphism $\delta_\pi : X \to X \times_Y X$ is an isomorphism. We may then take this as the definition of monomorphism.

Proof If π is a monomorphism. Consider the following diagram,

$$\begin{tikzcd} X \times_Y X \arrow[r, "\mathrm{pr}_2"] \arrow[d, "\mathrm{pr}_1"'] & X \arrow[d, "\pi"] \\ X \arrow[r, "\pi"] & Y. \end{tikzcd}$$

Since π is a monomorphism, $\mathrm{pr}_1 = \mathrm{pr}_2 := \mathrm{pr}$. Consider the diagram,

$$\begin{tikzcd} X \arrow[dr, dashed, "\theta"] \arrow[drr, "\mathrm{id}"] \arrow[ddr, "\mathrm{id}"'] & & \\ & X \times_Y X \arrow[r, "\mathrm{pr}"] \arrow[d, "\mathrm{pr}"'] & X \arrow[d, "\pi"] \\ & X \arrow[r, "\pi"] & Y \end{tikzcd}$$

there exists a unique morphism $\theta : X \to X \times_Y X$ such that $\mathrm{id} = \mathrm{pr}\,\theta$, by the universal property of fibered product. Let $\delta_\pi = \theta$, it is an isomorphism.

Conversely, if the fibered product $X \times_Y X$ exists, and the induced diagonal morphism $\delta_\pi : X \to X \times_Y X$

is an isomorphism. Consider the following diagram,

$$\begin{tikzcd}
W \arrow[dr, dashed, "\theta"] \arrow[drr, bend left, "\eta_2"] \arrow[ddr, bend right, "\eta_1"'] & & \\
& X \times_Y X \arrow[r, "\delta_\pi^{-1}"] \arrow[d, "\delta_\pi^{-1}"'] & X \arrow[d, "\pi"] \\
& X \arrow[r, "\pi"] & Y,
\end{tikzcd}$$

where W is any object with $\pi\eta_1 = \pi\eta_2$. We shall to show that $\eta_1 = \eta_2$. By the universal property of fibered product, there exists unique $\theta : W \to X \times_Y X$ such that $\eta_1 = \delta_\pi^{-1}\theta$ and $\eta_2 = \delta_\pi^{-1}\theta$, then $\eta_1 = \eta_2$, which implies that $\pi : X \to Y$ is an monomorphism. $\square$

✍ **Exercise 2.20** We use the notation of Exercise 2.14. Show that if $Y \to Z$ is a monomorphism, then the morphism $X_1 \times_Y X_2 \to X_1 \times_Z X_2$ you described in Exercise 2.14 is an isomorphism.

Proof Say $\pi : Y \to Z$. By Exercise 2.19, $\delta_\pi : Y \to Y \times_Z Y$ is an isomorphism. Consider the diagonal-base-change diagram (Exercise 2.15),

$$\begin{tikzcd}
X_1 \times_Z X_2 \arrow[dr, dashed, "\theta"] \arrow[drr, bend left, "\mathrm{id}"] \arrow[ddr, bend right, "\delta_\pi^{-1}\circ\varphi"'] & & \\
& X_1 \times_Y X_2 \arrow[r, "\eta"'] \arrow[d, "\psi"] & X_1 \times_Z X_2 \arrow[d, "\varphi"] \\
& Y \arrow[r, "\delta_\pi"] & Y \times_Z Y,
\end{tikzcd}$$

there exists unique $\theta : X_1 \times_Z X_2 \to X_1 \times_Y X_2$ such that above diagram commute, then we have $\eta\theta = \mathrm{id}$, that is, θ is an isomorphism. $\square$

The notion of an **epimorphism** is "dual" to the definition of monomorphism, where all the arrows are reversed. This concept will not be central for us, although it turns up in the defnition of an abelian category. Intuitively, it is the categorical version of a surjective map.

Definition 2.2.15 (Epimorphism)

A morphism $\pi : X \to Y$ is a ***epimorphism*** *if for any two morphisms $\mu_1 : Y \to Z$ and $\mu_2 : Y \to Z$ such that $\mu_1 \circ \pi = \mu_2 \circ \pi$ we have $\mu_1 = \mu_2$. In other words, there is at most one way of filling in the dotted arrow such that the diagram*

$$\begin{tikzcd}
X \arrow[r, "\pi"] \arrow[dr] & Y \arrow[d, dotted, "\leq 1"] \\
& Z
\end{tikzcd}$$

commutes — for any object Z, the natural map (induced by π) $\mathrm{Mor}(Y, Z) \to \mathrm{Mor}(X, Z)$ is an injection.

Be careful when working with categories of objects that are sets with additional structure, as epimorphisms need not be surjective.

Example 2.24 In the category **Rings**, $\mathbb{Z} \to \mathbb{Q}$ is an epimorphism, but not surjective.

Proof Say $\pi : \mathbb{Z} \to \mathbb{Q}$. Let $R \in$ **Rings**, and homomorphisms $\mu_1 : \mathbb{Q} \to R$ and $\mu_2 : \mathbb{Q} \to R$, let $\mu_1 \circ \pi = \mu_2 \circ \pi$, then $\mu_1 \circ \pi(n) = \mu_2 \circ \pi(n)$ for all $n \in \mathbb{Z}$. Since $\pi(n) = n\pi(1) = n$, we have $\mu_1(n) = \mu_2(n)$ for all $n \in \mathbb{Z}$, which implies that π is epimorphism. $\square$

2.2.6 Representable functors and Yoneda's Lemma

Much of our discussion about universal properties can be cleany expressed in termes of representable functors, under the rubric of "Yoneda's Lemma". Yoneda's lemma is an easy fact stated in a complicated way. Informally speaking, you can essentially recover an object in a category by knowing the maps into it. For example, we have seen that the data of maps to $X \times Y$ are naturally (canonically) the data of maps to X and to Y. Indeed, we have now taken this as the definition of $X \times Y$.

Recall Example 2.19 . Suppose A is an object of category $\mathscr{C}$. For any object $C \in \mathscr{C}$, we have a set of morphisms $\operatorname{Mor}(C, A)$. If we have a morphism $f : B \to C$, we get a map of sets

$$\operatorname{Mor}(C, A) \xrightarrow{\circ f} \operatorname{Mor}(B, A),$$

by composition: given a map from C to A, we get a map from B to A by precomposing with $f : B \to C$. Hence this gives a contravariant functor $h_A = \operatorname{Mor}(\square, A) : \mathscr{C} \to \mathbf{Sets}$ (or covariant functor $h_A : \mathscr{C}^{op} \to \mathbf{Sets}$). Yoneda's Lemma states that the functor h_A determines A up to unique isomorphism.

Note *If $F, G : A \to B$ are functors of the same variance, then*

$$\operatorname{Nat}(F, G) = \{\textit{natural transformations } F \to G\}.$$

Theorem 2.2.1 (Yoneda's Lemma)

Let $\mathscr{C}$ be a category, let $A \in \operatorname{obj}(\mathscr{C})$. For every functor $G : \mathscr{C}^{op} \to \mathbf{Sets}$, the map

$$y : \operatorname{Nat}(h_A, G) \to G(A), \quad \tau \mapsto \tau_A(\mathrm{id}_A)$$

is a bijection, where $h_A = \operatorname{Mor}(\square, A)$.

Proof If $\tau : \operatorname{Mor}_{\mathscr{C}}(\square, A) \to G$ is a natual transformation, then $y(\tau) = \tau_A(\mathrm{id}_A)$ lies in the set $G(A)$, for $\tau_A : \operatorname{Mor}_{\mathscr{C}}(A, A) \to G(A)$. Thus, y is a well-defined function.

Since $\tau \in \operatorname{Nat}(h_A, G)$ is a natural transformation, for each $B \in \operatorname{obj}(\mathscr{C})$ and $\varphi \in \operatorname{Mor}_{\mathscr{C}}(B, A)$, there is a commutative diagram

$$\begin{tikzcd}
\operatorname{Mor}_{\mathscr{C}}(A, A) \arrow[r, "\tau_A"] \arrow[d, "\square\circ\varphi"'] & G(A) \arrow[d, "G(\varphi)"] \\
\operatorname{Mor}_{\mathscr{C}}(B, A) \arrow[r, "\tau_B"] & G(B)
\end{tikzcd}$$

so that

$$(G\varphi)\tau_A(\mathrm{id}_A) = \tau_B(\square \circ \varphi)(\mathrm{id}_A) = \tau_B(\mathrm{id}_A \circ \varphi) = \tau_B(\varphi).$$

If $\sigma : \operatorname{Mor}_{\mathscr{C}}(\square, A) \to G$ is another natural transformation, then $\sigma_B(\varphi) = (G\varphi)\sigma_A(\mathrm{id}_A)$. Hence, if $\sigma_A(\mathrm{id}_A) = \tau_A(\mathrm{id}_A)$, then $\sigma_B = \tau_B$ for all $B \in \operatorname{obj}(\mathscr{C})$, and therefore $\sigma = \tau$, which implies y is an injection.

To see that y is a surjection, take $x \in G(A)$. For $B \in \operatorname{obj}(\mathscr{C})$ and $\psi \in \operatorname{Mor}(B, A)$, define

$$\tau_B(\psi) = (G\psi)(x)$$

(note that $G\psi : GA \to GB$, hence $(G\psi)(x) \in GB$). We claim that τ is a natural transformation, that is, if $\theta : C \to B$ is a morphism in $\mathscr{C}$, then the following diagram commutes:

$$\begin{tikzcd}
\operatorname{Mor}_{\mathscr{C}}(B, A) \arrow[r, "\tau_B"] \arrow[d, "\square\circ\theta"'] & GB \arrow[d, "G\theta"] \\
\operatorname{Mor}_{\mathscr{C}}(C, A) \arrow[r, "\tau_C"] & GC.
\end{tikzcd}$$

Going clockwise, we have $(G\theta)\tau_B(\psi) = G\theta G\psi(x)$; going counterclockwise, we have $\tau_C(\square \circ \theta)\psi = \tau_C(\psi \circ$

$\theta) = G(\psi\theta)(x)$. Since G is a contravariant functor, $G(\psi\theta) = (G\theta)(G\psi)$, hence $(G\theta)\tau_B = \tau_C(\square \circ \theta)$, thus, τ is a natural transformation. Now $y(\tau) = \tau_A(\mathrm{id}_A) = G(\mathrm{id}_A)(x) = x$, and so y is bijection. $\square$

✍ **Exercise 2.21 (Another form of Yoneda's Lemma)**

(a) There are two objects A and A' in a category $\mathscr{C}$, and the commutative diagram.

$$\begin{tikzcd} \mathrm{Mor}(C,A) \arrow[r, "i_C"] \arrow[d, "\mathrm{Mor}(\square,A)"'] & \mathrm{Mor}(C,A') \arrow[d, "\mathrm{Mor}(\square,A')"] \\ \mathrm{Mor}(B,A) \arrow[r, "i_B"] & \mathrm{Mor}(B,A') \end{tikzcd}$$

Then i_C are induced from a unique morphism $g : A \to A'$. More precisely, there is a unique morphsim $g : A \to A'$ such that for all $C \in \mathscr{C}$, i_C is $u \mapsto g \circ u$.

(b) If furthermore the i_C are all bijections, then the resulting g is an isomorphism.

Proof

(a) Since $i : \mathrm{Mor}(\square, A) \to \mathrm{Mor}(\square, A')$ is a natural transformation, for each $C \in \mathrm{obj}(\mathscr{C})$ and $u \in \mathrm{Mor}(C,A)$, there is a commutative diagram.

$$\begin{tikzcd} \mathrm{Mor}(A,A) \arrow[r, "i_A"] \arrow[d, "\square\circ u"'] & \mathrm{Mor}(A,A') \arrow[d, "\square\circ u"] \\ \mathrm{Mor}(C,A) \arrow[r, "i_C"] & \mathrm{Mor}(C,A) \end{tikzcd}$$

Hence, $(\square \circ u)i_A(\mathrm{id}_A) = (i_A\,\mathrm{id}_A) \circ u = i_C(\square \circ u)(\mathrm{id}_A) = i_C u$, let $g = i_A(\mathrm{id}_A)$, as we desired.

(b) i is natual isomorphism. By the Yoneda's Lemma 2.2.1, $y : i \mapsto i_A(\mathrm{id}_A) = g$ is bijection, g must be an isomorphism.

$\square$

Theorem 2.2.2 (Yoneda's Lemma for covariant functors)

Let $\mathscr{C}$ be a category, let $A \in \mathrm{obj}(\mathscr{C})$. For every covariant functor $G : \mathscr{C} \to \mathbf{Sets}$, the map

$$y : \mathrm{Nat}(h^A, G) \to G(A), \quad \tau \mapsto \tau_A(\mathrm{id}_A)$$

is a bijection, where $h^A = \mathrm{Mor}(A, \square)$.

✍ **Exercise 2.22** Suppose A and B are objects in a category $\mathscr{C}$. Give a bijection between the natural transformations $h^A \to h^B$ of covariant functors $\mathscr{C} \to \mathbf{Sets}$ and the morphisms $B \to A$.

Proof By Yoneda's Lemma 2.2.2, the map

$$\mathrm{Nat}(h^A, h^B) \to h^B(A) = \mathrm{Mor}(B, A), \quad \tau \mapsto \tau_A(\mathrm{id}_A)$$

is a bijection. $\square$

Definition 2.2.16 (Representable)

A covariant functor $F : \mathscr{C} \to \mathbf{Sets}$ is said to be ***representable*** *if there exists $A \in \mathrm{obj}(\mathscr{C})$ with natural isomorphism $F \cong \mathrm{Mor}_{\mathscr{C}}(A, \square) = h^A$.*

A contravariant functors $G : \mathscr{C} \to \mathbf{Sets}$ is said to be ***represantable*** *if there exists $A \in \mathrm{obj}(\mathscr{C})$ with natural isomorphism $G \cong \mathrm{Mor}_{\mathscr{C}}(\square, A) = h_A$.*

In fancy terms, Yoneda's lemma states the following. Given a category $\mathscr{C}$, we can produce a new category, called the **functor category of** $\mathscr{C}$ (denote by $\mathbf{Sets}^{\mathscr{C}^{op}}$), where the objects are contravariant functors $\mathscr{C} \to \mathbf{Sets}$, and the morphisms are natural transformations of such functors. We have a functor (which we can usefully call h) from $\mathscr{C}$ to its functor category, which sends A to h_A. Yoneda's Lemma states that this is a fully faithful functor, called the Yoneda embedding.

Definition 2.2.17 (Small category)

A category $\mathscr{C}$ is ***small*** *if the objects form a set and the morphisms form a set.*

Corollary 2.2.6 (Yoneda Imbedding)

If $\mathscr{C}$ is a small category, then there is a functor $h : \mathscr{C} \to \mathbf{Sets}^{\mathscr{C}^{op}}$ that is injective on objects and whose image is a full subcategory of $\mathbf{Sets}^{\mathscr{C}^{op}}$.

Proof Define h on objects by $h(A) = h_A = \mathrm{Mor}_{\mathscr{C}}(\square, A)$. If $A \neq A'$, then pairwise disjointness of Mor sets gives $\mathrm{Mor}(\square, A) \neq \mathrm{Mor}(\square, A')$, that is, $h_A \neq h_{A'}$, and so h is injective on objects. If $\psi : A \to B$ is a morphism in $\mathscr{C}$, then there is a natural transformation $h(\psi) : \mathrm{Mor}(\square, A) \to \mathrm{Mor}(\square, B)$ with $h(\psi)_C = \psi \circ \square$, for all $C \in \mathrm{obj}(\mathscr{C})$, by Yoneda's Lemma 2.2.1. Now, we have $h(\psi\eta) = h(\psi)h(\eta)$, and so h is covariant functor. Surjectivity of the Yoneda's function y in Theorem 2.2.1 shows that every natural transformation $h_A \to h_B$ arises as $h(\psi)$ for some ψ. Therefore, the image of h is a full subcategory of the functor category $\mathbf{Sets}^{\mathscr{C}^{op}}$. □

Corollary 2.2.7 (Useful corollary)

Let $\mathscr{C}$ be a category and let $A, B \in \mathrm{obj}(\mathscr{C})$.

(i) If $\tau : \mathrm{Mor}_{\mathscr{C}}(A, \square) \to \mathrm{Mor}_{\mathscr{C}}(B, \square)$ is a natural transformation, then for all $C \in \mathrm{obj}(\mathscr{C})$, we have $\tau_C = \psi^*$, where $\psi = \tau_A(\mathrm{id}_A) : B \to A$ (and ψ^* is the induced map $\mathrm{Hom}_{\mathscr{C}}(A, C) \to \mathrm{Hom}_{\mathscr{C}}(B, C)$ given by $\varphi \mapsto \varphi \circ \psi$). Moreover, the morphism ψ is unique: if $\tau_C = \theta^*$, then $\theta = \psi$.

(ii) Let

$$\mathrm{Mor}_{\mathscr{C}}(A, \square) \xrightarrow{\tau} \mathrm{Mor}_{\mathscr{C}}(B, \square) \xrightarrow{\sigma} \mathrm{Mor}_{\mathscr{C}}(B', \square)$$

be natural transformations. If $\sigma_C = \eta^*$ and $\tau_C = \psi^*$ for all $C \in \mathrm{obj}(\mathscr{C})$, then

$$(\sigma \circ \tau)_C = (\psi \circ \eta)^*.$$

(iii) If $\mathrm{Mor}_{\mathscr{C}}(A, \square)$ and $\mathrm{Mor}_{\mathscr{C}}(B, \square)$ are naturally isomorphic functor, then $A \cong B$. (The converse is clearly true.)

Proof

(i) Denote $\psi = \tau_A(\mathrm{id}_A) \in \mathrm{Mor}_{\mathscr{C}}(B, A)$. Since τ is a natural transformation, for all $C \in \mathrm{obj}(\mathscr{C})$, for any $\varphi \in \mathrm{Mor}_{\mathscr{C}}(A, C)$, we have the following commutative diagram.

$$\begin{tikzcd}
\mathrm{Mor}_{\mathscr{C}}(A, A) \arrow[r, "\mathrm{Mor}_{\mathscr{C}}(A,\varphi)"] \arrow[d, "\tau_A"'] & \mathrm{Mor}_{\mathscr{C}}(A, C) \arrow[d, "\tau_C"] \\
\mathrm{Mor}_{\mathscr{C}}(B, A) \arrow[r, "\mathrm{Mor}_{\mathscr{C}}(B,\varphi)"] & \mathrm{Mor}_{\mathscr{C}}(B, C)
\end{tikzcd}$$

Hence,

$$\begin{aligned}
\varphi_* \circ \tau_A(\mathrm{id}_A) =& \varphi_* \circ \psi = \varphi \circ \psi \\
=& \tau_C \circ \varphi_*(\mathrm{id}_A) = \tau_C(\varphi \circ \mathrm{id}_A) \\
=& \tau_C(\varphi)
\end{aligned}$$

Note that $\varphi \circ \psi = \psi^*(\varphi)$, we have

$$\tau_C = \psi^*.$$

By Yoneda Lemma (Theorem 2.2.2), $\tau \longleftrightarrow \tau_A(\mathrm{id}_A)$ is bijective, the uniqueness assertion follows from injectivity of the Yoneda function y.

(ii) By part (i), there are unique morphisms $\psi \in \mathrm{Mor}_{\mathscr{C}}(B,A)$ and $\eta \in \mathrm{Mor}_{\mathscr{C}}(B',B)$ with

$$\tau_C(\varphi) = \psi^*(\varphi) \quad \text{and} \quad \sigma_C(\varphi') = \eta^*(\varphi')$$

for all $\varphi \in \mathrm{Mor}_{\mathscr{C}}(A,C)$ and $\varphi' \in \mathrm{Mor}_{\mathscr{C}}(B,C)$. By definition, $(\sigma \circ \tau)_C = \sigma_C \circ \tau_C$, and therefore

$$(\sigma \circ \tau)_C(\varphi) = \sigma_C(\psi^*(\varphi)) = \eta^* \circ \psi^*(\varphi) = (\psi \circ \eta)^*(\varphi).$$

(iii) If $\tau : \mathrm{Mor}_{\mathscr{C}}(A,\square) \to \mathrm{Mor}_{\mathscr{C}}(B,\square)$ is a natural isomorphism, then there is a natural isomorphism $\sigma : \mathrm{Mor}_{\mathscr{C}}(B,\square) \to \mathrm{Mor}_{\mathscr{C}}(A,\square)$ with $\sigma \circ \tau = \mathrm{id}_{\mathrm{Mor}_{\mathscr{C}}(A,\square)}$ and $\tau \circ \sigma = \mathrm{id}_{\mathrm{Mor}_{\mathscr{C}}(B,\square)}$. By part (i), there are morphisms $\psi : B \to A$ and $\eta : A \to B$ with $\tau_C = \psi^*$ and $\sigma_C = \eta^*$ for all $C \in \mathrm{obj}(\mathscr{C})$. By part (ii), we have $\tau \circ \sigma = \psi^*\eta^* = (\eta \circ \psi)^* = \mathrm{id}_B^*$ and $\sigma \circ \tau = (\psi\eta)^* = \mathrm{id}_A^*$. The uniqueness in part (i) now gives $\psi\eta = \mathrm{id}_A$ and $\eta\psi = \mathrm{id}_B$, which implies that $A \cong B$.

$\square$

2.3 Limits and colimits

Definition 2.3.1 (Index category)

*Suppose $\mathscr{I}$ is any small category, and $\mathscr{C}$ is any category. Then a functor $F : \mathscr{I} \to \mathscr{C}$ (i.e., with an object $A_i \in \mathscr{C}$ for each element $i \in \mathscr{I}$, and appropriate commuting morphisms dictated by $\mathscr{I}$) is said to be a **diagram indexed by** $\mathscr{I}$. We call $\mathscr{I}$ an **index category**.*

Remark 2.16 Our index categories will usually be partially ordered sets, in which in particullar there is at most one morphism between any two objects.

Example 2.25 If $\square$ is the category

$$\begin{tikzcd} \bullet \arrow[r] \arrow[d] & \bullet \arrow[d] \\ \bullet \arrow[r] & \bullet \end{tikzcd}$$

and $\mathscr{A}$ is a category, then a functor $\square \to \mathscr{A}$ is precisely the data of a commuting square in $\mathscr{A}$.

Definition 2.3.2 (Limit)

*The **limit of the diagram** is an object $\lim\limits_{\mathscr{I}} A_i$ (or $\varprojlim\limits_{\mathscr{I}} A_i$) of $\mathscr{C}$ along with morphisms $f_j : \varprojlim\limits_{\mathscr{I}} A_i \to A_j$ for each $j \in \mathscr{I}$, such that if $m : j \to k$ is a morphism in $\mathscr{I}$, then*

$$\begin{tikzcd} \varprojlim\limits_{\mathscr{I}} A_i \arrow[d, "f_j"] \arrow[dr, "f_k"] & \\ A_j \arrow[r, "F(m)"] & A_k \end{tikzcd}$$

commutes, and this object and maps to each A_i are universal (final) with respect to this property. More precisely, given any other object W along with maps $g_i : W \to A_i$ commuting with the $F(m)$ (if $m : j \to k$ is a morphism in $\mathscr{I}$, then $g_k = F(m) \circ g_j$), then there is a unique map

$$g : W \to \varprojlim\limits_{\mathscr{I}} A_i$$

such that $g_i = f_i \circ g$ for all i, i.e,

$$\begin{tikzcd}
\varprojlim_{\mathscr{I}} A_i \arrow[dr, "f_j"] \arrow[ddr, bend right, "f_k"'] & & W \arrow[ll, dashed, "g"'] \arrow[dl, "g_j"'] \arrow[ddl, bend left, "g_k"] \\
& A_j \arrow[d, "F(m)"] & \\
& A_k. &
\end{tikzcd}$$

In some cases, the limit is sometimes called the ***inverse limit*** *or* ***projective limit****.*

Remark 2.17 By the universal property argument, if the limit exists, it is unique up to unique isomorphism.

Example 2.26 (Products.) For example, if $\mathscr{I}$ is the partially ordered set

$$\begin{tikzcd}
& \bullet \arrow[d] \\
\bullet \arrow[r] & \bullet
\end{tikzcd}$$

we obtain the fibered product (reveiw the definition of fibered product).

If $\mathscr{I}$ is

$$\bullet \qquad \bullet$$

we obtain the product (reveiw the definition of product).

If $\mathscr{I}$ is a set (i.e., the only morphisms are the identity maps), then the limit is called the **product** of the A_i, and is denoted $\prod_i A_i$.

✍ **Exercise 2.23** Suppose that the partially ordered set $\mathscr{I}$ has an initial object e. Show that the limit of any diagram indexed by $\mathscr{I}$ exists.

Proof Say $F : \mathscr{I} \to \mathscr{C}$. We claim that $\varprojlim F(i) = F(e)$. Since e is initial object, set $m : i \to j$, we have the following commutative diagram.

$$\begin{tikzcd}
F(e) \arrow[d, "f_i"'] \arrow[dr, "f_j"] & \\
F(i) \arrow[r, "F(m)"] & F(j)
\end{tikzcd}$$

Let W be an object in $\mathscr{C}$, which makes the following diagram commute.

$$\begin{tikzcd}
W \arrow[d, "g_i"'] \arrow[dr, "g_j"] & \\
F(i) \arrow[r, "F(m)"] & F(j)
\end{tikzcd}$$

Then there is a naturally morphism $g_e : W \to F(e)$, since the commutative diagram

$$\begin{tikzcd}
W \arrow[d, "g_e"'] \arrow[dr, "g_i"] & \\
F(e) \arrow[r] & F(i)
\end{tikzcd}$$

Hence,

$$\begin{tikzcd}
\varprojlim F(i) = F(e) \arrow[dr, "f_i"] \arrow[ddr, bend right, "f_j"'] & & W \arrow[ll, "g_e"'] \arrow[dl, "g_i"'] \arrow[ddl, bend left, "g_j"] \\
& F(i) \arrow[d, "F(m)"] & \\
& F(j), &
\end{tikzcd}$$

as we desired. $\square$

✍ **Exercise 2.24 (The diagonal-base-change diagram)** Solve Exercise 2.15 again by identifiying both $X_1 \times_Y X_2$ and $Y \times_{(Y\times_Z Y)} (X_1 \times_Z X_2)$ as the limit of the diagram.

$$(*):\quad \begin{tikzcd}
X_1 \arrow[dr] & & \\
& Y \arrow[r] & Z \\
X_2 \arrow[ur] & &
\end{tikzcd}$$

Proof Clearly, $X_1 \times_Y X_2$ can be identify with the limit of the diagram $(*)$. It suffices to show that $Y \times_{Y\times_Z Y} (X_1 \times_Z X_2)$ is the limit of above diagram. Say $T = Y \times_{Y\times_Z Y} (X_1 \times_Z X_2)$. Let W be any object with the following arrows:

$$\begin{tikzcd}
& W \arrow[rr] \arrow[dr] \arrow[ddl] \arrow[ddd] & & Z \\
& & Y \arrow[ur] & \\
X_1 \arrow[urr] & & & \\
& X_2 \arrow[uur] & &
\end{tikzcd}$$

Consider the following diagram,

$$\begin{tikzcd}
W \arrow[rrrrr, bend left] \arrow[drrrr, bend left] \arrow[ddrr, bend left] \arrow[dddrrr, bend right] \arrow[dr, dashed, "\exists!"] \arrow[ddddr, dashed, "\exists!"'] & & & & & Z \\
& T \arrow[rrr] \arrow[ddd] & & & Y \arrow[ur] \arrow[ddd, "\delta_{Y/Z}"] & \\
& & X_1 \arrow[urr] & & & \\
& & & X_2 \arrow[uur] & & \\
& X_1 \times_Z X_2 \arrow[uur] \arrow[urr] \arrow[rrr] & & & Y \times_Z Y &
\end{tikzcd}$$

where dashed arrows $W \to X_1 \times_Z X_2$ given by the universal property of $X_1 \times_Z X_2$ and $W \to T$ given by the universal property of T. Hence T is a limit of diagram $(*)$, and therefore $Y \times_{Y\times_Z Y} (X_1 \times_Z X_2) \cong X_1 \times_Y X_2$. $\square$

Example 2.27 (Formal power series) For a ring A, the **(formal) power series)**, $A[[x]]$, are often described informally (and somewhat unnaturally) as being the ring

$$A[[x]] = \{a_0 + a_1 x + a_2 x^2 + \cdots\}$$

where $a_i \in A$, and the ring operations are the "obvious" ones. It is an example of a limit in the category of rings:

$$\begin{tikzcd}
A[[x]] \arrow[dr] \arrow[drr] \arrow[drrr] & & & \\
\cdots \arrow[r] & A[x]/(x^3) \arrow[r] & A[x]/(x^2) \arrow[r] & A[x]/(x). \\
A[x] \arrow[ur] \arrow[urr] \arrow[urrr] \arrow[uu, dashed, bend left] & & &
\end{tikzcd}$$

The universal property of limits yields a natural ring morphism $A[x] \to A[[x]]$. If $A = \mathbb{R}$ or $\mathbb{C}$, this map factors through the ring of **convergent power series**.

Example 2.28 (The p-adic integers) For a prime number p, the **p-adic integers**, $\mathbb{Z}_p$, are often described informally, and somewhat unnaturally, as being of the form

$$a_0 + a_1 p + a_2 p^2 + \cdots$$

where $0 \leq \alpha_i < p$. They are an example of a limit in the category of rings:

$$\begin{tikzcd}
\mathbb{Z}_p \arrow[dr] \arrow[drr] \arrow[drrr] & & & \\
\cdots \arrow[r] & \mathbb{Z}/(p^3) \arrow[r] & \mathbb{Z}/(p^2) \arrow[r] & \mathbb{Z}/(p).
\end{tikzcd}$$

Remark 2.18 Limits do not always exist for any index category $\mathscr{I}$. However, we can often easily check that limits exists if the objects of your category can be interpreted as sets with additional structure, and arbitrary products exist.

✍ **Exercise 2.25** Show that in the category **Sets**,

$$\left\{ (a_i)_{i \in \mathscr{I}} \in \prod_i A_i : F(m)(a_j) = a_k \text{ for all } m \in \mathrm{Mor}_{\mathscr{I}}(j,k) \subseteq \mathrm{Mor}(\mathscr{I}) \right\},$$

along with the obvious projection maps to each A_i, is the limit $\varprojlim_{\mathscr{I}} A_i$.

Proof Say $L = \{(a_i)_{i \in \mathscr{I}} \in \prod_i A_i : F(m)(a_j) = a_k \text{ for all } m \in \mathrm{Mor}_{\mathscr{I}}(j,k) \subseteq \mathrm{Mor}(\mathscr{I})\}$. The definition of L gives a commutative diagram.

$$\begin{tikzcd}
L \arrow[d, "\mathrm{pr}_i"'] \arrow[dr, "\mathrm{pr}_j"] & \\
A_i \arrow[r] & A_j.
\end{tikzcd}$$

Consider the diagram,

$$\begin{tikzcd}
L \arrow[dr, "\mathrm{pr}_i"] \arrow[ddr, bend right, "\mathrm{pr}_j"'] & & W \arrow[ll, dashed, "\theta"'] \arrow[dl, "f_i"'] \arrow[ddl, bend left, "f_j"] \\
& A_i \arrow[d, "F(m)"] & \\
& A_j, &
\end{tikzcd}$$

where W is any object which satisfy $f_j = F(m) \circ f_i$.

Define $\theta : W \to L$ by setting $\theta(w) = (f_i(w))_{i \in \mathscr{I}}$, then θ is well-defined morphism and such that above diagram commute.

To show the uniqueness of θ, let $\psi : W \to L$ be another morphism such that above diagram commute. Note that

$$\mathrm{pr}_i \circ \psi(w) = f_i(w),$$

we have $\psi(w) = (f_i(w))_{i \in \mathscr{I}}$. Hence, $L = \varprojlim_{\mathscr{I}} A_i$ $\square$

Remark 2.19

(1) This clearly also works in the category $\mod_A$ of A-modules (in particular $\mathbf{Vec}_k$ and $\mathbf{Ab}$), as well as $\mathbf{Rings}$.

(2) From this point of view, $2 + 3p + 2p^2 + \cdots \in \mathbb{Z}_p$ can be understood as the sequence $(2, 2 + 3p, 2 + 3p + 2p^2, \cdots)$.

Definition 2.3.3 (Colimits)

*The **colimit of the diagram** is an object $\operatorname{colim}_{\mathscr{I}} A_i$ (or $\varinjlim_{\mathscr{I}} A_i$) of $\mathscr{C}$ along with morphisms $f_j : A_j \to \varinjlim_{\mathscr{I}} A_i$ for each $j \in \mathscr{I}$, such that if $m : j \to k$ is a morphism in $\mathscr{I}$, then*

$$\begin{tikzcd}
A_j \arrow[r, "F(m)"] \arrow[d, "f_j"] & A_k \arrow[dl, "f_k"] \\
\varinjlim_{\mathscr{I}} A_i &
\end{tikzcd}$$

commutes, and this object and maps from each A_i are universal (initial) with respect to this property. More precisely, given any other object W along with maps $g_i : A_i \to W$ commuting with the $F(m)$ (if $m : j \to k$ is a morphism in $\mathscr{I}$, then $g_j = g_k \circ F(m)$), then there is a unique map

$$g : \varinjlim_{\mathscr{I}} A_i \to W$$

such that $g_j = g \circ f_j$ for all i, i.e.,

$$\begin{tikzcd}
\varinjlim_{\mathscr{I}} A_i \arrow[rr, dashed, "g"] & & W \\
& A_j \arrow[ul, "f_j"] \arrow[ur, "g_j"'] \arrow[d, "F(m)"] & \\
& A_k. \arrow[uul, bend left, "f_k"] \arrow[uur, bend right, "g_k"'] &
\end{tikzcd}$$

*In some cases, the colimit is sometimes called the **direct limit**, **inductive limit**, or **injective limit**.*

Example 2.29 The abelian group $5^{-\infty}\mathbb{Z}$ of rational numbers whose denominators are powers of 5 is a colimit $\varinjlim_{i \in \mathbb{Z}^+} 5^{-i}\mathbb{Z}$. More precisely, $5^{-\infty}\mathbb{Z}$ is the colimit of the diagram

$$\begin{tikzcd}
5^{-\infty}\mathbb{Z} & & & \\
\mathbb{Z} \arrow[r] \arrow[u] & 5^{-1}\mathbb{Z} \arrow[r] \arrow[ul] & 5^{-2}\mathbb{Z} \arrow[r] \arrow[ull] & \cdots
\end{tikzcd}$$

in the category of abelian groups.

Example 2.30 The colimit over an index set I is called the **coproduct**, denoted $\coprod_i A_i$, and is the dual (arrow-reversed) notion to the product.

Exercise 2.26

(a) Interpret the statement "$\mathbb{Q} = \varinjlim \frac{1}{n}\mathbb{Z}$".

(b) Interpret the union of some subsets of a given set as a colimit. (Dually, the intersection can be interpreted as a limit.) The objects of the category in question are the subsets of given set.

Proof

(a) Note that $\frac{1}{m}\mathbb{Z} \subseteq \frac{1}{n}\mathbb{Z}$ if $m \mid n$. Then we have the commutative diagram.

$$\begin{tikzcd}
\mathbb{Q} \arrow[rr, dashed, "g"] & & W \\
& \frac{1}{m}\mathbb{Z} \arrow[ul, "\iota"'] \arrow[ur, "\alpha_m"] \arrow[d, "e"] & \\
& \frac{1}{n}\mathbb{Z}, \arrow[uul, bend left, "\iota"] \arrow[uur, bend right, "\alpha_n"'] &
\end{tikzcd}$$

where ι, e be inclusion and W is any object which satisfies $\alpha_m = \alpha_n \circ e$.

Define $g : \mathbb{Q} \to W$ by setting $g(p/q) = \alpha_q$, then above diagram commutes.

To show the uniqueness of g, suppose $f : \mathbb{Q} \to W$ is another morphism which such that above diagram commute. Note that $\alpha_q = f \circ \iota$, we have $f(p/q) = \alpha_q(p/q) = g(p/q)$. Hence, $f = g$, and therefore $\mathbb{Q} = \varinjlim \frac{1}{n}\mathbb{Z}$.

(b) Morphism in category (the subsets of given set) given by inclusion, i.e., $A_i \to A_j$ if $A_i \subseteq A_j$. Let $\{A_i\}$ be a family of subsets. We consider the following two commutative diagrams.

$$\begin{tikzcd}
\bigcup_i A_i \arrow[rr, dashed, "\theta"] & & W \\
& A_j \arrow[ul, "\iota_j"'] \arrow[ur, "\alpha_j"] \arrow[d, "i"] & \\
& A_k \arrow[uul, bend left, "\iota_k"] \arrow[uur, bend right, "\alpha_k"'] &
\end{tikzcd}
\qquad
\begin{tikzcd}
\bigcap_i A_i \arrow[dr, "e_j"] \arrow[ddr, bend right, "e_k"'] & & V \arrow[ll, dashed, "\varphi"'] \arrow[dl, "\beta_j"'] \arrow[ddl, bend left, "\beta_k"] \\
& A_j \arrow[d, "i"] & \\
& A_k &
\end{tikzcd}$$

Define $\theta : \bigcup_i A_i \to W$ by setting $\theta(a) = \alpha_t(a)$, where $a \in A_t$. Since $a \in \bigcup_i A_i$, $a \in A_t$ for some t, hence, θ is well defined and such that left diagram commutes. Define $\varphi : V \to \bigcap_i A_i$ by setting $\varphi(v) = \beta_t$ for all $t \in \mathscr{I}$. Since $\beta_k = i \circ \beta_j$, φ is well-defined and such that right diagram commutes.

To show the uniqueness of θ, suppose there is $\theta' : \bigcup_i A_i \to W$ such that left diagram commutes. Let $a_n \in \bigcup_i A_i$, then exists n such that $a_n \in A_n$. Hence, $\theta'(a_n) = \theta' \circ \iota_n(a_n) = \alpha_n(a_n) = \theta \circ \iota_n(a_n) = \theta(a_n)$, that is, $\theta = \theta'$.

To show the uniqueness of φ, suppose there is $\varphi' : V \to \bigcap_i A_i$ such that right diagram commutes. Let $v \in V$, $\varphi'(v) = e_n \circ \varphi(v) = \beta_n(v) = \varphi(v)$, hence, $\varphi = \varphi'$.

By above discussion, we have

$$\bigcup_i A_i = \varinjlim_{\mathscr{I}} A_i, \qquad \bigcap_i A_i = \varprojlim_{\mathscr{I}} A_i.$$

$\square$

Remark 2.20 Colimit do not always exist, but there are two useful large classes of examples for which they do.

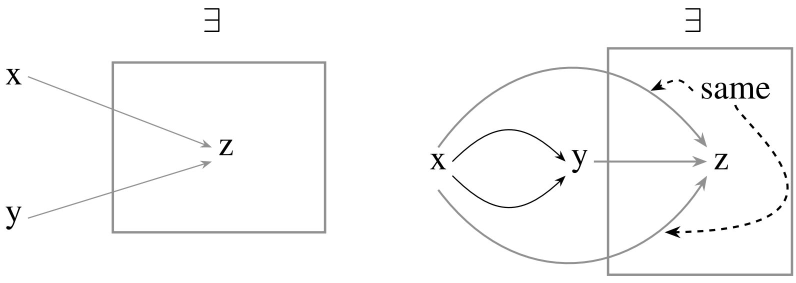
Definition 2.3.4 (Filtered)

*A nonempty partially ordered set $(S, \geq)$ is **filtered** (or is said to be a **filtered set**) if for each $x, y \in S$, there is a z such that $x \geq z$ and $y \geq z$. More generally, a nonempty category $\mathscr{I}$ is filtered if:*

(i) for each $x, y \in \mathscr{I}$, there is a $z \in \mathscr{I}$ and arrows $x \to z$ and $y \to z$,

(ii) for every two arrows $u : x \to y$ and $v : x \to y$, there is an arrow $w : y \to z$ such that $w \circ u = w \circ v$.

Remark 2.21 Other terminologies are also commonly used, such as "directed partially ordered set" and "fixed index category", respectively.



(i) (ii)

Figure 2.1: A filtered category (pictorial definition)

✍ **Exercise 2.27** Suppose $\mathscr{I}$ is filtered. Recall the symbol $\coprod$ for disjoint union of sets. Show that any diagram in **Sets** indexed by $\mathscr{I}$ has the following, with the obvious maps to it, as a colimit:

$$\left\{(a_i, i) \in \coprod_{i \in \mathscr{I}} A_i\right\} \Big/ \left(\begin{array}{l}(a_i, i) \sim (a_j, j) \text{ if and only if there are } f : A_i \to A_k \text{ and} \\ g : A_j \to A_k \text{ in the diagram for which } f(a_i) = g(a_j) \text{ in } A_k\end{array}\right)$$

Proof Say

$$\mathscr{S} = \left\{(a_i, i) \in \coprod_{i \in \mathscr{I}} A_i\right\} \Big/ \left(\begin{array}{l}(a_i, i) \sim (a_j, j) \text{ if and only if there are } f : A_i \to A_k \text{ and} \\ g : A_j \to A_k \text{ in the diagram for which } f(a_i) = g(a_j) \text{ in } A_k\end{array}\right).$$

Then for $A_j \to A_k$, the following diagram commutes,

$$\begin{tikzcd}
A_j \arrow[r] \arrow[d, "e_j"'] & A_k \arrow[dl, "e_k"] \\
\mathscr{S},
\end{tikzcd}$$

where e_j, e_k are embedding.

Cosider ther following diagram.

$$\begin{tikzcd}
\mathscr{S} \arrow[rr, dashed, "\theta"] & & W \\
& A_j \arrow[ul, "e_j"'] \arrow[ur, "\alpha_j"] \arrow[d, "g"] & \\
& A_k \arrow[uul, bend left, "e_k"] \arrow[uur, bend right, "\alpha_k"'] &
\end{tikzcd}$$

Define $\theta : W \to \mathscr{S}$ by setting $\theta(a_n, n) = \alpha_n(a_n)$. We need to show that θ is well-defined. Let $(a_i, i) \sim (a_j, j) \in \coprod_{i \in \mathscr{I}}$, then there are $f : A_i \to A_k$ and $g : A_j \to A_k$ in the diagram for which $f(a_i) = g(a_j)$. Note that $\alpha_k \circ f = \alpha_i$ and $\alpha_k \circ g = \alpha_j$, $\alpha_i(a_i) = \alpha_k \circ f(a_i) = \alpha_k \circ g(a_j) = \alpha_j(a_j)$, hence $\theta(a_i) = \theta(a_j)$. Hence, θ is well-defined and such that above diagram commutes.

To show the uniqueness of θ, suppose there is ψ such that above diagram is commutes. Let $(a_n, n) \in \coprod_{i \in \mathscr{I}} A_i$, then $\psi(a_n, n) = \psi \circ e_n(a_n) = \alpha_n(a_n) = \theta(a_n, n)$. Hence, $\psi = \theta$. □

For example, in Example 2.29 , each element of the colimit is an element of something upstairs, but you can't say in advance what it is an element of. For instance, $\frac{17}{125}$ is an element of the $5^{-3}\mathbb{Z}$ (or $5^{-4}\mathbb{Z}$, or later ones), but not $5^{-2}\mathbb{Z}$.

This idea applies to many categories whose objects can be interpreted as sets with additional structure (such as abelian groups, A-modules, groups, etc).

✍ **Exercise 2.28** The colimit $\varinjlim M_i$ in the category of A-modules $\mathbf{Mod}_A$ can be described as follows. The set underlying $\varinjlim M_i$ is defined as in Exercise 2.27. To add the elements $m_i \in M_i$ and $m_j \in M_j$, choose an $l \in \mathscr{I}$ with arrows $u : i \to l$ and $v : j \to l$, and then define the sum of m_i and m_j to be $F(u)(m_i) + F(v)(m_j) \in M_l$. The element $m_i \in M_i$ is 0 if and only if there is some arrow $u : i \to k$ for which $F(u)(m_i) = 0$, i.e., if it becomes 0 "later in the diagram". Last, multiplication by an element of A is defined in the obvious way.

Verify that the A-module described above is indeed the colimit.

Proof Let

$$\mathscr{S} = \left\{ (m_i, i) \in \prod_{i \in \mathscr{I}} M_i \right\} \Big/ \left(\begin{array}{l} (m_i, i) \sim (m_j, j) \text{ if and only if there are } f : M_i \to M_k \text{ and} \\ g : M_j \to M_k \text{ in the diagram for which } f(a_i) = g(a_j) \text{ in } M_k \end{array} \right).$$

We need to show that $\mathscr{S}$ indeed an A-module. Let $(m_i, i) \sim (m_{i'}, i')$, then there are $f : M_i \to M_k$, $g : M_{i'} \to M_k$ such that $f(m_i) = g(m_{i'})$.

(1) Addition is well-defined.

Notice that

$$m_i + m_j = f(m_i) + F(j \to k)(m_j) = g(m_{i'}) + F(j \to k)(m_j) = m_{i'} + m_j,$$

addition is independent of the choice of representatives m_i and $m_{i'}$.

Suppose we have another index l' with $u' : i \to l'$ and $v' : j \to l'$. Since $\mathscr{I}$ is filtered, exists $w : l \to k$ and $w' : l' \to k$ such that $w \circ u = w' \circ u'$ and $w \circ v = w' \circ v'$. Then we have

$$m_i + m_j = F(wu)(m_i) + F(wv)(m_j) = F(w'u')(m_i) + F(w'v')(m_j) = m_i + m_j,$$

which implies that addition is independent of the choice of l, and the choice of arrows u and v.

(2) Scalar multiplication is obviously well-defined.

Hence, $\mathscr{S}$, indeed, an A-module. Same as Exercise 2.27, $\varinjlim M_i = \mathscr{S}$. □

✍ **Exercise 2.29 (Localization as a colimit)** Generalize Exercise 2.26 (a) to interpret localization of an integral domain as a colimit over a filtered set: suppose S is a multipicative set of A, and interpret $S^{-1}A = \varinjlim A_s$ where the colimit is over $s \in S$, and in category of A-modules.

Proof In fact, $A_{s_1} \to A_{s_2}$ exists if $ks_1 = s_2$ for some $k \in A$. Consider the following diagram.

$$\begin{tikzcd}
S^{-1}A \arrow[rr, dashed, "\theta"] & & W \\
& A_{s_i} \arrow[ul, "\iota_i"] \arrow[ur, "f_i"'] \arrow[d, "e"] & \\
& A_{s_j} \arrow[uul, bend left, "\iota_j"] \arrow[uur, bend right, "f_j"'] &
\end{tikzcd}$$

Defined $\theta : S^{-1}A \to W$ by $\theta(\frac{a}{s_i^d}) = f_i(\frac{a}{s_i^d})$, then θ makes above diagram commutes. By the definition of θ, it is easy to see that θ is unique.

Hence, $S^{-1}A = \varinjlim A_s$. □

A variant of this construction works without the filtered condition, if you have another means of "connecting elements in different objects of your diagram". For example:

✍ **Exercise 2.30 (Colimits of A-modules without the filtered condition)** Suppose you are given a diagram of A-modules indexed by $\mathscr{I}$: $F : \mathscr{I} \to \mathbf{Mod}_A$, where we let $M_i = F(i)$. Show that the colimit is

$$\bigoplus_{i \in \mathscr{I}} M_i \Big/ \{m_i - F(i \to j)(m_i) \text{ for every } i \to j \text{ in } \mathscr{I}\}.$$

Proof Say

$$\mathscr{S} = \bigoplus_{i \in \mathscr{I}} M_i \Big/ \{m_i - F(n)(m_i) \text{ for every } n : i \to j \text{ in } \mathscr{I}\}.$$

Consider the following diagram.

$$\begin{tikzcd}
\mathscr{S} \arrow[rr, dashed, "\theta"] & & W \\
& M_i \arrow[ul, "\iota_i"'] \arrow[ur, "f_i"] \arrow[d, "F(i\to j)"] & \\
& M_j \arrow[uul, bend left, "\iota_j"] \arrow[uur, bend right, "f_j"'] &
\end{tikzcd}$$

Define $\theta : \mathscr{S} \to W$ by setting $\theta((m_k)_{k\in\mathscr{I}}) = f_i(m_i)$. θ is well-defined, since in $\mathscr{S}$, $m_i = F(i \to j)(m_i)$. By the definition of θ, the commmutativity and uniqueness is obvious.

Hence, $\mathscr{S} = \varinjlim M_i$. $\square$

Note *One useful thing to informally keep in mind is the following. In a category where the objects are "set-like", an element of a limit can be thought of as a family of elements of each object in the diagram, that are "compatible" (Exercise 2.25). And an elements of a colimit can be thought of as ("has a representative that is") an element of a single object in the diagram (Exercise 2.27). Even though the definitions of limit and colimit are the same, just with arrows reversed, these interpretations are quite different.*

Remark 2.22 In fact, colimits exist in the category of sets for all reasonable ("smal") index categories, but that won't matter to us.

2.4 Adjoints

Definition 2.4.1 (Adjoint)

*Two covariant functors $F : \mathscr{A} \to \mathscr{B}$ and $G : \mathscr{B} \to \mathscr{A}$ are **adjoint** if there is a natural bijection for all $A \in \mathscr{A}$ and $B \in \mathscr{B}$*

$$\tau_{AB} : \mathrm{Mor}_{\mathscr{B}}(F(A), B) \to \mathrm{Mor}_{\mathscr{A}}(A, G(B)).$$

*We say that (F, G) form an **adjoint pair**, and that F is **left-adjoint** to G (and G is **right-adjoint** to F). We say F is a **left adjoint** (and G is a **right adjoint**). By "natural" we mean the following. For all $f : A \to A'$ in $\mathscr{A}$, we require*

$$\begin{tikzcd}
\mathrm{Mor}_{\mathscr{B}}(F(A'), B) \arrow[r, "Ff^*"] \arrow[d, "\tau_{A'B}"'] & \mathrm{Mor}_{\mathscr{B}}(F(A), B) \arrow[d, "\tau_{AB}"] \\
\mathrm{Mor}_{\mathscr{A}}(A', G(B)) \arrow[r, "f^*"] & \mathrm{Mor}_{\mathscr{A}}(A, G(B))
\end{tikzcd}$$

to commute, where f^* is the map induced by $f : A \to A'$ and Ff^* is the map induced by $Ff : F(A) \to F(A')$. For all $g : B \to B'$ in $\mathscr{B}$ we want a similar commutative diagram to commute,

$$\begin{tikzcd}
\mathrm{Mor}_{\mathscr{B}}(F(A), B) \arrow[r, "g_*"] \arrow[d, "\tau_{AB}"'] & \mathrm{Mor}_{\mathscr{B}}(F(A), B') \arrow[d, "\tau_{AB'}"] \\
\mathrm{Mor}_{\mathscr{A}}(A, G(B)) \arrow[r, "Gg_*"] & \mathrm{Mor}_{\mathscr{A}}(A, G(B')),
\end{tikzcd}$$

where g_* is the map induced by $g : B \to B'$, and Gg_* is the map induced by $Gg : G(B) \to G(B')$.

✍ **Exercise 2.31** Show that the map τ_{AB} in Definition 2.4.1 has the following properties. For each A there is a map $\eta_A : A \to GF(A)$ so that for any $g : F(A) \to B$, the corresponding $\tau_{AB}(g) : A \to G(B)$ is given by the composition

$$A \xrightarrow{\eta_A} GF(A) \xrightarrow{Gg} G(B).$$

Similarly, there is a map $\varepsilon_B : FG(B) \to B$ for each B so that for any $f : A \to G(B)$, the corresponding map $\tau_{AB}^{-1}(f) : F(A) \to B$ is given by the composition

$$F(A) \xrightarrow{Ff} FG(B) \xrightarrow{\varepsilon_B} B.$$

Proof

(1) By Definition 2.4.1, we have the following diagram,

$$\begin{tikzcd} \mathrm{Mor}_{\mathscr{B}}(F(A),F(A)) \arrow[r, "g_*"] \arrow[d, "\tau_{A,F(A)}"'] & \mathrm{Mor}_{\mathscr{B}}(F(A),B) \arrow[d, "\tau_{A,B}"] \\ \mathrm{Mor}_{\mathscr{A}}(A,GF(A)) \arrow[r, "Gg_*"] & \mathrm{Mor}_{\mathscr{A}}(A,G(B)) \end{tikzcd}$$

then we have $\tau_{A,B} \circ g_* = Gg_* \circ \tau_{A,F(A)}$. Let $\mathrm{id}_{F(A)} \in \mathrm{Mor}_{\mathscr{B}}(F(A),F(A))$, then

$$\tau_{A,B} \circ g_*(\mathrm{id}_{F(A)}) = \tau_{A,B}(g \circ \mathrm{id}_{F(A)}) = \tau_{A,B}(g) = Gg \circ (\tau_{A,F(A)}(\mathrm{id}_{F(A)})).$$

Say $\eta_A = \tau_{A,F(A)}(\mathrm{id}_{F(A)}) : A \to GF(A)$, then we have $\tau_{A,B}(g) = Gg \circ \eta_A$.

(2) By Definition 2.4.1 again, we have the following diagram,

$$\begin{tikzcd} \mathrm{Mor}_{\mathscr{A}}(G(B),G(B)) \arrow[r, "f^*"] \arrow[d, "\tau^{-1}_{G(B),B}"'] & \mathrm{Mor}_{\mathscr{A}}(A,G(B)) \arrow[d, "\tau^{-1}_{A,B}"] \\ \mathrm{Mor}_{\mathscr{B}}(FG(B),B) \arrow[r, "Ff^*"] & \mathrm{Mor}_{\mathscr{B}}(F(A),B) \end{tikzcd}$$

then we have $\tau^{-1}_{A,B} \circ f^* = Ff^* \circ \tau^{-1}_{G(B),B}$. Let $\mathrm{id}_{G(B)} \in \mathrm{Mor}_{\mathscr{A}}(G(B),G(B))$, then

$$\tau^{-1}_{A,B} \circ f^*(\mathrm{id}_{G(B)}) = \tau^{-1}_{A,B}(f) = Ff^* \circ \tau^{-1}_{G(B),B}(\mathrm{id}_{G(B)}) = Ff(\tau^{-1}_{G(B),B}(\mathrm{id}_{G(B)})).$$

Say $\varepsilon_B = \tau^{-1}_{G(B),B}(\mathrm{id}_{G(B)})$, then we have $\tau^{-1}_{A,B}(f) = Ff \circ \varepsilon_B$.

□

Here is a key example of an adjoint pair.

Lemma 2.4.1

Suppose M, N, and P are A-modules (where A is a ring). There is a bijection $\mathrm{Hom}_A(M \otimes_A N, P) \leftrightarrow \mathrm{Hom}_A(M, \mathrm{Hom}_A(N,P))$.

Proof Let $f \in \mathrm{Hom}_A(M \otimes_A N, P)$. Define $\tau_{M,N,P} : \mathrm{Hom}_A(M \otimes_A N, P) \to \mathrm{Hom}_A(M, \mathrm{Hom}_A(N,P))$ (for simply, say $\tau = \tau_{M,N,P}$) as follow:

$$\tau : f \mapsto \tau(f), \tau(f) : m \mapsto \tau(f)_m, \tau(f)_m : n \mapsto f(m \otimes n).$$

We shall to τ is an A-module isomorphism.

(1) τ is an A-homomorphism.

Let $f, g \in \mathrm{Hom}_A(M \otimes_A N, P)$. The definition of $f + g$ gives, for all $a \in A$,

$$\begin{aligned} \tau(f+g)_m : n \mapsto & (f+g)(m \otimes n) \\ &= f(a \otimes b) + g(a \otimes b) \\ &= \tau(f)_a(b) + \tau(g)_a(b). \end{aligned}$$

Therefore, $\tau(f+g) = \tau(f) + \tau(g)$.

(2) τ is injective.
Suppose $\tau(f)_m = 0$ for all $m \in M$, then $0 = \tau(f)_m(n) = f(m \otimes n)$ for all $m \in M$ and $n \in N$. Hence, $f = 0$, which implies that τ is injective.

(3) τ is surjective.
If $F \in \mathrm{Hom}_A(M, \mathrm{Hom}_A(N, P))$ is an A-map, define $\varphi : M \times N \to P$ by $\varphi(m, n) = F_m(n)$. Now consider the diagram

$$\begin{tikzcd} M \times N \arrow[rr, "h"] \arrow[dr, "\varphi"'] & & M \otimes_A N \arrow[dl, dashed, "\widetilde{\varphi}"] \\ & P. & \end{tikzcd}$$

Since $F_m(n) = m \otimes n$, it is easy to see that φ is A-bilinear. By the universal property of tensor product, there exists unique A-homomorphism $\widetilde{\varphi} : M \otimes_A N \to P$ with $\widetilde{\varphi}(m \otimes n) = \varphi(m, n) = F_m(n)$ for all $m \in M$ and $n \in N$. Hence, $F = \tau(\widetilde{\varphi})$ and τ is surjective.

By above discussion, τ is an A-module isomorphism. □

Theorem 2.4.1 (Tensor-Hom adjunction)
$\square \otimes_A N$ and $\mathrm{Hom}_A(N, \square)$ are adjoint functors.

Proof Let $f : M \to M'$, $g : P \to P'$ in $\mathbf{Mod}_A$. We shall to show that the following diagram commute,

$$\begin{tikzcd} \mathrm{Hom}_A(M' \otimes_A N, P) \arrow[r, "(f \otimes_A N)^*"] \arrow[d, "\tau_{M',N,P}"'] & \mathrm{Hom}_A(M \otimes_A N, P) \arrow[r, "g_*"] \arrow[d, "\tau_{M,N,P}"] & \mathrm{Hom}_A(M \otimes_A N, P') \arrow[d, "\tau_{M,N,P'}"] \\ \mathrm{Hom}_A(M', \mathrm{Hom}(N, P)) \arrow[r, "f^*"] & \mathrm{Hom}_A(M, \mathrm{Hom}(N, P)) \arrow[r, "(g_*)_*"] & \mathrm{Hom}_A(M, \mathrm{Hom}(N, P')). \end{tikzcd}$$

where $(g_*)_* = (\mathrm{Hom}(N, f))_* = \mathrm{Hom}(M, \mathrm{Hom}(N, f))$ and $(f \otimes_A N)^* = \mathrm{Hom}_A(f \otimes \mathrm{id}_N, P)$.

Let $\varphi \in \mathrm{Hom}_A(M' \otimes_A N, P)$, we have

$$f^* \circ \tau_{M',N,P}(\varphi) = f^*(\tau_{M',N,P}(\varphi)) = \tau_{M',N,P}(\varphi) \circ f$$

and

$$\tau_{M,N,P} \circ (f \otimes_A N)^*(\varphi) = \tau_{M,N,P}(\varphi \circ (f \otimes \mathrm{id}_N)).$$

Let $m \in M$ and $n \in N$, by Lemma 2.4.1, we have $\tau_{M',N,P}(\varphi) \circ f(m) = \tau_{M',N,P}(\varphi)_{f(m)}$, and therefore

$$\tau_{M',N,P}(\varphi)_{f(m)}(n) = \varphi(f(m) \otimes n) = (\varphi \circ f \otimes \mathrm{id}_N)(m \otimes n) = \tau_{M,N,P}(\varphi \circ (f \otimes \mathrm{id}_N))_m(n),$$

which implies that $f^* \circ \tau_{M',N,P} = \tau_{M,N,P} \circ (f \otimes_A N)^*$.

Let $\psi \in \mathrm{Hom}_A(M \otimes_A N, P)$, we have

$$(g_*)_* \circ \tau_{M,N,P}(\psi) = g_* \circ \tau_{M,N,P}(\psi)$$

and

$$\tau_{M,N,P'} \circ g_*(\psi) = \tau_{M,N,P'}(g \circ \psi).$$

Let $m \in M$ and $n \in N$, by Lemma 2.4.1, we have

$$g_* \circ \tau_{M,N,P}(\psi)(m) = g_*(\tau_{M,N,P}(\psi)_m) = g \circ \tau_{M,N,P}(\psi)_m,$$

hence

$$g \circ \tau_{M,N,P}(\psi)_m(n) = g(\psi(m \otimes n)) = \tau_{M,N,P'}(g \circ \psi)_m(n),$$

which implies that $(g_*)_* \circ \tau_{M,N,P} = \tau_{M,N,P'} \circ g_*$.

By above dicussion, the diagram commutes, hence, $\square \otimes_A N$ and $\mathrm{Hom}_A(N, \square)$ are adjoint functors. □

Proposition 2.4.1 (This adjoint pair is very important!)

Suppose $B \to A$ is a morphism of rings. If M is an A-module, we can create a B-module M_B by considering it as a B-module. This gives a functor $\square_B : \mathbf{Mod}_A \to \mathbf{Mod}_B$. This functor is right-adjoint to $\square \otimes_B A$. In other words, there is a bijection

$$\mathrm{Hom}_A(N \otimes_B A, M) \cong \mathrm{Hom}_B(N, M_B)$$

functorial in both arguments.

Proof Say $f : B \to A$ is a ring homomorphism.

(1) There is a bijection $\mathrm{Hom}_A(N \otimes_B A, M) \leftrightarrow \mathrm{Hom}_B(N, M_B)$.

Let $\varphi \in \mathrm{Hom}_A(N \otimes_B A, M)$, define $\tau_{N,M} : \mathrm{Hom}_A(N \otimes_B A, M) \to \mathrm{Hom}_B(N, M_B)$ as follow:

$$\tau_{N,M} : \varphi \mapsto \tau_{N,M}(\varphi), \tau_{N,M}(\varphi) : n \mapsto \varphi(n \otimes_B 1_A),$$

define its inverse $\theta_{N,M_B} : \mathrm{Hom}_B(N, M_B) \to \mathrm{Hom}_A(N \otimes_B A, M)$ by setting

$$\theta_{N,M_B} : \psi \mapsto \theta_{N,M_B}(\psi), \theta_{N,M_B}(\psi) : n \otimes a \mapsto a\psi(n).$$

Since

$$\theta_{N,M_B} \circ \tau_{N,M}(\varphi)(n \otimes_B a) = a\tau_{N,M}(\varphi)(n) = a\varphi(n \otimes_B 1_A) = \varphi(n \otimes_B a)$$

and

$$\tau_{N,M} \circ \theta_{N,M_B}(\psi)(n) = \theta_{N,M_B}(\psi)(n \otimes_B 1_A) = \psi(n),$$

θ_{N,M_B} is, indeed, the inverse of $\tau_{N,M}$, and therefore $\tau_{N,M}$ is bijection.

(2) τ is natural isomorphism.

Consider the following diagram, we shall to show that this diagram is commutative.

$$\begin{tikzcd}
\mathrm{Hom}_A(N' \otimes_B A, M) \arrow[r, "\square\circ(h\otimes_B \mathrm{id}_A)"] \arrow[d, "\tau_{N',M}"'] & \mathrm{Hom}_A(N \otimes_B A, M) \arrow[r, "l\circ\square"] \arrow[d, "\tau_{N,M}"] & \mathrm{Hom}_A(N \otimes_B A, M') \arrow[d, "\tau_{N,M'}"] \\
\mathrm{Hom}_B(N', M_B) \arrow[r, "\square\circ h"] & \mathrm{Hom}_B(N, M_B) \arrow[r, "\widetilde{l}\circ\square"] & \mathrm{Hom}_B(N, M'_B),
\end{tikzcd}$$

where $h : N \to N'$ and $l : M \to M'$, $\widetilde{l} : M_B \to M'_B$ which looks l as B-module homomorphism.

(a) The diagram at the left hand side is commutative.

Let $\varphi \in \mathrm{Hom}_A(N' \otimes_B A, M)$, then $(\square \circ h) \circ \tau_{N',M}(\varphi) = \tau_{N',M}(\varphi) \circ h$. For any $n \in N$, we have

$$\tau_{N',M}(\varphi) \circ h(n) = \tau_{N',M}(\varphi)(h(n)) = \varphi(h(n) \otimes_B 1_A).$$

On the other hand, $\tau_{N,M} \circ (\square \circ (h \otimes_B \mathrm{id}_A))(\varphi) = \tau_{N,M}(\varphi \circ (h \otimes_B \mathrm{id}_A))$, then we have

$$\tau_{N,M}(\varphi \circ (h \otimes_B \mathrm{id}_A))(n) = (\varphi \circ (h \otimes_B \mathrm{id}_A))(n \times_B 1_A) = \varphi(h(n) \otimes_B 1_A).$$

Hence, $(\square \circ h) \circ \tau_{N',M} = \tau_{N,M} \circ (\square \circ (h \otimes_B \mathrm{id}_A))$, this implies that the diagram at left hand side is commutative.

(b) The diagram at the right hand side is commutative.

Let $\psi \in \mathrm{Hom}_A(N \otimes_B A, M)$, then $(\widetilde{l} \circ \square) \circ \tau_{N,M}(\psi)$. For any $n \in N$, we have

$$(\widetilde{l} \circ \square) \circ \tau_{N,M}(\psi)(n) = \widetilde{l} \circ \tau_{N,M}(\psi)(n) = \widetilde{l}(\psi(n \otimes_B 1_A)) = l(\psi(n \otimes_B 1_A)).$$

On the other hand, $\tau_{N,M'} \circ (l \circ \square)(\psi) = \tau_{N,M'}(l \circ \psi)$, for any n, we have

$$\tau_{N,M'}(l \circ \psi)(n) = l \circ \psi(n \otimes_B 1_A) = l(\psi(n \otimes_B 1_A)).$$

Hence, $(\widetilde{l} \circ \square) \circ \tau_{N,M} = \tau_{N,M'} \circ (l \circ \square)$, as we desired.

□

Remark 2.23 The maps η_A and ε_B of Exercise 2.31 are called the **unit** and **counit** of the adjunction. This leads to a different characterization of adjunction. Suppose functors $F:\mathscr{A}\to\mathscr{B}$ and $G:\mathscr{B}\to\mathscr{A}$ are given, along with natural transformations $\eta:\mathrm{id}_{\mathscr{A}}\to GF$ and $\varepsilon:FG\to\mathrm{id}_{\mathscr{B}}$ with the property that $G\varepsilon\circ\eta G=\mathrm{id}_G$ (for each $B\in\mathscr{B}$, the composition of $\eta_{G(B)}:G(B)\to GFG(B)$ and $G(\varepsilon_B):GFG(B)\to G(B)$ is the identity) and $\varepsilon F\circ F\eta=\mathrm{id}_F$. Then F is left-adjoint to G.

Proof Let $f:A\to A'$ in $\mathscr{A}$ and $g:B\to B'$ in $\mathscr{B}$. It is suffice to show that the following diagram is commutative,

$$\begin{tikzcd}
\mathrm{Mor}_{\mathscr{B}}(F(A'),B) \arrow[r, "Ff^*"] \arrow[d, "\tau_{A',B}"'] & \mathrm{Mor}_{\mathscr{B}}(F(A),B) \arrow[r, "g_*"] \arrow[d, "\tau_{A,B}"] & \mathrm{Mor}_{\mathscr{B}}(F(A),B') \arrow[d, "\tau_{A,B'}"] \\
\mathrm{Mor}_{\mathscr{A}}(A',G(B)) \arrow[r, "f^*"] & \mathrm{Mor}_{\mathscr{A}}(A,G(B)) \arrow[r, "Gg_*"] & \mathrm{Mor}_{\mathscr{A}}(A,G(B')),
\end{tikzcd}$$

where f^* induced by f, Ff^* induced by Ff, g_* induced by g, Gg_* induced by Gg, $\tau_{A,B}$ given by $\tau_{A,B}(\varphi)=G\varphi\circ\eta_A$, we give $\tau_{A'B}$ and $\tau_{A,B'}$.

(1) $\tau_{A,B}:\mathrm{Mor}_{\mathscr{B}}(F(A),B)\to\mathrm{Mor}_{\mathscr{A}}(A,G(B))$ is a bijection.

We give the inverse of $\tau_{A,B}$ directly, $\tau^{-1}_{A,B}(\psi)=\varepsilon_B\circ F\psi$. We shall to show that $\tau^{-1}_{A,B}$, indeed, is the inverse of $\tau_{A,B}$. Let $\varphi\in\mathrm{Mor}_{\mathscr{B}}(F(A),B)$, we have

$$\begin{aligned}
\tau^{-1}_{A,B}\circ\tau_{A,B}(\varphi)=&\tau^{-1}_{A,B}(G\varphi\circ\eta_A)=\varepsilon_B\circ(F(G\varphi\circ\eta_A))\\
=&\varepsilon_B\circ(F(G\varphi)\circ F(\eta_A))\\
=&\varepsilon_B\circ F(G\varphi)\circ F(\eta_A),
\end{aligned}\tag{2.15}$$

Since $\varepsilon:FG\to\mathrm{id}_{\mathscr{B}}$ is natural transformation, the following diagram commutes,

$$\begin{tikzcd}
FGF(A) \arrow[r, "FG\varphi"] \arrow[d, "\varepsilon_{F(A)}"'] & FG(B) \arrow[d, "\varepsilon_B"] \\
F(A) \arrow[r, "\varphi"] & B,
\end{tikzcd}$$

that is, $\varphi\circ\varepsilon_{F(A)}=\varepsilon_B\circ FG\varphi$. Hence, note that $\varepsilon F\circ F\eta=\mathrm{id}_F$,

$$\tau^{-1}_{A,B}\circ\tau_{A,B}(\varphi)=\varepsilon_B\circ F(G\varphi)\circ F(\eta_A)=\varphi\circ\varepsilon_{F(A)}\circ F(\eta_A)=\varphi\circ\mathrm{id}_{F(A)}=\varphi.$$

Similarly, for $\psi\in\mathrm{Mor}_{\mathscr{A}}(A,G(B))$, by the commuative diagram,

$$\begin{tikzcd}
A \arrow[r, "\psi"] \arrow[d, "\eta_A"'] & G(B) \arrow[d, "\eta_{G(B)}"] \\
GF(A) \arrow[r, "GF\psi"] & GFG(B),
\end{tikzcd}$$

we have $\tau_{A,B}\circ\tau^{-1}_{A,B}(\psi)=\psi$.

(2) Commutativity of diagram.

Let $\varphi\in\mathrm{Mor}_{\mathscr{B}}(F(A'),B)$, then

$$f^*\circ\tau_{A',B}(\varphi)=f^*\circ(G\varphi\circ\eta_{A'})=G\varphi\circ\eta_{A'}\circ f$$

and

$$\tau_{A,B}\circ Ff^*(\varphi)=\tau_{A,B}(\varphi\circ Ff)=G(\varphi\circ Ff)\circ\eta_A=G\varphi\circ GFf\circ\eta_A=G\varphi\circ\eta_{A'}\circ f.$$

Hence, the diagram at left hand side commutes.

By the same way, we can prove the diagram at right hand side is also commutative.

□

Here are some motivating examples:

Example 2.31 For those familiar with representation theory: Frobenius reciprocity may be understood in terms of adjoints. Suppose V is a finite-dimensional representation of a finite group G, and W is a representation of a subgroup $H < G$. Then induction and restriction are an adjoint pair $(\mathrm{Ind}_H^G, \mathrm{res}_H^G)$ between the category of G-modules and the category of H-modules.

Example 2.32 (Groupification of abelian semigroups) Getting an abelian group from an abelian semigroup. (An **abelian semigroup** is just like an abelian group, except we don't require an identity or an inverse. Morphisms of abelian semigroups are maps of sets preserving the binary operation. One example is the non-negative integers $\mathbb{Z}^{\geq 0} = \{0, 1, 2, \cdots\}$ under addition. Another is the positive integers $\mathbb{Z}^{+} = \{1, 2, 3, \cdots\}$ under addition. Yet another is the positive integers $\mathbb{Z}^{+}$ under multiplication.) From an abelian semigroup, we can create an abelian group. In our examples, from the nonnegative integers under addition $(\mathbb{Z}^{\geq \nvdash}, +)$, we create the integers $(\mathbb{Z}, +)$, and from the positive integers under multiplication $(\mathbb{Z}^{+}, \times)$, we create the positive rationals $(\mathbb{Q}^{+}, \times)$. Here is a formalization of that notion. A **groupification** of an abelian semigroup S is a map of abelian semigroups $\pi : S \to G$ such that G is an abelian group, and any map of abelian semigroups from S to an abelian group G' factors uniquely through G:

$$\begin{tikzcd} S \arrow[r, "\pi"] \arrow[dr] & G \arrow[d, dotted, "\exists!"] \\ & G' \end{tikzcd}$$

Exercise 2.32 (An Abelian group is groupified by itself) Show that if an abelian semigroup is already a group then the identity morphism is the groupification.

Proof Say G be an abelian group. Note that

$$\begin{tikzcd} G \arrow[r, "\mathrm{id}_G"] \arrow[dr, "\varphi"'] & G \arrow[d, dotted, "\varphi"] \\ & G', \end{tikzcd}$$

then we done. $\square$

Exercise 2.33 Construct the "groupification functor" H from the category of nonempty abelian semigroups to the category of abelian groups. One possible construction: given an abelian semigroup S, the elements of its groupification $H(S)$ are ordered pairs $(a, b) \in S \times S$, which may think of as $a - b$, with the equivalence that $(a, b) \sim (c, d)$ if $a + d + e = b + c + e$ for some $e \in S$. Describe addition in this group, and show that it satisfies the properties of an abelian group. Descride the abelian semigroup map $S \to H(S)$. Let F be the forgetful functor from the category of abelian groups $\mathbf{Ab}$ to the category of abelian semigroups. Show that H is left-adjoint to F.

Proof Say $\mathscr{S}$ be the category of nenempty abelian semigroups. We need describe addition in $H(S)$ such that $H(S)$ be a group. Let $(a, b) + (c, d) = (a + c, b + d)$, we need to check this addition is well-defined. Let $(a, b) \sim (a', b')$, then $a + b' + e = a' + b + e$, hence, $a + c + b' + d + e = a' + c + b + d + e$, that is, $(a + c, b + d) \sim (a' + c, b' + d)$, which implies that addition is well-defined. The identity in $H(S)$ is $(0, 0)$. The inverse of (a, b) in $H(S)$ is (b, a). Hence, $H(S)$ is an abelian group. Define the abelian semigroup map $\iota_S : S \to H(S)$ by setting $a \mapsto (a, 0)$.

Now, we shall to show that there is a bijection between $\mathrm{Mor}_{\mathbf{Ab}}(H(S), A)$ and $\mathrm{Mor}_{\mathscr{S}}(S, F(A))$. Define $\tau_{S,A} : \mathrm{Mor}_{\mathbf{Ab}}(H(S), A) \to \mathrm{Mor}_{\mathscr{S}}(S, F(A))$ by setting $\tau_{S,A} : \varphi \mapsto \tau_{S,A}(\varphi), \tau_{S,A}(\varphi) : a \mapsto \varphi(\iota_S(a))$. The inverse of $\tau_{S,A}$ defined by $\tau_{S,A}^{-1} : \psi \mapsto \tau_{S,A}^{-1}(\psi), \tau_{S,A}^{-1}(\psi) : (a, b) \mapsto \psi(a) - \psi(b)$. We shall to show that $\tau_{S,A}^{-1}$

indeed the inverse of $\tau_{S,A}$. Let $\varphi \in \mathrm{Mor}_{\mathbf{Ab}}(H(S), A)$, then

$$\tau_{S,A}^{-1} \circ \tau_{S,A}(\varphi)(a,b) = \tau_{S,A}(\varphi)(a) - \tau_{S,A}(\varphi)(b) = \varphi(a,0) - \varphi(b,0) = \varphi(a,b),$$

hence, $\tau_{S,A}^{-1} \circ \tau_{S,A} = \mathrm{id}_{\mathrm{Mor}_{\mathbf{Ab}}(H(S),A)}$. Let $\psi \in \mathrm{Mor}_{\mathscr{S}}(S, F(A))$, then

$$\tau_{S,A} \circ \tau_{S,A}^{-1}(\psi)(a) = \tau_{S,A}^{-1}(\psi)(a,0) = \psi(a) - \psi(0) = \psi(a),$$

hence, $\tau_{S,A} \circ \tau_{S,A}^{-1} = \mathrm{id}_{\mathrm{Mor}_{\mathscr{S}}(S,F(A))}$. By above discussion, $\tau_{S,A}$ is a bijection.

Let $f : S \to S'$ in $\mathscr{S}$, and $g : A \to A'$ in $\mathbf{Ab}$. Consider the following diagram,

$$\begin{array}{ccccc}
\mathrm{Mor}_{\mathbf{Ab}}(H(S'), A) & \xrightarrow{Hf^*} & \mathrm{Mor}_{\mathbf{Ab}}(H(S), A) & \xrightarrow{g_*} & \mathrm{Mor}_{\mathbf{Ab}}(H(S), A') \\
{\scriptstyle \tau_{S',A}}\downarrow & & \downarrow{\scriptstyle \tau_{S,A}} & & \downarrow{\scriptstyle \tau_{S,A'}} \\
\mathrm{Mor}_{\mathscr{S}}(S', F(A)) & \xrightarrow{f^*} & \mathrm{Mor}_{\mathscr{S}}(S, F(A)) & \xrightarrow{Fg_*} & \mathrm{Mor}_{\mathscr{S}}(S, F(A')),
\end{array}$$

where f^* induced by f, Ff^* induced by Ff, g_* induced by g, Fg_* induced by Fg. We shall to show that this diagram is commutative.

Let $\varphi \in \mathrm{Mor}_{\mathbf{Ab}}(H(S'), A)$, then $f^* \circ \tau_{S',A}(\varphi) = \tau_{S',A}(\varphi) \circ f$. For all $s \in S$, we have

$$\tau_{S',A}(\varphi) \circ f(s) = \varphi(f(s), 0).$$

On the other hand, $\tau_{S,A} \circ Hf^*(\varphi) = \tau_{S,A}(\varphi \circ Hf)$. For all $s \in S$, we have

$$\tau_{S,A}(\varphi \circ Hf)(s) = \varphi(Hf(s,0)) = \varphi(f(s),0).$$

Hence, $f^* \circ \tau_{S',A} = \tau_{S,A} \circ Hf^*$.

Similarly, we can shall that $\tau_{S,A'} \circ g_* = Fg_* \circ \tau_{S,A}$. Hence, H is left-adjoint to F. $\square$

Remark 2.24 We have a full subcategory. We want to "project" from the category to the subcategory. We have

$$\mathrm{Mor}_{\text{category}}(S, H) = \mathrm{Mor}_{\text{subcategory}}(G, H)$$

automatically; thus we are describing the left adjoint to the forgetful functor. How the argument worked: we constructed something which was in the smaller category, which automatically satisfies the universal property.

✍ **Exercise 2.34** Suppose A is a ring, and S is a multiplicative subset. Then $S^{-1}A$-modules are a full subcategory of the category of A-modules (via the obvious inclusion $\mathbf{Mod}_{S^{-1}A} \hookrightarrow \mathbf{Mod}_A$). Then $\mathbf{Mod}_A \to \mathbf{Mod}_{S^{-1}A}$ can be interpreted as an adjoint to the forgetful functor $\mathbf{Mod}_{S^{-1}A} \to \mathbf{Mod}_A$. State and prove the correct statement.

Proof Every $S^{-1}A$-module is an A-module, and this is an injective map, so we have a forgetful functor $F : \mathbf{Mod}_{S^{-1}A} \to \mathbf{Mod}_A$. In fact this is a fully faithful functor: it is injective on objects, and the morphisms between any two $S^{-1}A$-modules as A-modules are just the same when they are considered as $S^{-1}A$-modules. Define the localization functor $S^{-1} : \mathbf{Mod}_A \to \mathbf{Mod}_{S^{-1}A}$ by setting $S^{-1}(u : M \to N) = S^{-1}u : S^{-1}M \to S^{-1}N$, where $S^{-1}u : m/s \mapsto u(m)/s$. We shall to show that $\mathrm{Mor}(S^{-1}M, N)$ are in natural bijection with $\mathrm{Mor}(M, F(N))$.

Define $\tau_{M,N} : \mathrm{Mor}(S^{-1}M, N) \to \mathrm{Mor}(M, F(N))$ by setting $\tau_{M,N} : \varphi \mapsto \varphi|_M$. Since the morphisms between any two $S^{-1}A$-modules as A-modules are just the same when they are considered as $S^{-1}A$-modules, $\tau_{M,N}$ is well-defined. We give the inverse of $\tau_{M,N}$ directly, $\tau_{M,N}^{-1} : \psi \mapsto \tau_{M,N}^{-1}(\psi), \tau_{M,N}^{-1}(\psi) : m/s \mapsto \psi(m)/s$. We need to check that $\tau_{M,N}^{-1}$ indeed the inverse of $\tau_{M,N}$. Let $\varphi \in \mathrm{Mor}(S^{-1}M, N)$, then

$$\tau_{M,N}^{-1} \circ \tau_{M,N}(\varphi)(m/s) = \tau_{M,N}^{-1}(\varphi|_M)(m/s) = \varphi|_M(m)/s.$$

Let $\psi \in \mathrm{Mor}(M, F(N))$, then

$$\tau_{M,N} \circ \tau_{M,N}^{-1}(\psi)(m) = \tau_{M,N}^{-1}(\psi)|_M(m) = \psi(m).$$

Hence, there is a bijection $\tau_{M,N} : \mathrm{Mor}(S^{-1}M, N) \to \mathrm{Mor}(M, F(N))$.

To show the naturalness, let $f : M \to M'$ and $g : N \to N'$. Now, we consider the following diagram,

$$\begin{tikzcd} \mathrm{Mor}(S^{-1}M', N) \arrow[r, "S^{-1}f^*"] \arrow[d, "\tau_{M',N}"'] & \mathrm{Mor}(S^{-1}M, N) \arrow[r, "g_*"] \arrow[d, "\tau_{M,N}"] & \mathrm{Mor}(S^{-1}M, N') \arrow[d, "\tau_{M,N'}"] \\ \mathrm{Mor}(M', F(N)) \arrow[r, "f^*"] & \mathrm{Mor}(M, F(N)) \arrow[r, "Fg_*"] & \mathrm{Mor}(M, F(N')), \end{tikzcd}$$

where f^* induced by f, g_* induced by g, $S^{-1}f^*$ induced by $S^{-1}f$, Fg_* induced by Fg. We shall to show this diagram commutes.

Let $\varphi \in \mathrm{Mor}(S^{-1}M', N)$, then $f^* \circ \tau_{M',N}(\varphi) = \tau_{M',N}(\varphi) \circ f$. For all $m \in M$,

$$\tau_{M',N}(\varphi)(f(m)) = \varphi|_{M'}(f(m)).$$

On the other hand, $\tau_{M,N} \circ S^{-1}f^*(\varphi) = S^{-1}f^*(\varphi)|_M = (\varphi \circ S^{-1}f)|_M$. For all $m \in M$,

$$(\varphi \circ S^{-1}f)|_M(m) = \varphi_{M'}(f(m)).$$

Hence, the diagram at left hand side is commutative.

Let $\psi \in \mathrm{Mor}(S^{-1}M, N)$, then $\tau_{M,N'} \circ g_*(\psi) = \tau_{M,N'}(g \circ \psi)$. For all $m \in M$,

$$\tau_{M,N'}(g \circ \psi)(m) = (g \circ \psi)|_M(m) = g(\psi|_M(m)).$$

On the other hand, $Fg_* \circ \tau_{M,N}(\psi) = Fg \circ \psi|_M$, then we have

$$Fg \circ \psi|_M(m) = g(\psi|_M(m)).$$

Hence, the diagram at right hand side is commutative. $\square$

Table 2.1 gives most of the adjoints that will be come up for us.

Situation	Category $\mathscr{A}$	Category $\mathscr{B}$	Left adjoint $F : \mathscr{A} \to \mathscr{B}$	Right adjoint $G : \mathscr{B} \to \mathscr{A}$
A-modules	$\mathbf{Mod}_A$	$\mathbf{Mod}_A$	$\square \otimes_A N$	$\mathrm{Hom}_A(N, \square)$
Ring maps $B \to A$	$\mathbf{Mod}_B$	$\mathbf{Mod}_A$	$\square \otimes_B A$ (extension of scalars)	$M \mapsto M_B$ (restriction of scalars)
(Pre)Sheaves on a topological space X	Presheaves on X	Sheaves on X	Sheafification	Forgetful
(Semi)Groups	Semigroups	Groups	Groupification	Forgetful
Sheaves, $\pi : X \to Y$	Sheaves on Y	Sheaves on X	π^{-1}	π_*
Sheaves of abelian groups or $\mathscr{O}$-modules, open embeddings $\pi : U \hookrightarrow Y$	Sheaves on U	Sheaves on Y	$\pi_!$	π^{-1}
Quasicoherent sheaves, $\pi : X \to Y$	$\mathbf{QCoh}_Y$	$\mathbf{QCoh}_X$	π^*	π_*
Ring maps $B \to A$	$\mathbf{Mod}_A$	$\mathbf{Mod}_B$	$M \mapsto M_B$ (restriction of scalars)	$N \mapsto \mathrm{Hom}_B(A, N)$
Quasicoherent sheaves, affine $\pi : X \to Y$	$\mathbf{QCoh}_X$	$\mathbf{QCoh}_Y$	π_*	$\pi^{!?}$

Table 2.1: Some important adjoint pairs

Remark 2.25 If (F, G) is an adjoint pair of functors, then F commutes with colimits, and G commutes with limits. Also, limits commute with limits and colimits commute with colimits.

2.5 An introduction to abelian categories

2.5.1 Abelian category

We first define the notion of **additive category**. We will use it only as a stepping stone to the notion of an abelian category.

Definition 2.5.1 (Additive category)

*A category $\mathscr{C}$ is said to be **additive** if it satisfies the following properties.*

(1) $\operatorname{Mor}(A, B)$ *is an (additive) abelian group for every* $A, B \in \operatorname{obj}(\mathscr{C})$,

(2) the distributive laws hold: given morphisms

$$X \xrightarrow{a} A \overset{f}{\underset{g}{\rightrightarrows}} B \xrightarrow{b} Y,$$

where X and $Y \in \operatorname{obj}(\mathscr{C})$, then

$$b(f+g) = bf + bg \quad \text{and} \quad (f+g)a = fa + ga,$$

(3) $\mathscr{C}$ has a zero object, denoted 0. (recall that a zero object is an object that is both initial and final object),

(4) $\mathscr{C}$ has finite products and finite coproducts: for all objects A, B in $\mathscr{C}$, both $A \times B$ and $A \coprod B$ exist in $\operatorname{obj}(\mathscr{C})$.

Note *In an additive category, the morphism are often called **homomorphisms**, and* Mor *is denoted by* Hom. *In fact, this notation* Hom *is a good indication that you're working in an additive category.*

Definition 2.5.2 (Additive functor)

*If $\mathscr{C}$ and $\mathscr{D}$ are additive categories, a functor $T : \mathscr{C} \to \mathscr{D}$ (of either variance) is **additive** if, for all A, B and all $f, g \in \operatorname{Hom}(A, B)$, we have*

$$T(f+g) = Tf + Tg,$$

that is, the function $\operatorname{Hom}_{\mathscr{C}}(A, B) \to \operatorname{Hom}_{\mathscr{D}}(TA, TB)$, given by $f \mapsto Tf$, is a homomorphism of abelian groups.

Remark 2.26 If T is an additive functor, then $T(0) = 0$, where 0 is either a zero object or a zero morphism.

Proof Note that $T(0) = T(0+0) = T(0) + T(0)$, $T(0) = 0$, where 0 is 0-homomorphsim.

Let Z be a zero object. Since Z is both initial and final object, $\mathrm{id}_Z = 0_Z$, hence, $\mathrm{id}_{T(Z)} = 0_{T(Z)}$, which implies that $T(Z) = 0$. □

Lemma 2.5.1

Let $\mathscr{C}$ be an additive category, and let $M, A, B \in \operatorname{obj}(\mathscr{C})$. Then $M \cong A \times B$ if and only if there are morphisms $i : A \to M$, $j : B \to M$, $p : M \to A$, and $q : M \to B$ such that

$$pi = \mathrm{id}_A, qj = \mathrm{id}_B, pj = 0, qi = 0, \text{ and } ip + jq = \mathrm{id}_M .$$

Moreover, $A \times B$ is also a coproduct with injections i and j, and so

$$A \times B \cong A \coprod B.$$

Proof

"$\Rightarrow$": Since $A \times B$ is product, M is a product, then we have the following commutative diagrams.

$$\begin{tikzcd}
B \arrow[rrrdd, bend left, "\mathrm{id}_B"] \arrow[rrdd, dashed, "j"] \arrow[rrddd, bend right, "0"'] & & & \\
& A \arrow[rrd, bend left, "0"] \arrow[rd, dashed, "i"] \arrow[rdd, bend right, "\mathrm{id}_A"'] & & \\
& & M \arrow[r, "q"] \arrow[d, "p"] & B \\
& & A &
\end{tikzcd}$$

Hence, $p \circ i = \mathrm{id}_A$, $q \circ j = \mathrm{id}_B$, $pj = 0$, $qi = 0$. Note that both $ip + jq$ and id_M agree with the following commutative diagram.

$$\begin{tikzcd}
M \arrow[rd, dashed] \arrow[rrd, bend left] \arrow[rdd, bend right] & & \\
& M \arrow[r, "q"] \arrow[d, "p"] & B \\
& A &
\end{tikzcd}$$

By the universal property of product, we have $ip + jq = \mathrm{id}_M$.

"$\Leftarrow$": If there are morphisms $i : A \to M$, $j : B \to M$, $p : M \to A$, and $q : M \to B$ such that

$$pi = \mathrm{id}_A, qj = \mathrm{id}_B, pj = 0, qi = 0, \text{ and } ip + jq = \mathrm{id}_M\,.$$

It suffice to show that M is a product. Consider the following diagram.

$$\begin{tikzcd}
W \arrow[rd, dashed, "\theta"] \arrow[rrd, "g"] \arrow[rdd, "f"'] & & \\
& M \arrow[r, "q"'] \arrow[d, "p"] & B \\
& A &
\end{tikzcd}$$

We need to construct $\theta : W \to M$ such that above diagram commutes. Let $\theta = i \circ f + j \circ g$, by condition, above diagram commutes. To show uniqueness, if there is another ψ such that diagram commutes, we have $\psi = \mathrm{id}_M\, \psi = ip\psi + jq\psi = if + jg = \theta$. Hence, M is product, and therefore $M \cong A \times B$.

"$\cong$": Consider the following diagram.

$$\begin{tikzcd}
A \times B \arrow[r, "q"] \arrow[d, "p"'] & B \arrow[d, "j"] \arrow[rd, bend left, "f_2"] & \\
A \arrow[r, "i"] \arrow[rr, bend right, "f_1"'] & A \times B \arrow[r, dashed, "\eta"] & W
\end{tikzcd}$$

We need to construct a morphism $\eta : A \times B \to W$ such that above diagram commutes. Let $\eta = f_1 p + f_2 q$, then

$$\eta \circ i = f_1 pi + f_2 qi = f_1$$

and

$$\eta \circ j = f_1 pj + f_2 qj = f_2,$$

which make above diagram commuts. To show the uniqueness of η, suppose that exists η' such that above diagram commutes. Since $\eta' = \eta' \circ \mathrm{id}_{A\times B} = \eta' \circ (ip + jq) = f_1 p + f_2 q = \eta$, η is unique. Hence, $A \times B$ satisfies the universal property of coproduct, and therefore

$$A \times B \cong A \coprod B.$$

$\square$

Remark 2.27 It is a consequence of the definition of additive category that finite products are also finite coproducts (i.e. sums). The symbol $\oplus$ is used for this notion.

Proposition 2.5.1

If $\mathscr{C}$ and $\mathscr{D}$ are additive categories and $T : \mathscr{C} \to \mathscr{D}$ is an additive functor of either variance, then $T(A \oplus B) \cong T(A) \oplus T(B)$ for all $A, B \in \mathrm{obj}(\mathscr{C})$.

Proof Let $i : A \to A \oplus B$, $j : B \to A \oplus B$, $p : M \to A$, and $q : M \to B$, then we have

$$pi = \mathrm{id}_A, qj = \mathrm{id}_B, pj = 0, qi = 0, \text{ and } ip + jq = \mathrm{id}_M\,.$$

Then we have $T(i) : T(A) \to T(A \oplus B)$, $T(j) : T(B) \to T(A \oplus B)$, $T(p) : T(M) \to T(A)$, and $T(q) : T(M) \to T(B)$, since T is an aditive functor, we have

$$T(p)T(i) = \mathrm{id}_{T(A)}, T(q)T(j) = \mathrm{id}_{T(B)}, T(P)T(j) = 0, T(q)T(i) = 0, \text{ and } T(i)T(p) + T(j)T(q) = \mathrm{id}_{T(M)}\,.$$

By Lemma 2.5.1, we have $T(A \oplus B) \cong T(A) \oplus T(B)$. $\square$

One motivation for the name 0-object is that the 0-morphism in the abelian group $\mathrm{Hom}(A, B)$ is the composition $A \to 0 \to B$. (We also remark that the notion of 0-morphism thus makes sense in any category with a 0-obeject.)

Note *A cleaner axiomatization of additive categories that makes clear that the abelian group structure of $\mathrm{Mor}(A, B)$ is intrinsic to the category itself is the following.*

A0. *$\mathscr{C}$ has a zero object.*

A1. *$\mathscr{C}$ has products of any two objects, and coproducts of any two objects.*
By the universal property of product and coproduct, we have natural morphisms $\varphi_{A,B} : A \coprod B \to A \times B$.

A2. *$\varphi_{A,B}$ is an isomorphism.*
This allows us to define a binary operation on $\mathrm{Mor}(A, B)$, with $f + g$ (for $f, g \in \mathrm{Mor}(A, B)$) defined by the composition

$$A \xrightarrow{(f,g)} B \times B \xrightarrow{\varphi_{B,B}^{-1}} B \coprod B \longrightarrow B$$

*where the last map is the "codiagonal" defined by universal property of coproduct. A little work shows that this endows $\mathrm{Mor}(A, B)$ with the structure of a **commutative monoid**, i.e, an abelian semigroup with identity. The identity is the composition $A \to 0 \to B$.*

A3. *This commutative monoid $\mathrm{Mor}(A, B)$ is an abelian group.*

Example 2.33 The category of A-modules $\mathbf{Mod}_A$ is clearly an additive category, but it has even more structure, which we now formalize as an example of an abelian category.

Definition 2.5.3 (Kernel and cokernel)

*Let $\mathscr{C}$ be a category with a 0-object (and thus 0-morphisms). A **kernel** of a morphism $f : B \to C$ is defined to be a map $i : A \to B$ such that $f \circ i = 0$, and that is universal with respect to this property.*

Diagramatically:

$$\begin{tikzcd} Z \arrow[d, dashed, "\exists!"'] \arrow[dr, "g"] \arrow[drr, "0"] & & \\ A \arrow[r, "i"] \arrow[rr, bend right, "0"'] & B \arrow[r, "f"] & C \end{tikzcd}$$

Hence it is unique up to unique isomorphism by universal property. The kernel is written $\operatorname{Ker} f \to B$.

A ***cokernel*** *of a morphism* $f : B \to C$ *is defined to be a map* $\pi : C \to D$ *such that* $\pi \circ f = 0$*, and that is universal with respect to this property. Diagramatically:*

$$\begin{tikzcd} B \arrow[r, "f"] \arrow[rr, bend left, "0"] \arrow[drr, "0"'] & C \arrow[r, "\pi"] \arrow[dr, "h"] & D \arrow[d, dashed, "\exists!"] \\ & & Y \end{tikzcd}$$

Hence it is unique up to unique isomorphism by universal property. The cokernel is witten $C \to \operatorname{Coker} f$.

Remark 2.28

(1) Note that the kernel is not just an object, it is a morphism of an object to B. In practice, the term is often applied to just the object, and the intended interpretation is clear from the context.

(2) The kernel of $f : B \to C$ is the limit of the diagram

$$\begin{tikzcd} & 0 \arrow[d] \\ B \arrow[r, "f"] & C. \end{tikzcd}$$

(3) The cokernel of $f : B \to C$ is the colimit of the diagram

$$\begin{tikzcd} B \arrow[r, "f"] \arrow[d] & C \\ 0. & \end{tikzcd}$$

Definition 2.5.4 (Subobject and quotient object)

If $i : A \to B$ *is a monomorphism, then we say that* A *is a* ***subobject*** *of* B*, where the map* i *is implicit.*

If $\pi : A \to B$ *is epimorphism, then we say that* B *is a* ***quotient object*** *of* A*, where the map* π *is implicit.*

Definition 2.5.5 (Abelian category)

An ***abelian category*** *is an additive category satisfying three additional properties.*

(1) Every map has a kernel and cokernel.

(2) Every monomorphism is the kernel of its cokernel.

(3) Every epimorphism is the cokernel of its kernel.

Remark 2.29 It is a nonobvious (and imprecisely stated) fact that every property you want to be ture about kernals, cokernals, etc. follows from these three.

Definition 2.5.6 (Image)

Let $f : A \to B$ be a morphism in an abelian category, and let $\operatorname{Coker} f$ *be* $\tau : B \to C$ *for some object* C*. Then its* ***image*** *is*

$$\operatorname{Im} f = \operatorname{Ker}(\operatorname{Coker} f) = \operatorname{Ker} \tau.$$

In more suggestive notation,

$$\operatorname{Im}(A \to B) = \operatorname{Ker}(\operatorname{Coker}(A \to B)) = \operatorname{Ker} \tau.$$

Remark 2.30 $\operatorname{Ker}(\operatorname{Coker}(\operatorname{Ker}(f))) = \operatorname{Ker}(f)$, $\operatorname{Coker}(\operatorname{Ker}(\operatorname{Coker}(f))) = \operatorname{Coker}(f)$.

Proof

(1) $\operatorname{Ker}(\operatorname{Coker}(\operatorname{Ker}(f))) = \operatorname{Ker}(f)$.

Say $k_0 : \operatorname{Ker}(f) \to A$, $k : \operatorname{Ker}(\operatorname{Coker}(\operatorname{Ker}(f))) \to A$, and $c : A \to \operatorname{Coker}(\operatorname{Ker}(f))$. It suffice to show that there exists unique solution in following universal problem.

$$\begin{tikzcd}
X \arrow[d, dashed] \arrow[dr, "h"] \arrow[drr, "0"] & & \\
\operatorname{Ker}(\operatorname{Coker}(\operatorname{Ker}(f))) \arrow[r, "k"] \arrow[rr, bend right, "0"'] & A \arrow[r, "f"] & B
\end{tikzcd}$$

By the universal property of $\operatorname{Coker}(\operatorname{Ker}(f))$, there exists unique $\theta : \operatorname{Coker}(\operatorname{Ker}(f)) \to B$ such that the following diagram commute.

$$\begin{tikzcd}
\operatorname{Ker}(f) \arrow[r, "k_0"] \arrow[rr, bend left, "0"] \arrow[rrd, "0"'] & A \arrow[r, "c"] \arrow[rd, "f"] & \operatorname{Coker}(\operatorname{Ker}(f)) \arrow[d, dashed, "\theta"] \\
& & B
\end{tikzcd}$$

Then we have $f = \theta \circ c$. Note that $\operatorname{Ker}(\operatorname{Coker}(\operatorname{Ker}(f)))$ is the kernel of $\operatorname{Coker}(\operatorname{Ker}(f))$, $c \circ k = 0$, and therefore $f \circ k = \theta \circ c \circ k = 0$.

Consider the following diagram.

$$\begin{tikzcd}
X \arrow[d, dashed, "\varphi"'] \arrow[dr, "h"] \arrow[drr, "0"] \arrow[drrr, "0"] & & & \\
\operatorname{Ker}(\operatorname{Coker}(\operatorname{Ker}(f))) \arrow[r, "k"] \arrow[rr, bend right, "0"'] \arrow[rrr, bend right=40, "0"'] & A \arrow[r, "c"] & \operatorname{Coker}(\operatorname{Ker}(f)) \arrow[r, "\theta"] & B
\end{tikzcd}$$

By the universal property of $\operatorname{Ker}(\operatorname{Coker}(\operatorname{Ker}(f)))$, there exists $\varphi : X \to \operatorname{Ker}(\operatorname{Coker}(\operatorname{Ker}(f)))$ such that above diagram commutes, and note that φ agree with the definition of $\operatorname{Ker}(f)$, hence, $\operatorname{Ker}(\operatorname{Coker}(\operatorname{Ker}(f))) = \operatorname{Ker}(f)$.

(2) $\operatorname{Coker}(\operatorname{Ker}(\operatorname{Coker}(f))) = \operatorname{Coker}(f)$.

Say $c_0 : B \to \operatorname{Coker}(f)$, $k : \operatorname{Ker}(\operatorname{Coker}(f)) \to B$, and $c : B \to \operatorname{Coker}(\operatorname{Ker}(\operatorname{Coker}(f)))$. It suffice to show that there exists unique solution in following universal problem.

$$\begin{tikzcd}
A \arrow[r, "f"] \arrow[rr, bend left, "0"] \arrow[drr, "0"'] & B \arrow[r, "c"] \arrow[dr, "h"] & \mathrm{Coker}(\mathrm{Ker}(\mathrm{Coker}(f))) \arrow[d, dashed] \\
& & Y
\end{tikzcd}$$

By the universal property of $\mathrm{Ker}(\mathrm{Coker}(f))$, there exists unique $\theta : A \to \mathrm{Ker}(\mathrm{Coker}(f))$ such that the following diagram commute.

$$\begin{tikzcd}
A \arrow[d, dashed, "\theta"'] \arrow[dr, "f"] \arrow[drr, "0"] & & \\
\mathrm{Ker}(\mathrm{Coker}(f)) \arrow[r, "k"] \arrow[rr, bend right] & B \arrow[r, "c_0"] & \mathrm{Coker}(f)
\end{tikzcd}$$

Then we have $f = k \circ \theta$. Note that $\mathrm{Coker}(\mathrm{Ker}(\mathrm{Coker}(f)))$ is the cokernel of $\mathrm{Ker}(\mathrm{Coker}(f))$, $c \circ k = 0$, and therefore $c \circ f = c \circ k \circ \theta = 0$.

Consider the following diagram.

$$\begin{tikzcd}
A \arrow[r, "\theta"'] \arrow[rr, bend left, "0"] \arrow[rrr, bend left, "0"] \arrow[drrr, "0"'] & \mathrm{Ker}(\mathrm{Coker}(f)) \arrow[r, "k"'] \arrow[drr, "0"] & B \arrow[r, "c"'] \arrow[dr, "h"] & \mathrm{Coker}(\mathrm{Ker}(\mathrm{Coker}(f))) \arrow[d, dashed, "\psi"] \\
& & & Y
\end{tikzcd}$$

By the universal property of $\mathrm{Coker}(\mathrm{Ker}(\mathrm{Coker}(f)))$, there exists unique $\psi : \mathrm{Coker}(\mathrm{Ker}(\mathrm{Coker}(f))) \to Y$ such that above diagram commutes, and note that ψ agree with the definition of $\mathrm{Coker}(f)$, hence, $\mathrm{Coker}(\mathrm{Ker}(\mathrm{Coker}(f))) = \mathrm{Coker}(f)$.

□

Proposition 2.5.2

If $f : A \to B$ is a morphism in an abelian category, then we have a factorization $f = me$, where $m : \mathrm{Im}\, f \to B$ is monomorphism and $e : A \to \mathrm{Im}\, f$ is epimorphism. Also,

$$m = \mathrm{Ker}(B \to \mathrm{Coker}\, f), e = \mathrm{Coker}(\mathrm{Ker}\, f \to A).$$

Diagramatically,

$$\begin{tikzcd}
& \mathrm{Im}\, f \arrow[d, dashed, "m"] \\
A \arrow[ur, dashed, "e"] \arrow[r, "f"] & B.
\end{tikzcd}$$

If also $f = m'e'$, where m' is a kernal, then in the commutative square

$$\begin{tikzcd}
\bullet \arrow[r, "e"] \arrow[d, "e'"'] & \bullet \arrow[d, "m"] \arrow[dl, dashed, "t"'] \\
\bullet \arrow[r, "m'"] & \bullet
\end{tikzcd}$$

there is a unique diagonal arrow t with $m = m't$ and $e' = te$.

Proof See, Mac Lane, Categories for the working Mathematician, Chapter VIII, Section 1 and 3. □

Definition 2.5.7 (Quotient)

The cokernal of a monomorphism is called ***quotient****. The quotient of a monomorphism $A \to B$ is often denoted B/A (with the map from B implicit).*

Remark 2.31 (Small remark on chasing diagrams.) It is useful to prove facts (and solve exercise) about abelian categories by chasing elements. Unfortunately, some commonly used abelian categories, such as the category of complexes, do not have "elements" — they are not naturally "sets with additional structure" in any obvious way. Nonetheless, proof by element-chasing can be justified by the **Freyd-Mitchell Embedding Theorem**:

Theorem 2.5.1 (Freyd-Mitchell Embedding Theorem)

If $\mathscr{C}$ is an abelian category whose object form a set, then there is a ring A and an exact, fully faithful functor from $\mathscr{C}$ into $\mathbf{Mod}_A$, which embeds $\mathscr{C}$ as a full subcategory.

(Unfortunately, the ring A need not be commutative.) The upshot is that to prove something about a diagram in some abelian category, and we may then "diagram-chase" elements. Moreover, any fact about kernels, cokernels, and so on that holds in $\mathbf{Mod}_A$ holds in any abelian category.

In this entire discussion, we assume we are working in an abelian category.

2.5.2 Complexes

Definition 2.5.8 (Complex and exact)

We say a sequence

$$\cdots \longrightarrow A \xrightarrow{f} B \xrightarrow{g} C \longrightarrow \cdots$$

is a ***complex at*** *B if $g \circ f = 0$, and is* ***eact at*** *B if $\operatorname{Ker} g = \operatorname{Im} f$.*

A sequence is a ***complex*** *(resp.,* ***exact****) if it is a complex (resp., exact) at each term. A* ***short exact sequence*** *is an exact sequence with five terms, the first and last of which are zeros — in other words, an exact sequence of the form*

$$0 \longrightarrow A \longrightarrow B \longrightarrow C \longrightarrow 0.$$

Proposition 2.5.3

Some properties of exact sequence (in abelian category):

(1) Sequence

$$0 \longrightarrow A \longrightarrow 0$$

is exact if and only if $A = 0$.

(2) Sequence

$$0 \longrightarrow A \xrightarrow{f} B$$

is exact if and only if f is a monomorphism.

(3) Sequence

$$A \xrightarrow{f} B \longrightarrow 0$$

is exact if and only if f is a epimorphism.

(4) Sequence

$$0 \longrightarrow A \xrightarrow{f} B \longrightarrow 0$$

is exact if and only if f is an isomorphism.

(5) Sequence

$$0 \longrightarrow A \xrightarrow{f} B \xrightarrow{g} C$$

is exact if and only if f is kernel of g, and sequence

$$A \xrightarrow{f} B \xrightarrow{g} C \longrightarrow 0$$

is exact if and only if g is cokernel of f.

Remark 2.32 In additive category, Proposition 2.5.3 also holds.

Proof

(1) If sequence

$$0 \longrightarrow A \longrightarrow 0$$

exact, then $\operatorname{Ker}(A \to 0) = \operatorname{Im}(0 \to A)$. We shall to show that $\operatorname{Ker}(A \to 0) = \mathrm{id}_A : A \to A$ and $\operatorname{Im}(0 \to A) = 0 : 0 \to A$.

Note that $\mathrm{id}_A : A \to A$ is the solution of the following universal problem,

$$\begin{tikzcd}
X \arrow[d, dashed, "f"'] \arrow[dr, "f"] \arrow[drr, "0"] & & \\
A \arrow[r, "\mathrm{id}_A"] \arrow[rr, bend right, "0"'] & A \arrow[r, "0"] & 0,
\end{tikzcd}$$

hence, $\operatorname{Ker}(A \to 0) = \mathrm{id}_A : A \to A$. Since $\operatorname{Im}(0 \to A) = \operatorname{Ker}(\operatorname{Coker}(0 \to A))$, consider the following diagram,

$$\begin{tikzcd}
0 \arrow[r, "0"] \arrow[drr, "0"'] & A \arrow[r, "\mathrm{id}_A"] \arrow[dr, "g"] & A \arrow[d, dashed, "g"] \\
& & Y
\end{tikzcd}$$

$\mathrm{id}_A : A \to A$ is the solution of above universal problem, hence $\operatorname{Coker}(0 \to A) = \mathrm{id}_A : A \to A$. Now, it suffice to show that $\operatorname{Ker}(\mathrm{id} : A \to A) = 0 : 0 \to A$. Consider the following universal problem,

$$\begin{tikzcd}
Z \arrow[d, "0"'] \arrow[dr, "0"] \arrow[drr, "0"] & & \\
0 \arrow[r, "0"'] \arrow[rr, bend right] & A \arrow[r, "\mathrm{id}"'] & A,
\end{tikzcd}$$

$0 : 0 \to A$ is the solution, and therefore $\operatorname{Im}(0 \to A) = 0 : 0 \to A$. Since $\operatorname{Ker}(A \to 0) = \operatorname{Im}(0 \to A)$, we have $A = 0$.

Conversely, if $A = 0$, obviously, $\operatorname{Ker}(A \to 0) = \operatorname{Im}(0 \to A)$.

(2) If sequence

$$0 \longrightarrow A \xrightarrow{f} B$$

is exact, then $\operatorname{Im}(0 \to A) = \operatorname{Ker} f$. By part (1), $\operatorname{Im}(0 \to A) = 0 : 0 \to A$, hence, $\operatorname{Ker}(f) = 0 : 0 \to A$. Let two morphisms $\mu_1 : X \to A$ and $\mu_2 : X \to A$ with $f \circ \mu_1 = f \circ \mu_2$. Since in Abelian category,

$f \circ (\mu_1 - \mu_2) = 0$ and $\mu_1 - \mu_2 : X \to A$, consider the following diagram,

$$\begin{tikzcd}
X \arrow[d, dashed, "0"'] \arrow[dr, "\mu_1-\mu_2"] \arrow[drr, "0"] & & \\
0 \arrow[r, "0"'] \arrow[rr, bend right] & A \arrow[r, "f"'] & B,
\end{tikzcd}$$

by the universal property of $\operatorname{Ker} f$, $\mu_1 - \mu_2 = 0$, that is, f is monomorphism.
Conversely, if f is monomorphism, we shall to show that $\operatorname{Ker}(f) = 0 : 0 \to A$. It suffice to show that $0 : 0 \to A$ is the solution of the following universal problem.

$$\begin{tikzcd}
X \arrow[d, dashed] \arrow[dr, "g"] \arrow[drr, "0"] & & \\
0 \arrow[r, "0"'] \arrow[rr, bend right] & A \arrow[r, "f"'] & B
\end{tikzcd}$$

Since, f is monomorphism, g is the unique morphism such that $f \circ g = 0$, note that $0 = f \circ 0$, $g = 0$. Hence, $0 : 0 \to A$ is the solution of the universal problem, and therefore $\operatorname{Ker}(f) = 0 : 0 \to A = \operatorname{Im}(0 \to A)$, which implies the sequence

$$0 \longrightarrow A \xrightarrow{f} B$$

is exact.

(3) If sequence

$$A \xrightarrow{f} B \longrightarrow 0$$

is exact, then $\operatorname{Im} f = \operatorname{Ker}(B \to 0) = \mathrm{id}_B : B \to B$. Let two morphisms $\eta_1 : B \to Y$ and $\eta_2 : B \to Y$ with $\eta_1 \circ f = \eta_2 \circ f$, since in Abelian category, we have $(\eta_1 - \eta_2) \circ f = 0$. Note that $\operatorname{Im} f = \operatorname{Ker}(\operatorname{Coker}(f)) = \mathrm{id}_B : B \to B$, $\operatorname{Coker}(f) \circ \mathrm{id}_B = \operatorname{Coker}(f) = 0 : B \to 0$. Now consider the following diagram,

$$\begin{tikzcd}
A \arrow[r, "f"] \arrow[rr, bend left, "0"] \arrow[drr, "0"'] & B \arrow[r, "0"] \arrow[dr] & 0 \arrow[d, dashed] \\
& & Y
\end{tikzcd}$$

$\eta_1 - \eta_2$ agree with above diagram, hence, $\eta_1 = \eta_2$, which implies that f is epimorphism.
Conversely, if f is epimorphism, we shall to show that $\operatorname{Im}(f) = \operatorname{Ker}(\operatorname{Coker}(f)) = \operatorname{Ker}(B \to 0) = \mathrm{id}_B : B \to B$. It suffice to show that $\mathrm{id}_B : B \to B$ is the solution of the following universal problem.

$$\begin{tikzcd}
Z \arrow[d, dashed] \arrow[dr] \arrow[drr, "0"] & & \\
B \arrow[r] \arrow[rr, bend right, "0"'] & B \arrow[r] & \operatorname{Coker}(f)
\end{tikzcd}$$

We need to show that $\operatorname{Coker}(f) \circ \mathrm{id}_B = \operatorname{Coker}(f) = 0 : B \to 0$. Consider the following universal problem,

$$
\begin{tikzcd}
A \arrow[r, "f"] \arrow[rr, bend left, "0"] \arrow[drr, "0"'] & B \arrow[r] \arrow[dr] & 0 \arrow[d, dashed] \\
 & & W,
\end{tikzcd}
$$

since f is epimorphism, $B \to 0$ and $B \to W$ must be same, hence, $W = 0$, and the unique morphism is $0 : 0 \to 0$. Hence, $\mathrm{Coker}(f) = 0 : B \to 0$. It is clearly $\mathrm{id}_B : B \to B$ is the solution of the universal problem

$$
\begin{tikzcd}
Z \arrow[d, dashed, "g"'] \arrow[dr, "g"] \arrow[drr, "0"] & & \\
B \arrow[r] \arrow[rr, bend right, "0"'] & B \arrow[r] & \mathrm{Coker}(f),
\end{tikzcd}
$$

hence, $\mathrm{Im}(f) = \mathrm{id}_B : B \to B = \mathrm{Ker}(B \to 0)$.

(4) By part (2) and (3), immediately.

(5) If sequence

$$0 \longrightarrow A \xrightarrow{f} B \xrightarrow{g} C$$

is exact, then $\mathrm{Ker}\, g = \mathrm{Im}\, f$. We shall to show that $g \circ f = 0$. By Proposition 2.5.2, we have $f = me$, where epimorphism $e : A \to \mathrm{Im}\, f$ and monomorphism $m : \mathrm{Im}\, f \to B$. Since $\mathrm{Ker}\, g = \mathrm{Im}\, f$, we have $e : A \to \mathrm{Ker}\, g$ and $m : \mathrm{Ker}\, g \to B$. By the definition of kernal, we have $g \circ m = 0$. Hence, $g \circ f = g \circ m \circ e = 0$.

By Proposition 2.5.2, $f = me$, where $m : \mathrm{Im}\, f \to B$ monomorphism and $e : A \to \mathrm{Im}\, f$. We claim that e is an isomorphism. It suffice to show that e is monomorphism. Let $\eta_1 : X \to A$ and $\eta_2 : X \to A$ morphisms with $e \circ \eta_1 = e \circ \eta_2$, then $m \circ e \circ \eta_1 = m \circ e \circ \eta_2$, since $f = me$ is monomorphism, $\eta_1 = \eta_2$. Hence, e is isomorphism. Consider the following universal problem.

$$
\begin{tikzcd}
 & X \arrow[dl, dashed] \arrow[d, dashed, "t"] \arrow[dr, "h"] \arrow[drr, "0"] & & \\
A \arrow[r, "e"] \arrow[rrr, bend right, "0"'] & \mathrm{Im}\, f \arrow[r, "m"] \arrow[rr, bend right, "0"'] & B \arrow[r, "g"] & C
\end{tikzcd}
$$

Since $\mathrm{Im}\, f = \mathrm{Ker}\, g$, by the universal property of kernel, exists unique $t : X \to \mathrm{Im}\, f$. Let $\theta : X \to A$ by setting $\theta = e^{-1} \circ t$, then above diagram commutes. Note that $f = me$ is monomorphism, θ must be unique. Hence, $f = \mathrm{Ker}\, g$.

Conversely, if $f : A \to B = \mathrm{Ker}(g)$, then $g \circ f = 0$. By Proposition 2.5.2, $f = me$ uniquely, where $e : A \to \mathrm{Im}\, f$ epimorphism and $m : \mathrm{Im}\, f \to B$ monomorphism. Consider the following diagram,

$$
\begin{tikzcd}
A \arrow[d, dashed, "e"] \arrow[dr, "f"] \arrow[drr, "0"] & & \\
\mathrm{Im}\, f \arrow[r, "m"'] \arrow[rr, bend right, "0"'] & B \arrow[r, "g"'] & C,
\end{tikzcd}
$$

since e is epimorphism, $g \circ f = g \circ m \circ e = 0$ implies $g \circ m = 0$. Hence, e is the solution of above universal problem, and therefore $\operatorname{Im} f = \operatorname{Ker} g$. Let X be any object, $u : X \to A$ and $v : X \to A$ are two morphisms with $f \circ u = f \circ v$, then $f \circ (u - v) = 0$. Consider the following diagram,

$$\begin{tikzcd}
0 \arrow[d, dashed, "\exists!"'] \arrow[dr, "0"] \arrow[drr, "0"] & & \\
A \arrow[r, "f"] \arrow[rr, bend right, "0"'] & B \arrow[r, "g"] & C,
\end{tikzcd}$$

since $f = \operatorname{Ker}(g)$, by the universal property of kernel, $u - v = 0$, that is, $u = v$. Hence, f is monomorphism.

If sequence

$$A \xrightarrow{f} B \xrightarrow{g} C \longrightarrow 0$$

is exact, then $\operatorname{Im} f = \operatorname{Ker} g$ and g is epimorphism. We shall to show that $g = \operatorname{Coker} f$. Note that $\operatorname{Im} f = \operatorname{Ker}(\operatorname{Coker}(f))$, we have $\operatorname{Coker}(\operatorname{Ker}(\operatorname{Coker}(f))) = \operatorname{Coker}(f) = \operatorname{Coker}(\operatorname{Ker} g)$. It suffice to show that $g = \operatorname{Coker}(\operatorname{Ker}(g))$. By Proposition 2.5.2, $g = me$, where $m = \operatorname{Ker}(C \to \operatorname{Coker} g)$ monomorphism and $e = \operatorname{Coker}(\operatorname{Ker}(g) \to B)$ epimorphism. Since g is epimorphism, $\operatorname{Coker}(g) = 0 : C \to 0$, and therefore $m = \operatorname{Ker}(0 : C \to 0) = \mathrm{id}_C : C \to C$. Hence, $g = \operatorname{Coker}(\operatorname{Ker}(g))$.

Conversely, If $g = \operatorname{Coker} f$, then $\operatorname{Ker}(g) = \operatorname{Ker}(\operatorname{Coker}(f)) = \operatorname{Im}(f)$. We shall to show that g is epimorphism. Let morphisms $\mu_1 : C \to X$ and $\mu_2 : C \to X$ with $\mu_1 \circ g = \mu_2 \circ g$, then $(\mu_1 - \mu_2) \circ g = 0$. Since $g = \operatorname{Coker}(f)$, by the universal property of cokernel, $\mu_1 - \mu_2 = 0$, that is, $\mu_1 = \mu_2$. Hence, g is epimorphism. Consequently, sequence

$$A \xrightarrow{f} B \xrightarrow{g} C \longrightarrow 0$$

is exact.

□

Corollary 2.5.1

Let $u : A \to B$ be a morphism in an additive category $\mathscr{C}$.

(1) If $\operatorname{Ker} u$ exists, then u is monomorphism if and only if $\operatorname{Ker} u = 0$.

(2) If $\operatorname{Coker} u$ exists, then u is epimorphism if and only if $\operatorname{Coker} u = 0$.

Proof See the proof in Proposition 2.5.3. □

Definition 2.5.9 (Homology, cohomology, cycle, and boundary)

If sequence

$$\cdots \longrightarrow A \xrightarrow{f} B \xrightarrow{g} C \longrightarrow \cdots$$

*is a complex at B, then its **homology at** B (often denoted by H) is $\operatorname{Ker} g / \operatorname{Im} f$. More precisely, there is some monomorphism $\operatorname{Im} f \hookrightarrow \operatorname{Ker} g$, and that H is the cokernel of this monomorphism.*

*We say that elements of $\operatorname{Ker} g$ (assuming the objects of the category are sets with some additional structure) are the **cycles**, and elements of $\operatorname{Im} f$ are the **boundaries** (so homology is "cycles mod boundaries").*

*If the complex is indexed in decreasing order, the indices are often written subscripts, and H_i is the homology at $A_{i+1} \to A_i \to A_{i-1}$. If the complex is indexed in increasing order, the indices are often written as superscripts, and the homology H^i at $A^{i-1} \to A^i \to A^{i+1}$ is often called **cohomology**.*

Remark 2.33 Sequence

$$\cdots \longrightarrow A \xrightarrow{f} B \xrightarrow{g} C \longrightarrow \cdots$$

is exact at B if and only if its homology at B is 0.

Theorem 2.5.2

An exact sequence

$$A^\bullet: \qquad \cdots \longrightarrow A^{i-1} \xrightarrow{f^{i-1}} A^i \xrightarrow{f^i} A^{i+1} \xrightarrow{f^{i+1}} \cdots \tag{2.16}$$

can be "factored" into short exact sequences

$$0 \longrightarrow \mathrm{Ker}\, f^i \longrightarrow A^i \longrightarrow \mathrm{Ker}\, f^{i+1} \longrightarrow 0$$

which is helpful in proving facts about long exact sequences by reducing them to facts about short exact sequences.

More generally, if (2.16) is assumed only to be a complex, then it can be "factored" into short exact sequences.

$$\begin{gathered} 0 \longrightarrow \mathrm{Ker}\, f^i \longrightarrow A^i \longrightarrow \mathrm{Im}\, f^i \longrightarrow 0 \\ 0 \longrightarrow \mathrm{Im}\, f^{i-1} \longrightarrow \mathrm{Ker}\, f^i \longrightarrow H^i(A^\bullet) \longrightarrow 0 \end{gathered} \tag{2.17}$$

Proof If $A^\bullet$ is exact, we have

$$\begin{tikzcd}
& 0 \arrow[d] & & 0 \arrow[d] & 0 \\
& \mathrm{Ker}\, f^{i-1} \arrow[d, hook] & & \mathrm{Im}\, f^i = \mathrm{Ker}\, f^{i+1} \arrow[d, hook] \arrow[ur] & \\
\cdots \arrow[r] & A^{i-1} \arrow[r, "f^{i-1}"] \arrow[d, two heads] & A^i \arrow[r, "f^i"] \arrow[ur, two heads] & A^{i+1} \arrow[r, "f^{i+1}"] \arrow[d, two heads] & \cdots \\
& \mathrm{Im}\, f^{i-1} = \mathrm{Ker}\, f^i \arrow[d] \arrow[ur, hook] & & \mathrm{Im}\, f^{i+1} \arrow[d] & \\
0 \arrow[ur] & 0 & & 0. &
\end{tikzcd}$$

If $A^\bullet$ is complex, by Proposition 2.5.2 and Proposition 2.5.3, we have

$$\begin{tikzcd}
& & 0 & & & \\
& 0 \arrow[r] & \mathrm{Im}\, f^i \arrow[u] \arrow[r, hook] \arrow[dr, hook] & \mathrm{Ker}\, f^{i+1} \arrow[r, two heads] \arrow[d, hook] & H^{i+1}(A^\bullet) \arrow[r] & 0 \\
\cdots \arrow[r] & A^{i-1} \arrow[r, "f^{i-1}"] \arrow[d, two heads] & A^i \arrow[r, "f^i"] \arrow[u, two heads] & A^{i+1} \arrow[r, "f^{i+1}"] & \cdots & \\
0 \arrow[r] & \mathrm{Im}\, f^{i-1} \arrow[r, hook] \arrow[ur, hook] & \mathrm{Ker}\, f^i \arrow[r, two heads] \arrow[u, hook] & H^i(A^\bullet) \arrow[r] & 0 & \\
& & 0. \arrow[u] & & &
\end{tikzcd}$$

$\square$

Exercise 2.35 Describe exact sequence

$$
\begin{gathered}
0 \longrightarrow \operatorname{Im} f^i \longrightarrow A^{i+1} \longrightarrow \operatorname{Coker} f^i \longrightarrow 0 \\
0 \longrightarrow H^i(A^\bullet) \longrightarrow \operatorname{Coker} f^{i-1} \longrightarrow \operatorname{Im} f^i \longrightarrow 0
\end{gathered}
\tag{2.18}
$$

(These are somehow dual to (2.17). In fact in some mirror universe this might have been given as the standard definition of homology.) Assume the category is that of modules over a fixed ring for convenience, but be aware that the result is true for any abelian category.

Proof Note that

$$
\begin{tikzcd}
 & & & & 0 \\
 & & & \operatorname{Coker} f^{i-1} \arrow[ur] & \\
\cdots \arrow[r] & A^{i-1} \arrow[r, "f^{i-1}"] \arrow[d, two heads] & A^i \arrow[r, "f^i"] \arrow[ur] & A^{i+1} \arrow[r, "f^{i+1}"] & \cdots \\
 & \operatorname{Im} f^{i-1} \arrow[ur, hook] & & & \\
0, \arrow[ur] & & & &
\end{tikzcd}
$$

then we have the first sequence.

In $\mathbf{Mod}_A$, we have $\operatorname{Coker} f^{i-1} = A^i / \operatorname{Im} f^{i-1}$ and $H^i(A^\bullet) = \operatorname{Ker} f^i / \operatorname{Im} f^{i-1}$, then we have inclusion $H^i(A^\bullet) \hookrightarrow \operatorname{Coker} f^{i-1}$. Since $f^i : A^i \to \operatorname{Im} f^i$ is surjective, $\operatorname{Im} f^i \cong A^i / \operatorname{Ker} f^i$. By the homomorphism fundamental theorem, we have $\operatorname{Coker} f^{i-1} / H^i(A^\bullet) = (A^i / \operatorname{Im} f^{i-1}) / (\operatorname{Ker} f^i / \operatorname{Im} f^{i-1}) \cong A^i / \operatorname{Ker} f^i \cong \operatorname{Im} f^i$. Then sequence

$$0 \longrightarrow H^i(A^\bullet) \longrightarrow \operatorname{Coker} f^{i-1} \longrightarrow \operatorname{Im} f^i \longrightarrow 0$$

is exact. $\square$

Proposition 2.5.4

Suppose

$$0 \xrightarrow{d^0} A^1 \xrightarrow{d^1} \cdots \xrightarrow{d^{n-1}} A^n \xrightarrow{d^n} 0$$

is a complex of finite-dimensional k-vector spaces (often called $A^\bullet$ for short). Define $h^i(A^\bullet) = \dim H^i(A^\bullet)$. Then $\sum(-1)^i \dim A^i = \sum(-1)^i h^i(A^\bullet)$.
In particular, if $A^\bullet$ is exact, then $\sum(-1)^i \dim A^i = 0$.

Proof By Theorem 2.5.2, spilit up the sequence into short exact sequences

$$0 \longrightarrow \operatorname{Ker} d^i \longrightarrow A^i \longrightarrow \operatorname{Im} d^i \longrightarrow 0$$

$$0 \longrightarrow \operatorname{Im} d^{i-1} \longrightarrow \operatorname{Ker} d^i \longrightarrow H^i(A^\bullet) \longrightarrow 0.$$

Since dim is additive function (see, Atiyah-MacDonald, Introdution to Commutative Algebra, page 23), we have $\dim A^i = \dim \operatorname{Ker} d^i + \dim \operatorname{Im} d^i$ and $\dim H^i(A^\bullet) = h^i(A^\bullet) = \dim \operatorname{Ker} d^i - \dim \operatorname{Im} d^{i-1}$. Note that $\operatorname{Ker} d^0 = 0$, $\operatorname{Ker} d^n = \mathrm{id}_{A^n}$, $\operatorname{Im} d^0 = 0$, and $\operatorname{Im} d^n = 0$, we have

$$\begin{aligned}\sum(-1)^i \dim A^i &= \sum_{i=0}^{n}(-1)^i(\dim \operatorname{Ker} d^i + \dim \operatorname{Im} d^i) \\ &= \sum_{i=1}^{n}(-1)^i \dim \operatorname{Ker} d^i + \sum_{i=1}^{n-1}(-1)^i \dim \operatorname{Im} d^i,\end{aligned}$$

and

$$\begin{aligned}\sum(-1)^i h^i(A^\bullet) &= \sum_{i=1}^{n}(\dim \operatorname{Ker} d^i - \dim \operatorname{Im} d^{i-1}) \\ &= \sum_{i=1}^{n}(-1)^i \dim \operatorname{Ker} d^i + \sum_{i=1}^{n}(-1)^{i+1} \dim \operatorname{Im} d^{i-1} \\ &= \sum_{i=1}^{n}(-1)^i \dim \operatorname{Ker} d^i + \sum_{i=1}^{n-1}(-1)^i \dim \operatorname{Im} d^i.\end{aligned}$$

Hence, $\sum(-1)^i \dim A^i = \sum(-1)^i h^i(A^\bullet)$.

In particular, if $A^\bullet$ is exact, then $h^i(A^\bullet) = \dim \operatorname{Ker} d^i - \dim \operatorname{Im} d^{i-1} = 0$. Hence,

$$\sum(-1)^i \dim A^i = 0.$$

□

$\mathbf{Com}_{\mathscr{C}}$ is an abelian category

Definition 2.5.10 (Category $\mathbf{Com}_{\mathscr{C}}$ of complexes)

*Suppose $\mathscr{C}$ is an abelian category. Define the **category $\mathbf{Com}_{\mathscr{C}}$ of complexes** as follows. The objects are infinite complexes*

$$A^\bullet : \qquad \cdots \longrightarrow A^{i-1} \xrightarrow{d^{i-1}} A^i \xrightarrow{d^i} A^{i+1} \xrightarrow{d^{i+1}} \cdots$$

*in $\mathscr{C}$, each d^i in a complex called **differentials**, and the morphisms $A^\bullet \to B^\bullet$ (called **chain map**) are commuting diagrams.*

$$\begin{tikzcd} \cdots \arrow[r] & A^{i-1} \arrow[r, "d^{i-1}"] \arrow[d] & A^i \arrow[r, "d^i"] \arrow[d] & A^{i+1} \arrow[r, "d^{i+1}"] \arrow[d] & \cdots \\ \cdots \arrow[r] & B^{i-1} \arrow[r, "d'^{i-1}"] & B^i \arrow[r, "d'^i"] & B^{i+1} \arrow[r, "d'^{i+1}"] & \cdots . \end{tikzcd}$$

Our goal is to prove that $\mathbf{Com}_{\mathscr{C}}$ is an abelian category.

Lemma 2.5.2

Let $\mathscr{C}$ be an additive category and $\mathscr{S}$ be a subcategory if $\mathscr{S}$ is full, contains a zero object of $\mathscr{C}$, and contains the direct sum $A \oplus B$ (in $\mathscr{C}$) of all $A, B \in \operatorname{obj}(\mathscr{S})$.

Proof Since $\mathscr{S}$ is full,

(1) $\operatorname{Hom}(A, B)$ is an additive abelian group for every $A, B \in \operatorname{obj}(\mathscr{S})$,

(2) distributive laws given by $\mathscr{C}$.

Also, $\mathscr{S}$ contains a zero object of $\mathscr{C}$. By Lemma 2.5.1, product is isomorphic to coproduct, hence, it suffice to check for all $A, B \in \operatorname{obj}(\mathscr{S})$ the direct sum $A \oplus B \in \mathscr{S}$. By the hypothesis, $\mathscr{S}$ is an additive category. □

Lemma 2.5.3

Let $\mathscr{S}$ be a full subcategory of an abelian category $\mathscr{C}$. If for all $A, B \in \operatorname{obj}(\mathscr{S})$ and all $f : A \to B$,

(i) a zero object in $\mathscr{C}$ lies in $\mathscr{S}$,

(ii) the dierect sum $A \otimes B$ in $\mathscr{C}$ lies in $\mathscr{S}$,

(iii) both $\operatorname{Ker} f$ and $\operatorname{Coker} f$ lie in $\mathscr{S}$,

then $\mathscr{S}$ is an abelian category.

Moreover, if

$$A \xrightarrow{f} B \xrightarrow{g} C$$

is an exact sequence in $\mathscr{S}$, then it is an exact sequence in $\mathscr{C}$.

Proof The hypothesis gives $\mathscr{S}$ additive, by Lemma 2.5.2, hence, $\mathscr{S}$ is abelian if axiom (iii) in the definition of abelian category holds. If $f : A \to B$ is a monomorphism in $\mathscr{S}$, then $\operatorname{Ker} f = 0$. Since $\operatorname{Ker} f$ is the same in $\mathscr{C}$ as in $\mathscr{S}$, by hypothesis, f is monomorphism in $\mathscr{C}$. By hypothesis, $\operatorname{Coker} f$ is a morphism in $\mathscr{S}$. Since $\mathscr{C}$ is abelian, there is a morphism $g : B \to C$ with $f = \operatorname{Ker} g$, where $g = \operatorname{Coker} f$. Since S contains cokernels, g is a morphism in $\mathscr{S}$. Hence, $f = \operatorname{Ker} g$ in $\mathscr{S}$. The dual argument shows that epimorphisms in $\mathscr{S}$ are cokernels.

Moreover, since kernels and cokernels are the same in $\mathscr{S}$ as in $\mathscr{C}$, images are also the same, and so exactness in $\mathscr{S}$ implies exactness in $\mathscr{C}$. $\square$

Proposition 2.5.5

If $\mathscr{C}$ is an abelian category and $\mathscr{I}$ is a small category, then the functor category $\mathscr{C}^{\mathscr{I}}$ is an abelian category.

Proof We assume that $\mathscr{I}$ is small to gurantee that the morphism set $\operatorname{Mor}(F, G)$, where $F, G : \mathscr{I} \to \mathscr{C}$, are sets, not proper classes. The zero object in $\mathscr{C}^{\mathscr{I}}$ is the constant functor with value 0, where 0 is a zero object in $\mathscr{C}$. If $\tau, \sigma : F \to G$ by $(\tau + \sigma)_i = \tau_i + \sigma_i : F(i) \to G(i)$ for all $i \in \operatorname{obj}(\mathscr{I})$. Finally, define $F \oplus G$ by $(F \oplus G)(i) = F(i) \oplus G(i)$. Since $\mathscr{C}$ is abelian category, $F \oplus G \in \mathscr{C}^{\mathscr{I}}$. Hence, these definitions make $\mathscr{C}^{\mathscr{I}}$ an additive category.

If $\tau : F \to G$, define K by

$$K(i) = \operatorname{Ker}(\tau_i).$$

In the following commutative diagram with exact rows, where $f : i \to j$ in $\mathscr{I}$, there is a unique $Kf : K(i) \to K(j)$ making the augmented diagram commute (by the universal property of $\operatorname{Ker}(\tau_j)$).

$$\begin{tikzcd}
0 \arrow[r] & K(i) \arrow[r, "\iota_i"] \arrow[d, dashed, "Kf"'] & F(i) \arrow[r, "\tau_i"] \arrow[d, "Ff"] & G(i) \arrow[d, "Gf"] \\
0 \arrow[r] & K(j) \arrow[r, "\iota_j"] & F(j) \arrow[r, "\tau_j"] & G(j)
\end{tikzcd}$$

We shall to check that K is a functor, $\iota : K \to F$ is natural transformation, and $\iota = \operatorname{Ker} \tau$.

(1) K is a functor.

Let $i \in \mathscr{I}$, then $K(\mathrm{id}_i) : K(i) \to K(i)$, by the uniqueness of $K(\mathrm{id}_i)$, we have $K(\mathrm{id}_i) = \mathrm{id}_{K(i)}$. Let $i \to j \to k$ in $\mathscr{I}$, also by the uniqueness, we have $K(i \to j \to k) = K(j \to k) \circ K(i \to j)$. Hence, K is a functor.

(2) $\iota : K \to F$ is a natural transformation.

Since, $\tau : F \to G$ is a natural transformation, by the universal property of $\mathrm{Ker}(\tau_i)$, diagram

$$\begin{tikzcd} K(i) \arrow[r, "\iota_i"] \arrow[d, dashed, "Kf"'] & F(i) \arrow[d, "Ff"] \\ K(j) \arrow[r, "\iota_j"] & F(j) \end{tikzcd}$$

commutes, for all $i, j \in \mathrm{Im}(\mathscr{I})$. Hence, $\iota : K \to F$ is a natural transformation.

(3) $\iota = \mathrm{Ker}\,\tau$.

Note that we have exact sequence,

$$0 \longrightarrow K \longrightarrow F \longrightarrow G,$$

by Proposition 2.5.3, $\iota = \mathrm{Ker}\,\tau$.

Hence, kernels exists in $\mathscr{C}^{\mathscr{I}}$. Dually, cokernels exists in $\mathscr{C}^{\mathscr{I}}$. By Lemma 2.5.3, $\mathscr{C}^{\mathscr{I}}$ is an abelian category. ☐

Definition 2.5.11 (Subcomplex)

A complex $(A^\bullet, \delta^\bullet)$ is defined to be a ***subcomplex*** *of complex $(C^\bullet, d^\bullet)$ if there is a chain map $i : A^\bullet \to C^\bullet$ with each i^n is monomorphism.*

Example 2.34 In $\mathbf{Com}_{\mathbf{Mod}_A}$, we have that $(A^\bullet, \delta^\bullet)$ is a subcomplex of $(C^\bullet, d^\bullet)$ if A_n is a submodule of C_n and $\delta_n = d_n|_{A_n}$ for every $n \in \mathbb{Z}$.

Theorem 2.5.3

$\mathbf{Com}_{\mathscr{C}}$ *is an abelian category.*

Proof We only prove this when $\mathscr{C} = \mathbf{Mod}_A$. View $\mathbb{Z}$ first as a partially ordered set under reverse inequality and then as a small category (with morphisms $m \to n$ if $m \leq n$). By Proposition 2.5.5, the functor category $\mathbf{Mod}_A^{\mathscr{I}}$ is an abelian category. Lemma 2.5.3 says that $\mathbf{Com}_{\mathscr{C}}$ is abelian if $\mathbf{Com}_{\mathbf{Mod}_A}$ is a full category of $\mathbf{Mod}_A^{\mathscr{I}}$ containing a zero object, the direct sum $A^\bullet \oplus B^\bullet$ of $A^\bullet, B^\bullet \in \mathrm{obj}(\mathbf{Com}_{\mathbf{Mod}_A})$, and both $\mathrm{Ker}\, f^\bullet$ and $\mathrm{Coker}\, f^\bullet$, where $f^\bullet$ is a morphism in $\mathbf{Com}_{\mathbf{Mod}_A}$. All the steps are routine. Note that $\mathbf{Com}_{\mathbf{Mod}_A}$ is, by definition, a full subcategory of $\mathbf{Mod}_A^{\mathscr{I}}$. The **zero complex** is the complex each of whose terms is 0, while $(A^\bullet, d^\bullet) \oplus (B^\bullet, d'^\bullet)$ is, by definition, the complex each of whose n-th term is $A_n \oplus B_n$ and whose n-th differential is $d^n \oplus d'^n$. If $f^\bullet : (A^\bullet, d^\bullet) \to (B^\bullet, d'^\bullet)$ is a chain map, define

$$\mathrm{Ker}\, f^\bullet = \cdots \longrightarrow \mathrm{Ker}\, f^{n-1} \xrightarrow{\delta^{n-1}} \mathrm{Ker}\, f^n \xrightarrow{\delta^n} \mathrm{Ker}\, f^{n+1} \longrightarrow \cdots,$$

where $\delta^n = d^n|_{\mathrm{Ker}\, f^n}$, and

$$\mathrm{Im}\, f^\bullet = \cdots \longrightarrow \mathrm{Im}\, f^{n-1} \xrightarrow{\Delta^{n-1}} \mathrm{Im}\, f^n \xrightarrow{\Delta^n} \mathrm{Im}\, f^{n+1} \longrightarrow \cdots,$$

where $\Delta^n = d'^n|_{\mathrm{Im}\, f^n}$. Then $\mathrm{Ker}\, f^\bullet$ is a subcomplex of $A^\bullet$, and $\mathrm{Im}\, f^\bullet$ is a subcomplex of $B^\bullet$. If $A'^\bullet$ is a subcomplex of $A^\bullet$, define the **quotient complex** to be

$$A^\bullet / A'^\bullet = \cdots \longrightarrow A^n / A'^n \xrightarrow{\overline{d^n}} A^{n+1} / A'^{n+1} \longrightarrow \cdots,$$

where $\overline{d^n} : a_n + A'^n \mapsto d^n a_n + A'^{n+1}$. If $p^n : A^n \to A^n / A'^n$ is the natural map, then $p^\bullet : A^\bullet \to A^\bullet / A'^\bullet$ is a chain map. Finally, define

$$\mathrm{Coker}\, f^\bullet = \cdots \longrightarrow B^{n-1} / \mathrm{Im}\, f^{n-1} \xrightarrow{\overline{d'^{n-1}}} B^n / \mathrm{Im}\, f^n \xrightarrow{\overline{d'^n}} B^{n+1} / \mathrm{Im}\, f^{n+1} \xrightarrow{\overline{d'^{n+1}}} \cdots.$$

Now, we shall to show that above definitions agree with the categorical definitions of Ker and Coker in $\mathbf{Com}_{\mathbf{Mod}_A}$.

(1) Kernel.

Consider the following universal problem.

$$\begin{tikzcd}
C^\bullet \arrow[d, dashed] \arrow[dr, "g"'] \arrow[drr, "0"] & & \\
\mathrm{Ker}\, f^\bullet \arrow[r, "i"'] \arrow[rr, bend right, "0"'] & A^\bullet \arrow[r, "f"'] & B^\bullet
\end{tikzcd}$$

Since $f^n \circ i^n : \mathrm{Ker}\, f^n \to A^n \to B^n$ for all $n \in \mathbb{Z}$, indeed, $f^\bullet \circ i = 0$. By the universal property of $\mathrm{Ker}\, f^n$, we have the following commutative diagram.

$$\begin{tikzcd}
C^n \arrow[d, dashed, "\exists!\varphi^n"'] \arrow[dr, "g^n"] \arrow[drr, "0"] & & \\
\mathrm{Ker}\, f^n \arrow[r, "i^n"'] \arrow[rr, bend right, "0"'] & A^n \arrow[r, "f^n"'] & B^n
\end{tikzcd}$$

Then we can define a unique map $\varphi : C^\bullet \to \mathrm{Ker}\, f^\bullet$, since φ^n is unique. We need to show that φ is a chain map, it suffices to check the following diagram commutes.

$$\begin{tikzcd}
\cdots \arrow[r] & C^{n-1} \arrow[r, "d_C^{n-1}"] \arrow[d, "\varphi^{n-1}"'] & C^n \arrow[r, "d_C^n"] \arrow[d, "\varphi^n"] & C^{n+1} \arrow[r] \arrow[d, "\varphi^{n+1}"] & \cdots \\
\cdots \arrow[r] & \mathrm{Ker}\, f^{n-1} \arrow[r, "\delta^{n-1}"] & \mathrm{Ker}\, f^n \arrow[r, "\delta^n"] & \mathrm{Ker}\, f^{n+1} \arrow[r] & \cdots
\end{tikzcd}$$

Consider following diagram.

$$\begin{tikzcd}
& C^{n-1} \arrow[r, "d_C^{n-1}"] \arrow[dl, "g^{n-1}"'] \arrow[d, "\varphi^{n-1}"'] & C^n \arrow[d, "\varphi^n"] \arrow[dr, "g^n"] & \\
A^{n-1} \arrow[rrr, bend right, "d^{n-1}"'] & \mathrm{Ker}\, f^{n-1} \arrow[l, hook', "i^{n-1}"'] \arrow[r, "\delta^{n-1}"] & \mathrm{Ker}\, f^n \arrow[r, "i^n"] & A^n
\end{tikzcd}$$

Since $d^{n-1} \circ g^{n-1} = g^n \circ d_C^{n-1}$, and note that $g^k = i^k \circ \varphi^k$ and i^k is inclusion for all k,

$$\begin{aligned}
d^{n-1} \circ g^{n-1} =& d^{n-1} \circ i^{n-1} \circ \varphi^{n-1} \\
=& d^{n-1}|_{\mathrm{Ker}\, f^{n-1}} \circ \varphi^{n-1} \\
=& g^n \circ d_C^{n-1} = i^n \circ \varphi^n \circ d_C^{n-1} \\
=& \varphi^n \circ d_C^{n-1}.
\end{aligned}$$

Note that $\delta^{n-1} = d^{n-1}|_{\mathrm{Ker}\, f^{n-1}} \circ \varphi^{n-1}$, $\delta^{n-1} \circ \varphi^{n-1} = \varphi^n \circ d_C^{n-1}$. Hence, $\varphi : C^\bullet \to \mathrm{Ker}\, f^\bullet$ is a chain map, and therefore $\mathrm{Ker}\, f$ agree with the categorical definition of kernel in $\mathbf{Com}_{\mathbf{Mod}_A}$.

(2) Cokernel.

Similar to the proof of the kernel.

Then both $\mathrm{Ker}\, f^\bullet$ and $\mathrm{Coker}\, f^\bullet$ lie in $\mathbf{Com}_{\mathbf{Mod}_A}$, by Lemma 2.5.3, $\mathbf{Com}_{\mathbf{Mod}_A}$ is an abelian category.

□

We introduce some notations and reformulate Definition 2.5.9.

Definition 2.5.12 (Cochains, Cocycles, and Coboundaries)

If $(C^\bullet, d^\bullet)$ is a complex in $\mathbf{Com}_{\mathscr{C}}$, where $\mathscr{C}$ is an abelian category, define

$$\begin{aligned} n\textbf{-cochains} &= C^n, \\ n\textbf{-cocycles} &= Z^n(C^\bullet) = \operatorname{Ker} d^n, \\ n\textbf{-coboundaries} &= B^n(C^\bullet) = \operatorname{Im} d^{n-1}. \end{aligned}$$

Definition 2.5.13 (Cohomology)

*If $C^\bullet$ is a complex in $\mathbf{Com}_{\mathscr{C}}$, where $\mathscr{C}$ is an abelian category, and $n \in \mathbb{Z}$, its n-th **cohomology** is*

$$H^n(C^\bullet) = Z^n(C^\bullet)/B^n(C^\bullet).$$

*Now $H^n(C^\bullet)$ lies in $\operatorname{obj}(\mathscr{C})$ if quotients are viewed as objects (see Definition 2.5.7). However, if we recognize $\mathscr{C}$ as a full subcategory of $\mathbf{Ab}$ (or $\mathbf{Mod}_A$), then an element of $H^n(C^\bullet)$ is a coset $z + B^n(C^\bullet)$, we call this element a **cohomology class**, and often denote it by $\operatorname{cls}(z)$.*

Proposition 2.5.6

If $\mathscr{C}$ is an abelian category, then $H^n : \mathbf{Com}_{\mathscr{C}} \to \mathscr{C}$ is an additive covariant functor.

Proof By Freyd-Mitchell Embedding Theorem, it suffices to prove this proposition when $\mathscr{C} = \mathbf{Mod}_A$. We have just defined H^n on objects, it remains to define H^n on morphisms. If $f : C^\bullet \to C'^\bullet$ is a chain map, define $H^n(f) : H^n(C^\bullet) \to H^n(C'^\bullet)$ by

$$H^n(f) : \operatorname{cls}(z_n) \mapsto \operatorname{cls}(f_n z_n).$$

We must show that $f_n z_n$ is a cocycle and $H^n(f)$ is independent of the choice of cocycle z_n, both of these follow from f being a chain map, that is, from commutativity of the following diagram:

$$\begin{tikzcd} C^{n-1} \arrow[r, "d^{n-1}"] \arrow[d, "f^{n-1}"'] & C^n \arrow[r, "d^n"] \arrow[d, "f^n"'] & C^{n+1} \arrow[d, "f^{n+1}"] \\ C'^{n-1} \arrow[r, "d'^{n-1}"] & C'^n \arrow[r, "d'^n"] & C'^{n+1}. \end{tikzcd}$$

First, let z be an n-cocycle in $Z^n(C^\bullet)$, then $d^n z = 0$. The commutativity of the diagram gives $d'^n f^n z = f^{n+1} d^n z = 0$, hence, $f^n z$ is an n-cocycle.

Next, assume that $z + B^n(C^\bullet) = y + B^n(C^\bullet)$, hence, $z - y \in B^n(C^\bullet)$. We may assume that

$$z - y = d^{n-1} c$$

for some $c \in C^{n-1}$. Applying f^n gives

$$f^n z - f^n y = f^n d^{n-1} c = d'^{n-1} f^{n-1} c \in B^n(C'^\bullet).$$

Thus, $\operatorname{cls}(f^n z) = \operatorname{cls}(f^n y)$, and therefore $H^n(f)$ is well-defined.

Let us now see that H^n is a functor. It is obvious that $H^n(\mathrm{id}_{C^\bullet})$ is the identity. If f and g are chain maps whose composite gf is defined, then for every n-cocycle z, we have

$$\begin{aligned} H^n(gf) : \operatorname{cls}(z) &\mapsto \operatorname{cls}((gf)^n z) \\ &= \operatorname{cls}(g^n f^n z) \\ &= H^n(g) \operatorname{cls}(f^n z) \\ &= H^n(g) H^n(f) \operatorname{cls}(z). \end{aligned}$$

Finally, H^n is additive: if $f, g : C^\bullet \to C'^\bullet$ are chain maps, then

$$\begin{aligned} H^n(f+g) : \operatorname{cls}(z) &\mapsto \operatorname{cls}((f+g)^n z) \\ &= \operatorname{cls}(f^n z + g^n z) \\ &= (H^n(f) + H^n(g)) \operatorname{cls}(z). \end{aligned}$$

□

Note *We call $H^n(f)$ the* ***induced map****.*

Homotopic maps induce the same maps on homology

Definition 2.5.14 (Homotopic)

*(1) (**Morphism homotopic.**) A morphism $f : C^\bullet \to D^\bullet$ in $\mathbf{Com}_{\mathscr{C}}$ is **homotopic to zero** if for all n, there exists a morphism $s^n : C^n \to D^{n-1}$, such that*

$$f^n = s^{n+1} \circ d_C^n + d_D^{n-1} \circ s^n.$$

*Two morphisms $f, g : C^\bullet \to D^\bullet$ are **homotopic** if $f - g$ is homotopic to zero.*

(2) An object $C^\bullet \in \mathbf{Com}_{\mathscr{C}}$ is homotopic to zero if $\mathrm{id}_{C^\bullet}$ is homotopic to zero.

A morphism homotopic to zero is visualized by the diagram (which is not commutative):

$$\begin{tikzcd} \cdots \arrow[r] & C^{n-1} \arrow[r] & C^n \arrow[r, "d_C^n"] \arrow[dl, "s^n"'] \arrow[d, "f^n"] & C^{n+1} \arrow[r] \arrow[dl, "s^{n+1}"] & \cdots \\ \cdots \arrow[r] & D^{n-1} \arrow[r, "d_D^{n-1}"] & D^n \arrow[r] & D^{n+1} \arrow[r] & \cdots . \end{tikzcd}$$

Proposition 2.5.7

If $f : C^\bullet \to D^\bullet$ is homotopic to zero, then every map $f^n : H^n(C^\bullet) \to H^n(D^\bullet)$ is zero. If f and g are homotopic, then they induce the same maps $H^n(C^\bullet) \to H^n(D^\bullet)$.

Proof It is enough to prove the first assertion, suppose that $f : C^\bullet \to D^\bullet$ defined by $f^n = s^{n+1} \circ d_C^n + d_D^{n-1} \circ s^n$. Every element of $H^n(C^\bullet)$ is represented by an n-cocycle x. Then

$$f^n(x) = s^{n+1} \circ d_C^n(x) + d_D^{n-1} \circ s^n(x) = 0 + d_D^{n-1}(s^n(x)) = d_D^{n-1}(s^n(x)),$$

that is, $f^n(x)$ is an n-coboundary in $B^n(D^\bullet)$, and therefore $f^n(x) = 0$ in $H^n(D^\bullet)$. This implies that f^n is zero.

If f and g are homotopic, then $f - g$ is homotopic to zero, and therefore they induce the same maps $H^n(C^\bullet) \to H^n(D^\bullet)$. □

Long exact sequences

Lemma 2.5.4 (Snake Lemma)

Consider a commutative diagram in abelian category $\mathscr{C}$ with exact rows:

$$\begin{tikzcd} & X' \arrow[r, "f"] \arrow[d, "u'"'] & X \arrow[r, "g"] \arrow[d, "u"] & X'' \arrow[r] \arrow[d, "u''"] & 0 \\ 0 \arrow[r] & Y' \arrow[r, "f'"] & Y \arrow[r, "g'"] & Y''. \end{tikzcd}$$

There exists an exact sequence

$$\mathrm{Ker}(u') \longrightarrow \mathrm{Ker}(u) \longrightarrow \mathrm{Ker}(u'') \xrightarrow{\delta} \mathrm{Coker}(u') \longrightarrow \mathrm{Coker}(u) \longrightarrow \mathrm{Coker}(u'').$$

If, moreover, f is monomorphism, then so is $\mathrm{Ker}(u') \to \mathrm{Ker}(u)$*; if g' is epimorphism, then so is* $\mathrm{Coker}(u) \to \mathrm{Coker}(u'')$.

Proof We give a proof by using spectral sequence, Example 2.36 . □

Theorem 2.5.4

A short exact sequence of complexes

$$\begin{tikzcd}
0^\bullet \arrow[d] & : & \cdots \arrow[r] & 0 \arrow[r] \arrow[d] & 0 \arrow[r] \arrow[d] & 0 \arrow[r] \arrow[d] & \cdots \\
A^\bullet \arrow[d] & : & \cdots \arrow[r] & A^{n-1} \arrow[r, "d_A^{n-1}"] \arrow[d] & A^n \arrow[r, "d_A^n"] \arrow[d] & A^{n+1} \arrow[r, "d_A^{n+1}"] \arrow[d] & \cdots \\
B^\bullet \arrow[d] & : & \cdots \arrow[r] & B^{n-1} \arrow[r, "d_B^{n-1}"] \arrow[d] & B^n \arrow[r, "d_B^n"] \arrow[d] & B^{n+1} \arrow[r, "d_B^{n+1}"] \arrow[d] & \cdots \\
C^\bullet \arrow[d] & : & \cdots \arrow[r] & C^{n-1} \arrow[r, "d_C^{n-1}"] \arrow[d] & C^n \arrow[r, "d_C^n"] \arrow[d] & C^{n+1} \arrow[r, "d_C^{n+1}"] \arrow[d] & \cdots \\
0^\bullet & : & \cdots \arrow[r] & 0 \arrow[r] & 0 \arrow[r] & 0 \arrow[r] & \cdots
\end{tikzcd}$$

*induces a **long exact sequence in cohomology***

$$\begin{tikzcd}
 & \cdots \arrow[r] & H^{n-1}(C^\bullet) \arrow[r] & {} \\
H^n(A^\bullet) \arrow[r] & H^n(B^\bullet) \arrow[r] & H^n(C^\bullet) \arrow[r] & {} \\
H^{n+1}(A^\bullet) \arrow[r] & \cdots & &
\end{tikzcd}$$

Proof From the Snake Lemma and the diagram

$$\begin{tikzcd}
 & 0 \arrow[d] & 0 \arrow[d] & 0 \arrow[d] & \\
0 \arrow[r] & Z^{n-1}(A^\bullet) \arrow[r] \arrow[d] & Z^{n-1}(B^\bullet) \arrow[r] \arrow[d] & Z^{n-1}(C^\bullet) \arrow[d] & \\
0 \arrow[r] & A^{n-1} \arrow[r] \arrow[d, "d_A^{n-1}"'] & B^{n-1} \arrow[r] \arrow[d, "d_B^{n-1}"] & C^{n-1} \arrow[r] \arrow[d, "d_C^{n-1}"] & 0 \\
0 \arrow[r] & A^n \arrow[r] \arrow[d] & B^n \arrow[r] \arrow[d] & C^n \arrow[r] \arrow[d] & 0 \\
 & A^n/d_A^{n-1}(A^{n-1}) \arrow[r] \arrow[d] & B^n/d_B^{n-1}(B^{n-1}) \arrow[r] \arrow[d] & C^n/d_C^{n-1}(C^{n-1}) \arrow[r] \arrow[d] & 0 \\
 & 0 & 0 & 0 &
\end{tikzcd}$$

we see that the top and bottom rows are exact. Consider the following commutative diagram

$$\begin{tikzcd} & A^n/d_A^{n-1}(A^{n-1}) \arrow[r] \arrow[d, "\overline{d_A^n}"'] & B^n/d_B^{n-1}(B^{n-1}) \arrow[r] \arrow[d, "\overline{d_B^n}"] & C^n/d_C^{n-1}(C^{n-1}) \arrow[r] \arrow[d, "\overline{d_C^n}"] & 0 \\ 0 \arrow[r] & Z^{n+1}(A^\bullet) \arrow[r] & Z^{n+1}(B^\bullet) \arrow[r] & Z^{n+1}(C^\bullet) & \end{tikzcd}$$

where $\overline{d^n_{(\cdot)}}$ is induced by $d^n_{(\cdot)}$. Note that $H^n(\cdot)$ is the kernel of $\overline{d^n_{(\cdot)}}$ and $H^{n+1}(\cdot)$ is the cokernel of $\overline{d^n_{(\cdot)}}$, apply the Snake lemma again, we obtain

$$H^n(A^\bullet) \longrightarrow H^n(B^\bullet) \longrightarrow H^n(C^\bullet) \xrightarrow{\partial} H^{n+1}(A^\bullet) \longrightarrow H^{n+1}(B^\bullet) \longrightarrow H^{n+1}(C^\bullet).$$

Pasting these sequences together we obtain the result. □

2.5.3 Exactness

Definition 2.5.15 (Right-exact, left-exact, and exact)

If $F : \mathscr{A} \to \mathscr{B}$ is an additive covariant functor from one abelian category to another, we say that F is ***right-exact*** *if the exactness of*

$$A' \longrightarrow A \longrightarrow A'' \longrightarrow 0,$$

in $\mathscr{A}$ implies that

$$F(A') \longrightarrow F(A) \longrightarrow F(A'') \longrightarrow 0$$

is also exact. Dually, we say that F is ***left-exact*** *if the exactness of*

$$0 \longrightarrow A' \longrightarrow A \longrightarrow A''$$

implies

$$0 \longrightarrow F(A') \longrightarrow F(A) \longrightarrow F(A'')$$

is exact.

An additive contravariant functor is ***left-exact*** *if the exactness of*

$$A' \longrightarrow A \longrightarrow A'' \longrightarrow 0$$

implies

$$0 \longrightarrow F(A'') \longrightarrow F(A) \longrightarrow F(A')$$

is exact. Dually, we say that contravariant functor is ***right-exact*** *if the exactness of*

$$0 \longrightarrow A' \longrightarrow A \longrightarrow A''$$

implies

$$F(A'') \longrightarrow F(A) \longrightarrow F(A') \longrightarrow 0$$

An additive covariant or contravariant functor is ***exact*** *if it is both left-exact and right-exact.*

✍ **Exercise 2.36** Suppose F is an exact functor. Show that applying F to an exact sequence preserves exactness. For example, if F is covariant, and $A' \to A \to A''$ is exact, then $FA' \to FA \to FA''$ is exact.

Proof Since $A' \to A \to A''$ is exact, it can be decomposed into some exact sequences, that is, following diagram commutative.

$$\begin{tikzcd}
 & 0 \arrow[d] & & & 0 \\
 & \operatorname{Ker} f \arrow[d, hook] & & \operatorname{Im} g \arrow[d, hook] \arrow[ur] & \\
 & A' \arrow[r, "f"] \arrow[d, two heads] & A \arrow[r, "g"] \arrow[ur, two heads] & A'' & \\
 & \operatorname{Im} f = \operatorname{Ker} g \arrow[ur, hook] \arrow[d] & & & \\
0 \arrow[ur] & 0 & & &
\end{tikzcd}$$

Since F is an exact sequence, we have exact sequences

$$0 \longrightarrow F(\operatorname{Ker} f) \longrightarrow F(A') \longrightarrow F(\operatorname{Im} f) = F(\operatorname{Ker} g) \longrightarrow 0$$

$$0 \longrightarrow F(\operatorname{Im} f) = F(\operatorname{Ker} g) \longrightarrow F(A) \longrightarrow F(\operatorname{Im} g) \longrightarrow 0$$

and commutative diagram

$$\begin{tikzcd}
 & 0 \arrow[d] & & & 0 \\
 & F(\operatorname{Ker} f) \arrow[d, hook] & & F(\operatorname{Im} g) \arrow[d, hook] \arrow[ur] & \\
 & F(A') \arrow[r, "Ff"] \arrow[d, two heads] & F(A) \arrow[r, "Fg"] \arrow[ur, two heads] & F(A'') & \\
 & F(\operatorname{Im} f) = F(\operatorname{Ker} g) \arrow[ur, hook] \arrow[d] & & & \\
0 \arrow[ur] & 0. & & &
\end{tikzcd}$$

Note that

$$\begin{aligned}
\operatorname{Im} Ff &= \operatorname{Im}(F(A') \to F(\operatorname{Im} f) \to F(A)) \\
&\overset{a}{=} \operatorname{Im}(F(\operatorname{Im} f) \to F(A)) \\
&= \operatorname{Ker}(F(A) \to F(\operatorname{Im} g)) \\
&\overset{b}{=} \operatorname{Ker}(F(A) \to F(\operatorname{Im} g) \to F(A'')) \\
&= \operatorname{Ker} Fg,
\end{aligned}$$

where "a" holds since $F(A') \to F(\operatorname{Im} f)$ is epimorphism and "b" holds since $F(\operatorname{Im} g) \to F(A'')$ is monomorphism, hence $F(A') \to F(A) \to F(A'')$ is exact. □

✍ **Exercise 2.37** Suppose A is a ring, $S \subseteq A$ is a multiplicative subset, and M is an A-module.

(a) Show that localization of A-modules $\mathbf{Mod}_A \to \mathbf{Mod}_{S^{-1}A}$ is an exact covariant functor.

(b) Show that $\square \otimes_A M$ is a right-exact covariant functor $\mathbf{Mod}_A \to \mathbf{Mod}_A$.

(c) Show that $\operatorname{Hom}(M, \square)$ is a left-exact covariant functor $\mathbf{Mod}_A \to \mathbf{Mod}_A$. If $\mathscr{C}$ is any abelian category, and $C \in \mathscr{C}$, show that $\operatorname{Hom}(C, \square)$ is a left-exact covariant functor $\mathscr{C} \to \mathbf{Ab}$.

(d) Show that $\operatorname{Hom}(\square, M)$ is a left-exact contravariant functor $\mathbf{Mod}_A \to \mathbf{Mod}_A$. If $\mathscr{C}$ is any abelian category, and $C \in \mathscr{C}$, show that $\operatorname{Hom}(\square, C)$ is a left-exact contravariant functor $\mathscr{C} \to \mathbf{Ab}$.

Proof

(a) Let $M' \xrightarrow{f} M \xrightarrow{g} M''$, then $\operatorname{Im} f = \operatorname{Ker} g$. Hence, $g \circ f = 0$, and therefore $S^{-1}g \circ S^{-1}f = S^{-1}(0)$. Hence, $\operatorname{Im} S^{-1}f \subseteq \operatorname{Ker} S^{-1}g$. Conversely, let $m/s \in \operatorname{Ker}(S^{-1}g)$, then $g(m)/s = 0$ in $S^{-1}M''$, hence, there exists $t \in S$ such that $tg(m) = 0$ in M''. But $tg(m) = g(tm)$, since g is an A-module homomorphism. Hence, $tm \in \operatorname{Ker} g = \operatorname{Im} f$, there exists $m' \in M'$ such that $tm = f(m')$. Hence, in $S^{-1}M$ we have $m/s = f(m')/st = S^{-1}f(m'/st) \in \operatorname{Im} S^{-1}f$. Hence, $\operatorname{Im} S^{-1}f = \operatorname{Ker} S^{-1}g$, that is sequence

$$S^{-1}M' \xrightarrow{S^{-1}f} S^{-1}M \xrightarrow{S^{-1}g} S^{-1}M''$$

is exact. Hence, S^{-1} is an exact covariant functor.

(b) Say

$$M' \xrightarrow{i} M \xrightarrow{p} M'' \longrightarrow 0.$$

is an exact sequence.

We shall to show that the sequence

$$M' \otimes_A N \xrightarrow{i\otimes \mathrm{id}_N} M \otimes_A N \xrightarrow{p\otimes \mathrm{id}_N} M'' \otimes_A N \longrightarrow 0$$

is exact.

(i) $\operatorname{Im} i \otimes \mathrm{id}_N = \operatorname{Ker} p \otimes \mathrm{id}_N$.

For $\operatorname{Im} i \otimes \mathrm{id}_N \subseteq \operatorname{Ker} p \otimes \mathrm{id}_N$, it suffices to prove that the composite is 0. Note that

$$(p \otimes \mathrm{id}_N)(i \otimes \mathrm{id}_N) = pi \otimes \mathrm{id}_N = 0 \otimes \mathrm{id}_N = 0,$$

then we done.

Let $E = \operatorname{Im}(i \otimes \mathrm{id}_N)$, then $E \subseteq \operatorname{Ker} p \otimes \mathrm{id}_N$, and therefore $i \otimes \mathrm{id}_N$ induces a map $\widehat{p} : (M \otimes N)/E \to M'' \otimes N$ with

$$\widehat{p} : m \otimes n + E \mapsto pm \otimes n,$$

where $m \in M$ and $n \in N$. Now if $\pi : M \otimes N \to (M \otimes N)/E$ is the natural nap, then

$$\widehat{p}\pi = p \otimes \mathrm{id}_N,$$

for both send $a \otimes b \mapsto pa \otimes b$.

$$\begin{tikzcd} M \otimes N \arrow[rr, "\pi"] \arrow[dr, "p\otimes \mathrm{id}_N"'] & & (M \otimes N)/E \arrow[dl, "\widehat{p}"] \\ & M'' \otimes N & \end{tikzcd}$$

Suppose we show that $\widehat{p}$ is an isomorphism. Then

$$\operatorname{Ker}(p \otimes \mathrm{id}_N) = \operatorname{Ker} \widehat{p}\pi = \operatorname{Ker} \pi = E = \operatorname{Im}(i \otimes \mathrm{id}_N),$$

and we done. To see that $\widehat{P}$ is an isomorphism, we construct its inverse $M'' \otimes N \to (M \otimes N)/E$. Define

$$f : M'' \times N \to (M \otimes N)/E$$

as follows. If $m'' \in M''$, there is $m \in M$ such that $p(m) = m''$, since p is surjective, let

$$f : (m'', n) \mapsto m \otimes n + E.$$

Now f is well-defined: if $pm_1 = m''$, then $p(m - m_1) = 0$ and $m - m_1 \in \operatorname{Ker} p = \operatorname{Im} i$. Thus there is $m' \in M'$ with $i(m') = m - m_1$, and hence $(m - m_1) \otimes n = i(m') \otimes n \in \operatorname{Im}(i \otimes \mathrm{id}_N) = E$, that is, $m \otimes n + E = m_1 \otimes n + E$. Clearly, f is A-biadditive, and so the definition of tensor product gives a homomorphism $\widehat{f} : M'' \otimes N \to (M \otimes N)/E$ with $\widehat{f}(m'' \otimes n) = m \otimes n + E$. It is easy to check that $\widehat{f}$ is the inverse of $\widehat{p}$, as desired.

Hence, we have $\operatorname{Im} i \otimes \mathrm{id}_N = \operatorname{Ker} p \otimes \mathrm{id}_N$.

(ii) $p \otimes \mathrm{id}_N$ is surjective.

Since, p is surjective, if $\sum m_i'' \otimes n_i \in M'' \otimes N$, then there exist $m_i \in M$ with $pm_i = m_i''$ for all i. Hence, $(p \otimes \mathrm{id}_N)(\sum m_i \otimes n_i) = \sum m_i'' \otimes n_i$, which implies that p is surjective.

(c) We only prove the case when $\mathscr{C} = \mathbf{Mod}_A$. $\operatorname{Hom}(M, \square)$ is a covariant functor. Let

$$0 \longrightarrow N' \xrightarrow{\ i\ } N \xrightarrow{\ p\ } N''$$

be an exact sequence of A-module. Consider the sequence

$$0 \longrightarrow \operatorname{Hom}_A(M, N') \xrightarrow{\ i_*\ } \operatorname{Hom}_A(M, N) \xrightarrow{\ p_*\ } \operatorname{Hom}_A(M, N''),$$

we shall to show this sequence is exact.

(i) $\operatorname{Ker} i_* = 0$.

If $f \in \operatorname{Ker} i_*$, then $f : M \to N'$ and $i_*(f) = 0$, that is, $i(f(m)) = 0$ for all $m \in M$. Since i is injective, $f(m) = 0$ for all $m \in M$, and therefore $f = 0$. Hence, $\operatorname{Ker} i_* = 0$.

(ii) $\operatorname{Im} i_* \subseteq \operatorname{Ker} p_*$.

If $g \in \operatorname{Im} i_*$, then $g : M \to N$ and there is some $f : M \to N'$ with $g = i_*(f) = if$. Then $p_*(g) = pg = pif = 0$, since $\operatorname{Im} i = \operatorname{Ker} p$, and therefore $g \in \operatorname{Ker} p_*$, which implies that $\operatorname{Im} i_* \subseteq \operatorname{Ker} p_*$.

(iii) $\operatorname{Ker} p_* \subseteq \operatorname{Im} i_*$.

If $g \in \operatorname{Ker} p_*$, then $g : M \to N$ and $p_*(g) = pg = 0$. Hence, $pg(m) = 0$ for all $m \in M$, so that $g(m) \in \operatorname{Ker} p = \operatorname{Im} i$. Thus, $g(m) = i(n')$ for some $n' \in N'$, since i is injective, n' is unique. Hence the map $f : M \to N'$, given by $f(m) = n'$ if $g(m) = i(n')$, is well-defined. We shall to check that $f \in \operatorname{Hom}_A(M, N')$. Note that $g(m_1+m_2) = g(m_1)+g(m_2) = i(n_1')+i(n_2') = i(n_1'+n_2')$, we have $f(m_1+m_2) = n_1'+n_2' = f(m_1)+f(m_2)$. Also note that $g(am) = ag(m) = ai(n') = i(an')$, we have $f(am) = an' = af(m)$. Hence, f is A-map. Since $i_*(f) = if$ and $if(m) = i(n') = g(m)$ for all $m \in M$, that is, $i_*(f) = g$, and so $g \in \operatorname{Im} i_*$.

(d) The proof is same as part (c).

$\square$

Exercise 2.38 Suppose M is a **finitely presented A-module**: M has a finite number of generators, and with these generators it has a finite number of relations; or usefully equivalently, fits in an exact sequence

$$A^{\oplus q} \longrightarrow A^{\oplus p} \longrightarrow M \longrightarrow 0 \tag{2.19}$$

Use (2.19) and the left-exactness of Hom to describe an isomorphism

$$S^{-1} \operatorname{Hom}_A(M, N) \stackrel{\sim}{\longleftrightarrow} \operatorname{Hom}_{S^{-1}A}(S^{-1}M, S^{-1}N).$$

Proof Since S^{-1} is exact functor, we have following exact sequence.

$$S^{-1}A^{\oplus q} \longrightarrow S^{-1}A^{\oplus p} \longrightarrow S^{-1}M \longrightarrow 0 \tag{2.20}$$

Apply contravariant left-exact fuctor $\operatorname{Hom}_{S^{-1}A}(\square, S^{-1}N)$ to (2.20), sequence

$$0 \longrightarrow \operatorname{Hom}_{S^{-1}A}(S^{-1}M, S^{-1}N) \longrightarrow \operatorname{Hom}_{S^{-1}A}(S^{-1}A^{\oplus p}, S^{-1}N) \longrightarrow \operatorname{Hom}_{S^{-1}A}(S^{-1}A^{\oplus q}, S^{-1}N) \tag{2.21}$$

is exact. Note that localization commutes with direct sum, we have $S^{-1}A^{\oplus k} = (S^{-1}A)^{\oplus k}$ for $k = p, q$. Also, Hom commutes with direct sum (since p, q finite), hence,

$$\operatorname{Hom}_{S^{-1}A}(S^{-1}A^{\oplus k}, N) \cong \operatorname{Hom}_{S^{-1}A}((S^{-1}A)^{\oplus k}, S^{-1}N) \cong \bigoplus_{i=1}^{k} \operatorname{Hom}_{S^{-1}A}(S^{-1}A, S^{-1}N)$$

for $k = p, q$. Note that $\operatorname{Hom}_{S^{-1}A}(S^{-1}A, S^{-1}N)$ can be seen as $S^{-1}A$-module $S^{-1}N$, then we have exact sequence.

$$0 \longrightarrow \operatorname{Hom}_{S^{-1}A}(S^{-1}M, S^{-1}N) \longrightarrow (S^{-1}N)^{\oplus p} \longrightarrow (S^{-1}N)^{\oplus q} \tag{2.22}$$

On the other hand, apply $\operatorname{Hom}_A(\square, N)$ to (2.19), we have following exact sequence.

$$0 \longrightarrow \operatorname{Hom}_A(M, N) \longrightarrow \operatorname{Hom}_A(A^{\oplus p}, N) \longrightarrow \operatorname{Hom}_A(A^{\oplus q}, N) \tag{2.23}$$

Apply S^{-1} to (2.23), note that Hom commutes with direct sum and localization commutes with direct sum, we have exact sequence

$$0 \longrightarrow S^{-1}\operatorname{Hom}_A(M, N) \longrightarrow (S^{-1}\operatorname{Hom}_A(A, N))^{\oplus p} \longrightarrow (S^{-1}\operatorname{Hom}_A(A, N))^{\oplus q}. \tag{2.24}$$

Since N is A-module, $\operatorname{Hom}_A(A, N)$ can be seen as A-module N, hence, sequence

$$0 \longrightarrow S^{-1}\operatorname{Hom}_A(M, N) \longrightarrow (S^{-1}N)^{\oplus p} \longrightarrow (S^{-1}N)^{\oplus q} \tag{2.25}$$

is exact.

Hence, (2.22) implies that

$$\operatorname{Hom}_{S^{-1}A}(S^{-1}M, S^{-1}N) = \operatorname{Ker}((S^{-1}N)^{\oplus p} \to (S^{-1}N)^{\oplus q}),$$

and (2.25) implies that

$$S^{-1}\operatorname{Hom}_A(M, N) = \operatorname{Ker}((S^{-1}N)^{\oplus p} \to (S^{-1}N)^{\oplus q}).$$

By the universal property of kernel,

$$\operatorname{Hom}_{S^{-1}A}(S^{-1}M, S^{-1}N) \cong S^{-1}\operatorname{Hom}_A(M, N).$$

□

Example 2.35 (Hom **doesn't always commute with localization.**) In the langrage of Exercise 2.38, take $A = N = \mathbb{Z}$, $M = \mathbb{Q}$, and $S = \mathbb{Z} \setminus \{0\}$.

Proof In fact, $\operatorname{Hom}_{\mathbb{Z}}(\mathbb{Q}, \mathbb{Z}) = 0$. Suppose there is a nonzero morphism $f : \mathbb{Q} \to \mathbb{Z}$. Let $q \in \mathbb{Q}$, and $f(q) = n$ for some $n \in \mathbb{Z}$. For all $k \in \mathbb{Z}$, since f is $\mathbb{Z}$-map, we have

$$f(q) = f(k \cdot \frac{q}{k}) = kf(\frac{q}{k}) = n.$$

Hence, $f(\frac{q}{k}) = \frac{n}{k}$ for all $n \in \mathbb{Z}$. Let n large enough, then $n/k < 1$, a contradiction! Hence, $f = 0$, and therefore $\operatorname{Hom}_{\mathbb{Z}}(\mathbb{Q}, \mathbb{Z}) = 0$. Thus, $S^{-1}\operatorname{Hom}_{\mathbb{Z}}(\mathbb{Q}, \mathbb{Z}) = 0$. Note that $\operatorname{Hom}_{\mathbb{Q}}(\mathbb{Q}, \mathbb{Q}) \cong \mathbb{Q}$, hence,

$$S^{-1}\operatorname{Hom}_{\mathbb{Z}}(\mathbb{Q}, \mathbb{Z}) \ncong \operatorname{Hom}_{\mathbb{Q}}(\mathbb{Q}, \mathbb{Q}).$$

□

Two useful facts in homological algebra.

Note *Ravi Vakil: "We now come to two (sets of) facts I wish I had learned as a child, as they would have saved me lots of grief. They encapsulate what is best and worst of abstract nonsense. The statements are so general as to be nonintuitive. The proofs are very short. They generalize some specific behavior that is easy to prove on an ad hoc basis. Once they are second nature to you, many subtle facts will become obvious to you as special cases. And you will see that they will get used (implicitly or explicitly) repeatedly."*

Interaction of homology and (right/left-)exact functors.

Theorem 2.5.5 (The $FHHF$ Theorem)

Suppose $F : \mathscr{A} \to \mathscr{B}$ is a covariant functor of abelian categories, and $C^\bullet$ is a complex in $\mathscr{A}$.

*(a) (F **right-exact yields** $FH^\bullet \longrightarrow H^\bullet F$) If F is right-exact, there is a natural morphism $FH^\bullet \longrightarrow H^\bullet F$. More precisely, for each i, the left side is F applied to the cohomology at piece i of $C^\bullet$, while the right side is the cohomology at piece i of $FC^\bullet$.*

*(b) (F **left-exact yields** $FH^\bullet \longleftarrow H^\bullet F$) If F is left-exact, there is a natural morphism $H^\bullet F \longrightarrow FH^\bullet$.*

*(c) (F **exact yields** $FH^\bullet \overset{\sim}{\longleftrightarrow} H^\bullet F$) If F is exact, there is a natural isomorphism between $FH^\bullet$ and $H^\bullet F$.*

Interaction of adjoint, (co)limits, and (left- and right-)exactness.

A surprising number of arguments boil down to the statement:

Limits commute with limits and right adjoints. In particular, in an abelian category, because kernels are limits, both limits and right adjoints are left-exact.

as well as its dual:

Colimits commute with colimits and left adjoints. In particular, because cokernels are colimits, both colimits and left adjoints are right-exact.

The latter has a useful extension:

In Mod_A, colimits over filtered index categories are exact.

Proposition 2.5.8 (Kernels commute with limits)

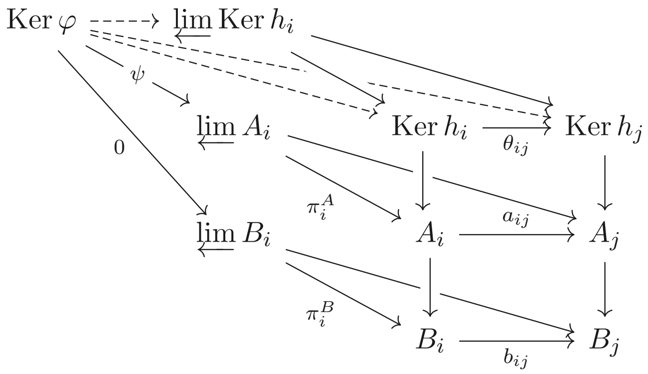
Suppose $\mathscr{C}$ is an abelian category, and $a : \mathscr{I} \to \mathscr{C}$ and $b : \mathscr{I} \to \mathscr{C}$ are two diagrams in $\mathscr{C}$ indexed by $\mathscr{I}$. For convenience, let $A_i = a(i)$ and $B_i = b(i)$ be the objects in those two diagrams. h is a natural transformation of functors $a \to b$. Then there is a cononcial isomorphism

$$\varprojlim \operatorname{Ker} h_i \overset{\sim}{\longleftrightarrow} \operatorname{Ker}(\varprojlim A_i \to \varprojlim B_i).$$

Proof Since $b_{ij} \circ h_i \circ \operatorname{Ker} h_i = 0$, by the universal property of $\operatorname{Ker} h_j$, there exists unique $\theta_{ij} : \operatorname{Ker} h_i \to \operatorname{Ker} h_j$. By the universal property of $\varprojlim B_i$, there exists unique $\varphi : \varprojlim A_i \to \varprojlim B_i$. By the universal property of $\varprojlim A_i$, there exists unique $\iota : \varprojlim \operatorname{Ker} h_i \to \varprojlim A_i$. It suffices to show that $\varprojlim \operatorname{Ker} h_i$ is the kernel of φ. Let W be any object with $\varphi \circ \psi = 0$. Consider the following commutative diagram.

$$\begin{tikzcd}
\operatorname{Ker}\varphi \arrow[dr, "\psi"] \arrow[ddr, "0"'] & \varprojlim \operatorname{Ker} h_i \arrow[l, dashed] \arrow[d, dashed, "\iota"] \arrow[dr, "\pi_i^{\operatorname{Ker}}"] \arrow[drr] & & \\
& \varprojlim A_i \arrow[d, dashed, "\varphi"] \arrow[dr, "\pi_i^A"] \arrow[drr] & \operatorname{Ker} h_i \arrow[r, dashed, "\theta_{ij}"] \arrow[d] & \operatorname{Ker} h_j \arrow[d] \\
& \varprojlim B_i \arrow[dr, "\pi_i^B"] \arrow[drr] & A_i \arrow[r, "a_{ij}"] \arrow[d, "h_i"] & A_j \arrow[d, "h_j"] \\
& & B_i \arrow[r, "b_{ij}"] & B_j
\end{tikzcd}$$

Note that $h_i \circ \operatorname{Ker} h_i = 0$, $\pi_i^B \circ \varphi \circ \iota = h_i \circ \operatorname{Ker} h_i \circ \pi_i^{\operatorname{Ker}} = 0 \circ \pi_i^{\operatorname{Ker}} = 0$ for all i. Hence, $\varphi \circ \iota = 0$. By the universal property of $\operatorname{Ker}\varphi$, there exists unique $\chi : \varprojlim \operatorname{Ker} h_i \to \operatorname{Ker}\varphi$. Now, consider the following diagram,



by the universal property of $\operatorname{Ker} h_i$ and $\operatorname{Ker} h_j$, there are unique morphism $\operatorname{Ker}\varphi \to \operatorname{Ker} h_i$ and $\operatorname{Ker}\varphi \to \operatorname{Ker} h_j$. By the universal property of $\varprojlim \operatorname{Ker} h_i$, there is unique morphism $\kappa : \operatorname{Ker}\varphi \to \varprojlim \operatorname{Ker} h_i$. Then $\chi\circ\kappa : \operatorname{Ker}\varphi \to \operatorname{Ker}\varphi$, by the universal property of $\operatorname{Ker}\varphi$ again, $\chi\circ\kappa = \mathrm{id}_{\operatorname{Ker}\varphi}$. Similarly, $\kappa\circ\chi = \mathrm{id}_{\varprojlim \operatorname{Ker} h_i}$. Hence, there is a cononcial isomorphism

$$\varprojlim \operatorname{Ker} h_i \cong \operatorname{Ker}(\varprojlim A_i \to \varprojlim B_i).$$

$\square$

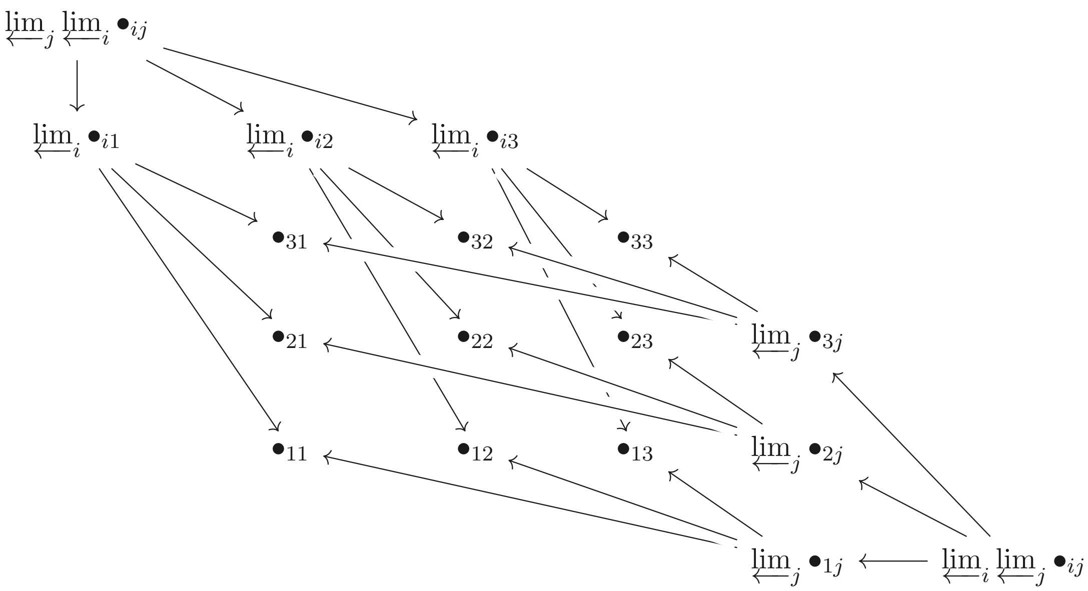
Remark 2.34 Implicit in the previous exercise is the idea that limits should somehow be understand as functor.

Proposition 2.5.9

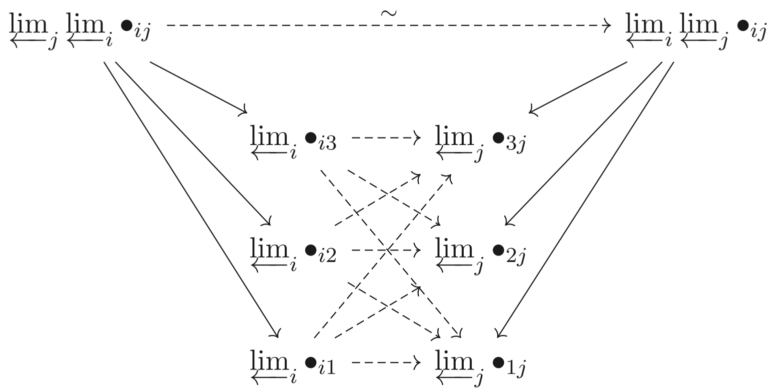
In any category, limits commute with limits. More precisely, let $F : I \times J \to \mathscr{C}$ be a functor, and suppose $\varprojlim F(i,j)$ exists for every $i \in I$. Then there is a cononcial isomorphism

$$\varprojlim_i \varprojlim_i F(i,j) \cong \varprojlim_j \varprojlim_i F(i,j).$$

Proof Consider the following diagram.



In the following diagram, all the dashed lines defined by the universal property of limits, and are all isomorphisms.



Hence,

$$\varprojlim_i \varprojlim_i F(i,j) \cong \varprojlim_j \varprojlim_i F(i,j).$$

$\square$

Proposition 2.5.10

Suppose $(F : \mathscr{C} \to \mathscr{D}, G : \mathscr{D} \to \mathscr{C})$ is a pair of adjoint functors. If $A = \varprojlim A_i$ is a limit in $\mathscr{D}$ of a diagram indexed by $\mathscr{I}$, then $GA = \varprojlim GA_i$ is a limit in $\mathscr{C}$.

Proof We must show that $GA \to GA_i$ satisfies the universal property of limits. Suppose we have maps $W \to GA_i$ commuting with the maps of $\mathscr{I}$. We wish to show that there exists a unique $W \to GA$ extending the $W \to GA_i$. By adjointness of F and G, we can restate this as: Suppose we have maps $FW \to A_i$ commuting with the maps of $\mathscr{I}$. We wish to show that there exists a unique $FW \to A$ extending the $FW \to A_i$. But this is precisely the universal property of the limit. $\square$

Dually, we have the following two propositions.

Proposition 2.5.11

In any category, colimits commute with colimits. More precisely, let $F : I \times J \to \mathscr{C}$ be a functor, and suppose $\varinjlim_j F(i,j)$ exists for ever $i \in I$. Then there is a cononcial isomorphism

$$\varinjlim_i \varinjlim_j F(i,j) \cong \varinjlim_j \varinjlim_i F(i,j).$$

Proposition 2.5.12

Suppose $(F : \mathscr{C} \to \mathscr{D}, G : \mathscr{D} \to \mathscr{C})$ is a pair of adjoint functors. If $A = \varinjlim A_i$ is a colimit in $\mathscr{C}$ of a diagram indexed by $\mathscr{I}$, then $FA = \varinjlim FA_i$.

Proposition 2.5.13

If F and G are additive functors between abelian categories, and (F,G) is an adjoint pair, then (as kernels are limits and cokernels are colimits) G is left-exact and F is right-exact.

Proof Let

$$0 \longrightarrow A \xrightarrow{\ i\ } B \xrightarrow{\ p\ } C \longrightarrow 0$$

be an exact sequence. By Proposition 2.5.3, $i = \operatorname{Ker} p$ and $p = \operatorname{Coker} i$.

By Proposition 2.5.10, G commutes with limits, and kernels are limits, hence, $Gi = G(\operatorname{Ker} p) = \operatorname{Ker} Gp$, and therefore sequence

$$0 \longrightarrow GA \xrightarrow{Gi} GB \xrightarrow{Gp} GC$$

is exact, which implies that G is left-exact.

By Proposition 2.5.12, F commutes with colimits, and cokernels are colimits, hence, $Fp = F(\operatorname{Coker} f) = \operatorname{Coker} Fi$, and therefore sequence

$$FA \xrightarrow{Fi} FB \xrightarrow{Fp} FC \longrightarrow 0$$

is exact, which implies that F is right-exact. □

Lemma 2.5.5

In $\mathbf{Mod}_A$, let $\{M_i, \varphi_{ij}\}$ be a direct system of R-modules over a filtered index categories $\mathscr{I}$, and let $\iota_i : M_i \to \bigoplus M_i$ be the i-th injection, then $\varinjlim M_i = (\bigoplus M_i)/S$, where $S = (\iota_j\varphi_{ij}m_i - \iota_i m_i : m_i \in M_i$ and $i \to j)$. Also,

(i) Each element of $\varinjlim M_i$ has a representative of the form $\iota_i m_i + S$.

(ii) $\iota_i m_i + S = 0$ if and only if $\varphi_{it}(m_i) = 0$ for some $i \to t$.

Proof

(i) It is easy to check that $(\bigoplus M_i)/S$ satisfies the universal property of colimit, and therefore $\varinjlim M_i = (\bigoplus M_i)/S$. Hence, each element $x \in \varinjlim M_i$ has the form $x = \sum \iota_i m_i + S$. Since, $\mathscr{I}$ is filtered, there exists $j \in \mathscr{I}$, such that $i \to j$ for all i occurring in the sum for x. For each such i, define $b^i = \varphi_{ij} m_i \in M_j$. Let $b = \sum_i b^i$, then $b \in M_j$. It follows that

$$\begin{aligned}\sum \iota_i m_i - \iota_j b =& \sum (\iota_i m_i - \iota_j b^i)\\ =& \sum (\iota_i m_i - \iota_j \varphi_{ij} m_i) \in S.\end{aligned}$$

Therefore, $x = \sum \iota_i m_i + S = \iota_j b + S$, as desired.

(ii) If $\varphi_{it} m_i = 0$ for some $i \to t$, then

$$\iota_i m_i + S = \iota_i m_i + (\iota_t \varphi_{it} m_i - \iota_i m_i) + S = S.$$

Conversely, if $\iota_i m_i + S = 0$, then $\iota_i m_i \in S$, and there is an expression

$$\iota_i m_i = \sum_j a_j(\iota_k \varphi_{jk} m_j - \iota_j m_j) \in S,$$

where $a_j \in A$. We are going to normalize this expression. First, we introduce the following notation for relators: if $j \to k$, define

$$r(j, k, m_j) = \iota_k \varphi_{jk} m_j - \iota_j m_j.$$

Since $a_j r(j, k, m_j) = r(j, k, a_j m_j)$, we may assume that the notion has been adjusted such that

$$\iota_i m_i = \sum_j r(j, k, m_j).$$

As $\mathscr{I}$ is filtered, we may choose an index $t \in \mathscr{I}$ larger than any of the indices i, j, k occuring in the last equation (i.e. $i \to t$, $j \to t$, and $k \to t$). Now

$$\begin{aligned}\iota_t \varphi_{it} m_i =& (\iota_t \varphi_{it} m_i - \iota_i m_i) + \iota_i m_i\\ =& r(i, t, m_i) + \iota_i m_i\\ =& r(i, t, m_i) + \sum_j r(j, k, m_j).\end{aligned}$$

Next,

$$\begin{aligned} r(j,k,m_j) =& \iota_k\varphi_{jk}m_j - \iota_j m_j \\ =& (\iota_t\varphi_{jt}m_j - \iota_j m_j) + [\iota_t\varphi_{kt}(-\varphi_{jk}m_j) - \iota_k(-\varphi_{jk}m_j)] \\ =& r(j,t,m_j) + r(k,t,-\varphi_{jk}m_j), \end{aligned}$$

because $\varphi_{kt}\circ\varphi_{ik} = \varphi_{it}$, by the definition of filtered. Hence,

$$\iota_t\varphi_{it}m_i = \sum_l r(l,t,x_l),$$

where $x_l \in M_l$. For $l \to t$,

$$\begin{aligned} r(l,t,m_l) + r(l,t,m_l') =& \iota_t\varphi_{lt}m_l - \iota_l m_l + \iota_t\varphi_{lt}m_I' - \iota_l m_l' \\ =& \iota_t\varphi_{lt}(m_l + m_l') - \iota_l(m_l + m_l') \\ =& r(l,t,m_l + m_l'). \end{aligned}$$

Therefore, we may amalgamate all relators with the same smaller index l and write

$$\begin{aligned} \iota_t\varphi_{it}m_i =& \sum_l r(l,t,x_l) \\ =& \sum_l (\iota_t\varphi_{lt}x_l - \iota_l x_l) \\ =& \iota_t(\sum_l \varphi_{lt}x_l) - \sum_l \iota_l x_l, \end{aligned}$$

where $x_\iota \in M_\iota$ and all the indices l are distinct. The unique expression of an element in a direct sum allows us to conclude, if $l \neq t$, that $\iota_l x_l = 0$, it follows that $x_l = 0$, for ι_l is an injection. The right side simplifies to $\iota_t\varphi_{tt}m_t - \iota_t m_t = 0$. Since φ_{tt} is the identity, right side is 0 and $\iota_t\varphi_{it}m_i = 0$. Since ι_t is an injection, we have $\varphi_{it}m_i = 0$, as we desired.

□

Proposition 2.5.14

In $\mathbf{Mod}_A$, colimits over filtered index categories are exact.

Proof Suppose sequence

$$0 \longrightarrow A_j \xrightarrow{r} B_j \xrightarrow{p} C_j \longrightarrow 0$$

is exact, for all $j \in \mathscr{I}$, then $p = \operatorname{Coker} r$. Since cokernel is colimit, colimit comutes with colimit, we have $\varinjlim p = \varinjlim \operatorname{Coker} r = \operatorname{Coker} \varinjlim r$, hence, the sequence,

$$\varinjlim_j A_j \xrightarrow{\varinjlim r} \varinjlim_j B_j \xrightarrow{\varinjlim p} \varinjlim_j C_j \longrightarrow 0$$

is exact. This implies that colimit is right-exact.

Now, we need to show that colimit is left adjoint. It suffices to show that $\varinjlim r$ is injection. Suppose that $\varinjlim r(x) = 0$, where $x \in \varinjlim A_j$. Since the index categories is filtered, Lemma 2.5.5 allow us to write $x = \iota_i^A a_i + S_A$. By definition, $\varinjlim r(x) = \iota_i^B r_i a_i + S_B$. Now Lemma 2.5.5 shows that $\iota_i^B r_i a_i + S_B$ in $\varinjlim B_j$ implies that there is an index k with $i \to k$ such that $\varphi_{ik}^B r_i a_i = 0$. Since r is a morphism of direct systems (i.e., r is natural transformation), we have

$$0 = \varphi_{ik}^B r_i a_i = r_k\varphi_{ik}^A a_i.$$

Since r_k is injection, we have $\varphi_{ik}^A a_i = 0$, and therefore,

$$x = \iota_i^A a_i + S_A = \iota_k \circ \varphi_{ik}^A a_i + S_A = S_A = 0.$$

Hence, $\varinjlim r$ is an injection. □

Corollary 2.5.2

Filtered colimits commute with homology in $\mathbf{Mod}_A$.

Proof By Proposition 2.5.14, filtered colimits are exact. By the $FHHF$ Theorem 2.5.5, there is a natural isomorphism between $FH^\bullet$ and $H^\bullet F$, which implies that filtered colimits commute with homology in $\mathbf{Mod}_A$. □

Remark 2.35 Just as colimits are exact (not just right-exact) in especially good circumstances, limits are exact (not just left-exact) too.

✍ **Exercise 2.39** Suppose

$$\begin{tikzcd}
& \vdots \arrow[d, two heads, "\alpha_{n+2}"'] & \vdots \arrow[d, "\beta_{n+2}"'] & \vdots \arrow[d, "\gamma_{n+2}"] & \\
0 \arrow[r] & A_{n+1} \arrow[r, "r_{n+1}"] \arrow[d, two heads, "\alpha_{n+1}"'] & B_{n+1} \arrow[r, "p_{n+1}"] \arrow[d, "\beta_{n+1}"'] & C_{n+1} \arrow[r] \arrow[d, "\gamma_{n+1}"] & 0 \\
0 \arrow[r] & A_n \arrow[r, "r_n"] \arrow[d, two heads, "\alpha_n"'] & B_n \arrow[r, "p_n"] \arrow[d, "\beta_n"'] & C_n \arrow[r] \arrow[d, "\gamma_n"] & 0 \\
& \vdots \arrow[d, two heads, "\alpha_1"'] & \vdots \arrow[d, "\beta_1"'] & \vdots \arrow[d, "\gamma_1"] & \\
0 \arrow[r] & A_0 \arrow[r, "r_0"] & B_0 \arrow[r, "p_0"] & C_0 \arrow[r] & 0
\end{tikzcd}$$

is an inverse system of exact sequence of modules over a ring, such that the maps $A_{n+1} \to A_n$ are surjective. (We say: "transition maps of the left term are surjective".) Show that the limit

$$0 \longrightarrow \varprojlim A_n \longrightarrow \varprojlim B_n \longrightarrow \varprojlim C_n \longrightarrow 0$$

is also exact.

Proof By Proposition 2.5.8, left-exactness is obviously. It suffices to show that $\varprojlim B_n \to \varprojlim C_n$ is surjective.

In fact, in $\mathbf{Mod}_R$, it is easy to check $\varprojlim A_i = \Big((a_i) \in \prod A_i : a_i = \psi^A_{ji}(a_j), j \geq i\Big)$ is a submodule of $\prod A_i$.

$$\begin{tikzcd}
0 \arrow[r] & \varprojlim A_i \arrow[r, dashed] \arrow[d, "\pi^A_{n+1}"] \arrow[dd, bend right, "\pi^A_n"'] & \varprojlim B_i \arrow[r, dashed] \arrow[d, "\pi^B_{n+1}"] \arrow[dd, bend right, "\pi^B_n"'] & \varprojlim C_i \arrow[d, "\pi^C_{n+1}"] \arrow[dd, bend right, "\pi^C_n"'] & \\
0 \arrow[r] & A_{n+1} \arrow[r, "r_{n+1}"] \arrow[d, "\alpha_{n+1}"] & B_{n+1} \arrow[r, "p_{n+1}"] \arrow[d, "\beta_{n+1}"] & C_{n+1} \arrow[r] \arrow[d, "\gamma_{n+1}"] & 0 \\
0 \arrow[r] & A_n \arrow[r, "r_n"] & B_n \arrow[r, "p_n"] & C_n \arrow[r] & 0
\end{tikzcd}$$

Define $\varprojlim p : \varprojlim B_i \to \varprojlim C_i$ by setting $\varprojlim p((b_i)) = (p_i(b_i))$, then it is easy to see $\varprojlim p$ is well-defined and agree with above diagram. We shall to show that $\varprojlim p$ is surjective. Let $(c_i) \in \varprojlim C_i$. We construct (b_i) by induction. Since p_0 is surjective, there exists b_0 such that $p_0(b_0) = c_0$. Suppose we defined b_i for all $i \leq n$ such that $\psi^B_{ji}(b_j) = b_i$ for all $n \geq j \geq i$. Let $b'_{n+1} \in B_{n+1}$ with $p_{n+1}(b'_{n+1}) = c_{n+1}$. Say $\beta_{n+1}(b'_{n+1}) = b'_n$, since $\gamma_{n+1} \circ p_{n+1} = p_n \circ \beta_{n+1}$ and $\gamma_{n+1}(c_{n+1}) = c_n$, $\gamma_{n+1}(c_{n+1}) = p_n(b'_n) = c_n$. Note that $\operatorname{Ker} p_n = \operatorname{Im} r_n$, we may assume $b'_n = b_n + r_n(a_n)$ for some $a_n \in A_n$. Since α_{n+1} is surjective, there exists a_{n+1} such that

$\alpha_{n+1}(a_{n+1}) = a_n$. Let $b_{n+1} = b'_{n+1} - r_{n+1}(a_{n+1})$, then

$$\begin{aligned}\beta_{n+1}(b_{n+1}) =&\beta_{n+1}(b'_{n+1}) - \beta_{n+1}r_{n+1}(a_{n+1})\\ =&\beta_{n+1}(b'_{n+1}) - r_n\alpha_{n+1}(a_{n+1})\\ =&b'_n - r_n(a_n) = b_n.\end{aligned}$$

Hence, we defined (b_i) with $\psi^B_{ji}(b_j) = b_i$, where $j \geq i$ and $\psi^B_{ji} = \beta_{i+1} \circ \cdot \circ \beta_j$, and therefore $\varprojlim p$ is surjective, which implies the right-exactness. □

Remark 2.36 Based on these ideas, we may suspect that right-exact functors always commute with colimits. The fact that tensor product commutes with infinite direct sums may reinforce this idea. Unfortunately, it is not ture — "double dual" $\square^{\vee\vee} : \mathbf{Vec}_k \to \mathbf{Vec}_k$ is covariant and right exact (in fact, exact), but does not commute with infinite direct sums, as $\bigoplus_{i=1}^{\infty}(k^{\vee\vee})$ is not isomorphic to $(\bigoplus_{i=1}^{\infty} k)^{\vee\vee}$.

Dreaming of derived functors

Remark 2.37 When we see a left-exact functor, you should always dream that you are seeing the end of a long exact sequence. If

$$0 \longrightarrow M' \longrightarrow M \longrightarrow M'' \longrightarrow 0$$

is an exact sequence in abelian category $\mathscr{A}$, and $F : \mathscr{A} \to \mathscr{B}$ is a left-exact functor, then

$$0 \longrightarrow FM' \longrightarrow FM \longrightarrow FM''$$

is exact, and you should always dream that it should continue in some natural way. For example, the next term should depend only on M', call it R^1FM', and if it is zero, then $FM \to FM''$ is an epimorphism. This remark holds true for left-exact and contravariant functors too. In god cases, such a continuation exists, and is incredibly useful.

2.6 ⋆ Spectral sequences

Spectral sequence are a powerful book-keeping tool for proving things involving complicated commutative diagrams. They were introduced by Leray in the 1940's at the same time as he introduced sheaves. They have a reputation for being abstruse and difficult. It has been suggested that the name "spectral" was given because, like specters, spectral sequences are terrifying, evil, and dangerous. I (Ravi Vakil) have heard no one disagree with this interpretation, which is perhaps not surprising since I just made it up.

Nonetheless, the goal of this section is to tell you enough that you can use spectral sequences without hesitation or fear, and why you shouldn't be frightened when they come up in a seminar. What is perhaps different in this presentation is that we will use spectral sequences to prove things that you may have already seen, and that you can prove easily in other ways. This will allow you to get some hands-on experience for how to use them. We will also see them only in the special case of double complexes (the version by far the most often used in algebraic geometry), and not in the general form usually presented (filtered complexes, exact couples, etc).

You should not read this section when you are reading the rest of Chapter 2. Instead, you should read it just before you need it for the first time.

For concreteness, we work in the category $\mathbf{Mod}_A$ of module over a ring A. However, everything we say will apply in any abelian category. (And if it helps you feel secure, work instead in the category $\mathbf{Vec}_k$ of vector spaces over a field k.)

2.6.1 Double complexes

Definition 2.6.1 (Double complex)

*A **double complex** is a collection of A-modules $E^{p,q}$ ($p, q \in \mathbb{Z}$), and "rightward" morphisms $d_{\to}^{p,q} : E^{p,q} \to E^{p+1,q}$ and "upward" morphisms $d_{\uparrow}^{p,q} : E^{p,q} \to E^{p,q+1}$. In the superscript, the first entry denotes the column number (the "x-coordinate"), and the second entry denotes the row number (the "y-coordinate"). (Warning: this is opposite to the convention for matrices.) The subscript is meant to suggest the direction of the arrows. We will always write these as $d_{\to}$ and $d_{\uparrow}$ and ignore the superscripts.*

$$\begin{tikzcd}
 & \vdots & \vdots & \\
\cdots \arrow[r] & E^{p,q+1} \arrow[u] \arrow[r, "d_{\to}^{p,q+1}"] & E^{p+1,q+1} \arrow[u] \arrow[r] & \cdots \\
\cdots \arrow[r] & E^{p,q} \arrow[u, "d_{\uparrow}^{p,q}"] \arrow[r, "d_{\to}^{p,q}"] & E^{p+1,q} \arrow[u, "d_{\uparrow}^{p+1,q}"'] \arrow[r] & \cdots \\
 & \vdots \arrow[u] & \vdots \arrow[u] &
\end{tikzcd}$$

We require that $d_{\to}$ and $d_{\uparrow}$ satisfy

(a) $d_{\to}^2 = 0$, $d_{\uparrow}^2 = 0$;

(b) either $d_{\to}d_{\uparrow} = d_{\uparrow}d_{\to}$ (all the sequence commute) or $d_{\to}d_{\uparrow} + d_{\uparrow}d_{\to} = 0$ (they all anti-commute).

Remark 2.38 In (b), both come up in nature, and you can switch from one to the other by replacing $d_{\uparrow}^{p,q}$ with $(-1)^p d_{\uparrow}^{p,q}$. So we will assume that all the squares anti-commute, but that you know how to turn the commuting case into this one. (You will see that there is no difference in the recipe, basically because the image and kernel of a homomorphism f equal the image and kernel respectively of $-f$.)

$$\begin{tikzcd}
E^{p,q+1} \arrow[rr, "d_{\to}^{p,q+1}"] & & E^{p+1,q+1} \\
 & \text{anti-commutes} & \\
E^{p,q} \arrow[uu, "d_{\uparrow}^{p,q}"] \arrow[rr, "d_{\to}^{p,q}"] & & E^{p+1,q} \arrow[uu, "d_{\uparrow}^{p+1,q}"']
\end{tikzcd}$$

There are variations on this definition, where for example the vertical arrows go downwards, or some subset of the $E^{p,q}$ is required to be zero.

Definition 2.6.2 (Total complex, Cohomology of the double complex)

From the double complex we construct a corresponding (single) complex $E^{\bullet}$ with $E^k = \bigoplus_i E^{i,k-i}$, with $d = d_{\to} + d_{\uparrow}$. In other words, when there is a single superscript k, we mean a sum of the k-th *anti-diagonal of the double complex. The single complex is sometimes called the **total complex**.*

$$\begin{tikzcd}
\vdots \\
E^{0,3} \arrow[u] \arrow[dr, phantom, "\oplus"] \\
E^{0,2} \arrow[u, "d_\uparrow"] \arrow[r, "d_\rightarrow"] \arrow[dr, phantom, "\oplus"] & E^{1,2} \arrow[dr, phantom, "\oplus"] \\
E^{0,1} \arrow[u, "d_\uparrow"] \arrow[r, "d_\rightarrow"] \arrow[dr, phantom, "\oplus"] & E^{1,1} \arrow[u, "d_\uparrow"] \arrow[r, "d_\rightarrow"] \arrow[dr, phantom, "\oplus"] & E^{2,1} \arrow[dr, phantom, "\oplus"] \\
E^{0,0} \arrow[u, "d_\uparrow"] \arrow[r, "d_\rightarrow"] \arrow[dr, equal] & E^{1,0} \arrow[u, "d_\uparrow"] \arrow[r, "d_\rightarrow"] \arrow[dr, equal] & E^{2,0} \arrow[u, "d_\uparrow"] \arrow[r, "d_\rightarrow"] \arrow[dr, equal] & E^{3.0} \arrow[r] \arrow[dr, equal] & \cdots \\
E^\bullet : & E^0 \arrow[r, "d^0"] & E^1 \arrow[r, "d^1"] & E^2 \arrow[r, "d^2"] & E^3 \arrow[r] & \cdots
\end{tikzcd}$$

*The cohomology of the single complex is sometimes called the **hypercohomology** of the double complex. We will instead use the phrase **cohomology of the double complex**.*

Remark 2.39 Note that $d^2 = (d_\rightarrow + d_\uparrow)^2 = d_\rightarrow^2 + (d_\rightarrow d_\uparrow + d_\uparrow d_\rightarrow) + d_\uparrow^2 = 0$, so $E^\bullet$ is indeed a complex.

Our initial goal will be to find the cohomology of the double complex. You will see later that we secretly also have other goals.

A spectral sequence is a recipe for computing some information about the cohomology of the double complex. I won't yet give the full recipe. Surprisingly, this fragmentary bit of information is sufficient to prove lots of things.

2.6.2 Approximate definition of spectral sequence

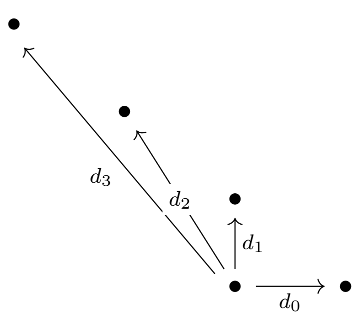
A **spectral sequence** with **rightward orientation** is a sequence of tables or **pages** ${}_\rightarrow E_0^{p,q}, {}_\rightarrow E_1^{p,q}, {}_\rightarrow E_2^{p,q}, \ldots$ ($p, q \in \mathbb{Z}$), where ${}_\rightarrow E_0^{p,q} = E^{p,q}$, along with a differential

$$ {}_\rightarrow d_r^{p,q} : {}_\rightarrow E_r^{p,q} \longrightarrow {}_\rightarrow E_r^{p-r+1,q+r} $$

($r \in \mathbb{Z}^{\geq 0}$) with ${}_\rightarrow d_r^{p,q} \circ {}_\rightarrow d_r^{p+r-1,q-r} = 0$, and with an isomorphism of the cohomology of ${}_\rightarrow d_r$ at ${}_\rightarrow E_r^{p,q}$ (i.e., $\operatorname{Ker} {}_\rightarrow d_r^{p,q} / \operatorname{Im} {}_\rightarrow d_r^{p+r-1,q-r}$) with ${}_\rightarrow E_{r+1}^{p,q}$, i.e., ${}_\rightarrow E_{r+1}^{p,q} \cong \operatorname{Ker} {}_\rightarrow d_r^{p,q} / \operatorname{Im} {}_\rightarrow d_r^{p+r-1,q-r}$.

The orientation indicates that our 0-th differential is the rightward one: $d_0 = d_\rightarrow$. The left subscript "$\longrightarrow$" is usually omitted.

The order of the morphisms is best understood visually:



(2.26)

(the morphisms each apply to different pages). Notice that the map always is "degree 1" in terms of the grading of the single complex $E^\bullet$.

Remark 2.40 Why we draw (2.26) in this way?

Denote $d_r := {}_{\rightarrow}d_r$, then we have

$$\begin{aligned} d_0^{p,q} :E_0^{p,q} &\longrightarrow \quad E_0^{p+1,q}\\ d_1^{p,q} :E_1^{p,q} &\longrightarrow \quad E_1^{p,q+1}\\ d_2^{p,q} :E_2^{p,q} &\longrightarrow \quad E_2^{p-1,q+2}\\ d_3^{p,q} :E_3^{p,q} &\longrightarrow \quad E_3^{p-2,q+3}. \end{aligned}$$

Hence the direction of d_0 is along the x-axis, d_1 is along the y-axis, d_2 points in the direction of $(-1,2)$, and d_3 points in the direction of $(-2,3)$.

The actual definition describes what $E_r^{\bullet,\bullet}$ and $d_r^{\bullet,\bullet}$ really are, in terms of $E^{\bullet,\bullet}$. We will describe d_0, d_1, and d_2 below, and you should for now take on faith that this sequence continues in some natural way.

Note that $E_r^{p,q}$ is always a subquotient of the corresponding term on the i-th page $E_i^{p,q}$ for all $i<r$. In particular, if $E^{p,q}=0$, then $E_r^{p,q}=0$ for all r.

Suppose now that $E^{\bullet,\bullet}$ is a first quadrant double complex, i.e., $E^{p,q}=0$ for $p<0$ or $q<0$ (so $E_r^{p,q}=0$ for all r unless $p,q\in\mathbb{Z}^{\geq 0}$). Then for any fixed p,q, once r is sufficiently large, $E_{r+1}^{p,q}$ is computed from $(E_r^{\bullet,\bullet}, d_r)$ using the complex

$$\begin{tikzcd} 0 & & \\ & E_r^{p,q} \arrow[ul, "d_r^{p,q}"'] & \\ & & 0 \arrow[ul, "d_r^{p+r-1,q-r}"'] \end{tikzcd}$$

(note that $d_r^{p,q}: E_r^{p,q}\to E_r^{p-r+1,q+r}$ and $E_r^{p-r+1,q+r}=0$ for sufficiently large r, similarly, $d_r^{p+r-1,q-r}$: $E_r^{p+r-1,q-r}=0\to E_r^{p,q}$ for sufficiently large r) and thus we have canonical isomorphisms

$$E_r^{p,q}\cong E_{r+1}^{p,q}\cong E_{r+2}^{p,q}\cong\cdots.$$

We denote this module $E_\infty^{p,q}$. The same idea works in other circumstances, for example if the double complex is only nonzero in a finite number of rows — $E^{p,q}=0$ unless $q_0<q<q_1$. This will come up for example in the mapping cone and long exact sequence discussion.

We now describe the first few pages of the spectral sequence explicitly. As stated above, the differential d_0 on $E_0^{\bullet,\bullet}=E^{\bullet,\bullet}$ is defined to be $d_{\rightarrow}$. The rows are complexes:

$$\text{The 0-th page } E_0: \qquad \begin{array}{ccccc} \bullet & \longrightarrow & \bullet & \longrightarrow & \bullet \\ \\ \bullet & \longrightarrow & \bullet & \longrightarrow & \bullet \\ \\ \bullet & \longrightarrow & \bullet & \longrightarrow & \bullet \end{array}$$

and so E_1 is just the table of cohomologies of the rows. There are now vertical maps $d_1^{p,q}: E_1^{p,q}\to E_1^{p,q+1}$ of the row cohomology groups, induced by $d_\uparrow$, and that these make the columns into complexes. (This is essentially the fact that a map of complexes induces a map on homology.) We have "used up the horizontal morphisms", but "the vertical differentials live on".

$$\text{The 1-st page } E_1 : \quad \begin{tikzcd} \bullet & \bullet & \bullet \\ \bullet \arrow[u] & \bullet \arrow[u] & \bullet \arrow[u] \\ \bullet \arrow[u] & \bullet \arrow[u] & \bullet \arrow[u] \end{tikzcd}$$

We take cohomology of d_1 on E_1, giving us a new table, $E_2^{p,q}$. It turns out that there are natural morphisms from each entry to the entry two above and one to the left, and that the composition of these two is 0. (It is a very worthwhile exercise to work out how this natural morphism d_2 should be defined. Your argument may be reminiscent of the connecting homomorphism in the Snake Lemma 2.5.4 or in the long exact sequence in cohomology arising from a short exact sequence of complexes, Theorem 2.5.4. This is no coincidence.)

$$\text{The 2-nd page } E_2 : \quad \begin{tikzcd} \bullet & \bullet & \bullet \\ \bullet & \bullet & \bullet \\ \bullet & \bullet \arrow[uul] & \bullet \arrow[uul] \end{tikzcd}$$

This is the beginning of a pattern.

Then it is a theorem that there is a filtration of $H^k(E^\bullet)$ by $E_\infty^{p,q}$ where $p+q=k$. (We can't yet state it as an official **Theorem** because we haven't precisely defined the pages and differentials in the spectral sequence.) More precisely, there is a filtration

$$E_\infty^{0,k} \xhookrightarrow{E_\infty^{1,k-1}} F_1 \xhookrightarrow{E_\infty^{2,k-2}} \cdots \xhookrightarrow{E_\infty^{k-1,1}} F_{k-1} \xhookrightarrow{E_\infty^{k,0}} H^k(E^\bullet) \tag{2.27}$$

where the quotients are displayed above each inclusion (i.e., $F_1/E_\infty^{0,k} \cong E_\infty^{1,k-1}$, $\dots$, $F_i/F_{i-1} \cong E_\infty^{i,k-i}$, $\dots$, $H^k(E^\bullet)/F_{k-1} \cong E_\infty^{k,0}$). (Here is a tip for remember which way the quotients are supposed to go. The differentials on later and later pages point deeper and deeper into the filtration. Thus the entries in the direction of the later arrowheads are the subobjects, and the entries in the direction of the later "arrowtails" are quotients. This tip has the advantage of being independent of the details of the spectral sequence, e.g, the "quadrant" or the orientation.)

We say that the spectral sequence ${}_{\rightarrow}E_\bullet^{\bullet,\bullet}$ **converges to** $H^\bullet(E^\bullet)$. We often say that ${}_{\rightarrow}E_2^{\bullet,\bullet}$ (or any other page) **abuts to** $H^\bullet(E^\bullet)$.

Although the filtration gives only partial information about $H^\bullet(E^\bullet)$, sometimes one can find $H^\bullet(E^\bullet)$ precisely. One example is if all $E_\infty^{i,k-i}$ are zero, or if all but one of them are zero (e.g., if $E_r^{\bullet,\bullet}$ has precisely one non-zero row or column, in which case one says that the spectral sequence **collapses** at the r-th step, although we will not use this term). Another example is in the category of vector spaces over a field, in which case we can find the dimension of $H^k(E^\bullet)$. Also, in lucky circumstances, E_2 (or some other small page) already equals E_∞.

Proposition 2.6.1 (Information from the second page)

Let $E^{\bullet,\bullet}$ be a first quadrant double complex, i.e., $E^{p,q} = 0$ for $p < 0$ or $q < 0$. Then we have $H^0(E^\bullet) = E_\infty^{0,0} = E_2^{0,0}$ and

$$0 \longrightarrow E_2^{0,1} \longrightarrow H^1(E^\bullet) \longrightarrow E_2^{1,0} \xrightarrow{d_2^{1,0}} E_2^{0,2} \longrightarrow H^2(E^\bullet)$$

is exact.

Proof Consider $H^0(E^\bullet)$, by (2.27), we have $H^0(E^\bullet)/E_\infty^{0,0} \cong E_\infty^{1,-1} = 0$, i.e., $H^0(E^\bullet) \cong E_\infty^{0,0}$. We next calculate $E_r^{0,0}$, from complex

$$\begin{tikzcd}
E_r^{-r+1,r} & & \\
& E_r^{0,0} \arrow[ul] & \\
& & E_r^{r-1,-r}, \arrow[ul]
\end{tikzcd}$$

we have $E_\infty^{0,0} = E_2^{0,0}$, i.e., $H^0(E^\bullet) = E_2^{0,0}$.

By filtration (2.27), we have

$$0 \longrightarrow E_\infty^{0,1} \xhookrightarrow{E_\infty^{1,0}} H^1(E^\bullet),$$

where $H^1(E^\bullet)/E_\infty^{0,1} \cong E_\infty^{1,0}$, hence we have exact sequence

$$0 \longrightarrow E_\infty^{0,1} \hookrightarrow H^1(E^\bullet) \twoheadrightarrow E_\infty^{1,0} \longrightarrow 0.$$

Calculate $E_\infty^{0,1}$ and $E_\infty^{1,0}$. Note that $E_{r+1}^{0,1}$ is computed from complex

$$\begin{tikzcd}
E_r^{-r+1,1+r} & & \\
& E_r^{0,1} \arrow[ul] & \\
& & E_r^{r+1,1-r} \arrow[ul]
\end{tikzcd}$$

and $E_r^{-r+1,1+r} = E_r^{r+1,1-r} = 0$ for all $r \geq 2$, we have $E_\infty^{0,1} = E_2^{0,1}$. Similarly, since $E_{r+1}^{1,0}$ is computed from complex

$$\begin{tikzcd}
E_r^{2-r,r} & & \\
& E_r^{1,0} \arrow[ul, "d_r^{1,0}"'] & \\
& & E_r^{r,-r} \arrow[ul, "d_r^{r,-r}"']
\end{tikzcd}$$

and $E_r^{-r+1,1+r} = E_r^{2-r,1+r} = 0$ for all $r \geq 3$, hence $E_\infty^{1,0} = E_3^{1,0}$. Note that $E_2^{2,-2} = 0$ and from

$$\begin{tikzcd}
E_2^{0,2} & & \\
& E_2^{0,1} \arrow[ul, "d_2^{1,0}"'] & \\
& & E_2^{2,-2} = 0, \arrow[ul, "d_2^{2,-2}"']
\end{tikzcd}$$

we have $E_3^{1,0} \cong \operatorname{Ker} d_2^{1,0} / \operatorname{Im} d_2^{2,-2} = \operatorname{Ker} d_2^{1,0}$. Hence we have exact sequence

$$0 \longrightarrow E_2^{0,1} \longrightarrow H^1(E^\bullet) \longrightarrow \operatorname{Ker} d_2^{1,0} \longrightarrow 0,$$

since $\operatorname{Ker} d_2^{1,0} \subseteq E_2^{0,1}$, we have exact sequence

$$0 \longrightarrow E_2^{0,1} \longrightarrow H^1(E^\bullet) \longrightarrow E_2^{1,0} \xrightarrow{d_2^{1,0}} E_2^{0,2},$$

where exactness at $E_2^{1,0}$ is given by $\operatorname{Im}(H^1(E^\bullet) \to E_2^{1,0}) = \operatorname{Ker} d_2^{1,0}$. By (2.27), we have a filtration

$$E_\infty^{0,2} \xhookrightarrow{E_\infty^{1,1}} F_1 \xhookrightarrow{E_\infty^{2,0}} H^2(E^\bullet),$$

where $F_1/E_\infty^{0,2} \cong E_\infty^{1,1}$ and $H^2(E^\bullet)/F_1 \cong E_\infty^{2,0}$. Now we compute $E_\infty^{0,2}$, note that

$$\begin{tikzcd}
E_r^{-r+1,2+r} & & \\
& E_r^{0,2} \arrow[ul, "d_r^{0,2}"'] & \\
& & E_r^{r-1,2-r} \arrow[ul, "d_r^{r-1,2-r}"']
\end{tikzcd}$$

and $E_r^{-r+1,2+r} = E_r^{r-1,2-r} = 0$ for all $r \geq 3$, we have

$$E_\infty^{0,2} = E_3^{0,2} = \operatorname{Ker} d_2^{0,2} / \operatorname{Im} d_2^{1,0} = E_2^{0,2} / \operatorname{Im} d_2^{1,0}.$$

Hence we have exact sequence

$$E_2^{1,0} \xrightarrow{d_2^{1,0}} E_2^{0,2} \longrightarrow E_3^{0,2} \longrightarrow 0.$$

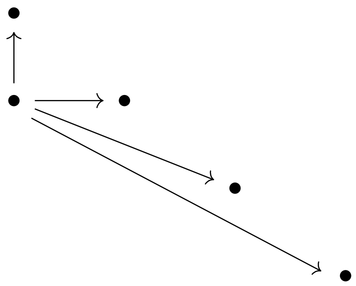
Since $E_3^{0,2} \hookrightarrow H^2(E^\bullet)$, the sequence

$$0 \longrightarrow E_2^{0,1} \longrightarrow H^1(E^\bullet) \longrightarrow E_2^{1,0} \xrightarrow{d_2^{1,0}} E_2^{0,2} \longrightarrow H^2(E^\bullet)$$

is exact. □

The other orientation

You may have observed that we could as well have done everything in the opposite direction, i.e., reversing the roles of horizontal and vertical morphisms. The the sequences of arrows giving the spectral sequence would look like this (compare to (2.26)).



This spectral sequence is denoted ${}_{\uparrow}E_{\bullet}^{\bullet,\bullet}$ ("with the **upward orientation**"). Then we would again get pieces of a filtration of $H^\bullet(E^\bullet)$ (where we have to be a bit careful with which ${}_{\uparrow}E_{\infty}^{p,q}$ corresponds to the subquotients — it is the opposite order to that of (2.27) for ${}_{\rightarrow}E_{\infty}^{p,q}$). **Warning:** in general there is no isomorphism between ${}_{\rightarrow}E_{\infty}^{p,q}$ and ${}_{\uparrow}E_{\infty}^{p,q}$.

In fact, this observation that we can start with either the horizontal or vertical maps was our secret goal all along. Both algorithms compute information about the same thing ($H^\bullet(E^\bullet)$), and usually we don't care about the final answer — we often care about the answer we get in one way, and we get at it by doing the spectral sequence in the other way.

2.6.3 Example

We are now ready to see how this is useful. The moral of these example is the following. In the past, you may have proved various facts involving various sorts of diagrams, by chasing elements around. Now, you will just plug them into a spectral sequence, and let the spectral sequence machinery do your chasing for you.

Example 2.36 (Proving the Snake Lemma) Consider the diagram

$$\begin{tikzcd} 0 \arrow[r] & D \arrow[r] & E \arrow[r] & F \arrow[r] & 0 \\ 0 \arrow[r] & A \arrow[r] \arrow[u, "\alpha"] & B \arrow[r] \arrow[u, "\beta"] & C \arrow[r] \arrow[u, "\gamma"] & 0 \end{tikzcd}$$

where the rows are exact in the middle (at A, B, C, D, E, F) and the squares commute. (Normally the Snake Lemma is described with the vertical arrows pointing downwards, but I want to fit this into my spectral sequence conventions.) We wish to show that there is an exact sequence

$$0 \longrightarrow \operatorname{Ker}\alpha \longrightarrow \operatorname{Ker}\beta \longrightarrow \operatorname{Ker}\gamma \longrightarrow \operatorname{Coker}\alpha \longrightarrow \operatorname{Coker}\beta \longrightarrow \operatorname{Coker}\gamma \longrightarrow 0. \tag{2.28}$$

We plug this into our spectral sequence machinery. We first compute the cohomology using the rightward orientation, i.e., using the order (2.26). Then because the rows are exact, $E_1^{p,q} = 0$, so the spectral sequence has already converged: $E_\infty^{p,q} = 0$.

We next compute this "0" in another way, by computing the spectral sequence using the upward orientation. Then ${}_{\uparrow}E_1^{\bullet,\bullet}$ (with its differentials) is:

$$\begin{array}{c} 0 \longrightarrow \operatorname{Coker}\alpha \longrightarrow \operatorname{Coker}\beta \longrightarrow \operatorname{Coker}\gamma \longrightarrow 0 \\ \\ 0 \longrightarrow \operatorname{Ker}\alpha \longrightarrow \operatorname{Ker}\beta \longrightarrow \operatorname{Ker}\gamma \longrightarrow 0. \end{array}$$

Then ${}_{\uparrow}E_2^{\bullet,\bullet}$ is of the form:

$$\begin{tikzcd}
0 \arrow[drr] & 0 \arrow[drr] & & & & & \\
& 0 \arrow[drr] & ?? \arrow[drr] & ? \arrow[drr] & ? & 0 & \\
& 0 & ? & ? \arrow[drr] & ?? \arrow[drr] & 0 & \\
& & & & & 0 & 0
\end{tikzcd}$$

We see that after ${}_{\uparrow}E_2$, all the terms will stabilize except for the double question marks — all maps to and form the single question marks are to and from 0-entires. And after ${}_{\uparrow}E_3$, even these two double-question-mark terms will stabilize. But in the end our complex must be the 0 complex (since ${}_{\rightarrow}E_\infty^{p,q} = 0$, by (2.27), we know that $H^\bullet(E^\bullet) = 0$). This means that in ${}_{\uparrow}E_2$, all the entries must be zero, except for the two double question marks (these two ?? are $\operatorname{Ker}(\operatorname{Coker}\alpha \to \operatorname{Coker}\beta)$ and $\operatorname{Coker}(\operatorname{Ker}\beta \to \operatorname{Ker}\gamma)$), and these two must be isomorphic. This means that $0 \to \operatorname{Ker}\alpha \to \operatorname{Ker}\beta \to \operatorname{Ker}\gamma$ and $\operatorname{Coker}\alpha \to \operatorname{Coker}\beta \to \operatorname{Coker}\gamma \to 0$ are both exact (that comes from the vanishing of the vanishing of the single question marks), and

$$\operatorname{Coker}(\operatorname{Ker}\beta \to \operatorname{Ker}\gamma) \cong \operatorname{Ker}(\operatorname{Coker}\alpha \to \operatorname{Coker}\beta)$$

is an isomorphism (that comes from the equality of the double question marks). Taken together, we have proved the exactness of (2.28), and hence the Snake Lemma! (**Notice:** in the end we didn't really care about the double complex. We just used it as a prop to prove the Snake Lemma.)

Spectral sequences make it easy to see how to generalize results further. For example, if $A \to B$ is no longer assumed to be injective, then ${}_{\rightarrow}E_1^{\bullet,\bullet}$ (with its differentials) is:

$$\begin{tikzcd}
0 & 0 & 0 \\
\operatorname{Ker}(A \to B) \arrow[u] & 0 \arrow[u] & 0 \arrow[u] \\
0 \arrow[u] & 0 \arrow[u] & 0. \arrow[u]
\end{tikzcd}$$

Hence ${}_{\rightarrow}E_\infty^{0,0} = H^0(E^\bullet) = \operatorname{Ker}(A \to B)$, by (2.27). Note that ${}_{\uparrow}E_\infty^{0,0} = H^0(E^\bullet)$ ((2.27) again), ${}_{\uparrow}E_2^{\bullet,\bullet}$ is of the form:

$$\begin{tikzcd}
0 \arrow[drr] & 0 \arrow[drr] & & & & & \\
0 \arrow[drr] & 0 \arrow[drr] & \operatorname{Ker}(\operatorname{Coker}\alpha \to \operatorname{Coker}\beta) \arrow[drr] & 0 \arrow[drr] & 0 & & \\
& 0 & \operatorname{Ker}(A \to B) \arrow[drr] & 0 \arrow[drr] & \operatorname{Coker}(\operatorname{Ker}\beta \to \operatorname{Ker}\gamma) \arrow[drr] & 0 & \\
& & & & 0 & 0 & 0.
\end{tikzcd}$$

Since $E_\infty^{p,q} = 0$ for all $p \neq 0$ or $q \neq 0$ and $\operatorname{Ker}(\operatorname{Ker}\alpha \to \operatorname{Ker}\beta) = \operatorname{Ker}(A \to B)$, we have exact sequence

$$\operatorname{Ker}\alpha \longrightarrow \operatorname{Ker}\beta \longrightarrow \operatorname{Ker}\gamma \longrightarrow \operatorname{Coker}\alpha \longrightarrow \operatorname{Coker}\beta \longrightarrow \operatorname{Coker}\gamma \longrightarrow 0.$$

✍ **Exercise 2.40 Unimportant exercise (Grafting exact sequence, a variant of the snake Lemma)** Extend the

Snake Lemma as follows. Suppose we have a commuting diagram

$$\begin{tikzcd}
 & 0 \arrow[r] & X' \arrow[r] & Y' \arrow[r] & Z' \arrow[r] & A' \arrow[r] & \cdots \\
\cdots \arrow[r] & W \arrow[r] \arrow[u] & X \arrow[r] \arrow[u, "a"] & Y \arrow[r] \arrow[u, "b"] & Z \arrow[r] \arrow[u, "c"] & 0. \arrow[u] &
\end{tikzcd}$$

where the top and bottom rows are exact. Show that the top and bottom rows can be "grafted together" to an exact sequence

$$\begin{tikzcd}
\cdots \arrow[r] & W \arrow[r] & \operatorname{Ker} a \arrow[r] & \operatorname{Ker} b \arrow[r] & \operatorname{Ker} c \\
\arrow[r] & \operatorname{Coker} a \arrow[r] & \operatorname{Coker} b \arrow[r] & \operatorname{Coker} c \arrow[r] & A' \arrow[r] & \cdots .
\end{tikzcd}$$

Proof We first compute the cohomology using the rightward orientation. Then because the rows are exact, ${}_{\rightarrow}E_1^{p,q} = 0$, so the spectral sequence has already converged: ${}_{\rightarrow}E_\infty^{p,q} = 0$. By (2.27), we have $H^\bullet(E^\bullet) = 0$.

We next compute the cohomology using the upward orientation. ${}_{\uparrow}E_1^{\bullet,\bullet}$ is

$$\begin{tikzcd}
 & 0 \arrow[r] & \operatorname{Coker} a \arrow[r] & \operatorname{Coker} b \arrow[r] & \operatorname{Coker} c \arrow[r] & A' \arrow[r] & \cdots \\
\cdots \arrow[r] & W \arrow[r] & \operatorname{Ker} a \arrow[r] & \operatorname{Ker} b \arrow[r] & \operatorname{Ker} c \arrow[r] & 0. &
\end{tikzcd} \tag{2.29}$$

And ${}_{\uparrow}E_2^{\bullet,\bullet}$ is:

$$\begin{tikzcd}
 & 0 \arrow[drr] & 0 \arrow[drr] & 0 \arrow[drr] & 0 \arrow[drr] & & & & \\
0 \arrow[drr] & 0 \arrow[drr] & 0 \arrow[drr] & ?? \arrow[drr] & ? \arrow[drr] & ? \arrow[drr] & ? \arrow[drr] & 0 & \\
 & 0 & ? \arrow[drr] & ? \arrow[drr] & ? \arrow[drr] & ?? \arrow[drr] & 0 & 0 & 0 \\
 & & & & 0 & 0 & 0 & 0. &
\end{tikzcd}$$

We see that after page ${}_{\uparrow}E_2^{\bullet,\bullet}$, all the terms will stabilize except for the double question marks. They are $\operatorname{Ker}(\operatorname{Coker} a \to \operatorname{Coker} b)$ and $\operatorname{Coker}(\operatorname{Ker} b \to \operatorname{Ker} c)$. Since $H^\bullet(E^\bullet) = 0$, ${}_{\uparrow}E_\infty^{p,q} = 0$. Hence ? are all 0, and after page ${}_{\uparrow}E_3^{\bullet,\bullet}$, ?? terms will stabilize, moreover, they must be isomorphism, i.e.,

$$\operatorname{Ker}(\operatorname{Coker} a \to \operatorname{Coker} b) \cong \operatorname{Coker}(\operatorname{Ker} b \to \operatorname{Ker} c),$$

note that we have

$$\begin{tikzcd}
\operatorname{Ker} b \arrow[r] & \operatorname{Ker} c \arrow[r, two heads] & \operatorname{Coker}(\operatorname{Ker} b \to \operatorname{Ker} c) \arrow[r, "\sim"] & \operatorname{Ker}(\operatorname{Coker} a \to \operatorname{Coker} b) \arrow[r, hook] & \operatorname{Coker} a,
\end{tikzcd}$$

so sequence $\operatorname{Ker} b \to \operatorname{Ker} c \to \operatorname{Coker} a$ is exact. Since ? are all 0, (2.29) exact. Taken together, sequence

$$\begin{tikzcd}
\cdots \arrow[r] & W \arrow[r] & \operatorname{Ker} a \arrow[r] & \operatorname{Ker} b \arrow[r] & \operatorname{Ker} c \\
\arrow[r] & \operatorname{Coker} a \arrow[r] & \operatorname{Coker} b \arrow[r] & \operatorname{Coker} c \arrow[r] & A' \arrow[r] & \cdots .
\end{tikzcd}$$

is exact. □

Example 2.37 (The five Lemma) Suppose

$$\begin{tikzcd}
F \arrow[r] & G \arrow[r] & H \arrow[r] & I \arrow[r] & J \\
A \arrow[r] \arrow[u, "\alpha"] & B \arrow[r] \arrow[u, "\beta"] & C \arrow[r] \arrow[u, "\gamma"] & D \arrow[r] \arrow[u, "\delta"] & E \arrow[u, "\varepsilon"]
\end{tikzcd} \tag{2.30}$$

where the rows are exact and squares commute.

Suppose $\alpha, \beta, \delta, \varepsilon$ are isomorphisms. Show that γ is an isomorphism.

Proof We first compute the cohomology of the total complex using the rightward orientation (2.26). We choose this because we see that we will get lots of zeros. Then ${}_{\rightarrow}E_1^{\bullet,\bullet}$ looks like this:

$$\begin{tikzcd}
? & 0 & 0 & 0 & ? \\
? \arrow[u] & 0 \arrow[u] & 0 \arrow[u] & 0 \arrow[u] & ?. \arrow[u]
\end{tikzcd}$$

Then ${}_{\rightarrow}E_2^{\bullet,\bullet}$ looks like this:

$$\begin{tikzcd}
0 & & & & 0 & & \\
0 & & & & 0 & & \\
& ? \arrow[uul] & 0 & 0 & 0 & ? \arrow[uul] & \\
& ? \arrow[uul] & 0 & 0 & 0 & ? \arrow[uul] & \\
& & 0 \arrow[uul] & & & & 0 \arrow[uul] \\
& & 0 \arrow[uul] & & & & 0 \arrow[uul]
\end{tikzcd}$$

and therefore the sequence will converge by E_2, as we will never get any arrows between two nonzero entires in a table thereafter. We can't conclude that the cohomology of the total complex vanishes, but we can note that it vanishes in all but four degrees — and most important, it vanishes in the two degrees corresponding to the entries C and H (the source and target of γ).

We next compute this using the upward orientation. Then ${}_{\uparrow}E_1$ looks like this:

$$\begin{tikzcd}
0 \arrow[r] & 0 \arrow[r] & ? \arrow[r] & 0 \arrow[r] & 0 \\
0 \arrow[r] & 0 \arrow[r] & ? \arrow[r] & 0 \arrow[r] & 0
\end{tikzcd}$$

and the spectral sequence converges at this step. We wish to show that those two question marks are zero. But they are precisely the cohomology groups of the total complex that we just showed were zero — so we are done! □

Theorem 2.6.1 (Five Lemma for modules)

Consider the following commutative diagram.

$$\begin{tikzcd}
B_1 \arrow[r] & B_2 \arrow[r] & B_3 \arrow[r] & B_4 \arrow[r] & B_5 \\
A_1 \arrow[r] \arrow[u, "f_1"] & A_2 \arrow[r] \arrow[u, "f_2"'] & A_3 \arrow[r] \arrow[u, "f_3"'] & A_4 \arrow[r] \arrow[u, "f_4"'] & A_5 \arrow[u, "f_5"']
\end{tikzcd}$$

where the top and bottom rows are exact sequences.

(1) If f_2 and f_4 are surjective and f_5 is injective, then f_3 is surjective.

(2) If f_2 and f_4 are injective and f_1 is surjective, then f_3 is injective.

(3) If f_2 and f_4 are isomorphisms, f_1 is surjective, and f_5 is injective, then f_3 is isomorphism.

Proof (2) is the dual statement of (1), we just prove (1), and (2) is similarly.

Suppose f_2 and f_4 are surjective and f_5 is injective, we want to show that f_3 is surjective.

We first compute the cohomology of the total complex using the rightward orientation. Then ${}_{\rightarrow}E_1^{\bullet,\bullet}$ looks like this:

$$\begin{tikzcd}
\operatorname{Ker}(B_1 \to B_2) & 0 & 0 & 0 & \operatorname{Coker}(B_4 \to B_5) \\
\operatorname{Ker}(A_1 \to A_2) \arrow[u] & 0 \arrow[u] & 0 \arrow[u] & 0 \arrow[u] & \operatorname{Coker}(A_4 \to A_5). \arrow[u]
\end{tikzcd}$$

Then ${}_{\rightarrow}E_2^{\bullet,\bullet}$ looks like this:

$$\begin{tikzcd}
0 & & & & 0 & & \\
0 & & & & 0 & & \\
& ? \arrow[uul] & 0 & 0 & 0 & ? \arrow[uul] & \\
& ? \arrow[uul] & 0 & 0 & 0 & ? \arrow[uul] & \\
& & 0 \arrow[uul] & & & & 0 \arrow[uul] \\
& & 0 \arrow[uul] & & & & 0 \arrow[uul]
\end{tikzcd}$$

and therefore the sequence will converge by ${}_{\rightarrow}E_2$. Hence by (2.27),

$$E_2^{0,3} \xhookrightarrow{E_2^{1,2}} F_1 \xhookrightarrow{E_2^{2,1}} F_2 \xhookrightarrow{E_2^{3,0}} H^3(E^\bullet),$$

we have $H^3(E^\bullet) = 0$.

We next compute this using the upward orientation. The ${}_{\uparrow}E_1$ looks like this:

$$\begin{tikzcd}
0 \arrow[r] & \operatorname{Coker} f_1 \arrow[r] & 0 \arrow[r] & \operatorname{Coker} f_3 \arrow[r] & 0 \arrow[r] & \operatorname{Coker} f_5 \arrow[r] & 0 \\
0 \arrow[r] & \operatorname{Ker} f_1 \arrow[r] & \operatorname{Ker} f_2 \arrow[r] & \operatorname{Ker} f_3 \arrow[r] & \operatorname{Ker} f_4 \arrow[r] & 0 \arrow[r] & 0.
\end{tikzcd}$$

And ${}_{\uparrow}E_2$ looks like this:

$$\begin{tikzcd}
0 \arrow[drr] & & 0 \arrow[drr] & & 0 \arrow[drr] & & & & \\
0 \arrow[drr] & 0 \arrow[drr] & \operatorname{Coker} f_1 \arrow[drr] & 0 \arrow[drr] & \operatorname{Coker} f_3 \arrow[drr] & 0 & \operatorname{Coker} f_5 \arrow[drr] & 0 & \\
0 & 0 & ? \arrow[drr] & ? \arrow[drr] & ? \arrow[drr] & ?? \arrow[drr] & 0 & 0 & 0 \\
& & & & 0 & 0 & 0 & 0. &
\end{tikzcd}$$

Hence ${}_{\uparrow}E_\infty^{2,1} = \operatorname{Coker} f_3$. By (2.27), i.e. filtration

$${}_{\uparrow}E_2^{3,0} \xhookrightarrow{{}_{\uparrow}E_2^{2,1}} {}_{\uparrow}F_1 \xhookrightarrow{{}_{\uparrow}E_2^{1,2}} {}_{\uparrow}F_2 \xhookrightarrow{{}_{\uparrow}E_2^{0,3}} H^3(E^\bullet),$$

we have $H^3(E^\bullet)/?? \cong \operatorname{Coker} f_3$, since $H^3(E^\bullet) = 0$, $\operatorname{Coker} f_3 = 0$, i.e., f_3 is surjective.

Now, we prove (3). Suppose f_2 and f_4 are isomorphism, and f_5 is injective, we want to show f_3 is isomorphism.

We first compute the cohomology of the total complex using the rightward orientation. Same as the proof of (1), we have $H^3(E^\bullet) = 0$ and $H^2(E^\bullet) = 0$.

We next compute this using the upward orientation. The ${}_\uparrow E_1$ looks like this:

$$\begin{array}{ccccccccc} 0 & \longrightarrow & 0 & \longrightarrow & \operatorname{Coker} f_3 & \longrightarrow & 0 & \longrightarrow & \operatorname{Coker} f_5 \\ \\ \operatorname{Ker} f_1 & \longrightarrow & 0 & \longrightarrow & \operatorname{Ker} f_3 & \longrightarrow & 0 & \longrightarrow & 0. \end{array}$$

Hence the spectral sequence converges at page 1. By (2.27), i.e. filtration

$${}_\uparrow E_1^{3,0} \xhookrightarrow{{}_\uparrow E_1^{2,1}} \bullet \xhookrightarrow{{}_\uparrow E_1^{1,2}} \bullet \xhookrightarrow{{}_\uparrow E_1^{0,3}} H^3(E^\bullet)$$

and

$${}_\uparrow E_1^{2,0} \xhookrightarrow{{}_\uparrow E_1^{1,1}} \bullet \xhookrightarrow{{}_\uparrow E_1^{0,2}} H^2(E^\bullet),$$

we have $H^3(E^\bullet) \cong {}_\uparrow E_1^{2,1} = \operatorname{Coker} f_3$ and $H^2(E^\bullet) \cong \operatorname{Ker} f_3$, hence $\operatorname{Coker} f_3 = 0$ and $\operatorname{Ker} f_3 = 0$, i.e., f_3 is isomorphism, we are done! ☐

Proposition 2.6.2 (The mapping cone)

Suppose $\mu : A^\bullet \to B^\bullet$ is a morphism of complexes. Suppose $C^\bullet$ is the single complex associated to the double complex $A^\bullet \to B^\bullet$. ($C^\bullet$ is called the ***mapping cone*** *of μ.) Then there is a long exact sequence of complexes:*

$$\cdots \longrightarrow H^{i-1}(B^\bullet) \longrightarrow H^i(C^\bullet) \longrightarrow H^i(A^\bullet) \longrightarrow H^i(B^\bullet) \longrightarrow H^{i+1}(C^\bullet) \longrightarrow \cdots.$$

In particular, μ induces an isomorphism on cohomology if and only if the mapping cone is exact.

Remark 2.41 There is a slight notational ambiguity here; depending on how you index your double complex, your long exact sequence might look slightly different.

Proof Consider the double complex as following.

$$\begin{array}{ccccccccc} \cdots & \longrightarrow & B^{n-1} & \longrightarrow & B^n & \longrightarrow & B^{n+1} & \longrightarrow & \cdots \\ & & \uparrow{\scriptstyle \mu^{n-1}} & & \uparrow{\scriptstyle \mu^n} & & \uparrow{\scriptstyle \mu^{n+1}} & & \\ \cdots & \longrightarrow & A^{n-1} & \longrightarrow & A^n & \longrightarrow & A^{n+1} & \longrightarrow & \cdots \end{array}$$

We compute the cohomology of the total complex $C^\bullet$ using the rightward orientation. Then ${}_\rightarrow E_1^{\bullet,\bullet}$ looks like this:

$$\begin{array}{ccc} 0 & 0 & 0 \\ \uparrow & \uparrow & \uparrow \\ H^{n-1}(B^\bullet) & H^n(B^\bullet) & H^{n+1}(B^\bullet) \\ \uparrow & \uparrow & \uparrow \\ H^{n-1}(A^\bullet) & H^n(A^\bullet) & H^{n+1}(A^\bullet) \\ \uparrow & \uparrow & \uparrow \\ 0 & 0 & 0 \end{array}$$

And ${}_{\to}E_2^{\bullet,\bullet}$ looks like this:

$$
\begin{tikzcd}
0 & & \\
0 & & \\
 & \operatorname{Coker}(H^n(A^\bullet) \to H^n(B^\bullet)) \arrow[uul] & \\
 & \operatorname{Ker}(H^n(A^\bullet) \to H^n(B^\bullet)) \arrow[uul] & \\
 & & 0 \arrow[uul] \\
 & & 0 \arrow[uul]
\end{tikzcd}
$$

Hence the spectral sequence converges at page 2, and therefore ${}_{\to}C_\infty^{i,0} = {}_{\to}C_2^{i,0} = \operatorname{Ker}(H^i(A^\bullet) \to H^i(B^\bullet))$, ${}_{\to}C_\infty^{i,1} = C_2^{i,1} = \operatorname{Coker}(H^i(A^\bullet) \to H^i(B^\bullet))$, and ${}_{\to}C^{i,j} = 0$ for all $j \neq 0, 1$. By (2.27), we have

$$
{}_{\to}C_2^{0,i} \xhookrightarrow{{}_{\to}C_2^{1,i-1}} \bullet \hookrightarrow \cdots \hookrightarrow \bullet \xhookrightarrow{{}_{\to}C_2^{i,0}} H^i(C^\bullet),
$$

hence $H^i(C^\bullet)/{}_{\to}C_2^{i-1,1} \cong {}_{\to}C_2^{i,0}$. Then we have exact sequence

$$
0 \longrightarrow \operatorname{Coker}(H^{i-1}(A^\bullet) \to H^{i-1}(B^\bullet)) \longrightarrow H^i(C^\bullet) \longrightarrow \operatorname{Ker}(H^i(A^\bullet) \to H^i(B^\bullet)) \longrightarrow 0. \tag{2.31}
$$

Consider the sequence

$$
\cdots \longrightarrow H^{i-1}(A^\bullet) \longrightarrow H^{i-1}(B^\bullet) \xrightarrow{\alpha_{i-1}} H^i(C^\bullet) \xrightarrow{\beta_i} H^i(A^\bullet) \longrightarrow H^i(B^\bullet) \longrightarrow \cdots, \tag{2.32}
$$

where $\alpha_{i-1} : H^{i-1}(B^\bullet) \to H^i(C^\bullet)$ is given by $H^{i-1}(B^\bullet) \twoheadrightarrow \operatorname{Coker}(H^{i-1}(A^\bullet) \to H^{i-1}(B^\bullet)) \hookrightarrow H^i(C^\bullet)$ and $\beta_i : H^i(C^\bullet) \to H^i(A^\bullet)$ is given by $H^i(C^\bullet) \twoheadrightarrow \operatorname{Ker}(H^i(A^\bullet) \to H^i(B^\bullet)) \hookrightarrow H^i(A^\bullet)$. We now check the exactness. Note that

$$
\operatorname{Ker} \alpha \cong \operatorname{Im}(H^{i-1}(A^\bullet) \to H^{i-1}(B^\bullet)),
$$

sequence (2.32) is exact at $H^{i-1}(B^\bullet)$. By exactness of (2.31)

$$
\begin{aligned}
\operatorname{Im} \alpha_{i-1} &\cong \operatorname{Im}(\operatorname{Coker}(H^{i-1}(A^\bullet) \to H^{i-1}(B^\bullet)) \to H^i) \\
&\cong \operatorname{Ker}(H^i(C^\bullet) \to \operatorname{Ker}(H^i(A^\bullet) \to H^i(B^\bullet))) \cong \operatorname{Ker} \beta_i,
\end{aligned}
$$

(2.32) is exact at $H^i(C^\bullet)$. Finally, note that

$$
\operatorname{Im} \beta \cong \operatorname{Ker}(H^i(A^\bullet) \to H^i(B^\bullet)),
$$

(2.32) is exact at $H^i(A^\bullet)$. Hence sequence (2.32) is exact, as we desired.

In particular, μ induces an isomorphism on cohomology if and only if $\operatorname{Ker}(H^n(A^\bullet) \to H^n(B^\bullet)) = 0$ and $\operatorname{Coker}(H^n(A^\bullet) \to H^n(B^\bullet)) = 0$ if and only if $H^n(C^\bullet) = 0$ for all n if and only if the mapping cone $C^\bullet$ is exact. □

✍ **Exercise 2.41** Use spectral sequences to show that a short exact sequence of complexes give a long exact sequence in cohomology (Theorem 2.5.4). (This is a generalization of Proposition 2.6.2.)

Proof Suppose we have a short exact sequence of complexes.

$$\begin{tikzcd}
0^\bullet & : & \cdots \arrow[r] & 0 \arrow[r] & 0 \arrow[r] & 0 \arrow[r] & \cdots \\
C^\bullet \arrow[u] & : & \cdots \arrow[r] & C^{n-1} \arrow[r] \arrow[u] & C^n \arrow[r] \arrow[u] & C^{n+1} \arrow[r] \arrow[u] & \cdots \\
B^\bullet \arrow[u] & : & \cdots \arrow[r] & B^{n-1} \arrow[r] \arrow[u] & B^n \arrow[r] \arrow[u] & B^{n+1} \arrow[r] \arrow[u] & \cdots \\
A^\bullet \arrow[u] & : & \cdots \arrow[r] & A^{n-1} \arrow[r] \arrow[u] & A^n \arrow[r] \arrow[u] & A^{n+1} \arrow[r] \arrow[u] & \cdots \\
0^\bullet \arrow[u] & : & \cdots \arrow[r] & 0 \arrow[r] \arrow[u] & 0 \arrow[r] \arrow[u] & 0 \arrow[r] \arrow[u] & \cdots
\end{tikzcd}$$

We want to show that it induces a long exact sequence in cohomology.

$$\begin{tikzcd}
 & \cdots \arrow[r] & H^{n-1}(C^\bullet) \arrow[r] & {} \\
H^n(A^\bullet) \arrow[r] & H^n(B^\bullet) \arrow[r] & H^n(C^\bullet) \arrow[r] & {} \\
H^{n+1}(A^\bullet) \arrow[r] & \cdots & &
\end{tikzcd}$$

We first compute the cohomology of the total complex $E^\bullet$ using the upward orientation. Then ${}_\uparrow E_1^{\bullet,\bullet}$ looks like this:

$$\begin{tikzcd}
\cdots \arrow[r] & 0 \arrow[r] & 0 \arrow[r] & 0 \arrow[r] & \cdots \\
\cdots \arrow[r] & 0 \arrow[r] & 0 \arrow[r] & 0 \arrow[r] & \cdots \\
\cdots \arrow[r] & 0 \arrow[r] & 0 \arrow[r] & 0 \arrow[r] & \cdots
\end{tikzcd}$$

Hence ${}_\uparrow E_\infty^{p,q} = {}_\uparrow E_1^{p,q} = 0$. By (2.27), we have $H^k(E^\bullet) = 0$ for all k.

We next compute the cohomology of total complex $E^\bullet$ using the rightward orientation. Denote

$$0^\bullet \longrightarrow A^\bullet \xrightarrow{\alpha} B^\bullet \xrightarrow{\beta} C^\bullet \longrightarrow 0.$$

Then ${}_\rightarrow E_1^{\bullet,\bullet}$ looks like this:

$$\begin{tikzcd}
0 & 0 & 0 \\
H^{n-1}(C^\bullet) \arrow[u] & H^n(C^\bullet) \arrow[u] & H^{n+1}(C^\bullet) \arrow[u] \\
H^{n-1}(B^\bullet) \arrow[u, "\beta^{n-1*}"] & H^n(B^\bullet) \arrow[u, "\beta^{n*}"] & H^{n+1}(B^\bullet) \arrow[u, "\beta^{n+1*}"] \\
H^{n-1}(A^\bullet) \arrow[u, "\alpha^{n-1*}"] & H^n(A^\bullet) \arrow[u, "\alpha^{n*}"] & H^{n+1}(A^\bullet) \arrow[u, "\alpha^{n+1*}"] \\
0 \arrow[u] & 0 \arrow[u] & 0 \arrow[u]
\end{tikzcd}$$

And ${}_{\rightarrow}E_2^{\bullet,\bullet}$ looks like:

$$\begin{tikzcd}
0 & 0 & & & \\
& 0 & 0 & 0 & \\
& \operatorname{Coker}\beta^{n-1*} \arrow[uul] & \operatorname{Coker}\beta^{n*} \arrow[uul] & \operatorname{Coker}\beta^{n+1*} & \\
& \operatorname{Ker}\beta^{n-1*}/\operatorname{Im}\alpha^{n-1*} & \operatorname{Ker}\beta^{n*}/\operatorname{Im}\alpha^{n*} \arrow[uul] & \operatorname{Ker}\beta^{n+1*}/\operatorname{Im}\alpha^{n+1*} \arrow[uul] & \\
& \operatorname{Ker}\alpha^{n-1*} & \operatorname{Ker}\alpha^{n*} \arrow[uul] & \operatorname{Ker}\alpha^{n+1*} \arrow[uul] & \\
& 0 & 0 \arrow[uul] & 0 \arrow[uul] & \\
& & & 0 \arrow[uul] & 0 \arrow[uul]
\end{tikzcd}$$

Hence ${}_{\rightarrow}E_\infty^{n,1} = {}_{\rightarrow}E_2^{n,1} = \operatorname{Ker}\beta^{n*}/\operatorname{Im}\alpha^{n*}$. In page 3, i.e., ${}_{\rightarrow}E_3^{\bullet,\bullet}$ looks like:

$$\begin{tikzcd}
0 & & \\
0 & & \\
0 & & \\
& \operatorname{Coker}(\operatorname{Ker}\alpha^{n+1*} \to \operatorname{Coker}\beta^{n*}) \arrow[uuul] & \\
& \operatorname{Ker}\beta^{n*}/\operatorname{Im}\alpha^{n*} \arrow[uuul] & \\
& \operatorname{Ker}(\operatorname{Ker}\alpha^{n*} \to \operatorname{Coker}\beta^{n-1*}) \arrow[uuul] & \\
& & 0 \arrow[uuul] \\
& & 0 \arrow[uuul] \\
& & 0 \arrow[uuul]
\end{tikzcd}$$

Hence ${}_{\rightarrow}E_\infty^{n,0} = {}_{\rightarrow}E_3^{n,0} = \operatorname{Ker}(\operatorname{Ker}\alpha^{n*} \to \operatorname{Coker}\beta^{n-1*})$, ${}_{\rightarrow}E_\infty^{n,2} = {}_{\rightarrow}E_3^{n,2} = \operatorname{Coker}(\operatorname{Ker}\alpha^{n+1*} \to \operatorname{Coker}\beta^{n*})$, and ${}_{\rightarrow}E_\infty^{n,j} = 0$ for all $j \neq 0, 1, 2$. By (2.27), we have a filtration

$${}_{\rightarrow}E^{0,i} \xhookrightarrow{{}_{\rightarrow}E^{1,i-1}} \bullet \xhookrightarrow{{}_{\rightarrow}E^{2,i-2}} \bullet \hookrightarrow \cdots \hookrightarrow \bullet \xhookrightarrow{{}_{\rightarrow}E^{i,0}} H^i(E^\bullet).$$

Since $H^i(E^\bullet) = 0$, we know that ${}_\rightarrow E_\infty^{n,0} = {}_\rightarrow E_\infty^{n,1} = {}_\rightarrow E_\infty^{n,2} = 0$, i.e.,

$$\begin{aligned} \operatorname{Ker}(\operatorname{Ker}\alpha^{n*} \to \operatorname{Coker}\beta^{n-1*}) &= 0 \\ \operatorname{Ker}\beta^{n*} / \operatorname{Im}\alpha^{n*} &= 0 \\ \operatorname{Coker}(\operatorname{Ker}\alpha^{n+1*} \to \operatorname{Coker}\beta^{n*}) &= 0 \end{aligned}$$

It follows that sequence

$$H^n(A^\bullet) \xrightarrow{\alpha^{n*}} H^n(B^\bullet) \xrightarrow{\beta^{n*}} H^n(C^\bullet)$$

is exact, and

$$\operatorname{Ker}\alpha^{n*} \cong \operatorname{Coker}\beta^{n-1*}.$$

Define $H^{n-1}(C^\bullet) \to H^n(A^\bullet)$ by setting

$$H^{n-1}(C^\bullet) \twoheadrightarrow \operatorname{Coker}\beta^{n-1*} \xrightarrow{\sim} \operatorname{Ker}\alpha^{n*} \hookrightarrow H^n(A^\bullet),$$

then sequence

$$H^{n-1}(C^\bullet) \longrightarrow H^n(A^\bullet) \longrightarrow H^n(B^\bullet)$$

is exact. Taken together, the sequence

$$\begin{aligned} \cdots \longrightarrow H^{n-1}(C^\bullet) \longrightarrow \\ H^n(A^\bullet) \longrightarrow H^n(B^\bullet) \longrightarrow H^n(C^\bullet) \longrightarrow \\ H^{n+1}(A^\bullet) \longrightarrow \cdots \end{aligned}$$

is exact. □

The Grothendieck composition-of-functors spectral sequence (Chapter 24) will be an important example of a spectral sequence that specializes in a number of useful ways.

You are now ready to go out into the world and use spectral sequences to your heart's content.

2.6.4 Complete definition of spectral sequences, and proof

You should most definitely not read the precise definition of a spectral sequence, and the proof that they works as advertised, any time soon after reading the introduction to spectral sequences above. But after a suitable interval, you should at least flip through a construction and proof to convince yourself that nothing fancy is involved. The ideal is not as bad as you might think.

It is useful to notice that the proof implies that spectral sequences are functorial in the 0-th page: the spectral sequence formalism has good functorial properties in the double complex. Unfortunately, Grothendieck's terminology "spectral functor" did not catch on.

Chapter 3 Sheaves

It is perhaps surprising that geometric spaces are often best understood in terms of (nice) functions on them. For example, a differentiable mainfold that is a subset of $\mathbb{R}^n$ can be studied in terms of its smooth (C^∞) functions. Because "geometric spaces" can have few (everywhere-defined) functions, a more precise version of this insight is that the structure of the space can be well understood by considering all functions on all open subsets of the space. This information is encoded and organized in something called a **sheaf**. We will define sheaves and describe useful facts about them. We will begin with a motivating example to convince you that the notion is not so foreign.

One reason sheaves are slippery to work with is that they keep track of a huge amount of information, and there are some subtle local-to-global issues. There are also three different ways of getting a hold of them:

1. in terms of open sets — intuitive but in some ways the least helpful;
2. in terms of **stalks**;
3. in terms of a base of a topology.

(Some people strongly prefer the espace etale interpretation, as well.)

3.1 Motivating example: The sheaf of smooth functions

Consider smooth (C^∞) functions on the topological space $X = \mathbb{R}^n$ (or more generally on a manifold X). The sheaf of smooth functions on X is the data of all smooth functions on all open subsets on X. We will see how to manage these data, and observe some of their properties. On each open set $U \subseteq X$, we have a ring of smooth functions. We denote this ring of functions $\mathscr{O}(U)$.

Given a smooth function on an open set, you can restrict it to a smaller open set, obtaining a smooth function there. In other words, if $U \subseteq V$ is an inclusion of open sets, we have a "restriction map" $\operatorname{res}_{V,U} : \mathscr{O}(V) \to \mathscr{O}(U)$.

Take a smooth function on a big open set, and restrict it to a medium open set, and then restrict that to a small open set. The result is the same as if you restrict the smooth function on the big open set directly to the small open set. In other words, if $U \hookrightarrow V \hookrightarrow W$, then the following diagram commutes:

$$\begin{tikzcd} \mathscr{O}(W) \arrow[rr, "\operatorname{res}_{W,V}"] \arrow[dr, "\operatorname{res}_{W,U}"'] & & \mathscr{O}(V) \arrow[dl, "\operatorname{res}_{V,U}"] \\ & \mathscr{O}(U) & \end{tikzcd}$$

Next take two smooth functions f_1 and f_2 on a big open set U, and an open cover of U by some collection of open subsets $\{U_i\}$. (We say $\{U_i\}$ **covers** U, or is an **open cover of** U, if $U = \bigcup U_i$.) Suppose that f_1 and f_2 agree on each of these U_i. Then they must have been the same function to begin with. In other words, if $\{U_i\}_{i\in I}$ is a cover of U, and $f_1, f_2 \in \mathscr{O}(U)$, and $\operatorname{res}_{U,U_i} f_1 = \operatorname{res}_{U,U_i} f_2$, then $f_1 = f_2$. Thus we can **identify** functions on an open set by looking at them on a covering by small open sets.

Finally, suppose you are given the same U and cover $\{U_i\}$, take a smooth function on each of the U_i — a function f_1 on U_1, a function f_2 on U_2, and so on — and assume they agree on the pairwise overlaps. Then they can be "glued together" to make one smooth function on all of U. In other words, given $f_i \in \mathscr{O}(U_i)$ for all i, such that $\operatorname{res}_{U_i,U_i\cap U_j} f_i = \operatorname{res}_{U_j,U_i\cap U_j} f_j$ for all i and j, then there is some $f \in \mathscr{O}(U)$ such that $\operatorname{res}_{U,U_i} f = f_i$ for all i.

The entire example above would have worked just as well with continuous functions, or real-analytic functions, or just plain real-valued functions. Thus all of these classes of "nice" functions share some common properties. We will soon formalize these properties in the notion of a sheaf.

Before we do, we first give another definition, that of the germ of smooth function at a point $p \in X$. Intuitively, it is a "shred" of a smooth function at p.

Definition 3.1.1 (The germ of a smooth function)

Germs *are objects of the form*

$$(f,\ \text{open set } U) \quad \text{such that} \quad p \in U, f \in \mathscr{O}(U)$$

modulo the relation that $(f,U) \sim (g,V)$ if there is some open set $W \subseteq U,V$ containing p where $f|_W = g|_W$. In other words, two functions that are the same in an open neighborhood of p (but may differ elsewhere) have the same germ.

We call this set of germs the ***stalk*** *at p, and denote it $\mathscr{O}_p$.*

Remark 3.1

(1) The stalk is a ring.

We can add two germs, and get another germ: if you have a function f defined on U, and a function g defined on V, then $f+g$ defined on $U \cap V$. Moreover, $f+g$ is well-defined; if $\widetilde{f}$ has the same germ as f, meaning that there is some open set W containing p on which they agree, and $\widetilde{g}$ has the same germ as g, meaning they agree on some open W' containing p, then $\widetilde{f}+\widetilde{g}$ is the same function as $f+g$ on $U \cap V \cap W \cap W'$.

(2) The stalk is a colimit.

Notice also that if $p \in U$, you get a map $\mathscr{O}(U) \to \mathscr{O}_p$. For $V \subseteq U$, we have

$$\begin{tikzcd} \mathscr{O}_p & \\ \mathscr{O}(U) \arrow[u] \arrow[r, "\mathrm{res}_{U,V}"'] & \mathscr{O}(V). \arrow[ul] \end{tikzcd}$$

Proposition 3.1.1

The stalk $\mathscr{O}_p$ is a local ring with maximal ideal $\mathfrak{m}_p = \{f \in \mathscr{O}_p : f(p) = 0\}$.

Proof We can see that $\mathscr{O}_p$ is a local ring as follows. Consider those germs vanishing at p, which we denote $\mathfrak{m}_p \subseteq \mathscr{O}_p$, that is, $\mathfrak{m}_p = \{f \in \mathscr{O}_p : f(p) = 0\}$.

(i) $\mathfrak{m}_p$ is an ideal.

Let $f, g \in \mathfrak{m}_p$, then $f(p) = g(p) = 0$, hence, $\mathfrak{m}_p$ is closed under addition. Let $f \in \mathfrak{m}_p$ and $h \in \mathscr{O}_p$, then $hf(p) = 0$, and therefore $hf \in \mathfrak{m}_p$, that is, $\mathscr{O}_p\mathfrak{m}_p \subseteq \mathfrak{m}_p$. Hence, $\mathfrak{m}_p$ is an ideal.

(ii) $\mathfrak{m}_p$ is maximal.

Cosider the following sequence,

$$0 \longrightarrow \mathfrak{m}_p \longrightarrow \mathscr{O}_p \xrightarrow{\ \varphi\ } \mathbb{R} \longrightarrow 0,$$

where $\varphi : f \mapsto f(p)$. Clearly, above sequence is exact, and therefore $\mathscr{O}_p/\mathfrak{m}_p \cong \mathbb{R}$. Hence, $\mathscr{O}_p/\mathfrak{m}_p$ is a field, $\mathfrak{m}_p$ is maximal.

(iii) $\mathfrak{m}_p$ is the only maximal ideal of $\mathscr{O}_p$.

It suffices to show that each element in $\mathscr{O}_p - \mathfrak{m}_p$ is unit. Let $g \in \mathscr{O}_p - \mathfrak{m}_p$, then $g(p) \neq 0$, hence, $1/g(p) \neq 0$, this imples that g is a unit in $\mathscr{O}_p$. Hence, every ideal$\neq (1)$ is contained in $\mathfrak{m}$. Hence, $\mathfrak{m}_p$ is the only maximal ideal of $\mathscr{O}_p$.

□

Remark 3.2 Note that we can interpret the value of a function at a point, or the value of germ at a point, as an element of the local ring modulo the maximal ideal. We will see that this doesn't work for more general sheaves, but does work for things behaving like sheaves of functions. This will be formalized in the notion of a **locally ringed space**.

Remark 3.3 Notice that $\mathfrak{m}_p/\mathfrak{m}_p^2$ is a module over $\mathscr{O}_p/\mathfrak{m}_p \cong \mathbb{R}$, i.e., it is a real vector space. It turns out be naturally (whatever that means) the cotangent space to the differentiable or analytic manifold at p. In differential geometry, the cotangent space T_p^*M at p is the dual of the tangent space T_pM, whose elements are differential forms (e.g. df). Elements of $\mathfrak{m}_p$ are function germs that vanish at p; elements of $\mathfrak{m}_p^2$ are function germs that vanish at p with zero derivatives. Therefore, equivalence classes in $\mathfrak{m}_p/\mathfrak{m}_p^2$ represent "first-order infinitesimal changes" of functions at p, that is, derivative information. As a real rector space, $\mathfrak{m}_p/\mathfrak{m}_p^2$ is naturally isomorphic to the cotangent space T_p^*M. Specifically:

$$T_p^*M \cong \mathfrak{m}_p/\mathfrak{m}_p^2,$$

where the differential df corresponds to the equivalence class of the function germs $f - f(p)$. This insight will prove handy later, when we define tangent and cotangent spaces of schemes.

✍ Exercise 3.1 **(Exercise for those with differential geometric backgroud.)** Prove this. Rhetorical question for experts: what goes wrong if the sheaf of continuous functions is substituted for the sheaf of smooth functions? What goes wrong if you use the sheaf of C^1 functions?

Solution

(i) The sheaf of continuous functions.

The maximal ideal $\mathfrak{m}_p$ for continuous functions contains germs vanishing at p, but these functions may not be differentiable. Hence, $\mathfrak{m}_p/\mathfrak{m}_p^2$ may become infinite-dimensional, failing to match the finite-dimensional cotangent space T_p^*M.

(ii) The sheaf of C^1 functions.

$C_p^1/\mathfrak{m}_p \cong \mathbb{R}$ also holds, and first derivatives are definable. However, higher-order derivatives are not guaranteed. The sheaf C^1 is not closed under repeated differentiation, limiting geometric constructions that require smoothness. For example, let $f(x) = x + x^2 sin(\frac{1}{x})$, then $f(x) \in \mathfrak{m}_0$, but $f'(0)$ not exist.

3.2 Definition of sheaf and presheaf

We now formalize these notions, by defining presheaves and sheaves. Presheaves are simpler to define, and notions such as kernel and cokernel are straightforward. Sheaves are more complicated to defined, and some notions such as cokernel require more thought. But sheaves are more useful because they are in some vague sense more geometric, we can get information about a sheaf locally.

3.2.1 Definition of sheaf and presheaf on a topological space X

To be concrete, we will define sheaves of sets. However, in the definiton the category $\mathbf{Sets}$ can be replaced by any reasonable category, and other important examples are abelian groups $\mathbf{Ab}$, k-vector spaces $\mathbf{Vec}_k$, rings $\mathbf{Rings}$, modules over a ring $\mathbf{Mod}_A$, and more. Sheaves (and presheaves) are often written in calligraphic font.

The fact that $\mathscr{F}$ is a sheaf on a topological space X is often written as

$$\begin{array}{c}\mathscr{F}\\ \Big| \\ X.\end{array}$$

Definition 3.2.1 (Presheaf)

*A **presheaf of sets** $\mathscr{F}$ on a topological space X is the following data.*

(1) *To each open set $U \subseteq X$, we have a set $\mathscr{F}(U)$. The elements of $\mathscr{F}(U)$ are called **sections of $\mathscr{F}$ over** U. By convention, if the "U" is omitted, it is implicitly taken to be X: "**sections of** $\mathscr{F}$" means "sections of $\mathscr{F}$ over X". These are also called **global sections**.*

(2) *For each inclusion $U \hookrightarrow V$ of open sets, we have a **restriction map***

$$\operatorname{res}_{V,U} : \mathscr{F}(V) \to \mathscr{F}(U).$$

If $f \in \mathscr{F}(V)$, we often write

$$f|_U$$

for $\operatorname{res}_{V,U}(f)$.

The data is required to satisfy the following two conditions.

(1) *The map $\operatorname{res}_{U,U}$ is the identity: $\operatorname{res}_{U,U} = \operatorname{id}_{\mathscr{F}(U)}$.*

(2) *If $U \hookrightarrow V \hookrightarrow W$ are inclusions of open sets, then the restriction maps compose as you would expect, i.e., the diagram*

$$\begin{tikzcd}
\mathscr{F}(W) \arrow[rr, "\operatorname{res}_{W,U}"] \arrow[dr, "\operatorname{res}_{W,U}"'] & & \mathscr{F}(V) \arrow[dl, "\operatorname{res}_{V,U}"] \\
& \mathscr{F}(U) &
\end{tikzcd}$$

commutes.

Proposition 3.2.1

A presheaf is the same as a contravariant functor.

Proof By definition of presheaf, it is obvious. □

We define the stalk of a presheaf at a point in two equivalent ways. One will be hands-on, and the other will be as a colimit.

Definition 3.2.2 (Stalks and germs)

*Define the **stalk** of a presheaf $\mathscr{F}$ at a point p to be the set of **germs** of $\mathscr{F}$ at p, denote $\mathscr{F}_p$. Germs correspond to sections over some open set containing p, and two of these sections are considered the same if they agree on some smaller open set containing p. More precisely: the stalk is*

$$\{(f, \text{ open } U) : p \in U, f \in \mathscr{F}(U)\}$$

modulo the relation that $(f, U) \sim (g, V)$ if there is some open set $W \subseteq U, V$ where $p \in W$ and $\operatorname{res}_{U,W} f = \operatorname{res}_{V,W} g$.

Definition 3.2.3 (Stalk)

A useful equivalent definition of a stalk is as a colimit of all $\mathscr{F}(U)$ over all open sets U containing p:

$$\mathscr{F}_p = \varinjlim \mathscr{F}(U).$$

*If $p \in U$, and $f \in \mathscr{F}(U)$, then the image of f in $\mathscr{F}_p$ is called the **germ of f at** p.*

Remark 3.4 The index category is filtered set (given any two such open sets, there is a third such set contained in both), so these two definitions are the same by Exercise 2.27. Hence we can define stalks for sheaves of sets, groups, rings, and other things for which colimits exist for directed sets. It is very helpful to keep both definitions of stalk in mind at the same time.

Remark 3.5 Unlike Proposition 3.1.1, in general, the value of a section at a point doesn't make sense.

Definition 3.2.4 (Sheaf)

*A presheaf is a **sheaf** if it satisfies two more axioms. Notice that these axioms use the additional information of when some open sets cover another.*

*(1) **Identity axiom.** For any open set U, if $\{U_i\}_{i\in I}$ is an open cover of U, and $f_1, f_2 \in \mathscr{F}(U)$, and $f_1|_{U_i} = f_2|_{U_i}$ for all i, then $f_1 = f_2$.*
*(A presheaf satisfying the identity axiom is called a **separated presheaf**, but we will not use that notation in any essential way.)*

*(2) **Gluability axiom.** If $\{U_i\}_{i\in I}$ is an open cover of U, then given $f_i \in \mathscr{F}(U_i)$ for all i, such that $f_i|_{U_i\cap U_j} = f_j|_{U_i\cap U_j}$ for all i, j, then there is some $f \in \mathscr{F}(U)$ such that $\mathrm{res}_{U,U_i} f = f_i$ for all i.*

Remark 3.6 In mathematics, definitions often come paired: "at most one" and "at least one". In this case, identity means there is at most one way to glue, and gluability means that there is at least one way to glue.

Remark 3.7 An additional axiom sometimes included is that $\mathscr{F}(\varnothing)$ is a one-element set, and in general, for a sheaf with values in a category, $\mathscr{F}(\varnothing)$ is required to be the final object in the category. This actually follows from the above definitions, assuming that the empty product is appropriately defined as the final object. For example, in the definition of sheaf of rings, $\mathscr{F}(\varnothing) = 0 - \text{ring}$.

Example 3.1 If U and V are disjoint, then $\mathscr{F}(U\cap V) = \mathscr{F}(U) \times \mathscr{F}(V)$. Here we use the fact that $\mathscr{F}(\varnothing)$ is the final object.

Definition 3.2.5 (Stalk of a sheaf, germs of a section of a sheaf)

*The **stalk of a sheaf** at a point is just its stalk as a presheaf — the same definition applies — and similarly for the **germs of a section of a sheaf**.*

✍ **Exercise 3.2** **(Presheaves that are not sheaves.)** Show that the following are presheaves on $\mathbb{C}$ (with the classical topology), but not sheaves:

(a) bounded functions,

(b) holomorphic functios admitting a holomorphic square root.

Proof

(a) Let open set $U \subseteq \mathbb{C}$, and define $\mathscr{B}(U) = \{f : U \to \mathbb{C}$:f bounded on $U\}$. The restriction maps are function restrictions. Hence, $\mathscr{B}$ is a presheaf. We give a counterexample to show that $\mathscr{B}$ is not a sheaf. Let $U_n = D(n, 1)$ be an open disk, then $\{U_n\}$ is a countable open cover over $\mathbb{C}$. Let $f_n(z) = z$, then $f_n(z)$ bounded on each U_n, and therefore $f_n(z) \in \mathscr{B}(U_n)$. If $\mathscr{B}$ is a sheaf, there exists $f \in \mathscr{B}(\mathbb{C})$ such that $\mathrm{res}_{U,U_i} f = f_i$. This f must be $f(z)$, but $f(z)$ unbounded on $\mathbb{C}$, which a contradiction. Hence, $\mathscr{B}$ is not a sheaf.

(b) For each open set $U \subseteq \mathbb{C}$, define

$$\mathscr{G}(U) = \{f : U \to \mathbb{C} : f \text{ is holomorphic and there exists a holomorphic } g \text{ such that } g^2 = f\}.$$

The restriction maps are function restrictions. Hence, $\mathscr{G}$ is a presheaf. We give a counterexample to show that $\mathscr{G}$ is not a sheaf.

Let $U = \mathbb{C} \setminus \{0\}$, U is an open set of $\mathbb{C}$. Let $U_1 = \mathbb{C} \setminus [0, +\infty)$ and $U_2 = \mathbb{C} \setminus (-\infty, 0]$, so $\{U_1, U_2\}$ is the open cover of U. On U_1, define $f_1(z) = z$, with $g_1(z) = e^{\frac{1}{2}\mathrm{Log}(z)}$ (branch cut along $[0, +\infty)$), so $g_1^2 = z$. On U_2, define $f_2(z) = z$, with $g_2(z) = e^{\frac{1}{2}\mathrm{Log}(z)}$ (branch cut along $(-\infty, 0]$), so $g_2^2 = z$. On $U_1 \cap U_2 = \mathbb{C} \setminus \mathbb{R}$, $f_1|_{U_1 \cap U_2} = f_2|_{U_1 \cap U_2} = z$ and $g_1^2 = g_2^2 = z$. If there existed $f \in \mathscr{G}(\mathbb{C} \setminus \{0\})$ such that $f|_{U_i} = f_i$, then $f(z) = z$. However, $\sqrt{z}$ cannot be single-valued on $\mathbb{C} \setminus \{0\}$ (due to branch cut necessity), so no global holomorphic square root exists, leading to a contradiction. Hence, $\mathscr{G}$ is not a sheaf.

□

Definition 3.2.6 (Equalizer)

*Given two morphism $f, g : B \to C$, then their **equalizer** is an ordered pair (A, e) with $fe = ge$ that is universal with this property: if $q : X \to B$ satisfies $fq = gq$, then there exists a unique $q' : X \to B$ with $eq' = q$.*

$$\begin{tikzcd} A \arrow[r, "e"] & B \arrow[r, shift left, "f"] \arrow[r, shift right, "g"'] & C \\ X \arrow[u, dashed, "q'"] \arrow[ur, "q"'] & & \end{tikzcd}$$

More generally, if $(f_i : B \to C)_{i \in I}$ is a family of morphisms, then the equalizer is an ordered pair (A, e) with $f_i e = f_j e$ for all $i, j \in I$ that is universal with this property.

Example 3.2 In **Sets**, the equalizer is (E, e), where $E = \{b \in B : fb = gb\} \subseteq B$ and $e : E \to C$ is the inclusion. In $\mathbf{Mod}_R$, the equalizer of f and g is $(\mathrm{Ker}(f - g), e)$, where $e : \mathrm{Ker}(f - g) \to B$ is the inclusion. Hence, $\mathrm{Ker}\, f$ is the equalizer of f and 0.

Example 3.3 (**Interpretation in terms of the equalizer exact sequence.**) The two axioms for a presheaf to be a sheaf can be interpreted as "exactness" of the "equalizer exact sequence":

$$\cdot \longrightarrow \mathscr{F}(U) \longrightarrow \prod \mathscr{F}(U_i) \rightrightarrows \prod \mathscr{F}(U_i \cap U_j).$$

Identity axiom ensures that $\mathscr{F}(U) \to \prod \mathscr{F}(U_i)$ is injective. Gluability axiom implies that the equalizer of the two maps $\prod \mathscr{F}(U_i) \rightrightarrows \prod \mathscr{F}(U_i \cap U_j)$ (two maps are $(s_i) \mapsto (s_i|_{U_i \cap U_j})$ and $(s_i) \mapsto (s_j|_{U_i \cap U_j})$) must coincide exactly with the image of $\mathscr{F}(U)$.

✍ **Exercise 3.3** The identity and gluability axioms may be interpreted as saying that $\mathscr{F}\left(\bigcup_{i \in I} U_i\right)$ is a certain limit. What is that limit?

Proof $\mathscr{F}(\bigcup_{i \in I} U_i)$ is the equalizer of $\prod \mathscr{F}(U_i) \rightrightarrows \prod \mathscr{F}(U_i \cap U_j)$, the equalizer is a limit (by the definition of equalizer, it is clear). □

3.2.2 Example of sheaves

Here are number of example of sheaves.

Sheaves of functions

Exercise 3.4

(a) Verify that the example of §3.1 are indeed sheaves (of smooth functions, or continuous functions, or real-analytic functionsm, or plain real-valued functions, on a manifold or $\mathbb{R}^n$).

(b) Show that real-valued continuous functions on (open sets of) a topological space X form a sheaf.

Proof

(a) Let M be a smooth mainifold. For any open set $U \subseteq M$, define:

$$C^\infty(U) = \{f : U \to \mathbb{R} : f \text{ is smooth}\}.$$

The restriction maps are the natural restrictions of functions: $\mathrm{res}_{U,V} : C^\infty(U) \to C^\infty(V)$ for $V \subseteq U$.
Let $U \subseteq M$ be an open set, $\{U_i\}$ is an open cover of U, and $f \in C^\infty(U)$ satisfies $f|_{U_i} = 0$ for all i, we need to show that $f = 0$. Let $x \in U$, there exists some U_i containing x. Since $f|_{U_i} = 0$, $f(x) = 0$. Hence, $f = 0$.
Let $f_i \in C^\infty(U_i)$ satisfies $f_i|_{U_i \cap U_j} = f_j|_{U_i \cap U_j}$ for all i, j, we shall to show that there exists a unique $f \in C^\infty(U)$ with $f|_{U_i} = f_i$.
For existence. Define $f : U \to \mathbb{R}$ by setting $f(x) = f_i(x)$ when $x \in U_i$. The condition $f_i|_{U_i \cap U_j} = f_j|_{U_i \cap U_j}$ for all i, j ensures f is well-defined. For any $x \in U$, there exists some $i \in I$ such that $x \in U_i$. By assumption, $f|_{U_i}$ is smooth. Thus, there exists a coordinate chart (V_x, φ_x) containing x (with $V_x \subseteq U_i$ and $\varphi_x : V_x \to \varphi_x(V_x) \subseteq \mathbb{R}^n$ homeomorphism) such that:

$$f|_{U_i} \circ \varphi_x^{-1} : \varphi_x(V_x) \to \mathbb{R}$$

is smooth. Since M is a smooth manifold, its coordinate chart over the entire manifold. for each $x \in U$, choose the chart (V_x, φ_x) as above, where $V_x \subseteq U_i$ and $x \in V_x$. These charts $\{V_x\}_{x \in U}$ form an open cover of U. For any $x \in U$, the local representation of f in the chart (V_x, φ_x) is:

$$f \circ \varphi_x^{-1} = (f|_{U_i}) \circ \varphi_x^{-1},$$

and the right-hand side is smooth, by the previous discussion. Therefore, $f \circ \varphi_x^{-1}$ is smooth on $\varphi_x(V_x)$. Note that a function is smooth on a manifold if and only if it is smooth in every coordinate chart, $f \in C^\infty(U)$.
For uniqueness. If there is another $g \in C^\infty(U)$ with $g|_{U_i} = f_i$. For all $x \in U$,

$$g(x) = g|_{U_i}(x) = f_i(x) = f|_{U_i}(x) = f(x),$$

hence $f = g$.
Consequently, C^∞ is indeed a sheaf.

(b) Let X be a topological space X. For any open set $U \subseteq X$, define:

$$\mathscr{F}(U) = \{f : U \to \mathbb{R} : f \text{ is continuous}\}.$$

The restriction maps are the natural restriction of functions: $\mathrm{res}_{U,V} : \mathscr{F}(U) \to \mathscr{F}(V)$ for $V \subseteq U$.
Let $U \subseteq X$ be an open set, $\{U_i\}$ is an open cover of U, and $f \in \mathscr{F}(U)$ satisfies $f|_{U_i} = 0$ for all i, we need to show that $f = 0$. Let $x \in U$, there exists some U_i containing x. Since $f|_{U_i} = 0$, $f(x) = 0$. Hence, $f = 0$.
Let $f_i \in \mathscr{F}(U_i)$ satisfies $f_i|_{U_i \cap U_j} = f_j|_{U_i \cap U_j}$ for all i, j, we shall to show that there exists a unique $f \in \mathscr{F}(U)$ with $f|_{U_i} = f_i$.
For existence. Define $f : U \to \mathbb{R}$ by setting $f(x) = f_i(x)$ when $x_i \in U_i$. The condition $f_i|_{U_i \cap U_j} = f_j|_{U_i \cap U_j}$ for all i, j ensures f is well-defined. Now, we need to show that f is continuous. Let $V \subseteq \mathbb{R}$

open set, note that $f^{-1}(V) = \bigcup_i f_i^{-1}(V)$, since each f_i is continuous, $f_i^{-1}(V)$ is open, and therefore $f^{-1}(V) = \bigcup_i f_i^{-1}(V)$ is open. Hence, f is continuous, i.e., $f \in \mathscr{F}(U)$.
For uniqueness. If there is another $g \in \mathscr{F}(U)$ with $g|_{U_i} = f_i$. For all $x \in U$,

$$g(x) = g|_{U_i}(x) = f_i(x) = f|_{U_i}(x) = f(x),$$

hence $f = g$.

□

Definition 3.2.7 (Restriction of a sheaf) *Suppose $\mathscr{F}$ is a sheaf on X, and U is an open subset of X. Define the* ***restriction of $\mathscr{F}$ to U****, denoted $\mathscr{F}|_U$, to be the collection $\mathscr{F}|_U(V) = \mathscr{F}(V)$ for all open subsets $V \subseteq U$. Clearly this is a sheaf on U.*

Remark 3.8 In fact, we can restrict sheaves to arbitrary subsets.

The skyscraper sheaf

Example 3.4 **(The skyscraper sheaf.)** Suppose X is a topological space, with $p \in X$, and S is a set. Let $i_p : p \to X$ be the inclusion. Then $i_{p,*}S$ defined by

$$i_{p,*}S(U) = \begin{cases} S & \text{if } p \in U, \\ \{e\} & \text{if } p \notin U \end{cases}$$

forms a sheaf.

Let $V \subseteq U$, the restriction map defined by

$$f|_V = \begin{cases} e & \text{if } p \notin U, \\ \text{if } p \in U & \begin{cases} f & \text{if } p \in V \subseteq U, \\ e & \text{if } p \notin V \subseteq U. \end{cases} \end{cases}$$

It is easy to see that $i_{p,*}S$ is a presheaf. For any open set $U \subseteq X$, if $\{U_i\}_{i \in I}$ is an open cover of U, and $f_1, f_2 \in i_{p,*}U$, and $f_1|_{U_i} = f_2|_{U_i}$ for all i, by the definition of restriction map, $f_1 = f_2$. If $f_i \in i_{p,*}(U_i)$ for all i with $f_i|_{U_i \cap U_j} = f_j|_{U_i \cap U_j}$ for all i, j, also by the definition of restriction map, there is some $f \in i_{p,*}S(U)$ such that $f|_{U_i} = f_i$. Hence, $i_{p,*}S$ is a sheaf.

This is called a **skyscraper sheaf** supported at p, because the informal picture of it looks like a skyscarper at p. (Mild caution: this informal picture suggests that the only nontrivial stalk of a skyscraper sheaf is at p, which isn't case.)

There is an analogous definition for sheaves of abelian groups, except $i_{p,*}S(U)$ should be a final object.

Constant presheaf and constant sheaf

Definition 3.2.8 (Constant presheaf) *Let X be a topological space, and S a set. Define $\underline{S}_{\text{pre}}(U) = S$ for all open sets U. $\underline{S}_{\text{pre}}$ forms a presheaf (the restriction map is the identity). This is called the* ***constant presheaf associated to*** *S.*

Remark 3.9 This isn't (in general) a sheaf.

Definition 3.2.9 (Constant sheaf)

*Let X be a topological space, and S a set. Let $\mathscr{F}(U)$ be the set of maps $U \to S$ that are **locally constant**, i.e., for any point p in U, there is an open neighborhood of p where the function is constant. $\mathscr{F}$ is a sheaf. This is called the **constant sheaf** (with values in S). We denote this sheaf $\underline{S}$.*

(A better description is this: endow S with the discrete topology, and let $\mathscr{F}(U)$ be the continuous maps $U \to S$.)

Remark 3.10 DO NOT CONFUSE IT WITH THE CONSTANT PRESHEAF!

✍ **Exercise 3.5** Show that constant sheaf is a sheaf.

Proof By the definition of constant sheaf, it is easy to see that constant sheaf is a presheaf. Let U be an open set of X, and $\{U_i\}$ is the open cover of U. If $s_i \in \mathscr{F}(U_i)$ for all i with $s_i|_{U_i \cap U_j} = s_j|_{U_i \cap U_j}$ for all i, j. Defined $s(p) = s_i(p)$ when $p \in U_i$, $s_i|_{U_i \cap U_j} = s_j|_{U_i \cap U_j}$ ensures that s is well-defined. We need to show that s is locally constant. For any $x \in U$, there is a U_i countaining x. Since $s_i \in \mathscr{F}(U_i)$, there exist $p \in V \subseteq U_i$ such that $s_i : V \to S$ is constant, and therefore $s : V \to S$ is constant. Hence, gluability axiom holds.

If $s_1|_{U_i} = s_2|_{U_i}$ for all i. For all $x \in U$ there exists U_i such that $x \in U_i$, and therefore $s_1(x) = s_1|_{U_i}(x) = s_2|_{U_i}(x) = s_2(x)$, which implies the identity axiom.

Hence, constant sheaf is a sheaf. □

Morphisms glue and sheaf of sections of a map

✍ **Exercise 3.6** **("Morphisms glue.")** Suppose Y is a topological space. Show that "continuous maps to Y" form a sheaf of sets on X. More precisely, to each open set U of X, we associate the set of continuous maps of U to Y. Show that this forms a sheaf.

Proof Let U be an open set of X, and $\{U_i\}_i$ is the open cover of U. Define

$$\mathscr{F}(U) = \{f : U \to Y : f \text{ is continuous}\}.$$

The restriction maps are the natural restriction of functions: $\mathrm{res}_{U,V} : \mathscr{F}(U) \to \mathscr{F}(V)$ for $V \subseteq U$.

Let $f \in \mathscr{F}(U)$ satisfies $f|_{U_i} = 0$ for all i, we need to show that $f = 0$. Let $x \in U$, there exists some U_i containing x. Since $f|_{U_i} = 0$, $f(x) = 0$. Hence, $f = 0$, which implies identity axiom.

Let $f_i \in \mathscr{F}(U_i)$ satisfies $f_i|_{U_i \cap U_j} = f_j|_{U_i \cap U_j}$ for all i, j, we shall to show that there exists $f \in \mathscr{F}(U)$ with $f|_{U_i} = f_i$.

Define $f : U \to Y$ by setting $f(x) = f_i(x)$ when $x_i \in U_i$. The condition $f_i|_{U_i \cap U_j} = f_j|_{U_i \cap U_j}$ for all i, j ensures f is well-defined. Now, we need to show that f is continuous. Let $V \subseteq Y$ be an open set, then $f^{-1}(V) = \bigcup_i f_i^{-1}(V)$. Since each f_i is continuous, each $f_i^{-1}(V)$ is open, and therefore $f^{-1}(V) = \bigcup_i f_i^{-1}(V)$ is open. Hence, f is continuous, i.e., $f \in \mathscr{F}(U)$. Hence, gluability axiom holds.

Hence, $\mathscr{F}$ is a sheaf. □

Remark 3.11 Exercise 3.4, with $Y = \mathbb{R}$, and Exercise 3.5, with $Y = S$ with the discrete topology, are both special cases.

This is a fancier version of the previous exercise.

✍ **Exercise 3.7**

(a) **(Sheaf of sections of a map.)** Suppose we are given a continuous map $\mu : Y \to X$. Show that "section of μ" form a sheaf. More precisely, to each open set U of X, associate the set of continuous maps $s : U \to Y$ such that $\mu \circ s = \mathrm{id}|_U$. Show that this forms a sheaf. This is motivation for the phrase "section of a sheaf". (vector bundles)

(b) Suppose that Y is a topological group. Show that continuous maps to Y form a sheaf of groups.

Proof

(a) Let $U \subseteq X$ be an open set, and $\{U_i\}_i$ is the open cover of U. Define $\mathscr{F}(U) = \{s : U \to Y : s \text{ continuous and } \mu \circ s = \mathrm{id}\,|_U\}$. The restriction maps are the natural restriction of functions.

Let $f \in \mathscr{F}(U)$ satisfies $f|_{U_i} = 0$ for all i, we need to show that $f = 0$. Let $x \in U$, there exists some U_i containing x, then $f(x) = f|_{U_i}(x) = 0$, which implies identity axiom.

Let $f_i \in \mathscr{F}(U_i)$ satisfies $f_i|_{U_i \cap U_j} = f_j|_{U_i \cap U_j}$ for all i, j, we shall to show that there exists $f \in \mathscr{F}(U)$ with $f|_{U_i} = f$.

Define $f : U \to Y$ by setting $f(x) = f_i(x)$ when $x_i \in U_i$. The condition ensures f is well-defined. Now, we need to show that $f \in \mathscr{F}(U)$, that is, f is continuous and $\mu \circ f = \mathrm{id}_U$. Same as the proof in Exercise 3.6, f is continuous. Let $x \in U$, then there exists U_i containing x. Hence, $\mu \circ f(x) = \mu \circ f|_{U_i}(x) = x$, since $\mu \circ f|_{U_i} = \mathrm{id}_{U_i}$. This implies that $\mu \circ f = \mathrm{id}_U$, and therefore $f \in \mathscr{F}(U)$.

Consequently, $\mathscr{F}$ is a sheaf.

(b) It suffices to show that $\mathscr{F}(U)$ is a group and the restriction maps are group homomorphism. And apply the same method of part (a), $\mathscr{F}$ is a sheaf.

Define the multiplication on $\mathscr{F}(U)$ by setting $(f \cdot g)(x) = f(x)g(x)$ for all $x \in U$. The inverse of $f \in \mathscr{F}(U)$ given by $f^{-1}(x) = (f(x))^{-1}$. Let $e_Y : U \to Y$ defined by $e_Y(x) = e$, where e is the unite element in Y. Since $(e_Y \cdot f)(x) = e_Y(x)f(x) = e \cdot f(x) = f(x)$ and $(f \cdot e_Y)(x) = f(x)e_Y(x) = f(x) \cdot e = f(x)$, e_Y is the unite element in $\mathscr{F}(U)$. Hence, $\mathscr{F}(U)$ is a group.

Let $V \subseteq U$ and $f, g \in \mathscr{F}(U)$, then

$$\mathrm{res}_{U,V}(f \cdot g)(x) = (f \cdot g)|_V(x) = (f \cdot g)(x) = f(x)g(x) = f|_V(x)g|_V(x)$$

for all $x \in V$. Hence, the restriction maps are the group homomorphism.

□

The space of sections (espace étalé) of a (pre)sheaf

Depending on your background, you may prefer the following perspective on sheaves.

Definition 3.2.10 (Space of sections (espace étalé))

Suppose $\mathscr{F}$ is a presheaf of sets (e.g., a sheaf) on a topological space X. Construct a topological space F along with a continuous map $\pi : F \to X$ as follows: as a set, F is the disjoint union of all the stalks of $\mathscr{F}$, i.e.,

$$F = \coprod_{x \in X} \mathscr{F}_x.$$

This naturally gives a map of sets $\pi : F \to X$ (see remark). Topologize F as follows. Each $s \in \mathscr{F}(U)$ determines a subset $\{(x, s_x) : x \in U\}$ of F. The topology on F is the weakest topology such that these subsets are open. The topological space F could be thought of as the ***space of sections of*** $\mathscr{F}$ *(and in French is called the* ***espace étalé****).*

Remark 3.12

(1) Each element $s_x \in F$ must belongs to an unique $\mathscr{F}_x$, since F is the disjoint union of $\mathscr{F}_x$. Formally, each $s_x \in F$ can be written as (x, s_x), then this naturally gives a map of sets $\pi : F \to X$ by setting $\pi((x, s_x)) = x$.

(2) Subsets $\{(x, s_x) : x \in U\}$ form a base of the topology. For each $y \in F$, there is an open neighborhood V of y and an open neighborhood U of $\pi(y)$ such that $\pi|_V$ is a homeomorphism from V to U.

Proof Let $y \in F$, then $y = (x, s_x)$ for some $x \in U \subseteq X$. Hence, $y \in \{(x, s_x) : x \in U\}$. Let $B_s = \{(x, s_x) : x \in U\}$, $B_t = \{(x, t_x) : x \in V\}$, and $B_s \cap B_t \neq \varnothing$. Say $y = (x, s_x) = (x, t_x) \in B_s \cap B_t$, then as germs $s_x = t_x$, hence, there exists $W \subseteq U \cap V$ such that $s|_W = t|_W$. Let $B = \{(x, (s|_W)_x) : x \in W\}$, then $y \in B \subseteq B_s \cap B_t$. Hence, subsets $\{(x, s_x) : x \in U\}$ form a base of the topology.

Let $y \in F$, then $y = (x_0, s_{x_0})$ for some $x_0 = \pi(y)$ and $s_{x_0} \in \mathscr{F}_{x_0}$. Then there exist an open set $U \subseteq X$ containg x_0 such that s_{x_0} is the image of s in $\mathscr{F}_{x_0}$. Let $V = \{(x, s_x) : x \in U\}$, then V is an open neighborhood of y. We claim that $\pi|_V : V \to U$ is a homeomorphism. By the construction of V, $\pi|_V$ is a bijection. Let $V_0 \subseteq V$ be an open subset, then V_0 is the union some bases of the topology. Hence, $\pi|_V(V_0)$ is the union of some open sets of X, and therefore is open. Let $W \subseteq U$ be an open set, then $\pi|_V^{-1}(W) = \{(x, s_x) : x \in W\}$, it is an open set. Hence, $\pi|_V : V \to U$ is a homeomorphism. □

Remark 3.13 We will not discuss this construction at any length, but it can have some advantages:

(a) It is always better to know as many ways as possible of thinking about a concept.

(b) "Inverse image" (informally, "pullback") has a natural interpretation in this language.

(c) Sheafification has a natural interpretation in this language.

The pushforward sheaf

Definition 3.2.11 (The pushforward sheaf or direct image sheaf)

*Suppose $\pi : X \to Y$ is a continuous map, and $\mathscr{F}$ is a presheaf on X. Then define a presheaf $\pi_*\mathscr{F}(V) = \mathscr{F}(\pi^{-1}(V))$, where V is an open subset of Y. $\pi_*\mathscr{F}$ is a presheaf on Y, and is a sheaf if $\mathscr{F}$ is. This is called the **pushforward (or direct image)** of $\mathscr{F}$. More precisely, $\pi_*\mathscr{F}$ is called the **pushforward of $\mathscr{F}$ by** π.*

✍ **Exercise 3.8** Show that $\pi_*\mathscr{F}$ is a presheaf on Y, and is a sheaf if $\mathscr{F}$ is.

Proof

(1) $\pi_*\mathscr{F}$ is a presheaf.

Let $V_1 \supseteq V_2$ be open sets of Y. We need to show that the restriction map $\mathrm{res}_{V_1,V_2} : \pi_*\mathscr{F}(V_1) \to \pi_*\mathscr{F}(V_2)$ is well-defined ($\mathrm{res}_{V_1,V_2}(f) = \mathrm{res}_{\pi^{-1}(V_1),\pi^{-1}(V_2)}(f)$, RHS as the restriction map of sheaf $\mathscr{F}$). Note that $\pi_*\mathscr{F}(V_1) = \mathscr{F}(\pi^{-1}(V_1))$ and $\pi_*\mathscr{F}(V_1) = \mathscr{F}(\pi^{-1}(V_2))$, since π is continuous, $\pi^{-1}(V_1)$ and $\pi^{-1}(V_2)$ are open, and $\pi^{-1}(V_1) \supseteq \pi^{-1}(V_2)$. Hence, the restriction map is well-defined, and therefore $\pi_*\mathscr{F}$ is a presheaf.

(2) If $\mathscr{F}$ is a sheaf, $\pi_*\mathscr{F}$ is a sheaf.

By part (1), we know $\pi_*\mathscr{F}$ is a presheaf. It suffices to check identity axiom and gluability axiom.

Let V be any open subset of Y, $\{V_i\}_{i\in I}$ is an open cover of V. Let $f_1, f_2 \in \pi_*\mathscr{F}(V)$ and $f_1|_{\pi^{-1}(V_i)} = f_2|_{\pi^{-1}(V_i)}$ for all i. Since $\mathscr{F}$ is a sheaf and $\{\pi^{-1}(V_i)\}_i$ is an open cover of $\pi^{-1}(V)$, $f_1 = f_2$ for $f_i \in \mathscr{F}\pi^{-1}(V) = \pi_*\mathscr{F}(V)$, which holds identity axiom.

Let $f_i \in \pi_*\mathscr{F}(V_i)$ for all i, such that $f_i|_{\pi^{-1}(V_i)\cap\pi^{-1}(V_j)} = f_j|_{\pi^{-1}(V_i)\cap\pi^{-1}(V_j)}$ for all i, j. Since $\mathscr{F}$ is a sheaf and $\{\pi^{-1}(V_i)\}_i$ is an open cover of $\pi^{-1}(V)$, there exists $f \in \mathscr{F}(\pi^{-1}(V)) = \pi_*\mathscr{F}(V)$ such that $\mathrm{res}_{\pi^{-1}(V),\pi^{-1}(V_i)}(f) = f_i$ for all i. Hence, $\pi_*\mathscr{F}$ is a sheaf.

□

Example 3.5 The skyscraper sheaf can be interpreted as the pushforward of the constant sheaf $\underline{S}$ on a one-point space p, under the inclusion morphism $i_p : \{p\} \to X$.

Proof Let $U \subseteq X$ be an open set, then

$$i_p^{-1}(U) = \begin{cases} \{p\} & \text{if } p \in U \\ 0 & \text{if } p \notin U. \end{cases}$$

Hence,

$$i_{p,*}\underline{S}(U) = \underline{S}(i_p^{-1}(U)) = \begin{cases} S & \text{if } p \in U \\ 0 & \text{if } 0 \notin U. \end{cases}$$

This is agree with the definition of the sky scraper sheaf. □

Remark 3.14 Once we endow sheaves with the structure of a category, we will see that the pushforward is a functor from sheaves on X to sheaves on Y.

Proposition 3.2.2 (Pushforward induces maps of stalks)

Suppose $\pi : X \to Y$ is a continuous map, and $\mathscr{F}$ is a sheaf of sets (or rings or A-modules) on X. If $\pi(p) = q$, there is a natural morphism of stalks $(\pi_*\mathscr{F})_q \to \mathscr{F}_p$.

Proof In fact, $(\pi_*\mathscr{F})_q = \varinjlim_{V\ni q} \pi_*\mathscr{F}(V) = \varinjlim_{V \ni q} \mathscr{F}(\pi^{-1}(V))$. For all $q \in V \subseteq Y$ open set, $\pi^{-1}(V)$ is an open set of X and containing p, since π is continuous and $\pi(p) = q$. The open subset of Y which containing q given a partial ordered set by inclusion, then $\{\mathscr{F}(\pi^{-1}(V)), V \ni q\}$ with restriction maps give a direct system. For all $V \ni q$, define $\iota_V : \pi_*\mathscr{F}(V) \to \mathscr{F}_p$ by setting $\iota_V(f) = \overline{(f, \pi^{-1}(V))}$, it is easy to see this map is well-defined. Now, let $V' \supseteq V$, and consider the following diagram.

$$\begin{tikzcd}
\varinjlim_{V\ni q} \pi_*\mathscr{F}(V) \arrow[rr, dashed, "\theta"] & & \mathscr{F}_p \\
& \pi_*\mathscr{F}(V') \arrow[ul] \arrow[ur, "\iota_{V'}"'] \arrow[d, "\mathrm{res}_{V',V}"'] & \\
& \pi_*\mathscr{F}(V) \arrow[uul, bend left] \arrow[uur, bend right, "\iota_V"'] &
\end{tikzcd}$$

By the universal property of colimit, there exists unique morphism $\theta : (\pi_*\mathscr{F})_q \to \mathscr{F}_p$ maps the image of section in $(\pi_*\mathscr{F})_q$ to the image of section in $\mathscr{F}_p$. □

Ringed spaces, and $\mathscr{O}_X$-modules

Definition 3.2.12 (Ringed spaces, structure sheaf)

Suppose $\mathscr{O}_X$ is a sheaf of rings on a topological space X (i.e., a sheaf on X with values in the category of Rings*). Then $(X, \mathscr{O}_X)$ is called a **ringed space**. The sheaf of rings is often denoted by $\mathscr{O}_X$. This sheaf is called the **structure sheaf** of the ringed space. Sections of the structure sheaf $\mathscr{O}_X$ over an open subset U are called **functions on** U. Functions on X are called **global functions**, or just **functions**.*

Remark 3.15 what we call "functions", others sometimes call "regular functions". Furthermore, we will later define "rational functions" on schemes, which are not precisely function in this sense, they are a particular type of "partially-defined function".

Note *The symbol $\mathscr{O}_X$ will always refer to the structure sheaf of a ringed space X. The restriction $\mathscr{O}_X|_U$ of $\mathscr{O}_X$ to an open subset $U \subseteq X$ is denoted $\mathscr{O}_U$. (We will later call $(U, \mathscr{O}_U) \to (X, \mathscr{O}_X)$ an open embedding of ringed* *spaces.) The stalk of* $\mathscr{O}_X$ *at a point* p *is written "*$\mathscr{O}_{X,p}$*".*

Just as we have modules over a ring, we have $\mathscr{O}_X$-modules over a sheaf of rings $\mathscr{O}_X$. There is only one possible definition that could go with the name $\mathscr{O}_X$-module (or often $\mathscr{O}$-module) — a sheaf of abelian groups $\mathscr{F}$ with the following additional structure.

Definition 3.2.13 ($\mathscr{O}_X$-module)

Let $(X, \mathscr{O}_X)$ *be a ringed space. An* $\mathscr{O}_X$*-module is a sheaf* $\mathscr{F}$ *of abelian groups on* X *together with maps*

$$\mathscr{O}_X(U) \times \mathscr{F}(U) \to \mathscr{F}(U)$$

for each open $U \subseteq X$*, giving each* $\mathscr{F}(U)$ *the structure of an* $\mathscr{O}_X(U)$*-module, and which are compatible with the restriction maps for open subsets* $U \subseteq V \subseteq X$*, i.e., diagram*

$$\begin{tikzcd}
\mathscr{O}_X(V) \times \mathscr{F}(V) \arrow[r, "action"] \arrow[d, "\mathrm{res}^{\mathscr{O}_X}_{V,U} \times \mathrm{res}^{\mathscr{F}}_{V,U}"'] & \mathscr{F}(V) \arrow[d, "\mathrm{res}^{\mathscr{F}}_{V,U}"] \\
\mathscr{O}_X(U) \times \mathscr{F}(U) \arrow[r, "action"] & \mathscr{F}(U)
\end{tikzcd} \tag{3.1}$$

commutes.

Recall that the notion of A-module generalizes the notion of abelian group, because an abelian group is the same thing as a $\mathbb{Z}$-module. Similarly, the notion of $\mathscr{O}_X$-module generalize the notion of sheaf of abelian groups, because the latter is the same thing as a $\underline{\mathbb{Z}}$-module. Hence when we are proving things about $\mathscr{O}_X$-modules, we are also proving things about sheaves of abelian groups.

✍ **Exercise 3.9** If $(X, \mathscr{O}_X)$ is a ringed space, and $\mathscr{F}$ is an $\mathscr{O}_X$-module, describe how for each $p \in X$, $\mathscr{F}_p$ is an $\mathscr{O}_{X,p}$-module.

Proof Let $U \ni p$ be an open set of X, $r \in \mathscr{O}_X(U)$, $m \in \mathscr{F}(U)$, and denote r_p be the image of r in $\mathscr{O}_{X,p}$ and m_p be the image of m in $\mathscr{F}_p$. Since $\mathscr{F}$ is an $\mathscr{O}_X$-module, this gives action of $\mathscr{O}_X(U)$ on $\mathscr{F}(U)$, say $r \cdot m = rm$. Define the action of $\mathscr{O}_{X,p}$ on $\mathscr{F}_p$ by setting $r_p \cdot m_p = (rm)_p$. We need to show this action is well-defined. Let $r_p \sim r'_p$ and $m_p \sim m'_p$, where $r' \in \mathscr{O}_X(U')$ and $m' \in \mathscr{F}(U')$, then there exist $W \subseteq U \cap U', K \subseteq U \cap U'$ open sets which containing p such that $r|_W = r'|_W$ and $m|_K = m'|_K$. Let $V = W \cap K$, then $r|_V = r'|_V$ and $m|_V = m'|_V$. Hence, $(rm)|_V = (r'm')|_V$, and therefore $(rm)_p = (r'm')_p$, which implies that action is well-defined. The module axiom given by $\mathscr{O}_X$-module $\mathscr{F}$, hence, $\mathscr{F}_p$ is an $\mathscr{O}_{X,p}$-module. □

Example 3.6 **(For those who know about vector bundles.)** The motivating example of $\mathscr{O}_X$-modules is the sheaf of sections of a vecter bundle. If $(X, \mathscr{O}_X)$ is a differentiable manifold (so $\mathscr{O}_X$ is the sheaf of smooth functions), and $\pi : V \to X$ is a vector bundle over X, then the sheaf of smooth sections $\sigma : X \to V$ is an $\mathscr{O}_X$-module.

Proof Let $U \subseteq X$ be any open set of X, the sheaf of sections of a vector bundle defined by

$$\mathscr{F}(U) = \{s : U \to V : \pi \circ s = \mathrm{id}_U, s \text{ is smooth}\}.$$

The restriction given by the restriction of section. By Exercise 3.7, $\mathscr{F}$ indeed a sheaf. Now, we give an $\mathscr{O}_X$-module structure on $\mathscr{F}$ as follow: for any open set $U \subseteq X$, define $\cdot : \mathscr{O}_X(U) \times \mathscr{F}(U) \to \mathscr{F}(U)$ by setting $(f, s) \mapsto f \cdot s$, where $(f \cdot s)(x) = f(x) \cdot s(x)$ (RHS, is the scalar multiplication in the fiber $\pi^{-1}(x)$). We need to show that $\cdot$ is well-defined. Consider $\pi \circ (f \cdot s)$, let $x \in U$, then $\pi \circ (f \cdot s)(x) = \pi(f(x)s(x)) = x$ (since $f(x)s(x) \in \pi^{-1}(x)$), hence, $f \cdot s \in \mathscr{F}(U)$ (f is smooth), which implies that "$\cdot$" is well-defined. Then it is easy to check that $\mathscr{F}$ is an $\mathscr{O}_X$-module. □

3.3 Morphisms of presheaves and sheaves

Whenever one defines a new mathematical object, category theory teaches to try to understand maps between them. We now define morphisms of presheaves, and similarly for sheaves. In other words, we will describe the **category of presheaves** (of sets, abelian groups, etc.) and the **category of sheaves**.

3.3.1 Definition of morphisms of (pre)sheaves

Definition 3.3.1 (Morphism of presheaves)

*A **morphism of presheaves** of sets (or indeed of presheaves with values in any category) on X, $\varphi : \mathscr{F} \to \mathscr{G}$, is the data of maps $\varphi(U) : \mathscr{F}(U) \to \mathscr{G}(U)$ for all open sets $U \subseteq X$ behaving well with respect to restriction: if $U \hookrightarrow V$, then*

$$\begin{tikzcd} \mathscr{F}(V) \arrow[r, "\varphi(V)"] \arrow[d, "\mathrm{res}_{V,U}"'] & \mathscr{G}(V) \arrow[d, "\mathrm{res}_{V,U}"] \\ \mathscr{F}(U) \arrow[r, "\varphi(U)"] & \mathscr{G}(U) \end{tikzcd}$$

commutes. (Notice: the underlying space of both $\mathscr{F}$ and $\mathscr{G}$ is X.)

Definition 3.3.2 (Morphism of sheaves)

*A **morphisms of sheaves** of sets on X, $\varphi : \mathscr{F} \to \mathscr{G}$, is the data of maps $\varphi(U) : \mathscr{F}(U) \to \mathscr{G}(U)$ for all open sets $U \subseteq X$ behaving well with respect to restriction: if $U \hookrightarrow V$, then*

$$\begin{tikzcd} \mathscr{F}(V) \arrow[r, "\varphi(V)"] \arrow[d, "\mathrm{res}_{V,U}"'] & \mathscr{G}(V) \arrow[d, "\mathrm{res}_{V,U}"] \\ \mathscr{F}(U) \arrow[r, "\varphi(U)"] & \mathscr{G}(U) \end{tikzcd}$$

commutes.

Definition 3.3.3 (Morphisms of $\mathscr{O}_X$-modules)

An $\mathscr{O}_X$-module homomorphism $\mathscr{F} \to \mathscr{G}$ between $\mathscr{O}_X$-modules $\mathscr{F}, \mathscr{G}$ on X is a sheaf morphism $\mathscr{F} \to \mathscr{G}$ such that for all open subsets $U \subseteq X$, the map $\mathscr{F}(U) \to \mathscr{G}(U)$ is a homomorphism of $\mathscr{O}_X(U)$-modules.

Example 3.7 An example of a morphism of sheaves is the map from the sheaf of smooth functions on $\mathbb{R}$ to the sheaf of continuous functions. This is a "forgetful map": we are forgetting that these functions are differentiable, and remembering only that they are continuous.

Note *We may as well set some notation:*

(1) $\mathbf{Sets}_X$, $\mathbf{Ab}_X$, etc. denote the category of sheaves of sets, abelian groups, etc. on a topological space X

(2) Let $\mathbf{Mod}_{\mathscr{O}_X}$ denote the category of $\mathscr{O}_X$-modules on a ringed space $(X, \mathscr{O}_X)$.

(3) Let $\mathbf{Sets}_X^{pre}$, etc. denote the category of presheaves of sets, etc. on X.

Remark 3.16 If we interpret presheaf on X as a contravariant functor (from the category of open sets), a morphism of presheaves on X is a natural transformation of functor.

Proposition 3.3.1 (Morphisms of (pre)sheaf induce morphisms of stalks)

*If $\varphi : \mathscr{F} \to \mathscr{G}$ is a morphism of (pre)sheaves on X, and $p \in X$, then φ induces a morphism of stalks $\varphi_p : \mathscr{F}_p \to \mathscr{G}_p$. Translation: taking the stalk at p induces a **stalkification functor** $\mathbf{Sets}_X \to \mathbf{Sets}$.*

Proof For all $U' \supseteq U$ open sets of X, consider the following diagram.

$$\begin{tikzcd}
\mathscr{F}_p \arrow[rrr, dashed] & & & \mathscr{G}_p \\
& \mathscr{F}(U') \arrow[ul, "\pi^{\mathscr{F}}_{U'}"'] \arrow[r, "\varphi_{U'}"] \arrow[d, "\mathrm{res}^{\mathscr{F}}_{U',U}"'] & \mathscr{G}(U') \arrow[ur, "\pi^{\mathscr{G}}_{U'}"] \arrow[d, "\mathrm{res}^{\mathscr{G}}_{U',U}"] & \\
& \mathscr{F}(U) \arrow[uul, bend left, "\pi^{\mathscr{F}}_{U}"] \arrow[r, "\varphi_U"] & \mathscr{G}(U) \arrow[uur, bend right, "\pi^{\mathscr{G}}_{U}"'] &
\end{tikzcd}$$

By the universal property of colimit, there exists unique morphism $\varphi_p : \mathscr{F}_p \to \mathscr{G}_p$. □

Remark 3.17 This proof will extend in obvious ways. For example, if φ is a morphism of $\mathscr{O}_X$-modules, then φ_p is a map of $\mathscr{O}_{X,p}$-modules.

✍ **Exercise 3.10** Suppose $\pi : X \to Y$ is a continuous map of topological spaces (i.e., a morphism in the category of topological spaces). Show that pushforward gives a functor $\pi_* : \mathbf{Sets}_X \to \mathbf{Sets}_Y$. Here **Sets** can be replaced by other categories.

Proof Let $\mathscr{F} \in \mathbf{Sets}_X$, then $\pi_*\mathscr{F} \in \mathbf{Sets}_Y$, by Exercise 3.8. Let $\varphi : \mathscr{F} \to \mathscr{G}$ be a morphism of sheaf in $\mathbf{Sets}_X$. Define $\pi_*\varphi : \pi_*\mathscr{F} \to \pi_*\mathscr{G}$ by setting $\pi_*\varphi_V = \varphi_{\pi^{-1}(V)}$, where $V \subseteq Y$ be open set in Y. We need to show that $\pi_*\varphi$ is a morphism of sheaf between $\pi_*\mathscr{F}$ and $\pi_*\mathscr{G}$. Let $V' \supseteq V$, then $\pi^{-1}(V') \supseteq \pi^{-1}(V)$. Since φ is the morphism of sheaf, we have the following commutative diagram.

$$\begin{tikzcd}
\mathscr{F}(\pi^{-1}(V')) \arrow[r, "\varphi_{\pi^{-1}(V')}"] \arrow[d, "\mathrm{res}^{\mathscr{F}}_{\pi^{-1}(V'),\pi^{-1}(V)}"'] & \mathscr{G}(\pi^{-1}(V')) \arrow[d, "\mathrm{res}^{\mathscr{G}}_{\pi^{-1}(V'),\pi^{-1}(V)}"] \\
\mathscr{F}(\pi^{-1}(V)) \arrow[r, "\varphi_{\pi^{-1}(V)}"] & \mathscr{G}(\pi^{-1}(V))
\end{tikzcd}$$

Hence, diagram

$$\begin{tikzcd}
\pi_*\mathscr{F}(V') \arrow[r, "\pi_*\varphi_{V'}"] \arrow[d, "\mathrm{res}^{\pi_*\mathscr{F}}_{V',V}"'] & \pi_*\mathscr{G}(V') \arrow[d, "\mathrm{res}^{\pi_*\mathscr{G}}_{V',V}"] \\
\pi_*\mathscr{F}(V) \arrow[r, "\pi_*\varphi_V"] & \pi_*\mathscr{G}(V)
\end{tikzcd}$$

commutes. This implies that $\pi_*\varphi$ is a morphism of sheaf. Let $\mathrm{id}_{\mathscr{F}} : \mathscr{F} \to \mathscr{F} \in \mathbf{Sets}_X$ be identity morphism of sheaf, then $\pi_* \mathrm{id}_{\mathscr{F}} = \mathrm{id}_{\pi_*\mathscr{F}}$ is also an identity morphism of sheaf. Finally, it is easy to check π_* preserves composition. Hence, pushforward gives a functor $\pi_* : \mathbf{Sets}_X \to \mathbf{Sets}_Y$. □

Definition 3.3.4 (Sheaf Hom)

Suppose $\mathscr{F}$ and $\mathscr{G}$ are two sheaves of sets on X. (In fact, it will suffice that $\mathscr{F}$ is a presheaf.) Let $\mathscr{H}om(\mathscr{F},\mathscr{G})$ be the collection of data

$$\mathscr{H}om(\mathscr{F},\mathscr{G})(U) := \mathrm{Mor}(\mathscr{F}|_U, \mathscr{G}|_U).$$

*This is a sheaf of sets on X. This sheaf is called "**sheaf** $\mathscr{H}om$".*

Remark 3.18

(1) Strictly speaking, we should reserve Hom for when we are in an additive category, so this should possibly be called "sheaf Mor". But the terminology "sheaf $\mathscr{H}om$" is too established to uproot.

(2) It will be clear from your construction that, like Hom, $\mathscr{H}om(\square, \mathscr{G})$ is contravariant functor and $\mathscr{H}om(\mathscr{F}, \square)$ is covariant functor.

Remark 3.19 $\mathscr{H}om$ does not commute with taking stalks. More precisely: it is not ture that $\mathscr{H}om(\mathscr{F},\mathscr{G})_p$ is isomorphic to $\mathrm{Mor}(\mathscr{F}_p, \mathscr{G}_p)$.

Example 3.8 (**Counterexample of above remark.**) Let H be a non-trivial abelian group. Let $\mathscr{G}$ be the constant sheaf H (i.e., $\mathscr{G} = \underline{H}$), and let $\mathscr{F}$ be the skyscraper sheaf at p with stalk H. We claim that $\mathscr{H}om(\mathscr{F}, \mathscr{G})$ is the zero sheaf. Let $U \subseteq X$ be any connected open set of X (if U is not connected, we only need to consider all connected components of U, as Exercise 3.12 (b)).

If $p \notin U$, for all $V \subseteq U$ open set, $p \notin V$, then $\mathscr{F}|_U(V) = \{0\}$. Hence,

$$\mathscr{H}om(\mathscr{F}, \mathscr{G})(U) = \mathrm{Mor}(\mathscr{F}|_U, \mathscr{G}|_U) = \{0\}.$$

If $p \in U$, we need to show that $\mathrm{Mor}_{\mathbf{Ab}}(\mathscr{F}|_U, \mathscr{G}|_U) = \{0\}$. Let $V \subseteq U$ be any open subset. If $V \ni p$, then $\mathscr{F}(V) = H$, $\mathscr{G}(V) = H$, and therefore $\varphi_V : H \to H$ is a group homomorphism. If $p \notin V$, then $\mathscr{F}(V) = 0$, φ_V must be zero homomorphism.

Let $p \in V \subseteq U$ and $p \notin W$ is an open subset of V. Consider the following diagram.

$$\begin{tikzcd}
\mathscr{F}(V) \arrow[r, "\varphi_V"] \arrow[d, "\mathrm{res}^{\mathscr{F}}_{V,W}"'] & \mathscr{G}(V) \arrow[d, "\mathrm{res}^{\mathscr{G}}_{V,W}"] \\
\mathscr{F}(W) \arrow[r, "\varphi_W"] & \mathscr{G}(W)
\end{tikzcd}$$

Pick $h \in \mathscr{F}(V) = H$, then

$$\mathrm{res}^{\mathscr{G}}_{V,W}(\varphi_V(h)) = \varphi_W(\mathrm{res}^{\mathscr{F}}_{V,W}(h)) = \varphi_W(0) = 0.$$

Since, $\mathscr{G}$ is a constant sheaf, restriction must be identity, hence, $\varphi_V(h) = 0$ for all $h \in H$, and therefore $\varphi_V = 0$ for all open subset $V \subseteq U$. Hence, $\mathscr{H}om(\mathscr{F}, \mathscr{G})(U) = 0$, for all $U \ni p$.

Taking stalk at p, we have $\mathscr{H}om(\mathscr{F}, \mathscr{G})_p = 0$. But $\mathrm{Mor}(\mathscr{F}_p, \mathscr{G}_p) = \mathrm{Mor}(H, H) \neq \{0\}$,i.e.,

$$\mathscr{H}om(\mathscr{F}, \mathscr{G})_p \ncong \mathrm{Mor}(\mathscr{F}_p, \mathscr{G}_p).$$

✍ **Exercise 3.11** Show that sheaf $\mathscr{H}om$ is a sheaf of sets on X.

Proof Let $U' \supseteq U \subseteq X$ be opensets. The restriction map $\mathrm{res}_{U',U} : \mathscr{H}om(\mathscr{F}, \mathscr{G})(U') \to \mathscr{H}om(\mathscr{F}, \mathscr{G})(U)$ given by $\varphi \in \mathrm{Mor}(\mathscr{F}|_{U'}, \mathscr{G}|_{U'}) \mapsto \varphi \in \mathrm{Mor}(\mathscr{F}|_U, \mathscr{G}|_U)$. The composition of the restriction maps is easy to check, hence, $\mathscr{H}om(\mathscr{F}, \mathscr{G})$ is a presheaf. To show $\mathscr{H}om(\mathscr{F}, \mathscr{G})$ is also a sheaf, it suffices to check identity axiom and gluability axiom.

To show identity axiom. For any open set $U \subseteq X$, let $\{U_i\}_{i \in I}$ is an open cover of U, and $\varphi_1, \varphi_2 \in \mathscr{H}om(\mathscr{F}, \mathscr{G})(U)$, and $\varphi_1|_{U_i} = \varphi_2|_{U_i}$ for all i, we need to check $\varphi_1 = \varphi_2$. It suffices to show $(\varphi_1)_V = (\varphi_2)_V$ for all open subset $V \subseteq U$. Since $\{U_i\}_{i \in I}$ is an open cover of U, $\{U_i \cap V\}$ is an open cover of V. Let $s \in \mathscr{F}(V)$, since $\varphi_1|_{U_i} = \varphi_2|_{U_i}$ for all i, we have $(\varphi_1|_{U_i})_{U_i \cap V}(s|_{U_i \cap V}) = (\varphi_2|_{U_i})_{U_i \cap V}(s|_{U_i \cap V})$ for all i. Since $\mathscr{G}$ is a sheaf, by the identity axiom of $\mathscr{G}$, we have $(\varphi_1)_V(s) = (\varphi_2)_V(s)$ for all $s \in \mathscr{F}(V)$ for all open subsets $V \subseteq U$. Hence, $\varphi_1 = \varphi_2$.

To show gluability axiom. Let $\{U_i\}$ be an open cover of open set $U \subseteq X$, and given $\varphi_i \in \mathscr{H}om(\mathscr{F}, \mathscr{G})(U_i)$ with $\varphi_i|_{U_i \cap U_j} = \varphi_j|_{U_i \cap U_j}$, we need to show there exists $\varphi \in \mathscr{F}(U)$ such that $\mathrm{res}_{U,U_i} \varphi = \varphi_i$ for all i.

We need to define φ. For all $V \subseteq U$ open set and section $s \in \mathscr{F}(V)$, $\varphi_V(s)$ must belong to $\mathscr{G}(V)$. Since $\{U_i \cap V\}$ is an open cover of V, by the condition, we have

$$\begin{aligned}
(\varphi_i|_{U_i \cap U_j})_{V \cap U_i \cap U_j}(s|_{V \cap U_i \cap U_j}) =& (\varphi_i)_{V \cap U_i \cap U_j}(s|_{V \cap U_i \cap U_j}) \\
=& ((\varphi_i)_{V \cap U_i}(s|_{V \cap U_i}))|_{V \cap U_i \cap U_j} \\
=& (\varphi_j|_{U_i \cap U_j})_{V \cap U_i \cap U_j}(s|_{V \cap U_i \cap U_j}) \\
=& (\varphi_j)_{V \cap U_i \cap U_j}(s|_{V \cap U_i \cap U_j}) \\
=& ((\varphi_j)_{V \cap U_j}(s|_{V \cap U_j}))|_{V \cap U_i \cap U_j},
\end{aligned}$$

where $(\varphi_i)_{V \cap U_i}(s|_{V \cap U_i}) \in \mathscr{G}(V \cap U_i)$ and $(\varphi_j)_{V \cap U_j}(s|_{V \cap U_j}) \in \mathscr{G}(V \cap U_j)$. By the gluability axiom of sheaf

$\mathscr{G}$, there exists $\widetilde{\varphi_V(s)} \in \mathscr{G}(V)$ such that $\widetilde{\varphi_V(s)}|_{V\cap U_i} = (\varphi_i)_{V\cap U_i}(s|_{V\cap U_i})$ for all i. Define $\varphi_V(s) = \widetilde{\varphi_V(s)} \in \mathscr{G}(V)$, for all $s \in \mathscr{F}(V)$ and for all $V \subseteq X$ open, then we defined φ. The gluability of $\mathscr{H}om(\mathscr{F}, \mathscr{G})$ given by the definition of φ.

Hence, $\mathscr{H}om$ is a sheaf of sets on X. □

Note *We will use many variants of the definition of $\mathscr{H}om$. For example:*

(1) if $\mathscr{F}$ and $\mathscr{G}$ are sheaves of abelian groups on X, then $\mathscr{H}om_{\mathbf{Ab}_X}(\mathscr{F}, \mathscr{G})$ is defined by taking $\mathscr{H}om_{\mathbf{Ab}_X}(\mathscr{F}, \mathscr{G})(U)$ to be the maps as sheaves of abelian groups $\mathscr{F}|_U \to \mathscr{G}|_U$ (Note that $\mathscr{H}om_{\mathbf{Ab}_X}(\mathscr{F}, \mathscr{G})$ has the structure of a sheaf of abelian groups in the natural way);

(2) if $\mathscr{F}$ and $\mathscr{G}$ are $\mathscr{O}_X$-modules, we define $\mathscr{H}om_{\mathbf{Mod}_{\mathscr{O}_X}}(\mathscr{F}, \mathscr{G})$ in the analogous way (and it is an $\mathscr{O}_X$-module).

Obnoxiously, the subscripts $\mathbf{Ab}_X$ and $\mathbf{Mod}_{\mathscr{O}_X}$ are often dropped (here and in the literature), so be careful which category you are working in!

Definition 3.3.5 (Dual of $\mathscr{O}_X$-module)

We call $\mathscr{H}om_{\mathbf{Mod}_{\mathscr{O}_X}}(\mathscr{F}, \mathscr{O}_X)$ the dual of the $\mathscr{O}_X$-module $\mathscr{F}$, and denote it $\mathscr{F}^\vee$.

Exercise 3.12

(a) If $\mathscr{F}$ is a sheaf of sets on X, then show that $\mathscr{H}om(\underline{\{p\}}, \mathscr{F}) \cong \mathscr{F}$, where $\underline{\{p\}}$ is the constant sheaf "with values in the one element set $\{p\}$".

(b) If $\mathscr{F}$ is a sheaf of abelian groups on X, then show that $\mathscr{H}om_{\mathbf{Ab}_X}(\underline{\mathbb{Z}}, \mathscr{F}) \cong \mathscr{F}$ (an isomorphism of sheaves of abelian groups).

(c) If $\mathscr{F}$ is an $\mathscr{O}_X$-module, then show that $\mathscr{H}om_{\mathbf{Mod}_{\mathscr{O}_X}}(\mathscr{O}_X, \mathscr{F}) \cong \mathscr{F}$ (an isomorphism of $\mathscr{O}_X$-modules).

Remark 3.20 (b) gives an example where taking stalks and sheaf $\mathscr{H}om$ commute, since

$$\mathscr{H}om_{\mathbf{Ab}_X}(\underline{\mathbb{Z}}, \mathscr{F})_p \cong \mathscr{F}_p \cong \mathrm{Hom}(\mathbb{Z}, \mathscr{F}_p) = \mathrm{Hom}(\underline{\mathbb{Z}}_p, \mathscr{F}_p).$$

Proof

(a) Let $U \subseteq X$ be any open set of X, then $\underline{\{p\}} = \{s : U \to \{p\} \text{ locally constant}\} = \{p\}$. Let $\varphi \in \mathscr{H}om(\underline{\{p\}}, \mathscr{F})(U)$. Let $W \subseteq V \subseteq U$ be any open sets of U, then $\varphi_V : \underline{\{p\}}(U) = \{p\} \to \mathscr{F}|_U(V) = \mathscr{F}(U)$, and we have the following commutative diagram,

$$\begin{CD} \{p\} @>{\varphi_V}>> \mathscr{F}(V) \\ @V{\mathrm{res}^{\{p\}}_{V,W}}VV @VV{\mathrm{res}^{\mathscr{F}}_{V,W}}V \\ \{p\} @>{\varphi_W}>> \mathscr{F}(W), \end{CD}$$

that is, $\varphi_V(p)|_W = \varphi_W(p)$. For any $V \subseteq U$, let $\{U_i\}$ be an open cover of U, then $\varphi_{U_i}(p)|_{U_i\cap U_j} = \varphi_{U_j}(p)|_{U_i\cap U_j}$. Since $\mathscr{F}$ is a sheaf, there exists unique global section $s \in \mathscr{F}(U)$ such that $s|_{U_i} = \varphi_{U_i}(p)$. This implies that for any $\varphi \in \mathscr{H}om(\underline{\{p\}}, \mathscr{F})(U)$, it determines unique global section $s \in \mathscr{F}(U)$. Conversely, for any section $s \in \mathscr{F}(U)$, we can define $\varphi \in \mathscr{H}om(\underline{\{p\}}, \mathscr{F})(U)$ by setting $\varphi_V(p) = s|_V$ for all $V \subseteq U$. Then we get a one-to-one correspondence between $\mathscr{H}om(\underline{\{p\}}, \mathscr{F})(U)$ and $\mathscr{F}(U)$ for all open set $U \subseteq X$. Formally, for any open set $U \subseteq X$, there is a bijection

$$\theta_U : \mathscr{H}om(\underline{\{p\}}, \mathscr{F})(U) \to \mathscr{F}(U)$$

by setting $\theta_U(\varphi) = s, \varphi_V(p) = s|_V$ for all open set $V \subseteq U$ (hence, $s = \varphi_U(p)$).

Now, we need to check that $\theta : \mathscr{H}om(\underline{\{p\}}, \mathscr{F})$ is a morphism of sheaves. Let $U' \supseteq U$ be two open sets of X, it suffice to check the following diagram

$$\begin{tikzcd} \mathscr{H}om(\underline{\{p\}}, \mathscr{F})(U') \arrow[r, "\theta_{U'}"] \arrow[d, "\mathrm{res}^{\mathscr{H}om}_{U',U}"'] & \mathscr{F}(U') \arrow[d, "\mathrm{res}^{\mathscr{F}}_{U',U}"] \\ \mathscr{H}om(\underline{\{p\}}, \mathscr{F})(U) \arrow[r, "\theta_U"] & \mathscr{F}(U) \end{tikzcd}$$

commutes. Let $\varphi \in \mathscr{H}om(\underline{\{p\}}, \mathscr{F})(U')$, then

$$\theta_U(\varphi|_U) = \varphi_U(p) = \varphi_{U'}(p)|_U = \theta_{U'}(\varphi)|_U,$$

which implies the diagram commutes. Hence, θ is a morphism of sheaves, and therefore,

$$\mathscr{H}om(\underline{\{p\}}, \mathscr{F}) \cong \mathscr{F}.$$

(b) Let $U \subseteq X$ be any open set of X, and $\pi_0(U)$ is the set of connected components of U, then

$$\underline{\mathbb{Z}}(U) = \prod_{C \in \pi_0(U)} \underline{\mathbb{Z}}(C).$$

If $V \subseteq U$, the restriction maps given by $\mathrm{res}^{\underline{\mathbb{Z}}}_{U,V}(s)|_{C\cap V} = s|_{C\cap V}$ for all $C \in \pi_0(U)$.
First, we define $\theta : \mathscr{H}om(\underline{\mathbb{Z}}, \mathscr{F}) \to \mathscr{F}$. For all $U \subseteq X$ be any open set of X, for all sheaf morphism $\varphi \in \mathscr{H}om(\underline{\mathbb{Z}}, \mathscr{F})(U)$, define

$$\theta_U(\varphi) = s, \quad s|_C = \varphi_C(1_C),$$

where $1_C \in \underline{\mathbb{Z}}(C) = \mathbb{Z}$. Now, we need to check θ_U is well-defined for all $U \subseteq X$. It suffices to check two things $s \in \mathscr{F}(U)$ and θ is a morphism of sheaves. $\pi_0(U)$ is an open cover of U, note that for any C_i, C_j, $C_i \cap C_j = \varnothing$, then we have $(s|_{C_i})|_{C_i\cap C_j} = (s|_{C_j})|_{C_i \cap C_j}$, by the gluability axiom of $\mathscr{F}$, there exists unique $s \in \mathscr{F}(U)$ be the global section. Let $U' \supseteq U$, consider the following diagram.

$$\begin{tikzcd} \mathscr{H}om(\underline{\mathbb{Z}}, \mathscr{F})(U') \arrow[r, "\theta_{U'}"] \arrow[d, "\mathrm{res}^{\mathscr{H}om}_{U',U}"'] & \mathscr{F}(U') \arrow[d, "\mathrm{res}^{\mathscr{F}}_{U',U}"] \\ \mathscr{H}om(\underline{\mathbb{Z}}, \mathscr{F})(U) \arrow[r, "\theta_U"] & \mathscr{F}(U) \end{tikzcd}$$

Let $\varphi \in \mathscr{H}om(\underline{\mathbb{Z}}, \mathscr{F})(U')$. In fact, for all $C' \in \pi_0(U')$, $\pi_0(U) = \{C' \cap U : C' \in \pi_0(U')\}$, then

$$\theta_U(\varphi|_U)|_{C'\cap U} = (\varphi|_U)_{C'\cap U}(1_{C'\cap U}) = \varphi_{C'\cap U}(1_{C'\cap U})$$

for all $C' \in \pi_0(U')$. On the other hand,

$$(\theta_{U'}(\varphi)|_{C'})|_U = \theta_{U'}(\varphi)|_{C'\cap U} = \varphi_{C'\cap U}(1_{C'\cap U})$$

for all $C' \in \pi_0(U')$. Hence, $\theta_U(\varphi|_U) = \theta_{U'}(\varphi)|_U$, and therefore θ is a sheaf morphism. Thus, θ is well-defined.
Next, we shall to check that θ is also a group homomorphism. For all open set $U \subseteq X$, let $\varphi, \psi \in \mathscr{H}om(\underline{\mathbb{Z}}, \mathscr{F})(U)$, then

$$\theta_U(\varphi + \psi)|_C = (\varphi + \psi)_C(1_C) = \varphi_C(1_C) + \psi_C(1_C)$$

for all $C \in \pi_0(U)$ (the last equality holds because $\mathscr{F}(U)$ is a group), that is, $\theta_U(\varphi+\psi) = \theta_U(\varphi)+\theta_U(\psi)$. Hence, θ is a group homomorphism.
We claim that θ_U is an isomorphism. Let $s \in \mathscr{F}(U)$, for all $C \in \pi_0(U)$, define $\varphi_C(1_C) = s|_C$, since each $C \in \pi_0(U)$ disjoint and $\mathscr{H}om(\underline{\mathbb{Z}}, \mathscr{F})$ is a sheaf, we can glue all local sections φ_C into a unique global section $\varphi \in \mathscr{H}om(\underline{\mathbb{Z}}, \mathscr{F})(U)$. Note that $\theta_U(\varphi)|_C = \varphi_C(1_C) = s|_C$, $\theta_U(\varphi) = s$, this implies that θ_U is a surjection. Let $\theta_U(\varphi) = 0$, then $\varphi_C(1_C) = 0$, hence, $\varphi_C = 0$, and therefore $\varphi = 0$, by the gluability axiom of $\mathscr{H}om(\underline{\mathbb{Z}}, \mathscr{F})$. Hence, θ_U is injection. Thus, θ_U is an isomorphism.

By above discussion, we know:

(i) θ is a well-defined sheaf homomorphism.

(ii) θ_U is a group isomorphism, for all open set $U \subseteq X$.

Hence, $\mathscr{H}om_{\mathbf{Ab}_X}(\mathbb{Z}, \mathscr{F}) \cong \mathscr{F}$.

(c) Note that $\mathscr{O}_X$ is a free $\mathscr{O}_X$-module with rank 1, and define $\theta_U : \mathscr{H}om_{\mathbf{Mod}_{\mathscr{O}_X}}(\mathscr{O}_X, \mathscr{F})(U) \to \mathscr{F}(U)$ by setting $\theta_U(\varphi) = \varphi_U(1_U)$, where $1_U \in \mathscr{O}_X(U)$, then similar to (b), we have $\mathscr{H}om_{\mathbf{Mod}_{\mathscr{O}_X}}(\mathscr{O}_X, \mathscr{F}) \cong \mathscr{F}$. $\square$

3.3.2 Presheaves of abelian groups (and even "presheaf $\mathscr{O}_X$-modules") form an abelian category

We can make module-like constructions using presheaves of abelian groups on a topological space X. (Throughout this section, all (pre)sheaves are of abelian groups.) For example, we can clearly add maps of presheaves and get another map of presheaves: if $\varphi, \psi : \mathscr{F} \to \mathscr{G}$, then we define the map $\varphi + \psi$ by $(\varphi + \psi)(V) = \varphi(V) + \psi(V)$. (There is something small to check here: that the result is indeed a map of presheaves.) In this way, presheaves of abelian groups form an additive category (Definition 2.5.1: the morphisms between any two presheaves of abelian groups from an abelian group; there is a 0-object; and one can take finite products). For exactly that same reasons, sheaves of abelian groups also form an additive category.

Definition 3.3.6 (Presheaf kernel)

If $\varphi : \mathscr{F} \to \mathscr{G}$ is a morphism of presheaves, define the ***presheaf kernel*** $\operatorname{Ker}_{\text{pre}} \varphi$ *by*

$$(\operatorname{Ker}_{\text{pre}} \varphi)(U) := \operatorname{Ker} \varphi_U,$$

for all open set $U \subseteq X$.

Proposition 3.3.2

$\operatorname{Ker}_{\text{pre}} \varphi$ *is a presheaf.*

Proof Let $U' \supseteq U$ be two open sets of X, consider the following diagram.

$$\begin{tikzcd}
0 \arrow[r] & \operatorname{Ker}_{\text{pre}} \varphi(U') \arrow[r, "\operatorname{Ker} \varphi_{U'}"] \arrow[d, dashed, "\exists!"] & \mathscr{F}(U') \arrow[r, "\varphi_{U'}"] \arrow[d, "\operatorname{res}^{\mathscr{F}}_{U',U}"] & \mathscr{G}(U') \arrow[d, "\operatorname{res}^{\mathscr{G}}_{U',U}"] \\
0 \arrow[r] & \operatorname{Ker}_{\text{pre}} \varphi(U) \arrow[r, "\operatorname{Ker} \varphi_U"'] & \mathscr{F}(U) \arrow[r, "\varphi_U"'] & \mathscr{G}(U)
\end{tikzcd}$$

Note that $\operatorname{res}^{\mathscr{G}}_{U',U} \circ \varphi_{U'} \circ \operatorname{Ker} \varphi_{U'} = 0$, by the universal property of $\operatorname{Ker} \varphi_U$, there exist unique group homomorphism from $\operatorname{Ker}_{\text{pre}} \varphi(U')$ to $\operatorname{Ker}_{\text{pre}} \varphi(U)$, denote $\operatorname{res}^{\operatorname{Ker}_{\text{pre}}}_{U',U}$. By the universal property of $\operatorname{Ker} \varphi_U$, $\operatorname{res}^{\operatorname{Ker}_{\text{pre}}}_{U,U} = \operatorname{id}_{\operatorname{Ker}_{\text{pre}} \varphi(U)}$. Now, consider the following diagram.

$$\begin{tikzcd}
0 \arrow[r] & \operatorname{Ker}_{\text{pre}} \varphi(U'') \arrow[r] \arrow[d, "\operatorname{res}^{\operatorname{Ker}_{\text{pre}}}_{U'',U'}"'] & \mathscr{F}(U'') \arrow[r, "\varphi_{U''}"] \arrow[d, "\operatorname{res}^{\mathscr{F}}_{U'',U'}"'] & \mathscr{G}(U'') \arrow[d, "\operatorname{res}^{\mathscr{G}}_{U'',U'}"] \\
0 \arrow[r] & \operatorname{Ker}_{\text{pre}} \varphi(U') \arrow[r] \arrow[d, "\operatorname{res}^{\operatorname{Ker}_{\text{pre}}}_{U',U}"'] & \mathscr{F}(U') \arrow[r, "\varphi_{U'}"] \arrow[d, "\operatorname{res}^{\mathscr{F}}_{U',U}"'] & \mathscr{G}(U') \arrow[d, "\operatorname{res}^{\mathscr{G}}_{U',U}"] \\
0 \arrow[r] & \operatorname{Ker}_{\text{pre}} \varphi(U) \arrow[r] & \mathscr{F}(U) \arrow[r, "\varphi_U"] & \mathscr{G}(U)
\end{tikzcd}$$

By the universal property of $\operatorname{Ker} \varphi_U$, $\operatorname{res}^{\operatorname{Ker}_{\text{pre}}}_{U',U} \circ \operatorname{res}^{\operatorname{Ker}_{\text{pre}}}_{U'',U'} = \operatorname{res}^{\operatorname{Ker}_{\text{pre}}}_{U'',U}$. Hence, $\operatorname{Ker}_{\text{pre}} \varphi$ is a presheaf. $\square$

Definition 3.3.7

If $\varphi : \mathscr{F} \to \mathscr{G}$ is a morphism of presheaf, define the ***presheaf cokernel*** $\operatorname{Coker}_{\text{pre}} \varphi$ *by*

$$(\operatorname{Coker}_{\text{pre}})(U) := \operatorname{Coker} \varphi_U,$$

for all open set $U \subseteq X$.

Proposition 3.3.3

$\operatorname{Coker}_{\text{pre}} \varphi$ *is a presheaf.*

Proof It is a presheaf by essentially the same (dual) argument. □

Exercise 3.13 (The cokernel deserves its name.) Show that the presheaf cokernel satisfies the universal property of cokernels in the category of presheaves.

Proof Let $\varphi : \mathscr{F} \to \mathscr{G}$ be a morphism of presheaf, let $\mathscr{K}$ be another sheaf with $\psi \circ \varphi = 0$, it suffices to show that there exists a unique presheaf morphism from $\operatorname{Coker}_{\text{pre}} \varphi$ to $\mathscr{K}$.

$$\begin{tikzcd}
\mathscr{F} \arrow[r, "\varphi"] \arrow[rrd, "0"'] & \mathscr{G} \arrow[r] \arrow[rd, "\psi"] & \operatorname{Coker}_{\text{pre}} \varphi \arrow[r] \arrow[d, dashed] & 0 \\
& & \mathscr{K} &
\end{tikzcd}$$

Let $U \subseteq X$ be any open set, note that each presheaf is presheaf of abelian group, $\operatorname{Coker}_{\text{pre}} \varphi(U) = \mathscr{G}(U)/\operatorname{Im} \varphi_U$. For all section $s_U \in \mathscr{G}(U)$, say $\overline{s_U}$ be the image of s_U in $\operatorname{Coker}_{\text{pre}} \varphi(U)$. Define

$$\theta_U : \operatorname{Coker}_{\text{pre}} \varphi(U) \to \mathscr{K}$$

by setting $\theta_U(\overline{s_U}) = \psi_U(s_U)$. We need to check that θ_U is well-defined for all $U \subseteq X$. Let $\overline{s_U} = \overline{t_U}$, then $s_U - t_U \in \operatorname{Im} \mathscr{F}(U)$. Since $\psi \circ \varphi = 0$, $\psi_U(s_U - t_U) = 0$, and therefore $\psi_U(s_U) = \psi_U(t_U)$, that is, $\theta_U(\overline{s_U}) = \theta_U(\overline{t_U})$. Hence, θ_U is well-defined for all open set $U \subseteq X$. By the definition of θ_U, uniqueness is clearly.

Finally, we need to check θ is a presheaf morphism. Let $U' \supseteq U$, consider the following diagram.

$$\begin{tikzcd}
\operatorname{Coker}_{\text{pre}} \varphi(U') \arrow[r, "\theta_{U'}"] \arrow[d, "\operatorname{res}^{\operatorname{Coker}_{\text{pre}}}_{U',U}"'] & \mathscr{K}(U') \arrow[d, "\operatorname{res}^{\mathscr{K}}_{U',U}"] \\
\operatorname{Coker}_{\text{pre}} \varphi(U) \arrow[r, "\theta_U"] & \mathscr{K}(U)
\end{tikzcd}$$

Let $\overline{s_{U'}} \in \operatorname{Coker}_{\text{pre}} \varphi(U')$, then

$$\theta_U(\overline{s_{U'}}|_U) = \theta_U(\overline{s_U}) = \psi_U(s_U) = \psi_{U'}(s_{U'})|_U = \theta_{U'}(\overline{s_{U'}})|_U$$

for all open set $U \subseteq X$. Hence, θ is a presheaf morphism.

By the arbitrary of $\mathscr{K}$, presheaf cokernel is indeed a cokernels in the category of presheaves. □

Remark 3.21 Similarly, $\operatorname{Ker}_{\text{pre}} \varphi \to \mathscr{F}$ satisfies the universal property for kernels in the category of presheaves.

It is not too tedious to verify that presheaves of abelian groups form an abelian category, we state the following result without proof here:

Theorem 3.3.1

$\mathbf{Ab}_X$ *is an abelian category.*

Remark 3.22 The key idea is that all abelian-categorical notions may be defined and verified "open set by open set". We needn't worry about restriction maps — they "come along for the ride". Hence, we can define terms such as **subpresheaf**, **image presheaf** (or **presheaf image**) and **quotient presheaf** (or **presheaf quotient**), and they behave as you would expect. Construct kernels, quotients, cokernels, and images open set by open set. Homological algebra (exact sequences and so forth) works, and also “works open set by open set”.

✍ **Exercise 3.14** Show that for a topological space X with open set U, $\mathscr{F} \mapsto \mathscr{F}(U)$ gives a functor from presheaves of abelian groups on X, $\mathbf{Ab}_X^{\text{pre}}$, to abelian groups, $\mathbf{Ab}$. Then show that this functor is exact.

Proof Let $\Theta_U : \mathbf{Ab}_X^{\text{pre}} \to \mathbf{Ab}$. Let $\varphi : \mathscr{F} \to \mathscr{G} \in \mathbf{Ab}_X^{\text{pre}}$, define $\Theta_U(\varphi) : \Theta_U(\mathscr{F}) \to \Theta_U(\mathscr{G})$ by setting $\Theta_U(\varphi) = \varphi_U$. Note that φ is a presheaf morphism, and therefore φ_U is a group homomorphism. Hence, $\Theta_U(\varphi) \in \operatorname{Hom}_{\mathbf{Ab}}(\mathscr{F}(U), \mathscr{G}(U))$. For $\mathscr{F} \in \mathbf{Ab}_X^{\text{pre}}$, $\Theta_U(\mathrm{id}_{\mathscr{F}}) = \mathrm{id}_{\mathscr{F}(U)}$. Let $\varphi : \mathscr{F} \to \mathscr{G}$ and $\psi : \mathscr{G} \to \mathscr{K}$, then $\Theta_U(\psi \circ \varphi) = (\psi \circ \varphi)_U = \psi_U \circ \varphi_U = \Theta_U(\psi) \circ \Theta_U(\varphi)$. Hence, Θ_U is a functor.

Since $\operatorname{Ker}_{\text{pre}}$, $\operatorname{Coker}_{\text{pre}}$ defined “open set by open set”, hence, Θ_U is exact. □

Proposition 3.3.4

A sequence of presheaves

$$0 \longrightarrow \mathscr{F}_1 \xrightarrow{f_1} \mathscr{F}_2 \xrightarrow{f_2} \cdots \xrightarrow{f_{n-1}} \mathscr{F}_n \longrightarrow 0$$

is exact if and only if

$$0 \longrightarrow \mathscr{F}_1(U) \xrightarrow{(f_1)_U} \mathscr{F}_2(U) \xrightarrow{(f_2)_U} \cdots \xrightarrow{(f_{n-1})_U} \mathscr{F}_n(U) \longrightarrow 0$$

is exact for all U.

Proof This is an obvious observation. □

Remark 3.23 The above discussion essentially carries over without change to presheaves with values in any abelian category.

However, we are interested in more geometric objects, sheaves, where things canbe understood in terms of their local behavior, thanks to the identity and gluing axioms. We will soon see that sheaves of abelian groups also form an abelian category, but a complication will arise that will force the notion of **sheafification** on us. Sheafification will be the answer to many of our prayers. We just haven’t yet realized what we should be praying for.

Proposition 3.3.5

$\mathbf{Ab}_X$ *form an additive category.*

Proof As description of presheaves. □

Kernels work just as with presheaves:

Proposition 3.3.6

Suppose $\varphi : \mathscr{F} \to \mathscr{G}$ is a morphism of sheaves. Then the presheaf kernel $\operatorname{Ker}_{\text{pre}} \varphi$ is in fact a sheaf, and it satisfies the universal property of kernels.

Proof $\operatorname{Ker}_{\text{pre}} \varphi$ is a presheaf, by Proposition 3.3.2. It suffices to check the identity axiom and gluability axiom of $\operatorname{Ker}_{\text{pre}} \varphi$. Let $U \subseteq X$ be open set, and $\{U_i\}_{i \in I}$ is an open cover of U. Let sections $s, t \in \operatorname{Ker}_{\text{pre}} \varphi(U)$ with $s|_{U_i} = t|_{U_i}$ for all i. Since each $\operatorname{Ker}_{\text{pre}} \varphi(U_i)$ is a subgroup of $\mathscr{F}(U_i)$, $s|_{U_i}, t|_{U_i} \in \mathscr{F}(U_i)$. Since $\mathscr{F}$ is a sheaf, we have $s = t \in \mathscr{F}(U)$, and therefore $s = t \in \operatorname{Ker}_{\text{pre}} \varphi(U)$, which implies the identity axiom. Let $s_i \in \operatorname{Ker}_{\text{pre}} \varphi(U_i)$ with $s_i|_{U_i \cap U_j} = s_j|_{U_i \cap U_j}$ for all i, j, since $\operatorname{Ker}_{\text{pre}} \varphi(U_i)$ is a subgroup of $\mathscr{F}(U_i)$ and $\mathscr{F}$ is a sheaf, by the gluability axiom of $\mathscr{F}$, there is a global section $s \in \mathscr{F}(U)$, such that $s|_{U_i} = s_i$. Now we need to check $s \in \operatorname{Ker}_{\text{pre}} \varphi(U)$. Let $x \in U$, then $x \in U_i$ for some i, note that $s(x) = s|_{U_i}(x) = s_i(x) = 0$ for all i, $s = 0$, that is, $s \in \operatorname{Ker}_{\text{pre}} \varphi(U)$, which implies the gluability axiom. Hence, $\operatorname{Ker}_{\text{pre}} \varphi$ is a sheaf.

The universal property of sheaf $\operatorname{Ker}_{\text{pre}} \varphi$ from the universal property of presheaf $\operatorname{Ker}_{\text{pre}} \varphi$. $\square$

Definition 3.3.8 (Sheaf Kernel)

In Definition of presheaf kernel, if $\varphi : \mathscr{F} \to \mathscr{G}$ is a morphism of sheaves, we define the ***sheaf of kernel*** $\operatorname{Ker} \varphi$ *by*

$$\operatorname{Ker} \varphi := \operatorname{Ker}_{\text{pre}} \varphi.$$

The problem arises with the cokernel:

Exercise 3.15 Let X be $\mathbb{C}$ with the classical topology, let $\mathscr{O}_X$ be the sheaf of holomorphic funtions, and $\mathscr{F}$ be the presheaf of functions admitting a holomorphic logarithm. Describe an exact sequence of presheaves on X:

$$0 \longrightarrow \underline{\mathbb{Z}} \longrightarrow \mathscr{O}_X \longrightarrow \mathscr{F} \longrightarrow 0$$

where $\underline{\mathbb{Z}} \to \mathscr{O}_X$ is the natural inclusion and $\mathscr{O}_X \to \mathscr{F}$ is given by $f \mapsto e^{2\pi i f}$. Show that $\mathscr{F}$ is not a sheaf.

Proof Let $U \subseteq X$ be any open set, then

$$\mathscr{F}(U) = \{g : U \to \mathbb{C} \setminus \{0\} : \exists \text{ holomorphic function } h : U \to \mathbb{C} \text{ such that } g = e^{2\pi i h}\}.$$

We need to check sequence of presheaf:

$$0 \longrightarrow \underline{\mathbb{Z}} \xrightarrow{i} \mathscr{O}_X \xrightarrow{p} \mathscr{F} \longrightarrow 0$$

is exact.

(1) $\underline{\mathbb{Z}} \to \mathscr{O}_X$ is a monomorphism.
Note that for locally constant function over U is also a holomorphic function over U, hence, for all $U \subseteq X$, we have $\underline{\mathbb{Z}}(U) \hookrightarrow \mathscr{O}_X(U)$. Let $f_i : \mathscr{K} \to \underline{\mathbb{Z}}$ be sheaf morphisms with $i \circ f_1 = i \circ f_2$. Let V be an open subset of U, then, locally, $(i \circ f_j)_V = i \circ (f_j)_V = (f_j)_V$, hence, $(f_1)_V = (f_2)_V$ for all open subset $V \subseteq U$, and therefore $f_1 = f_2$. This implies that $\underline{\mathbb{Z}} \to \mathscr{O}_X$ is a monomorphism.

(2) $\operatorname{Ker}(\mathscr{O}_X \to \mathscr{F}) = \operatorname{Im}(\underline{\mathbb{Z}} \to \mathscr{O}_X)$.
It suffices to check that for any open subset $U \subseteq X$, we have $\operatorname{Ker}(\mathscr{O}_X(U) \to \mathscr{F}(U)) = \operatorname{Im}(\underline{\mathbb{Z}}(U) \to \mathscr{O}_X(U))$, i.e., $\operatorname{Im} i_U = \operatorname{Ker} p_U$. Unit element in $\mathscr{F}(U)$ is 1, hence, $1 = e^{2\pi i f}$ if and only if f is locally constant, that is, $f \in \underline{\mathbb{Z}}$. Thus, $\operatorname{Ker}(\mathscr{O}_X(U) \to \mathscr{F}(U)) = \operatorname{Im}(\underline{\mathbb{Z}}(U) \to \mathscr{O}_X(U))$ for all $U \subseteq X$, and therefore $\operatorname{Ker}(\mathscr{O}_X \to \mathscr{F}) = \operatorname{Im}(\underline{\mathbb{Z}} \to \mathscr{O}_X)$.

(3) $\mathscr{O}_X \to \mathscr{F}$ is a epimorphism.
It suffices to show that $\operatorname{Coker}(\mathscr{O}_X \to \mathscr{F}) = 0$. For all open subset $U \subseteq X$, consider $\operatorname{Coker}(\mathscr{O}_X(U) \to \mathscr{F}(U))$. Let $g \in \mathscr{F}(U)$, then there exists some holomorphic function h such that $g = e^{2\pi i h} = p_U(h)$, this implies that p_U is a surjective, and therefore $\operatorname{Coker}(\mathscr{O}_X(U) \to \mathscr{F}(U)) = 0$ for all open subset $U \subseteq X$. Hence, $\operatorname{Coker}(\mathscr{O}_X \to \mathscr{F}) = 0$.

Hence, the sequence of presheaf

$$0 \longrightarrow \underline{\mathbb{Z}} \xrightarrow{i} \mathscr{O}_X \xrightarrow{p} \mathscr{F} \longrightarrow 0$$

is exact.

Now, we give a counterexample to show that $\mathscr{F}$ is not a sheaf. Let $U = \mathbb{C} \setminus \{0\}$, and consider the open cover $U_1 = \mathbb{C} \setminus (-\infty, 0]$ and $U_2 = \mathbb{C} \setminus [0, +\infty)$. Let $g(z) = z \in \mathscr{O}_X(U)$. Note that $g|_{U_1}(z) = e^{2\pi i \cdot \frac{\log z}{2\pi i}}$ and $g|_{U_2}(z) = e^{2\pi i \cdot (\frac{\log z}{2\pi i} + 1)}$, then $z \in \mathscr{F}(U_1)$ and $z \in \mathscr{F}(U_2)$. On $U_1 \cap U_2$, $(z|_{U_1})|_{U_1 \cap U_2} = (z|_{U_2})|_{U_1 \cap U_2}$, but $z \notin \mathscr{F}(U)$. This implies that $\mathscr{F}$ fails the gluing axiom. $\square$

We will have to put out hopes for understanding cokernels of sheaves on hold for a while. We will first learn to understand sheaves using stalks.

3.4 Properties determined at the level of stalks, and sheafification

3.4.1 Properties determined by stalks

We now come to the second way of understanding sheaves mentioned at the start of the chapter. In this section, we will see that lots of facts about sheaves can be checked "at the level of stalks". We call any property determined at the level of stalks **stalk-local**. This isn't true for presheaves, and reflects the local nature of sheaves. We will see that sections and morphisms are determined "by their stalks", and the property of a morphism being an isomorphism may be checked at stalks.

Proposition 3.4.1 (Sections are determined by germs)

A section of a sheaf of sets is determined by its germs, i.e., the natural map

$$\mathscr{F}(U) \longrightarrow \prod_{p\in U} \mathscr{F}_p$$

is injective.

Proof Consider the following diagram,

$$\begin{tikzcd}
\mathscr{F}(U) \arrow[dr, dashed, "\theta"] \arrow[drr, "\iota_i"] \arrow[ddr, "\iota_j"'] & & \\
& \prod_{p\in U}\mathscr{F}_p \arrow[r, "\mathrm{pr}_i"'] \arrow[d, "\mathrm{pr}_j"] & \mathscr{F}_{p_i} \\
& \mathscr{F}_{p_j}, &
\end{tikzcd}$$

where ι_i, ι_j given by the definition of $\mathscr{F}_{p_i}, \mathscr{F}_{p_j}$. By the universal property of product there exists unique morphism $\theta : \mathscr{F}(U) \to \prod_{p\in U} \mathscr{F}_p$ defined by $s \mapsto (\iota_i(s))$. Now, we need to check θ is injective. Let $(s_i) = (t_i) \in \prod_{p\in U} \mathscr{F}_p$, then for all i exist open subset $(p \in)W_i \subseteq U$ such that $s_i|_{W_i} = t_i|_{W_i}$. Note that $\{W_i\}$ is an open cover of U, hence, by the identity axiom of $\mathscr{F}$, $s = t$, and therefore θ is injective □

Remark 3.24 This proof will also apply to sheaves of groups, rings, etc. — to categories of "sets with additional structure". The same is true of many exercise in this section.

Proposition 3.4.1 suggests a question: which elements of the right side of $\mathscr{F}(U) \to \prod_{p\in U} \mathscr{F}_p$ are in the image of the left side?

Definition 3.4.1 (Compatible germs)

*We say that an element $(s_p)_{p\in U}$ of the right side $\prod_{p\in U} \mathscr{F}_p$ of $\mathscr{F}(U) \to \prod_{p\in U} \mathscr{F}_p$ consists of **compatible germs** if for all $p \in U$, there is some representative*

$$(\widetilde{s}_p \in \mathscr{F}(U_p), U_p \text{ open in } U)$$

for s_p (where $p \in U_p \subseteq U$) such that the germ of $\widetilde{s}_p$ at all $q \in U_p$ is s_q. Equivalently, there is an open cover $\{U_i\}$ of U, and section $f_i \in \mathscr{F}(U_i)$, such that if $p \in U_i$, then s_p is the germ of f_i at p.

Remark 3.25 Clearly any section s of $\mathscr{F}$ over U gives a choice of compatible germs for U.

Proof Let $U \subseteq X$ is an open set, let section $s \in \mathscr{F}(U)$, then $(s_p)_{p\in U}$ consists of compatible germs. U_p open in U, and let $\widetilde{s}_p = s|_{U_p}$, then the germs of $\widetilde{s}_p$ at $q \in U_q$ is s_q, which implies that $(s_p)_{p\in U}$ consists of compatible germs. □

Proposition 3.4.2

Any choice of compatible germs for a sheaf of sets $\mathscr{F}$ over U is the image of a section of $\mathscr{F}$

Proof Let $(s_p)_{p\in U} \in \prod_{p\in U} \mathscr{F}_p$ consists of compatible germs, then for all s_p there exists $\widetilde{s}_p \in \mathscr{F}(U_p)$ such that the germ of $\widetilde{s}_p$ at all $q \in U_p$ is s_q, for these q, there exists $\widetilde{s}_q \in \mathscr{F}(U_q)$ such that the germ of $\widetilde{s}_q$ is the germ of $\widetilde{s}_p$, hence, there exists an open subset $(p \in)W_p \subseteq U_p \cap U_q$ such that $\widetilde{s}_p|_{W_p} = \widetilde{s}_q|_{W_p}$ for all $q \in U_p$. Clearly, $\{W_p\}_{p\in U}$ is an open cover of U, and for each $\widetilde{s}_p \in \mathscr{F}(W_p)$ we have $\widetilde{s}_p|_{W_p\cap W_q} = \widetilde{s}_q|_{W_p\cap W_q}$, note that $\mathscr{F}$ is a sheaf, by gluability axiom of $\mathscr{F}$, there exists unique $s \in \mathscr{F}(U)$ such that $s|_{W_p} = \widetilde{s}_p$. Hence, $(s_p)_{p\in U}$ is the image of section $s \in \mathscr{F}(U)$. □

We have thus completely described the image of $\mathscr{F}(U) \to \prod_{p\in U} \mathscr{F}_p$, in a way that will prove useful.

Remark 3.26 This perspective motivates the agricultural terminology "sheaf": a sheaf is (the data of) a bunch of stalks, bundled together appropriately.

Now we throw morphisms into the mix. Recall Proposition 3.3.1: morphisms of (pre)sheaf induce morphisms of stalks.

Proposition 3.4.3 (Morphisms are determined by stalks)

If φ_1 and φ_2 are morphisms from a presheaf of sets $\mathscr{F}$ to a sheaf of sets $\mathscr{G}$ that induce the same maps on each stalk, then $\varphi_1 = \varphi_2$.

Proof Let $U \subseteq X$ be any open subset. Consider the following diagram.

$$\begin{tikzcd}
\mathscr{F}(U) \arrow[r] \arrow[d] & \mathscr{G}(U) \arrow[d, hook] \\
\prod_{p\in U} \mathscr{F}_p \arrow[r] & \prod_{p\in U} \mathscr{G}_p
\end{tikzcd}$$

Say $\widetilde{\varphi} : \prod_{p\in U} \mathscr{F}_p \to \prod_{p\in U} \mathscr{G}_p$ induced by φ_1 and φ_2, by condition,

$$\widetilde{\varphi}((s_p)_{p\in U}) = (\varphi_{1,U}(s)_p)_{p\in U} = (\varphi_{2,U}(s)_p)_{p\in U}. \tag{3.2}$$

It suffices to show that $\varphi_{1,U}(s) = \varphi_{2,U}(s)$ for all $s \in \mathscr{F}(U)$, for all open subset $U \subseteq X$.

Let $p \in U$ and section $s \in \mathscr{F}(U)$, by (3.2), $\varphi_{1,U}(s)_p = \varphi_{2,U}(s)_p$. Hence, there exists an open subset $(p \in)V_p \subseteq U$ such that $\varphi_{1,U}(s)|_{V_p} = \varphi_{2,U}(s)|_{V_p}$. Clearly, $\{V_p\}_{p\in U}$ is an open cover of U, by the identity axiom of sheaf $\mathscr{G}$, $\varphi_{1,U}(s) = \varphi_{2,U}(s)$, then we done. □

Proposition 3.4.4 (Isomorphisms are determined by stalks)

A morphism of sheaves of sets is an isomorphism if and only if it induces an isomorphisms of all stalks.

Proof "$\Rightarrow$": Let $V \subseteq U \subseteq X$ be open subsets of X. Consider the following diagram.

$$\begin{tikzcd}
\mathscr{F}_p \arrow[rrr, dashed, shift left] & & & \mathscr{G}_p \arrow[lll, dashed, shift left] \\
& \mathscr{F}(U) \arrow[ul] \arrow[r, hook, two heads] \arrow[d] & \mathscr{G}(U) \arrow[ur] \arrow[d] & \\
& \mathscr{F}(V) \arrow[uul, bend left] \arrow[r, hook, two heads] & \mathscr{G}(V) \arrow[uur, bend right] &
\end{tikzcd}$$

By the universal property of $\mathscr{F}_p$ and $\mathscr{G}_p$, there exists unique morphisms $\mathscr{F}_p \to \mathscr{G}_p$ and $\mathscr{G}_p \to \mathscr{F}_p$, and therefore $\mathscr{F}_p \cong \mathscr{G}_p$.

"$\Leftarrow$": Say $\varphi_p : \mathscr{F}_p \to \mathscr{G}_p$ induced by $\varphi : \mathscr{F} \to \mathscr{G}$ is an isomorphism for all $p \in X$. To show that φ is an isomorphism, it will be sufficient to show that $\varphi_U : \mathscr{F}(U) \to \mathscr{G}(U)$ is an isomorphism for all U. First, we show φ_U is injective. Let $s \in \mathscr{F}(U)$, and suppose $\varphi_U(s) = 0$ in $\mathscr{G}(U)$. Then for every point $P \in U$, the image $\varphi_U(s)_P$ of $\varphi_U(s)$ in the stalk $\mathscr{G}_p$ is 0. Since φ_p is injective for each p, we have $s_p = 0$ in $\mathscr{F}_p$ for each $P \in U$. Hence, there exists an open subset subset $(p \in)W_p \subseteq W$ such that $s|_{W_p} = 0$. Note that $\{W_p\}$ is an open cover of U, by the gluability axiom of sheaf $\mathscr{F}$, $s = 0$, which implies that φ_U is an injective.

Next, we show that φ_U is surjective. Suppose we have a section $t \in \mathscr{G}(U)$. For each $p \in U$, let $t_p \in \mathscr{G}_p$ be its germ at P. Since φ_p is surjective, we can find $s_p \in \mathscr{F}_p$ such that $\varphi_p(s_p) = t_p$. Let s_p be represented by a section $\widetilde{s}_p$ on a neighborhood V_p of P. Then $\varphi_{V_p}(\widetilde{s}_p)$ and $t|_{V_p}$ are two elements of $\mathscr{G}(V_p)$, whose germs at P are the same. Hence, replacing V_p by a smaller neighborhood of P if necessary, we may assume that $\varphi(\widetilde{s}_P) = t|_{V_p}$ in $\mathscr{G}(V_p)$. Now U is covered by the open sets V_p, and on each V_p we have a section $\widetilde{s}_p \in \mathscr{F}(V_p)$. If p, q are two points, then $\widetilde{s}_p|_{V_p \cap V_q}$ and $\widetilde{s}_q|_{V_p \cap V_q}$ are two sections of $\mathscr{F}(V_p \cap V_q)$, which are both sent by φ to $t|_{V_p \cap V_q}$. Hence, by the injective of φ proved above, $\widetilde{s}_p|_{V_p \cap V_q} = \widetilde{s}_q|_{V_p \cap V_q}$, by the gluability axiom of sheaf $\mathscr{F}$, there is a section $s \in \mathscr{F}(U)$ such that $s|_{V_p} = \widetilde{s}_p$. Finally, we have to check that $\varphi(s) = t$. Indeed, $\varphi_U(s)$, t are two sections of $\mathscr{G}(U)$, and for each $p \in U$, $\varphi_U(s)|_{V_p} = t|_{V_p}$, hence, by the identity axiom of sheaf $\mathscr{G}$, we conclude that $\varphi(s) = t$. □

Exercise 3.16

(a) Show that Proposition 3.4.1 is false for general presheaves.
(b) Show that Proposition 3.4.3 is false for general presheaves.
(c) Show that Proposition 3.4.4 is false for general presheaves.

Proof Let $X = \{p, q\}$ be a two points set with the discrete topology $\mathscr{T}_X = \{\{p\}, \{q\}, X, \varnothing\}$. Let $\mathscr{F}(X) = \{s, t\}$, $\mathscr{F}(\{p\}) = \{a\}$, and $\mathscr{F}(\{q\}) = \{b\}$, with restriction $s|_{\{p\}} = a$, $s|_{\{q\}} = b$, $t|_{\{p\}} = a$, and $t|_{\{q\}} = b$. Then $\mathscr{F}$ form a presheaf.

(a) Note that $s \mapsto (a, b)$ and $t \mapsto (a, b)$, $\mathscr{F}(X) \to \mathscr{F}_p \times \mathscr{F}_q$ is not injective.
(b) Let $\mathscr{G}(X) = \{m\}$, $\mathscr{G}(\{p\}) = \{a\}$, and $\mathscr{G}(\{q\}) = \{b\}$, then $\mathscr{G}$ form a presheaf, and its stalks are $\mathscr{G}_p = \{a\}$ and $\mathscr{G}_p = \{b\}$. Define $\varphi_1 : \mathscr{G} \to \mathscr{F}$ by setting $\varphi_{1,X}(m) = s$, $\varphi_{1,\{p\}}(a) = a$, and $\varphi_{1,\{q\}}(b) = b$. Define $\varphi_2 : \mathscr{G} \to \mathscr{F}$ by setting $\varphi_{2,X}(m) = t$, $\varphi_{2,\{p\}}(a) = a$, and $\varphi_{2,\{q\}}(b) = b$. Then φ_1 and φ_2 induced the same maps on each stalks, but $\varphi_1 \neq \varphi_2$.
(c) Let $\mathscr{G}(X) = \{m\}$, $\mathscr{G}(\{p\}) = \{a\}$, and $\mathscr{G}(\{q\}) = \{b\}$, then $\mathscr{G}$ form a presheaf, and its stalks are $\mathscr{G}_p = \{a\}$ and $\mathscr{G}_p = \{b\}$. Define $\varphi : \mathscr{G} \to \mathscr{F}$ by setting $\varphi_X(m) = s$, $\varphi_{\{p\}}(a) = a$, and $\varphi_{\{q\}}(b) = b$, then $\varphi_p : \mathscr{G}_p \to \mathscr{F}_p$ and $\varphi_q : \mathscr{G}_q \to \mathscr{F}_p$ are isomorphism, but φ is not isomorphism, since φ_X is not surjective.

□

3.4.2 Sheafification

Every sheaf is a presheaf (and indeed by definition sheaves on X form a full subcategory of the category of presheaves on X). Just as groupification (Example 2.32) gives an abelian group that best approximates a presheaf, with an analogous universal property. (One possible example to keep in mind is the sheafification of the pre sheaf of holomorphic functions admitting a square root on $\mathbb{C}$ with the classical topology.)

Definition 3.4.2 (Sheafification)

If $\mathscr{F}$ is a presheaf on X, then a morphism of presheaf $\mathrm{sh} : \mathscr{F} \to \mathscr{F}^{\mathrm{sh}}$ on X is a ***sheafification of $\mathscr{F}$*** *if $\mathscr{F}^{\mathrm{sh}}$ is a sheaf, and for any sheaf $\mathscr{G}$, and any presheaf morphism $g : \mathscr{F} \to \mathscr{G}$, there exists a unique* *morphism of sheaves $f : \mathscr{F}^{\text{sh}} \to \mathscr{G}$ making the diagram*

$$\begin{tikzcd} \mathscr{F} \arrow[r, "\text{sh}"] \arrow[dr, "g"'] & \mathscr{F}^{\text{sh}} \arrow[d, dashed, "f"] \\ & \mathscr{G} \end{tikzcd}$$

commute.

Note ***Construction of $\mathscr{F}^{\text{sh}}$.*** *Any presheaf of sets (or groups, rings, etc.) has a sheafification.*

Suppose $\mathscr{F}$ is a presheaf. Define $\mathscr{F}^{\text{sh}}$ by defining $\mathscr{F}^{\text{sh}}(U)$ as the set of "compatible (families of) germs" of the presheaf $\mathscr{F}$ over U. Explicitly:

$$\mathscr{F}^{\text{sh}}(U) := \left\{ (f_p \in \mathscr{F}_p)_{p \in U} \;\middle|\; \begin{array}{l} \textit{for all } p \in U, \textit{ there exists an open neighborhood } V \textit{ of } p, \\ \textit{contained in } U, \textit{ and } s \in \mathscr{F}(V) \textit{ with } s_q = f_q \textit{ for all } q \in V \end{array} \right\}.$$

Here s_q means the germ of s at q — the image of s in the stalk $\mathscr{F}_q$.

Proposition 3.4.5

If $\mathscr{F}$ is a presheaf on X, then $\mathscr{F}^{\text{sh}}$ is a sheaf.

Proof Let $U \subseteq U' \subseteq X$ be any open subset of X. The restriction map given by $\text{res}_{U',U}((f_p)_{p \in U'}) = (f_p)_{p \in U}$, where $(f_p)_{p \in U}$ consists of compatible germs. (This restriction is well-defined since it suffices to restrict s at $V' \cap U$ where $(p \in)V' \subseteq U'$.) By some easy check, $\mathscr{F}^{\text{sh}}$ is a presheaf.

Let $\{U_i\}$ be an open cover of U, and $(f_p)_{p \in U} \in \mathscr{F}^{\text{sh}}(U)$ with $(f_p)_{p \in U}|_{U_i} = (0)_{p \in U}|_{U_i}$, where $(f_p)_{p \in U}$ consists of compatible germs. We shall to show that $f_p = 0$ for all $p \in U$. Since $(f_p)_{p \in U_i} = (0)$, then exist $p \in W_i^p \in U_i$ such that $f|_{W_i^p} = 0$. Note that $\{W_i^p\}_{i,p}$ is an open cover of U, let $p \in U$, then $p \in W_i^p$ with $f|_{W_i^p} = 0$, hence, $f_p = 0$. This implies that $\mathscr{F}^{\text{sh}}$ satisfies identity axiom.

Let $(f_p^i)_{p \in U_i} \in \mathscr{F}^{\text{sh}}(U_i)$ with $(f_p^i)_{p \in U_i}|_{U_i \cap U_j} = (f_p^j)_{p \in U_j}|_{U_i \cap U_j}$, we shall to show there exists $(f_p)_{p \in U} \in \mathscr{F}^{\text{sh}}(U)$ such that $(f_p)_{p \in U}|_{U_i} = (f_p^i)_{p \in U_i}$. Since $(f_p^i)_{p \in U_i \cap U_j}$ consists of compatible germs, for all $p \in U_i \cap U_j$, there exists an open neighborhood $(p \in)V_p^i \in U_i \cap U_j$, and $s_p^i \in \mathscr{F}(V_p^i)$ with $(s_p^i)_q = f_q^i$ for all $q \in V_p^i$, we may assume that $s_p^i|_{V_p^i} = f^i|_{V_p^i}$ for all $p \in U_i \cap U_j$. Similarly, we have $s_p^j|_{V_p^j} = f^j|_{V_p^j}$. Note that $\{V_p^i \cap V_p^j\}$ is also an open cover of U and $f^i|_{V_p^i \cap V_p^j} = f^j|_{V_p^i \cap V_p^j}$ for all $p \in U_i \cap U_j$. Hence, $f_p^i = f_p^j$ for all $p \in U_i \cap U_j$. Let $(f_p)_{p \in U} = (f_p^1)_{p \in U_1} \times (f_p^2)_{p \in U_2 \setminus U_1} \times (f_p^3)_{p \in U_3 \setminus (U_1 \cap U_2)} \times \cdots$, then $(f_p)_{p \in U}$ consists of compatible germs with $(f_p)_{p \in U}|_{U_i} = (f_p^i)_{p \in U_i}$. This implies that gluability axiom holds.

Hence, $\mathscr{F}^{\text{sh}}$ is a sheaf. □

Proposition 3.4.6

(1) Sheafification is unique up to unique isomorphism, assuming it exists.

(2) If $\mathscr{F}$ is a sheaf, then the sheafification is $\text{id} : \mathscr{F} \to \mathscr{F}$.

Proof

(1) Let $\mathscr{F}$ is a presheaf on X. If there exists another morphism of presheaf $\theta : \mathscr{F} \to \mathscr{K}$ on X is a

sheafification of $\mathscr{F}$, the following diagram

$$\begin{tikzcd} \mathscr{F} \arrow[r] \arrow[dr] & \mathscr{K} \arrow[loop above, dashed, "\mathrm{id}_{\mathscr{K}}"] \arrow[d, dashed, shift right, "\exists!"'] \\ & \mathscr{F}^{\mathrm{sh}} \arrow[u, dashed, shift right, "\exists!"'] \arrow[loop below, dashed, "\mathrm{id}_{\mathscr{F}^{\mathrm{sh}}}"] \end{tikzcd}$$

commutes. Hence, $\mathscr{K} \cong \mathscr{F}^{\mathrm{sh}}$.

(2) Since $\mathscr{F}$ is a sheaf, clearly, any section s of $\mathscr{F}$ over U gives a choice of compatible germs for U, by Proposition 3.4.2, we have $\mathscr{F} \cong \mathscr{F}^{\mathrm{sh}}$. By part (1), the sheafification is $\mathrm{id} : \mathscr{F} \to \mathscr{F}$.

□

Proposition 3.4.7 (Sheafification is a functor)

For any morphism of presheaves $\varphi : \mathscr{F} \to \mathscr{G}$, there is a natural induced morphism of sheaves $\varphi^{\mathrm{sh}} : \mathscr{F}^{\mathrm{sh}} \to \mathscr{G}^{\mathrm{sh}}$. Moreover, sheafification is a functor from presheaves on X to sheaves on X.

Proof Consider the following diagram.

$$\begin{tikzcd} \mathscr{F} \arrow[r, "\mathrm{sh}"] \arrow[d, "\varphi"'] \arrow[dr, "\mathrm{sh}\circ\varphi"] & \mathscr{F}^{\mathrm{sh}} \arrow[d, dashed, "\exists!"] \\ \mathscr{G} \arrow[r, "\mathrm{sh}"'] & \mathscr{G}^{\mathrm{sh}} \end{tikzcd}$$

By the universal property of sheafification, there exists unique morphism from $\mathscr{F}^{\mathrm{sh}}$ to $\mathscr{G}^{\mathrm{sh}}$, denote it φ^{sh}.

Now, we shall to check $\square^{\mathrm{sh}}$ is, indeed, a functor from presheaves on X to sheaves on X. By the definition of $\square^{\mathrm{sh}}$, let $\varphi \in \mathrm{Mor}(\mathscr{F}, \mathscr{G})$, then $\varphi^{\mathrm{sh}} \in \mathrm{Mor}(\mathscr{F}^{\mathrm{sh}}, \mathscr{G}^{\mathrm{sh}})$. Let $\mathscr{F}$ be a presheaf, then $\square^{\mathrm{sh}}(\mathrm{id}_{\mathscr{F}}) = \mathrm{id}_{\mathscr{F}^{\mathrm{sh}}}$, that is, $\square^{\mathrm{sh}}$ preserves identity morphisms. Let $\varphi_1 \in \mathrm{Mor}(\mathscr{F}, \mathscr{G})$ and $\varphi_2 \in \mathrm{Mor}(\mathscr{G}, \mathscr{K})$, consider the following diagram.

$$\begin{tikzcd} \mathscr{F} \arrow[r] \arrow[d, "\varphi_1"'] \arrow[dr] & \mathscr{F}^{\mathrm{sh}} \arrow[d] \arrow[dd, dashed, bend left, "\exists!"] \\ \mathscr{G} \arrow[r] \arrow[d, "\varphi_2"'] \arrow[dr] & \mathscr{G}^{\mathrm{sh}} \arrow[d] \\ \mathscr{K} \arrow[r, "\mathrm{sh}"] & \mathscr{K}^{\mathrm{sh}} \end{tikzcd}$$

By the universal property of sheafification, we have $(\varphi_2 \circ \varphi_1)^{\mathrm{sh}} = \varphi_2^{\mathrm{sh}} \circ \varphi_1^{\mathrm{sh}}$. Hence, $\square^{\mathrm{sh}}$ is a functor. □

Proposition 3.4.8

$\mathrm{sh} : \mathscr{F} \to \mathscr{F}^{sh}$ *is a natural transformation.*

Proof Let $U \subseteq U' \subseteq X$ be two open subsets, consider following diagram,

$$\begin{tikzcd} \mathscr{F}(U') \arrow[r, "\mathrm{res}^{\mathscr{F}}_{U',U}"] \arrow[d, "\mathrm{sh}_{U'}"'] & \mathscr{F}(U) \arrow[d, "\mathrm{sh}_U"] \\ \mathscr{F}^{\mathrm{sh}}(U') \arrow[r, "\mathrm{res}^{\mathscr{F}^{\mathrm{sh}}}_{U',U}"] & \mathscr{F}^{\mathrm{sh}}(U), \end{tikzcd}$$

we shall to show this diagram is commute.

Define $\mathrm{sh}_U : \mathscr{F}(U) \to \mathscr{F}^{\mathrm{sh}}(U)$ by setting $s \mapsto (s_p)_{p\in U}$, where $s_p \in \mathscr{F}_p$. We need to check sh_U is well-defined. Let $(p \in)V \subseteq U$ be an open subset, consider $s|_V$, note that $(s|_V)_q = s_q$ for all $q \in V$. Hence, $(s_p)_{p\in U} \in \mathscr{F}^{\mathrm{sh}}(U)$. Let $s = t \in \mathscr{F}(U)$, then for all $p \in U$ exists $p \in V_s$ and $p \in V_t$ open subsets of U such that exists $\widetilde{s} \in \mathscr{F}(V_s)$ and $\widetilde{t} \in \mathscr{F}(V_t)$ with $\widetilde{s}_i = s_i$ for all $i \in V_s$ and $\widetilde{t}_j = t_j$ for all $j \in V_t$. Pick $\widetilde{s} = s|_{V_s}$ and $\widetilde{t} = t|_{V_t}$, then $\widetilde{s}|_{V_s\cap V_t} = \widetilde{t}|_{V_s\cap V_t}$, since $s = t \in \mathscr{F}(U)$. Hence, $s_i = t_i$ for all $i \in V_s \cap V_t$, and therefore $(s_p)_{p\in U} = (t_p)_{p\in U}$. This implies that sh_U is well-defined.

Let section $s \in \mathscr{F}(U')$, then

$$\mathrm{sh}_{U'}(s)|_U = (s_p)_{p\in U'}|_U = (s_p)_{p\in U}$$

and

$$\mathrm{sh}_U(s|_U) = ((s|_U)_p)_{p\in U} = (s_p)_{p\in U}.$$

Hence,

$$\mathrm{res}^{\mathscr{F}^{\mathrm{sh}}}_{U',U} \circ \mathrm{sh}_{U'} = \mathrm{sh}_U \circ \mathrm{res}^{\mathscr{F}}_{U',U}.$$

This implies that $sh : \mathscr{F} \to \mathscr{F}^{\mathrm{sh}}$ is a natural transformation. $\square$

Proposition 3.4.9

The natural transformation sh *satisfies the universal property of sheafification (Definition 3.4.2).*

Proof Let $\mathscr{G}$ be a sheaf with presheaf morphism $g : \mathscr{F} \to \mathscr{G}$. Consider the following diagram.

$$\begin{tikzcd} \mathscr{F} \arrow[r, "\mathrm{sh}"] \arrow[dr, "g"'] & \mathscr{F}^{\mathrm{sh}} \arrow[d, dashed] \\ & \mathscr{G} \end{tikzcd}$$

Now, we will define $f : \mathscr{F}^{\mathrm{sh}} \to \mathscr{G}$. Let $U \subseteq X$ be any open subset. Let $(s_p)_{p\in U} \in \mathscr{F}^{\mathrm{sh}}(U)$, then for all $p \in U$ exists an open neighborhood $p \in V_p \subseteq U$ such that exists a section $\widetilde{s}_p \in \mathscr{F}(V_p)$ with $(\widetilde{s}_p)_q = s_q$ for all $q \in V_p$. Then for all $p \in U$, we have $g(\widetilde{s}_p) \in \mathscr{G}(V_p)$ with $g_q((\widetilde{s}_p)_q) = (g(\widetilde{s}_p))_q = g_q(s_q) = (g(s))_q$ for all $q \in V_p$. This implies that $(g(s)_p)_{p\in U}$ consists of compatible germs. By Proposition 3.4.2, and note that $\mathscr{G}$ is a sheaf, $(g(s)_p)_{p\in U}$ determined a unique section t_U in $\mathscr{G}(U)$. Define $f_U : \mathscr{F}^{\mathrm{sh}}(U) \to \mathscr{G}(U)$ by setting $f_U((s_p)_{p\in U}) = t_U$. It is easy to see that f_U is well-defined and f is a sheaf morphism. And by the construction of f_U, above diagram is commutative. Hence, the natural transformation sh satisfies the universal property of sheafification. $\square$

Proposition 3.4.10

The sheafification functor is left-adjoint to the forgetful functor from sheaves on X to presheaves on X, i.e.,

$$\mathrm{Mor}(\mathscr{F}^{\mathrm{sh}}, \mathscr{G}) \cong \mathrm{Mor}(\mathscr{F}, F(\mathscr{G})),$$

where $\mathscr{F}$ is presheaf, $\mathscr{G}$ is sheaf, and $F(\square)$ is forgetful functor.

Proof Define $\tau_{\mathscr{F},\mathscr{G}} : \mathrm{Mor}(\mathscr{F}, F(\mathscr{G})) \to \mathrm{Mor}(\mathscr{F}^{\mathrm{sh}}, \mathscr{G})$ by setting $\tau_{\mathscr{F},\mathscr{G}}(\varphi) = \widetilde{\varphi}$, where $\widetilde{\varphi} \circ \mathrm{sh} = \varphi$. $\tau_{\mathscr{F},\mathscr{G}}$ is well-defined, since the universal property of sheafification gives unique $\widetilde{\varphi}$.

Consider the following diagram,

$$
\begin{tikzcd}
\mathrm{Mor}(\mathscr{F}', F(\mathscr{G})) \arrow[r, "g^*"] \arrow[d, "\tau_{\mathscr{F}',\mathscr{G}}"'] & \mathrm{Mor}(\mathscr{F}, F(\mathscr{G})) \arrow[r, "(Ff)_*"] \arrow[d, "\tau_{\mathscr{F},\mathscr{G}}"] & \mathrm{Mor}(\mathscr{F}, F(\mathscr{G}')) \arrow[d, "\tau_{\mathscr{F},\mathscr{G}'}"] \\
\mathrm{Mor}(\mathscr{F}'^{\mathrm{sh}}, \mathscr{G}) \arrow[r, "(g^{\mathrm{sh}})^*"] & \mathrm{Mor}(\mathscr{F}^{\mathrm{sh}}, \mathscr{G}) \arrow[r, "f_*"] & \mathrm{Mor}(\mathscr{F}^{\mathrm{sh}}, \mathscr{G}'),
\end{tikzcd}
\tag{3.3}
$$

where $f : \mathscr{G} \to \mathscr{G}'$ is sheaf morphism and $g : \mathscr{F} \to \mathscr{F}'$ is presheaf. We shall to check diagram (3.3) commutes.

Let $\varphi \in \mathrm{Mor}(\mathscr{F}', F(\mathscr{G}))$, then we have

$$(g^{\mathrm{sh}})^* \circ \tau_{\mathscr{F}',\mathscr{G}}(\varphi) = (g^{\mathrm{sh}})^*(\widetilde{\varphi}) = \widetilde{\varphi} \circ g^{\mathrm{sh}}$$

and

$$\tau_{\mathscr{F},\mathscr{G}} \circ g^*(\varphi) = \tau_{\mathscr{F},\mathscr{G}}(\varphi \circ g) = \widetilde{\varphi \circ g}.$$

Consider the following diagram.

$$
\begin{tikzcd}
\mathscr{F} \arrow[r, "g"] \arrow[d] & \mathscr{F}' \arrow[r, "\varphi"] \arrow[d, dashed] & \mathscr{G} \\
\mathscr{F}^{\mathrm{sh}} \arrow[r, "g^{\mathrm{sh}}"'] \arrow[urr, dashed] & \mathscr{F}'^{\mathrm{sh}} \arrow[ur, dashed] &
\end{tikzcd}
$$

Note that $\widetilde{\varphi} \circ g^{\mathrm{sh}} : \mathscr{F}^{\mathrm{sh}} \to \mathscr{G}$ and $\widetilde{\varphi \circ g} : \mathscr{F}^{\mathrm{sh}} \to \mathscr{G}$, by the universal property of sheafification, we have $\widetilde{\varphi} \circ g^{\mathrm{sh}} = \widetilde{\varphi \circ g}$.

Let $\psi \in \mathrm{Mor}(\mathscr{F}, F(\mathscr{G}))$, then we have

$$f_* \circ \tau_{\mathscr{F},\mathscr{G}}(\psi) = f_* \circ \widetilde{\psi} = f \circ \widetilde{\psi}$$

and

$$\tau_{\mathscr{F},\mathscr{G}'} \circ (Ff)_*(\psi) = \tau_{\mathscr{F},\mathscr{G}'}(Ff \circ \psi) = \widetilde{Ff \circ \psi}.$$

Consider the following diagram.

$$
\begin{tikzcd}
\mathscr{F} \arrow[r, "\psi"] \arrow[d] & \mathscr{G} \arrow[r, "f"] & \mathscr{G}' \\
\mathscr{F}^{\mathrm{sh}} \arrow[ur, dashed] \arrow[urr, dashed] & &
\end{tikzcd}
$$

Note that $f \circ \widetilde{\psi} : \mathscr{F}^{\mathrm{sh}} \to \mathscr{G}'$ and $\widetilde{Ff \circ \psi} : \mathscr{F}^{\mathrm{sh}} \to \mathscr{G}'$, by the universal property of sheafification, we have $f \circ \widetilde{\psi} = \widetilde{Ff \circ \psi}$.

Hence, diagram (3.3) commutes, and therefore τ is a natural transformation. Now, we give the inverse of τ. Let $\tau^{-1}_{\mathscr{F},\mathscr{G}} : \mathrm{Mor}(\mathscr{F}^{\mathrm{sh}}, \mathscr{G}) \to \mathrm{Mor}(\mathscr{F}, F(\mathscr{G}))$ defined by $\tau^{-1}_{\mathscr{F},\mathscr{G}}(\varphi) = \widehat{\varphi}$, where $\widehat{\varphi}^{\mathrm{sh}} = F(\varphi)$. Hence, τ is a natural isomorphism, and therefore

$$\mathrm{Mor}(\mathscr{F}^{\mathrm{sh}}, \mathscr{G}) \cong \mathrm{Mor}(\mathscr{F}, F(\mathscr{G})).$$

□

Proposition 3.4.11

$\mathrm{sh} : \mathscr{F} \to \mathscr{F}^{\mathrm{sh}}$ *induces an isomorphism of stalks.*

Proof Let $x \subseteq X$ be any point. Now, we define $\mathscr{F}_x \to \mathscr{F}^{\mathrm{sh}}_x$ as follow: let $s_x \in$, then $s_x = \overline{(s, U)}$, where $x \in U$ and $s \in \mathscr{F}(U)$; apply sh_U, we have $\mathrm{sh}_U(s) = (s_p)_{p \in U} \in \mathscr{F}^{\mathrm{sh}}(U)$; take stalk at x, we have $(\mathrm{sh}_U(s))_x = ((s_p)_{p \in U})_x$. Define $\mathrm{sh}_x : \mathscr{F}_x \to \mathscr{F}^{\mathrm{sh}}_x$ by setting $s_x \mapsto ((s_p)_{p \in U})_x$. We need to check sh_x is well-defined. Let $s_x = t_x$, then there exists open neighborhood U of x contained in X, such that $s|_U = t|_U$, hence, $((s|_U)_p)_{p\in U} = ((t|_U)_p)_{p\in U}$, and therefore $(((s|_U)_p)_{p\in U})_x = (((t|_U)_p)_{p\in U})_x$, this implies that sh_x is well-defined.

Let $((s_p)_{p\in U})_x = ((t_p)_{p\in U})_x$, then there exists an open neighborhood V of x contained in X such that $(s_p)_{p\in V} = (t_p)_{p\in V}$. This implies that $s_p = t_p$ for all $p \in V$. In particular, let $p = x \in V$, we have $s_x = t_x$. Hence, sh_x is an injective.

Let $((s_p)_{p\in U})_x \in \mathscr{F}_x^{\mathrm{sh}}$, then there exists an open neighborhood W of x such that $(s_p)_{p\in W} \in \mathscr{F}^{\mathrm{sh}}(W)$. Since $(s_p)_{p\in W}$ consists of compatible germs, there exists $t \in \mathscr{F}(W)$ such that $t_p = s_p$ for all $p \in W$. Hence, $\mathrm{sh}_x(t_x) = ((t_p)_{p\in W})_x = ((s_p)_{p\in W})_x$, which implies that sh_x is a surjective.

Hence, $\mathrm{sh} : \mathscr{F} \to \mathscr{F}^{\mathrm{sh}}$ induces an isomorphism of stalks,

$$\mathscr{F}_x \xleftrightarrow{\sim} \mathscr{F}_x^{\mathrm{sh}}.$$

□

Example 3.9 **The sheafification of a constant presheaf is the corresponding constant sheaf.** If X is a topological space and S is a set, then $(\underline{S}_{\mathrm{pre}})^{\mathrm{sh}}$ may be naturally identified with $\underline{S}$.

Proof Let $f \in \underline{S}_{\mathrm{pre}}(U)$, where $U \subseteq X$ be an open subset of X. Then for all $p \in U$, we have $s_p = S$, and therefore $\mathrm{sh}_U(s) = (s_p)_{p\in U}$. Hence, $(\underline{S}_{\mathrm{pre}})^{\mathrm{sh}}(U) = \prod S = \underline{S}(U)$ for all open subset $U \subseteq X$. □

Remark 3.27 The "space of sections" (or espace étalé) construction (Definition 3.2.10) yields a different-sounding description of sheafification. The main ideal is identical: if $\mathscr{F}$ is a presheaf, let $\mathscr{F}$ is a presheaf, let F be the espace étalé of $\mathscr{F}$. In fact, if $\mathscr{F}$ is a presheaf, the sheaf of sections of $F \to X$ (Exercise 3.7) is the sheafification of $\mathscr{F}$. Constant sheaf may be interpreted as an example of this construction. The "espace étalé" construction of the sheafification is essentially the same as Construction

$$\mathscr{F}^{\mathrm{sh}}(U) := \left\{ (f_p \in \mathscr{F}_p)_{p\in U} \;\middle|\; \begin{array}{l} \text{for all } p \in U, \text{ there exists an open neighborhood } V \text{ of } p, \\ \text{contained in } U, \text{ and } s \in \mathscr{F}(V) \text{ with } s_q = f_q \text{ for all } q \in V \end{array} \right\}.$$

3.4.3 Subsheaves and quotient sheaves

We now discuss subsheaves and quotient sheaves from the perspective of stalks.

Proposition 3.4.12

Suppose $\varphi : \mathscr{F} \to \mathscr{G}$ is a morphism of sheaves of sets on a topological space X. The following are equivalent.

(a) φ is a monomorphism in the category of sheaves.

(b) φ is injective on the level of stalks: $\varphi_p : \mathscr{F}_p \to \mathscr{G}_p$ is injective for all $p \in X$.

(c) φ is injective on the level of open sets: $\varphi_U : \mathscr{F}(U) \to \mathscr{G}(U)$ is injective for all open $U \subseteq X$.

Proof $(a) \Rightarrow (c)$. Since φ is a monomorphism in the category of sheaves, then we have sheaf of kernel $\operatorname{Ker}\varphi = 0$. Hence, for all open subset $U \subseteq X$, we have $\operatorname{Ker}\varphi(U) = \operatorname{Ker}\varphi_U = 0$. This implies that φ_U is injective.

$(c) \Rightarrow (b)$. Let $s_p \in \mathscr{F}_p$ with $\varphi_p(s_p) = 0$. We may assume that $s_p = \overline{(s, U)}$, where $(x \in)U \subseteq X$. Then $\varphi_p(s_p) = (\varphi_U(s))_p = 0$, hence, there exists open neighborhood V of p contained in U such that $\varphi_U(s)|_V = 0$. Note that $\varphi_V(s|_V) = \varphi_U(s)|_V = 0$ and φ_V is injective. We have $s|_V = 0$, and therefore $s_p = 0$, which implies that φ_p is injective for all $p \in X$.

$(b) \Rightarrow (a)$. Let $\psi_1 : \mathscr{E} \to \mathscr{F}$ and $\psi_2 : \mathscr{E} \to \mathscr{F}$ be two morphisms with $\varphi \circ \psi_1 = \varphi \circ \psi_2$, we shall to show that $\psi_1=\psi_2$. Let $U\subseteq X$ be any open subset, consider the following diagram.

$$\begin{tikzcd}
\mathscr{E}(U) \arrow[r, shift left, "\psi_1"] \arrow[r, shift right, "\psi_2"'] \arrow[d] & \mathscr{F}(U) \arrow[r, "\varphi"] \arrow[d] & \mathscr{G}(U) \arrow[d] \\
\prod_{p\in U}\mathscr{E}_p \arrow[r, shift left, "((\psi_1)_p)_{p\in U}"] \arrow[r, shift right, "((\psi_2)_p)_{p\in U}"'] & \prod_{p\in U}\mathscr{F}_p \arrow[r, "(\varphi_p)_{p\in U}"] & \prod_{p\in U}\mathscr{G}_p
\end{tikzcd}$$

Say $\widetilde{\varphi_1}=((\psi_1)_p)_{p\in U}$, $\widetilde{\varphi_2}=((\psi_2)_p)_{p\in U}$, and $\widetilde{\varphi}=(\varphi_p)_{p\in U}$. Since $\varphi\circ\psi_1=\varphi_1\circ\psi_2$, by the universal property of colimit, induced map satisfies $\widetilde{\varphi\circ\psi_i}=\widetilde{\varphi}\circ\widetilde{\psi_i}$ for $i=1,2$. Since each φ_p is injective, induced map $\widetilde{\varphi}$ is injective, and therefore $\widetilde{\psi_1}=\widetilde{\psi_2}$. By Proposition 3.4.3, $\psi_1=\psi_2$, which implies that φ is a monomorphism. $\square$

Definition 3.4.3 (Subsheaf)

If $\varphi:\mathscr{F}\to\mathscr{G}$ be a morphism of sheaves of sets on a topological space X which satisfies the conditions in Proposition 3.4.12, we say that $\mathscr{F}$ is a ***subsheaf*** *of $\mathscr{G}$ (where the "inclusion" φ is sometimes left implicit).*

Remark 3.28 In fact, for any morphism of presheaves, if all maps of sections are injective, then all stalk maps are injective. And furthermore, if $\varphi:\mathscr{F}\to\mathscr{G}$ is a morphism from a separated presheaf to an arbitrary presheaf, then injectivity on the level of stalks implies that φ is a monomorphism in the category of presheaves.

Proposition 3.4.13

If $\varphi:\mathscr{F}\to\mathscr{G}$ be a morphism of sheaves of a sets on a topological space X. The following are equivalent.

(a) φ is an epimorphism in the category of sheaves.

(b) φ is surjective on the level of stalks: $\varphi_p:\mathscr{F}_p\to\mathscr{G}_p$ is surjective for all $p\in X$.

Proof $(a)\Rightarrow(b)$. Proof by contradiction. Suppose there exists $p_0\in X$ such that $\varphi_{p_0}:\mathscr{F}_{p_0}\to\mathscr{G}_{p_0}$ is not surjective. Let $S=\{0,1\}$, and consider the skyscraper sheaf $i_{p_0,*}S$. For all open subset $U\subseteq X$, $i_{p_0,*}S(U)$ defined by

$$i_{p_0,*}S(U)=\begin{cases} S & \text{if } p_0\in U,\\ 0 & \text{if } p_0\notin U.\end{cases}$$

Let $U\subseteq X$ be an open neighborhood of p_0, then $i_{p_0,*}S(U)=\{0,1\}$. Let $s\in G(U)$, define

$$\alpha_U(s)=\begin{cases} 0 & \text{if } s_{p_0}\in\operatorname{Im}\varphi_{p_0}\\ 0 & \text{if } s_{p_0}\notin\operatorname{Im}\varphi_{p_0}\end{cases} \text{ and } \beta_U(s)=\begin{cases} 0 & \text{if } s_{p_0}\in\operatorname{Im}\varphi_{p_0}\\ 1 & \text{if } s_{p_0}\notin\operatorname{Im}\varphi_{p_0}\end{cases}.$$

Also, let $(p\notin)U\subseteq X$, for $s\in\mathscr{G}(U)$, define $\alpha_U(s)=\beta_U(s)=0$. It is easy to see that α and β is sheaf morphism.

Now, we need to check that $\alpha_U\circ\varphi_U=\beta_U\circ\varphi_U$ for all open subset $U\subseteq X$. If $p_0\in U$, note that $\varphi_{p_0}(t_{p_0})\in\operatorname{Im}\varphi_{p_0}$ for all $t\in\mathscr{F}(U)$, we have

$$\alpha_U\circ\varphi_U(t)=\beta_U\circ\varphi_U(t)=0;$$

if $p_0\notin U$, we also have

$$\alpha_U\circ\varphi_U(t)=\beta_U\circ\varphi_U(t)=0.$$

Hence, $\alpha_U\circ\varphi_U=\beta_U\circ\varphi_U$ for all open subset $U\subseteq X$.

But $\alpha\neq\beta$ as sheaf morphisms, which contradicts the fact that φ is an epimorphism.

$(b)\Rightarrow(a)$. Let $\alpha:\mathscr{G}\to\mathscr{E}$ and $\alpha:\mathscr{G}\to\mathscr{E}$ be two sheaf morphisms with $\alpha\circ\varphi=\beta\circ\varphi$, we shall to show that $\alpha = \beta$. Let $U \subseteq X$ be any open subset, consider the following diagram.

$$\begin{tikzcd}
\mathscr{F}(U) \arrow[r, "\varphi"] \arrow[d] & \mathscr{G}(U) \arrow[r, shift left, "\alpha"] \arrow[r, shift right, "\beta"'] \arrow[d] & \mathscr{E}(U) \arrow[d] \\
\prod_{p\in U} \mathscr{F}_p \arrow[r, "(\varphi_p)_{p\in U}"] & \prod_{p\in U} \mathscr{G}_p \arrow[r, shift left, "\tilde{\alpha}"] \arrow[r, shift right, "\tilde{\beta}"'] & \prod_{p\in U} \mathscr{E}_p
\end{tikzcd}$$

Use the same method as Proposition 3.4.12 $(b) \Rightarrow (a)$, we have $\alpha = \beta$. Hence, φ is an epimorphism. □

Definition 3.4.4 (Quotient sheaf)

*Suppose $\varphi : \mathscr{F} \to \mathscr{G}$ be a morphism of sheaves of a sets on a topological space X, if the conditions in Proposition 3.4.13 hold, we say that $\mathscr{G}$ is a **quotient sheaf** of $\mathscr{F}$.*

By Proposition 3.4.12 and Proposition 3.4.13, monomorphisms and epimorphisms — subsheafiness and quotient sheafiness — can be checked at the level of stalks.

Both propositions generalize readily to sheaves with values in any reasonable category, where "injective" is replaced by "monomorphism" and "surjective" is replaced by "epimorphism".

Notice that there was no part (c) to Proposition 3.4.13, and Example 3.10 shows why. (But there is a version of (c) that implies (a) and (b): surjective on all open sets in any base of a topology implies that the corresponding map of sheaves is an epimorphism.)

Example 3.10 Let $X = \mathbb{C}$ with the classical topology, and define $\mathscr{O}_X$ to be the sheaf of holomorphic functions, and $\mathscr{O}_X^*$ to be the sheaf of invertible (nowhere zero) holomorphic functions. This is a sheaf of abelian groups under multiplication. We have maps of sheaves

$$0 \longrightarrow \underline{\mathbb{Z}} \xrightarrow{\times 2\pi i} \mathscr{O}_X \xrightarrow{\exp} \mathscr{O}_X^* \longrightarrow 1.$$

(0 means the zero sheaf, for all open subset $U \subseteq X$, we have $0(U) = \{0\}$; 1 means the unit sheaf, for all open subset $U \subseteq X$, we have $1(U) = \{1\}$, where 1 is the unit of multiplication groups.) We will soon interpret this as an exact sequence of sheaves of abelian groups (the **exponential exact sequence**), although we don't yet have the language to do so.

Exercise 3.17 Show that $\exp : \mathscr{O}_X \to \mathscr{O}_X^*$ describes $\mathscr{O}_X^*$ as quotient sheaf of $\mathscr{O}_X$. Find an open set on which this map is not surjective.

Proof Let $\overline{(g, U)} \in \mathscr{O}_X^*$, then there exists a simply connected open set $(p \in) U \subseteq \mathbb{C}$ such that $g \in \mathscr{O}_X^*(U)$, that is, g is analytic and nowhere vanishing. Let

$$f(z) = \int_{z_0}^{z} \frac{g'(\xi)}{g(\xi)} d\xi + c,$$

where $c \in \mathbb{C}$ and $e^c = g(z_0)$. We need to show that $f(z)$ is analytic over U, and $\exp_U f(z) = g(z)$.

Since g nowhere zero and U is simply connected, $\frac{g'(z)}{g(z)}$ is holomorphic on U, and therefore $f'(z) = \frac{g'(z)}{g(z)}$ is holomorphic on U, hence, $f(z) \in H(D)$.

It is easy to see that $\exp_U f(z) = g(z)$. Hence, on the level of stalks $\exp_p : \mathscr{O}_{X,P} \to \mathscr{O}_X^*$ is a surjective for all $p \in \mathbb{C}$. By Proposition 3.4.13, $\exp : \mathscr{O}_X \to \mathscr{O}_X^*$ is an epimorphism.

Now, we give an example to show that there exists an open set U, such that $\mathscr{O}_X(U) \to \mathscr{O}_X^*(U)$ is not surjective. Let $U = \mathbb{C} \setminus \{0\}$, then $g(z) = z \in \mathscr{O}_X^*(U)$. If there exists $f \in \mathscr{O}_X(U)$ with $e^f = z$, then $f = \log z$ would be a single-valued branch of $\log z$, contradicting the multivalued nature of $\log z$ on U. □

Remark 3.29 This is a great example to get a sense of what "surjectivity" means for sheaves: nowhere vanishing holomorphic functions (such as the function z away from the origin) have logarithms locally, but they need not have logarithms globally.

3.5 Recovering sheaves from a "sheaf on a base"

Sheaves are natural things to want to think about, but hard to get our hands on. We like the identity and gluability axioms, but they make proving things tricker than for presheaves. We have discussed how we can understand sheaves using stalks (using "compatible germs"). We now introducing the notion of a sheaf on a base. **Warning:** this way of understanding an entire sheaf from limited information is confusing. It may help to keep sight of the central insight that this partial information is enough to understand germs, and the notion of when they are compatible (with nearby germs).

3.5.1 Sheaf on a base

Definition 3.5.1 (Base of topology)

Suppose X be a topological space with topology $\mathscr{T}_X$, a ***base of a topology*** *is a subcollection of the open sets $\{B_i\} \subseteq \mathscr{T}_X$, such that each $U \in \mathscr{T}_X$ is a union of the B_i. We say $\mathscr{B} = \{B_i\}$.*

Example 3.11 Suppose $X = \mathbb{R}^n$. Then the way the classical topology is often first defined is by defining open balls $B_r(x) = \{y \in \mathbb{R}^n : |y - x| < r\}$, and declearing that any union of open balls is open. So the balls form a base of the classical topology — we say they generate the classical topology.

As an application of how we use them, to check continuity of some map $\pi : X \to \mathbb{R}^n$, we need only thinking about the pullback of balls on $\mathbb{R}^n$ — part of the traditional $\varepsilon - \delta$ definition of continuity.

Now suppose we have a sheaf $\mathscr{F}$ on a topological space X, and a base $\{B_i\}$ of open sets on X. Then consider the information

$$\left(\{\mathscr{F}(B_i)\}, \{\mathrm{res}_{B_i,B_j} : \mathscr{F}(B_i) \to \mathscr{F}(B_j)\}\right),$$

which is a subset of the information contained in the sheaf — we are only paying attention to the information involving elements of the base, not all open sets.

We can recover the entire sheaf from this information. This is because we can determine the stalks from this information, and we can determine when germs are compatible.

Exercise 3.18 Make this precise. How can you recover a sheaf $\mathscr{F}$ from this partial information?

Proof Let $\mathscr{B} = \{B_i\}$ be a base of topology space X. Say $B = \bigcup_i B_i$.

Let $\mathscr{F}_x = \varinjlim_{x \in B_i} \mathscr{F}(B_i)$. If $s_p = t_p \in \mathscr{F}_x$, we may assume $s \in \mathscr{F}(B_i)$ and $t \in \mathscr{F}(B_j)$, then there exists $x \in B_k$ with $B_k \subseteq B_i \cap B_j$ such that $s|_{B_k} = t|_{B_k}$.

Let $U \subseteq B$ be any open set, define

$$\mathscr{F}(U) = \{(s_p)_{p \in U} : \forall p \in U, \exists (U \supseteq) B_i \ni p \text{ and } \widetilde{s} \in \mathscr{F}(B_i) \text{ s.t. } \widetilde{s}_q = s_q, \forall q \in B_i\}.$$

We shall to show that $\mathscr{F}$ form a sheaf on B.

(1) $\mathscr{F}$ is a presheaf.

Let $U \subseteq U'$, be two open subsets of X, the restriction map $\mathrm{res}_{U',U} : \mathscr{F}(U') \to \mathscr{F}(U)$ given by

$$\mathrm{res}_{U',U}((s_p)_{p \in U'}) = (s_p)_{p \in U}.$$

By the definition of base of topology, the restriction map is well-defined.

Note that for all $(s_p)_{p \in U} \in \mathscr{F}(U)$, we have $\mathrm{res}_{U,U}((s_p)_{p \in U}) = (s_p)_{p \in U}$, i.e., $\mathrm{res}_{U,U} = \mathrm{id}_{\mathscr{F}(U)}$.

The transitivity is clearly by the definition of base of topology, and therefore $\mathscr{F}$ form a presheaf on X.

(2) $\mathscr{F}$ is a sheaf.

Let U be any open subset of X. Since $\{B_i\}$ is a base of topology, U is covered by some $B_i \in \mathscr{B}$, i.e., $U = \bigcup_i B_i$.

For identity axiom: Let $(s_p)_{p\in U}, (t_p)_{p\in U} \in \mathscr{F}(U)$ with $(s_p)_{p\in U}|_{B_i} = (t_p)_{p\in U}|_{B_i}$, i.e., $(s_p)_{p\in B_i} = (t_p)_{p\in B_i}$. Hence, for all $p \in B_i$, there exists $(p \in)V_{p,i} \subseteq B_i$ such that $s|_{V_{p,i}} = t|_{V_{p,i}}$, hence $s_p = t_p$ for all $p \in U$, that is, $(s_p)_{p\in U} = (t_p)_{p\in U}$, which implies that identity axiom holds.

For gluability axiom: Let $s_i \in \mathscr{F}(B_i)$, and $s_i|_{B_i\cap B_j} = s_j|_{B_i\cap B_j}$. Define $(s_p)_{p\in U}$ as follow: $(s_p)_{p\in U}|_{B_i} = (s_{ip})_{p\in B_i}$. Since $s_i|_{B_i\cap B_j} = s_j|_{B_i\cap B_j}$, $(s_p)_{p\in U}$ is well-defined. It suffices to show that $(s_p)_{p\in U} \in \mathscr{F}(U)$. For all $p \in U$, since $U = \bigcup_i B_i, p \in B_i$ for some i. Note that $(s_q)_{q\in U}|_{B_i} = (s_{iq})_{q\in B_i}$, we have $s_q = s_{iq}$ for all $q \in B_i$. Also note that $s_i \in \mathscr{F}(B_i)$, we have $(s_p)_{p\in U} \in \mathscr{F}(U)$.

Hence, $\mathscr{F}$ is a presheaf, which satisfies identity axiom and gluability axiom, and therefore $\mathscr{F}$ is a sheaf.

□

Definition 3.5.2 (Sheaf on a base)

A sheaf of sets (or abelian groups, rings, $\cdots$) on a base $\{B_i\}$ *is the following:*

(i) For each B_i in the base, we have a set $F(B_i)$. If $B_i \subseteq B_j$, we have maps

$$\mathrm{res}_{B_j,B_i} : F(B_j) \to F(B_i),$$

with $\mathrm{res}_{B_i,B_i} = \mathrm{id}_{F(B_i)}$. If $B_i \subseteq B_j \subseteq B_k$, then

$$\mathrm{res}_{B_k,B_i} = \mathrm{res}_{B_j,B_i} \circ \mathrm{res}_{B_k,B_j} .$$

So far we have defined a ***presheaf on a base*** $\{B_i\}$.

(ii) ***The base identity axiom:*** *If $B = \bigcup B_i$, then if $f, g \in F(B)$ are such that $\mathrm{res}_{B,B_i} f = \mathrm{res}_{B,B_i} g$ for all i, then $f = g$.*

(iii) ***The base gluability axiom:*** *If $B = \bigcup B_i$, and we have $f_i \in F(B_i)$ such that f_i agrees with f_j on any basic open set contained in $B_i \cap B_j$ (i.e., $\mathrm{res}_{B_i,B_k} f_i = \mathrm{res}_{B_j,B_k} f_j$, for all $B_k \subseteq B_i \cap B_j$) then there exists $f \in F(B)$ such that $\mathrm{res}_{B,B_i} f = f_i$ for all i.*

Theorem 3.5.1

Suppose $\{B_i\}$ is a base on X, and F is a sheaf of sets on this base. Then there is a sheaf $\mathscr{F}$ extending F with isomorphisms $\mathscr{F}(B_i) \xleftrightarrow{\sim} F(B_i)$ agreeing with the restriction maps. This sheaf $\mathscr{F}$ is unique up to unique isomorphism.

Proof We will define $\mathscr{F}$ as the sheaf of families of compatible germs of F. Defined the **stalk** of a presheaf F on a base at $p \in X$ by

$$F_p = \varinjlim F(B_i)$$

where the colimit is over all B_i (in the base) containing p.

We will say a family of germs in an open set U is compatible near p if there is a section s of F over some B_i containing p such that the germs over B_i are precisely the germs of s. More formally, define

$$\mathscr{F}(U) := \{(f_p \in F_p)_{p\in U} : \forall p \in U, \exists B \text{ with } p \in B \subseteq U \text{ and } s \in F(B), \text{ s.t. } s_q = f_q, \forall q \in B\}$$

where each B is in our base.

Same as Exercise 3.18, this is a sheaf. Now, we claim that if B is in our base, the natural map $F(B) \to \mathscr{F}(B)$ is an isomorphism.

Natural map $\theta : F(B) \to \mathscr{F}(B)$ is given by $s \mapsto (s_p)_{p\in B}$. By the definition of $\mathscr{F}(U)$, θ is well-defined. Let $(f_p)_{p\in B} \in \mathscr{F}(B)$, then for all $p \in B$ exists $B_p \subseteq U$ and $\widetilde{s}_p \in F(B)$ such that $(\widetilde{s}_p)_q = f_q$

for all $q \in B_p$. Hence, for all $q \in B_p$, there exists $q \in B_{p,q}(\in \mathscr{B})$ such that $\widetilde{s}_p|_{B_{p,q}} = f|_{B_{p,q}}$, and therefore $(f|_{B_{p,q_1}})|_{B_{p,q_1}\cap B_{p,q_2}} = (f|_{B_{p,q_2}})|_{B_{p,q_1}\cap B_{p,q_2}}$ for all $q_1, q_2 \in B_p$ for all $p \in B$, moreover exists $(\mathscr{B} \ni) B_{p,q_1,q_2} \subseteq B_{p,q_1} \cap B_{p,q_2}$ such that $(f|_{B_{p,q_1}})|_{B_{p,q_1,q_2}} = (f|_{B_{p,q_2}})|_{B_{p,q_1,q_2}}$. Since F is sheaf on a base, by the base gluability axiom, there exists $\widetilde{f} \in F(B)$ such that $\widetilde{f}|_{B_i} = f|_{B_i}$. Also, apply θ to $\widetilde{f}$, we have $\widetilde{f}_p = f_p$ for all $p \in B$. Define $\tau : \mathscr{F}(B) \to F(B)$ by setting

$$(f_p \in \mathscr{F}_p)_{p\in B} \mapsto \widetilde{f},$$

by above discussion τ is well-defined and $\theta \circ \tau = \mathrm{id}_{\mathscr{F}(B)}$. Let $s \in F(B)$, then

$$\tau \circ \theta(s) = \tau((s_p \in \mathscr{F}_p)_{p\in B}) = \widetilde{s},$$

by the base identity axiom $s = \widetilde{s}$, this implies that $\tau \circ \theta = \mathrm{id}_{F(B)}$. Hence, we have an isomorphism:

$$\mathscr{F}(B) \overset{\sim}{\longleftrightarrow} F(B),$$

where $B \in \mathscr{B}$.

Clearly, the restriction maps for F are the same as the restriction maps of $\mathscr{F}$ (for elements of the base).

Finally, we shall to show that $\mathscr{F}$ is indeed unique up to unique isomorphism. Let $\mathscr{G}$ be a sheaf which extending F with isomorphisms $\mathscr{G}(B_i) \overset{\sim}{\longleftrightarrow} F(B_i)$. It suffices to show that for all open subset $U \subseteq X$, we have $\mathscr{F}(U) \overset{\sim}{\longleftrightarrow} \mathscr{G}(U)$. Let $f \in \mathscr{F}(U)$, since U is covered by some B_i and $\mathscr{G}(B_i) \overset{\sim}{\longleftrightarrow} F(B_i)$, $f|_{B_i} \in \mathscr{G}(B_i)$, by the identity axiom of $\mathscr{G}$ and the gluability axiom of $\mathscr{G}$, we have unique $g \in \mathscr{G}(U)$ such that $g|_{B_i} = f|_{B_i}$. This gives a one-to-one correspondence between $\mathscr{F}(U)$ and $\mathscr{G}(U)$ for all open subset $U \subseteq X$. □

Remark 3.30 Theorem 3.5.1 shows that sheaves on X can be recovered from their "restriction to a base". It is clear from the argument that if $\mathscr{F}$ is a sheaf and F is the corresponding sheaf on the base B, then for any $p \in X$, $\mathscr{F}_p$ is naturally isomorphic to F_p, i.e., $\mathscr{F}_p \cong F_p$.

Theorem 3.5.1 is a statement about objects in a category, so we should hope for a similar statement about morphisms.

Theorem 3.5.2 (Morphisms of sheaves correspond to morphisms of sheaves on a base)

Suppose $\{B_i\}$ is a base for the topology of X. A morphism $F \to G$ of sheaves on the base is a collection of maps $F(B_k) \to G(B_k)$ such that the diagram

$$\begin{tikzcd} F(B_i) \arrow[r] \arrow[d, "\mathrm{res}^F_{B_i,B_j}"'] & G(B_i) \arrow[d, "\mathrm{res}^G_{B_i,B_j}"] \\ F(B_j) \arrow[r] & G(B_j) \end{tikzcd}$$

commutes for all $B_j \hookrightarrow B_i$.

(a) A morphism of sheaves is determined by the induced morphism of sheaves on the base.

(b) A morphism of sheaves on the base gives a morphism of the induced sheaves.

Proof

(a) Let $\varphi : \mathscr{F} \to \mathscr{G}$ and $\psi : \mathscr{F} \to \mathscr{G}$ with $\varphi_{B_k} = \psi_{B_k}$. We shall to show that $\varphi = \psi$. It suffices to show that for all open subset $U \subseteq X$ we have $\varphi_U = \psi_U$.

Let U be any open subset of X, and therefore is covered by $\{B_k\}$. Consider the following commutative diagram.

$$\begin{tikzcd} \mathscr{F}(U) \arrow[r] \arrow[d, "\mathrm{res}^{\mathscr{F}}_{U,B_k}"'] & \mathscr{G}(U) \arrow[d, "\mathrm{res}^{\mathscr{G}}_{U,B_k}"] \\ \mathscr{F}(B_k) \arrow[r] & \mathscr{G}(B_k) \end{tikzcd}$$

Let $(f_p \in \mathscr{F}_p)_{p\in U} \in \mathscr{F}(U)$, then we have

$$\mathrm{res}^{\mathscr{G}}_{U,B_k} \circ \varphi_U((f_p)_{p\in U}) = \mathrm{res}^{\mathscr{G}}_{U,B_k}((\varphi_U(f)_p)_{p\in U}) = (\varphi_U(f)_p)_{p\in B_k},$$

similarly, we have $\mathrm{res}^{\mathscr{G}}_{U,B_k} \circ \psi_U((f_p)_{p\in U}) = (\psi_U(f)_p)_{p\in B_k}$. Since $\psi_{B_k} = \varphi_{B_k}$, we have

$$(\varphi_U(f)_p)_{p\in B_k} = (\psi_U(f)_p)_{p\in B_k},$$

hence, $\mathrm{res}^{\mathscr{G}}_{U,B_k} \circ \varphi_U((f_p)_{p\in U}) = \mathrm{res}^{\mathscr{G}}_{U,B_k} \circ \psi_U((f_p)_{p\in U})$. Since U is coved by $\{B_k\}$, by the identity axiom of $\mathscr{G}$, we have $\varphi_U((f_p)_{p\in U}) = \psi_U((f_p)_{p\in U})$, i.e., $\varphi_U = \psi_U$. Hence, $\varphi = \psi$.

(b) Let $\varphi : F \to G$ be a morphism of sheaves on the base. We need to show that there is a morphism of sheaves $\varphi^\sharp : \mathscr{F} \to \mathscr{G}$ induced by φ. Let U be any open set of X, and therefore U is covered by some base, i.e., $U = \bigcup_i B_i$.
Define $\varphi^\sharp : \mathscr{F} \to \mathscr{G}$ by setting

$$(f_p \in \mathscr{F}_p)_{p\in U} \mapsto (g_p \in \mathscr{G}_p)_{p\in U},$$

where $g_p = \varphi_{B_k}(f)_p$ if $p \in B_k$. We need to show that $\varphi^\sharp$ is well-defined. Let $(f_p)_{p\in U} \in \mathscr{F}(U)$, then for all $p \in U$, exists $B_k \subseteq U$ contained p and $\widetilde{f} \in \mathscr{F}(B_k)$ such that $\widetilde{f}_q = f_q$ for all $q \in B_k$. Hence, for all for all $p \in U$, exists $B_k \subseteq U$ contained p and $\varphi_{B_k}(\widetilde{f}) \in \mathscr{G}(B_k)$ such that $\varphi_{B_k}(\widetilde{f})_q = \varphi_{B_k}(f)_q$ for all $q \in B_k$. Note that $g_p = \varphi_{B_k}(f)_p$ if $p \in B_k$, $(g_p \in \mathscr{G}_p)_{p\in U} \in \mathscr{G}(U)$, which implies that $\varphi^\sharp$ is well-defined.
Now, we need to check $\varphi^\sharp$ is a morphism of sheaves. Let $U' \supseteq U$, and $U = \bigcup B_i$, $U' = \bigcup B_i'$, we may assume that $B_i' \supseteq B_i$. Consider the following diagram.

$$\begin{tikzcd}
\mathscr{F}(U') \arrow[r, "\varphi^\sharp_{U'}"] \arrow[d, "\mathrm{res}^{\mathscr{F}}_{U',U}"'] & \mathscr{G}(U') \arrow[d, "\mathrm{res}^{\mathscr{G}}_{U',U}"] \\
\mathscr{F}(U) \arrow[r, "\varphi^\sharp_{U}"] & \mathscr{G}(U)
\end{tikzcd}$$

Let $(f_p)_{p\in U'} \in \mathscr{F}(U')$, then

$$\varphi^\sharp_U(\mathrm{res}^{\mathscr{F}}_{U',U}((f_p)_{p\in U'})) = \varphi^\sharp_U((f_p)_{p\in U}) = (g_p)_{p\in U},$$

where $g_p = \varphi_{B_k}(f)_p$ if $p \in B_k$. On the other hand, we have

$$\mathrm{res}^{\mathscr{G}}_{U',U} \circ \varphi^\sharp_{U'}((f_p)_{p\in U'}) = \mathrm{res}^{\mathscr{G}}_{U',U}((g_p)_{p\in U'}) = (g_p)_{p\in U},$$

where $g_p = \varphi_{B_k'}(f)_p$ if $p \in B_k'$. Since $B_k' \supseteq B_k$, we have

$$\varphi^\sharp_U(\mathrm{res}^{\mathscr{F}}_{U',U}((f_p)_{p\in U'})) = \mathrm{res}^{\mathscr{G}}_{U',U} \circ \varphi^\sharp_{U'}((f_p)_{p\in U'}).$$

This implies that $\varphi^\sharp$ is a morphism of sheaves.

□

Remark 3.31 The above constructions and arguments describe an equivalence of categories (Definition 2.1.10) between sheaves on X and sheaves on a given base of X. (Functors are restriction $\mathbf{Sets}_X \to \mathbf{Sets}_{\mathscr{B}}$ and extension $\mathbf{Sets}_{\mathscr{B}} \to \mathbf{Sets}_X$. By Theorem 3.5.2, identity holds.)

Remark 3.32 It will be useful to extend these notion to $\mathscr{O}_X$-modules. It is easy to verify that there is a correspondence (really, equivalence of categories) between $\mathscr{O}_X$-modules and "$\mathscr{O}_X$-modules on a base". ($\mathscr{O}_X$-modules on a base: for each $B_i \in \mathscr{B}$, $\mathscr{F}(B_i)$ is an $\mathscr{O}_X(B_i)$-module, and for $B_j \subseteq B_i \in \mathscr{B}$, the restriction maps res_{B_i,B_j} are module homomorphisms satisfying the sheaf axioms.)

Proposition 3.5.1

Suppose a morphism of sheaves of sets $F \to G$ on a base $\{B_i\}$ is surjective for all B_i (i.e., $F(B_i) \to G(B_i)$ is surjective for all i). Then the corresponding morphism of sheaves (not on the base) is an epimorphism.

Proof Let $\mathscr{B} = \{B_i\}$ be a base of X. Say $\varphi^\sharp : \mathscr{F} \to \mathscr{G}$, where $\varphi : F \to G$ and $\mathscr{F}, \mathscr{G}$ corresponding sheaves of F, G respectively. By Proposition 3.4.13, it suffices to show that $\varphi^\sharp$ is surjective on the level of stalks, i.e., $\varphi^\sharp_q : \mathscr{F}_q \to \mathscr{G}_q$ is surjective for all $q \in X$. Let $s_q \in \mathscr{G}_q$, we may assume that s_q is the image of $(f_p)_{p\in U}$ for some U, then $(f_p)_{p\in U} \in \mathscr{G}(U)$. Hence, for all $p \in U$, exists $p \in B_p \subseteq U$ and $\widetilde{f_p} \in G(B_p)$ such that $(\widetilde{f_p})_q = f_q$ for all $q \in B_p$. Since $(\widetilde{f_p})_q = f_q$ for all $q \in B_p$, there exist $B_{p,q} \subseteq B_q (\in \mathscr{B})$ such that $\widetilde{f_p}|_{B_{p,q}} = f|_{B_{p,q}}$. Since $\varphi_{B_p} : F(B_p) \to G(B_p)$ is surjective, there exists $\widetilde{g}_p \in G(B_p)$ such that $\varphi_{B_p}(\widetilde{g}_p) = \widetilde{f_p}$. Hence, $\varphi_{B_p}(\widetilde{g}_p)|_{B_{p,q}} = \widetilde{f}|_{B_{p,q}}$. Similarly, there exists $g \in F(B_p)$ such that $\varphi_{B_p}(g)|_{B_{p,q}} = f|_{B_{p,q}}$. Hence, $\varphi_{B_p}(\widetilde{g}_p)|_{B_{p,q}} = \varphi_{B_p}(g)|_{B_{p,q}} = f|_{B_{p,q}}$, and therefore $f_q = \varphi_{B_p}(g)_q$ for all $q \in B_p$. This implies that $\varphi^\sharp_q$ is surjective for all $q \in U$ for any open subset $U \subseteq X$, then we done! $\square$

Remark 3.33 The converse is not true (Exercise 3.17), unlike the case of for injectivity. This gives a useful sufficient criterion for epimorphism: a morphism of sheaves is an epimorphism if it is surjective for sections on base.

3.5.2 Gluing sheaves

We will repeatedly see the theme of constructing some object by gluing, in many different contexts. Keep an eye out for it! In each case, we carefully consider what information we need in order to glue.

Theorem 3.5.3 (Gluing sheaves)

Suppose $X = \bigcup U_i$ is an open cover of X, and we have sheaves $\mathscr{F}_i$ on U_i along with isomorphisms

$$\varphi_{ij} : \mathscr{F}_i|_{U_i\cap U_j} \xrightarrow{\sim} \mathscr{F}_j|_{U_i\cap U_j}$$

(with φ_{ii} the identity) that agree on triple overlaps, i.e.,

$$\varphi_{jk} \circ \varphi_{ij} = \varphi_{ik}$$

*on $U_i \cap U_j \cap U_k$ (this is called the **cocycle condition**, for reasons we ignore). Then these sheaves can be glued together into a sheaf $\mathscr{F}$ on X (unique up to unique isomorphism), with isomorphisms $\mathscr{F}|_{U_i} \xrightarrow{\sim} \mathscr{F}_i$, and the isomorphisms over $U_i \cap U_j$ are the obvious ones.*

Proof Let V be any open subset of X, define

$$\mathscr{F}(V) = \left\{ (s_i) \in \prod \mathscr{F}_i(V \cap U_i) : \varphi_{ij}(s_i|_{V\cap U_i\cap U_j}) = s_j|_{V\cap U_i\cap U_j}, \text{ for all } i,j \right\}.$$

Now, we define restriction map as follow: let $V' \supseteq V$ be an open subset of X, define $\mathrm{res}_{V',V} : \mathscr{F}(V') \to \mathscr{F}(V)$ by setting

$$(s_i) \mapsto (s_i|_{V\cap U_i}).$$

Note that $s_i|_{V\cap U_i} \in \mathscr{F}_i(V \cap U_i)$ and

$$\begin{aligned}
\varphi_{ij}((s_i|_{V\cap U_i})|_{V\cap U_i\cap U_j}) =& \varphi_{ij}((s_i|_{V'\cap U_i\cap U_j})|_{V\cap U_i\cap U_j}) \\
=& (\varphi_{ij}(s_i|_{V'\cap U_i\cap U_j}))|_{V\cap U_i\cap U_j} \\
=& (s_j|_{V'\cap U_i\cap U_j})|_{V\cap U_i\cap U_j} \\
=& s_j|_{V\cap U_i\cap U_j},
\end{aligned}$$

the restriction map is well-defined. Also, it is easy to see that $\mathscr{F}$ is a presheaf.

(i) $\mathscr{F}$ is a sheaf.

Let V be any open subset of X, then V is covered by $U_i \cap V$, say $V_i = U_i \cap V$.

For identity axiom: Let $(s_i), (t_i) \in \mathscr{F}(V)$ with $(s_i)|_{V_k} = (t_i)|_{V_k}$ for all k. Since $(s_i)|_{V_k} = (s_i|_{V_k \cap U_i}) = (s_i|_{V \cap U_k \cap U_i}) = (s_i|_{V_i \cap U_k})$ and $(t_i)|_{V_k} = t_i|_{V_i \cap U_k}$, we have

$$s_i|_{V_i \cap U_k} = t_i|_{V_i \cap U_k}$$

for all k. Note that V_i is covered by $\{V_i \cap U_k\}_k$, by the identity axiom of $\mathscr{F}_i$, $s_i = t_i$, and therefore $(s_i) = (t_i)$, which implies that $\mathscr{F}$ satisfies the identity axiom holds.

For gluability axiom: Let $(s_i^{(n)}) \in \mathscr{F}(V_n)$ with $(s_i^{(n)})|_{V_n \cap V_m} = (s_i^{(m)})|_{V_n \cap V_m}$, then

$$\begin{aligned} s_i^{(n)}|_{V_i \cap V_n \cap V_m} &= s_i^{(n)}|_{V_i \cap (V_i \cap U_n) \cap (V_i \cap U_m)} \\ &= s_i^{(m)}|_{V_i \cap V_n \cap V_m} \\ &= s_i^{(m)}|_{V_i \cap (V_i \cap U_n) \cap (V_i \cap U_m)} \end{aligned}$$

for all i. Notice that V_i is covered by $\{V_i \cap U_n\}_n$, by the gluability axiom of $\mathscr{F}_i$, there exists unique $s_i \in \mathscr{F}_i(V_i)$ such that $s_i|_{V_i \cap U_n} = s_i^{(n)}$. Now, we need to check that $(s_i) \in \mathscr{F}(V)$. Note that

$$\varphi_{ij}(s_i|_{V \cap U_i \cap U_j}) = \varphi_{ij}(s_i^{(j)}) = s_j^{(j)} = s_j|_{V \cap U_i \cap U_j},$$

$(s_i) \in \mathscr{F}(V)$, which implies that $\mathscr{F}$ satisfies the gluability axiom.

(ii) $\mathscr{F}|_{U_i} \xrightarrow{\sim} \mathscr{F}_i$.

Let V be an open subset of U_k, then

$$\mathscr{F}|_{U_k}(V) = \mathscr{F}(V) = \left\{ (s_i) \in \prod \mathscr{F}_i(V \cap U_i) : \varphi_{ij}(s_i|_{V \cap U_i \cap U_j}) = s_j|_{V \cap U_i \cap U_j}, \text{ for all } i, j \right\}.$$

Define $\psi_k : \mathscr{F}|_{U_k} \to \mathscr{F}_k$ as follow:

$$(\psi_k)_V((s_i)) = s_k$$

for all V open subset of U_k. Since $\mathscr{F}_k(V \cap U_k) = \mathscr{F}_k(V)$, ψ_k is well-defined. We shall to check ψ_k is a morphism of sheaves for all k, and give the inverse of ψ_k^{-1}.

Let $V' \supseteq V$ be open subsets of U_k, consider the following diagram.

$$\begin{tikzcd} \mathscr{F}(V') \arrow[r, "(\psi_k)_{V'}"] \arrow[d, "\mathrm{res}_{V',V}"'] & \mathscr{F}_k(V') \arrow[d, "\mathrm{res}_{V',V}"] \\ \mathscr{F}(V) \arrow[r, "(\psi_k)_V"] & \mathscr{F}_k(V) \end{tikzcd}$$

Let $(s_i) \in \mathscr{F}(V')$, then

$$(\psi_k)_V \circ \mathrm{res}_{V',V}((s_i)) = (\psi_k)_V((s_i|_{V \cap U_i})) = s_k|_V$$

and

$$\mathrm{res}_{V',V} \circ (\psi_k)_{V'}(s_i) = \mathrm{res}_{V',V}(s_k) = s_k|_V.$$

Hence, $(\psi_k)_V \circ \mathrm{res}_{V',V} = \mathrm{res}_{V',V} \circ (\psi_k)_{V'}$. This implies that ψ_k is a morphism of sheaves.

Define $\psi_k^{-1} : \mathscr{F}_k \to \mathscr{F}|_{U_k}$ as follow:

$$(\psi_k^{-1})_V(s_k) = (\varphi_{k,1}(s_k|_{V \cap U_1}), \cdots, \varphi_{k,k-1}(s_k|_{V \cap U_{k-1}}), s_k, \varphi_{k,k+1}(s_k|_{V \cap U_{k+1}}), \cdots)$$

for all V is open subset of U_k. We need to check that ψ_k^{-1} is well-defined. Note that $\varphi_{j,i} \circ \varphi_{k,j} = \varphi_{k,i}$, hence, $(\varphi_{k,1}(s_k|_{V \cap U_1}), \cdots, \varphi_{k,k-1}(s_k|_{V \cap U_{k-1}}), s_k, \varphi_{k,k+1}(s_k|_{V \cap U_{k+1}}), \cdots) \in \mathscr{F}(V)$, i.e., ψ_k^{-1} is well-defined. It is easy to check that ψ_k^{-1} is a morphism of sheaves, and is the inverse of ψ_k. Hence,

$$\mathscr{F}|_{U_i} \xrightarrow{\sim} \mathscr{F}_i.$$

(iii) $\mathscr{F}$ is unique up to unique isomorphism.

Let $\mathscr{F}'$ be another sheaf with isomorphism $\mathscr{F}'|_{U_i} \xrightarrow{\sim} \mathscr{F}_i$, then $\mathscr{F}|_{U_i} \cong \mathscr{F}'|_{U_i}$ for all U_i, say isomorphism $\theta_i : \mathscr{F}|_{U_i} \to \mathscr{F}'|_{U_i}$. Let $p \in X$, we shall to show that $\mathscr{F}_p \cong \mathscr{F}'_p$ for all $p \in X$. Define $\theta_p : \mathscr{F}_p \to \mathscr{F}'_p$ as follow:

$$\theta_p(\overline{(s, V)}) = \overline{((\theta_i)_V(s), V)},$$

where $p \in U_i$, $V \subseteq U_i$, and $s \in \mathscr{F}|_{U_i}(V)$. It is easy to see θ_p is well-defined.

θ_p **is surjective:** Let $s'_p \in \mathscr{F}'_p$, since X is covered by $\{U_i\}$, $p \in U_i$ for some i, we may assume that $s'_p = \overline{(s', V)}$, where $V \subseteq U_i$ open subset. Since $(\theta_i)_V$ is isomorphism, there exists $s \in \mathscr{F}|_{U_i}(V)$ such that $(\theta_i)_V(s) = s'$, hence, $\theta_p(\overline{(s, V)}) = \overline{(s', V)} = s'_p$, which implies that $\mathscr{F}_p \to \mathscr{F}'_p$ is surjective.

θ_p **is injective:** Let $\theta_p(s_p) = \theta_p(t_p)$. We may assume that $p \in U_i$ and $s_p = \overline{(s, V_s)}$ where $V_s \subseteq U_i$, then $\theta_p(s_p) = \theta_p(\overline{(s, V_s)}) = \overline{((\theta_i)_{V_s}(s), V_s)}$. Similarly, $\theta_p(t_p) = \overline{((\theta_i)_{V_t}(t), V_t)}$, where $V_t \subseteq U_i$. Hence, there exists $V \subseteq V_s \cap V_t$ such that $(\theta_i)_{V_s}(s)|_V = (\theta_i)_{V_t}(t)|_V$. Since θ_i is an isomorphism, we have $s|_V = t|_V$, and therefore $s_p = t_p$, which implies that θ_p is injective.

Hence, $\mathscr{F}_p \cong \mathscr{F}'_p$ for all $p \in X$, by Proposition 3.4.4, $\mathscr{F} \cong \mathscr{F}'$.

□

Remark 3.34 Thus we can "glue sheaves together", using limited patching information. Small observation: the hypothesis φ_{ii} is the identity is extraneous, as it follows from the cocycle condition ($\varphi_{ii} \circ \varphi_{ii} = \varphi_{ii}$, φ_{ii} must be identity).

Remark 3.35 Theorem 3.5.3 almost says that the "set" of sheaves forms a sheaf itself, but not quite. Making this precise leads one to the notion of a **stack**.

3.6 Sheaves of abelian groups, and $\mathscr{O}_X$-modules, form abelian categories

We are now ready to see that sheaves of abelian groups, and their cousins, $\mathscr{O}_X$-modules, form abelian categories. In other words, we may treat them similarly to vector spaces, and modules over a ring. In the process of doing this, we will see that this is much stronger than an analogy; kernels, cokernels, exactness, and so forth can be understood at the level of stalks (which are just abelian groups), and the compatibility of the germs will come for free.

3.6.1 Sheaves of abelian groups form abelian category

The category of sheaves of abelian groups on a topological space X is clearly an additive category (Definition 2.5.1). In order to show that it is an abelian category, we must begin by showing that any morphism $\varphi : \mathscr{F} \to \mathscr{G}$ has a kernel and a cokernel. We have already seen that φ has a kernel (Proposition 3.3.6): the presheaf kernel is a sheaf, and is a kernel in the category of sheaves.

Proposition 3.6.1 (The stalk of the kernel is the kernel of the stalks)

For all $p \in X$, there is a natural isomorphism

$$(\mathrm{Ker}(\mathscr{F} \to \mathscr{G}))_p \xleftrightarrow{\sim} \mathrm{Ker}(\mathscr{F}_p \to \mathscr{G}_p),$$

where $\mathscr{F}, \mathscr{G}$ are sheaves.

Proof Say $\varphi : \mathscr{F} \to \mathscr{G}$, then

$$(\mathrm{Ker}(\mathscr{F} \to \mathscr{G}))_p = (\mathrm{Ker}\,\varphi)_p = \varinjlim_{U \ni p} (\mathrm{Ker}\,\varphi)(U).$$

Let $\overline{(s,U)} \in (\operatorname{Ker}\varphi)_p$, then $s \in (\operatorname{Ker}\varphi)(U) \subseteq \mathscr{F}(U)$, i.e., $\varphi_U(s) = 0$. Hence, $(\varphi_U(s))_p = 0$. Note that $(\varphi_U(s))_p = \varphi_p(s_p)$, we have $\varphi_p(s_p) = 0$. Define $\theta : (\operatorname{Ker}\varphi)_p \to \operatorname{Ker}(\varphi_p)$ by setting

$$\overline{(s,U)} \mapsto s_p.$$

We need to check θ is well-defined. Let $\overline{(s,U_1)} \sim \overline{(t,U_2)}$, then there exists an open subset $U \subseteq U_1 \cap U_2$ such that $s|_U = t|_U$ in $\operatorname{Ker}\varphi_U \subseteq \mathscr{F}(U)$, hence, $s_p = t_p$ in $\mathscr{F}_p$. Note that $s_p \in \operatorname{Ker}\varphi_p$, θ is well-defined.

Next, we give the inverse of θ. Let $s_p \in \operatorname{Ker}(\varphi_p)$, then $\varphi_p(s_p) = (\varphi_U(s))_p = 0$ for some open subset $(p \in)U \subseteq X$. Hence, there exists $V \subseteq U$ such that $\varphi_U(s)|_V = 0$. Note that $\varphi_U(s)|_V = \varphi_V(s|_V)$, then $s|_V \in \operatorname{Ker}\varphi_V = (\operatorname{Ker}\varphi)(V)$. Define $\theta^{-1} : \operatorname{Ker}(\varphi_p) \to (\operatorname{Ker}\varphi)_p$ by setting

$$s_p \mapsto \overline{(s|_V, V)}.$$

We need to check that θ^{-1} is well-defined. Let $s_p = t_p$ in $\operatorname{Ker}(\varphi_p)$, then there exists an open subset $(p \in)U \subseteq X$ such that $s|_U = t|_U$. Also there exists an open subset $(p \in)V$ such that $s|_V, t|_V \in \operatorname{Ker}\varphi_V = (\operatorname{Ker}\varphi)(V)$, hence we have $\overline{(s|_{V\cap U}, V\cap U)} = \overline{(t|_{V\cap U}, V\cap U)}$, this implies that θ^{-1} is well-defined. By the definition of θ^{-1}, it is easy to see that θ^{-1} is, indeed, the inverse of θ.

Consider the following diagram.

$$\begin{tikzcd}
(\operatorname{Ker}\varphi)_p \arrow[r, dashed] & \operatorname{Ker}(\varphi_p) \arrow[r, shift left] & \mathscr{F}_p \arrow[l, two heads, shift left] \arrow[rr, dashed, "\varphi_p"] & & \mathscr{G}_p \\
& (\operatorname{Ker}\varphi)(U') \arrow[ul] \arrow[r] \arrow[d] & \mathscr{F}(U') \arrow[u] \arrow[r, "\varphi_{U'}"] \arrow[d] & \mathscr{G}(U') \arrow[ur] \arrow[d] & \\
& (\operatorname{Ker}\varphi)(U) \arrow[uul] \arrow[r] & \mathscr{F}(U) \arrow[uu] \arrow[r, "\varphi_U"] & \mathscr{G}(U) \arrow[uur] &
\end{tikzcd}$$

θ is given by universal property of colimit, and therefore we have a natural isomorphism

$$(\operatorname{Ker}(\mathscr{F} \to \mathscr{G}))_p \xleftrightarrow{\sim} \operatorname{Ker}(\mathscr{F}_p \to \mathscr{G}_p).$$

$\square$

We next address the issue of the cokernel. Now $\varphi : \mathscr{F} \to \mathscr{G}$ has a cokernel in the category of presheaves, call it $\mathscr{H}_{\text{pre}}$ (where the subscript is meant to remind us that this is a presheaf). Let $\operatorname{sh} : \mathscr{H}_{\text{pre}} \to \mathscr{H}$ be its sheafification. Recall that the cokernel is defined using a universal property: it is the colimit of the diagram

$$\begin{tikzcd}
\mathscr{F} \arrow[r, "\varphi"] \arrow[d] & \mathscr{G} \\
0 &
\end{tikzcd} \tag{3.4}$$

in the category of presheaves.

Proposition 3.6.2

Let $\varphi : \mathscr{F} \to \mathscr{G}$ be morphism of sheaves, $\mathscr{H}_{\text{pre}}$ be the cokernel of φ in the category of presheaves. Let $\operatorname{sh} : \mathscr{H}_{\text{pre}} \to \mathscr{H}$ be the sheafification of $\mathscr{H}_{\text{pre}}$. Then the composition $\mathscr{G} \to \mathscr{H}$ is the cokernel of φ in the category of sheaves.

Proof We show that it satisfies the universal property of cokernel. Given any sheaf $\mathscr{E}$ and a commutative diagram

$$\begin{tikzcd}
\mathscr{F} \arrow[r, "\varphi"] \arrow[d] & \mathscr{G} \arrow[d] \\
0 \arrow[r] & \mathscr{E}.
\end{tikzcd}$$

We construct

$$\begin{tikzcd}
\mathscr{F} \arrow[r, "\varphi"] \arrow[d] & \mathscr{G} \arrow[d] \arrow[rrd, bend left] & & \\
0 \arrow[r] \arrow[rrrd, bend right] & \mathscr{H}_{\text{pre}} \arrow[r, "\text{sh}"] \arrow[rrd, dashed] & \mathscr{H} \arrow[rd, dashed] & \\
& & & \mathscr{E}.
\end{tikzcd}$$

We show that there is a unique morphism $\mathscr{H} \to \mathscr{E}$ making the diagram commute. As $\mathscr{H}_{\text{pre}}$ is the cokernel in the category of presheaves, there is a unique morphism of presheaves $\mathscr{H}_{\text{pre}} \to \mathscr{E}$ making the diagram commute. By the universal property of sheafification (Definition 3.4.2), there is a unique morphism of sheaves $\mathscr{H} \to \mathscr{E}$ making the diagram commute. □

Proposition 3.6.3 (The stalk of the cokernel is isomorphic to the cokernel of the stalk)

For all $p \in X$, there is a natural isomorphism

$$(\text{Coker}(\mathscr{F} \to \mathscr{G}))_p \xleftrightarrow{\sim} \text{Coker}(\mathscr{F}_p \to \mathscr{G}_p),$$

where $\mathscr{F}, \mathscr{G}$ are sheaves.

Proof Consider the following diagram.

$$\begin{tikzcd}
& \mathscr{F}_p \arrow[r] & \mathscr{G}_p \arrow[r] & \text{Coker}(\varphi_p) \arrow[r, dashed, shift left] & (\text{Coker}\,\varphi)_p \arrow[l, dashed, shift left] \\
\mathscr{F}(U') \arrow[r] \arrow[d] \arrow[ur] & \mathscr{G}(U') \arrow[r] \arrow[d] \arrow[ur] & \mathscr{H}_{\text{pre}}(U') \arrow[r, "\text{sh}_{U'}"] \arrow[d] & \mathscr{H}(U') \arrow[d] \arrow[ur] & \\
\mathscr{F}(U) \arrow[r] \arrow[uur] & \mathscr{G}(U) \arrow[r] \arrow[uur] & \mathscr{H}_{\text{pre}}(U) \arrow[r, "\text{sh}_U"] & \mathscr{H}(U) \arrow[uur] &
\end{tikzcd}$$

Since $\mathscr{F}$ and $\mathscr{G}$ are sheaves of abelian group, $\text{Coker}(\varphi_p)$ is an abelian group, moreover, $\text{Coker}(\varphi_p) = \mathscr{G}_p/\varphi_p(\mathscr{F}_p)$. Now, we shall to define $\theta_p : \text{Coker}(\varphi_p) \to (\text{Coker}\,\varphi)_p$. Let $\overline{s_p} \in \text{Coker}(\varphi_p)$ be the image of s_p in $\text{Coker}(\varphi_p)$, then there exists an open subset $U \ni p$ such that $s \in G(U)$. Let $[s]$ be the image of s in $\mathscr{H}_{\text{pre}}(U)$, then $\text{sh}_U([s]) = ([s]_q)_{q\in U} \in \mathscr{H}(U)$. Define $\theta_p : \text{Coker}(\varphi_p) \to (\text{Coker}\,\varphi)_p$ by setting

$$\overline{s_p} \mapsto (([s]_q)_{q\in U})_p.$$

(i) θ_p is well-defined.
Let $\overline{s_p} = \overline{t_p}$, then $s_p - t_p \in \varphi_p(\mathscr{F}_p)$. Let $s_p - t_p = \varphi_p(f_p)$, note that $\varphi_p(f_p) = (\varphi_U(f))_p$ for some open subset $U \ni p$, we have

$$s_p - t_p = (\varphi_U(f))_p.$$

Then there exist an open subset $(p \in)V \subseteq U$ such that $s|_V - t|_V = \varphi_U(f)|_V$. Hence, $[s] = [t]$ in $\mathscr{H}_{\text{pre}}(V)$, and therefore $(([s]_q)_{q\in V})_p = (([t]_q)_{q\in V})_p$, which implies that $\theta_p(\overline{s_p}) = \theta_p(\overline{t_p})$. Hence, θ_p is well-defined.

(ii) θ_p is injective.
Let $(([s]_q)_{q\in U})_p = (([t]_q)_{q\in U})_p$ in $(\text{Coker}\,\varphi)_p$, then there is an open subset of U, say $V \ni p$, such that $([s]_q)_{q\in V} = ([t]_q)_{q\in V}$. This implies that $[s]_q = [t]_q$ for all $q \in V$. We may assume that $[s] = [t]$ in $\mathscr{H}_{\text{pre}}(V)$, then $s - t \in \varphi_V(\mathscr{F}(V))$. Taking stalk at p, we have $s_p - t_p \in \varphi_p(\mathscr{F}_p)$, hence, $\overline{s_p} = \overline{t_p}$, and therefore θ_p is injective.

(iii) θ_p is surjective.

Let $((s_q)_{q\in U})_p \in (\operatorname{Coker}\varphi)_p$, then there is an open subset $U \ni p$ such that $(s_q)_{q\in U} \in \mathscr{H}(U)$. For $q = p$, there exists $(p \in)V \subseteq U$ and $\widetilde{s} \in \mathscr{H}_{\mathrm{pre}}(V)$ such that $\widetilde{s}_m = s_m$ for all $m \in V$. Consider $(s_q)_{q\in V} \in \mathscr{H}(V)$, we have $\mathrm{sh}_V(\widetilde{s}) = (s_q)_{q\in V}$. Let $\widetilde{s}$ be the image of $\widetilde{s}^* \in \mathscr{G}(V)$, then $\overline{\widetilde{s}^*_p} \in \operatorname{Coker}(\varphi_p)$, and therefore,

$$\theta_p(\overline{\widetilde{s}^*_p}) = (([\widetilde{s}^*]_q)_{q\in V})_p = ((s_q)_{q\in V})_p = ((s_q)_{q\in U})_p.$$

This implies that θ_p is a surjective.

Note that θ_p is given by universal property (see above diagram), θ_p is natural and is isomorphism. □

We have now defined the notions of kernel and cokernel, and verified that they may be checked at the level of stalks. We have also verified that the properties of a morphism being a monomorphism or epimorphism are also determined at the level of stalks (Proposition 3.4.12 and Proposition 3.4.13). Hence we have proved the following:

Theorem 3.6.1

Sheaves of abelian groups on a topological space X form an abelian category.

That's all there is to it — what needs to be proved has been shifted to the stalks, where everything works because stalks are abelian groups!

And we see more: all structures coming from the abelian nature of this category may be checked at the level of stalks. For example:

Proposition 3.6.4

Suppose $\varphi : \mathscr{F} \to \mathscr{G}$ is a morphism of sheaves of abelian groups.

(a) The image sheaf $\operatorname{Im}\varphi$ is the sheafification of the image presheaf.

(b) The stalk of the image is the image of the stalk.

Proof

(a) First we define the presheaf image. Since $\mathscr{F}$ and $\mathscr{G}$ are sheaves of abelian groups, define

$$\operatorname{Im}_{\mathrm{pre}}\varphi(U) := \operatorname{Im}\varphi_U = \varphi_U(\mathscr{F}(U)),$$

for any open subset U of X. It is easy to see that $\operatorname{Im}_{\mathrm{pre}}$ is a presheaf. Let $\mathrm{sh} : \operatorname{Im}_{\mathrm{pre}}\varphi \to \operatorname{Im}\varphi$ be sheafification.

We constructs a commutative diagram,

$$\begin{tikzcd} \mathscr{E} \arrow[r] \arrow[d] & \mathscr{G} \arrow[d] \\ 0 \arrow[r] & \operatorname{Coker}\varphi, \end{tikzcd}$$

where $\mathscr{E}$ be any sheaf with arrows $\mathscr{E} \to \mathscr{G}$ and $\mathscr{E} \to 0$.

Consider the following diagram.

$$\begin{tikzcd}
\mathscr{E} \arrow[dr, dashed] \arrow[ddr, dashed] \arrow[ddrr, bend left] \arrow[dddr, bend right] & & \\
& \operatorname{Im}\varphi \arrow[dr, dashed] & \\
& \operatorname{Im}_{\mathrm{pre}}\varphi \arrow[u, "\mathrm{sh}"] \arrow[r] \arrow[d] & \mathscr{G} \arrow[d] \\
& 0 \arrow[r] & \operatorname{Coker}\varphi
\end{tikzcd}$$

We show there is a unique morphism $\mathscr{E} \to \operatorname{Im}\varphi$ making the diagram commute. The universal property of presheaf kernel gives a unqiue morphism of presheaves $\mathscr{E} \to \operatorname{Im}_{\mathrm{pre}}\varphi$. And the universal property of sheafification gives a unique morphism of sheaves $\operatorname{Im}\varphi \to \mathscr{G}$. Define $\mathscr{E} \to \operatorname{Im}\varphi$ be the composition

$$\mathscr{E} \longrightarrow \operatorname{Im}_{\mathrm{pre}}\varphi \xrightarrow{\mathrm{sh}} \operatorname{Im}\varphi,$$

this morphism of sheaves make above diagram commute, and also it is unique.

(b) It suffices to show that

$$(\operatorname{Im}(\mathscr{F} \to \mathscr{G}))_p \xleftrightarrow{\sim} \operatorname{Im}(\mathscr{F}_p \to \mathscr{G}_p).$$

Note that $\operatorname{Im}(\mathscr{F} \to \mathscr{G}) = \operatorname{Ker}(\operatorname{Coker}(\mathscr{F} \to \mathscr{G})) = \operatorname{Ker}(\mathscr{G} \to \operatorname{Coker}(\mathscr{F} \to \mathscr{G}))$, we have

$$\begin{aligned}
(\operatorname{Im}(\mathscr{F} \to \mathscr{G}))_p &= \operatorname{Ker}(\mathscr{G} \to \operatorname{Coker}(\mathscr{F} \to \mathscr{G}))_p \\
&\cong \operatorname{Ker}(\mathscr{G}_p \to \operatorname{Coker}(\mathscr{F} \to \mathscr{G})_p) \\
&\cong \operatorname{Ker}(\mathscr{G}_p \to \operatorname{Coker}(\mathscr{F}_p \to \mathscr{G}_p)) \\
&= \operatorname{Ker}(\mathscr{G}_p \to (\mathscr{G}_p / \operatorname{Im}(\mathscr{F}_p \to \mathscr{G}_p))) \\
&= \operatorname{Im}(\mathscr{F}_p \to \mathscr{G}_p),
\end{aligned}$$

as we desired.

$\square$

As a consequence, **exactness of a sequence of sheaves may be checked at the level of stalks**.

Proposition 3.6.5

Suppose $\alpha : \mathscr{F} \to \mathscr{G}$ and $\beta : \mathscr{G} \to \mathscr{H}$ are two morphisms of sheaves of abelian groups on X. Then

$$\mathscr{F} \xrightarrow{\alpha} \mathscr{G} \xrightarrow{\beta} \mathscr{H}$$

is exact (at $\mathscr{G}$) if and only if for all $p \in X$,

$$\mathscr{F}_p \xrightarrow{\alpha_p} \mathscr{G}_p \xrightarrow{\beta_p} \mathscr{H}_p$$

is exact.

Proof Since sequence

$$\mathscr{F} \xrightarrow{\alpha} \mathscr{G} \xrightarrow{\beta} \mathscr{H}$$

is exact, we have $\operatorname{Ker}\beta = \operatorname{Ker}(\mathscr{G} \to \mathscr{H}) = \operatorname{Im}\alpha = \operatorname{Im}(\mathscr{F} \to \mathscr{G})$. By Proposition 3.6.4 and Proposition 3.6.1,

$$\operatorname{Im}(\mathscr{F}_p \to \mathscr{G}_p) \cong \operatorname{Im}(\mathscr{F} \to \mathscr{G})_p = \operatorname{Ker}(\mathscr{G} \to \mathscr{H})_p \cong \operatorname{Ker}(\mathscr{G}_p \to \mathscr{H}_p),$$

this implies that the sequence

$$\mathscr{F}_p \xrightarrow{\alpha_p} \mathscr{G}_p \xrightarrow{\beta_p} \mathscr{H}_p$$

is exact.

Above process is reversible, then we done! □

Proposition 3.6.6 (Taking the stalk of a sheaf of abelian groups is an exact functor)

If X is a topological space and $p \in X$ is a point, then taking the stalk at p defines an exact functor $\mathbf{Ab}_X \to \mathbf{Ab}$.

Proof By Proposition 3.6.5, clearly. □

✍ **Exercise 3.19** Check that the exponential exact sequence (Example 3.10)

$$0 \longrightarrow \underline{\mathbb{Z}} \xrightarrow{\times 2\pi i} \mathscr{O}_X \xrightarrow{\exp} \mathscr{O}_X^* \longrightarrow 1.$$

is indeed an exact sequence of sheaves of abelian groups.

Proof It suffice to check exactness at the level of stalk, i.e., check seqeunce

$$0 \longrightarrow \underline{\mathbb{Z}}_p \xrightarrow{(\times 2\pi i)_p} \mathscr{O}_{X,p} \xrightarrow{\exp_p} \mathscr{O}_{X,p}{}^* \longrightarrow 1$$

is exact.

In fact, $\underline{\mathbb{Z}}_p = \mathbb{Z}$.

(a) $(\times 2\pi i)_p : \mathbb{Z} \to \mathscr{O}_{X,p}$ is injective.

Let $(\times 2\pi i)_p(n) = 0$, then $(\times 2\pi i)_p(n) = (2n\pi i)_p = 0$, hence, $n = 0$, which implies that $(\times 2\pi i)_p$ is injective.

(b) $\operatorname{Im}(\times 2\pi i)_p = \operatorname{Ker}\exp_p$.

Let $(2n\pi i)_p \in \operatorname{Im} \times 2\pi i)_p$, then $\exp_p((2n\pi i)_p) = e_p^{2n\pi i} = 1$. Hence, $\operatorname{Im}(\times 2\pi i)_p \subseteq \operatorname{Ker}\exp_p$. Conversely, let $f_p \in \operatorname{Ker}\exp_p$, then $e_p^f = 0$. Hence, $f = 2n\pi i$, which implies that $f \in \operatorname{Im} \times 2\pi i)_p$, i.e., $\operatorname{Ker}\exp_p \subseteq \operatorname{Im} \times 2\pi i)_p$.

(c) $\exp_p : \mathscr{O}_{X,p} \to \mathscr{O}_{X,p}^*$ is surjective.

Let $g_p \in \mathscr{O}_{X,p}^*$, then there exists an open subset U such that g admits a holomorphic logarithm on U, i.e., $f = \log g \in \mathscr{O}_X(U)$. Hence, $f_p \in \mathscr{O}_{X,p}$, and $\exp_p(f_p) = g_p$, which implies that $\exp_p : \mathscr{O}_{X,p} \to \mathscr{O}_{X,p}^*$ is surjective.

Hence, sequence

$$0 \longrightarrow \underline{\mathbb{Z}}_p \xrightarrow{(\times 2\pi i)_p} \mathscr{O}_{X,p} \xrightarrow{\exp_p} \mathscr{O}_{X,p}{}^* \longrightarrow 1$$

is exact. By Proposition 3.6.5, we done. □

Proposition 3.6.7 (Left-exactness of the functor of "sections over U")

Suppose $U \subseteq X$ is an open set, and $0 \to \mathscr{F} \to \mathscr{G} \to \mathscr{H}$ is an exact sequence of sheaves of abelian groups. Then sequence

$$0 \longrightarrow \mathscr{F}(U) \longrightarrow \mathscr{G}(U) \longrightarrow \mathscr{H}(U)$$

is exact. Denote this functor by $\Gamma(U, \square)$. (i.e. $\Gamma(U, \mathscr{F}) = \mathscr{F}(U)$)

Proof Let $\square^\flat : \mathbf{Ab}_X \to \mathbf{Ab}_X^{\text{pre}}$ be forgetful functor, by Proposition 3.4.10, forgetful functor is right adjoint to the sheafification functor, i.e., $\square^\flat$ is left exact. Hence,

$$0 \longrightarrow \mathscr{F}^\flat \longrightarrow \mathscr{G}\flat \longrightarrow \mathscr{H}^\flat$$

By Proposition 3.3.4, for open subset U of X, sequence

$$0 \longrightarrow \mathscr{F}^\flat(U) \longrightarrow \mathscr{G}^\flat(U) \longrightarrow \mathscr{H}^\flat(U)$$

is exact.

Note that $\mathrm{Ker}_{\mathrm{pre}}(\mathscr{F}^\flat(U) \to \mathscr{G}^\flat(U)) = \mathrm{Ker}(\mathscr{F}(U) \to \mathscr{G}(U))$ and $\mathrm{Ker}_{\mathrm{pre}}(\mathscr{F}^\flat(U) \to \mathscr{G}^\flat(U)) = \mathrm{Im}_{\mathrm{pre}}(\mathscr{G}^\flat(U) \to \mathscr{H}^\flat(U)) = \mathrm{Im}_{\mathrm{pre}}(\mathscr{G} \to \mathscr{H}(U))$, by the universal property of sheafification, we have

$$\mathrm{Im}_{\mathrm{pre}}(\mathscr{G} \to \mathscr{H}(U)) \overset{\sim}{\longleftrightarrow} \mathrm{Im}(\mathscr{G} \to \mathscr{H}(U)),$$

and therefore $\mathrm{Ker}(\mathscr{F}(U) \to \mathscr{G}(U)) = \mathrm{Im}(\mathscr{G}(U) \to \mathscr{H}(U))$.

Also $\mathscr{F} \to \mathscr{G}$ is a monomorphism, by Proposition 3.4.12, $\mathscr{F}(U) \to \mathscr{G}(U)$ is injective. Hence, the sequence

$$0 \longrightarrow \mathscr{F}(U) \longrightarrow \mathscr{G}(U) \longrightarrow \mathscr{H}(U)$$

is exact. □

✍ **Exercise 3.20** **Show that the section functor need not be exact**: show that if $0 \to \mathscr{F} \to \mathscr{G} \to \mathscr{H} \to 0$ is an exact sequence of sheaves of abelian groups, then

$$0 \longrightarrow \mathscr{F}(U) \longrightarrow \mathscr{G}(U) \longrightarrow \mathscr{H}(U) \longrightarrow 0$$

need not be exact.

Solution *Consider the sequence,*

$$0 \longrightarrow \underline{\mathbb{Z}} \longrightarrow \mathscr{O}_X \longrightarrow \mathscr{O}_X^* \longrightarrow 1,$$

by Exercise 3.19, this sequence is exact. But, by Exercise 3.17,

$$0 \longrightarrow \underline{\mathbb{Z}}(U) \longrightarrow \mathscr{O}_X(U) \longrightarrow \mathscr{O}_X^*(U) \longrightarrow 1,$$

may not exact.

Proposition 3.6.8 (Left exactness of pushforward)

Suppose $0 \to \mathscr{F} \to \mathscr{G} \to \mathscr{H}$ is an exact sequence of sheaves of abelian groups on X. If $\pi : X \to Y$ is a continuous map, then sequence

$$0 \longrightarrow \pi_*\mathscr{F} \longrightarrow \pi_*\mathscr{G} \longrightarrow \pi_*\mathscr{H}$$

is exact.

Proof Let $V \subseteq Y$ be any open subset of Y, since $\pi : X \to Y$ is continuous, $\pi^{-1}(V)$ is open in X. Since the functor of "sections over $\pi^{-1}(V)$" is a left exact functor (Proposition), the sequence

$$0 \longrightarrow \mathscr{F}(\pi^{-1}(V)) \longrightarrow \mathscr{G}(\pi^{-1}(V)) \longrightarrow \mathscr{H}(\pi^{-1}(V))$$

is exact, i.e., the sequence

$$0 \longrightarrow \pi_*\mathscr{F}(V) \longrightarrow \pi_*\mathscr{G}(V) \longrightarrow \pi_*\mathscr{H}(V)$$

is exact for all open subset $V \subseteq Y$. Hence, for all $p \in Y$, we have the exact sequence (look above sequence as exact sequence of presheaves, and use Proposition 3.4.11).

$$0 \longrightarrow \pi_*\mathscr{F}_p \longrightarrow \pi_*\mathscr{G}_p \longrightarrow \pi_*\mathscr{H}_p$$

By Proposition 3.6.5, the sequence

$$0 \longrightarrow \pi_*\mathscr{F} \longrightarrow \pi_*\mathscr{G} \longrightarrow \pi_*\mathscr{H}$$

is exact. □

Proposition 3.6.9 (Left exactness of $\mathscr{H}om$)

Suppose $\mathscr{F}$ is a sheaf of abelian groups on a topological space X. Then

(i) $\mathscr{H}om(\mathscr{F}, \square)$ is a left-exact covariant functor $\mathbf{Ab}_X \to \mathbf{Ab}_X$;

(ii) $\mathscr{H}om(\square, \mathscr{F})$ is a left-exact contravariant functor $\mathbf{Ab}_X \to \mathbf{Ab}_X$.

Proof

(i) Let

$$0 \longrightarrow \mathscr{G} \xrightarrow{\alpha} \mathscr{G}' \xrightarrow{\beta} \mathscr{G}''$$

be an exact sequence of sheaves. We need to show that the sequence,

$$0 \longrightarrow \mathscr{H}om(\mathscr{F}, \mathscr{G}) \xrightarrow{\alpha_*} \mathscr{H}om(\mathscr{F}, \mathscr{G}') \xrightarrow{\beta_*} \mathscr{H}om(\mathscr{F}, \mathscr{G}''),$$

is exact.

Let U be any open subset of X, consider the sequence of abelian groups.

$$0 \longrightarrow \operatorname{Hom}(\mathscr{F}|_U, \mathscr{G}|_U) \xrightarrow{\alpha_*} \operatorname{Hom}(\mathscr{F}|_U, \mathscr{G}'|_U) \xrightarrow{\beta_*} \operatorname{Hom}(\mathscr{F}|_U, \mathscr{G}''|_U)$$

(1) $\alpha_* : \operatorname{Hom}(\mathscr{F}|_U, \mathscr{G}|_U) \to \operatorname{Hom}(\mathscr{F}|_U, \mathscr{G}'|_U)$ is injective.

Let $\varphi \in \operatorname{Hom}(\mathscr{F}|_U, \mathscr{G}|_U)$ with $\alpha_*(\varphi) = 0$, then $\alpha \circ \varphi = 0$. Since α is monomorphism, $\varphi = 0$, which implies that α_* is injective.

(2) $\operatorname{Ker} \beta_* = \operatorname{Im} \alpha_*$.

Let $\psi \in \operatorname{Ker} \beta_* \subseteq \operatorname{Hom}(\mathscr{F}|_U, \mathscr{G}'|_U)$, then $\beta \circ \psi = 0$. Since $\operatorname{Im} \alpha = \operatorname{Ker} \beta$, for any open subset V of U and section $s \in \mathscr{F}(V)$, we have $\psi_V(s) \in \operatorname{Im} \alpha_V$. Then there exists unique $\theta_V(s) \in \mathscr{G}(V)$ such that $\alpha_V(\theta_V(s)) = \psi_V(s)$, i.e., $\alpha_V \circ \theta_V = \psi_V$ for any open subset V of U. Hence, $\alpha \circ \theta = \psi$, which implies that $\psi \in \operatorname{Im} \alpha_*$.

Conversely, let $\psi \in \operatorname{Im} \alpha_*$, then there exists $\theta \in \operatorname{Hom}(\mathscr{F}|_U, \mathscr{G}|_U)$ such that $\alpha \circ \theta = \psi$. Then for any open subset V of U, we have $\alpha_V \circ \theta_V = \psi_V$. Note that $\operatorname{Ker} \beta_V = \operatorname{Im} \alpha_V$, we have $\beta_V \circ \alpha_V = 0$, and therefore $\beta_V \circ \psi_V = 0$, i.e., $\psi_V \in \operatorname{Ker} \beta_V$ for all $V \subseteq U$. Hence, $\beta \circ \psi = 0$, i.e., $\psi \in \operatorname{Ker} \beta_*$.

Hence, the sequence

$$0 \longrightarrow \operatorname{Hom}(\mathscr{F}|_U, \mathscr{G}|_U) \xrightarrow{\alpha_*} \operatorname{Hom}(\mathscr{F}|_U, \mathscr{G}'|_U) \xrightarrow{\beta_*} \operatorname{Hom}(\mathscr{F}|_U, \mathscr{G}''|_U)$$

is exact, i.e., the sequence

$$0 \longrightarrow \mathscr{H}om(\mathscr{F}, \mathscr{G})(U) \xrightarrow{\alpha_*} \mathscr{H}om(\mathscr{F}, \mathscr{G}')(U) \xrightarrow{\beta_*} \mathscr{H}om(\mathscr{F}, \mathscr{G}'')(U), \tag{3.5}$$

is exact for any open subset U of X. Look exact sequence 3.5 as exact sequence of presheaves, by Proposition 3.4.11. The sequence,

$$0 \longrightarrow \mathscr{H}om(\mathscr{F}, \mathscr{G})_p \xrightarrow{\alpha_{*p}} \mathscr{H}om(\mathscr{F}, \mathscr{G}')_p \xrightarrow{\beta_{*p}} \mathscr{H}om(\mathscr{F}, \mathscr{G}'')_p,$$

is exact for any $p \in X$. By Proposition 3.6.5, the sequence

$$0 \longrightarrow \mathscr{H}om(\mathscr{F}, \mathscr{G}) \xrightarrow{\alpha_*} \mathscr{H}om(\mathscr{F}, \mathscr{G}') \xrightarrow{\beta_*} \mathscr{H}om(\mathscr{F}, \mathscr{G}''),$$

is exact, and therefore $\mathscr{H}om(\mathscr{F}, \square)$ is a left-exact covariant functor $\mathbf{Ab}_X \to \mathbf{Ab}_X$.

(ii) Similar to (i).

$\square$

3.6.2 $\mathscr{O}_X$-modules form abelian category

Theorem 3.6.2

If $(X, \mathscr{O}_X)$ is a ringed space, then $\mathscr{O}_X$-modules form an abelian category.

Proof Let $(X, \mathscr{O}_X)$ be a ringed space. Let $\mathscr{F}, \mathscr{G}$ be sheaves of $\mathscr{O}_X$-module. Let $\varphi, \psi : \mathscr{F} \to \mathscr{G}$ be morphisms of sheaves of $\mathscr{O}_X$-modules. We define $\varphi + \psi : \mathscr{F} \to \mathscr{G}$ to be the map which on each open subset $U \subseteq X$ is the sum of the maps induced by φ, ψ, i.e.,

$$(\varphi + \psi)_U = \varphi_U + \psi_U.$$

This is clearly again a map of sheaves of $\mathscr{O}_X$-modules. It is also clear that composition of maps of $\mathscr{O}_X$-modules is bilinear with respect to this addition, i.e., for $\theta : \mathscr{E} \to \mathscr{F}$ and $\eta : \mathscr{G} \to \mathscr{H}$, we have

$$(\varphi + \psi) \circ \theta = \varphi \circ \theta + \psi \circ \theta$$

and

$$\eta \circ (\varphi + \psi) = \eta \circ \varphi + \eta \circ \psi.$$

We will denote 0 the sheaf of $\mathscr{O}_X$-modules which has constant value $\{0\}$ for all open subset $U \subseteq X$. Clearly this is both a final and an initial object of $\mathbf{Mod}_{\mathscr{O}_X}$.

Moreover, given a pair $\mathscr{F}, \mathscr{G}$ of sheaves of $\mathscr{O}_X$-modules we may define the direct sum as

$$\mathscr{F} \oplus \mathscr{G} = \mathscr{F} \times \mathscr{G}.$$

Thus $\mathbf{Mod}_{\mathscr{O}_X}$ is an additive category (Definition 2.5.1).

Let $\varphi : \mathscr{F} \to \mathscr{G}$ be a morphism of $\mathscr{O}_X$-modules. We may define $\operatorname{Ker} \varphi$ to be the subsheaf of $\mathscr{F}$ with sections

$$\operatorname{Ker} \varphi(U) := \{s \in \mathscr{F}(U) : \varphi(s) = 0 \in \mathscr{G}(U)\}$$

for all open subset $U \subseteq X$. It is easy to see that this is indeed a kernel in the category of $\mathscr{O}_X$-modules. Moreover, on the level of stalks we have

$$\operatorname{Ker}(\varphi)_p \cong \operatorname{Ker}(\varphi_p)$$

for all $p \in X$.

On the other hand, we define $\operatorname{Coker} \varphi$ as the sheaf of $\mathscr{O}_X$-modules associated to the presheaf of $\mathscr{O}_X$-modules defined by the rule

$$U \mapsto \operatorname{Coker}_{\mathrm{pre}}(\mathscr{G}(U) \to \mathscr{F}(U)) = \mathscr{F}(U)/\varphi_U(\mathscr{G}(U)),$$

i.e., $\operatorname{Coker}(\mathscr{F} \to \mathscr{G}) = (\operatorname{Coker}_{\mathrm{pre}}(\mathscr{F} \to \mathscr{G}))^{\mathrm{sh}}$. Since taking stalks commutes with taking sheafification (see Proposition 3.4.11 or Proposition 3.6.3), we see that

$$\operatorname{Coker}(\varphi)_p \cong \operatorname{Coker}(\varphi_p)$$

for all $p \in X$. Thus the map $\mathscr{G} \to \operatorname{Coker} \varphi$ is epimorphism, by Proposition 3.4.13. To show that this is a cokernel, consider the following diagram.

$$\begin{tikzcd}
\mathscr{F} \arrow[r, "\varphi"'] \arrow[rr, bend left, "0"] \arrow[rrr, bend left, "0"] \arrow[drr, "0"'] & \mathscr{G} \arrow[r] \arrow[dr, "\beta"] & \operatorname{Coker}_{\mathrm{pre}} \varphi \arrow[r, "\mathrm{sh}"] \arrow[d, dashed] & \operatorname{Coker} \varphi \arrow[dl, dashed] \\
 & & \mathscr{H} &
\end{tikzcd} \tag{3.6}$$

In diagram (3.6), $\beta : \mathscr{G} \to \mathscr{H}$ is a morphism of $\mathscr{O}_X$-modules such that $\beta \circ \varphi$ is zero, then we get for every open subset $U \subseteq X$ a map induced by β from $\operatorname{Coker}_{\text{pre}} \varphi(U)$ into $\mathscr{H}(U)$. By the universal property of sheafification we obtain a map $\operatorname{Coker} \varphi \to \mathscr{H}$ such that β is equal to the composition $\mathscr{G} \to \operatorname{Coker} \varphi \to \mathscr{H}$. The morphism $\operatorname{Coker} \varphi \to \mathscr{H}$ is unique because of $\mathscr{G} \to \operatorname{Coker} \varphi$ is surjective.

By the definition of abelian category (Definition 2.5.5) and above discussion, it suffice to show that every monomorphism is the kernel of its cokernel and every epimorphism is the cokernel of its kernel.

(1) Every monomorphism is the kernel of its cokernel.

Let $\varphi : \mathscr{F} \to \mathscr{G}$ be a monomorphism, it suffices to show that

$$\mathscr{F} \cong \operatorname{Ker}(\mathscr{G} \to \operatorname{Coker} \varphi).$$

By Proposition 3.4.4, it suffices to show that

$$\mathscr{F}_p \cong \operatorname{Ker}(\mathscr{G} \to \operatorname{Coker} \varphi)_p$$

for all $p \in X$.

Since $\varphi : \mathscr{F} \to \mathscr{G}$ is monomorphism, by Proposition 3.4.12, $\varphi_p : \mathscr{F}_p \to \mathscr{G}_p$ is injective for all $p \in X$, and therefore

$$\begin{aligned}\mathscr{F}_p &\cong \operatorname{Ker}(\mathscr{G}_p \to \operatorname{Coker}(\varphi_p)) \\ &\cong \operatorname{Ker}(\mathscr{G}_p \to (\operatorname{Coker} \varphi)_p) \\ &\cong (\operatorname{Ker}(\mathscr{G} \to \operatorname{Coker} \varphi))_p\end{aligned}$$

for all $p \in X$, then we done.

(2) Every epimorphism is the cokernel of its kernel.

Let $\psi : \mathscr{F} \to \mathscr{G}$ be an epimorphism, it suffices to show that

$$\mathscr{G} \cong \operatorname{Coker}(\operatorname{Ker} \psi \to \mathscr{F}).$$

By Propositon 3.4.4, it suffices to show that

$$\mathscr{G}_p \cong \operatorname{Coker}(\operatorname{Ker} \psi \to \mathscr{F})_p$$

for all $p \in X$.

Since $\psi : \mathscr{F} \to \mathscr{G}$ is epimorphism, by Proposition 3.4.13, $\varphi_p : \mathscr{F}_p \to \mathscr{G}_p$ is surjective for all $p \in X$, and therefore

$$\begin{aligned}\mathscr{G}_p &\cong \operatorname{Coker}(\operatorname{Ker}(\varphi_p) \to \mathscr{F}_p) \\ &\cong \operatorname{Coker}((\operatorname{Ker} \varphi)_p \to \mathscr{F}_p) \\ &\cong \operatorname{Coker}(\operatorname{Ker} \psi \to \mathscr{F})_p\end{aligned}$$

for all $p \in X$.

Hence, $\mathscr{O}_X$-modules form an abelian category. □

Remark 3.36 Many facts about sheaves of abelian groups carry over to $\mathscr{O}_X$-modules with out change, because a sequence of $\mathscr{O}_X$-modules is exact if and only if the underlying sequence of sheaves of abelian groups is exact. It is easily check that all of the statements of the earlier propositions in § 3.6 is also hold for $\mathscr{O}_X$-modules, when modified appropriately. For example:

Example 3.12 $\mathscr{H}om_{\mathscr{O}_X}(\square, \square)$ is a left-exact contravariant functor in its first argument and a left-exact covariant functor in its second argument.

Note ***Conclusion.*** *Just as presheaves of abelian groups on a topological space form an abelian category because all abelian-categorical notions make sense open set by open set, sheaves of abelian groups on a topological space form an abelian category because all abelian-categorical notions make sense stalk by stalk.*

3.6.3 Tensor product sheaf

We end with a useful construction using some of the idea in this section.

Definition 3.6.1 (Tensor product sheaf)

*Let $(X, \mathscr{O}_X)$ be a ringed space and let $\mathscr{F}$ and $\mathscr{G}$ be $\mathscr{O}_X$-modules. We define the **tensor product presheaf***

$$\mathscr{F} \otimes_{\mathrm{pre}, \mathscr{O}_X} \mathscr{G}$$

as the rule which assigns to $U \subseteq X$ open the $\mathscr{O}_X(U)$-module $\mathscr{F}(U) \otimes_{\mathscr{O}_X(U)} \mathscr{G}(U)$.
*Having defined this we define the **tensor product sheaf** as the sheafification of the above:*

$$\mathscr{F} \otimes_{\mathscr{O}_X} \mathscr{G} = (\mathscr{F} \otimes_{\mathrm{pre}, \mathscr{O}_X} \mathscr{G})^{\mathrm{sh}}.$$

Definition 3.6.2 (Bilinear maps of sheaves of $\mathscr{O}_X$-modules)

Let $(X, \mathscr{O}_X)$ be a ringed space. Let $\mathscr{F}$, $\mathscr{G}$, and $\mathscr{H}$ be $\mathscr{O}_X$-modules. A bilinear map $f : \mathscr{F} \times \mathscr{G} \to \mathscr{H}$ of sheaves of $\mathscr{O}_X$-modules is a map of sheaves of sets as indicated such that for every open subset $U \subseteq X$ the induced map

$$\mathscr{F}(U) \times \mathscr{G}(U) \to \mathscr{H}(U)$$

is an $\mathscr{O}_X(U)$-bilinear map of modules.

Proposition 3.6.10

Tensor product of two $\mathscr{O}_X$-modules satisfies categorical definition.

Proof Let $(X, \mathscr{O}_X)$ be a ringed space. Let $\mathscr{F}$, $\mathscr{G}$, and $\mathscr{H}$ be $\mathscr{O}_X$-modules. Let U be open subset of X, and consider the following diagram,

$$\begin{tikzcd}
\mathscr{F}(U) \times \mathscr{G}(U) \arrow[r, "t_U"] \arrow[dr, "f_U"'] & \mathscr{F}(U) \otimes_{\mathrm{pre}, \mathscr{O}_X(U)} \mathscr{G}(U) \arrow[d, dashed] \\
& \mathscr{H}(U),
\end{tikzcd}$$

where t_U and f_U are $\mathscr{O}_X(U)$-bilinear map. By the universal property of tensor product of modules, there exists unique $\mathscr{O}_X(U)$-map $\mathscr{F}(U) \otimes_{\mathrm{pre}, \mathscr{O}_X(U)} \mathscr{G}(U) \to \mathscr{H}(U)$. Then we have a unique $\mathscr{O}_X$-map (presheaf) $\mathscr{F} \otimes_{\mathrm{pre}, \mathscr{O}_X} \mathscr{G} \to \mathscr{H}$. By the universal property of sheafification, there exists a unique $\mathscr{O}_X$-map (sheaf) $(\mathscr{F} \otimes_{\mathscr{O}_X} \mathscr{G})^{\mathrm{sh}} \to \mathscr{H}$, such that the following diagram commutes.

$$\begin{tikzcd}
\mathscr{F} \otimes_{\mathrm{pre}, \mathscr{O}_X} \mathscr{G} \arrow[r, "\mathrm{sh}"] \arrow[dr] & \mathscr{F} \otimes_{\mathscr{O}_X} \mathscr{G} \arrow[d, dashed] \\
& \mathscr{H}
\end{tikzcd}$$

Hence the following diagram is commutative, and therefore tensor product of two $\mathscr{O}_X$-modules satisfies categorical definition.

$$\begin{tikzcd}
\mathscr{F} \times \mathscr{G} \arrow[r, "t"] \arrow[dr, "f"'] & \mathscr{F} \otimes_{\mathrm{pre}, \mathscr{O}_X} \mathscr{G} \arrow[r, "\mathrm{sh}"] \arrow[d] & \mathscr{F} \otimes_{\mathscr{O}_X} \mathscr{G} \arrow[dl] \\
& \mathscr{H} &
\end{tikzcd}$$

□

Proposition 3.6.11 (The tensor product of stalks is the stalk of the tensor product)

Let $(X, \mathscr{O}_X)$ be a ringed space. Let $\mathscr{F}, \mathscr{G}$ be $\mathscr{O}_X$-modules. Let $p \in X$. There is a canonical isomorphism of $\mathscr{O}_{X,p}$-modules

$$(\mathscr{F} \otimes_{\mathscr{O}_X} \mathscr{G})_p \cong \mathscr{F}_p \otimes_{\mathscr{O}_{X,p}} \mathscr{G}_p.$$

Proof Let $U' \supseteq U$ be open subsets of X. Consider the following diagram.

$$\begin{tikzcd}
\mathscr{F}(U') \otimes_{\mathrm{pre},\mathscr{O}_X(U')} \mathscr{G}(U') \arrow[r, "\mathrm{sh}_{U'}"] \arrow[drr, dashed] & \mathscr{F}(U') \otimes_{\mathscr{O}_X(U')} \mathscr{G}(U') \arrow[drrr] \arrow[drr, dashed] & & & \\
& & \mathscr{F}(U) \otimes_{\mathrm{pre},\mathscr{O}_X(U)} \mathscr{G}(U) \arrow[r, "\mathrm{sh}_U"] & \mathscr{F}(U) \otimes_{\mathscr{O}_X(U)} \mathscr{G}(U) \arrow[r] & (\mathscr{F} \otimes_{\mathscr{O}_X} \mathscr{G})_p \arrow[d, dashed, shift right] \\
\mathscr{F}(U') \times \mathscr{G}(U') \arrow[uu] \arrow[rrr] \arrow[d] \arrow[drr, dashed] & & & \mathscr{F}(U') \arrow[d] \arrow[dr] & \mathscr{F}_p \otimes_{\mathscr{O}_{X,p}} \mathscr{G}_p \arrow[u, dashed, shift right] \\
\mathscr{G}(U') \arrow[drrr] \arrow[drr] & & \mathscr{F}(U) \times \mathscr{G}(U) \arrow[uu] \arrow[r] \arrow[d] & \mathscr{F}(U) \arrow[r] & \mathscr{F}_p \\
& & \mathscr{G}(U) \arrow[r] & \mathscr{G}_p & \mathscr{F}_p \times \mathscr{G}_p \arrow[l] \arrow[u] \arrow[uu, bend right]
\end{tikzcd}$$

All dashed arrows given by the universal property, hence, there is a canonical isomorphism of $\mathscr{O}_{X,p}$-modules

$$(\mathscr{F} \otimes_{\mathscr{O}_X} \mathscr{G})_p \cong \mathscr{F}_p \otimes_{\mathscr{O}_{X,p}} \mathscr{G}_p.$$

□

3.7 The inverse image sheaf

We next describe a notion that is fundamental, but rather intricate. Suppose we have a continuous map $\pi : X \to Y$. If $\mathscr{F}$ is a sheaf on X, we have defined the pushforward (or direct image) sheaf $\pi_*\mathscr{F}$, which is a sheaf on Y. There is also a notion of **inverse image sheaf**. (We will not call it the pullback sheaf, reserving that name for a later construction for quasicoherent sheaves.) This is a covariant functor π^{-1} from sheaves on Y to sheaves on X. If the sheaves on Y have some additional structure (e.g., group or ring), then this structure is respected by π^{-1}.

3.7.1 Definition of the inverse image sheaf

Definition 3.7.1 (The inverse image sheaf)

We define the inverse image π^{-1} as the left adjoint to π_*.

Remark 3.37 This definition is given by adjoint: elegant but abstract.

This isn't really a definition; we need a construction to show that the adjoint exists. Note that we then get canonical maps $\pi^{-1}\pi_*\mathscr{F} \to \mathscr{F}$ (associated to the identity in $\mathrm{Mor}_Y(\pi_*\mathscr{F}, \pi_*\mathscr{F})$) and $\mathscr{G} \to \pi_*\pi^{-1}\mathscr{G}$ (associated to the identity in $\mathrm{Mor}_X(\pi^{-1}\mathscr{G}, \pi^{-1}\mathscr{G})$).

$$\begin{tikzcd} & \pi^{-1}\mathscr{G} \arrow[r] & \mathscr{F} \\ X \arrow[d, "\pi"'] & \mathscr{G} \arrow[r] & \pi_*\mathscr{F} \\ Y & & \end{tikzcd}$$

Remark 3.38 How to get canonical maps $\pi^{-1}\pi_*\mathscr{F} \to \mathscr{F}$ and $\mathscr{G} \to \pi_*\pi^{-1}\mathscr{G}$?

By Definition 3.7.1, π^{-1} is left adjoint to π_*. Then we have canonical isomorphisms

$$\mathrm{Mor}_X(\pi^{-1}\pi_*\mathscr{F}, \mathscr{F}) \cong \mathrm{Mor}_Y(\pi_*\mathscr{F}, \pi_*\mathscr{F})$$

and

$$\mathrm{Mor}_X(\pi^{-1}\mathscr{G}, \pi^{-1}\mathscr{G}) \cong \mathrm{Mor}_Y(\mathscr{G}, \pi_*\pi^{-1}\mathscr{G}).$$

Let $\pi^{-1}\pi_*\mathscr{F} \to \mathscr{F}$ be the image of $\mathrm{id}_{\pi_*\mathscr{F}}$ in $\mathrm{Mor}_X(\pi^{-1}\pi_*\mathscr{F}, \mathscr{F})$, and $\mathscr{G} \to \pi_*\pi^{-1}\mathscr{G}$ be the image of $\mathrm{id}_{\pi^{-1}\mathscr{G}}$ in $\mathrm{Mor}_Y(\mathscr{G}, \pi_*\pi^{-1}\mathscr{G})$.

Construction: concrete but ungly.

Proposition 3.7.1

Define the temporary notation

$$\pi_{\text{pre}}^{-1}\mathscr{G}(U) = \varinjlim_{V \supseteq \pi(U)} \mathscr{G}(V).$$

Then $\pi_{\text{pre}}^{-1}\mathscr{G}$ defines a presheaf on X.

Remark 3.39 Recall the explicit description of colimit: sections of π_{pre}^{-1} over U are sections on open sets containing $\pi(U)$, with an equivalence relation. Note that $\pi(U)$ won't be an open set in general.

Proof We first define the restriction map. Let $U' \supseteq U$ be open subsets of X. Define the restriction map $\mathrm{res}_{U',U} : \pi_{\text{pre}}^{-1}\mathscr{G}(U') \to \pi_{\text{pre}}^{-1}\mathscr{G}(U)$ by setting

$$s_{\pi(U')} \mapsto s_{\pi(U)}.$$

We shall to show the restriction map is well-defined. Let $s_{\pi(U')} = t_{\pi(U')}$ in $\pi_{\text{pre}}^{-1}\mathscr{G}(U')$, then $s_{\pi(U')} = \overline{(s, V_s)}$ and $t_{\pi(U')} = \overline{(t, V_t)}$. Hence, there exists an open subset $(\pi(U') \subseteq) W \subseteq V_s \cap V_t$ such that $s|_W = t|_W$. Since $U' \supseteq U$, we have $\pi(U') \supseteq \pi(U)$, and therefore $\pi(U) \subseteq \pi(U') \subseteq W \subseteq V_s \cap V_t$. Hence, $s_{\pi(U)} = t_{\pi(U)}$, which implies that the restriction map is well-defined.

Consider $\mathrm{res}_{U,U} : \pi_{\text{pre}}^{-1}\mathscr{G}(U) \to \pi_{\text{pre}}^{-1}\mathscr{G}(U)$, clearly, $\mathrm{res}_{U,U} = \mathrm{id}_{\pi_{\text{pre}}^{-1}\mathscr{G}(U)}$.

Let $U'' \supseteq U' \supseteq U$ be open sets of X, pick $s_{\pi(U'')} \in \pi_{\text{pre}}^{-1}\mathscr{G}(U'')$, then

$$\mathrm{res}_{U',U} \circ \mathrm{res}_{U'',U'}(s_{\pi(U'')}) = \mathrm{res}_{U',U}(s_{\pi(U')}) = s_{\pi(U)} = \mathrm{res}_{U'',U}(s_{\pi(U'')}),$$

i.e., $\mathrm{res}_{U',U} \circ \mathrm{res}_{U'',U'} = \mathrm{res}_{U'',U}$.

Hence, $\pi_{\text{pre}}^{-1}\mathscr{G}$ defines a presheaf on X. □

✍ **Exercise 3.21** Show that $\pi_{\text{pre}}^{-1}\mathscr{G}$ needn't form a sheaf.

Proof Let $X = \{x_1, x_2\}$ and $Y = \{y_1, y_2\}$. X endowed with the discrete topology, $\mathscr{P}(X) = \{\varnothing, X, \{x_1\}, \{x_2\}\}$, and Y endowed with the topology $\mathscr{T}_Y = \{\varnothing, \{y_1\}, Y\}$. Let $\pi : X \to Y$ by setting

$$\pi(x_1) = y_1, \quad \pi(x_2) = y_2.$$

Clearly, π is continuous.

Let $\mathscr{G} = \underline{\mathbb{Z}} \in \mathbf{Sets}_Y$ be the constant sheaves, then

$$\pi_{\text{pre}}^{-1}\mathscr{G}(\{x_1\}) = \varinjlim_{V \supseteq \{y_1\}} \mathscr{G}(V) = \mathbb{Z},$$
$$\pi_{\text{pre}}^{-1}\mathscr{G}(\{x_2\}) = \varinjlim_{V \supseteq \{y_2\}} \mathscr{G}(V) = \mathbb{Z},$$

and

$$\pi_{\text{pre}}^{-1}\mathscr{G}(X) = \varinjlim_{V \supseteq Y} \mathscr{G}(V) = \mathbb{Z}.$$

We verify that $\pi_{\text{pre}}^{-1}\mathscr{G}$ violates the gluability axiom. Choose $s_1 = 1 \in \pi_{\text{pre}}^{-1}\mathscr{G}(\{x_1\}) = \mathbb{Z}$ and $s_2 = 2 \in \pi_{\text{pre}}^{-1}\mathscr{G}(\{x_2\}) = \mathbb{Z}$. Since $\{x_1\} \cap \{x_2\} = \varnothing$, $s_1|_{\{x_1\}\cap\{x_2\}} = s_2|_{\{x_1\}\cap\{x_2\}}$. If $\pi_{\text{pre}}^{-1}\mathscr{G}$ is a sheaf, then there exist a global section $s \in \pi_{\text{pre}}^{-1}\mathscr{G}(X) = \mathbb{Z}$ such that $s|_{\{x_1\}} = 1$ and $s|_{\{x_2\}} = 2$. It is impossible!

Hence, $\pi_{\text{pre}}^{-1}\mathscr{G}$ needn't form a sheaf. □

Definition 3.7.2 (The inverse image sheaf)

*Suppose continuous map $\pi : X \to Y$, let $\mathscr{G}$ be a sheaf over Y, define the **inverse image of** $\mathscr{G}$ by*

$$\pi^{-1}\mathscr{G} := (\pi_{\text{pre}}^{-1}\mathscr{G})^{\text{sh}}.$$

Note that π^{-1} is a functor from sheaves on Y to sheaves on X. The next theorem shows that π^{-1} is indeed left-adjoint to π_*.

Theorem 3.7.1 ((π^{-1}, π_*) are adjoint)

If $\pi : X \to Y$ is a continuous map, and $\mathscr{F}$ is a sheaf on X and $\mathscr{G}$ is a sheaf on Y, then we have a natural bijection

$$\mathrm{Mor}_X(\pi^{-1}\mathscr{G}, \mathscr{F}) \overset{\sim}{\longleftrightarrow} \mathrm{Mor}_Y(\mathscr{G}, \pi_*\mathscr{F}).$$

Proof Let $\mathscr{F}$ and $\mathscr{F}'$ are sheaves on X, $\mathscr{G}$ and $\mathscr{G}'$ are sheaves on Y, $\varphi : \mathscr{F} \to \mathscr{F}'$ and $\psi : \mathscr{G} \to \mathscr{G}'$ are morphisms of sheaves. Define $\tau_{\mathscr{G},\mathscr{F}} : \mathrm{Mor}_X(\pi^{-1}\mathscr{G}, \mathscr{F}) \to \mathrm{Mor}_Y(\mathscr{G}, \pi_*\mathscr{F})$ as follow: Let $\alpha \in \mathrm{Mor}_X(\pi^{-1}\mathscr{G}, \mathscr{F})$, let $V \subseteq Y$ be an open subset, then $\pi^{-1}(V) \subseteq X$ open subset and $V \supseteq \pi(\pi^{-1}(V))$, consider the following sequence,

$$\begin{array}{l} \mathscr{G}(V) \xrightarrow{\eta_V} \pi_{\text{pre}}^{-1}\mathscr{G}(\pi^{-1}(V)) = \varinjlim_{V' \supseteq \pi(\pi^{-1}(V))} G(V') \\ \qquad\qquad\qquad \Big\downarrow \text{sh}_{\pi^{-1}(V)} \\ \qquad\qquad\qquad \pi^{-1}\mathscr{G}(\pi^{-1}(V)) \xrightarrow{\quad \alpha_{\pi^{-1}(V)} \quad} \mathscr{F}(\pi^{-1}(V)) = \pi_*\mathscr{F}(V), \end{array}$$

where η_V is natural embedding, let

$$\tau_{\mathscr{G},\mathscr{F}}(\alpha)_V = \alpha_{\pi^{-1}(V)} \circ \text{sh}_{\pi^{-1}(V)} \circ \eta_V.$$

Since $\alpha_{\pi^{-1}(V)}$, $\text{sh}_{\pi^{-1}(V)}$, and η_V are well-defined, $\tau_{\mathscr{G},\mathscr{F}}(\alpha)_V$ is well-defined. Hence we defined $\tau_{\mathscr{G},\mathscr{F}}$.

Next, we shall to define the inverse of $\tau_{\mathscr{F},\mathscr{G}}$. Let $U \subseteq X$ be any open subset of X, then $\pi(U) \subseteq Y$, for any

$V \supseteq \pi(U)$, we have $\pi^{-1}(V) \supseteq U$. Let $\beta \in \mathrm{Mor}_Y(\mathscr{G}, \pi_*\mathscr{F})$, consider the following diagram,

$$\begin{tikzcd}
\pi^{-1}_{\mathrm{pre}}\mathscr{G}(U) \arrow[rrr, dashed] & & & \mathscr{F}(U) \\
& \mathscr{G}(V') \arrow[r, "\beta_{V'}"] \arrow[d, "\mathrm{res}"'] \arrow[ul] & \pi_*\mathscr{F}(V') \arrow[d, "\mathrm{res}"] \arrow[ur, "\mathrm{res}"'] & \\
& \mathscr{G}(V) \arrow[r, "\beta_V"] \arrow[uul, bend left] & \pi_*\mathscr{F}(V) \arrow[uur, bend right, "\mathrm{res}"'] &
\end{tikzcd}$$

dashed arrow given by the universal property of colimit, say $\chi_{\mathrm{pre},\beta,U} : \pi^{-1}_{\mathrm{pre}}\mathscr{G}(U) \to \mathscr{F}(U)$. Then, we defined a morphism of presheaves $\chi_{\mathrm{pre},\beta} : \pi^{-1}_{\mathrm{pre}}\mathscr{G} \to \mathscr{F}$. Now, consider the following diagram,

$$\begin{tikzcd}
\pi^{-1}_{\mathrm{pre}}\mathscr{G} \arrow[r, "\mathrm{sh}"] \arrow[dr, "\chi_{\mathrm{pre},\beta}"'] & \pi^{-1}\mathscr{G} \arrow[d, dashed, "\chi_\beta"] \\
& \mathscr{F},
\end{tikzcd}$$

where $\chi_\beta : \pi^{-1}\mathscr{G} \to \mathscr{F}$ given by the universal property of sheafification. Define $\theta_{\mathscr{G},\mathscr{F}} : \mathrm{Mor}_Y(\mathscr{G}, \pi_*\mathscr{F}) \to \mathrm{Mor}_X(\pi^{-1}\mathscr{G}, \mathscr{F})$ by setting

$$\theta_{\mathscr{G},\mathscr{F}}(\beta) = \chi_\beta.$$

By above discussion, $\theta_{\mathscr{G},\mathscr{F}}$ is well-defined. Now, we need to prove $\theta_{\mathscr{G},\mathscr{F}}$ is, indeed, the inverse of $\tau_{\mathscr{G},\mathscr{F}}$.

Let $\alpha \in \mathrm{Mor}_X(\pi^{-1}\mathscr{G}, \mathscr{F})$, let U be any open subset of X, then $\pi(U) \subseteq Y$, for all open subset $V \supseteq \pi(U)$, we have

$$\tau_{\mathscr{G},\mathscr{F}}(\alpha)_V = \alpha_{\pi^{-1}(V)} \circ \mathrm{sh}_{\pi^{-1}(V)} \circ \eta_V.$$

Consider the following diagram.

$$\begin{tikzcd}
\pi^{-1}\mathscr{G}(U) \arrow[drrr, shift left, "\chi_{\tau_{\mathscr{G},\mathscr{F}}(\alpha),U}"] \arrow[drrr, shift right, "\alpha_U"'] & & & \\
\pi^{-1}_{\mathrm{pre}}\mathscr{G}(U) \arrow[u, "\mathrm{sh}_U"] \arrow[rrr, "\chi_{\mathrm{pre},\tau_{\mathscr{G},\mathscr{F}}(\alpha),U}"'] & & & \mathscr{F}(U) \\
& \mathscr{G}(V') \arrow[r, "\tau_{\mathscr{G},\mathscr{F}}(\alpha)_{V'}"] \arrow[d, "\mathrm{res}"'] \arrow[ul, "\eta_{V'}"] & \pi_*\mathscr{F}(V') \arrow[d, "\mathrm{res}"] \arrow[ur, "\mathrm{res}"'] & \\
& \mathscr{G}(V) \arrow[r, "\tau_{\mathscr{G},\mathscr{F}}(\alpha)_V"] \arrow[uul, bend left, "\eta_V"] & \pi_*\mathscr{F}(V) \arrow[uur, bend right, "\mathrm{res}"'] &
\end{tikzcd}$$

By the universal property of colimit, we have $\alpha_U \circ \mathrm{sh}_U = \chi_{\mathrm{pre},\tau_{\mathscr{G},\mathscr{F}}(\alpha),U}$, and therefore $\alpha \circ \mathrm{sh} = \chi_{\mathrm{pre},\tau_{\mathscr{G},\mathscr{F}}(\alpha)}$. By the universal property of sheafification, $\alpha = \chi_{\tau_{\mathscr{G},\mathscr{F}}(\alpha)}$. Hence,

$$\theta_{\mathscr{G},\mathscr{F}}(\tau_{\mathscr{G},\mathscr{F}}(\alpha)) = \chi_{\tau_{\mathscr{G},\mathscr{F}}(\alpha)} = \alpha,$$

i.e., $\theta_{\mathscr{G},\mathscr{F}} \circ \tau_{\mathscr{G},\mathscr{F}} = \mathrm{id}_{\mathrm{Mor}_X(\pi^{-1}\mathscr{G},\mathscr{F})}$.

Let $\beta \in \mathrm{Mor}_Y(\mathscr{G}, \pi_*\mathscr{F})$, then

$$\tau_{\mathscr{G},\mathscr{F}} \circ \theta_{\mathscr{G},\mathscr{F}}(\beta) = \tau_{\mathscr{G},\mathscr{F}}(\chi_\beta).$$

Let $V \subseteq Y$ be any open subset of Y, then $\pi^{-1}(V)$ open in X, and

$$\tau_{\mathscr{G},\mathscr{F}}(\chi_\beta)_V = \chi_{\beta,\pi^{-1}(V)} \circ \mathrm{sh}_{\pi^{-1}(V)} \circ \eta_V.$$

Let $s \in \mathscr{G}(V)$, then

$$\begin{aligned}\chi_{\beta,\pi^{-1}(V)} \circ \mathrm{sh}_{\pi^{-1}(V)} \circ \eta_V(s) =& \chi_{\beta,\pi^{-1}(V)} \circ \mathrm{sh}_{\pi^{-1}(V)}(s_{\pi^{-1}(V)}) \\ =& \chi_{\mathrm{pre},\beta,\pi^{-1}(V)}(s_{\pi^{-1}(V)}) \\ =& \beta_V(s),\end{aligned}$$

which implies that $\chi_{\beta,\pi^{-1}(V)} \circ \mathrm{sh}_{\pi^{-1}(V)} \circ \eta_V = \beta_V$. Hence,

$$\tau_{\mathscr{G},\mathscr{F}} \circ \theta_{\mathscr{G},\mathscr{F}}(\beta) = \beta,$$

i.e., $\tau_{\mathscr{G},\mathscr{F}} \circ \theta_{\mathscr{G},\mathscr{F}} = \mathrm{id}_{\mathrm{Mor}_Y(\mathscr{G},\pi_*\mathscr{F})}$. Hence, $\theta_{\mathscr{G},\mathscr{F}}$ is, indeed, the inverse of $\tau_{\mathscr{G},\mathscr{F}}$.

Consider the following diagram,

$$\begin{tikzcd} \mathrm{Mor}_X(\pi^{-1}\mathscr{G}',\mathscr{F}) \arrow[r, "(\pi^{-1}(\psi))^*"] \arrow[d, "\tau_{\mathscr{G}',\mathscr{F}}"'] & \mathrm{Mor}_X(\pi^{-1}\mathscr{G},\mathscr{F}) \arrow[r, "\varphi_*"] \arrow[d, "\tau_{\mathscr{G},\mathscr{F}}"'] & \mathrm{Mor}_X(\pi^{-1}\mathscr{G},\mathscr{F}') \arrow[d, "\tau_{\mathscr{G},\mathscr{F}'}"] \\ \mathrm{Mor}_Y(\mathscr{G}',\pi_*\mathscr{F}) \arrow[r, "\psi^*"] & \mathrm{Mor}_Y(\mathscr{G},\pi_*\mathscr{F}) \arrow[r, "(\pi_*(\varphi))_*"] & \mathrm{Mor}_Y(\mathscr{G},\pi_*\mathscr{F}), \end{tikzcd} \tag{3.7}$$

where $\pi_*\varphi_V = \varphi_{\pi^{-1}(V)}$ for all open subset $V \subseteq Y$, and $\pi^{-1}(\psi)$ given by the universal property of sheafification, i.e., the following commutative diagram

$$\begin{tikzcd} \pi^{-1}\mathscr{G} \arrow[rrr, dashed, "\pi^{-1}(\psi)"] & & & \pi^{-1}\mathscr{G}' \\ \pi^{-1}_{\mathrm{pre}}\mathscr{G} \arrow[u, "\mathrm{sh}"] \arrow[rrr, dashed, "\pi^{-1}_{\mathrm{pre}}(\psi)"] & & & \pi^{-1}_{\mathrm{pre}}\mathscr{G}' \arrow[u, "\mathrm{sh}"'] \\ & \mathscr{G}(V') \arrow[ul, "\eta^{\mathscr{G}}_{V'}"] \arrow[r, "\psi_{V'}"] \arrow[d, "\mathrm{res}"'] & \mathscr{G}'(V') \arrow[ur, "\eta^{\mathscr{G}'}_{V'}"'] \arrow[d, "\mathrm{res}"] & \\ & \mathscr{G}(V) \arrow[uul, bend left, "\eta^{\mathscr{G}}_V"] \arrow[r, "\psi_V"] & \mathscr{G}'(V). \arrow[uur, bend right, "\eta^{\mathscr{G}'}_V"'] & \end{tikzcd}$$

Let $\alpha \in \mathrm{Mor}_X(\pi^{-1}\mathscr{G}',\mathscr{F})$, then we have

$$\psi^* \circ \tau_{\mathscr{G}',\mathscr{F}}(\alpha) = \tau_{\mathscr{G}',\mathscr{F}}(\alpha) \circ \psi$$

and

$$\tau_{\mathscr{G},\mathscr{F}} \circ (\pi^{-1}(\psi))^*(\alpha) = \tau_{\mathscr{G},\mathscr{F}}(\alpha \circ \pi^{-1}(\psi)).$$

Let $V \subseteq Y$ be any open subset, then

$$\begin{aligned}(\psi^* \circ \tau_{\mathscr{G}',\mathscr{F}}(\alpha))_V =& \tau_{\mathscr{G}',\mathscr{F}}(\alpha)_V \circ \psi_V \\ =& \alpha_{\pi^{-1}(V)} \circ \mathrm{sh}_{\pi^{-1}(V)} \circ \eta^{\mathscr{G}'}_V \circ \psi_V\end{aligned}$$

and

$$\begin{aligned}(\tau_{\mathscr{G},\mathscr{F}} \circ (\pi^{-1}(\psi))^*(\alpha))_V =& \tau_{\mathscr{G},\mathscr{F}}(\alpha \circ \pi^{-1}(\psi))_V \\ =& (\alpha \circ \pi^{-1}(\psi))_{\pi^{-1}(V)} \circ \mathrm{sh}_{\pi^{-1}(V)} \circ \eta^{\mathscr{G}}_V \\ =& \alpha_{\pi^{-1}(V)} \circ \pi^{-1}(\psi)_{\pi^{-1}(V)} \circ \mathrm{sh}_{\pi^{-1}(V)} \circ \eta^{\mathscr{G}}_V \\ =& \alpha_{\pi^{-1}(V)} \circ \mathrm{sh}_{\pi^{-1}(V)} \circ \pi^{-1}_{\mathrm{pre}}(\psi) \circ \eta^{\mathscr{G}}_V \\ =& \alpha_{\pi^{-1}(V)} \circ \mathrm{sh}_{\pi^{-1}(V)} \circ \eta^{\mathscr{G}'}_V \circ \psi_V.\end{aligned}$$

Hence, $(\psi^* \circ \tau_{\mathscr{G}',\mathscr{F}}(\alpha))_V = (\tau_{\mathscr{G},\mathscr{F}} \circ (\pi^{-1}(\psi))^*(\alpha))_V$, for any open subset $V \subseteq Y$, for all $\alpha \in \mathrm{Mor}_X(\pi^{-1}\mathscr{G}',\mathscr{F})$, and therefore

$$\psi^* \circ \tau_{\mathscr{G}',\mathscr{F}} = \tau_{\mathscr{G},\mathscr{F}} \circ (\pi^{-1}(\psi))^*.$$

Hence, the left side of diagram (3.7) commutes.

Let $\alpha \in \mathrm{Mor}_X(\pi^{-1}\mathscr{G}, \mathscr{F})$, let $V \subseteq Y$ be any open subset of Y, then

$$\begin{aligned}((\pi_*(\varphi))_* \circ \tau_{\mathscr{G},\mathscr{F}}(\alpha))_V =&\pi_*(\varphi)_V \circ \tau_{\mathscr{G},\mathscr{F}}(\alpha)_V \\ =&\pi_*(\varphi)_V \circ \alpha_{\pi^{-1}(V)} \circ \mathrm{sh}_{\pi^{-1}(V)} \circ \eta_V^{\mathscr{G}}\end{aligned}$$

and

$$\begin{aligned}(\tau_{\mathscr{G},\mathscr{F}'} \circ \varphi_*(\alpha))_V =&\tau_{\mathscr{G},\mathscr{F}'}(\varphi \circ \alpha)_V \\ =&(\varphi \circ \alpha)_{\pi^{-1}(V)} \circ \mathrm{sh}_{\pi^{-1}(V)} \circ \eta_V^{\mathscr{G}} \\ =&\varphi_{\pi^{-1}(V)} \circ \alpha_{\pi^{-1}(V)} \circ \mathrm{sh}_{\pi^{-1}(V)} \circ \eta_V^{\mathscr{G}} \\ =&\pi_*(\varphi)_V \circ \alpha_{\pi^{-1}(V)} \circ \mathrm{sh}_{\pi^{-1}(V)} \circ \eta_V^{\mathscr{G}}.\end{aligned}$$

Hence, $((\pi_*(\varphi))_* \circ \tau_{\mathscr{G},\mathscr{F}}(\alpha))_V = (\tau_{\mathscr{G},\mathscr{F}'} \circ \varphi_*(\alpha))_V$, for any open subset $V \subseteq Y$, for all $\alpha \in \mathrm{Mor}_X(\pi^{-1}\mathscr{G}, \mathscr{F})$, and therefore

$$(\pi_*(\varphi))_* \circ \tau_{\mathscr{G},\mathscr{F}} = \tau_{\mathscr{G},\mathscr{F}'} \circ \varphi_*.$$

Hence, the right side of diagram (3.7) commutes.

Consequently, diagram (3.7) is commutative, and therefore there is a natural bijection

$$\mathrm{Mor}_X(\pi^{-1}\mathscr{G}, \mathscr{F}) \stackrel{\sim}{\longleftrightarrow} \mathrm{Mor}_Y(\mathscr{G}, \pi_*\mathscr{F}).$$

□

Remark 3.40 **Alternative Proof Method:** Show that both side agree with the following third construction, which we denote $\mathrm{Mor}_{YX}(\mathscr{G}, \mathscr{F})$. A collection of maps $\Phi_{V,U} : \mathscr{G}(V) \to \mathscr{F}(U)$ (as U runs through all open sets of X, and V runs through all open sets of Y containing $\pi(U)$) is said to be **compatible** if for all open $U' \subseteq U \subseteq X$ and all open open $V' \subseteq V \subseteq Y$ with $\pi(U) \subseteq V$, $\pi(U') \subseteq V'$, the diagram

$$\begin{array}{ccc} \mathscr{G}(V) & \xrightarrow{\Phi_{V,U}} & \mathscr{F}(U) \\ {\scriptstyle \mathrm{res}_{V,V'}}\downarrow & & \downarrow{\scriptstyle \mathrm{res}_{U,U'}} \\ \mathscr{G}(V') & \xrightarrow{\Phi_{V',U'}} & \mathscr{F}(U') \end{array} \tag{3.8}$$

commutes. Define $\mathrm{Mor}_{YX}(\mathscr{G}, \mathscr{F})$ to be the set of all compatible collections $\Phi = \{\Phi_{V,U}\}$.

Remark 3.41 **("Stalk and skyscraper are an adjoint pair").** As a special case, if X is a point $p \in Y$, we see that $\pi^{-1}\mathscr{G}$ is the stalk $\mathscr{G}_p$ of $\mathscr{G}$, and maps from the stalk $\mathscr{G}_p$ to a set S are the same as maps of sheaves on Y from $\mathscr{G}$ to the skyscraper sheaf with set S supported at p.

Proof By Theorem 3.7.1, we have

$$\mathrm{Mor}_X(\pi^{-1}\mathscr{G}, \mathscr{F}) = \mathrm{Mor}_X(\mathscr{G}_p, \mathscr{F}) \cong \mathrm{Mor}_Y(\mathscr{G}, i_{p,*}(\mathscr{F})),$$

for all $p \in Y$, then we done. □

Proposition 3.7.2 (The stalks of $\pi^{-1}\mathscr{G}$ are the same as the stalks of $\mathscr{G}$)

If $\pi(p) = q$, there is a natural isomorphism

$$\mathscr{G}_q \xrightarrow{\sim} (\pi^{-1}\mathscr{G})_p.$$

Proof Let $i_p : \{p\} \in X$ and $i_q : \{q\} \in Y$, then we get skyscraper sheaves $i_{p,*}S$ and $i_{q,*}S$, where S is any set. Since stalk and skyscraper are an adjoint pair, we have

$$\mathrm{Mor}_{\mathbf{Sets}}(\mathscr{G}_q, S) \cong \mathrm{Mor}_Y(\mathscr{G}, i_{q,*}S),$$

and

$$\mathrm{Mor}_{\mathbf{Sets}}((\pi^{-1}\mathscr{G})_p, S) \cong \mathrm{Mor}_X(\pi^{-1}\mathscr{G}, i_{p,*}S).$$

Since (π^{-1}, π^*) are adjoint, we have

$$\mathrm{Mor}_X(\pi^{-1}\mathscr{G}, i_{p,*}S) \cong \mathrm{Mor}_Y(\mathscr{G}, \pi_*(i_{p,*}S)).$$

Now, let's calculate $\pi_*(i_{p,*}S)$. Let V be any open subset of Y, then

$$\begin{aligned}(\pi_*(i_{p,*}S))(V) = (i_{p,*}S)(\pi^{-1}(V)) &= \begin{cases} S & \text{if } p \in \pi^{-1}(V) \\ \{e\} & \text{if } p \notin \pi^{-1}(V) \end{cases} \\ &= \begin{cases} S & \text{if } q \in V \\ \{e\} & \text{if } q \notin V \end{cases} \\ &= (i_{q,*}S)(V).\end{aligned}$$

Hence,

$$\mathrm{Mor}_{\mathbf{Sets}}((\pi^{-1}\mathscr{G})_p, S) \cong \mathrm{Mor}_X(\pi^{-1}\mathscr{G}, i_{p,*}S) \cong \mathrm{Mor}_Y(\mathscr{G}, i_{q,*}S) \cong \mathrm{Mor}_{\mathbf{Sets}}(\mathscr{G}_q, S),$$

for all $S \in \mathbf{Sets}$, and therefore $\mathrm{Mor}_{\mathbf{Sets}}((\pi^{-1}\mathscr{G})_p, \Box)$ and $\mathrm{Mor}_{\mathbf{Sets}}(\mathscr{G}_q, \Box)$ are naturally isomorphic functor. By corollary of Yoneda's Lemma (Corollary 2.2.7 (iii)), we have

$$(\pi^{-1}\mathscr{G})_p \overset{\sim}{\longleftrightarrow} \mathscr{G}_q.$$

$\square$

Remark 3.42 Proposition 3.7.2, along with the notion of compatible germs, may give you a simple way of thinking about (and perhaps visualizing) inverse image sheaves. Closely related: you can think of sections of the inverse image sheaf as, locally, inverse images of sections on the target. (Those preferring the "espace étalé" perspective, §3.2.2, can check that the "inverse image of the espace étalé" is the "espace étalé" of the inverse image.)

Proposition 3.7.3 (Useful)

If U is an open subset of Y, $i : U \to Y$ is the inclusion, and $\mathscr{G}$ is a sheaf on Y. Then $i^{-1}\mathscr{G}$ is isomorphic to $\mathscr{G}|_U$.

Proof Since $i : U \to Y$ is inclusion, by Proposition 3.4.4, it suffices to show that $(i^{-1}\mathscr{G})_p \cong (\mathscr{G}|_U)_p$. By Proposition 3.7.2, we done. $\square$

Proposition 3.7.4 (π^{-1} is an exact functor)

(i) π^{-1} is an exact functor from sheaves of abelian groups on Y to sheaves of abelian groups on X.

(ii) π^{-1} is an exact functor from $\mathscr{O}_Y$-modules (on Y) to $(\pi^{-1}\mathscr{O}_Y)$-modules (on X).

Proof We only prove (i), essentially the same argument will show (ii).

Let $\mathscr{F} \to \mathscr{G} \to \mathscr{H}$ be an exact sequence in $\mathbf{Ab}_Y$, we shall to show that

$$\pi^{-1}\mathscr{F} \longrightarrow \pi^{-1}\mathscr{G} \longrightarrow \pi^{-1}\mathscr{H} \tag{3.9}$$

is exact. By Proposition 3.6.5, sequence (3.9) is exact if and only if

$$(\pi^{-1}\mathscr{F})_p \longrightarrow (\pi^{-1}\mathscr{G})_p \longrightarrow (\pi^{-1}\mathscr{H})_p \tag{3.10}$$

is exact for any $p \in X$. By Proposition 3.7.2, sequence (3.10) is isomorphic to sequence

$$\mathscr{F}_q \longrightarrow \mathscr{G}_q \longrightarrow \mathscr{H}_q, \tag{3.11}$$

for all $q = \pi(p)$. Since $\mathscr{F} \to \mathscr{G} \to \mathscr{H}$ is exact, by Proposition 3.6.5, sequence (3.11) is exact. Hence, sequence (3.10) is exact, and therefore

$$\pi^{-1}\mathscr{F} \longrightarrow \pi^{-1}\mathscr{G} \longrightarrow \pi^{-1}\mathscr{H}$$

is exact. □

3.7.2 The push-pull map

Definition 3.7.3 (The push-pull map)

Suppose

$$\begin{tikzcd} W \arrow[r, "\beta'"] \arrow[d, "\alpha'"'] & X \arrow[d, "\alpha"] \\ Y \arrow[r, "\beta"] & Z \end{tikzcd} \tag{3.12}$$

is a commutative (not necessarily Cartesian (Definition 2.2.11)!) diagram, and $\mathscr{F}$ is a sheaf on X. Define the ***push-pull map***

$$\beta^{-1}\alpha_*\mathscr{F} \longrightarrow \alpha'_*(\beta')^{-1}\mathscr{F} \tag{3.13}$$

of sheaves on Y as follows:

i. *Start with the identity*

$$(\beta')^{-1}\mathscr{F} \xrightarrow{\sim} (\beta')^{-1}\mathscr{F}$$

on W.

ii. *By adjointness of $((\beta')^{-1}, \beta'_*)$, this is the same as the data of a morphism*

$$\mathscr{F} \longrightarrow (\beta'_*)(\beta')^{-1}\mathscr{F}$$

on X.

iii. *Apply α_* to get a map*

$$\alpha_*\mathscr{F} \longrightarrow \alpha_*(\beta'_*)(\beta')^{-1}\mathscr{F}$$

on Z

iv. *By the commutativity of diagram (3.12), this is the map*

$$\alpha_*\mathscr{F} \longrightarrow \beta_*(\alpha'_*)(\beta')^{-1}\mathscr{F}$$

on Z.

v. *By adjointness of (β^{-1}, β_*), this yields a map (3.13).*

We observe that this entire construction is functorial in $\mathscr{F}$, i.e., given a map $\mathscr{F} \to \mathscr{G}$ of sheaves on X, we get a certain commutative diagram of sheaves on Y.

$$\begin{tikzcd} \beta^{-1}\alpha_*\mathscr{F} \arrow[r] \arrow[d] & \beta^{-1}\alpha_*\mathscr{G} \arrow[d] \\ \alpha'_*(\beta')^{-1}\mathscr{F} \arrow[r] & \alpha'_*(\beta')^{-1}\mathscr{G} \end{tikzcd}$$

We will later extend this to $\mathscr{O}$-modules, quasicoherent sheaves, and cohomology.

✍ **Exercise 3.22** (Surprisingly hard exercise.) We could have defined the push-pull map in a "dual way" starting with the identity $\alpha_*\mathscr{F} \to \alpha_*\mathscr{F}$ on Z, then using adjointness of (α^{-1}, α_*), and continuing from there. Why does this give the same definition of the push-pull map?

Proof See onRiv, Canonicity of the push-pull map for sheaves of sets. $\square$

3.7.3 The support of a sheaf, and the support of a section of a sheaf

Proposition 3.7.6 below gives us an excuse to introduce the notion of **support**, which we use repeatedly later.

Definition 3.7.4 (Support of the section)

*Suppose $\mathscr{F}$ is a sheaf (or indeed separated presheaf) of abelian groups on X, and s is a global section of $\mathscr{F}$. Define the **support of the section** s, denoted $\operatorname{Supp} s$, to be the set of points p of X where s has a nonzero germ:*

$$\operatorname{Supp} s := \{p \in X : s_p \neq 0 \textit{ in } \mathscr{F}_p\}.$$

Remark 3.43

(1) We think of this as the subset of X where "the sections s lives" — the complement is the locus where s is the 0-section.

(2) We could define this even if $\mathscr{F}$ is a presheaf, but without the inclusion $\mathscr{F}(U) \hookrightarrow \prod_{p\in U} \mathscr{F}_p$ of Proposition 3.4.1, we could have the strange situation where we have a nonzero section that "lives nowhere", because it is 0 "near every point", i.e., is 0 in every stalk.
For example: let $U \subseteq X$ be any open subset of X, define $\mathscr{F}(U) = \mathbb{Z}$, and the restriction map defined by $\mathrm{res}_{U,V}(s) = s$ if $U = V$ and $\mathrm{res}_{U,V}(s) = 0$ if $U \supsetneq V$. Then $\mathscr{F}$ form a presheaf, also $\mathscr{F}(U) \to \prod_{p\in U} \mathscr{F}_p$ not injective. Let $s \in \mathscr{F}(X) = \mathbb{Z}$ be a global section, then $s_p = 0$ in $\mathscr{F}_p$. Hence $\operatorname{Supp} s = \varnothing$ for all $s \in \mathscr{F}(X)$.

Proposition 3.7.5 (The support of a section is closed)

$\operatorname{Supp} s$ *is a closed subset of* X.

Proof It suffices to show that $X \setminus \operatorname{Supp} s$ is an open subset of X. In fact,

$$X \setminus \operatorname{Supp} s = \{p \in X : s_p = 0 \text{ in } \mathscr{F}_p\},$$

say $U = X \setminus \operatorname{Supp} s$. Let $p \in U$, then there exists open subset $(p \in V_p) \subseteq X$ such that $s|_V = 0$. We shall to show that $V_p \subseteq U$. For any $q \in V_p$, there exists open subset $W \ni q$ such that $W \subseteq V_p$. Note that

$$\mathrm{res}_{V_p,W}(\mathrm{res}_{X,V_p}(s)) = 0 = \mathrm{res}_{X,W}(s) = s|_W,$$

we have $s_q = 0$. Hence, $y \in U$, and therefore $V_p \in U$. Since $U = \bigcup_{p\in U} V_p$ and each V_p is open in X, U is open in X, i.e., $\operatorname{Supp} s$ is a closed subset of X. $\square$

Remark 3.44 **Caution: the locus where a continuous function is nonzero is open; the locus where the germ of a function is nonzero is closed.**

Basically by the definition of continuity, the locus where the value of a continuous function is nonzero is open. (More generally, the locus where the value of a function on a locally ringed space is nonzero is open.) In contrast, Proposition 3.7.5 shows that the locus where the germ of a function is nonzero is closed. We will try to avoid misunderstanding by using phrases like "f is 0 at p" (the value of f is zero, i.e., $f(p) = 0$) and "f is 0 near p" (the germ of f is zero, i.e., $f = 0$ in $\mathscr{O}_{X,p}$, or equivalently, f is zero in some neighborhood of p).

Definition 3.7.5 (Support of s sheaf)

Define the ***support of a sheaf*** *of groups* $\mathscr{G}$*, denoted* $\operatorname{Supp}\mathscr{G}$*, as the locus where the stalks are nontrivial:*

$$\operatorname{Supp}\mathscr{G} := \{p \in X : |\mathscr{G}_p| \neq 1\}.$$

Equivalently, $\operatorname{Supp}\mathscr{G}$ *is the union of supports of sections over all open sets:*

$$\operatorname{Supp}\mathscr{G} = \bigcup_{\substack{U \subseteq X \text{ open} \\ s \in \mathscr{G}(U)}} \operatorname{Supp} s$$

Remark 3.45 Clearly support is a "stalk-local notion", and hence "commutes" with restriction to open sets, that is,

$$\operatorname{Supp}\mathscr{G}|_U = \operatorname{Supp}\mathscr{G} \cap U.$$

Irrelevant for us: more generally, if the sheaf has values in some category, such as the category of sets, the support can be defined as the points where the stalk is not the terminal object.

Proposition 3.7.6

(a) Suppose $Z \subseteq Y$ *is a closed subset, and* $i : Z \hookrightarrow Y$ *is the inclusion. If* $\mathscr{F}$ *is a sheaf of groups on* Z*, then the stalk* $(i_*\mathscr{F})_q$ *is the one-element group if* $q \notin Z$*, and* $(i_*\mathscr{F})_q = \mathscr{F}_q$ *if* $q \in Z$*.*

(b) Suppose $\operatorname{Supp}\mathscr{G} \subseteq Z$ *where* Z *is closed. Then the natural map* $\mathscr{G} \to i_*i^{-1}\mathscr{G}$ *is an isomorphism. Thus a sheaf supported on a closed subset can be considered a sheaf on that closed subset.*

Proof

(a) Let $V \subseteq Y$ be any open subset of Y, then $i^{-1}(V) = V \cap Z$. Hence,

$$i_*\mathscr{F}(V) = \mathscr{F}(i^{-1}(V)) = \mathscr{F}(V \cap Z).$$

Let $q \in Y$.

If $q \notin Z$, then $Y \setminus Z$ is an open neighborhood of q, and $(Y \setminus Z) \cap Z = \varnothing$, hence, $i_*\mathscr{F}(Y \setminus Z) = \mathscr{F}(\varnothing)$, and therefore $(i_*\mathscr{F})_q$ is the one-element group.

If $q \in Z$, let V be any open neighborhood of q, then $V \cap Z \neq \varnothing$. Note that the topology on Z can be induced by the topology on Y, each open neighborhood W of q in Z can be seen as $W = V \cap Z$, where V is the open neighborhood of q in Y. Hence, $(i_*\mathscr{F})_q = \mathscr{F}_q$.

(b) The natural map is given by the adjoint pair (i^{-1}, i_*), i.e., consider the identity $i^{-1}\mathscr{G} \xrightarrow{\sim} i^{-1}\mathscr{G}$, by adjointness of (i^{-1}, i_*), we get a natural map

$$\mathscr{G} \longrightarrow i_*i^{-1}\mathscr{G}.$$

Let $p \in Y$.

If $p \notin Z$, then $p \notin \operatorname{Supp}\mathscr{G}$, and therefore $|\mathscr{G}_p| = 1$, say $\mathscr{G}_p = \{e\}$ where e is unit element. Since $p \notin Z$, $Y \setminus Z$ is an open neighborhood of p, which disjoint Z, say $V = Y \setminus Z$. Hence,

$$i_*i^{-1}\mathscr{G}(V) = i^{-1}\mathscr{G}(\varnothing) = (\varinjlim_{W \supseteq i(\varnothing)} \mathscr{G}(W))^{\text{sh}} = \mathscr{G}(\varnothing).$$

Hence, $(i_*i^{-1}\mathscr{G})_p \cong G_p$.

If $p \in Z$, by part (a), we have $(i_*i^{-1}\mathscr{G})_p = (i^{-1}\mathscr{G})_p$. By Proposition 3.7.2, $\mathscr{G}_p \cong (i^{-1}\mathscr{G})_p$, and therefore $\mathscr{G}_p \cong (i_*i^{-1}\mathscr{G})_p$.

Hence, by Proposition 3.4.4, we have an isomorphism $\mathscr{G} \cong i_*i^{-1}\mathscr{G}$.

□

Extension by zero, an occasional left adjoint to the inverse image functor. In addition to always being a left adjoint, π^{-1} can sometimes be a right adjoint, when π is an inclusion of an open subset. We discuss this when we need it.

Part II

Schemes

Chapter 4 Toward affine schemes: the underlying set, and topological space

We are now ready to consider the notion of **scheme**, which is the type of geometric space central to algebraic geometry. We should first think through what we mean by "geometric space". You have likely seen the notion of manifold, and we wish abstract this notion so that it can be generalized to other settings, notably so that we can deal with nonsmooth and arithmetic objects.

4.1 Toward scheme

See Ravi Vakil, The Rising Sea: Foundations of Algebraic Geometry.

The key insight behind this generalization from the notion of something like a manifold to a more versatile notion of a "geometric space" is the following: we can understand a geometric space (such as manifold) well by understanding the functions on this space. More precisely, we will understand it though the sheaf of functions on the space. If we are interested in differentiable manifolds, we will consider smooth functions; if we are interested in analytic manifolds, we will consider real analytic functions; and so on.

Thus we will define a scheme to be the following data.

- The set: the points of the scheme
- The topology: the open sets of the scheme
- The structure sheaf: the sheaf of "algebraic functions" (a sheaf of rings) on the scheme.

Recall that a topological space with a sheaf of rings is called a ringed space.

We will try to draw picture throughout. Pictures can help develop geometric intuition, which can guide the algebraic development (and, eventually, vice versa).

We will try to make all three notions as intuitive as possible. For the set, in the key example of complex (affine) varieties (roughly, things cut out in $\mathbb{C}^n$ by polynomials), we will see that the points are the "traditional points" (n-tuples of complex numbers), plus some extra points that will be handy to have around. For the topology, we will require that "the subset where an algebraic function vanishes must be closed", and require nothing else. For the sheaf of algebraic functions (the structure sheaf), we will expect that in the complex plane $\mathbb{C}^2$, $(3x^2 + y^2)/(2x + 4xy + 1)$ should be an algebraic function on the open set consisting of points where the denominator doesn't vanish, and this will largely motivate our definition.

4.1.1 Example:Differentiable manifolds

Differentiable manifolds. As motivation, we return to our example of differentiable manifolds, reinterpreting them in this light. We will be quite informal in this discussion. Suppose X is a differentiable manifold. It is a topological space, and has a sheaf of smooth (C^∞) functions $\mathscr{O}_X$ (see §3.1). This gives X the structure of a ringed space. We have observed that evaluation at a point $p \in X$ gives a surjective map from the stalk to $\mathbb{R}$,

$$\mathscr{O}_{X,p} \twoheadrightarrow \mathbb{R}$$

so the kernel, the (germs of) functions vanish at p, is a maximal ideal $\mathfrak{m}_{X,p}$ (see Proposition 3.1.1).

We could define a differentiable real manifold as a topological space X with a sheaf of rings. We would require that there is a cover of X by open sets such that on each open set the ringed space is isomorphic to a ball around the origin in $\mathbb{R}^n$ (with the sheaf of smooth functions on that ball). With this definition, the ball is the basic patch, and a genera manifold is obtained by gluing these patches together. (Admittedly a great deal of geometry comes from how one chooses to patch the balls together!) In the algebraic setting, the basic bath is the notion of an affine scheme, which we will discuss soon. (In the definition of manifold, there is an additional requirement that the topological space be Hausdorff and second-countable, to avoid pathologies. Schemes are often required to be "separated" to avoid essentially the same pathologies.)

Functions are determined by their values at points. This is an obvious statement, but won't be true for schemes in general.

Morphisms of manifolds. How can we describe maps of differentiable manifolds $\pi : X \to Y$? They are certainly continuous maps — but which ones? We can pull back functions along continuous maps. Smooth functions pull back to smooth functions. More formally, we have a map $\pi^{-1}\mathscr{O}_Y \to \mathscr{O}_X$. (The inverse image sheaf π^{-1} was defined in §3.7.) Inverse image is left-adjoint to pushforward, so we also get a map $\pi^\sharp : \mathscr{O}_Y \to \pi_*\mathscr{O}_X$.

Certainly given a map of differentiable manifolds, smooth functions pull back to smooth functions. It is less obvious that **this is a sufficient condition for a continuous map to be smooth.**

Proposition 4.1.1

Suppose that $\pi : X \to Y$ is a continuous map of differentiable manifolds (as topological spaces — not a priori smooth). Then π is smooth if smooth functions pull back to smooth functions, i.e., if pullback by π gives a map $\mathscr{O}_Y \to \pi_*\mathscr{O}_X$.

Proof Let V be any open set of Y, and (V, ψ) is a chart of Y, we defined the pullback $\pi_V^* : \mathscr{O}_Y(V) \to \pi_*\mathscr{O}_X(V) = \mathscr{O}_X(\pi^{-1}(V))$ by setting

$$f \mapsto f \circ \pi|_{\pi^{-1}(V)}.$$

We shall to show that $f \circ \pi|_{\pi^{-1}(V)}$ is smooth. Say $(U, \varphi) = (\pi^{-1}(V), \varphi)$, without loss of generally, we may assume that (U, φ) be a chart of X.

Let U is homeomorphic to $\mathbb{R}^n$ and V is homeomorphic to $\mathbb{R}^m$. Note that

$$f \circ \pi|_{\pi^{-1}(V)} \circ \varphi^{-1} = (f \circ \psi^{-1}) \circ (\psi \circ \pi \circ \varphi^{-1}),$$

since $f \in \mathscr{O}_Y(V)$ (i.e., $f \circ \psi^{-1}$ is smooth), $f \circ \pi|_{\pi^{-1}(V)}$ is smooth if and only if π is smooth. Then we done. □

Proposition 4.1.2

A morphism of differentiable manifolds $\pi : X \to Y$ with $\pi(p) = q$ induces a morphism of stalks $\pi^\sharp : \mathscr{O}_{Y,q} \to \mathscr{O}_{X,p}$. Then $\pi^\sharp(\mathfrak{m}_{Y,q}) \subseteq \mathfrak{m}_{X,p}$. In other words, if you pull back a function that vanishes at q, you get a function that vanishes at p.

Proof In fact, $\mathfrak{m}_{Y,p} = \{f_q \in \mathscr{O}_{Y,q} : f_q(q) = 0\}$ and $\mathfrak{m}_{X,p} = \{f_p \in \mathscr{O}_{X,p} : f_p(p) = 0\}$. Define $\pi^\sharp : \mathscr{O}_{Y,q} \to \mathscr{O}_{X,p}$ by setting

$$\pi^\sharp(f_q) = (\pi^*(f))_p = (f \circ \pi)_p.$$

We shall to show that $\pi^\sharp$ is well-defined. If $f_q = g_q$, then there exists $(q \in)V \subseteq Y$ such that $f|_V = g|_V$, hence $f \circ \pi|_{\pi^{-1}(V)} = g \circ \pi|_{\pi^{-1}(V)}$, where $\pi^{-1}(V)$ is an open neighborhood of p. Hence, $(\pi^*(f))_p = (\pi^*(g))_p$, this implies that $\pi^\sharp$ is well-defined.

Let $f_q \in \mathscr{O}_{Y,q}$, then there exists open subset $(q \in)V \subseteq Y$ such that $f \in \mathscr{O}_Y(V)$ with $f(q) = 0$. Apply pullback π_V^* to f, then we have $\pi_V^*(f) = f \circ \pi|_{\pi^{-1}(V)}$. Since $\pi(p) = q$, $\pi_V^*(f)(p) = 0$, which implies that

$\pi_V^*(f)_p \in \mathfrak{m}_{X,p}$, i.e., $\pi^\sharp(\mathfrak{m}_{Y,q}) \subseteq \mathfrak{m}_{X,p}$. □

Remark 4.1 **Aside.** Notice that π induces a map on tangent space

$$(\mathfrak{m}_{X,p}/\mathfrak{m}_{X,p}^2)^\vee \longrightarrow (\mathfrak{m}_{Y,q}/\mathfrak{m}_{Y,q}^2)^\vee.$$

This is the tangent map you would geometrically expect. Again, it is interesting that the cotangent map

$$\mathfrak{m}_{Y,q}/\mathfrak{m}_{Y,q}^2 \longrightarrow \mathfrak{m}_{X,p}/\mathfrak{m}_{X,p}^2$$

is algebraically more natural than the tangent map (there are no "duals").

Remark 4.2 Manifolds are covered by disks that are all isomorphic. This isn't true for schemes (even for "smooth complex varieties"). There are examples of two "smooth complex curves" (the algebraic version of Riemann surfaces) X and Y so that no nonempty open subset of X is isomorphic to a nonempty open subset of Y. And there is a Riemann surface X such that no two open subsets of X are isomorphic. Informally, this is because in the Zariski topology on schemes, all nonempty open sets are "huge" and have more "structure".

4.1.2 Other examples

If you are interested in differential geometry, you might be interested in differentiable manifolds, on which the functions under consideration are smooth functions. Similarly, if you are interested in topology, you will be interested in topological spaces, on which you will consider the continuous functions. If you are interested in complex geometry, you will be interested in complex manifolds (or possibly "complex analytic varieties"), on which the functions are holomorphic functions. In each of these cases of interesting "geometric spaces", the topological space and sheaf of functions is clear. The notion of scheme fits naturally into this family.

4.2 The underlying set of an affine scheme

See Ravi Vakil, The Rising Sea: Foundations of Algebraic Geometry.

For any ring A, we are going to define something called $\operatorname{Spec} A$, the **spectrum of** A. In this section, we will define a sheaf of rings on it (the structure sheaf). Such an object is called an **affine scheme**. Later $\operatorname{Spec} A$ will denote the set along with the topology, and sheaf of functions. But for now, as there is no possibility of confusion, $\operatorname{Spec} A$ will just be the set.

Definition 4.2.1 (Spectrum)

The set $\operatorname{Spec} A$ *is the set of prime ideals of* A*. The prime ideal* $\mathfrak{p}$ *of* A *when consider as an element of* $\operatorname{Spec} A$ *will be denoted* $[\mathfrak{p}]$*, to avoid confusion. Elements* $a \in A$ *will be called functions on* $\operatorname{Spec} A$*, and their* ***value*** *at the point* $[\mathfrak{p}]$ *will be* $a \pmod{\mathfrak{p}}$*.*

Remark 4.3 This is weird: a function can take values in different rings at different points — the function 5 on $\operatorname{Spec} \mathbb{Z}$ takes the value $1 \pmod 2$ at $[(2)]$ and $2 \pmod 3$ at $[(3)]$.

Note *"An element* a *of the ring lying in a prime ideal* $\mathfrak{p}$*" translates to "a function* a *that is* 0 *at the point* $[\mathfrak{p}]$*" or "a function* a *vanishing at the point* $[p]$*", and we will use these phrases interchangeably.*

Notice that if you add or multiply two functions, you add or multiply their values at all points; this is a translation of the fact that $A \to A/\mathfrak{p}$ is a ring homomorphism.

If A is generated over a base field (or base ring) by elements $x_1, \ldots, x_r$, the elements $x_1, \ldots, x_r$ are often called **coordinates**, because we will later be able to reinterpret them as restrictions of "coordinates on r-space".

We are building up the beginning of a grand dictionary. At some point you will get the sense that we are slowly decoding some timeless Rosetta Stone whose etchings we struggle to make out.

Glimpses of the future. In Chapter 5: we will interpret functions on $\operatorname{Spec} A$ as global sections of the "structure sheaf", i.e., as a function on a ringed space. We repeat a caution from "ringed space": what we will call "functions", others may call "regular functions". And we will later define "rational functions", which are not precisely functions in this sense; they are a particular type of "partially-defined function".

The notion of "value of a function" will be later interpreted as a value of a function on a particular locally ringed space.

4.2.1 Some examples.

The complex affine line

The complex affine line: $\mathbb{A}^1_{\mathbb{C}} := \operatorname{Spec} \mathbb{C}[x]$. Let's find the prime ideals of $\mathbb{C}[x]$. As $\mathbb{C}[x]$ is an integral domain, 0 is prime. Also, $(x-a)$ is prime, for any $a \in \mathbb{C}$: it is even a maximal ideal, as the quotient by this ideal is a field:

$$0 \longrightarrow (x-a) \longrightarrow \mathbb{C}[x] \xrightarrow{f \mapsto f(a)} \mathbb{C} \longrightarrow 0$$

We now show that there are no other prime ideals. We use the fact that $\mathbb{C}[x]$ has a division algorithm, and is a UFD. Suppose $\mathfrak{p}$ is a prime ideal. If $\mathfrak{p} \neq (0)$, then suppose $f(x) \in \mathfrak{p}$ is a nonzero element of smallest degree. It is not constant, as prime ideals can't contain 1. If $f(x)$ is not linear, then factor $f(x) = g(x)h(x)$, where $g(x)$ and $h(x)$ have positive degree. (Here we use that $\mathbb{C}$ is algebraically closed.) Then $g(x) \in \mathfrak{p}$ or $h(x) \in \mathfrak{p}$, contradicting the minimality of the degree of f. Hence there is a linear element $x-a$ of $\mathfrak{p}$. Then we claim that $\mathfrak{p} = (x-a)$. Suppose $f(x) \in \mathfrak{p}$. Then the division algorithm would give $f(x) = g(x)(x-a) + m$ where $m \in \mathbb{C}$. Then $m = f(x) - g(x)(x-a) \in \mathfrak{p}$. If $m \neq 0$, then $1 \in \mathfrak{p}$, giving a contradiction.

Thus we can and should (and must!) make a picture of $\mathbb{A}^1_{\mathbb{C}} = \operatorname{Spec} \mathbb{C}[x]$ (see Figure 4.1).

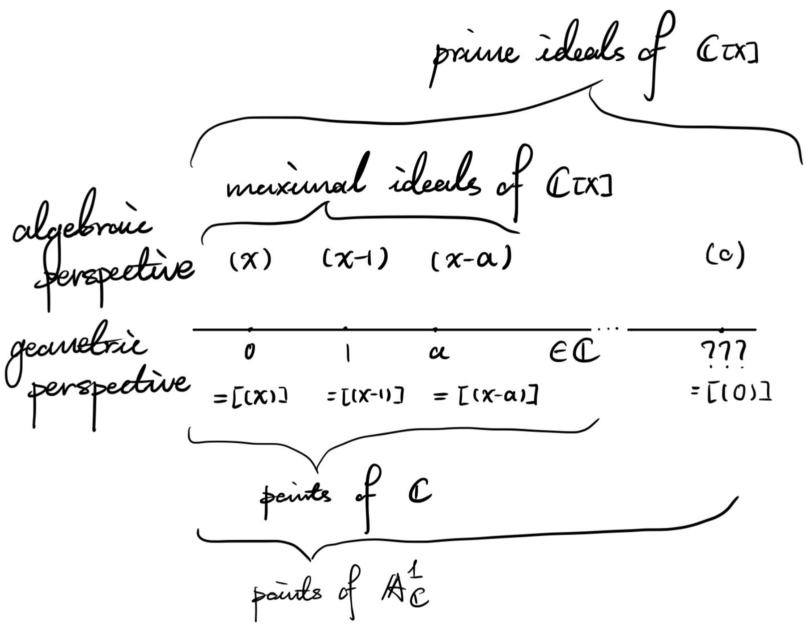


Figure 4.1: A picture of $\mathbb{A}^1_{\mathbb{C}} = \operatorname{Spec} \mathbb{C}[x]$

There is one "traditional" point for each complex number, plus one extra ("bonus") point $[(0)]$. We can mostly picture $\mathbb{A}^1_{\mathbb{C}}$ as $\mathbb{C}$: the point $[(x-a)]$ we will reasonably associate to $a \in \mathbb{C}$. Where should we picture the point $[(0)]$? The best way of thinking about it is somewhat zen. It is somewhere on the complex line, but nowhere in particular. Because (0) is contained in all of these prime ideals, we will somehow associate it with this line passing through all the other points. This new point $[(0)]$ is called the "generic point" of line. (We will formally define "generic point" in §3.6.) It is "generically on the line" but you can't pin it down any further than that. It is not at any particular place on the line. (This is misleading too — we will see that it is "near" every point. So it is near everything, but located nowhere precisely.) We will place it far to the right for lack of anywhere better to put it. Notice that we sketch $\mathbb{A}^1_{\mathbb{C}}$ as one-(real-)dimensional (even though we picture it as an enhanced version of $\mathbb{C}$); this is to later remind ourselves that this will be a one-dimensional space, where dimensions are defined in an algebraic (or complex-geometric) sense. (Dimension will be defined in Chapter 13.)

To give you some feeling for this space, we make some statements that are currently undefined, but suggestive. The functions on $\mathbb{A}^1_{\mathbb{C}}$ are the polynomials. So $f(x) = x^2 - 3x + 1$ is a function. $f(x) \pmod{x-1} = f(1)$ is $f(x)$'s value at point $[(x-1)]$. (What is its value at $[(0)]$? It is $f(x) \pmod 0$, which is just $f(x)$.)

Here is a more complicated example: $g(x) = (x-3)^3/(x-2)$ is a "rational function". It is defined everywhere but $x = 2$. (When we know what the structure sheaf is, we will be able to say that it is an element of the structure sheaf on the open set $\mathbb{A}^1_{\mathbb{C}} \setminus \{2\}$.) We want to say that $g(x)$ has a triple zero at 3, and a single pole at 2, and we will be able to after Chapter 14.

The affine line over $k = \overline{k}$

The affine line over $k = \overline{k}$: $\mathbb{A}^1_k := \operatorname{Spec} k[x]$ where k is an algebraically closed field. This is called the affine line over k. All of our discussion in the previous example carries over without change. We will use the same picture, which is after all intended to just be a metaphor.

$\operatorname{Spec} \mathbb{Z}$

$\operatorname{Spec} \mathbb{Z}$. An amazing fact is that from our perspective, this will look a lot like the affine line $\mathbb{A}^1_{\overline{k}}$. The integers, like $\overline{k}[x]$, form a UFD, with a division algorithm. The prime ideals are: (0), and (p) where p is prime. Thus everything from "The complex affine line" carries over without change, even the picture. Our picture of $\operatorname{Spec} \mathbb{Z}$ is shown in Figure 4.2.

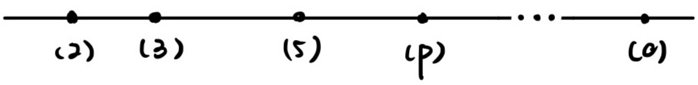


Figure 4.2: A "picture" of $\operatorname{Spec} \mathbb{Z}$, which looks suspiciously like Figure 4.1

Let's blithely carry over our discussion of functions to this space. 100 is a function on $\operatorname{Spec} \mathbb{Z}$. Its value at (3) is "$1 \pmod 3$". Its value at (2) is "$0 \pmod 2$", and in fact it has a double zero. $27/4$ is a "rational function" on $\operatorname{Spec} \mathbb{Z}$, defined away from (2). We want to say that it has a double pole at (2), and a triple zero at (3). Its value at (5) is

$$27 \times 4^{-1} \equiv 2 \times 4 \equiv 3 \pmod 5.$$

(We will gradually make this discussion precise over time.)

Silly but important examples, and the German word for bacon.

The set $\operatorname{Spec} k$ where k is any field is boring: one point. $\operatorname{Spec} 0$, where 0 is the zero-ring, is the empty set, as 0 has no prime ideals, i.e., $\operatorname{Spec} k = \varnothing$.

✍ **Exercise 4.1** A small exercise about small schemes.

(a) Describe the set $\operatorname{Spec} k[\varepsilon]/(\varepsilon^2)$. The ring $k[\varepsilon]/(\varepsilon^2)$ is called the ring of **dual numbers**, and will turn out to be quite useful. You should think of ε as a very small number, so small that its square is 0 (although it itself is not 0). It is a nonzero function whose value at all points is zero, thus giving our first example of functions not being determined by their values at points.

(b) Describe the set $\operatorname{Spec} k[x]_{(x)}$. We will see this scheme again repeatedly. You might later think of it as a shred of a particularly nice "smooth curve".

Proof

(a) Each element in $\operatorname{Spec} k[\varepsilon]/(\varepsilon^2)$ has form $a + b\varepsilon$, where $a, b \in k$. We claim that $a + b\varepsilon$ is invertible if and only if $a \neq 0$.
If $a \neq 0$, note that
$$(a + b\varepsilon)(a^{-1} - a^{-2}b\varepsilon) = 1,$$
hence $a + b\varepsilon$ is invertible.
If $a = 0$, note that $(b\varepsilon)^2 = 0$, hence $a + b\varepsilon$ is not invertible.
Hence, the prime ideal of $k[\varepsilon]/(\varepsilon^2)$ is only (ε), and therefore $\operatorname{Spec} k[\varepsilon]/(\varepsilon^2) = \{(\varepsilon)\}$.
Moreover, $\varepsilon \pmod{(\varepsilon)} = 0$ in $(k[\varepsilon]/(\varepsilon^2))/(\varepsilon)$, i.e., ε is a nonzero function whose value at all points is zero.

(b) Using the consequence in commutative algebra, the prime ideal in $k[x]_{(x)}$ is one-to-one correspondence to the prime ideal in $k[x]$ which contained in (x). Assume $(0) \subsetneq \mathfrak{p} \subsetneq (x)$, since $k[x]$ is a PID, we may assume $\mathfrak{p} = (xa)$ for some $a \in k[x]$. However, $\mathfrak{p}$ is prime and $x \notin \mathfrak{p}$, $a \in \mathfrak{p}$. Then $xa \mid a$, this implies that x is a unit, contradiction. Hence, for $\mathfrak{p} \in \operatorname{Spec} k[x]$ with $\mathfrak{p} \subseteq (x)$. $\mathfrak{p} = (x)$. Also, note that $k[x]_{(x)}$ is integral domain, (0) is the prime ideal of $k[x]_{(x)}$. Hence, $\operatorname{Spec} k[x]_{(x)} = \{(0), (x)k[x]_{(x)}\}$.

□

In example of "The affine line over $k = \overline{k}$", we restricted to the case of algebraically closed fields for a reason: things are more subtle if the field is not algebraically closed.

The affine line over $\mathbb{R}$

The affine line over $\mathbb{R}$: $\mathbb{A}^1_{\mathbb{R}} = \operatorname{Spec} \mathbb{R}[x]$. Using the fact that $\mathbb{R}[x]$ is a Euclidean domain, similar arguments to those of examples above show that the prime ideals are $(0), (x - a)$ where $a \in \mathbb{R}$, and $(x^2 + ax + b)$ is an irreducible quadratic. The latter two are maximal ideals, i.e., their quotients are field. For example: $\mathbb{R}[x]/(x - 3) \cong \mathbb{R}$, $\mathbb{R}[x]/(x^2 + 1) \cong \mathbb{C}$.

✍ **Exercise 4.2** Show that for the last type of prime, of the form $(x^2 + ax + b)$, the quotient is always isomorphic to $\mathbb{C}$.

Proof By division algorithm, for all $f(x) \in \mathbb{R}[x]$, $f(x) = g(x)(x^2 + ax + b) + cx + d$. Since $x^2 + ax + c$ is irreducible, the two roots of $x^2 + ax + c$ are both imaginary numbers, say the root $r = r_1 + ir_2$ where $r_2 \neq 0$. Define a ring homomorphism $\nu_r : \mathbb{R}[x]/(x^2 + ax + d) \to \mathbb{C}$ by setting $cx + d \mapsto cr + d$. Let $\nu_r(cx + d) = cr + d = 0$, then $(cr_1 + d) + icr_2 = 0$, hence $c = 0$ and $d = 0$, this implies that ν_r is injective. Let $\alpha + i\beta \in \mathbb{C}$, note that $\nu_r(\frac{\beta}{r_2}x + \alpha - \beta\frac{r_1}{r_2}) = \alpha + i\beta$, ν_r is surjective. Hence, ν_r is isomorphism, and therefore $\mathbb{R}[x]/(x^2 + ax + b) \cong \mathbb{C}$. □

So we have the points that we would normally expect to see on the real line, corresponding to real numbers; the generic point 0; and new points which we may interpret as conjugate pairs of complex numbers (the roots of the quadratic). This last type of point should be seen as more akin to the real numbers than to the generic point. We can picture $\mathbb{A}^1_{\mathbb{R}}$ as the complex plane, folded along the real axis (see Figure 4.3). But the key point is that Galois-conjugate points (such as i and $-i$) are consider glued.

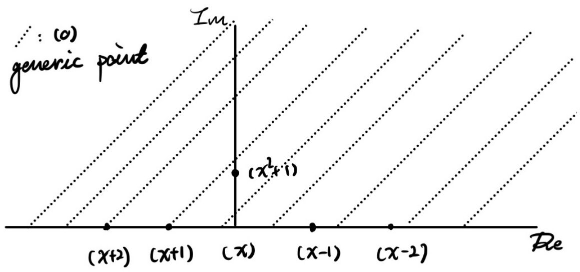


Figure 4.3: A "picture" of $\operatorname{Spec}\mathbb{R}[x]$

Let's explore functions on this space. Consider the function $f(x) = x^3 - 1$. Its value at the point $[(x-2)]$ is 7, or perhaps better, "$7 \pmod{x-2}$". How about at (x^2+1)? We get

$$x^3 - 1 \equiv -x - 1 \pmod{x^2+1},$$

which may be profitably interpreted as $-i - 1$.

One moral of this example is that we can work over a non-algebraically closed field if we wish. It is more complicated, but we can recover much of the information we care about.

Exercise 4.3 Describe the set $\mathbb{A}^1_{\mathbb{Q}}$.

Solution *$\mathbb{Q}[x]$ is an integral domain, hence (0) is prime. Since $\mathbb{Q}[x]$ is also a Euclidean domain (and therefore PID), an ideal $(f(x))$ is prime if and only if $f(x)$ is irreducible. Hence,*

$$\operatorname{Spec}\mathbb{Q}[x] = \{(0), (f(x)) : f(x) \textit{ is an irreducible polynomial}\}.$$

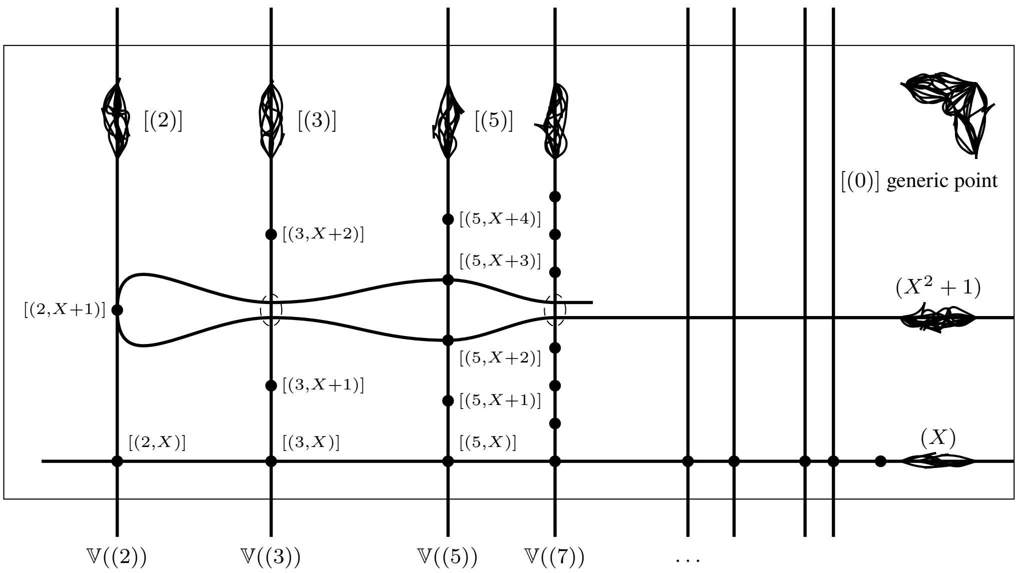


Figure 4.4: Picturing $\operatorname{Spec}\mathbb{Z}[x]$

The affine line over $\mathbb{F}_p$

The affine line over $\mathbb{F}_p$: $\mathbb{A}^1_{\mathbb{F}_p} = \operatorname{Spec} \mathbb{F}_p[x]$. As in the previous examples, $\mathbb{F}_p[x]$ is a Euclidean domain, so the prime ideals are of the form (0) or $(f(x))$ where $f(x) \in \mathbb{F}_p[x]$ is an irreducible polynomial, which can be of any degree. Irreducible polynomials correspond to sets of Galois conjugates in $\overline{\mathbb{F}_p}$.

Note that $\operatorname{Spec} \mathbb{F}_p[x]$ has p points corresponding to the elements of $\mathbb{F}_p$, but also many more (infinitely more, see 4.4). For example, consider $\mathbb{F}_2 = \{0, 1\}$, $(x), (x-1)$ in $\operatorname{Spec} \mathbb{F}_2$ corresponding to the elements of $\mathbb{F}_p$, (x^2+x+1) also in $\operatorname{Spec} \mathbb{F}_2$ but not corresponding to the elements of $\mathbb{F}_2$. This makes this space much richer than simply p points. For example, a polynomial $f(x)$ is not determined by its values at the p elements of $\mathbb{F}_p$, but it is determined by its values at the points of $\operatorname{Spec} \mathbb{F}_p[x]$. (As we have mentioned before, this is not ture for all schemes.)

You should think about this, even if you are a geometric person — this intuition will later turn up in geometric situations. Even if you think you are interested only in working over an algebraically closed field (such as $\mathbb{C}$), you will have nonalgebraically closed fields (such as $\mathbb{C}(x)$) forced upon you.

✍ **Exercise 4.4** If k is a field, show that $\operatorname{Spec} k[x]$ has infinitely many points.

Proof Since k is a field, the maximal ideals in $k[x]$ are principal, generated by irreducible polynomials. Hence, it suffices to show that there are infinitely many irreducible polynomials. We prove this by using Euclid's trick.

Suppose there are finitely many irreducible polynomials in $k[x]$, say $p_1, \ldots, p_r$. Consider a new polynomial

$$F(x) = p_1(x)p_2(x)\cdots p_r(x) + 1.$$

Since $k[x]$ is a UFD, there is an irreducible factor p, also, the irreducible polynomial p can't be any of $p_1, \cdots, p_r$. Then we get a new irreducible polynomial which is not in $p_1, \cdots, p_r$, a contradiction. □

The complex affine plane

The complex affine plane: $\mathbb{A}^2_{\mathbb{C}} = \operatorname{Spec} \mathbb{C}[x, y]$. (As with examples "The complex affine line" and "The affine line over $k = \overline{k}$", our discussion will apply with $\mathbb{C}$ replaced by any algebraically closed field.) Sadly, $\mathbb{C}[x, y]$ is not a PID: (x, y) is not a principal ideal. We can quickly name some prime ideals. One is (0), which has the same flavor as the (0) ideals in the previous examples. $(x-2, y-3)$ is prime, and indeed maximal, because $\mathbb{C}[x, y]/(x-2, y-3) \cong \mathbb{C}$, where this isomorphism is via $f(x, y) \mapsto f(2, 3)$. More generally, $(x-a, y-b)$ is prime for any $(a, b) \in \mathbb{C}^2$. Also, if $f(x, y)$ is an irreducible polynomial (e.g., $y - x^2$ or $y^2 - x^3$) then $(f(x, y))$ is prime.

✍ **Exercise 4.5** Show that we have identified all the prime ideal of $\mathbb{C}[x, y]$.

Proof $\mathbb{C}[x, y]$ is a UFD, and therefore an integral domain, hence, (0) and $(f(x, y))$ where $f(x, y)$ is an irreducible polynomial are prime ideals in $\mathbb{C}[x, y]$.

Let $\mathfrak{p} \in \operatorname{Spec} \mathbb{C}[x, y]$ that is not principal, and let $f(x, y), g(x, y) \in \mathfrak{p}$ with no common factor (if we can not find, $\mathfrak{p}$ must be principal). The greatest common divisor for $f(x, y)$ and $g(x, y)$ in $\mathbb{C}(x)[y]$ can be written in the form $\frac{a(x)c(x)}{b(x)}$, where $c(x)$ has no non-trivial factors in $\mathbb{C}[x, y]$. Since $\mathbb{C}[x, y]$ is a UFD, so this statement makes sense. Then for some $f'(x, y), p(x), g'(x, y)$, and $q(x)$, we have

$$f(x, y) = \frac{a(x)c(x)}{b(x)} \frac{f'(x, y)}{p(x)}$$

and

$$g(x, y) = \frac{a(x)c(x)}{b(x)} \frac{g'(x, y)}{q(x)},$$

which is to say that

$$a(x)c(x)f'(x,y) = b(x)p(x)f(x,y)$$
$$a(x)c(x)g'(x,y) = b(x)q(x)g(x,y).$$

Note that $c(x) \nmid b(x)$, $c(x) \nmid p(x)$, and $c(x) \nmid q(x)$, $c(x) \nmid f(x,y)$ and $c(x) \nmid g(x,y)$. Since, $f(x,y)$ and $g(x,y)$ have no common factor in $\mathbb{C}[x,y]$, $c(x)$ must be a constant, we may assume $c(x) = 1$. Hence, $gcd_{\mathbb{C}(x)[y]}(f(x,y), g(x,y)) = \frac{a(x)}{b(x)}$.

Using the Euclidean algorithm, it must be possible to write the gcd $\frac{a(x)}{b(x)}$ as follow:

$$\frac{a(x)}{b(x)} = \frac{u(x,y)}{t(x)} f(x,y) + \frac{v(x,y)}{w(x)} g(x,y),$$

clearing denominators,

$$a(x)t(x)w(x) = u(x,y)b(x)w(x)f(x,y) + v(x,y)b(x)t(x)g(x,y).$$

Define

$$h(x) := a(x)t(x)w(x),$$

we have a non-zero polynomial $h(x) \in (f(x,y), g(x,y)) \subseteq mfp$. Since $\mathbb{C}$ is algebraically closed, we may assume $h(x) = \prod(x - a_i)$. Since $\mathfrak{p}$ is prime ideal, there exists $x - a := x - a_i$ for some i such that $x - a \in \mathfrak{p}$. Similarly, there exists $y - b \in \mathfrak{p}$. Hence, $(x - a, y - b) \subseteq \mathfrak{p}$. Note that $\mathbb{C}[x,y]/(x-a, y-b) \cong \mathbb{C}$, $(x - a, y - b)$ is a maximal ideal, and therefore $\mathfrak{p} = (x - a, y - b)$. $\square$

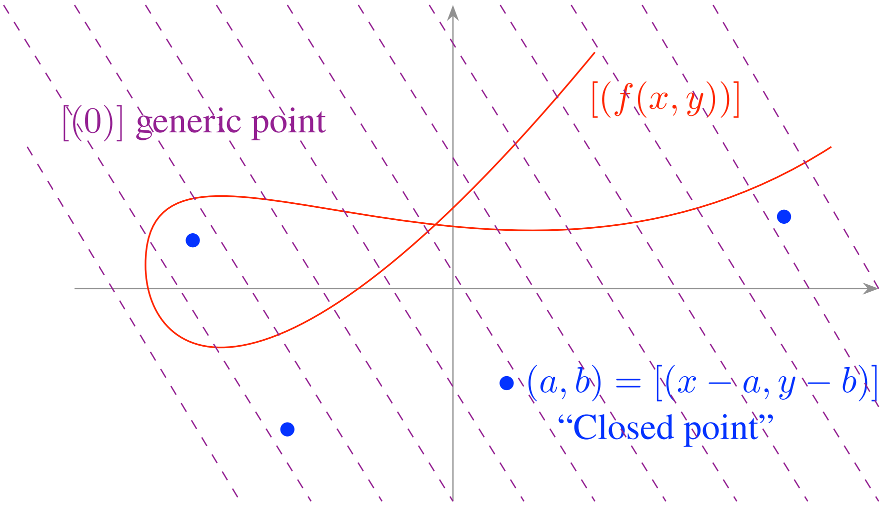


Figure 4.5: Picturing $\mathbb{A}^2_{\mathbb{C}} = \operatorname{Spec} \mathbb{C}[x,y]$

We now attempt to draw a picture of $\mathbb{A}^2_{\mathbb{C}}$ (see Figure 4.5). The maximal prime ideals of $\mathbb{C}[a,b]$ correspond to the traditional points in $\mathbb{C}^2$: $[(x-a, y-b)]$ corresponds to $(a,b) \in \mathbb{C}^2$. We now have to visualize the "bonus points". $[(0)]$ somehow lives behind all of the traditional points; it is somewhere on the affine plane, but nowhere in particular. So for example, it does not lie on the parabola $y = x^2$. The point $[(y - x^2)]$ lies on the parabola $y = x^2$, but nowhere in particular on it. (Figure 4.5 is a bit misleading. For example, the point $[(0)]$ isn't in the fourth quadrant; it is somehow near every other point, which is why it is depicted as a somewhat diffuse large dot.) You can see from this picture that we already are implicitly thinking about "dimension". The prime ideals $(x-a, y-b)$ are somehow of dimension 0, the prime ideals $(f(x,y))$ are of dimension 1, and (0) is of dimension 2. (All of our dimensions here are complex or algebraic dimensions. The complex plane $\mathbb{C}^2$ has real dimension 4, but complex dimension 2. Complex dimension are in general half of real dimensions.) We won't define dimension precisely until Chapter 13, but you should feel free to keep it in mind before then.

Note too that maximal ideals correspond to the "smallest" points. Smaller ideals correspond to "bigger" points. "One prime ideal contains another" means that the points "have the opposite containment." All of this will be made precise once we have a topology. This order-reversal is a little confusing, and will remain so even once we have made the notions precise.

Complex affine n-space

We now come to the obvious generalization of example "The complex affine plane". Let $\mathbb{A}^n_{\mathbb{C}} = \operatorname{Spec}\mathbb{C}[x_1,\ldots,x_n]$. (More generally, $\mathbb{A}^n_A$ is defined to be $\operatorname{Spec} A[x_1,\ldots,x_n]$, where A is an arbitrary ring. When the base ring is clear from context, the subscript A is often omitted. For pedants: the notation $\mathbb{A}^n_A$ implicitly includes the data of the n coordinate functions $x_1,\ldots,x_n$.) For concreteness, let's consider $n=3$. We now have an interesting question in what at first appears to be pure algebra: What are the prime ideals of $\mathbb{C}[x,y,z]$?

Analogously to before, $(x-a, y-b, z-c)$ is a prime ideal. This is a maximal ideal, because its residue ring is field $\mathbb{C}$; we think of these as "0-dimensional points". We will often write (a,b,c) for $[(x-a, y-b, z-c)]$ because of our geometric interpretation of these ideals. There are no more maximal ideals, by Hilbert's Weak Nullstellensatz.

Theorem 4.2.1 (Hilbert's Weak Nullstellensatz)

If k is an algebraically closed field, then the maximal ideals of $k[x_1,\ldots,x_n]$ are precisely those ideals of the form $(x_1-a_1,\ldots,x_n-a_n)$, where $a_i\in k$.

We may as well state a slightly stronger version now.

Theorem 4.2.2 (Hilbert's Nullstellensatz)

If k is any field, every maximal ideal $\mathfrak{m}$ of $k[x_1,\ldots,x_n]$ has residue field $k[x_1,\ldots,x_n]/\mathfrak{m}$ a finite extension of k.

Translation: any field extension of k that is finitely generated as a k-algebra is necessarily also finitely generated as a k-vector space (i.e., is a finite extension of fields).

Remark 4.4 This statement is also often called **Zariski's Lemma**.

✍ **Exercise 4.6** Show that the Nullstellensatz 4.2.2 implies the Weak Nullstellensatz 4.2.1.

Proof Let $\mathfrak{m}$ be the maximal ideal of $k[x_1,\ldots,x_n]$, then $k[x_1,\ldots,x_n]/\mathfrak{m}$ is a field extension of k. Let a_i be the image of x_i in $k[x_1,\ldots,x_n]/\mathfrak{m}$, then $k[x_1,\ldots,x_n]/\mathfrak{m} = k[a_1,\ldots,a_n]$ is finitely generated as a k-algebra. By Hilbert's Nullstellensatz, $k[a_1,\ldots,a_n]$ is also finite field extension of k. Since k is algebraically closed, $k\cong k[a_1,\ldots,a_n]$, and therefore we may assume each $a_i\in k$.

Define the k-algebra homomorphism

$$\begin{aligned}\varphi : k[x_1,\ldots,x_n] &\longrightarrow k[a_1,\ldots,a_n] = k\\ x_i &\longmapsto a_i,\end{aligned} \tag{4.1}$$

Clearly, $\mathfrak{m}$ is the kernel of φ. Note that $\operatorname{Ker}\varphi = (x_1-a_1,\ldots,x_n-a_n)$, we have

$$\mathfrak{m} = (x_1-a_1,\ldots,x_n-a_n).$$

□

The following fact is a useful accompaniment to the Nullstellensatz

Proposition 4.2.1

Any integral domain A which is a finite k-algebra (i.e., a k-algebra that is a finite-dimensional vector space over k) must be a field.

Proof Define k-algebraic homomorphism $\times f : A \to A$ by setting $g \mapsto gf$, where f be any nonzero element in A. It is easy to see that $\times f$ is well-defined and is a k-algebraic homomorphism. Note that A is a field if and only if k-algebra homomorphism $\times f$ is isomorphism. It suffices to show that $\times f$ is isomorphism.

Let $\times f(g) = gf = 0$, since A is integral domain and f is nonzero element, $g = 0$, which implies that $\times f$ is injective.

Since k is a field, A is also a finite k-vector space, then we have $\dim(A) = \dim \operatorname{Im}(\times f) + \dim \operatorname{Ker}(\times f) = \dim \operatorname{Im}(\times f)$, which implies that $\times f$ is surjective.

Hence, A is a field. $\square$

Remark 4.5 In combination the Nullstellensatz 4.2.2, we see that prime ideals of $k[x_1, \ldots, x_n]$ with finite-dimensional residue ring are the same as maximal ideals of $k[x_1, \ldots, x_n]$. More precisely, let $\mathfrak{p} \in \operatorname{Spec} k[x_1, \ldots, x_n]$, then $k[x_1, \ldots, x_n]/\mathfrak{p}$ is an integral domain, if $k[x_1, \ldots, x_n]/\mathfrak{p}$ is also a finite k-algebra, by Proposition 4.2.1, $k[x_1, \ldots, x_n]/\mathfrak{p}$ is a field, and therefore $\mathfrak{p}$ is maximal ideal.

There are other prime ideals of $\mathbb{C}[x, y, z]$ too. We have (0), which corresponds to a "3-dimensional point". We have $(f(x, y, z))$, where f is irreducible. To this we associate the "hypersurface" $f = 0$, so this is "2-dimensional" in nature. But we have not found them all! One clue: we have prime ideals of "dimension" 0, 2, and 3 — we are missing "dimension 1". Here is one such ideal: (x, y). We picture this as as the locus where $x = y = 0$, which is the z-axis. This is a prime ideal, as the corresponding quotient $\mathbb{C}[x, y, z]/(x, y) \cong \mathbb{C}[z]$ is an integral domain (and should be interpreted as the functions on the z-axis). There are lots of "1-dimensional prime ideals", and it is not possible to classify them in a reasonable way. It will turn out that they correspond to things that we think of as irreducible curves. Thus remarkably the answer to the purely algebraic question ("what are the prime ideals of $\mathbb{C}[x, y, z]$") is fundamentally geometric!

The fact that the points of $\mathbb{A}^1_{\mathbb{Q}}$ corresponding to maximal ideals of the ring $\mathbb{Q}[x]$ (what we will soon call "closed points") can be interpreted as points of $\overline{\mathbb{Q}}$ where Galois-conjugates are glued together extends to $\mathbb{A}^n_{\mathbb{Q}}$. For example, in $\mathbb{A}^2_{\mathbb{Q}}$, $(\sqrt{2}, \sqrt{2})$ is glued to $(-\sqrt{2}, -\sqrt{2})$ but not to $(\sqrt{2}, -\sqrt{2})$. The following exercise will give you some idea of how this works.

Exercise 4.7

(a) Describe the maximal ideal of $\mathbb{Q}[x, y]$ corresponding to $(\sqrt{2}, \sqrt{2})$ and $(-\sqrt{2}, -\sqrt{2})$.
(b) Describe the maximal ideal of $\mathbb{Q}[x, y]$ corresponding to $(\sqrt{2}, -\sqrt{2})$ and $(-\sqrt{2}, \sqrt{2})$.
(c) What are the residue fields in each case?

Solution

(a) Let $\mathfrak{m} = (x^2 - 2, x - y)$, note that

$$\mathbb{Q}[x, y]/(x^2 - 2, x - y) \cong \mathbb{Q}[x]/(x^2 - 2) \cong \mathbb{Q}(\sqrt{2}),$$

$\mathbb{Q}(\sqrt{2})$ is a field, and therefore $\mathfrak{m}$ is a maximal ideal. Note that $\operatorname{Gal}(\mathbb{Q}(\sqrt{2})/\mathbb{Q}) = \{\mathrm{id}, \tau\}$, where $\tau(\sqrt{2}) = -\sqrt{2}$. Hence, $(\sqrt{2}, \sqrt{2})$ and $(-\sqrt{2}, -\sqrt{2})$ are at the same orbit.

(b) Let $\mathfrak{m} = (x^2 - 2, x + y)$, then

$$\mathbb{Q}[x, y]/(x^2 - 2, x + y) \cong \mathbb{Q}[x]/(x^2 - 2) \cong \mathbb{Q}(\sqrt{2}),$$

hence, $\mathfrak{m}$ is maximal ideal. Note that $\operatorname{Gal}(\mathbb{Q}(\sqrt{2})/\mathbb{Q}) = \{\mathrm{id}, \tau\}$, where $\tau(\sqrt{2}) = -\sqrt{2}$. Hence, $(\sqrt{2}, -\sqrt{2})$ and $(-\sqrt{2}, \sqrt{2})$ are at the same orbit.

(c) *They have the same residue field:* $\mathbb{Q}(\sqrt{2})$.

The description of "closed points" of $\mathbb{A}^2_{\mathbb{Q}}$ (those points corresponding to maximal ideals of the ring $\mathbb{Q}[x,y]$) as Galois-orbits of points in $\overline{\mathbb{Q}}^2$ can even be extended to other "nonclosed" points, as follows.

✍ **Exercise 4.8** Consider the map of sets $\varphi : \mathbb{C}^2 \to \mathbb{A}^2_{\mathbb{Q}}$ defined as follows. (z_1, z_2) is sent to the prime ideal of $\mathbb{Q}[x,y]$ consisting of polynomials vanishing at (z_1, z_2).

(a) What is the image of (π, π^2)?

(b) Show that φ is surjective.

Proof

(a) We claim that the image of (π, π^2) is $(y-x^2)$. $y-x^2$ is vanishing at (π, π^2). Also, $\mathbb{Q}[x,y]/(y-x^2) \cong \mathbb{Q}[x]$ is an integral domain, hence, $(y-x^2)$ is prime. Thus, the image of (π, π^2) is $(y-x^2)$.

(b) In fact,

$$\begin{aligned}\mathbb{A}^2_{\mathbb{Q}} =& \operatorname{Spec} \mathbb{Q}[x,y] \\ =& \{(0), \operatorname{Max} \mathbb{Q}[x,y], (f(x,y)) \text{ where } f(x,y) \text{ is irreducible polynomial in } \mathbb{Q}[x,y]\}.\end{aligned}$$

For $(0) \in \operatorname{Spec} \mathbb{Q}[x,y]$. Consider $(\pi, e) \in \mathbb{C}^2$. Note that there are no irreducible polynomials vanish at (π, e), also, $\pi, e \notin \mathbb{Q}$, hence, $\varphi(\pi, e) = (0)$.

For $\mathfrak{m} \in \operatorname{Max} \mathbb{Q}[x,y]$. Note that $\mathbb{Q}[x,y]/\mathfrak{m}$ is a field, and is the extension field of k, then there exists $(a,b) \in \mathbb{Q}[x,y]/\mathfrak{m}$ such that each element in $\mathfrak{m}$ vanish at (a,b), i.e., $\varphi(a,b) = \mathfrak{m}$.

For $(f(x,y))$, where $f(x,y)$ is irreducible polynomial in $\mathbb{Q}[x,y]$. Since $\mathbb{C}$ is algebraically closed, $f(x,y)$ has roots in $\mathbb{C}$, they given by

$$\operatorname{Gal}\left(\mathbb{Q}[x,y]/(f(x,y)) : \text{ extension of } \mathbb{Q}\right) \curvearrowright (r_1, r_2) \in \mathbb{C}^2,$$

where $(f(x,y))$ vanish at (r_1, r_2). Hence, $\varphi(r_1, r_2) = (f(x,y))$.

By above discussion, φ is a surjective.

□

4.2.2 Quotients and localization

Two natural ways of getting new rings from old — quotient and localizations — have interpretations in terms of spectra.

Quotients

Lemma 4.2.1

Suppose A is a ring, and I an ideal of A. Let $\varphi : A \to A/I$. Then φ^{-1} gives an inclusion-preserving bijection between prime ideals of A/I and prime ideals of A containing I.

Proof The proof of this lemma can be found in any standard textbook on abstract algebra; since it is straightforward, we omit it here. □

By the Lemma 4.2.1, we get following immediately.

Proposition 4.2.2

$\operatorname{Spec} A/I$ *as a subset of* $\operatorname{Spec} A$.

Thus we can picture $\operatorname{Spec} A/I$ as a subset of $\operatorname{Spec} A$.

As an important motivational special case, you now have a picture of affine complex varieties. Suppose A is a finitely generated $\mathbb{C}$-algebra, generated by $x_1, \dots, x_n$, with relations $f_1(x_1, \dots, x_n) = \cdots = f_r(x_1, \dots, x_n) = 0$. Then this description in terms of generators and relations naturally gives us an interpretation of $\operatorname{Spec} A$ as a subset of $\mathbb{A}^n_{\mathbb{C}}$, which we think of as "traditional points" (n-tuples of complex numbers) along with some "bonus" points we haven't yet fully described. To see which of the traditional points are in $\operatorname{Spec} A$, we simply solve the equations $f_1 = \cdots = f_r = 0$. For example, $\operatorname{Spec} \mathbb{C}[x, y, z]/(x^2 + y^2 - z^2)$ may be pictured as shown in Figure 4.6. (Admittedly this is just a "sketch of the $\mathbb{R}$-points", but we will still find it helpful later.) This entire picture carries over (along with the Nullstellensatz) with $\mathbb{C}$ replaced by any algebraically closed field. Indeed, the picture of Figure 4.6 can be said to depict $\operatorname{Spec} k[x, y, z]/(x^2 + y^2 - z^2)$ for most algebraically closed fields k (although it is misleading in characteristic 2, because of the coincidence $x^2 + y^2 - z^2 = (x + y + z)^2$).

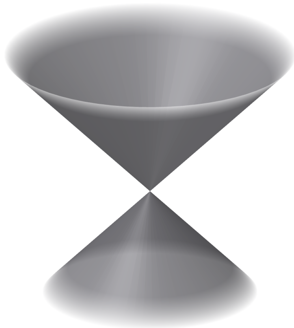


Figure 4.6: A "picture" of $\operatorname{Spec} \mathbb{C}[x, y, z]/(x^2 + y^2 - z^2)$

Localizations

The following Proposition shows how prime ideals behave under localization.

Proposition 4.2.3

Suppose S is a multiplicative subset of A. The prime ideal of $S^{-1}A$ are one-to-one correspondence with the prime ideals of A which don't meet S, i.e.,

$$\mathfrak{p} \longleftrightarrow S^{-1}\mathfrak{p},$$

where $\mathfrak{p} \cap S = \varnothing$.

Proof Let $f : A \to S^{-1}A$ be a ring homomorphism which is defined by $f(x) = x/1$.

If $\mathfrak{q} \in \operatorname{Spec} S^{-1}A$, then $f^{-1}(\mathfrak{q}) := \mathfrak{q} \cap A \in \operatorname{Spec} A$, by Lemma 4.2.1.

Conversely, if $\mathfrak{p} \in \operatorname{Spec} A$, then $A/\mathfrak{p}$ is an integral domain. On the other hand, let $\overline{S}$ be the image of S in $A/\mathfrak{p}$. Note that localization commutes with quotient, we have $S^{-1}A/S^{-1}\mathfrak{p} \cong \overline{S}^{-1}(A/\mathfrak{p})$, which $\overline{S}^{-1}(A/\mathfrak{p})$ is either 0 or else is contained in the field of fraction of $A/\mathfrak{p}$, and therefore $S^{-1}\mathfrak{p}$ is either $S^{-1}A$ or prime ideal. Since every ideal in $S^{-1}A$ is an extended ideal, $S^{-1}\mathfrak{p} = (1)$ if and only if $\mathfrak{p} \cap S \neq \varnothing$. Then we done. □

Hence, by Proposition 4.2.3, we get follow:

Proposition 4.2.4

$\operatorname{Spec} S^{-1}A$ *as a subset of* $\operatorname{Spec} A$.

Recall from §2.2.2 that there are two important flavors of localization. The first is $A_f = \{1, f, f^2, \dots\}^{-1}A$

where $f \in A$. A motivating example is $A = \mathbb{C}[x, y]$, $f = y - x^2$. The second is $A_{\mathfrak{p}} = (A \setminus \mathfrak{p})^{-1}A$, where $\mathfrak{p}$ is a prime ideal. A motivating example is $A = \mathbb{C}[x, y]$, $S = A \setminus (x, y)$.

If $S = \{1, f, f^2, \dots\}$, the prime ideals of $S^{-1}A$ are just those prime ideals not containing f — the points where "f doesn't vanish". (In §4.5, we will call this a distinguished open set, once we know what open sets are.) So to picture $\operatorname{Spec} \mathbb{C}[x, y]_{y-x^2}$, we picture the affine plane, and throw out those points on the parabola $y - x^2$ — the points (a, a^2) for $a \in \mathbb{C}$ (by which we mean $[(x - a, y - a^2)]$), as well as the "new kind of point" $[(y - x^2)]$.

It can be initially confusing to think about localization in the case where zero divisors are inverted, because localization $A \to S^{-1}A$ is not injective (Exercise 2.6). Geometric intuition can help. Consider the case $A = \mathbb{C}[x, y]/(xy)$ and $f = x$. What is the localization A_f? The space $\operatorname{Spec} \mathbb{C}[x, y]/(xy)$ "is" the union of the two axes in the affine plane. Localizing means throwing out the locus where x vanishes. So we are left with the x-axis, minus the origin, so we expect $\operatorname{Spec} \mathbb{C}[x]_x$. So there should be some natural isomorphism

$$(\mathbb{C}[x, y]/(xy))_x \xrightarrow{\sim} \mathbb{C}[x]_x.$$

Exercise 4.9 Show that these two rings are isomorphic.

Proof In fact,

$$(\mathbb{C}[x, y]/(xy))_x = \left\{ \frac{f(x, y)}{x^k} : f(x, y) \in \mathbb{C}[x, y], k \geq 0, xy = 0 \right\}$$

and

$$\mathbb{C}[x]_x = \left\{ \frac{f(x)}{x^k} : f(x) \in \mathbb{C}[x], k \geq 0 \right\}.$$

Define the ring homomorphism $\varphi : (\mathbb{C}[x, y]/(xy))_x \to \mathbb{C}[x]_x$ by setting $\frac{f(x,y)}{x^k} \mapsto \frac{f(x,0)}{x^k}$. Clearly, φ is well-defined ring homomorphism, and is a surjective.

Let $\varphi(\frac{f(x,y)}{x^k}) = \frac{f(x,0)}{x^k} = 0$. Let $f(x, y) = \sum_i a_i x^i + \sum_j b_j y^j$, then $x^n f(x, 0) = x^n \sum_i a_i x_i^i = 0$ for some n in $\mathbb{C}[x]$, hence, $f(x, 0) = \sum_i a_i x_i^i = 0$, which implies that $f(x, y) = \sum_j b_j y^j$. Note that $\frac{f(x,y)}{x^k} = \frac{\sum_j b_j y^j}{x^k} = \frac{\sum_j b_j x^d y^j}{x^{k+d}}$, where d is the maximal degree of y, since $xy = 0$, we have $\frac{f(x,y)}{x^k} = 0$, which implies that φ is injective.

Hence,

$$(\mathbb{C}[x, y]/(xy))_x \xrightarrow{\sim} \mathbb{C}[x]_x.$$

$\square$

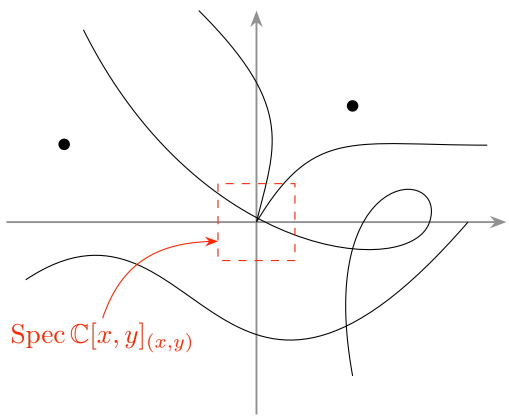


Figure 4.7: Picturing $\operatorname{Spec} \mathbb{C}[x, y]_{(x,y)}$ as a "shred of $\mathbb{A}^2_{\mathbb{C}}$". Only those points near the origin remain.

If $S = A \setminus \mathfrak{p}$, the prime ideals of $S^{-1}A$ are just the prime ideals of A contained in $\mathfrak{p}$. In our example $A = \mathbb{C}[x,y]$, $\mathfrak{p} = (x,y)$, we keep all those points corresponding to "things through the origin", i.e., the 0-dimension point (x,y), the 2-dimensional point (0), and those 1-dimensional points $(f(x,y))$ where $f(0,0) = 0$, i.e., those "irreducible curves through the origin". You can think of this being a shred of the plane near the origin; anything not actually "visible" at the origin is discarded (see Figure 4.7).

Another example is when $A = k[x]$, and $\mathfrak{p} = (x)$ (or more generally when $\mathfrak{p}$ is any maximal ideal). Then $A_{\mathfrak{p}}$ has only two prime ideals (Exercise 4.1 (b), $\operatorname{Spec} A_{\mathfrak{p}} = \{(0), (x)k[x]_{(x)}\}$). You should see this as the germ of a "smooth curve", where one point is the "classical point", and the other is the "generic point of the curve". This is an example of a DVR, and indeed all DVRs should be visualized in such a way. We will discuss DVRs in Chapter 13. By then we will have justified the use of the words "smooth" and "curve".

4.2.3 Maps of rings induce maps of spectra (as sets)

We now make an observation that will later grow up to be the notion of morphisms of schemes.

Proposition 4.2.5

If $\varphi : B \to A$ is a map of rings, and $\mathfrak{p}$ is a prime ideal of A, then $\varphi^{-1}(\mathfrak{p})$ is a prime ideal of B.

Proof Let $ab \in f^{-1}(\mathfrak{p})$, then $f(ab) = f(a)f(b) \in \mathfrak{p}$, since $\mathfrak{p}$ is prime, $f(a) \in \mathfrak{p}$ or $f(b) \in \mathfrak{p}$, i.e., $a \in f^{-1}(\mathfrak{p})$ or $b \in f^{-1}(\mathfrak{p})$, which implies that $f^{-1}(\mathfrak{p})$ is prime. □

Hence a map of rings $\varphi : B \to A$ induces a map of sets $\operatorname{Spec} A \to \operatorname{Spec} B$ "in the opposite direction". This gives a contravariant functor from the category of rings to the category of sets: the composition of two maps of rings induces the composition of the corresponding maps of spectra.

Proposition 4.2.6

Let B be a ring.

(a) Suppose $I \subseteq B$ is an ideal. Then the map $\operatorname{Spec} B/I \to \operatorname{Spec} B$ is the inclusion of Proposition 4.2.2.

(b) Suppose $S \subseteq B$ is a multiplicative set. Then the map $\operatorname{Spec} S^{-1}B \to \operatorname{Spec} B$ is the inclusion of Proposition 4.2.4.

Proof

(a) The elements in $\operatorname{Spec} B/I$ is one-to-one correspondence with the elements in $\operatorname{Spec} B$ which containing I, hence, the map $\operatorname{Spec} B/I \to \operatorname{Spec} B$ is the inclusion.

(b) The elements in $\operatorname{Spec} S^{-1}B$ is one-to-one correspondence with the elements in $\operatorname{Spec} B$ which do not meet S, and therefore, the map $\operatorname{Spec} S^{-1}B \to \operatorname{Spec} B$ is the inclusion.

□

Example 4.1 In the case of "affine complex varieties" (or indeed affine varieties over any algebraically closed field), the translation between maps given by explicit formulas and maps of rings is quite direct. For example, consider a map from the parabola in $\mathbb{C}^2$ (with coordinates a and b) given by $b = a^2$, to the "curve" in $\mathbb{C}^3$ (with coordinates x, y, and z) cut out by the equations $y = x^2$ and $z = y^2$. Suppose the map sends the point $(a,b) \in \mathbb{C}^2$ to the point $(a,b,b^2) \in \mathbb{C}^3$.

In our new language, we have a map

$$\operatorname{Spec} \mathbb{C}[a,b]/(b-a^2) \longrightarrow \operatorname{Spec} \mathbb{C}[x,y,z]/(y-x^2, z-y^2)$$

given by

$$\begin{aligned}\mathbb{C}[x,y,z]/(y-x^2,z-y^2) &\longrightarrow \mathbb{C}[a,b]/(b-a^2)\\ (x,y,z) &\longmapsto (a,b,b^2),\end{aligned}$$

i.e., $x \mapsto a$, $y \mapsto b$, and $z \mapsto b^2$.

✍ **Exercise 4.10** Consider the map of complex manifolds sending $\mathbb{C} \to \mathbb{C}$ via $x \mapsto y = x^2$. We interpret the "source" $\mathbb{C}$ as the "x-line", and the "target" $\mathbb{C}$ the "y-line". You can picture it as the projection of the parabola $y = x^2$ in the xy-plane to the y-axis (see Figure 4.8). Interpret the corresponding map of rings as given by $\mathbb{C}[y] \to \mathbb{C}[x]$ by $y \mapsto x^2$. Verify that the preimage (the fiber) above the point $a \in \mathbb{C}$ is the point(s) $\pm\sqrt{a} \in \mathbb{C}$, using definition given above (identifying a with $[(y-a)]$, and $\sqrt{a}$ with $[(x-\sqrt{a})]$).

Proof Define $\varphi : \mathbb{C}[y] \to \mathbb{C}[x]$ by setting $\varphi(y) = x^2$. Point $a \in \mathbb{C}$ is corresponding to the maximal ideal $(y-a) \in \operatorname{Spec}\mathbb{C}[y]$, then $\varphi((y-a)) = (x^2-a) = (x-\sqrt{a})(x+\sqrt{a})$, which is corresponding to two points $\pm\sqrt{a} \in \mathbb{C}$. Hence, the preimage (the fiber) above the point $a \in \mathbb{C}$ is the point(s) $\pm\sqrt{a} \in \mathbb{C}$. □

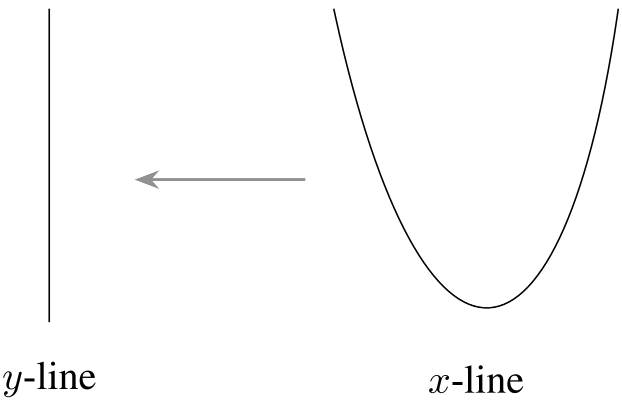


Figure 4.8: The map $\mathbb{C} \to \mathbb{C}$ given by $x \mapsto y = x^2$

✍ **Exercise 4.11** Suppose k is a field, and $f_1, \ldots, f_n \in k[x_1, \ldots, x_m]$ are given. Let $\varphi : k[y_1, \ldots, y_n] \to k[x_1, \ldots, x_m]$ be the morphism of k-algebras defined by $y_i \mapsto f_i$.

(a) Show that φ induces a map of sets $\operatorname{Spec} k[x_1, \ldots, x_m]/I \to \operatorname{Spec} k[y_1, \ldots, y_n]/J$ for any ideals $I \subseteq k[x_1, \ldots, x_m]$ and $J \subseteq k[y_1, \ldots, y_n]$ such that $\varphi(J) \subseteq I$.

(b) Show that the map of part (a) sends the point $(a_1, \ldots, a_m) \in k^m$ (or more precisely, $[(x_1 - a_1, \ldots, x_m - a_m)] \in \operatorname{Spec} k[x_1, \ldots, x_m]$) to

$$(f_1(a_1, \ldots, a_m), \ldots, f_n(a_1, \ldots, a_m)) \in k^n.$$

Proof

(a) Let $\mathfrak{p} \in \operatorname{Spec} k[x_1, \ldots, x_m]/I$, then $\mathfrak{p}$ is one-to-one correspondence with $\widetilde{\mathfrak{p}} \in \operatorname{Spec} k[x_1, \ldots, x_m]$ which is containing I, hence, $\widetilde{\mathfrak{p}} \supseteq I \supseteq \varphi(J)$, and therefore $\varphi^{-1}(\widetilde{\mathfrak{p}}) \in \operatorname{Spec} k[y_1, \ldots, y_n]$ with $\varphi^{-1}(\widetilde{\mathfrak{p}}) \supseteq J$. Denote $\mathfrak{q}$ be the image of $\varphi^{-1}(\widetilde{\mathfrak{p}})$ in $k[y_1, \ldots, y_n]/J$. Hence, φ induces a map of sets $\operatorname{Spec} k[x_1, \ldots, x_m]/I \to \operatorname{Spec} k[y_1, \ldots, y_n]/J$ which defined by $\mathfrak{p} \mapsto \mathfrak{q}$.

(b) Let $[(x_1 - a_1, \ldots, x_m - a_m)] \in \operatorname{Spec} k[x_1, \ldots, x_m]$. Denote $\overline{\varphi}$ be the induced map from φ. Then

$$\begin{aligned}\overline{\varphi}((x_1 - a_1, \ldots, x_m - a_m)) =& \varphi^{-1}((x_1 - a_1, \ldots, x_m - a_m))\\ =& \{h \in k[y_1, \ldots, y_n] : \varphi(h) \in (x_1 - a_1, \ldots, x_m - a_m)\}\\ \in& \operatorname{Spec} k[y_1, \ldots, y_n].\end{aligned}$$

We may assume $\varphi(h) = h(f_1, \ldots, f_n) = \sum_i h_i(x_i - a_i)$, where $h_i \in k[x_1, \ldots, x_m]$, then

$$h(f_1(a_1, \ldots, a_m), \ldots, f_n(a_1, \ldots, a_m)) = 0,$$

which implies that $h \in (y_1 - f_1(a_1, \ldots, a_m), \ldots, y_n - f_n(a_1, \ldots, a_m))$. Hence,

$$\overline{\varphi}((x_1 - a_1, \ldots, x_m - a_m)) \subseteq (y_1 - f_1(a_1, \ldots, a_m), \ldots, y_n - f_n(a_1, \ldots, a_m)).$$

Conversely, it is easy to see that

$$\varphi((y_1 - f_1(a_1, \ldots, a_m), \ldots, y_n - f_n(a_1, \ldots, a_m)) \subseteq (x_1 - a_1, \ldots, x_m - a_m),$$

and therefore

$$\overline{\varphi}((x_1 - a_1, \ldots, x_m - a_m)) \supseteq (y_1 - f_1(a_1, \ldots, a_m), \ldots, y_n - f_n(a_1, \ldots, a_m)).$$

Hence,

$$\overline{\varphi}((x_1 - a_1, \ldots, x_m - a_m)) = (y_1 - f_1(a_1, \ldots, a_m), \ldots, y_n - f_n(a_1, \ldots, a_m)),$$

i.e., the map of part (a) sends the point $(a_1, \ldots, a_m) \in k^m$ to

$$(f_1(a_1, \ldots, a_m), \ldots, f_n(a_1, \ldots, a_m)) \in k^n.$$

$\square$

Exercise 4.12 Picturing $\mathbb{A}^n_{\mathbb{Z}}$. Consider the map of sets $\pi : \mathbb{A}^n_{\mathbb{Z}} \to \operatorname{Spec} \mathbb{Z}$, given by the ring map $\mathbb{Z} \to \mathbb{Z}[x_1, \ldots, x_n]$. If p is prime, describe a bijection between the fiber $\pi^{-1}([(p)])$ and $\mathbb{A}^n_{\mathbb{F}_p}$. This exercise may give you a sense of how to picture maps (see 4.9), and in particular why you can think of $\mathbb{A}^n_{\mathbb{Z}}$ as an "$\mathbb{A}^n$-bundle" over $\operatorname{Spec} \mathbb{Z}$.

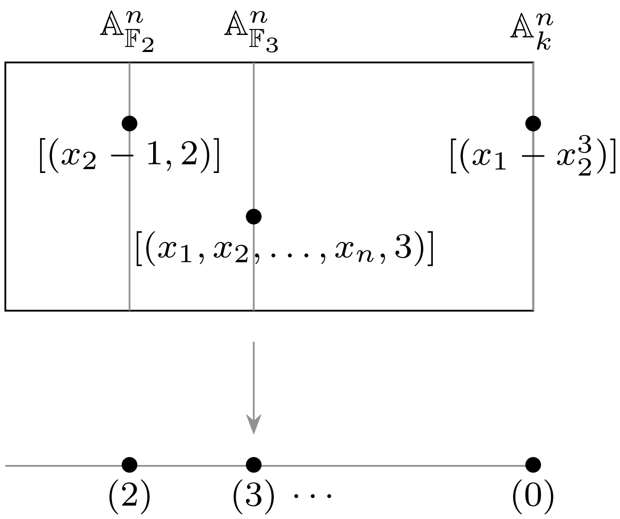


Figure 4.9: A picture of $\mathbb{A}^n_{\mathbb{Z}} \to \operatorname{Spec} \mathbb{Z}$ as a "family of $\mathbb{A}^n$'s", or an "$\mathbb{A}^n$-bundle over $\operatorname{Spec} \mathbb{Z}$".

Solution *$\pi : \mathbb{A}^n_{\mathbb{Z}} \to \operatorname{Spec} \mathbb{Z}$ is given by $\mathfrak{p} \mapsto \mathfrak{p} \cap \mathbb{Z}$.*

Let $(p) \in \operatorname{Spec} \mathbb{Z}$, we need to describe a bijection between the fiber $\pi^{-1}([(p)])$ and $\mathbb{A}^n_{\mathbb{F}_p}$. In fact,

$$\pi^{-1}([(p)]) = \{\mathfrak{p} \in \operatorname{Spec} \mathbb{Z}[x_1, \ldots, x_n] : \mathfrak{p} \cap \mathbb{Z} = (p)\},$$

this implies that $\mathfrak{p}$ is containing (p). Let $\overline{\mathfrak{p}}$ be the image of $\mathfrak{p}$ in $\mathbb{Z}[x_1, \ldots, x_n]/(p) = \mathbb{F}_p[x_1, \ldots, x_n]$. Define $\varphi : \pi^{-1}([(p)]) \to \mathbb{A}^n_{\mathbb{F}_p}$ by setting $\varphi(\mathfrak{p}) = \overline{\mathfrak{p}}$, where $\overline{\mathfrak{p}} = \mathfrak{p} \pmod{(p)}$. By above discussion φ is well-defined.

Let $\overline{\mathfrak{p}} = \overline{\mathfrak{q}}$. Suppose $\mathfrak{p} \neq \mathfrak{q}$, then there exists $f \in \mathfrak{p}$ but $f \notin \mathfrak{q}$. Note that $f \pmod{(p)} \in \overline{\mathfrak{p}} = \overline{\mathfrak{q}}$, we have $f \in \mathfrak{q}$, a contradiction. Hence, $\mathfrak{p} = \mathfrak{q}$, and therefore φ is injective.

Let $\mathfrak{p} \in \mathbb{A}^n_{\mathbb{F}_p}$. Since p is prime, $\mathbb{F}_p$ is a field, hence, $\mathfrak{p} \cap \mathbb{F}_p = (0)$. Let

$$\mathfrak{q} = \left\{ \sum a_{i_1,\ldots,i_n} x_1^{i_1} \cdots x_n^{i_n} \in \mathbb{Z}[x_1, \ldots, x_n] : \sum \overline{a_{i_1,\ldots,i_n}} x_1^{i_1} \cdots x_n^{i_n} \in \mathfrak{p} \right\},$$

then $\overline{\mathfrak{q}} = \mathfrak{p}$ and $\mathfrak{q} \cap \mathbb{Z} = (p)$, which implies that $\varphi(\mathfrak{q}) = \mathfrak{p}$. Hence, φ is surjective.

4.2.4 Functions are not determined by their values at points: the fault of nilpotents

We conclude this section by describing some strange behavior. We are developing machinery that will let us bring our geometric intuition to algebra. There is one serious point where your intuition will be false, so you should know now, and adjust your intuition appropriately. As noted by Mumford, "it is this aspect of schemes which was most scandalous when Grothendieck defined them."

Suppose we have a function (ring element) vanishing at all points. Then it is not necessarily the zero function! The translation of this question is: is the intersection of all prime ideals necessarily just 0? The answer is no, as is shown by the example of the ring of dual numbers $k[\varepsilon]/(\varepsilon^2) : \varepsilon \neq 0$, but $\varepsilon^2 = 0$. (We saw this ring in Exercise 4.1 (a).) Any function whose power is zero certainly lies in the intersection of all prime ideals.

Definition 4.2.2 (Nilpotent)

*Ring elements that have a power that is 0 are called **nilpotents**.*

Proposition 4.2.7

Let B be a ring, then:

*(a) the nilpotents of a ring B form an ideal. This ideal is called the **nilradical**, and is denoted $\mathfrak{N} = \mathfrak{N}(B)$;*

(b) if $I \subseteq \mathfrak{N}$ is an ideal of nilpotents, the inclusion $\operatorname{Spec} B/I \to \operatorname{Spec} B$ is a bijection, thus nilpotents don't affect the underlying set.

Proof

(a) If $f \in \mathfrak{N}$, then $f^m = 0$ for some m, for all $b \in B$, we have $(bf)^m = b^m f^m = 0$, and therefore $B\mathfrak{N} \subseteq \mathfrak{N}$. We next show that $\mathfrak{N}$ is a subgroup of B. For all $f \in \mathfrak{N}$, we have $f^m = 0$ for some m, hence, $(-f)^m = (-1)^m f^m = 0$. Let $g \in \mathfrak{N}$, then $g^n = 0$ for some n. By the binomial theorem, $(x+y)^{m+n-1}$ is a sum of integer multiples of products $x^r y^s$, where $r + s = m + n - 1$; we cannot have both $r < m$ and $s < n$, hence each of these products vanishes and therefore $(x+y)^{m+n-1} = 0$. Hence, $x + y \in \mathfrak{N}$, and therefore $\mathfrak{N}$ is a subgroup of B.
By above discussion, $\mathfrak{N} \in \operatorname{Spec} B$.

(b) It suffices to show that I is contained in any prime ideal of B. Let $f \in I$, then $f^m = 0$ for some m. Let $\mathfrak{p} \in \operatorname{Spec} B$, then $0 = f^m \in \mathfrak{p}$, hence, $f \in \mathfrak{p}$ or $f^{m-1} \in \mathfrak{p}$. If $f \notin \mathfrak{p}$, we have $f^{m-2} \in \mathfrak{p}$, repeat this process, we have $f^2 \in \mathfrak{p}$, then $f \in \mathfrak{p}$, a contradiction. Hence, $f \in \mathfrak{p}$, and therefore $I \subseteq \mathfrak{p}$. Since the element in $\operatorname{Spec} B/I$ is one-to-one correspondence to the element in $\operatorname{Spec} B$ which is containing I, we have bijection

$$\operatorname{Spec} B/I \longleftrightarrow \operatorname{Spec} B.$$

$\square$

Thus the nilradical is contained in the intersection of all the prime ideals. The converse is also true:

Theorem 4.2.3

The nilradical $\mathfrak{N}(A)$ is the intersection of all the prime ideals of A. Geometrically: a function on Spec A *vanishes at every point if and only if it is nilpotent.*

Proof Denote $\mathfrak{N}' = \bigcap_{\mathfrak{p}\in \operatorname{Spec} A} \mathfrak{p}$. By the proof in Proposition 4.2.7, the nilradical $\mathfrak{N}(A)$ is contained in all prime ideal of A, i.e., $\mathfrak{N} \subseteq \mathfrak{N}'$

Conversely, suppose that f is not nilpotent. Let

$$\Sigma = \{\mathfrak{a} \subseteq A : \mathfrak{a} \text{ is an ideal of } A \text{ and } f^n \notin \mathfrak{a} \text{ for all } n > 0\},$$

then Σ is not empty because $(0) \in \Sigma$. Apply Zorn's Lemma to the set Σ, ordered by inclusion, and therefore Σ has a maximal element. Let $\mathfrak{m}$ be the maximal element of Σ. Now, we shall to show that $\mathfrak{m}$ is a prime ideal.

Let $x, y \notin \mathfrak{m}$, then the ideals $(x) + \mathfrak{m}$, $(y) + \mathfrak{m}$ are containing $\mathfrak{m}$, and therefore $(x) + \mathfrak{m}, (y) + \mathfrak{m} \notin \Sigma$. Hence, exists n, m such that

$$f^m \in (x) + \mathfrak{m}, \quad f^n \in (y) + \mathfrak{m}.$$

It follows that $f^{m+n} = f^m f^n \in (xy) + \mathfrak{m}$, hence, $(xy) + \mathfrak{m} \notin \Sigma$, and therefore $xy \notin \mathfrak{m}$. Hence, $\mathfrak{m}$ is prime ideal. Note that $f \notin \mathfrak{m}$, $f \notin \mathfrak{N}'$, i.e., $\mathfrak{N}' \subseteq \mathfrak{N}$. Thus $\mathfrak{N} = \mathfrak{N}'$. □

In particular, although it is upsetting that functions are not determined by their values at points, we have precisely specified what the failure of this intuition is: two functions have the same values at points if and only if they differ by a nilpotent. You should think of this geometrically: a function vanishes at every point of the spectrum of a ring if and only if it has a power that is zero. And if there are no nonzero nilpotents — if $\mathfrak{N} = (0)$ — then functions are determined by their values at points.

Definition 4.2.3 (Reduced)

If a ring has no nonzero nilpotents, we say that it is ***reduced****.*

✍ **Exercise 4.13** **Derivatives without deltas and epsilons (or at least without deltas).** Suppose we have a polynomial $f(x) \in k[x]$. Instead, we work in $k[x, \varepsilon]/(\varepsilon^2)$. What then is $f(x + \varepsilon)$?

Solution *In fact,* $k[x, \varepsilon]/(\varepsilon^2) = \{a_0(x) + a_1(x)\varepsilon : a_i(x) \in k[x], i = 0, 1\}$*. Suppose* $f(x) = \sum_{i=0}^n a_i x^i$*, then in* $k[x, \varepsilon]/(\varepsilon^2)$ *we have*

$$\begin{aligned} f(x+\varepsilon) &= \sum_i a_i (x+\varepsilon)^i = \sum_{i=0}^n a_i (x^i + i x^{i-1}\varepsilon) \\ &= \sum_{i=0}^n a_i x^i + \varepsilon \sum_{i=0}^n i a_i x^{i-1} \\ &:= f(x) + \varepsilon f'(x). \end{aligned}$$

Remark 4.6 This is a hint that nilpotents will be important in defining differential information (Chapter 22).

4.3 Visualizing schemes: Generic points

A heavy warning used to be given that pictures are note rigorous; this has never had its bluff called and has permanently frightened its victims into playing for safety. Some pictures, of cause, are not rigorous, but I should say most are (and I use them whenever possible myself).

— J.E.Littlewood

We all know that Art is not truth. Art is a lie that makes us realize truth, at least the truth that is given us to understand. The artist must know the manner whereby to convince others of the truthfulness of his lies.

— P.Picasso

For years, you have been able to picture $x^2 + y^2 = 1$ in the plane, and you now have an idea of how to picture $\operatorname{Spec} \mathbb{Z}$. If we are claiming to understand rings as geometric objects (through the Spec functor), then we should wish to develop geometric insight into them. To develop geometric intuition about schemes, it is helpful to have pictures in your mind, extending your intuition about geometric spaces you are already familiar with. As we go along, we will empirically develop some idea of what schemes should look like. This summarizes what we have gleaned so far.

Some mathematicians prefer to think completely algebraically, and never think in terms of pictures. Others will be disturbed by the fact that this is an art, not a science. And finally, this hand-waving will necessarily never be used in the rigorous development of the theory. For these reasons, you may wish to skip these sections. However, having the right picture in your mind can greatly help understanding what facts should be true, and how to prove them. Fitgerald's exhortation on the importance of stretching one's mind is essential advice to a mathematician.

The test of a first-rate intelligence is the ability to hold two opposed ideas in the mind at the same time, and still retain the ability to hold two opposed ideas in the mind at the same time, and still retain the ability to function.

— F.Scott Fitzgerald

Our starting point is the example of "affine complex varieties" (things cut out by equations involving a finite number of variables over $\mathbb{C}$), and more generally similar examples over arbitrary algebraically closed fields. We begin with notions that are intuitive ("traditional" points behaving the way you expect them to), and then add in the two features which are new and disturbing, generic points and nonreduced behavior. You can then extend this notion to seemingly different spaces, such as $\operatorname{Spec} \mathbb{Z}$.

Hilbert's Weak Nullstellensatz 4.2.1 shows that the "traditional points" are present as points of the scheme, and this carries over to any algebraically closed field. If the field is not algebraically closed, the traditional points are glued together into clumps by Galois conjugation, as in Example "the real affine line" and "the affine line over $\mathbb{F}_p$" in §4.2.1. This is a geometric interpretation of Hilbert's Nullstellensatz 4.2.2.

But we have some additional points to add to the picture. You should remember that they "correspond" to "irreducible" "closed" (algebraic) subsets. As motivation, consider the case of the complex affine plane §4.2.1: we had one for each irreducible polynomial, plus one corresponding to the entire plane. We will make "closed" precise when we define the Zariski topology (in the next section). You may already have an ideal of what "irreducible" should mean; we make that precise at the start of §4.6. By "correspond" we mean that each closed irreducible subset has a corresponding point sitting on it, called its **generic point** (defined in §4.6). It is a new point, distinct from all the other points in the subset. We don't know precisely where to draw the generic point, so we may stick it arbitrarily anywhere, but you should think of it as being "almost everywhere", and in particular, near every other point in the subset.

In §4.2.2, we saw how the points of $\operatorname{Spec} A/I$ should be interpreted as subsets of $\operatorname{Spec} A$. So for example, when you see $\operatorname{Spec} \mathbb{C}[x,y]/(x+y)$, you should picture this not just as a line, but as a line in the xy-plane; the choice of generators x and y of the algebra $\mathbb{C}[x,y]$ implies an inclusion into affine space.

In §4.2.2, we saw how the points of $\operatorname{Spec} S^{-1}A$ should be interpreted as subsets of $\operatorname{Spec} A$. The two points of $\operatorname{Spec} A$ where f doesn't vanish; we will later (§4.5) interpret this as a distinguished open set.

If $\mathfrak{p}$ is a prime ideal, then $\operatorname{Spec} A_{\mathfrak{p}}$ should be seen as a "shred of the space $\operatorname{Spec} A$ near the subset corresponding to $\mathfrak{p}$". The simplest nontrivial case of this is $A = k[x]$ and $\mathfrak{p} = (x) \subseteq A$ (see Exercise 4.1).

"If any of them can explain it", said Alice, (she had grown so large in the last few minutes that she wasn't a bit afraid of interrupting him), "I'll give him sixpence. I don't believe there's an atom of meaning in it."...

"If there's no meaning in it," said the King, "that saves a world of trouble, you know, as we needn't try to find any."

— Lewis Carroll

4.4 The underlying topological space of an affine scheme

We next introduce the **Zariski topology** on the spectrum of a ring. When you first hear the definition, it seems odd, but with a little experience it becomes reasonable. As motivation, consider $\mathbb{A}^2_{\mathbb{C}} = \operatorname{Spec} \mathbb{C}[x, y]$, the complex (affine) plane (with a few extra points). In algebraic geometry, we will only be allowed to consider algebraic functions, i.e., polynomials in x and y. The locus where a polynomial vanishes should reasonably be a closed set, and the Zariski topology is defined by saying that the only sets we should consider closed should be these sets, and other sets forced to be closed by these. In other words, it is the coarsest topology where these sets are closed.

4.4.1 Definition of Zariski topology

In particular although topologies are often described using open subsets, it will be more convenient for us to define this topology in terms of closed subsets.

Definition 4.4.1 (Vanishing set)

*If S is a subset of a ring A, define the **vanishing set** of S by*

$$V(S) := \{[\mathfrak{p}] \in \operatorname{Spec} A : S \subseteq \mathfrak{p}\}.$$

It is the set of points on which all elements of S are zero. (It should now be second nature to equate "vanishing at a point" with "contained in a prime".) We declare that these — and no others — are the closed subsets.

For example, consider $V(xy, yz) \subseteq \mathbb{A}^3_{\mathbb{C}} = \operatorname{Spec} \mathbb{C}[x, y, z]$. Which points are contained in this locus? We think of this as solving $xy = yz = 0$. Of the "traditional" points (interpreted as ordered triples of complex numbers, thanks to the Hilbert's Nullstellensatz), we have the points where $y = 0$ or $x = z = 0$: the xz-plane and the y-axis respectively. Of the "new" points, we have the generic point of the xz-plane (also known as the point $[(y)]$), and the generic point of the y-axis (also known as the point $[(x, z)]$). You might imagine that we also have a number of "one-dimensional" points contained in the xz-plane.

✍ **Exercise 4.14** Check that the x-axis is contained in $V(xy, yz)$.

Proof The x-axis is defined by $y = z = 0$. It is corresponding to point $[(y, z)]$. Since $\mathbb{C}[x, y, z]/(y, z) \cong \mathbb{C}[x]$ is an integral domain, (y, z) is a prime ideal. Note that $xy \in (y, z)$ and $yz \in (y, z)$, we have $\{xy, yz\} \subseteq (y, z)$, and therefore $[(y, z)] \in V(xy, yz)$. □

Let's return to the general situation. The following proposition lets us restrict attention to vanishing sst of ideals.

Proposition 4.4.1

If (S) is the ideal generated by S, then $V(S) = V((S))$.

Proof Let $[\mathfrak{p}] \in V(S)$, then we have $S \subseteq \mathfrak{p}$, and therefore $(S) \subseteq \mathfrak{p}$, hence, $\mathfrak{p} \in V((S))$, i.e., $V(S) \subseteq V((S))$. Conversely, if $\mathfrak{p} \in V((S))$, then $(S) \subseteq \mathfrak{p}$. Note that $S \subseteq (S)$, $S \subseteq \mathfrak{p}$, and therefore $\mathfrak{p} \in V(S)$. □

Definition 4.4.2 (Zariski topology)

*We define the **Zariski topology** by declaring that $V(S)$ is closed for all S. (We may as well state here that **Zariski closure** means closure in the Zariski topology.)*

Let's check that the Zariski topology is indeed a topology.

Proposition 4.4.2 (Zariski topology is indeed a topology)

(a) $\varnothing$ and $\operatorname{Spec} A$ are both open subsets of $\operatorname{Spec} A$.

(b) If $\{I_i\}_i$ is a collection of ideals (as i runs over some index set), then $\bigcap_i V(I_i) = V(\sum_i I_i)$. Hence, the union of any collection of open sets is open.

(c) $V(I_1) \cup V(I_2) = V(I_1 I_2)$. Hence, the intersection of any finite number of open sets is open.

By (a), (b), and (c), Zariski topology is indeed a topology

Remark 4.7 The **product of two ideals** I_1 and I_2 of A are finite A-linear combinations of products of elements of I_1 and I_2, i.e., elements of the form $\sum_{j=1}^{n} i_{1,j} i_{2,j}$, where $i_{k,j} \in I_k$. Equivalently, it is the ideal generated by products of elements of I_1 and I_2. Also, products are associative, i.e., $(I_1 I_2) I_3 = I_1 (I_2 I_3)$.

Proof

(a) Note that $V(\varnothing) = \{[\mathfrak{p}] \in \operatorname{Spec} A : \mathfrak{p} \supseteq \varnothing\} = \operatorname{Spec} A$ and $V(\operatorname{Spec} A) = \{[\mathfrak{p}] \in \operatorname{Spec} A : \mathfrak{p} \supseteq \operatorname{Spec} A\} = \varnothing$, φ and $\operatorname{Spec} A$ are both closed set, and therefore φ and $\operatorname{Spec} A$ are both open.

(b) Since $\sum_i I_i$ is the smallest ideal which contains each I_i, $\mathfrak{p}$ contains $\sum_i I_i$ if and only if $\mathfrak{p}$ contains I_i for all i. It follows that $\bigcap_i V(I_i) = V(\sum_i I_i)$.

(c) Certainly, if $\mathfrak{p} \supseteq I_1$ or $\mathfrak{p} \supseteq I_2$, then $\mathfrak{p} \supseteq I_1 I_2$, which implies that $V(I_1) \cup V(I_2) \subseteq V(I_1 I_2)$. Conversely, if $\mathfrak{p} \supseteq I_1 I_2$ with $\mathfrak{p} \not\supseteq I_2$, then there exists $b \in I_2$ with $b \notin \mathfrak{p}$. Note that for any $a \in I_1$, we have $ab \in I_1 I_2 \subseteq \mathfrak{p}$, since $\mathfrak{p}$ is prime, $a \in \mathfrak{p}$, that is, $\mathfrak{p} \supseteq I_1$. Hence, $V(I_1) \cup V(I_2) \supseteq V(I_1 I_2)$. Consequently, $V(I_1) \cup V(I_2) = V(I_1 I_2)$.

□

4.4.2 Properties of the "vanishing set" function $V(\cdot)$

The function $V(\cdot)$ is obviously inclusion-reversing: if $S_1 \subseteq S_2$, then $V(S_2) \subseteq V(S_1)$. Warning: We could have equality in the second inclusion without equality in the first.

Definition 4.4.3 (Radical)

*If $I \subseteq A$ is an ideal, then define its **radical** by*

$$\sqrt{I} := \{r \in A : r^n \in I \text{ for some } n \in \mathbb{Z}^{>0}\}.$$

*We say an ideal is **radical** if it equals its own radical.*

Example 4.2 The nilradical $\mathfrak{N}$ is $\sqrt{(0)}$.

Proposition 4.4.3

(a) $\sqrt{I}$ is an ideal.

(b) Prime ideals are radical.

(c) $\sqrt{\sqrt{I}} = \sqrt{I}$.
(d) $V(\sqrt{I}) = V(I)$.

Proof

(a) Let $r \in \sqrt{I}$, then $r^n \in I$ for some n. For any $a \in A$, we have $a^n r^n = (ar)^n \in I$, since I is an ideal. Hence, $ar \in \sqrt{I}$. Note that $(-1)^n r^n = (-r)^n \in I$, $-r \in \sqrt{I}$. Let $r' \in \sqrt{I}$, then there exists m such that $r'^m \in I$, hence, $(r+r')^{n+m-1} \in I$, and therefore $r + r' \in \sqrt{I}$. Hence $\sqrt{I}$ is an ideal.

(b) It suffices to show that $\mathfrak{p} = \sqrt{\mathfrak{p}}$, where $\mathfrak{p} \in \operatorname{Spec} A$. Clearly, $\mathfrak{p} \subseteq \sqrt{\mathfrak{p}}$. Let $r \in \sqrt{\mathfrak{p}}$, then $r^n \in \mathfrak{p}$ for some n. Hence, $r \in \mathfrak{p}$, it follows that $\mathfrak{p} = \sqrt{\mathfrak{p}}$.

(c) Clearly, $\sqrt{I} \subseteq \sqrt{\sqrt{I}}$. Let $r \in \sqrt{\sqrt{I}}$, then $r^n \in \sqrt{I}$ for some n, by the definition of $\sqrt{I}$, $r^{nm} \in I$ for some m. Hence, $r \in \sqrt{I}$, it follows that $\sqrt{I} \supseteq \sqrt{\sqrt{I}}$.

(d) $V(\sqrt{I}) \subseteq V(I)$, since $I \subseteq \sqrt{I}$. Let $[\mathfrak{p}] \in V(I)$, then $\mathfrak{p} \supseteq I$, taking radical (use part (b)), we have $\mathfrak{p} \supseteq \sqrt{I}$. Hence, $[\mathfrak{p}] \in V(\sqrt{I})$, i.e., $V(\sqrt{I}) \supseteq V(I)$.

□

Remark 4.8 Here are two useful consequences.

(i) $(I \cap J)^2 \subseteq IJ \subseteq I \cap J$.

Proof Simple observation. □

(ii) $V(IJ) = V(I \cap J) = V(I) \cup V(J)$.

Proof By (i), we have $(I \cap J)^2 \subseteq IJ \subseteq I \cap J$, taking radical, we have

$$\sqrt{I \cap J} = \sqrt{(I \cap J)^2} \subseteq \sqrt{IJ} \subseteq \sqrt{I \cap J},$$

hence, $\sqrt{IJ} = \sqrt{I \cap J}$. By Proposition 4.4.3 (d),

$$V(\sqrt{I \cap J}) = V(I \cap J) = V(IJ) = V(\sqrt{IJ}).$$

□

(iii) $V(S) = V((S)) = V(\sqrt{(S)})$.

Proof By Proposition 4.4.1 and Proposition 4.4.3 (d). □

Proposition 4.4.4 (Radicals commute with finite intersections)

If $I_1, \dots, I_n$ are ideals of a ring A, then $\sqrt{\bigcap_{i=1}^n I_i} = \bigcap_{i=1}^n \sqrt{I_i}$.

Proof If we prove that $\sqrt{I_1 \cap I_2} = \sqrt{I_1} \cap \sqrt{I_2}$, then by induction $\sqrt{\bigcap_{i=1}^n I_i} = \bigcap_{i=1}^n \sqrt{I_i}$. Hence, it suffices to show that $\sqrt{I_1 \cap I_2} = \sqrt{I_1} \cap \sqrt{I_2}$

We already prove that $\sqrt{I_1 I_2} = \sqrt{I_1 \cap I_2}$. Let $f \in \sqrt{I_1 I_2}$, then $f^n \in I_1 \cap I_2$. Hence, $f^n \in I_1 \cap I_2 \subseteq I_1$ and $f^n \in I_1 \cap I_2 \subseteq I_2$, and therefore $f \in \sqrt{I_1}$ and $f \in \sqrt{I_2}$, i.e., $f \in \sqrt{I_1} \cap \sqrt{I_2}$. Thus, $\sqrt{I_1 I_2} \subseteq \sqrt{I_1} \cap \sqrt{I_2}$.

Conversely, let $f \in \sqrt{I_1} \cap \sqrt{I_2}$, then there exists n and m such that $f^n \in I_1$ and $f^m \in I_2$, and therefore $f^{nm} \in I_1 \cap I_2$. It follows that $f \in \sqrt{I_1 \cap I_2}$.

By above discussion, we have $\sqrt{I_1 I_2} = \sqrt{I_1} \cap \sqrt{I_2}$, hence

$$\sqrt{I_1 \cap I_2} = \sqrt{I_1} \cap \sqrt{I_2}.$$

□

Proposition 4.4.5

$\sqrt{I}$ is the intersection of all the prime ideals containing I.

Proof Let $\pi : A \to A/I$ be a natural homomorphism. We claim that $\pi(\sqrt{I}) = \mathfrak{N}(A/I)$. Let $f \in \sqrt{I}$, then $f^n \in I$ for some n, hence $\pi(f^n) = \pi(f)^n = 0$, it follows that $\pi(f) \in \mathfrak{N}(A/I)$, i.e., $\pi(\sqrt{I}) \subseteq \mathfrak{N}(A/I)$. Conversely, let $f \in \mathfrak{N}(A/I)$, then $f^n = 0$ for some n. Since π is surjective, $f = \pi(g)$ for some $g \in A$. Hence, $\pi(g)^n = \pi(g^n) = 0$, which implies that $g^n \in I$, and therefore $g \in \sqrt{I}$. Thus, $\pi(\sqrt{I}) = \mathfrak{N}(A/I)$.

Note that $\mathfrak{N}(A/I) = \bigcap_{\mathfrak{q}\in \operatorname{Spec} A/I} \mathfrak{q}$, say $\mathfrak{p} = \pi^{-1}(\mathfrak{q}) \in \operatorname{Spec} A$, we have

$$\sqrt{I} = \pi^{-1}\left(\bigcap_{\mathfrak{q}\in \operatorname{Spec} A/I} \mathfrak{q}\right) = \bigcap_{\mathfrak{q}\in \operatorname{Spec} A/I} \pi^{-1}(\mathfrak{q}) = \bigcap_{\mathfrak{p}\in \operatorname{Spec} A,\ \mathfrak{p}\supseteq I} \mathfrak{p},$$

as we desired. □

Example 4.3 Let's see how this meshes with our examples from the previous section.

Recall that $\mathbb{A}^1_{\mathbb{C}}$, as a set, was just the "traditional" points (corresponding to maximal ideas, in bijection with $a \in \mathbb{C}$), and one "new" point $[(0)]$. The Zarisiki topology on $\mathbb{A}^1_{\mathbb{C}}$ is not that exciting: the open sets are the empty set, and $\mathbb{A}^1_{\mathbb{C}}$ minus a finite number of maximal ideals. (It "almost" has the cofinite topology. Notice that the open sets are determined by their intersections with the "traditional points". The "new" point $[(0)]$ comes along for the ride, which is a good sign that it is harmless. Ignoring the "new" point, observe that the topology on $\mathbb{A}^1_{\mathbb{C}}$ is a coarser topology then the classical topology on $\mathbb{C}$.)

✍ **Exercise 4.15** Describe the topological space $\mathbb{A}^1_k$.

Proof In fact, $\mathbb{A}^1_k = \operatorname{Spec} k[x] = \{(0), \mathfrak{m}_a\ a \in k\}$, where $\mathfrak{m}_a$ is maximal ideal. Hence, the closed subsets of $\mathbb{A}^1_k$ are:

(i) the set of maximal ideals;
(ii) $\operatorname{Spec} k[x]$ and $\varnothing$.

The open subsets of $\mathbb{A}^1_k$ are:

(i) $\operatorname{Spec} k[x]$ minus a finite number of maximal ideals;
(ii) $\operatorname{Spec} k[x]$ and $\varnothing$.

□

Remark 4.9 Notice that the strange new point $[(0)]$ is "near every other point" — every neighborhood of every point contains $[(0)]$.

The case of $\operatorname{Spec}\mathbb{Z}$ is similar. The topology is "almost" the cofinite topology in the same way. The open sets are the empty set, $\operatorname{Spec}\mathbb{Z}$, and $\operatorname{Spec}\mathbb{Z}$ minus a finite number of "ordinary" ((p) where p is prime) primes.

Example 4.4 **Closed subsets of $\mathbb{A}^2_{\mathbb{C}}$.** The case $\mathbb{A}^2_{\mathbb{C}}$ is more interesting. You should think through where the "one-dimensional prime ideals" fit into the picture. In Exercise 4.5, we identified all the prime ideals of $\mathbb{C}[x, y]$ (i.e., the points of $\mathbb{A}^2_{\mathbb{C}}$) as the maximal ideals $[(x-a, y-b)]$ (where $a, b \in \mathbb{C}$ — "zeor-dimensional points"), the "one-dimensional points" $[(f(x, y))]$ (where $f(x, y)$ is irreducible), and the "two-dimensional point" $[(0)]$.

Then the closed subsets are of the following form:

(a) the entire space (the closure of the "two-dimensional" $[(0)]$),
(b) a finite number (possibly none) of "curves" (each the closure of a "one-dimensional point" — the "one-dimensional point" along with the "zero-dimensional points" "lying on it") and a finite number (possibly none) of "zero-dimensional" points (what we will soon call "closed points".)

We will soon know enough to verify this using general theory.

4.4.3 Maps of rings induce continuous maps of topological spaces

We saw in §4.2.3 that a homomorphism of rings $\varphi : B \to A$ induces a map of sets $\pi : \operatorname{Spec} A \to \operatorname{Spec} B$.

Proposition 4.4.6

Let $\varphi : B \to A$ be a homomorphism of rings, $\pi : \operatorname{Spec} A \to \operatorname{Spec} B$ is induced by φ, then

(a) π is a continuous map;

(b) $\operatorname{Spec}\square$ can be interpreted as a contravariant functor **Rings** $\to$ **Top**.

Proof

(a) Let $V(I) \subseteq \operatorname{Spec} B$ be a closed subset, where I is an ideal of B. In fact, $\pi : \operatorname{Spec} A \to \operatorname{Spec} B$ is given by $\pi(\mathfrak{p}) = \varphi^{-1}(\mathfrak{p})$. Consider $\pi^{-1}(V(I))$, let $\mathfrak{q} \in \pi^{-1}(V(I))$, then $\pi(\mathfrak{q}) = \varphi^{-1}(\mathfrak{q}) \in V(I)$. Since $V(I) = \{\mathfrak{p} \in \operatorname{Spec} B : \mathfrak{p} \supseteq I\}$, we have $\varphi^{-1}(\mathfrak{q}) \supseteq I$, and therefore $\mathfrak{q} \supseteq \varphi(I)$, it follows that $\pi^{-1}(V(I)) \subseteq V(\varphi(I)) \subseteq \operatorname{Spec} A$. We claim that $V(\varphi(I)) = \pi^{-1}(V(I))$. Let $\mathfrak{q} \in V(\varphi(I))$, then $\mathfrak{q} \supseteq \varphi(I)$, hence, $\varphi^{-1}(\mathfrak{q}) \supseteq I$. It follows that $\varphi^{-1}(\mathfrak{q}) \in V(I) \subseteq \operatorname{Spec} B$, i.e., $\pi(\mathfrak{q}) \in V(I)$, and therefore $\mathfrak{q} \in \pi^{-1}(V(I))$. Thus, $V(\varphi(I)) = \pi^{-1}(V(I)) \subseteq \operatorname{Spec} A$, and therefore $\pi^{-1}(V(I))$ is closed in $\operatorname{Spec} A$. Hence, π is a continuous map.

(b) Let $\varphi : A \to B \in$ **Rings**, then $\operatorname{Spec}(\varphi) = \varphi^{-1} : \operatorname{Spec} B \to \operatorname{Spec} A$. Let $A \in$ **Rings**, then $\operatorname{Spec}(\mathrm{id}_A) = \mathrm{id}_A^{-1} = \mathrm{id}_{\operatorname{Spec} A}$, which implies that $\operatorname{Spec}\square$ preserves identity morphisms. Let $\varphi : A \to B$ and $\psi : B \to C$, then $\psi \circ \varphi : A \to C$. Consider $\operatorname{Spec}(\psi \circ \varphi)$, we have

$$\operatorname{Spec}(\psi \circ \varphi) = (\psi \circ \varphi)^{-1} = \varphi^{-1} \circ \psi^{-1} = \operatorname{Spec}\varphi \circ \operatorname{Spec}\psi.$$

Hence, $\operatorname{Spec}\square$ is a contravariant functor.

□

Remark 4.10 Not all continuous maps arise in this way. Consider for example the continuous map on $\mathbb{A}^1_{\mathbb{C}}$ that is the identity except 0 and 1 (i.e., $[(x)]$ and $[(x-1)]$) are swapped; no polynomial can manage this marvelous feat.

Proof Let $\varphi : \mathbb{C}[x] \to \mathbb{C}[x]$ be a ring homomorphism, given by $x \mapsto p(x)$, where $p(x) \in \mathbb{C}[x]$. Then the induced map $\pi : \operatorname{Spec}\mathbb{C}[x] \to \operatorname{Spec}\mathbb{C}[x]$ is given by $\pi((x-a)) = \varphi^{-1}((x-a)) = (x - p(a))$.

If continuous map $\pi : \operatorname{Spec}\mathbb{C}[x] \to \operatorname{Spec}\mathbb{C}[x]$ is given by

$$\pi((x-a)) = \begin{cases} [(x-1)] & \text{if } a = 0 \\ [(x)] & \text{if } a = 1 \\ [(x-a)] & \text{others}, \end{cases}$$

then $p(0) = 1$, $p(1) = 0$, and $p(a) = a$ for $a \neq 0, 1$. Consider $q(x) = p(x) - x$, then $q(a) = 0$ for all $a \in \mathbb{C} \setminus \{0, 1\}$, i.e., polynomial $q(x)$ has infinite zeros, this contradicts the Fundamental Theorem of Algebra. Hence, not all continuous maps induced by a ring homomorphism. □

In §4.2.2, we saw that $\operatorname{Spec} B/I$ and $\operatorname{Spec} S^{-1}B$ are naturally subsets of $\operatorname{Spec} B$. It is natural to ask if the Zariski topology behaves well with respect to these inclusions, and indeed it does.

Proposition 4.4.7

Suppose that $I, S \subseteq B$ are an ideal and multiplicative subset respectively.

(a) $\operatorname{Spec} B/I$ is naturally a closed subset of $\operatorname{Spec} B$. If $S = \{1, f, f^2, \dots\}$ ($f \in B$), then $\operatorname{Spec} S^{-1}B$ is naturally an open subset of $\operatorname{Spec} B$.

(b) The Zariski topology on $\operatorname{Spec} B/I$ (resp., $\operatorname{Spec} S^{-1}B$) is the subspace topology induced by inclusion in $\operatorname{Spec} B$.

Proof

(a) Note that $\operatorname{Spec} B/I = \{\mathfrak{p} \in \operatorname{Spec} B : \mathfrak{p} \supseteq I\} = V(I) \subseteq \operatorname{Spec} B$, $\operatorname{Spec} B/I$ is a closed subset of $\operatorname{Spec} B$.

In fact,

$$\begin{aligned}\operatorname{Spec} B_f =&\{\mathfrak{p} \in \operatorname{Spec} B : \mathfrak{p} \cap \{1, f, f^2, \dots\} = \varnothing\}\\ =&\{\mathfrak{p} \in \operatorname{Spec} B : f \notin \mathfrak{p}\}\\ =&\operatorname{Spec} B - \{\mathfrak{p} \in \operatorname{Spec} B : (f) \subseteq \mathfrak{p}\}\\ =&\operatorname{Spec} B - V(f).\end{aligned}$$

Hence, $\operatorname{Spec} S^{-1}B$ is an open subset of $\operatorname{Spec} B$.

(b) Let $S \subseteq B/I$ be an ideal, then exists $\widetilde{S} \supseteq I$ in $\operatorname{Spec} B$ such that the image of $\widetilde{S}$ in B/I is S. Then

$$\begin{aligned}V(S) =&\{\mathfrak{p} \in \operatorname{Spec} B/I : \mathfrak{p} \supseteq S\}\\ =&\{\mathfrak{q} \in \operatorname{Spec} B : \mathfrak{q} \supseteq \widetilde{S}\} \cap \{\mathfrak{q} \in \operatorname{Spec} B : \mathfrak{q} \supseteq I\}\\ =&V(\widetilde{S}) \cap \operatorname{Spec} B/I.\end{aligned}$$

It follows that the Zariski topology on $\operatorname{Spec} B/I$ is the subspace topology induced by inclusion in $\operatorname{Spec} B$. Let $T \subseteq B_f$ be an ideal, then exists ideal $\widetilde{T} \subseteq B$ with $\widetilde{T} \cap S = \varnothing$ such that $T = S^{-1}\widetilde{T} := \widetilde{T}_f$. Then

$$\begin{aligned}V(T) =&\{\mathfrak{p} \in \operatorname{Spec} B_f : \mathfrak{p} \supseteq T\}\\ =&\{\mathfrak{q} \in \operatorname{Spec} B : \mathfrak{q} \cap S = \varnothing, \mathfrak{q}_f \supseteq T = \widetilde{T}_f\}\\ =&\{\mathfrak{q} \in \operatorname{Spec} B : \mathfrak{q} \cap S = \varnothing\} \cap \{\mathfrak{q} \in \operatorname{Spec} B : \mathfrak{q}_f \supseteq \widetilde{T}_f\}\\ =&\operatorname{Spec} B_f \cap \{\mathfrak{q} \in \operatorname{Spec} B : \mathfrak{q} \supseteq \widetilde{T}\}\\ =&\operatorname{Spec} B_f \cap V(\widetilde{T}).\end{aligned}$$

It follows that the Zariski topology on $\operatorname{Spec} S^{-1}B$ is the subspace topology induced by inclusion in $\operatorname{Spec} B$.

$\square$

Remark 4.11 For arbitrary S, $\operatorname{Spec} S^{-1}B$ need not be open or closed.

Proof Consider $\operatorname{Spec} \mathbb{Q} = \{(0)\} \subseteq \operatorname{Spec} \mathbb{Z}$. In fact, $\operatorname{Spec} \mathbb{Z} = \{(0), (p) \text{ where } p \text{ is prime}\}$. Let $I = (n)$ be any ideal of $\mathbb{Z}$, then

$$\begin{aligned}V(I) &= \begin{cases}\{\mathfrak{p} \in \operatorname{Spec} \mathbb{Z} : \mathfrak{p} \supseteq (n)\} & \text{if } n \neq 0\\ \{\mathfrak{p} \in \operatorname{Spec} \mathbb{Z} : \mathfrak{p} \supseteq (0)\} & \text{if } n = 0\end{cases}\\ &= \begin{cases}\{(p) : p \mid n\} & \text{if } n \neq 0\\ \operatorname{Spec} \mathbb{Z} & \text{if } n = 0.\end{cases}\end{aligned}$$

It follows that (0) is neither open nor closed. $\square$

In particular, if $I \subseteq \mathfrak{N}$ is an ideal of nilpotents, the bijection $\operatorname{Spec} B/I \to \operatorname{Spec} B$ (Proposition 4.2.7 gives bijection, Proposition 4.4.6 gives continuous) is a homeomorphism. Thus nilpotents don't affect the topological space.

Proposition 4.4.8

Suppose $I \subseteq B$ is an ideal, f vanishes on $V(I)$ if and only if $f \in \sqrt{I}$.

Proof If f vanishes on $V(I)$, then for all $\mathfrak{p} \in V(I)$, we have $f \in \mathfrak{p}$. Note that $\sqrt{I}$ is the intersection of all prime ideals containing I, hence, $f \in \sqrt{I}$.

Conversely, if $f \in \sqrt{I}$, since $\sqrt{I}$ is the intersection of all prime ideals containing I, hence $f \in \mathfrak{p}$ where $\mathfrak{p}$

contains I, and therefore f vanishes on $V(I)$. $\square$

Exercise 4.16 Describe the topological space $\operatorname{Spec} k[x]_{(x)}$.

Proof In fact,

$$\begin{aligned}\operatorname{Spec} k[x]_{(x)} =&\{\mathfrak{p} \in \operatorname{Spec} k[x] : \mathfrak{p} \cap (\operatorname{Spec} k[x] - (x)) = \varnothing\}\\ =&\{\mathfrak{p} \in \operatorname{Spec} k[x] : \mathfrak{p} \subseteq (x)\}\\ =&\{(0),(x)\}.\end{aligned}$$

Let I be any ideal of $k[x]_{(x)}$, then

$$\begin{aligned}V(I) =&\{\mathfrak{p} \in \operatorname{Spec} k[x]_{(x)} : \mathfrak{p} \supseteq I\}\\ =&\begin{cases}\{(x)\} & I=(x)\\ \operatorname{Spec} k[x]_{(x)} & I=(0)\end{cases}\end{aligned}$$

Hence, the closed set in $\operatorname{Spec} k[x]_{(x)}$ are $\varnothing$, (x), and $\operatorname{Spec} k[x]_{(x)}$. $\square$

4.5 A base of the Zariski topological on Spec A: Distinguished open sets

Definition 4.5.1 (Distinguished open set)

If $f \in A$, define the distinguished open set

$$\begin{aligned}D(f) :=& \{[\mathfrak{p}] \in \operatorname{Spec} A : f \notin \mathfrak{p}\}\\ =& \{[\mathfrak{p}] : \operatorname{Spec} A : f([\mathfrak{p}]) \neq 0\}.\end{aligned}$$

It is the locus where f doesn't vanish.

We have already seen this set when discussing $\operatorname{Spec} A_f$ as a subset of $\operatorname{Spec} A$. For example, we have observed that the Zariski-topology on the distinguished open set $D(f) \subseteq \operatorname{Spec} A$ coincides with the Zariski topology on $\operatorname{Spec} A_f$ (Proposition 4.4.7).

The reason these sets are important is that they form a particularly nice base for the (Zariski) topology:

Proposition 4.5.1

The distinguished open sets form a base for the Zariski topology.

Proof Let $S \subseteq A$, then $V(S) = \{\mathfrak{p} \in \operatorname{Spec} A : \mathfrak{p} \supseteq S\}$. Note that open subset $\operatorname{Spec} A - V(S) = \{\mathfrak{p} \in \operatorname{Spec} A : \mathfrak{p} \not\supseteq S\}$, we claim that $\{\mathfrak{p} \in \operatorname{Spec} A : \mathfrak{p} \not\supseteq S\} = \bigcup_{f \in S} D(f)$.

Let $\mathfrak{p} \in \{\mathfrak{p} \in \operatorname{Spec} A : \mathfrak{p} \not\supseteq S\}$, then $\mathfrak{p} \not\supseteq S$, hence, there exists $f \in S$ such that $f \notin \mathfrak{p}$. It follows that $[\mathfrak{p}] \in D(f)$, and therefore $\{\mathfrak{p} \in \operatorname{Spec} A : \mathfrak{p} \not\supseteq S\} \subseteq \bigcup_{f \in S} D(f)$.

Let $\mathfrak{p} \in \bigcup_{f \in S} D(f)$, then exists $f \in S$ such that $\mathfrak{p} \in D(f)$, hence, $f \notin \mathfrak{p}$, and therefore $\mathfrak{p} \in \{\mathfrak{p} \in \operatorname{Spec} A : \mathfrak{p} \not\supseteq S\}$. Hence, $\bigcup_{f \in S} D(f) \subseteq \{\mathfrak{p} \in \operatorname{Spec} A : \mathfrak{p} \not\supseteq S\}$.

Consequently, we have $\{\mathfrak{p} \in \operatorname{Spec} A : \mathfrak{p} \not\supseteq S\} = \bigcup_{f \in S} D(f)$. Hence, $D(f)$ form a base for the Zariski topology on $\operatorname{Spec} A$. $\square$

Proposition 4.5.2

Suppose $f_i \in A$ as i runs over some index set J, then $\bigcup_{i \in J} D(f_i) = \operatorname{Spec} A$ if and only if $(\{f_i\}_{i \in J}) = A$, or equivalently and very usefully, if there are a_i $(i \in J)$, all but finitely many 0, such that $\sum_{i \in J} a_i f_i = 1$.

Proof If $(\{f_i\}_{i \in J}) = A$, then exists $a_i \in A$ such that $\sum_i a_i f_i = 1$ with all but finitely many a_i being zero.

Let $\mathfrak{p} \in \operatorname{Spec} A$, if $\mathfrak{p}$ contains each f_i, then $\mathfrak{p} = (1) = A$, this contradicts to $\mathfrak{p}$ is a prime ideal. Hence, there exists f_i such that $f_i \notin \mathfrak{p}$, and therefore $[\mathfrak{p}] \in D(f_i)$. Hence, $\operatorname{Spec} A \subseteq \bigcup_{i \in J} D(f_i)$. Clearly, we have $\bigcup_{i \in J} D(f_i) \subseteq \operatorname{Spec} A$. Hence, $\bigcup_{i \in J} D(f_i) = \operatorname{Spec} A$.

Conversely, if $\bigcup_{i \in J} D(f_i) = \operatorname{Spec} A$, then for all $\mathfrak{p} \in \operatorname{Spec} A$, there exists f_i such that $\mathfrak{p} \in D(f_i)$, i.e., $f_i \notin \mathfrak{p}$. Hence, there are no prime ideal contains all f_i. Note that maximal ideal is prime ideal, hence, $(\{f_i\}_{i \in J}) = A$. □

Proposition 4.5.3

If $\operatorname{Spec} A$ is an infinite union of distinguished open sets $\bigcup_{j \in J} D(f_j)$, then in fact it is a union of a finite number of these, i.e., there is a finite subset J' so that $\operatorname{Spec} A = \bigcup_{j \in J'} D(f_j)$.

Proof Since $1 = \sum_{i \in J} a_i f_i$ where J is finite set, we have $A = (f_i)_{i \in J}$, by Proposition 4.5.2, we done. □

Proposition 4.5.4

$D(f) \cap D(g) = D(fg)$. In particular, $D(f^n) = D(f)$ for any $n \in \mathbb{Z}$.

Proof

$$
\begin{aligned}
D(f) \cap D(g) =& \{\mathfrak{p} \in \operatorname{Spec} A : f \notin \mathfrak{p}, g \notin \mathfrak{p}\} \\
\subseteq& \{\mathfrak{p} \in \operatorname{Spec} A : fg \notin \mathfrak{p}\} = D(fg).
\end{aligned}
$$

Conversely, let $[\mathfrak{p}] \in D(fg)$, then $fg \notin \mathfrak{p}$. If $f \in \mathfrak{p}$, since $\mathfrak{p}$ is an ideal, $fg \in \mathfrak{p}$, a contradiction. Hence, $f \notin \mathfrak{p}$, similarly, $g \notin \mathfrak{p}$. It follows that

$$D(fg) \subseteq D(f) \cap D(g).$$

□

Proposition 4.5.5

The following are equivalent:

(i) $D(f) \subseteq D(g)$.

(ii) $f^n \in (g)$ for some $n \geq 1$.

(iii) g is an invertible element of A_f.

Proof (i) $\Rightarrow$ (ii): Since $D(f) \subseteq D(g)$, then $V(f) \supseteq V(g)$. For any $\mathfrak{p} \in V(g)$, we have $\mathfrak{p} \ni f$, hence, $f \in \bigcap_{\mathfrak{p} \in V(g)} \mathfrak{p} = \sqrt{(g)}$. It follows that $f^n \in (g)$ for some $n \geq 1$.

(ii) $\Rightarrow$ (iii): Since $f^n \in (g)$ for some $n \geq 1$, exists $h \in A$ such that $f^n = hg$. Since f^n is an invertible element of A_f, we have $1 = (\frac{h}{f^n})g$. Hence, g is an invertible element of A_f.

(iii) $\Rightarrow$ (i): Since g is an invertible element of A_f, $g \in \{1, f, f^2, \dots\}$. We may assume that $g = f^n$ for some $n \geq 1$. Let $\mathfrak{p} \in V(g)$, then $g = f^n \in \mathfrak{p}$, and therefore $f \in \mathfrak{p}$, which implies that $\mathfrak{p} \in V(f)$. Hence, $V(g) \subseteq V(f)$, i.e., $D(f) \subseteq D(g)$. □

Remark 4.12 We will use Proposition 4.5.5 often. You can prove it thinking purely algebraically, but the following geometric interpretation may be helpful. Inside $\operatorname{Spec} A$, we have the closed subset $V(g) = \operatorname{Spec} A/(g)$, where g vanishes, and its complement $D(g)$, where g doesn't vanish. Then f is a function on this closed subset $V(g)$ (or more precisely, on $\operatorname{Spec} A/(g)$), and by assumption it vanishes at all points of the closed subset. Now any function vanishing at every point of the spectrum of ring must be nilpotent (Theorem 4.2.3). In other words, there is some n such that $f^n = 0$ in $A/(g)$, i.e., $f^n \equiv 0 \pmod g$ in A, i.e., $f^n \in (g)$.

Proposition 4.5.6

$D(f) = \varnothing$ if and only if $f \in \mathfrak{N}$.

Proof Since $D(f) = \varnothing$, we have $V(f) = \operatorname{Spec} A$, it follows that f vanishing at every point of Spec, by Theorem 4.2.3, we done. □

Proposition 4.5.7 (Injective ring homomorphisms induce maps of spectra with dense image)

If $B \hookrightarrow A$, then the induced map of topological spaces $\operatorname{Spec} A \to \operatorname{Spec} B$ has dense image.

Proof Say $\varphi : B \hookrightarrow A$, then $\pi : \operatorname{Spec} A \to \operatorname{Spec} B$ is given by $\pi(\mathfrak{p}) = \varphi^{-1}(\mathfrak{p})$. Suppose the image of $\operatorname{Spec} A$ not dense, then there exists a non-empty open subset of $\operatorname{Spec} B$, say U, such that $\pi(\operatorname{Spec} A) \cap U = \varnothing$. Since, each open subset of spectrum is the union of distinguished open set, we may assume $U = D(f)$, where $f \in B$.

Consider $\pi(\operatorname{Spec} A)$, we have $\pi(\operatorname{Spec} A) = \{\pi(\mathfrak{p}) : \mathfrak{p} \in \operatorname{Spec} A\} = \{\varphi^{-1}(\mathfrak{p}) : \mathfrak{p} \in \operatorname{Spec} A\}$. Since $D(f) \cap \{\varphi^{-1}(\mathfrak{p}) : \mathfrak{p} \in \operatorname{Spec} A\} = \varnothing$, for all $\mathfrak{p} \in \operatorname{Spec} A$, we have $\varphi^{-1}(\mathfrak{p}) \notin D(f)$, i.e., $\varphi^{-1}(\mathfrak{p}) \in V(f)$. Hence, $f \in \varphi^{-1}(\mathfrak{p})$ for all $\mathfrak{p} \in \operatorname{Spec} A$, i.e., $\varphi(f) \in \mathfrak{p}$ for all $\mathfrak{p} \in \operatorname{Spec} A$. By Theorem 4.2.3, $\varphi(f)$ is nilpotent, i.e., $\varphi(f)^n = \varphi(f^n) = 0$ for some n. Since φ is injective, we have $f^n = 0$. By Proposition 4.5.6, $D(f) = \varnothing$, a contradiction. □

Remark 4.13 **Caution about notation.** There will be two variations on the notation $D(f)$: the projective distinguished open set $D_+(f)$ (the locus on a projective scheme where a "form f of positive degree" doesn't vansh), and X_f (the locus on a scheme X where a function f doesn't vanish). These are almost the same thing as $D(f)$, but not quite, so we keep the notation separate for reasons of hygiene.

4.6 Topological (and Noetherian) properties

Many topological notions are useful when applied to the topological space $\operatorname{Spec} A$, and later, to schemes.

4.6.1 Possible topological attributes of $\operatorname{Spec} A$: Connectedness, Irreducibility, Quasicompactness

Connectedness

Definition 4.6.1 (Connected)

A topological space X is ***connected*** *if it cannot be written as the disjoint union of two nonempty open sets.*

Example 4.5 below gives an example of a non-connected $\operatorname{Spec} A$, and the subsequent remark explains that all examples are of this form.

Example 4.5 If $A = A_1 \times A_2 \times \cdots \times A_n$, then there is a homeomorphism $\operatorname{Spec} A_1 \coprod \operatorname{Spec} A_2 \coprod \cdots \coprod \operatorname{Spec} A_n \to \operatorname{Spec} A$ for which each $\operatorname{Spec} A_i$ is mapped onto a distinguished open subset $D(f_i)$ of $\operatorname{Spec} A$. Thus $\operatorname{Spec} \prod_{i=1}^n A_i = \coprod_{i=1}^n \operatorname{Spec} A_i$ as topological spaces.

Remark 4.14 $\operatorname{Spec} \prod_{i=1}^n A_i = \{A_1 \times A_2 \times \cdots \times \mathfrak{p}_i \times \cdots \times A_n : \mathfrak{p}_i \in \operatorname{Spec} A_i\}$.

The ideal of $\prod_{i=1}^n A_i$ is of the form $I_1 \times \cdots \times I_n$, where I_k is the ideal of A_k.

Proof Prove by induction. It suffices to show the case of $n = 2$, i.e., $\operatorname{Spec}(A_1 \times A_2) = \operatorname{Spec} A_1 \coprod \operatorname{Spec} A_2$. Define $\varphi_1 : \operatorname{Spec} A_1 \to \operatorname{Spec}(A_1 \times A_2)$ by setting $\mathfrak{p} \mapsto \mathfrak{p} \times A_2$, and define $\varphi_2 : \operatorname{Spec} A_2 \to \operatorname{Spec}(A_1 \times A_2)$

by setting $\mathfrak{p} \mapsto A_1 \times \mathfrak{p}$. Since φ_1 and φ_2 induced by projection $A \to A_i$, φ_i are both well-defined. Then we get a map:

$$\begin{aligned} \varphi : \operatorname{Spec} A_1 \coprod \operatorname{Spec} A_2 &\longrightarrow \operatorname{Spec}(A_1 \times A_2) \\ (\mathfrak{p}, i) &\longmapsto \varphi_i(\mathfrak{p}). \end{aligned}$$

Let $f_1 = (1, 0)$ and $f_2 = (0, 1)$ in $A_1 \times A_2$, then we have

$$D(f_1) = \{\mathfrak{p} \in \operatorname{Spec} A : f_1 \notin \mathfrak{p}\}, \quad D(f_2) = \{\mathfrak{p} \in \operatorname{Spec} A : f_2 \notin \mathfrak{p}\}.$$

Note that φ_i is a bijection between $\operatorname{Spec} A_i$ and $D(f_i)$. We claim that φ_i is also a homeomorphism.

Let I be any ideal of $A_1 \times A_2$ with $f_i \notin I$. Then the closed subset of $D(f_i)$ has the form $V(I) = \{\mathfrak{p} \in D(f_i) : \mathfrak{p} \supseteq I\}$. Consider $\varphi_i^{-1}(V(I))$,

$$\begin{aligned} \varphi_i^{-1}(V(I)) =& \varphi_i^{-1}(\{\mathfrak{p} \in D(f_i) : \mathfrak{p} \supseteq I\}) \\ =& \{\mathfrak{p}_i \in \operatorname{Spec} A_i : \mathfrak{p}_i \supseteq I_i, 1 \notin I_i\} \\ =& V(I_i). \end{aligned}$$

Hence, $\varphi_i^{-1}(V(I))$ closed in $\operatorname{Spec} A_i$. It follows that φ_i is continuous for all $i = 1, 2$.

Let I_i be an ideal of A_i, consider $\varphi_i(V(I_i))$,

$$\begin{aligned} \varphi_i(V(I_i)) =& \varphi_i(\{\mathfrak{p} \in \operatorname{Spec} A_i : \mathfrak{p} \supseteq I_i\}) \\ =& \{\varphi_i(\mathfrak{p}) \in \operatorname{Spec}(A_1 \times A_2) : \varphi_i(\mathfrak{p}) \supseteq \varphi_i(I_i)\} \\ =& V(\varphi_i(I_i)), \end{aligned}$$

which is closed in $\operatorname{Spec}(A_1 \times A_2)$. Hence, φ_i^{-1} is continuous for all $i = 1, 2$. It follows that each φ_i is homeomorphism.

Note that $\operatorname{Spec}(A_1 \times A_2) = D(f_1) \coprod D(f_2)$ and each φ_i is homeomorphism, φ is a homeomorphism between $\operatorname{Spec} A_1 \coprod \operatorname{Spec} A_2$ and $\operatorname{Spec}(A_1 \times A_2)$, as we desired. $\square$

Remark 4.15 The idempotent-connectedness package. An extesion of Example 4.5 is that $\operatorname{Spec} A$ is not connected if and only if A is isomorphic to the product of non-zero rings A_1 and A_2. The key idea is to show that both conditions are equivalent to there existing non-zero $a_1, a_2 \in A$ for which $a_1^2 = a_1$, $a_2^2 = a_2$, $a_1 + a_2 = 1$, and hence $a_1 a_2 = 0$. An element $a \in A$ satisfying $a^2 = a$ is called an **idempotent**.

Irreducibility

Definition 4.6.2 (Irreducible)

A topological space is said to be ***irreducible*** *if it is nonempty, and it is not the union of two proper closed subsets. In other words, a nonempty topological space X is irreducible if whenever $X = Y \cup Z$ with Y and Z closed in X, we have $Y = X$ or $Z = X$.*

Equivalently (and helpfully): any two nonempty open subsets of X intersect.

This is a less useful notion in classical geometry — $\mathbb{C}^2$ is reducible (endow classical topology, $\mathbb{C}^2 = \{(z_1, z_2) \in \mathbb{C}^2 : \operatorname{Re}(z_1) \geq 0\} \cup \{(z_1, z_2) \in \mathbb{C}^2 : \operatorname{Re}(z_1) \leq 0\} := A \cup B$, A, B are closed), but we will see that $\mathbb{A}^2_{\mathbb{C}}$ is irreducible.

Proposition 4.6.1

(a) In an irreducible topological space, any non-empty open set is dense.

(b) If X is a topological space, and Z is a subset of X (with the subspace topology), then Z is irreducible if and only if the closure $\overline{Z}$ in X is irreducible.

Proof

(a) Let X be an irreducible topological space. If there exists an non-empty open set is not dense, say U, then $X \setminus U$ is a non-empty proper closed subset of X. Note that $X = \overline{U} \cup X \setminus U$ where $\overline{U}$ and $X \setminus U$ are proper closed, which contradicts the fact that X is irreducible.

(b) If Z is irreducible. Suppose that $\overline{Z}$ is not irreducible, then there exists two proper closed subsets of $\overline{Z}$, say Z_1, Z_2, such that $\overline{Z} = Z_1 \cup Z_2$. Hence, $Z = Z \cap (Z_1 \cup Z_2) = (Z \cap Z_1) \cup (Z \cap Z_2)$. Note that $Z \cap Z_i$ is closed in Z and Z is irreducible, $Z = Z \cap Z_1$ or $Z = Z \cap Z_2$. We may assume that $Z = Z \cap Z_1$, then $Z \subseteq Z_1$, and therefore $\overline{Z} \subseteq Z_1$, which contradicts the fact that Z_1 is a proper closed subset of $\overline{Z}$.
Conversely, if the closure $\overline{Z}$ in X is irreducible. Suppose that Z is reducible, i.e., $Z = Z_1 \cup Z_2$ where Z_1 and Z_2 are proper closed in Z. Since Z endowed the subspace topology of X, there exists $A, B \subseteq X$ closed sets such that $Z_1 = Z \cap A$ and $Z_2 = Z \cap B$. Hence, $Z \subseteq A \cup B$, taking closure, we have $\overline{Z} \subseteq \overline{A \cup B} = A \cup B$. Thus, $\overline{Z} = (A \cap \overline{Z}) \cup (B \cap \overline{Z})$, since $\overline{Z}$ is irreducible, we may assume that $\overline{Z} = \overline{Z} \cap A$, i.e., $\overline{Z} \subseteq A$. Hence, $Z \subseteq A \cap Z = Z_1$, which contradicts the fact that Z_1 is proper closed in Z.

□

Remark 4.16 For (a), we see that unlike in the classical topology, in the Zariski topology, non-empty open sets are all "huge".

Proposition 4.6.2

(i) If A is an integral domain, then $\operatorname{Spec} A$ is irreducible.

(ii) $\operatorname{Spec} A$ is irreducible if and only if $\mathfrak{N}(A)$ is prime.

Proof

(i) Suppose that $\operatorname{Spec} A$ is reducible, then there exists non-empty proper closed subsets, say $V(I_1), V(I_2)$ where I_i is ideal of A, such that $\operatorname{Spec} A = V(I_1) \cup V(I_2)$. Note that
$$\begin{aligned} V(I_1) \cup V(I_2) =& V(I_1 I_2) \\ =& \{\mathfrak{p} \in \operatorname{Spec} A : \mathfrak{p} \supseteq I_1 I_2\} = \operatorname{Spec} A, \end{aligned}$$
i.e., any prime ideal of A must contain $I_1 I_2$. Hence, $I_1 I_2 \subseteq \mathfrak{N}(A)$. Since A is an integral domain, $\mathfrak{N}(A) = (0)$, hence, $I_1 I_2 = (0)$, and therefore $I_1 = (0)$ or $I_2 = (0)$. We may assume that $I_1 = (0)$, then $V(I_1) = \operatorname{Spec} A$, a contradiction.

(ii) If $\mathfrak{N}(A)$ is prime. Suppose $\operatorname{Spec} A = V(I_1) \cup V(I_2)$, then $\operatorname{Spec} A = V(I_1 I_2)$. Since $\mathfrak{N}(A)$ is prime, we have $I_1 I_2 \subseteq \mathfrak{N}(A)$, hence $I_1 \subseteq \mathfrak{N}(A)$ or $I_2 \subseteq \mathfrak{N}(A)$. By Theorem 4.2.3, we know that $V(\mathfrak{N}(A)) = \operatorname{Spec} A$. Hence $V(I_1) = \operatorname{Spec} A$ or $V(I_2) = \operatorname{Spec} A$, which implies that $\operatorname{Spec} A$ is irreducible.
Conversely, if $\operatorname{Spec} A$ is irreducible. Suppose $ab \in \mathfrak{N}(A)$, by Theorem 4.2.3, ab contained in any prime ideal of A. Consider $V(a) \cup V(b)$, since $V(a) \cup V(b) = V(ab) = \operatorname{Spec} A = V(\mathfrak{N}(A))$, by the irreducibility of $\operatorname{Spec} A$, we know that $V(a) = V(\mathfrak{N}(A))$ or $V(b) = V(\mathfrak{N}(A))$. It follows that $a \in \mathfrak{N}(A)$ or $b \in \mathfrak{N}(A)$, i.e., $\mathfrak{N}(A)$ is a prime ideal.

□

Proposition 4.6.3

An irreducible topological space is connected.

Proof If X is an irreducible topological space, then any two non-empty open subsets of X intersect, by the definition of Connected (Definition 4.6.1), X is connected. □

Exercise 4.17 Give an example of a ring A where $\operatorname{Spec} A$ is connected but reducible.

Solution *Let $A = \mathbb{C}[x,y]/(x^2-y^2)$, then*

$$\begin{aligned}\operatorname{Spec} A =&\{\mathfrak{p} \in \operatorname{Spec}\mathbb{C}[x,y] : \mathfrak{p} \supseteq (x^2-y^2)\}\\ =&V(((x-y)(x+y)))\\ =&V(x-y)\cup V(x+y),\end{aligned}$$

where $V(x-y)$ and $V(x+y)$ are proper closed subsets of $\operatorname{Spec} A$. Hence, $\operatorname{Spec} A$ is reducible.

Since $V(x-y) = \operatorname{Spec}\mathbb{C}[x,y]/(x-y) \cong \operatorname{Spec}\mathbb{C}[x]$ and $V(x+y) = \operatorname{Spec}\mathbb{C}[x,y]/(x+y) \cong \operatorname{Spec}\mathbb{C}[x]$, note that $\mathbb{C}[x]$ is an integral domain, by Proposition 4.6.2, $V(x-y)$ and $V(x+y)$ is irreducible. By Proposition 4.6.3, $V(x-y)$ and $V(x+y)$ are connected. Note that $V(x-y)\cap V(x+y) = V((x-y,x+y)) = [(x-y,x+y)] \neq \varnothing$, $V(x-y)\cap V(x+y) = \operatorname{Spec} A$ is connected.

Exercise 4.18

(a) Suppose $I = (wz-xy, wy-x^2, xz-y^2) \subseteq k[w,x,y,z]$. Show that $\operatorname{Spec} k[w,x,y,z]/I$ is irreducible, by show that $k[w,x,y,z]/I$ is an integral domain. (We will later see this as the cone over the twisted cubic curve.)

(b) Note that the generators of the ideal of part (a) may be rewritten as the equations ensuring that

$$\operatorname{rank}\begin{pmatrix} w & x & y\\ x & y & z\end{pmatrix} \leq 1,$$

i.e., as the determinants of the 2×2 submatrices. Generalize part (a) to the ideal of rank one $2\times n$ matrices. This notion will correspond to the cone over the degree n rational normal curve.

Proof

(a) Define $\varphi : k[w,x,y,z] \to k[s,t]$ as follow:

$$\varphi := \begin{cases} w & \mapsto s^3,\\ x & \mapsto s^2t,\\ y & \mapsto st^2,\\ z & \mapsto t^3.\end{cases}$$

It is easy to check φ is a ring homomorphism. Clearly, $\varphi(k[w,x,y,z]) = k[s^3,s^2t,st^2,t^3]$. We claim that $I = \operatorname{Ker}\varphi$. Note that $\varphi(wz-xy) = s^3t^3 - s^3t^3 = 0$, $\varphi(wy-x^2) = s^4t^2 - s^4t^2 = 0$, and $\varphi(xz-y^2) = s^2t^4 - s^2t^4 = 0$, we have $I \subseteq \operatorname{Ker}\varphi$.

Let $f(w,x,y,z) \in \operatorname{Ker}\varphi$, then $\varphi(f) = f(s^3,s^2t,st^2,t^3) = 0$. Say $g_1 = wz-xy$, $g_2 = wy-x^2$, and $g_3 = xz-y^2$. Hence, we may assume that $f(w,x,y,z) = f_1g_1+f_2g_2+f_3g_3+r$, where the monomial of r none of which is divisible by wz, wy, xz. Hence, r only has forms: w^lx^k with $k\geq 0, l\geq 1$, y^mz^n with $m\geq 0, n\geq 1$, and x^ay^b with $a\geq 0, b\geq 0$. Since $f(s^3,s^2t,st^2,t^3) = 0$, $r(s^3,s^2t,st^2,t^3) = 0$. The monomials (in s and t) of $r(s^3,s^2t,st^2,t^3)$ are of the following types: $s^{3l+2k}t^k$, s^mt^{2m+3n}, and

$s^{2a+b}t^{a+2b}$. Consider following equations,

$$\begin{cases} 3l + 2k & = m \\ k & = 2m + 3n \end{cases} \Longrightarrow \begin{cases} l & = -m - n \\ k & = 2m + 3n, \end{cases}$$

note that $l \geq 1$, hence above equations do not have solve. Hence $s^{3l+2k}t^k \neq s^m t^{2m+3n}$. Similarly, we have $s^{3l+2k}t^k \neq s^{2a+b}t^{a+2b}$ and $s^m t^{2m+3n} \neq s^{2a+b}t^{a+2b}$. Hence, there is no possible cancelation between these monomials, so $r = 0$, and therefore $f \in I$, i.e., $\operatorname{Ker}\varphi \subseteq I$.

Hence, $I = \operatorname{Ker}\varphi$, and therefore $k[s^3, s^2t, st^2, t^3] \cong k[w, x, y, z]/I$. Also, $k[s^3, s^2t, st^2, t^3]$ is a subring of integral domain $k[s,t]$, $k[s^3, s^2t, st^2, t^3]$ is integral domain. Hence, $k[w, x, y, z]/I$ is an integral domain, by Proposition 4.6.2, $\operatorname{Spec} k[w, x, y, z]/I$ is irreducible.

(b) Let

$$M = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \end{pmatrix}$$

be a $2 \times n$ matrix with $\operatorname{rank} M \leq 1$ and $a_{2,i} = a_{1,(i+1)}$, then the determinants of the 2×2 submatrices are 0, i.e.,

$$\det \begin{pmatrix} a_{1i_1} & a_{1i_2} \\ a_{2i_1} & a_{2i_2} \end{pmatrix} = a_{1i_1}a_{2i_2} - a_{1i_2}a_{2i_1} = 0$$

for all $1 \leq i_1 < i_2 \leq n$.

For $n = 2$, all determinants of the 2×2 submatrices are the generators of I, as part (a).

For $n > 2$, let

$$I = \left(\det \begin{pmatrix} a_{1i_1} & a_{1i_2} \\ a_{2i_1} & a_{2i_2} \end{pmatrix} = 0, \forall 1 \leq i_1 < i_2 \leq n \right)$$

Define $\varphi : k[a_{11}, \ldots, a_{1n}, a_{2n}] \to k[s,t]$ by setting

$$\varphi := \begin{cases} a_{1k} & \mapsto s^{n+1-k}t^{k-1}, \forall 1 \leq k \leq n \\ a_{2n} & \mapsto t^n. \end{cases}$$

It is easy to check that $I \subseteq \operatorname{Ker}\varphi$. Let $f \in \operatorname{Ker}\varphi$, then $f(s^n, s^{n-1}t, \ldots, t^n) = 0$. Note that $\{s^n, s^{n-1}t, \ldots, t^n\}$ agree with

$$\det \begin{pmatrix} a_{1i_1} & a_{1i_2} \\ a_{2i_1} & a_{2i_2} \end{pmatrix} = 0, \forall 1 \leq i_1 < i_2 \leq n.$$

Hence, $I = \operatorname{Ker}\varphi$ (use the same method as part (a)), and therefore

$$k[a_{11}, \ldots, a_{1n}, a_{2n}]/I \cong k[s^n, s^{n-1}t, \ldots, t^n] \subseteq k[s,t].$$

Since $k[s,t]$ is an integral domain, $k[a_{11}, \ldots, a_{1n}, a_{2n}]/I$ is integral domain, by Proposition 4.6.2, we done.

□

Quasi-compactness

Definition 4.6.3 (Quasi-compact)

A topological space X is ***quasi-compact*** *if given any cover $X = \bigcup_{i\in I} U_i$ by open sets, there is a finite subset S of the index set I such that $X = \bigcup_{i\in S} U_i$. Informally: every open cover has a finite subcover.*

We will like this condition, because we are afraid of infinity. Depending on your definition of "compactness", this is the definition of compactness, minus possibly a Hausdorff condition. However, this isn't really the algebra-geometric analog of "compact" (we certainly wouldn't want $\mathbb{A}^1_{\mathbb{C}}$ to be compact) — the right analog is "properness".

Proposition 4.6.4

Spec A *is quasi-compact.*

Proof Let $\{U_i\}_{i\in J}$ be an open cover of $\operatorname{Spec} A$, i.e., $\operatorname{Spec} A = \bigcup_{i\in J} U_i$. Since each open set is the union of some $D(f_i)$, we may assume $\operatorname{Spec} A = \bigcup_{i\in J} D(f_i)$. By Proposition 4.5.3, we can choose a finite subset $J' \subseteq J$ such that $\operatorname{Spec} A = \bigcup_{i\in J'} D(f_i)$, which implies that $\operatorname{Spec} A$ is quasi-compact. □

Remark 4.17 In general $\operatorname{Spec} A$ can have nonquasi-compact open sets.

Proof Let $A = k[x_1, x_2, x_3, \dots]$, $\mathfrak{m} = (x_1, x_2, x_3, \dots)$ is its maximal ideal. Consider $\operatorname{Spec} A \setminus V(\mathfrak{m})$, we have

$$\operatorname{Spec} A \setminus V(\mathfrak{m}) = \operatorname{Spec} A \setminus \{\mathfrak{m}\} = \bigcup_{i=1}^{\infty} D(x_i).$$

We shall to show that we can not choose finite numbers of $D(x_i)$, such that $\bigcup_i D(x_i) = \operatorname{Spec} A \setminus V(\mathfrak{m})$.

Suppose $\operatorname{Spec} A \setminus V(\mathfrak{m}) = \bigcup_{k=1}^n D(x_{i_k})$, where $\{x_{i_1}, \dots, x_{i_n}\} \subseteq \{x_1, x_2, \dots\}$ be a finite subset. Hence, there exists $x_t \in \{x_1, x_2, \dots\} \setminus \{x_{i_1}, \dots, x_{i_n}\}$. Let $\mathfrak{p} = (x_t, x_{i_1}, \dots, x_{i_n})$, then $\mathfrak{p}$ is a prime ideal, and $\mathfrak{p} \notin \bigcup_{k=1}^n D(x_{i_k})$, but $\mathfrak{p} \in \operatorname{Spec} A \setminus V(\mathfrak{m})$, a contradiction. Hence, $\operatorname{Spec} A \setminus V(\mathfrak{m})$ is not quasi-compact. □

Proposition 4.6.5

(a) If X is a topological space that is a finite union of quasi-compact spaces, then X is quasi-compact.

(b) Every closed subset of a quasi-compact topological space is quasi-compact.

Proof

(a) Let $\{U_j\}_{j\in J}$ be any open cover of X. Since X is a finite union of quasi-compact spaces, we may assume $X = \bigcup_{i=1}^n X_i$, where X_i is quasi-compact spaces. Hence, $\{U_j \cap X_i\}_{j\in J}$ is an open cover of X_i. Since X_i is quasi-compact, we may assume $X_i = \bigcup_{k=1}^{m_i} (U_{i_k} \cap X_i)$. Hence,

$$X = \bigcup_{i=1}^{n} \bigcup_{k=1}^{m_i} (U_{i_k} \cap X_i) \subseteq \bigcup_{i=1}^{n} \bigcup_{k=1}^{m_i} U_{i_k},$$

and therefore X is quasi-compact.

(b) Let F be a closed subset of X, endow F with subspace topology. Let $V_{j_{j\in J}}$ be an open over of F, then $V_j = F \cap U_j$ where U_j is open subset of X. Since $F = \bigcup_j V_j \subseteq \bigcup_j U_j$, we have

$$X = \bigcup_j U_j \cup (X \setminus \bigcup_j U_j) \subseteq \bigcup_j U_j \cup (X \setminus F),$$

hence, $\{U_j\ (j \geq 1), X \setminus F\}$ is an open cover of X. Since X is quasi-compact, we may assume $X = \bigcup_{j=1}^n U_j \cup (X \setminus F)$. Note that $F \cap (X \setminus F) = \varnothing$, F must be covered by $\{U_j\}_{j=1}^n$. Hence,

$F = \bigcup_{j=i}^{n}(U_i \cap F)$, and therefore F is quasi-compact.

□

Remark 4.18 Example 4.5 shows that $\coprod_{i=1}^{n} \operatorname{Spec} A_i \cong \operatorname{Spec} \prod_{i=1}^{n} A_i$, but this never holds if "n is infinite" and all A_i are nonzero, as Spec of any rings is quasi-compact (Proposition 4.6.4). This leads to an interesting phenomenon. We show that $\operatorname{Spec} \prod_{i=1}^{\infty} A_i$ is "strictly bigger" than $\coprod_{i=1}^{\infty} \operatorname{Spec} A_i$, where each A_i is isomorphic to the field k. First, we have an inclusion of sets $\coprod_{i=1}^{\infty} \operatorname{Spec} A_i \hookrightarrow \operatorname{Spec} \prod_{i=1}^{\infty} A_i$, as there is a maximal ideal of $\prod A_i$ correspongding to each i (precisely, those elements 0 in the i-th component). But there are other maximal ideals of $\prod A_i$. Cosider elements $\prod \mathfrak{a}_i$ that are "eventually zero", i.e., $\mathfrak{a}_i = 0$ for $i \gg 0$, $\prod \mathfrak{a}_i$ is a proper ideal not contained in any of these maximal ideals. This leads to the notion of **ultrafilters**, which are very useful, but irrelevant to our current discussion.

4.6.2 Possible topological properties of points of $\operatorname{Spec} A$

Definition 4.6.4 (Closed point)

*A point of a topological space $p \in X$ is said to be a **closed point** if $\{p\}$ is a closed subset.*

Example 4.6 In the classical topology on $\mathbb{C}^n$, all points are closed. In $\operatorname{Spec} \mathbb{Z}$ and $\operatorname{Spec} k[t]$, all the points are closed except for $[(0)]$.

Proposition 4.6.6

The closed points of $\operatorname{Spec} A$ *correspond to the maximal ideals. In particular, nonempty affine schemes have closed points, as non-zero rings have maximal ideals.*

Proof Let $[\mathfrak{m}] \in \operatorname{Spec} A$ be a closed point, then $\{[\mathfrak{m}]\} = V(I)$ for some ideal $I \subseteq A$. Since $V(I) = \{[\mathfrak{p}] \in \operatorname{Spec} A : \mathfrak{p} \supseteq I\} = \{[\mathfrak{m}]\}$, this implies $\mathfrak{m}$ is the only prime ideal which is containing I, note that I must contained in a maximal ideal, hence, $\mathfrak{m}$ must be maximal ideal. □

Connection to the classical theory of varieties

Hilbert's Nullstellensatz lets us interpret the closed points of $\mathbb{A}^n_{\mathbb{C}}$ as the n-tuples of complex numbers. More generally, the closed points of $\operatorname{Spec} \overline{k}[x_1, \ldots, x_n]/(f_1, \ldots, f_r)$ (where $\overline{k}$ is an algebraically closed field) are naturally interpreted as those points in $\overline{k}^n$ satisfying the equations $f_1 = \cdots = f_r = 0$. Hence from now on we will say "closed point" instead of "traditional point" and "non-closed point" instead of "bonus" point when discussing subsets of $\mathbb{A}^n_{\overline{k}}$. The following proposition is the reason that on algebraic varieties, the set of closed points is dense.

Proposition 4.6.7

(a) Suppose that k is a field, and A is a finitely generated k-algebra. Then the set of closed points of $\operatorname{Spec} A$ *is dense.*

(b) If A is a k-algebra that is not finitely generated, the set of closed points need not be dense.

Proof

(a) By Proposition 4.6.6, the set of closed points of $\operatorname{Spec} A$ is $\operatorname{Max} A$. To show that $\operatorname{Max} A$ is dense, it suffices to show that $\operatorname{Max} A \cap D(f) \neq \varnothing$ for any $f \in A$ (see Wikipedia, Dense set, Definition), i.e., every nonempty $D(f)$ contains a closed point.

To this end, let $f \in A$ such that $D(f) \neq \varnothing$. Then, f is not nilpotent, and therefore A_f is not zero-ring. Hence, A_f has a maximal ideal. This is of the form $\mathfrak{p}_f$ for some prime ideal $\mathfrak{p} \subseteq A$ not containing f. In other words, $[\mathfrak{p}] \in D(f)$. We show that $[\mathfrak{p}]$ is a closed point by showing that $\mathfrak{p}$ is maximal (Proposition 4.6.6). It suffices to show that $A/\mathfrak{p}$ is a field. Note that $A/\mathfrak{p}$ is an integral domain, recall Proposition 4.2.1, it suffices to show that $A/\mathfrak{p}$ is a finite k-algebra (i.e., a k-algebra that is a finite-dimensional vector space over k).

Note that $A_f = A[1/f]$ is the image of $A[x]$ under map $x \mapsto 1/f$, hence, A_f is a finitely generated k-algebra, by Hilbert's Nullstenllensatz 4.2.2, $A_f/\mathfrak{p}_f$ is finitely generated as a k-vector space. Note that $(A/\mathfrak{p})_f \cong A_f/\mathfrak{p}_f$, $(A/\mathfrak{p})_f$ is finitely generated as a k-vector space. Since $A/\mathfrak{p}$ is integral domain, canonical homomorphism $A/\mathfrak{p} \to (A/\mathfrak{p})_f$ is injective. This homomorphism is also a k-linear map, hence, $A/\mathfrak{p}$ is a finite-dimensional vector space over k.

(b) Let $A = k[x_1, x_2, x_3, \dots]$ with maximal ideal $\mathfrak{m} = (x_1, x_2, x_3, \dots)$. Consider $A_\mathfrak{m}$, then $A_\mathfrak{m}$ is a k-algebra and is not finitely generated. Since $A_\mathfrak{m}$ is local ring, its maximal ideal is $\mathfrak{m}A_\mathfrak{m}$. Hence, the closure of $\operatorname{Max} A = \{[\mathfrak{m}A_\mathfrak{m}]\}$ is $\{[\mathfrak{m}A_\mathfrak{m}]\}$, hence, $\operatorname{Max} A = \{[\mathfrak{m}A_\mathfrak{m}]\}$ is not dense.

□

Proposition 4.6.8

Suppose k is an algebraically closed field, and $A = k[x_1, \dots, x_n]/I$ is a finitely generated k-algebra with $\mathfrak{N}(A) = \{0\}$. Consider the set $X = \operatorname{Spec} A$ as a subset of $\mathbb{A}_k^n$. The space $\mathbb{A}_k^n$ contains the "classical" points k^n. Then the functions on X are determined by their values on the closed points.

Proof We want to show: if $f, g \in A$ have the same value on X, then $f = g$.

Suppose f and g are different functions on X, then $f - g \neq 0$. Since $\mathfrak{N}(A) = 0$, $f - g \notin \mathfrak{N}(A)$, it follows that $D(f-g) \neq \varnothing$. By Proposition 4.6.7 (a), we have $\operatorname{Max} A \cap D(f-g) \neq \varnothing$, say $\mathfrak{m} \in \operatorname{Max} A \cap D(f-g)$, then $f - g \notin \mathfrak{m}$, a contradiction. □

Once we know what a variety is, this will immediately imply that a function on a variety over an algebraically closed field is determined by its values on the "classical points". (Before the advent of scheme theory, functions on "classical" points, and Proposition 4.6.8 basically shows that there is no harm in thinking of "traditional" varieties as a particular flavor of schemes.)

Specialization and generization, and generic point

Definition 4.6.5 (Specialization and generization)

*Given two points x, y of a topological space X, we say that x is a **specialization** of y, and y is a **generization** of x, if $x \in \overline{\{y\}}$.*

This (and Proposition 4.6.9) now makes precise our hand-waving about "one point containing another". It is of closure nonsense for a point to contain another. But it is not nonsense to say that the closure of a point contains another.

Example 4.7 For example, in $\mathbb{A}_\mathbb{C}^2 = \operatorname{Spec} \mathbb{C}[x, y]$, $[(y - x^2)]$ is a generization of $[(x-2, y-4)] = (2, 4) \in \mathbb{C}^2$, and $(2, 4)$ is a specialization of $[(y - x^2)]$ (see Figure 4.10).

Lemma 4.6.1

Let $X = \operatorname{Spec} A$, $[\mathfrak{p}] \in X$, then $\overline{\{[\mathfrak{p}]\}} = V(\mathfrak{p})$.

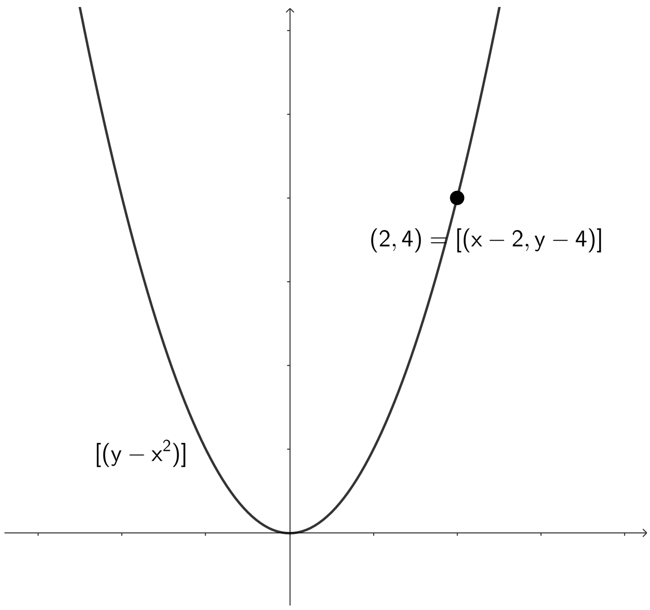


Figure 4.10: $(2,4) = [(x-2, y-4)]$ is a specialization of $[(y-x^2)]$. $[(y-x^2)]$ is a generization of $(2,4)$.

Proof In fact,

$$\overline{\{[\mathfrak{p}]\}} = \bigcap_{V(I)\supseteq\{[\mathfrak{p}]\}} V(I) \subseteq V(\mathfrak{p}).$$

Let $[\mathfrak{q}] \in V(\mathfrak{p})$, then $\mathfrak{q} \supseteq \mathfrak{p}$. For all $V(I) \subseteq \{[\mathfrak{p}]\}$, $I \subseteq \mathfrak{p}$, hence, $\mathfrak{q} \supseteq I$, and therefore $[\mathfrak{q}] \in V(I)$. It follows that $[\mathfrak{q}] \in \bigcap_{V(I)\supseteq\{[\mathfrak{p}]\}} V(I) = \overline{\{[\mathfrak{p}]\}}$. Hence, $\overline{\{[\mathfrak{p}]\}} = V(\mathfrak{p})$. □

Proposition 4.6.9

If $X = \operatorname{Spec} A$, then $[\mathfrak{q}]$ is a specialization of $[\mathfrak{p}]$ if and only if $\mathfrak{p} \subseteq \mathfrak{q}$.

Proof By Lemma 4.6.1, clearly. □

Definition 4.6.6 (Generic point)

*We say that a point $p \in X$ is a **generic point** for a closed subset K if $\overline{\{p\}} = K$.*

This important notion predates Grothendieck. The early twentieth-century Italian algebraic geometers had a notion of "generic points" of a variety, by which they meant points with no special properties, so that anything proved of "a generic point" was true of "almost all" the points on that variety. The modern "generic point" has the same intuitive meaning. If something is "generically" or "mostly" true for the points of an irreducible subset, in the sense of being true for a dense open subset (for "almost all points"), then it is true for the generic point, and vice versa. (This is a statement of principle, not of fact. An interesting case is "reducedness", for which this principle does not hold in general, but does hold for "reasonable" schemes such as varieties.) For example, a function has value zero at the generic point of an irreducible scheme if and only if it has the value zero at all points. You should keep an eye out for other examples of this.

The phrase **general point** does not have the same meaning. The phrase "the general point of X satisfies such-and-such a property" means "every point of some dense open subset of X satisfies such-and-such a property". Be careful not to confuse "general" and "generic". But be warned that accepted terminology does not always follow this convention; witness "generic smoothness" and "generic flatness".

✍ **Exercise 4.19** Verify that $[(y-x^2)] \in \mathbb{A}^2_{\mathbb{C}}$ is a generic point for $V(y-x^2)$.

Proof Note that $\overline{\{[(y-x^2)]\}} = V((y-x^2)) = V(y-x^2)$ where $V(y-x^2)$ is a closed subset, $[(y-x^2)] \in \mathbb{A}^2_{\mathbb{C}}$ is a generic point for $V(y-x^2)$. □

As more motivation for the terminology "generic": we think of $[(y-x^2)]$ as being some non-specific point on the parabola (with the closed points $(a, a^2) \in \mathbb{C}^2$, i.e., $(x-a, y-a^2)$ for $a \in \mathbb{C}$, being "specific points"); it is "generic" in the conventional sense of the word. We might "specialize it" to a specific point of the parabola; hence for example $(2,4)$ is a specialization of $[(y-x^2)]$. (Again, see Figure 4.10.) To make this somewhat more precise:

Proposition 4.6.10

Suppose p is a generic point for the closed subset K. p is "near every point q of K" (every neighborhood of q contains p), and "not near any point r not in K" (there is a neighborhood of r not containing p).

Proof Since p is a generic point for the closed subset K, $\overline{\{p\}} = K$, i.e., $\{p\}$ is dense in K. Let U_q be any open neighborhood of q, then $K \cap U_q$ is an open neighborhood of q in K (endow K with subspace topology). Since $\{p\}$ is dense in K, $\{p\} \cap (U_q \cap K) \neq \varnothing$, and therefore $\{p\} \in U_q$. It follows that every neighborhood of q contains p.

Let X be a topological space, K is a closed subset of X. Let $r \in X \setminus K$, and U_r be any open neighborhood of r, then $(X \setminus K) \cap U_r$ is also an open neighborhood of r, say $V_r = (X \setminus K) \cap U_r$. Note that $p \notin V_r$, as we desired. □

We will soon see that there is a natural bijection between points of $\operatorname{Spec} A$ and irreducible closed subsets of $\operatorname{Spec} A$, sending each point to its closure, and each irreducible closed subset to its (unique) generic point.

4.6.3 Irreducible and connected components, and Noetherian conditions

Irreducible component and connected components

Definition 4.6.7 (Irreducible component)

*An **irreducible component** of a topological space is a maximal irreducible subset (an irreducible subset not contained in any larger irreducible subset).*

Irreducible components are closed (*By Proposition 4.6.1 (b), the closure of irreducible component is irreducible, sine an irreducible subset not contained in any larger irreducible subset, they must be same, hence, closed*), and it can be helpful to think of irreducible components of a topological space X as maximal among the irreducible closed subsets of X. We think of these as the "pieces of X" (see Figure 4.11).

Definition 4.6.8 (Connected component)

*A subset Y of a topological space X is a **connected component** if it is a maximal connected subset (a connected subset not contained in any larger connected subset).*

Proposition 4.6.11 (Every topological space is the union of irreducible components)

Every point p of a topological space X is contained in an irreducible component of X.

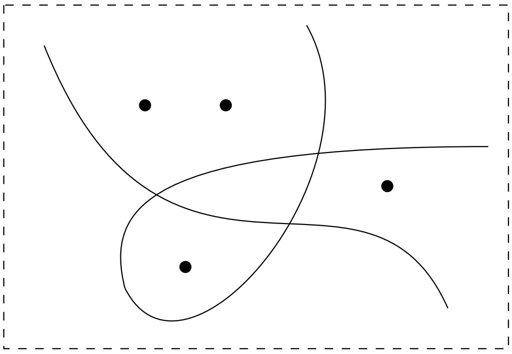


Figure 4.11: This closed subset of $\mathbb{A}^2_{\mathbb{C}}$ has six irreducible components

Proof Consider the partially ordered set

$$\mathscr{S}_p = \{\text{Irreducible subsets of } X \text{ containing } p\}.$$

We claim that $\mathscr{S}_p$ non-empty. Consider $\overline{\{p\}}$. If $\overline{\{p\}}$ is reducible, there exists proper closed subsets A and B of $\overline{\{p\}}$ such that $\overline{\{p\}} = A \cup B$. Hence, $\{p\} \subseteq A$ or $\{p\} \subseteq B$, we may assume that $\{p\} \subseteq A$. Note that $A \subsetneq \overline{\{p\}}$ proper closed, which contradicts to the fact that $\overline{\{p\}}$ is the smallest closed set containing $\{p\}$. Hence, $\overline{\{p\}}$ is irreducible, and therefore $\overline{\{p\}} \in \mathscr{S}_p$.

For any totally ordered subset $\{Z_\alpha\}$ of $\mathscr{S}_p$, Let $Z = \bigcup_\alpha Z_\alpha$, it is a upper bound of $\{Z_\alpha\}$, we want to show that $Z \in \mathscr{S}_p$. Consider the closure of Z, i.e., $\overline{Z}$. Suppose $\overline{Z}$ is reducible, i.e., exists proper closed subsets A and B of $\overline{Z}$ such that $\overline{Z} = A \cup B$, since $\{Z_i\}$ is totally ordered and each Z_i is irreducible, $Z \subseteq A$ and $Z \subseteq B$, we may assume that $Z \subseteq A$. Note that $A \subsetneq \overline{Z}$ proper closed, this contradicts to the fact that $\overline{Z}$ is the smallest closed set containing Z. Hence, $\overline{Z}$ is irreducible, by Proposition 4.6.1 (b), Z is irreducible, and therefore $Z \in \mathscr{S}$. Hence, every totally ordered subset has an upper bound belong to $\mathscr{S}_p$.

By Zorn's Lemma, there exists at least one maximal element in $\mathscr{S}_p$, say $X_p(\ni p)$, which is an irreducible component of X. □

Remark 4.19 Every point is contained in a connected component, and connected components are always closed. You can prove this now, but we deliberately postpone this until we need it.

On the other hand, connected components need not be open. An example of an affine scheme with connected components that are not open is $\operatorname{Spec}(\prod_1^\infty \mathbb{F}_2)$. ($\operatorname{Spec}(\prod_1^\infty \mathbb{F}_2) = \{[(0)], \text{maximal ideal}\}$, where each maximal ideal has form $\mathbb{F}_2 \times \mathbb{F}_2 \times \cdots \times \mathbb{F}_2 \times (0) \times \mathbb{F}_2 \times \cdots \times \mathbb{F}_2$. $\operatorname{Spec}(\prod_1^\infty \mathbb{F}_2) = \coprod_{i=1}^\infty \mathfrak{m}_i$, each connected component $\mathfrak{m}_i$ is closed.)

Noetherian topological space and Noetherian induction

In the examples we have considered, the spaces have naturally broken up into a finite number of irreducible components. For example, the locus $xy = 0$ in $\mathbb{A}^2_{\mathbb{C}}$ we think of as having two "pieces" — the two axes. The reason for this is that their underlying topological spaces (as we shall soon establish) are **Noetherian**.

Definition 4.6.9 (Noetherian topological space)

*A topological space X is called **Noetherian** if it satisfies the **descending chain condition (d.c.c.)** for closed subsets: any sequence*

$$Z_1 \supseteq Z_2 \supseteq \cdots \supseteq Z_n \supseteq \cdots$$

of closed subsets eventually stabilizes: there is an r such that $Z_r = Z_{r+1} = \cdots$.

Here is a first example:

Example 4.8 Show that $\mathbb{A}^2_{\mathbb{C}}$ is a Noetherian topological space: any decreasing sequence of closed subsets of $\mathbb{A}^2_{\mathbb{C}} = \operatorname{Spec}\mathbb{C}[x,y]$ must eventually stabilize. Show that $\mathbb{C}^2$ with the classical topology is not a Noetherian topological space.

Solution *In fact,* $\operatorname{Spec}\mathbb{C}[x,y] = \{(0), (x-a, y-b)$ *where* $a, b \in \mathbb{C}, (f(x,y))\ f(x,y)$ *is irreducible* $\}$. *Let*

$$V(I_1) \supseteq V(I_2) \supseteq \cdots \supseteq V(I_n) \supseteq \cdots$$

be any descending chain, by Hilbert's Nullstellensatz, then we have ascending chain of ideals

$$\sqrt{I_1} \subseteq \sqrt{I_2} \subseteq \cdots \subseteq \sqrt{I_n} \subseteq \cdots. \tag{4.2}$$

Since $\mathbb{C}[x,y]$ *is a Noetherian space, chain 4.2 is stationary. By Hilbert's Nullstellensatz, the chain of closed subset*

$$V(I_1) \supseteq V(I_2) \supseteq \cdots \supseteq V(I_n) \supseteq \cdots$$

is stationary. Hence, $\operatorname{Spec}\mathbb{C}[x,y]$ *is a Noetherian topological space.*

Next, we want show that $\mathbb{C}^2$ *with the classical topology is not a Noetherian space. Let* $F_n := \{z \in \mathbb{C}^2 : |z| \geq n$ *or* $|z| \leq \frac{1}{n}\}$*, clearly,* F_n *is closed in* $\mathbb{C}^2$*, but*

$$F_1 \supseteq F_2 \supseteq F_3 \supseteq \cdots$$

not stabilize. Hence, $\mathbb{C}^2$ *is not a Noetherian space.*

The following technique is called **Noetherian induction**.

Theorem 4.6.1 (Noetherian induction)

Let X be a Noetherian topological space, and let $\mathscr{P}$ be a property of closed subsets of X. Assume that for any closed subset Y of X, if $\mathscr{P}$ holds for every proper closed subset of Y, then $\mathscr{P}$ holds for Y. (In particular, $\mathscr{P}$ must hold for the empty set.) Then $\mathscr{P}$ holds for X.

Proof Suppose that property $\mathscr{P}$ not holds for X. Consider the following set:

$$\Sigma = \{Y \subseteq X : Y \text{ is closed}, \mathscr{P} \text{ not holds for } Y\},$$

Σ is not empty, since $X \in \Sigma$. Since X is a Noetherian space, Σ has minimal element, say Y_0. Let $Z \subsetneq Y_0$ be proper closed subset of Y_0, since Y_0 is minimal element in Σ, $Z \notin \Sigma$, and therefore Z satisfies property $\mathscr{P}$. Hence, $\mathscr{P}$ holds for every proper closed subset of Y_0, by hypothesis, $\mathscr{P}$ holds for Y_0, but $Y_0 \in \Sigma$, a contradiction. □

Proposition 4.6.12 (Any closed subset of Noetherian space has a finite number of "pieces")

Suppose X is a Noetherian topological space. Then every nonempty closed subset Z can be expressed uniquely as a finite union $Z = Z_1 \cup \cdots \cup Z_n$ of irreducible closed subsets, none contained in any other.

Proof Consider the collection of nonempty closed subsets of X that cannot be expressed as a finite union of irreducible closed subsets, say Σ. We will show that Σ is empty. Suppose Σ is not empty, let $Y_1 \in \Sigma$. If Y_1 contains another element in Σ, then choose one, and call it Y_2. If Y_2 properly contains another element in Σ, then choose one, and call it Y_3, and so on. By the descending chain condition, this must be stationary, and we must have some Y_r that cannot be written as a finite union of irreducible closed subsets. Since Y_r is not itself irreducible, so we may assume $Y_r = Y' \cup Y''$, where Y' and Y'' are both proper closed subsets. Since every closed subset properly contained in Y_r not belong to Σ, Y' and Y'' can be written as the union of a finite number of irreducible closed subsets, and hence so can Y_r, yielding a contradiction. Thus each closed subset can be written as a finite union of irreducible closed subsets. We can assume that none of these irreducible closed subsets contain any others, by discarding some of them.

We now show uniqueness. Suppose

$$Z = Z_1 \cup Z_2 \cup \cdots \cup Z_r = Z_1' \cup Z_2' \cup \cdots \cup Z_s'$$

are two such representations. Then $Z_1' \subseteq Z_1 \cup Z_2 \cup \cdots \cup Z_r$, so $Z_1' = (Z_1 \cap Z_1') \cup \cdots \cup (Z_r \cap Z_1')$. Now Z_1' is irreducible, so one of these is Z_1' itself, say (without loss of generality) $Z_1 \cap Z_1'$. Thus $Z_1 / \subseteq Z_1$. Similarly, $Z_1 \subseteq Z_a'$ for some a; but because $Z_1' \subseteq Z_1 \subseteq Z_a'$, and Z_1' is contained in no other Z_i', we must have $a = 1$, and $Z_1' = Z_1$. Thus each element of the list of Z's is in the list of Z''s, and vice versa. Hence, they must be the same list. □

Proposition 4.6.13

(a) Every connected component of a topological space X is the union of irreducible components of X.

(b) Any subset of X that is simultaneously open and closed must be the union of some of the connected components of X.

(c) If X is a Noetherian topological space, then the union of any subset of the connected components of X is always open and closed in X. In particular, connected components of Noetherian topological spaces are always open, which is not true for more general topological spaces (example: $\operatorname{Spec}(\prod_1^\infty \mathbb{F}_2)$).

Proof

(a) Every connected component of a topological space X is closed, endow connected component with subspace topology, by Proposition 4.6.11, connected component is the union of irreducible components of X.

(b) Let Y be a subset of X is simultaneously open and closed. Let C be any connected components of X. If $Y \cap C \neq \varnothing$ and $(X \setminus Y) \cap C \neq \varnothing$, since $X \setminus Y$ is open and C is connected, we have $C \subseteq X \setminus Y$ or $C \subseteq Y$. Hence, Y is the union of some of the connected components of X.

(c) By Proposition 4.6.12, X can be expressed uniquely as a finite union of irreducible closed subset, say $X = Z_1 \cup Z_2 \cup \cdots \cup Z_m$. Each Z_i belong to a connected component, and if $Z_i \cap Z_j \neq \varnothing$, Z_i and Z_j must belong to the same connected component, hence, X can be expressed uniquely as a finite disjoint union of connected components, say $X = \coprod_{i=1}^n C_i$, where each C_i is a finite union of some Z_j, hence, each C_i must be closed.

We claim that every connected component is open and closed in X. Let C_i be a connected component of X. Then $X \setminus C_i = \coprod_{\substack{j \neq i \\ 1 \leq j \leq n}} C_j$. Since each connected component is closed, $\coprod_{\substack{j \neq i \\ 1 \leq j \leq n}} C_j$ is open, i.e., C_i is open and closed in X.

Hence, the union of any subset of the connected components of X is always open and closed in X.

□

Noetherian rings

It turns out that all of the spectra we have considered (except $\operatorname{Spec} A[x_1, x_2, x_3, \ldots]$) are Noetherian topological spaces, but that isn't true for the spectra of all rings. The Key characteristic all of our examples have had in common is that the rings were **Noetherian**.

Definition 4.6.10 (Noetherian ring)

*A ring is **Noetherian** if every ascending sequence $I_1 \subseteq I_2 \subseteq \cdots$ of ideals eventually stabilizes: there is an r such that $I_r = I_{r+1} = \cdots$. (This is called the **ascending chain condition** on ideals.)*

Here are some quick facts about Noetherian rings.

Example 4.9 Fields are Noetherian. $\mathbb{Z}$ is Noetherian.

Proposition 4.6.14 (Noetherianness is preserved by quotients)

If A is Noetherian, and I is any ideal of A, then A/I is Noetherian.

Proof Let $I_1 \subseteq I_2 \subseteq \cdots$ be any ascending sequence in A/I. Then I_i is corresponding to ideal $\widetilde{I_i} \subseteq A$, which contains I_1, and therefore we have ascending sequence in A,

$$\widetilde{I_1} \subseteq \widetilde{I_2} \subseteq \cdots .$$

Since A is Noetherian ring, this sequence must be stationary. Since this sequence is corresponding to sequence $I_1 \subseteq I_2 \subseteq \cdots$, $I_1 \subseteq I_2 \subseteq \cdots$ is stationary, and therefore A/I is Noetherian. □

Proposition 4.6.15 (Noetherianness is preserved by localization)

if A is a Noetherian, and S is any multiplicative set, then $S^{-1}A$ is Noetherian.

Proof Let $S^{-1}I_1 \subseteq S^{-1}I_2 \subseteq \cdots$ be any ascending sequence in $S^{-1}A$. This sequence is corresponding to sequenec

$$\widetilde{I_1} \subseteq \widetilde{I_2} \subseteq \widetilde{I_3} \subseteq \cdots ,$$

where $\widetilde{I_i}$ is corresponding to I_i and $\widetilde{I_i} \cap S = \varnothing$. Since A is Noetherian ring, $S^{-1}I_1 \subseteq S^{-1}I_2 \subseteq \cdots$ is stationary, and therefore

$$\widetilde{I_1} \subseteq \widetilde{I_2} \subseteq \widetilde{I_3} \subseteq \cdots$$

is stationary. Hence, $S^{-1}A$ is Noetherian. □

Proposition 4.6.16

PIDs are Noetherian rings.

Proof Let A be a PID. Let $I_1 \subseteq I_2 \subseteq I_3 \subseteq \cdots$ be any ascending sequence in A, since A is PID, we may assume $I_i = (x_i)$. Since $(x_i) \subseteq (x_{i+1})$, we have $x_{i+1} \mid x_i$. Note that x_i can be decomposed into a product of finitely many irreducible elements, we may assume that $x_1 = up_1^{e_1} p_2^{e_2} \cdots p_m^{e_m}$, then sequence $I_1 \subseteq I_2 \subseteq I_3 \subseteq \cdots$ must be stationary, and therefore A is Noetherian. □

Proposition 4.6.17

A ring A is Noetherian if and only if every ideal of A is finitely generated.

Proof "⇒": Let $\mathfrak{a}$ be an ideal of A, and let Σ be the set of all finitely generated ideal of A. Since $0 \in \Sigma$, Σ is not empty, and therefore has a maximal element, say $\mathfrak{a}_0$. If $\mathfrak{a} \neq \mathfrak{a}_0$, consider the submodule $\mathfrak{a}_0 + Ax$, where $x \in \mathfrak{a}$ and $x \notin \mathfrak{a}_0$; this is finitely generated and strictly contains $\mathfrak{a}_0$, a contradiction. Hence, $\mathfrak{a} = \mathfrak{a}_0$, and therefore $\mathfrak{a}$ is finitely generated.

"$\Leftarrow$": Let $\mathfrak{a}_1 \subseteq \mathfrak{a}_2 \subseteq \mathfrak{a}_3 \subseteq \cdots$ be an ascending chain of ideals of A. Then $\mathfrak{a} = \sum_i \mathfrak{a}_i$ is an ideal of A, hence is finitely generated, say $\mathfrak{a} = (x_1, x_2, \dots, x_r)$. Say $x_i \in \mathfrak{a}_{n_i}$ and let $n = \max_{i=1}^r x_i$, then each $x_i \in \mathfrak{a}_n$, hence, $\mathfrak{a}_n = \mathfrak{a}$, and therefore the chain is stationary. Hecne, A is Noetherian. $\square$

Theorem 4.6.2 (The Hilbert Basis Theorem)

If A is Noetherian, then so is $A[x]$.

Proof We show that any ideal $I \subseteq A[x]$ is finitely generated. We inductively produce a set of generators $f_1, \dots$ as follows. For $n \geq 0$, if $I \neq (f_1, \dots, f_n)$, let f_{n+1} be any non-zero element of $I - (f_1, \dots, f_n)$ of lowest degree. Thus f_1 is any element of I of lowest degree, assuming $I \neq (0)$. If this procedure terminates, we are done. Otherwise, let $a_n \in A$ be the initial coefficient of f_n for all $n > 0$. As A is Noetherian, $(a_1, a_2, \dots) = (a_1, \dots, a_N)$ for some N. Say $a_{N+1} = \sum_{i=1}^N b_i a_i$. Then

$$f_{N+1} - \sum_{i=1}^{N} b_i f_i x^{\deg f_{N+1} - \deg f_i}$$

is an element of I that is not in $(f_1, \dots, f_N)$ (since $f_{N+1} \notin (f_1, \dots, f_N)$), and of lower degree than f_{N+1}, contradicting how we were supposed to have chosen f_{N+1}. $\square$

Remark 4.20 By the results described above, any polynomial ring in finitely many variables over any field, or over the integers, is Noetherian — and also any quotient or localization thereof. Hence for example any finitely generated algebra over k or $\mathbb{Z}$, or any localization thereof, is Noetherian. Most "nice" rings are Noetherian, but not all rings are Noetherian: $k[x_1, x_2, \dots]$ is not, because $(x_1) \subseteq (x_1, x_2) \subseteq (x_1, x_2, x_3) \subseteq \cdots$ is a strictly ascending chain of ideals.

We now connect Noetherian rings and Noetherian topological spaces.

Proposition 4.6.18

If A is Noetherian, then $\operatorname{Spec} A$ *is a Noetherian topological space.*

Proof Let $V(I_1) \supset V(I_2) \supset V(I_3) \supseteq \cdots$ be any descending chain of closed subsets in $\operatorname{Spec} A$, by Hilbert's Nullstellensatz, we have

$$\sqrt{I_1} \subseteq \sqrt{I_2} \subseteq \sqrt{I_3} \subseteq \cdots$$

be a ascending sequence of ideals in A. Since A is a Noetherian ring, the sequence of radical ideals is stationary, by Hilbert's Nullstellensatz again, the descending sequence of closed set

$$V(I_1) \supseteq V(I_2) \supseteq V(I_3) \supseteq \cdots$$

is stationary. Hence, $\operatorname{Spec} A$ is a Noetherian topological space. $\square$

✍ **Exercise 4.20** Describe a ring A such that $\operatorname{Spec} A$ is not a Noetherian topological space.

Proof Let $A = k[x_1, x_2, \dots]$, we claim that $\operatorname{Spec} A$ is not a Noetherian topological space. Consider the following descending chain of closed set,

$$V((x_1)) \supseteq V((x_1, x_2)) \supseteq V((x_1, x_2, x_3)) \supseteq \cdots \supseteq V((x_1, x_2, \dots, x_n)) \supseteq \cdots . \tag{4.3}$$

Suppose chain (4.3) is stationary, then $V((x_1, \dots, x_r)) = V((x_1, \dots, x_r, x_{r+1}))$ for some r. Note that $(x_1, \dots, x_r)$ is prime ideal, hence, $(x_1, \dots, x_r) \in V((x_1, \dots, x_r))$, but $(x_1, \dots, x_r) \subsetneq (x_1, \dots, x_r, x_{r+1})$, we have $(x_1, \dots, x_r) \notin V((x_1, \dots, x_r, x_{r+1}))$, a contradiction. Hence, chain of closed subset (4.3) is not stationary, and therefore $\operatorname{Spec} A$ is not a Noetherian topological space. □

Remark 4.21 If $\operatorname{Spec} A$ is a Noetherian topological space, A need not be Noetherian. One example is $A = k[x_1, x_2, x_3, \dots]/(x_1, x_2^2, x_3^3, \dots)$. Then $\operatorname{Spec} A = \{[(x_1, x_2, \dots)]\}$ is a one point set, so is Noetherian. But A is not Noetherian as $(x_1) \subsetneq (x_1, x_2) \subsetneq (x_1, x_2, x_3) \subsetneq \cdots$ in A.

Proposition 4.6.19

Let X be a topological space. Then X is Noetherian space if and only if every open subset of X is quasi-compact.

Hence if A is Noetherian, every open subset of $\operatorname{Spec} A$ *is quasi-compact.*

Proof If X is Noetherian, let $U \subseteq X$ be an open subset. Let $\{U_i\}$ be an open cover of U. Let $V_n = \bigcup_{i=1}^n U_i$, then we get an ascending chain of open subsets,

$$V_1 \subseteq V_2 \subseteq V_3 \subseteq \cdots, \tag{4.4}$$

hence, we have a descending chain of closed subsets,

$$X \setminus V_1 \supseteq X \setminus V_2 \supseteq X \setminus V_3 \cdots. \tag{4.5}$$

Since X is Noetherian, descending chain of closed subsets (4.5) is stationary, say $X \setminus V_r = X \setminus V_{r+1} = \cdots$, then $\bigcup_{i=1}^r U_i = \bigcup_{i=1}^{r+1} U_i = \cdots$. Hence, $U = \bigcup_{i=1}^r U_i$, which implies that U is quasi-compact.

Conversely, if every open subset of X is quasi-compact. Let $U_1 \subseteq U_2 \subseteq U_3 \subseteq \cdots$ be an ascending chain of open subsets of X, let $U = \bigcup_{i=1}^\infty U_i$ is open. Note that $\{U_i\}$ is an open cover of X, by hypothesis, U is quasi-compact, we may assume that $U = \bigcup_{i=1}^n U_i$. Hence, $\bigcup_{i=1}^n U_i = \bigcup_{i=1}^{n+1} U_i = \cdots$, and therefore $U_n = U_{n+1} = U_{n+2} = \cdots$. Thus, $U_1 \subseteq U_2 \subseteq U_3 \subseteq \cdots$ is stationary, and therefore X is Noetherian space. □

Noetherian conditions for modules

Definition 4.6.11 (Noetherian module)

If A is any ring, not necessarily Noetherian, we say an ***A-module is Noetherian*** *if it satisfies the ascending chain condition (a.c.c.) for submodules.*

Example 4.10 A ring A is Noetherian if and only if it is a Noetherian A-module.

Proposition 4.6.20

M is a Noetherian A-module if and only if every submodule of M is finitely generated.

Proof The proof same as Proposition 4.6.17, or see [AM18, Proposition 6.2]. □

Proposition 4.6.21

Let

$$0 \longrightarrow M' \xrightarrow{\alpha} M \xrightarrow{\beta} M'' \longrightarrow 0$$

be an exact sequence of A-modules. Then M is Noetherian if and only if M' and M'' are Noetherian.

Proof "⇒": If M is Noetherian A-module. Let $M_1' \subseteq M_2' \subseteq \cdots$ be an ascending chain of submodules of M'. Then we have a chain

$$\alpha(M_1') \subseteq \alpha(M_2') \subseteq \cdots \subseteq \alpha(M_n') \subseteq \cdots.$$

Since M' is Noetherian, above chain is stationary, i.e., exists n such that $\alpha(M_n') = \alpha(M_{n+1}') = \cdots$. Noting that α is injective, we have $M'_n = M'_{n+1} = \cdots$.

Let $M''_1 \subseteq M''_2 \subseteq \cdots$ be an ascending chain of submodules of M''. Then we have a chain

$$\beta^{-1}(M''_1) \subseteq \beta^{-1}(M''_2) \subseteq \cdots \subseteq \beta^{-1}(M''_n) \subseteq \cdots .$$

Since M'' is Noetherian, above chain is stationary, i.e., exists n such that $\beta^{-1}(M''_n) = \beta^{-1}(M''_{n+1}) = \cdots$. Note that β is a surjective, we have $M''_n = M''_{n+1} = \cdots$.

"$\Leftarrow$": Let $(L_n)_{n\geq 1}$ be an ascending chain of submodules of M. Then $(\alpha^{-1}(L_n))$ is a chain in M', and $(\beta(L_n))$ is a chain in M''. For large enough n both these chains are stationary, i.e.,

$$\alpha^{-1}(L_n) = \alpha^{-1}(l_{n+1}) = \cdots, \quad \beta(L_n) = \beta(l_{n+1}) = \cdots .$$

We shall to prove $L_n = L_{n+1}$. Let $x \in L_{n+1}$, then $\beta(y) = \beta(x)$ for some $y \in L_n$. Note that $x - y \in \operatorname{Ker}\beta = \operatorname{Im}\alpha$, hence exists $z \in L_n$ such that $\alpha(z) = x - y$, and therefore $z \in \alpha^{-1}(L_{n+1}) = \alpha^{-1}(L_n)$. Hence, $x - y \in \alpha(\alpha^{-1}(L_n)) \subseteq L_n$. This implies $x \in L_n$, that is, $L_n = L_{n+1}$, by induction, we have

$$L_n = L_{n+1} = L_{n+2} = \cdots .$$

Hence, M is Noetherian. □

Corollary 4.6.1

If M_i $(1 \leq i \leq n)$ are Noetherian A-modules, then $\bigoplus_{i=1}^n M_i$ is Noetherian.

Proof By induction, we reduce immediately to the case that $M_1 \oplus M_2$. Consider exact sequence,

$$0 \longrightarrow M_1 \xrightarrow{i} M_1 \oplus M_2 \xrightarrow{p} M_2 \longrightarrow 0,$$

which i is inclusion and p us projection. By Proposition 4.6.21, $M_1 \oplus M_2$ is Noetherian module. □

Corollary 4.6.2

If A is a Noetherian ring , then $A^{\oplus n}$ is a Noetherian A-module.

Proof By Corollary 4.6.1, clearly. □

Proposition 4.6.22

If A is a Noetherian ring and M is a finitely generated A-module, then M is a Noetherian module.

Proof Since M is a finitely generated A-module, then M is isomorphic to a quotient of $A^{\oplus n}$ for some integer $n > 0$. By Corollary 4.6.2, $A^{\oplus n}$ is a Noetherian A-module. We may assume that $M \cong A^{\oplus n}/K$, then we have a exact sequence,

$$0 \longrightarrow K \hookrightarrow A^{\oplus n} \twoheadrightarrow A^{\oplus n}/K \longrightarrow 0,$$

since $A^{\oplus n}$ is a Noetherian A-module, $M \cong A^{\oplus n}/K$ is Noetherian, by Proposition 4.6.21. □

Corollary 4.6.3

Any submodule of a finitely generated module over a Noetherian ring is Noetherian module.

Proof Let A be a Noetherian ring, M is a finitely generated A-module, let N be a submodule of M. By Proposition 4.6.20, N is finitely generated A-module, by Proposition 4.6.22, N is Noetherian. □

Why you should not worry about Noetherian hypotheses: Should you work hard to eliminate Noetherian hypotheses? Should you worry about Noetherian hypotheses? Should you stay up at night thinking about non-Noetherian rings? For the most past, the answer to these questions is "no". Most people will never need to worry about non-Noetherian rings, but there are reasons to be open to them. First, they can actually come up. For example, fibered products of Noetherian schemes over Noetherian schemes (and even fibered products of Noetherian points over Noetherian points!) can be non-Noetherian, and the normalization of Noetherian rings can be non-Noetherian. You can either work hard to show that the rings or schemes you care about don't have this pathology, or you can just relax and not worry about it. Second, there is often no harm in working with schemes in general. Knowing when Noetherian conditions are needed will help you remember why results are true, because you will have some sense of where Noetherian conditions enter into arguments. Finally, for some people, non-Noetherian rings naturally come up. For example, adeles are not Noetherian. And many valuation rings that naturally arise in arithmetic and tropical geometry are not Noetherian.

4.7 The function $I(\cdot)$, taking subsets of $\operatorname{Spec} A$ to ideals of A

We now introduce a notion that is in some sense "inverse" to the vanishing set function $V(\cdot)$.

Definition 4.7.1 (Function $I(\cdot)$)

Given a subset $S \subseteq \operatorname{Spec} A$, $I(S)$ is the set of functions vanishing on S. In other words,

$$I(S) = \{f \in A : f \in \mathfrak{p}, \forall [\mathfrak{p}] \in S\} = \bigcap_{[\mathfrak{p}] \in S} \mathfrak{p} \subseteq A.$$

We make three quick observations.

- $I(S)$ is clearly an ideal of A.
- $I(\cdot)$ is inclusion-reversing: if $S_1 \subseteq S_2$, then $I(S_1) \supseteq I(S_2)$.

  Proof Let $f \in I(S_2)$, then $f \in \mathfrak{p}$ for all $[\mathfrak{p}] \in S_2$. Since $S_1 \subseteq S_2$, $f \in \mathfrak{p}$ for all $[\mathfrak{p}] \in S_1$, i.e., $f \in I(S_1)$. Hence, $I(S_2) \subseteq I(S_1)$. $\square$
- $I(\overline{S}) = I(S)$.

  Proof In fact, $S \subseteq \overline{S}$, then $I(S) \supseteq I(\overline{S})$. Since $\overline{S}$ is the smallest closed subset of $\operatorname{Spec} A$ which contains S, we have $\overline{S} = \bigcap_{V(I) \supseteq S} V(I)$. Hence,

  $$I(\overline{S}) = I\left(\bigcap_{V(I)\supseteq S} V(I)\right) = \left\{f \in A : f \in \mathfrak{p}, \forall [\mathfrak{p}] \in \bigcap_{V(I)\supseteq S} V(I)\right\}$$

  Let $f \in I(S)$, then $f \in \mathfrak{p}$ for all $[\mathfrak{p}] \in S$. Hence, closed subset $V(f) \supseteq S$, and therefore $\bigcap_{V(I)\supseteq S} V(I) \subseteq V(f)$. It follows that f vanished at $\bigcap_{V(I)\supseteq S} V(I)$, and therefore $f \in I(\overline{S})$. Thus, $I(S) = I(\overline{S})$. $\square$

✍ **Exercise 4.21** Let $A = k[x, y]$. If $S = \{[(y)], [(x, y-1)]\}$ (see Figure 4.12), then $I(S)$ consists of those polynomials vanishing on the x-axis, and at the point $(0, 1)$. Give generators for this ideal.

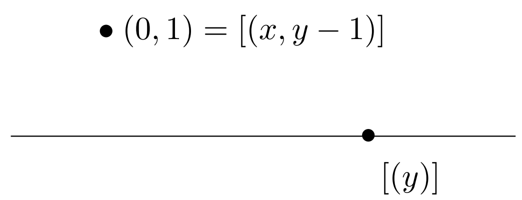


Figure 4.12: The set S of Exercise 4.21, pictured as a subset of $\mathbb{A}^2$

Proof By definition, $I(S) = \bigcap_{[\mathfrak{p}]\in S} \mathfrak{p} = (y) \cap (x, y-1)$. Let $f \in I(S)$, then $f \in (y) \cap (x, y-1)$, i.e. f

vanishing on points $[(y)]$ and $[(x, y-1)]$, i.e., f vanishing on the x-axis and at the point $(0,1)$. Note that $(y)+(x, y-1)=(1)$, $(y) \cap (x, y-1)=(y)(x, y-1)=(xy, y^2-y)$. Hence, the generators of $I(S)$ is xy and y^2-y. □

✍ **Exercise 4.22** Suppose $S \subseteq \mathbb{A}^3_{\mathbb{C}}$ is the union of the three axes. Give generators for the ideal $I(S)$. Be sure to prove it!

Proof $S \subseteq \mathbb{A}^3_{\mathbb{C}}$ is the union of the three axes, hence, $S = V((x,y)) \cup V((y,z)) \cup V((x,z))$. Then $I(S)=(x,y)\cap(y,z)\cap(x,z)$. We claim that $I(S)=(xy,xz,yz)$. Note that $xy,xz,yz \in (x,y)\cap(y,z)\cap(x,z)$, we have $(xy,xz,yz) \subseteq (x,y)\cap(y,z)\cap(x,z)$. Let $f \in (x,y)\cap(y,z)\cap(x,z)$. Hence, if the monomial of f has x, then this monomial must have y or z; if the monomial of f has y, then this monomial must have x or z; if the monomial of f has z, then this monomial must have x or y. This implies that f generated by xy, xz, and yz, and therefore $f \in (xy,xz,yz)$. Hence, $I(S)=(xy,xz,yz)$. □

Remark 4.22 We will see that this ideal is not generated by less than three elements.

Proposition 4.7.1 (Closure in Zariski topology)

Let A be a ring, and $S \subseteq \operatorname{Spec} A$, then $V(I(S))=\overline{S}$. Hence $V(I(S))=S$ for a closed set S.

Proof Let $[\mathfrak{p}] \in S$, then $f \in \mathfrak{p}$ for all $f \in I(S)$, hence, $\mathfrak{p} \supseteq I(S)$, and therefore $S \subseteq V(I(S))$. Since $V(I(S))$ is closed, we have $\overline{S} \subseteq V(I(S))$.

Let $V(J)$ be any closed subset which contains S, then for all $[\mathfrak{p}] \in S$ we have $\mathfrak{p} \supseteq J$. It follows that

$$J \subseteq \bigcap_{[\mathfrak{p}]\in S} \mathfrak{p} = I(S),$$

and therefore $V(J) \supseteq V(I(S))$. Hence,

$$\overline{S} = \bigcap_{V(J)\supseteq S} V(J) \supseteq V(I(S)),$$

which implies that $V(I(S))=\overline{S}$. □

Note that $I(S)$ is always a radical ideal — if $f \in \sqrt{I(S)}$, then $f^n \in I(S)$ for some n, this implies that f^n vanishes on S for some n, hence, f vanishes on S, so $f \in I(S)$. Hence, $I(S)=\sqrt{I(S)}$.

Proposition 4.7.2

If $J \subseteq A$ is an ideal, then $I(V(J))=\sqrt{J}$.

Proof $f \in I(V(J))$ if and only if f vanishes on $V(J)$ if and only if $f \in \sqrt{J}$, since $\sqrt{J}$ is the intersection of prime ideals which contains J. □

Proposition 4.7.1 and Proposition 4.7.2 show that V and I are "almost" inverse. More precisely:

Theorem 4.7.1 (Hilbert's Nullstellensatz)

$V(\cdot)$ and $I(\cdot)$ give an inclusion-reversing bijection between closed subsets of Spec A *and radical ideals of A (where a closed subset gives a radical ideal by $I(\cdot)$, and a radical ideal gives a closed subset by $V(\cdot)$).*

✍ **Exercise 4.23** Let $J=(x^2+y^2-1, y-1)$. Find, with proof, an element of $I(V(J)) \setminus J$.

Proof By Hilbert's Nullstellensatz, $I(V(J))=\sqrt{J}=\sqrt{(x^2+y^2-1, y-1)}$, we claim that

$$\sqrt{(x^2+y^2-1, y-1)}=(x, y-1).$$

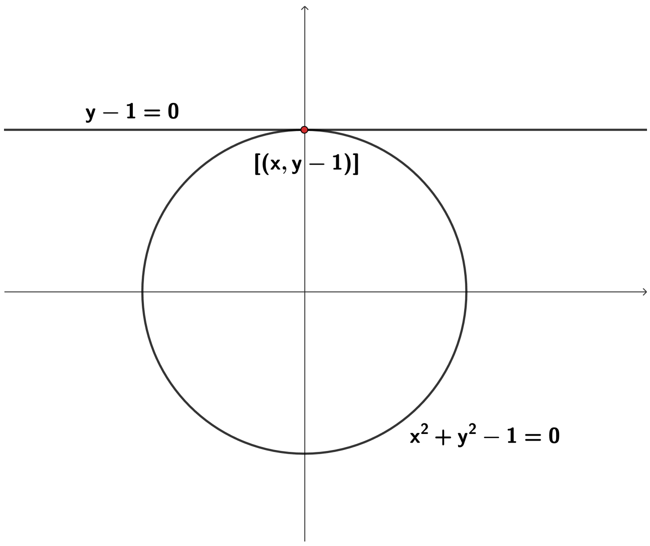


Figure 4.13: Picture of Exercise 4.23

Clearly, $y-1 \in (x^2+y^2-1, y-1)$, and therefore $y-1 \in \sqrt{(x^2+y^2-1, y-1)}$. Note that

$$x^2+y^2-1 = x^2+(y-1)^2+2(y-1),$$

$x^2 \in (x^2+y^2-1, y-1)$, hence, $(x, y-1) \subseteq \sqrt{(x^2+y^2-1, y-1)}$. Since $(x, y-1)$ is maximal ideal in $k[x,y]$, then $(x, y-1) = \sqrt{(x^2+y^2-1, y-1)}$.

Note that $x \notin (x^2+y^2-1, y-1)$, then we find an element $x \in I(V(J))$, but $x \notin J$, as we desired. □

Proposition 4.7.3

$V(\cdot)$ and $I(\cdot)$ give a bijection between irreducible closed subsets of Spec A *and prime ideals of A. Hence, in* Spec A *there is a bijection between points of* Spec A *and irreducible closed subsets of* Spec A

Proof Let $V(\mathfrak{a})$ be an irreducible closed subsets of Spec A. In fact, $V(\mathfrak{a}) = \operatorname{Spec} A/\mathfrak{a}$. By Proposition 4.6.2 (ii), $\mathfrak{N}(A/\mathfrak{a})$ is prime. Since $\mathfrak{N}(A/\mathfrak{a})$ is the intersection of prime ideals of A which contains $\mathfrak{a}$, there exists a prime ideal $\mathfrak{p} \supseteq \mathfrak{a}$, such that $\mathfrak{p} = \mathfrak{N}(A/\mathfrak{a})$. Note that $I(V(\mathfrak{a})) = \sqrt{\mathfrak{a}}$, we have $\sqrt{\mathfrak{a}} = \mathfrak{p}$ is a prime ideal.

Conversely, let $\mathfrak{p}$ be a prime ideal, consider $V(\mathfrak{p})$. Note that $V(\mathfrak{p}) = \{[\mathfrak{q}] \in \operatorname{Spec} A : \mathfrak{q} \supseteq \mathfrak{p}\} = \operatorname{Spec} A/\mathfrak{p}$, since $\mathfrak{p}$ is a prime ideal, $A/\mathfrak{p}$ is integral domain, hence, $\operatorname{Spec} A/\mathfrak{p} = V(\mathfrak{p})$ is irreducible closed subset of Spec A. □

Remark 4.23 From this conclude that in Spec A there is a bijection between points of Spec A and irreducible closed of Spec A (where a point determines an irreducible closed subset by taking the closure). Hence each irreducible closed subset of Spec A has precisely one generic point — any irreducible closed subset Z can be written uniquely as $\overline{\{z\}}$.

Definition 4.7.2 (Minimal prime ideal)

*A prime ideal of a ring A is a **minimal prime ideal** (or more simply, **minimal prime**) if it is minimal with respect to inclusion.*

Example 4.11 The only minimal prime ideal of $k[x,y]$ is (0).

Proposition 4.7.4

If A is any ring, then the irreducible components of $\operatorname{Spec} A$ are in bijection with the minimal prime ideals of A. In particular, $\operatorname{Spec} A$ is irreducible if and only if A has only one minimal prime ideal.

Proof Let Z be an irreducible component of $\operatorname{Spec} A$, then Z is an irreducible closed subset of $\operatorname{Spec} A$. By Proposition 4.7.3, $I(Z)$ is a prime ideal. If $I(Z)$ is not minimal prime ideal, there exists a prime ideal $\mathfrak{p}$ such that $\mathfrak{p} \subsetneq I(Z)$. By Proposition 4.7.3 again, $V(\mathfrak{p})(\supsetneq V(I(Z)) = Z)$ is an irreducible closed subset, contradicts to the fact that Z is irreducible component. Hence, $I(Z)$ is the minimal prime ideal.

Conversely, if $\mathfrak{p}$ is a minimal prime ideal, then $V(\mathfrak{p})$ is an irreducible closed subset. If $V(\mathfrak{p})$ is not irreducible component, then there exists an irreducible closed subset $V(\mathfrak{a})$ such that $V(\mathfrak{a}) \supsetneq V(\mathfrak{p})$, by Proposition 4.7.3, we have prime ideal $\sqrt{a}(\subsetneq \mathfrak{p})$, contradicts to the fact that $\mathfrak{p}$ is a minimal prime. Hence, $V(\mathfrak{p})$ is an irreducible component of $\operatorname{Spec} A$.

In particular, if $\operatorname{Spec} A$ is irreducible, then $I(\operatorname{Spec} A) = \bigcap_{[\mathfrak{p}]\in\operatorname{Spec} A} \mathfrak{p} = \mathfrak{N}(A)$ is prime ideal. Since it is contained in every prime ideal of A, $\mathfrak{N}(A)$ is the unique minimal ideal of A. Conversely, if A has only one minimal prime ideal, say $\mathfrak{a}$, then $\mathfrak{a}$ contains in all prime ideals of A. Hence, $\mathfrak{a} = \mathfrak{N}(A)$, and therefore $V(\mathfrak{a}) = V(\mathfrak{N}(A)) = \operatorname{Spec} A$ is irreducible. $\square$

Proposition 4.6.12, Proposition 4.6.18, and Proposition 4.7.4 imply that every Noetherian ring has a finite number of minimal prime ideals. An algebraic fact is now revealed to be really a "geometric" fact!

✍ **Exercise 4.24** In $\mathbb{A}_k^n = \operatorname{Spec} k[x_1, \dots, x_n]$, the subset cut out by $f(x_1, \dots, x_n) \in k[x_1, \dots, x_n]$ should certainly have irreducible components corresponding to the distinct irreducible factor of f. Prove this.

Proof Let $f = ap_1^{e1}p_2^{e2}\cdots p_n^{e_n}$, where p_i is irreducible polynomial and $a \in k$. Consider $V((f))$. Note that $(f) = (p_1^{e_1}p_2^{e_2}\cdots p_n^{e_n}) = (p_1^{e_1})(p_2^{e_2})\cdots(p_n^{e_n})$. Hence,

$$\begin{aligned} V((f)) = \operatorname{Spec} k[x_1, \dots, x_n]/(f) =& V((p_1^{e_1})) \cup V((p_2^{e_2})) \cup \cdots \cup V((p_n^{e_n})) \\ =& V(\sqrt{(p_1^{e_1})}) \cup \cdots \cup V(\sqrt{(p_n^{e_n})}) \\ =& V((p_1)) \cup \cdots \cup V((p_n)). \end{aligned}$$

We claim that each (p_i) is the minimal prime ideal in $\operatorname{Spec} k[x_1, \dots x_n]/(f)$. If there exists prime ideal $\mathfrak{p} \subseteq k[x_1, \dots x_n]/(f)$ such that $\mathfrak{p} \subseteq (p_i)$. Let $q \in \mathfrak{p}$ be an irreducible polynomial, then $p_i \mid q$, since p_i and q are irreducible polynomials, $q = p_i$, hence, $(p_i) = (q) \subseteq \mathfrak{p}$, and therefore $(p_i) = \mathfrak{p}$. Hence, each (p_i) is minimal ideal, by Proposition 4.7.4, $V((p_i))$ is an irreducible component. $\square$

✍ **Exercise 4.25** What are the minimal prime ideals of $k[x, y]/(xy)$, where k is a field?

Solution *By Exercise 4.24, (x) and (y) are the minimal prime ideals.*

Beginning of a grand dictionary between algebra and geometry.

We are now well on our away to building a grand dictionary between algebra and geometry.

Algebra	Geometry
Ring A	Affine scheme $\operatorname{Spec} A$
$\mathfrak{p} = I(\mathfrak{p})$ prime ideal of A	$\mathfrak{p} = [\mathfrak{p}]$ point of $\operatorname{Spec} A$
Element $f \in A$	Function f on $\operatorname{Spec} A$
$f \pmod{\mathfrak{p}}$	$f(\mathfrak{p})$; value of the function f at $\mathfrak{p}$
$f \in \mathfrak{p}$	The function f vanishes (is zero) at $\mathfrak{p}$
Nilradical ideal $\mathfrak{N} = \sqrt{0} = \bigcap_{[\mathfrak{p}] \in \operatorname{Spec} A} \mathfrak{p}$ of A	Functions vanishing at every point
Maximal ideal of A	Irreducible closed subset of $\operatorname{Spec} A$ that is a point (=closed point of $\operatorname{Spec} A$)
$\mathfrak{p} = I(Z)$ prime ideal of A	Irreducible closed subset $Z = V(\mathfrak{p})$ of $\operatorname{Spec} A$
Minimal prime ideal of A	Irreducible component of $\operatorname{Spec} A$
Radical ideal $I = \sqrt{I} = I(S)$ of A	Closed subset $S = V(I)$ of $\operatorname{Spec} A$

Table 4.1: The correspondence between algebra and geometry.

Chapter 5 The structure sheaf, and the definition of schemes in general

The final ingredient in the definition of an affine scheme is the **structure sheaf** $\mathscr{O}_{\operatorname{Spec} A}$, which we think of as the "sheaf of algebraic functions". You should keep in your mind the example of "algebraic functions" on $\mathbb{C}^n$, which you understand well. For example, for the plane $\mathbb{A}^2$, we expect that on the open set $D(xy)$ (away from the two axes), $(3x^4 + y + 4)/x^7y^3$ should be an algebraic function.

These functions will have values at points, but won't be determined by their values at points. But like all sections of sheaves, they will be determined by their germs.

5.1 The structure sheaf of an affine scheme

5.1.1 Definition of structure sheaf

We want now to give a ring of functions on any open set, but we know that it suffices to describe the functions on a base of the topology. We will see that the base of distinguished open sets will make things much more tractable than you might fear. So we now describe that structure sheaf as a sheaf of rings on the base of distinguished open sets (Theorem 3.5.1 and Proposition 4.5.1).

Definition 5.1.1 (Functions on distinguished open set)

Define $\mathscr{O}_{\operatorname{Spec} A}(D(f))$ to be the localization of A at the multiplicative set of all functions that do not vanish outside of $V(f)$ (i.e., those $g \in A$ such that $V(g) \subseteq V(f)$, or equivalently $D(f) \subseteq D(g)$.) Formally,

$$\mathscr{O}_{\operatorname{Spec} A}(D(f)) := S^{-1}A,$$

where $S = \{g \in A : V(g) \subseteq V(f)\} = \{g \in A : D(g) \supseteq D(f)\}$.

Remark 5.1

(i) This definition depends only on $D(f)$, and not on f itself.

(ii) $\mathscr{O}_{\operatorname{Spec} A}(\varnothing) = \{0\}$.

Proof The multiplicative set $S = \{g \in A : D(g) \supseteq \varnothing\} = A$. By Proposition 4.5.6, $D(f) = \varnothing$ if and only if $f \in \mathfrak{N}(A)$. Hence, $f^n = 0$ for some n, hence, $0 \in S$. It follows that $S^{-1}A = \{0\}$, i.e., $\mathscr{O}_{\operatorname{Spec} A}(\varnothing) = \{0\}$. □

Proposition 5.1.1

If A is a ring, $f \in A$, then there is an isomorphism of rings

$$\mathscr{O}_{\operatorname{Spec} A}(D(f)) \cong A_f.$$

In particular, if $f = 1$, then $\mathscr{O}_{\operatorname{Spec} A}(\operatorname{Spec} A) \cong A$.

Proof Say $\mathscr{O}_{\operatorname{Spec} A}(D(f)) = S^{-1}A$, where $S = \{g \in A : D(g) \supseteq D(f)\}$. Define $\varphi : A \to S^{-1}A$ by setting $f \mapsto f/1$, clearly, it is a well-defined ring homomorphism. Let $s \in \{1, f, f^2, \ldots\}$, we want to show that $\varphi(s)$ is the unit in $S^{-1}A$. We may assume $s = f^n$ for some n, then $\varphi(s) = f^n/1 = f^n$. Note that $D(f^n) \supseteq D(f)$, we have $f^n \in S$, i.e., $\varphi(s)$ is the unit in $S^{-1}A$. Consider the following diagram,

$$\begin{tikzcd} A \arrow[r, "\varphi"] \arrow[d, "\psi"'] & S^{-1}A \\ A_f, \arrow[ur, dashed, "\theta"'] & \end{tikzcd}$$

where $\psi : A \to A_f$ is the canonical ring homomorphism, by the universal property of localization, there exists unique ring homomorphism $\theta : A_f \to S^{-1}A$ such that above diagram commutes.

Next, we want to show that $\psi(s)$ is the unit in A_f, where $s \in S$. Let $s \in S$, then $D(s) \supseteq D(f)$, by Proposition 4.5.5, s is an invertible element of A_f. Consider the following diagram,

$$\begin{tikzcd} A \arrow[r, "\psi"] \arrow[d, "\varphi"'] & A_f \\ S^{-1}A, \arrow[ur, dashed, "\eta"'] & \end{tikzcd}$$

by the universal property of localization, there exists unique ring homomorphism $\eta : S^{-1}A \to A_f$. Hence, we have the isomorphism of rings

$$\mathscr{O}_{\operatorname{Spec} A}(D(f)) = S^{-1}A \cong A_f.$$

□

Definition 5.1.2 (The restriction map)

If $D(f') \subseteq D(f)$, define ***the restriction map***

$$\operatorname{res}_{D(f),D(f')} : \mathscr{O}_{\operatorname{Spec} A}(D(f)) \longrightarrow \mathscr{O}_{\operatorname{Spec} A}(D(f'))$$

in the obvious way: the latter ring is a further localization of the former ring.

The restriction maps obviously compose: this is a "presheaf on the distinguished base".

Theorem 5.1.1

The data just described give a sheaf on the distinguished base, and hence determine a sheaf on the topological space Spec A.

Definition 5.1.3 (Structure sheaf)

In Theorem 5.1.1, the sheaf on the topological space Spec A *is called the* ***structure sheaf****, and will be denoted $\mathscr{O}_{\operatorname{Spec} A}$, or sometimes $\mathscr{O}$ if the subscript is clear from the context.*

The notation Spec A *will hereafter denote the data of a topological space with a structure sheaf. Such a topological space, with structure sheaf, will be called an* ***affine scheme***.

Remark 5.2 We continue to use the language of ringed space: functions on open sets, global functions, and so forth. An important lesson of Theorem 5.1.1 is not just that $\mathscr{O}_{\operatorname{Spec} A}$ is a sheaf, but also that the distinguished base provides a good way of working with $\mathscr{O}_{\operatorname{Spec} A}$. Notice also that we have justified interpreting elements of A as functions on Spec A.

Now, let's prove Theorem 5.1.1.

Proof We must show that the base identity and base gluability axioms hold (Definition 3.5.2). Suppose Spec $A = \bigcup_{i\in I} D(f_i)$, or equivalently (Proposition 4.5.2) the ideal generated by the f_i is the entire ring A.

(Experts familiar with the equalizer exact sequence of Example 3.3 will realize that we are showing exactness of

$$0 \longrightarrow A \longrightarrow \prod_{i\in I} A_{f_i} \longrightarrow \prod_{i\neq j\in I} A_{f_i f_j}$$

where $\{f_i\}_{i\in I}$ is a set of functions with $(f_i)_{i\in I} = A$. Signs are involved in the right-hand map: the map $A_{f_i} \to A_{f_i f_j}$ is the localization map, and the map $A_{f_j} \to A_{f_i f_j}$ is the negative of the localization map. Base identity corresponds to injectivity at A, and gluability corresponds to exactness at $\prod_i A_{f_i}$.)

We check identity on the base. Suppose that $D(f) = \operatorname{Spec} A_f = \bigcup_{i\in I} D(f_i)$ where i runs over some index set I. Then there is some finite subset of I, which we name $\{1, 2, \dots, n\}$, such that $D(f) = \operatorname{Spec} A_f = \bigcup_{i=1}^n D(f_i)$, i.e., $(f_1, \dots, f_n) = A_f$ (quasi-compactness of $\operatorname{Spec} A_f$, Proposition 4.6.4; each f_i is the element in A_f). Suppose we are given $s \in A_f$ such that $\operatorname{res}_{D(f),D(f_i)} s = 0$ in $A_{f_i} (\cong \mathscr{O}_{\operatorname{Spec} A}(D(f_i)))$ for all i. We want to show that $s = 0$. The fact that $\operatorname{res}_{D(f),D(f_i)} s = 0$ in A_{f_i} implies that there is some m such that for each $i \in \{1, \dots, n\}$, $f_i^m s = 0$. Since $\operatorname{Spec} A_f = \bigcup_{i=1}^n D(f_i) = \bigcup_{i=1}^n D(f_i^m)$ (Proposition 4.5.4), we have $A_f = (f_1^m, f_2^m, \dots, f_n^m)$, so there are $r_i \in A_f$ with $\sum_{i=1}^n r_i f_i^m = 1$ in A_f, form which

$$s = \left(\sum r_i f_i^m\right) s = \sum r_i (f_i^m s) = 0.$$

Thus we have checked the base identity axiom.

We next show base gluability. (Serre has described this as a "partition of unity" argument, and if you look at it in the right way, his insight is very enlightening.) Suppose $D(f) = \bigcup_{i\in I} D(f_i)$, where I is an index set (possibly horribly infinite). Suppose we are given elements in each A_{f_i} that agree on the overlaps $A_{f_i f_j} (= \mathscr{O}_{\operatorname{Spec} A}(D(f_i f_j)) = \mathscr{O}_{\operatorname{Spec} A}(D(f_i) \cap D(f_j)))$ (Proposition 4.5.4).

Assume first that I is finite, say $I = \{1, \dots, n\}$. We have elements $a_i / f_i^{l_i} \in A_{f_i}$ agreeing on overlaps $A_{f_i f_j}$ (see Figure 5.1(a)). Let $g_i = f_i^{l_i}$, note that $D(f_i) = D(f_i^m) = D(g_i)$, we can simplify notion by considering our elements as of the form $a_i / g_i \in A_{g_i}$ (Figure 5.1(b)).

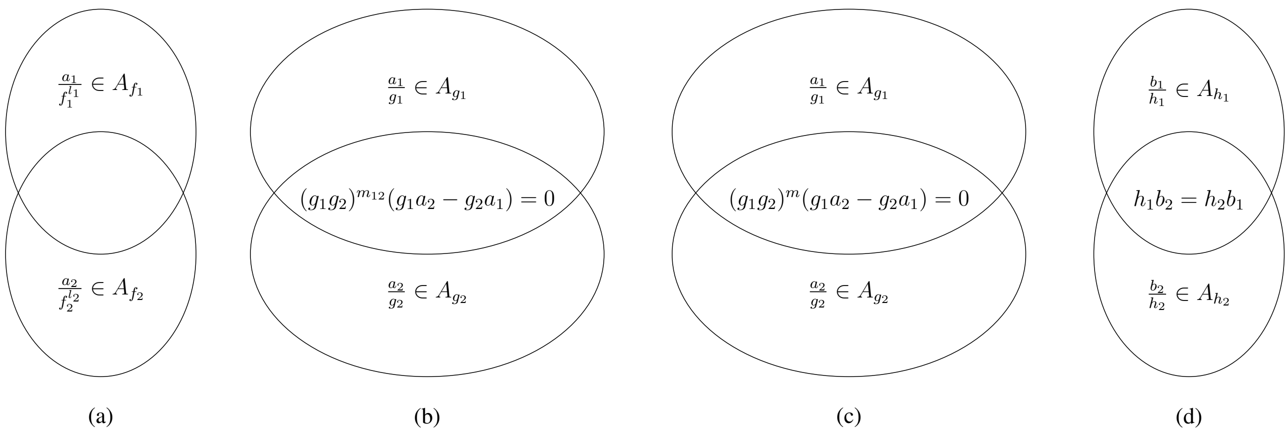


Figure 5.1: Base gluability of the structure sheaf

The fact that a_i/g_i and a_j/g_j "agree on the overlap" (i.e., in $A_{g_i g_j}$) means that for some m_{ij},

$$(g_i g_j)^{m_{ij}} (g_j a_i - g_i a_j) = 0$$

in A. By taking $m = \max m_{ij}$ (here we use the finiteness of I), we can simplify notation:

$$(g_i g_j)^m (g_j a_i - g_i a_j) = 0$$

for all i, j (Figure 5.1(c)). Let $b_i = a_i g_i^m$ for all i, and $h_i = g_i^{m+1}$ (so $D(h_i) = D(g_i)$). Then we can simplify notation even more (Figure 5.1(d)): on each $D(h_i)$, we have a function b_i/h_i, and the overlap condition is

$$h_j b_i = h_i b_j. \tag{5.1}$$

Now $\bigcup_i D(h_i) = \operatorname{Spec} A_f$, implies that $1 = \sum_{i=1}^n r_i h_i$ for some $r_i \in A_f$. Define

$$r = \sum r_i b_i. \tag{5.2}$$

This will be the element of A_f that restricts to each b_j/h_j. Indeed, from the overlap condition (5.1),

$$r h_j = \sum_i r_i b_i h_j = \sum_i r_i h_i b_j = b_j.$$

We next deal with the case where I is infinite. Choose a finite subset $\{1, \ldots, n\} \subseteq I$ with $(f_1, \ldots, f_n) = A_f$ (or equivalently, use quasi-compactness of $\operatorname{Spec} A_f$ to choose a finite subcover by $D(f_i)$). Construct r as above, using (5.2). We will show that for any $z \in I - \{1, \ldots, n\}$, r restricts to the desired element $a_z/f_z^{l_z}$ of A_{f_z}. Repeat the entire process above with $\{1, \ldots, n, z\}$ in place of $\{1, \ldots, n\}$, to obtain $r' \in A_f$ which restricts to $a_i/f_i^{l_i}$ for $i \in \{1, \ldots, n, z\}$. Then by the base identity axiom, $r' = r$. (Note that we use base identity to prove base gluability. This is an example of how the identity axiom is somehow "prior" to the gluability axiom.) Hence, r restricts to the desired element $a_z/f_z^{l_z}$ of A_{f_z}.

By Definition 3.5.2, $\mathscr{O}_{\operatorname{Spec} A}$ is a sheaf on a base, by Theorem 3.5.1, $\mathscr{O}_{\operatorname{Spec} A}$ extend to a sheaf on topological space $\operatorname{Spec} A$. □

Remark 5.3 Definition 5.1.1 and Theorem 5.1.1 suggests a potentially slick way of describing sections of $\mathscr{O}_{\operatorname{Spec} A}$ over any open subset: perhaps $\mathscr{O}_{\operatorname{Spec} A}(U)$ is the localization of A at the multiplicative set of all function that do not vanish at any point of U. This is not true. A counterexample (that you will later be able to make precise): let $\operatorname{Spec} A$ be two copies of $\mathbb{A}_k^2$ glued together at their origins and let U be the complement of the origin(s). Then the function which is 1 on the first copy of $\mathbb{A}_k^2 \setminus \{(0,0)\}$ and 0 on the second copy of $\mathbb{A}_k^2 \setminus \{(0,0)\}$ is not of this form.

5.1.2 $\mathscr{O}_{\operatorname{Spec} A}$-modules coming from A-modules

The following generalization of Theorem 5.1.1 will be essential in the definition of quasi-coherent sheaf in Chapter 7.

Definition 5.1.4 ($\mathscr{O}_{\operatorname{Spec} A}$-module)

Suppose M is an A-module. Define $\widetilde{M}(D(f))$ to be the localization of M at the multiplicative set of all functions that do not vanish outside of $V(f)$, i.e.,

$$\widetilde{M}(D(f)) := S^{-1}M,$$

where $S = \{g \in A : V(g) \subseteq V(f)\} = \{g \in A : D(g) \supseteq D(f)\}$.
Define restriction maps $\operatorname{res}_{D(f),D(g)}$ in the analogous way to $\mathscr{O}_{\operatorname{Spec} A}$. This defines a sheaf on the distinguished base, and hence a sheaf on $\operatorname{Spec} A$. Also, $\widetilde{M}$ is an $\mathscr{O}_{\operatorname{Spec} A}$-module.

Proof By [AM18, Proposition 3.5], we have $\widetilde{M}(D(f)) = S^{-1}M \cong S^{-1}A \otimes_A M$, by Proposition 5.1.1, $S^{-1}A \cong A_f$, hence,

$$\widetilde{M}(D(f)) \cong A_f \otimes_A M \cong M_f.$$

We check identity on the base. Suppose that $D(f) = \operatorname{Spec} A_f = \bigcup_{i\in I} D(f_i)$, where I is an index set. Since $\operatorname{Spec} A_f$ is quasi-compact (Proposition 4.6.4), we may assume that $D(f) = \bigcup_{i=1}^n D(f_i)$, i.e., $(f_1, \ldots, f_n) = A_f$. Suppose we are given $s \in \widetilde{M}(D(f)) = M_f$ such that $\operatorname{res}_{D(f),D(f_i)} s = 0$ in $\widetilde{M}(D(f_i)) = M_{f_i}$ for all i. We want to show that $s = 0$. The fact that $\operatorname{res}_{D(f),D(f_i)} s = 0$ in M_{f_i}, implies that there is some m such that for each $i \in \{1, \ldots, n\}$, $f_i^m s = 0$. Since $\operatorname{Spec} A_f = \bigcup_{i=1}^n D(f_i) = \bigcup_{i=1}^n D(f_i^m)$ (Proposition 4.5.4), we have $A_f = (f_1^m, \ldots, f_n^m)$, so there are $r_i \in A_f$ with $\sum_{i=1}^n r_i f_i^m = 1$ in A_f, since M_f is A_f-module, form which

$$s = \left(\sum r_i f_i^m\right) s = \sum r_i (f_i^m s) = 0.$$

Thus we have checked the base identity axiom.

We next show base gluability. Suppose $D(f) = \bigcup_{i \in I} D(f_i)$, where I is an index set. Suppose we are given elements in each $\widetilde{M}(D(f_i)) = M_{f_i}$ that agree on the overlaps $\widetilde{M}(D(f_i) \cap D(f_j)) = \widetilde{M}(D(f_i f_j)) = M_{f_i f_j}$.

Assume first that I is finite, say $I = \{1, \dots, n\}$. We have elements $m_i / f_i^{l_i} \in M_{f_i}$, agreeing on overlaps $M_{f_i f_j}$. Let $g_i = f_i^{l_i}$, note that $D(f_i) = D(f_i^{l_i}) = D(g_i)$, we can simplify notion by considering our elements as of the form $m_i / g_i \in M_{g_i}$. The fact that m_i / g_i and m_j / g_j agree on the overlap (i.e., in $M_{g_i g_j}$) means that for some k_{ij},

$$(g_i g_j)^{k_{ij}} (g_j m_i - g_i m_j) = 0$$

in M. By taking $k = \max k_{ij}$ (we use the finiteness of I), we can simplify notation:

$$(g_i g_j)^k (g_j m_i - g_i m_j) = 0$$

for all i, j. Let $n_i = m_i g_i^k$ for all i, and $h_i = g_i^{k+1}$ (so $D(h_i) = D(g_i)$). Then we can simplify notation even more: on each $D(h_i)$, we have a function n_i / h_i, and the overlap condition is

$$h_j n_i = h_i n_j. \tag{5.3}$$

Now $\bigcup_i D(h_i) = \operatorname{Spec} A_f$, implies that $1 = \sum_{i=1}^n r_i h_i$ for some $r_i \in A_f$. Define

$$r = \sum r_i n_i. \tag{5.4}$$

Note that, from the overlap condition (5.3),

$$r h_j = \sum_i r_i n_i h_j = \sum_i r_i h_i n_j = n_j,$$

which implies that

$$\operatorname{res}_{D(f), D(f_i)} r = n_j / h_j = m_j / f_j^{l_j}.$$

We next deal with the case where I is infinite. Choose a finite subset $\{1, \dots, n\} \subseteq I$ with $(f_1, \dots, f_n) = A_f$. Construct r as above, using (5.4). We will show that for any $z \in I - \{1, \dots, n\}$, r restricts to the desired element $m_z / f_z^{l_z}$ of M_{f_z}. Repeat the entire process above with $\{1, \dots, n, z\}$ in place of $\{1, \dots, n\}$, to obtain $r' \in M_f$ which restricts to $m_i / f_i^{l_i}$ for $i \in \{1, \dots, n, z\}$. Then by the base identity axiom, $r' = r$. Hence, r restricts to the desired element $m_z / f_z^{l_z}$ of M_{f_z}.

By Definition 3.5.2, $\widetilde{M}$ is a sheaf on a base, by Theorem 3.5.1, $\widetilde{M}$ extend to a sheaf on topological space $\operatorname{Spec} A$.

To show that $\widetilde{M}$ is an $\mathscr{O}_{\operatorname{Spec} A}$-module, it suffices to show that $\widetilde{M}$ is an $\mathscr{O}_{\operatorname{Spec} A}$-module on distinguished base. Note that $\widetilde{M}(D(f)) \cong M_f$ is A_f-module, it suffices to check the diagram (3.1) in Definition 3.2.13 commutes. Let $D(f) \supseteq D(g)$, consider the following diagram,

$$\begin{tikzcd}
\mathscr{O}_{\operatorname{Spec} A}(D(f)) \times \widetilde{M}(D(f)) \arrow[r, "\text{action}"] \arrow[d, "\operatorname{res}^{\mathscr{O}_{\operatorname{Spec} A}}_{D(f),D(g)} \times \operatorname{res}^{\widetilde{M}}_{D(f),D(g)}"'] & \widetilde{M}(D(f)) \arrow[d, "\operatorname{res}^{\widetilde{M}}_{D(f),D(g)}"] \\
\mathscr{O}_{\operatorname{Spec} A}(D(g)) \times \widetilde{M}(D(g)) \arrow[r, "\text{action}"] & \widetilde{M}(D(g)),
\end{tikzcd}$$

since $\mathscr{O}_{\operatorname{Spec} A}(D(h)) \cong A_h$ and $\widetilde{M}(D(h)) \cong A_h$ for any $h \in A$, it is equal to check that following diagram commutes.

$$\begin{array}{ccc} A_f \times M_f & \xrightarrow{\text{action}} & M_f \\ {\scriptstyle \mathrm{res}^{\mathscr{O}_{\mathrm{Spec}\,A}}_{D(f),D(g)} \times \mathrm{res}^{\widetilde{M}}_{D(f),D(g)}}\big\downarrow & & \big\downarrow{\scriptstyle \mathrm{res}^{\widetilde{M}}_{D(f),D(g)}} \\ A_g \times M_g & \xrightarrow{\text{action}} & M_g. \end{array}$$

Let $(a/f^l, m/f^k) \in A_f \times M_f$. Since $D(f) \supseteq D(g)$, by Proposition 4.5.5, f is an invertible element of A_g, hence,

$$\mathrm{res}^{\mathscr{O}_{\mathrm{Spec}\,A}}_{D(f),D(g)} \times \mathrm{res}^{\widetilde{M}}_{D(f),D(g)}(a/f^l, m/f^k) = (a/f^l, m/f^k)$$

and

$$\mathrm{res}^{\widetilde{M}}_{D(f),D(g)}(am/f^{l+k}) = am/f^{l+k},$$

which implies above diagram commutes. Hence, $\widetilde{M}$ is an $\mathscr{O}_{\mathrm{Spec}\,A}$-module on distinguished base, and therefore $\widetilde{M}$ is an $\mathscr{O}_{\mathrm{Spec}\,A}$-module. □

Remark 5.4 In the course of the proof in Definition 5.1.4, we show that if $(f_i)_{i\in I} \in A$,

$$0 \longrightarrow M \longrightarrow \prod_{i\in I} M_{f_i} \longrightarrow \prod_{i\neq j\in I} M_{f_i f_j} \tag{5.5}$$

is exact. In particular, M can be identified with a specific submodule of $M_{f_1} \times \cdots \times M_{f_r}$. Even though $M \to M_{f_i}$ may not be an inclusion for any f_i, $M \to M_{f_1} \times \cdots \times M_{f_r}$ is an inclusion. This will be useful later: we will want to show that if $(f_1, \ldots, f_n) = A$, and M_{f_i} have this property, then M does too. This ideal will be made precise in the Affine Communication Lemma in Chapter 6.

Note *By the proof in Theorem 3.5.1, we can provide precise descriptions of both $\mathscr{O}_{\mathrm{Spec}\,A}(U)$ and $\widetilde{M}(U)$, for $U \subseteq \mathrm{Spec}\,A$ open subset:*

$$\begin{aligned} \mathscr{O}_{\mathrm{Spec}\,A}(U) =& \{(\varphi_{[\mathfrak{p}]} \in \mathscr{O}_{\mathrm{Spec}\,A,[\mathfrak{p}]})_{[\mathfrak{p}]\in U} : \\ & \forall [\mathfrak{p}] \in U, \exists D(f) \text{ with } [\mathfrak{p}] \in D(f) \subseteq U \text{ and } s \in A_f \text{ s.t. } s_{[\mathfrak{q}]} = \varphi_{[\mathfrak{q}]}, \forall [\mathfrak{q}] \in D(f)\}, \end{aligned} \tag{5.6}$$

where $\mathscr{O}_{\mathrm{Spec}\,A,[\mathfrak{p}]} = \varinjlim_{D(h)\ni[\mathfrak{p}]} \mathscr{O}_{\mathrm{Spec}\,A}(D(f)) = \varinjlim_{h\in A-\mathfrak{p}} A_h$,

$$\begin{aligned} \widetilde{M}(U) =& \{(\varphi_{[\mathfrak{p}]} \in \widetilde{M}_{[\mathfrak{p}]})_{[\mathfrak{p}]\in U} : \\ & \forall [\mathfrak{p}] \in U, \exists D(f) \text{ with } [\mathfrak{p}] \in D(f) \subseteq U \text{ and } s \in M_f \text{ s.t. } s_{[\mathfrak{q}]} = \varphi_{[\mathfrak{q}]}, \forall [\mathfrak{q}] \in D(f)\}, \end{aligned} \tag{5.7}$$

where $\widetilde{M}_{[\mathfrak{p}]} = \varinjlim_{D(h)\ni[\mathfrak{p}]} \widetilde{M}(D(h)) = \varinjlim_{h\in A-\mathfrak{p}} M_h$.

Proposition 5.1.2

Suppose $\mathfrak{p}$ is a prime ideal of A. Then there is a canonical isomorphism

$$\widetilde{M}_{[\mathfrak{p}]} \overset{\sim}{\longleftrightarrow} M_{\mathfrak{p}}$$

between the stalk of $\widetilde{M}$ and the localization of M. The stalk of $\mathscr{O}_{\mathrm{Spec}\,A}$ at the point $[\mathfrak{p}]$ is the local ring $A_{\mathfrak{p}}$.

Proof In fact, $\widetilde{M}_{[\mathfrak{p}]} = \varinjlim_{D(h)\ni[\mathfrak{p}]} \widetilde{M}(D(h))$. Define $\theta : \varinjlim_{D(h)\ni[\mathfrak{p}]} \widetilde{M}(D(h)) \to M_{\mathfrak{p}}$ by setting

$$\overline{(m/f^k, D(f))} \mapsto m/f^k.$$

We shall to check that θ is well-defined $A_{\mathfrak{p}}$-map. If $\overline{(m/f^k, D(f))} = \overline{(n/g^l, D(g))}$, then $m/f^k = n/g^l$ on some $([\mathfrak{p}] \in)D(h) \subseteq D(f) \cap D(g)$. Since $\widetilde{M}(D(h)) \cong M_h$, $h^t(mg^l - nf^k) = 0$. Note that $h \notin \mathfrak{p}$, $h^t \notin \mathfrak{p}$, we have $m/f^k = n/g^i$ in $M_{\mathfrak{p}}$. Hence, θ is well-defined $A_{\mathfrak{p}}$-module homomorphism.

(a) Surjective.

Let $m/f \in M_{\mathfrak{p}}$, then $f \notin \mathfrak{p}$. Note that $\theta(\overline{(m/f, D(f))}) = m/f$, θ is surjective.

(b) Injective.

Let $m/f^k \in M_{\mathfrak{p}}$ such that $m/f^k = 0$. Note that $m/f^k \in \widetilde{M}(D(f))$, hence $\overline{(m/f^k, D(f))} = 0$. Thus θ is injective.

Hecne, we have the canonical isomorphism,

$$\widetilde{M}_{[\mathfrak{p}]} \xleftrightarrow{\sim} M_{\mathfrak{p}}.$$

Then we done.

In particular, $\mathscr{O}_{\operatorname{Spec} A,[\mathfrak{p}]}$ is $\mathscr{O}_{\operatorname{Spec} A,[\mathfrak{p}]}$-module itself, hence, we have isomorphism,

$$\mathscr{O}_{\operatorname{Spec} A,[\mathfrak{p}]} \xleftrightarrow{\sim} A_{\mathfrak{p}}.$$

$\square$

The following exercise shows how an important result can be understood quite differently from algebraic and geometric perspective.

Exercise 5.1 Suppose M is an A-module. Show that the natural map

$$M \hookrightarrow \prod_{\mathfrak{p} \in \operatorname{Spec} A} M_{\mathfrak{p}}$$

is an injection in two ways:

(a) by considering the kernel of the map, and show that any element of it must be 0.

(b) by applying Proposition 3.4.1 (sections of a sheaf are determined by germs) to the sheaf $\widetilde{M}$.

Thus an A-module is zero if and only if all its localizations at prime ideals are zero, and we see this as a simultaneously (a) algebraic and (b) geometric fact.

Proof

(a) Denote $\iota : M \to \prod_{\mathfrak{p} \in \operatorname{Spec} A} M_{\mathfrak{p}}$. Let $m \in \operatorname{Ker} \iota$, then $\iota(m) = (m, m, \dots) = (0, 0, \dots)$, where each $m \in M_{\mathfrak{p}}$. If $m \neq 0$, then $\operatorname{Ann}(m)$ is a proper ideal of A. By [AM18, Corollary 1.4], there exists a maximal ideal of A containing $\operatorname{Ann}(m)$, say $\mathfrak{m}$. Since $m = 0$ in $M_{\mathfrak{m}}$, there exists $h \in A - \mathfrak{m}$ such that $hm = 0$ in A, i.e., $h \in \operatorname{Ann}(m)$ but $h \notin \mathfrak{m}$, a contradiction. Hence, $m = 0$. It follows that $\operatorname{Ker} \iota = \{0\}$, and therefore $M \hookrightarrow \prod_{\mathfrak{p} \in \operatorname{Spec} A} M_{\mathfrak{p}}$.

(b) Note that $M = M_1 = \widetilde{M}(\operatorname{Spec} A) = \widetilde{M}(D(1))$ and $\prod_{\mathfrak{p} \in \operatorname{Spec} A} M_{\mathfrak{p}} = \prod_{\mathfrak{p} \in \operatorname{Spec} A} \widetilde{M}_{[\mathfrak{p}]}$, consider

$$\widetilde{M}(\operatorname{Spec} A) \longrightarrow \prod_{\mathfrak{p} \in \operatorname{Spec} A} \widetilde{M}_{[\mathfrak{p}]}.$$

Apply Proposition 3.4.1, we done.

$\square$

We write above exercise as proposition and corollary:

Proposition 5.1.3

Suppose M is an A-module. The natural map

$$M \hookrightarrow \prod_{\mathfrak{p} \in \operatorname{Spec} A} M_{\mathfrak{p}}$$

is an injection.

Corollary 5.1.1

A-module is zero if and only if all its localizations at prime ideals are zero.

Proposition 5.1.4

There is a bijection:

$$\{\text{maps of } A\text{-modules } M \to N\} \longleftrightarrow \left\{\text{maps of } \mathscr{O}_{\operatorname{Spec} A}\text{-modules } \widetilde{M} \to \widetilde{N}\right\}.$$

Fancy translation: $\mathbf{Mod}_A$ *as a full subcategory of* $\mathbf{Mod}_{\mathscr{O}_{\operatorname{Spec} A}}$.

Proof Let $\varphi : M \to N$ be a maps of A-modules. By Theorem 3.5.2, it suffice to consider the morphism of sheaves on distinguished base. Define $\widetilde{\varphi} : \widetilde{M} \to \widetilde{N}$ as follow: let $D(f)$ be distinguished open subset, define $\widetilde{\varphi}_f : M_f \to N_f$ by setting

$$m/f^k \mapsto \varphi(m)/f^k.$$

Clearly, it is a well-defined A_f-map. We need to check $\widetilde{\varphi}$ is a morphism of sheaves on distinguished base. Let $D(f) \supseteq D(g)$, consider following diagram.

$$\begin{tikzcd} M_f \arrow[r, "\widetilde{\varphi}_f"] \arrow[d, hook] & N_f \arrow[d, hook] \\ M_g \arrow[r, "\widetilde{\varphi}_g"] & N_g. \end{tikzcd}$$

(Since $D(f) \subseteq D(g)$, by Proposition 4.5.5, f is an invertible element of A_g, hence, we get the embedding $M_f \hookrightarrow M_g$ and $N_f \hookrightarrow N_g$)

Let $m/f^k \in M_f$, then

$$\widetilde{\varphi}_g(m/f^k) = \varphi(m)/f^k = \widetilde{\varphi}_f(m/f^k).$$

It follows that above diagram commutes, and therefore $\widetilde{\varphi}$ is a morphism of sheaves on distinguished base. By Theorem 3.5.2, $\widetilde{\varphi}$ can extend to a morphism of sheaves. Note that $\widetilde{\varphi}_f$ is $\mathscr{O}_{\operatorname{Spec} A}(D(f))$-module homomorphism, $\widetilde{\varphi}$ is an $\mathscr{O}_{\operatorname{Spec} A}$-module homomorphism over distinguished base, by Theorem 3.5.2, extended sheaves morphism is an $\mathscr{O}_{\operatorname{Spec} A}$-module homomorphism. We also denote $\widetilde{\varphi}$ as extended sheaves morphism.

Conversely, let $\widetilde{\varphi} : \widetilde{M} \to \widetilde{N}$ be a map of $\mathscr{O}_{\operatorname{Spec} A}$-modules. Consider $\widetilde{\varphi}_1 : \widetilde{M}(\operatorname{Spec} A) \to \widetilde{N}(\operatorname{Spec} A)$, note that $\widetilde{M}(\operatorname{Spec} A) = M$, $\widetilde{N}(\operatorname{Spec} A) = N$, and $\mathscr{O}_{\operatorname{Spec} A}(\operatorname{Spec} A) = A$, we have A-module homomorphism $\widetilde{\varphi}_1 : M \to N$, define $\varphi = \widetilde{\varphi}_1$.

Define

$$\tau : \{\text{maps of } A\text{-modules } M \to N\} \to \left\{\text{maps of } \mathscr{O}_{\operatorname{Spec} A}\text{-modules } \widetilde{M} \to \widetilde{N}\right\}$$

by setting

$$\varphi \mapsto \widetilde{\varphi}$$

and define inverse

$$\theta : \left\{\text{maps of } \mathscr{O}_{\operatorname{Spec} A}\text{-modules } \widetilde{M} \to \widetilde{N}\right\} \to \{\text{maps of } A\text{-modules } M \to N\}$$

by setting

$$\widetilde{\varphi} \mapsto \widetilde{\varphi}_{D(1)}.$$

Note that

$$\theta \circ \tau(\varphi) = \theta(\widetilde{\varphi}) = \widetilde{\varphi}_{D(1)} = \varphi$$

and

$$\tau \circ \theta(\widetilde{\varphi}) = \tau(\widetilde{\varphi}_{D(1)}) = \widetilde{\varphi},$$

there is a bijection:

$$\{\text{maps of } A\text{-modules } M \to N\} \longleftrightarrow \left\{\text{maps of } \mathscr{O}_{\operatorname{Spec} A}\text{-modules } \widetilde{M} \to \widetilde{N}\right\}.$$

□

Corollary 5.1.2

Let A be a ring and let $X = \operatorname{Spec} A$. The functor $\widetilde{\cdot} : M \to \widetilde{M}$ gives an exact, fully faithful functor from the category of A-modules to the category of $\mathscr{O}_X$-module.

Proof Consider the exact sequence of A-modules,

$$0 \longrightarrow M \longrightarrow N \longrightarrow K \longrightarrow 0,$$

by Proposition 5.1.4, we have sequence,

$$0 \longrightarrow \widetilde{M} \longrightarrow \widetilde{N} \longrightarrow \widetilde{K} \longrightarrow 0. \tag{5.8}$$

To show that sequence (5.9) is exact, by Proposition 3.6.5, it suffices to show the exactness at the level of stalks. Let $p \in X$, then exists affine open subset $\operatorname{Spec} A_i \hookrightarrow X$ such that $p \in \operatorname{Spec} A$. By Proposition 3.6.5, we have exact sequence

$$0 \longrightarrow M_p \longrightarrow N_p \longrightarrow K_p \longrightarrow 0.$$

By Proposition 5.1.2, we have $M_p = \widetilde{M}_p$, $N_p = \widetilde{N}_p$, and $K_p = \widetilde{K}_p$, hence sequence

$$0 \longrightarrow \widetilde{M}_p \longrightarrow \widetilde{M}_p \longrightarrow \widetilde{M}_p \longrightarrow 0$$

is exact. By Proposition 3.6.5, sequence (5.9) is exact.

To say $\widetilde{\cdot}$ is fully faithful means that for any A-modules M and N, we have $\operatorname{Hom}_A(M, N) = \operatorname{Hom}_{\mathscr{O}_X}(\widetilde{M}, \widetilde{N})$. By Proposition 5.1.4, it is clear. □

Definition 5.1.5 (Support of a module)

Motivated by Proposition 5.1.2, and the notion of support of a section of a sheaf (Definition 3.7.4), define the ***support of*** *$m \in M$ by*

$$\operatorname{Supp} m := \left\{[\mathfrak{p}] \in \operatorname{Spec} A : m_{\mathfrak{p}} \neq 0 \text{ in } \widetilde{M}_{[\mathfrak{p}]}\right\} \subseteq \operatorname{Spec} A,$$

which is the support of m considered as a section of $\widetilde{M}$.

Similarly (following Definition 3.7.5), define the ***support of*** *M by*

$$\operatorname{Supp} M := \operatorname{Supp} \widetilde{M} = \{[\mathfrak{p}] \in \operatorname{Spec} A : M_{\mathfrak{p}} \neq 0\} \subseteq \operatorname{Spec} A.$$

Remark 5.5 These notions will come up repeatedly. We will discuss support in more detail in Chapter 7.

Proposition 5.1.5

$\operatorname{Supp} m$ *is a closed subset of* $\operatorname{Spec} A$.

Proof By Proposition 3.7.5 and Proposition 5.1.2, clearly. □

5.1.3 Recurring counterexample

The example of two planes meeting at a point (§5.1.1) will appear many times as an example or a counterexample. It is important to have such examples (or counterexamples) at the back of your mind, as "boundary markers" reminding you of what is reasonable, and what to watch out for when you hear a new statement. The appearances of the following are listed in the index (under "recurring (counter)examples"), so you can see how often they come up, and where. You do not have the language to understand many of these yet, but you can come back to this later.

- two planes meeting at a point
- affine space minus the origin, and its inclusion in affine space
- affine space with doubled origin
- (the cone over) the quadric surface, $A = k[w,x,y,z]/(wz-xy)$
- an embedded point on a line, $k[x,y]/(y^2, xy)$
- x^2 in the projective plane
- infinite disjoint unions of schemes (especially $\coprod \operatorname{Spec} k[x]/(x^n)$)
- the morphism $\operatorname{Spec} \overline{\mathbb{Q}} \to \operatorname{Spec} \mathbb{Q}$
- $k(x) \otimes_k k(y)$
- $\prod^\infty \mathbb{F}_2$

5.2 Visualizing schemes: Nilpotents

In §4.3, we discussed how to visualize the underlying set of schemes, adding in generic points to our previous intuition of "classical" (or closed) points. Our later discussion of the Zariski Topology fit well with that picture. In our definition of the "affine scheme" $(\operatorname{Spec} A, \mathscr{O}_{\operatorname{Spec} A})$, we have the additional information of nilpotents, which are invisible on the level of points (§4.2.4), so now we figure out how to picture them. We will then readily be able to glue them together to picture schemes in general, once we have made the appropriate definitions. As we are building intuition, we cannot be rigorous of precise.

As motivation, note that we have inclusion-reversing bijections

$$\text{maximal ideals of } A \longleftrightarrow \text{closed points of } \operatorname{Spec} A \qquad \text{(Proposition 4.6.6)}$$

$$\text{prime ideals of } A \longleftrightarrow \text{irreducible closed subsets of } \operatorname{Spec} A \qquad \text{(Proposition 4.7.3)}$$

$$\text{radical ideals of } A \longleftrightarrow \text{closed subsets of } \operatorname{Spec} A \qquad \text{(Theorem 4.7.1)}$$

If we take the things on the right as "pictures", our goal is to figure out how to picture ideals that are not radical:

$$\text{ideals of } A \longleftrightarrow \ ???$$

(We will later fill this in rigorously in a different way with the notion of a closed subscheme, the scheme-theoretic version of closed subsets. But our goal now is create a picture.)

As motivation, when we see the expression $\operatorname{Spec} \mathbb{C}[x]/(x(x-1)(x-2)) = \{[(x)], [(x-1)], [(x-2)]\}$, we immediately interpret it as a closed subset of $\mathbb{A}^1_{\mathbb{C}}$, namely $\{0, 1, 2\}$. In particular, the map $\mathbb{C}[x] \to \mathbb{C}[x]/(x(x-1)(x-2))$ can be interpreted (via the Chinese Remainder Theorem, see [AM18, Proposition 1.10]) as: take a function on $\mathbb{A}^1_{\mathbb{C}}$, and restrict it to the three points 0, 1, and 2.

This will guide us in how to visualize a non-radical ideal. The simplest example to consider is $\operatorname{Spec} \mathbb{C}[x]/(x^2)$ (Exercise 4.1 (a)). As a subset of $\mathbb{A}^1_{\mathbb{C}}$, it is just the origin $0 = [(x)]$, which we are used to thinking of as $\operatorname{Spec} \mathbb{C}[x]/(x)$ (i.e., corresponding to the ideal (x), not (x^2)). We want to enrich this picture in some way. We should picture $\mathbb{C}[x]/(x^2)$ in terms of the information the quotient remembers. The image of a polynomial $f(x)$ is the information of its value at 0, and its derivative (Exercise 4.13). We thus picture this as being the point, plus a little bit more — a little bit of infinitesimal "fuzz" on the point (see Figure 5.2). The sequence of restrictions $\mathbb{C}[x] \to \mathbb{C}[x]/(x^2) \to \mathbb{C}[x]/(x)$ should be interpreted as nested pictures.

$$\begin{array}{ccccc} \mathbb{C}[x] & \twoheadrightarrow & \mathbb{C}[x]/(x^2) & \twoheadrightarrow & \mathbb{C}[x]/(x) \\ \\ f(x) & & \longmapsto & & f(0), \end{array}$$

Similarly, $\mathbb{C}[x]/(x^3)$ remembers even more information — the second derivative as well. Thus we picture this as the point 0 with even more fuzz.

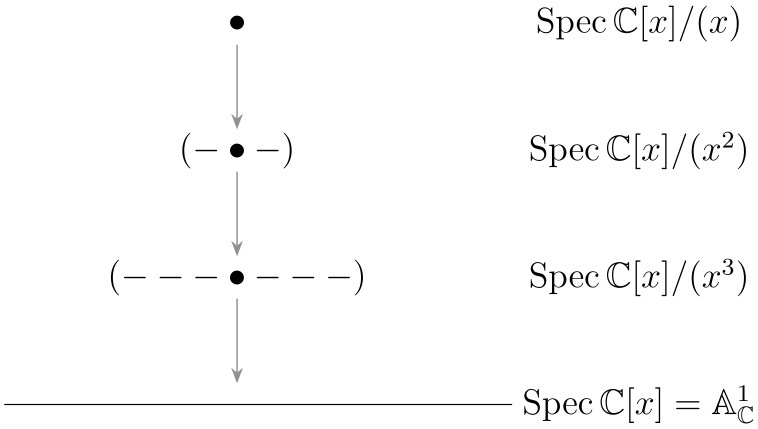


Figure 5.2: Picturing quotients of $\mathbb{C}[x]$

More subtleties arise in two dimensions (see Figure 5.3). Consider

$$\operatorname{Spec} \mathbb{C}[x,y]/(x,y)^2,$$

which is sandwiched between two rings we know well:

$$\begin{array}{ccccc} \mathbb{C}[x,y] & \twoheadrightarrow & \mathbb{C}[x,y]/(x,y)^2 & \twoheadrightarrow & \mathbb{C}[x,y]/(x,y) \\ \\ f(x,y) & & \longmapsto & & f(0,0). \end{array}$$

Again, taking the quotient by $(x,y)^2$ remembers the first derivative, "in all directions". We picture this as fuzz around the point, in the shape of a circle (no direction is privileged). Similarly, $(x,y)^3$ remembers the second derivative "in all directions" — bigger circular fuzz.

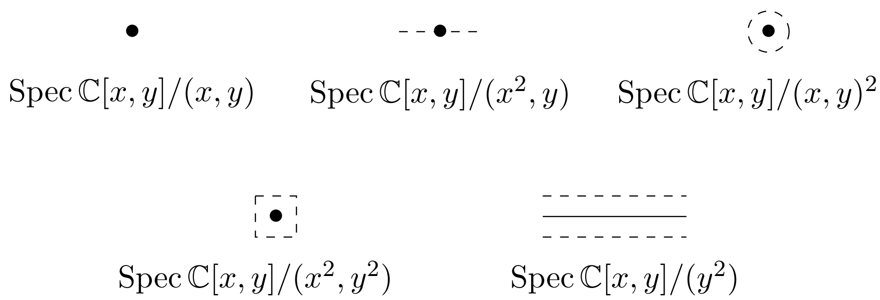


Figure 5.3: Picturing quotients of $\mathbb{C}[x,y]$

Consider instead the ideal (x^2, y). What it remembers is the derivative only in the x direction — given a polynomial, we remember its value at 0, and the coefficient of x. We remember this by the fuzz only in the x direction.

This gives us some handle on picturing more things of this sort, but now it becomes more an art than a science. For example, $\operatorname{Spec}\mathbb{C}[x,y]/(x^2,y^2)$ we might picture as a fuzzy square around the origin. Since $(x,y)^3 \subseteq (x^2,y^2) \subseteq (x,y)^2$, we have $\mathbb{C}[x,y]/(x,y)^3 \supseteq \mathbb{C}[x,y]/(x^2,y^2) \supseteq \mathbb{C}[x,y]/(x,y)^2$, hence, this square is circumscribed by the circular fuzz $\operatorname{Spec}\mathbb{C}[x,y]/(x,y)^3$, and inscribed by the circular fuzz $\operatorname{Spec}\mathbb{C}[x,y]/(x,y)^2$. One feature of this example is that given two ideals I and J of a ring A (such as $\mathbb{C}[x,y]$), your fuzzy picture of $\operatorname{Spec} A/(I,J)$ should be the "intersection" of picture of $\operatorname{Spec} A/I$ and $\operatorname{Spec} A/J$ in $\operatorname{Spec} A$. For example, $\operatorname{Spec}\mathbb{C}[x,y]/(x^2,y^2)$ should be the intersection of two thickened lines.

Example 5.1

(i) $\operatorname{Spec}\mathbb{C}[x,y]/(x^5,y^3) = \operatorname{Spec}\mathbb{C}[x,y]/(x^5) \cap \operatorname{Spec}\mathbb{C}[x,y]/(y^3)$.

(ii)
$$\begin{aligned}&\operatorname{Spec}\mathbb{C}[x,y]/(x^3,y^4,z^5,(x+y+z)^2)\\ =&\operatorname{Spec}\mathbb{C}[x,y]/(x^3) \cap \operatorname{Spec}\mathbb{C}[x,y]/(y^4) \cap \operatorname{Spec}\mathbb{C}[x,y]/(z^5) \cap \operatorname{Spec}\mathbb{C}[x,y]/((x+y+z)^2)\end{aligned}$$

(iii) $\operatorname{Spec}\mathbb{C}[x,y]/((x,y)^5,y^3) = \operatorname{Spec}\mathbb{C}[x,y]/(x,y)^5 \cap \operatorname{Spec}\mathbb{C}[x,y]/(y^3)$.

One final example that will motivate us in Chapter 7 is $\operatorname{Spec}\mathbb{C}[x,y]/(y^2,xy)$. Knowing what a polynomial in $\mathbb{C}[x,y]$ is modulo (y^2,xy) is the same as knowing its value on the x-axis, as well as first-order differential information around the origin (see Figure 5.4).

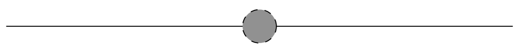


Figure 5.4: A picture of the scheme $\operatorname{Spec} k[x,y]/(y^2,xy)$. The fuzz at the origin indicates where "there is nonreducedness".

Our picture capture useful information that you already have some intuition for. For example, consider the intersection of the parabola $y = x^2$ and the x-axis (in the xy-plane), see Figure 5.5. You already have a sense that the intersection has multiplicity two. In terms of this visualization, we interpret this as intersecting (in $\operatorname{Spec}\mathbb{C}[x,y]$):

$$\begin{aligned}\operatorname{Spec}\mathbb{C}[x,y]/(y-x^2) \cap \operatorname{Spec}\mathbb{C}[x,y]/(y) &= \operatorname{Spec}\mathbb{C}[x,y]/(y-x^2,y)\\ &= \operatorname{Spec}\mathbb{C}[x,y]/(y,x^2)\\ &= \operatorname{Spec}\mathbb{C}[x]/(x^2)\end{aligned}$$

which we interpret as the fact that the parabola and line not just meet with multiplicity two, but that the "multiplicity 2" part is in the direction of the x-axis. We will make this example precise in Chapter 10.

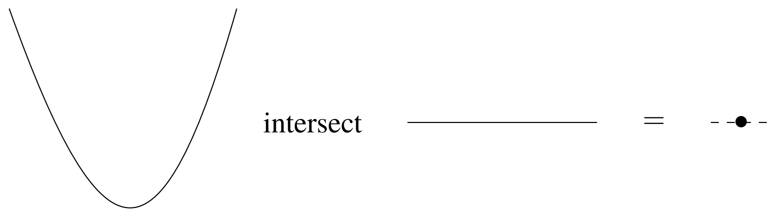


Figure 5.5: The "scheme-theoretic" intersection of the parabola $y = x^2$ and the x-axis is a nonreduced scheme (with fuzz in the x-direction)

We will later make the location of the fuzz more precise when we discuss associated points. We will see that in reasonable circumstances, the fuzz is concentrated on closed subsets.

5.3 Definition of schemes

5.3.1 Schemes

We can now define **scheme** in general.

Definition 5.3.1 (Isomorphism of ringed spaces)

*Define an **isomorphism of ringed spaces** $\varphi = (\pi, \pi^\sharp) : (X, \mathscr{O}_X) \to (Y, \mathscr{O}_Y)$ as*

(i) a homeomorphism $\pi : X \to Y$

(ii) an isomorphism of sheaves $\mathscr{O}_X$ and $\mathscr{O}_Y$, considered to be on the same space via π. More precisely, is an isomorphism

$$\pi^\sharp : \mathscr{O}_Y \xrightarrow{\sim} \pi_* \mathscr{O}_X$$

of sheaves on Y, or equivalently, an isomorphism

$$\pi^{-1} \mathscr{O}_Y \xrightarrow{\sim} \mathscr{O}_X$$

of sheaves on X.

Remark 5.6 In other words, we have a "correspondence" of sets, topologies, and structure sheaves. Every isomorphism $\alpha : (X, \mathscr{O}_X) \to (Y, \mathscr{O}_Y)$ clearly has an "inverse isomorphism" $\alpha^{-1} : (Y, \mathscr{O}_Y) \to (X, \mathscr{O}_X)$.

Definition 5.3.2 (Affine scheme, Scheme, and Zariski topology)

*An **Affine scheme** is a ringed space that is isomorphic to $(\operatorname{Spec} A, \mathscr{O}_{\operatorname{Spec} A})$ for some A. A **scheme** $(X, \mathscr{O}_X)$ is a ringed space such that any point of X has an open neighborhood U such that $(U, \mathscr{O}_X|_U)$ is an affine scheme. The topology on a scheme is called the **Zariski topology**. The scheme can be denoted $(X, \mathscr{O}_X)$, although it is often denoted X, with the structure sheaf $\mathscr{O}_X$ left implicit.*

Definition 5.3.3 (An isomorphism of schemes)

*An **isomorphism of schemes** $(X, \mathscr{O}_X) \to (Y, \mathscr{O}_Y)$ is an isomorphism as ringed spaces.*

Definition 5.3.4 (Functions)

*Let $(X, \mathscr{O}_X)$ be a scheme, if $U \subseteq X$ is an open subset, then the elements of $\Gamma(U, \mathscr{O}_X) = \mathscr{O}_X(U)$ are said to be the **functions on** U.*

Remark 5.7 From the definition of the structure sheaf on an affine scheme, several things are clear. First of all, if we are told that $(X, \mathscr{O}_X)$ is an affine scheme, we may recover its ring (i.e., find the ring A such that $\operatorname{Spec} A = X$) by taking the ring of global sections, as $X = D(1)$, so:

$$\begin{aligned}\Gamma(X, \mathscr{O}_X) =& \Gamma(D(1), \mathscr{O}_{\operatorname{Spec} A}) \quad \text{as } D(1) = \operatorname{Spec} A \\ =& A.\end{aligned}$$

You can verify that we get more, and can "recognize X as the scheme $\operatorname{Spec} A$": we get an isomorphism $\pi : (\operatorname{Spec} \Gamma(X, \mathscr{O}_X), \mathscr{O}_{\operatorname{Spec} \Gamma(X, \mathscr{O}_X)}) \xrightarrow{\sim} (X, \mathscr{O}_X)$. For example, if $\mathfrak{m}$ is a maximal ideal of $\Gamma(X, \mathscr{O}_X)$, then $\{\pi([\mathfrak{m}])\} = V(\mathfrak{m})$. The following propositions will make these ideas rigorous — they are subtler than they first appear.

Remark 5.8 **Caution.** It is not part of the definition that the overlap of two affine open subsets is also affine, although this is true in most cases of interest.

Proposition 5.3.1

There is a bijection between the isomorphisms $\operatorname{Spec} A \xrightarrow{\sim} \operatorname{Spec} A'$ *(of ringed spaces) and the ring isomorphisms* $A' \xrightarrow{\sim} A$*, i.e.,*

$$\left\{\textit{isomorphisms } \operatorname{Spec} A \xrightarrow{\sim} \operatorname{Spec} A'\right\} \longleftrightarrow \left\{\textit{the ring isomorphisms } A' \xrightarrow{\sim} A\right\}.$$

Proof Let $\varphi : A' \to A$ be a ring isomorphism. Define $\widetilde{\varphi} : \operatorname{Spec} A \to \operatorname{Spec} A'$ by setting $\widetilde{\varphi}(\mathfrak{p}) = \varphi^{-1}(\mathfrak{p})$. Clearly, it is a well-defined. $\widetilde{\varphi}^{-1} : \operatorname{Spec} A' \to \operatorname{Spec} A$ is given by $\widetilde{\varphi}^{-1}(\mathfrak{p}) = \varphi(\mathfrak{p})$. Since, φ is isomorphism of rings, it is easy to see that $\widetilde{\varphi}^{-1}$ is, indeed, the inverse of $\widetilde{\varphi}$.

First, we want to check this map is a homeomorphism. Let $V(I') = \operatorname{Spec} A'/I'$ be any closed subset of $\operatorname{Spec} A'$, then

$$\begin{aligned}\widetilde{\varphi}^{-1}(\operatorname{Spec} A'/I') =& \widetilde{\varphi}^{-1}\left(\left\{\mathfrak{p} \in \operatorname{Spec} A' : \mathfrak{p} \supseteq I'\right\}\right)\\ =&\{\varphi(\mathfrak{p}) \in \operatorname{Spec} A : \varphi(\mathfrak{p}) \supseteq \varphi(I')\}\\ =&\operatorname{Spec} A/\varphi(I'),\end{aligned}$$

where the last equal is given by φ is isomorphism, hence, $\widetilde{\varphi}^{-1}$ maps closed subset to closed subset, and therefore $\widetilde{\varphi}$ is continuous.

Conversely, let $V(I) = \operatorname{Spec} A/I$ be any close subset of $\operatorname{Spec} A$, then

$$\begin{aligned}\widetilde{\varphi}(\operatorname{Spec} A/I) =& \widetilde{\varphi}\left(\left\{\mathfrak{p} \in \operatorname{Spec} A' : \mathfrak{p} \supseteq I\right\}\right)\\ =&\left\{\varphi^{-1}(\mathfrak{p}) \in \operatorname{Spec} A' : \varphi^{-1}(\mathfrak{p}) \supseteq \varphi^{-1}(I)\right\}\\ =&\operatorname{Spec} A'/\varphi^{-1}(I)\end{aligned}$$

where the last equal is given by φ is isomorphism, hence, $\widetilde{\varphi}$ maps closed subset to closed subset, and therefore $\widetilde{\varphi}^{-1}$ is continuous. Hence, $\widetilde{\varphi}$ is a homeomorphism.

Next, we want to check that $\mathscr{O}_{\operatorname{Spec} A'} \xrightarrow{\sim} \widetilde{\varphi}_*\mathscr{O}_{\operatorname{Spec} A}$. It suffices to check this on the distinguished base. Define $\widetilde{\varphi}^\sharp : \mathscr{O}_{\operatorname{Spec} A'} \to \widetilde{\varphi}_*\mathscr{O}_{\operatorname{Spec} A}$ by setting $\widetilde{\varphi}^\sharp_{D(f)} : \mathscr{O}_{\operatorname{Spec} A'}(D(f)) \to \widetilde{\varphi}_*\mathscr{O}_{\operatorname{Spec} A}(D(f))$ as follow: since φ is isomorphism, we have

$$\begin{aligned}\widetilde{\varphi}^{-1}(D(f)) =& \widetilde{\varphi}^{-1}(\{\mathfrak{p} \in \operatorname{Spec} A' : f \notin \mathfrak{p}\})\\ =&\{\varphi(\mathfrak{p}) \in \operatorname{Spec} A : \varphi(f) \notin \varphi(\mathfrak{p})\}\\ =&D(\varphi(f)),\end{aligned}$$

hence, $\widetilde{\varphi}_*\mathscr{O}_{\operatorname{Spec} A}(D(f)) = \mathscr{O}_{\operatorname{Spec} A}(\widetilde{\varphi}^{-1}(D(f))) = \mathscr{O}_{\operatorname{Spec} A}(D(\varphi(f))) = A_{\varphi(f)}$, define $\widetilde{\varphi}^\sharp_{D(f)} : A'_f \to A_{\varphi(f)}$ with $\widetilde{\varphi}^\sharp_{D(f)}(a'/f^n) = \varphi(a')/\varphi(f)^n$. Clearly, it is a well-defined. Let $D(f) \supseteq D(g)$ in $\operatorname{Spec} A'$, consider the following diagram.

$$\begin{tikzcd} A'_f \arrow[r, "\widetilde{\varphi}^\sharp_{D(f)}"] \arrow[d, "\operatorname{res}^{\operatorname{Spec} A'}_{D(f),D(g)}"'] & A_{\varphi(f)} \arrow[d, "\operatorname{res}^{\operatorname{Spec} A}_{D(\varphi(f)),D(\varphi(g))}"] \\ A'_g \arrow[r, "\widetilde{\varphi}^\sharp_{D(g)}"] & A_{\varphi(g)} \end{tikzcd}$$

We want to check above diagram commutes. Let $a'/f^n \in A'_f$, then

$$\widetilde{\varphi}^\sharp_{D(g)} \circ \operatorname{res}^{\operatorname{Spec} A'}_{D(f),D(g)}(a'/f^n) = \widetilde{\varphi}^\sharp_{D(g)}(a'/f^n) = \varphi(a')/\varphi(f)^n$$

and

$$\operatorname{res}^{\operatorname{Spec} A}_{D(\varphi(f)),D(\varphi(g))} \circ \widetilde{\varphi}^\sharp_{D(f)}(a'/f^n) = \operatorname{res}^{\operatorname{Spec} A}_{D(\varphi(f)),D(\varphi(g))}(\varphi(a')/(\varphi(f))^n) = \varphi(a')/(\varphi(f))^n.$$

Hence, $\widetilde{\varphi}^\sharp_{D(g)} \circ \operatorname{res}^{\operatorname{Spec} A'}_{D(f),D(g)} = \operatorname{res}^{\operatorname{Spec} A}_{D(\varphi(f)),D(\varphi(g))} \circ \widetilde{\varphi}^\sharp_{D(f)}$, and therefore $\widetilde{\varphi}^\sharp$ is a morphism of sheaves. Since φ is an isomorphism, $\widetilde{\varphi}^\sharp_{D(f)}$ is isomorphism for all distinguished open subset $D(f)$, hence, $\widetilde{\varphi}^\sharp$ is an isomorphism of sheaves. Thus, $\varphi^\dagger := (\widetilde{\varphi}, \widetilde{\varphi}^\sharp)$ is an isomorphism of ringed spaces.

Let $\psi = (\pi, \pi^\sharp) : \operatorname{Spec} A \to \operatorname{Spec} A'$ be an isomorphism of ringed space. Then we have an isomorphism of sheaves,

$$\mathscr{O}_{\operatorname{Spec} A'} \xrightarrow{\sim} \psi_* \mathscr{O}_{\operatorname{Spec} A}.$$

Consider $D(1) \subseteq \operatorname{Spec} A'$, then we have an isomorphism of rings

$$\mathscr{O}_{\operatorname{Spec} A'}(D(1)) = A' \xrightarrow{\sim} \psi_* \mathscr{O}_{\operatorname{Spec} A}(D(1)).$$

In fact, $\psi_* \mathscr{O}_{\operatorname{Spec} A}(D(1)) = \mathscr{O}_{\operatorname{Spec} A}(\psi^{-1}(D(1)))$. Since ψ is also a homeomorphism between $\operatorname{Spec} A$ and $\operatorname{Spec} A'$, we have $\psi^{-1}(D(1)) = \psi^{-1}(\operatorname{Spec} A') = \operatorname{Spec} A$, and therefore $\psi_* \mathscr{O}_{\operatorname{Spec} A}(D(1)) = A$. Hence, there is an isomorphism of rings

$$\pi^\sharp_{D(1)} : A' \to A.$$

Denote $\psi^\natural = \pi^\sharp_{D(1)}$.

At last, we want to check

$$\mathscr{A} = \left\{\text{isomorphisms } \operatorname{Spec} A \xrightarrow{\sim} \operatorname{Spec} A'\right\} \longrightarrow \left\{\text{the ring isomorphisms } A' \xrightarrow{\sim} A\right\} = \mathscr{B}.$$

is a bijection.

Let $\varphi = (\pi, \pi^\sharp) \in \mathscr{A}$, by above discussion, φ induces an isomorphisms of rings $\varphi^\natural : A' \to A$, where $\varphi^\natural = \pi^\sharp_{D(1)}$, and $\varphi^\natural$ induces an isomorphism of ringed space $\varphi^{\natural\dagger} = (\eta, \eta^\sharp) : \operatorname{Spec} A \to \operatorname{Spec} A'$, where $\eta : \operatorname{Spec} A \to \operatorname{Spec} A'$ induced by $\varphi^\natural$ and $\eta^\sharp : \mathscr{O}_{\operatorname{Spec} A'} \to \eta_* \mathscr{O}_{\operatorname{Spec} A}$. We want to show that $\varphi = \varphi^{\natural\dagger}$ as isomorphism of ringed space.

(i) On the level of point.

Let $[\mathfrak{p}] \in \operatorname{Spec} A$, then

$$\varphi^{\natural\dagger}([\mathfrak{p}]) = \eta([\mathfrak{p}]) = (\varphi^\natural)^{-1}([\mathfrak{p}]) = (\pi^\sharp_{D(1)})^{-1}([\mathfrak{p}]) = \pi([\mathfrak{p}]) = \varphi([\mathfrak{p}]),$$

hence, on the level of point $\varphi = \varphi^{\natural\dagger}$.

(ii) As maps of topological space.

It is clear, since φ and $\varphi^{\natural\dagger}$ are homeomorphisms and they are the same on the level of point, hence, $\varphi = \varphi^{\natural\dagger}$ as maps of topological space.

(iii) On the level of structure sheaves.

We need to check $\pi^\sharp = \eta^\sharp$. It suffices to check on the distinguished base. Let $D(f) \subseteq \operatorname{Spec} A'$ be any distinguished open subset, then for all $a'/f^n \in \mathscr{O}_{\operatorname{Spec} A'}(D(f)) = A'_f$, we have

$$\eta^\sharp_{D(f)}(a'/f^n) = \varphi^\natural(a')/\varphi^\natural(f)^n = \pi^\sharp_{D(1)}(a')/\pi^\sharp_{D(1)}(f)^n = \pi^\sharp_{D(f)}(a'/f^n),$$

(the last equal from the universal property of localization) i.e., $\eta^\sharp_{D(f)} = \pi^\sharp_{D(f)}$. Hence, $\eta^\sharp = \pi^\sharp$.

Conversely, let $\varphi \in \mathscr{B}$, φ induces an isomorphisms of ringed space $\varphi^\dagger = (\widetilde{\varphi}, \widetilde{\varphi}^\sharp)$, and $\varphi^\dagger$ induces an isomorphisms of rings $\varphi^{\dagger\natural} : A' \to A$, where $\varphi^{\dagger\natural} = \widetilde{\varphi}^\sharp_{D(1)}$. We want to show that $\varphi^{\dagger\natural} = \varphi$. Note that $\varphi^{\dagger\natural} = \widetilde{\varphi}^\sharp_{D(1)} = \varphi$, then we done.

By above discussion, we have a bijection,

$$\left\{\text{isomorphisms } \operatorname{Spec} A \xrightarrow{\sim} \operatorname{Spec} A'\right\} \longleftrightarrow \left\{\text{the ring isomorphisms } A' \xrightarrow{\sim} A\right\}.$$

$\square$

More generally, given $f \in A, \Gamma(D(f), \mathscr{O}_{\operatorname{Spec} A}) \cong A_f$. Thus under the natural inclusion of sets $\operatorname{Spec} A_f \hookrightarrow$

Spec A, the Zariski topology on Spec A restricts to give the Zariski topology on Spec A_f (Proposition 4.4.7), and the structure sheaf of Spec A restricts to the structure sheaf of Spec A_f, as the next proposition shows.

Proposition 5.3.2

Suppose $f \in A$. Then under the identification of $D(f)$ in Spec A with Spec A_f, there is a natural isomorphism of ringed spaces

$$(D(f), \mathscr{O}_{\operatorname{Spec} A}|_{D(f)}) \stackrel{\sim}{\longleftrightarrow} (\operatorname{Spec} A_f, \mathscr{O}_{\operatorname{Spec} A_f}).$$

Proof Let $\mathrm{id} : D(f) \to \operatorname{Spec} A_f$, clearly, id is a homeomorphism.

Let $D_{\operatorname{Spec} A_f}(g) \subseteq \operatorname{Spec} A_f$ be a distinguished open subset, then

$$D_{\operatorname{Spec} A_f}(g) = \{[\mathfrak{p}] \in \operatorname{Spec} A_f : g \notin \mathfrak{p}\} = \{[\mathfrak{p}] \in \operatorname{Spec} A : g \notin \mathfrak{p}, f \notin \mathfrak{p}\} = D_{\operatorname{Spec} A}(fg),$$

hence,

$$\begin{aligned} \mathrm{id}_* \mathscr{O}_{\operatorname{Spec} A}|_{D_{\operatorname{Spec} A}(f)}(D_{\operatorname{Spec} A_f(g)}) =& \mathscr{O}_{\operatorname{Spec} A}(D_{\operatorname{Spec} A}(f) \cap D_{\operatorname{Spec} A}(g)) \\ =& \mathscr{O}_{\operatorname{Spec} A}(D_{\operatorname{Spec} A}(fg)) \\ =& A_{fg}. \end{aligned}$$

On the other hand, we have

$$\mathscr{O}_{\operatorname{Spec} A_f}(D_{\operatorname{Spec} A_f}(g)) = (A_f)_g \cong A_{fg}.$$

It follows that $\mathrm{id}^\sharp : \mathscr{O}_{\operatorname{Spec} A_f} \xrightarrow{\sim} \mathrm{id}_* \mathscr{O}_{\operatorname{Spec} A}|_{D_{\operatorname{Spec} A}(f)}$. Hence,

$$(D(f), \mathscr{O}_{\operatorname{Spec} A}|_{D(f)}) \stackrel{\sim}{\longleftrightarrow} (\operatorname{Spec} A_f, \mathscr{O}_{\operatorname{Spec} A_f}).$$

□

Proposition 5.3.3

If X is a scheme, and U is any open subset, then $(U, \mathscr{O}_X|_U)$ is also a scheme.

Proof It suffices to show that for all point of U has an open neighborhood V such that $(V, (\mathscr{O}_X|_U)|_V)$ is an affine scheme. Let $p \in U$, and $V \ni p$ is an open neighborhood of p in U. Note that $(\mathscr{O}_X|_U)|_V = \mathscr{O}_X|_{U \cap V}$ and $U \cap V$, we have $(V, (\mathscr{O}_X|_U)|_V) = (V, \mathscr{O}_X|_{U\cap V}) = (U \cap V, \mathscr{O}_X|_{U \cap V})$. Since X is a scheme, $(U \cap V, \mathscr{O}_X|_{U\cap V})$ is an affine scheme, and therefore $(V, (\mathscr{O}_X|_U)|_V)$ is an affine scheme. Hence, $(U, \mathscr{O}_X|_U)$ is a scheme. □

Definition 5.3.5 (Open subscheme)

Suppose X is a scheme, and U is any open subset. We say $(U, \mathscr{O}_X|_U)$ is an ***open subscheme of*** *X (Proposition 5.3.3, U is indeed a scheme). If U is also an affine scheme, we often say U is an* ***affine open subset****, or an* ***affine open subscheme****, or sometimes informally just an* ***affine open****.*

Example 5.2 $D(f)$ is an affine open subscheme of Spec A.

Remark 5.9 It is not true that every affine open subscheme of Spec A is of the form $D(f)$.

Proposition 5.3.4

If X is a scheme, then the affine open sets form a base for the Zariski topology.

Proof Let $U \subseteq X$ be any open subset of X, then U is a scheme, by the definition of scheme, for all $p \in U$, exists an open neighborhood $V_p \subseteq U$ such that V_p is an affine scheme. Note that $U = \bigcap\limits_{\substack{\text{open } V_p \ni p \\ p \in U}} V_p$, where each V_p is affine open set. It follows that the affine open sets form a base for the Zariski topology. □

Definition 5.3.6 (The disjoint union of schemes)

The ***disjoint union of schemes*** *is the disjoint union of sets, with the expected topology (thus it is the disjoint union of topological spaces), with the expected sheaf. We use the symbol $X \coprod Y$ for the disjoint union of X and Y (because once we know what morphisms of schemes are, this construction will turn out to be the coproduct in the category of schemes).*

Proposition 5.3.5

(a) *The disjoint union of a finite number of affine schemes is also an affine scheme.*

(b) *(A first example of a non-affine scheme) An infinite disjoint union of (non-empty) affine scheme is not an affine scheme.*

Proof

(a) Let $I = \{1, 2, \dots, n\}$ be a finite set, and $\{X_i\}_{i \in I}$ be a collection of affine schemes. Since each X_i is affine scheme, we may assume $X_i = \operatorname{Spec} A_i$ for all $i \in I$. Consider $\coprod_{i \in I} \operatorname{Spec} A_i$, by Example 4.5 , $\coprod_{i=1}^n \operatorname{Spec} A_i = \operatorname{Spec} \prod_{i=1}^n A_i$ as topological spaces, say $\pi : \coprod_{i=1}^n \operatorname{Spec} A_i \to \operatorname{Spec} \prod_{i=1}^n A_i$
Let $D(\prod_{i=1}^n f_i)$ be a distinguished open subset of $\operatorname{Spec} \prod_{i=1}^n A_i$. The structure of $D(\prod_{i=1}^n f_i)$ is,

$$\begin{aligned}
D\left(\prod_{i=1}^n f_i\right) &= \left\{[\mathfrak{p}] \in \operatorname{Spec} \prod_{i=1}^n A_i : \prod_{i=1}^n f_i \notin \mathfrak{p}\right\} \\
&= \left\{[\mathfrak{p}] \in \operatorname{Spec} \prod_{i=1}^n A_i : \mathfrak{p} = A_1 \times \cdots \times A_{i-1} \times \mathfrak{p}_i \times A_{i+1} \times \cdots \times A_n, f_i \notin \mathfrak{p}_i, \forall i \in I\right\} \\
&= \coprod_{i=1}^n \{A_1 \times \cdots \times A_{i-1} \times \mathfrak{p}_i \times A_{i+1} \times \cdots \times A_n : f_i \notin \mathfrak{p}_i\} \\
&\cong \coprod_{i=1}^n D_{\operatorname{Spec} A_i}(f_i) \subseteq \coprod_{i=1}^n \operatorname{Spec} A_i.
\end{aligned}$$

Hence, $\pi^{-1}(D(\prod_{i=1}^n f_i)) = \coprod_{i=1}^n D_{\operatorname{Spec} A_i}(f_i)$.
Note that

$$\begin{aligned}
\mathscr{O}_{\operatorname{Spec} \prod_{i=1}^n A_i}\left(D\left(\prod_{i=1}^n f_i\right)\right) &= \left(\prod_{i=1}^n A_i\right)_{\prod_{i=1}^n f_i} \\
&\cong \prod_{i=1}^n (A_i)_{f_i} = \prod_{i=1}^n \mathscr{O}_{\operatorname{Spec} A_i}(D_{\operatorname{Spec} A_i}(f_i)) \\
&= \left(\prod_{i=1}^n \mathscr{O}_{\operatorname{Spec} A_i}\right)\left(\coprod_{i=1}^n D_{\operatorname{Spec} A_i}(f_i)\right) \\
&= \left(\prod_{i=1}^n \mathscr{O}_{\operatorname{Spec} A_i}\right)\left(\pi^{-1}\left(D\left(\prod_{i=1}^n f_i\right)\right)\right) \\
&= \left(\pi_* \prod_{i=1}^n \mathscr{O}_{\operatorname{Spec} A_i}\right)\left(D\left(\prod_{i=1}^n f_i\right)\right),
\end{aligned}$$

and it is easy to check the compatibility of the ring isomorphism with the restriction maps, hence, we have an isomorphism of schemes

$$\coprod_{i=1}^n \operatorname{Spec} A_i \xrightarrow{\sim} \operatorname{Spec} \prod_{i=1}^n A_i.$$

It follows that the disjoint union of a finite number of affine schemes is also an affine scheme.

(b) By Proposition 4.6.4, affine scheme is quasi-compact, but infinite disjoint union of nonempty quasi-compact space is clearly not quasi-compact, and therefore infinite disjoint union of affine scheme is not an affine scheme.

□

Remark 5.10 **(A first glimpse of closed subschemes.)** Open subsets of a scheme come with a natural scheme structure (Definition 5.3.6). For comparison, closed subsets can have many "natural" scheme structures. We will discuss this later in Chapter 10, but for now, it suffices for you know that a closed subscheme of X is, informally, a particular kind of scheme structure on a closed subset of X. As an example: if $I \subseteq A$ is an ideal, then $\operatorname{Spec} A/I$ endows the closed subset $V(I) \subseteq \operatorname{Spec} A$ with a scheme structure; but note that there can be different ideals with the same vanishing set (for example (x) and (x^2) in $k[x]$).

5.3.2 Stalks of the structure sheaf: germs, values at a point, and the residue field of a point

Like every sheaf, the structure sheaf has stalks, and unsurprisingly, they are interesting from an algebraic point of view. In fact, we have seen them before.

Definition 5.3.7 (Locally ringed space)

We say a ringed space is a ***locally ringed space*** *if its stalks are local ring.*

Example 5.3 **(Schemes are locally ringed spaces.)** By Proposition 5.1.2, the stalk of $\mathscr{O}_{\operatorname{Spec} A}$ at the point $[\mathfrak{p}]$ is the local ring A_p. Hence Schemes are locally ringed spaces.

Example 5.4 **(Manifolds are locally ringed spaces.)** By Proposition 3.1.1, manifolds are locally ringed spaces.

In both cases, taking the quotient by the maximal ideal may be interpreted as evaluating at the point.

Definition 5.3.8 (Residue field, value of a function at a point)

The maximal of the local ring $\mathscr{O}_{X,p}$ is denoted by $\mathfrak{m}_{X,p}$ or $\mathfrak{m}_p$, and the ***residue field*** *$\mathscr{O}_{X,p}/\mathfrak{m}_p$ is denoted $\kappa(p)$. Each function f on an open subset U of a locally ringed space X has a value at each point p of U: the* ***value of*** *f* ***at*** *p (denoted $f(p)$) is the image of the germ of f at p under the canonical map $\mathscr{O}_{X,p} \to \kappa(p)$. We say that a function* ***vanishes*** *at a point p if its value at p is 0.*

This generalizes our notion of the value of a function on $\operatorname{Spec} A$ defined in Definition 4.2.1. Notice that we can't even make sense of the phrase "function vanishing" on ringed spaces in general.

Proposition 5.3.6

(a) If f is a function on a locally ringed space X, then the subset of X where f vanishes is closed.

(b) If f is a function on a locally ringed space that vanishes nowhere, then f is invertible.

Proof

(a) Let $f \in \mathscr{O}_X(X)$. Say $V = \{p \in X : f_p \in \mathfrak{m}_p\}$, we want to show that V is closed in X. It suffices to show that $X \setminus V = \{p \in X : f_p \notin \mathfrak{m}_p\} = \{p \in X : f_p \text{ is invertible in } \mathscr{O}_{X,p}\}$ is open. Let $p \in X \setminus V$, then there exists $g_p \in \mathscr{O}_{X,p}$ such that $f_p g_p = 1$. Hence, there exists an open subset $U_p \subseteq X$ such that $f|_{U_p} \cdot g = 1$, where $g \in \mathscr{O}_X(U_p)$. Hence, $f_q g_q = 1$ for all $q \in U_q$, it follows that for all $p \in X \setminus V$ exists an open neighborhood $U_p \ni p$ such that $U_p \subseteq X \setminus V$. Hence,

$$X \setminus V = \bigcup_{p \in X \setminus V} U_p,$$

which implies that $X \setminus V$ is an open set, and therefore V is closed in X.

(b) Let $f \in \mathscr{O}_X(X)$, if f vanishes nowhere, $f_p \notin \mathfrak{m}_p$ for all $p \in X$. It follows that for all p, there exists g_p such that $f_p g_p = 1$. Hence, there exists an open neighborhood U_p of p such that $f|_{U_p} \cdot \widetilde{g}_p = 1$, where $\widetilde{g}_p \in \mathscr{O}_X(U_p)$, so $\widetilde{g}_p = \frac{1}{f|_{U_p}}$.

Let $p, q \in X$, then we have $\widetilde{g}_p \in \mathscr{O}_X(U_p)$ and $\widetilde{g}_q \in \mathscr{O}_X(U_q)$ such that $\widetilde{g}_p = \frac{1}{f|_{U_p}}$ and $\widetilde{g}_q = \frac{1}{f|_{U_q}}$. Hence, we have

$$\widetilde{g}_p\big|_{U_p \cap U_q} = \frac{1}{f|_{U_p}}\bigg|_{U_p \cap U_q} = \frac{1}{f|_{U_q}}\bigg|_{U_p \cap U_q} = \widetilde{g}_q\big|_{U_p \cap U_q}.$$

Note that $\{U_p\}_{p \in X}$ is an open cover of X, by the gluability axiom of $\mathscr{O}_X$, there exists $g \in \mathscr{O}_X(X)$ such that $g|_{U_p} = \widetilde{g}_p \in \mathscr{O}_X(U_p)$. Hence, $f(x)g(x) = 1$ for all $x \in X$, it follows that f is invertible.

$\square$

Consider a point $[\mathfrak{p}]$ of an affine scheme $\operatorname{Spec} A$. (Of course, any point of a scheme can be interpreted in this way, as each point has an affine open neighborhood.) The residue field at $[\mathfrak{p}]$ is $A_{\mathfrak{p}}/\mathfrak{p}A_{\mathfrak{p}}$, which is isomorphic to $K(A/\mathfrak{p})$, the fraction field of quotient (since $A/\mathfrak{p}$ is integral domain, its localization is its fraction field). It is useful to note that localization at $\mathfrak{p}$ and taking the quotient by $\mathfrak{p}$ "commute", i.e., the following diagram commutes.

$$\begin{tikzcd}
 & A_{\mathfrak{p}} \arrow[dr, "\text{quotient}"] & & \\
A \arrow[ur, "\text{localize}"] \arrow[dr, "\text{quotient}"'] & & A_{\mathfrak{p}}/\mathfrak{p}A_{\mathfrak{p}} \arrow[r, leftrightarrow, "\sim"] & K(A/\mathfrak{p}) \\
 & A/\mathfrak{p} \arrow[ur, "\text{localize, i.e., } K(\cdot)"'] & &
\end{tikzcd}$$

For example, consider the scheme $\mathbb{A}^2_k = \operatorname{Spec} k[x, y]$, where k is a field of characteristic not 2. Then $(x^2 + y^2)/x(y^2 - x^5)$ is a function away from the y-axis and the curve $y^2 - x^5$. Its value at $(2, 4)$ (by which we mean $[(x - 2, y - 4)]$) is $(2^2 + 4^2)/(2(4^2 - 2^5))$, as

$$\frac{x^2 + y^2}{x(y^2 - x^5)} \equiv \frac{2^2 + 4^2}{2(4^2 - 2^5)}$$

in the residue field. And its value at $[(y)]$, the generic point of the x-axis, is $\frac{x^2}{-x^6} = -1/x^4$, which we see by setting y to 0. This is indeed an element of the fraction field of $k[x, y]/(y)$, i.e., $k(x)$. (If you think you care only about algebraically closed fields, let this example be a first waring: $A_{\mathfrak{p}}/\mathfrak{p}A_{\mathfrak{p}}$ won't be algebraically closed in general, even if A is a finitely generated $\mathbb{C}$-algebra!)

Example 5.5 27/4 is a function on $\operatorname{Spec} \mathbb{Z} \setminus \{[(2)], [(7)]\}$ or indeed on an even bigger open set. Its value at $[(5)]$ is $2/(-1) \equiv -2 \pmod 5$. Its value at the generic point $[(0)]$ is $27/4$. Its value vanishes at $[(3)]$.

Definition 5.3.9 (The fiber and the rank of an $\mathscr{O}_X$-module at a point)

*If $\mathscr{F}$ is an $\mathscr{O}_X$-module on a scheme X (or more generally, a locally ringed space), define the **fiber** (or fibre) **of $\mathscr{F}$ at a point** $p \in X$ by*

$$\mathscr{F}|_p := \mathscr{F}_p \otimes_{\mathscr{O}_{X,p}} \kappa(p).$$

The fiber is a vector space over $\kappa(p)$. The dimension of this vector space $\dim_{\kappa(p)} \mathscr{F}|_p$ is called the ***rank of*** $\mathscr{F}$ *at* p*, and is denoted* $\operatorname{rank}_p \mathscr{F}$*.*

Example 5.6 $\mathscr{O}_X|_p = \mathscr{O}_{X,p} \otimes_{\mathscr{O}_{X,p}} \kappa(p) = \mathscr{O}_{X,p} \otimes_{\mathscr{O}_{X,p}} (\mathscr{O}_{X,p}/\mathfrak{m}_p) \cong 1 \otimes_{\mathscr{O}_{X,p}} \kappa(p) \cong \kappa(p)$, and therefore $\operatorname{rank}_p \mathscr{O}_X = \dim_{\kappa(p)} \mathscr{O}_X|_p = 1$.

In the same way, we can cleanly and concisely define (various notions of) manifolds.

Definition 5.3.10 (Manifolds of various sorts)

A ***topological*** *(respectively,* ***differentiable****, C^∞(****smooth****)) (real)* ***manifold*** *is a ringed space $(X, \mathscr{O}_X)$, whose underlying topological space X is Hausdorff and second countable, such that any point of X has an open neighborhood U such that $(U, \mathscr{O}_X|_U)$ is isomorphic to an open ball in $\mathbb{R}^n$ with the sheaf of continuous (respectively, differentiable, C^∞) real-valued functions on it. The phrase* ***manifold*** *(without adjective) often means* ***smooth manifold****.*

A ***complex manifold*** *is a ringed space $(X, \mathscr{O}_X)$, whose underlying topological space X is Hausdorff and second countable, such that any point of X has an open neighborhood U such that $(U, \mathscr{O}_X|_U)$ is isomorphic to an open ball in $\mathbb{C}^n$ with the sheaf of holomorphic function on it.*

5.4 Three examples

We now give three extended examples. Our short-term goal is to see that we can really work with the structure sheaf, and can compute the ring of sections of interesting open sets that aren't just distinguished open sets of affine schemes. Our long-term goal is to meet interesting examples that will come up repeatedly in the future.

5.4.1 The (affine) plane minus the origin

This example will show you that the distinguished base is something that you can work with. Let $A = k[x, y]$, so $\operatorname{Spec} A = \mathbb{A}_k^2$. Let's work out the space of functions on the open set $U = \mathbb{A}_k^2 \setminus \{(0,0)\} = \mathbb{A}_k^2 \setminus \{[(x,y)]\}$.

It is not immediately obvious whether this is a distinguished open set. (In fact it is not, since (x, y) is not a principal ideal.) But in any case, we can describe it as the union of two things which are distinguished open sets: $U = D(x) \cup D(y)$. We will find the functions on U by gluing together functions on $D(x)$ and $D(y)$.

The functions on $D(x)$ are, $\mathscr{O}_{\mathbb{A}_k^2}(D(x)) = A_x = k[x, y, 1/x]$. The functions on $D(y)$ are $A_y = k[x, y, 1/y]$. Note that A injects into its localizations (if 0 is not inverted), as it is an integral domain (Exercise 2.6), so A injects into both A_x and A_y, and both inject in to A_{xy} (and indeed into $k(x, y) = K(A)$). So we are looking for functions on $D(x)$ and $D(y)$ that agree on $D(x) \cap D(y) = D(xy)$, i.e., we are interpreting $A_x \cap A_y$ in A_{xy} (or in $k(x, y)$). Clearly those rational functions with only powers of x in the denominator, and also with only powers of y in the denominator, are polynomials. Translations: $A_x \cap A_y = A$. Thus we conclude:

$$\Gamma(U, \mathscr{O}_{\mathbb{A}_k^2}) \equiv k[x, y]. \tag{5.9}$$

In other words, we get no extra functions by removing the origin.

Remark 5.11 Notice that any function on $\mathbb{A}_k^2 \setminus \{(0,0)\}$ extends over all of $\mathbb{A}_k^2$. This is an analog of **Hartogs's Lemma** in complex geometry: you can extend a holomorphic function defined on the complement of a set of codimension at least two on a complex manifold over the missing set. This will work more generally in the algebraic setting: you can extend over points in codimension at least 2 not only if they are "smooth", but also if they are mildly singular — what we will call normal. We will make this precise in Chapter 14. This fact will be very useful for us.

We now show an interesting fact: $(U, \mathscr{O}_{\mathbb{A}^2}|_U)$ is a scheme ($U = D(x) \cup D(y)$), but it is not an affine scheme. Here's why: otherwise, if $(U, \mathscr{O}_{\mathbb{A}^2}|_U) \cong (\operatorname{Spec} A', \mathscr{O}_{\operatorname{Spec} A'})$, then we can recover A' by taking global sections:

$$A' = \Gamma(U, \mathscr{O}_{\mathbb{A}^2}|_U),$$

which we have already identified in (5.9) as $k[x, y]$. So if U is affine, then we have the isomorphism

$$U \overset{\sim}{\longleftrightarrow} \operatorname{Spec} A' = \operatorname{Spec} k[x, y] = \mathbb{A}^2_k.$$

But this bijection between prime ideals in a ring and points of the the spectrum is more constructive than that: *given the prime ideal I, you can recover the point as the generic point of the closed subset cut out by I, i.e., $V(I)$, and given the point p, you can recover the ideal as those functions vanishing at p, i.e., $I(p)$.* In particular, the prime ideal (x, y) of A' should cut out a point of $\operatorname{Spec} A'$. But on U, $V(x) \cap V(y) = \varnothing$. Conclusion: U is not an affine scheme. (If you are ever looking for a counterexample to something, and you are excepting one involving a non-affine scheme, keep this example in mind!)

5.4.2 Gluing two copies of $\mathbb{A}^1_k$ together in two different ways

We have now seen two examples of non-affine schemes: an infinite disjoint union of nonempty schemes (Proposition 5.3.5 (b)) and $\mathbb{A}^2 \setminus \{(0,0)\}$. We will give two more examples. They are important because they are the first examples of fundamental behavior, the first pathological , and second central.

First, we need to discuss how to glue two schemes together. Before that, we should review how to glue topological spaces together along isomorphic open sets. Give two topological spaces X and Y, and open subsets $U \subseteq X$ and $V \subseteq Y$ along with a homeomorphism $U \overset{\sim}{\longleftrightarrow} V$, we can create a new topological space W, that we think of as gluing X and Y together along $U \overset{\sim}{\longleftrightarrow} V$. It is the quotient of the disjoint union $X \coprod Y$ by the equivalence relation $U \sim V$, where the quotient is given the quotient topology. Then X and Y are naturally (identified with) open subsets of W, and indeed cover W.

Now that we have discussed gluing topological spaces, let's glue schemes together. (This applies without change more generally to ringed spaces.) Suppose you have two schemes $(X, \mathscr{O}_X)$ and $(Y, \mathscr{O}_Y)$, and open subsets $U \subseteq X$ and $V \subseteq Y$, along with a homeomorphism $\pi : U \xrightarrow{\sim} V$, and an isomorphism of structure sheaves $\mathscr{O}_V \xrightarrow{\sim} \pi_* \mathscr{O}_U$ (i.e., an isomorphism of schemes $(U, \mathscr{O}_X|_U) \overset{\sim}{\longleftrightarrow} (V, \mathscr{O}_Y|_V)$). Then we can glue these together to get a single scheme. Reason: let W be X and Y glued together using the isomorphism $U \overset{\sim}{\longleftrightarrow} V$. Then Theorem 3.5.3 shows that the structure sheaves can be glued together to get a sheaf of rings. Note that this is indeed a scheme: any point has an open neighborhood that is an affine scheme.

Theorem 5.4.1 (Glue an arbitrary collection of scheme together)

We can glue an arbitrary collection of schemes together. Suppose we are given:

- *scheme X_i (as i runs over some index set I, not necessarily finite),*
- *open subschemes $X_{ij} \subseteq X_i$ with $X_{ii} = X_i$,*
- *isomorphisms $f_{ij} = (\pi_{ij}, \pi^\sharp_{ij}) : X_{ij} \xrightarrow{\sim} X_{ji}$ with f_{ii} the identity*

such that

- *(the cocycle condition) the isomorphisms "agree on triple intersections", i.e.,*

$$f_{ik}|_{X_{ij}\cap X_{ik}} = f_{jk}|_{X_{ji}\cap X_{jk}} \circ f_{ij}|_{X_{ij}\cap X_{ik}}$$

(so implicitly, to make sense of the right side, $f_{ij}(X_{ik} \cap X_{ij}) \subseteq X_{jk}$).

Then there is a unique scheme X (up to unique isomorphism) along with open subschemes isomorphic to the X_i respecting this gluing data in the obvious sense.

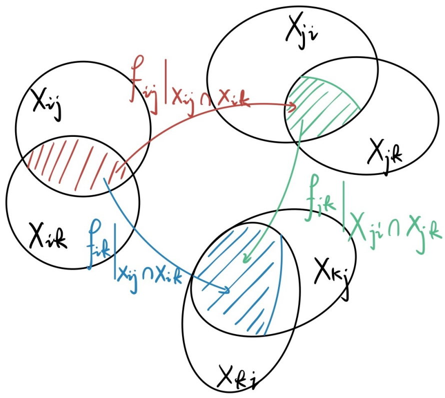


Figure 5.6: Picture of the cocycle condition

Remark 5.12 The cocycle condition ensures that f_{ij} and f_{ji} are inverses (take $k = i$ in the cocycle condition). In fact, the hypothesis that f_{ii} is the identity also follows from the cocycle condition.

Proof

(i) Construct the underlying topological space X.

Consider the disjoint union $\coprod_{i\in I} X_i$, we define a relation $\sim$ by setting $(x,i) \sim (y_j)$ if exists $k \in I$ such that $\pi_{ik}(x) = \pi_{jk}(y)$. We want to check relation $\sim$ is a equivalence relation.

- **Reflexivity:** Note that $\pi_{ii} = \text{id}$, so we have $(x,i) \sim (x,i)$.
- **Symmetry:** If $(x,i) \sim (y,j)$, then there exists $k \in I$ such that $\pi_{ik}(x) = \pi_{jk}(y)$. Hence, $(y,j) \sim (x,i)$.
- **Transitivity:** If $(x,i) \sim (y,j)$ and $(y,j) \sim (z,k)$, then there exists $m, n \in I$ such that $\pi_{im}(x) = \pi_{jm}(y)$ and $\pi_{jn}(y) = \pi_{kn}(z)$. Note that $\pi_{jn} = \pi_{mn} \circ \pi_{jm}$, we have $\pi_{mn}(\pi_{im}(x)) = \pi_{in}(x) = \pi_{mn}\pi_{jm}(y) = \pi_{jn}(y) = \pi_{kn}(z)$, i.e., $(x.i) \sim (z,k)$.

Hence, relation $\sim$ is a equivalence relation. Define

$$X = \left(\coprod X_i\right) \Big/_{\sim}$$

as the quotient space.

We now endow X with the quotient topology. Define $U \subseteq X$ be an open subset of X if and only if $q_i^{-1}(U)$ is open in X_i for all i, where $q_i : X_i \to X$ is the quotient map.

(ii) Construct the structure sheaf $\mathscr{O}_X$.

We claim that each quotient map q_i is open. Let $V \subseteq X_i$ be an open subset of X_i, consider $q_j^{-1}(q_i(V))$, note that

$$\begin{aligned} q_j^{-1}(q_i(V)) =& \{y \in X_j : q_j(y) = q_i(x), \text{ for some } x \in V\} \\ =& \{y \in X_j : \exists k \in I \text{ and } x \in V, \pi_{jk}(y) = \pi_{ik}(x)\} \\ =& \{y \in X_j : \exists k \in I \text{ and } x \in V, \pi_{kj}\pi_{jk}(y) = y = \pi_{kj}\pi_{ik}(x) = \pi_{ij}(x)\} \\ =& \{y \in X_j : \exists x \in V, y = \pi_{ij}(x)\} \\ =& \pi_{ij}(V \cap X_{ij}), \end{aligned} \tag{5.10}$$

and π_{ij} is a homeomorphism, we have $q_j^{-1}(q_i(V))$ is open in X_j. Hence, q_i is open.
Let $U_i = q_i(X_i)$, since q_i is open, U_i is open in X. Let $\overline{x} \in X$ be any point in X, then $x \in X_i$ for some i, and therefore $\overline{x} = q_i(x) \in q_i(X_i) \subseteq X$. Hence, $X = \bigcup_{i \in I} q_i(X_i) = \bigcup_{i \in I} U_i$.
Next, we define the structure sheaf on U_i. Since $q_i : X_i \to U_i$ is a homeomorphism, define

$$\mathscr{F}_{U_i} = q_{i,*}\mathscr{O}_{X_i}.$$

By (5.10), we have $q_j^{-1}(q_i(X_i)) = q_j^{-1}(U_i) = \pi_{ij}(X_i \cap X_{ij}) = \pi_{ij}(X_{ij}) = X_{ji}$, i.e.,

$$X_{ji} = q_j^{-1}(U_i).$$

Hence, we have the commutative diagram.

$$\begin{tikzcd} X_{ij} \arrow[rr, "\pi_{ij}"] \arrow[dr, "q_i|_{X_{ij}}"'] & & X_{ji} \arrow[dl, "q_j|_{X_{ji}}"] \\ & U_i \cap U_j & \end{tikzcd}$$

Consider $\mathscr{F}_{U_j}|_{U_i \cap U_j}$, let $V \subseteq U_i \cap U_j$, then

$$\begin{aligned} \mathscr{F}_{U_j}|_{U_i \cap U_j}(V) =& (q_{j,*}\mathscr{O}_{X_j})|_{U_i \cap U_j}(V) = \mathscr{O}_{X_{ji}}(q_j^{-1}(V)) \\ =& \mathscr{O}_{X_{ji}}(\pi_{ij} \circ q_i^{-1}(V)) = \mathscr{O}_{X_{ji}}(\pi_{ji}^{-1} \circ q_i^{-1}(V)) \\ =& \pi_{ji,*}\mathscr{O}_{X_{ji}}(q_i^{-1}(V)) \\ \cong& \mathscr{O}_{X_{ij}}(q_i^{-1}(V)) = (q_{j,*}\mathscr{O}_{X_j})|_{U_i \cap U_j}(V) \\ =& \mathscr{F}_{U_i}|_{U_i \cap U_j}(V), \end{aligned}$$

it follows that $\mathscr{F}_{U_i}|_{U_i \cap U_j} \overset{\sim}{\longleftrightarrow} \mathscr{F}_{U_j}|_{U_i \cap U_j}$.
Say $\varphi_{ij} : \mathscr{F}_{U_i}|_{U_i \cap U_j} \overset{\sim}{\longrightarrow} \mathscr{F}_{U_j}|_{U_i \cap U_j}$, it is easy to check that $\varphi_{jk} \circ \varphi_{ij} = \varphi_{ik}$ on $U_i \cap U_j \cap U_k$. By Theorem 3.5.3, $\mathscr{F}_{U_i}$ can be glued together into a sheaf on X, say $\mathscr{O}_X$, with isomorphism $\mathscr{O}|_{U_i} \overset{\sim}{\longrightarrow} \mathscr{F}_{U_i}$.

(iii) $(X, \mathscr{O}_X)$ is a scheme.
Since each $(U_i, \mathscr{O}_X|_{U_i}) \cong (X_i, \mathscr{O}_{X_i})$ and each $(X_i, \mathscr{O}_{X_i})$ is a scheme, $(U_i, \mathscr{O}_X|_{U_i})$ is therefore a scheme.

(iv) Uniqueness of $(X, \mathscr{O}_X)$.
Uniqueness given by the uniqueness of quotient topology and gluing sheaves (Theorem 3.5.3).

□

We will now give two non-affine schemes. Both are handy to know. In both cases, we will glue together two copies of the affine line $\mathbb{A}_k^1$. Let $X = \operatorname{Spec} k[t]$ and $Y = \operatorname{Spec} k[u]$. Let $U = D(t) = \operatorname{Spec} k[t, 1/t] \subseteq X$ and $V = D(u) = \operatorname{Spec} k[u, 1/u] \subseteq Y$. We will get both examples by gluing X and Y together along U and V. The difference will be in how we glue.

The affine line with the doubled origin

Consider the isomorphism $U \overset{\sim}{\longleftrightarrow} V$ via the isomorphism $k[t, 1/t] \overset{\sim}{\longleftrightarrow} k[u, 1/u]$ given by $t \longleftrightarrow u$ (Proposition 5.3.1). The resulting scheme is called the affine line with double origin. Figure 5.7 is a picture of it.

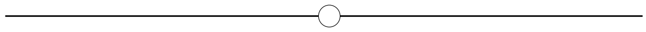


Figure 5.7: The affine line with doubled origin

As the picture suggests, intuitively this is an analog of a failure of Hausdorffness. Now $\mathbb{A}^1$ itself is not Hausdorff, so we can't say that it is a failure of Hausdorffness. We see this as weird and bad, so we will want to make a definition that will prevent this from happening. This will be the notion of **separatedness** (to be dicussed in Chapter 12). (In fact, in some sense, separatedness is the right definition of Hausdorfness.) This will answer other of our prayers as well. For example, on a separated scheme, the "affine base of the Zariski topology" is nice — the intersection of two affine open sets will be affine.

✍ **Exercise 5.2** Show that the affine line with doubled origin is not affine.

Proof Let $X = \operatorname{Spec} k[t]$, $Y = \operatorname{Spec} k[u]$, $U = D(t)$, and $V = D(u)$. Let $Z = (X \coprod Y)/\sim$ be the affine line with doubled origin, where $\sim$ given by isomorphism $U \overset{\sim}{\longleftrightarrow} V$ which induced by ring isomorphism $t \mapsto u$.

We first calculate the global section $\Gamma(Z, \mathscr{O}_Z)$. Let $s \in \Gamma(Z, \mathscr{O}_Z)$, then $(s|_X)|_{X\cap Y} = (s|_Y)|_{X\cap Y}$. In fact, $X \cap Y = U \cong V$, hence, $\Gamma(X \cap Y, \mathscr{O}_Z) \cong k[t]_t$. Since $(s|_X)|_{X\cap Y} = (s|_Y)|_{X\cap Y}$ in $k[t]_t$, exists n such that $t^n(s|_X - s|_Y) = 0$ in $k[t]$. Note that $\Gamma(X, \mathscr{O}_Z|_X) \cong k[t]$ and $\Gamma(Y, \mathscr{O}_Z|_Y) \cong k[t]$, also $k[t]$ is an integral domain, since $t \neq 0$, we have $s|_X = s|_Y$ in $k[t]$. It follows that $s \in k[t]$. Conversely, each element in $k[t]$ can be seen as the function over Z. Hence,

$$\Gamma(Z, \mathscr{O}_Z) \cong k[t].$$

Suppose Z is an affine scheme, then $(Z, \mathscr{O}_Z) \overset{\sim}{\longleftrightarrow} (A, \operatorname{Spec} A)$, hence,

$$\Gamma(Z, \mathscr{O}_Z) \cong k[t] \cong \mathscr{O}_{\operatorname{Spec} A}(\operatorname{Spec} A) = A.$$

Note that $[(u)] \in Z$, but $[(u)] \notin k[t]$, a contradiction! Hence, Z is not an affine scheme. □

✍ **Exercise 5.3** Do the same construction with $\mathbb{A}^1$ replaced by $\mathbb{A}^2$. You will have defined the **affine plane with doubled origin**. Describe two affine open subsets of this scheme whose intersection is not an affine open subset.

Proof Let $X = \operatorname{Spec} k[x_1, y_1]$ and $Y = \operatorname{Spec} k[x_2, y_2]$. Let $U = \operatorname{Spec} k[x_1, y_1] \setminus \{[(x_1, y_1)]\}$ and $V = \operatorname{Spec} k[x_2, y_2] \setminus \{[(x_2, y_2)]\}$, then $U = D(x_1) \cup D(y_1)$ and $V = D(x_2) \cup D(y_2)$. By §5.4.1, we have $\Gamma(U, \mathscr{O}_X) = k[x_1, y_1]$ and $\Gamma(V, \mathscr{O}_Y) = k[x_2, y_2]$. Let $\varphi : k[x_1, y_1] \to k[x_2, y_2]$ by setting $x_1 \mapsto x_2$ and $y_1 \mapsto y_2$. Clearly, φ is a ring isomorphism. It induced an isomorphism of schemes $(U, \mathscr{O}_X|_U) \overset{\sim}{\longleftrightarrow} (V, \mathscr{O}_Y|_V)$.

Now, we glue X and Y via $U \overset{\sim}{\longleftrightarrow} V$. Let $Z = (X \coprod Y)/\sim$. We want to describe two affine open subsets of this scheme whose intersection is not an affine open subset. Consider X and Y, they are affine open subschemes of Z. In Z, the intersection of X and Y is $Z \setminus \{[(x_1, y_1)], [(x_2, y_2)]\}$. Clearly, $Z \setminus \{[(x_1, y_1)], [(x_2, y_2)]\}$ is isomorphic to $\operatorname{Spec} k[x, y] \setminus \{[(x, y)]\}$, by §5.4.1, $\operatorname{Spec} k[x, y] \setminus \{[(x, y)]\}$ is not affine, this implies that $X \cap Y$ is not affine. □

The projective line

Consider the isomorphism $U \cong V$ via the isomorphism $k[t, 1/t] \cong k[u, 1/u]$ given by $t \longleftrightarrow 1/u$. Figure 5.8 is a suggestive picture of this gluing. The resulting scheme is called the **projective line over the field** k, and is denoted $\mathbb{P}^1_k$.

Notice how the points glue. We assume that k is algebraically closed for convenience. On the first affine line, we have the closed points $[(t - a)]$, which we think of as "a on the t-line", and we have the generic point $[(0)]$. On the second affine line, we have closed points that are "b on the u-line", and generic point. Then a on

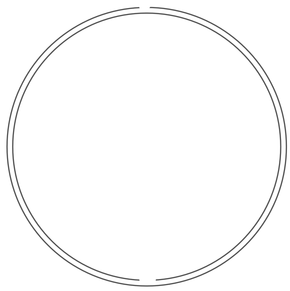


Figure 5.8: Gluing two affine lines together to get $\mathbb{P}^1$

the t-line is glued to $1/a$ on the u-line (if $a \neq 0$ of course), and the generic point is glued to the generic point (the ideal (0) of $k[t]$ becomes the ideal (0) of $k[t, 1/t]$ upon localization, and the ideal (0) of $k[u]$ becomes the ideal (0) of $k[u, 1/u]$. And (0) in $k[t, 1/t]$ is (0) in $k[u, 1/u]$ under the isomorphism $t \xleftrightarrow{\sim} 1/u$).

If k is algebraically closed, we can interpret the closed points of $\mathbb{P}^1_k$ in the following way, which may make this sound closer to the way you have seen projective space defined earlier. The points are of the form $[a, b]$, where a and b are not both zero, and $[a, b]$ is defined with $[ac, bc]$ where $c \in k^\times$. Then if $b \neq 0$, this is identified with a/b on the t-line, and if $a \neq 0$, this is identified with b/a on the u-line.

Proposition 5.4.1

$\mathbb{P}^1_k$ is not affine.

Proof Let $X = \operatorname{Spec} k[t]$ and $Y = \operatorname{Spec} k[u]$. $\mathbb{P}^1_k$ glued by X and Y via $t \longleftrightarrow 1/u$. We do this by calculating the ring of global sections. Let $s \in \Gamma(\mathbb{P}^1_k, \mathscr{O}_{\mathbb{P}^1_k})$ with $(s|_X)|_{X\cap Y} = (s|_Y)|_{X\cap Y}$. Since $\Gamma(X, \mathscr{O}_{\mathbb{P}^1_k}) \cong k[t]$ and $\Gamma(Y, \mathscr{O}_{\mathbb{P}^1_k}) \cong k[u]$, we may assume $s|_X = f(t)$ and $s|_Y = g(u)$ are polynomials. Note that $\Gamma(X \cap Y, \mathscr{O}_{\mathbb{P}^1_k}) = k[t]_t$, restrict $f(t)$ to $X \cap Y$, we get something we can still call $f(t)$, and similarly for $g(u)$. Since $t \longleftrightarrow 1/u$ and $(s|_X)|_{X\cap Y} = (s|_Y)|_{X\cap Y}$, we have $f(t) = g(1/t)$. But the only polynomials in t that are at the same time polynomials in $1/t$ are the constants k. Thus $\Gamma(\mathbb{P}^1, \mathscr{O}_{\mathbb{P}^1}) = k$.

If $\mathbb{P}^1$ were affine, then it would be $\operatorname{Spec}\Gamma(\mathbb{P}^1, \mathscr{O}_{\mathbb{P}^1}) = \operatorname{Spec} k$, i.e., one point. But it isn't — it has lots of points. $\square$

Remark 5.13 We have proved an analog of an important theorem: the only holomorphic functions on $\mathbb{CP}^1$ are the constants!

5.4.3 Projective space

We now make a preliminary definition of **projective n-space over a field** k, denoted $\mathbb{P}^n_k$, by gluing together $n+1$ open sets each isomorphic to $\mathbb{A}^n_k$. Judicious choice of notation for these open sets will make our life easier. Our motivation is as follows. In the construction of $\mathbb{P}^1$ above, we thought of points of projective space as $[x_0 : x_1]$, where (x_0, x_1) are only determined up to scalars, i.e., (x_0, x_1) is considered the same as $(\lambda x_0, \lambda x_1)$. Then the first patch can be interpreted by taking the locus where $x_0 \neq 0$, and then we consider the points $[1 : t]$, and we think of t as x_0/x_1; even though x_0 and x_1 are not well-defined, x_1/x_0 is. The second corresponds to where $x_1 \neq 0$, and we consider the points $[u : 1]$, and we think of u as x_0/x_1. It will be useful to instead use the notation $x_{1/0}$ for t and $x_{0/1}$ for u.

For $\mathbb{P}^n$, we glue together $n+1$ open sets, one for each of $i = 0, \ldots, n$. The i-th open set U_i will have coordinates $x_{0/i}, \ldots, x_{(i-1)/i}, x_{(i+1)/i}, \ldots, x_{n/i}$. It will be convenient to write this as

$$\operatorname{Spec} k[x_{0/i}, x_{1/i}, \ldots, x_{n/i}]/(x_{i/i} - 1) \tag{5.11}$$

(so we have introduced a "dummy variable" $x_{i/i}$ which we immediately set to 1). We glue the distinguished open set $D(x_{j/i})$ of U_i to the distinguished open set $D(x_{i/j})$ of U_j, by identifying these two schemes by describing the identification of rings

$$k[x_{0/i}, x_{1/i}, \ldots, x_{n/i}, 1/x_{j/i}]/(x_{i/i} - 1) \cong k[x_{0/j}, x_{1/j}, \ldots, x_{n/j}, 1/x_{i/j}]/(x_{j/j} - 1)$$

via $x_{k/i} = x_{k/j}/x_{i/j}$ and $x_{k/j} = x_{k/i}/x_{j/i}$ and $x_{k/j} = x_{k/i}/x_{j/i}$ (which implies $x_{i/j}x_{j/i} - 1$). We need check that this gluing information agrees over triple overlaps.

Exercise 5.4 Check this, as painlessly as possible.

Proof Say $f_{ij} : D(x_{j/i}) \to D(x_{i/j})$, f_{ij} is induced by ring isomorphism $\varphi_{ij} : \Gamma(D(x_{i/j}), \mathscr{O}_{U_j}) \to \Gamma(D(x_{j/i}), \mathscr{O}_{U_i})$ which is given by $x_{k/j} \mapsto x_{k/i}/x_{j/i}$, and therefore f_{ij} is an isomorphism of affine open schemes.

We want to show that $f_{ik}|_{D(x_{j/i}) \cap D(x_{k/i})} = f_{ik}|_{D(x_{i/j}) \cap D(x_{k/j})} \circ f_{ij}|_{D(x_{j/i}) \cap D(x_{k/i})}$. Consider the corresponding ring of $D(x_{j/i}) \cap D(x_{k/i})$, $D(x_{i/j}) \cap D(x_{k/j})$, and $D(x_{j/k}) \cap D(x_{i/k})$, they are all affine open subset, hence, we have

$$\begin{aligned}
\Gamma(D(x_{j/i}) \cap D(x_{k/i}), \mathscr{O}_{U_i}) &= k\left[x_{0/i}, \ldots, x_{n/i}, \frac{1}{x_{j/i}}, \frac{1}{x_{k/i}}\right] \Big/ (x_{i/i} - 1), \\
\Gamma(D(x_{i/j}) \cap D(x_{k/j}), \mathscr{O}_{U_j}) &= k\left[x_{0/j}, \ldots, x_{n/j}, \frac{1}{x_{i/j}}, \frac{1}{x_{k/j}}\right] \Big/ (x_{j/j} - 1), \\
\Gamma(D(x_{j/k}) \cap D(x_{i/k}), \mathscr{O}_{U_k}) &= k\left[x_{0/k}, \ldots, x_{n/k}, \frac{1}{x_{i/k}}, \frac{1}{x_{j/k}}\right] \Big/ (x_{k/k} - 1).
\end{aligned}$$

Define $\varphi_{ij}^e : \Gamma(D(x_{i/j}x_{k/j}), \mathscr{O}_{U_j}) \to \Gamma(D(x_{j/i}x_{k/i}), \mathscr{O}_{U_i})$ by setting

$$\overline{\sum_{i_0, \ldots, i_n, j_1, j_2} x_{0/j}^{i_0} \cdots x_{n/j}^{i_n} x_{i/j}^{-j_1} x_{k/j}^{-j_2}} \mapsto \overline{\sum_{i_0, \ldots, i_n, j_1, j_2} \varphi_{ij}(x_{0/j})^{i_0} \cdots \varphi_{ij}(x_{n/j})^{i_n} \varphi_{ij}(x_{i/j})^{-j_1} \varphi_{ij}(x_{k/j})^{-j_2}}.$$

Clearly, φ_{ij}^e is a ring homomorphism, since φ_{ij} is an isomorphism, it is easy to check that φ_{ij}^e is an isomorphism. Hence, we have the following commutative diagram.

$$\begin{tikzcd}
\Gamma(D(x_{j/i}) \cap D(x_{k/i}), U_i) & & \Gamma(D(x_{i/j}) \cap D(x_{k/j}), U_j) \arrow[ll, "\varphi_{ij}^e"'] \\
& \Gamma(D(x_{i/k}) \cap D(x_{j/k}), U_k) \arrow[ul, "\varphi_{ik}^e"] \arrow[ur, "\varphi_{ik}^e"'] &
\end{tikzcd}$$

By Proposition 5.3.1, it follows that

$$f_{ik}|_{D(x_{j/i}) \cap D(x_{k/i})} = f_{ik}|_{D(x_{i/j}) \cap D(x_{k/j})} \circ f_{ij}|_{D(x_{j/i}) \cap D(x_{k/i})}.$$

□

Note that our definition does not use the fact that k is a field. Hence we may as well define $\mathbb{P}_A^n$ for any ring A. This will be useful later.

Definition 5.4.1 ($\mathbb{P}_A^n$)

Let A be a ring. The projective n-space over a ring A is defined by gluing together $n+1$ open sets each isomorphic to $\mathbb{A}_A^k$, denoted $\mathbb{P}_A^n$.

Proposition 5.4.2

The only functions on $\mathbb{P}_k^n$ are constants, i.e., $\Gamma(\mathbb{P}_k^n, \mathscr{O}) \cong k$, and hence that $\mathbb{P}_k^n$ is not affine if $n > 0$.

Proof Let $U_i = \operatorname{Spec} k[x_{0/i}, \dots, x_{n/i}]/(x_{i/i} - 1)$ be affine open subscheme of $\mathbb{P}_k^n$. Let $f \in \Gamma(\mathbb{P}_k^n, \mathscr{O})$, then $(f|_{U_i})|_{U_i \cap U_j} = (f|_{U_j})|_{U_i \cap U_j}$ in $\Gamma(U_i \cap U_j, \mathscr{O}_{\mathbb{P}_k^n})$.

We may assume that

$$f|_{U_i} = f_i(x_{0/i}, \dots, x_{i-1/i}, x_{i+1/i}, \dots, x_{n/i})$$

and

$$f|_{U_j} = f_j(x_{0/j}, \dots, x_{j-1/j}, x_{j+1/j}, \dots, x_{n/j}).$$

On the $D(x_{j/i}) = U_i \cap U_j$, we have the coordinate change:

$$x_{k/i} = x_{k/j}/x_{i/j},$$

hence,

$$\begin{aligned} f_i(x_{0/i}, \dots, x_{i-1/i}, x_{i+1/i}, \dots, x_{n/i}) =& f_i\left(\frac{x_{0/j}}{x_{i/j}}, \dots, \frac{x_{i-1/j}}{x_{i/j}}, \frac{x_{i+1/j}}{x_{i/j}}, \dots, \frac{x_{n/j}}{x_{i/j}}\right) \\ =& \frac{1}{x_{i/j}^d} F(x_{0/j}, \dots, x_{i-1/j}, x_{i/j}, x_{i+1/j}, \dots, x_{n/j}) \\ =& f_j(x_{0/j}, \dots, x_{j-1/j}, x_{j+1/j}, \dots, x_{n/j}), \end{aligned}$$

where $F(x_{0/j}, \dots, x_{i-1/j}, x_{i/j}, x_{i+1/j}, \dots, x_{n/j})$ is a homogeneous polynomial with degree d. Since f_j is a polynomial, d must be 0, and therefore $f|_{U_i} = f|_{U_j}$ must be constants for all i, j. Hence, $\Gamma(\mathbb{P}_k^n, \mathscr{O}) \cong k$. □

Remark 5.14 There is even some geometric intuition behind this: the complement of the union of two open sets has codimension 2. But "Algebraic Hartogs' Lemma" says that any function defined on this union extends to be a function on all of projective space. Because we are expecting to see only constants as functions on all of projective space, we should already see this for this union of our two affine open sets.

Proposition 5.4.3

If k is algebraically closed, the closed points of $\mathbb{P}_k^n$ may be interpreted in the traditional way: the points are of the form $[a_0 : \cdots : a_n]$, where the a_i are not all zero, and $[a_0 : \cdots : a_n]$ is identified with $[\lambda a_0 : \cdots : \lambda a_n]$ where $\lambda \in k^\times$.

Proof $\mathbb{P}_k^n$ is covered by $U_i = \operatorname{Spec} k[x_{0/i}, \dots, x_{n/i}]/(x_{i/i} - 1)$. Let $p \in \mathbb{P}_k^n$ be a closed point, then $p \in U_i$ for some i. Since $\{p\}$ is closed in $\mathbb{P}_k^n$, $\{p\}$ is closed in U_i. Note that U_i is affine open subscheme, $\{p\}$ correspondence to a maximal ideal of $k[x_{0/i}, \dots, x_{n/i}]/(x_{i/i} - 1)$, say $p = [(x_{0/i} - a_{0/i}, \dots, x_{i-1/i} - a_{i-1/i}, x_{i+1/i} - a_{i+1/i}, \dots, x_{n/i} - a_{n/i})] := [a_0 : \cdots : a_i : \cdots : a_n]$.

Next, we want to show that $[a_0 : \cdots : a_n]$ is identified with $[\lambda a_0 : \cdots : \lambda a_n]$, where $\lambda \in k^\times$. It correspondence to maximal ideal $(x_{0/i} - \lambda a_{0/i}, \dots, x_{n/i} - \lambda a_{n/i})$. Note that

$$(x_{0/i} - \lambda a_{0/i}, \dots, x_{n/i} - \lambda a_{n/i}) = \left(\frac{x_{0/i}}{\lambda} - a_{0/i}, \dots, \frac{x_{n/i}}{\lambda} - a_{n/i}\right)$$

and

$$k[x_{0/i}, \dots, x_{n/i}]/(x_{i/i} - 1) \cong k\left[\frac{x_{0/i}}{\lambda}, \dots, \frac{x_{n/i}}{\lambda}\right] \Big/ \left(\frac{x_{i/i}}{\lambda} - 1\right),$$

$(x_{0/i} - \lambda a_{0/i}, \dots, x_{n/i} - \lambda a_{n/i})$ and $(x_{0/i} - a_{0/i}, \dots, x_{n/i} - a_{n/i})$ correspondence to the same point $[a_0 : \cdots : a_i : \cdots : a_n]$. Hence, $[a_0 : \cdots : a_i : \cdots : a_n]$ is identified with $[\lambda a_0 : \cdots : \lambda a_i : \cdots : \lambda a_n]$, where $\lambda \in k^\times$. □

Remark 5.15 Helpful translation: we think of the closed points of $\mathbb{P}_k^n$ (where $k = \overline{k}$) as $[x_0 : x_1 : \cdots : x_n]$, with $x_i \in k$ (not all x_i zero), and we identify $[x_0 : x_1 : \cdots : x_n]$ with $[\lambda x_0 : \lambda x_1 : \cdots : \lambda x_n]$ for all $\lambda \in k^\times$. Then $[x_0 : x_1 : \cdots : x_n]$ corresponds to a line through the origin, and $(x_{0/i}, x_{1/i}, \ldots, x_{n/i})$ is where this line meets the hyperplane $x_i = 1$.

We will later give other definitions of projective space, and the x_i will be called **projective coordinates** there. Our first definition here will often be handy for computing things. But there is something unnatural about it — projective space is highly symmetric, and that isn't clear from our current definition. Furthermore, as noted by Herman Weyl, " The introduction of numbers as coordinates is an act of violence."

5.4.4 The Chinese Remainder Theorem is a geometric fact

The Chinese Remainder Theorem is embedded in what we have done. We will see this in a single example, and will give the general statement.

Theorem 5.4.2 (Chinese Remainder Theorem)

Let A be a ring and $\mathfrak{a}_1, \ldots, \mathfrak{a}_n$ ideals of A. Define a homomorphism

$$\varphi : A \longrightarrow \prod_{i=1}^{n} (A/\mathfrak{a}_i)$$

by the rule $\varphi(x) = (x + \mathfrak{a}_1, \ldots, x + \mathfrak{a}_n)$.

(i) If $\mathfrak{a}_i$, $\mathfrak{a}_j$ are coprime whenever $i \neq j$, then $\prod \mathfrak{a}_i = \bigcap \mathfrak{a}_i$.

(ii) φ is surjective if and only if $\mathfrak{a}_i, \mathfrak{a}_j$ are coprime whenever $i \neq j$.

(iii) φ is injective if and only if $\bigcap \mathfrak{a}_i = (0)$.

Proof See [AM18, Proposition 1.10]. □

The Chinese Remainder Theorem says that knowing an integer modulo 60 is the same as knowing an integer modulo 3, 4, and 5. Here is how to see this in the language of schemes. What is $\operatorname{Spec} \mathbb{Z}/(60)$? What are the prime ideals of this ring? Answer: those prime ideals of $\mathbb{Z}$ containing (60), i.e., those primes dividing 60, i.e., (2), (3), and (5). Figure 5.9 is a sketch of $\operatorname{Spec} \mathbb{Z}/(60)$. They are all closed points, as these are all maximal ideals, so the topology is the discrete topology (each closed point is also open). What are the stalks? They are $(\mathbb{Z}/(60))_{(2)} = \mathbb{Z}/(4)$, $(\mathbb{Z}/(60))_{(3)} = \mathbb{Z}/(3)$ and $(\mathbb{Z}/(60))_{(5)} = \mathbb{Z}/(5)$. The nilpotents "at (2)" are indicated by the "fuzz" on that point. (We discussed visualizing nilpotents with "infinitesimal fuzz" in §5.2). So what are global sections on this scheme? They are sections on this open set (2), this other open set (3), and this third open set (5). In other words, we have a natural isomorphism of rings

$$\mathbb{Z}/(60) \xrightarrow{\sim} \mathbb{Z}/(2^2) \times \mathbb{Z}/(3) \times \mathbb{Z}/(5).$$

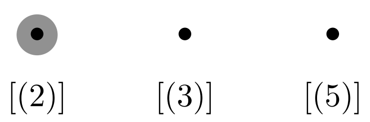


Figure 5.9: A picture of the scheme $\operatorname{Spec} \mathbb{Z}/(60)$

Example 5.7 Here is an example of a function on an open subset of a scheme with some surprising behavior. On $X = \operatorname{Spec} k[w, x, y, z]/(wz - xy)$, consider the open subset $D(y) \cap D(w)$. Clearly the function z/y on $D(y)$ agree with x/w on $D(w)$ on the overlap $D(y) \cap D(w)$. Hence they glue together to give a section. You may have seen this before when thinking about analytic continuation in complex geometry — we have a "holomorphic" function which has the description z/y on an open set, and this description breaks down elsewhere, but you can still "analytically continue" it by giving the function a different definition on different parts of the space.

This function has no "single description" as a well-defined expression in terms of w, x, y, z! There is a lot of interesting geometry here, and this scheme will be a constant source of (counter)example for us (§5.1.3). Here is a glimpse, in terms of words we have not yet defined. The space $\operatorname{Spec} k[w, x, y, z]$ is $\mathbb{A}^4$, and is, not surprisingly, 4-dimensional. We are working with the subset $X = \operatorname{Spec} k[w, x, y, z]/(wz - xy)$, which is a hypersurface, and is 3-dimensional. It is a cone over a smooth quadric surface in $\mathbb{P}^3$. The open subset $D(y) \subseteq X$ is X minus some hypersurface, sow we are throwing away codimension 1 locus. You might think that the intersection of these two discarded loci is then codimension 2, and that failure of extending this weird function to a global polynomial comes because of a failure of our Hartogs's Lemma-type theorem. But that's not true — $V(y) \cap V(w)$ is in fact codimension 1. Here is what is actually going on. The space $V(y)$ is obtained by throwing away the (cone over the) union of two lines l and m_1, one in each "ruling" of the surface, and $V(w)$ also involves throwing away the (cone over the) line l, which is a codimension 1 set. Remarkably, despite being "pure codimension 1" the cone over l is not cut out even set-theoretically by a single equation. This means that any expression $f(w, x, y, z)/g(w, x, y, z)$ for our function cannot correctly describe our function on $D(y) \cup D(w)$ — at some point of $D(y) \cup D(w)$ it must be $0/0$. Here's why. Our function can't be defined on $V(y) \cap V(w)$, so g must vanish here. But g can't vanish just on the cone over l — it must vanish elsewhere too.

5.5 Projective schemes, and the Proj construction

Projective schemes are important for a number of reasons. Here are a few. Schemes that were of "classical interest" in geometry — and those that you would have cared about before knowing about schemes — are all projective or an open subset thereof (basically, quasiprojective). Moreover, most "schemes of interest" tend to be projective or quasiprojective. In fact, it is very hard to even give an example of a scheme satisfying basic properties — for example, finite type and "Hausdorff" ("separated") over a field — that is provably not quasiprojective. For complex geometers: it is hard to find a compact complex variety that is provably not projective, and it is quite hard to come up with a complex variety that is provably not an open subset of a projective variety. So projective schemes are really ubiquitous. Also, the notion of "projective k-scheme" is a good approximation of the algebra-geometric version of compactness ("properness").

Finally, although projective schemes may be obtained by gluing together affine schemes, and we know that keeping track of gluing can be annoying, there is a simple means of dealing with them worrying about gluing. Just as there is a rough dictionary between rings and affine schemes, we will have a (slightly looser) analogous dictionary between graded rings and projective schemes. Just as one can work with affine schemes by instead working with rings, one can work with projective schemes by instead working with graded rings.

5.5.1 Motivation from classical geometry

For geometric intuition, we recall how one thinks of projective space "classically" (in the classical topology, over the real numbers). $\mathbb{P}^n$ can be interpreted as the lines through the origin in $\mathbb{R}^{n+1}$. Thus subsets of $\mathbb{P}^n$ correspond to unions of lines through the origin of $\mathbb{R}^{n+1}$, and closed subsets correspond to such unions which are closed. (The same is not true with "closed" replaced by "open"!)

One often pictures $\mathbb{P}^n$ as being the "points at infinite distance" in $\mathbb{R}^{n+1}$, where the points infinitely far in one direction are associated with the points infinitely far in the opposite direction. We can make this more precise using the decomposition

$$\mathbb{P}^{n+1} = \mathbb{R}^{n+1} \coprod \mathbb{P}^n$$

by which we mean that there is an open subset in $\mathbb{P}^{n+1}$ identified with $\mathbb{R}^{n+1}$ (the points with last "projective coordinate" nonzero), and the complementary closed subset identified with $\mathbb{P}^n$ (the points with last "projective coordinate" zero).

Then for example any equation cutting out sone set V of points in $\mathbb{P}^n$ will also cut out some set of points in $\mathbb{R}^{n+1}$ that will be a closed union of lines. We call this the **affine cone** of V. These equations will cut out some union of $\mathbb{P}^1$'s in $\mathbb{P}^{n+1}$, and we call this the **projective cone** of V. The projective cone is the disjoint union of the affine cone and V. For example, the affine cone over $x^2+y^2=z^2$ in $\mathbb{P}^2$ is just the "classical" picture of a cone in $\mathbb{R}^3$, see Figure 5.10. We will make this analogy precise in our algebraic setting in Chapter 10.

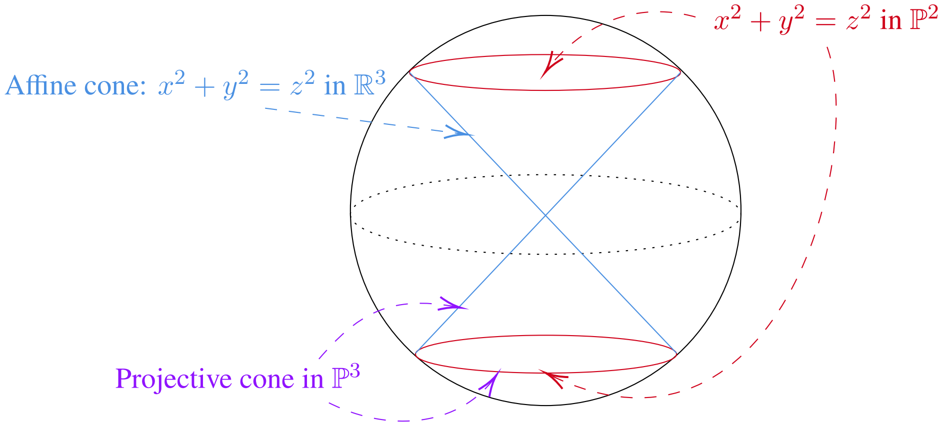


Figure 5.10: The affine and projective cone of $x^2+y^2=z^2$ in classical geometry

5.5.2 Projective schemes, a first description

We now describe a construction of projective schemes, which will help motivate the Proj construction. We begin by giving an algebraic interpretation of the cone just described, and more generally, getting some practice with transforming between projective coordinates, and coordinates on the standard affine open subsets. We switch coordinates from x, y, z to x_0, x_1, x_2 in order to use the notation of §5.4.3. For the next questions, $\mathbb{P}^2_k$, we take projective coordinates x_0, x_1, x_2. The big open set (or "coordinate chart") $U_0 = \{[x_0 : x_1 : x_2] : x_0 \neq 0\}$ has coordinates $x_{1/0}$ and $x_{2/0}$, which we interpret as x_1/x_0 and x_2/x_0. We have similar definitions for U_1 and U_2. It will be convenient to define $x_{0/0}$ as 1.

✍ **Exercise 5.5** Describe $(x_{0/1}, x_{2/1})$ in terms of $(x_{1/0}, x_{2/0})$.

Solution *Set $U_0 = \{[x_0 : x_1 : x_2] : x_0 \neq 0\}$ and $U_1 = \{[x_0 : x_1 : x_2] : x_1 \neq 0\}$. On $U_0 \cap U_1$, we have*

$$x_{0/1} = \frac{1}{x_{1/0}}, \quad x_{2/1} = \frac{x_{2/0}}{x_{1/0}}.$$

✍ **Exercise 5.6** Explain how to define a scheme that should be interpreted as $x_0^2 + x_1^2 - x_2^2 = 0$ "in $\mathbb{P}^2_k$".

Solution *Say X be a scheme which is interpreted as $x_0^2 + x_1^2 - x_2^2 = 0$ "in $\mathbb{P}^2_k$". Let*

$$X_2 := \operatorname{Spec}(k[x_{0/2}, x_{1/2}]/(x_{0/2}^2 + x_{1/2}^2 - 1)),$$
$$X_1 := \operatorname{Spec}(k[x_{0/1}, x_{2/1}]/(x_{0/1}^2 - x_{2/1}^2 + 1)),$$
$$X_0 := \operatorname{Spec}(k[x_{1/0}, x_{2/0}]/(x_{1/0}^2 - x_{2/0}^2 + 1)).$$

We claim that $X = (X_0 \coprod X_1 \coprod X_2) / \sim$, where $\sim$ is given by $x_{k/i} = x_{k/j}/x_{i/j}$ and $x_{k/j} = x_{k/i}/x_{j/i}$. By Exercise 5.4, gluing information agrees over triple overlaps. By Theorem 5.4.1, X_i can glue together. Clearly, each X_i is an affine scheme, and therefore X is a scheme.

✍ **Exercise 5.7** Consider the parabola $x_{2/0} = x_{1/0}^2$ (or, if you prefer, $y = x^2$). How does it meet the line at infinity? (What does this mean?) What is its description in the different coordinate patches U_1 and U_2?

Solution *The parabola $x_{2/0} = x_{1/0}^2$ lives in U_0. It can be written as $x_1^2 - x_0x_2 = 0$. The infinite line is $x_0 = 0$, consider the following equations.*

$$\begin{cases} x_1^2 - x_0x_2 &= 0, \\ x_0 &= 0. \end{cases}$$

We have $x_1 = 0$. Hence, $x_{2/0} = x_{1/0}^2$ meets line at infinity at point $[0:0:1] \in \mathbb{P}^2$. Moreover, $x_{2/0} = x_{1/0}^2$ is tangent to the line at infinity at $[0:0:1]$ which intersection multiplicity is 2 (by Bezout's Theorem).

On U_1, the parabola on U_0 can be written as $x_0x_2 = 1$, it is a hyperbola. On U_2, the parabola can be written as $x_1^2 = x_0$, it is a parabola.

Remark 5.16 (**Degree d hypersurfaces in $\mathbb{P}^n$**.) The degree d homogeneous polynomials in $n+1$ variables over a field form a vector space of dimension $\binom{n+d}{d}$. It is almost true that two polynomials cut out the same subset of $\mathbb{P}_k^n$ if one is a nonzero scalar multiple of the other. Unfortunately, the examples of $x^2y = 0$ and $xy^2 = 0$ show that this isn't quite right. We will late be able to check that two polynomials cut out the same **closed subscheme** (whatever that means) if and only if one is a nonzero scalar multiple of the other. (This is some evidence that the notion of a "closed subscheme" is better than that of a "closed subset".) The zero polynomial doesn't really cut out a hypersurface in any reasonable sense of the word. Thus we informally imagine that "degree d hypersurfaces in $\mathbb{P}^n$ are parametrized by $\mathbb{P}^{\binom{n+d}{d}-1}$". This intuition will come up repeatedly (in special cases), and we will give it precise meaning in Chapter 26. (We will properly define **hypersurfaces** in Chapter 10, once we have the language of closed subschemes. At that time we will also define line, hyperplane, quadric hypersurfaces, conic curves, and other wondrous notions.)

✍ **Exercise 5.8** Consider $\mathbb{P}_A^n$, with projective coordinates $x_0, \ldots, x_n$. Given a collection of homogeneous polynomials $f_i \in A[x_0, \ldots, x_n]$, make sense of the scheme "cut out in $\mathbb{P}_A^n$ by the f_i." (This will later be made precise as an example of a "vanishing scheme".)

Solution *Let $V_i = \operatorname{Spec}\left(A[x_{0/i}, \ldots, x_{n/i}] \middle/ \left(\frac{f_1}{x_i^{\deg f_1}}, \ldots, \frac{f_m}{x_i^{\deg f_m}}\right)\right)$, define the scheme "cut out in $\mathbb{P}_A^n$ by the f_i" by setting*

$$V = \left(\coprod_i V_i\right) \Big/ \sim,$$

where $\sim$ is given by $x_{k/i} = x_{k/j}/x_{i/j}$ and $x_{k/j} = x_{k/i}/x_{j/i}$. By the construction, V is indeed a scheme, and each f_i vanished on V.

Remark 5.17 This could be taken as the definition of a **projective A-scheme**.

5.5.3 Preliminaries on graded rings

The Proj construction produces a scheme out of a graded ring. We now give some background on graded rings.

Definition 5.5.1 ($\mathbb{Z}$-graded rings, Homogeneous ideals)

*A $\mathbb{Z}$-**graded ring** is a ring $S_\bullet = \bigoplus_{n\in\mathbb{Z}} S_n$ (the subscript is called the **grading**), where multiplication respects the grading, i.e., $S_m S_n \subseteq S_{m+n}$. Clearly S_0 is a subring, each S_n is an S_0-module, and $S_\bullet$ is a S_0-algebra. Those elements of some S_n are called **homogeneous elements** of $S_\bullet$.*
*An ideal I of $S_\bullet$ is a **homogeneous ideal** (or a **graded ideal**) if it is generated by homogeneous elements. Nonzero homogeneous elements have an obvious **degree**. An element of S_d is called a **form of degree** d.*

Proposition 5.5.1

(a) Ideal I is homogeneous if and only if it contains the degree n piece of each of its elements for each n. (Hence, we can decomposed into homogeneous pieces, $I = \bigoplus I_n$, and $S_\bullet/I$ has a natural $\mathbb{Z}$-grading. This is the reason for the name homogeneous ideal.)
(b) The set of homogeneous ideals of a given $\mathbb{Z}$-graded ring $S_\bullet$ is closed under sum, product, intersection, and radical.
(c) A homogeneous ideals $I \subseteq S_\bullet$ is prime if $I \neq S_\bullet$, and if for any homogeneous $a, b \in S_\bullet$, if $ab \in I$, then $a \in I$ or $b \in I$.

Proof

(a) Let I be an homogeneous ideal of graded ring $S_\bullet$, then I is generated by homogeneous elements, say $I = (f_i)_{i\in\mathscr{I}}$, where $\mathscr{I}$ is an index set and $f_i \in S_{d_i}$ for some d_i. Let $f \in I$, we may write $f = \sum_i g_i f_i$, where $g_i \in S_\bullet$. Since $g_i \in S_\bullet$, we may write $g_i = \sum_j g_{ij}$. Hence,

$$f = \sum_i \left(\sum_j g_{ij}\right) f_i = \sum_i \sum_j g_{ij} f_i,$$

clearly, each $g_{ij} f_i \in I$, then we done.
Conversely, if for all $f = \sum_i f_i \in I$ where $f_i \in S_i$, we have $f_i \in I$. We want to show that I is generated by homogeneous elements. Since each elements in I is generated by homogeneous elements which are all belong to I, I is generated by homogeneous elements, and therefore is a homogeneous ideal.
Say $I_n = I \cap S_n$, then we have $I = \bigoplus_n I_n$. Hence,

$$S_\bullet/I = \left(\bigoplus_n S_n\right) \Big/ \left(\bigoplus_n I_n\right) = \bigoplus_n (S_n/I_n).$$

Note that $(S_n/I_n)(S_m/I_m) \subseteq S_{n+m}/I_{n+m}$, $S_\bullet/I$ is a graded ring.

(b) Let $\mathscr{S}$ be the set of homogeneous ideals of a given $\mathbb{Z}$-graded ring $S_\bullet$. Let $I_1, I_2 \in \mathscr{S}$.

(i) $I_1 + I_2$ is homogeneous ideal.
Let $f + g \in I_1 + I_2$, where $f \in I_1$ and $g \in I_2$, then we may write $f = \sum_i f_i$ and $g = \sum_i g_i$, where $f_i, g_i \in S_i$. Hence, $f + g = \sum_i (f_i + g_i)$. Since I_1 and I_2 are homogeneous ideals, we have $f_i \in I_1$ and $g_i \in I_2$, and therefore $f_i + g_i \in I_1 + I_2$ for all i. By part (a), $I_1 + I_2$ is homogeneous ideal.

(ii) $I_1 I_2$ is homogeneous ideal.
Let $f = \sum_i f_i \in I_1$ and $g = \sum_i g_i \in I_2$, where $f_i, g_i \in S_i$, then

$$fg = \sum_{i,j} f_i g_j.$$

Since Since I_1 and I_2 are homogeneous ideals, we have $f_i \in I_1$ and $g_i \in I_2$, and therefore $f_i g_j \in I_1 I_2$ for all i. By part (a), $I_1 I_2$ is homogeneous ideal.

(iii) $I_1 \cap I_2$ is homogeneous ideal.

Let $f = \sum_i f_i \in I_1 \cap I_2$, since I_1 and I_2 are both homogeneous ideal $f_i \in I_1$ and $f_i \in I_2$. It follows that $f_i \in I_1 \cap I_2$. By part (a), $I_1 \cap I_2$ is homogeneous ideal.

(iv) $\sqrt{I}$ is homogeneous ideal, where $I \in \mathscr{S}$.

Let $f = \sum_i f_i \in \sqrt{I}$, then $f^n \in I$. Note that

$$f^n = \left(\sum_i f_i\right)^n = \sum_{i_0+i_1+\cdots=n} f_0^{i_0} f_1^{i_1} f_2^{i_2} \cdots,$$

we want to show that each $f_0^{i_0} f_1^{i_1} \cdots$ is belong to $\sqrt{I}$. Let f_{n_0} is the first term which is not zero in f, then the lowest term in f^n is $f_{n_0}^n$. Since $f^n \in I$, $f_{n_0}^n \in I$, and therefore $f_{n_0} \in \sqrt{I}$. Hence, $f - f_{n_0} \in \sqrt{I}$. By induction each term of f belong to $\sqrt{I}$, by part (a), it follows that $\sqrt{I}$ is a homogeneous ideal.

(c) If I is not a homogeneous ideal, then there exists $f \notin I$ and $g \notin I$ such that $fg \in I$. We may assume that

$$f = \sum_i f_i, \quad g = \sum_j g_j,$$

then

$$fg = \sum_i \sum_j f_i g_j.$$

Since $f \notin I$, there exists $f_{i_0} \notin I$, similarly, exists $g_{j_0} \notin I$. Since $fg \in I$, we have $f_i g_j \in I$ for all i, j, in particular, $f_{i_0} g_{j_0} \in I$. By the hypothesis, we have $f_{i_0} \in I$ and $g_{j_0} \in I$, which a contradiction.

□

Proposition 5.5.2

If T is a multiplicative subset of $S_\bullet$ containing only homogeneous elements, then $T^{-1}S_\bullet$ has a natural structure as a $\mathbb{Z}$-graded ring.

Proof For each $\frac{a}{t} \in T^{-1}S_\bullet$, since $a \in S_\bullet$, we may write $a = \sum_i a_i$ where $a_i \in S_i$, hence,

$$\frac{a}{t} = \sum_i \frac{a_i}{t}.$$

Let

$$(T^{-1}S_\bullet)_n = \left\{\frac{a}{t} : a \in S_{n+\deg t}, t \in T\right\},$$

then $T^{-1}S_\bullet$ has a natural structure as a $\mathbb{Z}$-graded ring:

$$T^{-1}S_\bullet = \bigoplus_{n\in\mathbb{Z}} (T^{-1}S_\bullet)_n.$$

□

5.5.4 $\mathbb{Z}^{\geq 0}$-graded rings, graded ring over A, and finitely generated graded rings

Definition 5.5.2 ($\mathbb{Z}^{\geq 0}$-graded rings)

*A $\mathbb{Z}^{\geq 0}$-**graded ring** is a $\mathbb{Z}$-graded ring with no elements of negative degree.*

Note ***For the remainder of this note, graded ring will refer to a $\mathbb{Z}^{\geq 0}$-graded ring. Warning: this convention is nonstandard (for good reason).***

Note *From now on, unless otherwise stated, $S_\bullet$ is assumed to be a graded ring.*

Definition 5.5.3 (Graded ring over base ring)

*Fix a ring A, which we call the **base ring**. If $S_0 = A$, we say that $S_\bullet$ is a **graded ring over** A.*

Example 5.8 A key example of Definition 5.5.3 is $A[x_0, \ldots, x_n]$, or more generally $A[x_0, \ldots, x_n]/I$ where I is a homogeneous ideal with $I \cap S_0 = 0$. Here we take the conventional grading on $A[x_0, \ldots, x_n]$, where each x_i has weight 1 (i.e., $\deg x_i = 1$).

Definition 5.5.4 (Irrelevant ideal, Finitely generated graded ring)

*The subset $S_+ := \bigoplus_{i>0} S_i \subseteq S_\bullet$ is an ideal, called the **irrelevant ideal**. (The reason for the name "irrelevant" will be clearer in a few paragraphs.) If the irrelevant ideal S_+ is a finitely generated ideal, we say that $S_\bullet$ is a **finitely generated graded ring over** A. If $S_\bullet$ is generated by S_1 as an A-algebra, we say that $S_\bullet$ is **generated in degree** 1.*

Remark 5.18 We will later find it useful to interpret "$S_\bullet$ is generated in degree 1" as "the natural map $\mathrm{Sym}^\bullet S_1 \to S_\bullet$ is a surjection". The symmetric algebra construction will be briefly discussed in Chapter 15.

Lemma 5.5.1

S_+ is a homogeneous ideal.

Proof Each element in S_+ can be written as $f = \sum_{i \geq 1} f_i$, note that each $f_i \in S_i$, hence, S_+ is a homogeneous ideal. □

Proposition 5.5.3

(a) A graded ring $S_\bullet$ over A is a finitely generated graded ring (over A) if and only if $S_\bullet$ is a finitely generated graded A-algebra, i.e., generated over $A = S_0$ by a finite number of homogeneous elements of positive degree.

(b) A graded ring $S_\bullet$ over A is Noetherian if and only if $A = S_0$ is Noetherian and $S_\bullet$ is a finitely generated graded ring.

Proof

(a) If graded ring $S_\bullet$ over A is a finitely generated graded ring, by definition, $S_+ \subseteq S_\bullet$ is an finitely generated ideal, say $S_+ = (f_{i_1}, f_{i_2}, \ldots, f_{i_n})$, where $f_{i_j} \in S_{i_j}$ is homogeneous element. Let $g \in S_\bullet$, then x can be written as

$$g = \sum_{i \geq 0} g_i = g_0 + \sum_{i \geq 1} g_i.$$

Note that $\sum_{i \geq 1} g_i$ belong to S_+, we may assume that

$$\sum_{i \geq 1} g_i = \sum_{k=1}^{n} c_k f_{i_k},$$

where $c_k \in S_\bullet$. Each c_k can be written as

$$c_k = c_{k,0} + \sum_{t \geq 0} c_{k,t},$$

then we have

$$\begin{aligned} g =& g_0 + \sum_{k=1}^{n} (c_{k,0} + \sum_{t \geq 0} c_{k,t}) f_{i_k} \\ =& g_0 + \sum_{k=1}^{n} c_{k,0} f_{i_k} + \sum_{k=1}^{n} \sum_{t \geq 0} c_{k,t} f_{i_k}. \end{aligned}$$

Repeat above process, since in $\{g_i\}_{i\geq 0}$ there are only finite terms not zero, the whole process stops after a finite number of steps. It follows that g can be written as a polynomial of $f_{i_1}, f_{i_2}, \ldots, f_{i_n}$ with coefficient in A. It follows that $S_\bullet$ is a finitely generated graded A-algebra.

Conversely, if $S_\bullet$ is a finitely generated graded A-algebra. We want to show that $S_\bullet$ over A is a finitely generated ring, it suffices to show that S_+ is an finitely generated ideal of $S_\bullet$. Each element in $S_\bullet$ can be written as an element in $A[f_{i_1}, f_{i_2}, \ldots, f_{i_n}]$, where $f_{i_k} \in S_{i_k}$. Let $g \in S_+$, then

$$g = \sum_{i \geq 1} g_i,$$

where $g_i \in S_i$ and almost every g_i is zero. Each g_i can by written as a polynomial in $A[f_{i_1}, \ldots, f_{i_n}]$ with no terms of g_i belong to A, i.e.,

$$g_i = \sum_{k_1,\ldots,k_n} a_{k_1,\ldots,k_n} f_{i_1}^{k_1} \cdots f_{i_n}^{k_n} = h_1 f_{i_1} + h_2 f_{i_2} + \cdots + h_n f_{i_n},$$

where $h_t \in S_\bullet$. Hence, $S_+ \in (f_{i_1}, \ldots, f_{i_n})$. Clearly, $(f_{i_1}, \ldots, f_{i_n}) \subseteq S_+$, we have $S_+ = (f_{i_1}, \ldots, f_{i_n})$, and therefore $S_\bullet$ over A is a finitely generated graded ring.

(b) If graded ring $S_\bullet$ over A is Noetherian, then S_+ is a finitely generated ideal, and therefore $S_\bullet$ is a finitely generated graded ring over A. Note that $A = S_0 \cong S_\bullet / S_+$, since $S_\bullet$ is Noetherian, $A = S_0$ is Noetherian (Proposition 4.6.14).

Conversely, if $S_\bullet$ is a finitely generated graded ring, by part (a), $S_\bullet$ is a finitely generated graded A-algebra, we may assume that $S_\bullet = A[f_{i_1}, \ldots, f_{i_n}]$, where each f_{i_k} is homogeneous element. Note that A is Noetherian, by Hilbert's Basis Theorem, $S_\bullet$ is Noetherian.

□

5.5.5 The Proj construction

We now define a scheme $\operatorname{Proj} S_\bullet$, where $S_\bullet$ is a ($\mathbb{Z}^{\geq 0}$-)graded ring. Here are two examples, to provide a light at the end of the tunnel. If $S_\bullet = A[x_0, \ldots, x_n]$, we will recover $\mathbb{P}^n_A$; and if $S_\bullet = A[x_0, \ldots, x_n]/(f(x_0, \ldots, x_n))$ where f is homogeneous, we will construct something "cut out in $\mathbb{P}^n_A$ by the equation $f = 0$".

As we did with Spec of a ring, we will build $\operatorname{Proj} S_\bullet$ first as a set, then as a topological space, and finally as a ringed space. In our preliminary definition of $\mathbb{P}^n_A$, we glued together $n+1$ well-chosen affine pieces, but we don't want to make any choices, but we don't want to make any choices, so we do this by simultaneously consider "all possible" affine open sets. Our affine building blocks will be as follows. For each homogeneous $f \in S_+$, note that the localization $(S_\bullet)_f$ is naturally a $\mathbb{Z}$-graded ring, where $\deg(1/f) = -\deg f$. Consider

$$\operatorname{Spec}((S_\bullet)_f)_0, \tag{5.12}$$

where $((S_\bullet)_f)_0$ means the 0-graded piece of the graded ring $(S_\bullet)_f$. (These will be our affine building blocks, as f varies over the homogeneous elements of S_+.) The notation $((S_\bullet)_f)_0$ is admittedly horrible — the first and third subscripts refer to the grading, and the second refers to localization. As motivation for considering this construction: applying this to $S_\bullet = k[x_0, \ldots, x_n]$, with $f = x_i$, we obtain the ring appearing in (5.11):

$$k[x_{0/i}, x_{1/i}, \ldots, x_{n/i}]/(x_{i/i} - 1).$$

(Before we begin the construction: another possible way of defining $\operatorname{Proj} S_\bullet$ is by gluing together affines of this form.)

Definition 5.5.5 (The points of $\operatorname{Proj} S_\bullet$)

*The **points** of $\operatorname{Proj} S_\bullet$ are the homogeneous prime ideals of $S_\bullet$ not containing the irrelevant ideal S_+ (the "relevant homogeneous prime ideals").*

Proposition 5.5.4

Suppose $f \in S_+$ is homogeneous.

(a) There is a bijection between $\operatorname{Spec}((S_\bullet)_f)_0$ and the homogeneous prime ideals of $(S_\bullet)_f$.

(b) $\operatorname{Spec}((S_\bullet)_f)_0$ is a subset of $\operatorname{Proj} S_\bullet$.

Proof

(a) Let $d = \deg f$. Let $\mathfrak{q}$ be the homogeneous prime ideals of $(S_\bullet)_f$, note that we have a natural embedding $((S_\bullet)_f)_0 \hookrightarrow (S_\bullet)_f$, then $\mathfrak{q} \cap ((S_\bullet)_f)_0$ is a prime ideal of $((S_\bullet)_f)_0$, i.e., $\mathfrak{q} \cap ((S_\bullet)_f)_0 \in \operatorname{Spec}((S_\bullet)_f)_0$. Conversely, let $\mathfrak{p} \in \operatorname{Spec}((S_\bullet)_f)_0$. Define $\mathfrak{q} = \bigoplus_i \mathfrak{q}_i$, where $\mathfrak{q}_i \subseteq ((S_\bullet)_f)_i$ and $a \in \mathfrak{q}_i$ if and only if $a^d/f^i \in \mathfrak{p}$. It is easy to see that $\mathfrak{p} = \mathfrak{q}_0$. We want to show that $\mathfrak{q} = \bigoplus_i \mathfrak{q}_i$ is a homogeneous prime ideals of $(S_\bullet)_f$.

(i) $\mathfrak{q}$ is an ideal of $(S_\bullet)_f$.

Let $a \in \mathfrak{q}_i$, then $a \in \mathfrak{q}_i$ iff $\frac{a^d}{f^i} \in \mathfrak{p}$ iff $\left(\frac{a^d}{f^i}\right)^2 \in \mathfrak{p}$ iff $a^2 \in \mathfrak{q}_{2i}$.

Let $a_1, a_2 \in \mathfrak{q}_i$, we want to show that $a_1 + a_2 \in \mathfrak{q}_i$, it suffices to show that $(a_1+a_2)^{2d}/f^{2i} \in \mathfrak{p}$. In fact,

$$\begin{aligned}\frac{(a_1+a_2)^{2d}}{f^{2i}} &= \frac{\sum_{k=0}^{2d} \binom{2d}{k} a_1^k a_2^{2d-k}}{f^{2i}} \\ &= \frac{a_2^{2d}}{f^{2i}} + \sum_{k=1}^{d} \binom{2d}{k} \frac{a_1^k a_2^{d-k}}{f^i} \cdot \frac{a_2^d}{f^i} + \sum_{k=d+1}^{2d-1} \binom{2d}{k} \frac{a_1^{k-d} a_2^{2d-k}}{f^i} \cdot \frac{a_1^d}{f^i} + \frac{a_1^{2d}}{f^{2i}}.\end{aligned}$$

For $1 \leq k \leq d$, $a_1^k a_2^{d-k} \in S_{ki} S_{(d-k)i} \subseteq S_{di}$, and therefore $\binom{2d}{k} \frac{a_1^k a_2^{d-k}}{f^i} \in (S_d)_f$. Similarly, for $d+1 \leq k \leq 2d-1$, $\binom{2d}{k} \frac{a_1^{k-d} a_2^{2d-k}}{f^i} \in (S_d)_f$. Note that $\frac{a_2^{2d}}{f^{2i}}, \frac{a_2^d}{f^i}, \frac{a_1^d}{f^i}, \frac{a_1^{2d}}{f^{2i}} \in \mathfrak{p}$, we have

$$\frac{(a_1+a_2)^{2d}}{f^{2i}} \in \mathfrak{p}.$$

Hence, $(a_1+a_2)^2 \in \mathfrak{q}_{2i}$, and therefore $a_1 + a_2 \in \mathfrak{q}_i$.

Let $a \in \mathfrak{q}_i$, then $\frac{a^d}{f^i} \in \mathfrak{p}$, hence, $(-1)^d \frac{a^d}{f^i} = \frac{(-a)^d}{f^i} \in \mathfrak{p}$, i.e., $-a \in \mathfrak{q}_i$.

By above discussion, $\mathfrak{q} = \bigoplus_i \mathfrak{q}_i$ is an ideal of $(S_\bullet)_f$.

(ii) $\mathfrak{q}$ is a homogeneous ideal of $(S_\bullet)_f$.

Since each $\mathfrak{q}_i \subseteq ((S_\bullet)_f)_i$, clearly, $\mathfrak{q}$ is a homogeneous ideal.

(iii) $\mathfrak{q}$ is prime ideal.

For any $a \in ((S_\bullet)_f)_i$ and $b \in ((S_\bullet)_f)_j$, suppose $ab \in \mathfrak{q}$. We want to show that $a \in \mathfrak{q}$ or $b \in \mathfrak{q}$. We may assume the $b \notin \mathfrak{q}$. Let $a = \frac{s_{i+n}}{f^n}$ and $b = \frac{s_{j+m}}{f^m}$, then $ab = \frac{s_{i+n} s_{j+m}}{f^{m+n}} \in \mathfrak{q}_{i+j}$, hence, $\frac{(ab)^d}{f^{i+j}} \in \mathfrak{p}$. Since $b \notin \mathfrak{q}$, we have $b \notin \mathfrak{q}_j$, i.e., $\frac{b^d}{f^j} \notin \mathfrak{p}$. Since $\mathfrak{p}$ is a prime ideal, we have $\frac{a^d}{f^i} \in \mathfrak{p}$, i.e., $a \in \mathfrak{q}_i \subseteq \mathfrak{q}$.

By Proposition 5.5.1, we done!

Define φ as follow:

$$\varphi : \operatorname{Spec}((S_\bullet)_f)_0 \longrightarrow \{\text{the homogeneous prime ideals of } (S_\bullet)_f\}$$
$$\mathfrak{p} \longmapsto \mathfrak{q} = \bigoplus_i \mathfrak{q}_i,$$

where $\mathfrak{q}_i \subseteq (S_{i+d})_f$ and $a \in \mathfrak{q}_i$ if and only if $\frac{a^d}{f^i} \in \mathfrak{p}$. Define ψ as follow:

$$\psi : \{\text{the homogeneous prime ideals of } (S_\bullet)_f\} \longrightarrow \operatorname{Spec}((S_\bullet)_f)_0$$
$$\mathfrak{q} \longmapsto \mathfrak{q} \cap ((S_\bullet)_f)_0.$$

By above discussion, φ and ψ are all well-defined.

Let $\mathfrak{p} \in \operatorname{Spec}((S_\bullet))_0$, then

$$\psi \circ \varphi(\mathfrak{p}) = \psi(\cdots \oplus \mathfrak{q}_0 \oplus \cdots) = \mathfrak{q}_0 = \mathfrak{p}.$$

Let $\mathfrak{q} = \bigoplus_i \mathfrak{p}_i \in \{\text{the homogeneous prime ideals of } (S_\bullet)_f\}$, then

$$\varphi \circ \psi(\mathfrak{q}) = \varphi(\mathfrak{q}_0),$$

we want to show that $\varphi(\mathfrak{q}_0) = \mathfrak{q}$. It suffices to check that for each $\mathfrak{q}_i$, $a \in \mathfrak{q}_i$ if and only if $\frac{a^d}{f^i} \in \mathfrak{q}_0$. Let $a \in \mathfrak{q}_i$, then $a^d \in \mathfrak{q}_{id}$, and therefore $\frac{a^d}{f^i} \in \mathfrak{q}_0$. Conversely, if $\frac{a^d}{f^i} \in \mathfrak{q}_0 \subseteq \mathfrak{q}$, since $f \notin \mathfrak{q}$, $a^d \in \mathfrak{q}$. Note that $\mathfrak{q}$ is prime ideal, we have $a \in \mathfrak{q}$. Since $d \cdot \deg a - di = 0$, i.e., $\deg a = i$, hence, $a \in \mathfrak{q}_i$. Thus,

$$\varphi \circ \psi(\mathfrak{q}) = \mathfrak{q}.$$

It follows that there is a bijection

$$\{\text{the homogeneous prime ideals of } (S_\bullet)_f\} \longleftrightarrow \operatorname{Spec}((S_\bullet)_f)_0.$$

(b) It suffices to check each element in

$$\mathscr{S} = \{\text{the homogeneous prime ideals of } (S_\bullet)_f\}$$

not containing the irrelevant ideal S_+. Let $\mathfrak{q} = \bigoplus_i \mathfrak{q}_i \in \mathscr{S}$. $\mathscr{S}$ as a subset of $\operatorname{Proj} S_\bullet$, $\mathfrak{q}$ must not containing $f \in S_+$, and therefore not containing S_+. It follows that

$$\operatorname{Spec}((S_\bullet)_f)_0 \longleftrightarrow \mathscr{S} \hookrightarrow \operatorname{Proj} S_\bullet,$$

as we desired.

□

The correspondence of the points of $\operatorname{Proj} S_\bullet$ with homogeneous prime ideals helps us picture $\operatorname{Proj} S_\bullet$. For example, if $S_\bullet = k[x, y, z]$ with the usual grading, then we picture the homogeneous prime ideal $(z^2 - x^2 - y^2)$ first as a subset of $\operatorname{Spec} S_\bullet$; it is a cone (see Figure 5.10). As in §5.5.1, we picture $\mathbb{P}^2_k$ as the "plane at infinity". Thus we picture this equation as cutting out a conic "at infinity" (in $\operatorname{Proj} S_\bullet$). We will make this intuition somewhat more precise in Chapter 10.

Motivated by the affine case, we can define **projective vanishing set** and **projective distinguished open set**:

Definition 5.5.6 (Projective vanishing set, projective distinguished open set)

If T is a set of homogeneous elements of $S_\bullet$, define the ***projective vanishing set of*** *T, $V(T) \subseteq \operatorname{Proj} S_\bullet$, to be those homogeneous prime ideals containing T but not S_+, i.e.,*

$$V(T) := \{[\mathfrak{p}] \in \operatorname{Proj} S_\bullet : \mathfrak{p} \supseteq T, \mathfrak{p} \not\supseteq S_+\}.$$

Define $V(f) = \{[\mathfrak{p}] \in \operatorname{Proj} S_\bullet : \mathfrak{p} \ni f, \mathfrak{p} \not\supseteq S_+\}$ if f is a homogeneous element of positive degree.

*Let the **projective distinguished open set** $D_+(f) := \operatorname{Proj} S_\bullet \setminus V(f)$ be the complement of $V(f)$. (The subscript is intended to distinguish this notation from the similar $D(f)$.)*

Remark 5.19 Once we define a scheme structure on $\operatorname{Proj} S_\bullet$, we will (without comment) use $D_+(f)$ to refer to the *open subscheme*, not just the open subset. Although the definition of $D_+(f)$ makes sense even if f has degree 0, we deliberately allow only f of positive degree. For example, we will want the $D_+(f)$ to form an affine cover, and if f has degree 0, then $D_+(f)$ needn't be affine.

Exercise 5.9 Show that $D_+(f)$ "is" (or more precisely, "corresponds to") the subset $\operatorname{Spec}((S_\bullet)_f)_0$ you described in Proposition 5.5.4 (b). For example, in §5.4.3, the $D_+(x_i)$ are the standard open sets covering projective space.

Proof By the definition,

$$D_+(f) = \operatorname{Proj} S_\bullet \setminus V(f) = \{[\mathfrak{p}] \in \operatorname{Proj} S_\bullet : f \notin \mathfrak{p}, \mathfrak{p} \not\supseteq S_+\}.$$

Since $f \notin \mathfrak{p}$, $\mathfrak{p}$ can be seen as a homogeneous prime ideal of $(S_\bullet)_f$, by Proposition 5.5.4 (a), $\mathfrak{p}$ corresponds to $\mathfrak{p} \cap ((S_\bullet)_f)_0$ in $\operatorname{Spec}((S_\bullet)_f)_0$, i.e.

$$D_+(f) \cong \operatorname{Spec}((S_\bullet)_f)_0 \hookrightarrow \operatorname{Proj} S_\bullet,$$

then we done. □

Definition 5.5.7 (Zariski topology on $\operatorname{Proj} S_\bullet$)

*As in the affine case, the $V(I)$'s satisfy the axioms of the closed sets of a topology and we call this **Zariski topology** on $\operatorname{Proj} S_\bullet$.*

Remark 5.20 Other definitions given in the literature may look superficially different, but can be easily shown to be the same.

Many statements about the Zariski topology on Spec of a ring carry over to this situation with little extra work.

Lemma 5.5.2

$D_+(f) \cap D_+(g) = D_+(fg)$.

Proof

$$\begin{aligned} D_+(f) \cap D_+(g) =& \{[\mathfrak{p}] \in \operatorname{Proj} S_\bullet : f \notin \mathfrak{p}, g \notin \mathfrak{p}, \mathfrak{p} \not\supseteq S_+\} \\ \subseteq& \{[\mathfrak{p}] \in \operatorname{Proj} S_\bullet : fg \notin \mathfrak{p}, \mathfrak{p} \not\supseteq S_+\} \\ =& D_+(fg). \end{aligned}$$

Conversely, let $[\mathfrak{p}] \in D(fg)$, then $fg \notin \mathfrak{p}$ and $\mathfrak{p} \not\supseteq S_+$. If $f \in \mathfrak{p}$, since $\mathfrak{p}$ is an ideal, $fg \in \mathfrak{p}$, a contradiction. Hence, $f \notin \mathfrak{p}$, similarly, $g \notin \mathfrak{p}$. It follows that

$$D_+(f) \cap D_+(g) = D_+(fg).$$

□

Proposition 5.5.5

The projective distinguished open sets $D_+(f)$ (as f runs through the homogeneous elements of S_+) form a base of the Zariski topology.

Proof Let $U = \operatorname{Proj} S_\bullet \setminus V(T)$ be any open set, where $T \subseteq \operatorname{Proj} S_\bullet$. Then

$$\begin{aligned} U =& \{[\mathfrak{p}] \in \operatorname{Proj} S_\bullet : \mathfrak{p} \not\supseteq T, \mathfrak{p} \not\supseteq S_+\} \\ =& \bigcup_{f \in T} \{[\mathfrak{p}] \in \operatorname{Proj} S_\bullet : f \notin \mathfrak{p}, \mathfrak{p} \not\supseteq S_+\} \\ =& \bigcup_{f \in T} D_+(f). \end{aligned}$$

It follows that the projective distinguished open sets $D_+(f)$ (as f runs through the homogeneous elements of S_+) form a base of the Zariski topology. □

Proposition 5.5.6

Let $S_\bullet$ be a graded ring. Suppose I is any homogeneous ideal of $S_\bullet$ contained in S_+, and f is a homogeneous element of positive degree. Then f vanishes on $V(I)$ (i.e., $V(I) \subseteq V(f)$) if and only if $f^n \in I$ for some n. In particular, $D_+(f) \subseteq D_+(g)$ if and only if $f^n \in (g)$ for some n. (Here g is also homogeneous of positive degree.)

Proof Let f be a homogeneous element of positive degree and $I \subseteq S_+$ is any homogeneous ideal of $S_\bullet$.

If f vanishes on $V(I)$. Suppose that $f^n \notin I$ for any n, i.e., $f \notin \sqrt{I}$, hence, there exist a prime ideal $\mathfrak{p} \supseteq I$ such that $f \notin \mathfrak{p}$. Let $\mathfrak{p}^* =$ (homogeneous elements of $\mathfrak{p}$), clearly, $\mathfrak{p}^*$ is a homogeneous prime ideal which contains I. We want to show that $\mathfrak{p}^*$ is prime. Let a, b be two homogeneous elements such that $a \notin \mathfrak{p}^*$ and $b \notin \mathfrak{p}^*$, then $a \notin \mathfrak{p}$ and $b \notin \mathfrak{p}$, since $\mathfrak{p}$ is prime, homogeneous element $ab \notin \mathfrak{p}$, and therefore $ab \notin \mathfrak{p}^*$, by Proposition 5.5.1 (c), $\mathfrak{p}^*$ is prime. Then we get a homogeneous prime ideal $\mathfrak{p}^*$ which contains I. Since f is a homogeneous element, $f \notin \mathfrak{p}^*$, and therefore $\mathfrak{p}^* \not\supseteq S_+$, hence, $\mathfrak{p}^* \in \operatorname{Proj} S_\bullet$. Since $\mathfrak{p}^* \supseteq I$, $[\mathfrak{p}^*] \in V(I)$. Since f vanishes on $V(I)$, we have $f \in \mathfrak{p}^*$, but $f \notin \mathfrak{p}^*$, a contradiction. Hence, $f \in \sqrt{I}$.

Conversely, if $f \in \sqrt{I}$. We want to show that $f \in \mathfrak{p}$ for all $[\mathfrak{p}] \in V(I)$. Note that

$$\sqrt{I} = \bigcap_{[\mathfrak{p}] \in \operatorname{Spec} I} \mathfrak{p} \subseteq \bigcap_{[\mathfrak{p}] \in V(I)} \mathfrak{p},$$

we have $f \in \mathfrak{p}$ for all $[\mathfrak{p}] \in V(I)$, i.e., f vanish on $V(I)$.

In particular, $D_+(f) \subseteq D_+(g)$ if and only if $V(f) \supseteq V(g)$ if and only if $f^n \in I$ for some n. □

Definition 5.5.8 (Function $I(\cdot)$)

Let $S_\bullet$ be a graded ring. Given a subset $Z \subseteq \operatorname{Proj} S_\bullet$, $I(Z)$ is the set of functions vanishing on S. In other words,

$$I(Z) := \{f \in S_\bullet : f \in \mathfrak{p}, \forall [\mathfrak{p}] \in Z\} = \bigcap_{[\mathfrak{p}] \in Z} \mathfrak{p} \subseteq S_\bullet.$$

Proposition 5.5.7

Let $S_\bullet$ be a graded ring.

(a) If $Z \subseteq \operatorname{Proj} S_\bullet$, then $I(Z)$ is a homogeneous ideal of $S_\bullet$.

(b) For any two subset, $I(Z_1 \cup Z_2) = I(Z_1) \cap I(Z_2)$.

(c) For any subset $Z \subseteq \operatorname{Proj} S_\bullet$, $V(I(Z)) = \overline{Z}$.

Proof

(a) Let $f \in I(Z)$, then $f \in \mathfrak{p}$ for all $[\mathfrak{p}] \in Z$. Write f as $f = \sum_{i \geq 0} f_i$. Since each $[\mathfrak{p}] \in Z$ as an ideal is homogeneous prime ideal, each $f_i \in \mathfrak{p}$ for all $[\mathfrak{p}] \in Z$. Hence, $f_i \in I(Z)$, it follows that $I(Z)$ is a homogeneous ideal.

(b) Let Z_1, Z_2 be any two subset. Let $f \in I(Z_1 \cup Z_2)$, then $f \in \mathfrak{p}$ for all $[\mathfrak{p}] \in Z_1 \cup Z_2$, hence $f \in \mathfrak{p}$ for all $[\mathfrak{p}] \in Z_1$ and $f \in \mathfrak{p}$ for all $[\mathfrak{p}] \in Z_2$, that is, $I(Z_1 \cup Z_2) \subseteq I(Z_1) \cap I(Z_2)$.
Conversely, let $f \in I(Z_1) \cap I(Z_2)$, then $f \in \mathfrak{p}$ for all $\mathfrak{p} \in Z_1$ and $f \in \mathfrak{p}$ for all $\mathfrak{p} \in Z_2$, i.e., $f \in \mathfrak{p}$ for all $\mathfrak{p} \in Z_1 \cup Z_2$. Hence, $I(Z_1) \cap I(Z_2) \subseteq I(Z_1 \cup Z_2)$.
By above discussion, we have $I(Z_1 \cup Z_2) = I(Z_1) \cap I(Z_2)$.

(c) Let $[\mathfrak{p}] \in Z$, then $f \in \mathfrak{p}$ for all $f \in I(Z)$, hence, $\mathfrak{p} \supseteq I(Z)$, and therefore $Z \subseteq V(I(Z))$. Since $V(I(Z))$ is closed, we have $\overline{Z} \subseteq V(I(Z))$.
Let $V(J)$ be any closed subset which contains Z, then for all $[\mathfrak{p}] \in Z$ we have $\mathfrak{p} \supseteq J$. It follows that
$$J \subseteq \bigcap_{[\mathfrak{p}] \in Z} \mathfrak{p} = I(Z),$$
and therefore $V(J) \supseteq V(I(Z))$. Hence,
$$\overline{Z} = \bigcap_{V(J) \supseteq Z} V(J) \supseteq V(I(Z)),$$
which implies that $V(I(Z)) = \overline{Z}$.

□

Proposition 5.5.8

Let $S_\bullet$ be a graded ring, and $I \subseteq S_+$ be a homogeneous ideal. The following are equivalent.

(a) $V(I) = \varnothing$.

(b) For any homogeneous f_i (as i runs through some index set) generating I, $\bigcup D_+(f_i) = \operatorname{Proj} S_\bullet$.

(c) $\sqrt{I} \supseteq S_+$.

Proof Since I is a homogeneous ideal, I is generated by homogeneous elements, say $I = (f_i)_{i \in \mathscr{I}}$, where $\mathscr{I}$ is an index set.

(a) $\Rightarrow$ (b): Clearly, $D_+(f_i) \subseteq \operatorname{Proj} S_\bullet$, then $\bigcup_{i; \in \mathscr{I}} D_+(f_i) \subseteq \operatorname{Proj} S_\bullet$. Conversely, let $[\mathfrak{p}] \in \operatorname{Proj} S_\bullet$. Since $V(I) = \varnothing$, $\mathfrak{p} \not\supseteq I$, then exists f_i for some $i \in \mathscr{I}$ such that $f_i \notin \mathfrak{p}$, and therefore $\mathfrak{p} \in D_+(f_i)$. It follows that $\operatorname{Proj} S_\bullet \subseteq \bigcup_i D_+(f_i)$. Hence, $\operatorname{Proj} S_\bullet = \bigcup_{i \in \mathscr{I}} D_+(f_i)$.

(b) $\Rightarrow$ (c): If there exists a homogeneous element $f \notin \sqrt{I}$, then exists a prime ideal $\mathfrak{p} \supset I$ such that $f \notin \mathfrak{p}$. By the proof in Proposition 5.5.6, we can construct a homogeneous prime ideal $\mathfrak{p}^*$ from $\mathfrak{p}$ such that $f \notin \mathfrak{p}^*$ and $\mathfrak{p}^* \supseteq I$, hence $\mathfrak{p}^* \not\supseteq S_+$, and therefore $[\mathfrak{p}^*] \in \operatorname{Proj} S_\bullet$. Note that $\operatorname{Proj} S_\bullet = \bigcup_{i \in \mathscr{I}} D_+(f_i)$, $[\mathfrak{p}^*] \in D(f_{i_0})$ for some i_0, which implies that $f_{i_0} \notin \mathfrak{p}^*$, but $I \subseteq \mathfrak{p}^*$, a contradiction! Hence, $\sqrt{I} \supseteq S_+$.

(c) $\Rightarrow$ (a): Since $\sqrt{I} \supseteq S_+$, we have $V(\sqrt{I}) \subseteq V(S_+) = \varnothing$. It suffices to show that $V(I) = V(\sqrt{I})$. Since $I \subseteq \sqrt{I}$, we have $V(\sqrt{I}) \subseteq V(I)$. Let $[\mathfrak{p}] \in V(I)$, then $\mathfrak{p} \supseteq I$. Let $f \in \sqrt{I}$, then $f^n \in I$ for some n, hence, $f^n \in \mathfrak{p}$. Since $\mathfrak{p}$ is a prime ideal, $f \in \mathfrak{p}$, it follows that $\sqrt{I} \subseteq \mathfrak{p}$, and therefore $[\mathfrak{p}] \in V(\sqrt{I})$. Hence, $V(I) = V(\sqrt{I})$. □

Remark 5.21 This is more motivation for the ideal S_+ being "irrelevant": any ideal whose radical contains it is "geometrically irrelevant".

We now construct $\operatorname{Proj} S_\bullet$ as a scheme.

Proposition 5.5.9

Let $f \in S_+$ is a homogeneous element. Via the inclusion

$$D_+(f) = \operatorname{Spec}((S_\bullet)_f)_0 \longhookrightarrow \operatorname{Proj} S_\bullet$$

of Exercise 5.9, then the Zariski topology on $\operatorname{Proj} S_\bullet$ *restricts to the Zariski topology on* $\operatorname{Spec}((S_\bullet)_f)_0$.

Proof $\operatorname{Spec}((S_\bullet)_f)_0 \hookrightarrow \operatorname{Proj} S_\bullet$ gives an induced topology on $\operatorname{Spec}((S_\bullet)_f)_0$. In fact, by Proposition 5.5.4,

$$\operatorname{Spec}((S_\bullet)_f)_0 \longleftrightarrow \{\text{the homogeneous prime ideals of } (S_\bullet)_f\},$$

the elements in $\operatorname{Spec}((S_\bullet)_f)_0$ can be seen as the homogeneous prime ideals of $(S_\bullet)_f$, moreover they can be seen as the elements in $\operatorname{Proj} S_\bullet$ which not meet f.

Let $V_{\mathrm{Spec}}(I)$ be any closed subset of $\operatorname{Spec}((S_\bullet)_f)_0$. We want to show exists a closed subset V_{Proj} of $\operatorname{Proj} S_\bullet$ such that $V_{\mathrm{Spec}}(I) = V_{\mathrm{Proj}} \cap \operatorname{Spec}((S_\bullet)_f)_0$. Consider $V_{\mathrm{Spec}}(I)$,

$$\begin{aligned} V_{\mathrm{Spec}}(I) =&\{[\mathfrak{p}] \in \operatorname{Spec}((S_\bullet)_f)_0 : \mathfrak{p} \supseteq I\} \\ =&\{[\mathfrak{p}] \in \operatorname{Proj}(S_\bullet)_f : \mathfrak{p} \supseteq \widetilde{I}\} \\ =&\{[\mathfrak{p}] \in \operatorname{Proj} S_\bullet : \mathfrak{p} \supseteq \widetilde{I} \cap S_\bullet, f \notin \mathfrak{p}\} \\ =&\operatorname{Proj} S_\bullet \cap V_{\mathrm{Proj}}(\widetilde{I} \cap S_\bullet), \end{aligned}$$

note that $V_{\mathrm{Proj}}(\widetilde{I} \cap S_\bullet)$ closed in $\operatorname{Proj} S_\bullet$, it follows that the topology on $\operatorname{Spec}((S_\bullet)_f)_0$ same as induced topology on $\operatorname{Spec}((S_\bullet)_f)_0$. □

Now that we have defined $\operatorname{Proj} S_\bullet$ as a topological space, we are ready to define the structure sheaf. On $D_+(f)$, we wish it to be the structure sheaf of $\operatorname{Spec}((S_\bullet)_f)_0$. We will glue these sheaves together using Theorem 3.5.3 on gluing sheaves.

Proposition 5.5.10

If $f, g \in S_+$ are homogeneous and nonzero, there is an isomorphism between $\operatorname{Spec}((S_\bullet)_{fg})_0$ *and the distinguished open subset $D(g^{\deg f}/f^{\deg g})$ of* $\operatorname{Spec}((S_\bullet)_f)_0$.

Proof Say $A = S_\bullet$, $k = \deg f$, and $l = \deg g$. By Proposition 5.3.2,

$$D\left(\frac{g^k}{f^l}\right) \cong \operatorname{Spec}((A_f)_0)_{\frac{g^k}{f^l}}.$$

By Proposition 5.3.1, it suffice to show that $(A_{fg})_0 \cong ((A_f)_0)_{\frac{g^k}{f^l}}$.

In fact, we have

$$(A_{fg})_0 = \left\{\frac{a}{(fg)^n} \in A_{fg} : a \text{ homogeneous elements in } A, \deg a = n(k+l), \forall n \geq 0\right\}$$

and

$$(A_f)_0 = \left\{\frac{a}{f^n} \in A_f : a \text{ homogeneous elements in } A, \deg a = nk, \forall n \geq 0\right\}.$$

Define $\varphi : (A_f)_0 \to (A_{fg})_0$ by setting

$$\frac{a}{f^n} \longmapsto \frac{ag^n}{(fg)^n}.$$

We shall to show that φ is well-defined. Let $\frac{a}{f^n} = \frac{b}{f^m}$ in $(A_f)_0$, then there exists t such that $f^t(af^m - bf^n) = 0$, hence, $f^t(af^m g^{n+m} - bf^n g^{n+m}) = 0$, which implies that $\frac{ag^n}{(fg)^n} = \frac{bg^m}{(fg)^m}$, i.e., $\varphi\left(\frac{a}{f^n}\right) = \varphi\left(\frac{b}{f^m}\right)$. It follows that φ is well-defined. Clearly, φ is a ring homomorphism.

Consider $\varphi\left(\frac{g^k}{f^l}\right)$, we have
$$\varphi\left(\frac{g^k}{f^l}\right)=\frac{g^{k+l}}{(fg)^l}=\frac{g^k}{f^l}.$$
Note that $\frac{f^l}{g^k}=\frac{f^{l+k}}{(fg)^k}\in(A_{fg})_0$ and $\frac{g^k}{f^l}\cdot\frac{f^l}{g^k}=1$, $\frac{g^k}{f^l}$ is a unit in $(A_{fg})_0$. By the universal property of localization, there exists unique ring homomorphism $\widetilde{\varphi}:((A_f)_0)_{\frac{g^k}{f^l}}\to(A_{fg})_0$, it is defined by
$$\frac{\frac{a}{f^n}}{(\frac{g^k}{f^l})^m}\longmapsto\frac{\varphi\left(\frac{a}{f^n}\right)}{\varphi\left(\frac{g^k}{f^l}\right)^m},$$
where $\frac{a}{f^n}\in(A_f)_0$. We want to show that $\widetilde{\varphi}$ is a ring isomorphism.

Let $\frac{\varphi\left(\frac{a}{f^n}\right)}{\varphi\left(\frac{g^k}{f^l}\right)^m}=\frac{\frac{ag^n}{(fg)^n}}{\frac{g^{(k+l)m}}{(fg)^{lm}}}=\frac{af^{lm}}{f^ng^{km}}=\frac{af^{(k+l)m}g^n}{(fg)^{km+n}}=0$ in $(A_{fg})_0$, then exists s such that $a(fg)^sf^{(k+l)m}g^n=af^{s+(k+l)m}g^{n+s}=0$, hence, $af^{s+(k+l)m}g^{k(n+s)}=0$. It follows that $\frac{g^{k(n+s)}a}{f^{l(n+s)+n}}=\left(\frac{g^k}{f^l}\right)^{n+s}\left(\frac{a}{f^n}\right)=0$ in A_f. Hence, $\frac{\frac{a}{f^n}}{\left(\frac{g^k}{f^l}\right)^m}=0$, which implies that $\widetilde{\varphi}$ is injective.

Let $\frac{a}{(fg)^p}\in(A_{fg})_0$, where a is homogeneous element in A and $\deg a=p(k+l)$. It suffices to show that exists $b\in A$ such that
$$\frac{a}{(fg)^p}=\frac{b}{f^n}\cdot\frac{1}{\left(\frac{g^k}{f^l}\right)^m},$$
i.e.,
$$b=af^{n-ml-p}g^{mk-p}.$$
Pick $m=\lceil\frac{p}{k}\rceil$ and $n=\lceil ml+p\rceil$, we have
$$n-ml-p\geq0,\quad mk-p\geq0,$$
and therefore $b\in A$. Hence, $\widetilde{\varphi}$ is surjective.

Consequently, $(A_{fg})_0\cong((A_f)_0)_{\frac{g^k}{f^l}}$. □

Similarly, $\operatorname{Spec}((S_\bullet)_{fg})_0$ is identified with a distinguished open subset of $\operatorname{Spec}((S_\bullet)_g)_0$. We then glue the various $\operatorname{Spec}((S_\bullet)_f)_0$ (as f varies) altogether, using these pair wise gluings.

By Proposition 5.5.5, we have $\operatorname{Proj}S_\bullet=\bigcup_{f\in S_\bullet}D_+(f)$ (as topological space). By Exercise 5.9, we have $D_+(f)\cong\operatorname{Spec}((S_\bullet)_f)_0$, hence, for each projective distinguished open subset we have structure sheaf $(D_+(f),\mathscr{O}_{\operatorname{Spec}((S_\bullet)_f)_0})$. Let $f_i,f_j\in S_+$, then $D_+(f_if_j)=\operatorname{Spec}((S_\bullet)_{f_if_j})_0$. By Proposition 5.5.10, we have isomorphisms
$$\operatorname{Spec}\left((S_\bullet)_{f_if_j}\right)_0\cong D\left(\frac{f_j^{\deg f_i}}{f_i^{\deg f_j}}\right)\subseteq\operatorname{Spec}((S_\bullet)_{f_i})_0=D_+(f_i)$$
and
$$\operatorname{Spec}\left((S_\bullet)_{f_if_j}\right)_0\cong D\left(\frac{f_i^{\deg f_j}}{f_j^{\deg f_i}}\right)\subseteq\operatorname{Spec}((S_\bullet)_{f_j})_0=D_+(f_j).$$
Hence we get an isomorphism,
$$\varphi_{ij}:D\left(\frac{f_j^{\deg f_i}}{f_i^{\deg f_j}}\right)\xrightarrow{\sim}D\left(\frac{f_i^{\deg f_j}}{f_j^{\deg f_i}}\right),$$
where $D\left(\frac{f_j^{\deg f_i}}{f_i^{\deg f_j}}\right)$ is an open subscheme of $D(f_i)$ and $D\left(\frac{f_i^{\deg f_j}}{f_j^{\deg f_i}}\right)$ is an open subscheme of $D(f_j)$.

Note *Say* $D_{ij} := D\left(\frac{f_j^{\deg f_i}}{f_i^{\deg f_j}}\right)$, *then*

$$\varphi_{ij} : D_{ij} \to D_{ji}.$$

Now, we shall to check the cocycle condition in the Theorem 5.4.1:

Exercise 5.10 Checking that these gluings behave well on the triple overlaps (Theorem 5.4.1).

Proof Let $f_i, f_j, f_k \in S_+$, then $D_+(f_i) \cap D_+(f_j) \cap D_+(f_k) = D_+(f_i f_j f_k) = \operatorname{Spec}((S_\bullet)_{f_i f_j f_k})_0$ by Proposition 5.5.10, we have isomorphisms

$$\operatorname{Spec}\left((S_\bullet)_{f_i f_j f_k}\right)_0 \cong D\left(\frac{(f_j f_k)^{\deg f_i}}{(f_i)^{\deg f_j + \deg f_k}}\right) \subseteq \operatorname{Spec}((S_\bullet)_{f_i})_0 = D_+(f_i),$$

$$\operatorname{Spec}\left((S_\bullet)_{f_i f_j f_k}\right)_0 \cong D\left(\frac{(f_i f_k)^{\deg f_j}}{(f_j)^{\deg f_i + \deg f_k}}\right) \subseteq \operatorname{Spec}((S_\bullet)_{f_j})_0 = D_+(f_j),$$

and

$$\operatorname{Spec}\left((S_\bullet)_{f_i f_j f_k}\right)_0 \cong D\left(\frac{(f_i f_j)^{\deg f_k}}{(f_k)^{\deg f_i + \deg f_j}}\right) \subseteq \operatorname{Spec}((S_\bullet)_{f_k})_0 = D_+(f_k).$$

Then we have

$$D\left(\frac{(f_j f_k)^{\deg f_i}}{(f_i)^{\deg f_j + \deg f_k}}\right) \cong D\left(\frac{(f_i f_k)^{\deg f_j}}{(f_j)^{\deg f_i + \deg f_k}}\right) \cong D\left(\frac{(f_i f_j)^{\deg f_k}}{(f_k)^{\deg f_i + \deg f_j}}\right),$$

that is,

$$D_{ij} \cap D_{ik} \cong D_{ji} \cap D_{jk} \cong D_{ki} \cap D_{kj},$$

which implies that the cocycle condition holds. □

By Theorem 5.4.1, each $D_+(f)$ can glue together, we call this scheme Proj $\mathbf{S}_\bullet$.

Proposition 5.5.11

Let $S_\bullet$ be a graded ring, for any $[\mathfrak{p}] \in \operatorname{Proj} S_\bullet$, there is an isomorphism,

$$\mathscr{O}_{\operatorname{Proj} S_\bullet, [\mathfrak{p}]} \xleftrightarrow{\sim} ((S_\bullet)_{\mathfrak{p}})_0.$$

Proof In fact, $\mathscr{O}_{\operatorname{Proj} S_\bullet,[\mathfrak{p}]} = \varinjlim_{D_+(f) \ni [\mathfrak{p}]} \mathscr{O}_{\operatorname{Proj} S_\bullet}(D_+(f)) = \varinjlim_{f \notin \mathfrak{p}} ((S_\bullet)_f)_0$. Define $\theta : \mathscr{O}_{\operatorname{Proj} S_\bullet,[\mathfrak{p}]} \to ((S_\bullet)_{\mathfrak{p}})_0$ by setting

$$\overline{\left(\frac{s}{f^n}, D_+(f)\right)} \longmapsto \frac{s}{f^n},$$

where $\deg s = n \cdot \deg f$. We shall to check that θ is well-defined. If $\overline{\left(\frac{s}{f^n}, D_+(f)\right)} = \overline{\left(\frac{t}{g^m}, D_+(g)\right)}$, then $\frac{s}{f^n} = \frac{t}{g^m}$ on some $([\mathfrak{p}] \in) D_+(h) \subseteq D_+(f) \cap D_+(g)$. Since $\mathscr{O}_{\operatorname{Proj} S_\bullet}(D_+(h)) \cong ((S_\bullet)_h)_0$, there exists l such that $h^l(sg^m - tf^n) = 0$. Note that $h \notin \mathfrak{p}$, $h^l \notin \mathfrak{p}$, hence, $\frac{s}{f^n} = \frac{t}{g^m}$ in $((S_\bullet)_{\mathfrak{p}})_0$, Hence, θ is well-defined ring homomorphism.

Now, we want to show that θ is an isomorphism.

(i) Surjective.

Let $\frac{s}{f} \in ((S_\bullet)_{\mathfrak{p}})_0$, then $f \notin \mathfrak{p}$. Note that $\theta\left(\overline{\left(\frac{s}{f}, D_+(f)\right)}\right) = \frac{s}{f}$, which implies that θ is surjective.

(ii) Infective.

Let $\frac{s}{f^n} \in ((S_\bullet)_{\mathfrak{p}})_0$ such that $\frac{s}{f^n} = 0$. Note that $\frac{s}{f^n} \in ((S_\bullet)_f)_0$, hence $\overline{\left(\frac{s}{f^n}, D_+(f)\right)} = 0$. Thus θ is injective.

Hence, we have an isomorphism,

$$\mathscr{O}_{\operatorname{Proj} S_\bullet, [\mathfrak{p}]} \xleftrightarrow{\sim} ((S_\bullet)_{\mathfrak{p}})_0,$$

as we desired. □

Exercise 5.11 **(Some will find this essential, while others will prefer to ignore it.)** (Re)interpret the structure sheaf of $\operatorname{Proj} S_\bullet$ in terms of compatible germs.

Solution *Let $U \subseteq \operatorname{Proj} S_\bullet$ be any open subset, by the proof in Theorem 3.5.1, we have*

$$\begin{aligned}\Gamma(U, \mathscr{O}_{\operatorname{Proj} S_\bullet}) =&\{\left(s_{[\mathfrak{p}]} \in \mathscr{O}_{\operatorname{Proj} S_\bullet,[\mathfrak{p}]}\right)_{[\mathfrak{p}]\in U} :\\ &\qquad \forall [\mathfrak{p}] \in U, \exists D_+(f) \text{ with } [\mathfrak{p}] \in D_+(f) \subseteq U \text{ and } t \in \mathscr{O}_{\operatorname{Proj} S_\bullet}(D_+(f)),\\ &\qquad \text{s.t. } t_{[\mathfrak{q}]} = s_{[\mathfrak{q}]}, \forall [\mathfrak{q}] \in D_+(f)\}\\ =&\{\left(s_{[\mathfrak{p}]} \in ((S_\bullet)_\mathfrak{p})_0\right)_{[\mathfrak{p}]\in U} :\\ &\qquad \forall [\mathfrak{p}] \in U, \exists f \notin \mathfrak{p} \text{ and } t/f^n \in ((S_\bullet)_f)_0, \text{ s.t. } s_{[\mathfrak{q}]} = t/f^n, \forall [\mathfrak{q}] \in \operatorname{Spec}((S_\bullet)_f)_0\}.\end{aligned}$$

Definition 5.5.9 (Projective space)

*We (re)definition **projective space** (over a ring A) by*

$$\mathbb{P}^n_A := \operatorname{Proj} A[x_0, \dots, x_n].$$

*This definition involves no messy gluing, or special choice of patches. Note that the variables $x_0, \dots, x_n$, which we call the **projective coordinates** on $\mathbb{P}^n_A$, are part of the definition. (They may have other names than x's, depending on the context.)*

Exercise 5.12 Check that this agrees with our earlier construction of $\mathbb{P}^n_A$ (§5.4.3).

Proof Let $S_\bullet = A[x_0, \dots, x_n]$ and $D_+(x_i)$ is distinguished open subset of $\operatorname{Spec} S_\bullet$, then

$$D_+(x_i) = \operatorname{Spec}((S_\bullet)_{x_i})_0 = \operatorname{Spec}(A[x_0, \dots, x_n]_{x_i})_0 = \operatorname{Spec} A[x_{0/i}, \dots, x_{n/i}]/(x_{i/i} - 1).$$

$D(x_{j/i})$ is an open subscheme of $D_+(x_i)$, and

$$D(x_{j/i}) = \operatorname{Spec} A[x_{0/i}, \dots, x_{n/i}, 1/x_{j/i}]/(x_{i/i} - 1).$$

Clearly, we have $D(x_{j/i}) \cong D(x_{i/j})$. Hence, $\operatorname{Proj} A[x_0, \dots, x_n]$ agree with construction of $\mathbb{P}^n_A$ in §5.4.3. □

Notice that with our old definition of projective space, it would have been a nontrivial exercise to show that $D_+(x^2 + y^2 - z^2) \subseteq \mathbb{P}^2_k$ (the complement of a plane conic) is affine; with our new perspective, it is immediate — it is $\operatorname{Spec}(k[x, y, z]_{x^2+y^2-z^2})_0$.

Exercise 5.13 Suppose that k is an algebraically closed field. We know from Proposition 5.4.3 that the closed points of $\mathbb{P}^n_k$, as defined in §5.4.3 (Proposition 5.4.3), are in the bijection with the points of "classical" projective space. By Exercise 5.12, the scheme $\mathbb{P}^n_k$ as defined in §5.4.3 is isomorphic to $\operatorname{Proj} k[x_0, \dots, x_n]$. Therefore, each point $[a_0, \dots, a_n]$ of classical projective space corresponds to a homogeneous prime ideal of $k[x_0, \dots, x_n]$. Which homogeneous prime ideal is it?

Solution *Let $[a_0 : \cdots : a_n]$ is closed point of $\mathbb{P}^n_k$, then the corresponding ideal in $\operatorname{Spec}(k[x_0, \dots, x_n]_{x_i})_0$ is $\left(x_{0/i} - a_{0/i}, \dots, x_{n/i} - a_{n/i}\right)$. In $S_\bullet = k[x_0, \dots, x_n]$, this corresponds to the homogeneous prime ideal $(a_i x_0 - a_0 x_i, \dots, a_i x_n - a_n x_i)$.*

We now figure out the "right definition" of the vanishing scheme, in analogy with the vanishing set $V(\cdot)$ (Definition 4.4.1). We will be defining a closed subscheme (properly defined in Chapter 10).

Definition 5.5.10 (Vanishing scheme)

Let $S_\bullet$ be a $\mathbb{Z}^{\geq 0}$-graded ring, which is generated as S_0-algebra by S_1, let $f \in S_+$ and let $V_+(f)$ the *vanishing scheme of f in* $\operatorname{Proj} S_\bullet$ *constructed as:*

$$V_+(f) := \operatorname{Proj} S_\bullet/(f).$$

Let I be any homogeneous ideal I of S_+. Define

$$V_+(I) := \operatorname{Proj} S_\bullet/I.$$

On every affine open set $U = \operatorname{Spec} R$, $V_+(f) \cap U$ is clearly a closed subset by Proposition 5.5.9, hence, $V_+(f) \cap U = \operatorname{Spec} R/J$ for some ideal J.

Remark 5.22 This is the same notation as the "vanishing set" $V(\cdot)$, but now means something richer than a mere set.

Warning: in general, f isn't a function on $\operatorname{Proj} S_\bullet$. We will later interpret it as something close: a section of a line bundle.

5.5.6 Projective and quasi-projective schemes

Definition 5.5.11 (Projective and quasi-projective schemes)

We call a scheme of the form (i.e., isomorphic to) $\operatorname{Proj} S_\bullet$, where $S_\bullet$ is a finitely generated graded ring over A, a ***projective scheme over*** *A, or a* ***projective*** *A**-scheme**.*

A ***quasi-projective*** *A-scheme is a quasi-compact open subscheme of a projective A-scheme.*

Remark 5.23 The "A" is omitted if it is clear from the context, often A is a field.

Remark 5.24

(i) Note that $\operatorname{Proj} S_\bullet$ makes sense even when $S_\bullet$ is not finitely generated. This can be useful.

(ii) The quasi-compact requirement in the definition of quasi-projectivity is of course redundant in the Noetherian case, which is all that matters to most.

Example 5.9 Note that $\mathbb{P}^0_A = \operatorname{Proj} A[T] = \{$the homogeneous prime ideal of $A[T]$ which not containing $(T)\}$, we have

$$\mathbb{P}^0_A = \operatorname{Proj} A[T] \cong \operatorname{Spec} A.$$

Thus "$\operatorname{Spec} A$ is a projective A-scheme".

We can make this definition of projective space even more choice-free as follows.

Definition 5.5.12 (Projectivization of a vector space)

Let V be an $(n+1)$-dimensional vector space over k. (Here k can be replaced by any ring A and V by a free module as usual.) Define

$$\operatorname{Sym}^\bullet V^\vee = k \oplus V^\vee \oplus \operatorname{Sym}^2 V^\vee \oplus \cdots.$$

If for example V is the dual of the vector space with basis associated to $x_0, \ldots, x_n$ (i.e., $V = \operatorname{Span}\{x_0, \ldots, x_n\}^\vee$), we would have $\operatorname{Sym}^\bullet V^\vee = k[x_0, \ldots, x_n]$. Then we can define

$$\mathbb{P}V := \operatorname{Proj}(\operatorname{Sym}^\bullet V^\vee).$$

Remark 5.25 In this language, we have an interpretation for $x_0, \ldots, x_n$: they are linear functionals on the underlying vector space V. **Warning:** some authors use the definition $\mathbb{P}V = \operatorname{Proj}(\operatorname{Sym}^\bullet V)$, so be cautious.

Proposition 5.5.12

Suppose k is algebraically closed. There is a natural bijection between one-dimensional subspaces of V and the closed points of $\mathbb{P}V$. Thus this construction canonically (in a basis-free manner) describes the one-dimensional subspaces of the vector space V.

Proof By Exercise 5.12 and Proposition 5.4.3, we done. □

Remark 5.26 You may be surprised at the appearance of the dual in the definition of $\mathbb{P}V$. This is partially explained by the previous discussion. Most normal (traditional) people define the projectivization of a vector space V to be the space of one-dimensional subspaces of V. Grothendieck considered the projectivization to be the space of one-dimensional quotients. One motivation for this is that it gets rid of the annoying dual in the definition above. There are better reasons, that we won't go into here. In a nutshell, quotients tend to be better-behaved than subobjects for coherent sheaves, which generalize the notion of vector bundle. (Coherent sheaves are discussed in Chapter 15.)

Remark 5.27 On another note related to Proposition 5.5.12: you can also describe a natural bijection between points of V and the closed points of $\operatorname{Spec}(\operatorname{Sym}^\bullet V^\vee)$. This construction respects the affine/projective cone picture of Chapter 10.

The Grassmannian. At this point, we could describe the fundamental geometric object known as the **Grassmannian**, and give the "wrong" (but correct) definition of it. We will instead wait until Chapter 8 to give the wrong definition, when we will know enough to sense that something is amiss. The right definition will be given in Chapter 17.

Chapter 6 Some properties of schemes

Now that we have defined the notion of a scheme, we can define some useful properties of schemes. As you see each definition, you should try it out in specific examples of your choice, such as your favorite schemes of the form $\operatorname{Spec}\mathbb{C}[x_1,\ldots,x_n]/(f_1,\ldots,f_r)$.

6.1 Topological properties

6.1.1 Some simple properties

The definition of connected (Definition 4.6.1), connected component (Definition 4.6.8), (ir)reducible (Definition 4.6.2), irreducible component (Definition 4.6.7), quasi-compact (Definition 4.6.3), generization (Definition 4.6.5), specialization (Definition 4.6.5), generic point (Definition 4.6.6), Noetherian topological space (Definition 4.6.9), and closed point (Definition 4.6.4), were given in §4.6.

Proposition 4.6.2 shows that $\mathbb{A}^n_k$ is irreducible. This argument "behaves well under gluing", yielding:

Proposition 6.1.1

$\mathbb{P}^n_k$ is irreducible.

Proof By Definition 4.6.2, it suffices to show that in $\mathbb{P}^n_k$ any two nonempty open subsets intersect. Let U_1,U_2 be any nonempty open subsets of $\mathbb{P}^n_k$, we want to show that $U_1\cap U_2\neq\varnothing$. Note that $\mathbb{P}^n_k=\bigcup_i D_+(x_i)$, where $D_+(x_i)$ is distinguished base, we have

$$\begin{aligned}U_1\cap U_2 =&\mathbb{P}^k_n\cap U_1\cap U_2\\ =&\left(\bigcup_i D_+(x_i)\right)\cap U_1\cap U_2\\ =&\bigcup_i\left((D_+(x_i)\cap U_1)\cap(D_+(x_i)\cap U_2)\right).\end{aligned}$$

Since $D_+(x_i)\cap U_j$ is open in $D_+(x_i)$ for $j=1,2$ and $D_+(x_i)\cong\mathbb{A}^n_k$ is irreducible (Proposition 4.6.2), we have

$$U_1\cap U_2=(D_+(x_i)\cap U_1)\cap(D_+(x_i)\cap U_2)\neq\varnothing$$

for some i. Hence, $U_1\cap U_2\neq\varnothing$, which implies that $\mathbb{P}^n_k$ is irreducible. □

Proposition 6.1.2

There is a bijection between irreducible closed subsets and points for any schemes (the map sending a point p to the closed subset $\overline{\{p\}}$ is a bijection).

Proof Let $(X,\mathscr{O}_X)$ be a scheme. Let $p\in X$ be any point of X, clearly, $\{p\}$ is irreducible in X, by Proposition 4.6.1 (b), $\overline{\{p\}}$ is irreducible closed subset of X.

Conversely, let Z be any irreducible closed subset of X. Let U,V be any two affine open subscheme of X which joint Z nonempty, then $U\cap Z$ is an irreducible closed subset of U and $V\cap Z$ is an irreducible closed subset of V. Since U is affine open subscheme, by Proposition 4.7.3, there exists a unique generic point p_U for $U\cap Z$, i.e., the closure of p_U in U is $U\cap Z$. We want to show that p_U is also generic point for $V\cap Z$.

Note that the closure of p_U in X must contain $U\cap Z$, we claim that the closure of p_U is Z. If not, $\overline{\{p_U\}}$ is a proper closed subset of Z. Note that $Z=(Z\setminus U\cap Z)\cup\overline{\{p_U\}}$ and $Z\setminus U\cap Z$ is a proper closed subset of Z,

Z is the union of two proper closed subset, it contradicts to the fact that Z is irreducible. Hence, the closure of p_U in X is Z.

Since $\overline{\{p_U\}} = Z$ in X, $\{p_U\}$ is a dense subset of Z, hence $\{p_U\}$ contained in every nonempty open subset of Z, and therefore $p_U \in V \cap Z$. Note that the closure of p_U in $V \cap Z$ is $V \cap Z$, p_U is the generic point for $V \cap Z$. By Proposition 4.7.3, p_U is the unique generic point for $V \cap Z$.

By above discussion we get a bijection between irreducible closed subsets and points for any schemes, i.e.,

$$\{\text{points in Scheme}\} \longleftrightarrow \{\text{irreducible closed subsets of scheme}\}$$

$$p \longmapsto \overline{\{p\}}$$

$$\begin{array}{c}\text{generic point for } Z \cap U \\ \text{where U is any affine open subscheme}\end{array} \longleftarrow\!\shortmid \; Z.$$

□

Proposition 6.1.3

If X is a scheme that has a finite cover $X = \bigcup_{i=1}^{n} \operatorname{Spec} A_i$ where A_i is Noetherian, then X is a Noetherian topological space.

Proof Since each A_i is Noetherian, by Proposition 4.6.18, $\operatorname{Spec} A_i$ is a Noetherian topological space. It suffices to show that a topological space that is a finite union of Noetherian subspaces is itself Noetherian.

Let $X = \bigcup X_i$ be a topological space, where each X_i is Noetherian topological space. Let

$$Z_1 \supseteq Z_2 \supseteq \cdots \supseteq Z_n \supseteq \cdots \tag{6.1}$$

be any sequence of closed subsets of X. We want to show this sequence is stationary. Note that

$$Z_1 \cap X_i \supseteq Z_2 \cap X_i \supseteq \cdots \supseteq Z_n \cap X_i \supseteq \cdots \tag{6.2}$$

is a sequence of closed subsets of X_i. Since X_i is Noetherian, sequence (6.2) is stationary, i.e., exists r_i such that $X_i \cap Z_{r_i} = X_i \cap Z_{r_i+1} = \cdots$. Let $r = \max_i r_i$. Note that $Z_j = \bigcup_i (Z_j \cap X_i)$, we have

$$Z_r = \bigcup_i (Z_r \cap X_i) = \bigcup_i (Z_{r+1} \cap X_i) = Z_{r+1},$$

which implies that sequence (6.1) is stationary. Hence, X is a Noetherian topological space. □

Corollary 6.1.1

$\mathbb{P}_k^n$ and $\mathbb{P}_{\mathbb{Z}}^n$ are Noetherian topological spaces.

Proposition 6.1.4

A scheme X is quasi-compact if and only if it can be written as a finite union of affine open subschemes (affine open subsets).

Proof Let X be a quasi-compact scheme. By the definition of scheme, we may assume that $X = \bigcup_{q \in X} U_q$ where U_q is affine open subscheme. Since X is a quasi-compact scheme, we can choice finite numbers of affine open subscheme $\{U_i\}_{i=1}^n$ in $\{U_q\}_{q \in X}$ such that $X = \bigcup_{i=1}^n U_i$.

Conversely, if X is a finite union of affine open subscheme, say $X = \bigcup_{i=1}^{n} U_i$, where U_i is affine open subscheme. Let $\{V_i\}$ is any open cover of X. By Proposition 4.6.4, each U_i is quasi-compact, we can choice finite numbers of V_i cover each U_i, and therefore X can be covered by finite numbers of V_i. Hence, X is quasi-compact.

□

Remark 6.1 We will soon call a scheme with such a cover a Noetherian scheme.

Proposition 6.1.5 (Quasi-compact schemes have closed points)

If X is a quasi-compact scheme, then

(a) every point has a closed point in its closure;

(b) every nonempty closed subset of X contains a closed point of X.

In particular, every nonempty quasi-compact scheme has a closed point.

Proof It is enough to show that every closed subset Z of X has a closed point. Since X is a quasi-compact scheme, by Proposition 6.1.4, we may assume that $X = \bigcup\limits_{i=1}^{n} \operatorname{Spec} A_i$. Then $Z = \bigcup\limits_{i=1}^{n} (Z \cap \operatorname{Spec} A_i)$. We may assume that each $Z \cap \operatorname{Spec} A_i \neq \varnothing$ and $Z \cap \operatorname{Spec} A_i \not\subseteq \bigcup_{j\neq i}(Z \cap \operatorname{Spec} A_j)$ (use [BS]), then $Z \cap \operatorname{Spec} A_i$ closed in $\operatorname{Spec} A_i$, and therefore $V(I_i) = Z \cap \operatorname{Spec} A_i$ for some ideal I_i. Since each $V(I_i)$ is open in Z, $Z \setminus (\bigcup_{i=2}^{n} V(I_i))$ is closed in $V(I_1)$, pick closed point $[\mathfrak{m}] \in Z \setminus (\bigcup_{i=2}^{n} V(I_i))$, then $\overline{\{[\mathfrak{m}]\}} = \{[\mathfrak{m}]\} \subseteq Z \setminus (\bigcup_{i=2}^{n} V(I_i)) \subseteq V(I_1)$. Since $Z \setminus (\bigcup_{i=2}^{n} V(I_i))$ is closed in Z, the closure of $\{[\mathfrak{m}]\}$ in Z is also $\{[\mathfrak{m}]\}$. □

Remark 6.2 **Warning:** there exist nonempty schemes with no closed points, see Chapter 16.

Remark 6.3 Proposition 6.1.5 will often be used in the following way. If there is some property $\mathscr{P}$ of points of a scheme that is "open" (if a point p has $\mathscr{P}$, then there is some open neighborhood U of p such that all the points in U have $\mathscr{P}$), then to check if all points of a quasi-compact scheme have $\mathscr{P}$, it suffices to check only the closed points. This provides a connection between schemes and the classical theory of varieties — the points of traditional varieties are the closed points of the corresponding scheme (essentially by the Nullstellensatz). In many good situations, the closed points are dense, but this is not true in some fundamental cases (Proposition 4.6.7 (b)).

Definition 6.1.1 (Open condition)

We call some property $\mathscr{P}$ of points of a scheme that is "open", if a point p has $\mathscr{P}$, then there is some open neighborhood U of p such that all the points in U have $\mathscr{P}$.

6.1.2 Quasi-separated schemes

Quasi-separatedness is a weird notion that comes in handy for certain people (**Warning:** we will later realize that this is a really a property of morphisms, not for schemes, Chapter 9.) Most people, however, can ignore this notion, as the schemes they will encounter in real life will all have this property.

Definition 6.1.2 (Quasi-separated)

A topological space is ***quasi-separated*** *if the intersection of any two quasi-compact open subsets is quasi-compact.*

Remark 6.4 The motivation for the "separatedness" in the name is that it is a weakened version of separated, for which the intersection of any two affine open sets is affine, see Chapter 12.

Proposition 6.1.6

A scheme is quasi-separated if and only if the intersection of any two affine open subsets is a finite union of affine open subsets.

Proof Let X be a quasi-separated scheme. Let U_1, U_2 be any two affine subschemes of X, since X is quasi-separated and each affine scheme is quasi-compact (Proposition 4.6.4), $U_1 \cap U_2$ is quasi-compact. Since $U_1 \cap U_2$ open in X, by Proposition 5.3.3, $U_1 \cap U_2$ is a scheme, by Proposition 6.1.4, $U_1 \cap U_2$ can be written as a finite union of affine open subschemes.

Conversely, let U_1, U_2 be any two quasi-compact open subsets of scheme X, we want to show that $U_1 \cap U_2$ is quasi-compact. Since $U_1 \cap U_2$ is open in X, by Proposition 5.3.3, $U_1 \cap U_2$ is a scheme, by Proposition 6.1.4, it suffices to show that $U_1 \cap U_2$ can be written as a finite union of affine open subschemes.

Since U_1 and U_2 are quasi-compact open subschemes of X, we may assume that

$$U_1 = \bigcup_{i=1}^{n} \operatorname{Spec} A_i, \quad U_2 = \bigcup_{j=1}^{m} \operatorname{Spec} B_j,$$

then

$$U_1 \cap U_2 = \left(\bigcup_{i=1}^{n} \operatorname{Spec} A_i \right) \cap \left(\bigcup_{j=1}^{m} \operatorname{Spec} B_j \right) = \bigcup_{i,j} (\operatorname{Spec} A_i \cap \operatorname{Spec} B_j).$$

By the hypothesis, the intersection of any two affine open subschemes is quasi-compact, each $\operatorname{Spec} A_i \cap \operatorname{Spec} B_j$ can be written as the finite union of affine open subschemes. Hence, $U_1 \cap U_2$ can be written as a finite union of affine open subschemes, as we desired. □

Corollary 6.1.2

Affine schemes are quasi-separated.

Proof Let X be an affine schemes, by Proposition 4.5.3, we may assume that $X = \bigcup_{i=1}^{n} D(f_i)$. By Proposition 6.1.6, it suffices to show that the intersection of any two affine open subsets of X is a finite union of affine open subsets. Let U_1, U_2 be any two affine open subsets of X, by Proposition 4.6.4, U_1 and U_2 are quasi-compact, hence,

$$U_1 = \bigcup_{k=1}^{s} D(f_{i_k}), \quad U_2 = \bigcup_{l=1}^{t} D(f_{j_l}),$$

where $\{f_{i_k}\}_{k=1}^{s} \subseteq \{f_i\}$ and $\{f_{j_l}\}_{l=1}^{t} \subseteq \{f_i\}$. Then we have

$$U_1 \cap U_2 = \bigcup_{k=1}^{s} \bigcup_{l=1}^{t} (D(f_{i_k}) \cap D(f_{j_l})) = \bigcup_{k=1}^{s} \bigcup_{l=1}^{t} D(f_{i_k} f_{j_l}).$$

By Proposition 5.3.2, each $D(f_{i_k} f_{j_l})$ is affine open subsets, hence, $U_1 \cap U_2$ is a finite union of affine open subsets. □

Corollary 6.1.3

A scheme X is quasi-separated if and only if there exists a cover of X by affine open subsets, any two of which have intersection also covered by a finite number of affine open subsets.

Proof If X is a quasi-separated scheme. Since X is a scheme, X can be covered by affine open subsets, by Proposition 6.1.6, the intersection of any two affine open subsets is a finite union of affine open subsets.

Conversely, suppose that $X = \bigcup_i \operatorname{Spec} A_i$, where $\operatorname{Spec} A_i \cap \operatorname{Spec} A_j$ covered by a finite number of affine open subsets. Let U_1, U_2 be any two affine open subsets of X, we want to show that $U_1 \cap U_2$ is a finite union of affine open subsets. By Proposition 4.6.4, U_1 and U_2 are quasi-compact, we may assume that

$$U_1 = \bigcup_{k=1}^{s} \operatorname{Spec} A_{i_k}, \quad U_2 = \bigcup_{l=1}^{t} \operatorname{Spec} A_{j_l},$$

then

$$U_1 \cap U_2 = \bigcup_{k=1}^{s} \bigcup_{l=1}^{t} (\operatorname{Spec} A_{i_k} \cap \operatorname{Spec} A_{j_l}).$$

By the hypothesis, $\operatorname{Spec} A_{i_k} \cap \operatorname{Spec} A_{j_l}$ covered by a finite number of affine open subsets, hence, $U_1 \cap U_2$ is a finite union of affine open subsets. By Proposition 6.1.6, X is quasi-separated. □

We will see that quasi-separatedness will be a useful hypothesis in theorems (in conjunction with quasi-compactness), and that various interesting kinds of schemes (affine, locally Noetherian, separated) are quasi-separated, and this will allow us to state theorems more succinctly (e.g., "if X is a quasi-compact and quasi-separated" rather than "if X is quasi-compact, and either this or that or the other thing hold").

"Quasi-compact and quasi-separated" means something concrete:

Corollary 6.1.4

A scheme X is quasi-compact and quasi-separated if and only if X can be covered by a finite number of affine open subsets, any two of which have intersection also covered by a finite number of affine open subsets.

Proof By Proposition 6.1.4 and Proposition 6.1.6, we done. □

Note ***Qcqs.*** *When you see "quasi-compact and quasi-separated" as hypotheses in a theorem, you should take this as a clue that you will use this interpretation, and that finiteness will be used in an essential way. This combination of hypotheses is so common that people often describe it as* ***qcqs*** *for short.*

Proposition 6.1.7

All projective A-schemes are qcqs.

Proof Let $X = \operatorname{Proj} S_\bullet$ be any projective A-scheme, where $S_\bullet$ is a finitely generated graded ring over A. Hence, S_+ is a finitely generated ideal, say $S_+ = (f_1, \dots, f_n)$. By Proposition 5.5.8, we have $\operatorname{Proj} S_\bullet = \bigcup_{i=1}^n D_+(f_i)$. By Proposition 5.5.9, each $D_+(f_i)$ is affine open subsets. By Lemma 5.5.2, $D_+(f_i) \cap D_+(f_j) = D_+(f_i f_j)$ is affine open subsets. By Corollary 6.1.4, X is qsqc. □

Example 6.1 **A nonquasi-separated scheme.** Let $X = \operatorname{Spec} k[x_1, x_2, \dots]$, and let U be $X \setminus [\mathfrak{m}]$ where $\mathfrak{m}$ is the maximal ideal $(x_1, x_2, \dots)$. Take two copies of X, glued along U ("affine ∞-space with a doubled origin"). The result is not quasi-separated.

Proof Let X_1 and X_2 denote the two copies of X and let Y denote the space obtained by gluing the two affine schemes along U. Then the image of X_1 and X_2 are still open in Y and they are all affine scheme, hence, they are quasi-compact (Proposition 4.6.4). Consider $X_1 \cap X_2$, we have

$$X_1 \cap X_2 \cong (\operatorname{Spec} k[x_1, x_2, \dots]) \setminus \{[\mathfrak{m}]\},$$

by remark after Proposition 4.6.4, $X_1 \cap X_2$ is not quasi-compact, and therefore X is not quasi-separated. □

6.1.3 Dimension

One very important topological notion is **dimension**. (It is amazing that this is a topological idea.) But despite being intuitively fundamental, it is more difficult, so we postpone it until Chapter 13.

6.2 Reducedness and integrality

6.2.1 Reducedness

Recall that one of alarming things about schemes is that functions are not determined by their values at points, and that was because of the presence of nilpotents (§4.2.4).

Definition 6.2.1 (Reduced scheme)

We say that a scheme X is ***reduced*** *if its stalks $\mathscr{O}_{X,p}$ have no nonzero nilpotents for all p.*

Proposition 6.2.1

If f and g are two functions (global sections of $\mathscr{O}_X$) on a reduced scheme that agree at all points, then $f = g$.

Proof Let X be a reduced scheme, let $X = \bigcup_i \operatorname{Spec} A_i$ be an affine open cover of X. Let $f, g \in \mathscr{O}_X(X)$ with $f(p) = g(p)$ for all $p \in X$, say $h = f - g$, then $h(p) = 0$ for all $p \in X$, we want to show that $h = 0$ in $\mathscr{O}_X(X)$. For any $U_i := \operatorname{Spec} A_i$, by Proposition 3.4.1, we have an injective,

$$\mathscr{O}_X|_{U_i}(U_i) = A_i \longleftrightarrow \textstyle\prod_{[\mathfrak{p}]\in U_i} \mathscr{O}_{\operatorname{Spec} A_i,[\mathfrak{p}]} = \prod_{[\mathfrak{p}]\in U_i} (A_i)_{\mathfrak{p}}. \tag{6.3}$$

Since $h(p) = 0$ for all $p \in X$, we have $(h|_{U_i})_{[\mathfrak{p}]} \in \mathfrak{p}(A_i)_{\mathfrak{p}}$ for all $[\mathfrak{p}] \in U_i$. Since (6.3) is injective, $h|_{U_i} \in \mathfrak{p}$ for all $[\mathfrak{p}] \in \operatorname{Spec} A_i$, i.e., $h|_{U_i} \in \mathfrak{N}(A_i)$. Note that X is reduced, $\mathfrak{N}((A_i)_{\mathfrak{p}}) = \mathfrak{N}(A_i)_{\mathfrak{p}} = 0$ for all $[\mathfrak{p}] \in U_i$. Hence, $(h|_{U_i})_{[\mathfrak{p}]} = 0$ for all $[\mathfrak{p}] \in U_i$, by the injective (6.3), $h|_{U_i} = 0$. By the identity axiom of sheaf $\mathscr{O}_X$, $h = 0$ in $\mathscr{O}_X(X)$, that is, $f = g$. □

Proposition 6.2.2

Let A be a ring, A is a reduced ring (Definition 4.2.3) if and only if $\operatorname{Spec} A$ *is reduced.*

Proof If A is a reduced ring, then $\mathfrak{N}(A) = 0$. For any $[\mathfrak{p}] \in \operatorname{Spec} A$, note that

$$\mathfrak{N}(A_{\mathfrak{p}}) = \bigcap_{[\mathfrak{q}]\in\operatorname{Spec} A_{\mathfrak{p}}} \mathfrak{q} = \bigcap_{\substack{[\mathfrak{q}]\in\operatorname{Spec} A\\ \mathfrak{q}\subseteq\mathfrak{p}}} \mathfrak{q} = \bigcap_{[\mathfrak{q}]\in\operatorname{Spec} A} \mathfrak{q}_{\mathfrak{p}} = \left(\bigcap_{[\mathfrak{q}]\in\operatorname{Spec} A} \mathfrak{q}\right)_{\mathfrak{p}} = \mathfrak{N}(A)_{\mathfrak{p}} = 0 \tag{6.4}$$

and $\mathscr{O}_{\operatorname{Spec} A,[\mathfrak{p}]} = A_{\mathfrak{p}}$, $\mathscr{O}_{\operatorname{Spec} A,[\mathfrak{p}]}$ has no nonzero nilpotents for all $[\mathfrak{p}] \in \operatorname{Spec} A$. Hence, $\operatorname{Spec} A$ is a reduced ring.

If $\operatorname{Spec} A$ is reduced, then $\mathscr{O}_{\operatorname{Spec} A,[\mathfrak{p}]} = A_{\mathfrak{p}}$ have no nonzero nilpotents for all $[\mathfrak{p}] \in \operatorname{Spec} A$, that is, $\mathfrak{N}(A_{\mathfrak{p}}) = 0$. We want to show that $\mathfrak{N}(A) = 0$. Let $f \in \mathfrak{N}(A)$, by (6.4), $f_{\mathfrak{p}} = 0$ in $A_{\mathfrak{p}}$ for all $[\mathfrak{p}] \in \operatorname{Spec} A$. By Proposition 3.4.1, we have an injection

$$\mathscr{O}_{\operatorname{Spec} A}(\operatorname{Spec} A) = A \longleftrightarrow \textstyle\prod_{[\mathfrak{p}]\in\operatorname{Spec} A} \mathscr{O}_{\operatorname{Spec} A,[\mathfrak{p}]} = \prod_{[\mathfrak{p}]\in\operatorname{Spec} A} A_{\mathfrak{p}},$$

hence, $f = 0$ in A, and therefore $\mathfrak{N}(A) = 0$. Hence, A is a reduced ring. □

Corollary 6.2.1

$\mathbb{A}_k^n$ and $\mathbb{P}_k^n$ are reduced.

Proof Note that $\mathbb{A}_k^n = \operatorname{Spec} k[x_1, \ldots, x_n]$, by Proposition 6.2.2, it suffices to show that $k[x_1, \ldots, x_n]$ is a reduced ring. Since $k[x_1, \ldots, x_n]$ is an integral domain, $\mathfrak{N}(k[x_1, \ldots, x_n]) = 0$, and therefore $k[x_1, \ldots, x_n]$ is a reduced ring.

$\mathbb{P}_k^n$ is covered by $n+1$ copies of $\mathbb{A}_k^n$ and therefore is reduced. □

Proposition 6.2.3

A scheme X is reduced if and only if $\mathscr{O}_X(U)$ is reduced for every open set U of X.

Proof If X is reduced. Let $f \in \mathscr{O}_X(U)$ be a section such that $f^n = 0$. Then the image of f^n in $\mathscr{O}_{X,p}$ is zero for all $p \in U$. Since X is reduced, $\mathfrak{N}(\mathscr{O}_{X,p}) = 0$, it follows that $f = 0$ in $\mathscr{O}_{X,p}$ for all $p \in U$. By Proposition 6.2.1, we have an injective,

$$\mathscr{O}_X(U) \hookrightarrow \textstyle\prod_{p \in U} \mathscr{O}_{X,p}.$$

Hence, $f = 0$ in $\mathscr{O}_X(U)$, which implies that $\mathscr{O}_X(U)$ is reduced.

Conversely, it suffices to show that $\mathscr{O}_{X,p}$ has no nonzero nilpotents for all p. Suppose there exists p_0 such that $\mathfrak{N}(\mathscr{O}_{X,p_0}) \neq 0$, let nonzero $f_{p_0} \in \mathfrak{N}(\mathscr{O}_{X,p_0})$, then exists n such that $f_{p_0}^n = 0$. Let $f_{p_0} = \overline{(f, U_{p_0})}$, where U_{p_0} is an open neighborhood of p_0. Since $f_{p_0}^n = 0$ in $\mathscr{O}_{X,p_0}$, there exists an open neighborhood of p_0, say V, such that $V \subseteq U$ with $(f|_V)^n = 0$. By the hypothesis, $\mathfrak{N}(\mathscr{O}_X(V)) = 0$, hence, $f|_V = 0$, a contradiction. Hence, $\mathscr{O}_{X,p}$ has no nonzero nilpotents for all p, and therefore X is reduced. □

Example 6.2 The scheme $\operatorname{Spec} k[x,y]/(y^2, xy)$ is nonreduced. When we sketched it in Figure 5.4, we indicated that the fuzz represented nonreducedness at the origin. The following exercise is a first stab at making this precise.

Exercise 6.1

(a) Show that $\big(k[x,y]/(y^2, xy)\big)_x$ has no nonzero nilpotent elements.

(b) Show that the only point of $\operatorname{Spec} k[x,y]/(y^2, xy)$ with a nonreduced stalk is the origin.

Proof

(a) We claim that

$$\big(k[x,y]/(y^2, xy)\big)_x \cong k[x,y]_x/(y^2, xy)_x \cong k[x]_x.$$

First isomorphism given by the localization commutes with quotient. In fact,

$$\begin{aligned} k[x,y]_x/(y^2, xy)_x &= \left\{ \frac{f(x,y)}{x^k} \in k[x,y]_x : \frac{y^2}{x^m} = 0, \frac{xy}{x^n} = 0, \forall m, n \right\} \\ &= \left\{ \frac{f(x,0)}{x^k} \in k[x]_x : \frac{f(x,y)}{x^k} \in k[x,y]_x \right\}, \end{aligned}$$

define $\varphi : k[x,y]_x/(y^2, xy)_x \to k[x]_x$ by setting

$$\overline{\frac{f(x,y)}{x^k}} \longmapsto \frac{f(x,0)}{x^k}.$$

Clearly, φ is well-defined ring homomorphism. By the construction of $k[x,y]_x/(y^2, xy)_x$, φ is surjective. Let $\frac{f(x,0)}{x^k} = 0$ in $k[x]_x$, since $k[x]$ is integral domain, we have $f(x,0) = 0$. In $k[x,y]_x/(y^2, xy)_x$, $\frac{y}{x^n} = \frac{xy}{x^{n+1}} = 0$, hence, $\frac{f(x,y)}{x^k} = \frac{f(x,0)+f(0,y)}{x^k} = \frac{f(x,0)}{x^k} = 0$ in $k[x,y]_x/(y^2, xy)_x$. It follows that φ is injective. Hence,

$$\big(k[x,y]/(y^2, xy)\big)_x \cong k[x,y]_x/(y^2, xy)_x \cong k[x]_x.$$

(b) Let $[\mathfrak{p}] \in \operatorname{Spec} k[x,y]/(y^2, xy)$, then $(y^2, xy) \subseteq \mathfrak{p} \subseteq k[x,y]$. Since $y^2 \in \mathfrak{p}$, note that $\mathfrak{p}$ is a prime ideal, we have $y \in \mathfrak{p}$. We have the following situations.

(i) $x \notin \mathfrak{p}$.
Consider $\left(k[x,y]/(y^2,xy)\right)_{\mathfrak{p}}$, since $x \notin \mathfrak{p}$, x is invertible in $\left(k[x,y]/(y^2,xy)\right)_{\mathfrak{p}}$. Hence

$$\left(k[x,y]/(y^2,xy)\right)_{\mathfrak{p}} \cong k[x]_{\mathfrak{p}}.$$

Since $\mathfrak{N}(k[x]_{\mathfrak{p}}) = \mathfrak{N}(k[x])_{\mathfrak{p}}$ and $\mathfrak{N}(k[x]) = 0$, we have $\mathfrak{N}(k[x]_{\mathfrak{p}}) = 0$. Hence, stalks $\left(k[x,y]/(y^2,xy)\right)_{\mathfrak{p}}$ are reduced.

(ii) $x \in \mathfrak{p}$, $y \in \mathfrak{p}$, i.e., $\mathfrak{p} = (x,y)$.
Consider $\left(k[x,y]/(y^2,xy)\right)_{(x,y)}$. Note that $\frac{y}{x} \in \left(k[x,y]/(y^2,xy)\right)_{(x,y)}$ and $\left(\frac{y}{x}\right)^2 = \frac{y^2}{x^2} = 0$, $\frac{y}{x}$ is a nilpotent in the stalk $\left(k[x,y]/(y^2,xy)\right)_{(x,y)}$, hence, $\left(k[x,y]/(y^2,xy)\right)_{(x,y)}$ is nonreduced.

By above discussion the only point of $\operatorname{Spec} k[x,y]/(y^2,xy)$ with a nonreduced stalk is the origin.

□

Proposition 6.2.4

If X is a quasi-compact scheme, then it suffices to check reducedness at closed point. More precisely, if $\mathscr{O}_{X,p}$ have no nonzero nilpotent for all closed point $p \in X$, then X is reduced.

Proof X is reduced if and only if $\mathscr{O}_{X,p}$ have no nonzero nilpotent for all $p \in X$. We want to show that if $\mathscr{O}_{X,p}$ have no nonzero nilpotent for all closed point $p \in X$, $\mathscr{O}_{X,p}$ have no nonzero nilpotent for all $p \in X$.

Suppose $[\mathfrak{q}]$ is not a closed point in X, since X is a quasi-compact scheme, by Proposition 6.1.5, $\overline{\{\mathfrak{q}\}}$ contains a closed point of X, say $[\mathfrak{p}]$. Let U be an affine open neighborhood of $[\mathfrak{p}]$, note that $[\mathfrak{q}]$ is a generic point for $\overline{\{\mathfrak{q}\}}$, by Proposition 4.6.10, U contains $[\mathfrak{q}]$. Since U is affine open subset, we may assume that $U = \operatorname{Spec} A$, then $\operatorname{Spec} A \cap \overline{\{\mathfrak{q}\}} = V(\mathfrak{q})$. Note that $\mathfrak{p} \in V(\mathfrak{q})$, we have $\mathfrak{p} \supseteq \mathfrak{q}$. Note that $\mathscr{O}_{X,[\mathfrak{q}]} = A_{\mathfrak{q}}$, we want to show $\mathfrak{N}(A_{\mathfrak{q}}) = 0$. Claim that $(A_{\mathfrak{p}})_{\mathfrak{q}A_{\mathfrak{p}}} = A_{\mathfrak{q}}$. Clearly, we have inclusions, $A_{\mathfrak{p}} \hookrightarrow A_{\mathfrak{q}}$ and $A \hookrightarrow (A_{\mathfrak{p}})_{\mathfrak{q}A_{\mathfrak{p}}}$. Since $\mathfrak{p} \supseteq \mathfrak{q}$, we have $A - \mathfrak{p} \subseteq A - \mathfrak{q}$ and $A - \mathfrak{q} \subseteq A_{\mathfrak{p}} - \mathfrak{q}A_{\mathfrak{p}}$. Hence, the following diagrams commute.

$$\begin{tikzcd} A_{\mathfrak{p}} \arrow[r, hook] \arrow[d, "\text{localization}"'] & A_{\mathfrak{q}} \\ (A_{\mathfrak{p}})_{\mathfrak{q}A_{\mathfrak{p}}} \arrow[ur, dashed] & \end{tikzcd} \qquad \begin{tikzcd} A \arrow[r, hook] \arrow[d, "\text{localization}"'] & (A_{\mathfrak{p}})_{\mathfrak{q}A_{\mathfrak{p}}} \\ A_{\mathfrak{q}} \arrow[ur, dashed] & \end{tikzcd}$$

It follows that $(A_{\mathfrak{p}})_{\mathfrak{q}A_{\mathfrak{p}}} = A_{\mathfrak{q}}$. Since $\mathscr{O}_{X,p}$ have no nonzero nilpotent for all closed point $p \in X$, we have $\mathfrak{N}(A_{\mathfrak{p}}) = 0$, hence,

$$\mathfrak{N}(A_{\mathfrak{q}}) = \mathfrak{N}((A_{\mathfrak{p}})_{\mathfrak{q}A_{\mathfrak{p}}}) = \mathfrak{N}(A_{\mathfrak{p}})_{\mathfrak{q}A_{\mathfrak{p}}} = 0.$$

It follows that $\mathscr{O}_{X,[\mathfrak{q}]}$ has no nonzero nilpotent, then we done. □

Remark 6.5 We will see in Chapter 7 that reducedness is an open condition for locally Noetherian schemes. But in general, reducedness is not an open condition. You may able to identify the underlying topological space of

$$X = \operatorname{Spec} \mathbb{C}[x, y_1, y_2, \ldots]/(y_1^2, y_2^2, y_3^2, \cdots, (x-1)y_1, (x-2)y_2, (x-3)y_3, \cdots)$$

with that of $\operatorname{Spec} \mathbb{C}[x]$, and then to show that the nonreduced points of X are precisely the closed points corresponding to the positive integers. The complement of this set is not Zariski open.

Proof

(i) X is homeomorphic to $\operatorname{Spec} \mathbb{C}[x]$.

Let $\mathfrak{p} \in X$, since $y_i^2 = 0 \in \mathfrak{p}$, each y_i must belong to X. Define $\varphi : \operatorname{Spec} \mathbb{C}[x] \to X$ by setting

$$\mathfrak{p} \longmapsto \mathfrak{p} + (y_1, y_2, \cdots)$$

and define $\psi : X \to \operatorname{Spec} \mathbb{C}[x]$ by setting

$$\mathfrak{q} \longmapsto \mathfrak{q} \cap \mathbb{C}[x].$$

Clearly, φ and ψ is well-defined. Let $\mathfrak{p} \in X$, then

$$\psi \circ \varphi(\mathfrak{p}) = \psi(\mathfrak{p} + (y_1, y_2, \cdots)) = \mathfrak{p}.$$

Let $\mathfrak{q} \in X$, then

$$\varphi \circ \psi(\mathfrak{q}) = \varphi(\mathfrak{q} \cap \mathbb{C}[x]) = \mathfrak{q} \cap \mathbb{C}[x] + (y_1, y_2, \cdots) = \mathfrak{q}.$$

It follows that φ is a bijection. Let $V_X(I)$ be a closed set of X, then

$$\begin{aligned}\varphi^{-1}(V_X(I)) =&\psi(V_X(I))\\ =&\psi(\{\mathfrak{q} \in X : \mathfrak{q} \supseteq I\})\\ =&\{\mathfrak{q} \cap \operatorname{Spec} \mathbb{C}[x] \in \operatorname{Spec} \mathbb{C}[x] : \mathfrak{q} \cap \mathbb{C}[x] \supseteq I \cap \mathbb{C}[x]\}\\ =&\{\mathfrak{p} \in \operatorname{Spec} \mathbb{C}[x] : \mathfrak{p} \supseteq I \cap \mathbb{C}[x]\}\\ =&V_{\operatorname{Spec} \mathbb{C}[x]}(I \cap \mathbb{C}[x]),\end{aligned}$$

is a closed subset of $\operatorname{Spec} \mathbb{C}[x]$. Conversely, let $V_{\operatorname{Spec} \mathbb{C}[x]}(J)$ be a closed subset, then

$$\begin{aligned}\psi^{-1}(V_{\operatorname{Spec} \mathbb{C}[x]}(J)) =&\varphi(V_{\operatorname{Spec} \mathbb{C}[x]}(J))\\ =&\varphi(\{\mathfrak{p} \in \operatorname{Spec} \mathbb{C}[x] : \mathfrak{p} \supseteq J\})\\ =&\{\mathfrak{p} + (y_1, y_2, \cdots) \in X : \mathfrak{p} + (y_1, y_2, \cdots) \supseteq J + (y_1, y_2, \cdots)\}\\ =&V_X(J + (y_1, y_2, \cdots)),\end{aligned}$$

is a closed subset of X. Hence, φ is a homeomorphism.

(ii) The nonreduced points of X are precisely the closed points corresponding to the positive integers.

By part (i), each element in X have form $(x - a, y_1, y_2, \cdots)$ where $a \in \mathbb{C}$. Consider $\mathscr{O}_{X,[\mathfrak{p}]}$, then

$$\mathscr{O}_{X,[\mathfrak{p}]} = \left(\mathbb{C}[x, y_1, y_2, \cdots]/(y_i^2, (x - n)y_n)_{n\geq 1}\right)_{(x-a,y_1,y_2,\cdots)}.$$

If $a \in \mathbb{Z}^{\geq 0}$, then each $x - n$, where $n \neq a$, is invertible. Since $(x - n)y_n = 0$, we have $y_n = 0$ for all $n \neq a$. Then we have

$$\left(\mathbb{C}[x, y_1, y_2, \cdots]/(y_i^2, (x - n)y_n)_{n\geq 1}\right)_{(x-a,y_1,y_2,\cdots)} \cong \left(\mathbb{C}[x, y_1, y_2, \cdots]/(y_a^2, (x - a)y_a)\right)_{(x-a,y_a)}.$$

Note that y_a is a nilpotent in $\left(\mathbb{C}[x, y_1, y_2, \cdots]/(y_a^2, (x - a)y_a)\right)_{(x-a,y_a)}$, hence, $\mathscr{O}_{X,[\mathfrak{p}]}$ is nonreduced for all $a \in \mathbb{Z}^{\geq 0}$.

If $a \notin \mathbb{Z}^{\geq 0}$, then each $x - n$, where $n \in \mathbb{Z}^{\geq 0}$, is invertible. Hence, each $y_n = 0$ in $\left(\mathbb{C}[x, y_1, y_2, \cdots]/(y_i^2, (x - n)y_n)_{n\geq 1}\right)_{(x-a,y_1,y_2,\cdots)}$, and therefore we have

$$\left(\mathbb{C}[x, y_1, y_2, \cdots]/(y_i^2, (x - n)y_n)_{n\geq 1}\right)_{(x-a,y_1,y_2,\cdots)} \cong \mathbb{C}[x]_{x-a}.$$

Note that $\mathfrak{N}(\mathbb{C}[x]_{x-a}) = \mathfrak{N}(\mathbb{C}[x])_{x-a} = 0$, $\mathscr{O}_{X,[\mathfrak{p}]}$ has no nonzero nilpotent, and therefore is a reduced ring.

(iii) The complement of the set of nonreduced point of X is not Zariski open.

It suffices to check the set of nonreduced point of X is not closed in X. Suppose this set is closed in X, say $V_X(I) = \{(x - n, y_1, y_2, \cdots)\}_{n\in\mathbb{Z}^{\geq 0}}$. Since $V_X(\sqrt{I}) = V_X(I)$, we may assume that I is a radical ideal. By Hilbert's Nullstellensatz, we have

$$I = I(V_X(I)) = \bigcap_{[\mathfrak{p}]\in V_X(I)} \mathfrak{p} = \bigcap_{n\in\mathbb{Z}^{\geq 0}} (x-n, y_1, y_2, \cdots) = (y_1, y_2, \dots).$$

But for all $a \notin \mathbb{Z}^{geq}$, we have $(x-a, y_1, y_2, \cdots) \supseteq (y_1, y_2, \dots)$, and therefore $(x-a, y_1, y_2, \cdots) \in V_X(I)$, a contradiction.

□

Proposition 6.2.5

A scheme X is reduced, then its ring of global section $\Gamma(X, \mathscr{O}_X)$ is reduced.

Proof Suppose $\Gamma(X, \mathscr{O}_X)$ is not reduced, then $\Gamma(X, \mathscr{O}_X)$ has nonzero nilpotent, say f, there exists some n such that $f^n = 0$ in $\Gamma(X, \mathscr{O}_X)$. Hence, $f_p^n = 0$ in $\mathscr{O}_{X,p}$ for all $p \in X$. Note that X is reduced, $\mathfrak{N}(\mathscr{O}_{X,p}) = 0$ for all $p \in X$, hence, $f_p = 0$. By Proposition, we have injection

$$\Gamma(X, \mathscr{O}_X) \hookrightarrow \textstyle\prod_{p\in X} \mathscr{O}_{X,p},$$

hence, $f = 0$ in $\Gamma(X, \mathscr{O}_X)$, a contradiction. □

However, the converse of Proposition 6.2.5 is not true:

Example 6.3 If X is the scheme cut out by $x^2 = 0$ in $\mathbb{P}_k^2$, then $\Gamma(X, \mathscr{O}_X) \cong k$, and that X is nonreduced.

Proof Since X is the scheme cut out by $x^2 = 0$ in $\mathbb{P}_k^2$, $X = \operatorname{Proj}\left(k[x,y,z]/(x^2)\right)$. In fact, $X = D_+(x) \cup D_+(y) \cup D_+(z)$, by Proposition 5.5.9, we have

$$D_+(x) = \operatorname{Spec}((k[x,y,z]/(x^2))_x)_0 = \operatorname{Spec} 0 = \varnothing,$$
$$D_+(y) = \operatorname{Spec}((k[x,y,z]/(x^2))_y)_0 = \operatorname{Spec} k[x/y, z/y]/((x/y)^2),$$

and

$$D_+(z) = \operatorname{Spec}((k[x,y,z]/(x^2))_z)_0 = \operatorname{Spec} k[x/z, y/z]/((x/z)^2),$$

hence, $X = \operatorname{Spec} k[x/y, z/y]/((x/y)^2) \cup \operatorname{Spec} k[x/z, y/z]/((x/z)^2)$. Consider the following sequence.

$$0 \longrightarrow \mathscr{O}_X(X) \longrightarrow \mathscr{O}_X(D_+(y)) \times \mathscr{O}_X(D_+(z)) \rightrightarrows \mathscr{O}_X(D_+(y) \cap D_+(z)) \tag{6.5}$$

By Proposition 5.5.10, we have

$$D_+(y) \cap D_+(z) = D_+(yz) \cong D_{D_+(y)}(z/y) \cong D_{D_+(z)}(y/z).$$

Hence, from sequence (6.5) we have

$$\begin{tikzcd}
 & & & \left(k\left[\frac{x}{y}, \frac{z}{y}\right] \Big/ \left(\left(\frac{x}{y}\right)^2\right)\right)_{\frac{z}{y}} \arrow[dd, leftrightarrow, "\cong"] \\
0 \arrow[r] & \mathscr{O}_X(X) \arrow[r] & k\left[\frac{x}{y}, \frac{z}{y}\right] \Big/ \left(\left(\frac{x}{y}\right)^2\right) \times k\left[\frac{x}{z}, \frac{y}{z}\right] \Big/ \left(\left(\frac{x}{z}\right)^2\right) \arrow[ur] \arrow[dr] & \\
 & & & \left(k\left[\frac{x}{z}, \frac{y}{z}\right] \Big/ \left(\left(\frac{x}{z}\right)^2\right)\right)_{\frac{y}{z}}
\end{tikzcd}$$

Let $s \in \mathscr{O}_X(X)$, let $s|_{D_+(y)} = f\left(\frac{x}{y}, \frac{z}{y}\right)$ and $s|_{D_+(z)} = g\left(\frac{x}{z}, \frac{y}{z}\right)$. On $D_+(y) \cap D_+(z)$, we have coordinate change

$$\frac{x}{y} = \frac{\frac{x}{z}}{\frac{y}{z}}, \qquad \frac{z}{y} = \frac{1}{\frac{y}{z}}.$$

Since $(s|_{D_+(y)})|_{D_+(y)\cap D_+(z)} = (s|_{D_+(z)})|_{D_+(y)\cap D_+(z)}$, we have

$$f\left(\frac{x}{y}, \frac{z}{y}\right) = f\left(\frac{\frac{x}{z}}{\frac{y}{z}}, \frac{1}{\frac{y}{z}}\right) = \frac{1}{\left(\frac{y}{z}\right)^d} F\left(\frac{x}{z}\right) = \frac{\left(\frac{x}{z}\right)^d}{\left(\frac{y}{z}\right)^d} = g\left(\frac{x}{z}, \frac{y}{z}\right),$$

where $d = \deg g$ and $F\left(\frac{x}{z}\right)$ is a homogeneous polynomial with variable $\frac{x}{z}$ (and therefore $F\left(\frac{x}{z}\right) = \left(\frac{x}{z}\right)^d$). Since $g\left(\frac{x}{z}, \frac{y}{z}\right)$ is a polynomial with variable $\frac{x}{z}, \frac{y}{z}$, d must be 0, i.e., $s|_{D_+(y)} = s|_{D_+(z)}$ is constant, by the identity axiom, s is constant. It follows that $\Gamma(X, \mathscr{O}_X) \cong k$. Since f is a field, k is reduced, and therefore $\Gamma(X, \mathscr{O}_X)$ is reduced.

Note that $\mathscr{O}_X(D_+(y)) \cong k\left[\frac{x}{y}, \frac{z}{y}\right] \Big/ \left(\left(\frac{x}{y}\right)^2\right)$ has nonzero nilpotent $\frac{x}{y}$, $\mathscr{O}_X(D_+(y))$ is not reduced ring. By Proposition 6.2.3, X is nonreduced. □

Proposition 6.2.6

Suppose X is quasi-compact, and f is a function that vanishes at all points of X, then there is some n such that $f^n = 0$.

Proof Since X is a quasi-compact scheme, we may assume that $X = \bigcup_{i=1}^n \operatorname{Spec} A_i$. In each $\operatorname{Spec} A_i$, since f vanishes at all points of X, we have $D(f) = \varnothing$, where $D(f)$ is distinguished open subset of $\operatorname{Spec} A_i$, by Proposition 4.5.6, there exists k_i such that $f^{k_i} = 0$. Let $k = \max_{1\leq i\leq n} k_i$, then $f^k = 0$, as we desired. □

Example 6.4 Proposition 6.2.6 may fail if X is not quasi-compact.

Proof Let $A_n = k[\varepsilon]/(\varepsilon^n)$, and $X = \coprod_{n=1}^\infty \operatorname{Spec} A_n$. Note that each $\operatorname{Spec} A_n = \{(\varepsilon)\}$, $X = \{((\varepsilon), n)\}_{n=1}^\infty$. Let $f = \varepsilon$, then f vanishes at all points of X. Suppose there exists n such that $f^n = \varepsilon^n = 0$, then $\varepsilon^n \in (\varepsilon^m)$ for all m, it is impossible. It follows that Proposition 6.2.6 fail. □

Remark 6.6 Proposition 6.2.6 is less important, but shows why we like quasi-compactness, and gives a standard pathology when quasi-compactness doesn't hold.

6.2.2 Integrality

Definition 6.2.2 (Integral)

*A scheme X is **integral** if it is nonempty, and $\mathscr{O}_X(U)$ is an integral domain for every nonempty open subset U of X.*

Proposition 6.2.7

A scheme X is integral if and only if it is irreducible and reduced.

Proof If X is a integral scheme, then $\mathscr{O}_X(U)$ is an integral domain for every nonempty open subsets U of X. Hence, $\mathscr{O}_X(U)$ is reduced ring for any open subset U of X, by Proposition 6.2.3, X is reduced. Suppose X is not irreducible (Definition 4.6.2), then there exists disjoint open subsets of X, say U, V. Consider $\mathscr{O}_X(U \cup V)$. Note that we have exact sequence

$$0 \longrightarrow \mathscr{O}_X(U\cup V) \longrightarrow \mathscr{O}_X(U) \times \mathscr{O}_X(V) \rightrightarrows \mathscr{O}_X(U\cap V) = \mathscr{O}_X(\varnothing) = 0,$$

it follows that $\mathscr{O}_X(U\cup V) \cong \mathscr{O}_X(U) \times \mathscr{O}_X(V)$. Let $(s, 0), (0, t) \in \mathscr{O}_X(U) \times \mathscr{O}_X(V)$, then $(s, 0)\cdot(0, t) = 0$, i.e., $(s, 0)$ and $(0, t)$ are zero divisors, which implies that $\mathscr{O}_X(U\cup V)$ is not a integral domain, a contradiction.

Conversely, if X is irreducible and reduced. Let U be any open subset of X, we want show $\mathscr{O}_X(U)$ is an integral domain. Let $f, g \in \mathscr{O}_X(U)$ such that $fg = 0$. Define the closed subsets of U (Proposition 5.3.6 (a))

$$Z_f = \{p \in U : f_p \in \mathfrak{m}_p \subseteq \mathscr{O}_{X,p}\}, \quad Z_g = \{p \in U : g_p \in \mathfrak{m}_p \subseteq \mathscr{O}_{X,p}\},$$

where $\mathfrak{m}_p$ is the maximal ideal of local ring $\mathscr{O}_{X,p}$. Since $fg = 0$, for all $p \in U$, we have $f_p g_p = 0$ in $\mathscr{O}_{X,p}$. We claim that $\mathscr{O}_{X,p}$ is an integral domain. Let $V = \operatorname{Spec} A$ be an affine open neighborhood of p, since X is irreducible and reduced, V is also irreducible and reduced. Let $ab = 0$ in A, then $V(0) = \operatorname{Spec} A = V(a) \cup V(b)$. Since $\operatorname{Spec} A$ is irreducible, $V(a) = \operatorname{Spec} A$ or $V(b) = \operatorname{Spec} A$, we may assume that $V(a) = \operatorname{Spec} A$, then $a \in \bigcap_{[\mathfrak{p}] \in \operatorname{Spec} A} \mathfrak{p} = \mathfrak{N}(A)$, hence, by Proposition 6.2.2, $\mathfrak{N}(A) = 0$, which implies that $a = 0$. Hence, A is an integral domain, and therefore $A_p = \mathscr{O}_{X,p}$ is an integral domain. Since $f_p g_p = 0$ in $\mathscr{O}_{X,p}$, we have $f_p = 0$ or $g_p = 0$ for all $p \in U$, that is, $U \subseteq Z_f \cup Z_g$. Clearly, we have $Z_f \cup Z_g \subseteq U$, hence, $U = Z_f \cup Z_g$. Since X is irreducible and reduced, so is U. Hence, $U = Z_f$ or $U = Z_g$. We may assume that $Z_f = U$, then f vanish at all points of U. By Proposition 6.2.1, we have $f = 0$, which implies that $\mathscr{O}_{X,p}$ is an integral domain. Hence, X is integral. □

Proposition 6.2.8

An affine scheme $\operatorname{Spec} A$ *is integral if and only if* A *is an integral domain.*

Proof If A is an integral domain, by Proposition 4.6.2, $\operatorname{Spec} A$ is irreducible. Since A is integral domain, we have $\mathfrak{N}(A) = 0$, that is, A is reduced ring, by Proposition 6.2.2, $\operatorname{Spec} A$ is reduced. By Proposition 6.2.7, $\operatorname{Spec} A$ is integral.

Conversely, if $\operatorname{Spec} A$ is integral, by Proposition 6.2.7, $\operatorname{Spec} A$ is irreducible and reduced. By Proposition 6.2.2, A is a reduced ring. We want to show A is integral domain. Let $fg = 0$ in A, then we have $\operatorname{Spec} A = V(0) = V(fg) = V(f) \cup V(g)$, since $\operatorname{Spec} A$ is irreducible, $\operatorname{Spec} A = V(f)$ or $\operatorname{Spec} A = V(g)$. We may assume that $\operatorname{Spec} A = V(f)$, then f belong to any prime ideal of A, it follows that $f \in \mathfrak{N}(A) = 0$, hence, $f = 0$, and therefore A is an integral domain. □

Proposition 6.2.9

Suppose X *is an integral scheme. Then* X *has a generic point* η. *Suppose* $\operatorname{Spec} A$ *is any nonempty affine open subset of* X. *Then* $\mathscr{O}_{X,\eta}$ *is naturally identified with* $K(A)$, *the fraction field of* A.

Proof By Proposition 6.1.2, there is a bijection between irreducible closed subsets and points for any scheme, hence, there exists a point $\eta \in X$ such that $\overline{\{\eta\}} = X$, that is, η is a generic point for X.

Let $\operatorname{Spec} A$ be any nonempty affine open subset of X, since η is a generic point of X, by Proposition 4.6.10, η must belong to $\operatorname{Spec} A$, hence, $\mathscr{O}_{X,\eta} = A_\eta$. Since X is integral, $\operatorname{Spec} A$ is integral, by Proposition 6.2.8, A is integral domain. Hence, $(0) \in \operatorname{Spec} A$, note that $V((0)) = \overline{\{(0)\}} = \operatorname{Spec} A$, (0) is the generic point for $\operatorname{Spec} A$. In fact, $\overline{\{(0)\}} = \overline{\{\eta\}} \cap \operatorname{Spec} A$, by Proposition 6.1.2, $\eta = (0)$, hence, $A_\eta = A_{(0)} = K(A)$, where $K(A)$ is the fraction field of A, that is,

$$\mathscr{O}_{X,\eta} \cong K(A).$$

□

By Proposition 6.2.9, we can give the definition of function field and rational functions:

Definition 6.2.3 (Function field, rational functions)

*Suppose X is an integral scheme. Say η is the generic point of X. Let $\operatorname{Spec} A$ be any nonempty affine open subset of X. The **function field** $K(X)$ of X is defined by*

$$K(X) := K(A) = \mathscr{O}_{X,\eta}.$$

*It can be computed on any nonempty open set of X, as any such open set contains the generic point. The elements of $K(X)$ are called **rational functions** on X.*

Proposition 6.2.10

Suppose X is an integral scheme, then the restriction maps $\operatorname{res}_{U,V} : \mathscr{O}_X(U) \to \mathscr{O}_X(V)$ are inclusions so long as $V \neq \varnothing$. Suppose $\operatorname{Spec} A$ is any nonempty affine open subset of X (so A is an integral domain), then the natural map $\mathscr{O}_X(U) \to \mathscr{O}_{X,\eta} = K(A)$ (where U is any nonempty open subset and η is the generic point) is an inclusion.

Proof Let $X = \bigcup_i \operatorname{Spec} A_i$, then $U = \bigcup_i (U \cap \operatorname{Spec} A_i)$ and $V = \bigcup_i (V \cap \operatorname{Spec} A_i)$. Since $U \supseteq V$, we have $U \cap \operatorname{Spec} A_i \supseteq V \cap \operatorname{Spec} A_i$. Since each $U \cap \operatorname{Spec} A_i$ is open in $\operatorname{Spec} A_i$, let $U \cap \operatorname{Spec} A_i = \bigcup_j D(f_{i,j})$, where $D(f_{i,j})$ is distinguished open subset of $\operatorname{Spec} A_i$. Similarly, we may assume that $V \cap \operatorname{Spec} A_i = \bigcup_k D(g_{i,k})$. Hence,

$$V \cap \operatorname{Spec} A_i = (U \cap \operatorname{Spec} A_i) \cap (V \cap \operatorname{Spec} A_i) = \bigcup_j \bigcup_k (D(f_{i,j}) \cap D(g_{i,k})).$$

By the definition of sheaf we have following diagram,

$$\begin{tikzcd}
0 \arrow[r] & \mathscr{O}_X(U) \arrow[r, hook] \arrow[d, "\operatorname{res}_{U,V}"'] & \prod_i \prod_j (A_i)_{f_{i,j}} \arrow[d, hook] \\
0 \arrow[r] & \mathscr{O}_X(V) \arrow[r, hook] & \prod_i \prod_j \prod_k (A_i)_{f_{i,j} g_{i,k}},
\end{tikzcd}$$

where $\prod_i \prod_j (A_i)_{f_{i,j}} \hookrightarrow \prod_i \prod_j \prod_k (A_i)_{f_{i,j} g_{i,k}}$ is given by $(A_i)_{f_{i,j}} \hookrightarrow (A_i)_{f_{i,j} g_{i,k}}$ (note that each A_i is integral domain, since X is integral). Hence, $\mathscr{O}_X(U) \to \mathscr{O}_X(V)$ is inclusion.

Let U be any nonempty open subset and η is the generic point for X, then $\eta \in U$, by Proposition 4.6.10. Note that $U \cap \operatorname{Spec} A$ is open subset of $\operatorname{Spec} A$, and therefore it is covered by some distinguished open subsets, hence, by above discussion, we have inclusion $\mathscr{O}_X(U) \hookrightarrow \mathscr{O}_X(D(f)) = A_f$. Since $D(f) \ni \eta$, we have inclusion $A_f \hookrightarrow A_\eta$, that is,

$$\mathscr{O}_X(U) \hookrightarrow \mathscr{O}_X(D(f)) = A_f \hookrightarrow A_\eta = \mathscr{O}_{X,\eta} = K(A),$$

as we desired. □

Thus irreducible varieties (an important example of integral schemes defined later) have the convenient property that sections over different open sets can be considered subsets of the same ring. In particular, restriction maps (except to the empty set) are always inclusions, and gluing is easy: functions f_i on a cover U_i of U (as i runs over an index set) glue if and only if they are the same element of $K(X)$ (since $\mathscr{O}_X(U_i) \hookrightarrow \mathscr{O}_X(U_i \cap U_j) \hookrightarrow K(A)$). This is one reason why irreducible varieties are usually introduced before schemes.

Remark 6.7 Caution. Integrality is not stalk-local (the disjoint union of two integral schemes is not integral, as $\operatorname{Spec} A \coprod \operatorname{Spec} B = \operatorname{Spec}(A \times A)$ by Example 4.5), but it almost is.

6.3 The Affine Communication Lemma, and properties of schemes that can be checked "affine-locally"

6.3.1 The Affine Communication Lemma

This section is intended to address something tricky in the definition of schemes. We have defined a scheme as a topological space with a sheaf of rings, that can be covered by affine schemes. Hence we have all of the affine open sets in the cover, but we don't know how to communicate between any two of them. Somewhat more explicitly, if I have an affine cover, and you have an affine cover, and we want to compare them, and I calculate something on my cover, there should be some way of us getting together, and figuring out how to translate my calculation over to your cover. The **Affine Communication Lemma** will provide a convenient machine for doing this.

Thanks to this lemma, we can define a host of important properties of schemes. All of these are "**affine-local**", in that they can be checked on any affine cover, i.e., a covering by open affine sets. (We state this more formally after the statement of the Affine Communication Lemma.) We like such properties because we can check them using any affine cover we like. If the scheme in question is quasi-compact, then we need only check a finite number of affine open sets.

Proposition 6.3.1

Suppose $\operatorname{Spec} A$ *and* $\operatorname{Spec} B$ *are affine open subschemes of a scheme* X*. Then* $\operatorname{Spec} A \cap \operatorname{Spec} B$ *is the union of open sets that are simultaneously distguished open subschemes of* $\operatorname{Spec} A$ *and* $\operatorname{Spec} B$*.*

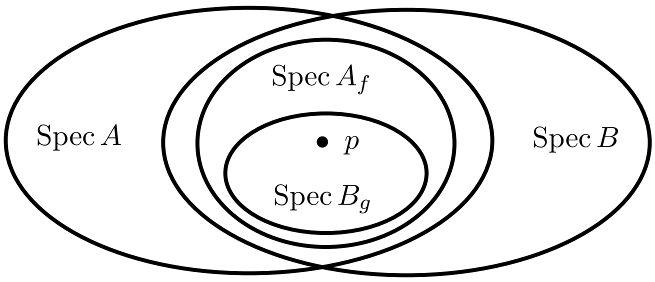


Figure 6.1: A trick to show that the intersection of two affine open sets may be covered by open sets that are simultaneously distinguished in both affine open sets.

Proof (See Figure 6.1.) Given any point $p \in \operatorname{Spec} A \cap \operatorname{Spec} B$, we produce an open neighborhood of p in $\operatorname{Spec} A \cap \operatorname{Spec} B$ that is simultaneously in both $\operatorname{Spec} A$ and $\operatorname{Spec} B$. Let $\operatorname{Spec} A_f$ be a distinguished open subset of $\operatorname{Spec} A$ contained in $\operatorname{Spec} A \cap \operatorname{Spec} B$ and containing p. Let $\operatorname{Spec} B_g$ be a distinguished open subset of $\operatorname{Spec} B$ contained in $\operatorname{Spec} A_f$ and containing p. Then $g \in \Gamma(\operatorname{Spec} B, \mathscr{O}_X)$ restricts to an element $g' \in \Gamma(\operatorname{Spec} A_f, \mathscr{O}_X) = A_f$. The points where g and g' vanish on $\operatorname{Spec} A_f$ are the same, so

$$\begin{aligned}\operatorname{Spec} B_g &= \operatorname{Spec} A_f \setminus \{[\mathfrak{p}] \in \operatorname{Spec} A_f : g' \in \mathfrak{p}\} \\ &= \operatorname{Spec}(A_f)_{g'}.\end{aligned}$$

If $g' = g''/f^n$ ($g'' \in A$) then $\operatorname{Spec}(A_f)_{g'} = \operatorname{Spec} A_{fg''}$, and we are done. ☐

Definition 6.3.1 (Affine-local property)

Suppose X is a scheme, an ***affine-local property*** *is a property $\mathscr{P}$ of affine open subsets of X satisfying the following axioms. For any affine open subset $\operatorname{Spec} A \hookrightarrow X$,*

(i) if $\operatorname{Spec} A \hookrightarrow X$ has property $\mathscr{P}$ then for any $f \in A$, $\operatorname{Spec} A_f \hookrightarrow X$ has property $\mathscr{P}$;

(ii) if $(f_1, \ldots, f_n) = A$ (i.e., the $\operatorname{Spec} A_{f_i}$ cover $\operatorname{Spec} A$, see Proposition 4.5.3 and Proposition 4.5.2), and $\operatorname{Spec} A_{f_i} \hookrightarrow X$ has $\mathscr{P}$ for all i, then $\operatorname{Spec} A \hookrightarrow X$ has property $\mathscr{P}$.

Example 6.5 Reducedness is an affine-local property.

Proof If $\operatorname{Spec} A$ is reduced, by Proposition 6.2.2, A is reduced ring, i.e., $\mathfrak{N}(A) = 0$, hence, $\mathfrak{N}(A_f) = \mathfrak{N}(A)_f = 0$, by Proposition 6.2.2 again, $\operatorname{Spec} A_f$ is reduced.

Since $\operatorname{Spec} A$ is quasi-compact, by Proposition 4.5.3, we may assume $A = (f_1, \ldots, f_n)$. If $\operatorname{Spec} A_{f_i}$ is reduced, then A_{f_i} is reduced ring. Let $a \in \mathfrak{N}(A)$, then $a^n = 0$ in A for some n, Hence, $a^n = 0$ in A_{f_i}. Since each A_{f_i} is reduced, $\mathfrak{N}(A_{f_i}) = 0$, we have $a = 0$ in A_{f_i}. By the identity axiom of sheaf $\mathscr{O}_{\operatorname{Spec} A}$, $a = 0$ in A. Hence, $\mathfrak{N}(A) = 0$, that is, $\operatorname{Spec} A$ is reduced. □

The following easy result will be crucial for us.

Theorem 6.3.1 (Affine Communication Lemma)

Let X be a scheme and let $\mathscr{P}$ be an affine-local property of affine open subsets of X. If $X = \bigcup_{i \in I} \operatorname{Spec} A_i$ where $\operatorname{Spec} A_i$ has property $\mathscr{P}$. Then every affine open subset of X has $\mathscr{P}$ too.

Proof Let $\operatorname{Spec} A$ be an affine subscheme of X. Cover $\operatorname{Spec} A$ with a finite number of distinguished open sets $\operatorname{Spec} A_{g_j}$, each of which is distinguished in some $\operatorname{Spec} A_i$. This is possible by Proposition 6.3.1 and Proposition 4.6.4. By (i), each $\operatorname{Spec} A_{g_j}$ has $\mathscr{P}$. By (ii), $\operatorname{Spec} A$ has $\mathscr{P}$. □

Note that if U is an open subscheme of X, then U inherits any affine-local property of X. Note that also that any property that is stalk-local (a scheme has property P if and only if all its stalks have property Q) is necessarily affine-local (a scheme has property P if and only if all of its affine open sets have property R, where an affine scheme has property R if and only if all its stalks have property Q). But it is sometimes not so obvious what the right definition of Q is; see for example the discussion of normality in the next section.

6.3.2 Properties of schemes that can be checked "affine-locally": (Locally) Noetherian schemes, A-schemes, (Locally of) finite type over A, Affine varieties, and Projective varieties

By choosing the property P appropriately, we define some important properties of schemes.

Proposition 6.3.2

Suppose A is a ring, and $(f_1, \ldots, f_n) = A$.

(a) If A is a Noetherian ring, then so is A_{f_i}. If each A_{f_i} is Noetherian, then so is A.

(b) Suppose B is a ring, and A is a B-algebra. (Hence A_g is a B-algebra for all $g \in A$.) If A is a finitely generated B-algebra, then so is A_{f_i}. If each A_{f_i} is a finitely generated B-algebra, then so is A.

We will prove these shortly. But let's first motivate you to read the proof by giving some interesting definitions and results assuming Proposition 6.3.2 is true.

Locally Noetherian schemes, Noetherian shceme

Definition 6.3.2 (Locally Noetherian scheme, Noetherian scheme)

*Suppose X is a scheme. If X can be covered by affine open sets $\operatorname{Spec} A$ where A is Noetherian, we say that X is a **locally Noetherian scheme**. If in addition X is quasi-compact, or equivalently can be covered by finitely many such affine open sets (Proposition 6.1.4), we say that X is a **Noetherian scheme**.*

Remark 6.8 By Proposition 6.3.2, being Noetherian is a local property. Hence, by Affine Communication Lemma 6.3.1, if X is locally Noetherian scheme, every affine open $U \subseteq X$ the ring $\mathscr{O}_X(U)$ is Noetherian.

Remark 6.9

(1) We will see a number of definitions of the form "if X has this property, we say that it is locally Q; if further X is quasi-compact, we say that it is Q."

(2) By Proposition 6.1.3, the underlying topological space of a Noetherian scheme is Noetherian. Hence by Proposition 4.6.19, all open subsets of a Noetherian scheme are quasi-compact.

Proposition 6.3.3

Locally Noetherian schemes are quasi-separated.

Proof Let X be a locally Noetherian scheme. We may assume that $X = \bigcup_i \operatorname{Spec} A_i$, where A_i is Noetherian. Let $\operatorname{Spec} B$ and $\operatorname{Spec} C$ be any two affine open subsets, we want to show that $\operatorname{Spec} B \cap \operatorname{Spec} C$ is a finite union of affine open subsets. In fact,

$$\operatorname{Spec} B = \operatorname{Spec} B \cap X = \bigcup_i (\operatorname{Spec} B \cap \operatorname{Spec} A_i),$$
$$\operatorname{Spec} C = \operatorname{Spec} C \cap X = \bigcup_i (\operatorname{Spec} C \cap \operatorname{Spec} A_i).$$

Note that $\operatorname{Spec} B \cap \operatorname{Spec} A_i$ and $\operatorname{Spec} C \cap \operatorname{Spec} A_i$ are open subsets of $\operatorname{Spec} A_i$ and affine scheme is quasi-compact, hence, they are finite union of distinguished open subsets of $\operatorname{Spec} A_i$. We may assume that $\operatorname{Spec} B \cap \operatorname{Spec} A_i = \operatorname{Spec}(A_i)_{f_i}$ and $\operatorname{Spec} C \cap \operatorname{Spec} A_i = \operatorname{Spec}(A_i)_{g_i}$, since $\operatorname{Spec} B$ and $\operatorname{Spec} C$ is quasi-compact, we may assume that $\operatorname{Spec} B = \bigcup_{i=1}^n \operatorname{Spec}(A_i)_{f_i}$ and $\operatorname{Spec} C = \bigcup_{j=1}^m \operatorname{Spec}(A_j)_{g_j}$. Hence,

$$\operatorname{Spec} B \cap \operatorname{Spec} C = \bigcup_{i=1}^n \bigcup_{j=1}^m (\operatorname{Spec}(A_i)_{f_i} \cap \operatorname{Spec}(A_j)_{g_j}).$$

By Proposition 6.3.1, each $\operatorname{Spec}(A_i)_{f_i} \cap \operatorname{Spec}(A_j)_{g_j}$ is the union of open sets that are simultaneously distinguished open subschemes of $\operatorname{Spec}(A_i)_{f_i}$ and $\operatorname{Spec}(A_j)_{g_j}$. Since A_i is Noetherian, by Proposition 4.6.15, each $(A_i)_{f_i}$ and $(A_j)_{g_j}$ are Noetherian. Note that $\operatorname{Spec}(A_i)_{f_i} \cap \operatorname{Spec}(A_j)_{g_j}$ is open subset of $\operatorname{Spec}(A_i)_{f_i}$, by Proposition 4.6.19, $\operatorname{Spec}(A_i)_{f_i} \cap \operatorname{Spec}(A_j)_{g_j}$ is quasi-compact. Hence, $\operatorname{Spec}(A_i)_{f_i} \cap \operatorname{Spec}(A_j)_{g_j}$ is the finite union of open sets that are simultaneously distinguished open subschemes of $\operatorname{Spec}(A_i)_{f_i}$ and $\operatorname{Spec}(A_j)_{g_j}$. Hence, $\operatorname{Spec} B \cap \operatorname{Spec} C$ is the finite union of some distinguished open sets, each distinguished open set is affine scheme, and therefore X is quasi-separated. □

Proposition 6.3.4

(a) A Noetherian scheme has a finite number of irreducible components.

(b) A Noetherian scheme has a finite number of connected components, each a finite union of irreducible components.

Proof

(a) Let X be a Noetherian scheme, by Proposition 6.1.3, then X is a Noetherian topological space, by Proposition 4.6.12 X can be expressed uniquely as a finite union of irreducible closed subsets, none contained in any other, then we done.

(b) By Proposition 4.6.13 (a), every connected component of X is the union of irreducible components of X. Since X is a finite union of irreducible components, hence, each connected components is the finite union of irreducible components of X. Since any two connected components disjoint, the number of connected components is finite.

□

Proposition 6.3.5

A Noetherian scheme X is integral if and only if X is nonempty and connected and all stalks $\mathscr{O}_{X,p}$ are integral domains.

Proof Let X be a Noetherian scheme. If X is integral, by Proposition 6.2.7, X is irreducible and reduced. By Proposition 4.6.3, X is connected. Each point $p \in X$ contained in some affine open subsets of X, say $\operatorname{Spec} A$. Since X is integral, A is integral domain, hence, localization $A_p = \mathscr{O}_{X,p}$ is integral domain.

Conversely, if a Noetherian ring X is nonempty and connected and all stalks $\mathscr{O}_{X,p}$ are integral domains, by Proposition 6.2.7, it suffices to show that X is reduced and irreducible. Since all stalks $\mathscr{O}_{X,p}$ are integral domain, $\mathfrak{N}(\mathscr{O}_{X,p}) = 0$, by the definition of reducedness, X is reduced.

If X is not irreducible, by Proposition 6.3.4, Noetherian scheme X is a finite number of irreducible components, say $X = \bigcup_{i=1}^n X_i$, where X_i is irreducible component. Since X is connected, the intersection of any two irreducible components not empty. Let $p \in X$, take all X_i which contains p, we obtain $\{X_{i_k}\}_{k=1}^m$ (i.e., no irreducible component X_i outside $\{X_{i_k}\}_{k=1}^m$ contains p), then $p \in \bigcap\limits_{k=1}^{m} X_{i_k}$. Since X is a scheme, exists an affine open subset $\operatorname{Spec} A \ni p$. In fact, $\operatorname{Spec} A = \bigcup_i (\operatorname{Spec} A \cap X_i)$, since each X_i is irreducible component, $\operatorname{Spec} A \cap X_i$ is also irreducible component of $\operatorname{Spec} A$. Since each X_i is closed (note that X_i is irreducible component), $\operatorname{Spec} A \cap X_i$ is irreducible closed subset of $\operatorname{Spec} A$. By Proposition 4.7.3, there exists prime ideal of A correspond to $\operatorname{Spec} A \cap X_i$, say $V(\mathfrak{p}_i) = \operatorname{Spec} A \cap X_i$. Then $p \in \bigcap\limits_{k=1}^{m} (X_{i_k} \cap \operatorname{Spec} A) = \bigcap\limits_{k=1}^{m} V(\mathfrak{p}_{i_k})$. Since there is no irreducible component X_i outside $\{X_{i_k}\}_{k=1}^m$ contains p, $\{\mathfrak{p}_{i_k}\}_{k=1}^m$ is all prime ideal which contained in p. Since $\operatorname{Spec} A \cap X_{i_k}$ is irreducible componet of $\operatorname{Spec} A$, $(\operatorname{Spec} A) \cap X_{i_k} \neq \operatorname{Spec} A$, it follows that each $\mathfrak{p}_{i_k}$ is not zero ideal. Hence, $(0) \notin \mathscr{O}_{X,p} = A_p$, and therefore A_p is not integral domain, it contradicts to $\mathscr{O}_{X,p}$ is integral domain. Hence, X is irreducible, and therefore is integral. □

Remark 6.10 Thus, by Proposition 6.3.5, in "good situations", integrality is the union of local (stalks are integral domains) and global (connected) conditions.

Remark 6.11 The ring of global section of a Noetherian scheme need not be Noetherian.

Schemes over a given field k, or more generally over a given ring A (A-schemes)

You may be particularly interested in working over a particular field, such as $\mathbb{C}$ or $\mathbb{Q}$, or over a ring such as $\mathbb{Z}$. Motivated by this we define the following notions:

Definition 6.3.3 (A-scheme (or scheme over A))

Suppose X is a scheme, A is a ring. We say X is A-scheme (or scheme over A), if $\mathscr{O}_X(U)$ is A-algebra for all open sets $U \subseteq X$, and all restriction maps are maps of A-algebras.

Remark 6.12 Like some earlier notions such as quasi-separatedness, this will later in Definition 8.3.4 be properly understood as a "relative notion"; it is the data of a morphism $X \to \operatorname{Spec} A$.

Definition 6.3.4 (Locally of finite type over A, finite type over A)

Suppose X is an A-scheme. If X can be covered by affine open sets $\operatorname{Spec} B_i$ where each B_i is a finitely generated A-algebra, we say that X is ***locally of finite type over*** *A, or that it is a* ***locally finite type*** *A**-scheme**. (This is admittedly cumbersome terminology; it will make more sense later, once we know about morphisms in Chapter 9.) If furthermore X is quasi-compact, X is (of)* ***finite type over*** *A, or a* ***finite type*** *A**-scheme**.*

Remark 6.13 Note that a scheme locally of finite type over k or $\mathbb{Z}$ (or indeed any Noetherian ring) is locally Noetherian (apply Hilbert's Basis Theorem), and similarly a scheme of finite type over any Noetherian ring is Noetherian.

Example 6.6 As our key "geometric" examples:

(i) $\operatorname{Spec} \mathbb{C}[x_1, \ldots, x_n]/I$ is a finite type $\mathbb{C}$-scheme,

(ii) $\mathbb{P}^n_{\mathbb{C}}$ is a finite type $\mathbb{C}$-scheme. (The field $\mathbb{C}$ may be replaced by an arbitrary ring A.)

Proposition 6.3.6

*(a) (**Quasi-projective implies finite type.**) If X is a quasi-projective A-scheme (Definition 5.5.11), then X is of finite type over A. If A is furthermore assumed to be Noetherian, then X is a Noetherian scheme, and hence has a finite number of irreducible components.*

(b) Suppose U is an open subscheme of a projective A-scheme. Then U is locally of finite type over A. If A is Noetherian, then U is quasi-compact, and hence quasi-projective over A, and hence by (a) of finite type over A.

Proof

(a) Since X is a quasi-projective A-scheme, then X is a quasi-compact open subscheme of $\operatorname{Proj} S_\bullet$, where $S_\bullet$ is a finitely generated graded ring over $S_0 = A$. By Proposition 5.5.5, we may assume that $\operatorname{Proj} S_\bullet = \bigcup_i D_+(f_i)$, then $X = \bigcup_i (X \cap D_+(f_i))$. Since X is open subset of $\operatorname{Proj} S_\bullet$, $X \cap D_+(f_i)$ is open subset of $D_+(f_i) = \operatorname{Spec}((S_\bullet)_{f_i})_0$. Since $S_\bullet$ is a finitely generated graded ring over A, by Proposition 5.5.3 (a), $S_\bullet$ is a finitely generated graded A-algebra, by Proposition 6.3.2, $(S_\bullet)_{f_i}$ is finitely generated A-algebra, and therefore $((S_\bullet)_{f_i})_0$ is finitely generated A-algebra. Since $X \cap D_+(f_i)$ is open subset of $D_+(f_i)$, $X \cap D_+(f_i)$ covered by some distinguished open subsets, we may assume that $X \cap D_+(f_i) = D(g)$, then $X \cap D_+(f_i) = \operatorname{Spec}(((S_\bullet)_{f_i})_0)_g$, by Proposition 6.3.2 again, $(((S_\bullet)_{f_i})_0)_g$ is finitely generated A-algebra. By above discussion, we know that X is quasi-compact, and X can be covered by affine open subsets $\operatorname{Spec}(((S_\bullet)_{f_i})_0)_g$, where each $(((S_\bullet)_{f_i})_0)_g$ is a finitely generated A-algebra, hence, X is of finite type over A.

If A is furthermore assumed to be Noetherian, by Hilbert's Basis Theorem, each $(((S_\bullet)_{f_i})_0)_g$ is Noetherian, and therefore X is locally Noetherian scheme, also X is quasi-compact, X is Noetherian scheme, by Proposition 6.3.4, X has a finite number of irreducible components.

(b) Similar to the proof of (a), by removing the quasi-compactness condition, we conclude that U is locally of finite type over A.

If A is Noetherian. By the proof in (a) and Proposition 6.1.7, we may assume that $U = \bigcup_{i=1}^n (U \cap D_+(f_i))$. Since $U \cap D_+(f_i)$ is open in $\operatorname{Spec}((S_\bullet)_{f_i})_0$, since A is Noetherian, $((S_\bullet)_{f_i})_0$ is Noetherian, by Proposition

4.6.19, $\operatorname{Spec}((S_\bullet)_{f_i})_0$ is quasi-compact, and therefore $U \cap D_+(f_i)$ is quasi-compact. It follows that U is quasi-compact, and therefore U is quasi-projective over A, by part (a), U is of finite type over A.

□

✍ **Exercise 6.2** Show that in Proposition 6.3.6 (b), U may not be finite type over A if A is not Noetherian. Give an example of an open subscheme of a projective A-scheme that is not quasi-compact, necessarily for some non-Noetherian A.

Solution *Let $A = k[x_0, x_1, \ldots]$, consider $\mathbb{P}^0_A$, by Example 5.9 , $\mathbb{P}^0_A \cong \operatorname{Spec} A$. Let $\mathfrak{m} = (x_0, x_1, \ldots)$. By remark after Proposition 4.6.4, $U = \mathbb{P}^0_A \setminus V(\mathfrak{m})$ is not quasi-compact open subset of $\mathbb{P}^0_A$, and therefore U not be finite type over A.*

Affine varieties, Projective varieties

We now make a connection to the classical language of varieties.

Definition 6.3.5 (Affine variety, Projective variety)

*An affine scheme that is reduced and of finite type over k is said to be an **affine variety (over k)**, or an **affine k-variety**. A reduced (quasi)projective k-scheme is a **(quasi)projective variety (over k)**, or a **(quasi)projective k-variety**. (By Proposition 6.3.6, (quasi)projective k-scheme is already of finite type over A.)*

Remark 6.14 **Warning:** in the literature, it is sometimes also assumed in the definition of variety that the scheme is irreducible, or that k is algebraically closed.

Proposition 6.3.7

(a) $\operatorname{Spec} k[x_1, \ldots, x_n]/I$ is an affine k-variety if and only if $I \subseteq k[x_1, \ldots, x_n]$ is a radical ideal.

(b) Suppose $I \subseteq k[x_0, \ldots, x_n]$ is a radical graded ideal. Then $\operatorname{Proj} k[x_0, \ldots, x_n]/I$ is a projective k-variety.

Proof

(a) In fact, $k[x_1, \ldots, x_n]$ is finitely generated k-algebra, and therefore $k[x_1, \ldots, x_n]/I$ is finitely generated k-algebra. By Proposition 4.6.4, $\operatorname{Spec} k[x_1, \ldots, x_n]/I$ is quasi-compact. Hence, $X = \operatorname{Spec} k[x_1, \ldots, x_n]/I$ is of finite type over k. It suffices to show that X is reduced if and only if I is a radical ideal.
If X is reduced, let $f\sqrt{I}$, then exists some n such that $f^n \in I$, hence, $f^n = 0$ in X. Since $\mathfrak{N}(X) = 0$, $f = 0$, which implies that $f \in I$, that is, $\sqrt{I} \subseteq I$. Hence I is radical ideal.
Conversely, if I is radical ideal, we want to show that X is reduced. Let $f \in \mathfrak{N}(\mathscr{O}_X(X)) = \mathfrak{N}(k[x_1, \ldots, x_n]/I)$, then $f^n \in I$, and therefore $f \in \sqrt{I} = I$, hence, $\mathfrak{N}(\mathscr{O}_X(X)) = 0$, it follows that $\mathscr{O}_X(X)$ is reduced ring. By Proposition 6.2.2, X is reduced.

(b) Let $S_\bullet = k[x_0, \ldots, x_n]/I$, then $S_\bullet$ is a finitely generated graded ring, by Proposition 5.5.3 (a), $S_\bullet$ is a finitely generated k-algebra. By Proposition 5.5.8 we have $\operatorname{Proj} S_\bullet = \bigcup_{i=0}^{n} D_+(x_i)$, where $D_+(x_i) = \operatorname{Spec}((S_\bullet)_{x_i})_0$. Since $S_\bullet$ is a finitely generated k-algebra, by Proposition 6.3.2 (b), $(S_\bullet)_{x_i}$ is finitely generated k-algebra, and therefore $((S_\bullet)_{x_i})_0$ is finitely generated k-algebra. By Proposition 6.1.7, $\operatorname{Proj} S_\bullet$ is quasi-compact, hence, $\operatorname{Proj} S_\bullet$ is of finite type over k. To show $\operatorname{Proj} S_\bullet$ is projective k-variety it suffices to show that $\operatorname{Proj} S_\bullet$ is reduced.
Since reducedness is affine-local property, it suffices to show that $D_+(x_i)$ is reduced. Note that $((S_\bullet)_{x_i})_0 \cong k[x_{1/i}, \ldots, x_{n/i}]/I_{x_i}$, it suffices to show that $k[x_{1/i}, \ldots, x_{n/i}]/I_{x_i}$ is reduced ring. We

claim that I_{x_i} is radical ideal. Let $\frac{f}{x_i^n} \in \sqrt{I_{x_i}}$, then $\left(\frac{f}{x_i^n}\right)^m \in I_{x_i}$ for some m. Hence, $f^m \in I$, since I is radical, $f \in I$, and therefore $\frac{f}{x_i^n} \in I_{x_i}$, which implies that I_{x_i} is radical ideal. Let $f \in \mathfrak{N}(\mathscr{O}_{\operatorname{Proj} S_\bullet}(D_+(x_i))) = \mathfrak{N}(k[x_{1/i}, \ldots, x_{n/i}]/I_{x_i})$, then $f^n \in I_{x_i}$ for some n. Since I_{x_i} is radical, $f \in I_{x_i}$. It follows that $\mathfrak{N}(\mathscr{O}_{\operatorname{Proj} S_\bullet}(D_+(x_i))) = 0$. Hence, $D_+(x_i)$ is reduced by Proposition 6.2.2.

□

We will not define varieties in general until Chapter 12; we will need the notion of separatedness first, to exclude abominations like the line with the doubled origin (§5.4.2). But many of the statements we will make in this section about affine k-varieties will automatically apply more generally to k-varieties.

Proposition 6.3.8

A point of a locally finite type k-scheme is a closed point if and only if the residue field of the structure sheaf at that point is a finite extension of k. On a locally finite type k-scheme, the set of all closed points of X is dense (it follows that every neighborhood of a point contains a closed point).

Proof Let X be a locally finite type k-scheme, then $X = \bigcup_i \operatorname{Spec} A_i$ where A_i is finitely generated k-algebra.

If p is a closed point of X. Exists $\operatorname{Spec} A_i$ be an affine open neighborhood of p, since A_i is a finitely generated k-algebra, we may assume that $A_i = k[x_1, \ldots, x_r]$. Since p is a closed point, p is a maximal ideal of A_i. Consider $\mathscr{O}_{X,p}/\mathfrak{m}_p$, then

$$\mathscr{O}_{X,p}/\mathfrak{m}_p = (A_i)_p/\mathfrak{m}_p = (A_i)_p/p(A_i)_p \cong (A_i/p)_{(0)} \cong \operatorname{Frac}(A_i/p),$$

since p is maximal ideal of A_i, $\operatorname{Frac}(A_i/p) = (A_i)/p$, by Zariski's Lemma 4.2.2, A_i/p is a finite extension of k, and therefore $\mathscr{O}_{X,p}/\mathfrak{m}_p$ is a finite extension of k.

Conversely, let $p \in X$ such that $\kappa(p) = \mathscr{O}_{X,p}/\mathfrak{m}_p$ is a finite extension of k. Let $\operatorname{Spec} A_i$ be any affine open neighborhood of p, then $\kappa(p) \cong (A_i)_p/p(A_i)_p \cong \operatorname{Frac}(A_i/p)$ is a finite extension of k. Since A_i is finitely generated k-algebra, A_i/p is also finitely generated k-algebra, and therefore $k \subseteq A_i/p$. Note that $k \subseteq A_i/p \subseteq \operatorname{Frac}(A_i/p)$ where k and $\operatorname{Frac}(A_i/p)$ are both fields, A_i/p must be a field (see [Bec]), hence, A_i/p is a field, and therefore p is maximal ideal of A_i, which implies that p is a closed point in each $\operatorname{Spec} A_i$. Let $\operatorname{cl}(\{p\})$ be the closure of $\{p\}$ in X, note that

$$\operatorname{cl}(\{p\}) \cap \operatorname{Spec} A_i = \overline{\{p\} \cap \operatorname{Spec} A_i} = \overline{\{p\}} = \{p\},$$

hence,

$$\operatorname{cl}(\{p\}) = \bigcup_i (\operatorname{cl}(\{p\}) \cap \operatorname{Spec} A_i) = \bigcup_i \{p\} = \{p\},$$

it follows that $\{p\}$ is closed in X.

Let $\mathscr{S}$ be a set of all closed points of X, we want to show that $\operatorname{cl}_X(\mathscr{S}) = X$. By Proposition 4.6.7, $\operatorname{cl}_X(\mathscr{S}) \cap \operatorname{Spec} A_i = \operatorname{cl}_{\operatorname{Spec} A_i}(\mathscr{S} \cap \operatorname{Spec} A_i) = \operatorname{Spec} A_i$, hence,

$$\operatorname{cl}_X(\mathscr{S}) = \bigcup_i (\operatorname{cl}_X(\mathscr{S}) \cap \operatorname{Spec} A_i) = \bigcup_i \operatorname{Spec} A_i = X,$$

which implies that $\mathscr{S}$ dense in X. □

Remark 6.15 Closed points need not be dense even on quite reasonable schemes, see Proposition 4.6.7 (b).

Definition 6.3.6

*The **degree of a closed point** p of a locally finite type k-scheme (e.g., a variety over k) is the degree of the field extension $\kappa(p)/k$.*

Example 6.7 In $\mathbb{A}_k^1 = \operatorname{Spec} k[t]$, the point $[(p(t))]$ ($p(t) \in k[t]$ irreducible) is $\deg p(t)$. If k is algebraically closed, the degree of every closed point is 1.

The proof of Proposition 6.3.2

Now, let's prove Proposition 6.3.2:

Proposition

Suppose A is a ring, and $(f_1, \ldots, f_n) = A$.

(a) *If A is a Noetherian ring, then so is A_{f_i}. If each A_{f_i} is Noetherian, then so is A.*

(b) *Suppose B is a ring, and A is a B-algebra. (Hence A_g is a B-algebra for all $g \in A$.) If A is a finitely generated B-algebra, then so is A_{f_i}. If each A_{f_i} is a finitely generated B-algebra, then so is A.*

Proof

(a) If A is a Noetherian ring, by Proposition 4.6.15, A_{f_i} is Noetherian ring for all f_i.
Conversely, if each A_{f_i} is Noetherian. Let $I_1 \subseteq I_2 \subseteq I_3 \subseteq \cdots$ be an increasing chain of ideals of A, then in each A_{f_i} we have correspondence increasing chain of ideals,

$$I_1 \otimes_A A_{f_i} \subseteq I_2 \otimes_A A_{f_i} \subseteq I_3 \otimes_A A_{f_i} \subseteq \cdots. \tag{6.6}$$

Since each A_{f_i} is Noetherian, chain (6.6) must be stationary, hence, for each f_i exists N_i such that for all $k \geq N_i$, $I_k \otimes_A A_{f_i} = I_{N_i} \otimes_A A_{f_i}$. Let $N = \max\limits_{1 \leq i \leq n} N_i$, we want to show that for all $k \geq N$, $I_k = I_N$.
Let $x \in I_k$, then $x \in I_k \otimes_A A_{f_i} = I_N \otimes_A A_{f_i}$, hence, exists m_i such that $f_i^{m_i} x \in I_N$. Let $m = \max\limits_{1 \leq i \leq n} m_i$.
Since $(f_1, \ldots, f_n) = A$, exists a_i such that

$$a_1 f_1 + \cdots + a_n f_n = 1,$$

and therefore

$$(a_1 f_1 + \cdots + a_n f_n)^{nm} := b_1 f_1^m + \cdots + b_n f_n^m = 1.$$

Hence,

$$x = (b_1 f_1^m + \cdots + b_n f_n^m)x = b_1 f_1^m x + \cdots + b_n f_n^m x \in I_N,$$

that is, $I_k \subseteq I_N$. Hence, $I_k = I_N$, it follows that A is Noetherian.

(b) If A is a finitely generated B-algebra, we may assume that $A = B[x_1, \ldots, x_n]$, then $A_{f_i} = B[x_1, \ldots, x_n]_{f_i} = B[x_1, \ldots, x_n, 1/f_i]$ is a finitely generated B-algebra.
Conversely, if each A_{f_i} is a finitely generated B-algebra, we may assume that $A_{f_i} = B[\frac{r_{i1}}{f_i^{k_1}}, \ldots, \frac{r_{in_i}}{f_i^{k_{n_i}}}]$.
Since $A = (f_1, \ldots, f_n)$, exists $a_i \in A$ such that $1 = \sum_{i=1}^n a_i f_i$. Let

$$\mathscr{S} = \{f_i\}_{i=1}^n \cup \{a_i\}_{i=1}^n \cup \{r_{ij}\}_{ij},$$

$\mathscr{S}$ is a finite set, we claim that $A = B[\mathscr{S}]$. Clearly, we have $B[\mathscr{S}] \subseteq A$. Let $r \in A$, then in A_{f_i} exists polynomial $p_i(\frac{r_{i1}}{f_i^{k_1}}, \ldots, \frac{r_{in_i}}{f_i^{k_{n_i}}}) \in A_{f_i}$ such that $r = p_i(\frac{r_{i1}}{f_i^{k_1}}, \ldots, \frac{r_{in_i}}{f_i^{k_{n_i}}})$, multiply both sides of the equation by a sufficiently large $f_i^{N_i}$, we obtain

$$f_i^{N_i} r = f_i^{N_i} p_i(\frac{r_{i1}}{f_i^{k_1}}, \ldots, \frac{r_{in_i}}{f_i^{k_{n_i}}}) \in B[\mathscr{S}].$$

Let $N = \max_{1\leq i\leq n} N_i$, similar to the part (a) we have $1 = \sum_{i=1}^n b_i f_i^N$ for some $b_i \in B[\mathscr{S}]$. Hence,

$$r = (\sum_{i=1}^n b_i f_i^N)r = \sum_{i=1}^n b_i f_i^N r \in B[\mathscr{S}],$$

which implies that $A \subseteq B[\mathscr{S}]$, and therefore $A = B[\mathscr{S}]$, i.e., A is finitely generated B-algebra.

□

6.4 Normality and factoriality

6.4.1 Normality

We can now define a property of schemes that says that they are "not too far from smooth", called **normality**, which will come in very handy. We will see later that "locally Noetherian normal schemes satisfy Hartogs's Lemma" (Algebraic Hartogs's Lemma for Noetherian normal schemes): functions defined away from a set of codimension 2 extend over that set. (We saw a first glimpse of this in §5.4.1.) As a consequence, rational functions (Definition 6.2.3) that have no poles (certain sets of codimension one where the function isn't defined) are defined everywhere. We need definitions of dimension and poles to make this precise. See Chapter 14 and Chapter 27 for the fact that "smoothness" (really, "regularity") implies normality.

Definition 6.4.1 (Integral)

Let $\varphi : R \to S$ be a ring map.

(i) An element $s \in S$ is integral over R if there exists a monic polynomial $P(x) \in R[x]$ such that $P^\varphi(s) = 0$, where $P^\varphi(x) \in S[x]$ is the image of P under $\varphi : R[x] \to S[x]$.

(ii) The ring map φ is integral if every $s \in S$ is integral over R.

In the following discussion, we will only consider integral domains.

Definition 6.4.2 (Integral closure, Integrally closed)

*Let $R \to S$ be a ring homomorphism. The ring $S' = \{s \in S : s \text{ is integral over } R\}$ is called the **integral closure** of R in S. If $R \subseteq S$ we say that R is **integrally closed** in S if $R = S'$.*

In this section we just use following definition of integrally closed:

Definition 6.4.3 (Integrally closed)

*An integral domain A is **integrally closed** if the only zeros in $K(A)$ (fraction field of A, i.e., $\mathrm{Frac}(A)$) to any monic polynomial in $A[x]$ must lie in A itself.*

Remark 6.16 Consider the ring homomorphism $A \to K(A)$, Definition 6.4.3 agree with Definition 6.4.2.

Definition 6.4.4 (Normal)

*We say a scheme X is **normal** if all of its stalks $\mathscr{O}_{X,p}$ are normal, i.e., are integral domains, and integrally closed in their fraction fields $K(\mathscr{O}_{X,p})$.*

Remark 6.17

(1) By definition, normality is a stalk-local property, and therefore is affine-local property. Normal scheme is also reduced (Definition 6.2.1), since each stalks are integral domain.

(2) One might say that a ring A is **normal ring** if $\operatorname{Spec} A$ is normal, thereby extending the notion of "integral closure" to rings that are not integral domains.

Proposition 6.4.1 (Integral closure commutes with localization)

If $A \to B$ is a ring homomorphism, and $S \subseteq A$ is a multiplicative subset, then the integral closure of $S^{-1}A$ in $S^{-1}B$ is $S^{-1}B'$, where $B' \subseteq B$ is the integral closure of A in B.

Proof Since localization is exact we know that $S^{-1}B' \subseteq S^{-1}B$. Let $x/f \in S^{-1}B'$ where $x \in B'$ and $f \in S$, then

$$x^d + \sum_{i=1}^{d} a_i x^{d-i} = 0$$

in B for some $a_i \in A$. Hence, we have

$$(x/f)^d + \sum_{i=1}^{d} (a_i/f^i)(x/f)^{d-i} = 0$$

in $S^{-1}B$, it follows that x/f in the integral closure of $S^{-1}A$ in $S^{-1}B$, and therefore, $S^{-1}B'$ is contained in the integral closure of $S^{-1}A$ in $S^{-1}B$.

Conversely, suppose that $x/f \in S^{-1}B$ is integral over $S^{-1}A$. Then we have

$$(x/f)^d + \sum_{i=1}^{d} (a_i/f_i)(x/f)^{d-i} = 0$$

in $S^{-1}B$ for some $a_i \in A$ and $f_i \in S$. This means that

$$(f' f_1 \cdots f_d x)^d + \sum_{i=1}^{d} b_i a_i (f' f_1 \cdots f_d x)^{d-i} = 0,$$

for a suitable $f' \in S$. Hence, $f' f_1 \cdots f_d x \in B'$, and therefore $x/f \in S^{-1}B'$, which implies that the integral closure of $S^{-1}A$ in $S^{-1}B$ contained in $S^{-1}B'$. □

Corollary 6.4.1 (Integrally closed domain behave well under localization)

If A is an integrally closed domain, and S is a multiplicative subset not containing 0, then $S^{-1}A$ is an integrally closed domain.

Proof Since A integrally closed, $K(A)' = A$, apply Proposition 6.4.1, then we done. □

Corollary 6.4.2

If A is an integrally closed domain, then $\operatorname{Spec} A$ *is normal.*

Proof Since A is integrally closed, by Corollary 6.4.1, $A_{\mathfrak{p}}$ in integrally closed for all $[\mathfrak{p}] \in \operatorname{Spec} A$. □

Corollary 6.4.3

For schemes that are quasi-compact or locally finite type over a field, normality can be checked at closed point. More precisely, if $\mathscr{O}_{X,p}$ is normal (integral domain and integrally closed in their fraction fields) for all closed point $p \in X$, then X is normal

Proof We want to show that $\mathscr{O}_{X,p}$ is normal for all $p \in X$. Suppose q is not a closed point in X, since X is quasi-compact or locally finite type over a field, by Proposition 6.1.5 or Proposition 6.3.8 (on a locally finite type k-scheme the set of all closed points of X is dense, which implies that every neighborhood of q intersects this set), $\overline{\{q\}}$ contains a closed point of X, say p. Let U be an affine open neighborhood of p, note that q is a generic point for $\overline{\{q\}}$, by Proposition 4.6.10, U contains q. Since U is affine open subset, we may assume that $U = \operatorname{Spec} A$, then $\operatorname{Spec} A \cap \overline{\{q\}} = V(q)$. Note that $p \in V(q)$, we have $p \supseteq q$. Note that $\mathscr{O}_{X,q} = A_q$, we want to show that $\mathscr{O}_{X,q} = A_q$ is integrally domain and is integrally closed in its fraction fields. Since $p \supseteq q$, we have $(A_p)_{qA_p} = A_q$ (see the proof in Proposition 6.2.4). Since $\mathscr{O}_{X,p} = A_p$ is normal, by Corollary 6.4.1, $(A_p)_{qA_p} = A_q$ is normal, as we desired. □

Example 6.8 **It is not true that normal schemes are integral.** For example, the disjoint union of two normal schemes is normal. Thus $\operatorname{Spec} k \coprod \operatorname{Spec} k \cong \operatorname{Spec}(k \times k) \cong \operatorname{Spec} k[x]/(x(x-1))$ is normal, but its ring of global sections is not an integral domain, and therefore is not integral scheme.

Proof We give a proof of the fact that the disjoint union of two normal schemes is normal. Let X, Y be two normal schemes. Let $p \in X \coprod Y$, then $p \in X$ or $p \in Y$. Since X and Y are open in $X \coprod Y$, we have $\mathscr{O}_{X \coprod Y,p} = \mathscr{O}_{X,p}$ or $\mathscr{O}_{X \coprod Y,p} = \mathscr{O}_{Y,p}$. Since X and Y are normal, $\mathscr{O}_{X,p}$ and $\mathscr{O}_{Y,p}$ are normal, and therefore $\mathscr{O}_{X \coprod Y,p}$ is normal, it follows that $X \coprod Y$ is normal. □

Proposition 6.4.2

A Noetherian scheme is normal if and only if it is the finite disjoint union of integral Noetherian normal schemes.

Proof Let X be a Noetherian scheme, if X is the finite disjoint union of integral Noetherian normal schemes, say $X = \coprod_{i=1}^{n} X_i$, where each X_i is integral Noetherian normal scheme. Let $p \in X$, then $p \in X_i$ for some i, hence, $\mathscr{O}_{X,p} = (\mathscr{O}_X|_{X_i})_p$ is normal, it follows that X is normal.

Conversely, if X is normal scheme, then X is reduced by the definition of normal scheme (Definition 6.4.4). Since X is Noetherian scheme, by Proposition 6.3.4, X has a finite number of irreducible components, say $X = \bigcup_{i=1}^{n} X_i$ where X_i is irreducible component. We claim that each X_i disjoint. Suppose that $X_i \cap X_j \neq \varnothing$, let $p \in X_i \cap X_j$, then there exists an affine open subset $\operatorname{Spec} A \ni p$. Hence, $p \in (\operatorname{Spec} A \cap X_i) \cap (\operatorname{Spec} A \cap X_j)$. Note that both $\operatorname{Spec} A \cap X_i$ and $\operatorname{Spec} A \cap X_j$ are irreducible components of $\operatorname{Spec} A$, by Hilbert's Nullstellensatz, $\operatorname{Spec} A \cap X_i = V(\mathfrak{q}_i)$ and $\operatorname{Spec} A \cap X_j = V(\mathfrak{q}_j)$ where $\mathfrak{q}_i$ and $\mathfrak{q}_j$ are nonzero minimal prime ideals of A. Also, since $p \in V(\mathfrak{q}_i) \cap V(\mathfrak{q}_j)$, we have $p \supseteq q_i$ and $p \supseteq q_j$, i.e., p has two different minimal prime ideals. It follows that (0) is not the prime ideal of $\mathscr{O}_{X,p} = A_p$ (since if $(0) \in \operatorname{Spec} A_p$, p must has unique minimal prime ideal (0)), hence, $\mathscr{O}_{X,p}$ is not domain, contradicts to the fact that $\mathscr{O}_{X,p}$ is normal domain. Thus each X_i is disjoint, and therefore $X = \coprod_{i=1}^{n} X_i$. Clearly, each X_i is a scheme, since X is normal, X_i is normal. It is easy to check X_i is Noetherian from X is Noetherian. Note that X_i is normal, irreducible, reduced, Noetherian scheme, by Proposition 6.2.7, X_i is integral Noetherian normal schemes. Hence, X is the finite disjoint union of integral Noetherian normal schemes. □

We are close to proving a useful result in commutative algebra, so we may as well go all the way.

Proposition 6.4.3

If A is an integral domain, then the following are equivalent.

(i) A is integrally closed.

(ii) $A_{\mathfrak{p}}$ is integrally closed for all prime ideals $\mathfrak{p} \subseteq A$.

(iii) $A_{\mathfrak{m}}$ is integrally closed for all maximal ideals $\mathfrak{m} \subseteq A$.

Proof Corollary 6.4.1 shows that integral closedness is preserved by localization, so (i) implies (ii). Clearly, (ii) implies (iii).

It remains to show that (iii) implies (i). Suppose A is not integrally closed. We show that there is some $\mathfrak{m}$ such that $A_{\mathfrak{m}}$ is also not integrally closed. Suppose

$$x^n + a_{n-1}x^{n-1} + \cdots + a_0 = 0 \tag{6.7}$$

(with $a_i \in A$) has a solution s in $K(A) \setminus A$. Let I be the **ideal of denominators of** s:

$$I := \{r \in A : rs \in A\}. \tag{6.8}$$

It is easy to see I is an ideal of A. Note that $1 \notin A$, we have $I \neq A$. Thus there is some maximal ideal $\mathfrak{m}$ containing I. We claim that $s \notin A_{\mathfrak{m}}$. If not, let $s = a/f$ where $a \in A$ and $f \in A - \mathfrak{m}$. Since $\mathfrak{m} \supseteq I$, we have $f \in A - I$, which implies that $fs \notin A$, a contradiction. Hence, $s \notin A_{\mathfrak{m}}$. So equation (6.7) in $A_{\mathfrak{m}}[x]$ shows that $A_{\mathfrak{m}}$ is not integrally closed as well, as we desired. □

Corollary 6.4.4

Spec A *is normal if and only if A is an integrally closed domain.*

Proof If A is an integrally closed domain, by Corollary 6.4.2, Spec A is normal.

Conversely, if Spec A is normal, then all of its stalks $\mathscr{O}_{X,[\mathfrak{p}]} = A_{\mathfrak{p}}$ is normal, by Proposition 6.4.3, A is normal. □

Proposition 6.4.4

If A is an integral domain, then $A = \bigcap_{\mathfrak{m}\in \operatorname{Max} A} A_{\mathfrak{m}}$.

Proof Since A is integral domain, we have inclusion $A \hookrightarrow A_{\mathfrak{m}}$. Hence, $A \subseteq \bigcap_{\mathfrak{m}\in \operatorname{Max} A} A_{\mathfrak{m}}$. Let $s \in \bigcap_{\mathfrak{m}\in \operatorname{Max} A} A_{\mathfrak{m}}$, suppose that $s \notin A$. Consider the ideal of denominators of s, i.e.,

$$I := \{r \in A : rs \in A\}.$$

By the hypothesis, $1 \notin I$, and therefore $I \neq A$. Thus there is some maximal ideal $\mathfrak{m}$ containing I. Similar to the proof of Proposition 6.4.3, $s \notin A_{\mathfrak{m}}$, a contradiction. Hence $s \in A$, which implies that $A = \bigcap_{\mathfrak{m}\in \operatorname{Max} A} A_{\mathfrak{m}}$. □

✍ **Exercise 6.3** **(Unimportant exercise relating to the ideal of denominators.)** One might naively hope from experience with unique factorization domains that the ideal of denominators is principal. This is not true. As a counterexample, consider our new friend $A = k[w, x, y, z]/(wz - xy)$ which we first met in Example 5.7 , and which we will later recognize as the cone over the quadric surface, and $w/y = x/z \in K(A)$. Show that the ideal of denominators of this element of $K(A)$ is (y, z).

Proof Let I be the ideal of denominators of $w/y = x/z$, clearly $y, z \in I$, and therefore $(y, z) \subseteq I$. Conversely, let $s \in I$, then $sw/y \in A$ or $sx/z \in A$. Hence exists $b \in A$ such that $sw = yb$ or $sx = zb$, i.e., $sw \in (y)$ or $sx \in (z)$. Since both (y) and (z) are prime ideals, note that w, x not in (y) and (z), $s \in (y)$ or $s \in (z)$, hence, $s \in (y, z)$. It follows that $I = (y, z)$. □

We see that the I in Exercise 6.3 is not principal (In Chpater 14, you may be able to show it directly, using the fact that I is a homogeneous ideal of a graded ring). But we will also see that in good situations (Noetherian, normal), the ideal of denominators is "pure codimension 1" — this is the content of Algebraic Hartog's Lemma. In its proof, Chapter 14, we give a geometric interpretation of the ideal of denominators.

6.4.2 Factoriality

Definition 6.4.5 (Factorial)

We say a scheme X is factorial, if the stalk $\mathscr{O}_{X,p}$ is UFD (unique factorization domain) for all $p \in X$.

Remark 6.18 The locus of points on an affine variety over an algebraically closed field that are factorial is an open subset.

Proposition 6.4.5

Any nonzero localization of a UFD is a UFD.

Proof Let A be a UFD, and $S \subseteq A$ be a multiplicative subset not containing 0. Let $\frac{a}{s} \in S^{-1}A$, where $a \in A$ and $s \in S$. Since A is UFD, we may assume that $a = up_1^{n_1} \cdots p_k^{n_k}$ and $s = vq_1^{m_1} \cdots q_r^{m_r}$ where u, v are units and each p_i, q_j are prime elements, then

$$\frac{a}{s} = \frac{u}{v} \cdot p_1^{n_1} \cdots p_k^{n_k} \cdot \frac{1}{q_1^{m_1}} \cdots \frac{1}{q_r^{m_r}}.$$

We claim that p_i and $\frac{1}{q_j}$ are prime element in $S^{-1}A$. Let $p_i = \frac{b_1}{t_1} \cdot \frac{b_2}{t_2} \in S^{-1}A$ where $b_1, b_2 \in A$ and $t_1, t_2 \in S$, then exists $l \in S$ such that $l(p_i t_1 t_2 - b_1 b_2) = 0$ in A, i.e., $lp_i t_1 t_2 = lb_1 b_2$, since A is UFD, $p_i \mid b_1$ or $p_i \mid b_2$. Without loss of generality, we may assume that $p_1 \mid b_1$, hence $b_1 = up_1$ for some unit u. Then $p_i = \frac{up_1}{t_1} \cdot \frac{b_2}{t_2}$, note that $\frac{up_1}{t_1}$ is unit in $S^{-1}A$, we have $p_i \mid \frac{b_2}{t_2}$. Hence, p_i is prime element. Similarly, it is easy to show that $\frac{1}{q_j}$ is prime element in $S^{-1}A$.

Next we want to show that the decomposition of $\frac{a}{s}$ is unique. Let

$$\frac{a}{s} = \frac{u}{v} \cdot p_1^{n_1} \cdots p_k^{n_k} \cdot \frac{1}{q_1^{m_1}} \cdots \frac{1}{q_r^{m_r}} = \frac{u'}{v'} \cdot p_1'^{n_1'} \cdots p_k'^{n_k'} \cdot \frac{1}{q_1'^{m_1'}} \cdots \frac{1}{q_r'^{m_r'}}. \tag{6.9}$$

where $\frac{u'}{v'}$ is unit in $S^{-1}A$, be another decomposition of $\frac{a}{s}$. Then exists w such that

$$wuv'p_1^{n_1} \cdots p_k^{n_k} q_1'^{m_1'} \cdots q_r'^{m_r'} = wu'vp_1'^{n_1'} \cdots p_k'^{n_k'} q_1^{m_1} \cdots q_r^{m_r},$$

by the unique factoriality of A, $p_i = p_i'$, $q_j = q_j'$, $n_i = n_i'$, and $m_j = m_j'$, it follows that the decomposition of $\frac{a}{s}$ is unique. Hence, $S^{-1}A$ is UFD. □

Corollary 6.4.5

If A is a UFD, then Spec A *is factorial.*

Proof By Proposition 6.4.5, the localization of A is UFD, hence, $\mathscr{O}_{\operatorname{Spec} A, [\mathfrak{p}]} = A_{\mathfrak{p}}$ is UFD for all $[\mathfrak{p}] \in \operatorname{Spec} A$, and therefore Spec A is factorial. □

Remark 6.19 The converse need not hold. In fact, we will see that elliptic curves are factorial, yet NO affine open set is the Spec of a unique factorization domain. Hence, one can show factoriality by finding an appropriate affine cover, but there need not be such a cover of a factorial scheme. (One might reasonably call a ring A such that Spec A is factorial, **a factorial ring**; this is a strictly weaker notion than UFD. We won't need this terminology.)

Remark 6.20 **How to check if a ring is a UFD ?** There are very few means of checking that a Noetherian integral domain is a UFD. Some useful ones are:

(i) By Gauss's Lemma, the polynomial rings over a UFD is UFD.

(ii) The localization of a unique factorization domain is also a unique factorization domain (Proposition 6.4.5).

(iii) Height 1 prime ideals are principal (Proposition 13.3.7).
(iv) Normal and $\mathrm{Cl} = 0$ (Proposition 16.6.8).
(v) Nagata's Lemma (Theorem 16.6.2).

Caution: even if A is a UFD, $A[[t]]$ need not be UFD, see [Mat89, page 165]. The first example, due to P.Salmon, was $A = k(u)[[x, y, z]]/(z^2 + x^3 + uz^6)$.

One reason we like factoriality is that it implies normality.

Proposition 6.4.6 (Factoriality is that it implies normality)

UFDs are integrally closed. Hence, factorial schemes are normal.

Proof Let A be a UFD, and $\frac{a}{b} \in K(A)$ agree with equation

$$x^n + c_{n-1}x^{n-1} + \cdots + c_1 x + c_0 = 0 \tag{6.10}$$

in $A[x]$. Without loss of generality, we may assume that a, b do not have common irreducible factor. By (6.10), we have

$$a^n + c_{n-1}a^{n-1}b + \cdots + c_1 ab^{n-1} + c_0 b^n = 0,$$

hence,

$$b(c_{n_1}a^{n-1} + \cdots + c_0 b^{n-1}) = -a^n.$$

Since a, b do not have common irreducible factor, $b \nmid a^n$, it follows that b is a unit, hence, $\frac{a}{b} = ab^{-1} \in A$, which implies that A is integrally closed. □

Remark 6.21 However, rings can be integrally closed without being UFD, as we will see in Exercise 6.9. Another example is given without proof in Exercise 6.11; in that example, Spec of the ring is factorial. A variation on Exercise 6.9 will show that schemes can be normal without being factorial.

Corollary 6.4.6

If A is UFD, then Spec A *is normal.*

Proof By Proposition 6.4.6, it is clearly. □

6.4.3 Examples

✍ **Exercise 6.4** The following schemes are normal: $\mathbb{A}^n_k$, $\mathbb{P}^n_k$, $\operatorname{Spec}\mathbb{Z}$.

Proof Since $k[x_1, \ldots, x_n]$ and $\mathbb{Z}$ are both UFDs, by Corollary 6.4.6, $\mathbb{A}^n_k$ and $\operatorname{Spec}\mathbb{Z}$ are normal. Let $p \in \mathbb{P}^n_k$, then $d \in D_+(x_i)$ for some x_i, since $D_+(x_i) \cong \mathbb{A}^n_k$, $\mathscr{O}_{\mathbb{P}^n_k, p} \cong \mathscr{O}_{\mathbb{A}^n_k, p}$ is UFD. Hence, $\mathbb{P}^n_k$ is normal. □

✍ **Exercise 6.5** **(Yielding many enlightening examples later.)** Suppose A is a UFD with 2 invertible, and $z^2 - f$ is irreducible in $A[z]$.

(a) Show that if $f \in A$ has no repeated prime factors, then $\operatorname{Spec} A[z]/(z^2 - f)$ is normal.
(b) Show that if $f \in A$ has repeated prime factors, then $\operatorname{Spec} A[z]/(z^2 - f)$ is not normal.

Proof

(a) Since A is UFD, then $A[z]$ is UFD. Since the irreducible element in UFD is prime element, and therefore $(z^2 - f)$ is prime ideal of $A[z]$. Hence, $B := A[z]/(z^2 - f)$ is indeed an integral domain. Suppose $\alpha \in K(B)$ integral over B, then exists a monic polynomial $F(T) \in B[T]$ such that $F(\alpha) = 0$. Replacing $F(T)$ by $\overline{F}(T)F(T)$, then $F(T) \in A[T]$. Note that $K(B) = K(A)(z)/(z^2 - f)$, we may assume that $\alpha = g + hz$ where $g, h \in K(A)$. Note that α is the solution of $Q(T) = T^2 - 2gT + (g^2 - h^2 f) \in K(A)[T]$, so we can factor $F(T) = P(T)Q(T)$ in $K(A)[T]$. By multiplying $P(T)$ and $Q(T)$ by suitable elements $c_1, c_2 \in A$ to clear denominators, we obtain $c_1P(T) \in A[T]$ and $c_2Q(T) \in A[T]$ such that they are primitive. Hence, by Gauss's Lemma (see [AM18, page 11], Exercises 2 (iv)), $c_1c_2P(T)Q(T) = c_1c_2F(T)$ is primitive, since $F(T)$ is primitive, $c_1c_2 = 1$, hence, $c_1 = c_2 = 1$. It follows that $Q(T) \in A[T]$, hence, $2g \in A$ and $g^2 - h^2 f \in A$. Since 2 invertible, $g \in A$, and therefore $h^2 f \in A$. Let $f = p_1p_2 \cdots p_n \in A$ with distinct prime element p_i and $h = \frac{s}{t}$ with $\gcd(s,t) = 1$, then

$$h^2 f = \frac{s^2}{t^2} \cdot p_1p_2 \cdots p_n \in A.$$

Hence, $t^2 \mid s^2 p_1p_2 \cdots p_n$, since each p_i distinct, t must be unit, and therefore $h \in A$. Consequently, $\alpha \in B$, which implies that $B = A[z]/(z^2 - f)$ is integrally closed, by Corollary 6.4.2, $\operatorname{Spec} A[z]/(z^2 - f)$ is normal.

(b) Without loss of generality, let $f = p^2 q$, where p is prime element and $p \nmid q$. Let $\alpha = \frac{z}{p} \in K(B)$, note that

$$\alpha^2 = \frac{z^2}{p^2} = \frac{f}{p^2} = q \in A,$$

hence, α agree with $t^2 - q = 0$ in $B[t]$, i.e., α. We want to show that $\alpha \notin B$.

If $\alpha \in B$, then exists $c, d \in A$ such that

$$\frac{z}{p} = c + dz \Longrightarrow z = pc + pdz \Longrightarrow (1 - pd)z = pc.$$

Hence, $1 - pd = 0$ and $pc = 0$, note that p is a prime element, it is impossible. It follows that $\alpha \notin B$. Hence, B is not integrally closed, by Corollary 6.4.4, $\operatorname{Spec} A[z]/(z^2 - f)$ is not normal.

□

✍ **Exercise 6.6** Show that the following schemes are normal:

(a) $\operatorname{Spec} \mathbb{Z}[x]/(x^2 - n)$ where n is a square-free integer congruent to 3 modulo 4. **Caution:** the hypotheses of Exercise 6.5 do not apply, so you will have to do this directly. (Your argument may also show the result when 3 is replaced by 2. A similar argument shows that $\mathbb{Z}[(1 + \sqrt{n})/2]$ is integrally closed if $n \equiv 1 \pmod 4$ is square-free.)

(b) $\operatorname{Spec} k[x_1, \ldots, x_n]/(x_1^2 + x_2^2 + \cdots + x_m^2)$ where $\operatorname{char} k \neq 2$, $n \geq m \geq 3$.

(c) $\operatorname{Spec} k[w, x, y, z]/(wz - xy)$ where $\operatorname{char} k \neq 2$. This is our cone over a quadric surface example from Example 5.7 and Exercise 6.3. (The result also holds for $\operatorname{char} k = 2$, but don't worry about this.)

Proof

(a) We claim that $x^2 - n$ is an irreducible element in $\mathbb{Z}[x]$. If $x^2 - n = (a_1x + b_1)(a_2x + b_2)$, then $a_1a_2 = 1$, $a_1b_2 + a_2b_1 = 0$, and $b_1b_2 = n$. Without loss of generality, we may assume that $a_1 = a_2 = 1$, then $n = b_1^2$, contradicts to the fact that n is a sequence. Hence, $x^2 - n$ is irreducible, since $\mathbb{Z}[x]$ is UFD, $(x^2 - n)$ is prime ideal. Hence, $\mathbb{Z}[x]/(x^2 - n)$ is integral domain.

Since n is a square-free number, we have $\mathbb{Z}[x]/(x^2 - n) \cong \mathbb{Z}[\sqrt{n}]$. We want to show that $\mathbb{Z}[\sqrt{n}]$ is integrally closed. Let $\gamma = \alpha + \beta\sqrt{n} \in K(\mathbb{Z}[\sqrt{n}]) = \mathbb{Q}(\sqrt{n})$ where $\alpha, \beta \in \mathbb{Q}$ and γ integral over $\mathbb{Z}[\sqrt{n}]$, then there exists a monic polynomial $F(T) \in \mathbb{Z}[\sqrt{n}][T]$ such that $F(\gamma) = 0$. Replacing $F(T)$ by $\overline{F}(T)F(T)$, then $F(T) \in \mathbb{Z}[T]$. Note that γ is the solution of $Q(T) = T^2 - 2\alpha T + (\alpha^2 - n\beta^2) \in \mathbb{Q}[T]$, so we can factor $F(T) = P(T)Q(T)$ in $\mathbb{Q}[T]$. By multiplying $P(T)$ and $Q(T)$ by suitable elements $c_1, c_2 \in \mathbb{Z}$ to clear denominators, we obtain $c_1P(T) \in \mathbb{Z}[T]$ and $c_2Q(T) \in \mathbb{Z}[T]$ such that they are primitive, by Gauss's Lemma ([AM18, page 11], Exercise 2 (iv)), $c_1c_2P(T)Q(T) = c_1c_2F(T)$ is primitive, since $F(T)$ is primitive, $c_1c_2 = 1$, hence, $c_1 = c_2 = 1$. It follows that $Q(T) \in \mathbb{Z}[T]$, hence,

$2\alpha \in \mathbb{Z}$ and $\alpha^2 - n\beta^2 \in \mathbb{Z}$. Suppose $\alpha = \frac{a}{2}$ and $\beta = \frac{c}{d}$ with $\gcd(c,d) = 1$, then

$$\alpha^2 - n\beta^2 = \frac{a^2}{4} - n \cdot \frac{c^2}{d^2} \Longrightarrow 4d^2(\alpha^2 - n\beta^2) = a^2d^2 - 4nc^2.$$

Note that $a^2d^2 - 4nc^2 \equiv a^2d^2 \pmod 4$ and $4d^2(\alpha^2 - n\beta^2) \equiv 0 \pmod 4$, we have

$$a^2d^2 \equiv 0 \pmod 4,$$

i.e., $4 \mid a^2d^2$, i.e., $2 \mid ad$.

If d is odd, then $2 \mid a$, let $a = 2k$, then

$$\alpha^2 - n\beta^2 = \frac{4k^2}{4} - \frac{nc^2}{d^2} = k^2 - \frac{nc^2}{d^2} \in \mathbb{Z}.$$

Hence, $d^2 \mid nc^2$, since $gcd(c,d) = 1$, $d^2 \mid n$. It is impossible, since n is square-free.

If d is even, then $2 \mid d$, let $d = 2k$, then

$$\alpha^2 - n\beta^2 = \frac{a^2}{4} - \frac{nc^2}{4k^2} \in \mathbb{Z}.$$

Hence, exists m such that $m = \frac{a^2}{4} - \frac{nc^2}{4k^2}$, and therefore $a^2 - \frac{nc^2}{k^2} = 4m$, it follows that

$$a^2 - \frac{nc^2}{k^2} \equiv 0 \pmod 4.$$

Since $n \equiv 3 \pmod 4$, we have $\frac{nc^2}{k^2} \equiv \frac{3c^2}{k^2} \pmod 4$, hence,

$$a^2 \equiv \frac{3c^2}{k^2} \pmod 4 \Longrightarrow k^2a^2 \equiv 3c^2 \pmod 4.$$

If a is even, then $k^2a^2 \equiv 0 \pmod 4$, hence $3c^2 \equiv 0 \pmod 4$, it is impossible, since $c^2 \equiv 0 \pmod 4$ or $c^2 \equiv 1 \pmod 4$.

If a is odd, then $a^2 \equiv 1 \pmod 4$. If k is odd, then $k^2 \equiv 1 \pmod 4$, hence, $k^2a^2 \equiv 1 \pmod 4$. But $3c^2 \equiv 0$ or $3 \pmod 4$, a contradiction. If k is even, $k^2a^2 \equiv 0 \pmod 4$, hence, $3c^2 \equiv 0 \pmod 4$. It follows that c is even. Note that $\gcd(c,d) = 1$ and $d = 2k$, a contradiction.

By above discussion, $\frac{a}{2}$ and $\frac{c}{d}$ must belong to $\mathbb{Z}$, and therefore $\gamma \in \mathbb{Z}[\sqrt{n}]$. It follows that $\mathbb{Z}[\sqrt{n}]$ is integrally closed in $\mathbb{Q}(\sqrt{n})$. By Corollary 6.4.2, $\operatorname{Spec} \mathbb{Z}[x]/(x^2 - n)$ is normal.

(b) Let $z = x_1$, $f = x_2^2 + \cdots + x_m^2$, and $A = k[x_2, \ldots, x_n]$, then $k[x_1, \ldots, x_n]/(x_1^2 + x_2^2 + \cdots + x_m^2) \cong A[z]/(z^2 + f)$. We want to show (i) A is UFD with 2 invertible; (ii) f has no repeated prime factors; (iii) $z^2 + f$ is irreducible in $A[z]$. Then by Exercise 6.5, $\operatorname{Spec} A[z]/(z^2 + f)$ is normal.

(i) A is UFD with 2 invertible.

Clearly, $A = k[x_2, \ldots, x_n]$ is a UFD. Since $\operatorname{char} k \neq 2$, 2 is invertible.

(ii) f has no repeated prime factors

We claim that f is irreducible in A. Note that $\deg f = 2$, if $f = gh$, then $\deg g = \deg h = 1$. Let $g = a_2x_2 + \cdots + a_mx_m$ and $h = b_2x_2 + \cdots + b_mx_m$, then

$$\begin{aligned} x_1^2 + \cdots + x_m^2 =& (a_2x_2 + \cdots + a_mx_m)(b_2x_2 + \cdots + b_mx_m) \\ =& \sum_{i=2}^{m} a_ib_ix_i^2 + \sum_{2 \le i < j \le m} (a_ib_j + a_jb_i)x_ix_j. \end{aligned}$$

Hence, $a_ib_i = 1$ and $a_ib_j + a_jb_i = 0$, thus, we have

$$\frac{a_i}{a_j} + \frac{a_j}{a_i} = \frac{a_i^2 + a_j^2}{a_ia_j} = 0 \Longrightarrow a_i^2 + a_j^2 = 0 \Longrightarrow a_i = \pm a_j\sqrt{-1}.$$

Sine $a_2 = \pm a_1\sqrt{-1}$ and $a_3 = \pm a_1\sqrt{-1}$, we have $a_2^2 + a_3^2 = -2a_1^2 = 0 \Rightarrow a_1 = 0$, by induction, $a_i = b_i = 0$ for all i, it is impossible. Hence f is irreducible in A, and therefore f has no repeated prime factor.

(iii) $z^2 + f$ is irreducible in $A[z]$.

By Gauss's Lemma, if $z^2 + f$ is irreducible in $K(A)[z]$ and $z^2 + f$ is primitive, then $z^2 + f$ is irreducible in $A[z]$. We claim that $z^2 + f$ is irreducible in $K(A)[z]$. If not, there exists $\frac{h}{g} \in K(A)$ where $h \in A$ and $g \in A$ such that $\frac{h^2}{g^2} = -f$, i.e., $h^2 = -fg^2$. Since f is irreducible, $f \mid h^2$, and therefore $f \mid h$. Let $h = fh'$, then $f^2h'^2 = -fg^2 \Rightarrow fh'^2 = -g^2$, by f is irreducible, $f \mid g^2 \Rightarrow f \mid g$. Let $g = fg'$, then we have $h'^2 = -fg'^2$. Repeat above discussion, we have ${h^{(n)}}^2 = -f{g^{(n)}}^2$ and

$$h = fh' = f^2h'' = \cdots = f^n h^{(n)} = \cdots, \quad g = fg' = f^2g'' = \cdots = f^n g^{(n)} = \cdots,$$

where $\deg h^{(n)} = \deg h^{(n-1)} - 1$ and $\deg g^{(n)} = \deg g^{(n-1)} - 1$. It is impossible, since h and g are finite degree. Hence, $z^2 + f$ is irreducible in $K(A)[z]$. Since it is primitive, by Gauss's Lemma, $z^2 + f$ is irreducible in $A[z]$.

By above discussion, apply Exercise 6.5, $\operatorname{Spec} A[z]/(z^2 + f) \cong \operatorname{Spec} k[x_1, x_2, \ldots, x_n]/(x_1^2 + \cdots + x_m^2)$ is normal.

(c) Note that

$$wz - xy = \left(\frac{w+z}{2}\right)^2 + \left(\frac{x-y}{2}\right)^2 + \left(\frac{i(w-z)}{2}\right)^2 + \left(\frac{i(x+y)}{2}\right)^2,$$

apply part (b), we done.

□

✍ **Exercise 6.7** **(Diagonalizing quadratic forms.)** Suppose k is an algebraically closed field of characteristic not 2. (The hypothesis that k is algebraically closed is not necessary, so feel free to deal with this more general case.)

(a) Show that any quadratic form in n variables can be "diagonalized" by changing coordinates to be a sum of at most n squares. (e.g., $uw - v^2 = \left(\frac{u+w}{2}\right)^2 + \left(\frac{i(u-w)}{2}\right)^2 + (iv)^2$, where the linear forms appearing in the squares are linearly independent.)

(b) Show that the number of squares appearing depends only on the quadratic form. For example, $x^2+y^2+z^2$ cannot be written as a sum of two squares. (Possible approach: given a basis $x_1, \ldots, x_n$ of the linear forms, write the quadratic form as

$$\begin{pmatrix} x_1 & \cdots & x_n \end{pmatrix} M \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}$$

where M is a symmetric matrix. Determine how M transforms under a change of basis, and show that the rank of M is independent of the choice of basis.)

Proof This result comes from linear algebra and is presented without proof. You may see [con25]. □

Remark 6.22 The **rank** of the quadratic form is the number of ("linearly independent") squares needed. If the number of squares equals the number of variables, the quadratic form is said to be **full rank** or (of) **maximal rank**.

✍ **Exercise 6.8** **(Rings can be integrally closed but not UFD, arithmetic version.)** Show that $\mathbb{Z}[\sqrt{-5}]$ is integrally closed but not a UFD.

Proof In fact, $K(\mathbb{Z}[\sqrt{-5}]) = \mathbb{Q}(\sqrt{-5})$. Let $\gamma = \alpha + \beta\sqrt{-5} \in \mathbb{Q}(\sqrt{-5})$ is integrally closed over $\mathbb{Z}[\sqrt{-5}]$, then exists $F(t) \in \mathbb{Z}[\sqrt{-5}][t]$ such that $F(\gamma) = 0$. Replacing $F(t)$ by $\overline{F}(t)F(t)$, then $F(t) \in \mathbb{Z}[t]$. Note that $\gamma = \alpha + \beta\sqrt{-5}$ agree with $Q(t) = t^2 - 2\alpha x + \alpha^2 + 5\beta^2 = 0$. Similar to the proof in Exercise 6.6 (a),

$Q(t) \in \mathbb{Z}[t]$. Hence, $2\alpha \in \mathbb{Z}$ and $\alpha^2 + 5\beta^2 \in \mathbb{Z}$. Let $\alpha = \frac{a}{c}$ and $\beta = \frac{b}{c}$ where $a, b, c \in \mathbb{Z}$ and $\gcd(a, b, c) = 1$, then

$$\frac{2a}{c} \in \mathbb{Z}, \quad \frac{a^2}{c^2} + \frac{5b^2}{c^2} = \frac{a^2 + 5b^2}{c^2} \in \mathbb{Z}.$$

We want to show $c = 1$. If $c > 1$, then exists prime number p such that $p \mid c$. Since $c \mid 2a$, we have $p \mid 2a$, and therefore $p = 2$ or $p \mid a$. Also, since $c^2 \mid (a^2 + 5b^2)$, we have $p^2 \mid (a^2 + 5b^2)$.

If $p = 2$, then we may assume that $c = 2m$, and therefore $m \mid a$. If $m \neq 1$. Let $a = km$, then we have

$$4m^2 \mid (k^2m^2 + 5b^2) \Longrightarrow 4 \mid (k^2 + 5 \cdot \left(\frac{b}{m}\right)^2).$$

Since $\gcd(a, b, c) = 1$, $m \nmid b$, hence, $\frac{b^2}{m^2} \notin \mathbb{Z}$, and therefore $\frac{5b^2}{m^2} \in \mathbb{Z}$. It is impossible, since 5 has no repeated prime factor. Hence, $m = 1$, it follows that $c = 2, a = k$, then $4 \mid (k^2 + 5b^2)$, i.e., $k^2 + 5b^2 \equiv 0 \pmod 4$. Thus,

$$k^2 + b^2 \equiv 0 \pmod 4,$$

which implies that $4 \mid k^2$ and $4 \mid b^2$, i.e., $2 \mid k$ and $2 \mid b$. Hence $\gcd(a, b, c) \geq 2 > 1$, a contradiction.

If $p \mid a$, then from $p^2 \mid (a^2 + 5b^2)$ we have $p^2 \mid 5b^2$. If $p \nmid 5$, then $p \mid b$, note that $p \mid c$, $\gcd(a, b, c) \geq p > 1$, a contradiction. Hence, $p \mid 5$, i.e., $p = 5$. Since $5 \mid b^2$, $5 \mid b$, i.e., $p \mid b$, hence, $\gcd(a, b, c) \geq p > 1$, a contradiction.

By above discussion, $c = 1$, hence, $\gamma = a + b\sqrt{-5} \in \mathbb{Z}[\sqrt{-5}]$, which implies that $\mathbb{Z}[\sqrt{-5}]$ is integrally closed. Consider $6 \in \mathbb{Z}[\sqrt{-5}]$, note that

$$6 = 2 \times 3 = (1 + \sqrt{-5})(1 - \sqrt{-5}),$$

$\mathbb{Z}[\sqrt{-5}]$ not a UFD. □

✍ **Exercise 6.9** **(Rings can be integrally closed but not UFD, geometric version.)** Suppose $\operatorname{char} k \neq 2$. Let $A = k[w, x, y, z]/(wz - xy)$, so $\operatorname{Spec} A$ is the cone over the smooth quadric surface (Example 5.7 and Exercise 6.3).

(a) Show that A is integrally closed.

(b) Show that A is not a UFD.

Proof

(a) By Exercise 6.6, $\operatorname{Spec} k[w, x, y, z]/(wz - xy)$ is normal, by Corollary 6.4.4, $k[w, x, y, z]/(wz - xy)$ is integrally closed domain.

(b) In $k[w, x, y, z]/(wz - xy)$, we have $wz = xy$. We want to show that w, z, x, y are irreducible elements. Without loss of generality, it suffices to show that w is irreducible, others are similar. Suppose $w = fg$, since $\deg w = 1$, $\deg f + \deg g = 1$, hence f or g must be constant, and therefore w is irreducible.

□

Remark 6.23 Exercise 6.8 and Exercise 6.9 look similar, but there is a difference. The cone over the quadric surface is normal but not factorial. On the other hand, $\operatorname{Spec} \mathbb{Z}[\sqrt{-5}]$ is factorial — all of its stalks are UFD. (Since $\mathbb{Z}[\sqrt{-5}]$ is a Dedekind domain).

✍ **Exercise 6.10** Suppose A is a k-algebra, and l is a finite field extension of k. (Your proof might not use finiteness; this hypothesis is included to avoid distraction by infinite-dimensional vector spaces.) Show that if $A \otimes_k l$ is a normal integral domain, then A is a normal integral domain as well. (Although we won't need this, a version of the converse is true if l is separable extension of k.)

Proof Since l is finite field extension of k, l is finitely generated k-module. Let $b_1 = 1, b_2, \ldots, b_d$ be the generators of l. Note that A is a k-module, hence, $A \otimes_k l$ is a finitely generated A-module, its generators are

$1 \otimes b_1, \ldots, 1 \otimes b_d$. Clearly, we have the following injections

$$\begin{tikzcd} A \arrow[r, hook] \arrow[d, hook] & K(A) \arrow[d, hook] \\ A \otimes_k l \arrow[r, hook] & K(A) \otimes_k l. \end{tikzcd} \tag{6.11}$$

We claim that $K(A) \otimes_k l = K(A \otimes_k l)$. We first show that $K(A) \otimes_k l$ is a field. Since l is finite filed extension of k, $K(A) \otimes_k l$ is finitely generated $K(A)$-module, also $k(A) \otimes_k l$ is a ring, $k(A) \otimes_k l$ is a finite $K(A)$-algebra. We shall to show that $K(A) \otimes_k l$ is integral domain. Let $S = k - \{0\}$, then $S^{-1}A = K(A)$, $S^{-1}k = k$, and $S^{-1}l = l$. Hence, we have k-module isomorphism

$$K(A) \otimes_k l = S^{-1}A \otimes_{S^{-1}k} S^{-1}l \cong S^{-1}(A \otimes_k l).$$

Since $A \otimes_k l$ is integral domain, $S^{-1}(A \otimes_k l)$ is integral domain, and therefore $K(A) \otimes_k l$ is integral domain. By Proposition 4.2.1, $K(A) \otimes_k l$ is a field. Since $K(A \otimes_k l)$ is the minimal field which contains $A \otimes_k l$, by diagram (6.11) we have $A \otimes_k l \hookrightarrow K(A) \otimes_k l$, hence, $K(A \otimes_k l) \subseteq K(A) \otimes_k l$. We next show that $K(A) \otimes_k l \hookrightarrow K(A \otimes_k l)$. Define map $\iota : K(A) \otimes_k l \to K(A \otimes_k l)$ by setting

$$\frac{a}{b} \otimes c \longmapsto \frac{a \otimes c}{b \otimes 1}.$$

Clearly, ι is well-defined field homomorphism, and therefore ι is injective. Hence, $K(A) \otimes_k l \subseteq K(A \otimes_k l)$. Consequently, $K(A) \otimes_k l = K(A \otimes_k l)$.

We claim that $(A \otimes_k l) \cap K(A) = A$. By diagram (6.11), we have $A \subseteq A \otimes_k l$ and $A \subseteq K(A)$, hence, $A \subseteq (A \otimes_k l) \cap K(A)$. Conversely, let $t = \sum_{i=1}^n a_i \otimes b_i \in (A \otimes_k l) \cap K(A)$, since $t \in K(A)$, we can multiply by some $u \in A$ such that $ut = u \cdot (\sum_{i=1}^n a_i \otimes b_i) = \sum_{i=1}^n ua_i \otimes b_i \in A$. Hence, there exists $v \otimes 1 \in A$ such that $\sum_{i=1}^n ua_i \otimes b_i = v \otimes 1$, it follows that $a_j = 0$ for all $j \geq 2$. Hence, $t = a_1 \otimes b_1 = a_1 \otimes 1 \in A$. Thus $(A \otimes_k l) \cap K(A) = A$.

Finally, we show that A is integrally closed. Let $\gamma \in K(A)$ integral over A, then exists monic polynomial $F(T) \in A[T]$ such that $F(\gamma) = 0$. By diagram (6.11), $K(A) \hookrightarrow K(A) \otimes_k l$, we may write γ as $\gamma \otimes 1$. Since $K(A) \otimes_k l = K(A \otimes_k l)$, $\gamma \otimes 1$ in $K(A \otimes_k l)$. Since $\gamma \otimes 1$ is the foot of $F(T) \in (A \otimes_k l)[T]$, note that $A \otimes_k l$ integrally closed, we have $\gamma \otimes 1 \in A \otimes_k l$. Hence, $\gamma \in (A \otimes_k l) \cap K(A) = A$, which implies that A is integrally closed. □

✍ **Exercise 6.11** **(UFD-ness is not affine-local.)** Let $A = \left(\mathbb{Q}[x,y]_{x^2+y^2}\right)_0$ denote the homogeneous degree 0 part of the ring $\mathbb{Q}[x,y]_{x^2+y^2}$. In other words, it consists of quotients $\frac{f(x,y)}{(x^2+y^2)^n}$, where f has pure degree $2n$.

(i) Show that the distinguished open sets $D\left(\frac{x^2}{x^2+y^2}\right)$ and $D\left(\frac{y^2}{x^2+y^2}\right)$ cover $\operatorname{Spec} A$.

(ii) Show that $A_{\frac{x^2}{x^2+y^2}}$ and $A_{\frac{y^2}{x^2+y^2}}$ are UFDs.

(iii) Finally, show that A is not a UFD.

Proof

(i) Note that $\frac{x^2}{x^2+y^2} + \frac{y^2}{x^2+y^2} = 1$, then $A = (\frac{x^2}{x^2+y^2}, \frac{y^2}{x^2+y^2})$. By Proposition 4.5.2, we have $\operatorname{Spec} A = D\left(\frac{x^2}{x^2+y^2}\right) \cup D\left(\frac{y^2}{x^2+y^2}\right)$, as we desired.

(ii) Define a ring homomorphism $\varphi : A \to \mathbb{Q}[t]_{t^2+1}$ by setting

$$\frac{f(x,y)}{(x^2+y^2)^n} \longmapsto \frac{f(x,tx)}{(x^2+t^2x^2)^n} = \frac{f(1,t)}{(1+t^2)^n}.$$

It is easy to check φ is well-defined. Consider $\varphi(\frac{x^2}{x^2+y^2})$,

$$\varphi(\frac{x^2}{x^2+y^2}) = \frac{x^2}{x^2+t^2x^2} = \frac{1}{1+t^2}$$

is the unit in $\mathbb{Q}[t]_{t^2+1}$. By the universal property of localization, there exists unique ring homomorphism $\varphi^\dagger : A_{\frac{x^2}{x^2+y^2}} \to \mathbb{Q}[t]_{t^2+1}$.

$$\begin{tikzcd}
A \arrow[r, "\varphi"] \arrow[d, "\text{localization}"'] & \mathbb{Q}[t]_{t^2+1} \arrow[dl, dashed, shift left, "\psi^\dagger"] \\
A_{\frac{x^2}{x^2+y^2}} \arrow[ur, dashed, shift left, "\varphi^\dagger"] & \mathbb{Q}[t] \arrow[l, "\psi"] \arrow[u, "\text{localization}"']
\end{tikzcd}$$

Define $\psi : \mathbb{Q}[t] \to A_{\frac{x^2}{x^2+y^2}}$ by setting

$$f(t) \longmapsto f\left(\frac{y}{x}\right) = \frac{\frac{x^{2d} f\left(\frac{y}{x}\right)}{(x^2+y^2)^d}}{\frac{x^{2d}}{(x^2+y^2)^d}},$$

where $d = \deg f$. It is easy to check ψ is a well-defined ring homomorphism. Consider $\psi(t^2+1)$, we have

$$\psi(t^2+1) = \frac{y^2}{x^2} + 1 = \frac{x^2+y^2}{x^2}$$

is the unit in $A_{\frac{x^2}{x^2+y^2}}$. By the universal property of localization, there exists unique ring homomorphism $\psi^\dagger : \mathbb{Q}[t]_{t^2+1} \to A_{\frac{x^2}{x^2+y^2}}$. Hence, we have

$$A_{\frac{x^2}{x^2+y^2}} \overset{\sim}{\longleftrightarrow} \mathbb{Q}[t]_{t^2+1}.$$

Similarly, we have $A_{\frac{y^2}{x^2+y^2}} \overset{\sim}{\longleftrightarrow} \mathbb{Q}[t]_{t^2+1}$.

(iii) In A, note that

$$\left(\frac{xy}{x^2+y^2}\right)^2 = \left(\frac{x^2}{x^2+y^2}\right)\left(\frac{y^2}{x^2+y^2}\right),$$

A is not a UFD.

□

Number theorists may prefer the example of Exercise 6.8: $\mathbb{Z}[\sqrt{-5}]$ is not a UFD, but it turns out that you can cover it with two affine open subsets $D(2)$ and $D(3)$, each corresponding to UFDs. (For number theorists: to show that $\mathbb{Z}[\sqrt{-5}]_2$ and $\mathbb{Z}[\sqrt{-5}]_3$ are UFDs, first show that the class group of $\mathbb{Z}[\sqrt{-5}]$ is $\mathbb{Z}/2$ using the geometry of numbers. Then show that the ideals $(1+\sqrt{-5}, 2)$ and $(1+\sqrt{-5}, 3)$ are not principal, using the usual norm in $\mathbb{C}$.) The ring $\mathbb{Z}[\sqrt{-5}]$ is an example of a Dedekind domain.

Remark 6.24 For an example of k-algebra A that is not a UFD, but becomes one after a particular field extension, see Chapter 16.

Chapter 7 Rings are to modules as schemes are to $\cdots$

When considering the notion of rings for the first time, one is quickly led to define the notion of module. For example, any ring morphism $R \to S$ expresses S as an R-module. An ideal I of a ring is also an R-module, as is the quotient R/I; and the exact sequence $0 \to I \to R \to R/I \to 0$ is a first example of the utility of the abelian category structure of $\mathbf{Mod}_R$.

With a scheme X, you may think that we already have the right analog of modules: the category of $\mathscr{O}_X$-modules $\mathbf{Mod}_{\mathscr{O}_X}$, which also forms an abelian category (§3.6.2). But the right analog for a scheme X turns out to be better-behaved subset (subcategory!) of $\mathbf{Mod}_{\mathscr{O}_X}$, called **quasi-coherent $\mathscr{O}_X$-modules**, or **quasi-coherent sheaves**.

7.1 Quasi-coherent sheaves

7.1.1 Definition of quasi-coherent sheaf

Given an A-module M, we defined an $\mathscr{O}$-module $\widetilde{M}$ on $\operatorname{Spec} A$ in §5.1.2 — the sections over $D(f)$ were M_f. These are our "local models" for quasi-coherent sheaves. In the same way that a scheme is defined by "gluing together rings", a quasi-coherent sheaf over that scheme is obtained by "gluing together modules over those rings".

Definition 7.1.1 (The category of quasi-coherent sheaves)

If X is a scheme, then an $\mathscr{O}_X$-module $\mathscr{F}$ is a ***quasi-coherent sheaf*** *if for every affine open subset $\operatorname{Spec} A \subseteq X$, $\mathscr{F}|_{\operatorname{Spec} A} \cong \widetilde{M}$ for some A-module M. We make quasi-coherent sheaves on a scheme X into category $\mathbf{QCoh}_X$ by taking the morphisms to be morphisms as $\mathscr{O}_X$-modules.*

Theorem 7.1.1

Let X be a scheme, and $\mathscr{F}$ an $\mathscr{O}_X$-module. Suppose $\mathscr{P}$ is the property of affine open subschemes $\operatorname{Spec} A \subseteq X$ that $\mathscr{F}|_{\operatorname{Spec} A} \cong \widetilde{M}$ for some A-module M (in other words, if affine open subscheme $\operatorname{Spec} A \subseteq X$ satisfies $\mathscr{F}|_{\operatorname{Spec} A} \cong \widetilde{M}$ for some A-module M, we say $\operatorname{Spec} A$ has property $\mathscr{P}$). Then $\mathscr{P}$ satisfies the two hypotheses of the Affine-local property (Definition 6.3.1). Thus to check quasi-coherence of an $\mathscr{O}$-module, it suffices to check a collection of affine open sets covering X, by Affine Communication Lemma 6.3.1.

Remark 7.1 For example, $\widetilde{M}$ is a quasi-coherent sheaf on $\operatorname{Spec} A$. Proposition 5.1.4 shows that the category $\mathbf{Mod}_A$ and the category $\mathbf{QCoh}_{\operatorname{Spec} A}$ are essentially the same (we have described an equivalence $\mathbf{Mod}_A \overset{\sim}{\longleftrightarrow} \mathbf{QCoh}_{\operatorname{Spec} A}$).

Proof As usual, the first hypothesis of the Affine-local property 6.3.1 is easier: if $\operatorname{Spec} A$ has property $\mathscr{P}$, then $\operatorname{Spec} A_f$ has property $\mathscr{P}$: if M is an A-module, then $\widetilde{M}|_{\operatorname{Spec} A_f} \cong \widetilde{M_f}$ as sheaves of $\mathscr{O}_{\operatorname{Spec} A_f}$-modules (both sides agree on the level of distinguished open sets and their restriction maps), i.e., $\mathscr{F}|_{\operatorname{Spec} A_f} \cong \widetilde{M_f}$, hence, $\operatorname{Spec} A_f$ has property $\mathscr{P}$.

We next show the second hypothesis of the Affine-local property 6.3.1. We are given:

- an $\mathscr{O}_{\operatorname{Spec} A}$-module $\mathscr{F}$,
- elements $f_1, \ldots, f_n$ generating A,
- and isomorphisms $\varphi_i : \mathscr{F}|_{D(f_i)} \xrightarrow{\sim} \widetilde{M_i}$.

The isomorphisms

$$\begin{tikzcd} & \mathscr{F}|_{D(f_i)\cap D(f_j)} \arrow[dl, "\sim"'] \arrow[dr, "\sim"] & \\ \widetilde{M_i}|_{D(f_i)\cap D(f_j)} & & \widetilde{M_j}|_{D(f_i)\cap D(f_j)} \end{tikzcd}$$

yield (by taking sections over $D(f_i) \cap D(f_j)$) isomorphisms $\varphi_{ij} : (M_i)_{f_j} \xrightarrow{\sim} (M_j)_{f_i}$ of $A_{f_if_j}$-modules, satisfying the cocycle condition (Theorem 3.5.3). As suggestive notation, let $M_{ij} := (M_i)_{f_j}$ (identified with $(M_j)_{f_i}$ via φ_{ij}).

We seek an A-module M with an isomorphism $\widetilde{M} \xleftrightarrow{\sim} \mathscr{F}$, and (5.5) suggests that we should define M by

$$0 \longrightarrow M \longrightarrow M_1 \times \cdots \times M_2 \xrightarrow{\gamma} M_{12} \times M_{13} \times \cdots \times M_{(n-1)n}. \tag{7.1}$$

(The discussion at (5.5) suggests how the map γ should be define, with appropriate signs.) Translation: $M = \operatorname{Ker}(\gamma)$. If we can describe an isomorphism $\widetilde{M}|_{D(f_i)} \xrightarrow{\sim} \widetilde{M_i}$ on each $D(f_i)$ (or equivalently, an isomorphism $M_{f_i} \xrightarrow{\sim} M_i$ of A_{f_i}-modules), such that triangle

$$\begin{tikzcd} & \widetilde{M}|_{D(f_i)\cap D(f_j)} \arrow[dl, "\sim"'] \arrow[dr, "\sim"] & \\ \widetilde{M_i}|_{D(f_i)\cap D(f_j)} \arrow[rr, "\widetilde{\varphi_{ij}}", "\sim"'] & & \widetilde{M_j}|_{D(f_i)\cap D(f_j)} \end{tikzcd}$$

of sheaves on $\operatorname{Spec} A_{f_if_j}$ commutes (for all i, j), then Theorem 3.5.3 (on gluing sheaves) gives our desired isomorphism $\widetilde{M} \xleftrightarrow{\sim} \mathscr{F}$.

By Proposition 5.1.4, we need only describe an isomorphism $M_{f_i} \xrightarrow{\sim} M_i$ such that the triangles

$$\begin{tikzcd} & M_{f_if_j} \arrow[dl, "\sim"'] \arrow[dr, "\sim"] & \\ (M_i)_{f_j} \arrow[rr, "\varphi_{ij}"] & & (M_j)_{f_i} \end{tikzcd} \tag{7.2}$$

commute.

For notational convenience, we assume $i = 1$. Because localization is exact, form (7.1) we have an exact sequence.

$$0 \longrightarrow M_{f_1} \longrightarrow (M_1)_{f_1} \times \cdots \times (M_n)_{f_1} \xrightarrow{\gamma_{f_1}} (M_{12})_{f_1} \times (M_{13})_{f_1} \times \cdots \times (M_{(n-1)n})_{f_1}$$

We will show that

$$0 \longrightarrow M_1 \xrightarrow{\beta} (M_1)_{f_1} \times \cdots \times (M_n)_{f_1} \xrightarrow{\gamma_{f_1}} (M_{12})_{f_1} \times (M_{13})_{f_1} \times \cdots \times (M_{(n-1)n})_{f_1} \tag{7.3}$$

is an exact sequence, where β defined by $m \mapsto (m/1, 0, \ldots, 0)$. If sequence 7.3 is exact, we have

$$M_{f_1} \cong \operatorname{Im}(M_{f_i} \to (M_1)_{f_1} \times \cdots \times (M_n)_{f_1}) = \operatorname{Ker} \gamma_{f_1} = \operatorname{Im} \beta \cong M_1,$$

as we desired. We rewrite (7.3) as

$$\begin{tikzcd} 0 \arrow[r] & M_1 \arrow[r] & (M_1)_{f_1} \times \cdots \times (M_n)_{f_1} \arrow[d, "\alpha"] \\ & & (M_1)_{f_1f_2} \times \cdots \times (M_1)_{f_1f_n} \times (M_1)_{f_2f_3} \times \cdots \times (M_1)_{f_{n-1}f_n}. \end{tikzcd} \tag{7.4}$$

But this is precisely the exact sequence (5.5), except the ring A is replaced by A_{f_1}, and the module M is replaced by M_1. Hence, we have

$$M_{f_i} \xrightarrow{\sim} M_i.$$

Note that $(M_{f_i})_{f_j} \cong (M_{f_j})_{f_i}$, triangle (7.2) commutes. □

Remark 7.2 The proof of Theorem 7.1.1, phrased slightly more carefully, shows that quasi-coherent sheaves satisfy faithfully flat descent.

7.1.2 Examples of quasi-coherent sheaves

We give some examples to show that not every $\mathscr{O}_X$-module is a quasi-coherent sheaf.

Example 7.1 Suppose $X = \operatorname{Spec} k[t]$. Let $\mathscr{F}$ be the skyscraper sheaf supported at the origin $[(t)]$, with group $k(t)$ and the usual $k[t]$-module structure. Then $\mathscr{F}$ is an $\mathscr{O}_X$-module that is not a quasi-coherent sheaf. (More generally, if X is an integral scheme, and $p \in X$ is not generic point, we could take the skyscraper sheaf at p with group the function field of X. Except in silly circumstances, this sheaf won't be quasi-coherent.)

Proof Clearly, the skyscraper sheaf is an $\mathscr{O}_X$-module. Consider $(\mathscr{F}|_{D(t+1)})_{[\mathfrak{p}]}$, note that

$$(\mathscr{F}|_{D(t+1)})_{[\mathfrak{p}]} = \begin{cases} k(t) & \text{if } [\mathfrak{p}] = [(t)] \\ 0 & \text{if } [\mathfrak{p}] \neq [(t)] \end{cases}$$

but $\widetilde{k(t)}_{[\mathfrak{p}]} = k(t)_{\mathfrak{p}}$ where $k(t)$ as $k[t]_{t+1}$-module, hence, $(\mathscr{F}|_{D(t+1)})_{[\mathfrak{p}]}$ not isomorphic to $\widetilde{k(t)}_{[\mathfrak{p}]}$, and therefore $\mathscr{F}|_{D(t+1)}$ not isomorphic to $\widetilde{k(t)}$. Hence, $\mathscr{F}$ is not a quasi-coherent sheaf. □

Example 7.2 Suppose $X = \operatorname{Spec} k[t]$. Let $\mathscr{F}$ be the skyscraper sheaf supported at the generic point $[(0)]$, with group $k(t)$ and the usual $k[t]$-module structure. Then $\mathscr{F}$ is a quasi-coherent sheaf

Proof Clearly, $\mathscr{F}$ is an $\mathscr{O}_X$-module. We claim that $\mathscr{F} \cong \widetilde{k(t)}$, by Proposition 3.4.4, it suffices to check $\mathscr{F}_{[\mathfrak{p}]} \cong \widetilde{k(t)}_{[\mathfrak{p}]}$. Let U be any open affine neighborhood of $[\mathfrak{p}]$, by Proposition 4.6.10, $[(0)] \in U$, $\mathscr{F}(U) = k(t)$, and therefore $\mathscr{F}_{[\mathfrak{p}]} = k(t)$. On the other hand, $\widetilde{k(t)} = k(t)_{\mathfrak{p}} \cong k(t)$, since $k(t)$ is a field. Hence, we have $\mathscr{F}_{[\mathfrak{p}]} \cong \widetilde{k(t)}_{[\mathfrak{p}]}$, as we desired. It follows that $\mathscr{F}$ is a quasi-coherent sheaf. □

Remark 7.3 Example 7.2 will apply more generally, for example when X is an integral scheme with generic point η, and $\mathscr{F}$ is the skyscraper sheaf $i_{\eta,*}K(X)$.

7.1.3 Torsion-free sheaves (a stalk-local condition) and torsion sheaves

Definition 7.1.2 (Torsion-free)

*An A-module M is said to be **torsion-free** if $am = 0$ implies that either a is a zero-divisor in A or $m = 0$.*

Remark 7.4 In the case where A is an integral domain, which is basically the only context in which we will use this concept, the definition of torsion-freeness can be restated as $am = 0$ only if $a = 0$ or $m = 0$.

Definition 7.1.3 (Torsion submodule)

*Let A be an integral domain, M is an A-module. The **torsion submodule** of M, denoted $\operatorname{Tor}(M)$, consists of those elements of M annihilated by some nonzero element of A, i.e.,*

$$\operatorname{Tor} M := \{m \in A : am = 0, \text{ for some nonzero } a \in A\}.$$

(If A is not an integral domain, this construction needn't yield an A-module.)

Remark 7.5 Clearly M is torsion-free if and only if $\operatorname{Tor} M = 0$.

Definition 7.1.4 (Torsion module)

We say a module M over an integral domain A is ***torsion*** *if $M = \operatorname{Tor} M$, this is equivalent to $M \otimes_A K(A) = 0$.*

Definition 7.1.5 (Torsion-free sheaf)

If X is a scheme, then an $\mathscr{O}_X$-module $\mathscr{F}$ is said to be ***torsion-free*** *if $\mathscr{F}_p$ is a torsion-free $\mathscr{O}_{X,p}$-module for all p.*

Motivated by the definition of $\operatorname{Tor} M$ above, we give the definition of **torsion quasi-coherent sheaf on a reduced scheme**:

Definition 7.1.6 (Torsion quasi-coherent sheaves on a reduced schemes)

We say that a quasi-coherent sheaf on a reduced scheme is ***torsion*** *if its stalk at the generic point of every irreducible component is 0.*

We will mainly use this for coherent sheaves on regular curves, where this notion is very simple indeed, but in the literature it comes up in more general situation.

7.2 Characterizing quasi-coherence using the distinguished affine base

Because quasi-coherent sheaves are locally of a very special form, in order to "know" a quasi-coherent sheaf, we need only know what the sections are over every affine open set, and how to restrict sections from an affine open set U to a distinguished affine open subset of U. We make this precise by defining what we will call the **distinguished affine base** of the Zariski topology — not a base in the usual sense. This is a refinement of the notion of a **sheaf on a base**. The point of this discussion is to give a useful characterization of quasi-coherence, but you may wish to just jump to §7.2.2.

7.2.1 Distinguished affine base

The open sets of the distinguished affine base are the affine open subsets of X. We have already observed that this forms a base. But forget that fact. We like distinguished open sets $\operatorname{Spec} A_f \hookrightarrow \operatorname{Spec} A$, and we don't really understand open embeddings of one random affine open subset in another. So we just remember the "nice" inclusions.

Definition 7.2.1 (Distinguished affine base)

The ***distinguished affine base*** *of a scheme X is the data of the affine open sets and the distinguished inclusion.*

In other words, we remember only some of the open sets (the affine open sets), and only some of the morphisms between them (the distinguished morphisms). If we think of a topology as a topology as a category (the category of open sets), we have described a subcategory.

We can define a sheaf on the distinguished affine base in the obvious way: we have a set (or abelian group, or ring) for each affine open set, and we know how to restrict to distinguished open sets.

Given a sheaf $\mathscr{F}$ on X, we get a sheaf on the distinguished affine base. We will show that all the information of the sheaf is contained in the information of the sheaf on the distinguished affine base.

As a warm-up, we can recover stalks as follows. (We will be implicitly using only the following fact. We have a collection of open subsets, and some inclusions among these subsets, such that if we have any $p \in U, V$ where U and V are in collection of open sets, there is some W containing p, and contained in U and V such that $W \hookrightarrow U$ and $W \hookrightarrow V$ are both in our collection of inclusions. In the case we are considering here, this is the key Proposition 6.3.1 that given any two affine open sets $\operatorname{Spec} A$, $\operatorname{Spec} B$ in X, $\operatorname{Spec} A \cap \operatorname{Spec} B$ could be covered by affine open sets that were simultaneously distinguished in $\operatorname{Spec} A$ and $\operatorname{Spec} B$. In fancy language: the category of affine open sets, and distinguished inclusion, forms a filtered set 2.3.4.)

The stalk $\mathscr{F}_p$ is the colimit $\varinjlim(f \in \mathscr{F}(U))$ where the colimit is over all open sets contained in X. We compare this to $\varinjlim(f \in \mathscr{F}(U))$ where the colimit is over all affine open sets, and all distinguished inclusions. It is easy to check that the elements of one correspond to elements of the other (check by definition of germs).

Lemma 7.2.1

Let $\mathscr{F}$ be a sheaf on a scheme X, it gives a sheaf $\mathscr{F}^b$ on the distinguished base of X. Let $p \in X$ be a point, then there is a one-to-one correspondence between the elements of stalk $\mathscr{F}_p$ and the elements of stalk $\mathscr{F}_p^b$.

Proof Let $(f, U) \in \mathscr{F}_p$, where $U \ni p$ is an open subset of X. Then there exists an affine open subsets $\operatorname{Spec} A \subseteq U$ where $\operatorname{Spec} A \ni p$ such that $(f, U) \sim (f|_{\operatorname{Spec} A}, \operatorname{Spec} A)$, note that $(f|_{\operatorname{Spec} A}, \operatorname{Spec} A) \in \mathscr{F}_p^b$, define a map from $\mathscr{F}_p$ to $\mathscr{F}_p^b$ by setting $(f, U) \mapsto (f|_{\operatorname{Spec} A}, \operatorname{Spec} A)$ where $\operatorname{Spec} A \ni p$ contained in U. Conversely, let $(f|_{\operatorname{Spec} A}, \operatorname{Spec} A) \in \mathscr{F}_p^b$, clearly, $(f|_{\operatorname{Spec} A}, \operatorname{Spec} A) \in \mathscr{F}_p$. Hence, we get a one-to-one correspondence between the elements of stalk $\mathscr{F}_p$ and the elements of stalk $\mathscr{F}_p^b$. □

Proposition 7.2.1

A section of a sheaf on the distinguished affine base is determined by the section's germs.

Proof By Proposition 3.4.1 and Lemma 7.2.1, the natural map

$$\mathscr{F}(\operatorname{Spec} A) = \mathscr{F}^b(\operatorname{Spec} A) \longrightarrow \prod_{p \in \operatorname{Spec} A} \mathscr{F}_p = \prod_{p \in \operatorname{Spec} A} \mathscr{F}_p^b$$

is injective, then we done. □

Theorem 7.2.1

(a) A sheaf on the distinguished affine base $\mathscr{F}^b$ determines a unique (up to unique isomorphism) sheaf $\mathscr{F}$ which when restricted to the affine base is $\mathscr{F}^b$. (Hence if you start with a sheaf, and take the sheaf on the distinguished affine base, and then take the induced sheaf, you get the sheaf you started with.)

(b) A morphism of sheaves on a distinguished affine base uniquely determines a morphism of sheaves.

(c) An $\mathscr{O}_X$-module "on the distinguished affine base" yields an $\mathscr{O}_X$-module.

Remark 7.6

(1) This proof is identical to our argument of §3.5 showing that sheaves are (essentially) the same as sheaves on a base, using the "sheaf of compatible germs" construction. The main reason for repeating it is to let you see that all that is needed is for the open sets to form a filtered set (or in the current case, that the category of open sets and distinguished inclusions is filtered).

(2) (a) and (b) are describing an equivalence of categories between sheaves on the Zariski topology of X and sheaves on the distinguished affine base of X.

Proof

(a) Suppose $\mathscr{F}^b$ is a sheaf on the distinguished affine base. Then we can define stalks.

For any open set U of X, define the sheaf of compatible germs

$$\mathscr{F}(U) := \{(f_p \in \mathscr{F}^b_p)_{p\in U} : \text{ for all } p \in U, \text{ there exists } U_p \text{ with } p \in U_p \subseteq U, \\ s \in \mathscr{F}^b(U_p) \text{ such that } s_q = f_q \text{ for all } q \in U_p\},$$

where each U_p is in our base, and s_q means "the germ of s at q".

This really is a sheaf: convince yourself that we have restriction maps, identity, and gluability, really quite easily.

We next claim that if U is in our base, that $\mathscr{F}(U) = \mathscr{F}^b(U)$. We clearly have a map $\mathscr{F}^b(U) \to \mathscr{F}(U)$. This is an isomorphism on stalks, and hence an isomorphism by Proposition 3.4.4.

(b) Theorem 3.5.2.

(c) Remark of Theorem 3.5.2.

□

7.2.2 A characterization of quasi-coherent sheaves in terms of distinguished inclusions

We use this perspective to give a useful characterization of quasi-coherent sheaves among $\mathscr{O}_X$-modules. Suppose $\mathscr{F}$ is an $\mathscr{O}_X$-module, and $\operatorname{Spec} A_f \hookrightarrow \operatorname{Spec} A \subseteq X$ is a distinguished open subscheme of an affine open subscheme of X. Let $\mathrm{res} : \Gamma(\operatorname{Spec} A, \mathscr{F}) \to \Gamma(\operatorname{Spec} A_f, \mathscr{F})$ be the restriction map. The source of res is an A-module, and the target is an A_f-module, so by the universal property of localization, res naturally factors as:

$$\begin{tikzcd} \Gamma(\operatorname{Spec} A, \mathscr{F}) \arrow[rr, "\mathrm{res}"] \arrow[dr, "\otimes_A A_f"'] & & \Gamma(\operatorname{Spec} A_f, \mathscr{F}) \\ & \Gamma(\operatorname{Spec} A, \mathscr{F})_f . \arrow[ur, "\alpha"'] & \end{tikzcd}$$

Theorem 7.2.2

An $\mathscr{O}_X$-module $\mathscr{F}$ is quasi-coherent if and only if for each such distinguished $\operatorname{Spec} A_f \hookrightarrow \operatorname{Spec} A$,

$$\begin{tikzcd} \Gamma(\operatorname{Spec} A, \mathscr{F}) \arrow[rr, "\mathrm{res}"] \arrow[dr, "\otimes_A A_f"'] & & \Gamma(\operatorname{Spec} A_f, \mathscr{F}) \\ & \Gamma(\operatorname{Spec} A, \mathscr{F})_f , \arrow[ur, "\alpha"'] & \end{tikzcd}$$

α is an isomorphism.

Proof If $\mathscr{F}$ is quasi-coherent, then $\mathscr{F}|_{\operatorname{Spec} A} \cong \widetilde{M}$ for some A-module M. Hence

$$\Gamma(\operatorname{Spec} A_f, \mathscr{F}) = \mathscr{F}|_{\operatorname{Spec} A}(\operatorname{Spec} A_f) = \widetilde{M}(\operatorname{Spec} A_f) = M_f = \Gamma(\operatorname{Spec} A, \mathscr{F})_f,$$

as we desired.

Conversely, if for each such distinguished $\operatorname{Spec} A_f \hookrightarrow \operatorname{Spec} A$, there is isomorphism $\Gamma(\operatorname{Spec} A, \mathscr{F})_f \cong \Gamma(\operatorname{Spec} A_f, \mathscr{F})$, we want to show that $\mathscr{F}$ is quasi-coherent. It suffices to check that for any affine open subset $\operatorname{Spec} A \subseteq X$ we have $\mathscr{F}|_{\operatorname{Spec} A} \cong \widetilde{M}$. Let $M = \Gamma(\operatorname{Spec} A, \mathscr{F})$, we claim that $\mathscr{F}|_{\operatorname{Spec} A} \cong \widetilde{M}$. Since $\Gamma(\operatorname{Spec} A, \mathscr{F})_f \cong \Gamma(\operatorname{Spec} A_f, \mathscr{F})$, we have $\mathscr{F}|_{\operatorname{Spec} A_f} \cong \widetilde{M_f}$ for all $f \in A$. By Theorem 7.1.1, $\mathscr{F}|_{\operatorname{Spec} A} \cong \widetilde{M}$, it follows that $\mathscr{F}$ is a quasi-coherent sheaf. □

Thus a quasi-coherent sheaf is (equivalent to) the data of one module for each affine open subset (a module over the corresponding ring), such that the module over a distinguished open set $\operatorname{Spec} A_f$ is given by localizing the module over $\operatorname{Spec} A$.

Example: the quasi-coherent sheaf of nilpotents

Proposition 7.2.2

If A is a ring, and $f \in A$, then $\mathfrak{N}(A_f) \cong \mathfrak{N}(A)_f$.

Proof In fact,

$$\mathfrak{N}(A_f) = \bigcap_{\mathfrak{p} \in \operatorname{Spec} A_f} \mathfrak{p} \cong \bigcap_{\substack{\mathfrak{p} \in \operatorname{Spec} A \\ f \notin \mathfrak{p}}} \mathfrak{p} \cong \left(\bigcap_{\mathfrak{p} \in \operatorname{Spec} A} \mathfrak{p} \right)_f = \mathfrak{N}(A)_f,$$

i.e., $\mathfrak{N}(A_f) \cong \mathfrak{N}(A)_f$. □

Definition 7.2.2 (The quasi-coherent sheaf of nilpotents)

$\mathfrak{N}(A)$ is an A-module. Define

$$\widetilde{\mathfrak{N}(A)}(D(f)) = \mathfrak{N}(A)_f,$$

the restriction maps $\operatorname{res}_{D(f),D(fg)}$ defined by $\mathfrak{N}(A)_f \hookrightarrow \mathfrak{N}(A)_{fg}$. This defines a sheaf on the distinguished base, and hence a sheaf on $\operatorname{Spec} A$. Also, $\widetilde{\mathfrak{N}(A)}$ is an $\mathscr{O}_{\operatorname{Spec} A}$-module.

Let X be a scheme, then an $\mathscr{O}_X$-module $\mathfrak{N}_X$ is the ***quasi-coherent sheaf of nilpotents*** *if for every affine open subset $\operatorname{Spec} A \subseteq X$, $\mathfrak{N}_X|_{\operatorname{Spec} A} \cong \widetilde{\mathfrak{N}(A)}$.*

Remark 7.7 This is an example of an ideal sheaf of $\mathscr{O}_X$.

Proposition 7.2.3

The quasi-coherent sheaf of nilpotents is indeed a quasi-coherent sheaf.

Proof Let $\operatorname{Spec} A \subseteq X$ be any affine open subscheme of X. Note that

$$\Gamma(\operatorname{Spec} A, \mathfrak{N}_X) = \Gamma(\operatorname{Spec} A, \mathfrak{N}_X|_{\operatorname{Spec} A}) = \widetilde{\mathfrak{N}(A)}(\operatorname{Spec} A) = \mathfrak{N}(A),$$

by Proposition 7.2.2, for $\operatorname{Spec} A_f \hookrightarrow \operatorname{Spec} A$ we have $\mathfrak{N}(A_f) \cong \mathfrak{N}(A)_f$. Note that $\Gamma(\operatorname{Spec} A_f, \mathfrak{N}_X) = \mathfrak{N}(A_f)$, hence,

$$\Gamma(\operatorname{Spec} A_f, \mathfrak{N}_X) \cong \Gamma(\operatorname{Spec} A, \mathfrak{N}_X)_f.$$

By Theorem 7.2.2, $\mathscr{O}_X$-module $\mathfrak{N}_X$ is quasi-coherent. □

Tensor product of quasi-coherent sheaves

Tensor product in the category of quasi-coherent sheaves on X can also be cleanly described in terms of affine open sets.

Proposition 7.2.4

If $\mathscr{F}$ and $\mathscr{G}$ are quasi-coherent sheaves on a scheme X, then $\mathscr{F} \otimes \mathscr{G}$ is a quasi-coherent sheaf described by the following information: If $\operatorname{Spec} A \subseteq X$ is an affine open subset, and $\Gamma(\operatorname{Spec} A, \mathscr{F}) = M$ and $\Gamma(\operatorname{Spec} A, \mathscr{G}) = N$, then $\Gamma(\operatorname{Spec} A, \mathscr{F} \otimes \mathscr{G}) = M \otimes_A N$, and the restriction map $\Gamma(\operatorname{Spec} A, \mathscr{F} \otimes \mathscr{G}) \to \Gamma(\operatorname{Spec} A_f, \mathscr{F} \otimes \mathscr{G})$ is precisely the localization map $M \otimes_A N \to (M \otimes_A N)_f \cong M_f \otimes_{A_f} N_f$.

Proof Let $\operatorname{Spec} A \subseteq X$ be any affine open subsets of X, for all $\operatorname{Spec} A_f \hookrightarrow \operatorname{Spec} A$, note that

$$\begin{aligned}\Gamma(\operatorname{Spec} A_f, \mathscr{F} \otimes_{\mathrm{pre}, \mathscr{O}_X} \mathscr{G}) &= \mathscr{F}(\operatorname{Spec} A_f) \otimes_{A_f} \mathscr{G}(\operatorname{Spec} A_f) \\ &\cong M_f \otimes_{A_f} N_f \\ &\cong (M \otimes N)_f \\ &= \widetilde{M \otimes_A N}(\operatorname{Spec} A_f),\end{aligned}$$

since $\widetilde{M \otimes_A N}$ is a sheaf, $\mathscr{F} \otimes_{\mathrm{pre}, \mathscr{O}_X} \mathscr{G}$ is already a sheaf on the distinguished affine base. Hence, $\mathscr{F} \otimes_{\mathscr{O}_X} \mathscr{G} = (\mathscr{F} \otimes_{\mathrm{pre}, \mathscr{O}_X} \mathscr{G})^{\mathrm{sh}} = \mathscr{F} \otimes_{\mathrm{pre}, \mathscr{O}_X} \mathscr{G}$ is a sheaf on the distinguished affine base. By Theorem 7.2.1, it determines unique sheaf on X. Also, $\mathscr{F} \otimes_{\mathscr{O}_X} \mathscr{G}$ is clearly an $\mathscr{O}_X$-module, since $(\mathscr{F} \otimes \mathscr{G})|_{\operatorname{Spec} A} \cong \widetilde{M \otimes_A N}$, $\mathscr{F} \otimes \mathscr{G}$ is quasi-coherent. □

Note that thanks to the machinery behind the distinguished affine base, sheafification is taken care of. This is a feature we will use often: *constructions involving quasi-coherent sheaves that involves sheafification for general sheaves don't require sheafification when consider on the distinguished affine base*. Along with the fact that injectivity, surjectivity, kernels, tensor products and so on may be computed on affine opens, this is the reason that it is particularly convenient to think about quasi-coherent sheaves in terms of affine open sets. (There is a slight caveat in the case of $\mathscr{H}om$.)

Remark 7.8 **Elegant side remark, but perhaps too fancy for now.** In Proposition 7.2.4, the tensor product is in the category of quasi-coherent sheaves. But in fact this is even the tensor product in the category of $\mathscr{O}_X$-modules. To show that "$\mathscr{F} \otimes_{\mathbf{QCoh}_X} \mathscr{G}$ satisfies the universal property of $\mathscr{F} \otimes_{\mathscr{O}_X} \mathscr{G}$", first show the following. Let $\mathscr{M}$ be a quasi-coherent sheaf on X, and $\mathscr{N}$ any $\mathscr{O}_X$-module. Let $\operatorname{Spec} A \subseteq X$ be an affine open subset, and $M = \mathscr{M}(\operatorname{Spec} A)$. Then the "global sections" map

$$\operatorname{Hom}_{\operatorname{Spec} A}(\mathscr{M}|_{\operatorname{Spec} A}, \mathscr{N}|_{\operatorname{Spec} A}) \longrightarrow \operatorname{Hom}_A(M, \mathscr{N}(\operatorname{Spec} A))$$

is an isomorphism. Our slogan: on an affine scheme, maps from a quasi-coherent sheaf to any $\mathscr{O}$-module are determined by global sections. You will hear an echo of this slogan in the parenthetical comment in Chapter 8.

Definition 7.2.3

Given a section s of $\mathscr{F}$ and a section t of $\mathscr{G}$, we have a section $s \otimes t$ of $\mathscr{F} \otimes \mathscr{G}$.

7.2.3 X_f and the Qcqs lemma

The next result will be useful in the future in showing certain constructions yield quasi-coherent sheaves.

Definition 7.2.4 (X_f)

If X is a scheme, and $f \in \Gamma(X, \mathscr{O}_X)$ is a function, let $X_f \subseteq X$ be the subset on which f doesn't vanish, i.e.,

$$X_f = \{p \in X : f(p) \neq 0\} = \{p \in X : f_p \notin \mathfrak{m}_p \subseteq \mathscr{O}_{X,p}\},$$

where f_p is the image of f in $\mathscr{O}_{X,p}$.

Theorem 7.2.3 (The Qcqs Lemma)

Suppose X is a quasi-compact and quasi-separated scheme, $\mathscr{F}$ is a quasi-coherent sheaf on X, and $f \in \Gamma(X, \mathscr{O}_X)$ is a function on X. Then the restriction map

$$\operatorname{res}_{X_f \subseteq X} : \Gamma(X, \mathscr{F}) \longrightarrow \Gamma(X_f, \mathscr{F})$$

is precisely localization: there is an isomorphism $\Gamma(X,\mathscr{F})_f \xrightarrow{\sim} \Gamma(X_f,\mathscr{F})$ making the following diagram commute.

$$\begin{tikzcd} \Gamma(X,\mathscr{F}) \arrow[rr, "\mathrm{res}_{X_f\subseteq X}"] \arrow[dr, "\otimes_{\Gamma(X,\mathscr{O}_X)}(\Gamma(X,\mathscr{O}_X)_f)"'] & & \Gamma(X_f,\mathscr{F}) \\ & \Gamma(X,\mathscr{F})_f \arrow[ur, "\sim"'] & \end{tikzcd}$$

Remark 7.9 We will use Five Lemma to prove the Qcqs Lemma:

Lemma 7.2.2 (Five Lemma)

Let $\mathscr{A}$ be an abelian category. Consider a commutative diagram in $\mathscr{A}$ of the form

$$\begin{tikzcd} A_1 \arrow[r] \arrow[d, "f_1"'] & A_2 \arrow[r] \arrow[d, "f_2"] & A_3 \arrow[r] \arrow[d, "f_3"] & A_4 \arrow[r] \arrow[d, "f_4"] & A_5 \arrow[d, "f_5"] \\ B_1 \arrow[r] & B_2 \arrow[r] & B_3 \arrow[r] & B_4 \arrow[r] & B_5 \end{tikzcd}$$

where the top and bottom rows are exact sequences.

(1) If f_2 and f_4 are epimorphisms and f_5 is monomorphism, then f_3 is epimorphism.

(2) If f_2 and f_4 are monomorphisms and f_1 is epimorphism, then f_3 is monomorphism.

(3) If f_2 and f_4 are isomorphisms, f_1 is epimorphism, and f_5 is monomorphism, then f_3 is isomorphism.

Proof Since X is qcqs, by the definition of sheaf, we have exact sequence,

$$0 \longrightarrow \Gamma(X,\mathscr{F}) \longrightarrow \bigoplus_{i=1}^{n}\Gamma(U_i,\mathscr{F}) \longrightarrow \bigoplus_{\substack{i,j=1\\ i\neq j}}^{n}\Gamma(U_i\cap U_j,\mathscr{F}), \tag{7.5}$$

where $X=\bigcup_{i=1}^{n}U_i$. $U_i\cap U_j$ is quasi-compact, since X is quasi-separated, by Proposition 6.3.1, we may assume that $U_i\cap U_j=\bigcup_{k_{ij}=1}^{m_{ij}}U_{ijk_{ij}}$, where $U_{ijk_{ij}}$ is simultaneously distinguished open subscheme of U_i and U_j. By the definition of sheaf again, we have exact sequence.

$$0 \longrightarrow \Gamma(U_i\cap U_j,\mathscr{F}) \longrightarrow \bigoplus_{k_{ij}=1}^{m_{ij}}\Gamma(U_{ijk_{ij}},\mathscr{F}) \tag{7.6}$$

Hence, we get a sequence from (7.5) and (7.6):

$$0 \longrightarrow \Gamma(X,\mathscr{F}) \xrightarrow{d^0} \bigoplus_{i=1}^{n}\Gamma(U_i,\mathscr{F}) \xrightarrow{d^1} \bigoplus_{\substack{i,j=1\\ i\neq j}}^{n}\Gamma(U_i\cap U_j,\mathscr{F}) \xrightarrow{\delta} \bigoplus_{\substack{i,j=1\\ i\neq j}}^{n}\bigoplus_{k_{ij}=1}^{m_{ij}}\Gamma(U_{ijk_{ij}},\mathscr{F}). \tag{7.7}$$

By exact sequence (7.6), δ is an injection, hence, $\operatorname{Ker}\delta\circ d^1=\operatorname{Ker}d^1$, and therefore sequence (7.7) is exact.

Since localization is exact (Exercise 2.37), apply the exact functor $\otimes_{\Gamma(X,\mathscr{O}_X)}\Gamma(X,\mathscr{O}_X)_f$ to exact sequence (7.7), note that tensor product commutes with direct product (Proposition 2.2.7), we have a new exact sequence.

$$0 \longrightarrow \Gamma(X,\mathscr{F})_f \longrightarrow \bigoplus_{i=1}^{n}\Gamma(U_i,\mathscr{F})_f \longrightarrow \bigoplus_{\substack{i,j=1\\ i\neq j}}^{n}\bigoplus_{k_{ij}=1}^{m_{ij}}\Gamma(U_{ijk_{ij}},\mathscr{F})_f. \tag{7.8}$$

Say $U_i=\operatorname{Spec}A_i$ and $U_{ijk_{ij}}=\operatorname{Spec}(A_i)_{g_{k_{ij}}}$, since $\mathscr{F}$ is quasi-coherent, by Theorem 7.2.2, we have

$$\Gamma(\operatorname{Spec}A_i,\mathscr{F})_f\cong\Gamma(\operatorname{Spec}(A_i)_f,\mathscr{F}),\quad \text{and } \Gamma(\operatorname{Spec}(A_i)_{g_{k_{ij}}},\mathscr{F})_f\cong\Gamma(\operatorname{Spec}(((A_i)_{g_{k_{ij}}})_f,\mathscr{F}). \tag{7.9}$$

Denote $(U_i)_f := \operatorname{Spec}(A_i)_f$ and $(U_{ijk_{ij}})_f := \operatorname{Spec}((A_i)_{g_{k_{ij}}})_f = \operatorname{Spec}(A_i)_{fg_{k_{ij}}}$.

On the other hand, note that

$$X_f = \bigcup_{i=1}^n (X_f \cap U_i) = \bigcup_{i=1}^n \operatorname{Spec}(A_i)_f = \bigcup_{i=1}^n (U_i)_f,$$

also note that

$$(X_f \cap U_i) \cap (X_f \cap U_j) = \bigcup_{k_{ij}=1}^{m_{ij}} (X_f \cap U_{ijk_{ij}}) = \bigcup_{k_{ij}=1}^{m_{ij}} \operatorname{Spec}(A_i)_{fg_{k_{ij}}} = \bigcup_{k_{ij}=1}^{m_{ij}} (U_{ijk_{ij}})_f,$$

by the definition of sheaf, we have exact sequence

$$0 \longrightarrow \Gamma(X_f, \mathscr{F}) \longrightarrow \bigoplus_{i=1}^n \Gamma((U_i)_f, \mathscr{F}) \longrightarrow \bigoplus_{\substack{i,j=1 \\ i\neq j}}^n \bigoplus_{k_{ij}=1}^{m_{ij}} \Gamma((U_{ijk_{ij}})_f, \mathscr{F}). \tag{7.10}$$

By (7.9) we have

$$\bigoplus_{i=1}^n \Gamma((U_i)_f, \mathscr{F}) \cong \bigoplus_{i=1}^n \Gamma(U_i, \mathscr{F})_f, \quad \text{and} \quad \bigoplus_{\substack{i,j=1 \\ i\neq j}}^n \bigoplus_{k_{ij}=1}^{m_{ij}} \Gamma((U_{ijk_{ij}})_f, \mathscr{F}) \cong \bigoplus_{\substack{i,j=1 \\ i\neq j}}^n \bigoplus_{k_{ij}=1}^{m_{ij}} \Gamma(U_{ijk_{ij}}, \mathscr{F})_f, \tag{7.11}$$

hence, we get a diagram from (7.8), (7.10) and (7.11),

$$\begin{tikzcd}
0 \arrow[r] & \Gamma(X, \mathscr{F})_f \arrow[r] \arrow[d, dashed, "\sim"'] & \bigoplus_{i=1}^n \Gamma(U_i, \mathscr{F})_f \arrow[r] \arrow[d, "\sim"] & \bigoplus_{\substack{i,j=1 \\ i\neq j}}^n \bigoplus_{k_{ij}=1}^{m_{ij}} \Gamma(U_{ijk_{ij}}, \mathscr{F})_f \arrow[d, "\sim"] \\
0 \arrow[r] & \Gamma(X_f, \mathscr{F}) \arrow[r] & \bigoplus_{i=1}^n \Gamma((U_i)_f, \mathscr{F}) \arrow[r] & \bigoplus_{\substack{i,j=1 \\ i\neq j}}^n \bigoplus_{k_{ij}=1}^{m_{ij}} \Gamma((U_{ijk_{ij}})_f, \mathscr{F}),
\end{tikzcd}$$

by Five Lemma 7.2.2, there is an isomorphism

$$\Gamma(X, \mathscr{F})_f \xrightarrow{\sim} \Gamma(X_f, \mathscr{F}).$$

By the universal property of localization, the following diagram commutes.

$$\begin{tikzcd}
\Gamma(X, \mathscr{F}) \arrow[rr, "\mathrm{res}_{X_f \subseteq X}"] \arrow[dr, "\otimes_{\Gamma(X, \mathscr{O}_X)}(\Gamma(X, \mathscr{O}_X)_f)"'] & & \Gamma(X_f, \mathscr{F}) \\
& \Gamma(X, \mathscr{F})_f \arrow[ur, "\sim"'] &
\end{tikzcd}$$

□

Example 7.3 **The Qcqs Lemma 7.2.3 is false without the quasi-compactness hypothesis.** Let $X = \coprod \mathbb{A}_k^1 = \coprod k[t]$, and let $f = (t, t, \dots) \in X$. In fact, $\Gamma(X_f, \mathscr{O}_X) = \prod k[t]_t$ and $\Gamma(X, \mathscr{O}_X)_f = (\prod k[t])_f$, note that $(\prod k[t])_f \subsetneq \prod k[t]_t$, hence the Qcqs Lemma 7.2.3 is false.

7.2.4 ⋆⋆ Grothendieck topologies

The distinguished affine base isn't a topology in the usual sense — the union of two affine sets isn't necessarily affine, for example. It is however a first new example of a generalization of a topology — the notion of a **site** or a **Grothendieck topology**. We give the definition to satisfy the curious, but we certainly won't use this notion. The idea is that we should abstract away only those notions we need to define sheaves. We need the notion of open set, but it turns out that we won't even need an underlying set, i.e., we won't even need the notion of points! Let's think through how little we need. For our discussion of sheaves to work, we needed to know what the open sets were, and what the (allowed) inclusions were, and these should "behave well", and in particular the data of the open sets and inclusions should form a category. (For example, the composition of an allowed inclusion with another allowed inclusion should be an allowed inclusion — in the distinguished affine base, a distinguished open set of a distinguished open set is a distinguished open set.) So we just require the data of this category. At this point, we can already define presheaf (as just a contravariant functor from this category of "open sets"). We saw this idea earlier in Proposition 3.2.1.

Definition 7.2.5 (Presheaf)

Let $\mathscr{C}$ be a category. A presheaf of sets on $\mathscr{C}$ is a contra-variant functor from $\mathscr{C}$ to **Sets**. *Morphisms of presheaves are transformations of functors. The category of presheaves of sets is denoted* $\mathrm{PSh}(\mathscr{C})$.

In order to extend this definition to that of a sheaf, we need to know more information. We want two open subsets of an open set to intersect in an open set, so *we want the category to be closed under fibered products (Exercise 2.11)*. For the identity and gluability axioms, we need to know *when some open sets cover another*, so we also remember this as part of the data of a Grothendieck topology. The data of the coverings satisfy some obvious properties. Every open set covers itself (i.e., *the identity map in the category of open sets is a covering*). Covering pull back: *if we have a map $Y \to X$, then any cover of X pulls back to a cover of Y*. Finally, *a cover of a cover should be a cover*. Such data (satisfying these axioms) is called a **Grothendieck topology** or a **site**. There are useful variants of this definition in the literature:

Note *Let $\mathscr{C}$ be a category. A family of morphisms with fixed target in $\mathscr{C}$ is given by an object $U \in \mathrm{obj}(\mathscr{C})$, a set I and for each $i \in I$ a morphism $U_i \to U$ of $\mathscr{C}$ with target U. We use the notation $\{U_i \to U\}_{i\in I}$ to indicate this.*

Remark 7.10 It can happen that the set I is empty! This notation is meant to suggest an open covering as in topology.

Definition 7.2.6 (Site (or Grothendieck topology))

A ***site*** *is given by a category $\mathscr{C}$ and a set $\mathrm{Cov}(\mathscr{C})$ of families of morphisms with fixed target $\{U_i \to U\}_{i\in I}$, called* ***coverings of*** $\mathscr{C}$, *satisfying the following axioms:*

(1) If $V \to U$ is an isomorphism then $\{V \to U\} \in \mathrm{Cov}(\mathscr{C})$.

(2) If $\{U_i \to U\}_{i\in I} \in \mathrm{Cov}(\mathscr{C})$ and for each i we have $\{V_{ij} \to U_i\}_{j\in J_i} \in \mathrm{Cov}(\mathscr{C})$, then $\{V_{ij} \to U\}_{i\in I, j\in J_i} \in \mathrm{Cov}(\mathscr{C})$.

(3) If $\{U_i \to U\}_{i\in I} \in \mathrm{Cov}(\mathscr{C})$ and $V \to U$ is a morphism of $\mathscr{C}$ then $U_i \times_U V$ exists for all i and $\{U_i \times_U V \to V\}_{i\in I} \in \mathrm{Cov}(\mathscr{C})$.

We can define the notion of a sheaf on a Grothendieck topology in the usual way, with no change:

Definition 7.2.7 (Sheaf on site)

Let $\mathscr{C}$ be a site, and let $\mathscr{F}$ be a presheaf of sets on $\mathscr{C}$. We say $\mathscr{F}$ is a sheaf if for every coving $\{U_i \to U\}_{i\in I} \in \mathrm{Cov}(\mathscr{C})$ the diagram

$$\mathscr{F}(U) \longrightarrow \prod_{i\in I} \mathscr{F}(U_i) \overset{\mathrm{pr}_0^*}{\underset{\mathrm{pr}_1^*}{\rightrightarrows}} \prod_{(i_0,i_1)\in I\times I} \mathscr{F}(U_{i_0} \times_U U_{i_1}) \tag{7.12}$$

represents the first arrow as the equalizer (Definition 3.2.6) of pr_0^* and pr_1^*.

A **topos** is a scary name for a category of sheaves of sets on a Grothendieck topology.

Definition 7.2.8 (Topos, Topoi)

A ***topos*** *is the category* $\mathbf{Sh}(\mathscr{C})$ *of sheaves on a site* $\mathscr{C}$.

(1) *Let* $\mathscr{C}$, $\mathscr{D}$ *be sites. A* ***morphism of topoi*** f *from* $\mathbf{Sh}(\mathscr{D})$ *to* $\mathbf{Sh}(\mathscr{C})$ *is given by a pair of functors* $f_* : \mathbf{Sh}(\mathscr{D}) \to \mathbf{Sh}(\mathscr{C})$ *and* $f^{-1} : \mathbf{Sh}(\mathscr{C}) \to \mathbf{Sh}(\mathscr{D})$ *such that*

   (a) *we have*

$$\mathrm{Mor}_{\mathbf{Sh}(\mathscr{D})}(f^{-1}\mathscr{G}, \mathscr{F}) = \mathrm{Mor}_{\mathbf{Sh}(\mathscr{C})}(\mathscr{G}, f_*\mathscr{F})$$

   bifunctorially, and

   (b) *the functor* f^{-1} *commutes with finite limits, i.e., is left exact.*

(2) *Let* $\mathscr{C}, \mathscr{D}, \mathscr{E}$ *be sites. Given morphisms of topoi* $f : \mathbf{Sh}(\mathscr{D}) \to \mathbf{Sh}(\mathscr{C})$ *and* $g : \mathbf{Sh}(\mathscr{E}) \to \mathbf{Sh}(\mathscr{D})$ *the composition* $f \circ g$ *is the morphism of topoi defined by the functors* $(f \circ g)_* = f_* \circ g_*$ *and* $(f \circ g)^{-1} = g^{-1} \circ f^{-1}$.

Grothendieck topologies are used in a wide variety of contexts in and near algebraic geometry. Étale cohomology (using the étale topology), a generalization of Galois cohomology, is a central tool, as are more general flat topologies, such as the smooth topology. The definition of a Deligne-Mumford or Artin stack uses the étale and smooth topology, respectively. Tate developed a good theory of non-archimedean analytic geometry over totally disconnected ground fields such as $\mathbb{Q}_p$ using a suitable Grothendieck topology. Work in K-theory (related for example to Voevodsky's work) uses exotic topologies.

7.3 Quasi-coherent sheaves form an abelian category

Morphisms from one quasi-coherent sheaf on a scheme X to another are defined to be just morphisms as $\mathscr{O}_X$-modules. In this way, the quasi-coherent sheaves on a scheme X form a category, denoted $\mathbf{QCoh}_X$. (By definition it is a full subcategory of $\mod_{\mathscr{O}_X}$.) We now show that quasi-coherent sheaves on X form an abelian category.

When you show that something is an abelian category, you have to check many things, because the definition has many parts. However, if the objects you are considering lie in some ambient abelian category, then it is much easier. You have seen this idea before: there are several things you have to do to check that something is a group. But if you have a subset of group elements, it is much easier to check that it forms a subgroup.

You can look at back at the definition of an abelian category, and you will see that in order to check that a subcategory is an abelian subcategory, it suffices to check only the following:

(i) 0 is in the subcategory;

(ii) the subcategory is closed under finite sums;

(iii) the subcategory is closed under kernels and cokernels.

In our case of $\mathbf{QCoh}_X \subseteq \mathbf{Mod}_{\mathscr{O}_X}$, the first two are cheap: 0 is certainly quasi-coherent, and the subcategory is closed under finite sums: if $\mathscr{F}$ and $\mathscr{G}$ are sheaves on X, and over $\operatorname{Spec} A$, $\mathscr{F} \cong \widetilde{M}$ and $\mathscr{G} \cong \widetilde{N}$, then $\mathscr{F} \oplus_{\text{pre}} \mathscr{G} \cong \widetilde{M \oplus N}$, note that $\widetilde{M \oplus N}$ is already a sheaf, hence $\mathscr{F} \oplus \mathscr{G} \cong \widetilde{M \oplus N}$, so $\mathscr{F} \oplus \mathscr{G}$ is a quasi-coherent sheaf.

We now check (iii), using the characterization of Theorem 7.2.2. Suppose $\alpha : \mathscr{F} \to \mathscr{G}$ is a morphism of quasi-coherent sheaves. Then on any affine open set U, where the morphism is given by $\beta : M \to N$, define $(\operatorname{Ker} \alpha)(U) = \operatorname{Ker} \beta$ and $(\operatorname{Coker} \alpha)(U) = \operatorname{Coker} \beta$. Let $U = \operatorname{Spec} A$ and $\operatorname{Spec} A_f \hookrightarrow \operatorname{Spec} A$, we want to show that $\operatorname{Ker} \alpha$ is a quasi-coherent sheaf, note that $(\operatorname{Ker} \alpha)|_U(\operatorname{Spec} A_f) = (\operatorname{Ker} \alpha)|_{\operatorname{Spec} A_f}(\operatorname{Spec} A_f) =$

$\operatorname{Ker}(\beta_f)$ and $((\operatorname{Ker}\alpha)|_U(\operatorname{Spec} A))_f = (\operatorname{Ker}\beta)_f$, it suffices to show that $\operatorname{Ker}(\beta_f) \cong (\operatorname{Ker}\beta)_f$. Similarly, to show $\operatorname{Coker}\alpha$ is a quasi-coherent sheaf, it suffices to show that $\operatorname{Coker}(\beta_f) \cong (\operatorname{Coker}\beta)_f$. Note that we have an exact sequence,

$$0 \longrightarrow \operatorname{Ker}\beta \longrightarrow M \xrightarrow{\beta} N \longrightarrow \operatorname{Coker}\beta \longrightarrow 0,$$

since localization is exact, apply Five Lemma 7.2.2, we have the following commutative diagram.

$$\begin{tikzcd}
0 \arrow[r] & \operatorname{Ker}\beta \arrow[r] \arrow[d, "\otimes_A A_f"'] & M \arrow[r, "\beta"] \arrow[d, "\otimes_A A_f"] & N \arrow[r] \arrow[d, "\otimes_A A_f"] & \operatorname{Coker}\beta \arrow[r] \arrow[d, "\otimes_A A_f"] & 0 \\
0 \arrow[r] & (\operatorname{Ker}\beta)_f \arrow[r] \arrow[d, dashed, "\sim"'] & M_f \arrow[r, "\beta_f"'] \arrow[d, "\sim"] & N_f \arrow[r] \arrow[d, "\sim"] & (\operatorname{Coker}\beta)_f \arrow[r] \arrow[d, dashed, "\sim"] & 0 \\
0 \arrow[r] & \operatorname{Ker}(\beta_f) \arrow[r] & M_f \arrow[r, "\beta_f"'] & N_f \arrow[r] & \operatorname{Coker}(\beta_f) \arrow[r] & 0
\end{tikzcd}$$

It follows that

$$(\operatorname{Ker}\beta)_f \cong \operatorname{Ker}(\beta_f), \quad \text{and } (\operatorname{Coker}\beta)_f \cong \operatorname{Coker}\beta_f,$$

i.e., $\operatorname{Ker}\alpha$ and $\operatorname{Coker}\alpha$ are quasi-coherent sheaves. Moreover, by checking stalks, they are indeed the kernel and cokernel of α (exactness can be checked stalk-locally). Thus the quasi-coherent sheaves indeed form an abelian category.

Proposition 7.3.1

A sequence of quasi-coherent sheaves $\mathscr{F} \to \mathscr{G} \to \mathscr{H}$ on X is exact if and only if it is exact on every open set in any given affine cover of X, i.e., if $X = \bigcup_i \operatorname{Spec} A_i$, sequence

$$\mathscr{F}|_{\operatorname{Spec} A_i} \longrightarrow \mathscr{G}|_{\operatorname{Spec} A_i} \longrightarrow \mathscr{H}|_{\operatorname{Spec} A_i}$$

is exact for any $\operatorname{Spec} A_i$.
(Thus we may check injectivity or surjectivity of a morphism of quasi-coherent sheaves by checking on an affine cover of our choice.)

Proof "$\Rightarrow$": Obviously.

"$\Leftarrow$": By Proposition 3.6.5, we may check the exactness of sheaves at the level of stalks. Let $X = \bigcup_i \operatorname{Spec} A_i$, say $U_i = \operatorname{Spec} A_i$, by the condition, we have exact sequence

$$\mathscr{F}|_{U_i} \longrightarrow \mathscr{G}|_{U_i} \longrightarrow \mathscr{H}|_{U_i}.$$

Let $p \in X$, then $p \in \operatorname{Spec} A_i$ for some i, by Proposition 3.6.5, sequence

$$(\mathscr{F}|_{U_i})_p \longrightarrow (\mathscr{G}|_{U_i})_p \longrightarrow (\mathscr{H}|_{U_i})_p$$

is exact. Note that $(\mathscr{F}|_{U_i})_p = \mathscr{F}_p$, $(\mathscr{G}|_{U_i})_p = \mathscr{G}_p$, and $(\mathscr{H}|_{U_i})_p = \mathscr{H}_p$, sequence

$$\mathscr{F}_p \longrightarrow \mathscr{G}_p \longrightarrow \mathscr{H}_p$$

is exact for all $p \in X$. By Proposition 3.6.5 again, sequence

$$\mathscr{F} \longrightarrow \mathscr{G} \longrightarrow \mathscr{H}$$

is exaxt. $\square$

Remark 7.11

(i) In particular, taking sections over an affine open $\operatorname{Spec} A$ is an functor from the category of quasi-coherent sheaves on X to the category of A-modules. Recall that taking sections is only left-exact in general, see Proposition 3.6.7.

(ii) **Caution:** If $0 \to \mathscr{F} \to \mathscr{G} \to \mathscr{H} \to 0$ is an exact sequence of quasi-coherent sheaves, then for any open set

$$0 \longrightarrow \mathscr{F}(U) \longrightarrow \mathscr{G}(U) \longrightarrow \mathscr{H}(U)$$

is exact, and exactness on the right is guaranteed to hold only if U is affine. (To set you up for cohomology: whenever you see left-exactness, you expect to eventually interpret this as a start of a long exact sequence. So we are expecting H^1's on the right, and now we expect that $H^1(\operatorname{Spec} A, \mathscr{F}) = 0$. This will indeed be the case.)

Proposition 7.3.2

An $\mathscr{O}_X$-module $\mathscr{F}$ on a scheme X is quasi-coherent if and only if there exists an open cover $\{U_i \hookrightarrow X\}_i$ such that on each U_i, $\mathscr{F}|_{U_i}$ is isomorphic to the cokernel of a map of two "free sheaves":

$$\mathscr{O}_{U_i}^{\oplus I} \longrightarrow \mathscr{O}_{U_i}^{\oplus J} \longrightarrow \mathscr{F}|_{U_i} \longrightarrow 0$$

is exact.

Proof If $\mathscr{O}_X$-module $\mathscr{F}$ on a scheme X is quasi-coherent, then we have an affine open cover $\{\operatorname{Spec} A_i \hookrightarrow X\}_i$ such that $\mathscr{F}|_{\operatorname{Spec} A_i} \cong \widetilde{M_i}$. Consider M_i, M_i is an A_i-module, since every A_i-module is a quotient of a free A_i-module $A_i^{\oplus J}$ (see [Rot09, Theorem 2.35 page 58]), hence we have an exact sequence

$$K_i \hookrightarrow A_i^{\oplus J} \longrightarrow M_i \longrightarrow 0, \tag{7.13}$$

where $K_i = \operatorname{Ker}(A_i^{\oplus J} \to M_i)$. Note that K_i is also an A_i-module, we have the following exact sequence.

$$A_i^{\oplus I} \longrightarrow K_i \longrightarrow 0 \tag{7.14}$$

Combining exact sequences (7.13) and (7.14) yields the following sequence.

$$\begin{tikzcd} A_i^{\oplus I} \arrow[r] \arrow[d] & A_i^{\oplus J} \arrow[r] & M_i \arrow[r] & 0 \\ K_i \arrow[ur, hook] & & & \end{tikzcd} \tag{7.15}$$

We want show that (7.15) is exact, it suffices to check that $\operatorname{Im}(A_i^{\oplus I} \to K_i \hookrightarrow A_i^{\oplus J}) = \operatorname{Ker}(A_i^{\oplus J} \to M_i)$. By the exactness of sequence (7.13), we have $\operatorname{Ker}(A_i^{\oplus J} \to M_i) = \operatorname{Im}(K_i \hookrightarrow A_i^{\oplus J}) = K_i$. Since $K_i \hookrightarrow A_i^{\oplus J}$, we have $\operatorname{Im}(A_i^{\oplus I} \to K_i) = \operatorname{Im}(A_i^{\oplus I} \to K_i \hookrightarrow A_i^{\oplus J})$, hence,

$$\operatorname{Im}(A_i^{\oplus I} \to K_i \hookrightarrow A_i^{\oplus J}) = \operatorname{Im}(A_i^{\oplus I} \to K_i) = K_i = \operatorname{Ker}(A_i^{\oplus J} \to M_i),$$

it follows that (7.15) is an exact sequence. By Corollary 5.1.2, $\widetilde{\cdot}$ is an exact functor, hence the sequence

$$\widetilde{A_i^{\oplus I}} \longrightarrow \widetilde{A_i^{\oplus J}} \longrightarrow \widetilde{M_i} \longrightarrow 0$$

is exact. Note that $\widetilde{A}(\operatorname{Spec} A_f)^{\oplus T} = A_f^{\oplus T} = (A^{\oplus T})_f = \widetilde{A^{\oplus T}}(\operatorname{Spec} A_f)$ (localization commutes with arbitrary direct sums), we have $\widetilde{A}^{\oplus T} \cong \widetilde{A^{\oplus T}}$ by Theorem 7.2.1, hence the following sequence,

$$\widetilde{A_i}^{\oplus I} = \mathscr{O}_{\operatorname{Spec} A_i}^{\oplus I} \longrightarrow \widetilde{A_i}^{\oplus J} = \mathscr{O}_{\operatorname{Spec} A_i}^{\oplus J} \longrightarrow \widetilde{M_i} = \mathscr{F}|_{\operatorname{Spec} A_i} \longrightarrow 0,$$

is exact, as we desired.

Conversely, suppose $\{U_i \hookrightarrow X\}_i$ be an open cover of X such that on each U_i, sequence

$$\mathscr{O}_{U_i}^{\oplus I} \longrightarrow \mathscr{O}_{U_i}^{\oplus J} \longrightarrow \mathscr{F}|_{U_i} \longrightarrow 0 \tag{7.16}$$

is exact. By Theorem 7.2.2, we may assume that $X = \operatorname{Spec} A$. Let $M = \Gamma(X, \mathscr{F})$. By the assumption, we may shrink U to some distinguished open subsets, say $D(f)$, such that sequence

$$\mathscr{O}_{D(f)}^{\oplus I} \longrightarrow \mathscr{O}_{D(f)}^{\oplus J} \longrightarrow \mathscr{F}|_{D(f)} \longrightarrow 0$$

is exact. Hence, sequence

$$A_f^{\oplus I} \longrightarrow A_f^{\oplus J} \longrightarrow \Gamma(D(f), \mathscr{F}) \longrightarrow 0$$

is exact. Let $M_{D(f)} = \Gamma(D(f), \mathscr{F})$, then

$$\mathscr{F}|_{D(f)} \cong \widetilde{M_{D(f)}}.$$

Let $\{D(f_i) \hookrightarrow \operatorname{Spec} A\}_i$ be an affine open cover of X, we may write $M_i = M_{D(f_i)}$, then $M_i = \Gamma(D(f_i), \mathscr{F})$. By Theorem 7.2.2, to show $\mathscr{F}$ is quasi-coherent, it suffices to show that $\Gamma(X, \mathscr{F})_{f_i} \cong \Gamma(D(f_i), \mathscr{F})$, i.e., $M_{f_i} \cong M_i$. Consider the restriction map $\operatorname{res}_{X,D(f_i)} : M \to M_i$, it induces an A_{f_i}-module homomorphism

$$\widehat{\operatorname{res}_{X,D(f_i)}} : M_{f_i} \longrightarrow M_i,$$

we want to show that $M_{f_i} \cong M_i$.

For injectivity. Suppose that $\widehat{\operatorname{res}_{X,D(f_i)}}\left(\frac{m}{f_i^n}\right) = 0$, then $\widehat{\operatorname{res}_{X,D(f_i)}}\left(\frac{m}{f_i^n}\right) = \frac{\operatorname{res}_{X,D(f_i)}(m)}{f_i^n} = 0$, hence $\operatorname{res}_{X,D(f_i)}(m) = 0$ in M_i. On $D(f_i) \cap D(f_j) = D(f_i f_j)$, we have

$$\mathscr{F}|_{D(f_if_j)} = (\mathscr{F}|_{D(f_i)})|_{D(f_if_j)} = (\mathscr{F}|_{D(f_j)})|_{D(f_if_j)} = \widetilde{M_j}|_{D(f_if_j)} = \widetilde{(M_j)_{f_if_j}} \tag{7.17}$$

Denote $\operatorname{res}_i := \operatorname{res}_{X,D(f_i)}$. Since $\operatorname{res}_j(m)|_{D(f_if_j)} = \operatorname{res}_i(m)|_{D(f_if_j)} = 0$ in $\Gamma(D(f_if_j), \mathscr{F}) = (M_j)_{f_if_j}$, there exists $n_j > 0$ such that $(f_if_j)^{n_j} \operatorname{res}_j(m) = 0$ in M_j. Since f_j is unit in M_j, we have $f_i^{n_j} \operatorname{res}_j(m) = \operatorname{res}_j(f_i^{n_j} m) = 0$ in M_j. We may pick $N \gg 0$ such that

$$\operatorname{res}_j(f_i^{n_j} m) = 0 \in M_j, \ \forall j.$$

By the identity axiom of $\mathscr{F}$, $f_i^{n_j} m = 0$ in M. Hence $\frac{m}{f_i^n} = 0$ in M_{f_i}.

For surjectivity. Let $s_i \in M_i$, then $s_i|_{D(f_if_j)} \in \Gamma(D(f_if_j), \mathscr{F})$, by (7.17),

$$\Gamma(D(f_if_j), \mathscr{F}) = \Gamma(D(f_if_j), \mathscr{F}|_{D(f_if_j)}) = \Gamma(D(f_if_j), \widetilde{(M_j)_{f_if_j}}) = (M_j)_{f_if_j},$$

hence exists n_j such that $(f_if_j)^{n_j} s_i \in M_j$. Pick $N \gg 0$, since f_j is unit in M_j, we have $f_i^N s_i \in M_j$ for all j. By the identity axiom of $\mathscr{F}$, exists $s \in M$ such that $\operatorname{res}_i(s) = f_i^N s_i$, hence $s_i = \frac{\operatorname{res}_i(s)}{f_i^N} = \widehat{\operatorname{res}_i}\left(\frac{s}{f_i^N}\right)$ in M_i. Hence, $\widehat{\operatorname{res}_i}$ is surjective.

Hence $M_{f_i} \cong M_i$, by Theorem 7.2.2, $\mathscr{F}$ is quasi-coherent sheaf. □

Another proof by [Yan18].

Proof For every $x \in X$ we have a neighborhood $U \subseteq X$ such that $\mathscr{F}|_U$ is the cokernel of a morphism between free $\mathscr{O}_X|_U$-modules. Shrinking U to make $U = \operatorname{Spec} A$ is affine, we know on U, free $\mathscr{O}_X|_U$-module is of the form $\widetilde{A}^n$ for any cardfinality n, i.e., free $\mathscr{O}_X|_U$-module is quasi-coherent, so the cokernel is quasi-coherent. Hence $\mathscr{F}|_U = \widetilde{M}$ for some A-module M. Hence $\mathscr{F}$ is quasi-coherent. Moreover, if $\mathscr{F}|_U$ is the cokernel of a morphism between free $\mathscr{O}_X|_U$-modules of finite rank, i.e., cokernel of a morphism between two coherent sheaves, it is clearly coherent. □

Proposition 7.3.2 is the definition of a quasi-coherent sheaf on a ringed space in general. It is useful in many circumstances, for example in complex analytic geometry.

Definition 7.3.1 (Quasi-coherent sheaves on ringed space)

Let $(X, \mathscr{O}_X)$ be a ringed space. Let $\mathscr{F}$ be sheaf of $\mathscr{O}_X$-modules. We say that $\mathscr{F}$ is a quasi-coherent sheaf of $\mathscr{O}_X$-modules if for every point $x \in X$ there exists an open neighborhood $x \in U \subseteq X$ such that *sequence*

$$\mathscr{O}_U^{\oplus I} \longrightarrow \mathscr{O}_U^{\oplus J} \longrightarrow \mathscr{F}|_U \longrightarrow 0$$

is exact, i.e., $\mathscr{F}|_U \cong \operatorname{Coker}(\mathscr{O}_U^{\oplus I} \to \mathscr{O}_U^{\oplus J})$.

7.4 Finite type quasi-coherent, finitely presented, and coherent sheaves

Here are three natural finiteness conditions on an A-module M. If A is a Noetherian ring, which is the case that almost all of you will ever care about, they are all the same.

The first is the simplest: a module could be **finitely generated**. In other words, there is a surjection $A^{\oplus p} \to M \to 0$.

The second is reasonable too. It could be finitely presented — it could have a finite number of generators with a finite number of relations. Translation: there exists a **finite presentation**, i.e., an exact sequence

$$A^{\oplus q} \longrightarrow A^{\oplus p} \longrightarrow M \longrightarrow 0.$$

The third notion is frankly a bit surprising:

Definition 7.4.1 (Coherent module)

*We say that an A-module M is **coherent** if*

(i) it is finitely generated;

(ii) for any map $A^{\oplus p} \to M$ (not necessarily surjective!), the kernel is finitely generated.

Proposition 7.4.1

If A is Noetherian ring, the following are the same:

(i) A-module M is finitely generated;

(ii) A-module M is finite presentation;

(iii) A-module M is coherent.

Proof Clearly coherent implies finitely presented, which in turn implies finitely generated. So suppose M is finitely generated. Take any $\alpha : A^{\oplus p} \to M$. Then $\operatorname{Ker} \alpha$ is a submodule of a finitely generated module over A, and thus finitely generated by Corollary 4.6.3. Thus M is coherent. $\square$

Hence most people can think of these three notions as the same.

Proposition 7.4.2

The coherent A-module form an abelian subcategory of the category of A-modules.

Remark 7.12 The proof in general is a series of short exercise in §7.7. You should try the only if you are particularly curious.

Proof If the ring A is Noetherian. It suffices to check three things:

(i) The 0-module is coherent.

(ii) The category of coherent modules is closed under finite sums (i.e., direct sums).

(iii) The category of coherent modules is closed under kernels and cokernels.

The first two are clear. For (iii), suppose that $f : M \to N$ is a map of finitely generated modules. Then $\operatorname{Coker} f$ is finitely generated (it is the image of N), and $\operatorname{Ker} f$ is too (it is a submodule of a finitely generated module over a Noetherian ring, Proposition 4.6.3). □

Remark 7.13 Finitely generated modules over a PID. We record for future reference that finitely generated modules over a PID are finite direct sums of cyclic modules. We only mention it here because it needs mentioning somewhere.

We now extend the definitions of these three classes of modules to quasi-coherent sheaves.

Proposition 7.4.3 (Finite generation is an affine-local property 6.3.1)

(a) If $f \in A$, and M is a finitely generated A-module, then M_f is a finitely generated A_f-module.

(b) If $(f_1, \ldots, f_n) = A$, and M_{f_i} is a finitely generated A_{f_i}-module for all i, then M is a finitely generated A-module.

Proof

(a) Suppose that M is generated by $x_1, \ldots, x_2$. Note that $M_f = M \otimes_A A_f$, M_f is generated by $x_1 \otimes 1, \ldots, x_n \otimes 1$, it follows that M_f is a finitely generated A_f-module.

(b) Since each M_{f_i} is finitely generated A_{f_i}-module, we may assume that M_{f_i} is generated by $x_{i1}, \ldots, x_{in_i}$. Let $m \in M$, then $m \in M_{f_i}$ for all i. Suppose that

$$m = \sum_{j=1}^{n_i} a_{ij} x_{ij},$$

where $a_{ij} \in A_{f_i}$. Eliminate the denominator using f_i, then we have

$$f_i^{k_i} m = \sum_{j=1}^{n_i} b_{ij} y_{ij},$$

where $b_{ij} \in A$ and $y_{ij} \in M$. Pick $k = \max_{1 \le i \le n} \{k_i\}$, then

$$f_i^k m = \sum_{j=1}^{n_i} b_{ij} y_{ij},$$

for all i. Since $A = (f_1, \ldots, f_n)$, we have $1 = c_1 f_1 + \cdots + c_n f_n$, hence

$$1 = (c_1 f_1 + \cdots + c_n f_n)^{nk} = d_1 f_1^k + d_2 f_2^k + \cdots + d_n f_n^k,$$

where $d_i \in A$. Let $m \in M$, then

$$\begin{aligned} m =& 1 \cdot m = (d_1 f_1^k + d_2 f_2^k + \cdots + d_n f_n^k) m \\ =& d_1 f_1^k m + d_2 f_2^k m + \cdots + d_n f_n^k m \\ =& \sum_{i=1}^{n} \sum_{j=1}^{n_i} d_i b_{ij} y_{ij}, \end{aligned}$$

it follows that any element of M is finitely generated by $\{y_{ij}\}$, and therefore M is a finitely generated A-module.

□

For the finite presentation case, the following result is handy, and versions apply in other situations.

Lemma 7.4.1 (Finitely presented implies always finitely presented (module version).)

Suppose M is a finitely presented A-module for ***some*** *surjection from a finite free module, the kernel is finitely generated. More precisely, if $\varphi : A^{\oplus p} \to M$ is* ***any*** *surjection, then* $\operatorname{Ker} \varphi$ *is finitely generated.*

Proof Choose a finite presentation of M:

$$\bigoplus_{k=1}^{r} Af_k \xrightarrow{\beta} \bigoplus_{i=1}^{n} Ax_i \xrightarrow{\alpha} M \longrightarrow 0$$

so $\alpha(x_1), \ldots, \alpha(x_n)$ generate M. Write the surjection of the hypothesis as $\varphi : \bigoplus_{j=1}^{p} Ay_i \to M$. Choose lifts $g_j \in \bigoplus Ax_i$ of $\varphi(y_j) \in M$, i.e., $\alpha(g_j) = \varphi(y_j)$. Then we can write

$$\begin{aligned} M &= \bigoplus Ax_i/(\beta(f_1), \ldots, \beta(f_r)) \\ &= \left(\left(\bigoplus_i Ax_i \right) \oplus \left(\bigoplus_j Ay_j \right) \right) \Big/ (\beta(f_1), \ldots, \beta(f_r), y_1 - g_1, \ldots, y_p - g_p), \end{aligned}$$

the last equal is given by homomorphism $(m, n) \mapsto \alpha(m) + \varphi(n)$ where $m \in \bigoplus_i Ax_i$ and $n \in \bigoplus_j Ay_j$.

As the $\{\varphi(y_j)\}$ generate M, for each $i = 1, \ldots, n$ we can choose $h_i \in \bigoplus_{j=1}^{p} Ay_j$ such that $\alpha(x_i) = \varphi(h_i)$. Define $h : \bigoplus_{i=1}^{n} Ax_i \to \bigoplus_{j=1}^{p} Ay_j$ by setting $h(x_i) = h_i$. Thus

$$\begin{aligned} M &= \left(\left(\bigoplus_i Ax_i \right) \oplus \left(\bigoplus_j Ay_j \right) \right) \Big/ (\beta(f_1), \ldots, \beta(f_r), y_1 - g_1, \ldots, y_p - g_p, x_1 - h_1, \ldots, x_n - h_n) \\ &= \bigoplus_j Ay_j/(h(\beta(f_1)), \ldots, h(\beta(f_r)), y_1 - h(g_1), \ldots, y_p - h(g_p)), \end{aligned}$$

which implies that $\operatorname{Ker} \varphi$ is finitely generated. □

Proposition 7.4.4 (Finite presentation is an affine-local property 6.3.1)

(a) If $f \in A$, and if M is a finitely presented A-module, then M_f is a finitely presented A_f-module.

(b) If $(f_1, \ldots, f_n) = A$, and M_{f_i} is a finitely presented A_{f_i}-module for all i, then M is finitely presented A-module.

Proof

(a) Since M is a finitely presented A-module, then there exists an exact sequence

$$A^{\oplus q} \longrightarrow A^{\oplus p} \longrightarrow M \longrightarrow 0.$$

Since localization is exact functor and it is commutes with arbitrary direct sums, we have exact sequence,

$$A_f^{\oplus q} \longrightarrow A_f^{\oplus p} \longrightarrow M_f \longrightarrow 0,$$

which implies that M_f is finitely presented.

(b) Since M_{f_i} is a finitely generated A_{f_i}-module for all i, hence each M_{f_i} is a finitely generated A_{f_i}-module. By Proposition 7.4.3, M is a finitely generated A-module, so we may write $M = \operatorname{Coker}(\alpha : N \hookrightarrow A^{\oplus n})$ for some submodule $N \subseteq A^{\oplus n}$. To show that M is finitely presentation, it suffices to show that N is finitely generated. Consider the following exact sequence.

$$N \longrightarrow A^{\oplus n} \longrightarrow M \longrightarrow 0$$

Localize at each f_i, we get exact sequence

$$N_{f_i} \longrightarrow A_{f_i}^{\oplus n} \longrightarrow M_{f_i} \longrightarrow 0.$$

Note that $N_{f_i} = \operatorname{Ker}(A_{f_i}^{\oplus n} \to M_{f_i})$ and each M_{f_i} is finitely presented, by Lemma 7.4.1, N_{f_i} is finitely generated for all i. Since finite generation is an affine-local property (Proposition 7.4.3), N is finitely generated, and therefore M is finitely presented.

$\square$

Proposition 7.4.5 (Coherence is an affine-local property 6.3.1)

(a) If $f \in A$, and M is a coherent A-module, then M_f is a coherent A_f-module.

(b) If $(f_1, \ldots, f_n) = A$, and M_{f_i} is a coherent A_{f_i}-module for all i, then M is a coherent module.

Proof

(a) Since M is coherent, M is finitely generated, by Proposition 7.4.3, M_f is finitely generated. Let $A_f^{\oplus p} \to M_f$ be any map, note that $\operatorname{Ker}(A_f^{\oplus p} \to M_f) = \operatorname{Ker}(A^{\oplus p} \to M)_f$ and $\operatorname{Ker}(A^{\oplus p} \to M)$ is finitely generated (since M is coherent), by Proposition 7.4.3, $\operatorname{Ker}(A^{\oplus p} \to M)_f$ is finitely generated. Hence M_f is coherent.

(b) By Proposition 7.4.3, M is finitely generated. Let $\varphi : A^{\oplus p} \to M$ be any map, note that $(\operatorname{Ker}\varphi)_{f_i} = \operatorname{Ker}\varphi_{f_i}$, since M_{f_i} is coherent, $\operatorname{Ker}\varphi_{f_i}$ is finitely generated, by Proposition 7.4.3 again, $\operatorname{Ker}\varphi$ is finitely generated, and therefore M is coherent.

$\square$

Definition 7.4.2 (Finite type, finitely presented, coherent)

*A quasi-coherent sheaf $\mathscr{F}$ is **finite type** (resp., **finitely presented**, **coherent**) if for every affine open $\operatorname{Spec} A$, $\Gamma(\operatorname{Spec} A, \mathscr{F})$ is a finitely generated (resp., finitely presented, coherent) A-module.*

Remark 7.14

(1) Note that coherent sheaves are always finite type, and that on a locally Noetherian scheme, all three notions are the same (by Proposition 7.4.1). Proposition 7.4.2 implies that the coherent sheaves on X form an abelian category, which we denote $\mathbf{Coh}_X$.

(2) By the Affine Communication Lemma 6.3.1, and Proposition 7.4.3, 7.4.4, and 7.4.5, it suffices to check "finite typeness" (resp., finite presentation, coherence) on the open sets in a single affine cover.

Remark 7.15 Warning. It is not uncommon in the later literature to incorrectly define coherent as finitely generated. Please only use correct definition, as the wrong definition cause confusion. Besides doing this for reasons of honesty, it will also help you see what hypotheses are actually necessary to prove things. And that always helps you remember what the proofs are — and hence why things are true.

Why coherence? Proposition 7.4.2 is a good motivation for the definition of coherence: it gives a small (in a non-techincal sense) abelian category in which we can think about vector bundles.

There are two sorts of people who should care about the details of this definition, rather than living in a Noetherian world where coherent means finite type. Complex geometers should care. They consider consider complex-analytic spaces with the classical topology. One can define the notion of coherent $\mathscr{O}_X$-module in a way analogous to this. Then Oka's Theorem states that the structure sheaf of $\mathbb{C}^n$ (hence of any complex manifold) is coherent, and this is very hard.

The second sort of people who should care are the sort of arithmetic people who may need to work with non-Noetherian rings, or work in non-achimedean analytic geometry.

Remark 7.16 Quasi-coherent and coherent sheaves on ringed spaces. We will discuss quasi-coherent and coherent sheaves on schemes, but they can be defined more generally (see Definition 7.3.1). Many of the results we state will hold in greater generality, but because the proofs look slightly different, we restrict ourselves to scheme to avoid distraction.

Example 7.4 **Coherence is not a good notion in smooth geometry.** The following example from B.Conrad shows that in quite reasonable (but less "rigid") situations, the structure sheaf is not coherent over itself. Consider the ring $\mathscr{O}_0$ of germs of smooth (C^∞) function at $0 \in R$, with coordinate x. Now $\mathscr{O}_0$ is a local ring. Its maximal ideal $\mathfrak{m}$ is generated by x. (Key idea: suppose $f \in \mathfrak{m}$, and suppose f has a representative defined on $(-\varepsilon, \varepsilon)$. Then for $t \in (-\varepsilon, \varepsilon)$, $f(t) = \int_0^t f'(u)du = t\int_0^1 f'(tv)dv$. By "differentiating under the integral sign" repeatedly, we may check that $\int_0^1 f'(tv)dv$ is smooth.)

Let $\varphi \in \mathscr{O}_0$ be the germ of a smooth function that is 0 for $x \leq 0$, and positive for $x > 0$ (such as $\varphi(x) = e^{1\frac{1}{x^2}}$ for $x > 0$). Consider the map $\times\varphi : \mathscr{O}_0 \to \mathscr{O}_0$. The kernel is the ideal I_φ of functions vanishing for $x \geq 0$. Clearly I_φ is nonzero (since $\varphi(-x) \in I_\varphi$), but as $\mathfrak{m} = (x)$, $I_\varphi = (x)\varphi$, by Nakayama's Lemma, I_φ cannot be finitely generated. (Essentially the same argument shows that the sheaf of smooth functions on $\mathbb{R}$ is not coherent.) This is why coherence has no useful meaning for smooth manifolds.

7.5 Algebraic interlude: The Jordan-Hölder package

7.5.1 Jordan-Hölder Theorem

The Jordan-Hölder Theorem in group theory is part of a more fundamental and somehow simpler story. The Jordan-Hölder "yoga" is why you can often factor some sort of algebraic object into primes or irreducibles, uniquely (in an appropriate sense), where each prime/irreducible appears the same number of times no matter how you factor. From this point of view, it generalizes unique factorization of integers; well-definedness of dimension of vector spaces; classification of finitely generated abelian groups; unique factorization of ideals in a Dedekind domain; and the traditional Jordan-Hölder Theorem in group theory.

Jordan-Hölder Theorem for abelian category

We will be mostly interested in modules over a ring, but there is no harm in working in a general abelian category $\mathscr{C}$. (This can be readily generalized further.)

Definition 7.5.1 (Simple (or irreducible) module, composition series, finite length)

*We say an object $M \in \mathscr{C}$ is **simple** (or **irreducible**) if its only subobjects are 0 and itself. A **composition series** for M is a (finite) filtration*

$$0 = X_0 \subsetneq X_1 \subsetneq \cdots \subsetneq X_{n-1} \subsetneq X_n = M \tag{7.18}$$

*such that the quotients X_{i+1}/X_i are all simple. If M has a composition series, we say that M has **finite length**.*

Theorem 7.5.1 (The Jordan-Hölder Theorem)

Let $\mathscr{C}$ be an abelian category. Let M be an object of $\mathscr{C}$ which has a finite composition series. Given any two filtrations

$$A_\bullet : 0 = A_0 \subsetneq A_1 \subsetneq A_2 \subsetneq \cdots \subsetneq A_n = M$$

and

$$B_\bullet : 0 = B_0 \subsetneq B_1 \subsetneq B_2 \subsetneq \cdots \subsetneq B_m = M$$

with $S_i = A_i/A_{i-1}$ and $T_j = B_j/B_{j-1}$ simple objects, we have $n = m$ and there exists a permutation σ of $\{1, \dots, n\}$ such that $S_i \cong T_{\sigma(i)}$ for all $i \in \{1, \dots, n\}$.

Definition 7.5.2 (Length of object)

*We call the length of any of the composition series for M the **length** of M, denoted $\ell(M)$. Length of object is well-defined by the Jordan-Hölder Theorem. But we even have a refined notion: we have the multiplicity with which each simple object appears in any composition series for M. If M is not of finite length, we say $\ell(M) = \infty$.*

Definition (Length of modules)

*Let R be a ring. For any R-module M we define the **length** of M over R by the formula*

$$\operatorname{length}_R(M) := \sup\{n : \exists 0 = M_0 \subseteq M_1 \subseteq \cdots \subseteq M_n = M, M_i \neq M_{i+1}\}.$$

Example 7.5 In the category of abelian groups, the finite-length objects are the finite ableian groups. The Jordan-Hölder Theorem in this case, applied to $\mathbb{Z}/n\mathbb{Z}$, can be used to give the unique factorization of n.

Proof of the Jordan-Hölder Theorem 7.5.1.

Proof Let j be the smallest index such that $A_1 \subseteq B_j$. Consider the map $S_1 = A_1 \hookrightarrow B_j \to B_j/B_{j-1} = T_j$, we claim that $S_1 \cong T_j$. Note that $\operatorname{Ker}(A_1 \hookrightarrow B_j \to B_j/B_{j-1}) = A_1 \cap B_{j-1}$, since $A_1 \nsubseteq B_{j-1}$ and A_1 is simple, $A_1 \cap B_{j-1} = 0$, which implies that $S_1 \to T_j$ is monomorphism. Since T_j is simple, we have $A_1 \cong T_j$. Moreover, the object $M/A_1 = A_n/A_1 = B_m/A_1$ has the two filtrations

$$A'_\bullet : 0 \subseteq A_2/A_1 \subseteq A_3/A_1 \subseteq \cdots \subseteq A_n/A_1$$

and

$$B'_\bullet : 0 \subseteq (B_1 + A_1)/A_1 \subseteq \cdots \subseteq (B_{j-1} + A_1)/A_1 \subseteq B_j/A_1 \subseteq B_{j+1}/A_1 \subseteq \cdots \subseteq B_m/A_1.$$

Note that

$$(B_{j-1} + A_1)/A_1 \cong B_{j-1}/(A_1 \cap B_{j-1}) \cong B_{j-1} \cong B_j/A_1,$$

we have

$$B'_\bullet : 0 \subseteq (B_1 + A_1)/A_1 \subseteq \cdots \subseteq (B_{j-1} + A_1)/A_1 = B_j/A_1 \subseteq B_{j+1}/A_1 \subseteq \cdots \subseteq B_m/A_1.$$

We want to show that $((B_i + A_1)/A_1)/((B_{i-1} + A_1)/A_1) \cong (B_i + A_1)/(B_{i-1} + A_1) \cong B_i/B_{j-1}$, for all $i \leq j-1$. Since $B_{i-1} \subseteq B_i$, we have $B_i + B_{i-1} + A_1 = B_i + A_1$. Note that $B_i \cap (B_{i-1} + A_1) = B_{i-1} + A_1 \cap B_i$, since A_1 is simple $A_1 \cap B_i$ must be 0, hence

$$B_i \cap (B_{i-1} + A_1) = B_{i-1} + A_1 \cap B_i = B_{i-1}.$$

By the Fundamental Theorem of Isomorphism, we have

$$\frac{B_i + A_1}{B_{i-1} + A_1} = \frac{B_i + (B_{i-1} + A_1)}{B_{i-1} + A_1} \cong \frac{B_i}{(B_{i-1} + A_1) \cap B_i} = B_i/B_{i-1}.$$

Hence $\ell(A_n/A_1) \leq n-1$ and $\ell(B_m/A_1) \leq m-1$. We may assume that $\ell(A_n/A_1) = n-1$ and $\ell(B_m/A_1) = m-1$. Let $A'_i = A_i/A_1$, $B'_i = (B_i + A_1)/A_1$ for $i \leq j-1$ and $B'_i = B_i/A_1$ for $i \geq j$, then we have two composition series for M/A_1

$$A'_\bullet : 0 \subsetneq A'_2 \subsetneq A'_3 \subsetneq \cdots \subsetneq A'_n$$

and

$$B'_\bullet : 0 \subsetneq B'_1 \subsetneq \cdots \subsetneq B'_{j-2} \subsetneq B'_j \subsetneq \cdots \subsetneq B'_m.$$

with $S_i' = A_i'/A_{i-1}' \cong S_i$ and $T_i' = B_i'/B_{i-1}' \cong T_i$

Let j' be the smallest index such that $A_2' \subseteq B_{j'}'$. Then $S_2 \cong T_{j'}$. By induction then we done. □

Proposition 7.5.1

Every subquotient of a finite-length object M is finite length.

Proof Since M is finite-length, say $\ell(M) = n$, let chain

$$0 = M_0 \subsetneq M_1 \subsetneq \cdots \subsetneq M_n = M \tag{7.19}$$

be a composition series for M. Let $M' \subseteq M'' \subseteq M$, we want to show that $\ell(M''/M') < \infty$. By chain (7.19), we have a chain for M''

$$0 = M_0 \cap M'' \subseteq M_1 \cap M'' \subseteq \cdots \subseteq M_n \cap M'' = M''.$$

We claim that $(M_i \cap M'')/(M_{i-1} \cap M'') \cong M_i/M_{i-1}$. Consider the map $M_i \cap M'' \hookrightarrow M_i \xrightarrow{\pi} M_i/M_{i-1}$, then

$$\mathrm{Ker}(M_i \cap M'' \hookrightarrow M_i \xrightarrow{\pi} M_i/M_{i-1}) = M_i \cap M'' \cap M_{i-1} = M'' \cap M_{i-1},$$

which implies that $(M_i \cap M'')/(M_{i-1} \cap M'') \cong M_i/M_{i-1}$.

Merge equal terms, we get a composition series for M'', say

$$0 = M_0'' \subsetneq M_1'' \subsetneq \cdots \subsetneq M_m'' = M'',$$

where $m \leq n$, it follows that M' and M'' are finite length. Let j be the smallest index such that $M' \subseteq M_j''$, since M' is finite length, we have composition series

$$0 = M_0' \subsetneq M_1' \subsetneq \cdots \subsetneq M_k' = M' \subseteq M_j'' \subseteq \cdots \subseteq M_m'' = M'',$$

where M_i'/M_{i-1}' and M_i''/M_{i-1}'' are simple. Since $M_k' \subseteq M'$,

$$0 = 0 = \cdots = 0 = M'/M' \subseteq M_j''/M' \subseteq \cdots \subseteq M_m''/M' = M''/M',$$

note that

$$(M_i''/M')/(M_{i-1}''/M') \cong M_i''/M_{i-1}''$$

is simple, we have

$$\ell(M''/M') \leq m - j,$$

which implies that subquotient of M is finite length. □

Proposition 7.5.2

Length is additive in exact sequences: if $0 \to M' \to M \to M'' \to 0$ is an exact sequence, then

$$\ell(M) = \ell(M') + \ell(M'').$$

Proof Let

$$0 = M_0' \subsetneq M_1' \subsetneq \cdots \subsetneq M_n' = M'$$

be a composition series for M' and

$$0 = M_0'' \subsetneq M_1'' \subsetneq \cdots \subsetneq M_m'' = M''$$

be a composition series for M''. Let

$$0 \longrightarrow M' \xrightarrow{\alpha} M \xrightarrow{\beta} M'' \longrightarrow 0$$

Since α is monomorphism, we have

$$0 = \alpha(M'_0) \subsetneq \alpha(M'_1) \subsetneq \cdots \subsetneq \alpha(M'_n) = \alpha(M').$$

Since β is epimorphism, we have

$$\beta^{-1}(0) = \beta^{-1}(M''_0) \subsetneq \beta^{-1}(M''_1) \subsetneq \cdots \subsetneq \beta^{-1}(M''_m) = \beta^{-1}(M'').$$

Since $\operatorname{Im}(M' \to M) = \operatorname{Ker}(M \to M'')$, we have a chain for M

$$\begin{aligned} 0 =& \alpha(M'_0) \subsetneq \cdots \subsetneq \alpha(M'_n) = \alpha(M') = \beta^{-1}(0) = \beta^{-1}(M''_0) \\ &\subsetneq \beta^{-1}(M'_1) \subsetneq \cdots \subsetneq \beta^{-1}(M''_m) = \beta^{-1}(M'') = M. \end{aligned} \tag{7.20}$$

We want to show that chain (7.20) is a composition series for M. Since α is monomorphism, we have $M'_i \cong \alpha(M'_i)$, hence $M'_i/M'_{i-1} \cong \alpha(M'_i)/\alpha(M'_{i-1})$ is simple. Consider $\beta|_{\beta^{-1}(M''_i)} : \beta^{-1}(M''_i) \to M''_i$, since β is epimorphism, by the Fundamental Theorem of Isomorphism, we have $\beta^{-1}(M''_i)/\operatorname{Ker}\beta|_{\beta^{-1}(M''_i)} \cong M''_i$. Note that $0 \in M_i$ for all i, we have $\operatorname{Ker}\beta = \operatorname{Ker}\beta|_{\beta^{-1}(M''_i)} = \beta^{-1}(0)$, hence $\beta^{-1}(M''_i)/\beta^{-1}(0) \cong M''_i$, and therefore

$$\beta^{-1}(M''_i)/\beta^{-1}(M''_{i-1}) \cong \frac{\beta^{-1}(M''_i)/\beta^{-1}(0)}{\beta^{-1}(M''_{i-1})/\beta^{-1}(0)} \cong M_i/M_{i-1}$$

is simple. Hence by chain (7.20) we have

$$\ell(M) = \ell(M') + \ell(M'').$$

□

Proposition 7.5.3

Any filtration of a finite length module can be refined into a composition series.

Proof Let

$$0 = N_0 \subseteq N_1 \subseteq \cdots \subseteq N_n = M$$

be any filtration of M. If N_i/N_{i-1} not simple, then exists a nonzero proper subobject, say $\widetilde{N'_{i-1}}$. Hence, exists $N'_{i-1} \subsetneq N_i$ which contains N_{i-1} such that $\widetilde{N'_{i-1}}$ be the image of N'_{i-1} in N_i/N_{i-1}. The we get a new filtration of M.

$$0 = N_0 \subseteq \cdots \subseteq N_{i-1} \subseteq N'_{i-1} \subsetneq N_i \subseteq \cdots \subseteq N_n = M$$

Repeat above process, since M is finite length, this process will be terminated. Then we get a filtration of M,

$$0 = M_0 \subsetneq M_1 \subsetneq \cdots \subsetneq M_{n-1} \subsetneq M_n = M,$$

where each M_i/M_{i-1} is simple, i.e., it is a composition series for M, as we desired. □

Proposition 7.5.4

The finite length objects in $\mathscr{C}$ form a full subcategory of $\mathscr{C}$.

Proof Clearly. □

Jordan-Hölder Theorem for groups

The category of groups does not form an abelian category, so Theorem 7.5.1 can't immediately imply the traditional Jordan-Hölder Theorem for groups. However, the same proof applies without change, with only one additional input.

Definition 7.5.3 (Subnormal series, Normal series)

*A **subnormal series** of a group G is a sequence of subgroups, each a normal subgroup of the next one. In a standard notation*

$$1 = A_0 \lhd A_1 \lhd \cdots \lhd A_n = G.$$

*There is no requirement made that A_i be a normal subgroup of G, only a normal subgroup of A_{i+1}. The quotient groups A_{i+1}/A_i are called the **factor groups** of the series.*
If in addition each A_i is normal in G, then the series is called a normal series.

Definition 7.5.4 (Isomorphic for subnormal series)

Suppose

$$1 = G_0 \lhd G_1 \lhd \cdots \lhd G_n = G, \quad 1 = H_0 \lhd H_1 \lhd \cdots \lhd H_m = G$$

are two subnormal series of G. We say they are isomorphic, if $n = m$ and there exists a permutation σ of $\{1, \ldots, n\}$ such that $G_i/G_{i-1} \cong H_{\sigma(i)}/H_{\sigma(i)-1}$ for all $i \in \{1, \ldots, n\}$.

Theorem 7.5.2 (Jordan-Hölder Theorem for groups)

If G is finite group, then any two composition series of G are isomorphic.

Proof Our proof is by induction on $|G|$. The base case, $|G| = 1$, is trivial. We use the symbol $\sim$ to denote "are rearrangements of each other".

Let

$$1 = G_0 \lhd G_1 \lhd \cdots \lhd G_{r-1} \lhd G_r = G, \quad 1 = H_0 \lhd H_1 \lhd \cdots \lhd H_{s-1} \lhd H_s = G$$

be any two composition series of G. If $G_{r-1} = H_{r-1}$ then we are done by induction, applying the inductive hypothesis to the group G_{r-1}. So we may assume that $G_{r-1} \neq H_{s-1}$. We set

$$M = G_{r-1} \qquad N = H_{s-1} \qquad K = M \cap N.$$

We can find a composition series

$$0 = K_0 \lhd K_1 \lhd \cdots \lhd K_t = K$$

for K.

Then $0 = G_0 \lhd \cdots \lhd G_{r-1} = M$ and $0 = K_0 \lhd \cdots \lhd K_t = K \lhd M$ are both composition series for M. By induction, we have $r - 1 = t + 1$ and

$$(G_1/G_0, \ldots, G_{r-1}/G_{r-2}) \sim (K_1/K_0, K_2/K_1, \ldots, K_t/K_{t-1}, M/K). \tag{7.21}$$

Similarly, we have $s - 1 = t + 1$ and

$$(H_1/H_0, \ldots, H_{s-1}/H_{s-2}) \sim (K_1/K_0, K_2/K_1, \ldots, K_t/K_{t-1}, N/K). \tag{7.22}$$

From the equalities $r - 1 = t + 1 = s - 1$ we deduce $r = s$. Taking (7.21) and appending the quotient G/M to both lists, we have

$$(G_1/G_0, \ldots, G_{r-1}/G_{r-2}, G/G_{r-1}) \sim (K_1/K_0, K_2/K_1, \ldots, K_t/K_{t-1}, M/K, G/M).$$

Similarly (7.22) gives,

$$(H_1/H_0, \ldots, H_{s-1}/H_{s-2}, G/H_{s-1}) \sim (K_1/K_0, K_2/K_1, \ldots, K_t/K_{t-1}, N/K, G/N).$$

Note that the right hand sides of the above equations are identical except for the last two elements, and by the Fundamental Theorem for Isomorphism, we have $(M/K, G/M) \sim (N/K, G/N)$, as we desired. $\square$

7.5.2 Additional facts particular to modules over a ring.

We now apply these concepts specifically to the category $\mathbf{Mod}_A$.

Proposition 7.5.5

The simple object of $\mathbf{Mod}_A$ are precisely the objects of the form $A/\mathfrak{m}$, where $\mathfrak{m}$ is a maximal ideal of A.

Proof Let M be a simple module. Pick nonzero element $x \in M$, then Ax is a submodule of M, since M is a simple module, $Ax = M$. We want to show that $A/\mathfrak{m}$ is simple. Let $\mathfrak{a} \subseteq A/\mathfrak{m}$ be an ideal of $A/\mathfrak{m}$, then $\mathfrak{a}$ corresponds to ideal of $A/\mathfrak{m}$ which contains $\mathfrak{m}$, since $\mathfrak{m}$ is maximal ideal, $\mathfrak{a} = (0)$. Hence $A/\mathfrak{m}$ is simple. Next, we want to show that $Ax \cong A/\mathfrak{m}$, where $\mathfrak{m} = \operatorname{Ann}(x)$. Define $\varphi : A \to Ax$ be setting

$$a \longmapsto ax.$$

Clearly φ is a surjective. By the Fundamental Theorem for Isomorphism, we have $Ax \cong A/\operatorname{Ann}(x)$. We claim that $\operatorname{Ann}(x)$ is a maximal ideal. If not, there exists an ideal, say I, such that $\operatorname{Ann}(x) \subsetneq I \subsetneq A$, hence $I/\operatorname{Ann}(x)$ is an ideal of $A/\operatorname{Ann}(x)$. Note that Ax is simple, $I = A$, a contradiction! Hence $\operatorname{Ann}(x)$ is a maximal ideal of A. $\square$

Proposition 7.5.6

Suppose M is a finite length A-module, and

$$0 = M_0 \subsetneq M_1 \subsetneq \cdots \subsetneq M_{n-1} \subsetneq M_n = M$$

is a composition series with $M_i/M_{i-1} \cong A/\mathfrak{m}_i$ (where the $\mathfrak{m}_i$ are maximal ideals, not necessarily distinct, Proposition 7.5.5). Then M is annihilated by $\mathfrak{m}_1 \cdots \mathfrak{m}_n$. Equivalent, M is an $A/(\mathfrak{m}_1 \ldots \mathfrak{m}_n)$-module.

Proof Prove by induction on $\ell(M)$. If $\ell(M) = 1$, then our composition series for M is

$$0 = M_0 \subsetneq M_1 = M,$$

i.e., M is simple, i.e., $M = Ax$ for some x. By Proposition 7.5.5, $M \cong A/\mathfrak{m}_1$ where $\mathfrak{m}_1 = \operatorname{Ann}(x)$ is maximal ideal. It follows that $M = Ax$ is annihilated by $\mathfrak{m}_1$, and also M is an $A/\mathfrak{m}_1$-module.

If Proposition 7.5.6 holds for $\ell(M) < n$, we want to show that Proposition 7.5.6 holds for $\ell(M) = n$. Consider the composition series for M,

$$0 = M_0 \subsetneq M_1 \subsetneq \cdots \subsetneq M_{n-1} \subsetneq M_n = M.$$

Note that

$$0 = M_0 \subsetneq M_1 \subsetneq \cdots \subsetneq M_{n-1}$$

is the composition for M_{n-1} with $\ell(M_{n-1}) = n - 1$, by the hypothesis, M_{n-1} is annihilated by $\mathfrak{m}_1 \cdots \mathfrak{m}_{n-1}$. We want to show that $M = M_n$ is annihilated by $\mathfrak{m}_1 \cdots \mathfrak{m}_{n-1}\mathfrak{m}_n$. Note that $M_n/M_{n-1} \cong A/\mathfrak{m}_n$, by the proof in Proposition 7.5.5, $M_n/M_{n-1} = Ax$ where $x \in M_n/M_{n-1}$ nonzero and $Ax \cong A/\operatorname{Ann}(x)$, hence $\operatorname{Ann}(x) \cdot M_n \subseteq M_{n-1}$, i.e., $\mathfrak{m}_n \cdot M_n \subseteq M_{n-1}$.

Let $a_1 \cdots a_{n-1}a_n \in \mathfrak{m}_1 \cdots \mathfrak{m}_{n-1}\mathfrak{m}_n$ where $a_i \in \mathfrak{m}_i$. For any $m \in M = M_n$, note that

$$a_1 \cdots a_{n-1}a_n m \in a_1 \cdots a_{n-1}M_{n-1} = 0,$$

it follows that M is annihilated by $\mathfrak{m}_1 \cdots \mathfrak{m}_{n-1}\mathfrak{m}_n$, as we desired. $\square$

Suppose now that the list $(\mathfrak{m}_1, \ldots, \mathfrak{m}_n)$ consists of the distinct maximal ideals $\mathfrak{n}_1, \ldots, \mathfrak{n}_s$, appearing with multiplicity $l_1, \ldots, l_s$. (There are the "refined" lengths mentioned in Definition 7.5.2.) Since each $\mathfrak{n}_i$ is maximal ideal, we have $\mathfrak{n}_i + \mathfrak{n}_j = A$ whenever $i \neq j$, then exists $n_i \in \mathfrak{n}_i$ and $n_j \in \mathfrak{n}_j$ such that $a_i n_i + a_j n_j = 1$, hence

$(a_in_i + a_jn_j)^{l_i+l_j} = b_in_i^{l_i} + b_jn_j^{l_j} = 1$, which implies that $\mathfrak{n}_i^{l_i} + \mathfrak{n}_j^{l_j} = A$, hence $\mathfrak{n}_i^{l_i}$, $\mathfrak{n}_j^{l_j}$ are coprime whenever $i \neq j$. By Chinese Remainder Theorem 5.4.2 (ii), we have a surjective

$$\varphi : A \twoheadrightarrow A/\mathfrak{n}_1^{l_1} \times \cdots \times A/\mathfrak{n}_s^{l_s}. \tag{7.23}$$

Consider $\operatorname{Ker}\varphi$, note that $\operatorname{Ker}\varphi = \bigcap_{i=1}^{s} \mathfrak{n}_i^{l_i}$, since $\mathfrak{n}_i^{l_i}$ pairwise coprime, by Chinese Remainder Theorem 5.4.2 (i), we have $\bigcap_{i=1}^{s} \mathfrak{n}_i^{l_i} = \mathfrak{n}_1^{l_1} \cdots \mathfrak{n}_s^{l_s}$. By the Fundamental Theorem for Isomorphism, we have an isomorphism

$$A/(\mathfrak{n}_s^{l_s} \cdots \mathfrak{n}_s^{l_s}) \xleftrightarrow{\sim} A/\mathfrak{n}_1^{l_1} \times \cdots \times A/\mathfrak{n}_s^{l_s}.$$

For $1 \leq i \leq s$, let $e_i \in A$ be an element of A such that $e_i \equiv 1 \pmod{n_i^{l_i}}$ and $e_i \equiv 0 \pmod{n_j^{l_j}}$ for $i \neq j$. The e_i exist by (7.23).

Proposition 7.5.7

Suppose M is a finite length A-module, then M is a direct sum of pieces, each with composition series with only one type of "simple factor", i.e.,

$$M \cong e_1M \oplus \cdots \oplus e_sM,$$

each e_iM is a finite-length module where all the simple quotients are $A/\mathfrak{n}_i$. Moreover, $\mathfrak{n}_i$ are distinct maximal ideal of A, and exists l_i such that $e_i \equiv 1 \pmod{\mathfrak{n}_i^{l_i}}$ and $e_i \equiv 0 \pmod{\mathfrak{n}_j^{l_j}}$ for $i \neq j$.

Proof Since M is a finite length A-module, let

$$0 = M_0 \subsetneq M_1 \subsetneq \cdots \subsetneq M_{n-1} \subsetneq M_n = M \tag{7.24}$$

be a composition series with $M_i/M_{i-1} \cong A/\mathfrak{m}_i$. By Proposition 7.5.6, M is annihilated by $\mathfrak{m}_1 \cdots \mathfrak{m}_n$, and therefore can be seen as $A/(\mathfrak{m}_1 \cdots \mathfrak{m}_n)$-module. Suppose the list $(\mathfrak{m}_1, \ldots, \mathfrak{m}_n)$ consists of the distinct maximal ideals $\mathfrak{n}_1, \ldots, \mathfrak{n}_s$, appearing with multiplicity $l_1, \ldots, l_s$. By the discussion before Proposition 7.5.7, $\mathfrak{n}_i^{l_i}$ and $\mathfrak{n}_j^{l_j}$ are coprime whenever $i \neq j$. Apply Chinese Remainder Theorem 5.4.2, we have surjective

$$\varphi : A \twoheadrightarrow A/\mathfrak{n}_1^{l_1} \times \cdots \times A/\mathfrak{n}_s^{l_s}.$$

For $1 \leq i \leq s$, let $e_i \in A$ be an element of A such that $e_i \equiv 1 \pmod{\mathfrak{n}_i^{l_i}}$ and $e_i \equiv 0 \pmod{\mathfrak{n}_j^{l_j}}$ for $i \neq j$, (by φ is surjective). Hence $e_i(e_i - 1) \equiv 0 \pmod{\mathfrak{n}_k^{l_k}}$, $e_ie_j \equiv 0 \pmod{\mathfrak{n}_k^{l_k}}$, and $e_1 + \cdots + e_s - 1 \equiv 0 \pmod{\mathfrak{n}_k^{l_k}}$ for all k. Hence $e_i(e_i - 1), e_ie_j, e_1 + \cdots + e_s - 1 \in \operatorname{Ker}\varphi = \bigcap_{i=1}^{s} \mathfrak{n}_i^{l_i} = \mathfrak{n}_1^{l_1} \cdots \mathfrak{n}_s^{l_s}$. In fact, $\mathfrak{n}_1^{l_1} \cdots \mathfrak{n}_s^{l_s} = \mathfrak{m}_1 \cdots \mathfrak{m}_n \subseteq \operatorname{Ann}(M)$. It follows that for any $m \in M$, we have

(i) $e_i(e_i - 1)m = 0 \Longrightarrow e_i^2m = e_im$;

(ii) $e_ie_jm = 0$ whenever $i \neq j$;

(iii) $(e_1 + \cdots + e_s - 1)m = 0 \Longrightarrow (e_1 + \cdots + e_s)m = m$.

By (iii) we know that

$$M \cong e_1M + \cdots e_sM_s.$$

Consider $e_jM \cap \left(\sum_{i\neq j} e_iM\right)$, let $m \in e_jM \cap \left(\sum_{i\neq j} e_iM\right)$ then $m = e_jm_j$ for some $m_j \in M$. Since $m \in \sum_{i\neq j} e_iM$, we may assume that

$$m = e_jm_j = e_1m_1 + \cdots + e_{j-1}m_{j-1} + e_{j+1}m_{j+1} + \cdots + e_sm_s. \tag{7.25}$$

Apply e_j to both sides of equation (7.25), by (i) and (ii),

$$e_j^2 m_j = e_j m_j = m = 0,$$

hence $e_j M \cap \left(\sum_{i \neq j} e_i M\right) = \{0\}$, which implies that

$$M \cong e_1 M \oplus \cdots \oplus e_s M.$$

Next we want to show that each $e_i M$ is finite length module where all the simple quotients are $A/\mathfrak{n}_i$. By (7.24) we have a filtration for $e_i M$,

$$0 = e_i M_0 \subsetneq e_i M_1 \subsetneq \cdots \subsetneq e_i M_{n-1} \subsetneq e_i M_n = e_i M. \tag{7.26}$$

We claim that $e_i M_k / e_i M_{k-1} \cong A/\mathfrak{m}_k$ or 0 for all $1 \leq k \leq n$ for some maximal ideal $\mathfrak{m}_k$. Note that $M_{k-1} = \bigoplus_{t=1}^{s} e_t M_{k-1}$, hence $e_i M_{k-1} \cap M_{k-1} = e_i M_{k-1}$. Since $e_i M_{k-1} \subseteq e_i M_k$, we have $e_i M_{k-1} \subseteq e_i M_k \cap M_{k-1}$. Conversely, let $e_i m_k \in e_i M_k \cap M_{k-1}$, then $e_i m_k \in M_{k-1}$. Suppose $e_i m_k = \sum_{t=1}^{s} e_t m_{k-1,t}$ where $m_{k-1,t} \in e_t M_{k-1}$. Apply e_i to both sides, we get $e_i m_k = e_i m_{k-1,i} \in e_i M_{k-1}$. Hence $e_i M_{k-1} = e_i M_k \cap M_{k-1}$. By Proposition 7.5.5, we have

$$\frac{e_i M_k}{e_i M_{k-1}} = \frac{e_i M_k}{e_i M_k \cap M_{k-1}} \cong \frac{M_{k-1} + e_i M_k}{M_{k-1}} \subseteq \frac{M_k}{M_{k-1}}.$$

Since M_k / M_{k-1} is simple, $e_i M_k / e_i M_{k-1}$ must be 0 or M_k/M_{k-1}. If $e_i M_k / e_i M_{k-1} \neq 0$, by Proposition 7.5.5, $e_i M_k / e_i M_{k-1} \cong M_k / M_{k-1} \cong A/\mathfrak{m}_k$ for some maximal ideal $\mathfrak{m}_k$. By combining the equal terms in (7.26), we obtain a composition series for $e_i M$, which implies that each $e_i M$ is of finite length.

Say

$$0 = N_0 \subsetneq N_1 \subsetneq \cdots \subsetneq N_{p-1} \subsetneq N_p = e_i M$$

be the composition series for $e_i M$, where $N_k / N_{k-1} \cong A/\mathfrak{m}_k$ for some maximal ideal $\mathfrak{m}_k$, moreover, by the proof of Proposition 7.5.5, $\mathfrak{m}_k = \operatorname{Ann}(N_k/N_{k-1})$. We want to show that $\mathfrak{m}_k$ are the same. Since $e_i \equiv 0 \pmod{\mathfrak{n}_j^{l_j}}$ for all $i \neq j$, $e_i \in \bigcap_{j \neq i} \mathfrak{n}_j^{l_j} = \prod_{j \neq i} \mathfrak{n}_j^{l_j}$, and note that M can be seen as $A/(\mathfrak{n}_1^{l_1} \cdots \mathfrak{n}_s^{l_s})$-module, hence $e_i M$ annihilated by $\mathfrak{n}_i^{l_i}$. Hence $\mathfrak{n}_i^{l_i}$ annihilates N_k, it follows that $\mathfrak{n}_i^{l_i} \subseteq \operatorname{Ann}(N_k/N_{k-1}) = \mathfrak{m}_k$. Taking radical both sides, we have $\mathfrak{n}_i = \sqrt{\mathfrak{n}_i^{l_i}} \subseteq \sqrt{\mathfrak{m}_k} = \mathfrak{m}_k$. Since $\mathfrak{n}_i$ is maximal ideal, we have $\mathfrak{m}_k = \mathfrak{n}_i$ for all k, as we desired. ☐

Lemma 7.5.1

Suppose M is an A-module. M has a composition series if and only if M satisfies both chain condition.

Proof If M has composition series, then any chain in M are of bounded length, hence both a.c.c. and d.c.c..

Conversely, if M satisfies both chain condition. Since $M = M_0$ satisfies the maximum condition. It has a maximal submodule $M_1 \subseteq M_0$. Similarly M_1 has a maximal submodule $M_2 \subseteq M_1$ and so on. Thus we have a strictly descending chain $M_0 \supseteq M_1 \supseteq \cdots$ which by d.c.c. must be finite, and hence is a composition series of M. ☐

Proposition 7.5.8

Suppose M is a finite length A-module. Then M is finitely generated, and $\operatorname{Supp} M$ (Definition 5.1.5) consists of finitely many points of $\operatorname{Spec} A$, all closed. We thus have a notion of the "length of M at each of these closed points".

Proof By Lemma 7.5.1, M satisfies a.c.c., hence M is Noetherian module, and therefore is finitely generated.

Let
$$0 = M_0 \subsetneq M_1 \subsetneq \cdots \subsetneq M_{n-1} \subsetneq M_n = M \tag{7.27}$$
be the composition series of M with $M_i/M_{i-1} \cong A/\mathfrak{m}_i$.

Let $[\mathfrak{p}] \in \operatorname{Supp} M$, then $M_\mathfrak{p} \neq 0$. By chain (7.27), we have a filtration for $M_\mathfrak{p}$
$$0 = (M_0)_\mathfrak{p} \subseteq (M_1)_\mathfrak{p} \subseteq \cdots \subseteq (M_{n-1})_\mathfrak{p} \subseteq (M_n)_\mathfrak{p} = M_\mathfrak{p}.$$
Since $M_\mathfrak{p} \neq 0$, there exists k such that $(M_k)_\mathfrak{p}/(M_{k-1})_\mathfrak{p} \neq 0$. Since localization commutes with quotient,
$$0 \neq (M_k)_\mathfrak{p}/(M_{k-1})_\mathfrak{p} \cong (M_k/M_{k-1})_\mathfrak{p} \cong (A/\mathfrak{m}_k)_\mathfrak{p} \cong A_\mathfrak{p}/(\mathfrak{m}_k)_\mathfrak{p}.$$
Since $(\mathfrak{m}_k)_\mathfrak{p} = \mathfrak{m}_k A_\mathfrak{p}$ is an ideal of $A_\mathfrak{p}$, we must have $\mathfrak{m}_k \subseteq \mathfrak{p}$, note that $\mathfrak{m}_k$ is maximal ideal, $\mathfrak{p} = \mathfrak{m}_k$. It follows that $\operatorname{Supp} M \subseteq \{\mathfrak{m}_1, \dots, \mathfrak{m}_n\}$, i.e., $\operatorname{Supp} M$ consists of finitely many points of $\operatorname{Spec} A$, all closed. □

Remark 7.17 A converse under Noetherian hypotheses will be proved in Proposition 7.6.29.

Definition 7.5.5 (Artinian ring, Artinian local ring)

*If A has finite length as a module over itself, we say A is an **Artinian ring**. If A is furthermore a local ring, we say A is an **Artin** or **Artinian local ring**.*

7.5.3 Applying to schemes

Now applying above language to schemes. We next consider the category $\mathbf{QCoh}_X$ of quasi-coherent sheaves on a scheme X. We have the notion of the length $\ell(\mathscr{F})$ of a finite-length quasi-coherent sheaf on X.

Proposition 7.5.9

Suppose X is quasi-compact (why need quasi-compact, see [Sta25, Lemma 01PE]. The simple objects of $\mathbf{QCoh}_X$ are those isomorphism to the structure sheaves of closed points.

Proof Let $\mathscr{F} \in \mathbf{QCoh}_X$ be a simple object, then for any affine open subsets $\operatorname{Spec} A \subseteq X$, $\mathscr{F}|_{\operatorname{Spec} A} = \widetilde{M}$ where M is simple ($\mathscr{F}|_{\operatorname{Spec} A}$ is simple, if X is quasi-compact, see [Inf]). By Proposition 7.5.5, we have $M \cong A/\operatorname{Ann}(M)$ where $\operatorname{Ann}(M)$ is maximal ideal. Say $\mathfrak{m} = \operatorname{Ann}(M)$, we have
$$\mathscr{F}|_{\operatorname{Spec} A} \cong \widetilde{A/\mathfrak{m}} \cong \mathscr{O}_{\operatorname{Spec}(A/\mathfrak{m})}.$$
Let $[\mathfrak{p}] \in \operatorname{Spec} A$, then
$$(\mathscr{F}|_{\operatorname{Spec} A})_{[\mathfrak{p}]} \cong (A/\mathfrak{m})_\mathfrak{p} = A_\mathfrak{p}/\mathfrak{m}A_\mathfrak{p} = \begin{cases} A_\mathfrak{m}/\mathfrak{m}A_\mathfrak{m} = \kappa([\mathfrak{m}]) & \text{if } \mathfrak{p} = \mathfrak{m} \\ 0 & \text{if } \mathfrak{p} \neq \mathfrak{m}. \end{cases}$$
It follows that $\mathscr{F}|_{\operatorname{Spec} A} \cong i_{[\mathfrak{m}],*}\underline{\kappa([\mathfrak{m}])}$ where $i : \{[\mathfrak{m}]\} \hookrightarrow X$. In other words, for all affine open subsets $\operatorname{Spec} A \subseteq X$ exists closed point $x \in \operatorname{Spec} A$ such that $\mathscr{F}|_{\operatorname{Spec} A} \cong i_{x,*}\underline{\kappa(x)}$ where $i : \{x\} \hookrightarrow X$. Also, $\operatorname{Supp} \mathscr{F}|_{\operatorname{Spec} A} = \{x\}$.

Claim that $\operatorname{Supp} \mathscr{F} = \{x\}$. Suppose there exists $y \in \operatorname{Supp} \mathscr{F}$ such that $y \neq x$, We first show that y is a closed point. Let V be an affine open subset which contains y, then there exists closed point $z \in V$ such that $\operatorname{Supp} \mathscr{F}|_V = \{z\}$. Since $\operatorname{Supp} \mathscr{F}|_V = \operatorname{Supp} \mathscr{F} \cap V$, we have $z = y$, and therefore y is a closed point. Consider $j : X \setminus \{x\} \hookrightarrow X$. Note that $\mathscr{F}$ is a quasi-coherent sheaf, we have an isomorphism
$$j^*\mathscr{F} \xrightarrow{\sim} j^*$$
on $X \setminus \{x\}$. By adjointness of (j^*, j_*), we have a morphism
$$\mathscr{F} \longrightarrow j_*j^*\mathscr{F}.$$

Consider $\mathscr{G} := \operatorname{Ker}(\mathscr{F} \to j_* j^* \mathscr{F}) \hookrightarrow \mathscr{F}$. We next calculate $\mathscr{G}_x$ and $\mathscr{G}_y$.

Let $U = \operatorname{Spec} A \subseteq X$ be any affine open neighborhood of x, then

$$(j_* j^* \mathscr{F})(U) = j^* \mathscr{F}(U \setminus \{x\}) = j^{-1}\mathscr{F}(U \setminus \{x\}) \otimes_{f^{-1}\mathscr{O}_X(U\setminus\{x\})} \mathscr{O}_{X\setminus\{x\}}(U \setminus \{x\}).$$

Note that

$$j_{\mathrm{pre}}^{-1}\mathscr{F}(U \setminus \{x\}) = \varinjlim_{\text{open } V \supseteq j(U\setminus\{x\})} \mathscr{F}(V) = \mathscr{F}(U \setminus \{x\}),$$

where last equal given by $\{x\}$ is a closed point and therefore $U \setminus \{x\}$ is open. Hence

$$\begin{aligned}(j_* j^* \mathscr{F})_x = ((j_* j^* \mathscr{F})|_U)_x &= ((j^{-1}\mathscr{F})|_{U\setminus\{x\}} \otimes_{(f^{-1}\mathscr{O}_X)|_{U\setminus\{x\}}} \mathscr{O}_{U\setminus\{x\}})_x \\ &= (\mathscr{F}|_{U\setminus\{x\}})_x \otimes_{((f^{-1}\mathscr{O}_X)|_{U\setminus\{x\}})_x} (\mathscr{O}|_{U\setminus\{x\}})_x \\ &= 0 \otimes (\mathscr{O}|_{U\setminus\{x\}})_x \\ &= 0.\end{aligned}$$

Thus

$$\mathscr{G}_x = \operatorname{Ker}(\mathscr{F} \to j_* j^* \mathscr{F})_x = \operatorname{Ker}(\mathscr{F}_x \to (j_* j^* \mathscr{F})_x) = \operatorname{Ker}(\mathscr{F}_x \to 0) = \mathscr{F}_x \neq 0.$$

Let U be an affine open neighborhood of y such that $U \subseteq X \setminus \{x\}$, then $(j_* j^* \mathscr{F})(U) = j^* \mathscr{F}(U)$, hence

$$\begin{aligned}(j_* j^* \mathscr{F})_y =& (j_* j^* \mathscr{F}|_U)_y = (j^* \mathscr{F}|_U)_y \\ =& (j^{-1}\mathscr{F}|_U)_y \otimes_{(j^{-1}\mathscr{O}_X|_U)_y} (\mathscr{O}_U)_y \\ =& \mathscr{F}_{j(y)} \otimes_{(\mathscr{O}_U)_y} (\mathscr{O}_U)_y \\ =& \mathscr{F}_y \otimes_{(\mathscr{O}_U)_y} (\mathscr{O}_U)_y \cong \mathscr{F}_y.\end{aligned}$$

Thus $\mathscr{G}_y = \operatorname{Ker}(\mathscr{F}_y \to (j_* j^* \mathscr{F})_y) = \operatorname{Ker}(\mathscr{F}_y \to \mathscr{F}_y) = 0$. Since $y \in \operatorname{Supp} \mathscr{F}$, $\mathscr{F}_y \neq 0$. It follows that $\mathscr{G} \hookrightarrow \mathscr{F}$ is a nonzero subobject of $\mathscr{F}$, contradicts to the fact that $\mathscr{F}$ is simple. Hence $\operatorname{Supp} \mathscr{F} = \{x\}$, and therefore $\mathscr{F} \cong \mathscr{O}_{\operatorname{Spec}(A/\mathfrak{m})}$, where $\mathfrak{m}$ is maximal ideal of A. □

Proposition 7.5.10

Suppose X is quasi-compact (why need quasi-compact, see [Sta25, Lemma 01PE]). Suppose that $\mathscr{F}$ is a finite length element of $\mathbf{QCoh}_X$, then $\mathscr{F}$ is finite type, and $\operatorname{Supp} \mathscr{F}$ consists of finitely many points of X, all closed. Thus we can define the length of a $\mathscr{F}$ at one of the points of $\operatorname{Supp} \mathscr{F}$.

Proof Let $\operatorname{Spec} A \subseteq X$ be any affine open subset of X, then $\mathscr{F}|_U \cong \widetilde{M}$ for some A-module M. Since $\mathscr{F}$ is finite length, $\mathscr{F}|_U$ must be finite length, hence M is finite length. By Proposition 7.5.8, M is finitely generated, and therefore $\mathscr{F}$ is finite type. Let

$$0 = \mathscr{F}_0 \hookrightarrow \mathscr{F}_1 \hookrightarrow \cdots \hookrightarrow \mathscr{F}_{n-1} \hookrightarrow \mathscr{F}_n = \mathscr{F} \tag{7.28}$$

be the composition series of $\mathscr{F}$. Since each $\operatorname{Coker}(\mathscr{F}_{i-1} \hookrightarrow \mathscr{F}_i)$ is simple, by Proposition 7.5.9, $\operatorname{Coker}(\mathscr{F}_{i-1} \hookrightarrow \mathscr{F}_i) \cong \widetilde{A_i/\mathfrak{m}_i}$ for some $\mathfrak{m}_i$ which is the maximal ideal of A_i. Let $x \in \operatorname{Supp} \mathscr{F}$, then $\mathscr{F}_x \neq 0$. By composition series (7.29), we have a filtration for $\mathscr{F}_x$,

$$0 = (\mathscr{F}_0)_x \hookrightarrow (\mathscr{F}_1)_x \hookrightarrow \cdots \hookrightarrow (\mathscr{F}_{n-1})_x \hookrightarrow (\mathscr{F}_n)_x = \mathscr{F}_x. \tag{7.29}$$

Since $\mathscr{F}_x \neq 0$, there exists k such that $\operatorname{Coker}((\mathscr{F}_{k-1})_x \hookrightarrow (\mathscr{F}_k)_x) \neq 0$. Since taking stalks commutes with Coker, we have

$$\operatorname{Coker}((\mathscr{F}_{k-1})_x \hookrightarrow (\mathscr{F}_k)_x) \cong \operatorname{Coker}(\mathscr{F}_{k-1} \hookrightarrow \mathscr{F}_k)_x \cong (\widetilde{A_k/\mathfrak{m}_k})_x \cong A_x/(\mathfrak{m}_k A_x).$$

Hence, $x = [\mathfrak{m}_k]$. It fullows that $\operatorname{Supp} \mathscr{F} \subseteq \{\mathfrak{m}_1, \ldots, \mathfrak{m}_n\}$ where each $\mathfrak{m}_k$ is closed. □

Remark 7.18 A converse under Noetherian hypotheses will be proved in Proposition 7.6.29.

Definition 7.5.6 (Length of scheme)

*The **length of a scheme** X is the length of the structure sheaf $\mathscr{O}_X$ (in $\mathbf{QCoh}_X$). A scheme X is **finite length** or **Artinian** if $\mathscr{O}_X$ is finite length.*

7.6 Visualizing schemes: Associated points and zero-divisors

The theory of **associated points** of a module refines the notion of support (Definition 5.1.5). Associated points will help us understand and visualize nilpotents, and generalize the notion of "rational functions" to non-integral schemes. They are useful in ways we won't use, for example through their connection to primary decomposition. They might be most useful for us in helping us understand and visualize (non-)zerodivisors, which will come up repeatedly, through effective Cartier divisors and line bundles, regular sequences, depth and Cohen-Macaulayness, and more.

7.6.1 Motivation

Figure 7.1 is a sketch of a scheme X. We see two connected components, and three irreducible components. The irreducible components of X have dimensions 2, 1, and 1, although we won't be able to make sense of "dimension" until Chapter 13. Both connected components are nonreduced.

We see a little more in this picture, which we will make precise in this section, in terms of "associated points". The reducible connected component seems to have different amounts of nonreduced behavior on different loci. The scheme X has six associated points, which are the generic points of the irreducible subsets "visible" in the picture. A function on X is a zerodivisor if its zero locus contains any of these six irreducible subvarieties.

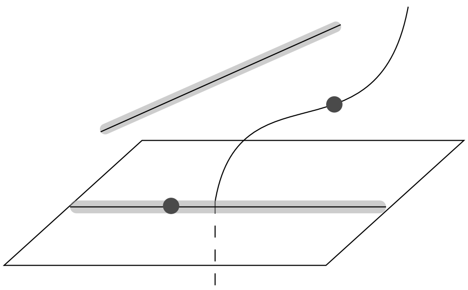


Figure 7.1: This scheme has six associated points, of which three are embedded points. A function is a zerodivisor if it vanishes at any of these six points.

Suppose M is a finitely generated module over a Noetherian ring A. For example, M could be A. Then there are some special points of $\operatorname{Spec} A$ that are particularly crucial to understanding M. These are the **associated points** of M (or equivalently, the **associated prime ideals** of M — we will use these terms interchangeably). As motivation, we give a zillion properties of associated points, and leave it to you to verify them from the theory developed in the rest of this section.

As you read this section, you may wish to keep in mind

$$M = A = k[x, y]/(y^2, xy)$$

(Figure 5.4) as a running example.

A zillion properties of associated points

Here are some of the properties of associated points that we will prove.

Introduction

- *There are finitely many associated points* $\operatorname{Ass}_A M \subseteq \operatorname{Spec} A$.
- *The support of M is the closure of the associated points of M: $\operatorname{Supp} M = \overline{\operatorname{Ass}_A M}$. The support of any submodule of M is the closure of some subset of the associated points of M. The support of any element of M is the closure of some subset of the associated points.*
- *The associated points of M are precisely the generic points of irreducible components of $\operatorname{Supp} m$ for all $m \in M$. The associated points of M are precisely the generic points of those $\operatorname{Supp} m$ which are irreducible (as m runs over the elements of M). The associated primes of M are precisely those prime ideals that are annihilators of some element of M.*
- *Taking "associated points" commutes with localization. Hence this notion is "geometric in nature", which will allow us to extend the notion to coherent sheaves on locally Noetherian schemes.*
- *Associate points behave **fairly** well in exact sequence. For example, the associated points of a submodule are a subset of the associated points of the module.*
- *If $I \subseteq A$ is an ideal, the associated primes $\mathfrak{p}$ of A/I are precisely those $\mathfrak{p}$ such that a $\mathfrak{p}$-primary ideal appears in the primary decomposition of I.*
- ***Important!** An element of A is a zerodivisor if and only if it vanishes at an associated point.*
- *An element of A is nilpotent if and only if it vanishes at every associated point. The locus of points $[\mathfrak{p}]$ of $\operatorname{Spec} A$ where the stalk $A_{\mathfrak{p}}$ is nonreduced is the closure of some subset of the associated points.*
- *An associated point that is in the closure of another associated point is said to be an **embedded point**. If A is reduced, then $\operatorname{Spec} A$ has no embedded points. Hypersurfaces in $\mathbb{A}_k^n$ have no embedded points. We will later see that complete intersections have no embedded points (Chapter 27).*
- *Elements of M are determined by their localization at the associated points. Sections of the corresponding sheaf $\widetilde{M}$ are determined by their germs at the associated points.*

This discussion immediately implies a notion of **associated point** for a coherent sheaf on a locally Noetherian scheme, with all the good properties described here. The phrase **associated point of a locally Noetherian scheme** X (without explicit mention of a coherent sheaf) means "associated point of $\mathscr{O}_X$", and similarly for **embedded points**.

We now establish these zillion facts.

7.6.2 More on the notion of support

Support and annihilator ideal

The notion of associated points of an A-module M refines the notion of support (in the case where M is finitely generated over a Noetherian ring A). (In what follows, we make no assumptions that A is Noetherian or that M is finitely generated until we need to.) To set this up, recall Definition 5.1.5 that the support of $m \in M$,

$$\operatorname{Supp} m = \{[\mathfrak{p}] \in \operatorname{Spec} A : m_{\mathfrak{p}} \neq 0\},$$

is a closed subset of $\operatorname{Spec} A$ (Proposition 5.1.5), and thus of the form $V(I)$ for some I. Proposition 7.6.1 gives the "best such" I.

Definition 7.6.1 (Annihilator ideal)

Define the ***annihilator ideal*** $\operatorname{Ann}_A m \subseteq A$ *of an element* m *of an* A*-module* M *by:*

$$\operatorname{Ann}_A m := \{a \in A : am = 0\} = \operatorname{Ker}(A \xrightarrow{\sim} M).$$

The subscript A *is omitted if it is clear from context.*

Proposition 7.6.1

Suppose M *is an* A*-module, then* $\operatorname{Supp} m = V(\operatorname{Ann} m)$ *for all* $m \in M$. *Moreover, if* M *is finitely generated* A*-module, then* $\operatorname{Supp}(M) = V(\operatorname{Ann}_A M)$.

Proof Note that

$$\begin{aligned}
\operatorname{Supp} m =& \{[\mathfrak{p}] \in \operatorname{Spec} A : m_{\mathfrak{p}} \neq 0\} \\
=& \{[\mathfrak{p}] \in \operatorname{Spec} A : (A - \mathfrak{p})m \neq 0\} \\
=& \{[\mathfrak{p}] \in \operatorname{Spec} A : (A - \mathfrak{p}) \cap \operatorname{Ann} m = \varnothing\} \\
=& \{[\mathfrak{p}] \in \operatorname{Spec} A : \mathfrak{p} \supseteq \operatorname{Ann} m\} \\
=& V(\operatorname{Ann} m),
\end{aligned}$$

as we desired.

Since M is finitely generated A-module, we may assume that $M = Ax_1 + \cdots Ax_n$, then

$$\begin{aligned}
V(\operatorname{Ann}_A M) =& V\left(\bigcap_{i=1}^n \operatorname{Ann}_A x_i\right) = V\left(\prod_{i=1}^n \operatorname{Ann}_A x_i\right) \\
&= \bigcup_{i=1}^n V(\operatorname{Ann}_A x_i) = \bigcup_{i=1}^n \operatorname{Supp}_A x_i \\
&= \operatorname{Supp}_A M,
\end{aligned}$$

(we use the fact that $\operatorname{Ann} x_i$ are pairwise coprime, and apply CRT) as we desired. □

Recall (Definition 3.7.5) that

$$\operatorname{Supp} \widetilde{M} = \{p \in \operatorname{Spec} A : \widetilde{M}_p \neq 0\},$$

and the analogous Definition 5.1.5 of the support of the module M,

$$\operatorname{Supp} M := \{\mathfrak{p} \in \operatorname{Spec} A : M_{\mathfrak{p}} \neq 0\}, \tag{7.30}$$

so $\operatorname{Supp} M = \operatorname{Supp} \widetilde{M}$. If M is a principal module generated by $m \in M$, then

$$\operatorname{Supp} M = \operatorname{Supp} Am = \operatorname{Supp} m = V(\operatorname{Ann} m).$$

The notions of support and associated points behave well in exact sequences, and under localization. We begin to explain this now.

The notion of support behaves well in exact sequences

Proposition 7.6.2

Suppose that $0 \to M' \to M \to M'' \to 0$ is a short exact sequence of A-modules.

(a) $\operatorname{Supp} M = \operatorname{Supp} M' \cup \operatorname{Supp} M''$.

(b) If M is a finitely generated module, then $\operatorname{Supp} M$ *is a closed subset of* $\operatorname{Spec} A$.

Proof

(a) Let $\mathfrak{p} \in \operatorname{Supp} M$, then $M_{\mathfrak{p}} \neq 0$. Since localization is exact, we have an exact sequence

$$0 \longrightarrow M'_{\mathfrak{p}} \longrightarrow M_{\mathfrak{p}} \longrightarrow M''_{\mathfrak{p}} \longrightarrow 0. \tag{7.31}$$

Since $M_{\mathfrak{p}} \neq 0$, $M'_{\mathfrak{p}} \neq 0$ or $M''_{\mathfrak{p}} \neq 0$, which implies that $\mathfrak{p} \in \operatorname{Supp} M' \cup \operatorname{Supp} M''$. Conversely, let $\mathfrak{p} \in \operatorname{Supp} M' \cup \operatorname{Supp} M''$, then $M'_{\mathfrak{p}} \neq 0$ or $M''_{\mathfrak{p}} \neq 0$. By the exactness of sequence (7.31), $M_{\mathfrak{p}} \neq 0$, i.e., $\mathfrak{p} \in \operatorname{Supp} M$. Hence $\operatorname{Supp} M = \operatorname{Supp} M' \cup \operatorname{Supp} M''$.

(b) Prove by induction on the number of generators of M. Suppose M is generated by $x_1, \cdots, x_n$. If $n = 1$, then M is generated by x_1, i.e., $M = Ax_1$. Hence $\operatorname{Supp} M = \operatorname{Supp} Ax_1 = \operatorname{Supp} x_1$, by Proposition 5.1.5, $\operatorname{Supp} M$ is a closed subset of $\operatorname{Spec} A$. Assume the result holds for the case of $n-1$ generators, we want to show that $\operatorname{Supp} M$ is closed in $\operatorname{Spec} A$, where $M = \bigoplus_{i=1}^{n} Ax_n$. Consider the following exact sequence.

$$0 \longrightarrow \bigoplus_{i=1}^{n-1} Ax_i \longrightarrow M \longrightarrow Ax_n \longrightarrow 0$$

By part (a), $\operatorname{Supp} M = \operatorname{Supp} \bigoplus_{i=1}^{n-1} Ax_i \cup \operatorname{Supp} Ax_n$. By the induction hypothesis, $\operatorname{Supp} \bigoplus_{i=1}^{n-1} Ax_i$ is closed in $\operatorname{Spec} A$, also note that $\operatorname{Supp} Ax_n = \operatorname{Supp} x_n$ is closed in $\operatorname{Spec} A$ (Proposition 5.1.5), hence $\operatorname{Supp} M$ is a closed subset of $\operatorname{Spec} A$.

□

Remark 7.19 **Warning:** $\operatorname{Supp} M$ need not be closed in general; consider $A = \mathbb{Z}$ and $M = \bigoplus_{q \text{ prime}} \mathbb{Z}/(q)$. Let $\mathfrak{p} \in \operatorname{Spec} \mathbb{Z}$, then

$$M_{\mathfrak{p}} = \bigoplus_{q \text{ prime}} (\mathbb{Z}/(q))_{\mathfrak{p}}.$$

If $\mathfrak{p} = (0)$, then $(\mathbb{Z}/(q))_{(0)} = \mathbb{Q} \otimes_{\mathbb{Z}} \mathbb{Z}/(q) = 0$ for all prime number q. If $\mathfrak{p} = (p) \neq (0)$, then $(\mathbb{Z}/(p))_{(p)} \cong \mathbb{Z}_{(p)}/(p)\mathbb{Z}_{(p)} \cong \operatorname{Frac}(\mathbb{Z}/(p)) = \mathbb{Z}/(p)$ and $(\mathbb{Z}/(q))_{(p)} = 0$ for all $q \neq p$ (since $q \notin (p)$, $q \in \mathbb{Z} - (p)$, but $q = 0$ in $\mathbb{Z}/(q)$, which implies that $(\mathbb{Z}/(q))_{(p)} = 0$). Hence $M_{\mathfrak{p}} \neq 0$ for all $\mathfrak{p} \in \operatorname{Spec} \mathbb{Z} \setminus \{(0)\}$, it follows that $\operatorname{Supp} M = \operatorname{Spec} \mathbb{Z} \setminus \{(0)\}$, it is not a closed subset of $\operatorname{Spec} \mathbb{Z}$ since the closure $\operatorname{cl}(\operatorname{Supp} M) = \operatorname{Spec} \mathbb{Z} \neq \operatorname{Supp} M$.

Proposition 7.6.3

Suppose M is a finitely generated A-module, and $x \in A$ has value 0 at all the points of $\operatorname{Supp} M$, *i.e., x is contained in all of the primes where M is supported. Then some power x^n of x annihilates every element of M.*

Proof Since M is finitely generated, we may assume that $M = \bigoplus_{i=1}^{n} Am_i$. Let $\mathfrak{p} \in \operatorname{Supp} M$, then $M_{\mathfrak{p}} \neq 0$, hence

$$M_{\mathfrak{p}} = \left(\bigoplus_{i=1}^{n} Am_i\right)_{\mathfrak{p}} = \bigoplus_{i=1}^{n}(A_{\mathfrak{p}} \otimes_A Am_i) \neq 0,$$

it follows that $A_{\mathfrak{p}} \otimes_A Am_i \neq 0$ for some i. Hence $\mathfrak{p} \in \operatorname{Supp} m_i$ for some i, which implies that $\operatorname{Supp} M \subseteq \bigcup_{i=1}^{n} \operatorname{Supp} m_i$. Clearly, we have $\bigcup_{i=1}^{n} \operatorname{Supp} m_i \subseteq \operatorname{Supp} M$. Hence $\operatorname{Supp} M = \bigcup_{i=1}^{n} \operatorname{Supp} m_i$. By Proposition 7.6.1, $\operatorname{Supp} M = \bigcup_{i=1}^{n} V(\operatorname{Ann} m_i) = V(\prod_{i=1}^{n} \operatorname{Ann} m_i)$. By the condition, we know that $x \in \bigcap_{[\mathfrak{p}]\in \operatorname{Supp} M} \mathfrak{p}$, hence

$$x \in \bigcap_{[\mathfrak{p}]\in V(\prod_{i=1}^{n} \operatorname{Ann} m_i)} \mathfrak{p} = \sqrt{\prod_{i=1}^{n} \operatorname{Ann} m_i}.$$

Next we want to show that $\prod_{i=1}^{n} \operatorname{Ann} m_i = \bigcap_{i=1}^{n} \operatorname{Ann} m_i$, it suffices to show that $\operatorname{Ann} m_i$ pairwise coprime. By the proof in Proposition 7.5.5, each $\operatorname{Ann} m_i$ is maximal ideal of A, and therefore $\operatorname{Ann} m_i$ pairwise coprime, so

$$x \in \sqrt{\bigcap_{i=1}^{n} \operatorname{Ann} m_i}.$$

Hence $x^n \in \bigcap_{i=1}^{n} \operatorname{Ann} m_i$ for some n, which implies that x^n annihilates every element of M. □

Proposition 7.6.4

Suppose M is a finitely generated A-module, and $x \in A$. Then $\operatorname{Supp}(M/xM) = (\operatorname{Supp} M) \cap V(x)$. *Here M/xM is defined by the exact sequence*

$$M \xrightarrow{\times x} M \longrightarrow M/xM \longrightarrow 0.$$

Proof Let $\mathfrak{p} \in \operatorname{Supp}(M/xM)$, then $(M/xM)_{\mathfrak{p}} \neq 0$. Since localization commutes with quotient, we have $(M/xM)_{\mathfrak{p}} = M_{\mathfrak{p}}/(xM)_{\mathfrak{p}} \neq 0$, hence $M_{\mathfrak{p}} \neq 0$ and $(xM)_{\mathfrak{p}} \neq M_{\mathfrak{p}}$. We claim that $x \in \mathfrak{p}$. If not, $x \notin \mathfrak{p}$, i.e. $x \in A - \mathfrak{p}$, then

$$(xM)_{\mathfrak{p}} = A_{\mathfrak{p}} \otimes_A (xM) = A_{\mathfrak{p}} \otimes M = M_{\mathfrak{p}},$$

contradicts to the fact that $(xM)_{\mathfrak{p}} \neq M_{\mathfrak{p}}$. Hence $\mathfrak{p} \in (\operatorname{Supp} M) \cap V(x)$.

Conversely, let $\mathfrak{p} \in (\operatorname{Supp} M) \cap V(x)$. Consider the exact sequence

$$0 \longrightarrow xM \longrightarrow M \longrightarrow M/xM \longrightarrow 0.$$

Since localization is exact, the sequence

$$0 \longrightarrow (xM)_{\mathfrak{p}} \longrightarrow M_{\mathfrak{p}} \longrightarrow (M/xM)_{\mathfrak{p}} \longrightarrow 0 \tag{7.32}$$

is exact. Since $\mathfrak{p} \in V(x)$, we know that $(xM)_{\mathfrak{p}} \neq M_{\mathfrak{p}}$. Since $\mathfrak{p} \in \operatorname{Supp} M$, $M_{\mathfrak{p}} \neq 0$. By the exactness of (7.32), we know that $(M/xM)_{\mathfrak{p}} \neq 0$, which implies that $\mathfrak{p} \in \operatorname{Supp}(M/xM)$, i.e., $(\operatorname{Supp} M)\cap V(x) \subseteq \operatorname{Supp}(M/xM)$.

By above discussion, we have $\operatorname{Supp}(M/xM) = (\operatorname{Supp} M) \cap V(x)$. □

7.6.3 Definition: Associated points and associated primes

Definition 7.6.2 (Associated points and associated primes)

Define the ***associated prime ideals*** *of an A-module M to be those prime ideals of A of the form $\operatorname{Ann}_A(m)$ for some $m \in M$.*

Define the ***associated points*** *of M to be the corresponding points of $\operatorname{Spec} A$; we use the terminology "associated points" and "associated primes" interchangeably.*

The set of associated points is denoted $\operatorname{Ass}_A M \subseteq \operatorname{Spec} A$. The subscript A is dropped if it is clear from the context.

Definition 7.6.3 (Associated primes of a ring)

The ***associated primes of a ring*** *A are the associated primes of A considered as an A-module (i.e., $M = A$ in Definition 7.6.2)*

Proposition 7.6.5 (Associated points of integral domains)

If A is an integral domain, then $\operatorname{Ass} A = \{[(0)]\}$ — the zero ideal is the only associated prime.

Proof Let $\operatorname{Ann}(a) \in \operatorname{Ass} A$, then $\operatorname{Ann}(a) \in \operatorname{Spec} A$ and $\operatorname{Ann}(a) \cdot a = 0$. Since A is an integral domain, A has nonzero divisor and $(0) \in \operatorname{Spec} A$, we have $\operatorname{Ann}(a) = (0)$, as we desired. $\square$

Proposition 7.6.6 (Associated points of hypersurfaces)

Given $f \in k[x_1, \dots, x_n]$, the associated primes of $k[x_1, \dots, x_n]/(f)$ are those principal ideals generated by the prime factors of f.

Proof Say $A = k[x_1, \dots, x_n]$ and $R = A/(f)$. Since A is UFD, we may assume that $f = f_1^{e_1} f_2^{e_2} \cdots f_n^{e_n}$ where each f_i is the prime factors of f. Let $\mathfrak{p} \in \operatorname{Ass} R$, then $\mathfrak{p} = \operatorname{Ann}(\overline{h})$ where $\overline{h}$ is the image of $h \in A$ in R. Say $\mathfrak{q} \supseteq (f)$ be the corresponding prime ideal of $\mathfrak{p}$ in A, in fact,

$$\mathfrak{q} = \{g \in A : gh \in (f)\}.$$

Since $f \in \mathfrak{q}$, exists f_i such that $f_i \in \mathfrak{q}$, then $f_i h \in (f)$. It follows that exists $c \in A$ such that $f_i h = cf$, then

$$h = cf_1^{e_1} \cdots f_i^{e_i - 1} \cdots f_n^{e_n},$$

where $f_i \nmid h$. Now, we calculate $\operatorname{Ann}(\overline{h})$,

$$\begin{aligned} \operatorname{Ann}(\overline{h}) =& \{\overline{g} \in R : \overline{g} \cdot \overline{h} = 0\} \\ =& \{g \in A : gh = bf \text{ for some } b \in A\} \\ =& \{g \in A : cg = bf_i \text{ for some } b \in A\}. \end{aligned}$$

Since $f_i \nmid h$, $c \mid b$, hence $g = b' f_i$, which implies that $\operatorname{Ann}(\overline{h}) = (\overline{f_i}) = (f_i)$, as we desired. $\square$

Remark 7.20 Above argument will apply more generally to any $f \in A$ where A is a UFD.

Lemma 7.6.1

Suppose M is an A-module, $[\mathfrak{p}] \in \operatorname{Ass}_A(M)$ if and only if there is an injection $A/\mathfrak{p} \hookrightarrow M$ of A-modules.

Proof If $[\mathfrak{p}] \in \operatorname{Ass}_A(M)$, then $\mathfrak{p} = \operatorname{Ann}(m)$ for some $m \in M$. Consider homomorphism $A \xrightarrow{\times m} Am$, then $A/\operatorname{Ann}(m) \cong Am \hookrightarrow M$.

Conversely, say $i : A/\mathfrak{p} \hookrightarrow M$ and $i(1) = m$. Consider $\times m : A \to Am$, we want to show that $A/\mathfrak{p} \cong Am$.

Clearly, $A/\operatorname{Ann}_A(m) \cong Am$, it suffices to show that $\mathfrak{p} = \operatorname{Ann}_A(m)$. Note that

$$\operatorname{Ann}_A(m) = \{a \in A : am = 0\} = \{a \in A : i(a) = 0\} = \mathfrak{p},$$

as we desired. □

Proposition 7.6.7

Let M and M' are both A-module.

(a) Suppose $M' \subseteq M$, then $\operatorname{Ass} M' \subseteq \operatorname{Ass} M$.

(b) $\operatorname{Ass} M \subseteq \operatorname{Supp} M$.

Proof

(a) Let $[\mathfrak{p}] \in \operatorname{Ass}_A(M')$, by Lemma 7.6.1, there is an injection $A/\mathfrak{p} \hookrightarrow M'$ of A-module. Since $M' \subseteq M$, then $A/\mathfrak{p} \hookrightarrow M$, by Lemma 7.6.1 again, $[\mathfrak{p}] \in \operatorname{Ass}_A(M)$. (The corresponding statement for "support" is implicit in Proposition 7.6.2 (a).)

(b) Let $[\mathfrak{p}] \in \operatorname{Ass} M$, then we have an exact sequence.

$$0 \longrightarrow A/\mathfrak{p} \hookrightarrow M$$

Since localization is exact, we have

$$0 \longrightarrow (A/\mathfrak{p})_{\mathfrak{p}} \hookrightarrow M_{\mathfrak{p}}.$$

Since $(A/\mathfrak{p})_{\mathfrak{p}} = A_{\mathfrak{p}}/\mathfrak{p}A_{\mathfrak{p}}$ and $\mathfrak{p}A_{\mathfrak{p}}$ is the maximal ideal of $A_{\mathfrak{p}}$, $(A/\mathfrak{p})_{\mathfrak{p}} \neq 0$, and therefore $M_{\mathfrak{p}} \neq 0$. it follows that $[\mathfrak{p}] \in \operatorname{Supp} M$.

□

If M is finitely generated, then $\operatorname{Supp} M$ is closed (Proposition 7.6.2), so $\overline{\operatorname{Ass} M} \subseteq \operatorname{Supp} M$. Equality will be shown later, when A is Noetherian.

7.6.4 Nonzero modules over Noetherian rings have associated points

Suppose m is a nonzero element of an A-module M. Observe that for any nonzero multiple of xm of m, $\operatorname{Ann} m \subseteq \operatorname{Ann} xm \subsetneq A$.

Proposition 7.6.8

Suppose A is Noetherian, then there is some multiple $n = xm$ such that any nonzero multiple $yn \neq 0$ of n satisfies $\operatorname{Ann} yn = \operatorname{Ann} n$. Moreover, $\operatorname{Ann} n$ is prime ideal.

Proof Clearly, we have $\operatorname{Ann}(m) \subseteq \operatorname{Ann}(xm)$ for all $x \in A$. Then we get a chain

$$\operatorname{Ann}(m) \subseteq \operatorname{Ann}(x_1 m) \subseteq \operatorname{Ann}(x_2 m) \subseteq \cdots \subseteq \operatorname{Ann}(x_k m) \subseteq \cdots.$$

Since A is Noetherian ring, above chain must be stationary, hence exists x_n such that $\operatorname{Ann}(x_n m) = \operatorname{Ann}(x_{n+1} m) = \cdots$. Let $n = x_n m$, then $\operatorname{Ann}(n) = \operatorname{Ann}(yn)$ for any nonzero multiple $yn \neq 0$.

Next, we want to show that $\operatorname{Ann}(n)$ is prime ideal. Suppose $ab \in \operatorname{Ann}(n)$, so $abn = 0$. Then either $bn = 0$ (in which case $b \in \operatorname{Ann}(b)$), or else $a \in \operatorname{Ann}(bn) = \operatorname{Ann}(n)$ by above discussion. Hence $\operatorname{Ann}(n)$ is prime ideal. □

We have thus proved the following.

Proposition 7.6.9 (Nonzero modules over Noetherian rings have associated primes)

If M is a nonzero module over a Noetherian ring A, then $\operatorname{Ass}_A M$ is nonempty. More precisely, for any $m \neq 0$ in M, there is an associated prime $\mathfrak{p}$ containing $\operatorname{Ann} m$, and $\mathfrak{p} = \operatorname{Ann} xm$ for some $x \in A$.

7.6.5 Localizations at the associated primes

Recall the useful fact that $M \to \prod_{\mathfrak{p} \in \operatorname{Spec} A} M_{\mathfrak{p}}$ is an injection (Proposition 5.1.3). Our current situation is much better: we can take the product over only the localization at associated primes.

Proposition 7.6.10

Suppose M is a module over a Noetherian ring A. Then the natural map

$$M \longrightarrow \prod_{\mathfrak{p} \in \operatorname{Ass} M} M_{\mathfrak{p}} \tag{7.33}$$

is an injection.

Proof If the kernel K is nonzero, by Proposition 7.6.9, K has an associated prime $\mathfrak{p}$, which is the annihilator of some $m \in K \subseteq M$, i.e., $\mathfrak{p} = \operatorname{Ann} m$. Consider $M_{\mathfrak{p}} = A_{\mathfrak{p}} \otimes_A M$, $m = 1 \otimes m \neq 0$ in $M_{\mathfrak{p}}$, a contradiction. □

Remark 7.21 Clearly we need only the maximal among the associated primes in (7.33).

7.6.6 Zerodivisors = elements of associated primes

Proposition 7.6.11

Suppose $f \in A$, with A Noetherian. Then f a zerodivisor on M if and only if f vanishes at an associated point of M. Translation: the set of zerodivisors is the union of the associated prime ideals.

Remark 7.22 Again, we need only the maximal among the associated primes. For example, if $(A, \mathfrak{m})$ is a local ring, then $\mathfrak{m}$ is an associated prime if and only if every element of $\mathfrak{m}$ is a zerodivisor.

Proof If f vanishes at an associated point of M, then $f \in \operatorname{Ann}(m)$ for some $m \in M$, hence $fm = 0$, i.e., f a zerodivisor on M.

Conversely, if f vanishes at no associated point, i.e., $f \notin \mathfrak{p}$ for all $\mathfrak{p} \in \operatorname{Ass} M$. Consider the commuting diagram

$$\begin{tikzcd} M \arrow[r, hook] \arrow[d, "\times f"'] & \prod_{\mathfrak{p} \in \operatorname{Ass} M} M_{\mathfrak{p}} \arrow[d, "\times f"] \\ M \arrow[r, hook] & \prod_{\mathfrak{p} \in \operatorname{Ass} M} M_{\mathfrak{p}} \end{tikzcd}$$

where the rows are the maps of (7.33). The vertical arrow on the right (multiplication by f) is an injection by hypothesis, so the vertical arrow on the left must be an injection, hence f is not a zerodivisor on M, a contradiction. □

7.6.7 Associated points behave fairly well in exact sequences

Proposition 7.6.12

Suppose

$$0 \longrightarrow M' \longrightarrow M \longrightarrow M'' \longrightarrow 0 \tag{7.34}$$

is a short exact sequence of A-modules. Then

$$\operatorname{Ass} M' \subseteq \operatorname{Ass} M \subseteq \operatorname{Ass} M' \cup \operatorname{Ass} M''. \tag{7.35}$$

Proof The first inclusion of (7.35) was shown in Proposition 7.6.7 (a).

Suppose $[\mathfrak{p}] \in \operatorname{Ass} M$, so there is some $m \in M$ with $\operatorname{Ann} m = \mathfrak{p}$, then $Am \cong A/\mathfrak{p}$. We wish to find a submodule of M' or M'' isomorphic to $A/\mathfrak{p}$ (Lemma 7.6.1). If this proposition were true, we would expect to find such a submodule in the "part of (7.34) spanned by m". So we consider instead the exact sequence

$$0 \longrightarrow Am \cap M' \longrightarrow Am \longrightarrow Am/(Am \cap M') \longrightarrow 0,$$

noting that the three modules appearing here are submodules of the corresponding modules in (7.34). So by Proposition 7.6.7 it suffices to prove the result in this "special case", which can be rewritten as ($Am \cong A/\mathfrak{p}$)

$$0 \longrightarrow I/\mathfrak{p} \longrightarrow A/\mathfrak{p} \longrightarrow A/I \longrightarrow 0$$

where I is the annihilator of m considered as an element of the module $Am/(Am \cap M')$. For convenience, let $B = A/\mathfrak{p}$ (an integral domain) and $J = I/\mathfrak{p}$, so we rewrite the exact sequence further as the top row of

$$\begin{tikzcd}
0 \arrow[r] & J \arrow[r] \arrow[d, hook] & B \arrow[r] \arrow[d, hook] & B/J \arrow[r] \arrow[d, hook] & 0 \\
0 \arrow[r] & M' \arrow[r] & M \arrow[r] & M'' \arrow[r] & 0.
\end{tikzcd}$$

Now localize the top row of B-modules at $(0) \subseteq B$, so it becomes an exact sequence of vector spaces over the fraction field $K(B)$, and the central element is one-dimensional:

$$0 \longrightarrow J \otimes K(B) \longrightarrow K(B) \longrightarrow (B/J) \otimes K(B) \longrightarrow 0.$$

Thus one of the outside terms $J \otimes K(B)$ and $(B/J) \otimes K(B)$ has a nonzero element.

If $\dim_{K(B)} J \otimes_B K(B) = 1$, then exists $x \in J$ such that $\operatorname{Ann}_B(x) = 0$. (if not, for all $x \in J$, $\operatorname{Ann}_B(x) \neq 0$. Let $x \otimes b \in J \otimes_B K(B)$ and $c \in \operatorname{Ann}_B(x)$, then $x \otimes_B b = cx \otimes (b/c) = 0$, a contradiction.) Note that $B = A/\mathfrak{p}$, then $\operatorname{Ann}_A(x) = \mathfrak{p}$, which implies that $\mathfrak{p} \in \operatorname{Ass} J \subseteq \operatorname{Ass} M'$.

If $\dim_{K(B)}(B/J) \otimes K(B) = 1$, then exists $y \in B/J$ such that $\operatorname{Ann}_B(y) = 0$. Note that $B = A/\mathfrak{p}$, then $\operatorname{Ann}_A(y) = \mathfrak{p}$, which implies that $\mathfrak{p} \in \operatorname{Ass} B/J \subseteq \operatorname{Ass} M''$.

Hence, $\mathfrak{p} \in \operatorname{Ass} M' \cup \operatorname{Ass} M''$, and therefore $\operatorname{Ass} M \subseteq \operatorname{Ass} M' \cup \operatorname{Ass} M''$. □

Example 7.6 **Cautionary example.** The short exact sequence of $\mathbb{Z}$-modules

$$0 \longrightarrow \mathbb{Z} \xrightarrow{\times 2} \mathbb{Z} \longrightarrow \mathbb{Z}/2 \longrightarrow 0.$$

Note that $\operatorname{Ass} \mathbb{Z} = \{(0)\}$ and $\operatorname{Ass}(\mathbb{Z}/2) = \{(2)\}$, in this case $\operatorname{Ass} M \subsetneq M = \operatorname{Ass} M' \cup \operatorname{Ass} M''$. However, sometimes we can still ensure some associated primes of M'' lift to associated primes of M.

7.6.8 Finitely generated modules over Noetherian rings have finitely many associated points (primes)

Proposition 7.6.13

Suppose that M is a finitely generated module over a Noetherian ring A. Then M has a finite filtration

$$0 = M_0 \subseteq M_1 \subseteq \cdots \subseteq M_n = M \tag{7.36}$$

where $M_{i+1}/M_i \cong A/\mathfrak{p}_i$ for some prime $\mathfrak{p}_i$. Also, every associated prime of M appears as one of the $\mathfrak{p}_i$.

Proof Since M is finitely generated module over a Noetherian ring A, M is Noetherian module. We proceed by induction on n. The $n = 0$ case is trivial. Assume we have chosen $0 = M_0 \subseteq \cdots \subseteq M_{k-1}$ with $M_i/M_{i-1} \cong A/\mathfrak{p}_i$ where $\mathfrak{p}_i \in \operatorname{Spec} A$. If $M_{k-1} \neq M$, then $M/M_{k-1} \neq 0$. Since M is Noetherian, M/M_{k-1} is Noetherian, by Proposition 7.6.9, $\operatorname{Ass}_A(M/M_{k-1})$ is not empty, so exists $\mathfrak{p}_k \in \operatorname{Ass}_A(M/M_{k-1}) \subseteq \operatorname{Spec} A$. By Lemma 7.6.1, $A/\mathfrak{p}_k \hookrightarrow M/M_{k-1}$. Pick a submodule of M, named M_k, such that $M_k/M_{k-1} \cong A/\mathfrak{p}_k$, then we get a chain in M:

$$0 = M_0 \subseteq M_1 \subseteq \cdots \subseteq M_{k-1} \subseteq M_k,$$

where $M_i/M_{i-1} \cong A/\mathfrak{p}_i$. Repeat this procedure, which terminates after a finite number of steps, since M is Noetherian.

By Proposition 7.6.12, from chain (7.36), we have

$$\begin{aligned}
\operatorname{Ass} M &\subseteq \operatorname{Ass}(M_{n-1}) \cup \operatorname{Ass}(M/M_{n-1}) \\
&\subseteq \operatorname{Ass}(M_{n-2}) \cup \operatorname{Ass}(M_{n-1}/M_{n-2}) \cup \operatorname{Ass}(M/M_{n-1}) \\
&\subseteq \cdots \subseteq \bigcup_{i=1}^{n} \operatorname{Ass}(M_i/M_{i-1}) \\
&= \bigcup_{i=1}^{n} \operatorname{Ass}(A/\mathfrak{p}_i).
\end{aligned}$$

We now calculate $\operatorname{Ass}(A/\mathfrak{p}_i)$. Let $\mathfrak{q} \in \operatorname{Ass}(A/\mathfrak{p}_i)$, then exists nonzero $a \in A/\mathfrak{p}_i$ such that $\mathfrak{q} = \operatorname{Ann}_A(a)$. $\mathfrak{q}$ can be seen as an ideal of A which contains $\mathfrak{p}_i$. Hence $\mathfrak{q} \cdot a \subseteq \mathfrak{p}_i$, since $\mathfrak{p}_i$ is prime, $\mathfrak{q} \subseteq \mathfrak{p}_i$, and therefore $\mathfrak{q} = \mathfrak{p}_i$. It follows that $\operatorname{Ass}(A/\mathfrak{p}_i) = \{\mathfrak{p}_i\}$. Hence $\bigcup_{i=1}^{n} \operatorname{Ass}(A/\mathfrak{p}_i) = \{\mathfrak{p}_1, \ldots, \mathfrak{p}_n\}$, and therefore $\operatorname{Ass} M \subseteq \{\mathfrak{p}_1, \ldots, \mathfrak{p}_n\}$, as we desired. □

Corollary 7.6.1

Suppose an A-module M has a finite filtration (7.36), with no assumptions of finite generation or Noetherianity. Then every associated prime of M appears as one of the $\mathfrak{p}_i$.

Proof Recall the proof of Proposition 7.6.13, we don't use assumptions of finite generation or Noetherianity, hence the proof works in this Corollary, then we done. □

Corollary 7.6.2

If M is finitely generated over a Noetherian ring, M has finitely many associated points (primes).

Proof Proposition 7.6.13. □

Remark 7.23 **Caution: Non-associated prime ideals may unavoidably appear among the quotients in (7.36).** Example 7.6 shows that the non-associated prime ideals may be among the quotients in (7.36), although in that case it is because the filtration was chosen unwisely. But better choices will not always remedy the problem:

Example 7.7 Consider the module $M = (x, y) \subseteq A = k[x, y]$ over the ring A. Then any filtration (7.36) of M contains a quotient $A/\mathfrak{p}_i$ where $\mathfrak{p}_i$ is not an associated prime.

Proof We first calculate $\operatorname{Ass} M$. Let nonzero element $m \in M$, consider $\operatorname{Ann}(m) \subseteq A$. Let $a \in \operatorname{Ann}(m)$, then $am = 0$. Since $k[x, y]$ is integral domain and $m \neq 0$, $a = 0$. Hence $\operatorname{Ann}(m) = (0)$ for all nonzero $m \in M$. Since $k[x, y]$ is integral domain, $(0) \in \operatorname{Spec} A$, and therefore $\operatorname{Ass}(M) = \{(0)\}$.

Let

$$0 = M_0 \subseteq M_1 \subseteq \cdots \subseteq M_n = M$$

be any filtration of M, where $M_i/M_{i-1} \cong A/\mathfrak{p}_i$. If for any quotient $A/\mathfrak{p}_i$ prime $\mathfrak{p}_i$ is associated point, we have $A/\mathfrak{p}_i = A/(0) = A$. Hence $M_1/M_0 = M_1 \cong A$. But $M_n = (x, y) \subsetneq A$, a contradiction. it follows that any filtration of M contains a quotient $A/\mathfrak{p}_i$ where $\mathfrak{p}_i$ is not an associated prime. $\square$

Remark 7.24 However, not all is lost: Proposition 7.6.19 will show that for any quotient $A/\mathfrak{p}_i$ in any filtration (7.36) of M, $\mathfrak{p}_i$ must contain an associated prime of M.

7.6.9 Associated points behave fairly well in exact sequences, continued

Proposition 7.6.14

Suppose A is a Noetherian ring. Consider the short exact sequence

$$0 \longrightarrow M' \longrightarrow M \longrightarrow M'' \longrightarrow 0$$

of A-modules. Suppose $\mathfrak{p} \in \operatorname{Ass} M''$, but $\mathfrak{p} \notin \operatorname{Supp} M'$ (a stronger hypothesis than $\mathfrak{p} \notin \operatorname{Ass} M'$). Then $\mathfrak{p} \in \operatorname{Ass} M$.

Proof Let $\mathfrak{p} \in \operatorname{Ass} M'$, then $\mathfrak{p} = \operatorname{Ann} m''$ for some $m'' \in M''$. Choose a lift of $m \in M$ of $m'' \in M''$. Consider the "inclusion of short exact sequences"

$$\begin{tikzcd}
0 \arrow[r] & \operatorname{Ker}(Am \to Am'') \arrow[r] \arrow[d, hook] & Am \arrow[r] \arrow[d, hook] & Am'' \arrow[r] \arrow[d, hook] & 0 \\
0 \arrow[r] & M' \arrow[r] & M \arrow[r] & M'' \arrow[r] & 0.
\end{tikzcd}$$

As $\operatorname{Supp}(\operatorname{Ker}(Am \to Am'')) \subseteq \operatorname{Supp} M'$, $\operatorname{Ass}(Am) \subseteq \operatorname{Ass} M$, and $\operatorname{Ass}(Am'') \subseteq \operatorname{Ass} M''$ (Proposition 7.6.2 and Proposition 7.6.7), we have reduced to considering the top row instead of the bottom row. The top row can be conveniently rewritten as

$$0 \longrightarrow \mathfrak{p}/I \longrightarrow A/I \longrightarrow A/\mathfrak{p} \longrightarrow 0$$

(here $I = \operatorname{Ann}(m)$) where our hypothesis translates to $[\mathfrak{p}] \notin \operatorname{Supp}(\mathfrak{p}/I)$. For convenience, let $B = A/I$ so we may now consider the sequence

$$0 \longrightarrow \mathfrak{q} \longrightarrow B \longrightarrow B/\mathfrak{q} \longrightarrow 0,$$

where $\mathfrak{q} = \mathfrak{p}/I$ is prime of B, with $[\mathfrak{p}] \notin \operatorname{Supp}_A(\mathfrak{q})$. Since $B_\mathfrak{q} \cong \frac{\mathfrak{p}A_\mathfrak{p}}{IA_\mathfrak{p}}$, we have $[\mathfrak{q}] \notin \operatorname{Supp}_B \mathfrak{q}$.

From $[\mathfrak{q}] \notin \operatorname{Supp}_B \mathfrak{q}$, there is an element b of B that vanishes on $\operatorname{Supp} \mathfrak{q}$ but doesn't vanish at $[\mathfrak{q}]$. Translation: (i) b lies in all the primes of $\operatorname{Supp} \mathfrak{q}$, but (ii) $b \notin \mathfrak{q}$. Then from (i) there is some power b^n of b that annihilates all elements of $\mathfrak{q}$ (Proposition 7.6.3, which applies because $\mathfrak{q}$ is finitely generated, which in turn follows from Noetherianity of A). From (ii), $b^n \notin \mathfrak{q}$.

Then $\operatorname{Ann}(b^n)$ contains $\mathfrak{q}$ from (i), i.e., $\mathfrak{q} \subseteq \operatorname{Ann}(b^n)$. If exists nonzero $x \in \operatorname{Ann}(b^n) \setminus \mathfrak{q}$, then $xb^n = 0$. Since $\mathfrak{q}$ is prime, note that $b^n \notin \mathfrak{q}$ (from (ii)) and $x \notin \mathfrak{q}$, we have $xb^n = 0 \notin \mathfrak{p}$, a contradiction. Hence

$\mathfrak{q} = \operatorname{Ann}(b^n)$, and therefore $\mathfrak{q}$ is an associated prime of B, which (unwinding our argument) shows that $\mathfrak{p}$ is an associated prime of M. □

7.6.10 Minimal primes are associated

Proposition 7.6.15 (Minimal primes ("irreducible components") are associated)

Suppose M is a finitely generated module over Noetherian A, and $\mathfrak{p} \subseteq A$ is a prime ideal corresponding to an irreducible component of $\operatorname{Supp} M \subseteq \operatorname{Spec} A$*. Then* $[\mathfrak{p}] \in \operatorname{Ass} M$.

Proof By Proposition 7.6.13, M has a finite filtration

$$0 = M_0 \subseteq M_1 \subseteq \cdots \subseteq M_n = M$$

where $M_{i+1}/M_i \cong A/\mathfrak{p}_i$. Then we have an exact sequence

$$0 \longrightarrow M_{i-1} \longrightarrow M_i \longrightarrow A/\mathfrak{p}_i \longrightarrow 0.$$

By Proposition 7.6.2, we have $\operatorname{Supp} M_i = \operatorname{Supp} M_{i-1} \cup \operatorname{Supp}(A/\mathfrak{p}_i)$. Hence $\operatorname{Supp} M = \bigcup\limits_{i=1}^{n} \operatorname{Supp}(A/\mathfrak{p}_i)$. Consider $\operatorname{Supp}(A/\mathfrak{p}_i)$, let $\mathfrak{q} \in \operatorname{Supp}(A/\mathfrak{p}_i)$, then $\frac{A_\mathfrak{q}}{\mathfrak{p}_i A_\mathfrak{q}} \neq 0$, it follows that $\mathfrak{p}_i A_\mathfrak{q}$ must be an ideal of $A_\mathfrak{q}$, hence $\mathfrak{p}_i \subseteq \mathfrak{q}$, which implies that $\operatorname{Supp}(A/\mathfrak{p}_i) \subseteq V(\mathfrak{p}_i)$. Conversely, let $\mathfrak{q} \in V(\mathfrak{p}_i)$, then $\mathfrak{p}_i \subseteq \mathfrak{q}$. Hence, $\mathfrak{p}_i A_\mathfrak{q}$ is an ideal of $A_\mathfrak{q}$, and therefore $(A/\mathfrak{p}_i)_\mathfrak{q} = \frac{A_\mathfrak{q}}{\mathfrak{p}_i A_\mathfrak{q}} \neq 0$. It follows that $\mathfrak{q} \in \operatorname{Supp}(A/\mathfrak{p}_i)$. Hence $\operatorname{Supp}(A/\mathfrak{p}_i) = V(\mathfrak{p}_i)$, and therefore $\operatorname{Supp} M = \bigcup\limits_{i=1}^{n} V(\mathfrak{p}_i)$. Since $\mathfrak{p} \in \operatorname{Spec} A$ corresponding to an irreducible component of $\operatorname{Supp} M \subseteq \operatorname{Spec} A$, by Proposition 4.7.4, $\mathfrak{p}$ is minimal prime in $\operatorname{Supp} M$, hence $\mathfrak{p} = \mathfrak{p}_i$ for some i. Let's assume that i is taken minimally. We have an exact sequence

$$0 \longrightarrow M_{i-1} \longrightarrow M_i \longrightarrow A/\mathfrak{p} \longrightarrow 0. \tag{7.37}$$

We want to show: (i) $\mathfrak{p} \in \operatorname{Ass}(A/\mathfrak{p})$; (ii) $\mathfrak{p} \notin \operatorname{Supp} M_{i-1}$.

For (i), let $\operatorname{Ann}(\overline{a}) \in \operatorname{Ass}(A/\mathfrak{p}) \subseteq \operatorname{Spec} A$, where $\overline{a} \in A/\mathfrak{p}$ nonzero, then $\operatorname{Ann}(\overline{a}) \cdot a \subseteq \mathfrak{p}$. Since $a \notin \mathfrak{p}$, and $\mathfrak{p}$ is prime, we have $\operatorname{Ann}(\overline{a}) \subseteq \mathfrak{p}$. Since $\mathfrak{p} \cdot a \subseteq \mathfrak{p}$, we have $\operatorname{Ann}(\overline{a}) = \mathfrak{p}$. Hence $\operatorname{Ass}(A/\mathfrak{p}) = \{\mathfrak{p}\}$. For (ii), by the assumption "i is taken minimally", we know that $\mathfrak{p} \notin \operatorname{Supp} M_{i-1}$.

By Proposition 7.6.14, $[\mathfrak{p}] \in \operatorname{Ass} M$, as we desired. □

Remark 7.25 Non-Noetherian Remark. By combining Proposition 7.6.11 with Proposition 7.6.15, we see that if A is a Noetherian ring, then any element of any minimal prime $\mathfrak{p}$ is a zerodivisor. This is true without Noetherian hypotheses: suppose $s \in \mathfrak{p}$. Then by minimality of $\mathfrak{p}$, $\mathfrak{p}A_\mathfrak{p}$ is the unique prime ideal in $A_\mathfrak{p}$, so the element $s/1$ of $A_\mathfrak{p}$ is nilpotent (because it is contained in all prime ideals of $A_\mathfrak{p}$, Theorem 4.2.3). Thus for some $t \in A \setminus \mathfrak{p}$, $ts^n = 0$, so s is a zerodivisor in A. We will use this in Chapter 13.

Proposition 7.6.16

Suppose M is a finitely generated module over a Noetherian ring A. Then $\operatorname{Supp} M = \overline{\operatorname{Ass} M}$.

Proof By Proposition 7.6.2 (b), $\operatorname{Supp} M$ is a closed subset of $\operatorname{Spec} A$. By Proposition 7.6.7 (b) and , we have $\overline{\operatorname{Ass} M} \subseteq \operatorname{Supp} M$. By Corollary 7.6.2, M has finitely many associated points, say $\operatorname{Ass} M = \{\mathfrak{p}_1, \dots, \mathfrak{p}_n\}$, hence

$$\overline{\operatorname{Ass} M} = \overline{\{\mathfrak{p}_1, \dots, \mathfrak{p}_n\}} = \bigcup_{i=1}^{n} \overline{[\mathfrak{p}_i]}.$$

Let $\mathfrak{q} \in \operatorname{Supp} M$, then $\mathfrak{q}$ must belong to an irreducible component of M, by Proposition 7.6.15, the corresponding minimal ideal of irreducible component is associated ideal, which implies that $\mathfrak{q} \in \overline{[\mathfrak{p}_i]}$ for some i. Hence $\mathfrak{q} \in \overline{\operatorname{Ass} M}$. Hence $\operatorname{Supp} M = \overline{\operatorname{Ass} M}$. $\square$

Proposition 7.6.17

The locus of points $[\mathfrak{p}]$ where $A_{\mathfrak{p}}$ is nonreduced (Definition 4.2.3) is the support of the nilradical $\operatorname{Supp} \mathfrak{N}$. Hence the "reduced locus" of a locally Noetherian scheme is open.

Proof Let $\mathfrak{R} = \{[\mathfrak{p}] \in \operatorname{Spec} A : A_{\mathfrak{p}} \text{ is nonreduced}\}$. Let $[\mathfrak{p}] \in \mathfrak{R}$, then $A_{\mathfrak{p}}$ has nonzero nilpotents, which implies that $\mathfrak{N}(A_{\mathfrak{p}}) = \mathfrak{N}(A)_{\mathfrak{p}} \neq 0$. Hence $[\mathfrak{p}] \in \operatorname{Supp} \mathfrak{N}(A)$. Conversely, if $[\mathfrak{p}] \in \operatorname{Supp} \mathfrak{N}(A)$, then $\mathfrak{N}(A)_{\mathfrak{p}} = \mathfrak{N}(A_{\mathfrak{p}}) \neq 0$, which implies that $A_{\mathfrak{p}}$ is reduced. Hence $\mathfrak{R} = \operatorname{Supp} \mathfrak{N}(A)$.

Let X be a locally Noetherian scheme, then $X = \bigcup_i \operatorname{Spec} A_i$ where A is Noetherian. Say $\mathfrak{R} = \{[\mathfrak{p}] \in X : \mathfrak{N}(\mathscr{O}_{X,[\mathfrak{p}]}) = 0\}$ be the reduced locus of X. Note that

$$\mathfrak{R} = \bigcup_i \{[\mathfrak{p}] \in \operatorname{Spec} A_i : \mathfrak{N}((A_i)_{\mathfrak{p}}) = 0\} := \bigcup_i \mathfrak{R}_i,$$

by above discussion, we know that $\mathfrak{R}_i = \operatorname{Spec} A_i - \operatorname{Supp} \mathfrak{N}(A_i)$ which is open in $\operatorname{Spec} A_i$, and therefore is open in X. Hence $\mathfrak{R}$ is open. $\square$

The following justifies a simple way of thinking about associated primes of a ring.

Proposition 7.6.18

A prime ideal $\mathfrak{p} \subseteq A$ is an associated prime of a Noetherian ring A if and only if there is $f \in A$ such that $\operatorname{Supp} f = V(\mathfrak{p}) = \overline{[\mathfrak{p}]}$.

Proof Let $\mathfrak{p}$ be an associated prime of A, then $\mathfrak{p} = \operatorname{Ann}(f)$ for some $f \in A$. By Proposition 7.6.1 and Lemma 4.6.1, $\operatorname{Supp} f = V(\operatorname{Ann}(f)) = V(\mathfrak{p}) = \overline{[\mathfrak{p}]}$. Conversely, if there is $f \in A$ such that $\operatorname{Supp} f = V(\mathfrak{p})$. By Proposition 7.6.1, we have $\operatorname{Supp} f = V(\operatorname{Ann}(f))$. By the proof in Proposition 7.5.5, $\operatorname{Ann}(f)$ is maximal ideal of A, and therefore is prime. Hence $\mathfrak{p} = \operatorname{Ann}(f)$, which implies that $\mathfrak{p} \in \operatorname{Ass} A$. $\square$

Proposition 7.6.19

For each quotient in the filtration (7.36) of M, $\operatorname{Supp} A/\mathfrak{p}_i = \overline{[\mathfrak{p}_i]} \subseteq \operatorname{Supp} M$, and that every $\mathfrak{p}_i$ contains a minimal ideal (associated ideal).

Proof Let $\mathfrak{q} \in \operatorname{Supp} A/\mathfrak{p}_i$, then $(A/\mathfrak{p}_i)_{\mathfrak{q}} = A_{\mathfrak{q}}/\mathfrak{p}_i A_{\mathfrak{q}} \neq 0$, hence $\mathfrak{p}_i A_{\mathfrak{q}}$ must be an ideal of $A_{\mathfrak{q}}$. Hence $\mathfrak{p}_i \subseteq \mathfrak{q}$, which implies that $\operatorname{Supp}(A/\mathfrak{p}_i) = V(\mathfrak{p}_i) = \overline{[\mathfrak{p}_i]}$. Since $(M_i/M_{i-1})_{\mathfrak{p}_i} \cong (M_i)_{\mathfrak{p}_i}/(M_{i-1})_{\mathfrak{p}_i} \cong A_{\mathfrak{p}_i}/\mathfrak{p}_i A_{\mathfrak{p}_i} \neq 0$, $(M_i)_{\mathfrak{p}_i} \neq 0$, it follows that $\mathfrak{p}_i \in \operatorname{Supp} M_i \subseteq \operatorname{Supp} M$. By Proposition 7.6.2 (b), $\operatorname{Supp} M$ is closed subset of $\operatorname{Spec} A$, hence $\overline{[\mathfrak{p}_i]} \subseteq \operatorname{Supp} M$. Since $\mathfrak{p}_i \in \operatorname{Supp} M$, $\mathfrak{p}_i$ must belong to an irreducible components of $\operatorname{Supp} M$, say $V(\mathfrak{q})$, by Proposition 7.6.15, $[\mathfrak{q}] \in \operatorname{Ass} M$. Since $\mathfrak{p}_i \in V(\mathfrak{q})$, we have $\mathfrak{q} \subseteq \mathfrak{p}_i$. Since $V(\mathfrak{q})$ is irreducible component of $\operatorname{Supp} M$, by Proposition 4.7.4, $\mathfrak{q}$ is also minimal prime in $\operatorname{Supp} M$. $\square$

7.6.11 "Support" and "associated points" commute with localization

Suppose S is a multiplicative subset of A, and $\mathfrak{p} \subseteq A$ is a prime ideal not meeting S, so (abusing notation slightly) $[\mathfrak{p}] \in \operatorname{Spec} S^{-1}A \subseteq \operatorname{Spec} A$ (Proposition 4.2.3).

Proposition 7.6.20 (Supp **commutes with localization)**

For any A-module M, $\operatorname{Supp}_A M \cap \operatorname{Spec} S^{-1}A = \operatorname{Supp}_{S^{-1}A} S^{-1}M$.

Proof Let $S^{-1}\mathfrak{p} \in \operatorname{Supp}_A M \cap \operatorname{Spec} S^{-1}A$. Consider $(S^{-1}M)_{S^{-1}\mathfrak{p}}$, since localization commutes with tensor product, we have

$$\begin{aligned}(S^{-1}M)_{S^{-1}\mathfrak{p}} &\cong (S^{-1}A)_{S^{-1}\mathfrak{p}} \otimes_{S^{-1}A} (S^{-1}A \otimes_A M) \\ &\cong ((S^{-1}A)_{S^{-1}\mathfrak{p}} \otimes_{S^{-1}A} S^{-1}A) \otimes_A M \\ &\cong (S^{-1}A)_{S^{-1}\mathfrak{p}} \otimes_A M.\end{aligned}$$

Note that $S \subseteq A - \mathfrak{p}$, the image of S in $A_\mathfrak{p}$ is also S. Hence $(S^{-1}A)_{S^{-1}\mathfrak{p}} \cong S^{-1}(A_\mathfrak{p}) \cong A_\mathfrak{p}$. It follows that

$$(S^{-1}M)_{S^{-1}\mathfrak{p}} \cong (S^{-1}A)_{S^{-1}\mathfrak{p}} \otimes_A M \cong A_\mathfrak{p} \otimes_A M \cong M_\mathfrak{p} \neq 0.$$

Hence $\operatorname{Supp}_A M \cap \operatorname{Spec} S^{-1}A \subseteq \operatorname{Supp}_{S^{-1}A} S^{-1}M$.

Conversely, let $\mathfrak{q} \in \operatorname{Supp}_{S^{-1}A} S^{-1}M$, we may assume that $\mathfrak{q} = S^{-1}\mathfrak{p}$ where $\mathfrak{p} \in \operatorname{Spec} A$. By above discussion,

$$M_\mathfrak{p} \cong (S^{-1}M)_{S^{-1}\mathfrak{p}} \neq 0,$$

which implies that $S^{-1}\mathfrak{p} \in \operatorname{Supp}_A M \cap \operatorname{Spec} S^{-1}A$. Hence $\operatorname{Supp}_A M \cap \operatorname{Spec} S^{-1}A = \operatorname{Supp}_{S^{-1}A} S^{-1}M$. □

Lemma 7.6.2

Let M be a finitely generated A-module, S is a multiplicatively closed subset of A. Then $S^{-1}(\operatorname{Ann} M) = \operatorname{Ann}(S^{-1}M)$.

Proof If this is true for two A-modules M, N. Then it is true for $M + N$:

$$\begin{aligned}S^{-1}(\operatorname{Ann}(M+N)) =& S^{-1}(\operatorname{Ann} M + \operatorname{Ann} N) \\ =& S^{-1}(\operatorname{Ann} M) \cap S^{-1}(\operatorname{Ann} N) \\ =& \operatorname{Ann}(S^{-1}M) \cap \operatorname{Ann}(S^{-1}N) \\ =& \operatorname{Ann}(S^{-1}M + S^{-1}N) \\ =& \operatorname{Ann}(S^{-1}(M+N))\end{aligned}$$

Since M is finitely generated A-module, M can be written as $M = Ax_1 + \cdots + Ax_n$. Hence it suffices to show that Lemma holds for M generated by a single element. By Proposition 7.5.5, we know that $M = Ax \cong A/\operatorname{Ann}(x)$. Hence

$$\operatorname{Ann}(S^{-1}M) = \operatorname{Ann}(S^{-1}A/S^{-1}\operatorname{Ann}(x)) = S^{-1}\operatorname{Ann}(x) = S^{-1}\operatorname{Ann}(M),$$

as we desired. □

Proposition 7.6.21 (Ass commutes with localization)

Let M be a finitely generated module over a Noetherian ring A. Then $\operatorname{Ass}_A M \cap \operatorname{Spec} S^{-1}A = \operatorname{Ass}_{S^{-1}A} S^{-1}M$.

Proof We first show that $\operatorname{Ass}_A M \cap \operatorname{Spec} S^{-1}A \subseteq \operatorname{Ass}_{S^{-1}A} S^{-1}M$. If $\mathfrak{p} \in \operatorname{Ass}_A M$, by Lemma 7.6.1, then we have an injection $A/\mathfrak{p} \hookrightarrow M$. Since $\mathfrak{p} \in \operatorname{Spec} S^{-1}A$, $\mathfrak{p} \cap S = \varnothing$, hence the image of S in $A/\mathfrak{p}$ is also S. Localizing by S (which preserves injectivity), we have $S^{-1}(A/\mathfrak{p}) \cong S^{-1}A/S^{-1}\mathfrak{p} \hookrightarrow S^{-1}M$. By Lemma 7.6.1, $S^{-1}\mathfrak{p} \in \operatorname{Ass}_{S^{-1}A} S^{-1}M$. Hence $\operatorname{Ass}_A M \cap \operatorname{Spec} S^{-1}A \subseteq \operatorname{Ass}_{S^{-1}A} S^{-1}M$.

We next show that $\operatorname{Ass}_{S^{-1}A} S^{-1}M \subseteq \operatorname{Ass}_A M \cap \operatorname{Spec} S^{-1}A$. Suppose $\mathfrak{q} := S^{-1}\mathfrak{p} \in \operatorname{Ass}_{S^{-1}A} S^{-1}M$, so $\mathfrak{q} = \operatorname{Ann}_{S^{-1}A}(m/s)$, for some $s \in S$, and $m \in M$. Since the elements of S are the units in $S^{-1}A$, we have that $\mathfrak{q} = S^{-1}\mathfrak{p} = \operatorname{Ann}_{S^{-1}A} m/1$, by Lemma 7.6.2, $\mathfrak{q} = S^{-1}(\operatorname{Ann}_A m)$, moreover, $\operatorname{Ann}_A m$ is prime. By Proposition 7.6.1, we have $V(\mathfrak{q}) = V(\operatorname{Ann}_{S^{-1}A} m/1) = \operatorname{Supp}_{S^{-1}A} m/1$. By Proposition 7.6.20,

$V(\mathfrak{q}) = \operatorname{Supp}_{S^{-1}A} m/1 = \operatorname{Supp}_A m \cap \operatorname{Spec} S^{-1}A$. Hence $\mathfrak{p} \in \operatorname{Supp}_A m$, we claim that $\mathfrak{p}$ is minimal prime in $\operatorname{Supp}_A m$. By Proposition 7.6.1, $\operatorname{Supp}_A m = V(\operatorname{Ann}_A m)$, then $\mathfrak{p} \in V(\operatorname{Ann}_A m)$. If exists $\mathfrak{p}' \subsetneq \mathfrak{p}$ in $V(\operatorname{Ann}_A m)$, then $\mathfrak{p}' \cap S \neq \varnothing$ (if not $\mathfrak{p}' \cap S = \varnothing$, then $S^{-1}\mathfrak{p} = S^{-1}(\operatorname{Ann}_A m) \subseteq S^{-1}\mathfrak{p}' \subsetneq S^{-1}\mathfrak{p}$, a contradiction). But $\mathfrak{p}' \cap S \subseteq \mathfrak{p} \cap S = \varnothing$, a contradiction. Hence $\mathfrak{p}$ is minimal prime in $\operatorname{Supp}_A m$. By Proposition 4.7.4, $\overline{[\mathfrak{p}]}$ is irreducible component of $\operatorname{Supp}_A m$. By Proposition 7.6.15, $[\mathfrak{p}] \in \operatorname{Ass} Am \subseteq \operatorname{Ass} M$, which implies that $\operatorname{Ass}_{S^{-1}A} S^{-1}M \subseteq \operatorname{Ass}_A M \cap \operatorname{Spec} S^{-1}A$. □

7.6.12 Embedded points/primes

Definition 7.6.4 (Embedded points)

The associated points that are not the generic points of irreducible components of $\operatorname{Supp} M$ *are called* ***embedded points****, and their corresponding primes are called* ***embedded primes****.*

The motivation for this language is their geometric incarnation as embedded points. For example, the scheme of Figure7.1 has three embedded points (the corresponding ring has three embedded primes).

Remark 7.26 Proposition 7.6.6 translate to "hypersurfaces in $\mathbb{A}_k^n$ have no embedded points". More generally, if A is a UFD, then $\operatorname{Spec} A/(f)$ has no embedded points for any $f \in A$. Generalizing in a different direction, we will see that "complete intersections have no embedded points".

Proposition 7.6.22

Suppose A is a reduced ring (i.e., $\mathfrak{N}(A) = 0$). Then $\operatorname{Spec} A$ *has no embedded primes.*

Proof Let $\mathfrak{p} = \operatorname{Ann} a \in \operatorname{Ass} A$ be an embedded point, then $\overline{[\mathfrak{p}]}$ is not an irreducible components of $\operatorname{Supp} A$, and therefore $\mathfrak{p}$ is not minimal prime of A (since $\operatorname{Supp} A = V(\operatorname{Ann} A) = V((0)) = \operatorname{Spec} A$). We want to show that there exists an element b of $\mathfrak{p}$ not contained in any of the minimal primes of A. It suffices to show that $\mathfrak{p} - \left(\bigcup_{\mathfrak{q}\text{ is minimal prime}} \mathfrak{q}\right) \neq \varnothing$. If not, we have $\mathfrak{p} - \left(\bigcup_{\mathfrak{q}\text{ is minimal prime}} \mathfrak{q}\right) = \varnothing$, i.e., $\mathfrak{p} \subseteq \bigcup_{\mathfrak{q}\text{ is minimal prime}} \mathfrak{q}$. By [AM18, Proposition 1.11], $\mathfrak{p} \subseteq \mathfrak{q}$ for some minimal prime $\mathfrak{q}$, a contradiction. Hence $\mathfrak{p} - \left(\bigcup_{\mathfrak{q}\text{ is minimal prime}} \mathfrak{q}\right) \neq \varnothing$. Pick $b \in \mathfrak{p} - \left(\bigcup_{\mathfrak{q}\text{ is minimal prime}} \mathfrak{q}\right)$, since $b \in \operatorname{Ann} a$, $ab = 0$. Since b not contained in any minimal primes $\mathfrak{q}$ of A, note that $ab = 0 \in \mathfrak{q}$, we have $a \in \mathfrak{p}$, which implies that $a \in \bigcap_{\mathfrak{q}\text{ is minimal prime}} \mathfrak{q}$. Since

$$\mathfrak{N}(A) = \bigcap_{\mathfrak{q}\in\operatorname{Spec} A} \mathfrak{q} = \bigcap_{\mathfrak{q}\text{ is minimal prime}} \mathfrak{q} = 0,$$

$a = 0$, and therefore $\mathfrak{p} = \operatorname{Ann} 0$, a contradiction. Hence $\mathfrak{p}$ is not an embedded point. □

Remark 7.27 Thus reduced rings have no embedded primes. In the proof of Proposition 7.6.22, if A is not reduced ring, let $\mathfrak{p} = \operatorname{Ann} a$ be an embedded point, then $a \in \mathfrak{N}(A)$. Hence, we have:

Corollary 7.6.3

The only elements of a ring that an embedded prime can annihilate are nilpotents. More precisely, if A is nonreduced ring, any embedded point $\mathfrak{p}$ is annihilator of a nilpotent, i.e., $\mathfrak{p} = \operatorname{Ann} a$ where $a \in \mathfrak{N}(A)$.

Remark 7.28 The converse to Proposition 7.6.22 is false. Rings without embedded primes can still have nilpotents — witness $k[x]/(x^2)$. Only associated points of $k[x]/(x^2)$ is $[(x)]$, note that $\overline{[(x)]} = [(x)]$, $[(x)]$ is not embedded point. But $\mathfrak{N}(k[x]/(x^2)) = (x^2) \neq 0$.

Proposition 7.6.23

If $\mathfrak{p}$ is an embedded prime of a ring A, then $A_\mathfrak{p}$ is nonreduced.

Proof If $A_\mathfrak{p}$ is reduced, by Proposition 7.6.22, $\operatorname{Spec} A_\mathfrak{p}$ has no embedded prime. In particular, $\mathfrak{p}A_\mathfrak{p}$ is minimal prime. Note that $\mathfrak{p}A_\mathfrak{p}$ is maximal ideal, $\operatorname{Spec} A_\mathfrak{p} = \{[\mathfrak{p}A_\mathfrak{p}]\}$. Hence $\mathfrak{p}$ must be minimal prime, contradicts to the fact that $\mathfrak{p}$ is an embedded prime of A. □

7.6.13 Get your hands dirty: Explicit algebraic exercises

✍ **Exercise 7.1** (see Figure 5.4)

(a) Suppose f is a function on $\operatorname{Spec} k[x,y]/(y^2,xy)$, i.e., $f \in k[x,y]/(y^2,xy)$. Show that $\operatorname{Supp} f$ is either the empty set, or the origin, or the entire space. Hence find the associated points of $\operatorname{Spec} k[x,y]/(y^2,xy)$.

(b) Show explicitly by hand that $f \in k[x,y]/(y^2,xy)$ is a zerodivisor if and only if $f(0,0)=0$.

Proof

(a) By Proposition 7.6.1, we have $\operatorname{Supp} f = V(\operatorname{Ann} f)$. Since $f \in k[x,y]/(y^2,xy)$, we may assume that $f = p + qy$ where $p \in k[x]$ and $q \in k$. Now, we calculate $\operatorname{Supp} f$,

(i) if $f=0$, then $\operatorname{Ann} f = k[x,y]/(y^2,xy)$, hence $\operatorname{Supp} f = V\left(k[x,y]/(y^2,xy)\right) = \varnothing$;

(ii) if $f = p + qy \neq 0$. Let $g \in \operatorname{Ann} f$, say $g = a + by$, then

$$fg = (p+qy)(a+by) = ap + (aq+bp)y = 0$$

in $k[x,y]/(y^2,xy)$. Hence $ap = 0$ and $aq + bp \in (x,y)$.

If $p = 0$, then $fg = aqy = 0$, since $q \in k$, we have $a \in (x)$. Hence

$$\operatorname{Ann} f = \{a + by \in k[x,y]/(y^2,xy) : a \in (x), b \in k\} = (x,y).$$

We have $\operatorname{Supp} f = V((x,y)) = (x,y)$.

If $p \neq 0$, then $a = 0$, hence $bpy = 0$. If $p \in (x)$, then $\operatorname{Ann}(f) = (y)$; if $p \notin (x)$, then $b = 0$, which implies that $\operatorname{Ann} f = (0)$.

Hence $\operatorname{Supp} f = V((0)) = \operatorname{Spec} k[x,y]/(y^2,xy)$ or $\operatorname{Supp} f = V((y)) = \operatorname{Spec} k[x,y]/(y^2,xy)$.

By above discussion, we know that $\operatorname{Ass} k[x,y]/(y^2,xy) = \{(y),(x,y)\}$.

(b) If f is zerodivisor, by Proposition 7.6.11, f vanishes at $[(y)]$ or $[(x,y)]$, i.e., $f \in (y)$ or $f \in (x,y)$. Hence $f(0,0) = 0$.

Conversely, if $f(0,0) = 0$. We may write $f = p(x) + qy$ where $q \in k$. Hence $f(0,0) = p(0) = 0$, it follows that $p(x) = xp'(x)$. Pick $y \in k[x,y]/(y^2,xy)$, then

$$yf = xyp'(x) + qy^2 = 0$$

in $k[x,y]/(y^2,xy)$, which implies that f is zerodivisor.

□

✍ **Exercise 7.2** Suppose $X = \operatorname{Spec} \mathbb{C}[x,y]/I$, and that the associated points of X are $[(y-x^2)]$, $[(x-1,y-1)]$, and $[(x-2,y-2)]$. (Exercise 7.2 will verify that such an X actually exists.)

(a) Sketch X as a subset of $\mathbb{A}^2_\mathbb{C} = \operatorname{Spec} \mathbb{C}[x,y]$, including fuzz.

(b) Do you have enough information to know if X is reduced?

(c) Do you have enough information to know if $x+y-2$ is a zerodivisor? How about $x+y-3$? How about $y-x^2$?

(d) Let $I = (y-x^2)^3 \cap (x-1, y-1)^{15} \cap (x-2, y-2)$. Show that $X = \operatorname{Spec} \mathbb{C}[x,y]/I$ satisfies the hypotheses of this exercise. (Rhetorical question: Is there a "smaller" example? Is there a "smallest"?)

Solution

(a) By the condition, we know that $\operatorname{Ass} \mathbb{C}[x,y]/I = \{[(y-x^2)], [(x-1,y-1)], [(x-2,y-2)]\}$. *By Proposition 7.6.16, we have*

$$\begin{aligned}\operatorname{Supp} \mathbb{C}[x,y]/I =& \overline{\{[(y-x^2)], [(x-1,y-1)], [(x-2,y-2)]\}} \\ =& \overline{[(y-x^2)]} \cup \overline{[(x-1,y-1)]} \cup \overline{[(x-2,y-2)]} \\ =& \overline{[(y-x^2)]} \cup \{[(x-2,y-2)]\},\end{aligned}$$

it follows that $\overline{[(y-x^2)]}$ *and* $\{[(x-2,y-2)]\}$ *are two irreducible component of* $\operatorname{Supp} \mathbb{C}[x,y]/I$. *Hence* $[(x-1,y-1)]$ *is the only embedded points of* X. *By Proposition 7.6.23,* $(\mathbb{C}[x,y]/I)_{(x-1,y-1)}$ *is nonreduced. Hence we draw fuzz at* $[(x-1,y-1)]$.

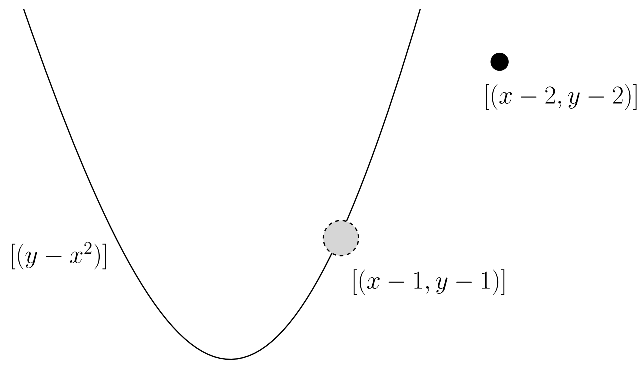


Figure 7.2: Picture of X

(b) If X *is reduced, by Proposition 6.2.2,* $\mathbb{C}[x,y]/I$ *is reduced ring, by Proposition 7.6.22,* X *has no embedded primes, contradicts to the fact that* $[(x-1,y-1)]$ *is embedded prime. Hence* X *is nonreduced.*

(c) By Proposition 7.6.11, $f \in \Gamma(X, \mathscr{O}_X)$ *is a zerodivisor if and only if* f *vanished at an associated point of* X. *Note that* $x+y-2 = x-1+y-1 \in (x-1,y-1)$, $y-x^2 \in (y-x^2)$, $x+y-2$ *and* $y-x^2$ *are zerodivisors of* $\Gamma(X, \mathscr{O}_X)$. *But* $x+y-3 \notin (x-1,y-1) \cup (x-2,y-2) \cup (y-x^2)$, $x+y-3$ *not zerodivisor.*

(d) Let $\mathfrak{p}_1 = (y-x^2)$, $\mathfrak{p}_2 = (x-1,y-1)$, $\mathfrak{p}_3 = (x-2,y-2)$, *then* $I = \mathfrak{p}_1^3 \cap \mathfrak{p}_2^{15} \cap \mathfrak{p}_3$. *Hence, in* $\mathbb{C}[x,y]/I$, *we have* $(0) = \mathfrak{p}_1^3 \cap \mathfrak{p}_2^{15} \cap \mathfrak{p}_3$, *it is a minimal primary decomposition of* (0) *in* $\mathbb{C}[x,y]/I$. *By* 1*st uniqueness theorem (see [AM18, Theorem 4.5]) and [AM18, Proposition 7.17] (note that* $\mathbb{C}[x,y]/I$ *is indeed Noetherian ring), each* $\mathfrak{p}_i$ *is of the form* $\operatorname{Ann}(f_i)$ *where* $f_i \in \mathbb{C}[x,y]/I$. *Hence* $\mathfrak{p}_i$ *is associated prime.*

$I' = \mathfrak{p}_1^2 \cap \mathfrak{p}_2 \cap \mathfrak{p}_3$ *is another example. But there is no smallest* I, *since* $I'' = \mathfrak{p}_1 \cap \mathfrak{p}_2^2 \cap \mathfrak{p}_3$ *also holds the hypotheses.*

7.6.14 Geometric definitions

Associated points

Definition 7.6.5 (Associated points of quasi-coherent sheaf)

Let X be a scheme. Let $\mathscr{F}$ be a quasi-coherent sheaf on X.

(1) *We say $x \in X$ is associated to $\mathscr{F}$ if the maximal ideal $\mathfrak{m}_x$ is associated to the $\mathscr{O}_{X,x}$-module $\mathscr{F}_x$, i.e., $\mathfrak{m}_x = \operatorname{Ann}(f)$ for some $f \in \mathscr{F}_x$.*

(2) *We denote $\operatorname{Ass}(\mathscr{F})$ or $\operatorname{Ass}_X(\mathscr{F})$ the set of associated points of $\mathscr{F}$.*

(3) *The associated points of X are the associated points of $\mathscr{O}_X$.*

Remark 7.29 These definitions are most useful when X is locally Noetherian and $\mathscr{F}$ of finite type. For example it may happen that a generic point of an irreducible component of X is not associated to X, see [Sta25, Example 05AI]. In the non-Noetherian case it may be more convenient to use weakly associated points, see [Sta25, Section 056K].

Let us link the scheme theoretic notion with the algebraic notion on affine opens; note that this correspondence works perfectly only for locally Noetherian schemes.

Proposition 7.6.24

Let X be a scheme. Let $\mathscr{F}$ be a quasi-coherent sheaf on X. Let $\operatorname{Spec}(A) = U \hookrightarrow X$ be an affine open, and set $M = \Gamma(U, \mathscr{F})$. Let $x \in U$, and let $\mathfrak{p} \in \operatorname{Spec} A$ be the corresponding prime.

(1) *If $\mathfrak{p}$ is associated to M, then x is associated to $\mathscr{F}$.*

(2) *If $\mathfrak{p}$ is finitely generated, then the converse holds as well.*

In particular, if X is locally Noetherian, then the equivalence

$$\mathfrak{p} \in \operatorname{Ass}(M) \Longleftrightarrow x \in \operatorname{Ass}(\mathscr{F})$$

holds for all pairs $(\mathfrak{p}, x)$ as above.

Proof If $\mathfrak{p}$ is associated to M, then $\mathfrak{p} = \operatorname{Ann}(m) = \operatorname{Ann}(Am)$. By Lemma 7.6.2, we have $\mathfrak{p}A_{\mathfrak{p}} = \operatorname{Ann}_A(Am)_{\mathfrak{p}} = \operatorname{Ann}_{A_{\mathfrak{p}}}(A_{\mathfrak{p}} \otimes_A Am) = \operatorname{Ann}_{A_{\mathfrak{p}}}(m/1)$. Note that $\mathfrak{p}A_{\mathfrak{p}}$ is the maximal ideal of $A_{\mathfrak{p}}$, $\mathfrak{p}A_{\mathfrak{p}} \in \operatorname{Ass}(M_{\mathfrak{p}}) \cong \operatorname{Ass}(\mathscr{F}_x)$, which implies that x is associated to $\mathscr{F}$.

Conversely, assume that x is associated to $\mathscr{F}$, and $\mathfrak{p}$ is finitely generated. As x is associated to $\mathscr{F}$ there exists an element $m' \in M_{\mathfrak{p}}$ such that $\mathfrak{p}A_{\mathfrak{p}} = \operatorname{Ann}_{A_{\mathfrak{p}}}(m')$. Write $m' = m/f$ for some $f \in A$, $f \notin \mathfrak{p}$. By Lemma 7.6.2, we have

$$\mathfrak{p}A_{\mathfrak{p}} = \operatorname{Ann}_{A_{\mathfrak{p}}}(A_{\mathfrak{p}}(m/f)) = \operatorname{Ann}_{A_{\mathfrak{p}}}(A_{\mathfrak{p}}(m/1)) = \operatorname{Ann}_A(m)_{\mathfrak{p}}.$$

Hence $\operatorname{Ann}_A(m) \subseteq \mathfrak{p}$, and $(\mathfrak{p}/I)_{\mathfrak{p}} = 0$. Since $\mathfrak{p}$ is finitely generated, there exists a $g \in A - \mathfrak{p}$ such that $g(\mathfrak{p}/I) = 0$. Hence $g\mathfrak{p} \subseteq I$, it is easy to check that $\mathfrak{p} = \operatorname{Ann}_A(gm)$, which implies that $\mathfrak{p} \in \operatorname{Ass}(M)$.

If X is locally Noetherian, then A is Noetherian and $\mathfrak{p}$ is always finitely generated. □

Proposition 7.6.25

Let X be a locally Noetherian scheme. Let $\mathscr{F}$ be a coherent $\mathscr{O}_X$-module. Then $\operatorname{Ass}(\mathscr{F}) \cap U$ is finite for every quasi-compact open $U \subseteq X$.

Proof Since $\Gamma(U, \mathscr{F})$ is a finitely generated $\mathscr{O}_X(U)$-module over Noetherian ring $\mathscr{O}_X(U)$. By Corollary 7.6.2, $\operatorname{Ass}(\Gamma(U, \mathscr{F}))$ is finite, by Proposition 7.6.24, $\operatorname{Ass}(\mathscr{F}) \cap U = \operatorname{Ass}(\Gamma(U, \mathscr{F}))$ is finite. □

By Proposition 7.6.24 and Proposition 7.6.25, we may define the **associated points of a coherent sheaf on a locally Noetherian scheme**.

Definition 7.6.6 (Associated points of a coherent sheaf on a locally Noetherian scheme)

Let X be a locally Noetherian scheme, $\mathscr{F}$ be a coherent $\mathscr{O}_X$-module. We say $x \in X$ is ***associated*** *if $x \in \operatorname{Ass}(\Gamma(U, \mathscr{F}))$ for any affine open $U \ni x$. Denote $\operatorname{Ass}(\mathscr{F})$ the set of associated points of $\mathscr{F}$.*
The ***associated points of*** *X are the associated points of $\mathscr{O}_X$, i.e., $\operatorname{Ass}(X) := \operatorname{Ass}(\mathscr{O}_X)$.*

Proposition 7.6.26

Suppose X is a locally Noetherian scheme, and $U \hookrightarrow X$ is an open subscheme. Then the natural map

$$\Gamma(U, \mathscr{O}_X) \longrightarrow \prod_{p \in \operatorname{Ass} X \cap U} \mathscr{O}_{X,p} \tag{7.38}$$

is an injection.

Proof Since $U \hookrightarrow X$ is an open subscheme, we may assume that $U = \bigcup_i \operatorname{Spec} A_i$ where A_i is Noetherian. Hence, by Proposition 7.6.10 and Definition of sheaves, we have sequecne

$$0 \longrightarrow \Gamma(U, \mathscr{O}_X) \hookrightarrow \prod_i A_i \hookrightarrow \prod_i \prod_{\mathfrak{p} \in \operatorname{Ass}(A_i)} (A_i)_{\mathfrak{p}} \text{ (remove the same items)} .$$

Consider $\prod_i \prod_{\mathfrak{p} \in \operatorname{Ass}(A_i)} (A_i)_{\mathfrak{p}}$ (remove the same items) , by Proposition 7.6.24, we know that $\operatorname{Ass}(A_i) = \operatorname{Ass}(\mathscr{O}_X) \cap \operatorname{Spec} A_i = \operatorname{Ass}(X) \cap \operatorname{Spec} A_i$. Hence

$$\begin{aligned} &\prod_i \prod_{\mathfrak{p} \in \operatorname{Ass}(A_i)} (A_i)_{\mathfrak{p}} \text{ (remove the same items)} \\ =&\prod_i \prod_{\mathfrak{p} \in \operatorname{Ass}(X) \cap \operatorname{Spec} A_i} (A_i)_{\mathfrak{p}} \text{ (remove the same items)} \\ \cong& \prod_{p \in \operatorname{Ass} X \cap U} \mathscr{O}_{X,p}, \end{aligned}$$

as we desired. □

Embedded points

Definition 7.6.7 (Embedded points of quasi-coherent sheaf)

Let X be a scheme. Let $\mathscr{F}$ be a quasi-coherent sheaf on X.

1. *An embedded associated point of $\mathscr{F}$ is an associated point which is not maximal among the associated points of $\mathscr{F}$, i.e., it is the specialization of another associated point of $\mathscr{F}$.*
2. *A point x of X is called an embedded point if x is an embedded associated point of $\mathscr{O}_X$.*
3. *An embedded component of X is an irreducible closed subset $Z = \overline{\{x\}}$ where x is an embedded point of X.*

In the Noetherian case when $\mathscr{F}$ is coherent we have the following.

Proposition 7.6.27

Let X be a locally Noetherian scheme. Let $\mathscr{F}$ be a coherent $\mathscr{O}_X$-module. Then

1. *the generic points of irreducible components of $\operatorname{Supp}(\mathscr{F})$ are associated points of $\mathscr{F}$, and*
2. *an associated point of $\mathscr{F}$ is embedded if and only if it is not a generic point of an irreducible component of $\operatorname{Supp}(\mathscr{F})$.*

In particular an embedded point of X is an associated point of X which is not a generic point of an *irreducible component of X*

Proof Note that $\mathrm{Supp}(\mathscr{F}) = \bigcup_{\mathrm{Spec}\, A_i \hookrightarrow X} \mathrm{Supp}(\mathscr{F}|_{\mathrm{Spec}\, A_i})$, since $\mathscr{F}$ is coherent, $\mathscr{F}|_{\mathrm{Spec}\, A_i} \cong \widetilde{M_i}$ where M_i is coherent A_i-module. Hence

$$\mathrm{Supp}(\mathscr{F}) \cong \bigcup_i \mathrm{Supp}(\widetilde{M_i}) \cong \bigcup_i \mathrm{Supp}(M_i)$$

Let η be a generic point, then $\overline{\{\eta\}} \subseteq \mathrm{Supp}(M_i)$ for some i. Hence $\overline{\{\eta\}}$ is irreducible component of $\mathrm{Supp}(M_i)$. Since M_i is coherent, M_i is finitely generated A_i-module, also since X is locally Noetherian, A_i is Noetherian. By Proposition 7.6.15, $\eta \in \mathrm{Ass}\, M_i$. By Proposition 7.6.24, $\eta \in \mathrm{Ass}(\mathscr{F})$. Part (2) is clearly from Part (1) and Definition 7.6.7. □

By Proposition 7.6.27, we may define the **embedded points of a coherent sheaf on a locally Noetherian scheme**.

Definition 7.6.8 (Embedded points of a coherent sheaf on a locally Noetherian scheme)

Let X be a locally Noetherian scheme, $\mathscr{F}$ be a coherent $\mathscr{O}_X$-module. An associated point $x \in X$ is called embedded if it is not generic point of an irreducible component of $\mathrm{Supp}\,\mathscr{F}$.

Scheme-theoretic density (and the adjective "scheme-theoretic" more generally).

Definition 7.6.9 (Scheme-theoretically dense)

An open subscheme U of a scheme X is said to be ***scheme-theoretically dense*** *if any function on any open set V is 0 if it restricts to 0 on $U \cap V$, i.e., $f|_{U\cap V} = 0 \Longrightarrow f = 0$ in V, for any V.*

Remark 7.30 Definition 7.6.9 is equivalent to [Sta25, Definition 01RB]: We say U is scheme theoretically dense in X if for every open $V \subset X$ the scheme theoretic closure of $U \cap V$ in V is equal to V.

This is stronger than density in the usual (topological) sense. For example:

Proposition 7.6.28

If X is locally Noetherian, then an open subscheme $U \subseteq X$ is scheme-theoretically dense if and only if it contains all the associated points of X.

Proof Let U be scheme-theoretically dense, suppose exists $[\mathfrak{p}] \in \mathrm{Ass}(X)$ such that $[\mathfrak{p}] \notin U$. Since $[\mathfrak{p}] \in \mathrm{Ass}(X)$, by definition, $[\mathfrak{p}] \in \mathrm{Ass}(A)$ for any affine open $\mathrm{Spec}\, A \ni [\mathfrak{p}]$. Hence $\mathfrak{p} = \mathrm{Ann}(f)$ for some $f \in A$. We claim that any point $\mathfrak{q} \in U \cap \mathrm{Spec}\, A$ does not contain $\mathfrak{p}$. Since $[\mathfrak{p}] \notin U$, we have $[\mathfrak{p}] \in X - U$, note that we have $\overline{\{\mathfrak{p}\}} \subseteq X - U$, since $X - U$ is closed. Hence any element in $U \cap \mathrm{Spec}\, A$ does not contain $\mathfrak{p}$. Let $a \in \mathfrak{p} - \mathfrak{q}$, then $af = 0 \in \mathfrak{q}$, since $a \notin \mathfrak{q}$ and $\mathfrak{q}$ is prime, we have $f \in \mathfrak{q}$, which implies that f vanishes on $U \cap \mathrm{Spec}\, A$, i.e., $f|_{U\cap \mathrm{Spec}\, A} = 0$. Since U is scheme-theoretically dense, $f = 0$ in $\mathrm{Spec}\, A$, hence $\mathfrak{p} = \mathrm{Ann}_A(0) = A$, a contradiction.

Conversely, let V be any open subset of X, then $V = \bigcup_i \mathrm{Spec}\, A_i$, hence $U \cap V = \bigcup_i (U \cap \mathrm{Spec}\, A_i)$. Suppose $f|_{U\cap V} = 0$, where $f \in \Gamma(V, \mathscr{O}_X)$. By Proposition 7.6.24, we have

$$\mathrm{Ass}\, X \cap V = \bigcup_i (\mathrm{Ass}\, X \cap \mathrm{Spec}\, A_i) = \bigcup_i \mathrm{Ass}\, A_i.$$

Since $U \supseteq \mathrm{Ass}\, X$, we have $f|_{\mathrm{Ass}\, X \cap V} = f|_{\bigcup_i \mathrm{Ass}\, A_i} = 0$, and therefore $f|_{\mathrm{Ass}\, A_i} = 0$, hence $f = 0$ in $\mathscr{O}_{X,p}$ for all $p \in \operatorname{Ass} A_i$ for all i. By Proposition 7.6.27 (7.38), the natural map

$$\Gamma(V, \mathscr{O}_X) \longrightarrow \prod_{p \in \operatorname{Ass} X \cap V} \mathscr{O}_{X,p} = \prod_{p \in \bigcup_i \operatorname{Ass} A_i} \mathscr{O}_{X,p}$$

is an injection. Hence $f|_V = 0$, which implies that U is scheme-theoretically dense. $\square$

Remark 7.31 **Why scheme-theoretically dense is stronger then topological dense?** If U is topological dense, U meet every irreducible components of X, hence U contains all generic point of X (Proposition 4.6.10), but U may not contains embedded point. If U is scheme-theoretically dense, U contains $\operatorname{Ass} X$, hence U contains all embedded points.

Lemma 7.6.3

Let A be a Noetherian ring, let $U \subseteq X = \operatorname{Spec} A$ be an open subset, and let $\mathfrak{a} \subseteq A$ be any ideal such that $U = X \setminus V(\mathfrak{a})$. Then the following assertions are equivalent.

(i) *U is scheme-theoretically dense in X.*

(ii) *U contains a principal open subset $D(t)$, where $t \in A$ is not a divisor.*

(iii) *$\mathfrak{a}$ contains an element $t \in A$ that is not a zero divisor.*

(iv) $\operatorname{Ann}(\mathfrak{a}) := \{s \in A : s\mathfrak{a} = 0\}$ *is zero.*

(v) *U contains* $\operatorname{Ass}(A)$.

Proof Clearly (ii) and (iii) are equivalent.

(ii) $\Rightarrow$ (i). It suffices to show that $D(t)$ is scheme-theoretically dense in X. Let $V \subseteq X$ be an open subset and let $s \in \Gamma(V, \mathscr{O}_X)$ such that $s|_{V \cap D(t)} = 0$. Then for every open affine subset $W \subseteq V \cap D(t)$ there exists an $n \geq 1$ such that $t^n s|_W = 0$. As $t|_W$ is not a zero divisor, we see that $s|_W = 0$ for all W and hence $s = 0$.

(i) $\Rightarrow$ (iv). Let $s \in \operatorname{Ann}(\mathfrak{a})$ and $x = [\mathfrak{p}] \in U$. As $\mathfrak{p} \notin V(\mathfrak{a})$, there exists an $a \in \mathfrak{a} \backslash \mathfrak{p}$. Then $a_x \in \mathscr{O}_{X,x}^\times = A_\mathfrak{p}^\times$ and $sa = 0$. This shows that $s_x = 0$ for all $x \in U$ and hence $s|_U = 0$. As U is scheme-theoretically dense, this implies $s = 0$.

(iv) $\Rightarrow$ (v). If we had $\operatorname{Ass}(A) \cap V(\mathfrak{a}) \neq \varnothing$, we would find $\mathfrak{p} \in \operatorname{Ass}(A)$ with $\mathfrak{p} \supseteq \mathfrak{a}$. But $\mathfrak{p} = \operatorname{Ann}(s)$ for some $0 \neq s \in A$ and hence we get $s \in \operatorname{Ann}(\mathfrak{a})$, so $s = 0$, a contradiction.

(v) $\Rightarrow$ (iii). If $\mathfrak{a}$ does not contain any non-zerodivisor of A, recall Proposition 7.6.11, $\mathfrak{a} \subseteq \bigcup_{\mathfrak{p} \in \operatorname{Ass}(A)} \mathfrak{p}$, by Prime avoidance 13.2.5, there exists an associated prime $\mathfrak{p}$ of A such that $\mathfrak{a} \subseteq \mathfrak{p}$, which implies that $\mathfrak{p} \in V(\mathfrak{a}) \cap \operatorname{Ass}(A)$. But $V(\mathfrak{a}) \cap \operatorname{Ass}(A) = \varnothing$, a contradiction! $\square$

More generally, the adjective **scheme-theoretic** typically indicates an ideal-theoretic definition, enriching and refining a more naive set-theoretic definition. Example will include:

- scheme-theoretically dense
- scheme-theoretic intersection
- scheme-theoretic support
- scheme-theoretic image
- scheme-theoretic closure
- scheme-theoretic preimage
- scheme-theoretic fiber

7.6.15 Revisiting the notion of length

Proposition 7.6.29

(a) (Noetherian converse to Proposition 7.5.8) Suppose A is a Noetherian ring. Then any finitely generated module whose support consists of finitely many points of $\operatorname{Spec} A$*, all closed, has finite length.*

(b) (Noetherian converse to Proposition 7.5.10) Suppose $\mathscr{F}$ is a finite type quasi-coherent sheaf on a locally Noetherian scheme X, supported at finitely many points of X, all closed. Then $\mathscr{F}$ has finite length.

Proof

(a) Note that a finitely generated module over Noetherian (resp. Artin) ring is Noetherian (resp. Artin) module, hence this module satisfies both a.c.c. and d.c.c., apply Lemma 7.5.1, this module has a composition series, and therefore has finite length. Hence, it suffices to show that finitely generated module M can be seen as a module over Artin ring.

Let M is a finitely generated module over Noetherian ring A. By condition, we may assume that $\operatorname{Supp}(M) = \{\mathfrak{m}_1, \ldots, \mathfrak{m}_n\}$ where $\mathfrak{m}_i \in \operatorname{Max} A$. By Proposition 7.6.1, $\operatorname{Supp} M = V(\operatorname{Ann}_A M)$, hence $\operatorname{Ann} M \subseteq \mathfrak{m}_i$ for all i. Note that M can be seen as an $A/\operatorname{Ann}(M)$-module, we want to show that $A/\operatorname{Ann}(M)$ is Artin ring. Consider $\operatorname{Spec} A/\operatorname{Ann}(M)$, $\operatorname{Spec} A/\operatorname{Ann}(M) = V(\operatorname{Ann}(M)) = \{\mathfrak{m}_1, \ldots, \mathfrak{m}_n\}$, hence Krull dimension $\dim A/\operatorname{Ann}(M) = 0$, and therefore $A/\operatorname{Ann}(M)$ is Artin ring, as we desired.

(b) Let $\operatorname{Supp} X = \{\mathfrak{m}_1, \ldots, \mathfrak{m}_n\}$, then $X \setminus \{\mathfrak{m}_1, \ldots, \mathfrak{m}_{i-1}, \mathfrak{m}_{i+1}, \ldots, \mathfrak{m}_n\}$ is open. Let $U_i \hookrightarrow X \setminus \{\mathfrak{m}_1, \ldots, \mathfrak{m}_{i-1}, \mathfrak{m}_{i+1}, \ldots, \mathfrak{m}_n\}$ be an affine open neighborhood of x_i. Let $U = \bigcup_{i=1}^n U_i$. Define $\mathscr{F}_i := j_{i,!}(\mathscr{F}|_{U_i})$, where $j_i : U_i \hookrightarrow U$. We claim that $\mathscr{F}|_U \cong \prod_{i=1}^n \mathscr{F}_i$. Check it at the stalk level, let $\mathfrak{m}_i \in \operatorname{Supp} X$, since $\mathfrak{m}_i \notin U_j$ for all $i \neq j$, we have

$$(\mathscr{F}|_U)_{\mathfrak{m}_i} \cong (\mathscr{F}|_{U_i})_{\mathfrak{m}_i} \cong \left(\prod_{i=1}^n \mathscr{F}_i\right)_{\mathfrak{m}_i},$$

hence $\mathscr{F}|_U \cong \prod_{i=1}^n \mathscr{F}_i$. We next show that $\mathscr{F}|_U$ is finite length. Since $\mathbf{QCoh}_X$ is abelian category, we have $\prod_{i=1}^n \mathscr{F}_i \cong \bigoplus_{i=1}^n \mathscr{F}_i$, hence $\ell(\mathscr{F}|_U) = \sum_{i=1}^n \ell(\mathscr{F}_i)$. Consider $\ell(\mathscr{F}_i)$, since $\mathscr{F}_i = j_{i,!}(\mathscr{F}|_{U_i})$, $\ell(\mathscr{F}|_{U_i}) = \ell(\mathscr{F}_i)$. Since $\mathscr{F}$ is finite type quasi-coherent sheaf on locally Noetherian scheme, $\mathscr{F}|_{U_i} \cong \widetilde{M_i}$ where M_i is finitely generated $\mathscr{O}_X(U_i)$-module and $\mathscr{O}_X(U_i)$ is Noetherian ring. By Part (1), $\ell(M_i) < \infty$, and therefore $\ell(\mathscr{F}|_{U_i}) < \infty$. Hence $\ell(M) = \sum_{i=1}^n \ell(M_i) < \infty$, which implies that $\mathscr{F}|_U$ is finite length.

Since $\operatorname{Supp}(X) \cap X \setminus U = \varnothing$, we know that $\mathscr{F}|_{X\setminus U} = 0$, and therefore $\mathscr{F} = j_!(\mathscr{F}|_U)$ where $j : U \hookrightarrow X$. Hence $\ell(\mathscr{F}) = \ell(\mathscr{F}|_U) < \infty$, which implies that $\mathscr{F}$ is finite length.

□

Proposition 7.6.30

(a) Each finite length quasi-compact scheme (Definition 7.5.6) has a finite number of points, with the discrete topology.

(b) If X is finite type over $\overline{k}$, of finite length l, then $\dim_{\overline{k}} \Gamma(X, \mathscr{O}_X) = l$.

Proof

(a) Let X be a finite length quasi-compact scheme. Since X is finite length, $\mathscr{O}_X$ is finite length, then we have a composition series

$$0=\mathscr{F}_0 \hookrightarrow \mathscr{F}_1 \hookrightarrow \cdots \hookrightarrow \mathscr{F}_n=\mathscr{O}_X$$

with $\operatorname{Coker}(\mathscr{F}_{i-1} \hookrightarrow \mathscr{F}_i)$ is simple object. By Proposition 7.5.9, $\operatorname{Coker}(\mathscr{F}_{i-1} \hookrightarrow \mathscr{F}_i) \cong \widetilde{A_i/\mathfrak{m}_i}$ where $\mathfrak{m}_i$ is maximal ideal of A_i. We claim that $X=\{\mathfrak{m}_1,\dots,\mathfrak{m}_n\}$.
Let $x \in X \setminus \{\mathfrak{m}_1,\dots,\mathfrak{m}_n\}$, take stalk at x, we have

$$0=(\mathscr{F}_0)_x \hookrightarrow (\mathscr{F}_1)_x \hookrightarrow \cdots \hookrightarrow (\mathscr{F}_n)_x=\mathscr{O}_{X,x}$$

with $\operatorname{Coker}((\mathscr{F}_{i-1})_x \hookrightarrow (\mathscr{F}_i)_x) \cong \operatorname{Coker}(\mathscr{F}_{i-1} \hookrightarrow \mathscr{F}_i)_x=0$. Hence $\mathscr{O}_{X,x}=0$, it is impossible, which implies that $X \setminus \{\mathfrak{m}_1,\dots,\mathfrak{m}_n\}=\varnothing$, and therefore $X=\{\mathfrak{m}_1,\dots,\mathfrak{m}_n\}$.
Since each $\mathfrak{m}_i$ is closed point, hence the topology on X is discrete topology.

(b) Since $\ell(X)=l$, by Part (a), we may assume that $X=\{\mathfrak{m}_1,\dots,\mathfrak{m}_l\}$, where each $\mathfrak{m}_i$ is closed point. We may write $X=\bigcup\limits_{i=1}^{l}\operatorname{Spec}(A_i/\mathfrak{m}_i)$ where $\mathfrak{m}_i$ is maximal ideal of A_i. Since each A_i is finitely generated $\overline{k}$-algebra, $A_i/\mathfrak{m}_i$ is a field which isomorphic to $\overline{k}$. By the definition pf sheaves, we have an exact sequence

$$0 \longrightarrow \Gamma(X,\mathscr{F}) \longrightarrow \prod_{i=1}^{l}\Gamma(\operatorname{Spec} A_i/\mathfrak{m}_i,\mathscr{O}_X) \longrightarrow \prod_{i\neq j}\Gamma((\operatorname{Spec} A_i/\mathfrak{m}_i)\cap(\operatorname{Spec} A_j/\mathfrak{m}_j),\mathscr{O}_X).$$

Note that each $(\operatorname{Spec} A_i/\mathfrak{m}_i)\cap(\operatorname{Spec} A_j/\mathfrak{m}_j)=\varnothing$, we get an isomorphism

$$\Gamma(X,\mathscr{F}) \cong \prod_{i=1}^{l}\Gamma(\operatorname{Spec}(A_i/\mathfrak{m}_i)) \cong \prod_{i=1}^{l}A_i/\mathfrak{m}_i \cong \bigoplus_{i=1}^{l}A_i/\mathfrak{m}_i \cong \overline{k}^{\oplus l},$$

it follows that $\dim_{\overline{k}}\Gamma(X,\mathscr{O}_X)=l$.

□

7.6.16 Generalizing the fraction field: the total fraction ring

Definition 7.6.10 (Rational functions on locally Noetherian schemes)

*A **rational function** on a locally Noetherian scheme is an element of the image of $\Gamma(U,\mathscr{O}_U)$ in (7.38) for some U containing all the associated points (by Proposition 7.6.28, U is scheme-theoretically dense). Equivalent, the set of rational functions is the colimit of $\mathscr{O}_X(U)$ over all open sets containing the associated points, i.e., $\varinjlim_{U\supseteq \operatorname{Ass} X}\mathscr{O}_X(U)$ (by Proposition 7.6.28, U is scheme-theoretically dense in X).*

Example 7.8 On $\operatorname{Spec} k[x,y]/(y^2,xy)$ (Figure 5.4), by Exercise 7.1, $\frac{x-2}{(x-1)(x-3)}$ is a rational function, since it belong to $(k[x,y]/(y^2,xy))_{(y)}$ and $(k[x,y]/(y^2,xy))_{(x,y)}$. But $\frac{x-2}{x(x-1)}$ is not, since $\frac{x-2}{x(x-1)} \notin (k[x,y]/(y^2,xy))_{(x,y)}$.

A rational function has a maximal **domain of definition**, because any two actual functions on an open set (i.e., sections of the structure sheaf over that open set) that agree as "rational functions" (i.e., on small enough open sets containing associated points) must be the same function, by the injectivity of (7.38).

Definition 7.6.11 (Regular at a point)

*We say that a rational function f is **regular at a point** $\mathfrak{p}$ if $\mathfrak{p}$ contained in this maximal domain of definition (or equivalently, there is some open set containing $\mathfrak{p}$ where f is defined).*

Example 7.9 On $\operatorname{Spec} k[x,y]/(y^2,xy)$, then rational function $\frac{x-2}{(x-1)(x-3)}$ has domain of definition consisting of everything but 1 and 3 (i.e., $[(x-1)]$ and $[(x-3)]$), and is regular away from those two points.

Definition 7.6.12 (Regular)

*A rational function is **regular** if it is regular at all points.*

Remark 7.32 Unfortunately, "regular" is a regularly overused word in mathematics, and in algebraic geometry in particular.

Definition 7.6.13 (Indeterminacy locus)

*The complement of the domain of definition of a rational function f is called the **indeterminacy locus of** f.*

Definition 7.6.14 (Total fraction ring)

*The rational functions form a ring, called the **total fraction ring** or **total quotient ring** of X. If $X = \operatorname{Spec} A$ is affine, then this ring is called the **total fraction (or quotient) ring** of A.*

Remark 7.33 If $X = \operatorname{Spec} A$ is affine, the total fraction ring is the function field $K(X)$ — the stalk at the generic point (since $\operatorname{Ass}(X) = \{(0)\}$, and apply Proposition 6.2.9) — so this extends our earlier Definition 6.2.3 of $K(\cdot)$.

Proposition 7.6.31

The ring of rational functions on a locally Noetherian scheme is the colimit of the functions over all open sets containing the associated points:

$$\varinjlim_{\{U:\operatorname{Ass} X\subseteq U\}} \mathscr{O}(U).$$

Slightly better (but slightly different): a rational function is the data of a function f defined on an open set containing all associated points, where two such data (U,f) and (U',f') define the same rational function if and only if $f|_{U\cap U'} = f'|_{U\cap U'}$.

If X is reduced, then this is the same as requiring that they are defined on an open set of each of the irreducible components.

Proof If X is reduced, reducedness is affine local property (Example 6.5), we may assume that $X = \operatorname{Spec} A$. By Proposition 7.6.22, X has no embedded prime, then we done. □

Remark 7.34 **Associated points of integral schemes.** In order for some of our discussion elsewhere to make sense in non-Noetherian settings, we note that the notion of associated points for integral schemes works perfectly, because it works for integral domains — only the generic point is associated. In particular, the definition above of rational functions on an integral scheme X agree with Definition 6.2.3, as precisely the elements of the function field $K(X)$.

7.7 ⋆⋆ Coherent modules over non-Noetherian rings

Part III

Morphisms of schemes

Chapter 8 Morphisms of Schemes

We now describe the morphisms between schemes. We will define some easy-to-state properties of morphisms, but leave more subtle properties for late.

Recall that a scheme is (i) a set, (ii) with a topology, (iii) and a (structure) sheaf of rings, and that it is some times helpful to think of the definition as having three steps. In the same way, the notion of morphism of schemes $X \to Y$ may be defined (i) as a map of sets, (ii) that is continuous, and (iii) with some further information involving the sheaf of functions. In the case of affine schemes, we have already seen the map of sets (§4.2.3) and later saw that this map is continuous (Proposition 4.4.6).

8.1 Motivations for the "right" definition of morphism of schemes

Here are two motivations for how morphisms should behave. The first is algebraic, and the second is geometric.

Algebraic motivation. We will want morphisms of affine schemes $\operatorname{Spec} A \to \operatorname{Spec} B$ to be precisely the ring maps $B \to A$. We have already seen that ring maps $B \to A$ induce maps of topological spaces in the opposite direction (Proposition 4.4.6); the main new ingredient will be to see how to add the structure sheaf of functions into the mix. Then a morphism of schemes should be something that "on the level of affine open sets, look like this".

Geometric motivation. Motivated by the theory of manifolds (§4.1.1), which like schemes are ringed spaces, we want morphisms of schemes at the very least to be morphisms of ringed spaces; we now motivate what these are. (We will formalize this in the next section.) Notice that if $\pi : X \to Y$ is a map of differentiable manifolds, then a smooth function on Y pulls back to a smooth function on X. More precisely, given an open subset $U \subseteq Y$, there is a natural map $\Gamma(U, \mathscr{O}_Y) \to \Gamma(\pi^{-1}(U), \mathscr{O}_X)$. This behaves well with respect to restriction (restricting a function to a smaller open set and pulling back yields the same result as pulling back and then restricting), so in fact we have a map of sheaves on Y, i.e., $\mathscr{O}_Y \to \pi_*\mathscr{O}_X$. Similarly a morphism of schemes $\pi : X \to Y$ should induce a map $\mathscr{O}_Y \to \pi_*\mathscr{O}_X$. But in fact in the category of differentiable manifolds a continuous map $\pi : X \to Y$ is a map of differentiable manifolds precisely when smooth functions on Y pullback to smooth functions on X (i.e., the pullback map from smooth functions on Y to functions on X in fact lies in the subset of smooth functions, i.e., the continuous map π induces a pullback of smooth functions, which can be interpreted as a map $\mathscr{O}_Y \to \pi_*\mathscr{O}_X$), so this map of sheaves characterizes morphisms in the differentiable category. So we could use this as the definition of morphism in the differentiable category (see Proposition 4.1.1).

But how do we apply this to the category of schemes? In the category of differentiable manifolds, a continuous map $\pi : X \to Y$ induces a pullback of (the sheaf of) functions, and we can ask when this induces a pullback of smooth functions. However, function are odder on schemes, and we can't recover the pullback map just from the map of topological spaces. The right path is to hardwire this into the definition of morphism, i.e., to have a continuous map $\pi : X \to Y$, along with a pullback map $\pi^\sharp : \mathscr{O}_Y \to \pi_*\mathscr{O}_X$. This leads to the definition of the category of ringed spaces.

On might hope to define morphisms of schemes as morphisms of ringed spaces. This isn't quite right, as then algebraic motivation isn't satisfied: as desired, to each morphism $A \to B$ there is a morphism $\operatorname{Spec} B \to \operatorname{Spec} A$, but there can be addition morphisms of ringed spaces $\operatorname{Spec} B \to \operatorname{Spec} A$ not arising in this way. A revised definition as morphisms of sorting out the details, we take a different approach, using locally ringed spaces, which corresponds to asking not just that functions pullback, but also that values of functions pullback. However, we will check that our eventual definition actually is equivalent to this.

We begin by formally defining morphisms of ringed space.

8.2 Morphisms of ringed spaces

8.2.1 Definition: morphisms of ringed spaces

Definition 8.2.1 (Morphism of ringed spaces)

*A **morphism of ringed spaces** from $(X, \mathscr{O}_X)$ to $(Y, \mathscr{O}_Y)$ is a pair $(\pi, \pi^\sharp)$ of a continuous morphism $\pi : X \to Y$ and a map $\pi^\sharp : \mathscr{O}_Y \to \pi_*\mathscr{O}_X$ of sheaves of rings on Y.*

Remark 8.1 By adjointness (Theorem 3.7.1), this is the same as a map $\pi^{-1}\mathscr{O}_Y \to \mathscr{O}_X$.

Remark 8.2 There is an obvious notion of composition of morphisms, so ringed spaces form a category. Hence we have notion of automorphisms and isomorphism. It is easy to verify that an isomorphism of ringed spaces means the same thing as it did before (Definition 5.3.1).

If $U \subseteq Y$ is an open subset, then there is a natural morphism of ringed spaces $(U, \mathscr{O}_Y|_U) \to (Y, \mathscr{O}_Y)$. More precisely:

Definition 8.2.2 (Open embedding (or open immersion))

*If $U \to Y$ is an isomorphism of U with an open subset V of Y, and we are given an isomorphism $(U, \mathscr{O}_U) \xrightarrow{\sim} (V, \mathscr{O}_Y|_V)$ (via the isomorphism $U \xrightarrow{\sim} V$), then the resulting map of ringed spaces is called an **open embedding** (or **open immersion**) of ringed spaces, and morphism $U \to Y$ is often written $U \hookrightarrow Y$.*

Proposition 8.2.1 (Morphisms of ringed spaces glue)

Suppose $(X, \mathscr{O}_X)$ and $(Y, \mathscr{O}_Y)$ are ringed spaces, $X = \bigcup_i U_i$ is an open cover of X, and we have morphisms of ringed spaces $\pi_i : U_i \to Y$ that "agree on the overlaps", i.e., $\pi_i|_{U_i \cap U_j} = \pi_j|_{U_i \cap U_j}$. Then there is a unique morphism of ringed spaces $\pi : X \to Y$ such that $\pi|_{U_i} = \pi_i$.

Proof Define $\pi : X \to Y$ by setting $\pi|_{U_i} = \pi_i$. We first check π is well-defined. Let $x \in U_i \cap U_j$, since $\pi_i|_{U_i \cap U_j} = \pi_j|_{U_i \cap U_j}$, we have $\pi(x) = \pi_i(x) = \pi_j(x)$. Hence π is well-defined. We next show that π is continuous. Let $V \subseteq Y$, then $\pi^{-1}(V) = \bigcup_i \pi_i^{-1}(V)$. Since each π_i is continuous $\pi_i^{-1}(V)$ is open in X, and therefore $\pi^{-1}(V)$ is open in X. Hence π is continuous.

We now check π along with a map $\mathscr{O}_Y \to \pi_*\mathscr{O}_X$. Let $V \subseteq Y$ be any open subset of Y, pick $f \in \mathscr{O}_Y(V)$, we want to show that $f \circ \pi \in \pi_*\mathscr{O}_X(V) = \mathscr{O}_X(\pi^{-1}(V))$. By the condition, we know that $(f \circ \pi)|_{U_i} = f \circ \pi_i \in (\pi_i)_*\mathscr{O}_X(V) = \mathscr{O}_X(\pi_i^{-1}(V))$, with $((f \circ \pi)|_{U_i})|_{U_i \cap U_j} = ((f \circ \pi)|_{U_j})|_{U_i \cap U_j}$. By the gluability axiom of $\mathscr{O}_X$, there exists $s \in \mathscr{O}_X(\pi^{-1}(V))$ such that $s|_{\pi_i^{-1}(V)} = (f \circ \pi)|_{U_i}$. By the identity axiom of $\mathscr{O}_X$, we know that $f \circ \pi = s \in \mathscr{O}_X(\pi^{-1}(V))$. It follows that π along with $\mathscr{O}_Y \to \pi_*\mathscr{O}_X$. □

Proposition 8.2.2

Open immersions of ringed spaces are monomorphisms (in the category of ringed space).

Proof Let $i : U \hookrightarrow Y$ be an open immersion. Suppose there exists two morphisms $\mu_1 : Z \to U$ and $\mu_2 : Z \to U$ with $i \circ \mu_1 = i \circ \mu_2$. We want to show that $\mu_1 = \mu_2$. Look i as a morphism of topological space, i is injection, hence, as morphisms of topological spaces, $i \circ \mu_1 = i \circ \mu_2$ implies $\mu_1 = \mu_2$. As morphism of sheaves, we have $\mathscr{O}_Y \to (i \circ \mu_1)_* \mathscr{O}_Z$ and $\mathscr{O}_Y \to (i \circ \mu_2)_* \mathscr{O}_Z$ are the same. Note that $(i \circ \mu_i)_* \mathscr{O}_Z = (i_* \circ (\mu_i)_*) \mathscr{O}_Z$, $\mathscr{O}_Y \to (i_* \circ (\mu_1)_*) \mathscr{O}_Z$ and $\mathscr{O}_Y \to (i_* \circ (\mu_2)_*) \mathscr{O}_Z$ are the same. By adjointness (Theorem 3.7.1), $i^{-1} \mathscr{O}_Y \to (\mu_1)_* \mathscr{O}_Z$ and $i^{-1} \mathscr{O}_Y \to (\mu_2)_* \mathscr{O}_Z$ are the same. Since $i : U \hookrightarrow Y$ is open immersion, then we get an isomorphism of ringed spaces $(U, \mathscr{O}_U) \xrightarrow{\sim} (V, \mathscr{O}_Y|_V)$. By Proposition 3.7.3, we know that $i^{-1} \mathscr{O}_Y \cong \mathscr{O}_Y|_V$, hence $i^{-1} \mathscr{O}_Y \cong \mathscr{O}_U$. Hence $\mathscr{O}_U \to (\mu_1)_* \mathscr{O}_Z$ and $\mathscr{O}_U \to (\mu_2)_* \mathscr{O}_Z$ are the same. It follows that $\mu_1 = \mu_2$ as morphisms of ringed spaces, and therefore $i : U \hookrightarrow Y$ is monomorphism in the category of ringed space. □

8.2.2 Pushing Forward $\mathscr{O}$-module, pulling back $\mathscr{O}$-module

Pushing forward $\mathscr{O}$-module

Proposition 8.2.3 (Pushing Forward $\mathscr{O}$-modules)

Given a morphism of ringed spaces $\pi : X \to Y$, then sheaf pushforward induces a functor $\pi_* : \mathbf{Mod}_{\mathscr{O}_X} \to \mathbf{Mod}_{\mathscr{O}_Y}$.

Proof Let $\mathscr{F} \in \mathbf{Mod}_{\mathscr{O}_X}$ be an $\mathscr{O}_X$-module. Then $\pi_* \mathscr{F}$ is an $\pi_* \mathscr{O}_X$-module. Since we have a morphism of ringed spaces $\pi : X \to Y$, there is a morphism of sheaves, $\mathscr{O}_Y \to \pi_* \mathscr{O}_X$, hence $\pi_* \mathscr{F}$ can be seen as $\mathscr{O}_Y$ module, i.e., define the action $\mathscr{O}_Y(V) \times \pi_* \mathscr{O}_X(V)$ by setting

$$(f, s) \longmapsto (f \circ \pi) \cdot s,$$

where $V \subseteq Y$ open subset.

Let $\mathscr{F}_1, \mathscr{F}_2 \in \mathbf{Mod}_{\mathscr{O}_X}$, and morphism $\varphi \in \mathrm{Mor}_{\mathbf{Mod}_{\mathscr{O}_X}}(\mathscr{F}_1, \mathscr{F}_2)$. Define $\pi_* \varphi$ by setting $(\pi_* \varphi)_V = \varphi_{\pi^{-1}(V)}$ for any open subset $V \subseteq Y$, it is easy to see that $\pi_* \varphi \in \mathrm{Mor}_{\mathbf{Mod}_{\mathscr{O}_Y}}(\pi_* \mathscr{F}_1, \pi_* \mathscr{F}_2)$. By the definition, it is easy to verify that π_* preserves identity morphisms and π_* preserves composition. Hence $\pi_* : \mathbf{Mod}_{\mathscr{O}_X} \to \mathbf{Mod}_{\mathscr{O}_Y}$ is a functor. □

Pulling back $\mathscr{O}$-module

Suppose $\pi : (X, \mathscr{O}_X) \to (Y, \mathscr{O}_Y)$ is a morphism of ringed spaces. A slight variation on the inverse image allows us to **pull back an $\mathscr{O}_Y$-module** $\mathscr{G}$ to get an $\mathscr{O}_X$-module $\pi^* \mathscr{G}$.

Lemma 8.2.1

Suppose $\pi : (X, \mathscr{O}_X) \to (Y, \mathscr{O}_Y)$ is a morphism of ringed spaces. Then $(X, \pi^{-1} \mathscr{O}_Y)$ is a ringed space, and that the morphism of ringed spaces π factors into

$$(X, \mathscr{O}_X) \longrightarrow (X, \pi^{-1} \mathscr{O}_Y) \longrightarrow (Y, \mathscr{O}_Y).$$

Proof Clearly $\pi = \pi \circ \mathrm{id}$. Since $\pi : X \to Y$ is a morphism of ringed spaces, we have a morphism of sheaves of rings on Y,

$$\mathscr{O}_Y \longrightarrow \pi_* \mathscr{O}_X.$$

By adjointness (Theorem 3.7.1), we have a morphism of sheaves on X,

$$\pi^{-1} \mathscr{O}_Y \longrightarrow \mathscr{O}_X.$$

Note that $\mathscr{O}_X$ can be seen as $\mathrm{id}_* \mathscr{O}_X$ where $\mathrm{id} : X \to X$. Then we get a morphism of ringed spaces from $(X, \mathscr{O}_X)$ to $(X, \pi^{-1}\mathscr{O}_Y)$. Consider isomorphism $\pi^{-1}\mathscr{O}_Y \xrightarrow{\sim} \pi^{-1}\mathscr{O}_Y$, by adjointness (Theorem 3.7.1), we have

$$\mathscr{O}_Y \longrightarrow \pi_* \pi^{-1}\mathscr{O}_Y.$$

Then we get a morphism of ringed spaces from $(X, \pi^{-1}\mathscr{O}_Y)$ to $(Y, \mathscr{O}_Y)$. Clearly, $(\pi \circ \mathrm{id})_* \mathscr{O}_X = (\pi_* \mathrm{id}_*)\mathscr{O}_X$, then π factors into

$$(X, \mathscr{O}_X) \longrightarrow (X, \pi^{-1}\mathscr{O}_Y) \longrightarrow (Y, \mathscr{O}_Y).$$

□

Lemma 8.2.2

Suppose $\pi : (X, \mathscr{O}_X) \to (Y, \mathscr{O}_Y)$ is a morphism of ringed spaces. Let $\mathscr{G}$ be an $\mathscr{O}_Y$-module, then $\pi^{-1}\mathscr{G}$ is a $\pi^{-1}\mathscr{O}_Y$-module. Moreover, $\pi^{-1}\mathscr{G} \otimes_{\pi^{-1}\mathscr{O}_Y} \mathscr{O}_X$ is an $\mathscr{O}_X$-module.

Proof Let $U \subseteq X$ be any open subset of X, then $\pi^{-1}\mathscr{G}(U) = \varinjlim_{V \supseteq \pi(U)} \mathscr{G}(V)$ and $\pi^{-1}\mathscr{O}_Y(U) = \varinjlim_{V \supseteq \pi(U)} \mathscr{O}_Y(V)$. Define the action $\pi^{-1}\mathscr{O}_Y(U) \times \pi^{-1}\mathscr{G}(U) \to \pi^{-1}\mathscr{G}(U)$ same as action $\mathscr{O}_Y(U) \times \mathscr{G}(U) \to \mathscr{G}(U)$, and therefore $\pi^{-1}\mathscr{G}$ is a $\pi^{-1}\mathscr{O}_Y$-module.

π gives a morphism of sheaves on Y, $\mathscr{O}_Y \to \pi_*\mathscr{O}_X$, by adjointness, we have $\pi^{-1}\mathscr{O}_Y \to \mathscr{O}_X$, hence $\mathscr{O}_X$ can be seen as $\pi^{-1}\mathscr{O}_Y$-module, and therefore $\pi^{-1}\mathscr{G} \otimes_{\pi^{-1}\mathscr{O}_Y} \mathscr{O}_X$ is an $\mathscr{O}_X$-module. □

Definition 8.2.3 (Pullback)

*Suppose $\pi : (X, \mathscr{O}_X) \to (Y, \mathscr{O}_Y)$ is a morphism of ringed spaces. Let $\mathscr{G}$ be an $\mathscr{O}_Y$-module, then $\pi^{-1}\mathscr{G}$ is a $\pi^{-1}\mathscr{O}_Y$-module. We define $\pi^*\mathscr{G}$ be the tensor product $\pi^*\mathscr{G} := \pi^{-1}\mathscr{G} \otimes_{\pi^{-1}\mathscr{O}_Y} \mathscr{O}_X$, call it the **pullback** of $\mathscr{G}$.*

Remark 8.3 Lemma 8.2.1 and Lemma 8.2.2 make the definition of pullback meaningful.

Proposition 8.2.4

Pullback π^* is a covariant functor $\mathbf{Mod}_{\mathscr{O}_Y} \to \mathbf{Mod}_{\mathscr{O}_X}$.

Proof Let $\mathscr{G}_1, \mathscr{G}_2 \in \mathbf{Mod}_{\mathscr{O}_Y}$, and morphism $\varphi \in \mathrm{Mor}_{\mathbf{Mod}_{\mathscr{O}_Y}}(\mathscr{G}_1, \mathscr{G}_2)$. Define $\pi^*\varphi : \pi^*\mathscr{G}_1 \to \pi^*\mathscr{G}_2$ by setting $\pi^{-1}\varphi \otimes \mathrm{id}_{\mathscr{O}_X}$, where $\pi^{-1}\varphi$ given by the universal property of colimit. Hence $\pi^*\varphi \in \mathrm{Mor}_{\mathbf{Mod}_{\mathscr{O}_X}}(\pi^*\mathscr{G}_1, \pi^*\mathscr{G}_2)$.

For any $\mathscr{G} \in \mathbf{Mod}_{\mathscr{O}_Y}$, consider $\pi^*(\mathrm{id}_{\mathscr{G}})$, we have $\pi^*(\mathrm{id}_{\mathscr{G}}) = \pi^{-1}(\mathrm{id}_{\mathscr{G}}) \otimes \mathrm{id}_{\mathscr{O}_X} = \mathrm{id}_{\pi^{-1}(\mathscr{G})} \otimes \mathrm{id}_{\mathscr{O}_X} = \mathrm{id}_{\pi^*\mathscr{G}}$. It follows that π^* preserves identity morphisms.

Let $\mathscr{G}_1 \xrightarrow{\varphi} \mathscr{G}_2 \xrightarrow{\psi} \mathscr{G}_3$. Note that

$$\pi^*(\psi \circ \varphi) = \pi^{-1}(\psi \circ \varphi) \otimes \mathrm{id}_{\mathscr{O}_X} = (\pi^{-1}(\psi) \otimes \mathrm{id}_{\mathscr{O}_X}) \circ (\pi^{-1}(\varphi) \otimes \mathrm{id}_{\mathscr{O}_X}) = \pi^*(\psi) \circ \pi^*(\varphi),$$

π^* preserves composition. Hence π^* is a functor. □

Proposition 8.2.5

Suppose $\pi : (X, \mathscr{O}_X) \to (Y, \mathscr{O}_Y)$ and $\rho : (W, \mathscr{O}_W) \to (X, \mathscr{O}_X)$ are morphisms of ringed spaces, then there is a natural isomorphism of functors

$$\rho^*\pi^* \xleftrightarrow{\sim} (\pi \circ \rho)^*.$$

Proof Let $\mathscr{G}$ be an $\mathscr{O}_Y$-module. Consider $(\rho^*\pi^*)\mathscr{G}$ and $(\pi \circ \rho)^*\mathscr{G}$, we want to show that $((\rho^*\pi^*)\mathscr{G})_p \cong$

$((\pi\circ\rho)^*\mathscr{G})_p$ for all $p\in W$. In fact,

$$\begin{aligned}(\pi\circ\rho)^*\mathscr{G} &=(\pi\circ\rho)^{-1}\mathscr{G}\otimes_{(\pi\circ\rho)^{-1}\mathscr{O}_Y}\mathscr{O}_W\\ &=(\rho^{-1}\circ\pi^{-1})\mathscr{G}\otimes_{(\rho^{-1}\circ\pi^{-1})\mathscr{O}_Y}\mathscr{O}_W,\end{aligned}$$

where last equal by $(\pi\circ\rho)^{-1}\xleftrightarrow{\sim}\rho^{-1}\circ\pi^{-1}$ (check at the level of stalks, use Proposition 3.4.3). On the other hand, we have

$$\begin{aligned}(\rho^*\pi^*)\mathscr{G} &=\rho^*(\pi^{-1}\mathscr{G}\otimes_{\pi^{-1}\mathscr{O}_Y}\mathscr{O}_X)\\ &=\rho^{-1}(\pi^{-1}\mathscr{G}\otimes_{\pi^{-1}\mathscr{O}_Y}\mathscr{O}_X)\otimes_{\rho^{-1}\mathscr{O}_X}\mathscr{O}_W.\end{aligned}$$

We claim that $\rho^{-1}(\pi^{-1}\mathscr{G}\otimes_{\pi^{-1}\mathscr{O}_Y}\mathscr{O}_X)\cong(\rho^{-1}\pi^{-1})\mathscr{G}\otimes_{(\rho^{-1}\pi^{-1})\mathscr{O}_Y}\rho^{-1}\mathscr{O}_X$. Check it at the level of stalks, let $p\in W$, then by Proposition 3.7.2 and Proposition 3.6.11, we have

$$\begin{aligned}\rho^{-1}(\pi^{-1}\mathscr{G}\otimes_{\pi^{-1}\mathscr{O}_Y}\mathscr{O}_X)_p &\cong(\pi^{-1}\mathscr{G}\otimes_{\pi^{-1}\mathscr{O}_Y}\mathscr{O}_X)_{\rho(p)}\\ &\cong(\pi^{-1}\mathscr{G})_{\rho(p)}\otimes_{(\pi^{-1}\mathscr{O}_Y)_{\rho(p)}}\mathscr{O}_{X,\rho(p)}\\ &\cong\mathscr{G}_{(\pi\circ\rho)(p)}\otimes_{\mathscr{O}_{Y,(\pi\circ\rho)(p)}}\mathscr{O}_{X,\rho(p)}\end{aligned}$$

and

$$\begin{aligned}((\rho^{-1}\pi^{-1})\mathscr{G}\otimes_{(\rho^{-1}\pi^{-1})\mathscr{O}_Y}\rho^{-1}\mathscr{O}_X)_p &\cong(\rho^{-1}\pi^{-1})\mathscr{G}_p\otimes_{(\rho^{-1}\pi^{-1})\mathscr{O}_{Y,p}}\rho^{-1}\mathscr{O}_{X,p}\\ &\cong\mathscr{G}_{(\pi\circ\rho)(p)}\otimes_{\mathscr{O}_{Y,(\pi\circ\rho)(p)}}\mathscr{O}_{X,\rho(p)},\end{aligned}$$

hence

$$\rho^{-1}(\pi^{-1}\mathscr{G}\otimes_{\pi^{-1}\mathscr{O}_Y}\mathscr{O}_X)_p\cong((\rho^{-1}\pi^{-1})\mathscr{G}\otimes_{(\rho^{-1}\pi^{-1})\mathscr{O}_Y}\rho^{-1}\mathscr{O}_X)_p,$$

which implies that (Proposition 3.4.3)

$$\rho^{-1}(\pi^{-1}\mathscr{G}\otimes_{\pi^{-1}\mathscr{O}_Y}\mathscr{O}_X)\cong(\rho^{-1}\pi^{-1})\mathscr{G}\otimes_{(\rho^{-1}\pi^{-1})\mathscr{O}_Y}\rho^{-1}\mathscr{O}_X.$$

Hence

$$(\rho^*\pi^*)\mathscr{G}\cong((\rho^{-1}\pi^{-1})\mathscr{G}\otimes_{(\rho^{-1}\pi^{-1})\mathscr{O}_Y}\rho^{-1}\mathscr{O}_X)\otimes_{\rho^{-1}\mathscr{O}_X}\mathscr{O}_W.$$

Note that $\mathscr{O}_W$ can be seen as $\rho^{-1}\mathscr{O}_X$-module, we have

$$\begin{aligned}(\rho^*\pi^*)\mathscr{G} &\cong((\rho^{-1}\pi^{-1})\mathscr{G}\otimes_{(\rho^{-1}\pi^{-1})\mathscr{O}_Y}\rho^{-1}\mathscr{O}_X)\otimes_{\rho^{-1}\mathscr{O}_X}\mathscr{O}_W\\ &\cong(\rho^{-1}\pi^{-1})\mathscr{G}\otimes_{(\rho^{-1}\pi^{-1})\mathscr{O}_Y}\mathscr{O}_W\\ &\cong(\pi\circ\rho)^*\mathscr{G},\end{aligned}$$

which implies that

$$\rho^*\pi^*\xleftrightarrow{\sim}(\pi\circ\rho)^*.$$

□

Theorem 8.2.1 ((π^*,π_*) is adjoint pair)

π^* left-adjoint to π_*. More precisely, for any $\mathscr{F}\in\mathbf{Mod}_{\mathscr{O}_X}$, there is a natural isomorphism

$$\mathrm{Hom}_{(X,\mathscr{O}_X)}(\pi^*\mathscr{G},\mathscr{F})\xleftrightarrow{\sim}\mathrm{Hom}_{(Y,\mathscr{O}_Y)}(\mathscr{G},\pi_*\mathscr{F})$$

that is functorial in both $\mathscr{F}\in\mathbf{Mod}_{\mathscr{O}_X}$ and $\mathscr{G}\in\mathbf{Mod}_{\mathscr{O}_Y}$.

Proof Since $\pi:(X,\mathscr{O}_X)\to(Y,\mathscr{O}_Y)$ is morphism of ringed spaces, we have a morphism of sheaves $\pi^\sharp:\mathscr{O}_Y\to\pi_*\mathscr{O}_X$. By adjointness, we have $\pi^{-1}\mathscr{O}_Y\to\mathscr{O}_X$, hence $\mathscr{F}$ can be seen as $\pi^{-1}\mathscr{O}_Y$-module, denote

$\mathscr{F}_{\pi^{-1}\mathscr{O}_Y}$. By Proposition 2.4.1 and adjointness of (π^{-1}, π_*), we have

$$\begin{aligned}\operatorname{Hom}_{(X,\mathscr{O}_X)}(\pi^*\mathscr{G}, \mathscr{F}) &= \operatorname{Hom}_{\mathbf{Mod}_{\mathscr{O}_X}}(\pi^{-1}\mathscr{G} \otimes_{\pi^{-1}\mathscr{O}_Y} \mathscr{O}_X, \mathscr{F}) \\ &\cong \operatorname{Hom}_{\mathbf{Mod}_{\pi^{-1}\mathscr{O}_Y}}(\pi^{-1}\mathscr{G}, \mathscr{F}_{\pi^{-1}\mathscr{O}_Y}) \\ &\cong \operatorname{Hom}_{(Y,\mathscr{O}_Y)}(\mathscr{G}, \pi^*\mathscr{F}).\end{aligned}$$

□

The push-pull map for $\mathscr{O}$-modules

Definition 8.2.4 (The push-pull map)

Suppose

$$\begin{tikzcd} W \arrow[r, "\beta'"] \arrow[d, "\alpha'"'] & X \arrow[d, "\alpha"] \\ Y \arrow[r, "\beta"] & Z \end{tikzcd} \tag{8.1}$$

is a commutative (not necessarily Cartesian (Definition 2.2.11)) diagram of ringed spaces, and $\mathscr{F}$ is an $\mathscr{O}_X$-module. Define the ***push-pull map***

$$\beta^*\alpha_*\mathscr{F} \longrightarrow \alpha'_*\beta'^*\mathscr{F} \tag{8.2}$$

of $\mathscr{O}_Y$-modules as follows:

i. *Start with the identity*

$$\beta'^*\mathscr{F} \xrightarrow{\sim} \beta'^*\mathscr{F}$$

on W.

ii. *By adjointness of (β'^*, β'_*), this is the same as the data of a morphism*

$$\mathscr{F} \longrightarrow \beta'_*\beta'^*\mathscr{F}$$

on X.

iii. *Apply α_* to get a map*

$$\alpha_*\mathscr{F} \longrightarrow \alpha_*\beta'_*\beta'^*\mathscr{F}$$

on Z.

iv. *By the commutative of diagram (8.1), this is the map*

$$\alpha_*\mathscr{F} \longrightarrow \beta_*\alpha'_*\beta'^*\mathscr{F}$$

on Z.

v. *By adjointness of (β^*, β_*), this yields a map (8.2).*

8.2.3 Properties

Proposition 8.2.6

Given a morphism of ringed spaces $\pi : X \to Y$ with $\pi(p) = q$, then there is a map of stalks $\mathscr{O}_{Y,q} \longrightarrow \mathscr{O}_{X,p}$.

Proof By $\pi : X \to Y$, we have a morphism of sheaves $\mathscr{O}_Y \to \pi_*\mathscr{O}_X$, by adjointness of (π^{-1}, π_*), we have $\pi^{-1}\mathscr{O}_Y \to \mathscr{O}_X$. By Proposition 3.3.1, we have $(\pi^{-1}\mathscr{O}_Y)_p \to \mathscr{O}_{X,p}$. By Proposition 3.7.2, we have

$\mathscr{O}_{Y,q} \cong (\pi^{-1}\mathscr{O}_Y)_p$, hence there is a map of stalks

$$\mathscr{O}_{Y,q} \longrightarrow \mathscr{O}_{X,p}.$$

□

Definition 8.2.5 (Key Definition)

Suppose $\pi^\sharp : B \to A$ is a morphism of rings. Define a morphism of ringed spaces $\pi : \operatorname{Spec} A \to \operatorname{Spec} B$ as follows. The map of topological spaces was given in Proposition 4.4.6. To describe a morphism of sheaves $\mathscr{O}_{\operatorname{Spec} B} \to \pi_*\mathscr{O}_{\operatorname{Spec} A}$ on $\operatorname{Spec} B$, it suffices to describe a morphism of sheaves on the distinguished base of $\operatorname{Spec} B$. On $D(g) \subseteq \operatorname{Spec} B$, we define

$$\mathscr{O}_{\operatorname{Spec} B}(D(g)) \longrightarrow \mathscr{O}_{\operatorname{Spec} A}(\pi^{-1}D(g)) = \mathscr{O}_{\operatorname{Spec} A}(D(\pi^\sharp g))$$

by $B_g \to A_{\pi^\sharp g}$.

✍ **Exercise 8.1** Verify Definition 8.2.5 makes sense (e.g., is independent of g).

Proof We first show that $B_g \to A_{\pi^\sharp g}$ is independent of choice of g. Suppose $D(g_1) = D(g_2)$, by Proposition 4.5.5, g_1 is an invertible element of B_{g_2} and g_2 is an invertible element of B_{g_1}. Define $B_{g_1} \to B_{g_2}$ by setting $\frac{h}{g_1^n} \mapsto \frac{g_1^{-n}h}{1}$. Conversely, define the inverse $B_{g_2} \to B_{g_1}$ by setting $\frac{h}{g_2^m} \mapsto \frac{g_2^{-m}h}{1}$. Hence $B_{g_1} \cong B_{g_2}$, which implies that $B_g \to A_{\pi^\sharp g}$ is independent of choice of g.

We next show that $\pi^{-1}D_{\operatorname{Spec} B}(g) = D_{\operatorname{Spec} A}(\pi^\sharp g)$. Let $\mathfrak{q} \in \pi^{-1}D(g) \subseteq \operatorname{Spec} A$, then $\pi(\mathfrak{q}) \in D(g) \subseteq \operatorname{Spec} B$. Hence $g \notin (\pi^\sharp)^{-1}(\mathfrak{q})$, and therefore $\pi^\sharp(g) \notin \mathfrak{q}$, which implies that $\mathfrak{q} \in D(\pi^\sharp g)$. Conversely, let $\mathfrak{q} \in D(\pi^\sharp g)$, then $\pi^\sharp g \notin \mathfrak{q}$, i.e., $g \notin (\pi^\sharp)^{-1}(\mathfrak{q}) = \pi(\mathfrak{q})$. Hence $\pi(\mathfrak{q}) \in D(g) \subseteq \operatorname{Spec} B$, and therefore $\mathfrak{q} \in \pi^{-1}D(g)$

By Theorem 5.1.1, $\mathscr{O}_{\operatorname{Spec} B}$ and $\mathscr{O}_{\operatorname{Spec} A}$ are sheaves on distinguished base. To describe a morphism of sheaves $\mathscr{O}_{\operatorname{Spec} B} \to \pi_*\mathscr{O}_{\operatorname{Spec} A}$, it suffices to show that $B_g \to A_{\pi^\sharp g}$ commutes with restriction. Let $D(f) \supseteq D(g)$. By Proposition 4.5.5, f is invertible element in B_g and $\pi^\sharp f$ is invertible element in $A_{\pi^\sharp g}$. Define $\varphi : B_f \to B_g$ by setting $\frac{h}{f^n} \mapsto \frac{f^{-n}h}{1}$, and define $\psi : A_{\pi^\sharp f} \to A_{\pi^\sharp g}$ by setting $\frac{t}{(\pi^\sharp f)^m} \mapsto \frac{(\pi^\sharp f)^{-m}t}{1}$. Also $\pi^\sharp_{D(f)} : B_f \to A_{\pi^\sharp f}$ is given by $\frac{h}{f^n} \mapsto \frac{\pi^\sharp(h)}{\pi^\sharp(f)^n}$. It is easy to see they are all well-defined. We want to show that the following diagram is commutes.

$$\begin{tikzcd} B_f \arrow[r, "\pi^\sharp_{D(f)}"] \arrow[d, "\varphi"'] & A_{\pi^\sharp f} \arrow[d, "\psi"] \\ B_g \arrow[r, "\pi^\sharp_{D(g)}"] & A_{\pi^\sharp g} \end{tikzcd}$$

Let $\frac{h}{f^n} \in B_f$, then we have

$$\pi^\sharp_{D(g)} \circ \varphi\left(\frac{h}{f^n}\right) = \pi^\sharp_{D(g)}\left(\frac{f^{-n}h}{1}\right) = \frac{\pi^\sharp(f)^{-n}\pi^\sharp(h)}{1}$$

and

$$\psi \circ \pi^\sharp_{D(f)}\left(\frac{h}{f^n}\right) = \psi\left(\frac{\pi^\sharp(h)}{\pi^\sharp(f)^n}\right) = \frac{\pi^\sharp(f)^{-n}\pi^\sharp(h)}{1},$$

hence $\pi^\sharp_{D(g)} \circ \varphi = \psi \circ \pi^\sharp_{D(f)}$, which implies that $B_g \to A_{\pi^\sharp g}$ commutes with restriction. □

Remark 8.4 Definition 8.2.5 describes a morphism of sheaves on the distinguished base. Definition 8.2.5 is the third in a series of exercise. We saw that a morphism of rings induces a map of set in §4.2.3, a map of topological spaces in Proposition 4.4.6, and now a map of ringed spaces here.

The map of ringed spaces of Key Definition 8.2.5 is really not complicated. Here is an example.

Example 8.1 Consider the ring map $\mathbb{C}[y] \to \mathbb{C}[x]$ given by $y \mapsto x^2$ (see Figure 4.8). We are mapping the affine line with coordinate x to the affine line with coordinate y, i.e., $\operatorname{Spec} \mathbb{C}[x] \to \operatorname{Spec} \mathbb{C}[y]$. The map is (on closed points) $a \mapsto a^2$. For example, where does $[(x-3)]$ go to? Answer: $[(y-3)]$, i.e., $3 \mapsto 9$. What is the preimage of $[(y-4)]$? Answer: those prime ideals in $\mathbb{C}[x]$ containing $[(x^2-4)]$, i.e., $[(x-2)]$ and $[(x+2)]$, so the preimage of 4 is indeed ± 2. This is just about the map of sets, which is old news (§4.2.3), so let's now think about functions pulling back. What is the pullback of the function $\frac{3}{y-4}$ on $D(y-4) = \mathbb{A}^1 = \{4\}$? Of course it is $\frac{3}{x^2-4}$ on $\mathbb{A}^1 \setminus \{-2, 2\}$.

The construction of Key Definition 8.2.5 will soon be an example of a morphism of schemes. In fact we could make that definition right now. Before we do, we point out (via the next example) that not every morphism of ringed spaces between affine schemes is of the form of Ker Definition 8.2.5. In the language of §8.3, this morphism of ringed spaces is not a morphism of locally ringed spaces. We won't use this example, but it answers a question that everyone will have.

Example 8.2 **Not every morphism of ringed spaces between affine schemes is of the form of Key Definition 8.2.5.** Recall (Exercise 4.16) that $\operatorname{Spec} k[y]_{(y)}$ has two points, $[(0)]$ and $[(y)]$, where the second point is closed, and the first is not. Describe a map of ringed spaces $\operatorname{Spec} k(x) \to \operatorname{Spec} k[y]_{(y)}$ sending the unique point of $\operatorname{Spec} k(x)$ to the closed point $[(y)]$, where the pullback map on global sections sends k to k by the identity, and sends y to x. Show that this map of ringed spaces is not of the form described in Key Definition 8.2.5.

Proof Say $X = \operatorname{Spec} k(x)$ and $Y = \operatorname{Spec} k[y]_{(y)}$. Define $\pi : X \to Y$ by setting $\pi((0)) = [(y)]$. Define $\pi^\sharp_Y : k[y]_{(y)} \to k(x)$ by setting $y \mapsto x$ and $k \mapsto k$. We want to show that π is not of the form described in Key Definition 8.2.5. Note that $(\pi^\sharp_Y)^{-1}((0)) = (0)$, but $\pi([(0)]) = [(y)]$, which implies that π is not of the form described in Key Definition 8.2.5. $\square$

8.2.4 Tentative Definition we won't use

A morphism of schemes $\pi : (X, \mathscr{O}_X) \to (Y, \mathscr{O}_Y)$ is a morphism of ringed spaces that "locally looks like" the maps of affine schemes described in Key Definition 8.2.5. Precisely, for each choice of affine open sets $\operatorname{Spec} A \subseteq X$, $\operatorname{Spec} B \subseteq Y$, such that $\pi(\operatorname{Spec} A) \subseteq \operatorname{Spec} B$, the induced map of ringed spaces should be of the form shown in Exercise 8.1.

We would like this definition to be checkable on any affine cover, and we might hope to use the Affine Communication Lemma to develop the theory in this way. This works, but it will be more convenient to use a clever trick: in the next section, we will use the notion of locally ringed spaces, and then once we have used it, we will discard it like yesterday's garbage.

8.3 From locally ringed spaces to morphisms of schemes

8.3.1 Morphisms of locally ringed spaces and morphisms of schemes

In order to prove that morphisms behave in a way we hope, we will use the notion of a locally ringed space. It will not be used later, although it is useful elsewhere in geometry. The notion of locally ringed spaces (and maps between them) is inspired by what we know about manifolds (see Proposition 4.1.2). If $\pi : X \to Y$ is a morphism of manifolds, with $\pi(p) = q$, and f is a function on Y vanishing at q, then the pulled back function $\pi^\sharp(f)$ on X should vanish on p. Put differently: germs of functions (at $q \in Y$) vanishing at should pull back to germs of functions (at $p \in X$) vanishing at p.

Recall (Definition 5.3.7) that a locally ringed space is a ringed space $(X, \mathscr{O}_X)$ such that the stalks $\mathscr{O}_{X,p}$ are all local rings.

Definition 8.3.1 (local homomorphism of local rings)

A ***local homomorphism of local rings*** *is a ring map $\varphi : R \to S$ such that R and S are local rings and such that $\varphi(\mathfrak{m}_R) \subseteq \mathfrak{m}_S$. If it is given that R and S are local rings, then the phrase "**local ring map** $\varphi : R \to S$" means that φ is a local homomorphism of local rings.*

Definition 8.3.2 (Morphism of locally ringed spaces)

A ***morphism of locally ringed spaces*** *$(\pi, \pi^\sharp) : (X, \mathscr{O}_X) \to (Y, \mathscr{O}_Y)$ is a morphism of ringed spaces such that for all $p \in X$ the induced ring map (Proposition 8.2.6) $\pi_p^\sharp : \mathscr{O}_{Y,\pi(p)} \to \mathscr{O}_{X,p}$ is a local ring map (Definition 8.3.1), i.e., $\pi_p^\sharp(\mathfrak{m}_{Y,\pi(p)}) \subseteq \mathfrak{m}_{X,p}$.*

Remark 8.5 In particular that locally ringed spaces form a category.

Definition 8.3.2 means something rather concrete and intuitive: "if $p \mapsto q$, and g is a function vanishing at q, then it will pull back to a function vanishing at p." You would also want: "if $p \mapsto q$, and g is a function not vanishing at q, then it will pull back to a function not vanishing at p." This is a consequence of the following Proposition.

Proposition 8.3.1

If $\pi : X \to Y$ is a morphism of locally ringed spaces, and $\pi(p) = q$, then π induces an inclusion $\kappa(q) \hookrightarrow \kappa(p)$ of residue fields. This captures that fact pullback sends the locus where a function is zero (resp., nonzero) to the locus where the pulled back function is zero (resp., nonzero). In particular, by Proposition 5.3.6 (b), the pullback of invertible functions are invertible functions.

Proof By Proposition 8.2.6, we have a map of stalks $\pi_p^\sharp : \mathscr{O}_{Y,q} \to \mathscr{O}_{X,p}$. Since $\pi : X \to Y$ is a morphism of locally ringed spaces, we have $\pi_p^\sharp(\mathfrak{m}_{Y,q}) \subseteq \mathfrak{m}_{X,p}$. It induces a morphism of residue fields $\widetilde{\pi_p^\sharp} : \kappa(q) \to \kappa(p)$ which is given by $f + \mathfrak{m}_{Y,q} \mapsto \pi_p^\sharp(f) + \mathfrak{m}_{X,p}$. We want to show that $\widetilde{\pi_p^\sharp}$ is injective. Let $\widetilde{\pi_p^\sharp}(f + \mathfrak{m}_{Y,q}) = 0$, then $\pi_p^\sharp(f) \in \mathfrak{m}_{Y,q}$. From $\pi_p^\sharp(\mathfrak{m}_{Y,q}) \subseteq \mathfrak{m}_{X,p}$, we have $\mathfrak{m}_{Y,q} \subseteq (\pi_p^\sharp)^{-1}(\mathfrak{m}_{X,p})$, since $\mathscr{O}_{Y,q}$ is local ring, $\mathfrak{m}_{Y,q} = (\pi_p^\sharp)^{-1}(\mathfrak{m}_{X,p})$, and therefore $f \in \mathfrak{m}_{Y,q}$, which implies that $f + \mathfrak{m}_{Y,q} = 0$ in $\kappa(q)$. Hence $\kappa(q) \to \kappa(p)$ is an inclusion, i.e., $\kappa(q) \hookrightarrow \kappa(p)$. $\square$

To summarize: we use the notion of locally ringed space only to define morphisms of schemes, and to show that morphisms have reasonable properties. **The main things you need to remember about locally ringed spaces are**

(i) that the functions have values at points,

(ii) that given a map of locally ringed spaces, values of functions "pull back", and in particular, zeros of functions "pull back".

Proposition 8.3.2 (Morphism of locally ringed spaces glue)

Suppose $(X, \mathscr{O}_X)$ and $(Y, \mathscr{O}_Y)$ are locally ringed spaces, $X = \bigcup_i U_i$ is an open cover of X, and we have morphisms of locally ringed spaces $\pi_i : U_i \to Y$ that "agree on the overlaps", i.e., $\pi_i|_{U_i \cap U_j} = \pi_j|_{U_i \cap U_j}$. Then there is a unique morphism of locally ringed spaces $\pi : X \to Y$ such that $\pi|_{U_i} = \pi_i$.

Proof By Proposition 8.2.1, we may define $\pi : X \to Y$ by setting $\pi|_{U_i} = \pi_i$, then we get a unique morphisms of ringed spaces from $(X, \mathscr{O}_X)$ to $(Y, \mathscr{O}_Y)$. To show π is a morphism of locally ringed spaces, it suffices to show that for all $p \in X$ the induced ring map $\pi_p^\sharp : \mathscr{O}_{Y,\pi(p)} \to \mathscr{O}_{X,p}$ is a local ring map.

Pick $p \in X$, since $X = \bigcup_i U_i$, then $p \in U_i$ for some i. Hence $\mathscr{O}_{X,p} \cong \mathscr{O}_{U_i,p}$ and $\pi_p^\sharp = (\pi_i)_p^\sharp$. By condition, $(\pi_i)_p^\sharp$ is a local ring map, we have $\pi_p^\sharp : \mathscr{O}_{Y,\pi(p)} \to \mathscr{O}_{X,p}$ is local ring map. Hence $\pi : X \to Y$ is morphism of locally ringed spaces. □

Proposition 8.3.3 (Important!)

Recall that Example 5.3 $\operatorname{Spec} A$ is a locally ringed space. Then the morphism of ringed spaces $\pi : \operatorname{Spec} A \to \operatorname{Spec} B$ defined by a ring morphism $\pi^\sharp_{\operatorname{Spec} B} : B \to A$ (Definition 8.2.5, for simply we write $\pi^\sharp = \pi^\sharp_{\operatorname{Spec} B}$) is a morphism of locally ringed spaces.

Proof Let $[\mathfrak{p}] \in \operatorname{Spec} A$ be any point of $\operatorname{Spec} A$, consider the induced map $\pi^\sharp_{[\mathfrak{p}]} : B_{(\pi^\sharp)^{-1}(\mathfrak{p})} \to A_\mathfrak{p}$, we want to show this map is local ring map. Say $\mathfrak{q} = (\pi^\sharp)^{-1}(\mathfrak{p})$.

Consider $\pi^\sharp_{[\mathfrak{p}]}(\mathfrak{q}B_\mathfrak{q})$, let $\frac{bf}{s} \in \mathfrak{q}B_\mathfrak{q}$ where $b \in B$, $f \in \mathfrak{q}$, and $s \in B \setminus \mathfrak{q}$, then

$$\pi^\sharp_{[\mathfrak{p}]}\left(\frac{bf}{s}\right) = \frac{\pi^\sharp(b)\pi^\sharp(f)}{\pi^\sharp(s)}.$$

Note that $\pi^\sharp(b) \in A$, $\pi^\sharp(f) \in \mathfrak{p}$ and $\pi^\sharp(s) \in \pi^\sharp(B \setminus \mathfrak{q}) \subseteq A \setminus \mathfrak{p}$, we have $\frac{\pi^\sharp(b)\pi^\sharp(f)}{\pi^\sharp(s)} \in \mathfrak{p}A_\mathfrak{p}$, i.e., $\pi^\sharp_{[\mathfrak{p}]}(\mathfrak{q}B_\mathfrak{q}) \subseteq \mathfrak{p}A_\mathfrak{p}$. It follows that $\pi : \operatorname{Spec} A \to \operatorname{Spec} B$ is a locally ringed space. □

Remark 8.6 **Small remark:** Why $\pi^\sharp(B \setminus \mathfrak{q}) \subseteq A \setminus \mathfrak{p}$? If not, there exists $s \in B \setminus \mathfrak{q}$ such that $\pi^\sharp(s) \in \mathfrak{p}$, hence $s \in (\pi^\sharp)^{-1}(\mathfrak{p}) = \mathfrak{q}$, a contradiction.

Proposition 8.3.4 (Key Proposition)

If $\pi : \operatorname{Spec} A \to \operatorname{Spec} B$ is a morphism of locally ringed spaces then it is the morphism of locally ringed spaces induced by the map $\pi^\sharp_{\operatorname{Spec} B} : B = \Gamma(\operatorname{Spec} B, \mathscr{O}_{\operatorname{Spec} B}) \to \Gamma(\operatorname{Spec} A, \mathscr{O}_{\operatorname{Spec} A}) = A$ as in Proposition 8.3.3.

Remark 8.7 Proposition 5.3.1 is a special case of Key Proposition 8.3.4.

Proof Suppose $\pi : \operatorname{Spec} A \to \operatorname{Spec} B$ is a morphism of locally ringed spaces. We wish to show that it is determined by its map on global sections $\pi^\sharp_{\operatorname{Spec} B} : B \to A$. We first need to check that the map of points is determined by global sections. Now a point p of $\operatorname{Spec} A$ can be identified with the prime ideal of global functions vanishing on it. By Proposition 8.3.1, we have inclusion $\kappa(\pi(p)) \hookrightarrow \kappa(p)$, note that $\kappa(\pi(p)) \cong K(B/\pi(p))$, we get a map

$$B \longrightarrow B/\pi(p) \hookrightarrow K(B/\pi(p)) \xrightarrow{\sim} \kappa(\pi(p)) \hookrightarrow \kappa(p), \tag{8.3}$$

hence $\operatorname{Ker}(B \to \kappa(p)) = \pi(p)$. (Here we use the fact that π is a map of locally ringed space.) Also note that

$$B \xrightarrow{\pi^\sharp_{\operatorname{Spec} B}} A \longrightarrow A/p \hookrightarrow K(A/p) \xrightarrow{\sim} \kappa(p), \tag{8.4}$$

we have $\operatorname{Ker}(B \to \kappa(p)) = (\pi^\sharp_{\operatorname{Spec} B})^{-1}(p)$. It is easy to check that (8.3) and (8.4) defined the same map, hence $\pi(p) = (\pi^\sharp_{\operatorname{Spec} B})^{-1}(p)$. It follows that π is the map of sets $\operatorname{Spec} A \to \operatorname{Spec} B$ induced by a ring map $B \to A$.

By the proof in Exercise 8.1, we have $\pi^{-1}(D(g)) = D(\pi^\sharp_{\operatorname{Spec} B} g)$ if $g \in B$.

It remains to show that $\pi^\sharp : \mathscr{O}_{\operatorname{Spec} B} \to \pi_*\mathscr{O}_{\operatorname{Spec} A}$ is the morphism of sheaves given by Definition 8.2.5. It suffices to check this on the distinguished base. We now want to check that for any map of locally ringed spaces inducing the map of sheaves $\mathscr{O}_{\operatorname{Spec} B} \to \pi_*\mathscr{O}_{\operatorname{Spec} A}$, the map of sections on any distinguished open set $D(g) \subseteq \operatorname{Spec} B$ $(g \in B)$ is determined by the map of global sections $B \to A$.

Consider the commutative diagram

$$\begin{tikzcd} B \arrow[r, equal] & \Gamma(\operatorname{Spec} B, \mathscr{O}_{\operatorname{Spec} B}) \arrow[r, "\pi^\sharp_{\operatorname{Spec} B}"] \arrow[d, "\operatorname{res}_{\operatorname{Spec} B, D(g)}"'] & \Gamma(\operatorname{Spec} A, \mathscr{O}_{\operatorname{Spec} A}) \arrow[r, equal] \arrow[d, "\operatorname{res}_{\operatorname{Spec} A, D(\pi^\sharp g)}"] & A \\ B_g \arrow[r, equal] & \Gamma(D(g), \mathscr{O}_{\operatorname{Spec} B}) \arrow[r, "\pi^\sharp_{D(g)}"'] & \Gamma(D(\pi^\sharp g), \mathscr{O}_{\operatorname{Spec} A}) \arrow[r, equal] & A_{\pi^\sharp g} = A \otimes_B B_g. \end{tikzcd}$$

The vertical arrows (restrictions to distinguished open sets) are localizations by g, so the lower horizontal map $\pi^\sharp_{D(g)}$ is determined by the upper map (it is just localization by g). □

We are ready for our definition.

Definition 8.3.3 (Morphism of schemes)

If X and Y are schemes, then a morphism $(\pi, \pi^\sharp) : X \to Y$ as locally ringed spaces is called a ***morphism of schemes****. We have thus defined the category of schemes, which we denote* **Sch***. We then have notions of* ***isomorphism*** *— just the same as Definition 5.3.3 — and* ***automorphism****. The target Y of π is sometimes called the* ***base scheme*** *or the* ***base*** *(this may become clearer once we have defined the fibers of morphisms in Chapter 11.)*

The definition in terms of locally ringed spaces easily implies Tentative Definition §8.2.4:

Proposition 8.3.5 (Important!)

A morphism of schemes $\pi : X \to Y$ is a morphism of ringed spaces that looks locally like morphisms of affine schemes. Precisely:

(1) If $\operatorname{Spec} A$ is an affine open subset of X and $\operatorname{Spec} B$ is an affine open subset of Y, and $\pi(\operatorname{Spec} A) \subseteq \operatorname{Spec} B$, then the induced morphism of ringed spaces is a morphism of affine schemes.

(2) Suppose $\pi : X \to Y$ is a morphism of ringed space, such that $X = \bigcup_i \operatorname{Spec} A_i$ and $Y = \bigcup_i \operatorname{Spec} B_i$ with the property that $\pi(\operatorname{Spec} A_i) \subseteq \operatorname{Spec} B_i$ for each i. Then π is a morphism of schemes if and only if the restriction of π to each $\operatorname{Spec} A_i$ gives a morphism of affine schemes from $\operatorname{Spec} A_i$ to $\operatorname{Spec} B_i$.

Proof

(1) Since $\pi : X \to Y$ is a morphism of schemes, then by definition it is a morphism of locally ringed space. If $\operatorname{Spec} A \subseteq X$ and $\operatorname{Spec} B \subseteq Y$ are affine opens with $\pi(\operatorname{Spec} A) \subseteq \operatorname{Spec} B$, then the restriction of π to $\operatorname{Spec} A$ gives a map of locally ringed space, i.e., $\pi|_{\operatorname{Spec} A} : \operatorname{Spec} A \to \operatorname{Spec} B$ is a morphism of affine schemes.

(2) If π is a morphism of schemes, by Part (1), the restriction of π to each $\operatorname{Spec} A_i$ gives a morphism of affine schemes from $\operatorname{Spec} A_i$ to $\operatorname{Spec} B_i$.

Conversely, if the restriction of π to each $\operatorname{Spec} A_i$ gives a morphism of affine schemes from $\operatorname{Spec} A_i$ to $\operatorname{Spec} B_i$. Pick $p \in X$, since $X = \bigcup_i \operatorname{Spec} A_i$, $p \in \operatorname{Spec} A_i$ for some i, since $\pi(\operatorname{Spec} A_i) \subseteq \operatorname{Spec} B_i$, we also have $\pi(p) \in \operatorname{Spec} B_i$. We want to show that $\pi^\sharp_p : \mathscr{O}_{Y,\pi(p)} \to \mathscr{O}_{X,p}$ is a local ring map. Note that $\mathscr{O}_{X,p} \cong \mathscr{O}_{\operatorname{Spec} A_i,p}$ and $\mathscr{O}_{Y,\pi(p)} \cong \mathscr{O}_{\operatorname{Spec} B_i,\pi(p)}$. Since $\pi|_{\operatorname{Spec} A} : \operatorname{Spec} A \to \operatorname{Spec} B$ is a morphism of affine schemes, $\mathscr{O}_{\operatorname{Spec} B_i,\pi(p)} \to \mathscr{O}_{\operatorname{Spec} A_i,p}$ is a local ring map, and therefore $\pi^\sharp_p : \mathscr{O}_{Y,\pi(p)} \to \mathscr{O}_{X,p}$ is a local ring map, which implies that π is a morphism of schemes.

□

In practice, we will use the affine cover interpretation, and forget completely about locally ringed spaces.

In particular, put imprecisely, the category of affine schemes is the category of rings with the arrows reversed. More precisely:

Proposition 8.3.6

The category of rings $\mathbf{Rings}$ *and the opposite category of affine schemes* $\mathbf{Aff.Sch}^{\mathrm{op}}$ *are equivalent.*

Proof Define the functor $F : \mathbf{Rings} \to \mathbf{Aff.Sch}^{\mathrm{op}}$ as follow. Let $A \in \mathbf{Rings}$, set $F(A) = \operatorname{Spec} A$. Let $\pi^\sharp_{\operatorname{Spec} B} \in \operatorname{Mor}_{\mathbf{Rings}}(B, A)$, set $F(\pi^\sharp_{\operatorname{Spec} B}) = \pi$ where $\pi : \operatorname{Spec} A \to \operatorname{Spec} B$ defined by $\pi(\mathfrak{p}) = (\pi^\sharp_{\operatorname{Spec} B})^{-1}(\mathfrak{p})$. By Proposition 8.3.4, this is well-defined. It is easy to see that $F : \mathbf{Rings} \to \mathbf{Aff.Sch}^{\mathrm{op}}$ is a functor.

Define the functor $G : \mathbf{Aff.Sch}^{\mathrm{op}} \to \mathbf{Rings}$ as follow. Let $\operatorname{Spec} A \in \mathbf{Aff.Sch}^{\mathrm{op}}$, set $G(\operatorname{Spec} A) = A$. Let $\pi \in \operatorname{Mor}_{\mathbf{Aff.Sch}^{\mathrm{op}}}(\operatorname{Spec} A, \operatorname{Spec} B)$, set $G(\pi) = \pi^\sharp_{\operatorname{Spec} B}$. By the definition of morphism of schemes, G is well-defined. Also, it is easy to see that $G : \mathbf{Aff.Sch}^{\mathrm{op}} \to \mathbf{Rings}$ is a functor.

Next, we shall to show that $F \circ G \cong \mathrm{id}_{\mathbf{Aff.Sch}^{\mathrm{op}}}$. Let $\operatorname{Spec} A, \operatorname{Spec} B \in \mathbf{Aff.Sch}^{\mathrm{op}}$, consider the following diagram,

$$\begin{array}{ccc} F\circ G(\operatorname{Spec} A) & \xrightarrow{F\circ G(\pi)} & F\circ G(\operatorname{Spec} B) \\ \downarrow & & \downarrow \\ \operatorname{Spec} A & \xrightarrow{\quad\pi\quad} & \operatorname{Spec} B, \end{array}$$

note that $F\circ G(\operatorname{Spec} A) = \operatorname{Spec} A$, $F\circ G(\operatorname{Spec} B) = \operatorname{Spec} B$, and $F\circ G(\pi) = F(\pi^\sharp_{\operatorname{Spec} B}) = \pi$ (by Proposition 8.3.3), clearly, the diagram commutes and $F \circ G$ is naturally isomorphic to $\mathrm{id}_{\mathbf{Aff.Sch}^{\mathrm{op}}}$.

Similarly, we can show that $G \circ F$ is naturally isomorphic to $\mathrm{id}_{\mathbf{Rings}}$. Hence $\mathbf{Rings}$ and $\mathbf{Aff.Sch}^{\mathrm{op}}$ are equivalent. □

In particular, there can be many different maps from one point to another! For example, here are two different maps from the point $\operatorname{Spec} \mathbb{C}$ to the point $\operatorname{Spec} \mathbb{C}$: the identity (corresponding to the identity $\mathbb{C} \to \mathbb{C}$), and complex conjugation. (There are even more such maps!)

It is clear (from the corresponding facts about locally ringed spaces) that morphisms of schemes glue (Proposition 8.3.2), and the composition of two morphisms is a morphism. Isomorphism in this category are precisely what we defined them to be earlier (Definition 5.3.3).

Exercise 8.2 **Enlightening Exercise. This exercise can give you some practice with understanding morphisms of schemes by cutting up into affine open sets.** Make sense of the following sentence: "$\mathbb{A}^{n+1}_k \setminus \{0\} \to \mathbb{P}^n_k$ given by

$$(x_0, x_1, \ldots, x_n) \mapsto [x_0 : x_1 : \cdots : x_n]$$

is a morphism of schemes." Caution: you can't just say where points go; you have to say where functions go. So you may have to divide these up into affines, and describe the maps, and check that they glue.

Solution *In fact,* $\mathbb{A}^{n+1}_k \setminus \{0\} = \bigcup_{i=0}^{n} D(x_i)$ *and* $\mathbb{P}^n_k = \bigcup_{i=0}^{n} D_+(x_i)$. *By §5.4.3, we have*

$$D_+(x_i) \cong \operatorname{Spec} k[x_{0/i}, \ldots, x_{n/i}]/(x_{i/i} - 1) \cong \operatorname{Spec} k[x_{0/i}, \ldots, x_{i-1/i}, x_{i+1/i}, \ldots, x_{n/i}].$$

Also we have $D(x_i) \cong \operatorname{Spec} k[x_0, \ldots, x_n]_{x_i}$. *Define the maps of ring*

$$(\pi_i)^\sharp_{D_+(x_i)} : k[x_{0/i}, \ldots, x_{i-1/i}, x_{i+1/i}, \ldots, x_{n/i}] \to k[x_0, \ldots, x_n]_{x_i}$$

by setting $x_{j/i} \mapsto \frac{x_j}{x_i}$, *then by Proposition 8.3.3, it gives a morphism of affine schemes* $\pi_i : D(x_i) \to D_+(x_i)$ *which defined by* $\pi_i(p) = ((\pi_i)^\sharp_{D_+(x_i)})^{-1}(p)$ *as continuous map and morphism of sheaves as Definition 8.2.5. In fact* $\pi_i|_{D(x_i x_j)} : D(x_i x_j) \to D_+(x_i x_j)$ *and* $\pi_j|_{D(x_i x_j)} : D(x_i x_j) \to D_+(x_i x_j)$, *we want to* *show that* $\pi_i|_{D(x_ix_j)} = \pi_j|_{D(x_ix_j)}$ *as morphisms of schemes. It suffices to check that* $(\pi_i|_{D(x_ix_j)})^\sharp_{D_+(x_ix_j)} = (\pi_j|_{D(x_ix_j)})^\sharp_{D_+(x_ix_j)}$ *as ring maps. By Proposition 5.5.10, we have*

$$\begin{aligned}(\pi_i|_{D(x_ix_j)})^\sharp_{D_+(x_ix_j)} : k[x_{0/i}, \ldots, x_{i-1/i}, x_{i+1/i}, \ldots, x_{n/i}]_{x_{j/i}} &\longrightarrow k[x_0, \ldots, x_n]_{x_ix_j} \\ \frac{f(x_{0/i}, \ldots, x_{i-1/i}, x_{i+1/i}, \ldots, x_{n/i})}{(x_{j/i})^n} &\longmapsto \frac{x_i^{2n} f(x_{0/i}, \ldots, x_{i-1/i}, x_{i+1/i}, \ldots, x_{n/i})}{(x_ix_j)^n}\end{aligned}$$

and

$$\begin{aligned}(\pi_j|_{D(x_ix_j)})^\sharp_{D_+(x_ix_j)} : k[x_{0/j}, \ldots, x_{j-1/j}, x_{j+1/j}, \ldots, x_{n/j}]_{x_{i/j}} &\longrightarrow k[x_0, \ldots, x_n]_{x_ix_j} \\ \frac{f(x_{0/j}, \ldots, x_{j-1/j}, x_{j+1/j}, \ldots, x_{n/j})}{(x_{i/j})^n} &\longmapsto \frac{x_j^{2n} f(x_{0/j}, \ldots, x_{j-1/j}, x_{j+1/j}, \ldots, x_{n/j})}{(x_ix_j)^n}.\end{aligned}$$

Note that on $D_+(x_i) \cap D_+(x_j)$ *we have coordinate change*

$$x_{j/i} = \frac{1}{x_{i/j}} \quad \text{and} \quad x_{k/i} = \frac{x_{k/j}}{x_{i/j}},$$

hence

$$k[x_{0/i}, \ldots, x_{i-1/i}, x_{i+1/i}, \ldots, x_{n/i}]_{x_{j/i}} \cong k[x_{0/j}, \ldots, x_{j-1/j}, x_{j+1/j}, \ldots, x_{n/j}]_{x_{i/j}},$$

and therefore $(\pi_i|_{D(x_ix_j)})^\sharp_{D_+(x_ix_j)}$ *and* $(\pi_j|_{D(x_ix_j)})^\sharp_{D_+(x_ix_j)}$ *define the same ring maps, i.e.,* $(\pi_i|_{D(x_ix_j)})^\sharp_{D_+(x_ix_j)} = (\pi_j|_{D(x_ix_j)})^\sharp_{D_+(x_ix_j)}$. *Hence* $\pi_i|_{D(x_ix_j)} = \pi_j|_{D(x_ix_j)}$ *as morphisms of affine schemes. Let* $\iota_i : D_+(x_i) \hookrightarrow \mathbb{P}^n_k$ *be open immersion, then* $(\iota_i \circ \pi_i)|_{D(x_ix_j)} = (\iota_j \circ \pi_j)|_{D(x_ix_j)}$. *By Morphism of locally ringed spaces glue 8.3.2, there is a unique morphism of schemes* $\pi : \mathbb{A}^{n+1}_k \setminus \{0\} \to \mathbb{P}^n_k$ *such that* $\pi|_{D(x_i)} = \iota_i \circ \pi_i$. *By above discussion, it is easy to see that* π *is indeed given by*

$$(x_0, x_1, \ldots, x_n) \mapsto [x_0 : x_1 : \cdots : x_n].$$

8.3.2 Morphisms to affine schemes

The following result shows that it is easy to describe morphisms to an affine scheme without working hard to cover the source with affine open sets.

Let X, Y be two schemes. We have a canonical map

$$\rho : \mathrm{Mor}(X, Y) \longrightarrow \mathrm{Hom}_{\mathbf{Rings}}(\Gamma(Y, \mathscr{O}_Y), \Gamma(X, \mathscr{O}_X)), \tag{8.5}$$

which to $(\pi, \pi^\sharp)$ associates $\pi^\sharp_Y : \mathscr{O}_Y(Y) \to \pi_*\mathscr{O}_X(Y) = \mathscr{O}_X(X)$. This map is "functorial" in X in the sense that for any morphism of schemes $\eta : Z \to X$, we have a commutative diagram

$$\begin{array}{ccc} \mathrm{Mor}(X, Y) & \longrightarrow & \mathrm{Hom}_{\mathbf{Rings}}(\mathscr{O}_Y(Y), \mathscr{O}_X(X)) \\ \downarrow & & \downarrow \\ \mathrm{Mor}(Z, Y) & \longrightarrow & \mathrm{Hom}_{\mathbf{Rings}}(\mathscr{O}_Y(Y), \mathscr{O}_Z(Z)) \end{array} \tag{8.6}$$

the vertical arrow on the left being the composition with η, i.e., $\square \circ \eta$; the vertical arrow on the right being the composition with $\eta^\sharp_X : \mathscr{O}_X(X) \to \mathscr{O}_Z(Z)$, i.e., $\eta^\sharp_X \circ \square$.

Lemma 8.3.1

Let X, Y *be affine schemes. Then the map (8.5) defined above is bijection.*

Proof Apply Proposition 8.3.6, we done. $\square$

Proposition 8.3.7

Let Y be an affine scheme. For any scheme X, the canonical map

$$\rho : \mathrm{Mor}(X,Y) \longrightarrow \mathrm{Hom}_{\mathbf{Rings}}(\Gamma(Y,\mathscr{O}_Y),\Gamma(X,\mathscr{O}_X))$$

is bijection.

Proof We may assume that $X = \bigcup_i U_i$ where the U_i are affine open subsets. Using the commutative diagram (8.6) above with each $U_i \hookrightarrow X$, we obtain a new commutative diagram:

$$\begin{array}{ccc} \mathrm{Mor}(X,Y) & \xrightarrow{\rho} & \mathrm{Hom}(\Gamma(Y,\mathscr{O}_Y),\Gamma(X,\mathscr{O}_X)) \\ {\scriptstyle\alpha}\downarrow & & \downarrow{\scriptstyle\beta} \\ \prod_i \mathrm{Mor}(U_i,Y) & \xrightarrow{\gamma} & \prod_i \mathrm{Hom}(\Gamma(Y,\mathscr{O}_Y),\Gamma(U_i,\mathscr{O}_X)). \end{array}$$

By Lemma 8.3.1, γ is bijective. As α is clearly injective (Morphism of locally ringed spaces glue 8.3.2), it follows that ρ is injective. Let $\varphi \in \mathrm{Hom}(\Gamma(Y,\mathscr{O}_Y),\Gamma(X,\mathscr{O}_X))$. Say $\iota_i : U_i \hookrightarrow X$, then we have ring maps $(\iota_i)^\sharp_X : \Gamma(X,\mathscr{O}_X) \to \Gamma(U_i,\mathscr{O}_X)$. In fact, we have $\beta(\varphi) = ((\iota_i)^\sharp_X \circ \varphi)_i$. Since $\gamma = (\gamma_i)_i$ is bijection, say $\pi_i := \gamma_i^{-1}((\iota_i)^\sharp_X \circ \varphi)$. Indeed, for every affine open $V \subseteq U_i \cap U_j$, $\pi_i|_V = \pi_j|_V$ have the same image in $\mathrm{Hom}(\mathscr{O}_Y(Y),\mathscr{O}_X(V))$, and therefore $\pi_i|_{U_i\cap U_j} = \pi_j|_{U_i\cap U_j}$. By Morphism of locally ringed spaces glue 8.3.2, there exists a unique $\pi \in \mathrm{Mor}(X,Y)$ such that $\pi|_{U_i} = \pi_i$. Hence $\rho(\pi) = \varphi$ by the injectivity of β, which proves that ρ is surjective. □

Remark 8.8 This is even true in the category of locally ringed spaces (see [Sta25, Lemma 01I1]).

Remark 8.9 In particular, we have a bijection

$$\mathrm{Mor}(X,\operatorname{Spec}\Gamma(X,\mathscr{O}_X)) \xrightarrow{\sim} \mathrm{Hom}(\Gamma(X,\mathscr{O}_X),\Gamma(X,\mathscr{O}_X)),$$

hence there is a canonical morphism from a scheme to Spec of its ring of global sections. (**Warning:** Even if X is a finite type k-scheme, the ring of global sections might be nasty! In particular, it might not be finitely generated.) The canonical morphism $X \to \operatorname{Spec}\Gamma(X,\mathscr{O}_X)$ is an isomorphism if and only if X is affine, and in this case it is the isomorphism hinted at in Remark after Definition 5.3.4.

✍ **Exercise 8.3** If $S_\bullet$ is a finitely generated graded A-algebra, describe a natural "structure morphism" $\operatorname{Proj} S_\bullet \to \operatorname{Spec} A$.

Proof Say $\operatorname{Proj} S_\bullet = \bigcup_i D_+(f_i)$, by Proposition 5.5.9, we have $D_+(f_i) \cong \operatorname{Spec}((S_\bullet)_{f_i})_0$. Define $(\pi_i)^\sharp_{\operatorname{Spec} A} : A \to ((S_\bullet)_{f_i})_0$ by setting

$$A = S_0 \longleftrightarrow S_\bullet \xrightarrow{\text{localization}} (S_\bullet)_{f_i} \longrightarrow ((S_\bullet)_{f_i})_0.$$

the image of $(\pi_i)^\sharp_{\operatorname{Spec} A}$ in $((S_\bullet)_{f_i})_0$. By Proposition 8.3.7, $(\pi_i)^\sharp_{\operatorname{Spec} A}$ induces a morphism of schemes $(\pi_i,\pi_i^\sharp) : D_+(f_i) \to \operatorname{Spec} A$. We want to show that $\pi_i|_{D_+(f_i)\cap D_+(f_j)} = \pi_j|_{D_+(f_i)\cap D_+(f_j)}$ as morphisms of schemes. It suffice to show that $(\pi_i|_{D_+(f_i)\cap D_+(f_j)})^\sharp_{\operatorname{Spec} A} = (\pi_j|_{D_+(f_i)\cap D_+(f_j)})^\sharp_{\operatorname{Spec} A}$. In fact, by diagram (8.6), $(\pi_i|_{D_+(f_i)\cap D_+(f_j)})^\sharp_{\operatorname{Spec} A}$ is given by

$$A \xrightarrow{(\pi_i)^\sharp_{\operatorname{Spec} A}} ((S_\bullet)_{f_i})_0 \xrightarrow{\text{localization}} ((S_\bullet)_{f_i})_{\frac{f_j^{\deg f_i}}{f_i^{\deg f_j}}} \cong ((S_\bullet)_{f_if_j})_0,$$

and $(\pi_i|_{D_+(f_i)\cap D_+(f_j)})^\sharp_{\operatorname{Spec} A}$ is given by

$$A \xrightarrow{(\pi_j)^\sharp_{\operatorname{Spec} A}} ((S_\bullet)_{f_j})_0 \xrightarrow{\text{localization}} ((S_\bullet)_{f_j})_{\frac{f_i^{\deg f_j}}{f_j^{\deg f_i}}} \cong ((S_\bullet)_{f_if_j})_0.$$

Hence $(\pi_i|_{D_+(f_i)\cap D_+(f_j)})^\sharp_{\operatorname{Spec} A}$ and $(\pi_j|_{D_+(f_i)\cap D_+(f_j)})^\sharp_{\operatorname{Spec} A}$ give the same ring map, and therefore

$$\pi_i|_{D_+(f_i)\cap D_+(f_j)} = \pi_j|_{D_+(f_i)\cap D_+(f_j)}$$

as morphisms of schemes. By Morphism of locally ringed spaces glue 8.3.2, there is a unique morphism of locally ringed spaces $\pi : \operatorname{Proj} S_\bullet \to \operatorname{Spec} A$ such that $\pi|_{D_+(f_i)} = \pi_i$. □

Remark 8.10 From Proposition 8.3.7, it is one small step to some products of schemes exists: if A and B are rings, then $\operatorname{Spec} A \times \operatorname{Spec} B = \operatorname{Spec}(A \otimes_{\mathbb{Z}} B)$; and if A and B are C-algebras, then $\operatorname{Spec} A \times_{\operatorname{Spec} C} \operatorname{Spec} B = \operatorname{Spec}(A \otimes_C B)$. But we are in no hurry, so we wait until Chapter 11 to discuss this properly.

Remark 8.11 **Side fact for experts: Γ and Spec are adjoints.** We have a contravariant functor Spec from rings to locally ringed spaces, and a contravariant functor Γ from locally ringed spaces to ring. In fact $(\Gamma, \operatorname{Spec})$ is an adjoint pair! Thus we could have defined Spec by requiring it to be right-adjoint to Γ. In fact, if we used ringed spaces rather than locally ringed spaces, Γ again has a right adjoint.

Corollary 8.3.1

Spec is a contravariant functor from rings to locally ringed spaces, Γ is a contravariant functor from locally ring spaces to rings, then $(\Gamma, \operatorname{Spec})$ is adjoint pair.

Proof Denote $\mathscr{C}$ be the category of locally ringed spaces. In fact, $\Gamma : \mathscr{C}^{\mathrm{op}} \to \mathbf{Rings}$ and $\operatorname{Spec} : \mathbf{Rings} \to \mathscr{C}^{\mathrm{op}}$. Let $X \in \mathscr{C}$ and $A \in \mathbf{Rings}$, by [Sta25, Lemma 01I1], we have a bijection

$$\operatorname{Hom}_{\mathbf{Rings}}(\Gamma(X, \mathscr{O}_X), A) \xrightarrow{\sim} \operatorname{Mor}_{\mathscr{C}^{\mathrm{op}}}(X, \operatorname{Spec} A),$$

which implies that $(\Gamma, \operatorname{Spec})$ is adjoint pair. □

Our ability to easily describe morphisms to affine schemes will allow us to revisit our earlier discussion of schemes over a given field or ring (§6.3.2).

8.3.3 The category of complex schemes (or more generally the category of k-schemes where k is a field, or more generally the category of A-schemes where A is a ring, or more generally the category of S-scheme where S is a scheme)

Definition 8.3.4 (The category of S-schemes Sch_S)

*The **category of S-schemes** $\mathbf{Sch}_S$ (where S is a scheme) defined as follows. The objects S-schemes are morphisms of the form*

$$\begin{tikzcd} X \arrow[d] \\ S \end{tikzcd}$$

*(The morphism to S is called the **structure morphism**. A motivation for this terminology is the fact that if $S = \operatorname{Spec} A$, the structure morphism gives the functions on each open set of X the structure of an A-algebra, cf. §6.3.2) The morphisms in the category of S-schemes are defined to be commutative diagrams*

$$\begin{tikzcd} X \arrow[r] \arrow[d] & Y \arrow[d] \\ S \arrow[r, "="] & S \end{tikzcd}$$

which is more conveniently written as a commutative diagram

$$\begin{tikzcd} X \arrow[rr] \arrow[dr] & & Y \arrow[dl] \\ & S & \end{tikzcd}$$

When there is no confusion (if the base scheme is clear), simply the top row of the diagram is given. In the case where $S = \operatorname{Spec} A$, where A is a ring, we get the notion of an A-scheme, which is the same as the same definition as in Definition 6.3.3, but in a more satisfactory form.

For example, complex geometers may consider the category of $\mathbb{C}$-schemes.

✍ **Exercise 8.4** Show that this definition of A-scheme given in Definition 8.3.4 agrees with the earlier definition with the earlier definition of Definition 6.3.3.

Proof Let X be an A-scheme, then we get a morphism of schemes $\pi : X \to \operatorname{Spec} A$, we want to show that for all open sets $U \subseteq X$, $\Gamma(U, \mathscr{O}_X)$ is A-algebra. By the definition of morphism of schemes, we have ring morphism $\pi^\sharp_{\operatorname{Spec} A} : A \to \Gamma(X, \mathscr{O}_X)$. Hence for all open subset $U \subseteq X$, we have ring map

$$A \longrightarrow \Gamma(X, \mathscr{O}_X) \longrightarrow \Gamma(U, \mathscr{O}_X).$$

It follows that $\Gamma(U, \mathscr{O}_X)$ is an A-algebra, and therefore X is A-scheme as Definition 6.3.3.

Conversely, if X is an A-scheme as Definition 6.3.3, then for all open $U \subseteq X$, $\Gamma(U, \mathscr{O}_X)$ is A-algebra. In particular, $\Gamma(X, \mathscr{O}_X)$ is an A-algebra, hence there is a ring map

$$A \longrightarrow \Gamma(X, \mathscr{O}_X),$$

by Proposition 8.3.7, there exists a unique morphism of schemes $X \to \operatorname{Spec} A$, and therefore X is an A-scheme as in Definition 8.3.4. □

Proposition 8.3.8

$\operatorname{Spec} \mathbb{Z}$ is the final object in the category of schemes. In other words, if X is any scheme, there exists a unique morphism to $\operatorname{Spec} \mathbb{Z}$. (Hence the category of schemes is isomorphic to the category of $\mathbb{Z}$-schemes.) If A is any ring (for example, a field k), then $\operatorname{Spec} A$ is the final object in the category of A-schemes.

Proof Note that $\mathbb{Z}$ is the initial object of **Rings**, by Proposition 8.3.7, $\operatorname{Spec} \mathbb{Z}$ is the final object of **Sch**.

Suppose A is any ring, and X is an A-scheme. Let $\pi : X \to \operatorname{Spec} A$ be any morphism of schemes from X to $\operatorname{Spec} A$, we want to show that π must be structure morphism. Say $\varphi : X \to \operatorname{Spec} A$ be structure morphism. We have the following commutative diagram.

$$\begin{tikzcd} X \arrow[rr, "\pi"] \arrow[dr, "\varphi"'] & & \operatorname{Spec} A \arrow[dl, "\mathrm{id}_{\operatorname{Spec} A}"] \\ & \operatorname{Spec} A & \end{tikzcd}$$

Then we have $\mathrm{id}_{\operatorname{Spec} A} \circ \pi = \pi = \varphi$, which implies that $\operatorname{Spec} A$ is the final object in the category $\mathbf{Sch}_{\operatorname{Spec} A}$. □

8.3.4 Coproducts of schemes

Proposition 8.3.9

Suppose X and Y are schemes. Then their disjoint union $X \coprod Y$ (Definition 5.3.6) is the coproduct of X and Y in the category of schemes, justifying the use of the coproduct symbol $\coprod$.

Proof Let W be any scheme, with morphisms $f_1 : X \to W$ and $f_2 : Y \to W$. Say $Z = X \coprod Y$. Consider the following diagram.

$$\begin{tikzcd} & Y \arrow[d, "i_Y"'] \arrow[ddr, "f_2"] & \\ X \arrow[r, "i_X"] \arrow[drr, "f_1"'] & Z \arrow[dr, dashed] & \\ & & W \end{tikzcd} \tag{8.7}$$

We want to show that there exists unique morphism from Z to W such that above diagram commutes. Define $\varphi : Z \to X$ as follow. As morphism between topological spaces, set

$$\varphi(z) = \begin{cases} f_1(z) & \text{if } z \in X \\ f_2(z) & \text{if } z \in Y. \end{cases}$$

Clearly, as morphism of topological spaces, diagram (8.7) commutes. As morphism of sheaves, consider $\varphi^\sharp : \mathscr{O}_W \to \varphi_* \mathscr{O}_Z$, for all open $U \subseteq W$, we have

$$\mathscr{O}_Z(\varphi^{-1}(U)) = \mathscr{O}_Z(f_1^{-1}(U) \coprod f_2^{-1}(U)) = \mathscr{O}_X(f_1^{-1}(U)) \times \mathscr{O}_Y(f_2^{-1}(U)).$$

Define $\varphi_U^\sharp : \mathscr{O}_W(U) \to \mathscr{O}_Z(\varphi^{-1}(U))$ by setting $s \mapsto ((f_1)_U^\sharp(s), (f_2)_U^\sharp(s))$. It is easy to show that $\varphi^\sharp$ is a morphism of sheaves. We next want to show that $\varphi_p^\sharp : \mathscr{O}_{W,\varphi(p)} \to \mathscr{O}_{Z,p}$ is a local ring map for all $p \in Z$. If $p \in X$, then $\mathscr{O}_{Z,p} \cong \mathscr{O}_{X,p}$, since $f_1 : X \to W$ is a local ring map, so is $\varphi_p^\sharp$. Similarly, if $p \in Y$, we have $\varphi_p^\sharp$ is a local ring map. Hence, $\varphi_W \to Z$ is a morphism of schemes.

Now we shall to show that diagram (8.7) commutes. As morphism of topological spaces, it is clear. We want to show that $f_1^\sharp = (\varphi \circ i_X)^\sharp = i_X^\sharp \circ h^\sharp$. Let $U \subseteq W$ open, then

$$(\varphi \circ i_X)_U^\sharp = (i_X)_{\varphi^{-1}(U)}^\sharp \circ \varphi_U^\sharp : \mathscr{O}_W(U) \to \mathscr{O}_X(i_X^{-1} \circ \varphi^{-1}(U)) = \mathscr{O}_X(f^{-1}(U)).$$

Pick $s \in \mathscr{O}_W(U)$, then

$$(\varphi \circ i_X)_U^\sharp(s) = (i_X)_{\varphi^{-1}(U)}^\sharp \circ \varphi_U^\sharp(s) = (i_X)_{\varphi^{-1}(U)}^\sharp((f_1)_U^\sharp(s), (f_2)_U^\sharp(s)) = (f_1)_U^\sharp(s),$$

which implies that $f_1^\sharp = (\varphi \circ i_X)^\sharp$. Similarly we have $f_2^\sharp = (\varphi \circ i_Y)^\sharp$.

By the definition of φ, clearly, φ is unique. Hence $X \coprod Y$ is the coproduct of X and Y in the category of schemes. $\square$

8.3.5 Morphisms from (some) affine schemes

Morphisms from affine schemes are not quite as simple as morphisms to affine schemes, but some cases are worth pointing out.

Proposition 8.3.10

Suppose p is a point of a scheme X.

(a) There exists a canonical (choice-free) morphism $\operatorname{Spec} \mathscr{O}_{X,p} \to X$.

(b) *There exists a canonical morphism* $\operatorname{Spec}\kappa(p) \to X$. *(This is often written* $p \to X$*; one gives* p *the obvious interpretation as a scheme,* $\operatorname{Spec}\kappa(p)$*.)*

Proof

(a) Let $p \in X$, then exists affine open $\operatorname{Spec} A \hookrightarrow X$ such that $p \in \operatorname{Spec} A$. In fact, we have a canonical ring homomorphism $A \to A_p$, by Proposition 8.3.6, we get a morphism of affine schemes, $\operatorname{Spec} A_p \to \operatorname{Spec} A$. Note that $\mathscr{O}_{X,p} \cong A_p$, then we have a morphism of schemes

$$\operatorname{Spec}\mathscr{O}_{X,p} \longrightarrow \operatorname{Spec} A \hookrightarrow X.$$

To show that the map is canonical, suppose $V = \operatorname{Spec} B$ is another affine neighborhood containing p so that there exists some prime ideal $\mathfrak{q} \subseteq$ with $\mathscr{O}_{X,p} = B_{\mathfrak{q}}$. Say $U = \operatorname{Spec} A$, and $\mathfrak{p} \in \operatorname{Spec} A$ such that $\mathscr{O}_{X,p} = A_{\mathfrak{p}}$. Since $p \in U \cap V$, by Proposition 6.3.1, there exists a distinguished open subscheme $W \subseteq U \cap V$ of the form $\operatorname{Spec} A_f \hookrightarrow U$ and $\operatorname{Spec} B_g \hookrightarrow V$. Since $p = [\mathfrak{p}] \in \operatorname{Spec} A_f$, $f \notin \mathfrak{p}$, i.e., f is invertible in $A_{\mathfrak{p}}$. By the universal property of localization, we get a morphism $A_f \to A_{\mathfrak{p}}$. Similarly, we get a morphism $B_g \to B_{\mathfrak{q}}$. In fact, we have $\mathscr{O}_{X,p} \cong A_{\mathfrak{p}} \cong B_{\mathfrak{q}}$ and $A_f = B_g$, applying functor Spec, we see that $\operatorname{Spec}\mathscr{O}_{X,p} \to U$ and $\operatorname{Spec}\mathscr{O}_{X,p} \to V$ must factor through $W \subseteq U \cap V$ and thus $\operatorname{Spec}\mathscr{O}_{X,p} \to X$ is independent of the choice of affine neighborhood.

(b) Suppose $U = \operatorname{Spec} A$ is an affine neighborhood of p then exists $[\mathfrak{p}] \in \operatorname{Spec} A$ such that $p = [\mathfrak{p}]$. Thus $\kappa(p) = A_{\mathfrak{p}}/\mathfrak{p}A_{\mathfrak{p}}$. Note that we have ring homomorphism $A_{\mathfrak{p}} \to A_{\mathfrak{p}}/\mathfrak{p}A_{\mathfrak{p}}$, by Proposition 8.3.6, we obtain a morphism of schemes $\operatorname{Spec}\kappa(p) \to \operatorname{Spec}\mathscr{O}_{X,p}$. By part (a), we get a canonical morphism $\operatorname{Spec}\kappa(p) \to \operatorname{Spec}\mathscr{O}_{X,p} \to X$.

□

Proposition 8.3.11 (Morphisms from Spec of a local ring to X)

Suppose X is a scheme, and $(A, \mathfrak{m})$ is a local ring. Suppose we have a scheme morphism $\pi : \operatorname{Spec} A \to X$ sending $[\mathfrak{m}]$ to p. Then

(a) *any open set containing p contains the image of π;*

(b) *there is a bijection between $\operatorname{Mor}(\operatorname{Spec} A, X)$ and*

$$\{(p \in X, \text{ some local homomorphism } \mathscr{O}_{X,p} \to A)\}.$$

Proof

(a) Let U be any open set which contains p, we want to show that $\pi(\operatorname{Spec} A) \subseteq U$. Since $\pi([\mathfrak{m}]) = p \in U$, we have $[\mathfrak{m}] \in \pi^{-1}(U)$. It suffices to show that $\pi^{-1}(U) = \operatorname{Spec} A$. If not, $\operatorname{Spec} A \setminus \pi^{-1}(U)$ is closed in $\operatorname{Spec} A$. Let $\mathfrak{p} \in \operatorname{Spec} A \setminus \pi^{-1}(U)$, since $(A, \mathfrak{m})$ is a local ring, $\mathfrak{p} \subseteq \mathfrak{m}$, and therefore $[\mathfrak{m}] \in \overline{\{\mathfrak{p}\}} \subseteq \operatorname{Spec} A \setminus \pi^{-1}(U)$, contradicts to the fact that $[\mathfrak{m}] \in \pi^{-1}(U)$. Hence $\pi^{-1}(U) = \operatorname{Spec} A$, which implies that $\pi(\operatorname{Spec} A) \subseteq U$.

(b) By part (a), let $\pi \in \operatorname{Mor}(\operatorname{Spec} A, X)$, π actually factors as $\operatorname{Spec} A \to U \hookrightarrow X$ where U is an open neighborhood of $\pi([\mathfrak{m}]) = p$. Say $\pi = i_U \circ f_U$, where $i_U : U \hookrightarrow X$ and $f_U : \operatorname{Spec} A \to U$. Thus a morphism $\pi : \operatorname{Spec} A \to X$ mapping $[\mathfrak{m}]$ to p corresponds to a equivalence class of morphisms $f_U : \operatorname{Spec} A \to U$ mapping $[\mathfrak{m}]$ to p, where U is an affine open neighborhood of p. More precisely, we have

$$\pi \longleftrightarrow (f_U : \operatorname{Spec} A \to U)_U$$

where $(f_U : \operatorname{Spec} A \to U)_U$ is equivalence class which defined by that $f_U \sim f_V$ if $\pi = i_U \circ f_U$ and

$\pi = i_V \circ f_V$. Note that such f_U corresponds to a homomorphism $f_U^\sharp : \mathscr{O}_X(U) \to A$ and $(f_U^\sharp)^{-1}(\mathfrak{m}) = p$, say $U = \operatorname{Spec} B$, we have $f_U^\sharp(B - p) \subseteq A - \mathfrak{m}$, by the universal property of localization, we get a homomorphism of rings $\mathscr{O}_{X,p} = B_p \to A$. Hence we have a bijection

$$\operatorname{Mor}(\operatorname{Spec} A, X) \longleftrightarrow \{(p \in X, \text{local homomorphism } \mathscr{O}_{X,p} \to A)\}.$$

□

Proposition 8.3.10 and Proposition 8.3.11 lead us to the notion of field-valued points, and more generally ring-valued points, and more generally still, scheme-valued points.

8.3.6 Definition:The functor of points, and scheme-valued points (and ring-valued points, and field-valued points) of a scheme

Definition 8.3.5 (Scheme-valued points)

*If Z is a scheme, then Z-**valued points** of a scheme X, denoted $X(Z)$, are defined to be maps $Z \to X$, i.e., $X(Z) = \operatorname{Mor}(Z, X)$. If A is a ring, then A-**valued points** of a scheme X, denoted $X(A)$, are defined to be the $(\operatorname{Spec} A)$-valued points of the scheme. (The most common case of this is when A is a field.)*

If you are working over a base scheme B — for example, complex algebraic geometers will consider only schemes and morphisms over $B = \operatorname{Spec} C$ — then in the above definition, there is an implicit structure map $Z \to B$ (or $\operatorname{Spec} A \to B$ in the case of $X(A)$). For example, for a complex geometer, if X is a scheme over $\mathbb{C}$, the $\mathbb{C}(t)$-valued points of X correspond to commutative diagrams of the form

$$\begin{tikzcd} \operatorname{Spec} \mathbb{C}(t) \arrow[rr] \arrow[dr, "\xi"'] & & X \arrow[dl, "\pi"] \\ & \operatorname{Spec} \mathbb{C} & \end{tikzcd}$$

where $\pi : X \to \operatorname{Spec} \mathbb{C}$ is the structure map for X, and ξ corresponds to the obvious inclusion of rings $\mathbb{C} \to \mathbb{C}(t)$.

Remark 8.12 **Warning:** A k-valued point of a k-scheme X is sometimes called a "rational point" of X, which is dangerous, as for most of the world, "rational" refers to $\mathbb{Q}$. We will use the safer phrase "k-valued point" of X. Another safe choice is "k-rational point" of X or k-point.

The terminology "Z-valued point" (and A-valued point) is unfortunate, because we earlier defined the notion of points of a scheme, and Z-valued points (and A-valued points) are not (necessarily) points! But these usages are well-established in the literature.

Proposition 8.3.12

(a) A morphism of schemes $X \to Y$ induces a map of Z-valued points $X(Z) \to Y(Z)$.

(b) Morphisms of schemes $X \to Y$ are determined by their induced maps of Z-valued points, as Z varies over all schemes.

Proof

(a) Let $\pi : X \to Y$ be a morphism of schemes, pick $\{Z \to X\} \in X(Z)$, then $Z \to X \to Y$ belong to $Y(Z)$, i.e., $\pi : X \to Y$ induced map $\pi \circ \square : X(Z) \to Y(Z)$.

(b) By part (a), any morphisms of schemes $X \to Y$ gives a natural transformation $\tau : \operatorname{Mor}(\square, X) \to$

$\mathrm{Mor}(\square, Y)$. Then by Yoneda's Lemma 2.2.1, we have a bijection

$$\begin{aligned} y : \mathrm{Nat}(\mathrm{Mor}(\square, X), \mathrm{Mor}(\square, Y)) &\longrightarrow \mathrm{Mor}(X, Y) \\ \tau &\longmapsto \tau_X(\mathrm{id}_X), \end{aligned}$$

as we desired.

□

Example 8.3 **Morphisms of schemes $X \to Y$ are not determined by their "underlying" map of points.** For example, let k be a field with $\mathrm{char}(k) = p$, let $X = Y = \operatorname{Spec} k[x]$. Consider Frobenius endomorphism $k[x] \to k[x]$ which maps x to x^p and identity $\mathrm{id}_{k[x]}$. They induce the same morphisms as morphisms of topological space, but induced morphisms are not the same as morphisms of schemes.

Furthermore, we will see that "products of Z-valued points" behave as you might hope (Chapter 11). A related reason this language is suggestive: the notation $X(Z)$ suggests the interpretation of X as a (contravariant) functor h_X from schemes to sets — the **functor of (scheme-valued) points** of the scheme X (cf. Example 2.19). More precisely:

Definition 8.3.6 (Functor of points)

Given a scheme X we can define a functor

$$h_X : \mathbf{Sch}^{\mathrm{op}} \longrightarrow \mathbf{Sets}, \qquad T \longmapsto \mathrm{Mor}(T, X).$$

h_X is called the ***functor of points of*** *X.*

Here is a low-brow reason A-valued points are a useful notion:

Proposition 8.3.13

The A-valued points of an affine scheme $\operatorname{Spec} \mathbb{Z}[x_1, \ldots, x_n]/(f_1, \ldots, f_r)$ (where $f_i \in \mathbb{Z}[x_1, \ldots, x_n]$ are relations) are precisely the solutions to the equations

$$f_1(x_1, \ldots, x_n) = \cdots = f_r(x_1, \ldots, x_n) = 0$$

in the ring A.

Proof Say $X = \operatorname{Spec} \mathbb{Z}[x_1, \ldots, x_n]/(f_1, \ldots, f_r)$. The A-valued points of an affine scheme X is $\mathrm{Mor}(\operatorname{Spec} A, X)$, by Proposition 8.3.7, it can be seen as $\mathrm{Hom}_{\mathbf{Rings}}(\mathbb{Z}[x_1, \ldots, x_n]/(f_1, \ldots, f_r), A)$. Let

$$\varphi : \mathbb{Z}[x_1, \ldots, x_n]/(f_1, \ldots, f_r) \longrightarrow A$$

be a ring map, then φ determined by $\varphi(x_i) = a_i$. Note that $\varphi(f_j) = f_j(a_1, \ldots, a_n) = 0$, we know that $\mathrm{Hom}_{\mathbf{Rings}}(\mathbb{Z}[x_1, \ldots, x_n]/(f_1, \ldots, f_r), A)$ is one-to-one corresponding to a solution of

$$f_1(x_1, \ldots, x_n) = \cdots = f_r(x_1, \ldots, x_n) = 0$$

in the ring A, then we done. □

Example 8.4 For example, the rational solutions to $x^2 + y^2 = 16$ are precisely the $\mathbb{Q}$-valued points of $\operatorname{Spec} \mathbb{Z}[x, y]/(x^2 + y^2 - 16)$. The integral solutions are precisely the $\mathbb{Z}$-valued points. So A-valued points of an affine scheme (finite type over $\mathbb{Z}$) can be interpreted simply. In the special case where A is local, A-valued points of a general scheme have a good interpretation too (Proposition 8.3.10 and Proposition 8.3.11).

Proposition 8.3.14

(a) Suppose B is a ring. If X is a B-scheme, and $f_0, \ldots, f_n$ are $n+1$ functions on X with no common zeros, then $[f_0 : \cdots : f_n]$ gives a morphism of B-schemes $X \to \mathbb{P}^n_B$.

(b) Suppose g is a nowhere vanishing function on X, and f_i are as in part (a). Then the morphisms $[f_0 : \cdots : f_n]$ and $[gf_0 : \cdots : gf_n]$ to $\mathbb{P}^n_B$ are the same.

Proof

(a) Let $\mathbb{P}^n_B = \bigcup_{i=0}^n D_+(x_i)$ where $D_+(x_i) \cong \operatorname{Spec}(B[x_0, \ldots, x_n]_{x_i})_0$. Pick $p \in X$, since $f_0, \ldots, f_n$ have no common zeros, there must have $p \in X_{f_i}$ for some i (X_{f_i} defined in Definition 7.2.4), hence $X = \bigcup\limits_{i=0}^{n} X_{f_i}$. Define $\pi_i^\sharp : (B[x_0, \ldots, x_n]_{x_i})_0 \to \Gamma(X_{f_i}, \mathscr{O}_X)$ by setting $x_{j/i} \mapsto \frac{f_j}{f_i}$, hence we get a morphism of schemes, $\pi_i : X_{f_i} \to D_+(x_i)$. We next show that $\pi_i|_{X_{f_i} \cap X_{f_j}} = \pi_j|_{X_{f_i} \cap X_{f_j}}$. It suffices to show that $\pi_i|^\sharp_{X_{f_i} \cap X_{f_j}} = \pi_j|^\sharp_{X_{f_i} \cap X_{f_j}}$. Note that $\pi_i|^\sharp_{X_{f_i} \cap X_{f_j}} : (B[x_0, \ldots, x_n]_{x_i x_j})_0 \to \Gamma(X_{f_i} \cap X_{f_j}, \mathscr{O}_X)$ maps $\frac{x_k x_l}{x_i x_j}$ to $\frac{f_k f_l}{f_i f_j}$, similar to $\pi_j|^\sharp_{X_{f_i} \cap X_{f_j}}$, we have $\pi_i|^\sharp_{X_{f_i} \cap X_{f_j}} = \pi_j|^\sharp_{X_{f_i} \cap X_{f_j}}$. Hence $\pi_i|_{X_{f_i} \cap X_{f_j}} = \pi_j|_{X_{f_i} \cap X_{f_j}}$. Say $\iota_k : D_+(x_i) \hookrightarrow \mathbb{P}^n_B$ be open immersion, then

$$\iota_i \circ \pi_i|_{X_{f_i} \cap X_{f_j}} = (\iota_i \circ \pi_i)|_{X_{f_i} \cap X_{f_j}} = (\iota_j \circ \pi_j)|_{X_{f_i} \cap X_{f_j}} = \iota_j \circ \pi_j|_{X_{f_i} \cap X_{f_j}},$$

by Morphisms of locally ringed spaces glue 8.3.2, we get a morphism of schemes $\pi : X \to \mathbb{P}^n_B$ such that $\pi|_{X_{f_i}} = \pi_i$. Hence π is of the form $[f_0 : f_1 : \cdots : f_n]$. Since all ring map is also B-algebra map, $\pi : X \to \mathbb{P}^n_B$ is a morphism of B-schemes.

(b) Since g is a nowhere vanishing function on X, we have $X_{f_i} = X_{gf_i}$. Say $\varphi = [gf_0 : \cdots : gf_n]$, consider $\varphi|_{X_{gf_i}}$, it induces a ring map $(B[x_0, \ldots, x_n]_{x_i})_0 \to \Gamma(X_{gf_i}, \mathscr{O}_X)$ mapping $x_{j/i} \mapsto \frac{gf_j}{gf_i} = \frac{f_j}{f_i}$. It is same as $\pi|_{X_{gf_i}} = \pi|_{X_{f_i}}$, hence $\pi = \varphi$.

□

Example 8.5 Consider the $n+1$ functions $x_0, \ldots, x_n$ on $\mathbb{A}^{n+1}$ (otherwise known as $n+1$ sections of the trivial bundle). They have no common zeros on $\mathbb{A}^{n+1} \setminus \{0\}$. Hence they determine a morphism $\mathbb{A}^{n+1} \setminus \{0\} \to \mathbb{P}^n$.

You might hope that Proposition 8.3.14 (a) gives all morphisms to projective space (over B). But this isn't the case. Indeed, even the identity morphism $X = \mathbb{P}^1_k \to \mathbb{P}^1_k$ isn't of this form, as the source $\mathbb{P}^1$ has no nonconstant global functions with which to build this map. (There are similar examples with an affine source.) However, there is a correct generalization (characterizing all maps from schemes to projective schemes) in Chapter 16. This result roughly states that this works, so long as the f_i are not quite functions, but sections of a line bundle. Our desire to understand maps to projective schemes in a clean way will be one important motivation for understanding line bundles.

We will see more ways to describe maps to projective space in the next section. A different description directly generalizing Proposition 8.3.14 (a) will be given in Chapter 16, which will turn out to be a "universal" description.

Incidentally, before Grothendieck, it was considered a real problem to figure out the right way to interpret points of projective space with "coordinates" in a ring. These difficulties were due to a lack of functorial reasoning. And the clues to the right answer already existed (the same problems arise for maps from a manifold to $\mathbb{RP}^n$) — if you ask such a geometric question (for projective space is geometric), the answer is necessarily geometric, not purely algebraic!

8.3.7 Visualizing morphisms: Picturing maps of schemes when nilpotents are present

You now know how to visualize the points of schemes (§4.3), and nilpotents (§5.2 and §7.6). The following imprecise exercise will give you some sense of how to visualize maps of schemes when nilpotents are involved.

✍ **Exercise 8.5** Suppose $a \in \mathbb{C}$. Consider the map of rings $\mathbb{C}[x] \to \mathbb{C}[\varepsilon]/(\varepsilon^2)$ given by $x \mapsto a\varepsilon$. Recall that $\operatorname{Spec}\mathbb{C}[\varepsilon]/(\varepsilon^2)$ may be pictured as a point with a tangent vector (§5.2). How would you picture this map if $a \neq 0$? How does your picture change if $a = 0$? (The tangent vector should be "crushed" in this case.)

Solution *Say $\pi^\sharp : \mathbb{C}[x] \to \mathbb{C}[\varepsilon]/(\varepsilon^2)$ given by $x \mapsto a\varepsilon$. It gives a morphism of schemes $\pi : \operatorname{Spec}\mathbb{C}[\varepsilon]/(\varepsilon^2) \to \operatorname{Spec}\mathbb{C}[x]$. In fact, $\operatorname{Spec}\mathbb{C}[\varepsilon]/(\varepsilon^2)$ has only one closed point $[(\varepsilon)]$ and $\pi([(\varepsilon)]) = (\pi^\sharp)^{-1}([(\varepsilon)]) = [(x)]$. By Proposition 8.2.6, π induces a map of stalks $\pi^\sharp_{[(\varepsilon)]} : \mathbb{C}[x]_{(x)} \to (\mathbb{C}[\varepsilon]/(\varepsilon^2))_{(\varepsilon)}$. Since $\mathbb{C}[\varepsilon]/(\varepsilon^2)$ is a local ring, $(\mathbb{C}[\varepsilon]/(\varepsilon^2))_{(\varepsilon)} \cong \mathbb{C}[\varepsilon]/(\varepsilon^2)$. The maximal ideal in $\mathbb{C}[x]_{(x)}$ is $(x)\mathbb{C}[x]_{(x)}$, say $\mathfrak{m}_{(x)} = (x)\mathbb{C}[x]_{(x)}$, then $\pi^\sharp_{[\varepsilon]}\big|_{\mathfrak{m}_{(x)}} : \mathfrak{m}_{(x)} \to (\varepsilon)(\mathbb{C}[\varepsilon]/(\varepsilon^2))_{(\varepsilon)}$. It induce a map of cotangent space $\widetilde{\pi^\sharp_{[(\varepsilon)]}} : \mathfrak{m}_{(x)}/\mathfrak{m}^2_{(x)} \to \mathfrak{m}_{(\varepsilon)}/\mathfrak{m}^2_{(\varepsilon)}$, where $\mathfrak{m}_{(\varepsilon)} = (\varepsilon)(\mathbb{C}[\varepsilon]/(\varepsilon^2))_{(\varepsilon)}$. In fact*

$$\mathfrak{m}_{(x)}/\mathfrak{m}^2_{(x)} = (x)\mathbb{C}[x]_{(x)}/(x^2)\mathbb{C}[x]_{(x)} \cong \mathbb{C}x$$

and

$$\mathfrak{m}_{(\varepsilon)}/\mathfrak{m}^2_{(\varepsilon)} = \frac{(\varepsilon)(\mathbb{C}[\varepsilon]/(\varepsilon^2))_{(\varepsilon)}}{(\varepsilon^2)(\mathbb{C}[\varepsilon]/(\varepsilon^2))_{(\varepsilon)}} \cong \mathbb{C}\varepsilon,$$

we have $\widetilde{\pi^\sharp_{[(\varepsilon)]}} : \mathbb{C}x \to \mathbb{C}\varepsilon$ maps x to $a\varepsilon$. If $a \neq 0$, $\widetilde{\pi^\sharp_{[(\varepsilon)]}}$ is an isomorphism, which implies that the cotangent space is isomorphic, take dual, their tangent space at origin are isomorphic. If $a = 0$, $\widetilde{\pi^\sharp_{[(\varepsilon)]}} = 0$, it follows that tangent vector should be "crushed" in this case.

8.3.8 ⋆⋆ Analytification of complex algebraic varieties

Any discussion of analytification is only for readers who are familiar with the notion of complex analytic varieties, or willing to develop it on their own in parallel with our development of schemes.

8.4 Maps of graded rings and maps of projective schemes

8.4.1 Graded ring morphisms and their induced scheme maps

As maps of rings correspond to maps of affine schemes in the opposite direction, maps of graded rings (over a base ring A) sometimes give maps of projective schemes in the opposite direction. This is an imperfect generalization: not every map of graded rings gives a map of projective schemes; not every map of projective scheme comes from a map of graded rings; and different maps of graded rings can yield the same map of schemes.

Definition 8.4.1 (Homomorphism of graded ring)

We say a ring map $\varphi : S_\bullet \to R_\bullet$ is a homomorphism of $\mathbb{Z}^{\geq 0}$-graded rings, if there exists an $r \geq 1$ such that for every $d \geq 0$, we have $\varphi(S_d) \subseteq R_{rd}$.

Proposition 8.4.1

Suppose that $\varphi : S_\bullet \to R_\bullet$ is a homomorphism of $\mathbb{Z}^{\geq 0}$-graded rings. This induces a morphism of schemes

$$(\operatorname{Proj} R_\bullet) \setminus V(\varphi(S_+)) \longrightarrow \operatorname{Proj} S_\bullet.$$

In particular, if

$$V(\varphi(S_+)) = \varnothing, \tag{8.8}$$

then we have a morphism $\operatorname{Proj} R_\bullet \to \operatorname{Proj} S_\bullet$*. If* φ *is furthermore a morphism of* A*-algebras, then the induced morphism* $(\operatorname{Proj} R_\bullet) \setminus V(\varphi(S_+)) \to \operatorname{Proj} S_\bullet$ *is a morphism of* A*-schemes.*

Proof Suppose $\operatorname{Proj} S_\bullet = \bigcup_{f\in S_+} \operatorname{Spec}((S_\bullet)_f)_0$. We claim that $(\operatorname{Proj} R_\bullet)\setminus V(\varphi(S_+)) = \bigcup_{f\in S_+} \operatorname{Spec}((R_\bullet)_{\varphi(f)})_0$. Pick $[\mathfrak{q}] \in (\operatorname{Proj} R_\bullet) \setminus V(\varphi(S_+))$, then $\mathfrak{q}$ is a homogeneous ideal with $\mathfrak{q} \not\supseteq R_+$ and $\mathfrak{q} \not\supseteq \varphi(S_+)$. Since φ is a morphism of graded ring, we have $\varphi^{-1}(\mathfrak{q})$ is a homogenous ideal with $\varphi^{-1}(\mathfrak{q}) \supseteq S_+$. Hence there exists $f \in S_+$ such that $f \notin \varphi^{-1}(\mathfrak{q})$, and therefore $\varphi(f) \notin \mathfrak{q}$, it follows that $\mathfrak{q} \in \operatorname{Spec}((R_\bullet)_{\varphi(f)})_0$. Conversely, pick $[\mathfrak{q}] \in \operatorname{Spec}((R_\bullet)_{\varphi(f)})_0$ for some $f \in S_+$, then $\varphi(f) \notin \mathfrak{q}$, so that $\mathfrak{q} \not\supseteq \varphi(S_+)$, which implies that $[\mathfrak{q}] \in (\operatorname{Proj} R_\bullet) \setminus V(\varphi(S_+))$. Hence $(\operatorname{Proj} R_\bullet) \setminus V(\varphi(S_+)) = \bigcup_{f\in S_+} \operatorname{Spec}((R_\bullet)_{\varphi(f)})_0$. Define $\pi_f^\sharp : ((S_\bullet)_f)_0 \to ((R_\bullet)_{\varphi(f)})_0$ by setting

$$\frac{h}{f^n} \longmapsto \frac{\varphi(h)}{\varphi(f)^n},$$

by definition of morphism of graded ring, $\pi_f^\sharp$ is well-defined. Hence this gives a morphism of affine schemes $\pi_f : \operatorname{Spec}((R_\bullet)_{\varphi(f)})_0 \to \operatorname{Spec}((S_\bullet)_f)_0$. Say $D_{R_\bullet,f} = \operatorname{Spec}((R_\bullet)_{\varphi(f)})_0$ and $D_{S_\bullet,f} = \operatorname{Spec}((S_\bullet)_f)_0$, let $\iota_f : D_{S_\bullet,f} \hookrightarrow \operatorname{Proj} S_\bullet$ be open immersion. We want to show that $(\iota_f\circ\pi_f)|_{D_{R_\bullet,f}\cap D_{R_\bullet,g}} = \iota_f\circ\pi_f|_{D_{R_\bullet,f}\cap D_{R_\bullet,g}} = \iota_g \circ \pi_g|_{D_{R_\bullet,f}\cap D_{R_\bullet,g}} = (\iota_g \circ \pi_g)|_{D_{R_\bullet,f}\cap D_{R_\bullet,g}}$. It suffices to check $\pi_f|^\sharp_{D_{R_\bullet,f}\cap D_{R_\bullet,g}} = \pi_g|^\sharp_{D_{R_\bullet,f}\cap D_{R_\bullet,g}}$ as ring maps. It is clear. By Morphisms of locally ringed spaces glue 8.3.2, we get a morphism of schemes

$$\pi : (\operatorname{Proj} R_\bullet) \setminus V(\varphi(S_+)) \longrightarrow \operatorname{Proj} S_\bullet.$$

In particular, if $V(\varphi(S_+)) = \varnothing$, then we have a morphism $\operatorname{Proj} R_\bullet \to \operatorname{Proj} S_\bullet$.

If φ is a morphism of A-algebras, above ring maps which induced by φ are all morphism of A-algebras, and therefore the induced morphism $(\operatorname{Proj} R_\bullet) \setminus V(\varphi(S_+)) \to \operatorname{Proj} S_\bullet$ is a morphism of A-schemes. □

Example 8.6 Let's see Proposition 8.4.1 in action. We will scheme-theoretically interpret the map of complex projective manifolds $\mathbb{CP}^1$ to $\mathbb{CP}^2$ given by

$$\begin{aligned} \mathbb{CP}^1 &\longrightarrow \mathbb{CP}^2 \\ [s:t] &\longrightarrow [s^{20} : s^9t^{11} : t^{20}] \end{aligned}$$

Notice first that this is well-defined: $[\lambda s, \lambda t]$ is sent to the same point of $\mathbb{CP}^2$ as $[s,t]$. The reason for it to be well-defined is that the three polynomials s^{20}, s^9t^{11}, and t^{20} are all homogeneous of degree 20.

Algebraically, this corresponds to a map of graded rings in the opposite direction

$$\varphi : \mathbb{C}[x,y,z] \longrightarrow \mathbb{C}[s,t]$$

given by $x \mapsto s^{20}$, $y \mapsto s^9t^{11}$, $z \mapsto t^{20}$.

Let $S_\bullet = \mathbb{C}[x,y,z]$ and $R_\bullet = \mathbb{C}[s,t]$, then $S_+ = (x,y,z)$, and therefore $\varphi(S_+) = (s^{20}, s^9t^{11}, t^{20})$. Hence $V(\varphi(S_+)) = \varnothing$. By Proposition 8.4.1, we have $\operatorname{Proj}\mathbb{C}[s,t] \to \operatorname{Proj}\mathbb{C}[x,y,z]$ which given by $[s:t] \to [s^{20} : s^9t^{11} : t^{20}]$.

Example 8.7 Notice that there is no map of complex manifolds $\mathbb{CP}^2 \to \mathbb{CP}^1$ given by $[x:y:z] \mapsto [x:y]$, because the map is not defined when $x = y = 0$. This corresponds to the fact that the map of graded rings $\mathbb{C}[s,t] \to \mathbb{C}[x,y,z]$ given by $s \mapsto x$ and $t \mapsto y$, doesn't satisfy hypothesis (8.8).

Proposition 8.4.2

If $\varphi : S_\bullet \to R_\bullet$ *satisfies* $\sqrt{(\varphi(S_+))} = R_+$*, then hypothesis (8.8) is satisfied.*

Proof Note that
$$V(\sqrt{(\varphi(S_+))}) = V((\varphi(S_+))) = V(\varphi(S_+)) = V(R_+) = \varnothing,$$
as we desired. □

Remark 8.13 This algebraic formulation of the more geometric hypothesis can some times be easier to verify.

✍ **Exercise 8.6** **This exercise shows that different maps of graded rings can given the same map of schemes.** Let $R_\bullet = k[x, y, z]/(xz, yz, z^2)$ and $S_\bullet = k[a, b, c]/(ac, bc, c^2)$, where every variable has degree 1. Show that $\operatorname{Proj} R_\bullet \cong \operatorname{Proj} S_\bullet \cong \mathbb{P}^1_k$. Show that the maps $S_\bullet \to R_\bullet$ given by $(a, b, c) \mapsto (x, y, z)$ and $(a, b, c) \mapsto (x, y, 0)$ given the same isomorphism $\operatorname{Proj} R_\bullet \to \operatorname{Proj} S_\bullet$. (The real reason is that all of these constructions are insensitive to what happens in a finite number of degrees. This will be made precise in a number of ways later.)

Proof We claim that $R_\bullet \cong k[x, y]$. In fact we have $k[x, y] \cong k[x, y, z]/(z)$. We want to show that $R_\bullet \cong k[x, y, z]/(z)$. Define $k[x, y, z] \to k[x, y, z]/(xz, yz, z^2)$ by setting $(x, y, z) \mapsto (x, y, z)$. Clearly, this map is a surjective. Also $\operatorname{Ker}(k[x, y, z] \to k[x, y, z]/(xz, yz, z^2)) = (z)$, by the Fundamental Theorem of Isomorphism, we have $k[x, y, z]/(z) \cong R_\bullet \cong k[x, y]$. Similarly, we have $k[a, b, c]/(c) \cong S_\bullet \cong k[a, b]$. Hence $\operatorname{Proj} R_\bullet \cong \operatorname{Proj} S_\bullet \cong \mathbb{P}^1_k$.

Say $\varphi^\sharp : S_\bullet \to R_\bullet$ given by $(a, b, c) \mapsto (x, y, z)$ and $\psi^\sharp : S_\bullet \to R_\bullet$ given by $(a, b, c) \mapsto (x, y, 0)$. As in the proof of Proposition 8.4.1, we may define isomorphism $\varphi^\sharp_f : ((S_\bullet)_f)_0 \to ((R_\bullet)_{\varphi(f)})_0$ by setting
$$\frac{h(a, b, c)}{f(a, b, c)^n} \longmapsto \frac{h(x, y, z)}{f(x, y, z)^n}$$
and define isomorphism $\psi^\sharp_f : ((S_\bullet)_f)_0 \to ((R_\bullet)_{\psi(f)})_0$ by setting
$$\frac{h(a, b, c)}{f(a, b, c)^n} \longmapsto \frac{h(x, y, 0)}{f(x, y, 0)^n}.$$
We claim that $((R_\bullet)_{\varphi(f)})_0 \cong ((R_\bullet)_{\psi(f)})_0$. Note that $R_\bullet \cong k[x, y, z]/(z)$, it suffices to show that
$$((k[x, y, z]/(z))_{\varphi(f)})_0 \cong ((k[x, y, z]/(z))_{\psi(f)})_0.$$
Consider $\frac{h(x,y)}{f(x,y)^n} \mapsto \frac{h(x,y)}{f(x,y)^n}$, this morphism is clearly isomorphism. Hence we have a commutative diagram.
$$\begin{tikzcd}
((S_\bullet)_f)_0 \arrow[r, "\varphi^\sharp_f"] \arrow[d, "\sim"'] & ((R_\bullet)_{\varphi(f)})_0 \arrow[d, "\sim"] \\
((S_\bullet)_f)_0 \arrow[r, "\psi^\sharp_f"] & ((R_\bullet)_{\psi(f)})_0
\end{tikzcd}$$
It follows $\varphi^\sharp$ and $\psi^\sharp$ induce same isomorphisms of affine subschemes as in the proof of Proposition 8.4.1, and therefore $\varphi^\sharp$ and $\psi^\sharp$ give the same isomorphism $\operatorname{Proj} R_\bullet \to \operatorname{Proj} S_\bullet$. □

Remark 8.14 In Chapter 16, we will show that not every morphism of schemes $\operatorname{Proj} R_\bullet \to \operatorname{Proj} S_\bullet$ comes from a map of graded rings $S_\bullet \to R_\bullet$, even in quite reasonable circumstances.

8.4.2 Veronese subrings

Here is a useful construction.

Definition 8.4.2 (Veronese subring)

Suppose $S_\bullet$ is a finitely generated graded ring. Define the n-th ***Veronese subring*** *of $S_\bullet$ by $S_{n\bullet} = \bigoplus_{j=0}^{\infty} S_{nj}$. (The "old degree" n is "new degree" 1.)*

The geometric interpretation of Veronese subring is the important **Veronese embedding**, discussed in

Chapter 10.

Proposition 8.4.3

The map of graded rings $S_{n\bullet} \hookrightarrow S_\bullet$ induces an isomorphism $\operatorname{Proj} S_\bullet \xrightarrow{\sim} \operatorname{Proj} S_{n\bullet}$.

Proof We claim that $\operatorname{Proj} S_\bullet = \bigcup_{g\in S_+} D_+(g) = \bigcup_{\substack{f\in S_m \\ n\mid m}} D_+(f)$. Pick $[\mathfrak{p}] \in \bigcup_{g\in S_+} D_+(g)$, then $[\mathfrak{p}] \in D_+(g)$ for some $g \in S_+$. Hence $g \notin \mathfrak{p}$, say $g = g_0 + g_1 + \cdots + g_r$ where g_i is homogeneous of degree i, then exists $g_i \notin \mathfrak{p}$, and therefore $g_i^n \notin \mathfrak{p}$. It follows that $[\mathfrak{p}] \in D_+(g_i^n)$. Note that $\deg g_i^n = ni$, $g_i^n \in S_{ni}$, hence $[\mathfrak{p}] \in \bigcup_{\substack{f\in S_m \\ n\mid m}} D_+(f)$. Consequently, we have $\operatorname{Proj} S_\bullet = \bigcup_{g\in S_+} D_+(g) = \bigcup_{\substack{f\in S_m \\ n\mid m}} D_+(f)$. In fact, $\operatorname{Proj} S_{n\bullet} = \bigcup_{f\in S_{n+}} D_+(f)$. Say $i : S_{n\bullet} \hookrightarrow S_\bullet$. Define ring map $i_f^\sharp : ((S_{n\bullet})_f)_0 \to ((S_\bullet)_f)_0$ by setting $i_f^\sharp = \mathrm{id}$ (since f is homogeneous with $n \mid \deg f$, $i_f^\sharp$ is well-defined), this gives an isomorphism of affine schemes $D_+(f) \xrightarrow{\sim} D_+(f)$, where left $D_+(f)$ is open subscheme of $\operatorname{Proj} S_\bullet$ and right $D_+(f)$ is open subscheme of $\operatorname{Proj} S_{n\bullet}$. Hence $i : S_{n\bullet} \hookrightarrow S_\bullet$ induces an isomorphism $\operatorname{Proj} S_\bullet \xrightarrow{\sim} \operatorname{Proj} S_{n\bullet}$. □

Proposition 8.4.4

If $S_\bullet$ is generated in degree 1*, then $S_{n\bullet}$ is also generated in degree* 1.

Proof It suffices to show that each S_{nk} is generated by S_n. Since $S_\bullet$ is generated in degree 1, S_1 generates S_k for all k. In particular, S_1 generates S_{nk} and S_1 generates S_n. In other words, we have surjective

$$\begin{tikzcd} S_1^{\otimes nk} = (S_1^{\otimes n})^{\otimes k} \arrow[r, two heads] \arrow[dr, two heads] & (S_n)^{\otimes k} \arrow[d, two heads] \\ & S_{nk}, \end{tikzcd}$$

where surjectivity of $(S_n)^{\otimes k} \twoheadrightarrow S_{nk}$ is given by $S_1^{\otimes nk} \twoheadrightarrow S_{nk}$ and $S_1^{\otimes nk} \twoheadrightarrow (S_n)^{\otimes k}$, it follows that S_{nk} is generated by S_n, i.e., $S_{n\bullet}$ is generated in degree 1. □

Proposition 8.4.5

If $R_\bullet$ and $S_\bullet$ are the same finitely generated graded rings except in a finite number of nonzero degrees; more precisely, there exists a finite set D such that for all $d \notin D$, $S_d = R_d$. Then $\operatorname{Proj} R_\bullet \cong \operatorname{Proj} S_\bullet$.

Proof Take $n > \max D$, then for all i we have $S_{ni} = R_{ni}$, hence $S_{n\bullet} = R_{n\bullet}$ for all i, hence $R_{n\bullet} = S_{n\bullet}$. By Proposition 8.4.3, we have isomorphism $\operatorname{Proj} S_\bullet \cong \operatorname{Proj} S_{n\bullet}$ and $\operatorname{Proj} R_\bullet \cong \operatorname{Proj} R_{n\bullet}$, and therefore $\operatorname{Proj} R_\bullet \cong \operatorname{Proj} S_\bullet$. □

Proposition 8.4.6

Suppose $S_\bullet$ is generated over S_0 by $f_1, \ldots, f_n$. There exists a d such that $S_{d\bullet}$ is finitely generated in "new" degree 1 *(= "old" degree d).*

Proof Suppose there are generators $f_1, \ldots, f_n$ of degrees $d_1, \ldots, d_n$ respectively. We claim that any monomial $f_1^{a_1} \cdots f_n^{a_n}$ of degree at least $nd_1 \cdots d_n$ has $a_i \geq (\prod_j d_j)/d_i$ for some i. If not, then $a_i < (\prod_j d_j)/d_i$ for all i, hence $a_1 d_1 + \cdots + a_n d_n < n \prod_j d_j$, a contradiction. Say $m = nd_1 \cdots d_n$. We shall to show that the m-th Veronese subring is generated by elements in S_m. It suffices to show that any monomial in S_{km} can be written as a monomial in elements of S_m. Prove by induction on k. If $k = 1$, then $S_{km} = S_m$, so this case is trivial.

Suppose we have shown this to hold for some $k_0 > 1$ and let $f_1^{a_1}\cdots f_n^{a_n}$ be a monomial in $S_{(k_0+1)m}$. By above discussion, there exists $a_i > (\prod_j d_j)/d_i$ for some i, then

$$f_1^{a_1}\cdots f_n^{a_n} = f_i^{(\prod_j d_j)/d_i}\cdot f_1^{a_1}\cdots f_i^{a_i-(\prod_j d_j)/d_i}\cdots f_n^{a_n},$$

where $\deg\left(f_1^{a_1}f_2^{a_2}\cdots f_i^{a_i-(\prod_j d_j)/d_i}\cdots f_n^{a_n}\right) = (k_0+1)nd_1\cdots d_n - d_1\cdots d_n = ((k_0+1)n-1)d_1\cdots d_n$. Consider $f_1^{a_1}\cdots f_i^{a_i-(\prod_j d_j)/d_i}\cdots f_n^{a_n}$, repeat above process $n-1$ times, we get a factorization of $f_1^{a_1}\cdots f_n^{a_n}$,

$$f_1^{a_1}\cdots f_n^{a_n} = (f_1^{b_1}\cdots f_n^{b_n})\cdot(f_1^{c_1}\cdots f_n^{c_n}),$$

where $\deg(f_1^{b_1}\cdots f_n^{b_n}) = nd_1\cdots d_n$ and $\deg(f_1^{c_1}\cdots f_n^{c_n}) = k_0nd_1\cdots d_n$. By the induction hypothesis, $f_1^{c_1}\cdots f_n^{c_n}$ can be written as a monomial in elements of S_m. Hence $S_{nd_1\cdots d_n\bullet}$ is finitely generated in $S_{nd_1\cdots d_n}$, as we desired. □

Proposition 8.4.6, in combination with Proposition 8.4.3, shows that there is little harm in assuming the finitely generated graded rings are generated in degree 1, as after a regrading (or more precisely, keeping only terms of degree a multiple of d, then dividing the degree by d), this is indeed the case. This is handy, as it means that, using Proposition 8.4.3, we can assume that any finitely generated graded ring is generated in degree 1. Chapter 10 will later imply as a consequence that we can embed every Proj in some projective space.

Proposition 8.4.7 (Less important)

Suppose $S_\bullet$ is a finitely generated graded ring. Then $S_{n\bullet}$ is a finitely generated graded ring.

Proof By Proposition 8.4.6 and Proposition 8.4.3, we may assume that $S_\bullet$ is generated in degree 1. By Proposition 5.5.3, $S_\bullet$ is a finitely generated S_0-algebra, i.e., there is a surjective $S_0[x_0,\ldots,x_n] \twoheadrightarrow S_\bullet$, where $\deg x_i = 1$. Clearly, we have $S_\bullet \twoheadrightarrow S_{n\bullet}$, hence

$$S_0[x_0,\ldots,x_n] \longrightarrow\!\!\!\!\!\rightarrow S_\bullet \longrightarrow\!\!\!\!\!\rightarrow S_{n\bullet},$$

i.e., $S_{n\bullet}$ is a finitely generated graded ring. □

8.5 Rational maps from reduced schemes

Informally speaking, a "rational map" is "a morphism defined almost everywhere", much as a rational function (Definition 7.6.10, Proposition 6.2.9, Definition 6.2.3) is a name for a function defined almost everywhere. We will later see that in good situations, just as with rational functions, where a rational map is defined (the Reduced-to-Separated Theorem), and has a largest "domain of definition". ***For this section only, we assume X to be reduced.*** A key example will be irreducible varieties, and the language of rational maps is most often used in this case.

8.5.1 Definition: Rational map

Definition 8.5.1 (Rational map of reduced schemes)

*A **rational map** π **of reduced schemes**, from X to Y, denoted $\pi : X \dashrightarrow Y$, is the data of a morphism $\alpha : U \to Y$ from a dense open set $U \subseteq X$, with the equivalence relation $(\alpha : U \to Y) \sim (\beta : V \to Y)$ if there is a dense open set $Z \subseteq U \cap V$ such that $\alpha|_Z = \beta|_Z$.*

Remark 8.15

(1) In Chapter 12, we will improve this to: if $\alpha|_{U\cap V} = \beta|_{U\cap V}$ in good circumstances — when Y is separated.

(2) People often use the word "map" for "morphism", which is quite reasonable, except that a rational map need not be a map. So to avoid confusion, when one means "rational map", one should never just say "map".

We will also discuss rational maps of S-schemes for a scheme S.

Definition 8.5.2 (Rational maps of S-schemes)

*A **rational map** π **of** S**-schemes**, from X to Y, denoted $\pi : X \dashrightarrow Y$, is the data of a morphism of S-schemes $\alpha : U \to Y$ from a dense open set $U \subseteq X$, with the equivalence relation $(\alpha : U \to Y) \sim (\beta : V \to Y)$ if there is a dense open set $Z \subseteq U \cap V$ such that $\alpha|_Z = \beta|_Z$.*

Remark 8.16 ⋆ **Rational maps more generally.** Just as with rational functions, Definition 8.5.1 can be extended to where X is not reduced, as is (using the same name, "rational map"), or in a version that imposes some control over what happens over the nonreduced locus (pseudo-morphisms, see [Sta25, Remark 01RX]). We will see in Chapter 12 that rational maps from reduced schemes to separated schemes behave particularly well, which is why they are usually considered in this context. the reason for the definition of pseudo-morphisms is to extend these results to when X is nonreduced. We will not use the notion of pseudo-morphism.

Example 8.8 An obvious example of a rational map is a morphism. Another important example is the projection $\mathbb{P}^n_A \dashrightarrow \mathbb{P}^{n-1}_A$ given by $[x_0 : \cdots : x_n] \mapsto [x_0 : \cdots : x_{n-1}]$. (Pick the dense open subset $D_+(x_n) \hookrightarrow \mathbb{P}^n_A$, then $D_+(x_n) = \bigcup\limits_{i=0}^{n-1} D_+(x_i x_n)$, define $D_+(x_i x_n) \to \operatorname{Spec}(k[x_0, \ldots, x_{n-1}]_{x_i})_0$ by $[x_0 : \cdots : x_n] \mapsto [x_0 : \cdots : x_{n-1}]$, we glue these morphisms together, then obtain a morphism of schemes $D_+(x_n) \to \mathbb{P}^{n-1}_A$. Hence projection is rational map.)

✍ **Exercise 8.7** Suppose X is an integral (irreducible and reduced) scheme. Describe how rational maps $X \dashrightarrow \mathbb{A}^1$ should be identified with rational functions on X.

Proof Let $\pi : X \dashrightarrow \mathbb{A}^1$ be a rational map, pick $(\alpha : U \to \mathbb{A}^1)$ be a representative morphism, then we get a ring map $\alpha^\sharp : k[t] \to \Gamma(U, \mathscr{O}_X)$, which is determined by $t \mapsto \alpha^\sharp(t)$. Define φ from rational maps $X \dashrightarrow \mathbb{A}^1$ to rational functions on X, by setting $\varphi(\alpha) = \alpha^\sharp(t)$. We next show φ is well defined. If $(\beta : V \to \mathbb{A}^1)$ be another representative, then $(\alpha : U \to \mathbb{A}^1) \sim (\beta : V \to \mathbb{A}^1)$, i.e., there exists a dense open set $Z \subseteq U \cap V$ such that $\alpha|_Z = \beta|_Z$, hence $(\alpha|_Z)^\sharp(t) = (\beta|_Z)^\sharp(t)$, which implies that $\alpha^\sharp(t)|_Z = \beta^\sharp(t)|_Z$. It follows that φ is a well-defined.

Let (U, f) be a rational function on X, where U is a dense open subset of X. Define ring map $\alpha^\sharp : k[t] \to \Gamma(U, \mathscr{O}_X)$ by setting $t \mapsto \alpha^\sharp(t)$. It induces a morphism of scheme $\alpha : U \to \operatorname{Spec} k[t] = \mathbb{A}^1$. Let $(U, f) \sim (V, g)$, then exists dense open subset $Z \subseteq U \cap V$ such that $f|_Z = g|_Z$. By above discussion $g|_Z$ induces a morphism of schemes $\beta : V \to \mathbb{A}^1$. We want to show that $(\alpha : U \to \mathbb{A}^1) \sim (\beta : V \to \mathbb{A}^1)$. Restrict α and β to Z, then they induced by the same ring maps, hence $(\alpha : U \to \mathbb{A}^1) \sim (\beta : V \to \mathbb{A}^1)$. Define ψ from rational functions on X to rational maps $X \dashrightarrow \mathbb{A}^1$ by setting $(U, f) \to (\alpha : U \to \operatorname{Spec} k[t])$ where $\alpha^\sharp(t) = f$.

By the construction of φ and ψ, there is a bijection between rational maps $X \dashrightarrow \mathbb{A}^1$ and rational functions on X, as we desired. □

Motivated by Exercise 8.7, we can define rational functions on schemes.

Definition 8.5.3 (Rational function)

Let X be a reduced scheme. A rational function on X is rational map $X \dashrightarrow \mathbb{A}^1$.

Note *We denote **the set of rational functions by** $R(X)$. As explained above we may consider $R(X)$ also* *the set of equivalence classed of pairs $(U, \widetilde{s})$, where U is an open scheme-theoretically dense subset of X and $\widetilde{s} \in \Gamma(U, \mathscr{O}_X)$ is a function on U. In other words, we have*

$$R(X) := \operatorname{colim}_U \Gamma(U, \mathscr{O}_X),$$

where U runs through the directed set of open scheme-theoretically dense subschemes of X.

Definition 8.5.4 (Dominant)

A rational map $\pi : X \dashrightarrow Y$ is ***dominant*** *(or in some sources, dominating) if for some (and hence every) representative $U \to Y$, the image is dense in Y. A morphism is a* ***dominant morphism*** *(or dominating morphism) if it is dominant as a rational map.*

Remark 8.17 **Why every image of representative $U \to Y$ is dense in Y?** Topology fact: if $U \subseteq U'$ are dense and $f(U)$ is dense then $f(U')$ must be dense. Let $(\alpha : U \to \mathbb{A}^1)$ and $(\beta : K \to \mathbb{A}^1)$ be two representative of rational map $X \dashrightarrow Y$, also let $\alpha(U)$ dense in Y. Then there exists dense open subset $Z \subseteq U \cap K$ such that $\alpha|_Z = \beta|_Z$. We next show that $\alpha(Z)$ is dense. If suffices to show that $\alpha(Z) \cap V \neq \varnothing$ for all open subset $V \subseteq Y$. Since $\alpha(U)$ is dense in Y, for all open subset $V \subseteq Y$, $\alpha(U) \cap V \neq \varnothing$, hence $U \cap \alpha^{-1}(V) \neq \varnothing$. Since Z is dense in X, $Z \cap (U \cap \alpha^{-1}(V)) = Z \cap \alpha^{-1}(V) \neq \varnothing$ for all open subset $V \subseteq Y$, hence $\alpha(Z) \cap V \neq \varnothing$ for all open subset $V \subseteq Y$. It follows that $\alpha(Z)$ is dense in Y. Note that $Z \subseteq K$, by the topology fact and $\alpha(Z) = \beta(Z)$, $\beta(K)$ is dense in Y.

Proposition 8.5.1

Suppose X, Y are reduced scheme. A rational map $\pi : X \dashrightarrow Y$ of irreducible schemes is dominant if and only if π sends the generic point of X to the generic point of Y. (In fact, by Proposition 6.2.7, X and Y are integral schemes.)

Proof Since X and Y are irreducible schemes, we may assume that $X = \overline{\{\xi\}}$ and $Y = \overline{\{\eta\}}$, where ξ and η are generic point. If $\pi : X \dashrightarrow Y$ is dominant. Let $\alpha : U \to Y$ be a representative of $\pi : X \dashrightarrow Y$. Since ξ is generic point of X, by Proposition 4.6.10, we have $\xi \in U$. Consider $\overline{\alpha(\xi)}$, we have $\overline{\alpha(\xi)} \subseteq \overline{\alpha(U)} = Y$, since Y is irreducible, $\overline{\alpha(\xi)} = Y$. By Proposition 6.1.2, we have $\alpha(\xi) = \eta$. It follows that π sends the generic point of X to the generic point of Y. Conversely, if $\pi(\xi) = \eta$, pick $\alpha : U \to Y$ be a representative of π, then $\alpha(\xi) = \eta$. Hence $\overline{\alpha(\xi)} = \overline{\{\eta\}} = Y$. Note that $\alpha(U) \supseteq \alpha(\xi)$, we have $\overline{\alpha(U)} \supseteq Y$, i.e., $\overline{\alpha(U)} = Y$. Hence $\pi : X \dashrightarrow Y$ is dominant. □

Remark 8.18 A little thought will convince you that you can compose (in a well-defined way) a dominant map $\pi : X \dashrightarrow Y$ from an integral scheme X to an integral scheme Y with a rational map $\phi : Y \dashrightarrow Z$. Furthermore, the composition $\rho \circ \pi$ will be dominant if ρ is dominant. Integral schemes and dominant rational maps between them form a category which is geometrically interesting.

Proposition 8.5.2

Dominant rational maps of integral schemes give morphisms of function fields in the opposite direction.

Proof Let $\pi : X \dashrightarrow Y$ be a dominant rational map, say $(\alpha : U \to Y)$ be a representative of π. Suppose ξ is the generic point of X and η is the generic point of Y, by Proposition 8.5.1, $\alpha(\xi) = \eta$. From $\alpha : U \to Y$, we get a morphism of sheaves $\alpha^\sharp : \mathscr{O}_Y \to \alpha_*(\mathscr{O}_X|_U)$. By Proposition 8.2.6, it induce a map of stalks $\alpha^\sharp_\xi : \mathscr{O}_{Y,\eta} \to \mathscr{O}_{X,\xi}$. By Proposition 6.2.9, we have $\mathscr{O}_{Y,\eta} \cong K(Y)$ and $\mathscr{O}_{X,\xi} \cong K(X)$, and therefore we get a morphism of function fields $\alpha^\sharp_\xi : K(Y) \to K(X)$, as we desired. □

In "suitably classical situations" (integral finite type k-schemes — and in particular, irreducible varieties, to be defined in Chapter 12) — this is reversible: dominant rational maps correspond to inclusions of function fields in the opposite direction. We make this precise in below. But it is not true that morphisms of function fileds *always* give dominant rational maps, or even rational maps. For example:

Example 8.9 $\operatorname{Spec} k[x]$ and $\operatorname{Spec} k(x)$ have the same function field $k(x)$, but there is no corresponding rational map $\operatorname{Spec} k[x] \dashrightarrow \operatorname{Spec} k(x)$ of k-schemes. Reason: such a rational map would correspond to a morphism from an open subset U of $\operatorname{Spec} k[x]$, say $\operatorname{Spec} k[x, 1/f(x)]$, to $\operatorname{Spec} k(x)$. But there is no map of rings $k(x) \to k[x, 1/f(x)]$ (sending k identically to k and x to x) for any one $f(x)$.

Next proposition gives a evidence that the topologically-defined notion of dominance is simultaneously algebraic.

Proposition 8.5.3

If $\varphi : A \to B$ is a ring morphism, then the corresponding morphism $\operatorname{Spec} B \to \operatorname{Spec} A$ is dominant if and only if φ has kernel contained in the nilradical of A.

Proof Say $\pi : \operatorname{Spec} B \to \operatorname{Spec} A$ induced by ring map $\varphi : A \to B$. Consider $\overline{\pi(\operatorname{Spec} B)}$, by Proposition 4.7.1,

$$\overline{\pi(\operatorname{Spec} B)} = V(I(\pi(\operatorname{Spec} B))) = V\left(\bigcap_{\mathfrak{q}\in\pi(\operatorname{Spec} B)} \mathfrak{q}\right).$$

π is dominant if and only if $V\left(\bigcap_{\mathfrak{q}\in\pi(\operatorname{Spec} B)} \mathfrak{q}\right) = \operatorname{Spec} A$, by Theorem 4.2.3, if and only if $\bigcap_{\mathfrak{q}\in\pi(\operatorname{Spec} B)} \mathfrak{q} \subseteq \mathfrak{N}(A)$. Note that each $\mathfrak{q} \in \pi(\operatorname{Spec} B)$ has the form $\mathfrak{q} = \varphi^{-1}(\mathfrak{p})$ where $\mathfrak{p} \in \operatorname{Spec} B$, it follows that

$$\bigcap_{\mathfrak{q}\in\pi(\operatorname{Spec} B)} \mathfrak{q} = \varphi^{-1}(\mathfrak{N}(B)).$$

Hence π is dominant if and only if $\varphi^{-1}(\mathfrak{N}(B)) \subseteq \mathfrak{N}(A)$.

We next show that $\varphi^{-1}(\mathfrak{N}(B)) \subseteq \mathfrak{N}(A)$ if and only if $\operatorname{Ker}\varphi \subseteq \mathfrak{N}(A)$. Pick $f \in \operatorname{Ker}\varphi$, we want to show that $f \in \mathfrak{N}(A)$. Since $f \in \operatorname{Ker}\varphi$, $\varphi(f) = 0$, and therefore $\varphi(f) \in \mathfrak{N}(B)$, i.e., $f \in \varphi^{-1}(\mathfrak{N}(B)) \subseteq \mathfrak{N}(A)$.

Conversely, if $\operatorname{Ker}\varphi \subseteq \mathfrak{N}(A)$, we want to show that $\varphi^{-1}(\mathfrak{N}(B)) \subseteq \mathfrak{N}(A)$. Pick $f \in \varphi^{-1}(\mathfrak{N}(B))$, then $\varphi(f) \in \mathfrak{N}(B)$. Hence $\varphi(f)^n = \varphi(f^n) = 0$ for some n, and therefore $f^n \in \operatorname{Ker}\varphi \subseteq \mathfrak{N}(A)$, thus $f \in \mathfrak{N}(A)$. $\square$

The following proposition is useful.

Proposition 8.5.4

Let $\pi : X \to Y$ be a morphism of integral schemes with respective generic points ξ_X, ξ_Y. The following properties are equivalent:

(i) π is dominant;

(ii) $\pi(\xi_X) = \xi_Y$;

(iii) for some nonempty affine opens $U \subseteq X$ and $V \subseteq Y$ with $\pi(U) \subseteq V$ the ring map $\mathscr{O}_Y(V) \to \mathscr{O}_X(U)$ is injective;

(iv) for all nonempty affine opens $U \subseteq X$ and $V \subseteq Y$ with $\pi(U) \subseteq V$ the ring map $\mathscr{O}_Y(V) \to \mathscr{O}_X(U)$ is injective;

(v) for some $x \in X$ with image $y = \pi(x) \in Y$ the local ring map $\mathscr{O}_{Y,y} \to \mathscr{O}_{X,x}$ is injective;

(vi) for all $x \in X$ with image $y = \pi(x) \in Y$ the local ring map $\mathscr{O}_{Y,y} \to \mathscr{O}_{X,x}$ is injective;

(vii) $\pi^\sharp : \mathscr{O}_Y \to \pi_*\mathscr{O}_X$ is monomorphism.

Proof (i) $\Leftrightarrow$ (ii): Proposition 8.5.1. Let $U \subseteq X$ and $V \subseteq Y$ be nonempty affine opens with $\pi(U) \subseteq V$. Since X, Y are integral domain, $A := \mathscr{O}_X(U)$ and $B := \mathscr{O}_Y(V)$ are integral domains. The morphism $\pi|_U : U \to V$ corresponds to a ring map $\pi|_U^\sharp : B \to A$. The generic points of X and Y correspond to the prime ideals $(0) \in \operatorname{Spec} A$ and $(0) \in \operatorname{Spec} B$. Thus (ii) is equivalent to the condition $(0) = (\pi|_U^\sharp)^{-1}((0))$, i.e., to the condition that $\pi|_U^\sharp$ is injective. In this way we see that (ii) $\Leftrightarrow$ (iii) $\Leftrightarrow$ (iv). Similarly, given x and y as in (v), the local rings $\mathscr{O}_{X,x}$ and $\mathscr{O}_{Y,y}$ are domains and the prime ideals $(0) \in \operatorname{Spec} \mathscr{O}_{X,x}$ and $(0) \in \operatorname{Spec} \mathscr{O}_{Y,y}$ correspond to the generic points of X and Y (Proposition 8.3.10). Thus we can argue in the same manner as above to see that (ii) $\Leftrightarrow$ (v) $\Leftrightarrow$ (vi). (vi) $\Leftrightarrow$ (vii) is given by Proposition 3.4.12. □

Definition 8.5.5 (Birational)

(i) A rational map $\pi : X \dashrightarrow Y$ is said to be ***birational*** *if it is dominant, and there is another rational map $\psi : Y \dashrightarrow X$ that is also dominant, such that $\pi \circ \psi$ is (in the same equivalence class as) the identity (as rational map) on Y, and $\psi \circ \pi$ is (in the same equivalence class as) the identity (as rational map) on X.*

(ii) A morphism is a ***birational morphism*** *if it is birational as a rational map. (Note that the "inverse" to a birational morphism may be only a rational map, not a morphism.)*

(iii) We say X and Y are ***birational*** *(to each other) if there exists a birational map $X \dashrightarrow Y$.*

Remark 8.19

(i) Birational map is the notion of isomorphism in the category of integral schemes and dominant rational maps. (In the differentiable category, this is not particularly interesting.)

(ii) If X and Y are irreducible, then birational maps induce isomorphisms of function fields. The fact that maps of function fields correspond to rational maps in the opposite direction for integral finite type k-schemes, to be proved in Proposition 8.5.5, shows that a map between integral finite type k-schemes that induces an isomorphism of function fields is birational.

Definition 8.5.6 (Rational)

An integral finite type k-scheme is said to be ***rational*** *if it is birational to $\mathbb{A}_k^n$ for some n.*

Proposition 8.5.5

Suppose X and Y are reduced schemes. Then X and Y are birational if and only if there is a dense open subscheme U of X and a dense open subscheme V of Y such that $U \cong V$.

Remark 8.20

(a) We have "integral" hypotheses can be dropped with sufficient care and modified definitions.

(b) Proposition 8.5.5 tells you how to think of birational maps. Just as a rational map is a "mostly-defined function", two birational reduced schemes are "mostly isomorphic". For example, a reduced finite type k-scheme (such as an affine k-variety) is rational if and only if it has a dense open subscheme isomorphic to an open subscheme of $\mathbb{A}_k^n$.

Proof The "if" direction is immediate, so we prove the "only if" direction. We basically follow our nose, and use what we are given.

f	$g \circ f$	$g \circ f$ $=\mathrm{id}$ as:		g	$f \circ g$	$f \circ g$ $=\mathrm{id}$ as:
rational map	rational map	rational map	$X \overset{\sim}{\dashleftarrow\dashrightarrow} Y$	rational map	rational map	rational map
morphism	"	"	$X_1 \quad Y_1$	morphism	"	"
"	morphism	"	X_2			
"	"	morphism	$X_3 \leftarrow Y_2$	"	morphism	"
			$X_4 \overset{\sim}{\leftrightarrow} Y_3$	"	"	morphism

We have inverse rational maps $F : X \dashrightarrow Y$ and $G : Y \dashrightarrow X$. Choose representative morphisms $f : X_1 \to Y$ (where $X_1 \subseteq X$) and $g : Y_1 \to X$ (where $Y_1 \subseteq Y$) for the rational maps F and G respectively. If $X_2 = (f|_{X_1})^{-1}(Y_1) \subseteq X_1$, then $g \circ f$ is a morphism from X_2 to X that is the identity as a rational map. Thus there is a dense open subset $X_3 \subseteq X_2$ such that the morphism $g \circ f : X_3 \to X$ is the identity morphism on X_3 (or more precisely, it is the open embedding $X_3 \hookrightarrow X$).

Similarly, let $Y_2 = (g|_{Y_1})^{-1}(X_3) \subseteq Y_1$, so $f \circ g$ is a morphism from $Y_2 \to Y$ that is the identity as a rational map. Then let $Y_3 \subseteq Y_2$ be a dense open subset such that $f \circ g : Y_3 \to Y$ is the inclusion (the "identity").

Finally, if $X_4 = (f|_{X_3})^{-1}(Y_3) \subseteq X_3$, then $(g \circ f)|_{X_4}$ is the identity morphism on X_4 (by way of Y_3), and $(f \circ g)|_{Y_3}$ is the identity morphism on Y_3 (by way of X_4), so we have found our isomorphism of open sets that is a "representative" for our birational map. $\square$

8.5.2 Rational maps of irreducible varieties

Proposition 8.5.6

Suppose X is an integral k-scheme and Y is an integral finite type k-scheme, and we are given an extension of function fields $\varphi^\sharp : K(Y) \hookrightarrow K(X)$ preserving k. Then there exists a dominant rational map of k-schemes $\varphi : X \dashrightarrow Y$ inducing $\varphi^\sharp$.

Hence, if X and Y are integral finite type k-schemes, then birational maps induce isomorphisms of function fields (by the proof of Proposition 8.5.6, we get it immediately).

Proof By replacing Y with an open subset, we may assume that Y is affine, say $\operatorname{Spec} B$, where B is generated over k by finitely many elements $y_1, \ldots, y_n$. Since we only need to define φ on an open subset of X, we may similarly assume that $X = \operatorname{Spec} A$ is affine. Then $\varphi^\sharp$ gives an inclusion $\varphi^\sharp : B \hookrightarrow K(A)$. Write $\varphi^\sharp(y_i)$ as f_i/g_i ($f_i, g_i \in A$), and let $g := \prod_{i=1}^{n} g_i$. Therefore $\varphi : \operatorname{Spec} A_g \to \operatorname{Spec} B$ induces $\varphi^\sharp$. The morphism φ is dominant because the inverse image of the zero ideal under the inclusion $B \hookrightarrow A_g$ is the zero ideal, so φ takes the generic point of X to the generic point of Y, by Proposition 8.5.1, φ is dominant. $\square$

Definition 8.5.7 (Finitely generated (field extension))

Let K be a finitely generated field extension of k, if there is a finite "generating set" $x_1, \ldots, x_n$ in K such that every element of K can be written as a rational function function in $x_1, \ldots, x_n$ with coefficients in k.

Proposition 8.5.7

Let K be a finitely generated field extension of k, then there exists an irreducible affine k-variety with function field K.

Proof Consider the map $\varphi : k[t_1, \ldots, t_n] \to K$ given by $t_i \mapsto x_i$. Since the image of $k[t_1, \ldots, t_n]$ is the subring of field K, by the Fundamental theorem of isomorphism, $\varphi(k[t_1, \cdots, t_n]) \cong k[t_1, \ldots, t_n]/\mathfrak{p}$, where $\mathfrak{p}$ is a prime ideal. Consider $K(k[t_1, \ldots, t_n]/\mathfrak{p}) \to K$, then φ induce a homomorphism $\varphi^\dagger : K(k[t_1, \cdots, t_n]/\mathfrak{p}) \to K$ given by $\overline{t_i} \mapsto x_i$ where $\overline{t_i}$ is the image of t_i in $K(k[t_1, \cdots, t_n]/\mathfrak{p})$. Clearly, $\varphi^\dagger$ is a surjective, also $K(k[t_1, \cdots, t_n]/\mathfrak{p}) \hookrightarrow K$, we have $K(k[t_1, \cdots, t_n]/\mathfrak{p}) \cong K$. It follows that $\operatorname{Spec} k[t_1, \ldots, t_n]/\mathfrak{p}$ is an irreducible affine k-variety with function field K. $\square$

Remark 8.21 Interpreted geometrically: consider the map $\operatorname{Spec} K \to \operatorname{Spec} k[t_1, \ldots, t_n]$ given by the ring map $t_i \mapsto x_i$, and take the closure of the one-point image (say p be the image of $[(0)]$ in $\operatorname{Spec} k[t_1, \ldots, t_n]$, then $\overline{\{p\}} = \operatorname{Spec}(k[t_1, \ldots, t_n]/\mathfrak{p})$ for some prime ideal $\mathfrak{p}$, also $K(k[t_1, \ldots, t_n]/\mathfrak{p}) \cong K$).

Proposition 8.5.8

The following two categories are equivalent.

(a) the category with objects "integral affine k-varieties", and morphisms "dominant rational maps defined over k"; and

(b) the opposite category with objects "finitely generated field extensions of k", and morphisms "inclusions extending the identity on k".

Proof Denote $\mathscr{C}$ be the category with objects "integral affine k-varieties", and morphisms "dominant rational maps defined over k"; and denote $\mathscr{D}$ be the category with objects "finitely generated field extensions of k", and morphisms "inclusions extending the identity on k". Define functor $F : \mathscr{C} \to \mathscr{D}^{\mathrm{op}}$ as follow. Let $X \in \operatorname{obj}(\mathscr{C})$, define $F(X) = K(X)$. Let $X \dashrightarrow Y$ be a dominant rational map defined over k. By Proposition 8.5.2, it induces a morphism of function fields $K(Y) \to K(X)$, hence define $F(X \dashrightarrow Y)$ by setting $K(Y) \to K(X)$. Then we defined a functor $F : \mathscr{C} \to \mathscr{D}^{\mathrm{op}}$.

Define functor $G : \mathscr{D}^{\mathrm{op}} \to \mathscr{C}$ as follow. Let $K \in \operatorname{obj}(\mathscr{D}^{\mathrm{op}})$, by Proposition 8.5.7, $K \cong K(X)$ where X is an integral k-variety, hence set $G(K(X)) = X$. Let $K(Y) \to K(X)$ be an extension of function fields preserving k, by Proposition 8.5.6, there exists a dominant rational map of k-schemes $X \dashrightarrow Y$ inducing $K(Y) \to K(X)$, so we define $G(K(Y) \to K(X))$ by setting $X \dashrightarrow Y$. Then we defined a functor $G : \mathscr{D}^{\mathrm{op}} \to \mathscr{C}$.

We next show that $F \circ G \cong \mathrm{id}_{\mathscr{D}^{\mathrm{op}}}$ and $G \circ F \cong \mathrm{id}_{\mathscr{C}}$. Let $K(Y) \xrightarrow{\varphi} K(X)$ in $\mathscr{D}^{\mathrm{op}}$. Then

$$F \circ G(K(Y) \xrightarrow{\varphi} K(X)) = F(X \dashrightarrow Y) = K(Y) \to K(X).$$

Hence the following diagram commutes.

$$\begin{tikzcd} F \circ G(K(Y)) \arrow[r] \arrow[d] & F \circ G(K(X)) \arrow[d] \\ K(Y) \arrow[r] & K(X) \end{tikzcd}$$

Let $X \overset{\pi}{\dashrightarrow} Y$ in $\mathscr{C}$. We may assume that $X = \operatorname{Spec} A$, then $G \circ F(X) = G(K(X)) = \operatorname{Spec} A'$ where $K(A') = K(X)$. Hence there exists a birational map such that

$$X \dashleftarrow\dashrightarrow \operatorname{Spec} A'.$$

We may assume that $A = A'$. Consider $G \circ F(\pi)$, then

$$G \circ F(\pi) = G(K(Y) \to K(X)) = \operatorname{Spec} A \dashrightarrow \operatorname{Spec} B = \pi$$

where $Y = \operatorname{Spec} B$. Hence the following diagram commutes.

$$\begin{tikzcd} G \circ F(X) = \operatorname{Spec} A \arrow[r, dashed, "G\circ F(\pi)"] \arrow[d, dashed, leftrightarrow, "\sim"'] & G \circ F(Y) = \operatorname{Spec} B \arrow[d, dashed, leftrightarrow, "\sim"] \\ X \arrow[r, dashed, "\pi"] & Y \end{tikzcd}$$

By above discussion, $\mathscr{C}$ and $\mathscr{D}$ are equivalent. $\square$

Remark 8.22 Once we define varieties in general in Chapter 12, you can add in: (c) the category with objects "integral k-varieties", and morphisms "dominant rational maps defined over k".

Remark 8.23 In particular, an integral affine k-variety X is rational if its function field $K(X)$ is a purely transcendental extension of k, i.e., $K(X) \cong k(x_1, \ldots, x_n)$ for some n. (This needs to be said more precisely: the map $k \hookrightarrow K(X)$ induced by $X \to \operatorname{Spec} k$ should agree with the "obvious" map $k \hookrightarrow k(x_1, \ldots, x_n)$ under this isomorphism.)

8.5.3 More examples of rational maps

A recurring theme in these examples is that domains of definition of rational maps to projective schemes extend over regular codimension one points. We will make this precise in the Curve-to-Projective Extension Theorem, when we discuss curves.

The classical formula for Pythagorean triples

The first example is the classical formula for Pythagorean triples, and its derivation by "stereographic projection". Suppose you are looking for rational points on the circle C given by $x^2 + y^2 = 1$ (Figure 8.1). One rational point is $p = (1, 0)$. If q is another rational point, then pq is a line of rational (non-infinite) slope. This gives a rational map from the conic C (now interpreted as $\operatorname{Spec} \mathbb{Q}[x, y]/(x^2 + y^2 - 1)$) to $\mathbb{A}^1_{\mathbb{Q}}$, given by $(x, y) \mapsto y/(x - 1)$. (Something subtle just happened: we were talking about $\mathbb{Q}$-points on a circle, and ended up with a rational map of schemes.)

Conversely, given a line of slope m through p, where m is rational, we can recover q by solving the equations

$$\begin{cases} m(x-1) - y & = 0, \\ x^2 + y^2 - 1 & = 0. \end{cases}$$

We substitute the first equation into the second, to get a quadratic equation in x. We know that we will have a solution $x = 1$ (because the line meets the circle at $(x, y) = (1, 0)$), so we expect to be able to factor this out, and find the other factor. This indeed works:

$$\begin{aligned} & x^2 + (m(x-1))^2 && = 1 \\ \Longrightarrow & (m^2+1)x^2 + (-2m^2)x + (m^2 - 1) && = 0 \\ \Longrightarrow & (x-1)((m^2+1)x - (m^2-1)) && = 0 \end{aligned}$$

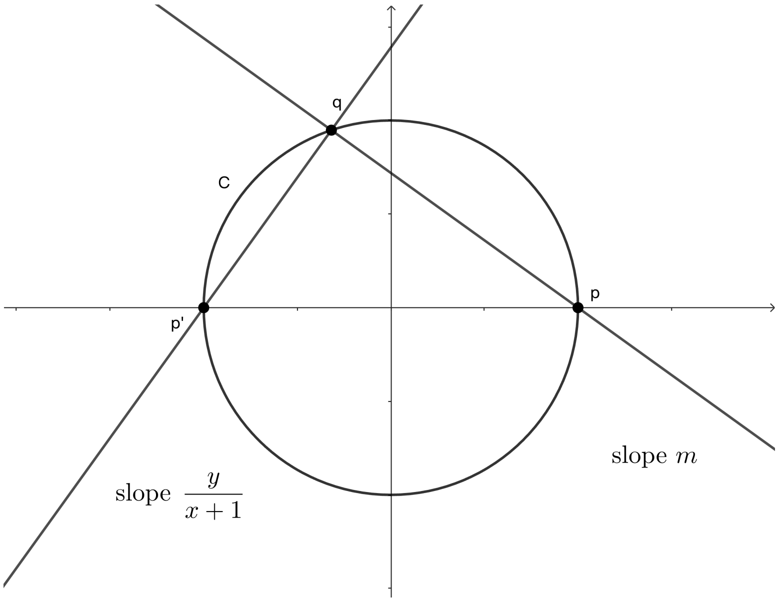


Figure 8.1: Finding primitive Pythagorean triples using geometry

The other solution is $x = \frac{m^2-1}{m^2+1}$, which gives $y = \frac{-2m}{m^2+1}$. Thus we get a birational map between the conic C and $\mathbb{A}^1_{\mathbb{Q}}$ with coordinate m, given by $f : (x, y) \mapsto \frac{y}{x-1}$ which is defined for $x \neq 1$, and with inverse rational map given by $m \mapsto \left(\frac{m^2-1}{m^2+1}, \frac{-2m}{m^2+1}\right)$ which is defined away from $m^2 + 1 = 0$.

We can extend to a rational map $C \dashrightarrow \mathbb{P}^1_{\mathbb{Q}}$ via the "inclusion" $\mathbb{A}^1_{\mathbb{Q}} \to \mathbb{P}^1_{\mathbb{Q}}$ which we later call an open embedding. Then f is given by $(x, y) \mapsto [y : x - 1]$. We then have an interesting question: *what is the domain of definition of f?* It appears to be defined everywhere except for where $y = x - 1 = 0$, i.e., everywhere but p. But in fact it can be extended over p! Note that $(x, y) \mapsto [x + 1 : -y]$ where $(x, y) \neq (-1, 0)$ agrees with f on their common domains of definition, as $[x + 1 : -y] = [y, x - 1]$. Hence this rational map can be extended farther than we at first thought. This will be a special case of the Curve-to-Projective Extension Theorem.

✍ **Exercise 8.8** Use the above to find a "formula" yielding all Pythagorean triples.

Proof $(0, 0, 0)$ is trivial. Consider equation $x^2 + y^2 = z^2$ with $z \neq 0$ and $\gcd(x, y, z) = 1$. Then we get an equation $\left(\frac{x}{z}\right)^2 + \left(\frac{y}{z}\right)^2 = 1$. By previous paragraph we know that we must have $\left(\frac{x}{z}, \frac{y}{z}\right) = (1, 0)$ or $\left(\frac{x}{z}, \frac{y}{z}\right) = \left(\frac{m^2-1}{m^2+1}, \frac{-2m}{m^2+1}\right)$. Hence $x = z = 1, y = 0$ or $x = m^2 - 1, y = -2m, z = m^2 + 1$ for some m. $\square$

✍ **Exercise 8.9** Show that the conic $x^2 + y^2 = z^2$ in $\mathbb{P}^2_k$ is isomorphic to $\mathbb{P}^1_k$ for any field k of characteristic not 2.

Proof In fact, $x^2 + y^2 = z^2$ in $\mathbb{P}^2_k$ can be written as $\operatorname{Proj}\left(k[x, y, z]/(x^2 + y^2 - z^2)\right)$. Define $\varphi^\sharp : k[x, y, z]/(x^2 + y^2 - z^2) \to k[u, v]$ by setting

$$x \mapsto u^2 - v^2,\ y \mapsto -2uv,\ z \mapsto u^2 + v^2.$$

Note that $\varphi^\sharp(\left(k[x, y, z]/(x^2 + y^2 - z^2)\right)_+) = \varphi^\sharp(x, y, z) = (u^2 - v^2, -2uv, u^2 + v^2) = (u^2, uv, v^2)$, hence

$$V(\varphi^\sharp(\left(k[x, y, z]/(x^2 + y^2 - z^2)\right)_+)) = V(u^2, uv, v^2) = \varnothing.$$

By Proposition 8.4.1, $\varphi^\sharp$ induces a morphism of schemes $\varphi : \mathbb{P}^1_k \to \operatorname{Proj}(k[x, y, z]/(x^2 + y^2 - z^2))$ which given by

$$[a, b] \longmapsto [a^2 - b^2 : -2ab : a^2 + b^2].$$

We next show that φ is an isomorphism. We claim that $\operatorname{Proj}(k[x, y, z]/(x^2+y^2-z^2)) = D_+(x-z) \cup D_+(x+z)$. By Proposition 5.5.8, it suffices to show that $V(x - z, x + z) = \varnothing$. If not, there exists homogeneous prime $[\mathfrak{p}] \in V(x - z, x + z)$, i.e., $x - z, x + z \in \mathfrak{p}$. Hence $x, z \in \mathfrak{p}$. Note that in $\operatorname{Proj}(k[x, y, z]/(x^2 + y^2 -$

$z^2)$), we have $y^2 = z^2 - x^2$, hence $y^2 \in \mathfrak{p}$. Since $\mathfrak{p}$ is prime ideal, we have $y \in \mathfrak{p}$. It follows that $\mathfrak{p} = (k[x,y,z]/(x^2+y^2-z^2))_+$, contradicts to the fact that $\mathfrak{p} \neq (k[x,y,z]/(x^2+y^2-z^2))_+$. Hence $\operatorname{Proj}(k[x,y,z]/(x^2+y^2-z^2)) = D_+(x-z) \cup D_+(x+z)$. In fact, we have $\mathbb{P}^1_k = D_+(u) \cup D_+(v)$. Consider $\varphi^\sharp_{x-z}$ and $\varphi^\sharp_{x+z}$. Note that

$$\begin{aligned}((k[x,y,z]/(x^2+y^2-z^2))_{x-z})_0 &\cong k\left[\frac{x}{x-z}, \frac{y}{x-z}, \frac{z}{x-z}\right] \Big/ \left(\frac{x^2}{(x-z)^2} + \frac{y^2}{(x-z)^2} - \frac{z^2}{(x-z)^2}\right) \\ &\cong k\left[\frac{x}{x-z}, \frac{y}{x-z}\right] \Big/ \left(\frac{y^2}{(x-z)^2} + \frac{2x}{x-z} - 1\right) \\ &\cong k\left[\frac{y}{x-z}\right]\end{aligned}$$

and

$$(k[u,v]_v)_0 \cong k\left[\frac{u}{v}\right],$$

we know that $\varphi^\sharp_{x-z} : ((k[x,y,z]/(x^+y^2-z^2))_{x-z})_0 \to (k[u,v]_v)_0$ which given by $\frac{y}{x-z} \mapsto \frac{u}{v}$ is an isomorphism. It induces an isomorphism of affine schemes $D_+(x-z) \xrightarrow{\sim} D_+(v)$. Similarly, we have $D_+(x+z) \xrightarrow{\sim} D_+(u)$. Hence

$$\mathbb{P}^1_k \xrightarrow{\sim} \operatorname{Proj}(k[x,y,z]/(x^2+y^2-z^2)),$$

as we desired. □

Remark 8.24 In fact, any conic in $\mathbb{P}^2_k$ with a k-valued point (i.e., a point with residue field k) of rank 3 (after base change to $\overline{k}$, so "rank" makes sense, see Exercise 6.7) is isomorphic to $\mathbb{P}^1_k$. (The hypothesis of having a k-valued point is certainly necessary: $x^2+y^2+z^2=0$ over $k=\mathbb{R}$ is a conic that is not isomorphic to $\mathbb{P}^1_k$.)

✍ **Exercise 8.10** Find all rational solutions to $y^2 = x^3 + x^2$, by finding a birational map to $\mathbb{A}^1_{\mathbb{Q}}$, mimicking what worked with the conic. (In Chapter 20, we will see that these points basically form a group, and that this is a degenerate elliptic curve.)

Proof Denote $C : y^2 = x^3 + x^2$. Let rational point $p = (0,0)$, q be another rational point on curve C. Then pq is a line of rational slope. This gives a rational map from the curve C to $\mathbb{A}^1_{\mathbb{Q}}$ given by $(a,b) \mapsto \frac{b}{a}$. Conversely, given a line of slope t through p, where t is rational, we can recover q by solving the following equations

$$\begin{cases} y - tx &= 0 \\ x^3 + x^2 - y^2 &= 0 \end{cases} \implies \begin{cases} x &= t^2 - 1 \\ y &= t(t^2-1). \end{cases}$$

Hence we get a birational map between the curve C and $\mathbb{A}^1_{\mathbb{Q}}$ with coordinate t, given by $(x,y) \mapsto \frac{y}{x}$ which defined for $x \neq 0$, and with inverse rational map given by $t \mapsto (t^2-1, t^3-t)$. By above discussion all rational solutions to $y^2 = x^3 + x^2$ have the form (t^2-1, t^3-t). □

You will obtain a rational map to $\mathbb{P}^1_{\mathbb{Q}}$ that is not defined over the "node" $x = y = 0$, and cannot be extended over this codimension 1 set. This is an example of the limits of our future result, the Curve-to-Projective Extension Theorem, showing how to extend rational maps to projective space over codimension 1 sets: the codimension 1 sets have to be regular.

✍ **Exercise 8.11** Use a similar idea to find a birational map form the quadric surface $Q = \{x^2+y^2 = w^2+z^2\} \subseteq \mathbb{P}^3_{\mathbb{Q}}$ to $\mathbb{P}^2_{\mathbb{Q}}$. Use this to find all rational points on Q.

Proof Consider $x^2+y^2 = w^2+z^2$, then $(x-w)(x+w) = (z-y)(z+y)$. Define a rational map from Q to $\mathbb{P}^2_{\mathbb{Q}}$ given by $[x:y:z:w] \mapsto [z-y : x-w : z+y]$ which defined for $x - w \neq 0$. Conversely, give the inverse rational map by setting $[u:v:s] \mapsto [v^2+us : v(s-u) : v(u+s) : us - v^2]$ which defined for $v \neq 0$. Hence we get a morphism of affine schemes $\operatorname{Spec}((\mathbb{Q}[x,y,z,w]/(x^2+y^2-z^2-w^2))_{x-w})_0 \to \operatorname{Spec}(\mathbb{Q}[u,v,s]_v)_0$

given by

$$[a:b:c:d] \mapsto [c-b:a-d:c+b].$$

It induce a ring map $(\mathbb{Q}[u,v,s]_v)_0 \to ((\mathbb{Q}[x,y,z,w]/(x^2+y^2-z^2-w^2))_{x-w})_0$ given by

$$\frac{u}{v} \longmapsto \frac{z-y}{x-w}, \quad \frac{s}{v} \longmapsto \frac{z+y}{x-w}.$$

Note that

$$((\mathbb{Q}[x,y,z,w]/(x^2+y^2-z^2-w^2))_{x-w})_0 \cong \mathbb{Q}\left[\frac{y}{x-w}, \frac{z}{x-w}\right],$$

above ring map is an isomorphism. Hence we get a birational map $Q \overset{\sim}{\dashrightarrow} \mathbb{P}^2_{\mathbb{Q}}$. We next use this map to find all rational points on Q. Let $[x:y:z:w] \in Q$. If $x-w \neq 0$, then $[x:y:z:w]$ must be the form $[v^2+us : v(s-u) : v(s+u) : us-v^2]$ for some u,v,s. If $x=w$, then $y=z$ or $y=-z$, hence solution of Q have the form $[x:y:y:x]$ or $[x:y:-y:x]$. □

Remark 8.25 This illustrates a good way of solving Diophantine equations. You will find a dense open subset of Q that is isomorphic to a dense open subset of $\mathbb{P}^2$, where you can easily find all the rational points. There will be a closed subset of Q where the rational is not defined, or not an isomorphism, but you can deal with this subset in an ad hoc fashion.

Definition 8.5.8 (The Cremona transformation)

The Cremona transformation is a rational map $\mathbb{P}^2_k \dashrightarrow \mathbb{P}^2_k$ given by

$$[x,y,z] \mapsto \left[\frac{1}{x}, \frac{1}{y}, \frac{1}{z}\right] = [yz : xz : xy].$$

The domain of Cremona transformation is $\mathbb{P}^2_k \setminus \{[1:0:0],[0:1:0],[0:0:1]\}$. (Since if $yz=xz=xy=0$, we get $[1:0:0]$ or $[0:1:0]$ or $[0:0:1]$. Now, $xyz \neq 0$, we get the domain.)

Remark 8.26 You will observe that you can extend it over "codimension 1 sets" (ignoring the fact that we don't yet know what codimension means). This again foreshadows the Curve-to-Projective Extension Theorem.

⋆ Complex curves that are not rational (fun but inessential)

We now describe two examples of curves C that do not admit a nonconstant rational map from $\mathbb{P}^1_{\mathbb{C}}$. (Admittedly, we do not yet know what "curve" means, but no matter.) Both proofs are by Fermat's method of infinite descent. These results can be interpreted as the fact that these curves have no "nontrivial" $\mathbb{C}(t)$-valued points, where by this, we mean that any $\mathbb{C}(t)$-valued point is secretly a $\mathbb{C}$-valued point. You may notice that if you consider the same examples with $\mathbb{C}(t)$ replaced by $\mathbb{Q}$ (and where C is a curve over $\mathbb{Q}$ rather then $\mathbb{C}$), you get two fundamental questions in number theory and geometry. The analog of Exercise 8.14 is the question of rational points on elliptic curves, and you may realize the analog of Exercise 8.12 is even more famous. Also, the arithmetic analog of Exercise 8.14 (a) is the "Four Squares Theorem" (there are not four integer squares in arithmetic progression), first stated by Fermat. These examples will give you a glimpse of how and why facts over number fields are often paralleled by facts over function fields of curves. This parallelism is a recurring deep theme in the subject.

✍ **Exercise 8.12** If $n > 2$, show that $\mathbb{P}^1_{\mathbb{C}}$ has no dominant rational maps to the "Fermat curve" $x^n + y^n = z^n$ in $\mathbb{P}^2_{\mathbb{C}}$.

Proof Suppose that there exists a dominant rational map from $\mathbb{P}^1_{\mathbb{C}}$ to $x^n+y^n=z^n$ in $\mathbb{P}^2_{\mathbb{C}}$. We may assume representative $U \to \operatorname{Proj}(\mathbb{C}[x,y,z]/(x^n+y^n-z^n))$ given by $t \mapsto [f(t):g(t):h(t)]$ where $f(t)^n + g(t)^n =$

$h(t)^n$. Since this rational map is dominant, the image of rational map is nonconstant (if not, the image is single point, contradicts to the fact that rational map is dominant). By clearing denominators, we may assume that $f(t), g(t), h(t)$ are relatively prime polynomials in $\mathbb{C}[t]$ and $(f(t), g(t), h(t))$ is the solution of $x^n + y^n = z^n$ with minimal degree, i.e., $\max\{\deg f(t), \deg g(t), \deg h(t)\}$ is minimal. Since $\mathbb{C}[t]$ is a UFD, we may assume that $h(t)^n - g(t)^n = \prod_{i=0}^{n-1}(h(t) - \zeta^i g(t)) = f(t)^n$, where $\zeta = e^{\frac{2\pi i}{n}}$. We claim that each factor $h(t) - \zeta^i g(t)$ is an n-th power in $\mathbb{C}[t]$. We first show each $h(t) - \zeta^i g(t)$ are pairwise coprime. Assume there exists an irreducible polynomial $p(t)$ dividing both $h(t) - \zeta^i g(t)$ and $h(t) - \zeta^j g(t)$, then $p(t)$ divides the linear combination $\zeta^j(h(t) - \zeta^i g(t)) - \zeta^i(h(t) - \zeta^j g(t)) = (\zeta^j - \zeta^i)h(t)$ and $(h(t) - \zeta^i g(t)) - (h(t) - \zeta^j g(t)) = (\zeta^j - \zeta^i)g(t)$. Since $\zeta^i \neq \zeta^j$, we have $p(t) \mid h(t)$ and $p(t) \mid g(t)$. Note that $f(t)^n = h(t)^n - g(t)^n$, we have $p(t)^n \mid f(t)^n$, since $p(t)$ is irreducible polynomial, $p(t) \mid f(t)$. We now show the claim. Let p be any irreducible factor of $h(t) - \zeta^k g(t)$ with multiplicity m_k. By pairwise coprimality, p does not divide any other $h(t) - \zeta^l g(t)$ $(l \neq k)$. Hence in $f(t)^n$, the multiplicity of p is a multiple of n, i.e., $m_k = n \cdot e_k$ for all k. Hence, each $h(t) - \zeta^i g(t)$ is an n-th power $h(t) - \zeta^i g(t) = k_i(t)^n$, where $k_i(t) \in \mathbb{C}[t]$. Pick $\alpha = -\zeta^{-1} - 1$, $\beta = -\zeta^{-1}$, then we have

$$(h(t) - g(t)) + \alpha(h(t) - \zeta g(t)) = \beta(h(t) - \zeta^2 g(t)),$$

and therefore we get

$$k_0(t)^n + \alpha k_1(t)^n = \beta k_2(t)^n.$$

Then $(k_0(t), \alpha^{\frac{1}{n}} k_1(t), \beta^{\frac{1}{n}} k_2(t))$ is another solution of $x^n + y^n = z^n$. Note that

$$\deg k_0(t), \deg \alpha^{\frac{1}{n}} k_1(t), \deg \beta^{\frac{1}{n}} k_2(t) < \max\{\deg f(t), \deg g(t), \deg h(t)\},$$

we have

$$\max\{\deg k_0(t), \deg \alpha^{\frac{1}{n}} k_1(t), \deg \beta^{\frac{1}{n}} k_2(t)\} < \max\{\deg f(t), \deg g(t), \deg h(t)\},$$

this contradicts to the fact that $\max\{\deg f(t), \deg g(t), \deg h(t)\}$ is minimal. Hence $\mathbb{P}^1_{\mathbb{C}}$ has no dominant rational maps to $x^n + y^n = z^n$ in $\mathbb{P}^2_{\mathbb{C}}$. □

✍ **Exercise 8.13** Give two smooth complex curves X and Y such that no nonempty open subset of X is isomorphic to a nonempty open subset of Y.

Proof Pick $X = \mathbb{P}^1_{\mathbb{C}}$ and $Y = \mathrm{Proj}(\mathbb{C}[x, y, z]/(x^n + y^n - z^n))$, by Exercise 8.12, we done. □

✍ **Exercise 8.14** Suppose a, b and c are distinct complex numbers. By the following steps, show that if $x(t)$ and $y(t)$ are two rational functions of t such that

$$y(t)^2 = (x(t) - a)(x(t) - b)(x(t) - c), \tag{8.9}$$

then $x(t)$ and $y(t)$ are constants $(x(t), y(t) \in \mathbb{C})$. (Here $\mathbb{C}$ may be replaced by any field K of characteristic not 2; slight extra care is needed if K is not algebraically closed.)

Proof We first show the following fact: **Suppose $P, Q \in \mathbb{C}[t]$ are relatively prime polynomials such that four linear combinations of them are perfect squares, no two of which are constant multiples of each other. Then P and Q are constant.**

Let $S_i = a_i P + b_i Q$ as above hypothesis for all $i = 1, 2, 3, 4$. Let $P' = S_1$ and $Q' = S_2$, then P' and Q' are coprime perfect squares. Since $\gcd(P, Q) = 1$, we may write $S_3 = cP' + dQ'$ and $S_4 = eP' + fQ'$ where $c, d, e, f \in \mathbb{C}$. Consider $\frac{S_3}{c}$ and $\frac{S_4}{e}$, they are also perfect squares and not constant multiples of each other. Hence we may assume that the perfect squares are $P, Q, P - Q, P - \lambda Q$ for some $\lambda \in \mathbb{C}$. Say $P = u^2$ and $Q = v^2$, then

$$P - Q = u^2 - v^2 = (u - v)(u + v)$$

and

$$P - \lambda Q = (u - \sqrt{\lambda} v)(u + \sqrt{\lambda} v)$$

are perfect squares. Let $d = \gcd(u - v, u + v)$, then $d \mid ((u - v) + (u + v)) = 2u$ and $d \mid ((u + v) - (u - v)) = 2v$, since $\gcd(u, v) = 1$, we have $d \mid 2$, it follows that $\gcd(u - v, u + v) = 1$. Similarly, we have $\gcd(u - \sqrt{\lambda} v, u + \sqrt{\lambda} v) = 1$. Since $P - Q$ and $P - \lambda Q$ are perfect squares, $u - v, u + v, u - \sqrt{\lambda} v, u + \sqrt{\lambda} v$ must be perfect squares and no two of which are constant multiples of each other.

Now we show the fact. If P and Q are not both constant and suppose $\max\{\deg P, \deg Q\}$ be the minimal. Say $P = u^2$ and $Q = v^2$, by above discussion, $u - v, u + v, u - \lambda v, u + \lambda v$ are perfect squares with no two of which are constant multiples of each other. Note that

$$0 < \max\{\deg u, \deg v\} < \max\{\deg P, \deg Q\},$$

this contradicts to the fact that $\max\{\deg P, \deg Q\}$ is minimal. Hence P and Q are constant.

We next show the Exercise 8.14. Suppose $(x(t), y(t)) = \left(\frac{p(t)}{q(t)}, \frac{r(t)}{s(t)}\right)$ is a solution of equation (8.9), where $p, q, r, s \in \mathbb{C}[t]$, also we may assume that p/q and r/s are in lowest term, i.e., $\gcd(p, q) = \gcd(r, s) = 1$. Clear denominators in (8.9), we have

$$r^2 q^3 = s^2 (p - aq)(p - bq)(p - cq).$$

Since $\gcd(r, s) = 1$, form $s^2 \mid r^2 q^3$, we get $s^2 \mid q^3$. Since $\gcd(p, q) = 1$, we have $q \nmid p - aq, p - bq, p - cq$, hence $q^3 \nmid (p - aq)(p - bq)(p - cq)$. Note that $q^3 \mid s^2 (p - aq)(p - bq)(p - cq)$, we get $q^3 \mid s^2$. Hence we may assume that $s^2 = \delta q^3$ for some $\delta \in \mathbb{C}$. Then we have

$$r^2 = \delta (p - aq)(p - bq)(p - cq).$$

Since $\gcd(p, q) = 1$, we know that $p - aq, p - bq, p - cq$ are pairwise coprime, hence they are all perfect squares with no two of which are constant multiples of each other. Note that $s^2 = \delta q^3$, we know that q must be perfect square. Hence $q, p - aq, p - bq, p - cq$ are all perfect squares with no two of which are constant multiples of each other. Apply above fact, we know that p and q are constants. So s and r are also constants, which implies that $x(t)$ and $y(t)$ are constants. □

A much better geometric approach to Exercise 8.12 and 8.14 is given in Chapter 22.

8.6 ⋆ Representable functors and group schemes

8.6.1 Maps to $\mathbb{A}^1$ correspond to functions

If X is a scheme, there is a bijection between the maps $X \to \mathbb{A}^1$ and global sections of the structure sheaf: by Proposition 8.3.7, maps $\pi : X \to \mathbb{A}^1_{\mathbb{Z}}$ correspond to maps of rings $\pi^\sharp : \mathbb{Z}[t] \to \Gamma(X, \mathscr{O}_X)$, and $\pi^\sharp(t)$ is a function on X; this is reversible.

This map is very natural in an informal sense: you can even picture this map to $\mathbb{A}^1$ as being given by the function. (By analogy, a function on a manifold is a map to $\mathbb{R}$.) But it is natural in a more precise sense: this bijection is functorial in X. We will ponder this example at length, and see that it leads us to two important sophisticated notions: representable functors and group schemes.

Proposition 8.6.1

Suppose X is a $\mathbb{C}$-scheme. Then there is a natural bijection between maps $X \to \mathbb{A}^1_{\mathbb{C}}$ in the category of $\mathbb{C}$-schemes and functions on X. (Here the base ring $\mathbb{C}$ can be replaced by any ring A.)

Proof Give a morphism of schemes $X \to \mathbb{A}^1_{\mathbb{C}}$, we get a ring map $\mathbb{C}[t] \to \Gamma(X, \mathscr{O}_X)$. This ring map is given by $t \mapsto f(t)$, and $f(t)$ is a function on X. Conversely, let $f(t) \in \Gamma(X, \mathscr{O}_X)$, define a ring map $k[t] \to X$ by setting $t \mapsto f(t)$. By Proposition 8.3.7, we get a morphism of schemes $X \to \mathbb{A}^1_{\mathbb{C}}$. Clearly, above process reversible, hence there is a natural bijection between maps $X \to \mathbb{A}^1_{\mathbb{C}}$ in the category of $\mathbb{C}$-schemes and functions on X. □

This interpretation can be extended to rational maps, as follows.

Proposition 8.6.2 (Unimportant)

Rational functions on an integral scheme as rational maps to $\mathbb{A}^1_{\mathbb{Z}}$.

Proof Let X be an integral scheme. Say $\eta \in X$ be the generic point of X. Then $K(X) = \mathscr{O}_{X,\eta}$. Let $f \in K(X)$, then $f \in \Gamma(U, \mathscr{O}_X)$ for some open subset $U \subseteq X$. Define the ring map $\mathbb{Z}[t] \to \Gamma(U, \mathscr{O}_X)$ by setting $t \mapsto f$. By Proposition 8.3.7, we get a morphism of schemes $U \to \operatorname{Spec} \mathbb{Z}[t] = \mathbb{A}^1_{\mathbb{Z}}$. Since X is irreducible, U is dense in X, hence we get a rational map $X \dashrightarrow \mathbb{A}^1_{\mathbb{Z}}$. Let $V \to \mathbb{A}^1_{\mathbb{Z}}$ be a representative of rational map $X \dashrightarrow \mathbb{A}^1_{\mathbb{Z}}$, by Proposition 8.3.7, we get a ring map $\mathbb{Z}[t] \to \Gamma(V, \mathscr{O}_X)$ which defined by $t \mapsto g$. Since $X \dashrightarrow \mathbb{A}^1_{\mathbb{Z}}$ is rational map, we can shrink U and V to a smaller open subset Z such that $(U \to \mathbb{A}^1_{\mathbb{Z}})|_Z = (V \to \mathbb{A}^1_{\mathbb{Z}})|_Z$. Hence the image of f in $\mathscr{O}_{X,\eta}$ is same as the image of g in $\mathscr{O}_{X,\eta}$.

Conversely, let $X \dashrightarrow \mathbb{A}^1_{\mathbb{Z}}$ be an rational map, then for dense open subset $U \subseteq X$ we have a morphism of schemes $U \to \mathbb{A}^1_{\mathbb{Z}} = \operatorname{Spec} \mathbb{Z}[t]$. By Proposition 8.3.7, we have a ring map $\mathbb{Z}[t] \to \Gamma(U, \mathscr{O}_X)$. This ring map is given by $t \mapsto f$. Consider the image of f in $\mathscr{O}_{X,\eta} = K(X)$, also denote it f, then f is a rational function on X. From f, apply Proposition 8.3.7, we may recover the rational map $X \dashrightarrow \mathbb{A}^1_{\mathbb{Z}}$. □

Remark 8.27 If you wish, you can extend your argument to rational maps on locally Noetherian schemes, see Definition 7.6.10.

8.6.2 Representable functors

We restate the bijection of §8.6.1 as follows.

Definition 8.6.1 (Representable functors)

We have two different contravariant functors from **Sch** *to* **Sets***: maps to $\mathbb{A}^1$ (i.e., $H : X \mapsto \operatorname{Mor}(X, \mathbb{A}^1_{\mathbb{Z}})$), and functions on X (i.e., $F : X \mapsto \Gamma(X, \mathscr{O}_X)$). The "naturality" of the bijection — the functoriality in X — is precisely the statement that the bijection gives a natural isomorphism of functors: given any $\pi : X \to X'$, the diagram*

$$\begin{array}{ccc} H(X') & \xrightarrow{H(\pi)} & H(X) \\ \downarrow & & \downarrow \\ F(X') & \xrightarrow{F(\pi)} & F(X) \end{array}$$

(where the vertical maps are the bijections given in §8.6.1) commutes.

More generally, if Y is an element of a category $\mathscr{C}$ (we care about the special case $\mathscr{C} =$ **Sch***), recall the contravariant functor $h_Y : \mathscr{C} \to$* **Sets** *defined by $h_Y(X) = \operatorname{Mor}(X, Y)$. We say a contravariant functor from $\mathscr{C}$ to* **Sets** *is* ***represented by*** *Y if it is naturally isomorphic to the functor h_Y. We say it is* ***representable*** *if it is represented by some Y.*

Remark 8.28 The bijection of §8.6.1 may now be restated as: **the global section functor is represented by $\mathbb{A}^1$.**

Proposition 8.6.3 (Representing objects are unique up to unique isomorphism)

If a contravariant functor F is represented by Y and by Z, then we have a unique isomorphism $Y \xrightarrow{\sim} Z$ induced by the natural isomorphism of functors $h_Y \xrightarrow{\sim} h_Z$.

Proof Since F is represented by Y and Z, F is naturally isomorphic to the functor h_Y and h_Z, and therefore h_Y is naturally isomorphic to h_Z. By Yoneda's Lemma 2.2.1, there is a bijection

$$\mathrm{Nat}(h_Y, h_Z) \overset{\sim}{\longleftrightarrow} \mathrm{Mor}(Y, Z),$$

since $h_Y \xrightarrow{\sim} h_Z$, we have a unique isomorphism $Y \xrightarrow{\sim} Z$. □

Remark 8.29 You have implicitly seen this notion before: **you can interpret the existence of products and fibered products in a category as examples of representable functors.**

Let $Y \times Z$ be a product, then for any object X there exists unique morphism such that the following diagram commutes.

$$\begin{tikzcd}
X \arrow[dr, dashed, "\exists!"] \arrow[drr, bend left, "f"] \arrow[ddr, bend right, "g"'] & & \\
& Y \times Z \arrow[r, "\mathrm{pr}_Y"'] \arrow[d, "\mathrm{pr}_Z"] & Y \\
& Z &
\end{tikzcd}$$

Define the contravariant functor F by setting

$$F(X) := h_Y(X) \times h_Z(X),$$

we shall to show that F is represented by $Y \times X$. Define the natural transformation $\tau : h_{Y\times Z} \to F$ by setting

$$\tau_X : \mathrm{Mor}(X, Y \times Z) \longrightarrow \mathrm{Mor}(X, Y) \times \mathrm{Mor}(X, Z), \quad \varphi \longmapsto (\mathrm{pr}_Y \circ \varphi, \mathrm{pr}_X \circ \varphi).$$

We want to show that τ_X is a bijection. For surjectivity, let $(f, g) \in \mathrm{Mor}(X, Y) \times \mathrm{Mor}(X, Z)$, by the universal property of product $Y \times Z$, there exists unique φ such that $\mathrm{pr}_Y \circ \varphi = f$ and $\mathrm{pr}_X \circ \varphi = g$, hence the surjectivity holds. For injectivity, if $\tau_X(\varphi) = \tau_X(\varphi')$, by the universal property again, we get $\varphi = \varphi'$. It follows that F is represented by $Y \times X$.

You may wish to work out how a natural isomorphism $h_{Y\times Z} \xrightarrow{\sim} h_Y \times h_Z$ induces the projection maps $Y \times Z \to Y$ and $Y \times Z \to Z$: Say $\tau : h_{Y\times Z} \xrightarrow{\sim} h_Y \times h_Z$ be the natural isomorphism. Consider

$$\tau_{Y\times Z} : \mathrm{Mor}(Y \times Z, Y \times Z) \to \mathrm{Mor}(Y \times Z, Y) \times \mathrm{Mor}(Y \times Z, Z),$$

say $\tau_{Y\times Z}(\mathrm{id}_{Y\times Z}) = (p, q)$, by the universal property of product, p, q are the projections map, i.e., $p = \mathrm{pr}_Y$ and $q = \mathrm{pr}_Z$.

✍ **Exercise 8.15** Suppose F is the contravariant functor $\mathbf{Sch} \to \mathbf{Sets}$ defined by $F(X) = \{\text{Grothendieck}\}$ for all schemes X. Show that F is representable.

Proof To show that F is representable, it suffices to show that there exists $Y \in \mathbf{Sch}$ such that there is a natural isomorphism $\tau : h_Y \xrightarrow{\sim} F$. Let $Y = \operatorname{Spec} \mathbb{Z}$, by Proposition 8.3.8, Y is the final object in the category of $\mathbf{Sch}$, hence $h_Y(X) = \mathrm{Mor}(X, Y)$ is a single point set, say $h_Y(X) = \{*_X\}$. Define $\tau_X : h_Y(X) \to F(X)$ by setting $*_X \mapsto \text{Grothendieck}$. It is easy to see that τ_X is a bijection for each $X \in \mathbf{Sch}$, and therefore τ is a natural isomorphism, i.e., F is representable. □

✍ **Exercise 8.16** In this exercise, $\mathbb{Z}$ may be replaced by any ring.

(a) (Affine n-space represents the functor of n functions.) Show that the contravariant functor from $\mathbb{Z}$-schemes to **Sets**

$$X \longmapsto \{(f_1, \dots, f_n) : f_i \in \Gamma(X, \mathscr{O}_X)\}$$

is represented by $\mathbb{A}^n_{\mathbb{Z}}$. Show that $\mathbb{A}^1_{\mathbb{Z}} \times_{\mathbb{Z}} \mathbb{A}^1_{\mathbb{Z}} \cong \mathbb{A}^2_{\mathbb{Z}}$, i.e., that $\mathbb{A}^2$ satisfies the functorial description of $\mathbb{A}^1 \times \mathbb{A}^1$. (You will undoubtedly be able to immediately show that $\prod \mathbb{A}^{m_i}_{\mathbb{Z}} \xleftrightarrow{\sim} \mathbb{A}^{\sum m_i}_{\mathbb{Z}}$.)

(b) (The functor of invertible functions is representable.) Show that the contravariant functor from $\mathbb{Z}$-schemes to **Sets** taking X to invertible functions on X is representable by $\operatorname{Spec} \mathbb{Z}[t, t^{-1}]$.

Proof

(a) By Proposition 8.3.7, there is a bijection between $\operatorname{Mor}(X, \mathbb{A}^n_{\mathbb{Z}})$ and $\operatorname{Hom}(\mathbb{Z}[x_1, \dots, x_n], \Gamma(X, \mathscr{O}_X))$. Say F is the contravariant functor by setting

$$X \longmapsto \{(f_1, \dots, f_n) : f_i \in \Gamma(X, \mathscr{O}_X)\}.$$

Define the natural transformation $\tau : h_{\mathbb{A}^n_{\mathbb{Z}}} \to F$ as follow: define $\tau_X : \operatorname{Mor}(X, \mathbb{A}^n_{\mathbb{Z}}) \to F(X)$ by setting

$$\pi \longleftrightarrow \pi^\sharp \longmapsto (\pi^\sharp(x_1), \dots, \pi^\sharp(x_n)).$$

It is easy to check that τ_X is bijection. Hence, F is represented by $\mathbb{A}^n_{\mathbb{Z}}$. In fact, we have

$$\mathbb{A}^1_{\mathbb{Z}} \times_{\mathbb{Z}} \mathbb{A}^1_{\mathbb{Z}} \cong \operatorname{Spec} \mathbb{Z}[x] \times_{\mathbb{Z}} \operatorname{Spec} \mathbb{Z}[y] \cong \operatorname{Spec}(\mathbb{Z}[x] \otimes \mathbb{Z}[y]) \cong \operatorname{Spec} \mathbb{Z}[x, y] = \mathbb{A}^2_{\mathbb{Z}}.$$

By the induction, we also have

$$\prod \mathbb{A}^{m_i}_{\mathbb{Z}} \xleftrightarrow{\sim} \mathbb{A}^{\sum m_i}_{\mathbb{Z}}.$$

(b) Let G be the contravariant functor from $\mathbb{Z}$-schemes to **Sets** taking X to invertible functions on X, i.e., $G(X) = \Gamma(X, \mathscr{O}_X)^\times$. Define the natural transformation $\tau : h_{\operatorname{Spec} \mathbb{Z}[t, t^{-1}]} \to G$ as follow: define $\tau_X : \operatorname{Mor}(X, \operatorname{Spec} \mathbb{Z}[t, t^{-1}]) \to \Gamma(X, \mathscr{O}_X)^\times$ by setting

$$\pi \longmapsto \pi^\sharp(t).$$

It is easy to check that τ_X is a bijection. Hence G is representable by $\operatorname{Spec} \mathbb{Z}[t, t^{-1}]$.

□

Definition 8.6.2 (Multiplicative group)

*The scheme $\operatorname{Spec} \mathbb{Z}[t, t^{-1}]$ is called the **multiplicative group** (or multiplicative group scheme) $\mathbb{G}_m$. "$\mathbb{G}_m$ over a field k" ("the multiplicative group over k") means $\operatorname{Spec} k[t, t^{-1}]$, with the same group operations.*
***Better:** it represents the group of invertible functions in the category of k-schemes.*
We can similarly define $\mathbb{G}_m$ over an arbitrary ring or even arbitrary scheme.

✍ **Exercise 8.17** (Less important exercise.) Fix a ring A. Consider the functor H from the category of locally ringed spaces to **Sets** given by $H(X) = \{A \to \Gamma(X, \mathscr{O}_X)\}$. Show that this functor is representable (by $\operatorname{Spec} A$). This gives another (admittedly odd) motivation for the definition of $\operatorname{Spec} A$, closed related to the fact that Γ and Spec are adjoints.

Proof By [Sta25, Lemma 01I1], there is a bijection between $\operatorname{Mor}(X, \operatorname{Spec} A)$ and $\operatorname{Hom}(A, \Gamma(X, \mathscr{O}_X))$ for all locally ringed space X. It follows that there exits a natural isomorphism $h_{\operatorname{Spec} A} \xrightarrow{\sim} H$, i.e., H is represented by $\operatorname{Spec} A$. □

8.6.3 ⋆⋆ Group schemes (or more generally, group objects in a category)

8.7 ⋆⋆ The Grassmannian: First construction

Chapter 9 Useful classes of morphisms of schemes

We now define an excessive number of types of morphisms. Some (often finiteness properties) are useful because most "reasonable" morphism has such properties, and they will be used in proofs in obvious ways. Others correspond to geometrically meaningful behavior, and you should have a picture of what each means.

Change of perspective: morphisms are more fundamental than objects. One of Grothendieck's lesson is that things that we often think of as properties of objects are better understood as properties of morphisms. One way of turning properties of objects into properties of morphisms is as follows.

Definition 9.0.1 (Morphism has property $\mathscr{P}$)

*If $\mathscr{P}$ is a property of schemes, we often (but not always) say that a **morphism** $\pi : X \to Y$ **has** $\mathscr{P}$ if for every affine open subset $U \subseteq Y$, $\pi^{-1}(U)$ has $\mathscr{P}$.*

We will see this for $\mathscr{P}$ = quasicompact, quasiseperated, affine, and more. (As you might hope, in good circumstance, $\mathscr{P}$ will satisfy the hypotheses of the Affine Communication Lemma 6.3.1, so we don't have to check every affine open subset.) Informally, you can think of such a morphism as one where all the fibers have $\mathscr{P}$. (You can quickly define the fiber of a morphism as a topological space, but once we define fibered product, we will define the **scheme-theoretic fiber**, and then this discussion will make sense.) But it means more than that: it means that "being $\mathscr{P}$" is really not just fiber-by-fiber, but behaves well as the fiber varies. (For comparison, "submersion of manifolds" means more than that the fibers are smooth.)

9.1 "Reasonable" classes of morphisms (such as open embeddings)

You will notice that almost all classes of morphisms that are useful have many properties in common, which follow from the following three basic properties.

Definition 9.1.1 ("Reasonable" class of morphisms of schemes)

*We call any class of morphisms satisfying these properties a **"reasonable" class of morphisms of schemes**. (To avoid any stupidities, we assume that the class includes all isomorphisms; but this requirement doesn't deserve a number.)*

*(i) The class is **preserved by composition**: if $\pi : X \to Y$ and $\rho : Y \to Z$ are both in this class, then so is $\rho \circ \pi$.*

*(ii) The class is **preserved by "base change" (or "pullback" or "fibered product")**. Precisely: if $\pi : X \to Y$ is in this class, then for any $Y' \to Y$, the induced map $X \times_Y Y' \to Y'$ is also in this class. (Implicit in this statement is that the fibered product $X \times_Y Y'$ exists; but we will soon see that all fibered products of schemes exist, Chapter 11.)*

*(iii) The class is **local on the target**. In other words,*

(LT1) if $\pi : X \to Y$ is in the class, then for any open subset V of Y, the restricted morphism $\pi^{-1}(V) \to V$ is in the class;

(LT2) for a morphism $\pi : X \to Y$, if there is an open cover $\{V_i\}$ of Y for which each restricted morphism $\pi^{-1}(V_i) \to V_i$ is in the class, then π is in the class.

In particular, as schemes are built out of affine schemes, properties are often easy to verify on any affine cover (the properties are "affine-local" on the target), as described in Definition 9.0.1.

(Stalk-local properties are automatically local on the target.)

Properties (i) and (ii) imply a useful additional property — (iv) "reasonable" classes of morphisms are **preserved by product**:

Proposition 9.1.1 (Preserved by product)

Suppose $\mathscr{P}$ is a property of morphisms preserved by composition and base change ((i) and (ii) in Definition 9.1.1), and $X \to Y$ and $X' \to Y'$ are two morphisms of S-schemes with property $\mathscr{P}$. Assume that $X \times_S X'$ and $Y \times_S Y'$ exist (which is indeed true, Chapter 11). Then $X \times_S X' \to Y \times_S Y'$ has property $\mathscr{P}$ as well.

Proof By the universal property of fiber product,

$$\begin{tikzcd} X \times_S X' \arrow[dr, dashed] \arrow[drr, bend left] \arrow[dd] & & \\ & X \times_S Y' \arrow[r] \arrow[d] & X \arrow[d] \\ X' \arrow[r] & Y' \arrow[r] & S \end{tikzcd} \qquad \begin{tikzcd} X \times_S Y' \arrow[dr, dashed] \arrow[drr, bend left] \arrow[dd] & & \\ & Y \times_S Y' \arrow[r] \arrow[d] & Y' \arrow[d] \\ X \arrow[r] & Y \arrow[r] & S \end{tikzcd}$$

we have a decomposition of $X \times_S X' \to Y \times_S Y'$,

$$X \times_S X' \longrightarrow X \times_S Y' \longrightarrow Y \times_S Y'.$$

In fact, we have canonical isomorphism

$$X \times_S X' \cong X \times_S (Y' \times_{Y'} X'), \quad X \times_S Y' \cong (X \times_Y Y) \times_S Y'.$$

Consider following diagram.

$$\begin{tikzcd} & X \times_S X' \arrow[r] \arrow[dl] & X \times_S Y' \arrow[r] \arrow[dl] \arrow[dr] & Y \times_S Y' \arrow[dr] & \\ X' \arrow[r] & Y' & & X \arrow[r] & Y \end{tikzcd}$$

Since $X' \to Y'$ has property $\mathscr{P}$ and $\mathscr{P}$ preserved by base change, $(X \times_S Y') \times_{Y'} X' \to X \times_S Y'$ has property $\mathscr{P}$, i.e., $X \times_S X' \to X \times_S Y'$ has property $\mathscr{P}$. Similarly, $X \times_S Y' \to Y \times_S Y'$ has property $\mathscr{P}$. Since $\mathscr{P}$ preserved by composition, $X \times_S X' \to Y \times_S Y'$ has property $\mathscr{P}$. ☐

(v) Another extremely important consequence of (i) and (ii) is the **Cancellation Theorem** for such morphism, see Chapter 12.

When you learn of a new class of morphisms, you should immediately ask whether these properties (i)-(iii) (and hence (iv) and (v)) hold. The answer will almost always be "yes"; and if it is "no", then you should exercise caution, and possibly figure out how to make the definition better.

To prepare you to think about these properties for schemes, you may want to verify that the analogous properties to (i)-(iii) holds in the category of topological spaces with the classes "injections", "surjections", "open embeddings" (open embedding of topological space = isomorphism with an open subset), and "closed embeddings" (closed embedding of topological spaces = isomorphism with a closed subset).

Proposition 9.1.2

Isomorphism of schemes are "reasonable" in this sense, i.e., that they satisfy the three properties (i)-(iii) in Definition 9.1.1.

Proof (i) is clearly. Let $\pi : X \xrightarrow{\sim} Y$ be an isomorphism of schemes. For any $Y' \to Y$, we have $X \times_Y Y' \cong Y \times_Y Y' \cong Y'$ is an isomorphism of schemes. Hence (ii) holds. Let V be any open subset of Y, since $\pi : X \to Y$ is an isomorphism, $\pi^{-1}(V) \to V$ is also an isomorphism. Let $\{V_i \hookrightarrow Y\}$ be an open cover of Y for which each restricted morphism $\pi^{-1}(V_i) \to V_i$ is isomorphism. For all $p \in X$, we have an isomorphism of stalks $\mathscr{O}_{Y,\pi(p)} \xrightarrow{\sim} \mathscr{O}_{X,p}$. Hence $\pi : X \xrightarrow{\sim} Y$ is isomorphism, i.e., (iii) holds. □

We are now ready to introduce a fundamental class of morphisms of schemes.

Definition 9.1.2 (Open embedding, open subscheme)

*A morphism $\pi : X \to Y$ of schemes is an **open embedding** (or **open immersion**) if it is an open embedding as ringed spaces (Definition 8.2.2). In other words, a morphism $\pi : (X, \mathscr{O}_X) \to (Y, \mathscr{O}_Y)$ of schemes is an open embedding if π factors as*

$$(X, \mathscr{O}_X) \xrightarrow[\sim]{\rho} (U, \mathscr{O}_Y|_U) \xhookrightarrow{\tau} (Y, \mathscr{O}_Y)$$

where ρ is an isomorphism, and $\tau : U \hookrightarrow Y$ is an inclusion of an open set.

*If X is actually a subset of Y (and π is the inclusion, i.e., ρ is the identity), then we say $(X, \mathscr{O}_X)$ is an **open subscheme** of $(Y, \mathscr{O}_Y)$.*

Remark 9.1

(1) It is immediate that isomorphisms are open embeddings. The symbol $\hookrightarrow$ is often used to indicate that a morphism is an open embedding (or more generally, a locally closed embedding).

(2) The difference between open embeddings and open subschemes is a bit confusing, and not too important: at the level of sets, open subschemes are subsets, while open embeddings are bijections onto subsets.

"Open subschemes" are scheme-theoretic analogs of open subsets. ("Closed subschemes" are scheme-theoretic analogs of closed subsets, but they have a surprisingly different flavor, as we will see in Chapter 10.)

The next two Propositions verify that the class of open embeddings is "reasonable" in the sense described above (Definition 9.1.1).

Proposition 9.1.3

The class of open embedding is preserved by composition and local on the target.

Proof We first show that the class of open embedding is local on the target. Suppose $\pi : X \to Y$ is an open embedding and $V \subseteq Y$ is an open subset. We want to show that $\pi^{-1}(V) \to V$ is an open embedding. Since $\pi : X \to Y$ is an open embedding, π factors as

$$(X, \mathscr{O}_X) \xrightarrow{\sim} (U, \mathscr{O}_Y|_U) \hookrightarrow (Y, \mathscr{O}_Y),$$

where U is an open subset of Y. Hence the restricted morphism factor as

$$(\pi^{-1}(V), \mathscr{O}_X|_{\pi^{-1}(V)}) \xrightarrow{\sim} (U \cap V, \mathscr{O}_Y|_{U \cap V}) \hookrightarrow (V, \mathscr{O}_Y|_V).$$

It follows that $\pi^{-1}(V) \to V$ is an open embedding.

Let V_i is an open cover of Y with each restricted morphism $\mathrm{res}_i : \pi^{-1}(V_i) \to V_i$ is open embedding, we want to show that $\pi : (X, \mathscr{O}_X) \to (Y, \mathscr{O}_Y)$ is an open embedding. Since res_i is an open embedding, res_i factors as

$$(\pi^{-1}(V_i), \mathscr{O}_X|_{\pi^{-1}(V_i)}) \xrightarrow{\sim} (\widetilde{V}_i, \mathscr{O}_Y|_{\widetilde{V}_i}) \hookrightarrow (V_i, \mathscr{O}_Y|_{V_i}).$$

By Gluing sheaves 3.5.3 and Morphism of locally ringed spaces glue 8.3.2, $\mathscr{O}_X|_{\pi^{-1}(V_i)}$ glue to $\mathscr{O}_X$ and $\mathscr{O}_Y|_{\widetilde{V}_i}$

glue to $\mathscr{O}_Y|_{\operatorname{Im}\pi}$ where $\operatorname{Im}\pi = \bigcup_i \widetilde{V}_i$, and we have

$$(X, \mathscr{O}_X) \xrightarrow{\sim} (\operatorname{Im}\pi, \mathscr{O}_Y|_{\operatorname{Im}\pi}) \hookrightarrow (Y, \mathscr{O}_Y).$$

Note that $\operatorname{Im}\pi$ is an open subset of Y, we know that $\pi : X \to Y$ is an open embedding. Hence the class of open embedding is local on the target.

Let $\pi_1 : X \to Y$ and $\pi_2 : Y \to Z$ are open embeddings, then π_1 factors as

$$(X, \mathscr{O}_X) \xrightarrow{\sim} (U, \mathscr{O}_Y|_U) \hookrightarrow (Y, \mathscr{O}_Y).$$

Since the class of open embedding is local one the target, the restriction morphism $\pi_2|_U$ is also open embedding, so $\pi_2|_U$ factors as

$$(U, \mathscr{O}_Y|_U) \xrightarrow{\sim} (V, \mathscr{O}_Z|_V) \hookrightarrow (Z, \mathscr{O}_Z).$$

Then we have

$$(X, \mathscr{O}_X) \xrightarrow{\sim} (U, \mathscr{O}_Y|_U) \xrightarrow{\sim} (V, \mathscr{O}_Z|_V) \hookrightarrow (Z, \mathscr{O}_Z),$$

it follows that $\pi_2 \circ \pi_1$ is an open embedding, i.e., the class of open embedding is preserved by composition. □

Proposition 9.1.4 (Fibered products with open embeddings exist)

The class of open embeddings is preserved by "base change". More specifically: suppose $i : U \to Z$ is an open embedding, and "$\rho : Y \to Z$" is any morphism. Then $U \times_Z Y$ exists and $U \times_Z Y \to Y$ is an open embedding. In particular, if $U \hookrightarrow Z$ and $V \hookrightarrow Z$ are open embeddings, $U \times_Z V \cong U \cap V$: "intersection of open embeddings is fiber product".

Proof We claim that $(\rho^{-1}(U), \mathscr{O}_Y|_{\rho^{-1}(U)})$ is the fiber product. Without loss of generality, we may assume that U is open subset of Z. Consider the following diagram,

$$\begin{tikzcd} (\rho^{-1}(U), \mathscr{O}_Y|_{\rho^{-1}(U)}) \arrow[r, hook] \arrow[d, "\rho|_{\rho^{-1}(U)}"'] & (Y, \mathscr{O}_Y) \arrow[d, "\rho"] \\ (U, \mathscr{O}_Z|_U) \arrow[r, "i"'] & (Z, \mathscr{O}_Z) \end{tikzcd} \tag{9.1}$$

where $(\rho^{-1}(U), \mathscr{O}_Y|_{\rho^{-1}(U)})$ is open subscheme of $(Y, \mathscr{O}_Y)$. Clearly diagram (9.1) commutes. Let $(W, \mathscr{O}_W)$ be any scheme with $i \circ \alpha = \rho \circ \beta$, we want to show that there exists unique morphism from $(W, \mathscr{O}_W)$ to $(\rho^{-1}(U), \mathscr{O}_Y|_{\rho^{-1}(U)})$ such that the following diagram commutes.

$$\begin{tikzcd} (W, \mathscr{O}_W) \arrow[drr, bend left, "\beta"] \arrow[dr, dashed] \arrow[ddr, bend right, "\alpha"'] & & \\ & (\rho^{-1}(U), \mathscr{O}_Y|_{\rho^{-1}(U)}) \arrow[r, hook] \arrow[d, "\rho|_{\rho^{-1}(U)}"] & (Y, \mathscr{O}_Y) \arrow[d, "\rho"] \\ & (U, \mathscr{O}_Z|_U) \arrow[r, "i"'] & (Z, \mathscr{O}_Z) \end{tikzcd}$$

Since $\rho^{-1}(U)$ is subscheme of Y, $W \to \rho^{-1}(U)$ must be β. Next we check $\beta : W \to \rho^{-1}(U)$ makes above diagram commute. Since $\rho \circ \beta = i \circ \alpha$ and $\beta(W) \subseteq \rho^{-1}(U)$, we have $\alpha = \rho|_{\rho^{-1}(U)} \circ \beta$. Hence $(\rho^{-1}(U), \mathscr{O}_Y|_{\rho^{-1}(U)}) = U \times_Z Y$, by the construction of $U \times_Z Y$, $U \times_Z Y \to Y$ is an open embedding.

In particular, if $U \hookrightarrow Z$ and $V \hookrightarrow Z$ are open embeddings, without loss of generality, we may assume $U \subseteq Z$ and $V \subseteq Z$. Say $i_1 : U \hookrightarrow Z$ and $i_2 : V \hookrightarrow Z$, by above discussion, we have

$$U \times_Z V = (i_2^{-1}(U), \mathscr{O}_Z|_{i_2^{-1}(U)}) = (U \cap V, \mathscr{O}_Z|_{U \cap V}),$$

as we desired. □

Corollary 9.1.1

The class of open embedding is "reasonable".

Proof By Proposition 9.1.3 and Proposition 9.1.4, we get immediately. □

Proposition 9.1.5

Open embeddings of schemes are monomorphisms (in the category of schemes).

Proof By Proposition 8.2.2, we done. □

✍ **Exercise 9.1** Suppose $\pi : X \to Y$ is an open embedding. If Y is locally Noetherian, then X is too. If Y is Noetherian, then X is too. However, if Y is quasicompact, X need not be.

Proof If Y is locally Noetherian, we may assume that $Y = \bigcup_i \operatorname{Spec} A_i$ where each A_i is Noetherian. Since $\pi : X \hookrightarrow Y$ is an open embedding, by Corollary 9.1.1, $\pi^{-1}(\operatorname{Spec} A_i) \hookrightarrow \operatorname{Spec} A_i$ is an open embedding. Hence $\pi^{-1}(\operatorname{Spec} A_i)$ can be seen as open subscheme of $\operatorname{Spec} A_i$, and therefore $\pi^{-1}(\operatorname{Spec} A_i)$ is the union of distinguished open subset of $\operatorname{Spec} A_i$. Since $\operatorname{Spec} A_i$ is Noetherian, each distinguished open subset is also Noetherian, hence $X = \bigcup_i \pi^{-1}(\operatorname{Spec} A_i)$ is locally Noetherian.

If Y is Noetherian, we may assume that $Y = \bigcup\limits_{i=1}^{n} \operatorname{Spec} A_i$, by above discussion $X = \bigcup\limits_{i=1}^{n} \operatorname{Spec} A_i$ is locally Noetherian, since this is a finite cover, X is Noetherian.

Let Y be $\operatorname{Spec} k[x_1, x_2, x_3, \dots]$, then Y is quasicompact. Say $\mathfrak{m} = (x_1, x_2, \dots)$, then $U = Y \setminus V(\mathfrak{m})$ is a nonquasicompact open subset of Y. Let $X = U$, then X is not quasicompact. □

Definition 9.1.3 (Local on the source, affine-local on the source, and affine-local on the target)

We call a property $\mathscr{P}$ of morphisms is ***local on the source****, if satisfies:*

(LS1) if $\pi : X \to Y$ has property $\mathscr{P}$, and $i : U \hookrightarrow X$ is an open subset, then $\pi \circ i : U \to Y$ has $\mathscr{P}$;

(LS2) for a morphism $\pi : X \to Y$, if for any open cover $\{\iota_i : U_i \hookrightarrow X\}$ of X each $\pi \circ \iota_i : U_i \to Y$ has property $\mathscr{P}$, then $\pi : X \to Y$ has property $\mathscr{P}$.

We call a property $\mathscr{P}$ of morphisms is ***affine-local on the source****, if satisfies:*

(ALS1) if $\pi : X \to Y$ has $\mathscr{P}$, and $i : \operatorname{Spec} A \hookrightarrow X$ is an affine open subset, then $\pi \circ i : \operatorname{Spec} A \to Y$ has $\mathscr{P}$;

(ALS2) for a morphism $\pi : X \to Y$, if for any affine open cover $\{\iota_i : \operatorname{Spec} A_i \hookrightarrow X\}$ of X each $\pi \circ \iota_i : \operatorname{Spec} A_i \to Y$ has property $\mathscr{P}$, then $\pi : X \to Y$ has property $\mathscr{P}$.

We call a property $\mathscr{P}$ of morphisms is ***affine-local on the target****, if satisfies:*

(ALT1) if $\pi : X \to Y$ has property $\mathscr{P}$, then for any affine open subset $\operatorname{Spec} B$ of Y, the restricted morphism $\pi^{-1}(\operatorname{Spec} B) \to \operatorname{Spec} B$ has property $\mathscr{P}$.

(ALT2) for a morphism $\pi : X \to Y$, if for any affine open cover $\{\operatorname{Spec} B_i \hookrightarrow Y\}$ of Y each restricted morphism $\pi^{-1}(\operatorname{Spec} B_i) \to \operatorname{Spec} B_i$ has property $\mathscr{P}$, then $\pi : X \to Y$ has property $\mathscr{P}$.

Remark 9.2 The use of "affine-local" rather than "local" is to emphasize that the criterion on affine schemes is simple to describe. (Clearly the Affine Communication Lemma 6.3.1 will be very handy.)

✍ **Exercise 9.2** The notion of "open embedding" is not local on the source.

Proof Consider the projection $\mathrm{pr} : \mathbb{A}_k^1 \coprod \mathbb{A}_k^1 \to \mathbb{A}_k^1$ given by $\mathfrak{p} \times \{i\} \mapsto \mathfrak{p}$. In fact, $\mathbb{A}_k^1 \coprod \mathbb{A}_k^1 = (\operatorname{Spec} k[x] \times \{1\}) \cup (\operatorname{Spec} k[y] \times \{2\})$ is an open cover. The restriction maps $\mathrm{pr}\,|_{\operatorname{Spec} k[x] \times \{1\}}$ and $\mathrm{pr}\,|_{\operatorname{Spec} k[x] \times \{2\}}$ are clearly isomorphism, and therefore is open embedding. But pr is not open embedding, since pr is not injection. Hence "open embedding" is not local on the source. □

9.2 Another algebraic interlude: Lying Over and Nakayama

To set up our discussion in the next section on integral morphisms, we develop some algebraic preliminaries. A clever trick we use can also be used to show Nakayama's Lemma, so we discuss this as well.

9.2.1 Integral

Definition 9.2.1 (Integral)

Suppose $\varphi : B \to A$ is a ring morphism. We say $x \in A$ is ***integral*** *over B if x satisfies some monic polynomial*

$$x^n + a_{n-1}x^{n-1} + \cdots + a_0 = 0$$

where the coefficients lie in $\varphi(B)$.

A ring morphism $\varphi : B \to A$ is ***integral*** *if every element of A is integral over $\varphi(B)$. An integral ring morphism φ is an* ***integral extension*** *if φ is an inclusion of rings.*

Remark 9.3 You should think of integral morphisms and integral extensions as ring-theoretic generalizations of the notion of algebraic extensions of fields.

Proposition 9.2.1

If $\varphi : B \to A$ is a ring morphism, $(b_1, \ldots, b_n) = 1$ in B, and $B_{b_i} \to A_{\varphi(b_i)}$ is integral for all i, then φ is integral.

Proof Replace B by $\varphi(B)$ to reduce to the case where B is a subring of A. Suppose $x \in A$, then $x \in A_{b_i}$ for all i. Since $B_{b_i} \to A_{b_i}$ is integral, there exists an equations

$$x^{m(i)} + k_{i,n-1}x^{m(i)-1} + \cdots + k_{i,0} = 0.$$

Clear the denominator, we have

$$b_i^{t(i)}x^{m(i)} + b_i^{t(i)}k_{i,n-1}x^{m(i)-1} + \cdots + b_i^{t(i)}k_{i,0} = 0,$$

where $b_i^{t(i)}k_{i,j} \in B$. Hence $b_i^{t(i)}x^{m(i)} \in B + Bx + \cdots + Bx^{m(i)-1}$. By Proposition 4.5.2 and Proposition 4.5.4, we have $\operatorname{Spec} B = \bigcup_{i=1}^n D(b_i) = \bigcup_{i=1}^n D(b_i^t)$, where $t = \max_i t(i)$. Hence $(b_1^t, b_2^t, \ldots, b_n^t) = 1$. Let $s(i) = t - t(i)$, then

$$b_i^{s(i)}x^{s(i)} \cdot b_i^{t(i)}x^{m(i)} = b_i^t x^t \in B + Bx + \cdots + Bx^{m(i)-1},$$

and therefore

$$\sum_{i=1}^n b_i^t x^t = x^t \in B + Bx + \cdots + Bx^{t-1}.$$

It follows that

$$x^t + k_{t-1}x^{t-1} + \cdots + k_0 = 0$$

for some $k_j \in B$. Hence $\varphi : B \to A$ is integral. $\square$

Proposition 9.2.2

(a) The property of a morphism $\varphi : B \to A$ being integral is always preserved by localization and quotient of B, and quotient of A, but not localization of A. More precisely: suppose φ is integral, the induced maps $T^{-1}B \to \varphi(T)^{-1}A$, $B/J \to A/\varphi(J)A$, and $B \to A/I$ are integral, where T

is a multiplicative subset of B, J is an ideal of B, and I is an ideal of A. But $B \to S^{-1}A$ need not be integral, where S is a multiplicative subset of A.

(b) The property of φ being an integral extension is preserved by localization of B, but not localization or quotient of A.

(c) In fact the property of φ being an integral extension (as opposed to integral morphism) is not preserved by taking quotients of B either. But it is in some cases. Suppose $\varphi : B \to A$ is an integral extension, and prime ideal $J \subseteq B$ is the restriction of a prime ideal $I \subseteq A$. The induced map $B/J \to A/JA$ is an integral extension.

Proof

(a) Let T be a multiplicative subset of B. Suppose $\frac{a}{\varphi(t)} \in \varphi(T)^{-1}A$. Since $\varphi : B \to A$ is integral, we have an equation
$$a^n + k_{n-1}a^{n-1} + \cdots + k_0 = 0,$$
where $k_i \in B$. Dividing the equation by $\varphi(t)^n$ given us
$$\frac{a^n}{\varphi(t)^n} + \frac{k_{n-1}}{\varphi(t)} \cdot \frac{a^{n-1}}{\varphi(t)^{n-1}} + \cdots + \frac{k_0}{\varphi(t)^n} = 0,$$
it follows that $T^{-1}B \to \varphi(T)^{-1}A$ is integral.
Next, let J be an ideal of B. Suppose $\overline{a} \in A/\varphi(J)A$. Since $\varphi : B \to A$ is integral, we have equation
$$a^n + \varphi(b_{n-1})a^{n-1} + \cdots + \varphi(b_0) = 0,$$
where $b_i \in B$. Hence, all $b_i \in J$ if and only if $\overline{a}^n = 0$ (i.e., $a \in \varphi(J)$). It follows that integrality is well-defined and preserved on the quotient of B.
Let I be an ideal of A. Suppose $\overline{a} \in A/I$. Since $\varphi : B \to A$ is integral, we have an equation
$$a^n + \varphi(b_{n-1})a^{n-1} + \cdots + \varphi(b_0) = 0,$$
where each $b_i \in B$. Take quotient, we have
$$\overline{a}^n + \overline{\varphi(b_{n-1})}\overline{a}^{n-1} + \cdots + \overline{\varphi(b_0)} = 0.$$
Hence $B \to A/I$ is integral.
We now give an counterexample to show that $B \to S^{-1}A$ need not be integral. Consider $\mathrm{id} : k[t] \to k[t]$, clearly it is integral. We want to show that $k[t] \to k[t]_{(t)}$ is not. Suppose $k[t] \to k[t]_{(t)}$ is integral. Consider $\frac{1}{t} \in k[t]_{(t)}$, then we have an equation
$$\frac{1}{t^n} + k_{n-1}(t)\frac{1}{t^{n-1}} + \cdots + k_0(t) = 0,$$
where $k_i(t) \in k[t]$. Multiplying the equation by t^n, we have
$$1 + k_{n-1}(t)t + \cdots + k_0(t)t^n = 0.$$
This is impossible. Hence $k[t] \to k[t]_{(t)}$ is not integral.

(b) Let $i : B \to A$ be an integral extension. Let T be a multiplicative subset of B. By part (a), $T^{-1}B \to T^{-1}A$ is integral. Note that $T^{-1}B$ is the subring of $T^{-1}A$, this map is an inclusion. Hence $T^{-1}B \to T^{-1}A$ is an integral extension.
We next give two counterexamples to show that the property of morphisms being an integral extension is not preserved by localization of A and quotient of A. Consider $\mathrm{id} : k[t] \to k[t]$, clearly, $\mathrm{id} : k[t] \to k[t]$ is an integral extension. By part (a), we know that induced map $k[t] \to k[t]_{(t)}$ is not integral. Now

consider $k[t] \to k[t]/(t)$. In fact, $k[t]/(t) \cong k$, map $k[t] \to k$ is clearly integral. But $k[t] \to k$ is not inclusion (since $k[t] \to k[t] \to k[t]/(t)$ is not injection), and therefore is not integral extension. Hence the property of morphisms being an integral extension is not preserved by localization or quotient of A.

(c) We first give a counterexample to show that the property of morphisms being an integral extension is not preserved by taking quotients of B either. Let $B = k[x,y]/(y^2)$ and $A = k[x,y,z]/(z^2, xz - y)$, then we have an obvious inclusion $B \hookrightarrow A$. It is easy to see that $B \hookrightarrow A$ is integral, hence $B \hookrightarrow A$ is integral extension. Consider $B/(x) \to A/(x)$. In fact, $B/(x) \cong k[y]/(y^2)$ and $A/(x) \cong k[x,y,z]/(z^2, xz - y, x) \cong k[z]/(z^2)$, and $B/(x) \to A/(x)$ is given by $a + by \mapsto a + bxz = a$. Map $B/(x) \to A/(x)$ is not injection.

Now suppose $\varphi : B \to A$ is an integral extension, and $J \subseteq B$ is the restriction of an ideal $I \subseteq A$. We want to show that $B/J \to A/JA$ is an integral extension. Since $\varphi : B \to A$ is an integral extension, by part (a), we know that $B/J \to A/JA$ is integral. Since $J = I \cap B$, $B/J \to A/I$ factors through A/JA, i.e., we have $B/J \to A/JA \to A/I$. Since $\varphi : B \to A$ is injective, B can be seen as the subring of A, hence $B/J \to A/I$ is injective. By [AM18, Corollary 5.9], we have $JA = I$. Hence $A/JA \cong A/I$, and therefore $B/J \to A/JA$ is injective.

$\square$

The following lemma uses a useful but sneaky trick.

Lemma 9.2.1

Suppose $\varphi : B \to A$ is a ring morphism. Then $a \in A$ is integral over B if and only if it is contained in a subalgebra of A that is a finitely generated B-module.

Proof If $a \in A$ is integral over B, then a satisfies a monic polynomial equation of degree n. Hence the B-submodule of A generated by $1, a, \dots, a^{n-1}$ is closed under multiplication, and hence a subalgebra of A.

Conversely, assume that a is contained in a subalgebra A' of A that is a finitely generated B-module. Choose a finite generating set $m_1, \dots, m_n$ of A' (as a B-module). Then $am_i = \sum b_{ij} m_j$ where $b_{ij} m_j := \varphi(b_{ij}) m_j$, for some $b_{ij} \in B$. Thus

$$(a \operatorname{id}_{n \times n} - (b_{ij})_{ij}) \begin{bmatrix} m_1 \\ \vdots \\ m_n \end{bmatrix} = \begin{bmatrix} 0 \\ \vdots \\ 0 \end{bmatrix}, \tag{9.2}$$

where id_n is the $n \times n$ identity matrix in A. We can't invert the matrix $(a \operatorname{id}_{n \times n} - (b_{ij})_{ij})$, but we almost can. Recall that an $n \times n$ matrix M has an adjugate matrix $\operatorname{adj}(M)$ such that $\operatorname{adj}(M) M = \det(M) \operatorname{id}_n$. Multiplying (9.2) by $\operatorname{adj}(a \operatorname{id}_n - (b_{ij})_{ij})$, we get

$$\det(a \operatorname{id}_n - (b_{ij})_{ij}) \begin{bmatrix} m_1 \\ \vdots \\ m_n \end{bmatrix} = \begin{bmatrix} 0 \\ \vdots \\ 0 \end{bmatrix}.$$

So $\det(a \operatorname{id}_n - (b_{ij})_{ij})$ annihilates the generating elements m_i, and hence every element of A', i.e., $\det(a \operatorname{id}_n - (b_{ij})) = 0$. Expanding the determinant yields an integral equation for a with coefficients in B. $\square$

Corollary 9.2.1 (Finite implies integral)

*If A is a **finite B-algebra** (a finitely generated B-module), then φ is an integral ring morphism.*

Proof Apply Lemma 9.2.1, we done. □

Remark 9.4 The converse is false: integral does not imply finite, as $\mathbb{Q} \hookrightarrow \overline{\mathbb{Q}}$ is an integral ring morphism, but $\overline{\mathbb{Q}}$ is not a finite $\mathbb{Q}$-module. (A field extension is integral if it is algebraic.)

Proposition 9.2.3

If $C \to B$ and $B \to A$ are both integral ring morphisms, then their composition $C \to A$ is integral ring morphism.

Proof Let $a \in A$, since $B \to A$ is integral, by Lemma 9.2.1 a is contained in a subalgebra R of A that is a finitely generated B-module. We may assume R is generated by $b_1, \ldots, b_n$. Since $C \to B$ is integral, by Lemma 9.2.1, each b_i is contained in a subalgebra R_i of B that is finitely generated C-module. Hence R is a finitely generated C-module. By Lemma 9.2.1 again, a is integral over C, and therefore $C \to A$ is integral. □

Proposition 9.2.4

Suppose $\varphi : B \to A$ is a ring morphism. The elements of A integral over B form a subalgebra of A, i.e.,

$$\widetilde{B} = \{a \in A : a \text{ is integral over } B\}$$

is an B-subalgebra of A.

Proof Suppose $a_1, a_2 \in A$ are integral over B, by Lemma 9.2.1, there exists R_1, R_2 be two subalgebras of A such that $a_i \in R_i$ and R_i is finitely generated B-module. Consider R_1R_2, then $a_1a_2 \in R_1R_2$ and $a_1 + a_2 \in R_1R_2$, hence a_1a_2 and $a_1 + a_2$ are integral over B. It follows that the elements of A integral over B form a subalgebra of A. □

Remark 9.5 **Transcendence theory.** These ideas lead to the main facts about transcendence theory we will need for a discussion of dimension of varieties, see Chapter 13.

9.2.2 The Lying Over and Going-Up Theorems

The Lying Over Theorem is a useful property of integral extensions.

Theorem 9.2.1 (The Lying Over Theorem)

Suppose $\varphi : B \to A$ is an integral extension. Then for any prime ideal $\mathfrak{q} \subseteq B$, there is a prime ideal $\mathfrak{p} \subseteq A$ such that $\mathfrak{p} \cap B = \mathfrak{q}$.

Remark 9.6 To clear on how weak the hypotheses are: B need not be Noetherian, and A need not be finitely generated over B.

Theorem 9.2.2 (The Lying Over Theorem (Geometric version))

If $B \to A$ is an integral extension, then $\operatorname{Spec} A \to \operatorname{Spec} B$ is surjective.

Remark 9.7 Although the Lying Over Theorem 9.2.1 is a theorem in algebra, the name can be interpreted geometrically: the theorem asserts that the corresponding morphism of schemes is surjective. (A map of schemes is **surjective** if the underlying map of sets is surjective.) Translation: "above" every prime $\mathfrak{q}$ "downstairs", there is a prime $\mathfrak{p}$ "upstairs", see Figure 9.1. (For this reason, it is often said that $\mathfrak{p}$ "lies over" $\mathfrak{q}$ if $\mathfrak{p} \cap B = \mathfrak{q}$.)

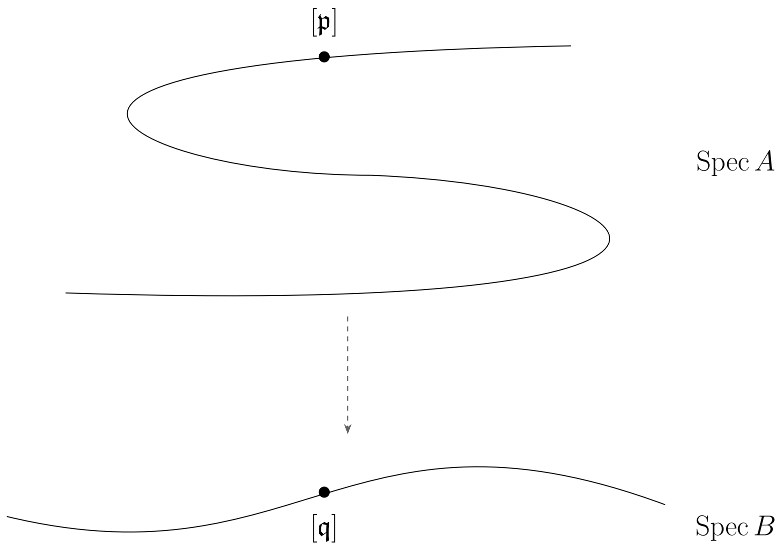


Figure 9.1: A picture of the Lying Over Theorem 9.2.1: if $\varphi : B \to A$ is an integral extension, then $\operatorname{Spec} A \to \operatorname{Spec} B$ is surjective

The following lemmas set up the proof.

Lemma 9.2.2 (The special case of the Lying Over Theorem 9.2.1 where A is a field)

Suppose A is a field. Let $B \subseteq A$ is a subring with A integral over B, then B is a field.

Proof Let $x \in B$, $x \neq 0$. Then $x^{-1} \in A$, since A is a field. Note that A integral over B, we have an equation

$$x^{-n} + b_{n-1}x^{-(n-1)} + \cdots + b_1 x^{-1} + b_0 = 0,$$

where $b_i \in B$. It follows that

$$x^{-1} = -(b_{n-1} + \cdots + b_1 x^{n-2} + b_0 x^{n-1}) \in A,$$

hence A is a field. □

Proof of the Lying Over Theorem 9.2.1:

Proof We first make a reduction: by localizing at $\mathfrak{q}$ and Proposition 9.2.2 (b), we can assume that $(B, \mathfrak{q})$ is a local ring. Then let $\mathfrak{p}$ be any maximal ideal of A. Consider the following diagram.

$$\begin{tikzcd} A \arrow[r, two heads] & A/\mathfrak{p} & \text{field} \\ B \arrow[u, hook] \arrow[r, two heads] & B/(\mathfrak{p} \cap B) \arrow[u, hook] & \end{tikzcd}$$

The right vertical arrow is an integral extension by Proposition 9.2.2 (c). By Lemma 9.2.2, $B/(\mathfrak{p} \cap B)$ is a field too, so $\mathfrak{p} \cap B$ is a maximal ideal, hence it is $\mathfrak{q}$. □

Theorem 9.2.3 (The Going-Up Theorem)

Suppose $\varphi : B \to A$ is an integral ring morphism (not necessarily an integral extension). If $\mathfrak{q}_1 \subseteq \mathfrak{q}_2 \subseteq \cdots \subseteq \mathfrak{q}_n$ is a chain of prime ideals of B, and $\mathfrak{p}_1 \subseteq \cdots \subseteq \mathfrak{p}_m$ is a chain of prime ideals of A such that $\mathfrak{p}_i$ "lies over" $\mathfrak{q}_i$ for all $1 \leq i \leq m$ (and $1 \leq m < n$), then the second chain can be extended to $\mathfrak{p}_1 \subseteq \cdots \subseteq \mathfrak{p}_n$ such that each $\mathfrak{p}_i$ "lies over" $\mathfrak{q}_i$ for all $1 \leq i \leq n$.

Proof

$$\begin{array}{ccccccccccccccc} A & & \operatorname{Spec} A & & \mathfrak{p}_1 & \subseteq & \cdots & \subseteq & \mathfrak{p}_m & \subseteq & \mathfrak{p}_{m+1} & \subseteq & \cdots & \subseteq & \mathfrak{p}_n \\ \uparrow & & \downarrow & & \downarrow & & & & \downarrow & & \downarrow & & & & \downarrow \\ B & & \operatorname{Spec} B & & \mathfrak{q}_1 & \subseteq & \cdots & \subseteq & \mathfrak{q}_m & \subseteq & \mathfrak{q}_{m+1} & \subseteq & \cdots & \subseteq & \mathfrak{q}_n \end{array}$$

By induction we reduce immediately to the case $m = 1$, $n = 2$. Let $\overline{B} = B/\mathfrak{q}_1$ and $\overline{A} = A/\mathfrak{p}_1$. Since $\varphi : B \to A$ is integral, by Proposition 9.2.2 (a), $\overline{B} \to \overline{A}$ is integral. Hence, by the Lying Over Theorem 9.2.1, there exists a prime ideal $\overline{\mathfrak{p}_2}$ of $\overline{A}$ such that $\overline{\mathfrak{p}_2} \cap \overline{B} = \overline{\mathfrak{q}_2}$, the image of $\mathfrak{q}_2$ in $\overline{B}$. Lift back $\overline{\mathfrak{p}_2}$ to A and we have a prime ideal $\mathfrak{p}_2$ with the required properties. □

There are analogous "Going-Down" results (requiring quite different hypotheses); see Chapter 13 and Chapter 25.

9.2.3 Nakayama's Lemma

The trick in the proof of Lemma 9.2.1 can be used to quickly prove Nakayama's Lemma, which we will use repeatedly in the future. This name is used for several different but related results, which we discuss here. (A geometric interpretation will be given in Chapter 15). We may as well prove it while the trick is fresh in our minds.

Lemma 9.2.3 (Nakayama's Lemma version 1)

Suppose A is a ring, I is an ideal of A, and M is a finitely-generated A-module, such that $M = IM$. Then there exists an $a \in A$ with $a \equiv 1 \pmod I$ with $aM = 0$. (Equivalently, there is some $i \in I$ for which multiplication by i induces the identity on M: $im = m$ for all $m \in M$.)

Proof Say M is generated by $m_1, \dots, m_n$. Then as $M = IM$, we have $m_i = \sum_j a_{ij} m_j$ for some $a_{ij} \in I$. Thus

$$(\mathrm{id}_n - Z)\begin{bmatrix} m_1 \\ \vdots \\ m_n \end{bmatrix} = 0 \tag{9.3}$$

where $Z = (a_{ij})$. Multiplying both sides of (9.3) on the left by $\operatorname{adj}(\mathrm{id}_n - Z)$, we obtain

$$\det(\mathrm{id}_n - Z)\begin{bmatrix} m_1 \\ \vdots \\ m_n \end{bmatrix} = 0.$$

Expand out $\det(\mathrm{id}_n - Z)$, as Z has entires in I, we get something that is $1 \pmod I$. □

Definition 9.2.2 (Jacobson radical)

Let A be a ring. The Jacobson radical of A is $\operatorname{rad}(A) = \bigcap_{\mathfrak{m} \in \operatorname{Max} A} \mathfrak{m}$ the intersection of all the maximal ideals of R.

Here is why you care. Suppose I is contained in all maximal ideals of A. (I contained in Jacobson radical, see Definition 9.2.2.) Then any $a \equiv 1 \pmod I$ is invertible. (We are not using Nakayama yet!) Reason: otherwise ($a \neq A$), so the ideal (a) is contained in some maximal ideal $\mathfrak{m}$ — but $a \equiv 1 \pmod{\mathfrak{m}}$, contradiction. As a is invertible, we have the following.

Lemma 9.2.4 (Nakayama's Lemma version 2)

Suppose A is a ring, I is an ideal of A contained in all maximal ideals, and M is a finitely generated A-module. (The most interesting case is when A is a local ring, and I is the maximal ideal.) Suppose $M = IM$. Then $M = 0$.

Proof By Nakayama's Lemma 9.2.3, there exists an $a \in A$ with $a \equiv 1 \pmod{I}$ with $aM = 0$. By the previous discussion, we know that a is invertible, and therefore $M = 0$. □

Lemma 9.2.5 (Nakayama's Lemma version 3)

Suppose A is a ring, and I is an ideal of A contained in all maximal ideals. Suppose M is a finitely generated A-module, and $N \subseteq M$ is a submodule. If $N/IN \to M/IM$ is surjective, then $M = N$.

Proof We claim that $N + IM = M$. Say $\varphi : N/IN \to M/IM$, and let $\varphi(n + IN) = m + IM$, then $n + IM = m + IM$, hence $m - n \in IM$. It follows that $M \subseteq N + IM$. Since $IM \subseteq M$ and $N \subseteq M$, we have $M = N + IM$. Consider the quotient module M/N, we have $M/N \cong (N + IM)/N \cong I(M/N)$. By Nakayama's Lemma 9.2.4, we done. □

Lemma 9.2.6 (Nakayama's Lemma version 4: generators of M/mM lift to generators of M)

Suppose $(A, \mathfrak{m})$ is a local ring. Suppose M is a finitely generated A-module, and $f_1, \ldots, f_n \in M$, with (the images of) $f_1, \ldots, f_n$ generating $M/\mathfrak{m}M$. Then $f_1, \ldots, f_n$ generate M. (In particular, taking $M = \mathfrak{m}$, if we have generators of $\mathfrak{m}/\mathfrak{m}^2$, they also generate $\mathfrak{m}$.)

Proof Let $N = (f_1, \ldots, f_n)$. Since the image of $f_1, \ldots, f_n$ generate M, map $N \to M/\mathfrak{m}M$ is surjective. Note that $N \to M/\mathfrak{m}M$ factors through $N/\mathfrak{m}N$, i.e., $N \to N/\mathfrak{m}N \to M/\mathfrak{m}M$. Hence $N/\mathfrak{m}N \to M/\mathfrak{m}M$ is surjective. Apply Nakayama's Lemma 9.2.5, we have $M = (f_1, \ldots, f_n)$. □

Remark 9.8 **Small remark:** In the proof of Lemma 9.2.6 we use the following fact:

Lemma

Let $f : M \to M'$ and $g : M \to N$ are A-module homomorphism, g is surjective and $\operatorname{Ker} g \subseteq \operatorname{Ker} f$, then exists unique homomorphism $h : N \to M'$ such that

$$f = hg.$$

Also $\operatorname{Ker} h = g(\operatorname{Ker} f)$, $\operatorname{Im} h = \operatorname{Im} f$. So h is injective iff $\operatorname{Ker} g = \operatorname{Ker} f$; h is surjective iff f is surjective.

$$\begin{tikzcd} M \arrow[r, "f"] \arrow[dr, two heads, "g"'] & M' \\ & N \arrow[u, dashed, "\exists! h"'] \end{tikzcd}$$

Definition 9.2.3 (Faithful module)

*A B-module N is said to be **faithful** if the only element of B acting on N by the identity is 1, i.e., if $b \cdot n = n$ for all $n \in N$ then $b = 1$ (or equivalently, if the only element of B acting as the 0-map on N is 0, i.e., $\operatorname{Ann}(N) = 0$).*

Lemma 9.2.7 (Generalizing Lemma 9.2.1)

Suppose S is a subring of a ring A, and $r \in A$. Suppose there is a faithful $S[r]$-module M that is finitely generated as an S-module. Then r is integral over S.

Proof Since M is finitely generated as an S-module, we may assume that $M = (m_1, \dots, m_n)$. Since $S[r]$-module M is finitely generated as an S-module, suppose $rm_i = \sum_j a_{ij}m_j$ where $a_{ij} \in S$, then we have

$$(r\,\mathrm{id}_n - (a_{ij})_{ij}) \begin{bmatrix} m_1 \\ \vdots \\ m_n \end{bmatrix} = 0. \tag{9.4}$$

Say $Z = (a_{ij})_{ij}$. Multiplying both sides of (9.4) on the left by $\operatorname{adj}(r\,\mathrm{id}_n - Z)$, we obtain

$$\det(r\,\mathrm{id}_n - Z) \begin{bmatrix} m_1 \\ \vdots \\ m_n \end{bmatrix} = 0.$$

Since M is faithful $S[r]$-module, $\det(r\,\mathrm{id}_n - Z) = 0$. Expanding the determinant yields an integral equation for r with coefficients in S. □

Proposition 9.2.5

Suppose B is a subring of a ring A. The following are equivalent:

(i) $x \in A$ is integral over B;

(ii) $B[x]$ is a finitely generated B-module;

(iii) $B[x]$ is contained in a subring S of A such that C is a finitely generated B-module.

(iv) There exists a faithful $B[x]$-module M which is finitely generated as an B-module.

Proof (i) $\Rightarrow$ (ii): Since $x \in A$ is integral over B, we have an equation of x,

$$x^n + b_{n-1}x^{n-1} + \cdots + b_0 = 0,$$

where $b_i \in B$. It follows that

$$x^{n+r} = -(b_{n-1}x^{n+r-1} + \cdots + b_0x^r)$$

for all $r \geq 0$. Hence, by induction, all positive powers of x lie in the B-module generated by $1, x, \dots, x^{n-1}$. Hence $B[x]$ is generated (as an B-module) by $1, x, \dots, x^{n-1}$.

(ii) $\Rightarrow$ (iii): Take $C = B[x]$.

(iii) $\Rightarrow$ (iv): Take $M = C$, which is a faithful $A[x]$-module (pick $y \in A[x]$, let $yC = 0$, since $1 \in C$, we have $y \cdot 1 = y = 0$, it follows that M is faithful $A[x]$-module).

(iv) $\Rightarrow$ (i): Lemma 9.2.7. □

Corollary 9.2.2

Let x_i $(1 \leq i \leq n)$ be elements of A, each integral over B. Then the ring $B[x_1, \dots, x_n]$ is a finitely generated B-module.

Proof By induction on n. The case $n = 1$ is part of Proposition 9.2.5. Assume $n > 1$, let $B_r = B[x_1, \dots, x_r]$, then by the inductive hypothesis B_{n-1} is a finitely generated B-module. $B_n = B_{n-1}[x_n]$ is finitely generated B_{n-1}-module by Proposition 9.2.5. Hence B_n is finitely generated B-module. □

Definition 9.2.4 (Integral closure)

*Let $B \to A$ be a ring map. The ring $\widetilde{B} \subseteq A$ of elements integral over B, see Proposition 9.2.4, is called the **integral closure** of B in A. If $B \subseteq A$, we say that B is **integrally closed** in A if $B = \widetilde{B}$.*

Proposition 9.2.6

Suppose A is an integral domain, and $\widetilde{A}$ is the integral closure of A in $K(A)$. Then $\widetilde{A}$ is integrally closed in $K(\widetilde{A}) = K(A)$.

Proof We first show that $K(\widetilde{A}) = K(A)$. Since $\widetilde{A} \subseteq K(A)$ and $K(\widetilde{A})$ is the minimal field which contains $\widetilde{A}$, we have $K(\widetilde{A}) \subseteq K(A)$. Since $A \subseteq \widetilde{A}$, we have $K(A) \subseteq K(\widetilde{A})$, hence $K(A) = K(\widetilde{A})$.

We next show that $\widetilde{A}$ is integrally closed in $K(A)$. Let $x \in K(A)$ and x is integral over $\widetilde{A}$, then we have an equation of x

$$x^n + \widetilde{a_{n-1}} x^{n-1} + \cdots + \widetilde{a_0} = 0, \tag{9.5}$$

where $\widetilde{a_i} \in \widetilde{A}$. Say $R = A[\widetilde{a_0}, \ldots, \widetilde{a_{n-1}}]$, since each $\widetilde{a_i}$ is integral over A, by the induction and Proposition 9.2.5 (ii), $A[\widetilde{a_0}, \ldots, \widetilde{a_{n-1}}]$ is a finitely generated A-module. From (9.5), we know that x is integral over R, by Proposition 9.2.5, $R[x]$ is a finitely generated R-module. Since R is finitely generated A-module, $R[x]$ is a finitely generated A-module. Let $M = R[x]$, then M can be seen as $A[x]$-module and also is a finitely generated A-module. Clearly, M is faithful, by Generalizing Lemma 9.2.7, x is integral over A. It follows that $\widetilde{A}$ is integrally closed in $K(\widetilde{A}) = K(A)$. □

9.3 A gazillion finiteness conditions on morphisms

By the end of this section, you will have seen the following types of morphisms: quasi-compact, quasi-separated, affine, finite, integral, closed, (locally) of finite type, quasi-finite — and possibly, (locally) of finite presentation.

9.3.1 Quasi-compact and quasi-separated morphisms.

Definition 9.3.1 (Quasi-compact morphism)

*A morphism $\pi : X \to Y$ of schemes is **quasi-compact** if for every affine open subset U of Y, $\pi^{-1}(U)$ is quasi-compact. (Equivalently, the preimage of any quasi-compact open subset is quasi-compact. This is the right definition in other parts of geometry.)*

Remark 9.9 We will like this notion because:

(i) finite sets have advantages over infinite sets (e.g., a finite set of integers has a maximum; also, things can be proved inductively),

(ii) most reasonable schemes will be quasi-compact.

Along with quasi-compactness comes the weird notion of quasi-separatedness.

Definition 9.3.2 (Quasi-separated morphism)

*A morphism $\pi : X \to Y$ is **quasi-separated** if for every affine open subset U of Y, $\pi^{-1}(U)$ is a quasi-separated scheme (Definition 6.1.2). (Equivalently, the preimage of any quasi-separated open subset is* *quasi-separated, although we won't worry about proving this. This the definition that extends to other parts of geometry.)*

Note *This will be a useful hypothesis in theorems, usually in conjunction with quasi-compactness. (For this reason, "quasi-compact and quasi-separated" is often abbreviated as **qcqs**.)*

Various interesting kinds of morphisms (locally Noetherian source, affine, separated) are quasi-separated, and having the word "quasi-separated" will allow us to state theorems more succinctly.

Remark 9.10 We will give an equivalent definition of quasi-separateness in Chapter 12 ("quasi-separated=quasi-compact diagonal"), which will be much simpler to use.

Proposition 9.3.1

The composition of two quasi-compact morphisms is quasi-compact.

Proof Let $\pi : X \to Y$ and $\rho : Y \to Z$ be two quasi-compact morphisms. Pick $U \subseteq Z$ be any open affine subset U of Z. We want to show that $\pi^{-1} \circ \rho^{-1}(U)$ is quasi-compact. Since ρ is quasi-compact, $\rho^{-1}(U)$ is quasi-compact. Since π is quasi-compact, the preimage of any quasi-compact open subset is quasi-compact. So $\pi^{-1} \circ \rho^{-1}(U)$ is quasi-compact. It follows that the composition of two quasi-compact morphisms is quasi-compact. □

Remark 9.11 It is also true — but not easy — that the composition of two quasi-separated morphisms is quasi-separated. You are free to show this directly, but will in any case follow easily once we understand it in a more sophisticated way, see Chapter 12.

Following Grothendieck's philosophy of thinking that the important notions are properties of morphisms, not of objects, we can restate the definition of a quasi-compact (resp., quasi-separated) scheme as a scheme that is quasi-compact (resp., quasi-separated) over the final object $\operatorname{Spec} \mathbb{Z}$ in the category of schemes (Proposition 8.3.8).

Proposition 9.3.2

(a) Any morphism from a Noetherian scheme is quasi-compact.

(b) Any morphism from a quasi-separated scheme is quasi-separated.

Proof

(a) Let X be a Noetherian scheme, and $\pi : X \to Y$ be a morphism from X to any scheme Y. We want to show that π is quasi-compact. Let U be any affine open subset of Y, then $\pi^{-1}(U)$ is open in X. Since X is Noetherian, by Proposition 4.6.19, $\pi^{-1}(U)$ is quasi-compact. It follows that $\pi : X \to Y$ is quasi-compact.

(b) Let X be a quasi-separated scheme, and $\pi : X \to Y$ be a morphism from X to any scheme Y. Let U be any affine open subset of Y, then $\pi^{-1}(U)$ is open. Let V_1, V_2 be any two affine open subsets of $\pi^{-1}(U)$, then V_1, V_2 are also affine open subsets of X. Since X is quasi-separated, by Proposition 6.1.6, $V_1 \cap V_2$ is a finite union of affine open subsets. By Proposition 6.1.6 again, X is quasi-separated.

□

Remark 9.12 **Caution.** Two parts of the Proposition 9.3.2 may lead you to suspect that any morphism $\pi : X \to Y$ with quasi-compact source and target is necessarily quasi-compact. **This is false**, and you may verify that the following is a counterexample.

Let Z be the nonquasi-separated scheme constructed in Example 6.1 , and let $X = \operatorname{Spec} k[x_1, x_2, \ldots]$

as in Example 6.1. The obvious open embedding $\pi : X \to Z$ (identifying X with one of the two pieces glued together to get Z) is not quasi-compact (remark after Proposition 4.6.4, pick $\operatorname{Spec} X/\{[\mathfrak{m}]\} \subseteq Z$, the inverse image of $\operatorname{Spec} X/\{[\mathfrak{m}]\}$ is also $\operatorname{Spec} X/\{[\mathfrak{m}]\}$, but $\operatorname{Spec} X/\{[\mathfrak{m}]\}$ is not quasi-separated). (But once you see the Cancellation Theorem, you will quickly see that any morphism from a quasi-compact source to a quasi-separated target is necessaily quasi-compact.)

Corollary 9.3.1

Any morphism from a locally Noetherian scheme is quasi-separated. Thus we working only with locally Noetherian schemes may take quasi-separateness as a standing hypothesis.

Proof By Proposition 6.3.3, we know that any locally Noetherian schemes are quasi-separated, by Proposition 9.3.1 (b), locally Noetherian scheme is quasi-separated. □

Proposition 9.3.3

(a) ***(Quasi-compactness is affine-local on the target.)*** *A morphism $\pi : X \to Y$ is quasi-compact if there is a cover of Y by affine open sets U_i such that $\pi^{-1}(U_i)$ is quasi-compact.*

(b) ***(Quasi-separateness is affine-local on the target.)*** *A morphism $\pi : X \to Y$ is quasi-separated if there is a cover of Y by affine open sets U_i such that $\pi^{-1}(U_i)$ is quasi-separated.*

Proof

(a) Let $\pi : X \to Y$ be a morphism of schemes. Consider the property "preimage of affine open subset of Y is quasi-compact", we want to show this property is affine-local property (Definition 6.3.1). It suffices to show two things:

 (i) Let $\operatorname{Spec} A \hookrightarrow Y$ be an affine open subset with $\pi^{-1}(\operatorname{Spec} A)$ is quasi-compact, then $\pi^{-1}(\operatorname{Spec} A_f)$ is quasi-compact for any $f \in A$;

 (ii) Let $(f_1, \ldots, f_n) = A$ with $\pi^{-1}(\operatorname{Spec} A_{f_i})$ is quasi-compact for all i, then $\pi^{-1}(\operatorname{Spec} A)$ is quasi-compact.

We first show (i). Pick $f \in A$. In fact, $\pi^{-1}(\operatorname{Spec} A_f) = \pi^{-1}(\operatorname{Spec} A) \setminus V(\pi^\sharp f)$. Since $\pi^{-1}(\operatorname{Spec} A)$ is quasi-compact, it can be covered by finite numbers of affine open subset, say $\pi^{-1}(\operatorname{Spec} A) = \bigcup_{i=1}^n \operatorname{Spec} B_i$. Then $\operatorname{Spec} B_i \setminus V(\pi^\sharp f)$ is also affine. Hence

$$\pi^{-1}(\operatorname{Spec} A_f) = \pi^{-1}(\operatorname{Spec} A) \setminus V(\pi^\sharp f) = \bigcup_{i=1}^n (\operatorname{Spec} B_i \setminus V(\pi^\sharp f)).$$

It follows that $\pi^{-1}(\operatorname{Spec} A_f)$ is quasi-compact.

Next we show (ii). Since $(f_1, \ldots, f_n) = A$, we have $\operatorname{Spec} A = \bigcup_{i=1}^n \operatorname{Spec} A_{f_i}$. Hence

$$\pi^{-1}(\operatorname{Spec} A) = \pi^{-1}\left(\bigcup_{i=1}^n \operatorname{Spec} A_{f_i}\right) = \bigcup_{i=1}^n \pi^{-1}(\operatorname{Spec} A_{f_i}).$$

Since each $\pi^{-1}(\operatorname{Spec} A_{f_i})$ is quasi-compact, we know that $\pi^{-1}(\operatorname{Spec} A)$ is quasi-compact. Hence "preimage of affine open subset of Y is quasi-compact" is affine-local property.

Let $\{U_i \hookrightarrow Y\}$ is an open cover of Y such that $\pi^{-1}(U_i)$ is quasi-compact, apply Affine Communication Lemma 6.3.1, the preimage of every affine open subset of Y is quasi-compact, i.e., $\pi : X \to Y$ is quasi-compact.

(b) Let $\pi : X \to Y$ be a morphism of schemes. Consider the property "preimage of affine open subset of Y

is quasi-separated", we want to show this property is affine-local property. It suffices to show two things:

(i) Let $\operatorname{Spec} A \hookrightarrow Y$ be an affine open subset with $\pi^{-1}(\operatorname{Spec} A)$ is quasi-separated, then $\pi^{-1}(\operatorname{Spec} A_f)$ is quasi-compact for any $f \in A$;

(ii) Let $(f_1, \ldots, f_n) = A$ with $\pi^{-1}(\operatorname{Spec} A_{f_i})$ is quasi-compact for all i, then $\pi^{-1}(\operatorname{Spec} A)$ is quasi-separated.

We first show (i). Pick $f \in A$. Since $\pi^{-1}(\operatorname{Spec} A)$ is quasi-separated, by Corollary 6.1.3, there exists an affine cover $\{\operatorname{Spec} B_i \hookrightarrow \pi^{-1}(\operatorname{Spec} A)\}$ such that any two of which have intersection covered by a finite number of affine open subsets. By part (a), we know that

$$\pi^{-1}(\operatorname{Spec} A_f) = \pi^{-1}(\operatorname{Spec} A) \setminus V(\pi^\sharp f) = \bigcup_i (\operatorname{Spec} B_i \setminus V(\pi^\sharp f)) = \bigcup_i (\operatorname{Spec}(B_i)_{\pi^\sharp f}).$$

Consider $(\operatorname{Spec} B_i \setminus V(\pi^\sharp f)) \cap (\operatorname{Spec} B_j \setminus V(\pi^\sharp f))$, since $\operatorname{Spec} B_i \cap \operatorname{Spec} B_j$ is the union of finite number of affine open subsets,

$$(\operatorname{Spec} B_i \setminus V(\pi^\sharp f)) \cap (\operatorname{Spec} B_j \setminus V(\pi^\sharp f)) = (\operatorname{Spec} B_i \cap \operatorname{Spec} B_j) \setminus V(\pi^\sharp f)$$

is the union of finite number of distinguished open subsets. By Corollary 6.1.3 again, $\pi^{-1}(\operatorname{Spec} A)$ is quasi-separated.

We next show (ii). Let $\operatorname{Spec} B, \operatorname{Spec} C$ be any two affine open subsets of $\pi^{-1}(\operatorname{Spec} A)$. We want to show that $\operatorname{Spec} B \cap \operatorname{Spec} C$ is a finite union of affine open subsets. In fact,

$$\operatorname{Spec} B \cap \operatorname{Spec} C = \bigcup_{i=1}^{n} (\operatorname{Spec} B \cap \operatorname{Spec} C \cap \pi^{-1}(\operatorname{Spec} A_{f_i})).$$

Since each $\pi^{-1}(\operatorname{Spec} A_{f_i})$ is quasi-separated, by Proposition 6.1.6, we know that

$$\operatorname{Spec} B \cap \operatorname{Spec} C \cap \pi^{-1}(\operatorname{Spec} A_{f_i}) = (\operatorname{Spec} B \cap \pi^{-1}(\operatorname{Spec} A_{f_i})) \cap (\operatorname{Spec} C \cap \pi^{-1}(\operatorname{Spec} A_{f_i}))$$

is finite union of affine open subsets. Hence $\operatorname{Spec} B \cap \operatorname{Spec} C$ is a finite union of affine open subsets, by Proposition 6.1.6, $\pi^{-1}(\operatorname{Spec} A)$ is quasi-separated. Hence "preimage of affine open subset of Y is quasi-separated"" is an affine-local property.

Let $\{U_i \hookrightarrow Y\}$ is an open cover of Y such that $\pi^{-1}(U_i)$ is quasi-separated, apply Affine Communication Lemma 6.3.1, the preimage of every affine open subset of Y is quasi-separated, i.e., $\pi : X \to Y$ is quasi-separated.

□

9.3.2 Affine morphisms

Definition 9.3.3 (Affine morphism)

*A morphism $\pi : X \to Y$ is **affine** if for every affine open set U of Y, $\pi^{-1}(U)$ (interpreted as an open subscheme of X) is an affine scheme.*

Proposition 9.3.4

The composition of two affine morphisms is affine.

Proof Clearly! □

Proposition 9.3.5

Affine morphisms are quasi-compact and quasi-separated.

Proof Let $\pi : X \to Y$ be an affine morphism, then for every affine open set U of Y, $\pi^{-1}(U)$ is an affine scheme, and therefore is quasi-compact. By Corollary 6.1.2, affine scheme is quasi-separated. Hence affine morphisms are quasi-compact and quasi-separated. □

Proposition 9.3.6 (The property of "affineness" of a morphism is affine-local on the target)

A morphism $\pi : X \to Y$ is affine if there is a cover of Y by affine open sets U such that $\pi^{-1}(U)$ is affine.

Proof We want to show that the condition "π is affine over" is an affine-local property. We check our two criteria. First, suppose $\pi : X \to Y$ is affine over $\operatorname{Spec} B$, i.e., $\pi^{-1}(\operatorname{Spec} B) = \operatorname{Spec} A$. Then for any $s \in B$,

$$\pi^{-1}(\operatorname{Spec} B_s) = \pi^{-1}(\operatorname{Spec} B \setminus V(s)) = \pi^{-1}(\operatorname{Spec} B) \setminus \pi^{-1}(V(s)) = \operatorname{Spec} A \setminus V(\pi^\sharp s) = \operatorname{Spec} A_{\pi^\sharp s}.$$

Second, suppose we are given $\pi : X \to \operatorname{Spec} B$ and $(s_1, \dots, s_n) = B$ with $X_{\pi^\sharp s_i} := \operatorname{Spec} A_i$. (Recall from Definition 7.2.4 that $X_{\pi^\sharp s_i}$ is the open subset of X where $\pi^\sharp s_i$ doesn't vanish.) We wish to show that X is affine too. Let $A = \Gamma(X, \mathscr{O}_X)$. Then $X \to \operatorname{Spec} B$ factors through the tautological map $\alpha : X \to \operatorname{Spec} A$ (arising from the (iso)morphism $A \to \Gamma(X, \mathscr{O}_X)$, Proposition 8.3.7).

$$\begin{tikzcd} X = \bigcup_{i=1}^n X_{\pi^\sharp s_i} \arrow[rr, "\alpha"] \arrow[dr, "\pi"'] & & \operatorname{Spec} A \arrow[dl, "\beta"] \\ & \operatorname{Spec} B = \bigcup_{i=1}^n D(s_i) & \end{tikzcd}$$

We want to show that α is an isomorphism. Now $\beta^{-1}(D(s_i)) = D(\beta^\sharp s_i) \cong \operatorname{Spec} A_{\beta^\sharp s_i}$ (the preimage of a distinguished open set is a distinguished open set), and $\pi^{-1}(D(s_i)) = \operatorname{Spec} A_i$. By Proposition 9.3.3, X is quasi-compact and quasi-separated, so the hypotheses of the Qcqs Lemma 7.2.3 are satisfied. Hence by the Qcqs Lemma 7.2.3, we have an induced isomorphism of $A_{\beta^\sharp s_i} = \Gamma(X, \mathscr{O}_X)_{\beta^\sharp s_i} \cong \Gamma(X_{\beta^\sharp s_i}, \mathscr{O}_X) = A_i$. Thus α induces an isomorphism $\operatorname{Spec} A_i \xrightarrow{\sim} \operatorname{Spec} A_{\beta^\sharp s_i}$ (an isomorphism of rings induces an isomorphism of affine schemes, Proposition 5.3.1). Thus α is an isomorphism over each $\operatorname{Spec} A_{\beta^\sharp s_i}$, which cover $\operatorname{Spec} A$, and hence α is an isomorphism. Therefore $X \to \operatorname{Spec} A$ is an isomorphism, so X is affine as desired. □

The affine-locality of affine morphisms (Proposition 9.3.6) has some nonobvious consequence, as shown in the next proposition.

Proposition 9.3.7 (Useful)

Suppose Z is a closed subset of an affine scheme $\operatorname{Spec} A$ locally cut out by one equation. (In other words, $\operatorname{Spec} A$ can be covered by smaller open sets, and on each such set Z is cut out by one equation.) Then the complement Y of Z is affine. (This is clear if Z is globally cut out by one function f, even set-theoretically; then $Y = \operatorname{Spec} A_f$. However, Z is not always of this form, see Chapter 20.)

Proof We may assume that $\operatorname{Spec} A = \bigcup_{i=1}^n D(f_i)$, say $U_i = D(f_i)$. By the condition, we know that $U_i \cap Z = V(g_i)$. Consider the inclusion $\iota : Y \to \operatorname{Spec} A$, we have

$$\iota^{-1}(U_i) = U_i \cap (\operatorname{Spec} A \setminus Z) = U_i \setminus Z = U_i \setminus (U_i \cap Z) = U_i \setminus V(g_i) = U_i \cap D(g_i) = D(f_i g_i).$$

Since $\operatorname{Spec} A$ is covered by U_i and $\iota^{-1}(U_i)$ is affine, by Proposition 9.3.6, the morphism $\iota : Y \to \operatorname{Spec} A$ is affine, and therefore Y is affine. □

9.3.3 Finite and integral morphisms

Before defining finite and integral morphisms, we give an example to keep in mind:

Example 9.1 If L/K is a field extension, then $\operatorname{Spec} L \to \operatorname{Spec} K$

(i) is always affine;
(ii) is integral if L/K is algebraic;
(iii) is finite if L/K is finite.

Finite morphisms

Definition 9.3.4 (Finite algebra)

*If we have a ring morphism $B \to A$ such that A is a finitely generated B-module then we say that A is a **finite** B-algebra.*

Remark 9.13 This is stronger than being a finitely generated B-algebra. (The similarity of the terminology "finite" and "finitely generated" B-algebra is unfortunate.)

Definition 9.3.5 (Finite morphism)

*We say a morphism $\pi : X \to Y$ is **finite** if for every affine open set $\operatorname{Spec} B$ of Y, $\pi^{-1}(\operatorname{Spec} B)$ is the spectrum of a finite B-algebra.*

Proposition 9.3.8

Finite morphisms are affine.

Proof By definition, it is clear. □

Proposition 9.3.9 (The property of finiteness is affine-local on the target)

A morphism $\pi : X \to Y$ is finite if there is a cover of Y by affine open sets $\operatorname{Spec} A$ such that $\pi^{-1}(\operatorname{Spec} A)$ is the spectrum of a finite A-algebra.

Proof Let $\pi : X \to Y$ be a morphism of schemes. We want to show that the property "the preimage of affine open sets of Y is the spectrum of a finite algebra" is an affine local property. If suffices to show the following two things:

(i) Let $\operatorname{Spec} A$ be an affine open subsets of Y and $\pi^{-1}(\operatorname{Spec} A)$ is the spectrum of finite A-algebra, than $\pi^{-1}(\operatorname{Spec} A_f)$ is the spectrum of a finite A_f-algebra for any $f \in A$.
(ii) Let $A = (f_1, \dots, f_n)$ with each $\pi^{-1}(\operatorname{Spec} A_{f_i})$ is the spectrum of finite A_{f_i}-algebra, then $\pi^{-1}(\operatorname{Spec} A)$ is the spectrum of finite A-algebra.

To show (i), we may assume $\pi^{-1}(\operatorname{Spec} A) = \operatorname{Spec} B$ where B is a finite A-algebra. In fact,

$$\pi^{-1}(\operatorname{Spec} A_f) = \pi^{-1}(\operatorname{Spec} A) \setminus V(\pi^\sharp f) = \operatorname{Spec} B_{\pi^\sharp f}.$$

From $\pi|_{\operatorname{Spec} B} : \operatorname{Spec} B \to \operatorname{Spec} A$, we get a ring morphism $\pi|^\sharp_{\operatorname{Spec} B} : A \to B$, it induces a map $A_f \to B_{\pi^\sharp f}$, hence $B_{\pi^\sharp f}$ is an A_f-algebra. Since B is a finite A-algebra, we know that $B_{\pi^\sharp f}$ is a finite A_f-algebra.

To show (ii), we may assume $\pi^{-1}(\operatorname{Spec} A_{f_i}) = \operatorname{Spec} B_i$ where B_i is a finite A_{f_i}-algebra. Say $Z = \pi^{-1}(\operatorname{Spec} A)$. As the proof in Proposition 9.3.6, we know that $Z = \operatorname{Spec} \Gamma(Z, \mathscr{O}_X)$ and $\Gamma(Z, \mathscr{O}_X)_{\pi^\sharp f_i} = B_i$, hence $\Gamma(Z, \mathscr{O}_X)_{\pi^\sharp f_i}$ is a finite A_{f_i}-algebra, and therefore is a finite A-algebra, by Proposition 6.3.2, $\Gamma(Z, \mathscr{O}_X)$ is a finite A-algebra, as we desired.

Hence the property "the preimage of affine open sets of Y is the spectrum of a finite algebra" is an affine-local property. Apply Affine Communication Lemma 6.3.1, we know that $\pi : X \to Y$ is finite. $\square$

The following four example will give you some feeling for finite morphisms. In each example, you will notice two things. In each case, you will notice two things. In each case, the maps are always finite-to-one (as maps of sets). We will verify this in general. You will also notice that the morphisms are **closed** as maps of topological spaces, i.e., the images of closed sets are closed. We will show that finite morphisms are always closed (and give a second proof in Chapter 10). Intuitively, you should think of finite as being closed plus finite fibers, although this isn't quite true. We will make this precise in Chapter 29.

Example 9.2 **Branched covers.** Consider the morphism $\operatorname{Spec} k[t] \to \operatorname{Spec} k[u]$ given by $u \mapsto p(t)$ where $p(t) \in k[t]$ is a degree n polynomial (see Figure 9.2). This is finite: $k[t]$ is generated as a $k[u]$-module by $1, t, t^2, \dots, t^{n-1}$ (we may assume $p(t) = t^n + a_{n-1}t^{n-1} + \cdots + a_0$, let $u = p(t)$, then $t^n \in k[u] + k[u]t + \cdots + k[u]t^{n-1}$, hence $k[t]$ is a finitely generated $k[u]$-module).

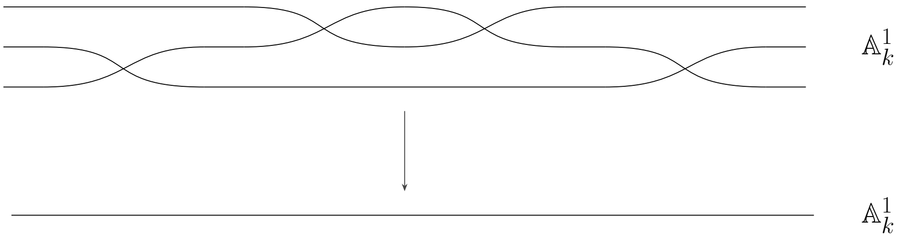


Figure 9.2: The "branched cover" $\mathbb{A}_k^1 \to \mathbb{A}_k^1$ of the "u-line" by the "t-line" given by $u \mapsto p(t)$ is finite

Example 9.3 **Closed embeddings** (to be defined soon, in Chapter 10). If I is an ideal of a ring A, consider the morphism $\operatorname{Spec} A/I \to \operatorname{Spec} A$ given by the obvious map $A \to A \setminus I$ (see Figure 9.3 for an example, with $A = k[t]$, $I = (t)$). This is a finite morphism ($A \setminus I$ is generated as an A-module by the element $1 \in A/I$).

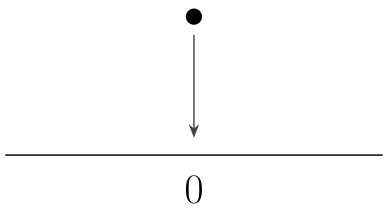


Figure 9.3: The "closed embedding" $\operatorname{Spec} k \to \operatorname{Spec} k[t]$ given by $t \mapsto 0$ is finite

Example 9.4 **Normalization (to be defined in Chapter 11).**

Consider the morphism $\operatorname{Spec} k[t] \to \operatorname{Spec} k[x,y]/(y^2-x^2-x^3)$ corresponding to $k[x,y]/(y^2-x^2-x^3) \to k[t]$ given by $x \mapsto t^2-1$, $y \mapsto t^3-t$, see figure 9.4. This is a finite morphism, as $k[t]$ is generated as a $(k[x,y]/(y^2-x^2-x^3))$-module by 1 and t (since $t^2 = x+1$, $t^3 = y+t$). The figure suggests that this is an isomorphism away from the "node" of the target. We can verify this, by checking that it induces an isomorphism between $D(t^2-1)$ in the source and $D(x)$ in the target. (Note that under the morphism $x \mapsto t^2-1, y \mapsto t^3-t^2$ we have isomorphism $(k[x,y]/(y^2-x^2-x^3))_x \cong k[t^2-1, t^3-t]_{t^2-1} \cong k[t]_{t^2-1}$). We will meet this example again repeatedly!

Proposition 9.3.10 (Finite morphism to $\operatorname{Spec} k$. Important!)

If $X \to \operatorname{Spec} k$ is a finite morphism, then X is a finite union of points with the discrete topology, each point with residue field a finite extension of k, see Figure 9.5.

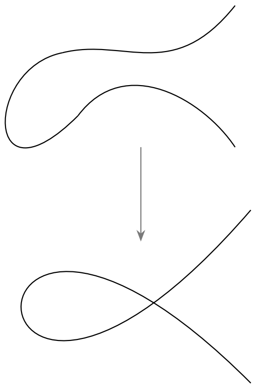


Figure 9.4: The "normalization" of the nodal cubic $\operatorname{Spec} k[t] \to \operatorname{Spec} k[x,y]/(y^2 - x^2 - x^3)$ given by $(x,y) \mapsto (t^2-1, t^3-t^2)$ is finite

Remark 9.14 An example is

$$\operatorname{Spec}(\mathbb{F}_8 \times \mathbb{F}_4[x,y]/(x^2,y^4) \times \mathbb{F}_4[t]/(t^9) \times \mathbb{F}_2) \longrightarrow \operatorname{Spec} \mathbb{F}_2.$$

Figure 9.5: A picture of a finite morphism to $\operatorname{Spec} k$. Bigger fields are depicted as bigger points.

Proof Suppose $\pi : X \to \operatorname{Spec} k$ is a finite morphism, then $\pi^{-1}(\operatorname{Spec} k) = X$ is the spectrum of finite k-algebra. We may assume that $X = \operatorname{Spec} A$, where A is a finite k-algebra. Pick $[\mathfrak{p}] \in \operatorname{Spec} A$, then $A/\mathfrak{p}$ is an integral domain and also is a finite k-algebra, by Proposition 4.2.1, we know that $A/\mathfrak{p}$ is a field, and therefore $\mathfrak{p}$ is maximal ideal. Hence $\operatorname{Spec} A = \operatorname{Max} A$. Since A is a finitely generated k-vector space, A is a finite dimension k-vector space, and therefore A is Noetherian ring. Note that $\operatorname{Spec} A = \operatorname{Max} A$, A is Artin ring. By [AM18, Proposition 8.3], we know that A has only a finite number of maximal ideal. Hence $\operatorname{Spec} A$ has finite closed points, i.e. $\operatorname{Spec} A$ is finite, and therefore discrete. Let $[\mathfrak{m}] \in \operatorname{Spec} A$, then the residue field is $\kappa([\mathfrak{m}]) \cong K(A/\mathfrak{m}) \cong A/\mathfrak{m}$. Since A is finite dimensional k-vector space, the residue field $A/\mathfrak{p}$ is a finite extension of k. $\square$

Proposition 9.3.11

The composition of two finite morphisms is also finite.

Proof Let $\pi : X \to Y$ and $\psi : Y \to Z$ be two finite morphisms. Let $\{\operatorname{Spec} C_i \hookrightarrow Z\}$ be a cover of Z. Since $\psi : Y \to Z$ is finite, we know that each $\psi^{-1}(\operatorname{Spec} C_i)$ is the spectrum of finite C_i-algebra. We may assume $\psi^{-1}(\operatorname{Spec} C_i) = \operatorname{Spec} B_i$ where B_i is finite C_i-algebra. Since $\{\psi^{-1}(\operatorname{Spec} C_i) \hookrightarrow Y\}$ is an open cover of Y, we know that $\{\pi^{-1} \circ \psi^{-1}(\operatorname{Spec} C_i) \hookrightarrow X\}$ is an open cover of X, since π is finite morphism, each

$\pi^{-1} \circ \psi^{-1}(\operatorname{Spec} C_i)$ is the spectrum of finite B_i-algebra, say $\pi^{-1} \circ \psi^{-1}(\operatorname{Spec} C_i) = \operatorname{Spec} A_i$ where A_i is finite B_i-algebra. Each A_i can be seen as finitely generated B_i-module, since B_i can be seen as finitely generated C_i-module, we know that A_i is a finitely generated C_i-module. Hence A_i is also a finite C_i-algebra. By Proposition 9.3.9, $\psi \circ \pi : X \to Y$ is finite morphism. $\square$

Proposition 9.3.12 (Finite morphisms to $\operatorname{Spec} A$ are projective)

If R is an A-algebra, define a graded ring $S_\bullet$ by $S_0 = A$, and $S_n = R$ for $n > 0$. Then there is an isomorphism $\operatorname{Proj} S_\bullet \xrightarrow{\sim} \operatorname{Spec} R$. Suppose $\pi : X \to \operatorname{Spec} A$ is a finite morphism, then X is a projective A-scheme (Definition 5.5.11).

Proof In fact, we have $\operatorname{Proj} S_\bullet = \bigcup_{f \in S_1} D_+(f)$ and $\operatorname{Spec} R = \bigcup_{f \in R} D(f)$. By Proposition 5.5.9, we know that $D_+(f) \cong \operatorname{Spec}((S_\bullet)_f)_0$ and $D(f) \cong \operatorname{Spec} R_f$. Define a ring map $R_f \to ((S_\bullet)_f)_0$ by setting

$$\frac{r}{f^k} \longmapsto \frac{r}{f^k},$$

where r can be seen as the element belong to $S_k = R$. Hence this ring map is well-defined, and clearly is an isomorphism. By Proposition 5.3.1, we obtain an isomorphism of schemes $\pi_f : D_+(f) \xrightarrow{\sim} D(f)$. Clearly π_f "agree on the overlaps", i.e. $\pi_f|_{D_+(f)\cap D_+(g)} = \pi_g|_{D_+(f)\cap D_+(g)}$, by Morphism of locally ringed spaces glue 8.3.2, there is an isomorphism $\operatorname{Proj} S_\bullet \xrightarrow{\sim} \operatorname{Spec} R$.

Suppose $\pi : X \to \operatorname{Spec} A$ is a finite morphism, then $X = \operatorname{Spec} R$ for some R where R is a finite A-algebra. By above discussion we know that $X \cong \operatorname{Proj} S_\bullet$, and $S_\bullet$ is a finitely generated graded ring over A, and hence that X is a projective A-scheme. $\square$

Remark 9.15 Hence finite morphisms are affine (by definition) and projective. The converse is also true, see Chapter 19.

Proposition 9.3.13 (Important)

The finite morphisms have finite fibers.

Remark 9.16 This is a useful proposition, because you will have to figure out how to get at points in a fiber of a morphism: given $\pi : X \to Y$, and $q \in Y$, what are the points of $\pi^{-1}(q)$? This will be easier to do once we discuss fibers in greater detail, but it will be enlightening to do it now.

Proof We reduce the problem via the following steps.

$$\begin{array}{ccccccc}
 & X & \xrightarrow{\pi} & Y & \ni & q & \\
 & \uparrow & & \uparrow & & & \\
\pi^{-1}(\operatorname{Spec} B) = \operatorname{Spec} A & \operatorname{Spec} A & \xrightarrow{\text{finite}} & \operatorname{Spec} B & \ni & q = [\mathfrak{q}] & \\
A' = A/\pi^\sharp(\mathfrak{q})A & \operatorname{Spec} A' & \xrightarrow{\text{finite}} & \operatorname{Spec} B' & \ni & [(0)] & B' = B/\mathfrak{q} \\
A'' = \pi^\sharp(B' \setminus \{(0)\})^{-1}A' & \operatorname{Spec} A'' & \xrightarrow{\text{finite}} & \operatorname{Spec} B'' & \ni & [(0)B'_{(0)}] & B'' = B'_{(0)}
\end{array}$$

Suppose the morphism of schemes $\pi : X \to Y$ is finite and $q \in Y$, then q contained in some affine open subsets of Y, say $q \in \operatorname{Spec} B$. Since $\pi : X \to Y$ is finite, we may assume $\pi^{-1}(\operatorname{Spec} B) = \operatorname{Spec} A$ where A is a finite B-algebra. Hence we may reduce the problem by setting $X = \operatorname{Spec} A$, $Y = \operatorname{Spec} B$, and $q = [\mathfrak{q}] \in \operatorname{Spec} B$. Clearly, the morphism of affine schemes $\operatorname{Spec} A \to \operatorname{Spec} B$ is finite.

Now we throw out everything in B outside $\overline{\{q\}}$ by modding out by $\mathfrak{q}$. From finite ring map $B \to A$, it induces a finite ring map $B/\mathfrak{q} \to A/\pi^\sharp(\mathfrak{q})A$, i.e., $A/\pi^\sharp(\mathfrak{q})A$ is a finite $B/\mathfrak{q}$-algebra. Then we obtain a finite morphism of schemes $\operatorname{Spec} A/\pi^\sharp(\mathfrak{q})A \to \operatorname{Spec} B/\mathfrak{q}$, and $q \in \operatorname{Spec} B$ is corresponding to $[(0)] \in \operatorname{Spec} B/\mathfrak{q}$. So we have reduced to the case where Y is the Spec of an integral domain B, and $[\mathfrak{q}] = [(0)]$ is the generic point.

Next, we can throw out the rest of the points of B by localizing at (0). From ring map $\pi^\sharp : B \to A$, we obtain ring map $B_{(0)} \to \pi^\sharp(B \setminus \{0\})^{-1}A$. Since A is finitely generated B-module, $(B \setminus \{0\})^{-1}A$ is finitely generated $B_{(0)}$-module, and therefore $(B \setminus \{0\})^{-1}A$ is finite $B_{(0)}$-algebra. Hence morphism of schemes $\operatorname{Spec}(\pi^\sharp(B \setminus \{0\})^{-1}A) \to \operatorname{Spec} B_{(0)}$ is finite. Note that B is an integral domain, $B_{(0)}$ is a field. Also q is corresponding to $[(0)B_{(0)}] \in \operatorname{Spec} B_{(0)}$. So we may reduce to the case that $Y = \operatorname{Spec} k$ and $q = [(0)] \in Y$.

By Proposition 9.3.10, $\pi^{-1}(q)$ is a finite union of points, i.e. fiber is finite. □

Proposition 9.3.12 and Proposition 9.3.13 show that finite morphisms are projective and have finite fibers. The converse is also true, see Chapter 19.

There is more to finiteness than finite fibers, and three examples to keep in mind are Example 9.5 , Example 9.6 , and figure in Chapter 20 (a variant of Figure 9.4 showing a morphism that is affine, closed, one-to-one on points and even a monomorphism, but not finite).

Example 9.5 The open embedding $\mathbb{A}^2_k \setminus \{(0,0)\} \to \mathbb{A}^2_k$ has finite fibers, but is not affine (as $\mathbb{A}^2_k \setminus \{(0,0)\}$ isn't affine, §5.4.1) and hence not finite.

Example 9.6 The open embedding $\mathbb{A}^1_{\mathbb{C}} \setminus \{0\} \to \mathbb{A}^1_{\mathbb{C}}$ has finite fibers and is affine, but is not finite.

Proof Note that $\mathbb{A}^1_{\mathbb{C}} \setminus \{0\} \cong \operatorname{Spec} \mathbb{C}[x]_x$, but $\mathbb{C}[x]_x$ is not finitely generated $\mathbb{C}[x]$-module, and therefore $\mathbb{C}[x]_x$ is not finite $\mathbb{C}[x]$-algebra. Hence open embedding $\mathbb{A}^1_{\mathbb{C}} \setminus \{0\} \to \mathbb{A}^1_{\mathbb{C}}$ is not finite. □

Integral morphisms

Definition 9.3.6 (Integral)

*A morphism $\pi : X \to Y$ of schemes in **integral** if π is affine, and for every affine open subset $\operatorname{Spec} B \subseteq Y$, with $\pi^{-1}(\operatorname{Spec} B) = \operatorname{Spec} A$, the induced map $B \to A$ is an integral ring morphism.*

Proposition 9.3.14

The property of integrality of morphisms is affine-local on the target.

Proof By Proposition 9.2.1 and Proposition 9.2.2, and Affine Communication Lemma 6.3.1. □

Proposition 9.3.15

The composition of two integral morphisms is also integral.

Proof By Proposition 9.2.3. □

Integral morphisms are mostly useful because:

Proposition 9.3.16

Finite morphisms are integral. But the converse implication doesn't hold.

Proof First part by Corollary 9.2.1. Consider $\operatorname{Spec} \overline{\mathbb{Q}} \to \operatorname{Spec} \mathbb{Q}$, since $\mathbb{Q} \to \overline{\mathbb{Q}}$ is integral, we know that $\operatorname{Spec} \overline{\mathbb{Q}} \to \operatorname{Spec} \mathbb{Q}$ is integral morphism. But $\overline{\mathbb{Q}}$ is not a finite $\mathbb{Q}$-module, and therefore $\operatorname{Spec} \overline{\mathbb{Q}} \to \operatorname{Spec} \mathbb{Q}$ is not finite, it follows that the converse implication doesn't hold. □

Proposition 9.3.17

Integral morphisms are closed, i.e., the image of closed subsets are closed.

Proof Reduce to the affine case. If $\pi^\sharp : B \to A$ is a ring map, inducing the integral morphism $\pi : \operatorname{Spec} A \to \operatorname{Spec} B$. Suppose $I \subseteq A$ be an ideal of A, then it cuts out a closed set $V(I)$ of $\operatorname{Spec} A$. Say $J = (\pi^\sharp)^{-1}(I)$, by Proposition 9.2.2, induced ring map $B/J \to A/I$ is integral. By The Lying Over Theorem 9.2.1, for any prime ideal $\mathfrak{q} \subseteq B/J$ there exists a prime ideal $\mathfrak{p} \supseteq A/I$ such that $\mathfrak{p} \cap B = \mathfrak{q}$. It follows that $\pi(V(I)) = V(J)$, i.e. the image of $V(I)$ is closed. $\square$

Corollary 9.3.2

Finite morphisms are closed. (A second proof will be given in §10.3.2.)

Proof Finite implies integral (Proposition 9.3.16), apply Proposition 9.3.17. $\square$

Proposition 9.3.18 (Unimportant)

Suppose $B \to A$ is integral. Then for any ring morphism $B \to C$, the induced map $C \to A \otimes_B C$ is integral.

Proof Suppose $\varphi : B \to A$ is integral and $\psi : B \to C$ be any ring morphism from B. Let $S = \{\sum_{i=1}^n a_i \otimes c_i \in A \otimes_B C : \sum_{i=1}^n a_i \otimes c_i \text{ is integral over } C\}$, by Proposition 9.2.4, we know that S is an C-subalgebra of $A \otimes_B C$. In fact, $A \otimes_B C$ is generated by $\{a \otimes 1 : a \in A\} \cup \{1 \otimes c : c \in C\}$. Clearly, $\{1 \otimes c : c \in C\} \subseteq S$. It suffices to show that $\{a \otimes 1 : a \in A\} \subseteq S$. Pick $a \otimes 1$. Since A is integral over B, a satisfies some monic polynomial,

$$p(x) = x^n + \varphi(b_{n-1})x^{n-1} + \cdots + \varphi(b_1)x + \varphi(b_0) = 0,$$

where $b_i \in B$. Induced map $\tau : C \to A \otimes_B C$ is given by $c \mapsto 1 \otimes c$. Let

$$\begin{aligned} q(x) =& x^n + \tau \circ \psi(b_{n-1})x^{n-1} + \cdots + \tau \circ \psi(b_1)x + \tau \circ \psi(b_0) \\ =& x^n + (1 \otimes \psi(b_{n-1}))x^{n-1} + \cdots + (1 \otimes \psi(b_1))x + 1 \otimes \psi(b_0) \end{aligned}$$

we want to check that $q(a \otimes 1) = 0$. Consider $q(a \otimes 1)$, we have

$$\begin{aligned} q(a \otimes 1) =& a^n \otimes 1 + a^{n-1} \otimes \psi(b_{n-1}) + \cdots + a \otimes \psi(b_1) + 1 \otimes \psi(b_0) \\ =& a^n \otimes 1 + \varphi(b_{n-1})a^{n-1} \otimes 1 + \cdots + \varphi(b_1)a \otimes 1 + \varphi(b_0) \otimes 1 \\ =& p(a) \otimes 1 \\ =& 0. \end{aligned}$$

It follows that $a \otimes 1$ is integral over C, hence $\{a \otimes 1 : a \in A\} \cup \{1 \otimes c : c \in C\} \subseteq S$, and therefore $S = A \otimes_B C$, which implies that induced map $C \to A \otimes_B C$ is integral. $\square$

Remark 9.17 Once we know what "base change" is, this will imply that the property of integrality of a morphisms is preserved by base change.

Fibers of integral morphisms

Unlike finite morphisms (Proposition 9.3.13), integral morphisms don't always have finite fibers. However, once we make sense of fibers as topological spaces (or even schemes) in §11.3.2, you can check (Proposition 13.1.5) that the fibers have the property that no point is in the closure of any other point.

Example 9.7 **Example of integral morphisms don't have finite fibers.** Consider finite field $\mathbb{F}_2$, and ring

$\mathbb{F}_2[x]/(x^2) = \{0, 1, x, x+1\}$. Then $\mathbb{F}_2 \to \mathbb{F}_2[x]/(x^2)$ is integral. Note that

$$p(t) = t^2(t-1)(t+1)$$

killing all of $\mathbb{F}_2[x]/(x^2)$, $p(t)$ also kills the element of $\prod_{i\in\mathbb{N}} \mathbb{F}_2[x]/(x^2)$. Hence $\mathbb{F}_2 \to \mathbb{F}_2[x]/(x^2)$ is integral. So we obtain an integral morphism of schemes

$$\pi : \operatorname{Spec} \prod_{i\in\mathbb{N}} \mathbb{F}_2[x]/(x^2) \longrightarrow \operatorname{Spec} \mathbb{F}_2,$$

the fiber $\pi^{-1}([(0)])$ is infinite.

9.3.4 Morphisms (locally) of finite type

Locally of finite type, finite type

Definition 9.3.7 (Locally of finite type)

*A morphism $\pi : X \to Y$ is **locally of finite type** if for every affine open set $\operatorname{Spec} B$ of Y, and every affine open subset $\operatorname{Spec} A$ of $\pi^{-1}(\operatorname{Spec} B)$, the induced morphism $B \to A$ express A as a finitely generated B-algebra. By the affine-locality of finite typeness of B-schemes (Proposition 6.3.2), this is equivalent to: $\pi^{-1}(\operatorname{Spec} B)$ can be covered by affine open subsets $\operatorname{Spec} A_i$ such that each A_i is a finitely generated B-algebra.*

Definition 9.3.8 (Finite type)

*A morphism π is **of finite type** if it is a locally of finite type and quasi-compact. Translation: for every affine open set $\operatorname{Spec} B$ of Y, $\pi^{-1}(\operatorname{Spec} B)$ can be covered with a finite number of open sets $\operatorname{Spec} A_i$ such that the induced morphism $B \to A_i$ expresses as A_i as a finitely generated B-algebra.*

Remark 9.18 **Linguistic curiosity.** It is a common practice to name properties as follows: $\mathscr{P}$ equals locally $\mathscr{P}$ plus quasi-compact. Two exceptions are "ringed space" (§8.3) and "finite presentation".

Example 9.8 The "structure morphism" $\mathbb{P}^n_A \to \operatorname{Spec} A$ is of finite type, as $\mathbb{P}^n_A$ is covered by $n+1$ open sets of the form $\operatorname{Spec} A[x_1, \ldots, x_n]$.

Our earlier definition of schemes of "finite type over k" (or "finite type k-schemes") from Definition 6.3.4 is now a special case of this more general notion: the phrase "a scheme X is of finite type over k" means that we are given a morphism $X \to \operatorname{Spec} k$ (the "structure morphism" Definition 8.3.4) that is of finite type.

Here are some properties enjoyed by morphisms of finite type.

Proposition 9.3.19 (The notions "locally finite type" and "finite type" are affine-local on the target)

A morphism $\pi : X \to Y$ is locally of finite type if there is a cover of Y by affine open sets $\operatorname{Spec} B_i$ such that $\pi^{-1}(\operatorname{Spec} B_i)$ is locally finite type over B_i (Definition 6.3.4). Hence the notions of "locally finite type" and "finite type" are both affine-local on the target.

Proof Suppose $\pi : X \to Y$ be a morphism. We shall to show that the property "preimage of affine open subset of Y is locally finite type" is affine-local property. It suffices to show the following two things:

(i) Let $\operatorname{Spec} B$ be an affine open subset of Y, with $\pi^{-1}(\operatorname{Spec} B)$ is locally finite type over B, then $\pi^{-1}(\operatorname{Spec} B_f)$ is locally finite type over B_f for all $f \in B$.

(ii) If $B = (f_1, \ldots, f_n)$ and $\pi^{-1}(\operatorname{Spec} B_{f_i})$ is locally finite type over B_{f_i} for all i, then $\pi^{-1}(\operatorname{Spec} B)$ is locally finite type over B.

We first show (i). Since $\pi^{-1}(\operatorname{Spec} B)$ is locally finite type over B, we may assume $\pi^{-1}(\operatorname{Spec} B) = \bigcup_i \operatorname{Spec} A_i$ where each A_i is finitely generated B-algebra. Localizing each A_i at $\pi^\sharp(f)$, we know that $(A_i)_{\pi^\sharp(f)}$ is finitely generated B_f-algebra. Note that

$$\begin{aligned}\pi^{-1}(\operatorname{Spec} B_f) =& \pi^{-1}(\operatorname{Spec} B) \setminus \pi^{-1}(V(f)) = \pi^{-1}(\operatorname{Spec} B) \setminus V(\pi^\sharp(f))\\ =& \bigcup_i (\operatorname{Spec} A_i \setminus V(\pi^\sharp(f)))\\ =& \bigcup_i \operatorname{Spec}(A_i)_{\pi^\sharp(f)},\end{aligned}$$

it follows that $\pi^{-1}(\operatorname{Spec} B_f)$ is locally finite type over B_f.

We next show (ii). Suppose $B = (f_1, \dots, f_n)$ and $\pi^{-1}(\operatorname{Spec} B_{f_i})$ is locally finite type over B_{f_i} for all i. We may assume $\pi^{-1}(\operatorname{Spec} B_{f_i}) = \bigcup_j \operatorname{Spec} A_{ij}$ where A_{ij} is finitely generated B_{f_i}-algebra. Hence A_{ij} is of the form $B_{f_i}[t_1, \dots, t_m]/I_{ij}$ where $B_{f_i}[t_1, \dots, t_m]$ is polynomial ring. Since localization commutes with quotient, we know that $A_{ij} \cong (C_{ij})_{f_i}$, hence $(C_{ij})_{f_i}$ is finitely generated B_{f_i}-algebra, and therefore is finitely generated B-algebra. Note that

$$\begin{aligned}\pi^{-1}(\operatorname{Spec} B) =& \pi^{-1}\left(\bigcup_{i=1}^n \operatorname{Spec} B_{f_i}\right) = \bigcup_{i=1}^n \pi^{-1}(\operatorname{Spec} B_{f_i})\\ =& \bigcup_{i=1}^n \bigcup_j \operatorname{Spec} A_{ij} \cong \bigcup_{i=1}^n \bigcup_j \operatorname{Spec}(C_{ij})_{f_i}\end{aligned}$$

and each $(C_{ij})_{f_i}$ is finitely generated B-algebra, we know that $\pi^{-1}(\operatorname{Spec} B)$ is locally finite type over B.

Hence the property "preimage of affine open subset of Y is locally finite type" is affine-local property. Apply Affine Communication Lemma 6.3.1, the notions "locally finite type" is affine-local on the target. □

Proposition 9.3.20 (Finite = Integral + Finite Type)

(a) Finite morphisms are of finite type.

(b) A morphism is finite if and only if it is integral and of finite type.

Proof

(a) Let $\pi : X \to Y$ be a finite morphism, then for any affine open subset $\operatorname{Spec} B \hookrightarrow Y$ preimage $\pi^{-1}(\operatorname{Spec} B)$ is the spectrum of a finite B-algebra, say $\operatorname{Spec} A$ where A is a finite B-algebra. Hence A is finitely generated B-module, and therefore is finitely generated B-algebra. It follows that $\pi : X \to Y$ is of finite type.

(b) If $\pi : X \to Y$ is a finite morphism, by Proposition 9.3.16 and part (a), we done.
Conversely, if $\pi : X \to Y$ is integral and of finite type, we want to show that $\pi : X \to Y$ is finite. Let $\operatorname{Spec} B \hookrightarrow Y$ be any affine open subset of Y, since $\pi : X \to Y$ is integral, $\pi^{-1}(\operatorname{Spec} B) = \operatorname{Spec} A$ where induced ring map $B \to A$ is integral. Since $\pi : X \to Y$ is also of finite type, A is finitely generated B-algebra. Then there is an evaluation homomorphism $\mathrm{val} : B[t_1, \dots, t_n] \twoheadrightarrow A$ given by $t_i \mapsto a_i$. Hence $A \cong B[t_1, \dots, t_n]/\operatorname{Ker} \mathrm{val}$. Since A is integral over B, each a_i satisfies polynomial equation

$$x^{m_i} + \pi^\sharp(b_{i,n-1})x^{m_i-1} + \cdots + \pi^\sharp(b_{i,1})x + \pi^\sharp(b_{i,0}) = 0,$$

where $b_{i,j} \in B$. Then we have

$$a_i^{m_i} = -\pi^\sharp(b_{i,n-1})a_i^{m_i-1} - \cdots - \pi^\sharp(b_{i,1})a_i - \pi^\sharp(b_{i,0}).$$

It follows that

$$\begin{aligned}A \cong& B[t_1, \ldots, t_n]/\operatorname{Ker} \mathrm{val} \\ \cong& Ba_1 + \cdots + Ba_1^{m_1-1} + Ba_2 + \cdots + Ba_2^{m_2-1} + \cdots + Ba_n^{m_n-1},\end{aligned}$$

i.e., A is finitely generated B-module, and therefore is finite B-algebra. Hence $\pi : X \to Y$ is finite.

□

Proposition 9.3.21 (Important!)

(a) Every open embedding is locally of finite type, and hence that every quasi-compact open embedding is of finite type. Every open embedding into a locally Noetherian scheme is of finite type.

(b) The composition of two morphisms locally of finite type is locally of finite type. The composition of two morphisms of finite type is of finite type.

(c) Suppose $\pi : X \to Y$ is locally of finite type, and Y is locally Noetherian. Then X is also locally Noetherian. If $\pi : X \to Y$ is a morphism of finite type, and Y is Noetherian, then X is Noetherian.

Proof

(a) Let $\iota : U \hookrightarrow Y$ be any open embedding. For any affine open subset $\operatorname{Spec} B$ of Y, $\iota^{-1}(\operatorname{Spec} B) = \operatorname{Spec} B \cap U$, hence $\iota^{-1}(\operatorname{Spec} B)$ is an open subset of $\operatorname{Spec} B$, and therefore it is covered by some distinguished subset $\operatorname{Spec} B_f$. Consider the canonical morphism $B \to B_f$, clearly, $B_f = B[f^{-1}]$ is finitely generated B-algebra. Hence open embedding $\iota : U \hookrightarrow Y$ is locally of finite type. Hence every quasi-compact open embedding is of finite type.

If Y is a locally Noetherian scheme, then B is a Noetherian ring. Since $U \cap \operatorname{Spec} B$ is open subset of $\operatorname{Spec} B$, by Proposition 4.6.19, $U \cap \operatorname{Spec} B$ is quasi-compact. So apply above discussion, we know that Every open embedding into a locally Noetherian scheme is of finite type.

(b) Let $\pi : X \to Y$ and $\psi : Y \to Z$ be two morphisms locally of finite type, we want to show that $\psi \circ \pi : X \to Z$ locally of finite type. Let $\operatorname{Spec} C$ be any affine open subset of Z, then $\psi^{-1}(\operatorname{Spec} C)$ is covered by affine open subsets $\operatorname{Spec} B_i$ where each B_i is a finitely generated C-algebra. Since $\pi : X \to Y$ locally of finite type, we know that each $\pi^{-1}(\operatorname{Spec} B_i)$ can be covered by $\operatorname{Spec} A_{ij}$ where A_{ij} is finitely generated B_i algebra, and therefore is finitely generated C-algebra. It follows that $(\psi \circ \pi)^{-1}(\operatorname{Spec} C)$ can be cover by the spectrums of C-algebra, i.e., $\psi \circ \pi$ locally of finite type.

Hence as the composition of two quasi-compact morphisms is quasi-compact, Proposition 9.3.1, the composition of two morphisms of finite type is of finite type.

(c) Since Y is locally Noetherian, we may assume $Y = \bigcup_{i=1}^{n} \operatorname{Spec} B_i$ where each B_i is Noetherian, then X is covered by $\pi^{-1}(\operatorname{Spec} B_i)$. Since $\pi : X \to Y$ is locally of finite type, we may assume $\pi^{-1}(\operatorname{Spec} B_i) = \bigcup_{j=1}^{m_i} \operatorname{Spec} A_{ij}$ where each A_{ij} is finitely generated B_i-algebra. Hence $A_{ij} \cong B_i[t_1, \ldots, t_l]/I_{ij}$, since B_i is Noetherian, by Hilbert's Basis Theorem, we know that A_{ij} is Noetherian. Thus X is covered by the spectrum of Noetherian ring which implies that X is locally Noetherian.

Similarly, if Y is Noetherian, from the quasi-compactness of Y, apply above discussion, we know that X can be covered by finite numbers of spectrums of Noetherian ring, it follows that X is Noetherian.

□

Absolute Frobenius morphism

Definition 9.3.9 (Absolute Frobenius morphism)

Suppose X is a scheme over $\mathbb{F}_p$ where p is a prime number. Define an endomorphism $F : X \to X$ such that for each affine open subset $\operatorname{Spec} A \subseteq X$, F corresponds to the map $A \to A$ given by $f \mapsto f^p$ for all $f \in A$. The morphism F is called the ***absolute Frobenius morphism****.*

Proposition 9.3.22

Suppose X is a scheme over $\mathbb{F}_p$ where p is a prime number. If X is locally of finite type over $\mathbb{F}_p$, then absolute Frobenius morphism is finite.

Proof Let $\operatorname{Spec} A$ be any affine open subset of X. We first show that F is an identity as map of topological spaces. Pick $[\mathfrak{p}] \in \operatorname{Spec} A$, then

$$F([\mathfrak{p}]) = (F^\sharp)^{-1}(\mathfrak{p}) = \{f \in A : f^p \in \mathfrak{p}\} = [\mathfrak{p}].$$

It follows that F is an identity as map of topological spaces.

Since X is locally of finite type over $\mathbb{F}_p$, X can be covered by affine open subsets $\operatorname{Spec} A_i$ such that each A_i is a finitely generated $\mathbb{F}_p$-algebra. Each $\operatorname{Spec} A_i$ is quasi-compact and A_i is obviously finitely generated A_i-algebra, by Proposition 9.3.19, F is of finite type. To show that F is finite, it suffices to show that F is integral. By Proposition 9.3.14, it suffices to show that each A_i is integral over $F^\sharp(A_i)$. Since A_i is finitely generated $\mathbb{F}_p$-algebra, suppose $a_1, \ldots, a_m$ be the generators of A_i. Note that each a_i agree with the equation

$$x^p - a_i^p = 0,$$

hence A_i is integral over $F^\sharp(A_i)$, which implies that $F : X \to X$ is integral. Apply Proposition 9.3.20, absolute Frobenius morphisms is finite. □

Quasi-finite morphisms

Definition 9.3.10 (Quasi-finite (less important))

A morphism $\pi : X \to Y$ is quasi-finite if it is of finite type, and for all $q \in Y$, $\pi^{-1}(q)$ is a finite set.

Remark 9.19 The main point of this notion is the "finite fiber" part; the "finite type" hypothesis will ensure that this notion is "preserved by fibered product", Chapter 11.

Proposition 9.3.23

Finite morphisms are quasi-finite.

Proof By Proposition 9.3.20, finite morphisms are of finite type. By Proposition 9.3.13, finite morphisms have finite fibers. Hence finite morphisms are of finite type. □

Example 9.9 **Example of quasi-finite morphisms which are not finite.** The open embedding $\mathbb{A}_k^2 \setminus \{(0,0)\} \to \mathbb{A}_k^2$ has finite fibers, and of finite type, hence is quasi-finite. But not finite, see Example 9.5 .

However, in section §9.4, we will soon see that quasi-finite morphism to $\operatorname{Spec} k$ are finite.

Example 9.10 **A morphism with finite fibers that is not quasi-finite.** A key example of a morphism with finite fibers that is not quasi-finite is $\operatorname{Spec} \mathbb{C}(t) \to \operatorname{Spec} \mathbb{C}$ ($\mathbb{C}(t)$ is function field). Another is $\operatorname{Spec} \overline{\mathbb{Q}} \to \operatorname{Spec} \mathbb{Q}$ (not of finite type, see Chapter 11).

How to picture quasi-finite morphisms?

Proposition 9.3.24

If $\pi : X \to Y$ is a finite morphism of locally Noetherian schemes, then for any quasi-compact open subset $U \subseteq X$, the induced morphism $\pi|_U : U \to Y$ is quasi-finite.

Proof It suffices to show that $\pi|_U : U \to Y$ is of finite type and finite fibers. We first show that $\pi|_U : U \to Y$ is of finite type. Let $\operatorname{Spec} B \subseteq Y$ be an affine open subset, since $\pi : X \to Y$ is a finite morphism, we may assume that $\pi^{-1}(\operatorname{Spec} B) = \operatorname{Spec} A$ where A is finite B-algebra. Consider $(\pi|_U)^{-1}(\operatorname{Spec} B)$, we have $(\pi|_U)^{-1}(\operatorname{Spec} B) = \operatorname{Spec} A \cap U$. Since X is locally Noetherian, A is Noetherian, by Proposition 4.6.18, $\operatorname{Spec} A$ is Noetherian space, by Proposition 4.6.19, $U \cap \operatorname{Spec} A$ is quasi-compact. Since $U \cap \operatorname{Spec} A$ is open subset of $\operatorname{Spec} A$, $U \cap \operatorname{Spec} A$ covered by finite numbers of distinguished open subsets, say

$$U \cap \operatorname{Spec} A = \bigcup_{i=1}^{n} \operatorname{Spec} A_{f_i}.$$

Since A is finite B-algebra, we know that A_{f_i} is finitely generated B-algebra. Hence $\pi|_U$ is of finite type.

We next show that $\pi|_U$ has finite fibers. Let $p \in Y$, since $\pi : X \to Y$ is a finite morphism, by Proposition 9.3.13, we know that $\pi^{-1}(p)$ is finite. Since $(\pi|_U)^{-1}(p) = U \cap \pi^{-1}(p)$, $(\pi|_U)^{-1}(p)$ is finite. Hence $\pi|_U : U \to Y$ is quasi-finite. □

In fact, every reasonable quasi-finite morphism arises in this way. (This simple-sounding statement is in fact a deep and important result — a form of Zariski's Main Theorem.) Thus the right way to visualize quasi-finiteness is as a finite map with some (closed locus of) points removed.

9.3.5 ⋆⋆ (Locally) finitely presented morphisms

The following variant of "locally of finite type" is useful in non-Noetherian situations.

9.4 Images of morphisms: Chevalley's Theorem and elimination theory

In this section, we will answer a question that you may have wondered about long before hearing the phrase "algebraic geometry". If you have a number of polynomial equations in a number of variables with indeterminate coefficients, you would reasonably ask what conditions there are on the coefficients for a (common) solution to exist. Give the algebraic nature of the problem, you might hope that the answer should be purely algebraic in nature — it shouldn't be "random", or involve bizarre functions like exponentials or cosines. You should expect the answer to be given by "algebraic conditions". This is indeed the case, and it can be profitably interpreted as a question about images of maps of varieties or schemes, in which guise it is answered by **Chevalley's Theorem**. As a consequence we get an immediate proof of the Nullstellensatz 4.2.2.

In certain cases, the image is nicer still. For example, the image of a finite morphism is always closed (Corollary 9.3.2). We will prove a classical result, the Fundamental Theorem of Elimination Theory, which essentially generalizes this to maps from projective space.

In a different direction, in the distant future we will see that in certain good circumstances ("flat" plus a bit more, see Chapter 25), morphisms are open (the image of open subsets is open); one example is $\mathbb{A}^n_B \to \operatorname{Spec} B$.

9.4.1 Chevalley's Theorem

If $\pi : X \to Y$ is a morphism of schemes, the notion of the image of π as sets is clear: we just take the points in Y that are the image of points in X. We know that the image can be open (open embeddings), and we have seen examples where it is closed, and more generally, locally closed. But it can be weirder still: consider the morphism $\mathbb{A}_k^2 \to \mathbb{A}_k^2$ given by $(x, y) \mapsto (x, xy)$. The image is the plane, with the y-axis removed, but origin put back in (Figure 9.6). We make a definition to capture this phenomenon.

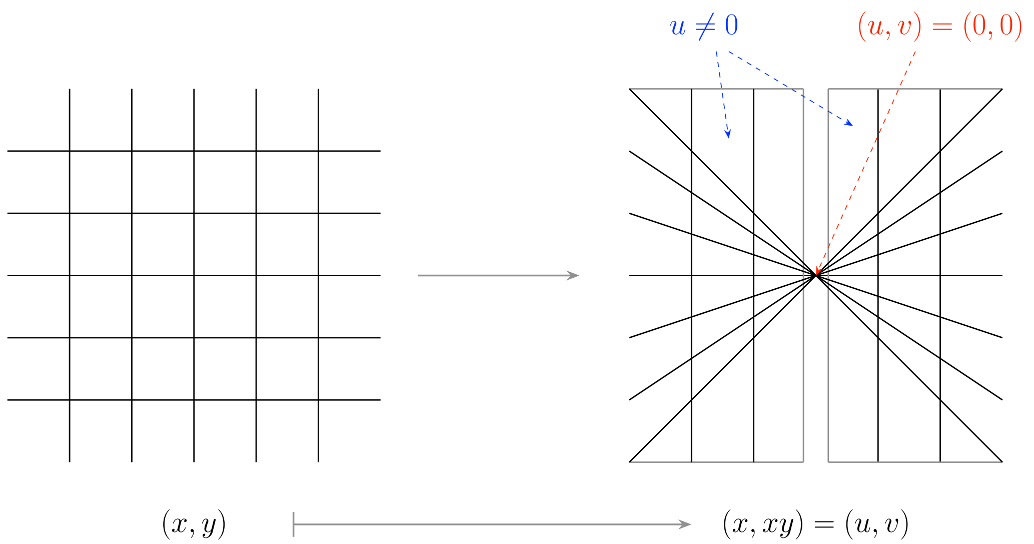


Figure 9.6: The image of $(x, y) \mapsto (x, xy) = (u, v)$

Definition 9.4.1 (Constructible subset)

*A **constructible subset** of a Noetherian topological space is a subset which belongs to the smallest family of subsets such that*

(i) every open set is in the family,

(ii) a finite intersection of family members is in the family,

(iii) the complement of a family member is also in the family.

Example 9.11 The image of $(x, y) \mapsto (x, xy)$ is constructible.

Proof Say $\pi : \mathbb{A}_k^2 \to \mathbb{A}_k^2$ given by $(x, y) \mapsto (x, xy)$, then as set

$$\begin{aligned}\pi(\mathbb{A}_k^2) &= \left\{(u, v) \in \mathbb{A}_k^2 : \text{exists } (a, b) \in \mathbb{A}_k^2, \text{ such that } (u, v) = (a, ab)\right\} \\ &= (\mathbb{A}_k^2 \setminus V(x)) \cup \{(0, 0)\} \\ &= (V(x) \cap (\mathbb{A}_k^2 \setminus \{(0, 0)\}))^c.\end{aligned}$$

Since $\mathbb{A}_k^2 \setminus \{(0, 0)\}$ is open and $V(x)$ is closed, $\pi(\mathbb{A}_k^2)$ is constructible subset. $\square$

Definition 9.4.2 (Locally closed)

*A subset of a topological space X is **locally closed** if it is the intersection of an open subset and a closed subset. Equivalently, it is an open subset of a closed subset, or a closed subset of an open subset.*

Remark 9.20 We will later have trouble extending this to open and closed and locally closed subschemes, see Chapter 10.

Proposition 9.4.1 (Constructible subsets are finite disjoint unions of locally closed subsets)

A subset of a Noetherian topological space X is constructible if and only if it is the finite disjoint union of locally closed subsets.

Proof Let $Z = \coprod_{i=1}^{n} (U_i \cap C_i)$ where each U_i is open and C_i is closed, then

$$Z^c = \bigcap_{i=1}^{n} (U_i \cap C_i)^c.$$

From U_i open and C_i closed, we know that each U_i and C_i are constructible, hence $U_i \cap C_i$ is constructible, and therefore $(U_i \cap C_i)^c$ is constructible. Since Z^c is the finite intersection of $(U_i \cap C_i)^c$, we know that Z^c is constructible, and therefore Z is constructible.

Conversely, let $\mathscr{G}$ be a collection of the finite disjoint union of locally closed subsets. We want to show that $\mathscr{G}$ holds the following conditions:

(i) every open sets of X in $\mathscr{G}$,

(ii) a finite intersection of members of $\mathscr{G}$ is in $\mathscr{G}$,

(iii) the complement of member of $\mathscr{G}$ is also in $\mathscr{G}$.

Let U be an open subset of X, then $U = U \cap X$, since X is closed, U is locally closed subset.

Let $\coprod_{i=1}^{n} (U_i \cap C_i)$, $\coprod_{j=1}^{m} (V_i \cap D_i) \in \mathscr{G}$, where U_i, V_i are open subsets and C_i, D_i are closed subsets. Consider $\left(\coprod_{i=1}^{n} (U_i \cap C_i) \right) \cap \left(\coprod_{j=1}^{m} (V_j \cap D_j) \right)$, then

$$\left(\coprod_{i=1}^{n} (U_i \cap C_i) \right) \cap \left(\coprod_{j=1}^{m} (V_j \cap D_j) \right) = \coprod_{i=1}^{n} \coprod_{j=1}^{m} (U_i \cap V_j) \cap (C_i \cap D_j).$$

Hence $\left(\coprod_{i=1}^{n} (U_i \cap C_i) \right) \cap \left(\coprod_{j=1}^{m} (V_j \cap D_j) \right) \in \mathscr{G}$.

Let $\coprod_{i=1}^{n} (U_i \cap C_i) \in \mathscr{G}$. Consider $\left(\coprod_{i=1}^{n} (U_i \cap C_i) \right)^c$, then

$$\left(\coprod_{i=1}^{n} (U_i \cap C_i) \right)^c = \bigcap_{i=1}^{n} (U_i \cap C_i)^c.$$

So, it suffices to show that $(U_i \cap C_i)^c \in \mathscr{G}$. Note that

$$(U_i \cap C_i)^c = U_i^c \cup C_i^c = U_i^c \coprod (C_i^c \setminus U_i^c) = (X \cap U_i^c) \coprod ((C_i^c \cap U_i) \cap X)$$

where U_i^c is closed and $C_i^c \cap U_i$ is open, hence $(U_i \cap C_i)^c$ belong to $\mathscr{G}$.

We say $\mathscr{F}$ be the smallest family of subsets of X such (i), (ii) and (iii) holds, i.e., $\mathscr{F}$ is the collection of constructible subsets. By above discussion, we know that $\mathscr{F} \subseteq \mathscr{G}$. Pick $A \in \mathscr{F}$, then A is the finite disjoint union of locally closed subsets. □

Remark 9.21 **Important remark:** the only reason for the hypothesis of the topological space in question being Noetherian is because this is the only setting in which we have defined constructible sets. An extension of the notion of constructibility to more general topological spaces is mentioned in Chapter 10.

Corollary 9.4.1

If $X \to Y$ is a continuous map of Noetherian topological spaces, then the preimage of a constructible set is a constructible set.

Proof Since $X \to Y$ is continuous, preimage of open set is open and preimage of closed set is closed. Hence preimage of locally closed subset is locally closed subset. By Proposition 9.4.1, constructible set is the finite disjoint union of locally closed subsets, and therefore the preimage of constructible set is constructible. $\square$

Exercise 9.3 Show that the generic point of $\mathbb{A}_k^1$ does not form a constructible subset of $\mathbb{A}_k^1$ (where k is a field).

Proof The generic point of $\mathbb{A}_k^1$ is $\eta = [(0)]$. If $\{\eta\}$ is a constructible subset, by Proposition 9.4.1, $\{\eta\}$ must be locally closed subset. So we may assume that $\eta = U \cap C$ where U is open and C is closed. Note that the closed subset which contains η is only $\mathbb{A}_k^1$, hence $\eta = U$. By Proposition 4.6.10, η is contained in any open subset of $\mathbb{A}_k^1$, but $\{\eta\}$ is not open (since closed subset of $\mathbb{A}_k^1$ is $\mathbb{A}_k^1$ and the set which constructed by finite closed points), a contradiction. $\square$

Proposition 9.4.2

(a) A constructible subset of a Noetherian scheme is closed if and only if it is "stable under specialization". More precisely, if Z is a constructible subset of a Noetherian scheme X, then Z is closed if and only if for every pair of points y_1 and y_2 with $y_1 \in \overline{\{y_2\}}$, if $y_2 \in Z$, then $y_1 \in Z$.

(b) A constructible subset of a Noetherian scheme is open if and only if it is "stable under generization". More precisely, if Z is a constructible subset of a Noetherian scheme X, then Z is open if and only if for every pair of points y_1 and y_2 with $y_1 \in \overline{\{y_2\}}$, if $y_1 \in Z$, then $y_2 \in Z$.

Proof Let X be a Noetherian scheme, Z be a constructible subset of X.

(a) If Z is closed. Consider pair of points y_1 and y_2 with $y_1 \in \overline{\{y_2\}}$. Let $y_2 \in Z$, then $\overline{\{y_2\}} \subseteq Z$, and therefore $y_1 \in Z$.

Conversely, if Z is "stable under specialization". Since Z is a constructible subset, we may assume $Z = \coprod_{i=1}^n U_i \cap Z_i$ where $U_i \subseteq X$ open and $Z_i \subseteq X$ closed. By Proposition 4.6.12, each Z_i is the finite union of irreducible closed subsets, none contained in any other, say $Z_i = \bigcup\limits_{j=1}^{m_i} Z_{ij}$. So

$$Z = \coprod_{i=1}^n (U_i \cap Z_i) = \bigcup_{i=1}^n \bigcup_{j=1}^{m_i} (U_i \cap Z_{ij}). \tag{9.6}$$

We next show that $Z_{ij} \subseteq Z$. By Proposition 6.1.2, we may assume $Z_{ij} = \overline{\{y_{ij}\}}$. It suffices to show that each y_{ij} is contained in Z. Since $U_i \cap Z_{ij}$ is open subset of Z_{ij}, by Proposition 4.6.10, $y_{ij} \in U_i \cap Z_{ij}$, and therefore $y_{ij} \in Z$. Since Z is "stable under specialization", we know that $Z_{ij} \subseteq Z$, hence $\bigcup\limits_{i=1}^n \bigcup\limits_{j=1}^{m_i} Z_{ij} \subseteq Z$.

From (9.6), $Z \subseteq \bigcup\limits_{i=1}^n \bigcup\limits_{j=1}^{m_i} Z_{ij}$, hence $Z = \bigcup\limits_{i=1}^n \bigcup\limits_{j=1}^{m_i} Z_{ij}$, and therefore Z is closed.

(b) If Z is open. Consider pair of points y_1 and y_2 with $y_1 \in \overline{\{y_2\}}$. Let $y_1 \in Z$. If $y_2 \notin Z$, then $y_2 \in X \setminus Z$, since $X \setminus Z$ is closed, $\overline{\{y_2\}} \subseteq X \setminus Z$, but $y_1 \notin X \setminus Z$, a contradiction.

Conversely, if Z is "stable under generization". Consider $X \setminus Z$, we claim that $X \setminus Z$ is "stable under specialization". For every pair of points y_1 and y_2 with $y_1 \in \overline{\{y_2\}}$, by the condition, we know that "if $y_1 \in Z$, then $y_2 \in Z$", i.e., "if $y_1 \notin X \setminus Z$, then $y_2 \notin X \setminus Z$", i.e., "if $y_2 \in X \setminus Z$, then $y_1 \in X \setminus Z$". Hence $X \setminus Z$ is "stable under specialization", by part (a), $X \setminus Z$ is closed, and therefore Z is open.

$\square$

The image of a morphism of schemes can be stranger than a constructible set. Indeed if S is any subset of a scheme Y, it can be the image of a morphism: let X be the disjoint union of spectra of the residue fields of all the points of S, and let $\pi : X \to Y$ be the natural map. This is quite pathological, but in any reasonable situation, the image is essentially no worse than what arose in the previous example of $(x, y) \mapsto (x, xy)$. This is made precise by Chevalley's Theorem.

Theorem 9.4.1 (Chevalley's Theorem)

If $\pi : X \to Y$ is a finite type morphism of Noetherian schemes, then the image of any constructible set is constructible. In particular, the image of π is constructible.

Proof Since Y is Noetherian scheme, suppose $Y = \bigcup_{i=1}^{n} \operatorname{Spec} B_i$, then $X = \bigcup_{i=1}^{n} \pi^{-1}(\operatorname{Spec} B_i)$. Since $\pi : X \to Y$ is a finite type morphism, each $\pi^{-1}(\operatorname{Spec} B_i) = \bigcup_{j=1}^{m_i} \operatorname{Spec} A_{ij}$ where A_{ij} is finitely generated B_i-algebra. So $X = \bigcup_{i=1}^{n} \bigcup_{j=1}^{m_i} \operatorname{Spec} A_{ij}$, hence

$$\pi(X) = \bigcup_{i=1}^{n} \bigcup_{j=1}^{m_i} \pi(\operatorname{Spec} A_{ij}).$$

By the definition of constructible subset, constructible preserves finite union of closed subset, so we can reduce to the case that $X = \operatorname{Spec} A$ and $Y = \operatorname{Spec} B$ where A is finitely generated B-algebra and A, B are Noetherian rings. Since A is finitely generated B-algebra, A can be written as $B[t_1, \dots, t_n]/I$. Note that $B \to B[t_1, \dots, t_n]/I$ factors through ring map $B \to B[t_1, \dots, t_n]$,

$$\begin{tikzcd} B \arrow[rr, "\pi^\sharp"] \arrow[dr] & & B[t_1, \dots, t_n]/I \\ & B[t_1, \dots, t_n] \arrow[ur, two heads] & \end{tikzcd}$$

it induces morphisms of affine schemes.

$$\begin{tikzcd} X = \operatorname{Spec} B[t_1, \dots, t_n]/I \arrow[rr, "\pi"] \arrow[dr, "\iota"'] & & Y = \operatorname{Spec} B \\ & \operatorname{Spec} B[t_1, \dots, t_n] \arrow[ur, "\psi"'] & \end{tikzcd}$$

We next show that if $\operatorname{Spec} B[t_1, \dots, t_n] \to Y$ holds Chevalley's Theorem then π also holds. Since ι is integral morphism, by Proposition 9.3.17, the image of ι is closed. Hence $\psi(\iota(X)) = \pi(X)$ is constructible. So we may reduce to the case that $X = \operatorname{Spec} B[t_1, \dots, t_n]$ and $Y = \operatorname{Spec} B$ where B is Noetherian. If we show that $\operatorname{Spec} B[t] \to \operatorname{Spec} B$ where B is Noetherian holds, by the induction, $X = \operatorname{Spec} B[t_1, \dots, t_n]$ and $Y = \operatorname{Spec} B$ where B is Noetherian holds. Hence we may reduce to the case that $X = \operatorname{Spec} B[t]$ and $Y = \operatorname{Spec} B$ where B is Noetherian.

We have simplified the task to showing that given $\pi : X = \operatorname{Spec} B[t] \to \operatorname{Spec} B = Y$, the image of any constructible subset of X is constructible in Y. Because constructible sets are finite unions of locally closed subsets (Proposition 9.4.1), we need only show that the image of any locally closed subset Z of X is constructible in Y.

(**An important special case.**) We begin with the case when Z is a closed subset of $X = \mathbb{A}^1_B = \operatorname{Spec} B[t]$, say

$$Z = V(f_1(t), \dots, f_r(t)), \qquad \text{where } f_i(t) = a_{id_i} t^{d_i} + \cdots + a_{i1} t + a_{i0},$$

with $a_{ij} \in B$, so $f_i(t) \in B[t]$ and $d_i = \deg f_i(t)$.

Let $q \in Y$, consider $\pi^{-1}(q)$. In fact,

$$\pi^{-1}(q) = X \times_Y \operatorname{Spec} \kappa(q) \cong \operatorname{Spec}(B[t] \otimes_B \kappa(q)) \cong \operatorname{Spec}(\kappa(q)[t]).$$

Let $Y_{\vec{e}}$ indexed by $\vec{e} = (e_1, \dots, e_r)$, where $e_i \in \mathbb{Z}$, $-1 \le e_i \le d_i$, and $Y_{\vec{e}}$ is the locus where $f_1(t), \dots, f_r(t)$ have "true degree" $e_1, \dots, e_r$ respectively. More precisely:

$$Y_{\vec{e}} = \left\{ q \in Y : \begin{smallmatrix} \forall 1,\dots,r, \text{ the coefficient of } t^{e_i} \text{ in } f_i(t) \in \kappa(q)[t], \\ \text{and the higher coefficients are zero.} \end{smallmatrix} \right\}$$
$$= \{ q \in Y : a_{ie_i} \neq 0 \in \kappa(q), a_{ij} = 0 \in \kappa(q) \text{ for } j > e_i \}$$

(The zero polynomial has "true degree" -1 under this convention.)

More precisely still, we define $Y_{\vec{e}} := \operatorname{Spec} B_{\vec{e}}$, where

$$B_{\vec{e}} := \frac{B_{\prod_i a_{ie_i}}}{(a_{ij})_{j > e_i}}, \qquad \text{so } Y_{\vec{e}} = \left(\bigcap_i D(a_{ie_i}) \right) \cap \left(\bigcap_{\substack{i \\ j > e_i}} V(a_{ij}) \right).$$

Since each point in Y unique determine index $\vec{e}$. So $Y = \operatorname{Spec} B$ is the finite disjoint union of locally closed subsets $Y_{\vec{e}}$, i.e., $Y = \coprod_{\vec{e}} Y_{\vec{e}}$.

If $\vec{e} = (-1, \dots, -1)$, then

$$\pi^{-1}(Y_{\vec{e}}) \cap Z \xrightarrow{\pi} Y_{\vec{e}} \tag{9.7}$$

is $\operatorname{Spec} B_{(-1,\dots,-1)}[t] \to \operatorname{Spec} B_{(-1,\dots,-1)}$, since corresponding ring map $B_{(-1,\dots,-1)} \to B_{(-1,\dots,-1)}[t]$ is integral extension, by The Lying Over Theorem 9.2.2, (9.7) is surjective.

If $\vec{e} \neq (-1, \dots, -1)$, (9.7) is a finite morphism corresponding to the map of rings

$$B_{\vec{e}} \longrightarrow \frac{B_{\vec{e}}}{\left(t^{e_1} + \frac{a_{1(e_1-1)}}{a_{1e_1}} t^{e_1-1} + \dots + \frac{a_{10}}{a_{1e_1}}, \dots, t^{e_r} + \frac{a_{r(e_r-1)}}{a_{re_r}} t^{e_r-1} + \dots + \frac{a_{r0}}{a_{re_r}} \right)}$$

(The ring on the right is generated as a $B_{\vec{e}}$-module by $1, t, \dots, t^{\max(e_i)-1}$.) Because the images of finite morphisms are closed (Proposition 9.3.17 and Proposition 9.3.20), $\pi(Z) \cap Y_{\vec{e}}$ (the image of (9.7)) is a locally closed subset of Y, so

$$\pi(Z) = \pi(Z) \cap Y = \coprod_{\vec{e}} (\pi(Z) \cap Y_{\vec{e}})$$

is indeed constructible when Z is closed. We have completed our important special case.

Remark

Now we proved the image of closed subset is constructible. We want to show that the image of locally closed subset is constructible. Suppose $Z \subseteq X$ is a locally closed subset, then $Z = \overline{Z} \setminus \delta Z$, where $\delta Z = \overline{Z} \setminus Z$. $\overline{Z}$ and δZ are both closed subsets of X (since $Z = U \cap \overline{Z}$ where U is open, hence $\delta Z = \overline{Z} \cap U^c$ is closed). We hope $\pi(Z) = \pi(\overline{Z}) \setminus \pi(\delta Z)$, then, by $\pi(\overline{Z})$ and $\pi(\delta Z)$ are constructible, $\pi(Z)$ is construction. This seems to complete the proof, but unfortunately this is **false**! In fact, $\pi(Z) \neq \pi(\overline{Z}) \setminus \pi(\delta Z)$. Let's take a look at what went wrong. Pick $q \in Y$, such that $\pi^{-1}(q) \subseteq \overline{Z}$. If $\pi^{-1}(q) \cap \delta Z \neq \varnothing$ and $\pi^{-1}(q) \cap Z \neq \varnothing$, then $q \in \pi(Z)$ and $q \in \pi(\delta Z)$, this is where the problem arises. So we need to define something to capture the situation where the fibers fall completely into a closed set, this is why we will define **fiber locus**.

Note that the locus of points $q \in Y$ where the entire fiber $\pi^{-1}(q)$ is contained in Z (Z is closed), $\{q \in Y : \pi^{-1}(q) \subseteq Z\}$, is a closed subset of Y: it is $\operatorname{Spec} B_{(-1,\dots,-1)} = \operatorname{Spec} B/(a_{ij})$. We call this $FL(Z) \subseteq Y$, the "fibral locus for Z", only for the rest of this proof.

Finally, if Z is locally closed, then $\overline{Z} \subseteq X$ is closed, so $\pi(\overline{Z})$ is a constructible subset of $Y = \operatorname{Spec} B$ (apply above consequence). Define $\delta Z := \overline{Z} \setminus Z$, a closed subset of X (since $Z = U \cap \overline{Z}$ where U is open, hence

$\delta Z = \overline{Z} \cap U^c$ is closed); Since $\overline{Z}$ and δZ are both closed, by above discussion (fibral locus for closed subset is closed), we know that $FL(\overline{Z})$ and $FL(\delta Z)$ are both closed subsets of Y. The point q in $FL(\delta Z)$ means the fiber $\pi^{-1}(q)$ contained in δZ, so $\pi^{-1}(q) \cap Z = \varnothing$, and therefore $FL(\delta Z) \cap \pi(Z) = \varnothing$. Hence we can cut $FL(\delta Z)$ out of Y, i.e., $Y \setminus FL(\delta Z)$. The point q in $FL(\overline{Z}) \setminus FL(\delta Z)$ means the fiber $\pi^{-1}(q) \subseteq \overline{Z}$ and $\pi^{-1}(q) \nsubseteq \delta Z$, so $\pi^{-1}(q) \cap \pi(Z) \neq \varnothing$, and therefore $q \in \pi(Z)$, i.e., $FL(\overline{Z}) \setminus FL(\delta Z) \subseteq \pi(Z)$. Note that $FL(\overline{Z}) \setminus FL(\delta Z)$ is locally closed, so we can cut $FL(\overline{Z}) \setminus FL(\delta Z)$ out of $Y \setminus FL(\delta Z)$, i.e., $Y \setminus FL(\overline{Z})$. It suffices to prove the result for the open subset $Y' := Y \setminus (FL(\overline{Z}) \cup FL(\delta Z)) \subseteq Y$. Let $\pi' : X' = \pi^{-1}(Y') \to Y'$ be the restriction of π above Y', and let $Z' = Z \cap X'$. Say $\delta Z' = \overline{Z'} \setminus Z' \subseteq X'$, where $\overline{Z'}$ is the closure of Z' in X'. In fact, $Z' \to Y'$ is quasi-finite. Since π is finite type, $Z' \to Y'$ is also finite type. Pick $q \in Y'$, then $\pi'^{-1}(q) = \operatorname{Spec} \kappa(q)[t] \cap Z'$. Since $\kappa(q)[t]$ is PID, any ideal in $\kappa(q)[t]$ is principal. Since $Z' \cap \operatorname{Spec} \kappa(q)[t]$ is closed in $\operatorname{Spec} \kappa(q)[t]$, then $Z' \cap \operatorname{Spec} \kappa(q)[t] = V(g(t))$. Since $\kappa(q)[t]$ is PID, we may assume

$$g(t) = p_1(t)^{k_1} \cdots p_m(t)^{k_m},$$

so $Z' \cap \operatorname{Spec} \kappa(q)[t]$ is finite, i.e., $X' \to Y'$ has finite fibers. Hence $X' \to Y'$ is quasi-finite.

We next show that the closed subset $\delta Z' \subseteq X'$ does not meet any generic fiber of π', i.e., the preimage of any generic point of Y'. Since Y' is open subscheme of Noetherian scheme Y, we may assume that $Y' = \bigcup \operatorname{Spec} B_f$. If there exists a generic point of Y', say η, such that $\pi'^{-1}(\eta) \cap \delta Z' \neq \varnothing$, then $\eta = [\mathfrak{q}] \in \operatorname{Spec} B_f$ for some f. Since η is generic point of Y', $\mathrm{cl}_{\operatorname{Spec} B_f}(\eta) = \operatorname{Spec} B_f$, so $\pi'^{-1}(\eta) = \operatorname{Spec} K(B_f/\mathfrak{q})[t]$. Since $K(B_f/\mathfrak{q})[t]$ is PID, by Proposition 6.2.8 and Proposition 6.2.7, $\pi'^{-1}(\eta)$ is irreducible. We claim that $\pi'^{-1}(\eta) \cap \mathrm{cl}_{X'}(Z')$ is proper closed in $\pi'^{-1}(\eta)$. If $\pi'^{-1}(\eta) \cap \mathrm{cl}_{X'}(Z') = \pi'^{-1}(\eta)$, then $\mathrm{cl}_{X'}(Z') \subseteq \pi^{-1}(\eta)$, it follows that $\eta \in FL(\mathrm{cl}_{X'}(Z'))$. Since $\mathrm{cl}_{X'}(Z') \subseteq \mathrm{cl}_X(Z)$, $FL(\mathrm{cl}_{X'}(Z')) \subseteq FL(\mathrm{cl}_X(Z))$, so $\eta \in FL(\mathrm{cl}_X(Z))$. But $Y' = Y \setminus (FL(\mathrm{cl}_X(Z)) \cap FL(\delta Z))$, $\eta \notin FL(\mathrm{cl}_X(Z))$, a contradiction. Hence $\pi'^{-1}(\eta) \cap \mathrm{cl}_{X'}(Z')$ is proper closed in $\pi'^{-1}(\eta)$. Since $\pi'^{-1}(\eta)$ is irreducible, the generic point of $\pi'^{-1}(\eta)$ does not belong to $\pi'^{-1}(\eta) \cap \mathrm{cl}_{X'}(Z')$, in the other words, $\pi'^{-1}(\eta) \cap \mathrm{cl}_{X'}(Z')$ is the finite union of closed point (finiteness because $Z' \to Y'$ is quasi-finite). Since $\pi'^{-1}(\eta) \cap Z' \subseteq \pi'^{-1}(\eta) \cap \mathrm{cl}_{X'}(Z')$, $\pi'^{-1}(\eta) \cap Z'$ is also the finite union of closed point, and therefore

$$\pi'^{-1}(\eta) \cap Z' = \mathrm{cl}_{\pi'^{-1}(\eta)}(\pi'^{-1}(\eta) \cap Z') = \pi'^{-1}(\eta) \cap Z'.$$

Hence

$$\begin{aligned}\delta Z' \cap \pi'^{-1}(\eta) =& (\mathrm{cl}_{X'}(Z') \setminus Z') \cap \pi'^{-1}(\eta) \\ =& (\mathrm{cl}_{X'}(Z') \cap \pi'^{-1}(\eta)) \setminus (Z' \cap \pi'^{-1}(\eta)) \\ =& \varnothing,\end{aligned}$$

a contradiction. Hence the closed subset $\delta Z' \subseteq X'$, does not meet any generic fiber of π', i.e., $\pi'(\delta Z')$ does not contain any generic point of Y'. So $\pi'(\delta Z')$ not dense in Y', it follows that $Y' \setminus \mathrm{cl}_{Y'}(\pi'(\delta Z')) \neq \varnothing$. We may assume $Y' = \bigcup_{i=1}^m Y_i$ where each Y_i is irreducible component of Y'. Clearly $\pi'(\delta Z') \cap Y_i$ not dense in Y_i, so there exists open subset W_i of Y_i such that $W_i \cap (\pi'(\delta Z') \cap Y_i) = \varnothing$. Let $V_i = W_i \setminus \bigcup_{j \neq i} cl_{Y'}(Y_j)$, then V_i is open in Y_i. Since Y_i is irreducible, V_i is dense in Y_i. Let $V = \bigcup_{i=1}^m V_i$, then open subset V is dense in Y' and $V \cap \pi'(\delta Z') = \varnothing$.

We finish the proof of Chevalley's Theorem by using Noetherian induction 4.6.1. Define the property $\mathscr{P}$ as follow: let C is a closed subset of Noetherian scheme Y', we say $\mathscr{P}$ holds for C if $\pi'(Z' \cap \pi'^{-1}(C))$ is constructible in Y'. Suppose $\mathscr{P}$ holds for every proper closed subset of C, we want to show that $\mathscr{P}$ holds for C. By our previous discussion, there exists a dense open subset $V \subseteq Y'$ such that $V \cap \pi'(\delta Z') = \varnothing$. Note that

$$C = (C \cap V) \coprod (C \cap (Y' \setminus V)),$$

we have

$$\begin{aligned}\pi'(Z'\cap\pi'^{-1}(C)) =&\pi'(Z'\cap(\pi'^{-1}(C\cap V)\coprod\pi'^{-1}(C\cap(Y'\setminus V)))) \\ =&\pi'(Z'\cap\pi'^{-1}(C\cap V))\cup\pi'(Z'\cap\pi'^{-1}(C\cap(Y'\setminus V))).\end{aligned}$$

Consider $\pi'(Z'\cap\pi^{-1}(C\cap V))$, since $\pi^{-1}(V)\cap\delta Z'=\varnothing$, we have

$$\pi'(Z'\cap\pi^{-1}(C\cap V))=\pi'(\mathrm{cl}_{X'}(Z')\cap\pi^{-1}(C\cap V)).$$

Since $\mathrm{cl}_{X'}(Z')\cap\pi^{-1}(C\cap V)$ is closed in $\pi^{-1}(C\cap V)$, $\pi'(\mathrm{cl}_{X'}(Z')\cap\pi^{-1}(C\cap V))$ is constructible in $C\cap V$, and therefore is constructible in Y'. Hence $\pi'(Z'\cap\pi'^{-1}(C\cap V))$ is constructible in Y'.

Consider $\pi'(Z'\cap\pi^{-1}(C\cap(Y'\setminus V)))$, since $C\cap(Y'\setminus V)$ is proper closed in C, by the induction hypothesis, $\pi'(Z'\cap\pi^{-1}(C\cap(Y'\setminus V)))$ is constructible in Y'.

Hence $\pi'(Z'\cap\pi'^{-1}(C))$ is constructible in Y'. By the Noetherian induction 4.6.1, $\mathscr{P}$ holds for Y', i.e., $\pi'(Z'\cap\pi'^{-1}(Y'))=\pi'(Z')$ is constructible in Y'. Note that

$$\pi(Z)=\pi'(Z)\cup(FL(\mathrm{cl}_X(Z))\setminus FL(\delta Z)),$$

we know that $\pi(Z)$ is constructible. We finished the proof of Chevalley's Theorem. □

Remark 9.22 For the minority who might care: see Chapter 11 for an extension to locally finitely presented morphisms.

We next give some important consequences that may seem surprising, in that they are algebraic corollaries of a seemingly quite geometric and topological theorem. The first is a proof of the Nullstellensatz. Recall the Nullstellensatz 4.2.2:

Theorem 9.4.2 (Hilbert's Nullstellensatz)

If k is any field, every maximal ideal of $k[x_1,\dots,x_n]$ has residue field $k[x_1,\dots,x_n]/\mathfrak{m}$ a finite extension of k.

Proof We wish to show that if K is a field extension of k that is finitely generated as a k-algebra, say by $x_1,\dots,x_n$, then it is a finite extension of fields. It suffices to show that each x_i is algebraic over k. But if x_i is not algebraic over k, then we have an inclusion of rings $k[x_i]\to K$, corresponding to a dominant morphism $\pi:\operatorname{Spec}K\to\mathbb{A}^1_k$ of finite type k-schemes. Of course $\operatorname{Spec}K$ is a single point, so the image of π is one point. By Chevalley's Theorem 9.4.1 and Exercise 9.3, the image of π is not the generic point of $\mathbb{A}^1_k$. Consider $\mathrm{cl}_{\mathbb{A}^1_k}(\mathrm{Im}(\pi))$, let $y\in\mathrm{cl}_{\mathbb{A}^1_k}(\mathrm{Im}(\pi))$, since $\mathrm{Im}(\pi)$ is constructible, by Proposition 9.4.2, we know that $y\in\mathrm{Im}(\pi)$, i.e., so $y=\mathrm{Im}(\pi)$. It follows that $\mathrm{Im}(\pi)$ is closed point, and therefore π is not dominant, a contradiction. □

A similar ideal can be used in the following Corollaries of Chevalley's Theorem 9.4.1.

Corollary 9.4.2 (Quasi-finite morphisms to a filed are finite)

Suppose $\pi:X\to\operatorname{Spec}k$ is a quasi-finite morphism. Then π is finite.

Proof Since $\pi:X\to\operatorname{Spec}k$ is a quasi-finite morphism, $\pi:X\to\operatorname{Spec}k$ is of finite type and has finite fibers. Since $\pi:X\to\operatorname{Spec}k$ is of finite type, suppose $\pi^{-1}(\operatorname{Spec}k)=\bigcup_{i=1}^{n}\operatorname{Spec}A_i$ where each A_i is a finitely generated k-algebra. Restrict π to $\operatorname{Spec}_{A_i}$, then we get a morphism $\pi|_{\operatorname{Spec}A_i}:\operatorname{Spec}A_i\to\operatorname{Spec}k$. Clearly, $\pi|_{\operatorname{Spec}A_i}$ is of finite type. We want to show that $\pi|_{\operatorname{Spec}A_i}$ is integral. It suffices to show that corresponding ring map $k\to A_i$ is integral. Suppose A_i contains an element x that is not algebraic over k, then we have an inclusion $k[x]\hookrightarrow A_i$, and A_i is finitely generated $k[x]$-algebra. Since A_i is finitely generated k-algebra, by Hilbert's Basis Theorem, A_i is Noetherian. Consider the corresponding dominant morphism $\varphi:\operatorname{Spec}A_i\to\operatorname{Spec}k[x]$, φ is a finite type morphism of Noetherian schemes, by Chevalley's Theorem 9.4.1, $\varphi(\operatorname{Spec}A_i)$ is constructible.

By the discussion in the proof of Hilbert's Nullstellensatz 9.4.2, we know that $\varphi(\operatorname{Spec} A_i)$ is closed point, and therefore φ is not dominant, a contradiction. Hence $\pi|^{\sharp}_{\operatorname{Spec} A_i} : k \to A_i$ is integral, by Proposition 9.3.20, $\pi|_{\operatorname{Spec} A_i} \operatorname{Spec} A_i \to \operatorname{Spec} k$ is integral for all i. By Proposition 9.3.10, $\operatorname{Spec} A_i$ is a finite union of points, each point with residue field a finite extension of k. So $\pi^{-1}(\operatorname{Spec} k)$ is a finite union of points, and therefore $\pi^{-1}(\operatorname{Spec} k)$ can be written as disjoint union of affine scheme, by Proposition 5.3.5 (a), $\pi^{-1}(\operatorname{Spec} k)$ is an affine scheme, say $\pi^{-1}(\operatorname{Spec} k) = \operatorname{Spec} A$, where A is finite product of residue field. Since each residue filed is a finite extension of k, A is finite generated k-vector space, i.e., A is finite k-algebra. Consequently, $\pi : X \to \operatorname{Spec} k$ is finite. □

Corollary 9.4.3

Suppose X is a scheme of finite type over $\operatorname{Spec} \mathbb{Z}$, and p is a closed point of X. Then the residue field $\kappa(p)$ is a finite field.

Proof Consider the structure morphism $\pi : X \to \operatorname{Spec} \mathbb{Z}$, since X is of finite type over $\operatorname{Spec} \mathbb{Z}$, $\pi^{-1}(\operatorname{Spec} \mathbb{Z})$ can be covered by finite number of open subscheme $\operatorname{Spec} A_i$ where A_i is finitely generated $\mathbb{Z}$-algebra. Since $\mathbb{Z}$ is Noetherian, by Hilbert's Basis Theorem, A_i is Noetherian, and therefore X is Noetherian scheme. Since p is a closed point of X, p must contained in $\operatorname{Spec} A \subseteq X$ for some A where A is Noetherian ring, say $p = [\mathfrak{m}] \in \operatorname{Spec} A$ where $\mathfrak{m}$ is maximal ideal of A, hence $\kappa(p) = A/\mathfrak{m}$. Consider the composition $\mathbb{Z} \to A \to A/\mathfrak{m}$. Since $\mathbb{Z}$ is PID, say $\operatorname{Ker}(\varphi : \mathbb{Z} \to A/\mathfrak{m}) = (p)$. Since $\operatorname{Ker}(\varphi : \mathbb{Z} \to A/\mathfrak{m}) = \varphi^{-1}(0)$ and (0) is prime ideal of $A/\mathfrak{m}$, $\operatorname{Ker}(\varphi : \mathbb{Z} \to A/\mathfrak{m})$ must be prime ideal, so p is prime. We obtain a field extension $\mathbb{F}_p \hookrightarrow A/\mathfrak{m}$. Since A is finitely generated $\mathbb{Z}$-algebra, $A/\mathfrak{m}$ is finitely generated $\mathbb{Z}$-algebra, and therefore $A/\mathfrak{m}$ is finitely generated $\mathbb{F}_p$-algebra. If $A/\mathfrak{m}$ not a finite field, then there exists $x \in A/\mathfrak{m}$ not algebraic over $\mathbb{F}_p$. Hence we obtain a ring map $\mathbb{F}_p[x] \hookrightarrow A/\mathfrak{m}$. Clearly, $A/\mathfrak{m}$ is finitely generated $\mathbb{F}_p[x]$-algebra, the corresponding dominant morphism $\psi : \operatorname{Spec} A/\mathfrak{m} \to \operatorname{Spec} \mathbb{F}_p[x]$ is of finite type. Note that $A/\mathfrak{m}$ and $\mathbb{F}_p[x]$ is Noetherian, apply Chevalley's Theorem 9.4.1, $\psi(\operatorname{Spec} A/\mathfrak{m})$ is constructible in $\operatorname{Spec} \mathbb{F}_p[x]$. Since $\operatorname{Spec} A/\mathfrak{m}$ is a single point, $\psi(\operatorname{Spec} A/\mathfrak{m})$ is also a single point, by Exercise 9.3, $\psi(\operatorname{Spec} A/\mathfrak{m})$ is not the generic point of $\operatorname{Spec} \mathbb{F}_p[x]$. It follows that $\psi(\operatorname{Spec} A/\mathfrak{m})$ is closed point (use Proposition 9.4.2 or see the proof of Hilbert's Nullstellensatz 9.4.2), hence ψ is not dominant, a contradiction. Hence $A/\mathfrak{m}$ is finite field. □

Corollary 9.4.4 (For maps of varieties, surjective can be checked on closed points)

A morphism of affine k-varieties $\pi : X \to Y$ is surjective if and only if it is surjective on closed points (i.e., if every closed point of Y is surjective if and only if it is surjective on closed point of X).

Proof If $\pi : X \to Y$ is surjective, obviously, it is surjective on closed points.

Conversely, if $\pi : X \to Y$ is surjective on closed points, we want to show that $\pi : X \to Y$ is surjective. If π is not surjective, there exists $p \notin \operatorname{Im}(\pi)$. By Chevalley's Theorem 9.4.1, $\operatorname{Im}(\pi)$ is constructible, and therefore the complement of $\operatorname{Im}(\pi)$ is also constructible, say $p \in U \cap C$ where U is open and C is closed. By Proposition 4.6.7, the set of closed points of C is dense, hence $U \cap C$ contains a closed point of Y. Since $\pi : X \to Y$ is surjective on closed point, $\operatorname{Im}(\pi)$ must contains all closed point of Y, a contradiction. □

Remark 9.23 Once we define varieties in general, in Chapter 12, we will see that our argument works without change with the adjective "affine" removed.

9.4.2 Elimination of quantifiers

A basic sort of question that arises in any number of contexts is when a system of equations has a solution. Suppose for example you have some polynomials in variables $x_1, \ldots, x_n$ over an algebraically closed field $\overline{k}$, some of which you set to be zero, and some of which you set to be nonzero. (This question is of fundamental interest even before you know any scheme theory!) Then there is an algebraic condition on the coefficients which will tell you if there is a solution. Define the **Zariski topology on** $\overline{k}^n$ in the obvious way: closed subsets are cut out by equations. (A mild generalization of this appears in Chapter 13.)

Proposition 9.4.3 (Elimination of quantifiers, over an algebraically closed field)

Fix an algebraically closed field $\overline{k}$. Suppose

$$f_1, \ldots, f_p, g_1, \ldots, g_q \in \overline{k}[W_1, \ldots, W_m, X_1, \ldots, X_n]$$

are given. There is a (Zariski-)constructible subset Y of $\overline{k}^m$ such that

$$f_1(w_1, \ldots, w_m, X_1, \ldots, X_n) = \cdots = f_p(w_1, \ldots, w_m, X_1, \ldots, X_n) = 0 \tag{9.8}$$

and

$$g_1(w_1, \ldots, w_m, X_1, \ldots, X_n) \neq 0 \quad \cdots \quad g_q(w_1, \ldots, w_m, X_1, \ldots, X_n) \neq 0 \tag{9.9}$$

has a solution $(X_1, \ldots, X_n) = (x_1, \ldots, x_n) \in \overline{k}^n$ if and only if $(w_1, \ldots, w_n) \in Y$.

Proof Let X be the locally closed subset of $\mathbb{A}^{m+n}$ cut out by the equalities and inequalities (9.8) and (9.9), i.e.,

$$X = V(f_1, \ldots, f_p) \cap \left(\bigcap_{j=1}^{q} D(g_j) \right).$$

Consider the projection $\mathbb{A}^{m+n} \to \mathbb{A}^m$, it is given by the ring map $\overline{k}[W_1, \ldots, W_m] \to \overline{k}[W_1, \ldots, W_m, X_1, \ldots, X_n]$, restrict the projection to X, i.e., $\pi : X \to \mathbb{A}^m$.

Remark

If Z is a finite type scheme over $\overline{k}$, and the closed points are denoted Z^{cl} ("cl" is for either "closed" or "classical"), then under the inclusion of topological spaces $Z^{\mathrm{cl}} \hookrightarrow Z$, the Zariski topology on Z, the Zariski topology on Z induces the Zariski topology on Z^{cl}.

By Hilbert's Nullstellensatz 4.7.1 and Proposition 6.3.8, we can identify $(\mathbb{A}^m_{\overline{k}})^{\mathrm{cl}}$ with $\overline{k}^m$. Consider the diagram

$$\begin{tikzcd}
X^{\mathrm{cl}} \arrow[r, hook] \arrow[d, "\pi^{\mathrm{cl}}"'] & X \arrow[r, hook, "\text{loc.cl.}"] \arrow[d, "\pi"] & \mathbb{A}^{m+n} \arrow[dl] \\
\overline{k}^m \arrow[r, hook] & \mathbb{A}^m &
\end{tikzcd}$$

where X^{cl} is the closed points of X, and $\pi^{\mathrm{cl}} : X^{\mathrm{cl}} \to \overline{k}^m$ is given by restrict π to X^{cl}, i.e.,

$$(w_1, \ldots, w_m, x_1, \ldots, x_n) \mapsto (w_1, \ldots, w_m).$$

Let $Y = \operatorname{Im} \pi^{\mathrm{cl}}$. Clearly, $\pi : X \to \mathbb{A}^m$ is a finite type morphism of Noetherian schemes, by Chevalley's Theorem 9.4.1, $\operatorname{Im} \pi$ is constructible, and hence so is $(\operatorname{Im} \pi) \cap \overline{k}^m$. It remains to show that $(\operatorname{Im} \pi) \cap \overline{k}^m = Y$.

Let $\pi^{\mathrm{cl}}(p) \in Y$, then $\pi^{\mathrm{cl}}(p) \in \operatorname{Im} \pi$, so $\pi^{\mathrm{cl}}(p) \in (\operatorname{Im} \pi) \cap \overline{k}^m$. Hence $Y \subseteq (\operatorname{Im} \pi) \cap \overline{k}^m$.

Conversely, let $q \in (\operatorname{Im} \pi) \cap \overline{k}^m$, then q is closed point in $\mathbb{A}^m$. Hence $\pi^{-1}(q)$ is a closed subset of X. By

Proposition 6.3.8, X^{cl} is dense in X, so $X^{\mathrm{cl}} \cap \pi^{-1}(q) \neq \varnothing$. Hence there exists $p \in X^{\mathrm{cl}}$ such that $\pi^{\mathrm{cl}}(p) = q$, it follows that $q \in \operatorname{Im} \pi^{\mathrm{cl}}$. Hence $Y = (\operatorname{Im} \pi) \cap \overline{k}^m$, and therefore Y is constructible subset. □

Remark 9.24 This is called "elimination of quantifiers" because it gets rid of the quantifier "there exists a solution". The analogous statement for real numbers, where inequalities are also allowed, is a special case of Tarski's celebrated theorem of elimination of quantifiers for real closed fields.

9.4.3 The Fundamental Theorem of Elimination Theory

In the case of projective space (and later, projective morphisms), one can do better than Chevalley (at least in describing image of closed sets).

Theorem 9.4.3 (Fundamental Theorem of Elimination Theory)
*The morphism $\pi : \mathbb{P}^n_A \to \operatorname{Spec} A$ is **closed** (sends closed sets to closed sets).*

Note that no Noetherian hypotheses are needed.

A great deal of classical algebra and geometry is contained in this theorem as special cases. Here are some example.

Example 9.12 Let $A = k[a, b, c, \ldots, i]$, and consider the closed subset of $\mathbb{P}^2_A$ (taken with coordinates x, y, z) corresponding to $ax + by + cz = 0, dx + ey + fz = 0, gx + hy + iz = 0$. Then we are looking for the locus in $\operatorname{Spec} A$ where these equations have a nontrivial solution. This indeed corresponds to a Zariski-closed set — where

$$\det \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} = 0.$$

Thus the ideal of the determinant is embedded in elimination theory.

Example 9.13 Let $A = k[a_0, a_1, \ldots, a_m, b_0, b_1, \ldots, b_n]$. Now consider the closed subset of $\mathbb{P}^1_A$ (taken with coordinates x and y) corresponding to $a_0x^m + a_1x^{m-1}y + \cdots + a_my^m = 0$ and $b_0x^n + b_1x^{n-1}y + \cdots + b_ny^n = 0$. Then there is a polynomial in the coefficients $a_0, \ldots, b_n$ (an element of A) which vanishes if and only if these two polynomials have a common nonzero root — this polynomial is called the **resultant**.

More generally, given a number of homogeneous equations in $n + 1$ variables with indeterminate coefficients, Theorem 9.4.3 implies that one can write down equations in the coefficients that precisely determine when the equations have a nontrivial solution.

Proof of the Fundamental Theorem of Elimination Theory 9.4.3.

Proof Suppose $Z \hookrightarrow \mathbb{P}^n_A$ is a closed subset. We wish to show that $\pi(Z)$ is closed.

By the definition of the Zariski topology on $\operatorname{Proj} A[x_0, \ldots, x_n]$ (Definition 5.5.6), Z is cut out (set-theoretically) by some homogeneous elements $f_1, f_2, \dots \in A[x_0, \ldots, x_n]$. We wish to show that the points $p \in \operatorname{Spec} A$ that are in $\pi(Z)$ form a closed subset. Unpacked, this means that $Z = V_+(\{f_i\}_{i \in I})$ where f_i is homogeneous elements of $A[x_0,\dots,x_n]$. We want to show that $\pi(Z)$ is closed in $\operatorname{Spec} A$. In fact,

$$\begin{aligned}\pi(Z) &= \left\{\pi(\mathfrak{p}) : \mathfrak{p} \in \bigcap_{i\in I} V_+(f_i) \text{ in } \operatorname{Proj} A[x_0,\dots,x_n]\right\} \\ &= \left\{\mathfrak{p}_0 \in \operatorname{Spec} A : \bigcap_{i\in I} \pi|_{V_+(f_i)}^{-1}(\mathfrak{p}_0) \neq \varnothing\right\} \\ &= \left\{\mathfrak{p}_0 \in \operatorname{Spec} A : \bigcap_{i\in I} (V_+(f_i)\cap \pi^{-1}(\mathfrak{p}_0)) \neq \varnothing\right\}.\end{aligned}$$

Define the ring map $\varphi_{\mathfrak{p}_0} : A[x_0,\dots,x_n] \to \kappa(\mathfrak{p}_0)[x_0,\dots,x_n]$ induced by $A \to \kappa(\mathfrak{p}_0)$ (for each prime ideal $\mathfrak{p}_0 \in \operatorname{Spec} A$), then we have $V_+(f_i)\cap\pi^{-1}(\mathfrak{p}_0) = V_+(\varphi_{\mathfrak{p}_0}(f_i)) \subseteq \operatorname{Proj} \kappa(\mathfrak{p}_0)[x_0,\dots,x_n]$. Hence,

$$\pi(Z) = \left\{\mathfrak{p}_0 \in \operatorname{Spec} A : \bigcap_{i\in I} V_+(\varphi_{\mathfrak{p}_0}(f_i)) \neq \varnothing \text{ in } \mathbb{P}^n_{\kappa(\mathfrak{p}_0)}\right\}.$$

So, equivalently, we want to show that those $p \in \operatorname{Spec} A$ for which $f_1, f_2, \dots$ have a common zero in $\operatorname{Proj} \kappa(p)[x_0,\dots,x_n]$ form a closed subset of $\operatorname{Spec} A$, i.e.,

$$\pi(Z) = \left\{p \in \operatorname{Spec} A : \bigcap_{i\in I} V_+(\varphi_p(f_i)) \neq \varnothing \text{ in } \mathbb{P}^n_{\kappa(p)}\right\} \tag{9.10}$$

form a closed subset of $\operatorname{Spec} A$.

Remark

To motivate our argument, we consider a related question. Suppose that $S_\bullet := k[x_0,\dots,x_n]$, and that $g_1, g_2, \dots \in k[x_0,\dots,x_n]$ are homogeneous polynomials. How can we tell if $g_1, g_2, \dots$ have a common zero in $\operatorname{Proj} S_\bullet = \mathbb{P}^n_k$?

They would have a common zero if and only if in

$$\mathbb{A}^{n+1}_k = \operatorname{Spec} S_\bullet,$$

they cut out (set-theoretically) more than the origin, i.e., if set-theoretically

$$V(g_1, g_2, \dots) \not\subseteq V(x_0,\dots,x_n).$$

But the inclusion-reversing bijection between closed subsets and radical ideals (Theorem 4.7.1), this is true if and only if

$$\sqrt{(g_1, g_2, \dots)} \not\supseteq \sqrt{(x_0,\dots,x_n)}.$$

Since $(x_0,\dots,x_n)$ is maximal ideal, so this is true if and only if

$$\sqrt{(g_1, g_2, \dots)} \not\supseteq (x_0,\dots,x_n).$$

This is true if and only if not all of the generators $x_0,\dots,x_n$ of the ideal $(x_0,\dots,x_n)$ are in $\sqrt{(g_1,g_2,\dots)}$, which is true if and only if there is some i such that no power of x_i is in $(g_1, g_2, \dots)$. This is true if and only if

$$(x_0,\dots,x_n)^N \not\subseteq (g_1, g_2, \dots) \quad \text{for all } N.$$

This is equivalent to

$$S_N \not\subseteq (g_1, g_2, \dots) \quad \text{for all } N,$$

which may be rewritten as

$$S_N \not\subseteq g_1 S_{N-\deg g_1} \oplus g_2 S_{N-\deg g_2} \oplus \cdots$$

for all N. In other words, this is equivalent to the statement that the k-linear map

$$S_{N-\deg g_1} \oplus S_{N-\deg g_2} \oplus \cdots \longrightarrow S_N$$

is not surjective. This map is given by a matrix with $\dim S_N$ rows. (It may have an infinite number of columns, but this will not bother us.) To check that this linear map is not surjective, we need only check that all the "maximal" ($\dim S_N \times \dim S_N$) determinants are zero. (Of cause, we need to check this for all N.) Thus the condition that $g_1, g_2, \ldots$ have common zeros in $\mathbb{P}_k^n$ is the same as checking some (admittedly infinite) number of equations. In other words, it is a Zariski-closed condition on the coefficients of the polynomials $g_1, g_2, \ldots$.

Say $S_\bullet = A[x_0, \ldots, x_n]$. $f_1, f_2, \ldots$ have a common zero in $\operatorname{Proj} S_\bullet$ if and only if $\varphi_p(f_1), \varphi_p(f_2), \ldots$ have common zeros in $\operatorname{Proj} \kappa(p)[x_0, \ldots, x_n]$ for some $p \in \operatorname{Spec} A$, by above remark, if and only if there exists $p \in \operatorname{Spec} A$ such that the $\kappa(p)$-linear map

$$\alpha_{N,p} : \bigoplus_{i \geq 1} \kappa(p)[x_0, \ldots, x_n]_{N-\deg \varphi_p(f_i)} \longrightarrow \kappa(p)[x_0, \ldots, x_n]_N \tag{9.11}$$

given by

$$(k_i)_{i \geq 1} \longmapsto \sum_{i \geq 1} k_i \varphi_p(f_i).$$

is not surjective, for all N. By above remark, (9.11) gives a matrix with coefficient in $\kappa(p)$, say $M_{N,p}$, and $\alpha_{N,p}$ is surjective if and only if all $\dim \kappa(p)[x_0, \ldots, x_n]_N \times \dim \kappa(p)[x_0, \ldots, x_n]_N$ minor of $M_{N,p}$ are zero. Say $\dim \kappa(p)[x_0, \ldots, x_n]_N = r$.

Obviously, $\deg f_i = \deg \varphi_p(f_i)$, so $N - \deg f_i = N - \deg \varphi_p(f_i)$. Consider the following diagram,

$$\begin{tikzcd}
\bigoplus_{i \geq 1} S_{N-\deg f_i} \arrow[r, "\alpha_N"] \arrow[d, "\text{induced by } \varphi_p"'] & S_N \arrow[d, "\varphi_p"] \\
\bigoplus_{i \geq 1} \kappa(p)[x_0, \ldots, x_n]_{N-\deg \varphi_p(f_i)} \arrow[r, "\alpha_{N,p}"] & \kappa(p)[x_0, \ldots, x_n]_N
\end{tikzcd}$$

we can lift $\alpha_{N,p}$ to a morphism from $\bigoplus_{i \geq 1} S_{N-\deg f_i}$ to S_N via φ_p. The A-module map α_N gives a matrix with coefficient in A, say M_N. Let $M_N = (m_{ij})$, define $\widetilde{\varphi_p}(M_N) = (\varphi_p(m_{ij}))_{ij}$, by the construction of (9.11), we know that $\widetilde{\varphi_p}(M_N) = M_{N,p}$, and the $r \times r$ minor of M_N in $\kappa(p)$ is the $r \times r$ minor of $M_{N,p}$. Hence, $\varphi_p(f_1), \varphi_p(f_2), \ldots$ have common zeros in $\operatorname{Proj} \kappa(p)[x_0, \ldots, x_n]$ for some $p \in \operatorname{Spec} A$ if and only if all $r \times r$ minor of M_N in $\kappa(p)$ are zero for some p.

Note that $\dim \kappa(p)[x_0, \ldots, x_n]_N = \operatorname{rank}(A[x_0, \ldots, x_n]_N)$, let I_N be an ideal which generated by $r \times r$ minor of M_N, then all $r \times r$ minor of M_N in $\kappa(p)$ are zero if and only if $p \in V(I_N)$. Hence, $\varphi_p(f_1), \varphi_p(f_2), \ldots$ have common zeros in $\operatorname{Proj} \kappa(p)[x_0, \ldots, x_n]$ for some $p \in \operatorname{Spec} A$ if and only if $p \in V(I_N)$ for all N.

By (9.10), we have

$$\begin{aligned}
\pi(Z) &= \left\{ p \in \operatorname{Spec} A : \bigcap_{i \in I} V_+(\varphi_p(f_i)) \neq \varnothing \text{ in } \mathbb{P}^n_{\kappa(p)} \right\} \\
&= \bigcap_{N \geq 0} V(I_N),
\end{aligned}$$

hence $\pi(Z)$ is closed. $\square$

Remark 9.25 Notice that projectivity was crucial to the proof: we used graded rings in an essential way. Notice also that the proof is essentially just linear algebra.

Chapter 10 Closed embeddings and related notions

The scheme-theoretic analog of closed subsets has a surprisingly different flavor from the analog of open sets (open subschemes/embeddings). However, just as open subschemes (the scheme-theoretic version of open set) are locally modeled on open sets $U \subseteq Y$, the analog of closed subsets also has a local model. This was foreshadowed by our understanding of closed subsets of $\operatorname{Spec} B$ as roughly corresponding to ideals. If $I \subseteq B$ is an ideal, then $\operatorname{Spec} B/I \hookrightarrow \operatorname{Spec} B$ is a morphism of schemes, and we have checked that on the level of topological spaces, this describes $\operatorname{Spec} B/I$ as a closed subset of $\operatorname{Spec} B$, with the subspace topology (Proposition 4.4.7). This morphism is our "local model" of a closed subscheme/embedding.

10.1 Closed embeddings and closed subschemes

10.1.1 Definition of closed embeddings and closed subschemes

Definition 10.1.1 (Closed embedding, closed subscheme)

A morphism $\pi : X \to Y$ is a ***closed embedding*** *(or* ***closed immersion****) if it is an affine morphism, and for every affine open subset $\operatorname{Spec} B \subseteq Y$, with $\pi^{-1}(\operatorname{Spec} B) \cong \operatorname{Spec} A$, the map $B \to A$ is surjective (i.e., of the form $B \to B/I$, our desired local model). The symbol $\hookrightarrow$ often is used to indicate that a morphism is a closed embedding (or more generally, a locally closed embedding, §10.2).*

If X is a subset of Y (and π on the level of sets is the inclusion), we say that X is a ***closed subscheme*** *of Y. A closed embedding is the same thing as an isomorphism with a closed subscheme.*

Remark 10.1 "Embedding" is preferable to "immersion", because the differential geometric notion of immersion is closer to what algebraic geometers call unramified, which we will define in Chapter 22.

Proposition 10.1.1

A closed embedding $\pi : X \hookrightarrow Y$ identifies the topological space of X with a closed subset of the topological space of Y, i.e., X is homeomorphic to a closed subset of Y.

Proof Let $Y = \bigcup_i \operatorname{Spec} B_i$, then $X = \bigcup_i \pi^{-1}(\operatorname{Spec} B_i)$. Since $\pi : X \to Y$ is closed embedding, we may assume $\pi^{-1}(\operatorname{Spec} B_i) \cong \operatorname{Spec} A_i$ where $\pi|^{\sharp}_{\operatorname{Spec} A_i} : B_i \to A_i$ is surjective. Hence, by the Fundamental Theorem of Isomorphism for rings, we have $A_i \cong B_i / \operatorname{Ker}(\pi|^{\sharp}_{\operatorname{Spec} A_i})$. Say $I_i = \operatorname{Ker}(\pi|^{\sharp}_{\operatorname{Spec} A_i})$, then as topological space

$$\pi : X \xrightarrow{\sim} \bigcup_i \operatorname{Spec} B_i/I_i.$$

By Proposition 4.4.7, each $\operatorname{Spec} B_i/I_i$ is closed in $\operatorname{Spec} B_i$. We want to show that $\pi(X) = \bigcup_i \operatorname{Spec} B_i/I_i$ is closed in Y. Consider $Y \setminus \pi(X)$,

$$\begin{aligned} Y \setminus \pi(X) &= \left(\bigcup_i \operatorname{Spec} B_i\right) \setminus \pi(X) \\ &= \bigcup_i (\operatorname{Spec} B_i \setminus (\operatorname{Spec} B_i \cap \pi(X))). \end{aligned}$$

It suffices to show that each $\operatorname{Spec} B_i \cap \pi(X)$ is closed in $\operatorname{Spec} B_i$. Note that

$$\begin{aligned}\operatorname{Spec} B_i \cap \pi(X) &= \operatorname{Spec} B_i \cap \bigcup_i \operatorname{Spec} B_i/I_i \\ &= \operatorname{Spec} B_i/I_i \cup \left(\bigcup_{j\neq i} (\operatorname{Spec} B_j/I_j \cap \operatorname{Spec} B_i)\right),\end{aligned}$$

consider $\operatorname{Spec} B_j/I_j \cap \operatorname{Spec} B_i$, we claim that

$$(\operatorname{Spec} B_j/I_j) \cap (\operatorname{Spec} B_i \cap \operatorname{Spec} B_j) = (\operatorname{Spec} B_i/I_i) \cap (\operatorname{Spec} B_i \cap \operatorname{Spec} B_j).$$

By Proposition 6.3.1, $\operatorname{Spec} B_i \cap \operatorname{Spec} B_j$ can be covered by simultaneously distinguished open subsets of $\operatorname{Spec} B_i$ and $\operatorname{Spec} B_j$, say

$$\operatorname{Spec} B_i \cap \operatorname{Spec} B_j = \bigcup_k \operatorname{Spec}(B_i)_{f_k} \cong \bigcup_k \operatorname{Spec}(B_j)_{g_k},$$

where $\operatorname{Spec}(B_i)_{f_k} \cong \operatorname{Spec}(B_j)_{g_k}$. Hence, we have

$$(\operatorname{Spec} B_j/I_j) \cap (\operatorname{Spec} B_i \cap \operatorname{Spec} B_j) = \bigcup_k (\operatorname{Spec} B_j/I_j \cap \operatorname{Spec}(B_j)_{g_k})$$

and

$$(\operatorname{Spec} B_i/I_i) \cap (\operatorname{Spec} B_i \cap \operatorname{Spec} B_j) = \bigcup_k (\operatorname{Spec} B_i/I_i \cap \operatorname{Spec}(B_i)_{f_k}).$$

It suffices to show that $\operatorname{Spec} B_j/I_j \cap \operatorname{Spec}(B_j)_{g_k} \cong \operatorname{Spec} B_i/I_i \cap \operatorname{Spec}(B_i)_{f_k}$ for all k. Note that $\operatorname{Spec} B_j/I_j \cap \operatorname{Spec}(B_j)_{g_k} \cong \operatorname{Spec}(B_j/I_j)_{\overline{g_k}}$ and $\operatorname{Spec} B_i/I_i \cap \operatorname{Spec}(B_i)_{f_k} \cong \operatorname{Spec}(B_i/I_i)_{\overline{f_k}}$. Since π is closed embedding, we have $\pi^{-1}(\operatorname{Spec}(B_i)_{f_k}) \cong \pi^{-1}(\operatorname{Spec} B_i/I_i)_{\pi^\sharp(f_k)} \cong \operatorname{Spec}(B_i/I_i)_{\overline{f_k}}$ and $\pi^{-1}(\operatorname{Spec}(B_j)_{g_k}) \cong \operatorname{Spec}(B_j/I_j)_{\overline{g_k}}$. Since $\operatorname{Spec}(B_i)_{f_k} \cong \operatorname{Spec}(B_j)_{g_k}$, we have $\pi^{-1}(\operatorname{Spec}(B_i)_{f_k}) \cong \pi^{-1}(\operatorname{Spec}(B_j)_{g_k})$, and therefore $\operatorname{Spec}(B_i/I_i)_{\overline{f_k}} \cong \operatorname{Spec}(B_j/I_j)_{\overline{g_k}}$, as we desired.

Hence we have proved

$$(\operatorname{Spec} B_j/I_j) \cap (\operatorname{Spec} B_i \cap \operatorname{Spec} B_j) = (\operatorname{Spec} B_i/I_i) \cap (\operatorname{Spec} B_i \cap \operatorname{Spec} B_j).$$

So

$$\begin{aligned}\operatorname{Spec} B_i \cap \pi(X) &= \operatorname{Spec} B_i/I_i \cup \left(\bigcup_{j\neq i} (\operatorname{Spec} B_j/I_j \cap \operatorname{Spec} B_i)\right) \\ &= \operatorname{Spec} B_i/I_i \cup \left(\bigcup_{j\neq i} (\operatorname{Spec} B_i/I_i \cap \operatorname{Spec} B_j)\right) \\ &= \operatorname{Spec} B_i/I_i,\end{aligned}$$

i.e., $\operatorname{Spec} B_i \cap \pi(X)$ is closed in $\operatorname{Spec} B_i$, and therefore $Y \setminus \pi(X)$ is open, we done! □

Remark 10.2 **Caution:** The closed embeddings $\operatorname{Spec} k[x]/(x) \hookrightarrow \operatorname{Spec} k[x]$ and $\operatorname{Spec} k[x]/(x^2) \hookrightarrow \operatorname{Spec} k[x]$ show that the closed subset does not determine the closed subscheme. The "infinitesimal" information, or "fuzz", is lost.

Proposition 10.1.2

Closed embeddings are finite morphisms, hence of finite type.

Proof Let $\pi : X \hookrightarrow Y$ be a closed embedding, say $Y = \bigcup_i \operatorname{Spec} B_i$, as in the proof of Proposition 10.1.1, we may assume

$$X = \bigcup_i \operatorname{Spec} B_i/I_i,$$

where I_i is an ideal of B_i, and $\pi^{-1}(\operatorname{Spec} B_i) \cong \operatorname{Spec} B_i/I_i$. Clearly, B_i/I_i is a finitely generated B_i-module, and therefore is finite B_i-algebra. By Proposition 9.3.9, closed embedding $\pi : X \hookrightarrow Y$ is finite morphism, hence of finite type. □

Proposition 10.1.3

The composition of two closed embeddings is a closed embedding.

Proof Let $\pi : X \hookrightarrow Y$ and $\psi : Y \hookrightarrow Z$ be two closed embeddings, we want to show that $\psi \circ \pi : X \to Z$ is closed embedding. Let $\operatorname{Spec} C \subseteq Z$, since ψ is closed embedding, we have $\psi^{-1}(\operatorname{Spec} C) \cong \operatorname{Spec} B$ where $C \to B$ is surjective. Since π is closed embedding, $\pi^{-1}(\psi^{-1}(\operatorname{Spec} C)) \cong \pi^{-1}(\operatorname{Spec} B) \cong \operatorname{Spec} A$, where $B \to A$ is surjective. Hence the composition $C \to B \to A$ is surjective, and therefore $\psi \circ \pi : X \to Z$ is closed embedding. □

Proposition 10.1.4 (The property of being a closed embedding is affine-local on the target)

A morphism $\pi : X \to Y$ is closed embedding if there is a cover of Y by affine open sets $\operatorname{Spec} B_i$ *such that* $\pi^{-1}(\operatorname{Spec} B_i) \cong \operatorname{Spec} A_i$ *where $B_i \to A_i$ is surjective.*

Proof Let $\pi : X \to Y$ be a morphism of schemes. By hypothesis and Proposition 9.3.6, $\pi : X \to Y$ is affine. We want to show that the property of being a closed embedding is affine local property. It suffices to show the following two things:

(i) let $\operatorname{Spec} B$ be an affine open subsets of Y and $\pi^{-1}(\operatorname{Spec} B) \cong \operatorname{Spec}\Gamma(\pi^{-1}(\operatorname{Spec} B), \mathscr{O}_X)$ where $B \to \Gamma(\pi^{-1}(\operatorname{Spec} B), \mathscr{O}_X)$ is surjective, then $\pi^{-1}(\operatorname{Spec} B_f) \cong \operatorname{Spec}\Gamma(\pi^{-1}(\operatorname{Spec} B_f), \mathscr{O}_X)$ where $B_f \to \Gamma(\pi^{-1}(\operatorname{Spec} B_f), \mathscr{O}_X)$ is surjective, for all $f \in B$;

(ii) if $B = (f_1, \ldots, f_n)$ and $\pi^{-1}(\operatorname{Spec} B_{f_i}) \cong \operatorname{Spec} A_i$ where $B_{f_i} \to A_i$ is surjective, for all i, then $\pi^{-1}(\operatorname{Spec} B) \cong \operatorname{Spec} A$ where $B \to A$ is surjective, for some A.

To show (i). Since $\pi : X \to Y$ is affine, we know that $\pi^{-1}(\operatorname{Spec} B_f) \cong \operatorname{Spec}\Gamma(\pi^{-1}(\operatorname{Spec} B_{f_i}), \mathscr{O}_X)$ for all $f \in B$. It suffices to show that $B_f \to \Gamma(\pi^{-1}(\operatorname{Spec} B_f), \mathscr{O}_X)$ is surjective, for all $f \in B$. In fact,

$$\begin{aligned}\pi^{-1}(\operatorname{Spec} B_f) =& \pi^{-1}(\operatorname{Spec} B \setminus V(f)) = \pi^{-1}(\operatorname{Spec} B) \setminus \pi^{-1}(V(f)) \\ =& \operatorname{Spec}\Gamma(\pi^{-1}(\operatorname{Spec} B), \mathscr{O}_X) \setminus V(\pi^\sharp f) \\ =& \operatorname{Spec}\Gamma(\pi^{-1}(\operatorname{Spec} B), \mathscr{O}_X)_{\pi^\sharp f}.\end{aligned}$$

Hence, it suffices to show that $B_f \to \Gamma(\pi^{-1}(\operatorname{Spec} B), \mathscr{O}_X)_{\pi^\sharp f}$ is surjective, this is obvious, since $B \to \Gamma(\pi^{-1}(\operatorname{Spec} B), \mathscr{O}_X)$ is surjective.

To show (ii). By the proof of Proposition 9.3.6, we know that there is a ring A such that $\pi^{-1}(\operatorname{Spec} B) \cong \operatorname{Spec} A$ and $A_{\pi^\sharp f_i} \cong A_i$. By the hypothesis, $B_{f_i} \to A_{\pi^\sharp f_i}$ is surjective for all i, we want to show that $B \to A$ is surjective. Note that $B_{f_i} = \Gamma(\operatorname{Spec} B_{f_i}, \mathscr{O}_Y)$, $B = \Gamma(\operatorname{Spec} B, \mathscr{O}_Y)$, $A_{\pi^\sharp f_i} = \Gamma(\operatorname{Spec} A_{\pi^\sharp f_i}, \mathscr{O}_X)$ and $A = \Gamma(\operatorname{Spec} A, \mathscr{O}_X)$, consider the following commutative diagram (definition of sheaves),

$$\begin{tikzcd} 0 \arrow[r] & \Gamma(\operatorname{Spec} B, \mathscr{O}_Y) \arrow[r, "\beta"] \arrow[d] & \prod_{i=1}^n \Gamma(\operatorname{Spec} B_{f_i}, \mathscr{O}_Y) \arrow[d, "\gamma"] \\ 0 \arrow[r] & \Gamma(\operatorname{Spec} A, \mathscr{O}_X) \arrow[r, "\alpha"] & \prod_{i=1}^n \Gamma(\operatorname{Spec} A_{\pi^\sharp f_i}, \mathscr{O}_X) \end{tikzcd}$$

where α, β induced by restriction and γ induced by $\pi|^\sharp_{\operatorname{Spec} A_{\pi^\sharp f_i}} : B_{f_i} \to A_{\pi^\sharp f_i}$. Let $f \in A$, then $\alpha(f) =$

$(f,\ldots,f)$. By the construction of γ, $\gamma((\pi|^{\sharp}_{\operatorname{Spec} A_{\pi^\sharp f_1}})^{-1}(f),\ldots,(\pi|^{\sharp}_{\operatorname{Spec} A_{\pi^\sharp f_n}})^{-1}(f)) = (f,\ldots,f)$, and each $(\pi|^{\sharp}_{\operatorname{Spec} A_{\pi^\sharp f_i}})^{-1}(f) \in B$, so $(\pi|^{\sharp}_{\operatorname{Spec} A_{\pi^\sharp f_i}})^{-1}(f) = (\pi^\sharp)^{-1}(f)$. Hence $\beta((\pi^\sharp)^{-1}(f)) = ((\pi^\sharp)^{-1}(f),\ldots,(\pi^\sharp)^{-1}(f))$, it follows that $B \to A$ is surjective.

By (i) and (ii), the property of being a closed embedding is affine local property, by Affine Communication Lemma 6.3.1, $\pi : X \hookrightarrow Y$ is closed embedding. □

In particular, if $B \to A$ is a surjection of rings, then the induced morphism $\operatorname{Spec} A \to \operatorname{Spec} B$ is a closed embedding.

10.1.2 Important: Closed subschemes correspond to quasi-coherent sheaves of ideals

Definition 10.1.2 (Ideal sheaf)

A closed subscheme $\pi : X \hookrightarrow Y$ gives a surjection of $\mathscr{O}_Y \to \pi_*\mathscr{O}_X$. We define the ***ideal sheaf*** *(or sheaf of ideals) on Y, denoted $\mathscr{I}_{X/Y}$, to be the kernel of $\pi^\sharp : \mathscr{O}_Y \to \pi_*\mathscr{O}_X$, i.e.,*

$$\mathscr{I}_{X/Y} = \operatorname{Ker}(\mathscr{O}_Y \to \pi_*\mathscr{O}_X).$$

Remark 10.3 On each open subset $U \subseteq Y$, it gives an ideal $\mathscr{I}_{X/Y}(U)$ of the ring $\mathscr{O}_Y(U)$; on the affine open subset $\operatorname{Spec} B \subseteq Y$, it gives the ideal I of B determining the surjection $B \to A = B/I$ in the definition of "closed subscheme" (Definition 10.1.1).

Proposition 10.1.5

$\pi : X \hookrightarrow Y$ is a closed embedding. Then $\pi_*\mathscr{O}_X$ is a quasi-coherent sheaf on Y.

Proof $\pi : X \hookrightarrow Y$ induces a morphism of sheaves $\mathscr{O}_Y \to \pi_*\mathscr{O}_X$, then $\pi_*\mathscr{O}_X$ can be seen as $\mathscr{O}_Y$-module. Let $\operatorname{Spec} B \subseteq Y$ be an affine open subset, since $\pi : X \hookrightarrow Y$ is a closed embedding, we may assume $\pi^{-1}(\operatorname{Spec} B) \cong \operatorname{Spec} B/I$ where I is an ideal of B. B/I can be seen as B-module, we claim that $(\pi_*\mathscr{O}_X)|_{\operatorname{Spec} B} \cong \widetilde{B/I}$. It suffices to check on the distinguished base. Let $\operatorname{Spec} B_f \hookrightarrow \operatorname{Spec} B$ be any distinguished open subset of $\operatorname{Spec} B$, then

$$\begin{aligned}\Gamma(\operatorname{Spec} B_f, (\pi_*\mathscr{O}_X)|_{\operatorname{Spec} B}) &= \Gamma(\pi^{-1}(\operatorname{Spec} B_f), \mathscr{O}_X) = \Gamma(\operatorname{Spec}(B/I)_{\pi^\sharp f}, \mathscr{O}_X)\\ &= (B/I)_{\pi^\sharp f} \cong B_f \otimes_B (B/I)\\ &= \widetilde{B/I}(\operatorname{Spec} B_f).\end{aligned}$$

Hence for any affine open subset $\operatorname{Spec} B \hookrightarrow Y$ we have $(\pi_*\mathscr{O}_X)|_{\operatorname{Spec} B} \cong \widetilde{B/I}$, and therefore $\pi_*\mathscr{O}_X$ is a quasi-coherent sheaf on Y. □

Quasi-coherent sheaves form an abelian category, so $\mathscr{I}_{X/Y} = \operatorname{Ker}(\mathscr{O}_Y \to \pi_*\mathscr{O}_X)$ is also a quasi-coherent sheaf on Y. Thus every closed subscheme of Y yields a quasi-coherent sheaf on Y, i.e., ideal sheaf on Y.

Definition 10.1.3 (Closed subscheme exact sequence)

We call the exact sequence (of quasi-coherent sheaves on Y)

$$0 \longrightarrow \mathscr{I}_{X/Y} \longrightarrow \mathscr{O}_Y \longrightarrow \pi_*\mathscr{O}_X \longrightarrow 0 \tag{10.1}$$

the ***closed subscheme exact sequence*** *for $\pi : X \hookrightarrow Y$.*

We may recover X (and its map π to Y) from the ideal sheaf $\mathscr{I}_{X/Y}$ via (10.1): for each affine open subset $\operatorname{Spec} B \subseteq Y$, the sections over $\operatorname{Spec} B$ give us an exact sequence

$$0 \longrightarrow I \longrightarrow B \longrightarrow B/I \longrightarrow 0,$$

and $\operatorname{Spec} B/I$ is $\pi^{-1}(\operatorname{Spec} B)$.

The converse holds:

Proposition 10.1.6 (Important)

Every quasi-coherent sheaf of ideals on Y produces a closed subscheme on Y.

Proof Suppose $Y = \bigcup_i \operatorname{Spec} B_i$. Let $\mathscr{I}$ be an ideal sheaf on Y, then $I(B_i) := \mathscr{I}(\operatorname{Spec} B_i)$ is an ideal of B_i. Consider quotient map $B_i \to B_i/I(B_i)$, it gives a closed subscheme $\operatorname{Spec} B_i/I(B_i)$ of $\operatorname{Spec} B_i$. We want to glue $\{\operatorname{Spec} B_i/I(B_i)\}$ to a closed subscheme $X \hookrightarrow Y$.

For any $\operatorname{Spec} B_i$ and $\operatorname{Spec} B_j$, by Proposition 6.3.1, we may assume that $\operatorname{Spec} B_i \cap \operatorname{Spec} B_j$ is covered by $\operatorname{Spec}(B_i)_{f_t} \cong \operatorname{Spec}(B_j)_{g_t}$. Hence

$$(B_i)_{f_t} \cong (B_j)_{g_t},$$

and therefore we have $I((B_i)_{f_t}) \cong I((B_j)_{g_t})$. Since $\mathscr{I}$ is a quasi-coherent sheaf, by Theorem 7.2.2, we have $I((B_i)_{f_t}) \cong I(B_i)_{f_t}$ and $I((B_j)_{g_t}) \cong I(B_j)_{g_t}$. Hence

$$(B_i)_{f_t}/I(B_i)_{f_t} \cong (B_i/I(B_i))_{f_t} \cong (B_j/I(B_j))_{g_t} \cong (B_j)_{g_t}/I(B_j)_{g_t},$$

sine each $\operatorname{Spec} B_i/I(B_i) \hookrightarrow \operatorname{Spec} B_i$ is closed subscheme, we have

$$\operatorname{Spec}(B_i)_{f_t} \cap \operatorname{Spec} B_i/I(B_i) = \operatorname{Spec}(B_i/I(B_i))_{f_t} \cong \operatorname{Spec}(B_j/I(B_j))_{g_t} = \operatorname{Spec}(B_j)_{g_t} \cap \operatorname{Spec} B_j/I(B_j),$$

it follows that we have an isomorphism

$$f_{ij} : \operatorname{Spec} B_i/I(B_i) \cap (\operatorname{Spec} B_i \cap \operatorname{Spec} B_j) \xrightarrow{\sim} \operatorname{Spec} B_j/I(B_j) \cap (\operatorname{Spec} B_i \cap \operatorname{Spec} B_j).$$

We next show the morphisms $\{f_{ij}\}_{ij}$ agree on triple intersections (Theorem 5.4.1). Say

$$X_{ij} = \operatorname{Spec} B_i/I(B_i) \cap (\operatorname{Spec} B_i \cap \operatorname{Spec} B_j),$$

then

$$\begin{aligned} X_{ij} \cap X_{ik} &= \operatorname{Spec} B_i/I(B_i) \cap (\operatorname{Spec} B_i \cap \operatorname{Spec} B_j \cap \operatorname{Spec} B_k) \\ X_{ji} \cap X_{jk} &= \operatorname{Spec} B_j/I(B_j) \cap (\operatorname{Spec} B_i \cap \operatorname{Spec} B_j \cap \operatorname{Spec} B_k) \\ X_{kj} \cap X_{ki} &= \operatorname{Spec} B_k/I(B_k) \cap (\operatorname{Spec} B_i \cap \operatorname{Spec} B_j \cap \operatorname{Spec} B_k). \end{aligned}$$

Consider $\operatorname{Spec} B_i \cap \operatorname{Spec} B_j \cap \operatorname{Spec} B_k$, since $\operatorname{Spec} B_i \cap \operatorname{Spec} B_j = \bigcup_t \operatorname{Spec}(B_i)_{f_t}$, we have

$$\begin{aligned} \operatorname{Spec} B_i \cap \operatorname{Spec} B_j \cap \operatorname{Spec} B_k &= \left(\bigcup_t \operatorname{Spec}(B_i)_{f_t}\right) \cap \operatorname{Spec} B_k \\ &= \bigcup_t (\operatorname{Spec}(B_i)_{f_t} \cap \operatorname{Spec} B_k). \end{aligned}$$

By Proposition 6.3.1, each $\operatorname{Spec}(B_i)_{f_t} \cap \operatorname{Spec} B_k$ can be covered by simultaneously distinguished open subset of $\operatorname{Spec}(B_i)_{f_t}$ and $\operatorname{Spec} B_k$. Since distinguished open subset of $\operatorname{Spec}(B_i)_{f_t}$ is also the distinguished open subset of $\operatorname{Spec} B_i$ and $\operatorname{Spec} B_j$, $\operatorname{Spec} B_i \cap \operatorname{Spec} B_j \cap \operatorname{Spec} B_k$, since $\operatorname{Spec} B_i \cap \operatorname{Spec} B_j = \bigcup_t \operatorname{Spec}(B_i)_{f_t}$ can be covered by simultaneously distinguished open subset of $\operatorname{Spec} B_i$, $\operatorname{Spec} B_j$, and $\operatorname{Spec} B_k$. Similar to the previous process, we have

$$f_{ij}|_{X_{ij} \cap X_{ik}} : \operatorname{Spec} B_i/I(B_i) \cap (\operatorname{Spec} B_i \cap \operatorname{Spec} B_j \cap \operatorname{Spec} B_k) \xrightarrow{\sim} \operatorname{Spec} B_j/I(B_j) \cap (\operatorname{Spec} B_i \cap \operatorname{Spec} B_j \cap \operatorname{Spec} B_k).$$

Hence,

$$f_{ik}|_{X_{ij} \cap X_{ik}} = f_{jk}|_{X_{ji} \cap X_{jk}} \circ f_{ij}|_{X_{ij} \cap X_{ik}}.$$

By Theorem 5.4.1, we can glue $\{\operatorname{Spec} B_i/I(B_i)\}$ together, say X. By the construction of X, clearly, X is a closed subscheme on Y. □

Example 10.1 **(Unimportant example: a sheaf of ideals (as $\mathscr{O}$-modules) that is not quasi-coherent.)** Let $X = \operatorname{Spec} k[x]_{(x)}$, the "germ of the affine line at the origin", which has two points, the closed point and the generic point η. Define $\mathscr{I}(X) = \{0\} \subseteq \mathscr{O}_X(X) = k[x]_{(x)}$, and $\mathscr{I}(\eta) = k(x) = \mathscr{O}_X(\eta)$ ($\{\eta\} = D(x)$ is open in X). Show that this sheaf of ideals does not correspond to a closed subscheme.

Proof We show that $\mathscr{I}$ is not quasi-coherent sheaf, and therefore not correspond to a closed subscheme.

If $\mathscr{I}$ is a quasi-coherent sheaf, then $\mathscr{I} \cong \widetilde{I}$ where $I \subseteq A$ is an ideal. By our definition, $\mathscr{I}(X) = \{(0)\}$, hence $\widetilde{I}(X) = I = (0)$. We now check $\widetilde{(0)}(\eta)$,

$$\widetilde{(0)}(\eta) = (k[x]_{(x)})_x \otimes_{k[x]_{(x)}} (0) = k(x) \otimes_{k[x]_{(x)}} (0) = 0 \neq k(x),$$

it follows that $\mathscr{I}$ is not quasi-coherent sheaf, and therefore not correspond to a closed subscheme. □

In the literature, the usual definition of a closed embedding is:

Definition 10.1.4 (Closed embedding)

A ***closed embedding*** *is a morphism $\pi : X \to Y$ such that π induces a homeomorphism of the underlying topological space of X onto a closed subset of the topological space of Y, and the induced map $\pi^\sharp : \mathscr{O}_Y \to \pi_*\mathscr{O}_X$ of sheaves on Y is surjective. (By "surjective" we mean that the ring morphism on stalks is surjective.)*

Remark 10.4 We not use Definition 10.1.4 later.

We next show Definition 10.1.4 agrees with Definition 10.1.1.

Lemma 10.1.1

Let $\pi : X \to Y$ be a morphism of schemes. If π is a homeomorphism onto a closed subset of Y, then π is affine.

Proof Let $y \in Y$ be a point. If $y \notin \pi(X)$, then there exists an affine neighborhood of y which is disjoint from $\pi(X)$. If $y \in \pi(X)$, since π is homeomorphism, let $x \in X$ be the unique point of X mapping to y. Let $V \subseteq Y$ be an affine open neighborhood of y. Let $U \subseteq X$ be an affine open neighborhood of x which maps into V. Since $\pi(U) \subseteq V \cap \pi(X)$ is open in the induced topology by our assumption on π we may choose a $h \in \Gamma(V, \mathscr{O}_Y)$ such that $y \in D(h)$ and $D(h) \cap \pi(X) \subseteq \pi(U)$. Denote $h' \in \Gamma(U, \mathscr{O}_X)$ the restriction of $\pi^\sharp(h)$ to U. Then we see that $D(h') \subseteq U$ is equal to $\pi^{-1}(D(h))$. In other words, every point of Y has an open neighborhood whose inverse image is affine. Thus π is affine. □

Proposition 10.1.7

Definition 10.1.4 agrees with Definition 10.1.1.

Proof To show that Definition 10.1.4 implies Definition 10.1.1. Let $\pi : X \to Y$ be a morphism such that π induces a homeomorphism of the underlying topological space of X onto a closed subset of the topological space Y, and the induced map $\pi^\sharp : \mathscr{O}_Y \to \pi_*\mathscr{O}_X$ of sheaves on Y is surjective. By hypothesis, apply Lemma 10.1.1, we know that π is affine. Let $\operatorname{Spec} B$ be any affine open subset of Y, we may assume that $\pi^{-1}(\operatorname{Spec} B) \cong \operatorname{Spec} A$, since $\mathscr{O}_Y \to \pi_*\mathscr{O}_X$ is surjective,

$$\mathscr{O}_Y(\operatorname{Spec} B) = B \longrightarrow A = \pi_*\mathscr{O}_X(\operatorname{Spec} B)$$

is surjective. It follows that Definition 10.1.4 implies Definition 10.1.1.

We next show that Definition 10.1.1 implies Definition 10.1.4. By Proposition 10.1.1, X is homeomorphic to a closed subset of Y. It suffices to show that the induced map $\pi^\sharp : \mathscr{O}_Y \to \pi_*\mathscr{O}_X$ of sheaves on Y is surjective. By Proposition 3.5.1, it suffices to show that $\mathscr{O}_Y \to \pi_*\mathscr{O}_X$ on a base $\operatorname{Spec} B_f$ is surjective for all $f \in B$. By the hypothesis, we may assume that $\pi^{-1}(\operatorname{Spec} B) \cong \operatorname{Spec} A$ where $\operatorname{Spec} B \subseteq Y$ is open and $B \twoheadrightarrow A$. Hence

$$\mathscr{O}_Y(\operatorname{Spec} B_f) = B_f \longrightarrow \pi_*\mathscr{O}_X(\operatorname{Spec} B_f) = \mathscr{O}_X(\pi^{-1}(\operatorname{Spec} B_f)) = A_{\pi^\sharp f}.$$

Since $B \twoheadrightarrow A$ is surjective, $B_f \to A_{\pi^\sharp f}$ is surjective. It follows that $\mathscr{O}_Y \to \pi_*\mathscr{O}_X$ on a base $\operatorname{Spec} B_f$ is surjective for all $f \in B$, then we done! $\square$

We have now defined the analog of open subsets and closed subsets in the land of schemes. Their definition is slightly less "symmetric" than in the classical topological setting: the "complement" of a closed subscheme is a unique open subscheme, but there are many "complementary" closed subschemes to a given open subscheme in general. (We will soon define one that is "best", that has a reduced structure.)

10.1.3 Examples and properties

Using the one-to-one corresponding between quasi-coherent sheaves of ideals and closed subschemes (§10.1.2) we can define scheme theoretic

Definition 10.1.5 (Finite (scheme-theoretic) union of closed subschemes)

Let Z be a scheme. Let $X \hookrightarrow Z$ and $Y \hookrightarrow Z$ be closed subschemes corresponding to quasi-coherent ideal sheaves $\mathscr{I}_{X/Z} \hookrightarrow \mathscr{O}_Z$ and $\mathscr{I}_{Y/Z} \hookrightarrow \mathscr{O}_Z$. The scheme theoretic union of X and Y is the closed subscheme of Z cut out by $\mathscr{I}_{X/Z} \cap \mathscr{I}_{Y/Z}$.

Definition 10.1.6 (Arbitrary (scheme-theoretic) intersection of closed subschemes)

Let Z be a scheme. Let $X \hookrightarrow Z$ and $Y \hookrightarrow Z$ be closed subschemes corresponding to quasi-coherent ideal sheaves $\mathscr{I}_{X/Z} \hookrightarrow \mathscr{O}_Z$ and $\mathscr{I}_{Y/Z} \hookrightarrow \mathscr{O}_Z$. The scheme theoretic intersection of X and Y is the closed subscheme of Z cut out by $\mathscr{I}_{X/Z} + \mathscr{I}_{Y/Z}$.

✍ **Exercise 10.1** Describe the scheme-theoretic intersection of $V(y-x^2)$ and $V(y)$ in $\mathbb{A}^2$. See Figure 5.5 for a picture. (For example, explain informally how this corresponds to two curves meeting at a single point with multiplicity 2 — notice how the 2 is visible in your answer. Alternatively, what is the non-reducedness telling you — both its "size" and its "direction"?) Describe their scheme-theoretic union.

Solution *Let $Z = \mathbb{A}^2 = \operatorname{Spec} k[x,y]$, $X = V(y-x^2) = \operatorname{Spec} k[x,y]/(y-x^2)$, and $Y = V(y) = \operatorname{Spec} k[x,y]/(y)$. Then the corresponding ideal sheaves of X and Y are $\widetilde{(y-x^2)}$ and $\widetilde{(y)}$, hence the corresponding ideal sheaves of $X \cap Y$ is $\widetilde{(y-x^2)} + \widetilde{(y)} = \widetilde{(y-x^2)+(y)} = \widetilde{(y,x^2)}$, and therefore $X \cap Y = \operatorname{Spec} k[x,y]/(y,x^2) \cong \operatorname{Spec} k[x]/(x^2)$. $\operatorname{Spec} k[x]/(x^2)$ is non-reduced scheme, its Krull dimension is $\dim_k \operatorname{Spec} k[x]/(x^2) = 2$, it means two curves meeting at a single point have "thickness". See Figure 5.3, $X \cap Y$ remembers the derivative only in the x direction.*

By Definition 10.1.5, the corresponding ideal sheaves of $X \cup Y$ is $\widetilde{(y-x^2)} \cap \widetilde{(y)} = \widetilde{(y-x^2)\cap(y)} = \widetilde{(y(y-x^2))}$, and therefore $X \cup Y = \operatorname{Spec} k[x,y]/(y(y-x^2)) = V(y(y-x^2))$.

Proposition 10.1.8

(a) The underlying set of a finite union of closed subschemes is the finite union of the underlying sets.

(b) The underlying set of intersection of closed subschemes is the intersection of the underlying sets.

Proof

(a) Let X be a scheme, and let $Z_1, \ldots, Z_n$ be a finite collection of closed subschemes of X. Then the corresponding ideal sheaf of $\bigcup_{j=1}^n Z_j$ is $\bigcap_{j=1}^n \mathscr{I}_{Z_j/X}$. Suppose $X = \bigcup_i \operatorname{Spec} A_i$, by Proposition 10.1.6,

$$\bigcup_{j=1}^n Z_j = \bigcup_i \operatorname{Spec}\left(A_i \Big/ \left(\bigcap_{j=1}^n \mathscr{I}_{Z_j/X}(\operatorname{Spec} A_i)\right)\right).$$

Note that

$$\begin{aligned}\bigcup_{j=1}^n \operatorname{Spec} A_i/\mathscr{I}_{Z_j/X}(\operatorname{Spec} A_i) = & \bigcup_{j=1}^n V_{\operatorname{Spec} A_i}(\mathscr{I}_{Z_j/X}(\operatorname{Spec} A_i)) \\ = & V_{\operatorname{Spec} A_i}\left(\bigcap_{j=1}^n \mathscr{I}_{Z_j/X}(\operatorname{Spec} A_i)\right) \\ = & \operatorname{Spec}\left(A_i \Big/ \left(\bigcap_{j=1}^n \mathscr{I}_{Z_j/X}(\operatorname{Spec} A_i)\right)\right),\end{aligned}$$

it follows that a finite union of closed subschemes is the finite union of the underlying sets.

(b) Let X be a scheme, and let $\{Z_j\}_{j\in\mathscr{J}}$ be a collection of closed subschemes of X. Then the corresponding ideal sheaf of $\bigcap_{j\in\mathscr{J}}$ is $\sum_{j\in\mathscr{J}} \mathscr{I}_{Z_j/X}$. Suppose $X = \bigcup_i \operatorname{Spec} A_i$, by Proposition 10.1.6,

$$\begin{aligned}\bigcap_{j\in\mathscr{J}} Z_j = & \bigcup_i \operatorname{Spec}\left(A_i \Big/ \left(\sum_{j\in\mathscr{J}} \mathscr{I}_{Z_j/X}(\operatorname{Spec} A_i)\right)\right) \\ = & \bigcup_i V_{\operatorname{Spec} A_i}\left(\sum_{j\in\mathscr{J}} \mathscr{I}_{Z_j/X}(\operatorname{Spec} A_i)\right) \\ = & \bigcup_i \bigcap_{j\in\mathscr{J}} V_{\operatorname{Spec} A_i}(\mathscr{I}_{Z_j/X}(\operatorname{Spec} A_i)) \\ = & \bigcap_{j\in\mathscr{J}} \bigcup_i V_{\operatorname{Spec} A_i}(\mathscr{I}_{Z_j/X}(\operatorname{Spec} A_i)),\end{aligned}$$

it follows that the underlying set of intersection of closed subschemes is the intersection of the underlying sets.

$\square$

Exercise 10.2 Describe the scheme-theoretic intersection of $V(y^2 - x^2)$ and $V(y)$ in $\mathbb{A}^2$. Draw a picture.

Solution *The scheme-theoretic intersection is* $V(y^2 - x^2) \cap V(y) = V(y, x^2) \cong \operatorname{Spec} k[x]/(x^2)$.

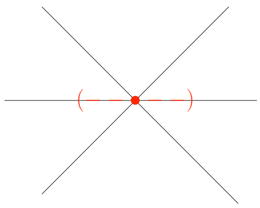


Figure 10.1: Exercise 10.2

Example 10.2 In general, if X, Y, and Z are closed subschemes of W, then $(X\cap Z)\cup(Y\cap Z) \neq (X\cup Y)\cap Z$. Here is an example. Consider $X = V(y-x)$, $Y = V(y+x)$, $Z = V(y)$, and $W = \mathbb{A}^2$. Then

$$(X\cap Z)\cup(Y\cap Z) = \{(0,0)\}\cup\{(0,0)\} = \{(0,0)\}$$

and

$$(X \cup Y) \cap Z = V(y^2 - x^2) \cap V(y) = V(x^2, y),$$

hence

$$(X \cap Z) \cup (Y \cap Z) \neq (X \cup Y) \cap Z.$$

Remark 10.5 In particular, not all properties of intersection and union carry over from sets to schemes.

Proposition 10.1.9

Closed embeddings are monomorphisms.

Proof Same as Proposition 8.2.2. □

In many cases, loci which are a priori closed sets often have enriched versions as closed subschemes. The remaining notions are examples of this. (As described in §7.6.14, we often indicate that we are considering scheme-theoretic enrichments by adding the adjective scheme-theoretic — for example, say "scheme-theoretic intersection", "scheme-theoretic union", and so forth.)

Definition 10.1.7 (Vanishing scheme)

Suppose $Y = \bigcup_i \operatorname{Spec} B_i$ *is scheme, and* $s \in \Gamma(Y, \mathscr{O}_Y)$. *Define the closed subscheme* ***cut out by*** s *as follow:*

$$V(s) := \bigcup_i \operatorname{Spec} B_i/(s_{B_i}),$$

where $s_{B_i} = s|_{\operatorname{Spec} B_i}$ *is the restriction of* s *to* $\operatorname{Spec} B_i$. *We call this the* ***vanishing scheme*** $V(s)$ *of* s. *If* S *is a set of functions, define* $V(S)$ *be the scheme-theoretic intersection*

$$V(S) = \bigcap_{s \in S} V(s).$$

We call this the ***vanishing scheme*** $V(S)$ *of* S.

Proposition 10.1.10

Suppose Y *is a scheme, and* $s \in \Gamma(Y, \mathscr{O}_Y)$. *If* u *is an invertible function, then* $V(s) = V(su)$.

Proof We may assume that $Y = \operatorname{Spec} B$, then $V(s) = \operatorname{Spec} B/(s)$ and $V(su) = \operatorname{Spec} B/(su)$. Since $u \in B$ is invertible, we have $B/(s) \cong B/(su)$, hence $V(s) = V(su)$. □

Definition 10.1.8 (Scheme-theoretic support)

If M *is an* A*-module, and* $m \in M$, *we call the closed subscheme* $\operatorname{Spec}(A/\operatorname{Ann} m)$ *of* $\operatorname{Spec} A$ *the* ***scheme-theoretic support*** *of* m.

Remark 10.6 This is often described as the "smallest closed subscheme on which m is defined", since $\operatorname{Spec}(A/\operatorname{Ann} m)$ is the smallest closed subscheme such that m is not zero divisor on global section.

Proposition 10.1.11

The underlying set of the scheme-theoretic support of m *is the usual "set-theoretic" support,* $\operatorname{Supp} m$.

Proof Let M is an A-module, and $m \in M$, we want to show that as set

$$\operatorname{Spec}(A/\operatorname{Ann} m) = \{[\mathfrak{p}] \in \operatorname{Spec} A : m_{\mathfrak{p}} \neq 0 \text{ in } M_{\mathfrak{p}}\}.$$

If $m_{\mathfrak{p}} \neq 0$ in $M_{\mathfrak{p}}$, then for all $s \in A - \mathfrak{p}$ we have $sm \neq 0$, hence $(A - \mathfrak{p}) \cap \operatorname{Ann} m = \varnothing$, and therefore

$\mathfrak{p} \supseteq \operatorname{Ann} m$, vice versa. Hence

$$\operatorname{Spec}(A/\operatorname{Ann} m) = \{[\mathfrak{p}] \in \operatorname{Spec} A : m_{\mathfrak{p}} \neq 0 \text{ in } M_{\mathfrak{p}}\}.$$

□

Definition 10.1.9 (Scheme-theoretic support of a finitely generated module)

Define the ***scheme-theoretic support of a finitely generated A-module*** *M to be the scheme-theoretic intersection of all closed subschemes $\operatorname{Spec} A/I \hookrightarrow \operatorname{Spec} A$ for which M is an (A/I)-module. More precisely,*

$$\operatorname{Supp} M := \bigcap_{I \subseteq \operatorname{Ann} M} \operatorname{Spec} A/I = \operatorname{Spec}(A/\operatorname{Ann}_A M).$$

Proposition 10.1.12

The scheme-theoretic support of a finitely generated A-module M is the scheme-theoretic union of the scheme-theoretic supports of any finite generating set of M.

Proof Let M be a finitely generated A-module which generated by $m_1, \dots, m_n$. We want to show that

$$\operatorname{Supp} M = \bigcup_{i=1}^{n} \operatorname{Spec}(A/\operatorname{Ann} m_i)$$

where the union is scheme-theoretic union. The corresponding ideal sheaf of $\operatorname{Supp} M$ is $\sum_{I \subseteq \operatorname{Ann} M} \widetilde{I} = \widetilde{\sum_{I \subseteq \operatorname{Ann} M} I}$. Consider $\sum_{I \subseteq \operatorname{Ann} M} I$. Note that

$$\sum_{I \subseteq \operatorname{Ann} M} I = \operatorname{Ann} M = \bigcap_{i=1}^{n} \operatorname{Ann} m_i,$$

we have

$$\sum_{I \subseteq \operatorname{Ann} M} \widetilde{I} = \widetilde{\bigcap_{i=1}^{n} \operatorname{Ann} m_i} = \bigcap_{i=1}^{n} \widetilde{\operatorname{Ann} m_i}.$$

Since $\bigcup_{i=1}^{n} \operatorname{Spec}(A/\operatorname{Ann} m_i)$ correspond to the ideal sheaf $\bigcap_{i=1}^{n} \widetilde{\operatorname{Ann} m_i}$, we have

$$\operatorname{Supp} M = \bigcup_{i=1}^{n} \operatorname{Spec}(A/\operatorname{Ann} m_i).$$

□

We next define the **scheme-theoretic support of finite type quasi-coherent sheaf**. Suppose $X = \bigcup_i \operatorname{Spec} A_i$ is a scheme and $\mathscr{F}$ is a finite type quasi-coherent sheaf on X. Since $\mathscr{F}$ is quasi-coherent sheaf, we may assume that $\mathscr{F}|_{\operatorname{Spec} A} \cong \widetilde{M}$, where M is finitely generated A-module. Define annihilate sheaf $\mathscr{A}\mathrm{nn}(\mathscr{F})$ as follow:

$$\mathscr{A}\mathrm{nn}(\mathscr{F})(\operatorname{Spec} A) := \operatorname{Ann}_A M.$$

Then $\mathscr{A}\mathrm{nn}(\mathscr{F})$ is a quasi-coherent sheaf on X. By Proposition 10.1.6, $\mathscr{A}\mathrm{nn}(\mathscr{F})$ corresponds to a closed subscheme on X, say $\operatorname{Supp} \mathscr{F}$. If $X = \bigcup_i \operatorname{Spec} A_i$, then

$$\operatorname{Supp} \mathscr{F} = \bigcup_i \operatorname{Spec}(A_i/\mathscr{A}\mathrm{nn}(\mathscr{F})(\operatorname{Spec} A_i)) = \bigcup_i \operatorname{Supp} M_i,$$

where $\mathscr{F}|_{\operatorname{Spec} A_i} \cong \widetilde{M_i}$.

Definition 10.1.10 (Scheme-theoretic support of finite type quasi-coherent sheaf)

Suppose $X = \bigcup_i \operatorname{Spec} A_i$ is a scheme and $\mathscr{F}$ is a finite type quasi-coherent sheaf on X. Say $\mathscr{F}|_{\operatorname{Spec} A_i} \cong \widetilde{M_i}$, define the ***scheme-theoretic support of finite type quasi-coherent sheaf*** *$\mathscr{F}$ be the scheme-theoretic union of the scheme-theoretic support of finitely generated A-module $\operatorname{Supp} M_i$, i.e.,*

$$\operatorname{Supp} \mathscr{F} := \bigcup_i \operatorname{Supp} M_i.$$

10.2 Locally closed embeddings and locally closed subschemes

Now that we have defined analogs of open and closed subsets, it is natural to define the analog of locally closed subsets. Recall that locally closed subsets are intersections of open subsets and closed subsets. Hence they are closed subsets of open subsets, or equivalently open subsets of closed subsets. The analog of these equivalences will be a little problematic in the land of schemes. (Similarly, the notion of subgroup and quotient group are not quite complementary. Rhetorical question: should "subquotient" of a group G be defined as a subgroup of a quotient of G, or a quotient of a subgroup of G, or are these the same? We will confront the same issue when defining locally closed embeddings.)

Definition 10.2.1 (Locally closed embedding, Locally closed immersion)

We say a morphism $\pi : X \to Y$ is a ***locally closed embedding*** *(or* ***locally closed immersion****) if π can be factored into*

$$X \xhookrightarrow[\text{closed}]{\rho} Z \xhookrightarrow[\text{open}]{\tau} Y$$

where (as indicated) ρ is a closed embedding and τ is an open embedding.
If X is a subset of Y (and π on the level of sets is the inclusion), we say X is a ***locally closed subscheme*** *of Y.*

Remark 10.7 The symbol $\hookrightarrow$ is often used to indicate that a morphism is a locally closed embedding. Because closed embeddings and open embeddings are monomorphisms (Proposition 10.1.9 and Proposition 8.2.2 respectively), and monomorphisms are preserved by composition, locally closed embeddings are also monomorphisms.

Remark 10.8 **Warning:** the naked term **subscheme** is often used to mean locally closed subscheme, but we will avoid this unhappy usage.

Example 10.3 The morphism $\operatorname{Spec} k[t,t^{-1}] \to \operatorname{Spec} k[x,y]$ given by $(x,y) \mapsto (t,0)$ is a locally closed embedding (Figure 10.2), since $\operatorname{Spec} k[t,t^{-1}] \cong D(x) \cap V(y)$.

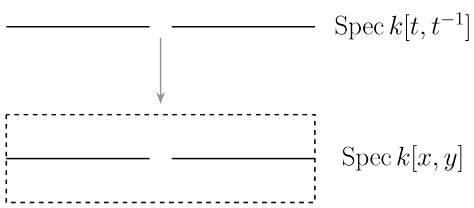


Figure 10.2: The locally closed embedding $\operatorname{Spec} k[t,t^{-1}] \to \operatorname{Spec} k[x,y]$ $(t \mapsto (t,0) = (x,y)$, i.e., $(x,y) \to (t,0))$

Remark 10.9 **Warning:** The term immersion is often used instead of locally closed embedding or locally closed immersion, but this is unwise terminology, for reasons that already arose for closed embedding in Definition

10.1.1: The differential geometric notion of immersion is closer to what algebraic geometers call unramified, which we will define in Chapter 22. Also, the naked term embedding should be avoided, because it is needlessly imprecise.

Proposition 10.2.1

Locally closed embedding are locally of finite type.

Proof Let $\pi : X \hookrightarrow Y$ be a locally closed embedding, then π can be factored into

$$X \xhookrightarrow{\rho} Z \xhookrightarrow{\tau} Y,$$

where ρ is closed embedding and τ is an open embedding. We may assume that $Y = \operatorname{Spec} B$, since τ is open embedding, suppose that $Z = \bigcup_i \operatorname{Spec} B_{f_i}$. Since ρ is closed embedding, we may assume that $X = \bigcup_i \operatorname{Spec} B_{f_i}/I_i$. Note that each B_{f_i}/I_i is finitely generated B-algebra, π is locally of finite type. $\square$

Proposition 10.2.2

Suppose $\pi : X \to Y$ is a locally closed embedding whose image is a closed subset of Y. Then π is a closed embedding.

Proof Since $\pi : X \to Y$ is a locally closed embedding, π can be factored into $X \hookrightarrow U \hookrightarrow Y$ where U is open subscheme of Y. Note that $\pi(X)$ is closed in Y, Y can be covered by $Y \setminus \pi(X)$ and U. Consider $\pi^{-1}(Y \setminus \pi(X))$ and $\pi^{-1}(U)$. In fact, $\pi^{-1}(Y \setminus \pi(X)) = \varnothing$, and therefore $\varnothing \hookrightarrow Y \setminus \pi(X)$ is closed embedding. By our assumption, $\pi^{-1}(U) \hookrightarrow U$ is closed embedding. By Proposition 10.1.4, $\pi : X \to Y$ is closed embedding. $\square$

At this point, you could define the intersection of two locally closed embeddings in a scheme X (which will also be a locally closed embedding in X). But it would be awkward, as you would have to show that your construction is independent of the factorizations of each locally closed embedding into a closed embedding and an open embedding. Instead, we wait until Chapter 11, when recognizing the intersection as a fibered product will make this easier.

Clearly an open subscheme U of a closed subscheme V of X can be interpreted as a closed subscheme of an open subscheme: as the topology on V is induced from the topology on X, the underlying set of U is the intersection of some open subset U' on X with V. We can take $V' = V \cap U'$, and then $V' \to U'$ is a closed embedding, and $U' \to X$ is an open embedding.

It is not clear that a closed subscheme V' of an open subscheme U' can be expressed as an open subscheme of a closed subscheme V. In the category of topological spaces, we would take V as the closure of V', so we are now motivated to define the analogous construction, which will give us an excuse to introduce several related ideas, in §10.4. We will then resolve this issue in goof cases (e.g., if X is Noetherian).

We formalize our discussion in a proposition.

Proposition 10.2.3

Suppose $V \to X$ is a morphism. Consider three conditions:

(i) V is the intersection of an open subscheme of X and a closed subscheme of X.

(ii) V is an open subscheme of a closed subscheme of X, i.e., it factors into an open embedding followed by a closed embedding.

(iii) V is a closed subscheme of an open subscheme of X, i.e., V is locally closed embedding.

Then (i) and (ii) are equivalent, and both imply (iii).

Proof Consider the following diagram.

$$\begin{tikzcd} V \arrow[r, hook, "\text{open emb.}"] \arrow[d, hook, "\text{closed emb.}"'] & K \arrow[d, hook, "\text{closed emb.}"] \\ U \arrow[r, hook, "\text{open emb.}"'] & X \end{tikzcd}$$

We define the intersection of an open subscheme of X and a closed subscheme of X be the fiber product of open subscheme and closed subscheme. More precisely, if $K \hookrightarrow X$ is closed subscheme of X and $U \hookrightarrow X$ is open subscheme of X, define

$$U \cap K := U \times_X K.$$

By Proposition 9.1.4, fiber product exists, and therefore well-defined.

(i) $\Rightarrow$ (ii): By Proposition 9.1.4, $U \times_X K \to K$ is an open embedding, hence $U \cap K$ is an open subscheme of closed subscheme K of X.

(ii) $\Rightarrow$ (i): Since the class of open embeddings is preserved by base change (Proposition 9.1.4), $U \times_X K$ exists.

(i) $\Rightarrow$ (iii): Say $\rho : K \hookrightarrow X$ be closed embedding, by Proposition 9.1.4, we know that $U \times_X K = (\rho^{-1}(U), \mathscr{O}_K|_{\rho^{-1}(U)})$. We want to show that $\rho|_{\rho^{-1}(U)} : U \times_X K \to U$ is closed embedding. Let $\operatorname{Spec} B \subseteq U$ be any affine open subset of U, then $\operatorname{Spec} B$ is also affine open subset of X. Since ρ is closed embedding,

$$\rho|_{\rho^{-1}(U)}^{-1}(\operatorname{Spec} B) = \rho^{-1}(\operatorname{Spec} B) \cong \operatorname{Spec} A$$

where $B \twoheadrightarrow A$. Hence $\rho|_{\rho^{-1}(U)} : U \times_X K \to U$ is closed embedding, and therefore $U \times_X K$ is a closed subscheme of open subscheme U of X.

(ii) $\Rightarrow$ (iii): Since (i) $\Leftrightarrow$ (ii), we done! □

Remark 10.10 (iii) does not always imply (i) and (ii), see the pathological example [Sta25, Example 01QW].

Proposition 10.2.4

The composition of two locally closed embeddings is a locally closed embedding.

Proof Let $X \hookrightarrow Y$ and $Y \hookrightarrow Z$ be two locally closed embeddings. Then we have

$$X \xhookrightarrow{\text{closed}} U \xhookrightarrow{\text{open}} Y \xhookrightarrow{\text{closed}} V \xhookrightarrow{\text{open}} Z.$$

By Proposition 10.2.3, we have the following commutative diagram.

$$\begin{tikzcd} & & Y \arrow[dr, hook, "\text{closed}"] & & \\ X \arrow[r, hook, "\text{closed}"] & U \arrow[ur, hook, "\text{open}"] \arrow[dr, hook, "\text{closed}"'] & & V \arrow[r, hook, "\text{open}"] & Z \\ & & W \arrow[ur, hook, "\text{open}"'] & & \end{tikzcd}$$

Since the composition of closed embeddings is closed embedding (Proposition 10.1.3) and the composition of open embeddings is open embeddings (Proposition 9.1.1), we have

$$X \xhookrightarrow{\text{closed}} W \xhookrightarrow{\text{open}} Z,$$

i.e., $X \to Z$ is locally closed embedding. □

Remark 10.11 (**Unimportant but fun remark.**) It may feel odd that in the definition of a locally closed embedding, we had to make a choice (as a composition of a closed embedding followed by an open embedding, rather than vice versa), but this type of issue comes up earlier: a subquotient of a group can be defined as the quotient of a subgroup, or a subgroup of a quotient. Which is the right definition? Or are they the same? (The subquotient of a subquotient should certainly be a subquotient, hence define subquotient is a quotient object of subobject.)

10.3 Important examples from projective geometry

We now interpret closed embeddings in terms of graded rings. Don't worry; most of the annoying foundational discussion of graded rings is complete, and we now just take advantage of our earlier work.

10.3.1 Example: Closed embeddings in projective space $\mathbb{P}^n_A$

Recall the definition of projective space $\mathbb{P}^n_A$ given in Definition 5.4.1 (and the terminology defined there). Any homogeneous polynomial f in $x_0, \dots, x_n$ defines a closed subscheme. (Thus even if f doesn't make sense as a function, its vanishing scheme still makes sense.) On the open set U_i, the closed subscheme is $V(f(x_{0/i}, \dots, x_{n/i}))$, which we privately think of as $V(f(x_0, \dots, x_n)/x_i^{\deg f})$. On the overlap

$$U_i \cap U_j = \operatorname{Spec} A[x_{0/i}, \dots, x_{n/i}, x_{j/i}^{-1}]/(x_{i/i} - 1),$$

these functions on U_i and U_j don't exactly agree, but they agree up to a nonvanishing scalar, and hence cut out the same closed subscheme of $U_i \cap U_j$ (Definition 10.1.7):

$$f(x_{0/i}, \dots, x_{n/i}) = x_{j/i}^{\deg f} f(x_{0/j}, \dots, x_{n/j}).$$

Similarly, a collection of homogeneous polynomials in $A[x_0, \dots, x_n]$ cuts out a closed subscheme of $\mathbb{P}^n_A$. (Chapter 16 will show that all closed subschemes of $\mathbb{P}^n_A$ are of the form.)

Definition 10.3.1 (Hypersurface, degree of hypersurface)

As usual, let k be a field. A closed subscheme of $\mathbb{P}^n_k$ cut out by a single (nonzero, homogeneous) equation is called a ***hypersurface*** *(in $\mathbb{P}^n_k$). (Be careful: the word "hypersurface" can be used to mean slightly different things in algebraic geometry, for entirely reasonable reasons, see for Chapter 13.)*

The ***degree of a hypersuface*** *is the degree of the polynomial. (Implicit in this is that this notion can be determined from the subscheme itself. You may have the tools to prove this now, but we won't formally prove it until Chapter 19.)*

A hypersurface of degree 1 (resp., degree $2, 3, \dots$) is called a ***hyperplane*** *(resp.,* ***quadric, cubic, quartic, quintic, sextic, septic, octic, ... hypersurface****). If $n = 2$, hypersufaces are called* ***curves****; a degree 1 hypersurface is called a* ***line****, and a degree 2 hypersurface is called a* ***conic curve****, or a* ***conic*** *for short. If $n = 3$, a hypersurface is called a* ***surface****. (In Chapter 13, we will justify the terms curve and surface.)*

Remark 10.12 Of cause, a hypersurface is not cut out by a single global function on $\mathbb{P}^n_k$: there are no nonconstant global functions (Proposition 5.4.2).

✍ **Exercise 10.3**

(a) Show that $wz = xy, x^2 = wy, y^2 = xz$ describes an irreducible closed subscheme in $\mathbb{P}^3_k$. In fact it is a curve, a notion we will define once we know what dimension is. This curve is called the **twisted cubic**. (The twisted cubic is a good nontrivial example of many things, so you should make friends with it as soon as possible. It implicitly appeared earlier in Exercise 4.18.)

(b) Show that the twisted cubic is isomorphic to $\mathbb{P}^1_k$.

Proof

(a) It suffices to show that $\operatorname{Proj} k[x,y,z,w]/(wz-xy, x^2-wy, y^2-xz)$ is irreducible closed subscheme. Note that

$$\begin{aligned}
&\operatorname{Proj} \frac{k[x,y,z,w]}{(wz-xy, x^2-wy, y^2-xz)}\\
&= \operatorname{Spec} \frac{k\left[\frac{y}{x}, \frac{z}{x}, \frac{w}{x}\right]}{\left(\frac{w}{x}\frac{z}{x}-\frac{y}{x}, 1-\frac{w}{x}\frac{y}{x}, \frac{y^2}{x^2}-\frac{z}{x}\right)} \cup \operatorname{Spec} \frac{k\left[\frac{x}{y}, \frac{z}{y}, \frac{w}{y}\right]}{\left(\frac{w}{y}\frac{z}{y}-\frac{x}{y}, \frac{x^2}{y^2}-\frac{w}{y}, 1-\frac{x}{y}\frac{z}{y}\right)}\\
&\cup \operatorname{Spec} \frac{k\left[\frac{x}{z}, \frac{y}{z}, \frac{w}{z}\right]}{\left(\frac{w}{z}-\frac{x}{z}\frac{y}{z}, \frac{x^2}{z^2}-\frac{w}{z}\frac{y}{z}, \frac{y^2}{z^2}-\frac{x}{z}\right)} \cup \operatorname{Spec} \frac{k\left[\frac{x}{w}, \frac{y}{w}, \frac{z}{w}\right]}{\left(\frac{z}{w}-\frac{x}{w}\frac{y}{w}, \frac{x^2}{w^2}-\frac{y}{w}, \frac{y^2}{w^2}-\frac{x}{w}\frac{z}{w}\right)},
\end{aligned}$$

we know that $\operatorname{Proj} \frac{k[x,y,z,w]}{(wz-xy,x^2-wy,y^2-xz)}$ is a closed subscheme of $\mathbb{P}^3_k$. Since

$$\operatorname{Spec} \frac{k\left[\frac{y}{x}, \frac{z}{x}, \frac{w}{x}\right]}{\left(\frac{w}{x}\frac{z}{x}-\frac{y}{x}, 1-\frac{w}{x}\frac{y}{x}, \frac{y^2}{x^2}-\frac{z}{x}\right)} = \operatorname{Proj} \frac{k[x,y,z,w]}{(wz-xy, x^2-wy, y^2-xz)} \cap \operatorname{Spec} k\left[\frac{y}{x}, \frac{z}{x}, \frac{w}{x}\right],$$

we know that

$$\operatorname{cl}_{\mathbb{P}^3_k}\left(\operatorname{Spec} \frac{k\left[\frac{y}{x}, \frac{z}{x}, \frac{w}{x}\right]}{\left(\frac{w}{x}\frac{z}{x}-\frac{y}{x}, 1-\frac{w}{x}\frac{y}{x}, \frac{y^2}{x^2}-\frac{z}{x}\right)}\right) = \operatorname{Proj} \frac{k[x,y,z,w]}{(wz-xy, x^2-wy, y^2-xz)}.$$

Since the closure of irreducible subset is also irreducible (Proposition 4.6.1 (b)), it suffices to show that $\operatorname{Spec} \frac{k\left[\frac{y}{x}, \frac{z}{x}, \frac{w}{x}\right]}{\left(\frac{w}{x}\frac{z}{x}-\frac{y}{x}, 1-\frac{w}{x}\frac{y}{x}, \frac{y^2}{x^2}-\frac{z}{x}\right)}$ is irreducible, by Proposition 4.6.2, it suffices to show that $\frac{k\left[\frac{y}{x}, \frac{z}{x}, \frac{w}{x}\right]}{\left(\frac{w}{x}\frac{z}{x}-\frac{y}{x}, 1-\frac{w}{x}\frac{y}{x}, \frac{y^2}{x^2}-\frac{z}{x}\right)}$ is integral domain. Say $a = \frac{y}{x}, b = \frac{z}{x}, c = \frac{w}{x}$, then

$$\frac{k\left[\frac{y}{x}, \frac{z}{x}, \frac{w}{x}\right]}{\left(\frac{w}{x}\frac{z}{x}-\frac{y}{x}, 1-\frac{w}{x}\frac{y}{x}, \frac{y^2}{x^2}-\frac{z}{x}\right)} = k[a,b,c]/(bc-a, 1-ac, a^2-b).$$

Note that

$$\frac{k[a,b,c]}{(bc-a, 1-ac, a^2-b)} \cong \frac{k[a, a^{-1}, b]}{(a^2-b)} \cong k[a, a^{-1}],$$

and $k[a,a^{-1}]$ is an integral domain, $\frac{k\left[\frac{y}{x}, \frac{z}{x}, \frac{w}{x}\right]}{\left(\frac{w}{x}\frac{z}{x}-\frac{y}{x}, 1-\frac{w}{x}\frac{y}{x}, \frac{y^2}{x^2}-\frac{z}{x}\right)}$ is integral domain, as we desired.

(b) Say $C = \operatorname{Proj} \frac{k[w,x,y,z]}{(wz-xy,x^2-wy,y^2-xz)}$, then there is a morphism

$$\begin{aligned}
\varphi : \mathbb{P}^1_k &\longrightarrow C \subseteq \mathbb{P}^3_k\\
[s:t] &\longmapsto [s^3 : s^2t : st^2 : t^3].
\end{aligned}$$

In fact, $C = (D_+(w) \cap C) \cup (D_+(z) \cap C)$. On $D_+(w)$ we have

$$\begin{aligned}
D_+(w) \cap C &\longrightarrow \mathbb{P}^1_k\\
[w:x:y:z] &\longmapsto [w:x]
\end{aligned}$$

is isomorphism. On $D_+(z)$ we have

$$\begin{aligned}
D_+(z) \cap C &\longrightarrow \mathbb{P}^1_k\\
[w:x:y:z] &\longmapsto [y:z]
\end{aligned}$$

is isomorphism. On the overlap $D_+(w) \cap D_+(z)$, we have $wz = xy$, so they glue together, and therefore we have $C \xrightarrow{\sim} \mathbb{P}^1_k$.

□

We now extend this discussion to projective schemes in general.

Proposition 10.3.1

Suppose that $S_\bullet \twoheadrightarrow R_\bullet$ is a surjection of graded rings. Then the domain of the induced morphism (Proposition 8.4.1) is $\operatorname{Proj} R_\bullet$, and that the induced morphism $\operatorname{Proj} R_\bullet \to \operatorname{Proj} S_\bullet$ is a closed embedding.

Proof Say $\pi^\sharp : S_\bullet \twoheadrightarrow R_\bullet$ be a surjection of graded rings, by Proposition 8.4.1, it suffices to show that $V(\pi^\sharp(S_+)) = \varnothing$. Since $S_\bullet \twoheadrightarrow R_\bullet$ is surjective, $\pi^\sharp(S_+) = R_+$, hence $V(\pi^\sharp(S_+)) = \varnothing$. By Proposition 8.4.1, $S_\bullet \to R_\bullet$ induces a morphism of projective schemes

$$\pi : \operatorname{Proj} R_\bullet \longrightarrow \operatorname{Proj} S_\bullet. \tag{10.2}$$

We next show that induced map is closed embedding. Since $\pi^\sharp(S_+) = R_+$, we may assume that $\operatorname{Proj} R_\bullet = \bigcup_{\pi^\sharp(f)\in\pi^\sharp(S_+)} \operatorname{Spec}((R_\bullet)_{\pi^\sharp(f)})_0$. On the other hand, $\operatorname{Proj} S_\bullet = \bigcup_{f\in S_+} \operatorname{Spec}((S_\bullet)_f)_0$. By Proposition 8.4.1, we know that (10.2) is glued by

$$\pi|_{D_+(\pi^\sharp(f))} : \operatorname{Spec}((R_\bullet)_{\pi^\sharp(f)})_0 \longrightarrow \operatorname{Spec}((S_\bullet)_f)_0,$$

hence $\pi^{-1}(\operatorname{Spec}((S_\bullet)_f)_0) = \operatorname{Spec}((R_\bullet)_{\pi^\sharp(f)})_0$. Since $S_\bullet \twoheadrightarrow R_\bullet$ is surjective, $((S_\bullet)_f)_0 \twoheadrightarrow ((R_\bullet)_{\pi^\sharp(f)})_0$ is surjective, and therefore $\pi : \operatorname{Proj} R_\bullet \hookrightarrow \operatorname{Proj} S_\bullet$ is closed embedding. $\square$

Proposition 16.7.2 will show the converse to Proposition 10.3.1: every closed embedding $X \hookrightarrow \operatorname{Proj} S_\bullet$ into a projective A-scheme ($S_\bullet$ is a finitely generated graded A-algebra) is of the form $\operatorname{Proj}(S_\bullet/I)$, where I is a homogeneous ideal, of "projective functions" vanishing on X.

Proposition 10.3.2

An injective linear map of k-vector spaces $V \hookrightarrow W$ induces a closed embedding $\mathbb{P}V \hookrightarrow \mathbb{P}W$. (This is another justification for the definition of $\mathbb{P}V$ in Definition 5.5.12 in terms of the dual of V.)

Proof By Definition 5.5.12, we have $\mathbb{P}V = \operatorname{Proj}(\operatorname{Sym}^\bullet V^\vee)$ and $\mathbb{P}W = \operatorname{Proj}(\operatorname{Sym}^\bullet W^\vee)$. By Proposition 10.3.1, it suffices to show that $\operatorname{Sym}^\bullet W^\vee \to \operatorname{Sym}^\bullet V^\vee$ is surjective. Say $\varphi : V \to W$ is injective k-linear map, it induces a dual k-linear map

$$\varphi^\vee : W^\vee \longrightarrow V^\vee, \qquad f \longmapsto f\circ\varphi,$$

we want to show that $\varphi^\vee$ is surjective. Let $V = \operatorname{Span}_k\{v_1,\dots,v_n\}$, since $\varphi : V \to W$ is injective, $\varphi(V)$ can be seen as subspace of W. Then $\varphi(V) = \operatorname{Span}_k\{\varphi(v_1),\dots,\varphi(v_n)\}$, by the Basis Extension Theorem, we have $W = \operatorname{Span}_k\{\varphi(v_1),\dots,\varphi(v_n),w_{n+1},\dots,w_m\}$. Let $V^\vee = \operatorname{Span}_k\{x_1,\dots,x_n\}$ where $x_i(v_j) = \delta_{ij}$. Let $g = \sum_{i=1}^n a_i x_i \in V^\vee$, define $f \in W^\vee$ as follow:

$$f(\varphi(v_i)) = a_i, \qquad f(w_j) = 0,$$

where $1 \le i \le n$ and $n+1 \le j \le m$. Hence $\varphi^\vee(f) = g$, it follows that $\varphi^\vee$ is surjective. Hence $\operatorname{Sym}^\bullet W^\vee \to \operatorname{Sym}^\bullet V^\vee$ is surjective, as we desired. $\square$

Definition 10.3.2 (Linear space)

*The closed subscheme defined in Proposition 10.3.2 is called a **linear space**. Once we know about dimension, we will call this closed subscheme a linear space of dimension $\dim V - 1 = \dim \mathbb{P}V$. More explicitly, a linear space of dimension n in $\mathbb{P}^N$ is any closed subscheme cut out by $N-n$ k-linearly independent homogeneous linear polynomials in $x_0,\dots,x_N$. A linear space of dimension 1 (resp., 2, n, $\dim\mathbb{P}V - 1$) is called a **line** (resp., **plane**, n**-plane**, **hyperplane**).*

Remark 10.13 If the linear map in the previous linear polynomials in Proposition 10.3.2 is not injective, then the hypothesis (8.8) in Proposition 8.4.1 fails.

✍ **Exercise 10.4** (**A special case of Bézout's Theorem.**) Suppose $X \subseteq \mathbb{P}_k^n$ is a degree d hypersurface cut out by $f = 0$, and l is a line not contained in X. A very special case of Bézout's Theorem implies that X and l meet with multiplicity d, "counted correctly". Make sense of this, by restricting the homogeneous degree d polynomial f to the line l, and using the fact that a degree d polynomial in $k[x]$ has d roots, counted properly. (If it makes you feel better, assume $k = \overline{k}$.)

Solution *Say $\mathbb{P}_k^n = \operatorname{Proj} k[x_0, \ldots, x_n]$. Let $X = V(f)$ where $f \in k[x_0, \ldots, x_n]$ and $\deg f = d$. Suppose line l is generated by two points $[a_0 : \cdots : a_n]$ and $[b_0 : \cdots : b_n]$, then*

$$l = \{[sa_0 + tb_0 : \cdots : sa_n + tb_n] : s \neq 0 \text{ or } t \neq 0\}.$$

Define $\varphi : \mathbb{P}^1 \to l \subseteq \mathbb{P}^n$ by setting

$$\varphi([s : t]) = [sa_0 + tb_0 : \cdots : sa_n + tb_n].$$

Hence we may restrict X to l by using $f \circ \varphi$:

$$g(s,t) := f(\varphi([s : t])) = f(sa_0 + tb_0 : \cdots : sa_n + tb_n).$$

Since f is homogeneous polynomial of degree d, g is homogeneous polynomial of degree d. Since $l \nsubseteq X$, g is not zero polynomial. In fact, geometric points of $X \cap l$ is one-to-one corresponding to points of $g(s,t) = 0$. In $k[s,t]$, we may factors $g(s,t)$ as

$$g(s,t) = c \prod_i (\alpha_i s - \beta_i t)^{m_i}$$

where $\sum_i m_i = d$. Each linear factor $\alpha_i s - \beta_i t$ cuts out a point $[\beta_i : \alpha_i] \in \mathbb{P}^1$, this point is corresponding to a point in $X \cap l$. The exponent m_i is exactly the intersection multiplicity of X and l at point $\varphi([\beta_i : \alpha_i])$. Also the sum of intersection multiplicities is d.

✍ **Exercise 10.5** Show that the map of graded rings $k[w,x,y,z] \to k[s,t]$ given by $(w,x,y,z) \mapsto (s^3, s^2t, st^2, t^3)$ induces a closed embedding $\mathbb{P}_k^1 \hookrightarrow \mathbb{P}_k^3$, which yields an isomorphism of $\mathbb{P}_k^1$ with the twisted cubic (defined in Exercise 10.3 — in fact, this will solve Exercise 10.3 (b)). Doing this in a hands-on way will set you up well for the general Veronese construction of §10.3.3.

Proof Let $S_\bullet = k[w,x,y,z]$ and $R_\bullet = k[s,t]$, say $\varphi^\sharp : S_\bullet \to R_\bullet$ which defined by

$$(w,x,y,z) \longmapsto (s^3, s^2t, st^2, t^3).$$

Consider $i^\sharp : R_{3\bullet} \hookrightarrow R_\bullet$, by the construction of $\varphi^\sharp$, we have $\varphi^\sharp = i^\sharp \circ \varphi'^\sharp$, where $\varphi'^\sharp : S_\bullet \to R_{3\bullet}$ defined by $\varphi^\sharp$. Clearly, $\varphi'^\sharp : S_\bullet \to R_{3\bullet}$ is a surjection of graded rings, by Proposition 10.3.1, it induces a closed embedding $\varphi' : \operatorname{Proj} R_{3\bullet} \hookrightarrow \operatorname{Proj} S_\bullet$. By Proposition 8.4.3, $R_{3\bullet} \hookrightarrow R_\bullet$ induces an isomorphism $\operatorname{Proj} R_\bullet \xrightarrow{\sim} \operatorname{Proj} R_{3\bullet}$. Hence we get a closed embedding $\mathbb{P}_k^1 \hookrightarrow \mathbb{P}_k^3$. □

10.3.2 A particularly nice case: when $S_\bullet$ is generated in degree 1

Suppose $S_\bullet$ is a finitely generated graded ring generated in degree 1 (Definition 5.5.4). Then S_1 is a finitely generated S_0-module, and the irrelevant ideal S_+ is generated in degree 1 (Proposition 5.5.3 (a)).

Proposition 10.3.3

If $S_\bullet$ is generated (as an A-algebra) in degree 1 by $n+1$ elements $x_0, \ldots, x_n$, then $\operatorname{Proj} S_\bullet$ may be described as a closed subscheme of $\mathbb{P}_A^n$ as follows. Consider $A^{\oplus(n+1)}$ as a free module with generators

$t_0, \dots, t_n$ associated to $x_0, \dots, x_n$. The surjection of

$$\begin{aligned} \operatorname{Sym}^\bullet(A^{\oplus(n+1)}) = A[t_0, t_1, \dots, t_n] &\longrightarrow\!\!\!\!\rightarrow S_\bullet \\ t_i &\longmapsto x_i \end{aligned}$$

implies $S_\bullet = A[t_0, t_1, \dots, t_n]/I$, where I is a homogeneous ideal.
In particular, $\operatorname{Proj} S_\bullet$ can always be interpreted as a closed subscheme of some $\mathbb{P}^n_A$ if $S_\bullet$ is finitely generated in degree 1. Then using Proposition 8.4.3 and Proposition 8.4.6, we can remove the hypothesis of generation in degree 1.

This is analogous to the fact that if R is a finitely generated A-algebra, then choosing n generators of R as an algebra is the same as describing $\operatorname{Spec} R$ as a closed subscheme of $\mathbb{A}^n_A$. In the affine case this is "choosing coordinates"; in the projective case this is "choosing projective coordinates".

Recall (Proposition 5.4.3) that if k is algebraically closed, then we can interpret the closed points of $\mathbb{P}^n$ as the lines through the origin in $(n+1)$-space. The following proposition states this more generally.

Proposition 10.3.4

Suppose $S_\bullet$ is a finitely generated graded ring over an algebraically closed field k, generated in degree 1 by $x_0, \dots, x_n$, inducing closed embeddings $\operatorname{Proj} S_\bullet \hookrightarrow \mathbb{P}^n$ and $\operatorname{Spec} S_\bullet \hookrightarrow \mathbb{A}^{n+1}$. Then there is a bijection between the closed points of $\operatorname{Proj} S_\bullet$ and the "lines through the origin" in $\operatorname{Spec} S_\bullet \subseteq \mathbb{A}^{n+1}$.

Proof Since $S_\bullet$ is a finitely generated graded ring over k, by Proposition 5.5.3, $S_\bullet$ is a finitely generated k-algebra, then we have surjective

$$\begin{aligned} k[t_0, t_1, \dots, t_n] &\longrightarrow\!\!\!\!\rightarrow S_\bullet \\ t_i &\longmapsto x_i. \end{aligned}$$

By the Fundamental Theorem of Isomorphism of rings, $S_\bullet \cong k[t_0, \dots, t_n]/I$, where I is a homogeneous ideal. Then $\operatorname{Proj} S_\bullet \cong \operatorname{Proj} k[t_0, \dots, t_n]/I$ and $\operatorname{Spec} S_\bullet \cong \operatorname{Spec} k[t_0, \dots, t_n]/I$. Pick closed point $[a_0 : \dots : a_n] \in \operatorname{Proj} S_\bullet$, since I is homogeneous ideal, $(\lambda a_0, \dots, \lambda a_n) \in \operatorname{Spec} S_\bullet$ for all $\lambda \in k$, this gives a line through the origin, say

$$L := \{(\lambda a_0, \dots, \lambda a_n) : \lambda \in k\} \subseteq \operatorname{Spec} S_\bullet.$$

Conversely, let L be a line through the origin in $\operatorname{Spec} S_\bullet$, we may assume this line through the point $(a_0, \dots, a_n) \in \operatorname{Spec} S_\bullet$, then

$$L = \{(\lambda a_0, \dots, \lambda a_n) : \lambda \in k\}.$$

This gives a projective coordinates $[a_0 : \dots : a_n]$ in $\operatorname{Proj} S_\bullet$. By our discussion, this correspondence is a bijection. □

A second proof that finite morphisms are closed

This interpretation of $\operatorname{Proj} S_\bullet$ as a closed subscheme of projective space (when it is generated in degree 1) yields the following second proof of the fact (shown in Corollary 9.3.2) that finite morphisms are closed.

Proposition

Finite morphisms are closed.

Proof Suppose $\pi : X \to Y$ is a finite morphism. The question is local on the target, so it suffices to consider the affine case $Y = \operatorname{Spec} B$, by the finiteness, we may assume that $X = \operatorname{Spec} A$ where A is finite B-algebra. It suffices to show that $\pi(X)$ is closed. Then by Proposition 9.3.12, X can be seen as a projective B-scheme, by Proposition 10.3.3, X can be described as a closed subscheme of some $\mathbb{P}^n_B$, and hence by the Fundamental Theorem of Elimination Theory 9.4.3

$$X \xrightarrow{\text{closed}} \mathbb{P}^n_B \xrightarrow{\text{closed}} \operatorname{Spec} B,$$

$\pi(X)$ is closed. $\square$

10.3.3 Important classical construction: The Veronese embedding

Suppose $S_\bullet = k[x,y]$, so $\operatorname{Proj} S_\bullet = \mathbb{P}^1_k$. Then $S_{2\bullet} = k[x^2, xy, y^2] \subseteq k[x,y]$ (see Definition 8.4.2). We identify this subring as follows.

Exercise 10.6 Let $u = x^2$, $v = xy$, $w = y^2$. Show that $S_{2\bullet} \cong k[u,v,w]/(uw - v^2)$, by mapping u, v, w to x^2, xy, y^2, respectively.

Proof Define $\varphi : k[u,v,w] \to k[x^2, xy, y^2]$ by setting

$$(u,v,w) \longmapsto (x^2, xy, y^2),$$

then φ is clearly a surjective. Consider $\operatorname{Ker} \varphi$, we claim that $\operatorname{Ker} \varphi = (uw - v^2)$. Note that $k[u,v,w]$ can be seen as $k[u,w][v]$, since $v^2 - uw$ is monic polynomial in $k[u,w][v]$, for all $f(u,v,w) \in k[u,w][v]$ we have

$$f(u,v,w) = (v^2 - uw)g(u,v,w) + r(u,v,w),$$

where $g(u,v,w), r(u,v,w) \in k[u,w][v]$ and $\deg_v r(u,v,w) < 2$. We may assume that

$$r(u,v,w) = a(u,w)v + b(u,w),$$

where $a(u,w), b(u,w) \in k[u,w]$. Now let $f(u,v,w) \in \operatorname{Ker} \varphi$, then

$$\varphi(f(u,v,w)) = r(x^2, xy, y^2) = a(x^2, y^2)xy + b(x^2, y^2) = 0.$$

Hence $a(x^2, y^2) = b(x^2, y^2) = 0$, since $k[u,w] \cong k[x^2, y^2]$ (given by $(u,w) \mapsto (x^2, y^2)$), $a(u,w) = b(u,w) = 0$, it follows that $\operatorname{Ker} \varphi = (uw - v^2)$. By the Fundamental Theorem of Isomorphism of rings, we have

$$S_{2\bullet} \cong k[u,v,w]/(uw - v^2).$$

$\square$

We have a graded ring generated by three element in degree 1. Thus we think of it as sitting "in" $\mathbb{P}^2$, via the construction of Proposition 10.3.3. This can be interpreted as "$\mathbb{P}^1$ as a conic in $\mathbb{P}^2$".

Lemma 10.3.1

$\mathbb{P}^1$ can be seen as a conic in $\mathbb{P}^2$.

Proof Say $\mathbb{P}^1 = \operatorname{Proj} k[x,y]$. By Proposition 8.4.3, we know that $\operatorname{Proj} S_\bullet \xrightarrow{\sim} \operatorname{Proj} S_{2\bullet}$, by Exercise 10.6, we have $\operatorname{Proj} S_{2\bullet} \cong \operatorname{Proj} k[u,v,w]/(uw - v^2)$, where $u = x^2, v = xy, w = y^2$. Hence

$$\mathbb{P}^1 \cong \operatorname{Proj} S_{2\bullet} \cong \operatorname{Proj} k[u,v,w]/(uw - v^2) \subseteq \mathbb{P}^2.$$

$\square$

Thus if k is algebraically closed of characteristic not 2, using the fact that we can diagonalize quadratics

(Exercise 6.7), the conics in $\mathbb{P}^2$, up to change of coordinates, come in only a few flavors: sums of three (and no fewer) squares (e.g., our conic of Exercise 10.6), sums of two (and no fewer) squares (e.g., $y^2 - x^2 = 0$, the union of two lines), a single (nonzero) square (e.g., $x^2 = 0$, which looks set-theoretically like a line, and is non-reduced), and 0 (perhaps not a conic at all). Thus we have proved: any plane conic (over an algebraically closed field of characteristic not 2) that can be written as the sum of three non-zero squares (and no fewer) is isomorphic to $\mathbb{P}^1$. (See Exercise 8.9 for a closely related fact.)

We now soup up this example.

Proposition 10.3.5

Take $S_\bullet = k[x, y]$. Then $\operatorname{Proj} S_{d\bullet}$ is given by the equations that

$$\begin{bmatrix} t_0 & t_1 & \cdots & t_{d-1} \\ t_1 & t_2 & \cdots & t_d \end{bmatrix}$$

is rank 1 (i.e., that all the 2×2 minors vanish). This is called the ***degree d rational normal curve*** *"in" $\mathbb{P}^d$.*

Remark 10.14 We did the twisted cubic case $d = 3$ in Exercise 10.3 and Exercise 10.5.

Proof In fact, $S_{d\bullet} = k[x^d, x^{d-1}y, \ldots, y^d]$. Consider the ring homomorphism

$$\begin{aligned} \varphi : \quad k[t_0, t_1, \ldots, t_d] &\twoheadrightarrow S_{d\bullet} = k[x^d, x^{d-1}y, \ldots, y^d] \\ (t_0, t_1, \ldots, t_d) &\longmapsto (x^d, x^{d-1}y, \ldots, y^d) \end{aligned}$$

it is clearly surjective. By Exercise 4.18, the kernel of φ is given by the equations (all 2×2 minors vanish) that

$$\begin{bmatrix} t_0 & t_1 & \cdots & t_{d-1} \\ t_1 & t_2 & \cdots & t_d \end{bmatrix}$$

is rank 1. $\square$

More generally:

Definition 10.3.3 (d-ulpe Veronese embedding)

If $S_\bullet = k[x_0, \ldots, x_n]$, then $v_d : \operatorname{Proj} S_{d\bullet} \to \mathbb{P}^{N-1}$ (where N is the dimension of the vector space of homogeneous degree d polynomials in $x_0, \ldots, x_n$) is called the ***d-uple embedding or the (d-uple) Veronese embedding****.*

Proposition 10.3.6

(d-uple) Veronese embedding $v_d : \operatorname{Proj} S_{d\bullet} \to \mathbb{P}^{N-1}$ is closed embedding, where $S_\bullet = k[x_0, \ldots, x_n]$.

Proof By Proposition 8.4.4, $\operatorname{Proj} S_{d\bullet}$ is generated in degree 1, by Proposition 10.3.3, $v_d : \operatorname{Proj} S_{d\bullet} \hookrightarrow \mathbb{P}^{N-1}$ is closed embedding. $\square$

Proposition 10.3.7

In Definition 10.3.3, $N = \binom{n+d}{d}$.

Proof By Definition 10.3.3, N is the dimension of the vector space of homogeneous degree d polynomials in $x_0, \ldots, x_n$. Hence N is the number of non-negative integer solution to equation

$$a_0 + a_1 + \cdots + a_n = d.$$

This is a classical "stars and bars" combinatorial problem. Imagine placing d identical stars (representing units of degree) and n identical bars (separating $n+1$ variables). Total positions are $d+n$. Choose n positions for bars, we get $N = \binom{n+d}{d}$. □

✍ **Exercise 10.7** (**Unimportant exercise.**) Find six linearly independent quadratic equations vanishing on the **Veronese surface** Proj $S_{2\bullet}$ where $S_\bullet = k[x_0, x_1, x_2]$, which sits naturally in $\mathbb{P}^5$. (You needn't show that these equations generate all the equations cutting out the Veronese surface, although this is in fact true.)

Proof The ring homomorphism $k[t_0, \ldots, t_5] \to S_{2d}$ is given by

$$(t_0, \ldots, t_5) \longmapsto (x_0^2, x_0x_1, x_0x_2, x_1^2, x_1x_2, x_2^2).$$

Then six linearly independent quadratic equations are all 2×2 minors of matrix

$$\begin{bmatrix} t_0 & t_1 & t_2 \\ t_1 & t_3 & t_4 \\ t_2 & t_4 & t_5 \end{bmatrix}.$$

□

10.3.4 Rulings on the quadric surface

We return to rulings on the quadric surface, which first appeared in the Example 5.7 .

✍ **Exercise 10.8** (**Useful geometric exercise: The rulings on the quadric surface $wz = xy$.**) This exercise is about the lines on the quadric surface X given by $wz - xy = 0$ in $\mathbb{P}_k^3$ (where the projective coordinates on $\mathbb{P}_k^3$ are ordered w, x, y, z). This construction arises all over the place in nature.

(a) Suppose a_0 and b_0 are given elements of k, not both zero. Make sense of the statement: as $[c : d]$ varies in $\mathbb{P}^1$, $[a_0c : b_0c : a_0d : b_0d]$ is a line in the quadric surface. (This describes "a family of lines parametrized by $\mathbb{P}^1$", although we can't yet make this precise.) Find another family of lines. These are the two **rulings** of the smooth quadric surface.

(b) Show that through every k-valued point of the quadric surface X, these passes one line from each ruling.

(c) Show there are no other lines. (There are many ways of proceeding. At risk of predisposing you to one approach, here is a germ of an idea. Suppose L is a line on the quadric surface, and $[1 : x : y : z]$ and $[1 : x' : y' : z']$ are distinct points on it. Because they are both on the quadric, $z = xy$ and $z' = x'y'$. Because all of L is on the quadric, $(1+t)(z+tz') - (x+tx')(y+ty') = 0$ for all t. After some algebraic manipulation, this translates into $(x - x')(y - y') = 0$. How can this be made watertight? Another possible approach uses Bézout's Theorem, in the form of Exercise 10.4.)

Proof

(a) Fix points $[a_0 : b_0] \in \mathbb{P}^1$, let $[c, d]$ varies in $\mathbb{P}^1$, then we get a line through points $[a_0 : b_0 : 0 : 0]$ and $[0 : 0 : a_0 : b_0]$, say

$$L_{[a_0:b_0]} := \{[a_0c : b_0c : a_0d : b_0d] \in \mathbb{P}^3 : [c : d] \in \mathbb{P}^1\}.$$

Clearly, line $L_{[a_0:b_0]}$ is contained in Proj $k[w, x, y, z]/(wz - xy)$. Hence collection

$$\mathscr{L} = \{L_{[a_0,b_0]} : [a_0 : b_0] \in \mathbb{P}^1\}$$

is a family of lines which contained in quadric surface.

Now, we fix points $[c_0 : d_0] \in \mathbb{P}^1$, let $[a : b]$ varies in $\mathbb{P}^1$, then we get a line through points $[c_0 : 0 : d_0 : 0]$ and $[0 : c_0 : 0 : d_0]$, say

$$M_{[c_0:d_0]} := \{[c_0a : c_0b : d_0a : d_0b] \in \mathbb{P}^3 : [a : b] \in \mathbb{P}^1\}.$$

$M_{[c_0:d_0]}$ is contained in quadric surface. And collection

$$\mathscr{M} = \{M_{[c_0:d_0]} : [c_0 : b_0] \in \mathbb{P}^1\}$$

is another family of lines which contained in quadric surface.

(b) Pick k-valued points $p = [p_0 : p_1 : p_2 : p_3]$ of X, where $p_0p_3 = p_1p_2$.

Small remark

What is the k-valued point of X?

k-valued points of the quadric surface $X = \operatorname{Proj} k[w, x, y, z]/(wz - xy)$ is defined by $X(k) = \operatorname{Mor}(\operatorname{Spec} k, X)$ (Definition 8.3.5). Say $p = \pi([(0)])$, by Proposition 8.3.1, π induces an inclusion $\kappa(p) \hookrightarrow k$, note that $k \subseteq \kappa(p)$, $\kappa(p) = k$, it follows that p is closed point of X.

We may assume $[p_0 : p_1] \in \mathbb{P}^1$, let $[a_0 : b_0] = [p_0 : p_1]$. We need to find $[c : d] \in \mathbb{P}^1$ such that

$$[p_0 : p_1 : p_2 : p_3] = [a_0c : b_0c : a_0d : b_0d].$$

Note that $c = \frac{p_0}{p_0} = \frac{p_1}{p_1} = 1$ and $p_0p_3 = p_1p_2$, take

$$d = \frac{p_2}{p_0} = \frac{p_3}{p_1}.$$

It follows that $p \in L_{[p_0:p_1]}$. Since $[p_0 : p_1] \in \mathbb{P}^1$, we may assume $p_0 \neq 0$, then $[p_0 : p_2] \in \mathbb{P}^1$. Let $[c_0 : d_0] = [p_0 : p_2]$. We want to find $[a : b] \in \mathbb{P}^1$ such that

$$[p_0 : p_1 : p_2 : p_3] = [c_0a : c_0b : d_0a : d_0b].$$

Note that $a = \frac{p_0}{p_0} = \frac{p_2}{p_2} = 1$ and $p_0p_3 = p_1p_2$, take

$$b = \frac{p_1}{p_0} = \frac{p_3}{p_2}.$$

Then $p \in M_{[p_0:p_2]}$. By uniqueness of $[c : d]$ and $[a : b]$, $L_{[p_0:p_1]}$ and $M_{[p_0:p_2]}$ are the unique lines passing through p in each ruling family.

(c) Let L be a line on the quadric surface, and $P = [1 : x : y : z]$ and $Q = [1 : x' : y' : z']$ are distinct points on it where $z = xy$ and $z' = x'y'$, then

$$L = \{[s + t : xs + x't : ys + y't : zs + z't] : [s : t] \in \mathbb{P}^1\}.$$

Let $s = 1$, since $L \subseteq X$, we have equation

$$(1 + t)(z + z't) - (x + x't)(y + y't) = 0,$$

by some calculations,

$$(x' - x)(y' - y)t = 0, \quad \forall t \in k,$$

hence $(x' - x)(y' - y) = 0$. It follows that $x' = x$ or $y' = y$. Suppose that $x' = x$, by part (b), $P, Q \in L_{[1,x]}$, i.e., $L = L_{[1,x]}$.

$\square$

Remark 10.15 Hence by Exercise 6.7, if we are working over an algebraically closed field of characteristic not 2, we have shown that all rank 4 quadric surfaces have two rulings of lines. (In Chapter 11, we will recognize this quadric as $\mathbb{P}^1 \times \mathbb{P}^1$.)

Remark 10.16 The quadric surface sketched in figure 10.3 is actually $a^2 + b^2 = c^2 + 1$. **Why this is an affine chart for S?**

This is a linear algebra problem. We need to transform the quadratic form $wz - xy$ into its canonical form.

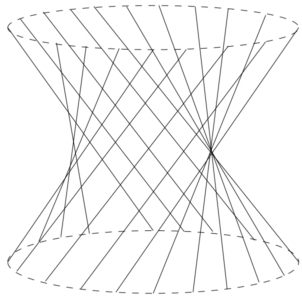


Figure 10.3: One of the two rulings on the quadric surface $S := V(wz - xy) \subseteq \mathbb{P}^3$. One ruling contains the line $V(w, x)$ and the other contains the line $V(w, y)$.

Let

$$\begin{cases} a' = & \frac{w+z}{2} \\ b' = & \frac{w-z}{2} \\ c' = & \frac{x+y}{2} \\ d' = & \frac{x-y}{2} \end{cases},$$

then we have $a'^2 + b'^2 - c'^2 - d'^2 = 0$. On the chart $D_+(d')$, we have $a^2 + b^2 = c^2 + 1$ where $a = \frac{a'}{d'}$, $b = \frac{b'}{d'}$ and $c = \frac{c'}{d'}$.

Remark 10.17 Side remark: Rulings on quadric hypersurfaces. The exercise of these two rulings is the first chapter of a number of important and beautiful stories. The second chapter is often the following. If k is an algebraically closed field, then a maximal rank (Exercise 6.7) quadric hypersurface X of dimension m contains no linear spaces of dimension greater than $m/2$. (We will see in Chapter 14 that the maximal rank quadric hypersurfaces are the "smooth" quadrics.) If $m = 2a + 1$, then X contains an irreducible $\binom{a+2}{2}$-dimensional family of a-planes. If $m = 2a$, then X contains two irreducible $\binom{a+2}{2}$-dimensional families of a-planes, and furthermore two a-planes Λ and Λ' are in the same family if and only if $\dim(\Lambda \cap \Lambda') \equiv a \pmod 2$. These families of linear spaces are also called **rulings**.

10.3.5 Affine and projective cones (Figure 10.5)

Definition 10.3.4 (Affine cone)

*If $S_\bullet$ is a finitely generated graded ring, then the **affine cone** of* $\operatorname{Proj} S_\bullet$ *is* $\operatorname{Spec} S_\bullet$.

Figure 10.4: The cone $x^2 + y^2 = z^2$ in $\mathbb{A}^3$.

Remark 10.18 Caution: this terminology is not ideal, as this construction depends on $S_\bullet$, not just on $\operatorname{Proj} S_\bullet$. As motivation, consider the graded ring $S_\bullet = \mathbb{C}[x, y, z]/(z^2 - x^2 - y^2)$. Figure 10.4 is a sketch of $\operatorname{Spec} S_\bullet$.

(Here we draw the "real picture" of $z^2 = x^2 + y^2$ in $\mathbb{R}^3$.) It is a cone in the traditional sense; the origin $(0,0,0)$ is the "cone point".

This gives a useful way of picture Proj (even over arbitrary rings, not just $\mathbb{C}$). Intuitively, you could imagine that if you discarded the origin, you would get something that would project onto $\operatorname{Proj} S_\bullet$. The following proposition makes that precise.

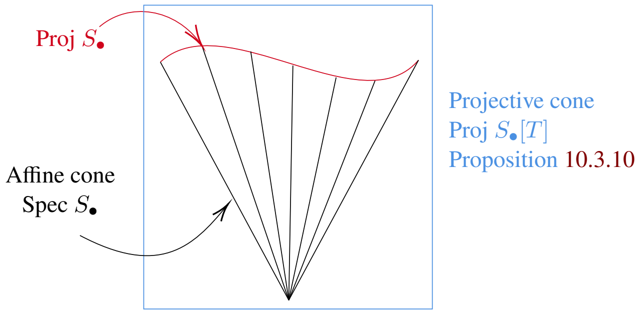


Figure 10.5: The affine and projective cones of $\operatorname{Proj} S_\bullet$

Proposition 10.3.8

If $\operatorname{Proj} S_\bullet$ *is a projective scheme over a field* k*, then there is a natural morphism* $\operatorname{Spec} S_\bullet \setminus V(S_+) \to \operatorname{Proj} S_\bullet$.

Proof In fact, $\operatorname{Spec} S_\bullet \setminus V(S_+) = \bigcup\limits_{f \in S_+} \operatorname{Spec}(S_\bullet)_f$ and $\operatorname{Proj} S_\bullet = \bigcup\limits_{f \in S_+} \operatorname{Spec}((S_\bullet)_f)_0$. From inclusion $\iota_f^\sharp : ((S_\bullet)_f)_0 \hookrightarrow (S_\bullet)_f$, by Proposition 8.3.6, we have a morphism of affine schemes

$$\iota_f : \operatorname{Spec}(S_\bullet)_f \longrightarrow \operatorname{Spec}((S_\bullet)_f)_0.$$

We now check ι_f agree on the overlaps, i.e., $\iota_f|_{D(f)\cap D(g)} = \iota_g|_{D(f)\cap D(g)}$ where $f, g \in S_+$. It is clear, since they are both given by inclusion

$$((S_\bullet)_{fg})_0 \longhookrightarrow (S_\bullet)_{fg},$$

by Morphism of locally ringed spaces glue 8.3.2, they glue together, then we get a natural morphism

$$\operatorname{Spec} S_\bullet \setminus V(S_+) \longrightarrow \operatorname{Proj} S_\bullet.$$

□

Remark 10.19 Why $V(S_+)$ is a single point, and should reasonably be called the origin?

Since $S_0 = k$ is a field, $S_+ = \bigoplus\limits_{i \geq 1} S_i$ is maximal ideal of $S_\bullet$, $V(S_+)$ is a single point. This point is called "the origin" because it is the location where all elements of positive degree, i.e., elements in S_+, vanish.

This readily generalizes to the following proposition, which again motivates the terminology "irrelevant".

Proposition 10.3.9

If $S_\bullet$ *is a finitely generated graded ring over a base ring* A*, then there is a natural morphism*

$$\operatorname{Spec} S_\bullet \setminus V(S_+) \to \operatorname{Proj} S_\bullet.$$

Proof Same as Proposition 10.3.8. □

In fact, it can be made precise that $\operatorname{Proj} S_\bullet$ is the quotient (by the multiplicative group of scalars) of the affine cone minus the origin.

Definition 10.3.5 (Projective cone)

*The **projective cone** of* $\operatorname{Proj} S_\bullet$ *is* $\operatorname{Proj} S_\bullet[T]$, *where* T *is a new variable of degree* 1. *The projective cone is sometimes called the **projective completion** of* $\operatorname{Spec} S_\bullet$. *(Note: this depends on* $S_\bullet$, *not just on* $\operatorname{Proj} S_\bullet$.)

Example 10.4 The cone corresponding to the conic $\operatorname{Proj} k[x,y,z]/(z^2-x^2-y^2)$ is $\operatorname{Proj} k[x,y,z,T]/(z^2-x^2-y^2)$.

Proposition 10.3.10 (Less important)

The "projective cone" $\operatorname{Proj} S_\bullet[T]$ *of* $\operatorname{Proj} S_\bullet$ *has a closed subscheme isomorphic to* $\operatorname{Proj} S_\bullet$ *(informally, corresponding to* $T=0$*), whose complement (the distinguished open set* $D_+(T) \hookrightarrow \operatorname{Proj} S_\bullet[T]$*) is isomorphic to the affine cone* $\operatorname{Spec} S_\bullet$.

Proof Clearly, $S_\bullet \cong S_\bullet[T]/(T)$, hence $\operatorname{Proj} S_\bullet \cong \operatorname{Proj} S_\bullet[T]/(T)$. Note that we have surjective $S_\bullet[T] \twoheadrightarrow S_\bullet[T]/(T)$, by Proposition 10.3.1, it induces closed embedding $\operatorname{Proj} S_\bullet \hookrightarrow \operatorname{Proj} S_\bullet[T]$. In fact, distinguished open set $D_+(T) \hookrightarrow \operatorname{Proj} S_\bullet[T]$ is $D_+(T) \cong \operatorname{Spec}((S_\bullet[T])_T)_0$. Define $\varphi : S_\bullet \to (S_\bullet[T]_T)_0$ by setting

$$f_d \longmapsto \frac{f_d}{T^d},$$

where $f_d \in S_d$. It is easy to check that φ gives an isomorphism of rings. Hence $\operatorname{Spec} S_\bullet \cong D_+(T)$, as we desired. □

This construction can be usefully pictured as the affine cone union some points "at infinity", and the points at infinity form the Proj. (The reader may wish to start with Figure 10.4, and try to visualize the conic curve "at infinity", and then compare this visualization to Figure 5.10.)

We have thus completely described the algebraic analog of the classical picture of §5.5.1.

Definition 10.3.6 (Weighted projective space)

If we put a nonstandard weighting on the variables of $k[x_0,\ldots,x_n]$ *— say we give* $\deg x_i = d_i$ *— then* $\operatorname{Proj} k[x_0,\ldots,x_n]$ *is called **weighted projective space*** $\mathbb{P}(d_0,d_1,\ldots,d_n)$.

Exercise 10.9

(a) Show that $\mathbb{P}(m,n)$ is isomorphic to $\mathbb{P}^1$.

(b) Show that $\mathbb{P}(1,1,2) \cong \operatorname{Proj} k[u,v,w,z]/(uw-z^2)$.

Proof

(a) Suppose $R_\bullet = k[t_0,t_1]$ where $\deg t_i = 1$. Let $x \in R_m$ and $y \in R_n$, say $S_\bullet = k[x,y]$, then $\mathbb{P}(m,n) = \operatorname{Proj} S_\bullet$. Denote $\gcd(m,n) = d$, then $m = dm'$ and $n = dn'$ where $\gcd(m',n') = 1$. Consider $S_{m'n'\bullet}$, note that $x^{n'}, y^{m'} \in S_{m'n'}$, we want to show that $S_{m'n'\bullet} \cong k[x^{n'}, y^{m'}]$. It suffices to show that any monomial in $S_{m'n'\bullet}$ is generated by $x^{n'}$ and $y^{m'}$. Suppose $x^a y^b \in S_{km'n'}$, then we have $am+bn = km'n'$. Since $\gcd(m',n') = 1$, $m'|b$ and $n'|a$, say $a = n'a'$ and $b = m'b'$, and therefore

$$x^a y^b = (x^{n'})^{a'} (y^{m'})^{b'}.$$

It follows that $x^a y^b$ is generated by $x^{n'}$ and $y^{m'}$. Hence $S_{m'n'\bullet} \cong k[x^{n'}, y^{m'}]$. Since $k[x^{n'}, y^{m'}]$ is finitely generated in degree 1, we have $k[x^{n'}, y^{m'}] \cong k[t_0,t_1]$, by Proposition 8.4.3, we have

$$\mathbb{P}(m,n) = \operatorname{Proj} S_\bullet \cong \operatorname{Proj} S_{m'n'\bullet} \cong \operatorname{Proj} R_\bullet = \mathbb{P}^1.$$

(b) Suppose $\mathbb{P}(1,1,2) = \operatorname{Proj} S_\bullet := \operatorname{Proj} k[r,s,t]$ where $\deg r = \deg s = 1$ and $\deg t = 2$. Consider Veronese subring $S_{2\bullet} = k[r^2, t, s^2, rs]$, then $S_{2\bullet}$ is finitely generated in degree 1. Define homomorphism of graded rings $\varphi : k[u,v,w,z] \to S_{2\bullet}$ by setting

$$(u,v,w,z) \longmapsto (r^2, t, s^2, rs).$$

Clearly, φ is surjective and $\operatorname{Ker}\varphi = (z^2 - uw)$. By the Fundamental Theorem of Isomorphism, we have $S_{2\bullet} \cong k[u,v,w,z]/(vw - z^2)$, and therefore

$$\mathbb{P}(1,1,2) = \operatorname{Proj} S_\bullet \cong \operatorname{Proj} S_{2\bullet} \cong \operatorname{Proj} k[u,v,w,z]/(uw - z^2).$$

$\square$

Remark 10.20 $\mathbb{P}(1,1,2)$ is a projective cone over a conic curve $\operatorname{Proj} k[u,v,w]/(uw - v^2)$. Over a field of characteristic not 2, it is isomorphic to the traditional cone $x^2 + y^2 = z^2$ in $\mathbb{P}^3$, see Figure 10.4.

Proposition 10.3.11

$\mathbb{P}(1,1,n)$ is isomorphic to the projective cone over the degree n rational normal curve.

Proof Let $\mathbb{P}(1,1,n) = \operatorname{Proj} S_\bullet := \operatorname{Proj} k[x,y,z]$ where $\deg x = \deg y = 1$ and $\deg z = n$. Consider Veronese subring $S_{n\bullet} = k[x^n, x^{n-1}y, \ldots, y^n, z]$, $S_{n\bullet}$ is finitely genenrated in degree 1, by Proposition 8.4.3, we have

$$\mathbb{P}(1,1,n) = \operatorname{Proj} S_\bullet \cong \operatorname{Proj} S_{n\bullet}.$$

Note that $S_{n\bullet} = k[x^n, x^{n-1}y, \ldots, y^n][z]$, say $R_\bullet = k[x,y]$, then $S_{n\bullet} = R_{n\bullet}[z]$. Hence $\mathbb{P}(1,1,n) \cong \operatorname{Proj} R_{n\bullet}[z]$, it follows that $\mathbb{P}(1,1,n)$ is isomorphic to the projective cone over the degree n rational normal curve (Proposition 10.3.5). $\square$

10.4 The (closed sub)scheme-theoretic image

We now define a series of notions that are all of the form "the smallest closed subscheme such that something or other is true". One example will be th enotion of scheme-theoretic closure of a locally closed embedding, which will allow us to interpret locally closed embeddings in three equivalent ways (open subscheme intersect closed subscheme; open subscheme of closed subscheme; and closed subscheme of open subscheme — cf. Proposition 10.2.3).

10.4.1 Scheme-theoretic image

We start with the notion of scheme-theoretic image. Set-theoretic images are badly behaved in general (§9.4.1), and even with reasonable hypotheses such as those in Chevalley's Theorem 9.4.1, things can be confusing. For example, there is no reasonable way to impose a scheme structure on the image of $\mathbb{A}^2_k \to \mathbb{A}^2_k$ given by $(x,y) \mapsto (x, xy)$ (the set-theoretic image of this map is $D(u) \cup \{(0,0)\}$). It will be useful (e.g., Proposition 10.4.3) to define a notion of a closed subscheme of the target that "best approximates" the image. This will incorporate the notion that the image of something with non-reduced structure ("fuzz") can also have non-reduced structure. As usual, we will need to impose reasonable hypotheses to make this notion behave well (see Theorem 10.4.1 and Corollary 10.4.1).

Definition 10.4.1 (Scheme-theoretic image)

Suppose $i : Z \hookrightarrow Y$ is a closed subscheme, giving an exact sequence

$$0 \longrightarrow \mathscr{I}_{Z/Y} \longrightarrow \mathscr{O}_Y \longrightarrow i_*\mathscr{O}_Z \longrightarrow 0.$$

*We say that **the image of** $\pi : X \to Y$ **lies in** Z if the composition $\mathscr{I}_{Z/Y} \to \mathscr{O}_Y \to \pi_* \mathscr{O}_X$ is zero.*
*If the image of π lies in some subschemes Z_j (as j runs over some index set), it clearly lies in their intersection (cf. Proposition 10.1.8 on intersections of closed subschemes). We then define the **scheme-theoretic image of** π, a closed subscheme of Y, as the "smallest closed subscheme containing the image", i.e., the intersection of all closed subschemes containing the image. In particular (and in our first example), if Y is affine, the scheme-theoretic image is cut out by functions on Y that are 0 when pulled back to X.*

Remark 10.21 Why we define "the image of $\pi : X \to Y$ lies in Z" this way?

Informally, locally, functions vanishing on Z pull back to the zero function on X.

Note *Other reasonable names for scheme-theoretic image are the **closed subscheme-theoretic image**, or **image closed subscheme**. They have the advantage that they remind the reader that the scheme-theoretic image is a closed subscheme, but we won't use these phrases.*

Example 10.5 Consider $\pi : \operatorname{Spec} k[\varepsilon]/(\varepsilon^2) \to \operatorname{Spec} k[x] = \mathbb{A}^1_k$ given by $x \mapsto \varepsilon$. Then the scheme-theoretic image of π is given by $\operatorname{Spec} k[x]/(x^2)$ (the polynomials pulling back to 0 are precisely (x^2)). Thus the image of the fuzzy point still has some fuzz.

Example 10.6 Consider $\pi : \operatorname{Spec} k[\varepsilon]/(\varepsilon^2) \to \operatorname{Spec} k[x] = \mathbb{A}^1_k$ given by $x \mapsto 0$. Then the scheme-theoretic image is given by $\operatorname{Spec} k[x]/(x)$: the image is reduced (Definition 4.2.3). In this picture, the fuzz is "collapsed" by π.

Example 10.7 Consider $\pi : \operatorname{Spec} k[t, t^{-1}] = \mathbb{A} - \{0\} \to \mathbb{A}^1 = \operatorname{Spec} k[u]$ given by $u \mapsto t$. Any function $g(u)$ which pulls back to 0 as a function of t must be the zero-function. Thus the scheme-theoretic image is everything. the set-theoretic image, on the other hand, is the distinguished open set $\mathbb{A}^1 \setminus \{0\}$. Thus in not -too-pathological cases, the underlying set of the scheme-theoretic image is not the set-theoretic image. But the situation isn't terrible: the underlying set of the scheme-theoretic image must be closed, and indeed it is the closure of the set-theoretic image. We might imagine that in reasonable cases this will be true, and in even nicer cases, the underlying set of the scheme-theoretic image will be the set-theoretic image. We will later see that this is indeed the case.

But sadly pathologies can sometimes happen in, well, pathological situations.

Example 10.8 (Figure 10.6) Let $X = \coprod \operatorname{Spec} k[\varepsilon_n]/((\varepsilon_n)^n)$ (a scheme which appeared in the hint to Example 6.4) and $Y = \operatorname{Spec} k[x]$, and define $X \to Y$ by $x \mapsto \varepsilon_n$ on the n-th component of X (in fact, $\Gamma(X, \mathscr{O}_X) = \prod_n k[\varepsilon_n]/((\varepsilon_n)^n)$). If a function $g(x)$ on Y pulls back to 0 on X, then its Taylor extension is 0 to order n (by examining the pullback to the n-th component of X) for all n, so $g(x)$ must be 0. (This argument will be vastly generalized in Chapter 14). Thus the scheme-theoretic image is $V(0)$ on Y, i.e., Y itself, while the set-theoretic image is easily seen to be just the origin (the closed point 0). (This morphism implicitly arises in Caution/Example 10.10 .)

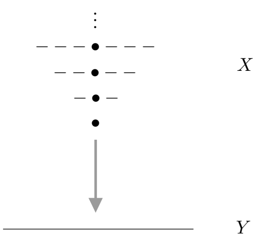


Figure 10.6: Yuck (Example 10.8)

10.4.2 Criteria for computing scheme-theoretic images affine-locally

Example 10.8 clearly is weird though, and we can show that in "reasonable circumstances" such pathology doesn't occur.

In the special case when the target Y is affine (pay attention to how this argument works in Example 10.5 through 10.7), say $Y = \operatorname{Spec} B$, almost by definition the scheme-theoretic image of $\pi : X \to Y$ is cut out by the ideal $I \subseteq B$ of functions on Y which pull back to zero on X.

It would be great to use this to compute the scheme-theoretic image affine by affine by affine (affine-locally). On the affine open set $\operatorname{Spec} B \subseteq Y$, define the ideal $I(B) \subseteq B$ of functions which pull back to 0 on X. Formally, $I(B) := \operatorname{Ker}(B \to \Gamma(\operatorname{Spec} B, \pi_* \mathscr{O}_X))$. If for each such B, and each $g \in B$, the map $\varphi : I(B)_g \to I(B_g)$ is an isomorphism, then we will have defined the scheme-theoretic image as a closed subscheme (see Proposition 10.1.6, i.e., show that the sheaf $\mathscr{I}(\operatorname{Spec} B) := I(B)$ form a quasi-coherent sheaf, Theorem 7.2.2). Injectivity of φ is straightforward: let $\varphi(\frac{r}{g^n}) = \frac{r}{g^n} = 0$ where $r \in I(B)$, then there exists some m such that $g^m r = 0$ in B, since $r \in I(B)$, we have $\frac{r}{g^n} = 0$ in $I(B)_g$, it follows that φ is injective.

So the question is the surjectivity of φ, which translates to: given a function $\frac{r}{g^n}$ on $D(g)$ that pulls back to zero on $\pi^{-1}(D(g))$, i.e., $\pi^\sharp(\frac{r}{g^n}) = \frac{\pi^\sharp(r)}{\pi^\sharp(g)^n} = 0$ in $\pi^{-1}(D(g))$, we want to show that $\frac{r}{g^n}$ in $I(B)_g$, i.e., show that $rg^m \in I(B)$ for some m, i.e., show that $\pi^\sharp(rg^m) = 0$ on $\pi^{-1}(\operatorname{Spec} B)$.

(i) We first show that the answer is yes if $\pi^{-1}(\operatorname{Spec} B)$ is reduced:
In this case we can take advantage of Proposition 6.2.1, that a function on a reduced scheme is zero if it has value zero at every point. We may take $m = 1$ (as $\pi^\sharp(r)$ vanishes on $\pi^{-1}(D(g))$ and g vanishes on $V(g)$, so $\pi^\sharp(rg)$ vanishes on $\pi^{-1}(\operatorname{Spec} B) = \pi^{-1}(D(g)) \cup \pi^{-1}(V(g))$.)

(ii) The answer is also yes if $\pi^{-1}(\operatorname{Spec} B)$ is affine, say $\operatorname{Spec} A$:
if $r' = \pi^\sharp(r)$ and $g' = \pi^\sharp(g)$ in A, then if $r'/g'^n = 0$ on $D(g')$, then there is an m such that $r'(g')^m = 0$ (as the statement $r'/g'^n = 0$ in $D(g')$ means precisely this fact — the functions on $D(g')$ are $A_{g'}$).

(iii) More generally, the answer is yes if $\pi^{-1}(\operatorname{Spec} B)$ is quasi-compact:
Cover $\pi^{-1}(\operatorname{Spec} B)$ with finitely many affine open sets U_i. For each one there will be some m_i such that $\pi|_{U_i}^\sharp(rg^{m_i}) = 0$ on U_i. Then let $m = \max\{m_i\}$, we have $\pi^\sharp(rg^m) = 0$ on $\pi^{-1}(\operatorname{Spec} B)$. (We see again that quasi-compactness is our friend!)

In conclusion, we have proved the following (subtle) theorem.

Theorem 10.4.1

Suppose $\pi : X \to Y$ is a morphism of schemes. If X is reduced or π is quasi-compact, then the scheme-theoretic image of π may be computed affine-locally: on $\operatorname{Spec} A \subseteq Y$, the scheme-theoretic image of X is cut out by the functions (elements of A) that pull back to the function 0 (on $\pi^{-1}(\operatorname{Spec} A)$).

Corollary 10.4.1

Suppose $\pi : X \to Y$ is a morphismof schemes. If X is reduced or π is quasi-compact, then the closure of the set-theoretic image of π is the underlying set of the scheme-theoretic image.

(Example 10.8 above shows how we need these hypotheses.)

Remark 10.22 In particular, if the set-theoretic image is closed (e.g., if π is finite (Corollary 9.3.2) or projective), the set-theoretic image is the underlying set of the scheme-theoretic image, as promised in Example 10.7

Proof of Corollary 10.4.1

Proof By definition of scheme-theoretic image (Definition 10.4.1), we know that the set-theoretic image is in the underlying set of the scheme-theoretic image, i.e., $\pi(X) \subseteq \pi(X)_{\mathbf{Sch}}$ where $\pi(X)_{\mathbf{Sch}}$ means scheme-theoretic image of X. The underlying set of the scheme-theoretic image is closed, so the closure of the set-theoretic image is contained in the underlying set of the scheme-theoretic image, i.e., $\mathrm{cl}_Y(\pi(X)) \subseteq \pi(X)_{\mathbf{Sch}}$. On the other hand, if U is the complement of the closure of the set-theoretic image, $\pi^{-1}(U) = \varnothing$. Under these hypotheses, the scheme theoretic image can be computed locally (Theorem 10.4.1), so the scheme-theoretic image is the empty set on U, i.e., $\pi(X)_{\mathbf{Sch}} \cap U = \varnothing$. Hence $\pi(X)_{\mathbf{Sch}} \subseteq \mathrm{cl}_Y(\pi(X))$, we done. $\square$

We conclude with a few stray remarks.

Proposition 10.4.1

If X is reduced, then the scheme-theoretic image of $\pi : X \to Y$ is also reduced.

Proof Since the scheme-theoretic image of π may be computed affine-locally (Theorem 10.4.1), we may assume that $Y = \operatorname{Spec} B$, then the scheme-theoretic image of π is $\mathrm{cl}_Y(\pi(X)) = \operatorname{Spec} B/I$, where $I = \operatorname{Ker}(B \to \Gamma(\operatorname{Spec} B, \pi_* \mathscr{O}_X))$. Since X is reduced, by Proposition 6.2.3, $\Gamma(\operatorname{Spec} B, \pi_* \mathscr{O}_X)$ is reduced. By the Fundamental Theorem of Isomorphism, we have

$$B/I \cong \operatorname{Im}(B \to \Gamma(\operatorname{Spec} B, \pi_* \mathscr{O}_X)) \subseteq \Gamma(\operatorname{Spec} B, \pi_* \mathscr{O}_X),$$

since $\Gamma(\operatorname{Spec} B, \pi_* \mathscr{O}_X)$ is reduced, B/I is reduced, by Proposition 6.2.2, the scheme-theoretic image $\mathrm{cl}_Y(\pi(X))$ is reduced. $\square$

More generally, you might expect there to be non unnecessary non-reduced structure on the image not forced by non-reduced structure on the source. The following makes this precise in the locally Noetherian case (when we can talk about associated points.)

Proposition 10.4.2 (Unimportant)

If $\pi : X \to Y$ is a quasi-compact morphism of locally Noetherian schemes, then the associated points of the image subscheme are a subset of the image of the associated points of X.

Proof Since the scheme-theoretic image of π may be computed affine-locally (Theorem 10.4.1), we may assueme that $Y = \operatorname{Spec} B$ where B is Noetherian ring. Since π is quasi-compact, we know that $X = \pi^{-1}(\operatorname{Spec} B)$ is quasi-compact, since X is locally Noetherian, X is Noetherian. Hence we may assume that $X = \operatorname{Spec} A$ where A is Noetherian, and therefore we reduced to the case that $\pi : \operatorname{Spec} A \to \operatorname{Spec} B$ is an affine morphism of Noetherian scheme.

By Theorem 10.4.1, the scheme-theoretic image of π is $\mathrm{cl}_Y(\pi(X)) = \operatorname{Spec} B/I$ where $I = \operatorname{Ker}(B \to A)$. We want to show that $\operatorname{Ass}_{B/I}(B/I) \subseteq \pi(\operatorname{Ass}_A(A))$. Let $\mathfrak{p} = \operatorname{Ann}_{B/I}(b) \in \operatorname{Ass}_{B/I}(B/I)$. By the Fundamental Theorem of Isomorphism, we have $B/I \cong \pi^\sharp(B)$, then $\pi^\sharp : B \to A$ factors as

$$\begin{tikzcd} B \arrow[rr, "\pi^\sharp"] \arrow[dr, two heads, "\mathrm{pr}"'] & & A \\ & B/I. \arrow[ur, hook, "i"'] & \end{tikzcd}$$

We may look B/I as a subring of A, then i is inclusion. Hence $\mathfrak{p} = \operatorname{Ann}_A(b) \cap B/I$. Inclusion $i : B/I \hookrightarrow A$ induces ring map $B/I \to A/\operatorname{Ann}_A(b)$, consider $\operatorname{Ker}(B/I \to A/\operatorname{Ann}_A(b))$, note that $\operatorname{Ker}(B/I \to A/\operatorname{Ann}_A(b)) = B/I \cap \operatorname{Ann}_A(b) = \mathfrak{p}$, then we get an injective

$$(B/I)/\mathfrak{p} \longhookrightarrow A/\operatorname{Ann}_A(b),$$

where $(B/I)/\mathfrak{p}$ is integral domain. We claim that there is a prime ideal of $A/\operatorname{Ann}_A(b)$ such that $(0) =$

$\mathfrak{q} \cap (B/I)/\mathfrak{p}$. Let
$$\Sigma = \{\text{ideal } J \subseteq A/\operatorname{Ann}_A(b) : J \cap (B/I)/\mathfrak{p} = (0)\}$$
ordered by the relation $\subseteq$. Since A is Noetherian, $A/\operatorname{Ann}_A(b)$ is Noetherian, hence Σ has maximal element, say $\mathfrak{q}$. We want to show that $\mathfrak{q}$ is prime. Suppose $ab \in \mathfrak{q}$ with $0 \neq a \notin \mathfrak{q}$ and $0 \neq b \notin \mathfrak{q}$, Consider $\mathfrak{q}+(a)$ and $\mathfrak{q}+(b)$, since $\mathfrak{q} \subsetneq \mathfrak{q}+(a), \mathfrak{q}+(b)$, $(\mathfrak{q}+(a)) \cap (B/I)/\mathfrak{p} \neq 0$ and $(\mathfrak{q}+(b)) \cap (B/I)/\mathfrak{p} \neq 0$. Let $r \in (\mathfrak{q}+(a)) \cap (B/I)/\mathfrak{p}$ and $r' \in (\mathfrak{q}+(b)) \cap (B/I)/\mathfrak{p}$. Since $ab \in \mathfrak{q}$, we have
$$rr' \in (\mathfrak{q}+(a))(\mathfrak{q}+(b)) \cap (B/I)/\mathfrak{p} = \mathfrak{q} \cap (B/I)/\mathfrak{p} = 0.$$
Since $(B/I)/\mathfrak{p}$ is integral domain, we have $r = 0$ or $r' = 0$, a contradiction. Hence $\mathfrak{q}$ is prime. We now lift $\mathfrak{q}$ to A, also say $\mathfrak{q}$, then $\mathfrak{q} \supseteq \operatorname{Ann}_A(b)$ and $\mathfrak{p} = \mathfrak{q} \cap B/I$. Since $\mathfrak{q} \in \operatorname{Spec} A$, $\mathfrak{q} \in \operatorname{Ass}_A(\operatorname{Ann}_A(b))$, by Proposition 7.6.7, we have $\operatorname{Ass}_A(\operatorname{Ann}_A(b)) \subseteq \operatorname{Ass}_A(A)$, hence $\mathfrak{q} \in \operatorname{Ass}_A(A)$.

Note that
$$\pi([\mathfrak{q}]) = (\pi^\sharp)^{-1}(\mathfrak{q}) = \operatorname{pr}^{-1}(\mathfrak{q} \cap B/I) = \operatorname{pr}^{-1}(\mathfrak{p}),$$
also note that point $[\operatorname{pr}^{-1}(\mathfrak{p})]$ can be seen as point $[\mathfrak{p}]$ in $\operatorname{Spec} B/I \hookrightarrow \operatorname{Spec} B$, we proved that $\operatorname{Ass}_{B/I}(B/I) \subseteq \pi(\operatorname{Ass}_A(A))$. □

Remark 10.23 Intuitively, this proposition shows that a quasi-compact morphism ”preserves” associated points: the geometric features of the scheme-theoretic image (described by its associated points) are entirely determined by those of the source scheme.

Example 10.9 (Example of Proposition 10.4.2 not work) Consider $\pi : \coprod_{a\in\mathbb{C}} \operatorname{Spec} \mathbb{C}[t]/(t-a) \to \operatorname{Spec} \mathbb{C}[t]$, by Definition 7.6.5, $\operatorname{Ass}(X) = \{((0), a) : a \in \mathbb{C}\}$. Since $\coprod_{a\in\mathbb{C}} \operatorname{Spec} \mathbb{C}[t]/(t-a)$ is reduced, by Corollary 10.4.1, the scheme-theoretic image of π is $\operatorname{Spec} \mathbb{C}[t]/(0) = \operatorname{Spec} \mathbb{C}[t]$, and $\operatorname{Ass}(\operatorname{Spec} \mathbb{C}[t]) = \{[(0)]\}$. Now consider $\pi(\operatorname{Ass}(X))$, in fact, $\pi(\operatorname{Ass}(X)) = \{[(t-a)] \in \operatorname{Spec} \mathbb{C}[t] : a \in \mathbb{C}\}$. But $\operatorname{Ass}(\operatorname{Spec} \mathbb{C}[t]) \nsubseteq \pi(\operatorname{Ass}(X))$. Hence this example shows that can go wrong if you give up quasi-compactness — note that reducedness of the source doesn't help.

10.4.3 Scheme-theoretic closure of a locally closed subscheme

Definition 10.4.2 (Scheme-theoretic closure of locally closed embedding)

*We define the **scheme-theoretic closure** of a locally closed embedding $\pi : X \hookrightarrow Y$ as the scheme-theoretic image of π.*

Remark 10.24 A shorter phrase for this is **schematic closure**, although this more elegant nomenclature has not caught on.

Proposition 10.4.3

If a locally closed embedding $V \hookrightarrow X$ is quasi-compact (e.g., if V is Noetherian, Proposition 9.3.2 (a)), or if V is reduced, then (iii) implies (i) and (ii) in Proposition 10.2.3. Thus in this fortunate situation, a locally closed embedding can be thought of in three different ways, whichever is convenient.

Proof Let Z be the scheme-theoretic image of $\pi : V \hookrightarrow X$, by Corollary 10.4.1, we have $Z = \operatorname{cl}_X(\pi(V))$. Since $\pi : V \hookrightarrow X$ is locally closed embedding, π factors as
$$V \xhookrightarrow[\text{closed}]{\tau} U \xhookrightarrow[\text{open}]{i} X,$$
where $i : U \hookrightarrow X$ is inclusion. Since $V \hookrightarrow U$ is closed embedding, we have $\pi(V) = U \cap \operatorname{cl}_X(\pi(V))$, which

implies that $\pi(V)$ is open in $\mathrm{cl}_X(\pi(V))$. Since $Z = \mathrm{cl}_X(\pi(V))$ is a scheme, $\pi(V)$ is an open subscheme of Z, hence $V \hookrightarrow Z = \mathrm{cl}_X(\pi(V))$ is open embedding, i.e., (iii) implies (ii). By Proposition 10.2.3, (i) $\Leftrightarrow$ (ii), we done. $\square$

Proposition 10.4.4 (Unimportant, but useful for intuition)

If $\pi : X \hookrightarrow Y$ is a locally closed embedding into a locally Noetherian scheme (so X is also locally Noetherian), then the associated points of the scheme-theoretic closure are (naturally in bijection with) the associated points of X.

Informally, the only non-reduced structure on the scheme-theoretic closure is that "forced by" non-reduced structure on X.

Proof We first show that locally closed embedding $\pi : X \hookrightarrow Y$ is quasi-compact. We may assume that π is inclusion. Since $\pi : X \hookrightarrow Y$ is locally closed embedding, π factors as

$$X \xhookrightarrow{\tau} U \xhookrightarrow{i} Y,$$

where τ is closed embedding and i is open embedding. Let $\operatorname{Spec} B \subseteq Y$ be an affine open subscheme of Y, then $\pi^{-1}(\operatorname{Spec} B) = X \cap \operatorname{Spec} B$. Since Y is locally Noetherian scheme, $\operatorname{Spec} B$ is Noetherian, by Proposition 4.6.19, $X \cap \operatorname{Spec} B$ is quasi-compact. Hence $\pi : X \hookrightarrow Y$ is quasi-compact. By Proposition 10.4.2, we have $\mathrm{Ass}(\mathrm{cl}_Y(\pi(X))) \subseteq \pi(\mathrm{Ass}(X))$, i.e., $\mathrm{Ass}(\mathrm{cl}_Y(X)) \supseteq \mathrm{Ass}(X)$. Since $X \subseteq \mathrm{cl}_Y(X)$, by Proposition 7.6.7, $\mathrm{Ass}(X) \subseteq \mathrm{Ass}(\mathrm{cl}_Y(X))$. Hence $\mathrm{Ass}(\mathrm{cl}_Y(X)) = \mathrm{Ass}(X)$, we done. $\square$

10.4.4 The (reduced) subscheme structure on a closed subset

Suppose X^{set} is a closed subset of a scheme Y. Then we can define a canonical scheme structure X on X^{set} that is reduced. We could describe it as being cut out by those functions whose values are zero at all the points of X^{set}. On the affine open set $\operatorname{Spec} B$ of Y, if the set X^{set} corresponds to the radical ideal $I = I(X^{\text{set}} \cap \operatorname{Spec} B)$ (recall the $I(\cdot)$ function from §4.7), the scheme X corresponds to $\operatorname{Spec} B/I$. We check that this behaves well with respect to any distinguished inclusion $\operatorname{Spec} B_f \hookrightarrow \operatorname{Spec} B$:

Lemma 10.4.1

Let X^{set} is a closed subset of a scheme Y, $\operatorname{Spec} B \subseteq Y$ be an affine open subset of Y, let $I = I(X^{set} \cap \operatorname{Spec} B)$, then for any distinguished open subset $\operatorname{Spec} B_f \hookrightarrow \operatorname{Spec} B$, there is an isomorphism

$$\operatorname{Spec}(B/I)_f \cong \operatorname{Spec} B_f/I',$$

where $I' = I(X^{set} \cap \operatorname{Spec} B_f)$.

Proof It suffices to show that $(B/I)_f \cong B_f/I'$. Since localization commutes with quotient, we have $(B/I)_f \cong B_f/IB_f$, hence it suffices to show that $I' = IB_f$.

Let $\frac{g}{f^n} \in IB_f$, where $g \in I$. Since $g \in I$, for all $p \in X^{\text{set}} \cap \operatorname{Spec} B$, $g(p) = 0$, in particular, $g(p) = 0$ for all $p \in X^{\text{set}} \cap \operatorname{Spec} B_f$. Since $p \in \operatorname{Spec} B_f$, we have $f \notin \operatorname{Spec} B_f$, i.e., $f(p) \neq 0$, hence $\frac{g}{f^n} \in I'$, i.e., $IB_f \subseteq I'$.

Conversely, let $\frac{g}{f^n} \in I'$. To show $\frac{g}{f^n} \in IB_f$, it suffices to show that $f^m g \in I = \bigcap_{[\mathfrak{p}] \in X^{\text{set}} \cap \operatorname{Spec} B} \mathfrak{p}$ for some m. If $[\mathfrak{p}] \in X^{\text{set}} \cap \operatorname{Spec} B_f$, then $g([\mathfrak{p}]) = 0$ and $f([\mathfrak{p}]) = 0$, hence $f^m g([\mathfrak{p}]) = 0$ for all m. If $[\mathfrak{p}] \in X^{\text{set}} \cap (\operatorname{Spec} B - \operatorname{Spec} B_f)$, then $f([\mathfrak{p}]) = 0$. Pick $m = 1$, we have $fg(p) = 0$ for all $p \in X^{\text{set}} \cap \operatorname{Spec} B$, hence $I' \subseteq IB_f$.

By above discussion, we know that $I' = IB_f$, and therefore $\operatorname{Spec}(B/I)_f \cong \operatorname{Spec} B_f/I'$. $\square$

By Lemma 10.4.1 and Theorem 7.2.2, $\mathscr{I}(\operatorname{Spec} B) = I(X^{\text{set}} \cap \operatorname{Spec} B)$ gives an ideal sheaf. We give the first definition of the reduced subscheme structure on the closed subset X^{set}.

Definition 10.4.3 (The reduced subscheme structure on the closed subset)

Let X^{set} be a closed subset of a scheme Y. We define ***the reduced subscheme structure*** *on X^{set} be the closed subscheme which produced by ideal sheaf*

$$\mathscr{I}(\operatorname{Spec} B) = I(X^{set} \cap \operatorname{Spec} B)$$

for any $\operatorname{Spec} B \subseteq Y$.

We also consider this construction as an example of a scheme-theoretic image in the following crazy way: let W be the scheme that is a disjoint union of all the points of X^{set}, where the point corresponding to p in X^{set} is Spec of the residue field of $\mathscr{O}_{Y,p}$, i.e., $W = \coprod_{p\in X^{\text{set}}} \operatorname{Spec} \kappa(p)$. Let $\rho : W \to Y$ be the "canonical" map sending "p to p", and giving an isomorphism on residue fields. Then the scheme structure on X is the scheme-theoretic image of ρ. Then we give the second definition of the reduced subscheme structure on the closed subset X^{set}.

Definition 10.4.4 (The reduced subscheme structure on the closed subset)

Let X^{set} be a closed subset of a scheme Y. Suppose $W = \coprod_{p\in X^{set}} \operatorname{Spec} \kappa(p)$, define $\rho : W \to Y$ by setting $p \mapsto p$. We define ***the reduced subscheme structure*** *X on X^{set} by setting X is the scheme-theoretic image of ρ.*

Finally, we give the third definition:

Definition 10.4.5 (The reduced subscheme structure on the closed subset)

We define ***the reduced subscheme structure*** *X on X^{set} by setting X is the smallest closed subscheme whose underlying set contains X^{set}.*

We next show that Definition 10.4.3, 10.4.4, and 10.4.5 are equivalent.

Proposition 10.4.5

Definition 10.4.3, 10.4.4, and 10.4.5 are equivalent.

Proof

(1) Definition 10.4.3 implies Definition 10.4.4.
It suffices to show that X is the scheme-theoretic image of ρ. Suppose $Y = \bigcup_i \operatorname{Spec} B_i$, then the scheme-theoretic image of ρ is $\bigcup_i \operatorname{Spec} B_i/I_i$, where $I_i = \operatorname{Ker}(B_i \to \Gamma(\operatorname{Spec} B_i, \rho_*\mathscr{O}_W))$. It suffices to show that $I_i = I(X^{\text{set}} \cap \operatorname{Spec} B_i)$. Consider $\Gamma(\operatorname{Spec} B_i, \rho_*\mathscr{O}_W)$, note that $\rho^{-1}(\operatorname{Spec} B_i) = \coprod_{p\in \operatorname{Spec} B_i \cap X^{\text{set}}} \operatorname{Spec} \kappa(p)$, we have

$$\Gamma(\operatorname{Spec} B_i, \rho_*\mathscr{O}_W) = \mathscr{O}_W(\rho^{-1}(\operatorname{Spec} B_i)) = \prod_{p\in \operatorname{Spec} B\cap X^{\text{set}}} \kappa(p).$$

We next consider $\operatorname{Ker}(B_i \to \prod_{p\in \operatorname{Spec} B\cap X^{\text{set}}} \kappa(p))$, $B_i \to \prod_{p\in \operatorname{Spec} B\cap X^{\text{set}}} \kappa(p)$ is given by $f \mapsto (f(p))_p$, hence $\operatorname{Ker}(B_i \to \prod_{p\in \operatorname{Spec} B\cap X^{\text{set}}} \kappa(p)) = I(X^{\text{set}} \cap \operatorname{Spec} B_i)$.

(2) Definition 10.4.4 implies Definition 10.4.5.
Note that $W = \coprod_{p\in X^{\text{set}}} \operatorname{Spec} \kappa(p)$ is reduced, by Corollary 10.4.1, the closure of the set-theoretic image

of ρ is the underlying set of the scheme-theoretic image, i.e., $X = X^{\text{set}}$. It is clearly the smallest closed subscheme whose underlying set contains X^{set}.

(3) Definition 10.4.5 implies Definition 10.4.3.

Let X be the smallest closed subscheme whose underlying set contains X^{set}. For any $\operatorname{Spec} B$, functions on $X \cap \operatorname{Spec} B$ must vanish on $X^{\text{set}} \cap \operatorname{Spec} B$, hence X must produced by ideal sheaf $\mathscr{I}(\operatorname{Spec} B) = I(X^{\text{set}} \cap \operatorname{Spec} B)$, we done.

□

Definition 10.4.6 (The reduced subscheme structure on the closed subset)

The construction as Definition 10.4.3 or 10.4.4 or 10.4.5 is called the (induced) reduced subscheme structure on the closed subset X^{set}.

Proposition 10.4.6

Let Y be a scheme. Let $X^{set} \subseteq Y$ be a closed subset. Then there exists unique reduced subscheme structure X on the closed subset X^{set} with the following properties:

(a) the underlying topological space of X is equal to X^{set};

(b) X is reduced.

Proof We use the first definition, i.e., Definition 10.4.3. For any affine open $\operatorname{Spec} B \subseteq Y$, we have $X \cap \operatorname{Spec} B = \operatorname{Spec} B/I$, where $I = I(X^{\text{set}} \cap \operatorname{Spec} B)$. We want to show that B/I is reduced. Consider $\mathfrak{N}(B/I)$, let $f + I \in \mathfrak{N}(B/I)$, then $f^n \in I$ for some n. Since I is radical ideal, we have $f \in I$. Hence $\mathfrak{N}(B/I) = 0$, B/I is reduced. By Proposition 6.2.2, $X \cap \operatorname{Spec} B$ is reduced. Hence X is reduced. By the construction of X, the underlying closed subset of X must be X^{set}.

We next show the uniqueness. Let $X' \subseteq Y$ be a second reduced subscheme structure on the closed subset X^{set}. By the definition, for any affine open $\operatorname{Spec} B \subseteq Y$, we have $X' \cap \operatorname{Spec} B = \operatorname{Spec} B/I'$ for some ideal $I' \subseteq B$. By Proposition 6.2.2, B/I' is reduced, and therefore I' is radical. Since $V(I') = X^{\text{set}} \cap \operatorname{Spec} B = V(I)$, by Hilbert's Nullstellensatz 4.7.1, $I = I'$. Hence X and X' are defined by the same ideal sheaf and hence are equal. □

10.4.5 Reduced version of a scheme

Consider the case where X^{set} is all of Y, we obtain a reduced closed subscheme which is called the reduction of Y:

Definition 10.4.7

*Let Y be a scheme, by Proposition 10.4.6, we obtain a reduced closed subscheme $Y^{\text{red}} \to Y$, called the **reduction** of Y. On the affine open subset $\operatorname{Spec} B \hookrightarrow Y$, $Y^{\text{red}} \hookrightarrow Y$ corresponds to (the quotient by) the nilradical $\mathfrak{N}(B) \subseteq B$.*

Remark 10.25 The reduction of a scheme is the "reduced version" of the scheme, and informally corresponds to "shearing off the fuzz".

Remark 10.26 **An alternative equivalent definition of the reduction:** on the affine open subset $\operatorname{Spec} B \hookrightarrow Y$, the reduction of Y corresponds to the nilradical $\mathfrak{N}(B) \subseteq B$. By Proposition 7.2.2, for any $f \in B$, $\mathfrak{N}(B)_f = \mathfrak{N}(B_f)$, by Theorem 7.2.2, $\mathscr{I}(\operatorname{Spec} B) := \mathfrak{N}(B)$ form a quasi-coherent sheaf. By Proposition 10.1.6 this defines a closed subscheme, which we call the reduction.

Example 10.10 (Caution/Example) It is not true that for *every open subset* $U \subseteq Y$, $\Gamma(U, \mathscr{O}_{Y^{\mathrm{red}}})$ is $\Gamma(U, \mathscr{O}_Y)$ modulo its nilpotents. For example, on $Y = \coprod \operatorname{Spec} k[x]/(x^n)$, $\Gamma(Y, \mathscr{O}_Y) = \prod k[x]/(x^n)$, function $(x, \dots)$ is not nilpotent. $Y^{\mathrm{red}} = \coprod \operatorname{Spec}(k[x]/(x^n))/\mathfrak{N}(k[x]/(x^n)) = \coprod \operatorname{Spec} k$, hence $(x, \dots)$ is 0 on Y^{red}, as it is "locally nilpotent". This may remind you of Example 10.8 .

10.5 Slicing by effective Cartier divisors, regular sequences and regular embeddings

We now introduce regular embeddings, an important class of locally closed embeddings, an important class of locally closed embeddings. Locally closed embeddings of regular schemes in regular schemes are one important example of regular embeddings (Chapter 13). Effective Cartier divisors, basically the codimension 1 case, will turn out to be repeatedly useful as well, with repeated "slicing by effective Cartier divisors" providing the inductive step in many arguments. (When we say **slicing** in many future contexts, we will mean slicing by effective Cartier divisors.)

10.5.1 Locally principle closed subschemes, and effective Cartier divisors

Definition 10.5.1 (Locally principal closed subscheme)

*A closed subscheme is **locally principal** if on each open set in a small enough open cover it is cut out by a single equation (i.e., by a principal ideal, hence the terminology). More specifically, a locally principal closed subscheme is a closed subscheme $\pi : X \hookrightarrow Y$ for which there is an open cover $\{U_i\}$ of Y for which for each i, $\pi^{-1}(U_i) \hookrightarrow U_i$ the closed subscheme $V(s_i)$ of U_i for some $s_i \in \Gamma(U_i, \mathscr{O}_X)$. (This open cover can clearly chosen to be by affine open subsets.)*

Example 10.11 Hypersurfaces in $\mathbb{P}^n_k$ (Definition 10.3.1) are locally principal: each homogeneous polynomial in $n+1$ variables defines a locally principal closed subscheme of $\mathbb{P}^n_A$. (Warnings: unlike "lcoal principality", "principality" is not an affine-local condition, see Chapter 20! Also, the example of a projective hypersurface, §10.3.1, shows that a locally principal closed subscheme need not be cut out by a globally-defined function. Finally, we unfortunately use the phrase "locally principal" in a different way in Chapter 16.)

Definition 10.5.2 (Effective Cartier divisor)

*If the ideal sheaf is locally generated near every point by a function that is not a zero-divisor, we call the closed subscheme an **effective Cartier divisor**. More precisely: if $\pi : X \to Y$ is a closed embedding, and there is a cover Y by affine open subsets $\operatorname{Spec} A_i \subseteq Y$, and there exists non-zerodivisors $t_i \in A_i$ with $V(t_i) = X|_{\operatorname{Spec} A_i}$ (scheme-theoretically — i.e., the ideal sheaf of X over $\operatorname{Spec} A_i$ is generated by t_i), then we say that X is **effective Cartier divisor on** Y. (We will not explain the origin of the phrase, as it is not relevant for this point of view.) We will often use the evocative phrase **slicing by an effective Cartier divisor** when we mean "restrict to an effective Cartier divisor".*

The notion of an effective Cartier divisor is central. The notion of a locally principal closed subscheme is much less so.

Proposition 10.5.1

Suppose $t \in A$ is a non-zerodivisor. Then t is a non-zerodivisor in $A_{\mathfrak{p}}$ for each prime $\mathfrak{p}$.

Proof Suppose t is a zerodivisor in $A_{\mathfrak{p}}$, then there exists $0 \neq \frac{a}{b} \in A_{\mathfrak{p}}$ such that $\frac{a}{b} \cdot \frac{t}{1} = \frac{at}{b} = 0$, hence there exists $s \in A - \mathfrak{p}$ such that $sat = 0$ in A. Since t is a non-zerodivisor in A, $sa = 0$. Hence $\frac{a}{b} = 0$ in $A_{\mathfrak{p}}$, a contradiction! □

Remark 10.27 Caution. If D is an effective Cartier divisor on an affine scheme $\operatorname{Spec} A$, it is not necessarily true that $D = V(t)$ for some $t \in A$ (see Chapter 16 — Chapter 20 gives a different flavor of example). In other words, the condition of a closed subscheme being an effective Cartier divisor can be verified on an affine cover, but cannot be checked on an arbitrary affine cover — it is not an affine-local condition in this obvious a way.

Proposition 10.5.2

Suppose X is a locally Noetherian scheme.

(a) *If $t \in \Gamma(X, \mathscr{O}_X)$ is a function on X, then t (or more precisely the closed subscheme $V(t)$) is an effective Cartier divisor if and only if it doesn't vanish on any associated point of X.*

(b) *A locally principal closed subscheme of X is an effective Cartier divisor if and only if it doesn't contain any associated points of X.*

Proof

(a) If $V(t)$ is an effective Cartier divisor. Suppose $X = \bigcup_i \operatorname{Spec} A_i$ where each A_i is Noetherian. Since $V(t)$ is an effective Cartier divisor, $V(t) = \bigcup_i \operatorname{Spec} A_i/(t_i)$ where $t_i = t|_{\operatorname{Spec} A_i}$ is non-zerodivisor in A_i. Suppose exists $p \in \operatorname{Ass}(X)$ such that $t(p) = 0$. Since X is covered by $\{\operatorname{Spec} A_i\}$, $p \in \operatorname{Spec} A_i$ for some i. Since $p \in \operatorname{Ass}(X)$, $\mathfrak{m}_p = pA_{ip} = \operatorname{Ann}(f)$ for some $f \in A_{ip}$ and the image of t in $A_{ip}/\operatorname{Ann}(f)$ is 0. Hence $\frac{t_i}{1} \in \operatorname{Ann}(f)$, this contradicts to the fact that t_i is non-zerodivisor.
Conversely, if t doesn't vanish on any associated point of X. Suppose $X = \bigcup_i \operatorname{Spec} A_i$ where A_i is Noetherian, then the vanish scheme can be written as $V(t) = \bigcup_i \operatorname{Spec} A_i/(t_i)$ where $t_i = t|_{\operatorname{Spec} A_i}$. Since t doesn't vanish on any associated point of X, t_i does't vanish on any associated point of $\operatorname{Spec} A_i$, note that A_i is Noetherian, by Proposition 7.6.11, t_i is not zerodivisor on B_i for all i. Hence vanish scheme $V(t)$ is effective Cartier divisor.

(b) Let locally principal closed subscheme D be an effective Cartier divisor on X. Suppose $X = \bigcup_i \operatorname{Spec} A_i$ where each A_i is Noetherian, then D can be written as $D = \bigcup_i \operatorname{Spec} A_i/(t_i)$ where t_i is non-zerodivisor in A_i, hence t_i doesn't vanishes at any associated points of A_i by Proposition 7.6.11, i.e., $\frac{t_i}{1} \notin p(A_i)_p$ for all $p \in \operatorname{Ass}(A_i)$, i.e., $t_i \notin p$ for all $p \in \operatorname{Ass}(A_i)$. Hence $\operatorname{Spec} A_i/(t_i)$ doesn't contain any any associated points of $\operatorname{Spec} A_i$, and therefore D doesn't contain any any associated points of X.
Conversely, if locally principal closed subscheme D doesn't contain any associated points of X. By Definition 10.5.1, there is an open cover $\{\operatorname{Spec} A_i \hookrightarrow X\}_i$ such that $D = \bigcup_i \operatorname{Spec} A_i/(s_i)$ where $s_i \in A_i$. By the hypothesis, each $\operatorname{Spec} A_i/(s_i)$ doesn't contain any associated points of $\operatorname{Spec} A_i$, hence s_i doesn't vanishes at any associated points of A_i, note that A_i is Noetherian, by Proposition 7.6.11, s_i is non-zerodivisor on A_i for all i. It follows that D is effective Cartier divisor on X.

□

Proposition 10.5.3

Suppose $V(t) = V(t') \hookrightarrow \operatorname{Spec} A$ is an effective Cartier divisor, with t and t' non-zerodivisors in A. Then t is an invertible function times t'.

Proof Since $V(t) = V(t')$ as closed subscheme, they must have the same ideal sheaf, i.e., $(t) = (t')$. Hence we may assume

$$t = at', \quad t' = bt, \tag{10.3}$$

where $a, b \in A$. From (10.3), we have

$$t = at' = abt \quad \Longrightarrow \quad (1 - ab)t = 0.$$

Since t is non-zerodivisor in A, $1 - ab = 0$, i.e., a, b are invertible functions, as we desired. ☐

The idea of effective Cartier divisors lead us to the notion of regular sequences. (We will close the loop in Exercise 9.5.H, where we will interpret effective Cartier divisors on any reasonable scheme as regular embedding of codimension 1.)

10.5.2 Regular sequences

The notion of "regular sequence", like so much else, is due to Serre. The definition of regular sequence is the algebraic version of the following geometric idea: locally, we take an effective Cartier divisor (a non-zerodivior); then an effective Cartier divisor on that; then an effective Cartier divisor on that; and so on, a finite number of times. A little care is necessary; for example, we might want this to be independent of the order of the equations imposed, and this is true only when we say this in the right way.

We make the definition of regular sequence for a ring A, and more generally for an A-module M.

Definition 10.5.3 (M-regular sequence)

*If M is an A-module, a sequence $x_1, \ldots, x_r \in A$ is called an M-**regular sequence** (or a **regular sequence for** M) if the following two conditions are satisfied.*

(i) For each i, x_i is not a zerodivisor for $M/(x_1, \ldots, x_{i-1})M$. (The case $i = 1$ should be interpreted as: "x_1 is not a zerodivisor of M.")

(ii) We have a proper inclusion $(x_1, \ldots, x_r)M \subsetneq M$.

In the case most relevant to us, when $M = A$, this should be seen as a reasonable approximation of a "complete intersection", and indeed we will use this as the definition (10.5.3).

*We say r is the **length of the regular sequence** $x_1, \ldots, x_r \in A$. An A-regular sequence is just called a **regular sequence**.*

Proposition 10.5.4

If M is an A-module, then an M-regular sequence continues to satisfy condition (i) of the Definition 10.5.3 of regular sequence upon any localization. More precisely, let $x_1, \ldots, x_r \in A$ be an M-regular sequence, then $\frac{x_i}{1}$ is not a zerodivisor for $S^{-1}M \big/ \left(\frac{x_1}{1}, \ldots, \frac{x_{i-1}}{1}\right) S^{-1}M$ for each i.

Proof Suppose exits $\frac{x_i}{1}$ is a zerodivisor for $S^{-1}M \big/ \left(\frac{x_1}{1}, \ldots, \frac{x_{i-1}}{1}\right) S^{-1}M$, then there exists $\frac{f}{g} \in S^{-1}M$ where $g \in S$ such that

$$\frac{x_i}{1} \cdot \frac{f}{g} \in \left(\frac{x_1}{1}, \ldots, \frac{x_{i-1}}{1}\right) S^{-1}M.$$

We may assume that

$$\frac{x_i}{1} \cdot \frac{f}{g} = \sum_{j=1}^{i-1} \frac{m_j x_j}{s_j}, \tag{10.4}$$

pick $s = \prod_{k=1}^{i-1} s_k$, we may write (10.4) as

$$\frac{x_i f}{g} = \frac{m}{s}$$

where $m \in (x_1, \ldots, x_{i-1})M$. Then there exists $t \in S$ such that $t(x_i fs - mg) = 0$, i.e., $x_i stf = tgm$. Note that $tgm \in (x_1, \ldots, x_{i-1})M$, $x_i stf \in (x_1, \ldots, x_{i-1})M$. But this is impossible, since x_i is not zerodivisor for $M/(x_1, \ldots, x_{i-1})M$. Hence M-regular sequence continues to satisfy condition (i) of the Definition 10.5.3 of regular sequence upon any localization. □

Remark 10.28 Once you know what flatness is, you will see that your argument shows that condition (i) is preserved by any flat base change.

Lemma 10.5.1

Suppose $x_1, x_2, \ldots, x_n$ is an M-regular sequence. If $x_1 m \in (x_2, \ldots, x_k)M$, then $m \in (x_2, \ldots, x_k)M$.

Proof We prove this by induction on n. If $n = 1$ and $x_1 m \in (0)$, since x_1 is not a zerodivisor of M, $m = 0$. Suppose consequence holds for the case $n-1$, i.e., if $x_1, \ldots, x_{n-1}$ is M-regular sequence and $x_1 m \in (x_2, \ldots, x_{n-1})M$ then $m \in (x_2, \ldots, x_{n-1})M$. We now check the case for n, suppose $x_1, x_2, \ldots, x_n$ is M-regular sequence and $x_1 m \in (x_2, \ldots, x_n)M$, want to show that $m \in (x_2, \ldots, x_n)M$. Define homomorphism $\times x_1 : M/(x_2, \ldots, x_n)M \to M/(x_2, \ldots, x_n)M$ by setting $a \mapsto ax_1$. It suffices to show that $\times x_1$ is injective. Let $ax_1 \equiv 0 \pmod{(x_2, \ldots, x_n)M}$, then

$$ax_1 = m_2 x_2 + \cdots + m_n x_n \tag{10.5}$$

for $m_i \in M$. Then $m_n x_n \in (x_1, \ldots, x_{n-1})M$. Since x_n is not a zerodivsor of $M/(x_1, \ldots, x_{n-1})M$, $m_n \in (x_1, \ldots, x_{n-1})M$. Hence exists $c_1, \ldots, c_{n-1} \in M$ such that

$$m_n = c_1 x_1 + \cdots + c_{n-1} x_{n-1}.$$

So by (10.6), we have

$$ax_1 = m_2 x_2 + \cdots + m_{n-1} x_{n-1} + x_n(c_1 x_1 + \cdots + c_{n-1} x_{n-1}),$$

rearranging terms, we have

$$x_1(a - c_1 x_n) = x_2(m_2 + c_2 x_n) + \cdots + x_{n-1}(m_{n-1} + c_{n-1} x_n) \in (x_2, \ldots, x_{n-1})M.$$

Since $x_1, x_2, \ldots, x_{n-1}$ is M-regular sequence, note that $x_1(a - c_1 x_n) \in (x_2, \ldots, x_{n-1})M$, by induction hypothesis, we know that $a - c_1 x_n \in (x_2, \ldots, x_{n-1})M$. Hence $a \in (x_2, \ldots, x_n)M$, it follows that $a \equiv 0 \pmod{(x_2, \ldots, x_n)M}$, i.e., $\times x_1$ is injective, we finished the proof. □

Proposition 10.5.5

If $x_1, \ldots, x_n$ is an M-regular sequence, and $a_1, \ldots, a_n \in \mathbb{Z}^+$, then $x_1^{a_1}, \ldots, x_n^{a_n}$ is a regular sequence.

Proof We first show that if $x_1, \ldots, x_n$ is an M-regular sequence then $x_1^N, x_2, \ldots, x_n$ is an M-regular sequence. We prove it by induction on N. If $N = 1$, then $x_1, \ldots, x_n$ is an M-regular sequence. Suppose $x_1^i, x_2, \ldots, x_n$ is an M-regular sequence for any $i < N$, we want to show that $x_1^N, x_2, \ldots, x_n$ is an M-regular sequence. Suppose $x_1^N, x_2, \ldots, x_n$ is not an M-regular sequence. Then there exists minimal $k \leq n$ such that $x_1^N, x_2, \ldots, x_k$ is not an M-regular sequence. Without loss of generally, we may assume $k = n$, i.e., $x_1^N, x_2, \ldots, x_n$ is not an M-regular sequence and $x_1^N, x_2, \ldots, x_{n-1}$ is an M-regular sequence. Since x_n is not an M-regular sequence, there exists $m \in M \setminus (x_1^N, x_2, \ldots, x_{n-1})M$ such that $x_n m \equiv 0 \pmod{(x_1^N, x_2, \ldots, x_{n-1})M}$. Since x_n is not a zerodivisor of $M/(x_1^{N-1}, x_2, \ldots, x_{n-1})M$, m must belong to $(x_1^{N-1}, x_2, \ldots, x_{n-1})M$, hence we may assume $m = m_1x_1^{N-1} + \sum_{i=2}^{n-1} m_ix_i$, then we have

$$x_nm = m_1x_1^{N-1}x_n + \sum_{i=2}^{n-1} m_ix_ix_n \in (x_1^N, x_2, \dots, x_{n-1})M. \tag{10.6}$$

On the other hand, since $x_nm \in (x_1^N, x_2, \dots, x_{n-1})M$, we may assume that

$$x_nm = k_1x_1^N + \sum_{i=2}^{n-1} k_ix_i. \tag{10.7}$$

Let (10.6) $-$ (10.7), rearranging terms, we have

$$x_1^{N-1}(m_1x_n - k_1x_1) + \sum_{i=2}^{n-1} x_i(m_ix_n - k_i) = 0.$$

Hence $x^{N-1}(m_1x_n - k_1x_1) \in (x_2, \dots, x_{n-1})M$, by Lemma 10.5.1, we know that $m_1x_n - k_1x_1 \in (x_2, \dots, x_{n-1})M$, and therefore $m_1x_n \in (x_1, x_2, \dots, x_{n-1})M$. Since x_n is not a zerodivisor of $M/(x_1, \dots, x_{n-1})M$, $m_1 \in (x_1, \dots, x_{n-1})M$, say $m_1 = \sum_{i=1}^{n-1} m_{1,i}x_i$. Then

$$\begin{aligned} m =& \left(\sum_{i=1}^{n-1} m_{1,i}x_i\right) x_1^{N-1} + \sum_{i=2}^{n-1} m_ix_i \\ =& m_{1,1}x_1^N + \sum_{i=2}^{n-1} x_i(m_{1,i}x_1^{N-1} + m_i) \in (x_1^N, x_2, \dots, x_{n-1})M, \end{aligned}$$

that contradicts to the fact that $m \in M \setminus (x_1^N, x_2, \dots, x_{n-1})M$. Hence $x_1^N, x_2, \dots, x_n$ is an M-regular sequence.

Now we prove this proposition, if $x_1, \dots, x_n$ is an M-regular sequence and $a_1, \dots, a_n \in \mathbb{Z}^+$, then $x_1^{a_1}, x_2, \dots, x_n$ is an M-regular sequence, by our above discussion. Hence $x_2, \dots, x_n$ is an $M/(x_1^{a_1}M)$-regular sequence, by above discussion again, $x_2^{a_2}, x_3, \dots, x_n$ is an $M/(x_1^{a_1}M)$-regular sequence. Repeat this process, we have x_n is an $M/(x_1^{a_1}, \dots, x_{n-1}^{a_{n-1}})M$-regular sequence, hence x_n is not a zerodivisor of $M/(x_1^{a_1}, \dots, x_{n-1}^{a_{n-1}})M$, and therefore $x_n^{a_n}$ is an $M/(x_1^{a_1}, \dots, x_{n-1}^{a_{n-1}})M$-regular sequence. Thus $x_1^{a_1}, \dots, x_n^{a_n}$ is an M-regular sequence. □

Example 10.12 (The regularity of a sequence depends on its order.) We now give an example showing that the order of a regular sequence matters. Suppose $A = k[x, y, z]/(x-1)z$, so $X = \operatorname{Spec} A$ is the union of the $z = 0$ plane and the $x = 1$ plane — X is reduced and has two components (see Figure 10.7). You can

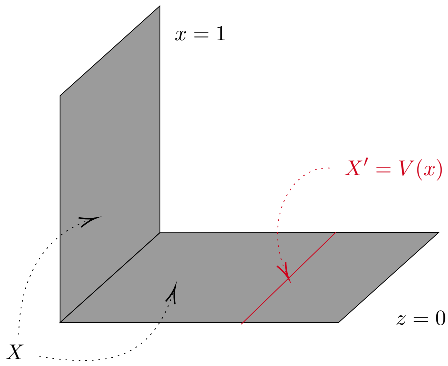


Figure 10.7: Order matters in a regular sequence (in the "non-local" situation), for unsurprising reasons

readily verify that x is a non-zerodivisor of A ($x = 0$ misses one component of X, and doesn't vanish entirely on the other, note that $\operatorname{Ass}(X) = \{[(z)], [(x-1)]\}$, recall Proposition 7.6.11 and Proposition 7.6.22), and that the corresponding effective Cartier divisor $X' = \operatorname{Spec} k[x,y,z]/(x,z)$ (on X) is integral. Then $(x-1)y$ gives an effective Cartier divisor on X' (it doesn't vanish entirely on X'), so $x, (x-1)y$ is a regular sequence for A. However, $(x-1)y$ is not a non-zerodivisor of A, as it does vanish entirely on $x = 1$. Thus $(x-1)y, x$ is not a regular sequence. The reason that reordering the regular sequence $x, (x-1)y$ ruins regularity is clear: there is a locus on which $(x-1)y$ isn't effective Cartier divisor, but it disappears if we enforce $x = 0$ first. This problem is one of "nonlocality" — "near" $x = y = z = 0$ there is no problem. This may motivate the fact that in the (Noetherian) local situation, this problem disappears. We now make this precise.

Theorem 10.5.1

Suppose $(A, \mathfrak{m})$ is a Noetherian local ring, and M is a finitely generated A-module. Then any M-regular sequence $(x_1, \dots, x_r)$ in $\mathfrak{m}$ remains a regular sequence upon any reordering.

Remark 10.29 In [Die66], Dieudonné gives a half-page example showing that Noetherian hypotheses are necessary in Theorem 10.5.1. Also, in Chapter 25, we give a different, somehow simpler proof in the case where A contains a field.

Proof Before proving Theorem 10.5.1, we prove the first nontrivial case, when $r = 2$. This discussion is secretly a baby case of the Koszul complex (which is briefly mentioned in Chapter 25).

Suppose $x, y \in \mathfrak{m}$, and x, y is an M-regular sequence. Translation: $x \in \mathfrak{m}$ is a non-zerodivisor on M, and $y \in \mathfrak{m}$ is a non-zerodivisor on M/xM.

Consider the double complex

$$\begin{array}{ccc} M & \xrightarrow{\times(-x)} & M \\ {\scriptstyle \times y}\uparrow & & \uparrow{\scriptstyle \times y} \\ M & \xrightarrow{\times x} & M \end{array} \tag{10.8}$$

where the bottom left is considered to be in position $(0,0)$. (The only reason for the minus sign in the top row is solely our arbitrary preference for anti-commuting rather than commuting squares in §2.6.1, but it really doesn't matter.)

We compute the cohomology of the total complex using a (simple) spectral sequence, beginning with the rightward orientation. On the first page, we have

$$\begin{array}{ccc} \operatorname{Ker}(M \xrightarrow{\times x} M) & & M/xM \\ {\scriptstyle \times y}\uparrow & & \uparrow{\scriptstyle \times y} \\ \operatorname{Ker}(M \xrightarrow{\times x} M) & & M/xM \end{array}$$

The entries $\operatorname{Ker}(M \xrightarrow{\times x} M)$ in the first column are 0, as x is a non-zerodivisor on M. Taking homology in the vertical direction to obtain the second page, we find

$$\begin{tikzcd}
0 & & \\
0 & M/(x,y)M \arrow[ul] & \\
0 & 0 & \\
& & 0 \arrow[uul]
\end{tikzcd} \tag{10.9}$$

using the fact that y is a non-zerodivisor on M/xM. The sequence clearly converges here. Thus the original double complex (10.8) only has nonzero cohomology in degree 2, where it is $M/(x,y)M$.

Now we run the spectral sequence on (10.8) using the upward orientation. The first page of the sequence is:

$$\begin{array}{c} M/yM \xrightarrow{\quad \times(-x) \quad} M/yM \\ \\ \mathrm{Ker}(M \xrightarrow{\times y} M) \xrightarrow{\times x} \mathrm{Ker}(M \xrightarrow{\times y} M) \end{array} \tag{10.10}$$

And the second page of the sequence is:

$$\begin{tikzcd}
0 \arrow[drr] & 0 \arrow[drr] & & & & \\
0 \arrow[drr] & 0 \arrow[drr] & A \arrow[drr] & B \arrow[drr] & & \\
& & C \arrow[drr] & D \arrow[drr] & 0 & 0 \\
& & & & 0 & 0
\end{tikzcd}$$

where $A = \mathrm{Ker}(M/yM \xrightarrow{\times(-x)} M/yM)$, $B = \mathrm{Coker}(M/yM \xrightarrow{\times(-x)} M/yM)$, $C = \mathrm{Ker}(\mathrm{Ker}(M \xrightarrow{\times y} M) \to \mathrm{Ker}(M \xrightarrow{\times y} M))$, and $D = \mathrm{Coker}(\mathrm{Ker}(M \xrightarrow{\times y} M) \to \mathrm{Ker}(M \xrightarrow{\times y} M))$. Since $H^0(E^\bullet) = H^1(E^\bullet) = 0$, the top row in (10.10) is injective and the bottom row in (10.10) is isomorphism. Since the top row in (10.10) is injective, x is a non-zerodivisor on M/yM. Since A is Noetherian and M is finitely generated A-module, we have M is Noetherian (M is the quotient of A^n, the quotient of Noetherian ring is also Noetherian), and therefore $\mathrm{Ker}(M \xrightarrow{\times y} M)$ is finitely generated A-module. As $x \in \mathfrak{m}$, by version 2 of Nakayama's Lemma 9.2.4, this implies that $\mathrm{Ker}(M \xrightarrow{\times y} M) = 0$, so y is non-zerodivisor on M. Thus we have shown that y, x is a regular sequence on M — the $r = 2$ case of Theorem 10.5.1.

We next prove Theorem 10.5.1. First we show that if $x_1, \ldots, x_i, \ldots, x_j, \ldots, x_r \in \mathfrak{m}$ is M-regular sequence, then $x_1, \ldots, x_j, \ldots, x_i, \ldots, x_r$ is M-regular sequence, i.e., regular sequences are invariant under transposition. By our above discussion, we know that $x_1, \ldots, x_i, \ldots, x_j, x_{j-1}, x_{j+1}, \ldots, x_r$ is regular sequence, repeat this process, we have

$$\begin{array}{cccccc} x_1 & \cdots & x_j & x_i & \cdots & x_r \\ \\ & & i\text{-th} & (i+1)\text{-th} & & \end{array}$$

is regular sequence. Now we perform transpositions of x_i with its right neighbor until x_i reaches the original position of x_j, then we have

$$\begin{array}{ccccccc} x_1 & \cdots & x_j & \cdots & x_i & \cdots & x_r \\ & & & & & & \\ & & i\text{-th} & & j\text{-th} & & \end{array}$$

is regular sequence. Since every permutation is a composition of transpositions, we have thus completed the proof of Theorem 10.5.1. □

10.5.3 Regular embeddings

Definition 10.5.4 (Regular embedding (of codimension r))

Suppose $\pi : X \hookrightarrow Y$ is a locally closed embedding. We say that π is ***a regular embedding (of condimension*** *r*) ***at a point*** *$p \in X$ if in the local ring $\mathscr{O}_{Y,p}$, the ideal of X is generated by a regular sequence (of length r). We say that π is a* ***regular embedding (of codimension*** *r*)* *if it is a regular embedding (of codimension r) at all $p \in X$.*

Remark 10.30 Another reasonable name for a regular embedding might be "local complete intersection". Unfortunately, "local complete intersection morphism", or "Ici morphism", is already used for a related notion, see [Sta25, Section 068E].

Our terminology uses the word "codimension", which we have not defined as a word on its own. The reason for using this word will become clearer once you meet Krull's Principal Ideal Theorem (Chapter 13) and Krull's Height Theorem (Chapter 13).

Lemma 10.5.2

If I and J are ideals of a Noetherian ring A, and $\mathfrak{p} \subseteq A$ is a prime ideal such that $I_\mathfrak{p} = J_\mathfrak{p}$, then there exists $a \in A \setminus \mathfrak{p}$ such that $I_a = J_a$ in A_a.

Proof Since A is Noetherian, I, J are finitely generated, say $I = (f_1, \ldots, f_n)$. Since $I_\mathfrak{p} = J_\mathfrak{p}$, we have $\frac{f_i}{1} \in I_\mathfrak{p} = J_\mathfrak{p}$. Hence exists $\frac{g_i}{s_i} \in J_\mathfrak{p}$ such that $\frac{g_i}{s_i} = \frac{f_i}{1}$, hence exists $t_i \in A \setminus \mathfrak{p}$ such that $t_i(s_i f_i - g_i) = 0$. Let $t = \prod_{i=1}^n t_i$ and $s = \prod_{i=1}^n s_i$, then $stf_i = g_i t \prod_{j \neq i} s_j \in J$ for all i. Say $b = st$, then we have $bI \subseteq J$. Since $s, t \in A - \mathfrak{p}$, $b \in A - \mathfrak{p}$. Localize A at b, we have $(bI)_b = I_b$, and therefore $((bI)_b)_\mathfrak{p} = I_\mathfrak{p} = J_\mathfrak{p} = (J_b)_\mathfrak{p}$. Hence we may reduce to the case that $I \subseteq J$.

Let $I \subseteq J$, denote $K = J/I$. Since $I_\mathfrak{p} = J_\mathfrak{p}$, we have $K_\mathfrak{p} = (J/I)_\mathfrak{p} \cong J_\mathfrak{p}/I_\mathfrak{p} = 0$. We want to show that $K_a = 0$ for some $a \in A \setminus \mathfrak{p}$. Since K is finitely generated, we may assume that $K = (k_1, \ldots, k_m)$. Since $K_\mathfrak{p} = 0$, for each k_i, there exists $a_i \in A \setminus \mathfrak{p}$ such that $a_i k_i = 0$. Let $a = \prod a_i$, then $ak_i = 0$ for all i, hence $aK = 0$. Now we localize aK at a, we have $(aK)_a = K_a = 0$, i.e., $I_a = J_a$. □

Proposition 10.5.6 (The condition of locally closed embedding being regular embedding is open)

If a locally closed embedding $\pi : X \hookrightarrow Y$ of locally Noetherian schemes is a regular embedding at p, then it is a regular embedding in some open neighborhood of p in X.

Proof Reduce to the case where π is a closed embedding, and then where Y (hence X) is affine, say $Y = \operatorname{Spec} B$, $X = \operatorname{Spec} B/I$, and $p = [\mathfrak{p}]$, where B is Noetherian. Since $\pi : X \hookrightarrow Y$ is a regular embedding at p, there are $f_1, \ldots, f_r \in B$ such that in $\mathscr{O}_{Y,p} = B_\mathfrak{p}$, the image of the f_i are a $B_\mathfrak{p}$-regular sequence generating

$I_\mathfrak{p} = IB_\mathfrak{p}$.

We first show there exists an open neighborhood U of p such that $(f_{1,q}, \ldots, f_{r,q}) = I_\mathfrak{q} = IB_\mathfrak{q}$ for all $q = [\mathfrak{q}] \in U$ where $f_{i,q} = f_i/1$ is the image of f_i in $\mathscr{O}_{Y,q}$, i.e., $(f_1, \ldots, f_r) = I$ "in an open neighborhood of p". Suppose for any open neighborhood U of p exists $q = [\mathfrak{q}] \in U$ such that $(f_{1,q}, \ldots, f_{r,q}) \neq IB_\mathfrak{q}$. Since $(f_{1,p}, \ldots, f_{r,p}) = I_\mathfrak{p}$, by Lemma 10.5.2, there exists $g \in B \setminus \mathfrak{p}$ such that $(f_{1,g}, \ldots, f_{r,g}) = I_g = IB_g$. Hence $p \in D(g)$, by our assumption, there exists $q = [\mathfrak{q}] \in D(g)$ such that $(f_{1,q}, \ldots, f_{r,q}) \neq IB_\mathfrak{q}$. Since $g \notin \mathfrak{q}$, we may localize B_g at $[\mathfrak{q}]$. Note that $(f_{1,g}, \ldots, f_{r,g}) = IB_g$, we have

$$(f_{1,g}, \ldots, f_{r,g})_\mathfrak{q} = (f_{1,q}, \ldots, f_{r,q}) = (IB_g)_\mathfrak{q} = IB_\mathfrak{q},$$

a contradiction! Hence there exists an open neighborhood U of p such that $(f_1, \ldots, f_r) = I$ in U.

We next show that there exists an open neighborhood $V \subseteq U$ of p such that $(f_{1,q}, \ldots, f_{r,q})$ is a $B_\mathfrak{q}$-regular sequence for all $q = [\mathfrak{q}] \in V$. Define the homomorphism as follow,

$$\begin{aligned} \varphi_i : B/(f_1, \ldots, f_{i-1}) &\longrightarrow B/(f_1, \ldots, f_{i-1}) \\ b &\longmapsto f_i b, \end{aligned}$$

denote $J_i := \operatorname{Ker} \varphi_i$, then $(f_{1,q}, \ldots, f_{r,q})$ is a $B_\mathfrak{q}$-regular sequence if and only if $(J_i)_\mathfrak{q} = 0$ for all i. Since $(f_{1,p}, \ldots, f_{r,p})$ is a $B_\mathfrak{p}$-regular sequence, $(J_i)_\mathfrak{p} = 0$ for all i, by Lemma 10.5.2, there exists $h \in B \setminus \mathfrak{p}$ such that $(J_i)_h = 0$ in B_h for all i. Consider $D(h) \cap U$, it is an open neighborhood of p. Note that for all $[\mathfrak{q}] \in D(h) \cap U$, $h \notin \mathfrak{q}$, hence we may localize $(J_i)_h$ at point $[\mathfrak{p}]$, then we have $((J_i)_h)_\mathfrak{q} = (J_i)_\mathfrak{q} = 0$ for all i and $[\mathfrak{q}] \in D(h) \cap U$. It follows that for all $q = [\mathfrak{q}] \in D(h) \cap U$, $(f_{1,q}, \ldots, f_{r,q})$ is a $B_\mathfrak{q}$-regular sequence. Let $V = D(h) \cap U$, we done! □

Corollary 10.5.1

If X is locally of finite type over a field (e.g., a variety), then to check that a closed embedding π is a regular embedding it suffices to check at closed points of X (Proposition 6.3.8).

Proof Let X be a locally of finite type over a field, then X is a locally Noetherian scheme. Suppose $\pi : X \hookrightarrow Y$ is a closed embedding of locally Noetherian schemes. By Proposition 10.5.6, set

$$Z = \{p \in X : \pi : X \hookrightarrow Y \text{ is a regular embedding at } p\}$$

is open. If we check $\pi : X \hookrightarrow Y$ is a regular embedding at all closed points of X, say C be the set of all closed points of X, then $C \subseteq Z$. By Proposition 6.3.8, we know that C is dense. We claim that $Z = X$ as topological space. If not $X \setminus Z$ is a non-empty closed subset, since C is dense, $C \cap (X \setminus Z) \neq \varnothing$, hence exists closed point not belong to Z, a contradiction! □

Remark 10.31 In Chapter 14, we will show that not all closed embeddings are regular embeddings.

Proposition 10.5.7

A closed embedding $X \hookrightarrow Y$ of locally Noetherian schemes is a regular embedding of codimension 1 if and only if X is an effective Cartier divisor on Y.

Proof If X is an effective Cartier divisor, then exists a cover of Y, say $\{\operatorname{Spec} B_i \hookrightarrow Y\}_i$, such that $X = \operatorname{Spec} B_i/(t_i)$, where t_i is not a zerodivisor of B_i. Pick p be any point of X, then exists affine piece $\operatorname{Spec} B_i \ni p = [\mathfrak{p}]$, we want to show that $t_{i,\mathfrak{p}} = \frac{t_i}{1}$ is a $(B_i)_\mathfrak{p}$-regular sequence, i.e., $\frac{t_i}{1}$ is not a zerodivisor of $(B_i)_\mathfrak{p}$. Since t_i is not a zerodivisor of B_i, by Proposition 10.5.4, $\frac{t_i}{1}$ is not a zerodivisor of $(B_i)_\mathfrak{p}$. Hence $\pi : X \hookrightarrow Y$ is a regular embedding of codimension 1.

If closed embedding $X \hookrightarrow Y$ is a regular embedding of codimension 1, we may reduce to the case that $Y = \operatorname{Spec} B$ where B is a Noetherian ring. Since $\pi : X \hookrightarrow Y$ is closed embedding, we may assume that $X = \operatorname{Spec} B/I$. Then for all $p = [\mathfrak{p}] \in X$, $I_\mathfrak{p}$ is generated by one element which is not a zerodivisor of $B_\mathfrak{p}$. Say $I_\mathfrak{p} = (f_\mathfrak{p})B_\mathfrak{p}$, then $I_\mathfrak{p} = (f)_\mathfrak{p}$, by Lemma 10.5.2, there exists $h_p \in B \setminus \mathfrak{p}$ such that $I_{h_p} = (f)_{h_p} = (f_{h_p})$. Consider $\operatorname{Ann}_{B_{h_p}}(f_{h_p})$, note that $\operatorname{Ann}_{B_{h_p}}(f_{h_p})_\mathfrak{p} = \operatorname{Ann}_{B_\mathfrak{p}}(f_\mathfrak{p}) = 0$, by Lemma 10.5.2, there exists $l_p \in B \setminus \mathfrak{p}$ such that $\operatorname{Ann}_{B_{h_p}}(f_{h_p})_{l_p} = \operatorname{Ann}_{B_{h_p l_p}}(f_{h_p l_p}) = 0$, say $s_p = h_p l_p$, then f_{s_p} is not a zerodivisor of B_{s_p} and $I_{s_p} = (f_{s_p})$. Hence $X = \bigcup\limits_{p \in X} \operatorname{Spec} B_{s_p}/(f_{s_p})$, where f_{s_p} is not a zerodivisor of B_{s_p}, i.e., X is an effective Cartier divisor on Y. □

Remark 10.32 (**Unimportant remark.**) The Noetherian hypotheses can be replaced by requiring $\mathscr{O}_Y$ to be coherent, and essentially the same argument applies. It is interesting to note that "effective Cartier divisor" implies "regular embedding of codimension 1" always, but that the converse argument requires Noetherian(-like) assumption.

Definition 10.5.5 (Codimension r complete intersection)

A ***codimension*** r ***complete intersection*** *in a scheme Y is closed subscheme X that can be written as the scheme-theoretic intersection of r effective Cartier divisors $D_1, \ldots, D_r$ such that at every point $p \in X$, the equations corresponding to $D_1, \ldots, D_r$ form a regular sequence. The phrase* ***complete intersection*** *means "codimension r complete intersection for some r".*

Chapter 11 Fibered products of schemes, and base change

Fibered products have an unexpectedly central place in algebraic geometry. Experience will gradually tech you why they play such an outsized role. Until you have acquired that experience, you should pay closer attention to them than you think they might deserve.

11.1 They exist

Before we get to products, we note that coproducts exists in the category of schemes: just as with the category of sets (Exercise 2.16), coproduct is disjoint union.

We will now construct the fibered product in the category of schemes.

Theorem 11.1.1 (Fibered products exist)

Suppose $\alpha : X \to Z$ and $\beta : Y \to Z$ are morphisms of schemes. Then the fibered product

$$\begin{array}{ccc} X \times_Z Y & \xrightarrow{\alpha'} & Y \\ {\scriptstyle\beta'}\downarrow & & \downarrow{\scriptstyle\beta} \\ X & \xrightarrow{\alpha} & Z \end{array}$$

exists in the category of schemes.

Note *If A is a ring, people often sloppily write $\times_A$ for $\times_{\operatorname{Spec} A}$. If B is an A-algebra, and X is an A-scheme, people often write X_B or $X \times_A B$ for $X \times_{\operatorname{Spec} A} \operatorname{Spec} B$.*

Remark 11.1 **Warning: products of schemes aren't products of sets.** Before showing existence, here is a warning: the product of schemes isn't a product of sets (and more generally for fibered products). We have made a big deal about schemes being sets, endowed with a topology, upon which we have a structure sheaf. So you might think that we will construct the product in this order: take the product set, endow it with the product topology, and then figure out what the structure sheaf must be. But we won't, because scheme products behave oddly on the level of sets. You may have checked (Exercise 8.16 (a)) that the product of two affine lines over your favorite algebraically closed field $\overline{k}$ is the affine plane: $\mathbb{A}^1_{\overline{k}} \times_{\overline{k}} \mathbb{A}^1_{\overline{k}} \cong \mathbb{A}^2_{\overline{k}}$. But the underlying set of the latter is not the underlying set of the former — we get additional points corresponding to curves in $\mathbb{A}^2$ that are not lines parallel to the axes!

Remark 11.2 On the other hand, W-valued points (where W is a scheme, Definition 8.3.5) do behave well under (fibered) products. This is just the universal property definition of fibered product: a W-valued point of a scheme X is defined as an element of $\operatorname{Hom}(W, X)$, and the fibered product is defined (via Yoneda Lemma, see §2.2.1) by

$$\operatorname{Hom}(W, X \times_Z Y) = \operatorname{Hom}(W, X) \times_{\operatorname{Hom}(W,Z)} \operatorname{Hom}(W, Y). \tag{11.1}$$

This is one justification for making the definition of scheme-valued point. For this reason, those classical people preferring to think only about varieties over an algebraically closed field $\overline{k}$ (or more generally, finite type scheme over $\overline{k}$), and preferring to understand them through their closed points — or equivalently, the $\overline{k}$-valued points, by the Nullstellensatz (Proposition 6.3.8) — needn't worry: the closed points of the product of two finite type $\overline{k}$-schemes over $\overline{k}$ are (naturally identified with) the product of the closed points of the factors. This will follow from the fact that the product is also finite type over $\overline{k}$, which we will verify in §11.2. This is one of the reasons that varieties over algebraically closed fields can be easier to work with. But over a non-algebraically closed field, things become even more interesting; An example in §11.2 is a first glimpse.

Remark 11.3 **Fancy side.** You may feel that (i) "products of topological spaces are products on the underlying sets" is natural, while (ii) "products of schemes are not necessarily products on the underlying sets" is weird. But really (i) is the lucky consequence of the fact that the underlying set of a topological space can be interpreted as set of p-valued points, where p is a point, so it is best seen as a consequence of above paragraph, which is the "more correct" — i.e., more general — fact.

Remark 11.4 **Warning on Noetherianness.** The fibered product of Noetherian schemes need not be Noetherian. You will later be able to verify that an example in §11.2, i.e., that $A := \overline{\mathbb{q}} \otimes_{\mathbb{q}} \overline{\mathbb{q}}$ is not Noetherian, as follows. By Chapter 13, $\dim A = 0$. A Noetherian dimension 0 scheme has a finite number of points. But by §11.2, $\operatorname{Spec} A$ has an infinite number of points.

On the other hand, the fibered product of finite type k-schemes over finite type k-schemes is a finite type k-scheme, so this pathology does not arise for varieties.

11.1.1 Philosophy behind the proof of Theorem 11.1.1

The proof of Theorem 11.1.1 can be confusing. The following comments may help a little.

We already basically know existence of fibered products in two cases: the case where X, Y, and Z are affine (stated explicitly below), and case where $\beta : Y \to Z$ is an open embedding (Proposition 9.1.4).

Proposition 11.1.1

Let $C \to A$ and $C \to B$ be two ring maps, then

$$\operatorname{Spec}(A \otimes_C B) \cong \operatorname{Spec} A \times_{\operatorname{Spec} C} \operatorname{Spec} B.$$

Hence the fibered product of affine schemes exists (in the category of schemes).

Proof Consider the following commutative diagram (Exercise 2.17).

$$\begin{array}{ccc} C & \longrightarrow & B \\ \downarrow & & \downarrow \\ A & \longrightarrow & A \otimes_C B \end{array} \tag{11.2}$$

By Proposition 8.3.7, we have the following diagram.

$$\begin{array}{ccc} \operatorname{Spec}(A \otimes_C B) & \longrightarrow & \operatorname{Spec} B \\ \downarrow & & \downarrow \\ \operatorname{Spec} A & \longrightarrow & \operatorname{Spec} C \end{array} \tag{11.3}$$

Since each morphisms induced by ring homomorphism, by the commutative of (11.2), diagram (11.3) is commutative.

We next check $\operatorname{Spec}(A \otimes_C B)$ agree with the universal property of fibered product. Let W be any scheme with maps to $\operatorname{Spec} A$ and $\operatorname{Spec} B$ whose compositions to $\operatorname{Spec} C$ agree commute, we want to check that these maps factor through some unique $W \to \operatorname{Spec} A \times_{\operatorname{Spec} Z} \operatorname{Spec} B$.

We first consider the case that $W = \operatorname{Spec} E$. By the universal property of fibered coproduct. There exists an unique morphism from $A \otimes_C B$ to E such that the following diagram commutes.

$$\begin{tikzcd}
C \arrow[r] \arrow[d] & B \arrow[d] \arrow[ddr, bend left] & \\
A \arrow[r] \arrow[drr, bend right] & A \otimes_C B \arrow[dr, dashed] & \\
& & E
\end{tikzcd}$$

By Proposition 8.3.7, we have the following commutative diagram.

$$\begin{tikzcd}
\operatorname{Spec} E \arrow[dr, dashed] \arrow[drr, bend left] \arrow[ddr, bend right] & & \\
& \operatorname{Spec}(A \otimes_C B) \arrow[r] \arrow[d] & \operatorname{Spec} B \arrow[d] \\
& \operatorname{Spec} A \arrow[r] & \operatorname{Spec} C
\end{tikzcd}$$

Now let W be any scheme, suppose $W = \bigcup_i \operatorname{Spec} E_i := \bigcup_i U_i$. By our above discussion, there exists unique morphism $\theta_i : \operatorname{Spec} E_i \to \operatorname{Spec}(A \otimes_C B)$ such that the following diagram commutes.

$$\begin{tikzcd}
U_i \arrow[dr, dashed, "\theta_i"] \arrow[drr, bend left] \arrow[ddr, bend right] & & \\
& \operatorname{Spec}(A \otimes_C B) \arrow[r] \arrow[d] & \operatorname{Spec} B \arrow[d] \\
& \operatorname{Spec} A \arrow[r] & \operatorname{Spec} C
\end{tikzcd}$$

By the gluability axiom, $\theta_i : U_i \to \operatorname{Spec}(A \otimes_C B)$ glue to unique $\theta : W \to \operatorname{Spec}(A \otimes_C B)$ such that the following diagram commutes.

$$\begin{tikzcd}
W \arrow[dr, dashed, "\theta"] \arrow[drr, bend left] \arrow[ddr, bend right] & & \\
& \operatorname{Spec}(A \otimes_C B) \arrow[r] \arrow[d] & \operatorname{Spec} B \arrow[d] \\
& \operatorname{Spec} A \arrow[r] & \operatorname{Spec} C
\end{tikzcd}$$

By the definition of fibered product, we have

$$\operatorname{Spec}(A \otimes_C B) \cong \operatorname{Spec} A \times_{\operatorname{Spec} C} \operatorname{Spec} B.$$

□

Remark 11.5 This generalizes the fact that the product of affine lines exist, Exercise 8.16 (a).

The main theme of the proof of Theorem 11.1.1 is that because schemes are built by gluing affine schemes along open subsets, these two special cases will be all that we need. The argument will repeatedly use the same ideas — roughly, that schemes glue (Theorem 5.4.1), and that morphisms of schemes glue (Proposition 8.3.2). This is a sign that something more structural is going on; §11.1.3 describes this for experts.

11.1.2 Proof of Theorem 11.1.1

Proof The key idea is this: we cut everything up into affine open sets, do fibered products there, and show that everything glues nicely. The conceptually difficult part of the proof comes from the gluing, and the realization that we have to check almost nothing. We divide the proof up into a number of bite-sized pieces.

Step 1: fibered products of affine with "almost-affine" over affine. We begin by combining the affine case with the open embedding case as follow. Suppose X and Z are affine, and $\beta : Y \to Z$ factor as

$$\begin{tikzcd} Y \arrow[r, hook, "\iota"] & Y' \arrow[r] & Z \end{tikzcd}$$

where ι is an open embedding and Y' is affine. Then $X \times_Z Y$ exists. This is because if the two small squares of

$$\begin{tikzcd} W \arrow[r] \arrow[d, hook, "\text{open}"'] & Y \arrow[d, hook, "\text{open}"] \\ W' \arrow[r] \arrow[d] & Y' \arrow[d] \\ X \arrow[r] & Z \end{tikzcd}$$

are Cartesian diagrams, then the "outside rectangle" is also a Cartesian diagram. (This was Exercise 2.13, although you should be able to see this on the spot.) It will be important to remember (from Important Proposition 9.1.4) that "open embeddings" are "preserved by fibered product": the fact that $Y \to Y'$ is an open embedding implies that $W \to W'$ is an open embedding.

Key Step 2: fibered product of affine with arbitrary over affine exists. We now come to the key part of the argument: if X and Z are affine, and Y is arbitrary. This is confusing when you first see it, so we first deal with a special case, when Y is the union of two affine open sets $Y_1 \cup Y_2$. Let $Y_{12} = Y_1 \cap Y_2$.

Now for $i = 1$ and 2, $X \times_Z Y_i$ exists by the affine cases, Proposition 11.1.1. Call this W_i. Also, $X \times_Z Y_{12}$ exists by **Step 1** (call it W_{12}), and comes with canonical open embeddings into W_1 and W_2 (by construction of fibered products with open embeddings, see the last sentence of **Step 1**). Thus we can glue W_1 to W_2 along W_{12}; call this resulting scheme W.

We check that the result is the fibered product by verifying that it satisfies the universal property. Suppose we have maps $\alpha'' : V \to X$, $\beta'' : V \to Y$ that compose (with α and β respectively) to the same map $V \to Z$. We need to construct a unique map $\gamma : V \to W$, so that $\alpha' \circ \gamma = \beta''$ and $\beta' \circ \gamma = \alpha''$.

$$\begin{tikzcd} V \arrow[dr, dashed, "\exists!\gamma?"] \arrow[drr, bend left, "\beta''"] \arrow[ddr, bend right, "\alpha''"'] & & \\ & W \arrow[r, "\alpha'"'] \arrow[d, "\beta'"] & Y \arrow[d, "\beta"'] \\ & X \arrow[r, "\alpha"'] & Z \end{tikzcd} \tag{11.4}$$

For $i = 1$ and 2, defined $V_i := (\beta'')^{-1}(Y_i)$. Define $V_{12} := (\beta'')^{-1}(Y_{12}) = V_1 \cap V_2$. Then there is a unique map $V_i \to W_i$ such that the composed maps $V_i \to X$ and $V_i \to Y_i$ are as desired (by the universal property of the fibered product $X \times_Z Y_i = W_i$), hence a unique map $\gamma_i : V_i \to W$. Similarly, there is a unique map $\gamma_{12} : V_{12} \to W$ such that the composed maps $V_{12} \to X$ and $V_{12} \to Y$ are as desired. But the restriction of γ_i to V_{12} is one such map, so it must be γ_{12}. Thus the maps γ_1 and γ_2 agree on V_{12}, and glue together to a unique map $\gamma : V \to W$. We have shown existence and uniqueness of desired γ.

We have thus shown that if Y is the union of two affine open sets, and X and Z are affine, then $X \times_Z Y$ exists.

We now tackle the general case. We now cover Y with open sets Y_i, as i runs over some index set (not necessarily finite!). As before, we define W_i and W_{ij}. We need to check the cocycle condition. Say

$f_{ij} : W_{ij} \xrightarrow{\sim} W_{ji}$, it suffices to show that $f_{ik}|_{W_{ij}\cap W_{ik}} = f_{jk}|_{W_{ji}\cap W_{jk}} \circ f_{ij}|_{W_{ij}\cap W_{ik}}$. In fact, we have

$$\begin{aligned} W_{ij} \cap W_{ik} =& \alpha_i'^{-1}(Y_i \cap Y_j) \cap \alpha_i'^{-1}(Y_i \cap Y_k) \\ =& \alpha_i'^{-1}(Y_i \cap Y_j \cap Y_k) \\ =& \alpha'^{-1}(Y_i \cap Y_j \cap Y_k) \\ \cong& W_{ji} \cap W_{jk} \cong W_{ki} \cap W_{kj}. \end{aligned}$$

Then we have the triple intersection cocycle conditions. By Theorem 5.4.1, we can glue these together to produce a scheme W along with open sets we identify with W_i.

As in the two-affine case, we show that W is the fibered product by showing that it satisfies the universal property. Suppose we have maps $\alpha'' : V \to X$, $\beta'' : V \to Y$ that compose to the same map $V \to Z$. We construct a unique map $\gamma : V \to W$, so that $\alpha' \circ \gamma = \beta''$ and $\beta' \circ \gamma = \alpha''$. Define $V_i = (\beta'')^{-1}(Y_i)$ and $V_{ij} := (\beta'')^{-1}(Y_{ij}) = V_i \cap V_j$. Then there is a unique map $V_i \to W_i$ such that the composed maps $V_i \to X$ and $V_i \to Y_i$ are as desired, hence a unique map $\gamma_i : V_i \to W$. Similarly, there is a unique map $\gamma_{ij} : V_{ij} \to W$ such that the composed maps $V_{ij} \to X$ and $V_{ij} \to Y$ are as desired. But the restriction of γ_i to V_{ij} is one such map, so it must be γ_{ij}. Thus the maps γ_i and γ_j agree on V_{ij}. Thus the γ_i glue together to a unique map $\gamma : V \to W$. We have shown existence and uniqueness of the desired γ, completing this step.

Step 3: Z affine, X and Y arbitrary. We next show that if Z is affine, and X and Y are arbitrary schemes, then $X \times_Z Y$ exists. We just follow **Step 2**, with the roles of X and Y reversed, using the fact that by the previous step, we can assume that the fibered product of an affine scheme with an arbitrary scheme over an affine scheme exists.

Step 4: Z "almost-affine", X and Y arbitrary. This is akin to **Step 1**. Let $Z \hookrightarrow Z'$ be an open embedding into an affine scheme. Then $X \times_Z Y$ satisfies the universal property of $X \times_{Z'} Y$, by Exercise 2.20 and that open embeddings are monomorphisms.

Step 5: the general case. We employ the same trick yet again. Suppose $\alpha : X \to Z$, $\beta : Y \to Z$ are two morphisms of schemes. Cover Z with affine open subschemes Z_i, and let $X_i = \alpha^{-1}(Z_i)$ and $Y_i = \beta^{-1}(Z_i)$. Define $Z_{ij} := Z_i \cap Z_j$, $X_{ij} := \alpha^{-1}(Z_{ij})$, and $Y_{ij} := \beta^{-1}(Z_{ij})$. Then $W_i := X_i \times_{Z_i} Y_i$ exists for all i (**Step 3**), and $W_{ij} := X_{ij} \times_{Z_{ij}} Y_{ij}$ exists for all i, j (**Step 4**), and for each i and j, W_{ij} comes with a canonical open embedding into both W_i and W_j (see the last sentence in **Step 1**). Since $X_i = \alpha^{-1}(Z_i)$, by Proposition 9.1.4, $X_i = X \times_Z Z_i$. Note that we have the tower of cartesian diagrams

$$\begin{tikzcd} W_i \arrow[r] \arrow[d] & Y_i \arrow[d] \\ X_i \arrow[r] \arrow[d] & Z_i \arrow[d] \\ X \arrow[r] & Z, \end{tikzcd}$$

by Exercise 2.13, we have $W_i = X \times_Z Y_i$. Similarly, we identify W_{ij} with $X \times_Z Y_{ij}$.

We then proceed exactly as in **Step 2**: the W_i's can be glued together along the W_{ij} (the cocycle condition can be readily checked to be satisfied), and W can be checked to satisfy the universal property of $X \times_Z Y$ (again, exactly as in **Step 2**). $\square$

11.1.3 ⋆⋆ Describing the existence of fibered products using the fancy language of representable functors

The proof above can be described more cleanly in the language of representable functors (§8.6). This will be enlightening only after you have absorbed the above argument and meditated on it for a long time. It may be most useful to shed light on representable functors, rather than on the existence of the fibered product.

11.2 Computing fibered products in practice

Before giving some examples, we first see how to compute fibered products in practice. There are four types of morphisms §11.2.1-§11.2.4 that it is particularly easy to take fibered products with, and all morphisms can be built from these atomic components. More precisely, §11.2.1 will imply that we can compute fibered products locally on the source and target. Thus to understand fibered products in general, it suffices to understand them on the level of affine sets, i.e., to be able to compute $A \otimes_B C$ given ring maps $B \to A$ and $B \to C$. Any map $B \to A$ (and similarly $B \to C$) may be expressed as $B \to B[t_1, \ldots]/I$, so if we know how to base change by "adding variables" §11.2.2 and "taking quotients" §11.2.3, we can "compute" any fibered product (at least in theory). The fourth type of morphism §11.2.4, corresponding to localization, is useful to understand explicitly as well.

11.2.1 Base change by open embeddings.

We have already done this (Proposition 9.1.4), and we used it in the proof hat fibered products of schemes exist.

11.2.2 Adding an extra variable.

Lemma 11.2.1

$A \otimes_B B[t] \cong A[t]$, so the following is a Cartesian diagram.

$$\begin{tikzcd} \operatorname{Spec} A[t] \arrow[r] \arrow[d] & \operatorname{Spec} B[t] \arrow[d] \\ \operatorname{Spec} A \arrow[r] & \operatorname{Spec} B \end{tikzcd}$$

Proof Since tensor product is fibered coproduct, we have the following Cartesian diagram.

$$\begin{tikzcd} B \arrow[r, hook] \arrow[d, "\varphi"'] & B[t] \arrow[d, hook] \\ A \arrow[r, hook] & A \otimes_B B[t] \end{tikzcd}$$

Consider ring homomorphism $i_A : A \hookrightarrow A[t]$ and $\widetilde{\varphi} : B[t] \to A[t]$ which induced by $\varphi : B \to A$, consider the following diagram.

$$\begin{tikzcd} B \arrow[r, hook, "i_B"] \arrow[d, "\varphi"'] & B[t] \arrow[d, hook] \arrow[ddr, bend left, "\widetilde{\varphi}"] & \\ A \arrow[r, hook] \arrow[drr, bend right, "i_A"'] & A \otimes_B B[t] \arrow[dr, dashed, "\theta"] & \\ & & A[t] \end{tikzcd} \tag{11.5}$$

Clearly, $i_A \circ \varphi = \widetilde{\varphi} \circ i_B$, by the universal property of fibered coproduct, there exists a unique morphism of rings $\theta : A \otimes_B B[t] \to A[t]$ such that diagram (11.5) commutes.

We next show that $A[t]$ is fibered coproduct. Consider the following diagram,

$$\begin{tikzcd} B \arrow[r, hook, "i_B"] \arrow[d, "\varphi"'] & B[t] \arrow[d, "\widetilde{\varphi}"'] \arrow[ddr, bend left, "\psi"] & \\ A \arrow[r, hook, "i_A"] \arrow[drr, bend right, "\eta"'] & A[t] \arrow[dr, dashed, "\theta'"] & \\ & & C \end{tikzcd} \tag{11.6}$$

where $\eta \circ \varphi = \psi \circ i_B$. We want to show that there exists a unique morphism $A[t] \to C$ such that diagram (11.6) commutes. Define $\theta' : A[t] \to C$ by setting

$$\sum_{i=0}^{n} a_i t^i \longmapsto \sum_{i=0}^{n} \eta(a_i)\psi(t)^i.$$

Let $a \in A$, then

$$\theta' \circ i_A(a) = \theta'(a) = \eta(a),$$

i.e., $\eta = \theta' \circ i_A$. Let $\sum\limits_{i=0}^{n} b_i t^i \in B[t]$, then by $\eta \circ \varphi = \psi \circ i_B$ we have

$$\theta' \circ \widetilde{\varphi}\left(\sum_{i=0}^{n} b_i t^i\right) = \theta'\left(\sum_{i=0}^{n} \varphi(b_i)t^i\right) = \sum_{i=0}^{n} (\eta \circ \varphi)(b_i)\psi(t)^i = \sum_{i=0}^{n} (\psi \circ i_B)(b_i)\psi(t)^i = \psi\left(\sum_{i=0}^{n} b_i t^i\right),$$

i.e., $\psi = \theta' \circ \widetilde{\varphi}$. By the definition of θ', uniqueness is obviously. Hence $A[t]$ is fibered coproduct. Let $C = A \otimes_B B[t]$, then there exists a unique morphism $A[t] \to A \otimes_B B[t]$. Hence

$$A \otimes_B B[t] \cong A[t].$$

By Proposition 8.3.7, we have the following Cartesian diagram.

$$\begin{tikzcd} \operatorname{Spec} A[t] \arrow[r] \arrow[d] & \operatorname{Spec} B[t] \arrow[d] \\ \operatorname{Spec} A \arrow[r] & \operatorname{Spec} B \end{tikzcd}$$

□

Definition 11.2.1 (Affine space over an arbitrary scheme)

If X is any scheme, we define $\mathbb{A}^n_X$ as $X \times_{\mathbb{Z}} \operatorname{Spec} \mathbb{Z}[x_1, \dots, x_n]$. Clearly, $\mathbb{A}^n_{\operatorname{Spec} A}$ is canonically the same as $\mathbb{A}^n_A = \operatorname{Spec} A[x_1, \dots, x_n]$.

11.2.3 Base change by closed embeddings.

Definition 11.2.2 (Extension ideal)

Suppose $\varphi : B \to A$ is a ring homomorphism, and $I \subseteq B$ is an ideal. Let $I^e := (\varphi(i))_{i \in I} = \varphi(I)A \subseteq A$ be the ***extension of*** *I* ***to*** *A.*

Lemma 11.2.2

Suppose $\varphi : B \to A$ is a ring homomorphism, and $I \subseteq B$ is an ideal. There is a natural isomorphism $A/I^e \xleftrightarrow{\sim} A \otimes_B (B/I)$.

Proof Consider the following exact sequence.

$$0 \longrightarrow I \longrightarrow B \longrightarrow B/I \longrightarrow 0$$

By the right exactness of $\square \otimes_B A$ (Exercise 2.37), the sequence

$$\begin{array}{ccccccc} I \otimes_B A & \longrightarrow & B \otimes_B A & \longrightarrow & B/I \otimes_B A & \longrightarrow & 0 \\ & & \| & & & & \\ & & A & & & & \end{array}$$

is exact, hence

$$B/I \otimes_B A \cong \mathrm{Coker}(I \otimes_B A \to A).$$

Consider $\mathrm{Im}(I \otimes_B A \to A)$, we have

$$\mathrm{Im}(I \otimes_B A \to A) = \varphi(I) \cdot A = I^e.$$

So,

$$B/I \otimes_B A \cong \mathrm{Coker}(I \otimes_B A \to A) \cong A/\,\mathrm{Im}(I \otimes_B A \to A) = A/I^e.$$

□

Proposition 11.2.1 (Closed embeddings are preserved by base change)

The fibered product with a closed subscheme is a closed subscheme of the fibered product. More precisely, if $X \hookrightarrow Z$ is a closed embedding, then for any $Y \to Z$, induced map $X \times_Z Y \hookrightarrow Y$ is closed embedding.

Proof By Lemma 11.2.2 and Proposition 11.1.1, we get this consequence immediately. □

Proposition 11.2.2

The intersection of two closed embeddings into X is their fibered product over X.

Proof It suffices to check this in the affine case. We may assume that $X = \mathrm{Spec}\, B$, and $Y = \mathrm{Spec}\, B/I \hookrightarrow \mathrm{Spec}\, B$ and $Z = \mathrm{Spec}\, B/J \hookrightarrow \mathrm{Spec}\, B$ be two closed embedding. Their fibered product over X is $\mathrm{Spec}\, B/I \times_B \mathrm{Spec}\, B/J$. By Proposition 11.1.1, we know that $\mathrm{Spec}\, B/I \times_B \mathrm{Spec}\, B/J \cong \mathrm{Spec}(B/I \otimes_B B/J)$. We claim that $B/I \otimes_B B/J \cong B/(I+J)$. Consider the following exact sequence.

$$0 \longrightarrow I \longrightarrow B \longrightarrow B/I \longrightarrow 0$$

By the right exactness of $\square \otimes_B (B/J)$ (Exercise 2.37), the sequence

$$0 \longrightarrow I \otimes_B (B/J) \longrightarrow B \otimes_B (B/J) \longrightarrow (B/I) \otimes_B (B/J) \longrightarrow 0$$

is exact. Note that $B \otimes_B (B/J) \cong B/J$, then we have exact sequence

$$0 \longrightarrow I \otimes_B (B/J) \longrightarrow B/J \longrightarrow (B/I) \otimes_B (B/J) \longrightarrow 0.$$

We now compute $I \otimes_B (B/J)$, claim that $I \otimes_B (B/J) \cong I/IJ$. Define $\varphi : I \otimes_B (B/J) \to I/IJ$ by setting

$$i \otimes b \longmapsto \overline{ib},$$

clearly this morphism is well-defined. We next define the inverse of φ, define $\psi : I/IJ \to I \otimes_B (B/J)$ by setting

$$\bar{i} \longmapsto i \otimes \bar{1}.$$

Let $i \in IJ$, then $i = \sum_k i_k j_k$ where $i_k \in I$ and $j_k \in J$, then $i \otimes \bar{1} = \sum_k i_k \otimes j_k = 0$. Hence ψ is well-defined, and therefore $I/IJ \cong I \otimes_B (B/J)$, then we get an exact sequence

$$0 \longrightarrow I/IJ \longrightarrow B/J \longrightarrow (B/I) \otimes_B (B/J) \longrightarrow 0,$$

i.e., $(B/I) \otimes_B (B/J) \cong (B/J)/\operatorname{Im}(I/IJ \to B/J)$. In fact, $\operatorname{Im}(I/IJ \to B/J) = (I+J)/J$, hence $(B/I) \otimes_B (B/J) \cong (B/J)/((I+J)/J)$, by the 3-rd Fundamental Theorem of Isomorphism, we have

$$(B/I) \otimes_B (B/J) \cong (B/J)/((I+J)/J).$$

Hence

$$Y \cap Z = \operatorname{Spec}(B/(I+J)) \cong \operatorname{Spec}(B/I \otimes_B B/J) \cong \operatorname{Spec} B/I \times_B \operatorname{Spec} B/J.$$

□

Proposition 11.2.3 (Locally closed embeddings are preserved by base change)

If $\pi : X \hookrightarrow Y$ is a locally closed embedding, then for any $Y' \to Y$, the induced map $X \times_Y Y' \to Y'$ is also a locally closed embedding.

Proof Since $\pi : X \hookrightarrow Y$ is a locally closed embedding, π can be factored into

$$X \longhookrightarrow Z \longhookrightarrow Y,$$

where $X \hookrightarrow Z$ is closed embedding and $Z \hookrightarrow Y$ is open embedding. Since open embedding is preserved by base change (Proposition 9.1.4), $Z \times_Y Y' \to Y'$ is an open embedding. By Proposition 11.2.1, $X \times_Z (Z \times_Y Y') \to Z \times_Y Y'$ is a closed embedding. Note that we have the following commutative diagram,

$$\begin{tikzcd}
X \times_Y Y' \arrow[dr, dashed, shift left] \arrow[drr, bend left] \arrow[ddr, bend right] & & \\
& X \times_Z (Z \times_Y Y') \arrow[ul, dashed, shift left] \arrow[r] \arrow[d, hook] & X \arrow[d, hook] \\
& Z \times_Y Y' \arrow[r] \arrow[d, hook] & Z \arrow[d, hook] \\
& Y' \arrow[r] & Y
\end{tikzcd}$$

by the universal property of $X \times_Y Y'$ and $X \times_Z (Z \times_Y Y')$, there is an isomorphism

$$X \times_Z (Z \times_Y Y') \cong X \times_Y Y'.$$

Hence $X \times_Y Y' \to Y'$ can be factored into

$$X \times_Y Y' \xhookrightarrow{\text{closed}} Z \times_Y Y' \xhookrightarrow{\text{open}} Y',$$

i.e., $X \times_Y Y' \hookrightarrow Y'$ is a locally closed embedding. □

Definition 11.2.3 (Intersection of n locally closed embeddings)

*We define the **intersection of** n **locally closed embeddings** $X_i \hookrightarrow Z$ ($1 \leq i \leq n$) by the fibered product of the X_i over Z, i.e., $X_1 \times_Z \cdots \times_Z X_n$.*

Proposition 11.2.4

The intersection of a finite number of locally closed embeddings is also a locally closed embedding.

Proof It suffices to show the case that $n = 2$, by Proposition 11.2.3, $X_1 \times_Z X_2 \hookrightarrow X_2$ is locally closed embedding, by Proposition 10.2.4, $X_1 \times_Z X_2 \hookrightarrow X_2 \hookrightarrow Z$ is locally closed embedding. By induction on n, we done. □

Proposition 11.2.5

Suppose $I \subseteq A[x_1, \ldots, x_m]$ and $J \subseteq A[y_1, \ldots, y_n]$ are ideals. Then there is an isomorphism

$$A[x_1, \ldots, x_m]/I \otimes_A A[y_1, \ldots, y_n]/J \xleftrightarrow{\sim} A[x_1, \ldots, x_m, y_1, \ldots, y_n]/(I, J).$$

Proof By Lemma 11.2.1, we have

$$\begin{aligned} A[x_1, \ldots, x_m, y_1, \ldots, y_n] \cong& A[x_1, \ldots, x_m, y_1, \ldots, y_{n-1}] \otimes_A A[y_n] \\ \cong& A[x_1, \ldots, x_m, y_1, \ldots, y_{n-2}] \otimes_A A[y_{n-1}, y_n] \\ \cong& \cdots \cong A[x_1, \ldots, x_m] \otimes_A A[y_1, \ldots, y_n]. \end{aligned}$$

Clearly, we have

$$\begin{aligned} A[x_1, \ldots, x_m, y_1, \ldots, y_n]/(J) \cong& (A[x_1, \ldots, x_m] \otimes_A A[y_1, \ldots, y_n])/(1 \otimes J) \\ \cong& A[x_1, \ldots, x_m] \otimes_A (A[y_1, \ldots, y_n]/J). \end{aligned}$$

Consider the ring map $A[x_1, \ldots, x_m] \to A[x_1, \ldots, x_m, y_1, \ldots, y_n]/(J)$, by Lemma 11.2.2, we have

$$(A[x_1, \ldots, x_m, y_1, \ldots, y_n]/(J))/I^e \cong (A[x_1, \ldots, x_m, y_1, \ldots, y_n]/(J)) \otimes_{A[x_1, \ldots, x_m]} (A[x_1, \ldots, x_m]/I).$$

Note that

$$(A[x_1, \ldots, x_m, y_1, \ldots, y_n]/(J))/I^e \cong A[x_1, \ldots, x_m, y_1, \ldots, y_n]/(I, J),$$

hence

$$\begin{aligned} A[x_1, \ldots, x_m, y_1, \ldots, y_n]/(I, J) \cong& (A[y_1, \ldots, y_n]/J) \otimes_A A[x_1, \ldots, x_m] \otimes_{A[x_1, \ldots, x_m]} (A[x_1, \ldots, x_m]/I) \\ \cong& (A[x_1, \ldots, x_m]/I) \otimes_A (A[y_1, \ldots, y_n]/J), \end{aligned}$$

we are done! □

As an application, we can compute tensor products of finitely generated k-algebras over k.

Example 11.1 For example we have

$$k[x_1, x_2]/(x_1^2 - x_2) \otimes_k k[y_1, y_2]/(y_1^3 + y_2^3) \cong k[x_1, x_2, y_1, y_2]/(x_1^2 - x_2, y_1^3 + y_2^3).$$

Proposition 11.2.6

Suppose X and Y are locally of finite type A-schemes, then $X \times_A Y$ is also locally of finite type over A. Also the same thing with "locally" removed from both the hypothesis and conclusion.

Proof Since X and Y are locally of finite type, we may assume $X = \bigcup_i \operatorname{Spec} B_i$ and $Y = \bigcup_j \operatorname{Spec} C_j$, where B_i and C_j are finitely generated A-algebras. Note that $X \times_A Y = \bigcup_{i,j} (\operatorname{Spec} B_i \times_A \operatorname{Spec} C_j)$, it suffices to show that each $\operatorname{Spec} B_i \times_A \operatorname{Spec} C_i$ is locally of finite type A-scheme. By Proposition 11.1.1, $\operatorname{Spec} B_i \times_A \operatorname{Spec} C_j \cong \operatorname{Spec}(B_i \otimes_A C_j)$, since B_i and C_j are finitely generated A-algebras, $B_i \otimes_A C_j$ is finitely generated A-algebra, hence $\operatorname{Spec} B_i \times_A \operatorname{Spec} C_j$ is locally of finite type A-scheme, as we desired. □

Example 11.2 We can use these ideas to compute $\mathbb{C} \otimes_{\mathbb{R}} \mathbb{C}$:

$$\begin{aligned}
\mathbb{C} \otimes_{\mathbb{R}} \mathbb{C} \cong & \mathbb{C} \otimes_{\mathbb{R}} (\mathbb{R}[x]/(x^2+1)) \\
\cong & (\mathbb{C} \otimes_{\mathbb{R}} \mathbb{R}[x])/(x^2+1) && \text{by Lemma 11.2.2} \\
\cong & \mathbb{C}[x]/(x^2+1) && \text{by Lemma 11.2.1} \\
\cong & \mathbb{C}[x]/((x-i)(x+i)) \\
\cong & \mathbb{C}[x]/(x-i) \times \mathbb{C}[x]/(x+i) && \text{by the Chinese Remainder Theorem 5.4.2} \\
\cong & \mathbb{C} \times \mathbb{C}.
\end{aligned}$$

Thus $\operatorname{Spec} \mathbb{C} \times_{\mathbb{R}} \operatorname{Spec} \mathbb{C} \cong \operatorname{Spec} \mathbb{C} \coprod \operatorname{Spec} \mathbb{C}$. This example is the first example of many different behaviors. Notice for example that two points somehow correspond to the Galois group of $\mathbb{C}$ over $\mathbb{R}$; for one of them, x (the "i" in one of the copies of $\mathbb{C}$) equals i (the "i" in the other copy of $\mathbb{C}$), and in the other, $x = -i$.

Remark 11.6 Here is a clue that there is something deep going on behind Example 11.2 . If L/K is a (finite) Galois extension with Galois group G, then $L \otimes_K L$ is isomorphic to L^G (the product of $|G|$ copies of L). This is turns out to be a restatement of the classical form of linear independence of characters! In the language of schemes, $\operatorname{Spec} L \times_K \operatorname{Spec} L$ is a union of a number of copies of $\operatorname{Spec} L$ that naturally form a torsor over the Galois group G; but we will not define torsor here.

✍ **Exercise 11.1 ⋆ Hard but fascinating exercise for those familiar with $\operatorname{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$.** Show that the points of $\operatorname{Spec}(\overline{\mathbb{Q}} \otimes_{\mathbb{Q}} \overline{\mathbb{Q}})$ are in natural bijection with $\operatorname{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$, and the Zariski topology on the former agrees with the profinite topology on the latter.

Proof Let $K/\mathbb{Q}$ be an finite Galois extension with Galois group $G = \operatorname{Gal}(K/\mathbb{Q})$. Consider the $K \otimes_{\mathbb{Q}} K$, we claim that $K \otimes_{\mathbb{Q}} K \cong \prod_{\sigma \in G} K$. Since $K/\mathbb{Q}$ is finite Galois extension, suppose $[K : \mathbb{Q}] = n$. Since $K/\mathbb{Q}$ is finite Galois extension, $K/\mathbb{Q}$ is finite separated, then there exists a primitive element of $K/\mathbb{Q}$, say α, with minimal polynomial $f(x) \in \mathbb{Q}[x]$ of degree n. Since $K/\mathbb{Q}$ is a Galois extension, $f(x)$ splits completely in K, i.e.,

$$f(x) = \prod_{\sigma \in G} (x - \sigma(\alpha)).$$

Consider the isomorphism $K \cong \mathbb{Q}[x]/(f(x))$. Then

$$\begin{aligned}
K \otimes_{\mathbb{Q}} K &\cong K \otimes_{\mathbb{Q}} \mathbb{Q}[x]/(f(x)) \\
&\cong K \otimes_{\mathbb{Q}} (\mathbb{Q}[x] \otimes_{\mathbb{Q}[x]} \mathbb{Q}[x]/(f(x))) \\
&\cong K[x] \otimes_{\mathbb{Q}[x]} \mathbb{Q}[x]/(f(x)) \\
&\cong K[x]/(f(x)).
\end{aligned}$$

by the Chinese Remainder Theorem, since $f(x)$ splits into distinct linear factors in K, we have

$$K[x]/(f(x)) \cong \prod_{\sigma \in G} K[x]/(x - \sigma(\alpha)) \cong \prod_{\sigma \in G} K.$$

Hence we have $K \otimes_{\mathbb{Q}} K \cong \prod_{\sigma \in G} K$, and therefore $\operatorname{Spec} K \otimes_{\mathbb{Q}} K \cong \operatorname{Spec} \prod_{\sigma \in G} K$. Note that the prime ideal of $\prod_{\sigma \in G} K$ is

$$\mathfrak{p}_\sigma = \{(x_\tau)_{\tau \in G} : x_\tau = K, \text{ if } \tau \neq \sigma, x_\sigma = (0)\}.$$

Then we get a one-to-one corresponding between $\operatorname{Spec} K \otimes_{\mathbb{Q}} K$ and $\operatorname{Gal}(K/\mathbb{Q})$.

Let $A = \overline{\mathbb{Q}} \otimes_{\mathbb{Q}} \overline{\mathbb{Q}}$. The following proof is given by [Slu].

Fact 1. All prime ideals of A are maximal and all its residue fields are canonically isomorphic with $\overline{\mathbb{Q}}$.

Proof of Fact 1. For the proof note that $\mathbb{Q} \hookrightarrow \overline{\mathbb{Q}}$ is an integral morphism. Hence $\overline{\mathbb{Q}} \hookrightarrow A$ is an integral morphism (as a base change of the previously considered morphism). Pick a prime ideal $\mathfrak{p} \subseteq A$, then the induced map $\overline{\mathbb{Q}} \hookrightarrow A/\mathfrak{p}$ is integral. Now if $K \hookrightarrow D$ is an integral morphism, where K is a field and D is an integral domain, then D is a field. Thus $A/\mathfrak{p}$ is a field algebraic over $\overline{\mathbb{Q}}$. Hence $\mathfrak{p}$ is maximal and $A/\mathfrak{p} \cong \overline{\mathbb{Q}}$.

Fact 2. Let R be a commutative ring such that all prime ideals of R are maximal. Then $\operatorname{Spec} R$ is a Hausdorff topological space. In particular, $\operatorname{Spec} R$ is compact.

Proof of Fact 2. By dividing R by nilradical, we may assume that R has no nilpotent elements. Since every prime ideal $\mathfrak{p}$ of R is maximal, we deduce that $R_{\mathfrak{p}}$ is a reduced local ring with single prime ideal. Hence $R_{\mathfrak{p}}$ is a field. Pick now distinct prime ideals $\mathfrak{p}$, $\mathfrak{q}$ of R. Since they are maximal, there exists $f \in \mathfrak{p} \setminus \mathfrak{q}$. Now $\frac{f}{1} \in R_{\mathfrak{p}}$ is zero (ideal $\mathfrak{p}R_{\mathfrak{p}} \subseteq R_{\mathfrak{p}}$ is zero as $R_{\mathfrak{p}}$ is a field). Thus there exists $g \notin \mathfrak{p}$ such that $g \cdot f = 0$. Since $g \cdot f = 0 \in \mathfrak{q}$ and $f \notin \mathfrak{q}$, we deduce that $g \in \mathfrak{q}$. Thus $g \in \mathfrak{q} \setminus \mathfrak{p}$ and $f \cdot g = 0$. Therefore, we have (for distinguished open subsets of $\operatorname{Spec} R$)

$$\mathfrak{p} \in D(g),\ \mathfrak{q} \in D(f),\ D(f) \cap D(g) = \emptyset$$

and thus $\operatorname{Spec} R$ is a Hausdorff space. Since it is quasi-compact, we deduce that it is compact.

Now we are ready to prove that $\operatorname{Spec} A$ is homeomorphic to $\operatorname{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$. Let $\mathcal{K}$ be a diagram consisting of all finite extensions of $\mathbb{Q}$ and inclusions between them. Consider the limit X in the category of topological spaces of the induced diagram

$$\left\{\operatorname{Spec}\left(\overline{\mathbb{Q}} \otimes_{\mathbb{Q}} K\right)\right\}_{K \in \mathcal{K}}$$

Clearly X is homeomorphic with $\operatorname{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ (this is just the definiton of a profinite topology on a Galois group). It suffices to check that X is homeomorphic with $\operatorname{Spec} A$. Note that

$$A = \operatorname{colim}_{K \in \mathcal{K}}\left(\overline{\mathbb{Q}} \otimes_{\mathbb{Q}} K\right)$$

in the category of $\overline{\mathbb{Q}}$-algebras. Hence $\operatorname{Spec} A$ gives rise to a cone over topological diagram $\left\{\operatorname{Spec}\left(\overline{\mathbb{Q}} \otimes_{\mathbb{Q}} K\right)\right\}_{K \in \mathcal{K}}$ and thus there exists a unique comparison map $f : \operatorname{Spec} A \to X$. Clearly f is continuous. We have

$$\begin{aligned}\text{Points of } \operatorname{Spec} A &= \operatorname{Mor}_{\overline{\mathbb{Q}}}\left(A, \overline{\mathbb{Q}}\right) = \operatorname{Mor}_{\overline{\mathbb{Q}}}\left(\operatorname{colim}_{K \in \mathcal{K}}\left(\overline{\mathbb{Q}} \otimes_{\mathbb{Q}} K\right), \overline{\mathbb{Q}}\right) = \\ &= \lim_{K \in \mathcal{K}} \operatorname{Mor}_{\overline{\mathbb{Q}}}\left(\overline{\mathbb{Q}} \otimes_{\mathbb{Q}} K, \overline{\mathbb{Q}}\right) = \text{Points of } X\end{aligned}$$

where the first identification is a consequence of **Fact 1**. Moreover, (as you can check carefully) this identification is induced by f. Thus f is bijective. Therefore, f is a continuous bijection with compact source (by **Fact 2**) and Hausdorff target. Hence f is a homeomorphism. □

At this point, we can compute any $A \otimes_B C$ (where A and C are B-algebras): any map of rings $\varphi : B \to A$ can be interpreted by adding variables (perhaps infinitely many) to B, and then imposing relations. But in practice §11.2.4 is useful, as we will see in examples.

11.2.4 Base change of affine schemes by localization.

Proposition 11.2.7 (Localizations are preserved by base change)

Suppose $\varphi : B \to A$ is a ring morphism, and $S \subseteq B$ is a multiplicative subset of B, which implies that $\varphi(S)$ is a multiplicative subset of A. Then there is a natural isomorphism

$$\varphi(S)^{-1}A \overset{\sim}{\longleftrightarrow} A \otimes_B (S^{-1}B).$$

Proof Define $\tau : A \otimes_B (S^{-1}B) \to \varphi(S)^{-1}A$ by setting

$$a \otimes \frac{b}{s} \longmapsto \frac{a\varphi(b)}{\varphi(s)}.$$

Let $\frac{a}{\varphi(s)} \in \varphi(S)^{-1}A$, then $\tau\left(a \otimes \frac{1}{s}\right) = \frac{a}{\varphi(s)}$, it follows that τ is surjective. Let $\tau\left(a \otimes \frac{b}{s}\right) = 0$ in $\varphi(S)^{-1}A$, then exists $\varphi(t) \in \varphi(S)$ such that $a\varphi(bt) = 0$. Note that

$$a \otimes \frac{b}{s} = a\varphi(bt) \otimes \frac{1}{st} = 0 \otimes \frac{1}{st} = 0,$$

we know that τ is injective. Then we have an isomorphism

$$\varphi(S)^{-1}A \xleftrightarrow{\sim} A \otimes_B (S^{-1}B).$$

□

Remark 11.7 Informal translation: "the fibered product with a localization is the localization of the fibered product in the obvious way." We say that "localizations are preserved by base change". This is handy if the localization is of the form $B \hookrightarrow B_f$ (corresponding to taking distinguished open sets) or (if B is an integral domain) $B \hookrightarrow K(B)$ (corresponding to taking the generic point), and various things in between.

11.2.5 Examples

Lemma 11.2.3 (The three important types of monomorphisms of schemes)

The following are monomorphisms: open embeddings, closed embeddings, and localization of affine schemes.

Proof We have already showed that open embeddings and closed embeddings are monomorphisms in Proposition 9.1.5 and Proposition 10.1.9. It suffices to show that localization of affine schemes are monomorphisms. Suppose $\iota : \operatorname{Spec} A_f \hookrightarrow \operatorname{Spec} A$, we want to show that ι is a monomorphism. Let $\mu_1 : Z \to \operatorname{Spec} A_f$ and $\mu_2 : Z \to \operatorname{Spec} A_f$ such that $\iota \circ \mu_1 = \iota \circ \mu_2$. Then $\iota^\sharp : A \to A_f$, $\mu_1^\sharp : A_f \to \Gamma(Z, \mathscr{O}_Z)$ and $\mu_2^\sharp : A_f \to \Gamma(Z, \mathscr{O}_Z)$. Since $\iota \circ \mu_1 = \iota \circ \mu_2$, we have $\mu_1^\sharp \circ \iota^\sharp = \mu_2^\sharp \circ \iota^\sharp$, by the universal property of localization, $\mu_1^\sharp = \mu_2^\sharp$, and therefore $\mu_1 = \mu_2$, which implies that ι is a monomorphism. □

Remark 11.8 As monomorphisms are preserved by composition compositions of the above are also monomorphisms — for example, locally closed embeddings, or maps from "Spec of stalks at points of X" to X.

Remark 11.9 **Caution:** if p is a point of a scheme X, the natural morphism $\operatorname{Spec} \mathscr{O}_{X,p} \to X$, cf. Proposition 8.3.10, is a monomorphism but is not in general an open embedding (since it is the composition of open embedding and localization of affine scheme).

✍ **Exercise 11.2** Recall that $\mathbb{A}_A^n \cong \operatorname{Spec} A \times_{\operatorname{Spec} \mathbb{Z}} \mathbb{A}_{\mathbb{Z}}^n$ (Definition 11.2.1 and Lemma 11.2.1). Prove similarly that $\mathbb{P}_A^n \cong \mathbb{P}_{\mathbb{Z}}^n \times_{\operatorname{Spec} \mathbb{Z}} \operatorname{Spec} A$. Thus affine space and projective space are pulled back from their "universal manifestation" over the final object $\operatorname{Spec} \mathbb{Z}$.

Proof Note that

$$\mathbb{P}_{\mathbb{Z}}^n = \operatorname{Proj} \mathbb{Z}[x_0, \ldots, x_n] = \bigcup_{i=0}^n \operatorname{Spec} \mathbb{Z}[x_{0/i}, \ldots, x_{n/i}]/(x_{i/i} - 1)$$

and each $\operatorname{Spec} \mathbb{Z}[x_{0/i}, \ldots, x_{n/i}]/(x_{i/i} - 1)$ is isomorphic to $\mathbb{A}_{\mathbb{Z}}^n$, hence by the proof of Theorem 11.1.1 (**Step**

2, they glue together) and Lemma 11.2.1 we have

$$\begin{aligned}
\mathbb{P}^n_{\mathbb{Z}} \times_{\operatorname{Spec}\mathbb{Z}} \operatorname{Spec} A &\cong \bigcup_{i=0}^{n} (\operatorname{Spec}\mathbb{Z}[x_{0/i},\dots,x_{n/i}]/(x_{i/i}-1) \times_{\operatorname{Spec}\mathbb{Z}} \operatorname{Spec} A) \\
&\cong \bigcup_{i=0}^{n} \operatorname{Spec} A[x_{0/i},\dots,x_{n/i}]/(x_{i/i}-1) \\
&\cong \mathbb{P}^n_A,
\end{aligned}$$

as we desired. □

Extending the base field.

One special case of base change is called **extending the base field**.

Definition 11.2.4 (Extending the base field)

If X is a k-scheme, and l is a field extension of k (often l is the algebraic closure of k), then $X \times_{\operatorname{Spec} k} \operatorname{Spec} l$ (sometimes informally written $X \times_k l$ or X_l) is an l-scheme. This special case of base change is called ***extending the base field****.*

Often properties of X can be checked by verifying them instead on X_l. This is the subject of descent — certain properties "descend" from X_l to X. We have already seen that the property of being the Spec of a normal integral domain descends in this way (Exercise 6.10). Proposition 11.2.8 and Proposition 11.2.9 give other examples of properties which descend: the property of two morphisms being equal, and the property of a morphism being a closed embedding, both descend in this way. Those interested in schemes over non-algebraically closed fields will use this repeatedly, to reduce results to the algebraically closed case.

Proposition 11.2.8

Suppose $\pi : X \to Y$ and $\rho : X \to Y$ are morphisms of k-schemes, l/k is a field extension, and $\pi_l : X \times_{\operatorname{Spec} k} \operatorname{Spec} l \to Y \times_{\operatorname{Spec} k} \operatorname{Spec} l$ and $\rho_l : X \times_{\operatorname{Spec} k} \operatorname{Spec} l \to Y \times_{\operatorname{Spec} k} \operatorname{Spec} l$ are the induced maps of l-schemes. If $\pi_l = \rho_l$, then $\pi = \rho$.

Remark 11.10 In the proof, we use the fact that $X \times_{\operatorname{Spec} k} \operatorname{Spec} l \to X$ is surjective, we will soon prove in §11.4.

Proof Consider the following diagram.

$$\begin{tikzcd}
X \times_{\operatorname{Spec} k} \operatorname{Spec} l \arrow[dr, dashed, shift left, "\pi_l"] \arrow[dr, dashed, shift right, "\rho_l"'] \arrow[drr, bend left] \arrow[dd, "\iota_X"'] & & \\
& Y \times_{\operatorname{Spec} k} \operatorname{Spec} l \arrow[r] \arrow[d, "\iota_Y"] & \operatorname{Spec} l \arrow[d] \\
X \arrow[r, shift left, "\pi"] \arrow[r, shift right, "\rho"'] & Y \arrow[r] & \operatorname{Spec} k
\end{tikzcd}$$

Then we get $\pi_l : X \times_{\operatorname{Spec} k} \operatorname{Spec} l \to Y \times_{\operatorname{Spec} k} \operatorname{Spec} l$ and $\rho_l : X \times_{\operatorname{Spec} k} \operatorname{Spec} l$ which are the induced maps of l-schemes, by the universal property of fibered product. Suppose $\pi_l = \rho_l$.

We first show that $\pi = \rho$ on the level of sets. Let $p \in X$, we want to show that $\pi(p) = \rho(p)$. Since $X \times_{\operatorname{Spec} k} \operatorname{Spec} l \to X$ is surjective, there exists $q \in X \times_{\operatorname{Spec} k} \operatorname{Spec} l$ such that $\iota_X(q) = p$, then we have

$$\pi(p) = \pi \circ \iota_X(q) = \iota_Y \circ \pi_l(q) = \iota_Y \circ \rho_l(q) = \rho \circ \iota_X(q) = \rho(p),$$

as we desired.

We next show that $\pi = \rho$ on the level of scheme. We may reduced the case where X and Y are affine. Suppose $X = \operatorname{Spec} A$ and $Y = \operatorname{Spec} B$, by Proposition 11.1.1, we have $X \times_{\operatorname{Spec} k} \operatorname{Spec} l \cong \operatorname{Spec}(A \otimes_k l)$ and $Y \times_{\operatorname{Spec} k} \operatorname{Spec} l \cong \operatorname{Spec}(B \otimes_k l)$. Since $\pi_l = \rho_l$, we have $\pi_l^\sharp = \rho_l^\sharp$. Consider the following diagram.

$$\begin{tikzcd}
B \arrow[r, shift left, "\pi^\sharp"] \arrow[r, shift right, "\rho^\sharp"'] \arrow[d, "\iota_Y^\sharp"'] & A \arrow[d, "\iota_X^\sharp"] \\
B \otimes_k l \arrow[r, shift left, "\pi_l^\sharp"] \arrow[r, shift right, "\rho_l^\sharp"'] & A \otimes_k l
\end{tikzcd}$$

we want to show that $\pi^\sharp = \rho^\sharp$. Let $b \in B$, then we have

$$\iota_X^\sharp \circ \pi^\sharp(b) = \pi_l^\sharp \circ \iota_Y^\sharp(b) = \rho_l^\sharp \circ \iota_Y^\sharp(b) = \iota_X^\sharp \circ \rho^\sharp(b),$$

since $\iota_X^\sharp$ is injective, we have $\pi^\sharp(b) = \rho^\sharp(b)$, i.e., $\pi^\sharp = \rho^\sharp$. We are done! □

Proposition 11.2.9

Suppose $\pi : X \to Y$ is a(n affine) morphism over k, and l/k is a field extension. Then π is a closed embedding if and only if $\pi \times_k l : X \times_k l \to Y \times_k l$ is closed embedding.

Remark 11.11 The affine hypothesis is not necessary for this result, but it makes the proof easier, and the this is the situation in which we will most need it.

Proof If $\pi : X \to Y$ is a closed embedding, we may suppose that $X = \operatorname{Spec} B/I$ and $Y = \operatorname{Spec} B$. By Proposition 11.1.1, we know that $X \times_k l \cong \operatorname{Spec}(B/I \otimes_k l)$ and $Y \times_k l \cong \operatorname{Spec}(B \otimes_k l)$. Clearly $B \otimes_k l \to B/I \otimes_k l$ is surjective, and therefore $\pi \times_k l : X \times_k l \to Y \times_k l$ is closed embedding.

Conversely, if $\pi \times_k l : X \times_k l \to Y \times_k l$ is a closed embedding,, we want to show that $\pi : X \to Y$ is closed embedding. Let $\operatorname{Spec} B \subseteq Y$ be an affine open subset of Y, since $\pi : X \to Y$ is affine, we may assume $\pi^{-1}(\operatorname{Spec} B) = \operatorname{Spec} A$. Since $\pi \times_k l$ is closed embedding, by Proposition 11.1.1, we know that $B \otimes_k l \to A \otimes_k l$ is surjective. Consider the following exact sequence.

$$B \longrightarrow A \longrightarrow \operatorname{Coker}(B \to A) \longrightarrow 0$$

By Exercise 2.37, we know that $\square \otimes_k l$ is a right-exact covariant functor, so we have the exact sequence

$$B \otimes_k l \longrightarrow A \otimes_k l \longrightarrow \operatorname{Coker}(B \to A) \otimes_k l \longrightarrow 0.$$

Since $B \otimes_k l \to A \otimes_k l$ is surjective, we know that $\operatorname{Coker}(B \to A) \otimes_k l = 0$, then $\operatorname{Coker}(B \to A) = 0$, i.e., $B \to A$ is surjective. It follows that $\pi : X \to Y$ is closed embedding. □

✍ **Exercise 11.3 Seemingly pathological behavior in nonpathological circumstances** Suppose k is a field, and $A = k(x) \otimes_k k(y)$. Show that A is a (nonzero) localization of $k[x, y]$, hence an integral domain. Thus (0) is the unique minimal prime ideal of A. Show that the remaining prime ideals of A correspond to ideals $(f(x, y)) \subseteq k[x, y]$, where $f(x, y)$ is an irreducible polynomial in $k[x, y]$ containing both the variables x and y.

Proof Say $S_x = k[x] \setminus (0)$ and $S_y = k[y] \setminus (0)$, then we have

$$k(x) \otimes_k k(y) = S_x^{-1}k[x] \otimes_k S_y^{-1}k[y] \cong (S_x \otimes S_y)^{-1}(k[x] \otimes_k k[y]),$$

Let $S = \{p(x)q(y) : p(x) \in S_x \text{ and } q(y) \in S_y\}$, by Lemma 11.2.1, we have $k[x] \otimes_k k[y] \cong k[x, y]$, and therefore we have

$$S^{-1}k[x, y] \cong (S_x \otimes S_y)^{-1}(k[x] \otimes_k k[y]).$$

Since $k[x, y]$ is integral domain, we know that $S^{-1}k[x, y]$ is integral domain, thus (0) is the unique minimal prime ideal of A. Let $\mathfrak{p}$ be an prime ideal of A, by our isomorphism, $\mathfrak{p}$ corresponds to the prime ideal of $k[x, y]$

which don't meet S, i.e., $(f(x,y)) \subseteq k[x,y]$, where $f(x,y)$ is an irreducible polynomial in $k[x,y]$ containing both the variables x and y. □

This example will come up again in remarks of Chapter 13 and Chapter 25. The idea in the solution to Exercise 11.3 also yields the following.

✍ **Exercise 11.4 Unimportant but fun exercise** Show that $\operatorname{Spec} \mathbb{Q}(t) \otimes_{\mathbb{Q}} \mathbb{C}$ has closed points in natural correspondence with the transcendental complex numbers. This scheme doesn't come up in nature, but it is certainly neat!

Proof Note that

$$\mathbb{Q}(t) \otimes_{\mathbb{Q}} \mathbb{C} \cong (\mathbb{Q}(t)_{(0)} \otimes_{\mathbb{Q}[t]} \mathbb{Q}[t]) \otimes_{\mathbb{Q}} \mathbb{C} \cong \mathbb{Q}[t]_{(0)} \otimes_{\mathbb{Q}[t]} \mathbb{C}[t] \cong S^{-1}C[t]$$

where $S = \mathbb{Q}[t] \setminus (0)$, hence $\operatorname{Spec}(\mathbb{Q}(t) \otimes_{\mathbb{Q}} \mathbb{C}) \cong \operatorname{Spec} S^{-1}\mathbb{C}[t]$. The prime ideal of $S^{-1}\mathbb{C}[t]$ is the prime ideal of $\mathbb{C}[t]$ which do not meet $\mathbb{Q}[t] \setminus (0)$. Let $\mathfrak{p} = (t-a)$ be the prime ideal of $\mathbb{C}[t]$ which do not meet $\mathbb{Q}[t] \setminus (0)$, then for all $f(t) \in \mathbb{Q}[t] \setminus (0)$, $f(a) \neq 0$, i.e., a is transcendental complex numbers. □

11.3 Interpretations: Pulling back families, and fibers of morphisms

11.3.1 Pulling back families

Before making any definitions, we give a motivating informal example. Consider the "family of curves"

$$y^2 = x^3 + tx$$

in the xy-plane parametrized by t. Translation: consider

$$\operatorname{Spec} k[x,y,t]/(y^2 - x^3 - tx) \longrightarrow \operatorname{Spec} k[t].$$

If we "pull back this family" to the uv-plane via $uv = t$, we get the family

$$y^2 = x^3 + uvx.$$

If instead we set t to 3, we get the curve

$$y^2 = x^3 + 3x,$$

which we interpret as the fiber of the original family above $t = 3$. You may have noticed the fibered products underlying in there constructions:

$$\begin{tikzcd}
\operatorname{Spec} k[x,y,u,v]/(y^2 - x^3 - uvx) \arrow[r] \arrow[d] & \operatorname{Spec} k[x,y,t]/(y^2 - x^3 - tx) \arrow[d] \\
\operatorname{Spec} k[u,v] \arrow[r, "uv \leftarrow t"] & \operatorname{Spec} k[t]
\end{tikzcd}$$

and

$$\begin{tikzcd}
\operatorname{Spec} k[x,y]/(y^2 - x^3 - 3x) \arrow[r] \arrow[d] & \operatorname{Spec} k[x,y,t]/(y^2 - x^3 - tx) \arrow[d] \\
\operatorname{Spec} k \arrow[r, "3 \leftarrow t"] & \operatorname{Spec} k[t]
\end{tikzcd}$$

We now formalize this.

Suppose $Y \to Z$ is a morphism. We interpret this as a "family of schemes parametrized by a **base scheme** (or just plain **base**) Z." Then if we have another another morphism $\psi : X \to Z$, we interpret the induced map

$X \times_Z Y \to X$ as the "pulled back family" (see Figure 11.1).

$$\begin{tikzcd} X \times_Z Y \arrow[r] \arrow[d, "\text{pulled back family}"'] & Y \arrow[d, "\text{family}"] \\ X \arrow[r, "\psi"] & Z \end{tikzcd}$$

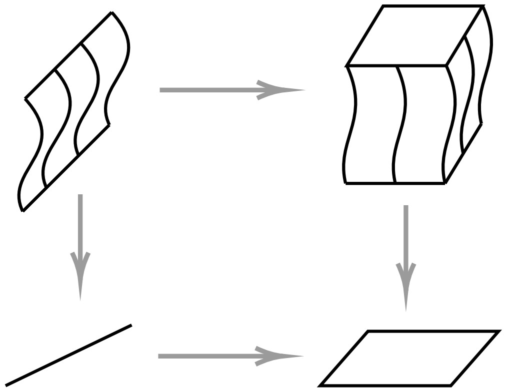


Figure 11.1: A picture of a pulled back family

Remark 11.12 **Informal remark:** imagine Z as a line (the parameter axis), and Y as a family of curves floating above Z. When we map another space X to Z via ψ, we pull Y back along ψ to above X, forming the new family $X \times_Z Y$. This is analogous to changing the coordinate system or the parameterization method.

We sometimes say that $X \times_Z Y$ is the **scheme-theoretic preimage** (or scheme-theoretic pullback, or scheme-theoretic inverse image, or inverse image scheme) of Y. (Our fourth-coming discussion of fibers may give some motivation for this.) For this reason, fibered product is often called **base change** or **change of base** or **pullback**. In addition to the various names for a Cartesian diagram given in §2.2.4, in algebraic geometry it is often called a **base change diagram** or a **pullback diagram**, and $X \times_Z Y \to X$ is called the **pullback** of $Y \to Z$ by ψ, and $X \times_Z Y$ is called the **pullback** of Y by ψ. One often uses the phrase "over X" or "above X" when discussing $X \times_Z Y$, especially if X is a locally closed subscheme of Z.

Remark 11.13 **Random side remark:** scheme theoretic preimage always makes sense, while the notion of scheme-theoretic image in somehow problematic, as discussed in §10.4.1.

11.3.2 Fibers of morphisms

Recall Proposition 9.3.13, that finite morphisms have finite fibers, you will not find the following discussion surprising. A special case of pull pack is the notion of a fiber of a morphism. We motivate this with the notion of fiber in the category of topological spaces.

Proposition 11.3.1

If $Y \to Z$ is a continuous map of topological spaces, and X is a point p of Z, then the fiber of Y over p (the set-theoretic fiber, with the induced topology) is naturally identified with $X \times_Z Y$.

Proof Say the continuous map $\pi : Y \to Z$, we want to show that $\pi^{-1}(p)$ is the fibered product of $\{p\}$ and Y

over Z. Consider the following diagram,

$$
\begin{tikzcd}
W \arrow[dr, dashed] \arrow[drr, bend left, "\alpha"] \arrow[ddr, bend right, "\beta"'] & & \\
& \pi^{-1}(p) \arrow[r, hook, "i_1"'] \arrow[d, "\pi|_{\{p\}}"] & Y \arrow[d, "\pi"] \\
& \{p\} \arrow[r, hook, "i_2"'] & Z
\end{tikzcd}
\tag{11.7}
$$

where $W \in \mathrm{obj}(\mathbf{Top})$ be any topological space. We want to show that there exists unique continuous map $\theta : W \to \pi^{-1}(p)$ such that diagram (11.7) commutes. Define $\theta : W \to \pi^{-1}(p)$ by setting $\theta = \pi^{-1} \circ \beta$. Pick $q \in W$, by $\pi \circ \alpha = i_2 \circ \beta$, we have

$$i_1 \circ \theta(q) = i_1 \circ \pi^{-1} \circ \beta(q) = \pi^{-1} \circ \pi \circ \alpha(q) = \alpha(q),$$

i.e., $i_1 \circ \theta = \alpha$. By the definition of θ, θ must be unique, so $\pi^{-1}(p) \cong \{p\} \times_Z Y$. □

More generally, we have:

Corollary 11.3.1

For any $\pi : X \to Z$, the fiber of $X \times_Z Y \to X$ over a point p of X is naturally identified with the fiber of $Y \to Z$ over $\pi(p)$.

Proof Consider the follows diagram.

$$
\begin{tikzcd}
& & \psi^{-1}(\pi(p)) \arrow[dr] \arrow[dd, "\psi"] \arrow[dll, dashed, shift left] & \\
\widetilde{\psi}^{-1}(p) \arrow[urr, dashed, shift left] \arrow[dr, hook] \arrow[dd, "\widetilde{\psi}"'] & & & Y \arrow[dd, "\psi"] \\
& X \times_Z Y \arrow[urr] \arrow[dd, "\widetilde{\psi}"] & \{\pi(p)\} \arrow[dr, hook] & \\
\{p\} \arrow[urr, "\pi"] \arrow[dr, hook] & & & Z \\
& X \arrow[urr, "\pi"'] & &
\end{tikzcd}
\tag{11.8}
$$

By Proposition 11.3.1, we know that $\widetilde{\psi}^{-1}(p)$ and $\psi^{-1}(\pi(p))$ are fibered products, by the universal property of fibered product, we give dashed arrows in diagram (11.8), and therefore $\widetilde{\psi}^{-1}(p) \cong \psi^{-1}(\pi(p))$, i.e., the fiber of $X \times_Z Y \to X$ over a point p of X is naturally identified with the fiber of $Y \to Z$ over $\pi(p)$. □

Motivated by topology, we return to the category of schemes.

Definition 11.3.1 (Scheme-theoretic fiber, generic fiber)

*Suppose $p \to Z$ is the inclusion of a point (not necessarily closed). More precisely, if p is a point with residue field K, consider the map $\operatorname{Spec} K \to Z$ sending $\operatorname{Spec} K$ to p, with the natural isomorphism of residue fields. Then if $g : Y \to Z$ is any morphism, the scheme-theoretic preimage of p is called the **scheme-theoretic fiber** of g above p, and is denoted $g^{-1}(p) := \operatorname{Spec} K \times_Z Y$.*

*If Z is irreducible, the scheme-theoretic fiber above the generic point of Z is called the **generic fiber** (of g).*

Remark 11.14 In an affine open subscheme $\operatorname{Spec} A$ containing p, p corresponds to some prime ideal $\mathfrak{p}$, and the morphism $\operatorname{Spec} K \to Z$ corresponds to the ring map $A \to A_{\mathfrak{p}}/\mathfrak{p}A_{\mathfrak{p}}$. This is the composition of localization and closed embedding, and thus can be computed by the tricks above. (Note that $p \to Z$ is a monomorphism, by Lemma 11.2.3.)

Proposition 11.3.2

The underlying topological space of the scheme-theoretic fiber of $X \to Y$ above a point p is naturally identified with the topological fiber $X \to Y$ above p.

Proof Say $\pi : X \to Y$. Let $p \in Y$, the scheme-theoretic fiber of π above p is $\operatorname{Spec}\kappa(p) \times_Y X$. As set, by Exercise 2.10,

$$\operatorname{Spec}\kappa(p) \times_Y X = \{(p, x) \in \operatorname{Spec}\kappa(p) \times X : \pi(x) = p\}.$$

Hence, as set, we have $\operatorname{Spec}\kappa(p) \times_Y X = \pi^{-1}(\operatorname{Spec}\kappa(p))$. Consider the following diagram (as topological spaces),

$$\begin{tikzcd}
\operatorname{Spec}\kappa(p) \times_Y X \arrow[dr, dashed, "\theta"] \arrow[drr, bend left, "\mathrm{pr}_X"] \arrow[ddr, bend right] & & \\
& \pi^{-1}(\operatorname{Spec}\kappa(p)) \arrow[r, hook, "i"'] \arrow[d, "\pi"] & X \arrow[d, "\pi"] \\
& \operatorname{Spec}\kappa(p) \arrow[r, hook] & Y
\end{tikzcd}$$

dashed arrow given by the universal property of $\pi^{-1}(\operatorname{Spec}\kappa(p))$ (as topological space), and $\theta : \operatorname{Spec}\kappa(p) \times_Y X \to \pi^{-1}(\operatorname{Spec}\kappa(p))$ is defined by $(p, x) \mapsto x$. Let V be an open subset of $\pi^{-1}(\operatorname{Spec}\kappa(p))$, its also an open subset of X. Say $\mathrm{pr}_X : \operatorname{Spec}\kappa(p) \times_Y X \to X$ and $i : \pi^{-1}(\operatorname{Spec}\kappa(p)) \hookrightarrow X$, by $i \circ \theta = \mathrm{pr}_X$, we have $\mathrm{pr}_X^{-1}(V) = \theta^{-1}(V)$ is an open subset of $\operatorname{Spec}\kappa(p) \times_Y X$, and therefore θ is continuous. We next show that θ is open, pick $U \subseteq \operatorname{Spec}\kappa(p) \times_Y X$ be an open subset, since $\operatorname{Spec}\kappa(p)$ only has two open set $\varnothing$ and $\operatorname{Spec}\kappa(p)$, hence the open subset of $\operatorname{Spec}\kappa(p) \times_Y X$ looks like $\mathrm{pr}_X^{-1}(V)$ where V is any open subset of X. Hence, by $i \circ \theta = \mathrm{pr}_X$, we know that θ is open. Since θ is continuous, open, and bijective, θ is homeomorphism. It follows that the underlying topological space of the scheme-theoretic fiber of $X \to Y$ above a point p is naturally identified with the topological fiber $X \to Y$ above p. $\square$

Proposition 11.3.3

Suppose that $\pi : Y \to Z$ and $\tau : X \to Z$ are morphisms, and $p \in X$ is a point. Then the fiber of $X \times_Z Y \to X$ over p is (isomorphic to) the base change to p of the fiber of $\pi : Y \to Z$ over $\tau(p)$.

Proof Consider the following diagram.

$$
\begin{tikzcd}
\psi^{-1}(p) \arrow[rrrr] \arrow[dr, dashed, shift left] \arrow[dddr] & & & & X \times_Z Y \arrow[dr] \arrow[dd, "\psi"] & \\
& p \times_{\tau(p)} (\tau(p) \times_Z Y) \arrow[ul, dashed, shift left] \arrow[dd] \arrow[drr] & & & & Y \arrow[dd, "\pi"] \\
& & & \tau(p) \times_Z Y \arrow[urr] \arrow[dd] & X \arrow[dr, "\tau"] & \\
& p \arrow[urrr, hook] \arrow[drr, "\tau"] & & & & Z \\
& & & \tau(p) \arrow[urr, hook] & &
\end{tikzcd}
$$

Each square is Cartesian square, and by the universal property of fibered product, we have

$$\psi^{-1}(p) \cong p \times_{\tau(p)} (\tau(p) \times_Z Y),$$

as we desired. □

Example 11.3 (Enlightening in several ways) Consider the projection of the parabola $y^2 = x$ to the x-axis over $\mathbb{Q}$, corresponding to the map of rings $\mathbb{Q}[x] \to \mathbb{Q}[y]$, with $x \mapsto y^2$. If $\mathbb{Q}$ alarms you, replace it with your favorite field and see what happens. (you should look at Figure 11.2, which is a flipped version of the parabola of Figure 4.8, and figure out how to edit it to reflect what we glean here.) Writing $\mathbb{Q}[y]$ as $\mathbb{Q}[x, y]/(y^2 - x)$ helps us interpret the morphism conveniently.

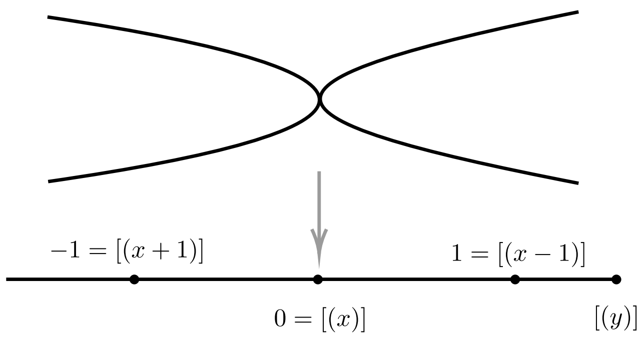


Figure 11.2: The map $\mathbb{C} \to \mathbb{C}$ given by $y \mapsto y^2$ (cf. Figure 4.8)

(i) Then the preimage of 1 is two points:

$$
\begin{aligned}
\operatorname{Spec} \mathbb{Q}[x, y]/(y^2 - x) \otimes_{\mathbb{Q}[x]} \mathbb{Q}[x]/(x-1) &\cong \operatorname{Spec} \mathbb{Q}[x, y]/(y^2 - x, x - 1) \\
&\cong \operatorname{Spec} \mathbb{Q}[y]/(y^2 - 1) \\
&\cong \operatorname{Spec} \mathbb{Q}[y]/(y-1) \coprod \operatorname{Spec} \mathbb{Q}[y]/(y+1).
\end{aligned}
$$

(ii) The preimage of 0 is one nonreduced point:

$$\operatorname{Spec} \mathbb{Q}[x, y]/(y^2 - x, x) \cong \operatorname{Spec} \mathbb{Q}[y]/(y^2).$$

(iii) The preimage of -1 is one reduced point, but of "size 2 over the base field".

$$\begin{aligned}\operatorname{Spec}\mathbb{Q}[x,y]/(y^2-x,x+1) &\cong \operatorname{Spec}\mathbb{Q}[y]/(y^2+1)\\ &\cong \operatorname{Spec}\mathbb{Q}[i]\\ &= \operatorname{Spec}\mathbb{Q}(i).\end{aligned}$$

(iv) The preimage of the generic point is again one reduced point, but of "size 2 over the residue field", as we verify now.

$$\operatorname{Spec}\mathbb{Q}[x,y]/(y^2-x)\otimes_{\mathbb{Q}[x]}\mathbb{Q}(x)\cong \operatorname{Spec}\mathbb{Q}[y]\otimes_{\mathbb{Q}[y^2]}\mathbb{Q}(y^2)$$

i.e., (informally) the Spec of the ring of polynomials in y divided by polynomials in y^2. A little thought shows you that in this ring you may invert any polynomial in y, as if $f(y)$ is any polynomial in y, then

$$\frac{1}{f(y)}=\frac{f(-y)}{f(y)f(-y)},$$

and the latter denominator is a polynomial in y^2. Thus

$$\mathbb{Q}[x,y]/(y^2-x)\otimes_{\mathbb{Q}[x]}\mathbb{Q}(x)\cong\mathbb{Q}(y)$$

which is a degree 2 field extension of $\mathbb{Q}(x)$ (note that $\mathbb{Q}(x)=\mathbb{Q}(y^2)$).

(You might want to work through the preimages of other points of $\mathbb{A}^1_{\mathbb{Q}}$, such as $[(x^3-4)]$ and $[(x^2-2)]$.)

Notice the following interesting fact: in each of the four cases, the number of preimage can be interpreted as 2, where you count to two in several ways: you can count points (as in the case of the preimage of 1); you can get nonreduced behavior (as in the case of the preimage of 0); or you can have a field extension of degree 2 (as in the case of the preimage of -1 or the generic point). In each case, the fiber is an affine scheme whose dimension as a vector space over the residue field of the point is 2. Number theoretic readers may have seen this behavior before. We will discuss this example again in Chapter 17. This is going to be symptomatic of a very important kind of morphism (a finite flat morphism, see Chapter 25).

Try to draw a picture of this morphism if you can, so you can develop a pictorial shorthand for what is going on. A good first approximation is the parabola of Figure 11.2, but you will want to somehow depict the peculiarities of (iii) and (iv).

✍ **Exercise 11.5 (Important for those with more arithmetic background)** What is the scheme-theoretic fiber of $\operatorname{Spec}\mathbb{Z}[i]\to\operatorname{Spec}\mathbb{Z}$ over the prime (p)? Your answer will depend on p, and there are four cases, corresponding to the four cases of Example 11.3 .

Proof

(i) If (p) is generic point, i.e., $p=0$, the scheme-theoretic fiber of $\operatorname{Spec}\mathbb{Z}[i]\to\operatorname{Spec}\mathbb{Z}$ over (0) is

$$\operatorname{Spec}\mathbb{Z}_{(0)}/(0)\mathbb{Z}_{(0)}\times_{\mathbb{Z}}\operatorname{Spec}\mathbb{Z}[i]\cong\operatorname{Spec}(\mathbb{Q}\otimes_{\mathbb{Z}}\mathbb{Z}[i])\cong\operatorname{Spec}\mathbb{Q}(i).$$

Hence the preimage of the generic point is $\operatorname{Spec}\mathbb{Q}(i)$, which is a reduced point of "size 2 over the residue field".

(ii) If $p=2$, the scheme-theoretic fiber of $\operatorname{Spec}\mathbb{Z}[i]\to\operatorname{Spec}\mathbb{Z}$ over (2) is

$$\begin{aligned}\operatorname{Spec}\mathbb{Z}_{(2)}/(2)\mathbb{Z}_{(2)}\times_{\mathbb{Z}}\operatorname{Spec}\mathbb{Z}[i] &\cong\operatorname{Spec}\mathbb{F}_2\otimes_{\mathbb{Z}}\mathbb{Z}[i]\\ &\cong\operatorname{Spec}\mathbb{F}_2[x]/(x^2+1)\\ &\cong\operatorname{Spec}\mathbb{F}_2[x]/(x+1)^2.\end{aligned}$$

Hence the preimage of (2) is nonreduced point.

(iii) If $p\equiv 1 \pmod 4$, assume that $p=4k+1$, by Euler's criterion, there exists an integer b such that

$b^2 \equiv -1 \pmod p$. Hence the scheme-theoretic fiber of $\operatorname{Spec}\mathbb{Z}[i] \to \operatorname{Spec} mbz$ over (p) is

$$\operatorname{Spec}\mathbb{Z}_{(p)}/(p)\mathbb{Z}_{(p)} \times_{\mathbb{Z}} \operatorname{Spec}\mathbb{Z}[i] \cong \operatorname{Spec}\mathbb{F}_p \coprod \operatorname{Spec}\mathbb{F}_p.$$

Then the preimage of (p) is two points.

(iv) If $p \equiv 3 \pmod 4$, by Euler's criterion, there are not integer b such that $b^2 \equiv -1 \pmod p$. Hence x^2+1 is irreducible over $\mathbb{F}_2[x]$, and therefore the scheme-theoretic fiber of $\operatorname{Spec}\mathbb{Z}[i] \to \operatorname{Spec}\mathbb{Z}$ over (p) is

$$\operatorname{Spec}\mathbb{Z}_{(p)}/(p)\mathbb{Z}_{(p)} \times_{\mathbb{Z}} \operatorname{Spec}\mathbb{Z}[i] \cong \operatorname{Spec}\mathbb{F}_p[x]/(x^2+1) \cong \operatorname{Spec}\mathbb{F}_p(i).$$

Hence the preimage of (p) is a reduced point of "size 2 over the residue field".

$\square$

Remark 11.15 We used following fact:

Lemma (Euler's criterion)

Let p be an odd prime and a be an integer coprime to p. Then

$$a^{\frac{p-1}{2}} \equiv 1 \pmod p$$

if and only if there is an integer x such that $x^2 \equiv a \pmod p$.

Exercise 11.6 (This exercise will give you practice in computing a fibered product over something that is not a field.) Consider the morphism of schemes $X = \operatorname{Spec} k[t] \to Y = \operatorname{Spec} k[u]$ corresponding to $k[u] \to k[t]$, $u \mapsto t^2$, where $\operatorname{char} k \neq 2$. Show that $X \times_Y X$ has two irreducible components. (What happens if $\operatorname{char} k = 2$? See Exercise in §11.5. A for a clue.)

Proof Clearly, $X \times_Y X = \operatorname{Spec}(k[t] \otimes_{k[u]} k[t])$. We claim the $k[t] \otimes_{k[u]} k[t] \cong k[x,y]/(x^2-y^2)$. Define $\theta : k[x,y] \to k[t] \otimes_{k[u]} k[t]$ by setting

$$x \longmapsto t \otimes 1, \qquad y \longmapsto 1 \otimes t.$$

Then θ is a surjective. We next calculate $\operatorname{Ker}\theta$. We claim that $\operatorname{Ker}\theta = (x^2-y^2)$. Let $f \in (x^2-y^2)$, since $\theta(x^2-y^2) = t^2 \otimes 1 - 1 \otimes t^2 = u \otimes 1 - 1 \otimes u = 0$, we have $(x^2-y^2) \subseteq \operatorname{Ker}\theta$. Conversely, let $f \in \operatorname{Ker}\theta$. Define $\psi : k[t] \otimes_{k[u]} k[t] \to k[x,y]/(x^2-y^2)$, by setting

$$t \otimes 1 \longmapsto x, \qquad 1 \otimes t \longmapsto y.$$

Then $\operatorname{Ker}(\psi \circ \theta) = (x^2-y^2)$. Since $\theta(f) = 0$, we have $\psi \circ \theta(f) = 0$, which implies that $f \in \operatorname{Ker}(\psi \circ \theta) = (x^2-y^2)$. Hence $\operatorname{Ker}\theta = (x^2-y^2)$. By the Fundamental Theorem of Isomorphism of rings, we have

$$k[t] \otimes_{k[u]} k[t] \cong k[x,y]/(x^2-y^2).$$

Hence

$$\begin{aligned} X \times_Y X &\cong \operatorname{Spec} k[x,y]/(x^2-y^2) \\ &\cong \operatorname{Spec}(k[x,y]/(x-y) \times k[x,y]/(x+y)) \\ &\cong \operatorname{Spec} k[x] \coprod \operatorname{Spec} k[x] \\ &\cong \mathbb{A}^1_k \coprod \mathbb{A}^1_k. \end{aligned}$$

If $\operatorname{char} k = 2$, we have

$$X \times_Y X \cong \operatorname{Spec} k[x,y]/(x^2-y^2) \cong \operatorname{Spec} k[x,y]/(y-x)^2,$$

which is nonreduced. $\square$

11.3.3 A first view of a blow-up

Exercise 11.7 (Important concrete exercise) (The discussion here immediately generalizes to $\mathbb{A}^n_A$.) Define a closed subscheme $\mathrm{Bl}_{(0,0)}\mathbb{A}^2_k$ of $\mathbb{A}^2_k \times_k \mathbb{P}^1_k$ as follows (see Figure 11.3). If the coordinates on $\mathbb{A}^2_k$ are x, y, and the projective coordinate on $\mathbb{P}^1_k$ are u, v, this subscheme is cut out in $\mathbb{A}^2_k \times_k \mathbb{P}^1_k$ by the single equation $xv = yu$. (You may wish to interpret $\mathrm{Bl}_{(0,0)}\mathbb{A}^2_k$ as follows. The $\mathbb{P}^1_k$ parametrizes lines through the origin. The blow-up corresponds to ordered pairs of (point p, line l) such that $(0,0)$ and p both lie on l.)

(i) Describe the fiber of the morphism $\mathrm{Bl}_{(0,0)}\mathbb{A}^2_k \to \mathbb{P}^1_k$ over each closed point of $\mathbb{P}^1_k$.

(ii) Show that the morphism $\mathrm{Bl}_{(0,0)}\mathbb{A}^2_k \to \mathbb{A}^2_k$ is an isomorphism away from $(0,0) \in \mathbb{A}^2_k$.

(iii) Show that the fiber of $\mathrm{Bl}_{(0,0)}\mathbb{A}^2_k \to \mathbb{A}^2_k$ over $(0,0)$ is an effective Cartier divisor (Definition 10.5.2, a closed subscheme that is locally cut out by a single equation, which is not a zero divisor). It is called the **exceptional divisor**.

We will discuss blow-ups in Chapter 23.

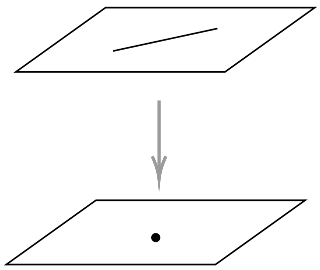


Figure 11.3: A first example of a blow-up

Proof

(i) Consider $\mathbb{A}^2_k \times_k \mathbb{P}^1_k$, we have

$$\begin{aligned}
\mathbb{A}^2_k \times_k \mathbb{P}^1_k &= \operatorname{Spec} k[x,y] \times_{\operatorname{Spec} k} \operatorname{Proj} k[u,v] \\
&\cong \left(\operatorname{Spec} k[x,y] \times_{\operatorname{Spec} k} \operatorname{Spec} k\left[\frac{u}{v}\right]\right) \cup \left(\operatorname{Spec} k[x,y] \times_{\operatorname{Spec} k} \operatorname{Spec} k\left[\frac{v}{u}\right]\right) \\
&\cong \operatorname{Spec}\left(k[x,y] \otimes_k k\left[\frac{u}{v}\right]\right) \cup \operatorname{Spec}\left(k[x,y] \otimes_k k\left[\frac{v}{u}\right]\right) \\
&\cong \operatorname{Spec} k\left[x,y,\frac{u}{v}\right] \cup \operatorname{Spec} k\left[x,y,\frac{v}{u}\right].
\end{aligned}$$

Since $\mathrm{Bl}_{(0,0)}\mathbb{A}^2_k$ is cut out by $xv = yu$, we know that

$$\mathrm{Bl}_{(0,0)}\mathbb{A}^2_k \cong \operatorname{Spec} \frac{k[x,y,\frac{u}{v}]}{(x - \frac{u}{v}y)} \cup \operatorname{Spec} \frac{k[x,y,\frac{v}{u}]}{(\frac{v}{u}x - y)} \cong \operatorname{Spec} k\left[y,\frac{u}{v}\right] \cup \operatorname{Spec} k\left[x,\frac{v}{u}\right]. \tag{11.9}$$

Let $P = [a:b]$ be a closed point of $\mathbb{P}^1_k$, we may assume that $b \neq 0$, then P is in the affine piece $\operatorname{Spec} k\left[\frac{u}{v}\right]$. The residue field of P is

$$\kappa(P) = \mathscr{O}_{\mathbb{P}^1,P}/\mathfrak{m}_P \cong k\left[\frac{u}{v}\right] \Big/ \left(\frac{u}{v} - \frac{a}{b}\right).$$

Hence the fiber of the morphism $\mathrm{Bl}_{(0,0)}\mathbb{A}^2_k \to \mathbb{P}^1_k$ is

$$\begin{aligned}
\operatorname{Spec} \kappa(P) \times_{\mathbb{P}^1_k} \mathrm{Bl}_{(0,0)}\mathbb{A}^2_k &\cong \operatorname{Spec} \kappa(P) \times_{k[\frac{u}{v}]} \operatorname{Spec} k\left[y,\frac{u}{v}\right] \\
&\cong \operatorname{Spec}\left(\kappa(P) \otimes_{k[\frac{u}{v}]} k\left[y,\frac{u}{v}\right]\right) \\
&\cong \operatorname{Spec} k\left[y,\frac{u}{v}\right] \Big/ \left(\frac{u}{v} - \frac{a}{b}\right) \\
&\cong \operatorname{Spec} k[y] \cong \mathbb{A}^1_k.
\end{aligned}$$

(ii) Note that $\mathbb{A}_k^2 \setminus \{(0,0)\} = \operatorname{Spec} k[x,y]_x \cup \operatorname{Spec} k[x,y]_y$, we claim that $\operatorname{Spec} k\left[x,y,\frac{u}{v}\right] / \left(x-\frac{u}{v}y\right) \cong \operatorname{Spec} k[x,y]_y$. Say $t=\frac{u}{v}$, define $\varphi : k[x,y]_y \to k[x,y,t]/(x-ty)$, by setting

$$x \longmapsto ty, \qquad y \longmapsto y.$$

Define $\psi : k[x,y,t]/(x-ty) \to k[x,y]_y$ by setting

$$y \longmapsto y, \qquad t \longmapsto \frac{x}{y}.$$

It is easy to see that $\psi\circ\varphi = \mathrm{id}_{k[x,y]_y}$ and $\varphi\circ\psi = \mathrm{id}_{k[x,y,t]/(x-ty)}$. Hence $\operatorname{Spec} k\left[x,y,\frac{u}{v}\right] / \left(x-\frac{u}{v}y\right) \cong \operatorname{Spec} k[x,y]_y$. Similarly, we have $\operatorname{Spec} k\left[x,y,\frac{v}{u}\right] / \left(\frac{v}{u}x-y\right) \cong \operatorname{Spec} k[x,y]_x$. Since they can glue together, we have $\mathrm{Bl}_{(0,0)} \mathbb{A}_k^2 \cong \mathbb{A}_k^2 \setminus \{(0,0)\}$

(iii) By (11.9), the fiber of $\mathrm{Bl}_{(0,0)} \mathbb{A}_k^2 \to \mathbb{A}_k^2$ over $(0,0)$ is

$$\begin{aligned}
\operatorname{Spec} \kappa((x,y)) \times_{\mathbb{A}_k^2} \mathrm{Bl}_{(0,0)} \mathbb{A}_k^2 &\cong \left(\operatorname{Spec} k \times_{k[x,y]} \operatorname{Spec} k\left[y,\frac{u}{v}\right]\right) \cup \left(\operatorname{Spec} k \times_{k[x,y]} \operatorname{Spec} k\left[x,\frac{v}{u}\right]\right) \\
&\cong \operatorname{Spec}\left(k[x,y]/(x,y) \otimes_{k[x,y]} k\left[x,y,t\right]/(x-yt)\right) \\
&\quad \cup \operatorname{Spec}\left(k[x,y]/(x,y) \otimes_{k[x,y]} k\left[x,y,s\right]/(sx-y)\right) \\
&\cong \operatorname{Spec}(k\left[x,y,t\right]/(x-yt))/(x,y)^e \cup \operatorname{Spec}(k\left[x,y,s\right]/(sx-y))/(x,y)^e \\
&\cong \operatorname{Spec} k[y,t]/(y) \cup \operatorname{Spec} k[x,s]/(x) \\
&\cong \mathbb{P}_k^1,
\end{aligned}$$

where $t=\frac{u}{v}$ and $s=\frac{v}{u}$, and they glue together via $t=\frac{1}{s}$. Hence the fiber of $\mathrm{Bl}_{(0,0)} \mathbb{A}_k^2 \to \mathbb{A}_k^2$ over $(0,0)$ is an effective Cartier divisor.

□

We haven't yet discussed regularity, but here is a hang-waving argument suggesting that the $\mathrm{Bl}_{(0,0)} \mathbb{A}_k^2$ is "smooth": the preimage above either standard open set $U_i \subseteq \mathbb{P}^1$ is isomorphic to $\mathbb{A}^2$. Thus "the blow-up is a surgery that takes the smooth surface $\mathbb{A}_k^2$, cuts out a point, and glues back in a $\mathbb{P}^1$, in such a way that the outcome is a another smooth surface."

11.3.4 General fibers, generic fibers, generically finite morphisms

The phrase "generic fiber" and "general fiber" parallel the phrases "generic point" and "general point" (Definition 4.6.6).

Definition 11.3.2 (General fiber)

*Suppose $\pi : X \to Y$ is a morphism of schemes. When one says the **general fiber** of π has a certain property, this means that there exists a dense open subset $U \subseteq Y$ such that the fibers above any point in U have the property.*

Remark 11.16 When one says the generic fiber of $\pi : X \to Y$ (Definition 11.3.1), this implicitly means that Y is irreducible, and the phrase refers to the fiber over the generic point. **General fiber and generic fiber are not the same thing!** Clearly if something holds for the general fiber, then it holds for the generic fiber, but the converse is not always true. However, in good circumstances, it can be — properties of the generic fiber extend to an honest open neighborhood.

Example 11.4 For example, if Y is irreducible and Noetherian, and π is finite type, then if the generic fiber of π is empty (resp., nonempty), then the general fiber is empty (resp., nonempty), by Chevalley's Theorem 9.4.1.

Proof By Chevalley's Theorem 9.4.1, we know that $\pi(X)$ is constructible set (Definition 9.4.1). Let η be the generic point of Y. Since the generic fiber of π is empty, we have $\eta \notin \pi(X)$, and therefore $\eta \in Y \setminus \pi(X)$. By Definition 9.4.1, $Y \setminus \pi(X)$ is constructible subset. By Proposition 9.4.1, there exists a locally closed subset $U \cap C$ such that $\eta \in U \cap C$ where U is a dense open subset of Y and C is closed. Since $\overline{\{\eta\}} = Y$, we may replace C by Y, then we have $\eta \in U \cap Y = U \subseteq Y \setminus \pi(X)$. Hence the fiber above any point in U is empty, as we desired. □

Definition 11.3.3 (Generically finite)

*If $\pi : X \to Y$ is finite type, we say π is **generically finite** is π is finite after base change to the generic point of each irreducible component (or equivalently, by Corollary 9.4.2, if the preimage of the generic point of each irreducible component of Y is a finite set.)*

Remark 11.17 The notion of generic finiteness can be defined in more general circumstances, see [Sta25, Remark 073A].

Proposition 11.3.4 ("Generically finite" usually means "Generally finite")

Suppose $\pi : X \to Y$ is an affine, finite type, generically finite morphism of locally Noetherian schemes, and Y is reduced. Then there is an open neighborhood of each generic point of Y over which π is actually finite. (The hypotheses can be weakened considerably, see [Sta25, Lemma 02NW].)

Proof Since $\pi : X \to Y$ is affine, we may reduced to the case that $Y = \operatorname{Spec} B$ and $X = \operatorname{Spec} A$. Since Y is reduced, B is reduced, i.e., $\mathfrak{N}(B) = (0)$. Since Y is irreducible, by Proposition 4.6.2 (ii), $\mathfrak{N}(B)$ is prime, and therefore B is integral domain. Since $\pi : X \to Y$ is of finite type, A is a finitely generated B-algebra, so we may write $A = B[x_1, \ldots, x_n]/I$ where I is an ideal of polynomial ring $B[x_1, \ldots, x_n]$. Since π is generically finite, the induced map $\operatorname{Spec} A \otimes_B K(B) \to \operatorname{Spec} K(B)$ is finite, hence $A \otimes_B K(B)$ is a finite $K(B)$ algebra. By Corollary 9.2.1, $K(B) \to A \otimes_B K(B)$ is integral, so there are monic polynomials $f_i(t) \in K(B)[t]$ such that $f_i(x_i) = 0$ in $A \otimes_B K(B)$. Let b be the product of the (finite number of) denominators appearing in the coefficients in the $f_i(x)$, then $f_i(t) \in B_b[t]$ is monic and $f_i(x_i) = 0$ in $A \otimes_B B_b \cong A_b$. Hence for each $f_i(x_i)$ there exists s_i such that $s_i f_i(x_i) = 0$ in A. Let $s = \prod_i s_i$, localize A_b at s, we have $f_i(x_i) = 0$ in A_{bs}. Hence $B_{bs} \to A \otimes_B B_{bs} \cong A_{bs}$ is integral, and therefore $\operatorname{Spec} A_{bs} \to \operatorname{Spec} B_{bs}$ is integral. Since π is of finite type, we know that $\operatorname{Spec} A_{bs} \to \operatorname{Spec} B_{bs}$ is of finite type. By Proposition 9.3.20, $\operatorname{Spec} A_{bs} \to \operatorname{Spec} B_{bs}$ is finite. □

11.3.5 ⋆⋆ Finitely presented families (morphisms) are locally pullbacks of particularly nice families

11.4 Properties preserved by base change

All "reasonable" properties of morphisms are preserved under base change (cf. Definition 9.1.1 (ii)). We discuss this, and in §11.5.1 we will explain how to fix those that don't fit this pattern.

We have already shown that the notion of "open embedding" is preserved by base change (Proposition 9.1.4). We did this by explicitly describing what the fibered product of an open embedding is: if $Y \hookrightarrow Z$ is an open embedding, and $\psi : X \to Z$ is any morphism, then we check that the open subscheme $\psi^{-1}(Y)$ of X satisfies the universal property of fibered products.

We have also shown that the notion of "closed embedding" is preserved by base change (Proposition

11.2.1). In other words, given a Cartesian diagram

$$\begin{tikzcd} W \arrow[r] \arrow[d] & Y \arrow[d, hook, "\text{cl. emb.}"] \\ X \arrow[r] & Z \end{tikzcd}$$

where $Y \hookrightarrow Z$ is a closed embedding, $W \to X$ is as well.

Proposition 11.4.1

The closed embeddings and locally closed embeddings are both "reasonable" class of morphisms of schemes (Definition 9.1.1).

Proof The composition of closed embeddings is also closed embedding (Proposition 10.1.3), i.e., closed embedding is preserved by composition. By Proposition 11.2.1, closed embeddings are preserved by base change. By Proposition 10.1.4, the property of being a closed embedding is affine-local on the target. Hence closed embeddings is reasonable class of morphisms of schemes.

By Proposition 10.2.4, the composition of locally closed embeddings is also a locally closed embedding. Since open embeddings and closed embeddings local on the target (Proposition 9.1.3), locally closed embeddings is local one the target. Consider the following cartesian diagrams

$$\begin{tikzcd} (X \times_Z U) \times_U Y \arrow[r] \arrow[d, hook, "\text{cl. emb.}"'] & Y \arrow[d, hook, "\text{cl. emb.}"] \\ X \times_Z U \arrow[r] \arrow[d, hook, "\text{op. emb.}"'] & U \arrow[d, hook, "\text{op. emb.}"] \\ X \arrow[r] & Z \end{tikzcd}$$

where $X \times_Z U \hookrightarrow X$ is open embedding given by Proposition 9.1.4 and $(X \times_Z U) \times_U Y \hookrightarrow X \times_Z U$ is closed embedding given by Proposition 11.2.1. Hence $(X \times_Z U) \times_U Y \hookrightarrow X$ is locally closed embedding. By Exercise 2.13, we have $(X \times_Z U) \times_U Y \cong X \times_Z Y$, so $X \times_Z Y \hookrightarrow X$ is locally closed embedding, i.e., locally closed embedding is preserved by base change. Hence locally closed embeddings is reasonable class of morphisms of schemes. □

Corollary 11.4.1

Suppose $X \to Z$ and $Y \to Z$ are both locally closed embeddings, then $X \times_Z Y \to Z$ is a locally closed embedding.

Proof In the proof of Proposition 11.4.1, we know that locally closed embedding is preserved by base change. So $X \times_Z Y \hookrightarrow X$ and $X \times_Z Y \hookrightarrow Y$ are both locally closed embeddings. The composition of locally closed embeddings is locally closed embedding. Hence $X \times_Z Y \hookrightarrow Z$ is a locally closed embedding. □

Apply Corollary 11.4.1, we may define the scheme-theoretic intersection of two locally closed embeddings.

Definition 11.4.1 (Scheme-theoretic intersection of two locally closed embeddings)

Let $X \hookrightarrow Z$ and $Y \hookrightarrow Z$ be two locally closed embeddings, we defined the scheme-theoretic intersection of two locally closed embedding by setting $X \times_Z Y \hookrightarrow Z$.

Proposition 11.4.2

The locally principal closed subschemes (Definition 10.5.1) pull back to locally principal closed subschemes.

Proof Let $\pi : X \hookrightarrow Z$ be a locally principal closed subschemes, and $\psi : Y \to Z$ be any morphism. We want to show that $X \times_Z Y \to Y$ is locally principal closed subscheme. Since $X \hookrightarrow Z$ is a locally principal closed subscheme, there exists an open cover of Z, say V_i such that $\pi^{-1}(V_i)$ is cut out by s_i where $s_i \in \Gamma(V_i, \mathscr{O}_Z)$. Say $V_i = \bigcup_j \operatorname{Spec} B_{ij}$, then $\pi^{-1}(V_i) = \bigcup_j \operatorname{Spec} B_{ij}/(s_i|_{B_{ij}})$. Say $\psi^{-1}(\operatorname{Spec} B_{ij}) = \bigcup_k \operatorname{Spec} A_{ijk}$, then $\psi^{-1}(V_i) = \bigcup_j \bigcup_k \operatorname{Spec} A_{ijk}$, and therefore $Y = \bigcup_{i,j,k} \operatorname{Spec} A_{ijk}$. We next calculate $X \times_Z Y$,

$$\begin{aligned} X \times_Z Y &= \bigcup_{i,j,k} \operatorname{Spec} B_{ij}/(s_i|_{B_{ij}}) \times_{\operatorname{Spec} B_{ij}} \operatorname{Spec} A_{ijk} \\ &= \bigcup_{i,j,k} \operatorname{Spec} A_{ijk} \otimes_{B_{ij}} B_{ij}/(s_i|_{B_{ij}}) \\ &\cong \bigcup_{i,j,k} \operatorname{Spec} A_{ijk}/(s_i|_{B_{ij}})^e, \end{aligned}$$

the last $\cong$ given by Lemma 11.2.2. By Definition 10.5.1, $X \times_Z Y$ is a locally principal closed subscheme. $\square$

Similarly, other important properties are preserved by base changes.

Proposition 11.4.3

The following properties of morphisms are preserved by base change.

(a) quasi-compact
(b) affine morphism
(c) finite
(d) integral
(e) locally of finite type
(f) finite type
$\star$ *(g) locally of finite presentation*

Proof Direct check. $\square$

Corollary 11.4.2

The following properties of morphisms are all "reasonable" class of morphisms of schemes (Definition 9.1.1).

(a) quasi-compact
(b) affine morphism
(c) finite
(d) integral
(e) locally of finite type
(f) finite type
$\star$ *(g) locally of finite presentation*

Proof Direct check. $\square$

Proposition 11.4.4

(a) *The notion of "quasifinite morphism" (finite type + finite fibers, Definition 9.3.10) is preserved by base change.*

(b) *The notion of "quasifinite morphism" is "reasonable" class of morphisms of schemes (Definition 9.1.1).*

Proof

(a) Let $\pi : X \to Z$ be a quasi-finite morphism, and $\varphi : Y \to Z$ be any morphism of schemes. By Proposition 11.4.3, $X \times_Z Y \to Y$ is of finite type. It suffices to check that $X \times_Z Y \to Y$ is finite fibers. Let p be a point in Y, we want to show that $(X \times_Z Y) \times_Y \operatorname{Spec} \kappa(p)$ is a finite set. By Proposition 11.3.3, we know that

$$(X \times_Z Y) \times_Y \operatorname{Spec} \kappa(p) \cong \operatorname{Spec} \kappa(p) \times_{\operatorname{Spec} \kappa(\varphi(p))} (\operatorname{Spec} \kappa(\varphi(p)) \times_Z X).$$

Since $X \to Z$ is quasi-finite morphism, $\operatorname{Spec} \kappa(\varphi(p)) \times_Z X$ is finite set, so $\operatorname{Spec} \kappa(p) \times_{\operatorname{Spec} \kappa(\varphi(p))} (\operatorname{Spec} \kappa(\varphi(p)) \times_Z X)$ is also finite set, i.e., $X \times_Z Y \to Y$ is finite fibers.

(b) By part (a), we know that quasi-finite morphism is preserved by base change.

We first show that quasi-finite morphism is preserved by composition. Let $\pi : X \to Y$ and $\varphi : Y \to Z$ be two quasi-finite morphism, since finite type is preserved by composition, it suffices to show that $\pi^{-1} \circ \varphi^{-1}(p)$ is finite set, for all $p \in Z$. Since $\varphi^{-1}(p)$ is finite, we may assume $\varphi^{-1}(p) = \{q_1, \ldots, q_r\}$. Since π is finite fibers, so $\pi^{-1}(q_i)$ is finite for all i, and therefore $\pi^{-1}(\varphi^{-1}(p))$ is a finite set. Hence $\varphi \circ \pi$ is a quasi-finite morphism, i.e., quasi-finite is preserved by composition.

We next show that quasi-finiteness is local on the target. Recall Definition 9.1.1, property (iii) (a) is straightforward. We now check (iii) (b). Let $\pi : X \to Z$ be a quasi-finite morphism, and $\{V_i \hookrightarrow Y\}_i$ is an affine open cover of Y for which each restricted morphism $\pi_i : \pi^{-1}(V_i) \to V_i$ is quasi-finite, we need to show that π is quasi-finite. By Corollary 11.4.2, finite type is local on the target, so it suffices to show that π is finite fibers. We only need to consider this issue at the set level. Pick $p \in Z$, then p in some V_i, so $\pi^{-1}(p) \subseteq \pi^{-1}(V_i)$. Note that $\pi_i^{-1}(p) = \pi^{-1}(p) \cap \pi^{-1}(V_i) = \pi^{-1}(p)$, since π_i is finite fibers morphism, we know that $\pi^{-1}(p)$ is finite set, and therefore π is a quasi-finite morphism. Hence quasi-finiteness is local on the target.

□

Remark 11.18 The notion of "finite fibers" is not preserved by base change. $\operatorname{Spec} \overline{\mathbb{Q}} \to \operatorname{Spec} \mathbb{Q}$ has finite fibers, but $\operatorname{Spec} \overline{\mathbb{Q}} \otimes_{\mathbb{Q}} \overline{\mathbb{Q}} \to \operatorname{Spec} \overline{\mathbb{Q}}$ has one point for each element of $\operatorname{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$, see Exercise 11.1.

Proposition 11.4.5

(a) *The surjectivity is preserved by base change. (**Surjectivity has its usual meaning: surjective as a map of sets.**)*

(b) *Surjective morphisms form a "reasonable" class of morphisms of schemes (Definition 9.1.1).*

Proof

(a) Let $\pi : X \to S$ be a surjective morphism, and $\varphi : Y \to S$ be any morphism. We want to show that $\psi : X \times_S Y \to Y$ is a surjective morphism. Let $p \in Y$, by Proposition 11.3.3, we want to show that

$$\psi^{-1}(p) \cong (X \times_S Y) \times_Y \operatorname{Spec} \kappa(p) \cong \operatorname{Spec} \kappa(p) \times_{\operatorname{Spec} \kappa(q)} (\operatorname{Spec} \kappa(q) \times_S X)$$

is not empty, where $q = \varphi(p)$. Since $\pi : X \to S$ is surjective, there exits $s \in X$ such that $\pi(s) = q$.

Consider the following diagram,

$$\begin{tikzcd}
\operatorname{Spec}\kappa(p)\times_{\operatorname{Spec}\kappa(q)}\operatorname{Spec}\kappa(s) \arrow[r] \arrow[dr] \arrow[ddr] & \operatorname{Spec}\kappa(s) \arrow[drr] \arrow[ddrr] & & \\
& \psi^{-1}(p) \arrow[rr, "\pi"] \arrow[d] & & \operatorname{Spec}\kappa(q)\times_S X \arrow[d, "\pi"] \\
& \operatorname{Spec}\kappa(p) \arrow[rr] & & \operatorname{Spec}\kappa(q)
\end{tikzcd}$$

it suffices to show that $\operatorname{Spec}\kappa(p)\times_{\operatorname{Spec}\kappa(q)}\operatorname{Spec}\kappa(s)$ is non-empty. Let $k_1=\operatorname{Spec}\kappa(p)$, $k_2=\operatorname{Spec}\kappa(s)$, and $k_3=\operatorname{Spec}\kappa(q)$, then

$$\operatorname{Spec}\kappa(p)\times_{\operatorname{Spec}\kappa(q)}\operatorname{Spec}\kappa(s)\cong\operatorname{Spec}k_1\otimes_{k_3}k_2.$$

Note that $k_1\otimes_{k_3}k_2\neq 0$, by Zorn's Lemma $k_1\otimes_{k_3}k_2$ must have a maximal ideal, so $\operatorname{Spec}k_1\otimes_{k_3}k_2$ is not empty, and therefore $\psi^{-1}(p)$ is not empty, as we desired.

(b) Clearly, surjectivity is preserved by composition and local on the target. By part (a), we know that surjective morphisms form a "reasonable" class of morphisms of schemes.

□

Remark 11.19 On the other hand, injectivity is not preserved by base change — witness the bijection $\operatorname{Spec}\mathbb{C}\to\operatorname{Spec}\mathbb{R}$, which loses injectivity upon base change by $\operatorname{Spec}\mathbb{C}\to\operatorname{Spec}\mathbb{R}$ (see Example 11.2). This can be rectified (see §11.5).

Proposition 11.4.6 (cf. Proposition 11.2.6)

Suppose X and Y are integral finite type $\overline{k}$-schemes. Then $X\times_{\overline{k}}Y$ is an integral finite type $\overline{k}$-scheme.

Remark 11.20 Once we define "variety", this will become the important fact that the product of irreducible varieties over an algebraically closed field is an irreducible variety. The fact that the base field $\overline{k}$ is algebraically closed is important, see §11.5.

Proof Reduce to the case where X and Y are both affine, say $X=\operatorname{Spec}A$ and $Y=\operatorname{Spec}B$ with A and B are integral finitely generated $\overline{k}$-algebra. Since A and B are finitely generated, we know that $A\otimes_{\overline{k}}B$ is finitely generated $\overline{k}$-algebra. So we only need to show that $A\otimes_{\overline{k}}B$ is an integral domain.

Suppose $(\sum a_i\otimes b_i)(\sum a'_j\otimes b'_j)=0$ in $A\otimes_{\overline{k}}B$ with $a_i,a'_j\in A$, $b_i,b'_j\in B$, where both $\{b_i\}_i$ and $\{b'_j\}_j$ are linearly independent over $\overline{k}$, and a_1 and a'_1 are nonzero (these condition also implies that $\sum a_i\otimes b_i\neq 0$ and $\sum a'_j\otimes b'_j$). Since A is integral domain and a_1,a'_1 are nonzero, we have $a_1a'_1\neq 0$, so $D(a_1a'_1)\subseteq\operatorname{Spec}A$ is nonempty. By the Weak Nullstellensatz 4.2.1, there is a maximal ideal $\mathfrak{m}\subseteq A$ in $D(a_1a'_1)$ with $A/\mathfrak{m}=\overline{k}$. Define $\varphi: A\otimes_{\overline{k}}B\to\overline{k}\otimes_{\overline{k}}B\cong B$ by setting

$$\sum_i a_i\otimes b_i\longmapsto\sum_i\overline{a_i}b_i.$$

It is easy to check that φ is a well-defined ring homomorphism. Then we have

$$\varphi\left(\left(\sum a_i\otimes b_i\right)\left(\sum a'_j\otimes b'_j\right)\right)=\varphi\left(\sum a_i\otimes b_i\right)\cdot\varphi\left(\sum a'_j\otimes b'_j\right)=\left(\sum\overline{a_i}b_i\right)\left(\sum\overline{a'_j}b'_j\right)=0,$$

it contradicts to the fact that B is integral domain. So $A\otimes_{\overline{k}}B$ is integral domain, combine that $A\otimes_{\overline{k}}B$ is finitely generated $\overline{k}$-algebra, we know that $X\times_{\overline{k}}Y$ is an integral finite type $\overline{k}$-scheme. □

11.5 ⋆ Properties not preserved by base change, and how to fix them

We saw in the previous section that many useful properties of morphisms are preserved by base change. Indeed, "preserved by base change" is one of the properties any "reasonable" class of morphisms should have (Definition 9.1.1). But some seemingly reasonable notions are not preserved by base change, and in this section we discuss how to modify them appropriately so that they are. We do this not for our own amusement, but because the resulting notions come up in nature, and perhaps in retrospect are the notions we should have started with.

The "universal" patch to this problem is as follows.

Definition 11.5.1 (Universally $\mathscr{P}$ morphism)

*If $\mathscr{P}$ is some property of schemes, then a morphism of schemes is said to be **universally** $\mathscr{P}$ if it remains $\mathscr{P}$ under any base change.*

Then the class of universally $\mathscr{P}$ morphism is preserved by base change.

An important example is the notion of **universally closed morphisms**. Another example is that of **universally injective morphisms**, which turns out to generalize (and "geometrize") purely inseparable field extensions.

One problem with "universally $\mathscr{P}$" morphisms is that it is a priori hard to determine whether a given morphism is universally $\mathscr{P}$ — you seem to have to check every single base change of the morphism. Finding other equivalent criteria is thus essential.

11.5.1 Geometric fiber

There are some notions that you should reasonably expect to be preserved by pullback based on your geometric intuition. Given a family in the topological category, fibers pull back in reasonable ways. So for example, any pullback of family in which all the fibers are irreducible will also have this property; ditto for connected. Unfortunately, both of these fail in algebraic geometry, as Example 11.2 shows:

$$\begin{array}{ccc} \operatorname{Spec}\mathbb{C}\coprod\operatorname{Spec}\mathbb{C} & \longrightarrow & \operatorname{Spec}\mathbb{C} \\ \downarrow & & \downarrow \\ \operatorname{Spec}\mathbb{C} & \longrightarrow & \operatorname{Spec}\mathbb{R} \end{array}$$

The family on the right (the vertical map) has irreducible and connected fibers (the fiber is $\operatorname{Spec}\mathbb{C}$, it is single point, so it is irreducible and connected), and the one on the left doesn't (not connected, since $\operatorname{Spec}\mathbb{C}\coprod\operatorname{Spec}\mathbb{C}$ is two not connected points). The same example shows that the notion of "integral fibers" doesn't behave well under pullback. And we used it in §11.4 to show that injectiveity isn't preserved by base change.

✍ **Exercise 11.8** Suppose k is a field of characteristic p, so $k(u)/k(u^p)$ is a purely inseparable extension. By considering $k(u)\otimes_{k(u^p)}k(u)$, show that the notion of "reduced fibers" does not necessarily behave well under pullback. (We will soon see that this happens only in characteristic p, in the presence of inseparability.)

Proof Consider $\pi:\operatorname{Spec}k(u)\to\operatorname{Spec}k(u^p)$, let $q\in\operatorname{Spec}k(u^p)$, then $\kappa(q)=k(u^p)$. So the fiber over q is

$$\begin{aligned}\operatorname{Spec}k(u^p)\times_{\operatorname{Spec}k(u^p)}\operatorname{Spec}k(u) &\cong \operatorname{Spec}k(u^p)\otimes_{k(u^p)}k(u)\\ &\cong \operatorname{Spec}k(u).\end{aligned}$$

Since $k(u)$ is a field, $k(u)$ is reduced, and therefore $\operatorname{Spec}k(u)$ is reduced, i.e., $\pi:\operatorname{Spec}k(u)\to\operatorname{Spec}k(u^p)$ has reduced fiber.

Consider $k(u) \otimes_{k(u^p)} k(u)$, we have

$$\begin{aligned} k(u) \otimes_{k(u^p)} k(u) \cong & k(u) \otimes_{k(u^p)} k(u^p)[x]/(x^p - u^p) \\ \cong & k(u) \otimes_{k(u^p)} (k(u^p)[x] \otimes_{k(u^p)[x]} k(u^p)[x]/(x^p - u^p)) \\ \cong & k(u)[x] \otimes_{k(u^p)[x]} k(u^p)[x]/(x^p - u^p) \\ \cong & k(u)[x]/(x^p - u^p). \end{aligned}$$

Pick $m \in \operatorname{Spec} k(u)$, then $\kappa(m) = k(u)$, so we have

$$\begin{aligned} \operatorname{Spec} \kappa(m) \times_{k(u)} (\operatorname{Spec} k(u) \times_{k(u^p)} \operatorname{Spec} k(u)) &\cong \operatorname{Spec} k(u) \times_{\operatorname{Spec} k(u)} \operatorname{Spec} k(u) \otimes_{k(u^p)} k(u) \\ &\cong \operatorname{Spec} k(u) \times_{k(u)} \operatorname{Spec} k(u)[x]/(x^p - u^p) \\ &\cong \operatorname{Spec} k(u) \otimes_{k(u)} k(u)[x]/(x^p - u^p) \\ &\cong \operatorname{Spec} k(u)[x]/(x^p - u^p). \end{aligned}$$

Hence $\operatorname{Spec} k(u) \times_{\operatorname{Spec} k(u^p)} \operatorname{Spec} k(u) \to \operatorname{Spec} k(u)$ do not have reduced fiber, i.e., the notion of "reduced fibers" does not necessarily behave will under pullback. □

We rectify this problem as follows.

Definition 11.5.2 (Geometric point)

*A **geometric point** of a scheme X is defined to be a morphism $\operatorname{Spec} k \to X$ where k is an algebraically closed field.*

Awkwardly, this is now the third kind of "point" of scheme! There are just plain points, which are elements of underlying set; there are Z-valued points (Z is a scheme), which are maps $Z \to X$, Definition 8.3.5; and there are geometric points. Geometric points are clearly a flavor of a scheme-valued point, but they are also an enriched version of a (plan) point: they are the data of a point with an inclusion of the residue field of the point in an algbraically closed field.

Definition 11.5.3 (Geometric fiber)

*A **geometric fiber** of a morphism $X \to Y$ is defined to be the fiber over a geometric point of Y, i.e., the fibered product with the geometric point $\operatorname{Spec} k \to Y$.*

*A morphism has **connected** (resp., **irreducible**, **integral**, **reduced**) **geometric fibers** if all its geometric fibers are connected (resp., irreducible, integral, reduced). One usually says that the morphism has **geometrically connected** (resp., **geometrically irreducible, geometrically integral, geometrically reduced**) fibers.*

*A k-scheme X is **geometrically connected** (resp., **geometrically irreducible**, **geometrically integral**, **geometrically reduced**) if the structure morphism $X \to \operatorname{Spec} k$ has geometrically connected (resp., irreducible, integral, reduced) fibers.*

We will soon see that to check any of these conditions, we need only base change to $\overline{k}$.

Remark 11.21 **Warning:** in some sources, in the definition of "geometric point", "algebraically closed" is replaced by "separably closed".

✍ **Exercise 11.9 (Exercise for the arithmetically-minded)** Show that for the morphism $\operatorname{Spec} \mathbb{C} \to \operatorname{Spec} \mathbb{R}$, all geometric fibers consist of two reduced points. (Cf. Example 11.2 .) Thus $\operatorname{Spec} \mathbb{C}$ is a geometrically reduced but not geometrically irreducible $\mathbb{R}$-scheme.

Proof The geometric point of $\operatorname{Spec} \mathbb{R}$ is morphism $\operatorname{Spec} k \to \operatorname{Spec} \mathbb{R}$, where k is algebraically closed field.

We shall to show that $\operatorname{Spec} k \times_{\mathbb{R}} \operatorname{Spec} \mathbb{C}$ consist of two reduced points. Consider $k \otimes_{\mathbb{R}} \mathbb{C}$,

$$\begin{aligned} k \otimes_{\mathbb{R}} \mathbb{C} \cong & k \otimes_{\mathbb{R}} \mathbb{R}[x]/(x^2+1) \\ \cong & k \otimes_{\mathbb{R}} (\mathbb{R}[x] \otimes_{\mathbb{R}[x]} \mathbb{R}[x]/(x^2+1)) \\ \cong & k[x] \otimes_{\mathbb{R}[x]} \mathbb{R}[x]/(x^2+1) \\ \cong & k[x]/(x^2+1) \\ \cong & k[x]/(x+i) \times k[x]/(x-i) \\ \cong & k \times k \end{aligned}$$

i.e., $\operatorname{Spec} k \times_{\mathbb{R}} \times \operatorname{Spec} \mathbb{C} \cong \operatorname{Spec} k \coprod \operatorname{Spec} k$, i.e., geometric fibers consist of two reduced points. Thus $\operatorname{Spec} \mathbb{C}$ is a geometrically reduced but not geometrically irreducible $\mathbb{R}$-scheme. □

Example 11.5 (k-scheme is reduced but not geometrically reduced) Let $k = \mathbb{F}_p(t)$, where p is a prime and t is an indeterminate. Consider the k-scheme $X = \operatorname{Spec} k[x]/(x^p - t)$, $x^p - t$ is irreducible on $k[x]$, so $k[x]/(x^p - t)$ is a field, and therefore X is reduced scheme.

Consider $X \times_k \overline{k} = \operatorname{Spec}(\overline{k} \otimes_k k[x]/(x^p - t))$,

$$\begin{aligned} \overline{k} \otimes_k k[x]/(x^p - t) \cong & \overline{k} \otimes_k (k[x] \otimes_{k[x]} k[x]/(x^p - t)) \\ \cong & \overline{k}[x] \otimes_{k[x]} k[x]/(x^p - t) \\ \cong & \overline{k}[x]/(x^p - t). \end{aligned}$$

Since $\overline{k}$ is algebraically closed, $x^p - t$ split on $\overline{k}[x]$, i.e., there exists $\alpha \in \overline{k}$ such that $\alpha^p = t$, and therefore $x^p - t = (x - \alpha)^p$, so $X \times_k \overline{k}$ is non-reduced scheme, i.e., X is not geometrically reduced.

Example 11.6 (k-scheme is connected but not geometrically connected) Consider $\mathbb{R}$-scheme $X = \operatorname{Spec} \mathbb{C}$. Since X is a single point, X is connected. Consider $X \times_{\mathbb{R}} \mathbb{C} = \operatorname{Spec}(\mathbb{C} \otimes_{\mathbb{R}} \mathbb{C})$, by Example 11.2 , we have $X \times_{\mathbb{R}} \mathbb{C} = \operatorname{Spec} \mathbb{C} \coprod \operatorname{Spec} \mathbb{C}$, which implies that $X \times_{\mathbb{R}} \mathbb{C}$ is not geometrically connected.

Example 11.7 (k-scheme is integral but not geometrically integral) Let $k = \mathbb{F}_p(t)$, where p is a prime and t is an indeterminate. Consider the k-scheme $X = \operatorname{Spec} k[x]/(x^p - t)$, by Example 11.5 , we know that $k[x]/(x^p - t)$ is a field, so $k[x]/(x^p - t)$ is an integral domain, and therefore X is an integral k-scheme.

Consider $X \times_k \overline{k}$, by Example 11.5 , we already know that

$$\overline{k} \otimes_k k[x]/(x^p - t) \cong \overline{k}[x]/(x^p - t) \cong \overline{k}[x]/((x - \alpha)^p),$$

where $\alpha \in \overline{k}$ and $\alpha^p = t$. Note that $\overline{k} \otimes_k k[x]/(x^p - t)$ not integral domain, so $X \times_k \overline{k}$ is not integral. It follows that X is not geometrically integral.

✍ **Exercise 11.10 (To convince geometers why geometric fibers are meaningful)** Recall Example 11.3 , the projection of the parabola $y^2 = x$ to the x-axis, corresponding to the map of rings $\mathbb{Q}[x] \to \mathbb{Q}[y]$, with $x \mapsto y^2$. Show that the geometric fibers of this map are always two points, except for those geometric fiber "over $0 = [(x)]$". (Note that $\operatorname{Spec} \mathbb{C} \to \operatorname{Spec} \mathbb{Q}[x]$ and $\operatorname{Spec} \overline{\mathbb{Q}} \to \operatorname{Spec} \mathbb{Q}[x]$, both corresponding to ring maps with $x \mapsto 0$, are both geometric points "above 0".)

Proof Say $X = \operatorname{Spec} \mathbb{Q}[x,y]/(x - y^2)$. Let $\operatorname{Spec} \overline{k} \to \operatorname{Spec} \mathbb{Q}[x]$, where $\overline{k}$ is an algebraically closed field. Consider $\overline{k} \otimes_{\mathbb{Q}[x]} \mathbb{Q}[x,y]/(x - y^2)$,

$$\begin{aligned} \overline{k} \otimes_{\mathbb{Q}[x]} \mathbb{Q}[x,y]/(x - y^2) \cong & \overline{k} \otimes_{\mathbb{Q}[x]} (\mathbb{Q}[x][y] \otimes_{\mathbb{Q}[x,y]} \mathbb{Q}[x,y]/(x - y^2)) \\ \cong & \overline{k}[y] \otimes_{\mathbb{Q}[x,y]} \mathbb{Q}[x,y]/(x - y^2) \\ \cong & \overline{k}[y]/(y^2 - a), \end{aligned}$$

where a is the image of x under the ring map $\mathbb{Q}[x] \to \overline{k}$.

If $a = 0$, then $\overline{k} \times_{\mathbb{Q}[x]} X = \operatorname{Spec} \overline{k}[y]/(y^2)$, i.e., a non-reduced point.

If $a \neq 0$, since $\overline{k}$ is algebraically closed, exits $b \in \overline{k}$ such that $b^2 = a$, by Chinese Remainder Theorem 5.4.2, then

$$\overline{k} \otimes_{\mathbb{Q}[x]} X \cong \overline{k}[y]/(y-b) \times \overline{k}[y]/(y+b) \cong \overline{k} \times \overline{k},$$

i.e., two points. □

Checking whether a k-scheme is geometrically connected etc. seems annoying: you need to check every single algebraically closed field containing k. However, in each of these four cases, the failure of nice behavior of geometric fibers can already be detected after a finite extension of fields. For example, $\operatorname{Spec}\mathbb{Q}(i) \to \operatorname{Spec}\mathbb{Q}$ is not geometrically connected, and in fact you only need to base change by $\operatorname{Spec}\mathbb{Q}(i)$ to see this (let $\overline{k}$ be an algebraically closed field with filed extension $\mathbb{Q} \to \overline{k}$, note that $\overline{k} \otimes_{\mathbb{Q}} \mathbb{Q}(i) \cong \overline{k}[x]/(x^2+1) \cong \overline{k} \times \overline{k}$). We make this precise as follows.

Note *Suppose X is a k-scheme. If K/k is a field extension, define $X_K = X \times_k \operatorname{Spec} K$. Consider the following twelve statements of the form "X_K is [property] for all fields [condition]":*

- $(C_K), (C_{K=\overline{K}}), (C_{\overline{k}}), (C_{k^s})$,
- $(I_K), (I_{K=\overline{K}}), (I_{\overline{k}}), (I_{k^s})$,
- $(R_K), (R_{K=\overline{K}}), (R_{\overline{k}}), (R_{k^p})$.

Here C means "connected", I means "irreducible", and R means "reduced". Also, k^s is the separable closure of k, and k^p is the perfect closure (see [Sta25, Lemma 046W], and purely inseparable extension, see [Mor12, page 45-49]) of k. If $\operatorname{char} k = 0$, then $\overline{k} = k^s$ and $k = k^p$, so life is simpler.

Thus (C_K) means "X_K is connected for all fields K", $(I_{\overline{k}})$ means "$X_{\overline{k}}$ is irreducible", and (R_{k^p}) means "X_{k^p} is reduced". Geometrically connected (resp., geometrically irreducible, geometrically reduced) translates to $(C_{K=\overline{K}})$ (resp., $(I_{K=\overline{K}})$, $(R_{K=\overline{K}})$).

Trivially (C_K) implies $(C_{K=\overline{K}})$ implies $(C_{\overline{k}})$, and (C_K) implies (C_{k^s}), and similarly (with appropriate changes) with "connected" replaced by "irreducible" and "reduced". See the following diagram,

$$\begin{tikzcd}
(C_K) \arrow[r, shift left] \arrow[d] \arrow[dr] & (C_{K=\overline{K}}) \arrow[l, dotted, shift left] \arrow[dl] \\
(C_{\overline{k}}) \arrow[r, dotted] & (C_{k^s})
\end{tikzcd}$$

connectedness conditions that will later turn out to be equivalent (solid arrows are trivial, dotted arrows are Proposition 11.5.1).

Proposition 11.5.1

(a) Suppose that E/F is a field extension, and A is an F-algebra. Then A is an F-subalgebra of $A \otimes_F E$.

(b) $(C_{K=\overline{K}})$ implies (C_K) and $(C_{\overline{k}})$ implies (C_{k^s}).

(c) $(I_{K=\overline{K}})$ implies (I_K) and $(I_{\overline{k}})$ implies (I_{k^s}).

(d) $(R_{K=\overline{K}})$ implies (R_K) and $(R_{\overline{k}})$ implies (R_{k^p}).

Proof

(a) It suffices to show that $A \to A \otimes_F E$ is injective. $\varphi : A \to A \otimes_F E$ is given by

$$a \longmapsto a \otimes 1.$$

Let $\varphi(a) = a \otimes 1 = 0$, we want to show that $a = 0$. If $a \neq 0$, since A can be seen as F-vector space (note that F is a field), there exists a F-linear map $f : A \to F$ such that $f(a) \neq 0$. Define $g : A \times E \to E$ by setting $g(x, e) = f(x)e$, g is an F-bilinear map, by the universal property of tensor product, there exits unique F-linear map $h : A \otimes_F E \to E$ such that $h \circ t = g$, where $t : A \times E \to A \otimes_F E$. In fact, $h : A \otimes_F E \to E$ is given by $h(x \otimes e) = g(x, e) = f(x)e$. Consider $h(a \otimes 1)$, since $a \otimes 1 = 0$, $h(a \otimes 1) = 0$. On the other hand, $h(a \otimes 1) = f(a) \cdot 1 = f(a)$, so $f(a) = 0$, a contradiction! Hence $a = 0$, and therefore $\varphi : A \to A \otimes_F E$ is an injective. It follows that A is an F-subalgebra of $A \otimes_F E$.

(b) If $(C_{K=\overline{K}})$ holds, let K/k be any field extension, we want to show that X_K is connected. Note that

$$X_{\overline{K}} = X \times_k \operatorname{Spec} \overline{K} = X \times_k (\operatorname{Spec} K \times_K \operatorname{Spec} \overline{K}) = X_K \times_K \operatorname{Spec} \overline{K},$$

consider the following diagram.

$$\begin{tikzcd} X_{\overline{K}} \arrow[r, two heads] \arrow[d] & X_K \arrow[r] \arrow[d] & X \arrow[d] \\ \operatorname{Spec} \overline{K} \arrow[r, two heads] & \operatorname{Spec} K \arrow[r] & \operatorname{Spec} k \end{tikzcd}$$

Since $K \to \overline{K}$ is an integral extension, by the Lying Over Theorem 9.2.2, $\operatorname{Spec} \overline{K} \twoheadrightarrow \operatorname{Spec} K$ is surjective. By Proposition 11.4.5, surjectivity is preserved by base change, we know that $X_{\overline{K}} \twoheadrightarrow X_K$ is surjective. By hypothesis, $X_{\overline{K}}$ is connected, note that $X_{\overline{K}} \to X_K$ is continuous and surjective as map of topological spaces, X_K is connected.

If $(C_{\overline{k}})$ holds, we want to show that X_{k^s} is connected. Note that

$$X_{\overline{k}} = X \times_k \overline{k} = X \times_k (k^s \times_{k^s} \overline{k}) = X_{k^s} \times_{k^s} \overline{k},$$

consider the following diagram.

$$\begin{tikzcd} X_{\overline{k}} \arrow[r, two heads] \arrow[d] & X_{k^s} \arrow[r] \arrow[d] & X \arrow[d] \\ \operatorname{Spec} \overline{k} \arrow[r, two heads] & \operatorname{Spec} k^s \arrow[r] & \operatorname{Spec} k \end{tikzcd}$$

Since $\overline{k}/k^s$ is purely inseparable extension (and therefore algebraic extension, see [Mor12, Proposition 4.20]), $k^s \to \overline{k}$ is an integral extension, by the Lying Over Theorem 9.2.2, $\operatorname{Spec} \overline{k} \twoheadrightarrow \operatorname{Spec} k^s$ is surjective. Surjectivity is preserved by base change (Proposition 11.4.5), we know that $X_{\overline{k}} \to X_{k^s}$ is surjective. By hypothesis, $X_{\overline{k}}$ is connected, since $X_{\overline{k}} \twoheadrightarrow X_{k^s}$ is continuous and surjective as map of topological spaces, X_{k^s} is connected.

(c) As part (b), we already know that $X_{\overline{K}} \to X_K$ and $X_{\overline{k}} \to X_{k^s}$ are surjective and continuous as maps of topological spaces, note that irreducibility is preserved under the continuous surjective map, so $(I_{K=\overline{K}})$ implies (I_K) and $(I_{\overline{k}})$ implies (I_{k^s}).

(d) Since reducedness is an affine-local property (Example 6.5), we may assume that $X = \operatorname{Spec} A$, where A is a k-algebra.

If $(R_{K=\overline{K}})$ holds, let K/k be any field extension, we want to show that X_K is reduced. Note that $K \otimes_k A \hookrightarrow \overline{K} \otimes_K (K \otimes_k A) = \overline{K} \otimes_k A$ (by part (a)), since $\overline{K} \otimes_k A$ is reduced ring, $K \otimes_k A$ is reduced ring, and therefore $X_K = \operatorname{Spec} A \otimes_k K$ is reduced.

If $(R_{\overline{k}})$ holds, note that $k^p \hookrightarrow \overline{k}$, similar to the case that "$(R_{K=\overline{K}})$ implies (R_K)", we have (R_{k^p}).

□

Thus for example a k-scheme is geometrically integral if and only if it remains integral under any field extension. Hence "geometrically integral" means "universally, integral fibers".

Proposition 11.5.2 (Harder fact)

(a) $(C_K) \Leftrightarrow (C_{K=\overline{K}}) \Leftrightarrow (C_{\overline{k}}) \Leftrightarrow (C_{k^s})$.
(b) $(I_K) \Leftrightarrow (I_{K=\overline{K}}) \Leftrightarrow (I_{\overline{k}}) \Leftrightarrow (I_{k^s})$.
(c) $(R_K) \Leftrightarrow (R_{K=\overline{K}}) \Leftrightarrow (R_{\overline{k}}) \Leftrightarrow (R_{k^p})$.

We defer the explanation to the end of this section, in a double-starred discussion (§11.5.3).

The following exercise may help even the geometrically-minded reader appreciate the utility of these notions. (There is nothing important about the dimension 2 and the degree 4 in this exercise!)

✍ **Exercise 11.11** ⋆ Recall that "degree d hypersurfaces in $\mathbb{P}^n$ are parametrized by $\mathbb{P}^{\binom{n+d}{d}-1}$", the quartic curves in $\mathbb{P}^2_k$ are parametrized by a $\mathbb{P}^{14}_k$. (This will be made much more precise in Chapter 26.) Show that the points of $\mathbb{P}^{14}_k$ corresponding to geometrically integral curves form an open subset. Explain the necessity of the modifier "geometrically" (even if k is algebraically closed).

11.5.2 Universally injective (= radical) morphisms

As remark in §11.4, injectivity is not preserved by base change. A batter notion is that of **universally injective morphisms**:

Definition 11.5.4 (Universally injective (radical))

*Let $\pi : X \to S$ be a morphism of schemes. We say that π is **universally injective** (or **radical**) if and only if for any morphism of schemes $S' \to S$ the base change $\pi_{S'} : X_{S'} \to S'$ is injective (on underlying topological spaces).*

Remark 11.22 In general, if $\pi : X \to S$ is a morphism, we say π is radical if π is injective as a map of topological spaces, and for every $x \in X$ the field extension $\kappa(x)/\kappa(\pi(x))$ is purely inseparable, see [Sta25, Definition 01S3]. We will prove this is same as Definition 11.5.4 in Proposition 11.5.4.

Example 11.8 As a first example: all locally closed embeddings are universally injective (as they are injective, and remain locally closed embeddings upon any base change). If you wish, you can show more generally that all monomorphisms are universally injective. (Hint: show that monomorphisms are injective, and that the class of monomorphisms is preseved by base change.)

Suppose you want to determine whether a given morphism is universally injective. A map of sets is injective if and only if each fiber contains at most one point. Now "the fiber of a base change is the base change of the fiber" (Proposition 11.3.3). Also the underlying set of the scheme-theoretic fiber is the fiber of the map on underlying sets (Proposition 11.3.1). Hence the question of universal injectivity turns into one about field theory. We make this precise in Proposition 11.5.4, after first making the field theory question precise in Proposition 11.5.3. En route, we will see why universal injectivity is the algebro-geometric generalization of the notion of purely inseparable extensions of fields.

Proposition 11.5.3

Suppose E/F is an extension of fields, inducing $\varphi : \operatorname{Spec} E \to \operatorname{Spec} F$.

(a) *If E contains an element x transcendental over F, then φ is not universally injective.*
(b) *If $E \setminus F$ contains an element x algebraic over F, and separable over F, then φ is not universally injective.*
(c) *Suppose E contains no elements of the above forms, i.e., E/F is a **purely inseparable extension***

(all elements of E are algebraic over F, and have some p^Nth power in F). If E'/F is any other field extension, then $\operatorname{Spec} E \otimes_F E'$ *contains precisely one point.*

Proof

(a) Let y be an transcendental element over F. Consider the following Cartesian diagrams.

$$\begin{tikzcd}
\operatorname{Spec} E \otimes_F F(y) \arrow[r, two heads] \arrow[d] & \operatorname{Spec} F(x) \otimes_F F(y) \arrow[r] \arrow[d] & \operatorname{Spec} F(y) \arrow[d] \\
\operatorname{Spec} E \arrow[r, two heads] & \operatorname{Spec} F(x) \arrow[r] & \operatorname{Spec} F
\end{tikzcd}$$

Since $\operatorname{Spec} E \to \operatorname{Spec} F(x)$ is a morphism from a point to a point, so it is surjective. Surjectivity is preserved by base change (Proposition 11.4.5), we know that $\operatorname{Spec} E \otimes_F F(y) \twoheadrightarrow \operatorname{Spec} F(x) \otimes_F F(y)$ is surjective. Consider $\operatorname{Spec} F(x) \otimes_F F(y)$, by Exercise 11.3, the prime ideal of $F(x) \otimes_F F(y)$ correspond to ideals $(f(x,y)) \subseteq F[x,y]$, where $f(x,y)$ is an irreducible polynomial in $F[x,y]$ containing both variables x and y. Since x is transcendental over F, $\operatorname{Spec} F(x) \otimes_F F(y)$ is not single point, by the surjectivity of $\operatorname{Spec} E \otimes_F F(y) \twoheadrightarrow \operatorname{Spec} F(x) \otimes_F F(y)$, we know that $\operatorname{Spec} E \otimes_F F(y)$ not single point. Since $\operatorname{Spec} E$ is a single point, $\operatorname{Spec} E \otimes_F F(y) \to \operatorname{Spec} E$ not injective. Hence φ is not universally injective.

(b) Since x is algebraic over F, let $\operatorname{Irr}(t,F) \in F[t]$ be a minimal polynomial of x, hence $F(x) \cong F[t]/(\operatorname{Irr}(t,F))$. Consider the following Cartesian diagrams.

$$\begin{tikzcd}
\operatorname{Spec} E \otimes_F \overline{F} \arrow[r] \arrow[d] & \operatorname{Spec} F[t]/(\operatorname{Irr}(t,F)) \otimes_F \overline{F} \arrow[r] \arrow[d] & \operatorname{Spec} \overline{F} \arrow[d] \\
\operatorname{Spec} E \arrow[r] & \operatorname{Spec} F[t]/(\operatorname{Irr}(t,F)) \arrow[r] & \operatorname{Spec} F
\end{tikzcd}$$

We may assume that $\deg \operatorname{Irr}(t,F) = n$, since x is separable over F, we may assume $\operatorname{Irr}(t,F) = \prod_{i=1}^{n}(t-x_i)$ in $\overline{F}$, where $x_i \neq x_j$ for all $i \neq j$. Consider $F[t]/(\operatorname{Irr}(t,F)) \otimes_F \overline{F}$,

$$\begin{aligned}
F[t]/(\operatorname{Irr}(t,F)) \otimes_F \overline{F} &\cong (F[t]/(\operatorname{Irr}(t,F)) \otimes_{F[t]} F[t]) \otimes_F \overline{F} \\
&\cong F[t]/(\operatorname{Irr}(t,F)) \otimes_{F[t]} \overline{F}[t] \\
&\cong \overline{F}[t]/(\operatorname{Irr}(t,F)) \\
&\cong \prod_{i=1}^{n} \overline{F}[t]/(t-x_i) \cong \prod_{i=1}^{n} \overline{F},
\end{aligned}$$

hence $\operatorname{Spec} F[t]/(\operatorname{Irr}(t,F)) \otimes_F \overline{F} \cong \coprod_{i=1}^{n} \operatorname{Spec} \overline{F}$, and therefore $\operatorname{Spec} F[t]/(\operatorname{Irr}(t,F))$ have n point. Note that $\operatorname{Spec} E \to \operatorname{Spec} F[t]/(\operatorname{Irr}(t,F))$ is a morphism from a point to a point, so it is surjective. Surjectivity is preserved by base change (Proposition 11.4.5), we know that $\operatorname{Spec} E \otimes_F \overline{F} \to \operatorname{Spec} F[t]/(\operatorname{Irr}(t,F)) \otimes_F \overline{F}$ is surjective. Hence $\operatorname{Spec} E \otimes_F \overline{F}$ at least n points, so $\operatorname{Spec} E \otimes_F \overline{F} \to \operatorname{Spec} E$ is not injective, and therefore φ is not universally injective.

(c) Suppose $\operatorname{char} F = p > 0$. Let $x \in E \otimes_F E'$, then x is of the form

$$x = \sum_{i=1}^{n} a_i \otimes b_i,$$

where $a_i \in E$ and $b_i \in E'$. Since E/F is purely inseparable extension, $a_i^{p^{m_i}} \in F$. Let $m =$

$\max\{m_1, \dots, m_n\}$. Consider x^{p^m},

$$x^{p^m} = \sum_{i=1}^{n} (a_i \otimes b_i)^{p^m} = \sum_{i=1}^{n} a_i^{p^m} \otimes b_i^{p^m} = \sum_{i=1}^{n} (1 \otimes a_i^{p^m} b_i^{p^m}) = 1 \otimes \sum_{i=1}^{n} a_i^{p^m} b_i^{p^m}.$$

Note that $1 \otimes E' \cong E'$, x^{p^m} can be seen as an element belong to E'. Since E' is a field, x^{p^m} must be 0 or unit. Hence $E \otimes_F E'$ is a local ring. Let $\mathfrak{p} \in \operatorname{Spec} E \otimes_F E'$, note that $0 \in \mathfrak{p}$, every nilpotent of $E \otimes_F E'$ must belong to $\mathfrak{p}$, so $\mathfrak{p} = \mathfrak{m}$. Hence $\operatorname{Spec} E \otimes_F E' = \{[\mathfrak{m}]\}$, as we desired.

□

Proposition 11.5.4

Suppose $\pi : X \to Y$ is a morphism of schemes. The following are equivalent.

(i) π is universally injective.

(ii) The morphism π is injective, and for each $p \in X$, the field extension $\kappa(p)/\kappa(\pi(p))$ is purely inseparable.

Proof (i) $\Rightarrow$ (ii): Note that $X \times_Y Y \cong X$, since $\pi : X \to Y$ is universally injective, $\pi : X \to Y$ must be injective.

If there exists $p \in X$ such that the $\kappa(p)/\kappa(\pi(p))$ is not purely inseparable. Hence $\kappa(p)$ contains an element x transcendental over F or $\kappa(p) \setminus \kappa(\pi(p))$ contains an element x algebraic and separable over $\kappa(\pi(p))$, by Proposition 11.5.3, we know that π is not universally injective, a contradiction!

(ii) $\Rightarrow$ (i): Let $Z \to Y$ be any morphism of schemes, we want to show that $\pi_Z : X \times_Y Z \to Z$ is injective. Let $p_1', p_2' \in X \times_Y Z$, with $\pi_Z(p_1') = \pi_Z(p_2') = q' \in Z$. Consider the following Cartesian diagram.

$$\begin{tikzcd}
X \times_Y Z \arrow[r] \arrow[d, "\pi_Z"'] & X \arrow[d, "\pi"] \\
Z \arrow[r] & Y
\end{tikzcd}$$

Let p_1, p_2 be the images of p_1', p_2' in X respectively, and q be the image of q' in Y. So we have $\pi(p_1) = \pi(p_2) = q$, by the Cartesian diagram. Since π is injective, we have $p := p_1 = p_2$, and therefore $p_1, p_2 \in \kappa(p) \times_X (X \times_Y Z) \cong \kappa(p) \times_Y Z$. Since $\pi_Z(p_1') = \pi_Z(p_2') = q'$, we have $p_1, p_2 \in \kappa(q') \times_Z (\kappa(p) \times_Y Z) \cong \kappa(q') \times_Y \kappa(p)$. Note that

$$\begin{tikzcd}
\kappa(p) \times_{\kappa(q)} \kappa(q') \arrow[dr, dashed, shift left] \arrow[drr, bend left] \arrow[ddr, bend right] & & & \\
& \kappa(p) \times_Y \kappa(q') \arrow[ul, dashed, shift left] \arrow[r] \arrow[d] & \operatorname{Spec} \kappa(p) \arrow[d, hook, "\pi"'] \arrow[ddr] & \\
& \operatorname{Spec} \kappa(q') \arrow[r, hook] \arrow[drr] & Y & \\
& & & \operatorname{Spec} \kappa(q) \arrow[ul]
\end{tikzcd}$$

where dashed arrows given by the universal property of fibered product, we have

$$p_1, p_2 \in \kappa(p) \times_Y \kappa(q') \cong \kappa(p) \times_{\kappa(q)} \kappa(q').$$

Since $\kappa(p)/\kappa(q)$ is purely inseparable, by Proposition 11.5.3, $\kappa(p) \times_{\kappa(q)} \kappa(q') = \operatorname{Spec} \kappa(p) \otimes_{\kappa(q)} \kappa(q')$ is a single point. So $p_1 = p_2$, i.e. π_Z is injective, and therefore π is universally injective. □

You may already see that the class of universally injective morphisms is also preserved by composition, and local on the target, and hence form a "reasonable" class (Definition 9.1.1). In any case, we will see this in a different way in Chapter 12.

11.5.3 ⋆⋆ Proof of Harder fact 11.5.2

We will use one fact we will not prove until much later. Recall that a map of topological spaces is an **open map** if the image of every open set is an open set.

Fact, we will proved in Chapter 25:

Theorem 11.5.1

*Suppose X is a k-scheme. Then $X \to \operatorname{Spec} k$ is **universally open**, i.e., remains open after any base change.*

we could prove Theorem 11.5.1 now without too much trouble, but it will follow much more easily with the magic of flatness.

Lemma 11.5.1

Suppose the field extension E/F is purely inseparable. Suppose X is any F-scheme. Then $\varphi : X_E \to X$ is a homeomorphism.

Proof By Proposition 11.5.3 (c), for any point $p \in X$, the scheme-theoretic fiber $\varphi^{-1}(p)$ is a single point, hence φ is a bijection, so we may identify the points of X and X_E. The morphism φ is continuous (so open sets in X are also in X_E), and by Theorem 11.5.1, φ is open (son open sets in X_E are also open in X). Hence φ is a homeomorphism. □

Connectedness

Recall that a connected component of a topological space X is a maximal connected subset of X (Definition 4.6.8).

Proposition 11.5.5

Every point of a topological space X is contained in a connected component, and that connected components are closed.

Proof Recall two facts in topology:

(1) the union of any collection of connected subsets of X that have a nonempty intersection is itself connected;

(2) If A is connected subset of X, then its closure, $\mathrm{cl}_X(A)$, is also connected.

We first show the statement that every point of X is contained in a connected component. Let $x \in X$. Say $S_x = \{U \subseteq X : x \in U \text{ and } U \text{ is connected}\}$. Consider the set $C_x = \bigcup_{U \in S_x} U$. Every set in collection S_x contains the point x, so their intersection is nonempty. By fact (1), their union C_x is a connected set. By the construction of C_x, C_x contains every connected subset of X that contains x. Therefore, it is the largest connected set containing x. Thus C_x is the connected component containing x. Since x was arbitrary, every point in X is contained in a connected component.

We next show the statement that connected components are closed. Let C be any connected component of X, then C is connected. By the fact (2), we know that $\mathrm{cl}_X(C)$ is also a connected set. Recall that connected component is the largest connected set, since $C \subseteq \mathrm{cl}_X(C)$ and C is connected component, we know that $C = \mathrm{cl}_X(C)$, i.e., every connected component is closed. □

Proposition 11.5.6

Let X, Y be topological spaces. Suppose $\varphi : X \to Y$ is open, and has nonempty connected fibers. Then φ induces a bijection of connected components.

Proof Let C_X, C_Y be the collection of connected components of X, Y respectively. We now define a map from C_X to C_Y. Let $C \in C_X$, since φ is continuous, $\varphi(C)$ is connected in Y. So there exists unique connected component $D \in C_Y$ such that $\varphi(C) \subseteq D$. Define $\Phi : C_X \to C_Y$ by setting $C \mapsto D$. We want to show Φ is a bijection.

We first show that Φ is surjective. Pick $D \in C_Y$. Since D is nonempty, pick $y \in D$. By hypotheses, $\varphi^{-1}(y)$ is nonempty connected fibers, so we can pick $x \in \varphi^{-1}(y)$, x must belong to a unique connected component of X, named C_x. We claim that $\Phi(C_x) = D$. Since $\varphi(x) = y \in \varphi(C_x)$ and $\varphi(C_x) \subseteq \Phi(C_x)$, $\Phi(C_x)$ is a connected component of y, and therefore $\Phi(C_x) = D$. It follows that Φ is surjective.

We next show that Φ is injective. Let $C_1, C_2 \in C_X$ be two connected components of X with $\Phi(C_1) = \Phi(C_2)$, we want to show that $C_1 = C_2$. Say $D = \Phi(C_1) = \Phi(C_2)$, then $\varphi(C_1) \subseteq D$ and $\varphi(C_2) \subseteq D$, hence $C_1 \cup C_2 \subseteq \varphi^{-1}(D)$. If $\varphi^{-1}(D)$ is connected, then $\varphi^{-1}(D) \subseteq C_D$ for some $C_D \in C_X$, so we have $C_1 \subseteq \varphi^{-1}(D) \subseteq C_D$, and therefore $C_1 = C_D$, similarly, we have $C_2 = C_D$, i.e., $C_1 = C_2$. Hence it suffices to show that $\varphi^{-1}(D)$ is connected.

Now we shall to show that $\varphi^{-1}(D)$ is connected. Suppose $\varphi^{-1}(D)$ is not connected, then $\varphi^{-1}(D) = U \cup V$ where U, V are nonempty open and $U \cap V = \varnothing$. Since U, V are open subsets of $\varphi^{-1}(D)$, we may assume $U = O_U \cap \varphi^{-1}(D)$ and $V = O_V \cap \varphi^{-1}(D)$ where O_U, O_V are open subsets of X and $O_U \cap O_V = \varnothing$. Let $D_U = \varphi(U)$ and $D_V = \varphi(V)$, then $D = \varphi(\varphi^{-1}(D)) = \varphi(U \cup V) = D_U \cup D_V$. We want to show D_U, D_V are open in D. Pick $y \in D_U$, it suffices to show that there exists an open neighborhood $N \subseteq Y$ of y such that $N \cap D \subseteq D_U$. Consider $\varphi^{-1}(y)$, note that

$$\varphi^{-1}(y) = \varphi^{-1}(y) \cap \varphi^{-1}(D) = (\varphi^{-1}(y) \cap U) \cup (\varphi^{-1}(y) \cap V)$$

and $\varphi^{-1}(y)$ is connected, we know that $\varphi^{-1}(y) \subseteq U$ or $\varphi^{-1}(y) \subseteq V$. Since $y \in D_U = \varphi(U)$, we know that $\varphi^{-1}(y) \subseteq U$. Since $U = O_U \cap \varphi^{-1}(D)$, we have $\varphi^{-1}(y) \subseteq O_U$. Since φ is an open map, $\varphi(O_U)$ is an open subset of Y. Let $N = \varphi(O_U)$, then $y \in N$. We next show that $N \cap D \subseteq D_U$. Let $y' \in N \cap D$, then there exists $x' \in O_U$ such that $\varphi(x') = y'$, since $y' \in D$, $x' \in \varphi^{-1}(D)$, hence $x' \in O_U \cap \varphi^{-1}(D) = U$. By the connectedness of $\varphi^{-1}(y')$ and $x' \in U$ and $\varphi(x') = y'$, we know that $\varphi^{-1}(y') \subseteq U$. Hence $y' = \varphi(\varphi^{-1}(y')) \subseteq \varphi(U) = D_U$, therefore $N \cap D \subseteq D_U$. Hence D_U is open in D, similarly, D_V is open in D. So D is the union of disjoint two open subsets, this contradicts to the fact that D is connected. Hence $\varphi^{-1}(D)$ is connected. Then we finished proof. □

Lemma 11.5.2

Suppose X is geometrically connected over k. Then for any scheme Y/k, $X \times_k Y \to Y$ induces a bijection of connected components.

Proof By Theorem 11.5.1, we know that $X \to \operatorname{Spec} k$ is universally open, and therefore $\pi : X \times_k Y \to Y$ is open. Since X is geometrically connected over k, $X_{\overline{k}}$ is connected. Let $p \in Y$, then $\pi^{-1}(p) = X_{\kappa(p)}$. Consider the following Cartesian squares.

$$\begin{tikzcd}
X_{\overline{k}} \arrow[r, two heads] \arrow[d] & X_{\kappa(p)} \arrow[r] \arrow[d] & X \times_k Y \arrow[r] \arrow[d, "\pi"'] & X \arrow[d] \\
\operatorname{Spec} \overline{k} \arrow[r, two heads] & \operatorname{Spec} \kappa(p) \arrow[r, hook] & Y \arrow[r] & \operatorname{Spec} k
\end{tikzcd}$$

Clearly $\operatorname{Spec} \overline{k} \to \operatorname{Spec} \kappa(p)$ is surjective, since surjectivity is preserved by base change (Proposition 11.4.5), $X_{\overline{k}} \twoheadrightarrow X_{\kappa(p)}$ is surjective, note that $X_{\overline{k}} \twoheadrightarrow X_{\kappa(p)}$ is continuous and $X_{\overline{k}}$ is connected, $X_{\kappa(p)}$ is connected. Apply Proposition 11.5.6, we done. □

We come next to a repeatedly useful algebra-geometry link, promised in §4.6.1: idempotent functions on X on the algebraic side correspond to a decomposition of X into two open and closed sets on the geometric side.

Proposition 11.5.7 (Idempotent-connectedness package)

A scheme X is disconnected if and only if there exists a function $e \in \Gamma(X, \mathscr{O}_X)$ that is an idempotent ($e^2 = e$) distinct from 0 and 1.

Proof If X is disconnected, for simply, we may assume $X = X_0 \cup X_1$ where X_0, X_1 are two nonempty disjoint open subsets. Let $e \in \Gamma(X, \mathscr{O}_X)$ by setting $e = 0$ on X_0 and $e = 1$ on X_1, then $e(e-1) = 0$ on X, i.e., $e = e^2$.

Conversely, let $e \in \Gamma(X, \mathscr{O}_X)$ be an idempotent, then $e(e-1) = 0$, i.e., $e(e-1)$ vanishing on entire X. Hence X can be written as vanishing scheme $V(e(e-1))$ which we defined in Definition 10.1.7. Hence $X = V(e) \cup V(1-e)$. Say $X_0 := V(e)$ and $X_1 := V(1-e)$. We want to show that X_0, X_1 are open. Let $x \in X_0$, then the germ of e at x is zero, i.e., $e_x = 0$. Hence there exists an open neighborhood $W \subseteq X_0$ of x such that $e|_W = 0$, and therefore $x \in W \subseteq X_0$. It follows that X_0 is open. Similarly, we can show that X_1 is open. Clearly $X_0 \cap X_1 = \varnothing$, hence X is disconnected. □

Proposition 11.5.8

Suppose k is separably closed, and A is a k-algebra with $\operatorname{Spec} A$ connected. Then $\operatorname{Spec} A$ is geometrically connected over k.

Proof We wish to show that $\operatorname{Spec} A \otimes_k K$ is connected for any field extension K/k. It suffices to assume that K is algebraically closed (as $\operatorname{Spec} A \otimes_k \overline{K} \to \operatorname{Spec} A \otimes_k K$ is surjective). By choosing an embedding $\overline{k} \hookrightarrow K$ and consider the diagram

$$\begin{tikzcd}
\operatorname{Spec} A \otimes_k K \arrow[r] \arrow[d] & \operatorname{Spec} A \otimes_k \overline{k} \arrow[r, "\text{homeo.}", "\text{by Lem. 11.5.1}"'] \arrow[d] & \operatorname{Spec} A \arrow[d] \\
\operatorname{Spec} K \arrow[r] & \operatorname{Spec} \overline{k} \arrow[r] & \operatorname{Spec} k
\end{tikzcd}$$

it suffices to assume k is algebraically closed.

If $\operatorname{Spec} A \otimes_k K$ is disconnected, then $A \otimes_k K$ contains an idempotent $e \neq 0, 1$ (by Proposition 11.5.7). Write $e = \sum_{i=1}^{N} a_i \otimes l_i$, where $a_i \in A$ and $l_i \in K$. Define B as the subalgebra of K generated over k by the l_i (as an algebra), so B is an integral domain of finite type over k. Then $K(B)$ is a finitely generated field extension of k.

$$\begin{tikzcd} \operatorname{Spec} A \otimes_k K \arrow[r] \arrow[d] & \operatorname{Spec} A \otimes_k B \arrow[r] \arrow[d] & \operatorname{Spec} A \arrow[d, "\text{universally open}"] \\ \operatorname{Spec} K \arrow[r] & \operatorname{Spec} B \arrow[r] & \operatorname{Spec} k \end{tikzcd}$$

Also, $e \in A \otimes_k B$, and we recall that e is an idempotent distinct from 0 and 1 (even in this ring). By the idempotent-connectedness package 11.5.7 applied to $A \otimes_k B$, $\operatorname{Spec} A \otimes_k B$ is disconnected, say with nonempty open subsets U and V with $U \coprod V = \operatorname{Spec} A \otimes_k B$.

Now $\varphi : \operatorname{Spec} A \otimes_k B \to \operatorname{Spec} B$ is an open map (because $\operatorname{Spec} A \to \operatorname{Spec} k$ is universally open, Theorem 11.5.1), so $\varphi(U)$ and $\varphi(V)$ are nonempty open set. Since $\operatorname{Spec} B$ is reduced and of finite type over k, $\operatorname{Spec} B$ is an affine k-variety. By Proposition 6.3.8, closed points of $\operatorname{Spec} B$ is dense. Hence there exists a closed point $p \in \varphi(U) \cap \varphi(V)$ with residue field k, as $k = \overline{k}$. But then $\varphi^{-1}(p) \cong \operatorname{Spec} A$, and we have covered $\operatorname{Spec} A$ with two disjoint open sets, yielding a contradiction. □

Remark 11.23 Sneaky "tensor-finiteness" trick that was just used. This argument used a particularly clever trick that we will use again: we conjured a finitely generated algebra B out of nowhere, and then used classical "geometry over algebraically closed fields" to prove something far-removed from classical algebraic geometry. What let us do that was that any single element of a tensor product (such as e in the argument above) is a finite sum of "elementary tensors", and so we worked instead in a subring generated by the "parts" of those "elementary tensors". We will call this **the tensor-finiteness trick** in order to help direct your attention to where it can be used again.

Corollary 11.5.1

If k is separably closed, and Y is a connected k-scheme, then Y is geometrically connected.

Proof We wish to show that for any field extension K/k, Y_K is connected. By Proposition 11.5.8, $\operatorname{Spec} K$ is geometrically connected over k. Then apply Lemma 11.5.2 with $X = \operatorname{Spec} K$. □

Irreducibility

Proposition 11.5.9

Suppose k is separably closed, A is a k-algebra with $\operatorname{Spec} A$ irreducible, and K/k is a field extension. Then $\operatorname{Spec} A \otimes_k K$ is irreducible.

Proof We follow the philosophy of the proof of Proposition 11.5.8. As in the first paragraph of that proof, it suffices to assume that k is algebraically closed.

If $A \otimes_k K$ is not irreducible, then we can find x and y with $V(x), V(y) \neq \operatorname{Spec} A \otimes_k K$ and $V(x) \cup V(y) = \operatorname{Spec} A \otimes_k K$. As in the second paragraph of the proof of Proposition 11.5.8, we use the "tensor-finiteness trick". We choose expressions $x = \sum a_i \otimes l_i$ and $y = \sum a'_j \otimes l'_j$ ($a_i, a'_j \in A$, $l_i, l'_j \in K$), and then define B to be the subring of K generated by the l_i and l'_j. Thus B is an integral domain finitely generated over $k = \overline{k}$, and $x, y \in A \otimes_k B$. Then $D(x)$ and $D(y)$ are nonempty open subsets of $\operatorname{Spec} A \otimes_k B$, whose image in $\operatorname{Spec} B$ are nonempty open sets (by universal openness of $\operatorname{Spec} A \to \operatorname{Spec} k$, Theorem 11.5.1), and thus their intersection is nonempty and contains a closed point p (with residue field $k = \overline{k}$). But then $\varphi^{-1}(p) \cong \operatorname{Spec} A$, and we have covered $\operatorname{Spec} A$ with two proper closed sets (the restrictions of $V(x)$ and $V(y)$), yielding a contradiction. □

Proposition 11.5.10

Suppose k is separably closed, and A and B are k-algebras, both with irreducible Spec *(i.e., with one minimal prime). Then $A \otimes_k B$ has irreducible* Spec *too.*

Proof Let $\overline{k}$ be the algebraic closure of k, since k is separably closed, $\overline{k}/k$ is a purely inseparable extension [Mor12, Proposition 4.20]. By Lemma 11.5.1, $\operatorname{Spec} A \otimes_k \overline{k} \to \operatorname{Spec} A$ and $\operatorname{Spec} B \otimes_k \overline{k} \to \operatorname{Spec} B$ are homeomorphisms, by the irreducibility of $\operatorname{Spec} A$ and $\operatorname{Spec} B$, we know that $\operatorname{Spec} A \otimes_k \overline{k}$ and $\operatorname{Spec} B \otimes_k \overline{k}$ are irreducible. If we can prove that the fibered product of two irreducible affine schemes over $k = \overline{k}$ is also irreducible, then $\operatorname{Spec}(A \otimes_k \overline{k}) \otimes_{\overline{k}} (B \otimes_k \overline{k})$ is irreducible. Note that

$$(A \otimes_k \overline{k}) \otimes_{\overline{k}} (B \otimes_k \overline{k}) \cong A \otimes_k (\overline{k} \otimes_{\overline{k}} (B \otimes_k \overline{k})) \cong A \otimes_k B \otimes_k \overline{k},$$

we have $\operatorname{Spec}(A \otimes_k B \otimes_k \overline{k})$ is irreducible. Use Lemma 11.5.1 again,

$$\operatorname{Spec}((A \otimes_k B) \otimes_k \overline{k}) \longrightarrow \operatorname{Spec}(A \otimes_k B)$$

is a homeomorphism, so $\operatorname{Spec}(A \otimes_k B)$ is irreducible. Hence, it suffices to show the following statement: **Let k be an algebraically closed field, A and B are k-algebra. If $\operatorname{Spec} A$ and $\operatorname{Spec} B$ are irreducible, then $\operatorname{Spec} A \otimes_k B$ is irreducible.**

Say $X = \operatorname{Spec} A \otimes_k B$. Let U, V be any two open subsets of X, we want to show that $U \cap V \neq \varnothing$. By Theorem 11.5.1, $\operatorname{Spec} B \to k$ is universally open, and therefore $\varphi : \operatorname{Spec} A \otimes_k B \to \operatorname{Spec} A$ is open, hence $\varphi(U)$ and $\varphi(V)$ are open in $\operatorname{Spec} A$. Since $\operatorname{Spec} A$ is irreducible, the generic point $\eta_A \in \varphi(U) \cap \varphi(V)$. Hence $\varphi^{-1}(\eta_A) \cap U \neq \varnothing$ and $\varphi^{-1}(\eta_A) \cap V \neq \varnothing$. Consider $\varphi^{-1}(\eta_A)$, note that

$$\varphi^{-1}(\eta_A) = \operatorname{Spec}(A \otimes_k B) \otimes_A \kappa(\eta_A) = \operatorname{Spec} B \otimes_k \kappa(\eta_A),$$

since $k \hookrightarrow \kappa(\eta_A)$, by Proposition 11.5.9, $\varphi^{-1}(\eta_A)$ is irreducible. Note that $\varphi^{-1}(\eta_A) \cap U$ and $\varphi^{-1}(\eta_A) \cap V$ are nonempty open subsets of $\varphi^{-1}(\eta_A)$, since $\varphi^{-1}(\eta_A)$ is irreducible, we have

$$(\varphi^{-1}(\eta_A) \cap U) \cap (\varphi^{-1}(\eta_A) \cap V) = \varphi^{-1}(\eta_A) \cap U \cap V \neq \varnothing,$$

hence $U \cap V \neq \varnothing$, which implies that X is irreducible. □

Proposition 11.5.11

A scheme X is irreducible if and only if there exists an open cover $X = \bigcup_i U_i$ with U_i irreducible for all i, and $U_i \cap U_j \neq \varnothing$ for all i, j.

Proof $\Rightarrow$. Clearly.

$\Leftarrow$. Let U, V be two nonempty open subsets of X, we want to show that $U \cap V \neq \varnothing$. Since $X = \bigcup_k U_k$, then $U \cap U_i \neq \varnothing$ and $V \cap U_i \neq \varnothing$ for some i, j. Note that U_i is irreducible, $U \cap U_i$ and $U_i \cap U_j$ are dense in U_i, so

$$(U \cap U_i) \cap (U_i \cap U_j) = U \cap U_i \cap U_j \neq \varnothing.$$

Similarly, we have $V \cap U_i \cap U_j \neq \varnothing$. Hence

$$(U \cap U_i) \cap (V \cap U_j) \neq \varnothing,$$

and therefore $U \cap V \neq \varnothing$, which implies that X is irreducible. □

Proposition 11.5.12

Suppose K/k is a field extension of a separated closed field and X_k is irreducible. Then X_K is irreducible.

Proof Take an open cover $X_k = \bigcup U_i$ by pairwise intersecting irreducible affine open subsets. The base change of each U_i to K is irreducible by Proposition 11.5.9, and they pairwise intersect. The result then follows from Proposition 11.5.11. $\square$

Proposition 11.5.13 (Geometrically integral $\times$ Integral $=$ Integral)

Suppose B is a k-algebra such that $B \otimes_k \overline{k}$ is an integral domain (Spec B is geometrically integral), and A is a k-algebra that is an integral domain (Spec A is integral). Then $A \otimes_k B$ is an integral domain (Spec $A \otimes_k B$ is integral).

Remark 11.24 Once we define "variety", this will imply that the product of a geometrically integral variety with an integral variety is an integral variety.

Proof Suppose $(\sum a_i \otimes b_i)(\sum a'_j \otimes b'_j) = 0$ in $A \otimes_k B$ with $a_i, a'_j \in A$, $b_i, b'_j \in B$, where both $\{b_i\}$ and $\{b'_j\}$ are linearly independent over k, and a_1, a'_1 are nonzero. Define A' be the subring of $\overline{k}$ generated by a_i, a'_j as k-algebra, then A' is finitely generated k-algebra. Since A is integral, $a_1 a'_1 \neq 0$, and therefore $D(a_1 a'_1) \subseteq \operatorname{Spec} A'$ is nonempty. By Zariski's Lemma 4.2.2, there is a maximal $\mathfrak{m} \subseteq A'$ in $D(a_1 a'_1)$ such that $A'/\mathfrak{m}$ is a field extension of k. Consider the following pushout square.

$$\begin{tikzcd}
k \arrow[r] \arrow[d] & A' \arrow[r] \arrow[d] & A'/\mathfrak{m} \arrow[r] \arrow[d] & \overline{k} \arrow[d] \\
B \arrow[r] & A' \otimes_k B \arrow[r] & A'/\mathfrak{m} \otimes_k B \arrow[r] & \overline{k} \otimes_k B
\end{tikzcd}$$

Let $\varphi : A' \otimes_k B \to \overline{k} \otimes_k B$ be the composition of $A' \otimes_k B \to A'/\mathfrak{m} \otimes_k B \to \overline{k} \otimes_k B$, i.e.,

$$\sum a_i \otimes b_i \longmapsto \sum \overline{a_i} \otimes b_i,$$

where $\overline{a_i}$ is the image of a_i in $A'/\mathfrak{m}$ and $\overline{a_i}$ can be seen as an element in $A'/\mathfrak{m}$ (since $A'/\mathfrak{m} \hookrightarrow \overline{k}$). Now we have

$$\varphi((\sum a_i \otimes b_i)(\sum a'_j \otimes b'_j)) = (\sum \overline{a_i} \otimes b_i)(\sum \overline{a'_j} \otimes b'_j) = 0.$$

Since $a_1, a'_1 \notin \mathfrak{m}$, $\overline{a_1}, \overline{a'_1} \neq 0$, also note that $\{b_i\}, \{b'_j\}$ are linearly independent over k, we have $\sum \overline{a_i} \otimes b_i \neq 0$ and $\sum \overline{a'_j} \otimes b'_j \neq 0$. But $\overline{k} \otimes_k B$ is integral domain, $(\sum \overline{a_i} \otimes b_i)(\sum \overline{a'_j} \otimes b'_j) \neq 0$, a contradiction! Hence $A' \otimes_k B$ must be integral domain. $\square$

Reducedness

We recall a statement from transcendence theory (the basics of which are developed in Chapter 13). Because this is a starred section, we content ourselves with a reference rather than a proof.

Theorem 11.5.2 (Finitely generated extensions of perfect fields are separably generated.)

Any finitely generated extension of a perfect field E/F can be factored into a finite separable part and a purely transcendental ([Sta25, Definition 030E]) part:

$$\begin{tikzcd}
E \arrow[d, "\text{finite separable}"] \\
F(t_1, \dots, t_n) \arrow[d, "\text{purely transcendental}"] \\
F.
\end{tikzcd}$$

Proof See [Eis13] or [Wae03] for a proof. $\square$

Definition 11.5.5 (Separably generated, separating transcendence basis)

Finitely generated extensions E/F that can be factored in

$$\begin{tikzcd} E \arrow[d, "\text{finite separable}"] \\ F(t_1,\dots,t_n) \arrow[d, "\text{purely transcendental}"] \\ F \end{tikzcd}$$

are said to be ***separably generated****. The transcendence basis $\{t_1,\dots,t_n\}$ is said to be a* ***separating transcendence basis****.*

Proposition 11.5.14 (Geometrically reduced $\times$ reduced $=$ reduced)

Suppose B is a geometrically reduced k-algebra, and A is a reduced k-algebra. Then $A\otimes_k B$ is reduced.

Proof Reduced to the case where A is finitely generated over k using the tensor-finiteness trick. (Suppose we have $x \in A\otimes_k B$ with $x^n = 0$. Suppose $x = \sum a_i \otimes b_i$. Let A' be the finitely generated subring of A generated by the a_i. Then $A'\otimes_k B$ is a subring of $A\otimes_k B$. Replace A by A'.) Then A is a subring of the product $\prod K_i$ of the function fields of its irreducible components. (One can see this as follows: the kernel of $A \to \prod A/\mathfrak{p}_i$ is the intersection of the minimal primes $\mathfrak{p}_i$, which is the ideal of nilpotents $\mathfrak{N}$ by Theorem 4.2.3; but $\mathfrak{N} = 0$ as A is reduced. And $A/\mathfrak{p}_i \hookrightarrow K(A/\mathfrak{p}_i) = K_i$ clearly. Alternatively, you may find this clear from our discussion of associated points (§7.6).) So it suffices to prove the result for A is a product of fields. Then it suffices to prove the result when A is a field. But then we are done, by the definition of geometric reducedness. □

Proposition 11.5.15

Suppose A is a reduced k-algebra. Then:

(a) $A\otimes_k k(t)$ is reduced.

(b) If E/k is a finite separable extension, then $A\otimes_k E$ is reduced.

Proof

(a) Clearly $A\otimes k[t]$ is reduced, and localization preserves reducedness (as reducedness is stalk-local, Proposition 6.2.1).

(b) Working inductively, we can assume E is generated by a single element, with minimal polynomial $p(t)$. By the tensor-finiteness trick, we can assume A is finitely generated over k. Then by the same trick as in the proof of Proposition 11.5.14, we can replace A by the product of its function fields of its components, and then we can assume A is a field. But then $A\otimes_k E \cong A[t]/(p(t))$ is reduced by the definition of separability of p.

□

Lemma 11.5.3

Suppose E/k is a field extension of a perfect field, and A is a reduced k-algebra. Then $A\otimes_k E$ is reduced.

Proof By the tensor-finiteness trick, we may assume E is finitely generated over k. By Theorem 11.5.2, we can factor E/k into $k \hookrightarrow F \hookrightarrow E$, where F/k is purely transcendental and E/F is finite separable. By Proposition 11.5.15, $A\otimes_k F$ is reduced, by Proposition 11.5.15, $A\otimes_k E \cong (A\otimes_k F)\otimes_F E$ is reduced. □

Proposition 11.5.16

Suppose E/k is an extension of a perfect field, and X is a reduced k-scheme. Then X_E is reduced.

Proof Reduce to the case where X is affine. Use Lemma 11.5.3. □

Corollary 11.5.2

Suppose k is perfect, and A and B are reduced k-algebras. Then $A \otimes_k B$ is reduced.

Proof By Lemma 11.5.3, A is a geometrically reduced k-algebra. Then apply Proposition 11.5.14. □

Theorem 11.5.3 (Hard fact)

(a) $(C_K) \Leftrightarrow (C_{K=\overline{K}}) \Leftrightarrow (C_{\overline{k}}) \Leftrightarrow (C_{k^s})$.
(b) $(I_K) \Leftrightarrow (I_{K=\overline{K}}) \Leftrightarrow (I_{\overline{k}}) \Leftrightarrow (I_{k^s})$.
(c) $(R_K) \Leftrightarrow (R_{K=\overline{K}}) \Leftrightarrow (R_{\overline{k}}) \Leftrightarrow (R_{k^p})$.

Proof By Proposition 11.5.1 and trivial fact that (C_K) implies $(C_{K=\overline{K}})$ implies $(C_{\overline{k}})$, and (C_K) implies (C_{k^s}), and similarly (with appropriate changes) with "connected" replaced by "irreducible" and "reduced", it suffices to show that (R_{k^p}) implies (R_K), (I_{k^s}) implies (I_K), and (C_{k^s}) implies (C_K). Suppose X be a k scheme.

If X_{k^p} is reduced. Let K/k be any field extension, if K/k^p, by Proposition 11.5.16, we done. For other case, there exists a field extension L/k such that L/K and L/k^p. Since reducedness is a local property, we may assume that $X = \operatorname{Spec} A$. By Proposition 11.5.16, we know that X_L is reduced, i.e., $L \otimes_k A$ is a reduced ring. Since we have injective $K \otimes_k A \hookrightarrow L \otimes_k A$, $K \otimes_k$ is reduced ring, i.e., X_K is reduced.

If X_{k^s} is irreducible. Let K/k be any field extension. If K/k^s, by Proposition 11.5.12, we know that X_K is irreducible. For other case, consider a field L such that L/k^s and L/K are field extensions. By Proposition 11.5.12, we know that X_L is irreducible. Since $\operatorname{Spec} L \to \operatorname{Spec} K$ is surjective, surjectivity preserved by base change, so $X_L \to X_K$ is surjective, moreover, this morphism is continuous, hence X_K is irreducible. Similarly, we can used the similar method to show that (C_{k^s}) implies (C_K) (use Corollary 11.5.1). Then we finished proof. □

Proposition 11.5.17

Suppose that A and B are two integral domains that are $\overline{k}$-algebras. Then $A \otimes_{\overline{k}} B$ is an integral domain.

Proof Compare Proposition 11.4.6. Suppose $(\sum a_i \otimes b_i)(\sum a'_j \otimes b'_j) = 0$ as the proof of Proposition 11.4.6. Let A' be a k-algebra which generated by a_i, a'_j, then A' is a finitely generated k-subalgebra of A. Replace A by A'. Next, follow the method of Proposition 11.4.6 to prove it, we done. □

11.6 Products of projective schemes: The Segre embedding

We next describe products of projective A-schemes over A. (The case of greatest initial interest is if $A = k$.) To do this, we need only describe $\mathbb{P}^m_A \times_A \mathbb{P}^n_A$, because any projective A-scheme has a closed embedding in some $\mathbb{P}^m_A$ (Proposition 10.3.3, or Proposition 10.3.1), and closed embeddings behave well under base change, so if $X \hookrightarrow \mathbb{P}^m_A$ and $Y \hookrightarrow \mathbb{P}^n_A$ are closed embeddings, then $X \times_A Y \hookrightarrow \mathbb{P}^m_A \times_A \mathbb{P}^n_A$ is also a closed embedding, cut out by the equations of X and Y (§11.2.3). We will describe $\mathbb{P}^m_A \times_A \mathbb{P}^n_A$, and see that it too is a projective A-scheme. (Hence if X and Y are projective A-schemes, then their product $X \times_A Y$ over A is also a projective A-scheme.)

Remark 11.25 Reasons of $X \times_A Y \hookrightarrow \mathbb{P}^m_A \times_A \mathbb{P}^n_A$ is a closed embedding: Consider the following diagram.

$$\begin{tikzcd}
X \times_A Y \arrow[r, hook, "\text{cl. emb}"] \arrow[d, hook, "\text{cl. emb.}"'] & Y \times_A \mathbb{P}^m_A \arrow[r] \arrow[d, hook, "\text{cl. emb.}"] & Y \arrow[d, hook, "\text{cl. emb.}"] \\
X \times_A \mathbb{P}^n_A \arrow[r, hook, "\text{cl. emb.}"] \arrow[d] & \mathbb{P}^A_k \times_A \mathbb{P}^n_A \arrow[r] \arrow[d] & \mathbb{P}^n_A \arrow[d] \\
X \arrow[r, hook, "\text{cl. emb.}"'] & \mathbb{P}^m_A \arrow[r] & \operatorname{Spec} A
\end{tikzcd}$$

Before we do this, we will get some motivation from classical projective spaces (nonzero vectors modulo nonzero scalars, Proposition 5.4.3) in a special case. Our map will send $[x_0 : x_1 : x_2] \times [y_0 : y_1]$ to a point in $\mathbb{P}^5$, whose coordinates we think of as being entires in the "multiplication table"

$$\begin{matrix} [& x_0y_0 : & x_1y_0 : & x_2y_0 : & \\ & x_0y_1 : & x_1y_1 : & x_2y_1 &]. \end{matrix}$$

This is indeed a well-defined map of sets. Notice that the resulting matrix is rank 1, and from the matrix, we can read off $[x_0 : x_1 : x_2]$ and $[y_0 : y_1]$ up to multiplication by nonzero scalars. For example, to read off the point $[x_0 : x_1 : x_2] \in \mathbb{P}^2$, we take the first row, unless it is all zero, in which case we take the second row. (They can't both be all zero.) In conclusion: in classical projective geometry, given a point of $\mathbb{P}^m$ and $\mathbb{P}^n$, we have produced a point in $\mathbb{P}^{mn+m+n}$, and from this point in $\mathbb{P}^{mn+m+n}$, we can recover the points of $\mathbb{P}^m$ and $\mathbb{P}^n$.

Suitably motivated, we return to algebraic geometry. We define a map

$$\mathbb{P}^m_A \times_A \mathbb{P}^n_A \longrightarrow \mathbb{P}^{mn+m+n}_A$$

by

$$\begin{aligned} ([x_0 : \cdots : x_m], [y_0 : \cdots : y_n]) \longmapsto & [z_{00} : z_{01} : \cdots : z_{ij} : \cdots : z_{mn}] \\ = & [x_0y_0 : x_0y_1 : \cdots : x_iy_j : \cdots : x_my_n]. \end{aligned}$$

More explicitly, we consider the map from the affine open set $U_i \times_A V_i$ (where $U_i = D_+(x_i)$ and $V_j = D_+(y_j)$) to the affine open set $W_{ij} = D_+(z_{ij})$ by

$$(x_{0/i}, \ldots, x_{m/i}, y_{0/j}, \ldots, y_{n/j}) \longmapsto (x_{0/i}y_{0/j}, \ldots, x_{i/i}y_{j/j}, \ldots, x_{m/i}y_{n/j})$$

or, in terms of algebras, $z_{ab/ij} \mapsto x_{a/i}y_{b/j}$.

✍ **Exercise 11.12** Check that these maps glue to give a well defined morphism $\mathbb{P}^m_A \times_A \mathbb{P}^n_A \to \mathbb{P}^{mn+m+n}_A$.

Proof Say $\pi_{ij} : U_i \times_A V_j \to W_{ij}$. Consider $(U_i \times_A V_j) \cap (U_k \times_A V_l)$,

$$(U_i \times_A V_j) \cap (U_k \times_A V_l) = (U_i \cap U_k) \times_A (V_j \cap V_l) := U_{ik} \times_A V_{jl},$$

where $U_{ik} = D_+(x_ix_k)$ and $V_{jl} = D_+(y_jy_l)$. We want to show that $\pi_{ij}|_{U_{ik} \times V_{jl}} = \pi_{kl}|_{U_{ik} \times V_{jl}}$. In fact,

$$\begin{aligned} \pi_{ij}|^{\sharp}_{U_{ik} \times_A V_{jl}} : (A[z_{00/ij}, \ldots, z_{mn/ij}]_{z_{kl/ij}})_0 \to & (A[x_{0/i}, \ldots, x_{m/i}]_{x_{k/i}})_0 \otimes_A (A[y_{0/j}, \ldots, y_{n/j}]_{y_{l/j}})_0 \\ \cong & A\left[\frac{x_{0/i}}{x_{k/i}}, \ldots, \frac{x_{m/i}}{x_{k/i}}, \frac{y_{0/j}}{y_{l/j}}, \ldots, \frac{y_{n/j}}{y_{l/j}}\right] \end{aligned}$$

is given by

$$\frac{z_{ab/ij}}{z_{kl/ij}} \longmapsto \frac{x_{a/i}}{x_{k/i}} \cdot \frac{y_{b/j}}{y_{l/j}},$$

and

$$\begin{aligned} \pi_{kl}|^{\sharp}_{U_{ik} \times_A V_{jl}} : (A[z_{00/kl}, \ldots, z_{mn/kl}]_{z_{ij/kl}})_0 \to & (A[x_{0/k}, \ldots, x_{m/k}]_{x_{i/k}})_0 \otimes_A (A[y_{0/l}, \ldots, y_{n/l}]_{y_{j/l}})_0 \\ \cong & A\left[\frac{x_{0/k}}{x_{i/k}}, \ldots, \frac{x_{m/k}}{x_{i/k}}, \frac{y_{0/l}}{y_{j/l}}, \ldots, \frac{y_{n/l}}{y_{j/l}}\right] \end{aligned}$$

is given by

$$\frac{z_{ab/kl}}{z_{ij/kl}} \longmapsto \frac{x_{a/k}}{x_{i/k}} \cdot \frac{y_{b/l}}{y_{j/l}},$$

On the $U_{ik} = U_i \cap U_k$, we have coordinate change:

$$x_{a/k} = \frac{x_{a/i}}{x_{k/i}};$$

on the $V_{jl} = V_j \cap V_l$, we have coordinate change:

$$y_{b/l} = \frac{y_{b/j}}{y_{l/j}};$$

and on the $W_{ij} \cap W_{kl}$, we have coordinate change:

$$z_{ab/kl} = \frac{z_{ab/ij}}{z_{kl/ij}}.$$

So pick $\frac{z_{ab/kl}}{z_{ij/kl}} \in (A[z_{00/kl}, \dots, z_{mn/kl}]_{z_{ij/kl}})_0$, we have

$$\begin{aligned} \pi_{kl}|^\sharp_{U_{ik}\times_A V_{jl}}\left(\frac{z_{ab/kl}}{z_{ij/kl}}\right) =& \frac{x_{a/k}}{x_{i/k}} \cdot \frac{y_{b/l}}{y_{j/l}} = \frac{\frac{x_{a/i}}{x_{k/i}} \cdot \frac{y_{b/j}}{y_{l/j}}}{\frac{x_{i/i}}{x_{k/i}} \cdot \frac{y_{j/j}}{y_{l/j}}} \\ =& \pi_{ij}|^\sharp_{U_{ik}\times_A V_{jl}}\left(\frac{z_{ab/ij}/z_{kl/ij}}{z_{ij/ij}/z_{kl/ij}}\right) = \pi_{ij}|^\sharp_{U_{ik}\times_A V_{jl}}\left(\frac{z_{ab/kl}}{z_{ij/kl}}\right), \end{aligned}$$

it follows that $\pi_{kl}|^\sharp_{U_{ik}\times_A V_{jl}} = \pi_{ij}|^\sharp_{U_{ik}\times_A V_{jl}}$, and therefore $\pi_{kl}|_{U_{ik}\times_A V_{jl}} = \pi_{ij}|_{U_{ik}\times_A V_{jl}}$. By Morphism of locally ringed spaces glue 8.3.2, they glue together, so that we defined morphism $\mathbb{P}^m_A \times_A \mathbb{P}^n_A \to \mathbb{P}^{mn+m+n}_A$. □

We next show that this morphism is a closed embedding. We can check this on an open cover of the target (the notion of being a closed embedding is affine-local, Proposition 10.1.4). Let's check this on the open set where $z_{ij} \neq 0$. The preimage of this open set in $\mathbb{P}^m_A \times_A \mathbb{P}^n_A$ is the locus where $x_i \neq 0$ and $y_j \neq 0$, i.e., $U_i \times_A V_j$. As described above, the map of rings is given by $z_{ab/ij} \mapsto x_{a/i} y_{b/j}$; this is clearly a surjection, as $z_{aj/ij} \mapsto x_{a/i}$ and $z_{ib/ij} \mapsto y_{b/j}$. (A generalization of this ad hoc description will be given in Chapter 16.)

This map is called the **Segre morphism** or **Segre embedding**. If A is a field, the image is called the **Segre variety**.

Proposition 11.6.1

The Segre scheme (the image of the Segre embedding) is cut out (scheme-theoretically) by the equations corresponding to

$$\operatorname{rank}\begin{bmatrix} z_{00} & \cdots & z_{0n} \\ \vdots & \ddots & \vdots \\ z_{m0} & \cdots & z_{mn} \end{bmatrix} = 1,$$

i.e., that all 2×2 minors vanish.

Proof Consider the ring homomorphism

$$\begin{aligned} \varphi_{ij} : A[z_{00/ij}, \dots, z_{mn/ij}]/(z_{ij/ij} - 1) &\longrightarrow A[x_{0/i}, \dots, x_{m/i}, y_{0/j}, \dots, y_{n/j}]/(x_{i/i} - 1, y_{j/j} - 1) \\ z_{ab/ij} &\longmapsto x_{a/i} y_{b/j}. \end{aligned}$$

By our above discussion, we know that φ_{ij} is surjective. Hence

$$A[x_{0/i}, \dots, x_{m/i}, y_{0/j}, \dots, y_{n/j}]/(x_{i/i} - 1, y_{j/j} - 1) \cong (A[z_{00/ij}, \dots, z_{mn/ij}]/(z_{ij/ij} - 1))/\operatorname{Ker}\varphi_{ij}.$$

We claim that $\operatorname{Ker}\varphi_{ij} = (z_{ab/ij} z_{cd/ij} - z_{ad/ij} z_{cb/ij})_{0 \le a < c \le m, 0 \le b < d \le n} := I$, i.e., kernel is generated by all

2×2 minors of

$$\begin{bmatrix} z_{00/ij} & \cdots & z_{0n/ij} \\ \vdots & \ddots & \vdots \\ z_{m0/ij} & \cdots & z_{mn/ij} \end{bmatrix}.$$

Note that

$$\varphi_{ij}(z_{ab/ij}z_{cd/ij} - z_{ad/ij}z_{cb/ij}) = (x_{a/i}y_{b/j})(x_{c/i}y_{d/j}) - (x_{a/i}y_{d/j})(x_{c/i}y_{b/j}) = 0,$$

we have $I \subseteq \operatorname{Ker}\varphi_{ij}$. Consider $S = (A[z_{00/ij}, \ldots, z_{mn/ij}]/(z_{ij/ij} - 1))/I$, define

$$\theta : A[x_{0/i}, \ldots, x_{m/i}, y_{0/j}, \ldots, y_{n/j}]/(x_{i/i} - 1, y_{j/j} - 1) \longrightarrow S$$
$$x_{a/i} \longmapsto z_{aj/ij} + I, \quad y_{b/j} \longmapsto z_{ib/ij} + I.$$

It is easy to see that θ is an isomorphism, so $I = \operatorname{Ker}\varphi_{ij}$. Hence on the affine piece W_{ij}, Segre scheme is cut out by all 2×2 minors of matrix

$$\begin{bmatrix} z_{00/ij} & \cdots & z_{0n/ij} \\ \vdots & \ddots & \vdots \\ z_{m0/ij} & \cdots & z_{mn/ij} \end{bmatrix}.$$

Since each affine piece glue together by coordinate change

$$z_{ab/kl} = \frac{z_{ab/ij}}{z_{kl/ij}},$$

the Segre scheme is cut out by the equations corresponding to

$$\operatorname{rank}\begin{bmatrix} z_{00} & \cdots & z_{0n} \\ \vdots & \ddots & \vdots \\ z_{m0} & \cdots & z_{mn} \end{bmatrix} = 1.$$

□

Example 11.9 (Important Example) Let's consider the first nontrivial example, when $m = n = 1$. We get a single equaition

$$\operatorname{rank}\begin{bmatrix} z_{00} & z_{01} \\ z_{10} & z_{11} \end{bmatrix} = 1,$$

i.e., $z_{00}z_{11} - z_{01}z_{10} = 0$. We again meet our old friend, quadric surface §10.3.4! Hence: the smooth quadric surface $wz - xy = 0$ (Figure 10.3) is isomorphic to $\mathbb{P}^1_k \times_k \mathbb{P}^1_k$. Recall from Exercise 10.8 that the quadric surface has two families (rulings) of lines. You may wish to check that one family of lines corresponds to the image of $\{x\} \times_k \mathbb{P}^1_k$ as x varies, and the other corresponds to the image $\mathbb{P}^1_k \times_k \{y\}$ as y varies.

If k is an algebraically closed field of characteristic not 2, then by diagonalizability of quadratics (Exercise 6.7), all rank 4 ("full rank") quadratics in 4 variables are isomorphic, so all rank 4 quadric surfaces over an algebraically closed field of characteristic not 2 are isomorphic to $\mathbb{P}^1_k \times_k \mathbb{P}^1_k$.

Note that this is not true over a field that is not algebraically closed. For example, over $\mathbb{R}$, $w^2+x^2+y^2+z^2 = 0$ (in $\mathbb{P}^3_{\mathbb{R}}$) is not isomorphic to $\mathbb{P}^1_{\mathbb{R}} \times_{\mathbb{R}} \mathbb{P}^1_{\mathbb{R}}$. Reason: the former has no real points, while the latter has lots of real points.

Proposition 11.6.2 (A coordinate-free description of products of projective A-scheme in general)

Suppose that $S_\bullet$ and $T_\bullet$ are finitely generated graded rings over A. There is an isomorphism

$$(\operatorname{Proj} S_\bullet) \times_A (\operatorname{Proj} T_\bullet) \xleftrightarrow{\sim} \operatorname{Proj}\left(\bigoplus_{n=0}^{\infty}(S_n \otimes_A T_n)\right)$$

where hopefully the definition of multiplication in the graded ring $\bigoplus_{n=0}^{\infty}(S_n \otimes_A T_n)$ is clear.

Proof Say $S_\bullet = \bigoplus_{n=0}^{\infty}(S_n \otimes_A T_n)$. Since $S_\bullet$ and $T_\bullet$ is finitely generated graded ring, by Proposition 8.4.6, we may assume that $S_\bullet$ and $T_\bullet$ are finitely generated in degree 1, say $x_1, \ldots, x_m$ and $y_1, \ldots, y_n$ are the generators of $S_\bullet$ and $T_\bullet$ respectively. Suppose $\operatorname{Proj} S_\bullet = \bigcup_{i=0}^{m} D_+(x_i)$ and $\operatorname{Proj} T_\bullet = \bigcup_{j=0}^{n} D_+(y_j)$, where $\deg x_i = \deg y_j = 1$. Define

$$\begin{aligned}\varphi_{ij}^\sharp : ((S_\bullet)_{x_i})_0 \otimes_A ((T_\bullet)_{y_j})_0 &\longrightarrow ((R_\bullet)_{x_i \otimes y_j})_0 \\ \frac{f}{x_i^k} \otimes \frac{g}{y_j^l} &\longmapsto \frac{f x_i^l \otimes g y_j^k}{(x_i \otimes y_j)^{k+l}}.\end{aligned}$$

Define the inverse of $\varphi_{ij}^\sharp$

$$\begin{aligned}\psi : ((R_\bullet)_{x_i \otimes y_j})_0 &\longrightarrow ((S_\bullet)_{x_i})_0 \otimes_A ((T_\bullet)_{y_j})_0 \\ \frac{r}{(x_i \otimes y_j)^n} &\longmapsto \sum_i \left(\frac{s_i}{x_i^n} \otimes \frac{t_i}{y_i^n}\right),\end{aligned}$$

where $r = \sum_i s_i \otimes t_i \in R_n = S_n \otimes T_n$. Let $\frac{f}{x_i^k} \otimes \frac{g}{y_j^l} \in ((S_\bullet)_{x_i})_0 \otimes_A ((T_\bullet)_{y_j})_0$, then

$$\psi \circ \varphi_{ij}^\sharp\left(\frac{f}{x_i^k} \otimes \frac{g}{y_j^l}\right) = \psi\left(\frac{f x_i^l \otimes g y_j^k}{(x_i \otimes y_j)^{k+l}}\right) = \frac{f}{x_i^k} \otimes \frac{g}{y_j^l},$$

so $\psi \circ \varphi_{ij}^\sharp = \mathrm{id}_{((S_\bullet)_{x_i})_0 \otimes_A ((T_\bullet)_{y_j})_0}$. Let $\frac{r}{(x_i \otimes y_j)^n} \in ((R_\bullet)_{x_i \otimes y_j})_0$, then

$$\varphi_{ij}^\sharp \circ \psi\left(\frac{r}{(x_i \otimes y_j)^n}\right) = \varphi^\sharp\left(\sum_i \left(\frac{s_i}{x_i^n} \otimes \frac{t_i}{y_i^n}\right)\right) = \sum_i \frac{s_i \otimes t_i}{(x_i \otimes y_j)^n} = \frac{r}{(x_i \otimes y_j)^n},$$

so $\varphi_{ij}^\sharp \circ \psi = \mathrm{id}_{((R_\bullet)_{x_i \otimes y_j})_0}$. Hence $\varphi_{ij}^\sharp$ is isomorphism, and therefore

$$\varphi_{ij} : D_+(x_i \otimes y_j) \xrightarrow{\sim} D_+(x_i) \times_A D_+(y_j).$$

It is easy to check that φ_{ij} glue together, hence there is an isomorphism

$$(\operatorname{Proj} S_\bullet) \times_A (\operatorname{Proj} T_\bullet) \xleftrightarrow{\sim} \operatorname{Proj}\left(\bigoplus_{n=0}^{\infty}(S_n \otimes_A T_n)\right).$$

□

Proposition 11.6.3 (A coordinate-free description of the Segre embedding)

The Segre embedding can be interpreted as $\mathbb{P}V \times \mathbb{P}W \to \mathbb{P}(V \otimes W)$ via the surjective map of graded rings

$$\operatorname{Sym}^\bullet(V^\vee \otimes W^\vee) \longrightarrow\!\!\!\!\!\rightarrow \bigoplus_{i=0}^{\infty}(\operatorname{Sym}^i V^\vee) \otimes (\operatorname{Sym}^i W^\vee)$$

"in the opposite direction".

Proof Let $V^\vee = \operatorname{Span}_k\{x_1, \ldots, x_m\}$ and $W^\vee = \operatorname{Span}_k\{y_1, \ldots, y_n\}$, then $V^\vee \otimes W^\vee = \operatorname{Spec}_k\{x_i \otimes$

$y_j\}_{1\le i\le m,1\le j\le n}$. Define graded ring map

$$\varphi:\operatorname{Sym}^\bullet(V^\vee\otimes W^\vee)\twoheadrightarrow\bigoplus_{i=0}^{\infty}(\operatorname{Sym}^i V^\vee)\otimes(\operatorname{Sym}^i W^\vee)$$

by setting

$$z_{ij}:=x_i\otimes y_j\longmapsto x_i\otimes y_j, \tag{11.10}$$

where $x_i\in\operatorname{Sym}^1V^\vee$ and $y_j\in\operatorname{Sym}^1W^\vee$. Let $(x_{i_1}\otimes\cdots\otimes x_{i_k})\otimes(y_{j_1}\otimes\cdots\otimes x_{j_k})\in(\operatorname{Sym}^kV^\vee)\otimes(\operatorname{Sym}^kW^\vee)$, note that

$$\varphi(z_{i_1j_1}\otimes\cdots\otimes z_{i_kj_k})=(x_{i_1}\otimes\cdots\otimes x_{i_k})\otimes(y_{j_1}\otimes\cdots\otimes x_{j_k}),$$

we know that φ is surjective. By Proposition 10.3.1, φ induces morphism

$$\operatorname{Proj}\left(\bigoplus_{i=0}^{\infty}(\operatorname{Sym}^i V^\vee)\otimes(\operatorname{Sym}^i W^\vee)\right)\longrightarrow\operatorname{Proj}\operatorname{Sym}^\bullet(V^\vee\otimes W^\vee),$$

by Proposition 11.6.2, $\operatorname{Proj}\left(\bigoplus_{i=0}^{\infty}(\operatorname{Sym}^i V^\vee)\otimes(\operatorname{Sym}^i W^\vee)\right)\cong\mathbb{P}V\times_k\mathbb{P}W$, so we have a morphism

$$\mathbb{P}V\times\mathbb{P}W\longrightarrow\mathbb{P}(V\otimes W),$$

by (11.10), we know that this morphism is same as Segre embedding which we defined above. □

11.7 Normalization

11.7.1 Basic of Nomalization

We discuss normalization now only because the central construction gives practice with the central ideal behind the construction in §11.1 of the fibered product (see Proposition 11.7.1 and Definition 11.7.3).

Normalization is a means of turning a reduced scheme into a normal scheme. (**Unimportant remark:** reducedness is not a necessary hypothesis, and is included only to avoid distraction. If you care, you can follow through the construction, and realize that the normalization of a scheme factors through the reduction, and is the normalization of the reduction.) A **normalization** of a reduced scheme X is a morphism $v:\widetilde{X}\to X$ from a normal scheme, where v induces a bijection of irreducible components of $\widetilde{X}$ and X, and in reasonable cases (cf. Proposition) v gives a birational morphism on each of the irreducible components. It will satisfy a universal property, and hence it will be unique upto unique isomorphism. Figure 9.4 is an example of a normalization.

We begin with the case where X is irreducible, and hence (by Proposition 6.2.7) integral. (We will then deal with a more general case, and also discuss normalization in a function extension.)

Definition 11.7.1 (Normalization of integral scheme)

*Let X be an integral scheme (irreducible and reduced), the **normalization** $v:\widetilde{X}\to X$ is a dominant morphism (not just dominant rational map!) from an integral normal scheme to X, such that any dominant morphism $f:X\to Y$ with Y integral normal factors through v:*

$$\begin{tikzcd} \text{normal} & Y \arrow[rr, dashed, "\exists!"] \arrow[dr, "\text{dominant}"'] & & \widetilde{X} \arrow[dl, "v\ \text{dominant}"] & \text{normal} \\ & & X & & \end{tikzcd} \tag{11.11}$$

Thus if the normalization exists, then it is unique up to unique isomorphism. We now have to show that it exists, and we do this in a way that will look familiar. We deal first with the case where X is affine, say $X = \operatorname{Spec} A$, where A is an integral domain. Then let $\widetilde{A}$ be the integral closure of A in its fraction field $K(A)$.

Lemma 11.7.1

Suppose A is an integral domain, let $\widetilde{A}$ be the integral closure of A in its fraction field $K(A)$. Then $v : \operatorname{Spec} \widetilde{A} \to \operatorname{Spec} A$ satisfies the universal property of normalization.

Proof Let Z be any integral normal scheme with dominant morphism $f : Z \to \operatorname{Spec} A$, we may assume $Z = \bigcup_i \operatorname{Spec} B_i$, where each B_i is normal (integrally closed integral domain). To prove that $v : \operatorname{Spec} \widetilde{A} \to \operatorname{Spec} A$ satisfies the universal property of normalization, we need to show that for any normal scheme Z and any dominant morphism $f : Z \to \operatorname{Spec} A$, there exits a unique morphism $g : Z \to \widetilde{X}$ such that $f = v \circ g$. Since this property is local, we can first construct the morphism on each affine open subset $\operatorname{Spec} B_i \hookrightarrow Z$. Since f is dominant and Z is irreducible, $\operatorname{Spec} B_i$ is dense in Z, and therefore $f(\operatorname{Spec} B_i)$ is dense in $f(Z)$, note that $f(Z)$ is dense in $\operatorname{Spec} A$, we know that $f(\operatorname{Spec} B_i)$ is dense in $\operatorname{Spec} A$, so $f_i := f|_{\operatorname{Spec} B_i}$ is dominant. If we prove the case that Z is affine scheme, there exists unique morphism $g_i : \operatorname{Spec} B_i \to \operatorname{Spec} \widetilde{A}$ such that $f_i = v \circ g_i$. By the universal property of normalization (affine case, if we proved), we have $g_i|_{\operatorname{Spec} B_i \cap \operatorname{Spec} B_j} = g_j|_{\operatorname{Spec} B_i \cap \operatorname{Spec} B_j}$, so they can glue together, then we get a morphism $g : Z \to \operatorname{Spec} \widetilde{A}$ such that $f = v \circ g$.

Now, we shall to prove the case that Z is an affine normal scheme. Suppose $Z = \operatorname{Spec} B$, where B is normal, i.e., B is an integral domain and integrally closed in $K(B)$. Since $f : \operatorname{Spec} B \to \operatorname{Spec} A$ is dominant, by Proposition 8.5.3, $\operatorname{Ker} f^\sharp \subseteq \mathfrak{N}(A)$. Since A is integral domain, $\mathfrak{N}(A) = 0$, so $f^\sharp : A \to B$ is injective, and therefore A can be seen as a subring of B. Hence $K(A) \subseteq K(B)$. We next define ring homomorphism $g^\sharp : \widetilde{A} \to B$. Let $a' \in \widetilde{A}$, since a' is integral over A, there exists a monic

$$p(x) = x^n + a_{n-1}x^{n-1} + \cdots + a_0 \in A[x]$$

such that $p(a') = 0$. Since A is a subring of B, $p(x) \in B$, so a' is also integral over B. Since B is integrally closed, $a' \in B$. Define $g^\sharp : \widetilde{A} \to B$ by setting $g^\sharp(a') = a'$, then we get a morphism of schemes $g : \operatorname{Spec} B \to \operatorname{Spec} \widetilde{A}$. Note that for $a \in A$ we have $g^\sharp \circ v^\sharp(a) = a = f^\sharp(a)$, so $v \circ g = f$. Finally, we show the uniqueness of g. If there exists another $g' : \operatorname{Spec} B \to \operatorname{Spec} \widetilde{A}$ such that $v \circ g' = f$, then we have $g'^\sharp \circ v^\sharp = f^\sharp$. For all $a \in A$, we have

$$g'^\sharp \circ v^\sharp(a) = g'^\sharp(a) = f^\sharp(a) = a = g^\sharp(a).$$

Hence $g = g'$, we proved uniqueness. □

Remark 11.26 En route, you might show that the global sections of an irreducible normal scheme are also "normal". i.e., integrally closed.

Proposition 11.7.1

The normalizations of integral schemes exist in general.

Proof Suppose X is an integral scheme. Suppose $X = \bigcup_i \operatorname{Spec} A_i$, where each A_i is integral domain. By Lemma 11.7.1, for each $\operatorname{Spec} A_i$ exists dominant morphism $v_i : \operatorname{Spec} \widetilde{A_i} \to \operatorname{Spec} A_i$ satisfies the universal property of normalization. Consider $\operatorname{Spec} A_i \cap \operatorname{Spec} A_j$, by Proposition 6.3.1, $\operatorname{Spec} A_i \cap \operatorname{Spec} A_j = \bigcup_k \operatorname{Spec}(A_i)_{f_k} = \bigcup_k \operatorname{Spec}(A_j)_{g_k}$ where $\operatorname{Spec}(A_i)_{f_k} = \operatorname{Spec}(A_j)_{g_k}$. Consider $v_i^{-1}(\operatorname{Spec} A_i \cap \operatorname{Spec} A_j)$, we have

$$v_i^{-1}(\operatorname{Spec} A_i \cap \operatorname{Spec} A_j) = v_i^{-1}\left(\bigcup_k \operatorname{Spec}(A_i)_{f_k}\right) = \bigcup_k v_i^{-1}(\operatorname{Spec}(A_i)_{f_k}) = \bigcup_k \operatorname{Spec} \widetilde{A_i}_{f_k},$$

similarly, we have $v_j^{-1}(\operatorname{Spec} A_i \cap \operatorname{Spec} A_j) = \bigcup_k \operatorname{Spec} \widetilde{A_j}_{g_k}$. By Proposition 6.2.9, $K(A_i) = K(A_j) = K(X)$, since $(A_i)_{f_k} \cong (A_j)_{g_k}$, we have $\widetilde{(A_i)_{f_k}} \cong \widetilde{(A_j)_{g_k}}$. By Proposition 6.4.1, $\widetilde{A_i}_{f_k}$ and $\widetilde{A_j}_{g_k}$ are integral closures of $(A_i)_{f_k}$ and $(A_j)_{g_k}$, so we have $\widetilde{(A_i)_{f_k}} \cong \widetilde{A_i}_{f_k}$ and $\widetilde{(A_j)_{g_k}} \cong \widetilde{A_j}_{g_k}$. Hence $v_i^{-1}(\operatorname{Spec} A_i \cap \operatorname{Spec} A_j) \cong v_j^{-1}(\operatorname{Spec} A_i \cap \operatorname{Spec} A_j)$. The cocycle condition of Theorem 5.4.1 is easy to check, by Theorem 5.4.1, we can glue $\operatorname{Spec} \widetilde{A_i}$ together via $\varphi_{ij} : v_i^{-1}(\operatorname{Spec} A_i \cap \operatorname{Spec} A_j) \xrightarrow{\sim} v_j^{-1}(\operatorname{Spec} A_i \cap \operatorname{Spec} A_j)$, say $\widetilde{X} = \bigcup_i \operatorname{Spec} \widetilde{A_i}$, then X is integral normal scheme. Consider $v_i|_{\operatorname{Spec} \widetilde{A_i} \cap \operatorname{Spec} \widetilde{A_j}}$ and $v_j|_{\operatorname{Spec} \widetilde{A_i} \cap \operatorname{Spec} \widetilde{A_j}}$, note that

$$\operatorname{Spec} \widetilde{A_i} \cap \operatorname{Spec} \widetilde{A_j} = v_i^{-1}(\operatorname{Spec} A_i \cap \operatorname{Spec} A_j) \cong v_j^{-1}(\operatorname{Spec} A_i \cap \operatorname{Spec} A_j),$$

there must be $v_i|_{\operatorname{Spec} \widetilde{A_i} \cap \operatorname{Spec} \widetilde{A_j}} = v_j|_{\operatorname{Spec} \widetilde{A_i} \cap \operatorname{Spec} \widetilde{A_j}}$. By the morphism of locally ringed spaces glue 8.3.2, we get a morphism $v : \widetilde{X} \to X$ such that $v|_{\operatorname{Spec} \widetilde{A_i}} = v_i$. Consider $v(\widetilde{X})$, since each v_i is dominant and X is irreducible (since X is integral), we have

$$v(\widetilde{X}) = v\left(\bigcup_i \operatorname{Spec} \widetilde{A_i}\right) = \bigcup_i v_i(\operatorname{Spec} \widetilde{A_i})$$

where $v_i(\operatorname{Spec} \widetilde{A_i})$ dense in $\operatorname{Spec} A_i$ and $\operatorname{Spec} A_i$ is dense in X. Hence $v(\widetilde{X})$ is dense in X, and therefore v is dominant. Thus we get a dominant morphism from integral normal scheme $\widetilde{X}$ to integral scheme X, i.e., $v : \widetilde{X} \to X$.

We next show that $v : \widetilde{X} \to X$ satisfies the universal property of normalization (Definition 11.7.1). Let Y be any integral normal scheme with dominant morphism $f : Y \to X$. Note that $Y = \bigcup_i f^{-1}(\operatorname{Spec} A_i)$ and each $f_i := f|_{\operatorname{Spec} A_i} : f^{-1}(\operatorname{Spec} A_i) \to \operatorname{Spec} A_i$ is dominant morphism from integral normal scheme to integral scheme. By the universal property of $v_i : \operatorname{Spec} \widetilde{A_i} \to \operatorname{Spec} A_i$ (Lemma 11.7.1), there exits a unique morphism $g_i : f^{-1}(\operatorname{Spec} A_i) \to \operatorname{Spec} \widetilde{A_i}$ such that $v_i \circ g_i = f_i$. Consider $g_i|_{f^{-1}(\operatorname{Spec} A_i) \cap f^{-1}(\operatorname{Spec} A_j)}$ and $g_j|_{f^{-1}(\operatorname{Spec} A_i) \cap f^{-1}(\operatorname{Spec} A_j)}$, we want to show that $g_i|_{f^{-1}(\operatorname{Spec} A_i) \cap f^{-1}(\operatorname{Spec} A_j)} = g_j|_{f^{-1}(\operatorname{Spec} A_i) \cap f^{-1}(\operatorname{Spec} A_j)}$. Recall that

$$\operatorname{Spec} A_i \cap \operatorname{Spec} A_j = \bigcup_k \operatorname{Spec}(A_i)_{f_k} = \bigcup_k \operatorname{Spec}(A_j)_{g_k},$$

we have

$$f^{-1}(\operatorname{Spec} A_i \cap \operatorname{Spec} A_j) = \bigcup_k f^{-1}(\operatorname{Spec}(A_i)_{f_k}) = \bigcup_k f^{-1}(\operatorname{Spec}(A_j)_{g_k}).$$

Recall that

$$v_i^{-1}(\operatorname{Spec} A_i \cap \operatorname{Spec} A_j) = \bigcup_k \operatorname{Spec} \widetilde{A_i}_{f_k}$$

and

$$v_j^{-1}(\operatorname{Spec} A_i \cap \operatorname{Spec} A_j) = \bigcup_k \operatorname{Spec} \widetilde{A_j}_{g_k},$$

note that $\widetilde{A_i}_{f_k} \cong \widetilde{A_j}_{g_k}$ and $A_{i f_k} = A_{j g_k}$, by the universal property of $\operatorname{Spec} \widetilde{A_i}_{f_k} \to \operatorname{Spec} A_{i f_k}$ (or $\operatorname{Spec} \widetilde{A_j}_{g_k} \to \operatorname{Spec} A_{j g_k}$), we have $g_i|_{f^{-1}(\operatorname{Spec}(A_i)_{f_k})} = g_j|_{f^{-1}(\operatorname{Spec}(A_i)_{f_k})}$ for all k. Hence

$$g_i|_{f^{-1}(\operatorname{Spec} A_i) \cap f^{-1}(\operatorname{Spec} A_j)} = g_j|_{f^{-1}(\operatorname{Spec} A_i) \cap f^{-1}(\operatorname{Spec} A_j)}.$$

By the morphism of locally ringed space glue 8.3.2, they glue together, then we get a unique morphism $g : Y \to \widetilde{X}$ such that $v \circ g = f$, where the uniqueness of g is given by the uniqueness of each g_i.

Hence we proved the existence of normalization of integral scheme. □

Proposition 11.7.2

Normalizations are integral and surjective.

Proof Let X be an integral scheme, by Proposition 11.7.1, we have the normalization morphism $v : \widetilde{X} \to X$. Suppose $X = \bigcup_i \operatorname{Spec} A_i$, where each A_i is integral, by the proof in Proposition 11.7.1, $\widetilde{X} = \bigcup_i \operatorname{Spec} \widetilde{A_i}$, where $\widetilde{A_i}$ is the integral closure of A_i in its fraction field. By Proposition 9.3.6, we know that v is affine, and by Proposition 9.3.14, v is integral.

Let $p \in X$, then there exists an affine open set $\operatorname{Spec} A_i \subseteq X$ such that $p = [\mathfrak{p}] \in \operatorname{Spec} A_i$. Consider $v|_{v^{-1}(\operatorname{Spec} A_i)} : \operatorname{Spec} \widetilde{A_i} \to \operatorname{Spec} A_i$, $v|_{v^{-1}(\operatorname{Spec} A_i)}$ is given by ring map $A_i \to \widetilde{A_i}$. Note that $A_i \to \widetilde{A_i}$ is integral extension, by the Lying Over Theorem 9.2.2, we have $\operatorname{Spec} \widetilde{A_i} \to \operatorname{Spec} A_i$ is surjective, and therefore $v : \widetilde{X} \to X$ is surjective. □

We will soon see that normalization of integral finite type k-schemes is always a birational morphism, in Proposition 11.7.5.

Definition 11.7.2 (Normalization of reduced scheme)

Let X be an reduced scheme such that every quasi-compact open has finitely many irreducible components, the normalization $v : \widetilde{X} \to X$ is a morphism from a normal scheme to X such that each irreducible component of $\widetilde{X}$ dominates an irreducible component of X, and satisfies the following universal property: for any morphism $Y \to X$ with Y a normal scheme such that each irreducible component of Y dominates an irreducible component of X there exists a unique factorization

$$\begin{tikzcd} Y \arrow[rr, dashed, "\exists!"] \arrow[dr] & & \widetilde{X} \arrow[dl] \\ & X. & \end{tikzcd}$$

Remark 11.27 If you wish, you can show that the normalization exists in this case. See [Sta25, Lemma 035Q] for more.

Here are some examples.

Example 11.10 Normalization of the nodal cubic Suppose that $\operatorname{char} k \neq 2$ (only so we can use the word "nodal"). $\operatorname{Spec} k[t] \to \operatorname{Spec} k[x,y]/(y^2 - x^2(x+1))$ given by $(x,y) \mapsto (t^2-1, t(t^2-1))$ (see Figure 9.4) is a normalization. The target curve is called the **nodal cubic curve**.

Proof We first show that $A := k[x,y]/(y^2-x^2(x+1))$ is integral domain. It suffices to show that $y^2-x^2(x+1)$ is irreducible in $k[x,y]$. Suppose $y^2 - x^2(x+1) = (y-a(x))(y-b(x))$, then we have

$$\begin{cases} a(x)+b(x) & = 0 \\ a(x)b(x) & = x^2(x+1), \end{cases}$$

it follows that $a(x)^2 = -x^2(x+1)$. Since $\operatorname{char} k \neq 2$, $x+1$ not a square, a contradiction. So $y^2 - x^2(x+1)$ is irreducible in $k[x,y]$.

We next show that $k[t]$ and A have the same fraction field. In fact, $K(k[t]) = k(t)$. Define $\varphi : k(t) \to K(A)$ by setting $t \mapsto \frac{y}{x}$. Since $y^2 = x^2(x+1)$ in $K(A)$, note that $\varphi(t^2-1) = x$ and $\varphi(t(t^2-1)) = y$, we know that φ is surjective. Since φ is a homomorphism of fields, φ is injective. Hence φ is isomorphism, i.e., $K(A) \cong k(t)$.

Since $k[t]$ is UFD, by Proposition 6.4.6, we know that $k[t]$ is integrally closed in its fraction field. We want to show that $k[t]$ is contained in $\widetilde{A}$ (the integral closure of A). Consider $F(T) = T^2 - (x+1) \in A[T]$, we have

$$F(t) = t^2 - (x+1) = 0,$$

hence t is integral over A, and therefore $t \in \widetilde{A}$. By Proposition 9.2.4, $k[t] \subseteq \widetilde{A}$. Since $k[t]$ is integrally closed, we know that $\widetilde{A} = k[t]$. □

We will see from the previous example that once we guess what the normalization is, it isn't hard to verify that it is indeed the normalization. Perhaps a few words are in order as to where the polynomials $t^2 - 1$ and $t(t^2 - 1)$ arose in the previous exercise. The Key idea is to guess $t = y/x$. (Then $t^2 = x + 1$ and $y = xt$ quickly.) This idea comes from three possible places. We begin by sketching the curve, and noticing the "node" at the origin. ("Node" will be formally defined in Chapter 29.)

(a) The function y/x is well-defined away from the node, and at the node, two branches have "values" $y/x = 1$ and $y/x = -1$.

(b) We can also note that if $t = y/x$, then t^2 is a polynomial, so we will need to adjoint t in order to obtain the normalization.

(c) The curve is cubic, so we except a general line to meet the cubic in three points, counted with multiplicity. (We will make this precise when we discuss Bézout's Theorem, Chapter 19, but in this case we have already gotten a hint of this in Exercise 8.10.)

There is a $\mathbb{P}^1$ parametrizing lines through the origin (with coordinate equal to the slope of the line, y/x), and most such lines meet the curve with multiplicity two at the origin, and hence meet the curve at precisely one other point of the curve. So this "coordinates" most of the curve, and we try adding in this coordinate.

Example 11.11 The normalization of the cusp $y^2 = x^3$ (see Figure 11.5). ("Cusp" will be formally defined in Chapter 29.)

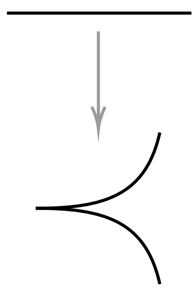


Figure 11.4: Normalization of a cusp

We claim that $\operatorname{Spec} k[t] \to \operatorname{Spec} k[x, y]/(y^2 - x^3)$ given by $(x, y) \mapsto (t^2, t^3)$ is a normalization. Say $A = k[x, y]/(y^2 - x^3)$. Clearly, $y^2 - x^3$ is irreducible in $k[x, y]$, so A is integral domain. We first show that $K(A) \cong k(t)$. Define $\varphi : k(t) \to K(A)$ by setting $t \mapsto \frac{y}{x}$. It is easy to see that φ is a well-defined homomorphism of fields. To show that φ is an isomorphism, it suffices to show that φ is surjective. Since $y^2 = x^3$ in $K(A)$, we have $\varphi(t^2) = x$ and $\varphi(t^3) = y$, which implies that φ is surjective. So A and $k[t]$ has the same fraction field $k(t)$. By Example 11.10 , we already know that $k[t]$ is integrally closed, hence it suffices to show that $k[t] \subseteq \widetilde{A}$. By Proposition 9.2.4, it suffices to show that $t \in \widetilde{A}$. Consider $F(T) = T^2 - x \in A[T]$, since

$$F(t) = t^2 - x = 0,$$

t is integral over $A[T]$, as we desired.

Example 11.12 Suppose $\operatorname{char} k \neq 2$. We next find the normalization of the **tacnode** $y^2 = x^4$. ("Tacnode" will be formally defined in Chapter 29.) Let $A = k[x, y]/(y^2 - x^4)$, by the Chinese Remainder Theorem 5.4.2,

$$A = k[x, y]/(y - x^2) \times k[x, y]/(y + x^2).$$

Say $X = \operatorname{Spec} A$, we have

$$\begin{aligned} X =& \operatorname{Spec}(k[x,y]/(y-x^2) \times k[x,y]/(y+x^2)) \\ \cong& \operatorname{Spec} k[x,y]/(y-x^2) \coprod \operatorname{Spec} k[x,y]/(y+x^2) \\ \cong& \mathbb{A}_k^1 \coprod \mathbb{A}_k^1. \end{aligned}$$

Each branch of X is already integral normal scheme, so $\widetilde{X} = \mathbb{A}_k^1 \coprod \mathbb{A}_k^1$. And normalization morphism is given by as follow: at the branch $\operatorname{Spec} k[x,y]/(y-x^2)$ given by the ring map $(x,y) \mapsto (t,t^2)$; at the branch $\operatorname{Spec} k[x,y]/(y+x^2)$ given by the ring map $(x,y) \mapsto (s,-s^2)$.

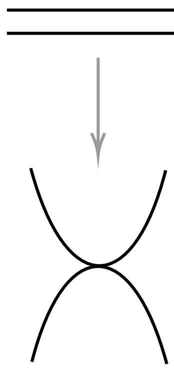


Figure 11.5: Normalization of a tacnode

(Although we haven't defined "singularity", "smooth", "curve", or "dimension", you should still read this.) Notice that in the previous examples, normalization "resolves" the singularities ("non-smooth" points) of the curve. In general, it will do so in dimension one (in reasonable Noetherian circumstances, as normal local Noetherian integral domains of dimension one are all discrete valuation rings, Chapter 14), but won't do so in higher dimension (the cone $z^2 = x^2 + y^2$ over a field k of characteristic not 2 is normal, Exercise 6.6 (b)).

Lemma 11.7.2 (Relation between integrally closed and conductor ideals)

Suppose A is an integral domain, $\widetilde{A}$ is the integral closure of A, and $\widetilde{A}$ is finitely generated A-module. Define ***the conductor ideal of*** A ***in*** $\widetilde{A}$ *by setting*

$$\mathfrak{f}(\widetilde{A}/A) := \operatorname{Ann}_A(\widetilde{A}/A) = \{a \in A : a\widetilde{A} \subseteq A\}.$$

Let $\mathfrak{p} \in \operatorname{Spec} A$, then $A_\mathfrak{p}$ is integrally closed if and only if $\mathfrak{p} \not\supseteq \mathfrak{f}(\widetilde{A}/A)$.

Proof Consider the following exact sequence

$$0 \longrightarrow A \longrightarrow \widetilde{A} \longrightarrow \widetilde{A}/A \longrightarrow 0.$$

Let $\mathfrak{p} \in \operatorname{Spec} A$, since localization is an exact covariant functor (Exercise 2.37 (a)), we have exact sequence (as A-module)

$$0 \longrightarrow A_\mathfrak{p} \longrightarrow \widetilde{A} \otimes_A A_\mathfrak{p} \longrightarrow \widetilde{A}/A \otimes_A A_\mathfrak{p} \longrightarrow 0.$$

Say $M = \widetilde{A}/A$, since $\widetilde{A}$ is finitely generated A-module, M is finitely generated A-module. $A_\mathfrak{p}$ is integrally closed if and only if $A_\mathfrak{p} \cong \widetilde{A_\mathfrak{p}} \cong \widetilde{A}_\mathfrak{p}$ if and only if $M_\mathfrak{p} = 0$. Consider $\operatorname{Supp} M$, since M is finitely generated A-module, by Proposition 7.6.1, we have $\operatorname{Supp} M = V(\operatorname{Ann}_A M) = V(\mathfrak{f}(\widetilde{A}/A))$. Hence $\mathfrak{p} \supseteq \mathfrak{f}(\widetilde{A}/A)$ if and only if $M_\mathfrak{p} \neq 0$, and therefore $A_\mathfrak{p}$ is integrally closed if and only if $\mathfrak{p} \not\supseteq \mathfrak{f}(\widetilde{A}/A)$. □

✍ **Exercise 11.13** Suppose $X = \operatorname{Spec} \mathbb{Z}[15i]$. Describe the normalization $\widetilde{X} \to X$.

Proof The fraction field of $\mathbb{Z}[15i]$ is $\mathbb{Q}(15i) = \mathbb{Q}(i)$, hence $\mathbb{Z}[i]$ and $\mathbb{Z}[15i]$ have the same fraction field. Note that $\mathbb{Z}[i]$ is a UFD, and hence is integrally closed by Proposition 6.4.6. Consider $\mathbb{Z}[15i] \hookrightarrow \mathbb{Z}[i]$, we want to show that $\mathbb{Z}[i] \subseteq \widetilde{\mathbb{Z}[15i]}$, where $\widetilde{\mathbb{Z}[15i]}$ is the integral closure of $\mathbb{Z}[15i]$. It suffices to show that $i \in \widetilde{\mathbb{Z}[15i]}$. Consider $f(X) = X^2 + 1 \in \mathbb{Z}[15i][X]$, we have $f(i) = -1 + 1 = 0$, hence i is integral over $\mathbb{Z}[15i][X]$, by Proposition 9.2.4, we know that $\mathbb{Z}[i] \subseteq \widetilde{\mathbb{Z}[15i]}$. Hence $\widetilde{X} = \operatorname{Spec} \mathbb{Z}[i]$.

To find which points of X is the normalization not an isomorphism, it suffices to find that which $\mathfrak{p} \in \operatorname{Spec} \mathbb{Z}[15i]$ such that $\mathbb{Z}[15i]_{\mathfrak{p}}$ not integrally closed. Clearly, $\mathbb{Z}[i]$ is finitely generated $\mathbb{Z}[15i]$-module, by Lemma 11.7.2, for $\mathfrak{p} \in \operatorname{Spec} \mathbb{Z}[15i]$, $\mathfrak{p} \supseteq \mathfrak{f}(\mathbb{Z}[i]/\mathbb{Z}[15i])$ if and only if $A_{\mathfrak{p}}$ is not integrally closed. We next compute the conductor ideal, claim that $\mathfrak{f}(\mathbb{Z}[i]/\mathbb{Z}[15i]) = (15, 15i)$. Clearly, $(15, 15i)\mathbb{Z}[i] \subseteq \mathbb{Z}[15i]$, then we have $(15, 15i) \subseteq \mathfrak{f}(\mathbb{Z}[i]/\mathbb{Z}[15i])$. Let $a \in \mathfrak{f}(\mathbb{Z}[i]/\mathbb{Z}[15i])$, then $a\mathbb{Z}[i] \subseteq \mathbb{Z}[15i]$, hence $a \in (15, 15i)$. then we have $\mathfrak{f}(\mathbb{Z}[i]/\mathbb{Z}[15i]) = (15, 15i)$. Let $\mathfrak{p} \in \operatorname{Spec} \mathbb{Z}[15i]$ with $\mathfrak{p} \supseteq (15, 15i)$, since $15 \in \mathfrak{p}$, we have $3 \in \mathfrak{p}$ or $5 \in \mathfrak{p}$. Note that $15i$ is irreducible in $\mathbb{Z}[15i]$, $\mathfrak{p}$ must contains $15i$. Hence $\mathfrak{p} \supseteq (3, 15i)$ or $\mathfrak{p} \supseteq (5, 15i)$. Note that $\mathbb{Z}[15i]/(3, 15i) \cong \mathbb{F}_3$ and $\mathbb{Z}[15i]/(5, 15i) \cong \mathbb{F}_5$, $(3, 15i)$ and $(5, 15i)$ are maximal ideals of $\mathbb{Z}[15i]$, hence $\mathfrak{p} = (3, 15i)$ or $\mathfrak{p} = (5, 15i)$. $\square$

Remark 11.28 Another example in a similar vein is the normalization of the "pinched plane", Chapter 14.

The following is an important generalization of normalization (of integral scheme):

Definition 11.7.3 (Normalization in a function field)

Suppose X is an integral scheme. The ***normalization*** $v : \widetilde{X} \to X$ ***of*** X ***in a given algebraic field extension*** L ***of the function field*** $K(X)$ ***of*** X *is a dominant morphism from a normal integral scheme $\widetilde{X}$ with function field L, such that v induces the inclusion $K(X) \hookrightarrow L$, and that is universal with respect to this property.*

$$\begin{tikzcd}
\operatorname{Spec} L \arrow[r, equal] \arrow[dr, equal] & \operatorname{Spec} K(Y) \arrow[r] \arrow[d, "\sim"'] & Y \arrow[d, dashed, "\exists!"'] \arrow[dd, bend left] & \textit{normal} \\
& \operatorname{Spec} K(\widetilde{X}) \arrow[r] \arrow[d] & \widetilde{X} \arrow[d, "v"'] & \textit{normal} \\
& \operatorname{Spec} K(X) \arrow[r] & X &
\end{tikzcd}$$

Note *Suppose A is an integral domain, $L/K(A)$ is arbitrary algebraic extension. Denote $\operatorname{Int.cl}_L(A)$ be the integral closure of A in L.*

Lemma 11.7.3 (Generalize Proposition 9.2.6)

Suppose A is an integral domain, $L/K(A)$ is arbitrary algebraic extension, and $\operatorname{Int.cl}_L(A)$ be the integral closure of A in L. Then $\operatorname{Int.cl}_L(A)$ is integrally closed in $K(\operatorname{Int.cl}_L(A)) = L$.

Proof Let $x \in L$ is integral over $\operatorname{Int.cl}_L(A)$, then there exists a monic

$$x^n + a_{n-1}x^{n-1} + \cdots + a_0 = 0,$$

where each $a_i \in \operatorname{Int.cl}_L(A)$. Since each $a_i \in L$ is integral over A, by Corollary 9.2.2, $A[a_0, a_1, \ldots, a_{n-1}]$ is finitely generated A-module. Now x is integral over $A[a_0, \ldots, a_{n-1}]$, by Proposition 9.2.5, $A[a_0, \ldots, a_{n-1}][x]$ is finitely generated $A[a_0, \ldots, a_{n-1}]$-module, and therefore $A[a_0, \ldots, a_{n-1}][x]$ is finitely generated A-module, by Proposition 9.2.5 again, $x \in L$ is integral over A, i.e., $x \in \operatorname{Int.cl}_L(A)$. It follows that $\operatorname{Int.cl}_L(A)$ is integrally closed.

We next show that $K(\operatorname{Int.cl}_L(A)) = L$. Say $K := K(\operatorname{Int.cl}_L(A))$. Since $\operatorname{Int.cl}_L(A) \subseteq L$ and L is field, we have $K \subseteq L$ (fraction field of $\mathrm{Int.\,cl}_L(A)$ is the smallest field which contains $\mathrm{Int.\,cl}_L(A)$). Let $y \in L$, since L/K is algebraic extension, there exists a non-zero polynomial in $K[t]$ such that

$$c_m y^m + c_{m-1} y^{m-1} + \cdots + c_0 = 0, \tag{11.12}$$

where each $c_j \in \mathrm{Int.\,cl}_L(A)$ and $c_m \neq 0$. Multiply both sides of (11.12) by c_m^{m-1}, we get

$$(c_m y)^m + c_{m-1}(c_m y)^{m-1} + \cdots + c_0 c_m^{m-1} = 0$$

Hence $c_m y$ is integral over $\mathrm{Int.\,cl}_L(A)$, since $\mathrm{Int.\,cl}_L(A)$ is integrally closed, $c_m y \in \mathrm{Int.\,cl}_L(A)$. Note that $y = \frac{c_m y}{c_m}$, we know that $y \in K$. Hence $K(\mathrm{Int.\,cl}_L(A)) = L$. □

Lemma 11.7.4

Suppose A is an integral domain, $L/K(A)$ is arbitrary algebraic extension. Then $v : \operatorname{Spec} \mathrm{Int.\,cl}_L(A) \to \operatorname{Spec} A$ satisfies the universal property in Definition 11.7.3.

Proof By Lemma 11.7.3, we know that $K(\mathrm{Int.\,cl}_L(A)) = L$, and same as the proof of Lemma 11.7.1, we done! □

Proposition 11.7.3

The normalization in a given extension of fields exists.

Proof Apply Lemma 11.7.4, use the same manner as in Proposition 11.7.1, we done! □

The following two examples, one arithmetic and one geometric, show that this is an interesting construction.

Example 11.13 Suppose $X = \operatorname{Spec} \mathbb{Z}$ (with function field $\mathbb{Q}$). We want to find its integral closure in the field extension $\mathbb{Q}(i)$, that is, find the integral closure of $\mathbb{Z}$ in $\mathbb{Q}(i)$. Claim that $\mathrm{Int.\,cl}_{\mathbb{Q}(i)}(\mathbb{Z}) = \mathbb{Z}[i]$.

Let $a+bi \in \mathbb{Z}[i]$, we shall to show that $a+bi$ is integral over $\mathbb{Z}$, consider $f(x) = x^2 - 2ax + (a^2+b^2) \in \mathbb{Z}[x]$, then $f(a+bi) = 0$, it follows that $a+bi$ is integral over $\mathbb{Z}$. Hence $\mathbb{Z}[i] \subseteq \mathrm{Int.\,cl}_{\mathbb{Q}(i)}(\mathbb{Z})$. Conversely, let $a+bi \in \mathrm{Int.\,cl}_{\mathbb{Q}(i)}(\mathbb{Z})$, i.e., $a+bi \in \mathbb{Q}(i)$ is integral over $\mathbb{Z}$. Hence there exists $g(x) = x^2 + px + q \in \mathbb{Z}[x]$ such that $g(a+bi) = 0$. Moreover by Vieta's formulas, we have

$$\begin{cases} p &= -2a \\ q &= a^2 + b^2. \end{cases}$$

If $a \in \mathbb{Z}$, since $q \in \mathbb{Z}$, $b \in \mathbb{Z}$. If $a = \frac{m}{2}$ with m is odd, from $q \in \mathbb{Z}$, we get $\frac{m^2}{4} + b^2 \in \mathbb{Z}$, and therefore $4b^2 + m^2 \in 4\mathbb{Z}$, hence $4b^2 + m^2 \equiv 0 \pmod 4$, it is impossible. So $a \in \mathbb{Z}$ and $b \in \mathbb{Z}$, and therefore $a+bi \in \mathbb{Z}[i]$. It follows that $\mathrm{Int.\,cl}_{\mathbb{Q}(i)}(\mathbb{Z}) = \mathbb{Z}[i]$.

Definition 11.7.4 (Number field, ring of integers)

A finite extension K of $\mathbb{Q}$ is called a ***number field****, and the integral closure of $\mathbb{Z}$ in K is called* ***the ring of integers in*** *K, denoted $\mathscr{O}_K$.*

$$\begin{tikzcd} \operatorname{Spec} K \arrow[r] \arrow[d] & \operatorname{Spec} \mathscr{O}_K \arrow[d] \\ \operatorname{Spec} \mathbb{Q} \arrow[r] & \operatorname{Spec} \mathbb{Z} \end{tikzcd}$$

Remark 11.29 By the previous discussion, $\mathscr{O}_K$ is a normal integral domain, and we will see soon (Theorem 11.7.1 (a)) that it is Noetherian, and later (Chapter 13) that it has "dimension 1". This is an example of a Dedekind domain, see Chapter 14. We will think of it as a "smooth" curve as soon as we define what "smooth" (really, regular) and "curve" mean.

✍ **Exercise 11.14** Find the ring of integers in $\mathbb{Q}(\sqrt{n})$, where n is square-free (a square-free integer is an integer which is divisible by no square number other than 1) and $n \equiv 3 \pmod 4$.

Solution *By Exercise 6.6, we know that* $\operatorname{Spec}\mathbb{Z}[x]/(x^2-n)$ *is normal, by Corollary 6.4.4,* $\mathbb{Z}[x]/(x^2-n)$ *is integrally closed in its fraction field* $K(\mathbb{Z}[x]/(x^2-n)) = \mathbb{Q}(\sqrt{n})$. *We want to show that* $\operatorname{Int.cl}_{\mathbb{Q}(\sqrt{n})}(\mathbb{Z}) = \mathbb{Z}[x]/(x^2-n) \cong \mathbb{Z}[\sqrt{n}]$. *Let* $a+b\sqrt{n} \in \mathbb{Z}[\sqrt{n}]$, *note that* $a+b\sqrt{n}$ *agree with the polynomial in* $\mathbb{Z}[x]$

$$x^2 - 2ax + a^2 - b^2 n = 0,$$

we know that $a+b\sqrt{n} \in \operatorname{Int.cl}_{\mathbb{Q}(\sqrt{n})}(\mathbb{Z})$. *Hence* $\mathbb{Z}[\sqrt{n}] \subseteq \operatorname{Int.cl}_{\mathbb{Q}(\sqrt{n})}(\mathbb{Z})$. *Since* $\mathbb{Z}[\sqrt{n}]$ *is integrally closed in* $\mathbb{Q}(\sqrt{n})$, *we know that* $\operatorname{Int.cl}_{\mathbb{Q}(\sqrt{n})}(\mathbb{Z}) = \mathbb{Z}[x]/(x^2-n) \cong \mathbb{Z}[\sqrt{n}]$. *Hence* $\mathscr{O}_{\mathbb{Q}(\sqrt{n})} \cong \mathbb{Z}[\sqrt{n}]$.

Example 11.14 Suppose $\operatorname{char} k \neq 2$ for convenience (although it isn't necessary).

(a) Suppose $X = \operatorname{Spec} k[x]$ (with function field $k(x)$). Find its normalization in the field extension $k(x)(y)$, where $y^2 = x^2 + x$. (Again we get a Dedekind domain.)

(b) Suppose $X = \mathbb{P}^1$, with distinguished open subscheme $\operatorname{Spec} k[x]$. Find its normalization in the field extension $k(x,y)$, where $y^2 = x^2 + x$.

Solution

(a) By Exercise 6.5, we know that $\operatorname{Spec} A := \operatorname{Spec} k[x,y]/(y^2-x^2-x)$ *is normal, and therefore* A *is integrally closed in its fraction field* $K(A) = k(x)[y]/(y^2-(x^2+x))$, *by Corollary 6.4.4. In fact, the field extension* $k(x)(y)$ *where* $y^2 = x^2+x$ *is isomorphic to* $k(x)[y]/(y^2-(x^2+x))$, *we have* $K(A) \cong k(x)(y)$ *where* $y^2 = x^2+x$. *Say* $L = k(x)[y]/(y^2-(x^2+x))$. *Claim that* $\operatorname{Int.cl}_L(k[x]) = A$. *Consider* $p(t) = t^2-(x^2+x) \in k[x][t]$, *note that* $p(y) = y^2-(x^2+x) = 0$ *in* A, y *is integral over* $k[x]$, *and therefore* $y \in \operatorname{Int.cl}_L(k[x])$. *By Proposition 9.2.4,* $A \subseteq \operatorname{Int.cl}_L(k[x])$. *Since* A *is integrally closed,* $A = \operatorname{Int.cl}_L(k[x])$. *It follows that* $v : \operatorname{Spec} A \to \operatorname{Spec} k[x]$ *is normalization in the field extension* $k(x)(y)$, *where* $y^2 = x^2+x$.

(b) Since $X = \mathbb{P}^1$, *we may write* $X = U_0 \cup U_\infty$, *where* $U_0 = \operatorname{Spec} k[x]$ *and* $U_\infty = \operatorname{Spec} k\left[\frac{1}{x}\right]$. *Denote* $t = \frac{1}{x}$, *then* $U_\infty = \operatorname{Spec} k[t]$. *The normalization of* U_0 *in the field extension* $k(x)[y]/(y^2-x^2-x)$ *is* $\operatorname{Spec} k[x,y]/(y^2-x^2-x) \to U_0$ *which is given by part (a). We now find the normalization of* U_∞ *in the field extension* $k(x)[y]/(y^2-x^2-x)$. *The fraction field of* $k[t]$ *is* $k\left(\frac{1}{x}\right) \cong k(x)$, *in the field extension* $k(x)[y]/(y^2-x^2-x)$ *we have* $y^2 = \frac{1}{t^2}+\frac{1}{t}$, *i.e.,* $t^2y^2 = 1+t$. *Let* $u = ty$, *then* $u^2 = 1+t$ *in* $k(x)[y]/(y^2-x^2-x) \cong k(t)[u]/(u^2-t-1)$. *Claim that* $\operatorname{Spec} k[t,u]/(u^2-t-1) \to U_\infty$ *is the normalization of* U_∞ *in the field extension* $k(t)[u]/(u^2-t-1)$. *By Exercise 6.5, we know that* $\operatorname{Spec} k[t,u]/(u^2-t-1)$ *is normal, by Corollary 6.4.4,* $k[t,u]/(u^2-t-1)$ *is integrally closed in its fraction field. Say* $K = k(t)[u]/(u^2-t-1)$, *we want to show that* $\operatorname{Int.cl}_K(k[t]) = k[t,u]/(u^2-t-1)$. *Consider* $p(T) = T^2-t+1 \in k[t][T]$, *since* $p(u) = u^2-t-1 = 0$, *we know that* u *is integral over* $k[t]$. *By Proposition 9.2.4,* $k[t,u]/(u^2-t-1) \subseteq \operatorname{Int.cl}_K(k[t])$, *since* $k[t,u]/(u^2-t-1)$ *is integrally closed,* $\operatorname{Int.cl}_K(k[t]) = k[t,u]/(u^2-t-1)$. *It follows that* $\operatorname{Spec} k[t,u]/(u^2-t-1) \to U_\infty$ *is the normalization of* U_∞ *in the field extension* $k(t)[u]/(u^2-t-1)$.

Next we glue $\widetilde{U_0} := \operatorname{Spec} k[x,y]/(y^2-x^2-x)$ *and* $\widetilde{U_\infty} := \operatorname{Spec} k[t,u]/(u^2-t-1)$ *to get a projective scheme. On the overlap* $\widetilde{U_0} \cap \widetilde{U_\infty}$, *we have the coordinate change*

$$t \longmapsto \frac{1}{x}, \qquad u \longmapsto \frac{y}{x}.$$

Hence $\widetilde{U_0}$ *and* $\widetilde{U_\infty}$ *glue together via above coordinate change, we obtain* $\operatorname{Proj} k[X,Y,Z]/(Y^2-X^2-XZ)$, *and therefore* $\operatorname{Proj} k[X,Y,Z]/(Y^2-X^2-XZ) \to \mathbb{P}^1$ *is the normalization of* $\mathbb{P}^1$ *in the field extension* $k(x,y)$, *where* $y^2 = x^2+x$.

11.7.2 Fancy fact: Finiteness of integral closure

The following fact is useful.

Theorem 11.7.1 (Finiteness of integral closure)

Suppose A is a Noetherian integral domain, $K = K(A)$, L/K is a finite extension of fields, and B is the integral closure of A in L ("the integral closure of A in the field extension L/K", i.e., those elements of L integral over A).

(a) If A is integrally closed and L/K is separable, then B is a finitely generated A-module.

(b) (E. Noether) If A is a finitely generated k-algebra, then B is a finitely generated A-module.

Remark 11.30 Eisenbud gives a proof in a page and a half: (a) is [Eis13, Proposition 13.14] and (b) is [Eis13, Corollary 13.13]. We give a sketch of proof.

Remark 11.31 **Warning.** Part (b) does not hold for Noetherian A in general — the integral closure of a Noetherian ring need not be Noetherian. (See [Rei95, §9.4] or [Eis13, §13.3] for some discussion.) This is alarming. The existence of such an example is a sign that Theorem 11.7.1 is not easy.

Proof Here is a sketch to show the structure of the argument. It use commutative algebra ideas from Chapter 13, so you should only glance at this to see that nothing fancy is going on.

Part (a): Reduce to the case where L/K is Galois, with group $\{\sigma_1, \dots, \sigma_n\}$. Choose $b_1, \dots, b_n \in B$ forming a K-vector space basis of L. Let $M = (\sigma_i b_j)_{ij}$ with ij-th entry $\sigma_i b_j$ (familiar from Galois theory), and let $d = \det M$. Show that the entries M lie in B, and that $d^2 \in K$ (as d^2 is Galois-fixed). Show that $d \neq 0$ using linear independence of characters. Then complete the proof by showing that $B \subseteq d^{-2}(Ab_1 + \cdots + Ab_n)$ (submodules of finitely generated modules over Noetherian rings are also Noetherian, Corollary 4.6.3) as follows. Suppose $b \in B$, and write $b = \sum c_i b_i, (c_i \in K)$. If c is the column vector with entries c_i, show that the i-th entry of the column vector Mc is $\sigma_i b \in B$. Multiplying Mc on the left by adj M (see the trick of the proof of Lemma 9.2.1), show that $dc_i \in B$. Thus $d^2 c_i \in B \cap K = A$ (as A is integrally closed), as desired.

Part (b): Use the Noetherian Normalization Lemma (Chapter 13) to reduce to the case $A = k[x_1, \dots, x_n]$. Reduce to the case where L is normally closed over K. Let L' be the subextension of L/K so that L/L' is Galois and L'/K is purely inseparable. Use part (a) to reduce to the case $L = L'$. If $L' \neq K$, then for some q, L' is generated over K by the q-th root of a finite set of rational functions. Reduce to the case $L' = k'(x_1^{1/q}, \dots, x_n^{1/q})$ where k'/k is a finite purely inseparable extension. In this case, show that $B = k'[x_1^{1/q}, \dots, x_n^{1/q}]$, which is indeed finite over $k[x_1, \dots, x_n]$. □

In the following four propositions, you can replace "integral finite type k-scheme" by "integral variety" once you know what a variety is.

Proposition 11.7.4

(a) If X is an integral finite type k-scheme, then its normalization $\nu : \widetilde{X} \to X$ is a finite morphism.

(b) Suppose X is a locally Noetherian integral scheme. If X is normal, then the normalization in a finite separable field extension is a finite morphism. If X is an integral finite type k-scheme, then the normalization in a finite extension of fields is a finite morphism.

Remark 11.32 In particular, once we define "variety" (Chapter 12), you will see that (b) implies that the normalization of a variety (including in a finite extension of the function field) is a variety.

Proof

(a) Since X is an integral finite type k-scheme, we may assume that $X = \bigcup_{i=1}^{n} \operatorname{Spec} A_i$ where A_i is finitely generated k-algebra and A_i is integral domain. By Proposition 11.7.2, we know that $\upsilon : \widetilde{X} \to X$ is integral, by Proposition 9.3.20, it suffices to show that $\upsilon : \widetilde{X} \to X$ is of finite type. υ is integral implies that υ is affine, moreover by the proof in Proposition 11.7.1, $\upsilon^{-1}(\operatorname{Spec} A_i) = \operatorname{Spec} \widetilde{A_i}$ for each i, where $\widetilde{A_i}$ is the integral closure of A_i in its fraction field $K(A_i) = K(X)$. Since each A_i is finitely generated k-algebra, by Theorem 11.7.1, $\widetilde{A_i}$ is a finitely generated A_i-module, and therefore $\widetilde{A_i}$ is a finitely generated A_i-algebra. By Proposition 9.3.19, we know that υ is of finite type, as we desired.

(b) Suppose $X = \bigcup_i \operatorname{Spec} A_i$, since X is a locally Noetherian integral scheme, each A_i is Noetherian integral domain.

If X is normal, by Proposition 6.4.3, each A_i is integrally closed Noetherian integral domain. Since A_i is integrally closed, for any $L/K(X)$ separable extension, by Theorem 11.7.1, $\operatorname{Int.cl}_L(A_i)$ is a finitely generated A_i-module, and therefore $\upsilon : \widetilde{X} \to X$ is finite.

If X is an integral finite type k-scheme, we may assume that $X = \bigcup_{i=1}^{n} \operatorname{Spec} A_i$ where A_i is finitely generated k-algebra. For any finite extension $L/K(X)$, by Theorem 11.7.1, each $\operatorname{Int.cl}_L(A_i)$ is finitely generated A_i-module, and therefore $\upsilon : \widetilde{X} \to X$ is finite.

□

Proposition 11.7.5

If X is an integral finite type k-scheme, then the normalization morphism is birational.

Proof By Proposition 11.7.4 (a), we know that $\upsilon : \widetilde{X} \to X$ is a finite morphism. Let $\operatorname{Spec} A \hookrightarrow X$ be an affine open subset, since X is integral finite type k-scheme, A is finitely generated k-algebra and also is an integral domain. Recall that $\upsilon^{-1}(\operatorname{Spec} A) = \operatorname{Spec} \widetilde{A}$, since υ is finite morphism, $\widetilde{A}$ is a finitely generated A-module, and therefore is a finitely generated k-module, hence $\widetilde{X}$ is an integral finite type k-scheme. Recall that we have the isomorphism $K(X) \cong K(\widetilde{X})$, by Proposition 8.5.6, we know that the normalization morphism $\upsilon : \widetilde{X} \to X$ is birational □

Proposition 11.7.6

Suppose that X is an integral finite type k-scheme. Then the normalization map of X is an isomorphism on an open dense subset of X. The normal points of X form a dense open subset.

Proof By Proposition 11.7.5, the normalization $\upsilon : \widetilde{X} \to X$ is birational. By Proposition 8.5.5, there is a dense open subscheme $\widetilde{U} \hookrightarrow \widetilde{X}$ and a dense open subscheme $U \hookrightarrow X$ such that $\widetilde{U} \cong U$.

Let $N = \{p \in X : \mathscr{O}_{X,p} \text{ is normal}\}$. Let $p \in X$, then $p = [\mathfrak{p}] \in \operatorname{Spec} A \hookrightarrow X$ where $\operatorname{Spec} A$ is an open subset of X with A is integral domain and A is finitely generated k-algebra. If p not normal point, then $A_{\mathfrak{p}}$ is not integrally closed in $K(A)$. Say $M = \widetilde{A}/A$. Then $p = [\mathfrak{p}] \in \operatorname{Spec} A \hookrightarrow X$ is not integrally closed if and only if $M_{\mathfrak{p}} = \widetilde{A}_{\mathfrak{p}}/A_{\mathfrak{p}} \neq 0$. Since X is integral finite type k-scheme, we may assume that $X = \bigcup_{i=1}^{n} \operatorname{Spec} A_i$ where each A_i is integral domain and also is finitely generated k-algebra. By Theorem 11.7.1, $\widetilde{A_i}$ is finitely generated A_i-module. Say $M_i = \widetilde{A_i}/A_i$, M_i is finitely generated A_i-module. By our above discussion, $X \setminus N = \bigcup_{i=1}^{n} \operatorname{Supp} M_i$. Since M_i is finitely generated A_i-module, by Proposition 7.6.1, we know that $\operatorname{Supp} M_i = V(\operatorname{Ann}_{A_i} M_i)$. Hence $X \setminus N = \bigcup_{i=1}^{n} V(\operatorname{Ann}_{A_i} M_i)$ is closed, and therefore N is open. Since

X is irreducible, N is dense in X. □

Remark 11.33 By Definition 11.7.2, the "integral" hypothesis can be relaxed to "reduced".

Proposition 11.7.7

Suppose $\rho : Z \to X$ is a finite birational morphism from an integral finite type k-scheme to a normal integral finite type k-scheme. Then ρ is an isomorphism.

Proof Let $\operatorname{Spec} B \hookrightarrow X$ be an affine open subset. Since X is a normal integral finite type k-scheme, B is integrally closed integral domain and B is finitely generated k-algebra. Since ρ is finite, ρ is integral and of finite type, we may assume that $\rho^{-1}(\operatorname{Spec} B) = \operatorname{Spec} A$ where $B \to A$ is integral. Since ρ is birational morphism, by Proposition 8.5.6, $K(X) \cong K(Z)$. Since ρ is dominant, by Proposition 8.5.4, $B \to A$ is injective, so B can be seen as a subring of A. Since $B \to A$ is integral and B is integrally closed in $K(X) \cong K(Z)$, we have $B \to A$ is an isomorphism, it follows that $\rho|_{\operatorname{Spec} A} : \operatorname{Spec} A \to \operatorname{Spec} B$ is an isomorphism. Note that $Z = \bigcup \rho^{-1}(U)$ where U is affine open of X and $\rho|_{\rho^{-1}(U)} : \rho^{-1}(U) \to U$ is isomorphism, $\rho : Z \to X$ is isomorphism. □

11.7.3 Reduction of schemes, revisited

It may be enlightening to revisit the "reduction" of a scheme using this point of view. The argument is clean and simple, and by working it out for yourself, you will better appreciate the various moving parts. We now define "reduction" from scratch, and then show that it agrees with the Definition 10.4.7.

Definition 11.7.5 (Reduction)

*Suppose X is a scheme. We say $\rho : X^{\mathrm{red}} \to X$ is **"the" reduction of** X if any morphism $W \to X$ from a reduced scheme W factors uniquely through ρ:*

$$\begin{tikzcd} \textit{reduced} & W \arrow[r, dashed, "\exists!"] \arrow[dr] & X^{\mathrm{red}} \arrow[d] \\ & & X. \end{tikzcd} \tag{11.13}$$

Remark 11.34 The "the" is in quotes because this is a slight abuse of notion.

In analogy with fibered products and normalization, we now check that such an X^{red} exists:

Lemma 11.7.5

Let A be a ring, then $\rho : \operatorname{Spec} A/\mathfrak{N}(A) \to \operatorname{Spec} A$ satisfies the universal property of reduction.

Proof We first show that for any affine reduced scheme, there exists unique morphism from W to $\operatorname{Spec} A/\mathfrak{N}(A)$ such that (11.13) holds. Let $W = \operatorname{Spec} B$ where B is reduced ring, and $\varphi : W \to \operatorname{Spec} A$ be a morphism. Define $\theta^\sharp : A/\mathfrak{N}(A) \to B$ by setting $\theta^\sharp(a + \mathfrak{N}(A)) = \varphi^\sharp(a)$. Let $a_1 + \mathfrak{N}(A) = a_2 + \mathfrak{N}(A)$ in $A/\mathfrak{N}(A)$, then $a_1 - a_2 \in \mathfrak{N}(A)$, i.e., $(a_1 - a_2)^n = 0$ for some n. Consider $\varphi^\sharp(a_1 - a_2)$, we have $\varphi^\sharp(a_1 - a_2)^n = \varphi^\sharp((a_1 - a_2)^n) = 0$, since B is reduced, $\varphi^\sharp(a_1 - a_2) = 0$, i.e., $\theta^\sharp(a_1 + \mathfrak{N}(A)) = \theta^\sharp(a_2 + \mathfrak{N}(A))$, which implies that $\theta^\sharp$ is well-defined. Note that we have $\theta^\sharp \circ \rho^\sharp(a) = \varphi^\sharp(a)$, the following diagram commutes.

$$\begin{tikzcd} B & A/\mathfrak{N}(A) \arrow[l, dashed, "\theta^\sharp"'] \\ & A \arrow[ul, "\varphi^\sharp"] \arrow[u, "\rho^\sharp"'] \end{tikzcd}$$

If there exists another $\theta_0^\sharp : A/\mathfrak{N}(A) \to B$ such that above diagram commutes. Then we have

$$\theta_0^\sharp \circ \rho^\sharp(a) = \theta_0^\sharp(a + \mathfrak{N}(A)) = \varphi^\sharp(a) = \theta^\sharp(a + \mathfrak{N}(A)),$$

for all $a \in A$, hence $\theta_0^\sharp = \theta^\sharp$, i.e., $\theta^\sharp$ is unique. By applying functor Spec, we have

$$\begin{tikzcd} \operatorname{Spec} B \arrow[r, dashed, "\exists!\theta"] \arrow[dr, "\varphi"'] & \operatorname{Spec} A/\mathfrak{N}(A) \arrow[d, "\rho"] \\ & \operatorname{Spec} A. \end{tikzcd}$$

Now we show that for any reduced scheme W with morphism $\varphi : W \to \operatorname{Spec} A$, there exists unique morphism such that (11.13) holds. Suppose $W = \bigcup_i \operatorname{Spec} B_i$ where B_i is reduced. Denote $\varphi_i := \varphi|_{\operatorname{Spec} B_i}$, by above discussion, there exists unique $\theta_i : \operatorname{Spec} B_i \to \operatorname{Spec} A/\mathfrak{N}(A)$ such that $\rho \circ \theta_i = \varphi_i$. Consider $\theta_i|_{\operatorname{Spec} B_i \cap \operatorname{Spec} B_j}$ and $\theta_j|_{\operatorname{Spec} B_i \cap \operatorname{Spec} B_j}$. By Proposition 6.3.1, we may assume that $\operatorname{Spec} B_i \cap \operatorname{Spec} B_j = \bigcup_f \operatorname{Spec}(B_i)_f = \bigcup_g \operatorname{Spec}(B_j)_g$ where $(B_i)_f \cong (B_j)_g$. By the uniqueness (we proved above), $\theta_i|_{\operatorname{Spec}(B_i)_f} = \theta_j|_{\operatorname{Spec}(B_j)_g}$ for all f, g, hence $\theta_i|_{\operatorname{Spec} B_i \cap \operatorname{Spec} B_j} = \theta_j|_{\operatorname{Spec} B_i \cap \operatorname{Spec} B_j}$. By Morphism of locally ringed spaces glue 8.3.2, θ_{ii} can glue together, then we obtain $\theta : W \to \operatorname{Spec} A/\mathfrak{N}(A)$ such that (11.13) commutes. Also θ must be unique, since each θ_i is unique. So $\rho : \operatorname{Spec} A/\mathfrak{N}(A) \to A$ satisfies the universal property of reduction. □

Proposition 11.7.8

The reduction of schemes exists in general.

Proof Recall Proposition 7.2.2, apply Lemma 11.7.5, use the same manner as Proposition 11.7.1, we done! □

To make sure we haven't contradicted our earlier selves:

Proposition 11.7.9

The construction of Definition 10.4.7 satisfies the universal property of reduction.

Proof By Definition 10.4.7, obviously. □

Definition 11.7.6 (Reduction of morphism)

*By the universal property of reduction, any morphism of schemes $\pi : X \to Y$ induces a morphism $X^{\text{red}} \to Y^{\text{red}}$, the **reduction of** π which we quite reasonably name π^{red}.*

Remark 11.35 "Reduction" gives a functor from the category of schemes to the category of reduced schemes.

Chapter 12 Separated and proper morphisms, and varieties

Separatedness is a fundamental notion. It is the analog of the Hausdorff condition for manifolds (see **??**), and as with Hausdorffness, this geometrically intuitive notion ends up being just the right hypothesis to make theorems work. Although the definition initially looks odd, in retrospect it is just perfect.

Let's review why we like Hausdorffness.

Definition 12.0.1 (Hausdorff topological space)

*A topological space is **Hausdorff** if for every two points x and y, there are disjoint open neighborhood of x and y.*

The real line is Hausdorff, but the "real line with doubled origin" (of which Figure 5.7 may be taken as a sketch) is not. Many proofs and results about manifolds use Hausdorffness in an essential way. For example, the classification of compact one-dimensional manifolds is very simple, but if the Hausdorff condition were removed, we would have a very wild set.

So once armed with this definition, we can cheerfully exclude the line with double origin from civilized discussion, and we can (finally) define the notion of a **variety**, in a way that corresponds to the classical definition.

With our motivation from manifolds, we shouldn't be surprised that all of our affine and projective schemes are separated. Subsets of Hausdorff topological spaces are automatically Hausdorff. Hence if Y is a manifold, and X is a subset that satisfies all the hypotheses of a manifold except possibly Hausdorffness, X is a manifold. Similarly, we will see that locally closed embeddings in something separated are also separated (combine Proposition 12.2.5 and Proposition 12.2.4).

Grothendieck taught us that one should try to define properties of morphisms, not of objects; then we can say that an object has that property if its morphism to the final object has that property. We discussed this briefly at the start of Chapter 9. In this spirit, separatedness will be a property of morphisms, not schemes.

As an unexpected bonus, a separated morphism to an affine scheme has the property that the intersection of two affine open sets in the source is affine (Proposition 12.2.18). This will make Čech cohomology work very easily on (quasicompact) schemes (Chapter 19). You might consider this an analog of the fact that in $\mathbb{R}^n$, the intersection of two convex sets is also convex. As affine schemes are trivial from the point of view of quasicoherent cohomology, just as convex sets in $\mathbb{R}^n$ have no cohomology, this metaphor is apt.

A lesson arising from the construction is the importance of the diagonal morphism (Definition 2.2.12). More precisely, given a morphism $\pi : X \to Y$, good consequence can be leveraged from good behavior of th **diagonal morphism** $\delta_\pi : X \to X \times_Y X$ (the product of the identity morphism $X \to X$ with itself), usually through fun diagram chases. This lesson applies across many fields of geometry. (Another nice gift of the diagonal morphism: it will give us a good algebraic definition of differentials, Chapter 22.)

So before we define separatedness, we make some observations about diagonal morphisms.

12.1 Fun with diagonal morphisms, and quasiseparatedness made easy

12.1.1 The Cancellation Theorem

Recall that most classes of morphisms are "reasonable" in the sense of Definition 9.1.1: they are (i) preserved by composition, (ii) preserved by base change, (iii) local on the target, and as a consequence they are (iv) preserved by product (Proposition 9.1.1).

Another useful consequence is that (v) "reasonable" classes of morphisms have a "Cancellation Property".

Definition 12.1.1 ($\mathscr{P}$-diagonal)

Let $\mathscr{P}$ be a class of morphisms. The "$\mathscr{P}$-diagonal" is a class of morphisms consisting of the morphisms $\pi : X \to Y$ which satisfy $\delta_\pi : X \to X \times_Y X$ lies in $\mathscr{P}$. Denote "$\mathscr{P}$-diagonal" by $\mathscr{P}\delta$.

Note *In this chapter, we will use the symbols δ_π and $\delta_{X/Y}$ to denote the diagonal morphism of $\pi : X \to Y$.*

The Cancellation Theorem is a strange-looking, but very useful, result. Like the Diagonal-Base-Change diagram (Exercise 2.15), this result is unexpectedly ubiquitous.

Theorem 12.1.1 (The Cancellation Theorem for a Property $\mathscr{P}$ of Morphisms)

Let $\mathscr{P}$ be a class of morphisms that is preserved by base change and composition. (Every "reasonable" class of morphisms satisfies this, Definition 9.1.1.) Suppose

$$\begin{tikzcd} X \arrow[rr, "\pi"] \arrow[dr, "\tau"'] & & Y \arrow[dl, "\rho"] \\ & Z & \end{tikzcd}$$

is a commuting diagram of schemes. Suppose ρ is in $\mathscr{P}\delta$ and τ is in $\mathscr{P}$. Then π is in $\mathscr{P}$.

Remark 12.1 The following diagram summaries this important theorem:

$$\begin{tikzcd} X \arrow[rr, "\therefore \in \mathscr{P}"] \arrow[dr, "\in \mathscr{P}"'] & & Y \arrow[dl, "\in \mathscr{P}\delta"] \\ & Z & \end{tikzcd}$$

When you plug in different $\mathscr{P}$, you get very different-looking (and nonobvious) consequences, the first of which are given in Proposition 12.2.8. (Here are some facts you can prove easily, but which can be interpreted as applications of the Cancellation Theorem in **Sets**, and which may thus shed light on how the Cancellation Theorem works. If $\pi : X \to Y$ and $\rho : Y \to Z$ are maps of sets, and $\rho \circ \pi$ is injective, then so is π; and if $\rho \circ \pi$ is surjective and ρ is injective, then π is surjective.)

Proof In fact, we have $X \cong X \times_Y Y$, $Y \cong Z \times_Z Y$, and $X \times_Z (Z \times_Z Y) \cong X \times_Z Y$. By the Diagonal-Base-Change diagram (Exercise 2.15), $X \times_Z Y \cong Z \times_Z (Z \times_Z Y)$. Consider the following diagram.

$$\begin{tikzcd} X \arrow[r, leftrightarrow, "\sim"] \arrow[rrrr, bend left, "\therefore \in \mathscr{P}"] & X \times_Y Y \arrow[r] \arrow[dl] & X \times_Z Y \arrow[r] \arrow[dl] \arrow[dr] & Z \times_Z Y \arrow[r, leftrightarrow, "\sim"] \arrow[dr] & Y \\ Y \arrow[r, "\in \mathscr{P}"'] & Y \times_Z Y & & X \arrow[r, "\in \mathscr{P}"'] & Z \end{tikzcd}$$

Since $\mathscr{P}$ is preserved by base change, we have $X \times_Y Y \to X \times_Z Y$ and $X \times_Z Y \to Z \times_Z Y$ in $\mathscr{P}$. Since $\mathscr{P}$ is preserved by composition, we know that $X \to Y$ belong to $\mathscr{P}$, we are done. $\square$

(vi) if $\mathscr{P}$ is a "reasonable" class, so is $\mathscr{P}\delta$:

Theorem 12.1.2

If $\mathscr{P}$ is a "reasonable" class of morphisms, then the class $\mathscr{P}\delta$ of morphisms whose diagonal is in $\mathscr{P}$ is also a "reasonable" class.

Proof We verify that $\mathscr{P}\delta$ satisfies the "three axioms of reasonableness".

(1) $\mathscr{P}\delta$ is preserved by composition.

Let $X \to Y$ and $Y \to Z$ are in $\mathscr{P}\delta$. We want to show that $X \to Z$ belongs to $\mathscr{P}\delta$. Consider the following diagram.

$$\begin{tikzcd}
X \arrow[r, "\in\mathscr{P}"'] \arrow[rr, bend left, "\in\mathscr{P}?"] & X \times_Y X \arrow[r] \arrow[d] & X \times_Z X \arrow[d] & \\
& Y \arrow[r, "\in\mathscr{P}"'] & Y \times_Z Y \arrow[r] & Y
\end{tikzcd}$$

By the Diagonal-Base-Change 2.15, we know that

$$X \times_Y X \cong Y \times_{Y \times_Z Y} (X \times_Z X),$$

since $\mathscr{P}$ is a reasonable class, $\mathscr{P}$ is preserved by base change, hence $X \times_Y X \to X \times_Z X$ belongs to $\mathscr{P}$. Since $\mathscr{P}$ is preserved composition, $X \to X \times_Z X$ belongs to $\mathscr{P}$, i.e., $X \to Z$ belongs to $\mathscr{P}\delta$.

(2) $\mathscr{P}\delta$ is preserved by base change.

Suppose $X \to Y$ belongs to $\mathscr{P}\delta$, we want to show that $X \times_Y W \to W$ belongs to $\mathscr{P}$ for any $W \to Y$. Say $V := X \times_Y W$. Since

$$\begin{tikzcd}
V \arrow[r] \arrow[d] & W \arrow[d] \\
X \arrow[r] & Y
\end{tikzcd}$$

is Cartesian square, by the universal property of $X \times_Y X$, we have a unique morphism $V \times_W V \to X \times_Y X$ such that the following diagram commutes.

$$\begin{tikzcd}
V \times_W V \arrow[rrr] \arrow[ddd] \arrow[dr, dashed] & & & V \arrow[dl] \arrow[ddd] \\
& X \times_Y X \arrow[r] \arrow[d] & X \arrow[d] & \\
& X \arrow[r] & Y & \\
V \arrow[ur] \arrow[rrr] & & & W \arrow[ul]
\end{tikzcd} \tag{12.1}$$

Consider the following diagram,

$$\begin{tikzcd}
V \arrow[r] \arrow[d] & V \times_W V \arrow[d] \\
X \arrow[r] & X \times_Y X
\end{tikzcd}$$

we want to show this diagram is Cartesian square. We first show that the following diagram is Cartesian square.

$$\begin{tikzcd}
V \times_W V \arrow[r] \arrow[d] & W \arrow[d] \\
X \times_Y X \arrow[r] & Y
\end{tikzcd} \tag{12.2}$$

By (12.1), diagram (12.2) commutes, then by the universal property of $(X \times_Y X) \times_Y W$ there exists unique morphism $V \times_W V \to (X \times_Y X) \times_Y W$ such that the following diagram commutes.

$$\begin{tikzcd}
V \times_W V \arrow[dr, dashed] \arrow[drr, bend left] \arrow[ddr, bend right] & & \\
& (X \times_Y X) \times_Y W \arrow[r] \arrow[d] & W \arrow[d] \\
& X \times_Y X \arrow[r] & Y
\end{tikzcd}$$

By the following diagram,

$$\begin{tikzcd}
(X \times_Y X) \times_Y W \arrow[rr] \arrow[dr, dashed] \arrow[ddr] & & X \times_Y X \arrow[d] \\
& V \arrow[r] \arrow[d] & X \arrow[d] \\
& W \arrow[r] & Y
\end{tikzcd} \tag{12.3}$$

there exists unique morphism $(X \times_Y X) \times_Y W \to V$ such that (12.3) commutes. Then by the universal property property of $V \times_W V$ there exists unique morphism $(X \times_Y X) \times_Y W \to V \times_W V$ such that the following diagram commutes.

$$\begin{tikzcd}
(X \times_Y X) \times_Y W \arrow[dr, dashed] \arrow[drr, bend left] \arrow[ddr, bend right] & & \\
& V \times_W V \arrow[r] \arrow[d] & V \arrow[d] \\
& V \arrow[r] & W
\end{tikzcd}$$

Hence $V \times_W V \cong (X \times_Y X) \times_Y W$, i.e., (12.2) is Cartesian square. Now, we consider the following diagram.

$$\begin{tikzcd}
& & & W \arrow[d] \\
V \arrow[r] \arrow[d] \arrow[urrr] & V \times_W V \arrow[d] \arrow[urr] & & Y \\
X \arrow[r] \arrow[urrr] & X \times_Y X \arrow[urr] & &
\end{tikzcd}$$

Since $V = X \times_Y W$ and $V \times_W V \cong (X \times_Y X) \times_Y W$, we have

$$X \times_{X \times_Y X} (V \times_W V) \cong X \times_{X \times_Y X} ((X \times_Y X) \times_Y W) \cong X \times_Y W \cong V.$$

Hence

$$\begin{tikzcd}
V \arrow[r] \arrow[d] & V \times_W V \arrow[d] \\
X \arrow[r] & X \times_Y X
\end{tikzcd}$$

is Cartesian square. Since $\delta_{X \to Y} \in \mathscr{P}$ and $\mathscr{P}$ is preserved by base change, we know that $\delta_{V \to W} \in \mathscr{P}$, as we desired.

(3) $\mathscr{P}\delta$ is local on the target.

Suppose $\pi : X \to Y$ belongs to $\delta\mathscr{P}$, we want to show that for any open subset $V \hookrightarrow Y$, $\pi^{-1}(V) \to V$

belongs to $\delta\mathscr{P}$. Say $U = \pi^{-1}(V) \hookrightarrow X$. Claim that $U \cong X \times_Y V$, consider the following diagram,

$$\begin{tikzcd} W \arrow[dr, dashed] \arrow[drr, bend left, "g"] \arrow[ddr, bend right, "f"'] & & \\ & U \arrow[r, "\pi|_U"'] \arrow[d, hook, "\iota_U"] & V \arrow[d, hook, "\iota_V"] \\ & X \arrow[r, "\pi"'] & Y \end{tikzcd} \tag{12.4}$$

where W be arbitrary scheme with $\pi \circ f = \iota_V \circ \pi|_U$. We shall to show there exists unique morphism $\theta : W \to U$ such that (12.4) commutes. Pick $\theta = f$, then $\pi|_U \circ \theta = g$ and $f = \iota_U \circ \theta$. Uniqueness is by our construction of θ, so $U \cong X \times_Y V$. By (2), we have the following Cartesian square.

$$\begin{tikzcd} U \arrow[r] \arrow[d, hook] & U \times_V U \arrow[d] \\ X \arrow[r] & X \times_Y X \end{tikzcd}$$

Since $\mathscr{P}$ is preserved by base change and $\delta_\pi \in \mathscr{P}$, we know that $\delta_{U \to V} \in \mathscr{P}$, i.e., $U \to V$ belongs to $\mathscr{P}\delta$. Then we finish the first part of the proof of (3).

Suppose $\{V_i \hookrightarrow Y\}$ is an open cover of Y for which each restricted morphism $\pi^{-1}(V_i) \to V_i$ in $\mathscr{P}\delta$, we want to show that $\pi : X \to Y$ is in $\mathscr{P}\delta$. Say $U_i = \pi^{-1}(V_i)$, then we have a Cartesian square.

$$\begin{tikzcd} U_i \arrow[r, "\in \mathscr{P}\delta"] \arrow[d, hook] & V_i \arrow[d, hook] \\ X \arrow[r] & Y \end{tikzcd}$$

Then same as part (2), we have a Cartesian square.

$$\begin{tikzcd} U_i \arrow[r, "\in \mathscr{P}"] \arrow[d, hook] & U_i \times_{V_i} U_i \arrow[d] \\ X \arrow[r] & X \times_Y X \end{tikzcd}$$

Since $\mathscr{P}$ is local on the target, we know that δ_π is in the $\mathscr{P}$, which implies that $\pi \in \mathscr{P}\delta$. Then we finish the proof of (3).

$\square$

(vi) Finally, if you have two morphisms from X that are of certain form, the next proposition shows what more you need to ensure that the map to the product is also of that form.

Proposition 12.1.1

Suppose $\mathscr{P}$ is a "reasonable" class of morphisms. Suppose $\rho : X \to Y$ and $\sigma : X \to Z$ are morphisms of S-schemes in class $\mathscr{P}$, and $X \to S$ is in $\mathscr{P}\delta$. Then $(\rho, \sigma) : X \to Y \times_S Z$ is in class $\mathscr{P}$ as well.

Proof We first show that $(\rho, \sigma) : X \to Y \times_S Z$ factors through as $X \xrightarrow{\delta_X} X \times_S X \longrightarrow Y \times_S Z$. Consider the following digram,

$$\begin{tikzcd}
X \arrow[dr, "\delta_{X\to S}"] \arrow[drr, bend left, "\mathrm{id}_X"] \arrow[ddr, bend right, "\mathrm{id}_X"'] \arrow[ddrr, dashed, bend right] & & \\
& X \times_S X \arrow[r] \arrow[d] \arrow[dr, dashed] & X \arrow[d] \\
& & Y \times_S Z \arrow[r] \arrow[d] & Z \arrow[d] \\
& X \arrow[r] & Y \arrow[r] & S
\end{tikzcd}$$

where dashed arrow $X \times_S X \to Y \times_S Z$ is given by the universal property of $Y \times_S Z$. By the universal property of $Y \times_S Z$ again, $X \to Y \times_S Z$ factors through $X \times_S X$.

Now consider $X \times_S X \to Y \times_S Z$, we want to show this morphism factor through $X \times_S Y$. Consider the following diagram.

$$\begin{tikzcd}
X \times_S X \arrow[drr] \arrow[dr, dashed] \arrow[ddr] & & \\
& X \times_S Y \arrow[r] \arrow[d] & X \arrow[d] \\
& Y \times_S Z \arrow[r] \arrow[d] & Z \arrow[d] \\
& Y \arrow[r] & S
\end{tikzcd}$$

By the universal property of $X \times_S Y \cong (Y \times_S Z) \times_Z X$, we have unique morphism $X \times_S X \to X \times_S Y$ such that above diagram commutes. Hence (ρ, σ) is the composition of

$$X \longrightarrow X \times_S X \longrightarrow X \times_S Y \longrightarrow Y \times_S Z.$$

Note that

$$\begin{tikzcd}
X \arrow[r, "\in \mathscr{P}"] \arrow[rrr, bend left, "\therefore \in \mathscr{P}"] & X \times_S X \arrow[r, "\therefore \in \mathscr{P}"] \arrow[dl] & X \times_S Y \arrow[r, "\therefore \in \mathscr{P}"] \arrow[dl] \arrow[dr] & Y \times_S Z \arrow[dr] & \\
X \arrow[r, "\in \mathscr{P}"] & Y & & X \arrow[r, "\in \mathscr{P}"] & Y
\end{tikzcd}$$

where each square is Cartesian square. Since $\mathscr{P}$ is preserved by base change, we know that $X \times_S X \to X \times_S Y$ and $X \times_S Y \to Y \times_S Z$ are in the class $\mathscr{P}$. Since $\mathscr{P}$ is preserved by composition, $(\rho, \sigma) : X \to Y \times_S Z$ is in $\mathscr{P}$, as we desired. □

Proposition 12.1.2

⋆ *Finitely presented morphisms form a "reasonable" class of morphisms.*

Example 12.1 (Monomorphism) Recall that a morphism $\pi : X \to Y$ is a monomorphism if and only if the diagonal $\delta_\pi : X \to X \times_Y X$ is an isomorphism (Exercise 2.19). Because isomorphisms form a "reasonable" class of morphisms, we have immediately shown that monomorphisms form a "reasonable" class as well (and accidentally solving Exercise 2.18, and proving that monomorphisms are preserved by base change.)

Side Remark: For an example, we give a proof of "monomorphism is preserved by base change". Suppose

$\pi : X \to Y$ is a monomorphism and $W \to Y$ is any morphism. Consider the following Cartesian square.

$$\begin{tikzcd} X \times_Y W \arrow[r] \arrow[d] & W \arrow[d] \\ X \arrow[r] & Y \end{tikzcd}$$

We want to show that $X \times_Y W \to W$ is a monomorphism. By Exercise 2.19, it suffices to show that $X \times_Y W \to (X \times_Y W) \times_W (X \times_Y W)$ is an isomorphism. Say $V := X \times_Y W$, by the proof of Theorem 12.1.2 part (2), we have the following Cartesian square.

$$\begin{tikzcd} V \arrow[r] \arrow[d] & V \times_W V \arrow[d] \\ X \arrow[r] & X \times_Y X \end{tikzcd}$$

Since isomorphism is preserved by base change, we know that $V \to V \times_W V$ is an isomorphism. By Exercise 2.19, we know that $V \to W$ is a monomorphism, i.e., monomorphism is preserved by base change.

12.1.2 Redefinition: Quasiseparated morphisms

Definition 12.1.2 (Quasiseparated morphisms)

We say a morphism $\pi : X \to Y$ is ***quasiseparated*** *if the diagonal morphism $\delta_\pi : X \to X \times_Y X$ is quasicompact.*

Proposition 12.1.3 (Definition 12.1.2 agrees with Definition 9.3.2)

$\pi : X \to Y$ is quasiseparated if and only if for any affine open $\operatorname{Spec} A$ of Y, and two affine open subsets U and V of X mapping to $\operatorname{Spec} A$, $U \cap V$ is a finite union of affine open sets.

Proof Suppose $\operatorname{Spec} A$ is an open subset of Y. Let $U = \operatorname{Spec} B, V = \operatorname{Spec} C$ be open subsets of X with two arrows $U \to \operatorname{Spec} A$ and $V \to \operatorname{Spec} A$. Note that we have the following commutative diagram,

$$\begin{tikzcd} U \times_Y V \arrow[dr, dashed, shift left] \arrow[drr, bend left] \arrow[ddr, bend right] & & & \\ & U \times_A V \arrow[ul, dashed, shift left] \arrow[r] \arrow[d] & V \arrow[d] \arrow[ddr, bend left] & \\ & U \arrow[r] \arrow[drr, bend right] & \operatorname{Spec} A \arrow[dr, hook] & \\ & & & Y \end{tikzcd}$$

where dashed arrows are given by the universal property of $U \times_A V$ and $U \times_Y V$, so $U \times_Y V \cong U \times_A V \cong \operatorname{Spec}(B \otimes_A C)$ is an affine open subset of $X \times_Y X$. Consider the following diagram.

$$\begin{tikzcd} U \times_X V \arrow[dr, dashed] \arrow[drr, bend left] \arrow[ddr, bend right] & & & \\ & U \times_Y V \arrow[r] \arrow[d] & V \arrow[d] \arrow[ddr, hook, bend left] & \\ & U \arrow[r] \arrow[drr, hook, bend right] & Y & \\ & & & X \arrow[ul] \end{tikzcd}$$

By the universal property of $U \times_Y V$, there exists unique morphism $U \times_X V \to U \times_Y V$. By the Diagonal-Base-Change 2.15, we have a cartesian square.

$$\begin{tikzcd} U \times_X V \arrow[r] \arrow[d, hook] & U \times_Y V \arrow[d, hook] \\ X \arrow[r] & X \times_Y X \end{tikzcd}$$

By the proof in Proposition 9.1.4, we know that $U \cap V \cong U \times_X V = \delta_\pi^{-1}(U \times_Y V)$.

If $\pi : X \to Y$ is quasiseparated, then δ_π is a quasicompact morphism, we know that $U \cap V$ is covered by finite numbers of affine open sets.

Conversely, if $U \cap V$ is a finite union of affine open sets, we know that $\pi : X \to X \times_Y X$ is quasicompact, and therefore $\pi : X \to Y$ is quasiseparated. □

Proposition 12.1.4

Quasiseparated morphisms are a reasonable class.

Proof By Proposition 11.4.2, quasicompact morphisms are a reasonable class. By the definition, $\pi : X \to Y$ is quasiseparated if and only if $\delta_\pi : X \to X \times_Y X$ is quasicompact, by using "quasicompact morphisms are a reasonable class", it is easy to check that "Quasiseparated morphisms are a reasonable class". □

All of this discussion was remarkably removed from geometry. To bring in the geometry, we now look more closely at the diagonal, which isn't just any old kind of morphism.

12.2 Separatedness, and varites

12.2.1 Separatedness I

Proposition 12.2.1

Let $\pi : X \to Y$ be a morphism of schemes.

(a) If X and Y are affine, then the diagonal morphism $\delta_\pi : X \to X \times_Y X$ is a closed embedding.

(b) In general, the diagonal morphism $\delta_\pi : X \to X \times_Y X$ is a locally closed embedding.

Definition 12.2.1 (Diagonal (locally closed) subscheme)

The corresponding locally closed subscheme of $X \times_Y X$ — the diagonal (locally closed) subscheme — will be denoted Δ, i.e., $\Delta = \delta_\pi(X)$.

Proof of Proposition 12.2.1:

Proof

(a) Suppose $X = \operatorname{Spec} A$ and $Y = \operatorname{Spec} B$. Then the diagonal morphism corresponds to the natural ring map $A \otimes_B A \to A$ given by $a_1 \otimes a_2 \mapsto a_1 a_2$, which is obviously surjective.

(b) We will describe a union of open subsets of $X \times_Y X$ covering the image of X, such that the image of X is a closed embedding in this union.

Say Y is covered with affine open sets V_i and X is covered with affine open sets U_{ij}, with $\pi : U_{ij} \to V_i$. Note that $U_{ij} \times_{V_i} U_{ij}$ is an affine open subscheme of the product $X \times_Y X$ (basically this is how we constructed the product, by gluing together affine building blocks). Then the diagonal is covered by these affine open subsets $U_{ij} \times_{V_i} U_{ij}$. (Any point $p \in X$ lies in some U_{ij}; then $\delta(p) \in U_{ij} \times_{V_i} U_{ij}$.

Figure 12.1 may be helpful.) Note that $\delta^{-1}(U_{ij} \times_{V_i} U_{ij}) = U_{ij}$: clearly $U_{ij} \subseteq \delta^{-1}(U_{ij} \times_{V_i} U_{ij})$, and because $\mathrm{pr}_1 \circ \delta = \mathrm{id}_X$ (where pr_1 is the first projection), $\delta^{-1}(U_{ij} \times_{V_i} U_{ij}) \subseteq U_{ij}$. Finally, by part (a), $U_{ij} \to U_{ij} \times_{V_i} U_{ij}$ is a closed embedding.

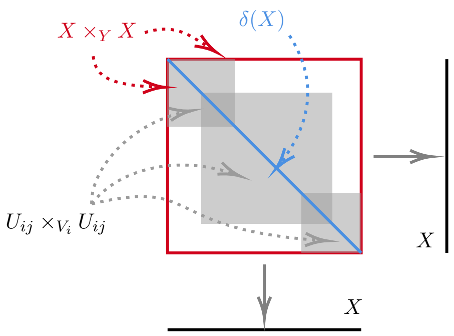


Figure 12.1: An open neighborhood of the diagonal is covered by $U_{ij} \times_{V_i} U_{ij}$.

□

Remark 12.2 The open subsets we described may not cover $X \times_Y X$, so we have not shown that δ is a closed embedding.

Proposition 12.2.2

Suppose $X \to Y$ and $X \to Z$ are both locally closed embeddings. Then $X \to Y \times Z$ is a locally closed embedding.

Proof Recall Proposition 11.4.1, we know that locally closed embeddings is "reasonable" class of morphisms of schemes. Since each scheme can be seen as $\mathbb{Z}$-scheme (Proposition 8.3.8), we have $Y \times_{\mathbb{Z}} Z \cong Y \times Z$. By Proposition 12.1.1, $X \to Y \times_{\mathbb{Z}} Z$ is locally closed embedding, i.e., $X \to Y \times Z$ is locally closed embedding.

□

Definition 12.2.2 (Separatedness)

*A morphism $\pi : X \to Y$ is **separated** if the diagonal morphism $\delta_\pi : X \to X \times_Y X$ is a closed embedding. An A-scheme X is said to be **separated over** A if the structure morphism $X \to \operatorname{Spec} A$ is separated.*

Remark 12.3 When people say the a **scheme** (rather than a morphism) X **is separated**, they mean implicitly that some "structure morphism" is separated. For example, if they are talking about A-schemes, they mean that X is separated over A.

Thanks to Proposition 12.2.1 (b) (and the fact that a locally closed embedding whose image is closed is actually a closed embedding, Proposition 10.2.2), we have following corollary immediately:

Corollary 12.2.1 (Separatedness is a topological condition on the diagonal)

A morphism is separated if and only if the diagonal Δ is a closed subset.

This is reminiscent of a definition of Hausdorff, as the next proposition shows:

Proposition 12.2.3

A topological space X is Hausdorff if and only if the diagonal is a closed subset.

Proof In the topology, $\Delta = \{(x, x) \in X \times X : x \in X\}$, where X is a topological space.

If Δ is closed in $X \times X$. To show that X is Hausdorff, it suffices to show that if x and y are any two points of X, then there are open sets $U, V \subseteq X$ such that $x \in U, y \in V$ and $U \cap V = \varnothing$. Let $p = (x, y) \in X \times X$ where $x \neq y$, then $p \notin \Delta$, hence $p \in (X \times X) \setminus \Delta$. Thus, there must be a basic open set B in the product topology such that $p \in B \subseteq (X \times X) \setminus \Delta$. Basic open sets in the product topology are open boxes, i.e., sets of the form $U \times V$, where U and V are open in X, so let $B = U \times V$ for such $U, V \subseteq X$. Since $U \times V \subseteq (X \times X) \setminus \Delta$, there must be $U \cap V = \varnothing$.

Now suppose that X is Hausdorff. To show that Δ is closed in $X \times X$, it suffices to show that $(X \times X) \setminus \Delta$ is open. To do this, just take any point $p \in (X \times X) \setminus \Delta$ and show that it has an open neighborhood disjoint from Δ. Let $p = (x, y) \in (X \times X) \setminus \Delta$, then $x \neq y$. By the Hausdorffness, there exists open subsets $U, V \subseteq X$ with $x \in U$ and $y \in V$ such that $U \cap V = \varnothing$. Then we get an open subset $U \times V \subseteq (X \times X) \setminus \Delta$ which contains p, and therefore $(X \times X) \setminus \Delta$ is open, we done. □

Proposition 12.2.4

Separated morphisms form a "reasonable" class in the sense of Definition 9.1.1 — they are by preserved by composition, base change, and the notion is local on the target.

Proof The class of closed embeddings is "reasonable" on this sense (Proposition 11.4.1), so by Theorem 12.1.2 the same is true of separated morphisms. □

Proposition 12.2.5

(a) Monomorphisms are separated.

(b) Locally closed embeddings (and in particular open and closed embeddings) are separated.

Proof Recall that $\pi : X \to Y$ is a monomorphism if and only if $\delta_\pi : X \to X \times_Y X$ is an isomorphism (Exercise 2.19). So $\Delta = X \times_Y X$ is a closed subset of $X \times_Y X$, by Corollary 12.2.1, $\pi : X \to Y$ is separated.

By Lemma 11.2.3, we know that locally closed embedding is monomorphism, by the first part of proof, locally closed embeddings are separated. □

Proposition 12.2.6

Separated morphisms are quasiseparated.

Proof Let $\pi : X \to Y$ be a separated morphism, then $\delta_\pi : X \to X \times_Y X$ is a closed embedding. Since closed embeddings are affine, δ_π is quasicompact, and therefore $\pi : X \to Y$ is quasiseparated. □

Proposition 12.2.7

Affine morphisms are separated. In particular, finite morphisms are separated.

Proof By Proposition 12.2.1 (a), every morphism of affine schemes is separated. By Proposition 12.2.4, the notion of separatedness is local on the target, so we are done. □

Many of the following propositions will drive home the importance of the Cancelation Theorem 12.1.1.

Proposition 12.2.8

An A-scheme is separated (over A) if and only if it is separated over $\mathbb{Z}$.

In particular, a complex scheme is separated over $\mathbb{C}$ if and only if it is separated over $\mathbb{Z}$, so complex geometers and arithmetic geometers can discuss separated schemes with each other without confusion.

Proof Suppose X is an A-scheme. Denote $\mathscr{P}$ be a class of separated morphism, by Proposition 12.2.4, $\mathscr{P}$ is a "reasonable" class.

If X is separated, then the structure morphism $X \to \operatorname{Spec} A$ is separated. By Proposition 12.2.1 (a), $\operatorname{Spec} A \to \operatorname{Spec} \mathbb{Z}$ is separated. Note that X and $\operatorname{Spec} A$ can be seen as $\mathbb{Z}$-schemes, $X \to \operatorname{Spec} \mathbb{Z}$ factors through $\operatorname{Spec} A$. Since separatedness is preserved by composition, $X \to \operatorname{Spec} \mathbb{Z}$ is separated.

Conversely, suppose $X \to \operatorname{Spec} \mathbb{Z}$ is separated. Since $\operatorname{Spec} A \times_{\mathbb{Z}} \operatorname{Spec} A \cong \operatorname{Spec} A \otimes_{\mathbb{Z}} A$, $\delta_A : \operatorname{Spec} A \to \operatorname{Spec} A \times_{\mathbb{Z}} \operatorname{Spec} A$ is a morphism of affine schemes, by Proposition 12.2.1, we know that δ_{δ_A} is a closed embedding, and therefore δ_A is separated. Hence $\operatorname{Spec} A \to \operatorname{Spec} \mathbb{Z}$ belongs to the class $\mathscr{P}\delta$. Since $X \to \operatorname{Spec} A$ is a morphism of $\mathbb{Z}$-schemes, we have a commutative diagram.

$$\begin{tikzcd} X \arrow[rr, "\therefore \in \mathscr{P}"] \arrow[dr, "\in \mathscr{P}"'] & & \operatorname{Spec} A \arrow[dl, "\in \mathscr{P}\delta"] \\ & \operatorname{Spec} \mathbb{Z} & \end{tikzcd}$$

By the Cancellation Theorem 12.1.1, we know that $X \to \operatorname{Spec} A$ is separated. □

Proposition 12.2.9

Suppose morphisms $X \xrightarrow{\pi} Y \xrightarrow{\rho} Z$.

(a) If $\rho \circ \pi$ is a locally closed embedding (resp., locally of finite type, separated), then so is π.

(b) If $\rho \circ \pi$ is quasicompact, and Y is Noetherian, the π is quasicompact.

(c) If $\rho \circ \pi$ is quasiseparated, then π is quasiseparated.

Proof Consider the following commutative diagram.

$$\begin{tikzcd} X \arrow[rr, "\pi"] \arrow[dr, "\rho\circ\pi"'] & & Y \arrow[dl, "\rho"] \\ & Z & \end{tikzcd}$$

(a) If $\rho \circ \pi$ is locally closed embedding. By Proposition 12.2.1 (b), we know that δ_ρ is locally closed embedding. Since $\rho \circ \pi$ is locally closed embedding, by the Cancellation Theorem 12.1.1, π is a locally closed embedding.

If $\rho \circ \pi$ is locally of finite type. Since δ_ρ is locally closed embedding, by Proposition 10.2.1, we know that δ_ρ is locally of finite type. By the Cancellation Theorem 12.1.1, π is a locally of finite type.

If $\rho \circ \pi$ is separated. By Proposition 12.2.5, locally closed embedding δ_ρ is separated, by the Cancellation Theorem 12.1.1, π is separated.

(b) Recall Proposition 9.3.2 (a), since Y is Noetherian, $\delta_\rho : Y \to Y \times_Z Y$ is quasicompact. Since $\rho \circ \pi$ is quasicompact, by the Cancellation Theorem 12.1.1, π is quasicompact.

(c) We claim that $\delta_\rho : Y \to Y \times_Z Y$ is quasiseparated. It suffices to show that δ_{δ_ρ} is quasicompact. Since δ_ρ is locally closed embedding, δ_ρ is a monomorphism, by Exercise 2.19, δ_{δ_ρ} is an isomorphism. Clearly any isomorphism is affine morphism, and therefore δ_{δ_ρ} is quasicompact. Hence δ_ρ is quasiseparated. Since $\rho \circ \pi$ is quasiseparated, by the Cancellation Theorem 12.1.1, π is quasiseparated.

□

Proposition 12.2.10 (Section of morphism)

Suppose $\mu : Z \to X$ is a morphism, and $\sigma : X \to Z$ is a ***section of*** *μ, i.e., $\mu \circ \sigma$ is the identity on X.*

$$\begin{tikzcd} Z \arrow[d, bend right, "\mu"'] \\ X \arrow[u, bend right, "\sigma"'] \end{tikzcd}$$

(a) σ is a locally closed embedding.

(b) If μ is separated, then σ is a closed embedding.

(c) If μ is quasiseparated, then σ is quasicompact.

Proof Consider the following commutative diagram.

$$\begin{tikzcd} X \arrow[rr, "\sigma"] \arrow[dr, "\mathrm{id}_X"'] & & Z \arrow[dl, "\mu"] \\ & X & \end{tikzcd}$$

(a) By Proposition 12.2.9 (a), σ is a locally closed embedding.

(b) Since μ is separated, δ_μ is closed embedding. Since id_X is closed embedding, by the Cancellation Theorem 12.1.1, σ is a closed embedding.

(c) Since μ is quasiseparated, δ_μ is quasicompact. Since id_X is quasicompact, by the Cancellation Theorem 12.1.1, σ is quasicompact.

$\square$

Example 12.2 (An example for Proposition 12.2.10: if μ is not separated, σ need not be a closed embedding) Suppose $k = \overline{k}$. Let $X = \mathbb{A}^1_k = \operatorname{Spec} k[t]$, and Z be an affine line with the double origin (which we defined in §5.4.2). Recall §5.4.2, $Z = (\operatorname{Spec} k[x_1] \coprod \operatorname{Spec} k[x_2]) / \sim$, where $\sim$ is defined by $\operatorname{Spec} k[x_1]_{x_1} \xleftrightarrow{\sim} \operatorname{Spec} k[x_2]_{x_2}$ which is given by ring isomorphism $x_1 \mapsto x_2$. Say $U = \operatorname{Spec} k[x_1]_{x_1}$ and $V = \operatorname{Spec} k[x_2]_{x_2}$, define two ring maps: $\mu_1^\sharp : k[t] \to k[x_1]_{x_1}$ by setting $t \mapsto \frac{x_1}{1}$ and $\mu_2^\sharp : k[t] \to k[x_2]_{x_2}$ by setting $t \mapsto \frac{x_2}{1}$. Then they induce morphisms of schemes $\mu_1 : U \to X$ and $\mu_2 : V \to X$. Consider $\mu_1|_{U\cap V} : U \cap V \to X$ and $\mu_2|_{U\cap V} : V \cap V \to X$, on the overlap $U \cap V$, we have coordinate change $x_1 \leftrightarrow x_2$, so $\mu_1|_{U\cap V} = \mu_2|_{U\cap V}$. By Morphism of locally ringed spaces glue 8.3.2, they glue together, then we obtain a morphism of schemes $\mu : Z \to X$.

We first show that μ is not separated. By Corollary 12.2.1, it suffices to show that Δ is not a closed subset of $Z \times_X Z$. Let $0_1, 0_2$ be two origin on Z, where $0_1 = [(x_1)] \in \operatorname{Spec} k[x_1]$ and $0_2 = [(x_2)] \in \operatorname{Spec} k[x_2]$. Say $p = (0_1, 0_2)$, it is easy to check that $p \in Z \times_X Z$ (as topological space). Since $0_1 \neq 0_2$, we know that $p \notin \Delta$. To show that Δ is not closed, it suffices to show that $p \in \mathrm{cl}_{Z\times_X Z}(\Delta)$ (closure of Δ in $Z \times_X Z$). As set,

$$\Delta = \{(q, q) \in Z \times_X Z : q \in Z\}.$$

Since $q \in Z$, $q \in \operatorname{Spec} k[x_i]$ for some i, if $q \neq 0_1, 0_2$, we have $q \in \operatorname{Spec} k[x_i]$ for all $i = 1, 2$. Say $q_a = [(x_1 - a)] = [(x_2 - a)]$ for $a \neq 0$, then

$$S := \{(q_a, q_a) \in Z \times_X Z : q_a \in Z\} = (U \times_X V) \cap \Delta.$$

Since

$$U \times_X V = \operatorname{Spec} k[x_1]_{x_1} \times_{k[t]} \operatorname{Spec} k[x_2]_{x_2} \cong \operatorname{Spec} k[x_1]_{x_1} \otimes_{k[t]} k[x_2]_{x_2} \cong \operatorname{Spec} k[x_i]_{x_i},$$

say $A_1 := \operatorname{Spec} k[x_1]$, $A_2 := \operatorname{Spec} k[x_2]$, we have

$$\mathrm{cl}_{A_1\times_X A_2}(S) \cong \operatorname{Spec} k[x_i] \cong \operatorname{Spec} k[x_1] \times_{k[t]} \operatorname{Spec} k[x_2].$$

Hence $p \in \mathrm{cl}_{A_1 \times_X A_2}(S) \subseteq \mathrm{cl}_{Z \times_X Z}(\Delta)$, it follows that Δ is not closed. Hence μ is not separated.

Let $\sigma : X \to Z$ by setting $X \xrightarrow{\sim} \operatorname{Spec} k[x_1]$, then $\sigma(X) \cong \operatorname{Spec} k[x_1]$. Consider $Z \setminus \sigma(X) = \{0_2\}$, since $\{0_2\}$ is closed in $\operatorname{Spec} k[x_2]$ and the preimage of $\{0_2\}$ under the quotient map $\operatorname{Spec} k[x_1] \coprod \operatorname{Spec} k[x_2] \to Z$ is $\{0_2\} \in \operatorname{Spec} k[x_2]$, we know that $\{0_2\}$ is closed in Z, and therefore $\sigma(X)$ is open. Hence σ is not closed embedding.

Definition 12.2.3 (The graph morphism)

*Suppose $\pi : X \to Y$ is a morphism of Z-schemes. The morphism $\gamma_\pi : X \to X \times_Z Y$ given by $\gamma = (\mathrm{id}, \pi)$ is called the **graph morphism** (see Figure 12.2). (It secretly appeared in Theorem 12.1.1.) Then π factors as $\mathrm{pr}_2 \circ \gamma_\pi$, where pr_2 is the second projection. The diagram of Figure 12.2 is often called the **graph of a morphism** π.*

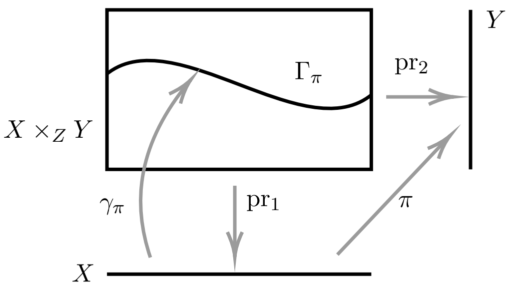


Figure 12.2: The graph morphism

Remark 12.4 We will discuss graphs of rational maps in §12.3.3.

Proposition 12.2.10 applied to $X \times_Z Y \to X$ yields the following.

Proposition 12.2.11

The graph morphism γ_π is always a locally closed embedding. If Y is a separated Z-scheme (i.e., $Y \to Z$ is separated), then γ_π is a closed embedding. If Y is a quasiseparated Z-scheme, then γ_π is quasicompact.

Proof Consider the following Cartesian square.

$$\begin{tikzcd}
X \arrow[dr, dashed, "\gamma_\pi"] \arrow[drr, bend left, "\pi"] \arrow[ddr, bend right, "\mathrm{id}_X"'] & & \\
& X \times_Z Y \arrow[r, "\mathrm{pr}_2"] \arrow[d, "\mathrm{pr}_1"] & Y \arrow[d] \\
& X \arrow[r] & Z
\end{tikzcd}$$

We have $\mathrm{pr}_1 \circ \gamma_\pi = \mathrm{id}_X$, i.e., γ_π is a section of pr_1. Note that separatedness and quasiseparatedness preserved by base change, apply Proposition 12.2.10, we are done. $\square$

Note *The image of γ_π (a locally closed subscheme of $X \times_Z Y$) is often also called the graph of π; we might denote it Γ_π.*

Proposition 12.2.12 (Unimportant, we won't use)

Suppose $\mathscr{P}$ is a class of morphisms such that closed embeddings are in $\mathscr{P}$, and $\mathscr{P}$ is preserved by fibered product and composition. Recall the Definition 11.7.5 of the "reduction" of a morphism. If $\pi : X \to Y$ is in $\mathscr{P}$ then $\pi^{\mathrm{red}} : X^{\mathrm{red}} \to Y^{\mathrm{red}}$ is in $\mathscr{P}$.

Remark 12.5 Two examples are the class of separated morphisms and quasiseparated morphisms.

Proof Recall Definition 11.7.5 and Proposition 11.7.9, $X^{\text{red}} \hookrightarrow X$ and $Y^{\text{red}} \hookrightarrow Y$ are closed embedding. Since closed embedding is monomorphism, by Exercise 2.19, we know that $Y^{\text{red}} \to Y^{\text{red}} \times_Y Y^{\text{red}}$ is an isomorphism, and therefore is a closed embedding, hence $Y^{\text{red}} \to Y$ belongs to $\mathscr{P}\delta$. Consider the following diagram.

$$\begin{tikzcd}
X^{\text{red}} \arrow[r, "\pi^{\text{red}}"] \arrow[d, "\in\mathscr{P}"'] \arrow[dr, "\in\mathscr{P}"] & Y^{\text{red}} \arrow[d, "\in\mathscr{P}\delta"] \\
X \arrow[r, "\pi\in\mathscr{P}"'] & Y
\end{tikzcd}$$

Since $\mathscr{P}$ preserved by composition, we know that $X^{\text{red}} \to Y$ belongs to $\mathscr{P}$, by the Cancellation Theorem 12.1.1, $\pi^{\text{red}} : X^{\text{red}} \to Y^{\text{red}}$ belongs to $\mathscr{P}$. □

12.2.2 Important examples

Example 12.3 (Line with doubled origin is not separated) The line with doubled origin is not separated (§5.4.2).

Proof See Example 12.2 . □

We next come to our first example of something separated but not affine. The following single calculation will imply that all quasiprojective A-schemes (recall Definition 5.5.11) are separated (once we know that the composition of separated morphisms are separated, Proposition 12.2.4).

Proposition 12.2.13

The morphism $\mathbb{P}^n_A \to \operatorname{Spec} A$ is separated.

Remark 12.6 We give two proofs. The first is by direct calculation. The second requires no calculation and just requires that you remember some classical constructions described earlier.

Proof 1 (Direct calculation):

Proof We cover $\mathbb{P}^n_A \times_A \mathbb{P}^n_A$ with open sets of the form $U_i \times_A U_j$, where $U_i = D_+(x_i)$ for all $i = 0, \dots, n$. The case $i = j$ was taken care of before, in the proof of Proposition 12.2.1 (a), i.e., $\delta_{U_i \to \operatorname{Spec} A}$ is closed embedding. If $i \neq j$ then

$$U_i \times_A U_j \cong \operatorname{Spec} A[x_{0/i}, \dots, x_{n/i}, y_{0/j}, \dots, y_{n/j}]/(x_{i/i} - 1, y_{j/j} - 1).$$

Consider the following diagram.

$$\begin{tikzcd}
\mathbb{P}^n_A \arrow[dr, "\delta"] \arrow[drr, bend left, "\mathrm{id}_{\mathbb{P}^n_A}"] \arrow[ddr, bend right, "\mathrm{id}_{\mathbb{P}^n_A}"'] & & \\
& \mathbb{P}^n_A \times_A \mathbb{P}^n_A \arrow[r, "\mathrm{pr}_2"'] \arrow[d, "\mathrm{pr}_1"] & \mathbb{P}^n_A \arrow[d] \\
& \mathbb{P}^n_A \arrow[r] & \operatorname{Spec} A
\end{tikzcd}$$

Now $\delta^{-1}(\mathrm{pr}_1^{-1}(U_i))$ is U_i (as $\mathrm{pr}_1 \circ \delta = \mathrm{id}_{\mathbb{P}^n_A}$), and similarly $U_j = \delta^{-1}(\mathrm{pr}_2^{-1}(U_j))$. Thus $\delta^{-1}(U_i \times_A U_j) = U_i \cap U_j$, so the diagonal morphism over $U_i \times_A U_j$ is $U_i \cap U_j \to U_i \times_A U_j$. This is a closed embedding, as the corresponding map of rings

$$A[x_{0/i}, \dots, x_{n/i}, y_{0/j}, \dots, y_{n/j}] \longrightarrow A[x_{0/i}, \dots, x_{n/i}, x_{j/i}^{-1}]/(x_{i/i} - 1)$$

(given by $x_{k/i} \mapsto x_{k/i}$, $y_{k/j} \mapsto x_{k/i}/x_{j/i}$) is clearly a surjection (as each generator of the ring on the right is clearly in the image — note that $x_{j/i}^{-1}$ is the image of $y_{i/j}$). ☐

Proof 2 (Classical geometry):

Proof Note that the diagonal morphism $\delta : \mathbb{P}_A^n \to \mathbb{P}_A^n \times_A \mathbb{P}_A^n$ followed by the Segre embedding $S : \mathbb{P}_A^n \times_A \mathbb{P}_A^n \to \mathbb{P}^{n^2+2n}$ (§11.6, a closed embedding) can also be factored as the second Veronese embedding $v_2 : \mathbb{P}_A^n \to \mathbb{P}^{\binom{n+2}{2}-1}$ (Definition 10.3.3) followed by a linear map $L : \mathbb{P}^{\binom{n+2}{2}-1} \to \mathbb{P}^{n^2+2n}$, both of which are closed embeddings (v_2 is closed embedding by Proposition 10.3.6, L is closed by Proposition 10.3.2).

Remark

Let $V^\vee = \operatorname{Span}_A\{x_0^2, x_0, x_1, \dots, x_{n-1}x_n, x_n^2\}$ and $W^\vee = \operatorname{Span}_A\{x_0^2, \dots, x_ix_j, \dots, x_n^2\}$, where in $W^\vee$ the subscripts run over all ordered pairs (i,j) where $0 \le i, j \le n$, then $V \hookrightarrow W$ is injective. By Proposition 10.3.2, we have a closed embedding $\mathbb{P}V \hookrightarrow \mathbb{P}W$. Recall Definition 5.5.12, we know that $\mathbb{P}V = \mathbb{P}_A^{\binom{n+2}{2}-1}$ and $\mathbb{P}W = \mathbb{P}_A^{n^2+2n}$, then L is a closed embedding.

$$\begin{tikzcd}
 & \mathbb{P}_A^n \times_A \mathbb{P}_A^n \arrow[dr, "S", "\text{cl.emb.}"'] & \\
\mathbb{P}_A^n \arrow[ur, "\delta", "\text{cl.emb. ?}"'] \arrow[dr, "\text{cl.emb.}", "v_2"'] & & \mathbb{P}_A^{n^2+2n} \\
 & \mathbb{P}_A^{\binom{n+2}{2}-1} \arrow[ur, "\text{cl.emb.}", "L"'] &
\end{tikzcd}$$

In formally, in coordinates:

$$\begin{tikzcd}
 & ([x_0 : x_1 : \cdots : x_n], [x_0 : x_1 : \cdots : x_n]) \arrow[dr, "S"] & \\
{[x_0 : x_1 : \cdots : x_n]} \arrow[ur, "\delta"] \arrow[dr, "v_2"'] & & \begin{bmatrix} x_0^2 : & x_0x_1 : & \cdots & x_0x_n : \\ x_1x_0 : & x_1^2 : & \cdots & x_1x_n : \\ \cdots & \cdots & \cdots & \cdots \\ x_nx_0 : & x_nx_1 : & \cdots & x_n^2 \end{bmatrix} \\
 & {[x_0^2 : x_0x_1 : \cdots : x_{n-1}x_n : x_n^2]} \arrow[ur, "L"'] &
\end{tikzcd}$$

The composed map $\mathbb{P}_A^n \to \mathbb{P}_A^{n^2+2n}$ may be written as

$$[x_0 : \cdots : x_n] \longmapsto [x_0^2 : x_0x_1 : x_1x_2 : \cdots : x_n^2],$$

where the subscripts on the right run over all ordered pairs (i,j) where $0 \le i, j \le n$. This forces δ to send closed sets to closed sets (or else $S \circ \delta$ won't, but $L \circ v_2$ does). ☐

Corollary 12.2.2

Any quasiprojective A-schemes are separated.

Proof Recall Definition 5.5.11, a quasiprojective A-scheme X is a quasicompact open subscheme of a projective A-scheme $\operatorname{Proj} S_\bullet$, where $S_\bullet$ is a finitely generated graded ring over A. By Proposition 10.3.3, $\operatorname{Proj} S_\bullet \hookrightarrow \mathbb{P}_A^n$ is a closed embedding for some n. By Proposition 12.2.5, $X \hookrightarrow \operatorname{Proj} S_\bullet \hookrightarrow \mathbb{P}_A^n$ is separated. Since separatedness preserved by composition, by Proposition 12.2.13, X is separated. ☐

12.2.3 Variety

Definition: Varieties

Next, we give an important definition (finally!):

Definition 12.2.4 (Variety)

*A **variety** over a field k, or k-variety, is a reduced, separated scheme of finite type over k.*

Example 12.4 By Proposition 12.2.1 (a), a reduced finite type affine k-scheme is a variety.

The notion of variety generalizes our earlier notion of affine variety (Definition 6.3.5) and projective variety (Definition 6.3.5). By Proposition 6.3.8, the closed points of a variety are dense. (**Notational caution:** In some sources, the additional condition of irreducibility is imposed. Also, it is often assumed that k is algebraically closed.)

Variety over k form a category: morphisms of k-varieties are just morphisms as k-schemes. (Of cause, the category of varieties over an algebraically closed field k long predates modern scheme theory.)

Proposition 12.2.14

The morphisms of k-schemes are finite type and separated.

Proof Let X, Y be two k-schemes with morphism $\pi : X \to Y$. Consider the following commutative diagram,

$$\begin{tikzcd} X \arrow[rr, "\pi"] \arrow[dr, "\varphi"'] & & Y \arrow[dl, "\rho"] \\ & \operatorname{Spec} k & \end{tikzcd}$$

where the structure morphism $\varphi = \rho \circ \pi$ is locally of finite type and separated. Apply Proposition 12.2.9, we know that $\pi : X \to Y$ is locally of finite type and separated. We next show that $\pi : X \to Y$ is quasicompact. By Proposition 12.2.6, we know that $\rho : Y \to \operatorname{Spec} k$ is quasiseparated, and therefore $Y \to Y \times_k Y$ is quasicompact. Note that φ is quasicompact (since φ is of finite type), by the Cancellation Theorem 12.1.1, we know that $\pi : X \to Y$ is quasicompact. Hence $\pi : X \to Y$ is of finite type and separated. $\square$

Definition 12.2.5 (Subvariety)

*A **subvariety** of a variety X is reduced locally closed scheme of X. An **open subvariety** of X is an open subscheme of X (reducedness is automatic in this case). A **closed subvariety** of X is a reduced closed subscheme of X.*

Proposition 12.2.15

A subvariety is a variety.

Proof Let X be a k-variety and $Y \hookrightarrow X$ be a subvariety of X. By Proposition 12.2.14, $Y \hookrightarrow X$ is of finite type and separated. Thus $Y \to \operatorname{Spec} k$ is of finite type and separated. Recall the definition of subvariety (Definition 12.2.5), Y is reduced. Hence Y is a reduced, separated scheme of finite type over k, which implies that Y is a variety. $\square$

For those with appropriate background:

Proposition 12.2.16 (⋆⋆ Complex algebraic varieties yield complex analytic varieties)

The analytification of a complex algebraic variety is a complex analytic variety.

Products of varieties

Proposition 12.2.17

(a) If k is perfect, the product of k-varieties (the fibered product over $\operatorname{Spec} k$) is a k-variety.

(b) If k is algebraically closed, the product of irreducible k-varieties is an irreducible k-variety.

(c) If k is algebraically closed, the product of connected k-varieties is a connected k-variety.

Proof

(a) The finite type and separated statements are straightforward, as both properties are preserved by base change and composition. For reducedness, since reducedness is affine-local property (Example 6.5), reduce to the affine case, then use Corollary 11.5.2.

(b) It only remains to show irreducibility. Reduce to the affine case using Proposition 11.5.11 (as in the proof of Proposition 11.5.12). Then use Proposition 11.5.10, which requires separable closure.

(c) It only remains to show connectedness. By Corollary 11.5.1, X is geometrically connected over k, and by Lemma 11.5.2, $X \times_k Y$ is connected.

□

12.2.4 Separatedness II

Here is a very hard consequence of separatedness.

Proposition 12.2.18

Suppose $X \to \operatorname{Spec} A$ is a separated morphism to an affine scheme, and U and V are affine open subsets of X. Then $U \cap V$ is an affine open subset of X.

Before proving this, we give an unexpected consequence.

Proposition

Let $\operatorname{Spec} A$ be an affine scheme, then the intersection of any two affine open subsets of $\operatorname{Spec} A$ is also an affine open subset.

This is certainly not obvious! We know the intersection of two distinguished affine open subsets is affine (from $D(f) \cap D(g) = D(fg)$), but we have little handle on affine open subsets in general.

Remark 12.7 **Warning:** this property does not characterize separatedness. (The correct property is "affine-diagonal", which we will not discuss, but which is useful in the theory of stacks.) For example, if X is the line with doubled origin over k, then X also this property (recall Example 12.3 or Example 12.2 , X is not separated).

In the proof of Proposition 12.2.18, we will use the following nifty result.

Lemma 12.2.1

Suppose U and V are open subsets of an A-scheme X. Then $\Delta \cap (U \times_A V) \cong U \cap V$ (Figure 12.3).

Proof Recall Definition 11.2.3, $U \cap V = U \times_X V$. We want to check that $\Delta \cap (U \times_A V)$ together the natural maps to U and V have the universal property for $U \times_X V$. (We think of $\Delta \cap (U \times_A V)$ as a locally closed

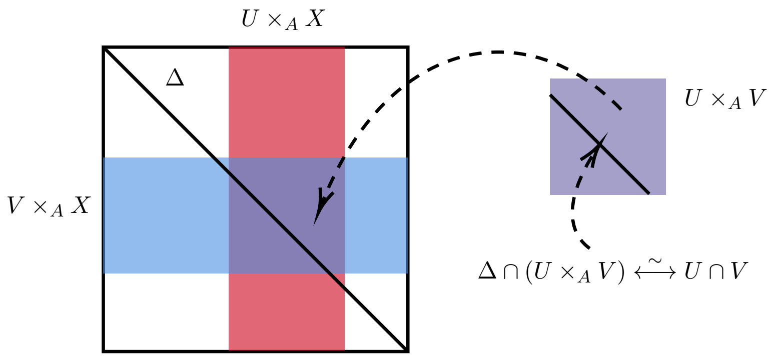


Figure 12.3: $\Delta \cap (U \times_A V) \cong U \cap V$ Lemma 12.2.1

subscheme of $X \times_A X$.) Consider the following diagram,

$$\begin{tikzcd}
W \arrow[r, "f"] \arrow[d, "g"'] & U \arrow[d, hook, "i_U"'] \\
V \arrow[r, hook, "i_V"] & X
\end{tikzcd}$$

where W be any A-scheme with $i_U \circ f = i_V \circ g$. Composing with $X \to \operatorname{Spec} A$, by the universal property of $U \times_A V$, there exists unique morphism $W \to U \times_A V$ such that the following diagram commutes.

$$\begin{tikzcd}
W \arrow[dr, dashed, "\exists!"] \arrow[drr, bend left, "f"] \arrow[ddr, bend right, "g"'] & & & \\
& U \times_A V \arrow[r] \arrow[d] & U \arrow[d, hook, "i_U"'] \arrow[ddr, bend left] & \\
& V \arrow[r, hook, "i_V"] \arrow[drr, bend right] & X \arrow[dr, "s"] & \\
& & & \operatorname{Spec} A
\end{tikzcd}$$

It follows that the image of W contained in $U \times_A V$. We next show that the image of W under $W \to X \times_A X$ contained in Δ. Consider the following diagram.

$$\begin{tikzcd}
W \arrow[drr, "i_U \circ f"] \arrow[ddrr, dashed, "\exists!"] \arrow[ddrrr, bend left, "i_U \circ f"] \arrow[dddrr, bend right, "i_V \circ g"'] & & & \\
& & X \arrow[d, "\delta_{X/A}"] & \\
& & X \times_A X \arrow[r, "\mathrm{pr}_X"] \arrow[d, "\mathrm{pr}_X"] & X \arrow[d, "s"] \\
& & X \arrow[r, "s"] & \operatorname{Spec} A
\end{tikzcd} \tag{12.5}$$

By the universal property of $X \times_A X$, there exists unique morphism $W \to X \times_A X$ such that diagram (12.5) commutes, moreover $W \to X \times_A X$ factor through X, i.e., $W \to X \times_A X$ is the composition of $W \longrightarrow X \xrightarrow{\delta_{X/A}} X \times_A X$. It follows that the image of $W \to X \times_A X$ contained in Δ. Consider the following diagram, by diagram (12.5), the composition of red arrows must agree with the composition of blue arrows.

$$\begin{tikzcd}
W \arrow[dr, dashed, "\exists!"] \arrow[drr, bend left] \arrow[ddr, bend right] & & \\
& \Delta \cap (U \times_A V) \arrow[r] \arrow[d] & U \times_A V \arrow[d, hook, "\text{open}"] \\
& \Delta \arrow[r, hook, "\text{loc.cl.}"'] & X \times_A X
\end{tikzcd}$$

Hence by the universal property of $\Delta \cap (U \times_A X)$, there exists a unique morphism from W to $\Delta \cap (U \times_A V)$. Then the following diagram commutes.

$$\begin{tikzcd}
W \arrow[dr, "\exists!"] \arrow[drrr, bend left] \arrow[dddr, bend right] & & & & \\
& \Delta \cap (U \times_A V) \arrow[rr] \arrow[dd] \arrow[dr] & & U \arrow[dd] & \\
& & U \times_A V \arrow[ur] \arrow[dl] & & \\
& V \arrow[rr] & & X \arrow[dr] & \\
& & & & \operatorname{Spec} A
\end{tikzcd}$$

It follows that $\Delta \cap (U \times_A V) \cong U \times_X V = U \cap V$. $\square$

Remark 12.8 You can interpret Lemma 12.2.1 by using the Diagonal-Base-Change diagram 2.15 (identify X as Δ, since $\delta_{X/A}$ is locally closed embedding).

$$\begin{tikzcd}
U \cap V \arrow[r] \arrow[d] & U \times_A V \arrow[d] \\
X \arrow[r, "\delta_{X/A}"] & X \times_A X
\end{tikzcd}$$

Proof of Proposition 12.2.18:

Proof By Lemma 12.2.1, $U \cap V \cong \Delta \cap (U \times_A V)$, where Δ is the diagonal. But $U \times_A V$ is affine (the fibered product of two affine schemes over san affine scheme is affine.), and $(U \times_A V) \cap \Delta$ is a closed subscheme of the affine scheme $U \times_A V$, and hence $U \cap V$ is affine. $\square$

Corollary 12.2.3

Every quasiprojective A-scheme is separated over A. In particular, every reduced quasiprojective k-scheme is a k-variety.

Proof Suppose $X \to \operatorname{Spec} A$ is a quasiprojective A-scheme. The structure morphism can be factored into an open embedding composed with a closed embedding followed by $\mathbb{P}^n_A \to \operatorname{Spec} A$. Open embeddings and closed embeddings are separated (Proposition 12.2.5), and $\mathbb{P}^n_A \to \operatorname{Spec} A$ is separated (Proposition 12.2.13) Compositions of separated morphisms are separated (Proposition 12.2.4), so we are done.

In particular, recall Proposition 6.3.6 (a), we know that reduced quasiprojective k-scheme is a k-variety. $\square$

Here is a proposition we won't use, but you may like. It includes a converse to Proposition 12.2.18.

Proposition 12.2.19

Let X be an A-scheme. Then the following properties are equivalent:

(i) X is separated over A.

(ii) for every pair of affine open subsets U, V of X, $U \cap V$ is affine the canonical homomorphism

$$\mathscr{O}_X(U) \otimes_A \mathscr{O}_X(V) \longrightarrow \mathscr{O}_X(U \cap V)$$

is surjective;

(iii) there exists a covering of X by affine open subsets U_i such that for all i, j, $U_i \cap U_j$ is affine and

$$\mathscr{O}_X(U_i) \otimes_A \mathscr{O}_X(U_j) \longrightarrow \mathscr{O}_X(U_i \cap U_j)$$

is surjective.

Proof Recall the 1st proof of Proposition 12.2.13, we have $\delta^{-1}(U \times_A V) = U \cap V$. The homomorphism $\mathscr{O}_X(U) \otimes_A \mathscr{O}_X(V) \to \mathscr{O}_X(U \cap V)$ corresponds to the restriction $\delta|_{U \cap V} : U \cap V = \delta^{-1}(U \times_A V) \to U \times_A V$.

If X is separated over A, $\delta|_{U \cap V}$ is closed embedding. By Proposition 12.2.18, $U \cap V$ is affine, then the homomorphism $\mathscr{O}_X(U) \otimes_A \mathscr{O}_X(V) \to \mathscr{O}_X(U \cap V)$ is surjective, hence (i) implies (ii). As (ii) implies (iii) is trivial, it remains to show that (iii) implies (i). Note that (iii) implies that $\delta|_{U_i \cap U_j} : \delta^{-1}(U_i \times_A U_j) \to U_i \times_A U_j$ is a closed embedding. As the open subsets $U_i \times_A U_j$ cover $X \times_A X$, δ is a closed embedding. □

12.2.5 ⋆⋆ Universally injective morphisms and the diagonal

Proposition 12.2.20

$\pi : X \to Y$ is universally injective if and only if the diagonal morphism $\delta_\pi : X \to X \times_Y X$ is surjective.

Proof If $\pi : X \to Y$ is universally injective, then natural projection $\mathrm{pr}_X : X \times_Y X \to X$ is injective. Note that $\mathrm{pr}_X \circ \delta_\pi = \mathrm{id}_X$, so δ_π is surjective.

Conversely, if $\delta_{X/Y}$ is surjective. Let $Y' \to Y$ be any morphism, say $X' = X \times_Y Y'$, as in the proof of Theorem 12.1.2 (2), we have a Cartesian square.

$$\begin{tikzcd}
X' \arrow[r, two heads, "\delta_{X'/Y'}"] \arrow[d] & X' \times_{Y'} X' \arrow[d] \\
X \arrow[r, two heads, "\delta_{X/Y}"'] & X \times_Y X
\end{tikzcd}$$

Since surjective preserved by base change we know that $\delta_{X'/Y'}$ is a surjective. Recall that $\mathrm{pr}_{X'} \circ \delta_{X'/Y'} = \mathrm{id}_{X'}$, $\mathrm{pr}_{X'}$ must be injective. It follows that $\pi : X \to Y$ is universally injective. □

Because surjective morphisms form a "reasonable" class (Proposition 11.4.5), the following corollary is immediately:

Corollary 12.2.4

Universally injective morphisms form a "reasonable" class.

Proposition 12.2.21

If $\pi : X \to Y$ and $\rho : Y \to Z$ are morphisms, and $\rho \circ \pi$ is universally injective, then π is universally injective.

Proof We want to apply the Cancellation Theorem 12.1.1, it suffices to show that $\delta_\rho = \delta_{X/Y}$ is universally injective. Consider the following Cartesian square.

$$\begin{tikzcd}
Y \arrow[dr, "\delta"] \arrow[drr, bend left, "\mathrm{id}_Y"] \arrow[ddr, bend right, "\mathrm{id}_Y"'] & & \\
& Y \times_{(Y\times_Z Y)} Y \arrow[r, "\mathrm{pr}_2"'] \arrow[d, hook, "\mathrm{pr}_1"] & Y \arrow[d, hook, "\delta_{Y/Z}"] \\
& Y \arrow[r, "\delta_{Y/Z}"'] & Y \times_Z Y
\end{tikzcd}$$

Since locally closed embedding preserved by base change $Y \times_{(Y\times_Z Y)} Y \to Y$ is locally closed embedding, and therefore $Y \times_{(Y\times_Z Y)} Y \to Y$ is injective as topological space map. Recall $\mathrm{pr}_1 \circ \delta = \mathrm{id}_Y$, δ must be surjective. By Proposition 12.2.20, we know that $\delta_\rho = \delta_{Y/Z}$ is universal injective. Apply the Cancellation Theorem 12.1.1, $\pi : X \to Y$ is universally injective. □

Proposition 12.2.22

(a) Universally injective morphisms are separated.

(b) A map between finite type schemes over an algebraically closed field k is universally injective if and only if it is injective on closed points.

Proof

(a) The diagonal morphism of a universally injective morphism is surjective (Proposition 12.2.20), recall that the diagonal morphism is locally closed embedding, by Proposition 10.2.2, diagonal morphism is a closed embedding, then universally injective morphism is separated.

(b) As $k = \overline{k}$ is algebraically closed, the closed points of X (resp., $X \times_Y X$) are the elements of k-valued points $X(k)$ (resp., $(X \times_Y X)_k$). Say $\pi : X \to Y$. Apply $\mathrm{Mor}(\operatorname{Spec} k, \square)$ to the following Cartesian diagram.

$$\begin{tikzcd}
X \arrow[dr, "\delta_{X/Y}"] \arrow[drr, bend left, "\mathrm{id}_X"] \arrow[ddr, bend right, "\mathrm{id}_X"'] & & \\
& X \times_Y X \arrow[r] \arrow[d] & X \arrow[d] \\
& X \arrow[r] & Y
\end{tikzcd}$$

Then we have a Cartesian diagram.

$$\begin{tikzcd}
X(k) \arrow[dr, "\delta_{X/Y}(k)"] \arrow[drr, bend left, "\mathrm{id}_{X(k)}"] \arrow[ddr, bend right, "\mathrm{id}_{X(k)}"'] & & \\
& (X \times_Y X)(k) \arrow[r] \arrow[d] & X(k) \arrow[d] \\
& X(k) \arrow[r] & Y(k)
\end{tikzcd}$$

It follows that $X(k) \times_{Y(k)} X(k) \cong (X \times_Y X)(k)$.

If $X \to Y$ is universally injective, by Proposition 12.2.20, δ_π is surjective, hence $\delta_\pi(k)$ is surjective, hence

$$X(k) \times_{Y(k)} X(k) = \{(f, g) : \pi(k)(f) = \pi(k)(g)\} = \{(h, h) : h \in X(k)\},$$

which implies that $f = g$ for all $(f, g) \in X(k) \times_{Y(k)} X(k)$, i.e., $\pi(k) : X(k) \to Y(k)$ is injective.

Conversely, if $\pi(k) : X(k) \to Y(k)$ is injective. Pick $(f, g) \in X(k) \times_{Y(k)} X(k)$, then $\pi(k)(f) =$

$\pi(k)(g)$, hence $f = g$, which implies that $\delta_{X/Y}(k)$ is surjective. By Corollary 9.4.4, $\delta_{X/Y}$ is surjective. □

12.3 The locus where two morphisms from X to Y agree, and the "Reduced-to-Separated" Theorem

When we introduced rational maps in §8.5, we promised that in good circumstances, a rational map has a "largest domain of definition". We are now ready to make precise what "good circumstances" means, in the Reduced-to-Separated Theorem 12.3.1.

12.3.1 The locus where two morphism from X to Y agree

We first introduce an important result making sense of locus where two morphisms with the same source and target "agree".

Proposition 12.3.1 (The locus where two morphisms agree)

Suppose $\pi : X \to Y$ and $\pi' : X \to Y$ are two morphisms over some scheme Z.

$$\begin{tikzcd} X \arrow[rr, bend left, "\pi"] \arrow[rr, bend right, "\pi'"'] \arrow[dr] & & Y \arrow[dl] \\ & Z & \end{tikzcd}$$

We can now give meaning to the phrase "the locus where π and π' agree", and that in particular there is a largest locally closed subscheme where they agree — which is closed if Y is separated over Z. Suppose $\mu : W \to X$ is some morphism (not assumed to be a locally closed embedding). We say that π ***and*** *π'* ***agree on*** *μ if $\pi \circ \mu = \pi' \circ \mu$.*

*(1) There is a locally closed subscheme $i : V \hookrightarrow X$ on which π and π' agree factors uniquely through i, i.e., there is a unique $j : W \to V$ such that $\mu = i \circ j$ (**side remark:** $V = \mathrm{Eq}(\pi, \pi')$, Definition 3.2.6).*

$$\begin{tikzcd} W \arrow[d, dashed, "\exists ! j"'] \arrow[dr, "\mu"] & & \\ V \arrow[r, hook, "i", "loc.cl."'] & X \arrow[r, shift left, "\pi"] \arrow[r, shift right, "\pi'"'] & Y \end{tikzcd}$$

(2) Further more, if $Y \to Z$ is separated, then $i : V \hookrightarrow X$ is a closed embedding.

Proof

(1) Define V to be the following fibered product:

$$\begin{tikzcd} V \arrow[r] \arrow[d, "i"] & Y \arrow[d, "\delta_{Y/Z}"] \\ X \arrow[r, "(\pi,\pi')"] & Y \times_Z Y \end{tikzcd}$$

As $\delta_{Y/Z}$ is a locally closed embedding (Proposition 12.2.1), and locally closed embedding preserved by base change, we know that $i : V \to X$ is locally closed embedding. Let $\mu : W \to X$ with $\pi \circ \mu = \pi' \circ \mu$. Consider the following diagram.

$$\begin{tikzcd}
W \arrow[dr, dashed, "\exists!?"] \arrow[drr, bend left, "\pi'\circ\mu"] \arrow[ddr, bend right, "\mu"'] & & \\
& V \arrow[r] \arrow[d, dashed, "i"] & Y \arrow[d, "\delta_{Y/Z}"] \\
& X \arrow[ur, "\pi"] \arrow[r, "(\pi,\pi')"'] & Y\times_Z Y
\end{tikzcd} \tag{12.6}$$

Since $\delta_{Y/Z}\circ\pi = (\pi,\pi')$, we have

$$(\pi,\pi')\circ\mu = \delta_{Y/Z}\circ\pi\circ\mu = \delta_{Y,Z}\circ(\pi'\circ\mu),$$

so by the universal property of V, there exists unique morphism $j : W \to V$ such that diagram (14.15) commutes, then we are done.

If further more $Y \to Z$ is separated, $\delta_{Y/Z} : Y \to Y \times_Z Y$ is closed, hence $i : V \to X$ is closed. $\square$

Example 12.5 (The locus where two maps agree can be nonreduced.) The fact that the locus where two maps agree can be nonreduced should not come as a surprise: consider two maps from $\mathbb{A}^1_k$ to itself, $\pi(x) = 0$ and $\pi'(x) = x^2$. They agree when $x = 0$, but the situation is (epsilonically) better than that — they should agree even on $\operatorname{Spec} k[x]/(x^2)$.

Remark 12.9 Minor Remarks:

(i) In the Proposition 12.3.1, we describe $V \hookrightarrow X$ by a universal property. Taking this as the definition, it is not a priori clear that V is a locally closed subscheme of X, or even that it exists.

(ii) **Warning:** consider two maps from $\operatorname{Spec}\mathbb{C}$ to itself over $\operatorname{Spec}\mathbb{R}$, the identity and complex conjugation. There are both maps from a point to a point, yet they do not agree despite agreeing as maps of sets. (If you are not convinced that they disagree as morphisms, this might help: after base change $\operatorname{Spec}\mathbb{C} \to \operatorname{Spec}\mathbb{R}$, they do not agree even as maps of sets.)

(iii) More generally, the (set-theoretic) locus where π and π' agree can be interpreted as follows: π and π' agree at p if $\pi(p) = \pi'(p)$ and the two maps of residue fields are the same.

Proposition 12.3.2 (Maps of $\overline{\mathrm{k}}$-varieties are determined by the maps on closed points)

Suppose $\pi : X \to Y$ and $\pi' : X \to Y$ are two morphisms of $\overline{\mathrm{k}}$-varieties that are the same at the level of closed points (i.e., for each closed point $p \in X$, $\pi(p) = \pi'(p)$). Then $\pi = \pi'$.

Remark 12.10 This implies that the functor from the category of "classical varieties over $\overline{\mathrm{k}}$, which we won't define here, to the category of $\overline{\mathrm{k}}$-schemes, is fully faithful.

Proof By Proposition 12.3.1, the locus where π and π' agree is defined by equalizer $\operatorname{Eq}(\pi,\pi') \hookrightarrow X$. Since $Y \to \operatorname{Spec}\overline{k}$ is separated, then $\operatorname{Eq}(\pi,\pi') \hookrightarrow X$ is a closed embedding. Since π and π' are the same at the level of closed point, we have $X(\overline{k}) \subseteq \operatorname{Eq}(\pi,\pi')$, by Proposition 6.3.8, $X(\overline{k})$ is dense in X, so $\operatorname{Eq}(\pi,\pi') = \operatorname{cl}_X(X(\overline{k})) = X$ as underlying set. Since X is reduced, $\operatorname{Eq}(\pi,\pi') = X$ as schemes, hence $\pi = \pi'$. $\square$

Generalize Proposition 12.3.2 to non-algebraically closed fields:

Proposition 12.3.3 (Generalize Proposition 12.3.2)

Let k be a field and X, Y be two k-varieties. Let $\pi, \pi' : X \to Y$ be two morphisms of k-schemes such that the induced morphisms $\pi(\overline{k}), \pi'(\overline{k}) : X(\overline{k}) \to Y(\overline{k})$ are equal, i.e., $\pi(\overline{k}) = \pi'(\overline{k})$, then $\pi = \pi'$.

Proof By Proposition 11.2.8, we have $\pi = \pi'$ iff $\pi_{\overline{k}} = \pi'_{\overline{k}}$ after base change, so we may assume that $k = \overline{k}$. By Proposition 12.3.2, we are done. □

Exercise 12.1 (Less important) Show that the line with doubled origin X (§5.4.2) is not separated, by finding two morphisms $\pi_1 : W \to X$, $\pi_2 : W \to X$ whose domain of agreement is not a closed subscheme (cf. Proposition 12.2.1 (b)). (Another argument was given in Example 12.3 . A fancy argument will be given in Chapter 14).

Proof Let $X = X_1 \coprod X_2 / \sim$, where $X_1 = \operatorname{Spec} k[x_1]$, $X_2 = \operatorname{Spec} k[x_2]$, and $\sim$ given by $\operatorname{Spec} k[x_1]_{x_1} \xleftrightarrow{\sim} \operatorname{Spec} k[x_2]_{x_2}$. Let $W = \mathbb{A} = \operatorname{Spec} k[t]$. Define $\pi_1 : W \to X$ by setting $W \xleftrightarrow{\sim} X_1 \hookrightarrow X$, and define $\pi_2 : W \to X$ by setting $W \xleftrightarrow{\sim} X_2 \hookrightarrow X$. The locus where two morphisms agree is given by equalizer of π_1 and π_2, as set-theoretic, we have

$$\operatorname{Eq}(\pi_1, \pi_2) = \{p \in W : \pi_1(p) = \pi_2(p)\} = W \setminus \{[(t)]\} = \operatorname{Spec} k[t]_t.$$

Then $\operatorname{Eq}(\pi_1, \pi_2)$ is open in W, and therefore X is not separated by Proposition 12.3.1. □

12.3.2 Reduced-to-Separated Theorem

Theorem 12.3.1 (Reduced-to-Separated Theorem)

Two S-morphisms $\pi : U \to Z$, $\pi' : U \to Z$ from a reduced scheme to a separated S-scheme agreeing on a dense open subset of U are the same, i.e., $\pi = \pi'$.

Proof Let V be the locus where π and π' agree. It is a closed subscheme of U (by Proposition 12.3.1), which contains a dense open subset of U. Hence the underlying set of V is entire U. Recall §10.4.4, the reduced subscheme structure on the V^{set} is the smallest closed subscheme whose underlying set contains V^{set}, since U is reduced, we must have $U = V$ as schemes. □

Example 12.6 (A counterexample if the reducedness hypothesis is removed in Theorem 12.3.1) Let $U = \operatorname{Spec} k[x, y]/(y^2, xy)$ (Figure 5.4, nonreduced) and $Z = \operatorname{Spec} k[t]$ (separated). Define $\pi_1 : U \to Z$ by setting ring map $t \mapsto x$, and define $\pi_2 : U \to Z$ by setting ring map $t \mapsto x + y$. Then $\pi_1 \neq \pi_2$. Let $O = U \setminus \{0\} = \operatorname{Spec}(k[x, y]/(y^2, xy))_x$, then O is dense in U. Note that $(k[x, y]/(y^2, xy))_x \cong k[x]_x$, $\pi_1|_O = \pi_2|_O$.

Corollary 12.3.1

Let X be a reduced scheme and Y be a separated scheme. If we have two morphisms from open subsets of X to Y, say $\pi : U \to Y$ and $\pi' : V \to Y$, and they agree on a dense open subset Z of $U \cap V$, then they necessarily agree on $U \cap V$.

Corollary 12.3.2

Let U be a reduced scheme and Y be a separated scheme. A rational map $U \dashrightarrow Y$ has a ***largest domain of definition*** *on which $\pi : U \dashrightarrow Y$ is a morphism, which is the union of all the domains of definition. In particular, a rational function (Definition 8.5.3) on a reduced scheme has a largest domain of definition.*

Note *From now on, we call the largest domain of definition in Corollary 12.3.1 as the domain of definition of rational map.*

Example 12.7 For example, the domain of definition of $\mathbb{A}^2_k \dashrightarrow \mathbb{P}^1_k$ given by $(x, y) \mapsto [x, y]$ has domain of definition $\mathbb{A}^2_k \setminus \{(0, 0)\}$ (cf. Example 8.8).

Corollary 12.3.2 partially extends the definition of the domain of a rational function on a locally Noetherian scheme (Definition 7.6.10).

Definition 12.3.1 (Indeterminacy locus)

Let X be a reduced scheme and Y be a separated scheme, $X \dashrightarrow Y$ be a rational map. The complement of the largest domain of definition is called the ***indeterminacy locus****.*

Remark 12.11 We will see in Chapter 23 that a rational map to a projective scheme can be upgraded to an honest morphism by "blow up" a scheme-theoretic version of the locus of indeterminacy.

Example 12.8 The Reduced-to-Separated Theorem 12.3.1 is false if we give up reducedness of source or separatedness of the target. For the first, this is Example 12.6 . For the second, consider the two maps from Spec $k[t]$ to the line with the double origin, on of which maps to the "upper half", and one of which maps to the "lower half". These two morphisms agree on the dense open set $D(t)$, but not agree on the entire affine line, see Figure 12.4.

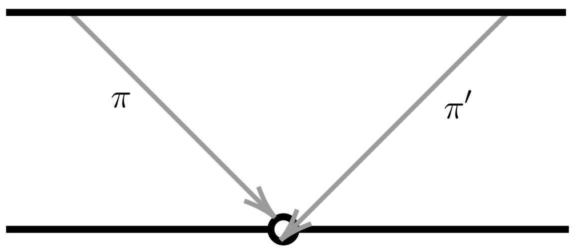


Figure 12.4: Two different maps to a nonseparated scheme agreeing on a dense open set

12.3.3 Graphs of rational maps

Graphs of morphisms were defined in Definition 12.2.3.

Definition 12.3.2 (Graph of a rational map)

If X is reduced and Y is separated, define the ***graph*** Γ_π ***of a rational map*** $\pi : X \dashrightarrow Y$ *as follows. Let (U, π') be any representative of this rational map (so $\pi' : U \to Y$ is a morphism). Let Γ_π be the scheme-theoretic closure of $\Gamma_{\pi'} \hookrightarrow U \times Y \hookrightarrow X \times Y$ ($\Gamma_{\pi'}$ is the image of $\gamma_{\pi'} : U \to U \times Y$), where the first map is a closed embedding (Proposition 12.2.11), and the second is an open embedding. The product here should be taken in the category you are working in. For example, if you are working with k-schemes, the fibered product should be taken over k.*

Proposition 12.3.4 (Definition 12.3.2 is well-defined)

The graph of a rational map $\pi : X \dashrightarrow Y$ is independent of the choice of representative of π.

Remark 12.12 **Notation:** In the proof of Proposition 12.3.4, we denote $\mathrm{cl}^{\mathrm{set}}(\square)$ be the topological closure in topological space $(X \times Y)^{\mathrm{set}}$ and denote $\mathrm{cl}^{\mathrm{sch}}(\square)$ be the scheme-theoretic closure in scheme $X \times Y$.

Proof Suppose $\xi' : U \to Y$ and $\xi : V \to Y$ are two representatives of π. Then there exists a dense open subset $W \subseteq U \cap V$ such that $\xi|_W = \xi'|_W$. If we will proved $\mathrm{cl}^{\mathrm{sch}}(\Gamma_{\xi|_W}) = \mathrm{cl}^{\mathrm{sch}}(\Gamma_\xi)$ and $\mathrm{cl}^{\mathrm{sch}}(\Gamma_{\xi'|_W}) = \mathrm{cl}^{\mathrm{sch}}(\Gamma_{\xi'})$, then we have $\mathrm{cl}^{\mathrm{sch}}(\Gamma_\xi) = \mathrm{cl}^{\mathrm{sch}}(\Gamma_{\xi'})$. Hence it suffices to show that $\mathrm{cl}^{\mathrm{sch}}(\Gamma_{\xi|_W}) = \mathrm{cl}^{\mathrm{sch}}(\Gamma_\xi)$ (similarly, $\mathrm{cl}^{\mathrm{sch}}(\Gamma_{\xi'|_W}) = \mathrm{cl}^{\mathrm{sch}}(\Gamma_{\xi'})$).

We may reduce to the case where V is the (largest) domain of π, and U is a dense open subset of V, then

$\xi' = \xi|_U$. We want to show that $\mathrm{cl}^{\mathrm{sch}}(\Gamma_{\xi|_U}) = \mathrm{cl}^{\mathrm{sch}}(\Gamma_\xi)$. Recall Definition 12.2.3 (or Figure 12.2), we have $\mathrm{pr}_1|_{\Gamma_\xi} \circ \gamma_\xi = \mathrm{id}_V$, where $\gamma_\xi : V \to \Gamma_\xi \subseteq V \times Y$ and $\mathrm{pr}_1 : \Gamma_\xi \to V$. Hence $\gamma_\xi : V \xrightarrow{\sim} \Gamma_\xi$ is an isomorphism. Similarly, we have $\gamma_{\xi|_U} : U \xrightarrow{\sim} \Gamma_{\xi|_U}$. Since $U \subseteq V$ is dense open subset, $\Gamma_{\xi|_U} \subseteq \Gamma_\xi$ is dense open. Hence as topological space, we have $\mathrm{cl}^{\mathrm{set}}(\Gamma_{\xi|_U}) = \mathrm{cl}^{\mathrm{set}}(\Gamma_{\xi|_U})$. Note that $\mathrm{cl}^{\mathrm{sch}}(\Gamma_{\xi|_U})$ is the smallest closed subscheme which contains $\mathrm{cl}^{\mathrm{set}}(\Gamma_{\xi|_U})$, recall Definition 10.4.5, $\mathrm{cl}^{\mathrm{sch}}(\Gamma_{\xi|_U})$ is the reduced subscheme structure on closed subset $\mathrm{cl}^{\mathrm{set}}(\Gamma_{\xi|_U})$. Similarly, $\mathrm{cl}^{\mathrm{sch}}(\Gamma_\xi)$ is the reduced subscheme structure on closed subset $\mathrm{cl}^{\mathrm{set}}(\Gamma_\xi)$. Since $\mathrm{cl}^{\mathrm{set}}(\Gamma_{\xi|_U}) = \mathrm{cl}^{\mathrm{set}}(\Gamma_{\xi|_U})$, by Proposition 10.4.6, we have $\mathrm{cl}^{\mathrm{sch}}(\Gamma_{\xi|_U}) = \mathrm{cl}^{\mathrm{sch}}(\Gamma_{\xi|_U})$, as we desired. □

Remark 12.13 The separatedness of Y is not necessary.

In analogy with graphs of morphisms, the following diagram of a graph of a rational map π can be useful (suppose X, Y are S-schemes, cf. Figure 12.2).

$$\begin{tikzcd}
U \arrow[d, "\gamma_\pi"'] \arrow[ddrrrr, "\pi"] & \subseteq & X \arrow[ddrr, dashed, "\pi"] & & \\
U \times_S Y \arrow[d] \arrow[drr, hook, "\text{op.emb.}"] & & & & \\
\Gamma_\pi \arrow[rr, hook, "\text{cl.emb.}"'] & & X \times_S Y \arrow[rr] \arrow[d] & & Y \arrow[d] \\
 & & X \arrow[rr] & & S
\end{tikzcd}$$

For simply,

$$\begin{tikzcd}
\Gamma_\pi \arrow[r, hook, "\text{cl.emb}"] & X \times_S Y \arrow[dl, "\mathrm{pr}_1"] \arrow[dr, "\mathrm{pr}_2"] & \\
X \arrow[u, dashed, "\gamma_\pi"] \arrow[rr, dashed, "\pi"] & & Y.
\end{tikzcd}$$

Exercise 12.2 (The Blow-up of the affine plane as the graph of a rational map.) Consider the rational map $\mathbb{A}^2_k \dashrightarrow \mathbb{P}^1_k$ given by $(x, y) \mapsto [x : y]$. Show that this rational map cannot be extended over the origin. (A similar argument arises in Definition 8.5.8 on the Cremona transformation.) Show that the graph of the rational map is the morphism (the blow-up) described in Exercise 11.7. (When we define blow-ups in general, we will see that they are often graphs of rational maps, see Chapter 23.)

Proof Say $\pi : \mathbb{A}^2_k \dashrightarrow \mathbb{P}^1_k$, then $\pi : \mathbb{A}^2_k \setminus \{(0,0)\} \to \mathbb{P}^1_k$ is a ture morphism. Suppose $F : \mathbb{A}^2_k \to \mathbb{P}^1_k$ be a morphism such that $F|_{\mathbb{A}^2_k \setminus \{(0,0)\}} = \pi$. In fact, $\mathbb{P}^1_k = \operatorname{Proj} k[s,t] = D_+(s) \cup D_+(t)$. Say $W_0 = F^{-1}(D_+(s))$ and $W_1 = F^{-1}(D_+(t))$, then $\mathbb{A}^2_k = W_0 \cup W_1$. Since $0 = [(x,y)] \in \mathbb{A}^2_k$ (say $\mathbb{A}^2_k = \operatorname{Spec} k[x,y]$), without loss of generality, we may assume $0 \in W_0$. Consider $F|_{W_0} : W_0 \to D_+(s)$, assume $F|_{W_0}$ is given by ring map

$$F|_{W_0}^\sharp : k[t/s] \longrightarrow \mathscr{O}_{\mathbb{A}^2_k}(W_0)$$

$$t/s \longmapsto g$$

where $g \in \mathscr{O}_{\mathbb{A}^2_k}(W_0)$. Since $\pi = F|_{\mathbb{A}^2_k \setminus \{(0,0)\}}$, note that $\pi|^\sharp_{D(x) \cap W_0}$ is given by $\frac{t}{s} \mapsto \frac{y}{x}$, we have $g = \frac{y}{x}$ on $D(x) \cap W_0$. Hence $xg - y = 0$ on $D(x) \cap W_0$, note that $D(x) \cap W_0$ is dense in W_0 and W_0 is reduced, by the Reduced-to-Separated Theorem 12.3.1 ($xg - y$ can be seen as a morphism form W_0 to $\mathbb{A}^1_k$, see Exercise 8.7 and Definition 8.5.3), we know that $xg - y = 0$ on W_0, i.e., $g = \frac{y}{x}$, this is impossible since $0 \in W_0$. So π cannot be extended over the origin.

We next show that $\Gamma_\pi = \mathrm{Bl}_{(0,0)} \mathbb{A}^2_k$. Let $\xi : \mathbb{A}^2_k \setminus \{(0,0)\} \to \mathbb{P}^1_k$ be the representative of π. Consider $\gamma_\xi : \mathbb{A}^2_k \setminus \{(0,0)\} \to (\mathbb{A}^2_k \setminus \{(0,0)\}) \times_k \mathbb{P}^1_k$. The underlying set of Γ_ξ is

$$\Gamma_\xi = \{((a,b), [c:d]) \in (\mathbb{A}^2_k \setminus \{(0,0)\}) \times_k \mathbb{P}^1_k : ad = bc\}.$$

Hence $\Gamma_\pi = \mathrm{cl}^{\mathrm{sch}}(\Gamma_\xi)$ is a closed subscheme of $\mathbb{A}^2_k \times_k \mathbb{P}^1_k$ which cut out by the single equation $xt = ys$, this is the definition of $\mathrm{Bl}_{(0,0)} \mathbb{A}^2_k$, so $\Gamma_\pi = \mathrm{Bl}_{(0,0)} \mathbb{A}^2_k$. □

12.3.4 Variations

Variations of the short proof of Reduced-to-Separated Theorem 12.3.1 yield other useful results. Proposition 12.3.2 is one example. Here is another.

Proposition 12.3.5

Maps to a separated scheme can be extended over an effective Cartier divisor in at most one way: *Suppose X is a Z-scheme (not necessarily reduced!), and Y is a separated Z-scheme. Suppose D is an effective Cartier divisor on X. Then any Z-morphism $X \setminus D \to Y$ can be extended in at most one way to a Z-morphism $X \to Y$.*

Proof Reduce to the case where $X = \operatorname{Spec} A$, since D is an effective Cartier divisor on X, we may assume that $D = V(t)$ where t is not zerodivisor of A. Then $X \setminus D = D(t)$. Let $\xi : D(t) \to Y$ be any Z-morphism. If there are two ways to extend ξ, say $\pi_1 : X \to Y$ and $\pi_2 : X \to Y$, then $\pi_1|_{D(t)} = \pi_{D(t)}$. The locus where π_1 and π_2 agree is the equalizer $E := \mathrm{Eq}(\pi_1, \pi_2)$, since Y is separated Z-scheme, by Proposition 12.3.1, E is a closed subscheme of X. Since π_1 and π_2 agree on $D(t)$, as set-theoretic, $D(t) \subseteq E$. Since E is a closed subscheme of X, we may assume that $E = \operatorname{Spec} A/I$, since $D(t)$ lie in E (recall Definition 10.4.1), functions vanishing on E pull back to the zero function on $D(t)$, hence the composition $I \hookrightarrow A \to A_t$ is zero. Let $f \in I$, then the image of f in A_t is 0, i.e., $f/1 = 0$, i.e., $t^n f = 0$ for some n. Since t is not zerodivisor of A, $f = 0$, and therefore $I = 0$ which implies that $E = \operatorname{Spec} A = X$, i.e., $\pi_1 = \pi_2$ □

As noted in remark of Definition 8.5.2, rational maps can be defined from any X that has associated points to any Y. The Reduced-to-Separated Theorem 12.3.1 can be extended to this setting, as follows.

Theorem 12.3.2 (Associated-to-Separated Theorem)

S-morphisms π and π' from a locally Noetherian scheme U to a separated S-scheme Z, agreeing on an open subset containing the associated points of U.

Proof By Proposition 12.3.1, $E := \mathrm{Eq}(\pi, \pi')$ is the locus of $\pi = \pi'$ and $E \hookrightarrow U$ is a closed embedding. Let $V \subseteq U$ be an open subset containing all associated points of U, by Proposition 7.6.28, $V \subseteq U$ is scheme-theoretically dense. Since $\pi|_V = \pi'|_V$, $V \subseteq E$, hence functions vanishing on E pull back to the zero function on V. Let W be any open subset of U. Pick $f \in \mathscr{I}_{E/U}(W)$, then $f|_{W \cap V} = 0$, since V is scheme-theoretically dense, recall Definition 7.6.9, $f = 0$ in $\mathscr{I}_{E/U}(W)$. Hence $\mathscr{I}_{E/U} = 0$, which implies that $E = U$. Hence $\pi = \pi'$. □

12.4 Proper morphisms

12.4.1 Properness

A map of topological spaces (also know as a continuous map!) is said to be **proper** if the preimage of any compact set is compact. **Properness of morphisms** is an analogous property. For example, a variety over $\mathbb{C}$ will be proper if it is compact in the classical topology (see [Ser56, §7]). Alternatively, we will see that projective A-schemes are proper over A — so this is a nice property satisfied by projective schemes, which also is convenient to work with.

Recall (§9.3.3) that *a (continuous) map of topological spaces $\pi : X \to Y$ is closed if for each closed subset $S \subseteq X$, $\pi(S)$ is also closed.* A morphism of schemes is closed if the underlying continuous map is closed.

Definition 12.4.1 (Universally closed)

*We say that a morphism of schemes $\pi : X \to Y$ is **universally closed** if for every morphism $Z \to Y$ the induced morphism $Z \times_Y X \to Z$ is closed. In other words, a morphism is universally closed if it remains closed under any base change.*

By now you will automatically expect the following proposition.

Proposition 12.4.1

The class of universally closed morphisms is "reasonable" in the sense of Definition 9.1.1.

Proof We first show that universally closed is preserved by composition. Suppose $\pi : X \to Y$ and $\rho : Y \to Z$ are two universally closed morphisms. Let $W \to Z$ be any morphisms, we want to show that $W \times_Z X \to W$ is closed. Consider the following Cartesian diagrams.

$$\begin{tikzcd} X \times_Z W \arrow[r] \arrow[d] & X \arrow[d, "\pi"] \\ Y \times_Z W \arrow[r] \arrow[d] & Y \arrow[d, "\rho"] \\ W \arrow[r] & Z \end{tikzcd}$$

Since π and ρ are universally closed, $X \times_Z W \to Y \times_Z W$ and $Y \times_Z W \to W$ are closed, the composition of closed morphisms is clearly closed, i.e., $X \times_Z W \to W$ is closed. Hence $\rho \circ \pi$ is universally closed.

We next show that universally closed morphism is preserved by base change. Let $\pi : X \to Y$ be a universally closed morphism, and let $Z \to Y$ be any morphism, we want to show that $X \times_Y Z \to Z$ is also universally closed. Let $W \to Z$ be any morphism. Consider the following diagram.

$$\begin{tikzcd} X \times_Y W \arrow[r] \arrow[d] & X \times_Y Z \arrow[r] \arrow[d] & X \arrow[d] \\ W \arrow[r] & Z \arrow[r] & Y \end{tikzcd}$$

Since $\pi : X \to Y$ is universally closed, we have $X \times_Z W \to W$ is closed, which implies $X \times_Y Z \to Z$ is also universally closed.

Lastly, we show that universally closed morphisms are local on the target. Let $\pi : X \to Y$ be a universally closed morphism, $V \subseteq Y$ be any open subset of X, we want to show that $\pi|_{\pi^{-1}(V)} : \pi^{-1}(V) \to V$ is universally closed morphism. Recall Proposition 9.1.4, we have $V \times_Y X \cong \pi^{-1}(V)$. Let $Z \to V$ be any morphism. Consider the following Cartesian squares.

$$\begin{tikzcd} \pi^{-1}(V) \times_V Z \arrow[r] \arrow[d] & \pi^{-1}(V) \arrow[r, hook] \arrow[d] & X \arrow[d] \\ Z \arrow[r] & V \arrow[r, hook] & Y \end{tikzcd}$$

Since π is universally closed, note that $\pi^{-1}(V) \times_V Z \cong X \times_Y Z$, so we have $\pi^{-1}(V) \times_V Z \to Z$ is closed, which implies that $\pi|_{\pi^{-1}(V)}$ is universally closed morphism.

Now let $\pi : X \to Y$ be a morphism, suppose there is an open cover $\{V_i\}$ of Y for which each restricted morphism $\pi|_{\pi^{-1}(V_i)} : \pi^{-1}(V_i) \to V_i$ is a universally closed morphism, we want to show that π is a universally closed morphism. Let $\rho : Z \to Y$ be any morphism, it suffices to check that $\widetilde{\pi} : X \times_Y Z \to Z$ is closed. Let $S \subseteq X \times_Y Z$ be a closed subset. Note that $X \times_Y Z = \bigcup_i (\pi^{-1}(V_i) \times_{V_i} \rho^{-1}(V_i))$, say $U_i = \pi^{-1}(V_i) \times_{V_i} \rho^{-1}(V_i)$, it suffices to show that $\widetilde{\pi}|_{U_i}(S \cap U_i)$ is closed in $\rho^{-1}(V_i)$. Since $\pi|_{\pi^{-1}(V_i)}$ is universally closed, $\widetilde{\pi}|_{U_i}$ is closed, note that $S \cap U_i$ is closed in U_i, we have $\widetilde{\pi}|_{U_i}(S \cap U_i)$ is closed in $\rho^{-1}(V_i)$. It follows that $X \times_Y Z \to Z$ is closed (recall the fact that a set of topological space is closed iff there exists an open cover and set is closed in each open subset). Hence universally closed morphisms are local on the target. □

To motivate the definition of properness for schemes, we remark that a continuous map $\pi : X \to Y$ of locally compact Hausdorff spaces that have countable bases for their topologies is universally closed if and only if it is proper (i.e., preimages of compact subsets are compact).

Definition 12.4.2 (Properness)

*(1) A **morphism** $\pi : X \to Y$ **is proper** if it is separated, finite type, and universally closed.*

*(2) A S-**scheme** X **is proper** if structure morphism $X \to S$ is proper.*

*(3) If A is a ring, an A-scheme is **proper over** A if it is proper over* Spec A.

Remark 12.14 A k-scheme is often said to be **complete** if it is proper. We will not use this terminology.

Let's try this idea out in practice. We expect that $\mathbb{A}^1_{\mathbb{C}} \to \operatorname{Spec} \mathbb{C}$ is not proper, because the complex manifold corresponding to $\mathbb{A}^1_{\mathbb{C}}$ is not compact. However, note that this map is separated (it is a map of affine schemes), finite type, and (trivially) closed. So the "universally" is what matters here.

Exercise 12.3 Show that $\mathbb{A}^1_{\mathbb{C}} \to \operatorname{Spec} \mathbb{C}$ is a closed map, but not universally closed. Hence π is not proper.

Proof Since $\mathbb{C}$ is a field, $\operatorname{Spec} \mathbb{C}$ only has single point, this point is closed, so $\mathbb{A}^1_{\mathbb{C}} \to \operatorname{Spec} \mathbb{C}$ is closed.

We next show that $\mathrm{pr} : \mathbb{A}^1_{\mathbb{C}} \times_{\mathbb{C}} \mathbb{P}^1_{\mathbb{C}} \to \mathbb{P}^1_{\mathbb{C}}$ is not closed. Consider

$$S = \{(t, [x : y]) \in \mathbb{A}^1_{\mathbb{C}} \times_{\mathbb{C}} \mathbb{P}^1_{\mathbb{C}} : tx - y = 0\},$$

then S is a well-defined subset of $\mathbb{A}^1_{\mathbb{C}} \times_{\mathbb{C}} \mathbb{P}^1_{\mathbb{C}}$. Say $\mathbb{A}^1_{\mathbb{C}} \times_{\mathbb{C}} \mathbb{P}^1_{\mathbb{C}} = \operatorname{Spec} \mathbb{C}[t] \times_{\mathbb{C}} \operatorname{Proj} \mathbb{C}[x, y]$. Then we have $\mathbb{A}^1_{\mathbb{C}} \times_{\mathbb{C}} \mathbb{P}^1_{\mathbb{C}} = \operatorname{Spec} \mathbb{C}[t, \frac{x}{y}] \cup \operatorname{Spec} \mathbb{C}[t, \frac{y}{x}]$. Note that S is cut out by $tx - y$, so S is a closed subscheme of $\mathbb{A}^1_{\mathbb{C}} \times_{\mathbb{C}} \mathbb{P}^1_{\mathbb{C}}$. Consider $\mathrm{pr}(S)$, we have

$$\mathrm{pr}(S) = \{[x : y] \in \mathbb{P}^1_{\mathbb{C}} : \exists t \in \mathbb{C}, s.t., tx - y = 0\}.$$

Clearly, $D_+(x) \subseteq \mathrm{pr}(S)$ and $V_+(x) \cap \mathrm{pr}(S) = \varnothing$, so $\mathrm{pr}(S) = D_+(x)$, it is an open set, which implies that $\mathrm{pr} : \mathbb{A}^1_{\mathbb{C}} \times_{\mathbb{C}} \mathbb{P}^1_{\mathbb{C}} \to \mathbb{P}^1_{\mathbb{C}}$ is not closed. Hence $\mathbb{A}^1_{\mathbb{C}} \to \operatorname{Spec} \mathbb{C}$ is not universally closed. □

As a first example of properness:

Proposition 12.4.2

Closed embeddings are proper.

Proof Closed embeddings are clearly separated, Proposition 12.2.5 (b). They are finite type, Proposition 10.1.2. After base change, they remain closed embedding, Proposition 11.2.1, and closed embeddings are always closed. □

This easily extends further as follows:

Proposition 12.4.3

Finite morphisms are proper.

Proof Finite morphisms are separated (as they are affine by definition, and affine morphisms are separated, Proposition 12.2.7), and finite type (basically because finite modules over a ring are automatically finitely generated, or see Proposition 9.3.20). To show that finite morphisms are closed after any base change, we note that they remain finite after any base change (finiteness is preserved by base change, Proposition 11.4.3), and finite morphisms are closed (Corollary 9.3.2). □

Proposition 12.4.4

(a) The notion of "proper morphism" is stable under base change.

(b) The notion of "proper morphism" is local on the target (i.e., $\pi : X \to Y$ is proper if and only if for any affine open cover $\{U_i \hookrightarrow Y\}_i$, $\pi^{-1}(U_i) \to U_i$ is proper).

(c) The notion of "proper morphism" is preserved by composition.

(d) The product of two proper morphisms is proper: if $\pi : X \to Y$ and $\pi' : X' \to Y'$ are proper, where all morphisms are morphisms of Z-schemes, then $\pi \times \pi' : X \times_Z X' \to Y \times_Z Y'$ is proper.

(e) Suppose

$$\begin{tikzcd} X \arrow[rr, "\pi"] \arrow[dr, "\tau"'] & & Y \arrow[dl, "\rho"] \\ & Z & \end{tikzcd} \tag{12.7}$$

is a commutative diagram, τ is proper, and ρ is separated. Then π is proper.

Proof Parts (a) through (c) say that proper morphisms form a "reasonable" class of morphisms in the sense of Definition 9.1.1. So they follow because separated morphisms, finite type morphisms and universal closed morphisms are all reasonable classes. Then (d) follows from (a) and (c) by Proposition 9.1.1. Since ρ is separated, δ_ρ is a closed embedding, by Proposition 12.4.2, δ_ρ is proper, by the Cancellation Theorem 12.1.1, $\pi : X \to Y$ is proper. □

A sample application of (e):

Corollary 12.4.1

A morphism (over $\operatorname{Spec} k$) from a proper k-scheme to a separated k-scheme is always proper.

Proposition 12.4.5 (Images of proper schemes are proper)

Suppose in the diagram

$$\begin{tikzcd} X \arrow[rr, "\pi"] \arrow[dr, "\tau"'] & & Y \arrow[dl, "\rho"] \\ & Z & \end{tikzcd}$$

that τ is proper and ρ is separated and finite type. Then the scheme-theoretic image of X under π is a proper Z-scheme.

(Thus properness in algebraic geometry behaves like properness in topology.)

Proof Let $\pi(X)$ be the scheme-theoretic image of X under π, consider the following commutative diagram,

$$\begin{tikzcd} X \arrow[r, "\pi"] \arrow[dr, "\tau"'] & \pi(X) \arrow[r, hook, "i", "\text{cl.emb.}"'] \arrow[d, dashed, "\rho\circ i"] & Y \arrow[dl, "\rho"] \\ & Z & \end{tikzcd} \tag{12.8}$$

where $i : \pi(Z) \hookrightarrow Y$ is a closed embedding. By Proposition 12.4.2, i is proper. Since ρ is separated and finite type, we know that $\rho \circ i$ is separated and of finite type. It suffices to show that $\rho \circ i$ is universally closed. Let

$Z' \to Z$ be any morphism. Base change diagram (12.8):

$$\begin{tikzcd}
& X \arrow[rr, "\pi"] \arrow[dl] \arrow[dd, "\text{prop.}"] & & \pi(X) \arrow[rr, hook, "\text{cl.emb.}"] \arrow[dl] \arrow[dd] & & Y \arrow[dl] \arrow[dd, "\text{sep.ft.}"] \\
X \times_Z Z' \arrow[rr, two heads] \arrow[dd, "\text{cl.}"] & & \pi(X) \times_Z Z' \arrow[rr, hook, "\text{cl.emb.}"] \arrow[dd, "\text{cl.?}"] & & Y \times_Z Z' \arrow[dd, "\text{sep.ft.}"] & \\
& Z \arrow[rr, equal] \arrow[dl] & & Z \arrow[rr, equal] \arrow[dl] & & Z \arrow[dl] \\
Z' \arrow[rr, equal] & & Z' \arrow[rr, equal] & & Z' &
\end{tikzcd}$$

Since π is universally closed (since π is proper, by Proposition 12.4.4), we know that $X \times_Z Z' \twoheadrightarrow \pi(X) \times_Z Z'$ is surjective. Since $\tau : X \to Z$ is universally closed (since τ is proper), $X \times_Z Z' \to Z'$ is closed. Note that $X \times_Z Z' \to Z'$ factor through $\pi(X) \times_Z Z'$ and $X \times_Z Z' \twoheadrightarrow \pi(X) \times_Z Z'$ is surjective, so $\pi(X) \times_Z Z' \to Z'$ must be closed. It follows that $\pi(X) \to Z$ is universally closed. Hence the structure morphism $\rho \circ i$ is proper, and therefore $\pi(X)$ is proper Z-scheme. □

We now come to the most important example of proper morphisms.

Theorem 12.4.1

Projective A-schemes (Definition 5.5.11) are proper over A.

Remark 12.15 As finite morphisms to $\operatorname{Spec} A$ are projective A-schemes, Proposition 9.3.12, Theorem 12.4.1 can be used to give a second proof that finite morphisms are proper, Proposition 12.4.3.

Proof The structure morphism of a projective A-scheme $X \to \operatorname{Spec} A$ factors as a closed embedding followed by $\mathbb{P}^n_A \to \operatorname{Spec} A$, Proposition 10.3.3. Closed embeddings are proper (Proposition 12.4.2), and compositions of proper morphisms are proper (Proposition 12.4.4), so it suffices to show that $\mathbb{P}^n_A \to \operatorname{Spec} A$ is proper. We have already seen that this morphism is finite type (Proposition 6.3.6) and separated (Proposition 12.2.13), so it suffices to show that $\mathbb{P}^n_A \to \operatorname{Spec} A$ is universally closed. As $\mathbb{P}^n_A = \mathbb{P}^n_{\mathbb{Z}} \times_{\mathbb{Z}} \operatorname{Spec} A$, it suffices to show that $\mathbb{P}^n_X := \mathbb{P}^n_{\mathbb{Z}} \times_{\mathbb{Z}} X \to X$ is closed for any scheme X. But the property of being closed is local on the target on X (Proposition 10.1.4), so by covering X with affine open subsets, it suffices to show that $\mathbb{P}^n_B \to \operatorname{Spec} B$ is closed for all rings B. This is the Fundamental Theorem of Elimination Theory 9.4.3. □

Remark 12.16 "Reasonable" proper schemes are projective. It is not easy to come up with an example of an A-scheme that is proper but not projective! Over a field, all proper curves are projective (Chapter 20), and all smooth surfaces are projective (see [Bad01, Theorem 1.28] for a proof of this theorem of Zariski; smoothness of course is not yet defined). We will meet a first example of a proper but not projective variety (a singular threefold) in Chapter 16. We will later see an example of a proper non-projective surface in Chapter 20, and a simpler one in Chapter 21. Once we know about flatness, we will see Hironaka's example of a proper non-projective irreducible smooth threefold over $\mathbb{C}$ (Chapter 25).

12.4.2 Functions on connected reduced proper $\overline{k}$-schemes must be constant.

As an enlightening application of these ideas, we show that if X is a connected reduced proper k-scheme where $k = \overline{k}$, then $\Gamma(X, \mathscr{O}_X) = k$. The analogous fact for connected compact complex manifolds uses the maximum principle. We saw this in the special case $X = \mathbb{P}^n$ in Proposition 5.4.2; it will be vastly generalized by Grothendieck's Coherence Theorem (Chapter 19).

Suppose $f \in \Gamma(X, \mathscr{O}_X)$ (f is a function on X). This is the same as a map $\pi : X \to \mathbb{A}_k^1$ (Proposition 8.3.7, discussed further in Proposition 8.6.1). Let π' be the composition of π with the open embedding $\mathbb{A}^1 \hookrightarrow \mathbb{P}^1$. By Corollary 12.4.1, π' is proper, and in particular, closed. As X is connected, the image of π' is also connected. Thus the set-theoretic image of π' must be either a closed point, or all of $\mathbb{P}^1$. But the set-theoretic image of π' lies in $\mathbb{A}^1$, so it must be a closed point p (which we identify with an element of k).

By Corollary 10.4.1, the support of the scheme-theoretic image of π is the closed point p (*support means the underlying set of scheme-theoretic image*). By Proposition 10.4.1, the scheme-theoretic image is precisely p (with the reduced structure). Thus π can be interpreted as the structure map to $\operatorname{Spec} k$, followed by a closed embedding to $\mathbb{A}^1$ identifying $\operatorname{Spec} k$ with p. So π is the map to $\mathbb{A}^1$ corresponding to the constant function $f = p$.

Example 12.9 **What are counterexamples if different hypotheses are relaxed?** The condition $k = \overline{k}$ cannot be ignored — if $k = \mathbb{R}$, then $X = \operatorname{Spec} \mathbb{C}$ is a connected reduced proper projective $\mathbb{R}$-scheme, with $\Gamma(X, \mathscr{O}_X) = \mathbb{R}^2$. Interesting fact: $x^2 = 0$ in $\mathbb{P}_{\mathbb{R}}^2$ shows that even nonreduced proper $\mathbb{R}$-schemes can have only constants as functions.

12.4.3 Facts (not yet proved) that may help you correctly think about finiteness

The following facts may shed some light on the notion of finiteness. We will prove them later.

A morphism of locally Noetherian schemes is finite if and only if it is proper and affine, if and only if it is proper and quasifinite (Chapter 29). We have proved parts of this statement, but we will only finish the proof once we know Zariski's Main Theorem; cf. Proposition 9.3.24.

As an application: quasifinite morphisms from proper schemes to separated schemes are finite. Here is why. Suppose $\pi : X \to Y$ is a quasifinite morphism over Z, where X is proper over Z. Then by the Cancellation Theorem 12.1.1 for proper morphisms, $X \to Y$ is proper. Hence as π is quasifinite and proper, π is finite.

As an explicit example, consider the map $\pi : \mathbb{P}_k^1 \to \mathbb{P}_k^1$ given by $[x, y] \mapsto [f(x, y) : g(x, y)]$, where f and g are homogeneous polynomials of the same degree with no common roots in $\mathbb{P}^1$. The fibers are finite, and π is proper (from the Cancellation Theorem 12.1.1 for proper morphisms, as Theorem 12.4.1), so π is finite. This could be checked directly as well, but now we can save ourselves the annoyance.

12.4.4 $\star\star$ Group Varieties

The rest of §12.4.4 is double-starred, and we will also throughout workin the category of *varieties over a field* k.

Part IV

“Geometric” properties of schemes

Chapter 13 Dimension

At this point, you know a fair bit about schemes, but there are some fundamental notions you cannot yet define. In particular, you cannot use the phrase "smooth surface", as it involves the notions of dimension and of smoothness. You may be surprised that we have gotten so far without using these ideas. You may also be disturbed to find that these notions can be subtle, but you should keep in mind that they are subtle in all parts of mathematics.

In this chapter, we will address the first notion, that of dimension of schemes. This should agree with, and generalize, our geometric intuition. Although we think of dimension as a basic notion in geometry, it is a slippery concept, as it is throughout mathematics. Even in linear algebra, the definition of dimension of a vector space is surprising the first time you see it, even though it quickly becomes second nature. The definition of dimension for manifolds is equally nontrivial. For example, how do we know that there isn't an isomorphism between some two-dimensional manifold and some three-dimensional manifold? Your answer will likely use topology, and hence you should not be surprised that the notion of dimension is often quite topological in nature.

A caution for those thinking over the complex numbers: our dimensions will be algebraic, and hence half that of the "real" picture. For example, we will see very shortly that $\mathbb{A}^1_{\mathbb{C}}$, which you may picture as the complex numbers (plus one generic point), has dimension 1.

13.1 Dimension and codimension

13.1.1 Dimension

Surprisingly, the right definition is purely topological — it depends just one the topological space, and not on the structure sheaf.

Definition 13.1.1 (Chain of irreducible closed subsets of topology space)

*Let X be a topological space. A **chain of irreducible closed subsets of** X is a strictly ascending sequence of irreducible closed subsets*

$$Z_0 \subsetneq Z_1 \subsetneq \cdots \subsetneq Z_n \subseteq X.$$

*The integer n is called **the length of the chain**.*

Definition 13.1.2 (Chain of prime ideal of a ring)

*Let A be a ring. A **chain of prime ideals of a ring** A is a finite strictly ascending sequence of prime ideals*

$$\mathfrak{p}_0 \subsetneq \mathfrak{p}_1 \subsetneq \cdots \subsetneq \mathfrak{p}_n.$$

*The **length of the chain** is n.*

Definition 13.1.3 (Krull Dimension)

*Let X be a topological space. We define the **Krull dimension of** X, which we denote by $\dim X$, to be the supremum of the lengths of the chain of irreducible closed subsets of X. This number is not necessarily finite. By convention, the empty set is of dimension $-\infty$.*

*The **Krull dimension** of the ring A is the supremum of the integers $n \geq 0$ such that A has a chain of prime ideals*

$$\mathfrak{p}_0 \subsetneq \mathfrak{p}_1 \subsetneq \cdots \subsetneq \mathfrak{p}_n.$$

of length n.

Remark 13.1

(1) An analogy from linear algebra: The dimension of a vector space is the supremum of the lengths of the chains of subspaces.

(2) A Noetherian ring has finite dimension because all chains of prime ideals are finite, but this isn't necessarily true — see Example **??**

Proposition 13.1.1

Let A be a ring. $\dim \operatorname{Spec} A = \dim A$.

Proof Suppose $\dim A = n$, then there exists a finite strictly ascending length n sequence of prime ideals

$$\mathfrak{p}_0 \subsetneq \mathfrak{p}_1 \subsetneq \cdots \subsetneq \mathfrak{p}_n.$$

Apply $V(\square)$, by Proposition 4.7.3, we have a strictly ascending length n sequence of irreducible closed subsets

$$V(\mathfrak{p}_n) \subsetneq V(\mathfrak{p}_{n-1}) \subsetneq \cdots \subsetneq V(\mathfrak{p}_1) \subsetneq V(\mathfrak{p}_0).$$

This length of chain must be maximum, if not, there exists irreducible closed subset V such that

$$V(\mathfrak{p}_n) \subsetneq \cdots \subsetneq V(\mathfrak{p}_1) \subsetneq V(\mathfrak{p}_0) \subsetneq V.$$

Apply $I(\square)$, by Proposition 4.7.3, we have chain of prime ideals

$$I(V) \subsetneq \mathfrak{p}_0 \subsetneq \cdots \subsetneq \mathfrak{p}_n.$$

This is impossible, since $\dim A = n$. Hence $\dim \operatorname{Spec} A = \dim A$. $\square$

Remark 13.2 The homeomorphism between $\operatorname{Spec} A$ and $\operatorname{Spec} A/\mathfrak{N}(A)$ implies that $\dim \operatorname{Spec} A = \dim \operatorname{Spec} A/\mathfrak{N}(A)$.

Example 13.1 We have identified all the prime ideals of $k[t]$ (they are 0, and $(f(t))$ for irreducible polynomials $f(t)$), $\mathbb{Z}$ ((0) and $(\mathfrak{p})$), k (only (0)), and $k[x]/(x^2)$ (only (x)), so we can quickly check that $\dim \mathbb{A}_k^1 = \dim \operatorname{Spec} \mathbb{Z} = 1$, $\dim \operatorname{Spec} k = 0$, $\dim \operatorname{Spec} k[x]/(x^2) = 0$.

We must careful with the notion of dimension for reducible spaces. If Z is the union of two closed subsets X and Y, then $\dim Z = \max\{\dim X, \dim Y\}$. Thus dimension is not a "local" characteristic of a space. This sometimes bothers us, so we try to only talk about dimensions of irreducible topological spaces.

Definition 13.1.4 (Pure dimensional (equidimensional))

*We say a topological space is **pure dimensional** or **equidimensional** (resp., **pure dimension** n or **equidimensional of dimension** n) if each of its irreducible components has the same dimension (resp., they are all of dimension n). A pure dimension 1 (resp., $2, \ldots, n$) topological space is said to be a **curve** (resp., **surface**, n-**fold**).*

Lemma 13.1.1 (Topological fact)

For any topological space X and open subset $U \subseteq X$, there is a bijection between irreducible closed subsets of U, and irreducible closed subsets of X that meet U.

Proof Let $Z \subseteq U$ be an irreducible closed subset of U. Consider the closure $\mathrm{cl}_X(Z) \subseteq X$, we claim that $\mathrm{cl}_X(Z)$ is an irreducible closed subset of X. Suppose $\mathrm{cl}_X(Z) = V_1 \cup V_2$, where each V_i closed in X. Then $Z = \mathrm{cl}_X(Z) \cap U = (V_1 \cap U) \cup (V_2 \cap U)$, since Z is irreducible, we may assume $Z = V_1 \cap U$. Hence $\mathrm{cl}_X(Z) = V_1$, which implies that $\mathrm{cl}_X(Z)$ is irreducible closed subset of X.

Conversely, let Z be an irreducible closed subset of X. Suppose $U \subseteq X$ open subset which meet Z, we want to show that $Z \cap U$ is irreducible. Suppose $Z \cap U = V_1 \cup V_2$ where each V_i is closed in U. In fact, we have $\mathrm{cl}_X(Z \cap U) = Z = \mathrm{cl}_X(V_1) \cup \mathrm{cl}_X(V_2)$, since Z is irreducible, we know that $\mathrm{cl}_X(V_1) = Z$ or $\mathrm{cl}_X(V_2) = Z$. We may assume $\mathrm{cl}_X(V_1) = Z$, then $V_1 = \mathrm{cl}_X(V_1) \cap U = Z \cap U$, which implies that $Z \cap U$ is irreducible closed subset of U. □

Proposition 13.1.2 (Useful)

A scheme has dimension n if and only if it admits an open cover by affine open subsets of dimension at most n, and at least one affine open subset has dimension exactly n.

Proof Suppose X be a scheme, and X is covered by $\{U_i \hookrightarrow X\}_i$.

If $\dim X = n$, then exists a length n chain of irreducible closed subsets, say

$$Z_0 \subsetneq Z_1 \subsetneq \cdots \subsetneq Z_n \subseteq X.$$

Since X is covered by $\{U_i\}_i$, there exists U_k such that $U_k \cap Z_0 \neq \varnothing$. By Lemma 13.1.1, we have an ascending sequence

$$U_k \cap Z_0 \subseteq U_k \cap Z_1 \subseteq \cdots \subseteq U_K \cap Z_n \subseteq U_k,$$

where each $U_k \cap Z_j$ is irreducible closed subset of U_k. We claim that $U_k \cap Z_{j-1} \subsetneq U_k \cap Z_j$ for all j. If not. Since $U_k \cap Z_j$ is dense in Z_j for each j, we have

$$\mathrm{cl}_X(U_k \cap Z_{j-1}) = \mathrm{cl}_X(U_k \cap Z_j) = Z_{j-1} = Z_j,$$

a contradiction! Hence we have a chain of irreducible closed subsets of U_k

$$U_k \cap Z_0 \subsetneq U_k \cap Z_1 \subsetneq \cdots \subsetneq U_K \cap Z_n \subseteq U_k.$$

It follows that $\dim U_k \geq n$. Clearly, $\dim U_i \leq n$ for all i, then we have $\dim U_k = n$.

If there exists $U_k \in \{U_i\}_i$ such that $\dim U_k = n$ and $\dim U_i \leq n$ for $i \neq k$, we want to show that $\dim X = n$. Clearly, $\dim X \geq n$. Pick any chain of irreducible closed subsets of X, this chain must meet some U_i, since $\dim U_i \leq n$, we have $\dim X \leq n$. So $\dim X = n$. □

Proposition 13.1.3

A Noetherian scheme of dimension 0 has a finite number of points.

Proof Let X be a Noetherian scheme with dimension 0. By Proposition 13.1.2, there exists $\{\operatorname{Spec} A_i \hookrightarrow X\}_{i=1}^n$ an open cover of X such that each A_i is Noetherian and $\dim \operatorname{Spec} A_i = 0$. By Proposition 13.1.1, $\dim A_i = 0$ for all i. By [AM18, Theorem 8.5], A_i is Artin, by [AM18, Proposition 8.3], each A_i has finite number of maximal ideals. Since $\dim A_i = 0$, the prime ideals of A_i are all maximal ideals, so $\operatorname{Spec} A_i$ is finite number of points, and therefore X is finite number of points. □

Proposition 13.1.4

A surjection of integral domains of the same finite dimension must be an isomorphism.

Proof Suppose A, B are integral domains with same dimension $\dim A = \dim B = n$. Suppose $\varphi : A \twoheadrightarrow B$ is a surjective. We want to show that φ is an isomorphism. it suffice to show that φ is injective, i.e., $\varphi^{-1}(0) = 0$. Since (0) is prime ideal of B and $\dim B = n$, each length n chain of prime ideals of B must contains (0), i.e., we have a finite strictly ascending sequence

$$(0) \subsetneq \mathfrak{q}_1 \subsetneq \cdots \subsetneq \mathfrak{q}_n \subsetneq B,$$

where each $\mathfrak{q}_i \in \operatorname{Spec} B$. Apply φ^{-1}, we have a chain of prime ideals of A,

$$\varphi^{-1}(0) \subsetneq \varphi^{-1}(\mathfrak{q}_1) \subsetneq \cdots \subsetneq \varphi^{-1}(\mathfrak{q}_n) \subsetneq A.$$

Note that A is integral domain, $(0) \in \operatorname{Spec} A$, so $(0) \subseteq \varphi^{-1}(0)$, since $\dim A = n$, we must have $(0) = \varphi^{-1}(0)$, it follows that φ is injective. Hence φ is an isomorphism. □

Proposition 13.1.5 (Fibers of integral morphisms)

Suppose $\pi : X \to Y$ is an integral morphism. Then every (nonempty) fiber of π has dimension 0.

Proof Let $p \in Y$ be a point, we want to show that $\pi^{-1}(p) = X \times_Y \operatorname{Spec} \kappa(p)$ has dimension 0. Recall Proposition 11.4.3, integral preserved by base change, so $\pi^{-1}(p) \to \operatorname{Spec} \kappa(p)$ is integral. Hence we may reduced to the show that if $\pi : X \to \operatorname{Spec} k$ is integral then $\dim X = 0$. Since π is integral, X must be affine, we may assume that $X = \operatorname{Spec} A$, then $\pi^\sharp : k \to A$ is an integral ring morphism, for simply, we may assume this is integral extension. Hence it suffices to show that $\dim A = 0$. Suppose $\mathfrak{p} \subseteq \mathfrak{m}$ are two prime ideals of A. Mod out by $\mathfrak{p}$, so we can assume that A is an integral domain. We claim that any nonzero element is invertible. Say $x \in A$, and $x \neq 0$. Since x is integral over k, there exists a minimal monic $p(X) \in k[X]$ such that

$$p(x) = x^n + c_{n-1}x^{n-1} + \cdots + c_1 x + c_0 = 0.$$

If $c_0 = 0$, then

$$x(x^{n-1} + c_{n-1}x^{n-2} + \cdots + c_1) = 0,$$

since A is integral domain,

$$x^{n-1} + c_{n-1}x^{n-2} + \cdots + c_1 = 0,$$

this contradicts to the minimalness of $p(X)$, so $c_0 \neq 0$. Hence, we have

$$x(x^{n-1} + c_{n-1}x^{n-2} + \cdots + c_1) = -c_0,$$

note that k is a field,

$$x\left(-\frac{1}{c_0}x^{n-1} - \frac{c_{n-1}}{c_0}x^{n-2} - \cdots - \frac{c_1}{c_0}\right) = 1,$$

which implies that x is invertible. Hence A is a field, clearly, $\dim A = 0$. □

Proposition 13.1.6 (Important)

If $\pi : \operatorname{Spec} A \to \operatorname{Spec} B$ corresponds to an integral extension of rings, then $\dim \operatorname{Spec} A = \dim \operatorname{Spec} B$.

Proof By hypothesis, $\pi^\sharp : B \to A$ is an integral extension. It suffices to show that $\dim A = \dim B$. Suppose $\dim B = n$, then a length n chain, say

$$\mathfrak{q}_0 \subsetneq \mathfrak{q}_1 \subsetneq \cdots \subsetneq \mathfrak{q}_n \subsetneq B,$$

where $\mathfrak{q}_i \in \operatorname{Spec} B$. By the Going-Up Theorem 9.2.3, there exists a sequence of prime ideals of A, say

$$\mathfrak{p}_0 \subseteq \mathfrak{p}_1 \subseteq \cdots \subseteq \mathfrak{q}_n \subseteq A,$$

where $\mathfrak{p}_i \cap B = \mathfrak{q}_i$. If there exists $\mathfrak{p}_i = \mathfrak{p}_{i+1}$ for some i, we have $\mathfrak{q}_i = \mathfrak{p}_{i+1}$, contradiction. So we have

$$\mathfrak{p}_0 \subseteq \mathfrak{p}_1 \subseteq \cdots \subseteq \mathfrak{q}_n \subseteq A.$$

Hence $\dim A \geq \dim B$. Suppose $\dim A = m$, then there exists a chain of prime ideals of A, say

$$\mathfrak{p}_0 \subsetneq \mathfrak{p}_1 \subsetneq \cdots \subsetneq \mathfrak{p}_m \subsetneq A,$$

where $\mathfrak{p}_i \in \operatorname{Spec} A$. Restrict above chain to B, we obtain a sequence of prime ideals of B,

$$(\pi^\sharp)^{-1}(\mathfrak{p}_0) \subseteq (\pi^\sharp)^{-1}(\mathfrak{p}_1) \subseteq \cdots \subseteq (\pi^\sharp)^{-1}(\mathfrak{p}_m) \subseteq B.$$

If there exists j such that $(\pi^\sharp)^{-1}(\mathfrak{p}_j) = (\pi^\sharp)^{-1}(\mathfrak{p}_{j+1}) = \mathfrak{q} \in \operatorname{Spec} B$. Since π is integral morphism, by Proposition 13.1.5, $\dim A \otimes_B (B_\mathfrak{q}/\mathfrak{q}B_\mathfrak{q}) = 0$. Note that

$$\begin{aligned} A \otimes_B (B_\mathfrak{q}/\mathfrak{q}B_\mathfrak{q}) &\cong A \otimes_B B_\mathfrak{q} \otimes_{B_\mathfrak{q}} (B_\mathfrak{q}/\mathfrak{q}B_\mathfrak{q}) \\ &\cong (A \otimes_B B_\mathfrak{q}) \otimes_{B_\mathfrak{q}} (B_\mathfrak{q}/\mathfrak{q}B_\mathfrak{q}) \\ &\cong (A \otimes_B B_\mathfrak{q})/(\mathfrak{q}B_\mathfrak{q}(A \otimes_B B_\mathfrak{q})) \\ &\cong A_\mathfrak{q}/\mathfrak{q}A_\mathfrak{q}, \end{aligned}$$

then $\dim A_\mathfrak{q}/\mathfrak{q}A_\mathfrak{q} = 0$. Note that the prime ideal of $A_\mathfrak{q}/\mathfrak{q}A_\mathfrak{q}$ is one-to-one corresponding to the prime ideal of A which contains $\mathfrak{q}A$, since $\mathfrak{p}_j$ and $\mathfrak{p}_{j+1}$ are two distinct prime ideals which contain $\mathfrak{q}A$, say P_j, P_{j+1} are the corresponding prime ideals of $\mathfrak{p}_j, \mathfrak{p}_{j+1}$ in the ring $A_\mathfrak{q}/\mathfrak{q}A_\mathfrak{q}$, from $\mathfrak{p}_j \subsetneq \mathfrak{p}_{j+1}$, we get a length 1 chain $P_j \subsetneq P_{j+1}$ in $A_\mathfrak{q}/\mathfrak{q}A_\mathfrak{q}$, a contradiction! So for all i we have $(\pi^\sharp)^{-1}(\mathfrak{p}_i) \subsetneq (\pi^\sharp)^{-1}(\mathfrak{p}_{i+1})$. Hence $\dim B \geq m = \dim A$, and therefore $\dim A = \dim B$. □

Proposition 13.1.7

If $v : \widetilde{X} \to X$ is the normalization of a scheme (possibly in a finite extension of the function field of X), then $\dim \widetilde{X} = \dim X$.

Proof Let X be an integral scheme, by Proposition 13.1.2, we may assume $X = \operatorname{Spec} A$, where A is integral domain. Let $K(A)$ is the fraction field of A, suppose $L/K(A)$ is an algebraic extension. Then, by Lemma 11.7.1, $\widetilde{X} = \operatorname{Spec} \operatorname{Int.cl}_L(A)$, where $\operatorname{Int.cl}_L(A)$ is integral closure of A in L. Note that $A \to \operatorname{Int.cl}_L(A)$ is an integral extension, by Proposition 13.1.6, we have $\dim \widetilde{X} = \dim X$. □

Definition 13.1.5 (Flat)

Let R be a ring.

(1) *An R-module M is called **flat** if whenever $N_1 \to N_2 \to N_3$ is an exact sequence of R-modules the sequence $M \otimes_R N_1 \to M \otimes_R N_2 \to M \otimes_R N_3$ is exact as well.*

(2) *An R-module M is called faithfully flat if the complex of R-modules $N_1 \to N_2 \to N_3$ is exact if and only if the sequence $M \otimes_R N_1 \to M \otimes_R N_2 \to M \otimes_R N_3$ is exact.*

(3) *A ring map $R \to S$ is called flat if S is flat as an R-module.*

(4) *A ring map $R \to S$ is called faithfully flat if S is faithfully flat as an R-module.*

Lemma 13.1.2 (Going down)

Let $\varphi : A \to B$ be a flat homomorphism of rings. Then

(GD1) for any $\mathfrak{p}, \mathfrak{p}' \in \operatorname{Spec} A$ such that $\mathfrak{p} \subseteq \mathfrak{p}'$, and for any $\mathfrak{q}' \in \operatorname{Spec} B$ lying over $\mathfrak{p}'$, there exists $\mathfrak{q} \in \operatorname{Spec} B$ lying over $\mathfrak{p}$ such that $\mathfrak{q} \subseteq \mathfrak{q}'$.

(GD2) for any $\mathfrak{p} \in \operatorname{Spec} A$, and for any minimal prime P which satisfies $P \supseteq \mathfrak{p}B$, we have $P \cap A = \mathfrak{p}$.

Moreover, (GD1) is equivalent to (GD2).

Proof Equivalence is given by [Mat70, (5.B)], and Going down is given by [Mat70, (5.D)]. □

Proposition 13.1.8

Suppose X is an affine k-scheme, and K/k is an algebraic field extension. Then X has pure dimension n if and only if $X_K := X \times_k K$ has pure dimension n.

Proof Since X is affine k-scheme, we may assume that $X = \operatorname{Spec} A$.

If X has pure dimension n, then for any minimal prime $\mathfrak{p}$ of A we have $\dim A/\mathfrak{p} = n$. Clearly $A \otimes_k K$ can be seen as a free A-module (K is a k-vector space, say $K = \operatorname{Span}_k\{e_i\}_i$, then $A \otimes_k K = \operatorname{Span}_A\{1 \otimes e_i\}_i$), so $A \otimes_k K$ is flat, and therefore $A \to A \otimes_k K$ is flat. Let $\mathfrak{q}$ be a minimal prime of $A \otimes_k K$, $\mathfrak{q} \cap A$ is a prime ideal of A, then there exists a minimal prime $\mathfrak{p}$ of A such that $\mathfrak{p} \subseteq \mathfrak{q} \cap A$. By Going Down Theorem 13.1.2, there exists $\mathfrak{q}_0 \in \operatorname{Spec} A \otimes_k K$ such that $\mathfrak{q}_0 \cap A = \mathfrak{p}$. Since $\mathfrak{q}$ is minimal prime, $\mathfrak{q} = \mathfrak{q}_0$, and therefore $\mathfrak{q} \cap A = \mathfrak{p}$. It follows that each irreducible component $\operatorname{Spec} A_K/\mathfrak{q}$ of X_K corresponds to an irreducible component $\operatorname{Spec} A/\mathfrak{p}$ of X. Hence we may reduce to the case where A is an integral domain. It suffices to show that $\dim A_K = n$. Let $\mathfrak{q}$ be a minimal ideal of A_K. Suppose $a \in \operatorname{Ker}(A \to A_K/\mathfrak{q})$, we want to show $a = 0$. Suppose $a \neq 0$. Since $a \otimes 1$ lies in $\mathfrak{p}$, and recall Proposition 7.6.15 $\mathfrak{q} \in \operatorname{Ass}_{A_K} A_K$, we know that $a \otimes 1$ is a zerodivisor. Recall A_K is a free A-module, consider ring homomorphism $\times a : A_K \to A_K$ given by $\sum_i a_i \otimes e_i \mapsto \sum_i aa_i \otimes e_i$. Since $\sum_i aa_i \otimes e_i = 0$ implies each $aa_i = 0$ ($\{1 \otimes e_i\}_i$ is linear independent), since $a \neq 0$ and A is an integral domain, we have $a_i = 0$ for all i, so $\times a$ is an injective, which implies that $a \otimes 1$ is not a zerodivisor, a contradiction! Hence $a = 0$, i.e., $A \to A_K/\mathfrak{q}$ is an injective. We next show that $A \hookrightarrow A_K$ is an integral extension. Since $A \otimes_k K$ is finitely generated A-module, by Corollary 9.2.1, $A \to A_K$ is integral. By [AM18, Proposition 5.6], $A/A \cap \mathfrak{q} = A \to A_K$ is integral extension (recall $A \to A_K/\mathfrak{q}$ is injective). By Proposition 13.1.6, $\dim A_K/\mathfrak{q} = \dim A = n$. Since each irreducible component of X_K corresponds to an irreducible component of X (we already proved), X_K has pure dimension n.

Conversely, if X_K is pure dimension n. Let $\mathfrak{p}$ is a minimal prime of A. Since $A \to A_K$ is integral extension, by Lying Over Theorem 9.2.1, there exists a prime ideal $\mathfrak{q} \in \operatorname{Spec} A_K$ such that $\mathfrak{q} \cap A = \mathfrak{p}$. There exists a minimal prime $\mathfrak{q}_0 \subseteq \mathfrak{q}$, restrict $\mathfrak{q}_0$ to A, we obtain a prime ideal $\mathfrak{q}_0 \cap A \subseteq \mathfrak{p}$, since $\mathfrak{p}$ is minimal prime, $\mathfrak{q}_0 \cap A = \mathfrak{p}$. By [AM18, Proposition 5.6], $A/\mathfrak{q}_0 \cap A = A/\mathfrak{p} \to A_K/\mathfrak{q}$ is an integral extension (injectivity is similar to our above discussion). By Proposition 13.1.6, we know that $\dim A/\mathfrak{p} = \dim A_K/\mathfrak{q} = n$ for all minimal prime $\mathfrak{p}$ of A. It follows that X has pure dimension n. □

Exercise 13.1 Show that $\dim \mathbb{Z}[x] = 2$.

Proof Clearly $\mathbb{Z}[x]$ is an integral domain, so we have a chain of prime ideals, $(0) \subsetneq (2) \subsetneq (2, x)$, so $\dim \mathbb{Z}[x] \geq 2$.

Suppose $\mathbb{Z}[x]$ has a length 3 chain of prime ideals, say $(0) \subsetneq \mathfrak{p}_1 \subsetneq \mathfrak{p}_2 \subsetneq \mathfrak{p}_3$. Then we get a sequence in $\operatorname{Spec} \mathbb{Z}$,

$$(0) \subseteq \mathfrak{p}_1 \cap \mathbb{Z} \subseteq \mathfrak{p}_2 \cap \mathbb{Z} \subseteq \mathfrak{p}_3 \cap \mathbb{Z}.$$

If $\mathfrak{p}_1 \cap \mathbb{Z} \neq (0)$, since $\dim \mathbb{Z} = 1$, we have $\mathfrak{p}_1 \cap \mathbb{Z} = \mathfrak{p}_2 \cap \mathbb{Z} = \mathfrak{p}_3 \cap \mathbb{Z} = (p)$ where $p \neq 0$ is a prime number. Consider the following chain of prime ideals in $\mathbb{F}_p[x]$,

$$\mathfrak{p}_1/(p) \subsetneq \mathfrak{p}_2/(p) \subsetneq \mathfrak{p}_2/(p),$$

since $\mathbb{F}_p[x]$ is PID, $\dim \mathbb{F}_p[x] = 1$, a contradiction.

If $\mathfrak{p}_1 \cap \mathbb{Z} = (0)$, since $\dim \mathbb{Z} = 1$, we have $\mathfrak{p}_2 \cap \mathbb{Z} = \mathfrak{p}_3 \cap \mathbb{Z} = (p)$ where $p \neq 0$ is a prime number. Then we get a chain in $\mathbb{F}_p[x]$, $(0) \subsetneq \mathfrak{p}_2/(p) \subsetneq \mathfrak{p}_3/(p)$, it contradicts to the fact that $\dim \mathbb{F}_2[x] = 1$.

Hence $\mathbb{Z}[x]$ does not have length 3 chain of prime ideals, so $\dim \mathbb{Z}[x] = 2$. □

13.1.2 Codimension

Because dimension behaves oddly for disjoint unions, we need some care when defining codimension, and in using the phrase. For example, if Y is a closed subset of X, we might define the codimension to be $\dim X - \dim Y$, but this behaves badly. For example, if X is the disjoint union of a point Y and a curve Z, then $\dim X - \dim Y = 1$, but this has nothing to do with the local behavior of X near Y. (Informally, Y is a point. Locally, Y looks like the entire space X, so the codimension of Y in X should be defined as 0. However, the presence of Z raises the dimension of Y, which is something we do not want to see.)

A better definition is as follows. In order to avoid excessive pathology, we define the codimension of Y in X only when Y is irreducible. (Use extreme caution when using this word in any other setting.)

Definition 13.1.6 (Codimension)

*(1) If $Z \subseteq X$ is an irreducible subset, then the **codimension** of Z, denoted by $\operatorname{codim}_X Z$, is defined as the supremum of the length of chains of irreducible closed subsets*

$$Z = Z_0 \subsetneq Z_1 \subsetneq \cdots \subsetneq Z_l.$$

(2) For any $Y \subseteq X$, define

$$\operatorname{codim}_X Y := \inf_{Z \subseteq Y} \operatorname{codim}_X Z$$

where the infimum is taken over all irreducible closed subset Z of X with $Z \subseteq Y$.

*(3) In particular, the **codimension of a point** is the codimension of its closure.*

Definition 13.1.7 (Codimension of an ideal (height))

*We say that a prime ideal $\mathfrak{p}$ in a ring has **codimension** equal to the supremum of the lengths of the chains of decreasing prime ideals starting at $\mathfrak{p}$, with indexing starting at 0. The codimension of $\mathfrak{p}$ is denoted by $\operatorname{ht}(\mathfrak{p})$. This notion is often called **height**.*

Remark 13.3 The codimension of the prime ideal $\mathfrak{p}$ in A is $\dim A_{\mathfrak{p}}$. Thus the codimension of $\mathfrak{p}$ in A is the codimension of $[\mathfrak{p}]$ in $\operatorname{Spec} A$.

Example 13.2 In an integral domain, the ideal (0) has codimension 0, and in $\mathbb{Z}$ the ideal (23) has codimension 1.

Remark 13.4 Continuing an analogy with linear algebra: the codimension of a vector subspace $Y \subseteq X$ is the supremum of the lengths of increasing chains of subspaces starting with Y. This is a better definition than $\dim X - \dim Y$, because it works even when $\dim X = \infty$. You might prefer to define $\operatorname{codim}_X Y$ as $\dim(X/Y)$; that is analogous to defining the codimension of $\mathfrak{p}$ in A as the dimension of $A_{\mathfrak{p}}$ — see the previous paragraph.

Proposition 13.1.9

If Y is an irreducible closed subset of a scheme X, and η is the generic point of Y, then

$$\operatorname{codim}_X Y = \dim \mathscr{O}_{X,\eta}.$$

Proof Suppose $\operatorname{codim}_X Y = n$. Given a chain $Y \subsetneq Y_1 \subsetneq \cdots \subsetneq Y_n$, the generic points $\eta_1, \ldots, \eta_n$ of the Y_i's will be contained in any affine open neighbourhood of η. For any affine neighbourhood U of η, we have a chain of irreducible closed subsets of U

$$Y \cap U \subsetneq Y_1 \cap U \subsetneq \cdots \subsetneq Y_n \cap U.$$

By Lemma 13.1.1, we have $\operatorname{codim}_X Y = \operatorname{codim}_X(Y \cap U)$ for any affine neighbourhood U of η. Therefore we may assume $X = \operatorname{Spec} A$, then $Y \subsetneq Y_1 \subsetneq \cdots \subsetneq Y_n$ gives a chain of prime ideals $\eta_n \subsetneq \cdots \subsetneq \eta_1 \subsetneq \eta$. Recall that $\operatorname{codim}_X(\operatorname{cl}_X(\eta)) = \operatorname{ht}(\eta) = \dim A_\eta$, we done. □

Notice that Y is codimension 0 in X if it is an irreducible component of X. Similarly, Y is codimension 1 if it is not an irreducible component, and for every irreducible component Y' which contains Y, there is no irreducible closed subset strictly between Y and Y'. (See Figure 13.1 for example.)

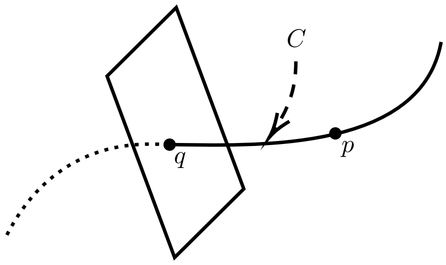


Figure 13.1: Behavior of codimension.

Definition 13.1.8 (Hypersurface)

*A closed subset all of whose irreducible components are codimension 1 in some ambient space X is said to be a **hypersurface** in X.*

Remark 13.5 **Be careful:** The word "hypersurface" can be used to mean slightly different things in algebraic geometry, for entirely reasonable reasons; see, for example, Definition 10.3.1.

Proposition 13.1.10

If Y is an irreducible closed subset of a scheme X, show that

$$\operatorname{codim}_X Y + \dim Y \leq \dim X. \tag{13.1}$$

Remark 13.6 We will soon see that equality always holds if X and Y are irreducible varieties (Theorem **??**), but equality doesn't hold in general (§**??**).

Proof Any chain of irreducible closed subsets of Y can be extended to a chain in X by inserting a chain between Y and X. Then we get

$$\operatorname{codim}_X Y + \dim Y \leq \dim X,$$

immediately. □

Remark 13.7 Warning. The notion of codimension still can behave slightly oddly. For example, consider Figure 13.1. (You should think of this as an intuitive sketch.) Here the total space X has dimension 2, but point p is dimension 0, and codimension 1. We also have an example of codimension 2 subset q contained in a codimension 0 subset C with no codimension 1 subset "in between".

Remark 13.8 Worse things can happen; we will soon see an example of a closed point in an irreducible surface that is nonetheless codimension 1, not 2, in §**??**. However, for irreducible varieties this can't happen, and inequality (14.15) must be an equality (Theorem **??**).

13.1.3 In UFDs, codimension 1 prime ideals are principal

We make a short observation, for future reference.

Lemma 13.1.3

In a UFD A, all codimension 1 prime ideals are principal.

Remark 13.9 This is a first glimpse of the fact that codimension 1 is rather special — this theme will continue in §13.3.2. We will see the converse of Lemma 13.1.3 holds as well (when A is a Noetherian integral domain, Proposition 13.3.7).

Proof Suppose $\mathfrak{p}$ is a codimension 1 prime. Choose any $f \neq 0$ in $\mathfrak{p}$, and let g be any irreducible factor of f that is in $\mathfrak{p}$ (there is at least one). Then (g) is a nonzero prime ideal contained in $\mathfrak{p}$, so $(0) \subsetneq (g) \subsetneq \mathfrak{p}$. As $\mathfrak{p}$ is codimension 1, we must have $\mathfrak{p} = (g)$, and thus $\mathfrak{p}$ is principal. □

13.1.4 Local rings of dimension 0

Dimension 0 objects aren't just points; they have further information, which might be thought of as some sort of infinitesimal "shape". We caught a glimpse of this earlier in §7.5 and Proposition 9.3.10.

Proposition 13.1.11 (Useful)

Suppose $(A, \mathfrak{m})$ is a Noetherian local ring. Then the following are equivalent.

(i) $\dim A = 0$*;*

(ii) $\mathfrak{m} = \mathfrak{N}(A)$*;*

(iii) for some $n > 1$, $\mathfrak{m}^n = 0$;

(iv) A has finite length;

(v) A satisfies d.c.c, i.e., every descending sequence of ideals of A is stationary.

Remark 13.10 Such a ring is called an **Artinian** or **finite length local ring** (Definition 7.5.5).

Proof (i) $\Rightarrow$ (ii). Since $\dim A = 0$, $\operatorname{ht}(\mathfrak{p}) = 0$ for all $\mathfrak{p} \in \operatorname{Spec} A$, it follows that every prime ideal of A is maximal. Since A is a local ring A has only one prime ideal, named $\mathfrak{m}$. Recall that $\mathfrak{N}(A)$ is the intersection of all prime ideals of A, we have $\mathfrak{N}(A) = \mathfrak{m}$.

(ii) $\Rightarrow$ (iii). Recall that $\mathfrak{N}(A) = \sqrt{(0)}$, we have $\mathfrak{m} = \sqrt{(0)}$. Since A is Noetherian, $\mathfrak{m}$ is finitely generated, we may assume $\mathfrak{m} = (f_1, \ldots, f_n)$. Since each $f_i \in \sqrt{(0)}$, we have $f_i^{n_i} = 0$ for some n_i. Let $n = \sum\limits_{i=1}^{n} n_i$, we have $f_i^n = 0$ for all i, so $\mathfrak{m}^n = 0$.

(iii) $\Rightarrow$ (iv). Since $\mathfrak{m}^n = 0$, we have a sequence of ideal

$$A := \mathfrak{m}^0 \supseteq \mathfrak{m}^1 \supseteq \mathfrak{m}^2 \supseteq \cdots \supseteq \mathfrak{m}^n = (0).$$

Say $M_i = \mathfrak{m}^i/\mathfrak{m}^{i+1}$, clearly, M_i is A-module. Note that $\mathfrak{m}M_i = 0$, M_i can be seen as an $A/\mathfrak{m}$-module, so it is $A/\mathfrak{m}$-vector space. Since $\mathfrak{m}$ is finitely generated, M_i is finite dimension vector space, so M_i has finite length. Note that

$$\ell(A) = \ell(A/\mathfrak{m}) + \ell(\mathfrak{m}) = \ell(A/\mathfrak{m}) + \ell(\mathfrak{m}/\mathfrak{m}^2) + \ell(\mathfrak{m}^2) = \cdots = \sum_{i=0}^{n-1} \ell(M_i),$$

since each $\ell(M_i) < \infty$, $\ell(A) < \infty$, i.e., A has finite length.

(iv) $\Rightarrow$ (v). This is Lemma 7.5.1.

(v) $\Rightarrow$ (i). It suffices to show that every prime ideal of A is maximal. Let $\mathfrak{p} \in \operatorname{Spec} A$, consider $A/\mathfrak{p}$. Let $x \in A/\mathfrak{p}$ nonzero, by d.c.c., $(x)^n = (x)^{n+1}$ for some n. Then there exists $y \in A/\mathfrak{p}$ such that $x^n = x^{n+1}y$, i.e., $x^n(1 - xy) = 0$. Since $A/\mathfrak{p}$ is integral domain and $x \neq 0$, $xy = 1$, which implies that x has inverse in $A/\mathfrak{p}$. Hence $A/\mathfrak{p}$ is a field, and therefore $\mathfrak{p}$ is maximal. □

13.1.5 A fun but unimportant counterexample

We end this introductory section with a fun pathology. As a Noetherian ring has no infinite chain of prime ideals, you may think that Noetherian rings must have finite dimension. Nagata, the master of counterexamples, show you otherwise with the following example.

Example 13.3 (An infinite-dimensional Noetherian Ring) Let $A = k[x_1, x_2, \dots]$. Choose an increasing sequence of positive integers $m_1, m_2, \dots$ whose differences are also increasing $m_{i+1} - m_i > m_i - m_{i-1}$. Let $\mathfrak{p}_i = (x_{m_i+1}, \dots, x_{m_{i+1}})$ and $S = A \setminus \bigcup_i \mathfrak{p}_i$.

- S is a multiplicative set.

  Clearly, $1 \notin \bigcup_i \mathfrak{p}_i$, so $1 \in S$. Let $a, b \in S$, we want to show that $ab \in S$. If $ab \notin S$, then $ab \in \bigcup_i \mathfrak{p}_i$, so $ab \in \mathfrak{p}_i$ for some i. Since $\mathfrak{p}_i$ is prime, $a \in \mathfrak{p}_i$ or $b \in \mathfrak{p}_i$. We may assume $a \in \mathfrak{p}_i$, then $a \notin S$, a contradiction. So S is a multiplicative set.
- $S^{-1}\mathfrak{p}_i$ in $S^{-1}A$ is the largest prime ideal in a chain of prime ideals of length $m_{i+1} - m_i$. Hence conclude that $\dim S^{-1}A = \infty$.

  Let $\mathfrak{q}$ be any prime ideal of $S^{-1}A$, then $\mathfrak{q}$ is one-to-one corresponding to $\mathfrak{p} \in \operatorname{Spec} A$ with $\mathfrak{p} \subseteq \bigcup_i \mathfrak{p}_i$. By [AM18, Proposition 1.11], $\mathfrak{p} \subseteq \mathfrak{p}_i$ for some i. Hence $S^{-1}\mathfrak{p}_i$ is maximal ideal of $S^{-1}A$. Consider $\mathfrak{p}_i = (x_{m_i+1}, \dots, x_{m_{i+1}})$, we get a composition series

  $$(0) \subsetneq (x_{m_i+1}) \subsetneq (x_{m_i+1}, x_{m_i+2}) \subsetneq \cdots \subsetneq \mathfrak{p}_i.$$

  Then we get a composition series of $S^{-1}\mathfrak{p}_i$,

  $$(0) \subsetneq S^{-1}(x_{m_i+1}) \subsetneq S^{-1}(x_{m_i+1}, x_{m_i+2}) \subsetneq \cdots \subsetneq S^{-1}\mathfrak{p}_i,$$

  hence $l(S^{-1}\mathfrak{p}_i) = m_{i+1} - m_i$. Note that $m_{i+1} - m_i > m_i - m_{i-1}$, we know that give any $N \in \mathbb{Z}$, there exists i such that $l(S^{-1}\mathfrak{p}_i) = m_{i+1} - m_i > N$, which implies that $\dim S^{-1}A = \infty$.
- Suppose B is a ring such that (i) for every maximal ideal $\mathfrak{m}$, $B_\mathfrak{m}$ is Noetherian, and (ii) every nonzero $b \in B$ is contained in finitely many maximal ideals. Then B is Noetherian.

  It suffices to show that any ideal of B is finitely generated. Let $I \subseteq B$ is a nonzero ideal. Now let's construct generators of I. Pick $x_1 \in I$, by (ii), x_1 belongs to finitely many maximal ideals, say $\{\mathfrak{m}_1, \dots, \mathfrak{m}_k\}$. Localize I at $\mathfrak{m}_j$, by (i) $I_{\mathfrak{m}_j}$ is Noetherian, and therefore finitely generated, say $I_{\mathfrak{m}_j} = (y_{j,1}, \dots, y_{j,r_j})$. We can clear the denominators of $y_{j,1}, \dots, y_{j,r_j}$, so that each $y_{j,1}, \dots, y_{j,r_j}$ can be viewed as elements of I. Let $J = (x_1, \{y_{j,1}, \dots, y_{j,r_j}\}_{j=1}^k)$, then $J \subseteq I$. Claim that $J = I$. It suffices to show that $J_\mathfrak{m} = I_\mathfrak{m}$ for any $\mathfrak{m} \in \operatorname{Max} B$.

  If $\mathfrak{m} \in \{\mathfrak{m}_1, \dots, \mathfrak{m}_k\}$, say $\mathfrak{m} = \mathfrak{m}_j$, since $I_{\mathfrak{m}_j} = (y_{j,1}, \dots, y_{j,r_j})$, we have $I_{\mathfrak{m}_j} \subseteq J_{\mathfrak{m}_j}$, note that $J_{\mathfrak{m}_j} \subseteq I_{\mathfrak{m}_j}$ which given by $J \subseteq I$, we have $I_{\mathfrak{m}_j} = I_{\mathfrak{m}_j}$. Hence $I_\mathfrak{m} = J_\mathfrak{m}$ for any $\mathfrak{m} \in \{\mathfrak{m}_1, \dots, \mathfrak{m}_k\}$.

  If $\mathfrak{m} \notin \{\mathfrak{m}_1, \dots, \mathfrak{m}_k\}$, then $x_1 \notin \mathfrak{m}$, so x_1 is a unit in $B_\mathfrak{m}$, note that $x_1 \in I_\mathfrak{m}, J_\mathfrak{m}$, hence $I_\mathfrak{m} = J_\mathfrak{m} = B_\mathfrak{m}$. Consequently, $I = J$, i.e., for any ideal I of B, I is finitely generated, hence B is Noetherian.
- $S^{-1}A$ is Noetherian.

  By our above discussion, it suffices to show that $S^{-1}A$ satisfies (i) and (ii) in previous paragraph.

  For (i). Recall that $S^{-1}\mathfrak{p}_i$ is maximal ideal of $S^{-1}A$. Localize $S^{-1}A$ at $S^{-1}\mathfrak{p}_i$, we have $(S^{-1}\mathfrak{p}_i)^{-1}S^{-1}A \cong A_{\mathfrak{p}_i}$. Let $V = \{x_1, x_2, \dots\} \setminus \{x_{m_i+1}, \dots, x_{m_{i+1}}\}$, and let $K := K(V)$ be the fraction field of $k[V]$. Note that $A = k[V][x_{m_i+1}, \dots, x_{m_{i+1}}]$, localize at $\mathfrak{p}_i$, we have

  $$A = (k[V][x_{m_i+1}, \dots, x_{m_{i+1}}])_{\mathfrak{p}_i} \cong K\left[\frac{x_{m_i+1}}{1}, \dots, \frac{x_{m_{i+1}}}{1}\right].$$

  K is a field, so it is Noetherian, by Hilbert's Base Theorem, $A_{\mathfrak{p}_i}$ is also Noetherian.

  For (ii). Let $a \in S^{-1}A$ be a nonzero element, we want to show that a contained in finitely many maximal ideals. Suppose $a = \frac{f}{g}$, where $f \in A$ and $g \in S$. Note that $a \in S^{-1}\mathfrak{p}_i$ iff $f \in \mathfrak{p}_i$, so it suffices to show that $f \in A$ belong to finitely many $\mathfrak{p}_i$. This is clear, since each element in A is generated by finite variables, so f must belong to finitely many $\mathfrak{p}_i$. Hence a contained in finitely many maximal ideals.

Remark 13.11 Noetherian local rings have finite dimension. However, we shall see in **??** that Noetherian local rings always have finite dimension. (This requires a surprisingly hard fact, Krull's Height Theorem **??**.) Thus points of locally Noetherian schemes always have finite codimension.

13.2 Dimension, transcendence degree, and Noether normalization

13.2.1 Transcendence degree = Dimension

We now give a powerful alternative interpretation for dimension for irreducible varieties, in terms of transcendence degree. The proof will involve a classical construction, Noether normalization, which will be useful in other ways as well. In case you haven't seen transcendence theory, here is a lightning introduction.

Recall that an element of a field extension E/F is an algebraic over F if it is integral over F. Recall also that a field extension E/F is an algebraic extension if it is an integral extension (if all elements of E are algebraic over F). The composition of two algebraic extensions is algebraic, by Proposition 9.2.3. If E/F is a field extension, and F' and F'' are two intermediate field extensions, then we write $F' \sim F''$ if $F'F''$ is algebraic over both F' and F''. Here $F'F''$ is the **compositum** of F' and F'', the smallest field extension in E containing F' and F''.

Lemma 13.2.1

$\sim$ is an equivalence relation on subextensions of E/F.

Proof Let E/F is a field extension. Let F', F'', F''' are intermediate field extensions, it is clearly $F' \sim F'$, i.e., reflexivity holds.

Suppose $F' \sim F''$, then $F'F''$ is algebraic over both F' and F'', hence $F'' \sim F'$. We proved symmetry.

We now show transitivity. Suppose $F' \sim F''$ and $F'' \sim F'''$, then $F'F''$ algebraic over F', F'' and $F''F'''$ algebraic over F'', F'''. We first show that $F'F'''$ is algebraic over F'. Consider the tower of field $F' \subseteq F'F'' \subseteq F'F''F'''$. Since $F''F'''/F''$ is algebraic, elements F''' are algebraic over F'', since F'' is a subfield of $F'F''$, we know that elements of F''' are algebraic over $F'F''$. Hence $F'F''F'''/F'F''$ is algebraic. Note that $F'F''/F'$ is algebraic, recall that the composition of algebraic extensions is algebraic, we know that $F'F''F'''/F'$ is algebraic. Since $F'F'''$ is a subfield of $F'F''F'''$, $F'F'''/F'$ is algebraic extension. We next show that $F'F'''$ is algebraic over F'''. Consider the tower of fields $F''' \subseteq F''F''' \subseteq F'F''F'''$. Since $F'F''/F''$ is algebraic, elements of F' are algebraic over F'', since F'' is a subfield of $F''F'''$, we know that elements of F' are algebraic over $F''F'''$, and therefore $F'F''F'''/F''F'''$ is algebraic. Hence $F'F''F'''/F'''$ is algebraic, since $F'F'''$ is subfield of $F'F''F'''$, $F'F'''/F'''$ is algebraic. Combine our above discussion, we have $F' \sim F'''$, i.e., transitivity holds. □

Definition 13.2.1 (Transcendence basis)

Let K/k be a field extensions.

*(1) A collection of elements $\{x_i\}_{i\in I}$ of K is called **algebraically independent** over k if the map*

$$k[X_i; i \in I] \longrightarrow K$$

which maps X_i to $x_i$is injective.

(2) The field of fraction of a polynomial ring $k[x_i; i \in I]$ is denoted $k(x_i; i \in I)$.

*(3) A field K is **purely transcendental** over a subfield k if K is isomorphic to a field of fractions of a polynomial ring over k.*

*(4) A **transcendence basis** of K/k is a collection of elements $\{x_i\}_{i\in I}$ which are algebraically independent over k and such that the extension $K/k(x_i; i \in I)$ is algebraic.*

Remark 13.12 We also can define the transcendence degree as follow (see [Rav22, Definition 12.2.A]): A transcendence basis of K/k is a set of elements $\{x_i\}$ that are algebraically independent over k such that $k(\{x_i\}) \sim K$. This definition is equivalent to Definition 13.2.1.

Definition 13.2.1 $\Rightarrow$ [Rav22, Definition 12.2.A]: If $K/k(\{x_i\})$ is algebraic, consider the compositum $k(\{x_i\})K$. Since $k(\{x_i\}) \subseteq K$, we have $k(\{x_i\})K = K$, hence $k(\{x_i\})K$ algebraic over K and $k(\{x_i\})$, i.e., $k(\{x_i\}) \sim K$.

[Rav22, Definition 12.2.A] $\Rightarrow$ Definition 13.2.1: If $k(\{x_i\}) \sim K$, then $k(\{x_i\})K = K$ algebraic over $k(\{x_i\})$, this is the Definition 13.2.1.

Lemma 13.2.2

Let K be a field extension of F, and let $S \subseteq K$ be algebraically independent over F. If $t \in K$ is transcendental over $F(S)$, then $S \cup \{t\}$ is algebraically independent over F.

Proof Suppose that the lemma is false. Then there is a nonzero polynomial $f \in F[x_1, \dots, x_n, y]$ with $f(s_1, \dots, s_n, t) = 0$ for some $s_i \in S$. This polynomial must involve y, since S is algebraically independent over F. Write $f = \sum_{j=0}^{m} g_j y^j$ with $g_j \in F[x_1, \dots, x_n]$. Since $g_m \neq 0$, the element t is algebraic over $F(S)$, a contradiction. Thus, $S \cup \{t\}$ is algebraically independent over F. □

We now prove the existence of a transcendence basis for any field extension.

Theorem 13.2.1

Let K be a field extension of F.

(1) There exists a transcendence basis for K/F.

(2) If $T \subseteq K$ such that $K/F(T)$ is algebraic, then T contains a transcendence basis for K/F.

(3) If $S \subseteq K$ is algebraically independent over F, then S is contained in a transcendence basis of K/F.

(4) If $S \subseteq T \subseteq K$ such that S is algebraically independent over F and $K/F(T)$ is algebraic, then there is a transcendence basis X for K/F with $S \subseteq X \subseteq T$.

Proof We first mention why statement (4) implies the first three statements. If statement (4) is true, then statements (2) and (3) are true by setting $S = \varnothing$ and $T = K$, respectively. Statement (1) follows from statement (4) by setting $S = \varnothing$ and $T = K$. To prove statement (4), let $\mathcal{S}$ be the set of all algebraically independent subsets of T that contain S. Then $\mathcal{S}$ is nonempty, since $S \in \mathcal{S}$. Ordering $\mathcal{S}$ by inclusion, a Zorn's Lemma argument shows that $\mathcal{S}$ contains a maximal element M. If K is not algebraic over $F(M)$, since K is algebraic over $F(T)$, we know that $F(T)$ is not algebraic over $F(M)$. Thus, there is a $t \in T$ with t transcendental over $F(M)$. But by Lemma 13.2.2, $M \cup \{t\}$ is algebraically independent over F and is a subset of T, contradicting maximality of M. Thus, K is algebraic over $F(M)$, so M is a transcendence basis of K/F contained in T. □

We now show that any two transcendence bases have the same size.

Lemma 13.2.3

Let K be field extension of F. If S and t are transcendence bases for K/F, then $|S| = |T|$.

Proof We first prove this in the case where $S = \{s_1, \dots, f_n\}$ is finite. Since S is a transcendence basis for K/F, the field K is not algebraic over $F(S \setminus \{s_1\})$. As K is algebraic over $F(T)$, some $t \in T$ must be transcendental over $F(S \setminus \{s_1\})$. Hence, by Lemma 13.2.2, $\{s_2, \dots, s_n, t\}$ is algebraically independent over F. Furthermore, s_1 is algebraic over $F(s_2, \dots, s_n, t)$, or else $\{s_1, \dots, s_n, t\}$ is algebraically independent, this is impossible (recall Definition 13.2.1). Since $K/F(s_1, \dots, s_n)$ is algebraic, we have $K/F(s_1, \dots, s_n, t)$ is algebraic, consider the tower of fields $F(s_2, \dots, s_n, t) \subseteq F(s_1, \dots, s_n, t) \subseteq K$, since $F(s_1, \dots, s_n, t)/F(s_2, \dots, s_n, t)$ is algebraic, $K/F(s_2, \dots, s_n, t)$ is algebraic. It follows that $\{s_2, \dots, s_n, t\}$ is a transcendence basis for K/F. Set $t_1 = t$. Assuming we have found $t_i \in T$ for all i with $1 \le i < m \le n$ such that $\{t_1, \dots, t_{m-1}, s_m, \dots, s_n\}$ is a transcendence basis for K/F, by replacing S by this set, the argument above shows that there is a $t' \in T$ such that $\{t_1, \dots, t_{m-1}, t', \dots, s_n\}$ is a transcendence basis for K/F. Setting $t_m = t'$ and continuing in this way, we get a transcendence basis $\{t_1, \dots, t_n\} \subseteq T$ of K/T. Since T is transcendence basis for K/T, we see that $\{t_1, \dots, t_n\} = T$, so $|T| = n$.

For the general case, by the previous argument we may suppose that S and T are both infinite. Each $t \in T$ is algebraic over $F(S)$, hence, there is a finite subset $S_t \subseteq S$ with t algebraic over $F(S_t)$. If $S' = \bigcup_{t \in T} S_t$, then each $t \in T$ is algebraic over $F(S')$. Since K is algebraic over $F(T)$, we see that K is algebraic over $F(S')$. Thus, $S = S'$, since $S' \subseteq S$ and S is a transcendence basis for K/F. We then have

$$|S| = |S'| = \left| \bigcup_{t \in T} S_t \right| \le |T \times \mathbb{N}| = |T|,$$

where the last equality is true since T is infinite. Reversing the argument, we see that $|T| \le |S|$, so $|S| = |T|$.

□

Definition 13.2.2 (Transcendence degree)

*Let K/k be a field extension. The **transcendence degree** of K over k is the cardinality of a transcendence basis of K over k. It is denoted* $\operatorname{tr.deg}_k(K)$.

Remark 13.13 Any finitely generated field extension necessarily has finite transcendence degree. See Theorem 11.5.2 for a related result.

Theorem 13.2.2 (Dimension = transcendence degree)

Suppose A is a finitely generated domain over a field k (i.e., a finitely generated k-algebra that is an integral domain). Then $\dim \operatorname{Spec} A = \operatorname{tr.deg}_k K(A)$. *Hence if X is an irreducible k-variety, then* $\dim X = \operatorname{tr.deg}_k K(X)$.

We will prove Theorem 13.2.2 shortly (§**??**). We first show that it is useful by giving some immediate consequences. We seem to have immediately $\dim \mathbb{A}_k^n = n$. However, our proof of Theorem 13.2.2 will go through this fact, so it isn't really a consequence.

A more substantive consequence is the following. If X is an irreducible k-variety (reduced, separated scheme of finite type over k), then $\dim X$ is the transcendence degree of the function field (the stalk at the generic point) $\mathscr{O}_{X,\eta}$ over k. Thus (as the generic point lies in all nonempty open sets) the dimension can be computed in any open set of X. (**Warning:** This is false without the finite type hypothesis, even in quite reasonable circumstances — let X be the two-point space $\operatorname{Spec} k[x]_{(x)}$, and let U consist of only generic point;

see Exercise 4.16.)

Another consequence is a second proof of the Nullstellensatz 4.2.2.

Exercise 13.2(The Nullstellensatz from dimension theory) Suppose $A = k[x_1, \ldots, x_n]/I$. Show that the residue field of any maximal ideal of A is a finite extension of k.

Proof Let $\mathfrak{m}$ be a maximal ideal of A, then $\kappa(\mathfrak{m}) = A/\mathfrak{m}$. Hence $\dim \operatorname{Spec} \kappa(\mathfrak{m}) = 0$, by Theorem 13.2.2, we have $\operatorname{tr.deg}_k K(\operatorname{Spec} \kappa(\mathfrak{m})) = \dim \operatorname{Spec} \kappa(\mathfrak{m}) = 0$. Thus $K(\operatorname{Spec} \kappa(\mathfrak{m})) = A/\mathfrak{m}$ is algebraic over k. Recall that finitely generated algebraic extension is finite (see [Sta25, Lemma 09GH]), note that $A/\mathfrak{m}$ is finite generated k-algebra, so $A/\mathfrak{m}$ is a finite extension of k, we are done. □

Yet another consequence is geometrically believable.

Proposition 13.2.1

If $\pi : X \to Y$ is a dominant morphism of irreducible k-varieties, then $\dim X \geq \dim Y$.

Proof Let η, ξ be generic points of X, Y, respectively. Since $\pi : X \to Y$ is a dominant morphism, by Proposition 8.5.4, we know that $\mathscr{O}_{Y,\xi} \to \mathscr{O}_{X,\eta}$ is injective. Since X, Y are irreducible, we have $K(X) \cong \mathscr{O}_{X,\eta}$ and $K(Y) \cong \mathscr{O}_{Y,\xi}$. Hence there is an injective $K(Y) \to K(X)$, and therefore $\operatorname{tr.deg}_k K(Y) \leq \operatorname{tr.deg}_k K(X)$. By Theorem 13.2.2, we have $\dim X \geq \dim Y$. □

Remark 13.14 Proposition is *false* more generally: consider the inclusion of the generic point into an irreducible curve.

For example, let $X = \operatorname{Spec} k[t] = \mathbb{A}_k^1$, and η be its generic point, so we have an inclusion $\operatorname{Spec} \kappa(\eta) \hookrightarrow X$ and $\dim \operatorname{Spec} \kappa(\eta) = 0 < 1 = \dim X$. In fact $\kappa(\eta) = k(t)$ is not finite generated k-algebra, so $\operatorname{Spec} \kappa(\eta)$ is not an k-variety.

Proposition 13.2.2

Suppose $f_1(x_1, \ldots, x_n), \ldots, f_m(x_1, \ldots, x_m) \in k[x_1, \ldots, x_n]$ are m polynomials in n variables over a field, with $m > n$. Then the polynomials $f_1, \ldots, f_m$ are algebraically dependent.

Proof If $f_1, \ldots, f_m$ are algebraically independent, by definition ,we know that $\operatorname{tr.deg}_k k(f_1, \ldots, f_m) = m$, since $k(f_1, \ldots, k_m) \subseteq k(x_1, \ldots, k_n)$, $\operatorname{tr.deg}_k k(x_1, \ldots, k_n) = n \geq \operatorname{tr.deg}_k k(f_1, \ldots, k_m) = m$, a contradiction! □

Remark 13.15 If $m = n$, how can you tell if the f_i are algebraically dependent? There is a remarkably simple answer in characteristic 0: they are algebraically dependent if and only if the department of the Jacobian matrix is the zero polynomial; see **??**! This is false in characteristic p: if $f_i(x_1, \ldots, x_n) = x_i^p$, then the f_i are algebraically independent, yet the Jacobian matrix is the zero matrix.

Exercise 13.3(Practice with the concept) Show that the three equations

$$wz - xy = 0, \quad wy - x^2 = 0, \quad xz - y^2 = 0$$

cut out an integral surface S in $\mathbb{A}_k^4$. (You may recognize these equations from Exercise 4.18 and 10.3.) You might expect S to be a curve, because it is cut out by three equations in four-space. One of many ways to proceed: cut S into pieces. For example, show that $D(w) \cong \operatorname{Spec} k[x, w]_w$. (You may recognize S as the affine cone over the twisted cubic. The twisted cubic was defined in Exercise 10.3.) It turns out that you need three equations to cut out this surface. The first equation cuts out a threefold in $\mathbb{A}_k^4$ (by Krull's Principal Ideal Theorem **??**, which we will meet soon). The second equation cuts out a surface. The second equation cuts out a surface: our surface, along with another surface. The third equation cuts out our surface, and removes the "extraneous component." One last aside: Notice once again that the cone over the quadric surface $k[w, x, y, z]/(wz - xy)$

makes an appearance.

Proof It suffices to show that S is dimension 2 integral scheme. Say $I = (wz - xy, wy - x^2, xz - y^2)$, then $S = \operatorname{Spec} k[w, x, y, z]/I$. Recall Exercise 4.18, S is integral scheme, hence we may compute $\dim S$ in any open subset of S. Consider $S \cap D(w)$. We want to show that $S \cap D(w) \cong \operatorname{Spec} k[x, w]_w$. Define $\varphi : (k[w, x, y, z]/I)_w \to k[x, w]_w$ by setting

$$(w, x, y, z) \longmapsto (w, x, x^2/w, x^3/w^2).$$

It is easy to check that φ is well-defined and is an isomorphism. So $S \cap D(w) \cong \operatorname{Spec} k[x, w]_w$. Since $K(\operatorname{Spec} k[x, w]_w) = k(x, w)$, we have $\operatorname{tr.deg}_k K(\operatorname{Spec} k[x, w]_w) = 2$, by Theorem 13.2.2, $\dim \operatorname{Spec} k[x, w]_w = 2$, hence $\dim S \cap D(w) = \dim S = 2$. □

Definition 13.2.3 (Degree of rational map of irreducible varieties)

*If $\pi : X \dashrightarrow Y$ is a dominant rational map of irreducible k-varieties (hence integral) of the same dimension, the degree of the field extension is called the **degree** of the rational map.*

Remark 13.16

(1) This readily extends if X is reducible: we add up the degrees on each of the components of X. If π is a rational map of integral affine k-varieties of the same dimension that is not dominant, we say the degree is 0. We will interpret this degree in terms of counting preimages of general points of Y in §**??**.

(2) Note that degree is multiplicative under composition: If $\rho : Y \dashrightarrow Z$ is a rational map of integral k-varieties of the same dimension, then $\deg(\rho \circ \pi) = \deg(\rho) \deg(\pi)$, as degrees of field extensions are multiplicative in towers.

13.2.2 Noetherian Normalization

Our proof of Theorem 13.2.2 will use another important classical notion, Noether normalization.

Theorem 13.2.3 (Noether Normalization Lemma)

Suppose A is an integral domain, finite generated over a field k (i.e., finitely generated k-algebra). If $\operatorname{tr.deg}_k K(A) = n$, then there are elements $x_1, \ldots, x_n \in A$, algebraically independent over k, such that A is finite (finitely generated as a module) over $k[x_1, \ldots, x_n]$ (hence integral by Corollary 9.2.1).

The geometric content behind this result is that given any integral finite type affine k-scheme X, we can find a surjective finite morphism $X \to \mathbb{A}^n_k$, where n is the transcendence degree of the function field of X (over k). Surjectivity follows from the Lying over Theorem 9.2.2, in particular, Proposition 13.1.6. This interpretation is sometimes called geometric Noether normalization. (See Chapter **??** for a projective version of Noether normalization.)

Corollary 13.2.1 (Geometric Noether Normalization)

Suppose X is an integral affine k-scheme of finite type. If $\dim X = n$, then we can find a surjective finite morphism

$$X \twoheadrightarrow \mathbb{A}^n_k.$$

Nagara's proof of the Noether Normalization Lemma 13.2.3:

Proof Suppose we can write $A = k[y_1, \ldots, y_m]/\mathfrak{p}$, i.e., that A can be chosen to have m generators. Note that $m \geq n$. We show the result by induction on m. The base case $m = n$ is immediate.

Assume now that $m > n$, and that we have proved the result for smaller m. We will find $m-1$ elements $z_1, \ldots, z_{m-1}$ of A such that A is finite over $k[z_1, \ldots, z_{m-1}]/\mathfrak{q}$ (i.e., the subring of A generated by $z_1, \ldots, z_{m-1}$, where z_i satisfy the relations given by the ideal $\mathfrak{q}$). Then by the induction hypothesis, $k[z_1, \ldots, z_{m-1}]/\mathfrak{q}$ is finite over some $k[x_1, \ldots, x_n]$, and A is finite over $k[z_1, \ldots, z_{m-1}]$, so by Proposition 9.3.11, A is finite over $k[x_1, \ldots, x_n]$.

$$\begin{array}{c} A = k[y_1, \ldots, y_m]/\mathfrak{p} \\ \Big| \text{finite} \\ k[z_1, \ldots, z_{m-1}]/\mathfrak{q} \\ \Big| \text{finite} \\ k[x_1, \ldots, x_n] \end{array}$$

As $y_1, \ldots, y_m$ are algebraically dependent in A, there is some nonzero algebraic relation $f(y_1, \ldots, y_m) = 0$ among them (where f is a polynomial in m variables over k).

Let $z_1 = y_1 - y_m^{r_1}, z_2 = y_2 - y_m^{r_2}, \ldots, z_{m-1} = y_{m-1} - y_m^{r_m-1}$, where $r_1, \ldots, r_{m-1}$ are positive integers to be chosen shortly. Then

$$f(z_1 + y_m^{r_1}, z_2 + y_m^{r_2}, \ldots, z_{m-1} + y_m^{r_m-1}, y_m) = 0.$$

Then upon extending this out, each monomial in f (as a polynomial in m variables) will yield a single term that is a constant times a power of y_m (with no z_i factors). By choosing the r_i so that $0 \ll r_1 \ll r_2 \ll \cdots \ll r_{m-1}$, we can ensure that the powers of y_m appearing are all distinct, so that, in particular, there is a leading term y_m^N, and all other terms (including those with factors of z_i) are of smaller degree in y_m. Thus we have described an integral dependence of y_m on $z_1, \ldots, z_{m-1}$, it follows that A is finitely generated $k[z_1, \ldots, z_{m-1}]$-module (in fact, the generators of A are $y_m^{N-1}, \ldots, y_m, 1$), hence A is finite over $k[z_1, \ldots, z_{m-1}]$, as we desired. □

Remark 13.17 The geometry behind Nagata's proof. Suppose we have an n-dimensional variety in $\mathbb{A}_k^m$ with $n < m$, for example, $xy = 1$ in $\mathbb{A}^2$. One approach is to hope the projection to a hyperplane is a finite morphism. In the case of $xy = 1$, if we projected to the x-axis, it wouldn't be finite, roughly speaking, because the asymptote $x = 0$ prevents the map from being closed (cf. Example 9.6). If we instead projected to a random line, we might hope that we would get rid of this problem, and indeed we usually can: this problem arises for only a finite number of directions. But we might have a problem if the field were finite: perhaps the finite number of directions in which to project each has a problem. (You can show that if k is an infinite field, then the substitution in the above proof $z_i = y_i - y_m^{r_i}$ can be replaced by the linear substitution $z_i = y_i - a_i y_m$, where $a_i \in k$, and that for a nonempty Zariski-open choice of a_i, we indeed obtain a finite morphism.) Nagata's trick in general is to "jiggle" the variables in a nonlinear way, and this jiggling kills the nonfiniteness of the map.

Proposition 13.2.3 (Dimension is additive for products of finite type k-schemes)

If X and Y are irreducible k-varieties, then $\dim X \times_k Y = \dim X + \dim Y$.

Proof Since X and Y are irreducible, $X \times_k Y$ is irreducible. Hence we may reduce to the case that X, Y are affine. Suppose $X = \operatorname{Spec} A$ and $Y = \operatorname{Spec} B$. Let $\dim A = n$ and $\dim B = m$, by Theorem 13.2.2, $\operatorname{tr.deg}_k K(A) = n$ and $\operatorname{tr.deg}_k K(B) = m$. By Noether Normalization 13.2.3, A is finite over $k[x_1, \ldots, x_n]$ for some $x_1, \ldots, x_n$ and B is finite over $k[y_1, \ldots, y_m]$ for some m. We want to construct a morphism $\theta : k[x_1, \ldots, x_n] \otimes_k k[y_1, \ldots, y_m] \to A \otimes_k B$ such that θ is an integral extension. Say $\varphi : k[x_1, \ldots, x_n] \to A$ and $\psi : k[y_1, \ldots, y_m] \to B$ be two finite morphism. Define $\theta : k[x_1, \ldots, x_n] \otimes_k k[y_1, \ldots, y_m] \to A \otimes_k B$ by setting

$$f \otimes g \longmapsto \varphi(f) \otimes \psi(g).$$

Clearly θ is well-defined. Since A, B are finitely generated $k[x_1, \dots, x_n], k[y_1, \dots, y_m]$-modules respectively, say A generated by $a_1, \dots, a_N$ and B generated by $b_1, \dots, b_M$, then $\varphi(f) = \sum_{i=1}^{N} f_i a_i$ and $\psi(g) = \sum_{j=1}^{M} g_i b_j$ where $f_i \in k[x_1, \dots, x_n]$ and $g_j \in k[y_1, \dots, y_m]$. Hence $\varphi(f) \otimes \psi(g) = \sum_{i,j} (f_i \otimes g_j)(a_i \otimes b_j)$, which implies that $A \otimes_k B$ is finitely generated $k[x_1, \dots, x_n] \otimes_k k[y_1, \dots, y_m]$-module, so θ is finite. Moreover, since φ and ψ are injective, θ is injective, so θ is an integral extension. By Proposition 13.1.6, we have

$$\dim X \times_k Y = \dim \mathbb{A}_k^n \times_k \mathbb{A}_k^m = \dim \mathbb{A}_k^{n+m} = n + m = \dim X + \dim Y.$$

$\square$

Remark 13.18 In particular, we have $\dim A[x] = \dim A + 1$ when A is finitely generated over a field.

This is true more generally when A is Noetherian, to prove it we will use following lemma.

Lemma 13.2.4

Let A be a Noetherian ring and let $\mathfrak{m}$ (resp. $\mathfrak{n}$) be a maximal ideal of A (resp. of $A[t_1, \dots, t_n]$) such that $\mathfrak{n} \cap A = \mathfrak{m}$. Then $\operatorname{ht}(\mathfrak{n}) = \operatorname{ht}(\mathfrak{m}) + n$.

Proof By Zariski's Lemma 4.2.2, $A[t_1, \dots, t_n]/\mathfrak{n}$ is a finite extension of $A/\mathfrak{m}$. If $\mathfrak{n}_1 = \mathfrak{n} \cap A[t_1, \dots, t_{n-1}]$, then $A[t_1, \dots, t_{n-1}]/\mathfrak{n}_1$ is a sub-$A/\mathfrak{m}$-algebra of $A[t_1, \dots, t_n]/\mathfrak{n}$ and is therefore a field. Thus $\mathfrak{n}_1$ is maximal. We then see that it is enough to prove the lemma in the case when $n = 1$.

Let $\mathfrak{p}_0 \subsetneq \cdots \subsetneq \mathfrak{p}_r \subseteq \mathfrak{m}$ be a chain of prime ideals of A. Then the $\mathfrak{p}_i A[t]$ form a chain of prime ideals of $A[t]$. Moreover, $\mathfrak{p}_r A[t] \subsetneq \mathfrak{n}$ because $A[t]/\mathfrak{p}_r A[t] \cong (A/\mathfrak{p}_r)[t]$ is not a field. Therefore $\operatorname{ht}(\mathfrak{n}) \geq \operatorname{ht}(\mathfrak{m}) + 1$.

We will show the inequality in the other direction by induction on $\operatorname{ht}(\mathfrak{m})$. If $\operatorname{ht}(\mathfrak{m}) = 0$, then $\mathfrak{m}$ is a minimal prime ideal of A, and any prime ideal of $A[t]$ contained in $\mathfrak{n}$ must contain $\mathfrak{m}$. Therefore $\operatorname{ht}(\mathfrak{n}) = \operatorname{ht}(\overline{\mathfrak{n}}) = 1$ where $\overline{\mathfrak{n}}$ is the image of $\mathfrak{n}$ in $A/\mathfrak{m}A[t]$. Let us suppose that $\operatorname{ht}(\mathfrak{m}) \geq 1$. As A has only a finite number of minimal prime ideals (Proposition 7.6.15 and Corollary 7.6.2) and $\mathfrak{m}$ is not contained in any of them, there exists an $f \in \mathfrak{m}$ which does not belong to any minimal prime ideal. Let $B = /fA$, and let $\mathfrak{n}'$ be the image of $\mathfrak{n}$ in $B[t]$. We have

$$\operatorname{ht}(\mathfrak{n}) = \dim A[t]_{\mathfrak{n}} \leq \dim A[t]_{\mathfrak{n}}/(f) + 1 = \dim B[t]_{\mathfrak{n}'} + 1 = \operatorname{ht}(\mathfrak{n}') + 1.$$

Let $\mathfrak{m}'$ be the image of $\mathfrak{m}$ in B. Then $\operatorname{ht}(\mathfrak{m}') = \dim B_{\mathfrak{m}'} = \dim A_{\mathfrak{m}}/(f) = \operatorname{ht}(\mathfrak{m}) - 1$. The induction hypothesis implies that $\operatorname{ht}(\mathfrak{n}') \leq \operatorname{ht}(\mathfrak{m}') + 1$. It follows that $\operatorname{ht}(\mathfrak{n}) \leq \operatorname{ht}(\mathfrak{m}) + 1$, which completes the proof. $\square$

Remark 13.19 In this proof, we use Krull's Principal Ideal Theorem 13.3.2.

Corollary 13.2.2

Let A be a Noetherian ring. Then we have

$$\dim A[t_1, \dots, t_n] = \dim A + n.$$

Proposition 13.2.4

(a) Suppose $X \subseteq \mathbb{P}^n$ is an irreducible projective k-variety. Then the affine cone and projective cone over X both have dimension $\dim X + 1$.

(b) A prime ideal of $k[x_0, \dots, x_n]$ that is minimal over a homogeneous ideal is also homogeneous.

(c) Suppose $\mathfrak{p} \subseteq \mathfrak{q}$ are two homogeneous prime ideals of $k[x_0, \dots, x_n]$ such that there is no ho-

mogeneous prime ideal strictly between the two. Then there is no prime ideal (not necessarily homogeneous) strictly between the two.

Proof

(a) Suppose $X = \operatorname{Proj} S_\bullet$, where $S_\bullet = k[t_1, \dots, t_n]/\mathfrak{p}$ (recall Proposition 10.3.3). Since X is irreducible, we have $\dim X = \dim D_+(t_1) = \operatorname{tr.deg}_k K(((S_\bullet)_{t_1})_0)$. The affine cone $\mathrm{C}(X)$ over X is $\mathrm{C}(X) = \operatorname{Spec} S_\bullet$, since $S_\bullet$ cut by a prime ideal, $\mathrm{C}(X)$ is irreducible, by Theorem 13.2.2, we have $\dim \mathrm{C}(X) = \operatorname{tr.deg}_k K(\mathrm{C}(X)) = \operatorname{tr.deg}_k K(S_\bullet)$. Let $t = t_1$. Pick $F \in K(S_\bullet)$, we may write $F = P/Q$ where $P = P_0 + \cdots + P_n, Q = Q_0 + \cdots + Q_m, P_i \in S_i, Q_j \in S_j$. Note that

$$F = \frac{P}{Q} = \frac{P_0 t^{-n} + \frac{P_1}{t} \cdot t^{-n+1} + \cdots + \frac{P_n}{t^n}}{Q_0 t^{-m} + \frac{Q_1}{t} \cdot t^{-m+1} + \cdots + \frac{Q_m}{t^m}} \cdot t^{n-m},$$

we have $K(((S_\bullet)_t)_0)(t) \supseteq K(S_\bullet)$. Clearly $K(((S_\bullet)_t)_0)(t) \subseteq K(S_\bullet)$, so $K(((S_\bullet)_t)_0)(t) = K(S_\bullet)$. It follows that

$$\operatorname{tr.deg}_k K(S_\bullet) = \operatorname{tr.deg}_k K(((S_\bullet)_t)_0) + 1,$$

which implies that $\dim \mathrm{C}(X) = \dim X + 1$.

The projective cone $\mathrm{PC}(X)$ over X is $\mathrm{PC}(X) = \operatorname{Proj} S_\bullet[T]$. Similar to the case of affine cone, we can prove that $K(((S_\bullet)_t)_0)(T) = K(((S_\bullet[T])_t)_0)$, so $\dim \mathrm{PC}(X) = \dim X + 1$.

(b) Let I be a homogeneous ideal, $\mathfrak{p}$ is a minimal prime ideal over I, i.e., $\mathfrak{p} \supseteq I$. Suppose $\mathfrak{p}^h$ is an ideal generated by all homogeneous elements of $\mathfrak{p}$, then $\mathfrak{p}^h$ is a homogeneous ideal. We want to show that $\mathfrak{p}^h$ is prime ideal. Let $ab \in \mathfrak{p}^h$ where $a, b \in S_\bullet$, we want to show that $a \in \mathfrak{p}^h$ or $b \in \mathfrak{p}^h$. Since $S_\bullet$ is graded, we may write $a = \sum_i a_i$ and $b = \sum_j b_j$ where $a_i \in S_i$ and $b_j \in S_j$. If $a \notin \mathfrak{p}^h$ and $b \notin \mathfrak{p}^h$, by Proposition 5.5.1, there exists a_i, b_j such that $a_i, b_j \notin \mathfrak{p}^h$. We claim that $a_i b_j \notin \mathfrak{p}^h$, if not, then $a_i b_j \in \mathfrak{p}^h$, by Proposition 5.5.1, $a_i, b_j \in \mathfrak{p}^h$, a contradiction! So $a_i b_j \notin \mathfrak{p}^h$, this contradicts to $ab \in \mathfrak{p}^h$, by Proposition 5.5.1 again. Hence $a \in \mathfrak{p}^h$ or $b \in \mathfrak{p}^h$, which implies that $\mathfrak{p}^h$ is prime ideal. Since $\mathfrak{p}$ is the minimal prime which contains I, there must be $\mathfrak{p} = \mathfrak{p}^h$, i.e., $\mathfrak{p}$ is a homogeneous ideal.

(c) Since there is no homogeneous prime ideal strictly between $\mathfrak{p}$ and $\mathfrak{q}$, we have

$$\dim \operatorname{Proj} k[x_0, \dots, x_n]/\mathfrak{p} = \dim \operatorname{Proj} k[x_0, \dots, x_n]/\mathfrak{q} + 1.$$

Say $S_\bullet = k[x_0, \dots, x_n]/\mathfrak{p}$ and $T_\bullet = \operatorname{Proj} k[x_0, \dots, x_n]/\mathfrak{q}$, by part (a), $\dim \operatorname{Spec} S_\bullet = \dim \operatorname{Proj} S_\bullet + 1$ and $\dim \operatorname{Spec} T_\bullet = \dim \operatorname{Proj} T_\bullet + 1$, so

$$\dim \operatorname{Spec} S_\bullet = \dim \operatorname{Spec} T_\bullet + 1.$$

It follows that there is no prime ideal strictly between $\mathfrak{p}$ and $\mathfrak{q}$.

$\square$

Proof of Theorem 13.2.2 on dimension and transcendence degree.

Proof Suppose X is an integral affine k-scheme of finite type. We show that $\dim X$ equals to the transcendence degree n of its function field, by induction on n. (The ideal is that we reduce from X to $\mathbb{A}^n$ to a hypersurface in $\mathbb{A}^n$ to $\mathbb{A}^{n-1}$.) Assume the result is known for all transcendence degree less than n.

By the Geometric Noether Normalization 13.2.1, there exists a surjective finite morphism $X \twoheadrightarrow \mathbb{A}^n_k$. By Proposition 13.1.6, $\dim X = \dim \mathbb{A}^n_k$. If $n = 0$, we are done, as $\dim \mathbb{A}^0_k = 0$.

We now show that $\dim \mathbb{A}^n_k = n$ for $n > 0$, by induction. Clearly, $\dim \mathbb{A}^n_k \geq n$, as we can describe a chain of irreducible subsets of length n: if $x_1, \dots, x_n$ are coordinates on $\mathbb{A}^n$, consider the chain of ideals

$$(0) \subsetneq (x_1) \subsetneq \cdots \subsetneq (x_1, \dots, x_n)$$

in $k[x_1, \dots, x_n]$. Suppose we have a chain of prime ideals of length at least $n+1$:

$$(0) = \mathfrak{p}_0 \subseteq \cdots \subseteq \mathfrak{p}_m.$$

Choose any nonzero element g of $\mathfrak{p}_1$, and let f be any irreducible factor of g lying in $\mathfrak{p}_1$. Then replace $\mathfrak{p}_1$ by (f). (Of cause, $\mathfrak{p}_1$ may have been (f) to begin with.) Since we have a chain of prime ideals of length at least $n+1$ in $k[x_1, \dots, x_n]$, $\dim k[x_1, \dots, x_n]/(f) \geq n$. On the other hand, $K(k[x_1, \dots, x_n]/(f))$ has transcendence degree $n-1$, so by induction,

$$\dim k[x_1, \dots, x_n]/(f) = n-1,$$

a contradiction! So $\dim \mathbb{A}_k^n \leq n$, completing the proof. $\square$

13.2.3 Codimension is the difference of dimensions for irreducible varieties

Noether normalization will help us show that codimension is the difference of dimensions for irreducible varieties, i.e., that the inequality (13.1) is always an equality.

Theorem 13.2.4

Suppose X is a pure-dimensional k-scheme locally of finite type (for example, an irreducible k-variety), Y is an irreducible closed subset, and η is the generic point of Y. Then

$$\dim Y + \dim \mathscr{O}_{X,\eta} = \dim X.$$

Hence by Proposition 13.1.9, $\dim Y + \operatorname{codim}_X Y = \dim X$ — inequality (13.1) is always an equality.

Proving this will give us an excuse to introduce some useful notions, such as the Going-down Theorem for finite extensions of integrally closed domains (Theorem **??**). Before we begin the proof, we give an algebraic translation.

Definition 13.2.4 (Catenary)

*A ring A is called **catenary** if for every nested pair of prime ideals $\mathfrak{p} \subseteq \mathfrak{q} \subseteq A$, every strictly increasing chain of prime ideals "from $\mathfrak{p}$ to $\mathfrak{q}$" is contained in a maximal such chain, and all maximal chains have the same (finite) length.*

Remark 13.20 We will not use this term in any serious way later.

Lemma 13.2.5

Any localization of a catenary ring is catenary.

Proof Let A be a catenary ring, $S \subseteq A$ be a multiplicative subset. Recall that the prime ideal of $S^{-1}A$ is one-to-one corresponding to the prime ideal of A which not meet S, pick nested pair of prime ideals $\mathfrak{p} \subseteq \mathfrak{q} \subseteq S^{-1}A$, it corresponding to $\mathfrak{p}' \subseteq \mathfrak{q}' \subseteq A$ where $\mathfrak{p}' \cap S = \varnothing$ and $\mathfrak{q}' \cap S = \varnothing$. Since A is a catenary ring, every strictly increasing chain of prime ideals from $\mathfrak{p}'$ to $\mathfrak{q}'$ is contained in a maximal such chain, and all maximal chains have the same length. Use the corresponding again, every strictly increasing chain of prime ideals from $\mathfrak{p}$ to $\mathfrak{q}$ is contained in a maximal such chain, and all maximal chains have the same length. It follows that $S^{-1}A$ is catenary. $\square$

Corollary 13.2.3

If A is a localization of a finitely generated ring over a field k (i.e., finitely generated k-algebra), then A is catenary.

Proof By Lemma 13.2.5, it suffices to show that finitely generated ring over a field k is catenary. Suppose R is a finitely generated k-algebra, let $\mathfrak{p} \subseteq \mathfrak{q}$ be any two prime ideals of R. Consider $R/(\mathfrak{p})$, then we have $(0) \subseteq \mathfrak{q}/\mathfrak{p}$, since the prime ideal which is contained in $\mathfrak{p} \subseteq \mathfrak{q}$ is one-to-one corresponding to the prime ideal which is contained in $(0) \subseteq \mathfrak{q}/\mathfrak{p}$. We may assume that R is an integral domain, and the problem is reduced to show that any maximal chain of prime ideals from (0) to $\mathfrak{p}$ has the same finite length.

Let $(0) = \mathfrak{p}_0 \subsetneq \mathfrak{p}_1 \subsetneq \cdots \subsetneq \mathfrak{p}_m = \mathfrak{p}$ be any maximal chain of prime ideals from (0) to $\mathfrak{p}$. Consider $\operatorname{Spec} R$, $\operatorname{Spec} R/\mathfrak{p}_1$ is a closed subset of $\operatorname{Spec} R$, by Theorem 13.2.4, we have

$$\dim R = \dim R/\mathfrak{p}_1 + \dim R_{\mathfrak{p}_1} = \dim R/\mathfrak{p}_1 + \mathrm{ht}_R(\mathfrak{p}_1) = \dim R/\mathfrak{p}_1 + 1.$$

Note that $\operatorname{Spec}(R/\mathfrak{p}_1)/(\mathfrak{p}_2/\mathfrak{p}_1) \cong \operatorname{Spec} R/\mathfrak{p}_2$ is a closed subset of $\operatorname{Spec} R/\mathfrak{p}_2$, by Theorem 13.2.4,

$$\dim R/\mathfrak{p}_1 = \dim R/\mathfrak{p}_2 + \dim(R/\mathfrak{p}_1)_{\mathfrak{p}_2/\mathfrak{p}_1} = \dim R/\mathfrak{p}_2 + \mathrm{ht}_{R/\mathfrak{p}_1}(\mathfrak{p}_2/\mathfrak{p}_1) = \dim R/\mathfrak{p}_2 + 1.$$

Repeat above process, we have

$$\dim R = \dim R/\mathfrak{p}_n + \sum_{i=0}^{m-1} \mathrm{ht}(\mathfrak{p}_{i+1}/\mathfrak{p}_i) = \dim R/\mathfrak{p}_n + m,$$

i.e., $m = \dim R - \dim R/\mathfrak{p}_n = \mathrm{ht}_R(\mathfrak{p}_n) = \mathrm{ht}_R(\mathfrak{p})$. Hence the length is only determined by $\mathrm{ht}(\mathfrak{p})$, and therefore any maximal chain of prime ideals from (0) to $\mathfrak{p}$ has the same finite length. So R is catenary, as we desired. □

Remark 13.21 Most rings arising naturally in algebraic geometry are catenary. Important examples include: localizations of finitely generated $\mathbb{Z}$-algebras; complete Noetherian local rings; Dedekind domains (§**??**); and Cohen-Macaulay rings (see §**??**). It is hard to give an example of a noncatenary ring; see, for example, [Sta25, Section 02JE].

Proof of Theorem 13.2.4.

Proof Since X is a pure-dimensional, we may reduce to the case that X is an irreducible k-scheme locally of finite type. By Proposition 13.1.2 and X is irreducible, there is an irreducible affine open subset of X such that this affine open subset has the same dimension as X. So we may reduced to the case that X is an irreducible affine k-scheme. Consider the reduction X^{red} of X, recall Proposition 10.4.6, the underlying topological space of X is equal to X^{red}. So we may reduce to the case that X is a reduced irreducible affine k-scheme, i.e., X is an irreducible affine k-variety. Let Y be an irreducible closed subset of X, say $\operatorname{codim}_X Y = l$, then there is a maximal chain of irreducible closed subsets

$$Y = Z_0 \subsetneq Z_1 \subsetneq \cdots \subsetneq Z_l \subseteq X.$$

If we prove $\dim Z_l = \dim X - 1$, inductively, we have $\dim X = \dim Z_0 + l = \dim Y + \operatorname{codim}_X Y$. So we may reduce the proof to the following problem: *If X is an irreducible affine k-variety and Z is a closed irreducible subset maximal among those smaller than X (the only larger closed irreducible subset is X), then* $\dim Z = \dim X - 1$.

Let $d = \dim X$ for convenience. By Geometric Noether Normalization 13.2.1, we have a surjective finite morphism $\pi : X \twoheadrightarrow \mathbb{A}^d$ corresponding to a finite extension of rings (i.e., $\Gamma(X, \mathscr{O}_X)$ is finitely generated $k[t_1, \ldots, t_d]$-module). Then $\pi(Z)$ is an irreducible closed subset of $\mathbb{A}^d$ (finite morphisms are closed, Corollary

9.3.2).

Regard $\pi(Z)$ as scheme theoretic image of Z. Consider $\pi|_{\pi^{-1}(\pi(Z))} : \pi^{-1}(\pi(Z)) \to \pi(Z)$, then we have the following Cartesian square.

$$\begin{tikzcd}
\pi^{-1}(\pi(Z)) \arrow[r, hook] \arrow[d, "\pi|_{\pi^{-1}(\pi(Z))}"'] & X \arrow[d, "\text{finite}"] \\
\pi(Z) \arrow[r, hook, "\text{cl.emb.}"'] & \mathbb{A}^d_k
\end{tikzcd}$$

By Proposition 11.4.3, finiteness is preserved by base change, $\pi|_{\pi^{-1}(\pi(Z))} : \pi^{-1}(\pi(Z)) \to \pi(Z)$ is finite. By Proposition 13.1.6, $\dim \pi^{-1}(\pi(Z)) = \dim \pi(Z)$. By the maximality of Z, we must have $\dim Z = \dim \pi^{-1}(\pi(Z))$. Hence it suffices to show that $\dim \pi(Z) = d - 1$. Recall that the dimension of any hypersurface is $d-1$ by Theorem 13.2.2 on dimension and transcendence degree. Hence it suffices to show that $\pi(Z)$ is a hypersurface.

Now if $\pi(Z)$ is not a hypersurface, then it is properly contained in an irreducible hypersurface H, so by the Going-Down Theorem 13.2.5 (polynomial rings are integrally closed) for finite extensions of integrally closed domains (which we shall now prove), there is some closed irreducible subset Z' of X properly containing Z, contradicting the maximality of Z. $\square$

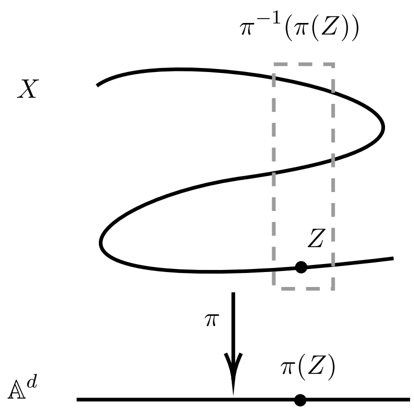


Figure 13.2: Key step in proof of Theorem 13.2.4.

Remark 13.22 We give a sketch of proof for the dimension of any hypersurface is the dimension of the total space -1: Reduced to the affine case. Let hypersurface $H = \operatorname{Spec} k[x_1, \ldots, x_n]/(f(x_1, \ldots, x_n))$ where f is irreducible and contains x_n. Since $f(\overline{x_1}, \ldots, \overline{x_n}) = 0$ in $k[x_1, \ldots, x_n]/(f)$, $\overline{x_1}, \ldots, \overline{x_n}$ is algebraic dependence over k, then $\overline{x_n}$ is algebraic over $k(\overline{x_1}, \ldots, \overline{x_{n-1}})$, so $\operatorname{tr.deg}_k K(H) \leq n-1$. If $\overline{x_1}, \ldots, \overline{x_{n-1}}$ is algebraic dependence over k, there exists $g \in k[x_1, \ldots, x_n]$ such that $g(\overline{x_1}, \ldots, \overline{x_{n-1}}) = 0$. Hence $g(x_1, \ldots, x_{n-1}) = h(x_1, \ldots, x_n) f(x_1, \ldots, x_n)$, note that f contains x_n but g not, we know that $g = 0$, a contradiction. So $\operatorname{tr.deg}_k K(H) = n-1$, and therefore $\dim H = n-1$.

Theorem 13.2.5 (Going-Down Theorem for finite extensions of integrally closed domains)

Suppose $\varphi : B \hookrightarrow A$ is a finite extension of rings (i.e., φ is injective and A is a finitely generated B-module), B is an integrally closed domain, and A is an integral domain. Then, given nested prime ideals $\mathfrak{q} \subseteq \mathfrak{q}'$ of B, and a prime $\mathfrak{p}'$ of A lying over $\mathfrak{q}'$ (i.e., $\mathfrak{p}' \cap B = \mathfrak{q}'$), there exists a prime $\mathfrak{p}$ of A contained in $\mathfrak{p}'$, lying over $\mathfrak{q}$.

As usual, you should sketch a geometric picture of the statement. This theorem is usually stated in the context of extending a chain of ideals, in the same way as the Going-Up Theorem 9.2.3, and you may want to think this through. (Another Going-Down Theorem, for flat morphisms, will be given in **??**.)

This theorem is true more generally with "finite" replaced by "integral"; see [AM18, Theorem 5.16] for completely different proofs. See [Eis13, Fig. 10.4] for an example (in the form of a picture) of why the "integrally closed" hypothesis on B cannot be removed.

In the course of the proof, we will use the following easy fact. We could prove it now, but we leave it until §**??**, because the proof uses a trick arising in Proposition **??**.

Proposition 13.2.5 (Prime avoidance)

Suppose $\mathfrak{p}_1, \ldots, \mathfrak{p}_n$ are prime ideals of a ring A, and I is another ideal of A not contained in any $\mathfrak{p}_i$. Then I is not contained in $\bigcup \mathfrak{p}_i$: there is an element $f \in I$ not in any of the $\mathfrak{p}_i$.

Remark 13.23 **Prime avoidance 13.2.5 has a nice geometric interpretation:** Consider the affine scheme $X = \operatorname{Spec} R$. Then $I \nsubseteq \mathfrak{p}_i$ for any i translates into $\mathfrak{p}_i \in X \setminus V(I)$ for all i, that is, the points $\mathfrak{p}_i$ are all contained in the open set $X \setminus V(I)$. The prime avoidance tells us that we can now find an element $x \in I$ which is not in any of the prime ideals $\mathfrak{p}_i$, which translates into saying that

$$\mathfrak{p}_i \in D(x) \subseteq X \setminus V(I).$$

So if we have finitely many points contained in an open set of an affine scheme, we can always find a smaller principal open set containing them.

Proof of Theorem 13.2.5 (Going-Down Theorem for finite extensions of integrally closed domains)

Proof The proof uses Galois theory. Let L be the normal closure of $K(A)/K(B)$ (the smallest subfield of $\overline{K(B)}$ containing $K(A)$ that is mapped to itself by any automorphism over $\overline{K(B)}/K(B)$, see [Sta25, Lemma 09DT, Definition 0BMF]). Let C be the integral closure of B in L (discussed in Definition 11.7.3, Lemma 11.7.3, Lemma 11.7.4, and Proposition 11.7.3), then $K(\operatorname{Int.cl}_L(B)) = K(C) = L$. So we have inclusions $B \hookrightarrow A \hookrightarrow \operatorname{Int.cl}_L(B) = C$. Since C is integral over B, C is of cause integral over A, then $A \hookrightarrow C$ is an integral extension. Because $B \hookrightarrow C$ is an integral extension, there is a prime Q of C lying over $\mathfrak{q} \subseteq B$ (by the Lying Over Theorem 9.2.1), and a prime Q' of C containing Q lying over $\mathfrak{q}'$ (by the Going-Up Theorem 9.2.3). Similarly, there is a prime P' of C lying over $\mathfrak{p}' \subseteq A$ (and thus over $\mathfrak{q}' \subseteq B$). We would be done if $P' = Q'$ (just take $\mathfrak{p} = Q \cap A$), but this needn't be the case. However, Lemma 13.2.6 below shows there is an automorphism σ of C over B that sends Q' to P', and then the image of $\sigma(Q)$ in A does the trick, completing the proof. The following diagram, in geometric terms, may help.

$$\begin{tikzcd}
\operatorname{Spec} C/P' \arrow[rrd] \arrow[dd, bend right] & & & \\
\operatorname{Spec} C/Q' \arrow[r, hook] \arrow[u, "\sigma"] & \operatorname{Spec} C/Q \arrow[r, hook] \arrow[dd, bend right] & \operatorname{Spec} C \arrow[d] & L \\
\operatorname{Spec} A/\mathfrak{p}' \arrow[r, hook] \arrow[d] \arrow[d, bend right] & ? \arrow[r, hook] \arrow[d] & \operatorname{Spec} A \arrow[d] & K(A) \arrow[u] \\
\operatorname{Spec} B/\mathfrak{q}' \arrow[r, hook] & \operatorname{Spec} B/\mathfrak{q} \arrow[r, hook] & \operatorname{Spec} B & K(B) \arrow[u]
\end{tikzcd}$$

□

Lemma 13.2.6

Suppose B is an integrally closed domain, $L/K(B)$ is a finite normal field extension, and C is the integral closure of B in L. If $\mathfrak{q}'$ is a prime ideal of B, then $\operatorname{Aut}(L/K(B))$ act transitively on the prime ideals of C lying over $\mathfrak{q}'$.

Remark 13.24 This result is often first seen in number theory, with $B = \mathbb{Z}$ and L a Galois extension of $\mathbb{Q}$.

Proof Let P and Q_1 be two prime ideals of C lying over $\mathfrak{q}'$, and let $Q_2, \ldots, Q_n$ be the prime ideals of C conjugate to Q_1 (the image of Q_1 under $\operatorname{Aut}(L/K(B))$). If P is not one of the Q_i, we claim that P is not contained in any of the Q_i. If there exists i such that $P \subseteq Q_i$, consider $\operatorname{Spec} C \to \operatorname{Spec} B$, since P, Q_i both lying over $\mathfrak{q}'$ (recall that $\sigma|_{K(B)} = \mathrm{id}_{K(B)}$), the fiber of $\mathfrak{q}'$ at least has dimension 1, this contradicts to Proposition 13.1.5. Hence P is not contained in any of the Q_i. Hence by prime avoidance 13.2.5, P is not contained in their union, so there is some $a \in P$ not contained in any Q_i. Thus no conjugate of a can be contained in Q_1, so the norm $\mathrm{N}_{L/K(B)}(a) = \prod_{\sigma \in \operatorname{Aut}(L/K(B))} \sigma(a) \in B$ is not contained in $Q_1 \cap B = \mathfrak{q}'$. (Recall: If $L/K(B)$ is separable, he norm is just the product of the conjugates. But even if $L/K(B)$ is not separable, the norm is the product of conjugates to the appropriate power, because we can factor $L/K(B)$ as a separable extension followed by a purely inseparable extension, or because we can take an explicit basis for the extension and calculate the norm of a.) But since $a \in P$, its norm lies in P, but also in B, and hence in $P \cap B = \mathfrak{q}'$, yielding a contradiction. □

Definition 13.2.5 (Dimension of a point)

*Suppose X is a finite-dimensional Noetherian scheme. Define a function $\dim_X(\cdot) : X \to \mathbb{Z}^{\geq 0}$ from the underlying set of X, by taking $\dim_X p$ to be the dimension of the largest irreducible component of X containing p. We call this **the dimension of X at p**.*

Remark 13.25 **Caution:** This definition differs from [Sta25, Definition 0055].

Proposition 13.2.6

$\dim_X(\cdot)$ *is an upper semicontinuous function.*

Remark 13.26 Recall that a function f from a topological space X to $\mathbb{R}$ is **upper semicontinuous** if for each $x \in \mathbb{R}$, $f^{-1}((-\infty, x))$ is open. Informally: the function can jump "up" upon taking limits. Our upper semicontinuous functions will map to $\mathbb{Z}$, so informally functions jump "up" on closed subsets. Similarly, f is **lower semicontinuous** if for each $x \in \mathbb{R}$, $f^{-1}((x, \infty))$ is open.

Proof We want to show that $\dim_X(\cdot)^{-1}((-\infty, k) \cap \mathbb{Z})$ is open in X for any $k \in \mathbb{Z}$, it is equivalent to show that $\dim_X(\cdot)^{-1}([k, +\infty) \cap \mathbb{Z})$ is closed in X for any $k \in \mathbb{Z}$. Let

$$S_k := \dim_X(\cdot)^{-1}([k, +\infty) \cap \mathbb{Z}) = \{p \in X : \dim_X(p) \geq k\}.$$

Since X is a finite-dimensional Noetherian scheme, by Proposition 4.6.12, we may write $X = \bigcup_{i=1}^{r} Z_i$ where Z_i is irreducible component of X. Say $d_i = \dim Z_i$, since $\dim X < \infty$, $d_i < \infty$ for all i. Recall Definition 13.2.5, $\dim_X(p) = \max_{1 \leq i \leq r}\{d_i\}$. Note that $\max_{1 \leq i \leq r}\{d_i\} \geq k$ means that there exists d_i such that $d_i \geq k$, hence

$$S_k = \bigcup_{d_i \geq k} Z_i,$$

since each Z_i is closed, S_k must be closed, as we desired. □

Recurring Theme: Semicontinuous

Semicontinuity is an important notion in algebraic geometry. Here are some examples.

(i) The rank of a matrix of functions (because rank drops on closed subsets, where various discriminants vanish),

(ii) the dimension of a variety at a point (Proposition 13.2.6),

(iii) fiber dimension (Theorem **??**),

(iv) the rank of a finite type quasicoherent sheaf (**??**),

(v) the degree of a finite morphism, as function of the target (§**??**),

(vi) the dimension of tangent space at closed points of a variety over an algebraically closed field (**??**), and

(vii) the rank of cohomology groups of coherent sheaves, in proper flat families (the Semicontinuity Theorem **??**).

All but (i) are upper semicontinuous; (i) is a lower semicontinuous function.

Proposition 13.2.7

Suppose p is a closed point of a locally finite type k-scheme X. Then the following three integers are the same:

(i) $\dim_X p$ *(Definition 13.2.5);*

(ii) $\dim \mathscr{O}_{X,p}$*;*

(iii) $\operatorname{codim}_X p$*.*

Remark 13.27 We define the **Zariski topology on the closed points of a finite type k-scheme** in the obvious way: Closed subsets are cut out by equations, as in §9.4.2.

Proof As p is a closed point, by Proposition 13.1.9, we have $\operatorname{codim}_X p = \dim \mathscr{O}_{X,p}$. Hence it suffices to show that $\dim_X p = \dim \mathscr{O}_{X,p}$. We may assume that X is affine, let $X = \operatorname{Spec} A$ where A is finitely generated k-algebra (moreover X is Noetherian). Say $p = [\mathfrak{m}]$, where $\mathfrak{m}$ is maximal ideal, then $\dim \mathscr{O}_{X,p} = \dim A_{\mathfrak{m}} = \operatorname{ht}(\mathfrak{m})$. Let $V(\mathfrak{p}_1), \ldots, V(\mathfrak{p}_r)$ be the irreducible components of X which contain $p = [\mathfrak{m}]$, then $\mathfrak{p}_i$ is the minimal prime belong to $\mathfrak{m}$. Hence $\operatorname{ht}(\mathfrak{m}) = \max_{1\leq i\leq r}\{\operatorname{ht}(\mathfrak{m}/\mathfrak{p}_i)\}$. Recall Theorem 13.2.4, we have

$$\dim A/\mathfrak{p}_i = \dim(A/\mathfrak{p}_i)/(\mathfrak{m}/\mathfrak{p}_i) + \dim(A/\mathfrak{p}_i)_{\mathfrak{m}/\mathfrak{p}_i} = \dim A/\mathfrak{m} + \operatorname{ht}(\mathfrak{m}/\mathfrak{p}_i).$$

Since $\mathfrak{m}$ is maximal ideal, $A/\mathfrak{m}$ is field, and therefore $\dim A/\mathfrak{m} = 0$, so $\dim A/\mathfrak{p}_i = \dim V(\mathfrak{p}_i) = \operatorname{ht}(\mathfrak{m}/\mathfrak{p}_i)$. Hence

$$\dim \mathscr{O}_{X,p} = \operatorname{ht}(\mathfrak{m}) = \max_{1\leq i\leq r}\{\operatorname{ht}(\mathfrak{m}/\mathfrak{p}_i)\} = \max_{1\leq i\leq r}\{\dim V(\mathfrak{p}_i)\} = \dim_X(p),$$

as we desired. □

Proposition 13.2.8 (Dimension of locally finite type k-scheme is preserved by any field extension)

Suppose X is a locally finite type k-scheme of pure dimension n, and K/k is a field extension (not necessarily algebraic). Then X_K has pure dimension n.

Proof Since X has pure dimension n, every irreducible component of X has the same dimension, so we may reduce to the case that X is an irreducible affine k-scheme of finite type. Suppose $X = \operatorname{Spec} A$ where A is finitely generated k-algebra. Clearly X is Noetherian, we may assume that $X = \bigcup_{i=1}^{m} \operatorname{Spec} A/\mathfrak{p}_i$ where each $\operatorname{Spec} A/\mathfrak{p}_i$ is irreducible component of X. We want to show that X_K has pure dimension n, it suffices to show that each $\operatorname{Spec} A/\mathfrak{p}_i \times_k \operatorname{Spec} K$ has pure dimension n. So we reduce to the case that A is an integral domain, i.e., $X = \operatorname{Spec} A$ is an irreducible affine k-variety.

Suppose $\mathcal{S} = \{S \subseteq K : S$ is algebraically independent over $k\}$, and define the partial order relation on $\mathcal{S}$ as set inclusion $\subseteq$. Clearly $\varnothing$ is algebraically independent (no elements exist to satisfy an algebraic relation), so $\varnothing \in \mathcal{S}$. Let $\{S_i\}_{i\in I}$ be a totally ordered subset in $\mathcal{S}$. We want to show that $\bigcup\limits_{i\in I} S_i \in \mathcal{S}$. If not, there exists $t_1, \dots, t_r$ algebraic dependence, since $\{S_i\}$ is a totally ordered subset there exists S_j (j sufficient large) such that $t_1, \dots, t_r \in S_j$, hence S_j not in $\mathcal{S}$, a contradiction. Hence any totally ordered subset in $\mathcal{S}$ has upper bound. By Zorn's Lemma, we know that there exists a maximal element $B \in \mathcal{S}$. Let $E = k(B)$, then E/k is purely transcendental and K/E is algebraic extension. Recall Proposition 13.1.8, it suffices to show the case that K/k is purely transcendental. Say $K = k(\{e_i\}_{i\in I})$ where $\{e_i\}_{i\in I}$ is transcendence basis for K/k. Note that

$$\begin{aligned} A \otimes_k K =& A \otimes_k k[\{e_i\}_{i\in I}]_{(0)} \\ \cong& A \otimes_k k[\{e_i\}] \otimes_{k[\{e_i\}]} k[\{e_i\}_{i\in I}]_{(0)} \\ \cong& A[\{e_i\}] \otimes_{k[\{e_i\}]} k[\{e_i\}_{i\in I}]_{(0)} \\ \cong& S^{-1}A[\{e_i\}], \end{aligned}$$

where S is the image of $k[\{e_i\}]\setminus\{0\}$ in $A[\{e_i\}]$. Since A is integral domain, $A_K := A\otimes_k K \cong S^{-1}A[\{e_i\}]$ is integral domain. Hence, by Theorem 13.2.2, we have $\dim X = \operatorname{tr.deg}_k K(A)$ and $\dim X_K = \operatorname{tr.deg}_K K(A_K)$. Let $\{t_j\}_{j=1}^n$ be the transcendence basis for $K(A)/k$, then $K(A)/k(\{t_j\})$ is algebraic extension. Note that $K(A_K) = K(A)(\{e_i\})$ and $K = k(\{e_i\})$, we have $K(A_K)/k(\{e_i\} \cup \{t_j\})$ is algebraic. We next show that $\{t_j\}$ is a transcendence basis for $K(A_K)/K$. By our above discussion $K(A_K)$ is algebraic over $K(\{t_j\}) = k(\{e_i\} \cup \{t_j\})$, it suffices to show that $\{t_j\}$ is algebraic independence over K. If not, there exists $g \in K[x_1, \dots, x_n]$ such that $g(t_1, \dots, t_n) = 0$, by clear the denominator, we may assume that $g(t_1, \dots, t_n) \in k[\{e_i\}][t_1, \dots, t_n] = k[\{e_i\} \cup \{t_j\}]$. Since $\{e_j\} \cup \{t_j\}$ is algebraic independence over k, g must be 0, a contradiction. So $\{t_j\}$ is algebraic independence over K. Hence $\{t_j\}$ is the transcendence basis for $K(A_K)/K$, and therefore $\operatorname{tr.deg}_K K(A_K) = n$, then we have $\dim X = \dim X_K$. $\square$

Remark 13.28 Proposition 13.2.8 is conceptually important. For example, if X is described by some questions with $\mathbb{Q}$-coefficients, the dimension of X doesn't depend on whether we consider it as a $\mathbb{Q}$-scheme or as a $\mathbb{C}$-scheme.

Remark 13.29 Unlike Proposition 13.1.8, Proposition 13.2.8 has finite type hypotheses on X. It is not true that if X is an arbitrary k-scheme of pure dimension n, and K/k is arbitrary extension, then X_K necessarily has pure dimension n. For example, Exercise 11.3 shows that if k is a field, then $\dim k(x) \otimes_k k(y) = 1$.

13.2.4 ⋆ Lines on hypersurfaces, part 1

Notice: Although dimension theory is not central to the following statement, it is essential to the proof.

✍ **Exercise 13.4 (Most surfaces in three-space of degree $d > 3$ have no lines)** In this exercise, we work over an algebraically closed field $\overline{k}$. For any $d > 3$, show that most degree d surfaces in $\mathbb{P}^3$ contain no lines. Here, "most" means "all closed points of a Zariski-open subset of the parameter space for degree d homogeneous polynomials in four variables, up to scalars." As there are $\binom{d+3}{3}$ such monomials, the degree d hypersurfaces are parametrized by $\mathbb{P}^{\binom{d+3}{3}-1}$.

Remark 13.30 If you do the previous exercise, your dimension count will suggest the true facts that degree 1 hypersurfaces — i.e., hyperplanes — have two-dimensional families of lines, and that most degree 2 hypersurfaces have one-dimensional families (rulings) of lines, as shown in Exercise 10.8. They will also suggest that most degree 3 hypersurfaces contain a finite number of lines, which reflects the celebrated fact that regular cubic surfaces over an algebraically closed field always contain 27 lines (Theorem **??**), and we will use this "incidence correspondence" or "incidence variety" to prove it (§**??**). The statement about quartic surfaces generalizes to the Noether-Lefschetz Theorem, implying that a very general surface of degree d at least 4 contains no curves that are not the intersection of the surface with a hypersurface. "Very general" means that in the parameter space (in this case, the projective space parametrizing surfaces of degree d), the statement is true away from a countable union of proper Zariski-closed subsets. Like "general" (which was defined in §11.3.4), "very general" is a weaker version of the phrase "almost every."

13.3 Krull's Theorems

We next introduce a number of results that we will repeatedly use, relating functions and the codimension 1 points where they vanish. The key result is Krull's Principal Ideal Theorem, which will be geometrically believable, and whose proof is short yet somehow nonintuitive and algebraic. This section will require Noetherian hypotheses.

13.3.1 Krull's Principal Ideal Theorem

In a vector space, a single linear equation always cuts out a subspace of codimension 0 or 1 (and codimension 0 occurs only when the equation is 0). The Principal Ideal Theorem, due to W. Krull, generalizes this linear algebra fact.

Theorem 13.3.1 (Krull's Principal Ideal Theorem (geometric version))

Suppose X is a locally Noetherian scheme, and f is a function. The irreducible components of $V(f)$ are codimension 0 or 1.

This is clearly a consequence of the following algebraic statement.

Theorem 13.3.2 (Krull's Principal Ideal Theorem (algebraic version))

Suppose A is a Noetherian ring, and $f \in A$. Then every prime $\mathfrak{p}$ minimal among those containing f has codimension at most 1. If furthermore f is not a zerodivisor, then every such prime $\mathfrak{p}$ containing f has codimension precisely 1.

For example, locally principal closed subschemes have "codimension 0 or 1",and effective Cartier divisors have "pure codimension 1". Here is another example.

Proposition 13.3.1

An irreducible homogeneous polynomial in $n+1$ variables over a field k describes an integral scheme of dimension $n-1$ in $\mathbb{P}_k^n$.

Proof Let $R = k[x_0, \ldots, x_n]$ and f is an irreducible homogeneous polynomial. Let $X = \operatorname{Proj} R/(f)$. Since f is irreducible, $R/(f)$ is an integral domain. Hence X is integral scheme. We next show that $\dim X = n-1$. Consider $\dim R/(f)$. Since R is Noetherian ring and f is not a zerodivisor, we know that $\operatorname{ht}((f)) = 1$, by Krull's Principal Ideal Theorem 13.3.2. By Theorem 13.2.4 and Corollary 13.2.2, $\dim R/(f) + \operatorname{ht}((f)) = \dim R$, so $\dim \operatorname{Spec} R/(f) = \dim R/(f) = \dim R - 1 = n + 1 - 1 = n$. Recall Proposition 13.2.4, we know that $\dim X = \dim \operatorname{Spec} R/(f) - 1 = n - 1$. $\square$

⋆⋆ Proof of Krull's Principal Ideal Theorem 13.3.2

Proof We dispose of the last sentence first: if $\mathfrak{p}$ is codimension 0, then it is an associated prime (Proposition 7.6.15), and hence any element of $\mathfrak{p}$ (including f) is a zerodivisor (Proposition 7.6.11).

We next wish to show that $\operatorname{ht}(\mathfrak{p}) \leq 1$. We reduce to the case. We reduce to the case of a local ring: replacing A by $A_{\mathfrak{p}}$ and $\mathfrak{p}$ by $\mathfrak{p}A_{\mathfrak{p}}$ does not change the problem, so we may assume that $\mathfrak{p}$ is a maximal ideal.

Suppose otherwise that $\mathfrak{q}$ is a strictly smaller prime ideal of A (so $f \notin \mathfrak{q}$ by the minimality of $\mathfrak{p}$ among prime ideals containing f). We wish to show that $\mathfrak{q}$ has codimension 0.

The key idea of the proof is the following clever construction:

Lemma 13.3.1

*Let $\varphi : A \to A_{\mathfrak{q}}$ be the localization, and define $\mathfrak{q}^{(n)} := \varphi^{-1}(\mathfrak{q}^n A_{\mathfrak{q}})$. (This is called the n-**th symbolic power of** $\mathfrak{q}$, but we will not use this terminology.) Note that $\mathfrak{q}^{(1)} \supseteq \mathfrak{q}^{(2)} \supseteq \cdots$ is a descending sequence of ideals of A. If $ag \in \mathfrak{q}^{(n)}$ and $g \notin \mathfrak{q}$, then $a \in \mathfrak{q}^{(n)}$ (i.e., $\mathfrak{q}^{(n)}$ is $\mathfrak{q}$-primary).*

Proof Let $ag \in \mathfrak{q}^{(n)}$ with $g \notin \mathfrak{q}$. Then $\varphi(ag) = \frac{ag}{1} \in \mathfrak{q}^n A_{\mathfrak{q}}$. Hence there exists $s \in A \setminus \mathfrak{q}$ such that $sga \in \mathfrak{q}^n$. Since $g \notin \mathfrak{q}$ and $s \notin \mathfrak{q}$, we have $sg \in A \setminus \mathfrak{q}$, and therefore $\frac{a}{1} \in \mathfrak{q}^n A_{\mathfrak{q}}$. Hence $a \in \mathfrak{q}^{(n)}$. □

Now $\sqrt{(f)} \subseteq A$ is the intersection of the minimal prime ideals containing f, which is precisely the maximal ideal $\mathfrak{p}$. Thus in the local ring $A/(f)$, the nilradical ideal is the maximal ideal, so by Proposition 13.1.11 (on dimension 0 Noetherian local rings), any descending chain of ideals of $A/(f)$ eventually stabilizes. In particular, the image of $\mathfrak{q}^{(1)} \supseteq \mathfrak{q}^{(2)} \supseteq \cdots$ in $A/(f)$ eventually stabilizes, so

$$\mathfrak{q}^n \equiv \mathfrak{q}^{n+1} \pmod{f} \quad \text{for } n \geq n_0 \text{ sufficiently large.} \tag{13.2}$$

We next claim that in fact $\mathfrak{q}^{(n)} = \mathfrak{q}^{(n+1)} + f\mathfrak{q}^{(n)}$ for $n \geq n_0$. Reason: Suppose $b_n \in \mathfrak{q}^{(n)}$. Then by (13.2), $b_n = b_{n+1} + af$ for $b_{n+1} \in \mathfrak{q}^{(n+1)} \subseteq \mathfrak{q}^{(n)}$ and $a \in A$. Hence $af \in \mathfrak{q}^{(n)}$; but $f \notin \mathfrak{q}$, by Lemma 13.3.1, $a \in \mathfrak{q}^{(n)}$, so $\mathfrak{q}^{(n)} = \mathfrak{q}^{(n+1)} + f\mathfrak{q}^{(n)}$ as claimed.

Thus by Nakayama's Lemma 9.2.4 (using A, $M = \mathfrak{q}^n/\mathfrak{q}^{n+1}$, $I = (f)$), we know that $\mathfrak{q}^{(n)}/\mathfrak{q}^{(n+1)} = 0$, i.e., $\mathfrak{q}^{(n)} = \mathfrak{q}^{(n+1)}$, from which $\mathfrak{q}^n A_{\mathfrak{q}} = \mathfrak{q}^{n+1} A_{\mathfrak{q}}$. By Nakayama's Lemma 9.2.4 (using $A_{\mathfrak{q}}$, $M = \mathfrak{q}^n A_{\mathfrak{q}}$, $I = \mathfrak{q}A_{\mathfrak{q}}$), $\mathfrak{q}^n A_{\mathfrak{q}} = 0$, so by Proposition 13.1.11, $A_{\mathfrak{q}}$ has dimension 0, so $\operatorname{ht}(\mathfrak{q}) = 0$ in A, as desired. □

We now use Krull's Principal ideal Theorem to prove some geometric facts. We start with a fun fact in preparation for Proposition 13.3.3 (a), which will also be very useful in other circumstances.

Proposition 13.3.2 (Important)

Suppose k is a field (possibly finite!), and $n \in \mathbb{Z}^+$. Let $\{p_1, \ldots, p_m\} \subseteq \mathbb{P}_k^n$ be a collection of points, not necessarily closed. Then there exists a nonempty hypersurface (of some unspecified degree) not containing any of the p_i.

Proof We show this by induction on m. If $m = 1$, we want to find a hypersurface that not containing p_1. Pick $f \notin p_1$, then f does not vanish at p_1, and therefore $V(f)$ is what we want. Let $\{p_1, \ldots, p_m\} \subseteq \mathbb{P}_k^n$ be $m + 1$ point, by inductive conditions, there exists a hypersurface does not vanish on $p_1, \ldots, p_m$. If it doesn't vanish on p_{m+1}, we are done. Otherwise, call this hypersurface f_{m+1}, say $d_{m+1} = \deg f_{m+1}$. Do something similar with $m + 1$ replaced by i for each $1 \leq i \leq m$, we obtain f_i which doesn't vanish on $p_1, \ldots, \widehat{p_i}, \ldots, p_m$, say $d_i = \deg f_i$. Let $N = \prod_i d_i$, then all $g_i := (f_i)^{N/d_i}$ have the same degree N, let homogeneous polynomial $g = \sum_{i=1}^{n} g_1 \cdots \widehat{g_i} \cdots g_{m+1}$, we want to show that $g \notin p_i$ for all i. If there exist p_j such that $g \in p_j$, since p_j is homogeneous ideal, $g_1 \cdots \widehat{g_j} \cdots g_{m+1} \in p_j$, this is impossible since each g_k $(k \neq j)$ doesn't vanish on p_j. So

$g \notin p_i$ for all i, i.e., there exists a nonempty hypersurface not containing any of the p_i. □

Proposition 13.3.3 (Important!)

(a) ***(Hypersurfaces meet everything of dimension at least*** 1 ***in projective space, unlike in affine space.)*** *Suppose X is a closed subset of $\mathbb{P}^n_k$ of dimension at least 1, and H is a nonempty hypersurface in $\mathbb{P}^n_k$. Then H meets X.*

(b) Suppose $X \hookrightarrow \mathbb{P}^n_k$ is a closed subset of dimension r. Then any codimension r linear space meets X.

(c) Suppose $X \hookrightarrow \mathbb{P}^n_k$ is a closed subset of dimension r. There is an intersection of $r+1$ nonempty hypersurfaces missing X. If k is infinite, there is a codimension $r+1$ linear subspace missing X.

(d) If k is an infinite field, there is an intersection of r hyperplanes meeting X in a finite number of points.

Proof

(a) Say $S = k[x_0, \ldots, x_n]$. Let $H = \operatorname{Proj} S/(f)$ be a nonempty hypersurface cut by a homogeneous polynomial f. Let $X = \operatorname{Proj} S/I$ be a closed subset of $\mathbb{P}^n_k$ where I is a homogeneous ideal and $\dim X \geq 1$. Denote the affine cones $C(H) = \operatorname{Spec} S/(f)$ and $C(X) = \operatorname{Spec} S/I$. Let Z be an irreducible component of $C(X)$ which dimension is $\dim C(X)$, then $\dim(C(H) \cap C(X)) \geq \dim(Z \cap C(H))$. Now, consider $Z \cap C(H)$, we may assume that $Z = \operatorname{Spec} S/\mathfrak{p}$ where $\mathfrak{p}$ is a prime ideal of S. Say $A = S/\mathfrak{p}$. Let see what the intersection is,

$$Z \cap C(H) = \operatorname{Spec} S/\mathfrak{p} \cap \operatorname{Spec} S/(f) \cong \operatorname{Spec} S/(\mathfrak{p}, f) \cong \operatorname{Spec} A/(\overline{f}),$$

where $\overline{f}$ is the image of f in $S/\mathfrak{p}$. If $Z \subseteq C(H)$, then

$$\dim Z \cap C(H) = \dim Z = \dim C(X) = \dim X + 1 \geq 2.$$

If $Z \nsubseteq C(H)$, then $f \notin \mathfrak{p}$, and therefore $\overline{f}$ is nonzero and not a zerodivisor. Let Y be an irreducible component of $Z \cap C(H)$ which dimension is $\dim Z \cap C(H)$, let $\mathfrak{q}$ be the corresponding minimal prime ideal in $\operatorname{Spec} A/(\overline{f})$, by Krull's Principal Ideal Theorem 13.3.2 and Theorem 13.2.4,

$$\dim Z \cap C(H) = \dim Y = \dim Z - \operatorname{codim}_Z Y = \dim Z - \operatorname{ht}(\mathfrak{q}) = \dim Z - 1 \geq 1,$$

where $\mathfrak{q}$ can be regarded as a prime ideal in $\operatorname{Spec} A$ so Y is also an irreducible closed subset of Z. Hence $\dim C(H) \cap C(X) \geq \dim Z \cap C(H) \geq 1$. Since f is homogeneous polynomial and I is homogeneous ideal, $C(H \cap X) = C(H) \cap C(X)$, so $\dim C(H \cap X) = \dim C(H) \cap C(X)$. If $H \cap X = \varnothing$, then $C(H \cap X) = C(H) \cap C(X)$ is just origin, hence $\dim C(H \cap X) = 0$, this contradicts to the fact that $\dim C(H \cap X) \geq 1$. Hence hypersurfaces meet everything of dimension at least 1 in projective space.

(b) Say $S = k[x_0, \ldots, x_n]$. Recall Definition 10.3.2, codimension r linear space L is a closed subscheme cut out by r k-linearly independent homogeneous linear polynomials in $x_0, \ldots, x_n$. By linear transformation, we may assume that $L = \operatorname{Proj} S/(x_0, \ldots, x_{r-1})$. Let $X = \operatorname{Proj} S/I$ be a closed subset of dimension r where I is a homogeneous polynomial. Denote the affine cones $C(L) = \operatorname{Spec} S/(x_0, \ldots, x_{r-1})$ and $C(X) = \operatorname{Spec} S/I$. Since origin is contained in $C(L)$ and $C(X)$, $L \cap X = \varnothing$ if and only if $\dim C(X \cap L) = \dim C(X) \cap C(L) = 0$. Hence it suffices to show that $\dim C(X) \cap C(L) \geq 1$. Let Z be an irreducible component of $C(X)$. Note that $\dim C(X) \cap C(L) \geq \dim Z \cap C(L)$, it suffices to show that $\dim Z \cap C(L) \geq 1$. Note that $C(L) = V(x_0) \cap \cdots \cap V(x_{r-1})$, say $H_i = V(x_i)$, then $C(L) = \bigcap_{i=0}^{r-1} H_i$. Say $Z_0 := Z$, denote $Z_{i+1} = Z_i \cap H_i$. By Krull's Principal Ideal Theorem 13.3.2, we have

$$\dim Z_{i+1} = \dim Z_i - \operatorname{ht}((f_i)) \geq \dim Z_i - 1,$$

so

$$\dim Z \cap C(L) = \dim Z_r \geq \dim Z_0 - r = \dim C(X) - r = \dim X + 1 - r = 1,$$

as we desired.

(c) Suppose $X = \operatorname{Proj} S/I$ where $S = k[x_0, \ldots, x_n]$ and I is a homogeneous ideal, then S/I is Noetherian, so X is Noetherian. We may assume that $Z_0 := X = \bigcup_{i=1}^{k_0} Z_{0,i}$. Let $\eta_{0,i}$ be the generic point of $Z_{0,i}$, by Proposition 13.3.2, there exists a nonempty hypersurface H_0 not containing any of the $\eta_{0,i}$. By the proof in part (a),

$$\dim Z_0 \cap H_0 = \dim Z_0 - 1 = r - 1.$$

Denote $Z_1 = Z_0 \cap H_0$. Repeat above process, we have $\dim Z_r = 0$. Note that $Z_r = X \cap H_0 \cap \cdots \cap H_{r-1}$, we know that the intersection of H_i cuts out finite number of points of X, namely $\{p_1, \ldots, p_l\}$. By Proposition 13.3.2, there exists a nonempty hypersurface H_r not containing any of p_i. So $X \cap \bigcap_{i=0}^{r} H_r = \varnothing$, i.e., there is an intersection of $r+1$ nonempty hypersurfaces missing X.

If k is infinite. If we show that there is a hyperplane not containing any generic point of a component of X, then we can get results by using the same process as above. Since X is Noetherian, we may assume that $X = \bigcup_{i=1}^{l} Z_i$ where Z_i is irreducible component of X. Note that a hyperplane

$$a_0x_0 + a_1x_1 + \cdots + a_nx_n = 0$$

in $\mathbb{P}_k^n$ is determined by a point $[a_0 : a_1 : \cdots : a_n] \in (\mathbb{P}_k^n)^\vee$. Let

$$V_i = \{[a_0 : a_1 : \cdots : a_n] \in (\mathbb{P}_k^n)^\vee : Z_i \subseteq V(a_0x_0 + a_1x_1 + \cdots + a_nx_n)\},$$

then V_i is linear subspace of $(\mathbb{P}_k^n)^\vee$, moreover $V_i \subsetneq (\mathbb{P}_k^n)^\vee$. Since k is infinite, $\bigcup_{i=1}^{l} V_i \subsetneq (\mathbb{P}_k^n)^\vee$, by linear algebra. Hence there exists $[k_0 : \cdots : k_n] \in (\mathbb{P}_k^n)^\vee \setminus \bigcup_{i=1}^{l} V_i$, it follows that there exists a hyperplane doesn't contain any Z_i, i.e., there is a hyperplane not containing any generic point of a component of X, as we desired.

(d) Use the same methods of (b) and (c), this is clear.

□

Remark 13.31

(1) **??** generalizes Proposition 13.3.3 (b) to show that any two things in projective space that you would expect to meet for dimensional reasons do in fact meet.

(2) We will see in **??** that if $k = \overline{k}$, for "most" choices of these r hyperplanes, this intersection is reduced, and in **??** that the number of points is the "degree" of X. But first of cause we must define "degree".

The proof of Prime avoidance 13.2.5 is similar to that of Proposition 13.3.3, and you may wish to solve it first. We will find it useful shortly.

Proof of Prime avoidance 13.2.5

Proof We prove this by induction. It is clearly true for $n = 1$, If $n > 1$ and the result is true for $n - 1$, then $I \nsubseteq \bigcup_{j \neq i} \mathfrak{p}_j$, so for each i there exists $x_i \in I$ such that $x_i \notin \mathfrak{p}_j$ whenever $j \neq i$. If for some i we have $x_i \notin \mathfrak{p}_i$, then we done. If not, then $x_i \in \mathfrak{p}_i$ for all i. Consider the element

$$y = \sum_{i=1}^{n} x_1 \cdots \widehat{x_i} \cdots x_n,$$

we have $y \in I$ and $y \notin \mathfrak{p}_i$ for all i. Hence $I \nsubseteq \bigcup_{i=1}^{n} \mathfrak{p}_i$. □

Proposition 13.3.4 (Bound on codimension of intersections in $\mathbb{A}_k^n$)

Let $k = \overline{k}$ be an algebraically closed field. Suppose X and Y are irreducible closed subvarieties (possibly singular) of codimension m and n respectively in $\mathbb{A}_k^d$. Then every component of $X \cap Y$ has codimension at most $m+n$ in $\mathbb{A}_k^n$.

Proof Recall Proposition 12.2.1, $\mathbb{A}_k^d \cong \Delta$ is a closed subscheme of $\mathbb{A}_k^d \times_k \mathbb{A}_k^d$. We first show that $\Delta \subseteq \mathbb{A}_k^d \times_k \mathbb{A}_k^d$ is a regular embedding of codimension d. Say $\mathbb{A}_k^d \times_k \mathbb{A}_k^d \cong \mathbb{A}_k^{2d} = k[x_1, \ldots, x_d, y_1, \ldots, y_d]$. Clearly, Δ is given by ideal $I = (x_1 - y_1, \ldots, x_d - y_d)$. Recall Definition 10.5.3 and Definition 10.5.4, for each point $\mathfrak{p} \in \mathbb{A}_k^{2d}$, in the local ring $k[x_1, \ldots, x_d, y_1, \ldots, y_d]_{\mathfrak{p}}$ is generated by a regular sequence $\frac{x_1 - y_1}{1}, \ldots, \frac{x_d - y_d}{1}$, so $\Delta \subseteq \mathbb{A}_k^d \times_k \mathbb{A}_k^d$ is a regular embedding of codimension d. By Lemma 12.2.1 (or direct check), we have $X \cap Y \cong \Delta \cap (X \times_k Y)$. Say $X = \operatorname{Spec} k[x_1, \ldots, x_d]/J_X$ and $Y = \operatorname{Spec} k[y_1, \ldots, y_d]/J_Y$, then $X \times_k Y \cong \operatorname{Spec} k[x_1, \ldots, x_d, y_1, \ldots, y_d]/(J_X, J_Y)$ where regard J_X, J_Y as ideals of $k[x_1, \ldots, x_d, y_1, \ldots, y_d]$ (in fact, $J_X = J_X k[x_1, \ldots, x_d, y_1, \ldots, y_d]$, $J_Y = J_Y k[x_1, \ldots, x_d, y_1, \ldots, y_d]$). Hence

$$\begin{aligned} X \cap Y \cong & \Delta \cap (X \times_k Y) \\ & \cong \operatorname{Spec} k[x_1, \ldots, x_d, y_1, \ldots, y_d]/(J_X, J_Y, (x_1 - y_1, \ldots, x_d - y_d)) \\ & \cong \operatorname{Spec}(k[x_1, \ldots, x_d, y_1, \ldots, y_d]/(J_X, J_Y))/(\overline{x_1 - y_1}, \ldots, \overline{x_d - y_d}), \end{aligned}$$

i.e., $X \cap Y$ is cut out in $X \times_k Y$ by d equations. By Krull's Principal Ideal Theorem 13.3.2, we know that $\operatorname{codim}_{X \times_k Y} X \cap Y \leq d$. By Theorem 13.2.4 and Proposition 13.2.3,

$$\begin{aligned} \dim X \cap Y \geq & \dim X \times_k Y - \operatorname{codim}_{X \times_k Y} X \cap Y \\ = & \dim X + \dim Y - \operatorname{codim}_{X \times_k Y} X \cap Y \\ \geq & d - m + d - n - d \\ = & d - (m+n). \end{aligned}$$

Apply Theorem 13.2.4 again, we have

$$\operatorname{codim}_{\mathbb{A}_k^d} X \cap Y \leq \dim \mathbb{A}_k^d - \dim X \cap Y \leq d - d + (m+n) = m+n.$$

□

Remark 13.32 The hypothesis of irreducibility can be replaced by pure dimensionality by appropriately generalizing Proposition 13.2.3. The hypothesis $k = \overline{k}$ can be removed by suitably invoking Proposition 13.1.8. See **??** for a generalization.

Proposition 13.3.5 (Generalize Proposition 13.3.3 (b))

Suppose X and Y are pure dimension closed subvarieties of $\mathbb{P}^n$ of codimensions d and e respectively, and $d + e \leq n$. Then X and Y intersect.

Proof Since X, Y are pure dimensional, so we may assume that X, Y are irreducible. Denote $C(X)$ and $C(Y)$ are the affine cones of X, Y respectively. Then

$$\dim C(X) = \dim X + 1 = \dim \mathbb{P}_k^n - \operatorname{codim}_{\mathbb{P}_k^n}(X) + 1 = n - d + 1,$$

hence $\operatorname{codim}_{\mathbb{A}_k^{n+1}} C(X) = \dim \mathbb{A}_k^{n+1} - (n-d+1) = d$. Similarly, $\operatorname{codim}_{\mathbb{A}_k^{n+1}} C(Y) = e$. By Proposition 13.3.4, we have

$$\operatorname{codim}_{\mathbb{A}_k^{n+1}} C(X) \cap C(Y) \leq d + e \leq n.$$

By Theorem 13.2.4,

$$\dim C(X) \cap C(Y) \geq \dim \mathbb{A}_k^{n+1} - \operatorname{codim}_{\mathbb{A}_k^{n+1}} C(X) \cap C(Y) \geq n + 1 - n = 1,$$

so

$$\dim X \cap Y = \dim C(X \cap Y) - 1 = \dim C(X) \cap C(Y) - 1 \geq 0,$$

i.e., X and Y intersect. □

Proposition 13.3.6 (Useful)

Suppose f is an element of a Noetherian ring A, contained in no codimension 0 or 1 prime ideal. Then f is invertible.

Proof Recall that Proposition 5.3.6, it suffices to show that f vanishes nowhere. By Krull's Principal Ideal Theorem 13.3.2, we know that every prime $\mathfrak{p}$ minimal among those containing f has codimension at most 1. By hypotheses, we know that f doesn't contained in any prime ideal of A, so f vanishes nowhere, as we desired. □

13.3.2 A useful characterization of unique factorization domains

We can use Krull's Principal Ideal Theorem to prove one of the four useful criteria for unique factorization domains, promised in §6.4.2.

Proposition 13.3.7

Suppose that A is a Noetherian integral domain. Then A is a UFD if and only if all codimension 1 prime ideals are principal.

Remark 13.33 This contains Lemma 13.1.3 and (in some sense) its converse.

Proof One direction is Lemma 13.1.3: if A is a unique factorization domain, then all codimension 1 prime ideals are principal. So we assume conversely that all codimension 1 prime ideals of A are principal, and we will show that A is a unique factorization domain. Recall that if each non-unit nonzero element of a ring can be factored into a finite number of irreducible elements and each irreducible elements is prime then this ring is UFD (you can find its proof in any algebra book on Earth), we need only to prove these two things. Now each codimension 1 prime ideal, being principal, is generated by a prime element, which is unique up to multiplication by a unit. All prime elements of an integral domain are irreducible, by the usual argument. (Suppose $p \in A$ is prime, and $p = ab$, with $a, b \in A \setminus \{0\}$. Then by primality of the ideal (p), either a or b is in (p), say, WLOG $a = kp$. Then $p = kpb$, from which $1 = bk$, from which b is unit. Thus p is irreducible.) Conversely, if $a \in A$ is irreducible, then by Krull's Principal Ideal Theorem 13.3.2, $V(a)$ contains some codimension 1 points $[(p)]$, so $a \in (p)$, say $a = kp$. But by irreducibility of a, as p is not a unit, k must be a unit, from which $(a) = (p)$, so a is prime. Thus we have shown that the prime elements are the irreducible elements (and, incidentally, that up to multiplicative units, they correspond to the codimension 1 prime ideals).

Since A is a Noetherian integral domain, each non-unit nonzero element can be factored into a finite number of irreducible elements, so we are done. (We give a short proof: Let $\mathcal{S}$ be a set of non-unit nonzero elements which can not factored into a finite number of irreducible elements, then there is a maximal element in $\mathcal{S}$, denote s. Consider (s), since s is reducible, we may assume $s = ab$ where a, b are not units, then $(s) \subsetneq (a)$, this contradicts to the maximality of (s). Hence $\mathcal{S} = \varnothing$.) $\square$

13.3.3 Using Krull's Principal Ideal Theorem inductively

The following result will allow us to use Krull's Principal Ideal Theorem inductively.

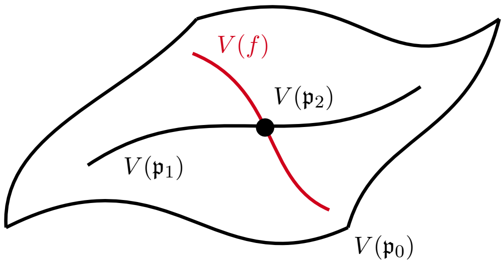
Lemma 13.3.2

Suppose $\mathfrak{p}_0 \subsetneq \mathfrak{p}_1 \subsetneq \cdots \subsetneq \mathfrak{p}_n$ is a strict chain of prime ideals in a Noetherian ring A, and $f \in \mathfrak{p}_n \setminus \mathfrak{p}_0$. Then there exists a strict chain of prime ideals

$$\mathfrak{p}_0 = \mathfrak{q}_0 \subsetneq \mathfrak{q}_1 \subsetneq \cdots \subsetneq \mathfrak{q}_{n-1} \subsetneq \mathfrak{q}_n = \mathfrak{p}_n.$$

in A, with $f \in \mathfrak{q}_1 \setminus \mathfrak{q}_0$.

Remark 13.34 We are given Spec A, and a nested sequence of irreducible closed subsets, as well as a "hypersurface" containing the smallest closed subset. We are going to create a new sequence of irreducible closed subsets, with the same start and finish, which are all (after the first) contained in the hypersurface.



Old chain: $V(\mathfrak{p}_0) \supsetneq V(\mathfrak{p}_1) \supsetneq V(\mathfrak{p}_2)$

New chain: $V(\mathfrak{q}_0) = V(\mathfrak{p}_0) \supsetneq V(\mathfrak{q}_1) = V(f) \supsetneq V(\mathfrak{q}_2) = V(\mathfrak{p}_2)$

Figure 13.3: The case $n = 2$, $V(\mathfrak{p}_0)$ is a surface, $V(\mathfrak{p}_1)$ is a curve, $V(\mathfrak{p}_2)$ is a point, and f is irreducible.

Proof Our plan is to create the ideals as follows:

$$\begin{tikzcd}
\mathfrak{p}_0 \arrow[r, hook, "\neq"] \arrow[dr, hook, "\neq"] & \mathfrak{p}_1 \arrow[r, hook, "\neq"] \arrow[dr, hook, "\neq"] & \cdots \arrow[r, hook, "\neq"] \arrow[dr, hook, "\neq"] & \mathfrak{p}_{n-2} \arrow[r, hook, "\neq"] \arrow[dr, hook, "\neq"] & \mathfrak{p}_{n-1} \arrow[r, hook, "\neq"] \arrow[dr, hook, "\neq"] & \mathfrak{p}_n \arrow[d, equal] \\
f \in & \mathfrak{q}_1 \arrow[r, hook, "\neq"'] & \cdots \arrow[r, hook, "\neq"'] & \mathfrak{q}_{n-2} \arrow[r, hook, "\neq"'] & \mathfrak{q}_{n-1} \arrow[r, hook, "\neq"'] & \mathfrak{q}_n
\end{tikzcd}$$

Unsurprisingly, we do this by descending induction. We prove the following. Suppose $\mathfrak{p}_{j-2} \subsetneq \mathfrak{p}_{j-1} \subsetneq \mathfrak{p}_j$ with $f \in \mathfrak{p}_j \setminus \mathfrak{p}_{j-2}$. We seek a prime ideal $\mathfrak{q}_{j-1}$ containing f, with $\mathfrak{p}_{j-2} \subsetneq \mathfrak{q}_{j-1} \subsetneq \mathfrak{p}_j$. Take $\mathfrak{q}_{j-1}$ to be a prime containing $(f, \mathfrak{p}_{j-2})$, and minimal among those contained in $\mathfrak{p}_j$. Now $\mathfrak{q}_{j-1}$ is codimension 1 in $A/\mathfrak{p}_{j-2}$ by Krull's Principal Theorem 13.3.2, and $\mathfrak{p}_j$ is codimension at least 2 (we have a chain $\mathfrak{p}_{j-2} \subsetneq \mathfrak{p}_{j-1} \subsetneq \mathfrak{p}_i$ of length 2). Thus $\mathfrak{q}_{j-1} \subsetneq \mathfrak{p}_j$. $\square$

Proposition 13.3.8

Suppose $(A, \mathfrak{m})$ is a Noetherian local ring, and $f \in \mathfrak{m}$. Then $\dim A/(f) \geq \dim A - 1$. *If f is not zero divisor (regular), then* $\dim A/(f) = \dim A - 1$.

Proof Since A is a Noetherian local ring, say $\dim A = n$, there exist a chain of prime ideals of length n, say

$$\mathfrak{p}_0 \subsetneq \mathfrak{p}_1 \subsetneq \mathfrak{p}_2 \subsetneq \cdots \subsetneq \mathfrak{p}_n = \mathfrak{m}.$$

If $f \in \mathfrak{p}_0$, then we have a strict chain of prime ideals

$$\mathfrak{p}_0/(f) \subsetneq \mathfrak{p}_1/(f) \subsetneq \mathfrak{p}_2/(f) \subsetneq \cdots \subsetneq \mathfrak{p}_n/(f) = \mathfrak{m}/(f).$$

So $\dim A/(f) \geq n = \dim A \geq \dim A - 1$.

If $f \notin \mathfrak{p}_0$, by Lemma 13.3.2, we may find a new chain of prime ideals

$$\mathfrak{p}_0 = \mathfrak{q}_0 \subsetneq \mathfrak{q}_1 \subsetneq \cdots \subsetneq \mathfrak{q}_{n-1} \subsetneq \mathfrak{q}_n = \mathfrak{m}$$

with $f \in \mathfrak{q}_1 \setminus \mathfrak{q}_0$. Consider $A/(f)$,

$$\mathfrak{q}_1 \subsetneq \cdots \subsetneq \mathfrak{q}_{n-1} \subsetneq \mathfrak{q}_n$$

is a strict chain of prime ideals in $A/(f)$, so $\dim A/(f) \geq n - 1 = \dim A - 1$, as we desired.

Let $\overline{\mathfrak{p}_0} \subsetneq \overline{\mathfrak{p}_1} \subsetneq \cdots \subsetneq \overline{\mathfrak{p}_r}$ be any chain of prime ideals in $A/(f)$, then it corresponds to a chain

$$\mathfrak{p}_0 \subsetneq \mathfrak{p}_1 \subsetneq \cdots \subsetneq \mathfrak{p}_r,$$

where $(f) \subseteq \mathfrak{p}_i$ for all $0 \leq i \leq r$. Since f is not zerodivisor, for each i, $\mathfrak{p}_i$ is not associated prime (Proposition 7.6.11). By Proposition 7.6.15, $\mathfrak{p}_0$ must containing a minimal prime, say $\mathfrak{p}$, then we have a chain

$$\mathfrak{p} \subsetneq \mathfrak{p}_0 \subsetneq \mathfrak{p}_1 \subsetneq \cdots \subsetneq \mathfrak{p}_r.$$

Hence, $r + 1 \leq \dim A = n$, i.e., $r \leq n - 1$, so $\dim A/(f) \leq n - 1$. It follows that $\dim A/(f) = \dim A - 1$, as we desired. □

The following generalization of Krull's Principal Ideal Theorem looks like it might follow by induction from Krull, but it is more subtle.

Theorem 13.3.3 (Krull's Height Theorem)

Suppose $X = \operatorname{Spec} A$, where A is Noetherian, and Z is an irreducible component of $V(f_1, \ldots, f_r)$, where $f_1, \ldots, f_r \in A$. Then the codimension of Z is at most r.

Proof Since Z is an irreducible component of $\operatorname{Spec} A/(f_1, \ldots, f_r)$, then $Z = \operatorname{Spec} A/\mathfrak{p}$ where $\mathfrak{p}$ is a minimal prime ideal contains $(f_1, \ldots, f_r)$. Note that $\operatorname{codim}_X Z = \operatorname{ht}(\mathfrak{p}) = \dim A_\mathfrak{p}$, it suffices to show that $\dim A_\mathfrak{p} \leq r$. Since A is Noetherian, $A_\mathfrak{p}$ is Noetherian local ring with maximal ideal $\mathfrak{p}A_\mathfrak{p}$. Since $\mathfrak{p}$ is a minimal prime ideal containing $(f_1, \ldots, f_r)$, we know that $\mathfrak{p}A_\mathfrak{p}$ is a minimal prime ideal containing $(f_1, \ldots, f_r)A_\mathfrak{p}$. So we may reduce to the case that $(A, \mathfrak{m})$ is a Noetherian local ring with $\mathfrak{m}$ is a minimal prime ideal containing $(f_1, \ldots, f_r)$, we want to show that $\dim A \leq r$. By Proposition 13.3.8, we have $\dim A/(f_1) \geq \dim A - 1$. Consider $A/(f_1)$, it is also a Noetherian local ring with maximal ideal $\mathfrak{m}/(f_1)$, and also $(\overline{f_2}, \ldots, \overline{f_r}) \subseteq \mathfrak{m}/(f_1)$. Repeating above process, we have

$$\dim A/(f_1, \ldots, f_r) \geq \dim A/(f_1, \ldots, f_{r-1}) - 1 \geq \cdots \geq \dim A - r.$$

Since $\mathfrak{m}$ is a minimal prime ideal containing $(f_1, \ldots, f_r)$, $\operatorname{Spec} A/(f_1, \ldots, f_r) = \{\mathfrak{m}/(f_1, \ldots, f_r)\}$, so

$$\dim A/(f_1, \ldots, f_r) = 0,$$

hence $\dim A \leq r$. □

Proposition 13.3.9 (Important)

Suppose $(A, \mathfrak{m})$ is a Noetherian local ring.

(a) (Noetherian local rings have finite dimension.) If there are $g_1, \ldots, g_l$ such that $V(g_1, \ldots, g_l) = \{[\mathfrak{m}]\}$, then $\dim A \leq l$.

(b) Let $d = \dim A$. Then there exist $g_1, \ldots, g_d \in A$ such that $V(g_1, \ldots, g_d) = \{[\mathfrak{m}]\}$.

Proof

(a) Clearly $V(\mathfrak{m})$ is an irreducible component of $V(g_1, \dots, g_l)$, by Krull's Height Theorem 13.3.3, we have $\operatorname{codim}_{\operatorname{Spec} A} V(\mathfrak{m}) = \operatorname{ht}(\mathfrak{m}) \leq l$. Note that in Noetherian local ring, we have $\dim A = \operatorname{ht}(\mathfrak{m})$, so $\dim A \leq l$.

(b) Prove it inductively. If $d = 0$, let $g_1 = 0$, then $V(g_1) = \{[\mathfrak{m}]\}$. Suppose $\dim A = d \geq 0$. Since A is Noetherian, then A has finite numbers of minimal prime, say $\mathfrak{p}_1, \dots, \mathfrak{p}_r$. Since $d > 0$, $\mathfrak{m} \supsetneqq \mathfrak{p}_i$ for all i, by Prime avoidance 13.2.5, there exists $g_1 \in \mathfrak{m}$ not belong to any $\mathfrak{p}_i$. Since A is Noetherian local ring, we may assume that

$$\mathfrak{p}_0 \subsetneqq \mathfrak{p}_1 \subsetneqq \cdots \subsetneqq \mathfrak{p}_d = \mathfrak{m}$$

is the maximal chain of prime ideals. Note that $g_1 \in \mathfrak{m} \setminus \mathfrak{p}_0$, apply Lemma 13.3.2, there exists a chain of prime ideals

$$\mathfrak{p}_0 = \mathfrak{q}_0 \subsetneqq \mathfrak{q}_1 \subsetneqq \cdots \subsetneqq \mathfrak{q}_d = \mathfrak{m}$$

with $g_1 \in \mathfrak{q}_1 \setminus \mathfrak{q}_0$. Consider $A/(g_1)$, it is also a Noetherian local ring with maximal ideal $\mathfrak{m}/(g_1)$. Since g_1 not contained in any minimal prime, each prime ideal which contains g_1 at least codimension 1, so $\dim A/(g_1) \leq \dim A - 1$. Since we already have a length $d-1$ strict chain

$$\mathfrak{q}_1/(g_1) \subsetneqq \cdots \subsetneqq \mathfrak{q}_d/(g_1) = \mathfrak{m}/(g_1)$$

in $A/(g_1)$, so $\dim A/(g_1) = d - 1$. By inductive hypotheses, there exists $g_2', \dots, g_d' \in A/(g_1)$ such that $V(g_2', \dots, g_d') = \{[\mathfrak{m}/(g_1)]\}$. Lift $g_2', \dots, g_d'$ to A, say $g_2, \dots, g_d$, then we have $V(g_1, \dots, g_d) = \{[\mathfrak{m}]\}$. $\square$

Remark 13.35 For comparison, Noetherian rings in general may have infinite dimension, see Example 13.3 .

The geometric translation of the Proposition 13.3.9 is the following. Given a d-dimensional "germ of a reasonable space" around a point p, p can be cut out set-theoretically by d equations, and we always need at least d equations. These d elements of A are called a **system of parameters at** p. Essentially equivalently:

Definition 13.3.1 (System of parameters)

*If $(A, \mathfrak{m})$ is a Noetherian local ring of dimension d, then $f_1, \dots, f_d$ is a **system of parameters** for $(A, \mathfrak{m})$ if $\mathfrak{m} = \sqrt{(f_1, \dots, f_d)}$.*

The following result will be central in the next chapter.

Theorem 13.3.4

For any Noetherian local ring $(A, \mathfrak{m}, k)$, $\dim A \leq \dim_k \mathfrak{m}/\mathfrak{m}^2$.

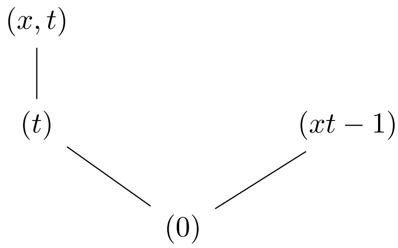
Proof Now $\mathfrak{m}$ is finitely generated , so $\mathfrak{m}/\mathfrak{m}^2$ is a finitely generated $(A/\mathfrak{m} = k)$-module, hence finite-dimensional. Say $\dim_k \mathfrak{m}/\mathfrak{m}^2 = n$. Choose a basis of $\mathfrak{m}/\mathfrak{m}^2$, and lift it to elements $f_1, \dots, f_n$ of $\mathfrak{m}$. By Nakayama's Lemma 9.2.6 (version 4), $(f_1, \dots, f_n) = \mathfrak{m}$. By Proposition 13.3.9, we have $\dim A \leq n$. $\square$

We conclude with two starred sections.

13.3.4 $\star$ Pathologies of the notion of "codimension"

We can use Krull's Principal Ideal Theorem to produce the example of pathology in the notion of codimension promised earlier in the chapter.

Let $A = k[x]_(x)[t]$. In other words, elements of A are polynomials in t, whose coefficients are quotients of polynomials in x, where no factors of x appear in the denominator. (**Warning:** A is not $k[x,t]_{(x)}$.) Clearly, A is an integral domain, so $xt-1$ is not a zero divisor. You can verify that $A/(xt-1) \cong k[x]_{(x)}[1/x] \cong k(x)$ — "in $k[x]_{(x)}$, we may divide by everything but x, and now we are allowed to divide by x as well" — so $A/(xt-1)$ is a field. Thus $(xt-1)$ is not just prime but also maximal. By Krull's Principal Ideal Theorem, $(xt-1)$ is codimension 1. Thus $(0) \subsetneq (xt-1)$ is a maximal chain. However, A has dimension at least 2: $(0) \subsetneq (t) \subsetneq (t,x)$ is a chain of prime ideals of length 2. (In fact, A has dimension precisely 2, although we don't need this fact in order to observe the pathology.) Thus we have a codimension 1 in a dimension 2 ring whose quotient is dimension 0, not 1. Here is a picture of this partially ordered set of ideals.



This example comes from geometry, and it is enlightening to draw a picture; see Figure 13.4. $\operatorname{Spec} k[x]_{(x)}$ corresponds to a "germ" of $\mathbb{A}^1_k$ near the origin, and $\operatorname{Spec} k[x]_{(x)}[t]$ corresponds to "this × the affine line". You may be able to see from the picture some motivation for this pathology — $V(xt-1)$ doesn't meet $V(x)$, so it can't have any specialization on $V(x)$, and there is nowhere else for $V(xt-1)$ to specialize. It is disturbing that this misbehavior turns up even in a relatively benign-looking ring.

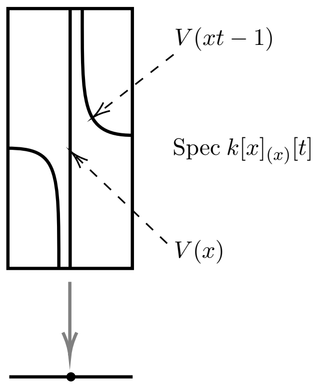


Figure 13.4: Dimension and codimension behave oddly on the surface $\operatorname{Spec} k[x]_{(x)}[t]$.

13.3.5 ⋆ Lines on hypersurfaces, part 2

(Part 1 was 13.2.4.) We now give a geometric application of Krull's Principal Ideal Theorem 13.3.2, applied through Proposition 13.3.3 (a). Throughout, we work over an algebraically closed field $\overline{k}$.

Proposition 13.3.10

(a) Suppose

$$f(x_0,\dots,x_n) = f_d(x_1,\dots,x_n) + x_0 f_{d-1}(x_1,\dots,x_n) + \cdots + x_0^{d-1} f_1(x_1,\dots,x_n) \qquad (13.3)$$

is a homogeneous degree d polynomial (so $\deg f_i = i$) cutting out a hypersurface X in $\mathbb{P}^n$ containing $p := [1:0:\cdots:0]$. Then there is a line through p contained in X if and only if $f_1 = f_2 = \cdots = f_d = 0$ has a common zero in $\mathbb{P}^{n-1} = \operatorname{Proj} \overline{k}[x_1,\dots,x_n]$.

(b) If $d \leq n-1$, then through any point $p \in X$, there is a line contained in X.

(c) If $d \geq n$, then for "most hypersurfaces" X of degree d in $\mathbb{P}^n$ (for all hypersurfaces whose corresponding point in the parameter space $\mathbb{P}^{\binom{n+1}{d}-1}$ — cf. Remark after Exercise 5.7 and Proposition 10.3.7 — lies in some nonempty Zariski-open subset), "most points $p \in X$" (all points in a nonempty dense Zariski-open subset of X) have no lines in X passing through them.

Proof

(a) Let $L \subseteq \mathfrak{p}^n$ be a line which contains p. Suppose L is determined by p and $q = [0 : a_1 : \cdots : a_n]$, then $L = \{[1 : \lambda a_1 : \cdots : \lambda a_n] : \lambda \in \overline{k}\}$. Substitute $[1 : \lambda a_1 : \cdots : \lambda a_n]$ into (13.3), we have

$$\begin{aligned} &f_d(\lambda a_1, \cdots, \lambda a_n) + f_{d-1}(\lambda a_1, \cdots, \lambda a_n) + \cdots + f_1(\lambda a_1, \cdots, \lambda a_n) \\ =&f_d(a_1, \ldots, a_n)\lambda^d + f_{d-1}(a_1, \ldots, a_n)\lambda^{d-1} + \cdots + f_1(a_1, \ldots, a_n)\lambda \end{aligned}$$

for all $\lambda \in \overline{k}$. Hence $L \subseteq X$ if and only if

$$f_1(a_1, \ldots, a_n) = f_2(a_1, \ldots, a_n) = \cdots = f_d(a_1, \ldots, a_n) = 0,$$

i.e., $f_1, \cdots, f_d$ has a common zero in $\mathbb{P}^{n-1}$.

(b) By linear transformation, we may assume $p = [1 : 0 : \cdots : 0]$ and X as part (a). By part (a), it suffices to show that $\dim V(f_1, \cdots, f_d) \geq 0$. Consider affine cone $CV(f_1, \cdots, f_d)$, by Krull's Height Theorem 13.3.3, we know that $\operatorname{codim}_{\mathbb{A}^n} CV(f_1, \cdots, f_d) \leq d$, so $\dim CV(f_1, \cdots, f_d) \geq n - d$, and therefore $\dim V(f_1, \cdots, f_d) \geq n - 1 - d$. Since $d \leq n - 1$, we have $\dim V(f_1, \cdots, f_d) \geq 0$, as we desired.

(c)

□

Remark 13.36 A projective (or proper) $\overline{k}$-variety X is **uniruled** if every point $p \in X$ is contained in some $\mathbb{P}^1 \subseteq X$. (We won't use this word beyond this remark.) Part (b) shows that all hypersurfaces of degree at most $n - 1$ are uniruled. One can show (using methods beyond what we know now; see [Kol13]) that if $\operatorname{char} \overline{k} = 0$, then every smooth hypersurface of degree at least $n + 1$ in $\mathbb{P}^n_{\overline{k}}$ is not uniruled (thus making the open set in (c) explicit). Furthermore, smooth hypersurfaces of degree n are uniruled, but covered by conics rather than lines. Thus there is a strong difference in how hypersurfaces behave depending on how the degree relates to $n+1$. This is true in many other ways as well. Smooth hypersurfaces of degree less than $n+1$ are examples of **Fano varieties**; smooth hypersurfaces of degree $n+1$ are examples of **Calabi-Yau varieties** (with the possible exception of $n = 1$, depending on the definition); and smooth hypersurfaces of degree greater than $n + 1$ are examples of **general type varieties**. We define these terms in §**??**.

13.4 Dimensions of fibers of morphisms of varieties

In this section, we show that the dimensions of fibers of morphisms of varieties behave in a way you might expect from our geometric intuition. The reason we have waited until now to discuss this because we will use Theorem 13.2.4 (for varieties, codimension is the difference of dimensions).

Before we begin, let's make sure we are on the same page with respect to our intuition. Elimination theory (Theorem 9.4.3) tells us that the projection $\pi : \mathbb{P}^n_A \to \operatorname{Spec} A$ is closed. We can interpret this as follows. A closed subset X of $\mathbb{P}^n_A$ is cut out by a bunch of homogeneous equations in $n+1$ variables (over A). The image of X is the subset of $\operatorname{Spec} A$ where these equations have a common nontrivial solution (cf. Example **??**). If we try hard enough, we can describe this by saying that the existence of a nontrivial solution (or the existence of a preimage of a point under $\pi : X \to \operatorname{Spec} A$) is an "upper semicontinuous" fact. More generally, you intuition might tell you that the locus where a number of homogeneous polynomials in $n+1$ variables over A have a solution space (in $\mathbb{P}^n_A$) of dimension at least d should be a closed subset of Spec A. (As a special case, consider linear equations. The condition for m linear equations in $n+1$ variables to have a solution space of dimension at least $d+1$ is a closed condition on the coefficients.) This intuition will be correct, and will use properness in a fundamental way (Theorem **??**). We will also make sense of upper semicontinuity in fiber dimension on the source (Theorem **??**). A useful example to think through is the map from the xy-plane to the xz-plane ($\operatorname{Spec} k[x,y] \to \operatorname{Spec} k[x,z]$), given by $(x,z) \mapsto (x,xy)$. (This example also came up in §9.4.1.)

We begin our substantive discussion with an inequality that holds more generally in the locally Noetherian setting.

Proposition 13.4.1 (Fiber dimensions can never be lower than expected)

Suppose $\pi : X \to Y$ is a morphism of locally Noetherian schemes, and $p \in X$ and $q \in Y$ are points such that $q = \pi(p)$. Then

$$\operatorname{codim}_X p \leq \operatorname{codim}_Y q + \operatorname{codim}_{\pi^{-1}(q)} p.$$

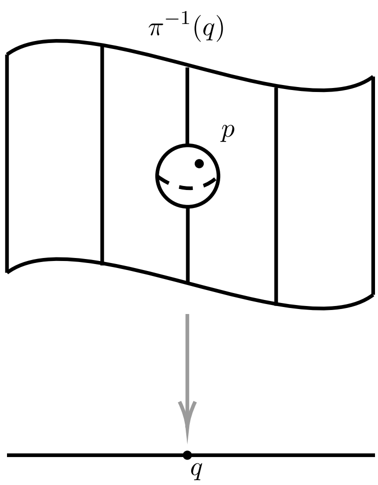


Figure 13.5: Proposition 13.4.1: The codimension of a point in the total space is bounded by the sum of the codimension of the point in the fiber and the codimension of the image in the target.

Proof By Proposition 13.1.9, we have $\operatorname{codim}_X p = \dim \mathscr{O}_{X,p}$, $\operatorname{codim}_Y q = \dim \mathscr{O}_{Y,q}$, and $\operatorname{codim}_{\pi^{-1}(q)} p = \dim \mathscr{O}_{\pi^{-1}(q),p}$. Suppose $p \in \operatorname{Spec} A \hookrightarrow X$ and $q \in \operatorname{Spec} B \hookrightarrow Y$, then $\mathscr{O}_{x,p} = A_p$ and $\mathscr{O}_{Y,q} = B_q$. Now we calculate $\mathscr{O}_{\pi^{-1}(q),p}$. Note that $p \in \operatorname{Spec} A \times_B \operatorname{Spec} \kappa(q) \hookrightarrow \pi^{-1}(q)$, we have $\mathscr{O}_{\pi^{-1}(q),p} \cong (A \otimes_B B_q/qB_q)_p$. Consider $(A \otimes_B B_q/qB_q)_p$,

$$\begin{aligned}(A \otimes_B B_q/qB_q)_p &\cong A_p \otimes_A (A \otimes_B B_q/qB_q)\\ &\cong A_p \otimes_B B_q/qB_q\\ &\cong A_p \otimes_B (B_q \otimes_{B_q} B_q/qB_q)\\ &\cong A_p \otimes_{B_q} B_q/qB_q\\ &\cong A_p/\varphi(qB_q)A_p\end{aligned}$$

where $A_p \otimes_B B_q \cong A_p$ and $\varphi : B_p \to A_q$ induced by $\pi : X \to Y$. Denote $R = A_p$, $\mathfrak{m} = pA_p$, $S = B_q$, and $\mathfrak{n} = qB_q$. Let $\dim \mathscr{O}_{Y,q} = n$. Since $\mathscr{O}_{Y,q} = S$ is a Noetherian local ring, by Proposition 13.3.9, there exits $f_1, \ldots, f_n \in S$ such that $\mathfrak{n} = \sqrt{(f_1, \ldots, f_n)}$. Let $\dim \mathscr{O}_{\pi^{-1}(q),p} = m$. Note that $\mathscr{O}_{\pi^{-1}(q),p} = R/\varphi(\mathfrak{n})R$, since R is Noetherian local ring, so is $R/\varphi(\mathfrak{n})R$, its maximal ideal is $\mathfrak{m}/\varphi(\mathfrak{n})R$. By Proposition 13.3.9, there exists $\overline{g_1}, \ldots, \overline{g_m} \in R/\varphi(\mathfrak{n})R$ such that $\mathfrak{m}/\varphi(\mathfrak{n})R = \sqrt{(\overline{g_1}, \ldots, \overline{g_m})}$, where $g_i \in R$. Consider $I = (\varphi(f_1), \ldots, \varphi(f_n), g_1, \ldots, g_m) \subseteq R$, we want to show that $V(I) = \{[\mathfrak{m}]\}$. Pick $\mathfrak{p} \in V(I)$, since $I \subseteq \mathfrak{p}$, $\varphi(f_i) \in \mathfrak{p}$, so $f_i \in \varphi^{-1}(\mathfrak{p})$, and therefore $(f_1, \ldots, f_n) \subseteq \varphi^{-1}(\mathfrak{p})$. Since $\varphi^{-1}(\mathfrak{p})$ is prime, take radical, we have $\mathfrak{n} = \sqrt{(f_1, \ldots, f_n)} = \varphi^{-1}(\mathfrak{p})$, which implies that $\varphi^{-1}(\mathfrak{n})R \subseteq \mathfrak{p}$. Let $\overline{\mathfrak{p}}$ be the image of $\mathfrak{p}$ in $R/\varphi(\mathfrak{n})R$, since $I \subseteq \mathfrak{p}$, we know that $(\overline{g_1}, \ldots, \overline{g_n}) \subseteq \overline{\mathfrak{p}}$, so $\overline{\mathfrak{p}} = \mathfrak{m}/\varphi(\mathfrak{n})R$. Hence $\mathfrak{p} = \mathfrak{m}$, which implies that $\{[\mathfrak{m}]\} = V(I)$. By Proposition 13.3.9, $\dim R = \dim \mathscr{O}_{X,p} \leq m + n$. That is,

$$\operatorname{codim}_X p \leq \operatorname{codim}_Y q + \operatorname{codim}_{\pi^{-1}(q)} p.$$

□

Remark 13.37 We will see that equality of Proposition 13.4.1 always holds for sufficiently nice — flat — morphisms; see Proposition **??**

We now show that the inequality of Proposition 13.4.1 is actually an equality over "most of Y".

Theorem 13.4.1

Suppose $\pi : X \to Y$ is a finite type dominant morphism of integral scheme such that the finitely generated field extension $K(X)/K(Y)$ has transcendence degree r. Then there exists a nonempty open subset $U \subseteq Y$ such that for all $q \in U$, the fiber over q is nonempty and has pure dimension r.

An interesting fact: We will use the Noetherian Normalization Lemma 13.2.3 in the proof, even though the statement doesn't involve varieties! If the generality of the statement bothers you, we have the following special case for varieties.

Corollary 13.4.1

Suppose $\pi : X \to Y$ is a (necessarily finite type) morphism of irreducible k-varieties, with $\dim X = m$ and $\dim Y = n$. Then there exists a nonempty open subset $U \subseteq Y$ such that for all $q \in U$, the fiber over q has pure dimension $m - n$, or is empty.

Remark 13.38 By convention, the empty set technically has pure dimension r — every irreducible component of the empty set has dimension r. This makes the statement of Corollary 13.4.1 a bit cleaner.

Proof If $\pi : X \to Y$ is not dominant. Let $U = Y \setminus \mathrm{cl}_Y(\pi(X))$, then for all $q \in U$, $\pi^{-1}(q) = \varnothing$.

If $\pi : X \to Y$ is dominant, by Proposition 8.5.4, π induces field extension $K(Y) \hookrightarrow K(X)$. By Theorem 13.2.2, we have $\dim X = \operatorname{tr.deg}_k K(X) = m$ and $\dim Y = \operatorname{tr.deg}_k K(Y) = n$, so

$$\operatorname{tr.deg}_{K(Y)} K(X) = \operatorname{tr.deg}_k K(X) - \operatorname{tr.deg}_k K(Y) = m - n.$$

By Theorem 13.4.1, there exists a nonempty open subset $U \subseteq Y$ such that for all $q \in U$, the fiber over q has pure dimension $m - n$. □

Proof of Theorem 13.4.1

Proof We begin with two quick reductions. (i) By shrinking Y if necessary, we may assume that Y is affine, say $\operatorname{Spec} B$. (ii) We may also assume that X is affine, say $\operatorname{Spec} A$. (Reason: Cover X with a finite number of affine open subsets $X_1, \ldots, X_a$, and take the intersection of the U's for each of the $\pi|_{X_i}$.)

In order to motivate the rest of the argument, we describe our goal. We will produce a nonempty distinguished open subset U of $\operatorname{Spec} B$ such that $\pi^{-1}(U) \to U$ factors through $\mathbb{A}^r_U$ via a finite surjective morphism:

$$\begin{tikzcd} X = & \operatorname{Spec} A & \pi^{-1}(U) \arrow[l, hook', "\text{open emb.}"'] \arrow[d, "\text{finite surj.}"] \\ & & \mathbb{A}^r_U \arrow[d] \\ Y = & \operatorname{Spec} B \arrow[from=1-2, "\pi"'] & U. \arrow[l, hook', "\text{open emb.}"'] \end{tikzcd} \tag{13.4}$$

Suppose $p \in U$ is a point, with residue field κ. Let Z be an irreducible component of $\pi^{-1}(p)$. Say $\varphi : \pi^{-1}(U) \to \mathbb{A}^r_U$, consider $\varphi|_Z : Z \to \mathbb{A}^r_\kappa$. Since φ is finite (thus integral), by Proposition 13.1.6, we have $\dim Z = \dim \varphi(Z) \leq \dim \mathbb{A}^r_\kappa = r$. We next show that $\dim Z = r$. Let $U = \operatorname{Spec} B_f$ where $f \in B$. Since $\varphi : \pi^{-1}(U) \to \mathbb{A}^r_U$ is finite (thus affine), we may assume $\pi^{-1}(U) = \operatorname{Spec} T$, and $B_f[t_1, \ldots, t_r] \to T$ is integral, since φ is surjective, $B_f[t_1, \ldots, t_r] \hookrightarrow T$ is integral extension. Consider the set $\mathcal{S} := \{\mathfrak{a} \in \operatorname{Spec} B : B_\mathfrak{a} \text{ is integrally closed}\}$, we shall to show that $\mathcal{S}$ is open. By Lemma 11.7.2, $B_\mathfrak{a}$ is integrally closed if and only if $\mathfrak{p} \not\supseteq \operatorname{Ann}_B(\widetilde{B}/B)$, it follows that $B_\mathfrak{a}$ is not integrally closed if and only if $\mathfrak{p} \in V(\operatorname{Ann}_B(\widetilde{B}/B))$. Hence $\mathcal{S}$ is open. Let $U \cap \mathcal{S}$, it is also open, so we may shrink U to a distinguished open subset U' such that $\Gamma(U', \mathscr{O}_{U'})$ is integrally closed. For simply, we may assume $U = \operatorname{Spec} B_f$, where B_f is integrally closed. Consider the following diagram.

$$\begin{tikzcd} \pi^{-1}(p) = & \operatorname{Spec} \kappa \otimes_{B_f} T \arrow[r] \arrow[d] & \operatorname{Spec} T \arrow[d] \\ & \operatorname{Spec} B_f[t_1, \ldots, t_r] \otimes_{B_f} \kappa \arrow[r] \arrow[d] & \operatorname{Spec} B_f[t_1, \ldots, t_r] \arrow[d] \\ p = & \operatorname{Spec} \kappa \arrow[r] & \operatorname{Spec} B_f \end{tikzcd}$$

In fact, $\kappa = (B_f)_p / p(B_f)_p$, so

$$\begin{aligned} \kappa \otimes_{B_f} T =& (B_f)_p / p(B_f)_p \otimes_{B_f} T \\ \cong& (B_f)_p / p(B_f)_p \otimes_{(B_f)_p} ((B_f)_p \otimes_{B_f} T) \\ \cong& (B_f)_p / p(B_f)_p \otimes_{(B_f)_p} T_p \\ \cong& T_p / p(B_f)_p T_p \\ \cong& T_p / pT_p. \end{aligned}$$

Hence the prime ideal of $\kappa \otimes_{B_f} T$ is one-to-one corresponding to prime ideal $\mathfrak{P}$ of T which satisfies $\mathfrak{P} \supseteq pT$ and $\mathfrak{P} \cap (B_f \setminus p) = \varnothing$. Let Z be an irreducible component of $\pi^{-1}(p)$, say $\eta \in \operatorname{Spec} T$ is corresponding to the generic point of Z, by above discussion, we know that $\eta \supseteq pT$. Claim that $\eta \cap B_f[t_1, \ldots, t_r] = p \cdot B_f[t_1, \ldots, t_r]$. Since $pT \subseteq \eta$, we have $p \cdot B_f[t_1, \ldots, t_r] \subseteq \eta \cap B_f[t_1, \ldots, t_r]$. If $p \cdot B_f[t_1, \ldots, t_r] \subsetneq \eta \cap B_f[t_1, \ldots, t_r]$, say $S = B_f[t_1, \ldots, t_r]$. Note that $S/pS \cong (B_f/p)[t_1, \ldots, t_r]$ is an integral domain, we know that pS is prime. Then we have a chain of prime ideals

$$(0) \subseteq pS \subsetneq \eta \cap S$$

in S. Since B_f is integrally closed, $S = B_f[t_1, \ldots, t_r]$ is integrally closed. Recall that $S \hookrightarrow T$ is integral extension. By Going-Down Theorem 13.2.5, there exists $\mathfrak{q} \in \operatorname{Spec} T$ such that

$$(0) \subseteq \mathfrak{q} \subseteq \eta$$

and $\mathfrak{q} \cap S = pS$. Since η is minimal prime which contains pT, we have $\eta = \mathfrak{q}$, thus $pS = \eta \cap S$, a contradiction.

So $\eta \cap S = pS$. Consider $S/S\cap\eta \to T/\eta$, clearly, $S\cap\eta$ is the kernel of $S \to T/\eta$, so $S/S\cap\eta \to T/\eta$ is injective. Since T is finite S-algebra, so T/η is finite $(S/S\cap\eta)$-algebra. Hence $S/S\cap\eta \to T/\eta$ is integral extension, and therefore $(S/S\cap\eta)\otimes_{B_f}\kappa \to (T/\eta)\otimes_{B_f}\kappa$ is integral extension. Hence

$$\begin{aligned}
\dim(T/\eta)\otimes_{B_f}\kappa &= \dim Z\\
&= \dim(S/S\cap\eta)\otimes_{B_f}\kappa\\
&= \dim(S/pS)\otimes_{B_f}\kappa\\
&= \dim\kappa[t_1,\dots,t_r] = r,
\end{aligned}$$

as we desired.

So we now work to build (13.4). We begin by noting that we have inclusions of B into both A and $K(B)$, and from both A and $K(B)$ into $K(A)$ (see Proposition 8.5.4). The maps from A and $K(B)$ into $K(A)$ both factor through $A\otimes_B K(B)$ (whose Spec is the generic fiber of π), so the maps from both A and $K(B)$ to $A\otimes_B K(B)$ must be inclusions.

$$\begin{tikzcd}
 & & K(A) \\
A \arrow[r, hook, "\therefore"'] \arrow[urr, hook, bend left] & A\otimes_B K(B) \arrow[ur, hook] & \\
B \arrow[u, hook] \arrow[r, hook] & K(B) \arrow[u, hook, "\therefore"] \arrow[uur, hook, bend right] &
\end{tikzcd} \tag{13.5}$$

Clearly, $K(A)\otimes_B K(B) = K(A)$ (as $A\otimes_B K(B)$ can be interpreted as taking A and inverting the elements that are nonzero elements of B), and $A\otimes_B K(B)$ is a finitely generated algebra over the field $K(B)$.

By Noetherian Normalization Lemma 13.2.3, we can find elements $t_1,\dots,t_r \in A\otimes_B K(B)$, algebraically independent over $K(B)$, such that $A\otimes_B K(B)$ is integral over $K(B)[t_1,\dots,t_r]$.

Now, we can think of the elements $t_i \in A\otimes_B K(B)$ as fractions, with numerators in A and (nonzero) denominators in B. If f is the product of the denominators appearing for each t_i, then by replacing B by B_f (replacing $\operatorname{Spec} B$ by its distinguished open subset $D(f)$), we may assume that the t_i are all in A. Thus (after sloppily renaming B_f as B, and A_f as A) we can trim and extend (13.5) to the following.

$$\begin{tikzcd}
A \arrow[r, hook] & A\otimes_B K(B) \\
B[t_1,\dots,t_r] \arrow[u, hook] \arrow[r, hook] & K(B)[t_1,\dots,t_r] \arrow[u, hook, "\text{integral}"'] \\
B \arrow[u, hook] \arrow[r, hook] & K(B) \arrow[u, hook]
\end{tikzcd}$$

Now A is finite generated over B, and hence over $B[t_1,\dots,t_r]$. But we cannot yet be sure that A is finite over $B[t_1,\dots,t_r]$, so we are not done. We will have to localize B further.

Suppose A is generated over B by $u_1,\dots,u_q$. Our Noether Normalization argument implies that each u_i satisfies some monic equation $f_i(u_i) = 0$, where $f_i \in K(B)[t_1,\dots,t_r][t]$ (i.e., monic considered as a polynomial in t with coefficients in $K(B)[t_1,\dots,t_r]$). The coefficients of f_i (considered as polynomials in $t_1,\dots,t_r,t$) are a priori fractions in B, but by multiplying by all those denominators, we can assume each $f_i \in B[t_1,\dots,t_r][t]$, at the cost of losing monicity of the f_i (this time considered as polynomials in t). Let $b\in B$ be the product of the leading coefficients (considered as polynomials in t) of all the f_i. If $U = D(b)$ (the locus where b is invertible), then, over U, the f_i (can be taken to) have leading coefficient 1, so the u_i (in A_b)

are integral over $B_b[t_1, \dots, t_r]$. Thus $\operatorname{Spec} A_b \to \operatorname{Spec} B_b[t_1, \dots, t_r]$ is finite and surjective (the latter by the Lying Over Theorem 9.2.2).

We have now constructed (13.5), as desired. □

Remark 13.39 There are a couple of things worth pointing out about the proof. First, although the result was originally motivated by classical varieties over a field k, the proof uses the theory of varieties over another field, the function field $K(B)$. This is an example of how the introduction of generic points to algebraic geometry is useful even for consider more "classical" question.

Second, the idea of the main part of the argument is that we have a result over the generic point ($\operatorname{Spec} A \otimes_B K(B)$ is finite and surjective over affine space over $K(B)$), and we want to "spread it out" to an open neighborhood of the generic point of $\operatorname{Spec} B$. We do this by realizing that "finitely many denominators" appear when correctly describing the problem, and inverting those denominators. This "spreading out" idea is a recurring theme.

Proposition 13.4.2 (Useful criterion for irreducibility)

Suppose $\pi : X \to Y$ is a proper morphism to an irreducible variety, and all the fibers of π are nonempty, and irreducible of the same dimension. Then X is irreducible.

Proof If X is reducible, we may write $X = Z_1 \cup Z_2$ where Z_1, Z_2 are irreducible components of X. Since all the fibers of π are nonempty, π is surjective, since π is proper (and thus closed), we have $Y = \pi(Z_1) \cup \pi(Z_2)$. Since Y is irreducible, we may assume that $\pi(Z_1) = Y$. Restrict π to Z_1. For any $y \in Y$, by Proposition 13.4.1 and Corollary 13.4.1, we know that $\dim \pi^{-1}(y) \cap Z_1 \geq \dim Z_1 - \dim Y$ with equality on some open subset of Y. Say $\dim \pi^{-1}(y) = d$, then $\dim \pi^{-1}(y) \cap Z_1 \geq d$. On the other hand, since $\pi^{-1}(y)$ is irreducible and $Z_1 \cap \pi^{-1}(y)$ is closed in $\pi^{-1}(y)$, we know that $\dim \pi^{-1}(y) \cap Z_1 \leq d$, hence $\pi^{-1}(y) \cap Z_1 = \pi^{-1}(y)$. Since

$$X = \bigcup_{y \in Y} \pi^{-1}(y) = \bigcup_{y \in Y} (\pi^{-1}(y) \cap Z_1) \subseteq Z_1,$$

we know that $X = Z_1$, a contradiction. Hence X is irreducible. □

Theorem 13.4.2 (Upper semicontinuity of fiber dimension)

Suppose $\pi : X \to Y$ is a morphism of finite type k-schemes.

(a) (upper semicontinuity on the source) The dimension of the fiber of π at $p \in X$ (the dimension of the largest component of $\pi^{-1}(\pi(p))$ containing p) is an upper semicontinuous function in p (i.e., on X).

(b) (upper semicontinuity on the target) If furthermore π is closed (e.g., if π is proper), then the dimension of the fiber of π over $q \in Y$ is an upper semicontinuous function in q (i.e., on Y).

Remark 13.40 You should be able to immediately construct a counterexample to part (b) if the closedness hypothesis is dropped. (We also remark that Theorem 13.4.2 (b) for projective morphisms is done, in a simple way, in **??**.)

Proof **For (a).** Let F_n be the subset of X consisting of points where the fiber dimension is at least n, i.e., $F_n = \{p \in X : \dim \pi^{-1}(\pi(p)) \geq n\}$. We wish to show that F_n is a closed subset for all n. We argue by induction on $\dim Y$. If $\dim Y = 0$, then Y consists of finite numbers of closed points, so $\pi^{-1}(y), y \in Y$ is closed. In fact, $F_n = \bigcup_{\dim \pi^{-1}(y) \geq n} \pi^{-1}(y)$, so F_n is closed. Now we fix Y, and assume the result for all smaller-dimensional targets.

Recall Proposition 10.4.6, the reductions of X and Y have the same underlying topological space, so we may assume that X, Y are reduced. Let $Y = \bigcup_{i=1}^{l} Y_i$ (since Y is Noetherian), where each Y_i is irreducible component of Y, then $X = \bigcup_{i=1}^{l} \pi^{-1}(Y_i)$. Hence

$$F_n = \left\{ p \in \bigcup_{i=1}^{l} \pi^{-1}(Y_i) : \dim \pi^{-1}(\pi(p)) \geq n \right\} = \bigcup_{i=1}^{l} \{p \in \pi^{-1}(Y_i) : \dim \pi^{-1}(\pi(p)) \geq n\},$$

so it suffices to check that each $\{p \in \pi^{-1}(Y_i) : \dim \pi^{-1}(\pi(p)) \geq n\}$ is closed. Hence we may reduced to the case that Y is integral. Let $X = \bigcup_{i=1}^{l} X_i$ where X_i is irreducible component of X, then $\pi^{-1}(\pi(p)) = \bigcup_{i=1}^{l}(X_i \cap \pi^{-1}(\pi(p)))$, and thus $\dim \pi^{-1}(\pi(p)) = \max_{1 \leq i \leq l}\{\dim(X_i \cap \pi^{-1}(\pi(p)))\}$. Hence

$$F_n = \bigcup_{i=1}^{l} \{p \in X_i : \dim(X_i \cap \pi^{-1}(\pi(p))) \geq n\},$$

it suffices to show that each $\{p \in X_i : \dim(X_i \cap \pi^{-1}(\pi(p))) \geq n\}$ is closed, so we may reduced to the case that X is integral. Consider the scheme theoretic image of π, say Z. Let $\pi' := \pi : X \to Z$, then π' is dominant. Clearly $\pi'^{-1}(\pi'(p)) = \pi^{-1}(\pi(p))$. If π is not dominant, then $\dim Z < \dim Y$, by induction hypothesis, we are done. So it suffices to show that the case π is dominant.

Now our setting is: X, Y are integral and $\pi : X \to Y$ dominant. Let $r = \dim X - \dim Y$ be the "relative dimension" of π. If $n \leq r$, then $F_n = X$ by Proposition 13.4.1 (combined with Theorem 13.2.4, that codimension is the difference of dimensions for varieties).

If $n > r$, then let $U \subseteq Y$ be the dense open subset of Theorem 13.4.1, where "the fiber dimension is exactly r." Then F_n does not meet the preimage of U. By replacing Y with $Y \setminus U$ (and X by $X \setminus \pi^{-1}(U)$), we are done by the inductive hypothesis.

For (b). Let $G_n = \{q \in Y : \dim \pi^{-1}(q) \geq n\}$. We want to show that G_n is closed. Note that $G_n = \{q \in \pi(X) : \dim \pi^{-1}(q) \geq n\} = \pi(F_n)$, since π is closed and F_n is closed by part (a), we know that G_n is closed, as we desired. $\square$

Proposition 13.4.3 (Generically finite implies generally finite; cf. Proposition 11.3.4)

Suppose $\pi : X \to Y$ is a generically finite morphism of irreducible k-varieties of dimension n. Then there is a dense open subset $V \subseteq Y$ above which π is finite.

Remark 13.41 If you wish, you can later relax the irreducibility hypothesis to simply requiring X and Y to be of pure dimension n.

Proof As in the proof of Theorem 13.4.1, we may assume that Y is affine, let η be the generic point of Y, since $\pi^{-1}(\eta) = X \times_Y \kappa(\eta) \to \operatorname{Spec} \kappa(\eta) = \eta$ is finite morphism, by Proposition 9.3.13, we know that $\dim \pi^{-1}(\eta) = 0$. By Proposition 13.4.1, we have $\dim \pi^{-1}(\eta) \geq \dim X - \dim \pi(X)$ where $\pi(X)$ is scheme theoretic image of π, so $\dim X = \dim \pi(X) = \dim Y$. Since $\pi(X)$ is closed in Y, we have $Y = \pi(X)$, that is, π is dominant. Suppose that $X = \bigcup_{i=1}^{n} U_i$, where the U_i are affine open subschemes of X. By Proposition 11.3.4, there are dense open subsets $V_i \subseteq Y$ over which $\pi|_{U_i}$ is finite. By replacing Y by an affine open subset of $\bigcap V_i$, we may assume that $\pi|_{U_i}$ is finite.

We next show that π is closed. Let $Z \subseteq X$ be a closed subset, then $Z \cap U_i$ closed in U_i, since finite morphisms are closed, $\pi|_{U_i}(Z \cap U_i)$ are closed. Hence $\pi(Z) = \bigcup_{i=1}^{n} \pi|_{U_i}(Z \cap U_i)$ is closed, and thus π is closed.

Consider $X \setminus U_1$, it is closed in X, so $\pi(X \setminus U_1)$ is closed. We want to show that $\pi(X \setminus U_1)$ is not all of Y. Note that $\dim \pi(X \setminus U_1) \leq \dim(X \setminus U_1) < \dim X = \dim Y$, we know that $\pi(X \setminus U_1)$ is not all of Y.

Define $V := Y \setminus \pi(X \setminus U_1)$. Then π is finite above V: it is the restriction of the finite morphism $\pi|_{U_1} : U_1 \to Y$ to the open subset V of the target Y. $\square$

Chapter 14 Regularity and smoothness

One natural notion we expect to see for geometric spaces is the notion of when an object is "smooth". In algebraic geometry, this notion, called **regularity**, is easy to define (Definition **??**) but a bit subtle in practice. This will lead us to a different related notion of when a variety is smooth (Definition **??**).

This chapter has many moving parts, of which §14.1-14.6 are the important ones. In §14.1, the **Zariski tangent space** is motivated and defined. In §14.2, we define **regularity** and **smoothness over a field**, the central topics of this chapter, and discuss some of their important properties. In §14.3, we give a number of important examples, mostly in the form of exercises. In §14.4, we discuss Bertini's Theorem, a fundamental classical result. In §14.5, we give many characterizations of **discrete valuation rings**, which play a central role in algebraic geometry. Having seen clues that "smoothness" is a "relative" notion rather than an "absolute" one, in §14.6 we define **smooth morphisms** (and in particular, **étale morphisms**), and give some of their properties. (We will revisit this definition in §**??**, once we know more.)

The remaining sections are less central. In §14.7, we discuss the **valuative criteria for separatedness and properness**. In §14.8, we mention some more sophisticated facts about **regular local rings** (the local rings at regular points). In §14.9, we prove the Artin-Rees Lemma, because it will have been invoked in §14.5 (and will be used later as well).

14.1 The Zariski tangent space

14.1.1 Definition: Zariski tangent spaces

We begin by defining the tangent space of a scheme at a point. It behaves like the tangent space you know and love at "smooth" points, but also makes sense at other points. In other words, geometric intuition at the "smooth" point guides the definition, and then the definition guides the algebra at all points, which in turn lets us refine our geometric intuition.

The definition is short but surprising. The main difficulty is convincing yourself that it deserves to be called the tangent space. This is tricky to explain, because we want to show that it agrees with our intuition, but our intuition is worse than we realize. So you should just accept this definition for now, and later convince yourself that it is reasonable.

Definition 14.1.1 (Zariski (co)tangent space)

*(1) The **Zariski cotangent space** of a local ring $(A, \mathfrak{m})$ is defined to be $\mathfrak{m}/\mathfrak{m}^2$; it is a vector space over the residue field $A/\mathfrak{m}$. The dual vector space is the **Zariski tangent space**.*

*(2) If X is a scheme, the **Zariski cotangent space** $T^\vee_{X,p}$ at a point $p \in X$ is defined to be the Zariski cotangent space of the local ring $\mathscr{O}_{X,p}$, and similarly for the **Zariski tangent space** $T_{X,p}$.*

*(3) Elements of the Zariski cotangent spaces are called **cotangent vectors** or **differentials**; elements of the tangent space are called **tangent vectors**.*

Remark 14.1 Note that for any maximal ideal $\mathfrak{m}$ of any ring A, we have canonically $\mathfrak{m}/\mathfrak{m}^2 = \mathfrak{m}A_\mathfrak{m}/\mathfrak{m}^2A_\mathfrak{m}$, this implies that the tangent space is a local concept.

The cotangent space is more algebraically natural than the tangent space, in that the definition is shorter. There is a moral reason for this: the cotangent space is more naturally determined in terms of functions on a space, and we are very much thinking about schemes in terms of "functions on them." This will come up later.

Here are two plausibility arguments that this is a reasonable definition. Hopefully one will catch your fancy.

In differential geometry, the tangent space at point is sometimes defined as the vector space of derivations at that point. A derivation at a point p of a manifold is an $\mathbb{R}$-linear operation that takes in functions f near p (i.e., elements of $\mathscr{O}_p$), and outputs elements $f'(p)$ of $\mathbb{R}$, and satisfies the **Leibniz rule**

$$(fg)' = f'g + g'f.$$

(We will later define derivations in a more general setting, §**??**) A derivation is the same as a map $\mathfrak{m} \to \mathbb{R}$, where $\mathfrak{m}$ is the maximal ideal of $\mathscr{O}_p$. (The map $\mathscr{O}_p \to \mathbb{R}$ extends this, via the map $\mathscr{O}_p \to \mathfrak{m}$ given by $f - f(p)$.) But $\mathfrak{m}^2$ maps to 0, as if $f(p) = g(p) = 0$, then

$$(fg)'(p) = f'(p)g(p) + g'(p)f(p) = 0.$$

Thus a derivation induces a map $\mathfrak{m}/\mathfrak{m}^2 \to R$, i.e., an element of $(\mathfrak{m}/\mathfrak{m}^2)^\vee$.

Lemma 14.1.1

Any linear map $\mathfrak{m}/\mathfrak{m}^2 \to \mathbb{R}$ gives a derivation which holds Leibniz rule.

Proof Say $v : \mathfrak{m}/\mathfrak{m}^2 \to \mathbb{R}$. Define $D : \mathscr{O}_p \to \mathbb{R}$ by setting $D(f) = v((f - f(p)) \pmod{\mathfrak{m}^2})$, we shall to show that D holds Leibniz rule. Let $f, g \in \mathscr{O}_p$, set $A = f - f(p)$ and $B = g - g(p)$ (hence $A, B \in \mathfrak{m}$), then $f = A + f(p)$ and $g = B + g(p)$. Consider $D(fg)$, we have

$$\begin{aligned} D(fg) =& v((fg - f(p)g(p)) \pmod{\mathfrak{m}^2}) \\ =& v((AB + f(p)B + g(p)A) \pmod{\mathfrak{m}^2}) \\ =& f(p)v(B \pmod{\mathfrak{m}^2}) + g(p)v(A \pmod{\mathfrak{m}^2}) \\ =& f(p)D(g) + g(p)D(f), \end{aligned}$$

i.e., Leibniz rule holds. □

Here is a second, vaguer, motivation that this definition is plausible for the cotangent space of the origin of $\mathbb{A}^n$. Functions on $\mathbb{A}^n$ should restrict to a linear function on the tangent space. What (linear) function does $x^2 + xy + x + y$ restrict to "near the origin"? You will naturally answer: $x + y$. Thus we "pick off the linear terms." Hence $\mathfrak{m}/\mathfrak{m}^2$ are the linear functionals on the tangent space, so $\mathfrak{m}/\mathfrak{m}^2$ is the cotangent space. In particular, you should picture functions vanishing at a point (i.e., lying in $\mathfrak{m}$) as giving functions on the tangent space in this obvious way.

Example 14.1 (Old-fashioned example) Computing the Zariski tangent space is actually quite hands-on, because you can compute it just as you did when you learned multivariable calculus. In $\mathbb{A}^3$, we have a curve cut out by $x + y + z^2 + xyz = 0$ and $x - 2y + z + x^2y^2z^3 = 0$. (You can use Krull's Principal Ideal Theorem 13.3.2 to check that this is a curve, but it is not important to do so.) What is the tangent line near the origin? (Is it even smooth there?) Answer: The first surface looks like $x + y = 0$ and the second surface looks like $x - 2y + z = 0$. The curve has tangent line cut out by $x + y = 0$ and $x - 2y + z = 0$. It is smooth (in the traditional sense). In multivariable calculus, the students work hard to get the answer, because we aren't allowed to tell them to just pick out the linear terms.

Let's make explicit the fact that we are using. If A is a ring, $\mathfrak{m}$ is a maximal ideal, and $f \in \mathfrak{m}$ is a function vanishing at the point $[\mathfrak{m}] \in \operatorname{Spec} A$, then the Zariski tangent space of $\operatorname{Spec} A/(f)$ at $\mathfrak{m}$ is cut out in the Zariski tangent space of $\operatorname{Spec} A$ (at $\mathfrak{m}$) by the single linear equation $f \pmod{\mathfrak{m}^2}$. The next proposition will force you to think this through.

Proposition 14.1.1 (Important: Krull's Principal Ideal Theorem for tangent spaces)

Suppose A is a ring, and $\mathfrak{m}$ a maximal ideal. If $f \in \mathfrak{m}$, then the Zariski tangent space of $A/(f)$ is cut out in the Zariski tangent space of A by $f \pmod{\mathfrak{m}^2}$. Hence the dimension of the Zariski tangent space of $\operatorname{Spec} A/(f)$ at $[\mathfrak{m}]$ is the dimension of the Zariski tangent space of $\operatorname{Spec} A$ at $[\mathfrak{m}]$, or one less.

Proof Reduced to the case that A is a local ring with maximal ideal $\mathfrak{m}$. The Zariski cotangent space of $A/(f)$ is $\frac{\mathfrak{m}/(f)}{(\mathfrak{m}/(f))^2} \cong \frac{\mathfrak{m}/(f)}{(\mathfrak{m}^2+(f))/(f)} \cong \frac{\mathfrak{m}}{\mathfrak{m}^2+(f)}$. In fact, $\frac{\mathfrak{m}}{\mathfrak{m}^2+(f)} \cong \frac{\mathfrak{m}/\mathfrak{m}^2}{(f) \pmod{\mathfrak{m}^2}}$, so we have an exact sequence

$$(f) \pmod{\mathfrak{m}^2} \longrightarrow \mathfrak{m}/\mathfrak{m}^2 \longrightarrow \mathfrak{m}/(\mathfrak{m}^2+(f)) \longrightarrow 0.$$

Say $k = A/\mathfrak{m}$. Since $\square^\vee = \operatorname{Hom}(\square, k)$ is left exact functor, by applying $\square^\vee$, we have the following exact sequence

$$0 \longrightarrow (\mathfrak{m}/(\mathfrak{m}^2+(f)))^\vee \longrightarrow (\mathfrak{m}/\mathfrak{m}^2)^\vee \longrightarrow ((f) \pmod{\mathfrak{m}^2})^\vee.$$

Hence $(\mathfrak{m}/(\mathfrak{m}^2+(f)))^\vee$ is kernel of $\varphi : (\mathfrak{m}/\mathfrak{m}^2)^\vee \to ((f) \pmod{\mathfrak{m}^2})^\vee$, which implies the Zariski tangent space of $A/(f)$ is cut out in the Zariski tangent space of A by $f \pmod{\mathfrak{m}^2}$.

By Linear Algebra, $\dim(\mathfrak{m}/\mathfrak{m}^2)^\vee = \dim \operatorname{Im} \varphi + \dim \operatorname{Ker} \varphi$, since $\operatorname{Im} \varphi$ is the subspace of $((f) \pmod{\mathfrak{m}^2})^\vee$, we know that $\dim \operatorname{Im} \varphi = 0$ or 1. If $f \in \mathfrak{m}^2$, $\varphi = 0$, so $\dim \operatorname{Im} \varphi = 0$, and therefore $\dim(\mathfrak{m}/(\mathfrak{m}^2+(f)))^\vee = \dim \operatorname{Ker} \varphi = \dim(\mathfrak{m}/\mathfrak{m}^2)^\vee$. If $f \notin \mathfrak{m}^2$, then $\dim \operatorname{Im} \varphi = 1$, therefore $\dim(\mathfrak{m}/(\mathfrak{m}^2+(f)))^\vee = \dim \operatorname{Ker} \varphi = \dim(\mathfrak{m}/\mathfrak{m}^2)^\vee - 1$. □

Remark 14.2 That last sentence should be suitably interpreted if the dimension is infinite, although it is less interesting in this case.

Here is another example to see this principle in action, extending Example 14.1 :

Example 14.2 $x+y+z^2 = 0$ and $x+y+x^2+y^4+z^5 = 0$ cut out a curve, which obviously pass through the origin. If I asked my multivariable calculus students to calculate the tangent line to the curve at the origin, they would do calculations which would boil down (without their realizing it) to picking off the linear terms. They would end up with the equations $x+y=0$ and $x+y=0$, which cut out a plane, not a line. They would be disturbed, and I would explain that this is because the curve isn't smooth at a point, and their techniques don't work. We, on the other hand, bravely declare that the tangent space is cut out by $x+y=0$, and (will soon) use this pathology as definition of what makes a point singular (or nonregular). (Intuitively, the curve near the origin is very close to lying in the plane $x+y=0$.)

Remark 14.3 The cotangent space jumped up in dimension from what it was "supposed to be", not down. (This is not a coincidence: see Theorem 13.3.4.)

Proposition 14.1.2

Suppose Y and Z are closed subscheme of X, both containing the point $p \in X$.

(a) $T_{Z,p}$ is naturally a sub-$\kappa(p)$-vector space of $T_{X,p}$.

(b) $T_{Y \cap Z,p} = T_{Y,p} \cap T_{Z,p}$. (On the left, $\cap$ means scheme-theoretic intersection. On the right, $\cap$ means intersection of subspaces of the vector space $T_{X,p}$.)

(c) $T_{Y \cup Z,p}$ contains the span of $T_{Y,p}$ and $T_{Z,p}$, where $\cup$, as usual in this context, is scheme-theoretic union.

(d) $T_{Y \cup Z,p}$ can be strictly larger than the span of $T_{Y,p}$ and $T_{Z,p}$.

Proof Since the tangent space is defined locally, we may assume X is affine. Let $X = \operatorname{Spec} A$, $Z = \operatorname{Spec} A/I$, and $Y = \operatorname{Spec} A/J$. Say $p = [\mathfrak{p}]$.

For (a). The cotangent space of Z at point $[\mathfrak{p}]$ is
$$\frac{\mathfrak{p}(A/I)_\mathfrak{p}}{(\mathfrak{p}(A/I)_\mathfrak{p})^2} \cong \frac{\mathfrak{p}A_\mathfrak{p}}{(\mathfrak{p}A_\mathfrak{p})^2 + I_\mathfrak{p}} \cong \left(\frac{\mathfrak{p}A_\mathfrak{p}}{(\mathfrak{p}A_\mathfrak{p})^2}\right) \Big/ (I_\mathfrak{p} \pmod{(\mathfrak{p}A_\mathfrak{p})^2}).$$
The cotangent space of X at $[\mathfrak{p}]$ is $\frac{\mathfrak{p}A_\mathfrak{p}}{(\mathfrak{p}A_\mathfrak{p})^2}$, hence we have an exact sequence
$$I_\mathfrak{p} \pmod{(\mathfrak{p}A_\mathfrak{p})^2} \longrightarrow \frac{\mathfrak{p}A_\mathfrak{p}}{(\mathfrak{p}A_\mathfrak{p})^2} \longrightarrow \left(\frac{\mathfrak{p}A_\mathfrak{p}}{(\mathfrak{p}A_\mathfrak{p})^2}\right) \Big/ (I_\mathfrak{p} \pmod{(\mathfrak{p}A_\mathfrak{p})^2}) \longrightarrow 0$$
Taking dual, we have exact sequence
$$0 \longrightarrow T_{Z,p} \longrightarrow T_{X,p} \longrightarrow (I_\mathfrak{p} \pmod{(\mathfrak{p}A_\mathfrak{p})^2})^\vee,$$
so $T_{Z,p}$ is naturally a sub-$\kappa(p)$-vector space of $T_{X,p}$. Moreover, $T_{Z,p}$ is kernel of $T_{X,p} \to (I_\mathfrak{p} \pmod{(\mathfrak{p}A_\mathfrak{p})^2})^\vee$, and therefore
$$T_{Z,p} = \{v \in T_{X,p} : v(f) = 0, \forall f \in I_\mathfrak{p} \pmod{(\mathfrak{p}A_\mathfrak{p})^2}\}.$$

For (b). As in the part (a), we have
$$\begin{aligned} T_{Z,p} =& \{v \in T_{X,p} : v(f) = 0, \forall f \in I_\mathfrak{p} \pmod{(\mathfrak{p}A_\mathfrak{p})^2}\}, \\ T_{Y,p} =& \{v \in T_{X,p} : v(f) = 0, \forall f \in J_\mathfrak{p} \pmod{(\mathfrak{p}A_\mathfrak{p})^2}\}, \\ T_{Y\cap Z,p} =& \{v \in T_{X,p} : v(f) = 0, \forall f \in (I+J)_\mathfrak{p} \pmod{(\mathfrak{p}A_\mathfrak{p})^2}\}. \end{aligned}$$
Note that $v(f) = 0$ for all $f \in (I+J)_\mathfrak{p} \pmod{(\mathfrak{p}A_\mathfrak{p})^2}$ if and only if $v(f) = 0$ for all $f \in I_\mathfrak{p} \pmod{(\mathfrak{p}A_\mathfrak{p})^2}$ and $v(g) = 0$ for all g for all $g \in I_\mathfrak{p} \pmod{(\mathfrak{p}A_\mathfrak{p})^2}$, we have $T_{Y\cap Z,p} = T_{Z,p} \cap T_{Y,p}$.

For (c). As in part (a), we have
$$T_{Y\cup Z,p} = \{v \in T_{X,p} : v(f), \forall f \in (I \cap J)_\mathfrak{p} \pmod{(\mathfrak{p}A_\mathfrak{p})^2}\}.$$
Clearly, we have $T_{Y,p} \subseteq T_{Y\cup Z,p}$ and $T_{Z,p} \subseteq T_{Y\cup Z,p}$, so $\operatorname{Span}_{\kappa(p)}\{T_{Y,p}, T_{Z,p}\} \subseteq T_{Y\cup Z,p}$.

For (d). We give an example to show that $T_{Y\cup Z,p}$ can be strictly larger than the span of $T_{Y,p}$ and $T_{Z,p}$. Let $X = \mathbb{A}^2_\mathbb{C}$, $Y = \operatorname{Spec}\mathbb{C}[x,y]/(y-x^2)$, and $Z = \operatorname{Spec}\mathbb{C}[x,y]/(y)$, set p is the origin $o = [(x,y)]$, then
$$T_{Y,o} = \{v \in T_{X,o} : v(f) = 0, \forall f \in (y)_o \pmod{(o\mathbb{C}[x,y]_o)^2}\} = T_{Z,o}.$$
Hence $\operatorname{Span}_\mathbb{C}\{T_{Y,o}, T_{Z,o}\} = T_{Y,o}$. But
$$T_{Y\cup Z,o} = \{v \in T_{X,o} : v(f) = 0, \forall f \in (y(y-x^2)) \pmod{(o\mathbb{C}[x,y]_o)^2}\} = \{v \in T_{X,o} : v(0) = 0\} = T_{X,o},$$
we know that $\operatorname{Span}_\mathbb{C}\{T_{Y,o}, T_{Z,o}\} = T_{Y,o} \subsetneq T_{X,o} = T_{Y\cup Z,o}$. □

Here is a pleasant consequence of the notion of Zariski tangent space.

Problem 14.1 Consider the ring $A = k[x,y,z]/(xy - z^2)$. Show that (x,z) is not a principal ideal.

Solution *As $\dim A = \operatorname{tr.deg}_k K(A) = 2$, and $A/(x,z) \cong k[y]$ has dimension 1, we see that (x,z) is height 1 (as codimension is the difference of dimensions for irreducible varieties, Theorem 13.2.4). Our geometric picture is that $\operatorname{Spec} A$ is a cone (we can diagonalize the quadric as $xy - z^2 = ((x+y)/2)^2 - ((x-y)/2)^2 - z^2$, at least if $\operatorname{char} k \neq 2$ — see Exercise 6.7), and that (x,z) is a line on the cone. (See Figure 14.1 for a sketch.) This suggests that we look at the cone point.*

Let $\mathfrak{m} = (x,y,z)$ be the maximal ideal corresponding to the origin. Then $\operatorname{Spec} A$ has Zariski tangent space of dimension 3 at the origin, and $\operatorname{Spec} A/(x,z)$ has Zariski tangent space of dimension 1 at the origin. But $\operatorname{Spec} A/(f)$ must have Zariski tangent space of dimension at least 2 at the origin by Proposition 14.1.1. (As an added bonus, we have shown that A is not a unique factorization, by Lemma 13.1.3.)

Remark 14.4 Another approach to solving the problem, not requiring the definition of the Zariski tangent space, is to use the fact that the ring is graded (where x, y, and z each have degree 1), and the ideal (x,z) is a homogeneous ideal. The advantage of using the tangent space is that it applies to more general situations where

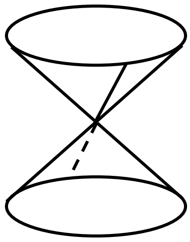


Figure 14.1: $V(x, z) \subseteq \operatorname{Spec} k[x, y, z]/(xy - z^2)$ is a line on a cone.

there is no grading. For example, (a) (x, z) is not a principal ideal of $k[x, y, z]/(xy - z^2 - z^3)$. As a different example, (b) (x, z) is not a principal ideal of the local ring $(k[x, y, z]/(xy - z^2))_{(x,y,z)}$ (the "germ of the cone"). However, we remark that the graded case is still very useful. The construction of replacing a filtered ring by its "associated graded" ring can turn more general rings into graded rings (and can be used to turn example (a) into the graded case). The construction of completion can turn local rings into graded local rings (and can be used to turn example (b) into, essentially, the graded case). Filtered rings will come up in §14.9, the associated graded construction will implicitly come up in our discussions of the blow-up in §**??**, and many aspects of completions will be described in Chapter **??**.

✍ **Exercise 14.1** Show that $(x, z) \subseteq k[w, x, y, z]/(wz - xy)$ is a codimension 1 ideal that is not principal. (See Figure 14.2 for the projectivization of this situation — a line on a smooth quadric surface.) This example was promised just after Exercise 6.3. An improvement is given in 16.17 : this closed subset $x = z = 0$ is not even set-theoretically cut out by one equation.

Proof Let $A = k[w, x, y, z]/(wz - xy)$, $\mathfrak{m} = (w, x, y, z)$. Since $wz - xy$ is not a zerodivisor on $k[w, x, y, z]$, by Krull's Principal Ideal Theorem 13.3.2, we know that $\dim A = \dim \mathbb{A}^4_k - 1 = 3$. Note that $A/(x, z) \cong k[w, y]$, we know that $\dim A/(x, z) = 2$. By Theorem 13.2.4, $\operatorname{ht}(x, z) = \dim A - \dim A/(x, z) = 1$, i.e., $(x, z) \subseteq k[w, x, y, z]/(wz - xy)$ is a codimension 1 ideal. We next show that (x, z) is not principal.

In fact, $T_{\operatorname{Spec} A,[\mathfrak{m}]} = (\mathfrak{m}/\mathfrak{m}^2)^\vee$, so $\dim_{A/\mathfrak{m}} T_{\operatorname{Spec} A,[\mathfrak{m}]} = 4$. Pick $f \in \mathfrak{m}$, by Proposition 14.1.1, we know that $\dim_{A/\mathfrak{m}} T_{\operatorname{Spec} A/(f),[\mathfrak{m}]} \geq 4 - 1 = 3$. Consider $T_{\operatorname{Spec} A/(x,z),[\mathfrak{m}]}$, since

$$\frac{\mathfrak{m}/(x, z)}{(\mathfrak{m}/(x, z))^2} \cong \frac{(w, y)}{(w, y)^2},$$

we know that $\dim_{A/\mathfrak{m}} T_{\operatorname{Spec} A/(x,z),[\mathfrak{m}]} = 2$. Hence $\dim T_{\operatorname{Spec} A/(x,z),[\mathfrak{m}]} < \dim T_{\operatorname{Spec} A/(f),[\mathfrak{m}]}$, which implies that (x, z) is not principal. □

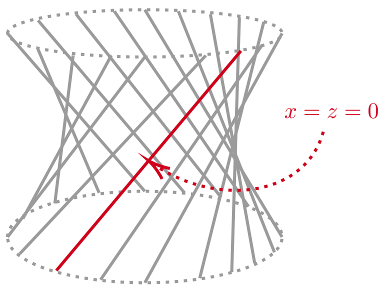


Figure 14.2: The line $V(x, z)$ on the smooth quadric surface $V(wz - xy) \subseteq \mathbb{P}^3$.

✍ **Exercise 14.2** Let $A = k[w, x, y, z]/(wz - xy)$. Show that $\operatorname{Spec} A$ is not factorial. (Exercise 6.9 shows that A is not a unique factorization domain, but this is not enough - why is the localization of A at the prime (w, x, y, z) not factorial? One possibility is to do this "directly," by trying to imitate the solution to Exercise 6.9, but this is hard. Instead, use the intermediate result that, in a unique factorization domain, any codimension 1 prime is principal, Lemma 13.1.3, and considering Exercise 14.1) As A is integrally closed if char $k \neq 2$ (Exercise 6.6

(c)), this yields an example of a scheme that is normal but not factorial, as promised in Proposition 6.4.6. A slight generalization will be given in **??**.

Proof Let $A = k[w,x,y,z]/(wz-xy)$, $\mathfrak{m} = (w,x,y,z)$. We claim that $A_{\mathfrak{m}}$ is not a UFD. Recall Exercise 14.1, we know that (x,z) is a codimension 1 prime ideal in $A_{\mathfrak{m}}$ and which is not principal. By Lemma 13.1.3, we know that $A_{\mathfrak{m}}$ is not a UFD, so $\operatorname{Spec} A$ is not factorial. $\square$

Exercise 14.3 (Less important exercise ("Higher-order data")) This exercise is fun, but we won't be used.

(a) In Exercise 4.22, you computed the equations cutting out the (union of the) three coordinate axes of $\mathbb{A}_k^3$. (Call this scheme X.) Your ideal should have had three generators. Show that the ideal cannot be generated by fewer than three elements. (Hint: Working modulo $\mathfrak{m} = (x,y,z)$ won't give any useful information, so work modulo a higher power of $\mathfrak{m}$.)

(b) Show that the coordinate axes in $\mathbb{A}_k^3$ are not regularly embedding in $\mathbb{A}_k^3$. (This was promised at the end of §10.5.)

Proof **(a)** Recall Exercise 4.22, we know that $X = \operatorname{Spec} k[x,y,z]/(xy,xz,yz)$. Say $I = (xy,xz,yz)$. Let $f_1,\dots,f_r$ be another generators of I, we want to show that $r \geq 3$. Say $\mathfrak{m} = (x,y,z)$, then $I \subseteq \mathfrak{m}^2$. Consider natural morphism $\varphi : I \to \mathfrak{m}^2/\mathfrak{m}^3$. Note that $\varphi(I) = \operatorname{Span}_k\{xy,xz,yz\}$, so $\dim_k \varphi(I) = 3$. Hence $\operatorname{rank}\{\varphi(f_1),\dots,\varphi(f_r)\} = 3$, it follows that $r \geq 3$.

(b). Since X is the union of three coordinates axes, we know that $\dim X = 1$. If $X \hookrightarrow \mathbb{A}_k^3$ is regularly embedding, it must be codimension $3-1=2$. By part (a), I generated by at least 3 elements, a contradiction. So the coordinate axes in $\mathbb{A}_k^3$ are not regularly embedding in $\mathbb{A}_k^3$. $\square$

14.1.2 Morphisms and tangent spaces

Suppose $\pi : X \to Y$, and $\pi(p) = q$. Then if we were in the category of differentiable manifolds, we would expect a tangent map, from the tangent space of p to the tangent space at q. Indeed that is the case; we have a map of stalks $\mathscr{O}_{Y,q} \to \mathscr{O}_{X,p}$, which sends the maximal ideal of the former $\mathfrak{n}$ to the maximal ideal of the latter $\mathfrak{m}$ (we checked that this is a "local morphism" when we briefly discussed locally ringed spaces; see §8.3.1). Thus $\mathfrak{n}^2$ maps to $\mathfrak{m}^2$, from which we see that $\mathfrak{n}/\mathfrak{n}^2$ maps to $\mathfrak{m}/\mathfrak{m}^2$. If $(\mathscr{O}_{X,p},\mathfrak{m})$ and $(\mathscr{O}_{Y,q},\mathfrak{n})$ have the same residue field κ, so that $\mathfrak{n}/\mathfrak{n}^2 \to \mathfrak{m}/\mathfrak{m}^2$ is a linear map of κ-vector spaces, then we have a natural map $(\mathfrak{m}/\mathfrak{m}^2)^\vee \to (\mathfrak{n}/\mathfrak{n}^2)^\vee$. This is the map from the tangent space of p to the tangent space at q that we sought.

Remark 14.5 **Aside:** Note that the *cotangent map always exists*, without requiring p and q to have the same residue field — a sign that cotangent spaces are more natural than tangent spaces in algebraic geometry.

Proposition 14.1.3

Let X be a k-scheme, and $p \in X$ is a k-point. Then there is a bijection between:

(i) The elements of $T_{X,p}$.

(ii) Morphisms of k-schemes

$$\operatorname{Spec} k[\varepsilon]/\varepsilon^2 \longrightarrow X$$

with image p.

Proof We may assume that X is affine, say $X = \operatorname{Spec} A$, where A is a k-algebra. Let $\mathfrak{m}$ be the corresponding maximal ideal of p, and let $\rho : A \to k = A/\mathfrak{m}$ be the quotient map corresponding to p. As p is a k-point, the composition of the structure map $k \to A$ and $\rho : A \to k$ is the identity map.

First of all, any map of k-algebras $A \to k[\varepsilon]/(\varepsilon^2)$ factors uniquely via $A/\mathfrak{m}^2 \to k[\varepsilon]/(\varepsilon^2)$ (since

$A \to k[\varepsilon]/(\varepsilon^2)$ is local ring map). Hence there is a natural bijection

$$\mathrm{Hom}_{\mathbf{Alg}/k}(A, k[\varepsilon]/(\varepsilon^2)) = \mathrm{Hom}_{\mathbf{Alg}/k}(A/\mathfrak{m}^2, k[\varepsilon]/(\varepsilon^2)).$$

Next, consider the exact sequence

$$0 \longrightarrow \mathfrak{m}/\mathfrak{m}^2 \longrightarrow A/\mathfrak{m}^2 \longrightarrow A/\mathfrak{m} \longrightarrow 0$$

This sequence splits, as the structure map $k \to A/\mathfrak{m}^2$ provides a section, so we may write $A/\mathfrak{m}^2 \cong k \oplus \mathfrak{m}/\mathfrak{m}^2$. Given a map of k-algebras $h : A/\mathfrak{m}^2 \to k[\varepsilon]/(\varepsilon^2)$, the restriction of h defines a k-linear map $\mathfrak{m}/\mathfrak{m}^2 \to k\varepsilon \cong k$. This defines a map of k-vector spaces

$$\mathrm{Hom}_{\mathbf{Alg}/k}(A/\mathfrak{m}^2, k[\varepsilon]/(\varepsilon^2)) \longrightarrow \mathrm{Hom}_k(\mathfrak{m}/\mathfrak{m}^2, k).$$

Conversely, given a linear map $\alpha : \mathfrak{m}/\mathfrak{m}^2 \to k$, define $h : A/\mathfrak{m}^2 \cong k \oplus \mathfrak{m}/\mathfrak{m}^2 \to k[\varepsilon]/(\varepsilon^2)$ by $h(a+t) = a + \alpha(t)\varepsilon$ where $a \in k$ and $t \in \mathfrak{m}/\mathfrak{m}^2$. It is easy to check that h is a map of k-algebras, and that $\alpha \mapsto h$ defines an inverse to the above assignment. □

Remark 14.6 This is important, for example in deformation theory. We won't use it, so you can treat it as a curiosity.

Exercise 14.4 (Fun arithmetic practice) Find the dimension of the Zariski tangent space at the point $[(2, 2i)] = [(2, x)]$ of $\mathbb{Z}[2i] \cong \mathbb{Z}[x]/(x^2 + 4)$. Find the dimension of the Zariksi tangent space at the point $[(2, x)]$ of $\mathbb{Z}[\sqrt{-2}] \cong \mathbb{Z}[x]/(x^2 + 2)$.

Solution *We first check $(2, x)$ is a maximal ideal of $\mathbb{Z}[x]/(x^2+4)$. Note $(\mathbb{Z}[x]/(x^2+4))/(2, x) \cong \mathbb{Z}[x]/(x^2 + 4, x, 2) \cong \mathbb{F}_2$, so $(2, x)$ is maximal ideal. We next calculate $(2, x)/(2, x)^2$,*

$$\frac{(2,x)}{(2,x)^2} \cong \frac{(2,x)}{(4,2x,x^2)} \cong \frac{(2,x)}{(4,2x)},$$

so $\dim_{\mathbb{F}_2}\left(\frac{(2,x)}{(4,2x)}\right)^\vee = \dim_{\mathbb{F}_2} \frac{(2,x)}{(4,2x)} = 2.$

Since $(\mathbb{Z}[x]/(x^2+2))/(2, x) \cong \mathbb{F}_2$, we know that $(2, x)$ is a maximal ideal of $\mathbb{Z}[x]/(x^2 + 2)$. Then

$$\frac{(2,x)}{(2,x)^2} \cong \frac{(2,x)}{(4,2x,x^2)} \cong \frac{(2,x)}{2} \cong (x),$$

so $\dim_{\mathbb{F}_2}\left(\frac{(2,x)}{(2,x)^2}\right)^\vee = \dim(x)^\vee = 1.$

14.1.3 Important interpretation in the finite type case: The Jacobian corank

Proposition 14.1.4 (The Jacobian computes the Zariski cotangent space at a rational point)

*Suppose X is a finite type k-scheme. Then locally it is of the form $\mathrm{Spec}\, k[x_1, \ldots, x_n]/(f_1, \ldots, f_r)$. The Zariski cotangent space at a k-valued point (a closed point with residue field k) is given by the cokernel of the "Jacobian map" $k^r \to k^n$ given by the **Jacobian matrix***

$$J = \begin{pmatrix} \frac{\partial f_1}{\partial x_1}(p) & \cdots & \frac{\partial f_r}{\partial x_1}(p) \\ \vdots & \ddots & \vdots \\ \frac{\partial f_1}{\partial x_n}(p) & \cdots & \frac{\partial f_r}{\partial x_n}(p) \end{pmatrix}. \tag{14.1}$$

Proof For convenience, we may assume that $X = \mathrm{Spec}\, k[x_1, \ldots, x_n]/(f_1, \ldots, f_r)$. Say $I = (f_1, \ldots, f_r)$. Let $p = [\mathfrak{m}] := [(x_1 - p_1, \ldots, x_n - p_n)] \in X$ be a k-valued point, then the cotangent space of X at p is

$$T_{X,p}^\vee = \frac{\mathfrak{m}/I}{(\mathfrak{m}^2 + I)/I} \cong \frac{\mathfrak{m}}{\mathfrak{m}^2 + I} \cong \frac{\mathfrak{m}/\mathfrak{m}^2}{(\mathfrak{m}^2 + I)/\mathfrak{m}^2}.$$

Note that $\mathfrak{m}/\mathfrak{m}^2 = \mathrm{Span}_k\{\overline{x_1} - p_1, \ldots, \overline{x_n} - p_n\}$ is k-vector space and its dimension is n, we have $\mathfrak{m}/\mathfrak{m}^2 \cong k^n$.

Since $I \subseteq \mathfrak{m}$, each f_i can be written as

$$f_i = \sum_{j=1}^{n} \frac{\partial f_i}{\partial x_j}(p)(x_j - p_j) + (\text{Higher terms}).$$

Define $\Phi : k^r \to k^n \cong \mathfrak{m}/\mathfrak{m}^2$ by setting

$$e_i \longmapsto \overline{f_i} = \sum_{j=1}^{n} \frac{\partial f_i}{\partial x_j}(p)(\overline{x_j} - p_j),$$

write as matrix we have

$$(e_1, \dots, e_r) \longmapsto (\overline{x_1} - p_1, \dots, \overline{x_n} - p_n) \begin{pmatrix} \frac{\partial f_1}{\partial x_1}(p) & \cdots & \frac{\partial f_r}{\partial x_1}(p) \\ \vdots & \ddots & \vdots \\ \frac{\partial f_1}{\partial x_n}(p) & \cdots & \frac{\partial f_r}{\partial x_n}(p) \end{pmatrix}.$$

The image of Φ is clearly $(\mathfrak{m}^2 + I)/\mathfrak{m}^2$, so the Zariski cotangent space at a k-valued point is given by the cokernel of the Φ, as we desired. □

Remark 14.7 This make precise our example of a curve in $\mathbb{A}^3$ cut out by a couple of equations, where we picked off the linear terms; see Example 14.1 .

Remark 14.8 You might be alarmed: what does $\frac{\partial f}{\partial x_1}$ mean? Do you need deltas and epsilons? No! Just define derivatives formally, e.g.,

$$\frac{\partial}{\partial x_1}(x_1^2 + x_1 x_2 + x_2^2) = 2x_1 + x_2.$$

Remark 14.9 This result can be extended to closed points of X whose residue field is separable over k (and, in particular, to all closed points if $\operatorname{char} k = 0$ or if k is finite); see **??**. The fact that wait until Chapter 22 to show this does not mean that it needs to be so complicated; a more elementary proof is possible.

Remark 14.10 **Warning:** It is more common in mathematics (but not universal) to define the Jacobian matrix as the transpose of (14.1).

Proposition 14.1.5 (The corank of the Jacobian is independent of the presentation)

Suppose A is a finitely generated k-algebra, generated by $x_1, \dots, x_n$, with ideal of relations I generated by $f_1, \dots, f_r$. Let p be a point of $\operatorname{Spec} A$.

(a) *Suppose $g \in I$. Appending the column of partials of g to the Jacobian matrix (14.1) does not change the **corank** (the dimension of the cokernel) at p. Hence the corank of the Jacobian matrix at p does not depend on the choice of generators of I.*

(b) *Suppose $q(x_1, \dots, x_n) \in k[x_1, \dots, x_n]$. Let h be the polynomial $y - q(x_1, \dots, x_n) \in k[x_1, \dots, x_n, y]$. Then the Jacobian matrix of $(f_1, \dots, f_r, h)$ with respect to the variables $(x_1, \dots, x_n, y)$ has the same corank at p as the Jacobian matrix of $(f_1, \dots, f_r)$ with respect to $(x_1, \dots, x_n)$. Hence the corank of the Jacobian matrix at p is independent of the choice of generators for A.*

Proof **(a)** Since $g \in I$, there exists $l_1, \dots, l_r \in k[x_1, \dots, x_n]$ such that $g = \sum_{i=1}^{r} l_i f_i$, so we have

$$\frac{\partial g}{\partial x_j} = \sum_{i=1}^{r} \frac{\partial l_i f_i}{\partial x_j} = \sum_{i=1}^{r} \left(\frac{\partial l_i}{\partial x_j} f_i + l_i \frac{\partial f_i}{\partial x_j} \right).$$

Hence $\frac{\partial g}{\partial x_j}(p) = \sum_{i=1}^{r} l_i(p)\frac{\partial f_i}{\partial x_j}(p)$. So

$$\begin{pmatrix} \frac{\partial g}{\partial x_1}(p) \\ \vdots \\ \frac{\partial g}{\partial x_r}(p) \end{pmatrix} = \sum_{i=1}^{r} l_i(p) \begin{pmatrix} \frac{\partial f_i}{\partial x_1}(p) \\ \vdots \\ \frac{\partial f_i}{\partial x_n}(p) \end{pmatrix}.$$

Hence $\operatorname{rank} J(f_1,\dots,f_r)(p) = \operatorname{rank} J(g, f_1,\dots,f_r)(p)$, and thus

$$\operatorname{corank} J(f_1,\dots,f_r)(p) = \operatorname{corank} J(g, f_1,\dots,f_r)(p).$$

(In fact, $\operatorname{corank} M = n - \operatorname{rank} M$ for $n \times m$ matrix M.)

(b) We first calculate $J(f_1,\dots,f_r,h)(p)$,

$$\begin{aligned} J(f_1,\dots,f_r,h)(p) &= \begin{pmatrix} \frac{\partial f_1}{\partial x_1}(p) & \cdots & \frac{\partial f_r}{\partial x_1}(p) & \frac{\partial h}{\partial x_1}(p) \\ \vdots & \ddots & \vdots & \vdots \\ \frac{\partial f_1}{\partial x_n}(p) & \cdots & \frac{\partial f_r}{\partial x_n}(p) & \frac{\partial h}{\partial x_n}(p) \\ 0 & \cdots & 0 & \frac{\partial h}{\partial y}(p) \end{pmatrix} \\ &= \begin{pmatrix} J(f_1,\dots,f_r)(p) & v_h(p) \\ 0 & 1 \end{pmatrix}, \end{aligned}$$

where $v_h(p) = (\frac{\partial h}{\partial x_1}(p), \cdots, \frac{\partial h}{\partial x_n}(p))^t$. Hence $\operatorname{rank} J(f_1,\dots,f_r,h)(p) = \operatorname{rank} J(f_1,\dots,f_r) + 1$, and thus

$$\begin{aligned} \operatorname{corank} J(f_1,\dots,f_r,h)(p) =&(n+1) - \operatorname{rank} J(f_1,\dots,f_r,h)(p) \\ =&n - \operatorname{rank} J(f_1,\dots,f_r) \\ =&\operatorname{corank} J(f_1,\dots,f_r), \end{aligned}$$

as we desired. □

Thus every finite type k-scheme X comes with an intrinsic "**Jacobian corank function**" from (the underlying set of) X to $\mathbb{Z}^{\geq 0}$, whose value at k-valued points p of X is $\dim_k T_{X,p}$. We temporarily call this function $JC : X \to \mathbb{Z}^{\geq 0}$, although we will later discover that it is really the rank of the coherent sheaf of differentials $\Omega_{X/k}$ (**??**).

Proposition 14.1.6

Suppose X is a finite type k-scheme. Then $JC : X(k) \to \mathbb{Z}^{\geq 0}, p \mapsto \dim_k T_{X,p}$ is an upper semicontinuous function.

Proof Let $F_m = \{p \in X : JC(p) \geq m\}$, we want to show that F_m is closed. For convenience, we may assume that $X = \operatorname{Spec} k[x_1,\dots,x_n]/(f_1,\dots,f_r)$. In fact,

$$JC(p) = \dim_k T_{X,p} = \operatorname{corank} J(f_1,\dots,f_r)(p) = n - \operatorname{rank} J(f_1,\dots,f_r)(p),$$

so $JC(p) \geq m$ implies that $\operatorname{rank} J(f_1,\dots,f_r)(p) \leq n-m$. Note that $\operatorname{rank} J(f_1,\dots,f_r)(p) \leq n-m$ if and only if $(n-m+1)\times(n-m+1)$ minors of $J(f_1,\dots,f_r)(p)$ are zero. Hence

$$F_m = \{p \in X : \operatorname{rank} J(f_1,\dots,f_r)(p) \leq n-m\}$$

is the zero locus of $(n-m+1)\times(n-m+1)$ minors $\det J(f_1,\dots,f_r)(p)_{n-m+1}$, so F_m is closed. □

Proposition 14.1.7

JC is preserved by field extension of k. More precisely, if X is a finite type k-scheme, and ℓ/k is a field extension, and under base change map $X \times_k \ell \to X, p \mapsto q$, we have $JC_{X_\ell}(p) = JC_X(q)$.

Proof We may assume that $X = \operatorname{Spec} k[x_1, \ldots, x_n]/I$, then $X_\ell = \operatorname{Spec} \ell[x_1, \ldots, x_n]/I$. Say $\pi : X_\ell \to X$, then $\pi(p) = q$, by Proposition 8.3.1, π induces an inclusion $\iota : \kappa(q) = k \hookrightarrow \kappa(p) = \ell$ of residue field. For any $g \in k[x_1, \ldots, x_n]$, $\iota(g(q)) = g(p)$ where we regard g as an element in $\ell[x_1, \ldots, x_n]$. Hence $J(I)(q)$ can be seen as a matrix with each elements belong to ℓ. Since $J(I)(q)$ and $J(I)(p)$ defined by same equations, we have $JC(p) = JC(q)$. □

Proposition 14.1.8

Suppose X is finite type over k, for any $p \in X$ (p need not be a closed point of X), $JC(p) \geq \dim_X p$. The dimension of X at p, $\dim_X p$, was defined in Definition 13.2.5.

Proof We first consider the case $k = \overline{k}$. For convenience, we may assume that $X = \operatorname{Spec} A$ where $A = k[x_1, \ldots, x_n]/I$ is a finitely generated k-algebra. Let $p = [\mathfrak{m}] \in X$ be a closed point, then $JC(p) = \dim_k \mathfrak{m}/\mathfrak{m}^2$. By Theorem 13.3.4, we know that $\dim A_\mathfrak{m} \leq \dim_k \mathfrak{m}/\mathfrak{m}^2$. By Proposition 13.2.7, we know that $\dim_X p = \dim A_\mathfrak{m}$, thus $\dim_X p \leq JC(p)$. Now let p be any point in X. Let V be largest irreducible component of X which containing p, then $\dim_X p = \dim V$. Say $\dim V = n$. By Proposition 14.1.6, JC is upper semicontinuous function, so $G_n = \{q \in X : JC(q) < \dim V = n\}$ is an open subset of X. If $JC(p) < \dim V$, then $p \in G_n \cap V$. Note that $G_n \cap V$ is open in V, recall Proposition 4.6.7 we know that closed points are dense, so there exists a closed point $c \in G_n \cap V$. Since V is an irreducible component which containing c, we have $\dim_X c \geq \dim V = n$. By above discussion, we already have $JC(c) \geq \dim_X c \geq n$, so $c \notin G_n$, a contradiction! Hence $JC(p) \geq \dim_X p$.

Next, let k be any field. Denote $X_{\overline{k}} = X \times_k \overline{k}$, and $\pi : X_{\overline{k}} \to X$. Let $p \in X$, then there exist $P \in X_{\overline{k}}$ such that $\pi(P) = p$. Hence, we have

$$\dim_X p = \dim_{X_{\overline{k}}} P \leq JC(P) = JC(p),$$

where first "=" given by Proposition 13.2.8, second "≤" given by our above discussion, third "=" given by Proposition 14.1.7. □

Lemma 14.1.2 (Euler's Homogeneous Function Theorem)

If R is a commutative ring and if $f \in (R[x_1, \ldots, x_n])_d$, then

$$d \cdot f = \sum_{i=1}^{n} x_i \cdot \frac{\partial f}{\partial x_i}.$$

Proof Suppose $f = \sum_{i_1+\cdots+i_n=d} a_{i_1,\ldots,i_n} x_1^{i_1} \cdots x_n^{i_n}$. Then

$$\frac{\partial f}{\partial x_k} = \sum_{i_1+\cdots+i_n=d} a_{i_1,\ldots,i_n} i_k x_1^{i_1} \cdots x_k^{i_k-1} \cdots x_n^{i_n},$$

so

$$\begin{aligned}\sum_{k=1}^{n} x_k \cdot \frac{\partial f}{\partial x_k} &= \sum_{k=1}^{n} \sum_{i_1+\cdots+i_n=d} a_{i_1,\ldots,i_n} i_k x_1^{i_1} \cdots x_n^{i_n} \\ &= \left(\sum_{k=1}^{n} i_k\right) \cdot \left(\sum_{i_1+\cdots+i_n=d} a_{i_1,\ldots,i_n} x_1^{i_1} \cdots x_n^{i_n}\right) \\ &= d \cdot f,\end{aligned}$$

as we desired. □

Proposition 14.1.9 (Computing the Jacobian corank function for projective k-schemes)

Suppose $X \subseteq \mathbb{P}^n_k$ is cut out by homogeneous polynomials $f_i \in k[x_0, \dots, x_n](1 \le i \le r)$. Then the Jacobian corank function of X is 1 less than the corank of the "projective Jacobian matrix":

$$JC(p) = \operatorname{corank} \begin{pmatrix} \frac{\partial f_1}{\partial x_0}(p) & \cdots & \frac{\partial f_r}{\partial x_0}(p) \\ \vdots & \ddots & \vdots \\ \frac{\partial f_1}{\partial x_n}(p) & \cdots & \frac{\partial f_r}{\partial x_n}(p) \end{pmatrix} - 1. \tag{14.2}$$

Proof By Proposition 14.1.7, we may assume that $k = \overline{k}$. Let $p \in X$. Without loss of generally, we may assume that $p \in D_+(x_0) = \operatorname{Spec} k[x_{1/0}, \dots, x_{n/0}]$ (for convenience, write $p = [1 : p_1 : \cdots : p_n]$), say $y_i = x_{i/0}$, then each f_i on $D_+(x_0)$ is $g_i(y_1, \dots, y_n) := f_i(1, y_1, \dots, y_n)$. Then

$$JC(p) = \operatorname{corank} \begin{pmatrix} \frac{\partial g_1}{\partial y_1}(p) & \cdots & \frac{\partial g_r}{\partial y_1}(p) \\ \vdots & \ddots & \vdots \\ \frac{\partial g_1}{\partial y_n}(p) & \cdots & \frac{\partial g_r}{\partial y_n}(p) \end{pmatrix} = \operatorname{corank} J(g_1, \dots, g_r)(p).$$

Note that $\frac{\partial f_j}{\partial x_i}(p) = \frac{\partial g_j}{\partial y_i}(p)$ for all $i > 0, 1 \le j \le r$, we have

$$\begin{pmatrix} \frac{\partial f_1}{\partial x_0}(p) & \cdots & \frac{\partial f_r}{\partial x_0}(p) \\ \vdots & \ddots & \vdots \\ \frac{\partial f_1}{\partial x_n}(p) & \cdots & \frac{\partial f_r}{\partial x_n}(p) \end{pmatrix} = \left(\begin{array}{c} \frac{\partial f_1}{\partial x_0}(p) \quad \cdots \quad \frac{\partial f_r}{\partial x_0}(p) \\ \hdashline J(g_1, \dots, g_r)(p) \end{array} \right).$$

We next show that $R_0 = \left(\frac{\partial f_1}{\partial x_0}(p), \cdots, \frac{\partial f_r}{\partial x_0}(p) \right)$ is linear combination of the row of $J(g_1, \dots, g_r)$. By Euler's Homogeneous Function Theorem 14.1.2, we have

$$x_0 \cdot \frac{\partial f_j}{\partial x_0} + \sum_{i=1}^n x_i \cdot \frac{\partial f_j}{\partial x_i} = (\deg f_j) \cdot f_j,$$

so

$$\frac{\partial f_j}{\partial x_0}(p) + \sum_{i=1}^n p_i \cdot \frac{\partial f_j}{\partial x_i}(p) = (\deg f_j) \cdot f_j(p) = 0,$$

which implies that

$$R_0 + \sum_{i=1}^n p_i \cdot R_i = 0,$$

where $R_i = \left(\frac{\partial f_1}{\partial x_i}(p), \cdots, \frac{\partial f_r}{\partial x_i}(p) \right)$. Hence

$$\operatorname{rank} J(f_1, \dots, f_r)(p) = \operatorname{rank} J(g_1, \dots, g_r)(p).$$

Hence

$$\begin{aligned} \operatorname{corank} J(f_1, \dots, f_r)(p) - 1 &= (n+1) - \operatorname{rank} J(f_1, \dots, f_r)(p) - 1 \\ &= n - \operatorname{rank} J(f_1, \dots, f_r)(p) \\ &= \operatorname{corank} J(f_1, \dots, f_r)(p) \\ &= JC(p), \end{aligned}$$

as we desired. □

14.2 Regularity, and smoothness over a field

The key idea in the definition of regularity is contained in Theorem 13.3.4 (the dimension of the Zariski tangent space is at least the dimension of the local ring).

14.2.1 Definition: regularity

Definition 14.2.1 (Regularity)

*(1) Suppose $(A, \mathfrak{m}, k)$ is a Noetherian local ring, we say A is a **regular local ring** if $\dim A = \dim_k \mathfrak{m}/\mathfrak{m}^2$.*

*(2) If a Noetherian ring A is regular at all of its prime ideals, i.e., $A_{\mathfrak{p}}$ is a regular local ring for all prime ideals $\mathfrak{p}$ of A, then A is said to be a **regular ring**.*

*(3) A locally Noetherian scheme X is **regular** at a point p if the local ring $\mathscr{O}_{X,p}$ is regular. It is **singular** at the point otherwise, and we say that the point is a **singularity**.*

*(4) A scheme is **regular** (or **nonsingular**) if it is regular at all points. It is **singular** otherwise (i.e., if it is singular at at least on point — if it has a singularity).*

Lemma 14.2.1

A dimension 0 Noetherian local ring is regular if and only if it is a field.

Proof Let $(A, \mathfrak{m}, k)$ be a Noetherian local ring with dimension 0. If A is a field, then $\mathfrak{m} = (0)$, so $\dim_k \mathfrak{m}/\mathfrak{m}^2 = 0 = \dim A$, by definition A is regular local ring.

Conversely, if A is regular local ring, we know that $\dim_k \mathfrak{m}/\mathfrak{m}^2 = 0 = \dim A$. Hence $\mathfrak{m}/\mathfrak{m}^2 = 0$, i.e., $\mathfrak{m} = \mathfrak{m}^2$, by Nakayama's Lemma 9.2.4, we know that $\mathfrak{m} = (0)$, which implies that A is a field. □

You will hopefully gradually become convinced that this is the right notion of "smoothness" of schemes. Remarkably, Krull introduced the notion of a regular local ring for purely algebraic reasons, some time before Zariski realized that it was a fundamental notion in geometry in 1947.

Lemma 14.2.2

Suppose $(A, \mathfrak{m})$ is a regular local ring of dimension $n > 0$, and $f \in A$. Then $A/(f)$ is a regular local ring of dimension $n - 1$ if and only if $f \in \mathfrak{m} \setminus \mathfrak{m}^2$.

Proof Since $(A, \mathfrak{m})$ is a regular local ring of dimension n, we know that $\dim A = \dim_k \mathfrak{m}/\mathfrak{m}^2 = n$ where $k = A/\mathfrak{m}$. If $(A/(f), \mathfrak{m}/(f))$ is a regular local ring of dimension $n-1$, then $\dim_k \frac{\mathfrak{m}}{\mathfrak{m}^2+(f)} = n - 1$, hence $f \in \mathfrak{m} \setminus \mathfrak{m}^2$. Conversely, if $f \in \mathfrak{m} \setminus \mathfrak{m}^2$. By Krull's Height Theorem 13.2.2, we know that $\dim A/(f) \geq n-1$. Recall the proof in Krull's Principal Ideal Theorem for tangent space 14.1.1, we know that $\dim_k \frac{\mathfrak{m}}{\mathfrak{m}^2+(f)} = n-1$, by Theorem 13.3.4, $\dim A/(f) \leq \dim_k \frac{\mathfrak{m}}{\mathfrak{m}^2+(f)} = n-1$, so $\dim A/(f) = \dim_k \frac{\mathfrak{m}}{\mathfrak{m}^2+(f)} = n-1$, which implies that $A/(f)$ is a regular local ring of dimension $n-1$. □

Proposition 14.2.1 (The slicing criterion for regularity)

Suppose X is a Noetherian scheme (such as a variety), D is an effective Cartier divisor on X (Definition 10.5.2), and $p \in D$. If p is a regular point of D then p is a regular point of X.

Proof Assume that $X = \operatorname{Spec} A$ where A is Noetherian, then $D = \operatorname{Spec} A/(f)$ where f is non-zerodivisor. Suppose $p = [\mathfrak{p}] \in D$. Since p is a regular point of D, we have $\dim(A/(f))_{\mathfrak{p}} = \dim_k T_{D,p}^{\vee}$. We want to show that $\dim A_{\mathfrak{p}} = \dim_k T^{\vee}_{X,p}$. Note that $(A/(f))_{\mathfrak{p}} \cong A_{\mathfrak{p}}/(f)A_{\mathfrak{p}}$, since f is non-zerodivisor, by Krull's Principal Ideal Theorem 13.2.2, we know that $\dim(A/(f))_{\mathfrak{p}} = \dim A_{\mathfrak{p}} - 1$. By Krull's Principal Ideal Theorem for tangent space 14.1.1, we know that $\dim_k T^{\vee}_{D,p} \geq \dim_k T^{\vee}_{X,p} - 1$. Hence

$$\dim(A/(f))_{\mathfrak{p}} = \dim A_{\mathfrak{p}} - 1 = \dim_k T^{\vee}_{D,p} \geq \dim_k T^{\vee}_{X,p} - 1,$$

i.e., $\dim A_{\mathfrak{p}} \geq \dim_k T^{\vee}_{X,p}$. By Theorem 13.3.4, we have $\dim A_{\mathfrak{p}} \leq \dim_k T^{\vee}_{X,p}$, so $\dim A_{\mathfrak{p}} = \dim_k T^{\vee}_{X,p}$. It follows that p is a regular point of X. □

Lemma 14.2.3

Suppose $(A, \mathfrak{m})$ is a regular local ring of dimension d, and $f_1, \ldots, f_d \in \mathfrak{m}$ are such that their images in the vector space $\mathfrak{m}/\mathfrak{m}^2$ form a basis. Then $f_1, \ldots, f_d$ are a system of parameters (Definition 13.3.1) for $(A, \mathfrak{m})$. (They are "particularly good" parameters.)

Proof Apply Nakayama's Lemma 9.2.6, we are done. □

14.2.2 The Jacobian criterion for regularity

A finite type k-scheme is locally of the form $\operatorname{Spec} k[x_1, \ldots, x_n]/(f_1, \ldots, f_r)$. The Jacobian criterion for regularity (Proposition 14.2.2) gives a hands-on method for checking for singularity at closed points, using the equations $f_1, \ldots, f_r$, if $k = \overline{k}$.

Proposition 14.2.2 (The Jacobian criterion for regularity)

Suppose $X = \operatorname{Spec} k[x_1, \ldots, x_n]/(f_1, \ldots, f_r)$ has pure dimension d. Then a k-valued point $p \in X(k)$ is regular if and only if the Jacobian corank function at p is d.

Proof Say $A = k[x_1, \ldots, x_n]$ and $I = (f_1, \ldots, f_r)$. Since X has pure dimension d, for k-valued point $p = [\mathfrak{m}]$, we have $\dim(A/I)_{\mathfrak{m}} = \operatorname{ht}(\mathfrak{m}) = \dim A/I = d$. So, $p = [\mathfrak{m}] \in X(k)$ is regular if and only if $(A/I)_{\mathfrak{m}}$ is regular local ring if and only if $\dim(A/I)_{\mathfrak{m}} = \dim_k T^{\vee}_{X,p} = \operatorname{corank} J(I) = d$. □

Proposition 14.2.3

Suppose $k = \overline{k}$. Then the singular closed points of the hypersurfaces $f(x_1, \ldots, x_n)$ are given by the equations

$$f = \frac{\partial f}{\partial x_1} = \cdots = \frac{\partial f}{\partial x_n} = 0.$$

(Translation: The singular points of $f = 0$ are where the gradient of f vanishes.)

Proof Recall Definition 13.1.8, $X = \operatorname{Spec} k[x_1, \ldots, x_n]/(f)$ has pure dimension $n-1$. By Jacobian criterion 14.2.2, a closed point $p \in X$ is singular if and only if $\operatorname{corank} J(f) \neq n-1$ if and only if $\operatorname{rank} J(f) \neq 1$. Note that $J(f) = \left(\frac{\partial f}{\partial x_1}(p), \ldots, \frac{\partial f}{\partial x_n}(p)\right)^t$, so $\operatorname{rank} J(f) \neq 1$ if and only if $\operatorname{rank} J(f) = 0$, i.e., $J(f) = 0$, i.e.,

$$f = \frac{\partial f}{\partial x_1} = \cdots = \frac{\partial f}{\partial x_n} = 0.$$

□

14.2.3 Smoothness over a field

There seem to be two serious drawbacks with the Jacobian criterion. For finite type schemes over $\overline{k}$, the criterion gives a necessary condition for regularity, but it is not obviously sufficient, as we need to check regularity at non-closed points as well. We can prove sufficiency by working hard to show Fact **??**, which implies that the non-closed points must be regular as well. A second failing is that the criterion requires k to be algebraically closed. These problems suggest that old-fashioned ideas of using derivatives and Jacobians are ill-suited to the fancy modern notion of regularity. But that is wrong - the fault is with the concept of regularity. There is a better notion of smoothness over a field. Better yet, this idea generalizes to the notion of a smooth morphism of schemes (to be discussed in §14.6, and again in Chapter 25), which behaves well in all possible ways (including in the sense of §9.1). This is another sign that some properties we think of as of objects ("absolute notions") should really be thought of as properties of morphisms ("relative notions"). We know enough to imperfectly (but correctly) define what it means for a scheme to be k**-smooth**, or **smooth over** k.

Definition 14.2.2 (Smoothness)

*(1) A k-scheme is k**-smooth of dimension** d, or **smooth of dimension** d **over** k, if it is pure dimension d, and there exists a cover by affine open sets $\operatorname{Spec} k[x_1, \ldots, x_n]/(f_1, \ldots, f_r)$ where the Jacobian matrix has corank d at all points (not just closed point).*
(In particular, smooth schemes are implicitly locally of finite type.)

*(2) A k-scheme is **smooth over** k if it is smooth of some dimension.*

Remark 14.11 The k is often omitted when it is clear from context. By the theory of the Jacobian corank function developed in §14.1.3, for any pure dimension d variety, this can be checked on *any* affine open cover.

✍ **Exercise 14.5** Show that $\mathbb{A}_k^n$ is smooth for any n and k. For which characteristics is the curve $y^2z = x^3 - xz^2$ in $\mathbb{P}_k^2$ smooth?

Solution *We first show that $\mathbb{A}_k^n$ is smooth for any n and any k. Clearly, $\mathbb{A}_k^n$ has pure dimension n, and the Jacobian matrix J is 0 for all points, so $\operatorname{corank} J = n - 0 = n$, which implies that $\mathbb{A}_k^n$ is smooth.*

Say $F = x^3 - xz^2 - y^2z$ and $X = \operatorname{Proj} k[x, y, z]/(x^3 - xz^2 - y^2z)$. Note that $y^2z = x^3 - xz^2$ is smooth in $\mathbb{P}_k^2$ if and only if $JC(p) = 1$ for any $p \in X$. Consider $\pi : X_{\overline{k}} \to X$, choose closed point P such that $\pi(P) = p$, by Proposition 14.1.7, we have $JC(P) = JC(p)$. So $y^2z = x^3 - xz^2$ is smooth in $\mathbb{P}_k^2$ if and only if $JC(P) = 1$ for any $P \in X_{\overline{k}}(\overline{k})$. By Proposition 14.1.9, $JC(P) = 1$ implies that $\operatorname{corank} J(F)(P) = 2$. Since $\operatorname{corank} J(F)(P) = 3 - \operatorname{rank} J(F)(P)$, we have $\operatorname{rank} J(F)(P) = 1$. Hence $y^2z = x^3 - xz^2$ is smooth in $\mathbb{P}_k^2$ if and only if $\operatorname{rank} J(F)(P) = 1$ for any $P \in X_{\overline{k}}$.

We next calculate $J(F)(P)$, say $P = [P_x : P_y : P_z]$,

$$J(F)(P) = \left(\frac{\partial F}{\partial x}(P), \frac{\partial F}{\partial y}(P), \frac{\partial F}{\partial z}(P)\right) = (3P_x^2 - P_z^2, -2P_yP_z, -2P_xP_z - P_y^2).$$

If $\operatorname{char} k \neq 2$ and $\operatorname{char} k \neq 3$, then $\operatorname{char} \overline{k} \neq 2$ and $\operatorname{char} \overline{k} \neq 3$. Let $J(F)(P) = 0$, then

$$\begin{cases} 3P_x^2 - P_z^2 & = 0 \\ -2P_yP_z & = 0 \\ -2P_xP_z - P_y^2 & = 0 \end{cases} \implies \begin{cases} P_x & = 0 \\ P_y & = 0 \\ P_z & = 0. \end{cases}$$

This is impossible, so in this case X is smooth.

If $\operatorname{char} k = 2$, then $\operatorname{char} \overline{k} = 2$. Let $J(F)(P) = 0$, then

$$\begin{cases} P_y & = 0 \\ 3P_x^2 - P_z^2 & = 0, \end{cases}$$

this equations has non-zero solution, so in this case X is not smooth.

If $\operatorname{char} k = 3$*, then* $\operatorname{char} \overline{k} = 3$*. Let* $J(F)(P) = 0$*, then*

$$\begin{cases} P_z^2 & = 0 \\ 2P_yP_z & = 0 \\ -2P_xP_z - P_y^2 & = 0, \end{cases} \implies \begin{cases} P_y & = 0 \\ P_z & = 0. \end{cases}$$

this equations has non-zero solution, so in this case X *is not smooth.*

Proposition 14.2.4

Suppose $f \in k[x_1, \ldots, x_n]$ *is a polynomial such that the system of equations*

$$f = \frac{\partial f}{\partial x_1} = \cdots = \frac{\partial f}{\partial x_n} = 0$$

has no solution in $\overline{k}$*. Then the hypersurface* $f = 0$ *in* $\mathbb{A}_k^n$ *is smooth.*

Proof Let $p \in X := \operatorname{Spec} k[x_1, \ldots, x_n]/(f)$, then X has pure dimension $n-1$, by Krull's Height Theorem 13.3.3. Consider $\pi : X_{\overline{k}} \to X$, choose $P \in X_{\overline{k}}$ such that $\pi(P) = p$, by Proposition 14.1.7, we know that $JC(P) = JC(p)$. Since equations

$$f = \frac{\partial f}{\partial x_1} = \cdots = \frac{\partial f}{\partial x_n} = 0$$

has no solution in $\overline{k}$. We know that $JC(P) = n - 1$ for all $P \in X_{\overline{k}}$, so $JC(p) = n - 1 = \dim X$, which implies that X is smooth. □

Proposition 14.2.5 (Smoothness of varieties is "independent of extension of base field")

Suppose X *is a locally finite type* k*-scheme, and* $k \subseteq \ell$ *is a field extension. Then* X *is smooth over* k *if and only if* $X \times_{\operatorname{Spec} k} \operatorname{Spec} \ell$ *is smooth over* ℓ*.*

Proof For convenience, we may assume that X is pure dimensional d, since dimension of locally finite type k-scheme is preserved by any field extension (Proposition 13.2.8), X_ℓ also has pure dimensional d. If X is smooth over k, then for any $p \in X$, we have $JC_X(p) = \dim X = d$. Consider $\pi : X_\ell \to X$. Let $Q \in X_\ell$, say $q := \pi(Q)$, by Proposition 14.1.7, we have $JC_{X_\ell}(Q) = JC_X(q) = d$. Hence X_ℓ is smooth.

Conversely, we may assume that X_ℓ has pure dimension d, then X has pure dimension d. Let $q \in X$, since $\pi : X_\ell \to X$ is surjective, there exists $Q \in X_\ell$ such that $\pi(Q) = q$, by Proposition 14.1.7, we have $JC_{X_\ell}(Q) = JC_X(q)$. Since X_ℓ is smooth, we have $JC_{X_\ell}(Q) = d$, so $JC_X(q) = d$, which implies that X is smooth over k. □

Remark 14.12 We give a short proof of X_ℓ has pure dimension d implies that X has pure dimension d.

Proof Let $Z \subseteq X$ be an irreducible component of X, by Proposition 13.2.8, we have $\dim Z_\ell = \dim Z$. Pick an irreducible component $V \subseteq Z_\ell$ such that $\dim V = \dim Z_\ell$, then $\pi(V) \subseteq Z$, moreover $\pi(V)$ is dense in Z (since π send generic point to generic point). Let W be an irreducible component of X_ℓ such that $W \supseteq V$. Since X_ℓ has pure dimension d, we have $\dim W = d$. Since $V \subseteq W$, we have $\operatorname{cl}_X(\pi(V)) = Z \subseteq \operatorname{cl}_X(\pi(W))$, where $\operatorname{cl}_X(\pi(W))$ is irreducible (since W is irreducible). Since Z is an irreducible component, we have $Z = \operatorname{cl}_X(\pi(W))$, note that $\dim W = \dim \operatorname{cl}_X(\pi(W)) = d$, $\dim Z = d$. □

The next proposition shows that we need only check closed points, thereby making a connection to classical geometry.

Proposition 14.2.6

If the Jacobian matrix for $X = \operatorname{Spec} k[x_1, \dots, x_n]/(f_1, \dots, f_r)$ has corank d at all closed points, then it has corank d at all points.

Proof Say $I = (f_1, \dots, f_r)$. Consider $Z_t = \{p \in X : \operatorname{rank} J(I)(p) \le t\}$. Note that $\operatorname{rank} J(I)(p) \le t$ if and only if every $(t+1) \times (t+1)$ minors of $J(I)(p)$ are zero, hence Z_t is given by the zero locus of some polynomials, and thus Z_t is closed in X. Since $\operatorname{rank} J(I)(p) = n - d$ at all closed points $p \in X$, we know that $X(k) \subseteq Z_{n-d}$. Recall Proposition 4.6.7, closed points X^{close} is dense in X, so $X = Z_{n-d}$, which implies $\operatorname{rank} J(I)(p) \le n - d$ for all $p \in X$. Now consider Z_{n-d-1}, since Z_{n-d-1} is closed in X, Z_{n-d-1} must containing a closed point or $Z_{n-d-1} = \varnothing$, note that $Z_{n-d-1} \cap X^{\text{close}} = \varnothing$, we know that $Z_{n-d-1} = \varnothing$. Hence $\operatorname{rank} J(I)(p) > n - d - 1$ for all $p \in X$. It follows that $\operatorname{rank} J(I)(p) = n - d$ for all $p \in X$, i.e., $\operatorname{corank} J(I)(p) = d$ for all $p \in X$. □

Remark 14.13 In defense of regularity. Having made a spirited case for smoothness, we should be clear that regularity is still very useful. For example, it is the only concept that makes sense in mixed characteristic. In particular, $\mathbb{Z}$ is regular at its points (it is a regular ring), and more generally, discrete valuation rings are incredibly useful examples of regular rings.

14.2.4 Regularity vs. smoothness

Proposition 14.2.7

Suppose X is a locally finite type scheme of pure dimension d over an algebraically closed field $k = \overline{k}$. Then X is regular at its closed points if and only if it is smooth.

Proof For convenience, we may assume that $X = \operatorname{Spec} k[x_1, \dots, x_n]/I$. If X is regular at its closed points, by Proposition 14.2.2, the corank of Jacobian at p is d for all $p \in X(k)$, by Proposition 14.2.6, we know that the corank of Jacobian at any point of X is d, by definition, X is smooth.

Conversely, if X is smooth, then the Jacobian has corank d at all points of X. In particular, the Jacobian has corank d at all closed points of X, by Proposition 14.2.2, X is regular at all closed points. □

Remark 14.14 We will soon learn that for locally finite type $\overline{k}$-schemes, regularity at closed points is the same as regularity everywhere, Theorem **??**.

More generally, if k is perfect (e.g., if $\operatorname{char} k = 0$ or k is a finite field), then smoothness is the same as regularity at closed points (see **??**). More generally still, we will later prove the following fact. We mention it now because it will make a number of statements cleaner long before we finally prove it. (There will be no circularity.)

Theorem 14.2.1 (Smoothness-Regularity Comparison Theorem)

(a) If k is perfect, every regular finite type k-scheme is smooth over k.

(b) Every smooth k-scheme is regular (with no hypotheses on perfection).

Part (a) will be proved in **??**. Part (b) will be prove in §**??**. The fact that Theorem 14.2.1 (b) will be proved so far in the future does not mean that we truly need to wait that long. But we deliberately postpone the proof until we have machinery that will do much of the work for us. If you wish for some insight right away, here is the outline of the argument for (b), which you can even try to implement after reading this chapter. We will soon show (in **??**) that $\mathbb{A}_k^n$ is regular for all fields k. Once we know the definition of étale morphism, we will realize that every smooth variety locally admits an étale morphism to $\mathbb{A}_k^n$ (Observation **??**). You can then show that if $\pi : X \to Y$ is an étale morphism of locally Noetherian schemes, and $p \in X$, then p is regular if $\pi(p)$ is regular. This will be shown in **??**, but you could reasonably do this after reading the definition of étaleness.

Example 14.3 (Caution: Regularity does not imply smoothness) If k is not perfect, then regularity does not imply smoothness, as demonstrated by the following example. Let $k = \mathbb{F}_p(u)$, and consider the hypersurface $X = \operatorname{Spec} k[x]/(x^p - u)$. Now $k[x]/(x^p - u)$ is a field, hence regular. But if $f(x) = x^p - u$, then $f(u^{1/p}) = \frac{\mathrm{d}f}{\mathrm{d}x}(u^{1/p}) = 0$, so the Jacobian criterion fails — X is not smooth over k.

(Never forget that smoothness requires a choice of field — it is a "relative" notion, and we will later define smoothness over an arbitrary scheme, in §14.6.)

Example 14.4 In case the previous example is too "small" to be enlightening (because the scheme in question is smooth over a different field, namely, $k[x]/(x^p - u)$), here is another. Let $k = \mathbb{F}_p(u)$ as above, with $p > 2$, and consider the curve $\operatorname{Spec} k[x, y]/(y^2 - x^p + u)$. Then the closed point $(y, x^p - u)$ is regular but not smooth.

Thus you should not use "regular" and "smooth" interchangeably.

14.2.5 Regular local rings are integral domains

You might expect from geometric intuition that a scheme is "locally irreducible" at a "smooth" point. Put algebraically:

Theorem 14.2.2

Suppose $(A, \mathfrak{m}, k)$ is a regular local ring of dimension n. Then A is an integral domain.

Before proving it, we give some consequences.

Proposition 14.2.8

Suppose p is a regular point of a locally Noetherian scheme X. Then there is only one irreducible component of X passes through p.

Proof Since p is a regular point of X, $\mathscr{O}_{X,p}$ is a regular local ring. By Theorem 14.2.2, we know that $\mathscr{O}_{X,p}$ is an integral domain. The minimal ideal of p is one-to-one corresponding to the minimal prime of $\mathscr{O}_{X,p}$. Since $\mathscr{O}_{X,p}$ has only one minimal prime (0), so there is only one minimal prime ideal contained in p, hence there is only one component of X pass through p. □

Proposition 14.2.9

A nonempty regular Noetherian scheme is irreducible if and only if it is connected.

Proof By Proposition 4.6.3, irreducible topological space is connected. Conversely, if a nonempty regular Noetherian scheme X is connected. By Proposition 4.6.13 (a), we may assume that $X = Z_1 \cup \cdots \cup Z_r$ where each Z_i is irreducible component of X. Claim that $Z_i \cap Z_j = \varnothing$. Let $p \in Z_i \cap Z_j$, by Proposition 14.2.8, we know that there is only one irreducible passes through p, so $Z_i \cap Z_j = \varnothing$. Since X is connected, $r = 1$, i.e., X is irreducible. □

Proposition 14.2.10 (Regular schemes in regular schemes are regular embeddings)

(a) Suppose $(A, \mathfrak{m}, k)$ is a regular local ring of dimension n, and $I \subseteq A$ is an ideal of A cutting out a regular local ring of dimension d. Then $\operatorname{Spec} A/I$ is a regular embedding in $\operatorname{Spec} A$.

(b) Suppose $\pi : X \to Y$ is a closed embedding of regular schemes. Then π is a regular embedding.

Proof **(a)** Let $\mathfrak{n}=\mathfrak{m}/I$. Consider $\pi: A\to A/I$, then we have a surjective $\mathfrak{m}\twoheadrightarrow\mathfrak{n}$, it induces a surjective k-linear map $\alpha:\mathfrak{m}/\mathfrak{m}^2\twoheadrightarrow\mathfrak{n}/\mathfrak{n}^2$. Then we have an exact sequence

$$0\longrightarrow \operatorname{Ker}\alpha\xrightarrow{\beta}\mathfrak{m}/\mathfrak{m}^2\xrightarrow{\alpha}\mathfrak{n}/\mathfrak{n}^2\longrightarrow 0.$$

In fact, we have $\operatorname{Ker}\alpha=\frac{\mathfrak{m}^2+I}{\mathfrak{m}^2}\cong\frac{I}{I\cap\mathfrak{m}^2}$, Since A, A/I are regular local ring, we have $\dim A=\dim_k\mathfrak{m}/\mathfrak{m}^2=n$ and $\dim A/I=\dim_k\mathfrak{n}/\mathfrak{n}^2=d$. By Linear Algebra, we have

$$\dim_k\mathfrak{m}/\mathfrak{m}^2=\dim_k\operatorname{Ker}\alpha+\dim_k\operatorname{Im}\alpha=\dim_k\frac{I}{I\cap\mathfrak{m}^2}+\dim_k\mathfrak{n}/\mathfrak{n}^2,$$

so

$$\dim_k\frac{I}{I\cap\mathfrak{m}^2}=\dim_k\mathfrak{m}/\mathfrak{m}^2-\dim_k\mathfrak{n}/\mathfrak{n}^2=n-d.$$

Let $r:=n-d$. Suppose $\frac{I}{I\cap\mathfrak{m}^2}=\operatorname{Span}_k\{\overline{f_1},\dots,\overline{f_r}\}$, where $f_i\in I$. Let $J=(f_1,\dots,f_r)$, then $J\subseteq I$. Consider the image of $f_1,\dots,f_r$ in $\mathfrak{m}/\mathfrak{m}^2$, extend $\overline{f_1},\dots,\overline{f_r}$ to a basis of $\mathfrak{m}/\mathfrak{m}^2$, say $\mathfrak{m}/\mathfrak{m}^2=\operatorname{Span}_k\{\overline{f_1},\dots,\overline{f_r},\overline{g_1},\dots,\overline{g_d}\}$ where $g_j\in\mathfrak{m}$. By Nakayama's Lemma version 4 9.2.6, we know that $\mathfrak{m}=(f_1,\dots,f_r,g_1,\dots,g_d)$. Hence $(g_1,\dots,g_d)$ is maximal ideal of local ring A/J, so $\dim A/J=d$. Since $J\subseteq I$, we have a surjective $A/J\twoheadrightarrow A/I$, since $\dim A/I=d$, by Proposition 13.1.4, $A/I\cong A/J$. So $I=J=(f_1,\dots,f_r)$. We next show that $f_1,\dots,f_r$ is a regular sequence. By Theorem 14.2.2, we know that A is an integral domain, so f_1 is not zero divisor in A. Suppose $f_1,\dots,f_{i-1}$ is a regular sequence, we want to show that f_i is not zero divisor in $A/(f_1,\dots,f_{i-1})$. By Krull's Principal Ideal Theorem 13.3.2, we know that $\dim A/(f_1,\dots,f_{i-1})=n-(i-1)$, note that $\dim_k\frac{\mathfrak{m}/(f_1,\dots,f_{i-1})}{(\mathfrak{m}/(f_1,\dots,f_{i-1}))^2}=n-(i-1)$, we know that $A/(f_1,\dots,f_{i-1})$ is a regular local ring, by Theorem 14.2.2, $A/(f_1,\dots,f_{i-1})$ is integral domain, so f_i is not zero divisor in $A/(f_1,\dots,f_{i-1})$, which implies that $f_1,\dots,f_i$ is regular sequence. By Corollary 10.5.1, $\operatorname{Spec}A/I\hookrightarrow\operatorname{Spec}A$ is a regular embedding.

(b) Pick $p\in X$, we have $\mathscr{O}_{Y,\pi(p)}\to\mathscr{O}_{X,p}$ where $\mathscr{O}_{X,p},\mathscr{O}_{Y,\pi(p)}$ are regular local rings. By part (a), in the local ring $\mathscr{O}_{Y,\pi(p)}$ the ideal of X is generated by a regular sequence, so $\pi:X\to Y$ is a regular embedding. $\square$

Proposition 14.2.10 has the following striking geometric consequence.

Proposition 14.2.11 (Generalizing Proposition 13.3.4)

Suppose W is a smooth variety of pure dimension d over a field k, and X and Y are pure dimensional subvarieties (possibly singular) of W codimension m and n, respectively. Then every component of $X\cap Y$ has codimension at most $m+n$ in W.

Proof Since W is separated, $W\cong\Delta\subseteq W\times_k W$ is a closed embedding, we want to show that $\Delta\subseteq W\times_k W$ is a regular embedding of codimension d. Recall Proposition 14.2.10 (b), we need to show that $W\times_k W$ is a regular scheme. By Smoothness-Regularity Comparison Theorem 14.2.1, it suffices to show that $W\times_k W$ is smooth. By Definition 14.2.2, we know that smoothness can be checked locally, so we may assume that $W=\operatorname{Spec}k[x_1,\dots,x_n]/(f_1,\dots,f_r)$ where corank $J(f_1,\dots,f_r)(p)=d$ for all $p\in W$. Since W has pure dimensional d, we may also reduced to the case that W is irreducible. Say $I=(f_1,\dots,f_r)$. Note that

$$W\times_k W\cong\operatorname{Spec}k[x_1,\dots,x_n,y_1,\dots,y_n]/(I_x,I_y)$$

where $I_x=(f_1(x_1,\dots,x_n),\dots,f_r(x_1,\dots,x_n))$ and $I_y=(f_1(y_1,\dots,y_n),\dots,f_r(y_1,\dots,y_n))$, for any

$P \in W \times_k W$ we have Jacobian

$$J(I_x, I_y)(P) = \begin{pmatrix} \frac{\partial f_1(S)}{\partial x_1}(P) & \cdots & \frac{\partial f_r(S)}{\partial x_1}(P) & \frac{\partial f_1(T)}{\partial x_1}(P) & \cdots & \frac{\partial f_r(T)}{\partial x_1}(P) \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ \frac{\partial f_1(S)}{\partial x_n}(P) & \cdots & \frac{\partial f_r(S)}{\partial x_n}(P) & \frac{\partial f_1(T)}{\partial x_n}(P) & \cdots & \frac{\partial f_r(T)}{\partial x_n}(P) \\ \frac{\partial f_1(S)}{\partial y_1}(P) & \cdots & \frac{\partial f_r(S)}{\partial y_1}(P) & \frac{\partial f_1(T)}{\partial y_1}(P) & \cdots & \frac{\partial f_r(T)}{\partial y_1}(P) \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ \frac{\partial f_1(S)}{\partial y_n}(P) & \cdots & \frac{\partial f_r(S)}{\partial y_n}(P) & \frac{\partial f_1(T)}{\partial y_n}(P) & \cdots & \frac{\partial f_r(T)}{\partial y_n}(P) \end{pmatrix}$$

$$= \begin{pmatrix} J(I_x)(P) & 0 \\ 0 & J(I_y)(P) \end{pmatrix}$$

where $S = (x_1, \ldots, x_n)$ and $T = (y_1, \ldots, y_n)$. Hence $\operatorname{rank} J(I_x, I_y)(P) = \operatorname{rank} J(I_x)(P) + \operatorname{rank} J(I_y)(P)$, so that $\operatorname{corank} J(I_x, I_y)(P) = 2n - \operatorname{rank} J(I_x, I_y)(P) = \operatorname{corank} J(I_x)(P) + \operatorname{corank} J(I_y)(P) = 2d$. By Proposition 13.2.3, we have $\dim W \times_k W = 2 \dim W = 2d$, hence $\operatorname{corank} J(I_x, I_y)(P) = \dim W \times_k W$ for all $P \in W \times_k W$. So $W \times_k W$ is smooth. Note that $\operatorname{codim}_{W \times_k W}(\Delta) = \dim W \times_k W - \dim \Delta = 2d - d = d$, we know that $\Delta \subseteq W \times_k W$ is a regular embedding of codimension d.

By Lemma 12.2.1, we have $X \cap Y \cong (X \times_k Y) \cap \Delta$. Since $\Delta \subseteq W \times_k W$ is regular embedding of codimension d, Δ is locally cutting by d equations, by Krull's Height Theorem 13.3.3,

$$\operatorname{codim}_{X \times_k Y} \Delta \cap (X \times_k Y) = \operatorname{codim}_{X \times_k Y} X \cap Y \leq d,$$

so

$$\dim X \cap Y \geq \dim X \times_k Y - d = (d - m) + (d - n) - d = d - (m + n),$$

where the first "=" since X, Y have pure dimensional m, n. Hence

$$\operatorname{codim}_W X \cap Y \leq \dim W - \dim X \cap Y \leq d - (d - (m + n)) = m + n.$$

□

Remark 14.15 Why is there a hypothesis of smoothness rather than of regularity on W?

Suppose $W = \operatorname{Spec} k[x, y]/(y^2 - x^p + u)$ where $k = \mathbb{F}_p(u)$. By Example 14.4 , we know that W is regular but not smooth. Claim that $W \times_k W$ is not regular. In fact,

$$W \times_k W \cong \operatorname{Spec} k[x_1, y_1, x_2, y_2]/(y_1^2 - x_1^p + u, y_2^2 - x_2^p + u).$$

Let $\mathfrak{m} = (x_1^p - u, x_1 - x_2, y_1, y_2) \subseteq k[x_1, y_1, x_2, y_2]$, $\mathfrak{m}$ is a maximal ideal of $k[x_1, y_1, x_2, y_2]$, since $x_1^p - u$ is irreducible over k and $k[x_1, y_1, x_2, y_2]/\mathfrak{m} \cong k(u^{1/p}) =: K$. It easy to check that $I \subseteq \mathfrak{m}$. Let $\mathfrak{p} = \mathfrak{m}/I$. Consider exact sequence

$$\frac{I + \mathfrak{m}^2}{\mathfrak{m}^2} \xrightarrow{\beta} \mathfrak{m}/\mathfrak{m}^2 \xrightarrow{\alpha} \mathfrak{p}/\mathfrak{p}^2 \longrightarrow 0$$

By Linear algebra, we have

$$\dim \mathfrak{p}/\mathfrak{p}^2 = \dim \mathfrak{m}/\mathfrak{m}^2 - \dim \frac{I + \mathfrak{m}^2}{\mathfrak{m}^2}.$$

Since $\mathbb{A}_k^4$ is regular, we have $\dim_K \mathfrak{m}/\mathfrak{m}^2 = 4$. Consider $\frac{I+\mathfrak{m}^2}{\mathfrak{m}^2}$. Note that

$$y_1^2 - x_1^p + u \equiv -x_1^p + u \pmod{\mathfrak{m}^2}$$

and

$$y_2^2 - x_2^p + u = y_2^2 - ((x_2 - x_1) + x_1)^p + u = y_2^2 - (x_2 - x_1)^p - x_1^p + u \equiv -x_1^p + u \pmod{\mathfrak{m}^2},$$

we know that $\frac{I+\mathfrak{m}^2}{\mathfrak{m}^2} = \mathrm{Span}_K\{-x_1^p + u \pmod{\mathfrak{m}^2}\}$, so $\dim_K \frac{I+\mathfrak{m}^2}{\mathfrak{m}^2} = 1$. Hence

$$\dim \mathfrak{p}/\mathfrak{p}^2 = \dim \mathfrak{m}/\mathfrak{m}^2 - \dim \frac{I+\mathfrak{m}^2}{\mathfrak{m}^2} = 4 - 1 = 3.$$

But $\dim \mathscr{O}_{W\times_k W,\mathfrak{p}} = 2$, so $W \times_k W$ is not regular at $\mathfrak{p}$.

The example above tells us that the regularity of a non-smooth variety W does not imply the regularity of $W \times_k W$, so the regularity condition alone is insufficient for Proposition 14.2.11.

The following example shows that the smoothness hypotheses in Proposition 14.2.11 cannot be (completely) dropped.

Example 14.5 Let $W = \operatorname{Spec} k[w,x,y,z]/(wz - xy)$ be the cone over the smooth quadric surface, which is an integral threefold (Proposition 14.2.3 shows that W is not smooth). Let X be the surface $w = x = 0$ and Y the surface $y = z = 0$; both lie in W. Then $X \cap Y$ is just the origin, so we have two codimension 1 subvarieties meeting in a codimension 3 subvariety. (It is no coincidence that X and Y are the affine cones over two lines in the same ruling; see Exercise 10.8. This example will arise again in **??**.)

Proposition 14.2.12

Smooth k-schemes are reduced.

Proof Suppose X is a smooth k-scheme, and let $X_{\overline{k}} := X \times_{\operatorname{Spec} k} \operatorname{Spec} \overline{k}$ as usual. Proposition 14.2.5 shows that $X_{\overline{k}}$ is smooth, so, by Proposition 14.2.7, $X_{\overline{k}}$ is regular at its closed points, by Theorem 14.2.2, the local rings of $X_{\overline{k}}$ at closed points are integral domains, and thus $X_{\overline{k}}$ is reduced at its closed points, by Proposition 6.2.4, $X_{\overline{k}}$ is reduced. Reducedness can be check locally, we may assume $X = \operatorname{Spec} A$ where A is k-algebra. Now $A \otimes_k \overline{k}$ is reduced, we want to show that A is reduced. Note that $A \to A \otimes_k \overline{k}$ is injective, A can be regarded as subring of $A \otimes_k \overline{k}$, hence A is reduced, and therefore X is reduced. ☐

Proposition 14.2.13

Suppose p is a closed point of a smooth k-scheme X of dimension n. Then $\mathscr{O}_{X,p}$ is an integral domain of dimension n.

Proof In fact, $\dim \mathscr{O}_{X,p} = \operatorname{codim}_X p = \dim X = n$. Since X is smooth, by Smoothness-Regularity Comparison Theorem 14.2.1, X is regular. So $\mathscr{O}_{X,p}$ is regular local ring of dimension n, by Theorem 14.2.2, $\mathscr{O}_{X,p}$ is an integral domain. ☐

Proof of Theorem 14.2.2

Proof We prove the result by induction on n. The case $n = 0$ is clear (dimension 0 regular local rings are fields, Lemma 14.2.1). So assume $n > 0$, and that the result holds for smaller dimension.

Suppose $\mathfrak{p}$ is prime, and $\dim A/\mathfrak{p} = n$. We wish to show that $\mathfrak{p} = (0)$. Now $A/\mathfrak{p}$ is a Noetherian local ring of dimension n, and its Zariski cotangent space is a quotient of that of A cut out by the "equations of $\mathfrak{p}$" (Proposition 14.1.1), so $\dim_k \mathfrak{m}/(\mathfrak{m}^2 + \mathfrak{p}) \geq n - r$ where r is the minimal numbers of generator of $\mathfrak{p}$. Recall Theorem 13.3.4, we have $n = \dim A/\mathfrak{p} \leq \dim_k \mathfrak{m}/(\mathfrak{m}^2 + \mathfrak{p})$, so $\dim_k \mathfrak{m}/(\mathfrak{m}^2 + \mathfrak{p}) = n$. But $\dim \mathfrak{m}/\mathfrak{m}^2 = n$, so $\mathfrak{p} \subseteq \mathfrak{m}^2$. Hence $A/\mathfrak{p}$ is a regular local ring of dimension n.

Choose any $f \in \mathfrak{m} \setminus \mathfrak{m}^2$ (so $f \notin \mathfrak{p}$). By Proposition 13.3.8 (applied to both $A/(f)$ and $A/(f, \mathfrak{p})$), we have $\dim A/(f) \geq n - 1$ and $\dim A/(f, \mathfrak{p}) \geq n - 1$, but both rings have Zariski tangent space of dimension $n - 1$ (Proposition 14.1.1 again), by Theorem 13.3.4, $A/(f)$ and $A/(f, \mathfrak{p})$ are both regular local rings of dimension $n - 1$, and by the inductive hypothesis both are integral domains. The only way one integral domain can be the quotient of another integral domain of the same dimension is if the quotient is an isomorphism (Proposition 13.1.4). Thus $(\mathfrak{p}, f) = (f)$, from which $\mathfrak{p} \subseteq (f)$, from which $\mathfrak{p} \subseteq f\mathfrak{p}$, from which $\mathfrak{p} = f\mathfrak{p}$. Then by Nakayama's Lemma 9.2.4, $\mathfrak{p} = (0)$, as desired. □

14.2.6 ⋆⋆ Checking regularity of k-schemes at closed points by base changing to $\overline{k}$

14.3 Examples

We now explore regularity and smoothness in practice through a series of examples and exercises.

14.3.1 Geometric examples

✍ **Exercise 14.6** Suppose k is a field. Show that $\mathbb{A}_k^1$ and $\mathbb{A}_{\overline{k}}^2$ are regular, by directly check the regularity of all points. Show that $\mathbb{P}_k^1$ and $\mathbb{P}_{\overline{k}}^2$ are regular.

Proof Let $\mathfrak{p} \in \mathbb{A}_k^1 = \operatorname{Spec} k[x]$, we want to show that $k[x]_{\mathfrak{p}}$ is a regular local ring. Since $k[x]$ is PID, $\mathfrak{p} = (0)$ or $\mathfrak{p} = (f)$ where f is irreducible. If $\mathfrak{p} = (0)$, then $k[x]_{(0)} \cong k(x)$, so $\dim k[x]_{(0)} = 0$. The maximal ideal of $k[x]_{(0)}$ is $(0)k[x]_{(0)}$, so $\dim_{k(x)} \frac{(0)k[x]_{(0)}}{((0)k[x]_{(0)})^2} = 0$. Hence $\dim k[x]_{(0)} = \dim_{k(x)} \frac{(0)k[x]_{(0)}}{((0)k[x]_{(0)})^2}$. If $\mathfrak{p} = (f)$ where f is irreducible. The prime ideal of $k[x]_{(f)}$ is one-to-one corresponding to the prime ideal which contained in $\mathfrak{p}$. Since (f) is maximal ideal, we know that $\dim k[x]_{(f)} = 1$. Note that $\dim_{k[x]/(f)} \frac{(f)k[x]_{(f)}}{((f)k[x]_{(f)})^2} = \dim_{k[x]/(f)} \frac{(f)}{(f^2)} = 1$, we have $\dim k[x]_{(f)} = \dim_{k[x]/(f)} \frac{(f)k[x]_{(f)}}{((f)k[x]_{(f)})^2}$. Hence, for any $\mathfrak{p} \in \operatorname{Spec} k[x]$, $\dim k[x]_{\mathfrak{p}} = \dim_{\kappa(\mathfrak{p})} \frac{\mathfrak{p}k[x]_{\mathfrak{p}}}{(\mathfrak{p}k[x]_{\mathfrak{p}})^2}$, it follows that $\mathbb{A}_k^1$ is regular.

We next show that $\mathbb{A}_{\overline{k}}^2 = \operatorname{Spec} \overline{k}[x, y]$ is a regular local ring. Let $p \in \mathbb{A}_{\overline{k}}^2$, then $p = (0)$ or $p = \mathfrak{m}$ or $p = (f(x, y))$ where $\mathfrak{m}$ is maximal ideal of $\overline{k}[x, y]$ and $f(x, y)$ is irreducible over $\overline{k}[x, y]$. If $p = (0)$, then $\overline{k}[x, y]_{(0)} \cong \overline{k}(x, y)$, so $\dim \overline{k}[x, y]_{(0)} = 0$. On the other hand, $\dim_{\overline{k}(x,y)} \frac{(0)k(x,y)}{((0)k(x,y))^2} = 0$. So $\dim_{\overline{k}(x,y)} \frac{(0)k(x,y)}{((0)k(x,y))^2} = \dim \overline{k}[x, y]_{(0)}$, i.e., $\mathbb{A}_{\overline{k}}^2$ is regular at generic point. If $p = \mathfrak{m}$. Without loss of generally, we may assume that $\mathfrak{m} = (x, y)$, then $\dim \overline{k}[x, y]_{\mathfrak{m}} = 2$. It is easy to see that $\dim_{\overline{k}} \frac{\mathfrak{m}}{\mathfrak{m}^2} = 2$, so $\dim \overline{k}[x, y]_{\mathfrak{m}}$, i.e., $\mathbb{A}_{\overline{k}}^2$ is regular at closed points. If $p = (f(x, y))$ where $f(x, y)$ irreducible. It is easy to check that $\operatorname{ht}(f) = 1$, hence $\dim \overline{k}[x, y]_{(f)} = 1$. f is irreducible implies that $f \notin (f^2)$, so $\dim_{[} \kappa(p)] \frac{(f)\overline{k}[x,y]_{(f)}}{((f)\overline{k}[x,y]_{(f)})^2} = 1$, hence $\mathbb{A}_{\overline{k}}^2$ is regular at $p = (f(x, y))$. Combine our above consequences, we obtain that $\mathbb{A}_{\overline{k}}^2$ is regular.

For $\mathbb{P}_k^1$ and $\mathbb{P}_{\overline{k}}^2$, note that $\mathbb{P}_k^1$ is glued by $\mathbb{A}_k^1$ and $\mathbb{P}_{\overline{k}}^1$ is glued by $\mathbb{A}_{\overline{k}}^1$, each affine piece is regular, then so are $\mathbb{P}_k^1$ and $\mathbb{P}_{\overline{k}}^2$. □

The generalization to arbitrary dimension is harder, so we leave it to **??**. By contrast, it is basically immediate that $\mathbb{A}_k^n$ and $\mathbb{P}_k^n$ are smooth.

Proposition 14.3.1 (The Jacobian criterion for regularity for projective hypersurfaces)

(a) The non-smooth points of the hypersurface $f = 0$ in $\mathbb{P}_k^n$ correspond to the locus

$$f = \frac{\partial f}{\partial x_0} = \cdots = \frac{\partial f}{\partial x_n} = 0.$$

If the degree of the hypersurface is not divisible by $\operatorname{char} k$ *(e.g., if* $\operatorname{char} k = 0$*), then it suffices to check* $\frac{\partial f}{\partial x_0} = \cdots = \frac{\partial f}{\partial x_n} = 0$.

(b) If furthermore $\mathrm{k} = \overline{k}$*, then the singular closed points are given by the same locus.*

Remark 14.16 In fact, (b) gives all the singular points, not just the singular closed points; cf. §14.2.3. We won't use this, so we won't prove it.

Proof **For (a).** Hypersurface $f=0$ in $\mathbb{P}_k^n$ is pure dimension $n-1$, let p be a non-smooth point on hypersurface, then $JC(p)\neq n-1$. Say $M=(\frac{\partial f}{\partial x_0},\dots,\frac{\partial f}{\partial x_n})^t$. By Proposition 14.1.9, $JC(p)=\operatorname{corank} M-1$, so $\operatorname{corank} M\neq n$, i.e., $\operatorname{rank} M<1$. It follows that

$$f=\frac{\partial f}{\partial x_0}=\cdots=\frac{\partial f}{\partial x_n}=0.$$

And vice versa.

If the degree of the hypersurface is not divisible by $\operatorname{char} k$. By Euler's Homogeneous Function Theorem 14.1.2,

$$(\deg f)\cdot f=\sum_{i=1}^n x_i\cdot\frac{\partial f}{\partial x_i}.$$

Since $\operatorname{char} k\nmid\deg f$, it is impossible for situation that $f(p)\neq 0$ and $\frac{\partial f}{\partial x_0}=\cdots=\frac{\partial f}{\partial x_n}=0$ to occur.

For (b). Let p be a singular closed point. Suppose $p\in\operatorname{Spec} k[x_{1/0},\dots,x_{n/0}]/(f(1,x_{1/0},\dots,x_{n/0}))\subseteq\operatorname{Proj} k[x_0,\dots,x_n]/(f)$, say $y_i=x_{i/0}$ and $h(y_1,\dots,y_n)=f(1,y_1,\dots,y_n)$. By Jacobian criterion 14.2.2, a closed point $p=[p_0:\cdots:p_n]\in X$ is singular if and only if $\operatorname{corank} J(h)(p)\neq n-1$ if and only if $\operatorname{rank} J(h)(p)=0$, i.e., $J(h)(p)=0$, i.e.,

$$h(p)=\frac{\partial h}{\partial y_1}(p)=\cdots=\frac{\partial h}{\partial y_n}(p)=0. \tag{14.3}$$

Note that $f(x_0,\dots,x_n)=x_0^{\deg f}f(1,y_1,dots,y_n)=x_0^{\deg f}h(y_1,\dots,y_n)$, we have

$$\frac{\partial f}{\partial x_0}(p)=(\deg f)\cdot p_0^{d-1}f(1,p_{1/0},\dots,p_{n/0})+p_0^{\deg f}\sum_{i=1}^n\frac{\partial h}{\partial y_i}(p)\frac{\mathrm{d}y_i}{\mathrm{d}x_0}(p)=0$$

and

$$\frac{\partial f}{\partial x_i}(p)=p_0^{\deg f}\frac{\partial h}{\partial y_i}(p)\cdot\frac{\mathrm{d}y_i}{\mathrm{d}x_i}(p)$$

for any $i\neq 0$. Hence (14.3) implies

$$f=\frac{\partial f}{\partial x_0}=\cdots=\frac{\partial f}{\partial x_n}=0.$$

□

✍ **Exercise 14.7** Suppose $\operatorname{char} k\neq 2$. Show that $y^2z=x^3-xz^2$ in $\mathbb{P}_k^2$ is an irreducible smooth curve. (Eisenstein's Criterion gives one way of showing irreducibility. Warning: We didn't specify $\operatorname{char} k\neq 3$, so be careful when using the Jacobian criterion.)

Proof We first show the irreducibility. Since $f(x,y,z)=x^3-y^2z-xz^2=0$ is a homogeneous polynomial, so we may check the irreducibility after de-homogenization. On $D_+(z)$, we have $x^3-x-y^2=0$, i.e., $y^2-x(x-1)(x+1)$. We may regard $k[x,y]$ as $k[x][y]$. Pick $p(x)=x$, then $p(x)\mid -x(x-1)(x+1), p(x)\nmid 1, p(x)^2\nmid -x(x-1)(x+1)$. By Eisenstein's Criterion, we know that $y^2-x(x-1)(x+1)$ is irreducible over $k(x)[y]$, and therefore irreducible over $k[x,y]$.

We next show that $f(x,y,z)=0$ is smooth. Recall Proposition 14.3.1, we only need to check that the following equations has no solution on $\overline{k}$

$$\begin{cases} f(x,y,z)=x^3-y^2z-xz^2 & =0\\ \dfrac{\partial f}{\partial x}=3x^2-z^2 & =0\\ \dfrac{\partial f}{\partial y}=-2yz & =0\\ \dfrac{\partial f}{\partial z}=-y^2-2xz & =0.\end{cases}$$

Use $\operatorname{char} k \neq 2$, it is easy to check that above equations has no solution on $\overline{k}$, we win. □

✍ **Exercise 14.8** Suppose $\operatorname{char} k \neq 2$. Show that a quadric hypersurface in $\mathbb{P}^n$ is smooth if and only if it is maximal rank (defined in Exercise 6.7).

Proof By Linear Algebra, we may assume quadric hypersurface $Q(x) = x^t M x = \sum_{i,j=0}^{n} M_{i,j} x_i x_j$, where $M = (M_{ij})$ is a symmetry matrix and $x = (x_0, \dots, x_n)^t$. Consider $\frac{\partial Q}{\partial x_k}$, we have

$$\frac{\partial Q}{\partial x_k} = \sum_{i=0}^{n} M_{ik} x_i + \sum_{j=0}^{n} M_{kj} x_j = 2 \sum_{i=1}^{n} M_{ik} x_i.$$

Let

$$\begin{cases} \dfrac{\partial Q}{\partial x_0} = 2M_{00}x_0 + \cdots + 2M_{n0}x_n & = 0 \\ \dfrac{\partial Q}{\partial x_1} = 2M_{01}x_0 + \cdots + 2M_{n1}x_n & = 0 \\ \quad\cdots & \\ \dfrac{\partial Q}{\partial x_n} = 2M_{0n}x_0 + \cdots + 2M_{nn}x_n & = 0, \end{cases}$$

that is,

$$2 \begin{pmatrix} M_{00} & \cdots & M_{n0} \\ \vdots & \ddots & \vdots \\ M_{0n} & \cdots & M_{nn} \end{pmatrix} \begin{pmatrix} x_0 \\ \vdots \\ x_n \end{pmatrix} = 2M^t x = 0. \tag{14.4}$$

Note that $\operatorname{char} k \nmid \deg Q = 2$, by Proposition 14.3.1, Q is smooth if and only if (14.4) does not have nonzero solution if and only if M maximal rank. □

✍ **Exercise 14.9** Suppose $k = \overline{k}$ has characteristic 0. Show that there exists a regular (projective) plane curve of degree d.

Remark 14.17 Feel free to weaken the hypotheses. Bertini's Theorem 14.4.1 will give another means of showing existence.

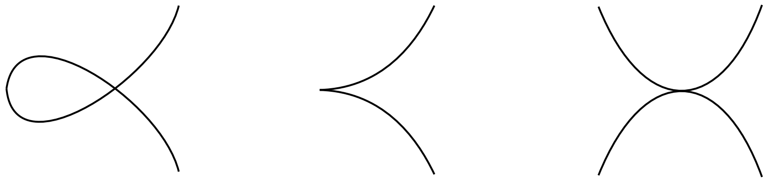
Proof Consider Fermat curve $f(x, y, z) = x^d + y^d + z^d = 0$. By Proposition 14.3.1, the singular closed points are given by

$$\begin{cases} f(x,y,z) = x^d + y^d + z^d & = 0 \\ \dfrac{\partial f}{\partial x} = dx^{d-1} & = 0 \\ \dfrac{\partial f}{\partial x} = dy^{d-1} & = 0 \\ \dfrac{\partial f}{\partial z} = dz^{d-1} & = 0. \end{cases}$$

Note that above equations does not have nonzero solution, so every closed point on Fermat curve is regular, by Proposition 14.2.7, Fermat curve is smooth, and therefore regular given by Smoothness-Regularity Comparison Theorem 14.2.1. □

✍ **Exercise 14.10 (See Figure 14.3)** Assume $k = \overline{k}$ and $\operatorname{char} k = 0$ to avoid distractions. Finding the singular closed points (which will be the non-smooth points) of the following plane curves.

(a) $y^2 = x^2 + x^3$. This is an example of a **node**, which we saw earlier in Exercise 8.10.
(b) $y^2 = x^3$. This is called a **cusp**; we met it earlier in Example 11.11 .
(c) $y^2 = x^4$. This is called a **tacnode**; we met it earlier in Example 11.12 .



(a) (b) (c)

Figure 14.3: Plane curve singularities.

Solution *For (a). Let* $f(x,y) = x^2 - y^2 + x^3$*. Consider equations*

$$f = \frac{\partial f}{\partial x} = \frac{\partial f}{\partial y} = 0,$$

then

$$\begin{cases} f(x,y) = x^2 - y^2 + x^3 & = 0 \\ \dfrac{\partial f}{\partial x} = 2x + 3x^2 & = 0 \\ \dfrac{\partial f}{\partial y} = -2y & = 0 \end{cases} \implies \begin{cases} x = 0 \\ y = 0. \end{cases}$$

It follows that $f = 0$ *is singular at* $(0,0)$.

For (b). Let $g(x,y) = y^2 - x^3$*. Consider equations*

$$g = \frac{\partial g}{\partial x} = \frac{\partial g}{\partial y} = 0,$$

then

$$\begin{cases} g(x,y) = y^2 - x^3 & = 0 \\ \dfrac{\partial g}{\partial x} = -3x^2 & = 0 \\ \dfrac{\partial g}{\partial y} = 2y & = 0 \end{cases} \implies \begin{cases} x = 0 \\ y = 0. \end{cases}$$

It follows that $g = 0$ *is singular at* $(0,0)$.

For (c). Let $h(x,y) = y^2 - x^4$*. Consider equations*

$$h = \frac{\partial h}{\partial x} = \frac{\partial h}{\partial y} = 0,$$

then

$$\begin{cases} h(x,y) = y^2 - x^4 & = 0 \\ \dfrac{\partial h}{\partial x} = -4x^3 & = 0 \\ \dfrac{\partial g}{\partial y} = 2y & = 0 \end{cases} \implies \begin{cases} x = 0 \\ y = 0. \end{cases}$$

It follows that $h = 0$ *is singular at* $(0,0)$.

Remark 14.18 There are different possible definitions of node, cusp, and tacnode. One reasonable definition in the case $k = \overline{k}$ is the follows. Suppose C is pure dimension 1, and $p \in C$ is a closed point. Let $(A, \mathfrak{m})$ be the local ring $\mathscr{O}_{C,p}$, so $A/\mathfrak{m} = k$. We say p is a **node** of C if $\dim_k \mathfrak{m}/\mathfrak{m}^2 = 2$, the kernel of

$$\mathrm{Sym}^2(\mathfrak{m}/\mathfrak{m}^2) \longrightarrow \mathfrak{m}^2/\mathfrak{m}^3$$

has dimension 1, and the kernel is not spanned by the square of an element in $\mathfrak{m}/\mathfrak{m}^2$. We won't need this definition, so you needn't memorize it.

Why it is reasonable? Suppose C is cut by $f = 0$. Since p is singular point, $\frac{\partial f}{\partial x_1}(p) = \cdots =$

$\frac{\partial f}{\partial x_n}(p) = 0$, so we may write $f(x_1, \ldots, x_n) = g(x_1, \ldots, x_n) + (\text{degree} \geq 3 \text{ terms})$ where $g(x_1, \ldots, x_n)$ is degree 2 homogeneous polynomial. Then $\operatorname{Ker}(\varphi : \operatorname{Sym}^2(\mathfrak{m}/\mathfrak{m}^2) \to \mathfrak{m}^2/\mathfrak{m}^3) = \operatorname{Span}\{g(x_1, \ldots, x_n)\}$. $\dim_k \operatorname{Ker} \varphi = 1$ implies $g \neq 0$. Note that on algebraic closed field k, $g = l_1 \cdot l_2$ or $g = l^2$ where l, l_1, l_2 are degree 1 homogeneous polynomial and $l_1 \neq l_2$. The kernel is not spanned by the square of an element in $\mathfrak{m}/\mathfrak{m}^2$ implies that $g = l_1 \cdot l_2$. A cusp means there are two different tangent lines passing through singular point p, so the definition above is reasonable.

We next give a guess for a definition of a cusp. From a geometric intuition, a cusp curve has two coincident tangent lines at the cusp, so we define a cusp as follow: we say p is a **cusp** of C if $\dim_k \mathfrak{m}/\mathfrak{m}^2 = 2$, the kernel of

$$\operatorname{Sym}^2(\mathfrak{m}/\mathfrak{m}^2) \longrightarrow \mathfrak{m}^2/\mathfrak{m}^3$$

has dimension 1, and the kernel is spanned by the square of an element in $\mathfrak{m}/\mathfrak{m}^2$.

Alternate definitions of node, cusp, and more will be given in §**??**.

Exercise 14.11 Show that the twisted cubic $\operatorname{Proj} k[w, x, y, z]/(wz - xy, wy - x^2, xz - y^2)$ is smooth.

Proof Recall Exercise 10.3, $\operatorname{Proj} k[w, x, y, z]/(wz - xy, wy - x^2, xz - y^2) \cong \mathbb{P}^1_k$, and $\mathbb{P}^1_k$ is smooth. □

Tangent planes and tangent lines

Definition 14.3.1 (Tangent planes, tangent lines)

*Suppose a scheme $X \subseteq \mathbb{A}^n$ is cut out by equations $f_1, \ldots, f_r$, and X is smooth of (pure) dimension d at the k-valued point $a = (a_1, \ldots, a_n)$. Then the **tangent d-plane to X at** a (sometimes denoted $T_a X$) is given by the r equations*

$$\left(\frac{\partial f_i}{\partial x_1}(a)\right)(x_1 - a_1) + \cdots + \left(\frac{\partial f_i}{\partial x_n}(a)\right)(x_n - a_n) = 0,$$

*where (as in (14.1)) $\frac{\partial f_i}{\partial x_j}(a)$ is the evaluation of $\frac{\partial f_i}{\partial x_j}$ at $a = (a_1, \ldots, a_n)$. If $d = 1$, $T_a X$ is called the **tangent line**.*

Exercise 14.12 Why is this independent of the choice of defining equations $f_1, \ldots, f_r$ of X?

Proof Let $I = (f_1, \ldots, f_r)$. Suppose $g \in I$, there exists $l_1, \ldots, l_r \in k[x_1, \ldots, x_n]$ such that $g = \sum_{i=1}^{r} l_i f_i$, so we have

$$\frac{\partial g}{\partial x_j} = \sum_{i=1}^{r} \frac{\partial l_i f_i}{\partial x_j} = \sum_{i=1}^{r} \left(\frac{\partial l_i}{\partial x_j} f_i + l_i \frac{\partial f_i}{\partial x_j}\right).$$

Since X is smooth at the k-valued point a, $\frac{\partial f_i}{\partial x_j}(a) = 0$, so $\frac{\partial g}{\partial x_j}(a) = \sum_{i=1}^{r} l_i(a) \frac{\partial f_i}{\partial x_j}(a)$. Let

$$L_i(x_1, \ldots, x_n) = \left(\frac{\partial f_i}{\partial x_1}(a)\right)(x_1 - a_1) + \cdots + \left(\frac{\partial f_i}{\partial x_n}(a)\right)(x_n - a_n)$$

and

$$L_g(x_1, \ldots, x_n) = \left(\frac{\partial g}{\partial x_1}(a)\right)(x_1 - a_1) + \cdots + \left(\frac{\partial g}{\partial x_n}(a)\right)(x_n - a_n),$$

then

$$\begin{aligned} L_g(x_1, \ldots, x_n) &= \left(\sum_{i=1}^{r} l_i(a) \frac{\partial f_i}{\partial x_1}(a)\right)(x_1 - a_1) + \cdots + \left(\sum_{i=1}^{r} l_i(a) \frac{\partial f_i}{\partial x_n}(a)\right)(x_n - a_n) \\ &= \sum_{i=1}^{r} l_i(a) L_i(x_1, \ldots, x_n). \end{aligned}$$

Hence $L_i = 0$ implies $L_g = 0$, it follows that tangent planes is independent of the choice of defining equations. □

The Jacobian criterion for regularity 14.2.2 ensures that these r equations indeed cut out a d-plane. This can be readily shown to be notion of tangent plane that we see in multivariable calculus, but note that here this is the definition, and thus we don't have to worry about δ's and ϵ's. Instead we will have to just be careful that it behaves the way we want it to.

✍ **Exercise 14.13** Compute the tangent line to the curve of Exercise 14.10 (b) at $(1,1)$.

Solution *Let $f(x,y) = y^2 - x^3$ and $p = (1,1)$, then $\frac{\partial f}{\partial x}(p) = -3x^2|_p = -3$ and $\frac{\partial f}{\partial y}(p) = 2y|_p = 2$, so the tangent line at $(1,1)$ is*

$$\left(\frac{\partial f}{\partial x}(p)\right)(x-1) + \cdots + \left(\frac{\partial f}{\partial y}(p)\right)(y-1) = -3(x-1) + 2(y-1) = 0.$$

Definition 14.3.2 (Projective tangent planes)

Suppose $X \subseteq \mathbb{P}^n_k$ (k, as usual, a field) is cut out by homogeneous equations $f_1, \dots, f_r$, and $p = [p_0 : \cdots : p_n] \in X$ is a k-valued point that is smooth of (pure) dimension d. The projective tangent d-plane to X at p (denote T_pX) is given by the r equations

$$\left(\frac{\partial f_i}{\partial x_0}(p)\right)(x_0 - p_0) + \cdots + \left(\frac{\partial f_i}{\partial x_n}(p)\right)(x_n - p_n) = 0$$

where $i = 1, \dots, r$.

✍ **Exercise 14.14** Compute the (projective) tangent line to the twisted cubic curve of Exercise 14.11 at $[1:1:1:1]$.

Solution *Let $f_1(w,x,y,z) = wz - xy$, $f_2(w,x,y,z) = wy - x^2$, and $f_3(w,x,y,z) = xz - y^2$, then*

$$\begin{cases} w - x - y + z = 0 \\ w - 2x + y = 0 \\ x - 2y + z = 0 \end{cases} \implies \begin{cases} w - 2x + y = 0 \\ x - 2y + z = 0 \end{cases}$$

is the projective tangent line of twisted cubic curve at $[1:1:1:1]$.

Remark 14.19 Side remark to help you think cleanly. We would want the definition of tangent k-plane to be natural in the sense that for any automorphism σ of $\mathbb{A}^n_k$ (or, $\mathbb{P}^n_k$), $\sigma(T_pX) = T_{\sigma(p)}\sigma(X)$. You could verify this by hand, but you can also say this in a cleaner way, by interpreting the equations cutting out the tangent space in a coordinate free manner. Informally speaking, we are using the canonical identification of n-space with the tangent space to n-space at p, and using the fact that the Jacobian "linear transformation" cuts out T_pX in $T_p\mathbb{A}^n$ in a way independent of the choice of coordinates on $\mathbb{A}^n$ or our defining equations of X. Your solution to Exercise 14.12 will help you start to think in this way.

✍ **Exercise 14.15** Suppose $X \subseteq \mathbb{P}^n_k$ is a degree d hypersurface cut by $f = 0$, and L is a line not contained in X. Exercise 10.4 (a case of Bézout's Theorem) showed that X and L meet at d points, counted "with multiplicity". Suppose L meets X "with multiplicity at least 2" at a k-valued point $p \in L \cap X$, and that p is a regular point of X. Show that L is contained in the tangent plane to X at p.

Proof Let $q \in L$ and $q \notin X$, say $p = [p_0 : \cdots : p_n]$ and $q = [q_0 : \cdots : q_n]$, then

$$L = \{[p_0 s + q_0 t : \cdots : p_n s + q_n t] : s \in k, t \in k\}.$$

Consider $f|_L = 0$, we have

$$F(s,t) := f(p_0 s + q_0 t, \cdots, p_n s + q_n t) = 0$$

Since f is homogeneous polynomial, say $\deg f = d$, we know that $F(s,t) \in (k[s,t]_{\bullet})_d$, so we may write

$$F(s,t) = a_0 s^d + a_1 s^{d-1} t + a_2 s^{d-2} t^2 + \cdots + a_n t^d$$

where $a_i \in k$. Since $p \in X$, $F(1,0) = 0$, hence $a_0 = 0$. Since L meets X "with multiplicity at least 2", we know that $t^2 \mid F(s,t)$, so $a_1 = 0$. Note that

$$\frac{\partial F}{\partial t} = a_1 s^{d-1} + 2a_2 s^{d-2} t + \cdots + d a_n t^{d-1},$$

we have $a_1 = \frac{\partial F}{\partial t}([1:0])$. On the other hand,

$$\frac{\partial F}{\partial t} = \frac{\partial f(sp + tq)}{\partial t} = \sum_{i=0}^{n} \frac{\partial f}{\partial x_i} \cdot \frac{\partial x_i}{\partial t},$$

so $\frac{\partial F}{\partial t}([1:0]) = \sum\limits_{i=0}^{n} \frac{\partial f}{\partial x_i}(p) \cdot \frac{\partial x_i}{\partial t}([1:0]) = \sum\limits_{i=0}^{n} \frac{\partial f}{\partial x_i}(p) q_i$. Hence, $\sum\limits_{i=0}^{n} \frac{\partial f}{\partial x_i}(p) q_i = 0$. By Euler's Homogeneous Function Theorem 14.1.2, we have $\deg(f) \cdot f = \sum\limits_{i=1}^{n} x_i \cdot \frac{\partial f}{\partial x_i}$, so $\deg(f) \cdot f(p) = \sum\limits_{i=1}^{n} \frac{\partial f}{\partial x_i}(p) p_i = 0$. Hence

$$\sum_{i=0}^{n} \frac{\partial f}{\partial x_i}(p)(q_i - p_i) = 0,$$

which implies q contained in a tangent plane to X at p. □

14.3.2 Arithmetic examples

✍ **Exercise 14.16** Show that $\operatorname{Spec} \mathbb{Z}$ is a regular curve.

Proof Let $p \in \mathbb{Z}$ be a prime number, consider Noetherian local ring $(\mathbb{Z}_{(p)}, (p)\mathbb{Z}_{(p)})$, clearly, $\dim \mathbb{Z}_{(p)} = 1$ and $\dim_{\mathbb{F}_p} \frac{(p)\mathbb{Z}_{(p)}}{((p)\mathbb{Z}_{(p)})^2} = \dim_{\mathbb{F}_p} \frac{(p)}{(p^2)} = 1$. So $\mathbb{Z}$ is regular at its closed points.

Consider generic point (0). We have $\mathbb{Z}_{(0)} \cong \mathbb{Q}$, so $\dim \mathbb{Z}_{(0)} = 0$. On the other hand, $\dim_{\mathbb{Q}} (0)/(0)^2 = 0$, so $\mathbb{Z}$ is regular at its generic points. Combine above results, we know that $\operatorname{Spec} \mathbb{Z}$ is regular. □

✍ **Exercise 14.17** (This tricky exercise is for those who know about the primes of the Gaussian integers $\mathbb{Z}[i]$.) There are several ways of showing that $\mathbb{Z}[i]$ is dimension 1. (For example: (i) It is a principal ideal domain; (ii) it is the normalization of $\mathbb{Z}$ in the field extension $\mathbb{Q}(i)/\mathbb{Q}$; (iii) using Kurll's Principal Ideal Theorem 13.3.2 and the fact that $\dim \mathbb{Z}[x] = 2$ by Exercise 13.1.) Show that $\operatorname{Spec} \mathbb{Z}[i]$ is a regular curve. (There are several ways to proceed. You could use Proposition 14.1.1. As an example to work through first, consider the prime $(1+i)$, which is cut out by the equation $1 + x$ in $\operatorname{Spec} \mathbb{Z}[x]/(x^2+1)$.) We will late (Example 14.7) have a simpler approach once we discuss discrete valuation rings.

Proof Recall Example 11.13, we know that $\mathbb{Z}[i] = \operatorname{Int.cl}_{\mathbb{Q}(i)}(\mathbb{Z})$, so $\mathbb{Z}[i]$ is the normalization of $\mathbb{Z}$ in the $\mathbb{Q}(i)/\mathbb{Q}$. By Proposition 13.1.7, $\dim \mathbb{Z}[i] = \dim \mathbb{Z} = 1$. Let $\mathfrak{p}$ be a prime ideal of $\mathbb{Z}[i]$, since $\dim \mathbb{Z}[i] = 1$, $\mathfrak{p} = (0)$ or $\mathfrak{p}$ is maximal ideal, so $\dim \mathbb{Z}[i]_{\mathfrak{p}} = 1$ for $\mathfrak{p} \neq 0$. In fact, $\mathbb{Z}[i]$ is PID (since $\mathbb{Z}[i]$ is ED, see [Rot15, E.g. A-3.117]). So $\dim_{\mathbb{Z}[i]/\mathfrak{p}} \mathfrak{p}/\mathfrak{p}^2 = 1$ for $\mathfrak{p} \neq (0)$, hence $\mathbb{Z}[i]$ is regular at its closed points. Since $\dim \mathbb{Z}[i]_{(0)} = \dim \mathbb{Q}(i) = \dim_{\mathbb{Q}(i)} (0)/(0)^2 = 0$, $\mathbb{Z}[i]$ is regular at its generic points. Hence $\operatorname{Spec} \mathbb{Z}[i]$ is a regular curve. □

✍ **Exercise 14.18** Show that $[(5, 5i)]$ is the unique singular point of $\operatorname{Spec} \mathbb{Z}[5i]$.

Proof Consider $\operatorname{Spec} \mathbb{Z}[5i] = D(5) \cup V(5)$. On $D(5) = \operatorname{Spec} \mathbb{Z}[5i]_5$, note that $\mathbb{Z}[5i]_5 \cong \mathbb{Z}[i]_5$, we have $D(5) \cong \operatorname{Spec} \mathbb{Z}[i]_5$. By Exercise 14.17, $\operatorname{Spec} \mathbb{Z}[i]$ is regular, so its open subset $\operatorname{Spec} \mathbb{Z}[i]_5$ is also regular. Hence $\operatorname{Spec} \mathbb{Z}[5i]$ is regular on $D(5)$. We next check $V(5)$. Note that

$$V(5) = \operatorname{Spec} \mathbb{Z}[5i]/(5) \cong \operatorname{Spec} \mathbb{Z}[x]/(x^2 + 25, 5) \cong \operatorname{Spec} \mathbb{F}_5[x]/(x^2).$$

Hence $V(5)$ has only one point $\mathfrak{m} = [(5, 5i)]$. Now, we want to show that $\mathfrak{m}$ is a singular point. Since

$R/(5) := \mathbb{Z}[5i]/(5) \cong \mathbb{F}_5[x]/(x^2) =: S$, we have $\mathrm{ht}_{R/(5)}(\mathfrak{m}/(5)) = \mathrm{ht}_S((x)) = 0$. So $\dim R_{\mathfrak{m}} = \mathrm{ht}_R(\mathfrak{m}) = \mathrm{ht}_{R/(5)}(\mathfrak{m}/(5)) + 1 = 1$ (Krull's Principal Ideal Theorem 13.3.2). Note that $\mathfrak{m}/\mathfrak{m}^2 = (5, 5i)/(25, 25i)$, $\dim_{\mathbb{F}_5} \mathfrak{m}/\mathfrak{m}^2 = 2$, hence $\dim R_{\mathfrak{m}} < \dim_{\mathbb{F}_5} \mathfrak{m}/\mathfrak{m}^2$, which implies that $\mathfrak{m}$ is singular. □

14.3.3 Back to geometry: $\mathbb{A}_k^n$ is regular

The key step to show that $\mathbb{A}_k^n$ is regular (where k is a field) is the following.

Proposition 14.3.2

Suppose $(B, \mathfrak{n}, k)$ is a regular local ring of dimension d. Let $\varphi : B \to B[x]$. Suppose $\mathfrak{p}$ is a prime ideal of $A := B[x]$ such that $\mathfrak{n}B[x] \subseteq \mathfrak{p}$. Then $A_{\mathfrak{p}}$ is a regular local ring.

Proof Geometrically: we have a morphism $\pi : X = \operatorname{Spec} B[x] \to Y = \operatorname{Spec} B$, and $\pi([\mathfrak{p}]) = [\mathfrak{n}]$. The fiber $\pi^{-1}([\mathfrak{n}])$ is $\operatorname{Spec} B/\mathfrak{n} \times_B \operatorname{Spec} B[x] = \operatorname{Spec}(B[x]/\mathfrak{n}B[x]) = \operatorname{Spec} k[x]$. Thus either (i) $\overline{[\mathfrak{p}]}$ is the fiber $\mathbb{A}_k^1$ above $[\mathfrak{n}]$, or (ii) is a closed point of the fiber $\mathbb{A}_k^1$ (since $k[x]$ is PID). (See Figure 14.4.)

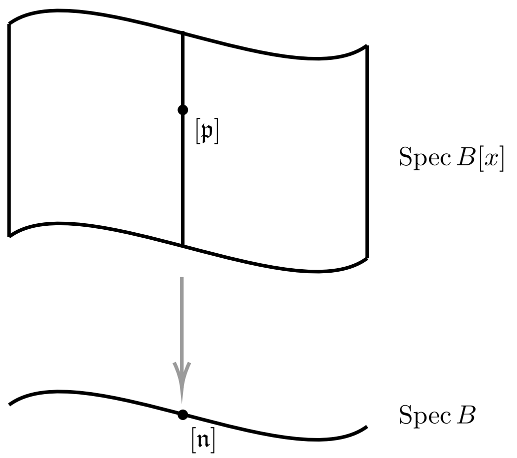


Figure 14.4: The geometry behind the proof of Proposition 14.3.2.

Before consider these two cases, we make two remarks. As $(B, \mathfrak{n})$ is a regular local ring of dimension d, $\mathfrak{n}$ is generated by d elements of B, say, $f_1, \ldots, f_d$ (Lemma 14.2.3, by Nakayama's Lemma 9.2.6, in fact, they are generators), and there is a chain of prime ideals

$$0 = \mathfrak{q}_0 \subsetneq \mathfrak{q}_1 \subsetneq \cdots \subsetneq \mathfrak{q}_d = \mathfrak{n}. \tag{14.5}$$

Case (i): $\overline{[\mathfrak{p}]}$ is the fiber $\mathbb{A}_k^1$. In this case, $\mathfrak{p} = \mathfrak{n}B[x]$. Thus by Krull's Height Theorem 13.3.3, the codimension of $\mathfrak{p} = \mathfrak{n}B[x]$ is at most d, because $\mathfrak{p}$ is generated by the d elements $f_1, \ldots, f_d$. But $\mathfrak{p}$ has codimension at least d: given the chain (14.5) ending with $\mathfrak{n}$, we have a corresponding chain of prime ideals of $B[x]$ ending with $\mathfrak{n}B[x]$. Hence the codimension of $\mathfrak{p}$ is precisely d, and $\mathfrak{p}A_{\mathfrak{p}}$ is generated by $f_1, \ldots, f_d$, implying that it is a regular local ring (because $\dim A_{\mathfrak{p}} = \mathrm{ht}(\mathfrak{p}) = d = \dim_{A_{\mathfrak{p}}/\mathfrak{p}A_{\mathfrak{p}}} \frac{\mathfrak{p}A_{\mathfrak{p}}}{(\mathfrak{p}A_{\mathfrak{p}})^2}$).

Case (ii): $\overline{[\mathfrak{p}]}$ is a closed point of the fiber $\mathbb{A}_k^1$. The closed point $\mathfrak{p}$ of $k[x]$ corresponds to some monic irreducible polynomial $g(x) \in k[x]$. Arbitrarily lift the coefficients of g to B; we sloppily denote the resulting polynomial in $B[x]$ by $g(x)$ as well. Then $\mathfrak{p} = (f_1, \ldots, f_d, g)$, so by Krull's Height Theorem 13.3.3, the codimension of $\mathfrak{p}$ is at most $d + 1$. But $\mathfrak{p}$ has codimension of $\mathfrak{p}$ is at least $d + 1$: given the chain (14.5) ending with $\mathfrak{n}$, we have a corresponding chain in $B[x]$ ending with $\mathfrak{n}B[x]$, and we can extend it by appending $\mathfrak{p}$. Hence the codimension of $\mathfrak{p}$ is precisely $d+1$, and $\mathfrak{p}A_{\mathfrak{p}}$ is generated by $f_1, \ldots, f_d, g$, implying that it is a regular local ring. $\square$

Theorem 14.3.1

If X is a regular (locally Noetherian) scheme, then so is $X \times \mathbb{A}^1$. In particular, $\mathbb{A}_k^n$ is regular.

Proof Since regularity is checked locally, we may assume $X = \operatorname{Spec} B$ where B is a Noetherian ring, then $X \times \mathbb{A}^1 = \operatorname{Spec} B[x]$. Let $[\mathfrak{p}] \in X \times \mathbb{A}^1$, we want to show that $B[x]_{\mathfrak{p}}$ is a regular local ring. Consider the following commutative diagram (for tensor product $B_{\mathfrak{p}} \otimes_B B[x] \cong B_{\mathfrak{p}}[x]$),

$$\begin{tikzcd} \mathfrak{q} \arrow[d, mapsto] & B \arrow[r, "\varphi"] \arrow[d] & B[x] \arrow[d] & \mathfrak{p} \arrow[d, mapsto] \\ \mathfrak{q}B_{\mathfrak{q}} & B_{\mathfrak{q}} \arrow[r] & B_{\mathfrak{q}}[x] & \mathfrak{p}B_{\mathfrak{q}}[x] \end{tikzcd}$$

where $\mathfrak{q} = \varphi^{-1}(\mathfrak{p}) = \mathfrak{p} \cap B$. We first show that $B_{\mathfrak{q}}[x]_{\mathfrak{p}B_{\mathfrak{q}}[x]}$ is a regular local ring. Recall Proposition 14.3.2, it is suffices to show that $\mathfrak{q}B_{\mathfrak{q}} \cdot B_{\mathfrak{q}}[x] = \mathfrak{q}B_{\mathfrak{q}}[x] \subseteq \mathfrak{p}B_{\mathfrak{q}}[x]$. Since $\mathfrak{p} \cap B = \mathfrak{q}$, we have $\mathfrak{q}B[x] \subseteq \mathfrak{p}$. So

$$\mathfrak{q}B_{\mathfrak{q}}[x] = \mathfrak{q}B[x] \otimes_{B[x]} B_{\mathfrak{q}}[x] \subseteq \mathfrak{p} \otimes_{B[x]} B_{\mathfrak{q}}[x] = \mathfrak{p}B_{\mathfrak{q}}[x],$$

by Proposition 14.3.2, $B_{\mathfrak{q}}[x]_{\mathfrak{p}B_{\mathfrak{q}}[x]}$ is a regular local ring. Claim that $B_{\mathfrak{q}}[x]_{\mathfrak{p}B_{\mathfrak{q}}[x]} \cong B[x]_{\mathfrak{p}}$. Consider $\psi : B[x] \to B_{\mathfrak{q}}[x]_{\mathfrak{p}B_{\mathfrak{q}}[x]}$, we have $\psi(B[x] \setminus \mathfrak{p}) \subseteq B_{\mathfrak{q}}[x] \setminus \mathfrak{p}B_{\mathfrak{q}}[x]$, by the universal property of localization, there exists a unique morphism $B[x]_{\mathfrak{p}} \to B_{\mathfrak{q}}[x]_{\mathfrak{p}B_{\mathfrak{q}}[x]}$ such that the following diagram commutes.

$$\begin{tikzcd} B[x] \arrow[r, "\psi"] \arrow[d, "\text{local.}"'] & B_{\mathfrak{q}}[x]_{\mathfrak{p}B_{\mathfrak{q}}[x]} \\ B[x]_{\mathfrak{p}} \arrow[ur, dashed, "\exists!"'] & \end{tikzcd}$$

Consider $\theta : B_{\mathfrak{q}}[x] \to B[x]_{\mathfrak{p}}$ given by $\frac{f}{g} \mapsto \frac{f}{g}$ where $f \in B[x]$ and $g \in B \setminus \mathfrak{q}$. Since $\mathfrak{q} = \mathfrak{p} \cap B$, we have $\theta(B_{\mathfrak{q}}[x] \setminus \mathfrak{p}B_{\mathfrak{q}}[x]) \subseteq B[x] \setminus \mathfrak{p}$ by the universal property of localization, there exists a unique morphism $B_{\mathfrak{q}}[x]_{\mathfrak{p}B_{\mathfrak{q}}[x]} \to B[x]_{\mathfrak{p}}$ such that the following diagram commutes.

$$\begin{tikzcd} B_{\mathfrak{q}}[x] \arrow[r, "\theta"] \arrow[d, "\text{local.}"'] & B[x]_{\mathfrak{p}} \\ B_{\mathfrak{q}}[x]_{\mathfrak{p}B_{\mathfrak{q}}[x]} \arrow[ur, dashed, "\exists!"'] & \end{tikzcd}$$

Hence, $B[x]_{\mathfrak{p}} \cong B_{\mathfrak{q}}[x]_{\mathfrak{p}B_{\mathfrak{q}}[x]}$, which implies that $B[x]_{\mathfrak{p}}$ is regular local ring, i.e., $X \times \mathbb{A}^1$ is regular.

In particular, by Exercise 14.6 , $\mathbb{A}_k^1$ is regular, by induction, so is $\mathbb{A}_k^n$. $\square$

14.4 Bertini's Theorem

14.4.1 Bertini-type theorems

We now discuss Bertini's Theorem, a fundamental classical result.

Definition 14.4.1 (Dual projective space)

*The **dual** (or **dual projective space**) to $\mathbb{P}_k^n$ (with projective coordinates $x_0, \ldots, x_n$) is informally the space of hyperplanes in $\mathbb{P}_k^n$. More precisely, it is a projective space $\mathbb{P}_k^n$ with projective coordinates $a_0, \ldots, a_n$ (which we denote $\mathbb{P}_k^{n\vee}$ in the futile hope of preventing confusion), along with the data of "incidence* *variety" or "incidence correspondence" $I \subseteq \mathbb{P}^n \times \mathbb{P}^{n\vee}$ cut out by the equation $a_0x_0 + \cdots + a_nx_n = 0$. (This is often called the* ***universal hyperplane.****)*

Remark 14.20 Note that the k-valued points of $\mathbb{P}^{n\vee}$ indeed correspond to hyperplanes in $\mathbb{P}^n$ defined over k, and this is also clearly a duality relation (there is a symmetry in the definition between the x-variables and the a-variables). This definition is concrete enough to use in practice, and extends over an arbitrary base (notably $\operatorname{Spec} \mathbb{Z}$). (But if you have a delicate and refined sensibility, you may want to come up with a coordinate-free definition.)

Theorem 14.4.1 (Bertini's Theorem)

Suppose X is a smooth closed subvariety of $\mathbb{P}^n_k$ of (pure) dimension d. Then there is a nonempty (=dense) open subset U of dual projective space $\mathbb{P}^{n\vee}_k$ such that for every point $p = [H] \in U$, the scheme $H \cap X$ is smooth over $\kappa(p)$ of (pure) dimension $d - 1$.

(You should perhaps think first of the case where $\kappa(p) = k$, i.e., H is a hyperplane cut out by a linear form with coefficients in k.) Any theorem of this flavor is often called a "Bertini Theorem". One example is the Kleiman-Bertini Theorem **??**, which was not proved jointly by Kleiman and Bertini.

As an application of Bertini's Theorem 14.4.1, a general degree $d > 0$ hypersurface in $\mathbb{P}^n_k$ intersects X in a smooth subvariety of codimension 1 in X: replace $X \hookrightarrow \mathbb{P}^n$ with the composition

$$X \longrightarrow \mathbb{P}^n \xhookrightarrow{v_d} \mathbb{P}^N,$$

where v_d is the d-th Veronese embedding (Definition 10.3.3). Here "general" has its usual meaning in algebraic geometry; see §11.3.4. (See Proposition 14.4.1 for a generalization.) A useful consequence of this, taking $X = \mathbb{P}^n$: we immediately see that there exists a smooth degree d hypersurface in $\mathbb{P}^n$ (over any infinite field k), without any meeting around with specific equations (as in the special case Exercise 14.9).

?? gives a useful improvement of Bertini's Theorem in characteristic 0 (see **??**).

Proof of Bertini's Theorem 14.4.1

Proof The central idea of the proof is quite naive. We will describe the hyperplanes that are "bad", and show that they form a closed subset of dimension at most $n - 1$ of $\mathbb{P}^{n\vee}_k$, and hence that the complement is a dense open subset. Somewhat more precisely, we will define a projective variety $Z \subseteq X \times \mathbb{P}^{n\vee}_k$ that can informally be described as:

$$Z = \{(p \in X, H \subseteq \mathbb{P}^n_k) : p \in H, \text{ and either } p \text{ is a singular point of } H \cap X, \text{ or } X \subseteq H\}. \tag{14.6}$$

We will see that the projection $\pi : Z \to X$ has fibers at closed points that are projective spaces of dimension $n - 1 - \dim X$, and hence that $\dim Z \leq n - 1$. Thus the image of Z in $\mathbb{P}^{n\vee}_k$ will be a closed subset (Fundamental Theorem of Elimination Theory 9.4.3), of dimension of at most $n - 1$, so its complement will be open and nonempty. We now put this strategy into action.

We first define Z more precisely, in terms of equations on $\mathbb{P}^n \times \mathbb{P}^{n\vee}$, where the coordinates on $\mathbb{P}^n$ are $x_0, \ldots, x_n$, and the dual coordinates on $\mathbb{P}^{n\vee}$ are $a_0, \ldots, a_n$. Suppose X is cut out by $f_1, \ldots, f_r$. Then we take these equations as the first of the defining equations of Z. (So far we have defined the subscheme $X \times \mathbb{P}^{n\vee}$.) We also add the equation $a_0x_0 + \cdots + a_nx_n = 0$. (So far we have described the subscheme of $\mathbb{P}^n \times \mathbb{P}^{n\vee}$

corresponding to point (p, H), where $p \in X$ and $p \in H$.) Note that the projective Jacobian matrix (14.2)

$$\begin{pmatrix} \frac{\partial f_1}{\partial x_0}(p) & \cdots & \frac{\partial f_r}{\partial x_0}(p) \\ \vdots & \ddots & \vdots \\ \frac{\partial f_1}{\partial x_n}(p) & \cdots & \frac{\partial f_r}{\partial x_n}(p) \end{pmatrix}$$

has corank precisely $d+1$ at all points of X by the projective Jacobian corank criterion (Proposition 14.1.9). We then require that the projective Jacobian matrix with a new column.

$$\begin{pmatrix} a_0 \\ \vdots \\ a_n \end{pmatrix}$$

appended has corank $\geq d+1$ (hence $= d+1$). This is precisely the notion that we wished to capture. This is cut out by equations (the determinants of certain minors). (Remark after the proof works through an example, which may help clarify how this works.)

Small remark

Why we need $\operatorname{corank} \geq d+1$**?**
Singular point means the dimension of tangent space at this point bigger than the local dimension of $X \cap H$. If H is not tangent to X, $\dim X \cap H = \dim X - 1$, so the corank of projective Jacobian matrix is d. If H is tangent to X, the corank of projective Jacobian matrix is $\geq d+1$. This is why we need $\operatorname{corank} \geq d+1$.

We next show that $\dim Z \leq n-1$. For each closed point $p \in X$, let W_p be the locus of hyperplanes containing p, such that $H \cap X$ is singular at p, or else contains all of X; what is the dimension of W_p? Suppose $\dim X = d$. Then the restrictions on the hyperplanes in the definition of W_p correspond to $d+1$ linear conditions (H contains p gives 1 conditions; H contains dimension d tangent space at p, this gives d conditions, more precisely, the normal vector of H must be perpendicular to the basis vectors of the tangent space along d distinct directions). This means that W_p is a codimension $d+1$, or dimension $n-d-1$, projective space. Thus the fiber of $\pi : Z \to X$ over each closed point has pure dimension $n-d-1$. By Proposition 13.4.1, $\dim Z \leq n-1$. □

Remark 14.21 If you wish, you can use Proposition 13.4.2 to show that $\dim Z = n-1$, and you can later show that Z is a projective bundle over X, once you know what a projective bundle is, Definition **??**. But we don't need this for the proof.

Remark 14.22 Here is an example that may help you convince you that the algebra of the proof of Bertini's Theorem 14.4.1 is describing the geometry we desire. Consider the plane conic $x_0^2 - x_1^2 - x_2^2 = 0$ over a field of characteristic not 2, which you might picture as the circle $x^2 + y^2 = 1$ from the real picture in the chart U_0. Consider the point $[1:1:0]$, corresponding to $(1, 0)$ on the circle. We expect the tangent line in the affine plane to be $x = 1$, which corresponds to $x_0 - x_1 = 0$. Let's see what the algebra gives us. The projective Jacobian matrix is

$$\begin{pmatrix} 2x_0 \\ -2x_1 \\ -2x_2 \end{pmatrix} = \begin{pmatrix} 2 \\ -2 \\ 0 \end{pmatrix},$$

which indeed has rank 1 as expected. Our recipe asks that the matrix

$$\begin{pmatrix} 2 & a_0 \\ -2 & a_1 \\ 0 & a_2 \end{pmatrix}$$

has rank 1 (i.e., $a_0 = -a_1$ and $a_2 = 0$), and also that $a_0 x_0 + a_1 x_1 + a_2 x_2 = 0$, which is precisely what we wanted.

Theorem 14.4.2 (Bertini's Theorem for finite non-smooth points)

Suppose X is a closed subvariety of $\mathbb{P}_k^n$ of (pure) dimension d, X has finitely many non-smooth points. Then there is a nonempty open subset U of dual projective space $\mathbb{P}_k^{n\vee}$ such that for every point $p = [H] \in U$, the scheme $H \cap X$ is smooth over $\kappa(p)$ of (pure) dimension $d - 1$.

Proof Suppose $\dim X = d$. Let X_{sing} be the set of non-smooth points of X. Since X has finitely many non-smooth points, $\dim X_{\text{sing}} = 0$. Let

$$Z_{\text{reg}} = \{(p \in X, H \subseteq \mathbb{P}_k^n) : p \in H, \text{ and either } p \text{ is a singular point of } H \cap X, \text{ or } X \subseteq H\}$$
$$Z_{\text{sing}} = \{(p \in X, H \subseteq \mathbb{P}_k^n) : p \in H, p \in X_{\text{sing}}\}.$$

We first show that Z_{sing} is closed. Pick $(p, H) \in Z_{\text{sing}}$, $p \in H$ gives a linear condition, $p \in X_{\text{sing}}$ means the corank of Jacobian at p is bigger than $\dim X$, so the rank of Jacobian at p is $< n - d$, and therefore all $n - d$ minors vanish. Hence Z_{sing} is cut by some equations, so Z_{sing} is closed in $X \times \mathbb{P}_n^k$. We next show that $\dim Z_{\text{sing}} \leq n - 1$. Consider the map $\varphi : Z_{\text{sing}} \to X_{\text{sing}}$. Let $p \in X_{\text{sing}}$, the fiber of p is the hypersurfaces which contains p, so $\dim \varphi^{-1}(p) = n - 1$, and thus

$$\dim Z_{\text{sing}} \leq \dim X_{\text{sing}} + \dim \pi^{-1}(p) = 0 + n - 1 = n - 1.$$

By the proof of Bertini's Theorem 14.4.1, we know that $\dim Z_{\text{reg}} \leq n - 1$. Denote $Z := Z_{\text{sing}} \cup Z_{\text{reg}}$, we have $\dim Z = \max\{\dim Z_{\text{sing}}, \dim Z_{\text{reg}}\} \leq n - 1$. Moreover, Z is closed. So, by Fundamental Theorem of Elimination Theory 9.4.3, the image of Z under $X \times \mathbb{P}_k^n \to \mathbb{P}_k^{n\vee}$ is closed, of dimension of at most $n - 1$, so its complement will be open and nonempty. □

Theorem 14.4.3 (Bertini's Theorem for locally closed embedding)

Suppose $X \hookrightarrow \mathbb{P}_k^n$ is a locally closed embedding (i.e., a subvariety), where X is a smooth scheme of (pure) dimension d. Then there is a nonempty open subset U of dual projective space $\mathbb{P}_k^{n\vee}$ such that for every point $p = [H] \in U$, the scheme $H \cap X$ is smooth over $\kappa(p)$ of (pure) dimension $d - 1$.

Proof We use the same notations in the proof of Bertini's Theorem 14.4.1. In fact, taking closure preserve dimension, $\dim Z = \dim \operatorname{cl}_{\operatorname{cl}_{\mathbb{P}_k^n}(X) \times \mathbb{P}_k^n}(Z)$. By the proof of Bertini's Theorem 14.4.1, we have $\dim Z \leq n - 1$, then $\dim \operatorname{cl}_{\operatorname{cl}_{\mathbb{P}_k^n}(X) \times \mathbb{P}_k^n}(Z) \leq n - 1$. Consider the image of $\operatorname{cl}_{\overline{X} \times \mathbb{P}_k^n}(Z)$ under $\operatorname{cl}_{\mathbb{P}_k^n}(X) \times \mathbb{P}_k^n \to \mathbb{P}_k^{n\vee}$, same argument as Bertini's Theorem 14.4.1, we win. □

Proposition 14.4.1 (Many hypersurface Bertini)

Suppose X is a smooth subvariety of $\mathbb{P}_k^n$ of (pure) dimension $d \geq r$. Suppose $e_1, \ldots, e_r$ are positive integers, so

$$\mathbb{P}^{\binom{e_1+n}{n}-1} \times \cdots \times \mathbb{P}^{\binom{e_r+n}{n}-1} \tag{14.7}$$

parametrizes r-tuples of hypersurfaces $(H_1, \ldots, H_r)$ of degrees $e_1, \ldots, e_r$, respectively. Then there is a dense open subset U of (14.7) such that for every point $p = [(H_1, \ldots, H_r)] \in U$, $H_1 \cap \cdots \cap H_r \cap X$ is

smooth over $\kappa(p)$ of (pure dimension $d-r$).
Informally: for a general choice of hypersurfaces $H_1, \ldots, H_r$ of degrees $e_1, \ldots, e_r$ respectively, $H_1 \cap \cdots \cap H_r \cap X$ is a smooth of (pure) dimension $d-r$.

Remark 14.23 The lesson of this proposition isn't that you should memorize this special result. It is that you can prove any result of this flavor that you need.

Proof Suppose $\dim X = d$. Let $\mathscr{H} = \mathbb{P}^{\binom{e_1+n}{n}-1} \times \cdots \times \mathbb{P}^{\binom{e_r+n}{n}-1}$, then $\dim \mathscr{H} = \sum_{i=1}^{r} \binom{e_i+n}{n} - r =: N$. Let $\mathrm{cl}_{\mathbb{P}^n_k}(X)$ is the scheme-theoretic closure of X. We define a projective variety $Z \subseteq X \times \mathscr{H}$ as follow: Suppose $\mathrm{cl}_{\mathbb{P}^n_k}(X)$ is cut by $f_1, \cdots f_l$. Then we take these equations as the first of the defining equation of Z. We add the equations $H_1 = \cdots = H_r = 0$ and the $n-d+r$ minors of Jacobian $J(H_1, \ldots, H_r, f_1, \ldots, f_l)$ vanish. Informally, we can described Z as:

$$Z = \left\{ (p, H_1, \ldots, H_r) \in X \times \mathscr{H} : p \in \bigcap_{i=1}^{r} H_i,\ p \text{ is a singular point of } \bigcap_{i=1}^{r} H_i \cap X, \text{ or } X \subseteq \bigcap_{i=1}^{r} H_i \right\}.$$

Hence Z is a closed subvariety of $X \times \mathscr{H}$.

We next show that $\dim Z \leq N-1$. Let $p \in X$ be a closed point. Say $\varphi : Z \to X$, consider fiber $\varphi^{-1}(p)$. Each H_i contains p gives r conditions. Since p is singular point of $\bigcap_i H_i \cap X$, the differentials $dH_1, \ldots, dH_r$ are linearly dependent in the cotangent space at p, this gives $rd - ((r-1)d + r - 1) = d - r + 1$ conditions (choose r linearly dependent vectors from a d-dimensional linear space). So the codimension of $\varphi^{-1}(p) = d - r + 1 + r = d + 1$, hence $\dim \varphi^{-1}(p) = N - d - 1$. Thus the fiber of $\varphi : Z \to X$ over each closed point has pure dimension $N - d - 1$. By Proposition 13.4.1, $\dim Z \leq N - 1$. Now consider the image of Z in $\mathscr{H}$, by Fundamental Theorem of Elimination Theory 9.4.3, the image of Z in $\mathscr{H}$ has dimension at most $N-1$, so its complement will be open and nonempty. □

Proposition 14.4.2

*Suppose that X is a projective variety of dimension d in $\mathbb{P}^n$, with a dense open subset that is smooth. (This turns out to be automatic if the base field is perfect, by Theorem **??** on generic smoothness of varieties.) Then the intersection of X with d general hyperplanes consists of a finite number of reduced points. More precisely: if $\mathbb{P}^{n\vee}$ is the dual projective space, then there is a nonempty Zariski-open subset $U \subseteq (\mathbb{P}^{n\vee})^d$ such that for each closed point $(H_1, \ldots, H_d)$ of U, the scheme-theoretic intersection $H_1 \cap \cdots \cap H_d \cap X$ consists of a finite number of reduced points. (The number of such points, counted correctly, is called the degree of the variety; see **??**.)*

Proof Let $V \subseteq X$ be a dense open subset which is smooth, then V is a smooth subvariety of $\mathbb{P}^n_k$. By Proposition 14.4.1, there is a dense open subset $U \subseteq \mathbb{P}^{n\vee}_k$ such that for every $[(H_1, \ldots, H_r)] \in U$, $H_1 \cap \cdots \cap H_r \cap V$ is smooth of dimension 0. Since V is dense in X, $\dim H_1 \cap \cdots \cap H_r \cap X = \dim H_1 \cap \cdots \cap H_r \cap V = 0$. So $H_1 \cap \cdots \cap H_r \cap X$ has finite number of points, since X is reduced, these points are all reduced points. □

14.4.2 Dual varieties

Our discussion of Bertini gives us an excuse to mention a classical construction.

Definition 14.4.2 (Dual variety)

*The image of (14.6) in $\mathbb{P}^{n\vee}_k$ is called the **dual variety of** X.*

As $\dim Z = n-1$, we "expect" the dual of X to be a hypersurface of $\mathbb{P}^{n\vee}$. It is a nonobvious fact that this in fact is a duality: the dual of the dual of X is X itself. The following proposition will give you some sense of the dual variety.

Proposition 14.4.3

The dual of a hyperplane in $\mathbb{P}^n$ is the corresponding point of the dual space $\mathbb{P}^{n\vee}$. In this way, the duality between $\mathbb{P}^n$ and $\mathbb{P}^{n\vee}$ is a special case of duality between projective varieties.

Proof Let X be a hyperplane in $\mathbb{P}^n$ cut by $f = k_0x_0 + \cdots + k_nx_n$. Recall (14.6), we have

$$Z = \left\{(p, H) \in X \times \mathbb{P}^{n\vee} : p \in X, p \text{ is singular point of } H \cap X, \text{ or } X \subseteq H\right\}.$$

Let $(p, H) = ([p_0 : \cdots : p_n], [a_0 : \cdots : a_n]) \in Z$. Since $p \in H$, we have $a_0p_0 + \cdots + a_np_n = 0$. Since p is singular at $X \cap H$, so the corank of the Jacobian

$$J := \begin{pmatrix} a_0 & k_0 \\ \vdots & \vdots \\ a_n & k_n \end{pmatrix}$$

is greater than $n-1$. So $\operatorname{rank} J < 2$, which implies that $[a_0 : \cdots : a_n] = [k_0 : \cdots : k_n]$. Hence the dual of a hyperplane in $\mathbb{P}^n$ is the corresponding point of the dual space $\mathbb{P}^{n\vee}$. $\square$

Proposition 14.4.4

Suppose $C \subseteq \mathbb{P}^2$ is a smooth conic over an algebraically closed field of characteristic not 2. Then the dual variety to C is also a smooth conic.

Remark 14.24 Thus, for example, through a general point in the plane (if $k = \overline{k}$), there are two tangents to C. (The points on a line in the dual plane correspond to those lines through a point of the original plane.)

Proof Say $\mathbb{P}^2 = \operatorname{Proj} k[x_0, x_1, x_n]$. By linear transformation, we may assume $C = \operatorname{Proj} k[x_0, x_1, x_n]/(x_0^2 + x_1^2 + x_2^2)$. Let $H = [a_0 : a_1 : a_2] \in \mathbb{P}^{2\vee}$, and $p = [p_0 : p_1 : p_2]$ is a singular point of $H \cap X$, then the corank of Jacobian is greater than 0, i.e.,

$$\operatorname{corank} J := \operatorname{corank} \begin{pmatrix} a_0 & \frac{\partial f}{\partial x_0}(p) \\ a_1 & \frac{\partial f}{\partial x_1}(p) \\ a_2 & \frac{\partial f}{\partial x_2}(p) \end{pmatrix} = \begin{pmatrix} a_0 & 2p_0 \\ a_1 & 2p_1 \\ a_2 & 2p_2 \end{pmatrix} - 1 > 0$$

which implies that $\operatorname{rank} J < 2$, hence $(p, H) = (p, [a_0 : a_1 : a_2]) = (p, [p_0 : p_1 : p_2])$. Since $p_0^2 + p_1^2 + p_2^2 = 0$, we know that $C^\vee$ is also a smooth conic. $\square$

Proposition 14.4.5 (There is one smooth conic tangent to five general lines, and generalizations)

Continuing the notation of the Proposition 14.4.4. Then the number of smooth conics C containing i generally chosen points and tangent to $5-i$ generally chosen lines is $1, 2, 4, 4, 2, 1$, respectively, for $i = 0, 1, 2, 3, 4, 5$.

Remark 14.25 You might interpret the symmetry of the sequence in terms of the duality between the conic and the dual conic. This fact was likely known in the paleolithic era.

Proof Since C is conic, we may parametrize C by $[a_0 : a_1 : \cdots : a_5]$. C containing i generally chosen points means $[a_0 : a_1 : \cdots : a_5]$ is the solution of i linear equations, which implies that $[a_0 : a_1 : \cdots : a_5]$ belong to the intersection of i hyperplanes in $\mathbb{P}^5$. C tangent to $5-i$ lines means the parametrize line (a point in $\mathbb{P}^{2\vee}$) belong to $C^\vee$, which implies $[a_0 : a_1 : \cdots : a_5]$ belong to a quadric hypersurface in $\mathbb{P}^n$ (if $C = x^tAx$, then

$C^\vee = u^t A^* u$, where $u = [u_0 : u_1 : u_2]$ is parametrize line). So the problem becomes to find the intersection number of $5-i$ quadric hypersurfaces in the intersection of i hyperplanes.

For $i = 5$. $[a_0 : a_1 : \cdots : a_5]$ is the solution of 5 linear equations. By linear algebra there is single $[a_0 : a_1 : \cdots : a_5]$

For $i = 4$. By Proposition 14.4.1, the intersection of 4 hyperplanes is pure dimension 1, so it is a line. Since $[a_0 : \cdots : a_5]$ contained in a conic, by Bézout's Theorem, a conic intersect a line contains $2 \times 1 = 2$ points. So there are two $[a_0 : a_1 : \cdots : a_5]$.

For $i = 3$. By Proposition 14.4.1, the intersection of 3 hyperplanes has pure dimension 2, so it is a plane. Restrict two quadric hypersurfaces to the plane, by Bézout's Theorem, intersection of two conics contains $2 \times 2 = 4$ points.

For $i = 0, 1, 2$, they are the dual problem of $i = 3, 4, 5$. So for $i = 0, 1, 2$ we have answers $1, 2, 4$. □

14.5 Discrete valuation rings, and Algebraic Hartogs's Lemma

14.5.1 Discrete valuation rings and Dedekind domain

The case of (co)dimension 1 is important, because if you understand how prime ideals behave that are separated by dimension 1, then you can use induction to prove facts in arbitrary dimension. This is one reason why Krull's Principal Ideal Theorem 13.3.2 is so useful.

A dimension 1 regular local ring can be thought of as a "germ of a smooth curve" (see Figure 14.5). Two examples to keep in mind are $k[x]_{(x)} = \{f(x)/g(x) : x \nmid g(x)\}$ and $\mathbb{Z}_{(5)} = \{a/b : 5 \nmid b\}$. The first example is "geometric" and the second is "arithmetic", but hopefully it is clear that they have something fundamental in common.

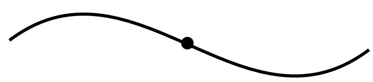


Figure 14.5: A germ of a curve.

The purpose of this section is to give a long series of equivalent definitions of these rings. Before beginning, we quickly sketch these seven definitions. There are a number of ways a Noetherian local ring can be "nice." It can be regular, or a principal ideal domain, or a unique factorization domain, or normal. In dimension 1, these are the same. Also equivalent are nice properties of ideals: if $\mathfrak{m}$ is principal; or if all ideals are either powers of the maximal ideal, or 0. Finally, the ring can have a discrete valuation, a measure of "size" of elements that behaves particularly well.

Theorem 14.5.1

Suppose $(A, \mathfrak{m})$ is a Noetherian local ring of dimension 1. Then the following are equivalent.

(DVR1) $(A, \mathfrak{m})$ *is regular.*

(DVR2) $\mathfrak{m}$ *is principal.*

Proof If A is regular, then $\mathfrak{m}/\mathfrak{m}^2$ is one-dimensional. Choose any element $t \in \mathfrak{m} \setminus \mathfrak{m}^2$. Then t generates $\mathfrak{m}/\mathfrak{m}^2$, so generates $\mathfrak{m}$ by Nakayama's Lemma 9.2.6 ($\mathfrak{m}$ is finitely generated by the Noetherian hypothesis). We call such an element a **uniformizer**.

Conversely, if $\mathfrak{m}$ is generated by one element t over A, then $\mathfrak{m}/\mathfrak{m}^2$ is generated by one element t over $A/\mathfrak{m} = k$. Since $\dim_k \mathfrak{m}/\mathfrak{m}^2 \geq 1$ by Theorem 13.3.4, we have $\dim_k \mathfrak{m}/\mathfrak{m}^2 = 1$, so $(A, \mathfrak{m})$ is regular. $\square$

We will soon use a useful fact, which is geometrically motivated, and is a special case of an important result, the Artin-Rees Lemma 14.9.1. We will prove it in §14.9 Theorem 14.9.2.

Proposition 14.5.1

If $(A, \mathfrak{m})$ is a Noetherian local ring, then $\bigcap_i \mathfrak{m}^i = 0$.

The geometric intuition for this is that any function that is analytically zero at a point (vanishes to all orders, i.e., Taylor series at a point is 0) actually vanishes in an open neighborhood of that point. (Proposition 14.9.2 will make this precise.) The geometric intuition also suggests an example showing that Noetherianness is necessary: consider e^{-1/x^2} in the ring of germs of C^∞-functions on $\mathbb{R}$ at the origin. (In particular, this implies that the ring of C^∞-functions on $\mathbb{R}$, localized at the origin, is not Noetherian!)

It is tempting to argue that

$$\mathfrak{m}\left(\bigcap_i \mathfrak{m}^i\right) = \bigcap_i \mathfrak{m}^i, \tag{14.8}$$

and then to use Nakayama's Lemma 9.2.4 to argue that $\bigcap_i \mathfrak{m}^i = 0$. Unfortunately, it is not obvious that this first equality is true: product does not commute with finite intersections in general. (Aside: Product also doesn't commute with finite intersections in general, as, for example, in $k[x, y, z]/(xz - yz)$, $(z)((x) \cap (y)) \neq (xz) \cap (yz)$.) We will establish Proposition 14.5.1 in Theorem 14.9.2 (b). (We could do it directly right now without too much effort.)

Theorem 14.5.2

Suppose $(A, \mathfrak{m})$ is a Noetherian local ring of dimension 1. Then (DVR1) and (DVR2) are equivalent to:

(DVR3) A is an integral domain, and all ideals are of the form $\mathfrak{m}^n$ (for $n \geq 0$) or (0).

Proof Assume (DVR1): suppose $(A, \mathfrak{m}, k)$ is a regular local ring of dimension 1. Claim that $\mathfrak{m}^n \neq \mathfrak{m}^{n+1}$ for any n. Otherwise, by Nakayama's Lemma, $\mathfrak{m}^n = 0$, from which $t^n = 0$. Theorem 14.2.2 shows that regular local ring is integral, so A is an integral domain, hence $t = 0$, from which $A = A/\mathfrak{m}$ is a field, which doesn't have dimension 1, contradiction.

We next claim that $\mathfrak{m}^n/\mathfrak{m}^{n+1}$ is dimension 1. Reason: $\mathfrak{m}^n = (t^n)$. So $\mathfrak{m}^n$ is generated as an A-module by one element, and $\mathfrak{m}^n/(\mathfrak{m}\mathfrak{m}^n)$ is generated as an $(A/\mathfrak{m} = k)$-module by one element (nonzero by the previous paragraph), so it is a one-dimensional vector space.

So we have a chain of ideals $A \supsetneq \mathfrak{m} \supsetneq \mathfrak{m}^2 \supsetneq \mathfrak{m}^3 \supsetneq \cdots$ with $\bigcap_i \mathfrak{m}^i = (0)$ (Proposition 14.5.1). We want to say that there is no room for any ideal besides these, because each pair is "separated by dimension 1", and there is "no room at the end". Proof: Suppose $I \subseteq A$ is an ideal. If $I \neq (0)$, then there is some n such that $I \subseteq \mathfrak{m}^n$ but $I \not\subseteq \mathfrak{m}^{n+1}$. Choose some $u \in I \setminus \mathfrak{m}^{n+1}$. Then $(u) \subseteq I$. But u generates $\mathfrak{m}^n/\mathfrak{m}^{n+1}$, hence by Nakayama's Lemma it generates $\mathfrak{m}^n$, so we have $\mathfrak{m}^n \subseteq I \subseteq \mathfrak{m}^n$, so we are done: (DVR3) holds.

We now show that (DVR3) implies (DVR1). Assume (DVR1) does not hold: suppose we have a dimension 1 Noetherian local integral domain that is not regular, so $\mathfrak{m}/\mathfrak{m}^2$ has dimension at least 2. Choose any $u \in \mathfrak{m} - \mathfrak{m}^2$. Then $(u, \mathfrak{m}^2)$ is an ideal, but $\mathfrak{m}^2 \subsetneq (u, \mathfrak{m}^2) \subsetneq \mathfrak{m}$, it contradicts to the assumption of (DVR3). $\square$

Theorem 14.5.3

Suppose $(A, \mathfrak{m})$ is a Noetherian local ring of dimension 1. Then (DVR1) - (DVR3) above are equivalent to:

(DVR4) A is a principal ideal domain.

Proof (DVR4) implies $\mathfrak{m}$ is principal, that is (DVR2).

(DVR3) implies that each ideal I is of the form $\mathfrak{m}^n$ for some n. Since $\mathfrak{m}$ is principal, say $\mathfrak{m} = (t)$, $I = (t^n)$ is principal, which implies (DVR4). □

Discrete valuation ring

We next define the notion of a discrete valuation ring.

Definition 14.5.1 (Discrete valuation, valuation)

(1) *A **discrete valuation** on K is a surjective homomorphism $v : K^\times \to \mathbb{Z}$ (in particular, $v(xy) = v(x) + v(y)$) satisfying*

$$v(x+y) \geq \min\{v(x), v(y)\}$$

*except if $x + y = 0$ (in which case the left sider is undefined). (Such a valuation is called **non-Archimedean**, although we will not use that term.) It is often convenient to say $v(0) = \infty$.*

(2) *More generally, a **valuation** is a surjective homomorphism $v : K^\times \to G$ to a totally ordered abelian group G. The symbol "val" will denote valuations.*

Here are three key examples.

Example 14.6

(i) (the 5-adic valuation) $K = \mathbb{Q}$, $v(r)$ is the "power of 5 appearing in r", e.g., $v(35/2) = 1$, $v(27/125) = -3$.

(ii) $K = k(x)$, $v(f)$ is the "power of x appearing in f".

(iii) $K = k(x)$, $v(f)$ is the negative of the degree. This is really the same as (ii), with x replaced by $1/x$.

Definition 14.5.2 (Valuation ring)

*Let $v : K^\times \to G$ be a valuation. Let $\mathscr{O}_v := 0 \cup \{x \in K^\times : v(x) \geq 0\}$, $\mathscr{O}_v$ is a ring, which is called the **valuation ring of** v.*

Remark 14.26 Not every valuation is discrete. Consider the ring of **Puiseux series** over a field k, $K = \bigcup_{n\geq 1} k((x^{1/n}))$, with $v : K^\times \to \mathbb{Q}$ given by $v(x^q) = q$.

✍ **Exercise 14.19** Describe the valuation rings in Example 14.6 . (You will notice that they are familiar-looking dimension 1 Noetherian local rings. What a coincidence!)

Solution *For (i).* $\mathscr{O}_v = 0 \cup \left\{\frac{p}{q} \cdot 5^n \in \mathbb{Q} : \gcd(p,q) = 1, 5 \nmid pq, n \geq 0\right\} = \left\{\frac{p}{q} \in \mathbb{Q} : 5 \nmid q\right\} = \mathbb{Z}_{(5)}$.

For (ii).

$$\begin{aligned}\mathscr{O}_v =& 0 \cup \left\{\frac{h(x)}{g(x)} \cdot x^n \in k(x) : \gcd(h(x), g(x)) = 1, x \nmid h(x)g(x), n \geq 0\right\} \\ =& \left\{\frac{h(x)}{g(x)} \in k(x) : x \nmid g(x)\right\} \\ =& k[x]_{(x)}.\end{aligned}$$

For (iii).

$$\begin{aligned}\mathscr{O}_v =& 0 \cup \left\{\frac{h(x)}{g(x)} \in k(x) : \deg h(x) - \deg g(x) \leq 0\right\} \\ =& \left\{\frac{H(t)}{G(t)} \in k(t) : t \nmid G(t)\right\} \\ =& k[t]_{(t)},\end{aligned}$$

where $t = \frac{1}{x}$.

Lemma 14.5.1

$\mathfrak{m} = \{0\} \cup \{x \in K^\times : v(x) > 0\}$ *is the unique maximal ideal of the valuation ring* $\mathscr{O}_v$*. Thus the valuation ring is a local ring.*

Proof We first show that $\mathfrak{m}$ is an ideal. It suffices to show that $\mathscr{O}_v\mathfrak{m} \subseteq \mathfrak{m}$. Let $a \in \mathscr{O}_v$ and $r \in \mathfrak{m}$, then $v(r) \geq 1$ and $v(a) \geq 0$, so $v(ar) \geq 1 + 0 = 1$, which implies that $ar \in \mathfrak{m}$. Hence $\mathfrak{m}$ is an ideal of $\mathscr{O}_v$.

We next show that every element in $\mathscr{O}_v \setminus \mathfrak{m}$ is invertible. Let $u \in \mathscr{O}_v \setminus \mathfrak{m}$, then $v(u) = 0$. Note that $v(u^{-1}) = v(1) - v(u) = 0$, we have $u^{-1} \in \mathfrak{m}$, i.e., every element in $\mathscr{O}_v \setminus \mathfrak{m}$ is invertible. By [AM18, Prop 1.6], $(\mathscr{O}_v, \mathfrak{m})$ is a local ring. □

Definition 14.5.3 (Discrete valuation ring (DVR))

An integral domain A is called a ***discrete valuation ring*** *(or* ***DVR****) if there exists a discrete valuation v on its fraction field $K = K(A)$ for which $\mathscr{O}_v = A$. Similarly, A is a* ***valuation ring*** *if there exists a valuation v on K for which $\mathscr{O}_v = A$.*

Now if A is a regular local ring of dimension 1, and t is a uniformizer (a generator of $\mathfrak{m}$ as an ideal, or, equivalently, of $\mathfrak{m}/\mathfrak{m}^2$ as a k-vector space), then any nonzero element r of A lies in some $\mathfrak{m}^n \setminus \mathfrak{m}^{n+1}$, so $r = t^n u$, where u is invertible (as t^n generates $\mathfrak{m}^n$ by Nakayama, and so does r), so $K(A) = A_t = A[1/t]$. So any element of $K(A)^\times$ can be written uniquely as ut^n, where u is invertible and $n \in \mathbb{Z}$. Thus we can define a valuation v by $v(ut^n) = n$.

Lemma 14.5.2

v defined above is a discrete valuation.

Proof Clearly, $v : K(A)^\times \to \mathbb{Z}$ is surjective homomorphism of abelian groups. Let $f = u_1 t^n$ and $g = u_2 t^m$ in $K(A)^\times$, we may assume that $n \geq m$. Consider $f + g$,

$$f + g = t^m(u_1 t^{n-m} + u_2) =: ht^m.$$

Consider h, since $n - m \geq 0$ and $u_1, u_2 \in A$, we have $h \in A$, then h can be written as $h = u_3 t^l$ for some $l \geq 0$, hence $v(h) \geq 0$. So

$$v(f + g) = v(ht^m) = v(h) + m \geq m = v(g),$$

as we desired. □

Lemma 14.5.3

Suppose $(A, \mathfrak{m})$ is a DVR. Then $(A, \mathfrak{m})$ is a regular local ring of dimension 1.

Proof We first show that the ideals of A are all of form (0) or $I_n = \{r \in A : v(r) \geq n\}$. Let I be a nonzero ideal of A. Let $n = \min_{r \in I}\{v(r)\}$, then $I \subseteq I_n$. Say $v(r_0) = n$. Pick $r \in I_n$, then $v(r) \geq n = v(r_0)$, so

$v(rr_0^{-1}) \geq 0$, which implies that $rr_0^{-1} \in A$. Say $u = rr_0^{-1}$, then $r = ur_0$, Since $u \in A$ and $r_0 \in I$, we have $I_n \subseteq I$. Hence $I = I_n$.

We next show that (0) and I_1 are the only prime ideals of A. Let I_n be a nonzero prime ideal of A. Let $t \in I_1$ such that $v(t) = 1$, then $v(t^n) = n$. So $t^n \in I_n$. For $n \geq 2$, $t \notin I_n$ and $t^{n-1} \notin I_n$. Hence I_n is not prime ideal, for $n \geq 2$. Since A is integral domain, (0) is prime ideal. Hence (0) and I_1 are the only prime ideals of A. Note that $v(0) = \infty$, we have $(0) \subsetneq I_1$, and thus (A, I_1) is a Noetherian local ring of dimension 1.

Claim that I_1 is principal. Let $t \in I_1$ such that $v(t) = 1$, we want to show that $I_1 = (t)$. Clearly, $(t) \subseteq I_1$. Let $x \in I_1$, then $v(x) = n \geq 1$ for some n. Consider $x \cdot t^{-n}$, then $v(x \cdot t^{-n}) = v(x) - nv(t) = 0$, hence $x = ut^n$ where u is unit, and thus $x \in (t)$, which implies that I_1 is principal. By Theorem 14.5.1, (A, I_1) is regular, we win. □

Combining Lemma 14.5.2 and Lemma 14.5.3, we have proved:

Theorem 14.5.4

An integral domain A is a Noetherian local ring of dimension 1 satisfying (DVR1) - (DVR4) if and only if

(DVR0) A is a discrete valuation ring.

Lemma 14.5.4

There is only one discrete valuation on a DVR.

Proof Let $(A, \mathfrak{m})$ be a DVR. By Theorem 14.5.4, $\mathfrak{m}$ is principal, say $\mathfrak{m} = (t)$. Suppose there are two discrete valuations v, w on $K(A)^\times$, we want to show that $v(t) = w(t) = 1$. Let $x_0 \in A$, such that $v(x_0) = 1$, then $x_0 = ut$ for some unit u, hence $t = u^{-1}x_0$. Consider $v(t)$,

$$v(t) = v(u^{-1}x_0) = v(u^{-1}) + v(x_0) = 0 + 1 = 1,$$

same argument apply to w, we have $w(t) = 1$. Let $x \in A$ be any element, then $x \in I$ for some ideal $I \subseteq A$. Theorem 14.5.4 shows that $I = \mathfrak{m}^n$ for some n. So $x = ut^m$ for some $m \geq n$ and unit u. Thus

$$v(x) = mv(t) = m = mw(t) = w(x),$$

which implies $v = w$. □

Thus any regular local ring of dimension 1 comes with a unique valuation on its fraction field. So it makes sense for us to discuss the order of zeros and the order of poles:

Definition 14.5.4 (Zero of order, pole of order)

If the valuation of an element is $n > 0$, we say that the element has a ***zero of order*** *n. If the valuation is $-n < 0$, we say that the element has a* ***pole of order*** *n.*

We will come back to this shortly, after dealing with (DVR5) and (DVR6)

Theorem 14.5.5

Suppose $(A, \mathfrak{m})$ is a Noetherian local ring of dimension 1. Then the following are equivalent:

(DVR0) A is a discrete valuation ring.

(DVR1) $(A, \mathfrak{m})$ is regular.

(DVR2) $\mathfrak{m}$ is principal.

(DVR3) A is an integral domain, and all ideals are of the form $\mathfrak{m}^n$ (for $n \geq 0$) or (0).

(DVR4) A is a principal ideal domain.

(DVR5) A is a UFD.

(DVR6) A is an integral domain, integrally closed in its fraction field.

Proof Equivalent of (DVR0) - (DVR4) are given by Theorem 14.5.4. (DVR0) - (DVR4) clearly imply (DVR5), because we have the following stupid unique factorization: each nonzero element of A can be written uniquely as ut^n, where $n \in \mathbb{Z}^{\geq 0}$ and u is invertible.

Now (DVR5) - (DVR6), because UFD are integrally closed in their fraction fields (Proposition 6.4.6).

It remains to check that (DVR6) implies (DVR0) - (DVR4). We will show that (DVR6) implies (DVR1).

Suppose $(A, \mathfrak{m})$ is a Noetherian local integral domain of dimension 1, integrally closed in its fraction field $K(A)$. Choose any nonzero $r \in \mathfrak{m}$. Then $S = A/(r)$ is a Noetherian local ring of dimension 0 — its only prime is the image of $\mathfrak{m}$, which we denote $\mathfrak{n}$ to avoid confusion. Then $\mathfrak{n}^N = 0$, where N is sufficiently large (Proposition 13.1.11). Hence there is some n such that $\mathfrak{n}^n = 0$ but $\mathfrak{n}^{n-1} \neq 0$.

Now comes the crux of the argument. Thus in A, $\mathfrak{m}^n \subseteq (r)$ but $\mathfrak{m}^{n-1} \nsubseteq (r)$. Choose $s \in \mathfrak{m}^{n-1} - (r)$. Consider $s/r \in K(A)$. As $s \notin (r)$, $s/r \notin A$, so as A is integrally closed, s/r is not integral over A.

Now (inside $K(A)$) $\frac{s}{r}\mathfrak{m} \nsubseteq \mathfrak{m}$ (or else $\frac{s}{r}\mathfrak{m} \subseteq \mathfrak{m}$ would imply that $\mathfrak{m}$ is a faithful $A\left[\frac{s}{r}\right]$-module, contradicting Lemma 9.2.7). But $s\mathfrak{m} \subseteq \mathfrak{m}^n \subseteq (r) = Ar$, so $\frac{s}{r}\mathfrak{m} \subseteq A$. Thus $\frac{s}{r}\mathfrak{m} = A$, from which $\mathfrak{m} = \frac{r}{s}A$, so $\mathfrak{m}$ is principal. □

Example 14.7 $\operatorname{Spec} \mathbb{Z}[i]$ **is regular.** We now have a simpler explanation of why $\operatorname{Spec} \mathbb{Z}[i]$ is regular (Exercise 14.17). First, $\mathbb{Z}[i]$ is a UFD (since $\mathbb{Z}[i]$ is ED), hence normal (Proposition 6.4.6). It is dimension 1, so its closed (codimension 1) points are regular by Theorem 14.5.5. Its generic point is also regular, as $\mathbb{Z}[i]$ is an integral domain.

Example 14.8 ("Curves are regular almost everywhere".) By Theorem 14.5.5, Proposition 13.1.7 and Proposition 11.7.6, every one-dimensional variety is regular on a dense open subset. Since the set of singular points does not contained generic point, so it has dimension 0, and therefore the set of singular points consists of finitely many closed points. Hence, every one-dimensional variety is regular away from finitely many closed points.

Aside: Dedekind domain

Dedekind domain is an important notion, but we won't use it much. We state some classical results below.

Theorem 14.5.6

Let A be a Noetherian domain of dimension 1. Then the following are equivalent:

(DD1) A is integrally closed;

(DD2) Every primary ideal in A is a prime power;

(DD3) Every local ring $A_{\mathfrak{p}}$ ($\mathfrak{p} \neq 0$) is a DVR.

Proof See [AM18, Thm. 9.3]. □

Definition 14.5.5 (Dedekind domain)

A Noetherian domain of dimension 1 satisfying the conditions (DD1) - (DD3) is called a ***Dedekind domain*** *(or,* ***DD****).*

Corollary 14.5.1

In a Dedekind domain every non-zero ideal has a unique factorization as a product of prime ideals.

Proof See [AM18, Cor. 9.4]. □

Theorem 14.5.7

The ring of integers (Definition 11.7.4) in an algebraic number field K is a Dedekind domain.

Proof See [AM18, Thm. 9.5]. □

14.5.2 Poles and zeros of functions on Noetherian schemes

We now generalize the sense in which $\frac{(x-2)^2}{(x-3)^4}$ on $\mathbb{A}^1_{\mathbb{C}}$ has a double zero at $x=2$ ad a quadruple pole at $x=3$.

Definition 14.5.6 (Regular in codimension 1 (R_1))

(1) *We say that a Noetherian ring A is **regular in codimension** 1 (or R_1 for short) if its localization at all codimension 1 primes are regular local rings.*

(2) *We say a locally Noetherian scheme A is **regular in codimension** 1 (or R_1) if it is regular at its codimension 1 points.*

Lemma 14.5.5

Noetherian normal schemes are regular in codimension 1.

Proof By Theorem 14.5.5, we get the result immediately. □

Definition 14.5.7 (Valuation at an R_1 point)

*Suppose X is a locally Noetherian scheme. Then for any regular codimension 1 point p (i.e., any point p where $\mathscr{O}_{X,p}$ is a regular local ring of dimension 1), we have a discrete valuation val_p. If f is any nonzero element of the fraction field of $\mathscr{O}_{X,p}$ (e.g., if X is integral, and f is a nonzero element of the function field of X), then if $\mathrm{val}_p(f)>0$, we say that f has a **zero of order** $\mathrm{val}_p(f)$ **at** p, and if $\mathrm{val}_p(f)<0$, we say that f has a **pole of order** $\mathrm{val}_p(f)$ **at** p.*

Remark 14.27 We are not yet allowed to discuss order of vanishing at a point that is not regular and codimension 1. One can make a definition, but it doesn't behave as well as it does when have you have a discrete valuation.

Example 14.9 $\frac{75}{34}$ has a double zero at 5, and a single pole at 2.

Solution *$\mathbb{Z}_{(5)}$ is a DVR, the uniformizer of $\mathbb{Z}_{(5)}$ is 5, so each element in $\mathbb{Q}^\times$ can be written as $u\cdot 5^n$ where u is invertible and $n\in\mathbb{Z}$. Define discrete valuation $v:\mathbb{Q}^\times\to\mathbb{Z}$ by setting $v(u5^n)=n$. Since $\frac{75}{34}=\frac{3}{34}\cdot 5^2$, $\frac{75}{34}$ has a double zero at 5.*

$\mathbb{Z}_{(2)}$ is a DVR, same argument as above, $\frac{75}{34}=\frac{75}{17}\cdot 2^{-1}$ implies that $\frac{75}{34}$ has a single pole at 2.

Example 14.10 Describe the zeros and poles of $\frac{x^3(x+y)}{(x^2+y)^3}$ on $\mathbb{A}^2_k$.

Proof Note that $k[x,y]_{(x)}, k[x,y]_{(x+y)}$ and $k[x,y]_{x^2+y}$ are all DVRs, so $\frac{x^3(x+y)}{(x^2+y)^3}$ has zero of order 3 at (x), zero of order 1 at $(x+y)$, and pole of order 3 at (x^2+y). □

Proposition 14.5.2 (Finiteness of zeros and poles on Noetherian schemes)

Suppose X is an integral Noetherian scheme, and $f\in K(X)^\times$ is a nonzero element of its function field. Then f has a finite number of zeros and poles.

Proof Since X is Noetherian scheme, X can be covered by finite number of affine open subsets. So it suffices to show that f has a finite number of zeros and poles on each affine open subsets. We may reduce to the case

$X = \operatorname{Spec} A$ where A is Noetherian domain.

In fact, $K(X) = K(A)$, suppose $f = \frac{f_1}{f_2}$ where $f_1 \in A$ and $f_2 \in A$. Let $p = [\mathfrak{p}]$ be any regular codimension 1 point, then $A_{\mathfrak{p}}$ is DVR. Then we have

$$\operatorname{val}_p(f) = \operatorname{val}_p\left(\frac{f_1}{f_2}\right) = \operatorname{val}_p(f_1) - \operatorname{val}_p(f_2).$$

If p is a zero or pole of f, we have $\operatorname{val}_p(f_1) \neq 0$ or $\operatorname{val}_p(f_2) \neq 0$. Hence, it suffices to show that f_i has finite number of regular codimension 1 zeros. So we reduced to the case that $f \in A$.

Now, we want to show f has finite zeros. Consider $V(f)$, by Proposition 4.6.12, we may write

$$V(f) = Z_1 \cup \cdots \cup Z_r,$$

where each Z_i is an irreducible component of $V(f)$. Say $Z_i = V(\mathfrak{p}_i)$. By Krull's Principal Ideal Theorem 13.3.2, $\operatorname{ht}(\mathfrak{p}_i) = 1$ for all i. So regular codimension 1 zeros of f contained in $\{\mathfrak{p}_1, \ldots, \mathfrak{p}_r\}$, which implies that f has finite number of regular codimension 1 zeros, as we desired. □

Remark 14.28 A Noetherian scheme can be singular in codimension 2 and still be normal. For example, we have shown that the cone $x^2 + y^2 = z^2$ in $\mathbb{A}^3$ in characteristic not 2 is normal (Exercise 6.6), but it is singular at the origin (the Zariski tangent space is visibly three-dimensional, and origin has codimension 2 is given by Theorem 13.2.4).

But singularities of normal schemes are not so bad in some ways: we will very shortly (14.5.3) discuss Algebraic Hartogs's Lemma 14.5.8 for Noetherian normal schemes, which states that you can extend functions over codimension 2 sets.

Summary

We know that for Noetherian rings we have implications:

$$\text{UFD} \Longrightarrow \text{Integrally closed} \Longrightarrow \text{Regular in codimension } 1.$$

Hence for locally Noetherian schemes, we have similar implications:

$$\text{Factorial} \Longrightarrow \text{Normal} \Longrightarrow \text{Regular in codimension } 1.$$

Here are two examples to show you that these inclusions are strict.

✍ **Exercise 14.20 (The pinched plane)** Let A be the subring $k[x^3, x^2, xy, y]$ of $k[x, y]$. (Informally, we allow all polynomials that don't include a nonzero multiple of the monomial x.)

(1) Show that $\operatorname{Spec} k[x, y] \to \operatorname{Spec} A$ is a normalization.
(2) Show that A is not integrally closed.
(3) Show that A is regular in codimension 1.

Proof For (1) and (2). It suffices to show that the integral closure of A in its fraction field is $k[x, y]$. First, we want to show that $k[x, y]$ is integral over A. Recall Lemma 9.2.1, it suffices to show that x, y is integral over A. Pick $f(t) = t^2 - x^2, g(t) = t - y \in A[t]$, then $f(x) = 0, g(y) = 0$, as we desired. So $k[x, y]$ is integral over A. Now, we want to show that $k[x, y]$ is integral closure of A. Note that $k[x, y]$ is UFD, by Proposition 6.4.6, $k[x, y]$ is integrally closed in its fraction field. Pick $h \in \operatorname{Int.cl}_{K(A)}(A)$, then h is integral over A, since $A \subseteq k[x, y]$, h is integral over $k[x, y]$, so $h \in k[x, y]$. Hence $\operatorname{Int.cl}_{K(A)}(A) = k[x, y]$, we win.

For (3). We first show that $\dim A = 2$. By part (1) and (2), we know that $\operatorname{Spec} k[x, y] \to \operatorname{Spec} A$ is a normalization, so by Proposition 13.1.7, $\dim A = \dim k[x, y] = 2$. Consider $V(x^2, y)$, we want to show any prime ideal in $V(x^2, y)$ has height at least 2. Let $\mathfrak{p} \in V(x^2, y)$, then $x^2 \in \mathfrak{p}$ and $y \in \mathfrak{p}$. It follows that $x^2y^2 = (xy)^2 \in \mathfrak{p}$, i.e., $xy \in \mathfrak{p}$; and $(x^2)^3 = (x^3)^2 \in \mathfrak{p}$, i.e., $x^3 \in \mathfrak{p}$. Hence the maximal ideal (x^3, x^2, xy, y) contained in $\mathfrak{p}$, which implies that $\mathfrak{p} = (x^3, x^2, xy, y) =: \mathfrak{m}$. It suffices to check that $\mathrm{ht}(\mathfrak{m}) = 2$. $\mathfrak{m}$ is closed point, so has dimension 0, by Theorem 13.2.4, we know that $\dim A = \mathrm{ht}(\mathfrak{m}) = 2$, so any prime ideal in $V(x^2, y)$ has codimension 2. So we may throw out $V(x^2, y)$, i.e., $\operatorname{Spec} A \setminus V(x^2, y)$. In fact $\operatorname{Spec} A \setminus V(x^2, y) = \operatorname{Spec} A_{x^2} \cup \operatorname{Spec} A_y$. Hence it suffices to show that A_{x^2} and A_y are R_1. Note that

$$A_{x^2} = k[x^3, x^2, xy, y]_{x^2} \cong k[x, y]_{x^2}$$

and

$$A_y = k[x^3, x^2, xy, y]_y \cong k[x, y]_y,$$

by Theorem 14.3.1, we know that $\mathbb{A}_k^2$ is regular, so A_{x^2} and A_y are regular, clearly, they holds R_1. □

Example 14.11 (The quadric surface once again) Suppose k is algebraically closed of characteristic not 2. Then $k[w, x, y, z]/(wz - xy)$ is integrally closed, but not a UFD; see Exercise 6.9 (and Exercise 14.2).

14.5.3 Algebraic Hartogs's Lemma for Noetherian normal schemes

Hartogs's Lemma in several complex variables states (informally) that a holomorphic function defined away from a codimension 2 set can be extended over that. We now describe an algebraic analog, for Noetherian normal schemes. (It may also be profitably compared to Second Riemann Extension Theorem.) We will use this repeatedly and relentlessly when connecting line bundles and divisors in §16.4.

Theorem 14.5.8 (Algebraic Hartogs's Lemma)

Suppose A is an integrally closed Noetherian integral domain. Then

$$A = \bigcap_{\mathrm{ht}(\mathfrak{p})=1} A_{\mathfrak{p}}.$$

The equality takes place in $K(A)$; recall that any localization of an integral domain A is naturally a subset of $K(A)$ (Exercise 2.6). **Warning:** Few people call this Algebraic Hartogs's Lemma. Vakil calls it this because it parallels the statement in complex geometry.

One might say that if $f \in K(A)$ does not lie in $A_{\mathfrak{p}}$, where $\mathfrak{p}$ has codimension 1, then f has a pole at $[\mathfrak{p}]$, and if $f \in K(A)$ lies in $\mathfrak{p}A_{\mathfrak{p}}$, where $\mathfrak{p}$ has codimension 1, then f has a zero at $[\mathfrak{p}]$.

It is worth geometrically interpreting Algebraic Hartogs's Lemma as saying that *a rational function on a Noetherian normal scheme with no poles is in fact regular* (an element of A). Informally: "Noetherian normal schemes have the Hartogs property."

One can state Algebraic Hartogs's Lemma more generally in the case that $\operatorname{Spec} A$ is a Noetherian normal scheme, meaning that A is a product of Noetherian normal integral domains.

Another generalization (and something closer to the "right" statement) is that if A is a subring of a field K, then the integral closure of A in K is the intersection of all valuation rings of K containing A; see [AM18, Cor. 5.22] for explanation and proof.

Proof of Algebraic Hartogs's Lemma 14.5.8

Proof The key clever construction in the proof is the following. If A is an integral domain, and $r \in K(A)$, then $r \in A$ if and only if $r \in A_{\mathfrak{p}}$ for all primes $\mathfrak{p}$ of A (Proposition 6.4.4), i.e., if r is regular at all points of $\operatorname{Spec} A$. For this reason, it makes sense to ask at what points $r \in K(A)$ is not regular. Our "ideal of denominators" construction (from the proof Proposition 6.4.3) shows that it is a closed subset: if $I = \{a \in A : ar \in A\}$, then the points $[\mathfrak{p}]$ of $V(I) = \operatorname{Supp} A/I$ are precisely the points where r is not regular. We get a more refined understanding of this closed subset from an easy-to-define module $M^{a/b}$.

Suppose $r = a/b \notin A$, where $a, b \in A$. Let $\overline{a}$ be the image of a in $A/(b)$. The fact that $r \notin A$ means that $\overline{a} \neq 0$ in $A/(b)$. Let $M^{a/b}$ be the A-submodule of $A/(b)$ generated by $\overline{a}$. We have constructed $M^{a/b}$ so that $M^{a/b}_{\mathfrak{q}} = 0$ if and only if $a/b \in A_{\mathfrak{q}}$. (Translation: a is not a multiple of b in $A_{\mathfrak{p}}$ if and only if $\overline{a} \neq 0$ in $A_{\mathfrak{p}}/(b)$.) Let's sort out the information we need:

- $M^{a/b}$ is the A-submodule of $A/(b)$ generated by $\overline{a}$;
- $M^{a/b}_{\mathfrak{q}} = 0$ if and only if $a/b \in A_{\mathfrak{q}}$.

We wish to show Algebraic Hartogs's Lemma, which says that if A is integrally closed, then it suffices to instead check only the height 1 prime ideals. We begin with the approximation of this fact, which works without the "integrally closed" hypothesis.

Lemma 14.5.6

Suppose A is a Noetherian integral domain, and $r \in K(A)$. Then $r \in A$ if and only if for every $\mathfrak{p} \in \operatorname{Ass}(A/(g))$ for every nonzero $g \in A$, we have $r \in A_{\mathfrak{p}}$.

Proof Clearly if $r \in A$, then $r \in A_{\mathfrak{p}}$ for all $\mathfrak{p}$. So assume that $r = a/b \notin A$. We will find such a $\mathfrak{p}$ such that $r \notin A_{\mathfrak{p}}$ as described in the statement of the lemma. By the construction of $M^{a/b}$, $a/b \notin A$ implies $M^{a/b} \neq 0$. So $\operatorname{Ass}(M^{a/b}) \neq \varnothing$, because A is Noetherian. Let $\mathfrak{p} \in \operatorname{Ass}(M^{a/b})$, as $M^{a/b}$ is a submodule of $A/(b)$, by Proposition 7.6.7, $\mathfrak{p} \in \operatorname{Ass}(A/(b))$. Hence, we have proved: if $r = a/b \notin A$ then there exists $\mathfrak{p} \in \operatorname{Ass}(A/(b))$ such that $r \notin A_{\mathfrak{p}}$. □

Proposition 14.5.3

If A is a Noetherian integrally closed integral domain, then

(i) A is regular in codimension 1 (Definition 14.5.6),

(ii) for any $b \in A$, the A-module $A/(b)$ has no embedding points.

Proof (i) Integral closure is preserved by localization (Proposition 6.4.1), so for each codimension 1 prime $\mathfrak{p}$, $A_{\mathfrak{p}}$ is integrally closed. Thus by Theorem 14.5.5, $A_{\mathfrak{p}}$ is regular.

(ii) Let $\mathfrak{p}$ be an associated prime of $A/(b)$. We wish to show that $\mathfrak{p}$ is codimension 1 in A, i.e., $\operatorname{ht}(\mathfrak{p}) = 1$. Since $\mathfrak{p} = \operatorname{Ass}(A/(b))$, there exists $a \in A$ such that $\mathfrak{p} = \operatorname{Ann}_A(\overline{a})$. Localize $A/(b)$ at $\mathfrak{p}$, then $(A/(b))_{\mathfrak{p}} \cong A_{\mathfrak{p}}/(b)$, so $\mathfrak{p}(A_{\mathfrak{p}}/(b)) = \operatorname{Ann}_{A_{\mathfrak{p}}}(\overline{a}/1)$ is an associated ideal of $A_{\mathfrak{p}}/(b)$. Hence, the truth of the statement is preserved by replacing A by its localization $A_{\mathfrak{p}}$, so it suffices to prove the following: "Suppose $(A, \mathfrak{m})$ is a Noetherian local ring, an integrally closed integral domain, and $b \in \mathfrak{m} \setminus \{0\}$. Suppose $\mathfrak{m} \in \operatorname{Ass}(A/(b))$. Then $\mathfrak{m}$ is principal (and thus codimension at most 1)."

We now prove this statement. Choose $\overline{a} \in A/(b)$ such that $\operatorname{Ann}(\overline{a}) = \mathfrak{m}$. Then $\overline{a} \neq 0$. Lift $\overline{a}$ to $a \in A$. Then $a/b \in K(A) \setminus A$, and $(a/b)\mathfrak{m} \subseteq A$ (since $\operatorname{Ann}(\overline{a}) = \mathfrak{m}$).

We show by contradiction that $(a/b)\mathfrak{m} \nsubseteq \mathfrak{m}$. Assume otherwise, i.e., $(a/b)\mathfrak{m} \subseteq \mathfrak{m}$. Recall Lemma 9.2.7: "Suppose A is a subring of K, $r \in K$, and N is a faithful $A[r]$-module that is finitely generated as an A-module. Then r is integral over A." We apply this here with $r = a/b$ and $N = \mathfrak{m}$ ($(a/b)\mathfrak{m} \subseteq \mathfrak{m}$ implies $\mathfrak{m}$ is $A[a/b]$-module, and $\operatorname{Ann}_{A[a/b]}(\mathfrak{m}) = 0$ because $0 \neq b \in \mathfrak{m}$ and $A[a/b] \subseteq K(A)$ is integral domain). Then a/b is integral over A, since A is integrally closed, $a/b \in A$, i.e., $\overline{a} = 0$ in $A/(b)$, contradiction.

Hence $(a/b)\mathfrak{m} = A$, it follows that $(b/a)A \subseteq \mathfrak{m}$. Let $x \in \mathfrak{m}$, then $(a/b)x \in A$, multiply both sides by b/a, we have $x \in (b/a)\mathfrak{m}$, which implies that $\mathfrak{m} = (b/a)A$, from which $\mathfrak{m}$ is principal, as desired. □

Theorem 14.5.9

Suppose A is a Noetherian integral domain, regular in codimension 1. Then (a) A is integrally closed if and only if (b) $A = \bigcap_{\mathrm{ht}(\mathfrak{p})=1} A_{\mathfrak{p}}$.

Remark 14.29 This implies Algebraic Hartogs's Lemma 14.5.8.

Proof (a) implies (b). We apply Lemma 14.5.6. From Proposition 14.5.3 (ii), the only prime we need to check are codimension 1.

(b) implies (a). The $A_{\mathfrak{p}}$ are dimension 1 Noetherian domains that are regular, and hence integrally closed by Theorem 14.5.5. Suppose $r \in K(A)$ is integral over A. Then $r \in K(A)$ is integral over $A_{\mathfrak{p}}$ for all codimension 1 $\mathfrak{p}$, so $r \in A_{\mathfrak{p}}$, so $r \in \bigcap_{\mathrm{ht}(\mathfrak{p})=1} A_{\mathfrak{p}} = A$. □

Thus, we have proven the Algebraic Hartogs's Lemma 14.5.8. □

We conclude with a fact that we will later (§**??**), along with Proposition 14.5.3, recognize as Serre's $R_1 + S_2$ criterion for normality.

Proposition 14.5.4

Suppose A is a Noetherian integral domain satisfying conditions (i) and (ii) of Proposition 14.5.3. Then A is integrally closed.

Proof Now $A = \bigcap_{\mathfrak{p}} A_{\mathfrak{p}}$ as $\mathfrak{p}$ runs over the primes described in Lemma 14.5.6. From condition (ii) of Proposition 14.5.3, there are all codimension 1 primes. The result then follows from Theorem 14.5.9, using condition (i) of Proposition 14.5.3. □

Consequence of Algebraic Hartogs's Lemma 14.5.8 we will use.

Corollary 14.5.2 (Useful)

If f is a nonzero rational function on a locally Noetherian normal scheme, and f has no poles, show that f is regular.

Proof Suppose X is a locally Noetherian normal scheme. Recall Definition 7.6.11 and Definition 7.6.12, it suffices to show that $f \in \mathscr{O}_{X,p}$ for all $p \in X$. Let $U = \operatorname{Spec} A$ be an affine open neighborhood of $p = [(\mathfrak{p})]$, where A is integrally closed Noetherian domain. Since f has no poles, $f \in A_{\mathfrak{q}}$ for any codimension 1 point $[(\mathfrak{q})]$. By Algebraic Hartogs's Lemma 14.5.8, we know that $f \in A$, which implies that $f \in A_{\mathfrak{p}} = \mathscr{O}_{X,p}$. Hence f is regular. □

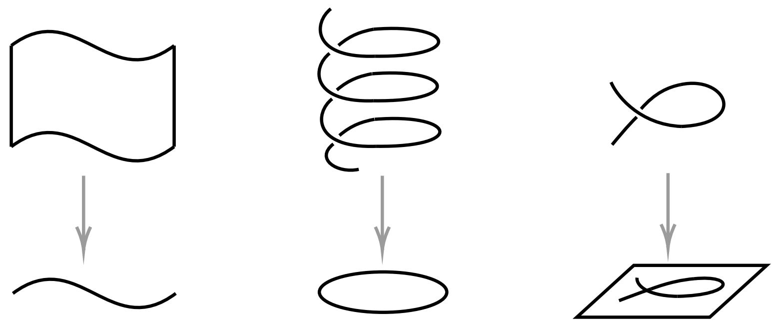
14.6 Smooth (and étale) morphisms: First definition

In §14.2.3, we defined smoothness over a field. We now know enough to define smooth morphisms (and, as special case, étale morphisms) in general. Our definition will be imperfect in a number of ways. For example, it will look a little surprising. As another sign, it will not be obvious that it actually generalizes smoothness over a field. A third flaw is that because it is of the form "there exist open covers satisfying some property," it is poorly designed to be used to show that some morphism is not smooth. But it has the major advantage that we can give the definition right now, and the basic properties of smooth morphisms will be straightforward to show.

14.6.1 Differential geometric motivation (see figure 14.6)

The notion of a smooth morphism is motivated by the following idea from differential geometry. For the purposes of this dicussion, we say a map $\pi : X \to Y$ of manifolds (or real analytic spaces, or any variant thereof) is **smooth of relative dimension** n if locally (on X) it looks like $Y' \times \mathbb{R}^n \to Y'$. (In differential geometry, this is usually called a **submersion**.) Translation: It is "locally on the source a smooth fibration." (If you have not heard the word fibration before, don't worry. Informally speaking, a **fibration** in some "category of geometric objects" is a map $\pi : X \to Y$ that locally on the source is a product, $V \times W \to V$; in some sort of category of manifolds, a fibration is "locally on the source a smooth fibration" if the W in the local description can be taken to be $\mathbb{R}^a$ for some a.)

In particular, "smooth of relative dimension 0" is the same as the important notion of "local isomorphism" ("isomorphism locally on the source" — not quite the same as "covering space"). Carrying this idea too naively into algebraic geometry leads to the "wrong" definition (see Exercise 14.23). We give a better definition now, and some equivalent definitions in §**??**.



(a) (b) (c)

Figure 14.6: Sketch of notions from differential geometry: (a) maps "locally on the source a smooth fibration", (b) local isomorphisms, and (c) immersions. (In algebraic geometry: (a) smooth morphisms, (b) étale morphisms, and (c) unramified morphisms.)

14.6.2 Smooth of relative dimension n, Étale morphism

Definition 14.6.1 (Smooth of relative dimension n, Étale morphism)

*(1) A morphism $\pi : X \to Y$ is **smooth of relative dimension** n if there exist open covers $\{U_i\}$ of X and $\{V_i\}$ of Y, with $\pi(U_i) \subseteq V_i$, such that for every i we have a commutative diagram*

$$\begin{tikzcd} U_i \arrow[r, leftrightarrow, "\sim"] \arrow[d, "\pi|_{U_i}"'] & W \arrow[r, hook, "open"] \arrow[d, "\rho|_W"] & \operatorname{Spec} B[x_1, \dots, x_{n+r}]/(f_1, \dots, f_r) \arrow[dl, "\rho"] \\ V_i \arrow[r, leftrightarrow, "\sim"] & \operatorname{Spec} B & \end{tikzcd}$$

where ρ is induced by the obvious map of rings in the opposite direction, and W is an open subscheme of $\operatorname{Spec} B[x_1, \dots, x_{n+r}]/(f_1, \dots, f_r)$, such that the determinant of the Jacobian matrix of the f_j's with respect to the first r x_i's,

$$\det \left(\frac{\partial f_j}{\partial x_i} \right)_{i,j \le r}, \tag{14.9}$$

is an invertible (=nowhere zero) function on W.

*(2) A morphism $\pi : X \to Y$ is **étale** if π is smooth of relative dimension 0.*

Note ***Quick observations.*** *Be sure to notice that if B is a field, then we have just the right number of equations (r of them) to hope to cut out something of dimension n in $\mathbb{A}_k^{n+r}$. More generally, intuitively and imprecisely, we have just the "right number of equations" to cut out W in $\mathbb{A}_B^{n+r}$, with no redundant ones to spare.*

From the definition, smooth morphisms are locally of finite presentation (hence locally of finite type). Also, $\mathbb{A}_B^n \to \operatorname{Spec} B$ is immediately seen to be smooth of relative dimension n. From the definition, the locus on X where π is smooth of relative dimension n is open. In particular, the locus where π is étale is open.

As easy examples, open embeddings are étale, and the projection $\mathbb{A}^n \times Y \to Y$ is smooth of relative dimension n. You may already have some sense of what makes this definition imperfect. For example, how do you know a morphism is smooth if it isn't given to you in a form where the "right" variables x_i and the "right" equations are clear?

Theorem 14.6.1 (Implicit function Theorem)

Let $f = (f_1, \ldots, f_m) : \mathbb{R}^{n+m} \to \mathbb{R}^m$ be a continuously differentiable function, and let $\mathbb{R}^{n+m}$ has coordinates $(\mathbf{x}, \mathbf{y}) = (x_1, \ldots, x_n, y_1, \ldots, y_m)$. Fix a point $(\mathbf{a}, \mathbf{b}) = (a_1, \ldots, a_n, b_1, \ldots, b_m)$ with $f(\mathbf{a}, \mathbf{b}) = \mathbf{0}$, where $\mathbf{0} \in \mathbb{R}^m$ is the zero vector. If the Jacobian matrix

$$J(f)(\mathbf{a}, \mathbf{b}) = \left(\frac{\partial f_i}{\partial y_j}(\mathbf{a}, \mathbf{b})\right)$$

is invertible, then there exists an open subset $U \subseteq \mathbb{R}^n$ containing $\mathbf{a}$ such that there exists a unique function $g : U \to \mathbb{R}^m$ such that $g(\mathbf{a}) = \mathbf{b}$, and $f(\mathbf{x}, g(\mathbf{x})) = 0$ for all $\mathbf{x} \in U$.

Exercise 14.21 (Motivating exercise) Show how Definition 14.6.1 gives the definition in differential geometry, as described in §14.6.1. (Your argument might use the implicit function theorem, which "doesn't work" in algebraic geometry; see Exercise 14.23.) Show that for complex varieties, analytification turns étale morphisms into "local biholomorphisms."

Proof For the first statement. Suppose X, Y are smooth manifolds which satisfy conditions in Definition 14.6.1, say $\pi : X \to Y$ is a smooth of relative dimension n morphism. Then there exist open covers $\{U_i\}$ of X and $\{V_i\}$ of Y, with $\pi(U_i) \subseteq V_i$. Let $p \in X$, then there exists $U_i \ni p$ and $V_i \ni \pi(p)$, such that $U_i \cong W \subseteq Z$ where $Z = \{(\mathbf{y}, \mathbf{x}) \in V_i \times \mathbb{R}^{n+r} : f_1(\mathbf{y}, \mathbf{x}) = \cdots = f_r(\mathbf{y}, \mathbf{x}) = 0\}$ and W is an open subset of Z. Moreover $\pi|_{U_i} : U_i \to V_i$ can be view as $\operatorname{pr}|_W : W \to V_i$ where $\operatorname{pr} : V_i \times \mathbb{R}^{n+r} \to V_i$. Say $\mathbf{u} := (x_1, \ldots, x_r)$ and $\mathbf{v} := (x_{r+1}, \ldots, x_{n+r})$. Let $F = (f_1, \ldots, f_r) : V_i \times \mathbb{R}^{n+r} \to \mathbb{R}^r$. Let $p \in W$, recall (14.9), by Implicit function Theorem 14.6.1, there exists an open subset $W_0 \ni p$ of $V_i \times \mathbb{R}^{n+r}$ such that there exists a unique function $g : W_0 \to \mathbb{R}^r$ such that $\mathbf{u} = g(\mathbf{y}, \mathbf{v})$. Since W is open subset of Z, we may shrink W_0 such that $W \cap W_0 = Z \cap W_0$. Now, $\pi|_{W_0 \cap W}$ is simply

$$(\mathbf{y}, \mathbf{v}) \longmapsto \mathbf{y}.$$

This is the definition of submersion which we defined in 14.6.1.

For the second statement. Suppose X, Y are complex varieties, and $\pi : X \to Y$ is étale which satisfies Definition 14.6.1 (2). We use the same symbol as above discussion, since étale, there are no $\mathbf{v}$ variables, the coordinates are just $(\mathbf{y}, \mathbf{u})$. By Complex Implicit function Theorem, locally, $\mathbf{u}$ is determined by $\mathbf{y}$ via a function $\mathbf{u} = h(\mathbf{y})$ where h is holomorphic. As discussion above, π restrict to an open neighborhood of a point p is the bijection $(\mathbf{y}, h(\mathbf{y})) \mapsto \mathbf{y}$, its inverse is $\mathbf{y} \mapsto (\mathbf{y}, h(\mathbf{y}))$, which is holomorphic. Thus, π is a local biholomorphism (a local isomorphism of complex manifolds). $\square$

Proposition 14.6.1

The notion of smoothness of relative dimension n (and, in particular, étaleness) is local on both source and the target.

Proof We first show that the notion of smoothness of relative dimension n is local on the source. For (LS1). Suppose $\pi : X \to Y$ is smooth of relative dimension n and $U \hookrightarrow X$ is an open subset, we want to show that $\pi|_U : U \to Y$ is smooth of relative dimension n. Use the notation in Definition 14.6.1, clearly, $\{U_i \cap U\}$ is an open cover of U. Then we have the following commutative diagram

$$\begin{tikzcd}
U_i \cap U \arrow[r, leftrightarrow, "\sim"] \arrow[dr, hook] \arrow[ddr, "\pi|_{U_i \cap U}"'] & W \cap U \arrow[dr, hook] \arrow[ddr] \arrow[drr, hook] & & \\
& U_i \arrow[r, leftrightarrow, "\sim"] \arrow[d, "\pi|_{U_i}"] & W \arrow[r, hook] \arrow[d, "\rho|_W"] & \operatorname{Spec} B[x_1, \ldots, x_{n+r}]/(f_1, \ldots, f_r) \arrow[dl, "\rho"] \\
& V_i \arrow[r, leftrightarrow, "\sim"] & \operatorname{Spec} B &
\end{tikzcd}$$

so it is clear that $\pi|_U : U \to Y$ is smooth of relative dimension n. For (LS2). For each U_i, there exists open covers $\{U_{ij}\}_j$ of U_i and $\{V_{ij}\}_j$ of Y with $\pi|_{U_i}(U_{ij}) \subseteq V_{ij}$ such that for every j we have the following commutative diagram and (14.9).

$$\begin{tikzcd}
U_{ij} \arrow[r, leftrightarrow, "\sim"] \arrow[d, "\pi|_{U_i}"'] & W_{ij} \arrow[r, hook, "\text{open}"] \arrow[d, "\rho|_W"] & \operatorname{Spec} B_{ij}[x_1, \ldots, x_{n+r}]/(f_1, \ldots, f_r) \arrow[dl, "\rho"] \\
V_{ij} \arrow[r, leftrightarrow, "\sim"] & \operatorname{Spec} B_{ij} &
\end{tikzcd}$$

Note that $\{V_{ij}\}_{ij}$ is also an open cover of Y and $\{U_{ij}\}_{ij}$ is an open cover of X, recall Definition 14.6.1, $\pi : X \to Y$ is smooth of relative dimension n.

We next show that the notation of smoothness of relative dimension n is local on the target. For (LT1). Suppose $\pi : X \to Y$ is smooth of relative dimension n, we want to show that for any open subset V of Y, the restricted morphism $\pi^{-1}(V) \to V$ is smooth of relative dimension n. Since $\pi : X \to Y$ is smooth of relative dimension n, there exists open covers $\{U_i\}$ of X and $\{V_i\}$ of Y with $\pi(U_i) \subseteq V_i$ such that for each i we have a commutative diagram and (14.9).

$$\begin{tikzcd}
U_i \arrow[r, leftrightarrow, "\sim"] \arrow[d, "\pi|_{U_i}"'] & W_i \arrow[r, hook, "\text{open}"] \arrow[d, "\rho_i|_{W_i}"] & \operatorname{Spec} B_i[x_1, \ldots, x_{n+r}]/(f_{i,1}, \ldots, f_{i,r}) \arrow[dl, "\rho_i"] \\
V_i \arrow[r, leftrightarrow, "\sim"] & \operatorname{Spec} B_i &
\end{tikzcd}$$

Consider $\{V_i \cap V\}$, it is an open cover of V, since $V_i \cap V$ is open subset of V_i, $V_i \cap V$ can be covered by distinguished open subsets of $\operatorname{Spec} B_i$, namely $V_i \cap V = \bigcup_j V_{ij}$ where $V_{ij} = \operatorname{Spec} B_{ij}$ is distinguished open subset of $\operatorname{Spec} B_i$. Then $\{V_{ij}\}_{ij}$ is an open cover of V. Set $U_{ij} = U_i \cap \pi^{-1}(V_{ij})$, then $\{U_{ij}\}_{ij}$ is an open cover of $\pi^{-1}(V)$. Note that for each i, j, we have the following commutative diagram and (14.9),

$$\begin{tikzcd}
U_{ij} \arrow[r, leftrightarrow, "\sim"] \arrow[d, "\pi|_{U_{ij}}"'] & W_{ij} \arrow[r, hook, "\text{open}"] \arrow[d, "\rho_i|_{W_{ij}}"] & \operatorname{Spec} B_{ij}[x_1, \ldots, x_{n+r}]/(f_{ij,1}, \ldots, f_{ij,r}) \arrow[dl, "\rho_{ij}"] \\
V_{ij} \arrow[r, leftrightarrow, "\sim"] & \operatorname{Spec} B_{ij} &
\end{tikzcd}$$

hence $\pi^{-1}(V) \to V$ is smooth of relative dimension n. For (LT2). For a morphism $\pi : X \to Y$, suppose there is an open cover $\{V_i\}$ of Y for which each restricted morphism $\pi^{-1}(V_i) \to V_i$ is smooth of relative dimension n. Then for each V_i there are open covers $\{V_{ij}\}_j$ of V_i and $\{U_{ij}\}_j$ of $\pi^{-1}(V_i)$ with $\pi(U_{ij}) \subseteq V_{ij}$ such that the following diagram commutes for each j and (14.9) holds.

$$\begin{tikzcd} U_{ij} \arrow[r, leftrightarrow, "\sim"] \arrow[d, "\pi|_{U_{ij}}"'] & W_{ij} \arrow[r, hook, "\text{open}"] \arrow[d, "\rho_i|_{W_{ij}}"] & \operatorname{Spec} B_{ij}[x_1, \ldots, x_{n+r}]/(f_{ij,1}, \ldots, f_{ij,r}) \arrow[dl, "\rho_{ij}"] \\ V_{ij} \arrow[r, leftrightarrow, "\sim"] & \operatorname{Spec} B_{ij} & \end{tikzcd}$$

Note that $\{U_{ij}\}_{ij}$ is open cover of X and $\{V_{ij}\}_{ij}$ is open cover of Y, we know that $\pi : X \to Y$ is smooth of relative dimension n. □

We can thus make sense of the phrase "$\pi : X \to Y$ is smooth of relative dimension n at $p \in X$"

Definition 14.6.2 (Smooth of relative dimension n at a point)

A morphism $\pi : X \to Y$ is smooth of relative dimension n at p if there is an open neighborhood U of p such that $\pi|_U$ is smooth of relative dimension n.

Note *The phrase **smooth morphism** (without reference to relative dimension n) often informally means "smooth morphism of some relative dimension n," but some times can mean "smooth of some relative dimension n in the neighborhood of every point."*

Proposition 14.6.2 (Smoothness (étaleness) preserved by base change)

The notion of smoothness of relative dimension n (and hence the notion of étaleness) is preserved by base change.

Proof Suppose $\pi : X \to Y$ is smooth of relative dimension n, we want to show for any $\varphi : Y' \to Y$, $\pi_{Y'} : X \times_Y Y' \to Y'$ is smooth of relative dimension n. Since smoothness is local on the target (Proposition 14.6.1), we may assume that $Y = \operatorname{Spec} B$ and $Y' = \operatorname{Spec} B'$. Also note that smoothness is local on the source (Proposition 14.6.1), we may assume that $W = X$ (W is given by Definition 14.6.1), then we have the following commutative diagram

$$\begin{tikzcd} X \arrow[r, hook, "\text{open}"] \arrow[d, "\pi"'] & \operatorname{Spec} B[x_1, \ldots, x_{n+r}]/(f_1, \ldots, f_r) \arrow[dl] \\ \operatorname{Spec} B & \end{tikzcd}$$

with the Jacobian matrix of the f_j's with respect to the first r x_i's, $\det \left(\frac{\partial f_j}{\partial x_i}\right)_{i,j \le r}$ is an invertible function on X. Consider the following diagram,

$$\begin{tikzcd} X_{B'} \arrow[rr] \arrow[d, hook, "\text{open}"] & & X \arrow[d, hook, "\text{open}"] \\ \operatorname{Spec} B'[x_1, \ldots, x_{n+r}]/(f_1', \ldots, f_r') \arrow[rr] \arrow[d] & & \operatorname{Spec} B[x_1, \ldots, x_{n+r}]/(f_1, \ldots, f_r) \arrow[d] \\ \operatorname{Spec} B' \arrow[rr] & & \operatorname{Spec} B \end{tikzcd}$$

where f_i' are the polynomials obtained by applying the ring homomorphism $B \to B'$ to the coefficients of the original polynomials f_i. Since open embedding preserved by base change, $X_{B'} \hookrightarrow \operatorname{Spec} B'[x_1, \ldots, x_{n+r}]/(f_1', \ldots, f_r')$ is open embedding. It suffices to show that $\det \left(\frac{\partial f_j'}{\partial x_i}\right)_{i,j \le r}$ is an invertible function on $X_{B'}$. Consider morphism $\mathrm{pr} : X_{B'} \to X$, since $\det \left(\frac{\partial f_j}{\partial x_i}\right)_{i,j \le r}$ is an invertible function on X, its pullback $\det \left(\frac{\partial f_j'}{\partial x_i}\right)_{i,j \le r}$ must be invertible. Hence $\pi_{Y'}$ is smooth of relative dimension n. □

Proposition 14.6.3 (Smoothness is preserved by composition)

Suppose $\pi : X \to Y$ is smooth of relative dimension m and $\rho : Y \to Z$ is smooth of relative dimension n. Then $\rho \circ \pi$ is smooth of relative dimension $m+n$. In particular, the composition of étale morphism is étale. Furthermore, this implies that smooth morphisms are closed under product, by Proposition 9.1.1.

Proof Recall Proposition 14.6.1, smoothness is local on the target, we may assume that $Z = \operatorname{Spec} C$. Then $Y = \rho^{-1}(\operatorname{Spec} C)$, recall that smoothness is local on the source, we may assume that $Y := \operatorname{Spec} B$. Then we have the following commutative diagram

$$\begin{tikzcd}
\operatorname{Spec} B \arrow[r, hook, "\text{open}"] \arrow[d, "\rho"'] & \operatorname{Spec} C[y_1, \dots, y_{n+l}]/(g_1, \dots, g_l) \arrow[dl] \\
\operatorname{Spec} C &
\end{tikzcd}$$

where $\det \left(\frac{\partial g_j}{\partial y_i}\right)_{i,j \le l}$ is an invertible function on $\operatorname{Spec} B$. Say $C' = C[y_1, \dots, y_{n+l}]/(g_1, \dots, g_l)$. Since $\operatorname{Spec} B$ is open subset of $\operatorname{Spec} C'$, $\operatorname{Spec} B$ can be covered by the union of some distinguished affine subset of $\operatorname{Spec} C'$, more precisely, $\operatorname{Spec} B := \bigcup_k \operatorname{Spec} C'_{h_k}$. Without loss of generally, we may assume $\operatorname{Spec} B = \operatorname{Spec} C'_h$. Now $X = \pi^{-1}(\operatorname{Spec} B)$, since smoothness is local on the source, we may assume that X is also affine, namely $X := \operatorname{Spec} A$. By the smoothness, we have the following commutative diagram,

$$\begin{tikzcd}
\operatorname{Spec} A \arrow[r, hook, "\text{open}"] \arrow[d, "\pi"'] & \operatorname{Spec} B[x_1, \dots, x_{m+r}]/(f_1, \dots, f_r) \arrow[dl] \\
\operatorname{Spec} B &
\end{tikzcd}$$

where $\det \left(\frac{\partial f_j}{\partial x_i}\right)_{i,j \le r}$ is an invertible function on $\operatorname{Spec} A$. Consider $B[x_1, \dots, x_{m+r}]/(f_1, \dots, f_r)$, we have

$$\begin{aligned}
B[x_1, \dots, x_{m+r}]/(f_1, \dots, f_r) &= C'_h[x_1, \dots, x_{m+r}]/(f_1, \dots, f_r) \\
&= \left(\frac{C[y_1, \dots, y_{n+l}]}{(g_1, \dots, g_l)}\right)_h [x_1, \dots, x_{m+r}]/(f_1, \dots, f_r) \\
&\cong \left(C[x_1, \dots, x_{m+r}, y_1, \dots, y_{n+l}]/(f_1, \dots, f_r, g_1, \dots, g_l)\right)_h .
\end{aligned}$$

Now, we have the following commutative diagram.

$$\begin{tikzcd}
\operatorname{Spec} A \arrow[r, hook, "\text{open}"] \arrow[d, "\pi"'] & \operatorname{Spec} \left(\frac{C[x_1, \dots, x_{m+r}, y_1, \dots, y_{n+l}]}{(f_1, \dots, f_r, g_1, \dots, g_l)}\right)_h \arrow[r, hook, "\text{open}"] & \operatorname{Spec} \frac{C[x_1, \dots, x_{m+r}, y_1, \dots, y_{n+l}]}{(f_1, \dots, f_r, g_1, \dots, g_l)} \arrow[ddll] \\
\operatorname{Spec} B \arrow[d, "\rho"'] & & \\
\operatorname{Spec} C & &
\end{tikzcd}$$

hence it remain to show that

$$\det J := \det \begin{pmatrix} \left(\frac{\partial f_j}{\partial x_i}\right)_{i,j \le r} & \left(\frac{\partial g_j}{\partial x_i}\right)_{i \le r, j \le l} \\ \left(\frac{\partial f_j}{\partial y_i}\right)_{i \le m, j \le r} & \left(\frac{\partial g_j}{\partial y_i}\right)_{i,j \le l} \end{pmatrix}$$

is an invertible function on $\operatorname{Spec} A$. Note that

$$\begin{aligned}\det J &= \det\begin{pmatrix} \left(\frac{\partial f_j}{\partial x_i}\right)_{i,j\le r} & 0 \\ \left(\frac{\partial f_j}{\partial y_i}\right)_{i\le m, j\le r} & \left(\frac{\partial g_j}{\partial y_i}\right)_{i,j\le l}\end{pmatrix} \\ &= \det\left(\frac{\partial f_j}{\partial x_i}\right)_{i,j\le r}\cdot\det\left(\frac{\partial g_j}{\partial y_i}\right)_{i,j\le l}.\end{aligned}$$

since $\det\left(\frac{\partial f_j}{\partial x_i}\right)_{i,j\le r}$ and $\det\left(\frac{\partial g_j}{\partial y_i}\right)_{i,j\le l}$ (pullback to $\operatorname{Spec} A$) are invertible functions on $\operatorname{Spec} A$, $\det J$ is an invertible functions on $\operatorname{Spec} A$. Thus $\rho\circ\pi$ is smooth of relative dimension $m+n$. □

Corollary 14.6.1

Smooth morphisms (and étale morphisms) form a "reasonable" class (Definition 9.1.1).

Proof This is Proposition 14.6.3, 14.6.2, and 14.6.1. □

Note ***Observation.*** *Suppose $\pi : X \to Y$ is smooth of relative dimension n. Then locally on X, π can be described as an étale cover of $\mathbb{A}^n_Y$. More precisely, for every $p\in X$, there is an open neighborhood U of p, such that $\pi|_U$ can be factored into*

$$U \xrightarrow{\alpha} \mathbb{A}^n_Y \xrightarrow{\beta} Y,$$

where α is étale and β is the obvious projection.

Proof We give a sketch of proof. Let $\{U_i\}$ be an open cover of X and $\{V_i\}$ be an affine cover of Y such that they hold the condition in Definition 14.6.1. Let $V\in\{V_i\}$, say $V=\operatorname{Spec} B$, then there exists $U\in\{U_i\}$ such that $\pi(U)\subseteq V$ and following diagram commutes

$$\begin{tikzcd} U \arrow[r, leftrightarrow, "\sim"] \arrow[d, "\pi|_U"'] & W \arrow[r, hook, "\text{open}"] \arrow[d] & \operatorname{Spec} B[x_1,\dots,x_{n+r}]/(f_1,\dots,f_r) \arrow[dl] \\ V \arrow[r, equal] & \operatorname{Spec} B & \end{tikzcd}$$

where $\det J := \det\left(\frac{\partial f_j}{\partial x_i}\right)_{i,j\le r}$ is a invertible function on W. Say $C:=B[x_1,\dots,x_{n+r}]/(f_1,\dots,f_r)$. Define $\varphi^\sharp : B[t_1,\dots,t_n]\to C$ by setting $t_i\mapsto\overline{x_{r+i}}$. Then we obtain a morphism $\varphi:\operatorname{Spec} C\to\mathbb{A}^n_V$. Define α and β as follow:

$$\begin{tikzcd} U \arrow[r, hook, "\text{open}"] \arrow[drr, "\alpha"'] & \operatorname{Spec} C \arrow[r, "\varphi"] & \mathbb{A}^n_V \arrow[r, "\text{pr}"] \arrow[d, hook, "\text{open}"'] & V \arrow[d, hook, "\text{open}"] \\ & & \mathbb{A}^n_Y \arrow[r, "\beta"] & Y \end{tikzcd}$$

where $\mathbb{A}^n_V\hookrightarrow\mathbb{A}^n_Y$ given by base change, α is the composition of $U\hookrightarrow\operatorname{Spec} C\xrightarrow{\varphi}\mathbb{A}^n_V\hookrightarrow\mathbb{A}^n_Y$ and β be the natural projection.

We next show that α is étale. Note that

$$\begin{aligned} C &= B[x_1,\dots,x_r,x_{r+1},\dots,x_{r+n}]/(f_1,\dots,f_r) \\ &\cong (B[\overline{x_{r+1}},\dots,\overline{x_{r+n}}])[x_1,\dots,x_r]/(f_1,\dots,f_r) \\ &\cong (B[t_1,\dots,t_n])[x_1,\dots,x_r]/(f_1,\dots,f_r), \end{aligned}$$

recall Definition 14.6.1, $U\to\mathbb{A}^n_V$ is étale, recall that open embeddings are étale, so α is étale. □

Exercise 14.22 Show that $\operatorname{Spec} k\to\operatorname{Spec} k[\varepsilon]/(\varepsilon^2)$ is not an étale morphism. (Does your mental picture of étale morphisms already capture this fact?)

Proof Suppose $\pi:\operatorname{Spec} k\to\operatorname{Spec} k[\varepsilon]/(\varepsilon^2)$ is étale, say $A:=k[\varepsilon]/(\varepsilon^2)$, then we have the following commutative diagram

$$\begin{tikzcd}
\operatorname{Spec} k \arrow[r, leftrightarrow, "\sim"] \arrow[d] & p \arrow[r, hook, "\text{open}"] \arrow[d] & \operatorname{Spec} A[x_1, \dots, x_r]/(f_1, \dots, f_r) \arrow[dl] \\
\operatorname{Spec} k[\varepsilon]/(\varepsilon^2) \arrow[r, equal] & \operatorname{Spec} k[\varepsilon]/(\varepsilon^2) &
\end{tikzcd}$$

where $\det\left(\frac{\partial f_j}{\partial x_i}\right)_{i,j \leq r}$ is invertible on p. Say $B = A[x_1, \dots, x_r]/(f_1, \dots, f_r), p = [\mathfrak{m} + I]$ where $\mathfrak{m}$ is maximal ideal of B and $I := (f_1, \dots, f_r)$. We claim that $\dim_k T_{\operatorname{Spec} A, \pi(p)} = \dim_k T_{\operatorname{Spec} B, p}$. Consider $\frac{\mathfrak{m}+I}{\mathfrak{m}^2+I}$, note that $\det\left(\frac{\partial f_j}{\partial x_i}\right)_{i,j \leq r}(p) \neq 0$, so each $f_i \notin \mathfrak{m}^2$, moreover $\{f_i \pmod{\mathfrak{m}^2}\}$ are linear independent, recall the proof of Proposition 14.1.1, we have

$$\dim_k(\mathfrak{m}/(\mathfrak{m}^2 + I))^\vee = \dim_k(\mathfrak{m}/\mathfrak{m}^2)^\vee - r.$$

Since p is k-valued point, $\mathfrak{m}$ is of the form $(\varepsilon, x_1 - p_1, \dots, x_r - p_r)$ where $p_i \in k$, so $\dim_k \mathfrak{m}/\mathfrak{m}^2 = r + 1$, and thus $\dim_k(\mathfrak{m}/\mathfrak{m}^2)^\vee = r + 1$. Hence

$$\dim_k T_{\operatorname{Spec} B, p} = \dim_k(\mathfrak{m}/(\mathfrak{m}^2 + I))^\vee = r + 1 - r = 1.$$

On the other hand $\pi(p) = [(\varepsilon)]$, so

$$\dim_k T_{\operatorname{Spec} A, \pi(p)} = \dim_k((\varepsilon)/(\varepsilon^2)) = 1.$$

Hence $\dim_k T_{\operatorname{Spec} A, \pi(p)} = \dim_k T_{\operatorname{Spec} B, p}$. But $\dim_k T_{\operatorname{Spec} B, p} = \dim_k T_{\operatorname{Spec} k, [(0)]} = 0$, a contradiction. It follows that π is not étale. $\square$

Remark 14.30 More sophisticated approaches to Exercise 14.22 may lead you to consider the following exercise: Classify the étale maps to $\operatorname{Spec} k$. Classify the étale maps to $\operatorname{Spec} \overline{k}[\varepsilon]/(\varepsilon^2)$. If $A = \overline{k}[x_1, \dots, x_n]/I$ is such that $\operatorname{Spec} A$ has only one point (an example of an Artinian local scheme, Definition 7.5.6, and A is an Artinian local ring, §13.1.4), classify the étale maps to $\operatorname{Spec} A$. These questions will lead you in different important directions, to deformation theory, Witt vectors, and more.

✍ **Exercise 14.23** Suppose k is a field of characteristic not 2. Let $Y = \operatorname{Spec} k[t]$, and $X = \operatorname{Spec} k[u, 1/u]$. Show that the morphism $\pi : X \to Y$ induced by $t \mapsto u^2$ is étale. Show that there is no nonempty open subset U of X on which π is an isomorphism. (This)

Proof Let $\rho^\sharp : k[t] \to k[t, u]/(t - u^2)$ by setting $t \mapsto t$, then we have the following commutative diagram,

$$\begin{tikzcd}
k[u]_u & k[t, u]/(t - u^2) \cong k[u] \arrow[l, "\text{local.}"'] \\
k[t] \arrow[u, "\pi^\sharp"] \arrow[ur, "\rho^\sharp"'] &
\end{tikzcd}$$

where $\pi^\sharp(t) = u^2$. Apply Spec, we have

$$\begin{tikzcd}
\operatorname{Spec} k[u]_u \arrow[r, hook, "\text{open}"] \arrow[d, "\pi"'] & \operatorname{Spec} k[t, u]/(t - u^2) \arrow[dl, "\rho"] \\
\operatorname{Spec} k[t] &
\end{tikzcd}$$

note that $\frac{\partial}{\partial u}(t - u^2) = -2u$ is invertible on open subset $\operatorname{Spec} k[u]_u$, we know that π is étale.

In fact, $k[u]_u$ and $k[t]$ are integral domain, so X, Y are integral scheme. If there exists $U \subseteq X, V \subseteq Y$ such that $\pi|_U : U \xrightarrow{\sim} V$, since $\pi^\sharp(t) = u^2$, π send the generic point of U to generic point of V, so we have the isomorphism of function fields $K(V) \xrightarrow{\sim} K(U)$. Note that $K(U) \cong K(X) \cong k(u)$ and $K(V) \cong K(Y) \cong k(t)$, we have $k(t) \xrightarrow{\sim} k(u)$. But this is impossible, since $k(t) = k(u^2)$ and $[k(u) : k(u^2)] = 2$. Hence there is no nonempty open subset U of X on which π is an isomorphism. □

Remark 14.31 Exercise 14.23 shows that étale cannot be defined as "local isomorphism," i.e., "isomorphism locally on the source." in particular, the naive extension of the differential intuition of §14.6.1 does not give the right definition of étaleness in algebraic geometry.

We now show that our new Definition 14.6.1 specializes to Definition 14.2.2 when the target is a field.

Theorem 14.6.2

Suppose X is a k-scheme. Then the following are equivalent.

(i) X is smooth of relative dimension n over $\operatorname{Spec} k$ *(Definition 14.6.1).*

(ii) X has pure dimension n, and is smooth over k (in the sense of Definition 14.2.2).

Note *This allow us to use the phrases "smooth over k" and "smooth over* $\operatorname{Spec} k$*" interchangeably, as we would expect. More generally,* ***"smooth over a ring*** *A**"** means "smooth over* $\operatorname{Spec} A$*".*

Proof ***(i) implies (ii).*** Suppose X is smooth of relative dimension n over $\operatorname{Spec} k$ (Definition 14.6.1). We will show that X has pure dimension n, and is smooth over k (Definition 14.2.2). Our goal is a Zariski-local statement, so we may assume that X is an open subset

$$W \subseteq \operatorname{Spec} k[x_1, \ldots, x_{n+r}]/(f_1, \ldots, f_r),$$

as described in Definition 14.6.1.

The central issue is showing that X has "pure dimension n," as the latter half of the statement would then follow because the corank of the Jacobian matrix is then n.

Let $X_{\overline{k}} := X \times_k \overline{k}$. By Proposition 13.1.8, $X_{\overline{k}}$ is pure dimension n if and only if X has pure dimension n. Let p be a closed point of the finite type scheme $X_{\overline{k}}$. By Krull's Height Theorem 13.3.3 and Theorem 13.2.4 (codimension is the difference of dimensions for varieties), we know that

$$\dim p + \dim \mathscr{O}_{X_{\overline{k}},p} = \dim \mathscr{O}_{X_{\overline{k}},p} = \dim_{X_{\overline{k}}} p \geq n,$$

where $\dim \mathscr{O}_{X_{\overline{k}},p}$ is the dimension of $X_{\overline{k}}$ at p (Definition 13.2.5). But the Zariski tangent space at a closed point of a finite type scheme over $\overline{k}$ is cut out by the Jacobian matrix (Proposition 14.1.4), by the Jacobian condition in the Definition 14.6.1, the corank of the Jacobian is n, and therefore $\dim_k (T_{X_{\overline{k}},p})^\vee = n$. As the dimension of the Zariski tangent space bounds the dimension of the scheme (Theorem 13.3.4), i.e., $\dim \mathscr{O}_{X_{\overline{k}},p} \leq \dim_k (T_{X_{\overline{k}},p})^\vee$. So $\dim_{X_{\overline{k}}} p = \dim \mathscr{O}_{X_{\overline{k}},p} = n$ for each closed point p of $X_{\overline{k}}$, recall Definition 13.2.5, $\dim X_{\overline{k}} = n$.

(ii) implies (i). It suffices to prove the result in a neighborhood of any given closed point p of X. (Here we use that any union of open neighborhoods of all closed points is all of X, by Proposition 6.3.8). We thus consider the following problem. Suppose we are given $X := \operatorname{Spec} k[x_1, \ldots, x_{n+r}]/(f_1, \ldots, f_N)$, and a point $p \in X$, and an open neighborhood U of p in X such that U is pure dimension n, and the Jacobian matrix of the f_j's with respect to the x_i's has corank n everywhere on U, We wish to show that the structure morphism $\pi : X \to \operatorname{Spec} k$ is smooth of relative dimension n at p (Definition 14.6.1). By permuting the f_i (and shrinking U), we may assume that the Jacobian matrix of the first $f_1, \ldots, f_r$ already has corank n (i.e., rank r). Then $Y := \operatorname{Spec} k[x_1, \ldots, x_{n+r}]/(f_1, \ldots, f_r)$ is smooth at p of relative dimension n over $\operatorname{Spec} k$ (Definition 14.6.1), by the proof that (i) implies (ii), Y has pure dimension n and is smooth over k (Definition 14.2.2), so the local ring $\mathscr{O}_{Y,p}$ is an integral domain by Proposition 14.2.13.

For the same reasons, $\mathscr{O}_{X,p}$ is also an integral domain of pure dimension n. We have a surjection $\mathscr{O}_{Y,p} \twoheadrightarrow \mathscr{O}_{Y,p}/(f_{r+1}, \ldots, f_N) = \mathscr{O}_{X,p}$ of integral domains of the same dimension, so by Proposition 13.1.4, $\mathscr{O}_{Y,p} = \mathscr{O}_{X,p}$, i.e., $f_{r+1} = \cdots = f_N = 0$ in $\mathscr{O}_{Y,p}$. Thus, near p, X is isomorphic to Y, so X is indeed smooth over k of relative dimension n (Definition 14.6.1), as desired. □

14.7 ⋆ Valuative criteria for separatedness and properness

(The only reason this section is placed here is that we need the theory of discrete valuation rings.)

In reasonable circumstances, it is possible to verify separatedness by checking only maps from spectra of valuation rings. There are four reasons you might like this (even if you never use it). First, it gives useful intuition for what separated morphisms look like. Second, given that we understand schemes by maps to them (the Yoneda philosophy), we might expect to understand morphisms by mapping certain maps of schemes to them, and this is how you can interpret the diagram appearing in the valuative criterion. And the third concrete reason is that one of the two directions in the statement is much easier (a special case of the Reduced-to-Separated Theorem 12.3.1; see Lemma 14.7.1), and this is the direction we will repeatedly use. Finally, the criterion is very useful!

Similarly, there is a valuative criterion for properness.

In this section, we will meet the valuative criteria, but aside from outlining the proof of one result (the DVR version of the valuative criterion of separatedness), we will not give proofs, and satisfy ourselves with references. There are two reasons for this controversial decision. First, the proofs require the development of some commutative algebra involving valuation rings that we will not otherwise need. Second, we will not use these results in any substantial or necessary way later in this book.

14.7.1 Valuative criterion for separatedness

We begin with a valuative criterion for separatedness that applies in a case that will suffice for the interests of most people, that of finite type morphisms of Noetherian schemes. We will then give a more general version for more general readers.

Theorem 14.7.1 (Valuative criterion for separatedness, DVR version; see Figure 14.7)
Suppose $\pi : X \to Y$ is a morphism of finite type of locally Noetherian schemes. Then π is separated if and only if the following condition holds: for any DVR A, and any diagram of the form

$$\begin{tikzcd} \operatorname{Spec} K(A) \arrow[r] \arrow[d, hook, "\textit{open emb.}"'] & X \arrow[d, "\pi"] \\ \operatorname{Spec} A \arrow[r] & Y \end{tikzcd} \tag{14.10}$$

(where the vertical morphism on the left corresponds to the inclusion $A \hookrightarrow K(A)$), there is at most one morphism $\operatorname{Spec} A \to X$ such that the diagram

$$\begin{tikzcd} \operatorname{Spec} K(A) \arrow[r] \arrow[d, hook, "\textit{open emb.}"'] & X \arrow[d, "\pi"] \\ \operatorname{Spec} A \arrow[r] \arrow[ur, dotted, "\leq 1"] & Y \end{tikzcd} \tag{14.11}$$

commutes.

We can show one direction right away, in the next lemma.

Lemma 14.7.1
If π is separated, then the valuation criterion holds.

Proof Suppose $\pi : X \to Y$ is separated, and suppose given a diagram as (14.10) where there are two morphisms h, h' of T to X making the whole diagram commutative.

$$\begin{tikzcd}
\operatorname{Spec} K(A) \arrow[r] \arrow[d, hook, "\text{open emb.}"'] & X \arrow[d, "\pi"] \\
\operatorname{Spec} A \arrow[r] \arrow[ur, shift left, "h"] \arrow[ur, shift right, "h'"'] & Y
\end{tikzcd}$$

Since A is DVR, by Theorem 14.5.5, A is an integral domain, and therefore $\operatorname{Spec} A$ is reduced. Note that $K(A)$ is the stalk of $\mathscr{O}_{\operatorname{Spec} A}$ at generic point of $\operatorname{Spec} A$, $\operatorname{Spec} K(A)$ is dense open subset of $\operatorname{Spec} A$, hence h, h' agreeing on a dense open subset of $\operatorname{Spec} A$, by Reduced-to-Separated Theorem 12.3.1, $h = h'$. □

Proposition 14.7.1

Suppose X is an integral Noetherian separated curve. If $p \in X$ is a regular closed point, then $\mathscr{O}_{X,p}$ is a DVR, so each regular point yields a discrete valuation on $K(X)$. Moreover, distinct regular closed points yield distinct discrete valuations.

Proof Since X is a integral curve, $\dim X = 1$, and therefore $\dim \mathscr{O}_{X,p} = \dim X = 1$. Since p is regular point, Noetherian local ring $\mathscr{O}_{X,p}$ is regular, by Theorem 14.5.5, $\mathscr{O}_{X,p}$ is a DVR, so each regular point yields a discrete valuation on $K(A)$.

We next show that distinct regular closed points yield distinct discrete valuations. Let p, q be two distinct regular closed points, if they yield the same discrete valuation on $K(X)$, by Lemma 14.5.4, $\mathscr{O}_{X,p} = \mathscr{O}_{X,q}$. Say $A = \mathscr{O}_{X,p}$, hence we have two morphisms from $\operatorname{Spec} A \to A$ such that the following diagram commutes:

$$\begin{tikzcd}
\operatorname{Spec} K(X) \arrow[r, equal] & \operatorname{Spec} K(A) \arrow[r] \arrow[d, hook, "\text{open emb.}"'] & X \arrow[d, "\text{sep.}"] \\
& \operatorname{Spec} A \arrow[r] \arrow[ur, shift left, "h_p"] \arrow[ur, shift right, "h_q"'] & \operatorname{Spec} k
\end{tikzcd}$$

where $h_p : \operatorname{Spec} \mathscr{O}_{X,p} \to X$ and $h_q : \operatorname{Spec} \mathscr{O}_{X,q} \to X$. But Lemma 14.7.1 tells us this impossible, so we must have $p = q$. □

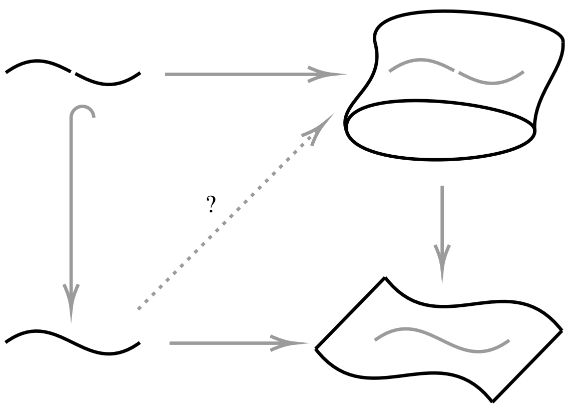


Figure 14.7: The valuative criterion for separatedness.

Here is the intuition behind the valuation criterion (see Figure 14.7). We think of Spec of a discrete valuation ring A as a "germ of a curve" (recall Proposition 14.7.1), and $\operatorname{Spec} K(A)$ as the "germ minus the origin" (even though it is just a point!). Then the valuative criterion says that if we have a map from a germ of a curve to Y, and have a lift of the map away from the origin to X, then there is at most one way to lift the map from the entire germ. In the case where Y is the spectrum of a field, you can think of this as saying that limits of one-parameter families are unique (if they exist).

Remark 14.32 The above paragraph illustrates that separatedness reflects the Hausdorffness. In topology, a defining feature of a Hausdorff space is that if a sequence (or path) has a limit, then that limit must be unique.

For example, this captures the idea of what is wrong with the map of the line with the doubled origin over k (Figure 14.8): we take $\operatorname{Spec} A$ to be the germ of the affine line at the origin, and consider the map of the germ minus the origin to line with doubled origin. Then we have two choice for how the map can extend over the origin.

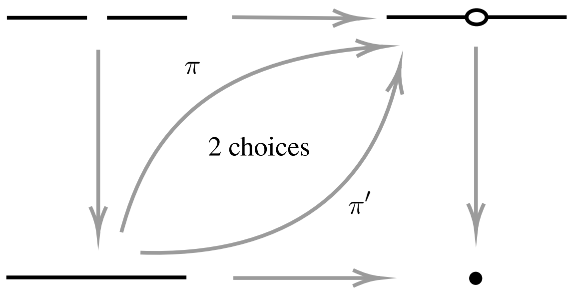


Figure 14.8: The line with the doubled origin fails the valuative criterion for separatedness. (You may notice Figure 12.4 embedded in this diagram.)

✍ **Exercise 14.24** Make this precise: show that map of the line with doubled origin over k to $\operatorname{Spec} k$ fails the valuative criterion for sparatedness. (Earlier arguments were given in Example 12.3 and Exercise 12.1.)

Proof Let X be the line with doubled origin over k, suppose X is glued by $\operatorname{Spec} k[x_1]$ and $\operatorname{Spec} k[x_2]$ via $\operatorname{Spec} k[x_1]_{x_1} \overset{\sim}{\leftrightarrow} \operatorname{Spec} k[x_2]_{x_2}$. Let $A = k[t]_{(t)}$, clearly A is DVR. Consider the following commutative diagram,

$$\begin{tikzcd} \operatorname{Spec} k(t) \arrow[r] \arrow[d, hook] & X \arrow[d] \\ \operatorname{Spec} k[t]_{(t)} \arrow[r] & \operatorname{Spec} k \end{tikzcd}$$

where $K(X) \cong k(t)$ since X is integral. There are two ways lift the map from $\operatorname{Spec} A$ to X: fist one is

$$\operatorname{Spec} k[t]_{(t)} \longrightarrow \operatorname{Spec} k[x_1] \xrightarrow{\text{open emb.}} X$$

where $\operatorname{Spec} k[t]_{(t)} \to \operatorname{Spec} k[x_1]$ induced by $k[x_1] \to k[x_1]_{(x_1)} \cong k[t]_{(t)}$; second one is

$$\operatorname{Spec} k[t]_{(t)} \longrightarrow \operatorname{Spec} k[x_2] \xrightarrow{\text{open emb.}} X$$

where $\operatorname{Spec} k[t]_{(t)} \to \operatorname{Spec} k[x_2]$ induced by $k[x_2] \to k[x_2]_{(x_2)} \cong k[t]_{(t)}$. By Valuative criterion for separated 14.7.1, X is not separated. □

Remark 14.33 ***For experts: Moduli spaces and the Valuative criterion for separatedness.*** If $Y = \operatorname{Spec} k$, and X is a (fine) moduli space (a term we won't define here) of some type of object, then the question of the separatedness of X (over $\operatorname{Spec} k$) has a natural interpretation: given a family of your objects parametrized by a "punctured discrete valuation ring", is there always at most one way of extending it over the closed point?

Idea behind the proof of Theorem 14.7.1 (the valuative criterion for separatedness, DVR version) in the case of varieties.

Proof One direction was done in Lemma 14.7.1. We will now prove the other direction.

If π is not separated, our goal is to produce a diagram (14.10) that can be completed to (14.11) in more than one way. If π is not separated, then $\delta_{X/Y} : X \to X \times_Y X$ is a locally closed embedding that is not a closed embedding.

Lemma 14.7.2

We can find points p not in the diagonal Δ of $X \times_Y X$ (recall Definition 12.2.1) and q in Δ such that $p \in \overline{q}$, and there are no points "between p and q" (no points r distinct from p and q with $p \in \overline{r}$ and $r \in \overline{q}$).

Proof Since X, Y are locally Noetherian schemes, and π is finite type, $X \times_Y X$ is locally Noetherian scheme. Since $X \times_Y X$ is locally Noetherian scheme, there exists a Noetherian affine open subscheme U such that $U \cap \Delta \neq \varnothing$. Say $\Delta_U = U \cap \Delta$, since Δ is locally closed and not closed, Δ_U is the constructible set of U and Δ_U is not closed in U. By Proposition 9.4.2, there exists $q_0 \in \Delta_U$ and $p_0 \in U$ such that $p_0 \in \mathrm{cl}_U(\{q_0\})$ and $p_0 \notin \Delta_U$. Suppose there exists a point r distinct from p_0 and q_0 with $p_0 \in \mathrm{cl}_U(r)$ and $r \in \mathrm{cl}_U(q_0)$ (clearly $r \in U$), then we have a strictly descending sequence,

$$\mathrm{cl}_U(q_0) \supsetneqq \mathrm{cl}_U(r) \supsetneqq \mathrm{cl}_U(p_0).$$

Since U is Noetherian, above chain must be finite length (d.c.c.), then we have a maximal strictly descending sequence:

$$\mathrm{cl}_U(q_0) =: \mathrm{cl}_U(r_n) \supsetneqq \cdots \supsetneqq \mathrm{cl}_U(r_0) := \mathrm{cl}_U(p_0).$$

Since $q_0 = r_n \in \Delta_U$ and $p_0 = r_0 \notin \Delta_U$, there exists a maximal j, such that $r_j \notin \Delta_U$, then $r_{j+1} \in \Delta_U$. Let $p = r_j$ and $q = r_{j+1}$, we next show that there are no points r distinct from p and q with $p \in \mathrm{cl}_{X\times_Y X}(r)$ and $r \in \mathrm{cl}_{X\times_Y X}(q)$. If not, we have a strictly descending sequence

$$\mathrm{cl}_{X\times_Y X}(q) \supsetneqq \mathrm{cl}_{X\times_Y X}(r) \supsetneqq \mathrm{cl}_{X\times_Y X}(p),$$

then

$$\mathrm{cl}_U(q) \supseteq \mathrm{cl}_U(r) \supseteq \mathrm{cl}_U(p),$$

by the maximality of j, $q = r$ or $p = r$, a contradiction. □

Let Q be the scheme obtained by giving the induced reduced subscheme structure to $\overline{q}$. Let $B = \mathscr{O}_{Q,p}$ be the local ring of Q at p.

Lemma 14.7.3

B is a Noetherian local integral domain of dimension 1.

Proof Clearly, B is Noetherian local ring. Since Q is integral (reduced and irreducible), B is Noetherian local integral domain. It suffices to show that $\dim B = 1$. If not, there exists a prime $q =: \mathfrak{q} \subsetneq \mathfrak{h} \subsetneq \mathfrak{p} := p$. Then $\mathfrak{p} \in \overline{\mathfrak{h}}$ and $\mathfrak{h} \in \overline{\mathfrak{q}}$, contradicts to Lemma 14.7.2. □

If B were regular, then we would be done: composing the inclusion morphism $Q \hookrightarrow X \times_Y X$ with the two projections,

$$\begin{tikzcd}
 & & & X \arrow[dr] & \\
\operatorname{Spec} B = \operatorname{Spec} \mathscr{O}_{Q,p} \arrow[r] & Q \arrow[r, hook, "i"] \arrow[urr, "\mathrm{pr}_1 \circ i"] \arrow[drr, "\mathrm{pr}_2 \circ i"'] & X \times_Y X \arrow[ur, "\mathrm{pr}_1"'] \arrow[dr, "\mathrm{pr}_2"] & & Y \\
 & & & X \arrow[ur] &
\end{tikzcd}$$

induces the same morphism $q \to X$ (i.e., $\operatorname{Spec} \kappa(q) \to X$, since q in diagonal; moreover $K(B) = \kappa(q)$) but different extensions extensions to B precisely because p is not in diagonal.

To complete the proof, one shows that the normalization of B is Noetherian; this is where need the hypothesis that we are working with varieties, to invoke Proposition 11.7.4, so the normalization of B is a Dedekind domain (recall Theorem 14.5.6). Then localizing at any prime above p (there is one by the Lying Over Theorem 9.2.2) yields the desired discrete valuation ring A (Theorem 14.5.6 again).

$$\begin{tikzcd}
\operatorname{Spec} K(B) \arrow[r, equal] \arrow[d, "K(A)=K(B)"'] & \operatorname{Spec} \kappa(q) \arrow[r] \arrow[d] & X \arrow[d] \\
\operatorname{Spec} A \arrow[r] \arrow[d, "\text{induced by local.}"'] & \operatorname{Spec} B \arrow[r] \arrow[ur, Rightarrow] & Y \\
\operatorname{Spec} \operatorname{Int.cl}_{K(B)}(B) \arrow[ur, two heads, "\text{normalization}"'] & &
\end{tikzcd}$$

For an actual proof, see [Sta25, Lemma 0207]. □

With a more powerful invocation of commutative algebra, we can prove a valuative criterion with much less restrictive hypotheses.

Theorem 14.7.2 (Valuative criterion for separatedness)

Suppose $\pi : X \to Y$ is a quasiseparated morphism. Then π is separated if and only if for any valuation ring A, and any diagram of the form (14.10), there is at most one morphism $\operatorname{Spec} A \to X$ *such that the diagram (14.11) commutes.*

Because I have already failed to completely prove the DVR version, I feel no urge to prove this harder fact. The proof of one direction, that π separated implies that the criterion holds, follows from an argument identical to the one in Lemma 14.7.1. For a complete proof, see [Sta25, Section 01KY, Lemma 01KZ].

14.7.2 Valuative criteria for (universal closedness and) properness

There is a valuative criterion for properness too. It is philosophically useful, and sometimes directly useful, although we won't need it. It naturally comes from the valuative criterion for separatedness combined with a valuative criterion for universal closedness.

Theorem 14.7.3 (Valuative criterion for universal closedness and properness, DVR version)

Suppose $\pi : X \to Y$ is a morphism of finite type of locally Noetherian schemes. Then π is universally closed (resp., proper) if and only if for any DVR A, and any diagram of the form

$$\begin{tikzcd}
\operatorname{Spec} K(A) \arrow[r] \arrow[d, hook, "\textit{open emb.}"'] & X \arrow[d, "\pi"] \\
\operatorname{Spec} A \arrow[r] & Y
\end{tikzcd} \tag{14.12}$$

there is at least one (resp., exactly one) morphism $\operatorname{Spec} A \to X$ *such that the diagram*

$$\begin{tikzcd}
\operatorname{Spec} K(A) \arrow[r] \arrow[d, hook, "\textit{open emb.}"'] & X \arrow[d, "\pi"] \\
\operatorname{Spec} A \arrow[r] \arrow[ur, dotted, "\geq 1\ (\textit{resp.}, =1)" description] & Y
\end{tikzcd} \tag{14.13}$$

commutes.

See [Har13, Thm. II.4.7, page 101-102] for proofs. A comparison with Theorem 14.7.1 will convince you that these three criteria naturally form a family.

In the case where Y is a field, you can think of the valuative criterion of properness as saying that limits of one-parameter families in proper varieties always exist, and are unique. This is a useful intuition for the notion of properness.

Theorem 14.7.4

$\mathbb{P}^n_A \to \operatorname{Spec} A$ is proper if A is Noetherian.

Proof Recall that properness is stable under base change (Proposition 12.4.4), it suffices to show that $\pi : \mathbb{P}^n_{\mathbb{Z}} \to \operatorname{Spec}\mathbb{Z}$ is proper. Clearly, $\mathbb{P}^n_{\mathbb{Z}}$ is of finite type over $\mathbb{Z}$ and is Noetherian. Say $\mathbb{P}^n_{\mathbb{Z}} = \operatorname{Proj}\mathbb{Z}[x_0, \ldots, x_n]$. Let R be any DVR with following diagram

$$\begin{tikzcd} \operatorname{Spec} K(R) \arrow[r] \arrow[d, hook, "\text{open emb.}"'] & \mathbb{P}^n_{\mathbb{Z}} \arrow[d, "\pi"] \\ \operatorname{Spec} R \arrow[r] & \operatorname{Spec}\mathbb{Z} \end{tikzcd}$$

Let $p \in \mathbb{P}^n_{\mathbb{Z}}$ be the image of the unique point of $\operatorname{Spec} K(R)$. Using the induction on n, suppose for $m < n$, $\mathbb{P}^m_{\mathbb{Z}}$ is proper over $\mathbb{Z}$. If $p \in \mathbb{P}^n_{\mathbb{Z}} \setminus D_+(x_i)$ for some i. Note that $\mathbb{P}^n_{\mathbb{Z}} \setminus D_+(x_i) \cong \mathbb{P}^{n-1}_{\mathbb{Z}}$, then, by inductive hypothesis, there exists unique morphism from $\operatorname{Spec} R$ to $\mathbb{P}^{n-1}_{\mathbb{Z}}$ such that the diagram of the form (14.13) commutes. By composite $\mathbb{P}^{n-1}_{\mathbb{Z}} \hookrightarrow \mathbb{P}^n_{\mathbb{Z}}$, we are done. So it suffices to show the case that $p \in \bigcap_i D_+(x_i)$.

Since $p \in \bigcap_i D_+(x_i)$, all of the functions $x_{i/j}$ are invertible elements of the local ring $\mathscr{O}_{\mathbb{P}^n_{\mathbb{Z}},p}$. We have an inclusion $\kappa(p) \subseteq K(R)$ given by the morphism $\operatorname{Spec} K(R) \to \mathbb{P}^n_{\mathbb{Z}}$. Let $f_{ij} \in K(R)$ be the image of $x_{i/j}$. Then the f_{ij} are nonzero elements of $K(R)$, and $f_{ik} = f_{ij} \cdot f_{jk}$ for all i, j, k. Let $v : K(R)^\times \to \mathbb{Z}$ be the discrete valuation associated to the DVR R. Let $g_i := v(f_{i0})$ for $i = 0, \ldots, n$. Choose k such that g_k is minimal among the set $\{g_0, \ldots, g_n\}$. Then for each i, we have

$$v(f_{ik}) = v(f_{i0}) - v(f_{k0}) = g_i - g_k \geq 0,$$

hence $f_{ik} \in R$ for $i = 0, \ldots, n$. Then we can define a homomorphism

$$\varphi^\sharp : \mathbb{Z}[x_{0/k}, \ldots, x_{n/k}] \longrightarrow R$$

by sending $x_{i/k}$ to f_{ik}. It is compatible with the given field inclusion $\kappa(p) \subseteq K(R)$. This homomorphism $\varphi^\sharp$ gives a morphism $\varphi : \operatorname{Spec} R \to D_+(x_k)$, and hence a morphism of $\operatorname{Spec} S$ to $\mathbb{P}^n_{\mathbb{Z}}$ which is one required. The uniqueness of this morphism follows from the construction and the way the $D_+(X_i)$ glue together. □

Proposition 14.7.2 (Cf. Proposition 14.7.1)

Suppose X is an irreducible regular (Noetherian) curve, proper over a field k (respectively, over $\mathbb{Z}$). There is a bijection

$$\{(\textit{discrete valuation } v : K(X)^\times \to \mathbb{Z}) : v|_{k^\times} = 0\} \xleftrightarrow{\sim} \{\textit{closed points of } X\}.$$

Proof Say $\mathrm{Val} = \{(v : K(X)^\times \to \mathbb{Z}) : v|_{k^\times} = 0\}$ and $\mathrm{Cl.pts} = \{\text{closed points of } X\}$. We first describe direction Val to Cl.pts. Define $\varphi : \mathrm{Cl.pts} \to \mathrm{Val}$ as follow: Pick p be a closed point of X, since X is irreducible regular curve, $\mathscr{O}_{X,p}$ is a regular local ring of dimension 1, by Theorem 14.5.5, $\mathscr{O}_{X,p}$ is a DVR. So this gives a discrete valuation $v_p : K(\mathscr{O}_{X,p})^\times = K(X)^\times \to \mathbb{Z}$. In order for φ to make sense, we need to check that $v_p|_{k^\times} = 0$. Note that $\mathscr{O}_{X,p}$ is the localization of some finitely generated k-algebra, there is a natural map $k \hookrightarrow \mathscr{O}_{X,p}$. Moreover, $k^\times$ is invertible in $\mathscr{O}_{X,p}$, so $v_p|_{k^\times} = 0$. Now, we set $\varphi(p) = v_p$, since distinct closed points yield distinct discrete valuations, φ is well-defined.

We next describe direction Cl.pts to Val. Define $\psi : \mathrm{Val} \to \mathrm{Cl.pts}$ as follow: Pick v be a discrete valuation in Val, then v yields a DVR, say $\mathscr{O}_v$. Since $v|_{k^\times} = 0$, we have an inclusion $k \subseteq \mathscr{O}_v$. By Valuative criterion for properness 14.7.3, there exists unique morphism $h : \operatorname{Spec} \mathscr{O}_v \to X$ such that the diagram

$$\begin{tikzcd} \operatorname{Spec} K(X) \arrow[r] \arrow[d, hook] & X \arrow[d] \\ \operatorname{Spec} \mathscr{O}_v \arrow[r] \arrow[ur, dotted, "\exists!"] & \operatorname{Spec} k \end{tikzcd} \tag{14.14}$$

commutes. Since $\mathscr{O}_v$ is a DVR, h must send the generic point of $\operatorname{Spec} \mathscr{O}_v$ to generic point of X. We claim that the only closed point $\mathfrak{m}$ in $\mathscr{O}_v$ does not send to the generic point η of X. If not, h induces a ring map $h^\sharp : \mathscr{O}_{X,\eta} = K(X) \to \mathscr{O}_{\mathscr{O}_v,\mathfrak{m}} \cong \mathscr{O}_v$. By (14.14), $i \circ h^\sharp = \mathrm{id}$ where $i : \mathscr{O}_v \to K(X)$ and $\mathrm{id} : K(X) \to K(X)$, hence $h^\sharp$ must be inclusion, i.e., $K(X) \subseteq \mathscr{O}_v$, this is impossible. So h send the closed point of $\mathscr{O}_v$ to the closed point p_v of X. We set $\psi(v) = p_v$, clearly, ψ is well-defined.

Consider $\varphi \circ \psi(v)$. Since the closed point of $\operatorname{Spec} \mathscr{O}_v$ send to the closed point p_v of X, it reduces a ring homomorphism $\mathscr{O}_{X,p_v} \to \mathscr{O}_v$. We want to show that $\mathscr{O}_{X,p_v} \cong \mathscr{O}_v$. By (14.14), we have the following commutative diagram

$$\begin{tikzcd} \mathscr{O}_{X,p_v} \arrow[r] \arrow[dr, hook] & \mathscr{O}_v \arrow[d, hook] \\ & K(X) \end{tikzcd}$$

hence, we may regard $\mathscr{O}_{X,p_v}$ as a subring of $\mathscr{O}_v$. Let $p_v = [\mathfrak{m}_v] = [(t)]$ where t is uniformizer of $\mathscr{O}_{X,p_v}$ and $\mathfrak{m}$ be the maximal ideal of $\mathscr{O}_v$. Since $t \in \mathfrak{m}_v \subseteq \mathfrak{m}$, so $v(t) > 0$. Since $\mathscr{O}_{X,p_v}$ is DVR, for any $f \in K(X)^\times$, $f = ut^n$ for some n and u is the unit of $\mathscr{O}_{X,p_v}$. If $f \in \mathscr{O}_v$, then $v(f) = nv(t) \geq 0$, so $n \geq 0$, which implies that $f \in \mathscr{O}_{X,p_v}$, and thus $\mathscr{O}_{X,p_v} = \mathscr{O}_v$. Hence they have the same discrete valuation, so $\varphi \circ \psi(v) = v$.

Consider $\psi \circ \varphi(p)$. Then $\mathscr{O}_{X,p}$ is DVR, by Valuative criterion for universal closedness and properness 14.7.3, there exists unique morphism $\operatorname{Spec} \mathscr{O}_{X,p} \to X$, it must send the closed point of $\operatorname{Spec} \mathscr{O}_{X,p}$ to p, so $\psi \circ \varphi(p) = p$. □

Remark 14.34 ***Remarks for experts.*** There is a moduli-theoretic interpretation similar to that for separatedness: X is proper if and only if there is always precisely one way of filling in a family over a punctured spectrum of a discrete valuation ring.

Finally, here is a fancier version of the valuative criterion for universal closedness and properness.

Theorem 14.7.5 (Valuative criterion for universal closedness and properness)

Suppose $\pi : X \to Y$ is a quasiseparated, finite type (hence quasicompact) morphism. Then π is universally closed (resp., proper) if and only if the following conditions holds. For any valuation ring A and any diagram of the form (14.12), there is at least one (resp., exactly one) morphism $\operatorname{Spec} A \to X$ *such that the diagram (14.13) commutes.*

Clearly the valuative criterion for properness is a consequence of the valuative criterion for separatedness 14.7.2 and the valuative criterion for universal closedness. For proofs, see [Sta25, Proposition 01KF].

One the importance of valuation rings in general

Although we have only discussed DVR in depth, general valuation rings should not be considered an afterthought. Serre makes the case to Grothendieck in a letter:

Tu es bien sévère pour les Valuations! Je persiste pourtant à les garder, pour plusieurs raisons: d'abord une pratique: n rédacteurs ont sué dessus, il n'y a rien à reprocher au fourbi, on ne doit pas le vider sans *raisons très sérieuses (et tu n'en as pas). . . Même un noethérien impénitent a besoin des valuations discrètes, et de leurs extension; en fait, Tate, Dwork, et tous les gen p-adiques te diront qu'on ne peut se limiter au cas discret et que le cas de rang 1 est indispensable; à ce moment, les méthodes noethériennes deviennent un vrai carcan, et on comprend beaucoup mieux en rédigeant le cas général que uniquement le cas de rang 1. . . . Il n'y a pas lieu d'en faire un plat, bien sûr, et c'est pourquoi j'avias vivement combattu le plan Weil initial qui en faisait le théorème central de l'Algèbre Commutative, mais d'autre part il faut le garder.*

You are very harsh on Valuations! I persist nonetheless in keeping them, for several reasons, of which the first is practical: n people have sweated over them, there is nothing wrong with the result, and it should not be thrown out without very serious reasons (which you do not have). . . . Even an unrepentant Noetherian needs discrete valuations and their extensions; in fact, Tate, Dwork and all the p-adic people will tell you that one cannot restrict oneself to the discrete case and the rank 1 case is indispensable; Noetherian methods then become a burden, and one understands much better if one considers the general case and not only the rank 1 case. . . . It is not worth making a mountain out of it, of course, which is why I energetically fought Weil's original plan to make it the central theorem of Commutative Algebra, but on the other hand it must be kept.

14.8 ⋆ More sophisticated facts about regular local rings

Regular local rings have essentially every good property you could want, but some of them require hard work. We now discuss a few fancier facts that may help you sleep well at night.

14.8.1 Localizations of regular local rings are regular local rings

Theorem 14.8.1 (Serre [Mat89, Thm. 19.3, page 157])

If $(A, \mathfrak{m})$ is a regular local ring, then any localization of A at a prime is also a regular local ring.

(We will not need this, and hence will not prove it.) This major theorem was an open problem in commutative algebra for a long time until settled by Serre and Auslander-Buchsbaum using homological methods.

Let $\operatorname{Spec} A$ be a Noetherian ring, to check $\operatorname{Spec} A$ is regular, we need to check the regularity at any prime ideal $\mathfrak{p}$ of A. By Proposition 6.1.5 (quasicompact schemes have closed points), there exists a maximal ideal $\mathfrak{m}$ in the closure of $\mathfrak{p}$. Then we have the follow:

$$A \xrightarrow{\text{local.}} A_{\mathfrak{m}} \xrightarrow{\text{local.}} A_{\mathfrak{p}},$$

by Serre's fact 14.8.1, to check the regular of $A_{\mathfrak{p}}$, it suffices to check the regularity at $\mathfrak{m}$. Hence to check if $\operatorname{Spec} A$ is regular, it suffices to check at closed points. Moreover, assuming Serre's fact 14.8.1 and using Proposition 6.1.5, you can check regularity of a Noetherian scheme by checking at closed points.

We will prove two important cases of Serre's fact 14.8.1. The first you can do right now.

Proposition 14.8.1

Suppose X is Noetherian dimension 1 scheme that is regular at its closed points. Then X is reduced. Moreover, X is regular.

Proof We first show that X is reduced. If not, there exists a point $p \in X$ such that $\mathfrak{N}(\mathscr{O}_{X,p}) \neq 0$. Suppose $\operatorname{Spec} A$ is an affine open subset of X which contains $p = [(\mathfrak{p})]$, then $\mathfrak{N}(\mathscr{O}_{X,p}) = \mathfrak{N}(A_{\mathfrak{p}}) \neq 0$. Let $\mathfrak{m}$ be a maximal ideal of A which contains $\mathfrak{p}$, since X is regular at its closed points, we know that $A_{\mathfrak{p}}$ is a regular local ring, by Theorem 14.2.2, $A_{\mathfrak{p}}$ is an integral domain, so $\mathfrak{N}(A_{\mathfrak{p}}) = 0$. Note that $A_{\mathfrak{p}}$ is the localization of $A_{\mathfrak{m}}$ at $\mathfrak{p}$, hence

$$\mathfrak{N}(A_{\mathfrak{p}}) = \mathfrak{N}((A_{\mathfrak{m}})_{\mathfrak{p}}) = \mathfrak{N}(A_{\mathfrak{m}})_{\mathfrak{p}} = 0,$$

a contradiction!

We next show that X is regular, it suffice to check that $\mathscr{O}_{X,p}$ is regular for p not closed point of X. As $\dim X = 1$, if η not closed point, then η is generic point of X. Suppose η contained in an affine open subset $\operatorname{Spec} A$ of X, since η is generic point, $\dim A_\eta = 0$. Since X is reduced, $\mathfrak{N}(A_\eta) = \eta = 0$, which implies that A_η is a field, and therefore a regular local ring. □

The second importance case will be proved in §**??**.

Theorem 14.8.2

If X is a finite type scheme over a perfect field k that is regular at its closed points, then X is regular.

More generally, **??** will show that Serre's fact 14.8.1 holds if A is the localization of a finite type algebra over a perfect field.

Proposition 14.8.2 (Generalizing Exercise 14.9)

Suppose k is an algebraically closed field of characteristic 0, then there exists a regular hypersurface of every positive degree d in $\mathbb{P}^n$.

Proof Let $f(x_0, \ldots, x_n) = \sum_{i=0}^{n} x_i^d$, and let $X = \operatorname{Proj} k[x_0, \cdots, x_n]/(f)$, then X is a finite type scheme with pure dimension $n-1$. We want to show that X is regular. By Theorem 14.8.2, it suffices to check it at its closed points. By the Jacobian criterion for regularity for projective hypersurfaces 14.3.1, the singular closed points are given by the locus

$$f = \frac{\partial f}{\partial x_0} = \cdots = \frac{\partial f}{\partial x_n} = 0. \tag{14.15}$$

It is easy to see that equations (14.15) has no nonzero solution, so X is regular at its closed points, by Theorem 14.8.2, we win. □

14.8.2 Regular local rings are UFDs, integrally closed, and Cohen-Macaulay

Theorem 14.8.3 (Auslander-Buchsbaum Theorem)

Regular local rings are UFDs.

(This is a hard theorem, so we will not prove it, and will therefore not use it. For a proof, see [Mat89, Thm. 20.3].) Thus regular scheme are factorial (Definition 6.4.5), and hence normal by Proposition 6.4.6 (UFDs are integrally closed).

Remark 14.35 (Factoriality is weaker than regularity) The implication "regular implies factorial" is strict. Here is an example showing this. Suppose k is an algebraically closed field of characteristic not 2. Let $A = k[x_1, \ldots, x_n]/(x_1^2 + \cdots + x_n^2)$ (cf. Exercise 6.6). Note that $\operatorname{Spec} A$ is clearly singular at the origin. Note that $\operatorname{Spec} A$ is clearly singular at the origin. In **??**, we will show that A is a UFD when $n \geq 5$, so $\operatorname{Spec} A$ is factorial. In particular, $A_{(x_1, \ldots, x_n)}$ is a Noetherian local ring that is a UFD, but not regular. (More generally, it is a consequence of Grothendieck's proof of a conjecture of P. Samuel that a Noetherian local ring that is a complete intersection — in particular, a hypersurface — that is factorial in codimension at most 3 must be factorial, [Gro68, Exp. XI, Cor. 3.14]. For a shorter and somewhat more elementary proof, see [CL94].)

Regular local rings are integrally closed

The Auslander-Buchsbaum Theorem 14.8.3 implies that regular local rings are integrally closed (by Proposition 6.4.6). We will prove this (without appealing to Serre's fact 14.8.1) in §**??**.

Regular local rings are Cohen-Macaulay

In §**??**, we will show that regular local rings are Cohen-Macaulay (a notion defined in Chapter **??**).

14.9 ⋆ Filtered rings and modules, and Artin-Rees Lemma

We conclude Chapter 14 by discussing the Artin-Rees Lemma 14.9.1, which was used to prove Proposition 14.5.1. The Artin-Rees Lemma generalizes the intuition behind Proposition 14.5.1, that any function that is analytically zero at a point actually vanishes in an open neighborhood of that point (paragraph after Proposition 14.5.1). Because we will use it later, and because it is useful to recognize it in other contexts, we discuss it in some detail.

Definition 14.9.1 (I-filtration, I-adic filtration, I-stable)

(1) Suppose I is an ideal of a ring A. A descending filtration of an A-module M

$$M = M_0 \supseteq M_1 \supseteq M_2 \supseteq \cdots \tag{14.16}$$

*is called an **I-filtration** if $I^d M_n \subseteq M_{n+d}$ for all $d, n \geq 0$.*

*(2) We call (14.16) is **I-adic filtration** if $M_k = I^k M$.*

*(3) We say an I-filtration is **I-stable** if for some s and all $d \geq 0$, $I^d M_s = M_{d+s}$.*

Example 14.12 The I-adic filtration is I-stable.

Definition 14.9.2 (Rees algebra)

Suppose I is an ideal of a ring A, and M is an A-module with I-filtration $M_\bullet$, (14.16).

*(1) Let $A(I)_\bullet$ be the graded ring $\bigoplus_{n\geq 0} I^n$ (with the convention $I^0 = A$). This is called the **Rees algebra** of the ideal I in A.*

(2) Define $M(I)_\bullet := \bigoplus_{n\geq 0} M_n$, it is naturally a graded module over $A(I)_\bullet$.

Proposition 14.9.1

If A is Noetherian, M is a finitely generated A-module, and (14.16) is an I-filtration, then $M(I)_\bullet$ is a finitely generated $A(I)_\bullet$-module if and only if the filtration (14.16) is I-stable.

Proof Note that $A(I)_\bullet$ is Noetherian (by Proposition 5.5.3, as A is Noetherian, and I is finitely generated A-module).

Assume first that $M(I)_\bullet$ is finitely generated over the Noetherian ring $A(I)_\bullet$, and hence is Noetherian. Consider the increasing chain of $A(I)_\bullet$-submodules whose k-th element L_k is

$$M \oplus M_1 \oplus M_2 \oplus \cdots \oplus M_k \oplus IM_k \oplus I^2 M_k \oplus \cdots$$

(which agrees with $M(I)_\bullet$ up until M_k, and then "I-stabilizes"). This chain must stabilize by Noetherianness. But $\bigcup_k L_k = M(I)_\bullet$, so for some $s \in \mathbb{Z}^{\geq 0}$, $L_s = M(I)_\bullet$, so $I^d M_s = M_{s+d}$ for all $d \geq 0$ — (14.16) is I-stable.

For the other direction, assume that $M_{d+s} = I^d M_s$ for fixed s and all $d \geq 0$. Then $M(I)_\bullet$ is generated over $A(I)_\bullet$ by $M \oplus M_1 \oplus \cdots \oplus M_s$. But each M_j is finitely generated, so $M(I)_\bullet$ is indeed a finitely generated $A(I)_\bullet$ by $M \oplus M_1 \oplus \cdots \oplus M_s$. But each M_j is finitely generated, so $M(I)_\bullet$ is indeed a finitely generated $A(I)_\bullet$-module. □

Theorem 14.9.1 (Artin-Ress Lemma)

Suppose A is a Noetherian ring, and (14.16) is an I-stable filtration of a finitely generated A-module M. Suppose that $L \subseteq M$ is a submodule, and let $L_n := L \cap M_n$. Then

$$L = L_0 \supseteq L_1 \supseteq L_2 \supseteq \cdots$$

is an I-stable filtration of L.

Proof Note that $L_\bullet$ is an I-filtration, as $IL_n \subseteq IL \cap IM_n \subseteq L \cap M_{n+1} = L_{n+1}$. Also $L(I)_\bullet$ is an $A(I)_\bullet$-submodule of the finitely generated $A(I)_\bullet$-module $M(I)_\bullet$, and hence finitely generated (as $A(I)_\bullet$ is Noetherian, see the proof in Proposition 14.9.1), by Proposition 14.9.1, we win. □

An important special case is the following.

Corollary 14.9.1

Suppose $I \subseteq A$ is an ideal of a Noetherian ring, and M is a finitely generated A-module, and L is a submodule. Then for some integer s, $I^d(L \cap I^s M) = L \cap I^{d+s} M$ for all $d \geq 0$.

Proof Applying the Artin-Rees Lemma 14.9.1 to the filtration $M_n = I^n M$. □

Remark 14.36 ***Warning:*** it need not be true that $I^d L = L \cap I^d M$ for all d. For a counterexample, let $A = M = k[x, y]$, $L = (x)$, $I = (x, y)$, then $IL = (x^2, xy)$ and $I \cap L = (x)$, they are clearly not the same.

Theorem 14.9.2 (Krull intersection theorem)

(1) Suppose I is an ideal of a Noetherian ring A, and M is a finitely generated A-module. There is some $a \equiv 1 \pmod{I}$ such that $a \bigcap_{j=1}^{\infty} I^j M = 0$.

(2) If A is a Noetherian integral domain or a Noetherian local ring, and I is a proper ideal, then $\bigcap_{j=1}^{\infty} I^j = 0$. In particular, if $(A, \mathfrak{m})$ is a Noetherian local ring, then $\bigcap_i \mathfrak{m}^i = 0$.

Proof ***For (1).*** Let $L = \bigcap_{j=1}^{\infty} I^j M$, then L is a submodule of A-module M. Since A is Noetherian and M is finitely generated A-module, M is Noetherian, so L is finitely generated A-module. We want to show that $IL = L$. Apply Artin-Rees Lemma 14.9.1 to the I-stable filtration $M_n := I^n M$, then I-filtration $L_n := L \cap M_n$ is an I-stable filtration of L. Hence there exists s such that $I^d(L \cap I^s M) = L \cap I^{d+s} M$ for all $d \geq 0$, note that $L \cap I^s M = L$ and $L \cap I^{d+s} M = L$, we obtain $I^d L = L$ for all $d \geq 0$, in particular, $IL = L$. Apply Nakayama's Lemma 9.2.3 to L, there is some $a \equiv 1 \pmod{I}$ such that $a \bigcap_{j=1}^{\infty} I^j M = 0$.

For (2). Apply part (1) to $M = A$, then there exists $a \equiv 1 \pmod{I}$ such that $a \bigcap_{j=1}^{\infty} I^j = 0$. If A is a Noetherian integral domain, since $a \neq 0$, $\bigcap_{j=1}^{\infty} I^j = 0$. If $(A, \mathfrak{m})$ is a Noetherian local ring, then $I \subseteq \mathfrak{m}$. Consider A/I, then the image of a in A/I is 1, so $a \notin \mathfrak{m}$, and thus a is a unit. Hence $\bigcap_{j=1}^{\infty} I^j = 0$. □

Proposition 14.9.2

If X is a locally Noetherian scheme, and f is a function on X that is analytically zero at a point $p \in X$ (paragraph after Proposition 14.5.1), then f vanishes in a (Zariski-)open neighborhood of p.

Proof Consider Noetherian local ring $(\mathscr{O}_{X,p}, \mathfrak{m}_p)$, since f is analytically at p, $f_p \in \bigcap_i \mathfrak{m}^i$ where f_p denotes the image of f in the stalk $\mathscr{O}_{X,p}$, by Krull intersection theorem 14.9.2, $f_p = 0$. Recall the definition of stalk (Definition 3.2.2), there is a (Zariski-)open neighborhood U of p such that $f|_U = 0$, as we desired. $\square$

Part V

Quasicoherent sheaves on schemes, and their uses

Chapter 15 More on quasicoherent and coherent sheaves

In Chapter 6, we discussed how, *algebraically*, quasicoherent and coherent sheaves generalized modules over a ring. In this chapter, we explore how, geometrically, quasicoherent and coherent sheaves generalize vector bundles. Informally, a vector bundle V on a geometric space X (such as a manifold) is a family of vector spaces continuously parametrized by points of X. In other words, for each point p of X, there is a vector space, and these vector spaces are glued into a space V so that, as p varies, the vector space above p varies continuously. Nontrivial examples to keep in mind are the tangent bundle to a differentiable manifold, and the Möbius strip over a circle (interpreted as a line bundle). We will make this somewhat more precise in §15.1, but if you have not seen this idea before, you shouldn't be concerned; the main thing we will use this notion for is motivation.

Definition 15.0.1 (Free sheaves)

*A **free sheaf** on a ringed space $(X, \mathscr{O}_X)$ is an $\mathscr{O}_X$-module isomorphic to $\mathscr{O}_X^{\oplus I}$, where the sum is over some index set I. The **rank** of the free sheaf is the cardinality of I.*

Remark 15.1 This agrees with its rank as an $\mathscr{O}$-module at a point; see Definition 5.3.9.

Definition 15.0.2 (Locally free sheaf)

Let $(X, \mathscr{O}_X)$ be a ringed space. Let $\mathscr{F}$ be a sheaf of $\mathscr{O}_X$-modules.

*(1) We say $\mathscr{F}$ is **locally free** if for every point $x \in X$ there exist a set I and an open neighborhood $x \in U \subseteq X$ such that $\mathscr{F}|_U$ is isomorphic to $\mathscr{O}_X|_U^{\oplus I} := \bigoplus_{i\in I} \mathscr{O}_X|_U$ as an $\mathscr{O}_X|_U$-module.*

*(2) We say $\mathscr{F}$ is **finite locally free** if we may choose the index sets I to be finite.*

*(3) We say $\mathscr{F}$ is **finite locally free of rank** r if we may choose the index sets I have cardinality r.*

This corresponds to the notion of a vector bundle (§15.1). Quasicoherent sheaves form a convenient abelian category containing the locally free sheaves (i.e., vector bundles) that is much smaller than the category of $\mathscr{O}$-modules. Quasicoherent sheaves generalize locally free sheaves in much the way that modules generalize free modules. Coherent sheaves are, roughly speaking, a finite rank version of quasicoherent sheaves, which form a well-behaved abelian category containing finite rank locally free sheaves (or, equivalently, finite rank vector bundles). Just as the notion of free modules leads us to the notion of modules in general, and finitely generated modules, the notion of free sheaves will lead us inevitably to the notion of quasicoherent sheaves and coherent sheaves. (There is a slight fib in comparing finitely generated modules to coherent sheaves; see §7.4.)

15.1 Vector bundles "=" locally free sheaves

15.1.1 Vector bundles on manifolds

We recall somewhat more precisely the notion of vector bundles on manifolds. Arithmetically minded readers shouldn't tune out: for example, fractional ideals of the ring of integers in a number field (Definition 11.7.4) turn out to be an example of a "line bundle on a smooth curve" (Proposition **??**).

Definition 15.1.1 (Vector bundle on a manifold)

A rank n vector bundle on a manifold M is a map $\pi : V \to M$ with the structure of an n-dimensional real vector space on $\pi^{-1}(p)$ for each point $p \in M$, such that for every $p \in M$, there is an open neighborhood U and a homeomorphism

$$\varphi : \pi^{-1}(U) \longrightarrow U \times \mathbb{R}^n$$

over U (so that the diagram

$$\begin{tikzcd} \pi^{-1}(U) \arrow[rr, "\varphi", "\sim"'] \arrow[dr, "\pi|_{\pi^{-1}(U)}"'] & & U \times \mathbb{R}^n \arrow[dl, "\textit{projection to first factor}"] \\ & U & \end{tikzcd} \tag{15.1}$$

*commutes) that is an isomorphism of vector spaces over each $q \in U$. An isomorphism (15.1) is called a **trivialization over** U. If $U = M$, we say that V is a **trivial bundle**.*
*We call n the **rank** of the vector bundle. A rank 1 vector bundle is called a **line bundle**.*

Remark 15.2 It can also be convenient to be agnostic about the rank of the vector bundle, so it can have different ranks on different connected components. It is also sometimes convenient to consider infinite-rank vector bundles.

Transition functions

Given trivializations over U_1 and U_2, over their intersection, the two trivializations must be related by an element T_{12} of GL_n with entires consisting of functions on $U_1 \cap U_2$. (There is an issue of convention as to whether T_{12} should be replaced by T_{12}^{-1}. We want this to be a matrix multiplying on the left, sending U_1-coordinates to U_2-coordinates. This translates to: $T_{ij} = \varphi_j \circ \varphi_i^{-1}$, where φ_i and φ_j are the trivializations for U_i and U_j respectively.) If $\{U_i\}$ is a cover of M, and we are given trivializations over each U_i, then the $\{T_{ij}\}$ must satisfy the **cocycle condition:**

$$T_{jk}|_{U_i \cap U_j \cap U_k} \circ T_{ij}|_{U_i \cap U_j \cap U_k} = T_{ik}|_{U_i \cap U_j \cap U_k}. \tag{15.2}$$

(This implies $T_{ij} = T_{ji}^{-1}$.) The data of the T_{ij} are called **transition functions** (or **transition matrices**) for the trivialization.

This is reversible: given the data of a cover $\{U_i\}$ and transition functions T_{ij}, we can recover the vector bundle (up to unique isomorphism) by "gluing together the various $U_i \times \mathbb{R}^n$ along $U_i \cap U_j$ using T_{ij}."

The sheaf of sections

Fix a rank n vector bundle $V \to M$. The sheaf of section $\mathscr{F}$ of V (Exercise 3.7) is an $\mathscr{O}_M$-module — given any open set U, we can multiply a section over U by a function on U and get another section.

Moreover, given a trivialization over U, the sections over U are naturally identified with n-tuples of functions of U.

$$\begin{tikzcd} U \times \mathbb{R}^n \arrow[d, shift right, "\pi"'] \\ U \arrow[u, shift right, "n\text{-tuple of functions}"'] \end{tikzcd}$$

Thus, given a trivialization, over each open set U_i, we have an isomorphism $\mathscr{F}|_{U_i} \xleftrightarrow{\sim} \mathscr{O}_{U_i}^{\oplus n}$. As mentioned in Definition 15.0.2, we say that such an $\mathscr{F}$ is a **locally free sheaf of rank** n.

Transition functions for the sheaf of sections

Suppose we have a vector bundle on M, along with a trivialization over an open cover U_i. Suppose we have a section of the vector bundle over M. (This discussion will apply with M replaces by any open subset.) Then over each U_i, the section corresponds to an n-tuple of functions over U_i, say $\vec{s}^{\,i}$.

Lemma 15.1.1

Suppose M is a manifold with open cover $\{U_i\}$, and $\pi : V \to M$ is a vector bundle over M. Let $\mathscr{F}$ be the sheaf of section of V. Over $U_i \cap U_j$, the vector-valued function $\vec{s}^{\,i}$ is related to $\vec{s}^{\,j}$ by the same transition functions: $T_{ij}\vec{s}^{\,i} = \vec{s}^{\,j}$.

Proof Consider the following diagram.

$$\begin{tikzcd}
\pi^{-1}(U_i \cap U_j) \arrow[rr, "\varphi_{ij}", "\sim"'] \arrow[dr, shift left, "\pi"] & & (U_i \cap U_j) \times \mathbb{R}^n \arrow[dl, "\mathrm{pr}_1"] \\
& U_i \cap U_j \arrow[ul, shift left, "s"] &
\end{tikzcd}$$

Let $p \in U_i \cap U_j$. Pick $s \in \mathscr{F}(M)$, then $\varphi_i(s(p)) = (p, \vec{s}^{\,i}(p))$ and $\varphi_j(s(p)) = (p, \vec{s}^{\,j}(p))$. Note that $T_{ij} = \varphi_j \circ \varphi_i^{-1}$, we have

$$(p, \vec{s}^{\,j}(p)) = \varphi_j(s(p)) = (\varphi_j \circ \varphi_i^{-1}) \circ \varphi_i(s(p)) = T_{ij}(p, \vec{s}^{\,i}(p)),$$

i.e., $T_{ij}\vec{s}^{\,i} = \vec{s}^{\,j}$. $\square$

Given a locally free sheaf with rank n, and a trivializing open neighborhood of $\mathscr{F}$ (an open cover $\{U_i\}$ such that over each U_i, $\mathscr{F}|_{U_i} \cong \mathscr{O}_{U_i}^{\oplus n}$ as $\mathscr{O}$-modules), we have transition functions $T_{ij} \in \mathrm{GL}_n(\mathscr{O}(U_i \cap U_j))$ satisfying the cocycle condition (15.2). Thus the data of a locally free sheaf of rang n is *equivalent* to the data of a vector bundle of rank n. This change of perspective is useful, and is similar to an earlier change of perspective when we introduced ringed spaces: understanding spaces is the same as understanding (sheaves of) functions on the spaces, and understanding vector bundles (a type of "space over M") is the same as understanding (sheaves of) sections.

Definition 15.1.2 (Invertible sheaf)

A rank 1 locally free sheaf is called an ***invertible sheaf***.

Remark 15.3 **Unimportant aside:** "Invertible sheaf" is a heinous term for something that is essentially a line bundle. The motivation is that if X is a locally ringed space, and $\mathscr{F}$ and $\mathscr{G}$ are $\mathscr{O}_X$-modules with $\mathscr{F} \otimes_{\mathscr{O}_X} \mathscr{G} \cong \mathscr{O}_X$, then $\mathscr{F}$ and $\mathscr{G}$ are invertible sheaves [bra]. Thus in the monoid of $\mathscr{O}_X$-modules under tensor product, invertible sheaves are the invertible elements. We will never use this fact. People often informally use the phrase "line bundle" when they mean "invertible sheaf". The phrase *line sheaf* has been proposed but has not caught on.

Note *Recall (Definition 7.2.3) that given a section s of $\mathscr{F}$ and a section t of $\mathscr{G}$, we have a section $s \otimes t$ of $\mathscr{F} \otimes \mathscr{G}$. If $\mathscr{F}$ is an invertible sheaf, this sections is often denoted "st".*

15.1.2 Locally free sheaves more generally

We can generalize the notion of locally free sheaves to schemes (or more generally, ringed spaces) without change.

Definition 15.1.3 (Locally free sheaf of rank n on a scheme)

*A **locally free sheaf of rank** n **on a scheme** X is defined as an $\mathscr{O}_X$-module $\mathscr{F}$ that is locally a free sheaf of rank n. Precisely, there is an open cover $\{U_i\}$ of X such that for each U_i, $\mathscr{F}|_{U_i} \cong \mathscr{O}_{U_i}^{\oplus n}$.*

Remark 15.4 This open cover (in Definition 15.1.3) determines transition functions — the data of a cover $\{U_i\}$ of X, and functions $T_{ij} \in \mathrm{GL}_n(\mathscr{O}(U_i \cap U_j))$ satisfying the cocycle condition (15.2) — which in turn determine the locally free sheaf. As before, given these data, we can find the sections over any open set U. Informally, for each i, they are

$$\vec{s}^{\,i} = \begin{pmatrix} s_1^i \\ \vdots \\ s_n^i \end{pmatrix} \in \Gamma(U \cap U_i, \mathscr{O}_X)^{\oplus n},$$

satisfying $T_{ij}\vec{s}^{\,i} = \vec{s}^{\,j}$ on $U \cap U_i \cap U_j$.

You should think of these as vector bundle, but just keep in mind that they are **not the "same"**, just equivalent notions. We will later define the "total space" of the vector bundle $V \to X$ (a scheme over X) in terms of the sheaf version of Spec. But the locally free sheaf perspective will prove to be more useful. As one example: The definition of a locally free sheaf is much shorter than that of a vector bundle.

As in our motivating discussion, it is sometimes convenient to let the rank vary among connected components, or to consider infinite rank locally free sheaves.

Pulling back vector bundles

Proposition 15.1.1 (Pulling back vector bundles)

Suppose $\pi : X \to Y$ is a morphism of ringed spaces. Recall the definition of $\pi^* : \mathbf{Mod}_{\mathscr{O}_Y} \to \mathbf{Mod}_{\mathscr{O}_X}$, pulling back $\mathscr{O}$-module (Definition 8.2.3). Suppose $n \in \mathbb{Z}^+$.

(a) There is a canonically $\pi^*(\mathscr{O}_Y^{\oplus n}) \cong \mathscr{O}_X^{\oplus n}$.

(b) If $\mathscr{G}$ is a locally free sheaf on Y of rank n, then $\pi^*\mathscr{G}$ is a locally free sheaf on X of rank n.

(c) If $\mathscr{G}$ is a locally free sheaf on Y of rank n, and $\{U_i\}$ are trivializing neighborhoods for $\mathscr{G}$ on Y, with transition matrices T_{ij}, then $\{\pi^{-1}(U_i)\}$ are trivializing neighborhoods for $\pi^*\mathscr{G}$ on X, with transition matrices π^*T_{ij}.

Proof For (a). In fact $\pi^*(\mathscr{O}_Y^{\oplus n}) = \pi^{-1}(\mathscr{O}_Y^{\oplus n}) \otimes_{\pi^{-1}\mathscr{O}_Y} \mathscr{O}_X$, since π^{-1} is exact functor (Proposition 3.7.4) and direct sum (in Abelian category, finite product = finite coproduct) is limit (exact functor commutes with limits and colimits), so we have $\pi^{-1}(\mathscr{O}_Y^{\oplus n}) \cong (\pi^{-1}\mathscr{O}_Y)^{\oplus n}$. Since $\square \otimes_{\pi^{-1}\mathscr{O}_Y} \mathscr{O}_X$ is right-exact, so commutes with the finite colimits, direct sum is finite colimits, hence

$$(\pi^{-1}\mathscr{O}_Y)^{\oplus n} \otimes_{\pi^{-1}\mathscr{O}_Y} \mathscr{O}_X \cong (\pi^{-1}\mathscr{O}_Y \otimes_{\pi^{-1}\mathscr{O}_Y} \mathscr{O}_X)^{\oplus n} \cong \mathscr{O}_X^{\oplus n},$$

i.e., $\pi^*(\mathscr{O}_Y^{\oplus n}) \cong \mathscr{O}_X^{\oplus n}$.

For (b). Since $\mathscr{G}$ is a locally free sheaf on Y of rank n, then there exists an open cover $\{U_i\}$ of Y such that $\mathscr{G}|_{U_i} \cong \mathscr{O}_{U_i}^{\oplus n}$. Consider $\pi^*\mathscr{G}|_{\pi^{-1}(U_i)}$. Clearly $\pi^*\mathscr{G}|_{\pi^{-1}(U_i)} \cong \pi|_{\pi^{-1}(U_i)}^*\mathscr{G}|_{U_i}$, then by part (a), we have

$$\pi|_{\pi^{-1}(U_i)}^*\mathscr{G}|_{U_i} \cong \pi|_{\pi^{-1}(U_i)}^*\mathscr{O}_{U_i}^{\oplus n} \cong \mathscr{O}_{\pi^{-1}(U_i)}^{\oplus n},$$

which implies that $\pi^*\mathscr{G}$ is a locally free sheaf on X of rank n.

For (c). By the discussion in part (b), $\{\pi^{-1}(U_i)\}$ are trivializing neighborhoods for $\mathscr{G}$ on Y. Say $\varphi_i : \mathscr{G}|_{U_i} \xrightarrow{\sim} \mathscr{O}_{U_i}^{\oplus n}$, then $T_{ij} = \varphi_j \circ \varphi_i^{-1}$. Since π^* is an exact functor (Proposition 3.7.4), we have an isomorphism $\pi^*\varphi_i : \pi^*\mathscr{G}|_{U_i} \xrightarrow{\sim} \mathscr{O}^{\oplus n}_{\pi^{-1}(U_i)}$, so $\pi^* T_{ij} = \pi^*(\varphi_j \circ \varphi_i^{-1}) = \pi^*(\varphi_j) \circ \pi^*(\varphi_i)^{-1}$. □

The problem with locally free sheaves (or, equivalently, vector bundles), as opposed to vector spaces.

Recall that $\mathscr{O}_X$-modules form an abelian category: we can talk about kernels, cokernels, and so forth, and we can do homological algebra. Similarly, vector spaces form an abelian category. But locally free sheaves (i.e., vector bundles), along with reasonably natural maps between them (those that arise as maps of $\mathscr{O}_X$-modules), don't form an abelian category. As a motivating example in the category of manifolds, consider the map of the trivial line bundle on $\mathbb{R}$ (with coordinate t) to itself, corresponding to multiplying by the coordinate t. Then this map jumps rank, and if you try to define a kernel or cokernel you will get confused.

This problem is resolved by enlarging our notion of nice $\mathscr{O}_X$-modules in a natural way, to quasicoherent sheaves.

$\mathscr{O}_X$-modules (abelian category)	$\supset$	quasicoherent sheaves (abelian category)	$\supset$	locally free sheaves (not an abelian category)

You can turn this into a definition of quasicoherent sheaves, equivalent to those we gave in §7.1. We want a notion that is local on X, of course. So we ask for the smallest abelian subcategory of $\mathbf{Mod}_{\mathscr{O}_X}$ that is "local" and includes vector bundles. It turns out that the main obstruction to vector bundles' being an abelian category is the failure of cokernels of maps of locally free sheaves — as $\mathscr{O}_X$-modules — to be locally free; we could define quasicoherent sheaves to be those $\mathscr{O}_X$-modules that are locally cokernels, yielding a description that works more generally on ringed spaces, as described in Definition 7.3.1. You may wish to later check that our definitions are equivalent to these.

Similarly, in the locally Noetherian setting, finite rank locally free sheaves will sit in a nice smaller abelian category, that of coherent sheaves.

quasicoherent sheaves (abelian category)	$\supset$	coherent sheaves (abelian category)	$\supset$	finite rank locally free sheaves (not an abelian category)

15.1.3 Useful constructions on locally free sheaves: $\mathscr{H}om$, dual, $\otimes$

We now give some useful constructions in the form of a series of lemmas about locally free sheaves on a scheme. They are useful, important, and surprisingly nontrivial!

Lemma 15.1.2

Suppose $\mathscr{F}$ and $\mathscr{G}$ are locally free sheaves on ringed space X of rank m and n respectively. Then $\mathscr{H}om_{\mathscr{O}_X}(\mathscr{F}, \mathscr{G})$ is a locally free sheaf of rank mn.

Proof Since $\mathscr{F}$ and $\mathscr{G}$ are locally free sheaves of rank m and rank n, there exists open cover $\{U_i\}$, $\{V_i\}$ of X such that for each U_i, V_i, $\mathscr{F}|_{U_i} \cong \mathscr{O}_{U_i}^{\oplus m}$ and $\mathscr{G}|_{V_i} \cong \mathscr{O}_{V_i}^{\oplus n}$. Setting $W_{ij} = U_i \cap V_j$, then $\{W_{ij}\}$ form a new open cover of X, moreover we have $\mathscr{F}|_{W_{ij}} \cong \mathscr{O}_{W_{ij}}^{\oplus m}$ and $\mathscr{G}|_{W_{ij}} \cong \mathscr{O}_{W_{ij}}^{\oplus n}$. Consider $\mathscr{H}om(\mathscr{F},\mathscr{G})(W_{ij})$,

$$\begin{aligned}\mathscr{H}om(\mathscr{F},\mathscr{G})(W_{ij}) &= \mathrm{Mor}(\mathscr{F}|_{W_{ij}},\mathscr{G}|_{W_{ij}})\\ &\cong \mathrm{Mor}(\mathscr{O}_{W_{ij}}^{\oplus m},\mathscr{O}_{W_{ij}}^{\oplus n})\\ &\cong \mathrm{Mor}(\mathscr{O}_{W_{ij}},\mathscr{O}_{W_{ij}}^{\oplus n})^{\oplus m}\\ &\cong \left(\mathrm{Mor}(\mathscr{O}_{W_{ij}},\mathscr{O}_{W_{ij}})^{\oplus n}\right)^{\oplus m}\\ &\cong (\mathscr{O}_{W_{ij}}^{\oplus n})^{\oplus m}\\ &\cong \mathscr{O}_{W_{ij}}^{\oplus mn}.\end{aligned}$$

Hence $\mathscr{H}om_{\mathscr{O}_X}(\mathscr{F},\mathscr{G})$ is a locally free sheaf of rank mn. □

Lemma 15.1.3 (Dual sheaves)

If $\mathscr{E}$ is a locally free sheaf on X of (finite) rank n, Lemma 15.1.2 implies that $\mathscr{E}^\vee := \mathscr{H}om(\mathscr{E},\mathscr{O}_X)$ is also a locally free sheaf of rank n. This is called the ***dual of*** *$\mathscr{E}$. Given transition functions for $\mathscr{E}$, say T_{ij}, then the transition functions for $\mathscr{E}^\vee$ is $(T_{ij}^t)^{-1}$. Moreover, $\mathscr{E} \cong \mathscr{E}^{\vee\vee}$.*

Remark 15.5 Note that if $\mathscr{E}$ is rank 1, i.e., invertible, the transition functions of the dual are the inverse of the transition functions of the original.

Proof Since $\mathscr{E}$ is a locally free sheaf one X of rank n, there exists an open cover $\{U_i\}$ of X such that $\varphi_i : \mathscr{E}|_{U_i} \xrightarrow{\sim} \mathscr{O}_{U_i}^{\oplus n}$. The transition functions for $\mathscr{E}$ is given by $T_{ij} = \varphi_j \circ \varphi_i^{-1}$. Consider $\mathscr{H}om(\varphi_i,\mathscr{O}_{U_i})$, we have $\varphi_i^* : \mathscr{H}om(\mathscr{O}_{U_i}^{\oplus n},\mathscr{O}_{U_i}) \cong \mathscr{O}_{U_i}^{\oplus n} \xrightarrow{\sim} \mathscr{E}|_{U_i}^\vee$, then the trivialization is $\psi_i := (\varphi_i^*)^{-1} : \mathscr{E}|_{U_i}^\vee \xrightarrow{\sim} \mathscr{O}_{U_i}^{\oplus n}$. Hence the transition functions for $\mathscr{E}^\vee$ is given by $S_{ij} = \psi_j \circ \psi_i^{-1}$. Now let's compute S_{ij}. We may assume $\mathscr{E}|_{U_i\cap U_j}$ is finitely generated by $e_1,\dots,e_n$ on piece U_i and $f_1,\dots,f_n$ on piece U_j, then we have

$$(e_1,\dots,e_n) = (f_1,\dots,f_n)T$$

where for simply denote $T := T_{ij} = (t_{ij})$. Suppose

$$(f^1,\dots,f^n) = (e^1,\dots,e^n)A,$$

where $A = (a_{ij})_{ij}$. Then we have $f^i = \sum\limits_{k=1}^n a_{ki}e^k$, hence $f^i(e_l) = \sum\limits_{k=1}^n a_{ki}\delta_l^k = a_{li}$. Since $e_l = \sum\limits_{p=1}^n t_{pl}f_p$, we have $f^i(e_l) = \sum\limits_{p=1}^n t_{pl}\delta_p^i = t_{il} = a_{li}$. Hence $A = T^t$, and therefore

$$(e^1,\dots,e^n) = (f^1,\dots,f^n)A^{-1} = (f^1,\dots,f^n)(T^t)^{-1},$$

i.e., $S_{ij} = (T_{ij}^t)^{-1}$.

We next show $\mathscr{E} \cong \mathscr{E}^{\vee\vee}$. First, we need to define a morphism of sheaves $\theta : \mathscr{E} \to \mathscr{E}^{\vee\vee}$. To do this, we must define, for every open set $U \subseteq X$, a map of $\mathscr{O}_X(U)$-modules $\theta_U : \mathscr{E}(U) \to \mathscr{E}^{\vee\vee}(U)$ such that these maps are compatible with restriction. Let analyze the target. An element of $\mathscr{E}^{\vee\vee}(U)$ is a morphism of sheaves on U, i.e., $\mathrm{Mor}_{\mathscr{O}_U}(\mathscr{E}^\vee|_U,\mathscr{O}_U)$. So, for any given section $s \in \mathscr{E}(U)$, we need to define a morphism $\theta_U(s) : \mathscr{E}^\vee|_U \to \mathscr{O}_U$. To define this morphism, we must specify how it acts on any section of $\mathscr{E}^\vee$ over any open subset $V \subseteq U$. Define $\theta_U(s)_V : \mathscr{E}^\vee(V) \to \mathscr{O}_U(V)$ by setting $\alpha \mapsto \alpha(s|_V)$. Combine our above discussion, define $\theta : \mathscr{E} \to \mathscr{E}^{\vee\vee}$ as follow: for any $U \subseteq X$ open subset, define

$$\begin{aligned}\theta_U : \mathscr{E}(U) &\longrightarrow \mathscr{E}^{\vee\vee}(U)\\ s &\longmapsto \theta_U(s),\end{aligned}$$

define $\theta_U(s) : \mathscr{E}^\vee|_U \to \mathscr{O}_U$ by setting for every $V \subseteq U$,

$$\begin{aligned}\theta_U(s)_V : \mathscr{E}^\vee(V) &\longrightarrow \mathscr{O}_U(V)\\ \alpha &\longmapsto \alpha(s|_V).\end{aligned}$$

Next, we want to show that θ is an isomorphism. It suffices to show that θ restricts to the open sets of some open cover of X are all isomorphisms. Since $\mathscr{E}$ is a locally free sheaf of rank n, there exists an open cover $\{U_i\}$ of X such that for each i, there is an isomorphism $\varphi_i : \mathscr{E}|_{U_i} \xrightarrow{\sim} \mathscr{O}_{U_i}^{\oplus n}$. Let's restrict our attention to one such open set $U = U_i$, we only need to show that $\theta|_U : \mathscr{E}|_U \to \mathscr{E}^{\vee\vee}|_U$ is an isomorphism. Using the trivialization φ_i, we have

$$\begin{aligned}\mathscr{E}|_U &\cong \mathscr{O}_U^{\otimes n},\\ \mathscr{E}^\vee|_U = \mathscr{H}om(\mathscr{E}|_U, \mathscr{O}_U) &\cong \mathscr{H}om(\mathscr{O}_U^{\otimes n}, \mathscr{O}_U) \cong \mathscr{O}_U^{\otimes n},\\ \mathscr{E}^{\vee\vee}|_U = \mathscr{H}om(\mathscr{E}^\vee|_U, \mathscr{O}_U) &\cong \mathscr{H}om(\mathscr{O}_U^{\otimes n}, \mathscr{O}_U) \cong \mathscr{O}_U^{\otimes n}.\end{aligned}$$

So, on the open set U, the morphism $\theta|_U$ corresponds to a morphism from a free sheaf to itself, i.e., $\theta|_U : \mathscr{O}_U^{\oplus n} \to \mathscr{O}_U^{\otimes n}$. Let's see what this morphism does in terms of a basis. Let $\{e_1, \dots, e_n\}$ be the standard basis of sections for $\mathscr{O}_U^{\otimes n}$ over any open subset $V \subseteq U$. Let $\{e^1, \dots, e^n\}$ be the corresponding dual basis for the dual sheaf $(\mathscr{O}_U^{\oplus n})^\vee$. Now, let's compute the image of a basis section e_k under the map θ_V. The image, $\theta_V(e_k)$, is an element of the double dual. Then for any open subset $W \subseteq V$, $\theta_V(e_k)_W(e^l|_W) = e^l|_W(e_k|_W) = \delta_k^l$. Hence $\{\theta_V(e_k)\}$ is the dual base of $\{e^k\}$ for all $V \subseteq U$, and therefore $\theta_V : \mathscr{E}(V) \to \mathscr{E}^{\vee\vee}(V)$ must be an isomorphism. So, we have $\mathscr{E} \cong \mathscr{E}^{\vee\vee}$. ☐

Remark 15.6 Caution: We will see an example in §16.1 of a locally free $\mathscr{F}$ that is not isomorphic to its dual: the invertible sheaf $\mathscr{O}(1)$ on $\mathbb{P}^n$.

Lemma 15.1.4

(1) If $\mathscr{F}$ and $\mathscr{G}$ are locally free sheaves, then $\mathscr{F} \otimes \mathscr{G}$ is a locally free sheaf. (Here $\otimes$ is tensor product as $\mathscr{O}_X$-modules, defined in Definition 3.6.1.)

(2) If $\mathscr{F}$ is an invertible sheaf, then $\mathscr{F} \otimes \mathscr{F}^\vee \cong \mathscr{O}_X$.

Proof For (1). Since $\mathscr{F}$ and $\mathscr{G}$ are locally free sheaves, there exists $\{U_i\}, \{V_j\}$ open covers of X such that $\mathscr{F}|_{U_i} \cong \mathscr{O}_{U_i}^{\oplus I}$ and $\mathscr{G}|_{V_j} \cong \mathscr{O}_{V_j}^{\oplus J}$ for some I, J. Note that $W_{ij} := U_i \cap V_j$ consists a open cover of X, so for simply, we may assume that $\{U_i\} = \{V_j\}$, so now we have $\mathscr{F}|_{U_i} \cong \mathscr{O}_{U_i}^{\oplus I}$ and $\mathscr{G}|_{U_i} \cong \mathscr{O}_{U_i}^{\oplus J}$ for some m, n. Claim that $(\mathscr{F} \otimes_{\mathscr{O}_X} \mathscr{G})|_{U_i} \cong \mathscr{O}_{U_i}^{\oplus(I\times J)}$. The tensor product sheaf $(\mathscr{F} \otimes_{\mathscr{O}_X} \mathscr{G})|_{U_i}$ is the sheafification of the presheaf $\mathscr{F}|_{U_i} \otimes_{\text{pre},\mathscr{O}_{U_i}} \mathscr{G}|_{U_i}$. Since $\mathscr{F}|_{U_i} \cong \mathscr{O}_{U_i}^{\oplus I}$ and $\mathscr{G}|_{U_i} \cong \mathscr{O}_{U_i}^{\oplus J}$, this presheaf isomorphic to the presheaf $\mathscr{K}$ on U_i defined by: for any $V \subseteq U_i$ open, $\mathscr{K}(V) := \mathscr{O}_{U_i}(V)^{\oplus I} \otimes_{\mathscr{O}_{U_i}(V)} \mathscr{O}_{U_i}(V)^{\oplus J} \cong \mathscr{O}_{U_i}(V)^{\oplus(I\times J)}$. The presheaf $\mathscr{K}$ is already a sheaf, so $(\mathscr{F} \otimes \mathscr{G})|_{U_i} \cong (\mathscr{F}|_{U_i} \otimes_{\text{pre},\mathscr{O}_{U_i}} \mathscr{G}|_{U_i})^{\text{sh}} \cong \mathscr{K}^{\text{sh}} \cong \mathscr{K}$. Hence $\mathscr{F} \otimes \mathscr{G}$ is a locally free sheaf.

For (2). Since $\mathscr{F}$ is invertible sheaf, $\mathscr{F}$ is locally free sheaf of rank 1. By Lemma 15.1.3, $\mathscr{F}^\vee$ is also locally free sheaf of rank 1. By part (1), we know that $\mathscr{F} \otimes \mathscr{F}^\vee$ is locally free sheaf of rank 1, hence there exists $\{U_i\}$, an open cover of X, such that $(\mathscr{F} \otimes \mathscr{F}^\vee)|_{U_i} \cong \mathscr{O}_{U_i}$. Define $\mathscr{F} \otimes \mathscr{F}^\vee \to \mathscr{O}_X$ as follow: for any open subset $U \subseteq X$, define $\theta_U : \mathscr{F}(U) \otimes_{\mathscr{O}_U} \mathscr{F}^\vee(U)$ by setting $s \otimes f \mapsto f(s)$, then we have a morphism of presheaves, $\theta : \mathscr{F} \otimes_{\text{pre}} \mathscr{G} \to \mathscr{O}_X$, by the universal property of sheafification θ factor through to a unique morphism $\theta^{\text{sh}} : \mathscr{F} \otimes \mathscr{G} \to \mathscr{O}_X$, this is a morphism of sheaves. We next show that θ^{sh} is an isomorphism. For any $p \in X$, p must belong to some U_i, θ^{sh} induces a morphism of stalks $\theta_p^{\text{sh}} : (\mathscr{F} \otimes_{\mathscr{O}_X} \mathscr{F}^\vee)_p \to \mathscr{O}_{X,p}$. By Proposition 3.6.11, $(\mathscr{F} \otimes_{\mathscr{O}_X} \mathscr{F}^\vee)_p \cong \mathscr{F}_p \otimes_{\mathscr{O}_{X,p}} \mathscr{F}_p^\vee$. Since $\mathscr{F}_p \cong \mathscr{O}_{X,p}$, say $\mathscr{F}_p$ is generated by s_p. Its dual stalk $\mathscr{F}_p^\vee$ is generated by f_p such that $f_p(s_p) = 1$. Now consider θ_p^{sh}, let $as_p \otimes bf_p \in \mathscr{F}_p \otimes \mathscr{F}_p^\vee$ where

$a, b \in \mathscr{O}_{X,p}$, then

$$\theta_p^{\mathrm{sh}}(as_p \otimes bf_p) = abf_p(s_p) = ab.$$

This is clearly an isomorphism. Hence $\mathscr{F} \otimes \mathscr{F}^\vee \cong \mathscr{O}_X$. □

Recall that tensor products tend to be only right-exact in general (Proposition 2.2.4), the following proposition tells us that tensoring by a locally free sheaf is exact.

Lemma 15.1.5 (Tensoring by a locally free sheaf is exact)

If $\mathscr{F}$ is a locally free sheaf, and $\mathscr{G}' \to \mathscr{G} \to \mathscr{G}''$ is an exact sequence of $\mathscr{O}_X$-modules, then so is $\mathscr{G}' \otimes \mathscr{F} \to \mathscr{G} \otimes \mathscr{F} \to \mathscr{G}'' \otimes \mathscr{F}$.

Proof It suffices to check exactness at the level of stalks. Pick $p \in X$, by Proposition 3.6.11, we have a sequence (not exact)

$$\mathscr{G}'_p \otimes_{\mathscr{O}_{X,p}} \mathscr{F}_p \longrightarrow \mathscr{G}_p \otimes_{\mathscr{O}_{X,p}} \mathscr{F}_p \longrightarrow \mathscr{G}''_p \otimes_{\mathscr{O}_{X,p}} \mathscr{F}_p. \tag{15.3}$$

Since $\mathscr{F}$ is locally free sheaf, there exists an open cover $\{U_i\}$ of X such that $\mathscr{F}|_{U_i} \cong \mathscr{O}_{U_i}^{\oplus I}$, then the sequence (15.3) change into

$$\mathscr{G}'_p \otimes_{\mathscr{O}_{X,p}} \mathscr{O}_{X,p}^{\oplus I} \longrightarrow \mathscr{G}_p \otimes_{\mathscr{O}_{X,p}} \mathscr{O}_{X,p}^{\oplus I} \longrightarrow \mathscr{G}''_p \otimes_{\mathscr{O}_{X,p}} \mathscr{O}_{X,p}^{\oplus I}.$$

Note that tensor products commutes with direct sum, we have $\mathscr{G}_p \otimes_{\mathscr{O}_{X,p}} \mathscr{O}_{X,p}^{\oplus I} \cong (\mathscr{G}_p \otimes_{\mathscr{O}_{X,p}} \mathscr{O}_{X,p})^{\oplus I} \cong \mathscr{G}_p^{\oplus I}$, similarly, we have $\mathscr{G}'_p \otimes_{\mathscr{O}_{X,p}} \mathscr{O}_{X,p}^{\oplus I} \cong \mathscr{G}'^{\oplus I}_p$ and $\mathscr{G}''_p \otimes_{\mathscr{O}_{X,p}} \mathscr{O}_{X,p}^{\oplus I} \cong \mathscr{G}''^{\oplus I}_p$. Since $\mathscr{G}' \to \mathscr{G} \to \mathscr{G}''$ is exact, $\mathscr{G}'^{\oplus I}_p \to \mathscr{G}_p^{\oplus I} \to \mathscr{G}''^{\oplus I}_p$ must be exact, and therefore sequence (15.3) is exact, as we desired. □

Remark 15.7 Proposition 15.1.5 implies that locally free sheaves are flat.

Lemma 15.1.6

Let $(X, \mathscr{O}_X)$ be a ringed space. Let $\mathscr{F}, \mathscr{G}, \mathscr{H}$ be $\mathscr{O}_X$-modules. There is a canonical isomorphism

$$\mathscr{H}om_{\mathscr{O}_X}(\mathscr{F} \otimes_{\mathscr{O}_X} \mathscr{G}, \mathscr{H}) \xrightarrow{\sim} \mathscr{H}om_{\mathscr{O}_X}(\mathscr{F}, \mathscr{H}om_{\mathscr{O}_X}(\mathscr{G}, \mathscr{H}))$$

which is functorial in all three entries (sheaf $\mathscr{H}om$ in all three spots). In particular, to give a morphism $\mathscr{F} \otimes_{\mathscr{O}_X} \mathscr{G} \to \mathscr{H}$ is the same as giving a morphism $\mathscr{F} \to \mathscr{H}om_{\mathscr{O}_X}(\mathscr{G}, \mathscr{H})$.

Proof This is the analogue of Algebra, Lemma 2.4.1. The proof is the same and is omitted. □

Lemma 15.1.7

If $\mathscr{E}$ is a locally free sheaf of finite rank, and $\mathscr{F}$ and $\mathscr{G}$ are $\mathscr{O}_X$-modules, then

$$\mathscr{H}om(\mathscr{F}, \mathscr{G} \otimes \mathscr{E}) \cong \mathscr{H}om(\mathscr{F} \otimes \mathscr{E}^\vee, \mathscr{G}) \cong \mathscr{H}om(\mathscr{F}, \mathscr{G}) \otimes \mathscr{E}.$$

In particular, if $\mathscr{D}$ is a locally free sheaf of finite rank, $\mathscr{H}om(\mathscr{D}, \mathscr{G}) \cong \mathscr{D}^\vee \otimes \mathscr{G}$.

Proof We first show the case where $\mathscr{E}$ is free, i.e., $\mathscr{E} \cong \mathscr{O}_X^{\oplus n}$ for some n. Then we have

$$\begin{aligned}\mathscr{H}om(\mathscr{F}, \mathscr{G} \otimes_{\mathscr{O}_X} \mathscr{O}_X^{\oplus n}) &\cong \mathscr{H}om(\mathscr{F}, (\mathscr{G} \otimes_{\mathscr{O}_X} \mathscr{O}_X)^{\oplus n}) \\ &\cong \mathscr{H}om(\mathscr{F}, \mathscr{G}^{\oplus n}) \\ &\cong \mathscr{H}om(\mathscr{F}, \mathscr{G})^{\oplus n}\end{aligned}$$

and, by $\mathscr{H}om(\mathscr{O}_X, \mathscr{O}_X) \cong \mathscr{O}_X$,

$$\begin{aligned}\mathscr{H}om(\mathscr{F} \otimes \mathscr{E}^\vee, \mathscr{G}) &\cong \mathscr{H}om(\mathscr{F} \otimes \mathscr{O}_X^{\oplus n}, \mathscr{G}) \\ &\cong \mathscr{H}om(\mathscr{F}^{\oplus n}, \mathscr{G}) \\ &\cong \mathscr{H}om(\mathscr{F}, \mathscr{G})^{\oplus n}.\end{aligned}$$

We also have

$$\mathscr{H}om(\mathscr{F}, \mathscr{G}) \otimes \mathscr{E} \cong \mathscr{H}om(\mathscr{F}, \mathscr{G}) \otimes \mathscr{O}_X^{\oplus n} \cong \mathscr{H}om(\mathscr{F}, \mathscr{G})^{\oplus n}.$$

Hence

$$\mathscr{H}om(\mathscr{F}, \mathscr{G} \otimes \mathscr{E}) \cong \mathscr{H}om(\mathscr{F} \otimes \mathscr{E}^\vee, \mathscr{G}) \cong \mathscr{H}om(\mathscr{F}, \mathscr{G}) \otimes \mathscr{E}.$$

For general, if $\mathscr{E}$ is locally free sheaf of finite rank, there exists an open cover $\{U_i\}$ of X such that $\mathscr{E}|_{U_i} \cong \mathscr{O}_{U_i}^{\oplus n}$ for some n. Note that

$$\begin{aligned}\mathscr{H}om(\mathscr{F}, \mathscr{G} \otimes \mathscr{E})|_{U_i} &\cong \mathscr{H}om(\mathscr{F}|_{U_i}, \mathscr{G}|_{U_i} \otimes \mathscr{E}|_{U_i}) \cong \mathscr{H}om(\mathscr{F}|_{U_i}, \mathscr{G}|_{U_i} \otimes \mathscr{O}_{U_i}^{\oplus n}), \\ \mathscr{H}om(\mathscr{F} \otimes \mathscr{E}^\vee, \mathscr{G})|_{U_i} &\cong \mathscr{H}om(\mathscr{F}, \mathscr{H}om(\mathscr{E}^\vee, \mathscr{G}))|_{U_i} \\ &\cong \mathscr{H}om(\mathscr{F}|_{U_i}, \mathscr{H}om(\mathscr{E}^\vee|_{U_i}, \mathscr{G}|_{U_i})) \\ &\cong \mathscr{H}om(\mathscr{F}|_{U_i}, \mathscr{H}om(\mathscr{O}_{U_i}^{\oplus n}, \mathscr{G}|_{U_i})),\end{aligned}$$

and

$$(\mathscr{H}om(\mathscr{F}, \mathscr{G}) \otimes \mathscr{E})|_{U_i} \cong \mathscr{H}om(\mathscr{F}|_{U_i}, \mathscr{G}|_{U_i}) \otimes \mathscr{E}|_{U_i},$$

these are our special case which we have proved above, so

$$\mathscr{H}om(\mathscr{F}, \mathscr{G} \otimes \mathscr{E})|_{U_i} \cong \mathscr{H}om(\mathscr{F} \otimes \mathscr{E}^\vee, \mathscr{G})|_{U_i} \cong (\mathscr{H}om(\mathscr{F}, \mathscr{G}) \otimes \mathscr{E})|_{U_i},$$

moreover, on the overlap $U_i \cap U_j$ they are also isomorphism, so we have

$$\mathscr{H}om(\mathscr{F}, \mathscr{G} \otimes \mathscr{E}) \cong \mathscr{H}om(\mathscr{F} \otimes \mathscr{E}^\vee, \mathscr{G}) \cong \mathscr{H}om(\mathscr{F}, \mathscr{G}) \otimes \mathscr{E}.$$

In particular. Clearly, $\mathscr{H}om(\mathscr{O}_X, \mathscr{F}) \cong \mathscr{F}$ for any sheaf $\mathscr{F}$, recall Lemma 15.1.3, we have

$$\mathscr{H}om(\mathscr{D}, \mathscr{G}) \cong \mathscr{H}om(\mathscr{O}_X \otimes \mathscr{D}^{\vee\vee}, \mathscr{G}) \cong \mathscr{H}om(\mathscr{O}_X, \mathscr{G} \otimes \mathscr{D}^\vee) \cong \mathscr{G} \otimes \mathscr{D}^\vee.$$

□

15.1.4 The Picard group

Definition 15.1.4 (Picard group)

The invertible sheaves on X, up to isomorphism, form an abelian group under tensor product. This is called the ***Picard group*** *of X, and is denoted* $\operatorname{Pic} X$.

Proof The group operation is tensor product of invertible sheaves, and tensor product commutes. Tensor products hold associativity. Identity element is the structure sheaf $\mathscr{O}_X$. By Lemma 15.1.3, tensor product of invertible sheaves is also invertible sheaves. By Lemma 15.1.4, each invertible sheaf $\mathscr{F}$ has inverse, it is its dual sheaf $\mathscr{F}^\vee$. So the invertible sheaves on X, up to isomorphism, form an abelian group under tensor product. □

Proposition 15.1.2

Pic *is a contravariant functor from the category of ringed spaces to the category* $\mathbf{Ab}$ *of abelian groups.*

Proof Let $\pi : X \to Y$ be a morphism of ringed spaces. Define $\operatorname{Pic} \pi := \pi^*$, by Proposition 15.1.1, the invertible sheaves on Y pullback to a invertible sheaves on X. Recall §8.2.2, it is easy to verify that Pic is a functor. □

✍ **Exercise 15.1 (⋆ For those with sufficient complex-analytic background)** Recall the analytification functor, that takes a complex finite type reduced scheme and produces a complex analytic space.

(a) If $\mathscr{L}$ is an invertible sheaf on a complex (algebraic) variety X, define (up to unique isomorphism) the corresponding invertible sheaf on the complex variety X_{an}.

(b) Show that the induced map $\operatorname{Pic} X \to \operatorname{Pic} X_{\text{an}}$ is a group homomorphism.

(c) Show that this construction is functorial, i.e., if $\pi : X \to Y$ is a morphism of complex varieties, the following diagram commutes:

$$\begin{tikzcd} \operatorname{Pic} Y \arrow[r, "\pi^*"] \arrow[d] & \operatorname{Pic} X \arrow[d] \\ \operatorname{Pic} Y_{\text{an}} \arrow[r, "\pi^*_{\text{an}}"] & \operatorname{Pic} X_{\text{an}}, \end{tikzcd}$$

where the vertical maps are the ones you have defined.

✍ **Exercise 15.2 (⋆ For those with sufficient arithmetic background)** Recall the definition of the ring of integers $\mathscr{O}_K$ in a number field K, Definition 11.7.4. A **fractional ideal** $\mathfrak{a}$ of $\mathscr{O}_K$ is a nonzero $\mathscr{O}_K$-submodule of K such that there is a nonzero $a \in \mathscr{O}_K$ such that $a\mathfrak{a} \subseteq \mathscr{O}_K$. Products of fractional ideals are defined analogously to products of ideals in a ring (defined in remark of Proposition 4.4.2): $\mathfrak{ab}$ consists of (finite) $\mathscr{O}_K$-linear combinations of products of elements of $\mathfrak{a}$ and elements of $\mathfrak{b}$. Thus fractional ideals form an abelian semigroup under multiplication, with $\mathscr{O}_K$ as the identity. In fact, fractional ideals of $\mathscr{O}_K$ form a group.

(a) Explain how a fractional ideal on a ring of integers in a number field yields an invertible sheaf. (Although we won't need this, it is worth noting that a fractional ideal is the same as an invertible sheaf with a trivialization at the generic point.)

(b) A fractional ideal is **principal** if it is of the form $r\mathscr{O}_K$ for some $r \in K^\times$. Show that any two fractional ideals that differ by a principal fractional ideal yield the same invertible sheaf.

(c) Show that two fractional ideals that yield the same invertible sheaf differ by a principal ideal.

(d) The **class group** is defined to be the group of fractional ideals modulo the principal ideals (i.e., modulo $K^\times$). Given an isomorphism of the class group with the Picard group of $\mathscr{O}_K$.

(This discussion applies to any Dedekind domain.)

15.2 Locally free sheaves on schemes in particular

We now discuss aspects of locally free sheaves more specific to schemes. Based on your intuition for line bundles on manifolds, you might hope that every point has a "small" open neighborhood on which all invertible sheaves (or locally free sheaves) are trivial. Sadly, this is not the case. We will eventually see (Chapter 20) that for the curve $y^2 - x^3 - x = 0$ in $\mathbb{A}^2_{\mathbb{C}}$, every nonempty open set has nontrivial invertible sheaves. (This will use the fact that it is an open subset of an elliptic curve.) This is a persistent problem in algebraic geometry: the Zariski topology is too blunt.

15.2.1 Basic

Proposition 15.2.1

(1) The locally free sheaves on a scheme are quasicoherent.

(2) Finite rank locally free sheaves are always finite type (Definition 7.4.2).

(3) If $\mathscr{O}_X$ is coherent, then finite rank locally free sheaves on X are coherent.

Proof For (1). Let $\mathscr{F}$ is a locally free sheaf on a scheme X, by definition, there exists an open cover $\{U_i\}_i$ of X such that for each U_i, $\mathscr{F}|_{U_i} \cong \mathscr{O}_{U_i}^{\oplus I}$. Since X is a scheme, we may assume that $X = \bigcup_j \operatorname{Spec} A_j$. Consider $\{U_i \cap \operatorname{Spec} A_j\}_j$, this is an open cover of U_i, since each $U_i \cap \operatorname{Spec} A_j$ is an open subset of $\operatorname{Spec} A_j$, it must covered by some distinguished affine open subsets, i.e., $U_i \cap \operatorname{Spec} A_j = \bigcup_k D_{ij}(f_{ijk})$ where $D_{ij}(f_{ijk}) = \operatorname{Spec}(A_j)_{f_{ijk}}$, hence $U_i = \bigcup_j \bigcup_k D_{ij}(f_{ijk})$. Since $\mathscr{F}|_{U_i} \cong \mathscr{O}_{U_i}^{\oplus I}$, we have $\mathscr{F}|_{D_{ij}(f_{ijk})} \cong \mathscr{O}_{D_{ij}(f_{ijk})}^{\oplus I}$. Note that $\{D_{ij}(f_{ijk})\}$ is an affine open cover of X, by Theorem 7.1.1, $\mathscr{F}$ is a quasicoherent sheaf.

For (2). Since $\mathscr{F}$ is a finite rank locally free sheaf, recall (1), there exists an affine open cover U_i such that $\mathscr{F}|_{U_i} \cong \mathscr{O}_{U_i}^{\oplus n}$, hence $\Gamma(U_i, \mathscr{F}|_{U_i}) \cong \Gamma(U_i, \mathscr{O}_{U_i})^{\oplus n}$ is a finite generated $\Gamma(U_i, \mathscr{O}_{U_i})$-module, by definition, $\mathscr{F}$ is finite type.

For (3). If $\mathscr{O}_X$ is coherent. Since direct sum of coherent sheaves is also coherent, we obtain that $\mathscr{F}$ is a coherent sheaf. □

Remark 15.8 Remember: If $\mathscr{O}_X$ is not coherent, then coherence is a pretty unless notion on X.

Example 15.1 (Not every quasicoherent sheaf is locally free) Suppose $X = \operatorname{Spec} k[t]$ where $k = \overline{k}$. Let $\mathscr{F}$ be the skyscraper sheaf supported at the generic point $[(0)]$, with group $k(t)$ and the usual $k[t]$-module structure. Recall Example 7.2 , $\mathscr{F} \cong \widetilde{k(t)}$ is a quasicoherent sheaf. We next show that $\mathscr{F}$ is not locally free. For any open subset U of X, by Proposition 4.6.10, $[(0)] \in U$, so $\mathscr{F}|_U \cong \widetilde{k(t)}$. Consider $(\mathscr{F}|_U)_p$. If $p \in U$ is not generic point, $\mathscr{O}_{X,p} \cong k[t]_p \ncong k(t) \cong (\mathscr{F}|_U)_p$. If $p = [(0)]$, then $\mathscr{O}_{X,p} \cong k[t]_{(0)} \cong k(t) \cong (\mathscr{F}|_U)_p$. Hence $\mathscr{F}|_U \ncong \mathscr{O}_U$, and therefore $\mathscr{F}$ is not locally free.

Lemma 15.2.1 (Structure theorem for finitely generated modules over a PID)

Let R be a PID and let M be a finitely generated R-module (generated by n elements).

(1) $M \cong R^{n-m} \oplus (R/(a_1)) \oplus (R/(a_2)) \oplus \cdots \oplus (R/(a_m))$ where nonzero elements $a_1, a_2, \ldots, a_m$ of R which are not units in R satisfy $(a_1) \supseteq (a_2) \supseteq \cdots \supseteq (a_m)$.

(2) M is torsion free if and only if M is free.

(3) In the decomposition in (1), $\operatorname{Tor} M \cong (R/(a_1)) \oplus (R/(a_2)) \oplus \cdots \oplus (R/(a_m))$.

Proof See [Wik25]. □

Proposition 15.2.2

Every rank n vector bundle on $\mathbb{A}_k^1$ is trivial of rank n.

Proof Let $\mathscr{F}$ be a rank n locally free sheaf on $\mathbb{A}_k^1 = \operatorname{Spec} k[t]$, we want to show that $\mathscr{F} \cong \mathscr{O}_X^{\oplus n}$. Since $\mathscr{F}$ is a quasicoherent sheaf (Proposition 15.2.1) and X is an affine scheme, we may assume that $\mathscr{F} \cong \widetilde{M}$ where M is a $k[t]$-module. It suffices to show that $M \cong k[t]^{\oplus n}$.

By Proposition 15.2.1, we know that $\mathscr{F}$ is finite type, recall Definition 7.4.2, we know that M is a finitely generated $k[t]$-module. By Lemma 15.2.1, we know that $M \cong k[t]^{\oplus r} \oplus \operatorname{Tor}(M)$ for some r. We claim that $\operatorname{Tor}(M) = 0$. Let $[\mathfrak{p}] \in \operatorname{Spec} k[t]$, Since $\mathscr{F} \cong \widetilde{M}$, we have $\mathscr{F}_{[\mathfrak{p}]} \cong M_{\mathfrak{p}}$, since $\mathscr{F}$ is also locally free, we have $M_{\mathfrak{p}} \cong \mathscr{O}_{X,\mathfrak{p}}^{\oplus n} \cong k[t]_{\mathfrak{p}}^{\oplus n}$, hence $\operatorname{Tor}(M_{\mathfrak{p}}) = 0$ for all $[\mathfrak{p}] \in \operatorname{Spec} k[t]$. Let $m \in \operatorname{Tor}(M)$, then there exists a nonzero $f \in k[t]$ such that $fm = 0$. Hence for any $\mathfrak{p} \in \operatorname{Spec} k[t]$, $fm/1 = 0$, where $fm/1$ is the image of fm in $M_{\mathfrak{p}}$. Note that $f/1 \in k[t]_{\mathfrak{p}}$, so $m/1 \in \operatorname{Tor}(M_{\mathfrak{p}})$, and therefore $m/1 = 0$ for any $\mathfrak{p} \in \operatorname{Spec} k[t]$. By [AM18, Proposition 3.8], $m = 0$, so $\operatorname{Tor}(M) = 0$. It follows that $M \cong k[t]^{\oplus n}$. $\square$

✍ **Exercise 15.3** Suppose s is a section of a rank n locally free sheaf $\mathscr{F}$ on a scheme X. Define the notion of the **closed subscheme cut out by** $s = 0$, denoted $V(s)$.

Solution *Since $\mathscr{F}$ is a locally free sheaf, we can find an open cover $\{U_i\}$ of X such that $\mathscr{F}|_{U_i} \cong \mathscr{O}_{U_i}^{\oplus n}$. Then we have a trivialization over each open subset U_i, named*

$$\varphi_i : \mathscr{F}|_{U_i} \xrightarrow{\sim} \mathscr{O}_{U_i}^{\oplus n}.$$

Now we restrict s to U_i, i.e., $s|_{U_i} \in \mathscr{F}(U_i)$, then

$$\varphi_{i,U_i}(s|_{U_i}) = (f_{i1}, \ldots, f_{in}),$$

where each $f_{ij} \in \mathscr{O}_X(U_i)$. On this open set U_i, the condition $s = 0$ is equivalent to

$$f_{i1} = 0, f_{i2} = 0, \ldots, f_{in} = 0.$$

Suppose $U_i = \bigcup_k \operatorname{Spec} A_{ik} = \bigcup_k V_{ik}$. Define the ideal sheaf $\mathscr{I}_i$ by setting

$$\mathscr{I}_i(V_{ik}) = (f_{i1}|_{V_{ik}}, f_{i2}|_{V_{ik}}, \ldots, f_{in}|_{V_{ik}}) \subseteq \mathscr{O}_{U_i}(V_{ik}) = A_{ik}.$$

By Proposition 10.1.6, this ideal sheaf $\mathscr{I}_i$ defines a closed subscheme Z_i of U_i.

Our definition of Z_i depended on the choice of the trivialization φ_i. We must show that if we choose a different trivialization, we get the same ideal sheaf and thus the same subscheme. Let $\psi_i : \mathscr{F}|_{U_i} \to \mathscr{O}_{U_i}^{\oplus n}$ be another trivialization. The map $\psi_i \circ \varphi_i^{-1}$ is an automorphism of $\mathscr{O}_{U_i}^{\oplus n}$. Any such automorphism is given by multiplication by an invertible $n \times n$ matrix A, whose entries are functions in $\mathscr{O}_X(U_i)$. Say $A = (a_{jk})$. Let $\psi(s|_{U_i}) = (g_{i1}, \ldots, g_{in})$ be the new coordinate functions. Then we have

$$(g_{i1}, \ldots, g_{in})^t = \psi_i \circ \varphi_i^{-1}((f_{i1}, \ldots, f_{in})^t) = A(f_{i1}, \ldots, f_{in})^t,$$

hence $g_{ij} = \sum_k a_{jk} f_{ik}$. This show that each g_{ij} is in the ideal generated by f_{ij}, so $(g_{i1}, \ldots, g_{in}) \subseteq (f_{i1}, \ldots, f_{fn})$. Conversely, since A is invertible, we have

$$(f_{i1}, \ldots, f_{in})^t = A^{-1}(g_{i1}, \ldots, g_{in})^t,$$

hence each f_{ij} can be written as g_{ij}, so we have $(f_{i1}, \ldots, f_{in}) \subseteq (g_{i1}, \ldots, g_{in})$. Hence the definition of ideal sheaf $\mathscr{I}_i$ (and the subscheme Z_i) is independent of the choice of trivialization on U_i.

Lastly, we glue Z_i to a closed subscheme of X. Let $Z_{ij} := Z_i \cap U_j$ be an open subscheme of Z_i, note that Z_{ij} is also a closed subscheme of U_i which is defined by ideal sheaf $\mathscr{I}_i|_{U_i \cap U_j}$. Similar to our previous discussion, we have $\mathscr{I}_i|_{U_i \cap U_j} = \mathscr{I}_j|_{U_i \cap U_j}$, so there is an isomorphism $\psi_{ij} : Z_{ij} \xrightarrow{\sim} Z_{ji}$. It is easy to check ψ_{ij} holds the cocycle condition, by Theorem 5.4.1, they glue together, then we obtain a closed subscheme of X, denoted $V(s)$.

We next give a definition that avoid coordinates:

Solution *Let $s \in \Gamma(X, \mathscr{F})$, define $\widetilde{s} : \mathscr{O}_X \to \mathscr{F}$ as follow: for any open subset U of X, define $\widetilde{s}_U : \mathscr{O}_X(U) \to \mathscr{F}(U)$ by setting $\widetilde{s}_U(f) = f \cdot s|_U$. $\widetilde{s}$ is clearly an $\mathscr{O}_X$-module morphism. Then we get a dual morphism $\widetilde{s}^\vee : \mathscr{F}^\vee \to \mathscr{O}_X^\vee \cong \mathscr{O}_X$. Consider the image of $\widetilde{s}^\vee$, $\operatorname{Im}(\widetilde{s}^\vee)$ is a subsheaf of $\mathscr{O}_X$ and it is a quasicoherent sheaf, by Proposition 10.1.6, this gives a closed subscheme on X.*

We next show this closed subscheme is same as our previous defined. Consider the trivialization on U, i.e., $\varphi : \mathscr{F}|_U \xrightarrow{\sim} \mathscr{O}_U^{\oplus n}$. Restrict s to U, we have $s|_U \in \mathscr{F}(U)$, say $\varphi_U(s|_U) = (f_1, \ldots, f_n)^t$ where $f_i \in \mathscr{O}_X(U)$. Let $e_1 = (1, 0, \ldots, 0)^t, \ldots, e_n = (0, 0, \ldots, 1)^t$, then $\varphi_U(s|_U) = \sum\limits_{i=1}^n f_i e_i$, hence $\varphi_U \circ \widetilde{s}_U(h) = \sum\limits_{i=1}^n h f_i e_i$ where $h \in \mathscr{O}_X(U)$. Take the dual of $\varphi_U \circ \widetilde{s}_U$, we obtain $\widetilde{s}_U^\vee \circ \varphi_U^\vee : (\mathscr{O}_U(U)^{\oplus n})^\vee \to \mathscr{O}_U(U)^\vee$ which given by

$\widetilde{s}_U^\vee \circ \varphi_U^\vee(g) = g \circ \varphi_U \circ \widetilde{s}_U$ *where* $g \in (\mathscr{O}_U(U)^{\oplus n})^\vee$. *Since* $g \circ \varphi_U \circ \widetilde{s}_U \in \mathscr{O}_U(U)^\vee$, $g \circ \varphi_U \circ \widetilde{s}_U$ *is determined by* $g \circ \varphi_U \circ \widetilde{s}_U(1) = g((f_1, \dots, f_n)^t) = \sum_{i=1}^n f_i g(e_i) \in \mathscr{O}_U(U)$. *Since* $\varphi_U^\vee$ *is an isomorphism, the image of* $\widetilde{s}_U^\vee$ *is genenrated by* $f_1, \dots, f_n$ *as an* $\mathscr{O}_U(U)$*-module, i.e., an ideal of* $\mathscr{O}_U(U)$. *So* $\operatorname{Im}(\widetilde{s}) = \mathscr{I}$*, i.e., agree with our previous definition.*

Definition 15.2.1 (Closed subscheme cut out by a section of a locally free sheaf on a scheme)

Suppose s *is a section of a rank* n *locally sheaf* $\mathscr{F}$ *on a scheme* X. *We define the closed subscheme cut out by* $s = 0$ *as Exercise 15.3, denoted* $V(s)$.

Proposition 15.2.3 (Hartogs for locally free sheaves on locally Noetherian normal scheme)

Locally free sheaves on locally Noetherian normal schemes satisfy "Algebraic Hartogs's Lemma": sections defined away from a set of codimension at least 2 extend over that set. More precisely, let X *be a locally Noetherian normal scheme,* $U \subseteq X$ *is an open subset. Let* Z *be a closed subset of* U *with* $\operatorname{codim}_U Z \geq 2$. *If* $\mathscr{F}$ *is a locally free sheaf on* X*, then the restriction map*

$$\operatorname{res}_{U,U\setminus Z} : \Gamma(U, \mathscr{F}) \xrightarrow{\sim} \Gamma(U \setminus Z, \mathscr{F})$$

is an isomorphism.

Proof Cover U by affine open subsets $W_i = \operatorname{Spec} A_i$ such that $\mathscr{F}|_{W_i} \cong \mathscr{O}_{W_i}^{\oplus I}$ where I is an index set and A_i is integrally closed Noetherian ring. Since W_i is Noetherian normal scheme, by Proposition 6.4.2, W_i is the finite union of integral Noetherian normal scheme, so we may shrink each W_i such that W_i is an integral Noetherian normal scheme, then each A_i is integrally closed Noetherian integral domain.

Let $f \in \Gamma(U \setminus Z, \mathscr{F}|_{U\setminus Z})$. Since $\operatorname{codim}_U Z \geq 2$, $U \setminus Z$ contains all codimension 1 points of U. Note that $U \setminus Z = \bigcup_i (W_i \setminus Z)$, we restrict f to each $V_i := W_i \setminus Z$, then

$$f_i := f|_{V_i} \in \Gamma(V_i, \mathscr{F}|_{V_i}) \cong \Gamma(V_i, \mathscr{O}_{V_i}^{\oplus I}) = \mathscr{O}_{V_i}(V_i)^{\oplus I}.$$

Since $U \setminus Z$ contains all codimension 1 points of U, V_i contains all codimension 1 points of W_i, by Algebraic Hartogs's Lemma 14.5.8, there exists $\widetilde{f}_i \in \Gamma(W_i, \mathscr{O}_{W_i})^{\oplus I} = \Gamma(W_i, \mathscr{O}_{W_i}^{\oplus I})$ such that $\widetilde{f}_i|_{V_i} = f_i$.

Now, we want to check that $\widetilde{f}_i|_{W_i \cap W_j} = \widetilde{f}_j|_{W_i \cap W_j}$. Since A_i, A_j are integral domains, the restriction map

$$\operatorname{res}_{W_i\cap W_j, V_i \cap V_j} : \Gamma(W_i \cap W_j, \mathscr{O}_{W_i\cap W_j}^{\oplus I}) \longrightarrow \Gamma(V_i \cap V_j, \mathscr{O}_{V_i \cap V_j}^{\oplus I}) \tag{15.4}$$

is injective (sketch of proof: A_i, A_j are integral, so the localizations are injective; use Proposition 6.3.1, this is easy to verify). By Algebraic Hartogs's Lemma 14.5.8, (15.4) is also a surjective, so (15.4) is an isomorphism, which implies that $\widetilde{f}_i|_{W_i \cap W_j} = \widetilde{f}_j|_{W_i \cap W_j}$. Glue $\{\widetilde{f}_i\}_i$ together, we obtain $\widetilde{f} \in \Gamma(U, \mathscr{F})$ such that $\widetilde{f}|_{U \setminus Z} = f$, which implies that the restriction map

$$\operatorname{res}_{U,U\setminus Z} : \Gamma(U, \mathscr{F}) \longrightarrow \Gamma(U \setminus Z, \mathscr{F})$$

is surjective.

For injectivity, this has already been included in our argument for gluing $\widetilde{f}_i$. □

15.2.2 Sheaves of meromorphic functions and rational functions

For arbitrary scheme we do not have a function field in general. Instead of working with fraction fields we use more generally the **total fraction ring** $\operatorname{Frac}(A)$ of a ring A.

Definition 15.2.2 (Regular element, total fraction ring)

Let A be a ring, $a \in A$ is called ***regular*** *if it is not a zero divisor in A, in other words, multiplication $x \mapsto ax$ is an injective homomorphism $A \to A$. The* ***total fraction ring of*** *A is defined as $\operatorname{Frac}(A) := R^{-1}A$, where R is the multiplicative set of regular elements in A. More generally, let M be an A-module. Then an element $m \in M$ is called* ***regular*** *if the A-linear map $A \to M$, $a \mapsto am$ is injective.*

Regular element of A-module can be globalized as follows:

Definition 15.2.3 (Regular section of quasicoherent sheaf)

Let X be a scheme and let $\mathscr{F}$ be a quasi-coherent $\mathscr{O}_X$-module. Then a section $s \in \Gamma(X, \mathscr{F})$ is called ***regular*** *if the following equivalent assertions hold.*

(I) The homomorphism of $\mathscr{O}_X$-modules $\mathscr{O}_X \to \mathscr{F}$, $f \mapsto fs$, is injective.

(II) For all $x \in X$ the image of s in $\mathscr{F}_x$ is a regular element of the $\mathscr{O}_{X,x}$-module $\mathscr{F}_x$.

(III) For any open affine $U = \operatorname{Spec} A \subseteq X$, the restriction $f|_U$ is a regular element of the A-module $\Gamma(U, \mathscr{F})$.

Remark 15.9 The equivalence in Definition 15.2.3 is immediate because injectivity can be checked on stalks for arbitrary sheaves (by definition) and on sections over open affine subschemes for quasi-coherent modules.

Definition 15.2.4 (Sheaf of meromorphic functions)

Let X be a scheme. For $U \subseteq X$ open, we let S_U be the subset of regular sections of $\Gamma(U, \mathscr{O}_X)$. This is a multiplicative subset of $\Gamma(U, \mathscr{O}_X)$. We define the sheaf $\mathscr{K}_X$ of $\mathscr{O}_X$-algebras as the sheaf associated to the presheaf

$$\mathscr{K}'_X : U \longmapsto S_U^{-1}\Gamma(U, \mathscr{O}_X). \tag{15.5}$$

This sheaf is called the ***sheaf of meromorphic functions on*** *X. A section of this sheaf is locally the quotient $\frac{f}{s}$ for a function f and a regular function s, and we therefore call such a section sometimes a* ***meromorphic function****.*

Remark 15.10 $\mathscr{K}'_X$ in general not a sheaf.

Remark 15.11 If X is integral, then $\Gamma(U, \mathscr{O}_X)$ is an integral domain for every non-empty open subset $U \subseteq X$ and S_U is the set of non-zero elements of $\Gamma(U, \mathscr{O}_X)$. Hence in this case $\mathscr{K}_X$ is the constant sheaf with value the function field $K(X)$. In other words, for every non-empty open $U \subseteq X$, we have $\mathscr{K}_X(U) = K(X)$. For every point $x \in X$, $\mathscr{K}_{X,x} = K(X)$.

As every element of S_U is a non-zero divisor of $\Gamma(U, \mathscr{O}_X)$, the homomorphism $\mathscr{O}_X \to \mathscr{K}_X$ is injective. This allows us to identify $\mathscr{O}_X$ with a subsheaf of rings of $\mathscr{K}_X$. We will see that if X is a locally Noetherian scheme, we have $\Gamma(U, \mathscr{K}_X) = \operatorname{Frac}(\Gamma(U, \mathscr{O}_X))$ for every open affine subscheme U. We mention that $\mathscr{K}_X$ is not always quasicoherent (evern if X is Noetherian), although it is quasi-coherent if X is Noetherian and reduced (Proposition 15.2.4) or if X is integral.

Proposition 15.2.4

Let X be a locally Noetherian scheme, if X is reduced, then $\mathscr{K}_X$ is a quasicoherent $\mathscr{O}_X$-module.

Proof Let $\operatorname{Spec} A \subseteq X$ be an affine open subset, then $\mathscr{K}_X(\operatorname{Spec} A) = S_A^{-1}A$ where S_A is the multiplicative set of regular elements in A, Since X is locally Noetherian and reduced, A is reduced Noetherian ring. Recall Proposition 7.6.11, the set of zero-divisors of A is the union of the associated prime ideals of A. Recall

Proposition 7.6.22, since A is reduced, A does not have embedded prime, so the associated ideals of A are all minimal prime ideals, say $\mathfrak{p}_1, \ldots, \mathfrak{p}_n$. Hence $S_A = A \setminus \bigcup_{i=1}^n \mathfrak{p}_i$, by Chinese Remainder Theorem 5.4.2

$$S_A^{-1}A \cong \prod_{i=1}^n (S_A^{-1}A/\mathfrak{p}_i S_A^{-1}A) \cong \prod_{i=1}^n S_A^{-1}(A/\mathfrak{p}_i) \cong \prod_{i=1}^n \kappa(\mathfrak{p}_i),$$

where last $\cong$ since each $\mathfrak{p}_i$ is minimal ideal so it is generic point of $\operatorname{Spec} A$. Recall Theorem 7.2.2, it suffices to check that $(S_A^{-1}A)_f \cong S_{A_f}^{-1}A_f$ for any $\operatorname{Spec} A_f \hookrightarrow \operatorname{Spec} A$. Note that

$$(S_A^{-1}A)_f \cong \prod_{f \notin \mathfrak{p}_i} \kappa(\mathfrak{p}_i) \cong \prod_{f \notin \mathfrak{p}_i} \kappa(\mathfrak{p}_i A_f) \cong S_{A_f}^{-1}A_f,$$

it follows that $\mathscr{K}_X$ is a quasicoherent $\mathscr{O}_X$-module. □

Example 15.2 (Counterexample for $\mathscr{K}_X$ may not be a quasi-coherent sheaf) Let B be a local Noetherian ring with $\dim B \geq 2$ and residue field k. Let $A = B \oplus k$ with multiplication $(b, x) \cdot (b', x') := (bb', bx' + b'x)$, and let $X = \operatorname{Spec} A$. Then $\mathscr{K}_X$ is not a quasicoherent sheaf.

Proof If $\mathscr{K}_X$ is a quasicoherent sheaf, then $\mathscr{K}_X \cong \widetilde{A} = \mathscr{O}_X$. Consider $\mathscr{K}_X(X) = S_X^{-1}A$, S_X is the regular elements of A. We claim that A is a local ring with maximal ideal $\mathfrak{m}_A = \mathfrak{m}_B \oplus k$ where $\mathfrak{m}_B$ is the maximal ideal of B. For any $(b, x) \notin \mathscr{M}_A$, then $b \notin \mathfrak{m}_B$, b is invertible in B. Then the inverse element of (b, x) is $(b^{-1}, -b^{-2}x)$. For any $(b, x) \in \mathscr{M}_A$, note that

$$(b, x) \cdot (0, 1) = (0, b) = (0, 0)$$

where last equal since $k = B/\mathfrak{m}_B$ and $b \in \mathfrak{m}_B$. Hence the set of zero-divisors of A is $\mathfrak{m}_A$, and thus $S_X = A \setminus \mathfrak{m}_A \ncong A$, a contradiction! So $\mathscr{K}_X$ is not a quasicoherent sheaf. □

Meromorphic functions $f \in \Gamma(X, \mathscr{K}_X)$ can also be consider as rational function on X as follows:

Definition 15.2.5 (Domain of definition of meromorphic function)

*Let X be a scheme, and meromorphic function $f \in \Gamma(X, \mathscr{K}_X)$. Let $\operatorname{dom}(f)$ be the set of $x \in X$ such that $f_x \in \mathscr{O}_{X,x} \subseteq \mathscr{K}_{X,x}$. $\operatorname{dom}(f)$ is called the **domain of definition** of f.*

Lemma 15.2.2

For every $f \in \Gamma(X, \mathscr{K}_X)$ its domain of definition $\operatorname{dom}(f)$ is a scheme-theoretically dense open subscheme of X.

Proof For any $x \in \operatorname{dom}(f)$, $f_x \in \mathscr{O}_{X,x}$, so there exists an open subset W of x and $g \in \Gamma(W, \mathscr{O}_X)$ such that $g_x = f_x$. Hence we can find an open neighborhood $W' \subseteq W$ of x such that $g|_{W'} = f|_{W'}$. This implies that $W' \subseteq \operatorname{dom}(f)$, and thus $\operatorname{dom}(f)$ is open in X.

We next show that $\operatorname{dom}(f)$ is scheme-theoretically dense. For any open subset $V \subseteq X$, let $g \in \Gamma(V, \mathscr{O}_X)$ with $g|_{\operatorname{dom}(f) \cap V} = 0$, we want to show that $g|_V = 0$. This may check locally, without loss of generally, we may assume that $V = \operatorname{Spec} A$. Restrict f to $\operatorname{dom}(f) \cap V$, we may write $f = a/s$ where $a \in A$ and s is regular in A. Consider $D(s) = \operatorname{Spec} A_f \subseteq V$, then $g|_{D(s)} = 0$, hence $s^n \cdot g = 0$ for some n. Since s is regular in A, $g = 0$, it follows that $\operatorname{dom}(f)$ is scheme-theoretically dense. □

For $f \in \Gamma(X, \mathscr{K}_X)$ the restriction $f|_{\operatorname{dom}(f)}$ defines an element in $\Gamma(\operatorname{dom}(f), \mathscr{O}_X)$ and the Lemma 15.2.2 shows that we obtain a rational function on X (Definition 8.5.3). For all open subsets $U \subseteq X$ we therefore have a homomorphism $\alpha_U : \Gamma(U, \mathscr{K}_X) \to R(U)$, where $R(U)$ denotes the ring of rational functions on U. If $\mathscr{R}_X$ denotes the sheaf $U \mapsto R(U)$ on X, the α_U defines a homomorphism

$$\alpha : \mathscr{K}_X \longrightarrow \mathscr{R}_X \tag{15.6}$$

of $\mathscr{O}_X$-algebra. As the canonical map $\Gamma(V, \mathscr{O}_X) \to R(U)$ is injective for every schematically dense open subset V of U, the homomorphism α is injective. Very often, the notions of meromorphic and of rational functions coincide:

Proposition 15.2.5 (The notions of meromorphic and of rational functions coincide)

If X is integral or a locally Noetherian scheme, then (15.6) is an isomorphism.

Proof In case X is integral, this is clear.

Now assume that X is locally Noetherian. We want to show that α_U is bijection for all open affine subsets $U = \operatorname{Spec} A$ of X. We have homomorphisms

$$\operatorname{Frac}(A) = \mathscr{K}'_X(U) \xrightarrow{\text{sh}} \mathscr{K}_X(U) \xrightarrow{\alpha_U} R(U),$$

where $\mathscr{K}'_X$ is the presheaf defined in (15.5). As we already know that α is injective, it suffices to show that $\operatorname{Frac}(A) \to R(U)$ is bijective. By definition we have

$$\operatorname{Frac}(A) = \varinjlim_{t \text{ regular in } A} \Gamma(D(t), \mathscr{O}_X), \qquad R(U) = \varinjlim_{V \supseteq \operatorname{Ass}(U)} \Gamma(V, \mathscr{O}_X),$$

where t runs through the set of regular elements of A and where V runs through the set of scheme-theoretically dense open subsets of U (recall Proposition 7.6.28). Pick an element s in $R(U)$, then there exists $V \supseteq \operatorname{Ass}(U)$ such that $s \in \Gamma(V, \mathscr{O}_X)$, by Lemma 7.6.3 an open subset V is scheme-theoretically dense if and only if there exists a regular element $t \in A$ such that $D(t) \subseteq V$, restrict s to $D(t)$, we have $s|_{D(t)} \in \Gamma(D(t), \mathscr{O}_X)$, it can be seen as an element in $\operatorname{Frac}(A)$, which implies that α_U is surjective. $\square$

Thus for locally Noetherian schemes we will often use the notions "meromorphic function" and "rational function" synonymously.

Remark 15.12 The proof of Proposition 15.2.5 shows in particular that for a Noetherian affine scheme $X = \operatorname{Spec} A$ we have $\Gamma(X, \mathscr{K}_X) = \operatorname{Frac}(A)$. This implies that for a locally Noetherian scheme X and a point $x \in X$ we have $\mathscr{K}_{X,x} = \operatorname{Frac}(\mathscr{O}_{X,x})$.

Proposition 15.2.6

Suppose s is a nonzero rational function of an invertible sheaf on a locally Noetherian normal scheme. If s has no poles, then s is regular.

Proof Let $\mathscr{L}$ be an invertible sheaf on a locally Noetherian normal scheme X, then there exists an open cover $\{U_i\}$ of X such that $\mathscr{L}|_{U_i} \cong \mathscr{O}_{U_i}$. Regularity can be checked locally, we may assume that $\mathscr{L} \cong \mathscr{O}_X$, then by Corollary 14.5.2, we win. $\square$

15.2.3 Locally free sheaves in short exact sequence

Proposition 15.2.7

(a) *Suppose*

$$0 \longrightarrow \mathscr{F}' \longrightarrow \mathscr{F} \longrightarrow \mathscr{F}'' \longrightarrow 0 \tag{15.7}$$

is a short exact sequence of quasicoherent sheaves on X. Suppose $U = \operatorname{Spec} A$ is an affine open set where $\mathscr{F}', \mathscr{F}''$ are free, say $\mathscr{F}'|_{\operatorname{Spec} A} = \widetilde{A}^{\oplus a}, \mathscr{F}''|_{\operatorname{Spec} A} = \widetilde{A}^{\oplus b}$. (Here a and b are assumed to be finite for convenience, but this is not necessary, so feel free to generalize to the infinite rank case.) Then $\mathscr{F}$ is also free on $\operatorname{Spec} A$, and that $0 \to \mathscr{F}' \to \mathscr{F} \to \mathscr{F}'' \to 0$ can be interpreted as coming from the tautological exact sequence $0 \to A^{\oplus a} \to A^{\oplus(a+b)} \to A^{\oplus b} \to 0$.

(As a consequence, given an exact sequence of quasicoherent sheaves (15.7) where $\mathscr{F}'$ and $\mathscr{F}''$ are locally free, $\mathscr{F}$ must also be locally free.)

(b) *In the finite rank case, given an open covering by trivializing affine open sets (of the form described in (a)), the transition functions (really, matrices) of $\mathscr{F}$ may be interpreted as block upper triangular matrices, where the top left $a \times a$ blocks are transition functions for $\mathscr{F}'$, and the bottom $b \times b$ blocks are transition functions for $\mathscr{F}''$.*

Proof ***For (a).*** We claim that $\mathscr{F}|_{\operatorname{Spec} A} \cong \widetilde{A}^{\oplus(a+b)}$. Since $\mathscr{F}$ is quasicoherent, suppose $\mathscr{F}|_{\operatorname{Spec} A} \cong \widetilde{M}$ where M is an A-module. Then we have a short exact sequence of quasicoherent sheaf on $\operatorname{Spec} A$,

$$0 \longrightarrow \widetilde{A}^{\oplus a} \longrightarrow \widetilde{M} \longrightarrow \widetilde{A}^{\oplus b} \longrightarrow 0$$

so we have an exact sequence of A-modules

$$0 \longrightarrow A^{\oplus a} \longrightarrow M \longrightarrow A^{\oplus b} \longrightarrow 0. \tag{15.8}$$

Since $A^{\oplus b}$ is free, and therefore projective, hence (15.8) is split, i.e., $M \cong A^{\oplus(a+b)}$. Hence

$$\mathscr{F}|_{\operatorname{Spec} A} \cong \left(A^{\oplus(a+b)}\right)^{\sim} \cong \widetilde{A}^{\oplus(a+b)},$$

as we desired.

For (b). Let $\{U_i\}_i$ be an open covering by trivializing affine open sets of the form described in (a). On each open set U_i, the short exact sequence splits, therefore we can choose a basis for the sections of $\mathscr{F}$ over U_i as follow: let

$$h_i = (e_{i,1}, \ldots, e_{i,a}, f_{i,1}, \ldots, f_{i,b})$$

such that the first a elements $\{e_{i,1}, \ldots, e_{i,a}\}$ form a basis for the subsheaf $\mathscr{F}'(U_i)$ and the image of the last b elements $\{f_{i,1}, \ldots, f_{i,b}\}$ in the quotient $\mathscr{F}''$ form a basis for $\mathscr{F}''(U_i)$. Let T_{ij} be the transition matrix sending U_i-coordinates to U_j-coordinates on the intersection U_{ij}, then we have

$$(e_{i,1}, \ldots, e_{i,a}, f_{i,1}, \ldots, f_{i,b}) = (e_{j,1}, \ldots, e_{j,a}, f_{j,1}, \ldots, f_{j,b})T_{ij}.$$

We write T_{ij} as a block matrix

$$T_{ij} = \begin{pmatrix} A_{ij} & B_{ij} \\ C_{ij} & D_{ij} \end{pmatrix}$$

where A_{ij} is $a \times a$, B_{ij} is $a \times b$, C_{ij} is $b \times a$, and D_{ij} is $b \times b$. Since $\mathscr{F}'$ is a subsheaf of $\mathscr{F}$, a section that belongs to $\mathscr{F}'$ on U_i must also belong to $\mathscr{F}'$ when restricted to U_{ij} and viewed in the coordinates of U_j. In our chosen basis, a section $s \in \mathscr{F}'$ is represented by a column vector where the bottom b components are zero:

$$v = \begin{pmatrix} v' \\ 0 \end{pmatrix}.$$

Applying the transition matrix, we have

$$T_{ij}v = \begin{pmatrix} A_{ij} & B_{ij} \\ C_{ij} & D_{ij} \end{pmatrix}\begin{pmatrix} v' \\ 0 \end{pmatrix} = \begin{pmatrix} A_{ij}v' \\ C_{ij}v' \end{pmatrix}.$$

For the resulting vector to remain in $\mathscr{F}'$, its bottom b components must be zero. Thus, $C_{ij}v' = 0$ for all vectors v'. This implies that $C_{ij} = 0$. Now the matrix of the form

$$T_{ij} = \begin{pmatrix} A_{ij} & B_{ij} \\ 0 & D_{ij} \end{pmatrix}$$

Restricting our attention to the subspace $\mathscr{F}'$ (the first a basis vectors), the transformation is given solely by

A_{ij}. Therefore, A_{ij} represents the transition functions for $\mathscr{F}'$. Consider the quotient sheaf $\mathscr{F}'' \cong \mathscr{F}/\mathscr{F}'$. The elements of $\mathscr{F}''$ correspond to vectors module the first a components. When the matrix acts on a vector $\begin{pmatrix} * \\ v'' \end{pmatrix}$:

$$\begin{pmatrix} A_{ij} & B_{ij} \\ 0 & D_{ij} \end{pmatrix} \begin{pmatrix} * \\ v'' \end{pmatrix} = \begin{pmatrix} * \\ D_{ij}v'' \end{pmatrix}.$$

The action on the quotient component v'' is determined entirely by D_{ij} represents the transition functions for $\mathscr{F}''$. Hence the transition matrices for $\mathscr{F}$ are block upper triangular

$$T_{ij} = \begin{pmatrix} T'_{ij} & B_{ij} \\ 0 & T''_{ij} \end{pmatrix}$$

where T'_{ij} are the transition matrices for $\mathscr{F}'$ and T''_{ij} are the transition matrices for $\mathscr{F}''$. □

Proposition 15.2.8

Suppose (15.7) is a short exact sequence of quasicoherent sheaves on X. By Proposition 15.2.7, if $\mathscr{F}'$ and $\mathscr{F}''$ are locally free, then $\mathscr{F}$ is too.

(a) If $\mathscr{F}$ and $\mathscr{F}''$ are locally free of finite rank, then $\mathscr{F}'$ is too.

(b) Even if $\mathscr{F}'$ and $\mathscr{F}$ are both locally free, $\mathscr{F}''$ need not be locally free.

Proof Let X be a scheme and suppose we have a short exact sequence of quasicoherent sheaves

$$0 \longrightarrow \mathscr{F}' \longrightarrow \mathscr{F} \xrightarrow{\varphi} \mathscr{F}'' \longrightarrow 0$$

For (a). Local freeness is a local property on X. Fix a point $p \in X$. Because $\mathscr{F}$ and $\mathscr{F}''$ are locally free of finite rank, there exists an open neighborhood $U \ni p$ such that $\mathscr{F}|_U \cong \mathscr{O}_U^{\oplus m}$ and $\mathscr{F}''|_U \cong \mathscr{O}_U^{\oplus n}$. Shrink further if necessary, we may assume $U = \operatorname{Spec} A$ is affine. Then the map $\varphi|_U$ corresponds to an A-map

$$A^{\oplus m} \xrightarrow{M} A^{\oplus n}$$

for some $n \times m$ matrix M with entires in A. S We want to show that $\mathscr{F}'|_U \cong (\operatorname{Ker}(M))^{\sim}$ is free. Let $p = [\mathfrak{p}] \in \operatorname{Spec} A$. Exactness of the sheaf sequence implies the stalk map

$$\varphi_p : \mathscr{F}_p \longrightarrow \mathscr{F}''_p$$

is surjective, using the chosen trivializations on U, this stalk map is the localization

$$M_{\mathfrak{p}} : A_{\mathfrak{p}}^{\oplus m} \longrightarrow A_{\mathfrak{p}}^{\oplus n},$$

which is surjective. Reduce mod the maximal ideal $\mathfrak{p}A_{\mathfrak{p}}$ to get a $\kappa(p)$-linear map

$$M(p) : \kappa(p)^{\oplus m} \longrightarrow \kappa(p)^{\oplus n}.$$

Surjectivity of M_p implies surjectivity of $M(p)$, hence $\operatorname{rank} M(p) = n$. Therefore, among the m column vectors of $M(p)$ in $\kappa(p)^n$, there exist n columns $c_1, \ldots, c_n$ that are linearly independent. Equivalently, there is an $n \times n$ minor of M whose determinant $\Delta_p \in A$ maps to a nonzero element of $\kappa(p)$, i.e., $\Delta_p \notin \mathfrak{p}$, i.e., $\mathfrak{p} \in D(\Delta_p)$. Thus, X can be covered by such distinguished opens corresponding to choices of n columns.

Now work on $D(\Delta_p)$. Reorder the basis of $A_{\Delta_p}^{\oplus m}$ if necessary so that the invertible $n \times n$ minor use the first n columns. Write $M = \begin{pmatrix} N & P \end{pmatrix}$, where N is $n \times n$ and invertible over A_{Δ_p}, and P is $n \times (m-n)$ block. Consider the invertible $m \times m$ matrix

$$Q = \begin{pmatrix} N^{-1} & -N^{-1}P \\ 0 & I_{m-n} \end{pmatrix},$$

then

$$MQ = \begin{pmatrix} N & P \end{pmatrix} \begin{pmatrix} N^{-1} & -N^{-1}P \\ 0 & I_{m-n} \end{pmatrix} = \begin{pmatrix} I_n & 0 \end{pmatrix}.$$

Consequently, this is a change of basis on the source free module $A_{\Delta_p}^{\oplus m}$ (i.e., change coordinates on $A^{\oplus m}$) making the map as simple as possible.

We next computes the kernel in this normal form. With respect to the new basis, the map is the projection

$$A_{\Delta_p}^{\oplus m} \oplus A_{\Delta_p}^{\oplus(m-n)} \longrightarrow A_{\Delta_p}^{\oplus n}, \qquad (x, y) \longmapsto x.$$

Hence

$$\operatorname{Ker}(M) \cong \{0\} \oplus A_{\Delta_p}^{\oplus(m-n)} \cong A_{\Delta_p}^{\oplus(m-n)},$$

which is free of rank $m-n$, and therefore $\mathscr{F}'|_{D(\Delta_p)}^{\oplus(m-n)}$. Since every point p lies in some such $D(\Delta_p)$, these opens cover U, and varying p covers X. Thus $\mathscr{F}'$ is locally free of finite rank.

For (b). Consider the following exact sequence

$$0 \longrightarrow (t) \longrightarrow k[t] \longrightarrow k[t]/(t) \longrightarrow 0,$$

then we have an exact sequence of quasicoherent sheaves on $\mathbb{A}_k^1$

$$0 \longrightarrow \widetilde{(t)} \longrightarrow \widetilde{k[t]} \longrightarrow (k[t]/(t))^{\sim} \longrightarrow 0.$$

Clearly, $\widetilde{(t)} = t\widetilde{k[t]} \cong \widetilde{k[t]}$, so $\widetilde{(t)}$ is locally free $\widetilde{k[t]}$-module. But $(k[t]/(t))^{\sim}$ is not locally free $\widetilde{k[t]}$-module, since $k[t]/(t)$ has torsion. ☐

Remark 15.13 If an injection of locally free sheaves $\alpha : \mathscr{F}' \to \mathscr{F}$ is such that the cokernel $\mathscr{F}''$ is also locally free, people sometimes say that α is an **injection of vector bundles**, not just of locally free sheaves. Implicitly, they are considering a category of vector bundles that is different from the category of locally free sheaves; the cokernel of a map of vector bundles is then always a vector bundle. See Definition 15.2.6 for further discussion.

Lemma 15.2.3

Let $(X, \mathscr{O}_X) = (\operatorname{Spec} R, \mathscr{O}_{\operatorname{Spec} R})$ be an affine scheme. If M is finitely presented R-module, then there is a canonical isomorphism $\mathscr{H}om_{\mathscr{O}_X}(\widetilde{M}, \widetilde{N}) \cong \operatorname{Hom}_R(M, N)^{\sim}$.

Proof See [Sta25, Lemma 01I8]. ☐

Proposition 15.2.9

(a) Suppose $0 \to \mathscr{F}' \to \mathscr{F} \to \mathscr{F}'' \to 0$ is a short exact sequence of finite rank locally free sheaves. Then

$$0 \longrightarrow \mathscr{F}''^{\vee} \longrightarrow \mathscr{F}^{\vee} \longrightarrow \mathscr{F}'^{\vee} \longrightarrow 0$$

(obtained by applying $\mathscr{H}om(\square, \mathscr{O}_X)$ to the original sequence) is also a short exact sequence of finite rank locally free sheaves.

(b) Suppose $\cdots \to \mathscr{F}_2 \to \mathscr{F}_1 \to \mathscr{F}_0 \to 0$ is an exact sequence of finite rank locally free sheaves. Then the induced sequence

$$0 \longrightarrow \mathscr{F}_0^{\vee} \longrightarrow \mathscr{F}_1^{\vee} \longrightarrow \mathscr{F}_2^{\vee} \longrightarrow \cdots$$

is also an exact sequence of finite rank locally free sheaves.

Proof ***For (a).*** Let $x \in X$, then there exists an affine open $U = \operatorname{Spec} A$ contains x such that $\mathscr{F}|_U \cong \mathscr{O}_U^r$,

$\mathscr{F}'|_U \cong \mathscr{O}_U^{\oplus r'}$ and $\mathscr{F}''|_U \cong \mathscr{O}_U^{\oplus r''}$, then we have a corresponding exact sequence of free A-modules,

$$0 \longrightarrow A^{r'} \longrightarrow A^{r} \longrightarrow A^{r''} \longrightarrow 0.$$

Apply $\mathrm{Hom}_A(\square, A)$, we have

$$0 \longrightarrow \mathrm{Hom}(A^{r''}, A) \longrightarrow \mathrm{Hom}(A^{r}, A) \longrightarrow \mathrm{Hom}(A^{r'}, A) \longrightarrow \mathrm{Ext}^1_A(A^{r''}, A) \longrightarrow \cdots,$$

since $A^{r''}$ is projective, $\mathrm{Ext}^1_A(A^{r''}, A) = 0$, then we have exact sequence

$$0 \longrightarrow \mathrm{Hom}(A^{r''}, A) \longrightarrow \mathrm{Hom}(A^{r}, A) \longrightarrow \mathrm{Hom}(A^{r'}, A) \longrightarrow 0.$$

Apply exact functor $\widetilde{\square}$, we have

$$0 \longrightarrow \mathrm{Hom}(A^{r''}, A)^{\sim} \longrightarrow \mathrm{Hom}(A^{r}, A)^{\sim} \longrightarrow \mathrm{Hom}(A^{r'}, A)^{\sim} \longrightarrow 0,$$

by Lemma 15.2.3, the sequence

$$0 \longrightarrow \mathscr{F}''|_U^{\vee} \longrightarrow \mathscr{F}|_U^{\vee} \longrightarrow \mathscr{F}'|_U^{\vee} \longrightarrow 0$$

is exact. Hence

$$0 \longrightarrow \mathscr{F}''^{\vee} \longrightarrow \mathscr{F}^{\vee} \longrightarrow \mathscr{F}'^{\vee} \longrightarrow 0$$

is exact.

For (b). We can break the given long exact sequence into a series of short exact sequences. Let $d_i : \mathscr{F}_i \to \mathscr{I}_{i-1}$ denote the boundary maps (with $\mathscr{F}_{-1} = 0$). Define the kernel sheaves as $K_i = \mathrm{Ker}\, d_i$. More precisely, for $i = 0$, since the sequence ends at 0, the map $\mathscr{F}_0 \to 0$ has kernel $K_0 = \mathscr{F}_0$; for $i = 1$, exactness at $\mathscr{F}_0$ implies d_1 is surjective. Thus, we have a short exact sequence

$$0 \longrightarrow K_1 \longrightarrow \mathscr{F}_1 \xrightarrow{d_1} K_0 \longrightarrow 0;$$

for $i \geq 2$, exactness at $\mathscr{I}_{i-1}$ implies that $\mathrm{Im}\, d_i = \mathrm{Ker}\, d_{i-1} = K_{i-1}$. Thus, for ever $i \geq 1$, we have a short exact sequence

$$0 \longrightarrow K_i \longrightarrow \mathscr{F}_i \xrightarrow{d_i} K_{i-1} \longrightarrow 0.$$

We next show that every K_i is locally free of finite rank. We prove it by induction on i. For $i = 0$, $K_0 = \mathscr{F}_0$ is locally free of finite rank. Assume K_{i-1} is locally free. Consider the sequence

$$0 \longrightarrow K_i \longrightarrow \mathscr{F}_i \xrightarrow{d_i} K_{i-1} \longrightarrow 0.$$

Since K_{i-1} is locally free of finite rank, by Proposition 15.2.8, K_i is locally free of finite rank.

Now we apply the dual functor $\mathscr{H}om_{\mathscr{O}_X}$ to short exact sequence

$$0 \longrightarrow K_i \longrightarrow \mathscr{F}_i \xrightarrow{d_i} K_{i-1} \longrightarrow 0.$$

by part (a), the sequence

$$0 \longrightarrow K_{i-1}^{\vee} \longrightarrow \mathscr{F}_i^{\vee} \longrightarrow K_i^{\vee} \longrightarrow 0$$

is exact. One check that $\mathrm{Ker}(\mathscr{F}_i^{\vee} \to K_i^{\vee} \to \mathscr{F}_{i+1}^{\vee}) = \mathrm{Im}(\mathscr{F}_{i-1}^{\vee} \to K_{i-1}^{\vee} \to \mathscr{F}_i^{\vee})$, so wo have long exact sequence

$$0 \longrightarrow \mathscr{F}_0^{\vee} \longrightarrow \mathscr{F}_1^{\vee} \longrightarrow \mathscr{F}_2^{\vee} \longrightarrow \cdots,$$

as we desired. $\square$

15.2.4 Vector bundle maps

Maps of vector bundles (in the more conventional sense) are more restrictive than morphisms of quasicoherent sheaves that happen to be locally free sheaves. A locally free subsheaf of a locally free sheaf does not always yield a "subvector bundle" of a vector bundle. The archetypal example is the exact sequence of quasicoherent sheaves on $\mathbb{A}^1_k = \operatorname{Spec} k[t]$ corresponding to the following exact sequence of $k[t]$-modules:

$$0 \longrightarrow tk[t] \longrightarrow k[t] \longrightarrow k[t]/(t) \longrightarrow 0 \tag{15.9}$$

The locally free sheaf $\widetilde{tk[t]}$ is a subsheaf of $\widetilde{k[t]}$, but it does not correspond to a "subvector bundle"; the cokernal is not a vector bundle. (This example was mentioned in §15.1.)

Because maps of locally free sheaves that are actually "maps of vector bundles" have some particularly good and useful behavior, it seems worthwhile to give them a name. There is no accepted name, so we will just choose one.

Definition 15.2.6 (Map of vector bundles)

Let X be a scheme (or, indeed, more generally, a ringed space), let $\mathscr{E}$ and $\mathscr{F}$ be two finite rank locally free sheaves on X, of rank $a+b$ and $a+c$, respectively.

(1) *We say that $\varphi : \mathscr{E} \to \mathscr{F}$ is a **rank** a **map of vector bundles** if for every $p \in X$, there is an open neighborhood U of p with a commutative diagram*

$$\begin{tikzcd} \mathscr{E}|_U \arrow[rr, "\varphi"] \arrow[d, leftrightarrow, "\sim"'] & & \mathscr{F}|_U \arrow[d, leftrightarrow, "\sim"] \\ \mathscr{O}_U^{\oplus(a+b)} \arrow[r, two heads] & \mathscr{O}_U^{\oplus a} \arrow[r, hook] & \mathscr{O}_U^{\oplus(a+c)}, \end{tikzcd}$$

where the map $\mathscr{O}_U^{\oplus(a+b)} \twoheadrightarrow \mathscr{O}_U^{\oplus a}$ is projection to the first a summands, and the map $\mathscr{O}_U^{\oplus a} \hookrightarrow \mathscr{O}_U^{\oplus(a+c)}$ is inclusion as the first a summands.

(2) *We say that φ is a **map of vector bundles** if near every point $p \in X$, φ is a rank a map of vector bundles for some a.*

(3) *If $b = 0$, we say φ is an **injection of vector bundles** (as in the remark of Proposition 15.2.8).*

Note ***Important but straightforward observations.***

(i) *The notion "rank a map of vector bundles" commutes with any base change $Y \to X$. More precisely, if $\varphi : \mathscr{E} \to \mathscr{F}$ is a rank a map of vector bundles over X, and $f : Y \to X$ is a morphism of schemes, then $f^*\varphi : f^*\mathscr{E} \to f^*\mathscr{F}$ is a rank a map of vector bundles over Y.*

(ii) *The quasicoherent sheaves $\operatorname{Ker}\varphi$, $\operatorname{Im}\varphi$, and $\operatorname{Coker}\varphi$ are locally free (of finite b, a, and c, respectively), and their construction commutes with any base change $Y \to X$ (say $f : Y \to X$, then $f^* \operatorname{Ker}\varphi \cong \operatorname{Ker} f^*\varphi$, $f^* \operatorname{Im}\varphi \cong \operatorname{Im} f^*\varphi$, and $f^* \operatorname{Coker}\varphi \cong \operatorname{Coker} f^*\varphi$).*

$$\begin{tikzcd} 0 \arrow[r] & \operatorname{Ker}\varphi \arrow[r] \arrow[d, leftrightarrow, "\sim"'] & \mathscr{E}|_U \arrow[r, "\varphi"] \arrow[d, leftrightarrow, "\sim"] & \mathscr{F}|_U \arrow[r] \arrow[d, leftrightarrow, "\sim"] & \operatorname{Coker}\varphi \arrow[r] \arrow[d, leftrightarrow, "\sim"] & 0 \\ 0 \arrow[r] & \mathscr{O}_U^{\oplus b} \arrow[r] & \mathscr{O}_U^{\oplus(a+b)} \arrow[r] & \mathscr{O}_U^{\oplus(a+c)} \arrow[r] & \mathscr{O}_U^{\oplus c} \arrow[r] & 0 \end{tikzcd}$$

(iii) *If $\varphi : \mathscr{E} \to \mathscr{F}$ is a rank a map of vector bundles, then at any point $p \in X$, the rank of $\varphi_p : \mathscr{E}|_p \to \mathscr{F}|_p$ is a (Definition 5.3.9).*

In the next propositions we give three criteria for when a morphism φ of finite rank locally free sheaves is a map of vector bundles.

Proposition 15.2.10 (Useful characterization of "map of vector bundles")

Suppose $\varphi : \mathscr{E} \to \mathscr{F}$ is a morphism of finite rank locally free sheaves, where $\mathscr{F}$ has rank $a + c$. Then φ is a rank a map of vector bundles if and only if $\operatorname{Coker}\varphi$ is locally free of rank c.
(In particular, every surjection of finite rank locally free sheaves is a map of vector bundles.)

Proof If φ is a rank a map of vector bundles. Let $p \in X$, then there is an open neighborhood U of p such that the following diagram commutes.

$$\begin{tikzcd}
\mathscr{E}|_U \arrow[rr, "\varphi|_U"] \arrow[d, leftrightarrow, "\sim"'] & & \mathscr{F}|_U \arrow[d, leftrightarrow, "\sim"] \\
\mathscr{O}_U^{\oplus(a+b)} \arrow[r, two heads] & \mathscr{O}_U^{\oplus a} \arrow[r, hook] & \mathscr{O}_U^{\oplus(a+c)},
\end{tikzcd}$$

Hence $(\operatorname{Coker}\varphi)|_U \cong \operatorname{Coker}(\mathscr{O}_U^{\oplus(a+b)} \to \mathscr{O}_U^{\oplus(a+c)}) \cong \mathscr{O}_U^{\oplus c}$, which implies that $\operatorname{Coker}\varphi$ is locally free of rank c.

Conversely, if $\operatorname{Coker}\varphi$ is locally free of rank c, then for each $x \in X$ there exists an open subset U such that $\mathscr{E}|_U \cong \mathscr{O}_U^{\oplus d}$, $\mathscr{F}|_U \cong \mathscr{O}_U^{\oplus(a+c)}$, and $(\operatorname{Coker}\varphi)|_U \cong \mathscr{O}_U^{\oplus c}$. Consider the following natural exact sequence.

$$\begin{tikzcd}
\mathscr{E}|_U \arrow[r, "\varphi|_U"] \arrow[d, leftrightarrow, "\sim"'] & \mathscr{F}|_U \arrow[r] \arrow[d, leftrightarrow, "\sim"] & (\operatorname{Coker}\varphi)|_U \arrow[r] \arrow[d, "\sim"] & 0 \\
\mathscr{O}_U^{\oplus d} \arrow[r] & \mathscr{O}_U^{\oplus a+c} \arrow[r] & \mathscr{O}_U^{\oplus c} \arrow[r] & 0
\end{tikzcd}$$

Say $\psi : \mathscr{F} \to \operatorname{Coker}\varphi$, then we have exact sequences

$$\begin{tikzcd}
0 \arrow[r] & (\operatorname{Ker}\psi)|_U \arrow[r] \arrow[d, leftrightarrow, "\sim"'] & \mathscr{F}|_U \arrow[r] \arrow[d, "\sim"] & (\operatorname{Coker}\varphi)|_U \arrow[r] \arrow[d, leftrightarrow, "\sim"] & 0 \\
0 \arrow[r] & \mathscr{O}_U^{\oplus a} \arrow[r] & \mathscr{O}_U^{\oplus a+c} \arrow[r] & \mathscr{O}_U^{\oplus c} \arrow[r] & 0
\end{tikzcd}$$

Since $\operatorname{Ker}\psi = \operatorname{Im}\varphi$, we know that $(\operatorname{Im}\varphi)|_U \cong \mathscr{O}_U^{\oplus a}$. Then, we obtain a decomposition of $\varphi|_U$, that is,

$$\begin{tikzcd}
\mathscr{E}|_U \arrow[r, two heads, "\varphi|_U"] \arrow[d, leftrightarrow, "\sim"'] & (\operatorname{Im}\varphi)_U \arrow[r, "\sim"] & (\operatorname{Ker}\psi)|_U \arrow[r, hook] \arrow[d, leftrightarrow, "\sim"'] & \mathscr{F}|_U \arrow[d, leftrightarrow, "\sim"] \\
\mathscr{O}_U^{\oplus d} \arrow[rr, two heads] & & \mathscr{O}_U^{\oplus a} \arrow[r, hook] & \mathscr{O}_U^{\oplus a+c}
\end{tikzcd}$$

it follows that φ is a rank a map of vector bundles. □

Proposition 15.2.11

Suppose $\varphi : \mathscr{E} \to \mathscr{F}$ is a morphism of finite rank locally free sheaves, and $p \in X$. Then the natural map $(\operatorname{Ker}\varphi)|_p \to \operatorname{Ker}(\varphi|_p)$ is surjective if and only if morphism φ is a map of vector bundles in some neighborhood of p.

Remark 15.14 The natural map is given by follow: consider the exact sequence

$$0 \longrightarrow \operatorname{Ker}\varphi \longrightarrow \mathscr{E} \longrightarrow \mathscr{F}$$

taking stalk at p, we obtain exact sequence

$$0 \longrightarrow (\operatorname{Ker}\varphi)_p \longrightarrow \mathscr{E}_p \longrightarrow \mathscr{F}_p.$$

Apply right exact functor $\square \otimes_{\mathscr{O}_{X,p}} \kappa(p)$, we have the following diagram

$$\begin{tikzcd} & (\operatorname{Ker}\varphi)|_p \arrow[d, dashed, "\exists!"'] \arrow[dr] & & \\ 0 \arrow[r] & \operatorname{Ker}\varphi|_p \arrow[r] & \mathscr{E}|_p \arrow[r, "\varphi|_p"] & \mathscr{F}|_p \end{tikzcd}$$

where dashed arrow is given by the universal property of the kernel of $\varphi|_p$.

Proof If φ is a map of vector bundles in some neighborhood of $p \in X$, by Proposition 15.2.10, $\operatorname{Coker}\varphi$ is locally free. Consider the exact sequence

$$0 \longrightarrow \operatorname{Im}\varphi \longrightarrow \mathscr{F} \longrightarrow \operatorname{Coker}\varphi \longrightarrow 0.$$

Since $\operatorname{Coker}\varphi$ is locally free, by Proposition 15.2.8, $\operatorname{Im}\varphi$ is locally free. Consider the exact sequence

$$0 \longrightarrow \operatorname{Ker}\varphi \longrightarrow \mathscr{E} \xrightarrow{\varphi} \operatorname{Im}\varphi \longrightarrow 0.$$

Proposition 15.2.8 again, $\operatorname{Ker}\varphi$ is locally free. Taking stalk at p, we have exact sequence

$$0 \longrightarrow (\operatorname{Ker}\varphi)_p \longrightarrow \mathscr{E}_p \longrightarrow (\operatorname{Im}\varphi)_p \longrightarrow 0.$$

Apply $\square \otimes_{\mathscr{O}_{X,p}} \kappa(p)$, we have long exact sequence

$$\cdots \longrightarrow \operatorname{Tor}_1^{\mathscr{O}_{X,p}}((\operatorname{Im}\varphi)_p, \kappa(p)) \longrightarrow (\operatorname{Ker}\varphi)|_p \longrightarrow \mathscr{E}|_p \longrightarrow (\operatorname{Im}\varphi)|_p \longrightarrow 0,$$

since $(\operatorname{Im}\varphi)_p$ is free $\kappa(p)$-module, $\operatorname{Tor}_1^{\mathscr{O}_{X,p}}((\operatorname{Im}\varphi)_p, \kappa(p)) = 0$, and therefore we obtain the following exact sequence

$$0 \longrightarrow (\operatorname{Ker}\varphi)|_p \longrightarrow \mathscr{E}|_p \longrightarrow (\operatorname{Im}\varphi)|_p \longrightarrow 0.$$

Hence $\operatorname{Ker}(\varphi|_p) \cong (\operatorname{Ker}\varphi)|_p$, in particular, the natural map $(\operatorname{Ker}\varphi)|_p \to \operatorname{Ker}(\varphi|_p)$ is surjective.

Conversely, if the natural map $(\operatorname{Ker}\varphi)|_p \to \operatorname{Ker}(\varphi|_p)$ is surjective, we want to show that φ is a map of vector bundles in some neighborhood of p. Since $\mathscr{E}, \mathscr{F}$ are locally free, say $\operatorname{rank}\mathscr{E} = e$ and $\operatorname{rank}\mathscr{F} = f$, there exists an open neighborhood $U \ni p$ such that $\mathscr{E}|_U \cong \mathscr{O}_U^{\oplus e}$ and $\mathscr{F}|_U \cong \mathscr{O}_U^{\oplus f}$. Hence $\varphi|_U$ can be regarded as a matrix $A \in M_{f\times e}(\mathscr{O}_X(U))$. Take stalk at p, we get a map of free modules over $\mathscr{O}_{X,p}$, i.e.,

$$A_p : \mathscr{O}_{X,p}^{\oplus e} \longrightarrow \mathscr{O}_{X,p}^{\oplus f},$$

where $A_p \in M_{f\times e}(\mathscr{O}_{X,p})$. By modulo $\mathfrak{m}_p$, we obtain

$$\overline{A_p} : \kappa(p)^{\oplus e} \longrightarrow \kappa(p)^{\oplus f},$$

where $\overline{A_p} \in M_{f\times e}(\kappa(p))$. In fact, $\operatorname{Ker}(\varphi|_p) = \operatorname{Ker}\overline{A_p}$ and $(\operatorname{Ker}\varphi)|_p = (\operatorname{Ker}A_p) \otimes_{\mathscr{O}_{X,p}} \kappa(p)$. Let $r = \operatorname{rank}_{\kappa(p)}(\overline{A_p})$, then some $r \times r$ minor Δ_p of $\overline{A_p}$ is nonzero, hence the corresponding minor in $\mathscr{O}_{X,p}$ is not in $\mathfrak{m}_p$, so it is a unit in the local ring $\mathscr{O}_{X,p}$. After permuting basis vectors (row/column permutations), we may assume $\Delta_p = \det(A_{11})$ where A_{11} is the upper-left $r \times r$ block of A_p, Because $\det(A_{11}) \in \mathscr{O}_{X,p}^{\times}$, apply Gaussian elimination, multiply on the left by invertible block matrices to obtain

$$A_p' = \begin{pmatrix} I_r & B \\ 0 & C \end{pmatrix}.$$

Let multiplication by an invertible matrix does not change the kernel, so we may assume that $A_p = A_p'$.

We next compute $\operatorname{Ker}(A_p)$. Write $\mathscr{O}_{X,p}^{\oplus e} = \mathscr{O}_{X,p}^{\oplus r} \oplus \mathscr{O}_{X,p}^{\oplus(e-r)}$, then

$$A_p \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x + By \\ Cy. \end{pmatrix}$$

Hence $(x, y)^t \in \operatorname{Ker} A_p$ if and only if $x = -By$ and $Cy = 0$. Therefore the map

$$\operatorname{Ker} C \longrightarrow \operatorname{Ker} A_p, \qquad y \longmapsto (-By, y)^t$$

is an isomorphism, so $\operatorname{Ker} A_p \cong \operatorname{Ker} C \subseteq \mathscr{O}_{X,p}^{\oplus(e-r)}$. Tensoring $\kappa(p)$ (i.e., modulo $\mathfrak{m}_p$), since $\overline{A_p} = \begin{pmatrix} I_r & \overline{B} \\ 0 & \overline{C} \end{pmatrix}$ has rank r, we have $\overline{C} = 0$. Hence $\operatorname{Ker} \overline{A_p} \cong \kappa(p)^{\oplus(e-r)}$. Under the identifications $\operatorname{Ker} A_p \cong \operatorname{Ker} C \subseteq \mathscr{O}_{X,p}^{\oplus(e-r)}$, the natural map

$$\operatorname{Ker} A_p \otimes_{\mathscr{O}_{X,p}} \kappa(p) \longrightarrow \operatorname{Ker} \overline{A_p}$$

becomes

$$\operatorname{Ker} C \otimes_{\mathscr{O}_{X,p}} \kappa(p) \longrightarrow \mathscr{O}_{X,p}^{\oplus(e-r)} \otimes_{\mathscr{O}_{X,p}} \kappa(p).$$

By condition above map is surjective, by Nakayama's Lemma 9.2.5 (version 3),

$$\operatorname{Ker} A_p \cong \operatorname{Ker} C = \mathscr{O}_{X,p}^{\oplus(e-r)}.$$

Hence $C = 0$, so $A_p = \begin{pmatrix} I_r & B \\ 0 & 0 \end{pmatrix}$, and therefore $\operatorname{Coker} A_p \cong \mathscr{O}_{X,p}^{\oplus(f-r)}$, which implies there exists an open neighborhood U of p, such that $\varphi|_U$ is a map of vector bundles (Proposition 15.2.10). $\square$

15.2.5 Tensor algebra constructions

For the next proposition, recall the following.

Definition 15.2.7 (Tensor algebra)

*Let A be a ring. Let M be an A-module. We define the **tensor algebra of M over A** to be the non-commutative A-algebra*

$$T^\bullet(M) := \bigoplus_{n \geq 0} T^n(M)$$

with $T^0(M) = A$, $T^1(M) = M$, $T^2(M) = M \otimes_A M$, $T^3(M) = M \otimes_A M \otimes_A M$, and so on. Multiplication is defined by the rule that on pure tensors we have

$$(x_1 \otimes x_2 \otimes \cdots \otimes x_n) \cdot (y_1 \otimes y_2 \otimes \cdots \otimes y_m) = x_1 \otimes x_2 \otimes \cdots \otimes x_n \otimes y_1 \otimes y_2 \otimes \cdots \otimes y_m$$

and we extend this by linearity.

Definition 15.2.8 (Symmetric algebra)

*The **symmetric algebra** $\operatorname{Sym}^\bullet M$ is a symmetric algebra, graded by $\mathbb{Z}^{\geq 0}$, defined as the quotient of $T^\bullet(M)$ by the two-side ideal generated by all elements of the form $x \otimes y - y \otimes x$ for all $x, y \in M$. Thus $\operatorname{Sym}^n M$ is the quotient of $M \otimes \cdots \otimes M$ by the relations of the form $m_1 \otimes \cdots \otimes m_n - m_1' \otimes \cdots \otimes m_n'$, where $(m_1', \cdots, m_n')$ is a rearrangement of $(m_1, \cdots, m_n)$.*

Definition 15.2.9 (Exterior algebra)

*The **exterior algebra** $\bigwedge^\bullet M$ is defined to be the quotient of $T^\bullet(M)$ by the two-sided ideal generated by all elements of the form $x \otimes x$ for all $x \in M$. Expanding $(a+b) \otimes (a+b)$, we see that $a \otimes b = -b \otimes a$ in $\bigwedge^2 M$. This implies that if 2 is invertible in A (e.g., if A is a field of characteristic not 2), $\bigwedge^n M$ is the quotient of $M \otimes \cdots \otimes M$ by the relations of the form $m_1 \otimes \cdots \otimes m_n - (-1)^{\operatorname{sgn}(\sigma)} m_{\sigma(1)} \otimes \cdots \otimes m_{\sigma(n)}$, where σ is a permutation of $\{1, \ldots, n\}$. The exterior algebra is a "skew-commutative" A-algebra.*

Remark 15.15 ***Better:*** both Sym and $\bigwedge$ can be defined by universal properties. For example, any map of A-modules $M^{\otimes n} \to N$ that is symmetric in the n entires factors uniquely through $\mathrm{Sym}_A^n(M)$.

Remark 15.16 It is most correct to write $T_A^\bullet(M)$, $\mathrm{Sym}_A^\bullet(M)$, and $\bigwedge_A^\bullet(M)$, but the "base ring" A is usually omitted for convenience.

Definition 15.2.10 (Sheaf of tensor algebra, symmetric algebra, and exterior algebra)

Let $(X, \mathscr{O}_X)$ be a ringed space. Let $\mathscr{F}$ be an $\mathscr{O}_X$-module.

*(1) We define the tensor algebra of $\mathscr{F}$ to be the **sheaf of noncommutative $\mathscr{O}_X$-algebras***

$$T^\bullet \mathscr{F} := \bigoplus_{n \geq 0} T^n \mathscr{F}.$$

Here $T^0\mathscr{F} = \mathscr{O}_X$, $T^1\mathscr{F} = \mathscr{F}$ and for $n \geq 2$ we have

$$T^n \mathscr{F} = \mathscr{F} \otimes_{\mathscr{O}_X} \cdots \otimes_{\mathscr{O}_X} \mathscr{F} \quad (n \text{ factors})$$

*(2) We define $\bigwedge^\bullet \mathscr{F}$ to be the quotient of $T^\bullet\mathscr{F}$ by the two sided ideal generated by local sections $s \otimes s$ of $T^2\mathscr{F}$ where s is a local section of $\mathscr{F}$. This is called the **exterior algebra of $\mathscr{F}$**.*

(3) We define $\mathrm{Sym}^\bullet \mathscr{F}$ to be the quotient of $T^\bullet\mathscr{F}$ by the two sided ideal generated by local section of the form $s \otimes t - t \otimes s$ of $T^2\mathscr{F}$.

Lemma 15.2.4

The sheaf $\bigwedge^n \mathscr{F}$ is the sheafification of the presheaf

$$U \longmapsto \bigwedge\nolimits^n_{\mathscr{O}_X(U)}(\mathscr{F}(U)).$$

Similarly, the sheaf $\mathrm{Sym}^n \mathscr{F}$ is the sheafification of the presheaf

$$U \longmapsto \mathrm{Sym}^n_{\mathscr{O}_X(U)}(\mathscr{F}(U)).$$

Proof Omitted. □

Lemma 15.2.5

Let $x \in X$, where $(X, \mathscr{O}_X)$ is a ringed space. There are canonical isomorphisms of $\mathscr{O}_{X,x}$-modules $T^\bullet(\mathscr{F})_x = T^\bullet(\mathscr{F}_x)$, $\mathrm{Sym}^\bullet(\mathscr{F})_x = \mathrm{Sym}^\bullet(\mathscr{F}_x)$, and $\bigwedge^\bullet(\mathscr{F})_x = \bigwedge^\bullet(\mathscr{F}_x)$.

Proof See [Sta25, Lemma 01CH]. □

Lemma 15.2.6

If $\mathscr{F}$ is locally free of rank m, then $T^n\mathscr{F}$, $\mathrm{Sym}^n \mathscr{F}$, and $\bigwedge^n \mathscr{F}$ are locally free of m^n, $\binom{n+m-1}{n}$, $\binom{m}{n}$, respectively.

Proof Let X be a scheme and $\mathscr{F}$ is a locally free sheaf of rank m on X. Since $\mathscr{F}$ is locally free of rank m, there exists an affine open cover $\{U_i\}$ such that $\mathscr{F}|_{U_i} \cong \mathscr{O}_{U_i}^{\oplus m}$. To prove that $T^n\mathscr{F}$, $\mathrm{Sym}^n \mathscr{F}$, and $\wedge^n\mathscr{F}$ are locally free, it suffices to check this property locally. Let $U = \operatorname{Spec} A$ be an affine open subset contained in one of $\{U_i\}$. Then $\mathscr{F}|_U \cong \mathscr{O}_U^{\oplus m}$. Let $\mathscr{F}|_U = \mathrm{Span}\{e_1, \ldots, e_m\}$.

For the tensor power $T^n\mathscr{F}$. Note that

$$T^n\mathscr{F}|_U \cong \mathscr{F}|_U^{\otimes n} \cong (\mathscr{O}_U^{\oplus m})^{\otimes n} \cong \mathscr{O}_U^{\oplus m^n},$$

it follows that $T^n\mathscr{F}$ is locally free of rank m^n.

For the symmetric power $\mathrm{Sym}^n \mathscr{F}$. A basis for $\mathrm{Sym}^n \mathscr{F}|_U$ consists of, monomials of degree n in the basis elements $e_1, \ldots, e_m$. We can represent these basis element as $e_{i_1}e_{i_2}\cdots e_{i_n}$. To ensure uniqueness, since order doesn't matter, we require the indices to be non-decreasing $1 \leq i_1 \leq i_2 \leq \cdots \leq i_n \leq m$. The problem of counting the number of such monomials is equivalent to the combinatorial problem of "stars and bars". We are choosing n items from $\{e_1, \ldots, e_m\}$, allowing replacement. Then the rank is

$$\operatorname{rank} \operatorname{Sym}^n \mathscr{F}|_U = \binom{n+m-1}{n},$$

so $\operatorname{Sym}^n \mathscr{F}$ is locally free of rank $\binom{n+m-1}{n}$.

For the exterior power $\bigwedge^n \mathscr{F}$. A basis for $\bigwedge^n \mathscr{F}|_U$ is given by wedge products of the form $e_{i_1} \wedge e_{i_2} \wedge \cdots \wedge e_{i_n}$. Due to the anti-symmetry property and the fact that $e_i \wedge e_i = 0$, a basis element is non-zero only if all indices are distinct. Furthermore, we can reorder them to be strictly increasing to obtain a unique basis $1 \leq i_1 \leq i_2 \leq \cdots \leq i_n \leq m$. We are choosing n distinct indices from the set $\{1, \ldots, m\}$. The number of ways to do this is given by the standard binormal coefficient

$$\operatorname{rank}\left(\bigwedge^n \mathscr{F}|_U\right) = \binom{m}{n}.$$

It follows that $\bigwedge^n \mathscr{F}$ is locally free of rank $\binom{m}{n}$. ☐

Definition 15.2.11 (Determinant bundle)

$\bigwedge^{\operatorname{rank} \mathscr{F}} \mathscr{F}$ is denoted $\det \mathscr{F}$, and is called the ***determinant (line) bundle*** *or the* ***determinant locally free sheaf****.*

Proposition 15.2.12

Suppose $0 \to \mathscr{F}' \to \mathscr{F} \to \mathscr{F}'' \to 0$ is an exact sequence of locally free sheaves. Then for any r, there is a filtration of $\operatorname{Sym}^r \mathscr{F}$,

$$\operatorname{Sym}^r \mathscr{F} = \mathscr{G}^0 \supseteq \mathscr{G}^1 \supseteq \cdots \supseteq \mathscr{G}^r \supseteq \mathscr{G}^{r+1} = 0,$$

with subquotients

$$\mathscr{G}^p / \mathscr{G}^{p+1} \cong (\operatorname{Sym}^p \mathscr{F}') \otimes (\operatorname{Sym}^{r-p} \mathscr{F}'').$$

Proof Since $\mathscr{F}', \mathscr{F}$, and $\mathscr{F}''$ are locally free, we can cover the ringed space $(X, \mathscr{O}_X)$ with sufficiently small affine open subsets $U = \operatorname{Spec} A$ such that $\mathscr{F}'|_U \cong \mathscr{O}_U^{\oplus p}$, $\mathscr{F}''|_U \cong \mathscr{O}_U^{\oplus q}$, and $\mathscr{F}|_U \cong \mathscr{O}_U^{\oplus (p+q)}$. Then on such U, we have exact sequence

$$0 \longrightarrow A^{\oplus p} \longrightarrow A^{\oplus (p+q)} \longrightarrow A^{\oplus q} \longrightarrow 0.$$

Let $e_1, \ldots, e_p$ be the standard basis for $A^{\oplus p}$ and $f_1, \ldots, f_q$ be the standard basis for $A^{\oplus q}$. Let $e_1', \ldots, e_p'$ be the image of $e_1, \ldots, e_p$ in $A^{\oplus (p+q)}$ and $f_1', \ldots, f_q'$ be any lift of $f_1, \ldots, f_q$ in $A^{\oplus (p+q)}$. Hence $e_1', \ldots, e_p', f_1', \ldots, f_q'$ forms a basis for $\mathscr{F}|_U$. The sheaf $\operatorname{Sym}^r \mathscr{F}|_U$ corresponds to the module of homogeneous polynomials of degree r in the variables $e_1', \ldots, e_p', f_1', \ldots, f_q'$ with coefficients in A. We define the submodule $\mathscr{G}^p(U) \subseteq \operatorname{Sym}^r \mathscr{F}(U)$ by setting

$$\mathscr{G}^p(U) := \bigoplus_{i=p}^{r} \operatorname{Sym}^i \mathscr{F}'(U) \otimes_{\mathscr{O}_X(U)} \operatorname{Sym}^{r-i} \mathscr{F}''(U).$$

This clearly defines a filtration

$$\operatorname{Sym}^r \mathscr{F}(U) = \mathscr{G}^0(U) \supseteq \mathscr{G}^1(U) \supseteq \cdots \supseteq \mathscr{G}^r(U) \supseteq \mathscr{G}^{r+1}(U) = 0,$$

moreover, by the construction, we have

$$\mathscr{G}^p(U)/\mathscr{G}^{p+1}(U) \cong (\operatorname{Sym}^p \mathscr{F}'(U)) \otimes_{\mathscr{O}_X(U)} (\operatorname{Sym}^{r-p} \mathscr{F}''(U)).$$

It remain to show that $\mathscr{G}^p$ is a well-defined locally free subsheaf that is independent of the choices of base. We first show that $\mathscr{G}^p$ is independent of the choices of base. If we change the base of $\mathscr{F}'$, the new basis is the linear combination of old basis, similarly, if we change the base of $\mathscr{F}''$, the new basis is the linear combination of old basis. Hence the number of generators in $\mathscr{F}$ do not change. Suppose we choose a different set of lifts $h_1, \dots, h_q$, then $h_1 - f_1', \dots, h_q - f_q'$ belong to the kernel of $A^{\oplus(p+q)} \to A^{\oplus q}$. By the exactness, we know that

$$h_j = f_j' + \sum_{k=1}^{p} a_{jk} e_k',$$

for all $j = 1, \dots, q$. Let $g \in \mathscr{G}^p(U)$ be a generator, then $g = e'^{l_1}_{i_1} \cdots e'^{l_s}_{i_s} f'^{d_1}_{j_1} \cdots f'^{d_t}_{j_t}$ where $\sum i_k \geq p$ and $\sum_k i_k + \sum_k j_k = r$. Then we have

$$g = e'^{l_1}_{i_1} \cdots e'^{l_s}_{i_s} (h_{j_1} - \sum_{m=1}^{p} a_{j_1 m} e_m')^{d_1} \cdots (h_{j_t} - \sum_{m=1}^{p} a_{j_t m} e_m'),$$

so g can be represented by the new generator. Notice that in the expansion, the degree of e' in each term is at least equal to the original degree of e' (which is $\geq p$). Therefore, the element remains in the filtration defined by the new basis. Similarly, if $g = e'^{l_1}_{i_1} \cdots e'^{l_s}_{i_s} h^{d_1}_{j_1} \cdots h^{d_t}_{j_t}$, then g can be represented by the monomial $e'^{l_1}_{i_1} \cdots e'^{l_s}_{i_s} f'^{d_1}_{j_1} \cdots f'^{d_t}_{j_t}$ where $\sum i_k \geq p$ and $\sum_k i_k + \sum_k j_k = r$. Hence $\mathscr{G}^p$ is independent of the choices of basis. We next show that $\mathscr{G}^p$ is a quasicoherent sheaf, recall Theorem 7.2.2, it suffices to show that for any $\operatorname{Spec} A_f \hookrightarrow \operatorname{Spec} A$ we have $\Gamma(\operatorname{Spec} A_f, \mathscr{G}^p) \cong \Gamma(\operatorname{Spec} A, \mathscr{G}^p)_f$. Define ring morphism from $\Gamma(\operatorname{Spec} A, \mathscr{G}^p)_f$ to $\Gamma(\operatorname{Spec} A_f, \mathscr{G}^p)$ as follow:

$$\begin{aligned} \varphi : \Gamma(\operatorname{Spec} A, \mathscr{G}^p)_f &\longrightarrow \Gamma(\operatorname{Spec} A_f, \mathscr{G}^p) \\ \frac{\sum_{i=p}^{r} m_i}{f^k} &\longmapsto \sum_{i=p}^{r} \frac{m_i}{f^k} \end{aligned}$$

where $m_i \in \operatorname{Sym}^i A^{\oplus p} \otimes_A \operatorname{Sym}^{r-i} A^{\oplus q}$. It is easy to see that φ is an isomorphism, which implies that $\mathscr{G}^p$ is quasicoherent sheaf, we are done. □

Proposition 15.2.13 (Useful)

Suppose $0 \to \mathscr{F}' \to \mathscr{F} \to \mathscr{F}'' \to 0$ is an exact sequence of locally free sheaves. Then for any r, there is a filtration of $\bigwedge^r \mathscr{F}$,

$$\bigwedge^r \mathscr{F} = \mathscr{G}^0 \supseteq \mathscr{G}^1 \supseteq \cdots \supseteq \mathscr{G}^r \supseteq \mathscr{G}^{r+1} = 0,$$

with subquotients

$$\mathscr{G}^p / \mathscr{G}^{p+1} \cong \left(\bigwedge^p \mathscr{F}'\right) \otimes \left(\bigwedge^{r-p} \mathscr{F}''\right)$$

for each p.
In particular, if the sheaves have finite rank, then

$$\det \mathscr{F} = (\det \mathscr{F}') \otimes (\det \mathscr{F}'').$$

Proof The proof is similar as Proposition 15.2.12. □

Proposition 15.2.14 (Perfect pairing of vector bundles)

Suppose $\mathscr{F}$ is locally free of rank n, there is an isomorphism

$$\bigwedge^r \mathscr{F} \xrightarrow{\sim} \left(\bigwedge^{n-r} \mathscr{F}\right)^{\vee} \otimes \bigwedge^n \mathscr{F}.$$

*This is called a **perfect pairing of vector bundles**.*

Proof Let X be a ringed space. Let $\mathscr{F}$ is a locally free sheaf of rank n on X. For any open $U \subseteq X$, the exterior powers $\bigwedge^r \mathscr{F}(U)$ and $\bigwedge^{n-r} \mathscr{F}(U)$ are $\mathscr{O}_X(U)$-modules. Define $\varphi_U^\flat : \bigwedge^r \mathscr{F}(U) \times \bigwedge^{n-r} \mathscr{F}(U) \to \bigwedge^n \mathscr{F}(U)$ as follow: for $U \subseteq X$ open, define

$$(s_{i_1} \wedge \cdots \wedge s_{i_r}) \times (t_{j_1} \wedge \cdots \wedge t_{j_{n-r}}) \longmapsto s_{i_1} \wedge \cdots \wedge s_{i_r} \wedge t_{j_1} \wedge \cdots \wedge t_{j_{n-r}}$$

as an element of $\bigwedge^n \mathscr{F}(U)$, linear extend this to all section of $\bigwedge^r \mathscr{F}(U) \times \bigwedge^{n-r} \mathscr{F}(U)$. Moreover, it is easy to see that φ is an $\mathscr{O}_X(U)$-bilinear function. By the universal property of tensor product, there exists unique $\mathscr{O}_X(U)$-linear map $\varphi_U : \bigwedge^r \mathscr{F}(U) \otimes \bigwedge^{n-r} \mathscr{F}(U) \to \bigwedge^n \mathscr{F}(U)$ such that the following diagram commutes.

$$\begin{tikzcd}
\bigwedge^r \mathscr{F}(U) \times \bigwedge^{n-r} \mathscr{F}(U) \arrow[r, "\varphi_U^\flat"] \arrow[d] & \bigwedge^n \mathscr{F}(U) \\
\bigwedge^r \mathscr{F}(U) \otimes_{\mathscr{O}_X(U)} \bigwedge^{n-r} \mathscr{F}(U) \arrow[ur, dashed, "\exists ! \varphi_U"'] &
\end{tikzcd}$$

It is easy to see φ_U is compatible with restriction maps of the sheaves, this yield a natural map of sheaves

$$\varphi : \bigwedge^r \mathscr{F} \otimes_{\mathscr{O}_X} \bigwedge^{n-r} \mathscr{F} \longrightarrow \bigwedge^n \mathscr{F}.$$

Recall Lemma 15.1.6, we have an isomorphism

$$\mathscr{H}om_{\mathscr{O}_X}\left(\bigwedge^r \mathscr{F} \otimes_{\mathscr{O}_X} \bigwedge^{n-r} \mathscr{F}, \bigwedge^n \mathscr{F}\right) \xrightarrow{\sim} \mathscr{H}om_{\mathscr{O}_X}\left(\bigwedge^r \mathscr{F}, \mathscr{H}om_{\mathscr{O}_X}\left(\bigwedge^{n-r} \mathscr{F}, \bigwedge^n \mathscr{F}\right)\right),$$

so there is a unique morphism $\psi : \bigwedge^r \mathscr{F} \to \mathscr{H}om_{\mathscr{O}_X}(\bigwedge^{n-r} \mathscr{F}, \bigwedge^n \mathscr{F})$ corresponds to φ, such that

$$\psi_U(\alpha)(\beta) = \alpha \wedge \beta,$$

where $\alpha \in \bigwedge^r \mathscr{F}(U)$ and $\beta \in \bigwedge^{n-r} \mathscr{F}(U)$.

We want to show that ψ is an isomorphism of sheaves. Since $\mathscr{F}$ is locally free of rank n, for every $x \in X$ there is an open neighborhood U such that $\mathscr{F}|_U \cong \mathscr{O}_U^{\oplus n}$. Let $\mathscr{F}(U) = \mathrm{Span}_{\mathscr{O}_X(U)}\{e_1, \ldots, e_n\}$, then $\bigwedge^r \mathscr{F}(U)$ is generated by $e_I := e_{i_1} \wedge \cdots \wedge e_{i_r}$ where $I = (i_1 < \cdots < i_r)$, $\bigwedge^{n-r} \mathscr{F}(U)$ is generated by $e_J := e_{j_1} \wedge \cdots \wedge e_{j_{n-r}}$ for $J = (j_1 < \cdots < j_{n-r})$, and $\bigwedge^n \mathscr{F}(U)$ is generated by $\omega := e_1 \wedge \cdots \wedge e_n$. We now calculate $\psi_U(e_I)(e_J)$. If $J = I^c$, the complement of I, then there is a unique permutation $\sigma_I \in S_n$ that reorders index $(i_1, \ldots, i_r, j_1, \ldots, j_{n-r})$ into $(1, 2, \ldots, n)$ and $e_I \wedge e_{I^c} = \mathrm{sgn}(\sigma_I)\omega$. If $J \neq I^c$, then $e_I \wedge e_J = 0$. Thus, we have

$$\psi_U(e_I)(e_J) = \begin{cases} \mathrm{sgn}(\sigma_I)\omega & \text{if } J = I^c, \\ 0 & \text{otherwise.} \end{cases} \tag{15.10}$$

It remain to show that $\psi|_U$ is an isomorphism. By Lemma 15.1.7, we have an isomorphism

$$\mathscr{H}om_{\mathscr{O}_X}\left(\bigwedge^{n-r} \mathscr{F}, \bigwedge^n \mathscr{F}\right) \xrightarrow{\sim} \bigwedge^{n-r} \mathscr{F}^\vee \otimes_{\mathscr{O}_X} \bigwedge^n \mathscr{F}.$$

Let $\{e_J^*\}$ be the dual basis of $\{e_J\}$, i.e., $e_J^*(e_{J'}) = \delta_{J,J'}$, then $\{e_J^* \otimes \omega\}$ is a basis of $\bigwedge^{n-r} \mathscr{F}(U)^\vee \otimes_{\mathscr{O}_X(U)} \bigwedge^n \mathscr{F}(U)$, so $\{e_J^* \otimes \omega\}$ is a basis of $\mathscr{H}om_{\mathscr{O}_X}\left(\bigwedge^{n-r} \mathscr{F}, \bigwedge^n \mathscr{F}\right)(U)$. Moreover, $e_J^* \otimes \omega$ correspond to morphism $e_{J'} \mapsto e_J^*(e_{J'})\omega$. By (15.10), for each I,

$$\psi_U(e_I)(e_J) = \mathrm{sgn}(\sigma_I)(e_{I^c}^* \otimes \omega)$$

so the matrix representing ψ_U is a signed permutation matrix, hence invertible. Therefore ψ_U is an isomorphism of $\mathscr{O}_X(U)$-module. It is easy to check that above discussion compatible with the restriction map, so $\psi|_U$ is an isomorphism. It follows that ψ is an isomorphism of $\mathscr{O}_X$-module, as we desired. $\square$

Proposition 15.2.15

Suppose $0 \to \mathscr{F}_1 \to \cdots \to \mathscr{F}_n \to 0$ is an exact sequence of finite rank locally free sheaves on X. Then "the alternating product of determinant bundles is trivial":

$$\det(\mathscr{F}_1) \otimes \det(\mathscr{F}_2)^\vee \otimes \det(\mathscr{F}_3) \otimes \det(\mathscr{F}_4)^\vee \otimes \cdots \otimes \det(\mathscr{F}_n)^{(-1)^n} \cong \mathscr{O}_X.$$

Proof Let

$$0 \longrightarrow \mathscr{F}_1 \xrightarrow{f^1} \mathscr{F}_2 \xrightarrow{f^2} \mathscr{F}_3 \xrightarrow{f^3} \cdots \xrightarrow{f^{n-1}} \mathscr{F}_n \longrightarrow 0$$

be a long exact sequence. Break it into short exact sequences, we have

$$0 \longrightarrow \operatorname{Ker} f^i \longrightarrow \mathscr{F}_i \longrightarrow \operatorname{Im} f^i \longrightarrow 0,$$

for $2 \leq i \leq n-1$ and $\mathscr{F}_1 \cong \operatorname{Ker} f^1$ and $\mathscr{F}_n \cong \operatorname{Im} f^{n-1}$. Note that $\operatorname{Im} f^n \cong \mathscr{F}_n$, we know that $\operatorname{Im} f^n$ is locally free sheaf, by Proposition 15.2.8, $\operatorname{Ker} f^i$ is locally free. By Proposition 15.2.13, we have

$$\det \mathscr{F}_i = (\det \operatorname{Ker} f^i) \otimes (\det \operatorname{Im} f^i),$$

notice that $\det \operatorname{Ker} f^i$, $\det \operatorname{Im} f^i$, and $\det \mathscr{F}_i$ are locally free sheaves of rank 1, recall Lemma 15.1.4, we have

$$\begin{aligned}
&\det(\mathscr{F}_1) \otimes \det(\mathscr{F}_2)^\vee \otimes \det(\mathscr{F}_3) \otimes \det(\mathscr{F}_4)^\vee \otimes \cdots \otimes \det(\mathscr{F}_n)^{(-1)^n} \\
\cong &\det \mathscr{F}_1 \otimes (\det \mathscr{F}_1)^\vee \otimes (\det \operatorname{Im} f^2)^\vee \otimes \det \operatorname{Ker} f^3 \otimes \det \operatorname{Im} f^3 \otimes \cdots \\
&\cdots \otimes (\det \operatorname{Ker} f^{n-1})^{(-1)^{n-1}} \otimes (\det \mathscr{F}_n)^{(-1)^{n-1}} \otimes (\det \mathscr{F}_n)^{(-1)^n} \\
\cong &\mathscr{O}_X \otimes \mathscr{O}_X \otimes \cdots \otimes \mathscr{O}_X \\
\cong &\mathscr{O}_X,
\end{aligned}$$

we win. $\square$

15.3 More pleasant properties of finite type and coherent sheaves

We now describe a number of important facts about quasicoherent sheaves when we have some finiteness conditions.

Proposition 15.3.1 ($\mathscr{H}om$ behaves reasonably if the source is finitely presented; cf. Exercise 2.38)

(a) *($\mathscr{H}om$(**fpr**, **qcoh**) **is qcoh**) Suppose $\mathscr{F}$ is a finitely presented (quasicoherent) sheaf on X, and $\mathscr{G}$ is a quasicoherent sheaf on X. Then $\mathscr{H}om(\mathscr{F}, \mathscr{G})$ is a quasicoherent sheaf.*

(b) *($\mathscr{H}om$(**fpr**, **coh**) **is coh**) If, further, $\mathscr{G}$ is coherent, then $\mathscr{H}om(\mathscr{F}, \mathscr{G})$ is also coherent.*

(c) *Suppose $\mathscr{F}$ is a finitely presented sheaf on X, and $\mathscr{G}$ is a quasicoherent sheaf on X. Then $\mathscr{H}om(\mathscr{F}, \square)$ gives a left-exact covariant functor $\mathbf{QCoh}_X \to \mathbf{QCoh}_X$ and $\mathbf{Coh}_X \to \mathbf{Coh}_X$, and that $\mathscr{H}om(\square, \mathscr{G})$ gives a left-exact contravariant functor from finitely presented sheaves on X to $\mathbf{QCoh}_X$.*

Proof ***For (a).*** Let $\mathscr{F}$ be a finitely presented sheaf on X, recall Definition 7.4.2, there exists an open cover $\{U_i\}$ of X such that the sequence

$$\mathscr{O}_{U_i}^{\oplus m} \longrightarrow \mathscr{O}_{U_i}^{\oplus n} \longrightarrow \mathscr{F}|_{U_i} \longrightarrow 0$$

is exact. Apply left-exact contravariant functor $\mathscr{H}om_{\mathscr{O}_{U_i}}(\square, \mathscr{G}|_{U_i})$, we have a exact sequence

$$\begin{tikzcd}
0 \arrow[r] & \mathscr{H}om_{\mathscr{O}_{U_i}}(\mathscr{F}|_{U_i}, \mathscr{G}|_{U_i}) \arrow[r] & \mathscr{H}om_{\mathscr{O}_{U_i}}(\mathscr{O}_{U_i}^{\oplus n}, \mathscr{G}|_{U_i}) \arrow[r] & \mathscr{H}om_{\mathscr{O}_{U_i}}(\mathscr{O}_{U_i}^{\oplus m}, \mathscr{G}|_{U_i}) \\
0 \arrow[r] & \mathscr{H}om_{\mathscr{O}_X}(\mathscr{F}, \mathscr{G})|_{U_i} \arrow[r] \arrow[u, leftrightarrow, "\sim"] & \mathscr{G}|_{U_i}^{\oplus n} \arrow[r] \arrow[u, leftrightarrow, "\sim"'] & \mathscr{G}|_{U_i}^{\oplus m}. \arrow[u, leftrightarrow, "\sim"']
\end{tikzcd}$$

Since $\mathscr{G}|_{U_i}$ is quasicoherent sheaf, so are $\mathscr{G}|_{U_i}^{\oplus n}$ and $\mathscr{G}|_{U_i}^{\oplus m}$. Recall that $\mathbf{QCoh}_X$ is Abelian category, we know that $\mathscr{H}om_{\mathscr{O}_X}(\mathscr{F}, \mathscr{G})|_{U_i}$ is quasicoherent sheaf, and therefore $\mathscr{H}om_{\mathscr{O}_X}(\mathscr{F}, \mathscr{G})$ is quasicoherent sheaf.

For (b). The category of coherent sheaves $\mathbf{Coh}_X$ is abelian (§7.4), by the proof of part (a), $\mathscr{H}om_{\mathscr{O}_X}(\mathscr{F}, \mathscr{G})|_{U_i}$ is coherent sheaf, and therefore $\mathscr{H}om_{\mathscr{O}_X}(\mathscr{F}, \mathscr{G})$ is coherent sheaf.

For (c). By part (a) and part (b), we get this immediately. □

Example 15.3 ($\mathscr{H}om$(qcoh, qcoh) is not quasicoherent) For an example of quasicoherent sheaves $\mathscr{F}$ and $\mathscr{G}$ on a scheme X such that $\mathscr{H}om(\mathscr{F}, \mathscr{G})$ is not quasicoherent, let A be a DVR with uniformizer t, let $X = \operatorname{Spec} A$, and let $\mathscr{F} = \widetilde{M}$ and $\mathscr{G} = \widetilde{N}$ with $M = \bigoplus_{i=1}^{\infty} A$ and $N = A$. Then $M_t = \bigoplus_{i=1}^{\infty} A_t$, and, of cause, $N_t = A_t$. Consider the homomorphism $\varphi : M_t \to N_t$ sending 1 in the i-th factor of M_t to $1/t^i$. Let $f \in \operatorname{Hom}_A(M, N)$ and $M = \operatorname{Span}\{e_i\}$, then $f(e_i) = 0$ for large i. Since $\operatorname{Hom}_A(M, N)_t \cong \operatorname{Hom}_A(M, N) \otimes_A A_t$, the element in $\operatorname{Hom}_A(M, N)_t$ is of form $f \otimes \frac{a}{t^{-k}}$, then $\left(f \otimes \frac{a}{t^{-k}}\right)(e_i) = 0$ for large i, hence φ is not the localization of any element of $\operatorname{Hom}_A(M, N)$, by Theorem 7.2.2, $\mathscr{H}om(\mathscr{F}, \mathscr{G})$ is not quasicoherent.

Duals of coherent sheaves

From Proposition 15.3.1, if $\mathscr{F}$ is coherent, its dual $\mathscr{F}^\vee := \mathscr{H}om(\mathscr{F}, \mathscr{O}_X)$ is too. This generalizes the notion of duals of vector bundles in Lemma 15.1.3. Your argument there generalizes to show that there is always a natural "double dual" morphism $\mathscr{F} \to (\mathscr{F}^\vee)^\vee$. Unlike in the vector bundle case, there is not always an isomorphism.

Example 15.4 ("double dual" morphism of coherent sheaves may not be an isomorphism) Let $\mathscr{F}$ be the coherent sheaf associated to $k[t]/(t)$ on $\mathbb{A}^1 = \operatorname{Spec} k[t]$, then $\mathscr{F}^\vee(\mathbb{A}^1) = \mathscr{H}om(\mathscr{F}, \mathscr{O}_X)(\mathbb{A}^1) \cong \operatorname{Hom}(k, k[t])^\sim = \widetilde{0}$, which implies that $\mathscr{F}^\vee = 0$. It is clearly $\mathscr{F} \ncong \mathscr{F}^{\vee\vee}$.

Definition 15.3.1 (Reflexive sheaves)

*Let $(X, \mathscr{O}_X)$ be a ringed space. Let $\mathscr{F}$ be a coherent sheaf on X. We call $\mathscr{F}$ is a **reflexive sheaf** if canonical morphism $\mathscr{F} \to (\mathscr{F}^\vee)^\vee$ is an isomorphism.*

Definition 15.3.2 (Trace)

*Let $(X, \mathscr{O}_X)$ be a ringed space. Let $\mathscr{F}$ be a coherent sheaf on X. The **trace** is defined as follows*

$$\begin{aligned} \operatorname{tr} : \mathscr{H}om_{\mathscr{O}_X}(\mathscr{F}, \mathscr{F}) \cong \mathscr{F}^\vee \otimes_{\mathscr{O}_X} \mathscr{F} &\longrightarrow \mathscr{O}_X, \\ \lambda \otimes s &\longmapsto \lambda(s), \end{aligned} \tag{15.11}$$

*where λ and s are sections of $\mathscr{F}^\vee$ and of $\mathscr{F}$, respectively, over some open subset $U \subseteq X$. If u is an endomorphism of $\mathscr{F}|_U$, we call its image $\operatorname{tr}(u) \in \Gamma(U, \mathscr{O}_X)$ the **trace of** u.*

Note *Definition 15.3.2 coincides with the usual notion of trace if X is affine and $\mathscr{F}$ is free: Let R be a ring, $n \geq 1$ an integer, and let $(e_1, \ldots, e_n)$ be the standard basis of R^n. We first show that under the isomorphism*

$$(R^n)^\vee \otimes_R R^n \xrightarrow{\sim} \operatorname{End}_R(R^n) = M_n(R)$$

the base vector $e_i^\vee \otimes e_j$ is sent to the matrix all of whose coefficients are zero except for the (j,i)-th coefficient which is 1. In fact, $\Phi : (R^n)^\vee \otimes_R R^n \to \mathrm{End}_R(R^n)$ is given by

$$\Phi(\varphi \otimes v)(x) = \varphi(x) \cdot v, \qquad \forall x \in R^n,$$

where $\varphi \in (R^n)^\vee$ and $v \in R^n$. Say $u_{ij} := \Phi(e_i^\vee \otimes e_j)$, for each base $e_k \in R^n$, we have

$$u(e_k) = (e_i^\vee \otimes e_j)(e_k) = \delta_{ik} \cdot e_j.$$

Hence

$$u_{ij}(e_k) = \begin{cases} e_j & \text{if } k = i \\ 0 & \text{if } k \neq i \end{cases}$$

It follows that u_{ij} correspond to matrix E_{ji} (the matrix at the j-th row and i-th column is 1, and other coefficients are all 0).

We next show that the trace of $\tau \in (R^n)^\vee \otimes_R R^n$ in the sense of (15.11) is equal to the trace of corresponding matrix. Write τ as

$$\tau = \sum_{i,j} c_{ij}(e_i^\vee \otimes e_j),$$

let u be the linear map which correspond to τ under above isomorphism, apply u to e_k, we have

$$u(e_k) = \sum_{i,j} c_{ij} e_i^\vee(e_k) e_j = \sum_{j=1}^{n} c_{kj} e_j.$$

Let $A = (a_{ij})$ be the corresponding matrix of linear map u, then $a_{ij} = c_{ji}$. Consider $\mathrm{tr}(u)$,

$$\mathrm{tr}(u) = \mathrm{tr}\left(\sum_{i,j} c_{ij}(e_i^\vee \otimes e_j)\right) = \sum_{i,j} c_{ij} \cdot \mathrm{tr}(e_i^\vee \otimes e_j) = \sum_{i,j} c_{ij}\delta_{i,j} = \sum_{i} c_{ii} = \mathrm{tr}(A),$$

as we desired.

Proposition 15.3.2

Suppose

$$0 \longrightarrow \mathscr{F} \longrightarrow \mathscr{G} \longrightarrow \mathscr{H} \longrightarrow 0 \tag{15.12}$$

is an exact sequence of quasicoherent sheaves on a scheme X, where $\mathscr{H}$ is a locally free quasicoherent sheaf, and suppose $\mathscr{E}$ is a quasicoherent sheaf. By left-exactness of $\mathscr{H}om$, the sequence

$$0 \longrightarrow \mathscr{H}om(\mathscr{H}, \mathscr{E}) \longrightarrow \mathscr{H}om(\mathscr{G}, \mathscr{E}) \longrightarrow \mathscr{H}om(\mathscr{F}, \mathscr{E})$$

is exact. Moreover, it is also exact on the right, i.e., the sequence

$$0 \longrightarrow \mathscr{H}om(\mathscr{H}, \mathscr{E}) \longrightarrow \mathscr{H}om(\mathscr{G}, \mathscr{E}) \longrightarrow \mathscr{H}om(\mathscr{F}, \mathscr{E}) \longrightarrow 0$$

is exact.

In particular, if $\mathscr{F}, \mathscr{G}$, and $\mathscr{H}$ are coherent, and $\mathscr{H}$ is locally free, then we have an exact sequence of coherent sheaves

$$0 \longrightarrow \mathscr{H}^\vee \longrightarrow \mathscr{G}^\vee \longrightarrow \mathscr{F}^\vee \longrightarrow 0.$$

Proof Since exactness can be checked locally, we may assume that $X = \operatorname{Spec} A$. Since $\mathscr{H}$ is a locally free sheaf, we may assume that $\mathscr{H} = \widetilde{A^{\oplus I}}$, so exact sequence (15.12) can be written as

$$0 \longrightarrow M \longrightarrow N \longrightarrow A^{\oplus I} \longrightarrow 0. \tag{15.13}$$

Note that $A^{\oplus I}$ is free, so is projective, the sequence (15.13) is split, and therefore $N \cong A^{\oplus I} \oplus M$, by applying

$\widetilde{\square}$, we have $\mathscr{G} \cong \mathscr{F} \oplus \mathscr{H}$. Since $\mathscr{H}om(\square, \mathscr{E})$ is an additive functor, we have isomorphism

$$\mathscr{H}om(\mathscr{G}, \mathscr{E}) \cong \mathscr{H}om(\mathscr{F} \oplus \mathscr{H}, \mathscr{E}) \cong \mathscr{H}om(\mathscr{F}, \mathscr{E}) \oplus \mathscr{H}om(\mathscr{H}, \mathscr{E}).$$

It follows that the sequence

$$0 \longrightarrow \mathscr{H}om(\mathscr{H}, \mathscr{E}) \longrightarrow \mathscr{H}om(\mathscr{G}, \mathscr{E}) \longrightarrow \mathscr{H}om(\mathscr{F}, \mathscr{E}) \longrightarrow 0$$

is exact.

In particular, if $\mathscr{F}, \mathscr{G}$, and $\mathscr{H}$ are coherent, and $\mathscr{H}$ is locally free, recall Proposition 15.3.1, we have an exact sequence of coherent sheaves

$$0 \longrightarrow \mathscr{H}^\vee \longrightarrow \mathscr{G}^\vee \longrightarrow \mathscr{F}^\vee \longrightarrow 0.$$

$\square$

Proposition 15.3.3 (Supp(FT $\mathscr{F}$) is closed)

Suppose $\mathscr{F}$ is a sheaf of abelian groups.

(a) The support of a finite type quasicoherent sheaf on a scheme X is a closed subset.

(b) The support of a quasicoherent sheaf need not be closed.

Proof ***For (a).*** Recall Definition 3.7.5, $\operatorname{Supp} \mathscr{F} = \{p \in X : \mathscr{F}_p \neq 0\}$. Since $\mathscr{F}$ is a finite type quasicoherent sheaf, for each affine open $U \subseteq X$, $\mathscr{F}|_U \cong \widetilde{M_U}$, where M_U is a finitely generated $\Gamma(U, \mathscr{O}_X)$-module. By Proposition 7.6.2, we know that $\operatorname{Supp} M_U$ is a closed subset of U. Note that

$$\operatorname{Supp} \mathscr{F} = \bigcup_{\text{affine } U \subseteq X} \operatorname{Supp} \mathscr{F}|_U = \bigcup_{\text{affine } U \subseteq X} \operatorname{Supp} M_U,$$

recall a topology fact that $Z \subseteq Y$ is closed if and only if there exists an open cover $\{V_i\}_i$ of Y such that $Z \cap V_i$ is closed in V_i for all i, hence $\operatorname{Supp} \mathscr{F}$ is closed in X.

For (b). We give a counterexample to show (b). Let $A = \mathbb{C}[t]$ and $M = \bigoplus_{a \in \mathbb{C}} \mathbb{C}[t]/(t-a)$, then M is an A-module. Let $X = \operatorname{Spec} A$ and $\mathscr{F} = \widetilde{M}$, then $\mathscr{F}$ is a quasicoherent sheaf on X, we claim that $\operatorname{Supp} \mathscr{F}$ is not closed in X. Let $[\mathfrak{p}] \in X$. If $\mathfrak{p} = (t-b)$ is a closed point, then

$$\begin{aligned} M_{\mathfrak{p}} &= \bigoplus_{a \in \mathbb{C}} (\mathbb{C}[t]/(t-a) \otimes_{\mathbb{C}[t]} \mathbb{C}[t]_{(t-b)}) \\ &= (\mathbb{C}[t]/(t-b) \otimes_{\mathbb{C}[t]} \mathbb{C}[t]_{(t-b)}) \oplus \left(\bigoplus_{\substack{a \in \mathbb{C} \\ a \neq b}} (\mathbb{C}[t]/(t-a) \otimes_{\mathbb{C}[t]} \mathbb{C}[t]_{(t-b)}) \right) \\ &\cong \mathbb{C} \oplus \left(\bigoplus 0 \right) \\ &\cong \mathbb{C}. \end{aligned}$$

If $\mathfrak{p} = (0)$, then

$$M_{(0)} = \bigoplus_{a \in \mathbb{C}} (\mathbb{C}[t]/(t-a) \otimes_{\mathbb{C}[t]} \mathbb{C}[t]_{(0)}) \cong \bigoplus_{a \in \mathbb{C}} (\mathbb{C}[t]/(t-a) \otimes_{\mathbb{C}[t]} \mathbb{C}(t)) \cong 0.$$

Hence $\operatorname{Supp} \mathscr{F} = \operatorname{Supp} M$ is all closed points of X. The closed subset of X are either the whole space or finite sets of points, so $\operatorname{Supp} \mathscr{F}$ is not closed in X. $\square$

We next come to a geometric interpretation of Nakayama's Lemma, which is why Nakayama's Lemma should be considered a geometric fact (with an algebraic proof). Informally: if some sections of a finite type quasicoherent sheaf generate the sheaf "at p" (the fiber at p), then they generate "near p" (in an actual open neighborhood of p).

Proposition 15.3.4 (Geometric Nakayama)

Suppose X is a scheme, and $\mathscr{F}$ is a finite type quasicoherent sheaf. If $U \subseteq X$ is an open neighborhood of $p \in X$ and $a_1, \ldots, a_n \in \mathscr{F}(U)$ so that their images $a_1|_p, \ldots, a_n|_p$ generate the fiber $\mathscr{F}|_p$ (defined as $\mathscr{F}_p \otimes_{\mathscr{O}_{X,p}} \kappa(p)$), then there is an affine open neighborhood $\operatorname{Spec} A \subseteq U$ of p such that "$a_1|_{\operatorname{Spec} A}, \ldots, a_n|_{\operatorname{Spec} A}$ generate $\mathscr{F}|_{\operatorname{Spec} A}$" in the following senses:

(i) $a_1|_{\operatorname{Spec} A}, \ldots, a_n|_{\operatorname{Spec} A}$ generate $\mathscr{F}(\operatorname{Spec} A)$ as an A-module;

(ii) for any $q \in \operatorname{Spec} A$, $a_1, \ldots, a_n$ generate the stalk $\mathscr{F}_q$ as an $\mathscr{O}_{X,q}$-module (and hence for any $q \in \operatorname{Spec} A$, the fibers $a_1|_q, \ldots, a_n|_q$ generate the fiber $\mathscr{F}|_q$ as a $\kappa(q)$-vector space).

In particular, if $\mathscr{F}|_p = 0$, then there exists an open neighborhood V of p such that $\mathscr{F}|_V = 0$.

Proof Clearly, there exists an affine open $\operatorname{Spec} A \subseteq U$ such that $\operatorname{Spec} A \ni p$. Since $\mathscr{F}$ is finite type quasicoherent sheaf, we may assume $\mathscr{F}|\operatorname{Spec} A \cong \widetilde{M}$ where M is a finitely generated A-module. Say $p = [\mathfrak{p}] \in \operatorname{Spec} A$. By condition, $\mathscr{F}|_p = \mathscr{F}_p \otimes \kappa(p) \cong M_\mathfrak{p}/\mathfrak{p}M_\mathfrak{p}$ is generated by $a_1|_p, \ldots, a_n|_p$ where $a_1|_{\operatorname{Spec} A}, \ldots, a_1|_{\operatorname{Spec} A} \in M$. By Nakayama's Lemma 9.2.6 (version 4), $a_1|_p, \ldots, a_1|_p$ generate $M_\mathfrak{p}$. Let N be an A-module generated by $a_1|_{\operatorname{Spec} A}, \ldots, a_1|_{\operatorname{Spec} A}$. Consider M/N, localize at $\mathfrak{p}$, we have $M_\mathfrak{p}/N_\mathfrak{p} = 0$, hence $\mathfrak{p} \notin \operatorname{Supp} M/N$. It is clearly M/N a finitely generated A-module. By Proposition 7.6.2, $\operatorname{Supp} M/N$ is closed in $\operatorname{Spec} A$, and therefore there exists $f \in A$ such that $\mathfrak{p} \in \operatorname{Spec} A_f$ and for each $\mathfrak{q} \in \operatorname{Spec} A_f$ we have $\mathfrak{q} \notin \operatorname{Supp} M/N$. It follows that M_f is generated by $a_1|_{\operatorname{Spec} A_f}, \ldots, a_1|_{\operatorname{Spec} A_f}$, this is (i). Moreover, for any $\mathfrak{q} \in \operatorname{Spec} A_f$, $M_\mathfrak{q}/N_\mathfrak{q} = 0$, i.e., $a_1, \ldots, a_n$ generate the stalk $\mathscr{F}_\mathfrak{q} \cong M_\mathfrak{q}$, this is (ii). □

Proposition 15.3.5 (For a finitely presented sheaf, local freeness is a stalk-local property)

Suppose $\mathscr{F}$ is a finitely presented sheaf on a scheme X. Then $\mathscr{F}$ is locally free if and only if $\mathscr{F}_p$ is a free $\mathscr{O}_{X,p}$-module for all $p \in X$.

In particular, if $\mathscr{F}_p$ is a free $\mathscr{O}_{X,p}$-module for some $p \in X$, then $\mathscr{F}$ is locally free in some open neighborhood of p.

Translation: For a finitely presented sheaf, local freeness is a stalk-local property, and free stalks imply local freeness nearby.

Proof If $\mathscr{F}$ is locally free, it is clearly that $\mathscr{F}_p$ is a free $\mathscr{O}_{X,p}$-module for all $p \in X$.

Conversely, suppose $\mathscr{F}_p$ is a free $\mathscr{O}_{X,p}$-module for all $p \in X$, we want to show that $\mathscr{F}$ is locally free. Let $p \in X$, by condition, $\mathscr{F}_p$ is finitely generated, then $\mathscr{F}_p \cong \mathscr{O}_{X,p}^{\oplus n}$ for some n. Hence, there exits an open neighborhood U of p and $a_1, \ldots, a_n \in \mathscr{F}(U)$ such that $\mathscr{F}_p$ is generated by $a_{1,p}, \ldots, a_{n,p}$, so their images $a_1|_p, \ldots, a_n|_p$ generate the fiber $\mathscr{F}|_p$. By Geometric Nakayama 15.3.4, there is an affine open neighborhood $Y := \operatorname{Spec} A \subseteq U$ of p such that $a_1|_Y, \ldots, a_n|_Y$ generate $\mathscr{F}|_Y$ and for any $q \in Y$, $a_{1,q}, \ldots, a_{n,q}$ generate $\mathscr{F}_q$. Hence, $\mathscr{O}_Y^{\oplus n} \to \mathscr{F}|_Y$ is surjection. Let $\mathscr{K} := \operatorname{Ker}(\mathscr{O}_Y^{\oplus n} \to \mathscr{F}|_Y)$ be the kernel, then we have the following exact sequence

$$0 \longrightarrow \mathscr{K} \hookrightarrow \mathscr{O}_Y^{\oplus n} \twoheadrightarrow \mathscr{F}|_Y \longrightarrow 0.$$

Since $\mathscr{F}$ is finitely presented, by Lemma 7.4.1, we know that $\mathscr{K}$ is finite type. Recall Proposition 15.3.3, $Z := \operatorname{Supp} \mathscr{K}$ is closed in Y. Since $\mathscr{F}_p \cong \mathscr{O}_{X,p}^{\oplus n}$, we know that $\mathscr{K}_p = 0$, so $p \notin Z$. Hence $\mathscr{O}_{Y\setminus Z}^{\oplus n} \to \mathscr{F}|_{Y\setminus Z}$ is an isomorphism, which implies that $\mathscr{F}$ is locally free. □

Corollary 15.3.1

Let X be an integral scheme. Then any finitely presented sheaf $\mathscr{F}$ over X is automatically free over the generic point, moreover, $\mathscr{F}$ is locally free over a dense open subset.

Proof Let η be the generic point of X, then $\mathscr{O}_{X,\eta} \cong K(X)$ is a field. Since $\mathscr{F}$ is finitely presented, $\mathscr{F}_\eta$ is finitely presented, in particular, $\mathscr{F}_\eta$ is a finitely generated $K(X)$-module, so $\mathscr{F}_\eta$ is a finite dimensional vector space, this shows the first statement. By Proposition 15.3.5, $\mathscr{F}$ is locally free in some neighborhood of η, since η is generic point and X is irreducible, this neighborhood is clearly dense in X. $\square$

Corollary 15.3.2

Any finitely presented sheaf that is 0 at the generic point of an irreducible scheme is 0 on a dense open subset.

Proof This is a straightforward conclusion of Proposition 15.3.5. $\square$

Corollary 15.3.1 and Corollary 15.3.2 show utility of generic points; such statements would have been more mysterious in classical algebraic geometry. An example showing the necessity of finiteness hypotheses: the $\mathbb{Z}$-module $A \subseteq \mathbb{Q}$, where A consists of fractions with square-free denominator, see [Sta25, Section 065J].

Proposition 15.3.6

(a) Torsion-free coherent sheaves (Definition 7.1.5) on a regular curve (hence implicitly locally Noetherian) are locally free.

(b) Torsion coherent sheaves (Definition 7.1.6) on a quasi-compact regular integral curve are supported at a finite number of closed points.

(c) Suppose $\mathscr{F}$ is a coherent sheaf on a quasicompact (for convenience) regular curve. There is a canonical short exact sequence

$$0 \longrightarrow \mathscr{F}_{\text{tors}} \longrightarrow \mathscr{F} \longrightarrow \mathscr{F}_{\text{lf}} \longrightarrow 0,$$

where $\mathscr{F}_{\text{tors}}$ is a torsion sheaf, and $\mathscr{F}_{\text{lf}}$ is locally free.

Proof ***For (a).*** Let X be a regular curve and $\mathscr{F}$ is a torsion-free coherent sheaf, we want to show that $\mathscr{F}$ is a locally free sheaf. Recall Proposition 15.3.5, it suffices to show that $\mathscr{F}_p$ is a free $\mathscr{O}_{X,p}$-module for all $p \in X$. Pick $p \in X$, since X is dimension one, p is closed point or generic point of X. If p is generic point of X, we know that $\mathscr{O}_{X,p}$ is a field. Since $\mathscr{F}_p$ is coherent module, $\mathscr{F}_p$ is finitely generated $\mathscr{O}_{X,p}$-module, so $\mathscr{F}_p$ is a finite dimensional vector space, which implies $\mathscr{F}_p$ is free $\mathscr{O}_{X,p}$-module. If p is a closed point, since X is regular curve, $\mathscr{O}_{X,p}$ is a DVR, by Theorem 14.5.5, $\mathscr{O}_{X,p}$ is a PID. By Structure Theorem for finitely generated modules over PID 15.2.1, we know that $\mathscr{F}_p$ is torsion free if and only if $\mathscr{F}_p$ is free.

For (b). Let Y be a quasi-compact regular integral curve and $\mathscr{G}$ is a torsion coherent sheaf. Let η be the generic point of X, since $\mathscr{G}$ is a torsion sheaf, we have $\mathscr{G}_\eta = 0$. It follows that $\eta \notin \operatorname{Supp}\mathscr{G}$. By Proposition 15.3.3, $\operatorname{Supp}\mathscr{G}$ is closed in X, so $\operatorname{Supp}\mathscr{G}$ is proper closed in X. Note that $\dim X = 1$, $\dim \operatorname{Supp}\mathscr{G} = 0$. Since X is Noetherian and irreducible, $\operatorname{Supp}\mathscr{G}$ consists of finite number of closed points.

For (c). Suppose $\mathscr{F}$ is a coherent sheaf on a quasi-compact regular curve X. Define torsion presheaf as follow: for any open subset $U \subseteq X$,

$$\mathscr{F}_{\text{tors,pre}}(U) := \{s \in \mathscr{F}(U) : \exists \text{ nonzero-divisor } a \in \mathscr{O}_X(U), as = 0\}.$$

Let torsion sheaf $\mathscr{F}_{\text{tors}} := \mathscr{F}^{\text{sh}}_{\text{tors,pre}}$ be the sheafification of torsion presheaf $\mathscr{F}_{\text{tors,pre}}$. Let $\mathscr{F}_{\text{lf}}$ be the quotient sheaf $\mathscr{F}/\mathscr{F}_{\text{tors}}$, then we have an exact sequence

$$0 \longrightarrow \mathscr{F}_{\text{tors}} \longrightarrow \mathscr{F} \longrightarrow \mathscr{F}_{\text{lf}} \longrightarrow 0.$$

We first show that $\mathscr{F}_{\text{tors}}$ is a torsion sheaf. Let η be a generic point of X, consider $(\mathscr{F}_{\text{tors}})_\eta$, we have

$$(\mathscr{F}_{\text{tors}})_\eta = \{s \in \mathscr{F}_\eta : \exists \text{ nonzero-divisor } a \in \mathscr{O}_{X,\eta}, as = 0\}.$$

Note that $\mathscr{O}_{X,\eta}$ is a field, $(\mathscr{F}_{\text{tors}})_\eta = 0$, which implies that $\mathscr{F}_{\text{tors}}$ is a torsion sheaf.

We next show that $\mathscr{F}_{\text{lf}}$ is locally free. Recall Proposition 15.3.5, it suffices to show that $(\mathscr{F}_{\text{lf}})_p$ is a free $\mathscr{O}_{X,p}$-module for all $p \in X$. Pick $p \in X$, since X is regular curve, p is generic point of X or closed point of X. If p is generic point, we have $(\mathscr{F}_{\text{lf}})_p = \mathscr{F}_p/\mathscr{F}_{\text{tors},p} = \mathscr{F}_p$. Since $\mathscr{O}_{X,p}$ is a field and $\mathscr{F}_p$ is a coherent $\mathscr{O}_{X,p}$-module, so $\mathscr{F}_p$ is free $\mathscr{O}_{X,p}$-module. If p is a closed point of X, then $(\mathscr{F}_{\text{lf}})_p = \mathscr{F}_p/\mathscr{F}_{\text{tors},p}$. By construction, we have $\mathscr{F}_{\text{tors},p} = \operatorname{Tor}\mathscr{F}_p$. Note that $\mathscr{O}_{X,p}$ is a DVR, and thus $\mathscr{O}_{X,p}$ is a PID, by Structure Theorem for finitely generated modules over PID 15.2.1, we know that $(\mathscr{F}_{\text{lf}})_p = \mathscr{F}_p/\operatorname{Tor}\mathscr{F}_p$ is free, as we desired. □

Remark 15.17 Structure theorem for finitely generated modules over a PID 15.2.1 is false without the finite generation hypothesis: consider $M = K(A)$ for a suitably general ring A (for example, $A = \mathbb{Z}$ and $M = \mathbb{Q}$).

It is also false if we give up the principal ideal domain hypothesis: consider $(x, y) \subseteq \mathbb{C}[x, y]$ and $(x, y) \subseteq \mathbb{C}[x, y]$. For $(x, y) \subseteq \mathbb{C}[x, y]$, $\mathbb{C}[x, y]$ is integral domain, so (x, y) is torsion-free. If (x, y) is free, there must $\operatorname{rank}(x, y) = 1$, which implies that (x, y) is principal ideal, this is impossible. Similarly, in the case $(x, y) \subseteq \mathbb{C}[x, y]/(xy)$, one check that (x, y) is torsion free $\mathbb{C}[x, y]/(xy)$-module and it is not free.

These example show that Proposition 15.3.6 (a) is false if we give up the "dimension 1" or "regular" hypothesis.

15.3.1 The fiber and rank of a quasicoherent sheaf at a point

We now apply the definition of fiber and rank of an $\mathscr{O}_X$-module at a point $p \in X$ (Definition 5.3.9) to a quasicoherent sheaf $\mathscr{F}$ on a scheme X.

Definition 15.3.3 (The fiber and the rank of an quasicoherent sheaf at a point)

*Suppose X is a scheme and $\mathscr{F}$ is a quasicoherent sheaf on X. The **fiber** of $\mathscr{F}$ at p is the vector space*

$$\mathscr{F}|_p := \mathscr{F}_p/\mathfrak{m}\mathscr{F}_p = \mathscr{F}_p \otimes_{\mathscr{O}_{X,p}} \kappa(p),$$

where $\mathfrak{m}$ is the maximal ideal of $\mathscr{O}_{X,p}$, and $\kappa(p) = \mathscr{O}_{X,p}/\mathfrak{m}$ is residue field at p.

*A **section** of $\mathscr{F}$ over an open set containing p can be said to take on a value at p, which is an element of this vector space.*

*The **rank** of a quasicoherent sheaf $\mathscr{F}$ at a point p is $\operatorname{rank}_p \mathscr{F} := \dim_{\kappa(p)} \mathscr{F}_p/\mathfrak{m}\mathscr{F}_p$ (possibly infinite). More explicity, on any affine set $\operatorname{Spec} A$ where $p = [\mathfrak{p}]$ and $\mathscr{F}(\operatorname{Spec} A) = M$, the rank is $\dim_{K(A/\mathfrak{p})} M_\mathfrak{p}/\mathfrak{p}M_\mathfrak{p}$.*

Remark 15.18 Note that this definition of rank is consistent with the notion of rank of a locally free sheaf. In the locally free case, the rank is a (locally) constant function of the point. The converse is sometimes true; see Proposition 15.3.8.

✍ **Exercise 15.4** Consider the coherent sheaf $\mathscr{F}$ on $\mathbb{A}^1_k = \operatorname{Spec} k[t]$ corresponding to the module $k[t]/(t)$. Find the rank of $\mathscr{F}$ at every point of $\mathbb{A}^1$. Don't forget the generic point!

Solution *Let $p = [\mathfrak{p}] \in \operatorname{Spec} k[t]$, consider $\mathscr{F}|_p$.*

If $(t) \subseteq \mathfrak{p}$, since (t) is maximal ideal of $k[t]$, we have $\mathfrak{p} = (t)$, then $(t) \cap (k[t] - \mathfrak{p}) = \varnothing$. So,

$$\mathscr{F}|_p = (k[t]/(t))_{(t)}/(t)(k[t]/(t))_{(t)} \cong \frac{(k[x]_{(t)}/(t)k[t]_{(t)})}{(t)(k[x]_{(t)}/(t)k[t]_{(t)})} \cong k,$$

and therefore that the rank of $\mathscr{F}$ at (t) is $\dim_{\kappa((t))} k = 1$.

If $(t) \nsubseteq \mathfrak{p}$, then $(t) \cap (k[t] - \mathfrak{p}) \neq \varnothing$ and $\mathfrak{p} = (f)$ for some $f \neq t$ (since $k[t]$ is PID). Hence,

$$(k[t]/(t))_{\mathfrak{p}} \cong k[t]/(t) \otimes_{k[t]} k[t]_{\mathfrak{p}} \cong 0,$$

and thus $\mathscr{F}|_p \cong 0$. It follows that that the rank of $\mathscr{F}$ at p is $\dim_{\kappa(p)} 0 = 0$.

If $p := [(0)]$ is a generic point of $\mathbb{A}^1$, then $(k[t]/(t))_{(0)} \cong (k[t]/(t)) \otimes_{k[t]} k(t) \cong 0$. So, $\mathscr{F}|_p = 0$, it follows that the rank of $\mathscr{F}$ at (0) is $\dim_{k(t)} 0 = 0$.

In conclusion, we have

$$\operatorname{rank}_p \mathscr{F} = \begin{cases} 1 & p = [(t)] \\ 0 & p = [(f)] \text{ for some } f \neq t \\ 0 & p \text{ is generic point of } \mathbb{A}^1. \end{cases}$$

Lemma 15.3.1

Suppose X is a scheme and $\mathscr{F}, \mathscr{G}$ are quasicoherent sheaves on X. Then for any point $p \in X$, we have

$$\operatorname{rank}_p(\mathscr{F} \oplus \mathscr{G}) = \operatorname{rank}_p \mathscr{F} + \operatorname{rank}_p \mathscr{G}$$

and

$$\operatorname{rank}_p(\mathscr{F} \otimes \mathscr{G}) = \operatorname{rank}_p \mathscr{F} \cdot \operatorname{rank}_p \mathscr{G}.$$

Proof Let $p := [\mathfrak{p}] \in \operatorname{Spec} A \subseteq X$ such that $\mathscr{F}|_U \cong \widetilde{M}$ and $\mathscr{G}|_U \cong \widetilde{N}$ where M, N are A-modules. Note that

$$(M \oplus N)_{\mathfrak{p}} \otimes_{A_{\mathfrak{p}}} \kappa(p) \cong (M_{\mathfrak{p}} \otimes_{A_{\mathfrak{p}}} \kappa(p)) \oplus (N_{\mathfrak{p}} \otimes_{A_{\mathfrak{p}}} \kappa(p)),$$

we have

$$\begin{aligned} \dim_{\kappa(p)}(M \oplus N)_{\mathfrak{p}} \otimes_{A_{\mathfrak{p}}} \kappa(p) &= \dim_{\kappa(p)}(M_{\mathfrak{p}} \otimes_{A_{\mathfrak{p}}} \kappa(p)) \oplus (N_{\mathfrak{p}} \otimes_{A_{\mathfrak{p}}} \kappa(p)) \\ &= \dim_{\kappa(p)}(M_{\mathfrak{p}} \otimes_{A_{\mathfrak{p}}} \kappa(p)) + \dim_{\kappa(p)}(N_{\mathfrak{p}} \otimes_{A_{\mathfrak{p}}} \kappa(p)), \end{aligned}$$

which implies that

$$\operatorname{rank}_p(\mathscr{F} \oplus \mathscr{G}) = \operatorname{rank}_p \mathscr{F} + \operatorname{rank}_p \mathscr{G}.$$

Now, consider $(M \otimes AN)_{\mathfrak{p}} \otimes_{A_{\mathfrak{p}}} \kappa(p)$. We have

$$\begin{aligned} (M \otimes_A N)_{\mathfrak{p}} \otimes_{A_{\mathfrak{p}}} \kappa(p) =& (M_{\mathfrak{p}} \otimes_{A_{\mathfrak{p}}} N_{\mathfrak{p}}) \otimes_{A_{\mathfrak{p}}} \kappa(p) \\ =& (M_{\mathfrak{p}} \otimes_{A_{\mathfrak{p}}} \kappa(p)) \otimes_{\kappa(p)} (N_{\mathfrak{p}} \otimes_{A_{\mathfrak{p}}} \kappa(p)), \end{aligned}$$

so

$$\begin{aligned} \dim_{\kappa(p)}(M \otimes_A N)_{\mathfrak{p}} \otimes_{A_{\mathfrak{p}}} \kappa(p) =& \dim_{\kappa(p)}(M_{\mathfrak{p}} \otimes_{A_{\mathfrak{p}}} \kappa(p)) \otimes_{\kappa(p)} (N_{\mathfrak{p}} \otimes_{A_{\mathfrak{p}}} \kappa(p)) \\ =& (\dim_{\kappa(p)}(M_{\mathfrak{p}} \otimes_{A_{\mathfrak{p}}} \kappa(p))) \cdot \dim_{\kappa(p)}(N_{\mathfrak{p}} \otimes_{A_{\mathfrak{p}}} \kappa(p)), \end{aligned}$$

which implies that

$$\operatorname{rank}_p(\mathscr{F} \otimes \mathscr{G}) = \operatorname{rank}_p \mathscr{F} \cdot \operatorname{rank}_p \mathscr{G}.$$

□

Note *If X is irreducible, and $\mathscr{F}$ is a quasicoherent (usually coherent) sheaf on X, then $\operatorname{rank} \mathscr{F}$ (with no mention of a point) by convention means at the generic point. (For example, a rank 0 quasicoherent sheaf on an integral scheme is torsion quasicoherent sheaf; see Definition 7.1.6.)*

Lemma 15.3.2 (Rank is additive function)

The rank of coherent sheaves on an integral scheme X is additive in exact sequences: if

$$0 \longrightarrow \mathscr{F} \longrightarrow \mathscr{G} \longrightarrow \mathscr{H} \longrightarrow 0$$

is an exact sequence of coherent sheaves, then

$$\operatorname{rank} \mathscr{F} - \operatorname{rank} \mathscr{G} + \operatorname{rank} \mathscr{H} = 0.$$

Proof Let η be the generic point of integral scheme X. Recall that localization is an exact functor, we have the following exact sequence of $\mathscr{O}_{X,\eta}$-modules ($K(X)$-vector spaces)

$$\begin{tikzcd} 0 \arrow[r] & \mathscr{F}_\eta \arrow[r] \arrow[d, equal] & \mathscr{G}_\eta \arrow[r] \arrow[d, equal] & \mathscr{H}_\eta \arrow[r] \arrow[d, equal] & 0 \\ & \mathscr{F}_\eta \otimes_{K(X)} K(X) & \mathscr{G}_\eta \otimes_{K(X)} K(X) & \mathscr{H}_\eta \otimes_{K(X)} K(X). & \end{tikzcd}$$

By Linear Algebra, we have

$$\dim_{K(X)} \mathscr{G}_\eta = \dim_{K(X)} \mathscr{F}_\eta + \dim_{K(X)} \mathscr{H}_\eta,$$

that is,

$$\operatorname{rank} \mathscr{F} - \operatorname{rank} \mathscr{G} + \operatorname{rank} \mathscr{H} = 0,$$

as we desired. □

Remark 15.19 Above argument use the fact that the rank is at the generic point; the example

$$0 \longrightarrow \widetilde{k[t]} \xrightarrow{\times t} \widetilde{k[t]} \longrightarrow \widetilde{k[t]/(t)} \longrightarrow 0.$$

on $\mathbb{A}^1_k$ shows that rank at a closed point is not additive in exact sequences.

If $\mathscr{F}$ is a finite type quasicoherent sheaf, then the rank of $\mathscr{F}$ at p is finite, and by Nakayama's Lemma 9.2.6, the rank is the minimal number of generators of $M_{\mathfrak{p}}$ as an $A_{\mathfrak{p}}$-module.

Proposition 15.3.7 (Rank is upper semicontinuous for finite type quasicoherent sheaves)

If $\mathscr{F}$ is a finite type quasicoherent sheaf on a scheme X, then $\operatorname{rank} \mathscr{F}$ *is an upper semicontinuous function on X.*

Proof Let $V_n = \{x \in X : \operatorname{rank}_x \mathscr{F} \leq n\}$ where n is zero or positive integer. We want to show V_n is open subset for each n. Let $p \in V_n$, since $\mathscr{F}$ is a finite type quasicoherent sheaf, say $\operatorname{rank}_p \mathscr{F} = r \leq n$. By the definition, we know that $\mathscr{F}|_p$ is a dimension r vector space, suppose $a_1|_p, \dots, a_r|_p \in \mathscr{F}|_p$ is a base of $\mathscr{F}|_p$, where $a_1, \dots, a_r \in \Gamma(U, \mathscr{F})$ for some open neighborhood U of p. By Geometric Nakayama 15.3.4, there is an affine open neighborhood $\operatorname{Spec} A \subseteq U$ of p such that for any $q \in \operatorname{Spec} A$, $a_1|_q, \dots, a_r|_q$ generate the fiber $\mathscr{F}|_q$ as a $\kappa(q)$-vector space. Hence, for any $q \in \operatorname{Spec} A$, we have $\operatorname{rank}_q \mathscr{F} \leq r$. It follows that every point $p \in V_n$ has an open neighborhood contained in V_n, and therefore V_n is open. Consequently, rank is an upper semicontinuous function. □

Remark 15.20 Proposition 15.3.3 (b) shows the necessity of the finite type hypothesis (consider V_0, V_0 is not open).

Proposition 15.3.8 (Constant rank = vector bundle, finite type qcoh-sheaves on reduced schemes)

(a) If X is reduced, $\mathscr{F}$ is a finite type quasicoherent sheaf on X, and the rank of $\mathscr{F}$ at any point is constant, then $\mathscr{F}$ is locally free.

(b) Finite type quasicoherent sheaves on a reduced scheme are locally free on a dense open set.

Proof ***For (a).*** Let $p \in X$, we want to show there exists an open neighborhood $U \ni p$ such that $\mathscr{F}|_U \cong \mathscr{O}_U^{\oplus n}$ for some n. Since $\mathscr{F}$ is a finite type quasicoherent sheaf, suppose $\operatorname{rank}_p \mathscr{F} = n$ for each $p \in X$. Choose n generators of the fiber $\mathscr{F}|_p$ (a basis as a $\kappa(p)$-vector space). By Geometric Nakayama's Lemma 15.3.4, we can find an affine open neighborhood $\operatorname{Spec} A \subseteq X$ of p, with $\mathscr{F}|_{\operatorname{Spec} A} = \widetilde{M}$, so that the chosen generators $\mathscr{F}|_p$ lift to generators $m_1, \ldots, m_n$ of M. Let $\varphi : A^{\oplus n} \twoheadrightarrow M$ by setting

$$(r_1, \ldots, r_n) \longmapsto \sum_i r_i m_i$$

be the corresponding surjection. Suppose $\operatorname{Ker} \varphi \neq 0$, say $(r_1, \ldots, r_n) \in \operatorname{Ker} \varphi$, with $r_i \neq 0$ for some i. As X is reduced, we know that the intersection of prime ideals of A is zero (Theorem 4.2.3), so there is some $\mathfrak{p}$ where $r_i \notin \mathfrak{p}$. Then r_i is invertible in $A_{\mathfrak{p}}$. From $\sum_i r_i m_i = 0$, we have

$$m_i = -\sum_{j \neq i} \frac{r_j}{r_i} \cdot m_j,$$

so $M_{\mathfrak{p}}$ has fewer than n generators, contradicting the constancy of rank. Hence $M \cong A^{\oplus n}$. Let $U = \operatorname{Spec} A$, then we have $\mathscr{F}|_U \cong \mathscr{O}_U^{\oplus n}$.

For (b). Let

$$L := \bigcup_{\text{open } U \subseteq X} \{U \subseteq X : \mathscr{F}|_U \text{ is locally free}\},$$

then L is open and $\mathscr{F}$ is locally free on L. We want to show that L is dense. Let $V \subseteq X$ be any open subset, suppose $V = \bigcup_\alpha V_\alpha$ where V_α is irreducible component of V for each α. Let η_α be the generic point of V_α, say $n_\alpha := \operatorname{rank}_{\eta_\alpha} \mathscr{F}$. Consider $U_{n_\alpha} = \{p \in V_\alpha : \operatorname{rank}_p \mathscr{F} \leq n_\alpha\}$, by the upper semicontinuity of rank (Proposition 15.3.7), U_{n_α} is open in V_α. Moreover, since η_α contained in each open subset of V_α, we have $U_{n_\alpha} = \{p \in V_\alpha : \operatorname{rank}_p \mathscr{F} = n_\alpha\}$. By part (a), we know that $\mathscr{F}|_{U_{n_\alpha}}$ is locally free. Note that $U_{n_\alpha} \subseteq V_\alpha \subseteq V$, we have $V \cap L \neq \varnothing$, which implies that L is dense in X, as we desired. □

Example 15.5 Part (a) in Proposition 15.3.8 can be false without the condition of X being reduced.

Let $A = k[x]/(x^2)$. Let $X = \operatorname{Spec} A$ and $\mathscr{F} = \widetilde{M}$ where $M = k$ is an A-module. Clearly, X is a single point, so the rank of $\mathscr{F}$ is constant. Specifically,

$$\begin{aligned} \operatorname{rank}_{(x)} \mathscr{F} &= \dim_{\kappa((x))} M_{(x)} \otimes_{A_{(x)}} \kappa((x)) \\ &= \dim_k k \otimes_{A_{(x)}} k \\ &= \dim_k k \\ &= 1. \end{aligned}$$

We now show that $\mathscr{F}$ is not locally free sheaf. If $\mathscr{F}$ is locally free, since X is single point, it follows that $M \cong A$. But it is impossible, since $x \cdot M = 0$ and $x \cdot A \neq 0$, a contradiction. Hence $\mathscr{F}$ is not locally free sheaf.

Proposition 15.3.9

Suppose X is reduced, and $\varphi : \mathscr{E} \to \mathscr{F}$ is a morphism of finite rank locally free sheaves. Then φ is a rank a map of vector bundles (Definition 15.2.6) if and only if $\varphi|_p : \mathscr{E}|_p \to \mathscr{F}|_p$ is rank a for all $p \in X$.

Proof Suppose $\mathscr{E}$ and $\mathscr{F}$ be two finite rank locally free sheaves on X, of rank $a + b$ and $a + c$, respectively.

If φ is a rank a map of vector bundles. Then for every $p \in X$, there is an open neighborhood U of p with a commutative diagram

$$\begin{tikzcd}
\mathscr{E}|_U \arrow[rr, "\varphi|_U"] \arrow[d, leftrightarrow, "\sim"'] & & \mathscr{F}|_U \arrow[d, leftrightarrow, "\sim"] \\
\mathscr{O}_U^{\oplus(a+b)} \arrow[r, two heads] & \mathscr{O}_U^{\oplus a} \arrow[r, hook] & \mathscr{O}_U^{\oplus(a+c)}.
\end{tikzcd}$$

By taking fiber at $p \in X$, we have

$$\begin{tikzcd}
\mathscr{E}|_p \arrow[rr, "\varphi|_p"] \arrow[d, leftrightarrow, "\sim"'] & & \mathscr{F}|_p \arrow[d, leftrightarrow, "\sim"] \\
\mathscr{O}_{X,p}^{\oplus(a+b)} \otimes_{\mathscr{O}_{X,p}} \kappa(p) \arrow[r, two heads] \arrow[d, leftrightarrow, "\sim"'] & \mathscr{O}_{X,p}^{\oplus a} \otimes_{\mathscr{O}_{X,p}} \kappa(p) \arrow[r, hook] \arrow[d, leftrightarrow, "\sim"'] & \mathscr{O}_{X,p}^{\oplus(a+c)} \otimes_{\mathscr{O}_{X,p}} \kappa(p) \arrow[d, leftrightarrow, "\sim"] \\
\kappa(p)^{\oplus(a+b)} \arrow[r, two heads] & \kappa(p)^{\oplus a} \arrow[r, hook] & \kappa(p)^{\oplus(a+c)}.
\end{tikzcd}$$

It follows that $\varphi|_p : \mathscr{E}|_p \to \mathscr{F}|_p$ is a rank a linear map for all $p \in X$.

Conversely, if $\varphi|_p : \mathscr{E}|_p \to \mathscr{F}|_p$ is a rank a linear map for all $p \in X$. Recall Proposition 15.2.10, it sufficient to show that $\operatorname{Coker}\varphi$ is locally free of rank c. Say $\mathscr{C} := \operatorname{Coker}\varphi$, then we have the following exact sequence

$$\mathscr{E} \xrightarrow{\varphi} \mathscr{F} \twoheadrightarrow \mathscr{C} \longrightarrow 0.$$

Taking stalk is exact, by taking stalk at $p \in X$, the following sequence is also exact.

$$\mathscr{E}_p \xrightarrow{\varphi_p} \mathscr{F}_p \twoheadrightarrow \mathscr{C}_p \longrightarrow 0.$$

Apply right exact functor $\square \otimes_{\mathscr{O}_{X,p}} \kappa(p)$, the sequence

$$\mathscr{E}_p \otimes_{\mathscr{O}_{X,p}} \kappa(p) \xrightarrow{\varphi|_p} \mathscr{F}_p \otimes_{\mathscr{O}_{X,p}} \kappa(p) \twoheadrightarrow \mathscr{C}_p \otimes_{\mathscr{O}_{X,p}} \kappa(p) \longrightarrow 0.$$

is exact. Hence $\mathscr{C}|_p \cong \operatorname{Coker}\varphi|_p$, since $\operatorname{Coker}\varphi|_p$ is $\kappa(p)$-vector space with dimension $(a+c)-c = c$, we know that $\mathscr{C}|_p$ is $\kappa(p)$-vector space with dimension c. It follows that the rank of $\mathscr{C}$ at any point is c, by Proposition 15.3.8, $\mathscr{C}$ is locally free sheaf of rank c, as we desired. □

You can use the notion of rank to help visualize finite type quasicoherent sheaves, or even quasicoherent sheaves. For example, finite type quasicoherent sheaves generalize finite rank vector bundles as follows: to each point there is an associated vector space, and although the ranks can jump, they fit together in families as well as one might hope. For Exercise 15.4, $\mathscr{F}$ is a finite type quasicoherent sheaf whose fiber at the origin is a one-dimensional vector space, while its fibers at all other points are zero. Clearly, $\mathscr{F}$ is not a finite rank vector bundle.

Nonreducedness can fit into the picture as well: For Example 15.5 , $\widetilde{k}$ and $k[\widetilde{x]/(x^2})$ have the same fiber at $[(x)]$, but they are different sheaves, $k[\widetilde{x]/(x^2})$ is rank 1 vector bundle but $\widetilde{k}$ is not. However, you also can regard $\widetilde{k}$ at a point with dimension 1 vector space but it is completely insensitive to the thickness direction. Consider the following example: $\operatorname{Spec} k[\varepsilon]/(\varepsilon^2)$ corresponding to $(k[\varepsilon]/(\varepsilon^2) \oplus k[\varepsilon]/(\varepsilon))^{\sim}$. The fiber at single point is clearly $k \oplus k$, it is the same as the fiber of $\widetilde{k[\varepsilon]/(\varepsilon^2)}^{\oplus 2}$ at this point. But they are different sheaves. It looks like: a two-dimensional fiber in which one direction extends into the nilpotent thickness, while the other direction exists only on the reduced point.

15.3.2 Degree of a finite morphism at a point

Definition 15.3.4 (The degree of the finite morphism at a point)

Suppose $\pi : X \to Y$ is a finite morphism. Then $\pi_* \mathscr{O}_X$ is a finite type (quasicoherent) sheaf on Y, and the rank of this sheaf at a point p is called the ***degree*** *of the finite morphism at p.*

Remark 15.21 By Proposition 15.3.7, the degree of π is an upper semicontinuous function on Y. The degree can jump: consider the closed embedding of a point into a line corresponding to $k[t] \to k$ given by $t \mapsto 0$. It can also be constant in cases that you might initially find surprising — see Example 11.3 , where the degree is always 2, but the 2 is obtained in a number of different ways.

Proposition 15.3.10

Suppose $\pi : X \to Y$ is a finite morphism. Then the degree of π at p is the dimension of the space of functions on the scheme-theoretic preimage of p, considered as a vector space over the residue field $\kappa(p)$. In particular, the degree is zero if and only if $\pi^{-1}(p)$ is empty.

Proof We first compute the degree of π at p. Suppose $p = [\mathfrak{p}] \in \operatorname{Spec} B \subseteq Y$. Since $\pi : X \to Y$ is finite morphism, say $\pi^{-1}(\operatorname{Spec} B) \cong \operatorname{Spec} A$ where A is finitely generated B-module. Recall Definition 15.3.4, consider $(\pi_* \mathscr{O}_X)|_p$,

$$\begin{aligned}(\pi_* \mathscr{O}_X)|_p =& (\pi_* \mathscr{O}_X)_p \otimes_{\mathscr{O}_{Y,p}} \kappa(p) \\ =& (A \otimes_B B_{\mathfrak{p}}) \otimes_{B_{\mathfrak{p}}} \kappa(p) \\ =& A \otimes_B \kappa(p)\end{aligned}$$

so $\operatorname{rank}_p(\pi_* \mathscr{O}_X) = \dim_{\kappa(p)}(A \otimes_B \kappa(p))$.

On the other hand, note that

$$\begin{aligned}\pi^{-1}(p) =& \operatorname{Spec} \kappa(p) \times_Y X \\ \cong& \operatorname{Spec} \kappa(p) \times_{\operatorname{Spec} B} \operatorname{Spec} A \\ \cong& \operatorname{Spec}(\kappa(p) \otimes_B A),\end{aligned}$$

so $\Gamma(\pi^{-1}(p), \mathscr{O}_{\pi^{-1}(p)}) = \kappa(p) \otimes_B A$. Hence the degree of π at p is the dimension of the space of functions on the scheme-theoretic preimage of p, considered as a vector space over the residue field $\kappa(p)$.

In particular, the degree is zero if and only if $\pi^{-1}(p)$ is empty. □

15.4 Pushforwards of quasicoherent sheaves

We conclude the chapter by discussing pushforward and pullbacks of quasicoherent sheaves, their properties, and some applications.

Suppose $B \to A$ is a morphism of rings. Recall (from Proposition 2.4.1) that $\square \otimes_B A \dashv \square_B$ is an adjoint pair between the categories of A-modules and B-modules: we have a bijection

$$\operatorname{Hom}_A(N \otimes_B A, M) \xleftrightarrow{\sim} \operatorname{Hom}_B(N, M_B)$$

functorial in both arguments. These constructions behave well with respect to localization (in an appropriate sense), and hence work (often) in the category of quasicoherent sheaves on schemes (and, indeed, always in the category of $\mathscr{O}$-modules on ringed spaces — see Definition **??** — although we won't particularly care). The easier construction $(M \mapsto M_B)$ will turn into our old friend pushforward. The other $(N \mapsto A \otimes_B N)$ will be a relative of pullback, whom I'm reluctant to call an "old friend".

We begin with the pushforwards, for which we have already done much of the work.

The main moral of this section is that in "reasonable" situations, the pushforward of a quasicoherent sheaf is quasicoherent, and that this can be understood in terms of one of the module constructions defined above. We begin with a motivating example:

Lemma 15.4.1 (Pushforward of a qcoh sheaf under an affine morphism is also qcoh)

Let $\pi : \operatorname{Spec} A \to \operatorname{Spec} B$ *be a morphism of affine schemes, and suppose* M *is an* A*-module, so* $\widetilde{M}$ *is a quasicoherent sheaf on* $\operatorname{Spec} A$*. Then there is an isomorphism*

$$\pi_* \widetilde{M} \xrightarrow{\sim} \widetilde{M_B}.$$

Proof Let $\operatorname{Spec} B_f \hookrightarrow \operatorname{Spec} B$ be an affine distinguished open subset. Then

$$\pi_* \widetilde{M}(\operatorname{Spec} B_f) = \widetilde{M}(\pi^{-1}(\operatorname{Spec} B_f)) = \widetilde{M}(\operatorname{Spec} A_{\pi^\sharp(f)}) = M \otimes_A A_{\pi^\sharp(f)}$$

and

$$\widetilde{M_B}(\operatorname{Spec} B_f) = M_B \otimes_B B_f.$$

Define $\varphi_{D(f)} : M_B \otimes_B B_f \to M \otimes_A A_{\pi^\sharp(f)}$ by setting

$$m \otimes \frac{b}{f^n} \longmapsto m \otimes \frac{\pi^\sharp(b)}{\pi^\sharp(f)^n},$$

we want to show this is an isomorphism. Define the inverse $\psi_{D(f)} : M \otimes_A A_{\pi^\sharp(f)} \to M_B \otimes_B B_f$ by setting

$$m \otimes \frac{a}{\pi^\sharp(f)^n} \longmapsto am \otimes \frac{1}{f^n}.$$

Hence, we have

$$\psi_{D(f)} \circ \varphi_{D(f)} \left(m \otimes \frac{b}{f^n} \right) = \psi_{D(f)} \left(m \otimes \frac{\pi^\sharp(b)}{\pi^\sharp(f)^n} \right) = \pi^\sharp(b) m \otimes \frac{1}{f^n} = m \otimes \frac{b}{f^n},$$

similarly, one check that $\varphi_{D(f)} \circ \psi_{D(f)} = \mathrm{id}$, hence $\varphi_{D(f)}$ is an isomorphism. We next check compatibility with restriction maps. Suppose $\operatorname{Spec} B_g \hookrightarrow \operatorname{Spec} B_f$. Clearly, the following diagram

$$\begin{tikzcd} B_f \arrow[r, "\pi^\sharp"] \arrow[d] & A_{\pi^\sharp(f)} \arrow[d] \\ B_g \arrow[r, "\pi^\sharp"] & A_{\pi^\sharp(g)} \end{tikzcd}$$

commutes, from this it is easy to check that the diagram

$$\begin{tikzcd} M_B \otimes_B B_f \arrow[r, "\pi^\sharp"] \arrow[d] & M \otimes_A A_{\pi^\sharp(f)} \arrow[d] \\ M_B \otimes_B B_g \arrow[r, "\pi^\sharp"] & M \otimes_A A_{\pi^\sharp(g)} \end{tikzcd}$$

commutes, which implies that $\varphi : \pi_* \widetilde{M} \to \widetilde{M_B}$ is a morphism of $\mathscr{O}_{\operatorname{Spec} B}$-modules. Since the distinguished opens form a basis for the topology of $\operatorname{Spec} B$ and $\varphi_{D(f)}$ is an isomorphism for each $f \in B$, it follows that φ is an isomorphism

$$\pi_* \widetilde{M} \xrightarrow{\sim} \widetilde{M_B}.$$

□

Remark 15.22 In particular, $\pi_* \widetilde{M}$ is quasicoherent. Perhaps more important, this implies that the pushforward of a quasicoherent sheaf under an affine morphism is also quasicoherent.

Proposition 15.4.1

If $\pi : X \to Y$ is an affine morphism, then π_* is an exact functor $\mathbf{QCoh}_X \to \mathbf{QCoh}_Y$.

Proof By Lemma 15.4.1 and Proposition 8.2.3, we know that $\pi_* : \mathbf{QCoh}_X \to \mathbf{QCoh}_Y$ is a functor. Recall Proposition 3.6.8, π_* is left exact functor. It remains to show that π_* is right exact functor.

Suppose $\varphi : \mathscr{F} \to \mathscr{G}$ be an epimorphism where $\mathscr{F}, \mathscr{G} \in \mathbf{QCoh}_X$. We want to show that $\pi_*\varphi : \pi_*\mathscr{F} \to \pi_*\mathscr{G}$ is also an epimorphism. Let $\{\operatorname{Spec} B_i \hookrightarrow Y\}_i$ be any open cover of Y, say $\operatorname{Spec} A_i := \pi^{-1}(\operatorname{Spec} B_i)$, then $\{\operatorname{Spec} A_i \hookrightarrow X\}_i$ is an open cover of X. Since φ is an epimorphism, $\varphi_{\operatorname{Spec} A_i} : \mathscr{F}(\operatorname{Spec} A_i) \to \mathscr{G}(\operatorname{Spec} A_i)$ is a surjective. So, $(\pi_*\varphi)_{\operatorname{Spec} B_i} : (\pi_*\mathscr{F})(\operatorname{Spec} B_i) \to (\pi_*\mathscr{G})(\operatorname{Spec} B_i)$ is a surjective for each i. By Proposition 7.3.1, we know that $(\pi_*\varphi)_{\operatorname{Spec} B_i}$ is surjection for each $\operatorname{Spec} B_i \hookrightarrow Y$, which implies that $\pi_*\varphi$ is an epimorphism. Hence, π_* is an exact functor. □

The following result generalizes the fact that the pushforward of a quasicoherent sheaf under an affine morphism is also quasicoherent.

Corollary 15.4.1 (Corollary of Qcqs Lemma 7.2.3: "$\pi_* : \mathrm{QCoh}_X \to \mathrm{QCoh}_Y$ for qcqs π")

Suppose $\pi : X \to Y$ is a quasicompact quasiseparated morphism, and $\mathscr{F} \in \mathbf{QCoh}_X$. Then $\pi_*\mathscr{F} \in \mathbf{QCoh}_Y$.

Proof The question is local on Y and hence we may assume that Y is affine. Since $\pi : X \to Y$ is quasicompact, X is quasicompact, say $X = \bigcup_{i=1}^{n} U_i$ where each U_i is affine open. Since π is quasiseparated we may write $U_i \cap U_j = \bigcup_{k=1}^{n_{ij}} U_{ijk}$ for some affine open U_{ijk}. Denote $\pi_i : U_i \to Y$ and $\pi_{ijk} : U_{ijk} \to Y$ be the restrictions of π, by Proposition 9.3.6, π_i, π_{ijk} are affine morphism. For any open $V \hookrightarrow Y$ and any $\mathscr{F} \in \mathbf{QCoh}_X$, we have

$$\begin{aligned}
\pi_*\mathscr{F}(V) &= \mathscr{F}(\pi^{-1}(V)) \\
&= \operatorname{Ker}\left(\bigoplus_i \mathscr{F}(\pi^{-1}(V) \cap U_i) \to \bigoplus_{i,j,k} \mathscr{F}(\pi^{-1}(V) \cap U_{ijk})\right) \\
&= \operatorname{Ker}\left(\bigoplus_i \pi_{i,*}(\mathscr{F}|_{U_i})(V) \to \bigoplus_{i,j,k} \pi_{ijk,*}(\mathscr{F}|_{U_{ijk}})(V)\right) \\
&= \operatorname{Ker}\left(\bigoplus_i \pi_{i,*}(\mathscr{F}|_{U_i}) \to \bigoplus_{i,j,k} \pi_{ijk,*}(\mathscr{F}|_{U_{ijk}})\right)(V).
\end{aligned}$$

In other words, there is an exact sequence of sheaves

$$0 \longrightarrow \pi_*\mathscr{F} \longrightarrow \bigoplus \pi_{i,*}\mathscr{F}_i \longrightarrow \bigoplus \pi_{ijk,*}\mathscr{F}_{ijk}$$

where $\mathscr{F}_i, \mathscr{F}_{ijk}$ denotes the restriction of $\mathscr{F}$ to the corresponding open. Since $\mathscr{F} \in \mathbf{QCoh}_X$, $\mathscr{F}_i, \mathscr{F}_{ijk} \in \mathbf{QCoh}_X$. Recall $\mathbf{QCoh}_Y$ is an abelian category, $\pi_*\mathscr{F}$ is a quasicoherent sheaf on Y. □

Coherent sheaves do not always push forward to coherent sheaves. For example:

Example 15.6 (Push forward a coh sheaf may not be a coh sheaf) Consider the structure morphism $\pi : \mathbb{A}^1_k \to \operatorname{Spec} k$, corresponding to $k \to k[t]$. Then $\pi_*\mathscr{O}_{\mathbb{A}^1_k}$ is the quasicoherent sheaf corresponding to $k[t]$, which is not a finitely generated k-module.

But in good situations, coherent sheaves do push forward. For example:

Proposition 15.4.2

Suppose $\pi : X \to Y$ is a finite morphism of locally Noetherian schemes. If $\mathscr{F} \in \mathbf{Coh}_X$, then $\pi_*\mathscr{F} \in \mathbf{Coh}_Y$.

Proof From $\pi : X \to Y$, we have a morphism $\pi^\sharp : \mathscr{O}_Y \to \pi_*\mathscr{O}_X$, so $\pi_*\mathscr{O}_X$ is an $\mathscr{O}_Y$-module. Let $\operatorname{Spec} B \subseteq Y$ be an affine open, since $\pi : X \to Y$ is finite, say $\operatorname{Spec} A = \pi^{-1}(\operatorname{Spec} B)$, where A is finite B-algebra, hence A is a finite generated B-module.

Since $\pi : X \to Y$ is a finite morphism, so it is affine, by Proposition 15.4.1, $\pi_*\mathscr{F} \in \mathbf{QCoh}_Y$. We shall to show that $\pi_*\mathscr{F} \in \mathbf{Coh}_Y$, it suffices to show that for each affine open $\operatorname{Spec} B \hookrightarrow_Y$, $(\pi_*\mathscr{F})|_{\operatorname{Spec} B} \cong \widetilde{M}$ for some coherent B-module M. Clearly, $\pi_*\mathscr{F}$ is a $\pi_*\mathscr{O}_X$-module. Consider $\pi_*\mathscr{F}(\operatorname{Spec} B)$, we have

$$\pi_*\mathscr{F}(\operatorname{Spec} B) = \mathscr{F}(\pi^{-1}(\operatorname{Spec} B)) = \mathscr{F}(\operatorname{Spec} A).$$

Since $\mathscr{F} \in \mathbf{Coh}_X$ and X is locally Noetherian scheme, recall Proposition 7.4.1, there exists $\mathscr{F}|_{\operatorname{Spec} A} \cong \widetilde{M}$, where M is a finitely generated A-module. Since A is finitely generated B-module, we know that M is finitely generated B-module. Since Y is locally Noetherian scheme, by Proposition 7.4.1 again, we have $\pi_*\mathscr{F} \in \mathbf{Coh}_Y$. $\square$

Once we define cohomology of quasicoherent sheaves, we will quickly prove that if $\mathscr{F}$ is a coherent sheaf on $\mathbb{P}^n_k$, then $\Gamma(\mathbb{P}^n_k, \mathscr{F})$ is a finite-dimensional k-module, and more generally, if $\mathscr{F}$ is a coherent sheaf on $\operatorname{Proj} S_\bullet$, then $\Gamma(\operatorname{Proj} S_\bullet, \mathscr{F})$ is a coherent A-module (where $S_0 = A$). This is a special case of the fact that the "pushforwards of coherent sheaves by projective morphisms are also coherent sheaves" (Theorem **??**). (The notion of projective morphism, a relative version of $\operatorname{Proj} S_\bullet \to \operatorname{Spec} A$, will be defined in §**??**.)

More generally, given Noetherian hypotheses, pushforwards of coherent sheaves by proper morphisms are also coherent sheaves (Grothendieck's Coherence Theorem **??**).

Remark 15.23 Vector bundles do not always pushforward to vector bundles, which is another reason to work in the larger category of (quasi)coherent sheaves. (For example: Set $X = \operatorname{Spec} k[t]/(t), Y = \operatorname{Spec} k[t]$. Let $i : X \hookrightarrow Y$ closed embedding induced by $k[t] \twoheadrightarrow k[t]/(t)$. Then $i_*\mathscr{O}_X$ is a coherent sheaf, but not a locally free sheaf.) But in particularly good circumstances they can, which is one of the purposes of the Cohomology and Base Change Theorem **??**.

15.5 Pullbacks of quasicoherent sheaves: Three different perspectives

We next discuss the pullback of a quasicoherent sheaf: if $\pi : X \to Y$ is a morphism of schemes, π^* is a covariant functor $\mathbf{QCoh}_Y \to \mathbf{QCoh}_X$. (In fact, our discussion will apply readily to pulling back $\mathscr{O}$-modules on ringed spaces; see §15.5.2 and §15.5.3.) The notion of the pullback of a quasicoherent sheaf can be confusing on first (and second) glance. (For example, it is not the inverse image sheaf, although we will see that it is related to it.)

Here are three contexts in which you have seen the pullback, or can understand it quickly. It may be helpful to keep these in mind, to keep you anchored in the long discussion that follows.

(i) ***(restriction to open subsets)*** If $i : U \hookrightarrow Y$ is an open embedding, then $i^*\mathscr{G}$ is $\mathscr{G}|_U$, the restriction of $\mathscr{G}$ to U (Definition 3.2.7, Proposition 3.7.3).

(ii) ***(restriction to points)*** If $i : p \hookrightarrow Y$ is the "inclusion" of a point p in Y (for example, the closed embedding of a closed point; see Proposition 8.3.10 (b)), then $i^*\mathscr{G}$ is $\mathscr{G}|_p$, the fiber of $\mathscr{G}$ at p (Definition 5.3.9).

Remark 15.24 The similarity of the notation $\mathscr{G}|_U$ and $\mathscr{G}|_p$ is precisely because both are pullbacks.

Pullbacks (especially to locally closed subschemes or generic points) are often called **restriction**. For this reason, if $\pi : X \to Y$ is some sort of inclusion (such as a locally closed embedding, or an "inclusion of a generic point"), then $\pi^*\mathscr{G}$ is often written as $\mathscr{G}|_X$ and called **the restriction of $\mathscr{G}$ to X**, when π can be interpreted as some type of "inclusion".

(iii) ***(pulling back vector bundles)*** Suppose $\mathscr{G}$ is a locally free sheaf on Y, and $\pi : X \to Y$ is any morphism. Then $\pi^*\mathscr{G}$ is the pullback of $\mathscr{G}$ as a vector bundle, as defined in Proposition 15.1.1 (b). (This will be established in §15.5.3.)

Strategy

We will see three different ways of thinking about the pullback. Each has significant disadvantages, but together they give a good understanding.

(a) Because we are understanding quasicoherent sheaves in terms of affine open sets, and modules over the corresponding rings, we begin with an interpretation in this vein. This will be very useful for proving and understanding facts. The disadvantage is that it is annoying to make a definition out of this (when the target is not affine), because gluing arguments can be tedious.

(b) As we saw with fibered product, gluing arguments can be made simpler using universal properties, so our second "definition" will be by universal property. This is elegant, but has the disadvantage that it still needs a construction, and because it works in the larger category of $\mathscr{O}$-modules, it isn't clear from the universal property that it takes quasicoherent sheaves to quasicoherent sheaves. But if the target is affine, our construction of (a) is easily seen to satisfy the universal property. Furthermore, the universal property is "local on the target": if $\pi : X \to Y$ is any morphism, $i : U \hookrightarrow Y$ is an open embedding, and $\mathscr{G}$ is a quasicoherent sheaf on Y, then if $\pi^*\mathscr{G}$ exists, then its restriction to $\pi^{-1}(U)$ is (canonically identified with) $(\pi|_U)^*(\mathscr{G}|_U)$. Thus if the pullback exists in general (even as an $\mathscr{O}$-module), affine-locally on Y it looks like the construction of (a) (and thus is quasicoherent).

(c) The third definition is one that works on ringed spaces in general. It is short, and is easily seen to satisfy the universal property. It doesn't obviously take quasicoherent sheaves to quasicoherent sheaves (at least in the way that we have defined quasicoherent sheaves) — a priori it takes quasicoherent sheaves to $\mathscr{O}$-modules. But thanks to the discussion at the end of (b) above, which used (a), this show that the pullback of a quasicoherent sheaf is indeed quasicoherent.

15.5.1 First attempt at describing the pullback, using affines

Suppose $\pi : X \to Y$ is a morphism of schemes, and $\mathscr{G}$ is a quasicoherent sheaf on Y. We want to define the pullback quasicoherent sheaf $\pi^*\mathscr{G}$ on X in terms of affine open sets on X and Y. Suppose $\operatorname{Spec} A \subseteq X$, $\operatorname{Spec} B \subseteq Y$ are affine open sets, with $\pi(\operatorname{Spec} A) \subseteq \operatorname{Spec} B$. Suppose $\mathscr{G}|_{\operatorname{Spec} B} \cong \widetilde{N}$. Perhaps motivated by the fact that pullback should relate to tensor product, we want

$$\Gamma(\operatorname{Spec} A, \pi^*\mathscr{G}) = N \otimes_B A. \tag{15.14}$$

More precisely, we would like $\Gamma(\operatorname{Spec} A, \pi^*\mathscr{G})$ and $N \otimes_B A$ to be identified. This could mean that we use this to construct a definition of $\pi^*\mathscr{G}$, by "gluing all this information together" (and showing it is well-defined). Or it could mean that we define $\pi^*\mathscr{G}$ in some other way, and then find a natural identification (15.14). The first can be made to work (and the following paragraph is the first step), but we will follow the second.

We begin this project by fixing an affine open subset $\operatorname{Spec} B \subseteq Y$. To avoid confusion, let $\varphi = \pi|_{\pi^{-1}(\operatorname{Spec} B)} : \pi^{-1}(\operatorname{Spec} B) \to \operatorname{Spec} B$. We will define a quasicoherent sheaf on $\pi^{-1}(\operatorname{Spec} B)$ that will turn

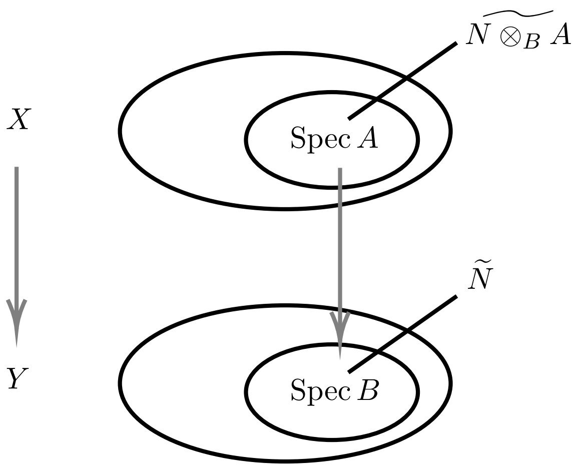


Figure 15.1: The pullback of a quasicoherent sheaf (module-theoretically).

out to be $\varphi^*(\mathscr{G}|_{\operatorname{Spec} B})$ (and will also be the restriction of $\pi^*\mathscr{G}$ to $\pi^{-1}(\operatorname{Spec} B)$).

If $\operatorname{Spec} A_f \hookrightarrow \operatorname{Spec} A$ is a distinguished open set, then

$$\Gamma(\operatorname{Spec} A_f, \varphi^*\mathscr{G}) = N \otimes_B A_f = (N \otimes_B A)_f = \Gamma(\operatorname{Spec} A, \varphi^*\mathscr{G})_f,$$

where "=" means "canonically isomorphic." Define the restriction map $\Gamma(\operatorname{Spec} A, \varphi^*\mathscr{G}) \to \Gamma(\operatorname{Spec} A_f, \varphi^*\mathscr{G})$,

$$\Gamma(\operatorname{Spec} A, \varphi^*\mathscr{G}) \longrightarrow \Gamma(\operatorname{Spec} A, \varphi^*\mathscr{G}) \otimes_A A_f, \tag{15.15}$$

by $\alpha \mapsto \alpha \otimes 1$. Thus $\varphi^*\mathscr{G}$ is (or: extends to) a quasicoherent sheaf on $\pi^{-1}(\operatorname{Spec} B)$ (by Theorem 7.2.2).

We have now defined a quasicoherent sheaf on $\pi^{-1}(\operatorname{Spec} B)$, for every affine open subset $\operatorname{Spec} B \subseteq Y$. We want to show that this construction, as $\operatorname{Spec} B$ varies, glues into a single quasicoherent sheaf on X.

You are welcome to do this gluing appropriately, for example, using the distinguished affine base of Y (§7.2.1). This works, but can be confusing, so we take another approach.

15.5.2 Universal property definition of pullback

Suppose $\pi : X \to Y$ is a morphism of ringed spaces, and $\mathscr{G}$ is an $\mathscr{O}_Y$-module. (We are of cause interested in the case where π is a morphism of schemes, and $\mathscr{G}$ is quasicoherent. Even once we specialize our discussion to schemes, much of our discussion will extend without change to this more general situation.) We "define" the pullback $\pi^*\mathscr{G}$ as an $\mathscr{O}_X$-module, using the following adjointness universal property: for any $\mathscr{O}_X$-module $\mathscr{F}$, there is a bijection $\operatorname{Hom}_{\mathscr{O}_X}(\pi^*\mathscr{G}, \mathscr{F}) \overset{\sim}{\longleftrightarrow} \operatorname{Hom}_{\mathscr{O}_Y}(\mathscr{G}, \pi_*\mathscr{F})$, and these bijections are functorial in $\mathscr{F}$. By universal property nonsense, this determines $\pi^*\mathscr{G}$ up to unique isomorphism; we just need to make sure that it exists (which is why the word "define" is in quotes). Notice that we avoid worrying about when the pushforward of a quasicoherent sheaf $\mathscr{F}$ is quasicoherent by working in the larger category of $\mathscr{O}$-modules.

Definition 15.5.1 (Universal property definition of pullback)

*Let $\pi : (X, \mathscr{O}_X) \to (Y, \mathscr{O}_Y)$ be a morphism of ringed spaces, and let $\mathscr{G}$ be an $\mathscr{O}_Y$-module. The **pullback of $\mathscr{G}$ along** π, denoted by $\pi^*\mathscr{G}$, is defined to be an $\mathscr{O}_X$-module satisfies the following universal property: for every $\mathscr{O}_X$-module $\mathscr{F}$, there is a natural bijection*

$$\operatorname{Hom}_{\mathscr{O}_X}(\pi^*\mathscr{G}, \mathscr{F}) \overset{\sim}{\longleftrightarrow} \operatorname{Hom}_{\mathscr{O}_Y}(\mathscr{G}, \pi_*\mathscr{F}),$$

functorial in $\mathscr{F}$ (i.e., $\operatorname{Hom}_{\mathscr{O}_X}(\pi^*\mathscr{G}, \square) \to \operatorname{Hom}_{\mathscr{O}_Y}(\mathscr{G}, \pi_*\square)$ form a natural isomorphism). If such an $\mathscr{O}_X$-module exists, then it is unique determined up to unique isomorphism by this universal property.

Lemma 15.5.1 (Quasicoherent sheaves on affines)

Let $X = \operatorname{Spec} A$ be an affine scheme and M is an A-module. Then, for any $\mathscr{F} \in \mathbf{Mod}_{\mathscr{O}_X}$, there exists a natural bijection

$$\operatorname{Hom}_{\mathscr{O}_X}(\widetilde{M}, \mathscr{F}) \overset{\sim}{\longleftrightarrow} \operatorname{Hom}_A(M, \Gamma(X, \mathscr{F})),$$

functorial in $\mathscr{F}$.

Proof Let $\alpha \in \operatorname{Hom}_{\mathscr{O}_X}(\widetilde{M}, \mathscr{F})$, then α gives a morphism between global sections $\alpha_X : M \to \Gamma(X, \mathscr{F})$. Define $\tau_{\mathscr{F}} : \operatorname{Hom}_{\mathscr{O}_X}(\widetilde{M}, \mathscr{F}) \to \operatorname{Hom}_A(M, \Gamma(X, \mathscr{F}))$ by setting

$$\alpha \longmapsto \alpha_X.$$

Conversely, let $\beta \in \operatorname{Hom}_A(M, \Gamma(X, \mathscr{F}))$. We want to define a morphism from $\widetilde{M}$ to $\mathscr{F}$. Define $\varphi(\beta)_{\operatorname{Spec} A_f} : M_f \to \Gamma(\operatorname{Spec} A_f, \mathscr{F})$ by setting

$$\frac{m}{f^n} \longmapsto \frac{\beta(m)|_{\operatorname{Spec} A_f}}{f^n}.$$

One checks that $\varphi(\beta)_{\operatorname{Spec} A_f}$ is well-defined. We next show that for any $\operatorname{Spec} A_{fg} \hookrightarrow \operatorname{Spec} A_f$, the following diagram is commutative.

$$\begin{tikzcd} M_f \arrow[r, "\varphi(\beta)_{\operatorname{Spec} A_f}"] \arrow[d, "\mathrm{res}"'] & \Gamma(\operatorname{Spec} A_f, \mathscr{F}) \arrow[d, "\mathrm{res}"] \\ M_{fg} \arrow[r, "\varphi(\beta)_{\operatorname{Spec} A_{fg}}"] & \Gamma(\operatorname{Spec} A_{fg}, \mathscr{F}) \end{tikzcd}$$

Let $\frac{m}{f^n} \in M_f$, then

$$\varphi(\beta)_{\operatorname{Spec} A_{fg}} \circ \mathrm{res}\left(\frac{m}{f^n}\right) = \varphi(\beta)_{\operatorname{Spec} A_{fg}}\left(\frac{g^n m}{f^n g^n}\right) = \frac{g^n \beta(m)|_{\operatorname{Spec} A_{fg}}}{f^n g^n}$$

and

$$\mathrm{res} \circ \varphi(\beta)_{\operatorname{Spec} A_f}\left(\frac{m}{f^n}\right) = \mathrm{res}\left(\frac{\beta(m)|_{\operatorname{Spec} A_f}}{f^n}\right) = \frac{g^n \beta(m)|_{\operatorname{Spec} A_{fg}}}{f^n g^n},$$

which implies that $\varphi(\beta)_{\operatorname{Spec} A_{fg}} \circ \mathrm{res} = \mathrm{res} \circ \varphi(\beta)_{\operatorname{Spec} A_f}$. So, $\varphi(\beta)_{\operatorname{Spec} A_f}$ glue to a morphism of $\mathscr{O}_X$-modules $\varphi(\beta) : \widetilde{M} \to \mathscr{F}$. Define $\varphi_{\mathscr{F}} : \operatorname{Hom}_A(M, \Gamma(X, \mathscr{F})) \to \operatorname{Hom}_{\mathscr{O}_X}(\widetilde{M}, \mathscr{F})$ by setting $\beta \mapsto \varphi(\beta)$.

It is easy to check that $\varphi_{\mathscr{F}} \circ \tau_{\mathscr{F}} = \mathrm{id}$ and $\tau_{\mathscr{F}} \circ \varphi_{\mathscr{F}} = \mathrm{id}$. We next check that τ is a natural transformation. Let $\theta : \mathscr{F} \to \mathscr{F}'$, consider the following diagram,

$$\begin{tikzcd} \operatorname{Hom}_{\mathscr{O}_X}(\widetilde{M}, \mathscr{F}) \arrow[r, "\tau_{\mathscr{F}}"] \arrow[d, "h^{\widetilde{M}}(\theta)"'] & \operatorname{Hom}_A(M, \Gamma(X, \mathscr{F})) \arrow[d, "h^M(\Gamma(X,\theta))"] \\ \operatorname{Hom}_{\mathscr{O}_X}(\widetilde{M}, \mathscr{F}') \arrow[r, "\tau_{\mathscr{F}'}"'] & \operatorname{Hom}_A(M, \Gamma(X, \mathscr{F}')) \end{tikzcd}$$

we want to show that above diagram commutes. Pick $\alpha \in \operatorname{Hom}_{\mathscr{O}_X}(\widetilde{M}, \mathscr{F})$, then

$$h^M(\Gamma(X, \theta)) \circ \tau_{\mathscr{F}}(\alpha) = h^M(\Gamma(X, \theta))(\alpha_X) = \Gamma(X, \theta) \circ \alpha_X = \theta_X \circ \alpha_X$$

and

$$\tau_{\mathscr{F}'} \circ h^{\widetilde{M}}(\theta)(\alpha) = \tau_{\mathscr{F}'}(\theta \circ \alpha) = \theta_X \circ \alpha_X,$$

hence, $h^M(\Gamma(X, \theta)) \circ \tau_{\mathscr{F}} = \tau_{\mathscr{F}'} \circ h^{\widetilde{M}}(\theta)$, which implies that $\tau_{\mathscr{F}}$ is a natural transformation. We win. $\square$

Lemma 15.5.2

If Y is affine, say $Y = \operatorname{Spec} B$, then the construction of the quasicoherent sheaf in §15.5.1 satisfies the universal property of pullback of $\mathscr{G}$. Thus call this sheaf $\pi^*\mathscr{G}$ is justified.

Proof Let $\mathscr{G} = \widetilde{N}$ is a quasicoherent sheaf on $Y = \operatorname{Spec} B$. Denote $\mathscr{H}$ be the sheaf which defined in §15.5.1. Let $\{U_i = \operatorname{Spec} A_i \hookrightarrow X\}$ be an open cover of X, say $\mathscr{H}|_{U_i} \cong \widetilde{N \otimes_B A_i}$. We want to show that for any $\mathscr{O}_X$-module $\mathscr{F}$, there is a natural bijection

$$\operatorname{Hom}_{\mathscr{O}_X}(\mathscr{H}, \mathscr{F}) \xleftrightarrow{\sim} \operatorname{Hom}_{\mathscr{O}_Y}(\mathscr{G}, \pi_*\mathscr{F}).$$

Recall Lemma 15.5.1, we have a natural bijection

$$\operatorname{Hom}_{\mathscr{O}_Y}(\mathscr{G}, \pi_*\mathscr{F}) \xleftrightarrow{\sim} \operatorname{Hom}_B(N, \Gamma(Y, \pi_*\mathscr{F})) = \operatorname{Hom}_B(N, \Gamma(X, \mathscr{F})).$$

Hence, it suffices to show that there exists a bijective

$$\operatorname{Hom}_{\mathscr{O}_X}(\mathscr{H}, \mathscr{F}) \xleftrightarrow{\sim} \operatorname{Hom}_B(N, \Gamma(X, \mathscr{F})).$$

By the construction of $\mathscr{H}$, for each $U_i \hookrightarrow X$, we have

$$\begin{tikzcd}
\operatorname{Hom}_{\mathscr{O}_{U_i}}(\mathscr{H}|_{U_i}, \mathscr{F}|_{U_i}) \arrow[r, equal] & \operatorname{Hom}_{\mathscr{O}_{U_i}}(\widetilde{N \otimes_B A_i}, \mathscr{F}|_{U_i}) \arrow[d, leftrightarrow, "\sim"', "\text{Lem. 15.5.1}"] \\
& \operatorname{Hom}_{A_i}(N \otimes_B A_i, \Gamma(U_i, \mathscr{F})) \arrow[d, leftrightarrow, "\sim"', "\text{Prop. 2.4.1}"] \\
& \operatorname{Hom}_B(N, \Gamma(U_i, \mathscr{F}))
\end{tikzcd} \tag{15.16}$$

Hence, given $\alpha \in \operatorname{Hom}_{\mathscr{O}_X}(\mathscr{H}, \mathscr{F})$, we obtain $\alpha_i := \alpha_{U_i} \in \operatorname{Hom}_{A_i}(\mathscr{H}|_{U_i}, \mathscr{F}|_{U_i})$, by (15.16), α_i one-to-one corresponding to $\widetilde{\alpha_i} \in \operatorname{Hom}_B(N, \Gamma(U_i, \mathscr{F}))$. For each $n \in N$, $\widetilde{\alpha_i}(n) \in \Gamma(U_i, \mathscr{F})$. We want to show that $\widetilde{\alpha_i}(n)|_{U_i \cap U_j} = \widetilde{\alpha_j}(n)|_{U_i \cap U_j}$. This is clearly, since $\alpha_i|_{U_i \cap U_j} = \alpha_i|_{U_i \cap U_j}$. Hence, $\widetilde{\alpha_i}(n)$ glue to a global section $\widetilde{\alpha}(n) \in \Gamma(X, \mathscr{F})$. Define $\varphi_{\mathscr{F}} : \operatorname{Hom}_{\mathscr{O}_X}(\mathscr{H}, \mathscr{F}) \to \operatorname{Hom}_B(N, \Gamma(X, \mathscr{F}))$ by setting

$$\alpha \longmapsto \widetilde{\alpha}.$$

Conversely, let $\beta \in \operatorname{Hom}_B(N, \Gamma(X, \mathscr{F}))$. Denote $\beta_i := \operatorname{res}_{X, U_i} \circ \beta : N \to \Gamma(U_i, \mathscr{F})$. By (15.16), we obtain $\widehat{\beta_i} : \mathscr{H}|_{U_i} \to \mathscr{F}|_{U_i}$. We now want to check that $\widehat{\beta_i}|_{U_i \cap U_j} = \widehat{\beta_j}|_{U_i \cap U_j}$. By Proposition 6.3.1, $U_i \cap U_j$ can be covered by affine opens, for each affine open $V \subseteq U_i \cap U_j$, we have $\widehat{\beta_i}|_V = \widehat{\beta_j}|_V$, since they all come from $N \to \Gamma(V, \mathscr{F})$. Hence, $\widehat{\beta_i}|_{U_i \cap U_j} = \widehat{\beta_j}|_{U_i \cap U_j}$, so they can glue together, i.e., $\widetilde{\beta} : \mathscr{H} \to \mathscr{F}$. Define $\tau_{\mathscr{F}} : \operatorname{Hom}_B(N, \Gamma(X, \mathscr{F})) \to \operatorname{Hom}_{\mathscr{O}_X}(\mathscr{H}, \mathscr{F})$ by setting

$$\beta \longmapsto \widehat{\beta}.$$

One checks that $\varphi_{\mathscr{F}} \circ \tau_{\mathscr{F}} = \mathrm{id}$ and $\tau_{\mathscr{F}} \circ \varphi_{\mathscr{F}} = \mathrm{id}$. Hence, for any $\mathscr{O}_X$-module $\mathscr{F}$, there is a natural bijection

$$\operatorname{Hom}_{\mathscr{O}_X}(\mathscr{H}, \mathscr{F}) \xleftrightarrow{\sim} \operatorname{Hom}_{\mathscr{O}_Y}(\mathscr{G}, \pi_*\mathscr{F}).$$

Also, by (15.16), the construction of $\mathscr{H}$ must be unique.

Finally, we need to check that φ is a natural transformation. Let $\theta : \mathscr{F} \to \mathscr{F}'$. Consider the following diagram,

$$\begin{tikzcd}
\operatorname{Hom}_{\mathscr{O}_X}(\mathscr{H}, \mathscr{F}) \arrow[r, "\varphi_{\mathscr{F}}"] \arrow[d, "h^{\mathscr{H}}(\theta)"'] & \operatorname{Hom}_B(N, \Gamma(X, \mathscr{F})) \arrow[r, leftrightarrow, "\sim"] \arrow[d, "h^N(\Gamma(X,\theta))"] & \operatorname{Hom}_{\mathscr{O}_Y}(\mathscr{G}, \pi_*\mathscr{F}) \\
\operatorname{Hom}_{\mathscr{O}_X}(\mathscr{H}, \mathscr{F}') \arrow[r, "\varphi_{\mathscr{F}'}"] & \operatorname{Hom}_B(N, \Gamma(X, \mathscr{F}')) \arrow[r, leftrightarrow, "\sim"] & \operatorname{Hom}_{\mathscr{O}_Y}(\mathscr{G}, \pi_*\mathscr{F}')
\end{tikzcd}$$

we want to show above diagram is commutative. Pick $\alpha : \mathscr{H} \to \mathscr{F}$ in $\mathbf{Mod}_{\mathscr{O}_X}$, then

$$h^N(\Gamma(X,\theta)) \circ \varphi_{\mathscr{F}}(\alpha) = \theta_X \circ \widetilde{\alpha}$$

and

$$\varphi_{\mathscr{F}'} \circ h^{\mathscr{H}}(\theta)(\alpha) = \varphi_{\mathscr{F}'}(\theta \circ \alpha) = \theta_X \circ \widetilde{\alpha},$$

hence, $h^N(\Gamma(X,\theta)) \circ \varphi_{\mathscr{F}} = \varphi_{\mathscr{F}'} \circ h^{\mathscr{H}}(\theta)$, which implies τ is a natural transformation. Moreover, τ is a natural isomorphism, since $\tau_{\mathscr{F}}$ is an isomorphism for each $\mathscr{F} \in \mathbf{Mod}_{\mathscr{O}_X}$. Therefore, the construction of the quasicoherent sheaf in §15.5.1 satisfies the universal property of pullback of $\mathscr{G}$. □

Proposition 15.5.1

Suppose $i : U \hookrightarrow X$ is an open embedding of ringed spaces, and $\mathscr{F}$ is an $\mathscr{O}_X$-module. Then $\mathscr{F}|_U$ satisfies the universal property of $i^*\mathscr{F}$ (and thus deserves to be called $i^*\mathscr{F}$). In other words , for each $\mathscr{O}_U$-module $\mathscr{E}$, there is a bijection

$$\operatorname{Hom}_{\mathscr{O}_U}(\mathscr{F}|_U, \mathscr{E}) \xleftrightarrow{\sim} \operatorname{Hom}_{\mathscr{O}_X}(\mathscr{F}, i_*\mathscr{E}),$$

functorial in $\mathscr{E}$.

Proof Let $\alpha \in \operatorname{Hom}_{\mathscr{O}_U}(\mathscr{F}|_U, \mathscr{E})$. Define $\Phi_{\mathscr{E}} : \operatorname{Hom}_{\mathscr{O}_U}(\mathscr{F}|_U, \mathscr{E}) \to \operatorname{Hom}_{\mathscr{O}_X}(\mathscr{F}, i_*\mathscr{E})$ by setting $\alpha \mapsto \Phi_{\mathscr{E}}(\alpha)$ where define $\Phi_{\mathscr{E}}(\alpha)$ as following: let $V \hookrightarrow X$ be an open subset of X, define $\Phi_{\mathscr{E}}(\alpha)_V : \Gamma(V,\mathscr{F}) \to \Gamma(U\cap V, \mathscr{E})$ by setting $\Phi_{\mathscr{E}}(\alpha)_V = \alpha_{U\cap V}$. $\Phi_{\mathscr{E}}(\alpha)$ is indeed a morphism of sheaves, since it is defined by α and α is a morphism of sheaves. Hence, we defined a map $\Phi_{\mathscr{E}} : \operatorname{Hom}_{\mathscr{O}_U}(\mathscr{F}|_U, \mathscr{E}) \to \operatorname{Hom}_{\mathscr{O}_X}(\mathscr{F}, i_*\mathscr{E})$.

We next show $\Phi_{\mathscr{E}}$ is an isomorphism. Define $\Psi_{\mathscr{E}} : \operatorname{Hom}_{\mathscr{O}_X}(\mathscr{F}, i_*\mathscr{E}) \to \operatorname{Hom}_{\mathscr{O}_U}(\mathscr{F}|_U, \mathscr{E})$ as following: for each open subset $V \hookrightarrow U$, define $\Psi_{\mathscr{E}}(\beta) : \mathscr{F}|_U \to \mathscr{E}$ by setting $\Psi_{\mathscr{E}}(\beta)_V = \beta_V$. Since β is a morphism of sheaves, so is $\Psi_{\mathscr{E}}(\beta)$. Therefore, we defined a map $\Psi_{\mathscr{E}} : \operatorname{Hom}_{\mathscr{O}_X}(\mathscr{F}, i_*\mathscr{E}) \to \operatorname{Hom}_{\mathscr{O}_U}(\mathscr{F}|_U, \mathscr{E})$. Consider $\Phi_{\mathscr{E}} \circ \Psi_{\mathscr{E}}$ and $\Psi_{\mathscr{E}} \circ \Phi_{\mathscr{E}}$. Pick $\alpha \in \operatorname{Hom}_{\mathscr{O}_U}(\mathscr{F}|_U, \mathscr{E})$ and $\beta \in \operatorname{Hom}_{\mathscr{O}_X}(\mathscr{F}, i_*\mathscr{E})$. For any open subset $V \hookrightarrow U$, we have

$$(\Psi_{\mathscr{E}} \circ \Phi_{\mathscr{E}}(\alpha))_V = \Phi_{\mathscr{E}}(\alpha)_V = \alpha_{U\cap V} = \alpha_V,$$

which implies $\Psi_{\mathscr{E}} \circ \Phi_{\mathscr{E}}(\alpha) = \alpha$, hence, $\Psi_{\mathscr{E}} \circ \Phi_{\mathscr{E}} = \mathrm{id}$. On the other hand, for any open subset $V \hookrightarrow X$, we have

$$(\Phi_{\mathscr{E}} \circ \Psi_{\mathscr{E}}(\beta))_V = \Psi_{\mathscr{E}}(\beta)_{U\cap V} = \beta_{U\cap V} = \beta_V|_{U\cap V} = \beta|_V,$$

where last equal since $\beta_V(s)|_{U\cap V} \in \Gamma(U\cap V, i_*\mathscr{E}) = \Gamma(V, i_*\mathscr{E})$. Hence, $\Phi_{\mathscr{E}} \circ \Psi_{\mathscr{E}} = \mathrm{id}$. Consequently, $\Phi_{\mathscr{E}}$ is an isomorphism for each $\mathscr{E} \in \mathbf{Mod}_{\mathscr{O}_U}$.

Finally, we want to show that Φ is a natural transformation. For $\theta : \mathscr{E} \to \mathscr{E}'$, consider the following diagram,

$$\begin{tikzcd}
\operatorname{Hom}_{\mathscr{O}_U}(\mathscr{F}|_U, \mathscr{E}) \arrow[r, "\Phi_{\mathscr{E}}"] \arrow[d, "h^{\mathscr{F}|_U}(\theta)"'] & \operatorname{Hom}_{\mathscr{O}_X}(\mathscr{F}, i_*\mathscr{E}) \arrow[d, "h^{\mathscr{F}}(i_*\theta)"] \\
\operatorname{Hom}_{\mathscr{O}_U}(\mathscr{F}|_U, \mathscr{E}') \arrow[r, "\Phi_{\mathscr{E}'}"] & \operatorname{Hom}_{\mathscr{O}_X}(\mathscr{F}, i_*\mathscr{E}')
\end{tikzcd}$$

we want to check above diagram is commutative. Pick $\alpha \in \operatorname{Hom}_{\mathscr{O}_U}(\mathscr{F}|_U, \mathscr{E})$, then, for each $V \hookrightarrow X$ open,

$$(h^{\mathscr{F}}(i_*\theta) \circ \Phi_{\mathscr{E}}(\alpha))_V = (i_*\theta \circ \Phi_{\mathscr{E}}(\alpha))_V = \theta_{U\cap V} \circ \alpha_{U\cap V}$$

and

$$(\Phi_{\mathscr{E}'} \circ h^{\mathscr{F}|_U}(\theta)(\beta))_V = (h^{\mathscr{F}|_U}(\theta)(\beta))_{V\cap U} = \theta_{U\cap V} \circ \alpha_{U\cap V}.$$

Therefore, $h^{\mathscr{F}}(i_*\theta) \circ \Phi_{\mathscr{E}} = \Phi_{\mathscr{E}'} \circ h^{\mathscr{F}|_U}(\theta)$, which implies that Φ is a natural transformation, by each $\Phi_{\mathscr{E}}$ is an isomorphism, we know that Φ is a natural isomorphism. $\square$

Proposition 15.5.2

Suppose $\pi : X \to Y$ is a morphism of ringed spaces, and $\mathscr{G}$ is an $\mathscr{O}_Y$-module. If $\pi^*\mathscr{G}$ satisfies the universal property, then if $j : V \hookrightarrow Y$ is any open subset and $i : U = \pi^{-1}(V) \hookrightarrow X$ (see (15.17)), then $(\pi^*\mathscr{G})|_U$ satisfies the universal property for $\pi|_U : U \to V$. Thus $(\pi^*\mathscr{G})|_U$ deserves to be called $\pi|_U^*(\mathscr{G}|_V)$.

Remark 15.25 You will notice that we really need to work with $\mathscr{O}$-modules, not just with quasicoherent sheaves.

Proof Denote:

$$\begin{tikzcd} \pi^{-1}(V) \arrow[r, equal] & U \arrow[r, hook, "i"] \arrow[d, "\pi|_U"'] & X \arrow[d, "\pi"] \\ & V \arrow[r, hook, "j"] & Y \end{tikzcd} \tag{15.17}$$

If $\mathscr{F}'$ is an $\mathscr{O}_U$-module, we have a series of bijections (using Proposition 15.5.1 and adjointness of pullback and pushforward):

$$\begin{aligned} \mathrm{Hom}_{\mathscr{O}_U}((\pi^*\mathscr{G})|_U, \mathscr{F}') &\overset{\sim}{\longleftrightarrow} \mathrm{Hom}_{\mathscr{O}_U}(i^*(\pi^*\mathscr{G}), \mathscr{F}') \\ &\overset{\sim}{\longleftrightarrow} \mathrm{Hom}_{\mathscr{O}_X}(\pi^*\mathscr{G}, i_*\mathscr{F}') \\ &\overset{\sim}{\longleftrightarrow} \mathrm{Hom}_{\mathscr{O}_Y}(\mathscr{G}, \pi_* i_*\mathscr{F}') \\ &\overset{\sim}{\longleftrightarrow} \mathrm{Hom}_{\mathscr{O}_Y}(\mathscr{G}, j_*(\pi|_U)_*\mathscr{F}') \\ &\overset{\sim}{\longleftrightarrow} \mathrm{Hom}_{\mathscr{O}_V}(j^*\mathscr{G}, (\pi|_U)_*\mathscr{F}') \\ &\overset{\sim}{\longleftrightarrow} \mathrm{Hom}_{\mathscr{O}_V}(\mathscr{G}|_V, (\pi|_U)_*\mathscr{F}'). \end{aligned}$$

We have thus described a bijection

$$\mathrm{Hom}_{\mathscr{O}_U}((\pi^*\mathscr{G})|_U, \mathscr{F}') \overset{\sim}{\longleftrightarrow} \mathrm{Hom}_{\mathscr{O}_V}(\mathscr{G}|_V, (\pi|_U)_*\mathscr{F}'),$$

which is (by construction) functorial in $\mathscr{F}'$. $\square$

Hence the discussion in the first paragraph of §15.5.2 is justified. For example, thanks to Lemma **??**, the pullback π^* exists if Y is an open subset of an affine scheme.

At this point, we could show that the pullback exists, following the ideal behind the construction of the fibered product: we would start with the definition when Y is affine, and "glue". We will instead take another route.

15.5.3 Third definition: Pullback of $\mathscr{O}$-modules via explicit construction

Suppose $\pi : X \to Y$ is a morphism of ringed spaces, and $\mathscr{G}$ is an $\mathscr{O}_Y$-module. Of cause, our case of interest is: π is a morphism of schemes, and $\mathscr{G}$ is quasicoherent. Now $\pi^{-1}\mathscr{G}$ is a $\pi^{-1}\mathscr{O}_Y$. (Notice that we are using the ringed space $(X, \pi^{-1}\mathscr{O}_Y)$, not $(X, \mathscr{O}_X)$. The inverse image construction π^{-1} was discussed in §3.7.) Furthermore, $\mathscr{O}_X$ is also a $\pi^{-1}\mathscr{O}_Y$-module, via the map $\pi^{-1}\mathscr{O}_Y \to \mathscr{O}_X$ that is part of the data of the morphism π.

Definition 15.5.2 (Pullback)

Suppose $\pi : X \to Y$ is a morphism of ringed spaces, and $\mathscr{G}$ is an $\mathscr{O}_Y$-module. Define the ***pullback of $\mathscr{G}$ by π as the $\mathscr{O}$-module***

$$\pi^*\mathscr{G} := \pi^{-1}\mathscr{G} \otimes_{\pi^{-1}\mathscr{O}_Y} \mathscr{O}_X. \tag{15.18}$$

It is immediate that pullback is a covariant functor $\pi^* : \mathbf{Mod}_{\mathscr{O}_Y} \to \mathbf{Mod}_{\mathscr{O}_X}$, and that $\pi^*\mathscr{O}_Y$ is canonically isomorphic to $\mathscr{O}_X$. We have now established §15.5 (iii), that pullback of quasicoherent sheaves subsumes our older definition of pullback of locally free sheaves.

Lemma 15.5.3 (The restriction and change of rings functors are adjoint to each other)

Let X be a ringed space. Let $\mathscr{O}_1 \to \mathscr{O}_2$ be a morphism of sheaves of rings on X. Let $\mathscr{F} \in \mathbf{Mod}_{\mathscr{O}_2}$ and $\mathscr{G} \in \mathbf{Mod}_{\mathscr{O}_1}$ (the restriction $\mathscr{F}_{\mathscr{O}_1}$ of $\mathscr{F}$ is clearly a sheaf of $\mathscr{O}_1$-modules), then there exists a canonical bijection

$$\operatorname{Hom}_{\mathscr{O}_1}(\mathscr{G}, \mathscr{F}_{\mathscr{O}_1}) \xleftrightarrow{\sim} \operatorname{Hom}_{\mathscr{O}_2}(\mathscr{O}_2 \otimes_{\mathscr{O}_1} \mathscr{G}, \mathscr{F}).$$

In other words, ***the restriction and change of rings functors are adjoint to each other****.*

Proof See [Sta25, Lemma 008A]. □

Lemma 15.5.4

Definition 15.5.2 satisfies the universal property in Definition 15.5.1. Thus the pullback exists, at least as a functor $\mathbf{Mod}_{\mathscr{O}_Y} \to \mathbf{Mod}_{\mathscr{O}_X}$.

Proof By Lemma 15.5.3 and Theorem 3.7.1, for any $\mathscr{F} \in \mathbf{Mod}_{\mathscr{O}_X}$, there exists canonical bijections

$$\begin{aligned}\operatorname{Hom}_{\mathscr{O}_X}(\pi^{-1}\mathscr{G} \otimes_{\pi^{-1}\mathscr{O}_Y} \mathscr{O}_X, \mathscr{F}) &\xleftrightarrow{\sim} \operatorname{Hom}_{\pi^{-1}\mathscr{O}_Y}(\pi^{-1}\mathscr{G}, \mathscr{F}) \\ &\xleftrightarrow{\sim} \operatorname{Hom}_{\mathscr{O}_Y}(\mathscr{G}, \pi_*\mathscr{F}),\end{aligned}$$

as we desired. □

Proposition 15.5.3 (Pullback is a covariant functor on quasicoherent sheaves)

If $\pi : X \to Y$ is a morphism of schemes, then π^* gives a covariant functor $\mathbf{QCoh}_Y \to \mathbf{QCoh}_X$.

Proof We already construct π^* as a functor between $\mathbf{Mod}_{\mathscr{O}_Y}$ and $\mathbf{Mod}_{\mathscr{O}_X}$ (Lemma 15.5.4). It remains to show that π^* send quasicoherent sheaves on Y to quasicoherent sheaves on X.

Let $\mathscr{G} \in \mathbf{QCoh}_Y$, recall the definition of quasicoherent sheaf (Definition 7.3.1), for every point $y \in Y$ there exists an open neighborhood $y \in U \hookrightarrow Y$ such that sequence

$$\mathscr{O}_U^{\oplus I} \longrightarrow \mathscr{O}_U^{\oplus J} \longrightarrow \mathscr{G}|_U \longrightarrow 0$$

is exact. Say $V := \pi^{-1}(U)$. By the universal property of pullback, we know that π^* is a right exact functor (and thus preserves small colimits), so the following sequence is exact.

$$(\pi|_V^*\mathscr{O}_U)^{\oplus I} \longrightarrow (\pi|_V^*\mathscr{O}_U)^{\oplus J} \longrightarrow \pi|_V^*\mathscr{G}|_U \longrightarrow 0.$$

Since $\pi|_V^*\mathscr{O}_U$ is canonically isomorphic to $\mathscr{O}_V$ and $\pi|_V^*\mathscr{G}|_U$ is canonical isomorphic to $(\pi^*\mathscr{G})|_V$ (Proposition 15.5.2), the sequence

$$\mathscr{O}_V^{\oplus I} \longrightarrow \mathscr{O}_V^{\oplus J} \longrightarrow (\pi^*\mathscr{G})_V \longrightarrow 0$$

is exact, which implies that $\pi^*\mathscr{G}$ is a quasicoherent sheaf. □

Definition 15.5.3 (The pullback of the section)

*If $\mathscr{G}$ is a quasicoherent sheaf on Y, note that we can interpret a global section of Y as a map $s : \mathscr{O}_Y \to \mathscr{G}$ (the section of $\mathscr{G}$ is the image of the section 1 of $\mathscr{O}_Y$). Note that a section of $\mathscr{G}$ is the data of a map $\mathscr{O}_Y \to \mathscr{G}$. By Proposition 15.5.3, if $s : \mathscr{O}_Y \to \mathscr{G}$ is a section of $\mathscr{G}$, then there is a natural section $\pi^* s : \mathscr{O}_X \to \pi^*\mathscr{G}$ of $\pi^*\mathscr{G}$. We call $\pi^* s$ **the pullback of the section** s.*

The following is immediate from the universal property, and Proposition 15.5.3. The "quasicompact and quasiseparated" hypotheses are solely to ensure that π_* indeed sends $\mathbf{QCoh}_X$ to $\mathbf{QCoh}_Y$ (Corollary 15.4.1).

Theorem 15.5.1

Suppose $\pi : X \to Y$ is a quasicompact, quasiseparated morphism. Then $\pi^* : \mathbf{QCoh}_Y \to \mathbf{QCoh}_X$ and $\pi_* : \mathbf{QCoh}_X \to \mathbf{QCoh}_Y$ are an adjoint pair, i.e., $\pi^* \dashv \pi_*$: there is an isomorphism

$$\operatorname{Hom}_{\mathscr{O}_X}(\pi^*\mathscr{G}, \mathscr{F}) \overset{\sim}{\longleftrightarrow} \operatorname{Hom}_{\mathscr{O}_Y}(\mathscr{G}, \pi_*\mathscr{F}), \tag{15.19}$$

functorial in both $\mathscr{F} \in \mathbf{QCoh}_X$ and $\mathscr{G} \in \mathbf{QCoh}_Y$.

Corollary 15.5.1 (Pullback is a right-exact functor)

Suppose $\pi : X \to Y$ is a morphism of schemes. Then $\pi^* : \mathbf{QCoh}_Y \to \mathbf{QCoh}_X$ is a right-exact functor.

Proof Recall Category Theory, left adjoint implies right exact. □

Remark 15.26 For an important class of morphisms, called **flat morphisms**, we will see that pullback is an exact functor (**??**).

Proposition 15.5.4 (Pullback preserves finite type and finitely presented quasicoherent sheaves)

Suppose that $\pi : X \to Y$ is a morphism of schemes, and $\mathscr{G}$ is a quasicoherent sheaf on Y. If $\mathscr{G}$ is finite type (resp., finitely presented), then $\pi^*\mathscr{G}$ is finite type (resp., finitely presented).

Proof Suppose $\mathscr{G}$ is a finite type quasicoherent sheaf on Y. Let $\{U_i \hookrightarrow Y\}$ be an affine cover of Y, recall the definition of finite type quasicoherent sheaf, we have surjection

$$\mathscr{O}_{U_i}^{\oplus n_i} \longrightarrow \mathscr{G}|_{U_i} \longrightarrow 0.$$

for some n_i. Say $V_i := \pi^{-1}(U_i)$. Apply right-exact functor π^*, and by Proposition 15.5.2, we have

$$\begin{tikzcd}
(\pi|_{V_i}^*\mathscr{O}_{U_i})^{\oplus n_i} \arrow[r] \arrow[d, leftrightarrow, "\sim"'] & \pi|_{V_i}^*\mathscr{G}|_{U_i} \arrow[r] \arrow[d, leftrightarrow, "\sim"'] & 0 \\
\mathscr{O}_{V_i}^{\oplus n_i} & (\pi^*\mathscr{G})|_{V_i}. &
\end{tikzcd}$$

It follows that $\pi^*\mathscr{G}$ is also finite type.

For the case which $\mathscr{G}$ is finitely presented quasicoherent sheaf. Apply the right-exact functor π^* to exact sequence

$$\mathscr{O}_{U_i}^{\oplus m_i} \longrightarrow \mathscr{O}_{U_i}^{\oplus n_i} \longrightarrow \mathscr{G}|_{U_i} \longrightarrow 0,$$

then we done. □

Proposition 15.5.5

Suppose $\xi : W \to X$ and $\pi : X \to Y$ are morphisms of schemes, and $\mathscr{G}$ is a quasicoherent sheaf on Y. Then there is a canonical isomorphism $\xi^*\pi^*\mathscr{G} \overset{\sim}{\longleftrightarrow} (\pi \circ \xi)^*\mathscr{G}$.

Proof It suffices to check that $\xi^*\pi^*\mathscr{G}$ satisfies the universal property of pullback. For any $\mathscr{F} \in \mathbf{Mod}_{\mathscr{O}_W}$, notice that

$$\begin{aligned}\mathrm{Hom}_{\mathscr{O}_W}(\xi^*\pi^*\mathscr{G},\mathscr{F}) &\xleftrightarrow{\sim} \mathrm{Hom}_{\mathscr{O}_X}(\pi^*\mathscr{G},\xi_*\mathscr{F})\\ &\xleftrightarrow{\sim} \mathrm{Hom}_{\mathscr{O}_Y}(\mathscr{G},\pi_*\xi_*\mathscr{F})\\ &= \mathrm{Hom}_{\mathscr{O}_Y}(\mathscr{G},(\pi\circ\xi)_*\mathscr{F})\\ &\xleftrightarrow{\sim} \mathrm{Hom}_{\mathscr{O}_W}((\pi\circ\xi)^*\mathscr{G},\mathscr{F}),\end{aligned}$$

we obtain a canonical isomorphism

$$\xi^*\pi^*\mathscr{G} \xleftrightarrow{\sim} (\pi\circ\xi)^*\mathscr{G}.$$

□

Proposition 15.5.6

Suppose $\pi : X \to Y$ is a morphism of schemes, with $\pi(p) = q$, and $\mathscr{G}$ is a quasicoherent sheaf on Y.

*(a) **(pullback on stalks)** The pullback induces an isomorphism*

$$(\pi^*\mathscr{G})_p \xrightarrow{\sim} \mathscr{G}_q \otimes_{\mathscr{O}_{Y,q}} \mathscr{O}_{X,p}. \tag{15.20}$$

*(b) **(pullback on fibers)** The pullback induces an isomorphism*

$$(\pi^*\mathscr{G})|_p \xrightarrow{\sim} \mathscr{G}|_q \otimes_{\kappa(q)} \kappa(p).$$

Proof ***For (a).*** Recall Proposition 3.6.7, we have

$$(\pi^*\mathscr{G})_p = (\pi^{-1}\mathscr{G} \otimes_{\pi^{-1}\mathscr{O}_Y} \mathscr{O}_X)_p \xleftrightarrow{\sim} (\pi^{-1}\mathscr{G})_p \otimes_{(\pi^{-1}\mathscr{O}_Y)_p} \mathscr{O}_{X,p} \xleftrightarrow{\sim} \mathscr{G}_q \otimes_{\mathscr{O}_{Y,q}} \mathscr{O}_{X,p}.$$

For (b). Without loss of generally, we may assume $Y = \operatorname{Spec} B$, suppose there $\operatorname{Spec} A \hookrightarrow X$ is an affine neighborhood of p such that $\pi(\operatorname{Spec} A) \subseteq \operatorname{Spec} B$. Since $\mathscr{G}$ is a quasicoherent sheaf on Y, say $\mathscr{G} = \widetilde{N}$. Denote $p = [\mathfrak{p}] \in \operatorname{Spec} A$ and $q = [\mathfrak{q}] \in \operatorname{Spec} B$, then, by part (a),

$$(\pi^*\mathscr{G})_p \cong N_{\mathfrak{q}} \otimes_{B_{\mathfrak{q}}} A_{\mathfrak{p}}.$$

So,

$$(\pi^*\mathscr{G})|_p \cong (\pi^*\mathscr{G})_p \otimes_{\mathscr{O}_{X,p}} \kappa(p) \cong (N_{\mathfrak{q}} \otimes_{B_{\mathfrak{q}}} A_{\mathfrak{p}}) \otimes_{A_{\mathfrak{p}}} A_{\mathfrak{p}}/\mathfrak{p}A_{\mathfrak{p}} \cong N_{\mathfrak{q}} \otimes_{B_{\mathfrak{q}}} A_{\mathfrak{p}}/\mathfrak{p}A_{\mathfrak{p}}.$$

On the other hand, notice that

$$\mathscr{G}|_q \otimes_{\kappa(q)} \kappa(p) \cong (N_{\mathfrak{q}} \otimes_{B_{\mathfrak{q}}} B_{\mathfrak{q}}/\mathfrak{q}B_{\mathfrak{q}}) \otimes_{B_{\mathfrak{q}}/\mathfrak{q}B_{\mathfrak{q}}} A_{\mathfrak{p}}/\mathfrak{p}A_{\mathfrak{p}} \cong N_{\mathfrak{q}} \otimes_{B_{\mathfrak{q}}} A_{\mathfrak{p}}/\mathfrak{p}A_{\mathfrak{p}}.$$

Hence,

$$(\pi^*\mathscr{G})|_p \xleftrightarrow{\sim} \mathscr{G}|_q \otimes_{\kappa(q)} \kappa(p).$$

□

Proposition 15.5.7 (Pullback preserves tensor product)

Suppose $\pi : X \to Y$ is a morphism of schemes, and $\mathscr{G}$ and $\mathscr{G}'$ are quasicoherent sheaves on Y. Then

$$\pi^*(\mathscr{G} \otimes_{\mathscr{O}_Y} \mathscr{G}') = \pi^*\mathscr{G} \otimes_{\mathscr{O}_X} \pi^*\mathscr{G}'.$$

Proof We check it at the level of stalks. Let $q \in Y$ with $\pi(p) = q$, then

$$\begin{aligned}(\pi^*(\mathscr{G} \otimes_{\mathscr{O}_Y} \mathscr{G}'))_p &\xleftrightarrow{\sim} (\mathscr{G} \otimes_{\mathscr{O}_Y} \mathscr{G}')_q \otimes_{\mathscr{O}_{Y,q}} \mathscr{O}_{X,p}\\ &\xleftrightarrow{\sim} \mathscr{G}_q \otimes_{\mathscr{O}_{Y,q}} \mathscr{G}'_q \otimes_{\mathscr{O}_{Y,q}} \mathscr{O}_{X,p}.\end{aligned}$$

On the other hand,

$$\begin{aligned}(\pi^*\mathscr{G} \otimes_{\mathscr{O}_X} \pi^*\mathscr{G}')_p &\xleftrightarrow{\sim} (\pi^*\mathscr{G})_p \otimes_{\mathscr{O}_{X,p}} (\pi^*\mathscr{G}')_p \\ &\xleftrightarrow{\sim} (\mathscr{G}_q \otimes_{\mathscr{O}_{Y,q}} \mathscr{O}_{X,p}) \otimes_{\mathscr{O}_{X,p}} (\mathscr{G}'_q \otimes_{\mathscr{O}_{Y,q}} \mathscr{O}_{X,p}) \\ &\xleftrightarrow{\sim} \mathscr{G}_q \otimes_{\mathscr{O}_{Y,q}} \mathscr{G}'_q \otimes_{\mathscr{O}_{Y,q}} \mathscr{O}_{X,p},\end{aligned}$$

so,

$$(\pi^*(\mathscr{G} \otimes_{\mathscr{O}_Y} \mathscr{G}'))_p \xleftrightarrow{\sim} (\pi^*\mathscr{G} \otimes_{\mathscr{O}_X} \pi^*\mathscr{G}')_p$$

for each $p \in X$. It follows that

$$\pi^*(\mathscr{G} \otimes_{\mathscr{O}_Y} \mathscr{G}') \xleftrightarrow{\sim} \pi^*\mathscr{G} \otimes_{\mathscr{O}_X} \pi^*\mathscr{G}',$$

as we desired. □

Example 15.7 (Pullback is not left-exact) Consider the exact sequence of sheaves on $\mathbb{A}^1_k$, where $i : p \hookrightarrow \mathbb{A}^1_k$ is the origin:

$$0 \longrightarrow \mathscr{O}_{\mathbb{A}^1_k}(-p) \longrightarrow \mathscr{O}_{\mathbb{A}^1_k} \longrightarrow i_*\mathscr{O}|_p \longrightarrow 0.$$

(This is the closed subscheme exact sequence (10.1) for $p \in \mathbb{A}^1_k$, and corresponds to the exact sequence of $k[t]$-modules $0 \to tk[t] \to k[t] \to k \to 0$. Warning: Here $\mathscr{O}|_p$ is not the stalk $\mathscr{O}_p$; it is the structure sheaf of the scheme p.) Restrict to p.

Proof We claim that pullback to p is not a left-exact sequence. The closed subscheme exact sequence gives an exact sequence of $k[t]$-modules

$$0 \longrightarrow (t) \longrightarrow k[t] \longrightarrow k \longrightarrow 0.$$

Apply right-exact functor $\square \otimes_{k[x]} k[t]/(t)$, we obtain an exact sequence

$$\begin{tikzcd} (t) \otimes_{k[t]} (k[t]/(t)) \arrow[r] \arrow[d, equal] & k[t] \otimes_{k[t]} (k[t]/(t)) \arrow[r] \arrow[d, equal] & k \otimes_{k[t]} (k[t]/(t)) \arrow[r] \arrow[d, equal] & 0 \\ (t)/(t^2) & k & k & \end{tikzcd}$$

Clearly, $(t)/(t^2) \to k[t]/(t)$ is not injective, since it has kernel $(t)/(t^2)$. It follows that in this case pullback is not left-exact. □

Proposition 15.5.8 (The push-pull map for quasicoherent sheaves)

Suppose that

$$\begin{tikzcd} X' \arrow[r, "\psi'"] \arrow[d, "\pi'"'] & X \arrow[d, "\pi"] \\ Y' \arrow[r, "\psi"] & Y \end{tikzcd}$$

is a commutative diagram of schemes, and π and π' are both quasicompact and quasiseparated (so pushforward by both sends quasicoherent sheaves to quasicoherent sheaves). Then there is a natural map (functorial in $\mathscr{F} \in \mathbf{QCoh}_X$) of quasicoherent sheaves on Y',

$$\psi^*\pi_*\mathscr{F} \longrightarrow \pi'_*(\psi')^*\mathscr{F}.$$

Proof Our main tool is using Theorem 15.5.1. We start with the identity $(\psi')^*\mathscr{F} \xrightarrow{\sim} (\psi')^*\mathscr{F}$ on X'. By $(\psi')^* \dashv \psi'_*$, we obtain $\mathscr{F} \to (\psi')_*(\psi')^*\mathscr{F}$ on X. Apply π_* to get a map $\pi_*\mathscr{F} \to \pi_*(\psi')_*(\psi')^*\mathscr{F}$ on Y. By $\pi \circ \psi' = \psi \circ \pi'$, we have $\pi_*\mathscr{F} \to \psi_*(\pi'_*)(\psi')^*\mathscr{F}$ on Y. By adjointness $\psi^* \dashv \psi_*$, we obtain

$$\psi^*\pi_*\mathscr{F} \longrightarrow (\pi'_*)(\psi')^*\mathscr{F}$$

on Y'. $\square$

By applying Proposition 15.5.8 in the special case where Y' is a point q of Y,

$$\begin{tikzcd} \pi^{-1}(q) \arrow[r, "i'"] \arrow[d, "\pi|_{\pi^{-1}(q)}"'] & X \arrow[d, "\pi"] \\ q \arrow[r, "i"] & Y \end{tikzcd}$$

we see that there is a natural map from the fiber of the pushforward to the sections over the fiber:

$$(\pi_*\mathscr{F})_q \otimes_{\mathscr{O}_{Y,q}} \kappa(q) \longrightarrow \Gamma(\pi^{-1}(q), \mathscr{F}|_{\pi^{-1}(q)}), \tag{15.21}$$

where $(\pi_*\mathscr{F})_q \otimes_{\mathscr{O}_{Y,q}} \kappa(q)$ is given by

$$i^*\pi_*\mathscr{F} = (\pi_*\mathscr{F})|_q = (\pi_*\mathscr{F})_q \otimes_{\mathscr{O}_{Y,q}} \kappa(q)$$

and $\Gamma(\pi^{-1}(q), \mathscr{F}|_{\pi^{-1}(q)})$ is given by

$$(\pi|_{\pi^{-1}(q)})_*(i')^*\mathscr{F} = (\pi|_{\pi^{-1}(q)})_*\mathscr{F}|_{\pi^{-1}(q)} = \Gamma(\pi^{-1}(q), \mathscr{F}|_{\pi^{-1}(q)}).$$

One might hope that (15.21) is an isomorphism, i.e., that $\pi_*\mathscr{F}$ "glues together" the fibers $\Gamma(\pi^{-1}(q), \mathscr{F}|_{\pi^{-1}(q)})$. This is too much to ask, but at least (15.21) gives a map. (In fact, under just the right circumstances, (15.21) is an isomorphism; see §**??**.)

Proposition 15.5.9 (The projection formula, to be generalized in ??)

Suppose $\pi : X \to Y$ is a quasicompact and quasiseparated, and $\mathscr{F}$ and $\mathscr{G}$ are quasicoherent sheaves on X and Y respectively.

(a) There is a natural morphism $(\pi_*\mathscr{F}) \otimes \mathscr{G} \to \pi_*(\mathscr{F} \otimes \pi^*\mathscr{G})$.

(b) If $\mathscr{G}$ is locally free, then this natural morphism is an isomorphism.

(c) If π is affine, then show that this natural morphism is an isomorphism.

Remark 15.27 Sadly, the phrase "projection formula" is used for a number of facts, not all related.

Proof ***For (a).*** We start with the identity $\pi_*\mathscr{F} \xrightarrow{\sim} \pi_*\mathscr{F}$, by adjointness of $\pi^* \dashv \pi_*$, we obtain $\pi^*\pi_*\mathscr{F} \to \mathscr{F}$. Apply functor $\square \otimes_{\mathscr{O}_X} \pi^*\mathscr{G}$, we obtain

$$\pi^*\pi_*\mathscr{F} \otimes_{\mathscr{O}_X} \pi^*\mathscr{G} \longrightarrow \mathscr{F} \otimes_{\mathscr{O}_X} \pi^*\mathscr{G}.$$

Notice that $\pi^*\pi_*\mathscr{F} \otimes_{\mathscr{O}_X} \pi^*\mathscr{G} \cong \pi^*(\pi_*\mathscr{F} \otimes_{\mathscr{O}_X} \mathscr{G})$, by using $\pi^* \dashv \pi_*$ again, there is a natural morphism

$$(\pi_*\mathscr{F}) \otimes_{\mathscr{O}_X} \mathscr{G} \longrightarrow \pi_*(\mathscr{F} \otimes_{\mathscr{O}_X} \pi^*\mathscr{G}).$$

For (b). We first show a special case, suppose Y is affine and $\mathscr{G} \cong \mathscr{O}_Y$. We start with the identity $\mathrm{id}_{\mathscr{F}} : \pi_*\mathscr{F} \xrightarrow{\sim} \pi_*\mathscr{F}$, by adjointness of $\pi^* \dashv \pi_*$, we obtain $\varepsilon_{\mathscr{F}} : \pi^*\pi_*\mathscr{F} \to \mathscr{F}$ which is one-to-one corresponding to $\mathrm{id}_{\mathscr{F}}$. Apply functor $\square \otimes_{\mathscr{O}_X} \pi^*\mathscr{G} = \square \otimes_{\mathscr{O}_X} \mathscr{O}_X$, we have $\varepsilon_{\mathscr{F}} \otimes 1 = \varepsilon_{\mathscr{F}} : \pi^*\pi_*\mathscr{F} \to \mathscr{F}$, by using adjointness of $\pi^* \dashv \pi_*$ again, we obtain $\mathrm{id}_{\mathscr{F}} : \pi_*\mathscr{F} \to \pi_*\mathscr{F}$, which implies $(\pi_*\mathscr{F}) \otimes_{\mathscr{O}_X} \mathscr{G} \to \pi_*(\mathscr{F} \otimes_{\mathscr{O}_X} \pi^*\mathscr{G})$ is an isomorphism.

In general case, since $\mathscr{G}$ is locally free sheaf, there exists an open cover $\{U_i \hookrightarrow Y\}_i$ such that $\mathscr{G}|_{U_i} \cong \mathscr{O}_{U_i}^{\oplus I_i}$. Say $V_i := \pi^{-1}(U_i)$. Notice that

$$(\pi_*\mathscr{F})|_{U_i} \otimes_{\mathscr{O}_{U_i}} \mathscr{G}|_{U_i} \cong (\pi_*\mathscr{F})|_{U_i} \otimes_{\mathscr{O}_{U_i}} \mathscr{O}_{U_i}^{\oplus I_i} \cong (\pi_*\mathscr{F})|_{U_i}^{\oplus I_i} \cong \mathscr{F}|_{V_i}^{\oplus I_i}$$

and

$$\pi_*(\mathscr{F} \otimes_{\mathscr{O}_X} \pi^*\mathscr{G})|_{U_i} \cong (\mathscr{F} \otimes_{\mathscr{O}_X} \pi^*\mathscr{G})|_{V_i} \cong \mathscr{F}|_{V_i} \otimes_{\mathscr{O}_{V_i}} \pi|_{V_i}^*\mathscr{G}|_{U_i} \cong \mathscr{F}|_{V_i} \otimes_{\mathscr{O}_{V_i}} \mathscr{O}_{V_i}^{\oplus I_i} \cong \mathscr{F}|_{V_i}^{\oplus I_i},$$

since morphism $\pi^*\pi_*\mathscr{F} \otimes_{\mathscr{O}_X} \pi^*\mathscr{G} \longrightarrow \mathscr{F} \otimes_{\mathscr{O}_X} \pi^*\mathscr{G}$ comes from counit $\varepsilon : \pi^*\pi_* \to \mathrm{id}$, morphism $\pi^*\pi_*\mathscr{F} \otimes_{\mathscr{O}_X} \pi^*\mathscr{G} \longrightarrow \mathscr{F} \otimes_{\mathscr{O}_X} \pi^*\mathscr{G}$ must be an isomorphism, as we desired.

For (c). Without lose of generally, we may assume $Y = \operatorname{Spec} B$, since π is affine, say $X = \operatorname{Spec} A$. Let $\mathscr{F} \cong \widetilde{M}$ and $\mathscr{G} \cong \widetilde{N}$. Since the morphism comes from counit, we only need to check both sides are isomorphism. One the left hand side,

$$(\pi_*\mathscr{F}) \otimes_{\mathscr{O}_Y} \mathscr{G} \cong \widetilde{M_B} \otimes_{\widetilde{B}} \widetilde{N} \cong \widetilde{M \otimes_B N},$$

on the other hand,

$$\pi_*(\mathscr{F} \otimes_{\mathscr{O}_X} \pi^*\mathscr{G}) \cong \pi_*(\widetilde{M} \otimes_{\widetilde{A}} \widetilde{N \otimes_B A}) \cong M \otimes_A \widetilde{(A \otimes_B N)} \cong \widetilde{M \otimes_B N}.$$

Hence $(\pi_*\mathscr{F}) \otimes_{\mathscr{O}_Y} \mathscr{G} \to \pi_*(\mathscr{F} \otimes_{\mathscr{O}_X} \pi^*\mathscr{G})$ is an isomorphism, we win. □

Remark: Pulling back ideal sheaves

There is an important subtlety in pulling back quasicoherent ideal sheaves. Suppose $i : X \hookrightarrow Y$ is a closed embedding, and $\mu : Y' \to Y$ is an arbitrary morphism. Let $X' := X \times_Y Y'$. As "closed embeddings", the pullback to closed embedding.

$$\begin{tikzcd} X' \arrow[r] \arrow[d, hook, "i'"'] & X \arrow[d, hook, "i"] \\ Y' \arrow[r, "\mu"] & Y \end{tikzcd}$$

Now μ^* induces canonical isomorphisms $\mu^*\mathscr{O}_Y \stackrel{\sim}{\longleftrightarrow} \mathscr{O}_{Y'}$ and $\mu^*(i_*\mathscr{O}_X) \stackrel{\sim}{\longleftrightarrow} (i'_*\mathscr{O}_{X'})$. (Check affine-locally.) But it is not necessarily true that $\mu^*\mathscr{I}_{X/Y} = \mathscr{I}_{X'/Y'}$. (Example 15.7 .) This is because the application of μ^* to the closed subscheme exact sequence

$$0 \longrightarrow \mathscr{I}_{X/Y} \longrightarrow \mathscr{O}_Y \longrightarrow i_*\mathscr{O}_X \longrightarrow 0$$

yields something that is a priori only right-exact:

$$\mu^*\mathscr{I}_{X/Y} \longrightarrow \mathscr{O}_{Y'} \longrightarrow i'_*\mathscr{O}_{X'} \longrightarrow 0.$$

Thus as $\mathscr{I}_{X'/Y'}$ is the kernel of $\mathscr{O}_{Y'} \to i'_*\mathscr{O}_{X'}$, we see that $\mathscr{I}_{X'/Y'}$ is the image of $\mu^*\mathscr{I}_{X/Y}$ in $\mathscr{O}_{Y'}$. We can also see this explicitly from Lemma 11.2.2: affine-locally, the ideal of the pullback is generated by the pullback of the ideal.

Note also that if μ^* is an exact functor (which we will later see as a consequence of flatness, in **??**), then $\mu^*\mathscr{I}_{X/Y} \to \mathscr{I}_{X'/Y'}$ is an isomorphism.

15.6 The quasicoherent sheaf corresponding to a graded module

15.6.1 Describe quasicoherent sheaves on a projective A-scheme

We now describe quasicoherent sheaves on a projective A-scheme. Recall that a projective A-scheme is produced from the data of a $\mathbb{Z}^{\geq 0}$-graded ring $S_\bullet$, with $S_0 = A$, and $S_\bullet$ is a finitely generated ideal (a "finitely generated graded ring over A", Definition 5.5.4). The resulting scheme is denoted $\operatorname{Proj} S_\bullet$.

Suppose $M_\bullet$ is a graded $S_\bullet$-module, *graded by* $\mathbb{Z}$. (While reading the next section, you may wonder why we don't grade by $\mathbb{Z}^{\geq 0}$. You will see that it doesn't matter. A $\mathbb{Z}$-grading will make things cleaner when we produce an $M_\bullet$ from a quasicoherent sheaf on $\operatorname{Proj} S_\bullet$.) We define the quasicoherent sheaf $\widetilde{M_\bullet}$ as follows. (I will avoid calling it $\widetilde{M}$, as this might cause confusion with the affine case; but $\widetilde{M_\bullet}$ is not graded in any way.) For each homogenous f of positive degree, we define a quasicoherent sheaf $\widetilde{M_\bullet(f)}$ on the distinguished open $D(f) = \{p \in \operatorname{Proj} S_\bullet : f(p) \neq 0\}$ by setting $\widetilde{M_\bullet(f)} = (((M_\bullet)_f)_0)^\sim$ — note that $((M_\bullet)_f)_0$ is an $((S_\bullet)_f)_0$-

module, and recall that $D(f)$ is identified with $\operatorname{Spec}((S_\bullet)_f)_0$ (Exercise 5.9). As in (5.12), the subscript 0 means "the 0-graded piece". We have obvious isomorphisms of the restriction of $\widetilde{M_\bullet(f)}$ and $\widetilde{M_\bullet(g)}$ to $D(fg)$, satisfying the cocycle conditions. By Gluing sheaves 3.5.3, these sheaves glue together to a single sheaf $\widetilde{M_\bullet}$ on $\operatorname{Proj} S_\bullet$. We then discard the temporary notation $\widetilde{M_\bullet(f)}$.

The $\mathscr{O}$-module $\widetilde{M_\bullet}$ is clearly quasicoherent, because it is quasicoherent on each $D(f)$, and quasicoherence is local.

Lemma 15.6.1

There is an isomorphism between the stalk of $\widetilde{M_\bullet}$ at a point corresponding to homogeneous prime $\mathfrak{p} \subseteq S_\bullet$ and $((M_\bullet)_\mathfrak{p})_0$. Here $((M_\bullet)_\mathfrak{p})_0$ means "the degree 0 homogeneous fractions in $(M_\bullet)_\mathfrak{p}$," and $\mathfrak{p}$ should be a "relevant" homogeneous prime (i.e., $[\mathfrak{p}]$ is a point of $\operatorname{Proj} S_\bullet$).

Remark 15.28 You can use this lemma to give an alternate definition of $\widetilde{M_\bullet}$ in terms of "compatible stalks"; cf. Exercise 5.11.

Proof Let $\mathfrak{p} \in \operatorname{Proj} S_\bullet$, then there exists $\operatorname{Spec}((S_\bullet)_f)_0 \ni \mathfrak{p}$. Since $\widetilde{M_\bullet}|_{\operatorname{Spec}((S_\bullet)_f)_0} = ((M_\bullet)_f)_0^\sim$, we have

$$\widetilde{M_\bullet}_\mathfrak{p} = (\widetilde{M_\bullet}|_{\operatorname{Spec}((S_\bullet)_f)_0})_\mathfrak{p} = (((M_\bullet)_f)_0)_\mathfrak{p}.$$

We want to show that $(((M_\bullet)_f)_0)_\mathfrak{p} \cong ((M_\bullet)_\mathfrak{p})_0$. Define $\varphi : (((M_\bullet)_f)_0)_\mathfrak{p} \to ((M_\bullet)_\mathfrak{p})_0$ by setting

$$\frac{m/f^s}{a/f^t} \longmapsto \frac{mf^t}{af^s},$$

where $a \notin \mathfrak{p}$, $\deg m = s \deg f$, $\deg a = t \deg f$. We first show that φ is a surjective. Take $\frac{m}{s} \in ((M_\bullet)_\mathfrak{p})_0$, where $m \in M_\bullet$, $s \in S_\bullet \setminus \mathfrak{p}$, and $\deg m = \deg s$. Consider $\frac{ms^{d-1}/f^e}{s^d/f^e}$, where $e := \deg s = \deg m$ and $d := \deg f$, then

$$\deg \frac{ms^{d-1}}{f^e} = e + e(d-1) - ed = 0 = ed - ed = \deg \frac{s^d}{f^e},$$

which implies that $\frac{ms^{d-1}/f^e}{s^d/f^e} \in (((M_\bullet)_f)_0)_\mathfrak{p}$. Notice that

$$\varphi\left(\frac{ms^{d-1}/f^e}{s^d/f^e}\right) = \frac{ms^{d-1}f^e}{s^df^e} = \frac{m}{s},$$

we find that φ is a surjective. Now, we want to find $\operatorname{Ker}\varphi$. Let $\varphi\left(\frac{m/f^s}{a/f^t}\right) = 0$, then $\frac{mf^t}{af^s} = 0$. Hence, there exists $b \in S_\bullet \setminus \mathfrak{p}$ such that $bmf^t = 0$. Notice that

$$\frac{b^{\deg f}}{f^{\deg b}} \cdot \frac{m}{f^s} = \frac{b^{\deg f}m}{f^{s+\deg b}} = 0$$

(since $b^{\deg f}mf^t = 0$), we know that $\frac{m/f^s}{a/f^t} = 0$, which implies that $\operatorname{Ker}\varphi = 0$. Therefore,

$$\varphi : (((M_\bullet)_f)_0)_\mathfrak{p} \xrightarrow{\sim} ((M_\bullet)_\mathfrak{p})_0.$$

$\square$

Given a map of graded modules $\varphi : M_\bullet \to N_\bullet$ induced map of sheaves $\widetilde{M_\bullet} \to \widetilde{N_\bullet}$. Explicitly, over $D(f)$, the map $M_\bullet \to N_\bullet$ induces $(M_\bullet)_f \to (N_\bullet)_f$, which induces $\varphi_f : ((M_\bullet)_f)_0 \to ((N_\bullet)_f)_0$; and this behaves well with respect to restriction to smaller distinguished open sets, i.e., the following diagram commutes.

$$\begin{tikzcd}
((M_\bullet)_f)_0 \arrow[r, "\varphi_f"] \arrow[d] & ((N_\bullet)_f)_0 \arrow[d] \\
((M_\bullet)_{fg})_0 \arrow[r, "\varphi_{fg}"] & ((N_\bullet)_{fg})_0
\end{tikzcd}$$

Thus $\sim$ is a functor from the category of graded $S_\bullet$-modules to the category of quasicoherent sheaves on

$\operatorname{Proj} S_\bullet$.

Lemma 15.6.2

$\widetilde{\square}$ is an exact functor from the category of graded $S_\bullet$-modules to the category of quasicoherent sheaves on $\operatorname{Proj} S_\bullet$.

Proof Consider the exact sequence of graded $S_\bullet$-modules

$$0 \longrightarrow L_\bullet \longrightarrow M_\bullet \longrightarrow N_\bullet \longrightarrow 0$$

Localize at $\mathfrak{p} \in \operatorname{Proj} S_\bullet$, since localization is exact, we obtain an exact sequence of $(S_\bullet)_\mathfrak{p}$-modules

$$0 \longrightarrow (L_\bullet)_\mathfrak{p} \longrightarrow (M_\bullet)_\mathfrak{p} \longrightarrow (N_\bullet)_\mathfrak{p} \longrightarrow 0.$$

We next take the 0 degree part of above sequence, since the maps are graded homomorphisms of degree 0, exactness may be checked degreewise; hence taking the degree 0 part preserves exactness. So, we have an exact sequence of $((S_\bullet)_\mathfrak{p})_0$-modules

$$0 \longrightarrow ((L_\bullet)_\mathfrak{p})_0 \longrightarrow ((M_\bullet)_\mathfrak{p})_0 \longrightarrow ((N_\bullet)_\mathfrak{p})_0 \longrightarrow 0.$$

By Lemma 15.6.1, we obtain

$$0 \longrightarrow \widetilde{L_\bullet}_\mathfrak{p} \longrightarrow \widetilde{M_\bullet}_\mathfrak{p} \longrightarrow \widetilde{N_\bullet}_\mathfrak{p} \longrightarrow 0,$$

which is exact. Since $\mathfrak{p} \in \operatorname{Proj} S_\bullet$ is arbitrary, the sequence of quasicoherent sheaves on $\operatorname{Proj} S_\bullet$

$$0 \longrightarrow \widetilde{L_\bullet} \longrightarrow \widetilde{M_\bullet} \longrightarrow \widetilde{N_\bullet} \longrightarrow 0,$$

is exact. $\square$

Lemma 15.6.3

If $M_\bullet$ and $M'_\bullet$ agree in high enough degrees, then $\widetilde{M_\bullet} \cong \widetilde{M'_\bullet}$.

Proof Let $Q_\bullet$ be graded $S_\bullet$-module with $Q_n = 0$ for large enough $n \gg 0$. We claim that $\widetilde{Q_\bullet} = 0$. It suffices to show that $\Gamma(D_+(f), \widetilde{Q_\bullet}) = 0$ for any $D_+(f) \hookrightarrow \operatorname{Proj} S_\bullet$. By the definition, we have $\Gamma(D_+(f), \widetilde{Q_\bullet}) = ((Q_\bullet)_f)_0$, take $\frac{q}{f^k} \in ((Q_\bullet)_f)_0$. Consider qf^N for N large enough, then $\deg qf^N \gg 0$, so $qf^N \in Q_{\deg qf^N} = 0$, which implies that $\frac{q}{f^k} = 0$. Hence, $((Q_\bullet)_f)_0 = 0$, and thus $\widetilde{Q_\bullet} = 0$.

Now, let M be a graded $S^\bullet$-module. Fix N, denote $M_{\bullet \geq N} = \bigoplus_{n \geq N} M_n$, then we obtain an exact sequence of $S_\bullet$-module

$$0 \longrightarrow M_{\bullet \geq N} \longrightarrow M_\bullet \longrightarrow M_\bullet / M_{\geq N} \longrightarrow 0.$$

Apply exact functor $\widetilde{\square}$, we obtain an exact sequence of quasicoherent sheaves on $\operatorname{Proj} S_\bullet$

$$0 \longrightarrow \widetilde{M_{\bullet \geq N}} \longrightarrow \widetilde{M_\bullet} \longrightarrow M_\bullet / \widetilde{M_{\bullet \geq N}} \longrightarrow 0.$$

Notice that for $n \geq N$ degree terms in $M_\bullet / M_{\bullet \geq N}$ are all zero, by the statement above, we find that $M_\bullet / \widetilde{M_{\bullet \geq N}} = 0$, which implies that $\widetilde{M_{\bullet \geq N}} \cong \widetilde{M_\bullet}$.

Suppose M' be a graded $S_\bullet$-module agree with M in high enough degree $N \gg 0$. Then we have $\widetilde{M_{\bullet \geq N}} \cong \widetilde{M'_{\bullet \geq N}}$, so $\widetilde{M_\bullet} \cong \widetilde{M'_\bullet}$, as we desired. $\square$

Corollary 15.6.1

The category of graded $S_\bullet$-modules is not equivalence to the category $\mathbf{QCoh}_{\operatorname{Proj} S_\bullet}$.

Proof By Lemma 15.6.3, we know that the map from graded $S_\bullet$-modules (up to isomorphism) to quasicoherent sheaves on $\operatorname{Proj} S_\bullet$ (up to isomorphism) is not a bijection, which implies the category of graded $S_\bullet$-modules is not equivalence to the category $\mathbf{QCoh}_{\operatorname{Proj} S_\bullet}$. $\square$

Exercise 15.5 Describe a map of S_0-modules $M_0 \to \Gamma(\operatorname{Proj} S_\bullet, \widetilde{M_\bullet})$.

Remark 15.29 This foreshadows the "saturation map" of §that takes a graded module to its saturation; see **??**.

Proof We describe this map locally. For each $D_+(f) \hookrightarrow \operatorname{Proj} S_\bullet$, define $\varphi_f : M_0 \to ((M_\bullet)_f)_0$ by setting $m \mapsto \frac{m}{1}$. Let $D_+(g) \hookrightarrow \operatorname{Proj} S_\bullet$ such that $D_+(f) \cap D_+(g) \neq \varnothing$. Notice that

$$\varphi_f(m)|_{D_+(g)} = \frac{m}{1} = \varphi_g(m)|_{D_+(f)},$$

by the gluability axiom there exist $s_m \in \Gamma(\operatorname{Proj} S_\bullet, \widetilde{M_\bullet})$ such that $s_m|_{D_+(f)} = \frac{m}{1}$ for each homogeneous $f \in S_\bullet$. Define $M_0 \to \Gamma(\operatorname{Proj} S_\bullet, \widetilde{M_\bullet})$ by setting $m \mapsto s_m$. $\square$

Lemma 15.6.3 shows that $\widetilde{\square}$ isn't an equivalence of categories, but it is close. The relationship is akin to that between presheaves and sheaves, and the sheafification functor (see §16.7). One convenient way of understanding this is by starting with category $\mathbf{f.g.Mod}_{S_\bullet}$ of finitely generated $S_\bullet$-modules, and creating a new category $\mathscr{C}$ with the same objects, but different morphisms: the morphisms from $M_\bullet$ to $N_\bullet$ in this new category are

$$\operatorname{Hom}_{\mathscr{C}}(M_\bullet, N_\bullet) := \lim_{m\to\infty}(\operatorname{Hom}_{\mathbf{f.g.Mod}_{S_\bullet}}(M_{\bullet \geq m}, N_{\bullet \geq m})).$$

(Here $M_{\bullet\geq m} = \bigoplus_{n\geq m} M_\bullet$.) Then you can show that $\mathscr{C}$ is equivalent to the category of finite type quasicoherent sheaves on $\operatorname{Proj} S_\bullet$, and this equivalence corresponds to the map $M_\bullet \mapsto \widetilde{M_\bullet}$. You are welcome to prove these facts, but we will not use them.

15.6.2 Example: Homogeneous ideals of $S_\bullet$ give closed subschemes of $\operatorname{Proj} S_\bullet$.

Recall that a homogeneous ideal $I_\bullet \subseteq S_\bullet$ yields a closed subscheme $\operatorname{Proj} S_\bullet/I_\bullet \hookrightarrow \operatorname{Proj} S_\bullet$ (Proposition 10.3.1). For example, suppose $S_\bullet = k[w,x,y,z]$, so $\operatorname{Proj} S_\bullet \cong \mathbb{P}^3$. The ideal $I_\bullet = (wz-xy, x^2-wy, y^2-xz)$ yields our old friend, the twisted cubic (defined in Exercise 10.3).

Exercise 15.6 Show that if the functor $\widetilde{\square}$ is applied to the exact sequence of graded $S_\bullet$-modules

$$0 \longrightarrow I_\bullet \longrightarrow S_\bullet \longrightarrow S_\bullet/I_\bullet \longrightarrow 0. \tag{15.22}$$

we obtain the closed subscheme exact sequence (10.1) for $\operatorname{Proj} S_\bullet/I_\bullet \hookrightarrow \operatorname{Proj} S_\bullet$.

Proof Apply $\widetilde{\square}$ to (15.22), we obtain an exact sequence

$$0 \longrightarrow \widetilde{I_\bullet} \longrightarrow \widetilde{S_\bullet} \longrightarrow (S_\bullet/I_\bullet)^\sim \longrightarrow 0.$$

Denote $X = \operatorname{Proj} S_\bullet/I_\bullet$, $Y = \operatorname{Proj} S_\bullet$, and $i : X \hookrightarrow Y$. We want to show that $\mathscr{O}_Y \cong \widetilde{S_\bullet}$ and $i_*\mathscr{O}_X \cong (S_\bullet/I_\bullet)^\sim$. For each $D_+(f) \hookrightarrow Y$, we have $\Gamma(D_+(f), \mathscr{O}_Y) = ((S_\bullet)_f)_0 = \Gamma(D_+(f), \widetilde{S_\bullet})$, so $\mathscr{O}_Y \cong \widetilde{S_\bullet}$. Next, let $D_+(g) \hookrightarrow Y$, then

$$\Gamma(D_+(g), i_*\mathscr{O}_X) = \Gamma(X \cap D_+(g), \mathscr{O}_X) = ((S_\bullet/I_\bullet)_g)_0$$

and

$$\Gamma(D_+(g), (S_\bullet/I_\bullet)^\sim) = ((S_\bullet/I_\bullet)_g)_0,$$

which implies $i_*\mathscr{O}_X \cong (S_\bullet/I_\bullet)^\sim$. Notice that

$$\widetilde{I_\bullet} = \operatorname{Ker}(\widetilde{S_\bullet} \to (S_\bullet/I_\bullet)^\sim) = \operatorname{Ker}(\mathscr{O}_Y \to i_*\mathscr{O}_X),$$

we know that $\widetilde{I_\bullet}$ is the ideal sheaf $\mathscr{I}_{X/Y}$. Hence, we obtain the closed subscheme exact sequence (10.1). $\square$

All closed subschemes of $\operatorname{Proj} S_\bullet$ arise in this way (see Proposition 16.7.2).

Remark 15.30 If $M_\bullet$ is finitely generated (resp., finitely presented, coherent), then so is $\widetilde{M_\bullet}$, **Rohrer2014Quasicoherent**.

We will not need this fact. (Quick proof of the first two: if $M_\bullet$ is finitely generated, resp., finitely generated, so is the quasicoherent sheaf $\widetilde{M_\bullet}$ on the affine cone $\operatorname{Spec} S_\bullet$. These two notions are preserved by pullback, and the standard open subsets of $\operatorname{Proj} S_\bullet$ are closed subschemes of the affine cone $\operatorname{Spec} S_\bullet$.)

Chapter 16 Line bundles, maps to projective space, and divisors

When learning algebraic geometry for the first time, it is surprising to hear that line bundles (invertible sheaves) are of such vital importance.

(i) They are, of course, the simplest vector bundles.

(ii) We can often classify them, using "codimension 1" information. In particular, we can build them and manipulate them.

(iii) Knowing line bundles on a variety X helps us classify X, and understand maps to projective space, and to other varieties.

(iv) And this is only the beginning.

In this chapter, we next describe convenient and powerful ways of working with and classifying line bundles. We begin with a fundamental example, the line bundles $\mathscr{O}(m)$ on projective space, §16.1. We then discuss how line bundles and some sections give maps to projective space, and indeed how all maps to projective schemes arise in this way, §16.2. This is simultaneously elementary, subtle, and powerful. In §16.3, we describe a concretization of the idea that "oneparameter families to projective space have unique limits." We then come to the most challenging part of the chapter, in §16.4. We introduce Weil divisors, and use them to determine all the line bundles on X in a number of circumstances. Our hard work leads to many fun examples we can work out by hand, in §16.5. We then discuss a different means of easily describing some line bundles as sheaves of ideals that happen to be invertible (effective Cartier divisors, §16.6). We conclude with a brief section once again connecting quasicoherent sheaves on projective schemes to graded modules, §16.7.

16.1 Some Line Bundles on Projective Space

We now describe an important family of invertible sheaves on projective space over a field k.

As a warm-up, we begin with the invertible sheaf $\mathscr{O}_{\mathbb{P}^1_k}(1)$ on $\mathbb{P}^1_k = \operatorname{Proj} k[x_0, x_1]$. The subscript $\mathbb{P}^1_k$ refers to the space on which the sheaf lives, and is often omitted when it is clear from the context. We describe the invertible sheaf $\mathscr{O}(1)$ using transition functions. It is trivial on the usual affine open sets $U_0 = D_+(x_0) = \operatorname{Spec} k[x_{1/0}]$ and $U_1 = D_+(x_1) = \operatorname{Spec} k[x_{0/1}]$. (We continue to use the convention $x_{i/j}$ for describing coordinates on patches of projective space; see §5.4.3.) Thus the data of a section over U_0 is a polynomial in $x_{1/0}$. The transition function from U_0 to U_1 is multiplication by $x_{0/1} = x_{1/0}^{-1}$. The transition function from U_1 to U_0 is hence multiplication by $x_{1/0} = x_{0/1}^{-1}$.

This information is summarized below:

open cover	$U_0 = \operatorname{Spec} k[x_{1/0}]$	$U_1 = \operatorname{Spec} k[x_{0/1}]$
trivialization and transition functions	$k[x_{1/0}] \underset{\times x_{1/0}=x_{0/1}^{-1}}{\overset{\times x_{0/1}=x_{1/0}^{-1}}{\rightleftarrows}} k[x_{0/1}]$	

To test our understanding, let's compute the global sections of $\mathscr{O}(1)$. This will generalize our hands-on caculation that $\Gamma(\mathbb{P}^1_k, \mathscr{O}_{\mathbb{P}^1_k}) \cong k$ (Proposition **??**). A global section is a polynomial $f(x_{1/0}) \in k[x_{1/0}]$ and a polynomial $g(x_{0/1}) \in k[x_{0/1}]$ such that $f(1/x_{0/1})x_{0/1} = g(x_{0/1})$. A little thought will show that f must be linear: $f(x_{1/0}) = ax_{1/0} + b$, and hence $g(x_{0/1}) = a + bx_{0/1}$. Thus

$$\dim \Gamma(\mathbb{P}^1_k, \mathscr{O}(1)) = 2 \neq 1 = \dim \Gamma(\mathbb{P}^1_k, \mathscr{O}).$$

Thus $\mathscr{O}(1)$ is not isomorphic to $\mathscr{O}$, and we have constructed our first (proved) example of a nontrivial line bundle!

We next define more generally $\mathscr{O}_{\mathbb{P}^1_k}(m)$ on $\mathbb{P}^1_k$. It is defined in the same way, except that the transition functions are the m-th power of those for $\mathscr{O}(1)$.

open cover	$U_0 = \operatorname{Spec} k[x_{1/0}]$	$U_1 = \operatorname{Spec} k[x_{0/1}]$
trivialization and transition functions	$k[x_{1/0}] \underset{\times x_{1/0}^m = x_{0/1}^{-m}}{\overset{\times x_{0/1}^m = x_{1/0}^{-m}}{\rightleftarrows}} k[x_{0/1}]$	

In particular, thanks to the explicit transformation functions, we see that $\mathscr{O}(m) = \mathscr{O}(1)^{\otimes m}$ (with the obvious meaning if m is negative: $(\mathscr{O}(1)^{\otimes -m})^\vee$). Clearly also $\mathscr{O}(m) \otimes \mathscr{O}(m') = \mathscr{O}(m + m')$.

Proposition 16.1.1 (Important!)

$\dim \Gamma(\mathbb{P}^1_k, \mathscr{O}(m)) = m + 1$ if $m \geq 0$, and 0 otherwise.

Proof If $m \geq 0$. Let $s \in \Gamma(\mathbb{P}^1_k, \mathscr{O}(m))$ be a global section, suppose $s|_{U_0}$ has the form

$$s|_{U_0} = a_0 + a_1 x_{1/0} + \cdots + a_n x_{1/0}^n,$$

where $a_n \neq 0$ and $n > m$, then $s|_{U_1} = x_{0/1}^m f(x_{0/1}^{-1})$. Hence,

$$s|_{U_1} = a_0 x_{0/1}^m + a_1 x_{0/1}^{m-1} + \cdots + a_n x_{0/1}^{m-n}.$$

Since $s|_{U_1}$ has no negative terms, we obtain $a_i = 0$ for all $m + 1 \leq i \leq n$, and therefore

$$s|_{U_0} = a_0 + a_1 x_{1/0} + \cdots + a_m x_{1/0}^m,$$

which implies that $\dim \Gamma(\mathbb{P}^1_k, \mathscr{O}(m)) = m + 1$.

If $m < 0$, use the symbols as above, since

$$s|_{U_1} = a_0 x_{0/1}^m + a_1 x_{0/1}^{m-1} + \cdots + a_n x_{0/1}^{m-n}.$$

has no negative terms, but the powers are all negative, we know that $a_0 = \cdots = a_n = 0$. It follows that $\dim \Gamma(\mathbb{P}^1_k, \mathscr{O}(m)) = 0$. ☐

Example 16.1 Long ago (in §3.6.3), we warned that sheafification was necessary when tensoring $\mathscr{O}_X$-modules: if $\mathscr{F}$ and $\mathscr{G}$ are two $\mathscr{O}_X$-modules on a ringed space, then it is not necessarily true that $\mathscr{F}(X) \otimes_{\mathscr{O}_X(X)} \mathscr{G}(X) \cong (\mathscr{F} \otimes \mathscr{G})(X)$. We now have an example: let $X = \mathbb{P}^1_k$, $\mathscr{F} = \mathscr{O}(1)$, $\mathscr{G} = \mathscr{O}(-1)$, and use the fact that $\mathscr{O}(-1)$ has no nonzero global sections.

Proposition 16.1.2

If $m \neq m'$, then $\mathscr{O}(m) \ncong \mathscr{O}(m')$. Hence conclude that we have an injection of groups $\mathbb{Z} \hookrightarrow \operatorname{Pic} \mathbb{P}^1_k$ given by $m \mapsto \mathscr{O}(m)$.

Proof Let $m \neq m'$. Suppose $\mathscr{O}(m) \cong \mathscr{O}(m')$, tensoring $\mathscr{O}(m')$ both side, we obtain $\mathscr{O}(m - m') \cong \mathscr{O}$. By Proposition 16.1.1, since $m \neq m'$, $\dim \Gamma(\mathbb{P}^1_k, \mathscr{O}(m - m')) \geq 2$ or 0. But $\dim \Gamma(\mathbb{P}^1_k, \mathscr{O}) = 1$, contradiction! Hence, $\mathscr{O}(m) \ncong \mathscr{O}(m')$, which implies that $\mathbb{Z} \hookrightarrow \operatorname{Pic} \mathbb{P}^1_k, m \mapsto \mathscr{O}(m)$ is an injection. ☐

It is useful to identify the global sections of $\mathscr{O}(m)$ with the homogeneous polynomials of degree m in x_0 and x_1, i.e., with the degree m part of $k[x_0, x_1]$ (cf. §**??** for the generalization to $\mathbb{P}^n$). We will see that this identification is natural in many ways. For example, you can show that the definition of $\mathscr{O}(m)$ doesn't depend on a choice of affine cover, and this polynomial description is also independent of cover. (For this, see Definition 5.5.12; you can later compare this to **??**.) As an immediate check of the usefulness of this point of view, ask yourself: where does the section $x_0^3 - x_0x_1^2$ of $\mathscr{O}(3)$ vanish? ($[0:1],[1:1],[1:-1]$.) The section x_0+x_1 of $\mathscr{O}(1)$ can be multiplied by the section x_0^2 of $\mathscr{O}(2)$ to get a section of $\mathscr{O}(3)$. Which one? ($x_0^3+x_0^2x_1$.) Where does the rational section $\frac{x_0^4(x_1+x_0)}{x_1^7}$ of $\mathscr{O}(-2)$ have zeros and poles, and to what order? (Zeros: $[0:1]$, order 4 and $[1:-1]$, order 1; poles: $[1:0]$, order 7.) (We saw the notions of zeros and poles in Definition 14.5.4, and will meet them again in §16.4, but you should intuitively be able to answer these questions already.)

We now define the invertible sheaf $\mathscr{O}_{\mathbb{P}_k^n}(m)$ on the projective space $\mathbb{P}_k^n$. On the usual affine open set $U_i = \operatorname{Spec} k[x_{0/i},\dots,x_{n/i}]/(x_{i/i}-1) = \operatorname{Spec} A_i$, it is trivial, so sections (as an A_i-module) are isomorphic to A_i. The transition function from U_i to U_j is multiplication by $x_{i/j}^m = x_{j/i}^{-m}$.

$$U_i = \operatorname{Spec} k[x_{0/i},\dots,x_{n/i}]/(x_{i/i}-1) \qquad\qquad U_j = \operatorname{Spec} k[x_{0/j},\dots,x_{n/j}]/(x_{j/j}-1)$$

$$k[x_{0/i},\dots,x_{n/i}]/(x_{i/i}-1) \underset{\times x_{j/i}^m = x_{i/j}^{-m}}{\overset{\times x_{i/j}^m = x_{j/i}^{-m}}{\rightleftarrows}} k[x_{0/j},\dots,x_{n/j}]/(x_{j/j}-1)$$

Note that these transition function clearly satisfy the cocycle condition.

Proposition 16.1.3 (CF. Proposition 10.3.7, Theorem ??)

$\Gamma(\mathbb{P}_A^n,\mathscr{O}_{\mathbb{P}_A^n}(m))$ can be identified with the free A-module of homogeneous polynomials of degree m in the variables $x_0,\dots,x_n$. In particular, $\dim_k\Gamma(\mathbb{P}_k^n,\mathscr{O}_{\mathbb{P}_k^n}(m)) = \binom{n+m}{n}$.

Proof Let $U_i \hookrightarrow \mathbb{P}_A^k$ be the usual affine open subset. A global section of $\mathscr{O}(m)$ consists of sections $s_i \in \Gamma(U_i,\mathscr{O}(m))$ that agree on the overlaps $U_i\cap U_j$. Over U_i, we have $\Gamma(U_i,\mathscr{O}(m)) = A[x_{0/i},\dots,x_{n/i}]/(x_{i/i}-1)$ and a section s_i is an element of the form $g_i(x_{0/i},\dots,x_{n/i}) \in A[x_{0/i},\dots,x_{n/i}]/(x_{i/i}-1)$. The compatibility condition on overlaps implies that

$$g_j(x_{0/j},\dots,x_{n/j})x_j^m = g_i(x_{0/i},\dots,x_{n/i})x_i^m. \tag{16.1}$$

If $m<0$, then the left hand side contains only terms with negative powers of x_j, whereas the right-hand side contains only negative powers of x_i. There is therefore only one tuple satisfying (16.1), namely where $g_0=\cdots=g_n=0$.

If $m\geq 0$, we see that the g_j are determined by g_0. By comparing degrees in (16.1), we see that g_0 must have degree $\leq m$ as a polynomial in the $x_{1/0},\dots,x_{n/0}$. This means that $F = g_0x_0^m = g_jx_j^m$ is a homogeneous polynomial of degree m in $x_0,\dots,x_n$ with coefficients in A for all $j=1,\dots,n$. Conversely, any homogeneous polynomial F of degree m determines a valid $(n+1)$-tuple $\left(\frac{F}{x_0^m},\dots,\frac{F}{x_n^m}\right)$, they glue to a global section of $\mathscr{O}(m)$. ☐

As in the case of $\mathbb{P}^1$, sections of $\mathscr{O}(m)$ on $\mathbb{P}_k^n$ are naturally identified with homogeneous degree m polynomials in our $n+1$ variables. (The connection to homogeneous polynomials will be put on a more precise footing in the next proposition.) Thus $x+y+2z$ is a section of $\mathscr{O}(1)$ on $\mathbb{P}^2$. It isn't a function, but we know where this section vanishes — precisely where $x+y+2z=0$.

Also, notice that for fixed n, $\binom{n+m}{n}$ is a polynomial in the variable m, of degree n, but only when $m\geq 0$ (or better: when $m\geq -n$). This should be telling you that this function "wants to be a polynomial," but won't succeed without assistanc. We will later define $h^0(\mathbb{P}_k^n,\mathscr{O}(m)) := \dim_k\Gamma(\mathbb{P}_k^n,\mathscr{O}(m))$, and later still we will define higher cohomology groups, and we will define the Euler characteristic $\chi(\mathbb{P}_k^n,\mathscr{O}(m)) :=$

$\sum_{i=0}^{\infty}(-1)^i h^i(\mathbb{P}^n_k, \mathscr{O}(m))$ (cohomology will vanish in degree higher than n). We will discover the moral that the Euler characteristic is better behaved than h^0, and so we should now suspect (and later prove; see Theorem **??**, **??**, and **??**) that this polynomial is in fact the Euler characteristic, and the reason that it agrees with h^0 for $m \geq 0$ is that all the other cohomology groups vanish.

We finally note that we can define $\mathscr{O}(m)$ on $\mathbb{P}^n_A$ for any ring A: the above definition applies without change, and Proposition 16.1.3 immediately generalizes to show that $\Gamma(\mathbb{P}^n_A, \mathscr{O}_{\mathbb{P}^n_A}(m))$ is a free A-module of rank $\binom{n+m}{n}$, with the same interpretation in terms of homogeneous polynomials.

Proposition 16.1.4

Suppose $n > 0$. Let $\pi : \mathbb{A}^{n+1}_A \setminus V(x_0, \dots, x_n) \to \mathbb{P}^n_A$ be the morphism $(x_0, \dots, x_n) \mapsto [x_0, \dots, x_n]$ (described in Exercise 8.2 and Example 8.5). For any $m \in \mathbb{Z}$, there is an isomorphism

$$\pi^* \mathscr{O}_{\mathbb{P}^n_A}(m) \xleftrightarrow{\sim} \mathscr{O}_{\mathbb{A}^{n+1}_A \setminus V(x_0,\dots,x_n)}.$$

This induces a map $\pi^* : \Gamma(\mathbb{P}^n_A, \mathscr{O}_{\mathbb{P}^n_A}(m)) \to \Gamma(\mathbb{A}^{n+1}_A \setminus V(x_0, \dots, x_n), \mathscr{O}_{\mathbb{A}^{n+1}_A})$. The pullback

$$\alpha : A[x_0, \dots, x_n] = \Gamma(\mathbb{A}^{n+1}_A, \mathscr{O}_{\mathbb{A}^{n+1}_A}) \longrightarrow \Gamma(\mathbb{A}^{n+1}_A \setminus V(x_0, \dots, x_n), \mathscr{O}_{\mathbb{A}^{n+1}_A})$$

is an isomorphism, and the composition

$$\alpha^{-1} \circ \pi^* : \Gamma(\mathbb{P}^n_A, \mathscr{O}_{\mathbb{P}^n_A}(m)) \longrightarrow A[x_0, \dots, x_n]$$

interprets $\Gamma(\mathbb{P}^n_A, \mathscr{O}_{\mathbb{P}^n_A}(m))$ as the homogenous degree m in $A[x_0, \dots, x_n]$.

Proof Let $V_i := D_+(x_i)$ and $U_i := D(x_i)$. Clearly, we have $\pi^{-1}(V_i) = U_i$. Recall Proposition 15.5.2, we have

$$(\pi^* \mathscr{O}_{\mathbb{P}^n_A}(m))|_{U_i} \cong (\pi|_{U_i})^* \mathscr{O}_{\mathbb{P}^n_A}(m)|_{V_i} \cong (\pi|_{U_i})^* \mathscr{O}_{V_i} \cong \mathscr{O}_{U_i}$$

Define $\varphi_i : (\pi^* \mathscr{O}_{\mathbb{P}^n_A}(m))|_{U_i} \to \mathscr{O}_{U_i}$ by setting $s_i \mapsto s_i x_i^m$. One checks that φ_i is well-defined. Since x_i^m is inverse on U_i, φ_i is an isomorphism. We want to show that $\varphi_i|_{U_i \cap U_j} = \varphi_j|_{U_i \cap U_j}$. Let $s_i \in \Gamma(U_i \cap U_j, (\pi^* \mathscr{O}_{\mathbb{P}^n_A}(m))|_{U_i})$, then

$$\varphi_i|_{U_i \cap U_j}(s_i) = x_i^m s^i = x_i^m x_{j/i}^m s_j = x_j^m s_j = \varphi_j|_{U_i \cap U_j}(s_j).$$

Hence, $\{\varphi_i\}_i$ glue to an isomorphism

$$\varphi : \pi^* \mathscr{O}_{\mathbb{P}^n_A}(m) \xrightarrow{\sim} \mathscr{O}_{\mathbb{A}^{n+1}_A \setminus V(x_0,\dots,x_n)}.$$

Denote $X := \mathbb{A}^{n+1}_A \setminus V(x_0, \dots, x_n)$. This induces a map $\pi^* : \Gamma(\mathbb{P}^n_A, \mathscr{O}_{\mathbb{P}^n_A}(m)) \to \Gamma(X, \mathscr{O}_{\mathbb{A}^{n+1}_A})$ by setting the composition

$$\Gamma(\mathbb{P}^n_A, \mathscr{O}_{\mathbb{P}^n_A}(m)) \longrightarrow \Gamma(X, \pi^* \mathscr{O}_{\mathbb{P}^n_A}(m)) \xrightarrow{\varphi} \Gamma(X, \mathscr{O}_{\mathbb{A}^{n+1}_A})$$

Consider the restriction

$$\alpha : \Gamma(\mathbb{A}^{n+1}_A, \mathscr{O}_{\mathbb{A}^{n+1}_A}) \longrightarrow \Gamma(\mathbb{A}^{n+1}_A \setminus V(x_0, \dots, x_n), \mathscr{O}_{\mathbb{A}^{n+1}_A}),$$

we claim that α is an isomorphism. In fact, $X = \bigcup_{i=1}^{n} D(x_i)$, a global regular function on X is a collection $h_i \in \Gamma(D(x_i), \mathscr{O}_X) = A[x_0, \dots, x_n]_{x_i}$ such that $h_i = h_j$ on $D(x_i x_j)$. Clearly, we have $\Gamma(\mathbb{A}^{n+1}_A, \mathscr{O}_{\mathbb{A}^{n+1}_A}) \subseteq \Gamma(X, \mathscr{O}_{\mathbb{A}^{n+1}_A})$. Let $f \in \Gamma(X, \mathscr{O}_{\mathbb{A}^{n+1}_A})$, since $f \in \Gamma(D(x_i), \mathscr{O}_{\mathbb{A}^{n+1}_A}) \cap \Gamma(D(x_j), \mathscr{O}_{\mathbb{A}^{n+1}_A})$, we may write, $f = \frac{a}{x_i^r} = \frac{b}{x_j^s}$ for some $a, b \in A[x_0, \dots, x_n]$. Then there exists N such that $(x_i x_j)^N (x_j^s a - x_i^r) = 0$, since $x_i x_j$ is not zero divisor, we have $x_j^s a = x_i^r b$. It follows that $x_i^r \mid a$ and $x_j^s \mid b$, which implies $f \in A[x_0, \dots, x_n]$. Hence, α is an isomorphism.

Consider the composition

$$\alpha^{-1}\circ\pi^*:\Gamma(\mathbb{P}^n_A,\mathscr{O}_{\mathbb{P}^n_A}(m))\longrightarrow\Gamma(\mathbb{A}^{n+1}_A,\mathscr{O}_{\mathbb{A}^{n+1}_A})=A[x_0,\dots,x_n].$$

Using the similar discussion in Proposition 16.1.3, we know that $\Gamma(\mathbb{P}^n_A,\mathscr{O}_{\mathbb{P}^n_A}(m))$ as the homogeneous degree m in $A[x_0,\dots,x_n]$. □

16.1.1 These are the only line bundles on $\mathbb{P}^n_k$

Suppose that k is a field. We will see in §?? that these $\mathscr{O}(m)$ are the only invertible sheaves on $\mathbb{P}^n_k$. The next proposition shows this when $n=1$.

Proposition 16.1.5

Every invertible sheaf on $\mathbb{P}^1_k$ is of the form $\mathscr{O}(m)$ for some m.

Proof Let $\mathscr{L}$ be an invertible sheaf on $\mathbb{P}^1_k$. Say $\mathbb{P}^1_k=\operatorname{Proj}k[x_0,x_1]$. By Proposition 15.2.2, we know that $\mathscr{L}|_{D_+(x_i)}$ is trivial of rank 1 for $i=0,1$. Let $\varphi_{ij}\in\Gamma(D_+(x_0x_1),\mathscr{O}_{\mathbb{P}^1_k})^\times$ be the transition function, then $\varphi_{ij}\in k[x_{1/0},x_{0/1}]^\times$. Say $t=x_{1/0}$, then $k[x_{1/0},x_{0/1}]^\times=k[t,t^{-1}]^\times$. Write $\varphi_{ij}=a_mt^m+a_{m+1}t^{m+1}+\cdots+a_nt^n$ where $m\le n$, and write its inverse $\varphi_{ij}^{-1}=b_pt^p+b_{p+1}t^{p+1}+\cdots+b_qt^q$ where $p\le q$. Since $\varphi_{ij}\varphi_{ij}^{-1}=1$, there must be

$$\begin{cases} m+p=0\\ n+q=0\end{cases}$$

Hence, $m=n$ and $p=q$, which implies that $\varphi_{ij}=a_mt^m$ for some n. It follows that $\mathscr{L}\cong\mathscr{O}(m)$. □

Remark 16.1 ***Caution:*** There can exist invertible sheaves on $\mathbb{P}^1_A$ not of the form $\mathscr{O}(m)$. Let $A=\mathbb{Z}[\sqrt{-5}]$, $I=(2,1+\sqrt{-5})\subseteq A$. Consider structure morphism $p:\mathbb{P}^1_A\to\operatorname{Spec}A$, define $\mathscr{L}:=p^*\widetilde{A}$, then $\mathscr{L}$ is an invertible sheaf but not of the form $\mathscr{O}(m)$ for any m.

16.1.2 Line bundles on projective A-schemes

Definition 16.1.1

Suppose $M_\bullet$ is a graded $S_\bullet$-module. Define the graded module $M(m)_\bullet$ by $M(m)_d:=M_{m+d}$. Thus the quasicoherent sheaf $\widetilde{M(m)_\bullet}$ satisfies

$$\Gamma(D_+(f),\widetilde{M(m)_\bullet})=((M_\bullet)_f)_m,$$

where here the subscript means we take the m-th graded piece.

Proposition 16.1.6

Let $S_\bullet=A[x_0,\dots,x_n]$, then $\widetilde{S(m)_\bullet}\cong\mathscr{O}_{\mathbb{P}^n_A}(m)$.

Proof We first show that $\widetilde{S(m)_\bullet}$ is trivial over $D_+(x_i)$. Define $\varphi_i:\widetilde{S(m)_\bullet}|_{D_+(x_i)}\to\mathscr{O}_{D_+(x_i)}$ by setting $s\mapsto sx_i^{-m}$. Notice that $x_i^m\in\Gamma(D_+(x_i),\mathscr{O})$ is invertible, we know that φ_i is an isomorphism, which implies that $\widetilde{S(m)_\bullet}$ is a line bundle. The transition function on overlap $D_+(x_ix_j)$ is clearly $\{\times x_{i/j}^m\}$, which is agree with the transition function of $\mathscr{O}(m)$. It follows that $\widetilde{S(m)_\bullet}\cong\mathscr{O}(m)$. □

Motivated by Proposition 16.1.6, we have the following:

Definition 16.1.2

If $S_\bullet$ is a graded ring generated in degree 1, we define $\mathscr{O}_{\operatorname{Proj} S_\bullet}(m)$ (or simply $\mathscr{O}(m)$, where $S_\bullet$ is implicit) by $\widetilde{S(m)_\bullet}$.

Proposition 16.1.7

If $S_\bullet$ is generated in degree 1, then $\mathscr{O}_{\operatorname{Proj} S_\bullet}(m)$ is an invertible sheaf.

Proof Say $X := \operatorname{Proj} S_\bullet$. Since $S_\bullet$ is generated in degree 1, we have $\operatorname{Proj} S_\bullet = \bigcup_{f \in S_1} D_+(f)$. We claim that $\mathscr{O}_X(m)$ is trivial over $D_+(f)$. Define $\mathscr{O}_X(m)|_{D_+(f)} \to \mathscr{O}_{D_+(f)}$ by setting $s \mapsto sf^{-m}$. On checks this gives a well-defined isomorphism. Hence, $\mathscr{O}_{\operatorname{Proj} S_\bullet}(m)$ is an invertible sheaf. □

Definition 16.1.3 (Twisting sheaf)

*If $S_\bullet$ is generated in degree 1, and $\mathscr{F}$ is a quasicoherent sheaf on $\operatorname{Proj} S_\bullet$, define $\mathscr{F}(m) := \mathscr{F} \otimes \mathscr{O}(m)$. This is often called **twisting** $\mathscr{F}$ **by** $\mathscr{O}(m)$, or **twisting** $\mathscr{F}$ **by** m. More generally, if $\mathscr{L}$ is an invertible sheaf, then $\mathscr{F} \otimes \mathscr{L}$ is often called **twisting** $\mathscr{F}$ **by** $\mathscr{L}$.*

Proposition 16.1.8

If $S_\bullet$ is generated in degree 1, then $\widetilde{M_\bullet}(m) \cong \widetilde{M(m)_\bullet}$.

Remark 16.2 Hereafter, we can be cavalier with the placement of the "dot" in such situations.

Proof Let $f \in S_1$. Notice that

$$\Gamma(D_+(f), \widetilde{M_\bullet}(m)) = ((M_\bullet)_f)_0 \otimes_{((S_\bullet)_f)_0} ((S_\bullet)_f)_m$$

and

$$\Gamma(D_+(f), \widetilde{M(m)_\bullet}) = ((M_\bullet)_f)_m,$$

define $\varphi_f : \Gamma(D_+(f), \widetilde{M_\bullet}(m)) \to \Gamma(D_+(f), \widetilde{M(m)_\bullet})$ by setting

$$\frac{a}{f^s} \otimes \frac{b}{f^t} \longmapsto \frac{ab}{f^{s+t}},$$

where $\deg a = s \deg f = s$ and $\deg b = t \deg f + m = m + t$. One checks that φ_f is well-defined. Define $\psi_f : \Gamma(D_+(f), \widetilde{M(m)_\bullet}) \to \Gamma(D_+(f), \widetilde{M_\bullet}(m))$ by setting

$$\frac{a}{f^s} \longmapsto \frac{a}{f^{s+m}} \otimes f^m,$$

where $\deg a = s + m$. One checks that ψ_f is well-defined and is inverse of φ_f. These local isomorphisms are compatible with restriction to overlaps, since they are given by the same multiplication formula in the localized graded ring. Hence they glue to a global isomorphism, i.e., $\widetilde{M_\bullet}(m) \cong \widetilde{M(m)_\bullet}$, we win. □

Proposition 16.1.9

If $S_\bullet$ is generated in degree 1, then $\mathscr{O}(m_1 + m_2) \cong \mathscr{O}(m_1) \otimes \mathscr{O}(m_2)$.

Proof Say $U_f := D_+(f)$, where $f \in S_1$. On the overlap $U_f \cap U_g$, the transition function of $\mathscr{O}(m_1 + m_2)$ is $(\frac{f}{g})^{m_1+m_2}$ and the transition function of $\mathscr{O}(m_1) \otimes \mathscr{O}(m_2)$ is $(\frac{f}{g})^{m_1} \cdot (\frac{f}{g})^{m_2} = (\frac{f}{g})^{m_1+m_2}$, so $\mathscr{O}(m_1 + m_2) \cong \mathscr{O}(m_1) \otimes \mathscr{O}(m_2)$. □

Remark 16.3 (Unimportant remark.) Even if $S_\bullet$ is not generated in degree 1, then (if $S_\bullet$ is finitely generated) by Proposition 8.4.6, $S_{d\bullet}$ is generated in degree 1 for some d. In this case, we may define the invertible sheaves $\mathscr{O}(dm)$ for $m \in \mathbb{Z}$. This does not mean that we can't define $\mathscr{O}(1)$; this depends on $S_\bullet$. For example, if $S_\bullet$ is the polynomial ring $k[x,y]$ with the usual grading, except without linear terms (so $S_\bullet = k[x^2, xy, y^2, x^3, x^2y, xy^2, y^3]$), then $S_{2\bullet}$ and $S_{3\bullet}$ are both generated in degree 1, meaning that we may define $\mathscr{O}(2)$ and $\mathscr{O}(3)$. There is good reason to call their "difference" $\mathscr{O}(1)$. (This algebra may remind of you of the pinched plane, Exercise **??**.)

16.2 Line Bundles and Maps to Projective Space

Recall that affine space $\mathbb{A}^n$ had a short description as a representable functor: maps (from X) to $\mathbb{A}^n$ are exactly the same data as a choice of n functions (on X), Exercise 8.16 (a). In this section, we accidentally reinterpret projective space in the same way.

16.2.1 Line bundles and maps to projective space

Suppose x_0, x_1, and x_2 are projective coordinates on $\mathbb{P}^2_k$. Do you see how

$$[x_0^3 : x_0^2x_2 : x_0x_1x_2 + x_1^3 : x_2^3] \tag{16.2}$$

gives a morphism $\mathbb{P}^2_k \to \mathbb{P}^3_k$? (Since the solution of $x_0^3 = x_0^2x_2 = x_0x_1x_2 + x_1^3 = x_2^3 = 0$ is $x_0 = x_1 = x_2 = 0$, notice that $[0 : 0 : 0]$ is not a point in $\mathbb{P}^2_k$, so this morphism is well-defined.) The entries of (16.2) aren't functions, so we can't make sense of them individually. But for any "point of $\mathbb{P}^2_k$", their ratios can make sense nonetheless. For example, the point $[1 : 1 : 1]$ is sent to $[1 : 1 : 2 : 1]$, and the same point $[2 : 2 : 2]$ is sent to $[8 : 8 : 16 : 8]$, so there is no contradiction — the entries of (16.2) are all homogeneous of degree 3, so the quotient of one by another is a rational function. On the other hand,

$$[x_0^3 : x_0^2x_2 : x_0x_1x_2 + x_1^3 : x_1^3] \tag{16.3}$$

doesn't give a morphism $\mathbb{P}^2_k \to \mathbb{P}^3_k$, because the point $[0 : 0 : 1]$ "doesn't know where to go." But you might be able to interpret (16.3) as giving a morphism $\mathbb{P}^2_k \setminus \{[0 : 0 : 1]\} \to \mathbb{P}^3_k$, or a rational map $\mathbb{P}^2_k \dashrightarrow \mathbb{P}^3_k$. This may even remind you of Proposition 8.3.14. We now make this precise, and state it much more generally. In particular, we won't need to work over a field.

Proposition 16.2.1 (A vitally important construction)

Suppose $s_0, \ldots, s_n$ are $n+1$ global sections of an invertible sheaf $\mathscr{L}$ on a scheme X, with no common zero. Then there is a corresponding map to $\mathbb{P}^n$:

$$X \xrightarrow{[s_0:\cdots:s_n]} \mathbb{P}^n.$$

Proof Let $U \hookrightarrow X$ be an open subset such that $\mathscr{L}$ is trivial over U, i.e., $\mathscr{L}|_U \cong \mathscr{O}_U$. By Proposition 8.3.14 (a), we obtain a map $\varphi_U : U \to \mathbb{P}^n$ by setting $x \mapsto [s_0|_U(x) : \cdots : s_n|_U(x)]$. Now, we want to show that φ_U glue to a global morphism $X \to \mathbb{P}^n$, it suffices to show that $\varphi_U|_{U\cap V} = \varphi_V|_{U\cap V}$. Let V be another open subset such that $\mathscr{L}$ is trivial over V. Consider the overlap $U \cap V$. Let $g_{UV} \in \Gamma(U\cap V, \mathscr{O}_{U\cap V})^\times$ be the transition function, by Proposition 8.3.14 (b), we have

$$\begin{aligned}\varphi_U = [s_0|_U : \cdots : s_n|_U] &= [g_{UV}^{-1}s_0|_V : \cdots : g_{UV}^{-1}s_n|_V] \\ &= [s_0|_V : \cdots : s_n|_V] \\ &= \varphi_V.\end{aligned}$$

It follows that φ_U glue to a global map

$$X \xrightarrow{[s_0:\cdots:s_n]} \mathbb{P}^n.$$

$\square$

Remark 16.4 (In Theorem **??**, we will see that this yields all maps to projective space.) Note that Proposition 16.2.1 as written "works over $\mathbb{Z}$" (as all morphisms are "over" the final object in the category of schemes), although many readers will just work over a field k.

Definition 16.2.1 (Base-point-free)

If $\mathscr{L}$ is an invertible sheaf on X, then those points where all global sections of $\mathscr{L}$ vanish are called the ***base points*** *of $\mathscr{L}$, and the set of base points is called the* ***base locus*** *of $\mathscr{L}$; it is a closed subset of X. (We can refine this to a closed subscheme: by taking the scheme-theoretic intersection of the vanishing loci of the sections of $\mathscr{L}$, we obtain the* ***scheme-theoretic base locus****.) The complement of the* ***base locus*** *is the* ***base-point-free locus****. If $\mathscr{L}$ has no base points, it is* ***base-point-free****. The base-point-free locus is an open subset of X.*

More generally, if we have a bunch of global sections of $\mathscr{L}$ (perhaps not all of them), we can define the ***base points*** *(and all the related notions, of the previous paragraph) of these sections.*

Example 16.2 For example, Proposition 16.2.1 states that if $s_0, \ldots, s_n$ are a base-point-free family of global sections of an invertible sheaf $\mathscr{L}$ on X, then they define a map to $\mathbb{P}^n$.

Definition 16.2.2 (Linear series (or linear system))

A ***linear series*** *(or* ***linear system****) on a k-scheme X is a k-vector space V (usually finite-dimensional), an invertible sheaf $\mathscr{L}$, and a linear map $\lambda : V \to \Gamma(X, \mathscr{L})$. Such a linear series is often called "V", with the rest of the data left implicit. If the map λ is an isomorphism, it is called a* ***complete linear series****, and is often written $|\mathscr{L}|$.*

The language of base points (base-point-free, base locus, scheme-theoretic base locus, ...) readily applies to linear series (base-point-free linear series, ...).

Example 16.3 As a reality check, Proposition 16.2.1 states that an $(n+1)$-dimensional linear series on a k-scheme X, with choice of basis, with base-point-free locus U, defines a morphism $U \to \mathbb{P}^n_k$.

Proposition 16.2.2

If $\mathscr{L}$ and $\mathscr{M}$ are base-point-free invertible sheaves, then $\mathscr{L} \otimes \mathscr{M}$ is also base-point-free.

Proof Let $x \in X$. Since $\mathscr{L}$ is base-point-free, there exists $s \in \Gamma(X, \mathscr{L})$ such that $s(x) \neq 0$. Since $\mathscr{M}$ is base-point-free, there exists $t \in \Gamma(X, \mathscr{M})$ such that $t(x) \neq 0$. Consider the global section $s \otimes t \in \Gamma(X, \mathscr{L} \otimes \mathscr{M})$. Its value at x is $(s \otimes t)(x) = s(x) \otimes t(x) \in (\mathscr{L} \otimes \mathscr{M})|_x = \mathscr{L}|_x \otimes \mathscr{M}|_x$, since $s(x) \neq 0$ and $t(x) \neq 0$, we conclude that $(s \otimes t)(x) \neq 0$. Hence, for every $x \in X$, there is a global section of $\mathscr{L} \otimes \mathscr{M}$ nonvanishing at x. Therefore $\mathscr{L} \otimes \mathscr{M}$ is also base-point-free. $\square$

Definition 16.2.3 (Degenerate)

Suppose $\pi : X \to \mathbb{P}^n_k$ is a morphism, corresponding to the base-point-free linear series $\Gamma(\mathbb{P}^n_k, \mathscr{O}(1)) \to \Gamma(X, \pi^*\mathscr{O}(1))$. If the scheme-theoretic image of X in $\mathbb{P}^n_k$ lies in a hyperplane, we say that the linear series (or X itself) is ***degenerate*** *(and otherwise,* ***nondegenerate****).*

Proposition 16.2.3

A base-point-free linear series $V \to \Gamma(X, \pi^*\mathscr{O}(1))$ is nondegenerate if and only if the map $V \to \Gamma(X, \pi^*\mathscr{O}(1))$ is an inclusion. In particular, a complete linear series is always nondegenerate.

Proof Let $V = \mathrm{Span}\{s_0, \dots, s_n\}$. If the map $V \to \Gamma(X, \pi^*\mathscr{O}(1))$ is an inclusion, then $s_0, \dots, s_n$ can be regard as the global sections of $\pi^*\mathscr{O}(1)$. Since V is base-point-free, by Proposition 16.2.1, there is a corresponding map to $\mathbb{P}^n$ which given by

$$[s_0 : \cdots : s_n] : X \longrightarrow \mathbb{P}^n_k.$$

If the scheme-theoretic image of X in $\mathbb{P}^n_k$ is lying in a hyperplane, say $H = V(a_0x_0 + \cdots + a_nx_n)$, then $a_0s_0 + \cdots + a_ns_n = 0$, it is impossible, since $s_0, \dots, s_n$ are linear independent. Hence, the linear series $V \to \Gamma(X, \pi^*\mathscr{O}(1))$ is nondegenerate.

Conversely, if base-point-free linear series $V \to \Gamma(X, \pi^*\mathscr{O}(1))$ is nondegenerate. Suppose $\varphi : V \to \Gamma(X, \pi^*\mathscr{O}(1))$ is not an inclusion, then there exists $a_0, \dots, a_n \in k$ not all zero such that

$$a_0\varphi(s_0) + \cdots + a_n\varphi(s_n) = 0.$$

Let $H = V(a_0x_0 + \cdots + a_nx_n)$ be a hyperplane, then the scheme-theoretic image of X is contained in H. Hence the linear series is degenerate, contradicting our assumption. □

Theorem 16.2.1, the converse or completion to Proposition 16.2.1, will give one reason why line bundles are crucially important: They tell us about maps to projective space, and more generally, to quasiprojective A-schemes, Given that we have had a hard time naming any non-quasiprojective schemes, they tell us about maps to essentially all interesting schemes.

Theorem 16.2.1

For a fixed scheme X, morphisms $X \to \mathbb{P}^n$ are in bijection with the data $(\mathscr{L}, s_0, \dots, s_n)$, where $\mathscr{L}$ is an invertible sheaf and $s_0, \dots, s_n$ are sections of $\mathscr{L}$ with no common zero (i.e., $(s_0, \dots, s_n)$ is base-point-free), up to isomorphism of these data.

(This works over $\mathbb{Z}$, or indeed any base.) Informally: morphisms to $\mathbb{P}^n$ correspond to $n+1$ sections of a lien bundle, not all vanishing at any point, modulo global sections of $\mathscr{O}_X^\times$, as multiplication by an invertible function gives an automorphism of $\mathscr{L}$. This is one of those important theorems in algebraic geometry that is easy to prove, but quite subtle in its effect on how one should think. It takes some time to properly digest. A "coordinate-free" version is given in Proposition 16.2.6.

Theorem 16.2.1 describes all morphisms to projective space, and hence by the Yoneda philosophy, this can taken as the definition of projective space: It defines projective space up to unique isomorphism. *Projective space $\mathbb{P}^n$ (over $\mathbb{Z}$) is the moduli space of line bundles $\mathscr{L}$ along with $n+1$ sections of $\mathscr{L}$ with no common zeros.* (Can you give an analogous definition of projective space over X, denoted $\mathbb{P}^n_X$? Answer: $\mathbb{P}^n_X := X \times_{\mathrm{Spec}\,\mathbb{Z}} \mathbb{P}^n_{\mathbb{Z}}$.) Or, if you prefer, *maps $X \to \mathbb{P}^n_k$ of k-schemes correspond to $(n+1)$-dimensional base-point-free linear series on X, along with choice of basis*. (If k is replaced by another ring, then maps $X \to \mathbb{P}^n_A$ correspond to quotient line bundles of $\mathscr{O}_X^{\oplus(n+1)}$.)

Every time you see a map to projective space, you should immediately simultaneously keep in mind the invertible sheaf and sections (or, if you prefer, the linear series).

Maps to projective schemes can be described similarly. For example, if $Y \hookrightarrow \mathbb{P}^2_k$ is the curve $x_2^2x_0 = x_1^3 - x_1x_0^2$, then maps from a scheme X to Y are given by an invertible sheaf on X along with three sections s_0, s_1, s_2, with no common zeros, satisfying $s_2^2s_0 - s_1^3 + s_1s_0^2 = 0$.

Proof of Theorem 16.2.1

The central idea of the proof of Theorem 16.2.1 is straightforward: we need to show a bijection, so we just figure out what the maps are from left to right, and from right to left, and show that they are indeed inverse. The key will be thinking about transition functions between the trivializing neighborhoods.

Here more precisely is the correspondence of Theorem 16.2.1. Any $n+1$ sections of $\mathscr{L}$ with no common zeros determine a morphism to $\mathbb{P}^n$, by Proposition 16.2.1. Conversely, if you have a map to projective space $\pi : X \to \mathbb{P}^n$, then we have $n+1$ sections of $\mathscr{O}_{\mathbb{P}^n}(1)$, corresponding to the hyperplane sections, $x_0, \dots, x_n$. Then $\pi^* x_0, \dots, \pi^* x_n$ are sections of $\pi^*\mathscr{O}_{\mathbb{P}^n}(1)$, and they have no common zero. (Reminder: It is helpful to think of pulling back invertible sheaves in terms of pulling back transition functions.)

So to prove Theorem 16.2.1, we just need to show that these two constructions compose to give the identity in either direction.

Proof Suppose we are given $n+1$ sections $s_0, \dots, s_n$ of an invertible sheaf $\mathscr{L}$, with no common zeros, that (via Proposition 16.2.1) induce a morphism $\pi : X \to \mathbb{P}^n$. For each s_i, we get a trivialization of $\mathscr{L}$ on the open set X_{s_i}, where s_i doesn't vanish. The transition functions for $\mathscr{L}$ are precisely s_i/s_j on $X_{s_i} \cap X_{s_j}$. As $\mathscr{O}(1)$ is trivial on the standard affine open sets $D_+(x_i)$ of $\mathbb{P}^n$, $\pi^*(\mathscr{O}(1))$ is trivial on $X_{s_i} = \pi^{-1}(D_+(x_i))$. Moreover, $s_i/s_j = \pi^*(x_i/x_j)$ (directly from the construction of π in Proposition 16.2.1). This gives an isomorphism $\mathscr{L} \xleftrightarrow{\sim} \pi^*\mathscr{O}(1)$, since the two invertible sheaves have the same transition functions. We next show that this isomorphism can be chosen so that

$$(\mathscr{L}, s_0, \dots, s_n) \xleftrightarrow{\sim} (\pi^*\mathscr{O}(1), \pi^* x_0, \dots, \pi^* x_n).$$

Since s_i is nowhere vanishing on X_{s_i}, it gives a trivialization

$$\alpha_i \mathscr{O}_{X_{s_i}} \xrightarrow{\sim} \mathscr{L}|_{X_{s_i}}, \qquad 1 \longmapsto s_i.$$

On the other hand, $\mathscr{O}(1)$ is trivial on $D_+(x_i) \subseteq \mathbb{P}^n$, with trivialization

$$\beta_i : \mathscr{O}_{D_+(x_i)} \xrightarrow{\sim} \mathscr{O}(1)|_{D_+(x_i)}, \qquad 1 \longmapsto x_i.$$

Pulling back by π, and using $X_{s_i} = \pi^{-1}(D_+(x_i))$, we obtain a trivialization

$$\pi^*\beta_i : \mathscr{O}_{X_{s_i}} \xrightarrow{\sim} \pi^*\mathscr{O}(1)|_{X_{s_i}}, \qquad 1 \longmapsto \pi^* x_i.$$

Define $\Phi_i := \alpha_i \circ (\pi^*\beta_i)^{-1}$, then $\Phi_i(\pi^* x_i) = s_i$. We claim that that Φ_i agree on overlaps. Indeed, on $X_{s_i} \cap X_{s_j}$, the transition function for $\mathscr{L}$ is s_i/s_j, while the transition function for $\pi^*\mathscr{O}(1)$ is $\pi^*(x_i/x_j)$. But by the construction of π, we have $s_i/s_j = \pi^*(x_i/x_j)$. Hence the local isomorphism Φ_i are compatible on overlaps, and therefore glue to a global isomorphism $\Phi : \pi^*\mathscr{O}(1) \xrightarrow{\sim} \mathscr{L}$. Hence, we obtain

$$(\mathscr{L}, s_0, \dots, s_n) \xleftrightarrow{\sim} (\pi^*\mathscr{O}(1), \pi^* x_0, \dots, \pi^* x_n).$$

For the other direction, suppose we are given a map $\pi : X \to \mathbb{P}^n$. Let $s_i = \pi^* x_i \in \Gamma(X, \pi^*\mathscr{O}(1))$. As the x_i's have no common zeros on $\mathbb{P}^n$, the s_i's have no common zeros on X. The map $[s_0 : \cdots : s_n]$ is the same as the map π. We see this as follows. The preimage of $D(x_i)$ (by the morphism $[s_0 : \cdots : s_n]$) is $D(s_i) = D(\pi^* x_i) = \pi^{-1} D(x_i)$, so "the right open sets go to the right open sets". To show the two morphisms $D(s_i) \to D(x_i)$ (induced from $(s_1, \dots, s_n)$ and π) are the same, we use the fact that maps to an affine scheme $D(x_i)$ are determined by their maps of global sections in the opposite direction (Lemma 8.3.1). Both morphisms $D(s_i) \to D(x_i)$ correspond to the ring map $\pi^\sharp : x_{j/i} = x_j/x_i \mapsto s_j/s_i$. □

Remark 16.5 (***Extending Theorem 16.2.1 to rational maps.***) Suppose $s_0, \dots, s_n$ are sections of an invertible sheaf $\mathscr{L}$ on a scheme X. Then Theorem 16.2.1 yields a morphism $X \setminus V(s_0, \dots, s_n) \to \mathbb{P}^n$. In particular, if X is integral, and the s_i are not all 0, these data yield a rational map $X \dashrightarrow \mathbb{P}^n$.

16.2.2 Examples and applications

Example 16.4 (The Veronese embedding is $|\mathscr{O}_{\mathbb{P}_k^n}(d)|$.) Consider the line bundle $\mathscr{O}_{\mathbb{P}_k^n}(d)$ on $\mathbb{P}_k^n$. We have checked that the sections of this line bundle form a vector space of dimension $\binom{n+d}{d}$, with a basis corresponding to homogeneous degree d polynomials in the projective coordinates for $\mathbb{P}_k^n$ (Proposition 16.1.3 and Proposition 16.1.4). Also, they have no common zeros (as, for example, the subset of sections $x_0^d, x_1^d, \ldots, x_n^d$ have no common zeros). Thus the complete linear series is base-point-free, and determines a morphism $v_d : \mathbb{P}^n \to \mathbb{P}^{\binom{n+d}{d}-1}$. This is the Veronese embedding (Definition 10.3.3). For example, if $n = 2$ and $d = 2$, we get a map $\mathbb{P}_k^2 \to \mathbb{P}_k^5$.

In §10.3.3, we saw that this is a closed embedding. The following is a more general method of checking that maps to projective space are closed embeddings.

Proposition 16.2.4

Suppose $\pi : X \to \mathbb{P}_A^n$ corresponds to an invertible sheaf $\mathscr{L}$ on X, and sections $s_0, \ldots, s_n$. Then π is a closed embedding if and only if

(i) each open set X_{s_i} is affine, and

(ii) for each i, the map of rings $A[y_0, \ldots, y_n] \to \Gamma(X_{s_i}, \mathscr{O})$ given by $y_j \mapsto s_j/s_i$ is surjective.

Proof If π is a closed embedding, recall Definition 10.1.1, π is affine, so $X_{s_i} = \pi^{-1}(D_+(x_i))$ is affine and $k[x_{0/i}, \ldots, x_{n/i}]/(x_{i/i}-1) \to \Gamma(X_{s_i}, \mathscr{O})$ is surjective. More precisely, this surjective is given by $x_{j/i} \mapsto s_j/s_i$. Let $y_j := x_{j/i}$, consider the composition

$$A[y_0, \ldots, y_n] \twoheadrightarrow A[y_0, \ldots, y_n]/(y_i - 1) \twoheadrightarrow \Gamma(X_{s_i}, \mathscr{O}),$$

we obtain a surjective, which maps y_j to s_j/s_i.

Conversely, suppose (i) and (ii) hold. Notice that $y_i - 1$ maps to $s_i/s_i - 1 = 0$, consider

$$\begin{tikzcd} A[y_0, \ldots, y_n] \arrow[rr, two heads] \arrow[dr, two heads] & & \Gamma(X_{s_i}, \mathscr{O}) \\ & A[y_0, \ldots, y_n]/(y_i - 1), \arrow[ur] & \end{tikzcd}$$

since $A[y_0, \ldots, y_n] \twoheadrightarrow \Gamma(X_{s_i}, \mathscr{O})$ is a surjective, we know that $A[y_0, \ldots, y_n]/(y_i - 1) \to \Gamma(X_{s_i}, \mathscr{O})$ is surjective. In fact, $\Gamma(D_+(x_i), \mathscr{O}_{\mathbb{P}_A^n}) \cong A[y_0, \ldots, y_n]/(y_i - 1)$, we know that $\pi|_{X_{s_i}} \to D_+(x_i)$ is a closed embedding. Since $X_{s_i i}$ cover X, by Proposition 10.1.4, $\pi : X \to \mathbb{P}_A^n$ is closed embedding. □

Example 16.5 (Special case of Example 16.4) Recall that the image of the Veronese embedding when $n = 1$ is called a ***rational normal curve of degree*** d (Proposition 10.3.5). Our map is $\mathbb{P}_k^1 \to \mathbb{P}_k^d$, given by $[x : y] \mapsto [x^d : x^{d-1}y : \cdots : xy^{d-1}, y^d]$.

Proposition 16.2.5

*Suppose $\pi : \mathbb{P}_k^1 \to \mathbb{P}_k^n$ is a map where the corresponding invertible sheaf on $\mathbb{P}_k^1$ is $\mathscr{O}(d)$. (This can be reasonably be called a **degree** d **map**; cf **??** and **??**.) If $d < n$, then the image is degenerate (defined in Definition 16.2.3). If $d = n$ and the image is nondegenerate, then the image is isomorphism (via an automorphism of projective space) to a rational normal curve.*

Proof By Theorem 16.2.1, π is corresponding to $\mathscr{O}(d)$ with base-point-free section $s_0, \ldots, s_n$. Recall Proposition 16.1.3, $\dim_k \Gamma(\mathbb{P}_k^1, \mathscr{O}(d)) = \binom{d+1}{1} = d + 1$.

If $d < n$, then $d + 1 < n + 1$, which implies that $s_0, \ldots, s_n$ are linear dependent, hence there exists $a_i \in k$

such that

$$a_0 s_0 + a_1 s_1 + \cdots + a_n s_n = 0.$$

Consider $V(a_0x_0 + \cdots + a_nx_n) \hookrightarrow \mathbb{P}_k^n$, it is a hyperplane in $\mathbb{P}_k^n$, also the image of π lying in this hyperplane, it follows that $\mathbb{P}_k^1$ is degenerate.

If $d = n$ and the image is nondegenerate, then $s_0, \ldots, s_n$ are linear independent. Hence, it is a base of $\Gamma(\mathbb{P}_k^1, \mathscr{O}(d))$. Let $\mathbb{P}_k^1 = \operatorname{Proj} k[x,y]$. Notice that $\Gamma(\mathbb{P}_k^1, \mathscr{O}(d)) = \operatorname{Span}_k\{x^d, x^{d-1}y, \ldots, y^d\}$, there exists $M = (m_{ij}) \in \mathrm{GL}_{d+1}(k)$ such that

$$(s_0, \ldots, s_n) = (x^d, x^{d-1}y, \ldots, y^d)M.$$

The matrix defines an automorphism $M : \mathbb{P}^n \to \mathbb{P}^n, [\cdots] \mapsto [\cdots] \cdot M$, denote $v_n : \mathbb{P}^1 \to \mathbb{P}^n$ as in Example 16.5 , then $M \circ v_n = \pi$, which implies that the image is isomorphic to a rational normal curve. □

Proposition 16.2.6

Suppose X is a k-scheme (although this statement can readily generalize). Then a finite-dimensional base-point-free linear series V on X corresponding to $\mathscr{L}$ induces a morphism to projective space

$$\Phi_V : X \longrightarrow \mathbb{P}V^\vee.$$

Remark 16.6 This should be seen as a coordinate-free version of Theorem 16.2.1.

Proof Suppose $V = \operatorname{Span}_k\{s_0, \ldots, s_n\}$, then $\mathbb{P}V^\vee = \operatorname{Proj}(\operatorname{Sym}^\bullet V)$. Denote $S = \operatorname{Sym}^\bullet V$, then $S_\bullet$ is a graded ring which generated in degree 1. Since V is base-point-free linear series, $X = \bigcup_{i=0}^n X_{s_i}$. On each X_{s_i}, the section s_i is nowhere zero. Since $\mathscr{L}$ is invertible, s trivializes $\mathscr{L}$ over X_{s_i}. More precisely,

$$\tau_i : \mathscr{O}_{X_{s_i}} \xrightarrow{\sim} \mathscr{L}|_{X_{s_i}}, \qquad f \mapsto f s_i.$$

Fix s_i. We define a morphism

$$\Phi_i : X_{s_i} \longrightarrow D_+(s_i) = \operatorname{Spec}(S_{s_i})_0$$

as follow: recall Corollary 8.3.1, define

$$\Phi_i^\sharp : (S_{s_i})_0 \longrightarrow \Gamma(X_{s_i}, \mathscr{O}_{X_{s_i}})$$

by setting

$$\frac{f}{s_i} \longmapsto \frac{f}{s_i},$$

where $f \in V$. We need to check $\frac{f}{s_i}$ is indeed a rational function on X_{s_i}. Consider the following composition

$$\begin{array}{ccccccc} V & \longrightarrow & \Gamma(X, \mathscr{L}) & \xrightarrow{\text{res}} & \Gamma(X_{s_i}, \mathscr{L}|_{X_{s_i}}) & \xrightarrow[\sim]{\tau_i^{-1}} & \Gamma(X_{s_i}, \mathscr{O}_{X_{s_i}}) \\ \\ f & \longmapsto & f & \longmapsto & \frac{f}{s_i} \cdot s_i & \longmapsto & \frac{f}{s_i}, \end{array}$$

we know that $\frac{f}{s_i}$ is a rational function. One checks that $\Phi_i^\sharp$ is well-defined, then we obtain $\Phi_i : X_{s_i} \to D_+(s_i)$.

We next show that Φ_i is compatibility on overlaps, i.e., Φ_i, Φ_j are agree on $X_{s_i} \cap X_{s_j}$. On $D_+(s_i) \cap D_+(s_j)$, let $\frac{f}{s_i} \in (S_{s_i s_j})_0$, we have

$$\Phi_i^\sharp\left(\frac{f}{s_i}\right) = \frac{f}{s_i} = \frac{f/s_j}{s_i/s_j} = \frac{\Phi_j^\sharp(f/s_j)}{\Phi_j^\sharp(s_i/s_j)} = \Phi_j^\sharp\left(\frac{f/s_j}{s_i/s_j}\right) = \Phi_j^\sharp\left(\frac{f}{s_i}\right),$$

which implies that $\Phi_i = \Phi_j$ agree on $D_+(s_i) \cap D_+(s_j)$. So, Φ_i glue to a global map $\Phi_V : X \to \mathbb{P}V^\vee$. □

Remark 16.7 Write $\mathbb{P}V$= $\operatorname{Proj}\operatorname{Sym}^\bullet V^\vee$, then $\Gamma(\mathbb{P}V, \mathscr{O}(1)) = \operatorname{Sym}^1 V^\vee$ and $\Gamma(\mathbb{P}V, \mathscr{O}(2)) = \operatorname{Sym}^2 V^\vee$.

Lemma 16.2.1

Suppose X is a k-scheme (this statement can readily generalize) and $\mathscr{L}$ is a line bundle on X. Then a linear system $V \to \Gamma(X, \mathscr{L})$ is base-point-free if and only if the natural morphism

$$V \otimes_k \mathscr{O}_X \longrightarrow \mathscr{L}$$

is surjective.

Proof Suppose $\dim_k V = n+1$, choose a basis $V = \operatorname{Span}_k\{s_0, \dots, s_n\}$. If linear system V is base-point-free, then for each $x \in X$ there exists $s \in V$ such that $s(x) \neq 0$ in the fiber $\mathscr{L}|_x = \mathscr{L}_x \otimes_{\mathscr{O}_{X,x}} \kappa(x)$. We now show that this is equivalent to the surjectivity of

$$e : V \otimes_k \mathscr{O}_X \longrightarrow \mathscr{L}.$$

Fix $x \in X$. On stalks-level, we have

$$e_x : V \otimes_k \mathscr{O}_{X,x} \longrightarrow \mathscr{L}_x.$$

Since $\mathscr{L}$ is line bundle, $\mathscr{L}_x$ is a free $\mathscr{O}_{X,x}$-module of rank 1. Choose a local generator e, so $\mathscr{L}_x \cong \mathscr{O}_{X,x}e$. If $s \in V$, then $s_x = ae$ for some $a \in \mathscr{O}_{X,x}$. Its value in the fiber is

$$s(x) = \overline{a}e_x \in \mathscr{L}_x \otimes_{\mathscr{O}_{X,x}} \kappa(x).$$

Thus $s(x) \neq 0$ is equivalent to $\overline{a} \neq 0$, which implies that $a \notin \mathfrak{m}_x$. Since $\mathscr{O}_{X,x}$ is local ring, a is a unit, so $\mathscr{L}_x$ is generated by s_x. Hence, e_x is surjective, it follows that $e : V \otimes_k \mathscr{O}_X \to \mathscr{L}$ is surjective.

Conversely, if e is surjective, then e_x is surjective for each $x \in X$. By tensoring $\kappa(x)$ to e_x, we obtain a surjective $V \otimes_k \kappa(x) \to \mathscr{L}_x \otimes_{\mathscr{O}_{X,x}} \kappa(x)$. The target is one-dimensional over $\kappa(x)$, so this map is nonzero. Therefore there exists $s \in V$ such that $s(x) \neq 0$. Hence, V is base-point-free. □

Proposition 16.2.7

$\mathbb{P}V^\vee$ represents the functor $F : (\mathbf{Sch}_k)^{\mathrm{op}} \to \mathbf{Sets}$ which is defined by

$$F(X) = \{V \otimes_k \mathscr{O}_X \twoheadrightarrow \mathscr{L} : \mathscr{L} \text{ is an invertible sheaf on } X\}/\sim$$

where $q \sim q'$ if there exits an isomorphism of line bundles $\alpha : \mathscr{L} \xrightarrow{\sim} \mathscr{L}'$ such that $q' = \alpha \circ q$.

Proof On $\mathbb{P}V^\vee = \operatorname{Proj} \operatorname{Sym}^\bullet V$, there is a natural surjective

$$V \otimes_k \mathscr{O}_{\mathbb{P}V^\vee} \longrightarrow\!\!\!\!\rightarrow \mathscr{O}_{\mathbb{P}V^\vee}(1).$$

If $f : X \to \mathbb{P}V^\vee$ is a morphism, pulling back the natural surjective gives

$$V \otimes_k \mathscr{O}_X \longrightarrow\!\!\!\!\rightarrow f^*\mathscr{O}_{\mathbb{P}V^\vee}(1).$$

Thus f determines an element of $F(X)$.

Conversely, gives $q : V \otimes_k \mathscr{O}_X \twoheadrightarrow \mathscr{L}$ in $F(X)$. Recall Lemma 16.2.1, $V \to \Gamma(X, \mathscr{L})$ is a base-point-free linear system, by Proposition 16.2.6, it induces a morphism to projective space

$$\Phi_V : X \longrightarrow \mathbb{P}V^\vee.$$

Hence, we have a bijection

$$h_{\mathbb{P}V^\vee}(X) = \operatorname{Mor}_{\mathbf{Sch}_k}(X, \mathbb{P}V^\vee) \xleftrightarrow{\sim} F(X).$$

Also, this bijection induce a natural isomorphism $h_{\mathbb{P}V^\vee} \cong F$, we omit this check. □

✍ **Exercise 16.1** Define the graded rings $R_\bullet = k[u, v, w]/(uw - v^2)$ and $S_\bullet = k[x, y]$ (with all variables having degree 1). By Exercise 10.6, we have an isomorphism $\operatorname{Proj} R_\bullet \xleftrightarrow{\sim} \operatorname{Proj} S_\bullet$ (via the Veronese embedding v_2). Show that this isomorphism is not induced by a map of graded rings $S_\bullet \to R_\bullet$.

Proof Let $C = \operatorname{Proj} R_\bullet$. In fact, the isomorphism $\varphi : \mathbb{P}^1_k \to C$ is given by $[x : y] \mapsto [x^2 : xy : y^2]$. Suppose there exists a map of graded rings $(\varphi^{-1})^\sharp : S_\bullet \to R_\bullet$ induce $\varphi^{-1} : C \to \mathbb{P}^1$. Since $(\varphi^{-1})^\sharp$ is a graded ring map and $\deg x = \deg y = 1$, $(\varphi^{-1})^\sharp(x), (\varphi^{-1})^\sharp(y) \in R_1$. Notice that $R_1 = \operatorname{Span}_k\{u, v, w\}$, we may assume $(\varphi^{-1})^\sharp(x) = au + bv + cw = F(u, v, w)$ and $(\varphi^{-1})^\sharp(y) = eu + fv + gw = G(u, v, w)$. Now, consider the composition

$$\mathbb{P}^1 \xrightarrow{\varphi} C \xrightarrow{\varphi^{-1}} \mathbb{P}^1.$$

Since φ is an isomorphism, we obtain $\varphi^{-1} \circ \varphi = \mathrm{id}$, that is,

$$[F(x^2, xy, y^2) : G(x^2, xy, y^2)] = [x : y].$$

Hence, $F(u, v, w) = au + bv$ and $G(u, v, w) = av + bw$. Notice that $F(x^2, xy, y^2) = x(ax + by)$ and $G(x^2, xy, y^2) = y(ax + by)$, hence on $ax + by = 0$, $F = G = 0$. It contradicts to Theorem 16.2.1. $\square$

Remark 16.8 You may be to show that, after "regrading," the isomorphism $\operatorname{Proj} R_\bullet \xleftrightarrow{\sim} \operatorname{Proj} S_\bullet$ does arise from a map of graded rings $(S_{2\bullet} \to R_\bullet)$. **??** will give an example where it is not possible to remedy the lack of maps of graded rings by just regrading.

An early look at intersection theory, related to Bézout's Theorem

A classical definition of the degree of a curve in projective space is as follows: Intersect it with a "general" hyperplane, and count the number of points of intersection, with appropriate multiplicity. We interpret this in the case of $\pi : \mathbb{P}^1_k \to \mathbb{P}^n_k$.

Exercise 16.2 Show that there is a hyperplane H of $\mathbb{P}^n_k$ not containing $\pi(\mathbb{P}^1_k)$. Equivalently, $\pi^* H \in \Gamma(\mathbb{P}^1, \mathscr{O}_{\mathbb{P}^1}(d))$ (interpreted as the pullback of a section of $\mathscr{O}_{\mathbb{P}^n}(1)$) is not 0. Show that the number of zeros of $\pi^* H$ is precisely d. (You will have to define "appropriate multiplicity." What does it means it mean geometrically if π is a closed embedding, and $\pi^* H$ has a double zero? Aside: Can you make sense of this even π is not a closed embedding?)

Proof Suppose k is algebraically closed.

We first show that the statement "a hyperplane $H = V(\ell)$ of $\mathbb{P}^n_k$ not containing $\pi(\mathbb{P}^1_k)$" is equivalent to "$\pi^*\ell \in \Gamma(\mathbb{P}^1, \mathscr{O}_{\mathbb{P}^1}(d))$ is not 0". Let $H = V(\ell)$ be a hyperplane, then $\ell \in \mathscr{O}_{\mathbb{P}^n_k}(1)$. If $\pi(\mathbb{P}^1_k) \subsetneq H$, there exists $x \in \mathbb{P}^1_k$ such that $\ell(\pi(x)) \neq 0$. Consider $\pi^*\ell(x)$, notice that

$$\begin{aligned}
\pi^* \mathscr{O}_{\mathbb{P}^n_k}(1)_x \otimes_{\mathscr{O}_{\mathbb{P}^1_k, x}} \kappa(x) &= \mathscr{O}_{\mathbb{P}^n_k}(1)_{\pi(x)} \otimes_{\mathscr{O}_{\mathbb{P}^n_k, \pi(x)}} \mathscr{O}_{\mathbb{P}^1_k, x} \otimes_{\mathscr{O}_{\mathbb{P}^1_k, x}} \kappa(x) \\
&= \mathscr{O}_{\mathbb{P}^n_k}(1)_{\pi(x)} \otimes_{\mathscr{O}_{\mathbb{P}^n_k, \pi(x)}} \kappa(x) \\
&= \mathscr{O}_{\mathbb{P}^n_k}(1)_{\pi(x)} \otimes_{\mathscr{O}_{\mathbb{P}^n_k, \pi(x)}} \kappa(\pi(x)) \otimes_{\kappa(\pi(x))} \kappa(x),
\end{aligned}$$

since ℓ is not zero in $\mathscr{O}_{\mathbb{P}^n_k}(1)_{\pi(x)} \otimes_{\mathscr{O}_{\mathbb{P}^n_k, \pi(x)}} \kappa(\pi(x))$, we know that $\pi^*\ell(x) \neq 0$, which implies section $\pi^*\ell \in \Gamma(\mathbb{P}^1_k, \pi^* \mathscr{O}_{\mathbb{P}^n_k}(1))$ is not 0. Conversely, if $\pi^*\ell \in \Gamma(\mathbb{P}^1_k, \pi^*\mathscr{O}_{\mathbb{P}^n_k}(1))$ is not 0, then there exists $x \in \mathbb{P}^1_k$ such that $\pi^*\ell(x) \neq 0$, i.e., the image of $\pi^*\ell$ in $\pi^*\mathscr{O}_{\mathbb{P}^n_k}(1)_x \otimes_{\mathscr{O}_{\mathbb{P}^1_k, x}} \kappa(x)$ is not 0, which implies $\pi^*\ell$ can be seen as a nonzero element in $\pi^*\mathscr{O}_{\mathbb{P}^n_k}(1)_x$. Therefore the image of $\pi^*\ell$ in $\mathscr{O}_{\mathbb{P}^n_k}(1)_{\pi(x)} \otimes_{\mathscr{O}_{\mathbb{P}^n_k, \pi(x)}} \kappa(\pi(x))$ is not 0, which implies that $\ell(\pi(x)) \neq 0$. It follows that hyperplane $H = V(\ell)$ of $\mathbb{P}^n_k$ not containing $\pi(\mathbb{P}^1_k)$. Now, recall Proposition 16.1.5, we know that $\pi^*\mathscr{O}_{\mathbb{P}^n_k}(1) = \mathscr{O}_{\mathbb{P}^1_k}(d)$ for some d.

We next show that for any $\pi : \mathbb{P}^1_k \to \mathbb{P}^n_k$, there exists a hyperplane H of $\mathbb{P}^n_k$ not containing $\pi(\mathbb{P}^1_k)$. Consider the sections $x_0, \ldots, x_n \in \Gamma(\mathbb{P}^n, \mathscr{O}_{\mathbb{P}^n}(1))$, by the proof of Theorem 16.2.1, their pullback $\pi^*x_0, \ldots, \pi^*x_n \in \Gamma(\mathbb{P}^1, \pi^*\mathscr{O}_{\mathbb{P}^n}(1))$ is base-point-free. Hence, for each $p \in \mathbb{P}^1$, there exists i such that $\pi^*x_i(p) \neq 0$, hence there must has a π^*x_i which is nonzero. Take $H_i = V(x_i)$, by the statement we proved above, we know that H_i not containing $\pi(\mathbb{P}^1_k)$.

Define the appropriate multiplicity. Let $s \in \Gamma(\mathbb{P}^1, \mathscr{O}_{\mathbb{P}^1_k}(d))$ be a nonzero section. We define the multiplicity of a zero of s as follows. Let $p \in \mathbb{P}^1$. Choose a local trivialization of $\mathscr{O}_{\mathbb{P}^1}(d)$ near p. With respect to this trivialization, the section s can be written locally as $s = fe$ where e is a local generator of $\mathscr{O}_{\mathbb{P}^1}(d)$. Then the multiplicity of s at p is defined by

$$\operatorname{mult}_p(s) := \operatorname{ord}_p(f).$$

This is independent of the chosen trivialization, because another trivialization changes f by multiplying it by a unit, and multiplying by a unit does not change the order of vanishing. Since $\mathbb{P}^1$ is a smooth curve, the local ring $\mathscr{O}_{\mathbb{P}^1,p}$ is a DVR, so $\operatorname{ord}_p(f)$ is well-defined. If k is algebraically closed, the number of zero of s, counted with multiplicity, is

$$\sum_{p\in\mathbb{P}^1(k)} \operatorname{mult}_p(s).$$

A nonzero section of $\mathscr{O}_{\mathbb{P}^1}(d)$ has d zeros counted with multiplicity. Choose homogeneous coordinates $[x : y]$ on $\mathbb{P}^1$. A global section of $\mathscr{O}_{\mathbb{P}^1}(d)$ is represented by a homogeneous polynomial $F(x, y)$ of degree d. Therefore a nonzero section s corresponds to a nonzero homogeneous polynomial $F(x, y) \in k[x, y]_d$. Consider $D_+(x)$. On this open set, set $t = y/x$. The line bundle $\mathscr{O}_{\mathbb{P}^1}(d)$ is trivialized by x^d on $D_+(x)$. Hence, on $D_+(x)$, $F(x, y) = x^d F(1, t)$. Thus the zeros of s on $D_+(x)$ are the zeros of the polynomial $f(t) := F(1, t)$. Suppose $\deg f = m$, then the zeros of $f(t)$ in the affine line, counted with multiplicity, contribute m. Now, we must also check the point at infinity. On $D_+(y)$, set $z = x/y$. Then $\mathscr{O}_{\mathbb{P}^1}(d)$ is trivialized by y^d, and $F(x, y) = y^d F(z, 1)$. Notice that $F(z, 1) = z^d F(1, 1/z) = z^d f(1/z)$, say $f(t) = a_m t^m +$ lower degree terms, $a_m \neq 0$, we have

$$z^d f(1/z) = a_m z^{d-m} + \text{higher powers of } z.$$

Therefore the order of vanishing at $z = 0$ is $d - m$. So the total number of zeros, counted with multiplicity is $m + (d - m) = d$. Hence, ever nonzero section $s \in \Gamma(\mathbb{P}^1, \mathscr{O}_{\mathbb{P}^1}(d))$ has exactly d zeros, counted with multiplicity. *Since π^*H is a section in $\Gamma(\mathbb{P}^1, \mathscr{O}_{\mathbb{P}^1}(d))$, we know that the number of zeros of π^*H is precisely d.*

If π is a closed embedding. Let $C = \pi(\mathbb{P}^1) \subseteq \mathbb{P}^n$ is a closed subscheme. In other words, $\pi : \mathbb{P}^1 \xrightarrow{\sim} C$ is an isomorphism. Let $H = V(\ell) \subseteq \mathbb{P}^n$ be a hyperplane, where $\ell \in \Gamma(\mathbb{P}^n, \mathscr{O}_{\mathbb{P}^n}(1))$. Pulling back ℓ along π, we get a section $\pi^*\ell \in \Gamma(\mathbb{P}^1, \pi^*\mathscr{O}_{\mathbb{P}^n}(1)) = \Gamma(\mathbb{P}^1, \mathscr{O}_{\mathbb{P}^1}(d))$. The zero locus of $\pi^*\ell$ is exactly the inverse image of the hyperplane section $V(\pi^*\ell) = \pi^{-1}(C \cap H)$. Take a point $p \in \mathbb{P}^1$, and write $q = \pi(p) \in C$. Because $\pi : \mathbb{P}^1 \to C$ is an isomorphism, the local rings are isomorphism $\mathscr{O}_{C,q} \cong \mathscr{O}_{\mathbb{P}^1,p}$. The hyperplane H cuts out a function on C, namely the restriction $\ell|_C$. Under the isomorphism $C \cong \mathbb{P}^1$, this restriction corresponds exactly to the pullback section $\pi^*\ell$. Therefore the order of vanishing of $\pi^*\ell$ at p is the same things as the local intersection multiplicity of C and H at q. More precisely,

$$I_q(C, H) = \operatorname{mult}_q(\ell|_C) = \operatorname{mult}_p(\pi^*\ell).$$

So, if π^ℓ has a double zero at p, that means $\operatorname{mult}_p(\pi^*\ell) = 2$. Geometrically, this means that the hyperplane H is tangent to the curve C at the point $q = \pi(p)$. Choose a local parameter t for $\mathbb{P}^1$ at p, so that $t(p) = 0$. Locally, the curve C is parametrized by $t \mapsto \pi(t)$. The hyperplane H is defined by the equation $\ell = 0$. If we restrict this equation to the curve, we get the local function $\ell(\pi(t))$. This local function is precisely a local expression for the pullback section $\pi^*\ell$. Now, saying that $\pi^*\ell$ has a double zero at p means that $\ell(\pi(t))$ vanishes to order 2 at $t = 0$. Thus locally we can write

$$\ell(\pi(t)) = a_2 t^2 + a_3 t^3 + \cdots$$

with $a_2 \neq 0$. In particular, the constant term and the linear term vanish. This says that the tangent vector of the curve at q is killed by the linear form ℓ. Equivalently, the tangent direction of C at q lies inside the hyperplane H. Therefore a double zero of π^*H at p means that H is tangent to the embedded curve $C = \pi\left(\mathbb{P}^1\right)$ at the point $\pi(p)$. □

Thus this classical notion of degree agrees with the notion of degree in Proposition 16.2.5. (See Exercise 10.4 for another case of Bézout's Theorem. Here we intersect a degree d curve with a degree 1 hyperplane; there we intersect a degree 1 curve with a degree d hypersurface. **??** will give a common generalization.)

The Segre embedding revisited

The Segre embedding can also be interpreted in this way. This is a useful excuse to define some notation. Suppose $\mathscr{F}$ is a quasicoherent sheaf on a Z-scheme X, and $\mathscr{G}$ is a quasicoherent sheaf on a Z-scheme Y. Let π_X, π_Y be the projections from $X \times_Z Y$ to X and Y, respectively. Then $\mathscr{F} \boxtimes \mathscr{G}$ (pronounced "$\mathscr{F}$ box-times $\mathscr{G}$") is defined to be $\pi_X^*\mathscr{F} \otimes \pi_Y^*\mathscr{G}$. In particular, $\mathscr{O}_{\mathbb{P}^m \times \mathbb{P}^n}(a,b)$ is defined to be $\mathscr{O}_{\mathbb{P}^m}(a) \boxtimes \mathscr{O}_{\mathbb{P}^n}(b)$ (over any base $\mathbb{Z}$). The Segre embedding $\mathbb{P}^m \times \mathbb{P}^n \to \mathbb{P}^{mn+m+n}$ corresponds to the complete linear series for the invertible sheaf $\mathscr{O}(1,1)$.

When we first saw the Segre embedding in §11.6, we saw (in different language) that this complete linear series is base-point-free. We also checked by hand (§11.6) that it is a closed embedding, essentially by Proposition 16.2.4.

Recall that if $\mathscr{L}$ and $\mathscr{M}$ are both base-point-free invertible sheaves on a scheme X, then $\mathscr{L} \otimes \mathscr{M}$ is also base-point-free (Proposition 16.2.2). We may interpret this using the Segre embedding (under reasonable hypotheses on X). If $\Phi_{\mathscr{L}} : X \to \mathbb{P}^M$ is a morphism corresponding to a (base-point-free) linear series based on $\mathscr{L}$, and $\Phi_{\mathscr{M}} : X \to \mathbb{P}^N$ is a morphism corresponding to a linear series on $\mathscr{M}$, then the Segre embedding yields a morphism corresponding to a linear series on $\mathscr{M}$, then the Segre embedding yields a morphism $X \to \mathbb{P}^M \times \mathbb{P}^N \to \mathbb{P}^{(M+1)(N+1)-1}$, which corresponds to a base-point-free series of sections of $\mathscr{L} \otimes \mathscr{M}$.

16.2.3 ⋆⋆ A proper nonprojective k-variety — and gluing schemes along closed subschemes

16.3 The Curve-to-Projective Extension Theorem

We now use the main theorem of the previous section, Theorem 16.2.1, to prove something useful and concrete. (In fact, we could have proved this far earlier — with a little cleverness you can replace the invocation of Theorem 16.2.1 by Proposition 8.3.11, and prove this as soon as you know about DVRs.)

16.3.1 The Curve-to-Projective Extension Theorem

As motivation, we consider a variation of the example at the start of §16.2: (16.3) was a failed attempt at a morphism $\mathbb{P}^2_k \to \mathbb{P}^3_k$. There was a reason the example used $\mathbb{P}^2_k$ rather than $\mathbb{P}^1_k$ as the domain. For example, at first it appears as though the rational map $\mathbb{P}^1_k \dashrightarrow \mathbb{P}^3_k$ given by

$$[x^7 : x^3y^4 : x^4y^3 + 4x^5y^2 : x^5y^2 + x^7]$$

doesn't give a morphism $\mathbb{P}^1_k \to \mathbb{P}^3_k$, because it is not defined at $[x : y] = [0 : 1]$. But we can extend the map over $[0 : 1]$, using

$$[x^7 : x^3y^4 : x^4y^3 + 4x^5y^2 : x^5y^2 + x^7] = [x^4 : y^4 : xy^3 + 4x^2y^2 : x^2y^2 + x^4] \tag{16.4}$$

— we can just divide through by x^3 to make the morphism make sense. Similarly, it appears as though $[x^2 : x^{-2}y^4 : x^{-1}y^3 + 4y^2 : y^2 + x^2]$ also is not defined at $[x : y] = [0 : 1]$, but you may see what to do (and why it can be extended to the same map as (16.4)). This will generalize to the following.

Theorem 16.3.1 (The Curve-to-Projective Extension Theorem)

Suppose C is a pure dimension 1 Noetherian scheme over an affine base $S = \operatorname{Spec} A$, and $p \in C$ is a regular closed point of it. Suppose Y is a projective S-scheme. Then any morphism $C \setminus \{p\} \to Y$ (of S-schemes) extends to all of C.

To preempt any confusion: C is "absolute pure dimension 1", not relative pure dimension 1.

In particular, we will use this theorem when $S = \operatorname{Spec} k$, and C is a k-variety. The only reason we assume S is affine is because we won't know the meaning of "projective S-scheme" until we know what a projective morphism is (§??). But the proof below extends immediately to general S once we know the meaning of the statement.

Note that if such an extension exists, then it is unique: the nonreduced locus of C is a closed subset (Proposition 7.6.17). So after replacing C by a sufficiently small open neighborhood U of p, we may assume that U is reduced. Since C has pure dimension 1 and p is a closed point, the open subset $U \setminus \{p\}$ must contain the generic point, so $U \setminus \{p\}$ must dense in U. Now, let $F, G : C \to Y$ be two extensions of the same morphism $C \setminus \{p\} \to Y$. Then $F|_{U\setminus\{p\}} = G|_{U\setminus\{p\}}$. Since Y is projective over S, it is separated over S. Hence, by the Reduced-to-Separated Theorem 12.3.1, we conclude that $F|_U = G|_U$, and since they already agree on $C \setminus \{p\}$, we get $F = G$ on all of C. Another proof: maps to a separated scheme can be extended over an effective Cartier divisor in at most one way (Proposition 12.3.5).

The following exercise shows that the hypotheses are necessary.

✍ **Exercise 16.3** In each of the following cases, prove that the morphism $C \setminus \{p\} \to Y$ cannot be extended to a morphism $C \to Y$.

(a) ***(Projective of Y is necessary.)*** Suppose $C = \mathbb{A}^1_k$, $p = 0$, $Y = \mathbb{A}^1_k$, and $C \setminus \{p\} \to Y$ is given by "$t \mapsto 1/t$".

(b) ***(One-dimensionality of C is necessary.)*** Suppose $C = \mathbb{A}^2_k$, $p = (0, 0)$, $Y = \mathbb{P}^1_k$, and $C \setminus \{p\} \to Y$ is given by $(x, y) \mapsto [x : y]$.

(c) ***(Non-singularity of C is necessary.)*** Suppose $C = \operatorname{Spec} k[x, y]/(y^2 - x^3)$, $p = (0, 0)$, $Y = \mathbb{P}^1_k$, and $C \setminus \{p\} \to Y$ is given by $(x, y) \mapsto [x, y]$.

Proof

(a) Let $C = \operatorname{Spec} k[t]$ and $Y = \operatorname{Spec} k[s]$, then $C \setminus \{p\} = \operatorname{Spec} k[t]_t$. Suppose $\pi : C \setminus \{p\} \to Y$ given by the ring map
$$k[s] \longrightarrow k[t]_t, \qquad s \longmapsto \frac{1}{t}.$$
If $C \setminus \{p\} \to Y$ can be extended to $\widetilde{\pi} : C \to Y$, say $\widetilde{\pi}^\sharp(s) = f(t)$. Since $\widetilde{\pi}|_{C\setminus\{p\}} = \pi$, in $k[t]_t$, there must be
$$\frac{f(t)}{1} = \frac{1}{t} \implies tf(t) = 1.$$
But, this is impossible, a contradiction.

(b) Let $\mathbb{P}^1_k = \operatorname{Proj} k[x, y]$. Suppose $C \setminus \{p\} \to Y$ can be extended to $\widetilde{\pi} : C \to Y$. Consider $\widetilde{\pi}(p)$, $\widetilde{\pi}(p)$ must belong to $D_+(x)$ or $D_+(y)$. If $\widetilde{\pi}(p) \in D_+(x)$. Since $\widetilde{\pi}|_{C\setminus\{p\}} = \pi$, so $\widetilde{\pi}^{-1}(D_+(x)) \cap (\mathbb{A}^1 \setminus \{p\}) = \pi^{-1}(D_+(x)) = D(x)$, and thus $\widetilde{\pi}^{-1}(D_+(x)) = D(x) \cup \{p\}$. But $D(x) \cup \{p\}$ is not open, since $(D(x) \cup \{p\}) \cap V(x) = \{p\}$ is closed in $V(x)$. Contradiction! Therefore, $\widetilde{\pi}(p) \notin D_+(x)$. Similarly, $\widetilde{\pi}(p) \notin D_+(y)$, this contradicts to the fact that $\widetilde{\pi}(p)$ must belong to $D_+(x)$ or $D_+(y)$.

(c) Suppose $\pi : C \setminus \{p\} \to Y$ can be extended to $\widetilde{\pi} : C \to Y$, then $\widetilde{\pi}(p) \in D_+(x)$ or $D_+(y)$. If $\widetilde{\pi}(p) \in D_+(x)$. Let $V = \widetilde{\pi}^{-1}(D_+(x))$. Consider the corresponding ring map $\widetilde{\pi}|_V^\sharp : k[\frac{y}{x}] \to \Gamma(V, \mathscr{O}_C)$. Since $\widetilde{\pi}|_{C\setminus\{p\}} = \pi$, on $V \cap (C \setminus \{p\})$, $\widetilde{\pi}|^\sharp_{V\cap(C\setminus\{p\})}(\frac{y}{x}) = \frac{y}{x}$. Because $\widetilde{\pi}$ is defined on p, so $\frac{y}{x} \in \mathscr{O}_{C,p}$. Notice that there is an isomorphism

$$k[x,y]/(y^2 - x^3) \xrightarrow{\sim} k[t^2,t^3], \qquad (x,y) \longmapsto (t^2,t^3),$$

we know that $\mathscr{O}_{C,p} \cong k[t^2,t^3]_{(t^2,t^3)}$. Under the isomorphism $\frac{y}{x}$ becomes $t^3/t^2 = t$. But $t \notin k[t^2,t^3]_{(t^2,t^3)}$, contradiction! Hence, $\pi : C \setminus \{p\} \to Y$ cannot be extended to $\widetilde{\pi} : C \to Y$.

□

We remark that by combining this (easy) theorem with the (hard) valuative criterion for properness (Theorem 14.7.3), one obtains a proof of the properness of projective space bypassing the (tricky but not hard) Fundamental Theorem of Elimination Theory 9.4.3 (see Theorem 14.7.4). Fancier remark: The valuative criterion of properness can be used to show that Theorem 16.3.1 remains true if Y is only required to be proper, but it requires some thought.

Note ***Central idea of proof.*** *The central idea of the proof may be summarized as "clear denominators", and is illustrated by the following motivating example. Suppose you have a morphism from $\mathbb{A}^1 \setminus \{0\}$ to projective space, and you wanted to extend it to $\mathbb{A}^1$. Suppose the map was given by $t \mapsto [t^4 + t^{-3} : t^{-2} + 4t]$. Then of cause you would "clear the denominators", and replace the map by $t \mapsto [t^7 + 1 : t + t^4]$. Similarly, if the map was given by $t \mapsto [t^2 + t^3 : t^2 + t^4]$, you would divide by t^2, to obtain the map $t \mapsto [1 + t : 1 + t^2]$.*

Proof of the Curve-to-Projective Extension Theorem 16.3.1

Proof Our plan is to maneuver our selves into the situation where we can apply the idea of above "*Central idea of proof*". We begin with some quick reductions. The nonreduced locus of C is closed (Proposition 7.6.17) and doesn't contain p, so replacing C by an appropriate open neighborhood of p, we may assume that C is reduced and affine.

We next reduce to the case where $Y = \mathbb{P}^n_A$. Choose a closed embedding $Y \to \mathbb{P}^n_A$. If the result holds for $\mathbb{P}^n$, and we have a morphism $C \to \mathbb{P}^n$ with $C \setminus \{p\}$ mapping to Y as well. Reason: Say $F : C \to \mathbb{P}^n_A$, pick $V = \mathbb{A}^n_A \subseteq \mathbb{P}^n_A$, let $U := F^{-1}(V)$, then U is open in C and $F|_U : U \to V$. To show that $F(C) \subseteq Y$. It suffices to show that, for each $q \in U_q \subseteq F^{-1}(V)$ where U_q affine open, there is a morphism $F|_{U_q} : U_q \to V \cap Y$. Hence, we can reduce to the case where the source is an affine open subset, and the target is $\mathbb{A}^n_A \subseteq \mathbb{P}^n_A$, i.e., $F : U \to \mathbb{A}^n_A$. Notice that $F(U) \subseteq Y \cap \mathbb{A}^n_A$ if and only if the functions vanishing on $Y \cap \mathbb{P}^n_A$ pull back to functions that vanish at U. Since $F(U \setminus \{p\}) \subseteq Y \cap \mathbb{A}^n_A$, functions vanishing on $Y \cap \mathbb{A}^n_A$ pull back to functions vanishing on $U \setminus \{p\}$. Since C is dimension one and $\{p\}$ is a closed point, $U \setminus \{p\}$ must contain all generic points, hence the functions vanishing on $Y \cap \mathbb{A}^n_A$ pull back to functions that vanish at the generic points of the irreducible components of C. Since C is reduced, for any affine pice $U \subseteq C$, we have $\mathfrak{N}(\Gamma(U, \mathscr{O}_C)) = \bigcap_{\text{minimal prime}} \mathfrak{p} = 0$, so these functions vanishing on U, i.e., C maps to Y.

Choose a uniformizer $t \in \mathfrak{m} \setminus \mathfrak{m}^2$ in the local ring of C at p. As t is a function in some neighborhood of p, we may assume that t is a function on C (by replacing C by this neighborhood of p). Then $V(t)$ contains p (as $t \in \mathfrak{m}$). Now, we claim that $V(t)$ does not contain the irreducible component of C containing p. Let C_0 be the irreducible component of C containing p. If $V(t)$ contained C_0, then t would vanish on C_0, that is, the image of t in $\mathscr{O}_{C,p}$ would belong to the minimal prime corresponding to C_0. But since p is regular, $\mathscr{O}_{C,p}$ is a domain. Hence, its only minimal prime is (0). Therefore t would have to be zero in $\mathscr{O}_{C,p}$. This is impossible, because t is a uniformizer. In particular, $t \notin \mathfrak{m}^2$, whereas $0 \in \mathfrak{m}^2$. Hence, $V(t)$ cannot contain the irreducible component of C passing through p. Replacing C by an affine open neighborhood of p in $(C \setminus V(t)) \cup \{p\} = C \setminus (V(t) \setminus \{p\})$, this is obtained from C by removing all zeros of t except p. We claim that this is open. Since C is a one-dimensional Noetherian scheme, any proper closed subset which does not contain an irreducible component is a finite set of closed points (i.e., C has the cofinite topology), now p is the only zero of the function t. Moreover, the local ring at p has not changed after replacing C by $(C \setminus V(t)) \cup \{p\}$ (it only cuts finite closed points). Therefore t is still a uniformizer in $\mathscr{O}_{C,p}$. Hence, $\mathrm{val}_p(t) = 1$ (t vanishes at p with multiplicity 1).

We have a map $C \setminus \{p\} \to \mathbb{P}_A$, which by Theorem 16.2.1 corresponds to a line bundle $\mathscr{L}$ on $C \setminus \{p\}$ and $n+1$ sections of it with no common zeros in $C \setminus \{p\}$. Let U be a nonempty open set of $C \setminus \{p\}$ on which $\mathscr{L}|_U \cong \mathscr{O}_U$. Then by replacing C by $U \cup \{p\}$, we interpret the map to $\mathbb{P}^n$ as $n+1$ rational functions $f_0, \dots, f_n$, defined away from p, with no common zeros away from p. Let $N = \min_i\{\mathrm{val}_p f_i\}$. Then $t^{-N} f_0, \dots, t^{-N} f_n$ are $n+1$ functions with no common zeros. Thus they determine a morphism $C \to \mathbb{P}^n_A$ extending $C \setminus \{p\} \to \mathbb{P}^n_A$ as desired. □

Proposition 16.3.1 (Useful)

Suppose X is a Noetherian k-scheme, and Z is an irreducible codimension 1 subvariety whose generic point η is a regular point of X (so the local ring $\mathscr{O}_{X,\eta}$ is a DVR). Suppose $\pi : X \dashrightarrow Y$ is a rational map to a projective k-scheme. Then the domain of definition of the rational map includes a dense open subset of Z. In other words, rational maps from Noetherian k-schemes to projective k-schemes can be extended over regular codimension 1 sets.

Remark 16.9 We have already seen this principle in action — see Definition 8.5.8 on the Cremona transformation.

Proof Suppose $\pi : X \dashrightarrow Y$ is a rational map, then there exists a dense open subset $U \subseteq X$ such that $\pi|_U : U \to Y$ is a morphism. Let η be a generic point of Z, since $\mathrm{codim}_X Z = \dim \mathscr{O}_{X,\eta} = 1$ and η is a regular point of X, we know that $\mathscr{O}_{X,\eta}$ is a DVR. To show that the domain of definition of π includes a dense open subset of Z, it suffices to show that $\eta \in \mathrm{dom}(\pi)$ (since $\mathrm{dom}(\pi) \cap Z \ni \eta$ is open in Z). Denote $C = \mathrm{Spec}\, \mathscr{O}_{X,\eta}$, then C is a curve and there is a canonical morphism $C \to X$ (Proposition 8.3.10). Since $\mathscr{O}_{X,\eta}$ is DVR, there is only one minimal prime, so there is only one irreducible components of X which containing p, say corresponding generic point is ξ. Since U is dense in X, it must contain all generic points of X, so $\xi \in U$. Notice that ξ is also the generic point of C, we can restrict $\pi|_U : U \to Y$ to $C \setminus \{\eta\}$, then we obtain a morphism $\pi|_{C\setminus\{\eta\}} : C \setminus \{\eta\} \to Y$. By the Curve-to-Projective Extension Theorem 16.3.1, we obtain a morphism $\pi_C : C \to Y$.

Now, we want to extend $C \to Y$ to $U \to Y$ where $U \subseteq X$ is an open neighborhood of η. Let $\mathrm{Spec}\, A \subseteq X$ be an affine open which containing η, then $\mathscr{O}_{X,\eta} = A_\eta$. Let $W := \mathrm{Spec}\, B \subseteq Y$ be an affine open neighborhood of $\pi_C(\eta)$. Since $\pi_C^{-1}(W)$ is open, $\pi_C^{-1}(W) = C$, so we obtain a morphism $\pi_C : C \to W$, and the corresponding ring map is $\pi_C^\sharp : B \to A_\eta$. Since Y is projective k-scheme, we may assume $B = k[x_1, \dots, x_n]/I$. Since $k[x_1, \dots, x_n]$ is Noetherian, I is finitely generated, we may assume $I = (f_1, \dots, f_m)$. Let a_i be the image of x_i in A_η, then a_i can be written as $\frac{b_i}{s_i}$ where $s_i \notin \eta$. Let $s = s_1 \cdots s_n$, then $\pi_C^\sharp$ factors through A_s (this is simple, we omit the check), that is,

$$\begin{tikzcd} B \arrow[rr, "\pi_C^\sharp"] \arrow[dr] & & A_\eta \\ & A_s. \arrow[ur, "i"'] & \end{tikzcd}$$

Therefore, we obtain a ring map $B \to A_s$, then we get a morphism $\pi_V : V \to W$ and $\pi_V|_C = \pi_C$ where $V = D(s)$.

Next, we want to check that $\pi_V|_{U\cap V} = \pi|_U$. Recall Proposition 12.3.1, the equalizer $\mathrm{Eq}(\pi_V|_{U\cap V}, \pi|_{U\cap V}) \hookrightarrow U \cap V$ is a closed embedding. Shrink V such that $V \cap U$ is an irreducible open subset containing generic point ξ. Hence, $V \cap U = \mathrm{Eq}(\pi_V|_{U\cap V}, \pi|_{U\cap V})$, i.e., $\pi_V|_{U\cap V} = \pi|_{U\cap V}$. Glue π_V and π, we obtain a morphism (also denote π) $\pi : U \cup V \to Y$, it is also a representation of $\pi : X \dashrightarrow Y$. Moreover, $\eta \in U \cap V$, so $\eta \in \mathrm{dom}(\pi)$, we win! $\square$

16.3.2 ⋆ Extended example: Tangents to regular plane curves are limits of secants.

We now discuss an extended example that may give insight into many different things. When Learning calculus, we're sometimes taught that tangent lines to plane curves are limits of secants. We can now see this in terms of the Curve-to-Projective Extension Theorem 16.3.1, and deltas and epsilons are turned into easy algebra. This discussion also foreshadows the importance of the diagonal morphism in understanding differentials.

Let k be a field, which you should initially take to be algebraically closed. (You can later think about how this makes sense over arbitrary fields.) Suppose $C \subseteq \mathbb{A}^2_k$ is a plane curve, given by the equation $f(x, y) = 0$. Fix a point $p = (x_0, y_0)$ of C. Given a point $q = (x_1, y_1)$ of C, with $x_1 \neq x_0$, the slope of the line joining them is $m = \frac{y_1-y_0}{x_1-x_0}$, and the equation of the line joining them is

$$(y - y_0) = \frac{y_1 - y_0}{x_1 - x_0}(x - x_0).$$

More generally (even if $x_1 = x_0$, but still with $y_1 \neq y_0$), the line joining has "slope" $[x_1-x_0 : y_1-y_0] \in \mathbb{P}^1_k$, and the equation of the line joining them is $(x_1 - x_0)(y - y_0) = (y_1 - y_0)(x - x_0)$. We thus have a map

$$\begin{aligned} C \setminus \{p\} &\longrightarrow \mathbb{P}^1_k \\ q = (x_1, y_1) &\longmapsto [x_1 - x_0 : y_1 - y_0] \end{aligned} \tag{16.5}$$

sending q to the "slope" of the secant line $\overline{pq}$. The Curve-to-Projective Extension Theorem 16.3.1 states that this will extend over p, where p is a smooth point of C. Let's work out the algebra. keep in mind that now $x_0, y_0 \in k$, as p is a fixed point.

Define $F(x_1, y_1) := f(x_1, y_1) - f(x_0, y_0) \in k[x_1, y_1]$.

✍ **Exercise 16.4** Show that $F \in (x_1 - x_0, y_1 - y_0)$, where $(x_1 - x_0, y_1 - y_0) \subseteq k[x_1, y_1]$ is an ideal.

Proof Clearly, since $F(x_0, y_0) = 0$. $\square$

Hence, we can write $F = g(x_1, y_1)(x_1 - x_0) + h(x_1, y_1)(y_1 - y_0)$ for some (nonunique) choice of $g, h \in k[x_1, y_1]$. Choose such a g and h.

✍ **Exercise 16.5** Show that if $p = (x_0, y_0)$ is a smooth point of C with nonvertical tangent vector, then $h(x_0, y_0) \neq 0$.

Proof Let $R = k[x_1, y_1]$, $A = R/(f(x_1, y_1))$, and $C = \operatorname{Spec} A$ (for simply, A is a local ring). Denote the maximal ideal $\mathfrak{n} = (x_1 - x_0, y_1 - y_0)$. Then the corresponding maximal ideal of $\mathfrak{n}$ in A is $\mathfrak{m} = \mathfrak{n}/(f)$. So the Zariski cotangent space is $T^\vee_{C,p} = \mathfrak{m}/\mathfrak{m}^2 \cong \mathfrak{n}/(\mathfrak{n}^2 + (f))$.

The image of F in $\mathfrak{n}/\mathfrak{n}^2$ is

$$g(p)\overline{(x_1 - x_0)} + h(p)\overline{(y_1 - y_0)}.$$

Denote $X = x_1 - x_0$ and $Y = y_1 - y_0$. Notice that $f(x_1, y_1) = F(x_1, y_1)$, we know that

$$\mathfrak{m}_p/\mathfrak{m}_p^2 \cong \mathfrak{n}/(\mathfrak{n}^2 + (f)) \cong \frac{k\overline{X} \oplus k\overline{Y}}{(g(p)\overline{X} + h(p)\overline{Y})}.$$

We now calculate $T_{C,p}$. Let $v \in T_{C,p}$. Suppose $v(\overline{X}) = a$ and $v(\overline{Y}) = b$, then

$$v(g(p)\overline{X} + h(p)\overline{Y}) = g(p)a + h(p)b = 0.$$

Hence, $T_{C,p} = \{(a,b) \in k^2 : g(p)a + h(p)b = 0\}$.

Since p is a smooth point,

$$\dim C = \dim A = \dim \mathfrak{m}/\mathfrak{m}^2 = \dim T_{C,p}^{\vee} = 1,$$

we know that $(g(p), h(p)) \neq (0,0)$. Suppose $h(p) = 0$, then $g(p) \neq 0$. Therefore, the tangent space at p is

$$T_{C,p} = \{(a,b) \in k^2 : g(p)a = 0\} = \{(0,b) : b \in k\},$$

which is a vertical direction. This contradicts the assumption that the tangent direction is nonvertical. Hence, $h(x_0, y_0) \neq 0$. □

✍ **Exercise 16.6** Show that if $p = (x_0, y_0)$ is a smooth point of C, then the slope of the tangent line at p is $[-h(x_0,y_0) : g(x_0,y_0)] \in \mathbb{P}^1_k$, and the equation of the tangent line is

$$g(x_0,y_0)(x - x_0) + h(x_0,y_0)(y - y_0) = 0.$$

Proof Since p is smooth, by the proof in Exercise 16.5, we obtain that

$$T_{C,p} = \{(a,b) \in k^2 : g(p)a + h(p)b = 0\}.$$

Hence, the equation of tangent line at p is

$$g(p)(x - x_0) + h(p)(y - y_0) = 0,$$

and the slop of the tangent line is $[-h(p) : g(p)] \in \mathbb{P}^1_k$. □

16.4 Hard but important: Line Bundles and Weil Divisors

The notion of Weil divisors gives a great way of understanding and classifying line bundles. Before we get started, you should be warned: this is one of those topics in algebraic geometry that is hard to digest — learning it changes the way in which you think about line bundles. But once you become comfortable with the imperfect dictionary to divisors, it becomes second nature.

In this sections, we consider only ***Noetherian normal schemes***. We do this because we will use finite decomposition into irreducible components (Proposition 6.3.4), properties of DVR (§14.5), and Algebraic Hartogs's Lemma 14.5.8. Some of what we discuss will apply in more general circumstances, and the expert is invited to consider generalization by judiciously weakening hypotheses in various statements.

16.4.1 Weil divisor

Definition 16.4.1 (Weil divisor)

Let X be a Noetherian normal scheme. Define a ***Weil divisor*** *on X as a formal $\mathbb{Z}$-linear combination of codimension 1 irreducible closed subsets of X. In other words, a Weil divisor on X is defined to be an object of the form*

$$\sum_{Y \subseteq X, \operatorname{codim}_X Y = 1} n_Y [Y],$$

where the n_Y are integers, all but a finite number of which are zero. Weil divisors obviously form an abelian group, denoted $\operatorname{Weil} X$.

Example 16.6 For example, if X is a curve, the Weil divisors are linear combinations of closed points.

Definition 16.4.2 (Irreducible divisor, effective, support, and restriction)

(1) *We say that* $[Y]$ *is an* ***irreducible (Weil) divisor (or prime divisor)***.

(2) *A Weil divisor* $D = \sum n_Y[Y]$ *is said to be* ***effective*** *if* $n_Y \geq 0$ *for all* Y. *In this case, we say* $D \geq 0$, *and by* $D_1 \geq D_2$ *we mean* $D_1 - D_2 \geq 0$.

(3) *The* ***support*** *of a Weil divisor* D, *denoted* $\operatorname{Supp} D$, *is the subset* $\bigcup_{n_Y \neq 0} Y$.

(4) *If* $U \subseteq X$ *is an open set, we define the* ***restriction map*** $\operatorname{Weil} X \to \operatorname{Weil} U$ *by* $\sum n_Y[Y] \mapsto \sum_{Y \cap U \neq \varnothing} n_Y[Y \cap U]$.

Proposition 16.4.1

The prime divisors on $\mathbb{P}^n_k$ *correspond to irreducible homogeneous polynomials in* $k[x_0, \dots, x_n]$, *up to multiplication by nonzero scalars* $k^\times$.

Proof Let $S = k[x_0, \dots, x_n]$. A prime divisor $D \subseteq \mathbb{P}^n_k$ corresponds to a homogeneous prime ideal $\mathfrak{p} \subseteq S$ of height 1. Since S is UFD, by Proposition 13.3.7, all height 1 homogeneous prime ideals are precisely principal ideals generated by irreducible homogeneous polynomials, i.e., $\mathfrak{p} = (F)$ for some irreducible homogeneous $F \in S$. Hence, $D = V_+(F)$. Conversely, if $F \in S$ is an irreducible homogeneous polynomial of positive degree, then (F) is homogeneous prime ideal. By Krull's Principal Ideal Theorem 13.3.2, (F) has height 1, hence $V_+(F)$ is an irreducible divisor on $\mathbb{P}^n_k$. Finally, $V_+(F) = V_+(G)$ for irreducible homogeneous F, G iff $(F) = (G)$, which holds iff $G = cF$, $c \in k^\times$. □

Definition 16.4.3 (Rational section)

Let X *be a normal Noetherian scheme, and let* $\mathscr{E}$ *be a locally free* $\mathscr{O}_X$*-module of finite rank. Let* $\mathscr{K}_X$ *be the sheaf of meromorphic function on* X *(Definition 15.2.4). A* ***rational section*** *of* $\mathscr{E}$ *is a global section of* $\mathscr{E} \otimes_{\mathscr{O}_X} \mathscr{K}_X$.

Equivalently, a rational section of $\mathscr{E}$ *is an equivalence class of pairs* (U, s_U), *where* U *is a dense open subset containing the generic points of all irreducible components of* X, *and* $s_U \in \Gamma(U, \mathscr{E}|_U)$. *Two pairs* (U, s_u) *and* (V, s_V) *are equivalent if there exists a dense open subset* $W \subseteq U \cap V$ *containing the generic points of all irreducible components of* X, *such that* $s_U|_W = s_V|_W$.

Suppose that $\mathscr{L}$ is an invertible sheaf, and s a rational section not vanishing everywhere on any irreducible component of X. (Rational sections are given by a section over a dense open subset of X, with the obvious equivalence.)

We can make sense if the **valuation of s along an irreducible Weil divisor** Y (denoted $\operatorname{val}_Y(s)$) as follows. Take any open set U containing the generic point of Y where $\mathscr{L}$ is trivializable, along with any trivialization over U; under this trivialization, s is a nonzero rational function on U, which thus has a valuation (Definition 14.5.7). Any two such trivializations differ by an invertible function (transition functions are invertible), so this valuation is well-defined. Note that $\operatorname{val}_Y(s) = 0$ for all but finitely many Y, by Proposition 14.5.2.

Thus s determines a Weil divisor

$$\operatorname{div}(s) := \sum_Y \operatorname{val}_Y(s)[Y].$$

The summation runs over all irreducible divisors Y of X. We call $\operatorname{div}(s)$ the **divisor of zeros and poles** of the rational sections s (cf. Definition 14.5.4).

16.4.2 The important group of "line bundles with rational sections".

Now consider the set $\{(\mathscr{L}, s)\}$ of pairs of line bundles $\mathscr{L}$ with rational sections s of $\mathscr{L}$, not the zero section on any irreducible component of X, up to isomorphism. (An isomorphism $(\mathscr{L}, s) \cong (\mathscr{L}', s')$ means an isomorphism of line bundles under which s is sent to s'.) This set (after taking quotient by isomorphism) forms an abelian group under tensor product $\otimes$, with identity $(\mathscr{O}_X, 1)$. (The inverse of $(\mathscr{L}, s)$ is $(\mathscr{L}^\vee, s^{-1})$.)

It is important to notice that if t is an invertible function on X, then multiplication by t gives an isomorphism

$$(\mathscr{L}, s) \overset{\sim}{\longleftrightarrow} (\mathscr{L}, st).$$

Similarly, $(\mathscr{L}, s)/(\mathscr{L}, u) = (\mathscr{O}, s/u)$. Here s/u is a rational function.

The map div yields a group homomorphism

$$\operatorname{div} : \{(\mathscr{L}, s)\}/\text{isomorphism} \longrightarrow \operatorname{Weil} X. \tag{16.6}$$

Exercise 16.7 Let k be an algebraically closed field.

(a) (divisors of rational functions) Verify that on $\mathbb{A}^1_k$, $\operatorname{div}(x^3/(x+1)) = 3[(x)] - [(x+1)]$ ("=3[0]-[-1]").

(b) (divisor of rational sections of a nontrivial invertible sheaf) On $\mathbb{P}^1_k$, there is a rational section of $\mathscr{O}(1)$ "corresponding to" $x^2/(x+y)$. Figure out what this means, and calculate $\operatorname{div}(x^2/(x+y))$.

Proof ***For (a).*** Since k is algebraically closed field, codimension one irreducible closed subsets of $\mathbb{A}^1_k$ are closed points. In $\mathscr{O}_{\mathbb{A}^1_k,[(x)]} = k[x]_{(x)}$, $\frac{x^3}{x+1} = \frac{1}{x+1} \cdot x^3$, where x is uniformizer and $\frac{1}{x+1}$ is a unit in $\mathscr{O}_{\mathbb{A}^1_k,[(x)]}$. So, $\operatorname{val}_{[(x)]}(\frac{x^3}{x+1}) = 3$. Similarly, we have $\operatorname{val}_{[(x+1)]}(\frac{x^3}{x+1}) = -1$. Hence,

$$\operatorname{div}(x^3/(x+1)) = 3[(x)] - [(x+1)].$$

For (b). Recall Definition 16.4.3, $\frac{x^2}{x+y} = x \cdot \frac{x}{x+y}$ can be regarded as a global section of $\mathscr{O}(1) \otimes \mathscr{K}_{\mathbb{P}^1_k}$. So, $\frac{x^2}{x+y}$ is a rational section of $\mathscr{O}(1)$.

We next calculate $\operatorname{div}(\frac{x^2}{x+y})$. On $D_+(y)$, set $t = \frac{x}{y}$, then

$$\frac{x^2}{x+y} = \frac{t^2}{t+1} \cdot y.$$

Hence, $\operatorname{val}_{[0:1]}(\frac{x^2}{x+y}) = 2$ and $\operatorname{val}_{[1:-1]}(\frac{x^2}{x+y}) = -1$. On $D_+(x)$, set $v = \frac{y}{x}$, then

$$\frac{x^2}{x+y} = \frac{1}{1+v} \cdot x.$$

Hence, $\operatorname{val}_{[1:0]}(\frac{x^2}{x+y}) = 0$. Therefore,

$$\operatorname{div}(\tfrac{x^2}{x+y}) = \operatorname{val}_{[0:1]}(\tfrac{x^2}{x+y})[0:1] + \operatorname{val}_{[1:-1]}(\tfrac{x^2}{x+y})[1:-1] = 2[0:1] - [1:-1].$$

□

The homomorphism (16.6) will be the key to determining all the line bundles on many X. (Note that any invertible sheaf will have such a rational section. For each irreducible component, take a nonempty open set not meeting any other irreducible component; then shrink it so that $\mathscr{L}$ is trivial; choose a trivialization; then take the union of all these open sets, and choose the section on this union corresponding to 1 under the trivialization.) We will see that in reasonable situation, this map div will be injective, and often an isomorphism. Thus by forgetting the rational section (i.e., taking an appropriate quotient), we will have described the Picard group of all line bundles. Let's put this strategy into action.

Proposition 16.4.2

If X is normal and Noetherian then the map div *(16.6) is injective.*

Proof Suppose $\operatorname{div}(\mathscr{L}, s) = 0$. Then s has no poles. By Proposition 15.2.6, s is a regular section. We now show that the morphism $\times s : \mathscr{O}_X \to \mathscr{L}$ is in fact an isomorphism; this will prove the proposition, as it will give an isomorphism $(\mathscr{O}_X, 1) \xleftrightarrow{\sim} (\mathscr{L}, s)$.

It suffices to show that $\times s$ is an isomorphism on an open neighborhoods of $\mathscr{L}$ (as $\mathscr{L}$ is locally trivial). Choose an isomorphism $i : \mathscr{L}|_U \xrightarrow{\sim} \mathscr{O}_U$. Composing $\times s$ with i yields a map $\times s' : \mathscr{O}_U \to \mathscr{O}_U$ that is multiplication by a rational function $s' = i(s)$ that has no zeros and no poles (since $\operatorname{div}(s) = 0$). The rational function s' is regular because it has no poles (Corollary 14.5.2), and $1/s'$ is regular for the same reason. Thus s' is an invertible function on U, so $\times s'$ is an isomorphism. Hence, $\times s$ is an isomorphism over U. □

Motivated by this, we try to find an inverse to div, or at least to determine the image of div.

16.4.3 Definition: $\mathscr{O}_X(D)$

Assume now that X is normal — this will be a standing assumption for the rest of this section.

Assume also that X is irreducible (purely to avoid making (16.7) look uglier — but feel free to relax this; see Definition 16.4.5). In case it helps: recall that a normal scheme is disjoint union of irreducible normal schemes (Proposition 6.4.2) — so this assumption truly is harmless.

Definition 16.4.4 (Sheaf $\mathscr{O}_X(D)$)

*Let X be an irreducible Noetherian normal scheme (i.e., Noetherian integral normal scheme). Suppose D is a Weil divisor. Define the **sheaf** $\mathscr{O}_X(D)$ by*

$$\Gamma(U, \mathscr{O}_X(D)) := \{t \in K(X)^\times : \operatorname{div}|_U t + D|_U \geq 0\} \cup \{0\}. \tag{16.7}$$

Here $D|_U$ is the restriction of D to U (defined in Definition 16.4.2), and $\operatorname{div}|_U t$ means the divisor of t considered as a rational function on U (i.e., consider just the irreducible divisors of U). The subscript X in $\mathscr{O}_X(D)$ is omitted when it is clear from context.

The sections of $\mathscr{O}_X(D)$ over U are the rational functions on U that have poles and zeros "constrained by D": a positive coefficient in D allows a pole of that order; a negative coefficient demands a zero of that order. Away from the support of D, this is (isomorphic to) the structure sheaf (by Algebraic Hartogs's Lemma 14.5.8).

Remark 16.10 ***Important remark.*** It will be crucial to note that $\mathscr{O}_X(D)$ comes along with a canonical nonzero "rational section" corresponding to $1 \in K(X)^\times$. (It is a rational section in the sense that it is a section over a dense open set, namely the complement of $\operatorname{Supp} D$.)

Definition 16.4.5 (Generalize sheaf $\mathscr{O}_X(D)$)

*Let X be a normal Noetherian scheme (not necessarily irreducible). Suppose D is a Weil divisor. Define the **sheaf** $\mathscr{O}_X(D)$ by*

$$\Gamma(U, \mathscr{O}_X(D)) = \{t \in \Gamma(U, \mathscr{K}_X) : \operatorname{div}|_U t + D|_U \geq 0\} \cup \{0\},$$

where $\operatorname{div}|_U t$ is computed componentwise on U.

Lemma 16.4.1

Let X be a Noetherian integral normal scheme. Suppose D is a Weil divisor, then $\mathscr{O}_X(D)$ is a coherent sheaf.

Proof We may assume that $X = \operatorname{Spec} A$ is affine where A is Noetherian normal domain. We first show that $\mathscr{O}_X(D)$ is quasicoherent by showing that for any $g \in A$, there is an equivalence

$$\Gamma(X, \mathscr{O}_X(D))_g = \Gamma(D(g), \mathscr{O}_X(D)).$$

If $t \in \Gamma(X, \mathscr{O}_X(D))$, then $\operatorname{div}(t) + D \geq 0$. Since g is invertible over $D(g)$, we have $\operatorname{div}|_{D(g)}(t/g^m) = \operatorname{div}|_{D(g)}(t)$ for any m, so $\Gamma(X, \mathscr{O}_X(D))_g \subseteq \Gamma(D(g), \mathscr{O}_X(D))$.

Conversely, take any $t \in \Gamma(D(g), \mathscr{O}_X(D))$ such that $\operatorname{div}|_{D(g)}(t) + D|_{D(g)} \geq 0$. Then the negative part of the divisor $\operatorname{div}(t) + D$ must be supported in the closed set $V(g) \subseteq \operatorname{Spec} A$. Since X is Noetherian, $V(g)$ is Noetherian, and thus has finitely many components. Hence, there exists an $m \geq 0$ such that

$$\operatorname{div}(g^m t) + D \geq 0.$$

Then $g^m t \in \Gamma(X, \mathscr{O}_X(D))$ which implies that $t \in \Gamma(X, \mathscr{O}_X(D))_g$. It follows that $\Gamma(X, \mathscr{O}_X(D))_g \subseteq \Gamma(D(g), \mathscr{O}_X(D))$. Hence, $\mathscr{O}_X(D)$ is a quasicoherent sheaf.

To prove that $\mathscr{O}_X(D)$ is coherent, it suffices to show that $\Gamma(X, \mathscr{O}_X(D))$ is a finitely generated A-module. Write $D = \sum n_i Z_i$ where Z_i is prime divisor, and let $g \in \Gamma(X, \mathscr{O}_X) = A$ be a nonzero element which vanishes on all the Z_i, i.e., $\operatorname{val}_{Z_i}(g) > 0$. Then we may choose an m such that $\operatorname{val}_{Z_i}(g^m) \geq n_i$ for all i, i.e., $\operatorname{div}(g^m) \geq D$. This implies that for any $f \in \Gamma(X, \mathscr{O}_X(D))$ we have $\operatorname{div}(g^m f) = \operatorname{div}(g^m) + \operatorname{div}(f) \geq 0$, and hence $g^m f \in \Gamma(X, \mathscr{O}_X) = A$. Hence, $g^m \Gamma(X, \mathscr{O}_X(D))$ is an ideal of A. As we assume that A is Noetherian, this ideal is finitely generated, and we deduce that $\Gamma(X, \mathscr{O}_X(D))$ is finitely generated as well. □

Corollary 16.4.1

Let X be a Noetherian normal scheme. Suppose D is a Weil divisor, then $\mathscr{O}_X(D)$ is a coherent sheaf.

Proof By Proposition 6.4.2, X is the disjoint union of Noetherian integral normal scheme, applying Lemma 16.4.1 to each components, we conclude consequence. □

In good situations, $\mathscr{O}_X(D)$ is an invertible sheaf. For example:

Example 16.7 Let $X = \mathbb{A}^1_k$ and Weil divisor $D = -2[(x)] + [(x-1)] + [(x-2)]$ (often written $-2[0]+[1]+[2]$ for convenience.) Consider $\mathscr{O}_X(D)$. Then $\frac{3x^2}{x-1}$ is a global section; it has the required two zeros at $x = 0$ (and even one to spare), and takes advantage of the allowed pole at $x = 1$, and doesn't have a pole at $x = 2$, even though one is allowed. (Unimportant aside: The statement remains true in characteristic 2, although the explanation requires editing.)

We next show that $\mathscr{O}(D)$ is indeed an invertible sheaf. Let $h = \frac{(x-1)(x-2)}{x^2}$, then $D = \operatorname{div} h$. Hence,

$$\begin{aligned}\Gamma(X, \mathscr{O}(D)) &= \{f \in K(X)^\times : \operatorname{div}(fh) \geq 0\} \\ &= \{f \in K(X)^\times : f \in \tfrac{1}{h} \cdot \Gamma(X, \mathscr{O}_X)\} \\ &= \frac{1}{h} \cdot \Gamma(X, \mathscr{O}_X),\end{aligned}$$

which implies that $\mathscr{O}(D)$ is an invertible sheaf.

✍ **Exercise 16.8**

(a) Show that any global section of $\mathscr{O}_{\mathbb{A}^1_k}(-2[(x)] + [(x-1)] + [(x-2)])$ is a $k[x]$-multiple of $\frac{x^2}{(x-1)(x-2)}$.

(b) Extend the argument of (a) to give an isomorphism

$$\mathscr{O}_{\mathbb{A}^1_k}(-2[(x)] + [(x-1)] + [(x-2)]) \xleftrightarrow{\sim} \mathscr{O}_{\mathbb{A}^1_k}.$$

Solution *See Example 16.7 .*

The next exercise is seemingly small, but is absolutely crucial to lessen confusion as the complications add up in the next few page.

✍ **Exercise 16.9** We continuous the example of Exercise 16.8, letting $k = \mathbb{C}$ for concreteness. Let $X = \mathbb{A}^1_{\mathbb{C}}$, and $D = -2[(x)] + [(x-1)] + [(x-2)]$.

(a) Consider the section of $\mathscr{O}_X(D)$ corresponding to $\frac{x^2(x-3)}{(x-1)(x-2)} \in K(X)$ (under the correspondence of

(16.7)). This section vanishes at precisely one point. What point is it?

(b) Consider the section of $\mathscr{O}_X(D)$ corresponding to $x^2(x-1) \in K(X)$. This section vanishes at precisely two pints. What are those points, and to what orders does it vanish at those points?

(c) Consider the rational section of $\mathscr{O}_X(D)$ corresponding to $1 \in K(X)$ (never forget, via the correspondence of (16.7)). Where are its zeros and poles? What are their orders? What is the most common wrong answer to this problem?

Solution ***For (a).*** *Note that* $\mathscr{O}_X(D) \cong \mathscr{O}_X \cdot \frac{1}{h}$ *where* $h = \frac{(x-1)(x-2)}{x^2}$, *section* $\frac{x^2(x-3)}{(x-1)(x-2)} = (x-3)\cdot\frac{1}{h}$ *vanishes at point* $[(x-3)]$.

For (b). $x^2(x-1) = (x-1)^2(x-2)\cdot\frac{1}{h}$. *So, the zeros are* $[(x-1)]$ *and* $[(x-2)]$. *Moreover,* $\mathrm{val}_{[(x-1)]}\, x^2(x-1) = 2$ *and* $\mathrm{val}_{[(x-2)]}\, x^2(x-1) = 1$.

For (c). $1 = h\cdot\frac{1}{h}$. *So, the zeros are* $[(x-1)]$ *(order* 1*) and* $[(x-2)]$ *(order* 1*); the pole is* $[(x)]$ *(order* 2*).*

We next show that in good circumstances, $\mathscr{O}_X(D)$ is an invertible sheaf. (In fact the $\mathscr{O}_X(D)$ construction can occasionally be useful even if $\mathscr{O}_X(D)$ is not an invertible sheaf, but this won't concern us here. An example of an $\mathscr{O}_X(D)$ that is not an invertible sheaf is given in **??**)

Proposition 16.4.3

Let X be a Noetherian (integral) normal scheme. Suppose $\mathscr{L}$ is an invertible sheaf, and s is a nonzero rational section of $\mathscr{L}$.

(a) There is an isomorphism $\mathscr{O}_X(\operatorname{div} s) \xleftrightarrow{\sim} \mathscr{L}$.

(b) Let σ be the map from $K(X)$ to the rational sections of $\mathscr{L}$, where $\sigma(t)$ is the rational section of $\mathscr{O}_X(\operatorname{div} s) \cong \mathscr{L}$ defined via (16.7). Then σ sends the canonical section "1" of $\mathscr{O}_X(D)$ to s. More concisely: $\sigma(1) = s$.

Proof ***For (a).*** Since $\mathscr{L}$ is an invertible sheaf, consider the trivialization open cover $\mathfrak{U} = \{U \hookrightarrow X\}$ where $\mathscr{L}|_U \cong \mathscr{O}_U$. Assume that $\mathscr{L}|_U = \mathscr{O}_U \cdot e_u$, since $s|_U$ is a rational section on U, we may write

$$s|_U = f_U \cdot e_u$$

where $f_U \in K(X)^\times$. Because for every regular codimension 1 point $p \in U$ (Definition 14.5.6), we have $\mathscr{O}_{U,p} = \mathscr{O}_{X,p}$, so the valuation val_p used to define the coefficient of $\operatorname{div}(s)$ is the same as the one used to define the coefficient of $\operatorname{div}(s|_U)$; hence the two divisors have the same coefficients at all regular codimension 1 points of U, and therefore

$$(\operatorname{div} s)|_U = \operatorname{div}|_U(s|_U).$$

Define $\varphi_U : \Gamma(U, \mathscr{O}(\operatorname{div} s)) \to \mathscr{L}(U) = \mathscr{O}_U(U)\cdot e_u$ by setting

$$t \longmapsto t f_U \cdot e_u.$$

We need to check that this morphism is well-defined. Since $t \in \Gamma(U, \mathscr{O}(\operatorname{div} s))$, we have

$$\operatorname{div}|_U(t) + (\operatorname{div} s)|_U = \operatorname{div}|_U(t) + \operatorname{div}|_U s|_U = \operatorname{div}|_U(t) + \operatorname{div}|_U f_U = \operatorname{div}|_U(t f_U) \geq 0,$$

which implies that $t f_U \in \Gamma(U, \mathscr{O}_X)$ (Corollary 14.5.2, since X is Noetherian integral normal scheme). It follows that φ_U is well-defined. By the construction, φ_U is also an isomorphism (with the obvious inverse map, division by $s|_U$).

We next show that $\mathfrak{U}$ is a topological basis of X. For any $V \subseteq X$ open and $x \in V$, we want to show that there exists $U \in \mathfrak{U}$ such that $x \in U \subseteq V$. Since $\mathscr{L}$ is invertible, there exists $W \ni x$ such that $\mathscr{L}|_W \cong \mathscr{O}_W$. Let $U = V \cap W$, then $\mathscr{L}|_U = (\mathscr{L}|_W)|_U \cong (\mathscr{O}_W)|_U = \mathscr{O}_U$, so $U \in \mathfrak{U}$. It follows that $\mathfrak{U}$ is a basis of X.

Finally, we want to $\{\varphi_U\}_{U\in\mathfrak{U}}$ give a sheaf isomorphism φ. By the construction of φ_U, for each $U \supseteq V$ in

$\mathfrak{U}$, we have

$$\operatorname{res}^{\mathscr{L}}_{U,V} \circ \varphi_U = \varphi_V \circ \operatorname{res}^{\mathscr{O}(\operatorname{div}(s))}_{U,V}.$$

Hence, $\{\varphi_U\}_{U\in\mathfrak{U}}$ give a sheaf isomorphism φ, since each φ_U is isomorphism, we know that φ is an isomorphism.

For (b). Consider $1 \in K(X)^\times$, suppose 1 define on $V \subseteq X$, then $1 = (V, 1)$. By the isomorphism in part (a), we have $\varphi_V(1) = 1 \cdot s|_V = s|_V$. So, as rational section $(V, 1) \mapsto (V, s|_V)$, i.e., $\sigma(1) = s$. □

In conclusion, we can identify which Weil divisors D are in the image of div (16.6): they are the D for which $\mathscr{O}_X(D)$ is a line bundle, and then we can construct the image — $(\mathscr{O}(D), \sigma(1))$ the unique $(\mathscr{L}, s)$ (up to isomorphism) such that $\operatorname{div}(\mathscr{L}, s) = D$.

Proposition 16.4.4 (The example of §16.1)

Suppose $X = \mathbb{P}^n_k$, $\mathscr{L} = \mathscr{O}(1)$, s is the section of $\mathscr{O}(1)$ corresponding to x_0, and $D = \operatorname{div}(s)$. Then $\mathscr{O}(mD) \cong \mathscr{O}(m)$, and the canonical rational section of $\mathscr{O}(mD)$ is precisely s^m.

Remark 16.11 For this reason, $\mathscr{O}(1)$ is sometimes called the **hyperplane class** in $\operatorname{Pic} X$. (Of cause, x_0 can be replaced by any linear form.)

Proof Clearly, $mD = m\operatorname{div}(s) = \operatorname{div}(s^m)$ and s^m is a regular (if $m \geq 0$) or rational (if $m < 0$) section of $\mathscr{O}(m)$. By Proposition 16.4.3, we obtain that $\mathscr{O}(m) \cong \mathscr{O}_X(\operatorname{div}(s^m)) = \mathscr{O}_X(mD)$, and the canonical rational section of $\mathscr{O}(mD)$ is s^m. □

Proposition 16.4.5 (The Picard group of projective space)

$\operatorname{Pic}\mathbb{P}^n_k \cong \mathbb{Z}$.

Proof Let $F \in k[x_0, \ldots, x_n]$ be an irreducible homogeneous polynomial, say $d = \deg F$. We first show that $\mathscr{O}(V(F)) \cong \mathscr{O}(d)$. F is a section in $\mathscr{O}(d)$, by Proposition 16.4.3, we obtain that $\mathscr{O}(\operatorname{div} F) \cong \mathscr{O}(d)$. Note that $\operatorname{div} F = \operatorname{val}_{V(F)} F[V(F)] = [V(F)]$ (divisor of rational section), we have $\mathscr{O}(d) \cong \mathscr{O}(V(F))$.

Now, let $D \in \operatorname{Weil}\mathbb{P}^n_k$, recall Proposition 16.4.1, we may assume $D = \sum_i a_i[V(F_i)]$ where $F_i \in k[x_0, \ldots, x_n]$ is irreducible homogeneous polynomial. Denote $d_i = \deg F$. Then

$$\begin{aligned}\mathscr{O}(D) &= \mathscr{O}\left(\sum_i a_i[V(F_i)]\right) \cong \bigotimes_i \mathscr{O}(a_i[V(F_i)]) \\ &\cong \bigotimes_i \mathscr{O}(\deg(F_i))^{\otimes a_i} \\ &\cong \mathscr{O}\left(\sum_i a_i \deg(F_i)\right).\end{aligned}$$

Denote $\deg D := \sum_i a_i \deg(F_i)$. Hence, each Weil divisor D on $\mathbb{P}^n_k$ gives an invertible sheaf $\mathscr{O}(\deg D)$ on $\mathbb{P}^n_k$.

Next, let $\mathscr{L}$ be any invertible sheaf on $\mathbb{P}^n_k$, let s be a rational section of $\mathscr{L}$, by Proposition 16.4.3, $\mathscr{L} \cong \mathscr{O}(\operatorname{div} s)$, by above discussion, $\mathscr{L} \cong \mathscr{O}(\deg(\operatorname{div} s))$. Hence, every invertible sheaf on $\mathbb{P}^n_k$ gives a $\mathscr{O}(d)$ for some d. Moreover, d is independent of the choice of the rational section. Let s and t be two rational sections of $\mathscr{L}$. Recall the definition of rational section (Definition 16.4.3), rational section of $\mathscr{L}$ consists a 1-dimensional $K(X)$-vector space, so there exists a rational function $f \in K(X)^\times$ such that $s = ft$. Hence,

$$\deg\operatorname{div}(s) = \deg\operatorname{div}(f) + \deg\operatorname{div}(t) = \deg\operatorname{div}(t),$$

which implies different rational section give the same $\mathscr{O}(d)$.

Consequently, we conclude that

$$\begin{aligned} \operatorname{Pic}\mathbb{P}^n_k &\xleftrightarrow{\sim} \mathbb{Z} \\ \mathscr{L} \longmapsto \mathscr{O}(d) &\longmapsto d \\ \mathscr{O}(d) &\longleftarrow\!\shortmid d \end{aligned}$$

which implies that $\operatorname{Pic}\mathbb{P}^n_k \cong \mathbb{Z}$. □

Definition 16.4.6 ((Locally) Principal)

Let X be a Noetherian integral normal scheme.

*(a) If D is a Weil divisor on X such that $D = \operatorname{div} f$ for some rational function f, we say that D is **principal**. Principal divisors clearly form a subgroup of $\operatorname{Weil} X$; denote this group of principal divisors $\operatorname{Prin} X$. Note that div induces a group homomorphism $K(X)^\times \to \operatorname{Prin} X$.*

*(b) If X can be covered with open sets U_i such that on U_i, D is principal, we say that D is **locally principal**. Locally principal divisors form a subgroup of $\operatorname{Weil} X$, which we denoted $\operatorname{LocPrin} X$.*

Remark 16.12

(1) **Caution:** We used the phrase "locally principal" in a slightly different way, when referring to closed subschemes.

(2) The notion of locally principal is not standard, and we only use it in (16.8).

Note ***Important observation: "$\mathscr{O}$(principal) $\cong \mathscr{O}$, $\mathscr{O}$(locally principal) $=$ line bundle".***

As a consequence of Proposition 16.4.3 (a) (taking $\mathscr{L} = \mathscr{O}$), if D is principal (and X is normal, our standing hypothesis), then (i) $\mathscr{O}(D) \cong \mathscr{O}$. (Diagram (16.8) will imply that the converse holds: if $\mathscr{O}(D) \cong D$, then D is principal.) Thus (ii) if D is locally principal, $\mathscr{O}_X(D)$ is locally isomorphic to $\mathscr{O}_X$ — $\mathscr{O}_X(D)$ is an invertible sheaf.

Lemma

Let X be a Noetherian (integral) normal scheme. For a nonzero rational function $h \in K(X)^\times$, we have

$$\mathscr{O}_X(\operatorname{div}(h)) = \frac{1}{h} \cdot \mathscr{O}_X.$$

Moreover, a Weil divisor D is principal if and only if $\mathscr{O}_X(D) \cong \mathscr{O}_X$

Proof Let $U \subseteq X$ be an open set. We claim that

$$\Gamma(U, \mathscr{O}_X(\operatorname{div} h)) = \frac{1}{h} \cdot \Gamma(U, \mathscr{O}_X).$$

Note that a rational function $f \in K(X)$ belongs to $\Gamma(U, \mathscr{O}_X(D))$ if and only if $\operatorname{div}|_U(f) + \operatorname{div}(h)|_U \geq 0$, or equivalently, $\operatorname{div}(fh) \geq 0$. By Corollary 14.5.2, $r \in \Gamma(U, \mathscr{O}_X)$, hence $f = \frac{r}{h} \in \frac{1}{h} \cdot \mathscr{O}_X$, i.e., $\Gamma(U, \mathscr{O}_X(\operatorname{div} h)) = \frac{1}{h} \cdot \Gamma(U, mso_X)$. The opposite containment is clear.

For the last statement: If $D = \operatorname{div}(h)$ is principal, then we have an isomorphism

$$\begin{aligned} \Gamma(U, \mathscr{O}_X(D)) &\longrightarrow \Gamma(U, \mathscr{O}_X) \\ f &\longmapsto fh. \end{aligned}$$

This is compatible with the restriction maps, and so $\mathscr{O}_X(D) \cong \mathscr{O}_X$.

Conversely, suppose that $\mathscr{O}_X(D) \cong \mathscr{O}_X$. Then there is a nonzero $g \in K(X)$ such that $\mathscr{O}_X(D) = \mathscr{O}_X \cdot g$.

Define $h = 1/g$, then

$$\mathscr{O}_X(\operatorname{div}(h)) = \frac{1}{h}\mathscr{O}_X = g\mathscr{O}_X = \mathscr{O}_X(D).$$

Hence, $D = \operatorname{div}(h)$. $\square$

Proposition 16.4.6 (Important!)

Let X be a Noetherian (integral) normal scheme. D be a Weil divisor on X.

(a) $\mathscr{O}_X(D)$ is invertible if and only if D is locally principal.

(b) If D is a locally principal divisor given by $D|_{U_i} = \operatorname{div}(f_i)|_{U_i}$, then

$$\mathscr{O}_X(D)|_{U_i} = \frac{1}{f_i} \cdot \mathscr{O}_{U_i},$$

for each $i \in I$.

Proof By Important observation above, one direction is straightforward.

Since $\mathscr{O}_X(D)$ is an invertible sheaf, for any trivialization open subset U, we have $\mathscr{O}_X(D)|_U \cong \mathscr{O}_U$. Let s be the generator of $\mathscr{O}_X(D)|_U$, then $\mathscr{O}_X(D)|_U = \mathscr{O}_U \cdot s$. Let $D|_U = \sum_Y n_Y[Y]$ where $Y \subseteq U$ be codimension 1 irreducible closed subset of U, say η_Y is generic point of Y. Localization at η_Y, we obtain

$$\begin{aligned}(\mathscr{O}_X(D)|_U)_{\eta_Y} &= \{t \in K(X)^\times : \operatorname{val}_{\eta_Y}(t) + n_Y \geq 0\} \\ &= \{t \in K(X)^\times : \operatorname{val}_{\eta_Y}(t) \geq \operatorname{val}_{\eta_Y}(s)\} \\ &= \mathscr{O}_{X,\eta_Y} \cdot s,\end{aligned}$$

hence $\operatorname{val}_{\eta_Y}(s) = -n_Y$. It follows that $D|_U = -\operatorname{div}(s)$, and thus D is locally principal. $\square$

Remark 16.13 In definition (16.7), it may seem cleaner to consider those s such that $\operatorname{div} s \geq D|_U$. The reason for the convention comes from our desire that $\operatorname{div} \sigma(1) = D$. (Taking the "opposite" convention would yield the dual bundle, in the case where D is a locally principal.)

Example 16.8 (A Weil divisor that is not locally principal) Let $X = \operatorname{Spec} k[x,y,z]/(xy - z^2)$, a cone, and let D be the line $z = x = 0$ (see Figure 14.1).

(a) Show that D is not a locally principal divisor. In particular, $\mathscr{O}_X(D)$ is not an invertible sheaf. (**Caution:** We earlier saw that it was not a locally principal closed subscheme (Definition 10.5.1) in Problem 14.1. It is unfortunate that "locally principal" has two different but very similar meanings.)

(b) Show that $\operatorname{div}(x) = 2D$. This corresponds to the fact that the plane $x = 0$ is tangent to the cone X along D. Hence $2D$ is a locally principal divisor.

Proof ***For (a).*** X is Noetherian integral normal scheme, since $k[x,y,z]/(xy - z^2)$ is Noetherian integral domain and normality by applying Serre's normality criterion [Sta25, Lemma 031S]. Let $Y = \operatorname{Spec} k[x,y,z]/(xy - z^2, x, z)$, then $Y \cong \mathbb{A}^1_k$, so $\dim Y = 1$, and therefore $\operatorname{codim}_X Y = 1$. Let $D = [Y]$. We shall to show that D is not locally principal. If D is locally principal, then there exists an open neighborhood U of $O = [(x,y,z)]$ such that $D|_U = \operatorname{div}(f)|_U$ for some $f \in K(X)^\times$. In particular, in $\mathscr{O}_{X,O}$, $\operatorname{div}(f)|_U$ is exactly the prime divisor, thus $\operatorname{val}_Y(f) = 1$, by Corollary 14.5.2, we obtain $f \in \mathscr{O}_{X,O}$. Since $\operatorname{val}_Y(f) = 1 > 0$, f vanishes along D. Equivalently, $f \in (x,z)\mathscr{O}_{X,O}$, so $(f) \subseteq (x,z)\mathscr{O}_{X,O}$. Now take any element $a \in (x,z)\mathscr{O}_{X,O}$. Then a is regular and $\operatorname{val}_Y(a) \geq 1$. Hence, $\operatorname{val}_Y(\frac{a}{f}) \geq 0$. Let E be a prime divisor on U such that $E \neq D|_U$, then $\operatorname{val}_E(f) = 0$, since $\operatorname{div}(f)|_U = D|_U$. Since $a \in \mathscr{O}_{X,O}$, we also have $\operatorname{val}_E(a) \geq 0$. Thus $\operatorname{val}_E(\frac{a}{f}) \geq 0$. It follows that $\frac{a}{f}$ has no poles on U, by Corollary 14.5.2, $\frac{a}{f}$ is regular, thus $\frac{a}{f} \in \mathscr{O}_{X,O}$. Hence, $a = f \cdot \frac{a}{f} \in (f)$. Together with $(f) \subseteq (x,z)\mathscr{O}_{X,O}$, this gives $(x,z)\mathscr{O}_{X,O} = (f)$. This is impossible, recall Problem 14.1, (x,z) is not principal, a contradiction!

In particular, by Proposition 16.4.6, $\mathscr{O}_X(D)$ is not invertible.

For (b). We now compute the principal divisor of $x \in K(X)^\times$. Let $\mathfrak{q} \in \operatorname{Spec} k[x,y,z]/(xy-z^2)$ be a height 1 prime ideal such that $x \in \mathfrak{q}$. In $k[x,y,z]/(xy-z^2)$, we have $xy = z^2$, and thus $z^2 = xy \in \mathfrak{q}$. Since $\mathfrak{q}$ is prime, $z \in \mathfrak{q}$, and therefore $(x,z) \subseteq \mathfrak{q}$. Since $\operatorname{ht}(x,z) = 1$, we obtain $\mathfrak{q} = (x,z)$. Hence, $\operatorname{div}(x) = \operatorname{val}_Y(x)[Y] = \operatorname{val}_Y(x)D$. Consider $\mathscr{O}_{X,[(x,z)]} = (k[x,y,z]/(xy-z^2))_{(x,z)}$, since y is invertible in $(k[x,y,z]/(xy-z^2))_{(x,z)}$, $x = \frac{z^2}{y}$, so the maximal ideal is

$$(x,z)(k[x,y,z]/(xy-z^2))_{(x,z)} = (z)(k[x,y,z]/(xy-z^2))_{(x,z)},$$

i.e., the uniformizer is z. Hence, $\operatorname{val}_Y(x) = \operatorname{val}_Y(\frac{z^2}{y}) = 2$, and thus $\operatorname{div}(x) = 2D$. $\square$

It is not true in general that every divisor is locally principal (Cartier). However, if X is factorial (Definition 6.4.5), the two notions are the same.

Theorem 16.4.1

Let X be a Noetherian factorial scheme (Proposition 6.4.6: factorial implies normality). Then every Weil divisor is locally principal.

Proof "Being locally principal" is a local condition, so we may assume $X = \operatorname{Spec} A$ is affine.

Let $Z = V(\mathfrak{p})$ be a prime divisor on X, where $\mathfrak{p}$ is a height 1 prime ideal. Note that Z is automatically principal over the open set $U = X \setminus Z$, because $Z|_U = 0$. If $x \in Z$, then x corresponds to a prime ideal $\mathfrak{q}$ containing $\mathfrak{p}$. As $\mathfrak{p}$ has height 1 in A, then $\mathfrak{p}A_\mathfrak{q}$ remains of height 1. As $A_\mathfrak{q} = \mathscr{O}_{X,x}$ is assumed to be a UFD (since X is factorial), by Lemma 13.1.3, $\mathfrak{p}A_\mathfrak{q}$ is a principal ideal. We may assume $\mathfrak{p}A_\mathfrak{q} = (\frac{a}{b})$ where $b \notin \mathfrak{q}$, since b is invertible in $A_\mathfrak{q}$, we see that in fact $\mathfrak{p}A_\mathfrak{q} = (a)A_\mathfrak{q}$.

If $\mathfrak{p} = (f_1, \dots, f_r)$, then $f_iA_\mathfrak{q} \in (a)A_\mathfrak{q}$ for each $i = 1, \dots, r$, so we may write $\frac{f_i}{1} = \frac{a_i}{b_i} \cdot a$. Setting $b = b_1 \cdots b_r$, we have $\mathfrak{p}A_b = (a)A_b$. Therefore, $U = D(b) \subseteq X$ is an open neighborhood of x such that $Z \cap U$ is defined by a principal ideal in $\mathscr{O}_X(U)$, i.e., $Z \cap U = V(\mathfrak{p}A_b) = V(a)$. In other words, $\operatorname{div}(a)|_U = V(\mathfrak{p})|_U = Z|_U$. Hence, each prime divisor is locally principal, which implies that every Weil divisor is locally principal. $\square$

Corollary 16.4.2 (Important)

If X is Noetherian and factorial, then for any Weil divisor D, $\mathscr{O}(D)$ is an invertible sheaf.

Proof By Theorem 16.4.1, we know that $\operatorname{Weil} X = \operatorname{LocPrin} X$. By Proposition 16.4.6, we win. $\square$

16.4.4 The class group

We can now get a handle on the Picard group of a normal Noetherian scheme.

Definition 16.4.7 (Class group)

Let X be a Noetherian normal scheme. Define the ***class group*** *of X, $\operatorname{Cl} X := \operatorname{Weil} X / \operatorname{Prin} X$.*

By taking the quotient of the inclusion (16.6) by $\operatorname{Prin} X$, we have the inclusion $\operatorname{Pic} X \hookrightarrow \operatorname{Cl} X$. This is summarized in the convenient and enlightening diagram.

$$\begin{tikzcd}
 & \{(\mathscr{L}, s)\}/\mathrm{iso.} \arrow[r, "\sim", "\operatorname{div}"'] \arrow[d] & \operatorname{LocPrin} X \arrow[l, bend right, "(\mathscr{O}(D),\sigma(1)) \leftarrow\!\shortmid D"'] \arrow[r, hook, "= \text{if } X", "\text{factorial}"'] \arrow[d, "/\operatorname{Prin} X"] & \operatorname{Weil} X \arrow[d, "/\operatorname{Prin} X"] \\
\operatorname{Pic} X \arrow[r, equal] & \{\mathscr{L}\}/\mathrm{iso.} \arrow[r, "\sim"] & \operatorname{LocPrin} X/\operatorname{Prin} X \arrow[l, bend left, "\mathscr{O}(D) \leftarrow\!\shortmid D"] \arrow[r, hook, "= \text{if } X", "\text{factorial}"'] & \operatorname{Cl} X
\end{tikzcd} \tag{16.8}$$

This diagram is very important, and dense with meaning. It takes time to digest.

In particular, if A is a UFD, then all Weil divisors on $\operatorname{Spec} A$ *are principal by Lemma 13.1.3, so* $\operatorname{Cl}(\operatorname{Spec} A) = 0$*, and hence* $\operatorname{Pic}(\operatorname{Spec} A) = 0$.

Remark 16.14 If A is a UFD, then every codimension 1 prime ideal of A is generated by one irreducible element (Lemma 13.1.3). That is, for each $\operatorname{ht}\mathfrak{p} = 1$, $\mathfrak{p} = (f_{\mathfrak{p}})$. Then $\operatorname{div}(f_{\mathfrak{p}}) = V(\mathfrak{p})$. Give a Weil divisor $D = \sum_i n_i V(\mathfrak{p}_i)$, say $\operatorname{div}(f_i) = V(\mathfrak{p}_i)$, $D = \sum_i n_i \operatorname{div}(f_i) = \operatorname{div}\left(\prod_i f_i^{n_i}\right)$ is principal. It follows that $\operatorname{Cl}(\operatorname{Spec} A) = 0$.

As $k[x_1, \ldots, x_n]$ has unique factorization, $\operatorname{Cl}(\mathbb{A}_k^n) = 0$, so $\boxed{\operatorname{Pic}(\mathbb{A}_k^n) = 0}$. Geometers might find this believable — "$\mathbb{C}^n$ is a contractible complex manifold, and hence should have no nontrivial line bundles" — even if some caution is in order, as the kinds of line bundles being considered are entirely different: holomorphic vs. topological or C^∞. (Aside: For this reason, you might expect that $\mathbb{A}_k^n$ also has nontrivial vector bundles. This is the Quillen-Suslin Theorem, formerly known as Serre's Conjecture, part of Quillen's work leading to his 1978 Fields Medal. The case $n = 1$ was Proposition 15.2.2. For a short proof by Vaserstein, see **lang2002algebra**.)

Removing a closed subset of X of codimension greater than 1 doesn't change the class group, as it doesn't change the Weil divisor group or the principal divisors. (**Warning: It can affect the Picard group; see ??.**)

Removing an irreducible closed subset of pure dimension 1 changes the Weil divisor group in a controllable way. Suppose Z is an irreducible codimension 1 subset (a prime divisor) of X. Then we clearly have an exact sequence:

$$0 \longrightarrow \mathbb{Z} \xrightarrow{1 \mapsto [Z]} \operatorname{Weil} X \longrightarrow \operatorname{Weil}(X \setminus Z) \longrightarrow 0$$

Proposition 16.4.7

*Let X be a Noetherian normal scheme and Z be a prime divisor on X, then we have an exact sequence (often call **excision exact sequence for class groups**):*

$$\mathbb{Z} \xrightarrow{1 \mapsto [Z]} \operatorname{Cl} X \longrightarrow \operatorname{Cl}(X \setminus Z) \longrightarrow 0. \tag{16.9}$$

Proof For simply, we may assume X is irreducible. By our above discussion we already have an exact sequence

$$0 \longrightarrow \mathbb{Z} \xrightarrow{1 \mapsto [Z]} \operatorname{Weil} X \longrightarrow \operatorname{Weil}(X \setminus Z) \longrightarrow 0$$

Now take the quotient by the subgroup of principal divisors, we want to check

- $\operatorname{Weil} X \to \operatorname{Weil}(X \setminus Z)$ induces a well-defined surjective $\operatorname{Cl} X \to \operatorname{Cl}(X \setminus Z)$, and
- $\operatorname{Ker}(\operatorname{Cl} X \to \operatorname{Cl}(X \setminus Z)) = \operatorname{Im}(\mathbb{Z} \to \operatorname{Cl} X)$.

Denote $U = X \setminus Z$. $\operatorname{Cl} X \to \operatorname{Cl}(X \setminus Z)$ is well-defined, since for any $\operatorname{div}(f) \in \operatorname{Prin} X$, we have $\operatorname{div}(f)|_U \in \operatorname{Prin}(X \setminus Z)$. Surjectivity is given by the surjectivity of $\operatorname{Weil} X \to \operatorname{Weil}(X \setminus Z)$.

Let $[D] \in \operatorname{Ker}(\operatorname{Cl} X \to \operatorname{Cl}(X \setminus Z))$, then $D|_U = \operatorname{div}(f)|_U$ for some $f \in K(U)^\times = K(X)^\times$. Hence, $D|_U - \operatorname{div}(f)|_U = 0$, which implies that $D - \operatorname{div}(f) = nZ$ for some $n \in \mathbb{Z}$. Therefore, $[D] \in \operatorname{Im}(\mathbb{Z} \to \operatorname{Cl} X)$.

Conversely, if $[D] = [nZ]$, then it is clearly that $[D]$ belong to the kernel. Consequently, (16.9) exact at $\operatorname{Cl} X$, we win. □

Remark 16.15 ***Why we may lose exactness on the left in (16.9)?*** Answer: Since there may exists $f \in K(X)^\times$ such that $\operatorname{div}(f) = nZ$ and $[nZ] = 0$ in $\operatorname{Cl} X$. (This seems like nonsense.)

Example 16.9 For example, if U is an open subscheme of $X = \mathbb{A}^n$, $\operatorname{Pic} U = \{0\}$.

Example 16.10 As another application, let $X = \mathbb{P}_k^n$, and Z be the hyperplane $x_0 = 0$, by (16.9), we have

$$\mathbb{Z} \longrightarrow \operatorname{Cl}(\mathbb{P}_k^n) \longrightarrow \operatorname{Cl}(\mathbb{A}_k^n) \longrightarrow 0$$

from which $\operatorname{Cl}(\mathbb{P}_k^n)$ is generated by the class $[Z]$ (since $\operatorname{Cl}(\mathbb{A}_k^n) = 0$, we obtain that $\mathbb{Z} \to \operatorname{Cl}(\mathbb{P}_k^n)$ is a surjective), and $\operatorname{Pic}(\mathbb{P}_k^n)$ is a subgroup of this.

We next show that $\operatorname{Cl}(\mathbb{P}_k^n) \cong \mathbb{Z}$. It suffices to show that $[Z]$ is not torsion in $\operatorname{Cl} \mathbb{P}_k^n$, i.e., $m[\mathbb{Z}] \neq 0$ for $m \neq 0$. If there exists $m \neq 0$ such that $m[Z] = 0$ in $\operatorname{Cl}(\mathbb{P}_k^n)$, i.e., there exists $f \in K(\mathbb{P}_k^n)^\times$ such that $mZ = \operatorname{div}(f)$. By Proposition 16.4.4,

$$\mathscr{O}(mZ) \cong \mathscr{O}(m) \cong \mathscr{O}(\operatorname{div}(f)) \cong \mathscr{O}.$$

This is impossible, Proposition 16.1.1 says $\mathscr{O}(m)$ is nontrivial for $m \neq 0$. Hence, $\operatorname{Cl}(\mathbb{P}_k^n) \cong \mathbb{Z}$. Note that $\mathscr{O}(Z) \cong \mathscr{O}(1) \in \operatorname{Pic} \mathbb{P}_k^n$ and $[Z]$ generate $\operatorname{Cl}(\mathbb{P}_k^n)$, the image of $\operatorname{Pic}(\mathbb{P}_k^n)$ is a subgroup of $\operatorname{Cl}(\mathbb{P}_k^n)$, and it contains the generator $[Z]$, therefore, this image is the entire class group, i.e., $\boxed{\operatorname{Pic}(\mathbb{P}_k^n) \cong \mathbb{Z}}$, with generator $\mathscr{O}(1)$.

The **degree** of an invertible sheaf on $\mathbb{P}_k^n$ is defined using this: define $\deg \mathscr{O}(d)$ to be d. (You will have already proved that $\operatorname{Pic}(\mathbb{P}_k^n) \cong \mathbb{Z}$ if you did Proposition 16.4.5; but we will use the strategy here to great effect in §**??**.)

We have gotten good mileage from the fact that the Picard group of the spectrum of a unique factorization domain is trivial. More generally, Theorem 16.4.1 gives us.

Proposition 16.4.8

If X is Noetherian and factorial, then for any Weil divisor D, $\mathscr{O}(D)$ is invertible, and hence the map $\operatorname{Pic} X \to \operatorname{Cl} X$ is an isomorphism.

This can be used to make the connection to the class group in number theory precise; see Exercise 15.2; see also §**??**.

16.4.5 Mild but important generalization: Twisting line bundles by divisors

The above constructions can be extended, with $\mathscr{O}_X$ replaced by an arbitrary invertible sheaf, as follows.

Definition 16.4.8 (Twisting line bundles by divisors)

Let $\mathscr{L}$ be an invertible sheaf on a Noetherian normal scheme X. Then define $\mathscr{L}(D)$ by $\mathscr{O}_X(D) \otimes \mathscr{L}$.

Lemma 16.4.2

If D is a principal, then $\mathscr{L}(D)$ is a line bundle.

Proof An obvious observation. □

Notice that in this case there are two different ways of interpreting sections of $\mathscr{L}(D)$ over an open set, each with different advantages:

- as a section of the new line bundle $\mathscr{L}(D)$, and
- as rational sections of $\mathscr{L}$ with constraints on poles and zeros given by the divisor D.

Proposition 16.4.9

(a) *Let X be a Noetherian integral normal scheme. Then the sections of $\mathscr{L}(D)$ can be interpreted as rational sections of $\mathscr{L}$ with zeros and poles constrained by D, just as in (16.7):*

$$\Gamma(U, \mathscr{L}(D)) := \{t \text{ nonzero rational section of } \mathscr{L} : \operatorname{div}|_U t + D|_U \geq 0\} \cup \{0\}.$$

(b) *Suppose D_1 and D_2 are locally principal. Then*

$$(\mathscr{O}(D_1))(D_2) \cong \mathscr{O}(D_1 + D_2)$$

Proof ***For (a).*** Let $U \hookrightarrow X$ be an open subset of X, since $\mathscr{L}$ is invertible, there exists an open cover $\mathfrak{U} = \{U_\alpha \hookrightarrow U\}_\alpha$ such that $\mathscr{L}|_{U_\alpha} \cong \mathscr{O}_{U_\alpha}$. Choose a local generator $e_\alpha \in \Gamma(U_\alpha, \mathscr{L})$ for each U_α. Then every rational section of $\mathscr{L}$ over U_α can be written as

$$s = t_\alpha e_\alpha$$

for some rational function $t_\alpha \in K(X)$. On U_α, we have

$$\mathscr{L}(D)|_{U_\alpha} = \mathscr{O}_X(D)|_{U_\alpha} \otimes_{\mathscr{O}_{U_\alpha}} \mathscr{L}|_{U_\alpha} \cong \mathscr{O}_X(D)|_{U_\alpha} \cdot e_\alpha.$$

Therefore a section in $\Gamma(U_\alpha, \mathscr{L}(D))$ is $t_\alpha e_\alpha$ where $t_\alpha \in \Gamma(U_\alpha, \mathscr{O}_X(D))$. By the construction of $\mathscr{O}_X(D)$, $t_\alpha \in \Gamma(U_\alpha, \mathscr{O}_X(D))$ is equivalent to

$$\operatorname{div}|_{U_\alpha}(t_\alpha) + D|_{U_\alpha} \geq 0.$$

By the definition of rational section, we have $\operatorname{div}|_{U_\alpha}(s) = \operatorname{div}|_{U_\alpha}(t_\alpha)$ for s is a rational section of $\mathscr{L}$. Thus the condition $t_\alpha \in \Gamma(U_\alpha, \mathscr{O}_X(D))$ is equivalent to

$$\operatorname{div}|_{U_\alpha}(s) + D|_{U_\alpha} \geq 0.$$

Therefore, for U_α,

$$\Gamma(U_\alpha, \mathscr{L}(D)) := \{s \text{ nonzero rational section of } \mathscr{L} : \operatorname{div}|_{U_\alpha} s + D|_{U_\alpha} \geq 0\} \cup \{0\}.$$

By the definition of the divisor of rational section, different representations of rational section differ an invertible regular function, so $\Gamma(U_\alpha, \mathscr{L}(D))$ glue to $\Gamma(U, \mathscr{L}(D))$, hence,

$$\Gamma(U, \mathscr{L}(D)) := \{t \text{ nonzero rational section of } \mathscr{L} : \operatorname{div}|_U t + D|_U \geq 0\} \cup \{0\}.$$

For (b). By definition, $(\mathscr{O}_X(D_1))(D_2) = \mathscr{O}_X(D_1) \otimes \mathscr{O}_X(D_2)$. Since D_1, D_2 are locally principal, there exists an open subset $U \hookrightarrow X$ such that $D_i|_U = \operatorname{div}(f_i)|_U$ for some $f_i \in K(X)^\times$. By Proposition 16.4.6,

$$\begin{aligned}(\mathscr{O}_X(D_1) \otimes \mathscr{O}_X(D_2))|_U \cong& \mathscr{O}_X(D_1)|_U \otimes \mathscr{O}_X(D_2)|_U \\ \cong& \frac{1}{f_1}\mathscr{O}_U \otimes \frac{1}{f_2}\mathscr{O}_U \cong \frac{1}{f_1 f_2}\mathscr{O}_U \cong \mathscr{O}(D_1 + D_2)|_U.\end{aligned}$$

These local isomorphisms are compatible on overlaps, so they glue to global isomorphism

$$(\mathscr{O}_X(D_1))(D_2)\mathscr{O}_X(D_1) \otimes \mathscr{O}_X(D_2) \cong \mathscr{O}(D_1 + D_2).$$

□

16.4.6 A variation of the Qcqs Lemma

The Qcqs Lemma 7.2.3, proved in §7.2.3, has the following generalization.

Lemma 16.4.3

Let X be a ringed space. Assume that each stack $\mathscr{O}_{X,x}$ is a local ring with maximal ideal $\mathfrak{m}_x$. Let $\mathscr{L}$ be an invertible $\mathscr{O}_X$-module. For any section $s \in \Gamma(X, \mathscr{L})$, the set

$$X_s := \{x \in X : \text{image } s \notin \mathfrak{m}_x \mathscr{L}_x\}$$

is open in X. The map $s : \mathscr{O}_{X_s} \to \mathscr{L}|_{X_s}$ is an isomorphism, and there exists a section s' of $\mathscr{L}^\vee$ over X_s such that $s'(s|_{X_s}) = 1$.

Proof Suppose $x \in X_s$. By Lemma 15.1.4 (2), we have an isomorphism

$$\mathscr{L}_x \otimes_{\mathscr{O}_{X,x}} (\mathscr{L}^\vee)_x \longrightarrow \mathscr{O}_{X,x}.$$

Both $\mathscr{L}_x$ and $(\mathscr{L}^\vee)_x$ are free $\mathscr{O}_{X,x}$-modules of rank 1. Then the image of s in $\mathscr{L}_x$ is a basis for $\mathscr{L}_x$, denote it s_x. Hence there exists a basis element $t_x \in (\mathscr{L}^\vee)_x$ such that $s_x \otimes t_x$ maps to 1. Choose an open neighborhood U of x such that t_x comes from a section t of $\mathscr{L}^\vee$ over U and such that $s \otimes t$ maps to $1 \in \mathscr{O}_X(U)$. By Geometric Nakayama 15.3.4, for every $x' \in U$, we see that s generates the module $\mathscr{L}_{x'}$. Hence $U \subseteq X_s$. This proves that X_s is open. Moreover, the section t constructed over U above is unique, and hence these glue to give the section s' of the Lemma 16.4.3. $\square$

Theorem 16.4.2 (Generalized Qcqs Lemma)

Suppose X is a quasicompact quasiseparated scheme, $\mathscr{L}$ is an invertible sheaf on X with section s, and $\mathscr{F}$ is a quasicoherent sheaf on X. We interpret s as a degree 1 element of the graded ring $R(\mathscr{L})_\bullet := \bigoplus_{n\geq 0} \Gamma(X, \mathscr{L}^{\otimes n})$. Note that $\bigoplus_{n\geq 0} \Gamma(X, \mathscr{F} \otimes_{\mathscr{O}_X} \mathscr{L}^{\otimes n})$ is a graded $R(\mathscr{L})_\bullet$-module. Then there is a natural isomorphism

$$\left(\left(\bigoplus_{n\geq 0} \Gamma\left(X, \mathscr{F} \otimes_{\mathscr{O}_X} \mathscr{L}^{\otimes n}\right)\right)_s\right)_0 \xrightarrow{\sim} \Gamma\left(X_s, \mathscr{F}\right). \tag{16.10}$$

Proof Since $\mathscr{L}$ is invertible, by Lemma 16.4.3, a section $s : \mathscr{O}_X \to \mathscr{L}$ gives an isomorphism $s|_{X_s} : \mathscr{O}_{X_s} \xrightarrow{\sim} \mathscr{L}|_{X_s}$. Hence, we get a section in $\mathscr{L}|_{X_s}^\vee$, i.e., $s|_{X_s}^{-1} : \mathscr{L}|_{X_s} \to \mathscr{O}_{X_s}$. Notice that $\mathscr{F} \cong (\mathscr{F} \otimes \mathscr{L}^{\otimes n}) \otimes (\mathscr{L}^\vee)^{\otimes n}$, for any $t \in \Gamma(X, \mathscr{F} \otimes \mathscr{L}^{\otimes n})$, we have

$$t|_{X_s} \otimes (s|_{X_s}^{-1})^{\otimes n} = t|_{X_s} \otimes s|_{X_s}^{-n} \in \Gamma(X_s, \mathscr{F}).$$

Define

$$\begin{aligned} \Phi : \left(\left(\bigoplus_{n\geq 0} \Gamma\left(X, \mathscr{F} \otimes_{\mathscr{O}_X} \mathscr{L}^{\otimes n}\right)\right)_s\right)_0 &\xrightarrow{\sim} \Gamma\left(X_s, \mathscr{F}\right) \\ t/s^n &\longmapsto t|_{X_s} \otimes s|_{X_s}^{-n} \end{aligned}$$

Since X is quasicompact. Choose a finite affine open covering $X = \bigcup_{i=1}^m$ with U_j affine and $\mathscr{L}|_{U_i} \cong \mathscr{O}_{U_j}$. Via this isomorphism, the image $s|_{U_j}$ corresponds to some $f_j \in \Gamma(U_j, \mathscr{O}_{U_j})$. Then $X_s \cap U_j = D(f_j)$.

Let $\frac{t}{s^n}$ be an element in the kernel of (16.10). Then $t|_{X_s} = 0$. Hence $(t|_{U_i})|_{D(f_j)} = 0$, by Qcqs Lemma 7.2.3, we conclude that $f_j^{e_j} t|_{U_j} = 0$ for some $e_j \geq 0$. Let $e = \max\{e_j\}$. Then we see that $t \otimes s^e$ restricts to zero on U_j for all j, hence is zero. Since $\frac{t}{s^n} = \frac{t\otimes s^e}{s^{n+e}}$ in $\left(\left(\bigoplus_{n\geq 0} \Gamma\left(X, \mathscr{F} \otimes_{\mathscr{O}_X} \mathscr{L}^{\otimes n}\right)\right)_s\right)_0$, we conclude that $\frac{t}{s^n} = 0$, which implies that Φ is injective.

Since X is quasiseparated. Then $U_j \cap U_{j'}$ is quasicompact for all pairs j, j'. Let $t' \in \Gamma(X_s, \mathscr{F})$. For every j, there exists an integer $e_j \geq 0$ and $t'_j \in \Gamma(U_j, \mathscr{F})$ such that $t'|_{D(f_i)}$ corresponds to $t'_j/f_j^{e_j}$ via the isomorphism of Qcqs Lemma 7.2.3. Set $e = \max\{e_j\}$ and

$$t_j = f_j^{e-e_j} t'_j \otimes q_j^e \in \Gamma(U_j, \mathscr{F}) \otimes \Gamma(U_j, \mathscr{L})^{\otimes e} = \Gamma(U_j, \mathscr{F} \otimes_{\mathscr{O}_X} \mathscr{L}^{\otimes e}),$$

where $q_j \in \Gamma(U_j, \mathscr{L})$ is the trivializing section coming from the isomorphism $\mathscr{L}|_{U_i}$. In particular, we have $s|_{U_i} = f_j q_j$. So,

$$\Phi(\tfrac{t_j}{s|_{U_j}^e}) = t_j \otimes s|_{U_j}^{-e} = (f_j^{e-e_j} t'_j \otimes q_j^e) \otimes (f_j q_j)^{-e} = \frac{t'_j}{f_j^{e_j}} = t'|_{D(f_i)}.$$

Now we need to glue the t_j's. On an overlap $U_j \cap U_{j'}$, $\frac{t'_j}{f_j^{e_j}}$ and $\frac{t'_{j'}}{f_{i'}^{e_{j'}}}$ both restrict to the same section t' on $(U_j \cap U_{j'}) \cap X_s$. Therefore

$$\Phi(\tfrac{t_j}{s^e} - \tfrac{t_{j'}}{s^e}) = 0 \in \Gamma((U_j \cap U_{j'}) \cap X_s, \mathscr{F}).$$

Because X is quasi-separated and $U_j, U_{j'}$ are affine opens, the intersection $U_j \cap U_{j'}$ is quasi-compact. By applying the injectivity part to $U_j \cap U_{j'}$, there exists an integer $e'_{jj'} \geq 0$ such that $(t_j - t_{j'}) \otimes s^{e'_{jj'}} = 0$ on $U_i \cap U_j$. Since there are only finitely many pairs (j, j'), choose one integer $e' \geq 0$ large enough to work for all pairs. Then for all j, j',

$$t_j \otimes s^{e'} = t_{j'} \otimes s^{e'}$$

as sections of $\mathscr{F} \otimes_{\mathscr{O}_X} \mathscr{L}^{\otimes(e+e')}$ over $U_j \cap U_{j'}$. Thus $\{t_j \otimes s^{e'}\}$ agrees on overlaps, so they glue to a section $t \in \Gamma(X, \mathscr{F} \otimes_{\mathscr{O}_X} \mathscr{L}^{\otimes(e+e')})$ such that $t|_{U_j} = t_j \otimes s^{e'}$ for any j. Now consider $\frac{t}{s^{e+e'}} \in \left(\left(\bigoplus_{n\geq 0} \Gamma\left(X, \mathscr{F} \otimes_{\mathscr{O}_X} \mathscr{L}^{\otimes n}\right)\right)_s\right)_0$. On U_j,

$$\Phi(\tfrac{t}{s^{e+e'}}) = \frac{t_j \otimes f_j^{e'}}{f_j^{e+e'}} = \frac{t_j}{f_j^e} = \frac{t'_j}{f_j^{e_j}} = t'|_{D(f_i)}.$$

Since $D(f_i)$ cover X_s, it follows that

$$\phi\left(\frac{t}{s^{e+e'}}\right) = t'.$$

Therefore Φ is surjective. □

Remark 16.16 ***Translation:*** Any section of $\mathscr{F}$ over X_s can be extended to a section over X after multiplying by some some appropriate power of s. And if we have two such extensions, they become equal after multiplying by another appropriate power of s.

16.5 Cartier Divisors

In this section, we mainly refer to **goertz2020algebraicgeometry1**. The notion of Weil divisors works only if suitable assumptions (Noetherian normal) on the underlying scheme X are made. The notion which works better in the general case is the notion of **Cartier divisor**. Here we just define a “configuration” as an equivalence class of rational functions where we call rational functions f, g on some open subset U equivalence if $f = ug$ for some $u \in \Gamma(U, \mathscr{O}_X)^\times$ — clearly since u is a unit, f and g should be thought of having the same zero or pole configuration in this case.

If X is an integral scheme, then many of the definitions simplify considerably, and therefore we treat this case separately as a first step.

16.5.1 Cartier divisor on integral schemes

Let X be an integral scheme. We denote by $\mathscr{K}_X$ the constant sheaf with value the function field $K(X)$ of X. In other words, for every non-empty open $U \subseteq X$, we have $\mathscr{K}_X(U) = K(X)$. For every point $x \in X$, $\mathscr{K}_{X,x} = K(X)$.

Since any non-empty open subset $U \subseteq X$ is scheme-theoretically dense, the ring $R(U)$ of rational functions coincides with $K(X)$. In particular, the sheaves $U \mapsto R(U)$ and $\mathscr{K}_X$ are equal.

Definition 16.5.1 (Cartier divisor on integral scheme)

*A **Cartier divisor** D **on the integral scheme** X is given by a tuple (U_i, f_i) where the U_i form an open covering of X and where $f_i \in K(X)^\times$ are elements with $f_i f_j^{-1} \in \Gamma(U_i \cap U_j, \mathscr{O}_X^\times)$ for all i, j. Two tuples $(U_i, f_i), (V_i, g_i)$ give rise to the same Cartier divisor, if $f_i g_j^{-1} \in \Gamma(U_i \cap V_j, \mathscr{O}_X^\times)$ for all i, j.*

Definition 16.5.2 ($\mathrm{Div}(X)$)

The set of Cartier divisors is denoted by $\mathrm{Div}(X)$. It is an abelian group: in fact, given divisors D, E represented by families $(U_i, f_i), (V_i, g_i)$, we define the sum $D + E$ as the divisor given by $(U_i \cap V_j, f_i g_j)$.

Definition 16.5.3 (Principal Cartier divisor)

*A Cartier divisor is called **principal**, if it is equal to a divisor given by (X, f).*

Definition 16.5.4 (Linearly equivalent)

*Two Cartier divisors D, E are called **linearly equivalent**, if their difference $D - E$ is a principal Cartier divisor.*

Note *We denote by $\mathrm{DivCl}(X)$ the quotient of $\mathrm{Div}(X)$ by the subgroup of principal divisors. We obtain an exact sequence*

$$1 \longrightarrow \Gamma(X, \mathscr{O}_X)^\times \longrightarrow K(X)^\times \longrightarrow \mathrm{Div}(X) \longrightarrow \mathrm{DivCl}(X) \longrightarrow 0.$$

There is a close relationship between divisors and line bundles.

Definition 16.5.5 (Associated line bundle of a Cartier divisor)

Let D be a Cartier divisor, we attach the line bundle $\mathscr{O}_X(D)$, given by

$$\Gamma(V, \mathscr{O}_X(D)) = \{f \in K(X) : f_i f \in \Gamma(U_i \cap V, \mathscr{O}_X), \text{ for all } i\}$$

for $V \subseteq X$ open.

Remark 16.17 Over U_i, $\mathscr{O}_X(D)$ is isomorphic to the free $\mathscr{O}_{U_i}$-submodule of rank 1 of $\mathscr{K}_{U_i}$ generated by f_i^{-1}.

We will see that:

Proposition

The map $\mathrm{Div}(X) \to \mathrm{Pic}(X)$, $D \mapsto \mathscr{O}_X(D)$, induces an isomorphism $\mathrm{DivCl}(X) \xrightarrow{\sim} \mathrm{Pic}(X)$ of abelian groups.

Thinking of a Cartier divisor as an object which locally models the sets of zeros and poles of meromorphic functions, we would like to understand Cartier divisors as geometric objects of their own. The notion one introduces first in this regard is the notion of support of a Cartier divisor.

Definition 16.5.6 (Support of Cartier divisor)

Let D be a Cartier divisor. We define support of D by setting

$$\operatorname{Supp}(D) := \{x \in X : (f_i)_x \notin \mathscr{O}_{X,x}^\times, \text{ for some (or all) } i \text{ with } x \in U_i\}.$$

Remark 16.18 $\operatorname{Supp}(D)$ is a closed subset of X.

Example 16.11 Let $X = \operatorname{Spec} k[t]$ and $D = (X, t)$ is a principal Cartier divisor. At point $[(t)]$, we have $t \in \mathfrak{m}_{[(t)]}$, so $t \notin \mathscr{O}_{X,[(t)]}^\times$. Hence, $[(t)] \in \operatorname{Supp}(D)$. For other point $p = [\mathfrak{p}] \neq [(t)]$, $t \notin \mathfrak{p}$, so $t(p) \neq 0$, which implies that $t \in \mathscr{O}_{X,p}^\times$. Hence, $p \notin \operatorname{Supp}(D)$, which implies that $\operatorname{Supp}(D) = \{[(t)]\} = \{0\}$.

Under suitable assumptions on X (like normality or regularity), one can extend this to viewing Cartier divisors as "subschemes of codimension 1 with multiplicities attached to the irreducible components", i.e., Weil divisors.

16.5.2 Cartier divisors on schemes

Recall the notions of 15.2.2. Let X be any scheme, $\mathscr{K}_X$ be the sheaf of meromorphic functions (Definition 15.2.4).

Definition 16.5.7 (Cartier divisors)

Let X be a scheme.

*(1) A **Cartier divisor** on X is an element of the group*

$$\operatorname{Div}(X) := \Gamma(X, \mathscr{K}_X^\times / \mathscr{O}_X^\times).$$

The group law is noted additively.

*(2) A Cartier divisor is called **principal** if it is in the image of the image of the canonical homomorphism $\Gamma(X, \mathscr{K}_X^\times) \to \operatorname{Div}(X)$. The principal divisor defined by $f \in \Gamma(X, \mathscr{K}_X^\times)$ is denoted by $\operatorname{div}(f)$.*

*(3) Two divisors D_1 and D_2 are called **linearly equivalent**, denoted $D_1 \sim D_2$, if $D_1 - D_2$ is principal.*

*(4) A divisor D is called **effective** if it is in the submonoid $\operatorname{Div}_+(X) := \Gamma(X, (\mathscr{K}_X^\times \cap \mathscr{O}_X)/\mathscr{O}_X^\times)$. In this case we write $D \geq 0$. For two divisors D, D' $\operatorname{Div}(X)$ we write $D \geq D'$ if $D - D' \geq 0$.*

We can describe divisors more concretely. For $f \in \Gamma(U, \mathscr{K}_X)$ and $g \in \Gamma(V, \mathscr{K}_X)$ we write simply fg instead of $f|_{U\cap V} g|_{U\cap V}$. Then a divisor is given by a tuple $(U_i, f_i)_i$, where $X = \bigcup_i U_i$ is an open covering and where $f_i \in \Gamma(U_i, \mathscr{K}_X^\times)$ is a meromorphic function that is the quotient of two regular functions such that $f_i f_j^{-1} \in \Gamma(U_i \cap U_j, \mathscr{O}_X^\times)$ for all i, j.

Let D and E be divisors given by tuples $(U_i, f_i)_i$ and $(V_j, g_j)_j$, respectively. Then $D = E$ if and only if $f_i g_j^{-1} \in \Gamma(U_i \cap V_j, \mathscr{O}_X^\times)$ for all i, j. The sum $D + E$ is given by the tuple $(U_i \cap V_j, f_i g_j)_{i,j}$ and the inverse $-D$ is given by (U_i, f_i^{-1}). The divisor D is effective if and only if $f_i \in \Gamma(U_i, \mathscr{O}_X)$ for all i. A divisor D is principal if and only if it can be given by (X, f).

Definition 16.5.8 (Cartier divisor classes)

The principal divisors form a subgroup $\operatorname{Div}_{\text{princ}}(X)$ of $\operatorname{Div}(X)$ and the quotient

$$\operatorname{DivCl}(X) := \operatorname{Div}(X)/\operatorname{Div}_{\text{princ}}(X) \tag{16.11}$$

*is called **the group of Cartier divisor classes** on X. We obtain an exact sequence*

$$1 \longrightarrow \Gamma(X, \mathscr{O}_X)^\times \longrightarrow \Gamma(X, \mathscr{K}_X)^\times \longrightarrow \operatorname{Div}(X) \longrightarrow \operatorname{DivCl}(X) \longrightarrow 0. \tag{16.12}$$

16.5.3 Cartier divisors and line bundles

Let X be a scheme. We will now attach to a divisor D on X an $\mathscr{O}_X$-submodule $\mathscr{O}_X(D)$ of $\mathscr{K}_X$ which is an invertible $\mathscr{O}_X$-module and then examine the question when two such invertible modules are isomorphic and if all invertible $\mathscr{O}_X$-modules (up to isomorphism) can be obtained in this way.

Divisors and Invertible Fractional Ideals

Definition 16.5.9 (Invertible fractional ideal)

We call an invertible $\mathscr{O}_X$-submodule (clearly, it is invertible sheaf) of $\mathscr{K}_X$ an ***invertible fractional ideal*** *of $\mathscr{O}_X$.*

Definition 16.5.10 (Product of invertible fractional ideals)

If $\mathscr{I}_1$ and $\mathscr{I}_2$ are two invertible fractional ideals, their ***product*** *is defined to be the $\mathscr{O}_X$-submodule*

$$\mathscr{I}_1\mathscr{I}_2 \subseteq \mathscr{K}_X$$

locally generated by products of local sections. More precisely, if locally $\mathscr{I}_1|_U = \mathscr{O}_U \cdot f$ and $\mathscr{I}_2|_U = \mathscr{O}_U \cdot g$ with $f, g \in \mathscr{K}_X^\times(U)$, then

$$(\mathscr{I}_1\mathscr{I}_2)|_U = \mathscr{O} \cdot fg.$$

In particular, the product of two invertible fractional ideals is again an invertible fractional ideal of $\mathscr{O}_X$.

Let D be a Cartier divisor on X, represented by $(U_i, f_i)_i$. As $f_i f_j^{-1} \in \Gamma(U_i \cap U_j)$ for all i, j, there exists a (unique) invertible fractional ideal $\mathscr{I}_X(D)$ such that

$$\boxed{\mathscr{I}_X(D)|_{U_i} = \mathscr{O}_{U_i} f_i.}$$

Moreover $\mathscr{O}_X(D)$ does not depend on the choice of the representation $(U_i, f_i)_i$.

Lemma 16.5.1

$\mathscr{O}_X(D)$ does not depend on the choice of the representation $(U_i, f_i)_i$.

Proof Suppose that the same Cartier divisor D is also represented by another tuple $(V_j, g_j)_j$. By the equivalence relation in the definition of Cartier divisors, for every i, j,

$$f_i g_j^{-1} \in \Gamma(U_i \cap V_j, \mathscr{O}_X^\times).$$

Hence, on $U_i \cap V_j$, there exists a unit $u_{ij} \in \Gamma(U_i \cap V_j, \mathscr{O}_X^\times)$ such that $f_i = u_{ij} g_j$. Therefore

$$\mathscr{O}_{U_i \cap V_j} f_i = \mathscr{O}_{U_i \cap V_j} u_i j g_j = \mathscr{O}_{U_i \cap V_i} g_j,$$

because multiplication by a unit does not change the generated $\mathscr{O}_X$-submodule. Hence, the fractional ideals obtained from $(U_i, f_i)_i$ and $(V_j, g_j)_j$ agree on the open cover $\{U_i \cap V_j\}_{i,j}$ of X. Therefore they are equal as submodules of $\mathscr{K}_X$. □

Conversely, if $\mathscr{I}$ is an invertible fractional ideal, there exists an open covering $X = \bigcup_i U_i$ and sections $f_i \in \Gamma(U_i, \mathscr{K}_X^\times)$ such that $\mathscr{I}|_{U_i} = \mathscr{O}_{U_i} \cdot f_i$. Then we get a Cartier divisor $D = (U_i, f_i)_i$. We obtain an isomorphism of abelian groups

$$\mathrm{Div}(X) \xrightarrow{\sim} \{\text{Invertible fractional ideals on } X\}, \qquad D \longmapsto \mathscr{I}_X(D). \tag{16.13}$$

Lemma 16.5.2

For two divisors $D_1, D_2 \in \mathrm{Div}(X)$, we have $D_1 \leq D_2$ if and only if $\mathscr{I}_X(D_1) \supseteq \mathscr{I}_X(D_2)$.

Proof Suppose $D_1 = (U_i, f_i)_i$ and $D_2 = (V_j, g_j)_j$. If $D_1 \leq D_2$, then $D_2 - D_1 \geq 0$, i.e., $g_j f_i^{-1} \in \Gamma(U_i \cap V_j, \mathscr{O}_X)$. Since $g_j = g_j f_i^{-1} \cdot f_i$, we have

$$\mathscr{I}_X(D_2)|_{U_i \cap V_j} = \mathscr{O}_{U_i \cap V_j} g_j = (\mathscr{O}_{U_i \cap V_j} g_j f_i^{-1}) f_i \subseteq \mathscr{O}_{U_i \cap V_j} f_i = \mathscr{I}_X(D_1)|_{U_i \cap V_j}.$$

Hence, $\mathscr{I}_X(D_1) \supseteq \mathscr{I}_X(D_2)$.

Conversely, if $\mathscr{I}_X(D_1) \supseteq \mathscr{I}_X(D_2)$, then $\mathscr{I}_X(D_1)|_{U_i \cap V_j} = \mathscr{O}_{U_i \cap V_j} f_i \supseteq \mathscr{O}_{U_i \cap V_j} g_j$. Then $g_j \in \mathscr{O}_{U_i \cap V_j} f_i$, so there exists $a_{ji} \in \mathscr{O}_{U_i \cap V_j}$, such that $g_j = a_{ji} f_i$, i.e., $a_{ji} = g_j f_i^{-1} \in \mathscr{O}_{U_i \cap V_j}$. It follows that $D_2 - D_1 \geq 0$, i.e., $D_2 \geq D_1$. □

We set

$$\boxed{\mathscr{O}_X(D) := \mathscr{I}_X(D)^{-1},}$$

i.e., for ant open subset $U \subseteq X$ its sections are given by

$$\boxed{\Gamma(U, \mathscr{O}_X(D)) = \{f \in \mathscr{K}_X(U) : f \mathscr{I}_X(D)(U) \subseteq \mathscr{O}_X(U)\}.}$$

Thus if D is represented by $(U_i, f_i)_i$, we have $\mathscr{O}_X(D)|_{U_i} = \mathscr{O}_{U_i} \cdot f_i^{-1}$. We obtain

$$D_1 \leq D_2 \Leftrightarrow \mathscr{O}_X(D_1) \subseteq \mathscr{O}_X(D_2). \tag{16.14}$$

For every $f \in \Gamma(X, \mathscr{K}_X^\times)$ we find

$$f \in \Gamma(X, \mathscr{O}_X(D)) \Leftrightarrow \mathrm{div}(f) \geq -D. \tag{16.15}$$

One checks that $D \mapsto \mathscr{O}_X(D)$ is a homomorphism of groups, i.e.,

$$\boxed{\begin{aligned} \mathscr{O}_X(0) &= \mathscr{O}_X, \\ \mathscr{O}_X(D_1 + D_2) &= \mathscr{O}_X(D_1)\mathscr{O}_X(D_2) \cong \mathscr{O}_X(D_1) \otimes_{\mathscr{O}_X} \mathscr{O}_X(D_2), \end{aligned}} \tag{16.16}$$

where the product is taken in $\mathscr{K}_X$.

Lemma 16.5.3

A Cartier divisor D is effective if and only if $\mathscr{I}_X(D)$ is an ideal of $\mathscr{O}_X$.

Proof Let $D = (U_i, f_i)_i$ be a Cartier divisor. If D is effective, then $f_i \in \Gamma(U_i, \mathscr{O}_X)$, hence, $\mathscr{I}_X(D)|_{U_i} = \mathscr{O}_{U_i} f_i$, which is an ideal of $\mathscr{O}_{U_i}$. So, $\mathscr{I}_X(D)$ is an ideal of $\mathscr{O}_X$.

Conversely, if $\mathscr{I}_X(D)$ is an ideal of $\mathscr{O}_X$, then $\mathscr{O}_{U_i} f_i$ is an ideal of $\mathscr{O}_{U_i}$, which implies that $f_i \in \Gamma(U_i, \mathscr{O}_X)$, i.e., D is effective Cartier divisor. □

By Lemma 16.5.3, effective Cartier divisor D gives an ideal sheaf $\mathscr{I}_X(D)$ of X, so it induces a closed subscheme of X, we denoted by D as well. If D is represented by $(U_i, f_i)_i$, then $D \cap U_i = V(f_i)$ (as subschemes of U_i). By definition, $\mathscr{I}_X(D) = \mathscr{O}_X(-D)$, so there is an exact sequence of $\mathscr{O}_X$-modules

$$0 \longrightarrow \mathscr{O}_X(-D) \longrightarrow \mathscr{O}_X \longrightarrow \mathscr{O}_D \longrightarrow 0. \tag{16.17}$$

Conversely, assume that Y is a closed subscheme of X such that there exists an open affine covering $\mathfrak{U} = \{U_i \hookrightarrow X\}_i$ and regular elements $f_i \in \Gamma(U_i, \mathscr{O}_X)$ such that $Y \cap U_i = V(f_i)$ for all i (i.e., $Y \hookrightarrow X$ is regular embedding of codimension 1 (Definition 10.5.4; i.e., slicing by an effective Cartier divisor 10.5.2). Then $(U_i, f_i)_i$ represents an effective Cartier divisor. We obtain a bijective correspondence between regular embedding closed subschemes of codimension 1 and effective Cartier divisors and will often use both notions synonymously.

Invertible fractional ideals and line bundles

We have just seen that attaching to a divisor D the invertible submodule $\mathscr{O}_X(D)$ of $\mathscr{K}_X$ yields a group isomorphism of $\operatorname{Div}(X)$ with the group of invertible fractional ideals of X (see (16.13)). We now study the following two questions.

(1) For two divisors D_1 and D_2, when are two invertible fractional ideals $\mathscr{O}_X(D_1)$ and $\mathscr{O}_X(D_2)$ isomorphic as $\mathscr{O}_X$-modules?

(2) Which invertible $\mathscr{O}_X$-modules are isomorphic to an invertible fractional ideal?

The first question is answered by the following proposition.

Proposition 16.5.1

Let D_1 and D_2 be two Cartier divisors. Then the $\mathscr{O}_X$-modules $\mathscr{O}_X(D_1)$ and $\mathscr{O}_X(D_2)$ are isomorphic if and only if D_1 and D_2 are linearly equivalent.

Proof We first show that a Cartier divisor D is principal if and only if $\mathscr{O}_X(D) \cong \mathscr{O}_X$. If $D = (X, f)$, i.e., D is principal, then $\mathscr{O}_X(D) = \mathscr{O}_X \cdot f^{-1} \cong \mathscr{O}_X$. Conversely, if $u : \mathscr{O}_X \xrightarrow{\sim} \mathscr{O}_X(D)$ is an isomorphism, the image of $1 \in \Gamma(X, \mathscr{O}_X)$ is an element $f \in \Gamma(X, \mathscr{O}_X(D)) \subseteq \Gamma(X, \mathscr{K}_X)$. Then $\mathscr{O}_X(D) = \mathscr{O}_X \cdot f$. As $\mathscr{O}_X(D)$ is invertible, f is not zero-divisor, we have $f \in \Gamma(X, \mathscr{K}_X)^\times$ and $D = \operatorname{div}(f^{-1})$.

Now we prove the statement. $D_1 \sim D_2$ iff $D_1 - D_2$ is principal iff $\mathscr{O}_X(D_1 - D_2) \cong \mathscr{O}_X$, recall (16.16),

$$\mathscr{O}_X(D_1 - D_2) \cong \mathscr{O}_X(D_1) \otimes_{\mathscr{O}_X} \mathscr{O}_X(D_2)^\vee \cong \mathscr{O}_X,$$

iff $\mathscr{O}_X(D_1) \cong \mathscr{O}_X(D_2)$, we win. $\square$

Proposition 16.5.1 can be also expressed by saying that the following is an exact sequence of abelian groups

$$\begin{aligned} 1 \longrightarrow \Gamma(X, \mathscr{O}_X)^\times \longrightarrow \Gamma(X, \mathscr{K}_X)^\times \longrightarrow \operatorname{Div}(X) &\xrightarrow{\delta} \operatorname{Pic}(X), \\ D &\longmapsto \mathscr{O}_X(D). \end{aligned} \tag{16.18}$$

It is easy to see that this exact sequence is the beginning of the exact cohomology sequence attached to the short exact sequence of abelian sheaves

$$1 \longrightarrow \mathscr{O}_X^\times \longrightarrow \mathscr{K}_X^\times \longrightarrow \mathscr{K}_X^\times / \mathscr{O}_X^\times \longrightarrow 1.$$

The second question is about the image of δ in $\operatorname{Pic}(X)$. Let $D \in \operatorname{Div}(X)$ be represented by $D = (U_i, f_i))i$, then $\delta(D) = [\mathscr{O}_X(D)] \in \operatorname{Pic}(X)$. Notice that $\mathscr{O}_X(D)|_{U_i} = \mathscr{O}_{U_i} \cdot f_i^{-1} \subseteq \mathscr{K}_X|_{U_i}$, so $\mathscr{O}_X(D)$ is an invertible $\mathscr{O}_X$-submodule of $\mathscr{K}_X$. Hence, there is a natural embedding $\mathscr{O}_X(D) \hookrightarrow \mathscr{K}_X$. Conversely, suppose $\mathscr{L}$ is an invertible $\mathscr{O}_X$-module and suppose there exists an embedding $\mathscr{L} \hookrightarrow \mathscr{K}_X$. Choose an open covering $\mathfrak{U} = \{U_i \hookrightarrow X\}$ such that $\mathscr{L}$ is trivial on each U_i. Then $\mathscr{L}|_{U_i} = \mathscr{O}_{U_i} e_i$ for some $e_i \in \Gamma(U_i, \mathscr{L})$. Let $f_i \in \Gamma(U_i, \mathscr{K}_X)$ be the image of e_i in $\mathscr{K}_X|_{U_i}$ under $\mathscr{L}|_{U_i} \hookrightarrow \mathscr{K}_X|_{U_i}$. By injectivity, in fact, $f_i \in \Gamma(U_i, \mathscr{K}_X^\times)$. Let D be a Cartier divisor given by $(U_i, f_i^{-1})_i$, then $\mathscr{O}_X(D) \cong \mathscr{L}$, i.e., $[\mathscr{L}]$ lies in the image of δ. Consequently,

$$\boxed{\operatorname{Im}(\delta) = \{[\mathscr{L}] \in \operatorname{Pic}(X) : \text{there exists an embedding } \mathscr{L} \hookrightarrow \mathscr{K}_X\}.}$$

In general, δ is not surjective, even for noetherian schemes. But very often this will be the case:

Proposition 16.5.2

Let X be a scheme. We assume that

(1) X is integral, or

(2) X is a locally Noetherian scheme that contains an affine open subscheme U that is scheme-theoretic

dense in X.

Then the homomorphism $\operatorname{Div}(X) \to \operatorname{Pic}(X), D \mapsto \mathscr{O}_X(D)$ *is surjective and therefore induces an isomorphism*

$$\operatorname{DivCl}(X) \xrightarrow{\sim} \operatorname{Pic}(X).$$

Proof ***First step.*** Let U be an affine open subscheme that is scheme-theoretic dense in X (under hypothesis (1) we can take for U any non-empty open affine subscheme). Let $\mathscr{L}$ be an invertible $\mathscr{O}_X$-module. We first claim that every embedding $s : \mathscr{L}|_U \hookrightarrow \mathscr{K}_X|_U$ can be extended to a unique embedding $\widetilde{s} : \mathscr{L} \hookrightarrow \mathscr{K}_X$. The open subsets $V \subseteq X$ with $\mathscr{L}|_V \cong \mathscr{O}_V$ form a basis for the topology on X and $U \cap V$ is scheme-theoretic dense in V for all V. Therefore we may assume that $\mathscr{L} = \mathscr{O}_X$. Then $s : \mathscr{O}_U : \mathscr{K}_X|_U$ can be considered as a section $t \in \Gamma(U, \mathscr{K}_X)$. The injectivity of s is equivalent to $t \in \Gamma(U, \mathscr{K}_X)^{\times}$. By Proposition 15.2.5, t may be considered as a rational function on U. But a rational function on U is the same as a rational function on X (since U is scheme-theoretic dense) and therefor restriction defines an isomorphism $\Gamma(X, \mathscr{K}_X) \xrightarrow{\sim} \Gamma(U, \mathscr{K}_X)$. This shows our claim.

Second step. Our hypothesis implies that we may assume that $X = \operatorname{Spec} A$ is affine, where A is a domain or a Noetherian ring. Let $\mathscr{L}$ be an invertible $\mathscr{O}_X$-module. Then $\mathscr{L} = \widetilde{M}$ where M is locally free A-module of rank 1 ([Sta25, Definition 00NW]). Let $S \subseteq A$ be the set of regular elements. We have $\Gamma(X, \mathscr{K}_X) = S^{-1}A$. We have to show that there exists a linear injective homomorphism $M \to S^{-1}A$. Recall [Sta25, Lemma 02M9], it suffices to show that $S^{-1}A$ is a semi-local ring (a ring with finitely many maximal ideals). If A is an integral domain, then $S^{-1}A = K(A)$, it is clearly semi-local ring. If A is Noetherian, S is the complement of the union of the finitely many associated ideals of A (Proposition 7.6.11 and Corollary 7.6.2). The prime ideals of $S^{-1}A$ are the prime ideals which contained in $\bigcup_{\mathfrak{p} \in \operatorname{Ass}(A)} \mathfrak{p}$, by Prime avoidance 13.2.5, the prime ideal of $S^{-1}A$ must contained in a associated ideal, which implies $S^{-1}A$ has finitely many maximal ideals, i.e., $S^{-1}A$ is a semi-local ring. Hence, M is a free A-module, so we have $M \hookrightarrow S^{-1}M \cong S^{-1}A$, which implies that $\mathscr{L} \hookrightarrow \mathscr{K}_X$. □

There are plenty of situations where Proposition 16.5.2 may be applied.

Corollary 16.5.1

Let X is a Noetherian and reduced scheme. Then $D \mapsto \mathscr{O}_X(D)$ induces an isomorphism $\operatorname{DivCl}(X) \xrightarrow{\sim} \operatorname{Pic}(X)$.

Proof As X is Noetherian, there are only finitely many irreducible components $Z_1, \dots, Z_r$. Choose within each irreducible component Z_i a non-empty open affine subset $U_i \subseteq Z_i$ that does not meet any of the other irreducible components. Then the union $U := \bigcup_i U_i = \coprod_i U_i$ is affine and dense in X. We now show that U is scheme-theoretic dense in X. We may assume $X = \operatorname{Spec} A$ and U is dense in X. Let $f \in \Gamma(V, \mathscr{O}_X)$ such that $f|_{U \cap V} = 0$. We may assume that $V = \operatorname{Spec} A$ where A is reduced (since X is reduced). Since $f|_{U \cap V} = 0$, we obtain $U \cap V \subseteq V(f) = \operatorname{Spec} A/(f)$. Since U is dense in X, $U \cap V$ is dense in V, hence,

$$\operatorname{cl}_V(U \cap V) = V = \operatorname{Spec} A \subseteq V(f),$$

which implies that $V(f) = \operatorname{Spec} A$. Hence, $f \in \mathfrak{N}(A) = 0$, i.e., $f|_V = 0$, which implies that U is scheme-theoretic dense. By Proposition 16.5.2, we win. □

Support of a Divisor

The support of a divisor is simply the support as a section of the abelian sheaf $\mathscr{K}_X^\times/\mathscr{O}_X^\times$:

Definition 16.5.11

The support of a divisor D on a scheme X is the closed subspace

$$\operatorname{Supp}(D) := \{x \in X : D_x \neq 1\}.$$

Here, D_x denotes the image of $D \in \Gamma(X, \mathscr{K}_X^\times/\mathscr{O}_X^\times)$ in the stalk $(\mathscr{K}_X^\times/\mathscr{O}_X^\times)_x$.

Remark 16.19 In other words, if D is represented by $(U_i, f_i)_i$, then a point $x \in X$ lies in $\operatorname{Supp}(D)$ if and only if $(f_i)_x \notin \mathscr{O}_{X,x}^\times$ for all (or, equivalently, for one) index i with $U_i \ni x$. As the support of any section of an abelian sheaf is closed, $\operatorname{Supp}(D)$ is closed in X.

Proposition 16.5.3

Let D be an effective Cartier divisor, then $\operatorname{Supp}(D)$ is the underlying closed subspace of the closed subscheme corresponding to D.

Proof Let D be an effective Cartier divisor on X, represented by $(U_i, f_i)_i$. Then $D \cap U_i = V(f_i)$. For each $x \in U_i$, $x \in D^{\text{top}}$ iff $\mathscr{I}_X(D)_x \subseteq \mathfrak{m}_x$ where $\mathfrak{m}_x$ is maximal ideal of $\mathscr{O}_{X,x}$. Note that $\mathscr{I}_X(D)_x = \mathscr{O}_{X,x}(f_i)_x$, $(f_i)_x \in \mathfrak{m}_x$, $\mathscr{I}_X(D)_x \subseteq \mathfrak{m}_x$ iff $(f_i)_x \in \mathfrak{m}_x$ iff $(f_i)_x \notin \mathscr{O}_{X,x}^\times$ iff $x \in \operatorname{Supp}(D)$. $\square$

Let D be a Cartier divisor on a scheme X, represented by $(U_i, f_i)_i$. Let V be any open subset of X. Then $\Gamma(V, \mathscr{O}_X(D))$ consists of those sections $s \in \Gamma(V, \mathscr{K}_X)$ such that $sf_i \in \Gamma(U_i \cap V, \mathscr{O}_X)$ for all i by the construction of $\Gamma(V, \mathscr{O}_X(D))$. Therefore if U_{eff} is the locus of points where D is effective (i.e., the open subset of $x \in X$ such that $(f_i)_x \in \mathscr{O}_{X,x}$ for those i with $U_i \ni x$), then $1 \in \Gamma(U_{\text{eff}}, \mathscr{K}_X)$ is a section $s_D \in \Gamma(U_{\text{eff}}, \mathscr{O}_X(D))$. This section is called the **canonical section of (the line bundle attached to)** D.

If $U = X \setminus \operatorname{Supp}(D)$, then $U \subseteq U_{\text{eff}}$ and $s_D|_U$ defines an isomorphism $\mathscr{O}_X(D)|_U \xrightarrow{\sim} \mathscr{O}_X|_U$.

Proposition 16.5.4

Let X be a scheme, let $\mathscr{L}$ be a line bundle on X, and let $R_{\mathscr{L}}$ be the set of $s \in \Gamma(X, \mathscr{L})$ that are regular. Then there is a natural bijection

$$\left\{\begin{matrix}\textit{effective Cartier divisors } D\\ \textit{such that } \mathscr{O}_X(D) \cong \mathscr{L}\end{matrix}\right\} \longleftrightarrow R_{\mathscr{L}}/\sim,$$

where $s \sim s'$ if there exists an $a \in \Gamma(X, \mathscr{O}_X^\times)$ with $s' = as$.

Proof For an effective divisor D and an isomorphism $\alpha : \mathscr{O}_X(D) \xrightarrow{\sim} \mathscr{L}$, $1 \in \Gamma(X, \mathscr{K}_X)$ is the canonical section of D, set $s = \alpha(1)$. We want to show that $s \in \Gamma(X, \mathscr{L})$ is a regular element. Note that 1 can be regarded as $1 : \mathscr{O}_X \to \mathscr{O}_X(D)$, since 1 is regular, $1 : \mathscr{O}_X \hookrightarrow \mathscr{O}_X(D)$ is injective. Hence, $\alpha(1) : \mathscr{O}_X \hookrightarrow \mathscr{O}_X(D) \xrightarrow{\sim} \mathscr{L}$ is injective, which implies $s = \alpha(1)$ is regular, i.e., $s \in R_{\mathscr{L}}$. Now, if there exists another isomorphism $\alpha' : \mathscr{O}_X(D) \xrightarrow{\sim} \mathscr{L}$, denote $s' = \alpha'(1)$, then $\alpha' \circ \alpha^{-1}$ is an automorphism of $\mathscr{L}$, hence $s' = as$ for some $a \in \Gamma(X, \mathscr{O}_X^\times)$, i.e., $s \sim s'$. Hence, this yields a well-defined map from the left hand side to the right hand side.

Conversely, let $s \in R_{\mathscr{L}}$. Choose an open covering $\mathfrak{U} = \{U_i \hookrightarrow X\}$ of X and isomorphisms $\mathscr{L}|_{U_i} \cong \mathscr{O}_X|_{U_i}$. Then the image of $s|_{U_i}$ define local equations f_i for an effective divisor D, which depends only on the equivalence class of s. The section s then defines a monomorphism $\mathscr{O}_X \hookrightarrow \mathscr{L}$ which extends by definition to an isomorphism $\mathscr{O}_X(D) \xrightarrow{\sim} \mathscr{L}$. $\square$

Lemma 16.5.4

Let X be a locally Noetherian scheme and let D be a Cartier divisor on X. Then $\operatorname{codim}_X(\operatorname{Supp}(D)) \geq 1$, i.e., for all $z \in \operatorname{Supp}(D)$, we have $\dim(\mathscr{O}_{X,z}) \geq 1$.

Proof Let $\eta \in X$ be a point with $\dim(\mathscr{O}_{X,\eta}) = 0$. Then $\mathscr{O}_{X,\eta}$ is a local Artin ring and thus every regular element in $\mathscr{O}_{X,\eta}$ is invertible [AM18, Proposition 8.6]. Thus $\mathscr{O}_{X,\eta} = \operatorname{Frac}(\mathscr{O}_{X,\eta}) = \mathscr{K}_{X,\eta}$. This shows that the germ $D_\eta \in \mathscr{K}_{X,\eta}^\times / \mathscr{O}_{X,\eta}^\times = 1$, a contradiction! □

Normal (line) bundles to effective Cartier divisors

If D is an effective Cartier divisor on a scheme X, then the restriction of the invertible sheaf $\mathscr{O}_X(D)$ to D it self is often called the **normal (line) bundle to** D (although perhaps it should be called the normal invertible sheaf). We denote it

$$\boxed{\mathscr{N}_{D/X} := \mathscr{O}_X(D)|_D}.$$

The motivation for this language is that if X and D are smooth complex varieties, the normal bundle in this algebro-geometric sense agrees with the normal bundle in the differential-geometric sense. This interpretation breaks down in more "singular" situations, but the intuition is useful even then. We will define the normal bundle (and normal sheaf) in a more general setting in §**??**.

16.5.4 Cartier Divisors and Weil divisors

Below we will discuss the relationship between Cartier divisors and Weil divisors (§16.4). In this part, we will first redefine some notation that appeared in Section 16.4, and then generalize Weil divisors in more general case.

Let X be a *Noetherian scheme*. Let $C \subseteq X$ be a closed irreducible subset with generic point ξ. We sometimes call $\mathscr{O}_{X,C} := \mathscr{O}_{X,\xi}$ the **local ring of** X **at** C. Recall that

$$\operatorname{codim}_X C = \dim \mathscr{O}_{X,C} \tag{16.19}$$

is called the codimension of C in X. More generally, for an arbitrary closed subset Z of X, we have $\operatorname{codim}_X(Z) = \inf_C \operatorname{codim}_X(C)$, where C runs through the set of irreducible components of Z. Thus Z is of codimension zero if and only if Z contains an irreducible component of X.

Note *For an integer $k \geq 0$, we denote by X^k the set of closed integral subschemes of codimension k. The free abelian group $\mathbb{Z}^{X^k}$ is denoted by $Z^k(X)$. Thus elements of $Z^k(X)$ are finite linear combinations $\sum_C n_C[C]$, where C runs through X^k.*

Definition 16.5.12 (Weil divisor, Prime Weil divisor, effective Weil divisor)

*An element of $Z^1(X)$ is called a **Weil divisor**. The elements of X^1 are called **prime Weil divisor**. We denote by $Z_+^1(X)$ the set of $\sum_{C \in X^1} n_C[C] \in Z^1(X)$ such that $n_C \geq 0$ for all C, and call its elements **effective Weil divisors**.*

We will now define a group homomorphism

$$\operatorname{cyc} : \operatorname{Div}(X) \longrightarrow Z^1(X). \tag{16.20}$$

For this we will have to define for a prime Weil divisor C the order $\operatorname{ord}_C(f)$ of a meromorphic function $f \in \Gamma(U, \mathscr{K}_X)$ along C. Thus assume that U contains the generic point of C. If $\mathscr{O}_{X,C}$ is a discrete valuation ring (e.g., if X is normal; see Theorem 14.5.5), the germ f_C is a nonzero element in $K := \mathrm{Frac}(\mathscr{O}_{X,C}) = K(\mathscr{O}_{X,C})$ and we set

$$\mathrm{ord}_C(f) := \mathrm{val}_C(f_C),$$

where val defined in Definition 14.5.7.

In general, $\mathscr{O}_{X,C}$ is only a local Noetherian ring of dimension 1 (may not be a DVR) and we use the following lemma to define the order of an element $f \in \mathrm{Frac}(\mathscr{O}_{X,C})^\times$.

Lemma 16.5.5

Let A be a Noetherian local ring of dimension 1. Write $f \in \mathrm{Frac}(A)^\times$ as $f = ab^{-1}$ for regular elements $a, b \in A$ and set

$$\mathrm{ord}_A(f) := \mathrm{length}_A(A/(a)) - \mathrm{length}_A(A/(b)).$$

Then $\mathrm{ord} : \mathrm{Frac}(A)^\times \to \mathbb{Z}$ is a well defined group homomorphism and $A^\times \subseteq \mathrm{Ker}(\mathrm{ord})$.

Remark 16.20

(1) Recall the length of object (Definition 7.5.2).

(2) If A is a DVR, this definition is agree with discrete valuation. More precisely, for $a \in A$, we may write $a = u\pi^n$ where π is uniformizer, then

$$\mathrm{length}_A A/(a) = \mathrm{length}_A A/(\pi^n) = n = \mathrm{val}\, a.$$

Proof Let $a \in A$ be a regular element, then a is not zero-divisor. By Proposition 13.3.8, $\dim A/(a) = \dim A - 1 = 0$ and thus $A/(a)$ is a local Artin ring and hence of finite length [AM18, Prop. 6.8]. If $b \in A$ is a second regular element, multiplication with b yields an isomorphism $A/(a) \cong bA/(ab)$. The exact sequence

$$0 \longrightarrow bA/(ab) \longrightarrow A/(ab) \longrightarrow A/(b) \longrightarrow 0$$

therefore shows that $\mathrm{length}\, A/(ab) = \mathrm{length}\, A/(a) + \mathrm{length}\, A/(b)$.

Now we show that ord is well defined. If $\frac{a}{b} = \frac{c}{d}$ in $\mathrm{Frac}(A)^\times$, then $ad = cb$ where a, b, c, d are regular elements. Hence, $A/(ad) = A/(cb)$, so

$$\mathrm{ord}_A(\tfrac{a}{b}) = \mathrm{length}_A A/(a) - \mathrm{length}_A A/(b) = \mathrm{length}_A A/(c) - \mathrm{length}_A A/(d) = \mathrm{ord}_A(\tfrac{c}{d}),$$

which implies that ord is a well-defined group homomorphism. The fact that $\mathrm{ord}(u) = 0$ for all units $u \in A^\times$ is then clear. □

To define cyc (16.20), let $D \in \mathrm{Div}(X)$ be a Cartier divisor represented by (U_i, f_i). For $C \in X^1$ a prime Weil divisor we choose an index i such that the generic point η_C of C is contained in U_i. Let $f \in \mathscr{K}_{X,\eta_C} = \mathrm{Frac}(\mathscr{O}_{X,C})$ be the germ of f_i at η_C. Then f does not depend on the choice of the presentation (U_i, f_i) (choose U_i which contained η_C) or on i up to a unit of $\mathscr{O}_{X,C}$. Therefore

$$\mathrm{ord}_C(D) := \mathrm{ord}_{\mathscr{O}_{X,C}}(f) \in \mathbb{Z} \tag{16.21}$$

depends only on D and C. The integer $\mathrm{ord}_C(D)$ is called the **vanishing order of D at C**. If $\mathrm{ord}_C(D) \geq 0$, we also say that D has a **zero of order** $\mathrm{ord}_C(D)$ in C, and if $\mathrm{ord}_C(D) < 0$, we say that D has a **pole of order** $-\mathrm{ord}_C(D)$ in C.

If C is not contained in the support of D, then we have $\mathrm{ord}_C(D) = 0$. By Lemma 16.5.4 one has $\mathrm{codim}_X(\mathrm{Supp}\, D) \geq 1$. Therefore, every $C \in X^1$ with $C \subseteq \mathrm{Supp}\, D$ is an irreducible component. As X is Noetherian, and hence $\mathrm{Supp}\, D$ is a Noetherian topological space, there are (for a given divisor D) only finitely many $C \in X^1$ such that $\mathrm{ord}_C(D) \neq 0$. Now we define

$$\mathrm{cyc} : \mathrm{Div}(X) \longrightarrow Z^1(X), \qquad D \longmapsto \sum_{C \in X^1} \mathrm{ord}_C(D)[C]. \tag{16.22}$$

Lemma 16.5.5 shows that cyc is a group homomorphism.

If $f \in \Gamma(X, \mathscr{K}_X)^\times$ is an invertible rational function, we call $\mathrm{ord}_C(f) := \mathrm{ord}_C(\mathrm{div}(f))$ the **vanishing order of f at C** and write $\mathrm{cyc}(f)$ instead of $\mathrm{cyc}(\mathrm{div}(f))$.

Definition 16.5.13 (Principal, class group)

*A Weil divisor is called **principal** if it is of the form $\mathrm{cyc}(f)$ for some $f \in \Gamma(X, \mathscr{K}_X)^\times$. The principal Weil divisors form a subgroup $Z^1_{\mathrm{princ}}(X)$ of $Z^1(X)$. We denote the quotient by*

$$\mathrm{Cl}(X) := Z^1(X)/Z^1_{\mathrm{princ}}(X). \tag{16.23}$$

The homomorphism cyc induces a homomorphism $\mathrm{DivCl}(X) \to \mathrm{Cl}(X)$.

In general cyc is neither injective nor surjective.

Theorem 16.5.1

Let X be a Noetherian scheme.

(1) If X is normal, the homomorphism $\mathrm{cyc} : \mathrm{Div}(X) \to Z^1(X)$ is injective. In particular, it induces an injective homomorphism $\mathrm{DivCl}(X) \hookrightarrow \mathrm{Cl}(X)$.

(2) If X is factorial, the homomorphism $\mathrm{cyc} : \mathrm{Div}(X) \to Z^1(X)$ is bijective. In particular, it induces an isomorphism $\mathrm{DivCl}(X) \xrightarrow{\sim} \mathrm{Cl}(X)$.

Proof This is diagram (16.8). □

Remark 16.21 Theorem 16.5.1 also has a converse, see **goertz2020algebraicgeometry1**.

Remark 16.22 Let X be an integral Noetherian factorial scheme with function field $K(X)$. Combining Theorem 16.5.1 and §16.5.3 we have an isomorphism of Abelian groups

$$Z^1(X) \xrightarrow{\sim} \{\text{Invertible fractional ideals of } \mathscr{O}_X\} \tag{16.24}$$

which induces by Corollary 16.5.1 an isomorphism

$$\mathrm{Cl}(X) \xrightarrow{\sim} \mathrm{Pic}(X). \tag{16.25}$$

The isomorphism (16.24) can be described as follows. Let $\sum_C n_C[C] \in Z^1(X)$. Then the corresponding invertible fractional ideal $\mathscr{L}$ is given by (for any $U \subseteq X$ open)

$$\Gamma(U, \mathscr{L}) := \{f \in K(X) : \mathrm{val}_C(f) \geq -n_C, \forall C \in X^1 \text{ with } C \cap U \neq \varnothing\}.$$

Here val_C is the normalized discrete valuation given by the DVR $\mathscr{O}_{X,C}$ on $\mathrm{Frac}(\mathscr{O}_{X,C}) = K(X)$. Thus $\Gamma(X, \mathscr{L})$ consists of those meromorphic functions whose order at C at least $-n_C$.

If $U \subseteq X$ is an open subscheme of a Noetherian scheme X, we define a restriction homomorphism $Z^1(X) \to Z^1(U)$ by $\sum_C n_C[C] \mapsto \sum_C n_C[C \cap U]$, where the second sum is only indexed by those $C \in X^1$ such that $C \cap U \neq \varnothing$. For those C, we have $\mathscr{O}_{U, C\cap U} = \mathscr{O}_{X,C}$. This shows that if D is a Cartier divisor, the restriction of $\mathrm{cyc}(D)$ to U is equal to $\mathrm{cyc}(D|_U)$. As the restriction of principal Cartier divisors are again principal Cartier divisors, this homomorphism induces a restriction homomorphism $\mathrm{Cl}(X) \to \mathrm{Cl}(U)$. If $U \subseteq X$ is scheme-theoretic dense, this restriction is surjective and one can describe its kernel:

Proposition 16.5.5 (Generalize Proposition 16.4.7)

Let X be a Noetherian scheme and let Z be a closed subscheme that does not contain any irreducible component of X. Let $Z_1, \ldots, Z_r$ be the irreducible components of Z that have codimension 1 in X. Set $U = X \setminus Z$. Suppose that U is scheme-theoretic dense in X. Then we have an exact sequence

$$\bigoplus_{i=1}^{r} \mathbb{Z} \cdot [Z_i] \longrightarrow \mathrm{Cl}(X) \longrightarrow \mathrm{Cl}(U) \longrightarrow 0.$$

Remark 16.23 The hypothesis on Z imply that U is always dense in X. Hence U is automatically scheme-theoretic dense in X if X is reduced (reduced schemes not have embedding points) or, more generally, if no embedded component of X is contained in Z.

Proof If C_U is a prime Weil divisor on U, its closure in X is a prime Weil divisor on X. This shows that $Z^1(X) \to Z^1(U)$ is surjective, in particular $\mathrm{Cl}(X) \to \mathrm{Cl}(U)$ is surjective.

The kernel of $Z^1(X) \to Z^1(U)$ is equal to $\bigoplus_{i=1}^{r} \mathbb{Z}[Z_i]$. In particular, $[Z_i]$ is in the kernel of $\mathrm{Cl}(X) \to \mathrm{Cl}(U)$.

Conversely, let $\sum_C n_C[C] \in Z^1(X)$ such that $\sum_C n_C[C \cap U]$ is the divisor of an invertible meromorphic function f_U on U. As U is scheme-theoretic dense, f_U can be uniquely extended to a meromorphic function f on X (Proposition 15.2.5). Then $(\mathrm{div}(f) - \sum_C n_C[C])_U = \mathrm{div}(f_U) - \sum_C n_C[C \cap U]$ is in the kernel of $Z^1(X) \to Z^1(U)$, which implies that $\sum_C n_C[C] = \mathrm{div}(f) - \sum_i k_i[Z_i]$. Hence, in $\mathrm{Cl}(X)$, $\sum_C n_C[C] = \sum_i k_i[Z_i]$. Therefore,

$$\mathrm{Ker}\,(\mathrm{Cl}(X) \to \mathrm{Cl}(U)) = \bigoplus_{i=1}^{r} \mathbb{Z}[Z_i].$$

$\square$

Under the hypotheses of Proposition 16.5.5, we obtain a commutative diagram

$$\begin{tikzcd}
\mathrm{Pic}(X) \arrow[d] & \mathrm{DivCl}(X) \arrow[l, "\mathscr{O}_X(-)"'] \arrow[r, "\mathrm{cyc}"] \arrow[d] & \mathrm{Cl}(X) \arrow[d] \\
\mathrm{Pic}(U) & \mathrm{DivCl}(U) \arrow[l, "\mathscr{O}_U(-)"'] \arrow[r, "\mathrm{cyc}"] & \mathrm{Cl}(U),
\end{tikzcd} \tag{16.26}$$

where the vertical arrows are the restriction maps. If X is factorial, then the horizontal maps are isomorphisms by Corollary 16.5.1 and Theorem 16.5.1. Therefore we obtain the following corollary.

Corollary 16.5.2

Let X be a Noetherian factorial scheme (e.g., if X is regular), and let $Z \subsetneq X$ be a closed subset of codimension at least 1. Let $U = X \setminus Z$ be the complement. Then $\mathrm{Pic}(X) \to \mathrm{Pic}(U), \mathscr{L} \mapsto \mathscr{L}|_U$ is surjective. If $\mathrm{codim}_X(Z) \geq 2$, this map is an isomorphism.

Proof By (16.26) and Proposition 16.5.5, we get the consequence immediately. $\square$

16.5.5 Inverse image of divisors

In general, although we can always form the inverse image of a line bundle alone a morphism of schemes (Proposition 15.1.1), the inverse image of a Cartier divisor on Y along a morphism $\pi : X \to Y$ of schemes is not well-defined. For example:

Example 16.12 Let $X = \operatorname{Spec} k$ and $Y = \operatorname{Spec} k[t]$, and $\pi : X \to Y$ given by $\pi^\sharp : k[t] \to k, t \mapsto 0$. Let $D = (X, t)$ be a principal Cartier divisor on Y, in naive viewpoint, the pullback $\pi^* D$ is given by $\pi^\sharp(t) = 0$.

But $0 \notin \Gamma(X, \mathscr{K}_X^\times)$, which implies, in the naive viewpoint, $\pi^* D$ is not well-defined.

The key question, given $\pi : X \to Y$, is whether the homomorphism $\mathscr{O}_Y \to \pi_* \mathscr{O}_X$ given by π extends to a homomorphism $\mathscr{K}_Y \to \pi_* \mathscr{K}_X$. Assume for a moment that this is the case. We obtain a homomorphism

$$\mathscr{K}_Y^\times / \mathscr{O}_Y^\times \longrightarrow \pi_* \mathscr{K}_X^\times / \pi_* \mathscr{O}_X^\times \longrightarrow \pi_* (\mathscr{K}_X^\times / \mathscr{O}_X^\times)$$

where second arrow given by the universal property of $\pi_* \mathscr{K}_X^\times \to \pi_* \mathscr{K}_X^\times / \pi_* \mathscr{O}_X^\times$, and hence a homomorphism

$$\pi^* : \operatorname{Div}(Y) = \Gamma(Y, \mathscr{K}_Y^\times / \mathscr{O}_Y^\times) \longrightarrow \Gamma(X, \mathscr{K}_X^\times / \mathscr{O}_X^\times) = \operatorname{Div}(X). \tag{16.27}$$

Definition 16.5.14 (Pullback a Cartier divisor)

*Let $\pi : X \to Y$ be a morphism of schemes such that the homomorphism $\pi^\sharp : \mathscr{O}_Y \to \pi_* \mathscr{O}_X$ given by π extends to a homomorphism $\mathscr{K}_Y \to \pi_* \mathscr{K}_X$. Let D be a Cartier divisor on Y. Then its image $\pi^* D$ under the map (16.27) is called **the pullback of** D **or the inverse image of** D.*

Clearly, π^* is a group homomorphism, i.e., $\pi^*(D_1 + D_2) = \pi^*(D_1) + \pi^*(D_2)$. One easily checks that $\boxed{\pi^* \mathscr{O}(D) \cong \mathscr{O}(\pi^* D)}$, i.e., that pullback of divisors is compatible with the pullback morphism $\pi^* : \operatorname{Pic}(Y) \to \operatorname{Pic}(X)$.

The following proposition gives two criteria when it is possible to form the inverse image of divisors.

Proposition 16.5.6

Suppose X is a reduced and locally Noetherian scheme. Let $\pi : X \to Y$ be a morphism of schemes such that each irreducible component of X dominates an irreducible component of Y. Then the homomorphism $f^\sharp : \mathscr{O}_Y \to \pi_* \mathscr{O}_X$ extends to a homomorphism $\mathscr{K}_Y \to \pi_* \mathscr{K}_X$.

Remark 16.24 One can show that the Proposition 16.5.6 still holds if the assumption that X is locally Noetherian is replaced by the assumption that the set of connected components of X is locally finite.

Proof Recall that $\mathscr{K}_X$ is the sheaf associated with the presheaf $U \mapsto S_U^{-1} \Gamma(U, \mathscr{O}_X)$, where $S_U \subseteq \Gamma(U, \mathscr{O}_X)$ is the multiplicative subset of regular sections (Definition 15.2.4). It is enough to show that the homomorphism $\mathscr{O}_Y \to \pi_* \mathscr{O}_X$ extends to a morphism between these presheaves, i.e., that for every open subset $V \subseteq Y$, the homomorphism $\pi_V^\sharp : \Gamma(V, \mathscr{O}_Y) \to \Gamma(\pi^{-1}(V), \mathscr{O}_X)$ induces a homomorphism between the corresponding rings of total fractions. This means that we have to show that $\pi_V^\sharp$ maps regular sections to regular sections. As being a regular section can be checked on stalks (Definition 15.2.3), it suffices to show that $\pi_x^\sharp : \mathscr{O}_{Y, f(x)} \to \mathscr{O}_{X,x}$ maps regular elements to regular elements for all $x \in X$.

By assumption, $\mathscr{O}_{X,x}$ is reduced, by Proposition 7.6.22, $\mathscr{O}_{X,x}$ has no embedded primes. Recall Proposition 7.6.11, the regular elements of $\mathscr{O}_{X,x}$ are precisely those which are not contained in a minimal prime ideal. Since every irreducible component of X dominates an irreducible component of Y, by Proposition 8.5.4, the inverse image of a minimal prime ideal under $\pi_x^\sharp$ is again a minimal prime ideal. We win. $\square$

For general, morphisms one can still define the pullback of Cartier divisors in a suitable subgroup of $\operatorname{Div}(Y)$ — roughly speaking, the subgroup where the above procedure works.

Corollary 16.5.3

Let $\pi : X \to Y$ be a morphism of schemes satisfying one of the hypotheses of Proposition 16.5.6. Let $Z \subseteq Y$ be an effective Cartier divisor. Then the inverse image $\pi^{-1}(Z)$ (as a subscheme) is an effective Cartier divisor in X and this divisor is the inverse image $\pi^*(Z)$ (as a divisor).

Proof We may assume that $Y = \operatorname{Spec} A$ and $X = \operatorname{Spec} B$ are affine and that $Z = V(t)$ for a regular element

$t \in A$. By Proposition 16.5.6, $\pi^\sharp(t)$ is regular in B. So $\pi^{-1}(Z) = V(\pi^\sharp(t))$ is an effective Cartier divisor which by definition equals the inverse image $\pi^* Z$. □

16.6 The Payoff: Many Fun Examples

16.6.1 Fun examples: Projective transformation

Proposition 16.6.1 (Automorphisms of projective space)

Let k be a field, and $n \geq 0$. Then there is a group isomorphism

$$\mathrm{PGL}_{n+1}(k) \xrightarrow{\sim} \mathrm{Aut}(\mathbb{P}^n_k),$$

where $\mathrm{PGL}_{n+1}(k) := \mathrm{GL}_{n+1}(k)/(k^\times \cdot I_{n+1})$.

Proof Clearly, every invertible matrix $A \in \mathrm{GL}_{n+1}(k)$ induces an automorphism of $\mathbb{P}^n_k$. Since scalar matrices induce the identity morphism, we obtain a map

$$\mathrm{PGL}_{n+1}(k) \longrightarrow \mathrm{Aut}(\mathbb{P}^n_k). \tag{16.28}$$

This is a group homomorphism, and it is clearly injective.

Recall 16.10 , we already know that $\mathrm{Pic}(\mathbb{P}^n_k) \cong \mathbb{Z}$ with generator $\mathscr{O}(1)$ or $\mathscr{O}(-1)$. Now, pick $\pi \in \mathrm{Aut}(\mathbb{P}^n_k)$, it induces an isomorphism $\pi^* : \mathrm{Pic}(\mathbb{P}^n_k) \to \mathrm{Pic}(\mathbb{P}^n_k)$. Hence, $\pi^*\mathscr{O}(1) = \mathscr{O}(1)$ or $\pi^*\mathscr{O}(1) \cong \mathscr{O}(-1)$. If $\pi^*\mathscr{O}(1) \cong \mathscr{O}(-1)$, we obtain that $\Gamma(\mathbb{P}^n_k, \pi^*\mathscr{O}(1)) = \Gamma(\mathbb{P}^n_k, \mathscr{O}(-1))$. But, as k-vector space, we have $\Gamma(\mathbb{P}^n_k, \pi^*\mathscr{O}(1)) \cong \Gamma(\mathbb{P}^n_k, \mathscr{O}(1))$. By Proposition 16.1.3, $\dim \Gamma(\mathbb{P}^n_k, \pi^*\mathscr{O}(1)) = \dim \Gamma(\mathbb{P}^n_k, \mathscr{O}(1)) = n+1$, but $\dim \Gamma(\mathbb{P}^n_k, \mathscr{O}(-1)) = 0$, a contradiction! So, $\pi^*\mathscr{O}(1) \cong \mathscr{O}(1)$. Hence, we obtain a map

$$\mathrm{Aut}(\mathbb{P}^n_k) \longrightarrow \mathrm{GL}(\Gamma(P, \mathscr{O}_{\mathbb{P}^1_k}(1)))/\mathrm{Aut}_k(\mathscr{O}_{\mathbb{P}^n_k}(1)). \tag{16.29}$$

By Lemma 15.1.7,

$$\mathrm{Hom}(\mathscr{O}(1), \mathscr{O}(1)) = \Gamma(\mathbb{P}^n_k, \mathscr{H}om(\mathscr{O}(1), \mathscr{O}(1))) = \Gamma(\mathbb{P}^n_k, \mathscr{O}(-1) \otimes \mathscr{O}(1)) = \Gamma(\mathbb{P}^n_k, \mathscr{O}) = k,$$

and therefore we obtain $\mathrm{Aut}_k(\mathscr{O}(1)) = k^\times \cdot \mathrm{id}$. Choose homogeneous coordinates $X_0, \ldots, X_n$ on $\mathbb{P}^n_k$ and identify $\Gamma(\mathbb{P}^n_k, \mathscr{O}(1))$ with $k[X_0, \ldots, X_n]_1 = \bigoplus_{i=0}^{n} kX_i$. Thus we can identify $\mathrm{GL}(\Gamma(P, \mathscr{O}_{\mathbb{P}^1_k}(1)))/\mathrm{Aut}_k(\mathscr{O}_{\mathbb{P}^n_k}(1))$ with $\mathrm{PGL}_{n+1}(k)$, and it is a straightforward computation to check that the map (16.29) is the inverse of the map (16.28). □

Automorphism of projective space are often called **projective transformations**. Because of Proposition 16.6.1, in their incarnation of metrices modulo scalars, projective transformations are also called **projective change of coordinates**. Proposition 16.6.1 will be useful later, especially for the case $n = 1$. In this case, these automorphisms are called **fractional linear transformations**. (For experts: Why was Proposition 16.6.1 not stated over an arbitrary base ring A ? Where does the argument go wrong in that case? For what rings A does the result still work?)

Proposition 16.6.2

$\mathrm{Aut}(\mathbb{P}^1_k)$ *is strictly* 3*-transitive on* k*-valued points, i.e., given two triplets* (p_1, p_2, p_3) *and* (q_1, q_2, q_3) *each of distinct* k*-valued points of* $\mathbb{P}^1$*, there is precisely one automorphism of* $\mathbb{P}^1$ *sending* p_i *to* q_i *(*$i = 1, 2, 3$*).*

Proof Each k-valued point can be regarded as a 1-dimensional subspace of k^2. Hence, p_1, p_2, p_3 are linear dependent. Assume that $p_3 = \alpha p_1 + \beta p_2$ where $k^2 = \mathrm{Span}_k\{p_1, p_2\}$, and let $p_i = [b_{i0} : b_{i1}]$. First we want to construct a linear transformation T such that $T(p_1) = e_1 = (1,0)$ and $T(p_2) = e_2 = (0,1)$. Let

$$T = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}$$

where $a_{ij} \in k$ such that $T(p_1) = e_1$ and $T(p_2) = e_2$. Then we obtain

$$\begin{pmatrix} b_{10} & b_{12} & 0 & 0 \\ 0 & 0 & b_{10} & b_{11} \\ b_{20} & b_{21} & 0 & 0 \\ 0 & 0 & b_{20} & b_{21} \end{pmatrix} \begin{pmatrix} a_{11} \\ a_{12} \\ a_{21} \\ a_{22} \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 1 \end{pmatrix}.$$

As p_1, p_2 are linear independent, above 4×4 matrix is invertible, hence T is solvable. Let $D = \operatorname{diag}\{\beta, \alpha\}$, then

$$DT(p_1) = \beta e_1, \quad DT(p_2) = \alpha e_2, \quad DT(p_3) = \alpha\beta(e_1 + e_2).$$

Hence, in $\mathbb{P}^1_k$ and regard $D, T \in \mathrm{PGL}_2(k)$, we have

$$DT(p_1) = [1:0], \quad DT(p_2) = [0:1], \quad DT(p_3) = [1:1].$$

Say $\varphi_p := DT \in \mathrm{PGL}_2(k)$. Similarly, we can construct $\varphi_q \in \mathrm{PGL}_2(k)$ such that $\varphi_q(q_1) = [1:0]$, $\varphi_q(q_2) = [0:1]$ and $\varphi_q(q_3) = [1:1]$. Let $\Phi := \varphi_q^{-1} \circ \varphi_p$, then $\Phi(p_i) = q_i$. It follows that there exists an automorphism of $\mathbb{P}^1$ sending p_i to q_i.

We next show uniqueness. If there exists another $\Psi \in \mathrm{Aut}(\mathbb{P}^1)$ such that $\Psi(p_i) = q_i$, then $\Psi^{-1}\Phi(p_i) = p_i$. Since $k^2 = \mathrm{Span}\{p_1, p_2\}$, we obtain $\Psi^{-1}\Phi = \mathrm{id}$, i.e., $\Phi = \Psi$, which implies uniqueness. □

✍ **Exercise 16.10** Solve these problems over an arbitrary field k.

(1) Find a linear fractional transformation $f(t) \in \mathrm{PGL}_2(k)$ that has order precisely 3 in $\mathrm{PGL}_2(k)$.

(2) Show that any two order 3 elements of $\mathrm{PGL}_2(k)$ are conjugate.

Proof ***For (1).*** We may regard $\mathbb{P}^1_k$ as $k \cup \{\infty\}$. Let $f(t) = \frac{1}{1-t}$ be a fractional transformation, then the corresponding matrix in $\mathrm{PGL}_2(k)$ is

$$A = \begin{pmatrix} 0 & 1 \\ -1 & 1 \end{pmatrix}.$$

Note that $f^3(t) = t$, it follows that $f(t) \in \mathrm{PGL}_2(k)$ has order precisely 3 in $\mathrm{PGL}_2(k)$.

For (b). Let $\sigma, \tau \in \mathrm{PGL}_2(k)$ be two elements of order 3. We wan to show there exists $g \in \mathrm{PGL}_2(k)$ such that $g\sigma g^{-1} = \tau$. Since $\sigma^3 = \mathrm{id}$, for any $p \in \mathbb{P}^1_k$ consider the orbit $\{p, \sigma(p), \sigma^2(p)\}$ where they are different points. Similarly, for $q \neq p$ in $\mathbb{P}^1_k$, consider orbit $\{q, \tau(q), \tau^2(q)\}$. By Proposition 16.6.2, there exits $g \in \mathrm{PGL}_2(k)$ such that

$$g(p, \sigma(p), \sigma^2(p)) = (q, \tau(q), \tau^2(q)).$$

One checks that $g\sigma g^{-1}$ and τ agree on the three distinct points $q, \tau(q), \tau^2(q)$, so $g\sigma g^{-1} = \tau$, we win. □

Proposition 16.6.3

Suppose $p_0, \ldots, p_{n+1}$ are $n+2$ distinct k-valued points of $\mathbb{P}^n_k$, no $n+1$ of which lie on a hyperplane. Then there is a unique projective transformation taking p_i $(0 \le i \le n)$ to $[0 : \cdots : 0 : 1 : 0 : \cdots : 0]$ (where the 1 is in the i-th position), and taking p_{n+1} to $[1 : \cdots : 1]$.

Proof Let $V = \mathrm{Span}_k\{e_0, \ldots, e_n\}$ be a k-vector space where $e_0, \ldots, e_n$ is the standard basis of V, then $\mathbb{P}^n_k = \mathbb{P}(V)$ The condition that no $n+1$ of the points lie on a hyperplane means any $n+1$ of the vectors $p_0, \ldots, p_{n+1}$ are linear independent. So, we may assume that $p_{n+1} = a_0 p_0 + \cdots + a_n p_n$ where $a_i \in k$.

Moreover, each $a_i \in k^\times$. If there exists $a_j = 0$, then $p_{n+1} \in \mathrm{Span}_k\{p_0, \dots, \widehat{p_j}, \dots, p_n\}$ which contradicts to the fact that $p_0, \dots, \widehat{p_j}, \dots, p_n, p_{n+1}$ are linear independent. Define a k-linear map $T : V \to V$ on the basis $p_0, \dots, p_n$ by

$$T(p_i) = a_i^{-1} e_i, \qquad 0 \le i \le n,$$

then $T(p_{n+1}) = e_0 + \dots + e_n$. Regard $T \in \mathrm{PGL}_{n+1}(k)$, then

$$\begin{aligned} T(p_i) &= [e_i] = [0 : \dots : 0 : 1 : 0 : \dots : 0] \quad 0 \le i \le n \\ T(p_{n+1}) &= [e_0 + \dots + e_n] = [1 : \dots : 1], \end{aligned}$$

as we desired.

We next show the uniqueness. If there exists another $S \in \mathrm{PGL}_{n+1}(k)$ such that

$$\begin{aligned} S(p_i) &= [e_i] = [0 : \dots : 0 : 1 : 0 : \dots : 0] \quad 0 \le i \le n \\ S(p_{n+1}) &= [e_0 + \dots + e_n] = [1 : \dots : 1], \end{aligned}$$

then

$$S^{-1}T(p_i) = p_i, \qquad 0 \le i \le n.$$

It follows that $S^{-1}T = I$, i.e., $S = T$. □

Proposition 16.6.4

Suppose X is a quasiprojective k-scheme (Definition 5.5.11), and $\pi : \mathbb{P}_k^n \to X$ is any morphism (over k). Then either the image of π has dimension n, or π contracts $\mathbb{P}_k^n$ to a point. In particular, there are no nonconstant maps from projective space to a smaller-dimensional quasi-projective variety.

Proof Let $\pi : \mathbb{P}_k^n \to X$ be a morphism over k, where X is quasiprojective over k. Let $\overline{\mathrm{k}}$ be the algebraic closure of k. After base change from k to $\overline{\mathrm{k}}$, we obtain a morphism $\pi_{\overline{\mathrm{k}}}\mathbb{P}_{\overline{\mathrm{k}}}^n \to X_{\overline{\mathrm{k}}}$. We claim the it is enough to prove the consequence for $\pi_{\overline{\mathrm{k}}}$. The main point is that passing from k to $\overline{\mathrm{k}}$ does not change the dimension of the image. Let $Z = \pi(\mathbb{P}_k^n) \subseteq X$ be the image of π, equipped with the reduced induced closed subscheme structure. Since $\mathbb{P}_k^n$ is proper over k (Theorem 12.4.1), and X is separated over k (Corollary 12.2.3), by Corollary 12.4.1, $\pi : \mathbb{P}_k^n \to X$ is a proper morphism. Proper morphisms are closed maps, and therefore the image $Z = \pi(\mathbb{P}_k^n)$ is a closed subset of X. Thus we may regard π as a surjective morphism onto its image $\mathbb{P}_k^n \to Z$. Now base change this morphism to $\overline{\mathrm{k}}$. We obtain a surjective $\mathbb{P}_{\overline{\mathrm{k}}}^n = \mathbb{P}_k^n \times_k \overline{\mathrm{k}} \twoheadrightarrow Z_{\overline{\mathrm{k}}} = Z \times_k \overline{\mathrm{k}}$. Therefore that image of $\pi_{\overline{\mathrm{k}}} : \mathbb{P}_{\overline{\mathrm{k}}}^n \to X_{\overline{\mathrm{k}}}$ is exactly $Z_{\overline{\mathrm{k}}}$. So, we have reduced the question to comparing the dimension of Z and $Z_{\overline{\mathrm{k}}}$. By Proposition 13.1.8, the dimension of the image is unchanged after base change from k to $\overline{\mathrm{k}}$. Now, suppose the desired result has been proved over algebraically closed fields. Applying it to $\pi_{\overline{\mathrm{k}}} : \mathbb{P}_{\overline{\mathrm{k}}}^n \to X_{\overline{\mathrm{k}}}$, we know that either $\dim \mathrm{Im}(\pi_{\overline{\mathrm{k}}}) = n$ or $\pi_{\overline{\mathrm{k}}}$ is constant. In the first case,

$$\dim \mathrm{Im}\, \pi = \dim \mathrm{Im}\, \pi_{\overline{\mathrm{k}}} = n.$$

In the second case, $\dim \mathrm{Im}\, \pi_{\overline{\mathrm{k}}} = 0$. Therefore $\dim \mathrm{Im}\, \pi = 0$. Since $\mathbb{P}_k^n$ is irreducible, its image under π is also irreducible. A zero-dimensional irreducible closed subset is a single point. Hence, π contracts $\mathbb{P}_k^n$ to a point.

Suppose $k = \overline{\mathrm{k}}$ is algebraically closed field. Since X is a quasiprojective k-scheme, then

$$X \xhookrightarrow{\text{open}} \mathrm{Proj}\, S_\bullet \xhookrightarrow{\text{closed}} \mathbb{P}_k^N$$

(by Proposition 10.2.3, it is a locally closed embedding), denote π be the composition

$$\pi : \qquad \mathbb{P}_k^n \xrightarrow{\pi} X \xhookrightarrow{\text{open}} \mathrm{Proj}\, S_\bullet \xhookrightarrow{\text{closed}} \mathbb{P}_k^N.$$

As $\mathrm{Pic}\, \mathbb{P}_k^n \cong \mathbb{Z}$, we may assume $\pi^* \mathscr{O}_{\mathbb{P}_k^N}(1) \cong \mathscr{O}_{\mathbb{P}_k^n}(d)$ for some d. As $\mathscr{O}_{\mathbb{P}_k^N}(1)$ is generated by global section ([Sta25, Definition 01AM]), since π^* is right exact, $\mathscr{O}_{\mathbb{P}_k^n}(d)$ is also generated by global section, hence $d \ge 0$.

If $d = 0$, then $\pi^*\mathscr{O}_{\mathbb{P}_k^N} \cong \mathscr{O}_{\mathbb{P}_k^n}$. Let $X_0, \dots, X_N$ be the standard homogeneous coordinate sections of $\mathscr{O}_{\mathbb{P}_k^N}(1)$, their pullback $\pi^*X_0, \dots, \pi^*X_N$ are the global section of $\pi^*\mathscr{O}_{\mathbb{P}_k^N}(1)$. As $\Gamma(\mathbb{P}_k^n, \mathscr{O}_{\mathbb{P}_k^n}) = k$, each section $\pi^*X_i \in k$. By Theorem 16.2.1, since $\pi^i X_i$ for all $0 \le i \le N$ define a morphism to $\mathbb{P}_k^N$, they have no common zero, i.e., not all π^*X_i are zero. Say $c_i := \pi^*X_i \in k$. Therefore, every point of $\mathbb{P}_k^n$ is mapped to $[c_0 : \cdots : c_N] \in \mathbb{P}_k^N$. So, π is constant. Thus, if π is nonconstant, we must have $d > 0$.

Assume $d > 0$. Let $Z = \pi(\mathbb{P}_k^n) \subseteq \mathbb{P}_k^N$. Let $r = \dim Z$, as $\dim \mathbb{P}_k^n = n$, we always have $r \le n$. We want to show that $r = n$. Assume $0 < r < n$. By Proposition 13.3.3, there exists $r + 1$ hypersurfaces H_i such that $Z \cap (H_1 \cap \cdots \cap H_{r+1}) = \varnothing$. Set $D_i = \pi^{-1}(H_i) \subseteq \mathbb{P}_k^n$. By Proposition 13.3.3, $Z \cap H_i \ne \varnothing$, so $D_i \ne \varnothing$. It may happen that some H_i contains Y, i.e., $D_i = \mathbb{P}_k^n$. But such H_i's are irrelevant, because if $Y \subseteq H_i$, then intersecting with H_i does not cut down Y. Hence, we can simply discard all hypersurfaces H_i that contain Y. So, we may assume that $D_i \ne \mathbb{P}_k^n$ for all i. Next, we want to show that each D_i is a hypersurface in $\mathbb{P}_k^n$. As H_i is a hypersurface in $\mathbb{P}_k^N$, say $H_i = V(F_i)$ where $\deg F_i = e_i$, then F_i can be regarded as a global section of $\mathscr{O}_{\mathbb{P}_k^N}(e)$, then $\pi^{-1}(H_i) = V(\pi^*F_i)$ where $\pi^*F_i \in \Gamma(\mathbb{P}_k^n, \pi^*\mathscr{O}_{\mathbb{P}_k^N}(e))$. Note that

$$\pi^*\mathscr{O}_{\mathbb{P}_k^N}(e) \cong \pi^*(\mathscr{O}_{\mathbb{P}_k^N}(1)^{\otimes e}) \cong \pi^*(\mathscr{O}_{\mathbb{P}_k^N}(1))^{\otimes e} \cong \mathscr{O}_{\mathbb{P}_k^n}(d)^{\otimes e} \cong \mathscr{O}_{\mathbb{P}_k^n}(de),$$

π^*F_i is a global section of $\mathscr{O}_{\mathbb{P}_k^n}(de)$, which implies that D_i is indeed a hypersurface. As $Z \cap (H_1 \cap \cdots \cap H_{r+1}) = \varnothing$, we have $D_1 \cap \cdots \cap D_{r+1} = \pi^{-1}(Z \cap (H_1 \cap \cdots \cap H_{r+1})) = \varnothing$. Since $r < n$, $r + 1 \le n$, by Proposition 13.3.3, $D_1 \cap \cdots \cap D_{r+1} \ne \varnothing$, a contradiction! Hence, $r = n$, which implies that $\dim \pi(\mathbb{P}_k^n) = n$, we win. □

✍ **Exercise 16.11 (⋆ For who have read §??, the double-stared section on group schemes)** Explain how GL_n acts (nontrivially!) on $\mathbb{P}^{n-1}$ (over $\mathbb{Z}$, or over a field of your choice).

Remark 16.25 Over an algebraically closed field $\overline{k}$, GL_n acts transitively on the closed points of $\mathbb{P}_{\overline{k}}^n$, and the stabilizer of the point $[1 : 0 : \cdots : 0]$ consists of the subgroup P of matrices with 0's in the first column below the first row. (*Side remark:* The point is better written as a column vector, so the GL_n-action can be interpreted as matrix multiplication in the usual way.) This suggests that $\mathbb{P}_{\overline{\mathrm{k}}}^{n-1}$ is the quotient GL_n / P. This is largely true; but we first would have to make sense of the notion of group quotient.

16.6.2 More fun examples

We can now actually calculate some class groups. First, a useful observation: Notice that you can restrict invertible sheaves on Y to any subscheme X, and this can be a handy way of checking that an invertible sheaf is not trivial. Effective Cartier divisors (Definition 16.5.7) sometimes restrict too: if you have an effective Cartier divisor on Y, then it restricts to a closed subscheme on X (need locally Noetherian), locally cut out scheme-theoretically by one equation. If you are fortunate and this equation doesn't vanish on any associated point of X (Proposition 7.6.11), then you get an effective Cartier divisor on X. You can check that the restriction of effective Cartier divisors corresponds to restriction of invertible sheaves.

Proposition 16.6.5 (A torsion Picard group)

Suppose that Y is a hypersurface in $\mathbb{P}_k^n$ corresponding to an irreducible degree d polynomial. Then $\mathrm{Pic}(\mathbb{P}_k^n \setminus Y) \cong \mathbb{Z}/(d)$.

Remark 16.26 For differential geometers: this is related to the fact that $\pi_1(\mathbb{P}_k^n \setminus Y) \cong \mathbb{Z}/(d)$.

Proof By Proposition 16.4.8, as $\mathbb{P}_k^n$ and $\mathbb{P}_k^n \setminus Y$ are factorial, $\mathrm{Pic}(\mathbb{P}_k^n) \cong \mathrm{Cl}(\mathbb{P}_k^n)$ and $\mathrm{Pic}(\mathbb{P}_k^n \setminus Y) \cong \mathrm{Cl}(\mathbb{P}_k^n \setminus Y)$. By Proposition 16.4.7, we obtain an exact sequence

$$0 \longrightarrow \{\mathscr{O}(n[Y]) : n \in \mathbb{Z}\} \longrightarrow \mathrm{Pic}(\mathbb{P}_k^n) \longrightarrow \mathrm{Pic}(\mathbb{P}_k^n \setminus Y) \longrightarrow 0.$$

As $Y = V(F)$ where F is an irreducible degree d polynomial, so F can be regarded as a global section of $\mathscr{O}(d)$, by Proposition 16.4.3 (or Proposition 16.5.4), we have $\mathscr{O}_{\mathbb{P}^n_k}([Y]) \cong \mathscr{O}_{\mathbb{P}^n_k}(d)$, hence

$$\{\mathscr{O}(n[Y]) : n \in \mathbb{Z}\} \cong \mathbb{Z} \cdot d,$$

so

$$\mathrm{Pic}(\mathbb{P}^n_k \setminus Y) \cong \mathrm{Pic}(\mathbb{P}^n_k)/\{\mathscr{O}(n[Y]) : n \in \mathbb{Z}\} \cong \mathbb{Z}/(d).$$

□

The next two propositions explore consequence of Proposition 16.6.5, and provide us with some examples promised in Exercise 6.11.

Proposition 16.6.6 (Generalizing Exercise 6.11)

Suppose that Y is a hypersurface in $\mathbb{P}^n$ corresponding to an irreducible degree $d > 1$ polynomials (so $\mathrm{Pic}(\mathbb{P}^n \setminus Y) \neq 0$), and let $H_0, \dots, H_n$ be the $n+1$ coordinate hyperplanes on $\mathbb{P}^n$. Then

(1) $\mathbb{P}^n \setminus Y$ is affine, and $\mathbb{P}^n - Y - H_i$ is a distinguished open subset of it;

(2) $\mathbb{P}^n - Y - H_i$ form an open cover of $\mathbb{P}^m \setminus Y$;

(3) $\mathrm{Pic}(\mathbb{P}^n - Y - H_i) = 0$;

(4) each $\mathbb{P}^n - Y - H_i$ is the Spec of a unique factorization domain, but $\mathbb{P}^n \setminus Y$ is not.

Thus the property of being a unique factorization domain is not an affine-local property — it satisfies only one of the two hypotheses of the Affine Communication Lemma 6.3.1.

Proof Let $S = k[x_0, x_1, \dots, x_n]$ and $Y = V_+(F) \subseteq \mathbb{P}^n$ where $F \in S$ is an irreducible homogeneous polynomial of degree $d > 1$. Let $H_i = V_+(x_i)$ be the i-th coordinate hyperplane. Put $U = \mathbb{P}^n \setminus Y$ and $U_i = \mathbb{P}^n \setminus (Y \cup H_i)$. So $U = D_+(F)$ and $U_i = D_+(F) \cap D_+(x_i)$. By Exercise 5.9 and Proposition 5.5.10, we obtain that $U = \mathrm{Spec}(S_F)_0$ is affine and $U_i = \mathrm{Spec}(S_{x_i F})_0$ is a distinguished open subset of U.

Note that $\mathbb{P}^n = \bigcup_i D_+(x_i)$, $\mathbb{P}^n \setminus Y = \bigcup_i D_+(F) \cap D_+(x_i) = \bigcup_i U_i$, which implies that U_i form an open cover of U.

On the affine chart $D_+(x_i) \cong \mathbb{A}^n$, let f_i be the dehomogenization of F with respect to x_i, i.e., $f_i = \frac{F}{x_i^d}$. Then $U_i = D_+(x_i) \setminus Y = D(f_i)$, it is an open subscheme of $\mathbb{A}^n$, by Example 16.9 , $\mathrm{Pic}(U_i) = 0$.

As $U_i = D(f_i)$, by Proposition 6.4.5, $(S_{x_i F})_0$ is UFD. Now, it remains to show that $\mathbb{P}^n \setminus Y$ is not a Spec of UFD. Suppose $\mathbb{P}^n \setminus Y$ is a Spec of UFD, say $\mathbb{P}^n \setminus Y = \mathrm{Spec}\, R$. Since R is UFD, $\mathrm{Pic}(\mathrm{Spec}\, R) = 0$ (see Remark 16.14), it contradicts to the fact that $\mathrm{Pic}(U) \cong \mathbb{Z}/(d)$ (Proposition 16.6.5). □

Proposition 16.6.7

Suppose Y is a hypersurface in $\mathbb{P}^n$ corresponding to an irreducible degree $d > 1$ polynomials, and let $H_0, \dots, H_n$ be the $n+1$ coordinate hyperplanes on $\mathbb{P}^n$. On $\mathbb{P}^n \setminus Y$, H_i (restricted to this open set) is an effective Cartier divisor that is not cut out by a single equation.

Proof We use the same notation in Proposition 16.6.6. On $\mathbb{P}^n$, the Cartier data of H_i is $(D_+(x_j), \frac{x_i}{x_j})_i$. As $\frac{x_i}{x_j} \in \Gamma(D_+(x_j), \mathscr{O}_{\mathbb{P}^n})$, H_i is an effective Cartier divisor. Restrict H_i to $U = \mathbb{P}^n \setminus Y$, H_i is also an effective Cartier divisor. Suppose, on U, H_i is given by Cartier data (U, f) where $f \in \Gamma(U, \mathscr{O}_U)$, i.e., H_i is principal effective Cartier divisor. Hence, $\mathscr{O}_U(H_i) \cong \mathscr{O}_U$. But, on $\mathbb{P}^n$, $\mathscr{O}_{\mathbb{P}^n}(H_i) = \mathscr{O}_{\mathbb{P}^n}(1)$, so $\mathscr{O}_U(H_i) = \mathscr{O}_{\mathbb{P}^n}(H_i)|_U = \mathscr{O}(1)|_U \not\cong \mathscr{O}_U$ (since $\mathrm{Pic}(U) = \mathbb{Z}/(d)$, $\mathscr{O}(1)|_U \in \mathrm{Pic}(U)$ corresponding to 1 and $\mathscr{O}_U$ corresponding to 0), a contradiction! □

✍ **Exercise 16.12** Show that $A := \mathbb{R}[x, y]/(x^2 + y^2 - 1)$ is not a unique factorization domain, but $A \otimes_{\mathbb{R}} \mathbb{C}$ is.

Proof Let $\mathbb{P}^1_{\mathbb{R}} = \operatorname{Proj} \mathbb{R}[s,t]$. From geometric intuition, we claim that $\operatorname{Spec} A \cong \mathbb{P}^1_{\mathbb{R}} \setminus V_+(s^2+t^2)$. In fact,

$$\begin{aligned}\mathbb{P}^1_{\mathbb{R}} \setminus V_+(s^2+t^2) =& D_+(s^2+t^2)\\ \cong& \operatorname{Spec}(\mathbb{R}[s,t]_{s^2+t^2})_0\\ =& \operatorname{Spec} \mathbb{R}[\tfrac{s^2}{s^2+t^2}, \tfrac{st}{s^2+t^2}, \tfrac{t^2}{s^2+t^2}].\end{aligned}$$

Set $a = \frac{s^2}{s^2+t^2}, b = \frac{st}{s^2+t^2}, c = \frac{t^2}{s^2+t^2}$, then $b^2 = ac$ and $a + c = 1$. Define $\varphi : \mathbb{R}[x,y] \to \mathbb{R}[a,b,c]$ by setting

$$(x,y) \longmapsto (a-c, 2b).$$

As $a = \frac{1+x}{2}, b = \frac{y}{2}$ and $c = \frac{1-x}{2}$, φ is surjective. Moreover, $\operatorname{Ker}\varphi = (x^2+y^2-1)$, we obtain $\mathbb{R}[x,y]/(x^2+y^2-1) \cong \mathbb{R}[\frac{s^2}{s^2+t^2}, \frac{st}{s^2+t^2}, \frac{t^2}{s^2+t^2}]$. Hence $\operatorname{Spec} A \cong \mathbb{P}^1_{\mathbb{R}} \setminus V_+(s^2+t^2)$, by Proposition 16.6.6, we know that A is not UFD.

Next, we want to show that $A \otimes_{\mathbb{R}} \mathbb{C}$ is UFD. In fact, $A \otimes_{\mathbb{R}} \mathbb{C} \cong \mathbb{C}[x,y]/(x^2+y^2-1)$. Let $u = x+iy$ and $w = x - iy$, then $uw = 1$ and

$$x = \frac{u+w}{2}, \qquad y = \frac{u-w}{2i}.$$

Hence,

$$\mathbb{C}[x,y]/(x^2+y^2-1) \cong \mathbb{C}[u,w]/(uw-1) \cong \mathbb{C}[u,u^{-1}].$$

As $\mathbb{C}[u,u^{-1}]$ is localization of $\mathbb{C}[u]$ and $\mathbb{C}[u]$ is UFD, we obtain that $\mathbb{C}[u,u^{-1}]$ is UFD. So, $A \otimes_{\mathbb{R}} \mathbb{C}$ is UFD. □

Example 16.13 (Picard group of $\mathbb{P}^1 \times \mathbb{P}^1$) Consider

$$X = \mathbb{P}^1_k \times_k \mathbb{P}^1_k \cong \operatorname{Proj} k[w,x,y,z]/(wz-xy)$$

a smooth quadric surface (see Figure 10.3 and Example 11.9). Then $\operatorname{Pic} X \cong \mathbb{Z} \oplus \mathbb{Z}$.

Proof Let $L = \{\infty\} \times_k \mathbb{P}^1_k \subseteq X$ and $M = \mathbb{P}^1_k \times_k \{\infty\} \subseteq X$, then $\operatorname{codim}_X L = \operatorname{codim}_X M = 1$. By Proposition 16.5.5, we obtain an exact sequence

$$\mathbb{Z}[L] \oplus \mathbb{Z}[M] \longrightarrow \operatorname{Cl}(X) \longrightarrow \operatorname{Cl}(U) \longrightarrow 0,$$

where $U = X \setminus (L \cup M)$. Note that

$$U = (\mathbb{P}^1_k \setminus \{\infty\}) \times_k (\mathbb{P}^1_k \setminus \{\infty\}) \cong \mathbb{A}^1_k \times_k \mathbb{A}^1_k \cong \mathbb{A}^2_k,$$

by Remark 16.14, $\operatorname{Cl}(U) = 0$. So, we obtain a surjective

$$\mathbb{Z}[L] \oplus \mathbb{Z}[M] \longrightarrow\!\!\!\!\!\rightarrow \operatorname{Cl}(X) \longrightarrow 0,$$

By Proposition 14.3.1, X is smooth, by Smoothness-Regularity Comparison Theorem 14.2.1, X is regular, by Auslander-Buchsbaum Theorem 14.8.3, we know that X is factorial, by (16.8), we know that $\operatorname{Pic} X \cong \operatorname{Cl} X$.

Next, we want to show that $\mathbb{Z}^2 \twoheadrightarrow \operatorname{Pic} X$ is an isomorphism. In fact, $\varphi : \mathbb{Z}^2 \twoheadrightarrow \operatorname{Pic} X$ given by

$$(a,b) \longmapsto \mathscr{O}_X(a[L] + b[M]).$$

Let $\varphi(a,b) = 0$, then

$$\mathscr{O}_X(a[L] + b[M]) \cong \mathscr{O}_X(L)^{\otimes a} \otimes \mathscr{O}_X(M)^{\otimes b} \cong \mathscr{O}_X.$$

We want to show that $a = b = 0$. Set $p_1 : X \to \mathbb{P}^1_k$ and $p_2 : X \to \mathbb{P}^1_k$, then $L = p_1^{-1}(\infty)$ and $M = p_2^{-1}(\infty)$. Say $\mathbb{P}^1_k = \operatorname{Proj} k[x_0, x_1]$. As $\infty = V_+(x_0)$, by Proposition 16.4.3, $\mathscr{O}_{\mathbb{P}^1_k}(1) \cong \mathscr{O}_{\mathbb{P}^1_k}(\operatorname{div} x_0) \cong \mathscr{O}_{\mathbb{P}^1_k}(V_+(x_0))$. Now, we want to compute $\mathscr{O}_X(L)$ and $\mathscr{O}_X(M)$. Since $L = p_1^{-1}(\infty)$ and $M = p_2^{-1}(\infty)$, we obtain that

$$\begin{aligned}\mathscr{O}_X(L) =& \mathscr{O}_X(p_1^{-1}(\infty)) \cong p_1^*\mathscr{O}_X(\infty) \cong p_1^*\mathscr{O}_{\mathbb{P}^1_k}(1),\\ \mathscr{O}_X(M) =& \mathscr{O}_X(p_2^{-1}(\infty)) \cong p_2^*\mathscr{O}_X(\infty) \cong p_2^*\mathscr{O}_{\mathbb{P}^1_k}(1).\end{aligned}$$

(Pullback of divisors is compatible with the pullback morphism.) Restrict $\mathscr{O}_X(L)$ to L, by Proposition 15.5.2,

$$\mathscr{O}_X(L)|_L \cong (p_1^*\mathscr{O}_{\mathbb{P}^1_k}(1))|_L \cong (p_1|_L)^*\mathscr{O}_{\mathbb{P}^1_k}(1)|_\infty \cong (p_1|_L)^*\mathscr{O}_{\operatorname{Spec}\kappa(\infty)} \cong \mathscr{O}_L.$$

Similarly, we have $\mathscr{O}_X(M)|_M \cong \mathscr{O}_M$. Restrict $\mathscr{O}_X(L)|_M$, as $p_1|_M : M \to \mathbb{P}^1_k$ is an isomorphism,

$$\mathscr{O}_X(L)|_M \cong (p_1^*\mathscr{O}_{\mathbb{P}^1_k}(1))|_M \cong (p_1|_M)^*\mathscr{O}_{\mathbb{P}^1_k}(1)|_{\mathbb{P}^1_k} \cong \mathscr{O}_M(1).$$

Similarly, we have $\mathscr{O}_X(M)|_L \cong \mathscr{O}_L(1)$. We next calculate a and b. Restrict $\mathscr{O}_X(a[L]+b[M])$ to L, we obtain

$$\mathscr{O}_X(a[L]+b[M])|_L \cong \mathscr{O}_X(L)|_L^{\otimes a} \otimes \mathscr{O}_X(M)|_L^{\otimes b} \cong \mathscr{O}_L^{\otimes a} \otimes \mathscr{O}_L(1)^{\otimes}b \cong \mathscr{O}_L(b) \cong \mathscr{O}_L,$$

so $b=0$. Similarly, $a=0$. Hence, φ is an isomorphism, i.e.,

$$\operatorname{Pic} X \cong \mathbb{Z} \otimes \mathbb{Z}.$$

□

Example 16.14 (Irreducible smooth project surjective surfaces (over k) can be birational but not isomorphism) $\mathbb{P}^2$ is birational equivalent to $\mathbb{P}^1 \times \mathbb{P}^1$ but not isomorphic to $\mathbb{P}^1 \times \mathbb{P}^1$.

Proof As $\mathbb{P}^1_k \times_k \mathbb{P}^1_k \cong \operatorname{Proj} k[w,x,y,z]/(wz-xy)$, $\mathbb{P}^1_k \times_k \mathbb{P}^1_k$ is integral. Since $\mathbb{A}^1_k \times_k \mathbb{A}^1_k$ is dense in $\mathbb{P}^1_k \times_k \mathbb{P}^1_k$, $\mathbb{A}^2_k$ is dense in $\mathbb{P}^2_k$, and $\mathbb{A}^1_k \times \mathbb{A}^1_k \cong \mathbb{A}^2_k$, by Proposition 8.5.5, $\mathbb{P}^2_k$ is birational equivalent to $\mathbb{P}^1_k \times_k \mathbb{P}^1_k$.

But $\mathbb{P}^2_k$ is not isomorphic to $\mathbb{P}^1_k \times_k \mathbb{P}^1_k$, since $\operatorname{Pic}(\mathbb{P}^2_k) \cong \mathbb{Z} \ncong \mathbb{Z} \oplus \mathbb{Z} \cong \operatorname{Pic}(\mathbb{P}^1_k \times_k \mathbb{P}^1_k)$. □

Remark 16.27 We will see in **??** that the Picard group of the "blown up plane" is $\mathbb{Z}^2$, but in **??** we will see that the blown up plane is not isomorphic to $\mathbb{P}^1 \times \mathbb{P}^1$, using a little more information in the Picard group.

This is unlike the case for curves: birational irreducible smooth projective curves (over k) must be isomorphic, as we will see in Theorem **??**. Nonetheless, any two surfaces are related in a simple way: if X and X' are projective, regular, and birational, then X can be sequentially blown up at judiciously chosen points, and X' can too, such that the two results are isomorphic (see [Har13, Thm. V.5.5]; blowing up will be discussed in Chapter **??**).

Example 16.15 (Picard group of the cone) Let $X = \operatorname{Spec} k[x,y,z]/(xy-z^2)$ a cone, where $\operatorname{char} k \neq 2$. (The characteristic hypothesis is not necessary for the result, but is included so you can use Exercise 6.5 to show normality of X.) Then $\operatorname{Pic} X = 0$ and $\operatorname{Cl} X \cong \mathbb{Z}/2$.

Proof We first show that $\operatorname{Cl} X \cong \mathbb{Z}/2$. By Exercise 6.5, X is normal. Set $Z = V(x) \subseteq X$, then Z is a prime divisor on X. By Proposition 16.4.7, we have an exact sequence

$$\mathbb{Z} \xrightarrow{1 \mapsto Z} \operatorname{Cl} X \longrightarrow \operatorname{Cl}(X \setminus Z) \longrightarrow 0.$$

We now calculate $\operatorname{Cl}(X \setminus Z)$. In fact, $X \setminus Z = D(x) \subseteq X$ and

$$\begin{aligned} D(x) &= \operatorname{Spec}(k[x,y,z]/(xy-z^2))_x \\ &\cong \operatorname{Spec} k[x,y,z\tfrac{1}{x}]/(y-\tfrac{z^2}{x}) \\ &\cong \operatorname{Spec} k[x,z,\tfrac{1}{x}] \\ &\cong \operatorname{Spec} k[x,z]_x \hookrightarrow \mathbb{A}^2_k. \end{aligned}$$

Hence, $\operatorname{Cl}(X \setminus Z) = 0$. It follows that $\mathbb{Z} \twoheadrightarrow \operatorname{Cl}(X)$ is surjective. Let $nZ = 0$ in $\operatorname{Cl}(X)$, by Example 16.8 , $\operatorname{div}(x) = 2Z$, so $\operatorname{Ker}(\mathbb{Z} \twoheadrightarrow \operatorname{Cl}(X)) = (2)$, which implies that $\operatorname{Cl}(X) = \mathbb{Z}/(2)$.

We next show that $\operatorname{Pic}(X) = 0$. Recall (16.8), we have group injective $\operatorname{Pic}(X) \hookrightarrow \operatorname{Cl}(X)$. Therefore, $\operatorname{Pic}(X) = 0$ or $\mathbb{Z}/(2)$. If $\operatorname{Pic}(X) = \mathbb{Z}/(2)$, then Z is a Cartier divisor (locally principal), but recall Example 16.8 Z is not, a contradiction! Hence, $\operatorname{Pic}(X) = 0$. □

Remark 16.28 Let $U = X \setminus \{(0,0,0)\}$, then U is factorial. Indeed, we have $U = D(x) \cup D(y)$. On $D(x)$, since x is invertible, we have $y = \frac{z^2}{x}$. Hence, $\Gamma(D(x), \mathscr{O}_X) \cong k[x,z]_x$, which is a UFD. Similarly, $\Gamma(D(y), \mathscr{O}_X) \cong k[y,z]_y$ is also UFD. Hence, $D(x)$ and $D(y)$ are factorial, and thus U is factorial. The point

$(0,0,0)$ has codimension 2 in the surface X. Since the class group of a normal integral scheme is unchanged after removing a closed subset of codimension at least 2, we have $\mathrm{Cl}(X) \cong \mathrm{Cl}(U) \cong \mathbb{Z}/2$. However, because U is factorial, by (16.8), $\mathrm{Pic}(U) \cong \mathrm{Cl}(U)$. Thus $\mathrm{Pic}(U) \cong \mathbb{Z}/2$. On the other hand, Example 16.15 shows that $\mathrm{Pic}(X) = 0$. Therefore, removing the vertex of the cone does not change the divisor class group, but it does change the Picard group:

$$\mathrm{Pic}(X) = 0, \qquad \mathrm{Pic}(X \setminus \{(0,0,0)\}) \cong \mathbb{Z}/2.$$

This illustrate that, although the class group is insensitive to removing closed subsets of codimension greater than 1, the Picard group is not. The reason is that the Picard group only sees Cartier divisors (Corollary 16.5.1), and the Cartier property is a local condition which can fail at singular points. In Example 16.15 , the vertex of the cone is precisely where the non-Cartier behavior occurs (see the proof of Example 16.8). After removing this singular point, the cone becomes factorial, so the nontrivial Weil divisor becomes Cartier.

Definition 16.6.1 ($\mathbb{Q}$-Cartier)

*A Weil divisor (on a normal scheme) with a nonzero multiple corresponding to a line bundle is called $\mathbb{Q}$-**Cartier**.*

Example 16.16 Recall Example 16.15 , $2Z = \mathrm{div}(x)$, as $\mathrm{div}(x)$ corresponds to line bundle $\mathscr{O}_X(\mathrm{div}(x))$, D is a $\mathbb{Q}$-Cartier. This gives an example of a Weil divisor that does not correspond to a line bundle, but is nonetheless $\mathbb{Q}$-Cartier.

We now give an example of a Weil divisor that is **not** $\mathbb{Q}$-Cartier.

Example 16.17 (A non-$\mathbb{Q}$-Cartier divisor) On the cone over the smooth quadric surface $X = \operatorname{Spec} k[w,x,y,z]/(wz - xy)$, let Z be the Weil divisor cut out by $w = x = 0$. Then Z is not $\mathbb{Q}$-Cartier.

Proof Recall Exercise 14.1, $\mathrm{ht}(x,z) = 1$ and (x,z) is not principal (which implies that Z is not cut out scheme-theoretically by a single equation), we know that Z is codimension 1. Moreover, as $k[w,x,y,z]/(wz - xy, x, z) \cong k[w,y]$, Z is integral, which implies that Z is a prime divisor. Suppose there exists $n \neq 0$ such that $n[Z]$ is Cartier divisor. Moreover, we may assume that $n > 0$ since if $n < 0$ and $n[Z]$ is Cartier divisor then so is $(-n)[Z]$. Let $n[Z]$ given by Cartier data $(U_i, f_i)_i$, as $n > 0$, $f_i \in \Gamma(U_i, \mathscr{O}_X)$ for all i, i.e., $n[Z]$ is an effective Cartier divisor.

We claim that $X \setminus Z$ is affine. Since $X = \bigcup_i U_i$, we have $X \setminus Z = \bigcup_i (U_i \setminus V(f_i)) = \bigcup_i D(f_i)$, by Proposition 9.3.6, $X \setminus Z \hookrightarrow X$ is affine. In particular, $X \setminus Z$ is affine. Consider the closed subscheme of $X \setminus Z$ cut out by $y = z = 0$, say V, as any closed subscheme of an affine scheme is affine, V must be affine. In fact, V is the scheme $y = z = 0$ (also known as the wx-plane) minus the point $w = x = 0$, but §5.4.1 tells us V is not affine, a contradiction! $\square$

16.6.3 Dedekind schemes

Invertible sheaves on Dedekind schemes

Recall that a Dedekind ring is a Noetherian integral domain A such that for each maximal ideal $\mathfrak{m}$ of A the local ring $A_{\mathfrak{m}}$ is a discrete valuation ring (and thus a principal ideal domain, Theorem 14.5.5). In other words, a Dedekind ring is a Noetherian regular (or equivalently normal) domain of dimension 1.

Definition 16.6.2 (Dedekind scheme)

A Noetherian integral scheme X is called a ***Dedekind scheme*** *if $\Gamma(U, \mathscr{O}_X)$ is a Dedekind ring for every open affine subscheme $U \subseteq X$. In other words, a Dedekind scheme is a Noetherian integral regular scheme of dimension $= 1$.*

Remark 16.29 If X is an integral domain which has a finite open covering $X = \bigcup_{i=1}^{n} U_i$ by Dedekind schemes U_i, then X is a Dedekind scheme.

Examples for Dedekind schemes are of course affine schemes $X = \operatorname{Spec} A$ where A is a Dedekind ring, e.g., if A is the integral closure of $\mathbb{Z}$ in a finite field extension of $\mathbb{Q}$. If k is field, the affine line $\mathbb{A}_k^1 = \operatorname{Spec} k[T]$ is a Dedekind scheme. Moreover, as the projective line $\mathbb{P}_k^1$ has an open covering by two copies of the affine line, $\mathbb{P}_k^1$ is a Dedekind scheme. More generally any regular integral curve C over k (i.e., C is a regular integral k-scheme of finite type with $\dim C = 1$) is a Dedekind scheme.

From now on we denote by X a Dedekind scheme of dimension 1. Then $\mathscr{O}_{X,x}$ is a DVR for every closed point $x \in X$. We denote by X_0 its set of closed points. Let $K := K(X) = \mathscr{O}_{X,\eta}$ be the function field. We may consider $\mathscr{O}_{X,x}$ (for $x \in X$) and $\Gamma(U, \mathscr{O}_X)$ (for $U \subseteq X$ non-empty open) as subrings of $K(X)$. If U is affine, K is the field of fractions of $\Gamma(U, \mathscr{O}_X)$. We denote by $v_x : K^\times \to \mathbb{Z}$ the discrete valuation and by π_x a uniformizing element of the discrete valuation ring $\mathscr{O}_{X,x}$. By Proposition 6.4.4, for any open set $U \subseteq X$

$$\Gamma(U, \mathscr{O}_X) = \bigcap_{x \in X_0 \cap U} \mathscr{O}_{X,x} = \{a \in K : v_x(a) \geq 0 \text{ for all } x \in X_0 \cap U\}.$$

Let $D := (n_x)_{x \in X_0} \in \operatorname{Div}(X) = \mathbb{Z}^{(X_0)}$ be a tuple of integers n_x (for $x \in X_0$). We attach to D an $\mathscr{O}_X$-module $\mathscr{L}_D$ as follows: If $U \subseteq X$ is open, we set

$$\Gamma(U, \mathscr{L}_D) := \{s \in K : v_x(s) \geq n_x \text{ for all } x \in X_0 \cap U\}.$$

For $V \subseteq U$, we define the restriction maps to be the inclusions $\Gamma(U, \mathscr{L}_D) \hookrightarrow \Gamma(V, \mathscr{L}_D)$ and this defines a sheaf $\mathscr{L}_D$ of abelian groups on X. Multiplication within K defines a scalar multiplication $\Gamma(U, \mathscr{O}_X) \times \Gamma(U, \mathscr{L}_D) \to \Gamma(U, \mathscr{L}_D)$ that makes $\mathscr{L}_D$ into an $\mathscr{O}_X$-module.

We think of elements in $\Gamma(U, \mathscr{L}_D)$ as "meromorphic functions on U" that have at x a zero of order at least n_x (or, for $n_x < 0$, a pole of order at most $-n_x$).

If $D' := (n'_x)_{x \in X_0} \in \operatorname{Div}(X)$ is a second divisor with $n'_x \leq n_x$ for all $x \in X_0$, we have a natural inclusion

$$\mathscr{L}_D \hookrightarrow \mathscr{L}_{D'}. \tag{16.30}$$

By Corollary 16.4.2, $\mathscr{L}_D$ is a locally free sheaf of rank 1. Let $\operatorname{Supp}(D)$ be the set if $x \in X_0$ such that $n_x \neq 0$. For each $x \in \operatorname{Supp}(D)$, let U_x be an open affine neighborhood of x which does not contain any other point of $\operatorname{Supp}(D)$ and such that there exists a section $s_x \in \Gamma(U_x, \mathscr{O}_X)$ whose germ at x is the chosen uniformizing element $\pi_x \in \mathscr{O}_{X,x}$. By shrinking U_x we can further assume that $v_y(s_x) = 0$ for all $y \in (U_x \cap X_0) \setminus \{x\}$. We also set $V := X \setminus \operatorname{Supp}(D)$. Then V and the U_x for $x \in \operatorname{Supp}(D)$ form an open covering of X. By definition we have $\mathscr{L}_D|_V \cong \mathscr{O}_X|_V$ and over U_x multiplication by $s_x^{n_x} \in K$ defines an isomorphism

$$\mathscr{O}_X|_{U_x} \xrightarrow{\sim} \mathscr{L}_D|_{U_x}.$$

For two divisors $D, D' \in \operatorname{Div}(X)$, we have $\mathscr{L}_D \otimes_{\mathscr{O}_X} \mathscr{L}_{D'} = \mathscr{L}_{D+D'}$, see (16.16). For every $f \in K^\times$, set $\operatorname{div}(f) = (v_x(f))_{x \in X_0} \in \operatorname{Div}(X)$. The multiplication by f defines an isomorphism $\mathscr{L}_D \xrightarrow{\sim} \mathscr{L}_{D+\operatorname{div}(f)}$. By Proposition 16.5.1, if two invertible sheaves $\mathscr{L}_D$ and $\mathscr{L}_{D'}$ are isomorphic, there exists an $f \in K^\times$ such that $D + \operatorname{div}(f) = D'$.

Divisors on Dedekind schemes

Let X be a Dedekind scheme that is not the Spec of a field, X_0 is the set of closed points of X, and for all $x \in X_0$, we denote by v_x the discrete valuation on the function field $K(X)$ given by the DVR $\mathscr{O}_{X,x}$. Clearly, we have $Z^1(X) = \mathbb{Z}^{(X_0)}$. As X is regular, we find $\mathrm{Div}(X) = Z^1(X)$ as already suggested by the notation of §16.5.4. For $D \in \mathrm{Div}(X)$, the invertible sheaf $\mathscr{L}_D$ is the sheaf $\mathscr{I}_X(D)$ which defined in Definition 16.5.10.

Let $X = \mathrm{Spec}\,\mathscr{O}_K$, where $\mathscr{O}_K$ is the ring of integers in a number field K (defined in Definition 11.7.4). We next identify the (ideal) class group of the ring of integers $\mathscr{O}_K$, with the class group of $\mathrm{Spec}\,\mathscr{O}_K$.

Definition 16.6.3 (Fractional ideal)

*Let A be an integral domain, K its field of fractions. An A-submodule M of K is a **fractional ideal** of A (or say it fractional A-ideal M) if $xM \subseteq A$ for some $x \neq 0$ in A. In particular, the "ordinary" ideals (now called **integral ideals**) are fractional ideals (take $x = 1$). Any element $u \in K$ generated a fractional ideal, denoted by (u) or Au, and called **principal**. If M is a fractional ideal, the set of all $x \in K$ such that $xM \subseteq A$ is denoted by $(A : M)$.*

Definition 16.6.4 (Invertible ideal)

An A-submodule M of K is an invertible ideal if there exists a submodule N of K such that $MN = A$.

The module N in definition 16.6.4 is unique and equal to $(A : M)$, for we have $N \subseteq (A : M) = (A : M)MN \subseteq AN = N$. It follows that M is finitely generated, and therefore a fractional ideal: for since $M \cdot (A : M) = A$ there exist $x_i \in M$ and $y_i \in (A : M)$ $(1 \le i \le n)$ such that $\sum x_i y_i = 1$, and hence for any $x \in M$ we have $x = \sum (y_i x) x_i$ where each $y_i x \in A$, so that M is generated by $x_1, \dots, x_n$.

The invertible ideals form a group with respect to multiplication, whose identity element is $A = (1)$, we denote it $\boxed{\mathscr{I}(A).}$ Clearly every non-zero principal fractional ideal (u) is invertible, its inverse being (u^{-1}). Hence, the set $\boxed{\mathscr{P}(A)}$ form a subgroup of $\mathscr{I}(A)$, moreover, $\mathscr{P}(A)$ is isomorphic to $K^\times / A^\times$.

Definition 16.6.5 (Picard group of a domain, class group)

*The **Picard group** of a domain A is defined as $\mathrm{Pic}(A) = \mathscr{I}(A)/\mathscr{P}(A)$. If A is Dedekind, $\mathrm{Pic}(A)$ is often called the **class group** of A and denoted by $\mathrm{Cl}(A)$.*

In other words, the Picard group of a domain A is defined by the long exact sequence

$$1 \longrightarrow A^\times \longrightarrow K^\times \xrightarrow{\varphi} \mathscr{I}(A) \longrightarrow \mathrm{Pic}(A) \longrightarrow 1,$$

where $\varphi(x) = Ax$ for $x \in K^\times$.

Theorem 16.6.1

Consider the ring of integers $\mathscr{O}_K$ in a number field K. There is an isomorphism

$$\mathscr{I}(\mathscr{O}_K) \xrightarrow{\sim} \bigoplus_{\mathfrak{p}} \mathbb{Z} = \mathrm{Div}(\mathrm{Spec}\,\mathscr{O}_K)$$

$$I \longmapsto (\mathrm{ord}_{\mathfrak{p}}(I))_{\mathfrak{p}},$$

and every $I \in \mathscr{I}(\mathscr{O}_K)$ factors uniquely as a product $I = \prod_{\mathfrak{p}} \mathfrak{p}^{\mathrm{ord}_{\mathfrak{p}}(I)}$.

Now, the exact sequence (16.12) becomes

$$\begin{tikzcd}
1 \arrow[r] & \mathscr{O}_K^\times \arrow[r] \arrow[d, equal] & K^\times \arrow[r] \arrow[d, equal] & \mathrm{Div}(\operatorname{Spec} \mathscr{O}_K) \arrow[r] \arrow[d, leftrightarrow, "\text{Thm. 16.6.1}"', "\sim"] & \mathrm{Cl}(\operatorname{Spec} \mathscr{O}_K) \arrow[r] \arrow[d, leftrightarrow, dashed, "\sim"] & 1 \\
1 \arrow[r] & \mathscr{O}_K^\times \arrow[r] & K^\times \arrow[r] & \mathscr{I}(\mathscr{O}_K) \arrow[r] & \mathrm{Cl}(\mathscr{O}_K) \arrow[r] & 1
\end{tikzcd}$$

In this case it is a basic result from number theory that the divisor class group $\mathrm{Pic}(\mathscr{O}_K)$ is a finite group and that $\mathscr{O}_K^\times \cong \mu(K) \times \mathbb{Z}^{r+s-1}$, where $\mu(K)$ is the finite cyclic group of roots in unity in K and where r (resp. s) is the number of real (resp. half the number of complex) embeddings of K.

16.6.4 More on class groups and unique factorization

As mentioned in Remark 6.20, there are few commonly used means of checking that a ring is a UFD. The next proposition is one of them, and it is useful. For example, it implies the classical fact that for rings of integers in number fields, the class group is the obstruction to unique factorization (see Exercise 15.2 and Proposition 16.4.8).

Proposition 16.6.8

Suppose A is a Noetherian integral domain. Then A is a unique factorial domain if and only if A is integrally closed and $\mathrm{Cl}(\operatorname{Spec} A) = 0$.

Proof Let A is a UFD, we want to show that A is integrally closed and $\mathrm{Cl}(\operatorname{Spec} A) = 0$. Recall Proposition 6.4.6, A is integrally closed. Also, Lemma 13.1.3 tells us all codimension 1 prime ideals of A are principal, hence, $\mathrm{Cl}(\operatorname{Spec} A) = 0$.

It remains to show that if A is integrally closed and $\mathrm{Cl}(\operatorname{Spec} A) = 0$ then A is UFD. Let $\mathfrak{p}$ be a codimension 1 prime ideal of A. Since $\mathrm{Cl}(\operatorname{Spec} A) = 0$, there exists $f \in K(A)^\times$ such that $\mathrm{div}(f) = [\mathfrak{p}]$. Hence, $\mathrm{val}_{[\mathfrak{p}]}(f) = 1$ and $\mathrm{val}_{[\mathfrak{q}]}(f) = 0$ for any $\mathfrak{q} \neq \mathfrak{p}$. It follows that $f \in A_{\mathfrak{q}}^\times$ for $\mathfrak{q} \neq \mathfrak{p}$ and $f \in \mathfrak{p}A_{\mathfrak{p}}$. Hence, $(f) \subseteq \mathfrak{p}$. Let $a \in \mathfrak{p}$, we want to show that $a \in (f)$, equivalently, it suffices to show that $\frac{a}{f} \in A$. Consider $\mathrm{val}_{[\mathfrak{p}]}(\frac{a}{f})$. As $a \in \mathfrak{p}$, we have $\mathrm{val}_{[\mathfrak{p}]}(a) \geq 1$, and thus

$$\mathrm{val}_{[\mathfrak{p}]}(\tfrac{a}{f}) = \mathrm{val}_{[\mathfrak{p}]}(a) - \mathrm{val}_{[\mathfrak{p}]}(f) \geq 1 - 1 = 0,$$

which implies that $\frac{a}{f} \in A_{\mathfrak{p}}$. For $\mathfrak{q} \neq \mathfrak{p}$, consider $\mathrm{val}_{[\mathfrak{q}]}(\frac{a}{f})$. As $a \in A$, $\mathrm{val}_{[\mathfrak{q}]}(a) \geq 0$, note that $\mathrm{val}_{[\mathfrak{q}]}(f) \geq 0$, we obtain that

$$\mathrm{val}_{[\mathfrak{q}]}(\tfrac{a}{f}) = \mathrm{val}_{[\mathfrak{q}]}(a) - \mathrm{val}_{[\mathfrak{q}]}(f) \geq 0 - 0 = 0,$$

i.e., $\frac{a}{f} \in A_{\mathfrak{q}}$ for all $\mathfrak{q} \neq \mathfrak{p}$. By Algebraic Hartogs's Lemma 14.5.8, $\frac{a}{f} \in A$, which implies that $\mathfrak{p} = (f)$. Now, by Proposition 13.3.7, A is a UFD, we win. □

Nagata's Lemma gives a method of checking that a ring is a UFD. It is useful.

Theorem 16.6.2 (Nagata's Lemma)

Suppose A is a Noetherian domain, $x \in A$ an element such that (x) is prime and $A_x = A[1/x]$ is a unique factorization domain. Then A is a unique factorization domain.

Proof As A is domain, (0) is prime ideal, since $(0) \subsetneq (x)$, $\mathrm{ht}(x) \geq 1$. As A is Noetherian, by Krull's Principal Ideal Theorem 13.3.2, $\mathrm{ht}(x) \leq 1$. Hence, we know that $\mathrm{ht}(x) = 1$, which implies that $V(x)$ is a prime divisor. By Proposition 16.5.5, there exists an exact sequence

$$\mathbb{Z}[V(x)] \longrightarrow \mathrm{Cl}(\operatorname{Spec} A) \longrightarrow \mathrm{Cl}(\operatorname{Spec} A_x) \longrightarrow 0.$$

As A_x is UFD, by Remark 16.14, $\mathrm{Cl}(\operatorname{Spec} A_x) = 0$. Note that $\operatorname{div}(x) = [V(x)]$, we obtain that $\mathbb{Z}[V(x)] \subseteq Z^1_{\text{princ}}(\operatorname{Spec} A)$, which implies that $\mathrm{Cl}(\operatorname{Spec} A) = 0$.

We next show that A is integrally closed in its field of fraction. Suppose $T^n + a_{n-1}T^{n-1} + \cdots + a_0 = 0$, where $a_i \in A$ and $T \in K(A)$. We want to show that $T \in A$. As A_x is integrally closed, $T \in A_x$, we may assume that $T = \frac{r}{x^m}$ where $r \in A$ and $m > 0$ is minimal. Then, we have

$$(\tfrac{r}{x^m})^n + a_{n-1}(\tfrac{r}{x^m})^{n-1} + \cdots + a_1(\tfrac{r}{x^m}) + a_0 = 0.$$

Suppose $m > 0$, multiplying both sides by x^{mn}, we get

$$r^n + a_{n-1}r^{n-1}x^m + a_{n-2}r^{n-2}x^{2m} + \cdots + a_0 x^{mn} = 0.$$

Hence,

$$r^n = -(a_{n-1}r^{n-1}x^m + a_{n-2}r^{n-2}x^{2m} + \cdots + a_0 x^{mn}).$$

As $m > 0$, $x^m \mid r^n$, so $r^n \in (x)$, since (x) is prime, we obtain that $r \in (x)$. But $m > 0$, by the minimality of m, $r \notin (x)$, a contradiction. Hence, $m = 0$, which implies that $T = r \in A$. It follows that A is integrally closed.

Finally, by Proposition 16.6.8, A is UFD. □

This leads to a fun algebra fact promised in Remark 14.35:

Example 16.18 Suppose k is an algebraically closed field of characteristic not 2. Let $A = k[x_1, \ldots, x_n]/(x_1^2 + \cdots + x_m^2)$ where $m \leq n$. When $m \leq 2$, we get some special behavior. (If $m = 0$, we get affine space; if $m = 1$, we get a nonreduced scheme; if $m = 2$, we gat a reducible scheme that is the union of two affine spaces, since $x_1^2 + x_2^2 = (x_1 + ix_2)(x_1 - ix_2)$.)

If $m \geq 3$, we have verified that $\operatorname{Spec} A$ is normal, in Exercise 6.6 (b). In fact, if $m \geq 3$, then A is UFD unless $m = 3$ or $m = 4$ (Exercise 6.9; Exercise 14.2). For the case $m = 3$:

$$A = k[x, y, z, w_1, \ldots, w_{n-3}]/(x^2 + y^2 - z^2)$$

is not a unique factorization domain, as it is has nonzero class group (by essentially the same argument as for Example 16.15).

The failure at 4 comes from the geometry of the quadric surface: we have checked that in $\operatorname{Spec} k[w, x, y, z]/(wz - xy)$, there is a codimension 1 irreducible subset — the cone over a line in a ruling (Exercise 14.1) — that is not principal.

Example 16.19 (The case $m \geq 5$) Suppose that k is algebraically closed of characteristic not 2. If $\ell \geq 3$, then

$$A = k[a, b, x_1, \ldots, x_n]/(ab - x_1^2 - \cdots - x_\ell^2)$$

is a UFD.

Proof Since $A/(a) \cong k[b, x_1, \ldots, x_n]/(x_1^2 + \cdots + x_\ell^2)$, by Exercise 6.6, $A/(a)$ is normal, and therefore (a). We next show that A_a is a UFD. In fact,

$$A_a \cong k[a, a^{-1}, b, x_1, \ldots, x_n]/(b - \tfrac{x_1^2 - \cdots - x_\ell^2}{a}) \cong k[a, a^{-1}, x_1, \ldots, x_n],$$

which implies that A_a is a UFD. By Nagata's Lemma 16.6.2, we win. □

16.7 The Graded Module Corresponding to a Quaiscoherent Sheaf

This section answers some fundamental questions. Throughout this section, $S_\bullet$ is a finitely generated graded algebra *generated in degree* 1, so, in particular we have the invertible sheaf $\mathscr{O}(m)$ for all m by Proposition 16.1.7. Also, throughout, $M_\bullet$ is a graded $S_\bullet$-module, and $\mathscr{F}$ is a quasicoherent sheaf on $\operatorname{Proj} S_\bullet$.

Note *In this section, $S_\bullet$ is a finitely generated graded algebra generated in degree* 1. *$M_\bullet$ is a graded $S_\bullet$-module, and $\mathscr{F}$ is a quasicoherent sheaf on* $\operatorname{Proj} S_\bullet$.

We know how to get quasicoherent sheaves on $\operatorname{Proj} S_\bullet$ from graded $S_\bullet$-modules. We will now see that we can get them all in this way. We will define a functor $\Gamma_\bullet$ from the category of quasicoherent sheaves on $\operatorname{Proj} S_\bullet$ to the category of graded $S_\bullet$-modules that will attempt to reverse the $\widetilde{\square}$ construction. They are not quite inverse, as $\widetilde{\square}$ can turn two different graded modules into the same quasicoherent sheaf (see, for example, Lemma 15.6.3). But we will see a natural isomorphism $\Gamma_\bullet(\mathscr{F})^\sim \longleftrightarrow \mathscr{F}$. In fact, $\Gamma(\widetilde{M_\bullet})$ is a better ("saturated") version of $M_\bullet$, and there is a saturation functor $M_\bullet \to \Gamma_\bullet(\widetilde{M_\bullet})$ that is akin to groupification and sheafification — it is adjoint to the forgetful functor from saturated graded modules to graded modules. And thus we come to the fundamental relationship between $\widetilde{\square}$ and $\Gamma_\bullet$: they are an adjoint pair.

$$\begin{tikzcd}
 & \text{graded } S_\bullet\text{-modules} \arrow[dl, "\widetilde{\square}"'] \arrow[dd, bend right, "\text{saturate}"'] \\
\mathbf{QCoh}_{\operatorname{Proj} S_\bullet} \arrow[dr, "\Gamma_\bullet"', "\text{equivalence}"] & \\
 & \text{saturated graded } S_\bullet\text{-modules} \arrow[uu, bend right, "\text{forget}"']
\end{tikzcd}$$

We now make some of this precise, but as little as possible to move forward. In particular, we will show that every quasicoherent sheaf on a projective A-scheme arises from a graded module (Theorem 16.7.1), and we will see that every closed subscheme of $\operatorname{Proj} S_\bullet$ arises from a homogeneous ideal $I_\bullet \subseteq S_\bullet$ (**??**).

When you do Proposition 16.1.3 (on global sections of $\mathscr{O}_{\mathbb{P}^n_k}(m)$), you will suspect that, in good situations,

$$M_m \cong \Gamma(\operatorname{Proj} S_\bullet, \widetilde{M}(m)_\bullet).$$

Motivated by this, we define:

$$\boxed{\Gamma_\bullet(\mathscr{F}) := \bigoplus_{m\in\mathbb{Z}} \Gamma_m(\mathscr{F}) := \bigoplus_{m\in\mathbb{Z}} \Gamma(\operatorname{Proj} S_\bullet, \mathscr{F}(m)).}$$

Exercise 16.13 Describe a morphism of S_0-modules $M_m \to \Gamma(\operatorname{Proj} S_\bullet, \widetilde{M(m)_\bullet})$, extending the $m = 0$ case of Exercise 15.5.

Proof As $S_\bullet$ is generated in degree 1, we know that $\operatorname{Proj} S_\bullet = \bigcup_{f\in S_1} D_+(f)$. For each $D_+(f) \hookrightarrow \operatorname{Proj} S_\bullet$, define $\varphi_f : M_m \to ((M_\bullet)_f)_m$ by setting $h \mapsto \frac{h}{1}$. Let $D_+(g) \hookrightarrow \operatorname{Proj} S_\bullet$ such that $D_+(f) \cap D_+(g) \neq \varnothing$, then

$$\varphi_f(h)|_{D_+(g)} = \frac{h}{1} = \varphi_g(h)|_{D_+(f)}.$$

By the gluability axiom, there exists $s_h \in \Gamma(\operatorname{Proj} S_\bullet, \widetilde{M(m)_\bullet})$ such that $s_h|_{D_+(f)} = \frac{h}{1}$ for each $f \in S_1$. Define $M_m \to \Gamma(\operatorname{Proj} S_\bullet, \widetilde{M(m)_\bullet})$ by setting $h \mapsto s_h$. □

Lemma 16.7.1
$\Gamma_\bullet(\mathscr{F})$ is a graded $S_\bullet$-module.

Proof Let $X := \operatorname{Proj} S_\bullet$ and let $\mathscr{F}$ be a quasicoherent sheaf on X. Recall that $\Gamma_\bullet(\mathscr{F}) := \bigoplus_{m\in\mathbb{Z}} \Gamma(X, \mathscr{F})$. We want to show that $\Gamma_\bullet(\mathscr{F})$ is a graded $S_\bullet$-module.

For every integer $m \geq 0$, Exercise 16.13 provides a natural S_0-map

$$\iota_m : S_m \longrightarrow \Gamma(X, \mathscr{O}_X(m)).$$

Now, let $a \in S_m$ and $t \in \Gamma(X, \mathscr{F}(n))$. Then

$$\iota_m(a) \otimes t \in \Gamma(X, \mathscr{O}_X(m) \otimes \mathscr{F}(n)),$$

note that

$$\mathscr{O}_X(m) \otimes \mathscr{F}(n) \cong (\mathscr{O}_X(m) \otimes \mathscr{O}_X(n)) \otimes \mathscr{F} \cong \mathscr{O}_X(m+n) \otimes \mathscr{F} \cong \mathscr{F}(m+n),$$

we define

$$a \cdot t := \iota_m(a) \otimes t \in \Gamma(X, \mathscr{F}(m+n)).$$

Thus, for every m, n, we obtain a bilinear map

$$S_m \times \Gamma(X, \mathscr{F}(n)) \longrightarrow \Gamma(X, \mathscr{F}(m+n)).$$

The module axioms is easy to verify, we omit the proof. Hence, $\Gamma_\bullet(\mathscr{F})$ is a graded $S_\bullet$-module. □

Lemma 16.7.2 (Saturation map)

*$\eta^M : M_\bullet \to \Gamma_\bullet(\widetilde{M_\bullet})$ is a map of $S_\bullet$-modules. We call this the **saturation map**.*

Proof For each $m \in \mathbb{Z}_\bullet$, Exercise 16.13 gives an S_0-map

$$\eta_m : M_m \longrightarrow \Gamma(X, \widetilde{M_\bullet(m)}).$$

We want to show that their direct sum

$$\eta := \bigoplus_{m \in \mathbb{Z}} \eta_m : M_\bullet \longrightarrow \Gamma_\bullet(\widetilde{M_\bullet}) = \bigoplus_{m \in \mathbb{Z}} \Gamma(X, \widetilde{M_\bullet}(m))$$

is a homomorphism of graded $S_\bullet$-modules. Note that

$$\eta(M_m) = \Gamma_m(M_m) \subseteq \Gamma(X, \widetilde{M_\bullet}(m)) = \Gamma_m(\widetilde{M_\bullet})$$

for all $m \geq 0$, it follows that

$$\eta : M_\bullet \longrightarrow \Gamma_\bullet(\widetilde{M_\bullet})$$

is a homomorphism of graded $S_\bullet$-modules. □

Example 16.20 (The saturation map need not be injective, nor need it be surjective.) Let $S_\bullet = k[x]$, $M_\bullet = k[x]/(x^2)$, and $N_\bullet = xk[x]$. As $S_\bullet$ is finitely generated graded algebra generated in degree 1, $\operatorname{Proj} S_\bullet = D_+(x)$. Consider $\Gamma_\bullet(\widetilde{M_\bullet})$,

$$\Gamma_\bullet(\widetilde{M_\bullet}) = \bigoplus_{m \in \mathbb{Z}} \Gamma(D_+(x), \widetilde{M_\bullet}(m)) = \bigoplus_{m \in \mathbb{Z}} ((k[x]/(x^2))_x)_m = 0,$$

it follows that $M_\bullet \to \Gamma_\bullet(\widetilde{M_\bullet})$ is not injective.

We next consider $\Gamma_\bullet(\widetilde{N_\bullet})$,

$$\Gamma_\bullet(\widetilde{N_\bullet}) = \bigoplus_{m \in \mathbb{Z}} \Gamma(D_+(x), \widetilde{N_\bullet}(m)) = \bigoplus_{m \in \mathbb{Z}} ((xk[x])_x)_m \cong \bigoplus_{m \in \mathbb{Z}} (k[x]_x)_m.$$

Consider $1 \in (k[x]_x)_0$, note that $1 \notin N_0 = \{0\}$, it follows that 1 has no preimage, which implies that $N_\bullet \to \Gamma_\bullet(\widetilde{N_\bullet})$ is not surjective.

Proposition 16.7.1 (Saturation functor)

*$\Gamma_\bullet$ is a functor from $\mathbf{QCoh}_{\operatorname{Proj} S_\bullet}$ to the category of graded $S_\bullet$-modules. Thus the saturation map can be better called the **saturation functor**.*

Proof By the discussion above, we already know that $\Gamma_\bullet$ sends a quasicoherent sheaf to a graded $S_\bullet$-module (Lemma 16.7.1). We first show that given a morphism of quasicoherent sheaves $\mathscr{F} \to \mathscr{G}$, $\Gamma_\bullet$ gives a morphism of graded $S_\bullet$-modules $\Gamma_\bullet\mathscr{F} \to \Gamma_\bullet\mathscr{G}$. Say $\varphi : \mathscr{F} \to \mathscr{G}$, then by tensoring $\mathscr{O}(m)$ and take global section, we obtain $\Phi_m : \Gamma(\operatorname{Proj} S_\bullet, \mathscr{F}(m)) \to \Gamma(\operatorname{Proj} S_\bullet, \mathscr{G}(m))$. This induces

$$\Gamma_\bullet(\varphi) := \bigoplus_m \Phi_m : \Gamma_\bullet(\mathscr{F}) = \bigoplus_m \Gamma(\operatorname{Proj} S_\bullet, \mathscr{F}(m)) \longrightarrow \Gamma_\bullet(\mathscr{G}) = \bigoplus_m \Gamma(\operatorname{Proj} S_\bullet, \mathscr{G}(m)).$$

We next show that $\Gamma_\bullet(\varphi)$ is a graded $S_\bullet$-map. Take $s \in S_m, t \in \Gamma(\operatorname{Proj} S_\bullet, \mathscr{F}(n))$. Let $\sigma_s \in \Gamma(\operatorname{Proj} S_\bullet, \mathscr{O}(m))$ is the image of s, then

$$\Gamma_\bullet(\varphi)(s \cdot t) = \Phi_{m+n}(\sigma_s \otimes t) = \sigma_s \otimes \Phi_n(t) = s \cdot \Gamma_\bullet(\varphi)(t),$$

which implies that $\Gamma_\bullet(\varphi)$ is a graded $S_\bullet$-map, and therefore $\Gamma_\bullet(\varphi) : \Gamma_\bullet\mathscr{F} \to \Gamma_\bullet\mathscr{G}$ is a graded $S_\bullet$-morphism.

By the construction above, clearly, $\Gamma_\bullet$ preserves identity and composition. Hence, $\Gamma_\bullet$ is functor. $\square$

The special case $M_\bullet = S_\bullet$

We have a saturation map $S_\bullet \to \Gamma_\bullet\widetilde{S_\bullet}$, which is a map of $S_\bullet$-modules. But $\Gamma_\bullet\widetilde{S_\bullet}$ has the structure of a graded ring (basically because we can multiply sections of $\mathscr{O}(m_1)$ by sections of $\mathscr{O}(m_2)$ to get sections of $\mathscr{O}(m_1 + m_2)$; see Proposition 16.1.9).

Lemma 16.7.3

(1) The saturation map $S_\bullet \to \Gamma_\bullet\widetilde{S_\bullet}$ is in fact a ring morphism.

(2) The saturation map $S_\bullet \to \Gamma_\bullet\widetilde{S_\bullet}$ is an isomorphism when $S_\bullet$ is a polynomial ring (with the usual grading), in at least one variable.

Proof Let $X = \operatorname{Proj} S_\bullet$. Say $\eta_m : S_m \to \Gamma(X, \widetilde{S_\bullet}(m))$ and $\eta := \bigoplus_m \eta_m : S_\bullet \to \Gamma_\bullet(\widetilde{S_\bullet})$. We want to show that $\eta(ab) = \eta(a)\eta(b)$ for any $a, b \in S_\bullet$. By the construction of η, it suffices to show that $\eta_{n+m}(a) = \eta_n(a)\eta_m(b)$ for all $a \in S_n$ and $b \in S_m$. Since $S_\bullet$ is generated in degree 1, $X = \bigcup_{f \in S_1} D_+(f)$. It is therefore enough to check the equality on each $D_+(f)$. On $D_+(f)$, we have $\eta_n(a)|_{D_+(f)} = \frac{a}{1} \in ((S_\bullet)_f)_n$ and $\eta_m(b)|_{D_+(f)} = \frac{b}{1} \in ((S_\bullet)_f)_m$. So, their product in $((S_\bullet)_f)_{m+n}$ is $\frac{a}{1} \cdot \frac{b}{1} = \frac{ab}{1}$. But $\eta_{m+n}(ab)|_{D_+(f)} = \frac{ab}{1}$. Hence,

$$(\eta_n(a)\eta_m(b))|_{D_+(f)} = \eta_{n+m}(ab)|_{D_+(f)}.$$

By gluability axiom of $\widetilde{S_\bullet}(n+m)$, we obtain

$$\eta_n(a)\eta_m(b) = \eta_{n+m}(ab).$$

Consider $\eta_0(1)$ is the section of $\mathscr{O}_X$ which restricts to $\frac{1}{1} = 1$ on every $D_+(f)$. Thus $\eta_0(1) = 1_{\Gamma(X, \mathscr{O}_X)}$. Therefore η preserves multiplication and the identity element. Hence, $\eta : S_\bullet \to \Gamma_\bullet(\widetilde{S_\bullet})$ is a homomorphism of graded rings.

Now, suppose $S_\bullet = R[x_0, \dots, x_n]$ is a polynomial ring where $\deg x_i = 1$. We want to show that $\eta : S_\bullet \to \Gamma_\bullet(\widetilde{S_\bullet})$ is an isomorphism. It suffices to show that $\eta_m : S_m \to \Gamma_m(\widetilde{S_\bullet})$ is an isomorphism for every $m \geq 0$. Note that $\Gamma_m(\widetilde{S_\bullet}) = \Gamma(X, \widetilde{S_\bullet}(m)) \cong R[x_0, \dots, x_n]_m$ where $R[x_0, \dots, x_n]_m$ means degree m polynomials by Proposition 16.1.3, η_m is indeed an isomorphism, and therefore η is an isomorphism. $\square$

Example 16.21 (The saturation map may not be an isomorphism.) Let $S_\bullet = k[x, y]/(xy)$, then the saturation map is not an isomorphism.

Proof Let $X = \operatorname{Proj} S_\bullet$. In fact, $S_\bullet = D_+(x) \cup D_+(y)$. Consider $D_+(x) \cap D_+(y) = D_+(xy)$, we have $D_+(xy) = D_+(0) = \varnothing$. Hence, by the equalizer exact sequence 3.3 , we obtain

$$\Gamma(X, \mathscr{O}_X) \cong \Gamma(D_+(x), \mathscr{O}_X) \times \Gamma(D_+(y), \mathscr{O}_X) \cong (k[x]_x)_0 \times (k[y]_y)_0 \cong k \times k.$$

Let $a \in S_0 = k$, say $\eta : S_\bullet \to \Gamma_\bullet(\widetilde{S_\bullet})$, then $\eta_0(a)|_{D_+(x)} = \frac{a}{1}$ and $\eta_0(a)|_{D_+(y)} = \frac{a}{1}$, hence $\eta(a) = (a, a)$. But η is not an isomorphism, since $(1, 0) \in k \times k$ has no preimage. $\square$

The reverse map

Now that we have defined the saturation map $M_\bullet \to \Gamma_\bullet \widetilde{M_\bullet}$, we will describe a map $(\Gamma_\bullet \mathscr{F})^\sim \to \mathscr{F}$. While subtler to define, it will have the advantage of being an isomorphism.

We now define $\tau : (\Gamma_\bullet \mathscr{F})^\sim \to \mathscr{F}$. Let $\mathscr{F}$ be a quasicoherent sheaf on $X := \operatorname{Proj} S_\bullet$. Let $f \in S_d$ $(d > 0)$ and define a homomorphism of $(S_\bullet)_f$-modules

$$\begin{aligned} \tau_{D_+(f)} : \Gamma(D_+(f), (\Gamma_\bullet \mathscr{F})^\sim) = ((\Gamma_\bullet \mathscr{F})_f)_0 &\longrightarrow \Gamma(D_+(f), \mathscr{F}), \\ \frac{x}{f^n} &\longmapsto x|_{D_+(f)} \cdot (\eta_{dn}(f^n)|_{D_+(f)})^{-1}. \end{aligned} \tag{16.31}$$

where $x \in (\Gamma_\bullet \mathscr{F})_{dn} = \Gamma(X, \mathscr{F}(nd))$ and $\eta_{dn} : S_{dn} \to \Gamma(X, \mathscr{O}_X(dn))$. We next show that $\{\tau_{D_+(f)}\}_{f \in S_\bullet}$ is compatible with restriction from $D_+(f)$ to $D_+(fg)$ for all $g \in S_e$ $(e > 0)$. Let $g \in S_e$ such that $D_+(f) \cap D_+(g) \neq \varnothing$. Suppose $\frac{x}{(fg)^n} \in \Gamma(D_+(fg), (\Gamma_\bullet \mathscr{F})^\sim) = ((\Gamma_\bullet \mathscr{F})_{fg})_0$. By Proposition 5.5.10, we know that

$$((S_\bullet)_{fg})_0 \cong (((S_\bullet)_f)_0)_{\frac{g^d}{f^e}}, \qquad ((S_\bullet)_{fg})_0 \cong (((S_\bullet)_g)_0)_{\frac{f^e}{g^d}}.$$

Let $\alpha = \frac{g^d}{f^e}$, then $\frac{x}{(fg)^n} \in ((\Gamma_\bullet \mathscr{F})_{fg})_0$ can be written as $\frac{(fg)^{n(d-1)}x/f^{n(d+e)}}{\alpha^n}$. Hence,

$$\tau_{D_+(f)}|_{D_+(fg)}(\tfrac{x}{(fg)^n}) = ((fg)^{n(d-1)}x)|_{D_+(fg)} \cdot (\eta_{dn(d+e)}(f^{n(d+e)})|_{D_+(fg)})^{-1} \cdot \alpha^{-n}.$$

Note that $\alpha^{-n} = \eta_{den}(f^{en})|_{D_+(fg)} \cdot (\eta_{den}(g^{dn})|_{D_+(fg)})^{-1}$,

$$\begin{aligned} \tau_{D_+(f)}|_{D_+(fg)}(\tfrac{x}{(fg)^n}) &= ((fg)^{n(d-1)}x)|_{D_+(fg)} \cdot (\eta_{dn(d+e)}(f^{n(d+e)})|_{D_+(fg)})^{-1} \\ &\quad \cdot \eta_{den}(f^{en})|_{D_+(fg)} \cdot (\eta_{den}(g^{dn})|_{D_+(fg)})^{-1} \\ &= ((fg)^{n(d-1)}x)|_{D_+(fg)} \cdot (\eta(f^{nd})|_{D_+(fg)})^{-1} \cdot (\eta(g^{nd})|_{D_+(fg)})^{-1} \\ &= ((fg)^{n(d-1)}x)|_{D_+(fg)} \cdot (\eta((fg)^{nd})|_{D_+(fg)})^{-1} \\ &= x|_{D_+(fg)} \cdot \eta(fg)^{n(d-1)}|_{D_+(fg)} \cdot (\eta(fg)^{nd}|_{D_+(fg)})^{-n} \\ &= x|_{D_+(fg)} \cdot \eta(fg)|_{D_+(fg)}^{-n} \\ &= \tau_{D_+(fg)}(\tfrac{x}{(fg)^n}). \end{aligned}$$

Similarly, we have $\tau_{D_+(g)}|_{D_+(fg)}(\frac{x}{(fg)^n}) = \tau_{D_+(fg)}(\frac{x}{(fg)^n})$, hence

$$\tau_{D_+(f)}|_{D_+(fg)} = \tau_{D_+(g)}|_{D_+(fg)}.$$

Then we obtain a morphism of quasicoherent sheaves on X,

$$\tau : (\Gamma_\bullet \mathscr{F})^\sim \longrightarrow \mathscr{F}. \tag{16.32}$$

Lemma 16.7.4

The natural morphism $\tau : (\Gamma_\bullet \mathscr{F})^\sim \to \mathscr{F}$ (16.32) is an isomorphism.

Proof Let $X = \operatorname{Proj} S_\bullet$ and $N_\bullet := \Gamma_\bullet \mathscr{F}$. It suffices to show that $\tau_{D_+(f)} : ((N_\bullet)_f)_0 \to \Gamma(D_+(f), \mathscr{F})$ is an isomorphism where $f \in S_d$. Let $\mathscr{L} := \mathscr{O}_X(d)$, as $S_\bullet$ is generated in degree 1, by Proposition 16.1.7, $\mathscr{L}$ is an invertible sheaf. Say $\eta_d : S_d \to \Gamma(X, \mathscr{O}_X(d))$, then $s := \eta_d(f)$ is a section of $\Gamma(X, \mathscr{O}_X(d))$ and the nonvanishing locus of s is precisely $X_s = D_+(f)$. By Generalized Qcqs Lemma 16.4.2, we obtain an isomorphism

$$\left(\left(\bigoplus_{r\geq 0}\Gamma(X,\mathscr{F}(rd))\right)_s\right)_0 \xrightarrow{\sim} \Gamma(D_+(f),\mathscr{F}). \tag{16.33}$$

We first show the surjectivity of $\tau_{D_+(f)}$. Let $u \in \Gamma(D_+(f),\mathscr{F})$, by (16.33), there exists an integer $n \geq 0$ and a section $z \in \Gamma(X,\mathscr{F}(nd))$ such that

$$u = z|_{D_+(f)} \cdot (s^n|_{D_+(f)})^{-1}.$$

Since $\eta : S_\bullet \to \Gamma_\bullet \widetilde{S_\bullet}$ is a ring morphism (Lemma 16.7.3), $s^n = \eta_d(f)^n = \eta_{nd}(f^n)$, hence,

$$u = z|_{D_+(f)} \cdot (\eta_{nd}(f^n)|_{D_+(f)})^{-1} = \tau_{D_+(f)}(\tfrac{z}{f^n}),$$

which implies that $\tau_{D_+(f)}$ is surjective.

We next show the injectivity of $\tau_{D_+(f)}$. Suppose $\frac{x}{f^n} \in ((N_\bullet)_f)_0$ such that

$$\tau_{D_+(f)}(\tfrac{x}{f^n}) = x|_{D_+(f)} \cdot (\eta_{nd}(f^n)|_{D_+(f)})^{-1} = 0.$$

Since $\eta_{nd}(f^n)|_{D_+(f)}$ is an invertible section of $\mathscr{O}_X(nd)|_{D_+(f)}$, multiplication by its inverse is an isomorphism $\mathscr{F}(nd)|_{D_+(f)} \xrightarrow{\sim} \mathscr{F}|_{D_+(f)}$. Hence, $x|_{D_+(f)} = 0$ in $\Gamma(D_+(f),\mathscr{F}(nd))$. Apply Generalized Qcqs Lemma 16.4.2, there is an isomorphism

$$\left(\left(\bigoplus_{r\geq 0}\Gamma(X,\mathscr{F}((n+r)d))\right)_s\right)_0 \xrightarrow{\sim} \Gamma(D_+(f),\mathscr{F}(nd)). \tag{16.34}$$

Under this map, $\frac{x}{1}$ is sent to $x|_{D_+(f)}$, since (16.34) is injective, $\frac{x}{1} = 0$ in the localized module. Then there exists some $r \geq 0$ such that $s^r \cdot x = \eta_{rd}(f^r)\cdot x = 0$ in $\Gamma(X,\mathscr{F}((n+r)d)) = N_{(n+r)d}$. Hence, in $((N_\bullet)_f)_0$, $\frac{x}{f^n} = 0$, which implies that $\tau_{D_+(f)}$ is injective. □

Recall Corollary 15.6.1, the category of graded $S_\bullet$ is not equivalence to the category $\mathbf{QCoh}_{\operatorname{Proj} S_\bullet}$. But if we restrict to a subcategory of the category of graded $S_\bullet$-modules (and to the case that $X = \operatorname{Proj} S_\bullet$ is quasicompact), we obtain an equivalence.

Definition 16.7.1 (Saturated $S_\bullet$-modules)

*We call a graded $S_\bullet$-module $M_\bullet$ **saturated** if the saturation map $\eta : M_\bullet \to \Gamma_\bullet \widetilde{M_\bullet}$ is an isomorphism. Denote $\mathbf{GrMod}^{\mathrm{sat}}_{S_\bullet}$ be the full subcategory of saturated graded $S_\bullet$-modules.*

Remark 16.30 This "saturation" map $M_\bullet \to \Gamma_\bullet \widetilde{M_\bullet}$ is analogous to the sheafification map, taking presheaves to sheaves. For example, the saturation of the saturation equals the saturation.

Remark 16.31 See the end of §15.6.1 for another description of the category of saturated $S_\bullet$-modules.

Theorem 16.7.1

Let $S_\bullet$ be a finitely generated graded ring generated in degree 1 and let $X = \operatorname{Proj} S_\bullet$. Then the functors give an equivalence of categories:

$$\mathbf{GrMod}^{\mathrm{sat}}_{S_\bullet} \underset{\Gamma_\bullet\mathscr{F} \leftarrow\!\shortmid \mathscr{F}}{\overset{M\mapsto \widetilde{M}}{\rightleftarrows}} \mathbf{QCoh}_X\,.$$

Proof Let $M_\bullet \in \mathbf{GrMod}^{\mathrm{sat}}_{S_\bullet}$, as $M_\bullet$ is saturated, $M_\bullet$ is isomorphic to $\Gamma_\bullet \widetilde{M_\bullet}$. By Lemma 16.7.4, $\tau : (\Gamma_\bullet\mathscr{F})^\sim \to \mathscr{F}$ is an isomorphism for all quasicoherent $\mathscr{O}_X$-modules $\mathscr{F}$. It remains to show that $M_\bullet := \Gamma_\bullet\mathscr{F}$ is a saturated $S_\bullet$-module. But it is immediate that the composition

$$M_\bullet \xrightarrow{\eta^M} \Gamma_\bullet(\widetilde{M_\bullet}) \xrightarrow{\Gamma_\bullet(\tau)} M_\bullet$$

is the identity. More precisely, for $x \in M_{dn} = \Gamma(X, \mathscr{F}(dn))$, let $f \in S_d$ with $d > 0$. On $D_+(f)$ (for simply, we denote $f := \eta^{S_\bullet}(f)$), the saturation map gives

$$\eta^{M_\bullet}_{dn}(x)|_{D_+(f)} = \frac{x}{1} = \frac{x}{f^n} \cdot f^n \in \Gamma(D_+(f), \widetilde{M_\bullet}(dn)).$$

By the definition of $\tau^{\widetilde{M_\bullet}}$,

$$\tau^{\widetilde{M_\bullet}}_{D_+(f)}(\tfrac{x}{f^n}) = x|_{D_+(f)} \cdot f^{-n}.$$

Therefore,

$$(\Gamma_\bullet(\tau) \circ \eta^{M_\bullet})(x)|_{D_+(f)} = \tau_{D_+(f)}(\tfrac{x}{f^n}) \cdot f^n = x|_{D_+(f)}.$$

Since $D_+(f)$ cover X, it follows that

$$\Gamma_\bullet(\tau) \circ \eta^{M_\bullet} = \mathrm{id}_{M_\bullet}.$$

As we have already seen that $\Gamma_\bullet(\tau)$ is an isomorphism, η is an isomorphism. □

Proposition 16.7.2 (Converse to Proposition 10.3.1)

Each closed subscheme of $\operatorname{Proj} S_\bullet$ *arise from a homogeneous ideal* $I_\bullet \subseteq S_\bullet$.

Proof Let $X = \operatorname{Proj} S_\bullet$ and let $i : Z \hookrightarrow X$ be a closed subscheme. Denoted by $\mathscr{I}_Z$ is the ideal sheaf, then there is an exact sequence

$$0 \longrightarrow \mathscr{I}_Z \longrightarrow \mathscr{O}_X \xrightarrow{q} \mathscr{O}_Z \longrightarrow 0. \tag{16.35}$$

For each $n \in \mathbb{Z}$, as tensoring by a locally free sheaf is exact (Lemma 15.1.5), we obtain an exact sequence

$$0 \longrightarrow \mathscr{I}_Z(n) \longrightarrow \mathscr{O}_X(n) \xrightarrow{q(n)} \mathscr{O}_Z(n) \longrightarrow 0. \tag{16.36}$$

Recall taking global section is left-exact (Proposition 3.6.7), we obtain an exact sequence of graded $S_\bullet$-modules

$$0 \longrightarrow \Gamma_\bullet \mathscr{I}_Z \longrightarrow \Gamma_\bullet \mathscr{O}_X \xrightarrow{\Gamma_\bullet(q)} \Gamma_\bullet \mathscr{O}_Z \tag{16.37}$$

Consider the following diagram,

$$\begin{tikzcd}
0 \arrow[r] & \Gamma_\bullet \mathscr{I}_Z \arrow[r] & \Gamma_\bullet \mathscr{O}_X \arrow[r, "\Gamma_\bullet(q)"] & \Gamma_\bullet \mathscr{O}_Z \\
& & S_\bullet \arrow[u, "\eta^{S_\bullet}"] \arrow[ur, "\varphi:=\Gamma_\bullet(q)\circ\eta^{S_\bullet}"'] & \\
& I_\bullet := \operatorname{Ker}\varphi \arrow[ur] & & \\
0 \arrow[ur] & & &
\end{tikzcd} \tag{16.38}$$

since φ is graded, $I_\bullet \subseteq S_\bullet$ is a homogeneous ideal. By Lemma 15.6.2, sheafification $\widetilde{\square}$ is exact, so from (16.38), we obtain an exact sequence

$$0 \longrightarrow \widetilde{I_\bullet} \longrightarrow \widetilde{S_\bullet} \xrightarrow{\widetilde{\varphi}} (\Gamma_\bullet \mathscr{O}_Z)^\sim. \tag{16.39}$$

In fact, we have canonical isomorphisms $\mathscr{O}_X \cong \widetilde{S_\bullet}$ and $(\Gamma_\bullet \mathscr{O}_Z)^\sim \cong \mathscr{O}_Z$, and thus we have an exact sequence

$$0 \longrightarrow \widetilde{I_\bullet} \longrightarrow \mathscr{O}_X \xrightarrow{\widetilde{\varphi}} \mathscr{O}_Z.$$

Therefore, $\mathrm{Ker}(\widetilde{\varphi}) \cong \widetilde{I_\bullet}$ By Theorem 16.7.1, we have a commutative diagram,

$$\begin{tikzcd} \widetilde{\varphi}: & \widetilde{S_\bullet} \arrow[r, "\sim"] & (\Gamma_\bullet \mathscr{O}_X)^\sim \arrow[r, "\widetilde{\Gamma_\bullet(q)}"] \arrow[d, "\sim"'] & (\Gamma_\bullet \mathscr{O}_Z)^\sim \arrow[d, "\sim"] \\ & & \mathscr{O}_X \arrow[r, "q"] & \mathscr{O}_Z \end{tikzcd}$$

hence, $\mathscr{I}_Z = \mathrm{Ker}\, q \cong \mathrm{Ker}\, \widetilde{\varphi} = \widetilde{I_\bullet}$. As isomorphic ideal sheaves give the isomorphism of closed subscheme, we have $Z \cong \mathrm{Proj}(S_\bullet/I_\bullet)$. Therefore every closed subscheme of $\mathrm{Proj}\, S_\bullet$ arises from a homogeneous ideal of $S_\bullet$. □

Remark 16.32 Proposition 16.7.2 fulfills promises made in Proposition 16.7.2 and Exercise **??**.

For the first time, we see that every closed subscheme of a projective scheme is cut out by homogeneous equations. This is the analog of the fact that every closed subscheme of an affine scheme is cut out by equations. It is disturbing that it is so hard to prove this fact. (For comparison, this was easy on the level of the Zariski topology — see Proposition 5.5.7.)

Theorem 16.7.2 ($\Gamma_\bullet$ and $\widetilde{\square}$ are adjoint functors)

Suppose $S_\bullet$ is a finitely generated graded ring generated in degree 1. $M_\bullet$ is a graded $S_\bullet$-module and $\mathscr{F}$ is a quasicoherent sheaf on $\mathrm{Proj}\, S_\bullet$. Then there is a natural bijection

$$\mathrm{Hom}(\widetilde{M_\bullet}, \mathscr{F}) \overset{\sim}{\longleftrightarrow} \mathrm{Hom}(M_\bullet, \Gamma_\bullet \mathscr{F}).$$

That is, $\widetilde{\square} \dashv \Gamma_\bullet$ form an adjoint pair.

Proof Let $X = \mathrm{Proj}\, S_\bullet$. For every homogeneous $f \in S_+$, write $U_f := D_+(f)$. We first construct a map

$$\mathrm{Hom}(M_\bullet, \Gamma_\bullet \mathscr{F}) \longrightarrow \{\text{compatible families } (\varphi_f : ((M_\bullet)_f)_0 \to ((\Gamma_\bullet \mathscr{F})_f)_0)_{f \in S_+}\},$$

where "compatible families" means if $D_+(g) \subseteq D_+(f)$, there is a commutative diagram

$$\begin{tikzcd} ((M_\bullet)_f)_0 \arrow[r] \arrow[d] & ((\Gamma_\bullet \mathscr{F})_f)_0 \arrow[d] \\ ((M_\bullet)_g)_0 \arrow[r] & ((\Gamma_\bullet \mathscr{F})_g)_0. \end{tikzcd}$$

Let $\varphi : M_\bullet \to \Gamma_\bullet \mathscr{F}$ be a graded $S_\bullet$-module homomorphism. Localizing at f and taking the degree 0 part gives

$$\varphi_f : ((M_\bullet)_f)_0 \longrightarrow ((\Gamma_\bullet \mathscr{F})_f)_0.$$

Since localization is functorial, these maps commute with all restriction maps. Hence, $(\varphi_f)_{f \in S_+}$ is compatible. Conversely, suppose that a compatible family $(\varphi_f : ((M_\bullet)_f)_0 \to ((\Gamma_\bullet \mathscr{F})_f)_0)_{f \in S_+}$ is given. Note that $\Gamma(U_f, \widetilde{M_\bullet}) = ((M_\bullet)_f)_0$ and $\Gamma(U_f, (\Gamma_\bullet \mathscr{F})^\sim) = ((\Gamma_\bullet \mathscr{F})_f)_0$, each φ_f can be regarded as

$$\varphi_f : \Gamma(U_f, \widetilde{M_\bullet}) \longrightarrow \Gamma(U_f, (\Gamma_\bullet \mathscr{F})^\sim).$$

On an intersection $U_f \cap U_g = U_{fg}$, the restriction of φ_f and φ_g to U_{fg} are both equal to φ_{fg}. Therefore, $(\varphi_f)_{f \in S_+}$ glue to a unique morphism of sheaves $\varphi : \widetilde{M_\bullet} \to (\Gamma_\bullet \mathscr{F})^\sim$. Apply $\Gamma_\bullet$ to φ, since $\Gamma_\bullet \mathscr{F}$ is saturated, we obtain

$$\Gamma_\bullet(\varphi) : \Gamma_\bullet(\widetilde{M_\bullet}) \longrightarrow \Gamma_\bullet((\Gamma_\bullet \mathscr{F})^\sim) \cong \Gamma_\bullet \mathscr{F}.$$

Define Φ be the composition of $M_\bullet \xrightarrow{\eta^{M_\bullet}} \Gamma_\bullet(\widetilde{M_\bullet}) \xrightarrow{\Gamma_\bullet(\varphi)} \Gamma_\bullet \mathscr{F}$. We claim that the localizations of Φ are precisely the originally given maps φ_f. Pick $\frac{x}{f^n} \in ((M_\bullet)_f)_0$, then

$$\Phi_f(\tfrac{x}{f^n}) = \frac{\Phi(x)}{f^n} = \frac{(\Gamma_\bullet(\varphi) \circ \eta^{M_\bullet})(x)}{f^n} = \frac{\varphi_X(x)}{f^n} = \varphi_f(\tfrac{x}{f^n}).$$

Thus we have a bijection

$$\operatorname{Hom}(M_\bullet, \Gamma_\bullet \mathscr{F}) \xrightarrow{\sim} \{\text{compatible families } (\varphi_f : ((M_\bullet)_f)_0 \to ((\Gamma_\bullet \mathscr{F})_f)_0)_{f \in S_+}\}.$$

Recall $\tau : (\Gamma_\bullet \mathscr{F})^\sim \to \mathscr{F}$ is an isomorphism (Lemma 16.7.4), it gives isomorphism on each $U_f = D_+(f)$,

$$\tau_{U_f} : \Gamma(U_f, (\Gamma_\bullet \mathscr{F})^\sim) \xrightarrow{\sim} \Gamma(U_f, \mathscr{F}).$$

Hence, we obtain a bijection

$$\begin{gathered}\{\text{compatible families } (((M_\bullet)_f)_0 \to ((\Gamma_\bullet \mathscr{F})_f)_0)_{f \in S_+}\} \\ \big\downarrow{\scriptstyle\sim} \\ \{\text{compatible families } (\Gamma(U_f, \widetilde{M_\bullet}) \to \Gamma(U_f, \mathscr{F}))_{f \in S_+}\}.\end{gathered}$$

In fact, there is a bijection

$$\operatorname{Hom}(\widetilde{M_\bullet}, \mathscr{F}) \xrightarrow{\sim} \{\text{compatible families } (\Gamma(U_f, \widetilde{M_\bullet}) \to \Gamma(U_f, \mathscr{F}))_{f \in S_+}\},$$

hence, there is a natural bijection

$$\operatorname{Hom}(\widetilde{M_\bullet}, \mathscr{F}) \xleftrightarrow{\sim} \operatorname{Hom}(M_\bullet, \Gamma_\bullet \mathscr{F}).$$

$\square$

Chapter 17 Maps to projective space, and properties of line bundles

Line bundles on a variety X are important in part because they can tell a lot about maps from X to other projective space (§16.2). The crucial concept of "ampleness" of a line bundle should be seen as some sort of "positivity" (§**??**), which will help us know when the line bundle has lots of sections and might even give a map to projective space. The fundamental facts of ampleness will take some work, but will be powerfully useful from then on. For example, in §**??**, we will see many applications to curves.

17.1 Globally generated quasicoherent sheaf

We begin with a result letting us "get at" any finite type quasicoherent sheaf on a projective scheme in terms of line bundles. Throughout this section, $S_\bullet$ will be a finitely generated graded ring over A, generated in degree 1. We will prove the following result.

Theorem 17.1.1

Any finite type quasicoherent sheaf $\mathscr{F}$ on $\operatorname{Proj} S_\bullet$ can be presented in the form

$$\bigoplus\nolimits_{\textit{finite}} \mathscr{O}(-m) \longrightarrow \mathscr{F} \longrightarrow 0.$$

Because we can work with the line bundles $\mathscr{O}(-m)$ in a hands-on way, this result will give us great control over all coherent sheaves (and, in particular, vector bundles) on $\operatorname{Proj} S_\bullet$. As just a first example, it will allow us to show that every coherent sheaf on a projective k-scheme has a finite-dimensional space of global sections (Corollary **??**). (This fact will grow up to be the fact that the higher pushforwards of coherent sheaves under proper morphisms are also coherent; see Theorem **??** and Grothendieck's Coherence Theorem **??**.)

Rather than proceeding directly to a proof, we use this as an excuse to introduce notions that are useful in wider circumstances (*global generation*, *base-point-freeness*, *ampleness*), and their interrelationships. But first we use it as an excuse to mention an important classical result.

17.1.1 The Hilbert Syzygy Theorem

Given any coherent sheaf $\mathscr{F}$ on $\mathbb{P}^n_k$, Theorem 17.1.1 gives the existence of a surjection $\varphi : \bigoplus_{\text{finite}} \mathscr{O}(-m) \to \mathscr{F} \to 0$. The kernel of the surjection is also coherent (as $\mathbb{P}^n_k$ is Noetherian, recall Proposition 7.4.2, $\mathbf{Coh}_X$ is an Abelian category), so iterating this construction, we can construct an infinite resolution of $\mathscr{F}$ by a direct sum of line bundles:

$$\cdots \longrightarrow \bigoplus\nolimits_{\text{finite}} \mathscr{O}(m_{2,j}) \longrightarrow \bigoplus\nolimits_{\text{finite}} \mathscr{O}(m_{1,j}) \longrightarrow \bigoplus\nolimits_{\text{finite}} \mathscr{O}(m_{0,j}) \longrightarrow \mathscr{F} \longrightarrow 0.$$

Theorem 17.1.2 (The Hilbert Syzygy Theorem)

Let $\mathscr{F}$ be a coherent sheaf on $\mathbb{P}^n_k$. Then there exists a finite resolution of $\mathscr{F}$ by direct sums of twisting sheaves:

$$0 \longrightarrow \bigoplus_{j=1}^{r_\ell} \mathscr{O}(m_{\ell,j}) \longrightarrow \cdots \longrightarrow \bigoplus_{j=1}^{r_1} \mathscr{O}(m_{1,j}) \longrightarrow \bigoplus_{j=1}^{r_0} \mathscr{O}(m_{0,j}) \longrightarrow \mathscr{F} \longrightarrow 0,$$

where $\ell \le n+1$, each r_i is finite, and every $m_{i,j}$ is an integer.

Remark 17.1 See the comments after Theorem 4.6.2 for the original history of this result. We won't use the theorem, but a proof will be given in the optional section §**??**.

17.1.2 Globally generated sheaves

Definition 17.1.1 (Globally generated sheaf)

Suppose X is a scheme and $\mathscr{F}$ is an $\mathscr{O}_X$-module. We say that $\mathscr{F}$ is ***globally generated (or generated by global sections)*** *if it admits a surjection from a free sheaf on X:*

$$\mathscr{O}_X^{\oplus I} \twoheadrightarrow \mathscr{F}.$$

Here I is some index set. The global sections in the question are the images of the $|I|$ sections coresponding to 1 in the various summands of $\mathscr{O}_X^{\oplus I}$; those images generate the stalks of $\mathscr{F}$. We say $\mathscr{F}$ is ***finitely globally generated (or generated by a finite number of global sections)*** *if the index set I is finite.*
More definitions in more detail: We say that $\mathscr{F}$ is ***globally generated at a point*** *p* ***(or some times generated by globel sections at*** *p) if we can find $\varphi : \mathscr{O}_X^{\oplus I} \to \mathscr{F}$ that is surjective on stalks at p,*

$$\mathscr{O}_{X,p}^{\oplus I} \xrightarrow{\varphi_p} \mathscr{F}_p.$$

(It would be more precise to say that the stalk of $\mathscr{F}$ at p is generated by global sections of $\mathscr{F}$.) The key insight is that global generation at p means that every germ at p is a linear combination (over the local ring $\mathscr{O}_{X,p}$) of germs of global sections.

Remark 17.2 Note that $\mathscr{F}$ is *globally generated* if it is globally generated at all points p. (Reason: Proposition 3.4.13.) Notice that we can take a single index set for all of X, by taking the union of all the index sets for each p.

Lemma 17.1.1

An invertible sheaf $\mathscr{L}$ on X is globally generated if and only if for any point $p \in X$, there is a global section of $\mathscr{L}$ not vanishing at p, i.e., $\mathscr{L}$ is base-point-free (Definition 16.2.3).

Proof If an invertible sheaf $\mathscr{L}$ on X is globally generated, then for any point $p \in X$ we have a surjection of $\mathscr{O}_{X,p}$-modules $\mathscr{O}_{X,p}^{\oplus I} \twoheadrightarrow \mathscr{L}_p$. Tensoring $\kappa(p)$ to $\mathscr{O}_{X,p}^{\oplus I} \twoheadrightarrow \mathscr{L}_p$ both side, we obtain a surjective

$$\varphi_p : \kappa(p)^{\oplus I} \twoheadrightarrow \mathscr{L}_p \otimes_{\mathscr{O}_{X,p}} \kappa(p).$$

Since $\mathscr{L}_p \otimes_{\mathscr{O}_{X,p}} \kappa(p)$ is a one-dimensional vector space, by the sujectivity, φ_p is not a zero map. Note that the composition

$$\mathscr{O}_X \xrightarrow{\iota_i} \mathscr{O}_X^{\oplus I} \xrightarrow{\varphi} \mathscr{L}$$

corresponds to a global section $s_i := \varphi(\iota_i(1)) \in \Gamma(X, \mathscr{L})$. As φ_p is not a zero map, there exists i such that $s_i(p) \neq 0$, it follows that $\mathscr{L}$ is base-point-free.

Conversely, suppose that for every point $p \in X$, there exists a global section $s \in \Gamma(X, \mathscr{L})$ such that $s(p) \neq 0$. Let $I := \Gamma(X, \mathscr{L})$ be the set of all global sections of $\mathscr{L}$. Define the evaluation morphisms

$$\mathrm{ev} : \mathscr{O}_X^{\otimes I} \longrightarrow \mathscr{L}$$

by sending the distinguished section 1 in the summand indexed by s to the section s. At each point $p \in X$, as $s(p) \neq 0$ for som $s \in \Gamma(X, \mathscr{L})$, s_p generates the rank-one free module $\mathscr{L}_p$, hence, the map

$$\mathrm{ev}_p : \mathscr{O}_{X,p}^{\otimes I} \longrightarrow \mathscr{L}_p$$

is surjective. By Proposition 3.4.13, we obtain that

$$\mathrm{ev} : \mathscr{O}_X^{\otimes I} \longrightarrow \mathscr{L}$$

is surjective, which implies that $\mathscr{L}$ is globally generated. □

Lemma 17.1.2

(1) Every quasicoherent sheaf on every affine scheme is globally generated.

(2) Every finite type quasicoherent sheaf on every affine scheme is generated by a finite number of global sections.

Proof **For (1).** Let $\mathscr{F}$ be a quasicoherent sheaf on $X = \operatorname{Spec} A$, then $\mathscr{F} \cong \widetilde{M}$ where M is an A-module. Note that for any A-module M, there is a surjection $A^{\oplus I} \twoheadrightarrow M$ where I is an index set. Recall Corollary 5.1.2, sheafification $\widetilde{\square}$ gives a surjection

$$\mathscr{O}_X^{\oplus I} \longrightarrow\!\!\!\!\rightarrow \mathscr{F},$$

which implies that $\mathscr{F}$ is globally generated.

For (2). Let $\mathscr{F}$ be a finite type quasicoherent sheaf on $X = \operatorname{Spec} A$. Similar to (1), now, M is finitely generated A-module, the index set I can be chosen finite. Hence, $\mathscr{F}$ is generated by finite number of global sections. □

Clearly, if $\mathscr{F}$ and $\mathscr{G}$ are globally generated, then so is $\mathscr{F} \oplus \mathscr{G}$ (and similarly for finitely generated, generated at a point p, etc). Similarly for tensor product:

Lemma 17.1.3 (Globally generated $\otimes$ globally generated is globally generated)

If quasicoherent sheaves $\mathscr{F}$ and $\mathscr{G}$ are globally generated at a point p, then so is $\mathscr{F} \otimes \mathscr{G}$.

Remark 17.3 If $\mathscr{F}$ and $\mathscr{G}$ are invertible sheaves, this is Proposition 16.2.2.

Proof Since $\mathscr{F}$ is globally generated at point p, we find $\mathscr{O}_X^{\oplus I} \to \mathscr{F}$ with an exact sequence of $\mathscr{O}_{X,p}$-modules,

$$\mathscr{O}_{X,p}^{\oplus I} \longrightarrow\!\!\!\!\rightarrow \mathscr{F}_p \longrightarrow 0$$

where I is some index set. As tensor product is right exact, by applying $\square \otimes_{\mathscr{O}_{X,p}} \mathscr{G}_p$, we obtain an exact sequence

$$\mathscr{O}_{X,p}^{\oplus I} \otimes_{\mathscr{O}_{X,p}} \mathscr{G}_p \longrightarrow\!\!\!\!\rightarrow \mathscr{F}_p \otimes_{\mathscr{O}_{X,p}} \mathscr{G}_p \longrightarrow 0,$$

which implies that $\mathscr{O}_{X,p}^{\oplus I} \otimes_{\mathscr{O}_{X,p}} \mathscr{G}_p \twoheadrightarrow \mathscr{F} \otimes_{\mathscr{O}_{X,p}} \mathscr{G}_p$ is surjective. Similarly, we have a surjective

$$\mathscr{O}_{X,p}^{\oplus I} \otimes_{\mathscr{O}_{X,p}} \mathscr{O}_{X,p}^{\oplus J} \longrightarrow\!\!\!\!\rightarrow \mathscr{O}_{X,p}^{\oplus I} \otimes_{\mathscr{O}_{X,p}} \mathscr{G}_p.$$

By composition, we have a surjective

$$\mathscr{O}_{X,p}^{\oplus (I\times J)} \cong \mathscr{O}_{X,p}^{\oplus I} \otimes_{\mathscr{O}_{X,p}} \mathscr{O}_{X,p}^{\oplus J} \longrightarrow\!\!\!\!\rightarrow \mathscr{O}_{X,p}^{\oplus I} \otimes_{\mathscr{O}_{X,p}} \mathscr{G}_p \longrightarrow\!\!\!\!\rightarrow \mathscr{F}_p \otimes_{\mathscr{O}_{X,p}} \mathscr{G}_p,$$

which implies that $\mathscr{F} \otimes \mathscr{G}$ is generated at point p. □

Lemma 17.1.4

Suppose $\mathscr{F}$ is a finite type quasicoherent sheaf on X.

(a) $\mathscr{F}$ is globally generated at p if and only if "the fiber of $\mathscr{F}$ is generated by global sections at p", i.e., the global sections span the fiber $\mathscr{F}_p/\mathfrak{m}\mathscr{F}_p$ as a verctor space, where $\mathfrak{m}$ is the maximal ideal of $\mathscr{O}_{X,p}$.

(b) If $\mathscr{F}$ is globally generated at p, then "$\mathscr{F}$ is globally generated near p": there is an open neigh-

borhood U of p such that $\mathscr{F}$ is globally generated at every point of U.

(c) *Suppose further that X is a quasicompact scheme. If $\mathscr{F}$ is globally generated at all closed point of X, then $\mathscr{F}$ is globally generated at all poins of X.*

Proof ***For (a).*** Suppose $\mathscr{F}$ is globally generated at p, then we can find $\mathscr{O}_X^{\oplus I} \to \mathscr{F}$ such that $\mathscr{O}_{X,p}^{\oplus I} \twoheadrightarrow \mathscr{F}_p$ is a surjection. Since $\mathscr{F}$ is a finite type quasicoherent sheaf on X, the index set I can be chosen finite. Tensoring $\kappa(p)$ bothside, as tesoring an $\mathscr{O}_{X,p}$-module is right exact, we obtain a surjection $\kappa(p)^{\oplus I} \twoheadrightarrow \mathscr{F}|_p$, which implies that the global sections span the fiber $\mathscr{F}|_p$ as a verctor space.

Conversely, if $\mathscr{F}|_p = \mathrm{Span}\{s_i|_p\}_{i\in I}$ where I is finite and $s_i|_p$ is the image of $s_i \in \Gamma(X, \mathscr{F})$ in $\kappa(p)$-vector space $\mathscr{F}|_p$. By Geometric Nakayama 15.3.4, there exist an affine open neighborhood $\operatorname{Spec} A \subseteq X$ of p such that $\{s_{i,p}\}_{i\in I}$ generated the stalk $\mathscr{F}_p$. Define $\mathscr{O}_X^{\oplus I} \to \mathscr{F}$ by setting $e_i \mapsto s_i$, then at the stalk-level, we have a surjection $\mathscr{O}_{X,p}^{\oplus I} \twoheadrightarrow \mathscr{F}_p$. Hence, $\mathscr{F}$ is globally generated at p.

For (b). By part (a) and Geometric Nakayama 15.3.4, we conclude the consequence.

For (c). Let

$$G(\mathscr{F}) := \{p \in X : \mathscr{F} \text{ is globally generated at } p\},$$

by part (b), we know that $G(\mathscr{F})$ is open in X. Let $Z = X \setminus G(\mathscr{F})$ be a closed subset of X. Suppose $Z \neq \varnothing$. Endow Z with the reduced closed subscheme structure. Since X is quasicompact, Z is also quasicompact. By Proposition 6.1.5, Z contains a closed point of X. But $\mathscr{F}$ is globally generated at all closed points of X, a contradiction! Hence, $Z = \varnothing$, i.e., $G(\mathscr{F}) = X$. □

Lemma 17.1.5

If $\mathscr{F}$ is a finite type quasicoherent sheaf on X, and X is quasicompact, then $\mathscr{F}$ is globally generated if and only if it is generated by a finite number of global sections.

Proof If $\mathscr{F}$ is generated by a finite number of global sections, it is clearly globally generated.

Conversely, if $\mathscr{F}$ is globally generated. Since $\mathscr{F}$ is finite type quasicoherent sheaf, for each point $p \in X$, $\mathscr{F}_p$ is a finitely generated $\mathscr{O}_{X,p}$-module. Since $\mathscr{F}$ is globally generated, let $s_{1,p}, \ldots, s_{n,p}$ be the generators of $\mathscr{F}_p$ where each $s_{i,p}$ is the image of $s_i \in \Gamma(X, \mathscr{F})$ in the stalk $\mathscr{F}_p$. Hence, $s_1|_p, \ldots, s_n|_p$ generate the fiber $\mathscr{F}|_p$, by Geometric Nakayama 15.3.4, there is an affine open neighborhood $U_p \subseteq X$ of p such that for any $q \in U_p$, $s_1, \ldots, s_n$ generate the stalk $\mathscr{F}_q$ as an $\mathscr{O}_{X,q}$-module. Note that $X = \bigcup_{p\in X} U_p$, as X is quasicompact, we may assume that $X = \bigcup_{i=1}^m U_{p_i}$. Hence, by above discussion, we may choose finite global sections to generate the stalk of $\mathscr{F}$ at any point of X. It follows that $\mathscr{F}$ is generated by a finite number of global sections. □

We are now able to state a celebrated result of Serre.

Theorem 17.1.3 (Serre's Theorem A)

Suppose $S_\bullet$ is generated in degree 1, and finitely generated over $A = S_0$. Let $\mathscr{F}$ be any finite type quasicoherent sheaf on $\operatorname{Proj} S_\bullet$*. Then there exists some m_0 such that for all $m \geq m_0$, $\mathscr{F}(m)$ is finitely globally generated.*

We could now prove Serre's Theorem A directly, but will continue to use it as an excuse to introduce more ideas; it will be a consequence of Theorem **??**.

Proof of Theorem 17.1.1 assuming Serre's Theorem A 17.1.3

Proof Suppose we have n global sections $s_1, \ldots, s_n$ of $\mathscr{F}(m)$ that generate $\mathscr{F}(m)$. This gives a map

$$\mathscr{O}_{\operatorname{Proj} S_\bullet}^{\oplus n} \longrightarrow \mathscr{F}(m)$$

given by $(f_1, \ldots, f_n) \mapsto f_1 s_1 + \cdots + f_n s_n$ on any open set. Because these global sections generate $\mathscr{F}(m)$, this is surjection. Tensoring with $\mathscr{O}(-m)$ (which is exact, as tensoring with any locally free sheaf is exact, Lemma 15.1.5) gives the desired result. □

17.2 Ample and Very Ample Line Bundles

We now introduce ampleness, a central "positivity" property of line bundles that will prove important for a number of disparate reasons — geometric, algebraic, topological, computational, and more.

Definition 17.2.1 (Very Ample)

*Suppose $\pi : X \to \operatorname{Spec} A$ is a morphism, and $\mathscr{L}$ is an invertible sheaf on X. (The case when A is a field is the one of most immediate interest.) We say that $\mathscr{L}$ is **very ample over** A **or** π**-very ample, or relatively very ample** if $X \cong \operatorname{Proj} S_\bullet$, where $S_\bullet$ is a finitely generated graded ring over A generated in degree 1 (Definition 5.5.4), and $\mathscr{L} \cong \mathscr{O}_{\operatorname{Proj} S_\bullet}(1)$. One often just says **very ample** if the structure morphism is clear from the context.*

Remark 17.4 Note that the existence of a very ample line bundle implies that X is projective.

Proposition 17.2.1 (Equivalent Definition of Very Ample Over A)

Suppose $\pi : X \to \operatorname{Spec} A$ is proper, and $\mathscr{L}$ is an invertible sheaf on X. Then $\mathscr{L}$ is very ample if and only if there exist a finite number of global sections $s_0, \ldots, s_n$ of $\mathscr{L}$, with no common zeros, such that the morphism

$$[s_0 : \cdots : s_n] : X \longrightarrow \mathbb{P}_A^n$$

is a closed embedding.

Proof Suppose $\mathscr{L}$ is very ample, then $X \cong \operatorname{Proj} S_\bullet$ where $S_\bullet$ is a finitely generated graded ring over A generated in degree 1, moreover there is an isomorphism $\varphi : \mathscr{O}_{\operatorname{Proj} S_\bullet}(1) \xrightarrow{\sim} \mathscr{L}$. Let $t_0, \ldots, t_n$ be the generators of $S_\bullet$ where $\deg t_i = 1$. Let $s_i = \varphi(t_i)$. In fact, $s_0, \ldots, s_n$ have no common zeros, since $S_+ = (t_0, \ldots, t_n)$.By Theorem 16.2.1, $s_0, \ldots, s_n$ define a morphisms

$$[s_0 : \cdots : s_n] : X \longrightarrow \mathbb{P}_A^n.$$

By Proposition 16.2.4, $[s_0 : \cdots : s_n] : X \to \mathbb{P}_A^n$ is a closed embedding.

Conversely, by the proof of Theorem 16.2.1, there is a natural isomorphism $\mathscr{L} \cong [s_0 : \cdots : s_n]^* \mathscr{O}_{\mathbb{P}_A^n}(1)$. Say $\mathbb{P}_A^n = \operatorname{Proj} A[t_0, \ldots, t_n]$, as $[s_0 : \cdots : s_n] : X \to \mathbb{P}_A^n$ is a closed embedding, by Proposition 16.7.2, there exists a homogeneous $I_\bullet \subseteq A[t_0, \ldots, t_n]$ such that $X = \operatorname{Proj} A[t_0, \ldots, t_n]/I_\bullet$. Say $S_\bullet = A[t_0, \ldots, t_n]/I_\bullet$, then

$$\mathscr{L} \cong [s_0 : \cdots : s_n]^* \mathscr{O}_{\mathbb{P}_A^n}(1) \cong \mathscr{O}_{\operatorname{Proj} S_\bullet}(1),$$

which implies that $\mathscr{L}$ is very ample. □

Part VI

More cohomological tools

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